MGP/BP/GS3/001

ECONOMIC GROWTH

1. PHASES OF ECONOMIC GROWTH IN INDIA

"Today the country is moving at a fast pace towards becoming the third largest economy." -PM Narendra Modi

India has surpassed Japan to become the **world's fourth-largest economy**. According to NITI Aayog, India is now poised to displace Germany from the third rank by 2027-2028.

Phases of Indian Economic Growth

Phase 1- Pre-Liberalization Period (1947–1991) "Hindu Rate of Growth"

GDP Growth: ~3.5% annually (1950–1990).

Sector Composition change during (1950-51) to (1990-91)

Sector	1950-51 (% of GDP)	1990-91 (% of GDP)
Agriculture	51.9%	29.5%
Industry	11.1%	20.6%
Services	34.6%	49.6%

Key Features of Pre-liberalisation Economic Growth Phasew

- **Planned Economic Framework-** Adoption of Five-Year Plans (starting 1951), with **strong state control (license Raj)**, focus on **self-reliance** and **import substitution** (IS).
- **Agriculture-led Economy-** Agriculture contributed ~50% of GDP (1950s), and employed over 70% of the workforce.
- **Industrial Policy-** Emphasis on heavy industries (steel, machinery, power). Restriction of the private sector to certain areas.
- **Foreign Trade & Exchange Controls-** Exports were only ~5% of GDP (1960s–1980s). High tariffs and import restrictions - "closed economy." Chronic balance-of-payments crises.

Phase 2- Liberalisation Reforms and Growth Take-Off (1991-2003)

GDP Growth: Accelerated to 5–6% annually.

1991 Economic Crisis- Foreign reserves were less than 3 weeks' imports cover, necessitating IMF bailout. India enunciated several economic reforms in 1991.

Economic Reforms of 1991

Industrial Policy Reforms

- Repeal of The **Monopolies & Restrictive Trade Practices (MRTP) Act**
- **Abolition of Industrial licensing** for all projects except in 18 industries.

Trade Policy Reforms

- Freer imports and exports (Regulation of imports by **limited negative list**)
- **Rationalization of tariff structure** and removal of quantitative restrictions (Reduction in import duty in consonance with Chelliah committee recommendations)
- **Establishment of Trading Houses** for promotion of exports.

Promoting Foreign Investment

- Automatic permission for **FDI equity up to 51%** in specified list of high technology and high-investment priority industries
- Establishment of **Foreign Investment Promotion Board (FIPB)** for FDI promotion.

Rationalization of Exchange Rate Policy- Devaluation of rupee by **around 20 percent on account of rising inflation** and making export competitive.

Fiscal Reforms

Major fiscal reforms in the Budget 1991-92

- Reduction in fertilizer subsidy
- Abolition of subsidy on sugar
- Disinvestment in select public sector undertakings
- Acceptance of major recommendations of the Tax Reforms Committee headed by Raja Chelliah.

Monetary and Financial Sector Reforms

- Reduction in **Statutory Liquidity Ratio** (SLR) and the **Cash Reserve Ratio** (CRR) in line with the recommendations of the **Narasimham Committee Report, 1991**.
- Greater competition among public sector, private sector and foreign banks and elimination of administrative constraints.
- Statutory recognition to **Securities & Exchange Board of India (SEBI)** in 1992 on the basis of recommendations of the Narasimham Committee.

Sectoral Shifts after the Liberalisation Reforms

- Rapid growth of the service sector (IT, BPO). IT exports rose from $150mn in 1991 to $12bn in 2003. The service sector has expanded from ~**37-44% of GDP in 1990-91** to around **55% by 2024-2025**.
- The share of the agricultural sector in GDP declined significantly, from **~30% in 1991** to **~16-17%** in **2024-25**.
- The share of the manufacturing and industrial sector to the GDP has remained constant at around **~30% of the GDP**. The manufacturing sector's contribution to **GDP is ~16-17% of the GDP**.

Phase 3- High Growth & Rising Inequality (2003–2011)

GDP Growth: Averaged **8–9% (2003–08),** second only to China.

Sectoral Growth During the Period

- Services contributed to **~57% of GDP**, but employed only **~25% of workers**.

- The manufacturing sector remained stagnant (**~16–17% of GDP**).
- Agriculture shrank to **~15% of GDP**, still employed **~55% of the workforce**.
- There was substantial growth in FDI during the period, as it grew from $4bn (2000) to $45bn (2008).

Rising inequality in Growth

- There was a rapid decline in urban areas. However, the Urban-rural income divide widened.
- States like Gujarat, Maharashtra, Tamil Nadu surged while the BIMARU states (Bihar, MP, Rajasthan, UP) lagged.

Phase 4- Growth Slowdown & Policy Paralysis (2012–2014)

GDP Growth: Fell to ~ 5% (2012–13).

Major Factors of Slowdown

- Policy gridlock (MoEFCC's NOC for environmental projects)
- Corruption scandals (Coal Gate, 2G scam),
- Twin Balance Sheet Problem (Increase in Bank's NPAs).

Phase 5- Post-2014 Growth Divergence under Structural Shocks (2014–2019)

GDP Growth: Initially ~7–8%, but slowed to ~4% in 2019–20.

Major Structural Shocks

- **Demonetization (2016)-** Growth of the informal & rural economy was hardest hit.
- **GST rollout (2017)-** The rollout was aimed at a long-term reform. However, it caused short-term disruption in the Indian economic growth.

Sectoral Growth Story

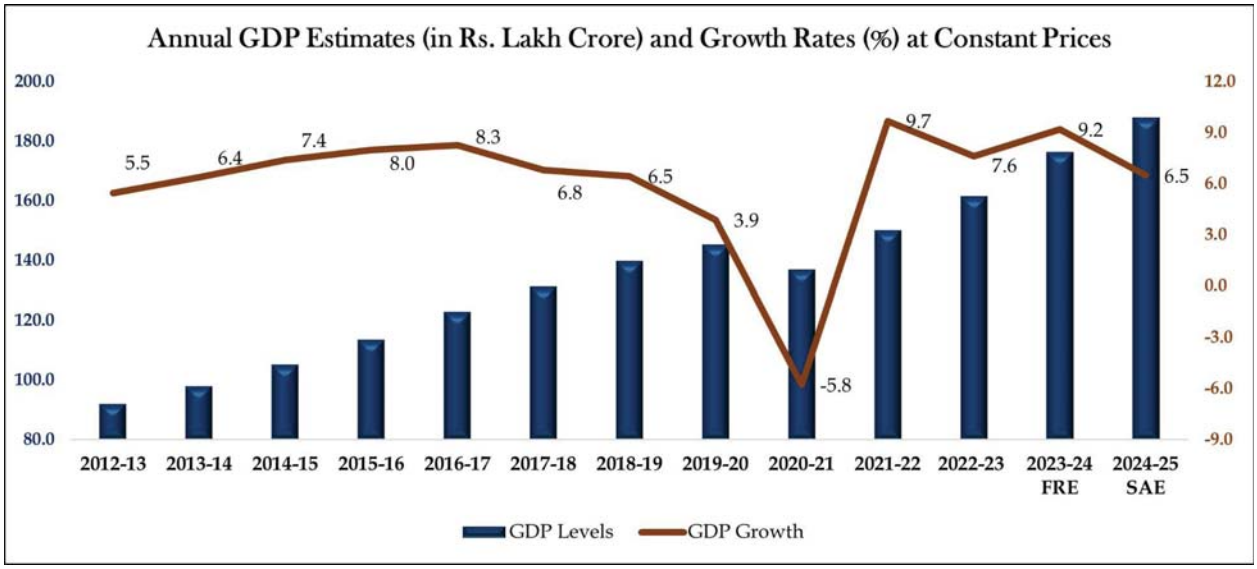
- Services remained strong (IT exports rose to **~$150bn by 2019**).
- Agriculture growth stagnated at **~2–3% annual growth**.
- Manufacturing was stuck at **~16–17% of GDP**

"Missing Middle" phenomenon has been observed in the manufacturing sector due to **scarcity of medium-sized enterprises**. The manufacturing landscape **remains dominated by informal**, **low-productivity micro-firms** and **a few large players**, limiting growth and employment.

Phase 6- Pandemic Slowdown & Post-Pandemic Recovery (2020–2022)

GDP Contraction- India's GDP witnessed its worst contraction since Independence (around **-7% in FY 2020–21**). This contraction was primarily caused by the **COVID-19 pandemic** and **subsequent nationwide lockdowns**. However, the GDP recovered in FY 2021-22 and FY 2021-22.

Pandemic Driven Poverty- Pew Research estimated that around **75mn Indians were pushed into poverty as a result of the Pandemic** in 2020.

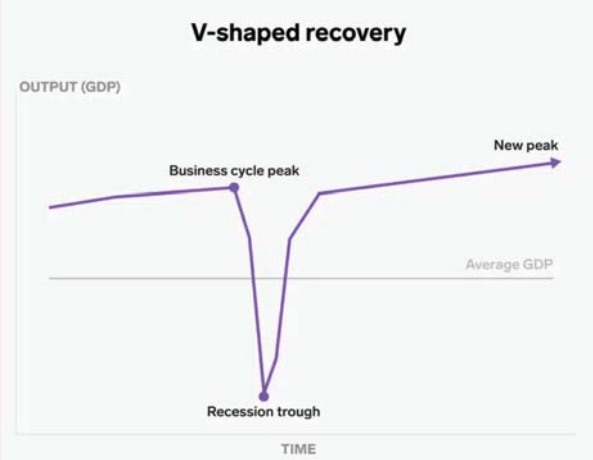


Post-Pandemic Economic Growth Recovery Debate
(V shaped recovery vs K Shaped Recovery)

V Shaped Recovery (On the Basis of Macro-economic Trends)

V-shaped recovery refers to a situation where an economy experiences a sharp decline followed by a rapid and strong recovery, resulting in a V-shaped pattern.

Macro-economic Evidences supporting V-Shaped recovery



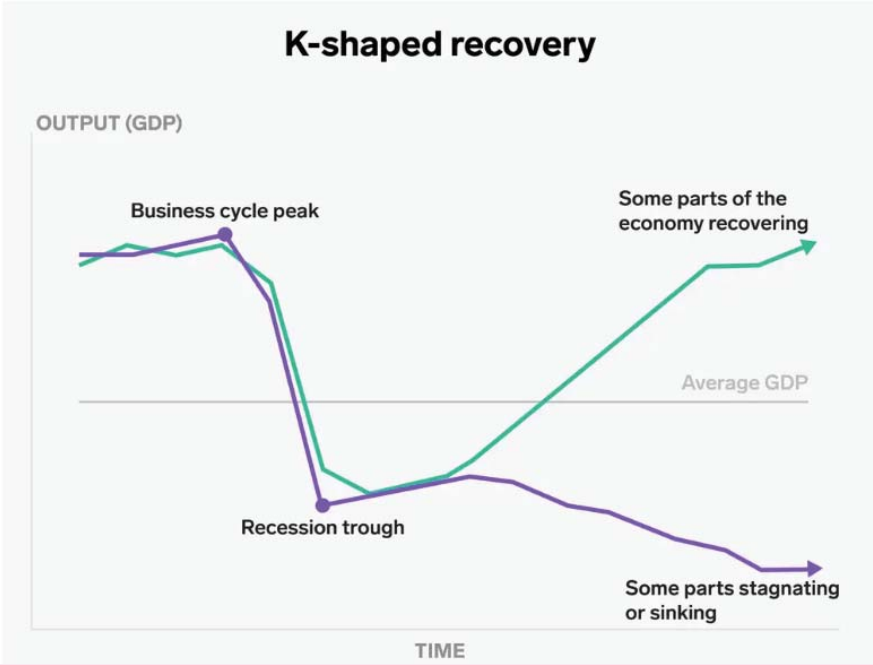
1. **GDP Growth Rebound**- After contraction of 7.3% in FY 2020-21, GDP growth rebounded with ~9% growth in FY 2021-22, followed by ~7% in FY 2022-23.
2. **Rebound in the GST Collections-** The GST collections dropped to a record low of Rs. **32,172 crore in April 2020** following the national lockdown. However, the GST collections hit a record high of **Rs. 1,15,174 crore** by **December 2020**. (This marked the first time collections crossed the ₹1.15 lakh crore mark since the introduction of GST, representing a 12% growth year-on-year).
3. **Rebound of the Index of Industrial Production (IIP)-** The IIP dipped to **-57.3% in April 2020** but **recovered significantly to -1.9%** by November 2020.
4. **Stock Market-** The BSE Sensex fell from over **41,000 in early 2020** to a low of approximately **25,000 in March**. However, the sensex rose to over **47,000 by December 2020**, perfectly tracing a V-shaped path.
5. **Foreign Direct Investment (FDI)-** Despite the pandemic, India attracted its highest-ever total FDI inflow of **$81.72 billion in FY 2020-21**, a 10% increase over the previous year. Net FPI Inflows Recorded an All-Time Monthly High of **$9.8 Billion** in November 2020.
6. **Services led revival-** IT, digital services, and financial sector led the growth revival. Further, there was an increase in Corporate investment and profitability post pandemic.

V-Shaped Economic Recovery was driven by **Mega Vaccination Drive**, **Robust Recovery in the Services Sector** and **Robust Growth in Consumption and Investment**.

However, critics argue that the recovery has been K-shaped, as formal sectors and high-income groups rebounded faster, while informal sectors, small businesses , rural demand lagged behind, worsening inequality.

K- Shaped Recovery

K-shaped Recovery- It is an uneven economic rebound following a recession, where different sectors, industries, or income groups recover at drastically different rates.



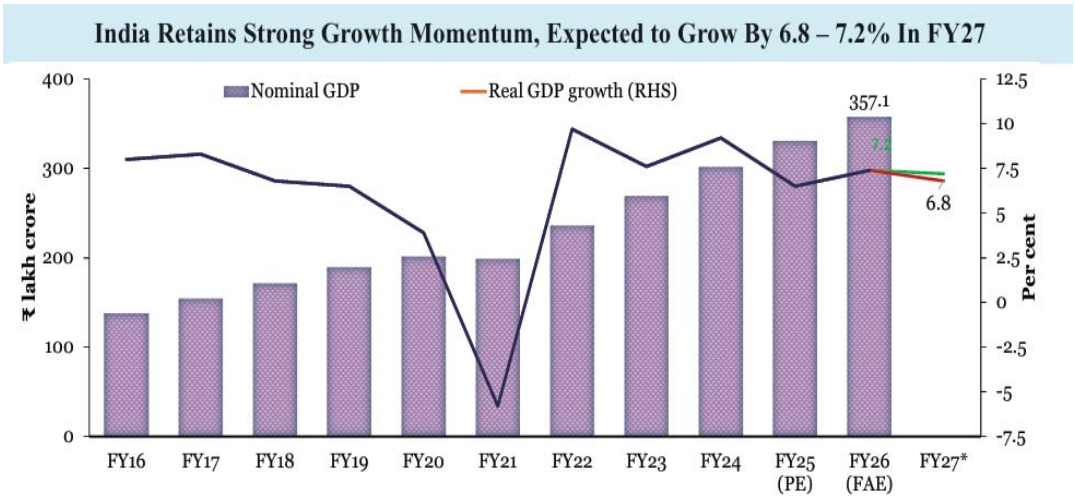
Arguments in support of K-shaped Recovery

1. **Divergence between Formal and Informal Sector-** Large listed companies saw record profits by cutting costs and adopting digital technologies (BSE Sensex rose from **~25,000 in March 2020** to **over 47,000 by December 2020**). Small and Medium Enterprises (MSMEs), which **lacked the cash reserves of larger firms**, faced a vicious cycle of debt (According to some estimates **9% of MSMEs may have shut down permanently** due to COVID Impact).
2. **Slower recovery of Contact-intensive sector-** While IT and Pharma boomed, sectors like hospitality, tourism, and aviation **remained significantly below pre-pandemic levels in FY 2021-22**.
3. **Widening Income and Wealth Inequality-** The annual income of the **poorest 20%** of Indian households, constantly rising since 1995, **plunged 53%** in the pandemic year 2020-21 from their levels in 2015-16. The **richest 20%** saw their annual household income **grow 39%** reflecting the sharp contrast Covid's economic impact.

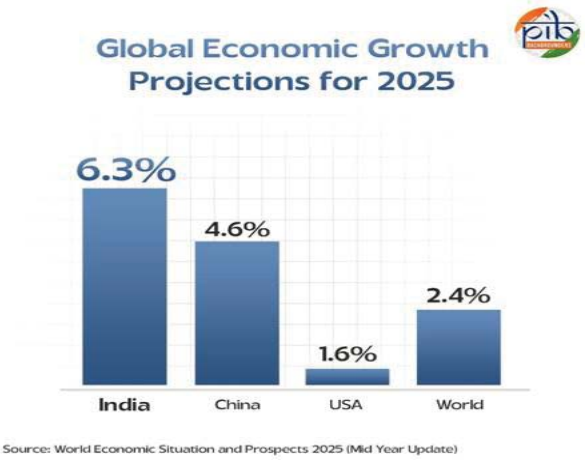
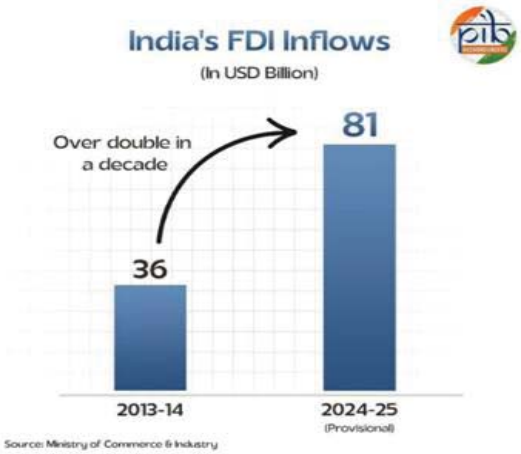
Conclusion- While macro-economic indicators suggest a V-shaped recovery, uneven growth across sectors hint at K-shaped recovery

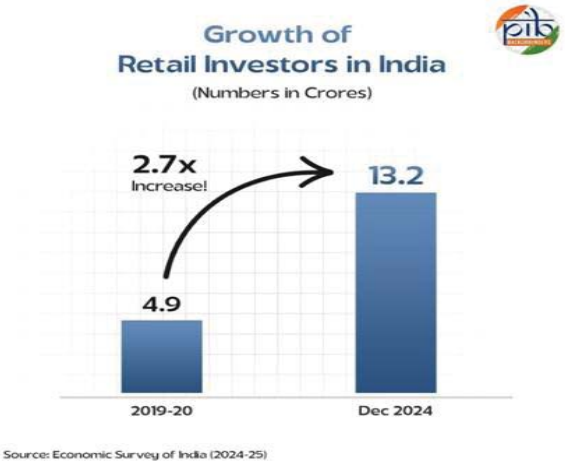
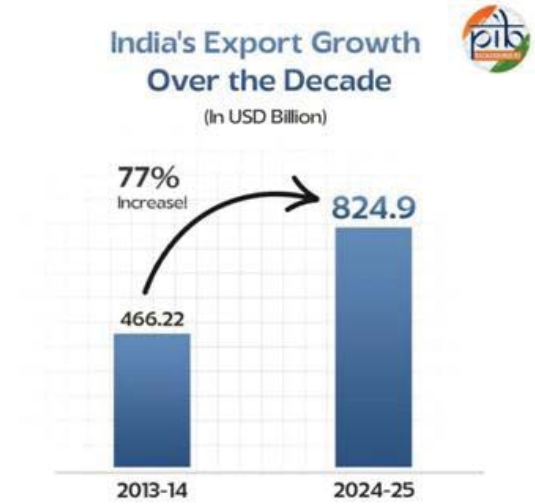
Phase 7- Current Phase (2023–2025) – Uneven Growth, Rising Aspirations

GDP Growth: India's real GDP has maintained strong momentum from 2023 to 2025, consistently positioning it as the world's fastest-growing major economy.



Macro-economic growth parameters

GDP growth	India's GDP growth continues to remain the highest amongst the major economies of the world. 
Foreign Direct Investment (FDI)	FDI inflows rose to **$81.04 bn** in **FY 2024–25**. This is more than double the **$36.05 bn** received in **FY 2013–14**, showing long-term progress. 

Market Confidence at Record levels	India's capital markets have become a powerful engine for economic growth by turning household savings into investments.  Source: Economic Survey of India (2024-25)
Manufacturing and Exports	India's total exports touched a new high of **$824.9 bn** in 2024–25, marking a sharp rise from **$466.22 bn** in **2013–14**.  Source: Ministry of Commerce & Industry
Startup Boom	India became the **3rd largest startup ecosystem** (~100+ unicorns). Today there are **2 lakh DPIIT-recognised startups in India**, a **massive increase from roughly 500** in 2016.

2. DRIVERS OF THE INDIAN ECONOMIC GROWTH

Infrastructure led Growth

1. **National Infrastructure Pipeline (NIP)-** The National Infrastructure Pipeline (NIP) is India's Rs. 111 lakh crore ($1.4+ trillion) plan (2020–2025) to boost economic growth by investing in world-class infrastructure, including energy, roads, railways, and urban projects.
2. **PM Gati Shakti (2021)-** It is an initiative to boost multimodal infrastructure across India's economic hubs. It coordinates development across seven core sectors - rail, road, ports, waterways, airports, mass transit, and logistics.

3. **National Industrial Corridor Development Programme (NICDP) (2024)-** Development of 12 new industrial areas across 10 states to promote manufacturing hubs and planned urbanization.
4. **National Logistics Policy (2022) (NLP)-** Designed to streamline logistics services, create efficiencies, and reduce logistics costs.
5. **Enhanced Capex-** The Union Budget 2026–27, has proposed a **9% increase** in public capital expenditure (capex) to **Rs. 12.2 lakh crore**. This marks a massive increase from **Rs. 2 lakh crore** in **FY 2014-15**.
6. **Bharatmala Pariyojana (2017)-** It aims to develop about 26,000 km of economic corridors, along with the Golden Quadrilateral (GQ) and North-South and East-West (NS-EW) Corridors, to carry the majority of the freight traffic on roads.
7. **Sagarmala Project (2015)-** Aims at holistic port infrastructure development along the 7,516-km long coastline through modernisation, mechanisation and computerisation.
8. **UDAN-** Aims to boost air connectivity and infrastructure in remote/underserved areas & making flying affordable.
9. **Dedicated Freight Corridors (DFC)-** Specialized freight routes that boost transport capacity through faster transit, double-stack container trains, and heavy-haul trains, enhancing supply chains for industries.

Infrastructural Growth in India

The National Highways network has expanded from **91,287 km in 2014** to **1,46,342 km in 2025**. **7.8 lakh km rural roads** were completed from **2014–2025**.

46,900 km of rail electrification completed since 2014. Over **78% of Railway tracks upgraded for sectional speed of 110 kmph and above**. 144 Vande Bharat trains running as of 31st July, 2025. India's metro network grew from **248 km (2014)** to **1,013 km (2025)**.

India's port capacity doubled to **2,762 MMTPA** since 2014. Inland waterways cargo rose by **710% since 2014** (from **18 MMT to 146 MMT**). 277 projects have been completed under the Sagarmala project and Sagarmala 2.0 has been launched.

92 airports operationalized under UDAN as on July, 2025. Over 1.53 crore passengers were flown under regional connectivity till 30th June, 2025. Digi Yatra has been adopted in 24 airports and has been used over 5 crore times.

Digitalised Growth

1. **Digital India Programme-** The Digital India programme has accelerated connectivity and digital inclusion.

   There has been an expansion in the internet penetration from **25 crore in 2014** to nearly **97 crore in 2024**.

2. **Internet Connectivity Push-** With BharatNet Project, 5G rollout and Bharat 6G vision, India is now the world's third-most digitalised economy, driving growth through technology-led transformation.
3. **UPI-** The adoption of UPI has allowed seamless real-time payments.

   UPI transactions in India reached record highs. December 2025 recorded **21.63 billion transactions**, which was a **29% increase year-on-year**. UPI volumes exceeded **640 million daily**, making it the world's largest real-time payment system.

Manufacturing Push

1. **Production Linked Incentive (PLI) Push-** The Production Linked Incentive (PLI) Scheme, launched

in 2020 with a **Rs. 1.97 lakh crore** (**~$24 billion**) outlay, has become a cornerstone of India's manufacturing transformation.

> Covering **14 key sectors**, it has attracted over **Rs. 2.16 lakh crore in investments** and generated over **14.39 lakh jobs by December 2025**, driving exponential export growth in electronics, pharma, and automobiles.

2. **Make in India Push-** The Make in India initiative, launched in 2014, has positioned India as a manufacturing hub with notable successes in railways, defence, toys, and electronics.

> India is now the **world's second-largest mobile phone manufacturer**, with over **300 factories** compared to **just 2 in 2014**.

Banking and Financial Inclusion Reforms

1. **Recapitalisation & PSB Merger-** Consolidation of the banking sector so as to enhance the overall stability and performance of the banking system.
2. **Insolvency and Bankruptcy Code (IBC (2016))-** The IBC has significantly transformed India's insolvency framework.

> IBC has rescued over **1,194 companies** and enabled creditors to recover **Rs. 3.89 lakh crore** with a **32.8% recovery rate** as of March 2025.

3. **Pradhan Mantri Jan Dhan Yojana-** It has led to opening of over **56 crore accounts**, with **55% held by women**. It provides access to banking, credit, insurance, and pensions, forming the backbone for Direct Benefit Transfers and inclusive growth.
4. **JAM Trinity-** It aims seamless Direct Benefit transfer and reducing leakage of government subsidies by linking the **Jan Dhan Accounts**, **Aadhar Cards**, and **Mobile numbers** of Indian citizens.

Employment and Skill Development Push

1. **PM Viksit Bharat Rozgar Yojana-** It targets **3.5 crore youth** by incentivising new employment between 2025 and 2027.
2. **Skill India Mission-** This mission has trained over **6 crore citizens** through PMKVY, JSS, and apprenticeship schemes, ensuring India's workforce is future-ready.

Reforms Push

1. **Tax Reforms-** GST and corporate tax rationalisation have **simplified compliance** and **reduced costs**, fostering a more business-friendly environment.
2. **RERA Reforms-** Increased transparency in real estate projects, with more than **1 lakh+ projects** registered under RERA.

3. CHALLENGES BEFORE THE INDIAN ECONOMY

The Indian economy faces significant structural and cyclical challenges. The major challenges before the Indian economy are presented below:

Global Headwinds

The Economic Survey 2025 warns of rising economic fragmentation, where trade barriers and bloc-based alignments pose challenges to India's growth, competitiveness, and global market access.

1. **Intensification of geopolitical conflicts and Supply Chain Disruptions-** Russia-Ukraine War, Middle East Crisis (Gulf of Hormuz crisis), and increased frequency of cyber and non-state actors attacks (Red sea crisis).

2. **Energy Weaponisation-** The competition for **critical minerals**, Rare Earth Elements (**REEs**) and export controls on critical technologies such as **semiconductors**, **AI** pose challenges for Indian economic growth.
3. **US-imposed tariffs-** The 50% tariffs on Indian exports could cut growth by **0.5–0.6%** in **2025–26**, affecting up to **55%** of **India's $87billion exports** to the US.
4. **Monetary tightening-** Synchronised monetary tightening by the central banks, has caused capital flight, leading to currency depreciation.

Slowing Private Investment & Weak Domestic Demand

1. **Declining investment ratio-** Gross fixed capital formation (**investment-to-GDP**) has dropped from **~39–42% (2006–10)** to **~33% in FY24.** Private investment which was around **18% of the GDP** between (**2006-2010)**, has dropped to **9.8%** in **FY23-24**.
2. **Subdued domestic demand-** The share of private consumption in India's GDP fell from **58.1%** in **FY22** to **55.8%** in **FY24**.
3. **Decline in Household Savings-** Household financial savings dropped from **7.3% of GDP** (FY22) to **5.3% of GDP** (FY23). This is well below the decadal average of **~8% of GDP**.

Fiscal Stress & Low Tax Base

1. **High debt servicing burden-** As per the Budget 2026-27, roughly 20% of government revenue goes toward servicing debt. This affects public investment and also leads to crowding out of private investments.
2. **High Debt Distress of States-** Several states in India like **Punjab**, **Himachal Pradesh** and **West Bengal**, have **high debt-to-GSDP ratio**. A significant revenue of these states are diverted to interest payments.
3. **Low tax-to-GDP ratio-** India's tax-to-GDP ratio **~11% in FY 25-26**, remains relatively low compared to OECD averages (OECD countries have a tax-GDP ratio of over **30%**).

Infrastructure, Credit, and Institutional Constraints

1. **High logistics costs-** India's higher logistics cost is a major bottleneck for the development of the Indian economy.
2. **NBFC crisis-** The NBFC crisis in India (**IL&FS collapse in 2018**), triggered a massive liquidity crunch and widespread defaults among shadow banks.
3. **Regulatory inefficiencies-** Delays in land acquisition, approvals, and corruption hinder business environment and speed of execution.

> **85 under-construction National Highways (NH) projects** are delayed by more than three years due to **bottlenecks relating to land acquisition**, **statutory clearances** and **permissions**, **shifting of utilities** and **removal of encroachments**.

Inequality, Poverty, & Social Imbalances

1. **Skewed asset distribution-** According to the Oxfam Report, the **top 10% of the Indian population holds ~77% of the total national wealth**. In contrast, the **bottom 50%** owns merely **3% to 6.4% of the national wealth**.
2. **High income disparities-** According to the **Economic Survey of 2024-25,** the **average income** of **casual workers** has been Rs. 418 a day or INR 12,540 per month, **self-employed person** has been **Rs. 13,279** and **regular salaried workers** has been **Rs. 20,702** in 2023-24.

3. **Hunger and malnutrition-** In the **Global Hunger Index (GHI) 2025**, India ranks **102nd** out of **123 countries** and **India's hunger level** has been classified as **serious**.

> **Data of Global Hunger Index (GHI) 2025**
> Undernourished Population- **12%**
> Stunted Children under 5 years- **32.9%**
> Wasted Children under 5 years- **18.7%**
> Children Death before 5th Birthday- **2.8%**

Climate Vulnerabilities & Resource Pressures

1. **Looming Water Scarcity-** India's water shortage has exacerbated **Zero day risks** and threaten food security.

> **Climate Risk Index (CRI) 2026-** India is ranked **9th** globally in long-term climate risk by Germanwatch. India has lost **$170 bn** with **80,000 deaths**.

2. **Inflation from weather shocks-** Food inflation often spikes due to weather-linked production disruptions, fueling broader inflationary pressures.

> Kitchen staples like **tomatoes**, **onions**, **potatoes** are highly susceptible to weather shocks and are major drivers of food inflation.

Employment & Labor Market Stresses

1. **AI based automation threats-** AI-driven automation threatens job security particularly the entry level jobs in India.

> According to IMF projections, **26% of the entry level jobs in India** are at risk due to **AI-based technological disruptions**. The AI based disruptions have been dubbed as a "**tsunami hitting labour markets**".

2. **Increasing Informalisation-** There has been **slow down in the formal job sector** and increase in **gig economy employment**.

> According to the **Economic Survey 2025-26**, Gig workers currently represent **2% of the total workforce**. This is projected to increase to **6.7% by 2029-30**. About **40% of gig workers earn less than Rs 15,000 per month**.

4. INDIA AND THE RISK OF MIDDLE INCOME TRAP

According to the Economic survey 2025-26, India could cross from **Low Middle Income (LMI) group** to join the **Upper Middle Income (UMI) status by 2037** and aspire to join the higher income (HI) status by 2047, the 100th year of Independence.

India has persisted in the **Lower Middle Income (LMI) group** since **2007**. India's peers such as **Brazil**, **China**, **Indonesia**, **Malaysia**, and **South Africa**, have moved into the Upper Middle Income **(UMI status)**.

World Bank's Classification of Countries based on Income

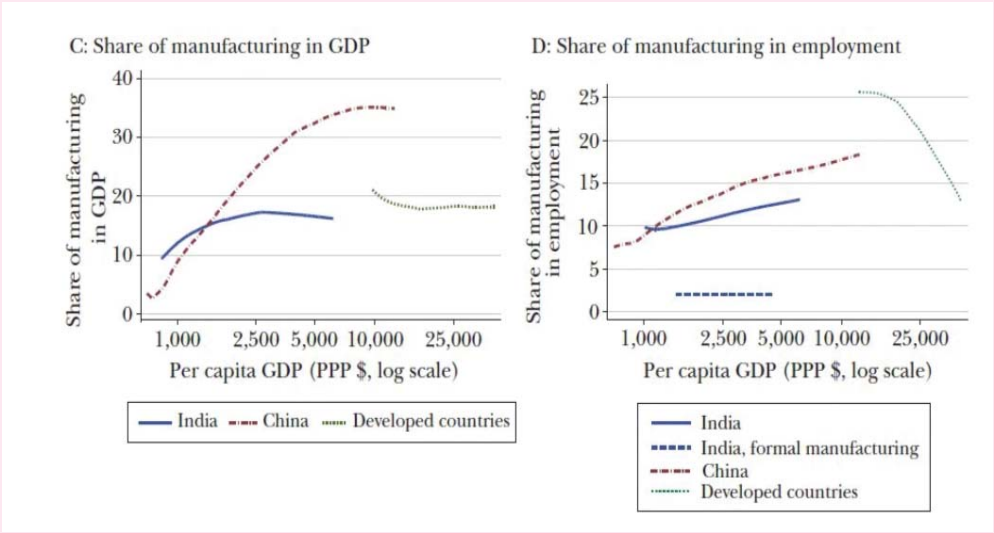
Low-Income Countries (LI) Countries	Annual per capita **Gross National Income (GNI)** of **$1,136** or **less as low-income (LI) countries**.
Low- Middle Income Countries (LMI)	Countries with a GNI of **$1,136-$4,465** are categorised as LMI countries. **India's current per capita GNI is around $2,600-$2,700.*

Upper-Middle Income Countries (UMI)	Countries with a GNI of **$4,465-$13,845** are categorised as UMI countries.
High Income (HI) Countries	Countries with incomes at **$13,845 and above** are designated as HI countries.

Factors which exacerbate the risks of India being stuck in the Middle Income Trap

1. **Premature De-industrialisation-** India skipped the low-skilled manufacturing phase and directly ventured into the services. India's manufacturing's share in output and employment has remained generally below 20%.

The **East Asian economies** started from **low-skilled manufacturing**, then moved to **high skilled-manufacturing** before moving on to services.



2. **The Missing Middle in Enterprise-** India's economy is dominated by a few massive conglomerates. There is a lack of mid-sized companies that typically drive innovation and stable jobs.

India has over 63 million MSMEs (Micro, Small and Medium Enterprises). **99% of the enterprises** fall into the **micro category**, **0.5%** is classified as **small** and **0.01%** are classified as **medium**.

3. **Precocious Development Model-** A poor and a socially fractured country like India took an audacious step of carrying out economic development, political development and redistribution all at the same time.

While **Korea started redistribution** when the **per capita GDP** was **$20,000**, India started **redistribution** when the **per capita GDP** was around **$5,000**.

4. **Lack of Innovation Capability-** India's R&D expenditure **as a percentage of GDP** remained between **0.6% to 0.7%**. India lags in **deep-tech patents** in **semiconductors**, **biotech**, and **green energy**, which are the engines of high-income growth.

India's **Gross Expenditure on Research and Development (GERD)** as a **percentage of GDP** is relatively much lower than other countries like **China (2.4%)**, **USA (3.5%)**, and **Israel (5.4%)**.

5. **Human Capital Stagnation-** India spends just 3.5% of its GDP on education, far below Brazil (6.5%) and Malaysia (4.5%).

South Korea's 'Asian Miracle' was built on **technical education** and **manufacturing**, enabling its leap to **high-income status** in **1995**.

6. **Skill Gap-** According to the **Economic Survey 2023-24**, only **51.25%** of India's graduates are considered immediately employable. **One in two graduates** are not yet readily employable straight out of college.
7. **Environmental Shocks-** Climate change is a **growth-killer** for India. Extreme weather **diverts billions of dollars** from **productive investment** into **disaster relief and healthcare**.
8. **Wealth Concentration and Low Domestic Demand-** In India, wealth is **heavily concentrated at the top 1%**, while **real wages for the bottom 50%** have remained largely **stagnant**.
9. **Global Headwinds-** Geopolitical tensions constrict foreign trade and investment. Intensification of geopolitical conflicts like the **Middle East War**, **Russia-Ukraine War** have disrupted the supply chains and pushed back India's economic growth.

Way Forward for India to avoid the Middle Income Trap

In recent decades, only a few dozen countries have grown from middle-income countries into high-income countries. This includes Saudi Arabia, Latvia, Bulgaria and South Korea.

The World Bank outlines a three-pronged approach that countries can follow to avoid the middle-income trap. This strategy is dubbed as the **3i strategy**-

Investment + Infusion + Innovation (3i)- Low-income countries should focus primarily on **increasing investment** in the economy. Once a country has reached lower-middle-income status, **policies should be adjusted to facilitate** a mix of **investment** and **infusion**. After reaching upper-middle-income status, countries should complement investment and infusion with innovation.

South Korea as a leading example of avoiding the Middle Income Trap

South Korea as an example of a country that has effectively implemented the **3i strategy**. In the **1970s** and **1980s**, **South Korea pursued reforms** to **encourage private investments** and **industrial policies** that boosted technology use and production efficiency. According to the **World Bank**, South Korea's per capita income rose from **$1,200 in 1960** to **$33,000 in 2023**.

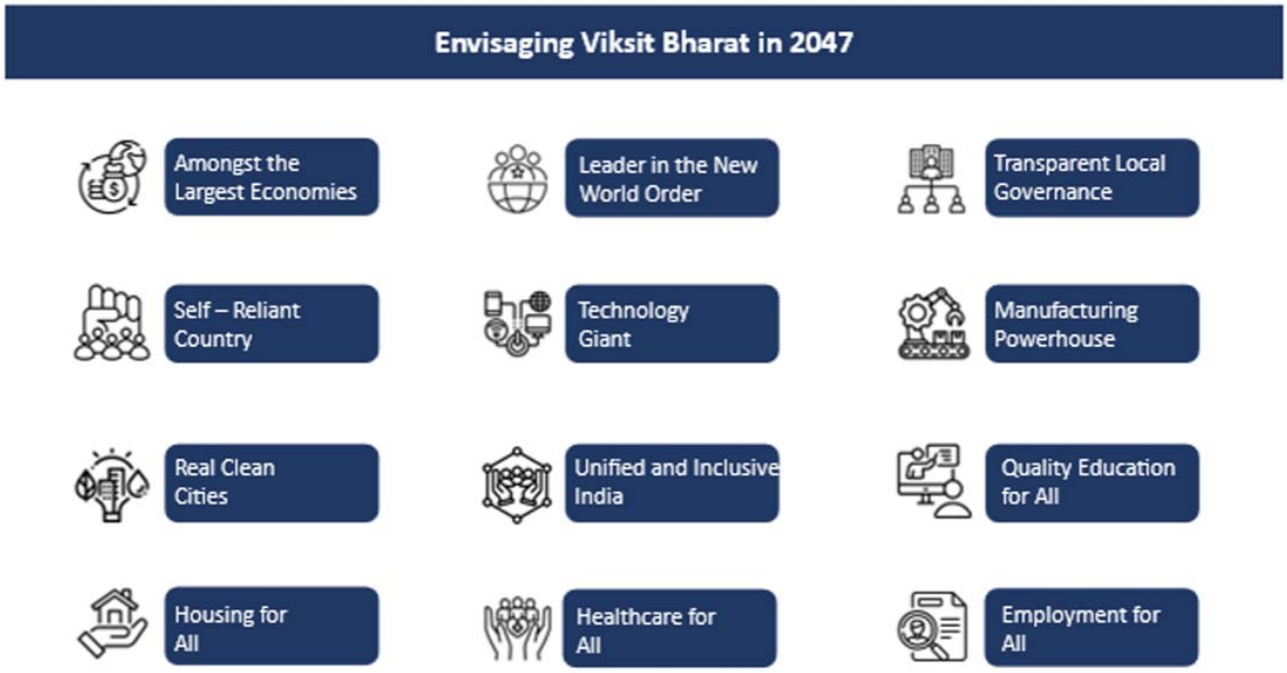
5. VISION OF VIKSIT BHARAT 2047

Viksit Bharat 2047 is the Government of India's flagship vision to transform the country into a fully developed and prosperous nation by 2047, coinciding with the 100th year of India's independence.

Economic Goal- Transforming India into a **$30-trillion economy** with a per capita income between **$20,000** and **$25,000**.

Key Pillars of Viksit Bharat

Youth (Yuva)	Harnessing the demographic advantage by empowering young talent through education and employment.
Poor (Garib)	Ensuring inclusive growth so that no one is left behind.
Women (Mahilayen)	Promoting equal rights and opportunities for women
Farmers (Annadata)	Strengthening the agriculture sector and livelihood security.



India's strength for Vikshit Bharat

1. **High Economic Growth-** India is the 4th largest global economy and the fastest growing economy of the world.

   Economic survey (2025-26) has projected 7.4% growth for FY 25-26 and 6.8% – 7.2% for FY 26-27.

2. **Demographic Dividend-** India's working-age cohort (15–59 years) is expected to reach its absolute peak of **1.01 billion people** in **2041**.
3. **Large and Expanding Consumer Market-** According to the Economic Survey 2025-26, the Indian economy is currently in its most consumption-driven phase since 2012.

   In fiscal year (FY) 2025-26, private final consumption expenditure (PFCE) accounted for **61.5 per cent** of gross domestic product (GDP), the **highest level since FY 2011-12**.

4. **Dynamic Services Sector-** India's services sector continues to be the primary engine of economic growth, contributing approximately **55% to 56.4%** of the Gross Value Added (GVA) (**Economic Survey 2025-26**).

   The Economic Survey 2025-26 identifies **data centres**, **space technology**, and **medical tourism** as high-potential niche areas for future expansion.

5. **FDI Potential-** India is increasingly viewed as a top global investment destination, with gross FDI inflows reaching a **cumulative $1.12 trillion** since April 2000.

Significance of Viksit Bharat

1. **Unprecedented Economic Development-** Aspires to make India one of the richest countries globally with a $30 trillion economy, creating abundant jobs and eradicating poverty.
2. **Better Infrastructure and Urban-Rural Development-** Focuses on improved roads, smart cities, clean villages, robust transportation, and digital infrastructure to enhance connectivity and quality of life.

3. **Farmer Welfare and Sustainable Agriculture-** Implements schemes to increase farmer income, adopt sustainable farming practices, develop irrigation, and ensure food security.
4. **World-Class Education and Healthcare-** Aims for 100% access to quality education and affordable healthcare for all citizens, including digital healthcare initiatives reaching remote areas.
5. **Women's Empowerment-** Promotes inclusivity and equal opportunities across sectors, enhancing women's participation in the economy, politics, and education.
6. **Environmental Sustainability-** Emphasizes clean energy (solar, wind), pollution control, forest conservation, water management, and greener urban environments to ensure a safe environment for future generations.
7. **Good Governance and Transparency-** Focuses on honest, efficient, and corruption-free governance to support rapid and inclusive development.

Challenges to Viksit Bharat

1. **Inadequate Growth Rate-** The current growth rate of around **8%** may be inadequate to raise the **Gross National Income (GNI)** of around **$2,700** presently to **$13,000** by **2047**.
2. **Middle Income Trap Risk-** India's growth risks stagnating if it cannot move beyond being a low-cost assembly hub.

   India's R&D spend is stuck at **0.64%–0.7%** of **GDP**, whereas high-income nations typically spend over **2.5%**.
3. **Premature Deindustrialization-** Manufacturing has been stagnant at **~14–17% of GDP** for two decades. The services sector (IT, Finance) is highly productive but cannot absorb the **42% of the workforce** still stuck in low-productivity agriculture.
4. **Human Capital & Learning Crisis-** India faces a significant skill gap, with only **2.3%–5.4% of its workforce** having **formal vocational training**.

   Developed nations have higher vocational training like **68% in the UK**, **75% in Germany**, and **96% in South Korea**.
5. **Inadequate Private Investment-** The bond market size (Bond Market as % of GDP) is around 70% of its GDP with a total market size of around $2.81 trillion.

   In the US, the total bond market size is **around $50 trillion** and the **bond market as a % of GDP is around 225%**.

Way Forward for Viksit Bharat

To achieve the vision of Viksit Bharat by 2047, India must shift from "business as usual" to a path of accelerated structural reforms.

1. **Infrastructure at Scale-** Sustained public capital expenditure is critical for growth. This includes expanding high-speed rail corridors and national waterways to reduce logistics costs.
2. **Manufacturing & MSME Growth-** Moving from an assembly hub to a global manufacturing leader by supporting MSMEs and creating competitive markets for private innovation.
3. **Global Trade Integration-** Reducing trade barriers and lowering tariffs to **integrate deeper into Global Value Chains (GVCs)** and compete with peers like **Vietnam** and **Thailand**.
4. **Closing the Learning Gap-** Improving primary and secondary education outcomes and scaling vocational training to align with Industry 4.0 requirements (AI, Robotics, Cybersecurity).
5. **Workforce Participation-** Raising overall labour force participation from **~56% to 65%,** with a specific focus on increasing female participation to **50%** by 2047**.**

6. **Red Tape to Red Carpet-** Simplifying regulations, reducing compliance burdens, and modernising the bureaucracy and judiciary for higher accountability.
7. **Land & Asset Records-** Digitalising land ownership titles and integrating blockchain to ensure security and ease of credit.
8. **Attract FDI by China+1 strategy-** Attracting **$100 billion** in **annual FDI by 2030**, leveraging "China Plus One" strategies and massive structural reforms.

Conclusion

Viksit Bharat 2047 envisions India as a developed, inclusive, and sustainable nation by the time it celebrates its 100 years of independence. Achieving this vision requires active participation of government, private sector, and citizens alike, with focus on innovation, education, skill development, and infrastructure.

6. ECONOMIC GROWTH VS DEVELOPMENT DEBATE

Differences between Economic Growth and Development

Parameters	Economic Growth	Economic Development
Definition	Increase in the **quantitative outputs** of the economy (GDP, GNP, Per Capita Income)	**Qualitative improvement** in standard of living, equity, and human welfare.
Theoretical Premise	**Trickle-down** approach (classical economists), which holds that tax breaks and benefits for corporations and the wealthy will trickle down and eventually benefit everyone.	**Development is freedom** (Amartya Sen), which argues that true development lies in expanding people's choices and capabilities, and not just national income.
Measurement	GDP, GNP, National Income, Per Capita Income.	HDI, MPI, Gini Coefficient, Literacy Rate, Life Expectancy.
Scope	**Narrow** as it focuses only on economic indicators.	**Broad** as it includes health, education, equity, freedom, sustainability.
Inclusiveness	**May benefit only certain sections** (can cause inequality).	**Inclusive**, aims to reduce poverty, unemployment, and inequality.
Sustainability	Can be **resource-depleting** and environmentally harmful.	**Emphasizes sustainable**, equitable use of resources.
Employment Impact	Can lead to "**jobless growth**" (output ↑ without jobs ↑).	Prioritizes **productive employment generation**.

Economic Growth → Without equity → No real development (Increase in Social Disparity)

Rising GDP vs Inequality:

The top 1% own 40%+ of wealth (Oxfam 2023).

Per capita income has improved, but 45% youth are unemployed (CMIE 2023–24).

Urban–Rural Divide:

Services sector - 57% of GDP, but employs only ~25% workforce.

Agriculture - 18% of GDP, yet supports 44% workforce → underemployment.

Regional Disparities:

Southern/western states lead in HDI while the BIMARU states lag.

Social Indicators:

India's Global Hunger Index rank = 111/125 (2023) despite being a major food producer.

Economic Growth → More resources → If inclusive policies → Eco Development

Kerala Model- High literacy, health outcomes, social development despite moderate GDP.

Digital India + UPI- Enabled financial inclusion for rural/urban poor → development via growth.

Green Revolution- Boosted agriculture growth and reduced famine → development gains.

7. SERVICE LED GROWTH VS INDUSTRY-LED GROWTH

Parameters	Service-Led Growth (India's Path)	Industry-Led Growth (East Asian Path)
Definition	Growth driven by expansion of the service sector like IT, finance, telecom, healthcare, education, tourism.	Growth driven by manufacturing & industrialization (textiles, electronics, automobiles, steel).
India's Experience	India has experienced service led growth with the Services sector contributing to **~57% of GDP**. However, it employs around **~30% of the workforce**. **Services exports (~340bn)** has surpassed merchandise exports	India skipped the **"manufacturing boom"** that East Asia experienced. Manufacturing remains stagnant at **16–17% of GDP**. Export-led manufacturing (in **China, South Korea)** drives sustained growth.
Employment Impact	High-skilled jobs in IT and finance, but limited job creation for semi-skilled/ unskilled workers. This often leads to the **phenomenon of jobless growth**.	Manufacturing has higher labor absorption. It has high potential to employ rural migrants and semi-skilled workers.
Inclusiveness	Benefits concentrated in **urban, educated middle class**. Rural and low-skilled workers are largely excluded.	**Broader employment opportunities** across social groups, which fosters inclusive growth.
Regional Spread	Limited to Services hubs concentrated in metros (**Bengaluru, Hyderabad, Gurugram, Pune**)	The manufacturing sector **spreads jobs more regionally and equitably**, particularly in tier-2/3 cities and semi-urban areas.

India's Model of Service-Led Growth

India's economic growth witnessed **premature de-industrialization** or the **skipped stage** of development. While most nations follow a linear path (**Farm → Factory → Services**), India's services sector grew to over **50% of GDP** while manufacturing remained stagnant at **14–17%**.

Reasons behind Services Outpacing Industry in India

1. **The 1991 Reforms at the Right Time-** India's **economic liberalization of 1991** took place just at the right time, as the **global telecom** and **internet revolution began**.
   - India's large pool of **English-speaking**, **technically skilled graduates** was perfectly positioned to meet the global demand for IT and BPO services.

2. **Regulatory Hurdles in Industry- License Raj**, **rigid labor laws**, and **complex land acquisition rules** made setting up factories difficult.
   - In contrast, the **service sector (software, consultancy)** required **less land**, **less electricity**, and **faced fewer "inspector raj" hurdles**.
3. **By-passing of Infrastructure Gaps-** Service sector being digital in nature bypassed India's poor physical infrastructure.
4. **Rise of IT and knowledge-based industries-** Growth of IT hubs like **Bangalore**, **Hyderabad**, and **Pune boosted sectors** such as **software services**, **BPO (Business Process Outsourcing)** and **consulting**.
5. **Lack of Vocational Skills-** India lacked the middle-tier vocational training needed for mass-scale manufacturing.

> Developed nations have higher vocational training like **68% in the UK**, **75% in Germany**, and **96% in South Korea**.

However, India cannot rely solely on service-led growth to become a developed Nation by 2047. There are several Challenges with service-led growth in India-

1. **The Jobs Gap-** Services like IT are **elite** and require high education. They **cannot absorb the 42% of India's workforce currently in agriculture**. Only manufacturing industries like **textiles**, **electronics**, **toys**, can provide millions of jobs for semi-skilled workers.
2. **Balanced economic structure-** Developed economies (like **Germany, South Korea, China**) all have strong manufacturing sectors. Developed industry supports services and agriculture (machines, logistics).

> China accounts for nearly **29% of global manufacturing output**, more than double that of the United States.

3. **Strategic independence-** Strong industry ensures domestic production of electronics like semiconductors, and critical defense equipment.

> China indigenously **controls the entire EV supply chain**, from raw material processing to finished battery cells. It produces an estimated **70–80% of the world's EV battery cells**.

4. **Income Inequality-** A service-led economy exhibits a **K-shaped growth pattern**. It **enriches the top 10%** (tech workers, bankers) but **leaves the bottom 50% behind**.
5. **Economic Resilience-** Relying only on services makes the **economy vulnerable to global tech shifts or AI automation**. A physical manufacturing base provides a **stable economic floor** for the economy.

India must push for **manufacturing-led growth** to become a developed nation by 2047. It must push industry-led growth alongside services by:

1. **Strengthening Manufacturing-** Expansion of Production Linked Incentive Scheme (PLI), promoting logistics corridors, and ease of doing business.
2. **Skill Development-** Train workforce for both IT and industry (Skill India Mission).
3. **MSME Support-** Credit and digital tools for 63 million small units.
4. **Integration with Service sector-** Combine IT strength with manufacturing (e.g., Industry 4.0, smart factories).

MGP/BP/GS3/002

GDP - INFLATION

GDP

1. GDP ESTIMATION IN INDIA

GDP measures the **total monetary value of all final goods and services** produced **within a country's borders** in a **specific period**. It helps gauge the **size and health** of the economy.

Types of GDP

GDP can be categorized into various forms like **Nominal GDP, Real GDP, GDP Per Capita, and GDP based on Purchasing Power Parity (PPP),** each serving distinct purposes for analysis.

Terms	Definition	Significance
Nominal GDP	The total value of all goods and services produced, measured at **current market prices without adjusting for inflation**.	Reflects India's economic size using current prices. However it is **affected by inflation.**
Real GDP	The total value of goods and services produced, **adjusted for inflation,** measured at constant **base-year** prices.	Shows true growth in production by removing price changes. India now uses **2022–23 as the base year** for calculating the real GDP.
GDP per capita	Average economic output or income per person in the country, calculated by **dividing GDP by population**.	Indicates average living standards and productivity in India.
GDP based on PPP	GDP adjusted to **reflect differences** in **price levels** and **cost of living** between countries.	Used to **compare India's economic output** and **living standards internationally** by neutralizing currency differences.

Methods of GDP Calculation

Gross Domestic Product (GDP) is calculated using three primary methods- the Production Method, Expenditure Method, and Income Method.

Method	Description	Key Formula/ Components	Use and Importance
Production Method or the Value Added Method	Calculates GDP by adding value added at each stage of production across all sectors.	**GDP= GVA at Basic Prices + Production Taxes - Production Subsidies**. GVA= Value of Output- Value of Intermediate consumption	Measures sectoral contributions. Commonly used for sector-wise analysis. Based on production data from industries, agriculture, and services.
Expenditure Method	Calculates GDP by summing expenditures on final goods and services by households, businesses, government and net exports.	**GDP = C + I + G + (X - M)** where C=consumption I=investment G=government spending X=exports M=imports	Highlights demand-side drivers of the economy. Reflects consumers' spending, investments, govt. expenditures, and trade balance. Highlights consumption patterns, investment trends (capital formation), and for tracking the impact of government policies.
Income Method	Calculates GDP by summing incomes earned by factors of production: wages, rents, interest, profits plus taxes minus subsidies.	**GDP = Compensation of employees + Rents + Interest + Profits + Taxes + Depreciation - Subsidies**	Captures the distribution of national income generated in the economy. Useful for policy on wages, taxation, profits.

Differences in GDP Computing Methodology: Pre-2015 vs. Post-2015

In **2015**, the **Central Statistical Office (CSO)** made changes to computing methodology of GDP based upon recommendations of **K Sundaram Committee.** This has brought India's methodology in line with international standards of **System of National Accounting 2008.**

Aspect	Pre-2015 Methodology	Post-2015 Methodology
GDP Base Year	**2004-05**	**2022-23**
GDP Calculation Metric	**GDP was calculated at factor cost**. It measured the cost of production (wages, rent, interest) but excluded indirect taxes and subsidies.	**GDP at market price**. It includes the impact of indirect taxes (like GST) and excludes subsidies. This reflects what the consumers actually pay.
Manufacturing Sector Data	Based on Index of **Industrial Production (IIP)** and **Annual Survey of Industries (ASI)** covering around **2 lakh factories**.	Uses of data from **MCA 21 (Ministry of Corporate Affairs)**, covering around **5 lakh companies** for a comprehensive view.
Financial Sector Coverage	Limited to a few mutual funds and RBI estimates for Non-Banking Finance Companies (NBFCs).	Expanded to include stockbrokers, exchanges, asset managers, regulators (**SEBI, PFRDA, IRDA**)

Labour Income Calculation	Treated all labour input equally	Introduced "**effective labour input**" with weights based on worker role (owner, professional, helper)
Agricultural Value Addition	Limited to farm produce	Broadened to include livestock and other agricultural activities
Global Standard	Aligned with the UN System of National Accounts 1993 (**UN SNA 1993**)	Aligned with the UN System of National Accounts 2008 **(UN SNA 2008)**

Reasons Behind the Base year updation of GDP to 2022-23

1. **Capturing Structural Shifts-** Updating the base year to 2022-23, allows for the inclusion of emerging sectors such as **digital services**, **renewable energy**, and the **gig economy** that were non-existent or negligible in 2011–12.
2. **Incorporation of major economic reforms-** The change of the base year to 2022-23 incorporates the effect of important reforms after 2011–12 such as **GST implementation**, **Insolvency and Bankruptcy Code (IBC)** and **Digitalization push**.
3. **Impact of COVID-19 and recovery phase-** The economy underwent major disruption during the COVID-19 pandemic. Using **2022–23 (a relatively stable recovery year)** provides a more **"normal" benchmark** for better trend analysis of GDP.
4. **Enhancement of Global Credibility-** Regular rebasing aligns India with international standards, such as the **UN System of National Accounts (SNA)**, and addresses concerns from bodies like the **IMF regarding data quality**.
5. **Improved Data Accuracy-** New administrative datasets, like **GST returns**, **e-Vahan** (vehicle registration), and the **Public Financial Management System (PFMS)**, have been integrated to replace outdated proxy indicators and provide more granular information.

Issues with GDP Calculation in India

1. **Data Quality and Coverage-** GDP data often relies on **outdated surveys**, **incomplete data from informal sectors**, and **gaps in capturing unorganized economic activities**.
2. **Dual GDP Calculation Methods-** The calculation of India's GDP using income and expenditure methods often mismatch leading to "statistical discrepancies".
3. **Limitations of the MCA21 data-** Use of MCA21 data has improved coverage. However, many firms are inactive and file the data irregularly. This often leads to estimation errors and wrong projections.
4. **Use of approximation and extrapolation-** The output of agriculture and small-scale manufacturing sector is often estimated using indicators such as **rainfall**, **area sown**. These often lead to approximation and extrapolation errors in the calculation of GDP.
5. **Double Counting Risk-** Complexity in calculations can lead to double counting of goods/services in intermediate stages if not adjusted properly.

Issues with GDP as an indicator of growth

1. **Exclusion of Non-Market Activities-** GDP ignores **unpaid domestic work**, **volunteer work**, **childcare**, **growing personal food**, and **informal economic activities** impacting real economic well-being.
2. **Income Distribution is ignored-** GDP does not reflect income inequality or distribution among different population groups and regions. India's per capita income is $2,610 vs world average $13,300.

3. **Non-accounting of Social and Environmental Costs-** Economic growth measured by GDP includes activities that may degrade the environment or social welfare.
4. **Lag in Reflecting Economic Shocks-** GDP Quarterly estimates often fail to accurately capture sudden economic shocks or changes due to reliance on annualized data and projections.
5. **Incomplete Well-being-** GDP measures the overall economic size but fails to capture unemployment, inequality, poverty and hunger.

2. GDP VS GVA

The **key difference** between GDP and GVA lies in their **approach to measuring the economy- GDP** captures the total monetary value of all final goods and services produced within the country, while **GVA** quantifies the value added by producers after deducting input costs, offering a sector-wise perspective on economic activity.

Basics of GDP vs GVA

Gross Domestic Product (GDP)

GDP is the monetary measure of all "final" goods and services produced in a country during a specified period.

It is computed as:

- **GDP** = Private Consumption + Gross Investment + Government Investment + Government Spending + (Exports – Imports)
- It provides a macroscopic view, useful for cross-country comparisons and tracking overall economic performance.

Gross Value Added (GVA)

GVA is the value of **output minus the value of intermediate inputs across sectors**, i.e. it reflects the **net value generated by the production process**.

- It calculates the **same national income from the supply side**.

The **sectoral GVA** offers insights into **productivity and efficiency**, supporting targeted policy interventions.

It allows policymakers to **assess which sectors need incentives or stimulus** and **track sector-wise economic growth.**

GDP and GVA: Relationship

GDP is **derived from GVA data** with adjustments for taxes and subsidies.

The GDP and GVA are related by the following equation:

GDP = GVA + Net Taxes (Indirect Taxes - Subsidies)

If **net taxes** are **high**, **GDP exceeds GVA**.

- When **subsidies exceed taxes**, **GVA may exceed GDP**.

Importance: Consequently, when wanting to **analyze the real economic activity by sector or industry**, **GVA is preferable**, whereas **GDP complements this with a broader overview incorporating government fiscal measures.**

Core Differences between GVA and GDP

Aspect	GVA (Gross Value Added)	GDP (Gross Domestic Product)
Definition	Value of output minus intermediate consumption (costs of inputs/materials)	Total market value of all final goods and services produced within a country
Adjustment for Taxes and Subsidies	Calculated at **basic prices** (excludes taxes and includes subsidies on products)	Calculated at **market prices** (includes indirect taxes minus subsidies)
Focus	Production side; sector-wise contributions to economy	Aggregate economic output including tax effects
Measurement Purpose	Measures actual value addition by sectors, giving a clear view of productivity and sectoral health	Measures overall size and growth of economy incorporating tax-driven price effects
Policy Utility	Helps identify which sectors contribute most/value patterns among sectors	Guides macroeconomic policy broadly by reflecting overall economic size and demand
Scope of Price Measurement	Basic prices (prices received by producers)	Market prices (prices paid by consumers, including taxes)

3. POTENTIAL GDP

Potential GDP is the **level of economic output an economy can sustain** at **full capacity without triggering inflation**. At this level, **resources are efficiently utilized**, **employment is near full**, and **the economy operates smoothly**.

It **reflects India's long-term productive potential**, shaped by its labor force, capital, technology, natural resources, policies, and demographics.

Actual vs. Potential GDP

- **Actual GDP-** What the economy is producing right now.
- **Potential GDP-** What it could produce under normal conditions

If Actual > Potential → Overheating → Inflation rises
If Actual < Potential → Underutilization → Unemployment rises

Key drivers and Determinants of Potential GDP in India

1. **Labor Force-** A skilled, healthy workforce is the backbone of productivity. India's demographic advantage could drive growth and lead to the realisation of potential GDP.
2. **Capital Stock-** Investments in infrastructure, machinery, and equipment expand the economy's productive capacity. India's MSME sector, which **contributes around 30% to GDP**, is a key determinant of potential GDP in India.
3. **Technological Progress-** Adoption of **advanced technologies** boosts efficiency and quality.
4. **Natural Resources-** Access to **land**, **energy**, and **minerals** is essential for growth.
5. **Policy Environment and Demographics-** Supportive policies encourage investment and innovation and enable the achievement of potential GDP.

Challenges limiting India's Potential GDP

1. **Skill and Productivity Deficits**- A large portion of the workforce lacks employable skills to raise the GDP.

   According to the Mercer report, only **42.6% Indian graduates are employable**; and their **non-technical skills**, **creativity remains low**.

2. **Infrastructure Gaps**- Inadequate roads, power shortages, and weak logistics networks raise business costs and slow both industrial and agricultural growth.
3. **Technological shortfalls**- Slow adoption of advanced technologies and low investment in research and development restrict competitiveness and innovation.

   However, **India's R&D spending (~0.6–0.7% of GDP)** lags behind global peers such as **China (2.4%)** and **Israel (4.9%)**, constraining competitiveness.

4. **Sectoral Imbalances in employment-** Around **45% of India's workforce is in agriculture**, which **contributes only 15% to GDP**. High **informal employment (~90%)** further limits productivity, social security, and the economy's potential output.
5. **Political and Regulatory Hurdles- Unclear policies**, **complex labor laws**, and **bureaucratic delays** make **formal job creation difficult** and discourage foreign investment.

Conclusion

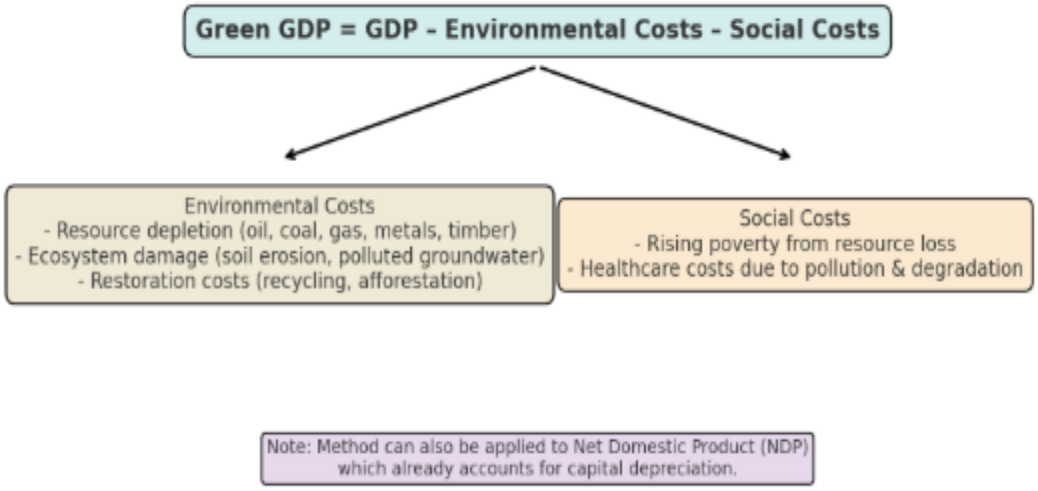
Accurately estimating potential GDP is challenging, and different methods may yield different results. However, it is widely recognized as a valuable tool for understanding the underlying trends and dynamics of an economy and making informed decisions that can positively impact the overall well-being of a society.

4. GREEN GDP

Green Gross Domestic Product (Green GDP) is an **enhanced measure of GDP** that **incorporates the environmental costs** of economic growth. Unlike conventional GDP, it accounts for factors such as **natural resource depletion**, **pollution**, and **ecological degradation**.

Green GDP provides a **more comprehensive assessment of a nation's overall well-being**, reflecting not only economic output but also the sustainability and long-term health of its environment.

Objectives of Green GDP: To integrate environmental costs into economic measurement by linking growth with ecological balance, highlighting resource depletion, guiding sustainable policies, encouraging green technologies, and ensuring intergenerational equity.



Difference between GDP and Green GDP

Aspect	GDP	Green GDP
Environmental Impact	Focuses **solely on economic activity**. It ignores environmental degradation and resource depletion.	Accounts for **negative environmental impacts of economic activities**, reflecting the sustainability of growth.
Resource Depletion	**Does not consider the consumption** or **exhaustion of natural resources**.	Deducts the **economic costs of natural resource depletion**, giving a sustainable perspective on progress
Pollution and Waste	**Ignores the economic cost of pollution, waste**, and environmental damage.	Subtracts losses due to **pollution and environmental harm**, providing a more accurate picture of well-being.
Economic Indicators	Serves as the **primary measure of economic performance** and growth	Assesses the **long-term sustainability of growth** by incorporating environmental health.
Policy Implications	Can **encourage growth-oriented policies** that overlook ecological limits.	**Promotes policies that balance economic development** with environmental preservation
Public Perception	Widely **recognized and understood as the standard measure** of economic performance	Adjusts **market value by factoring in environmental costs** for a holistic view of economic progress

Significance of Green GDP

1. **Preservation of Ecosystem Services-** Green GDP takes into account the loss of forests, clean water, and fertile soil resources which are life-support systems for future generations.

   > According to Global Forest Watch, **India lost 2.33 million hectares of forest** cover (2001–2023) , directly affecting ecosystem services.

2. **Addressing Climate Change-** By counting the costs of emissions, it nudges economies toward renewables and low-carbon growth.
3. **Conserving Biodiversity-** Green GDP ensures that rapid development does not come at the cost of species extinction and fragile habitats.

   > India's **10 biodiversity hotspots are under threat**, especially in Aravallis and Western Ghats due to mining.

4. **Promoting Resource Efficiency-** Encourages smarter use of materials, reducing waste and pollution.
5. **Building Economic Resilience-** It helps economies prepare for climate shocks like droughts, floods, and heatwaves.

   > According to the Asian Development Bank's (ADB) Asia-Pacific Climate Report, India faces a **potential 24.7% loss in GDP by 2070** due to climate change if no action is taken

6. **Safeguarding Human Health-** By accounting for pollution-linked costs, Green GDP highlights the link between economy and public health.

Air pollution caused **1.67 million deaths in India in 2020 (Lancet)**.

Challenges with Green GDP

1. **Complex calculation-** Assigning a monetary value to biodiversity, clean air, water, and ecosystem services remains difficult. **No universally accepted method exists** for **valuing forest carbon sequestration**, **wetlands**, or **urban green spaces**.
2. **Data Limitations-** Environmental data is **often fragmented**, **outdated**, and **lacks regional granularity**.

   The State of India's Environment in Figures 2025 report, highlights that **data gaps remain a major concern** in assessing the **country's environmental**, **agricultural**, and **public health landscape**.
3. **Methodological Ambiguity-** Different valuation approaches lead to inconsistent Green GDP estimates.

   China's **Green GDP pilot Program (2004–07) was discontinued** due to inconsistent results and policy resistance.
4. **Institutional Gaps-** India lacks a dedicated mechanism to compile environmental-economic data.

   The National Statistical Office (NSO) **primarily focuses on conventional GDP metrics**.
5. **Challenges in Long-Term Integration-** Integrating Green GDP with natural, human, and social capital indicators is complex, as **India's Natural Capital Accounting (NCA) framework** is **still under development** and **has not been fully implemented**.

Green GDP Global Practices

- **New Zealand**: Developed the Living Standards Framework to integrate social, environmental, and economic well-being into policy decisions.
- **Bhutan**: Uses the Gross National Happiness (GNH) framework, making ecological sustainability a central aspect of overall development.
- **United Nations Environment Programme (UNEP):** The Green Economy Initiative (GEI) encourages policymakers worldwide to invest in environmentally sustainable projects.

INFLATION

1. INFLATION MANAGEMENT

Inflation management refers to how governments and central banks contain the rise in prices to maintain economic stability, protect purchasing power, and support sustainable growth.

In India, inflation is mainly managed by the Reserve Bank of India (RBI) through monetary policy, supplemented by fiscal, administrative, and supply-side interventions.

Instruments of inflation management

Monetary Policy (by RBI)

- **Repo Rate** ↑ (borrowing cost ↑) → reduces money supply → controls inflation.
- **Repo Rate** ↓ (borrowing cost ↓) → increases money supply → boosts growth.

- **CRR (Cash Reserve Ratio)** & **SLR (Statutory Liquidity Ratio)-** Adjust liquidity with banks.
- **Open Market Operations (OMO)-** RBI buys/sells government securities to regulate money supply.
- **Inflation Targeting-** India follows a **flexible inflation targeting regime** (4% ± 2%).

Supply Side Measures

- **Boost agricultural output** (irrigation, storage, MSP reforms).
- **Reduce supply bottlenecks** (transport, logistics, imports of essential goods).
- **Export restrictions** on essential commodities (like wheat, rice, onions).
- **Strengthen supply chains** to control food inflation.

Administrative Measures

- **Price controls & anti-hoarding laws** (check black marketing).
- **Buffer stock management** by FCI for food grains.
- **Regulating essential commodities** under Essential Commodities Act.

Objectives of inflation management

1. **Price Stability-** Maintain a **predictable** and **moderate rate of inflation** to avoid excessive price fluctuations, which is crucial for household budgeting and business planning.
2. **Safeguarding Purchasing Power-** Prevent **erosion of real income** for salaried employees, pensioners, and low-income groups, as **unchecked inflation disproportionately impacts the poor** and **vulnerable**.
3. **Supporting Sustainable Economic Growth-** Balance **inflation control with the need for investment and job creation**. Extremely **tight policies can stifle growth**, while **loose policies can trigger runaway inflation**.
4. **Ensuring Financial Stability- Stabilize domestic currency** and **control interest rates to limit risks** such as asset bubbles and harmful volatility in financial markets.
5. **Policy Coordination-** Coordinate **monetary**, **fiscal**, and **trade policies to address structural supply shocks** and **volatile commodity and food prices**, which are frequently significant factors in India.
6. **Economic reality-** In India, food accounts for nearly 50% of Indian household expenditure. It is crucial to most people's cost of living. Ignoring food prices in inflation targeting would lead to neglect of a major economic concern for a large portion of the population.

Challenges in inflation management

1. **Monetary Policy singular focus on demand side-** RBI's monetary policy **targets only demand side constraints**. It faces the **problem of tackling supply shocks** originating from **food** and **oil**.
2. **Flawed Model of Inflation targeting-** Monetary Policy model used for Inflation management in India is not statistically validated for Indian data. The current model of Inflation targeting is **based on the assumption that inflation means overheating the economy**- that is increased output greater than natural level output.
   - However, In India it is **impossible to observe the actual level of output in an economy**. Hence, setting policy rates based on the assumption that the economy has overheated is unscientific.
3. **Failure in addressing supply shocks-** Adoption of a myopic vision in inflation management by **focussing on export ban of agricultural products (like wheat, rice, onions), leads to increased inflation**.
   - Export bans induce fear and panic in the domestic market, leading to rise in stock holdings, which ultimately result in price rise.

4. **Ignoring the Global Nature of inflation-** Inflation is global in nature, as the **price level of a good is determined by millions of producers across the world**. Hence, solely targeting inflation management is not good for the health of the economy, as certain prices of goods are beyond our control.
5. **Exclusive Focus on Inflation slows down growth-** RBI's current mandate of inflation management is **too singularly focused on controlling inflation**. Inflation management has **negatively impacted GDP growth**. High policy rates (repo) maintained to control inflation affected the cost of domestic capital. It led to a decline in investment rate, thereby resulting in less GDP.

> Since 2016 **(after inflation rate targeting was institutionalised)**, there has been a **steady increase in repo rates**, and a **steady decline in GDP growth**.

6. **Interdependence of Food and Core Inflation- Food prices influence wages**, which in turn **impact core inflation**. Hence, it would be difficult to control core inflation independently of food prices.
7. **Ineffectiveness of Interest Rate Adjustments-** Raising interest rates **have not led to curbing of core inflation** but **has instead exacerbated it by increasing the costs for firms**. This has led to higher prices of the products and higher inflation in the economy.
8. **Misguided Policy-** Exclusion of food prices from the inflation target could leave India vulnerable to rising food costs. This would undermine the standard of living for a large segment of the population.

Way forward for Inflation Management in India

1. **Increasing agricultural production-** Focus on improving agricultural productivity and controlling food prices through supply-side measures to address inflation in India.
2. **Release Excess Buffer Stocks-** The government holds more than **40 million tonnes of rice**, much above the **buffer stock norms of 13.5 MT**. This excess stock should be unloaded by the Food Corporation of India in the open market at reasonable prices. This will cool down food inflation.
3. **Enhancement of Processing Capacity-** About 10-15% of **perishable items like tomatoes and onions** should be processed. The availability of alternatives like tomato paste and onion powder will help to stabilize prices.
4. **Adjustment of Import Duties-** Import duties on items like wheat should be reduced, as cheaper imports can help control domestic prices.
5. **Greater Tolerance of Higher levels of Inflation-** Since inflation is a global issue, there must be greater tolerance for higher levels of inflation either by adjusting the acceptable range of inflation upwards, or by extending the period over which the MPC has to meet its inflation target.

Conclusion

Inflation management is a very important policy tool which should not only be limited to price control but shall be central to social well-being, macroeconomic stability, and achieving India's long-term development targets.

2. MONETARY POLICY

Monetary policy

Monetary policy refers to the actions and strategies used by a central bank (RBI) to manage money supply, interest rates, and credit in the economy, ensuring price stability, supporting economic growth, and ensuring financial stability.

The Reserve Bank of India (RBI) is vested with the responsibility of conducting monetary policy. This responsibility is explicitly mandated under the Reserve Bank of India Act, 1934.

Types of monetary policy

There are two types of monetary policy

- **Contractionary/Dear money policy:** It seeks to reduce the money supply in the economy. Under this policy:
  - These rates are **increased**: CRR, SLR, Bank Rate, Repo Rate, Reverse Repo Rate, MSF.
  - Under OMO, RBI will sell g-secs.
- **Expansionary/Cheap money policy:** It aims to increase the money supply in the economy. It is usually followed when RBI wants to push economic growth or counter recession. Under this policy:
  - These rates are **decreased**: CRR, SLR, Bank Rate, Repo Rate, Reverse Repo Rate, MSF.
  - Under OMO, RBI shall purchase G-secs.

Objectives of monetary policy

- **Price Stability:** Target headline inflation at 4% (±2%) to maintain stable prices for goods and services
- **Growth and Employment:** Support conditions for sustainable economic growth and job creation by adjusting interest rates, money supply, and credit flows.
- **Financial Stability:** Prevent excessive volatility in financial markets and banking systems. Manage liquidity, supervise banks, and avoid asset bubbles.
- **Exchange Rate Stability:** Manage the external value of the rupee through intervention in the foreign exchange market, mitigating excessive volatility, and ensuring competitive export/import sectors.
- **Interest Rate Stability:** Curb undue fluctuations in interest rates, ensuring certainty for businesses and consumers in planning investments and purchases.
- **Support Priority Sector:** Ensure affordable credit to agriculture, small industries, housing, and other priority sectors to promote inclusive growth and poverty reduction.

Monetary Policy Tools

Quantitative tools	**1. Reserve Ratio**: a. Cash Reserve Ratio (CRR) b. Statutory Liquidity Ratio (SLR) **2. Open Market Operation (OMO)** **3. Rates**: **a. Bank rate** **b. Liquidity Adjustment Facility (LAF)** ▪ Repo Rate ▪ Reverse Repo Rate **c. Marginal Standing Facility (MSF)** **4. Market Stabilisation Scheme (MSS)**
Qualitative Tools	• Credit rationing • Margin requirements • Other measures ▪ Moralsuasion ▪ Direct Action

Monetary Policy Committee

- The Monetary Policy Committee (MPC) is constituted by the Central Government under Section 45ZB of the RBI Act 1934.
- The Reserve Bank's Monetary Policy Department (MPD) assists the MPC in formulating the monetary policy.
- **Situation before MPC**- Before the constitution of the MPC, a **Technical Advisory Committee (TAC) on monetary policy** with experts from monetary economics, central banking, financial markets, and public finance, used to **advise** the Reserve Bank on monetary policy. With the formation of MPC, the TAC on Monetary Policy **ceased to exist.**
- **Structure of MPC**- 6 members
  - Governor (Chairperson of MPC)
  - Deputy Governor (in-charge of monetary policy)
  - One officer of the Reserve Bank of India to be nominated by the Central Board – Member, ex officio
  - 3 members to be nominated by the government (These members will hold office for a period of **four years or until further orders**, whichever is earlier)
- Each member of the MPC has **one vote**, and in the event of an equality of votes, the Governor has a second or **casting vote**.
- Under the flexible inflation targeting (FIT) framework, the "repo rate" is the policy rate. It is determined by the MPC based on the assessment of the macroeconomic condition and with the aim to keep CPI inflation near 4%+/-2%, as discussed above
- Currently, MPC is only responsible for setting policy (repo) rate and performing the liquidity operations to operationalize the monetary policy lies with the RBI.

Significance of monetary policy

1. **Price Stability & Inflation Control:** By targeting inflation and managing money supply through rates, the RBI preserves the purchasing power of citizens and prevents economic disruptions.
2. **Supports Sustainable Economic Growth:** The RBI ensures adequate credit availability for productive sectors, enabling investment, entrepreneurship, and employment generation, which drive India's development.
3. **Financial System Stability:** Monetary policy helps prevent financial crises, liquidity shortages, and systemic risks by actively overseeing and regulating banks, NBFCs, and financial institutions.
4. **Exchange Rate Stability-** The RBI manages the value of the rupee to avoid wild fluctuations, supporting international trade, controlling import/export costs, and maintaining investor confidence.
5. **Interest Rate & Credit Flow Management:** By influencing lending rates and credit conditions, the RBI supports business planning, housing markets, consumption, and investment.

Challenges or limitations of monetary policy

1. **Monetary Policy Transmission-** Reductions or hikes in the RBI's policy rates (repo rate) do not always reflect quickly or fully in the lending rates offered by banks due to inflexible deposit costs, high small savings interest rates, and administered pricing.
2. **Inflation Targeting Constraints-** Monetary policy is effective mainly for demand-driven inflation, but much of India's inflation is supply-side (especially food and fuel), which the RBI cannot address directly.
3. **Incoherence between Fiscal & Monetary Policy**
   - High fiscal deficit and government borrowing **can crowd out private investment** and **reduce the effectiveness of monetary policy**.

- **Price controls, subsidies, and MSP (Minimum Support Price) interventions** can conflict with policy aims, limiting the RBI's independence and flexibility.

4. **Policy Rate Rigidity & Administered Rates-** Administered rates on small-savings greatly influence household deposit decisions, causing banks to hold their deposit and lending rates steady even when RBI signals rate changes.
5. **Structural Rigidities & Market Constraints-**
   - India's financial markets are still evolving. Limited integration, underdeveloped bond markets, and preference for cash transactions reduce the impact of policy tools.
   - The large informal sector, which operates outside formal banking, means monetary policy changes might not reach a significant portion of the economy.
6. **External Factors-** Factors such as capital flows, global recession, US monetary policy, and geopolitical risks can impact India's currency, capital markets, and inflation, often beyond RBI's control.
7. **Conflicting Objectives-** The need to simultaneously control inflation and promote growth can create trade-offs. Expansionary policies boost growth but risk inflation, while contractionary policies fight inflation but slow growth.

Way forward

1. **Strengthen Monetary Policy Transmission:**
   - Improve the transmission mechanism so changes in the RBI's policy rates reflect promptly and effectively into bank lending and deposit rates.
   - Encourage banking sector reforms to reduce NPAs, improve competition, and lower fixed costs so banks can adjust rates dynamically.
2. **Enhance Coordination between Fiscal & Monetary Policies:**
   - Improve synchronization between RBI and government budgetary policies to reduce fiscal dominance and its adverse effects on inflation and interest rates.
   - Rationalize subsidies and move towards better-targeted support to limit distortions in price signals.
3. **Target Broader Inflation Measure Responsibly:**
   - Balance headline and core inflation targeting, ensuring policymakers do not overlook essential goods impacting the poor.
   - Build public understanding and set expectations for acceptable inflation bands tailored to India's developmental context.
4. **Strengthen Regulatory Framework & Digital Infrastructure:**
   - Expand digital payments infrastructure to improve monetary policy transmission through formal financial channels.
   - Strengthen regulation and oversight of NBFCs and microfinance institutions to ensure credit flow and financial stability.
5. **Enhance Global Risk Management:** Develop frameworks to better manage capital flow volatility, exchange rate fluctuations, and external shocks through a mix of monetary, fiscal, and macroprudential tools.
6. **Foster Transparency & Communication:**
   - Improve RBI's communication strategies to build market confidence and effectively anchor inflation expectations.
   - Publish detailed monetary policy reports, market surveys, and data releases for greater accountability.

Conclusion

Monetary policy in India thus aims to maintain stable price levels, promote growth, and support broad-based, inclusive financial sector development, ensuring that its benefits reach every section of society.

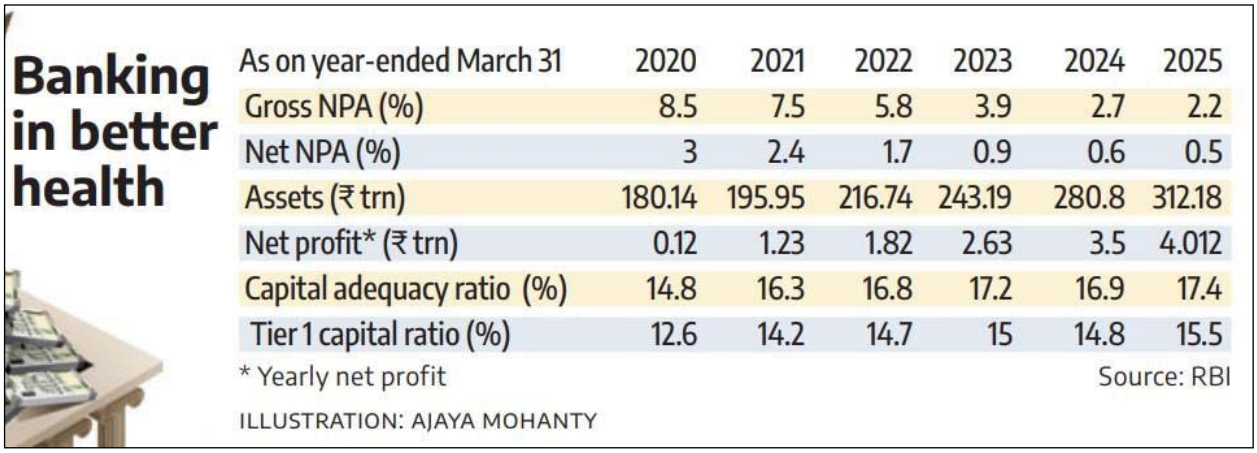
MGP/BP/GS3/003

BANKING AND FINANCIAL SECTOR

ISSUES OF BANKING SECTOR

1. ISSUES WITH SCHEDULED COMMERCIAL BANKS (SCBS)

The health of the Scheduled Commercial Banks have improved in the past decade. SCBs have been able to overcome the **twin-balance sheet problem**. There has been improvement in their NPAs, Net profit and Capital Adequacy Rations (CARs).



Banking in better health

As on year-ended March 31	2020	2021	2022	2023	2024	2025
Gross NPA (%)	8.5	7.5	5.8	3.9	2.7	2.2
Net NPA (%)	3	2.4	1.7	0.9	0.6	0.5
Assets (₹ trn)	180.14	195.95	216.74	243.19	280.8	312.18
Net profit* (₹ trn)	0.12	1.23	1.82	2.63	3.5	4.012
Capital adequacy ratio (%)	14.8	16.3	16.8	17.2	16.9	17.4
Tier 1 capital ratio (%)	12.6	14.2	14.7	15	14.8	15.5

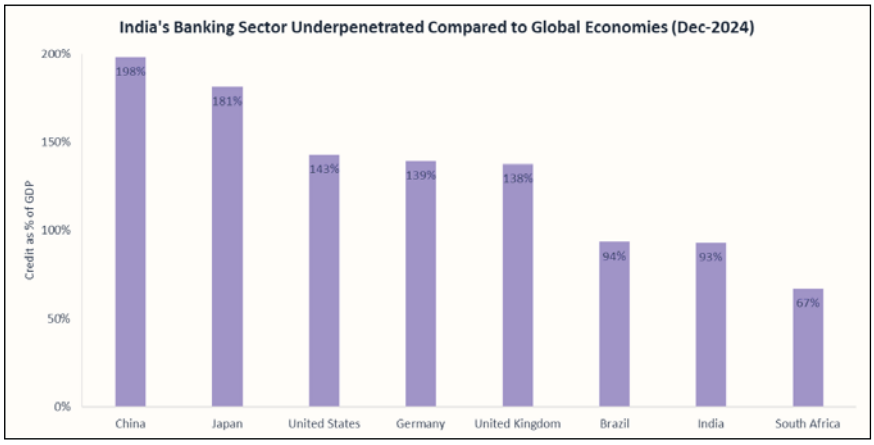
* Yearly net profit

Source: RBI

ILLUSTRATION: AJAYA MOHANTY

Challenges faced by Scheduled Commercial Banks

1. **Low Credit to GDP Ratio-** India had a **Credit-to-GDP ratio of 93%**, which is one of the lowest among global peers. A low Credit-to-GDP ratio indicates underpenetration of financial credit in the country.



2. **Increasing Credit-Deposit Ratio-** Commercial banks are facing increasing challenges in mobilising resources as credit growth continues to outpace deposit growth, pushing the **credit-deposit ratio beyond 80 per cent**.
3. **Changing Household Savings-** Depositors are increasingly **shifting funds from traditional bank deposits** to **capital markets**, **mutual funds**, and **insurance**. This has created a **structural liquidity gap in the banking sector.**
4. **Interest Rate Sensitivity-** Scheduled commercial banks are relying on **short-term borrowings** and **certificate of deposits (CD)**, which has **increased their vulnerability to interest rate fluctuations**.
5. **Over-exposure to Retail loans-** There is growing regulatory concern regarding **unsecured retail credit**, particularly small-ticket consumer loans.

   Consumer durables and **credit card receivables** have shown **weakening asset quality** in 2025.

6. **Bad Bank Delays-** The **National Asset Reconstruction Company Ltd. (NARCL) continues to face hurdles in asset valuation** and **transfer delays for the ₹2 trillion target** of stressed assets.
7. **Unmet Credit Demand for the MSME Sector-** According to a 2022 Lok Sabha Standing Committee report, the unmet credit demand of MSMEs is ₹25 trillion, which constitutes 47% of their borrowing needs.
   - However, banks meet only 15% of MSME credit requirements, leaving a significant gap in funding.
8. **Dependence on Government Support-** Public sector banks rely heavily on government capital infusion to meet Basel norms, which hampers their autonomy and efficiency.
   - The Government has infused **Rs. 3,10,997 crore** to **recapitalise banks** during the last five financial years from **2016-17** to **2020-21**.
9. **Digital Frauds-** Indians lost **Rs 22,845.73** crore to cyber criminals in **2024**, marking a sharp **206% surge** from the **Rs. 7,465.18 crore** in 2023.
10. **Increase in Mule accounts-** There has been tremendous increase in mule accounts (**24 lakh mule accounts**) to siphon fraudulent funds.
11. **Staff Vacancies-** PSBs have roughly **33,000 vacancies as of mid-2025**. There has been an increasing trend of contractualisation of banking jobs. Banks have employed **1.01 lakh contractual workers for non-core activities**.
12. **Board Vacancies-** Approximately **42% of director positions across 12 PSBs** remained vacant impacting top-level governance.

Committee Recommendations for Improving the functioning of SCBs

P. J. Nayak Committee (2014)	**Reduce government interference** in Public Sector Banks (PSBs) Create a **Bank Investment Company** (BIC) **Professionalize bank boards**.
First Narasimhan Committee Report (1998)	Recommended **'Narrow Banking Concept'** where weak banks will be allowed to place their funds only in the short term and risk-free assets. **Faster computerization**, technology up gradation, training of staff, **depoliticizing of banks**, **professionalism in banking**, reviewing bank recruitment.
First Narasimhan Committee Report (1991)	Establishment of **4 tier hierarchy for banking structure** with **3 to 4 large banks (including SBI) at the top** and **at bottom rural banks** engaged in agricultural activities.

	The **supervisory functions over banks and financial institutions** can be assigned to a **quasi-autonomous body sponsored by RBI**. Setting up an **Asset Reconstruction fund** to **take over a portion of the loan portfolio of banks** whose recovery has become difficult.

Government Initiatives for Banking Sector Reforms

Indradhanush Plan (2015)	Aimed at improving the performance of Public Sector Banks (PSBs) through reforms in areas such as appointments, capital infusion, and governance. • Introduced the **Bank Boards Bureau (BBB)** to recommend appointments for top bank positions and provide strategic advice
Recapitalization of Banks	Significant capital infusion by the government. This included the **₹2.11 lakh crore recapitalization plan** in 2017 to strengthen PSBs' balance sheets.
IBC (Insolvency and Bankruptcy Code) 2016	It was introduced for faster resolution of stressed assets. It enabled banks to recover bad loans efficiently and strengthen credit discipline.
PSB Amalgamation Drive	Consolidation of weaker PSBs with stronger ones to create larger, more efficient entities **(for ex- the merger of Bank of Baroda, Vijaya Bank, and Dena Bank in 2019).** • It aimed at improving scale, efficiency, and reducing redundancies.
Bad Bank (National Asset Reconstruction Company Limited)	Established to take over large stressed assets and facilitate their resolution, enabling banks to focus on core activities like lending. Complements existing Asset Reconstruction Companies (ARCs).
Digital Banking Push	Initiatives like the Unified Payments Interface (UPI), e-RUPI, and digital-only banking units to promote cashless transactions and financial inclusion.

Futuristic Banking Sector Reforms

1. **Strengthen Asset Quality and Risk Management**
   - **Improve credit appraisal-** Deploy AI/analytics for early warning signals, especially in retail/MSME segments.
   - **Improve Counter-cyclical provisioning-** Build buffers in good years to withstand downturns.
   - **Deepen resolution frameworks-** Strengthen Insolvency & Bankruptcy Code (IBC) timelines and enforcement to reduce haircuts.
2. **Enhance Capital and Governance in PSBs**
   - **Professionalise PSB boards-** Reduce political interference, strengthen independent oversight.
   - **Capital infusion linked to performance-** Recapitalisation support should be conditional on reforms (efficiency, tech adoption, governance).
   - **Privatisation/consolidation-** Gradual reduction of government stake in stronger PSBs to enhance market discipline. Reduce government stakes below 50%, offering greater autonomy, and ensuring professional management.
   - **Implement provisions from the Banking Laws (Amendment) Act, 2025-** This act tightens governance, mandates better audit quality, enhances depositor protection, and improves board structure in public sector and cooperative banks.
3. **Improve Profitability and Efficiency**
   - **Cost rationalisation-** Adopt shared services, digital-first operations, and branch rationalisation where feasible.

- **Diversify income-** Expand fee-based services (wealth management, insurance, payments). Strengthen treasury and risk products to manage interest-rate risks and volatility.

4. **Digital & Cybersecurity Transformation**
   - **Upgrade core IT systems-** Move away from legacy infrastructure to cloud-native, secure platforms.
   - **National cyber defence centre for banks-** Real-time threat intelligence sharing across institutions.
   - **Strong regulation for fintech partnerships-** standardised security and consumer-protection norms.
5. **Competition and Collaboration with Fintechs**
   - **Open banking framework-** Allow secure data-sharing with customer consent (like Europe's PSD2).
   - **Sandbox expansion-** RBI to support more pilots in AI-driven credit, blockchain-based settlements, and green finance.
   - **Partnership models-** Co-lending with fintechs for MSME/retail loans to combine scale and innovation.
6. **Financial Inclusion and MSME Credit Access**
   - **Data-driven MSME lending-** Leverage GST, UPI, and digital footprints for credit scoring.
   - **Priority sector reforms-** Rationalise targets and incentivise innovative lending models. Strengthen credit guarantee schemes to reduce lender risk in small-ticket loans.
   - Institutionalize the **Grameen Credit Score** for rural and underserved borrowers to boost formal credit outreach.
7. **Climate & Environment Social and Governance (ESG)-Related Reforms**
   - **Incentives for green lending-** Lower risk weights or refinancing lines for renewable projects.
   - **Climate stress testing-** Mandate banks to assess exposure to climate risks (floods, droughts, carbon transition).
8. **Regulatory and Supervisory Reforms**
   - **Unified regulatory framework-** Harmonise norms across banks, NBFCs, and fintechs to prevent regulatory arbitrage.
   - **Prompt corrective action (PCA) 2.0-** Make intervention rules more dynamic, focusing not only on capital/NPA but also governance and tech risks.
   - **Strengthen fraud detection-** Mandatory AI-based anomaly detection for large transactions.
9. **Human Capital and Culture**
   - **Skill development-** Continuous training in digital banking, risk analytics, ESG finance.
   - **Performance-linked incentives-** Reduce bureaucratic rigidity in PSBs.
   - **Ethics and governance culture-** Embed accountability and transparency across hierarchies.

Conclusion

Reforms in India's banking sector are critical to improving financial stability, enhancing credit accessibility, and fostering economic growth. By addressing governance challenges, expanding financial inclusion, and leveraging technology, the sector can become more resilient and competitive in a globalized economy.

2. ISSUES WITH PAYMENT BANKS (PBS)

Payment Banks in India were established in 2014 based on the recommendations of the **Nachiket Mor Committee**. It was set up to operate on a smaller scale with minimal credit risk.

The main objective is to **advance financial inclusion** by **offering banking** and **financial services to the unbanked** and **under-banked areas**. It also caters to the needs of **uncovered masses in the banking sector** like the **migrant labour force**, **low-income households** and **small entrepreneurs**.

Significance of Payments Banks

1. **Promotion of Financial inclusion-** These banks have **promoted financial inclusion** by catering to financial services to the **unbanked sections** of the society like **migrant workers**, **low-income households** and **small scale entrepreneurs**.
2. **Expanded Geographical reach-** These banks have a **wider geographical reach** due to its **technology oriented services**, unlike traditional banks whose geographical outreach is constrained by the requirements of physical infrastructure.
3. **Zero balance accounts and No minimum balance accounts-** These banks offer a **zero balance account** or a no minimum balance account without any extra or hidden charge, unlike a commercial bank who levy charges if the customer doesn't hold a minimum balance in their account.
4. **Complements the financial efforts of traditional banks-** Payments banks complement the financial efforts of traditional banks by **partnering to sell mutual funds, pension products, and insurance products** through their platform. **For ex- SBI Life Insurance product through Paytm.**
5. **Low Value, High Volume Transactions-** They have provided effective infrastructure to deal with low value, high volume transactions. **For Ex- Use of Paytm QR codes by the vegetable Vendors to grocery shop owners.**
6. **Higher rates of Interest-** The rates of interest being offered by these banks is higher in comparison to the traditional banks. **For Ex- ROI of Payment Bank is generally around 7%** whereas as **ROI of commercial bank ~3.5 and 6 per cent**.

Challenges with Payment Banks in India

1. **Low avenues of profitability-** These banks **are not allowed to lend money** and earn interest income unlike traditional banks. Further, the **stringent SLR requirements of 75% demand liabilities** to be invested in G-secs have impinged on the avenues of profitability of these banks.
   - **For ex-** The operational payments banks showed **net losses of Rs 516.5 crore** for the financial year 2018.
2. **Low return on equity-** The **cap on the amount of demand deposits at Rs 2,00,000** and the **15% capital to Risk Weighted Assets ratio**, has severely impacted the returns on equities of payment banks in India. Their **return on equity is less than 5%**.
3. **Digital Divide and slow internet connectivity-** These banks have no physical presence and their banking operations are solely reliant on internet connectivity. However, the **rural-urban divide in internet connectivity** has impacted their expansion and penetration.
4. **Large number of dormant accounts-** The large number of **dormant zero balance** accounts have impacted the operations of payment banks in India. They have also been used as conduits for personal loan scams and Money Laundering.
   - **For ex-** Out of the **35 crore Paytm payment bank accounts**, **31cr remained dormant** and were misused.
5. **Cut-throat Competition-** These banks are facing cut-throat competition from the payment wallets like **Phone pe**, **Google Pay** and conventional commercial banks payment bank services like SBI yono, ICICI i mobile pay.
6. **Increasing number of Defunct Payment Bank-** The over-regulated functioning and huge losses, have led to increase in the number of banks surrendering their licences and halting their operations.
   - **For ex-** Cholamandalam Distribution Services, Sun Pharmaceuticals, Tech Mahindra and Aditya Birla Payment Bank have surrendered their licences.

Way Forward

1. **Open up more avenues of profit generation-** RBI must increase deposit limits of payment banks. Also, a mechanism must be worked out to let these banks transfer the surplus money in the demand deposit accounts to the universal banks.
2. **Facilitate Infrastructure sharing-** RBI should take measures to facilitate infrastructure sharing among the traditional banks and Payment Bank. **For ex-** Payment bank desks in traditional bank branches.
3. **Increased Internet connectivity-** The internet connectivity in rural areas must be increased for the entry of new players in the payment bank market as the payment bank sector is dominated by telecom giants like Airtel and Jio which have their own network.
4. **Increase the scope of operations-** The **payment bank should be allowed to offer their own mutual fund and insurance products** to enhance their source of revenue generation and profitability.
5. **Increase regulatory vigilance-** The compliance of e-KYC and no frill accounts must be regularly undertaken by the RBI to prevent future crises like the Paytm crisis.

> The recommendations of the **Anand Sinha committee** must be implemented to **ring-fence the banking operations of the payment bank from the ownership structure**.

3. ISSUES WITH COOPERATIVE BANKS (CBS)

Cooperative banks in India are financial institutions that operate on a cooperative basis, where the customers are also the owners of the bank. Cooperative banks operate on the principle of '**no profit, no loss**' and '**one person, one vote**'. According to the Reserve Bank of India (RBI), India had **1,502 urban co-operative banks** as of March 2023.

Structure of Cooperative Banks in India

<table>
<tr><th colspan="4">Cooperative Banks In India</th></tr>
<tr><th colspan="2">Urban Cooperative Banks</th><th colspan="2">Rural Cooperative Banks</th></tr>
<tr><th>Scheduled</th><th>Non-Scheduled</th><th>Short Term</th><th>Long Term</th></tr>
<tr>
<td colspan="2">1. UCBs Operating in a single state
a. Registered- State Cooperative Societies Act.
b. Dual Control
Banking functions– RBI under Banking Regulation Act 1949 (BRA 1949).
Management functions– State registrar of Cooperative societies
2. Multistate UCBs
a. Registered- Multi State Cooperative Societies Act, 2002.</td>
<td>1. Primary Agricultural Credit Societies (PACS)- Village level- Not under RBI. No BRA 1949 applies.
2. Central Cooperative Banks (CCBs)- District level &
State Cooperative Banks (StCBs)-State level
a. Registered- State Cooperative Societies Act
b. Dual Control
Banking functions– RBI under</td>
<td>1. Primary cooperative agriculture and Rural Development Banks (PCARDBs)
2. State Cooperative agriculture and Rural Development Banks (SCARDBs)</td>
</tr>
</table>

b. **Dual Control** **Banking functions**– RBI under Banking Regulation Act 1949. **Management functions-** Central Registrar of Cooperative societies.	Banking Regulation Act 1949. The inspection function has been delegated to NABARD by RBI under BRA 1949. **Management function-** State registrar of Cooperative societies	

Significance of Cooperative Banks

1. **Extension of Credit to Agriculture and Rural Sectors-** PACS, District Central Cooperative Banks (DCCBs) and State Cooperative Banks (SCBs) play a vital role in providing credit to farmers and supporting agricultural activities. They contribute significantly to the development of the rural economy.
2. **Spurring local MSME and SHG growth-** Urban Cooperative Banks (UCBs) cater to the financial needs of small and medium-sized businesses and individuals in urban and semi-urban areas.
   - **For Ex-** Lijjat papad funding by cooperative banks.
3. **Ownership and Control-** Members of a credit cooperative society collectively decide on policies and elect the board of directors. This results in democratic ownership and control.
   - **For Ex- Greater Political empowerment** in areas with strong cooperatives like Maharashtra, Kerala and Gujarat.
4. **Customer-Centric Approach-** Cooperative banks prioritize the needs of their members rather than maximizing profits.
   - **For ex- Personalized loan products** like customized savings products, affordable housing loans tailored to the specific needs of their members.
5. **Community Development-** Cooperative banks support local infrastructure projects, educational initiatives, or healthcare programs. This involvement strengthens the socio-economic fabric of the community.
   - **For ex- Role of Cooperative Banks in development** o**f Sikkim**.
6. **Risk Mitigation-** Cooperative banks are slightly more resilient to economic downturns as they are not exposed to high-risk and high-value assets in comparison to commercial banks.
   - **For Ex-** Resilience shown by UCBs during Global Financial Crisis 2008.

Challenges of Cooperative Banks in India

1. **Financial Scams-** Large numbers of big Cooperative banks have failed due to financial scams.
   - **For Ex-** PMC Bank, Guru Raghavendra Cooperative Bank and Maharashtra State Cooperative (MSC) Bank have failed due to financial frauds.
2. **Board members as borrowers-** Cooperative bank board members can borrow from the banks, unlike the commercial bank board members. The board members in several cases have misused their borrowing powers to siphon off large sums of money, resulting in major cooperative bank failures.
   - **For Ex-** PMC Bank Failure due to misuse of power by board.
3. **Political Connivance and corruption-** The board of the cooperative banks are dominated by the local politicians, who connive with local private financiers and real estate players to issue illegal loans and aid their black money transactions.
4. **Financial Vulnerabilities-** Cooperative Banks face financial vulnerabilities such as low capitalization, high levels of NPAs, low Capital Adequacy Ratio (CAR).
5. **Challenges posed by other financial sector companies-** Changes in the financial sector and evolution of Microfinance Institutions (MFIs), FinTech companies, Payment gateways, Non-Banking

Financial Companies (NBFCs) have posed challenges to the Cooperative bank growth in terms of deposit and lending capabilities.

6. **Regulatory Challenges-** The dual control of cooperative banks- RBI in banking functions and the state government in management functions, has created regulatory ambiguities and uncertainties, in proper monitoring of cooperative banks.
7. **Improper audit mechanism-** Audits conducted by state department officials are irregular and lack comprehensiveness.
8. **Infrastructural Challenges-** The cooperative banks face logistical challenges like sub-standard softwares and sub-standard booking keeping systems, which has made these banks susceptible to frauds.
9. **Governance Challenges-** The small size, scattered nature and the lack of a unified brand and policy for cooperative banks makes it difficult for the watchdogs to effectively govern these banks.

Way Forward

1. **Strict Regulatory Action by RBI-** RBI must be stringent in its regulatory oversight by focussing on regular delicensing and the shrinkage or compulsory amalgamation of loss making cooperative banks.
2. **Cooperative Federation-** A cooperative federation must be formed to conduct regular comprehensive audits of the cooperative banks.
3. **Upgradation of infrastructure-** The provisions must be made for a common, standardized software, standardized bookkeeping systems. These must be linked to a central database for proper financial monitoring using artificial intelligence and pattern recognition.
4. **Remove political influence-** There is a need to bring in new people, young people and professionals in managerial roles, who will take cooperative banks forward.

Committee Recommendations for Urban Cooperative Banks in India

Implement the recommendations of N.S. Vishwanathan Committee-

a. Four tier classification (Tier 1 (having deposits up to Rs 100 crore), Tier 2 (deposits between Rs 100 crore and Rs 1,000 crore), Tier 3 (deposits between Rs 1,000 crore and Rs 10,000 crore), Tier 4 (deposits more than Rs 10,000 crore)),

b. Formation of board-level Committees, constitution of professional board of management and fixed tenure of board members recommended by the committee must be implemented at the earliest.

Implement recommendation of R Gandhi Committee

Conversion of Urban Cooperative Banks (UCBs) with business size of 20,000 crore rupees or more into regular banks.

4. DECLINING DEPOSITS

Indian banks are facing a significant **deposit growth slowdown**, even as **credit growth remains robust**. According to RBI data (June 2024), deposits grew **11.7%**, while bank credit rose **15%**, creating a widening **credit-deposit gap**. This imbalance raises concerns about **asset-liability mismatches**, **liquidity risks**, and the ability of banks to sustain credit expansion — crucial for India's growth story.

Reasons for Slowing Deposit Growth

1. **Shift of Household Savings to Capital Markets**
   - **Retail investor participation** in equity markets has surged post-pandemic due to higher returns and ease of access.

- **Demat accounts** rose from **11.45 crore (FY23)** to **15.14 crore (FY24)**.
- **Mutual Fund Assets under Management (AUM)** reached a record **₹64.97 lakh crore (July 2024)**; SIP accounts at **9.33 crore**.
- This diversion of savings away from bank deposits has slowed deposit mobilisation.

2. **Negative Real Returns in the Past**
   - For decades, savings deposits earned **3–5%**, while inflation averaged **6.6%**, yielding **negative real returns**.
   - Savers have now experienced better returns from equity, debt mutual funds, and other instruments — making the shift partly structural.
3. **CASA Decline**
   - **Current Account + Savings Account (CASA)** share dropped to **41%** (from 43% YoY).
   - CASA deposits are **low-cost, stable funding sources**. A decline forces banks to rely on costlier term deposits or market borrowings.
4. **Rising Competition & Technological Advances**
   - NBFCs, fintechs, and small finance banks are competing aggressively, offering higher deposit rates and innovative products.
   - Digital platforms make switching investments effortless, increasing competitive pressure on banks.

Implications of Slower Deposit Growth

1. **Asset-Liability Mismatch**
   - A widening credit-deposit gap creates **liquidity pressure** for banks.
   - May lead to higher reliance on bulk deposits or borrowings at higher cost, raising **cost of funds** and squeezing **net interest margins**.
2. **Impact on Credit Growth**
   - With limited deposit growth, banks may be forced to slow down credit expansion.
   - Risk of **tightened underwriting standards**, affecting **MSMEs**, NBFC funding, and last-mile credit delivery.
3. **Interest Rate & Profitability Risk**
   - If deposit mobilisation lags while credit demand remains strong, banks face:
     - Higher **interest rate risk** (asset-liability maturity mismatch)
     - Greater **profitability pressure** as they raise deposit rates to compete.
4. **Wider Economic Impact**
   - Slower credit availability could hurt **consumption and investment demand**.
   - The industrial sector, when credit demand rises beyond working capital, could face **funding constraints**, impacting GDP growth.

Policy & Strategic Interventions (Way Forward)

1. **Strengthen Small Deposit Mobilisation**
   - Re-focus on mobilising **small retail deposits** to ensure stability.
   - Build on financial inclusion efforts like **PM Jan Dhan Yojana (PMJDY)** to bring new depositors into the formal system.
2. **Offer Competitive & Innovative Products**
   - Flexible deposits, goal-based saving plans (ICICI's **iWish**).
   - Higher interest on savings accounts to attract sticky deposits.

3. **Monitor Credit-Deposit (C-D) Ratio**
   - RBI to ensure prudent balance between credit expansion and deposit mobilisation to avoid liquidity crunch.
4. **Tax Incentives & Regulatory Support**
   - Consider **tax breaks on bank deposit interest** to make deposits attractive relative to other asset classes.
5. **Deepen Bond & Money Markets**
   - Provide banks with **alternative funding avenues**, reducing over-reliance on deposits.

> **South Korea's deep corporate bond market** has reduced deposit dependence.

6. **Leverage Technology**
   - Use digital platforms (**SBI YONO**) to reduce operational costs, improve customer engagement, and make deposit products more appealing.

Conclusion

The deposit slowdown is not a short-term aberration but could be **structural**, driven by changing saver preferences and rising competition from alternative asset classes. If unaddressed, it can constrain credit flow and affect economic growth

A **multi-pronged strategy**- blending competitive deposit products, tax incentives, financial inclusion, and bond market deepening- is essential to maintain the delicate balance between deposit mobilisation and

ISSUES WITH FINANCIAL SECTOR

1. NBFC CRISIS IN INDIA

Non-Banking Financial Companies (NBFCs) are **financial institutions that provide banking-like services such as loans**, **credit facilities**, and **investment products** but do not hold a full banking license and cannot accept demand deposits from the public. They mobilize funds through bonds, debentures, market borrowings, and other instruments.

Size of Sector- As of FY 2024–25, NBFCs held assets worth approximately **Rs. 41 lakh crore**, accounting for around **15% of India's financial system** (RBI).

Regulation- NBFCs are primarily governed by the Companies Act, 2013. They are regulated by different authorities based on their functions: the **Reserve Bank of India (RBI)** for most NBFCs, the **Securities and Exchange Board of India (SEBI)** for investment-focused NBFCs, and the **National Housing Bank (NHB)** for housing finance companies.

Significance of NBFCs in India

1. **Credit Supplementation-** NBFCs complement bank lending, especially in underserved sectors. MSME loans made up **9.1%** of NBFCs' total portfolio in March 2024, growing at **32% CAGR** between FY21–FY24.
2. **Financial Inclusion-** They provide credit to rural and semi-urban populations. NBFC-Microfinance Institution held **Rs. 1.5 lakh crore** in micro-credit as of December 2024, about **39%** of the total industry.
3. **Market Efficiency-** By mobilizing private capital, NBFCs support infrastructure, industrial growth, and employment.
4. **Innovation-** NBFCs offer flexible financing like **microfinance**, **peer-to-peer lending**, and **structured products**. The microfinance sector reached **Rs. 3.77 lakh crore** by March 2024, with NBFCs growing **45% YoY**.

5. **Engine of Growth-** NBFCs finance MSMEs and infrastructure, driving rural and small-scale growth. Total NBFC credit stood at **Rs. 52 trillion** in December 2024, projected to exceed **₹60 trillion by FY2026**.
6. **Credit Access-** They provide loans to individuals, SMEs, and large firms, often to those who do not meet banks' strict requirements.
7. **Mobilization of Savings-** NBFCs raise funds from retail investors, high-net-worth individuals (HNIs), and institutions to finance various sectors.
8. **Investment & Payment Services-** They offer portfolio management, investment advisory, and distribution of financial products, along with payment services like debit/credit cards, fund transfers, and mobile banking.

Background of the NBFC Crisis

The dominance of banks in lending in India began to wane post-2017 due to their high non-performing assets (NPAs). Consequently, NBFCs gained prominence. However, the sector faced a significant crisis in 2018–19, initiated by the collapse of major NBFCs such as Infrastructure Leasing & Financial Services (IL&FS), Dewan Housing Finance Corporation (DHFL), and Reliance Capital.

Causes of the Crisis

1. **Asset-Liability Mismatch**: Many NBFCs borrowed short-term funds to finance long-term loans, creating liquidity risks. The IL&FS default highlighted this flawed business model.
2. **Over-Reliance on Mutual Funds and Banks**: NBFCs depended heavily on debt mutual funds and bank borrowings. During market downturns, fund outflows exacerbated liquidity stress.
3. **Sectoral Concentration**: NBFCs, especially housing finance companies, were overexposed to the struggling real estate sector, with projects like Amrapali, Unitech, and Supertech collapsing.
4. **Governance and Credit Risks**: Poor due diligence, corporate governance lapses, and exposure to high-risk infrastructure projects amplified defaults.

Challenges Facing the NBFC Sector

1. **Interconnectedness with Banks and Mutual Funds**- NBFCs rely heavily on banks and mutual funds for funding. As of April 2025, **bank credit to NBFCs fell 2.9% YoY to Rs. 16.1 lakh crore**, reflecting cautious lending despite RBI restoring risk weights to 100%.

   IMF Financial Stability Assessment 2025: The International Monetary Fund (IMF) highlighted NBFC vulnerabilities due to high exposure to the power and infrastructure sectors, where the top three NBFCs account for 63% of total power loans.

2. **Limited Liquidity Support**- Unlike banks, NBFCs cannot access direct RBI liquidity windows, making them vulnerable during market shocks. RBI assistance is considered only during crises, highlighting reliance on external funding.

   Approximately 56% of NBFC lending is financed through market instruments, making them sensitive to investor sentiment and liquidity crunches.

3. **Regulatory Gaps**- State-owned NBFCs dominate the sector and are not subject to large exposure limits, creating an uneven playing field compared to private NBFCs.
4. **Market Volatility**- Rising interest rates and bond market fluctuations directly increase NBFC funding costs. In August 2025, mutual funds saw outflows from corporate bond schemes due to rising yields.
5. **Multiple Regulatory Bodies**- NBFCs are overseen by RBI, SEBI, and NHB based on type, leading to regulatory fragmentation and potential inconsistencies.
6. **Less Cautious Credit Habits**- NBFCs have expanded their portfolios of small and micro loans, where there are risks of high NPAs, and the unsecured loan segment is on the rise.

7. **Delayed Projects**- Delays in statutory approvals, land acquisition, and environmental clearances have led to project delays.
8. **Unsecured Deposits**- Unlike banks, NBFCs often rely on unsecured deposits, increasing their vulnerability.

Key RBI Measures for NBFCs

1. **Increase in Net Owned Fund (NOF) Requirements:**
   - **NBFC-Investment and Credit Company (NBFC-ICC):** Raised from ₹2 crore to ₹10 crore.
   - **NBFC-Microfinance Institution (NBFC-MFI):** Raised from ₹5 crore to ₹10 crore.
2. **Standardization of Non-Performing Asset (NPA) Classification:**
   - All NBFCs must classify loans as **NPA** if they are overdue for more than **90 days**.
3. **Introduction of Prompt Corrective Action (PCA) Framework:**
   - Applicable to **deposit-taking NBFCs** and **systemically important NBFCs** to monitor financial health and enforce regulatory corrective measures.
4. **Regulatory Reforms**: The RBI introduced Scale-Based Regulation (SBR), categorizing NBFCs into Base, Middle, Upper, and Top layers based on size and risk.

Way Ahead

1. **Strengthen Liquidity Regulations**- NBFCs must maintain a higher liquidity coverage ratio to manage asset-liability mismatches.
2. **Uniform Regulatory Oversight**- Apply similar prudential norms to both state-owned and private NBFCs to ensure a level playing field.
3. **Diversify Funding Sources**- Reduce dependency on mutual funds and bank borrowings by expanding the retail and institutional investor base.
4. **Enhanced Risk Monitoring**- The RBI should ensure early warning systems for high-risk sectors like real estate and infrastructure.
5. **Corporate Governance Reforms**- NBFC boards must implement stronger risk management, audit mechanisms, and transparency norms.
6. **Macroprudential Coordination**- Close coordination between the RBI, SEBI, and the Ministry of Finance to prevent systemic contagion.

2. SEBI- ROLE AND LIMITATIONS

SEBI was established in **April 1988** as a non-statutory body. SEBI became a statutory on April 12, 1992, under the SEBI Act, 1992.

It acts as the **principal regulator** of the **securities market**, and **watchdog** of the **Indian capital market** functioning under the Ministry of Finance. Its **headquarters** is in Mumbai, with regional offices in Ahmedabad, Kolkata, Chennai, and Delhi.

The Board of Directors has **9 members** including a **chairperson** nominated by the Government of India, **2 members** from the Union Ministry of Finance, **1 member** from the Reserve Bank of India and **5 other members** appointed by the Government, including at least **3 full-time Members**.

Functions of SEBI

- **Market Regulation:** Sets **capital-raising rules**, and ensures compliance through inspections and investigations.

- **Registration of Intermediaries-** Registers and regulates the functioning of intermediaries like **stock brokers, sub-brokers**, **share transfer agents**, **merchant bankers**, **portfolio managers**, **investment advisers**, and others associated with the securities market.
- **Market Development:** Expand the market with **electronic trading** and **demat systems**.
- **Investor Protection:** Educates **investors**, **provides compensation**, and **combats fraudulent practices**.
- **Regulatory Actions:** Drafts **regulations**, conducts inquiries and imposes penalties.
- **Operational Powers:** Manages **money pooling schemes**, **conducts searches**, and **oversees mutual** and **venture capital funds**.

Powers of SEBI

SEBI holds extensive powers to regulate and control the securities market in India. These powers are equivalent to those of a **civil court** under the **Code of Civil Procedure, 1908.** This allows SEBI to investigate and take necessary actions to protect investor interests and ensure market integrity.

- **Discovery and production** of books of accounts and other documents at a specified place and time.
- **Summoning and enforcing** attendance of individuals and examining them under oath.
- **Inspection** of any books, registers, or other documents of any person or company involved in the securities market.
- **Suspend** trading of any security on a recognized stock exchange.
- **Restrain** individuals or entities from accessing the securities market and prohibit them from buying, selling, or dealing in securities.
- **Suspend** office-bearers of any stock exchange or self-regulatory organization from holding their positions.
- **Impound and retain** proceeds or securities related to transactions under investigation.

Initiatives of SEBI for Effective Functioning

- **Investor Education and Protection Fund (IEPF):** Promotes investor awareness and protection.
- **SCORES Portal:** A web-based system for lodging and tracking complaints.
- **Investor Education and Financial Literacy:** Initiatives like awareness campaigns and counseling centers.
- **Paperless Trading:** Elimination of physical share certificates through the **Depositories Act, 1996**, to prevent delays and fraud.
- **Handling Financial Shocks:** Swift response to the global financial crisis and the Satyam scandal with effective regulatory actions.
- **Promoting Transparency:** Increased corporate transparency by enhancing disclosure requirements for Indian promoters.
- **Takeover Code:** Liberalization of the takeover code to encourage investment by removing restrictive regulations.
- **T+1 settlement cycle:** The Securities and Exchange Board of India introduced the T+1 settlement cycle in India at the beginning of 2023. The T+1 settlement cycle means that trades are settled in one business day, and the investor receives the money the following day.

Limitations/Challenges of SEBI

1. **Financial Scandals-** Failure to effectively monitor companies involved in financial scandals, such as the **Satyam scam**, **IL&FS crisis**, **Punjab National Bank scam**, and **NSE co-location scam**.

2. **Organizational Apathy-** Lack of timely action in detecting and addressing fraudulent activities by market intermediaries, leading to investor losses.
3. **Insider Trading-** Inability to prevent insider trading, with many cases going undetected for extended periods.
4. **Inadequate Regulation of IPOs-** Resulting in overpriced offerings that harm retail investors.
5. **Market Manipulation-** Susceptibility of small-cap and mid-cap stocks to market manipulation.
6. **Hindenburg Allegations**- Allegations of conflicts of interest,when Hindenburg Research accused SEBI's chairperson of being involved in offshore entities used to inflate Adani Group stocks.

Recommendations for Improvement

1. **Strengthen regulatory approach-** Draw lessons from responses to scandals like Wirecard's misappropriation.

> **Germany's BaFin** reformed its financial regulations and oversight mechanisms.

2. **Increase staff and attract top talent-** Draw lessons from **UK's Financial Conduct Authority (FCA)**.
3. **Unified Financial Regulator-** Create a **unified financial regulator** to tackle overlaps, mirroring the UK's FCA model.
4. **Enhanced Market Surveillance-** Improve market monitoring to detect and prevent fraudulent activities more effectively. Shift focus to prudential oversight with enhanced intelligence similar to the **US post-2008 crisis**.
5. **Regular review of policies-** Act promptly on allegations, following the example of global regulators like the **European Securities and Markets Authority (ESMA)**.
6. **Strengthen Insider Trading Regulations-** Tighten insider trading rules and increase penalties for violations.
7. **Investor Education-** Develop programs to educate investors about market risks and fraud prevention.
8. **Leveraging Technology-** Use advanced technology and data analytics to improve market monitoring and enforcement.

Conclusion

As markets grow more complex and intertwined globally, SEBI's role as a regulator will only become more critical. Necessary steps should be taken to make its functioning more effective.

3. INSOLVENCY & BANKRUPTCY CODE

The Insolvency and Bankruptcy Code (IBC), enacted in 2016, is a landmark reform that consolidated and modernized India's insolvency framework.

Its main aim is to provide a unified, time-bound process for resolving insolvency and bankruptcy of corporates, individuals, and partnership firms, thereby improving overall financial sector health and creditor confidence.

> **Insolvency** refers to a situation where individuals or companies cannot repay back their outstanding debt obligations.
>
> **Bankruptcy** refers to a legal status declared by a court of competent jurisdiction for a person or entity that is insolvent i.e. unable to pay off debts.

Key Features of IBC

- IBC covers individuals, companies, LLPs, and partnership firms under a single, consolidated framework, merging multiple earlier laws.

- **Strict timelines:** Resolution must be completed in 180 days (extendable to 330 days in special cases, including litigation). Special provisions for small companies and startups (90 days + 45 days).
- **Committee of Creditors (CoC):** Takes charge of decision-making during insolvency; professional insolvency resolution professionals are appointed to manage proceedings.
- **Insolvency and Bankruptcy Board of India (IBBI):** Regulates professionals and agencies, sets standards, and oversees proceedings.
- **Adjudicating Authorities:**
  - National Company Law Tribunal (NCLT) for companies/LLPs.
  - Debt Recovery Tribunal (DRT) for individuals/partnerships.
- **Cross-border Insolvency Provisions:** Includes enabling provisions for handling insolvency of multinational companies operating in India.

Objectives of IBC

1. **Consolidation & Amendment of Insolvency Laws-** Merge and streamline multiple, outdated insolvency and bankruptcy laws under a single, comprehensive code for individuals, companies, LLPs, and partnership firms.
2. **Facilitate time-bound resolution-** Ensure fast and predictable outcomes for insolvency cases (180–330 days), minimizing value erosion and maximizing asset recovery for creditors.
3. **Maximise Value of Assets-** Prevent value depletion for stressed companies or individuals by encouraging restructuring, sale, or liquidation in a manner that realizes maximum possible returns.
4. **Promote Entrepreneurship-** By making exit easy and non-punitive, IBC encourages risk-taking, business innovation, and investment, fostering a dynamic entrepreneurial ecosystem.
5. **Protect interests of Creditors & other Stakeholders-** Structure processes to balance the interests of financial creditors, operational creditors, employees, government dues, and other stakeholders fairly.
6. **Improve Ease of Doing Business-** By offering clarity, predictability, and a speedy resolution process, IBC elevates India's reputation for contract enforcement and dispute management, making it more attractive for investment.
7. **Reduce NPAs & Boost Credit Supply-** Provide an effective mechanism for addressing bad loans and stressed assets, strengthening the financial system and allowing for more responsible credit creation.
8. **Establishment of Regulatory Mechanism-** Establish the **Insolvency and Bankruptcy Board of India (IBBI)** to regulate insolvency professionals and agencies, ensuring ethical, efficient, and accountable practice.

Need for IBC

1. **Fragmented & Outdated Insolvency Laws-** Prior to IBC, insolvency and bankruptcy were governed by multiple, overlapping laws and forums, causing confusion, delays, and high costs for resolution. Lack of clarity led to conflicting legal interpretations and inefficiency in resolving business distress.
2. **Prolonged Resolution Time & Value Erosion**
   - Average insolvency resolution in India took over **4 years**, in contrast to **1–1.5 years** in developed countries.
   - Long proceedings caused value erosion of assets and discouraged genuine business restructuring.
3. **Mounting NPAs & Stressed Assets-** Banks and financial institutions suffered from rising non-performing assets (NPAs) and mounting bad debts. Ineffective recovery mechanisms left creditors with little recourse and led to growing economic risks.
4. **Poor Ease of Doing Business:**

- India's low ranking in the **World Bank's Ease of Doing Business index** was partly due to cumbersome exit processes for distressed firms.
- Investors and entrepreneurs were deterred by unpredictable and costly insolvency procedures.

5. **Strengthening Credit Discipline & Market Confidence-** The absence of strong recovery laws allowed for poor credit discipline and willful defaulting, harming India's banking sector and overall business climate.
6. **Need for Modern, Unified, Creditor-friendly Framework:**
   - Global best practices demanded a unified, quick, and transparent framework that empowers creditors and ensures fair outcomes for all stakeholders.
   - Encouraging entrepreneurship, risk-taking, and a robust financial market needed time-bound exits and non-punitive resolution processes.

Achievements of IBC

1. **Improved Recovery Rates & Resolution Timelines:**
   - Average recovery rate for creditors has increased to 32–45%, up from ~20% before IBC.
   - Strict timelines (180–330 days) have accelerated stressed asset resolution and asset value preservation.
2. **Rescue & Settlement of Companies:**
   - As of March 2025, IBC has helped rescue **1,194 companies** through resolution plans and led to creditors recovering **Rs. 3.89 lakh crore**.
   - Over 30,000 cases (worth **Rs. 13.8 lakh crore**) were settled before admission, showing IBC's deterrent effect against defaults.
3. **Dominant Recovery Channel-** IBC contributed **48% of all bank recoveries in FY 2023–24**, outperforming SARFAESI (32%), DRTs (17%), and Lok Adalats (3%).
4. **Behavioral Change & Improved Credit Discipline**
   - The threat of losing control over businesses via IBC has **prompted debtors to repay dues** and **improved overall credit discipline** in the banking system.
   - **Pre-admission settlements** indicate companies now act early to resolve distress.
5. **Strengthening Banking Sector & Asset Quality-** Significant contribution to reducing non-performing assets (NPAs): **gross NPAs declined from 11.2% in March 2018 to 2.8% in March 2024**. Banks have freed up locked capital for redeployment and healthier lending.
6. **More Robust Credit Market:** IBC has made the Indian market more attractive to investors, especially in distressed asset deals, creating greater transparency and predictability.

Challenges/Limitations of IBC

1. **Case Backlog & Delays**
   - Despite strict timelines (**180–330 days**), real-world resolution frequently exceeds deadlines due to **limited judicial capacity, procedural complexities**, and **multiple legal challenges**.
   - Limited benches at **National Company Law Tribunal (NCLT)** and appellate tribunals create bottlenecks.
2. **Low Recovery for Certain Assets-** Although average recovery improved, some sectors and cases deliver lower-than-expected recoveries, especially where asset quality is poor or liquidation proceeds are limited. **For ex- Agricultural and service-based enterprises often face distinct liquidation challenges.**
3. **Structural Issues in Credit Markets**

- Banks exhibit risk aversion, **preferring secured over unsecured lending**, and delay initiating insolvency to maximize loan lifetimes, reducing early resolution incentives.
- **Non-Banking Financial Companies (NBFCs)** and informal creditors remain less integrated into the insolvency framework.

4. **Delay in Resolution Plan Approvals-** Resolution plans sometimes face resistance, litigation, or withdrawal, causing significant delays and uncertainty for creditors and employees.
5. **Impact on MSMEs & Startups-** The insolvency process can be costly and intimidating for small and medium enterprises and startups, which may opt for informal settlements or closure.
6. **Incomplete Creditor Participation-** Non-financial creditors often feel underrepresented or sidelined in decision-making by the **Committee of Creditors (CoC)**, impacting consensus and fair resolutions.

Way Forward

1. **Strengthen Judicial & Institutional Capacity**
   - Expand the number of dedicated NCLT benches and National Company Law Appellate Tribunal (NCLAT) members to reduce case backlog and expedite resolution.
   - Enhance training for judges, insolvency professionals, and related authorities to handle complex insolvency cases efficiently.
2. **Simplify Legal & Procedural Framework**
   - Streamline multiple legal provisions and reduce overlaps with sector-specific laws to minimize litigation and conflicting appeals.
   - Introduce fast-track insolvency resolution mechanisms for MSMEs and startups with simplified procedures.
3. **Improve Credit Participation & Transparency**
   - Ensure fair representation and participation of operational creditors and minority stakeholders in the Committee of Creditors (CoC).
   - Promote transparency in resolution processes and encourage stakeholder consultations to build trust and consensus.
4. **Enhance Awareness & Capacity Building**
   - Launch nationwide awareness campaigns for debtors, creditors, and businesses about the benefits and processes of IBC.
   - Increase the pool of qualified insolvency professionals and upgrade their skill sets through continuous professional development programs.
5. **Foster Early Insolvency Detection & Resolution**
   - Encourage early identification of financial distress through improved credit monitoring systems.
   - Facilitate pre-insolvency frameworks and corporate debt restructuring schemes to prevent insolvency where possible.
6. **Leverage Technology for Efficiency**
   - Strengthen information utilities and digital platforms for collecting and authenticating financial data.
   - Implement technology-enabled case management and monitoring systems to ensure transparency and real-time updates.

Conclusion

IBC is a crucial piece of economic legislation that has modernized India's bankruptcy laws, providing a much-needed framework for resolving financial distress and promoting a more robust and transparent credit market.

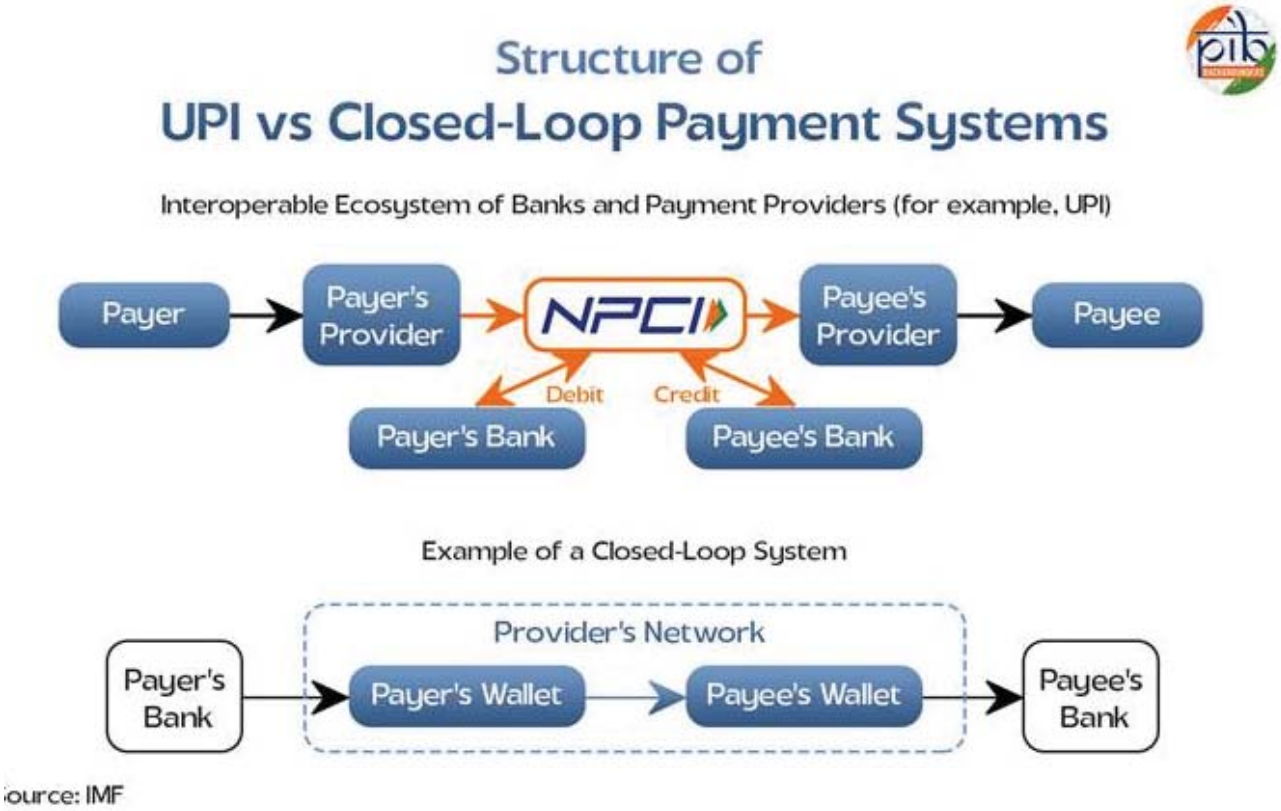
DIGITAL BANKING

1. UNIFIED PAYMENT INTERFACE (UPI)

India's Unified Payments Interface (UPI) has become the backbone of the country's digital payments ecosystem. As of June 2025, **UPI processes over 18 billion transactions monthly, worth Rs. 24.03 lakh crore**, making India a global leader in real-time and interoperable payments.

About Unified Payments Interface (UPI)

- **Launch**: Introduced in **2016** by the **National Payments Corporation of India (NPCI)** under the guidance of the **RBI** and **Indian Banks' Association (IBA)**.
- **Nature**: A **mobile-first, real-time payment system** that integrates multiple bank accounts into a single application.
- **Functionality**: Enables **instant peer-to-peer (P2P)** and **person-to-merchant (P2M)** transactions **24×7, 365 days**, with settlement directly between banks.
- **Security**: Based on **two-factor authentication** – (i) UPI PIN (**something you know**) and (ii) registered mobile device (**something you have**).
- **Interoperability**: Works seamlessly across all **UPI-enabled banks and applications**, making it universally accessible.
- **Global Recognition**: The **IMF** and **World Bank** recognize India as a **global leader in digital and fast payments**, crediting UPI's **innovation, scalability, and interoperability**.



ource: IMF

UPI at a Glance: Scale and Success

Aspect	Description
June 2025 Data	Rs. 24.03 lakh crore transacted through 18.39 billion transactions, a 32% YoY increase from June 2024.
Users & Merchants	491 million individuals, 65 million merchants, and 675 banks integrated.
Share of Transactions	UPI accounts for 85% of India's digital payments and nearly 50% of global real-time payment transactions
Comparison with Visa	UPI handles 640 million daily transactions, surpassing Visa's global daily average of 639 million
Usage Trends	Around 63% of UPI transactions are P2M, reflecting mainstream adoption for retail and small purchases

Significance of UPI in India's Digital Economy

1. **Financial Inclusion**
   - Brings **small vendors, micro-businesses, and rural households** into the formal digital economy.
   - **Zero/minimal transaction costs** and **instant settlements** drive adoption among the underbanked.
   - Integrated with **Jan Dhan–Aadhaar–Mobile (JAM) trinity**, enabling direct benefit transfers (DBT).
2. **Technological Innovation and Interoperability**
   - Overcomes **closed-loop system limitations**, enabling payments across banks and apps.
   - Encourages healthy competition among fintechs, while ensuring adherence to common technical standards and security protocols.
3. **Convenience and User Experience**
   - Consolidates multiple bank accounts into one app.

- Simple identifiers (UPI IDs, QR codes) avoid the need to share bank details.
- **99.2% success rate** with **two-factor authentication** ensures security and user trust.

4. **Economic Growth and Digital Infrastructure**
   - Boosts cashless commerce and formalizes economic activity.
   - Strengthens India's digital-first economy, supported by 5G expansion and low data costs.
5. **Global Expansion and Digital Diplomacy**
   - Accepted in **UAE, Singapore, Bhutan, Nepal, Sri Lanka, France, Mauritius, Cyprus, and the Maldives**.
   - India is pushing for UPI integration within BRICS and regional frameworks, improving remittance efficiency and bolstering India's fintech leadership.

Challenges Facing UPI

1. **Cybersecurity & Fraud Risks**- Rising phishing, fraud, and cyberattacks demand **continuous upgrades** and **user awareness**.

   Over **13 lakh fraud incidents** were reported in **FY 2024–25**, with losses exceeding **Rs. 1,000 crore**.

2. **Single Point of Failure-** The system's centralized reliance on NPCI creates a **Single Point of Failure (SPoF)**. If NPCI's systems go down, all associated banks and apps like PhonePe and Google Pay are affected.
3. **Lack of Incentives-** Since banks and payment providers cannot charge fees for most UPI transactions, they have limited revenue to reinvest in better infrastructure and security.
4. **Duopoly Risk-** Two players, PhonePe and Google Pay, **command over 80-85%** of the market share.
5. **Global Hurdles-** While expanding to countries like **Singapore, UAE,** and **France, UPI** faces challenges in **currency exchange markups** and a lack of reciprocal usage for foreign nationals in India.

The Digital Foundation Behind UPI's Success

- **Pradhan Mantri Jan Dhan Yojana (PMJDY):** Over **558 million bank accounts** opened by July 2025, driving financial inclusion and DBT.
- **Aadhaar Digital Identity:** Over **1.42 billion Aadhaar cards** issued by mid-2025, enabling secure biometric authentication.
- **Connectivity and 5G Revolution:** With **4.74 lakh 5G base stations** and **116 crore mobile subscribers** in 2025, supported by sharply reduced data costs, digital services have become widely accessible.

Way Forward

1. **Strengthening Cybersecurity-** Adoption of **AI-based fraud detection**, advanced encryption, and global best practices.
2. **Bridging the Digital Divide-** Expanding rural internet coverage, promoting affordable smartphones, and **digital literacy programs**.
3. **System Resilience-** Development of **alternative payment rails** to reduce systemic dependence on UPI.
4. **Global Integration-** Aligning UPI with **international standards** for smoother cross-border adoption.
5. **Balanced Monetization-** Designing **sustainable revenue models** for fintechs while keeping consumer access affordable.

2. CENTRAL BANK DIGITAL CURRENCY (CBDC)

A CBDC is a digital version of sovereign currency issued and regulated by a nation's central bank (RBI for India), available in electronic form alongside paper currency as legal tender.

The **Bahamas** was one of the **first countries** to introduce its Sand Dollar CBDC officially. **e-Rupee** is a legal tender issued by **RBI** in digital form. It is the same as a fiat currency and is no different from cash, and is exchangeable one-to-one with the fiat currency (bank notes/cash) at par.

Types of CBDC

Retail CBDC (CBDC-R)- For public use (households, businesses) in daily transactions; promotes financial inclusion. They are of two types -

- **Token-based retail CBDCs:** These currencies can be accessed using private and public keys, allowing for anonymous transactions.
- **Account-based retail CBDCs:** These require digital identification for users to access and use their accounts.

Wholesale CBDC (CBDC-W)- For inter-bank settlements; enhances liquidity and efficiency in financial markets.

Key Features of CBDC

- **Centralized Control**: Issued by RBI; no mining (unlike cryptocurrencies like Bitcoin).
- **Legal Tender**: Mandatory acceptance; traceable for anti-money laundering (AML).
- **Two-Tier System**: RBI issues to banks (Tier 1), which distribute to users (Tier 2) to leverage existing infrastructure.
- **Offline Functionality**: Supports P2P offline transfers to aid rural areas.
- It is **programmable** (can restrict usage for specific purposes), can operate **offline**, and is **different from cryptocurrencies** (which are private and decentralized).

Difference between cryptocurrency and CBDCs

Cryptocurrencies and CBDCs are **both blockchain-based digital currencies.**

Feature	CBDC (Central Bank Digital Currency)	Cryptocurrency
Issuer	Central bank (RBI)	Private, decentralized network
Legal Tender	Yes (government-backed, recognized)	No (not legal tender in most countries)
Regulation	Regulated by government and central bank	Not regulated or partially regulated
Value Stability	Stable, linked to fiat currency	Highly volatile, market-driven
Technology	Centralized ledger; can use blockchain	Decentralized blockchain
Anonymity/ Privacy	Low to moderate; traceable by authorities	High (depending on crypto type)
Transaction	Speed is Fast, designed for scalability	Fast to moderate; depends on network

Acceptance	Broad, by citizens for all payments	Limited; mainly investment, select use
Risk/Trust	Trusted (state-backed, less risk)	Risk of hacking, scams, ownership issues
Example	Digital Rupee, RBI)	Bitcoin, Ethereum, Ripple

Advantages/Significances of CBDC

1. **Financial Inclusion-** By providing access to digital money, **CBDCs can bring unbanked populations into the formal financial system**, improving financial access for all.
2. **Reduced Transaction Costs-** CBDCs can lower the costs of transactions by **eliminating intermediaries**, thus making payments more efficient and affordable.
3. **Enhanced Monetary Policy-** With direct control over the digital currency, central banks can more effectively implement monetary policy, such as **controlling inflation** and **managing interest rates**.
4. **Security-** CBDCs offer greater security compared to physical currency, as they are **resistant to counterfeiting** and **fraud**, with transactions being traceable.
5. **Eliminating Third-Party Risk-** CBDCs can reduce risks associated with events like **bank failures** or **bank runs**.
6. **Lower Cross-Border Transaction Costs-** By simplifying the distribution systems and fostering greater cooperation between jurisdictions.
7. **Direct Consumer-Central Bank Connection-** CBDCs can establish a direct connection between consumers and central banks, eliminating the need for costly intermediary infrastructure and improving the efficiency of payment systems.

Rationale for Introduction of CBDC in India

- Addresses cash dominance (86% of transactions, RBI 2023), reduces printing costs (Rs 4,000-5,000 crore annually), and counters private cryptos amid 30 crore crypto users (2024).
- Aligns with G20 priorities and SDGs (Goal 8: Decent Work, Goal 9: Innovation).
- **Economic Survey 2024-25:** CBDC could boost GDP by 0.5-1% via enhanced efficiency.

Challenges of CBDC

1. **Privacy concerns-** The digital nature of CBDCs makes it easier for governments to track and monitor transactions, raising concerns about user privacy and surveillance.
2. **Digital divide-** The complex technical requirements and need for digital literacy in using CBDCs could widen the gap between tech-savvy and less technologically adapted populations.
3. **Cybersecurity concerns-** Robust security measures are crucial to protect the e-rupee system from cyberattacks.
4. **UPI preference and interoperability-** Despite efforts to promote the CBDC, there is a continued preference for UPI among retail users.
5. **Threat to monetary sovereignty-** People preferring a foreign CBDC (Digital Dollar or Digital Yuan) over their national currency, could weaken the local monetary system.
6. **Impact on the banking system-** The widespread adoption of CBDCs could disrupt traditional banking models, potentially reducing the role of commercial banks in credit distribution.

Way Forward

1. **Short-Term**: Integrate the CBDC with UPI 2.0
2. **Medium-Term**: Enact CBDC Act (privacy safeguards); RBI-led awareness campaigns.

3. **Long-Term**- Full rollout by 2027; international standards (BIS mBridge); green CBDC for sustainable finance.
4. **Policy Linkages**- Align with Digital Rupee strategy; complement with Jan Dhan (50 crore accounts).
5. **Cross-border collaboration and standardization-** Global community can work with international financial institutions (IMF, BIS) to establish global standards for CBDC interoperability and regulation.

Conclusion

CBDC holds promise for India's payments architecture and targeted policy delivery; careful, phased scaling with strong privacy & stability safeguards is essential to reap benefits while minimizing systemic risks.

3. CRYPTOCURRENCY

A cryptocurrency is a **digital** or **virtual currency** that uses cryptography for security. It is a **decentralized currency**. It is neither controlled by any government or any institution. Some examples of cryptocurrencies include **Bitcoin**, **Ethereum**, and **Litecoin**.

Working of Cryptocurrency

Transactions with cryptocurrency are recorded on a public digital ledger called blockchain.

Blockchain technology is a **decentralized**, **digital ledger** that records transactions across a network of computers.

- Each block in the chain **contains a number of transactions**, and **every time a new transaction occurs** on the blockchain, a record of that transaction is added to every participant's ledger.
- The decentralized nature of technology ensures that **no single entity can alter** or **delete previous transactions**, providing a high degree of security and transparency.

Status of Cryptocurrency regulation in India

Legal and Regulatory Status	Cryptocurrencies **are not banned** in India. Buying, holding, selling and trading in them is legal. Cryptocurrencies **are not recognized** as legal tender. The Indian government classifies cryptocurrencies and similar assets as **Virtual Digital Assets** (**VDAs**). Union Budget 2022-23, imposed 30% tax on Cryptocurrency (VDAs).
Anti-Money Laundering (AML) / Financial Intelligence Oversight	Virtual Digital Asset Service Providers (VDASPs) (crypto exchanges, wallet providers etc.) must register with India's Financial Intelligence Unit (FIU-IND) and comply with KYC / AML rules.
RBI's Concerns on Cryptocurrencies	In 2013, RBI issued a circular warning to the public against using virtual currencies. In 2018, RBI issued a circular preventing commercial and cooperative banks, payments banks, small finance banks, NBFCs, and payment system providers from dealing in virtual currencies or providing services to all entities which deal with crypto exchanges. In 2020, SC declared the 2013 RBI circular unconstitutional.

RBI Issued Central Bank Digital Currency (CBDC) based on block-chain technology.

Aspect	Cryptocurrency	CBDC
Issuer	Private entities, open-source communities, or decentralized networks (for ex- Bitcoin, Ethereum).	Central Bank of a country (**for ex-** RBI in India).
Legal Status	Not legal tender in most countries (including India). Can be traded/held but not legal tender.	Legal tender backed by the government. It has the same status as physical cash.
Control	Decentralized – no central authority; governed by blockchain protocols and consensus mechanisms.	Centralized – controlled and regulated by the central bank.
Anonymity	Transactions can be pseudonymous, depending on the crypto.	Traceable and regulated. User data can be monitored by the central bank.

Challenges with Cryptocurrencies

1. **Volatility-** Cryptocurrency prices are highly volatile, which makes it difficult for businesses to accept it as a form of payment.

   In December 2017, Bitcoin was valued at around **USD 20,000 per coin**. However by November end, bitcoin's value toppled to **approximately 80 percent of its peak value** and bitcoin was trading at a **price of USD 3800**.

2. **Lack of sovereign backing-** Unlike fiat currencies, crypto currencies do not have sovereign backing.
3. **Financial Frauds-** Non-official virtual currencies can be used to defraud consumers. **For ex-** Rs 2000 cr GainBitcoin scam in India.
4. **Promotes Criminal activities-** According to the report of Financial Action Task Force (FATF) virtual currencies provide greater anonymity than mainstream non-cash payment methods. This makes them vulnerable to money laundering and use in terrorist financing activities.

   Use of Bitcoin by the **illegal website Silk Road** in 2014 to finance hacking, drug trafficking, and illegal weapon sales.

5. **Time consuming-** Processing transactions is time-consuming due to validation procedures and network latency, as crypto currencies are backed by trust and consensus-based algorithms.
   - The large gap in transaction processing speed between crypto currencies (especially Bitcoin), and other electronic payment methods, hinders their ability to be used as medium of exchange.
6. **Environmental Impact-** Bitcoin mining and proof-of-work consumes large amounts of energy, which release large amounts of carbon emissions.
7. **Low Adoption:** Crypto-complexities like wallets, private keys, gas fees, limit the adoption of wide scale adoption of crypto currencies.

Recommendations of the Inter-Ministerial committee with respect to crypto currencies

- Private crypto currencies are inconsistent with the essential functions of money/currency. Hence private crypto currencies cannot replace fiat currencies.
- The Committee recommended that all private crypto currencies, except any crypto currency issued by the State, be banned in India.
- The Committee recommended a law banning the crypto currencies in India and criminalizing carrying on of any activities connected with crypto currencies in India.

Way Forward

1. **Provide Legal Clarity and Recognition-** Introduce a **comprehensive crypto law** to replace piecemeal taxation and ad-hoc measures.
2. **Risk-Based Regulatory Framework-** Differentiate between types of crypto assets: **Payment tokens** (used like money), **Utility tokens** (access to a service) and **Security tokens** (investment-like instruments). Regulation should be proportionate to the risk level like **EU's MiCA regulation**.
3. **Taxation Reforms-** Rationalise tax rates as current 30% flat tax and 1% TDS discourage genuine investors. Allow loss set-off and lower TDS rates to make compliance easier.
4. **AML/KYC Compliance-** Enforce strict KYC and Anti-Money Laundering (AML) standards for exchanges and wallet providers. Collaborate with global watchdogs like FATF, G20 for cross-border monitoring.
5. **Encourage Innovation and Sandbox Models-** Allow startups to test blockchain/crypto products in **regulatory sandboxes** under supervision. Promote **Web3**, **tokenization**, and **DeFi experiments** while keeping user protection in mind.
6. **Environmental and Ethical Considerations-** Encourage green crypto practices like **prioritization** of **Proof-of-Stake** over **Proof-of-Work**.
   - Incorporate ethical frameworks to avoid exploitation, fraud, and unfair concentration of wealth.

4. FINTECH SECTOR

The term "FinTech" is a contraction of the words "Finance" and "Technology". It refers to businesses that use technology to enhance or automate financial services and processes.

According to the **Financial Stability Board** (FSB), of the BIS, "FinTech is **technologically enabled financial innovation** that could result in **new business models**, **applications**, **processes**, or **products** with an associated material **effect on financial markets** and **institutions** and the **provision of financial services**".

Different Types of Fintechs in India, classified according to their functions

Payment Fintechs	These offer digital payment solutions, such as **mobile wallets**, **online payment gateways**, and **peer-to-peer (P2P) payments**. **Ex-Phonepe**
Lending Fintechs	These offer digital lending solutions, such as **personal loans**, **business loans**, and **credit cards**. **Ex-Lending Kart**
Insurance Fintechs	These offer digital insurance solutions, such as **health insurance**, **life insurance**, and **car insurance**. **Ex-Policy bazaar**
Investment Fintechs	These offer digital investment solutions, such as **stock trading**, **mutual funds**, and **cryptocurrency trading**. **Ex-Zerodha**

Status of Fintechs in India

1. **World's 3rd largest FinTech ecosystem-** India is the 3rd largest FinTech ecosystem, with around 25 fintech companies having attained the Unicorn Status (Valuation of more than $1bn).
   - India's fintech ecosystem is expected to reach US$70-75 billion annual revenue by FY30.
2. **Highest Fintech Adoption rate-** India's Fintech adoption rate of 87% is highest in the world. The world's average Fintech adoption rate is around 64%. (Economic Survey 22-23)

3. **Growth of Unified Payments Interface (UPI)-** UPI has become the world's leading real-time payment platform, surpassing Visa. UPI accounts for **85 per cent** of all digital transactions in India, and powers nearly **50 per cent** of global real-time digital payments.

Government Initiatives Driving the Growth of Fintech Sector in India

Digital Identity Infrastructure (JAM Trinity)	**J- Jan Dhan Yojana (PMJDY):** This world's largest financial inclusion program has provided bank accounts to over **450 million people** creating a massive base for FinTech companies to offer new financial products and services like remittances, credit, insurance, and pensions directly through these accounts. **A- Aadhaar :** Aadhaar has facilitated bank account opening for over **570 million** previously unbanked adults in India. **Aadhaar Enabled Payment System (AePS)** has allowed Aadhaar card holders to conduct financial transactions using their Aadhaar number and biometric authentication. **M- Unified Payments Interface:** UPI has pushed India away from cash and card-based payments towards a digital-first economy.
Other Government Initiatives	**India Stack**, **Open Network for Digital Commerce** (ONDC), and **Central Bank Digital Currency** (CBDC), have propelled the growth of Fintechs in India.
Regulatory Support and Innovation	RBI granted recognition to **Peer-to-Peer** (P2P) lending platforms as **Non-Banking Financial Companies** (NBFCs) providing legitimacy and facilitated growth within the P2P lending segment, expanding credit access for individuals and small businesses.
Regulatory Sandbox (RS) and Fintech Repository	RBI established a **Regulatory Sandbox in 2017** to help Fintech players to **live test their products or solutions**, before getting the necessary regulatory approvals for a mass launch, saving start-ups time and cost. **Fintech Repository** launched in **2021** serves as a centralised information hub for fintech companies, promoting transparency and streamlining regulatory compliance.
RBI Guidelines on Digital Lending 2022	It aims to **bring unregulated digital lending players within the RBI's ambit** and create a comprehensive framework to protect consumers' data.

Benefits of the Fintech Sector

1. **Promotion of Financial Inclusion-** Innovation driven by Fintech has widened citizen's access to financial services. **For ex-** Use of UPI has improved the financial inclusion of small vendors in India.
2. **Improved Credit availability-** Lending Fintechs have improved credit availability options for the financially marginalised sections in India. **For ex-** Fintechs providing hassle-free loan options to the MSMEs.
3. **Improved Customer Experience-** Fintechs have improved the customer service experience in the financial sector by employing big data, machine learning tools. **For ex-** Ease of investment in share markets by investment fintechs like Zerodha.
4. **Increased FDI inflows and employment opportunities-** The growth of Fintech sector in India has attracted huge FDI inflows in India, with around 25 fintech companies crossing the valuation of $1 bn mark. These fintech startups have enhanced formal sector employment opportunities in the Country.
5. **Social Change and Justice-** Fintech startups have been used as a tool to bring social change and justice. **For ex-** Crowdfunding apps like Keto have helped in raising funds for various social causes like health care treatment of poors.

6. **Supply Chain Financing-** Blockchain-based fintech solutions can **streamline payments, improve traceability**, and **enhance working capital management** for businesses within the supply chain.

Challenges with the Fintech Sector In India

1. **Increase in Cyber-Attacks on Fintech-** The Fintech startups are vulnerable to cyber-attacks and hacking. According to the CERT-In data, a total of **13.91 lakh cases** of cyber-attacks were reported in India.
2. **Data Privacy Issues-** Fintechs face the challenge of maintaining data privacy to prevent the **misuse of personal information** and **financial data**. Fintech startups have not fully complied with the provisions of Digital Data Protection Act.
3. **Money Laundering-** There have been reports of Fintechs indulging in **Money Laundering activities**. **For ex-** Abu Dhabi-based payments app called Pyppl, operated by the Chinese investment scamsters was used for money laundering in India.
4. **Indulgence in Unethical Financial practices-** The fintech sector in India has been found to be indulged in unethical financial practices like **Illegal Digital Lending**, **mis-selling of financial products**, **opaque lending practices**, **brutal collection methods** and customer harassment.
5. **Infrastructural Issues-** The sector is marred by infrastructural inadequacies like **slow internet connectivity** in rural areas.
6. **Lack of Comprehensive regulatory guideline-** Due to the diverse and dynamic nature of FinTech Sector, the formulation of a comprehensive regulatory guideline for its regulation and development, has become a major challenge. **For ex-** Lack of comprehensive regulatory guidelines for cryptocurrencies based fintechs.
7. **Dominance of foreign entities owned fintech companies, apps and platforms- Walmart-backed PhonePe** and **Google backed Google Pay**, dominate the Indian fintech sector. PhonePe commands **46.91% of the market share** in volume terms, while Google Pay commands **36.39% of the market share** in volume terms.
8. **Low adoption of Indian fintech apps-** Indian fintech apps are not preferred by the customers for payments. NCPI's BHIM UPI's market share in terms of volume stands at a mere 0.22%.
9. **Fraud to Sales Ratio-** Fraud to Sales ratio represents the total number of fraudulent transactions in comparison to the total number of transactions in a financial year. This ratio has remained around **0.0015%**, despite the increase in volume of transactions over the past five years.
10. **Regulatory ambivalence-** Foreign fintech apps operate in multiple jurisdictions, which makes their regulation in India a challenging affair.

Way forward for regulation of the Fintech Sector in India

Parliamentary Standing Committee on Communications and Information Technology has suggested the following measures for complete overhaul of the Fintech Sector in India.

1. **30% volume cap on transactions-** The total number of transactions initiated by any third-party app (like **PhonePe** and **Amazon Pay**) individually, **should not exceed 30% of the overall transactions made using the interfaces cumulatively over three preceding months**.
2. **Effective regulation of Digital Payment apps-** The committee has recommended effective regulation of Digital payment apps due to the rise of digital platforms to make payments in India.
3. **Multifold penetration of the Digital Payment Market-** The committee has recommended that existing and new players (banks and non-banks) **must scale-up their consumer outreach** for the growth of UPI payments through their platforms. This will help us to achieve overall market equilibrium.

4. **Expansion of Fintech Services-** Encourage the use of Fintech for bolstering **cybersecurity**, **fraud control**, and **money laundering prevention**. Explore virtual banking and dematerialisation of financial instruments (converting physical certificates to electronic form).
5. **Financial Inclusion-** Develop a credit registry for farmers using **AI/ML based credit scoring** to enable easier loan access. Utilise Fintech for managing claims and premium payments in agricultural crop insurance schemes.
6. **Comprehensive legal framework for Fintech Regulation-** RBI must adopt a comprehensive approach to Fintech regulation. Fragmented and reactionary approach to regulation stifles development of the fintech sector.
7. **Address the Infrastructural issue of slow internet connectivity-** Fintech companies should enable offline access to financial services so that users can perform essential transactions with limited or no internet connectivity.
8. **Compliance Program for Fintechs-** Fintechs should develop a compliance program to ensure their regular compliance with all applicable laws and regulations like the **Personal Data Protection Act**. This will help them avert any crisis in the future like the PayTM crisis.
9. **Increase in customer awareness-** Customer awareness and digital literacy must be increased to help the customers in making informed choices.

Conclusion

In the last seven and half decades, India has made a tremendous leap to become a robust digital economy. Winning the **techade**, India must build on its digital advantage and pivot towards a more data-driven structure of governance to create a more inclusive, sustainable, and impact-led innovation in the Fintech sector.

RBI

1. ECONOMIC CAPITAL FRAMEWORK (ECF)

The Economic Capital Framework (ECF) is a **structured mechanism used by the Reserve Bank of India (RBI)** to determine the **appropriate level of risk provisions** and **the surplus (profit) that can be transferred to the Government of India**. It balances the need for RBI's financial buffers and prudent surplus distribution to maintain monetary and financial stability.

About Economic Capital Framework (ECF)

The Economic Capital Framework (ECF) is a policy that determines how the Reserve Bank of India (RBI) manages its financial reserves, risk provisioning, and surplus transfers to the central government.

ECF provides a methodology for determining the appropriate level of risk provisions and profit distribution to be made under **Section 47 of the RBI Act, 1934**. As per this provision, the **central bank is required to pay the balance of its profits to the central government** after **making provision for bad and doubtful debts**, **depreciation in assets**, and **contributions to staff**.

Historically, the RBI **retained a considerable portion of its profits** to strengthen its internal reserves. Over the years, several expert committees have examined the adequacy of RBI's capital buffers and the quantum of surplus to be transferred:

- **V. Subrahmanyam Committee (1997)**

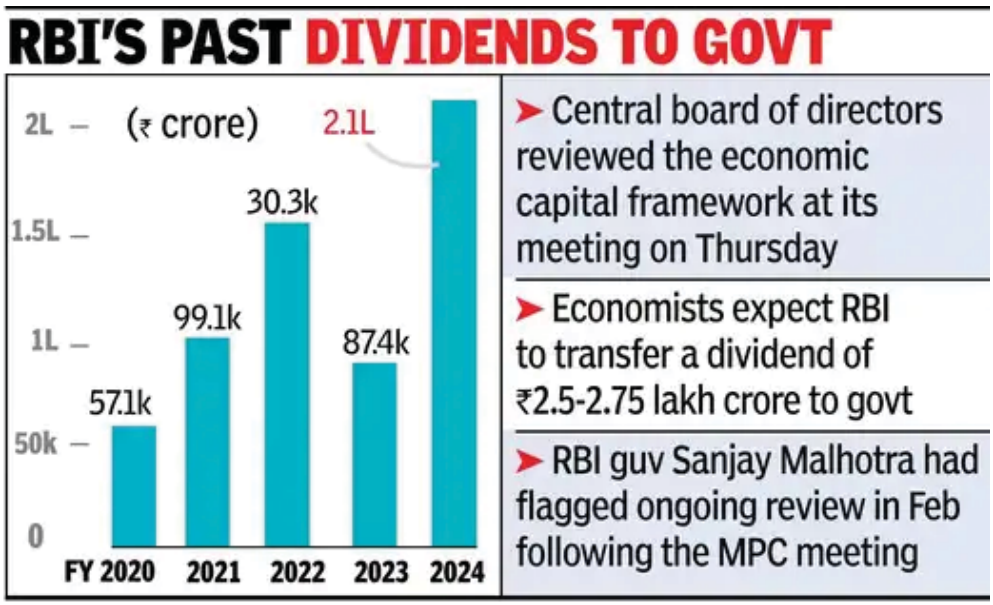
- **Usha Thorat Committee (2004)**
- **YH Malegam Committee (2013)** recommended **higher transfer**s to the government while maintaining prudent reserves.
- **Bimal Jalan Committee (2018)** introduced the revised **Economic Capital Framework (ECF),** balancing risk provisioning with fiscal needs.

In 2018, disputes arose between RBI and Government (Govt wanted higher transfers, RBI wanted to keep more for stability). To resolve this, the **Bimal Jalan Committee (2019)** was set up.

Bimal Jalan Committee recommended that the CRB (Contingency Risk Buffer) should be maintained between 5.5% to 6.5% of the RBI's balance sheet. This buffer acts as a safeguard for economic stability, ensuring that the RBI can function effectively as the Lender of Last Resort (LoLR) during financial crises.

Review Mechanism: As per the committee's recommendations**,** ECF is **reviewed every 5 years** to adjust for evolving economic conditions and risks, with the latest review done in August 2024.

Trends in Dividend Transfer: RBI's **dividend transfers to the government** have increased from **Rs 30,307 crore in FY22** to an **estimated Rs 2.5-3 lakh crore in FY25**.



Need/Importance of ECF/Surplus Transfer to Government

1. **Risk Mitigation and Financial Stability-** Central banks face multiple risks including credit risk, market risk, operational risk, and currency risk. ECF provides a structured approach to quantify and provision for these risks, ensuring RBI's resilience against financial shocks and economic uncertainties.
2. **Reduces Fiscal Deficit-** Assists the government in meeting its **fiscal deficit target** by augmenting non-tax revenue.
3. **Enhances Revenue Generation-** Acts as a major non-tax revenue source, enabling higher public expenditure and supporting economic growth.
4. **Lowers Government Borrowing-** May reduce gross borrowings by up to Rs 1 trillion, freeing fiscal space for capital expenditure.
5. **Maintaining Market Confidence**
   - A well-capitalized central bank upholds confidence among investors, commercial banks, and the international financial community.

- Maintaining adequate capital reserves through ECF reassures markets about RBI's ability to manage crises.

6. **Sustainable Surplus Distribution-** The ECF helps balance RBI's need to maintain financial buffers with legitimate surplus transfers to the government, ensuring fiscal policy requirements without compromising central bank independence or stability
7. **Autonomy and Transparency-** By having a clear, rule-based framework for capital adequacy and surplus transfers, RBI improves accountability and transparency, reducing arbitrariness in dividend distribution and strengthening institutional autonomy.
8. **Enabling Effective Monetary and Financial Policy-** Adequate capital under ECF enables RBI to act decisively as a lender of last resort, implement monetary policy without financial constraints, and support financial market interventions when needed.
9. **Aligning with Global Best Practices-** The ECF aligns RBI's financial management with global central banking standards and risk management practices, positioning India's central bank among the resilient global financial institutions.

Conclusion

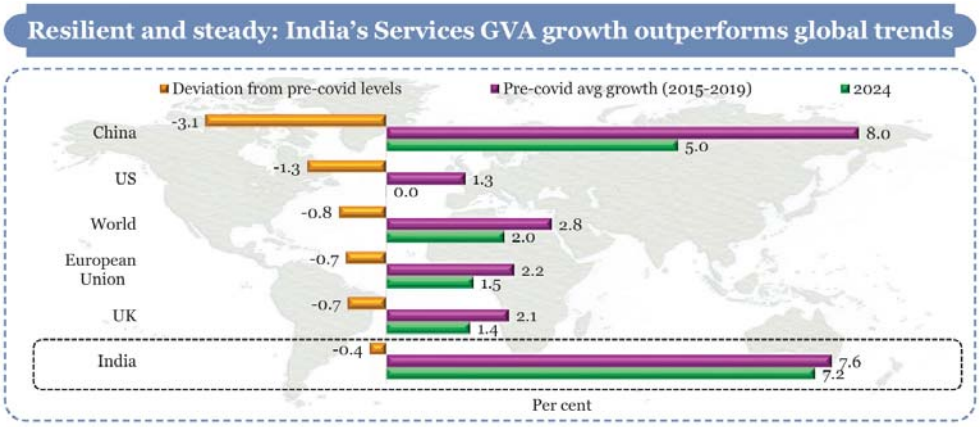
The Economic Capital Framework ensures a **balanced approach between RBI's autonomy and the government's fiscal needs**. By setting **clear risk buffers and surplus transfer rules**, it promotes **financial stability and fiscal prudence**. This framework supports **sustainable economic growth** and **strengthens macroeconomic resilience.**

MGP/BP/GS3/004

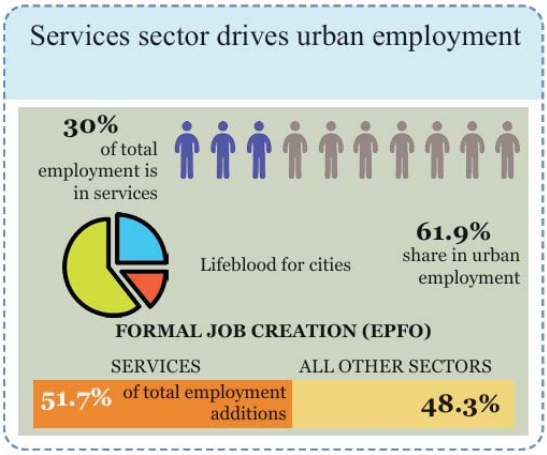
SERVICE SECTOR

1. TRENDS IN SERVICE SECTOR (ECONOMIC SURVEY 2025-26)

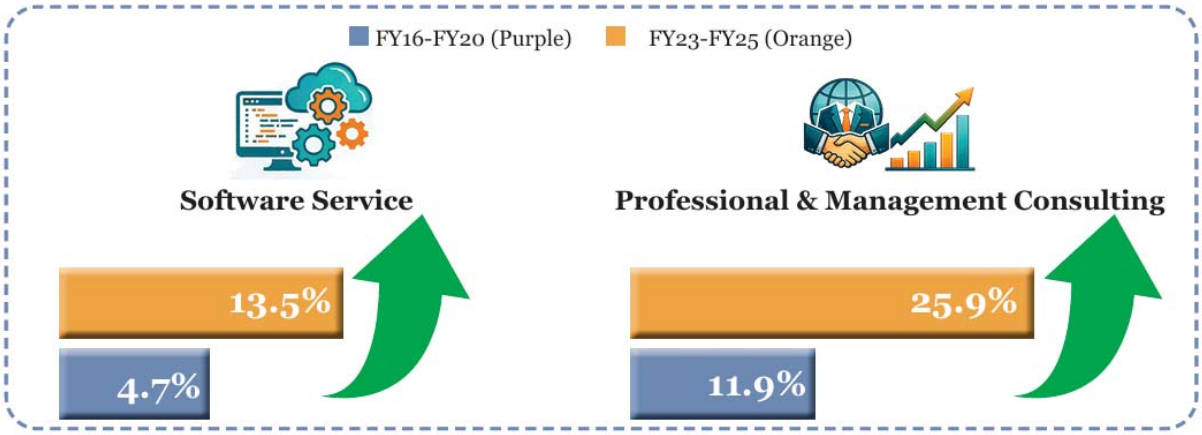
According to the Economic Survey 2025-26, India's services sector has transitioned from **post-pandemic stability** to a **phase of New Frontiers**, solidifying its role as the primary driver of national growth.



- **Service Export Milestone-** Services exports reached an all-time high of **USD 387.6 billion in FY25**, growing by **13.6%**. India is now the **world's 7th largest services exporter**, with a **4.3% share of global services trade**.
- **FDI Dominance-** The sector remains the largest recipient of foreign investment, accounting for **80.2% of total FDI inflows** during **FY23–FY25**. Investment is highly concentrated in **information and communication services (25.8%)** and **professional services (23.8%)**.
- **Service sector as a major driver of Employment-** Service sector employs **30% of the population** in India. Services drive **61.9% of urban employment** in India.



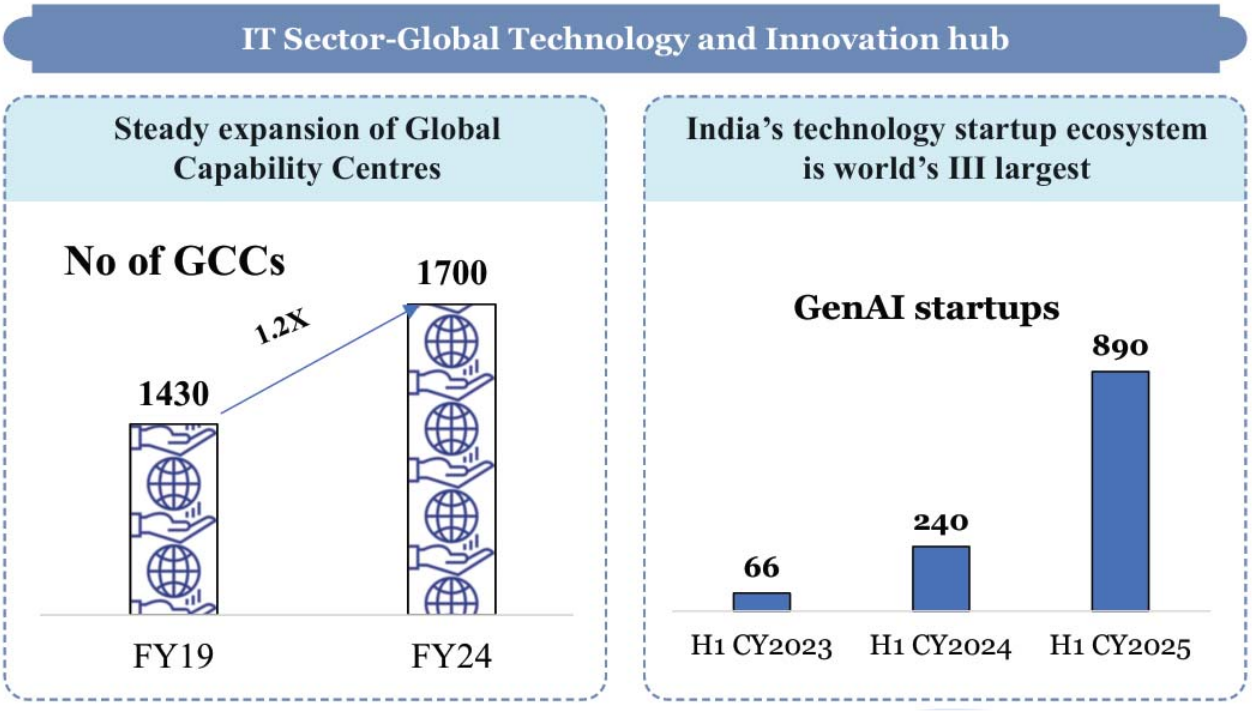
Services exports have **expanded on the back of high-value software** and **professional services**.



- **Stability Factor-** The sector has emerged as a "stabilising force," with a significantly **lower growth standard deviation (1.22)** compared to **manufacturing (4.20)** over the **FY05–FY25** period.

2. IT SECTOR IN INDIA

The Indian Information Technology (IT) sector is a cornerstone of the country's economy, renowned for its strong export-oriented services and a rapidly evolving domestic market. The industry has shown remarkable resilience and adaptability, transitioning from a focus on traditional services to a hub for high-value, digital-first solutions.



Introduction

The IT industry mainly encompasses IT services, IT-enabled services (ITES), e-commerce (online business), Software and Hardware products. This industry is also instrumental in creating infrastructure to store, process and exchange information for important business operations and other organisations.

Scale of IT Industry in India according to Economic Survey 2025-26

- IT revenues reached **USD 283 billion in FY25**, with IT services exports growing 3.7%.
- Total IT spending in India is forecast to reach **$160 billion in 2025**, with major boosts in software, cloud, cybersecurity, and digital infrastructure.
- India hosts **1,700+ Global Capability Centres employing over 19 lakh professionals**- the largest global hub.
- Data centre capacity is projected to reach 8 GW by 2030. India hosts only **~3% of global data centres** despite generating **20% of world's data**.
- The technology startup ecosystem comprises **32,000-35,000 startups**. GenAI startups have increased to 890.
- India's cybersecurity market is estimated at USD 6.0 billion, growing at 30%, with India achieving Tier-1 ranking on **ITU's Global Cybersecurity Index with a score of 98.49**.

Evolution of Indian IT Sector

Early Foundations (1960s–1980s)	• **1960s–70s**: Computers introduced in government & research institutions (ECIL, TCS founded in 1968). • **1980s**: Entry of private players (Infosys 1981, Wipro 1980s pivot to IT). • Policies liberalized for computer imports and software development. • Focus was on **mainframes, hardware maintenance, and custom software**.
Export & Outsourcing Era (1990s)	• 1991 Economic Reforms → liberalization, deregulation, foreign investment. • Rise of software services exports to the US & Europe. • The **Y2K problem (1999–2000) boosted Indian IT firms (Infosys, Wipro, TCS, Satyam)** as global clients outsourced fixes. • Creation of STPI (Software Technology Parks of India) for tax incentives and infrastructure. • IT exports became a major foreign exchange earner.
Global Delivery & BPO Boom (2000s)	• **IT-enabled services (ITES)** and **BPO/KPO** sectors grew (call centers, back-office, financial services). • India became known as the **"back office of the world."** • Emergence of **NASSCOM** as industry voice. • IT hubs like **Bangalore, Hyderabad, Pune, Gurgaon** expanded rapidly. • Multinationals (IBM, Accenture, Microsoft) set up Indian R&D centers.
Digital Transformation Era (2010s)	• Move from low-cost coding → end-to-end IT solutions, consulting, analytics, and R&D. • Growth in cloud computing, AI, machine learning, mobility, IoT, and cybersecurity. • Indian IT companies became global giants: TCS, Infosys, Wipro, HCL. • The startup ecosystem boom (Flipkart, Paytm, Ola, Zomato) strengthened IT innovation. • Government initiatives like Digital India (2015) pushed e-governance, fintech, and digital infrastructure.

New Age IT (2020s–Present)	• Focus shifted to **automation, AI, blockchain, 5G, quantum computing, green IT**. • COVID-19 pandemic accelerated **remote work, cloud adoption, and digital services demand**. • Indian IT firms now deliver **high-value consulting & product engineering**, not just outsourcing. • Expanding into **semiconductors, SaaS, and platform-based models**. • India has become the **world's largest sourcing destination**, with ~55% market share of global outsourcing.
Current Scenario	• IT & BPM sector = **~7.5% of India's GDP**. • IT services exports > **$190 billion** annually. • India is now moving from being a **service hub** → to a **global innovation and product development hub**.

Distribution of IT Sector

Significance of IT Sector

1. **GDP Contribution-** The IT sector contributed about 7.5% to India's GDP in FY23 and has reached 10% by FY25.
2. **Economic Growth-** The IT-based services and products have become indispensable for flourishing any business enterprise and accomplishing success. This industry has a conspicuous impact in improving the productivity of almost every other sector of the economy, it also has huge potential for further accelerating the growth and economic development.

3. **Export Revenue-** IT and IT-enabled services (ITeS) exports have reached **$283 billion in FY25**, and could surpass **$300 billion in FY26**.
4. **Employment Generation-** The IT sector supports over **five million direct jobs** and **drives indirect employment** through startups and digital platforms.
5. **Global Leadership-** India remains the top offshoring destination worldwide with over **67 Software Technology Parks of India (STPIs)** and **100% FDI support**.
6. **Innovation & Start-up Ecosystem**
   - The IT sector is at the core of India's start-up boom and innovation ecosystem, powering rapid advances in **AI**, **fintech**, **healthtech**, and other emerging fields.
   - India's robust **Digital Public Infrastructure (DPI)**, including platforms like the **Unified Payments Interface (UPI)** and **Aadhaar**, is creating new opportunities. IT companies are leveraging this infrastructure to build scalable solutions for fintech, e-commerce, and other sectors within the country.

Government initiatives for IT Sector

Digital India Mission	Launched in 2015, **Digital India transforms governance** and **public services** through digital infrastructure, e-governance, digital payments, and broadband for all.
Production Linked Incentive (PLI) Scheme	To reduce India's reliance on imports and boost domestic manufacturing, the **Production Linked Incentive (PLI) scheme** for IT hardware and electronics was introduced. The scheme provides financial incentives to companies based on their incremental sales from products manufactured in India.
National Policy on Software Products	Promotes domestic software product development, innovation, start-up growth, and export competitiveness.
Startup India Initiative	The Startup India initiative, while not exclusively for the IT sector, has provided a significant boost to technology startups. It offers various benefits, including tax exemptions, streamlined registration processes, and funding support.

Challenges faced by the IT Sector

1. **Talent and Skills Gap-** There is a constant demand for skilled professionals, particularly in **emerging technologies** like **AI**, **machine learning**, and **data science**. Companies are actively focusing on upskilling their existing workforce and hiring for specialized roles to meet this demand.
2. **Low on Innovation**
   - Predominant focus on outsourcing rather than creating intellectual property (IP).
   - Indian IT giants (TCS, Infosys, Wipro) prioritize human resource deployment over technological innovation.
   - **Despite a 55% share in global outsourcing**, India lacks global software products comparable to Google or OpenAI.
3. **Absence of long-term technology vision**
   - Companies **prioritise dividends over investing** in long-term research and development (R&D).
   - **Risk aversion** leads to missed opportunities in **emerging fields like AI, blockchain, and automation.** In contrast, countries like **China** invest heavily in cutting-edge technologies.
4. **AI blind spot**

- Limited investment in AI, with industry leaders **underestimating the feasibility of building cost-effective Large Language Models (LLMs)**. **DeepSeek's** success proves AI models can be developed for under **$7 million.**
- **Underrepresentation of Indian languages in AI datasets** (only 3% of global web data), hindering AI models from effectively serving Indian users.

5. **Overreliance on Global Capability Centres (GCCs)**
   - GCCs, despite hiring Indian engineers, **retain innovation and IP abroad.**
   - These centres **diminish revenues for Indian IT firms** and may **exploit tax loopholes** through transfer pricing.
6. **Dependence on American digital firms:**
   - India failed to develop **indigenous tech giants** despite having early industry leaders.
   - Unlike **China,** which fostered **domestic alternatives (Baidu, WeChat, Tencent),** India's market is dominated by foreign platforms.
   - **Weak policy support for homegrown companies** has exacerbated this reliance.
7. **Low Data centre capacity**
   - India's data centre capacity stands at **1GW,** significantly lagging behind the **US's 20GW.**
   - Reliance on foreign data hosting **limits India's sovereignty over data-driven innovations.**
8. **Weak Start-up culture**
   - Start-ups focus **mainly on services (e.g., Zomato, Swiggy)** rather than deep-tech solutions.
   - Investors prefer **low-risk ventures,** unlike in the US or China where venture capital supports high-risk, innovative technologies.

Way Forward

1. **Accelerate Skill Development:**
   - Expand advanced training in AI, data science, cybersecurity, cloud, and other in-demand skills through industry-academia-government collaboration.
   - Promote continuous learning, reskilling, and upskilling to bridge the talent gap and reduce attrition.
2. **Fostering innovation and risk-taking:**
   - Shift from a service-oriented model to **product-based innovation.**
   - Encourage companies to **reinvest profits into R&D** rather than distributing them solely as dividends.
3. **Promoting indigenous AI and deep-tech development:**
   - Develop **AI models in Indian languages** to cater to local users.
   - Invest in **cost-effective solutions** like open-source models (e.g., DeepSeek).
4. **Building a robust Start-up ecosystem:**
   - Government to incentivise deep-tech start-ups through **funding and procurement policies.**
   - Support for venture capital investment in **high-risk, high-reward ventures.**
5. **Strengthening data sovereignty:**
   - Expand India's data centre capacity to reduce dependence on foreign infrastructure.
   - Implement policies to **retain data within national borders** for security and innovation benefits.
6. **Ease of Doing Business:**
   - **Simplify regulations, incentivize R&D, streamline FDI policies**, and reduce compliance burdens to attract investment and outcomes-based innovation.
   - Foster coordination between central and state governments for uniform digital policy implementation.

Conclusion

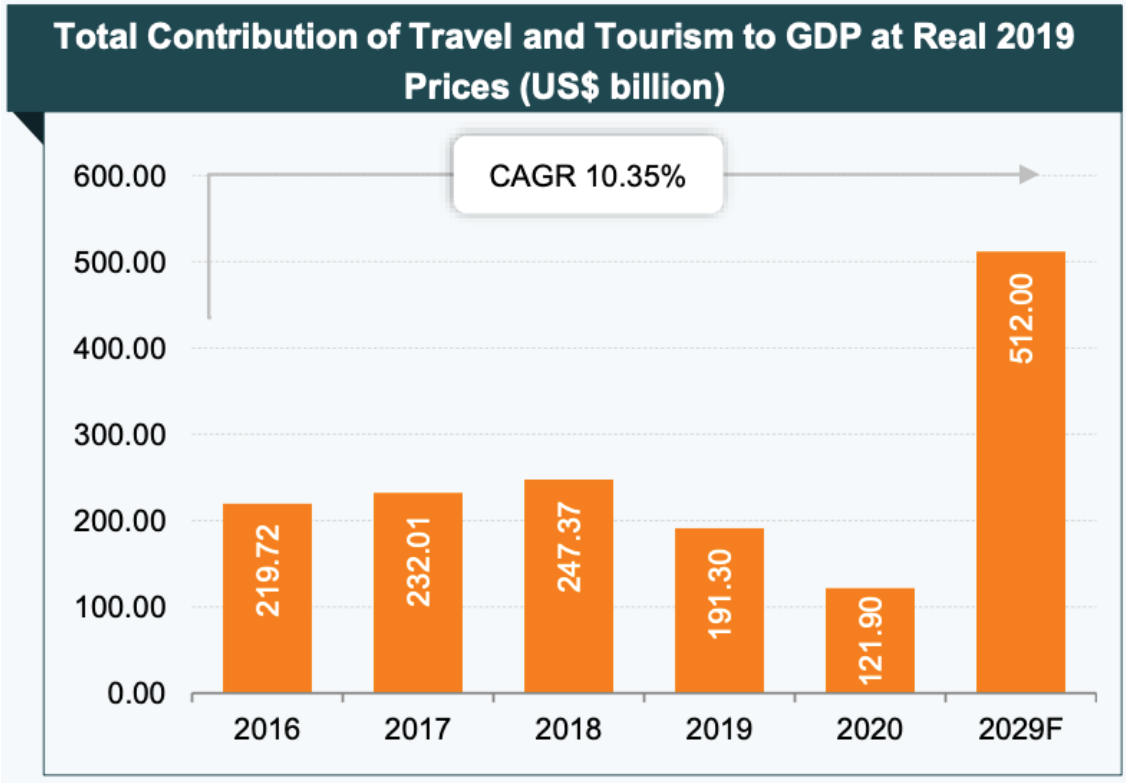
True leadership in IT will only come when India embraces a **culture of innovation, risk-taking, and long-term vision**.

3. TOURISM SECTOR

The tourism sector is a broad and dynamic industry encompassing all the businesses and services that facilitate travel, leisure, and recreation. It's one of the world's largest economic sectors, connecting people to new places, cultures, and experiences.

Status of the Tourism Sector in India (Economic Survey)

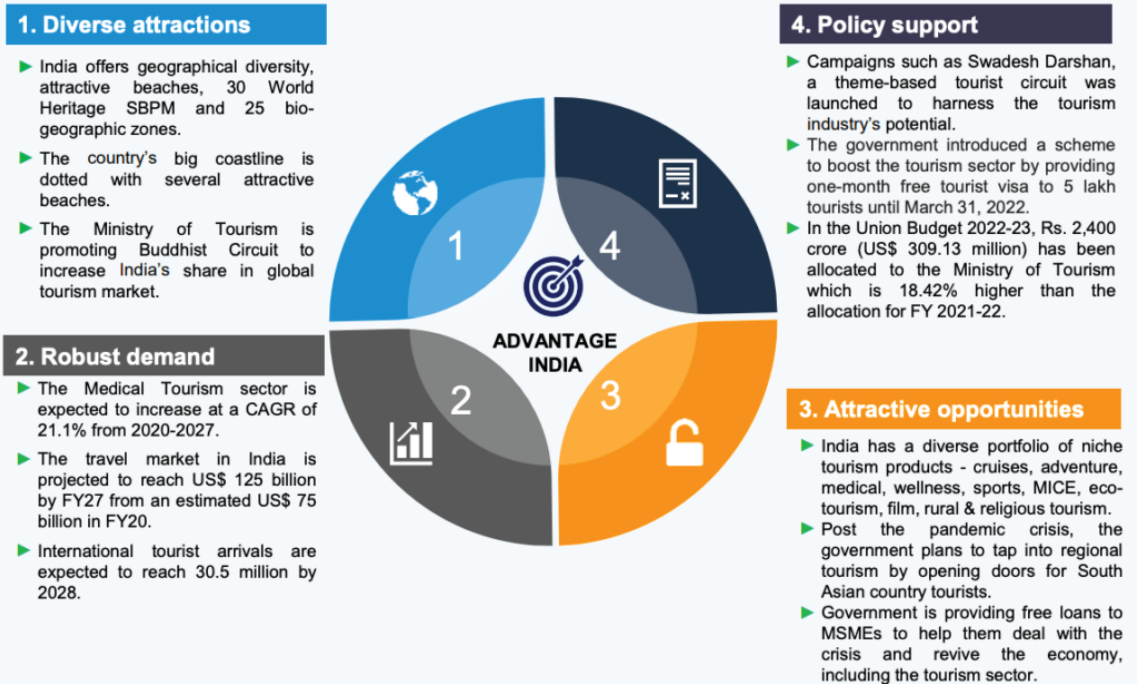
- Before the onset of the pandemic, the contribution of the tourism sector to India's GDP had reached **~$250 billion** in 2018. However, the contribution had fallen to **$122 billion** in 2020 due to the pandemic.
- In 2025, tourism contributed about **$231.6 billion to GDP**, making **India the world's 8th largest tourism economy**. The sector is projected to reach 4th position globally within the next decade. With post-pandemic recovery, the tourism industry is expected to reach US$ 512 billion by 2028.
- In FY24, travel and tourism contributed **5.22% to GDP**, supporting an estimated **8.46 crore jobs (13.3% of total employment)**, with foreign exchange earnings reaching **$35.0 billion**.
- The share of Tourism to GDP has hovered around **~5-6%**.



Source: IBEF

Driving factors of the Tourism Sector in India

The Tourism Sector in India is driven by various factors like diverse attractions, robust demand (like for medical tourism) and attractive opportunities.



Source: IBEF

Significance of the Tourism Sector

1. **Economic Contribution**:
   - Tourism contributes approximately 7% to India's GDP, generating foreign exchange earnings of around $43.7 billion in 2025.
   - It spurs growth in allied sectors such as hospitality, transport, handicrafts, and retail, amplifying economic impact.
2. **Employment Generation:**
   - The sector provides direct and indirect employment to about 42 million people, making it a vital source of livelihood, especially in rural and semi-urban areas.
   - It supports diverse skill levels—from unskilled workers to high-end professionals.
3. **Cultural & Heritage Preservation:**
   - Tourism incentivizes the conservation of India's rich cultural heritage, historic monuments, art forms, and traditional crafts by generating revenue and global interest.
   - Pilgrimage and heritage tourism help sustain religious traditions and local communities.
4. **Regional Development & Inclusive Growth:**
   - Tourism drives infrastructure development in tier-2 and tier-3 cities, promoting balanced regional growth.
   - Community-based and rural tourism empower local populations, improve quality of life, and reduce urban migration.
5. **Forex Earning:** Tourism is a significant source of foreign exchange, supporting India's balance of payments.

Challenges faced by the Tourism Sector in India

1. **Low Awareness**
   - Despite promotional campaigns by the Government, the awareness regarding India as a tourist destination remains low. Even among domestic tourists, the choice is limited to few popular

destinations which remain overcrowded, while many other potential places receive low footfalls of tourists.

- The information portals and centres are poorly managed. There is a lack of promotional campaigns in foreign countries. The absence of online branding campaigns fail to provide information to attract tourists.

2. **Infrastructure and Safety**
   - Many popular destinations lack air connectivity, especially in the hilly regions.
   - There is a lack of proper hygienic facilities in many places. Lack of cleanliness is off-putting to many tourists.
   - There are safety concerns especially among foreign visitors because of few cases of harassment. Poor experience of some tourists leads to bad word-of-mouth information impacting perception of potential tourists.
3. **Communication**- Many tourists face communication problems while in India. This makes them dependent on tourist guides or travel operators to curate their travel in India.
4. **Lack of Skilled Manpower**- There is dearth of skilled manpower especially multi-lingual tour guides or hotel staff. The sector is dominated by small unorganized players who can't spend on skilling their employees or sensitising them to cultural values of the foreign tourists. This impacts tourist experience.
5. **Visa Process**- The Government started the e-visa process (online) which has led to an increase in foreign tourists. However, the visa-on-arrival facility is limited to very few countries, limiting foreign tourists.
6. **Currency Fluctuations**- The inability to know the value of a currency means that long-range tourism prices are especially hard to predict and the fallout from this monetary instability is already impacting multiple tourism support systems.
7. **Environmental & Sustainability Concerns-** Overtourism causes environmental degradation, pollution, resource depletion, and waste management challenges at popular sites.
8. **Seasonality of Tourism Demand-** Demand fluctuates seasonally, leading to underutilized infrastructure and unstable income for providers during off-peak periods.

Steps have been taken for the development of the Tourism Sector in India

Infrastructure	The Government has been increasing investments in strengthening the country's road and rail networks and promoting port development is a significant driver for the growth of the Tourism sector. The **Adarsh Station Scheme** is helping modernize railway stations. The Regional Connectivity Scheme - **UDAN** (Ude Desh ka Aam Nagrik), is helping make air travel more economical and widespread to hitherto unserved routes. The Swadesh Darshan and **PRASHAD** schemes aim to stimulate growth in niche tourism segments such as religious, heritage, wellness, medical, adventure, MICE, wildlife. Under the Swadesh Darshan Scheme, the Government has launched several theme based circuits like Buddhist circuit which covers destinations associated with the life of Lord Buddha.
Promotional Campaign	Promotional activities such as the **Incredible India 2.0** campaign focuses on niche tourism products including yoga, wellness, luxury, cuisine wildlife among others. "**Find the Incredible You**" Campaign focuses on the promotion of niche tourism products of the Country on digital and social media.

Information Helpline	The government has introduced the concept of **e-tourist and e-medical visas** which has helped increase inbound tourists to the country. Additional initiatives such as **Atithi Devo Bhava,** a 24x7 multi-lingual Tourist Helpline, among others have helped improve the **safety and security** of tourists. On a pilot basis, an '**Incredible India Helpline**' has been set up to guide the tourists.
Safety	The Ministry of Tourism has adopted a **code of conduct for safe tourism**, which contains a set of guidelines to encourage tourism activities to be undertaken with respect to basic rights like dignity, and safety of both tourists and local residents, in particular women and children.
Investment	The government allows **100% Foreign Direct Investment in the Travel and Tourism sector** through the automatic route to increase investments across the sector. More recently, the **GST rate cut on hotel room tariffs** across the board has been a positive move for the industry and is expected to boost the sector's competitiveness globally.
Cleanliness and Hygiene	A major cleanliness campaign has been launched under the **Swachh Bharat movement** for protecting and preserving the sanctity of monuments of national heritage. The Ministry of Tourism has also launched an **awareness campaign to ensure cleanliness of surroundings** and help create a **Swachh Bharat, Swachh Smarak**.
Digital Database	In September 2021, the Government launched **NIDHI 2.0 (National Integrated Database of Hospitality Industry)**, a scheme which will maintain a hospitality database comprising accommodation units, travel agents, tour operators and others. NIDHI 2.0 will **facilitate digitalisation of the tourism sector** by encouraging hotels to register themselves on the platform.
Skill Development	The Ministry of Tourism has introduced the **Incredible India Tourist Facilitator (IITF)** and **Incredible India Tourist Guide (IITG)** Certification Programme to create an online learning platform of well-trained tourist facilitators and guides across the country. The Ministry of Tourism had launched an initiative called **SAATHI (System for Assessment, Awareness & Training for Hospitality Industry)** by partnering with the Quality Council of India (QCI) in October 2020. The initiative was focused on effective implementation of guidelines/SOPs issued with reference to COVID-19 for safe operations of hotels, restaurants, and other units.

Way Forward for Tourism Sector in India

1. **Infrastructure Expansion & Modernisation:**
   - Accelerate development of transport connectivity (roads, railways, airports) to underserved and emerging tourist destinations.
   - Upgrade accommodation, sanitation, and visitor facilities, focusing on quality and accessibility.
2. **Promotional Campaigns:** The government should continue to promote India's diversity and rich heritage to re-establish its position as a tourist paradise. The promotional campaigns should target both domestic and foreign tourists. Similarly, the extent of theme-based tourist circuits can be expanded.
3. **Skill development & Capacity Building:**
   - The **skilling initiatives should be scaled-up**. The tourism sector has a potential to provide a lot of livelihood opportunities in smaller cities/towns (below tier-2 level). It can help address the issue of jobless growth.

- Expand training programs for frontline workers, hospitality professionals, and local guides to uplift service standards.

4. **Sustainable Tourism:** There is a need to balance the promotion of tourism with safeguarding the physical, social, and cultural environment in the destination areas. The government should also **promote green and sustainable tourism** to tackle issues relating to water crisis, pollution, waste management.
5. **Policy & Regulatory Reforms:** The Government should further reform the tourist visa norms and processes to facilitate tourism. The Government should also explore the possibility of expanding the visa-on-arrival facility.
6. **Emergent Segments of Tourism:** The focus should also be on supporting and promoting the emerging segments of tourism.

Source: IBEF. Emerging Segments of Tourism.

Conclusion:

The tourism sector is a powerful force for both economic development and cultural connection. It has huge untapped potential in India. The multiplier effect associated with the tourism sector can help raise the income levels and ensure inclusive growth. A burgeoning tourism industry can prove to be vital in ensuring India's transition to a high income economy.

4. CARE ECONOMY

In India, the care economy is gaining prominence as a vital sector for economic growth and social development. It **encompasses both paid activities (like nursing, domestic work, and professional caregiving)** and **unpaid activities (such as childcare and elderly care, largely performed by women within households)**.

Data on Care Economy

International Labour Organisation suggests that increasing investments in the care services sector has the potential to **generate 475 million jobs globally by 2030**.

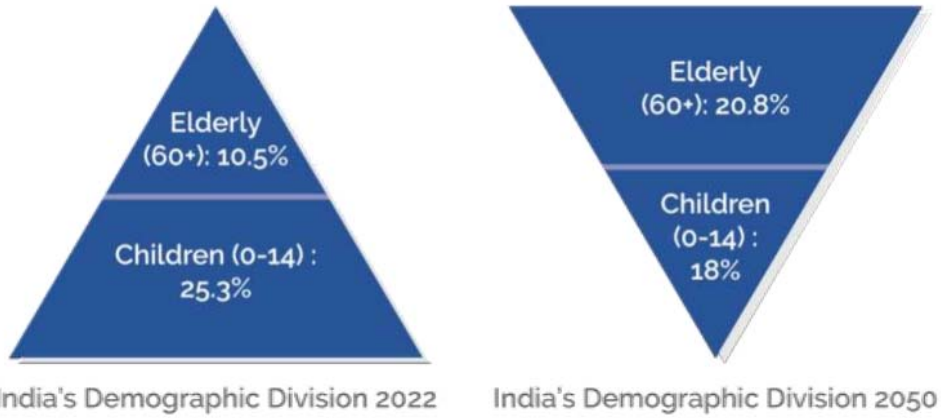
For India specifically, **direct public investment equivalent to 2% of GDP in the care economy**, can **potentially generate 11 million jobs**, nearly **70% of which will go to women**.

The economic value of women's unpaid care and domestic work is estimated to range between **15% and 17% of India's GDP**. Using the minimum wage method, this unpaid labour is valued at approximately **Rs. 30.7 lakh crore ($383.75 billion)**.

Need for Formalisation of Care Economy in India

1. **Changing Demographic-** Elderly population is **expected to increase between 2020 to 2050** although the fertility rates are decreasing thereby indicating a need for more care workers.

Figure 1: Inverted care pyramid in India



Source: *UNFPA, Our World in Data, Census, NITI Aayog, PIB*

2. **Wide gender gaps in unpaid domestic and care work-** Most of the unpaid work is being done by women resulting in increased gender divides. Women of working age spend **5.6 hours on unpaid work daily**, versus **30 minutes for men**.
3. **Double Burden of Work-** Working women in India have to face an even **higher burden of work due to unpaid care work**. But literate men have to spend less work on unpaid care work when compared to non-literate males.
4. **Support GDP growth-** The economic value of women's unpaid domestic and care work can contribute to **15-17% of GDP if accounted for** and will help in improving the economic status.
5. **Business potential-** Increasing investments in the Care Economy can bring a win-win situation as it can boost employment as well as supply caregivers for the needy.

Challenges with Care Economy in India

1. **Limited Policy Coverage-** Existing policies related to the care economy, such as maternity benefits and childcare leave, often have limited coverage and applicability, particularly in small-scale enterprises and the informal sector.
   - The **Maternity Benefit Act, 1961** is applicable only to establishments with **10+ employees**.
   - **98% of Indian enterprises are "micro" with less than 10 workers**, as per the Economic Census data.

- Even in registered manufacturing, **30% of establishments have under 10 employees**.
- This leaves many women without adequate support or protection in balancing work and caregiving responsibilities.

2. **Limited Workforce Participation-** The unequal burden of care work often hinders women's workforce participation and career advancement opportunities.
   - The Female Labour Force Participation Rate in India was **37% in 2023, according to PLFS 2022-23.** Despite this improvement, it remains below the desired level.
   - Many women are forced to prioritize caregiving over paid employment, leading to a lower representation of women in formal sectors and decision-making roles.
3. **Lack of Accessibility of Care Services-** Access to affordable and quality care services, such as childcare facilities and eldercare support, remains a challenge in many parts of India.
   - Limited availability and high costs of care services further exacerbate the caregiving burden on families, especially for low-income households.
   - It estimates that **women's unpaid care and domestic work represents almost 15%-17% of India's GDP**.
4. **Social Stigma and Cultural Norms-** Societal expectations and cultural norms often reinforce the perception that caregiving is primarily a woman's responsibility.
   - This stigma prevents men from actively participating in caregiving duties and perpetuates the cycle of unequal distribution of care work within households.

Government Initiatives for Care Economy in India

Saksham Anganwadi Poshan 2.0	Saksham Anganwadi and Poshan 2.0 is an **Integrated Nutrition Support Programme**. It seeks to address the challenges of malnutrition in children, adolescent girls, pregnant women, and lactating mothers through a strategic shift in nutrition content and delivery. It works for the creation of a **convergent ecosystem** to develop and promote practices that nurture health, wellness, and immunity.
Palna Ghar	**Palna Ghar Yojana** is a scheme of the Central Government to provide daycare facilities for the children of working mothers. It is also known as the **National Creche Scheme**. This care is given to children between the ages of **6 months to 6 years**.
Rashtriya Vayoshri Yojana	The senior citizens from BPL families **get physical aids and assistive devices free of cost**. The objective of this scheme is to **provide equipment according to the needs of senior citizens who are disabled due to old age**.
Childcare Leave Policy	It allows **730 days of paid leave for female employees to care for up to two children under 18**, over and above maternity leave.

Strategy for Strengthening Care Economy in India

1. **Recognise and mainstream the care economy-** Policymaking must explicitly **recognise unpaid work** as part of the economy. Shift from a purely market-based GDP view to a **broader economic framework.**
2. **Develop Satellite Accounts for unpaid work-** Creating satellite accounts to measure **household production** and **care work**.

3. **Identification-** The caregivers need to be identified and then brought into the formal system through identification mechanisms like **ID Cards**, **online portals**.
4. **Increase Government Spending-** The Government spending on the Care Economy needs to **improve to create more infrastructure** and **services** for both paid and unpaid workers.
5. **The 3R Framework**

- **Recognize** the extensive childcare responsibilities currently borne by mothers.
- **Reduce** the load on mothers through redistribution of childcare:
  - Within households by greater involvement of fathers
  - Outside households through affordable, quality neighborhood childcare options
- **Redistribute** childcare as a social responsibility, not just an individual burden on mothers.

Conclusion

The unequal distribution of unpaid care work responsibilities, high cost of child and elderly care, and insufficient social and public infrastructure together act as a significant barrier to female economic empowerment. **Support for parental leave policies**, **subsidies for care services**, **public and private investments in building care infrastructure**, **mechanisms for skill training for care workers**, and **mechanisms for monitoring service quality and benchmarks** offers a comprehensive framework for addressing the gender disparities in care and domestic work and boosting women's labour force participation.

5. ORANGE ECONOMY

The **creative economy (also known as the orange economy)** refers to knowledge-based activities that integrate culture, creativity, technology, and IP to drive economic, social, and cultural development, with strong potential to support sustainable development goals.

India's creator economy is projected to grow at an **18% Compound annual growth rate**, rising from **Rs 19 billion in 2023** to **Rs 34 billion by 2026**, according to Ernst & Young.

Current Status of India's Creative Economy

1. **Media and entertainment scale-** The sector was valued at **Rs. 2.5 trillion in 2024** with output near **Rs. 3 lakh crore**. It supports over 10 million livelihoods directly and indirectly.
2. **Digital revenue shift-** Digital media forms about one-third of sector revenues, reshaping production and distribution systems. Platform-based delivery expands reach across regions and markets.
3. **High-growth segments-** Animation and visual effects is valued around **₹103 billion**, gaming is valued about **₹232 billion**, and live entertainment is valued over **₹100 billion**. These sectors show strong scaling within the media economy.
4. **Growth trajectory-** Revenues may grow about **7 percent annually** until 2027, rising from ₹2,502 billion in 2024 to ₹3,067 billion in 2027. This confirms a stable services-sector growth engine.
5. **Global production integration-** India operates as a globally connected creative production base. Indian teams contribute to international films, streaming content, advertising, and immersive media through integrated global workflows supported by a layered and scalable talent pool.

Issues with India's Creative Economy

1. **Financing and infrastructure gaps-** Enterprises struggle to access credit and face infrastructure limits that restrict scaling of operations. Limited capacity slows expansion.
2. **Weak intellectual property protection-** Enforcement gaps make it difficult to protect original work. Infringement risks reduce incentives for innovation and investment levels.
3. **Digital divide impact-** Rural and semi-urban regions lack high-speed internet and digital literacy needed for participation. This limits market reach and inclusion.
4. **Fragmented informal markets-** Many creative sectors lack formal recognition and remain unstructured. Traditional artisans receive low income and limited institutional support.
5. **Declining traditional craft continuity-** Younger generations avoid traditional art forms because income potential remains low. This threatens preservation of cultural heritage.
6. **Artificial intelligence challenges-** AI creates opportunities but raises concerns over copyright, privacy, and content monopolisation. Rapid technological change challenges regulatory and ethical frameworks.
7. **Operational barriers-** Bureaucratic procedures, lack of mentorship, and non-transparent financial support systems create uncertainty for creators. Recognition and assistance processes remain uneven.

Initiatives Taken to Promote the Creative Economy

1. **National AVGC-XR roadmap-** Policy supports talent development, intellectual property creation, industry collaboration, and global market access. The sector may generate nearly **20 lakh jobs** over the next decade.
2. **Indian Institute of Creative Technologies- Indian Institute of Creative Technologies (IICT)** functions as a National Centre of Excellence for AVGC-XR and gaming. It provides training, infrastructure, and industry collaboration linked globally.
3. **Regional creative clusters-** Hubs in **Bengaluru, Hyderabad, Mumbai, Pune, Chennai,** and **Thiruvananthapuram** anchor growth. Emerging cities build new clusters that expand participation beyond major metropolitan centres.

4. **Event infrastructure systems-** Large venues and premium performance spaces support global touring circuits. Production companies, technicians, logistics teams, and hospitality services operate at international standards today.
5. **Orange Economy market platforms- World Audio Visual and Entertainment summit** (**WAVES)** connects creators, startups, industry leaders, and policymakers for collaboration and deal-making. WaveX supports startup incubation, while WAVES Bazaar enables trade in scripts, music, and audiovisual rights.
6. **Creator discovery initiatives-** The Create in India Challenge identifies emerging talent and links them to international platforms. It enables local ideas to compete, collaborate, and commercialise globally.
7. **Education and skilling pipeline-** Animation, Visual Effects, Gaming, Comics (AVGC) Content Creator Labs are proposed in **15,000 schools and 500 colleges** to build early skills. Demand may reach **nearly two million professionals by 2030**.

Conclusion

Creative industries now function as strategic economic infrastructure shaping employment, exports, and global cultural presence. India's expanding ecosystem shows strong market growth and institutional support.

Continued investment in skills, platforms, and access will determine how effectively creativity becomes lasting economic strength and sustained global engagement in coming years and across emerging digital value chains future worldwide.

MGP/BP/GS3/005

EXTERNAL SECTOR

INDIA'S EXTERNAL SECTOR

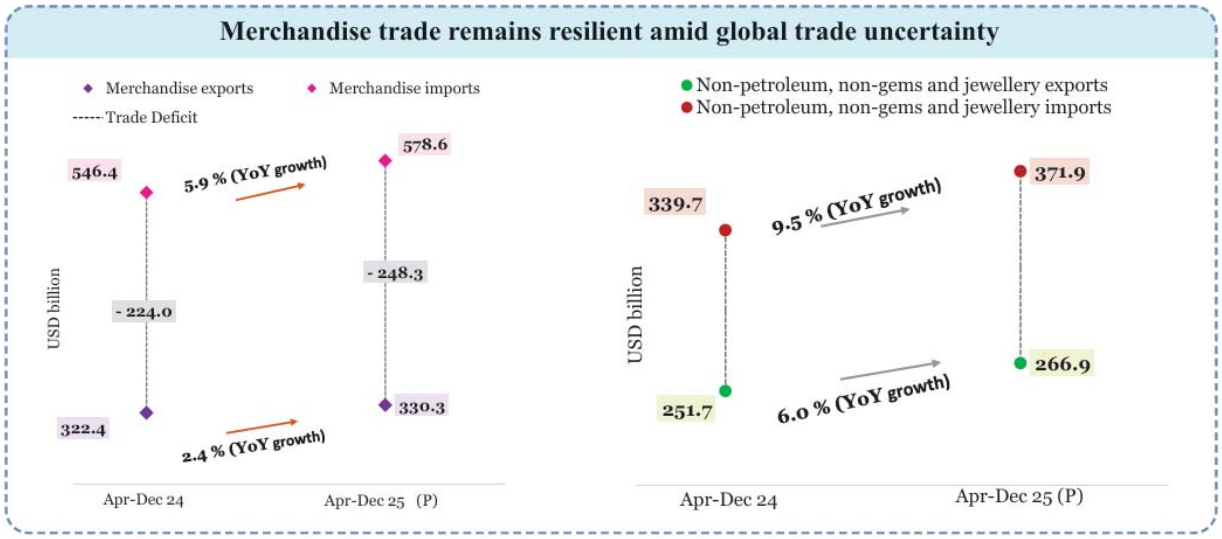
1. TRENDS IN EXTERNAL SECTOR (ECONOMIC SURVEY 2025-26)

According to the Economic Survey 2025-26, India's external sector has demonstrated remarkable resilience despite a fragile global environment.

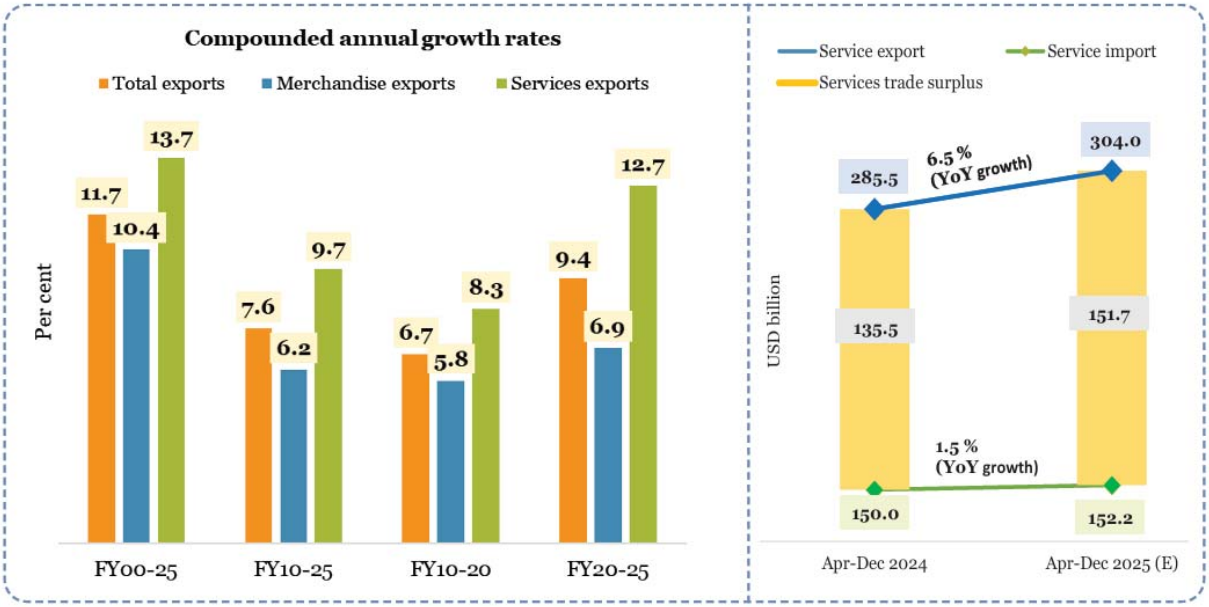
Indian Exports and Trade Performance

- **Record Export Growth-** India's total exports reached a historic high of USD 825.3 billion in FY25, growing by 6.1%.
- **Services Exports-** Reached an all-time high of USD 387.6 billion in FY25, growing at 13.6%.
- **Non-Petroleum Exports-** Non-petroleum exports achieved a historic high of USD 374.3 billion in FY25.
- **Global Share-** India's share in global commercial services exports rose to 4.3% in 2024, more than doubling from 2% in 2005.

India's trade performance in FY26 (April-Dec 2025)

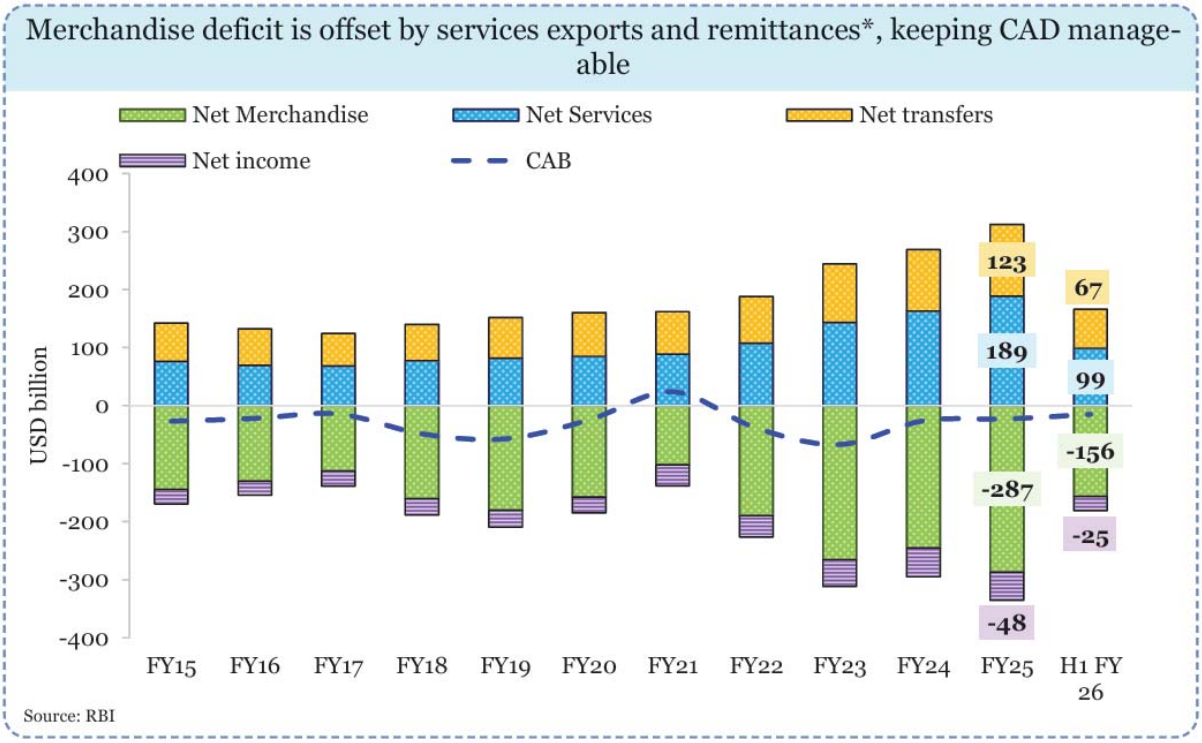


Services trade continues to be a key driver of India's trade performance

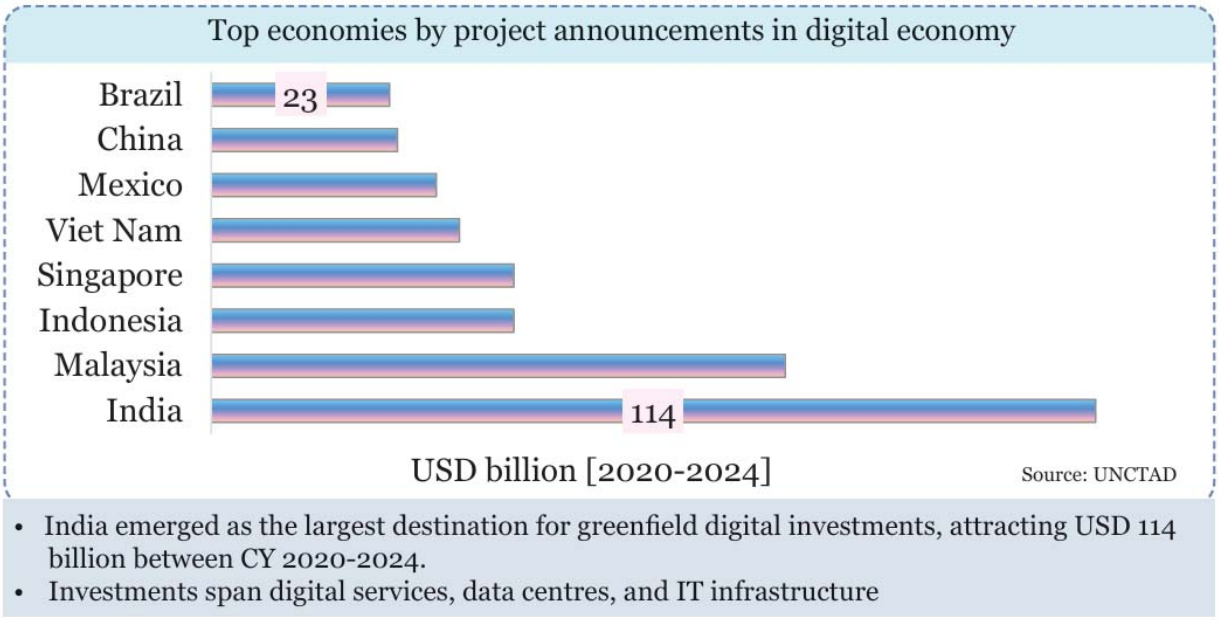


- **Foreign Exchange Reserves-** Increased significantly, strengthening the import cover to approximately 11 months, alongside a low Current Account Deficit.

Comfortable current account deficit (CAD) position

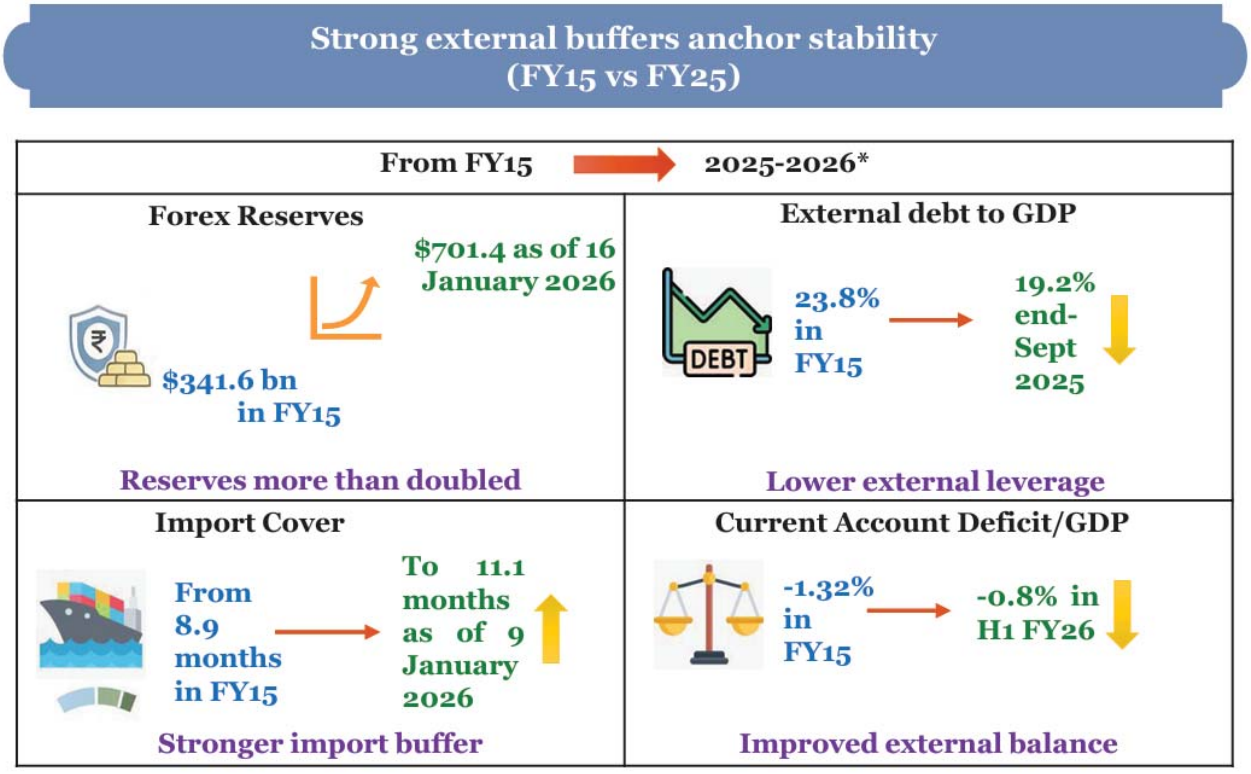


- **Greenfield Digital Investments-** India leads the world in greenfield digital investment between 2020-2024.



- India emerged as the largest destination for greenfield digital investments, attracting USD 114 billion between CY 2020-2024.
- Investments span digital services, data centres, and IT infrastructure

- **Strong external buffers anchor stability (FY15 vs FY25)-** India is entering 2026 with **strong buffers** against external shocks.



2. FOREIGN TRADE POLICY OF INDIA - 2023

India's Foreign Trade Policy (FTP) 2023, by the Directorate General of Foreign Trade (DGFT), marks a transformative step in the country's approach to international trade. Unlike the traditional five-year policy cycle, FTP 2023 follows a dynamic, open-ended framework, allowing timely updates in response to emerging global trade trends.

Objectives of FTP 2023

The primary goals of the policy are:

1. **Enhancing Global Integration-** Position India as a trusted and reliable trade partner in global markets.
2. **Creating a Supportive Ecosystem-** Foster a business-friendly environment aligned with the principles of **'Atma Nirbhar Bharat'** and **'Local goes Global.'**
3. **Future Preparedness-** Equip India to face future trade challenges, with the goal of becoming a **top exporting nation** during the **'Amrit Kaal'** period.
4. **Collaborative Efforts-** Promote partnerships with state governments to enhance exports at the grassroots level.
5. **Export Target-** Aim to **triple India's goods and services exports to USD 2 trillion by 2030**, up from the current USD 760 billion.

Four-Pillars Framework

FTP 2023 is built around four strategic pillars

Incentive to Remission	Transition from incentive-based schemes to remission-based benefits, including duty refunds and the Export Growth Capital Equipment Plan.
Export Promotion through Collaboration	Foster cooperation among exporters, states, districts, and Indian missions abroad to address challenges and promote exports.
Ease of Doing Business	Reduce transaction costs through simplified processes, digital initiatives, and streamlined documentation.
Emerging Areas	Focus on e-commerce exports, development of district export hubs, and streamlining SCOMET policies (Special Chemicals, Organisms, Materials, Equipment, and Technologies).

Key Features of FTP 2023

1. **Process Re-Engineering and Automation**
   - Emphasizes IT-driven automation, reducing paperwork and streamlining procedures.
   - Advance Authorisation and **Export Promotion Capital Goods (EPCG)** schemes continue, benefiting MSMEs with reduced fees and simplified processes.
2. **Towns of Export Excellence (TEE)**
   - **Faridabad, Mirzapur, Moradabad, and Varanasi** have been added to TEEs to boost exports, particularly in **handlooms, handicrafts, and carpets**.
   - These TEEs get first-hand access to export promotion funds and hands-on support through the EPCG scheme's **Common Service Provider (CSP) framework.**
3. **Recognition of Exporters**
   - Exporters recognized for performance will engage in capacity-building initiatives.
   - **2-star and above status exporters** will provide trade-related training, enhancing India's skilled workforce.
4. **Promoting Export from Districts**
   - Collaboration with states to promote district-level exports through the **Districts as Export Hubs (DEH)** initiative.
   - Export opportunities and challenges will be addressed by **State and District Export Promotion Committees**.

5. **Streamlining SCOMET Policy**
   - Improved compliance with international treaties ensures access to **advanced goods and technologies**.
   - Facilitates controlled exports under **SCOMET** from India.
6. **Facilitating E-Commerce Exports**
   - FTP 2023 targets e-commerce exports of **USD 200–300 billion by 2030** and lays out a roadmap for setting up e-commerce hubs, covering payment, bookkeeping, returns, and export entitlements.
   - **Courier export caps** increased and **Dak Niryat Facilitation Centers were established** for artisans, weavers, and MSMEs.
7. **Facilitation under EPCG Scheme**
   - Rationalization includes **dairy sector exemptions** and reduced obligations for **green technology products**.
   - Prime Minister Mega Integrated Textile Region and Apparel Parks **(PM MITRA) scheme** benefits are now included under CSP(Common Service Provider) Scheme of EPCG.
8. **Facilitation under Advance Authorization Scheme (AAS)**
   - Offers **duty-free imports** for manufacturing export items with provisions for **prompt execution** of export orders.
   - Extended benefits for **2-star and above status holders**.
9. **Amnesty Scheme**
   - A **one-time scheme** to address defaults on export obligations, providing relief from duties and interest.
   - Regularization of pending cases gives exporters a **fresh start with capped interest payments**.

Impact of FTP 2023

On Exports

- Focus on sectors like **electronics, engineering, gems and jewellery, petroleum, and pharmaceuticals**.
- Efforts to diversify markets with increased exports to the Middle **East and Africa**.

On Imports

- Streamlined **documentation** reduces processing times and improves customs clearance efficiency.
- Exceptions include **importing gifts via courier** unless essential (life-saving medicines, drugs, and rakhi).

On the Economy

- Promotes **innovation, collaboration, and ease of doing business**.
- Strengthens India's **global trade presence** while stimulating economic growth.

Conclusion

India's FTP 2023 reflects a progressive approach to global trade, emphasizing collaboration, technology, and sectoral diversification. By promoting 'Atma Nirbhar Bharat' and focusing on emerging sectors, the policy positions India as a strong player in global trade.

3. FTA 2.0 APPROACH

Free Trade Agreements (FTAs) are arrangements between two or more countries or trading blocs that agree to reduce or eliminate customs, tariff and non tariff barriers on substantial trade between them.

Failures/Challenges of India's Earlier FTAs which led to adoption of new FTA 2.0 approach

1. **Uneven distribution of gains-** The margin of preference (actual tariff cuts) given by India to its FTA partners in the old FTAs were higher than the margin of preference given by partner countries.
2. **India's low FTA utilisation-** India's FTA utilisation remained very low at around 25 per cent. On the other hand, developed countries utilize the FTAs to the maximum (~ 70–80 per cent).
3. **Increase in Indian imports to the detriment of Indian domestic industries-** The earlier signed FTAs, resulted in the increase of import of FTA partner's goods in India to the detriment of Indian domestic markets.

   For ex- Japan and ASEAN countries, exports to India increased tremendously while Indian export to them did not increase commensurately (**In the period between 2017 and 2022, India's exports to its FTA partners increased by 31 per cent, while its imports increased by 82 per cent**).
4. **Misuse of Rules of Origin-** The earlier negotiated FTAs had relaxed Rules of Origin(ROO) provisions, which were misused by the FTA partners to dump their products in the Indian market.

   For ex- Chinese electronics via ASEAN FTA. However, the **complexity of certification requirements of ROO**, under these FTAs hindered India's ability to streamline processes for its exporters.
5. **Lack of adequate industry and stakeholder consultation-** During the previous FTA negotiations, negotiators failed to involve representatives from relevant industries, businesses and associations. This led to **market access being granted to FTA partners without considering critical views** and **concerns from domestic industries**.
6. **Stringent Non-Tariff barrier provisions in the Old FTAs-** Previous FTA negotiation focus of India was more on Tariff barrier reduction. The non-tariff barriers provisions were not negotiated properly, which led to its mis utilisation against India.
   - **For ex- Indian exports have remained stagnant due to Japan's high import standards, despite India-Japan FTA**.
7. **Disparity in the performance of the manufacturing sectors in India and its FTA partner economies-** South Korea and ASEAN's manufacturing sectors outperformed India's in critical industries such as electronics, automobiles, leather and textile products as their industries were more competitive than the Indian counterparts. This led to a surge in Indian imports due to the FTA between South-Korea and ASEAN.
8. **Lack of Government's efforts to popularise FTAs among industry stakeholders-** Limited outreach activities and inadequate marketing to create awareness about the benefits of the FTAs resulted in the underutilisation of these agreements.
9. **Failure of India to take advantage of its service sector-** India failed to take advantage of its strong service sector capability by failing to include the service sector in some of the earlier FTAs.

Difference between FTA and FTA 2.0

Feature	Traditional FTA	FTA 2.0 (New Generation)
Primary Focus	Tariff reduction on **merchandise (goods)**.	Includes Services, investments, and digital trade.
Pace of Change	Gradual, phased tariff reductions.	Fast-track negotiation with immediate market access (e.g., India-UAE).
Strategic Goal	Market access for exports.	Strategic alignment and supply-chain security.

Newer Areas	Generally excluded.	Includes Digital Trade, Labour, and Environment chapters.
Regulatory Scope	Focused on border measures (tariffs).	Mutual Recognition Agreements (MRAs) for professionals.

We must focus on taking the following steps to use the FTAs 2.0 to our advantage by taking the following measures:

1. **Development of Trade infrastructure-** Govt must take efforts to give a boost to trade infrastructure, set up **fast track dispute settlement mechanisms** and **digitise trade procedures**.
2. **Increase global competitiveness of Domestic industries-** Increased competitiveness of Indian industries will help us increase exports to FTA partners and **achieve our target of $5 trillion economy**.
3. **Regular industry consultation and use of data analytics-** The regular feedback of industrial bodies like **CII, FICCI** and creation of data analytics division in the commerce ministry to regularly analyse the trade data to take corrective measures.
4. **WTO Plurilateral Negotiations-** India could further increase its exports by participating in the major plurilateral negotiations on services, environmental goods, and government procurement now taking place at the WTO.

4. INDIA-UK FTA

India and the European Union have finalized the long-pending Free Trade Agreement, officially known as the Bilateral Trade and Investment Agreement (BTIA).

Important provisions of India-EU FTA

Trade in Goods & Tariffs	**a. EU Access for India**: The EU will eliminate tariffs on over **99% of Indian exports** by value. This is a massive boost for labor-intensive sectors like **textiles, apparel, leather, gems and jewellery, and marine products**, which will now enter the EU duty-free. **b. Indian Access for EU**: India will eliminate or reduce tariffs on roughly **92–97% of EU goods**. • **Automobiles**: Duties on luxury EU cars (above €15,000) will drop from 110% to 10% over five years under a quota system (250,000 vehicles/year). • **Alcohol**: Tariffs on premium wines will be cut from 150% to 20%, and spirits to 40%. • **Machinery & Chemicals**: Duties of up to 44% on machinery and 22% on chemicals will be mostly eliminated.
Services and Professional Mobility	• **Market Access**: India secured access to **144 EU services sub-sectors** (including IT, finance, and education), while India opened 102 sub-sectors to the EU. • **Professional Mobility**: The FTA provides a predictable framework for business travelers, intra-corporate transferees, and independent professionals. • **Students**: A new framework guarantees **post-study work rights** (up to 9 months) and eases the path for Indian students to study in the EU. • **Social Security**: Both sides agreed to work toward a Social Security Agreement within five years to ensure Indian professionals don't lose their pension contributions while working in the EU.

Sustainability and Climate	• **CBAM (Carbon Border Adjustment Mechanism):** India secured "Most Favored Nation" (MFN) assurance. While not fully exempt, India will receive technical cooperation and financial assistance (€500 million) to help industries like steel and aluminum reduce their carbon footprint. • **Labor Rights**: Both parties committed to core **International Labour Organization** (ILO) principles, including the abolition of child labor and non-discrimination at work
Intellectual Property (IPR) & Technology	• **Traditional Knowledge**: The EU has formally recognized India's Traditional Knowledge Digital Library (TKDL), protecting Indian heritage (like Yoga or Ayurveda) from being patented by foreign firms. • **Tech Cooperation**: The deal goes beyond trade to include collaboration on Artificial Intelligence (AI), semiconductors, and clean energy.
Sensitive Sector Safeguards	• **Agriculture & Dairy**: To protect small-scale farmers, India has **completely excluded** sensitive items like **dairy, wheat, rice, and sugar from liberalization**. • **Geographical Indications (GIs):** A separate agreement is being finalized to protect iconic products (like Darjeeling Tea or Champagne) from imitations.

• WHO GAINS WHAT, WHAT NEXT

FOR INDIA

	Before FTA	After FTA
Marine sector	upto 26%	0%
Chemical	upto 12.8%	0%
Leather & footwear	upto17%	0%
Plastic & rubber	upto 6.5%	0%
Textile & apparel	upto 12%	0%
Base metal	upto 10%	0%
Gems & jewellery	upto 4%	0%
Railway, aircraft, ships	upto 7.7%	0%
Furniture & light consumer goods	upto 10.5%	0%
Toys	upto 4.7%	0%
Sports goods	upto 4.7%	0%

FOR EU

	Before FTA	After FTA
Motor vehicles	110%	10% (quota of 250k/yr)
Wine	150%	20% (premium); 30% (medium)
Spirits	up to 150%	40%
Beer	110%	50%
Olive Oil	up to 45%	0%
Kiwis & pears	33%	10% (in-quota)
Machinery & electrical equipment	up to 44%	0% for most products
Aircraft & spacecraft	up to 11%	0% for most items
Iron and steel	up to 22%	0% for most items
Pharma	11%	0% for most items
Processed food (bread, biscuit, pasta, chocolate, pet-food)	up to 50%	0%

- India's labour-intensive annual exports worth $35 billion benefit immediately
- Signing of the deal expected later this year
- Legal scrubbing to take 4-5 months
- Deal expected to come into effect by early next year
- India will receive same flexibility as US on EU's carbon tax

SOURCE: COMMERCE & INDUSTRY MINISTRY, EUROPEAN UNION

Benefits of India-EU FTA for India

1. **Boost to Exports & Economic Growth:** India would gain greater and more predictable market access for its competitive export sectors. This includes:
   - **Textiles and Apparel**: Reduced EU tariffs (currently up to 9-12%) would make Indian garments, home textiles, more competitive against rivals like Bangladesh and Vietnam (who already enjoy duty-free access to the EU under Everything But Arms (EBA scheme).
   - **Agriculture & Processed Foods**: Products like **basmati rice, mangoes, grapes**, **spices, and seafood** could see higher exports with simplified regulations and recognition of India's Geographical Indications (GIs).
   - **Services**: Major gains for IT/ITeS, business services, engineering, and R&D. Easier cross-border data flow provisions would be a huge boost.
   - Overall, 6% of India's exports will see tariff reduction – mostly sourced from the labour-intensive sectors – which will help in improving the trade competitiveness of Indian products in the EU market.
2. **Movement of Skilled Professionals:**
   - The FTA will help Indian professionals (engineers, IT experts, consultants, nurses, chefs) to get easier & quota-based temporary work visas & permits.
   - **Mutual Recognition Agreements (MRAs)** for professional qualifications (e.g., degrees, certifications for accountants, architects). This would allow Indian professionals to work in the EU without retraining.
   - This helps India leverage its demographic dividend, secure high-value remittances, and reduce friction in a traditionally restrictive area for the EU.
3. **Attracting FDI:** The agreement would boost the investor confidence, encouraging EU companies to set up more manufacturing and R&D hubs in India to serve both the Indian and wider Asian markets ("China+1" strategy). This aligns with Make in India and can create jobs and facilitate technology transfer.
4. **Consumer Benefits:** Indian consumers and industries would get access to high-quality EU goods at lower prices due to tariff reductions.
5. **Technology Transfer & Upgradation:** Increased competition and collaboration would push Indian industry to improve quality and innovation. Partnerships in green tech, renewable energy, and digital infrastructure (areas where the EU is a leader) could accelerate India's sustainable development goals.
6. **Recognition of Indian Heritage:** The EU has officially recognized India's Traditional Knowledge Digital Library (TKDL). This prevents foreign companies from "biopiracy" – patenting traditional Indian assets like **Yoga, Neem, or Ayurveda** – and ensures these remain protected Indian intellectual property.

Benefit of the India-EU FTA benefit for EU

1. **Access to a Massive, Fast-Growing Consumer Market:** India, with its 1.4 billion people and a rapidly growing middle class, represents one of the world's last untapped large consumer markets. The EU seeks guaranteed, preferential access for its premium goods -especially in sectors like – Automobile & Auto parts, Wines & Spirits, Cheese & Dairy, Machinery, Chemicals etc.
2. **Countering China's Influence (China-Plus-One):** The EU has a clear "**de-risking**" and diversification strategy. Deepening ties with democratic India is a geopolitical imperative to reduce economic over-dependence on China and build a reliable strategic partner in the Indo-Pacific.
3. **Access to Raw Materials & Intermediates:** Reliable access to Indian materials and processed goods (like generic pharmaceutical ingredients, textiles, leather) is crucial for EU industries.
4. **Boosting EU's Services & Digital Trade:**
   - **Financial, Legal, and Business Services**: EU banks, insurance companies, and professional service firms seek better access to India's vast and protected services market.

- **Digital Trade**: New rules protect software source code from mandatory disclosure and establish high standards for data privacy, providing legal certainty for EU tech firms.

5. **Sustainability Standards:** By including chapters on climate (aligned with the Paris Agreement) and labor rights, the EU ensures that its trade with India adheres to the high environmental and ethical standards demanded by European voters.

Challenges that remain despite the India-EU FTA

1. **India's Demands on Agri & Textiles:** India wants greater access for its textiles, garments, and agricultural products (like rice, fruits, shrimp). The EU resists due to sensitive domestic producers (e.g., textile makers in Italy/Portugal, farmers in France/Poland) and non-tariff barriers like stringent Sanitary and Phytosanitary (SPS) measures.
2. **Non-Tariff Barriers:** For Indian exporters, the biggest challenge isn't the tariff anymore; it's the Non-Tariff Barriers like:
   - **CBAM (Carbon Tax):** The EU's Carbon Border Adjustment Mechanism is set to enter its full definitive phase. Even with the FTA, Indian steel and aluminum exporters may still have to pay a "carbon cost" or face 15–22% price cuts to stay competitive unless they drastically green their production.
   - **EUDR (Deforestation Regulation):** New EU rules (phasing in through late 2026) require proof that products like coffee, rubber, and leather did not come from deforested land. This is incredibly difficult for India's millions of small-scale farmers to document.
   - **SPS Norms**: The EU has some of the world's strictest Sanitary and Phytosanitary rules. Many Indian agri-products often face rejection due to pesticide residue levels that are acceptable in India but banned in the EU.
3. **Data Privacy and Digital Friction:** While there is alignment between **India's DPDP Act (2023)** and the EU's GDPR, full "data adequacy" has not yet been granted. Without a formal "adequacy status" from the EU, Indian IT firms still face a "compliance tax"—expensive legal and technical hurdles to move and process European data in Indian servers.
4. **Ratification Hurdles:** The agreement must be ratified by the European Parliament. Some EU nations with strong agricultural lobbies (like France) may still raise objections to specific provisions, even though sensitive items like beef and sugar were excluded.

Way Forward

1. **Ratification and Implementation:** Ensure timely legal approval and ratification by the European Parliament, EU member states, and India so that the FTA can be operationalized at the earliest.
2. **Address Non-Tariff Barriers:** Collaborate to resolve non-tariff challenges such as CBAM, strict Sanitary and Phytosanitary norms, and regulatory standards to protect the competitiveness of Indian exports.
3. **Support Indian Industries:** Strengthen capacity-building measures, compliance assistance, and technical support for MSMEs to help them meet EU quality, sustainability, and carbon standards.
4. **Enhance Services and Mobility Gains:** Conclude agreements on professional mobility, mutual recognition of qualifications, and social security to ease movement of Indian professionals and students.
5. **Deepen Strategic Cooperation:** Use the FTA as a platform to expand collaboration in emerging areas like digital trade, clean energy, advanced technology, and climate action for long-term strategic alignment.

Conclusion

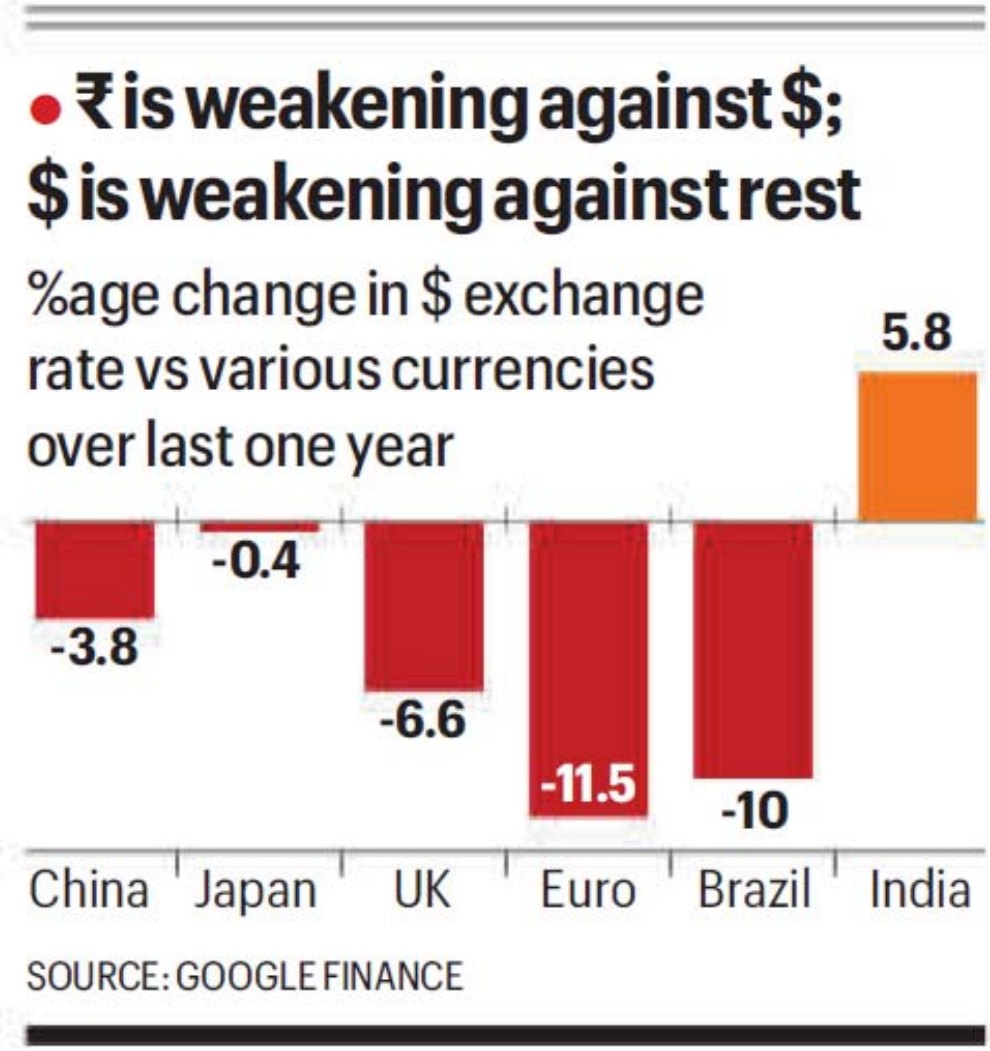
The India–EU FTA represents a major step in India's economic and strategic engagement with Europe. While it promises export growth, mobility, and investment, its success will depend on managing non-tariff

barriers and green transition challenges. Ultimately, it goes beyond trade to shape a long-term strategic partnership in a multipolar world.

EXTERNAL TRADE VULNERABILITIES

1. WEAKENING OF RUPEE – CAUSES & CONSEQUENCES

The Indian rupee has continuously been weakening against the US dollar in the recent past. The rupee has lost almost 6% of its value against the dollar over the past year. There are many factors which are responsible for weakening of the Indian rupee even when the currencies of other emerging market economies are strengthening against the dollar.



Factors responsible for weakening of the Rupee against the Dollar

1. **Domestic Factors:**
   - **Widening Current Account Deficit (CAD):** The CAD is expected to rise significantly in the Fiscal Year 2025-26, with forecasts indicating a rise to ~1.7% of GDP. This widening of the deficit is mainly driven by a large merchandise trade deficit.

- **Sustained FII (Foreign Institutional Investors) Outflows:** 2025 has been characterised by a sustained net selling trend by FIIs in both Indian equity & debt markets. This capital flight reduces the supply of foreign currency & directly pressures the rupee.
- **Inflation & Growth Headwinds:** Even though, the long-term growth outlook remains strong, but, the slower near-term GDP growth & very low inflation level have acted as negative economic indicators, dampening investor confidence in Rupee's short term stability.

2. **External & Global Factors:**
   - **India-USA Trade Tensions & Tariffs:** USA is India's one of the top trading partners. However, the imposition of 50% tariff on Indian goods by the Trump administration has severely impacted the export competitiveness of Indian goods & has increased the market risk perception - making Indian rupee the worst performing currency in Asia for the year.
   - **Strengthening of the US Dollar:** Despite the US Fed Reserve beginning its rate-cut cycle, the US Dollar has maintained persistent strength, reflecting its status as global reserve currency & a safe haven asset during a period of geopolitical uncertainty. Strengthening of US dollar against major currencies puts pressure on INR.
   - **High Crude Oil Demand & Import Bill:** India imports almost 80-85% of its crude oil. The rise in the crude oil prices & that of the important commodities imported by India like gold - lead to widening of India's trade deficit & weakening of INR.
2. **Monetary Policy Factors:**
   - **Unfavourable Interest Rate Differentials:** Even though the US Fed Reserve has begun its rate-cut cycle at a modest rate (e.g. 25 basis point cut in the late 2025), the cumulative interest rate differential remains attractive for the US Dollar relative to Rupee's real yield - driving the capital away from India.
   - **RBI's Stance:** The RBI has chosen a **Neutral Policy Stance** & kept the repo rate unchanged for most part of the year 2025 - prioritizing domestic liquidity management & growth over an aggressive defense of the Rupee.

Consequences of weakening of the Rupee

1. **Impact on Consumers:**
   - **Inflationary Pressure:** As the INR weakens, the Oil Marketing Companies have to pay more Rupees for the same barrel of oil. This increased cost is eventually passed on to the consumers through higher prices for petrol, diesel, and natural gas. This high fuel cost then triggers a cascading effect - contributing to broader consumer price inflation.
   - **Cost of Goods:** The price of other key imports, such as electronics, gold, industrial chemicals, and fertilisers, also rise - intensifying the inflationary pressure & eroding the purchasing power & savings of the average household.
   - **Foreign Travel & Education:** Foreign travel & education will become significantly expensive.
2. **Impact on Trade (Imports/Exports)**

The WINNERS	**a. Increased Competitiveness:** A weaker rupee makes the Indian goods & services cheaper for foreign buyers who pay in Dollars. This can boost the competitiveness of Indian exports in global market. **b. High Profitability for Exporters:** Indian exporters, particularly the IT Service Sector, benefit significantly. Weakening of the INR directly boosts their profit margin & revenue growth. **c. Boost to Domestic Investment:** Rise in export revenue can lead to increased domestic investment as exporters look to expand capacity to meet the higher demand.

The LOSERS	**a. Higher Import Bill:** Weakening of the Rupee against Dollar puts upward pressure on the net import bill. **b. Wider Trade Deficit:** The cost of essential imports outweighs the revenue gain from exports. A significant rise in import bill can lead to a widening of the Trade deficit.

3. **Impact on Corporates (External Debt)**
   - **Increase in Debt Servicing Cost:** The Indian Corporates who have taken ECBs denominated in USD & have not fully hedged their exposure, face a major risk. A weaker rupee means that a company has to pay more amount of INR for the USD-denominated debt.
   - **Divergent Fortunes:** The corporate sector witnesses a divergence - while the export-oriented Companies see higher profits, the import-dependent Companies & highly indebted cos. face significant financial strain.
4. **Macroeconomic Impact**
   - **Forex Reserves Drawdown:** The RBI often intervenes (spot intervention) in the forex market to prevent excessive depreciation of the Rupee. The RBI sells USD to absorb the excessive Rupee liquidity. However, it leads to reduction in the national reserve buffer.
   - **Capital Flight:** Withdrawal of funds by FIIs is one the causes for the weakening of the INR. If the Rupee continues to weaken, it could signal greater macroeconomic instability which may increase the rate of capital flight from India - creating a self-perpetuating cycle of depreciation.

Way forward

1. **Strengthen Macro-fundamentals:**
   - Contain **fiscal and current account deficits** through better tax mobilisation, rationalised subsidies, and export diversification so external financing needs remain credible.
   - Keep **inflation durably low and stable** via prudent monetary-fiscal coordination, which reduces the long-run tendency of the rupee to lose purchasing power versus the dollar.
2. **Manage External Vulnerability:**
   - Reduce import dependence in **oil, gas, electronics, fertilisers and defence** through domestic capacity, renewables, and energy efficiency, lowering structural dollar demand.
   - Maintain **adequate forex reserves and flexible exchange rates** so the RBI can smooth volatility without defending unsustainable levels, reassuring markets about India's shock-absorbing capacity.
3. **Upgrade Export Competitiveness:** Upgrade export competitiveness with reforms in **logistics, trade facilitation, skilling, and industrial policy**, focusing on high-value manufacturing and services instead of low-margin commodities.
4. **Attract Capital Inflows:** Attract stable **FDI and long-term portfolio flows** by improving contract enforcement, regulatory predictability, and financial-sector depth, reducing reliance on short-term "hot money".
5. **Safeguard Vulnerable Sectors & Households:**
   - Encourage firms with foreign-currency liabilities to **prudently hedge** and improve disclosure, limiting balance-sheet stress from sharp currency moves.
   - Use **targeted support** (e.g., fuel tax calibration, transport and fertilizer support, food security buffers) to protect poorer households from imported-inflation spikes without large, open-ended subsidies.
6. **Long-term Solution:** Continue structural reforms (land, labour, financial inclusion, digital public infrastructure) that raise **productivity and long-term growth**, making India more attractive to capital and easing pressure on the rupee over time.

Conclusion

The weakening of the Indian rupee reflects a mix of global pressures like a strong US dollar and domestic challenges such as trade deficits and capital outflows. While it raises import costs and inflation risks, it can improve export competitiveness if managed prudently.

2. TRUMP'S TARIFFS- IMPACT ON INDIA

President Donald Trump imposed an **additional ad valorem duty of 25%** on Indian imports starting from August 27 for "directly or indirectly importing Russian oil". This was in addition to the 25% reciprocal tariff that took effect on August 7.

Status of India-US Bilateral Trade (in FY24) before the imposition of Punitive Tariffs

Bilateral Trade and Trade Surplus	In FY24, the bilateral trade between India and the US stood at a record **US$ 118.2 billion**. In FY24, India had a trade surplus of **US$ 36.8 billion** with the US.
Indian Exports to the US	Of the total trade in FY24, Indian exports to the US stood at **US$ 77.5 billion**, while American exports to India stood at **US$ 40.7 billion**.
Major exports from India to US	Major exported items from India to the US include **Engineering goods** (US$ 17.6 billion), **Electronic goods** (US$ 10.0 billion), **Gems and Jewellery** (US$ 9.90 billion), **drug formulations and biologicals** (US$ 8.72 billion), **petroleum products** (US$ 5.83 billion), and **RMG cotton including accessories** (US$ 4.71 billion)

Features of Trump's Tariffs

50% Tariff Rate: The tariff rate has been doubled to a flat 50% on most Indian goods, making this one of the highest general tariffs imposed by the US in recent years.

Major sectors Impacted by Trump's Tariffs- Almost 66% of Indian exports to the US face 50% tariff threat. Major affected sectors include **textiles, gems & jewelry, footwear, furniture, chemicals, leather, and seafood.** These are labor-intensive industries, accounting for significant portions of Indian exports to the USA.

Sectors Exempted- Steel, **aluminum**, **copper**, **passenger vehicles**, **pharmaceuticals**, and **electronics** are largely exempted from 50% tariffs. Shipments in transit before the cut-off, humanitarian donations, and informational materials like books are also spared under specific conditions.

Reasons behind the 50% tariff imposed by the Trump Administration

1. **India's Acquisition of Russian Oil-** The US administration explicitly stated that the punitive tariffs are a **response to India's large-scale importation of Russian crude oil**.
   - Indian refineries, including government-backed refiners, have continued oil purchases despite US calls for sanctions and Russia's economic isolation.
2. **Addressing the US Trade Deficit Concerns-** The tariffs aim to nullify the US's $38 bn trade deficit with India.
3. **Protection of domestic US industries-** The administration holds that the tariffs will encourage domestic US manufacturing by making imported goods more expensive. This is in line with Trump's **Buy American**, **Hire American** and **Making America Great Again (MAGA) policy**.
4. **Gaining Indian Market Access-** The tariffs aim to put **pressure on India** to lower Indian tariffs on

automobiles, **medical devices**, and **agricultural goods**. US companies aim to gain Indian market access.

5. **Acceleration of Indo-US trade deal-** The punitive tariffs aim to accelerate the pace of Indo-US trade deal, which aim to reduce **the tariff** and **non-tariff barriers** imposed by India.

Likely impact of Trump's tariffs on India's exports

Economic Impact:

- **Decline in Exports-** The most immediate and direct impact will be a sharp decline in India's exports to the U.S. in the affected sectors.
  - With tariffs as high as 50%, Indian goods will become **prohibitively expensive** for U.S. consumers and retailers. Global Trade Research Initiative (GTRI) projects a **43% drop** in US-bound Indian exports.
- **Job Losses-** The tariffs **disproportionately target labor-intensive industries** like **textiles**, **gems**, and **jewelry**. This could lead to massive **job losses** in these sectors, affecting millions of workers and their families, particularly in rural and semi-urban areas.
- **Reduced GDP Growth-** A fall in exports and shrinking markets may dampen economic growth, with ripple effects through supply chains and supporting services.
  - **For ex-** The textiles & apparel manufacturers in **Tirpur, NOIDA** & **Surat** have reportedly stopped production. GTRI estimates that the punitive tariffs could cost the Indian economy **0.2–0.3%** of its GDP.
- **Competitive advantage for Competitors-** Higher priced Indian goods may force U.S. buyers to shift to products of lower tariffed nations, such as Vietnam, Bangladesh, and Mexico.

Geopolitical Impact:

- **Strained Indo-US Relationship-** The tariffs signal a growing Indo-US **economic** and **diplomatic rift** over issues like trade imbalance and energy policy.
- **Reduced Influence in Indo-US trade negotiations-** The tariffs could reduce India's leverage in future trade negotiations with the U.S. The U.S. may use the tariffs as a bargaining chip to push for a more favorable trade agreement on its terms.

Domestic Impact:

- **Pressure on Domestic Market-** With export markets becoming less viable, affected industries will be forced to sell their products domestically. It could lead to **oversupply** and a **further squeeze on profit margins** for businesses.
- **Fiscal Pressure on Government-** It could increase the fiscal pressure on the Indian government as **new financial packages**, **subsidies** would be needed to provide support to the affected industries.

Way forward

1. **Diplomatic Engagement & Non-Retaliation-** India should focus on maintaining open channels of communication with the U.S. to convey its national interest in purchasing Russian oil and to argue that the tariffs are not an effective or fair way to address the issue.
2. **Market Diversification-** Recognizing the risk of being overly dependent on a single market, India must accelerate its efforts to diversify its export destinations. This can be done by:
   - **Targeting New Markets-** The government should expedite work on "**40-country outreach**" to push exports to new and existing markets in **Europe**, the **Middle East**, **Latin America**, and other regions.
   - **Leveraging FTAs:** India should actively pursue and finalize Free Trade Agreements (FTAs), with blocs like the **European Union** (EU), to make its exports more competitive in these regions.

3. **Boosting domestic resilience-** To cushion the impact of the tariffs on affected industries, the Indian government should focus on domestic reforms and support:
   - **GST Rationalization:** The simplification of tax slabs and reduction of tax burden on businesses, would help lower the cost of goods and stimulate domestic demand, offsetting some of the losses from exports.
   - **Financial and Policy Support:** Lines of Credit, regulatory support, and extension of duty-free raw material imports will maintain the cost-competitiveness of the sector.
4. **Sector-Specific Cooperation**
   - **Agriculture:** India could allow more U.S. imports in exchange for greater U.S. openness to Indian spices, rice, and seafood.
   - **Energy:** Expand India's purchase of U.S. LNG and crude oil to reduce trade imbalance.
   - **Digital Trade:** Cooperate on e-commerce regulations, data flows, and IT services market access.
5. **Balanced Tariff Reductions-** India can consider **phased reduction of tariffs** on sensitive imports (for ex- whiskey, automobiles) while protecting vulnerable domestic industries through safeguards.
   - The U.S. can restore **GSP benefits** or provide India some **preferential access again**, balancing the tariff cuts.

Conclusion

A balanced & phased approach to address issues stalling the trade deal, boosting domestic resilience & bringing domestic reforms should be followed by India while protecting the interest of Indian businesses especially of small & micro enterprises.

3. TRADE WAR AND CURRENCY WAR

Trade War, Currency War and Trade Weaponization

1. **Trade War-** A trade war is a **situation** when **two or more countries** impose **tariffs (import taxes)**, **quotas**, or **other restrictions** on each other's goods and services.

   The goal is usually to protect domestic industries, reduce trade deficits, or pressure the other country into changing policies. However, retaliation by other countries usually leads to collapse of the international trade order.

Examples of Trade War

- **U.S.–China Trade War (2018–present)-** The U.S. accused China of unfair trade practices, such as intellectual property theft and trade imbalances. The U.S. imposed tariffs on Chinese goods worth hundreds of billions of dollars, and China retaliated with its own tariffs on American exports (soybeans, cars, etc.).
- **Smoot-Hawley Tariff Act (1930s, U.S.)-** During the Great Depression, the U.S. raised tariffs on thousands of imports. The retaliation by other countries, led to collapse of international trade, worsening the depression.

2. **Currency War-** A currency war happens when countries **compete to devalue their own currency** in order to make exports cheaper and imports more expensive. It is also known as "**competitive devaluation**".

   The devaluation can boost exports and domestic industries but often sparks retaliation, creating instability in global markets.

Examples of Currency Wars

- **China and the Yuan (2010s)-** China was often accused of deliberately keeping the yuan undervalued to make its exports more competitive. The U.S. claimed this hurt American manufacturing.

- **Brazil (2010–2011)-** Brazil declared a "Currency war" as countries like the U.S. and Japan used monetary easing (printing money) after the 2008 financial crisis, which hurt Brazilian currency and made their exports less competitive.

3. **Trade Weaponization-** Trade weaponization refers to the **strategic use of trade policies** as **geopolitical tools** rather than purely economic measures. It includes economic sanctions, export controls, supply chain restrictions, and currency manipulation to exert pressure on rival nations.

Examples of Trade Weaponization

- **U.S. sanctions on Huawei** to curb China's technological advancements.
- **Russia's gas supply cuts** to Europe amid geopolitical tensions.
- **China's rare earth export controls** to maintain dominance in technology manufacturing.
- **Recent U.S. tariffs:** A 25% tariff on all imports from Canada and Mexico and retaliatory tariffs on India.

Negative Impacts of the Trade War

1. **Supply Chain Disruptions-** Global businesses rely on complex supply chains. Tariffs and restrictions make production costlier and less efficient.
   - **For ex- U.S. tech companies** like **Apple** faced higher costs because many of their components were manufactured in China.
2. **Weaponization of Commodities-** Oil, gas and food grain exports are leveraged in geopolitical conflicts (Russia- Ukraine Crisis has impacted the **Black sea grain initiative**).
3. **Technological Wars-** AI and semiconductor supply chains become battlegrounds during Trade wars.
   - **For ex-** The U.S.-China chip war over semiconductor dominance, and the **rise of alternative payment systems** (**for ex-** Russia's Mir, India's rupee-ruble trade).
4. **Slower Economic Growth-** Trade wars reduce global trade, leading to slower GDP growth for both countries and negatively impact the global economic growth.
   - **For ex-** IMF's warning of a cut down of the Global GDP by 0.8% by 2020.
5. **Shifts in Global Alliances-** The trade war results in the realignment of the global trade order. (**Friendshoring:** Countries shifting trade towards geopolitical allies).
   - **For ex-** Formation of New Trade Alliances like the **Mineral Security Partnership** (MSP) for diversification and stabilization of critical mineral supply chains.
6. **Undermining of the Global Trade System-** Trade wars undermine the **WTO** (**World Trade Organization**) and the **rules-based international system**.
   - **For Ex-** The **U.S.–China trade war** led many countries to **adopt more protectionist measures**, weakening global cooperation.
7. **Risk of Escalation to Currency Wars-** Countries respond to tariffs by devaluing their currency, leading to further instability.
   - **For Ex-** The U.S. labelling China as "currency manipulator", after the escalation of trade war and devaluation of the currency.
8. **Reduced Business Confidence & Investment-** Uncertainty about future tariffs discourages companies from expanding, hiring, or investing.

Measures to Counter the External Trade Vulnerability

Countering Trade Weaponization & Supply Chain Risk

1. **Platform Power Strategy-** Moving beyond just exporting goods to building the "digital plumbing" of global trade- such as **payment infrastructure (UPI)**, **digital identity**, and **automated compliance systems**- to make it too costly for adversaries to exclude India.

2. **Strategic Diversification-** Reducing over-dependence on any single market (like the US or China). India's FTA network is projected to c**over 71% of its export basket by 2026**, up from 22% in 2019.
3. **China Plus One & Friend-Shoring-** Positioning India as a reliable alternative hub through the National Manufacturing Mission and PLI 2.0.

De-Dollarization

1. **Vostro Accounts & Digital Rupee-** Operationalizing over **22 Special Rupee Vostro Accounts (SRVAs)** globally and **expanding CBDC (Central Bank Digital Currency)** for cross-border settlements to bypass US-led payment sanctions.
2. **Foreign Exchange Shield-** Maintaining record forex reserves (**$709.75 billion as of March 2026**) to absorb external shocks and manage rupee volatility against a strengthening dollar.

Strategic Trade Diplomacy

1. **Coalition Building-** Working with the **Global South (G-33, BRICS)** to resist unilateral sanctions and pushing for WTO reforms to restore dispute resolution mechanisms.
2. **Investment-First Pacts-** Shifting toward deals like the TEPA with EFTA nations, which trade market access for a $100 billion investment commitment over 15 years, locking in long-term capital and technology transfers.

Conclusion

To counter the triple threat of trade wars, currency volatility, and the weaponization of global finance, India's strategy must transition from **passive resilience** to **active indispensability.**

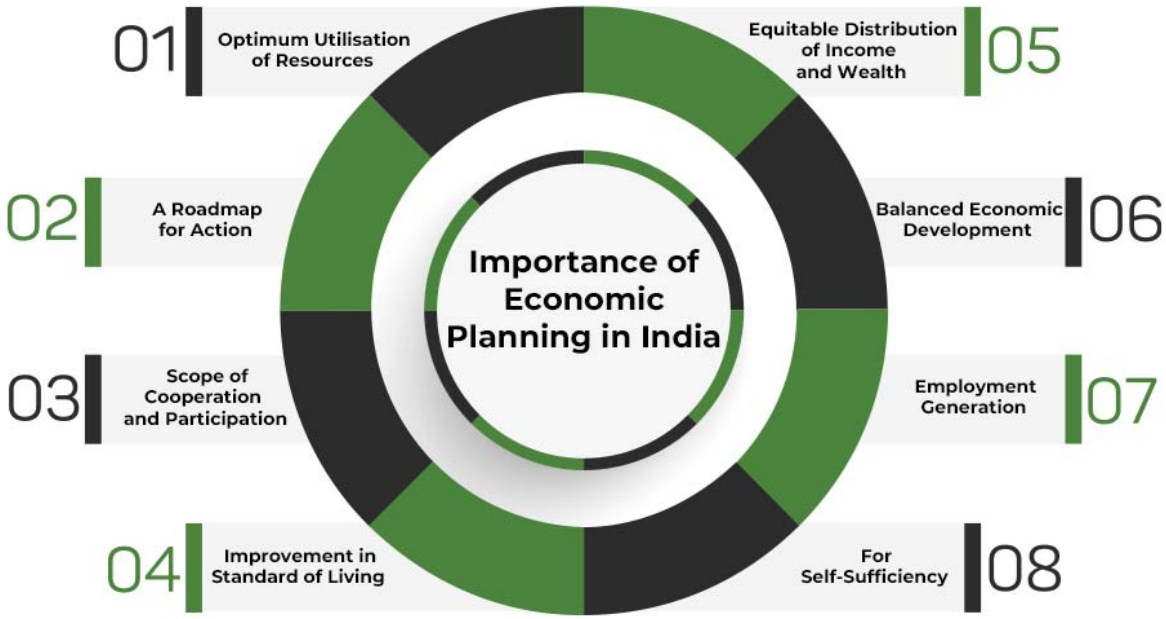
MGP/BP/GS3/006

PLANNING

1. ECONOMIC PLANNING IN INDIA - HISTORY AND CHALLENGES

Economic Planning is the **process** by which a government or central authority makes decisions and formulates **policies to allocate resources efficiently, set economic goals,** and guide the overall development of an economy.

Objectives of Economic Planning



History Of Planning in India & Origin of Five Year Plans

Though the planned economic development in India began in 1951 with the inception of the First Five Year Plan , **theoretical efforts had begun much earlier** , even prior to independence.

Pre-Independence Planning in India	
Visvesvaraya Plan (1934)	Proposed by **M. Visvesvaraya**, this plan focused on shifting workers from **agriculture to industrial sectors**.
FICCI Proposal (1934)	The Federation of Indian Chambers of Commerce and Industry (**FICCI**) stressed the need for planning and the **role of the state** in the economy.

Congress Plan (1938)	The **National Planning Committee** was formed **by S.C. Bose** under the chairmanship of Jawaharlal Nehru. It aimed to establish an economic **plan ensuring a decent standard of living** for the public.
Bombay Plan (1944)	Proposed by **industrialists like Tata and Birla**, this plan focused on industrial growth, state intervention in strategic sectors, support for MSMEs, full employment, and reducing income inequality and regional disparities.
Gandhian Plan (1944)	Authored by **Sriman Narayan Agarwal**, focusing on **small-scale production** and **agriculture** in line with **Gandhian** economic principles**.**
The People's Plan (1945)	Drafted by **M.N. Roy**, this plan aimed to meet the immediate basic needs of the Indian people within a 10-year timeframe.

After independence, India **launched its First FYP in 1951**, under socialist influence of first Prime Minister Jawaharlal Nehru. The process began with the setting up of the **Planning Commission in March 1950** in pursuance of declared objectives of the Government.

Planning Commission: India's Five year plans

Five Year Plans	Features
First Five Year Plan: 1951-1955	• **Harrod Domar Model** which led to high investment in heavy industries • Focus areas were **Irrigation** (Bhakra Nangal, Damodar, Hirakund); **Land Reforms**; Higher Education
Second Five Year Plan: 1956-1960	• **Mahalanobis Strategy** emphasized on strengthening heavy industry based plants (Durgapur, Bhilai, Rourkela Steel Plants) • Financial and Technical assistance boost from **USSR, UK, West Germany.**
Third Five Year Plan: 1961-1966	• In the backdrop of Indo-Pak War, Sino-India War, a **2-year continuous drought and 30% fall in food grain** • **PL480 aid** from USA and followed Gadgil formula to allocate grants.
Plan holidays: 1966-1969	• Included **3 Annual plan**s; Focus on **increasing exports** • **New Agriculture Strategy** with **HYV seeds and Irrigation**
Fourth Five Year Plan: 1969-1974	• **Socialist tilt** under Indira Gandhi; 14 Banks **nationalised**; Abolition of **Privy purses** • **Green revolution**; Indo-Pak war and Refugee crisis from Bangladesh
Fifth Five Year Plan: 1974-1979	• Focused on Poverty Alleviation Programme; Slogan was '**Garibi Hatao'** • Launched **Minimum Needs Programme** and **Twenty-point programme** • Carried out in the backdrop of **Emergency**

Rolling Plan: 1978-1979	• Appraisal of **Plan Performance** before next year plan
Sixth Five Year Plan: 1980-1985	• Gradual **Liberalisation** of the economy; **National Rural Employment Programme**; Integrated Rural Development Programme (**IRDP**); Shift from Heavy industrialisation to Infrastructure
Seventh Five Year Plan: 1985-1990	• Focus on **Service Sector** and **liberalisation** continued • **Technological Reforms** by Rajiv Gandhi government
Eighth Fiver Year Plan: 1992-1997	• **First plan after LPG reforms**, **High** GDP growth-6.8%; fiscal deficit and CAD were very low; **Investment** in infrastructure; Restructuring **subsidies**; **Decentralised** planning; Problem of **Jobless growth**
Ninth Five Year Plan: 1997-2002	• **Exports declined** due to **East Asian Crisis** and foreign capital inflows reduced • **Coalition** government created political instability • **Sanctions** were imposed due to Pokhran Nuclear Test • **Focus areas of the plan**: Population control; Generating employment; poverty reduction ; self-reliance in agriculture.
Tenth Five Year Plan: 2002-2007	• **Focus areas of the plan**: Doubling per capita income in 10 years; Ensuring proper 'Governance' and public service delivery; Agriculture as Prime Moving Force (PMF) • **Features**: Higher Foreign capital inflows and highest growth in services (savings and investment rate crossed 30%)
Eleventh Five Year Plan: 2007-2012	• '**Fast and inclusive growth**' 2008 Global financial crisis • **Tendulkar Committee** on Poverty Measurement • **TFR** (Total Fertility Rate) target of 2.1%
Twelfth Five Year Plan: 2012-2017	• '**Faster, Sustainable and More Inclusive Growth**' • 25 Core monitorable targets • **GDP** growth at 8%; Agriculture growth at 4% and Manufacturing growth at 10%.

Challenges Related to Economic Planning in India

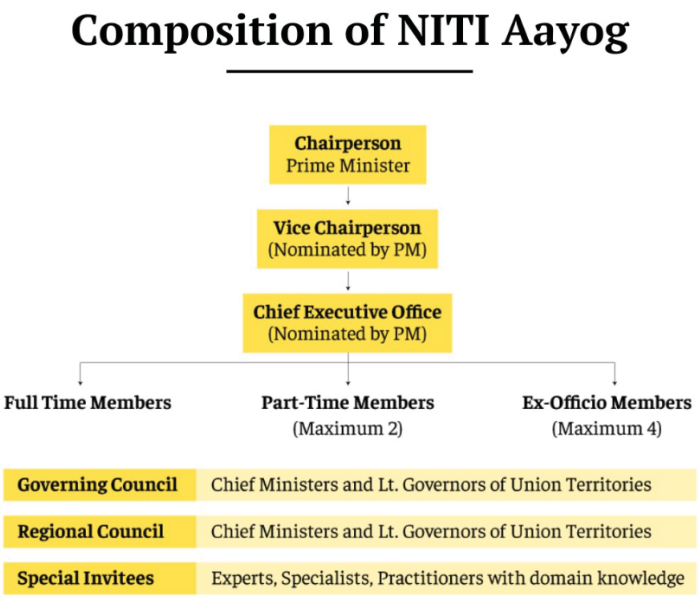
Challenges related to Economic Planning in India	
Top-down/One-size-fits-all Plans	5 Year Plans lacked state/regional customization and flexibility
Diverse and Large Population	Managing varying developmental needs across diverse social, cultural, and economic groups.
Data Limitations	Incomplete, outdated, or inaccurate data hampers precise planning and monitoring.
Resource Constraints	Limited financial, human, and natural resources challenge implementation of ambitious plans.

Coordination Issues	Lack of effective coordination among Union, State governments, and local bodies affects plan execution.
Implementation Gaps	Weak institutional capacity, bureaucratic delays, and corruption impede plan fulfillment.
Unorganized Sector	A large informal economy is difficult to quantify and integrate into formal planning.
Bureaucratic Inefficiencies	Slow decision-making and delays in execution
Persistence of Inequality	Economic and regional disparities persisted

2. NITI AAYOG VS PLANNING COMMISSION

NITI Aayog (National Institute for Transforming India) was established as a **successor** of the **Planning Commission** as an **extra-constitutional body** created by an **executive resolution** on **1st January, 2015.** . It has been created as a **premier policy think tank** of the government providing both directional and policy inputs.

Composition of Niti Aayog



Major Differences between Planning Commission and Niti Aayog

Aspect	Planning Commission	Niti Aayog
Established In	1950	2015
Nature	Centralized planning body	Policy think tank/advisory body
Approach	Top-down, control-oriented	Bottom-up, cooperative federalism
Fund Allocation Powers	Had powers to allocate funds to states	No fund allocation power (done by Finance Ministry)

Planning Model	Five-Year Plans with fixed targets	Medium and long-term strategies, no Five-Year Plans
State Involvement	Limited, states consulted later	Active, states involved in policy formulation
Structure	Hierarchical, rigid, more full-time members	Dynamic, flexible, leaner, experts/part-time members
Focus Area	Industrial growth, infrastructure	Innovation, technology, sustainable development
Key Initiatives	Five-Year Plans, Annual Plans	SDG Index, Atal Innovation Mission, India@75
Chairperson	Prime Minister	Prime Minister

Initiatives by the Niti Aayog

The Niti Aayog's activities can be divided into four main heads:

1. Policy and Programme Framework
2. Cooperative Federalism
3. Monitoring and Evaluation
4. Think Tank and Knowledge and Innovation Hub

Initiatives	Brief Description
Aspirational Districts Programme (ADP)	Targets 112 underdeveloped districts for improvements in health, education, nutrition via monitoring and partnerships.
Atal Innovation Mission (AIM)	Builds an innovation ecosystem through tinkering labs in schools and startup incubation centres.
Poshan Abhiyaan	Reduces malnutrition in children, women via community interventions and tracking.
India Chain	Nationwide blockchain for e-governance to enhance transparency and subsidy delivery.
Transforming India's Gold Market	Recommends reforms for domestic gold production, financialization, and exports.
Digital India Pragramme	Three core components - digital infra, delivering services digitally and digital literacy.

Indices by Niti Aayog: NITI Aayog has launched several **important indexes** to track and promote developmental progress in India. Some of the major indexes include:

Index Name	Brief Description
SDG India Index	Ranks states/UTs on progress toward 17 UN Sustainable Development Goals, assessing health, education, and sustainability.
Health Index	Measures state performance in health outcomes, governance, and key health indicators.

School Education Quality Index (SEQI)	Evaluates states/UTs on school education outcomes, access, infrastructure, and equity.
Export Preparedness Index	Assesses states' readiness for exports based on policy, infrastructure, and business environment.
Women's Social Security Index	Tracks state performance in providing social security to women, focusing on schemes and coverage.

Conclusion

NITI Aayog's model aligns better with the dynamic, complex, and diverse challenges of a rapidly evolving economy, enabling adaptive governance and inclusive growth. This transition signifies India's move from a directed economy to a facilitated economy — where planning is consultative, data-driven, and inclusive, leveraging technology and partnerships for transformative national progress.

3. ACHIEVEMENTS AND CHALLENGES OF NITI AAYOG

Achievements of NITI Aayog

1. **Enhanced Cooperative Federalism-** NITI Aayog served as a linking bridge between central and state governments, fostering collaboration to align regional priorities with national goals.
   - **For ex-** NITI Aayog's '**Team India Hub**' involves all states working towards a national development agenda. The **Aspirational Districts Programme (ADP)**, aiming to transform 112 under-developed districts across the country with close collaboration between line Ministries and development partners.
2. **Strengthened Competitive Federalism-** Encouraged healthy competition among states through data-driven and transparent indexes and ranking systems.
   - **For ex-** Fiscal Health Index, Aspirational District Programme, Composite Water Management Index, State Energy and Climate Index.
3. **Governance and Policy Advice-** As a think tank, NITI Aayog advised on long-term strategic policies and shifted focus from financial allocation (priority of the erstwhile Planning Commission) to policy advisory, promoting a decentralized governance approach.
   - **For ex-** It assisted states in setting up **State Institutions of Transformation (SITs)** for better governance and policy implementation.
4. **Inculcated Innovation, Entrepreneurship, and Digital Transformation-** Through initiatives like **Atal Innovation Mission (Atal Tinkering Labs, Atal Incubation Centre)**, **Knowledge and Innovation Hub**, **National Data and Analytics Platform (NDAP)**, and **the roadmap for digital payments**, NITI Aayog promoted innovation and digital transformation.
5. **Regional and Inter-Sectoral Social Interventions-** NITI Forum for North East, SATH-E initiative, Poshan Abhiyan, State Health Index, School Education Quality Index, focused development on regional and social interventions across various sectors.
6. **Sustainable Development Goals (SDGs) Monitoring-** Tasked with monitoring and adopting SDGs in India, NITI Aayog played a crucial role in synchronizing the nation's development programs in line with the SDG targets. **For ex- SDG India Index.**

Challenges of NITI Aayog

Structural Challenges

1. **Lack of Budgetary Powers-** Unlike the Planning Commission, NITI Aayog has no financial authority to approve state plans or influence resource allocation directly.

2. **Advisory Role-** Its recommendations are **not binding on ministries or state governments**, which often limits their implementation and overall effectiveness.
3. **Centralised Structure-** Despite its mandate for cooperative federalism, NITI Aayog's structure remains centralised, and state cooperation varies, which hampers effective policy implementation.
4. **Politicisation Risks-** There have been concerns about politicisation, with the body sometimes seen as a "**glorified recommendatory body**" lacking teeth to enforce change.

Institutional Challenges

1. **Perceived Constraints on Autonomy-** NITI Aayog's integration within the government can lead to perceptions of limited independence in providing unbiased policy advice. **For Ex-** Concerns over its recommendations aligning too closely with prevailing political agendas.
2. **Inter-Ministerial Coordination-** Ensuring that NITI Aayog's strategic directions are effectively integrated and aligned with the diverse priorities of various government ministries.
   - **For Ex-** Difficulty in aligning specific ministry budgets with NITI Aayog's long-term vision documents.
3. **Stakeholder Engagement-** The challenge of building consensus and ensuring broad-based support for initiatives across state governments, industry, and civil society.
   - **For Ex-** Resistance from some **state governments on proposed agricultural or land reforms.**

Functional Challenges

1. **Prioritization and Resource Allocation (No Financial Mandate)-** NITI Aayog must prioritize policy areas and influence resource allocation without direct budgetary control. **For Ex-** Lack of direct authority over GST slab revisions, which limits its influence on fiscal policy.
2. **Communicating Impact-** Effectively articulating its achievements and value proposition to the public to foster trust and support. **For ex-** Public confusion about NITI Aayog's specific role compared to traditional ministries with direct funding responsibilities.

Way Forward for strengthening NITI Aayog

1. **Grant Allocation Authority-** Grant NITI Aayog, the power to allocate development or transformational capital (**estimated at 1.5% to 2% of GDP**) to states. This would help mitigate regional infrastructure deficits that an advisory-only body cannot address.
2. **Cabinet Integration-** To ensure it sits at the **high table of decision-making**, it is suggested that the **Vice-Chairman be a permanent invitee** to the **Cabinet Committee on Economic Affairs (CCEA)**.
3. **Regional Offices-** Establish regional NITI Aayog to allow the body to understand local ground realities better and move away from "one-size-fits-all" planning.
4. **Specialist Bureaucracy-** Shift from **generalist administrators** to **subject-matter experts (economists, technologists, and scientists)** within the organization to handle complex global economic shifts.
5. **Statutory Status**- Granting NITI Aayog a **statutory or constitutional status** would ensure its political neutrality and prevent it from being seen as a tool of the central government, thereby building more trust with opposition-led states.

Conclusion

NITI Aayog acts as India's strategic **"action tank,"** replacing centralized planning with a collaborative federalist model to drive innovation and sustainable growth. Its future success depends on bridging the gap between policy design and local execution to achieve the vision of a developed India by 2047.

4. REGIONAL PLANNING IN INDIA

Regional planning in India is more than just policy—it's about ensuring that opportunities, infrastructure, and prosperity are not confined to a few pockets but spread across the entire nation. It has become one of the country's most important tools to bridge economic divides, promote sustainability, and deliver holistic growth.

Evolution of Regional Planning

Early Independent Decades	**First Five-Year Plan (1951–56):** Prioritized agriculture and rural upliftment, aiming to lift millions out of poverty. **Second Five-Year Plan (1956–61):** Laid the foundation for industrialization, while emphasizing that growth must reach all regions—not just industrial hubs.
Area-Specific Interventions (1970s Onwards)	Programs like the **Hill Area Development Programme** and the **Desert Development Programme** recognized that geographically disadvantaged areas need special attention. The **Command Area Development Programme (CADP)**, launched in 1974, focused on using irrigation more efficiently and fairly. It boosts agricultural productivity and reduces regional differences.
Reforms Era (Post-1991)	With liberalization, globalization, and privatization, regional planning gained a new dimension by encouraging private participation and market-led development, while still balancing social and environmental concerns. Key initiatives include: • **Economic Corridors** – East Coast Economic Corridor (ECEC) • **Decentralized Planning** – Kerala's People's Planning Campaign (1996) • **Urban Renewal** – Jawaharlal Nehru National Urban Renewal Mission (JNNURM, 2005)

Strategic Goals, Initiatives, and Outcomes of Regional Planning

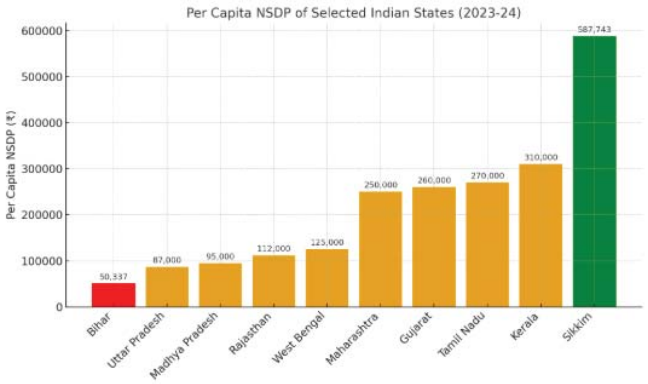
Goal	How It was Implemented	What it achieved
Balanced regional growth	Establishment of SEZs, targeted policies for backward areas	Reduced rural–urban disparities
Optimal resource use	GIS-based mapping, scientific resource allocation	More efficient utilization of land, minerals, and human resources
Infrastructure enhancement	Development of industrial corridors, Smart Cities Mission, PM Gati-Shakti	Improved regional connectivity and trade opportunities
Sustainability	Renewable energy initiatives and large-scale afforestation	Strengthened ecological balance and resilience
Job creation	Promotion of MSMEs, focused skill-development programs	Expanded employment opportunities and entrepreneurship

Policies and Strategies for Regional Planning in India

Macro-Level Strategies	• **Bharatmala:** A nationwide highway network connecting 550 district headquarters, revolutionizing freight movement. • **Sagarmala:** Boosting port-led growth and integrating multimodal logistics. • **AMRUT Mission:** Nearly Rs. 79,000 crore invested to improve urban water supply, sanitation, and mobility. • **HRIDAY Scheme:** Heritage towns rejuvenated by blending cultural preservation with modern services.
Micro-Level & Area-Specific Strategies	• **Backward Regions Grant Fund (BRGF):** Flexible support for 250 lagging districts. • **Border Area Development Programme (BADP):** Strengthening both security and infrastructure in frontier regions. • **Rajan Committee's Multidimensional Development Index:** Introduced a more holistic yardstick than per capita income for fairer resource allocation. • **North East Development Initiatives:** Dedicated focus on trade connectivity, roads, and infrastructure in one of India's most strategically important but underdeveloped regions.

Recent Landmark Examples of Regional Planing

- **AURIC (Aurangabad Industrial City):** A planned industrial smart city under DMIC, spread across 10,000 acres. With Rs. 71,343 crore of investment potential, it promises over 62,000 jobs and has attracted major players like Toyota Kirloskar and Ather Energy.
- **Nagpur Hi-Tech City:** A Rs. 6,500-crore modern township equipped with underground utilities, financial hubs, and digital services, expected to create more than 500,000 jobs in IT and knowledge sectors.
- **Vision Viksit UP-2047:**
  - Envisions 60% urbanization and transforming 10 cities into global-class metros.
  - Promotes "One Division, One International Airport," including the Jewar Airport, and new highways and airstrips.
  - Plans to make UP India's leading agricultural contributor by 2030 and a global agri-leader by 2047.
  - Sets an ambitious target of a $6 trillion economy and Rs. 26 lakh per capita income by 2047, driven by the three pillars of Arth Shakti, Srijan Shakti, and Jeevan Shakti.



Challenges in Regional Planning

1. **Persistent gaps:** Bihar's per capita Net State Domestic Product (NSDP) is still around **Rs. 50,000**, while states like **Maharashtra and Gujarat** have crossed Rs. **2.5 lakh**.
2. **Northeast infrastructure deficit:** Connectivity and geography still act as bottlenecks.
3. **Data deficiency:** Real-time, granular data is limited, though initiatives like the Open Government Data portal are a step forward.
4. **Climate stress:** Rising groundwater depletion and climate shocks could threaten hundreds of millions by 2050.
5. **Fiscal imbalances:** Changes in fiscal transfer formulas have weakened support for less-developed states.

Steps for Advancing Regional Planning in India

1. **Integrated planning across sectors-** Align economic, social, and environmental goals, as emphasized by the World Bank's Towards Resilient and Prosperous Cities in India report.
2. **Empowering local governance-** Strengthen local bodies using AI tools under the Bhashini initiative in eGramSwaraj for better planning and implementation.
3. **Leveraging technology-** Use GIS, remote sensing, and AI for data-driven planning, as in Hyderabad's AI-GIS predictive urban expansion model.
4. **Promoting renewable energy and conservation-** Prioritize sustainability, **e.g.,** Tamil Nadu's 122 GW wind and solar capacity promoting carbon-neutral manufacturing.
5. **Capacity building-** Train personnel for effective plan execution, illustrated by the Mission Karmayogi civil service capacity-building programme.

Conclusion

Regional planning in India addresses **spatial disparities** by shifting focus from national aggregates to location-specific strategies that leverage local resources and strengths. It serves as a vital bridge between macro-level goals and **micro-level execution**, ensuring that development is balanced across diverse geographical landscapes.

5. DECENTRALIZED PLANNING IN INDIA

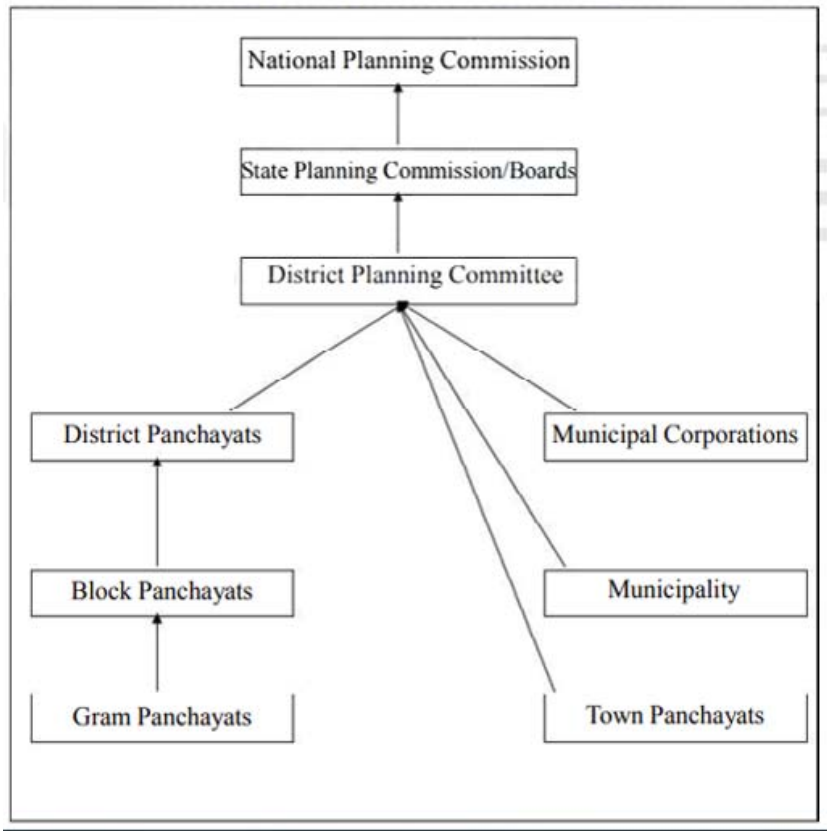
Decentralised planning in India refers to the delegation of planning, decision-making, and resource allocation from central authorities to local bodies, aiming to foster inclusive, participatory, and context-specific development. This approach is enshrined in the Indian Constitution and has evolved through various reforms and initiatives.

Constitutional and Institutional Framework

Constitutional Amendments	The 73rd and 74th Constitutional Amendments of 1992 were pivotal in institutionalising decentralised planning: • **73rd Amendment**: Established Panchayati Raj Institutions (PRIs) in rural areas, empowering them with authority over local planning and development. • **74th Amendment**: Constituted Urban Local Bodies (ULBs) in urban areas, granting them similar powers for urban planning and governance.

District Planning Committees (DPCs)	**Article 243ZD** mandates the formation of DPCs at the district level to consolidate plans prepared by Panchayats and Municipalities and prepare a draft development plan for the district as a whole. These committees play a crucial role in ensuring that development is aligned with local needs and priorities.

Governance Units involved in Planning



Key Dimensions of Decentralised Planning

Dimension	Description	Recent Initiative / Example
Political Decentralisation	Empowers elected local representatives to make decisions for their communities.	In 2025, a **DISHA** meeting in Gorakhpur reviewed local scheme implementation, highlighting cooperation between officials and elected representatives.
Administrative Decentralisation	Delegates administrative functions to local authorities, improving efficiency and responsiveness.	In 2024, over **4,000 Panchayat Bhawans** were sanctioned to strengthen grassroots governance.
Fiscal Decentralisation	Allocates resources and revenue powers to local bodies, promoting financial autonomy.	**eGramSwaraj** integrated with GeM allows Panchayats to use Finance Commission grants seamlessly for procurement and payments.
Functional Decentralisation	Assigns specific functions and services to local entities, tailoring services to local needs.	Kerala's Development Plan empowers Panchayats to implement SDGs and community-driven initiatives in health, education, and infrastructure.

Significance of Decentralized Planning

1. **Enhanced local development-** Decentralized planning enables solutions tailored to local needs. **For example**, Gram Panchayat Help Desks in rural blocks like Patamda and Gurabandha (Jharkhand) improve access to services such as job cards, pensions, and housing, employing local staff, particularly women.
2. **Improved service delivery-** It ensures efficient and timely delivery of public services. The "Viksit UP-2047" plan aims to urbanize 60% of Uttar Pradesh, develop 10 globally competitive cities, and rank three cities among the top 50 most livable globally.
3. **Empowerment of communities-** Local governance strengthens community participation. Gram Panchayat Plans include village- and ward-specific programs, giving residents a sense of ownership in decision-making.
4. **Better resource management-** Decentralized planning optimizes the use of local resources. Kerala's State Planning Board issues guidelines for district-level planning and conducts studies to enhance decision-making at the local level.
5. **Enhanced Climate Resilience & Disaster Preparedness** – Decentralized planning empowers local bodies to respond quickly to disasters. **For example**, Pune district provided Rs. 1 crore to 179 high-risk gram panchayats for disaster kits and volunteer training to ensure faster village-level response before monsoons.

Challenges in Decentralized Planning

1. **Capacity constraints-** Limited administrative and technical skills at the local level hinder effective planning and implementation.
2. **Financial limitations-** Dependence on state and central funds restricts autonomy. Local bodies need to generate more own-source revenue, for example, through building and professional taxes.
3. **Political interference-** Local decisions are sometimes shaped by political interests or short-term gains instead of community needs. This can delay projects, favor some areas over others, and weaken the effectiveness of decentralized planning.
4. **Coordination issues-** Poor coordination between government levels can cause inefficiencies. **For example**, the 2025 creation of the Greater Bengaluru Authority aims to better coordinate urban planning, transport, and environmental management.
5. **Equity Gaps & Elite Capture** – Local governance often benefits dominant groups rather than marginalized communities. NITI Aayog (2022) found District Planning Committees to be weak, while studies (**Indian Institute of Public Administration 2023**) show that funds and schemes like MGNREGA are frequently captured by elites, leaving SC/ST communities neglected.

Conclusion

Decentralised planning in India empowers local bodies like Panchayats and Municipalities to create ground-up solutions that directly address specific community needs and regional diversities. By shifting from a "one-size-fits-all" model to participatory governance, it ensures more equitable resource distribution and inclusive development at the grassroots level.

MGP/BP/GS3/007

MOBILISATION OF RESOURCES

DOMESTIC RESOURCE MOBILISATION

1. SLUGGISH PRIVATE INVESTMENT AND GFCF

India's investment story today reflects a clear divergence between **headline recovery and underlying weakness**. While GDP growth has remained robust (around **8% in recent quarters**), private investment has not kept pace. Even after a decade of reforms—corporate tax cuts, IBC, and PLI schemes- **private corporate capex remains subdued**, raising concerns about the sustainability of growth. This makes **Gross Fixed Capital Formation (GFCF)** a critical variable to assess the true health of the economy.

Introduction

Gross Fixed Capital Formation (GFCF) refers to the **net addition to fixed productive assets** in an economy over a given period. In simple terms, it captures how much of the economy's current output is being **invested to create future productive capacity**, rather than consumed.

Importantly, GFCF forms the **core component of investment (Gross Capital Formation)** in GDP and is therefore a key indicator of an economy's **growth potential and future output capacity**.

What does GFCF include?

GFCF covers creation and addition of **produced assets** that enhance productive capacity:

- **Physical assets:** plant & machinery, transport equipment, roads, railways, buildings
- **Infrastructure investment:** energy, logistics, urban infrastructure
- **Intangible assets:** software, R&D, intellectual property
- **Land improvements:** irrigation, drainage, land development
- **Major upgrades/repairs:** that extend the life or efficiency of assets

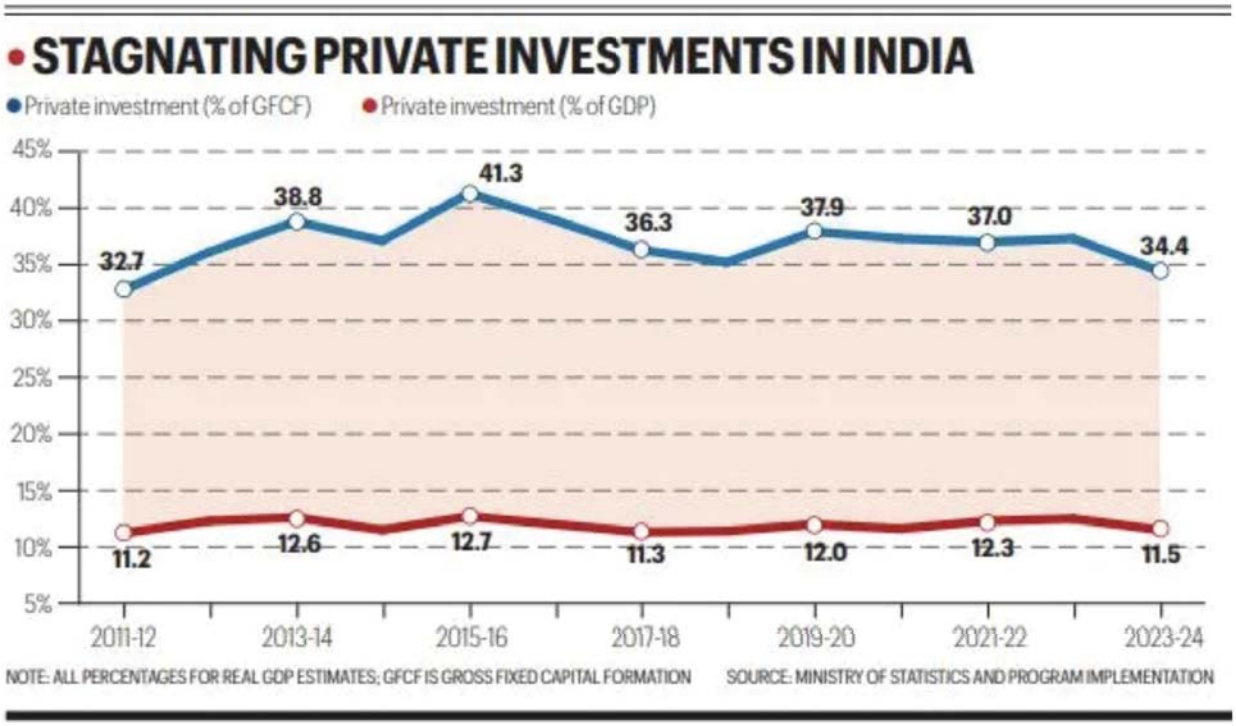
India's GFCF Trends

Private consumption accounts for **60% of GDP**, while **GFCF contributes only around 31%**.

In investment-led models like China, where **GFCF has consistently been above 40%**, while **India's GFCF has never crossed the 35% mark**.

India's investment trajectory over the past decade shows a clear **boom–bust–partial recovery cycle**.

During the high-growth phase of the early 2010s, the investment rate (GFCF) peaked at around **34.3% of GDP**, reflecting strong private sector participation. However, this momentum could not be sustained. Following the **twin balance sheet crisis** and subsequent economic disruptions, the investment rate declined sharply to about **27% in FY21**.



In recent years, there has been a recovery, with GFCF rising to around **30–31% in FY23–24**. However, this improvement appears to be losing strength, with a slight moderation to **~29.6% in FY25**.

Reasons for Weakness of Private Capex

1. **Low Consumer Demand-** High private consumption (**61.8% of the GDP**) reduces savings for investment. Weak export demand and competition from cheap imports discourage capacity expansion.
2. **Policy Uncertainty and Regulatory Bottlenecks-** Issues like **tax disputes, delayed approvals,** and **land acquisition challenges** reduce investor confidence and delay projects.
3. **Shift in Corporate Behaviour-** Despite improved balance sheets, firms are **hoarding cash, deleveraging, and investing in financial assets** instead of physical capital.
4. **High Cost and Input Pressures-** Elevated **raw material costs, interest rates (earlier), and logistics inefficiencies** reduce profitability of new investments.
5. **"Crowding Out" by Public Investment-** High government capex (**Rs. 11.11 trillion**) can absorb available capital, increasing competition for resources and limiting private sector access to funds.
6. **Inequality-led Demand Constraints-** Consumption is skewed towards higher-income groups, limiting **broad-based market demand**, essential for large-scale capacity expansion.
7. **Fragmented Investment Recovery-** Listed firms show capex growth, but **unlisted/MSMEs (bulk of economy) show stagnation**, preventing a broad-based investment cycle.
8. **External and Competitive Pressures-** Global uncertainty, **cheap imports, and trade disadvantages** reduce incentives for domestic capacity creation.
9. **Legacy Issues of Twin Balance Sheet Crisis-** Although improved, past stress in banks and corporates continues to influence **risk aversion and cautious investment behaviour**.

Importance of GFCF

1. **Key Driver of Economic Growth-** GFCF directly contributes to GDP, with **30–35% investment rate** necessary for sustaining **high growth (8%+)**.

2. **Enhances Productivity and Efficiency-** Investment in machinery and infrastructure leads to **capital deepening**, improving output per worker.
3. **Employment Generation-** Infrastructure and manufacturing investments create **strong multiplier effects**, generating both direct and indirect employment.
4. **Indicator of Business Confidence-** Rising GFCF signals **optimism and future demand expectations**, while stagnation reflects economic uncertainty.
5. **Facilitates Structural Transformation-** Enables shift from **low-productivity sectors (agriculture) to high-productivity sectors (manufacturing/services)**.
6. **Promotes Self-Reliance-** Domestic capital formation strengthens **industrial base and reduces import dependence**.

Way Forward

1. **Strengthening Demand Base-** Boost **rural incomes, employment, and wages** to create a broad-based consumption market that incentivises capacity expansion.
2. **Ensuring Policy Stability-** Provide a **predictable tax regime, faster clearances, and contract enforcement** to reduce uncertainty and improve investor confidence.
3. **Deepening Financial Sector-** Enhance **credit access for MSMEs, develop corporate bond markets**, and reduce cost of capital to support investment.
4. **Reforming PPP Framework-** Strengthen risk-sharing mechanisms through instruments like the **Infrastructure Risk Guarantee Fund** to crowd in private capital.
5. **Broad-basing Private Investment-** Move beyond large corporates by supporting **MSMEs, unlisted firms, and informal sector**, which form the bulk of the economy.
6. **Reducing Cost of Doing Business-** Address **logistics costs, input prices, and infrastructure gaps** to improve competitiveness of domestic firms.
7. **Addressing Structural Inequality-** Promote **inclusive growth and income redistribution** to expand domestic demand and sustain investment cycles.
8. **Accelerating Structural Reforms-** Implement **land, labour, and judicial reforms** to ease project execution and reduce delays.

Conclusion

India's investment challenge is not merely restoring **GFCF to ~30% of GDP**, but ensuring a **broad-based, private-led investment revival**. Sustained growth will depend on resolving **demand constraints, policy uncertainty, and structural bottlenecks** inhibiting private capex.

2. DECLINING HOUSEHOLD SAVINGS

Household savings in India, long considered the backbone of domestic capital formation, are undergoing a **significant structural transformation**. Traditionally, Indian households relied heavily on **safe and liquid instruments such as bank deposits**, reflecting a risk-averse saving culture.

However, over the past decade, this pattern has begun to change. **The** *Economic Survey 2025–26* notes that the share of deposits in household financial savings has fallen sharply from around **58% in FY12 to nearly 35% in FY25**, indicating a gradual move away from conventional instruments.

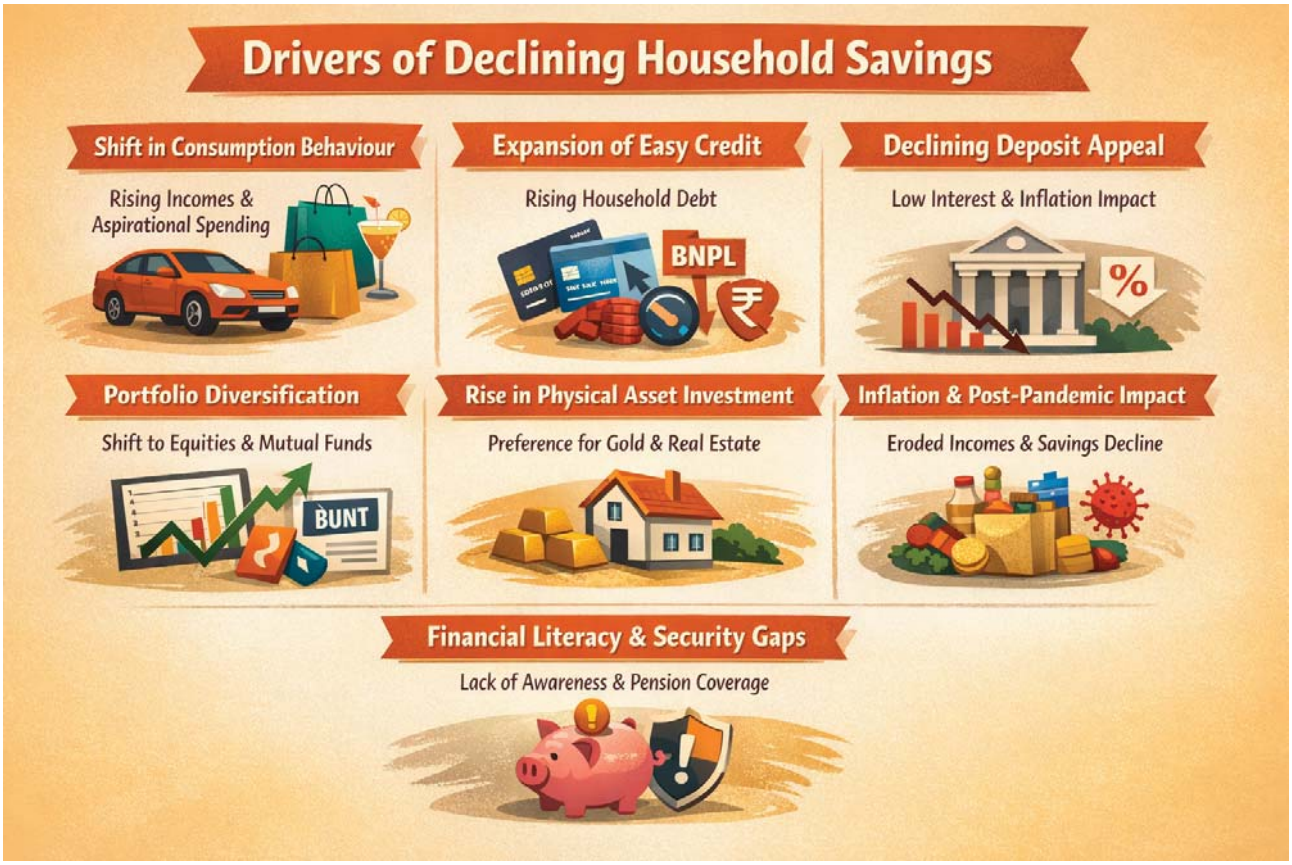
Key Trends in Household Savings

- **Declining Savings Rate:** Gross domestic savings fell from **34.6% of GDP (2011–12) to ~29.7% (2022–23)**, while **net financial savings dropped to 5.3% of GDP**, the lowest in ~50 years.
- **Sharp Fall in Net Financial Savings:** Net household financial savings declined from **₹23.3 lakh crore (2020–21) to ₹14.2 lakh crore (2022–23)**, reflecting erosion of financial buffers.
- **Rising Household Debt:** Household liabilities rose to **~6.4% of GDP (FY24)**, nearing pre-2008 crisis levels, indicating increasing dependence on **credit-led consumption**.
- **Decline in Bank Deposits:** Share of deposits in financial savings fell from **~58% (FY12) to ~35% (FY25)**, signalling reduced reliance on traditional instruments.
- **Rise of Market-linked Instruments:** Equity and mutual fund share increased from **~2% to over 15%**, supported by SIP flows rising
- **Growing Financialisation of Assets:** Share of equity & investment funds in household assets rose from **15.7% (2019) to ~23% (2025)**, with MF assets reaching **~23% of GDP**.

- **Increase in Physical Savings:** Physical assets (gold, real estate) rose from **~60% to over 70% of total savings**, reflecting preference for **inflation hedges and tangible assets**.
- **Rising Retail Participation in Markets:** Individual equity ownership increased from **~11% (FY14) to ~18.8% (2025)**, with total holdings reaching **₹84 lakh cror**e.

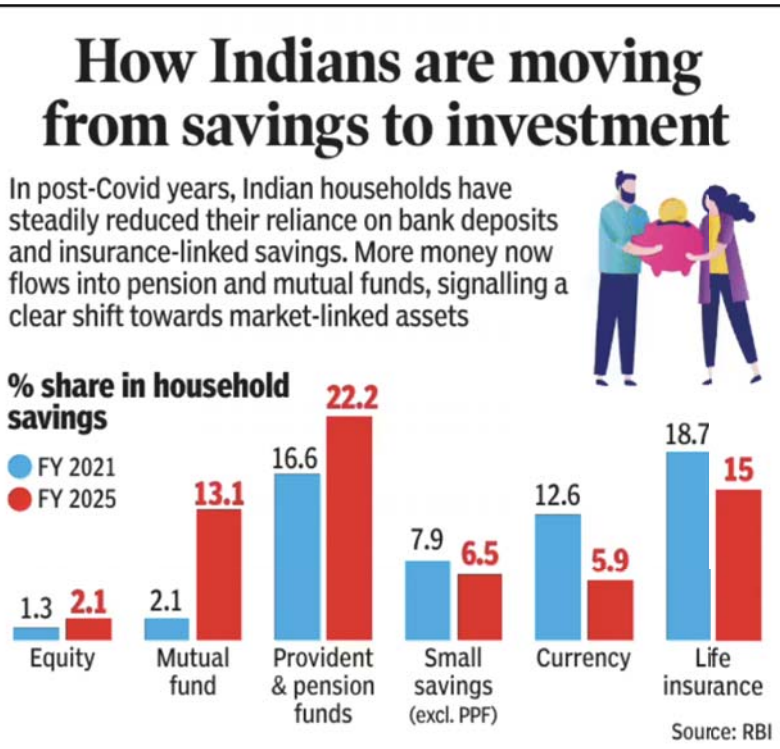
Significance of Household Savings in India:

1. **Primary Source of Domestic Capital-** Household savings contribute nearly **60% of total savings**, forming the backbone of **investment and capital formation** in the economy.
2. **Driver of Economic Growth-** Higher savings enable greater **investment in infrastructure, industry, and services**, supporting sustained GDP growth.
3. **Macro-economic Stability-** Strong household savings reduce dependence on **foreign capital**, thereby limiting exposure to external shocks and capital flow volatility.
4. **Support to Financial System-** Bank deposits and financial savings provide **low-cost funds to banks**, facilitating credit flow to businesses and households.
5. **Household Financial Security-** Savings act as a **buffer against income shocks, health emergencies, and unemployment**, ensuring social and economic resilience.
6. **Inter-generational Equity and Welfare-** Savings, especially in pensions and insurance, ensure **retirement security and reduce future fiscal burden** on the state.



Drivers of Declining Household Savings

1. **Shift in Consumption Behaviour-** Rising incomes, urbanisation, and aspirational lifestyles have led to a transition from *precautionary saving to discretionary spending*, reducing savings propensity.



2. **Expansion of Easy Credit-** Growth of **credit cards, BNPL, and personal loans** has enabled consumption without prior savings, pushing household liabilities to **~6.4% of GDP (FY24)**.
3. **Declining Attractiveness of Deposits-** Low real returns and inflation have reduced the appeal of traditional instruments, leading to a fall in deposit share.
4. **Portfolio Diversification-** Households are reallocating savings towards **equities and mutual funds**, reflecting higher risk appetite and financial awareness.
5. **Rise in Physical Asset Investment-** Increased preference for **gold and real estate** (physical savings ~71.5%) diverts funds from productive financial channels.
6. **Inflationary Pressures-** Rising costs of food, healthcare, and education have eroded disposable incomes, limiting the ability to save.
7. **Post-Pandemic Normalisation-** The temporary spike in savings during COVID-19 reversed sharply, with **net financial savings falling from ₹23.3 lakh crore (FY21) to ₹14.2 lakh crore (FY23)**.
8. **Financial Literacy and Social Security Gaps-** Limited awareness and weak pension coverage lead to **poor savings planning and over-reliance on credit**.

Implications of Declining Household Savings

1. **Reduced Domestic Capital Formation-** Lower household savings—traditionally ~60% of total savings—constrains investment, increasing reliance on **FDI and external borrowing**.
2. **Shift to Consumption-Led Growth-** Higher consumption boosts short-term demand but weakens long-term investment capacity, risking **unsustainable growth patterns**.
3. **Pressure on Banking System-** Declining deposits limit banks' ability to lend, raising the **cost of capital** for infrastructure and private investment.
4. **Rising Financial Fragility-** Increasing household debt raises risks of **defaults and NPAs**, especially with growth in unsecured loans.
5. **Macroeconomic Vulnerability-** Greater dependence on foreign capital increases exposure to **global shocks and capital flow volatility**.

6. **Household Financial Insecurity-** Lower savings weaken **emergency buffers and retirement security**, particularly in rural and informal sectors.
7. **Policy Trade-offs-** The RBI faces a dilemma—**higher interest rates encourage savings but dampen growth**, while lower rates do the opposite.
8. **Inequality and Social Risks-** Limited savings among lower-income households leads to **debt traps and long-term vulnerability**, widening inequality.

Is the Decline of Household Savings Trend Entirely Negative?

The trend is **not entirely negative**, as the shift towards equities and mutual funds reflects increasing **financialisation of savings**, leading to deeper capital markets and more efficient allocation of resources. The growing dominance of **domestic investors**, with inflows now exceeding foreign investments, has also enhanced **market stability and reduced external vulnerability**. Further, the rise of **Systematic Investment Plans (SIPs)** indicates a move towards **disciplined, long-term investment behaviour** among households.

Way Forward

1. **Strengthen Financial Literacy-** Expand financial education through schools, SHGs, and digital platforms to promote **saving discipline, risk awareness, and long-term planning**.
2. **Incentivise Long-Term Savings-** Enhance tax benefits for **PPF, NPS, and SIPs**, and introduce **inflation-indexed instruments** to protect real returns.
3. **Regulate Household Borrowing-** Implement stricter norms on unsecured lending, including **debt-to-income caps and transparent pricing**, to prevent debt-driven consumption.
4. **Revamp Traditional Savings Instruments-** Improve returns and accessibility of **small savings schemes and bank deposits** to restore their attractiveness.
5. **Deepen Financial Markets-** Develop the **corporate bond market** (currently ~16–17% of GDP) to provide diversified and stable investment options.
6. **Strengthen Social Security Systems-** Expand coverage of **pension schemes (NPS, APY) and health insurance** to reduce precautionary uncertainty.
7. **Leverage Digital Ecosystem-** Use **UPI, fintech apps, and micro-saving tools** to encourage small-ticket, regular savings among households.
8. **Encourage Responsible Financialisation-** Promote **disciplined investment (SIPs)** while regulating speculative retail participation in equity markets.

Conclusion

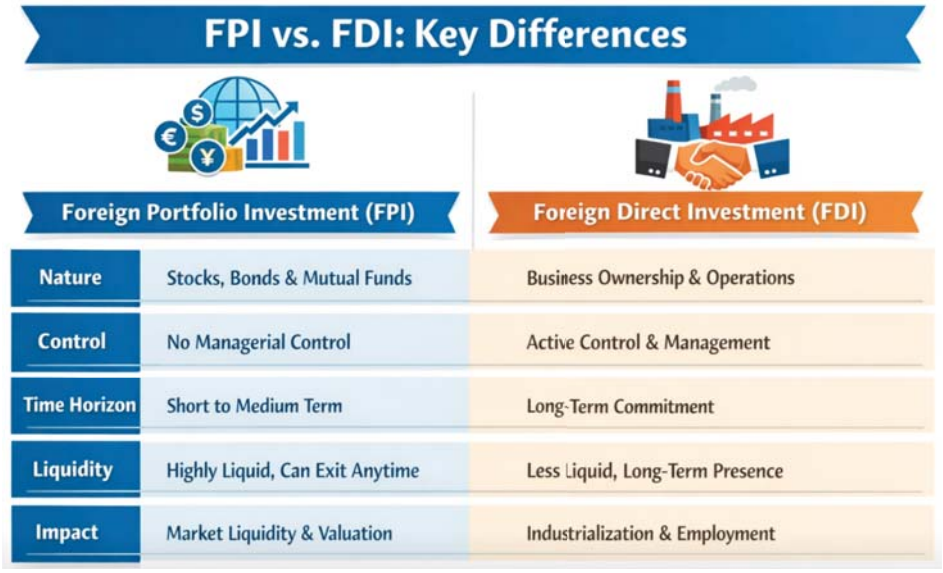
The decline in household savings reflects a **shift driven by rising consumption, higher leverage, and changing investment preferences**, rather than a simple erosion of thrift. While greater financialisation and domestic capital formation are positives, the fall in **net financial savings and liquid buffers** raises concerns about household resilience. The key challenge is to **balance growth with financial prudence**, ensuring

EXTERNAL RESOURCE MOBILISATION

1. FOREIGN INVESTMENT – FDI & FPI

India's foreign investment landscape primarily comprises two key streams: Foreign Direct Investment (FDI) and Foreign Portfolio Investment (FPI). Both impact economic growth, but differ markedly in nature, stability, and sectoral influence.

- **FDI (Foreign Direct Investment)-** Long-term cross-border investment where a foreign investor establishes/controls a lasting interest in an enterprise in India (equity, reinvested earnings, and other capital). It typically entails technology transfer, management control and real activity (plants, jobs).
- **FPI (Foreign Portfolio Investment)-** Short/medium-term investment in listed securities (equity, debt), mutual funds and other financial instruments. It is typically passive, liquid and price-sensitive.



1.1. FOREIGN DIRECT INVESTMENT

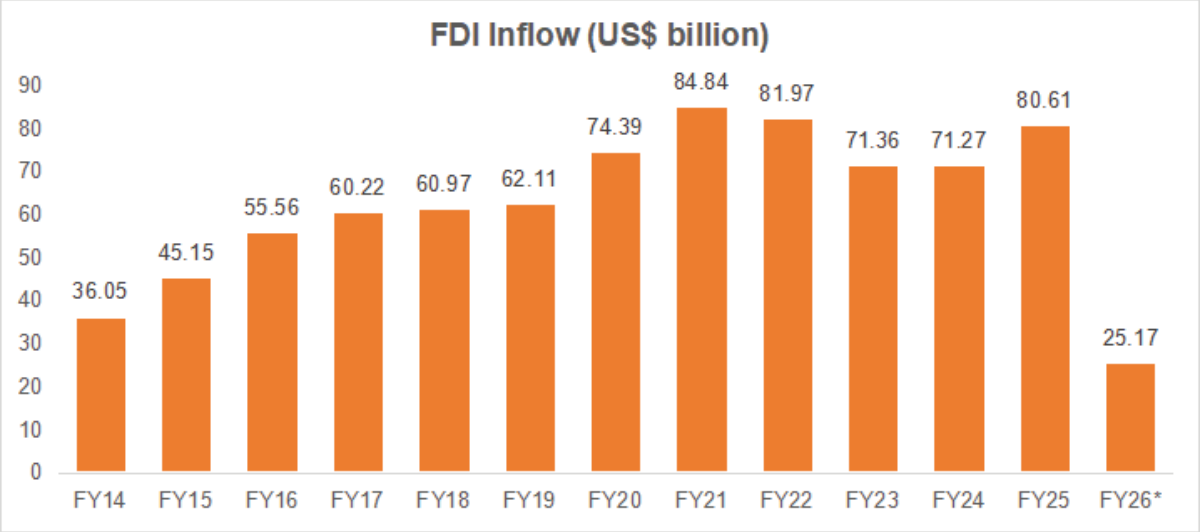
Foreign Direct Investment (FDI) has become a **central pillar of India's growth strategy**, especially in the post-liberalisation era. Over the past two decades, India has steadily positioned itself as a **preferred destination for global capital**, supported by policy liberalisation, macroeconomic stability, and a large and growing domestic market. The crossing of the **$1 trillion cumulative FDI milestone since 2000** reflects not just scale, but also the **deepening maturity of India's investment ecosystem**.

Meaning of FDI

Foreign Direct Investment is the investment made by a non-resident entity in an Indian company, either in an unlisted firm or by acquiring at least **10% equity in a listed company**. Its defining features are a **long-term horizon** and **active managerial control**, making it more stable than portfolio flows. FDI is considered a **quality capital inflow**, contributing significantly to industrialisation, employment, and overall economic development

Recent Trends in FDI in India

1. **Resilient Inflows Amid Global Uncertainty:** Gross FDI rose to **$81 billion (2024–25)**, showing India's ability to attract stable capital despite global volatility.
2. **Shift Towards High-Value Sectors:** FDI is increasingly concentrated in **manufacturing, semiconductors, fintech, AI, and renewable energy**, indicating movement up the value chain.
3. **FTA-Driven Investment Expansion:** Recent FTAs (EU, UK, UAE, New Zealand) providing **near 100% market access** are boosting export-oriented FDI.
4. **Rise of Digital and Green Investments:** Strong inflows into **data centres, AI infrastructure, and renewable energy (44.5 GW added in 2025)** reflect new-age investment drivers.
5. **Emergence of a Mature Investment Ecosystem:** High **repatriation (~$51.5 bn)** and rising outward FDI (**$29.2 bn**) indicate ease of exit and growing globalisation of Indian firms.



Importance of FDI for India:

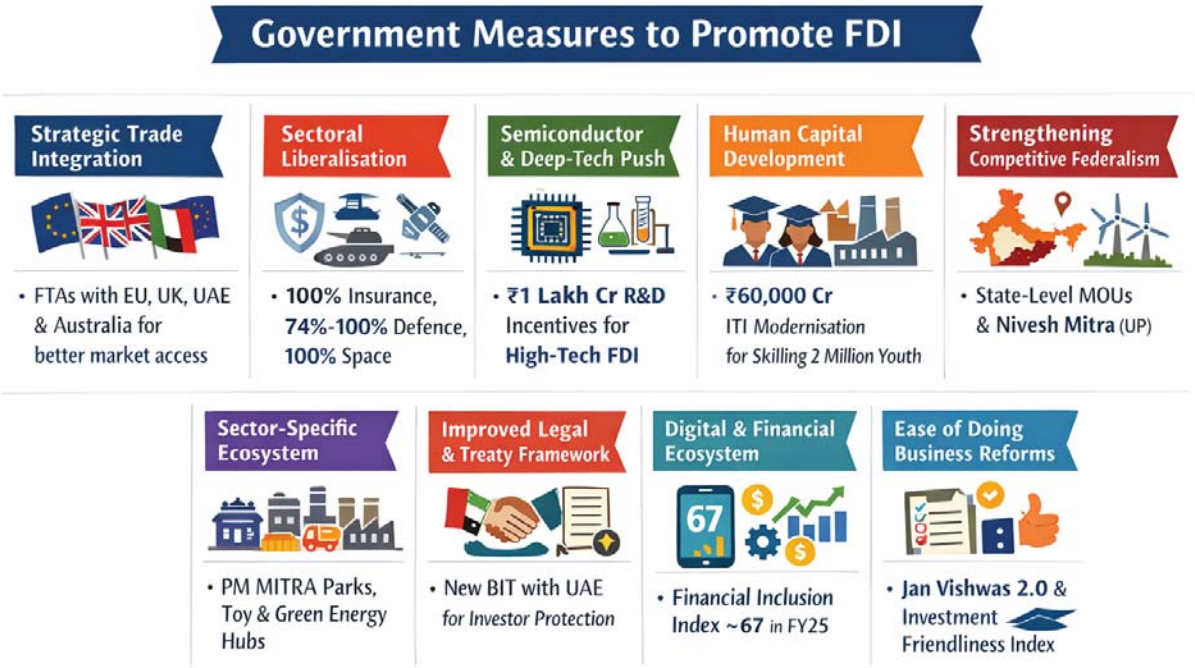
1. **Capital Formation and Growth-** FDI supplements domestic savings, supporting investment and infrastructure. India has attracted over **$1 trillion FDI since 2000**, aiding sustained GDP growth.
2. **Employment Generation-** Establishment of factories, startups, and R&D centres creates large-scale jobs. India's startup ecosystem alone has generated **over 1.6 million jobs**.
3. **Technology Transfer and Innovation-** FDI brings advanced technology, automation, and managerial expertise, improving productivity and global competitiveness (for ex- electronics manufacturing).
4. **Integration into Global Value Chains (GVCs)-** Multinational firms use India as a production base, boosting exports (e.g., **iPhone exports ~ $12 billion in 2023–24**).
5. **Infrastructure Development-** Foreign investments finance large projects in transport, logistics, and urban infrastructure (bullet train, highways, ports).
6. **Human Capital Development-** Training, skill transfer, and managerial exposure enhance workforce productivity, creating long-term developmental spillovers.
7. **Regional Development:** FDI promotes industrialisation of backward areas (**Sriperumbudur cluster in Tamil Nadu**), reducing regional disparities.
8. **Foreign Exchange Stability-** Continuous inflows improve forex reserves and help maintain exchange rate stability.
9. **Enhanced Competition and Efficiency-** Entry of global firms forces domestic companies to improve quality, innovation, and cost efficiency (e.g., e-commerce, food services).

Challenges in Attracting FDI in India

1. **Regulatory Complexity and Policy Uncertainty-** Complex tax laws, transfer pricing norms, and past issues like **retrospective taxation (Vodafone case)** reduce investor confidence.
2. **Infrastructure Deficits-** High logistics costs (**14–18% of GDP**) and inadequate connectivity increase operational costs and reduce competitiveness.
3. **Institutional Weaknesses-** Ineffective enforcement by regulators like **CCI** allows anti-competitive practices, affecting fair market conditions.
4. **Uneven Distribution of FDI-** Heavy concentration in **services and select states (Maharashtra, Karnataka)** widens regional and sectoral imbalances.
5. **Market Distortions and MSME Pressure-** Large multinational firms can engage in **predatory pricing**, impacting small domestic businesses.

6. **Labour Market Concerns-** Gig economy expansion raises issues of **job security, wages, and working conditions** (platform economy disputes).
7. **Environmental and Sustainability Concerns-** Weak enforcement of environmental norms leads to conflicts (e.g., **Niyamgiri mining controversy**).
8. **Intellectual Property and Technology Gaps-** Inadequate technology absorption and risks like **bio-piracy** limit long-term benefits.
9. **Global Economic Vulnerability-** FDI inflows are sensitive to global shocks (**Russia-Ukraine war, recession fears**) leading to volatility.
10. **Dependency on Foreign Capital-** Excess reliance may expose the economy to sudden capital withdrawal, impacting **BoP and currency stability**.

Government Measures to Promote FDI



Strategic Trade Integration	India is **expanding FTAs with the EU, UK, UAE**, and **Australia**, improving market access and attracting export-oriented FDI linked to global value chains.
Sectoral Liberalisation	Progressive easing of caps—**Insurance (100%), Defence (74% automatic, 100% govt route), Space (100% in sub-sectors)**—has widened entry for foreign investors.
Semiconductor & Deep-Tech Push	Government incentives and **Rs. 1 lakh crore R&D support** aim to attract high-tech and strategic FDI in critical sectors.
Human Capital Development	**Rs. 60,000 crore ITI modernisation** to train **2 million youth enhances labour quality**, making India more attractive for manufacturing FDI.
Strengthening Competitive Federalism	State-level initiatives like **Nivesh Mitra (UP)** and **large investment MoUs (Rs. 14.5 lakh crore in Maharashtra)** improve sub-national investment ecosystems.

Sector-Specific Ecosystem Development	Initiatives like **PM MITRA Parks (textiles), toy manufacturing push**, and **green energy investments** create integrated industrial clusters.
Improved Legal & Treaty Framework	**New BIT with UAE** enhances investor protection, legal certainty, and dispute resolution mechanisms.
Digital & Financial Ecosystem Strengthening	Rising **Financial Inclusion Index (~67 in FY25)** reflects deeper formalisation, boosting transparency and investor confidence.
Ease of Doing Business Reforms	Measures like **Jan Vishwas 2.0** and proposed **Investment Friendliness Index** aim to simplify regulations and improve the business climate.

Recent Investments / Developments

1. **Mega Trade Agreements Boost FDI:** FTAs with **EU, UK, Oman, New Zealand** provide **near 100% duty-free access**, expanding export markets and incentivising investment.
2. **Infrastructure Investment Leadership:** India ranks among **top 5 globally in private infrastructure investment**, accounting for **~90% of South Asia's share** (World Bank).
3. **Semiconductor Push:** Under India Semiconductor Mission (**$10 billion scheme**), **10 projects worth ₹1.6 lakh crore** approved, positioning India in global chip supply chains.
4. **Digital & AI Infrastructure Boom:** Data centres expected to attract **₹1.6–2 lakh crore investment**, alongside AI hubs (e.g., Airtel–Google), strengthening digital FDI inflows.
5. **Green and Sustainable Investments:** Renewable capacity addition of **44.5 GW in 2025** and green warehousing expansion reflect rising ESG-driven FDI.
6. **Fintech and Startup Ecosystem:** India ranked **3rd globally in fintech funding (2025)**, reinforcing its position as a global digital investment hub.
7. **Global Capability Centres (GCCs):** Expected leasing of **~180 million sq. ft (2025–30)** indicates India's emergence as a services and innovation hub.

Way Forward to Strengthen FDI Inflows (Economic Survey 2025-2026)

1. **Targeted GVC Integration Strategy-** Identify key **Global Value Chain (GVC) anchors** and build sector-specific strategies to integrate India into global production networks.
2. **Direct Investor Engagement Mechanism-** Create institutional systems for **continuous government–investor interface** to resolve cross-agency issues with **time-bound, customised solutions**.
3. **Single Empowered Accountability Framework-** Establish a **centralised nodal authority** to ensure coordination, policy clarity, and faster decision-making for large investors.
4. **Policy Stability & Predictability-** Ensure **consistent, reliable implementation of incentives**; every policy change must pass the test of **predictability and long-term stability**.
5. **Strategic Sectoral Prioritisation-** Focus on **priority sectors** (manufacturing, infrastructure, greenfield projects) with a calibrated mix of **reforms + incentives**.
6. **Global Investor Outreach & Diplomacy-** Create a **dedicated task force** to engage top global firms and promote India's strengths—**market size, stability, and growth potential**.
7. **Competing with Emerging Economies-** Match proactive efforts of countries like Vietnam/Malaysia through **high-level political engagement and faster clearances**.
8. **Leverage Macroeconomic Strengths-** Use India's **strong fundamentals, fiscal discipline, and credit rating upgrades** to attract long-term foreign capital.

9. **Enhance Ease of Doing Business-** Continue **regulatory simplification, reduced bureaucracy, and faster approvals** to sustain investment momentum.
10. **Trade Policy & FTA Alignment-** Streamline trade policy and leverage **trade & investment agreements** to improve India's position in global supply chains.

1.2. FOREIGN PORTFOLIO INVESTORS

Foreign Portfolio Investors (FPIs) have been net sellers in Indian equities for much of early 2026, offloading over **₹1.5 lakh crore (approx. $18 billion)** from February to mid-March, per NSE data. This marks a sharp **reversal from 2025's inflows**, driven by a confluence of global headwinds and domestic vulnerabilities.

Meaning of FPI

Foreign Portfolio Investment (FPI) refers to investments by non-residents in **Indian financial assets** like equities, debt, and hybrids, routed through SEBI-registered Foreign Portfolio Investors (FPIs). Unlike FDI (which involves ownership/control), FPI is passive, liquid, and short-term oriented—**often driven by global flows.**

Reasons for Recent FPI Withdrawal from India

1. **Global Risk Aversion (Risk-Off Sentiment)-** Geopolitical tensions (**West Asia conflict**) and global uncertainty push investors toward safer assets, reducing exposure to emerging markets like India.
2. **Currency Depreciation Risk-** Falling rupee **(₹92.48/$)** erodes returns in dollar terms, discouraging FPIs from staying invested in Indian assets.
3. **Rising Crude Oil Prices-** Elevated oil prices worsen India's external balance and inflation outlook, making markets less attractive to foreign investors.
4. **Attractive Opportunities Elsewhere-** Capital is shifting to **AI-driven markets (US, South Korea, Taiwan)** offering higher growth prospects and better returns.
5. **Global Interest Rate Dynamics-** Higher or uncertain interest rates in developed economies (**US, Europe**) make advanced markets more attractive relative to India.
6. **Weak Currency–Flow Feedback Loop-** FPI selling weakens rupee, which in turn triggers further outflows—creating a **self-reinforcing cycle**.
7. **Export Competitiveness Concerns-** Rising competition from China and Southeast Asia (especially in manufacturing sectors like autos) reduces investor confidence.
8. **Sectoral Headwinds (IT & Exports)-** AI disruption in IT services and stress in export-oriented sectors dampen earnings outlook, leading to portfolio rebalancing.

Impact of FPI Withdrawal from India:

1. **Rupee Depreciation-** Continuous outflows reduce demand for rupee, pushing it to a **record low of ₹93/$**, increasing external vulnerability.
2. **Stock Market Decline-** Heavy selling led to sharp corrections, with **Sensex and Nifty falling** heavily since late Feb 2026, eroding investor wealth.
3. **Pressure on Balance of Payments (BoP)-** Capital outflows weaken the capital account, contributing to deficits and forcing reliance on forex reserves.
4. **Forex Reserve Depletion Risk-** To stabilise currency and manage outflows, authorities may draw down reserves, reducing the external buffer.
5. **Higher Imported Inflation-** Rupee depreciation **raises cost of imports, especially crude oil, worsening inflation** and widening the current account deficit.

6. **Macroeconomic & Investment Uncertainty-** Persistent outflows signal reduced investor confidence, increasing market volatility and affecting future capital inflows and growth prospects.

Way Forward:

1. **Strengthen Macro Stability-** Maintain fiscal prudence and ensure inflation remains within RBI targets to boost investor confidence.
2. **Manage Currency Volatility-** Use calibrated interventions, currency hedging instruments, and forward markets to reduce sharp rupee swings.
3. **Enhance Domestic Financial Market Depth-** Develop deeper debt and equity markets, including derivatives and bond instruments, to absorb shocks from volatile FPI flows.
4. **Promote Long-Term FDI-** Encourage stable, long-term foreign investments in strategic sectors to reduce reliance on volatile portfolio flows.
5. **Sectoral Competitiveness & Policy Support-** Support IT, export, and manufacturing sectors through skill development, incentives, and trade facilitation to maintain investor interest.
6. **Global Engagement & Risk Diversification-** Strengthen trade, FTA ties, and international investor outreach to attract diversified capital and reduce dependence on any single region.

Conclusion

India's FPI outflows highlight structural vulnerabilities in a **high-risk global environment,** but they also present opportunities for resilience-building. Overall, this phase tests India's macro buffers, but proactive reforms could position it for a **V-shaped recovery by mid-2026.**

2. INDIA'S CAPITAL MARKET IN RESOURCE MOBILISATION FOR INVESTMENTS

India's capital market serves as a **critical engine for resource mobilisation**, transitioning the economy from a **bank-led model** to a **market-linked one**.

By 2024, India's **market capitalisation-to-GDP ratio crossed 130%,** reflecting deep financial integration and an increasing capacity to fund industrial and infrastructure growth.

Mechanisms of Resource Mobilisation in Capital Markets

1. **Primary Market (New Issues)-** Acts as the **direct link between savers** and the **corporate sector**.
   - **IPOs & FPOs-** Corporates use public offers to raise long-term risk capital for expansion.
   - **SME Platforms-** Dedicated platforms by BSE and NSE allow small and medium enterprises to raise funds despite having a lower paid-up capital base (typically up to ₹25 crore).
2. **Debt Market & Private Placements-** Private placement of corporate bonds remains the most preferred route for mobilization due to quicker access and competitive pricing compared to bank loans.

   According to RBI, fresh corporate bond issuances reached a record **Rs. 9.9 lakh crore** in **2024-25**.
3. **Mutual Funds & SIPs-** Retail savings are **aggregated through Mutual Funds**, with assets under management (AUM) growing fivefold between **2013 and 2023** to reach **Rs. 39.46 trillion**.
4. **Equity Market Growth-** India's equity market has shown significant growth and resilience.

   The **market capitalization to GDP ratio stood at a robust 136%** at the end of **December 2024**, which is significantly higher than that of other major economies like **China (65%)**. This indicates a **relatively deep** and **developed equity market**.

5. **Growth in retail Investors-** There has been a **positive trend of growing domestic retail participation**, especially in equities and mutual funds, facilitated by digital platforms.

> Total Demat Accounts has increased to **21.6 crore by December 2025**, up from just **3.6 crore in 2019**.

Challenges with India's capital markets in Resource Mobilisation

India's capital market faces several structural and regulatory hurdles that limit its efficiency in mobilizing resources for long-term investments.

1. **Shallow Corporate Bond Market**
   - **Size Gap-** Corporate bonds account for only **15–16% of India's GDP**, significantly lower than counterparts like South Korea or China.
   - **Concentrated Quality- 97% of issuances** are restricted to **top-tier AAA, AA+, and AA credit ratings**, effectively barring smaller or lower-rated firms from accessing capital.
   - **Low Secondary Liquidity-** Daily secondary market trading volumes are approximately **Rs. 7,000–Rs. 10,000 crore**, which is considered inadequate for an aspiring global financial hub.
2. **Retail Participation and Literacy**
   - **Geographic Concentration-** While unique investor **accounts hit 12.2 crore in late 2025**, over **73% of these accounts are concentrated in just the top 10 states**, led by Maharashtra (17%).
   - **Speculative Risks-** A significant portion of retail activity is **speculative with 91% of individual traders in the equity derivatives (F&O) segment** incurred net losses in FY25, totaling **Rs. 1.05 lakh crore**.
3. **External and Macroeconomic Volatility**
   - **Record Capital Outflows-** Foreign Portfolio Investors (FPIs) withdrew **Rs. 35,962 crore in January 2026 alone**, driven by global trade uncertainties and weak corporate earnings.
   - **Currency Depreciation-** Persistent FPI selling contributed to a nearly **5% depreciation of the Indian Rupee** against the US Dollar throughout 2025.
4. **Regulatory and Structural Bottlenecks**
   - **Private Placement Dominance-** Over **99% of corporate bond deals in 2025 were conducted via private placements**, which limits transparency and excludes retail participation.
   - **Lengthy Issuance-** Public debt issuance in India remains complex.

> According to NITI Aayog reports it often takes **significantly longer than the 1–5 days** seen in developed markets like the **UK or US**.

Way forward for India's Capital Market

1. **Deepening the Corporate Bond Market-** Encourage companies to raise funds via bonds instead of relying heavily on banks.

> Large firms like **Reliance Industries regularly raise funds through corporate bonds**, showing the potential scale.

2. **Improving Liquidity in Secondary Markets-** Increase trading activity in bonds. Encourage market-makers and institutional participation.
3. **Expanding Investor Base-** Allow greater participation from retail investors, pension funds and insurance companies. Promote the use of **retail direct bond platforms** introduced by the NSE.
4. **Promoting Infrastructure Bonds-** Develop long-term financing instruments for infrastructure projects.

> **National Highways Authority of India (NHAI)** has issued bonds to finance highway construction.

5. **Developing Green and ESG Bonds-** Encourage bonds for sustainable projects like renewable energy.

> **Indian Renewable Energy Development Agency (IREDA) issues** green bonds to fund clean energy.

6. **Strengthening Credit Rating Framework-** Improve transparency and accountability of rating agencies.

> The **Infrastructure Leasing & Financial Services (IL&FS) crisis** exposed weaknesses in credit ratings and risk assessment.

Conclusion

India's capital market lacks depth, particularly in the corporate bond segment. **Deepening the capital market** requires reforms to **enhance the secondary market liquidity for corporate bonds**, **encourage public issuances**, and **expand the investor base** for debt instruments.

MGP/BP/GS3/008

EMPLOYMENT

EMPLOYMENT

1. DEMOGRAPHIC DIVIDEND & EMPLOYABILITY CONCERNS IN INDIA

India has become the **most populous nation with ~1.44 billion population**, which is slightly more than China. The exponentially increasing population levels in the 1970s, predicted a **doom for India and the World**. However, the Indian population growth story has belied the prediction of doom. With drop fertility rates (which is now below the replacement levels today), significant reductions in maternal and child mortality rates, **India's demography has entered into a phase of 'demographic dividend'.**

Meaning:

According to the **United Nations Population Fund (UNFPA)**, demographic dividend means the **economic growth potential that can result from shifts in a population's age structure**. The demographic dividend leads to an increased labour supply that increases the production of goods and boosts savings and investment. India has **one of the youngest populations** (62.5% of its population in the age group 15-59) in an aging world.

India's Demographic Status:

- According to the State of World Population Report 2023, published by the UNFPA, India has **surpassed China as the most populous country**, with a population of 142.86 crores compared to China's 142.57 crores.
- India's population growth has slowed down significantly in the past 10 years, with the **total fertility rate (TFR) declining to 2 in 2020-2021** from about 3.4 in the early 1990s. A TFR of 2.1 is necessary for a country to attain population stability.
- India's population is forecast to grow to **1.67 billion in 2050** and peak at 1.7 billion in 2064 before settling at 1.53 billion in 2100.
- **Two-thirds of India's total population are between the ages of 15 and 64**. It presents a potential demographic dividend if education, skill development, and opportunities are provided, particularly for youth from disadvantaged sections and women.

Significance of India's huge demographic dividend:

1. **Increased Supply of Labour:** The rapidly rising young population results in the increased labour supply, as more people reach working age.

2. **Economic growth:** Demographic Dividend results in better economic growth brought about by increased economic activities due to higher working age population and lower dependent population. Demographic dividend has historically contributed up to 15 % of the overall growth in advanced economies.
3. **Capital formation:** The propensity of saving increases with the decrease in the number of dependents. This increases national savings rates, increases the stock of capital in developing countries and provides an opportunity for enhanced capital formation through investment.
4. **Creation of Infrastructure:** Increased fiscal space created by the demographic dividend enables the government to divert resources from spending on children to investing in physical and human infrastructure.
5. **Increase in Female Human capital**: Decrease in fertility rates result in healthier women and fewer economic pressures at home. This provides an opportunity to engage more women in the workforce and enhance human capital.
6. **Innovation and entrepreneurship:** A young population can lead to increased innovation and entrepreneurship, with more startups and unicorns emerging in various sectors like healthcare, education, agriculture, and financial services.
7. **Climate action and sustainability:** A young and educated population can drive sustainable development by prioritizing climate action and adopting environmentally friendly practices.
8. **Increase in Global influence:** India's rising population, combined with its position as the world's largest democracy and a major economy, can help it become a global manufacturing hub, startup capital, and exporter of skilled manpower.

Demographic Dividend: India's Window of Opportunity

McKinsey Global Institute's report *"Dependency and Depopulation? Confronting the Consequences of a New Demographic Reality"* highlights that **India has only 33 years to fully leverage its demographic dividend** before population dynamics shift unfavourably.

India-Specific Findings:

Fertility Trends

1. **National fertility rate**: 1.98, below replacement level of 2.1, signalling future workforce contraction.
2. **Regional variations**: Sikkim has the lowest fertility (1.05), Bihar the highest (2.98), requiring **subnational policies** to address declining fertility.

Population and Workforce

1. **Support ratio**: Current workers per senior = 9.8; projected to **halve by 2050**, increasing dependency.
2. **Demographic dividend contribution**: Historically added 0.7% per year to GDP per capita growth; projected to drop to **0.2% per year by 2050**.
3. **Shift in global consumption**: India and Emerging Asia will account for **30% of global consumption by 20**50, up from 12% in 1997, reflecting rising economic importance despite slower per-capita growth.

Challenges that can turn India's 'Demographic Dividend' into 'Demographic Disaster:

1. **Lack of employability:** Poor human capital formation is reflected in low employability among India's graduates and postgraduates. According to ASSOCHAM, only 20-30 % of engineers find a job suited to their skills. Thus, a low human capital base and lack of skills is a big challenge.
2. **Low human development:** India ranks 134 out of 189 countries in UNDP's Human Development Index. The life expectancy at birth, and the mean years of schooling is much lower than other developing countries.

3. **Hunger and Malnutrition:** In the Global Hunger Index (2023), India was ranked at 111 out of 125 countries. The nutrition, stunting, wasting and underweight among children below five years and anemia among women pose serious challenges. According to India's epidemiological trajectory, India faces the double burden of communicable and non-communicable diseases (NCD).
4. **Informal economy:** Informal nature of economy in India is another challenge in reaping the benefits of demographic transition in India. The workers in the Informal economy are underpaid and devoid of social security benefits.
5. **Jobless growth:** As per the NSSO Periodic Labour Force Survey 2017-18, India's labour force participation rate for the age-group 15-59 years is around 53%. This means that around half of the working age population is jobless. There are future concerns of further jobless growth due to deindustrialization, de-globalization, the fourth industrial revolution and technological progress.
6. **Low Women workforce Participation rate:** According to the latest Periodic Labour Force Survey (2022-23) female LFPR is around 37.0%. This poses a serious challenge in effectively reaping the demographic dividend.

Reasons for low employability of India's workforce:

1. **Lack of employable skills**: One of the main reasons behind the high youth unemployment rate in India is that schools even today are training students from a young age in skills that have no use in the market in the future. The World Economic Forum Report also highlighted the need for skills of the future, especially those that involve technology, such as programming, data science, big data, machine learning, AI, web development, etc.
2. **Obsession with "honourable" jobs**: Indian society puts a high value on graduates and white-collar jobs and often ends up neglecting those with adequate vocational skills or those who have completed on-the-job training.

3. **Poor communication skill**: In today's era, good communication skill is necessary, as our economy is expanding to other countries also. Employers want to recruit people who possess good communication skills so he/she can deal with foreign clients also.
4. **Dynamic nature of job market**: The job market is changing faster, candidates who do not equip themselves with new technology or adapt themselves to changing nature of jobs find it challenging to get employment.
5. **Rising trend of voluntary unemployment:** While the lack of sufficient job creation could lead to resentment due to people's high aspirations, NITI Aayog member Bibek Debroy, flagged a dramatic rise in voluntary unemployment across the country, where people choose not to work below a certain income level after 'investing' in education.
6. **Low labour force participation of women**: After recording some improvement in recent years, India's notoriously low female labour force participation rate (LFPR) seems to have stagnated, data from the Periodic Labour Force Survey (PLFS July 2021-June 2022) shows. 29.4% of women (aged 15-59) were part of India's labour force in 2021-22, as compared to 29.8% in the preceding year. In contrast, men's LFPR improved from 80.1% in 2020-21 to 80.7% in 2021-22.
7. **Less growth of the employment-intensive industry**: Less growth of employment-intensive industries like manufacturing, textile etc are a major problem. The skill-intensive service industry is unable to absorb the workforce,as these jobs require high-level skills.

Way Forward:

1. **Strengthen Industry-Academia Linkage:**
   a. Align curricula of schools, universities, and training institutes with real-world industry needs, ensuring the imparting of job-relevant skills through partnerships, internships, and apprenticeships.
   b. Embed soft skills, digital literacy, and adaptability within both technical and general education to boost workforce flexibility and professional conduct.
2. **Scale & Modernise Skill Development:**
   a. Expand vocational training and apprenticeship opportunities, emphasizing emerging sectors such as AI, robotics, renewable energy, healthcare, and tourism.
   b. Encourage modular skill courses, blended learning models, and experimentation with digital platforms to increase access, especially in rural and excluded communities.
3. **Promote Formalisation of Economy:** Formalize informal employment by leveraging digital platforms like e-Shram for benefits, skilling, health and retirement coverage, and dignified employment contracts.
4. **Foster Innovation, Startups & Entrepreneurship:** Streamline regulatory processes, offer startup incubation, and build access to credit to unleash self-employment and micro-enterprises, especially for youth and women.
5. **Public-Private Partnerships (PPPs):** The government can work with the private sector to fund and manage skill development initiatives on a large scale.

Conclusion:

"Solving the employability crisis is less about the skills that our youth need to build and more about the opportunities that need to be built to skill our youth,".

2. JOBLESS GROWTH

Meaning:

Jobless growth refers to a **macroeconomic situation where the GDP of an economy grows** at a robust rate but this growth does NOT **translate into proportional employment creation**. In other words, the

economy expands but the benefits of this expansion **remain confined to capital owners**, the highly skilled, and technology-driven sectors — leaving the broad mass of the labour force without adequate jobs.

Key Terms to Know:

Term	Definition	India Context
Employment Elasticity	% change in employment for 1% change in GDP	Fell from 0.52 (1970s) → 0.4 (1990s) → ~0.01–0.1 (2019)
Labour Force Participation Rate (LFPR)	% of working-age population actively working or seeking work	~55.2% (ILO 2022); low participation, especially among women
Worker Population Ratio (WPR)	Employed persons as % of working-age population	Increased by 9 percentage points from 2017–2023 (PLFS)
Disguised Unemployment	More people employed than productively needed (esp. in agriculture)	~46% of workforce in agriculture; most are disguised unemployed
Underemployment	Employed in jobs below skill/time capacity	Vast majority of informal sector workers
Structural Unemployment	Mismatch between worker skills and job requirements	~80% of engineering grads lack employable skills
Premature Deindustrialization	Manufacturing share falls before industrialization is complete	India's manufacturing stuck at 14–17% of GDP since 1991
Demographic Dividend	Economic benefit when working-age population > dependents	India's window: ~2041–2055; need to create productive jobs
Informalisation	Shift of workers to informal/ unorganized sector	~90% of India's workforce in informal sector

The Debate — Is India Really Experiencing Jobless Growth?

There are **two contrasting views** on India's employment scenario.

SIDE A — Yes, India IS experiencing Jobless Growth (Traditional View)

1. **Employment Elasticity collapse**: Employment elasticity fell from ~0.52 in the 1970s to near 0.001 by 2019 (Economic Survey), showing GDP growth is barely creating jobs.
2. **Job deficit**: India needs about 20 million new jobs annually (CMIE), but only ~4 million are generated each year, leaving a significant employment gap.
3. **GDP vs job growth gap**: While GDP grew at 6–7% per annum over the last decade, job growth remained just around 1% per annum, highlighting limited labour absorption.

4. **Manufacturing employment decline**: Jobs in manufacturing fell from 51 million (2016–17) to 27.6 million (2020–21), reflecting shrinking labour-intensive sectors.
5. **IT sector paradox**: The IT/ITeS sector contributes significantly to GDP but employs only ~4.5 million people, showing high-value growth with low direct employment.
6. **Graduate and youth unemployment**: Less than 7% of male graduates secure permanent salaried jobs within a year, while urban youth (15–29 years) unemployment is ~14.7% (CMIE, 2026).

SIDE B — India is NOT experiencing Jobless Growth (Counter-Narrative)

1. **Jobs added**: Between 2016–17 and 2022–23, India created 170 million new jobs, a 36% increase, indicating that economic growth is generating employment (PIB/ORF).
2. **Recent high employment elasticity**: ORF reports 1.11% increase in jobs for every 1% rise in value added (2017–2023), suggesting growth is translating into job creation.
3. **Declining unemployment**: Official PLFS data shows unemployment at 3.2% in 2022–23 and 4.9% in 2024–25, reflecting improving labour market conditions.
4. **Rising Worker Population Ratio (WPR)**: WPR rose by ~9 percentage points between 2017–2023, a 26% increase, indicating more people are entering the workforce.
5. **Youth unemployment improving**: Youth unemployment (15–29 years) fell from 17.8% in 2017–18 to 10% in 2022–23 (PLFS), showing better opportunities for young workers.
6. **Consumption-driven & services sector g**rowth: Rising private consumption and a labour-capital ratio of 1.17 in services indicate that India's services-led growth is generating jobs.

Causes Of Jobless Growth In India:

There are multiple factors driving jobless growth in India, broadly classified into **supply-side, demand-side, and structural/systemic causes**.

Supply-Side Causes (Production Structure)

1. **Capital-Intensive Growth Model**: Post-1991 growth is driven by machinery, technology, and automation rather than expanding the workforce. Capital substitutes labour, meaning GDP growth creates fewer jobs. Example: the automobile sector uses robotics → high output, fewer workers.
2. **Stagnant / Premature Deindustrialization**: Manufacturing contributes only 14–17% of GDP and ~12% of employment, far below East Asian peers. India bypassed labour-intensive manufacturing, moving directly from agriculture to services. CMIE: manufacturing jobs fell from 51 million (2016–17) to 27.6 million (2020–21).
3. **Service Sector Paradox**: Growth is services-led (IT, Finance, ITES, Telecom), which contribute significantly to GDP but require high skills and absorb few low-skilled workers (~4.5 million employed in IT).
4. **Technology, Automation & Industry 4.0**: AI, robotics, and automation reduce demand for low- and medium-skill workers, polarizing employment towards very high or very low skills. 2023 tech layoffs: 4.25 lakh globally.

Demand-Side Causes (Labour Market Failures)

1. **Skill Mismatch**: Only 4.7% of India's workforce has formal training (2015), while 80% of engineering graduates lack employable skills (NASSCOM). Rapid tech changes make existing skills obsolete.
2. **Labour Law Rigidities**: Historically complex compliance discouraged hiring, especially in manufacturing. The Industrial Disputes Act made firing difficult, pushing firms toward capital-intensive solutions. MSMEs face high compliance burdens.
3. **MSME / 'Dwarf Firm' Problem**: Micro and small firms dominate India's industrial structure but cannot scale due to regulatory and compliance barriers, limiting productivity and employment potential.

4. **Agricultural Overhang**: ~46% of the workforce in agriculture contributes only ~18% of GDP. Seasonal employment, slow rural-to-urban transition, and lack of alternative jobs drive informal sector growth.

Structural & Systemic Causes

1. **Education System Mismatch**: Outdated curricula and overemphasis on degrees over skills produce educated unemployment. Less than 7% of male graduates get permanent salaried jobs within a year (State of Working India 2026).
2. **Informalisation of the Economy**: ~90% of the workforce is informal with low wages and no social security. Gig economy jobs (Zomato, Swiggy, Ola) add to informalization without employment security.
3. **Female Labour Force Participation (FLFPR)**: FLFPR is low (~25–37% vs 47% world avg). Care work, cultural norms, safety concerns, and inflexible employment suppress women's participation; urban female youth unemployment is 20.1% (PLFS 2024).
4. **Global Factors**: US tariffs and post-COVID supply chain disruptions hurt labour-intensive exports like textiles. Protectionism limits India's ability to grow via export-led manufacturing.

Consequences & Implications:

Economic Consequences

1. **Demographic dividend squandered**: India's youth bulge (50% population under 30) risks becoming a liability if mass employment is not created before the window closes (~2041–2055).
2. **Weak domestic demand**: Low employment leads to low wages, reducing consumption and stagnating the domestic market; growth cannot rely on investment and exports alone.
3. **Middle-Income Trap risk**: Without widespread employment, India may remain stuck at middle-income levels, unable to transition to a high-income economy.
4. **Low savings and investment**: Unemployed or underemployed households cannot save, depressing domestic capital formation and long-term growth potential.
5. **Informalization spiral**: Absence of formal jobs forces workers into the informal sector, perpetuating low productivity, low wages, and job insecurity.
6. **Long-term GDP stagnation**: Low employment dampens consumption, weakening the multiplier effect of economic growth across sectors.

Social Consequences

1. **Rising inequality (K-shaped growth)**: Economic gains concentrate among capital owners, highly skilled workers, and formal sector employees, worsening income disparity and the Gini coefficient.
2. **Erosion of social mobility**: Without quality jobs, educated youth struggle to climb the economic ladder, leading to frustration, alienation, and social stagnation.
3. **Social unrest**: Youth unemployment correlates with crime, radicalization, and protests — e.g., Agnipath protests (2022) and caste-based reservation agitations (Jats, Patidars, Marathas).
4. **Mental health crisis**: Long-term unemployment leads to stress, depression, and loss of self-worth, affecting both individuals and communities.
5. **Rural distress**: Farm suicides, distress migration, and indebtedness are partly due to agriculture failing to provide adequate livelihoods.
6. **Cultural and societal frustration**: Persistent lack of employment opportunities undermines social cohesion and erodes trust in institutions.

Political Consequences

1. **Populism and welfarism**: Job scarcity pushes governments toward relief-based policies (MGNREGA, free rations) rather than productive employment generation.
2. **Reservation demands**: Communities like Jats, Gujjars, Patidars, and Marathas demand OBC reservations as a response to limited job opportunities.
3. **Political instability**: Large unemployed youth populations are vulnerable to political mobilization, populist promises, and ethnic or communal tensions.

4. **Brain drain**: Educated youth migrating to countries like the US, Canada, Australia, and the Gulf represents a loss of human capital funded by Indian taxpayers.
5. **Policy paralysis**: Persistent unemployment may force short-term populist measures instead of structural reforms for sustainable job creation.
6. **Erosion of democratic trust**: Chronic joblessness undermines citizens' confidence in governance, fueling dissatisfaction and political volatility.

Way Forward & Solutions:

A. Reviving Manufacturing — The Cornerstone

1. **Labour-intensive manufacturing**: Shift from capital-heavy production to labour-intensive sectors like textiles, leather, footwear, toys, and food processing to capture export niches like Bangladesh and Vietnam.
2. **'Assemble in India for the World'**: Integrate India into global assembly value chains (electronics, components), potentially creating **8 crore jobs by 2030**.
3. **Cluster-based manufacturing**: Promote industrial clusters (e.g., Tamil Nadu textile clusters) for shared infrastructure, enabling MSMEs to scale efficiently.
4. **MSME reforms**: Tackle the 'dwarf firm' problem via scale-neutral incentives and lower compliance thresholds to encourage growth.
5. **Reorient SEZs as EEEs**: Transform SEZs into Economic & Employment Enclaves (Baba Kalyani Committee recommendation) to boost both production and jobs.

B. Labour Market Reforms

1. **Implement Labour Codes effectively**: Consolidate 44 laws into 4 codes (Wages, Social Security, Industrial Relations, Occupational Safety) to formalize employment.
2. **Flexible employment contracts**: Promote fixed-term contracts to allow hiring flexibility without compromising worker protection.
3. **Employment-linked incentives**: Tie PLI-style production subsidies to job creation to ensure economic growth translates into employment.
4. **Incentivize formalisation**: Offer tax benefits and social security co-contributions to employers who hire workers formally.

C. Human Capital Development

1. **Align education with market needs**: Reform ITIs, polytechnics, and universities in partnership with industry; emphasise apprenticeship-heavy curricula.
2. **Digital and emerging sector skilling**: Scale Skill India via the National Digital Skilling Mission, focusing on AI, green jobs, logistics, care economy, and digital infrastructure.
3. **Invest in health and early development**: Strengthen early childhood education, nutrition, and primary healthcare to improve workforce productivity.
4. **Address graduate unemployment**: Introduce employability assessments, bridge courses, and operationalize NEP 2020's vocational integration.

D. Agriculture-to-Non-Agriculture Transition

1. **Secondary agriculture promotion**: Support rural small enterprises providing agri-inputs and processing outputs (Dalwai Panel recommendation).
2. **Agri-processing clusters**: Create rural non-farm employment while keeping workers close to home, reducing urban migration pressures.
3. **Rural infrastructure development**: Improve roads, electricity, and digital connectivity to attract industries to rural and semi-urban areas.

E. Gender & Social Inclusion

1. **Increase female labour force participation**: Provide safe public transport, crèche/daycare facilities, and flexible formal employment opportunities.
2. **Professionalise care economy**: Formalize domestic and care work to create dignified, paid employment for women.
3. **Boost urban employment**: Strengthen construction, housing, logistics, and municipal services as immediate job multipliers for migrant workers.

F. Social Protection & Safety Nets

1. **Strengthen social security**: Expand universal healthcare, pension schemes, and insurance to reduce vulnerability during economic transitions.
2. **Avoid excessive welfarism**: Short-term welfare (free rations, farm loan waivers) cannot replace long-term structural employment creation.
3. **Bridge low-wage transitions**: Provide wage subsidies and basic income supplements for workers moving from agriculture to industry.

Conclusion:

Jobless growth remains the **single biggest challenge to the Indian economy.** To create jobs on a mass scale there is a need to boost manufacturing that hinges on creating infrastructure, removing red tape and reviving investor sentiment. **A robust manufacturing sector** is a must for absorbing the excessive workforce of the agriculture sector as well as enabling the nation to reap its demographic dividend.

3. GIG ECONOMY

Introduction:

The gig economy is **reshaping India's labor market,** offering short-term, flexible work outside the traditional employer-employee relationship. As **defined under the Labour Codes of 2019:** According to the Labour Codes of 2019, a gig worker is: "A person who works and earns through activities outside traditional employer-employee relationships."

Gig work is largely **task-based, facilitated through digital platforms**. Workers, including freelancers and independent contractors, are paid per task rather than through full-time contracts. Common roles range from food delivery to online freelancing and digital services.

Key Statistics & Data:

77 lakh Gig workers in India (NITI Aayog / PLFS 2020-21)	**1.2 cr+** Estimated gig workers by FY 2024-25	**2.35 cr** Projected by 2029-30 (Economic Survey 2024)	**61.9 mn** Projected by 2047 (Govt. Vision 2047)
6.7% Share of non-agricultural workforce by 2029-30	**2.5%** Projected contribution to GDP by 2030	**17% CAGR** Growth rate of India's gig sector (ASSOCHAM)	**27%** India's share of global online gig/freelancers (2024)

Income Reality (Fair Work India Study, 2023)

- Delivery & ride-hailing workers earn **₹15,000–₹20,000/month** — often below minimum wage for hours worked.
- Over **70%** of gig workers report difficulty managing household expenses due to irregular earnings.
- NITI Aayog 2024: **90% of gig workers lack savings** and face high vulnerability to emergencies.
- Women constitute **~28%** of India's gig economy workforce.
- NITI Aayog (2022): **Medium-skilled jobs** dominate (~47%), followed by low-skill (~26%) and high-skill (~27%) gig roles.

Opportunities & Significance of the Gig Economy

A. Economic Significance

- **Job creation:** Provides flexible employment, absorbing surplus labour and acting as a **safety valve for formal sector deficits**.
- **GDP contribution:** Expected to contribute **2.5% of GDP by 2030**, primarily boosting **services, logistics, and digital platforms**.
- **Entrepreneurship:** Enables **micro-entrepreneurship**, allowing workers to start small businesses with **minimal capital and risk**.
- **Regional decentralisation:** Expansion of **Q-commerce, delivery, and logistics** into Tier-2/3 cities promotes **balanced regional economic growth**.
- **Global competitiveness:** India supplies **27% of global online freelancers**, earning **significant foreign exchange** and enhancing **digital workforce reputation**.

B. Social Significance

- **Women's empowerment:** Women in gig platforms (e.g., **Urban Company**) earn **40–50% higher** than traditional roles, improving **financial independence and FLFPR**.
- **Youth absorption:** Offers short-term income opportunities for youth during **job search or skill development**, reducing **NEET prevalence**.
- **Marginalised groups:** Provides accessible work for **SC/ST, migrants, persons with disabilities**, lowering barriers to economic participation.
- **Resilience:** Gig incomes acted as a **financial buffer** during the **2008 global crisis** and **COVID-19 pandemic**, reducing vulnerability.
- **Skill monetisation:** Converts **informal skills** (driving, cooking, carpentry) into **formal, monetised income streams**, enhancing human capital value.

Challenges & Concerns in the Gig Economy

A. Social Security Deficit

- Most gig workers are classified as **independent contractors**, excluding them from **ESI, PF, gratuity, paid leave, health insurance, and pensions**.
- **90% of gig workers lack savings** (NITI Aayog, 2024), leaving them with **negligible retirement security**.

B. Income Instability

- Monthly earnings of **₹15,000–20,000** often fall **below effective minimum wage** for hours worked.
- Income is **highly volatile**, depending on platform demand, weather, and **customer ratings**.
- **High platform commissions** significantly reduce net earnings; workers resort to **multi-apping** to survive, causing exhaustion.

C. Algorithmic Control

- **Opaque algorithms** dictate task allocation, pay, and deactivation with little **transparency or accountability**.
- Customer ratings can **terminate livelihoods arbitrarily**, with minimal grievance redressal.
- Workers cannot refuse tasks without **earning penalties**, making the supposed **flexibility illusory**.

D. Occupational Health & Safety

- Delivery and location-based workers face **extreme heat, traffic accidents, and long working hours**.
- High rates of **anxiety, depression, and burnout** observed among food delivery riders (Tamil Nadu, 2025).
- No mandatory **safety gear, insurance, or OHS regulations** exist for gig workers.

E. Other Challenges

- **Legal ambiguity:** Existing laws (Contract Labour Act 1970, Workmen's Compensation Act 1923) **inadequate** for platform work.
- **Gender gap:** Safety concerns and social norms restrict **women's participation** despite 28% involvement.
- **Digital divide:** Rural and elderly workers are often excluded from the **digital gig ecosystem**.
- **Skill mismatch:** Low-skilled workers face **wage suppression** and limited mobility to higher-value roles.
- **No collective bargaining:** Current law prevents gig workers from **joining traditional unions** or forming associations.

Global Comparison & International Frameworks:

Country / Body	Approach	Key Feature
EU (Platform Work Directive, 2024)	Presumption of employment	Reverses burden of proof — platforms must prove workers are truly self-employed; addresses algorithmic management.
UK	Third category: 'Worker' (between employee and contractor)	Hybrid status; ensures rights like minimum wage and holiday pay. Uber ruled by Supreme Court to provide these benefits.

California, USA (AB5)	ABC test for employment	Platforms must meet a 3-part test to classify workers as contractors; faced resistance from gig companies (Prop 22).
Netherlands	Reclassification with full social security rights	Moving towards treating many gig workers as employees.
ILO (June 2025)	Binding global standards	ILO 113th Conference agreed to develop a binding convention on decent work in the platform economy (final adoption expected 2026); India opposed.
India	Recognition without strong protection yet	Gig workers defined in Code 2020; state-level welfare funds; no reclassification or minimum wage guarantee.

Way Forward — Multi-Dimensional Solutions

A. Regulatory Reforms

- Enact a **National Gig Workers Welfare & Rights Framework** with minimum wages, working hour limits, and grievance redressal, aligned with NITI Aayog's **RAISE framework**.
- Scale the **Rajasthan model** — mandatory registration, welfare boards, and **cess-based funding** — across all states with **national minimum standards**.
- Revisit **'independent contractor' classification**; adopt a **hybrid/third-category status** (like UK) to bridge protection gaps.
- Draw lessons from **EU Platform Work Directive** for algorithmic accountability and transparency.

B. Social Security

- Implement **portable benefits systems** — health insurance, pensions, and disability benefits that **move with the worker** across platforms.
- Expand **AB-PMJAY** to all registered gig workers and operationalise **PM-SYM** fully for social protection.
- Mandate **platform contributions** to welfare funds (1–2% per transaction), as in Rajasthan.
- Encourage **gig worker mutual aid societies** — worker-owned cooperatives pooling **insurance and welfare resources**.

C. Data & Statistics

- Introduce a **dedicated gig worker category in PLFS**, capturing **income, hours, platform type, and demographics**.
- Adopt **ILO 2028 ICLS standards** for measuring **digital platform employment**.
- Expand **e-Shram data** to include **platform-specific details** for targeted welfare and policy delivery.

D. Skill & Upward Mobility

- Integrate gig workers into **PMKVY** for **reskilling and upskilling**, enhancing income potential.
- Promote transitions from **low-skill gigs** (delivery) to **higher-skill roles** (tech, consulting) via **recognition of prior learning**.
- Foster **platform-linked skilling partnerships** with NSDC and vocational institutes for career progression.

E. Algorithmic Fairness & Worker Voice

- Mandate **algorithmic transparency** — platforms must disclose **task allocation, rating systems, and deactivations**.

- Establish **tripartite welfare boards** (workers + platforms + government) at national and state levels for **grievance redressal**.
- Enable **collective bargaining rights** through new legal mechanisms like **worker associations**.
- Leverage **Digital Personal Data Protection Act, 2023** to safeguard **gig worker data** from exploitation by platforms.

Conclusion:

The gig economy in India presents **significant opportunities** for job creation, women's empowerment, and economic inclusion, particularly in Tier-2 and Tier-3 cities. However, **structural challenges** limit its potential. **Targeted regulatory reforms, social protection, skill upgradation, and data-driven policies** are essential to transform gig work into **secure, equitable, and sustainable employment**. A balanced approach can ensure that the gig economy **supports inclusive growth** while safeguarding worker rights.

4. AI THREAT TO EMPLOYMENT

Introduction:

Artificial Intelligence (AI) has evolved from a **speculative concept into a transformative economic force**, fundamentally altering production systems and redefining the nature of work. **For India**, which **possesses the world's largest pool of young digital talent and over 9 million technology professionals**, this transition presents a critical paradox characterized by the **Economic Survey 2025-26 as a "quiet, steady drift" rather than a sudden shock**. While the AI market is projected to reach $28.36 billion by 2030, the country faces a dichotomy: **a potential loss of 1.5 million low-skill jobs versus the creation of 4 million** new high-value opportunities.

AI Ecosystem in India at Present

- India's technology sector is expanding rapidly, with annual revenues projected to cross **USD 280 billion** this year.
- Over. **6 million** people are employed in the tech and AI ecosystem.
- The country hosts **1,800+** Global Capability Centres, including more than **500** focused on AI.
- India has around **1.8 lakh startups**, and nearly **89%** of new startups launched last year used AI in their products or services.
- On the NASSCOM AI Adoption Index, India scores 2.45 out of 4, showing that **87%** of enterprises are actively using AI solutions.
- Leading sectors in AI adoption include industrial and automotive, consumer goods and retail, banking, financial services and insurance, and healthcare. Together they contribute around **60 percent** of AI's total value.
- About 26% of Indian companies have achieved AI maturity at scale, according to a recent BCG survey.

Understanding the Threat: The Nature of Disruption

1. **Task-Based vs. Job-Based Displacement:** AI typically automates specific **tasks** within a role (e.g., **data entry, basic coding**) rather than eliminating the entire **job**. This transforms the **nature of work**—shifting human focus to **supervision and strategy**—rather than causing **outright extinction of roles**.
2. **The "Non-Linear" Trajectory:** The **Economic Survey** 2025-2026 warns that the impact will be **uneven across the economy**; while some sectors adapt gradually, others like **IT services may face sudden disruption**, making it difficult to **predict the pace of change**.

3. **The "Quiet" Hiring Freeze:** The most immediate threat is not mass layoffs but a **stagnation in recruitment**. Firms are **pausing the hiring of entry-level staff** (junior developers, analysts) to evaluate **AI capabilities**, effectively **blocking the ladder for fresh workforce entrants**.
4. **Cognitive vs. Manual Automation:** Unlike the **Industrial Revolution** which automated **physical labor**, AI targets **cognitive, non-routine tasks**. This creates a **unique paradox** where **white-collar professionals** (lawyers, coders) are at higher risk than **blue-collar workers** (plumbers, electricians) whose jobs require **physical dexterity**.

Challenges Specific to India:

1. **Vulnerability of the IT Sector:** Over 60% of formal jobs in India's IT and BPO **sectors** are susceptible to **automation by 2030**, threatening the **$250 billion services export model** that relies heavily on **"low-cost labor" arbitrage**.
2. **Erosion of Entry-Level Roles:** The **"hiring freeze" phenomenon** has led to a 14% decline in recruitment for workers aged 22–25 (Anthropic Study), destroying the **"experience ladder"** freshers need to become senior experts.
3. **Structural Skill Mismatch:** The current education system emphasizes **rote learning over critical thinking**, leading to a situation where the **supply of AI talent meets only 50% of the industry demand**.
4. **Brain Drain of Talent:** India faces a **net negative migration of AI professionals (-1.55 per 10,000)**, losing **high-value researchers to global competitors** due to better research ecosystems and compensation abroad.
5. **Exclusion of the Informal Sector:** With over **90% of the workforce in the informal sector**, a lack of **digital literacy and verifiable identities** risks excluding millions from AI-driven productivity gains, widening the **inequality gap**.
6. **Deepening Inequality:** AI threatens to **polarize the workforce**, benefiting **high-skilled elites** while displacing **mid-level administrative roles**, disproportionately affecting **women (54% in high-exposure roles)** and widening the **wage gap**.
7. **R&D and Innovation Deficit:** India's share of **global AI patents has dropped to under 5% in 2023**, reflecting **chronic underinvestment in R&D** compared to the **US and China**, stifling indigenous job creation in AI manufacturing.

Sectoral Analysis (India Specific)

Sector	Vulnerability	Impact Analysis
IT/ITeS	High	Entry-level software engineers and BPO voice support are most at risk. Shift from "low-cost labor" model to "high-value AI solutions."
Manufacturing	Medium	Industry 4.0 (Robots + AI) reduces the need for assembly line workers. India's "Make in India" goal of creating mass manufacturing jobs may be hindered.
Agriculture	Low to Medium	AI helps in crop prediction and pricing (benefit), but automation in supply chains may reduce labor demand in processing units.

Healthcare	Low (Hybrid)	AI aids diagnosis (telemedicine), but the human touch in nursing/caregiving remains irreplaceable.
Financial Services	High	Algorithmic trading, automated auditing, and AI loan officers reduce the need for back-office staff.

Advantages of AI Adoption in Employment:

1. **Job Creation Potential: NITI Aayog** projects the creation of **4 million new jobs by 2031** in emerging fields like **AI ethics, data annotation, and prompt engineering**, effectively offsetting displacement risks if managed well.
2. **Productivity Enhancement:** Widespread adoption of **Generative AI** is estimated to raise **labor productivity by 15–20%**, significantly **reducing service delivery costs** and boosting the global competitiveness of Indian exports.
3. **Wage Premium and Upskilling:** AI-focused roles command a 28% wage premium (World Bank), creating a strong **financial incentive** for the workforce to **acquire advanced skills and move up the value chain**.
4. **Democratization of Services:** AI tools like **BHASHINI** bridge **linguistic barriers**, allowing non-English speakers to access digital opportunities and significantly improving **workforce participation in rural and semi-urban areas**.
5. **Transformation of Public Services:** In sectors like **healthcare and education**, AI can bridge **infrastructure deficits** through **telemedicine and personalized learning**, creating "high-skill service jobs" in underserved regions.
6. **Shift to High-Value Tasks:** By automating **routine cognitive tasks (data entry, basic coding)**, AI forces a **structural shift** where professionals focus on creative, strategic, and empathy-driven work that machines cannot replicate.

Key Government Initiatives and Policy Push:

1. **Centres of Excellence (CoEs)**: The government has established dedicated CoEs in **Healthcare, Agriculture, Sustainable Cities, and Education (Budget 2025)** to foster collaboration between **academia and industry** for developing scalable AI solutions.
2. **AI Competency Framework**: This framework provides **structured training for government officials**, ensuring the bureaucracy acquires essential **AI skills** for effective policymaking and remains agile in an AI-driven future.
3. **IndiaAI Startups Global Acceleration Programme**: Launched in partnership with **Station F, Paris**, this initiative supports promising **Indian startups** by providing access to **global networks**, helping them compete internationally.
4. **Sovereign AI Ecosystem (Sarvam AI)**: Partnering with **UIDAI** for smarter **Aadhaar services**, Sarvam AI is building India's **Sovereign LLM Ecosystem**—an **open-source model** to enhance **public service delivery** and **digital trust**.
5. **Bhashini Initiative**: An **AI-led platform** promoting **digital inclusion** by offering **translation in 20+ Indian languages**; it has integrated with **railway platforms (CRIS)** to bridge **linguistic divides** for citizens.
6. **BharatGen AI**: Launched in **June 2025**, it is **India's first government-funded multimodal Large Language Model**, supporting **22 languages**, built on **domestic datasets** to reflect **India's cultural diversity**.
7. **India AI Impact Summit 2026**: India is set to host this **global summit** to showcase its **AI capabilities** and encourage **cross-sector innovation**, reinforcing its position as a **global AI leade**r.

Way Forward:

1. **National AI Talent Mission:** Establish a dedicated mission to embed **AI fluency across schools and universities**, creating a **national reskilling engine** to bridge the **talent supply-demand gap at scale.**
2. **Mission Digital ShramSetu:** Implement a digital identity system for informal workers **using blockchain and AI** to grant them access to **skilling programs, formal credit, and secure employment opportunities**.
3. **Curriculum Overhaul:** Shift the education focus from **static knowledge to STEM and critical thinking**, aligning university courses with modern concepts like Retrieval Augmented Generation (RAG) and data science.
4. **AI Talent Visa and Incentives:** Introduce a **fast-track residency visa** and **attractive research grants** to position India as a **"Global AI Talent Magnet"** to retain domestic experts and attract international researchers.
5. **Strengthening Social Safety Nets:** Develop **transitional support mechanisms**, such as **unemployment insurance**, specifically for workers displaced by **technological shocks during the reskilling transition phase**.
6. **Ethical and Regulatory Framework:** Establish an **AI Economic Council** to ensure **transparent deployment**, address **Intellectual Property (IP) rights for AI content**, and prevent **discriminatory hiring practices**.
7. **Pace the Diffusion:** Follow the **Economic Survey's advice** to manage the **pace of AI adoption**, ensuring that labor augmentation is prioritized over outright replacement through supportive policy nudges.

Conclusion:

The threat of AI to employment in India **is real but nuanced**. It is not a simple story of job loss, but a structural transformation of work. The policy challenge lies in **managing the "quiet drift"—**ensuring that the pace of AI adoption matches the pace of workforce reskilling. By shifting **focus from job preservation to productivity enhancement with equity**, India can convert this technological disruption into a **"Technological Dividend."**

5. LOW FEMALE LABOUR FORCE PARTICIPATION IN INDIA

Introduction:

The Female Labour Force Participation Rate (FLFPR) is defined as the **percentage of women in the population who are either employed or actively seeking employment,** serving as a critical barometer for economic inclusion. Recently, **the Economic Advisory Council to the Prime Minister (EAC-PM)** has released a working paper highlighting a dramatic increase in the Female Labor Force Participation Rate (LFPR).

Current Trends:

According to the **Periodic Labour Force Survey (PLFS) 2023–24**, India's female LFPR has seen a gradual increase:

- Overall female LFPR rose from **23.3% in 2017–18 to 41.7% in 2023–24**..
- **Rural female LFPR** experienced a sharp increase from **24.6% in 2017–18 to 47.6% in 2023–24**, representing nearly **69% growth** (Economic Advisory Council to the Prime Minister). This growth is largely attributed to women's increasing engagement in agriculture and informal sector employment.

- **Urban female LFPR**, in contrast, increased modestly from **20.4% to 25.4%** over the same period, highlighting persistent barriers to workforce participation in urban areas, including safety, household responsibilities, and societal norms.

There is also notable **regional variation**: southern and northeastern states record higher female LFPR, whereas Bihar, Punjab, and Haryana report much lower rates due to cultural and structural factors.

Determinants of Female LFPR:

According to the EAC-PM Working Paper, three key demographic factors influence participation:

Factor	Trend	Analysis
Age	Bell-Shaped Curve	Participation peaks between ages 30–40 and declines sharply afterward. In contrast, male participation remains near 100% for ages 30–50.
Marriage	Negative Correlation	Marriage significantly reduces LFPR, especially in urban areas, due to added household responsibilities and mobility restrictions.
Parenthood	High Impact	The presence of children under 14 drastically reduces participation (ages 20–35), highlighting the lack of childcare support.

Significance of Improving Female Labor Force Participation:

1. **Economic Growth Engine:** Increasing **FLFPR** can add significantly to the **GDP growth rate**, with **2023 World Bank–linked review estimates** that closing gender gaps in labour-force participation could **raise GDP per capita** by around 20% on average across countries
2. **Demographic Dividend Realization:** To truly reap the benefits of **India's youth bulge**, the female workforce must be productively employed, otherwise the demographic dividend may turn into a **demographic disaster**.
3. **Poverty Reduction: Women's earnings act as a shock absorber** for households during economic distress, significantly **lowering the incidence of poverty** and improving **household nutrition**.
4. **Social Empowerment:** Economic independence grants women **greater agency** in household decision-making, **challenging patriarchal structures** and improving their **status within the family**.
5. **Inter-generational Benefits:** Working women tend to invest more in their **children's health and education**, creating a **virtuous cycle of human capital development** for the future workforce.
6. **Achieving SDG Targets:** Higher participation is critical for achieving **SDG 5 (Gender Equality)** and **SDG 8 (Decent Work)**, reflecting a country's progress on **inclusive development**.
7. **Fiscal Sustainability:** A larger female workforce **expands the tax base** and **reduces the dependency ratio**, strengthening the **government's fiscal capacity to fund social welfare schemes**.

Challenges Contributing to Low Female LFPR

Safety Concerns

- Limited safety in public spaces and workplaces restricts women's mobility and employment opportunities.
- **NCRB data (2022)**: Workplace sexual harassment cases rose to **422 from 402 in 2018**, highlighting persistent safety challenges.

Cultural and Social Norms

- Traditional expectations prioritize women as homemakers, reducing engagement in paid work.
- Marriage significantly lowers workforce participation, particularly in urban areas where household responsibilities intensify.

Caregiving Responsibilities and Unpaid Work

- Women disproportionately shoulder unpaid domestic and caregiving work, restricting availability for formal employment.
- **Economic Survey 2024**: Women's unpaid care work contributes **3.1% of GDP**, compared to **0.4% by men**, illustrating the double burden.
- Parenthood further reduces participation, especially for women aged 20–35 with children under 14.

Education-Work Trade-off

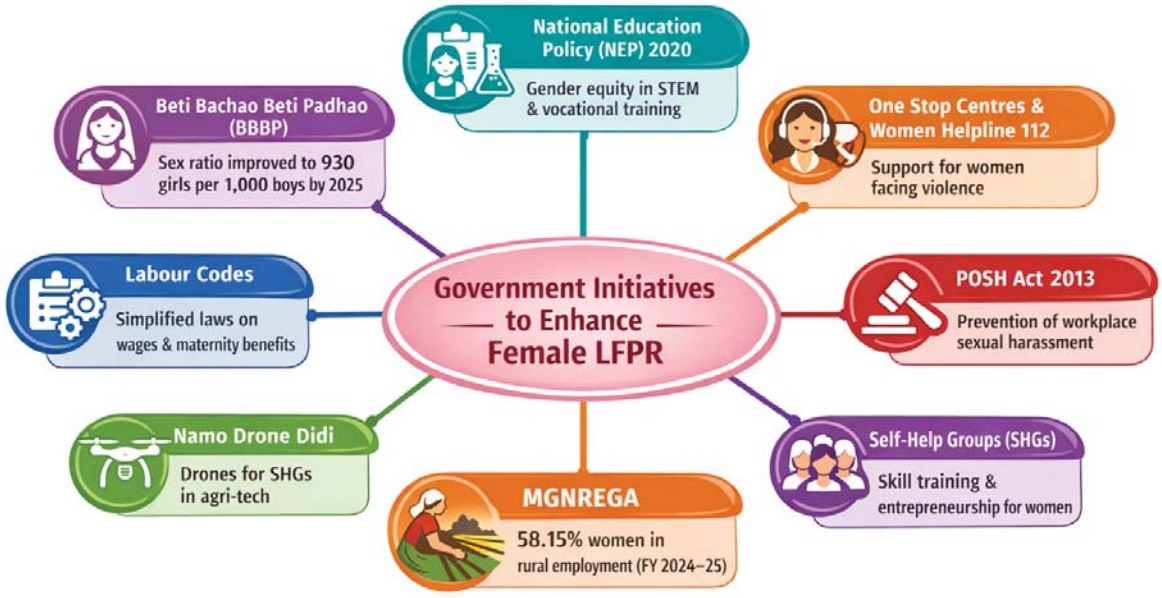
- Many women remain out of the labour force to pursue education.
- **PLFS 2023–24**: Approximately **37.9% of women** do not participate in the labour market for educational reasons.
- While higher education improves skills, it often delays workforce entry.

Digital Divide

- Limited access to digital tools reduces opportunities for online work or digital entrepreneurship.
- **NFHS-5 (2019–21)**: Only **33% of women** reported using the internet, compared to **57% of men**.

Informal Employment and Lack of Social Protection

- Women are concentrated in informal jobs with low wages and insecure conditions.
- **eShram database (March 2022)**: Women account for **52.7% of 287 million registered unorganized workers**.



Way Forward:

Redefine Employment Metrics

- Move beyond **FLFPR numbers** to assess **earnings, job quality, and working hours**, capturing real economic empowerment.

- Use **Time Use Surveys (NSO)** to account for **unpaid care work** in GDP, highlighting hidden female labour contributions.

Targeted Job Creation

- Leverage **PLI schemes** to incentivize **women's hiring in sectors like textiles, electronics, and manufacturing**, promoting formal employment.
- Expand **MGNREGA** with **women-focused tasks** such as afforestation, water management, and care-related work to boost participation and income.

Care Infrastructure (Social Infrastructure)

- Establish **community childcare centers** and **elderly care facilities** to reduce women's unpaid domestic responsibilities.
- Draw lessons from **Burkina Faso's mobile creches**, which provide safe, accessible childcare enabling women to work.

Bridging the Digital Divide

- Scale programs like **Google Internet Saathi** to improve **digital literacy** for rural and marginalized women.
- Enables access to the **gig economy, online jobs, digital banking**, and financial inclusion.

Changing Social Norms

- Promote **shared domestic responsibilities** through media, campaigns, and school curricula to shift household gender roles.
- Encourage women in **non-traditional sectors** like STEM, defense, driving, and entrepreneurship to expand economic opportunities.

Conclusion

The recent rise in India's Female LFPR is a double-edged sword. While it **reflects the resilience of women in the face of rural distress,** it highlights the **failure of the formal economy to absorb female labor.** For India to achieve its **goal of Viksit Bharat and SDG 5 (Gender Equality)**, policy focus must shift from mere **participation to productive, remunerative, and dignified employment.**

EMPLOYMENT AND SKILL DEVELOPMENT SCHEMES

1. LABOUR CODES

Context: Recently, the Ministry of Labour and Employment brought into force the 4 Labour Codes almost 5 years after being cleared in Parliament.

Objectives: This reform aims to:

- Simplify labour regulations, reducing compliance burdens for businesses.
- Protect and enhance workers' rights.
- Improve the ease of doing business while accommodating changing work patterns, including gig and platform-based employment.

The four Labour Codes are:

1. **Code on Wages, 2019**
2. **Industrial Relations Code, 2020**
3. **Occupational Safety, Health and Working Conditions (OSHWC) Code, 2020**
4. **Code on Social Security, 2020**

These Codes replace multiple overlapping laws, reducing confusion caused by **multiple definitions and authorities**.

Labour and Constitutional Rights in India

The **Constitution of India** provides several safeguards and rights concerning labour and workers:

1. **Article 19(1)(c):** Grants all citizens the **right to form associations or unions**, enabling collective bargaining and organisation of workers.
2. **Article 23:** Prohibits **forced labour**, making it unconstitutional to compel any person to work against their will.
3. **Article 24:** Prohibits **child labour**, specifically banning the employment of children below **14 years in hazardous occupations or processes**.
4. **Article 38(1) & 38(2):**
   - (1) Directs the state to **promote the welfare of the people**.
   - (2) Seeks to **reduce economic inequalities**, including income disparities.
5. **Article 43(A):** Provides for **workers' participation in the management of industrial and other undertakings**, fostering cooperative industrial relations.

Overview of Labour Codes:

Labour Codes	Major Acts Subsumed	Key Provisions
Code on Wages, 2019	- Payment of Wages Act, 1936 - Minimum Wages Act, 1948 - Payment of Bonus Act, 1965 - Equal Remuneration Act, 1976	• Establishes a **uniform definition of 'wages'** for all workers. • Mandates **timely payment of wages** to every employee. • Introduces a **national minimum wage**, adjustable by states above the central floor. • Ensures **equal remuneration** for men and women performing the same or similar work.
Industrial Relations Code, 2020	-Trade Unions Act, 1926 - Industrial Employment (Standing Orders) Act, 1946 - Industrial Disputes Act, 1947	• Simplifies **trade union registration and recognition**. • Introduces **negotiation unions or councils** for dispute resolution. • Sets conditions for strikes, including a 60-day notice period for strikes in public utility services. • Establishes **grievance redressal mechanisms**. • Raises threshold for prior government permission for closure, lay-off, or retrenchment from **100 to 300 workers**. • Introduces fixed-term employment, providing flexibility for employers while ensuring protection for workers.

Code on Social Security, 2020	- Employees' Provident Funds and Miscellaneous Provisions Act, 1952; - Employees' State Insurance Act, 1948; - Maternity Benefit Act, 1961; - Payment of Gratuity Act, 1972; - Unorganised Workers Social Security Act, 2008	• Expands **employee definition** to include **gig and platform workers**. • Introduces **social security schemes** for gig, platform, and unorganised workers. • Mandates creation of a **social security fund** for unorganised workers. • Provides **maternity benefits and gratuity** to women workers, including those in the unorganised sector.
Occupational Safety, Health and Working Conditions (OSHWC) Code, 2020	-Factories Act, 1948 -Mines Act, 1952 -Contract Labour (Regulation and Abolition) Act, 1970 -Inter-State Migrant Workmen (Regulation of Employment and Conditions of Service) Act, 1979	• Ensures **health, safety, and welfare** for all workers. • Mandates **annual health check-ups** for employees. • Regulates **working hours and conditions** for different sectors. • Introduces a **single licensing mechanism** for contractors and employers. • Increases **thresholds for contract labour applicability** and prohibits contract labour in core activities, with certain exceptions. • Shifts primary responsibility for welfare facilities from **contractors to principal employers**.

Need for Labour Reform:

1. **Simplifying compliance**: India's large number of labour laws made compliance complex and discouraged formal employment. The **2nd National Commission on Labour** recommended consolidating them into **4–5 comprehensive labour codes**.
2. **Modernisation of archaic laws**: Many pre-independence and early post-independence laws were outdated for contemporary economic realities. *Example*: **Payment of Wages Act, 1936** applied only to employees earning ≤ ₹24,000/month and did not cover modern work structures.
3. **Universal coverage gap**: Around **90% of workers** are in the unorganised sector, lacking access to social security and basic labour protections.
4. **Promoting economic objectives**: The new labour codes aim to **boost investment, growth, and job creation** by simplifying compliance and improving Ease of Doing Business.
5. **Addressing new forms of work**: Reforms formally recognise and provide social security to **gig/platform workers**and **inter-state migrants**, catering to evolving employment patterns.
6. **Gig economy significance**: As per **NITI Aayog (2020–21)**, 7.7 million workers (1.5% of India's workforce) were engaged in gig work, highlighting the need for tailored labour protections.

Expected Benefits Across Key Sectors:

1. **Fixed-Term Employees (FTEs) & Contract Workers**: FTEs receive **all benefits equal to permanent employees**, including leave, medical, and social security. Gratuity eligibility is reduced from five to one year, promoting direct hiring and reducing contractualisation.

2. **Gig & Platform Workers**: Gig, platform work, and aggregators are **formally recognised** for the first time. Aggregators contribute 1–2% of turnover (capped at 5% of payments to workers). **Aadhaar-linked Universal Account Numbers** ensure portable welfare benefits across states.
3. **Women Workers**: Equal pay for equal work is legally guaranteed, and gender discrimination is prohibited. Women can work night shifts and in all jobs, including mining and heavy machinery, with consent and safety measures. **Mandatory representation in grievance committees** ensures better protections.
4. **Youth Workers**: Minimum wage is guaranteed for all, with **mandatory appointment letters** ensuring formal employment. Payment of wages during leave is mandatory, and wages follow the **floor wage set by the Central Government**.
5. **MSME Workers**: All MSME employees are covered under the **Social Security Code 2020** based on enterprise size. Workers gain access to **canteens, drinking water, rest areas, standard working hours, double overtime pay, and paid leave**.
6. **IT, ITES & Audio-Visual/Digital Media Workers**: Mandatory appointment letters specify designation, wages, and social security benefits. Full benefits cover **journalists, stunt persons, and dubbing artists**, with **equal pay for equal work** and timely salary payments. Women can work night shifts safely.
7. **Beedi, Cigar, Mine, Dock & Hazardous Industry Workers**: Minimum wages are guaranteed, with working hours of 8–12 hours/day and 48 hours/week. **Consent-based double overtime, free health check-ups, and safety committees** are provided. Dock workers gain formal recognition and social security, while women can work in hazardous jobs.
8. **Plantation & Textile Workers**: Plantation workers are covered under **OSHWC and Social Security Codes**, with ESI medical facilities for families. Migrant textile workers receive **equal wages, welfare benefits, and PDS portability**, and can raise claims for pending dues up to 3 years.
9. **Export Sector Workers**: Fixed-term workers receive **gratuity, PF, and social security**, with annual leave after 180 days. Unauthorised wage deductions are prohibited, and workers are provided **consent-based double overtime, safe transportation, CCTV, and security measures**.

Challenges with the New Labour Codes:

1. **Uneven implementation across States**
   - Labour is in the Concurrent List, so states frame their own rules.
   - By mid-2025, **30+ states/UTs had published draft rules**, but some (e.g., West Bengal, Meghalaya) were still lagging (Ministry of Labour).
   - This has created a patchwork of protections → workers in Karnataka may enjoy better safeguards than those in Assam.
2. **Excessive delegation of Powers**
   - Key provisions like the **definition of "wages"** or **thresholds for retrenchment** are left to government notifications.
   - This reduces parliamentary oversight and opens scope for arbitrary decisions.
   - The **Standing Committee on Labour (2023)** flagged this as a major concern.
3. **Inadequate Protection for Gig & Informal Workers**
   - Despite **30+ crore unorganised workers registered on e-Shram**, benefits are still unclear.
   - Positive state efforts: **Karnataka's 2025 law** mandating platform firms to contribute to a welfare fund; **Maharashtra mapping 10 lakh gig workers**; **Telangana study** found gig workers earn ~₹20,000/month but face long hours & high commissions.
   - Union Budget 2025–26 extended **PM-JAY health coverage** to gig workers, but a comprehensive national framework is missing.

4. **Weakening of collective bargaining**
   - The **Industrial Relations Code** requires **75% approval for strikes**, nearly impossible in large firms.
   - In 2024, trade unions held **nationwide protests in Delhi, Lucknow, and Kolkata**, calling this provision "anti-labour."
5. **Reduced Job Security**
   - Retrenchment/closure threshold raised from **100 to 300 workers**.
   - Critics say this tilts towards "Ease of Doing Business" over worker security.
   - Example: **Gurugram auto-component firms (2024)** downsized without government approval due to the new threshold.
6. **Precarious Fixed-Term Contracts**
   - Employers use fixed-term contracts to avoid permanency.
   - **The Supreme Court (2024)** criticised exploitative contract renewals.
   - **Telangana HC (2025)** ordered regularisation of ad hoc staff, calling repeated contracts "unfair labour practice."
7. **Broad exemptions diluting protections**
   - Governments can exempt industries in "public interest," a vague term prone to misuse.
   - This could let factories bypass rules on wages, safety, and working hours.
8. **Low awareness & compliance (MSMEs and Informal Sector)**
   - Many MSMEs struggle with digital compliance.
   - **CII's 2024 survey**: only 46% of small firms were fully aware of labour code requirements.
   - Informal workers remain under-informed about e-Shram benefits.

Way Forward:

1. **Uniform Implementation Across States-** Establish a national benchmark framework for minimum wages, occupational safety, and social security. **For Example**- A **GST Council–like institutional mechanism** for labour could promote inter-state harmonisation while preserving federal flexibility.
2. **Strengthening Legislative Oversight-** Ensure **greater legislative clarity** by embedding critical definitions (e.g., "wages," retrenchment thresholds) within the Codes themselves.
3. **Comprehensive Social Security for Gig & Informal Workers-** Draft a **National Gig and Platform Workers Policy** mandating aggregator contributions to welfare funds. The **e-Shram portal** should be integrated with **DBT-enabled schemes** covering health (PM-JAY), pensions (PM-SYM), and accident insurance for effective last-mile delivery.
4. **Reviving Collective Bargaining Rights-** Rationalise the **strike approval threshold** (from 75% to around 51%) to make it practical while ensuring industrial peace. Promote **tripartite consultations** (Government–Employers–Workers) for dispute resolution.
5. **Balancing Flexibility with Worker Security-** Couple greater flexibility for firms with mandatory unemployment insurance and reskilling programmes for retrenched workers. This would align with global best practices in labour transition support.
6. **Regulating Fixed-Term Contracts-** Introduce a **cap on contract renewals** (e.g., 2–3 cycles). Beyond this, workers should either be granted permanency or strengthened social security entitlements to avoid "permanent temporariness."
7. **Restricting Exemptions-** Narrow down the scope of **"**public interest" exemptions. Make them time-bound, criteria-based, and subject to judicial or legislative review to prevent misuse.
8. **Improving Awareness and Compliance-** Launch **multilingual awareness campaigns, labour helplines, and simplified digital compliance portals**. Special digital support cells for MSMEs should be set up to ease compliance without imposing excessive costs.

Conclusion:

Establishment of the new Labour Codes marks a **transformative step in India's labour landscape**- one that balances the welfare of workers with the efficiency of enterprises. These reforms lay the foundation for a more equitable, transparent, and growth-oriented economy. They reaffirm India's commitment to fostering a modern labour ecosystem that empowers both workers and industry, paving th**e way for inclusive and sustainable progress.**

2.1. PM VISHKARMA SCHEME

The PM Vishwakarma Scheme was launched on 17th September 2023. It is a **Central Sector initiative** aimed at recognizing and empowering India's traditional artisans and craftspeople. The **Ministry of MSME implements the scheme** with a budget of Rs. 13,000 crore for 2023–28.

Objective- The scheme seeks to uplift India's artisans by giving them skills, modern tools, credit, digital support, and market access.

Eligibility

- **Who qualifies?**
  - Artisans and craftsmen engaged in **18 family-based trades** such as carpenters, blacksmiths, goldsmiths, potters, cobblers, masons, barbers, tailors, toy makers, and others.
- **Key conditions:**
  - Must be **above 18 years of age** and actively working with hands and tools.
  - **One member per family** (husband, wife, unmarried children) can benefit.
  - Applicants **should not have taken loans** under the Prime Minister's Employment Generation Programme (**PMEGP), MUDRA, or PM SVANidhi in the last 5 years** (unless repaid).
  - **Government employees and their families are excluded.**

Major Benefits

1. **Recognition**
   - Beneficiaries receive a **PM Vishwakarma Certificate and ID Card**, symbolising formal recognition and enabling access to all scheme benefits.
2. **Skill Development**
   - **Basic Training:** 5–7 days (40 hours) with a **Rs. 500/day stipend**; introduces modern tool use, marketing skills, and digital payments.
   - **Advanced Training:** At least 15 days with **Rs. 500/day stipend**; focuses on entrepreneurship, new technologies, and scaling up enterprises.
3. **Toolkit Incentive**
   - A grant of **Rs. 15,000 (through DBT/e-RUPI vouchers)** for modern tool purchase, helping artisans upgrade productivity and quality.
4. **Credit Support**
   - **Collateral-free Enterprise Development Loans**:
     - 1st tranche: Up to Rs. 1 lakh, repayable in 18 months.
     - 2nd tranche: Up to Rs. 2 lakh, repayable in 30 months.
   - **Concessional Interest:** 5% per annum, with the Government covering up to 8% interest subvention.
   - **Credit Guarantee Fee:** Borne entirely by the Government.
5. **Digital & Market Integration**
   - **Digital Empowerment:** Beneficiaries receive Rs. 1 for each digital transaction, up to 100 transactions per month, directly credited to their accounts for every digital payment or receipt.

- **Market Support:** A dedicated Rs. 250 crore corpus is provided for quality certification, branding, advertising, publicity, and other marketing activities, helping artisans access wider markets and improve visibility.

Institutional Mechanism for Implementation of the Scheme

- **National Steering Committee (NSC):** Apex body under MSME; approves inclusion of new trades, reviews policies, meets twice a year.
- **State Monitoring Committees (SMC):** Ensure smooth state-level rollout.
- **District Implementation Committees (DIC):** Directly manage field-level execution.
- **Credit Oversight Committee:** Chaired by Secretary (Department of Financial Services) with representatives from RBI, SIDBI, CGTMSE, and MSME; ensures assured credit flow and monitors interest subvention.
- **Executing Agencies:** MSME, Ministry of Skill Development & Entrepreneurship (MSDE), and Department of Financial Services (DFS).
- **Monitoring Tools:** Programme Management Units (PMUs) and an Online Monitoring System (OMS) with dashboards for real-time beneficiary tracking.

Achievements of the Scheme- As of January 2025, 26.87 lakh beneficiaries have been successfully registered under the scheme.

2.2. SKILL INDIA PROGRAMME

The Union Cabinet, chaired by Prime Minister Shri Narendra Modi, has recently approved the continuation and **restructuring of the Central Sector Scheme "Skill India Programme (SIP)" till 2026**. The scheme has been allocated an outlay of ₹8,800 crore for the period 2022-23 to 2025-26.

The restructuring merges three major skill development initiatives—**Pradhan Mantri Kaushal Vikas Yojana 4.0 (PMKVY 4.0), Pradhan Mantri National Apprenticeship Promotion Scheme (PM-NAPS),** and **Jan Shikshan Sansthan (JSS) Scheme**—into a composite scheme. It aims to provide structured skill development, on-the-job training, and community-based learning.

Background

- The Skill India Programme (**previously known as the Skill India Mission**) was initially launched in 2015 under the Ministry of Skill Development and Entrepreneurship.
- All courses and certifications under SIP are aligned with the **National Skills Qualification Framework (NSQF)** and integrated with **DigiLocker** and the **National Credit Framework (NCrF).** This ensures formal recognition of skills, seamless employment integration, and opportunities for higher education.
- Skill development under SIP is delivered through:
  - **Short-term training:** via Pradhan Mantri Kaushal Vikas Yojana (PMKVY) and Jan Shikshan Sansthan (JSS)
  - **Long-term training:** via the Craftsmen Training Scheme conducted through Industrial Training Institutes (ITIs)
- The programme aimed to train 40 crore individuals by 2022 across various skill sectors through key initiatives, including:
  - National Skill Development Mission (NSDM)
  - National Policy for Skill Development & Entrepreneurship (2015)
  - Pradhan Mantri Kaushal Vikas Yojana (PMKVY)
  - Skill Loan Scheme

Key Components of Restructured Skill India Programme

1. **Pradhan Mantri Kaushal Vikas Yojana 4.0 (PMKVY 4.0)**
   - Provides short-term training, reskilling, and upskilling.
   - Introduces 400+ new courses in emerging technologies, including AI, 5G, cybersecurity, green hydrogen, and drone technology.
   - Promotes **on-the-job training (OJT)** and **recognition of prior learning** to equip workers with globally recognized skills.
   - Aligns with other government initiatives such as **PM Vishwakarma, PM Surya Ghar Muft Bijli Yojana, National Green Hydrogen Mission**, and NAL JAL Mitra, enabling cross-sector skill impact.
   - **Target Beneficiaries:** Individuals aged **15–59 years**.
2. **Pradhan Mantri National Apprenticeship Promotion Scheme (PM-NAPS)**
   - Aims to expand apprenticeship training across industries.
   - Provides **25% of the stipend (up to Rs. 1,500 per apprentice per month)** through **Direct Benefit Transfer (DBT)**.
   - Expands apprenticeship opportunities in AI, robotics, blockchain, green energy, and Industry 4.0.
   - Focuses on small establishments, MSMEs, aspirational districts, and the North-East Region.
   - **Target Beneficiaries:** Individuals aged **14–35 years**.
3. **Jan Shikshan Sansthan (JSS) Scheme**
   - A community-driven vocational training program aimed at women, rural youth, and economically weaker sections.
   - Provides low-cost, flexible skill development programs.
   - Linked with initiatives like **PM JANMAN** and **Understanding of Lifelong Learning for All in Society (ULLAS)** to ensure **inclusive skilling**.
   - **Target Beneficiaries:** Individuals aged **15–45 years**.

Achievements of the Skill India Programme- The government confirms that over 2.27 crore individuals have benefited from these flagship schemes to date.

Challenges in Employment & Skill Development Schemes

1. **Loan Access & Financial Inclusion**
   - Many artisans cannot access financial support due to strict eligibility criteria, poor credit history, or documentation gaps.
   - **Example:** Under PM Vishwakarma, ~72% of loan applications were rejected, leaving vulnerable artisans without funds to enhance skills or businesses.
2. **Registration & Infrastructure Bottlenecks**
   - Technical issues like website crashes, OTP errors, and Aadhaar mismatches delay beneficiary registration and training.
   - **Example:** In Pune, only 20% of registered artisans completed training, and loan disbursals were slow, reflecting systemic inefficiencies.
3. **Training Quality & Industry Mismatch**
   - Outdated curricula and inadequate infrastructure in ITIs and other training centers result in skill sets that do not meet industry requirements.
   - **Example:** Under Skill India/PMKVY, only ~15% of trained individuals secured employment.
4. **Weak Placements & Employment Conversion**

- Poor industry linkages and limited market-oriented training lead to low placement rates.
- **Example:** Despite large-scale skill development efforts, most trainees remain unemployed or underemployed.

5. **Gender Disparities & Rural Exclusion**
   - Women and rural populations face social, mobility, and infrastructure barriers.
   - **Example:** Although over 90% of rural workers are in the informal sector, only ~10% have received formal training.
6. **Fragmented Implementation & Oversight**
   - Multiple schemes managed by different agencies cause duplication, confusion, and weak accountability.
   - **Example:** Beneficiaries often face difficulties understanding which program to join, leading to inefficiencies and low impact.
7. **Market Access & Branding Weakness**
   - Artisans lack e-commerce support and branding opportunities, reducing competitiveness and fair pricing.
   - **Example:** Many traditional craftsmen cannot reach wider markets, limiting income potential and sustainability.

What should be the Way forward?

- **Simplify Loan Access** – Make loans easier for artisans by relaxing eligibility rules and using digital platforms for quick approvals.
- **Enhance Registration & Infrastructure** – Upgrade digital systems and provide basic digital literacy to ensure smooth registration.
- **Align Training with Industry Needs** – Update training curricula and partner with industries for practical exposure and internships.
- **Strengthen Placement Mechanisms** – Build strong industry linkages and organize job fairs to improve employment opportunities.
- **Promote Gender Inclusivity & Rural Participation** – Encourage women and rural populations to join programs with transport and childcare support.
- **Streamline Implementation & Oversight** – Use centralized monitoring and regular audits to track progress and address challenges.
- **Facilitate Market Access & Branding** – Provide online platforms and marketing training to help artisans sell and promote their products.

MGP/BP/GS3/009

INCLUSIVE GROWTH

INCLUSIVE GROWTH AND DISPARITIES

1. INCLUSIVE GROWTH

India's guiding philosophy, often described as **Sabka Sath, Sabka Vikas (striving together for inclusive growth)**, operates on the conviction that the needs and aspirations of every human being are equally important. - **PM Narendra Modi**

Meaning:

Inclusive growth refers to a pattern of economic development that is both **broad-based and equitable in its outcomes**. The OECD defines it as growth that is **distributed fairly across society while creating opportunities for all individuals**. Similarly, the UNDP views inclusive growth as both a **process and an outcome**, wherein **all sections of society participate in economic progress and benefit from it equitably**.

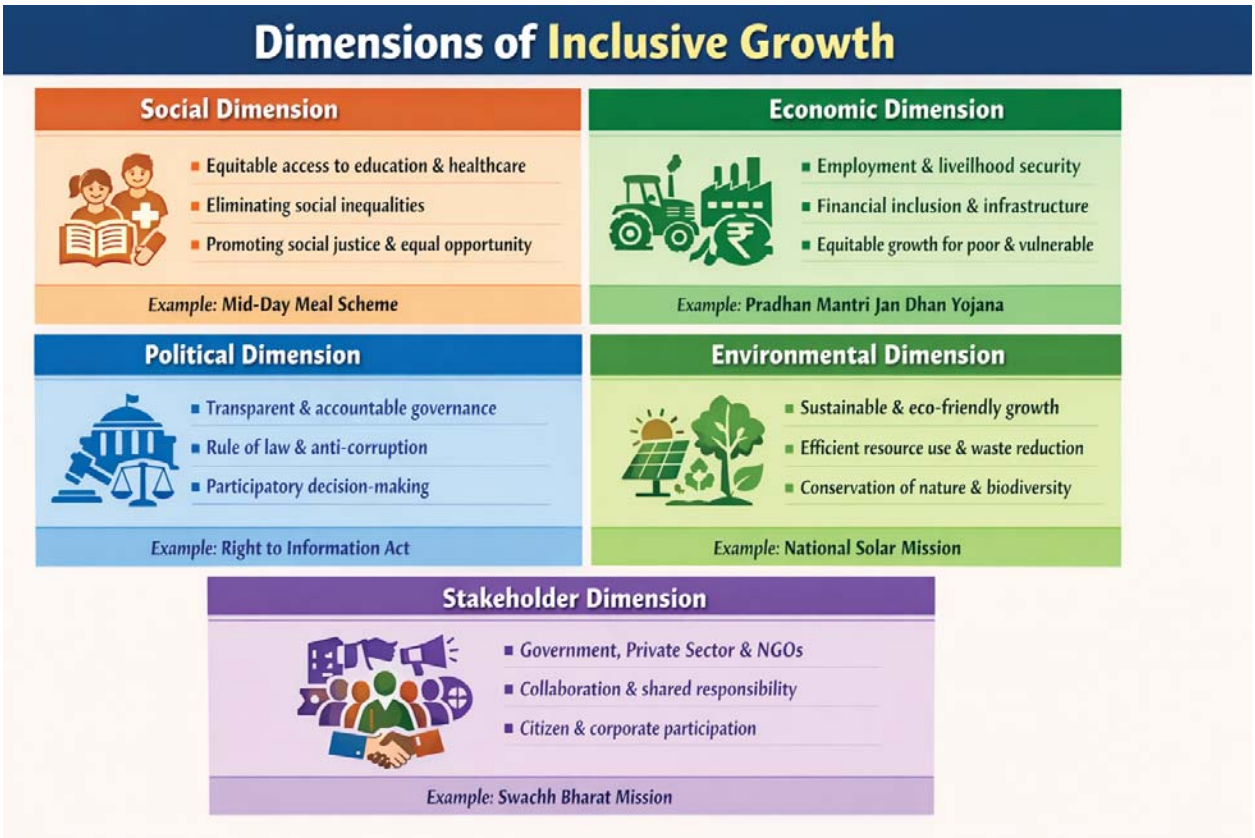
Dimensions of Inclusive Growth

1. **Social Dimension**
   - Focuses on **equitable access** to **education** and **healthcare**, thereby enhancing **human capabilities**.
   - Seeks to eliminate **social inequalities** based on **gender**, **caste**, and **region**.
   - Promotes **social justice**, **dignity**, and **equal opportunity** for all sections.
   - **Example: Mid-Day Meal Scheme** improving **school enrolment** and **nutrition** among children.
2. **Economic Dimension**
   - Ensures **broad-based economic growth** through **employment generation** and **livelihood security**.
   - Promotes **financial inclusion**, **agricultural development**, and **infrastructure expansion**.
   - Aims at **equitable distribution** of growth benefits, especially for the **poor and vulnerable**.
   - **Example: Pradhan Mantri Jan Dhan Yojana (PMJDY)** enabling **banking access** to the unbanked.
3. **Political Dimension**
   - Emphasizes **transparent**, **accountable**, and **efficient governance**.
   - Ensures rule of law, protection of rights, and absence of corruption.
   - Encourages **participatory decision-making** and **inclusive institutions**.
   - **Example: Right to Information (RTI) Act, 2005** strengthening transparency and accountability.
4. **Environmental Dimension**

- Integrates **sustainability** into the **growth process** to prevent **ecological degradation**.
- Promotes **efficient resource use** and **reduction of waste**.Focuses on **conservation of biodiversity** and **natural ecosystems**.
- **Example: National Solar Mission** promoting **clean energy** and reducing **carbon footprint**.

5. Stakeholder Dimension

- Recognizes the role of multiple actors in achieving **inclusive development**.
- Involves **government**, **private sector**, **NGOs**, **civil society**, and **media**.
- Promotes **collaboration**, **accountability**, and **shared responsibility**.
- **Example: Swachh Bharat Mission** involving **citizens**, **government**, and **corporates** in sanitation efforts.



Inclusive Growth: Current Status in India

A. Core Development & Inequality Indicators

- **HDI & Inequality:** India's **HDI is 0.685 (2025)**, but it falls by **30.7% after adjusting for inequality**, revealing deep disparities in human development outcomes.
- **Income Inequality (Gini Coefficient):** At **0.410 (2023)**, it reflects **rising income inequality**, indicating uneven distribution of growth benefits.
- **Wealth Inequality:** The **top 10% hold 77% of wealth**, while the **bottom 60% own just 4.8%**, highlighting **extreme concentration of assets**.
- **Multidimensional Poverty:** Around **234 million people** still face deprivations in health, education, and living standards (UNDP).

B. Global Indices Reflecting Inclusiveness

- **Commitment to Reducing Inequality (CRI) Index 2024:** India ranks **127/164**, indicating weak performance in public services, taxation, and labour rights.

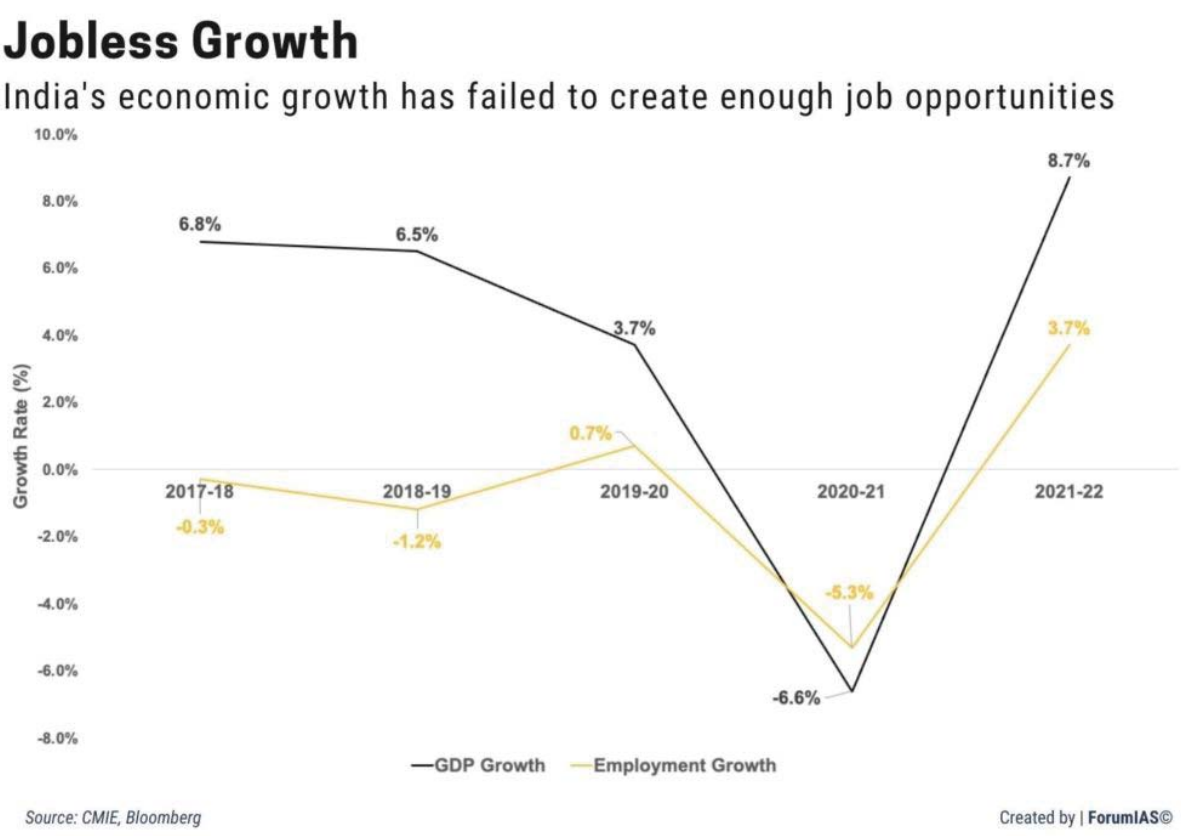
- **Social Progress Index (SPI) 2024:** India stands at **110th rank**, showing gaps in basic needs, well-being, and opportunity access.
- **Global Hunger Index (GHI) 2025:** Ranked **102/123**, still in the **"serious" category**, reflecting persistent **malnutrition and food insecurity**.

Importance of Inclusive Growth:

1. **Economic Stability and Sustainability:** Inclusive growth contributes to **economic stability** by creating a more **balanced distribution of wealth**. When more people have access to **resources and opportunities**, it can lead to a more sustainable economic development trajectory.
2. **Social Cohesion and Stability:** Addressing **disparities** fosters a more **cohesive society**. It reduces **social tensions arising from inequality**, which can lead to a more **stable and harmonious community**.
3. **Increased Human Capital Development:** When **marginalized groups** have access to **education and skill development**, it enhances the overall human capital of the nation. Educated and skilled individuals contribute more effectively to **economic growth and innovation**.
4. **Expanded Consumer Base:** Inclusive growth expands the **consumer base** by bringing more people into the **formal economy**. This leads to increased demand for goods and services, spurring **economic growth**.
5. **Enhanced Political Stability:** When people feel included in the economic and social fabric of society, it often translates to more **stable political environments**, reducing the risk of **social unrest or conflicts**.
6. **Greater Innovation and Creativity:** Inclusive growth brings together **diverse perspectives**, fostering **innovation and creativity**. A **diverse workforce and society** can generate new ideas and solutions to complex problems.

Reasons behind lack of inclusive growth in India:

1. **Historical reasons: Discrimination** against certain sections of the society since historic times has affected their access to **opportunities and to education, employment and health**. Affirmative policies have been restricted to a few beneficiaries.
2. **Gender Inequality:** Females are treated as **subordinate** to males. **Girl education** is considered a **burden on the family** and women have limited choices in employment.
3. **Large-scale informal employment: 80% of the Indian labour force** is employed in the **informal sector**. Informal sector jobs are more insecure without regular pay and social security benefits. This increases the **wage gap** between formal and informal sectors.
4. **Overdependence on Agriculture:** A huge proportion of the population is still dependent on **agriculture (~45% of the workforce)** but the share of agriculture to the total GDP is falling.
5. **Regional Disparities in Growth:** Growth has been **different across sectors and regions**. For example, **Green Revolution** has disproportionately benefitted **North-Western India** when compared to **Eastern India**.
6. **Globalisation:** Studies show that globalization and opening up the economy has benefited the **rich more than the poor**, thus **raising inequality**.
7. **Jobless Growth:** India experienced **job growth of 3% p.a in the 1970s** at a time when the economy grew at 3-3.5% p.a but over the last **3 decades** the economy grew at over 5-8% p.a but job growth has been close to 1% p.a.



8. **Government Inefficiencies:** This includes **corruption, maladministration, red tapism etc.** This prevents the optimum utilization of resources and creation of jobs.

Challenges in ensuring inclusive growth in India:

1. **Poverty:** The **Multidimensional Poverty Index 2024** places **India first**, with an estimated **23 crore people** living in **multidimensional poverty**.
2. **Uneven Distribution of Resources:** Unfair resource allocation can thwart efforts to create **inclusive growth**. Examples of unfair resource distribution include **wealth, land, and education**.
3. **Social and cultural obstacles: Marginalized groups** may encounter **social and cultural obstacles** that restrict them from accessing **opportunities and services**, such as **discrimination, exclusion, and gender inequity**. For instance, **women often face barriers** in accessing **education, employment, and other opportunities**.
4. **Weak institutional capacity:** Policies and programmes aimed at promoting **equitable growth** may be less effective if there is a lack of proper implementation, infrastructure, functionaries, etc.
5. **Rural-Urban Divide:** Disparities between urban and rural areas persist in terms of **infrastructure, access to education, healthcare, and employment opportunities**. This gap has still not been bridged.
6. **Unemployment and Underemployment:** Creating enough **jobs** for the **growing workforce**, especially in sectors that absorb a large number of people like **agriculture and manufacturing**, remains a challenge. Many are employed in the **informal sector** without **job security or proper benefits**.
7. **Education and Skills Gap:** Access to **quality education and skill development** is uneven across **regions and socio-economic groups**. This leads to a lack of **skilled workers** and limits **opportunities** for many individuals.

Global Perspective on Inclusive Growth & Lessons for India

1. **Nordic Model (Scandinavian countries):** Built on a **universal welfare system**, **high taxation**, and **high-quality public services**, supported by **active labour market policies**. This has resulted in **high HDI**, **low inequality (Gini ~0.25)**, and global leadership in inclusive development.
    a. **Lesson for India:** Prioritise **universal basic services**, **robust social safety nets**, and **trust-based governance frameworks**.
2. **Chinese Experience:** Characterised by **state-led infrastructure expansion**, **SEZ-driven export growth**, and **gradual market reforms**. It has lifted **~500 million people out of poverty**, but also led to **rising inequality (Gini ~0.47)** and **environmental degradation**.
    a. **Lesson for India:** Ensure that **growth is complemented with equity**, adopt **progressive redistribution**, and maintain **environmental sustainability**.
3. **Vietnamese Approach:** Focuses on **export-led growth**, **FDI attraction**, and **MSME support**, alongside a shift towards **high-tech manufacturing**. This has enabled **~7% GDP growth (2024)** and strong **global value chain integration**, though **skilling gaps persist**.
    a. **Lesson for India:** Promote **adaptive industrial policy**, strengthen **institutional capacity**, and invest in **human capital alongside infrastructure**.

Way Forward

1. **Education and Skill Development:** Improving access to **quality education for all**, especially in **rural and marginalized areas**, is crucial. Emphasizing **skill development programs** that align with market needs can enhance **employability**.
2. **Employment Generation:** Encouraging sectors that create more **jobs**, such as **manufacturing, services, and entrepreneurship**, can help reduce unemployment and underemployment.
3. **Social Protection Programs:** Strengthening and expanding **social safety nets** like the Mahatma Gandhi National Rural Employment Guarantee Act **(MGNREGA)**, subsidized food programs, and healthcare initiatives can provide a **safety net for vulnerable populations**.
4. **Financial Inclusion:** Expanding access to **banking and financial services**, especially in **rural areas**, can empower **individuals and small businesses**. Initiatives like **Jan Dhan Yojana** and **Microfinance Institutions** can play a **significant role** here.
5. **Gender Equality:** Promoting **gender equality** through policies that support education, employment, and entrepreneurship for women can unlock significant economic potential.
6. **Rural Development:** Focusing on **agriculture** by **modernizing techniques**, providing better **access to markets**, and improving **infrastructure** can spur **growth in rural areas**.
7. **Policy Reforms:** Implementing policies that promote **inclusive growth**, such as **progressive taxation, land reforms, labour reforms**, and policies targeting marginalized communities, can play a pivotal role.

Conclusion

A **holistic approach** that addresses these **multiple dimensions** while considering **regional disparities** and the **diverse needs of the population** will be key in moving India toward a path of inclusive growth. Collaboration between government, private sector, civil society, and international organisations is crucial in executing these strategies effectively.

2. INEQUALITY IN INDIA

Meaning:

Inequality refers to the **uneven distribution of income, wealth, power, and opportunities** among individuals and groups. The **United Nations** defines it as 'the state of not being equal, especially in status, rights, and opportunities.'

Dimensions of Inequality:

Economic Inequality

- **90/10 Income Ratio** rose from **6.7 ▯ 6.9 (2017-18 to 2022-23)**, showing the top 10% earners are widening their gap over the bottom 10%.
- **Self-employed workers** show the fastest rising inequality within occupations, reflecting uneven income growth in informal sectors.
- **Hidden hunger persists:** India ranked **102/185 in per capita calorie intake** (UN FAO 2019), indicating malnutrition despite economic growth.

Caste-Based Inequality

- **SC Consumption Share:** Only **7%** of total consumption vs **16–17% population** (PLFS 2022-23), highlighting under-consumption.
- **ST Poverty Share: 45.9%** of STs and **26.6%** of SCs in lowest wealth quintile (NFHS-4).

- **Upper-caste households** earn ~**47% more than the national average**, reflecting historical resource advantages (WID).
- **SC/ST underrepresentation:** Limited access to **formal, high-paying jobs** perpetuates structural disadvantage.
- **OBCs:** Consumption share (~41%) aligns with population, but **inequality within OBCs** remains high.

Gender Inequality

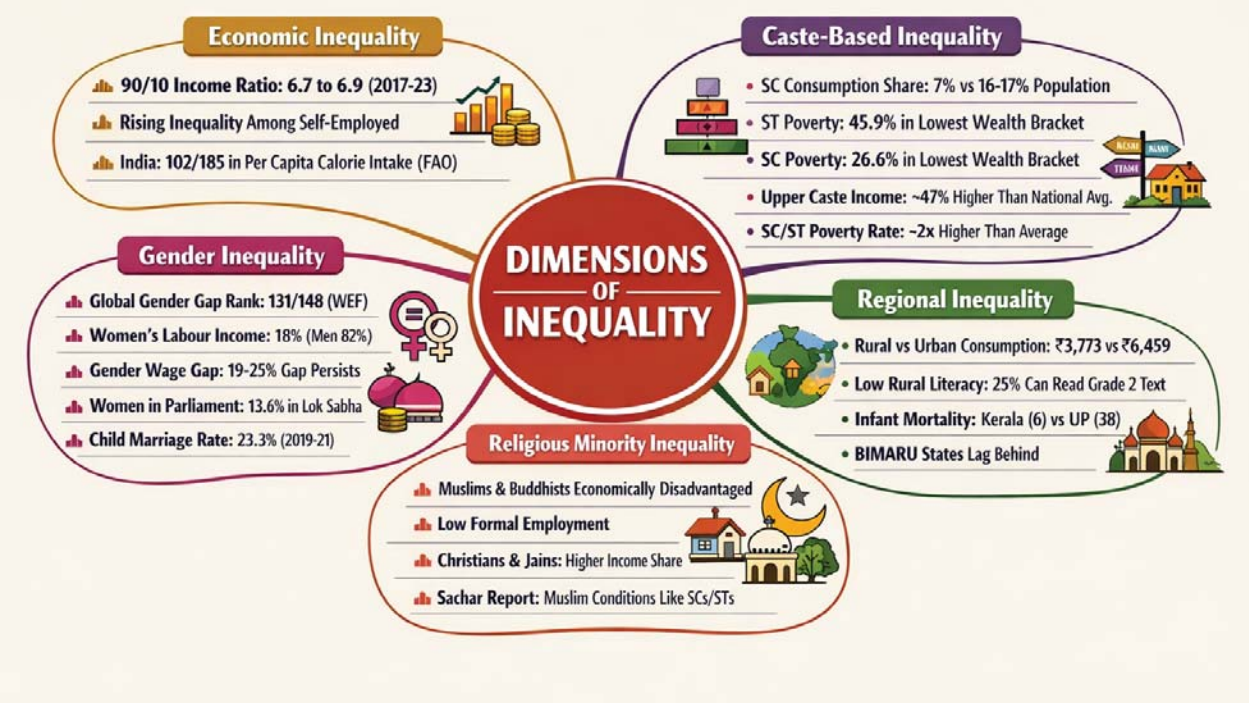
- **Global Gender Gap 2025:** India ranked **131/148**, with parity score **64.1%** (WEF).
- **Gender Inequality Index 2022:** Ranked **108/193**, score **0.437** (UNDP HDR).
- **Women's labour income:** Only **18%**, while men earn 82%, indicating a **wide gender income gap**.
- **Female LFPR:** Rose from **49.8%** → **60.1% (2017-18 to 2023-24)**, but participation remains below potential.
- **Gender wage gap** persists at **19–25%**; political representation remains low (**13.6% MPs, 5.56% ministers**).
- **Child marriage** reduced from **47% (2006)** to **23.3% (2019-21)** (NFHS-5), improving girls' social outcomes.

Regional Inequality

- **Rural-Urban Divide:** Monthly per capita consumption — **Rural Rs. 3,773**, Urban Rs. 6,459 (HCES 2022-23).
- **Rural incomes ~40% lower** than urban due to low diversification, low agri returns, and poor education — only **25% grade-3 can read grade-2 text** (ASER 2023).
- **Inter-state disparities:** IMR — Kerala **6** vs UP **38** per 1,000 live births; health and education outcomes vary sharply.
- **Southern & Western states** outperform Eastern and Central states in **income, education, and health**, while BIMARU states lag.
- **Digital Divide:** Urban areas have higher **internet penetration and access to digital financial services**.

Religious Minority Inequality

- **Muslims & Buddhists** have lower access to **income, employment, and resources** relative to their population share.
- **Christians, Parsis, and Jains** enjoy higher income shares than population size indicates.
- **Muslims underrepresented** in **formal, high-paying employment** (PLFS data).
- **Sachar Committee Report 2006:** Socio-economic conditions of Muslims **comparable to S**Cs/STs in education, employment, and assets.



Key Measurement Tools:

Gini Coefficient	0 = perfect equality; 1 = perfect inequality. Most used measure globally.
Consumption Gini (India)	Fell from 0.288 (2011-12) to 0.255 (2022-23) — World Bank claim
Income Gini (India)	0.61 as per WID 2023 — among highest globally
Wealth Gini (India)	~0.75-0.83, indicating extreme wealth concentration
P90/P10 Ratio	Income of top 10% vs bottom 10%; rose from 6.7 (2017-18) to 6.9 (2022-23)
Lorenz Curve	Visual representation of income distribution; flatter = more equal
Human Development Index	Combines income, health, education; India ranks 134/193 (HDR 2023-24)
Multidimensional Poverty Index	Tracks deprivation across 10 indicators in 3 dimensions

Current State Of Inequality In India — Key Data

Income Inequality:

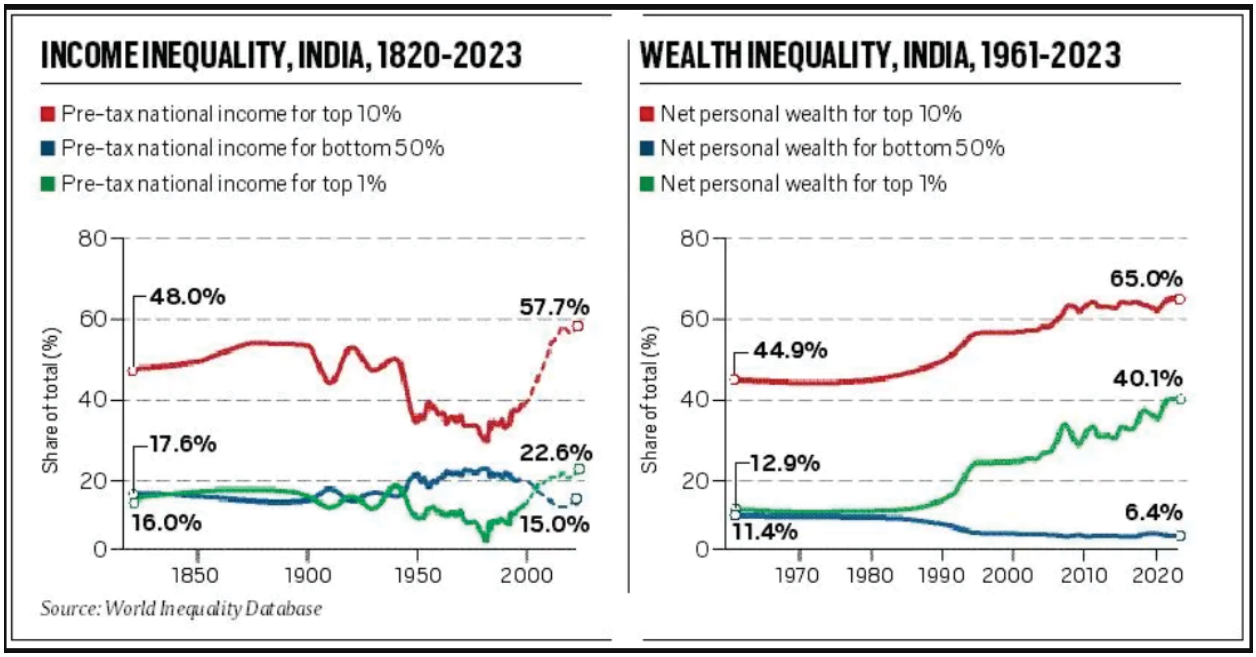
Top 1% Income Share	22.6% of national income (WID 2022-23) — higher than even the USA
Bottom 50% Income Share	Only 15% of national income (WID)
Income-based Gini	Rose from 0.52 in 2004 to 0.62 in 2023 — among most unequal globally
India's Rank	176 out of 216 countries in income equality (2019)
Top 10% vs Bottom 10%	Median earnings of top 10% were 13x higher than bottom 10% (2023-24)
CEO Pay Gap	CEO pay jumped 50% (2019-24); worker wages grew less than 1% (ILO)

Wealth Inequality:

Top 1% Wealth Share	Owns 40.1% of total national wealth (WID/Oxfam 2024)
Top 10% Wealth Share	Holds 77% of total national wealth (Oxfam)
Bottom 50% Wealth Share	Only 6.4% of national wealth (WID)
Wealth Gini Coefficient	~0.75-0.83, one of highest globally
Labour Income Share	Declined from 32% (1990s) to ~22% in 2024 (ILO)
Capital Income Concentration	56% of countries saw rise in capital income share since 1990 (Global Inequality Report 2025)

Consumption Inequality (Government/World Bank Claim)

Consumption Gini	Fell from 0.288 (2011-12) to 0.255 (2022-23) — claimed 4th most equal
Extreme Poverty Rate	Fell from 16.2% to 2.3% (2011-23) using $2.15/day threshold
People Lifted Out of Poverty	~17.1 crore over past decade (World Bank Spring 2025 Brief)
Rural Monthly Per Capita Exp.	Rs. 3,773 vs Urban Rs. 6,459 (HCES 2022-23)



Causes of Inequality in India:

A. Historical & Structural Causes

- **Colonial legacy:** Land was concentrated in **zamindars** and artisans were deindustrialised, creating wide **class divides pre-Independence**.
- **Inadequate land reforms: Ceiling laws poorly enforced**, leaving large landholdings intact; **landlessness** fuels rural poverty among **SCs/STs**.
- **Caste system:** Historically barred certain groups from **education, jobs, and resource access**, perpetuating social hierarchies.
- **Informal economy:** ~**90% of the workforce** lacks social security, working in low-wage, insecure **informal sector** jobs.

B. Economic & Policy Causes

- **Skewed sectoral benefits:** Reforms in **IT, telecom, and aviation** mostly benefited urban elites, neglecting **agriculture and small industries**.
- **Regressive taxation:** Heavy reliance on **indirect taxes (GST)** burdens the poor; **tax evasion by the wealthy** worsens income inequality.
- **Wealth concentration:** Absence of **inheritance and wealth taxes** allows accumulation of wealth across generations in select groups.
- **Crony capitalism & financialisation:** Policy favoritism enriches elites, while emphasis on **financial markets** reduces investments in productive sectors.
- **Weak labour protections:** Limited **minimum wage enforcement** and lack of social security, especially in **gig economy** jobs, worsen inequality.

C. Social Causes

- **Limited education access:** Only **25% of rural grade-3 children** can read grade-2 texts (**ASER 2023**); quality education is mostly urban- and elite-focused.
- **Healthcare gaps:** Poor healthcare access traps low-income groups in **cycles of poverty, illness, and disability**.
- **Gender discrimination:** Social norms and policies restrict **women's workforce participation and earning potential**.
- **Caste & religious discrimination:** Marginalised communities face barriers to **employment, credit, and basic resources**.

D. Political & Governance Causes

- **Elite capture:** Top **1% hold 40% of wealth**, influencing policymaking to preserve economic advantages.
- **Corruption & leakages:** Welfare schemes often fail due to **diversion of funds and bureaucratic inefficiencies**.
- **Inadequate social safety nets:** Programs like **MGNREGA** remain **underfunded and insufficient** relative to demand.
- **Lack of wealth census:** Absence of detailed wealth data makes **targeted redistribution and policy design** difficult.

Amartya Sen's 7 Types of Gender Inequality:

- **Mortality Inequality**: Higher female infant mortality; women receive less healthcare
- **Natality Inequality:** Son preference; sex-selective abortions despite PCPNDT Act
- **Basic Facility Inequality:** Girls deprived of education, nutrition, medical attention
- **Special Opportunity Inequality:** Restricted access to higher education and professional careers
- **Professional Inequality:** Glass ceiling, occupational segregation
- **Ownership Inequality:** Women hold far less property, land, and financial assets
- **Household Inequality:** Unequal power within family; domestic violence, unpaid care work

Government Measures & Schemes:

A. Financial Inclusion

- **PM Jan Dhan Yojana (PMJDY):** Over **55.69 crore accounts** opened (June 2025), providing **universal banking access** and enabling **Direct Benefit Transfers (DBT)**.
- **Direct Benefit Transfer (DBT):** Reduced leakages, saved **₹3.48 lakh crore** (March 2023), and ensures subsidies reach intended beneficiaries directly.
- **Aadhaar:** Over **142 crore cards issued**, forming the **backbone of welfare delivery** and enabling digital authentication.

- **PM Mudra Yojana:** Provides **collateral-free loans** for micro-enterprises, boosting **self-employment and entrepreneurship**.

B. Social Protection & Welfare

- **PM Garib Kalyan Anna Yojana (PMGKAY):** Free food grains for **80.67 crore beneficiaries** (Dec 2024), ensuring **food security for vulnerable populations**.
- **MGNREGA:** Guarantees **100 days of rural employment**, providing **income support to rural households** during lean periods.
- **Ayushman Bharat (PM-JAY):** Offers **₹5 lakh per family per year** for secondary and tertiary healthcare, reducing **out-of-pocket expenses**.
- **PM Awas Yojana (PMAY):** Provides **housing for low-income urban and rural families**, supporting **dignified living conditions**.
- **PM KISAN:** Direct **₹6,000/year income support** to small and marginal farmers, improving **rural livelihoods**.

C. Skill & Employment

- **Skill India Mission:** Trains and upskills workforce to meet **market demands**, enhancing **employability and productivity**.
- **PM Vishwakarma Yojana:** Supports **traditional artisans and craftspeople**, preserving **heritage skills and livelihoods**.
- **Stand-Up India:** Provides **loans from ₹10 lakh to ₹1 crore** to **SC/ST and women entrepreneurs**, with **2.75 lakh+ loans sanctioned** till July 2025.

D. Education & Empowerment

- **National Education Policy (NEP) 2020:** Promotes **equitable access, foundational literacy**, and **flexible educational pathways**.
- **Beti Bachao Beti Padhao:** Addresses **skewed sex-ratio**, improves **girls' education**, and strengthens **gender equality**.
- **SC/ST Scholarship Schemes:** Pre- and post-matric scholarships ensure **educational inclusion** for **marginalised students**.

E. Regional Development

- **PM Gati Shakti:** Integrated infrastructure planning reduces **regional disparities** and promotes **economic convergence**.
- **PURA (Providing Urban Amenities in Rural Areas):** Bridges **rural-urban gaps** in infrastructure, services, and economic opportunities.
- **Aspirational Districts Programme:** Focuses on **112 backward districts**, promoting **convergent development and social upliftment**.

Challenges & Limitations

A. Data & Measurement Challenges

- **Consumption vs Income Inequality:** Government's claim of being 4th most equal nation is based on **consumption Gini**, hiding **true income and wealth disparities**.
- **Informal economy:** ~**90% workforce** in informal sector makes accurate **measurement of income and wealth** very difficult.
- **No wealth census:** Reliance on **Forbes lists, SEBI filings**, and property prices **underestimates top-end wealth concentration**.
- **Income vs consumption Gini: Income Gini 0.61 (WID)** shows **much higher inequality** than **consumption Gini 0.255**, highlighting disparity.

B. Policy Implementation Challenges

- **Leakages:** Despite **DBT**, ghost beneficiaries and **identification errors** continue to divert welfare funds.
- **Exclusion errors:** Many genuinely poor households are **left out of social benefit schemes** due to procedural or data issues.
- **Poor public service quality:** Even when schemes exist, **schools and hospitals** often fail to provide adequate services.
- **Targeting failures:** Universal schemes reach non-poor, while **means-tested schemes** face high **exclusion errors**.
- **Resistance to structural reforms: Vested interests** block **land reforms, progressive taxation**, and labour policy changes.

C. Structural Limitations of Growth

- **Jobless growth:** High **GDP growth** has not translated into **sufficient quality employment** for the workforce.
- **K-shaped recovery post-COVID: Upper-income segments** recovered quickly, while **lower-income groups** remain stagnant.
- **Agricultural stagnation:** Over **45% of workforce** in agriculture earns low incomes due to **poor productivity and outdated practices**.

Way Forward & Policy Recommendations

A. Fiscal & Taxation Reforms

- **Reintroduce Wealth and Inheritance Tax:** Levy **1% wealth tax on ultra-rich** to generate funds for major **social schemes and poverty reduction**.
- **Progressive Taxation:** Ensure **top earners pay proportionate taxes**, while closing loopholes that enable **tax evasion**.
- **Reduce Indirect Tax Burden:** Rationalise **GST on essential goods** and shift **tax burden toward direct taxation** to reduce regressivity.
- **Integrate Tax, Survey & Asset Records:** Combine **tax, survey, and asset databases** to publish **accurate inequality metrics** beyond consumption data.

B. Labour Market & Employment

- **Strengthen Collective Bargaining:** Reverse the decline in **labour income share** (32% in 1990s → 22% in 2024) to improve **worker bargaining power**.
- **Formalise Gig Economy:** Provide **social security, minimum wages, and health coverage** for gig and platform workers to reduce precarity.
- **Job-Rich Growth:** Prioritise **labour-intensive sectors** like textiles, construction, and food processing to generate **quality employment**.
- **Raise Minimum Wages:** Align **minimum wages with living standards**, especially for informal sector and rural workers.

C. Social Sector Investment

- **Increase Health Budget:** Expand **universal health coverage** and strengthen **public hospitals and primary care**.
- **Education Equity:** Implement **NEP 2020 in mission mode**, addressing **foundational literacy gaps** revealed by ASER data.
- **Universal Basic Services:** Guarantee **quality education, healthcare, nutrition, and housing**, making them **rights-based entitlements**.

D. Targeted Inclusion Programmes

- **Expand MGNREGA:** Increase **workdays, wages, and coverage**, integrating **skill development opportunities** for rural workers.
- **Land Redistribution:** Strictly enforce **land ceiling laws** and provide **formal titles to the landless**, reducing rural inequality.
- **Deepen Financial Inclusion:** Ensure **Jan Dhan accounts and other financial tools** provide real access to **credit, insurance, and savings**.
- **Women's Economic Empowerment:** Enforce **Equal Remuneration Act**, expand **creche/childcare facilities**, and improve **female labour force participation**.

E. Data & Governance

- **Improve NSSO/PLFS Methodology:** Capture **informal sector and high-income data** to ensure **accurate representation of inequality**.
- **Post-Tax Inequality Measurement:** Include **post-tax and post-transfer effects** in reporting to reflect **true economic disparities**.
- **Wealth Census:** Conduct a **systematic national wealth census** for policy targeting and **redistribution planning**.
- **Aspirational Districts Model:** Scale **convergent development programmes** to all backward districts to **reduce regional inequality**.

Conclusion

Reducing inequalities aligns with **SDG 10** of the 2030 Agenda. India's **constitutional promise** as a socialist, secular, and democratic republic demands growth that is not only high but just. As "Inequality is not just an economic gap; it is a governance fault line." **Sustainable, equitable development** is key to realizing **Viksit Bharat 2047**.

3. GENERATIONAL JUSTICE IN GROWTH

"We do not inherit the earth from our ancestors; we borrow it from our children." – Native American Proverb

Meaning:

Intergenerational justice refers to the **concept of fairness and equity between different generations.** It posits that the current generation has a **moral obligation** to ensure that future generations inherit a planet and a society that is no worse off than the one we inhabit today.

Theoretical Framework:

- **John Rawls' "Just Savings" Principle:** Rawls argued that while we do not know which generation we will belong to (Veil of Ignorance), rational people would agree to a "just savings" principle—saving resources for successors to maintain just institutions over time.
- **Edith Brown Weiss's "Planetary Trust":** She proposed that the human species holds the earth in trust for future generations. We are not owners, but trustees. This implies three duties:
  - **Conservation Options:** Keep options open for the future.
  - **Conservation Quality:** Maintain the quality of the planet.
  - **Conservation Access:** Ensure equitable access to resources.
- **Indian Philosophy: The concept of "Vasudhaiva Kutumbakam" (the world is one family)** inherently extends moral consideration to future members of this global family.

Dimensions of Intergenerational Justice:

A. Environmental Dimension (Ecological Debt)

- **Resource Depletion:** Over-extraction of non-renewable resources (coal, oil, groundwater) deprives future generations of their utility.
  - *Example:* India's groundwater crisis; 21 Indian cities are expected to run out of groundwater by 2030 (NITI Aayog).
- **Climate Change:** The **"Carbon Budget" is finite.** Current industrialization consumes the carbon budget, leaving future generations with a hostile climate (extreme weather, rising sea levels).
- **Biodiversity Loss:** Extinction of species is irreversible. Future generations lose potential medicinal resources and ecosystem services.

B. Economic Dimension (Fiscal Sustainability)

- **Public Debt:** High fiscal deficits and sovereign debt borrowed today must be repaid by future taxpayers who had no say in the borrowing decisions. This is **often termed "fiscal child abuse."**
- **Infrastructure:** A balance is needed. While borrowing for infrastructure (roads, renewable energy) benefits future generations, borrowing for current consumption (subsidies, salaries) burdens them.
- **Pension Liabilities:** With aging populations (demographic dividend turning into a burden), the social security costs of the older generation will fall on a smaller working-age youth population.

C. Social Dimension (Human Capital & Demographics)

- **Education and Health:** Under-investment in education (human capital) today results in a **less skilled workforce** for the future.
- **Demographic Imbalance:** In countries with aging populations (like Japan, and eventually India), the burden of care falls on the youth, leading to "intergenerational inequity."

Indian Context & Constitutional Provisions:

Constitutional Mandate: While the term "intergenerational equity" isn't explicit, the spirit is embedded in the Constitution:

- **Article 48A (Directive Principles):** State shall endeavor to protect and improve the environment and to safeguard the forests and wildlife of the country.
- **Article 51A(g) (Fundamental Duties):** It shall be the duty of every citizen to protect and improve the natural environment.
- **Article 21 (Right to Life):** The Supreme Court has interpreted the Right to Life to include the right to a healthy environment (Virender Gaur v. State of Haryana).

Judicial Interpretation:

- **Virender Gaur Case (1995):** The Supreme Court held that the State is the trustee of all natural resources, which are by nature meant for public use and enjoyment. The State cannot convert them into private ownership.
- **MC Mehta v. Kamal Nath (2000):** Established the **"Public Trust Doctrine,"** stating that resources like air, sea, waters, and forests are held in trust by the government for the general public and future generations.

Significance of Intergenerational Justice:

1. **Sustainability of Resources (Survival Issue):** It acts as a check against the **"slash-and-burn" economic model.** Without intergenerational justice, we face irreversible "tipping points" (e.g., exceeding the 1.5°C warming limit). It ensures the **continuity of the human race** by preserving the "life-support systems" (air, water, soil).

2. **Economic Stability and Fiscal Prudence:** It is significant for **macroeconomic stability**. If the current generation burdens the future with excessive sovereign debt (fiscal profligacy) or depleted capital, it leads to intergenerational poverty traps. It **justifies policies like the FRBM Act** (Fiscal Responsibility and Budget Management) in India.
3. **Ethical Governance (Trusteeship):** It elevates **governance from "populist" to "visionary."** It reminds policymakers that they are "trustees" of the planet, not owners. This aligns with the Gandhian principle of "Antyodaya" and "Trusteeship," serving as a moral compass for state action.
4. **Social Cohesion:** It bridges the gap between the "Baby Boomer" generation (often seen as asset-rich) and "Gen Z/Millennials" (facing high unemployment and climate anxiety). Addressing this justice **reduces social friction and conflict over resources.**
5. **Legal and Constitutional Interpretation:** It provides a tool for the Judiciary to **interpret the Right to Life (Article 21) expansively.** It allows courts to strike down projects that cause irreversible environmental damage, establishing the Precautionary Principle and Polluter Pays Principle.

Mechanisms for Ensuring Intergenerational Justice

A. Global Mechanisms:

- **Sustainable Development Goals (SDGs):** A blueprint for shared prosperity and peace for people and the planet, now and into the future.
- **Paris Agreement (2015):** Focuses on **"Common but Differentiated Responsibilities"** to limit global warming, acknowledging that developed nations have consumed the carbon budget of developing nations.
- **Intergenerational Ombudsperson:** Countries like Wales have established a **"Future Generations Commissioner"** to scrutinize laws for their long-term impact. Hungary has an Ombudsman for Future Generations.

B. Indian Policy Initiatives:

- **National Action Plan on Climate Change (NAPCC):** Focuses on sustainable lifestyles.
- **Green National Accounting:** Moving beyond GDP to include environmental costs in national accounts (Green GDP).
- **National Monetization Pipeline:** Monetizing brownfield assets to fund greenfield infrastructure, transferring the benefit of assets to the next generation while ensuring they are not sold off permanently.
- **Compensatory Afforestation Fund Act (CAMPA):** Ensures that when forest land is diverted for industry, money is collected to create forests elsewhere (though implementation is debated).

Challenges to Intergenerational Justice:

1. **Temporal Tragedy (Short-termism): Democratic electoral cycles (5 years)** incentivize politicians to focus on immediate gains (freebies, subsidies) rather than long-term gains (reforestation, fiscal consolidation).
2. **Uncertainty of Future Needs:** We assume future generations will value the same things we do (e.g., fossil fuels vs. green energy transition).
3. **Lack of Representation:** Future generations cannot vote or lobby; they are voiceless in current policy-making.
4. **Technological Optimism:** The belief that **"future technology will fix the mess"** often serves as an excuse for inaction today.

Way Forward:

1. **Shift from "Growth" to "Green Growth":** Economic models must internalize environmental costs. The transition to a **Circular Economy is essential** to reduce waste for future generations.
2. **Fiscal Responsibility:** Strict adherence to Fiscal Responsibility and Budget Management (FRBM) targets to ensure debt sustainability.

3. **Institutionalizing the Voice of the Future:** India could consider establishing a **"Commission for Future Generations" within NITI Aayog** to audit policies for their long-term impact (a recommendation often made by environmental economists).
4. **Investing in Human Capital:** The best legacy we can leave is an educated, healthy, and skilled population (Demographic Dividend).
5. **Intragenerational Equity as a Pre-condition:** We cannot achieve justice between generations if there is massive inequality within the current generation. The poor are the most vulnerable to environmental degradation.

Conclusion:

The path to justice lies in **institutionalizing long-term thinking.** As the **economist Wilfred Beckerman argued,** the best gift we can give to future generations is not a specific set of resources, but the capacity to adapt and innovate. Therefore, **India's development trajectory must harmonize "Sabka Saath, Sabka Vikas" with "Vasudhaiva Kutumbakam",** ensuring that the pursuit of prosperity today does not become the precursor to poverty tomorrow.

4. INCLUSIVE GROWTH UNDER MARKET ECONOMY

Context/Introduction:

Inclusive growth means economic growth that **creates opportunities for everyone** and ensures that the **gains of prosperity are fairly shared** across society. In a market economy, where private ownership, supply-demand forces, and profit motives drive activity, wealth is often created efficiently but not always distributed equitably. While the **market idea—"a rising tide lifts all boats"**—assumes that growth benefits all, in reality, it **often produces a "K-shaped" pattern:** the **rich grow wealthier**, while the poor see **little or no improvement** in their living standards.

The Negative Side: How Market Economy Hinders Inclusion

1. **Concentration of Wealth:** Markets tend to reward capital more than labor, leading to extreme inequality. The **Oxfam 2023 report** revealed that the top 1% of Indians own more than 40% of the nation's wealth, highlighting this skewed distribution.
2. **Regional Disparity:** Profit-seeking capital flows to developed regions, leaving others behind. This creates **"islands of prosperity,"** where **states like Maharashtra and Tamil Nadu surge ahead,** while states like Bihar and Odisha struggle to attract comparable private investment.
3. **Exploitative Labor Practices:** The race for efficiency often leads to the **"informalization" of labor to cut costs.** Gig economy workers (like delivery agents for Zomato/Swiggy) often lack job security, health benefits, or minimum wage guarantees.
4. **Neglect of Unprofitable Areas:** Markets fail to provide services where **purchasing power is low**. Private hospitals are heavily concentrated in Tier-1 cities, leaving rural populations dependent on an overburdened public health system.
5. **Information Asymmetry:** Producers with less market information are often exploited by intermediaries. **Small onion farmers in Maharashtra frequently sell at distress prices** to traders due to a lack of real-time data on market rates.
6. **Negative Externalities:** Profit motives frequently **ignore environmental and social costs**, which disproportionately hurt the poor. Industrial pollution in the Yamuna river destroys the livelihoods of fishermen and farmers who depend on its water.

Why the Market Fails Inclusion?

Market Failures:

- **Public Goods:** Markets **under-provide public goods** like primary education, healthcare, and sanitation because they are not profitable. The poor suffer most due to **lack of purchasing power.**
- **Information Asymmetry:** Poor farmers or informal workers often **lack access to market information (prices, weather, rights)**, leading to exploitation.
- **Externalities:** Industrial growth creates **negative externalities (pollution)** affecting the poor disproportionately, while **positive externalities (education) are undervalued by the market.**

The "Piketty" Argument (r>g):

- **Thomas Piketty (in Capital in the Twenty-First Century)** argues that in a market economy, the return on capital (r) is historically greater than the rate of economic growth (g).
- **Implication: Wealth concentrates at the top;** inherited wealth grows faster than earned wages, leading to structural inequality.

Skill-Biased Technological Change (SBTC):

- **Modern markets reward high-skilled labor (tech, finance)** while automating or stagnating wages for low-skilled labor, widening the wage gap.

The Positive Side: How Market Economy Fosters Inclusion

1. **Democratization of Consumption**: Competition drives firms to target the "Bottom of the Pyramid," making goods affordable. **For instance, the Indian telecom revolution** reduced data costs to rock-bottom prices, digitally empowering millions of rural citizens.
2. **Breaking Social Hierarchies:** Markets prioritize merit over caste or lineage, enabling upward mobility. **Platforms like Amazon Karigar allow tribal artisans** to sell directly to global buyers, bypassing discriminatory local middlemen.
3. **Scale of Employment Generation:** The private sector creates jobs at a scale the state cannot match. The **IT and Services boom in cities like Bengaluru and Hyderabad** lifted millions of lower-middle-class families out of poverty through high-value employment.
4. **Asset Creation for the Masses:** Financial markets allow ordinary citizens to build wealth beyond land ownership. The **rise of Systematic Investment Plans (SIPs)** has enabled small Indian savers to participate in the country's capital growth story.
5. **Integration with Global Value Chains:** Markets connect local producers to global buyers, increasing income potential. **Grape farmers in Maharashtra** exporting to Europe earn significantly higher margins than they would in local *mandis*.
6. **Innovation for Affordability**: Profit motives spur "frugal innovation" for cost-sensitive consumers. The **Jaipur Foot and low-cost solar pumps** are prime examples of market-driven solutions specifically designed for the needs of the poor.

Case Studies & Data Points:

Aspect	Market Failure	Government Intervention (Inclusive Step)	Impact/Data
Banking	Banks ignored rural poor (unprofitable).	PM Jan Dhan Yojana.	Over 50 crore accounts opened; women account holders increased significantly.

Health	Private hospitals unaffordable.	Ayushman Bharat (PM-JAY).	World's largest health insurance scheme; merging market efficiency (private hospitals) with public funding.
Entrepreneurship	MSMEs lack collateral for loans.	Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE).	Collateral-free loans enabled small businesses to grow.
Digital	Digital Divide (Urban vs Rural).	Digital India / BharatNet.	Rural internet penetration rising; enabled e-learning and telemedicine.

The Role of the State: Making Markets Inclusive

In a market economy, the State must play the role of an Enabler and Regulator.

A. Interventions for "Broad-basing" Growth (Supply Side)

Human Capital Formation:

- **Rationale:** To participate in the market, one needs skills.
- **Examples:** Samagra Shiksha Abhiyan, Skill India Mission, Ayushman Bharat (Health). A healthy, educated workforce is a prerequisite for inclusive growth.

Infrastructure Development:

- Rationale: Connects remote areas to markets.
- Examples: PM Gati Shakti, BharatNet, Ujjwala Yojana (Energy access).

B. Interventions for Redistribution (Demand Side)

- **Direct Benefit Transfers (DBT):** Uses the "JAM Trinity" (Jan Dhan, Aadhaar, Mobile) to plug leakages and ensure resources reach the marginalized. This is a market-friendly welfare model (efficient subsidy delivery).
- **Progressive Taxation:** Taxing the "super-rich" to fund social welfare schemes. This corrects the r>g anomaly.
- **Reservation/Affirmative Action:** Mandating representation in jobs and education to correct historical market exclusion.

C. Interventions for Market Access

- **Financial Inclusion:**
  - **PM Jan Dhan Yojana:** Zero-balance accounts brought the unbanked into the formal financial market.
  - **MUDRA Yojana:** Providing credit to micro-enterprises (non-corporate small business sector).
- **Farmer Market Reforms:**
  - **e-NAM** (Electronic National Agriculture Market) aims to remove middlemen, ensuring farmers get market prices.

Way Forward: The "Samaveshi Vikas" Model

- **From "Trickle Down" to "Bottom Up":** Focus investment in **rural infrastructure and agriculture**, which has a higher multiplier effect on the economy.
- **Skill-Intensive Manufacturing:** Promote **labor-intensive sectors (textiles, tourism, construction)** rather than just capital-intensive sectors to absorb the demographic dividend.
- **Formalization of Economy:** Use technology to bring informal workers under social security nets

- **Stakeholder Capitalism: Encouraging Corporate Social Responsibility (CSR)** not just as charity, but as a business model where companies consider the welfare of employees, suppliers, and the environment, not just shareholders.
- **SDG Alignment:** Align India's inclusive growth **goals with Sustainable Development Goals**, particularly Goal 1 (No Poverty), Goal 5 (Gender Equality), and Goal 8 (Decent Work).

Conclusion:

The **dichotomy** between "Market Economy" and "Inclusive Growth" is a **false one**. The **market is a tool**, and like any tool, its impact depends on how it is wielded. We need a **market economy with a "human face."** This implies a model where the **State acts as a Facilitator** (building roads, digital infra, and law & order) **and a Regulator** (checking monopolies and ensuring fair play), while the **Market acts as the Engine of efficiency** and innovation.

5. REGIONAL DISPARITIES IN DEVELOPMENT

Context:

India has witnessed rapid economic growth over the past decades, emerging as one of the world's largest economies. However, this growth has **not been uniform across regions**, giving rise to significant disparities in income, development, and opportunities. While states like **Delhi, Karnataka, Telangana, and Maharashtra** have achieved high per capita income and advanced infrastructure, many central, eastern, and north-eastern states continue to lag behind.

Meaning:

Regional disparity refers to the **uneven distribution of economic, social, and infrastructural development** across different states or regions. Recently, a working paper **"Relative Economic Performance of Indian States: 1960-61 to 2023-24" by the Economic Advisory Council** to the Prime Minister (EAC-PM) highlights the uneven growth across Indian states.

Key Trends and Patterns

Income and GDP Contribution

- **High-income states:** Delhi, Telangana, Karnataka, Haryana – PCI well above national average (Delhi at 250.8%).
- **Declining states:** West Bengal dropped from 27% above national average in 1960–61 to 83.7% now.
- **Emerging states:** Odisha improved from 55.8% (2000–01) to 88.5% of national average.
- **Southern states dominate GDP contribution:** Karnataka, Andhra Pradesh, Telangana, Kerala, Tamil Nadu >30% of national GDP (2023–24).
- **Maritime states** generally outperform landlocked states, except West Bengal.
- **Divergence example:** Punjab vs Haryana – Punjab's PCI stagnated (agriculture-led, possible 'Dutch disease'), Haryana industrialized, PCI rose.

Multidimensional Deprivation

- Central and North-Eastern regions have **highest deprivation** (housing, education, employment, healthcare).
- Least deprived regions: Kerala, Goa, Punjab, Haryana.

- Deprivation is **multidimensional**, not just income-based – includes social, material, and infrastructural deficits.
- Use of **Composite Deprivation Index (CDI)** highlights spatial inequality across 28 indicators.

Industrial Development:

- Uneven industrial growth – prosperous urban hubs attract more investment.
- **Underdeveloped regions (North-East, central India)** face poor connectivity, weak banking, and power shortages.
- **Location-specific developm**ent benefits regions near capitals (Delhi NCR: Gurugram, Noida).

Causes of Regional Disparity:

1. **Historical Factors:** Colonial policies concentrated economic and administrative resources in cities like **Kolkata, Mumbai, and Chennai**, leaving hinterlands underdeveloped, a pattern that continues to influence current disparities.
2. **Geographical Constraints:** Difficult terrains in **Himalayan and North-Eastern states**, and flood-prone areas like **Bihar and Assam**, raise project and administrative costs, slowing development.
3. **Economic Structure:** States dependent on **services and manufacturing** (e.g., Maharashtra, Gujarat, Karnataka) enjoy higher income, whereas **agriculture-focused states** (Bihar, Odisha) lag.
4. **Policy Disparities:** Programs like the **Green Revolution** benefited select states such as **Punjab and Haryana**, exacerbating regional gaps.
5. **Governance & Political Instability:** Weak institutions, law-and-order challenges, and unstable governments discourage investment in lagging regions.
6. **Infrastructure Deficit:** Poor transport, power, banking, and digital connectivity limit growth, especially in **North-Eastern and central India**.
7. **Location-Specific Effects:** Industrial hubs cluster around metro regions, e.g., **Delhi NCR, Gurugram, Bengaluru**, concentrating growth while leaving remote areas behind.
8. **Resource Allocation Inefficiency:** Underinvestment in social and physical infrastructure perpetuates deprivation in central and north-eastern states.

Consequences of Regional Disparity:

1. **Economic Impacts:** Slower national growth due to underutilization of resources; income gaps remain stark, e.g., **Karnataka PCI Rs 2,04,605 vs Madhya Pradesh Rs 70,434**.
2. **Social Inequalities:** Marginalized communities—including **SC/ST/OBCs, women, and youth**—face limited access to education, health, and employment.
3. **Political Fragmentation:** Neglect fuels demands for new states, e.g., **Telangana, Bodoland, Vidarbha**, reflecting regional discontent.
4. **Security Challenges:** Neglected regions witness insurgency or Left-Wing extremism, particularly in **Central and North-East India**.
5. **Environmental & Urban Stress:** Concentrated industrial growth causes pollution and overburdened cities, e.g., **Delhi, Mumbai, Chennai**.
6. **Unequal Human Development:** Differences in **HDI, literacy, and healthcare** remain significant, with Kerala and Goa at the top, Bihar and Chhattisgarh at the bottom.

Indicators of Regional Disparity:

1. **Per Capita Income (PCI):** Reflects the average income of individuals in a state. Higher PCI indicates prosperity, while lower PCI highlights income deficits. For example, **Delhi's PCI is 2.5 times the national average**, Odisha is at **88.5%**, and Bihar lags at **less than 50%**.

2. **Composite Deprivation Index (CDI):** Captures multidimensional deprivation across **housing, basic amenities, social, and material aspects**. States like **Odisha, Chhattisgarh, and Jharkhand** score highest, indicating severe deficits in living standards and access to essential services.
3. **Industrial Distribution:** Shows how industrial development is concentrated in select regions. **Southern and maritime states** dominate industrial output, while **central and north-eastern states** remain under-industrialized, limiting employment and income opportunities.
4. **Human Development Index (HDI):** Measures overall well-being through **health, education, and living standards**. Kerala exhibits the highest HDI, reflecting near-universal literacy and healthcare access, whereas Bihar has the lowest, showing gaps in basic services.
5. **Infrastructure & Connectivity:** Assesses availability of **roads, electricity, banking, and telecom services**, which directly influence investment and growth. Poor connectivity in **north-eastern states and central India** hampers industrial and socio-economic progress.
6. **Social Indicators:** Include **literacy rates, birth registration, access to media, and healthcare facilities**, highlighting social inequalities. For instance, adult illiteracy remains high in **Chhattisgarh and Jharkhand**, while Kerala and Goa show minimal social deprivation.

Government Initiatives & Strategies:

1. **Aspirational Districts Programme (ADP):** Targets 112 underdeveloped districts for **rapid improvements in health, education, and infrastructure**.
2. **Aspirational Block Programme (ABP):** Focuses on 500 backward blocks to enhance **governance and human development outcomes**.
3. **'Vocal for Local' Initiative:** Promotes **grassroots entrepreneurship and self-reliance** in backward regions.
4. **Border Area Development Programme (BADP):** Meets special developmental needs of **remote and border areas**.
5. **Performance-Based Funding:** Rewards states for **reducing inequalities and meeting development targets**.
6. **Sectoral Investments:** Enhances agriculture, MSMEs, services, banking, and insurance to drive **inclusive growth**.
7. **Infrastructure & Connectivity Development:** Roads, power, telecom, irrigation, and digital networks attract investment.
8. **Strengthening Governance:** Empower local authorities for **efficient administration and better implementation**.
9. **Cooperative & Competitive Federalism:** Encourage states to **compete and collaborate** for balanced regional development.

Strategies for Bridging Disparities:

1. **Tailored Regional Approaches:** Design **region-specific development plans** targeting hilly, flood-prone, and drought-affected areas rather than using one-size-fits-all policies. For instance, **Hilly Area Development Programmes in Himachal Pradesh** address terrain-related challenges effectively.
2. **Infrastructure Development:** Strengthen **roads, electricity, telecom, irrigation, and digital connectivity** to attract investments and generate employment. Example: **Bharatmala and Sagarmala projects** enhance connectivity in underdeveloped states, boosting trade and industrial growth.
3. **Human Capital Enhancement:** Invest in **education, vocational training, digital skills, and healthcare** to improve employability and well-being. For example, **Skill India and digital literacy initiatives in rural Odisha** aim to bridge the skills gap.

4. **Strengthening Governance:** Empower **local authorities**, ensure transparent administration, and minimize bureaucratic delays to implement policies effectively. **Aspirational Districts Programme (ADP)** demonstrates success by improving governance at the district level in backward regions.
5. **Sustainable Livelihoods:** Promote **rural employment schemes, micro-enterprises, and enhanced agricultural productivity** for long-term economic stability. Example: **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)**provides guaranteed work and income in tribal and backward districts.
6. **Cooperative & Competitive Federalism:** Encourage **healthy competition among states** to improve performance while fostering **inter-state collaboration**. For instance, **"Ease of Doing Business" rankings** motivate states like Andhra Pradesh and Telangana to attract investments, benefiting both economic growth and social outcomes.

Conclusion:

Regional disparities in India are **deep-rooted, multidimensional, and geographically widespread**, affecting economic, social, and human development. Bridging them requires **targeted policies, investments in infrastructure and human capital, and strong governance**, tailored to local realities. The ultimate goal is **inclusive and sustainable growth**, ensuring that no state or region is left behind and every citizen can access opportunities for a better standard of living.

POVERTY

1. POVERTY IN INDIA

"*Poverty is the Worst Form of Violence*"- Mahatma Gandhi

Meaning:

According to UNHRC- Poverty is a human condition characterised by the **sustained or chronic deprivation** of the **resources**, **capabilities**, **choices**, **security** and **power** necessary for the enjoyment of an **adequate standard of living** and other **civil**, **cultural**, **economic**, **political** and **social rights**.

Types of Poverty:

Aspect	Absolute Poverty	Relative Poverty
Definition	Lack of basic necessities (fixed and universal)	Income/resource inadequacy compared to societal standards
Measurement	Specific income threshold (e.g., $2.15/day per capita)- World Bank	Comparative to the average or median income in society.
Focus	Essential for survival and subsistence	Disparities and social inequality
Policy Implications	Focus on providing basic needs and essential services.	Address income inequality and improve overall conditions
Long-Term Trends	Relatively stable unless significant changes occur	Can change with economic growth, social policies, and income distribution shifts.

Poverty Estimation in India

Income and Consumption Expenditure-Based Measurement	India primarily follows the **consumption expenditure approach,** which estimates **household spending** on **essential goods and services**. It determines whether individuals fall below the poverty line based on their **monthly per capita expenditure** (**MPCE**). • The **Tendulkar Committee** (2009) recommended shifting from **calorie-based poverty estimation** to **expenditure-based poverty estimation**. • The **Rangarajan Committee** (**2014**) revised **poverty thresholds**, suggesting higher expenditure limits for rural and urban areas. *Every five years, NSSO conducts surveys to collect household consumption expenditure (MPCE). Monthly per capita consumption expenditure (MPCE) is used to determine the poverty line.*
Poverty Line Basket (PLB) Approach	The **Poverty Line Basket (PLB)** is a set of essential goods and services required to maintain a minimum standard of living. It includes: • Food items (based on caloric intake recommendations). • Clothing, shelter, education, healthcare, and transportation costs. • Different poverty committees (such as **Planning Commission estimates**, **World Bank thresholds**, and **NITI Aayog's MPI-based assessment**) have varied benchmarks for PLB. The PLB helps governments determine poverty lines and structure welfare programs like free ration schemes (PDS), MNREGA, and direct cash transfers.

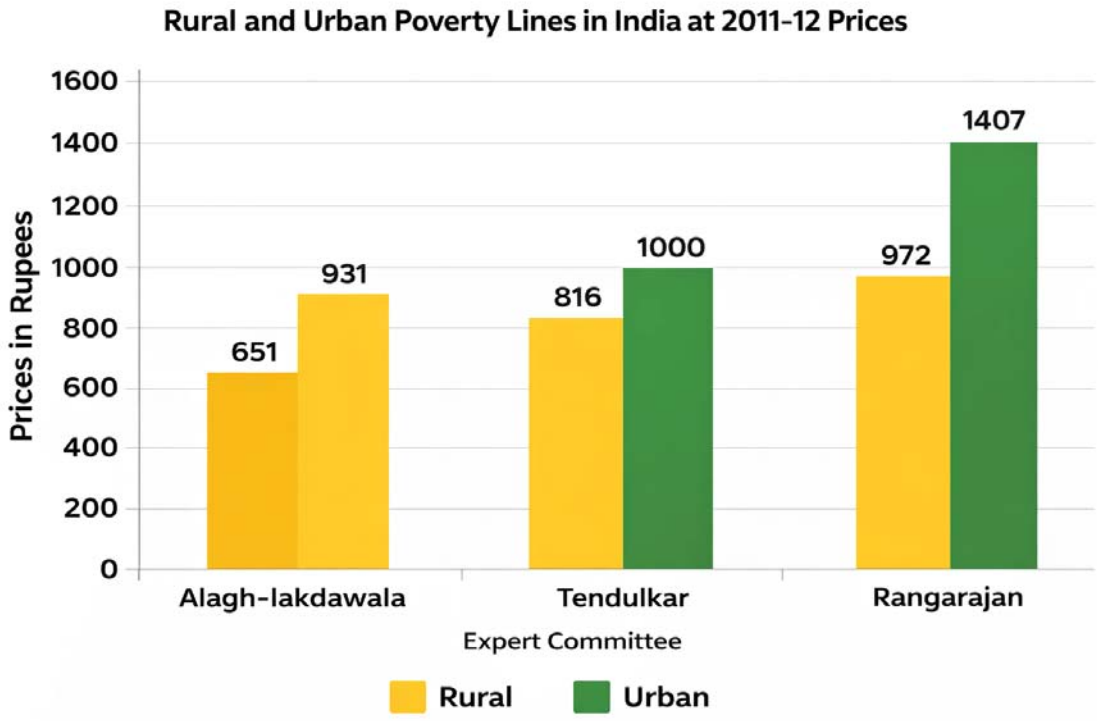
Poverty Estimation Exercises in India

Pre-Independence Poverty Estimation

- Dadabhai Naoroji in his book 'Poverty and Un-British Rule in India,' made the earliest estimate of the poverty line of ₹16 to ₹35 per capita per year at 1867-68 prices.
- The Bombay Plan (1944) suggested a poverty line of ₹75 per capita per year.

Post-Independence Poverty Estimation

Committee	Year	Methodology	Poverty Line (Monthly Per Capita Expenditure)
Alagh Committee	1979	Based on calorie consumption	**Rural:** 2100 calories, **Urban:** 2400 calories
Lakdawala Committee	1993	Modified mixed approach	**Rural:** Rs. 328 (1993-94 prices), **Urban:** Rs. 454 (1993-94 prices)
Tendulkar Committee	2009	Modified mixed approach	**Rural:** Rs. 672**, Urban:** Rs. 859 (2004-05 prices)
Rangarajan Committee	2012	Modified mixed approach	**Rural:** Rs. 972**, Urban:** Rs. 1407 **(2011-12 prices)**



Limitations of poverty line estimation

- **Inadequate measurement:** It is based on a **consumption-based approach**, hence may not capture all aspects of poverty, such as **access to healthcare**, **education**, and other social services.
- **PLB components:** Determining components of Poverty Line Basket (PLB) has been one of the challenges due to price differentials (of constituents of basket) which vary from state to state and period to period.
- **Limited coverage:** It only covers those households who fall below the poverty line, leaving out households who are just above the poverty line but still face significant challenges in meeting their basic needs.
- **Lack of regional variation:** Since it is based on a national average, which may not reflect the differences in the cost of living and poverty levels across different regions in India.
- **Lack of consideration for vulnerability:** It does not take into account vulnerable groups, such as women, children, and the elderly, who may face additional challenges in meeting their basic needs.

Status of Poverty in India

According to the World Bank, India has made significant progress in **reducing poverty,** with extreme poverty. Extreme poverty (**measured at $2.15 a day in purchasing power parity (PPP) terms**) has fallen from **16% in 2011-12** to **2.3% in 2022-23**. This decline has lifted **171 million** people out of extreme poverty.

Economic Survey 2025-2026 on Poverty: Key Insights

1. **Pro-poor shift in consumption patterns:** The Survey notes a **clear improvement in income distribution**, with **consumption inequality declining**. Growth is becoming more inclusive rather than concentrated.

2. **Fastest gains among the poorest sections:** Data from **HCES 2023–24** shows that the **bottom 5–10% experienced the highest increase in MPCE**, indicating that **the poorest are benefiting the most from recent growth**.
3. **Welfare-led poverty reduction model:** A combination of **subsidies, pensions, Direct Benefit Transfers (DBTs), and social sector spending** has played a decisive role in **lifting vulnerable populations out of poverty**.
4. **Centrality of rural transformation:** The Survey underlines that **revitalising the rural economy**—through **local employment, skill-building, and sustainable practices**—is essential for **durable poverty reduction**.
5. **Governance reforms for sustained impact:** It stresses the need for **continuous, data-driven policymaking**, supported by **technology-enabled surveys and community participation**, to ensure **effective targeting and long-term poverty alleviation**.

Factors behind Poverty in India

1. **Historical Reasons: British colonial rules** turned India into an exporter of raw materials and importer of finished goods impacting income of farmers, artisans.
2. **Low Agricultural Productivity:** Fragmented land holdings, lack of capital, and reliance on traditional farming methods limit yields. **For ex-** Lower yield of crops in India than developed economies affects income of farmers.
3. **Population Explosion:** India's rapid population growth has led to increased pressure on resources and services exacerbating the headcount ratio of poverty. **For ex-** India's population is expected to peak in the early 2060s (at 1.7 billion) and remain the **world's most populous country** throughout the century (UNDESA).
4. **Economic Inequality:** Disparities in income and wealth distribution concentrate resources among a few giving rise to relative poverty. **For ex- Top 10% of the Indian population** holds **77% of the total national wealth** (Oxfam).
5. **Social Inequalities:** Caste discrimination and gender disparities lead to social exclusion. For ex- India has 53% of women outside the labour force due to care responsibilities(ILO)
   a. Moreover, **rigid caste systems restrict access to resources** and opportunities for marginalized groups, perpetuating poverty across generations leading to Intra-generational and Intergenerational issues of equity.
6. **Geographical Disparities:** Dense forests, hilly terrains or areas vulnerable to natural disasters contribute to higher poverty rates. **For ex- Recurrent floods in Assam** and **Bihar** displaces millions of people and pushes them into absolute poverty every year.

Consequences of Poverty

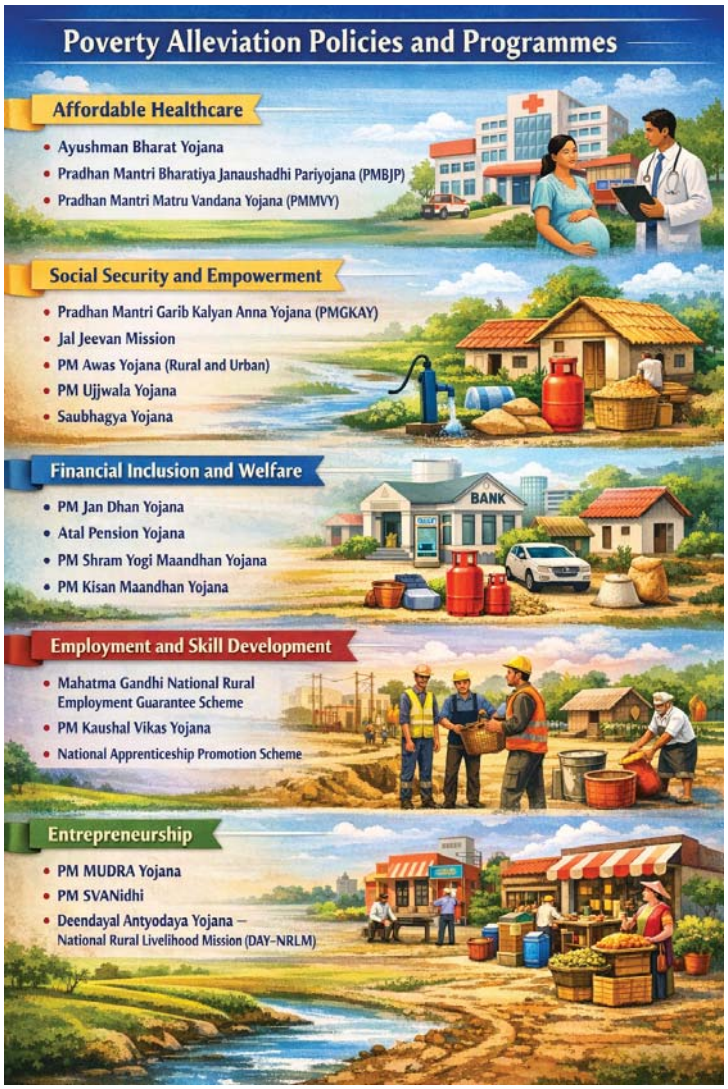
1. **Culture of Poverty:** Poverty can become a generational phenomenon, creating **hopelessness and powerlessness**. For example, children in long-term slum communities may view low-paying work as their only option, discouraging efforts to improve their situation.
2. **Limited Education and Lost Childhoods:** Children in poor households often face **inadequate schooling**, are forced into **child labor**, or, in the case of girls, experience **early marriage**, curtailing their education and development.
3. **Health Challenges:** Poor nutrition, unsafe living conditions, and limited healthcare access result in **higher infant mortality** and **greater disease vulnerability**. For instance, children in rural Madhya Pradesh without access to proper sanitation face frequent diarrheal diseases, affecting growth and learning.
4. **Social Exclusion and Stigma:** The poor often experience **marginalization**, facing prejudice and exclusion from community activities. For example, low-income families may be denied participation in local cooperative societies or school committees, weakening social support.

5. **Economic Vulnerability:** Limited financial resources make the poor prone to **debt traps** and exploitation. A street vendor borrowing at high interest to expand his small business may quickly fall into a cycle of debt when income is irregular.
6. **Intergenerational Impact:** Children raised in poverty often inherit **restricted opportunities**, perpetuating systemic inequality. For example, a child from a low-income household in Bihar may leave school early to work, continuing the cycle despite India's overall economic growth.

Vicious Cycle of Poverty

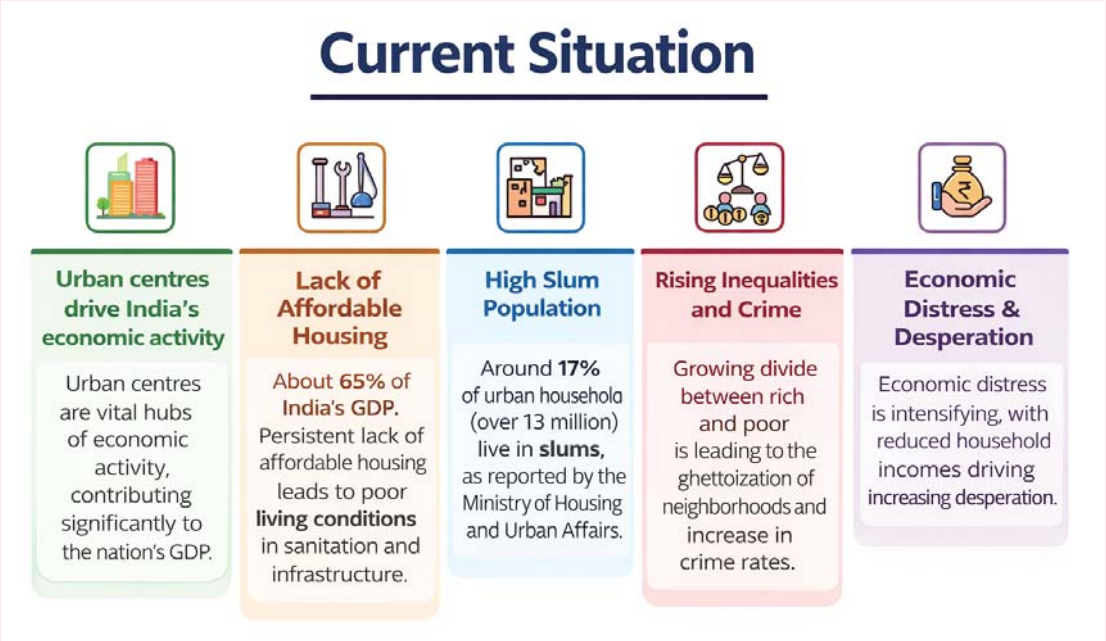
The **vicious cycle of poverty** highlights how interconnected challenges trap people in long-term deprivation. Malnutrition during pregnancy and protein deficiency in early childhood can stunt brain development and reduce energy, affecting school performance and future work opportunities. Children living in overcrowded, unsanitary conditions face frequent illness, further limiting learning and growth. For example, a child in an urban slum may miss school repeatedly due to illness, reducing chances of skill development.

Poverty also **imposes higher cost**s on the poor. Buying small quantities leads to higher per-unit prices, and borrowing comes with high interest rates. Employment is often low-paying and risky, increasing vulnerability to health hazards. For instance, a low-income construction worker may face repeated injuries and medical expenses while struggling to support family nutrition and education. Together, these factors reinforce one another, keeping poverty persistent across generations.



Urban Poverty in India

The **India Employment Report (IER) 2024** highlights a critical urban employment crisis, noting that youth (aged 15-29) unemployment is significantly higher in urban areas compared to rural areas, with the 15-19 age group experiencing the highest rates. Despite a 5.4% average real economic growth rate, the report **raises concerns about a weak "trickle-down" effect of economic benefits** to the urban working class



Challenges and Limitations of Current Poverty Alleviation Policies

1. **Identification and Targeting Inefficiencies**- Many genuinely poor households are excluded due to outdated data and bureaucratic loopholes. **For Ex- Errors** in the **Socio-Economic Caste Census** (SECC) database have led to **misidentification of beneficiaries** for schemes like NFSA.
2. **Overlaps in Schemes and Corruption Issues-** Multiple schemes with similar objectives lead to duplication of efforts. Further the mismanagement of funds, creates opportunities for corruption. **For Ex-** Overlapping of Rural housing and employment schemes causing inefficiencies in fund allocation.
3. **Lack of Adaptability to Regional Variations-** A one-size-fits-all approach ignores local economic, geographic, and social differences, making some schemes ineffective in certain areas. **For Ex- MGNREGA** is **less effective in certain rural areas** where daily wage rates are higher than the guaranteed wage.
4. **Bureaucratic Delays and Red Tape-** Lengthy administrative processes slow down the implementation of welfare programs, and reduce their immediate impact. **For Ex-** Delays in **PMAY (Pradhan Mantri Awaas Yojana)** have resulted in beneficiaries waiting for years to receive housing assistance.
5. **Leakages and Middlemen in Welfare Distribution-** Corruption and inefficiencies in the Public Distribution System (PDS) lead to the diversion of essential goods, depriving intended beneficiaries.
6. **Low Awareness and Accessibility-** Many eligible beneficiaries, especially in rural areas, lack awareness and access to government programs due to complex application processes. **For Ex-** Women and marginalized groups struggling to avail benefits due to lack of digital literacy and documentation.
7. **Inadequate Budget Allocation and Resource Constraints-** Many schemes face fund shortages, limiting their reach and effectiveness. **For Ex-** Insufficient healthcare spending under Ayushman Bharat to comprehensively cover all beneficiaries.

8. **Political Interference and Policy Discontinuity-** Frequent changes in governments lead to shifts in priorities, disrupting long-term poverty alleviation strategies. For Ex- Discontinuation of poverty alleviation schemes after changes in govt.

Way forward for poverty alleviation

1. **Strengthen Data-Driven Policy Making-** Conduct large-scale socio-economic surveys to assess the impact of existing schemes and modify policies accordingly.
   - Implement a centralized and dynamic database to track real-time progress and minimize exclusion errors.
2. **Enhance Employment Opportunities and Skill Development-** Expand **MGNREGA to cover urban unemployment,** ensuring a steady income for the informal workforce.
   - Strengthen vocational training and digital literacy programs under Pradhan Mantri Kaushal Vikas Yojana (PMKVY) to align with the job market demands.
3. **Reform Social Welfare Distribution Systems-** Shift to a **Direct Benefit Transfer (DBT) model** to reduce leakages and corruption in welfare programs like PDS and LPG subsidies.
   - Strengthen the **One Nation, One Ration Card scheme** to allow seamless food security benefits across states, aiding migrant workers.
4. **Bridge Regional Disparities and Customize Policies-** Adopt a **region-specific approach** by tailoring the poverty alleviation programs to local economic and environmental conditions.
   - Provide **special assistance** for backward states such as **Bihar, Jharkhand, and Odisha**, where poverty rates remain high.
5. **Strengthen Public Healthcare and Education- Expand Ayushman Bharat** to cover outpatient care, diagnostics, and preventive healthcare services.
   - **Increase public spending** on **education** and **skilling programs**, particularly in rural areas, to break the intergenerational cycle of poverty.
6. **Leverage Technology for Financial Inclusion- Strengthen Jan Dhan Yojana** by linking it with microcredit facilities and social security schemes.
   - **Promote FinTech solutions** for better financial planning and credit access to small businesses and farmers.
7. **Climate Resilience and Sustainable Development- Implement green job initiatives** to employ while tackling climate change.
   - Promote climate-resilient agriculture through efficient irrigation techniques, organic farming, and agroforestry.

Suggestions by NITI Ayog to combat poverty

- **Employment-intensive Sustained Rapid Growth-** Create better jobs and generate additional revenues for the government by expanding social expenditures at a faster pace.
  - Push for **Second green revolution in agriculture** particularly in eastern India.
- **Enhance effectiveness of anti-poverty programs-** Better targeting of schemes with use of technologies to reduce inclusion and exclusion errors. **For ex-** Use of Jan Dhan Yojana, Aadhaar, Mobile (**JAM**) **trinity**.
- **Focus on "Amartya Sen's 'Development as Freedom' and the Capability Approach"-** Investment in **people's capabilities** with a focus on **education**, **skills to expand their opportunities** and **freedoms** thereby leading to citizen's empowerment and reduction in poverty.
  - **For ex-** Enhance slum dwellers capabilities influences their entrepreneurship development and contributes to urban poverty reduction.

Conclusion

For India to achieve sustainable and inclusive poverty eradication, it must move beyond short-term relief measures and **focus on structural reforms**. A multi-sectoral, data-driven, and **region-specific approach** will be key in making poverty history.

2. MULTIDIMENSIONAL POVERTY

Introduction:

Traditional poverty assessments focus on income: "Who is below the poverty line?" However, **poverty is multidimensional**. It is not just about insufficient income but about **deprivations across health, education, living standards, and social capabilities**.

Multidimensional poverty captures the **true lived experiences of people** by considering overlapping deprivations such as malnutrition, lack of sanitation, inadequate housing, poor access to education, and exclusion from financial services.

In India, this shift is crucial because **income poverty alone underestimates deprivation**, especially in rural and marginalized communities. Recognizing this, the **Multidimensional Poverty Index (MPI)** has become a key tool for policymakers.

Understanding the Multidimensional Poverty Index (MPI)

Global MPI

- Developed by **OPHI (Oxford Poverty and Human Development Initiative)** in collaboration with **UNDP**, first published in 2010.
- Measures poverty **beyond monetary terms** across **three dimensions**:
  - **Health** – child mortality, nutrition
  - **Education** – years of schooling, school attendance
  - **Standard of Living** – cooking fuel, sanitation, drinking water, electricity, housing, assets
- A person is considered **MPI poor** if they are deprived in **one-third or more** of these indicators.

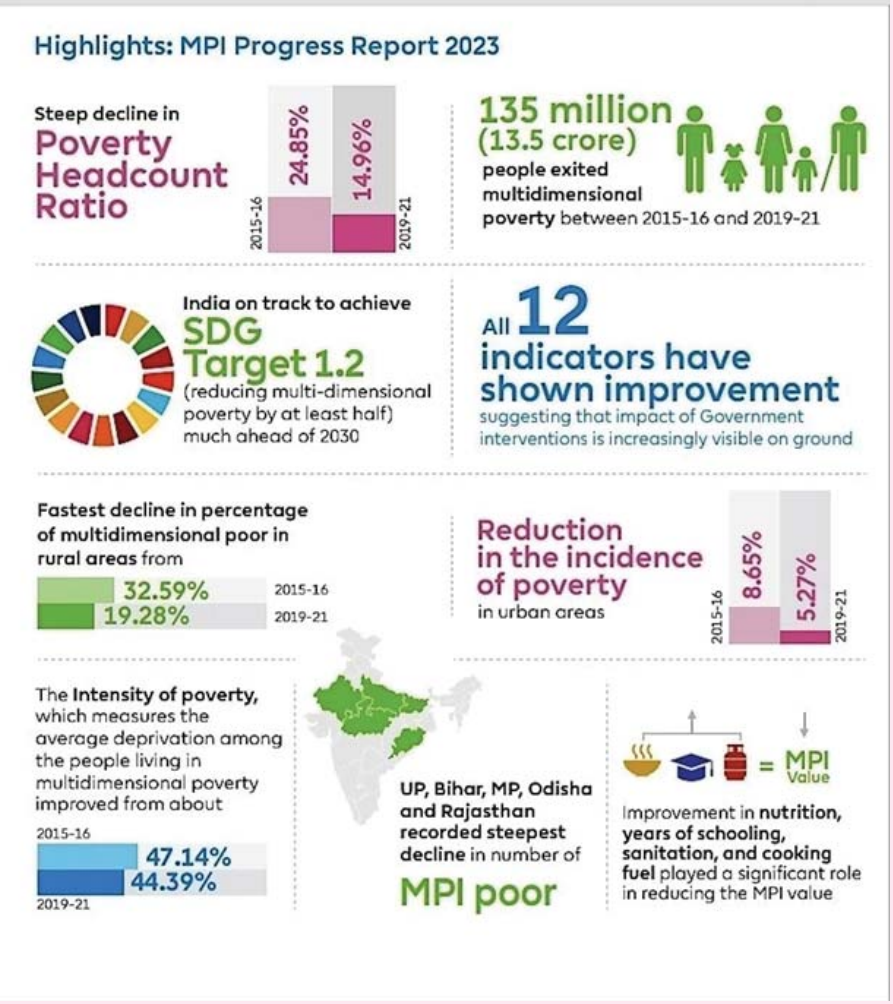
National MPI (India)

- Launched by **NITI Aayog in 2021**, aligned with **SDG Target 1.2**: reducing multidimensional poverty by half by 2030.
- India's MPI retains the **10 global indicators** and adds **2 national priorities**: *maternal health* and *bank account ownership*.
- Based on **NFHS-4 and NFHS-5**, it enables **district-level assessments**, guiding targeted interventions.
- **Computati**on: MPI = Headcount ratio (H) × Intensity of poverty (A).
  - H = proportion of people who are multidimensionally poor
  - A = average proportion of deprivations experienced by the poor

Trends in India's Multidimensional Poverty

- **MPI headcount declined** from **29.17% in 2013-14** to **11.28% in 2022-23**.
- Approximately **24.82 crore people escaped poverty** in the last nine years.
- **Intensity of poverty** also fell, meaning the extent of deprivation among the poor is decreasing.
- **Regional trends:**
  - States like **Bihar, Uttar Pradesh, Madhya Pradesh, Rajasthan, Odisha** witnessed the fastest decline.

- **Indicator-level improvements:**
  - Greatest gains: sanitation, cooking fuel, electricity, banking access
  - Persistent deprivations: housing and clean cooking fuel in some regions



Factors Driving the Decline

- **PM Ujjwala Yojana:** Clean cooking fuel access
- **Saubhagya Scheme:** Household electricity
- **Swachh Bharat Mission:** Sanitation coverage
- **Jal Jeevan Mission**: Safe drinking water
- Other health, education, and social welfare interventions

Significance of Multidimensional Poverty Measurement

1. **Holistic Poverty Perspective:** Multidimensional poverty captures deprivation across **health, education, and living standards**, reflecting the actual lived conditions of people. It goes beyond income to show real-life hardships.
2. **Targeted Policy Action:** By identifying **specific areas of deprivation** at state and district levels, MPI enables policymakers to design **focused interventions** for the poor.
3. **SDG Monitoring and Accountability:** MPI helps track India's progress toward **SDG 1.2** (reducing multidimensional poverty by half), providing a **quantitative tool to assess social outcomes**.

4. **Effective Resource Allocation:** Data on MPI guides government spending to **critical sectors** like sanitation, clean fuel, electricity, housing, and maternal health.
5. **Measures Depth of Poverty:** Unlike income-based metrics, MPI captures both **the proportion of people who are poor (headcount)** and **the intensity of deprivation (severity of poverty)**.
6. **Inclusive Planning:** Highlights **overlapping deprivations among marginalized groups**, ensuring interventions reach **those most left behind**.

Limitations of MPI

1. **Data Dependence:** India's MPI is based on **NFHS surveys**, which are periodic, limiting the ability to **track poverty in real time** and respond to sudden changes.
2. **Household-Level Focus:** MPI treats households as units and ignores **intra-household inequalities**, including gender and age disparities among members.
3. **Indicator Complexity:** Multiple dimensions and 12 indicators can make **implementation, monitoring, and prioritization difficult** for policymakers.
4. **Threshold Sensitivity:** Households must be deprived in **≥1/3 of indicators** to be classified as MPI poor, which can **overlook minor but important deprivations**.
5. **Limited Shock Coverage:** Short-term crises like **COVID-19, reverse migration, or wage stagnation** are not fully captured in MPI calculations.
6. **Weighting Subjectivity:** Assigning **relevance and weights to indicators** can be arbitrary, reducing comparability across states and regions.

Way Forward

1. **Expand Dimensions:** Include factors like **learning outcomes, social discrimination, unorganized workers, and environmental risks** to make MPI more comprehensive.
2. **Integrate Data Sources:** Combine **NFHS, NSS surveys, and administrative data** to improve accuracy, reliability, and geographic coverage.
3. **Simplify Indicators:** Focus on **high-impact, locally relevant indicators** to reduce complexity and enhance policy implementation.
4. **Real-Time Monitoring:** Use **digital tools and community involvement** to continuously track poverty and identify gaps in interventions.
5. **Link MPI with Schemes:** Connect MPI data to programs like **PDS, MGNREGA, Swachh Bharat, and PM Ujjwala** to ensure targeted delivery of benefits.
6. **Awareness and Social Audit:** Promote **community-level social audits and awareness campaigns** to ensure schemes reach intended beneficiaries and reduce exclusion errors.

Conclusion

Multidimensional poverty measurement provides a **comprehensive lens** for understanding deprivation in India. By focusing on **health, education, and living standards**, MPI allows policymakers to target interventions effectively. While India has made **remarkable progress**, persistent deprivations and data limitations highlight the need for **continuous monitoring, better data, and locally relevant interventions**.

INITIATIVES OF INCLUSIVE GROWTH

1. UNIVERSAL BASIC INCOME (UBI)

Context:

Recently, Chief Economic Advisor V Anantha Nageswaran said that the **concept of Universal Basic Income (UBI) was not necessary for India** since natural economic growth would take care of the country's many aspirations.

Meaning:

Universal Basic Income (UBI) is a type of **guaranteed income** in which **all citizens** or residents of a country receive an **unconditional** sum of money on a regular basis, either from the government or another public institution, in addition to any other income received. It **aims** to ensure a minimum standard of living and provide economic security.

It is rooted in liberal thought (e.g., **Thomas Paine's 1797 agrarian justice**) and **modern economics** (e.g., **John Maynard Keynes** on technological unemployment).

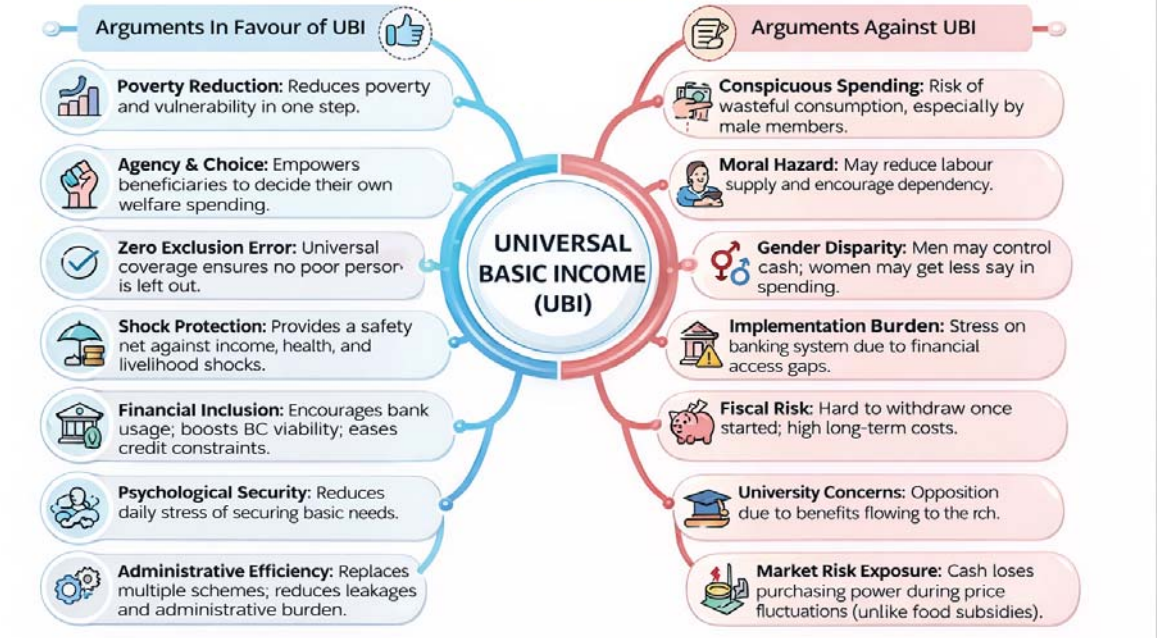
Core Principles:

- **Universality**: Available to everyone (adults/children, citizens/non-citizens in some models).
- **Unconditionality**: No strings attached (e.g., no obligation to work or train).
- **Individuality**: Paid per person, not per household.
- **Sufficiency**: Amount should cover basic needs (e.g., food, shelter).

Rationale for UBI Introduction in India

1. **Relevance to Poverty and Welfare:** With **21% multidimensional poverty** (NITI Aayog 2023), **45% of the workforce in the informal sector**, and rising inequality (Gini 0.35–0.40), UBI can **simplify welfare delivery** across 800+ fragmented schemes.
2. **Constitutional Alignment:** UBI aligns with **Directive Principles**—**Articles 39 and 41**—ensuring the **right to work** and **adequate livelihood**, and supports **SDG 1 (No Poverty)** and **SDG 10 (Reduced Inequalities)**.
3. **Economic Rationale:** In a **dual economy** (formal 10%, informal 90%), UBI can **boost aggregate demand** via the Keynesian multiplier, reduce **administrative costs** (potentially 1–2% of GDP saved in subsidy delivery), and facilitate **labor mobility** from low-productivity agriculture (45% workforce, 18% GDP).
4. **Social Rationale:** UBI can **address gender gaps** (female LFPR ~37%, PLFS 2023–24) and **empower marginalized groups** such as SC/ST communities and women by providing **direct cash autonomy**.

Arguments For and Against UBI
Universal Basic Income (UBI)



INDIA'S TRYST WITH INCOME SUPPORT

1950 — 2000

1938 A minimum income standard discussed

1964 Plan to start such a programme was on the horizon

2011-12 2 pilot projects conducted in 8 villages of Madhya Pradesh

2016-17 Economic Survey suggests scheme for bottom 75% of population, while rolling back some subsidies

2022 Sikkim Democratic Front plans to roll out UBI

UBI ACROSS THE WORLD

Alaska | The Permanent Fund distributes part of the state's oil revenues to all residents on per-capita basis

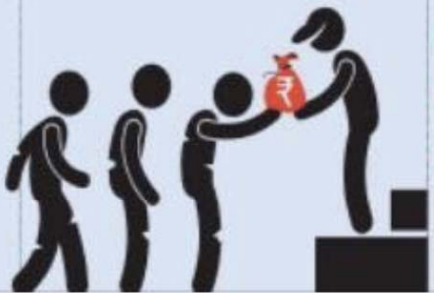
Stockton, California | Secured funding from private non-profits to launch a small project with about 100 participants receiving $500 a month for about 18 months

Finland | Scheme started in 2017 to pay 2,000 jobless people assistance of €560 a month stopped last year

Kenya | Largest experiment underway with some villages receiving $0.50-1 a day

Brazil | Has run experiments

Canada | Ontario plans to test a basic income scheme



France | A senate committee has recommended an experiment

UK & Germany | Studies have been conducted

Scotland | Committed funds to conduct an experiment

Barcelona, British Columbia | Plans to start experiments

Switzerland | Plan to give everyone right to basic income defeated in 2016

Pilots and Experiments of UBI in India

1. **Madhya Pradesh Pilot (2010–13):** Conducted by **SEWA and UNDP**, Rs 200–300/month was given to **8,000 villagers across 20 villages** for 18 months, providing practical insights on rural cash transfers.
   a. **Outcomes of the Pilot: School enrollment rose by 68%**, **nutrition improved by 32%**, and **savings increased by 52%**. Labor hours remained stable, disproving fears of work reduction.
   b. **Social Impacts:** Women reported **reduced drudgery**, and household **debt fell by 30%**, showing UBI's potential to empower women and reduce financial stress.
   c. **Cost and Scalability:** The program cost **Rs 5,000 per person per year**, offering valuable **scalable insights** for rural India despite small sample size.
2. **Quasi-UBI Schemes:** Initiatives like **PM-KISAN (2019, Rs 6,000/year to farmers)** and **PM Garib Kalyan Anna Yojana** (free grains to 80 crore people) test **DBT mechanisms**, though they remain targeted, not universal.

Challenges and Feasibility of UBI in the Indian Context:

1. **Fiscal Constraints:** India's tax-to-GDP ratio is around **11%**, low by global standards. Funding UBI through subsidy cuts (fertilizers/PDS: ~₹5 lakh crore) is possible but **politically sensitive**, as it may trigger **farmer protests**.
2. **Inflation and Moral Hazard:** Disbursing cash without parallel reforms in **agriculture, employment, and productivity** could exacerbate **inflationary pressures**, reducing the scheme's effectiveness.
3. **Implementation Hurdles: Digital infrastructure gaps** exist, with ~20% of the population unbanked. Identity verification issues affect **migrants and NRIs**, complicating seamless cash transfers.
4. **Corruption Risks:** Cash transfers may face **leakages and misappropriation**, especially in rural areas. Additionally, the **urban-rural divide** implies higher rural needs due to cost differences.
5. **Socio-Economic Issues:** Cultural preferences often favor **in-kind aid over cash**, and inflation in essential items (food basket accounts for ~45% of CPI) can erode real benefits.
6. **Gender Considerations:** Directing UBI to women could **empower them financially**, but intra-household dynamics may **divert funds**, limiting intended impact.
7. **Legal and Constitutional Aspects: Article 21** (Right to Life) supports citizens' entitlement to basic income, while **Article 14** (Equality) raises questions on **universality versus affirmative action** such as reservations.
8. **Regional Disparities:** A **uniform UBI** may not account for varying **costs of living across states**, risking inequitable benefits for people in high-cost regions.

Way Forward

1. **Linkages to Broader Policies:** UBI should align with **Atmanirbhar Bharat** goals, promoting self-reliance, and complement schemes like **MGNREGA** and **PMJJBY** to enhance coverage and impact.
2. **Quasi-UBI via DBT:** Expand existing **Direct Benefit Transfers (DBT)**, such as **One Nation One Ration Card**, to create a scalable **quasi-UBI framework** without full universal rollout.
3. **Phased Implementation:** Begin with **vulnerable groups**—women, elderly, and marginalized communities. Pilot programs can be launched in fiscally capable states like **Kerala** before nationwide scaling.
4. **Funding Sources:** Explore alternative financing through **carbon taxes** and a **wealth tax** on ultra-rich individuals (1% could raise ~₹1 lakh crore), reducing dependence on subsidy cuts.
5. **Policy Recommendations:** As suggested by **NITI Aayog** and the **Economic Survey**, adopt a **gradual rollout**, monitor outcomes using **Randomized Controlled Trials (RCTs)**, and integrate **gender-responsive budgeting**. International institutions like the **World Bank** can support pilot programs.

6. **Fiscal Reforms:** Rationalize subsidies on **food, fuel, and fertilizers**, while improving **tax compliance**, to generate the **fiscal space** required for sustainable UBI financing.

Conclusion

Universal Basic Income in India presents a **promising tool** to simplify welfare, reduce poverty, and empower marginalized groups, especially women. **Pilot studies and quasi-UBI schemes** demonstrate positive social and economic impacts without reducing labor effort. However, **fiscal constraints, regional disparities, and implementation challenges** necessitate a **phased, well-monitored rollout**, integrated with broader policy reforms for long-term sustainability.

2. JAM TRINITY

Introduction:

JAM Trinity refers to the **strategic integration** of Jan Dhan Yojana (bank accounts), Aadhaar (biometric identity), and Mobile numbers — a landmark **governance architecture introduced by the Government of India** to transform the delivery of welfare benefits. It is the **backbone of India's Direct Benefit Transfer (DBT) framework**, replacing intermediary-ridden, leakage-prone legacy systems with direct, targeted, and transparent digital transfers. It was first **proposed in the Economic Survey 2014-15.**

Components of JAM trinity:

- **Jan Dhan (PMJDY):** Provides every citizen with access to a bank account, integrating those previously outside the formal financial system.
- **Aadhaar:** Offers unique biometric identification, allowing accurate targeting and authentication of beneficiaries under welfare schemes, eliminating fraud and duplication.
- **Mobile:** Ensures ubiquitous digital connectivity to access financial services, receive subsidy payments, and communicate with government programs.

Need of JAM:

Before **JAM**, India's welfare system had **serious structural flaws** that wasted public funds and left genuine beneficiaries without entitlements.

Pre-JAM Welfare System — Key Problems

- **Leakages & ghost beneficiaries:** Nearly **85% of welfare funds** failed to reach intended recipients (**Economic Survey 2017-18**).
- **Subsidy misallocation:** The **richer half of India consumed 75% of subsidized LPG cylinders**; **PDS leakages** reached **46.7% in 2011**.
- **Heavy fiscal burden:** Subsidies took up **16% of government expenditure (~₹2.1 lakh crore annually)** but produced minimal social benefits.
- **Middlemen & corruption:** Multiple intermediaries diverted funds; **bogus ration cards, fake beneficiaries, and market diversion** were widespread.
- **Delays & inefficiency: Paper-based disbursals** caused slow transfers, worsened by **multi-level approval processes** under the federal system.
- **Financial exclusion:** Even **67 years after Independence**, millions lacked **bank accounts**, denying them access to savings, credit, or formal economic identity.
- **PDS corruption:** Over **3.99 crore duplicate or fake ration cards** existed, undermining the effectiveness of welfare programmes.

How Jam Trinity Works — The Mechanism

Step 1: IDENTIFY	Aadhaar — biometric authentication uniquely identifies the beneficiary, eliminating duplicates and ghost beneficiaries
Step 2: ROUTE PAYMENT	PMJDY account linked to Aadhaar via Aadhaar Payment Bridge (APB) — enables direct fund transfer
Step 3: NOTIFY & CONFIRM	Mobile — SMS/app notification confirms transfer; beneficiary can withdraw from BC/ATM

Multidimensional Significance:

A. Economic & Fiscal Governance

- **Reduces leakages: JAM** ensures subsidies reach **intended beneficiaries**, cutting fiscal wastage without reducing welfare.
- **Better fiscal management:** Subsidy burden **halved** despite **16× expansion** in coverage, improving spending efficiency.
- **Eliminates ghost beneficiaries: Aadhaar authentication** removes fake and duplicate entries in subsidy programmes.
- **Optimises procurement:** Digital records help accurately estimate **buffer stocks**, lowering **food wastage**.
- **Creates fiscal space:** Savings from efficiency allow higher **capital expenditure** and investment in **productive sectors**.

B. Social Justice & Inclusion

- **Women's empowerment: 55.7% of Jan Dhan accounts** held by women; **DBT transfers** strengthen their control over household finances.
- **Inclusion of marginalized groups: Tribals, Dalits, migrants, and informal sector employees** gain direct access to benefits.
- **Poverty reduction:** JAM accelerated decline of **extreme poverty from 47% (1991) to ~5% (2019)**.
- **Restores dignity:** Entitlements delivered as **legal rights** under **Article 21**, ensuring fair access.
- **Wide coverage: DBT reaches 85% of rural** and **69% of urban households**, expanding social inclusion.

C. Governance & Anti-Corruption

- **Reduces intermediaries:** Direct transfers cut opportunities for **rent-seeking** by officials or middlemen.
- **Transparency:** Every transaction is traceable; **SMS alerts** allow beneficiaries to detect **fraud** immediately.
- **Accountability: PFMS** maintains a **digital audit trail** of all transfers, ensuring responsible governance.
- **Universality:** Single platform for **central and state schemes** reduces coordination failures.
- **Prevents duplication: Aadhaar identity** ensures one beneficiary, eliminating multiple fraudulent claims.

D. Digital India & Fintech Ecosystem

- **India Stack foundation:** JAM underpins **Aadhaar, eKYC, UPI, DigiLocker, and APBS** for a unified digital infrastructure.
- **UPI revolution: ₹200 lakh crore** transacted in **FY 2023-24** demonstrates JAM's role in digital payments.
- **Fintech innovations:** Enabled **neo-banking, BNPL services, and digital microfinance** products for the underserved.

- **Global recognition:** India and **Kenya top World Bank's JAM Index** for digital delivery systems.
- **Praised internationally: IMF and World Bank** call India's **DBT system a "logistical marvel"** for scale and efficiency.

E. Crisis Response & Resilience

- **COVID-19 relief: ₹500/month** transferred to women's Jan Dhan accounts in **three installments** within weeks.
- **Rapid support:** JAM enabled **immediate financial aid**, unmatched globally during the pandemic.
- **Government acknowledgment: PM Modi** highlighted JAM as critical for **fast, direct transfers** to the needy.

Fiscal Impact — Key Data Points:

Parameter	Achievement
Cumulative DBT savings (2013-2024)	₹3.48 lakh crore (Finance Ministry)
Reduction in subsidy allocation	16% → 9% of total govt expenditure
Beneficiary coverage expansion	11 crore → 176 crore (16-fold)
Total DBT transferred since 2013	₹43+ lakh crore
DBT in FY 2021-22 alone	₹6.3 lakh crore
LPG subsidy savings (by Apr 2016)	₹17,000 crore
PDS savings (2013-2020)	₹1.85 lakh crore (53% of total DBT savings)
MGNREGS savings via DBT	₹42,534 crore (98% wages on time)
Fake/duplicate ration cards deleted	3.99 crore
Fake LPG connections eliminated	4.11 crore
Daily DBT payments processed (FY22)	90 lakh+ per day
Welfare Efficiency Index (WEI)	0.32 (2014) → 0.91 (2023)

INDIA'S JAM vs Global Comparisons

JAM Trinity is unique in its scale and integration. No country has simultaneously achieved biometric ID coverage, mass bank account opening, and mobile connectivity at 1.4 billion scale.

Country/System	Comparison with India's JAM
Kenya (M-Pesa)	Mobile-first financial inclusion; lacks biometric ID integration at scale; JAM is broader
Brazil (Bolsa Familia)	Conditional cash transfer; uses CPF national ID; lacks biometric authentication breadth
Mexico (Oportunidades)	DBT for conditional transfers; limited digital backbone compared to India's Stack
China (Alipay/WeChat Pay)	Private-sector led digital payments; state-controlled but not integrated welfare delivery at JAM scale
Estonia (e-ID system)	Small country digital governance model; India's achievement at 140 crore scale is uniquely transformative
World Bank JAM Index	India and Kenya rank as top countries for using ID+mobile+financial accounts for welfare

Challenges & Criticisms of JAM:

Exclusion Errors:

- **Biometric failures:** Fingerprints of **elderly citizens, manual labourers, or people with skin conditions** often fail **Aadhaar authentication**, denying rations or wages.
- **Jharkhand starvation deaths (2017):** Several deaths were linked to **Aadhaar-based PDS denial**, raising national concern.
- **Data entry errors:** Mistakes in **Aadhaar or bank records** prevent transfers, and beneficiaries often lack the skills to fix them.
- **Outdated beneficiary lists:** Reliance on the **2011 Census** fails to reflect current population or poverty status.
- **Inactive Aadhaar accounts:** Accounts marked **inactive** cannot receive transfers through **APBS**, blocking entitlements.

Inclusion Errors

- **Wrong enrolment:** Non-eligible individuals, like **four-wheeler owners**, can be enrolled, and fake Aadhaar IDs may be created.
- **Opacity of APBS:** Errors in **account mapping or software failures at NPCI** can silently block transfers, leaving beneficiaries without recourse.

Digital Divide & Infrastructure Gaps

- **Distance barriers:** Banks or **CSCs** are over **6 km away** for more than 20% of households, costing beneficiaries ~**13 hours/month** to collect benefits.
- **Low connectivity: Mobile and internet penetration** is weak in tribal, hilly, and remote regions, limiting digital authentication.
- **Power and network outages:** Frequent disruptions prevent **real-time verification**, especially in last-mile locations.
- **Digital literacy gap:** Many **women, elderly, and marginalized groups** cannot use digital self-service tools effectively.

Data Privacy & Surveillance Concerns

- **Centralised database risk: Aadhaar** is the world's largest biometric repository, making it a prime target for hacking.
- **Profiling potential:** Metadata of transactions can track **when, where, and how citizens use Aadhaar**, raising **Article 21 privacy concerns**.
- **Legal vacuum:** Delay in implementing **Personal Data Protection** laws (DPDP Act 2023) left scope for misuse.
- **Civil liberties challenges:** Groups like **INSAF** and **CFCL** have consistently raised concerns about Aadhaar-enabled surveillance.

Dormant Accounts & Low Financial Activity

- **Dormant Jan Dhan accounts:** Many accounts are opened for **compliance** and are rarely used for active banking.
- **Limited financial literacy:** Accounts mainly serve as **DBT pass-through**, not for savings, credit, or insurance.
- **Banking strain:** Handling **low-ticket, high-volume transactions** from crores of low-income accounts challenges infrastructure.

Gender & Social Exclusion Paradox

- **Intra-household control:** Despite **55% women account holders**, men may exercise control over funds.
- **Indirect exclusion:** Women without **Aadhaar or mobile access** in conservative households may miss benefits.
- **Documentation barriers:** Marginalized groups like **nomadic tribes, homeless, and transgender**

persons face hurdles in Aadhaar enrolment.

Technological Dependence & System Risk

- **Systemic risk:** Centralised **UIDAI database** creates vulnerability for **141 crore identities** in case of breach.
- **Pipeline vulnerability:** The **DBT ecosystem** (PFMS, NPCI, APBS, banks) is sensitive; failure at any node disrupts transfers.
- **Infrastructure dependence:** Over-reliance on **technology** in areas with weak **power or internet** is a structural weakness.

Way Forward — Reforms & Recommendations

Addressing Exclusion Errors

- **Offline alternatives:** Introduce **manual verification** and **Aadhaar-free bypass** for biometric failures.
- **Re-enrolment camps:** Regularly update **elderly, labourers, and others** with degraded biometrics.
- **Local grievance redressal:** Empower **Gram Panchayat officers** to restore benefits immediately.
- **Automated restoration:** Time-bound benefit recovery when **biometric failure** is flagged.

Strengthening Data Protection

- **DPDP Act 2023:** Ensure swift, full **operationalisation with strong enforcement**.
- **Data minimisation: Aadhaar metadata** should not be stored beyond authentication needs.
- **Independent regulator:** Empower a **transparent data authority** to oversee UIDAI operations.
- **Virtual ID system:** Make **VID default** to reduce exposure of actual Aadhaar numbers.

Bridging the Digital Divide

- **Connectivity expansion:** Extend **broadband and 4G/5G** to every Gram Panchayat under **BharatNet Phase 2**.
- **Bank Mitras density:** Increase **Business Correspondents**, prioritising **women BCs** in rural areas.
- **Offline authentication:** Enable **stored credentials** with periodic sync for low-connectivity regions.
- **Dedicated CSCs:** Provide **Aadhaar update and grievance resolution services** locally.

Enhancing Financial Depth (Beyond Inclusion)

- **Financial literacy campaigns:** Promote active use of **savings, insurance, and pensions**, not just account opening.
- **Activate dormant accounts:** Target **micro-products** like recurring deposits and micro-insurance.
- **Link SHGs/MFIs:** Connect **Self Help Groups** and **Microfinance Institutions** with Jan Dhan accounts to enable credit flow.
- **DBT 2.0 vision:** Move from cash transfers to **purchasing power vouchers**, giving beneficiaries more autonomy.

Updating Beneficiary Databases

- **Database refresh:** Update welfare records with **2021 Census** and **SECC data**.
- **Real-time tracking:** Use **JAM-linked surveys** and self-declaration for dynamic beneficiary updates.
- **Portability of benefits:** Ensure **migrant workers** can access entitlements across states.

Conclusion:

India has achieved remarkable financial inclusion in just six years (around 2017–2023) using **its Digital Public Infrastructure (DPI)**, a feat that would have taken **47 years without such technology, according to a World Bank G20 document.** The JAM Trinity (Jan Dhan-Aadhaar-Mobile) powered this transformation, which the **IMF lauded as a "logistical marvel"**

5. FINANCIAL INCLUSION

Meaning & Objective:

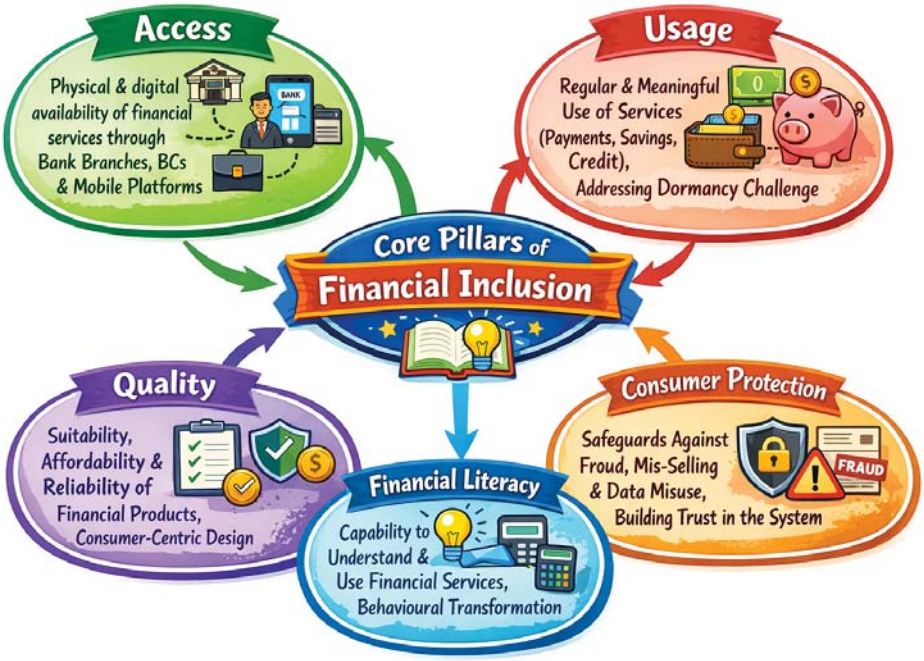
It refers to ensuring **affordable, timely and adequate access** to financial services like banking, credit, insurance, and pensions for all sections, especially vulnerable groups.

Core Objective: The aim is to integrate the **unbanked and underbanked population into the formal financial system**, reducing dependence on informal and exploitative sources of finance.

Financial inclusion reduces poverty, enhances **income generation, financial resilience, and social security**, making it central to inclusive growth. Moreover, it is a core pillar of India's vision of **Viksit Bharat 2047**, enabling welfare delivery, market participation, and economic resilience

Core Pillars of Financial Inclusion:

1. **Access:** Physical and digital availability of financial services through **bank branches, BCs, and mobile platforms**.
2. **Usage:** Regular and meaningful use of services (payments, savings, credit), addressing the **dormancy challenge**.
3. **Quality:** Suitability, affordability, and reliability of financial products, ensuring **consumer-centric design**.
4. **Financial Literacy:** Capability to understand and use financial services, crucial for **behavioural transformation**.
5. **Consumer Protection:** Safeguards against fraud, mis-selling, and data misuse, building **trust in the system**.



Need for Financial Inclusion in India:

1. **Banking the Unbanked:** Universal access to bank accounts ensures entry into the formal financial system, enabling individuals to access savings, credit, insurance, and payments.

2. **Poverty Reduction & Social Equity:** Access to financial tools allows poor households to **build assets, manage risks, and smooth consumption**, helping break the cycle of poverty.
3. **Support for Vulnerable Groups:** Migrants, small farmers, daily wage workers, and informal sector participants gain **economic stability and participation** through formal financial access.
4. **Women's Empowerment:** Financial inclusion enhances **financial independence, decision-making power, and access to credit**, improving household welfare and gender equality.
5. **Insurance & Social Security Coverage:** Inclusion ensures access to **life, accident insurance, and pensions**, reducing vulnerability to shocks and old-age insecurity.
6. **Asset Diversification:** Enables households to move beyond cash savings to **formal investments (mutual funds, pensions, capital markets)**, improving long-term wealth creation.
7. **Bridging Credit Gaps:** India's credit penetration remains uneven—only **154 loan accounts per 1000 adults (2016, NITI Aayog)**—highlighting need for affordable formal credit access.
8. **Boost to Economic Growth:** Inclusion mobilises savings and channels them into productive investment, thereby **enhancing GDP growth and capital formation**.
9. **Formalisation of Economy:** Encourages shift from informal to formal systems, improving **transparency, tax compliance, and financial traceability**.
10. **Efficient Welfare Delivery:** Financial inclusion underpins DBT and JAM, ensuring **targeted subsidy delivery, reduced leakages, and improved governance efficiency**.
11. **Resilience Against Shocks:** Access to savings, credit, and insurance enhances **household financial resilience** during crises like pandemics or crop failures.

Financial Inclusion in India: Evolution

1. **Evolution from Access to Usage:** Financial inclusion in India began with **bank nationalisation, priority sector lending, and SHG–Bank linkage**, focusing on expanding institutional reach and rural credit access.
2. **JAM-led Universalisation (Post-2014):** The **PMJDY + JAM Trinity (Jan Dhan–Aadhaar–Mobile)** enabled **near-universal bank account access** and scaled up **Direct Benefit Transfers (DBT)** for efficient welfare delivery.
3. **Digital Transformation Phase:** The rise of **UPI and fintech innovations** shifted the focus from mere account ownership to **real-time, low-cost digital financial transactions and active usage**.
4. **Improved Measurement & Impact:** Introduction of the **Financial Inclusion (FI) Index** supports **data-driven monitoring**, while outcomes include **reduced leakages, better subsidy targeting, and enhanced financial resilience**, though **regional disparities persist**.

Progress of Financial Inclusion in India:

1. **Expansion of Bank Accounts:** Over **55 crore PMJDY accounts (2025)** with deposits exceeding ₹2 lakh crore indicate massive outreach of formal banking.
2. **Rise in Account Ownership:** India's account ownership rose from **~53% (2014) to ~78% (2021)**, showing rapid financial deepening (Global Findex).
3. **Digital Payments Growth:** UPI transactions crossed **18,000+ crore annually (2024–25)**, reflecting a global-leading digital payments ecosystem.
4. **Financial Inclusion Index:** Reserve Bank of India FI Index improved to **~64 (2024)**, reflecting progress in access, usage, and quality.
5. **Gender Gap Reduction:** Female account ownership has significantly increased under PMJDY, though **usage gaps persist**.
6. **Expansion of Banking Infrastructure:** Growth of **Banking Correspondents (BCs), ATMs, and digital banking units** has improved last-mile access.

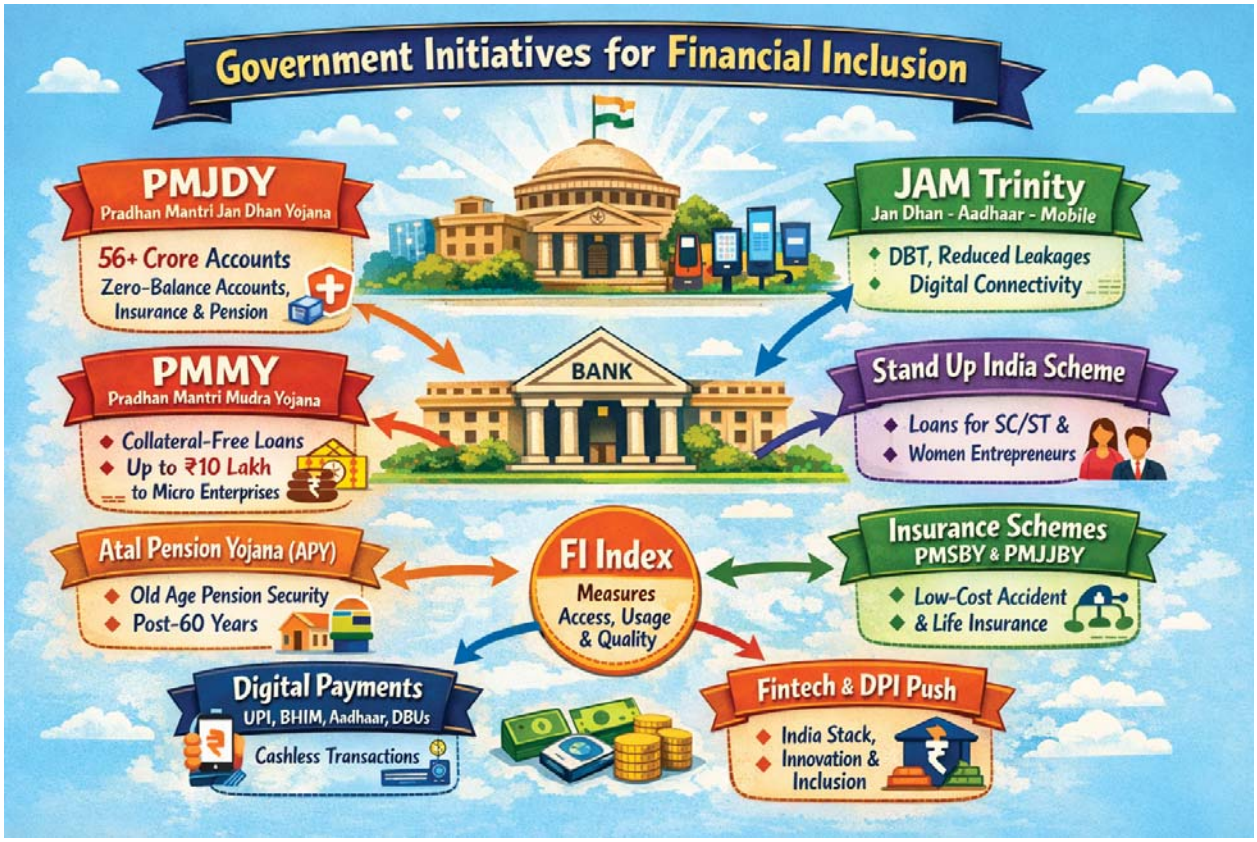
7. **Digital Public Infrastructure (DPI):** India Stack (Aadhaar, UPI, DigiLocker) has enabled **scalable, low-cost financial inclusion globally recognised as a model**.

Sectoral Impact of Financial Inclusion:

1. **Agriculture:** Improves access to **institutional credit, crop insurance, and DBT subsidies**, reducing dependence on moneylenders.
2. **MSMEs:** Enhances **formal credit flow and digital payments adoption**, supporting entrepreneurship and job creation.
3. **Households:** Promotes **savings, consumption smoothing, and risk mitigation** via insurance and pensions.
4. **Government Welfare:** Strengthens **targeted delivery and fiscal efficiency** through DBT, reducing leakages.
5. **Women Empowerment:** Financial access enhances **decision-making power and economic participation**, though usage gaps persist.

Government Initiatives for Financial Inclusion:

1. **Pradhan Mantri Jan Dhan Yojana (PMJDY):** Flagship scheme providing **zero-balance bank accounts with insurance, credit, and pension access**; over **56 crore accounts opened**, driving mass inclusion.
2. **JAM Trinity (Jan Dhan–Aadhaar–Mobile):** Integrates identity, banking, and mobile connectivity to enable **seamless DBT, reduced leakages, and digital access to welfare schemes**.
3. **Pradhan Mantri Mudra Yojana (PMMY):** Provides **collateral-free loans up to ₹10 lakh** to micro-enterprises, boosting entrepreneurship in the informal and rural sectors.
4. **Stand Up India Scheme:** Facilitates bank loans (₹10 lakh–₹1 crore) for **SC/ST and women entrepreneurs**, promoting inclusive business ownership.



5. **Atal Pension Yojana (APY):** Targets unorganised sector workers by offering **guaranteed pension post-60 years**, ensuring old-age income security.
6. **Insurance Schemes (PMSBY & PMJJBY):** Provide **low-cost accident and life insurance**, expanding risk coverage among low-income households.
7. **Digital Payments Ecosystem:** UPI, BHIM, Aadhaar-enabled payments, and Digital Banking Units have enabled **low-cost, real-time transactions**, strengthening digital financial inclusion.
8. **Financial Literacy Initiatives:** Reserve Bank of India's Project Financial Literacy and SEBI's "Pocket Money" programme promote **awareness, savings behaviour, and financial planning skills**.
9. **Financial Inclusion Index (FI Index):** Developed by RBI to measure inclusion across **access (35%), usage (45%), and quality (20%)**, enabling data-driven policy monitoring.
10. **Fintech & DPI Push:** Expansion of India Stack (UPI, Aadhaar, Account Aggregator) supports **scalable, low-cost, and innovation-driven inclusion**.

Challenges to Financial Inclusion in India:

1. **Access–Usage Gap:** While bank account penetration is high due to PMJDY, nearly **35% accounts remain inactive (2021)**. This is due to low trust, irregular income, and lack of confidence in operating accounts independently.
2. **Gender Disparities:** A significant gender gap persists in digital finance—**45% men vs 32% women** use digital payments. Social norms, lower literacy, and limited device access restrict women's financial participation.
3. **Rural & Geographic Barriers:** Remote areas still lack adequate banking infrastructure and connectivity. Low **rural smartphone penetration (~28%)** and poor network coverage limit access to digital financial services.
4. **Digital Divide:** Around **45% of Indians lack internet access**, creating a major barrier to fintech adoption. This divide is sharper among rural, elderly, and low-income populations.
5. **Low Financial Literacy:** Only **~27% Indians are financially literate (SEBI)**, leading to poor awareness about savings, credit, insurance, and digital safety, thus limiting effective usage of financial services.
6. **Limited Access to Formal Credit:** Formal institutions meet only **~19% of MSME credit demand**, while small farmers rely on moneylenders due to lack of collateral and credit history.
7. **Income & Informality Constraints:** With **~90% workforce in informal sector**, irregular and low incomes reduce the ability to save, invest, or consistently use formal financial services.
8. **High Transaction Costs:** Banking charges such as minimum balance requirements, ATM fees, and processing costs discourage low-income users from actively engaging with formal systems.
9. **Technological Barriers:** Frequent network issues, device limitations, and system failures reduce reliability of digital platforms, especially in rural and underserved regions.
10. **Trust Deficit & Documentation Issues:** Lack of trust in banks, fear of fraud, and complex KYC/ documentation processes discourage people from using financial services regularly.
11. **Regulatory Fragmentation:** Multiple regulators (RBI, NABARD, SIDBI) create coordination challenges, often termed **"regulatory cholesterol"**, affecting policy coherence and implementation efficiency.
12. **Regional Disparities:** Financial inclusion varies widely across states—regions like Bihar lag behind Kerala and Maharashtra due to lower income levels and weaker infrastructure.
13. **Cybersecurity Risks:** Rising digital transactions have increased exposure to fraud and cybercrime, weakening trust in digital financial systems among new users.

Way Forward:

1. **From Access to Usage:** The focus must shift from merely opening bank accounts to ensuring **active usage of savings, credit, insurance, and pension services** for meaningful inclusion.

2. **Strengthen Digital Infrastructure:** Expanding broadband connectivity through initiatives like **BharatNet** and improving mobile networks is essential for bridging the rural–urban digital divide.
3. **Enhance Financial Literacy:** Financial education should be integrated into school curricula and delivered through **SHGs, Panchayats, and NGOs** to build awareness and trust at the grassroots level.
4. **Targeted Policy Design:** Financial products must be tailored for **women, farmers, MSMEs, and informal workers**, addressing their specific income patterns and risk profiles.
5. **Gender-Focused Inclusion:** Promoting women-centric schemes, increasing female banking correspondents, and improving digital access can reduce gender gaps in financial participation.
6. **Innovative Credit Mechanisms:** Leveraging fintech for **collateral-free loans and digital credit scoring** can expand credit access for underserved groups like MSMEs and small farmers.
7. **Leverage Technology & AI:** User-friendly, multilingual apps and AI-driven solutions can simplify financial services, making them accessible to low-literacy and first-time users.
8. **Strengthen Consumer Protection:** Robust **cybersecurity frameworks, data protection laws, and grievance redress mechanisms** are essential to build confidence in digital finance.
9. **Public-Private Partnerships:** Collaboration between banks, fintech firms, and telecom companies can combine scale with innovation, improving last-mile service delivery.
10. **Credit-Plus Approach:** Financial inclusion should go beyond credit to include **skill development, entrepreneurship support, and community engagement** for sustainable impact.

Conclusion:

India's financial inclusion journey has achieved scale, but the next phase must ensure **depth, equity, and trust**, transforming access into genuine financial empowerment.

6. DECADE OF PM JAN DHAN YOJANA: BANKING THE UNBANKED

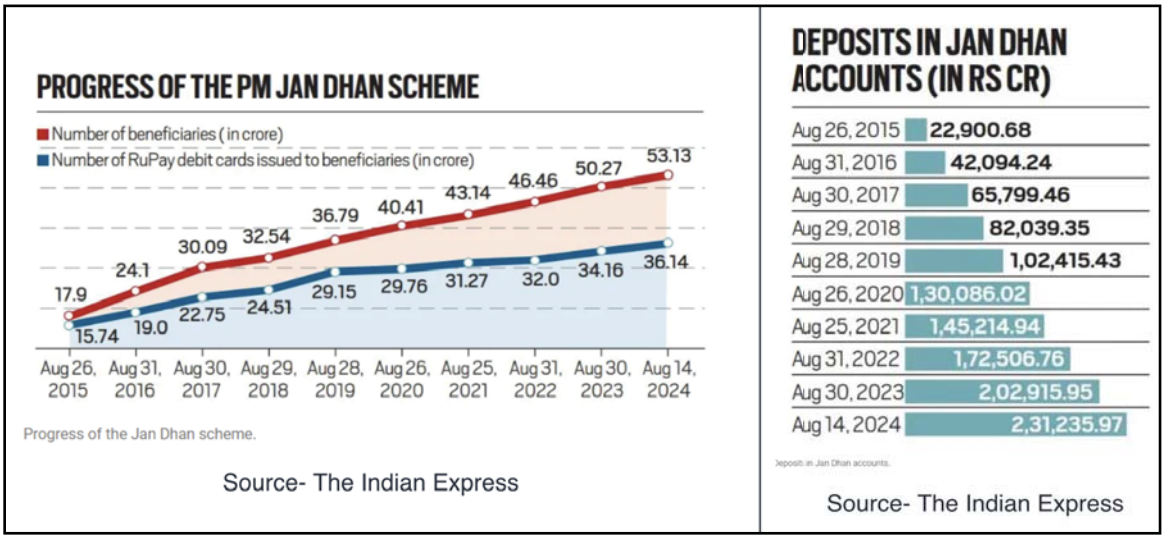
"Pradhan Mantri Jan Dhan Yojana has transformed access to financial services for the poorest. It has bridged the gap between banks and the unbanked, promoting dignity, self-reliance and economic inclusion."– PM Narendra Modi

Context:

The launch of the Pradhan Mantri Jan Dhan Yojana (PMJDY) on August 28, 2014 was a significant step by the government in its **concerted attempt to bring the unbanked sections of the population** into the ambit of the **formal financial system**. The scheme this year has completed 11 years. The Prime Minister held that the s**cheme was paramount in boosting financial inclusion** and giving dignity to crores of people, especially women, youth, and the marginalized communities.

Progress of Pradhan Mantri Jan Dhan Yojana (PMJDY)

As of the latest data, **53.1 crore accounts** have been opened under PMJDY, including **29.56 crore women beneficiaries**. The number of accounts opened exceeds the population of the European Union and is almost equal to the population of the United States.



Impact and Achievements:

1. **Banking the Unbanked:** Over **53.1 crore accounts** opened, increasing **financial inclusion**. As per **World Bank Findex (2021)**, **78% of Indian adults** had a bank account, up from 53% in 2014. Between 2014–17, account growth was **26%**, **4× global** and **3× developing countries' rate**.
2. **Narrowing Rural–Urban Gap: 67% of new accounts** are in **rural/semi-urban areas**, reducing disparity in **formal banking access**.
3. **Bridging Gender Gap:** Around **56% of new account holders are women**, improving **financial access for women**.
4. **Promotion of Direct Benefit Transfer (DBT):** As part of the **JAM Trinity (Jan Dhan–Aadhaar–Mobile)**, cumulative transfers of **₹38.5 lakh crore** have been enabled.
5. **Prevention of Financial Leakages:** DBT through PMJDY accounts has reduced fraud and exclusion errors, generating a gain of **₹3.48 lakh crore** (RBI report).
6. **Support in Times of Distress:** JAM architecture enabled **₹500 transfers to 20 crore women** during early pandemic days.
7. **Promotion of Digital Payments:** Integration with **UPI** has facilitated **low-value digital transactions**, promoting participation in the **digital economy**.
8. **Positive Externalities:** Access to banking has **smoothed rural consumption**, **reduced crime**, and lowered **alcohol/tobacco consumption** in states with higher PMJDY penetration.

Challenges:

1. **Account Dormancy & Inactivity:** Despite **53 crore accounts**, **~86.3% are operational**, leaving many accounts **dormant**.
2. **Use as Mule Accounts:** Risk of **money laundering** or **black money storage**, especially during **post-demonetization period**.
3. **Infrastructural Issues: Inadequate physical & digital infrastructure** in rural areas limits transactions (e.g., **Bihar & UP**).
4. **Technological Barriers: Poor internet connectivity** and weak banking technology impede effective account usage.
5. **Lack of Financial Literacy:** Beneficiaries are often **unaware of overdraft facilities** and **insurance cover**.
6. **Duplication of Accounts:** Multiple accounts under different schemes **complicate data management**.

7. **Exclusion of Certain Populations: Tribal and remote populations** remain underserved (e.g., **Chhattisgarh & Jharkhand**).
8. **Gender Disparity:** Women in **conservative rural areas** are less likely to **use accounts independently**, limiting **financial autonomy**.

Way Forward:

1. **Enhance Financial Literacy:** Launch **widespread campaigns** in partnership with **community leaders, NGOs, and schools** to increase awareness of **account features, overdraft facilities, and insurance coverage**.
2. **Promote Active Account Usage:** Incentivize **regular transactions** by linking accounts to **government schemes, subsidies, interest on savings**, or **digital cashback benefits**.
3. **Integrate Financial Services:** Combine **microcredit, pension, insurance**, and other **financial products** with Jan Dhan accounts to encourage **meaningful usage**.
4. **Strengthen Banking Infrastructure:** Expand **bank branches, ATMs, digital touchpoints**, and leverage **Business Correspondents (BCs) and mobile banking units** to reach **underserved rural and remote areas**.
5. **Address Technological Barriers:** Improve **internet connectivity** and deploy **robust digital banking solutions** to ensure **seamless access** for all account holders.
6. **Implement Monitoring & Feedback:** Establish mechanisms to **track account activity, detect misuse**, and collect **beneficiary feedback** for **continuous scheme improvement**.

Conclusion:

PMJDY has been **transformative** in promoting **financial inclusion**, **democratizing access to banking**, and enabling millions to participate in the **digital economy**. Sustaining its impact requires **ongoing efforts** to improve **literacy, infrastructure, and account usage**, ensuring the scheme evolves from **mere access** to **active, inclusive, and empowering participation** in India's **financial ecosystem**.

■■■

MGP/BP/GS3/010

INDUSTRIAL GROWTH

1. INDUSTRIAL POLICY AND NEW INDUSTRIAL POLICY IN INDIA

Industrial policy encompasses government measures—regulations, incentives, and investments—to shape industrial structure, promote sectors, and address market failures. In India, it balances public sector dominance (pre-1991) with private sector-led growth (post-1991), aiming for employment generation, technology transfer, and export competitiveness.

Evolution of Industrial Policy in India

Year/Policy	Focus & Features	Outcome/Limitations
Industrial Policy Resolution (IPR) - 1948	Mixed economy: Co-existence of public & private sector	Foundation for state intervention; limited growth
Industrial Policy Resolution (IPR) - 1956	Expansion of public sector, heavy industry focus; licensing; Small & Village Industries	Base for public sector; stifled private, created "license raj"
Industrial Policy Statement (IPS) - 1977, 1980	Increased stress on small-scale sector, some liberalization	Partial decontrol; slow industrial growth
1991 (New Industrial Policy)	Delicensing, liberalization, reduced public sector dominance, FDI facilitation	Major structural transformation, faster growth, global integration

Key Features of the New Industrial Policy (NIP), 1991

- **Abolition of License Raj:** Industrial licensing was eliminated for all but 18 industries, freeing private enterprises from bureaucratic controls.

Presently**, only 5 industries** compulsorily require **industrial licensing**:

- Distillation and brewing of alcoholic drinks
- Cigars, cigarettes, and manufactured tobacco substitutes
- Electronic aerospace and defense equipments of all types
- Specified Hazardous Chemicals such as Hydrocyanic acid, Phosgene, and Isocyanates.
- Industrial explosives such as detonating fuses, gun powder, nitrocellulose, and matches.

- **De-reservation of Public Sector:** Drastically reduced the number of industries reserved for the public sector (from 17 to only 8).

Presently, **only 2 sectors** are **reserved** for the public sector: **Atomic Energy and Railway operations.**

- **Private sector role enhanced:** Limits on size and expansion for big companies (MRTPA) removed
- **Foreign Direct Investment (FDI):** Automatic approval up to high thresholds, sector-wise FDI caps liberalized
- **Foreign Technology Agreements:** Allowed automatically in many sectors
- **Public sector reforms:** Reduced list of reserved industries, disinvestment policy initiated
- **Ease of Doing Business:** Regulatory simplification, unification of tax regimes over time (e.g., GST)
- **Export Promotion:** Trade liberalization; rupee devaluation (18-21% vs. USD) to boost exports; EXIM policy simplified.

Post-2014 "New Industrial Policy" Framework

The Post-2014 "New Industrial Policy" Framework of India is **market-driven with state support; PLI schemes (2020) as core,** emphasizing self-reliance amid COVID and global supply chain shifts.

Objectives of the Industrial Policy of India

- Stimulate Industrial Growth
- Generate Investments in the Private Sector
- Create a Propulsion for Technological Advancement
- Generation of EmploymentBalance of Regional Development

Some of the Policies interventions undertaken by India

- **Make in India (2014):** Aimed to transform India into a global manufacturing hub.
- **Startup India (2016):** Focused on promoting innovation and entrepreneurship.

Startup India

Launched on **January 16, 2016**, the **Startup India Initiative** has rolled out several programs **aimed** at supporting entrepreneurs, building a robust startup ecosystem, and transforming India into a country of job creators instead of job seekers.

India boasts the **third-largest startup ecosystem** in the world, with 1.91 lakh DPIIT-recognized startups as of 9 September 2025, which have created over 17.69 lakh direct jobs as of January 31, 2025.

This remarkable growth underscores the initiative's success in fostering innovation, enhancing employment opportunities, and driving economic development across the nation.

- **Production Linked Incentive (PLI) Scheme (2020):** Offers incentives to boost domestic manufacturing in key sectors.

- **Atmanirbhar Bharat (Self-Reliant India) initiative (2020):** Emphasizes self-reliance in various sectors.
- **National Manufacturing Mission (NMM):** Crown jewel of recent industrial policy, focusing on coordination, innovation, and sectoral clusters for manufacturing acceleration.

National Manufacturing Mission (NMM)

The **National Manufacturing Mission (NMM),** announced in the Union Budget 2025–26, is the crown jewel of India's industrial policy. It is a **long-term strategic roadmap** that integrates policy, execution, and governance into a single, unified vision. Unlike previous fragmented approaches, the NMM is **designed as a mission-mode framework** that cuts across ministries and states to create a synchronized manufacturing push.

Crucially, the NMM **places sustainability at the heart of industrialization**. It prioritizes clean-tech manufacturing from solar PV modules and EV batteries to green hydrogen and wind turbines ensuring **India's rise in global supply chains** aligns with its net-zero 2070 commitment.

- **National Logistics Policy**

National Logistics Policy

Launched in **September 2022**, it **aims** to overhaul India's logistics ecosystem by reducing costs, enhancing efficiency, and driving digital integration.

Working in tandem with the **PM GatiShakti National Master Plan,** the NLP sets ambitious **goals**: bringing down logistics costs, improving India's position in the World Bank's **Logistics Performance Index (LPI**) to among the **top 25 countries** by 2030, and building a robust, data-driven decision support mechanism.

To operationalize these objectives, the government has rolled out the **Comprehensive Logistics Action Plan (CLAP),** focusing on digital systems, standardization, human resource development, state-level coordination, and the creation of logistics parks.

Together, these initiatives are designed to **provide seamless multi-modal connectivity and strengthen India's competitiveness in global value chains.**

- **National Industrial Corridor Development Programme**

Industrialization and Urbanization

The **National Industrial Corridor Development Programme** is India's most ambitious infrastructure initiative, aiming to create new industrial cities as "**Smart Cities".**

This program **focuses** on developing integrated industrial corridors with robust multi-modal connectivity, promoting growth in manufacturing and systematic urbanization.

Last year, the government approved **12 new project proposals**, involving an estimated ₹28,602 crore investment, marking a significant step in this transformative effort, positioning India as a leading global destination for manufacturing and investment.

- **Districts as Export Hubs Initiative:** Cluster-based approach linking local strengths to global markets.

Impacts of Industrial Policies

- **Pre-1991 policies** achieved 5.6% industrial growth (1950-90) but stifled efficiency (public sector 70% output).
- **Post-1991:** Growth averaged 7% (1991-2010); **manufacturing exports** from $6 billion (1991) to $450 billion (FY25).
- **Make in India (MII) impacts:** After 10 years of the '**Make in India**', an 85% reduction in mobile imports and a 200% increase in manufacturing jobs from 2022 to 2024 is seen.

Challenges and Issues

Stagnant manufacturing share: Target of 25% by 2025-2030 unmet.
Skill Gap: Workforce-skills mismatch; need for greater skilling and STEM push.
Infrastructure: Logistical and power bottlenecks, land acquisition delays.
Innovation: R&D spending remains low (<1% of GDP, vs. 2–3% in E/SE Asia).
Regulatory bottlenecks: Despite improvements, still higher costs of doing business than global peers.
Regional imbalances: Some states/regions industrialize rapidly; others lag.
Slow technology adoption: Indian industries have been slow in adopting new technologies, leading to lower productivity and inefficiencies.
Sustainability & circular economy: Waste management, energy intensity, need for green transition.
Global Headwinds: Geo-economic fragmentation (GEF) reduced exports 3% FY24; protectionism (US tariffs) costs $1-2 billion annually.

Way Forward

- **Accelerate Infrastructure & Digital Transformation:** Complete corridor projects, massively expand digital manufacturing and supply chain monitoring.
- **Deepen Skill Ecosystem:** Universalize formal skilling, link training with technology adoption.
- **Boost R&D and Innovation:** Raise R&D spending to 2% of GDP, foster high-tech sectors and academia-industry collaboration.
- **Mainstream Green Industry:** Incentivize energy efficiency, waste recycling, and renewable energy adoption.
- **Sustainability Measures:** Green hydrogen mission (5 MMT by 2030); carbon credits for CBAM compliance (MoEFCC, 2025).
- **Export Promotion:** Implement targeted export promotion schemes for high-potential sectors.
- **Small and Medium Enterprise (SME) Support:** Expand credit guarantee schemes and promote alternative financing options like venture capital.
- **Global Benchmarks:** Strive for top-25 Logistics Performance Index, top-5 global exporter status by 2047: monitor competitive advantage and invest in new sectors (semiconductors, EVs, biotech).

India's trade policies in 2025

India revamped its **Foreign Trade Policy (FTP) in 2023** with an **aim** to increase its **exports to $2 trillion by 2030.** The FTP creates a framework that supports ease of doing business (**EoDB**), **reduces procedural bottlenecks**, facilitates **digitalisation** in trade and **boosts competitiveness** of Indian firms. The **goal** is to build a flexible and responsive structure that can be adjusted without waiting for fixed policy cycles.

Conclusion

Industrial policy for Viksit Bharat combines dynamic schemes (PLI, Make in India 2.0), smart infrastructure, regulatory reform, skill building, and sustainability thrust. Moving from fragmented and low-productivity structures to a resilient, job-rich, innovation-led manufacturing ecosystem, India is on track to become a global industrial leader, but must urgently address challenges in logistics, skills, regulations, and green transition to realize this ambition by 2047.

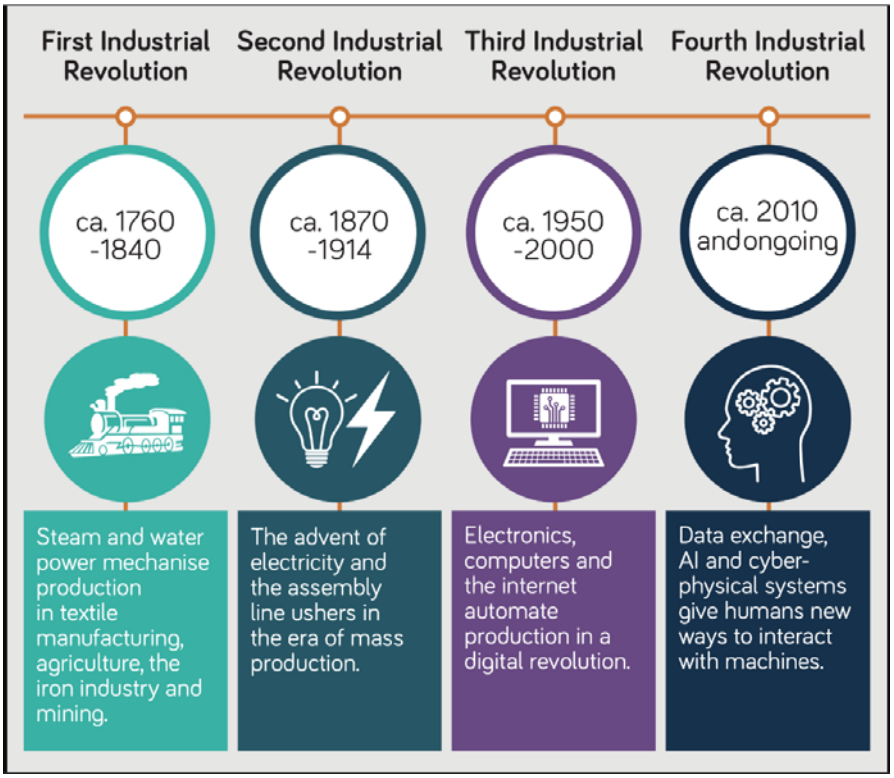
2. INDUSTRIAL REVOLUTION 4.0

Industrial Revolution 4.0 refers to the fourth industrial revolution related to manufacturing and chain production. It is driven by breakthroughs in digital technologies, such as artificial intelligence, robotics, 3D printing, the IoT, Big Data etc. It's not just about a single technology, but the fusion of digital, physical, and biological worlds, blurring the lines between them.

Introduction:

- The term was coined by Klaus Schwab, founder of the World Economic Forum (WEF) in 2016.
- 4IR means the digital transformation of the manufacturing industry by new technologies such as artificial intelligence, additive manufacturing, augmented/virtual reality, and the Internet of Things (IoT).
- It is also used to refer to the concept of "smart factories"–which are fully connected cyber-physical systems that merge the physical and digital aspects.

Stages of Industrial Revolution:



Technologies driving 4IR:

1. **AI & Machine Learning:** Artificial intelligence (AI) is a set of technologies that can reason and learn to solve problems or perform tasks that traditionally require human intelligence. It powers innovations like autonomous vehicles, advanced healthcare diagnostics and personalized customer experiences.

2. **Big Data & Analytics:** A significant underlying driver of the revolution, Big Data and analytics refers to collecting and analysing vast volumes of data to reveal trends and insights. Big Data is improving operational efficiencies and changing decision-making in business and wider society.
3. **IoT:** The Internet of Things (IoT) refers to the network of interconnected devices that exchange data over the internet. A key 4IR technology, IoT enables advances like industrial automation, smart homes and real-time health monitoring.
4. **Cloud Computing:** Cloud computing is the on-demand delivery of computing services like apps, storage and networking over the internet. Cloud services allow businesses to rapidly innovate and scale operations without investing in their infrastructure.
5. **Robotics & Automation:** Advanced robotics enables sophisticated robots, including humanoid assistants, to perform complex high-precision tasks independently with high precision. These robots can play vital roles in manufacturing, healthcare and customer service, often enhancing productivity and safety.
6. **AR & VR:** Augmented reality (AR) overlays digital images on the world around you. In contrast, virtual reality (VR) creates an alternative, computer-generated environment that you interact with. These technologies are set to provide immersive experiences for education, remote collaboration and customer service.
7. **Biotech & Nanotech:** Biotechnology uses living organisms, or their derivatives, to develop products or technologies. Nanotechnology refers to manipulating materials at a molecular level, creating new materials for medicine and engineering.
8. **Quantum computing:** Using the principles of quantum mechanics, quantum computers can operate at exponentially faster speeds than traditional computers. This enables us to solve complex problems in fields like material science and cryptography.
9. **Blockchain:** Blockchain is a decentralised ledger technology that ensures transactions aren't tampered with. Underpinning cryptocurrencies like Bitcoin, blockchain is also used for digital ID verification, smart contracts and supply chain management.
10. **3D Printing:** 3D printing uses computer-aided design to create three-dimensional objects by layering materials like plastics, composites or biomaterials. Also known as additive manufacturing, 3D printing revolutionises production by enabling rapid prototyping and customised manufacturing, reducing waste and costs.

Impact of 4IR:

Economy	• Integrating tech like AI, robotics and automation enables businesses to boost productivity and efficiency. This can lead to higher production, faster product innovation, enhanced customer service and reduced costs. • 4IR is set to drive innovation in products and services, transforming existing industries or creating new business models. Advanced manufacturing techniques will allow greater personalisation of products and services. • 4IR is set to displace some traditional manufacturing and routine white-collar jobs. However, they'll also create new jobs in high-tech industries, data analysis and cybersecurity.
Society	• To meet the demands of this new economy, education will have to adapt. Schools and higher education must focus on STEM education and lifelong learning to enable individuals for future jobs. • Advances in healthcare, personalised medicine and smart infrastructure will likely improve older people's life expectancy and quality of life. • Proliferation of internet-connected mobile devices and the rise of IoT gives us unprecedented connectivity. This enables near real-time data exchange globally, fostering collaboration and competition.

	• However, access to advanced technologies varies across communities and countries. Owners and investors in new technologies are poised to gain significant wealth. As a result, the rapid pace of technological change could exacerbate existing wealth and digital divides.
Government	• Governments will have new ways of interacting with citizens to address these challenges. For e.g. e-government services and smart cities are already used in the UK to enhance public services. • Emerging technologies can threaten fundamental freedoms. Across the globe, dictators find new ways to monitor and control their populations. And rapidly evolving technological innovation increases the risk of war and the means to wage it.
Environment	• Renewable energy, smart manufacturing and new sustainable practices can mitigate environmental impacts. • Together with new technologies like carbon emissions management, they can help us tackle climate change. • Increase in energy consumption from AI and cloud computing, electronic waste and the depletion of resources could degrade the environment if not managed properly.

Various government initiatives taken by India to promote 4IR:

1. In 2020, the Union Ministry of Heavy Industries launched the **Smart Advanced Manufacturing and Rapid Transformation Hub (SAMARTH)** scheme, which brings together manufacturers, vendors, and customers to make them aware of 4IR technologies.
2. In the 2022 Budget Speech, Finance minister Nirmala Sitharaman announced new 4IR-driven projects, including Drone Shakti, to encourage start-ups that will facilitate the use of drone services.
3. In the field of education, the **DESH Stack ecosystem** which will use blockchain technology to make the process of skill acquisition transparent and efficient would be announced.
4. India even has a 4IR centre in Mumbai run by WEF, which is closely working with several state governments.
5. The Centre has recently come up with the F**ourth Industrial Revolution for Sustainable Transformation (FIRST)** Cancer Care model in which 4IR technologies would be used to provide better healthcare for cancer patients.
6. India is also exploring digital twin technology (Creating a complex virtual model that is the exact counterpart of a physical thing) for creating models.
7. The pan-India 3D maps programme was launched for 100 smart cities so that 4IR-based projects, such as driverless cars, will become easier to implement.

Challenges of 4IR:

1. **Job Displacement & Workforce Disruption:**
   - Automation and AI threaten to displace routine and manual jobs, creating employment shifts and skill mismatches.
   - The need for reskilling and upskilling is urgent but challenging to implement at scale globally.
2. **Digital Divide & Inequality:**
   - Unequal access to technology and digital infrastructure risks exacerbating socio-economic and regional inequalities.
   - Vulnerable populations may be left behind due to lack of digital literacy and connectivity.

3. **Cybersecurity & Privacy Concern:**
   - Increased interconnectivity raises vulnerabilities to cyberattacks, data breaches, and misuse of personal data.
   - Balancing innovation with privacy rights and data protection regulations is a complex challenge.
4. **Ethical & Governance Issues:**
   - AI decision-making, surveillance technologies, and bioengineering raise profound ethical questions.
   - Existing legal and regulatory frameworks struggle to keep pace with technological advances, creating governance gaps.
5. **Environmental Impact:** Energy-intensive data centers, electronic waste, and demand for rare minerals raise sustainability issues.

Way Forward:

1. **Focus on Skill Development & Education:**
   - Launch large-scale reskilling and upskilling programs emphasizing digital literacy, AI, robotics, and emerging technologies for workers across all sectors.
   - Promote STEM education and lifelong learning to prepare future generations for the changing job market.
2. **Bridge the Digital Divide:**
   - Expand affordable broadband and digital infrastructure to underserved rural and marginalized communities.
   - Provide access to digital tools and training to ensure equitable participation in the digital economy.
3. **Develop robust cybersecurity frameworks:**
   - Strengthen national cybersecurity policies and capabilities to protect critical infrastructure and personal data.
   - Enforce data privacy laws aligned with global best practices while fostering innovation.
4. **Establish Ethical & Legal Frameworks:**
   - Create multi-stakeholder platforms to develop ethical codes of conduct for AI, biotechnology, and data use.
   - Update regulatory frameworks to keep pace with rapid technological advances and mitigate risks.
5. **Prioritise Environmental Sustainability:**
   - Integrate green technologies and circular economy principles in 4IR deployment.
   - Encourage energy-efficient data centers, sustainable sourcing of materials, and e-waste recycling.

Conclusion:

The 4IR is a complex and transformative period. It's a journey that will require adaptability and a focus on responsible innovation to ensure its benefits are shared widely.

3. MSME - SIGNIFICANCE & CHALLENGES

The Micro, Small, and Medium Enterprises (MSME) sector is a crucial pillar of the Indian economy, contributing significantly to industrial output, employment generation, and exports. In 2024-25, MSME-related products accounted for around **45.73% of India's total exports**. Often termed the backbone of the economy, MSMEs are instrumental in realizing the vision of Atmanirbhar Bharat (Self-Reliant India).

- **MSMEs**: These are enterprises engaged in the production, processing, and preservation of goods and commodities. These businesses are classified based on their investment in plant and machinery (for manufacturing) or equipment (for service enterprises), along with their annual turnover.
- **MSME Regulation in India**: MSMEs are regulated under the Micro, Small & Medium Enterprises Development (MSMED) Act, 2006. In 2007, the Ministry of Small Scale Industries and the Ministry of Agro

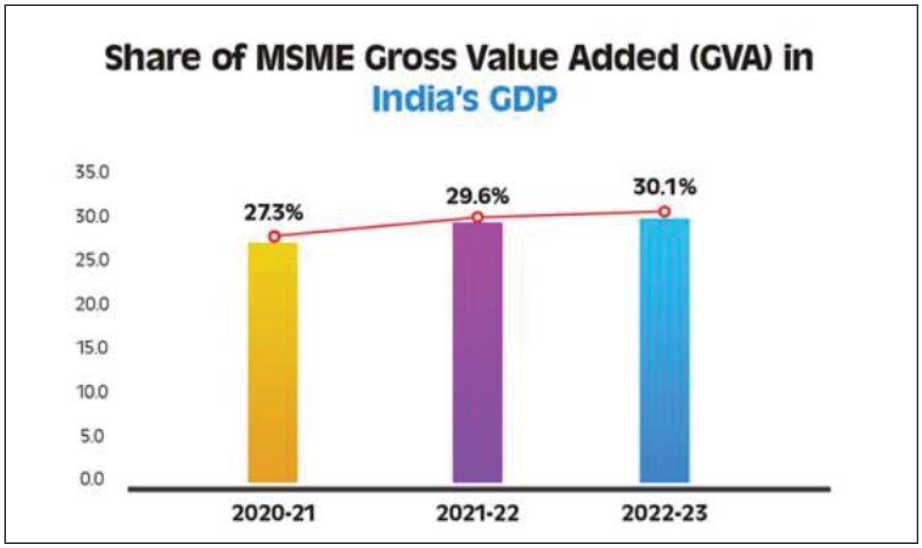
and Rural Industries were merged to form the Ministry of Micro, Small & Medium Enterprises.

Key Data on MSMEs in India

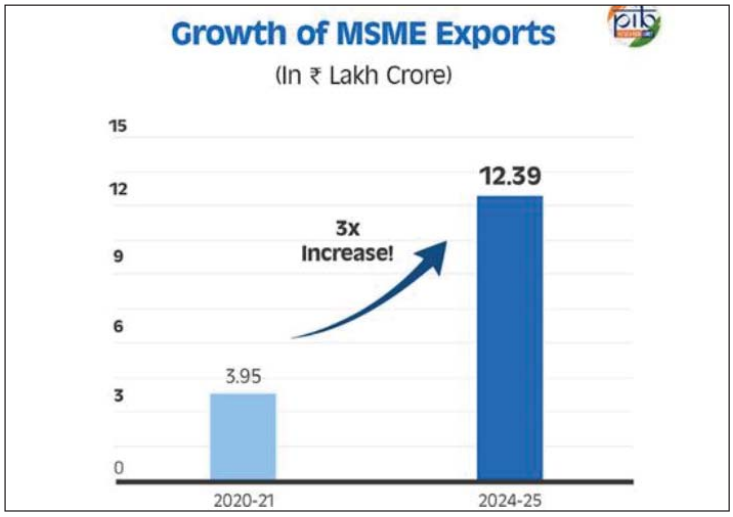
- **GDP Contribution:** Approximately 31.1% of India's total GDP.
- **Manufacturing Output:** Accounts for roughly 35.4% of the country's total manufacturing output.
- **Export Share:** Contributes 48.58% of total exports.
- **Employment:** Employs over 32.82 crore (328.2 million) people, making it the second-largest employer after agriculture.

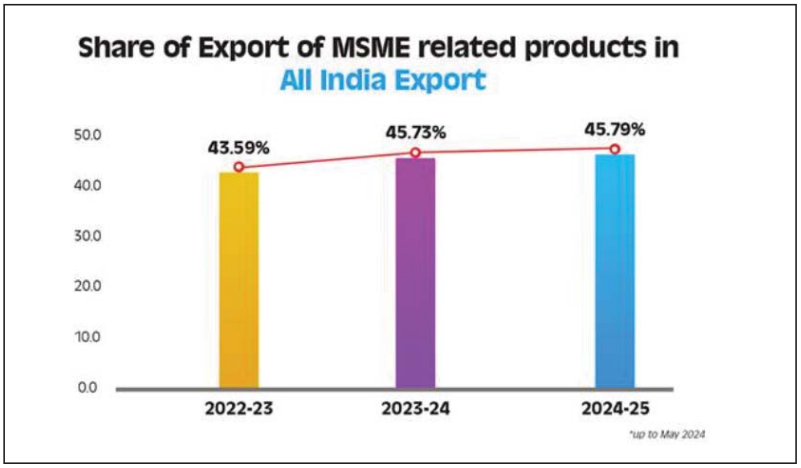
Role of MSMEs in India's economic growth

1. **Job Creation & Livelihood Opportunities:**
   - MSMEs are the largest source of non-agricultural employment in India, providing jobs primarily to semi-skilled and unskilled workers.
   - Government programs like PM Vishwakarma and Mudra Yojana have expanded self-employment prospects.
   - Currently, more than 1 crore registered MSMEs employ nearly 7.5 crore people across the country.
2. **Contribution to GDP & Industrial Development:**
   - MSMEs play a pivotal role in India's economic framework, accounting for around 30% of the GDP and 45% of the total exports.
   - They play a crucial role in industrial clusters by supplying raw materials and intermediates to large industries.
   - The Udyam portal has streamlined the formalization of MSMEs, promoting a more structured and organized industrial sector.



3. **Rural Development:** 51% of MSMEs are located in rural areas. In contrast to large corporations, MSMEs have aided in the industrialization of rural areas at a low capital cost. The sector has made significant contributions to rural socio-economic growth while also supplementing major industries.
4. **Enhancing Exports & Foreign Exchange Reserves:**
   - In 2024-25, MSMEs contributed around 45% of India's total exports, reinforcing their vital role in international trade.
   - Schemes such as the Government e-Marketplace (GeM) and the Production-Linked Incentive (PLI) program have strengthened MSMEs' integration into global supply chains.





5. **Advancing Digital & Technological Innovations:**
   - Initiatives like the Open Network for Digital Commerce (ONDC) are fostering digital transformation within the MSME sector.
   - Currently, 72% of MSME transactions are conducted digitally, enhancing financial inclusion and operational efficiency.
6. **Social Inclusion:** According to the Annual Report of The Ministry of MSMEs (2021-22), the socially backward groups owned almost 66.27% of MSMEs. In rural areas, almost 73.67% of MSMEs were owned by socially backward groups.

Statement No. 2.4: Percentage Distribution of enterprises by social group of owner in rural and urban Areas.

Sector	SC	ST	OBC	Others	Not known	All
Rural	15.37	6.70	51.59	25.62	0.72	100.00
Urban	9.45	1.43	47.80	40.46	0.86	100.00
All	12.45	4.10	49.72	32.95	0.79	100.00

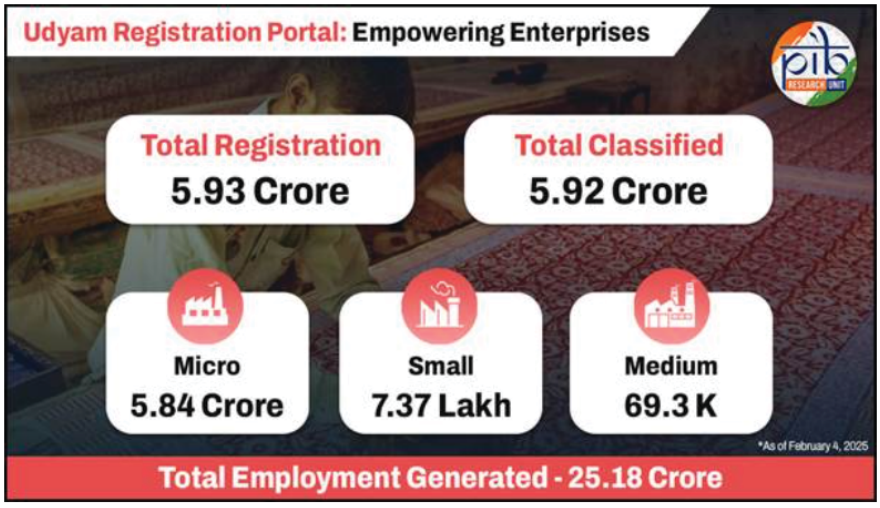
7. **Promoting Women & Social Entrepreneurship:**
   - Women-led MSMEs are contributing to gender equality and economic empowerment.
   - Under the Mudra Yojana, loans worth ₹32.36 lakh crore have been sanctioned, benefiting 51.41 crore entrepreneurs, with 68% of them being women.

Various government initiatives for MSMEs

1. **Revised Classification Criteria:** To help MSMEs scale operations and access better resources, the investment and turnover limits for classification have been increased by 2.5 times and 2 times, respectively.

Rs. in Crore	Investment		Turnover	
	Current	Revised	Current	Revised
Micro Enterprises	1	2.5	5	10
Small Enterprises	10	25	50	100
Medium Enterprises	50	125	250	500

2. **Udyam Registration Portal:** Provides simplified, online registration giving MSMEs formal recognition, essential for availing benefits across schemes.



1. **Enhanced Credit Availability:**
   - **Mudra Loan Scheme:** It was launched in April, 2015 for providing loans up to INR 10 lakh to the non-corporate, non-farm small/micro enterprises. It encompasses 3 financing loans: *Tarun* (loans up to INR 10 Lakhs), *Kishore* (loan up to INR 5 Lakhs), *Shishu* (loan up to INR 50,000).
   - **Credit Guarantee:** The credit guarantee cover for micro and small enterprises has been increased from ₹5 crore to ₹10 crore, enabling an additional credit infusion of ₹1.5 lakh crore over five years.
   - **Credit Cards for Micro Enterprises:** A customized Credit Card scheme to provide ₹5 lakh in credit to micro enterprises registered on the Udyam portal.
   - **Credit Guarantee Trust Fund for Micro and Small Enterprises (CGTMSE)**: It provides collateral-free credit to the micro and small enterprise sector.
   - **Special Credit Linked Capital Subsidy Scheme (SCLCSS)**: This scheme was launched to help the enterprises in the services sector meet various technology requirements. It also has a provision to grant 25% capital subsidy for procurement of plant & machinery and service equipment through institutional credit to MSMEs owned by SC/ST entrepreneurs without any sector specific restrictions on technology upgradation.
2. **Raising and Accelerating MSME Performance (RAMP)**: The scheme aims at strengthening institutions and governance at the Centre and State, improving Centre-State linkages and partnerships

and improving access of MSMEs to market and credit, technology upgradation and addressing issues of delayed payments and greening of MSMEs.

3. **Skill Development:**
   - **A Scheme for Promotion of Innovation, Rural Industry & Entrepreneurship (ASPIRE)**: The objectives of this scheme are to create new jobs, promote entrepreneurship culture in the country, and promote innovation in the MSME sector.
   - **Entrepreneurship and Skill Development Programmes (ESDP)**: Under this, the Ministry of MSME has been organising several programmes focussing on the process of improving skills and knowledge of entrepreneurs, and enhancing the capacity to develop, manage and organise a business venture.
4. **Scheme of Fund for Regeneration of Traditional Industries (SFURTI)**: The objectives are to organise traditional industries and artisans into clusters to make them competitive and provide support for their long-term sustainability, enhance marketability of products of such clusters, build innovative products, improve technologies etc.
5. **Entrepreneurship & Inclusion:**
   - **Stand-Up India:** Loans of ₹10 lakh–1 crore to support women and SC/ST entrepreneurs in setting up new businesses.
   - **First-Time Entrepreneurs Scheme:** Loans up to ₹2 crore and capacity-building for disadvantaged groups.
   - **PM Vishwakarma:** Focuses on enabling traditional artisans and craftspeople through training and access to markets, with special financial support.
6. **Technology & Quality Upgradation:**
   - **MSME Sustainable (ZED) Certification Scheme:** Encourages sustainable and high-quality manufacturing with incentives for adopting energy efficiency and zero-defect practices.
   - **Credit Linked Capital Subsidy Scheme (CLCSS):** Supports technology upgradation for MSMEs to make them globally competitive.
7. **Public Procurement Policy for MSMEs:** Mandates 25% of government and PSU procurement from MSMEs, giving them market access.

Digital Initiatives in the MSME Sector

- **Udyog Aadhar Memorandum**: It is a 1-page **online registration system** for MSMEs based on **self-certification**.
- **MSME Databank**: It enables the Ministry of MSME to **streamline and monitor the schemes** and pass on the benefits directly to MSMEs. MSMEs can update their enterprise information as and when required.
- **MY MSME**: It is a web-based application module in the form of a mobile app to facilitate the MSMEs to enjoy benefits of various schemes.
- **MSME *Sampark***: It is a digital platform wherein jobseekers (students or trainees of MSME Technology Centres) and recruiters can register themselves for mutually beneficial interactions.
- **MSME *Sambandh***: For effective implementation of the Public Procurement Policy, Central Ministries and Public Sector Enterprises (CPSEs) must procure 25% annual procurement from MSEs. The Ministry of MSME has launched MSME Sambandh.
- **MSME *Samadhaan***: This portal gives information about pending payments with the Central Ministries, State Governments, with respect to micro and small enterprises (MSEs).
- **MSME *Sambhav***: It is a national-level awareness programme to push economic growth by promoting entrepreneurship and domestic manufacturing.
- **Grievance Monitoring**: The Ministry of MSME has started an **MSME internet grievance monitoring system** (e-Samadhan) to help track and monitor the grievances and suggestions.

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Challenges faced by the MSME Sector

1. **Lack of Formalisation**: Almost 86% of the country's manufacturing MSMEs are unregistered. Only about 1.1 crore of the 6.3 crore MSMEs are registered with the Goods and Services Tax (GST) regime, and the number of income tax filers is even lower. As a result of limited availability and access to data, as well as legacy underwriting methods, the credit requirements of Indian MSMEs have largely gone unmet.
2. **Access to Technology**:
   - The majority of MSMEs use outdated technology that prevents them from keeping up with the modern world. Adoption of new technology and training employees is difficult and expensive, especially in manufacturing where both physical equipment and software are involved.
   - Lack of access to IT education contributes to the technological gap.
   - Another significant factor is a **lack of awareness**, which **reduces willingness to invest in advanced technology solutions**.
3. **Cumbersome Registration Process:** Lengthy registration procedures and inefficient single-window clearance systems create bureaucratic hurdles for MSMEs, delaying their formalization and operational setup.
4. **Lack of Awareness About Government Schemes:** Many MSMEs struggle to navigate the complex landscape of government schemes due to inadequate awareness and poor coordination between the Centre and States.
5. **Financial Constraints:** Limited access to institutional finance, stringent collateral requirements, and high interest rates make it difficult for MSMEs to secure the funds necessary for expansion and modernization.
6. **Export-Related Challenges:** Poor infrastructure, inadequate trade facilitation, and difficulty in meeting Environmental, Social, and Governance (ESG) standards restrict MSME participation in global markets.
7. **Labour Shortages and Skill Gaps:** A lack of skilled workers, wage disparities across regions, and inefficient training centers result in low productivity and reduced competitiveness for MSMEs.

Way forward

1. **Strengthening Credit Access & Financial Support:**
   - Expand collateral-free lending via fintech platforms and alternative credit assessment models.
   - Mandate strict payment timelines under the MSME Samadhan portal to ensure timely dues from large corporations and government agencies.
   - Enhance the role and reach of MUDRA and SIDBI to facilitate easier credit access.
   - Develop tailored fintech solutions and promote financial literacy programs for MSMEs.
2. **Reducing Compliance Burden & Regulatory Bottlenecks:**
   - Implement single-window clearance for faster approvals and streamlined processes.
   - Establish state-level MSME facilitation councils for quicker grievance redressal.
   - Simplify GST registration and reduce the frequency of regulatory amendments to ensure ease of doing business.
   - Set up a dedicated MSME Coordination Council to streamline policy implementation and address sector-specific challenges.
3. **Enhancing Digital & Technological Adoption:**
   - Facilitate adoption of AI, IoT, and automation through MSME technology hubs.
   - Launch Digital MSME 2.0 to improve cybersecurity, cloud computing access, and digital marketing capabilities.
   - Strengthen the Credit Linked Capital Subsidy Scheme (CLCSS) to support MSMEs investing in advanced technologies.

- Promote MSME Innovation & Incubation Hubs in collaboration with academic and research institutions.

4. **Workforce Development & Skilling Initiatives:**
   - Ensure uniform labor wages and introduce affordable labor insurance schemes for MSME employees.
   - Align skill training programs under Skill India, PMKVY**,** and NAPS to address MSME workforce needs.
   - Promote women entrepreneurs through targeted vocational training and credit linkage schemes.
5. **Infrastructure Development & Cluster-Based Growth:**
   - Strengthen MSME clusters under the Micro & Small Enterprises - Cluster Development Programme (MSE-CDP).
   - Encourage Public-Private Partnerships (PPPs) to set up Common Facility Centers (CFCs) for shared resources.
   - Develop MSME Industrial Parks with subsidized utilities, logistics support, and better last-mile connectivity.

Conclusion

MSMEs can play a vital role in growth of the economy as India enters the *Amrit Kaal* phase. They can help in inclusive and balanced development and make India a global manufacturing hub. The Government has been supporting the MSMEs through various initiatives, the need is to focus on the implementation and realizing the outcomes.

4. SEMICONDUCTOR INDUSTRY IN INDIA

Semiconductors- Semiconductors are materials which have a conductivity between conductors and insulators. They can either be pure elements, like **silicon or germanium** or compounds, like **gallium, arsenide, or cadmium selenide**.

Properties- Semiconductors have less resistivity than insulators and more than conductors. The resistance of the semiconductor decreases with an increase in temperature and vice versa.

Semiconductor Industry in India

Current Valuation (2025-26): Estimated between USD 45 billion and USD 55 billion.

2030 Projection: Expected to reach USD 100 billion – USD 110 billion.

2035 Projection: Forecasted to hit USD 300 billion.

Growth Rate (CAGR): Projected at 13% to 13.05% through 2030.

Importance of Semiconductor chips

1. Heart and brain of modern technology products- Semiconductor chips are the heart and brain of all **modern electronics and communications technology products**, **contemporary automobiles**, **household gadgets** such as refrigerators, and essential medical devices such as ECG machines.
2. **Propeller of emerging technologies-** The development of **emerging technologies like AI, 5G, or driverless cars** is dependent upon a fast and cheap semiconductor industry.
3. **Most traded products after petroleum-** Semiconductors are globally the most-traded products after petroleum and cars, with an **annual turnover** of **$500 billion**.
4. **Aiding the further development of electronic devices-** Semiconductors make the devices more compact, less expensive, and more powerful. **For ex-** Semiconductor chips have enhanced the features of smartphones with **powerful processing**.
5. **Transformative potential-** Semiconductors continue to enable the world's greatest breakthroughs and transformation in industries, ranging from **aerospace and consumer electronics** to **energy and medicine**.

Defining features of Semiconductor manufacturing

Semiconductor manufacturing has the following defining features-

1. **Front-end manufacturing and back-end assembly-** Semiconductor manufacturing comprises the **front-end fab manufacturing and the back-end assembly**, including packaging and testing. However, the front-end fab manufacturing is a complex process, with only a handful of companies involved in large scale fab-manufacturing.
2. **Complexity in manufacturing-** The semiconductor manufacturing is a complex process requiring at least **300 different high-technology inputs**.
3. **Highly concentrated global supply chain-** Globally, the entire semiconductor value chain has seeped in the interdependence between a handful of countries like the USA, Taiwan, Japan, China, and some European nations. Within this chain, there's an extraordinary degree of specialisation that makes it vulnerable to shocks.
   - **For ex-100% of the world's most advanced (below 10 nanometres) semiconductor manufacturing facilities** are located in just two countries, Taiwan and South Korea.
4. **Large investment-** The semiconductor product development requires the largest investment on both R&D and manufacturing. It is estimated that over the next decade about **$3 trillion in investment will be needed**.
5. **Revenue sharing b/w few companies-** The top three companies in each stage of the semiconductor supply chain take in about **80-90% of the revenue**.

Semiconductor manufacturing status in India

1. **Semiconductor R&D footprint in India-** India has an existing base for design and verification for the semiconductor industry. **Most of the global semiconductor companies have an R&D footprint in India**.
2. **Fabrication facilities limited to strategic semiconductor development-** India has two fabs — **SITAR**, a unit of the Defence Research and Development Organisation (DRDO) in Bengaluru, and a semiconductor laboratory in Chandigarh. These build **silicon chips for strategic purposes like defence and space**, and not for commercial use.

However, 100% of our semiconductor chips, memory, and display cards are imported into the country. In 2020, India spent $15bn on electronic imports, with 37% coming from China.

Global race to diversify semiconductor supply chains

1. **Over concentration of manufacturing units-** Semiconductor chip manufacturing is concentrated in a few countries. **For ex- Taiwan produces over 60% of the world's semiconductors**, and along with South Korea makes 100% of the most advanced chips (below 10 nanometers).
2. **Supply chain disruption due to epidemic-** The semiconductor industry experienced significant supply shortages during the pandemic due to **disruptions in China's production.**
3. **Geo-political conflicts-** The ongoing Russia-Ukraine conflict has resulted in shortages of raw materials for domestic industries. **For ex- Ukraine plays a critical role as a key supplier of neon,** which is an essential input in semiconductor manufacturing.
4. **Potential of another round of shortages due to emerging geo-political contestations-** The European Union and the United States has refused to sell advanced manufacturing equipment to China. In retaliation, **China imposed export controls on crucial inputs such as gallium and germanium**. This may lead to another potential round of shortages in the semiconductor industry.
   - **For ex-** US has restricted its firms and its allies from assisting the Chinese production of 16 nanometers or smaller chips.
   - Therefore, many countries are looking to diversify their supply chain. India seeks to take advantage of countries eyeing for diversification of supply chains.

Significance of development of indigenous semiconductor industry in India

1. **Tackling supply shocks-** The pandemic and the subsequent lockdowns impacted the supply of chips to India. **For ex- Automobile manufacturers like Mahindra & Mahindra and Tata group** were **compelled to reduce their production** due to the **shortage.**
2. **Meeting the rising demand-** Experts estimate that around **50 crore people will join the internet in the next decade, w**hich will increase the demand for more **phones and laptops.**
   - Similarly, the **post-pandemic world is showing a greater inclination towards work from home culture.** This warrants an enhanced demand for servers, internet connectivity, and cloud usage.
   - Hence, indigenous semiconductor industry is needed to meet the rising demands. There is a huge domestic market for semiconductors which could exceed **$60 billion by 2026.**
3. **Employment Creation-** Indigenous manufacturing of chips will build its smartphone assembly industry and strengthen its electronics supply chain. This will create **numerous employment opportunities for the Indian youth.**
4. **Revenue boost-** Indigenous capacity would attract local taxes and boost the export potential. Further, India would be required to import fewer semiconductor chips, which would decrease the import bill.
   - **For ex-** India imports almost all semiconductors to meet its demand, which is estimated to **reach around $100 billion by 2025.**
5. **Enhanced Security-** Chips made locally will be designated as **"trusted sources"** and can be used in products ranging from CCTV cameras to 5G equipment. This would improve the **national cybersecurity profile.**
6. **Geopolitical Benefits-** Countries having a sufficient supply of chips would be in a better position to mould the future course of geopolitics, driven by data and the digital revolution.
   - Further self-sufficiency will decrease reliance on Chinese chip imports, especially during hard times like the **recent Galwan Valley border clash.**
7. **Increased competitiveness-** Indian manufacturers **globally competitive** to attract investment in the areas of core competency and cutting-edge technology.

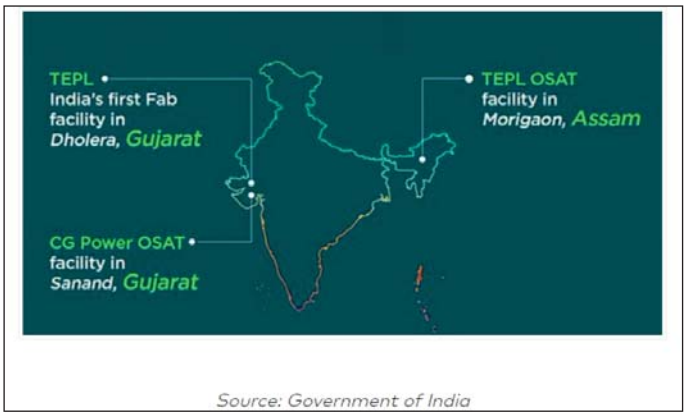
Government efforts towards development of semiconductor industry in India

National Policy on Electronics, 2019	It envisions positioning India as a **global hub for the Electronics System Design and Manufacturing (ESDM) sector.** It aims to **encourage** the development of **core components (including chipsets)** and create an enabling environment for the industry to compete globally.
Semicon India Programme	The government has approved the Semicon India programme with a total outlay of INR 76,000 crore for the **development of semiconductor and display manufacturing ecosystem in the country.**
'Modified Scheme for setting up of Semiconductor Fabs in India'	It aims to attract large investments for setting up semiconductor wafer fabrication facilities in the country. The Scheme extends a fiscal support of **50% of the project cost on an equal footing basis for setting up of Silicon complementary metal-oxide semiconductor (CMOS)** based Semiconductor fabrication in India.
Modified Electronics Manufacturing Clusters (EMC 2.0) Scheme	Under this, the government will **provide support for the setting up of Electronics Manufacturing Clusters (EMCs)** and **Common Facility Centres (CFCs).**
Foreign Direct Investment	The Government of India has allowed **100 percent (FDI)** under the **automatic route in the Electronics Systems Design & Manufacturing sector (ESDM).**

Production Linked Incentive Scheme(PLI)	Under this, the government will provide an **incentive of 4% to 6% on goods manufactured in India** and **covered under target segments to eligible companies for a period of five years.**

Semiconductor Plants in India

The semiconductor plants have been announced in **Sanand, Gujarat** (two units by Kaynes Semicon Pvt Ltd and CG Power), **Dholera, Gujarat** (one unit by Tata Electronics), and **Morigaon, Assam** (one unit by Tata Electronics).



Source: Government of India

Challenges in the semiconductor industry in India

1. **High Cost of establishment-** As per a government estimate, it would cost roughly **$5-$7 billion** to set up a chip fabrication unit in India.
2. **Bureaucratic inefficiencies-** The process of establishing an indigenous semiconductor facility requires clearances and approvals from multiple government departments. Further, there exists a **considerable degree of bureaucratic delays** at each stage that discourages the establishment of manufacturing units.
3. **Unstable power supply-** The smooth production of semiconductors requires the availability of an **uninterrupted 24*7 power supply**. However, this requirement is not fulfilled by many regions in the country. This restricts production to very few locations.
4. **Technological Constraint-** The indigenous manufacturing of semiconductors requires the **use of high-end technologies**. These technologies are licensed from patent holders at a very high price.
5. **Structural Flaws-** FDI in electronics is less than **1% of the total FDI inflow** because of the dearth of skilled labor, delays in land acquisition, and the uncertain tax regime.

Focus area for Indian Semiconductor Industry

R&D-intensive activities	R&D-intensive activities like **electronic design automation** (EDA), **core intellectual property** (IP), and **chip design**. The US is the leader in this segment. India can get part of the business by **supporting its existing chip-design experts** and **funding technology and innovation centres**, including top engineering colleges.
FABS facilities for advanced chips	India should focus on setting up of semiconductor fabrication (FABS) facilities for making of advanced chips.

Focus on medium and low-end chips	Due to the pandemic related **supply disruptions and tensions between the US and China**, the US, Japan, and many other countries have announced plans for **setting up local Fabs.** This might lead to a surplus capacity for high-end Fabs in the near future. Hence, India should focus on making medium and low-end chips.
Assembly, testing and packaging (ATP) segment	This segment captures **10% of the value**. China is the current leader. With low-cost skilled technical manpower, India is a natural choice to take some part of the business.

Way Forward

1. **Provision of adequate funding-** Adequate funding must be provided to augment the research and development potential of technical institutes. **For ex- IIT Madras developed a microprocessor named 'Moushik'** with funding support from the Ministry of Electronics and Information Technology.
2. **Expeditious execution of Sovereign Patent Fund (SPF)-** The **Sovereign Patent Fund (SPF)** under **National Policy on electronics** should be established expeditiously. It is a wholly or partly Government-backed entity that aims to bolster domestic businesses through the acquisition and licensing of patented technology.
3. **Domestic procurement assurances-** The manufacturers need to be given an assurance of minimum domestic procurement by the government and the private sector. The focus should be on **manufacturing economical and technically viable options like 28nm chips.**
4. **Support of businesses-** The government should also support businesses in the **acquisition of semiconductor manufacturing units in other countries.** This is easier than setting up a domestic facility and can be done swiftly for ensuring a continuous supply of chips. Hand-holding startups of entrepreneurial engineers can also produce large payoffs.
5. **Intial Focus on back-end of manufacturing- Semiconductor foundries are the world's most expensive factories, accounting for 65% of industry capital expenditure but only 25% of the value addition.** Therefore, to lower the risks of investment, India should especially look at back-end of manufacturing such as assembly, packaging and testing. Once it stabilises and an ecosystem develops, front-end of manufacturing will follow.
6. **Proactive cooperation of states-** Areas like **stable power, large quantities of pure water** and **land, are state subjects,** and the state governments should also create the right climate for easy implementation of semiconductor projects.

Conclusion

The 21st century will be an era of Digital revolution signifying an increased use of mobile phones and computer devices. This enhanced usage can be met only with a robust availability of semiconductor chips that sustains their functioning. Therefore India needs to focus on the indigenous development of semiconductors in order to realize its digital potential and emerge as a strong power in the present era.

5. PLI SCHEME

The Production Linked Incentive (PLI) Scheme is a flagship government programme designed to boost India's manufacturing sector by offering financial incentives tied directly to incremental production, sales, and investments. Its goal is to attract global and domestic manufacturers, promote self-reliance, and catalyze innovation and jobs.

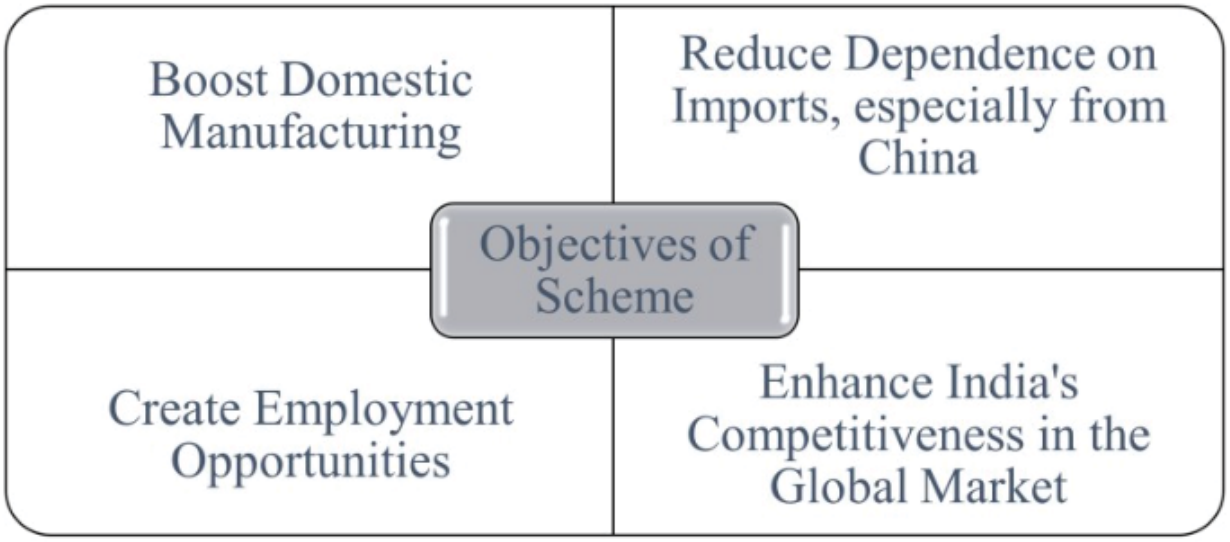
The PLI scheme is a crucial part of the Indian government's "Aatmanirbhar Bharat" (Self-Reliant India) mission, which aims to reduce the country's reliance on imports and establish it as a global manufacturing hub.

Introduction

- The PLI scheme was **launched in March 2020** to **strengthen India's domestic manufacturing sector** and improve its role in the global supply chain.
- The scheme **initially focused on three industries**: mobile and allied component manufacturing, electrical component manufacturing, and medical devices. It was **later expanded to cover 14 sectors.**
- In the PLI scheme, domestic and foreign companies receive financial rewards for manufacturing in India, based on a percentage of their revenue **over up to five years**.

Targeted sectors

Objectives of PLI Scheme:



Key Features of the PLI Scheme:

- **Incentives under PLI Scheme:**
  - Under the PLI scheme, eligible companies are provided financial incentives. The incentives given, are **calculated on the basis** of incremental sales.
  - Incentives range from 4% to 6% (or more in some sectors) of incremental sales, and are disbursed based on measurable production/output increases over base-year levels.
  - These incentives motivate companies to invest in improving their manufacturing capabilities, adopting advanced technologies, and expanding production capacities.
- **Incremental Production:** The incentive is not given on a company's total production but on the **increase in production** over a base year. This encourages companies to expand their manufacturing capacity and scale up their operations.
- **Investment Thresholds:** To be eligible, companies must meet certain investment and employment generation targets. This ensures that the scheme attracts serious players who are committed to long-term investment in India.
- **Global and Domestic Companies:** The scheme is open to both domestic and foreign companies, which has helped attract significant investments from global corporations like Apple's contract manufacturers (Foxconn and Wistron) and Samsung.
- As of 2025, over ₹1.97 lakh crore of incentives are sanctioned, with committed investments exceeding ₹1.76 lakh crore and actual investments touching ₹1.61 lakh crore.

Need for PLI Scheme:

1. **Low Share of Manufacturing in GDP:** Prior to PLI, India's manufacturing sector contributed only about 16% to GDP, much lower than leading economies like China, which limited India's ability to create jobs and build global competitiveness.
2. **Need to reduce import dependency:** India depended heavily on imported electronics, solar modules, APIs (pharma), and telecom equipment, posing risks to economic security and increasing the trade deficit.
3. **Lack of Scale, Technology, and R&D:** Domestic firms struggled to achieve world-class scale and lacked incentives to adopt advanced technologies, invest in R&D, and plug into global supply chains.

4. **Weak Value Chain Integration & MSME Linkages:** Manufacturing clusters and MSME networks in India were fragmented, restricting local value addition and diminishing multiplier benefits for jobs and regional growth.
5. **Need for Job Creation:** India's young population and rising workforce required large-scale job creation in productive sectors—manufacturing offers one of the highest employment multipliers.
6. **Vision for Aatmanirbhar Bharat:** The pandemic and global supply chain disruptions highlighted the need for robust self-reliance in critical goods—PLI aimed to turn India into a global manufacturing hub while reducing strategic vulnerabilities.

Achievements of PLI Scheme:

1. Over 764 companies approved under the PLI program, including 176 MSMEs, expanding manufacturing capacity and technological capabilities.
2. The scheme has resulted in clusters for electronics, semiconductors, medical devices, and textiles and spurred over 12 lakh jobs–direct and indirect.
3. Exports surpassed ₹4–5.3 lakh crore from sectors such as electronics, pharmaceuticals, food processing, and telecom—making India a global supplier of manufactured goods.
4. In telecom, India achieved 60% import substitution, and became a major exporter of 4G/5G equipment.
5. India is now a global exporter of mobile phones, solar modules, and pharmaceuticals under PLI, with substantial import substitution in telecom and electronics.
6. Benefits to MSMEs: MSMEs gain through anchor-multiplier effects—large approved units develop local supplier networks, benefiting MSME ecosystems in Tier II/III cities.

Challenges/Limitations of the PLI Scheme:

1. **Low Value Addition:**
   a. Even in successful sectors like mobile manufacturing, the percentage of local value addition remains in single digits.
   b. Critical components such as semiconductor chips and subassemblies are still imported.
2. **Limited Domestic Market Size:**
   a. Sectors like telecom and electronics do not generate sufficient demand to encourage large-scale local manufacturing.
   b.. Export-led growth is essential to drive economies of scale.
3. **Foreign vs. Domestic Players:**
   a. Indian manufacturers struggle with cost competitiveness compared to their Chinese and Vietnamese counterparts.
   b. Limited access to international markets and lower technological capability reduce India's bargaining power in the global supply chain.
4. **Dependency on Foreign OEMs (Original Equipment Manufacturers):**
   a. Currently, global players drive the manufacturing ecosystem, while Indian firms remain dependent on them for critical technologies.
   b. Lack of indigenous R&D investment in semiconductors, specialty steel, and advanced chemical cells remains a key hurdle.
5. **Regulatory & Implementation Bottlenecks:**
   a. Excessive bureaucracy, complex eligibility criteria, and slow fund disbursement have delayed scheme impact, especially for smaller firms and in certain states.
   b. Lack of standardization in incentive disbursal and a centralized database for tracking outputs and job creation reduces transparency.
6. **MSME Exclusion:** Disbursement favors large companies, with MSMEs struggling to compete and access benefits, even though they are critical for employment and supply chains.

7. **External Risks:** Geopolitical tensions, global supply chain disruptions, and trade restrictions could undermine export-led growth and import substitution objectives.

Way Forward:

1. **Deepen Domestic Value Addition:**
   a. Incentivize and monitor deeper local supply chain development, supporting component manufacturing, technology transfer, and R&D rather than just assembly operations.
   b. Introduce phased manufacturing programs and conditional incentives linked to value-added benchmarks.
2. **MSME Integration & Support:**
   a. Simplify eligibility, streamline application/disbursement processes, and provide technical hand-holding to MSMEs to ensure broader participation.
   b. Facilitate anchor-supplier relationships and create dedicated MSME clusters in high-impact sectors.
3. **Sector-specific Design & Course Correction:**
   a. Move from uniform incentive structures to sector-sensitive, graded designs—for example, higher initial support in capital-intensive or tech-deficient sectors, and sunset clauses for mature industries.
   b. Regularly review sector performance, prune non-performing areas, and scale up successful ones.
4. **Infrastructure & Skill Development:**
   a. Accelerate investment in logistics, energy, digital infrastructure, and establish common facilities/skill centres under the scheme.
   b. Promote industry-academia partnerships to build talent pipelines for emerging sectors.
5. **Reduce Regulatory Bottlenecks:**
   a. Digitize monitoring, create centralized dashboards for progress tracking, and automate fund disbursement to improve transparency and speed.
   b. Align policies across ministries and states for coordinated implementation.
6. **Prepare for External Risks:**
   a. Diversify export/geopolitical risk through product and market expansion.
   b. Build resilience in supply chains and foster strategic autonomy for critical sectors.

Conclusion:

The PLI scheme is a powerful example of a government policy that uses financial incentives to drive private sector investment and accelerate economic growth in strategically important sectors. It is considered a key policy for India's long-term economic transformation.

6. MAKE IN INDIA

On **September 25, 2014,** Prime Minister Narendra Modi launched the Make in India initiative to turn India into a global manufacturing hub. At a time of slow economic growth, the program aimed to boost manufacturing, attract foreign investment, promote innovation, and create jobs for India's workforce. Over the years, it has evolved into Make in India 2.0, covering 27 key sectors and focusing on stronger industrial capacity, modern infrastructure, and easier business processes.

Core Objectives

The initiative set out ambitious targets:

- Boost manufacturing growth to **12–14% per year**.

- Create **100 million industrial jobs by 2022**.
- Increase **manufacturing's share in GDP to 25% by 2022** (later revised to 2025).

These goals reflected a clear strategy to balance economic growth with employment generation and industrial modernization.

Make in India 2.0: Expansion and Focus

By 2021, Make in India evolved into **Make in India 2.0**, covering **27 sectors** in manufacturing and services. These sectors were chosen to use India's strengths and meet global demand. They include **aerospace and defense, automotive, electronics, textiles, chemicals, biotechnology, renewable energy, and pharmaceuticals**. The aim is not only to produce more but also to **promote innovation and high-value manufacturing**.

Pillars of Make in India

Make in India rests on four fundamental pillars:

New Processes	**Ease of Doing Business**: Reforms such as the Insolvency and Bankruptcy Code, GST, and the Jan Vishwas Act streamlined business operations and reduced bureaucratic hurdles. **National Single Window System**: Over 75,000 approvals have been granted, simplifying investment procedures and providing a seamless digital interface for entrepreneurs.
New Infrastructure	**Industrial Corridors & Smart Cities**: The government is developing industrial corridors and smart cities with advanced technology integration and high-speed connectivity. **PM GatiShakti**: A comprehensive infrastructure framework designed to accelerate logistics, connectivity, and industrial efficiency, supporting India's $5 trillion economy goal.
New Sectors	**FDI Liberalization**: Significant policy reforms have opened sectors such as defense production, insurance, medical devices, construction, and railways to international investment.
New Mindset	The state has shifted from a regulator to a partner, collaborating with industry to drive economic development and foster a culture of innovation.

Major Initiatives Enabling Make in India

1. **Production Linked Incentive (PLI) Schemes**: Covering 14 sectors with an outlay of ₹1.97 lakh crore, these schemes aim to boost domestic manufacturing and exports.
2. **PM GatiShakti**: Integrates multimodal connectivity and last-mile logistics to enhance industrial efficiency.
3. **Semiconductor Ecosystem Development**: The Semicon India Programme (2021), with a ₹76,000 crore investment, focuses on building a sustainable semiconductor and display ecosystem.
4. **National Logistics Policy (2022)**: Aims to streamline logistics networks for higher economic competitiveness.
5. **Startup India**: Supports entrepreneurship, job creation, and a vibrant startup ecosystem.
6. **Tax Reforms**: GST has unified India's tax structure, lowering production costs and improving compliance efficiency.
7. **Unified Payments Interface (UPI)**: With India handling 46% of global real-time payment transactions, it has positioned the country as a leader in digital payments.
8. **National Industrial Corridor Programme**: Developing 32 projects across 11 industrial corridors to create competitive, modern industrial regions.

Major Achievements under Make in India

1. **Defence and Aerospace**
   - **INS Vikrant**: India's first indigenously built aircraft carrier.
   - **Light Combat Aircraft (LCA) Tejas**: Successfully developed and exported to multiple countries, showcasing India's growing defense manufacturing capabilities.
2. **Electronics and Mobile Manufacturing**
   - India has become the second-largest mobile phone manufacturer globally, **producing 99% of smartphones domestically**.
   - Electronics manufacturing has grown from USD 48 billion in FY17 to over USD 155 billion in FY23.
3. **Pharmaceuticals and Medical Devices**
   - During the COVID-19 pandemic, India supplied nearly 60% of the world's vaccines, solidifying its role as the "pharmacy of the world".
   - The medical device sector has also seen significant growth due to domestic production expansion.
4. **Textiles and Apparel**
   - The sector has generated an estimated 14.5 crore jobs, making it a critical engine of employment and industrial growth.
5. **Toys and Consumer Goods**
   - India **produces 400 million toys annually**, with innovation driving 10 new toys every second, reflecting the potential of creative manufacturing.
6. **Export Growth**
   - **Merchandise exports reached $437.42 billion in FY 2024-25**, highlighting India's growing integration into global trade.

Challenges Related to the Make in India Programme

1. **Low Manufacturing Contribution to GDP**: India's manufacturing sector contributed approximately 12.53% to GDP in 2024, a decline from 13.02% in 2023. This remains significantly below the government's target of 25% by 2025.
2. **Declining Employment in Manufacturing**: Jobs in manufacturing fell from 51 million in 2017 to 35 million in 2023, showing that growth has not translated into sufficient employment opportunities.
3. **Falling Investment Rates**: India's productive investment rate dropped from 39.1% in 2008 to 32.2% in 2023, indicating that more investment is needed to expand industrial capacity.
4. **Uneven FDI Distribution**: Most FDI since 2017 has gone to just 9 sectors, mostly services, leaving 53 sectors with only 30% of FDI. Diversifying investment is crucial for balanced growth.
5. **Weak Export Performance**: Merchandise exports as a share of GDP fell from ~10% in 2013-14 to ~8% in 2022-23, highlighting the need to strengthen export competitiveness.
6. **Costly PLI Schemes**: High government spending on projects like the $2.75 billion Micron semiconductor plant in Gujarat raises questions about the long-term sustainability of such incentive schemes.
7. **Global Competitiveness**: India ranks 5th in the Global Manufacturing Index, behind countries like China and the USA, showing the need for better innovation, infrastructure, and skills.
8. **Skill Gaps**: Around 60% of India's workforce lacks the necessary skills for modern manufacturing jobs, limiting sector growth.

The Way Forward

1. **Improving Capabilities**
   - Adopt advanced technologies such as AI, machine learning, robotics, and **Industry 4.0 solutions** to enhance productivity and competitiveness.

- Strengthen the skill base through targeted training programs to ensure a workforce capable of handling modern manufacturing demands.

2. **Encouraging Entrepreneurial Spirit**
   - Create supportive ecosystems that empower educated workers and young entrepreneurs to start, scale, and sustain businesses.
   - Facilitate access to capital, mentorship, and incubation centers to nurture innovation-driven enterprises.
3. **Enhancing the Innovation Ecosystem**
4. Promote "triple-helix" collaboration among academia, industry, and government to drive research and development.
   - Support research initiatives through institutions such as the **Anushandhan National Research Foundation (ANRF).**
4. **Strengthening Domestic Supply Chains**
   - Develop resilient domestic supply chains to reduce import dependency and buffer against global shocks.
   - Encourage local sourcing and manufacturing clusters to improve efficiency and cost-effectiveness.
5. **Global Integration**
   - Foster international trade and investment partnerships, leveraging opportunities such as Chinese FDI, as highlighted in the **Economic Survey 2023-24.**
   - Promote India as a reliable global manufacturing hub through strategic diplomatic and economic engagement.
6. **Investment in Infrastructure**
   - Upgrade transportation networks, ports, logistics systems, and industrial corridors to facilitate smoother manufacturing operations and reduce supply chain bottlenecks.
7. **Monitoring and Evaluation**
   - Regularly assess program outcomes to identify bottlenecks, replicate successful initiatives, and recalibrate policies to achieve maximum impact.

7. SPECIAL ECONOMIC ZONES (SEZS)

Special Economic Zones (SEZs) are designated areas within a country where business and trade regulations differ from those in the rest of the country. In India, SEZs have been instrumental in promoting exports, attracting foreign direct investment (FDI), creating employment opportunities, and fostering industrial and economic growth.

As of March 31, 2024, India had 280 operational SEZs, contributing significantly to the country's export earnings.

What Are Special Economic Zones (SEZs)?

An SEZ is a specifically delineated duty-free enclave treated as foreign territory for the purposes of trade operations, customs duties, and tariffs. In India, SEZs are established under the **Special Economic Zones Act, 2005,** and governed by the **SEZ Rules, 2006**. These zones are designed to facilitate exports, attract foreign investment, generate employment, and improve infrastructure.

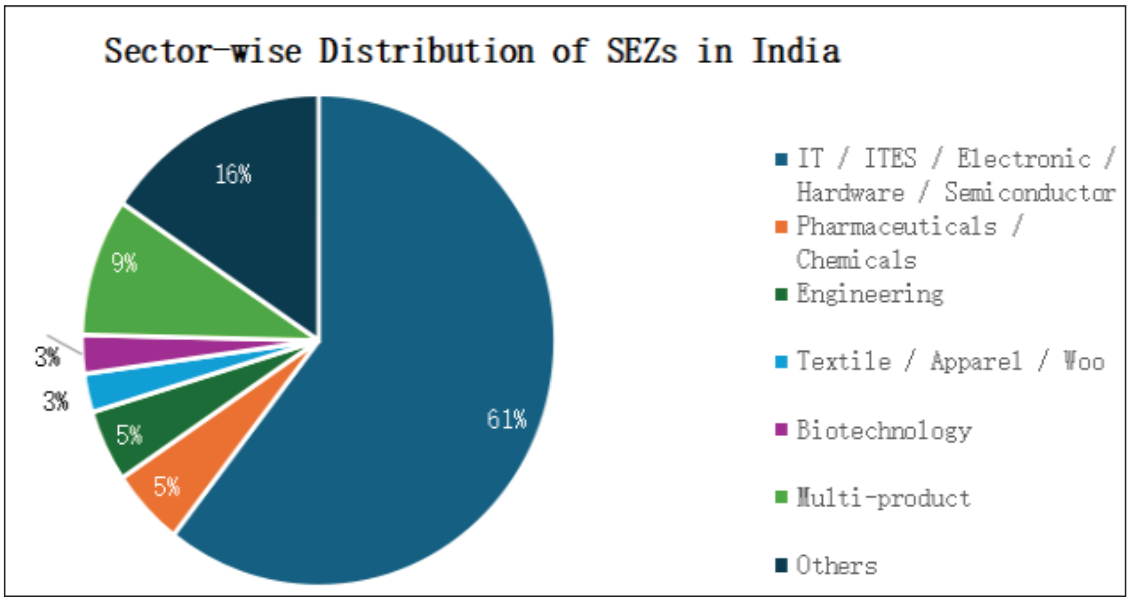
Objectives of SEZs

- **Promotion of Exports**: Providing a conducive environment for export-oriented industries to enhance India's global trade.
- **Attraction of Foreign Investment**: Offering fiscal incentives and infrastructure support to attract FDI.
- **Employment Generation**: Industrial development in SEZs creates job opportunities, contributing to economic growth.

- **Infrastructure Development**: State-of-the-art infrastructure, including roads, ports, power, and logistics facilities, supports industrial expansion.

Types of SEZs in India

The SEZ framework encompasses various zone types, including Export Processing Zones (EPZs), Free Zones (FZs), Industrial Estates (IE), Free Trade Zones (FTZs), Free Ports, and Urban Enterprise Zones. An example is the **Gujarat International Finance Tec-City (GIFT City).**



Historical Background of SEZs in India

- **First EPZ**: Asia's first Export Processing Zone (EPZ) was established in Kandla, Gujarat, in 1965, followed by seven more EPZs across the country.
- **SEZ Policy**: Introduced in April 2000, with the aim of making SEZs drivers of economic growth through quality infrastructure and fiscal incentives.
- **Conversion of EPZs**: The eight pre-existing EPZs in Kandla, Surat, Mumbai, Cochin, Chennai, Visakhapatnam, Falta, and Noida were converted into SEZs under this policy.

Incentives Provided to SEZ Units

- **Duty-free Imports**: Exemption from customs duties for imports used in the production of goods for export.
- **Tax Exemptions**: Income tax benefits on export income under Section 10AA of the Income Tax Act.
- **Exemption from Taxes**: Central Sales Tax, Service Tax, and State Sales Tax (now under GST).
- **External Commercial Borrowing (ECB)**: SEZ units can raise ECB up to USD 500 million per year through recognised banking channels.
- **Single-window Clearance**: Simplified approval processes from multiple government agencies.
- **No Import License Required**: Both manufacturing and service activities are permitted without additional import licensing.

Recent Reforms in SEZ Rules for Semiconductor and Electronics Manufacturing

To strengthen domestic manufacturing, especially in high-tech sectors like semiconductors and electronics, the government has introduced key reforms:

1. **Reduced Land Requirement**: Minimum land requirement for SEZs in semiconductor and electronics manufacturing has been reduced from 50 hectares to 10 hectares, lowering entry barriers.
2. **Relaxation in Land Encumbrance Rules**: The Board of Approval (BoA) can now relax the encumbrance-free land requirement if the land is mortgaged or leased to the Central/State governments or their agencies.
3. **Inclusion of Free-of-Cost Goods in Net Foreign Exchange (NFE)**: Free-of-cost goods can now be included in NFE calculations, improving the SEZ unit's performance metrics.
4. **Allowance for Domestic Sales**: SEZ units can supply goods to the Domestic Tariff Area (DTA) after paying applicable duties, reducing dependence on exports.

Challenges Faced by SEZs in India

1. **Declining Cost Competitiveness**: Rising costs of raw materials, energy, and logistics compared to countries like China and Vietnam
   - **Example**: Manufacturing costs in India are 20–30% higher than in Vietnam and China, affecting sectors like electronics.
2. **Land Acquisition and Infrastructure Issues**: High land prices, poor connectivity, and irregular utilities hinder development.
   - **Example**: The Dholera SEZ in Gujarat faced delays due to land acquisition disputes and lack of last-mile connectivity.
3. **Regulatory and Compliance Burdens**: Delays in approvals and high compliance costs impact startups and R&D firms.
   - **Example**: A report by the Ministry of Commerce (2022) highlighted that over 40% of approved SEZ projects faced regulatory delays exceeding two years.
4. **Limited Domestic Market Access**: High duties on DTA sales and competition from PLI-benefitted non-SEZ firms.
   - **Example**: Electronics SEZ units find it harder to compete with domestic PLI-supported mobile manufacturing units in Tamil Nadu.
5. **Underutilization of SEZs**: Gaps between approved, notified, and operational SEZs indicate poor viability assessments.
   - **Example**: Out of 578 approved SEZs, only 280 were operational by 2024, with some zones lying vacant for over a decade.
6. **Environmental and Sustainability Concerns**: Industrialisation often leads to pollution and resource depletion.
   - **Example**: The Falta SEZ in West Bengal faced environmental protests due to industrial effluents affecting the Hooghly river.

Government Measures to Revamp Special Economic Zones

1. Baba Kalyani Committee Report (2018)

The Ministry of Commerce and Industry constituted a committee chaired by **Mr. Baba Kalyani** to review India's SEZ policy and recommend strategic reforms. Key recommendations included:

- **Renaming SEZs as Employment and Economic Enclaves (3Es):** Shift focus from export-centric zones to employment- and growth-oriented hubs.
- **Delinking Incentives from Net Foreign Exchange (NFE):** Allow domestic supplies and payments in Indian Rupees without restricting operational flexibility.
- **Separate Frameworks for Manufacturing and Services SEZs:** Develop sector-specific policies to cater to the distinct needs of manufacturing and services units.
- **Infrastructure Development:** Create high-quality infrastructure, including expressways, ports, airports, warehouses, and integrated industrial-urban zones.

- **Ease of Doing Business (EoDB):** Implement integrated online portals for approvals, investment facilitation, operational management, and exit procedures.

2. Development of Enterprise and Services Hub (DESH) Bill, 2022

To make SEZs more flexible, inclusive, and globally competitive, the **DESH Bill, 2022** was introduced to replace the SEZ Act, 2005. Key objectives of the DESH framework include

- **Transforming SEZs into Development Hubs:** Support both domestic and international trade, integrating manufacturing, services, and research & development.
- **Reducing Regulatory Constraints:** Simplify approvals, compliance requirements, and operational procedures for SEZ units.
- **Promoting Integrated Zones:** Facilitate zones that combine industrial, service, and R&D activities for greater efficiency and competitiveness.
- **Encouraging Domestic Consumption alongside Expor**ts: Allow SEZ units to cater to domestic markets while maintaining export-oriented incentives.

Measures to Strengthen SEZs

1. **Policy and Regulatory Reforms**:
   - Replace SEZ Act, 2005, with the Development of Enterprise and Services Hub (DESH) framework.
   - Implement Baba Kalyani Committee recommendations for 3Es and sector-specific rules.
2. **Infrastructure and Connectivity Upgrades**:
   - Integration with industrial corridors and enhancement of last-mile connectivity to ports and airports.
   - Ensure 24/7 power supply, water recycling, and sector-specific cluster development.
3. **Enhancing Competitiveness and Global Integration**:
   - Provide R&D tax incentives and link SEZs with the National Research Foundation.
   - Strengthen export infrastructure through customs hubs and e-commerce zones.
4. **Skill Development and Employment**:
   - Establish Skill Parks near SEZs with ITIs/NSDC collaboration.
   - Incentivise local employment and provide special visas for foreign experts.
5. **Adoption of Global Best Practices**:
   - Emulate China's mega cluster Shenzhen model and Dubai's tax-free zones to enhance SEZ competitiveness.

8. DISCOM SECTOR IN INDIA

News- Power distribution companies (DISCOMs) in India have recently shown an improvement in their performance- recording a positive turnaround marked by **reduced Aggregate Technical & Commercial (AT&C) losses**, a **narrowed ACS-ARR gap** (**Average Cost of Supply** (ACS) and **Average Realizable Revenue (ARR))** & improved financial discipline. However, still many utilities continue to rely on tariff subsidies & loss takeover by the state governments – underscoring the scope for further improvement.

Brief Overview of Power Sector in India

Power generation, **transmission**, and **distribution** are the three main processes involved in the power sector.

Distribution is done by the **Distribution Companies (DISCOMs)** which **connect power producers to the households**. They are the interface between utilities and the consumers. There are a total **72 DISCOMs in India** (**44 State-owned**, **16 Private sector entities** & **12 Power departments**).

- Under the Indian Constitution, **power is a concurrent subject** and the responsibility for **distribution** and **supply of power** to rural and urban consumers **rests with the states**. Hence, DISCOMs are predominantly owned by the state governments.

- The Government of India provides assistance to states through various Central Sector / centrally sponsored schemes for improving the distribution sector. For many years, most power distribution companies in India have been incurring losses every year. Due to these accumulated losses and various other reasons the condition of DISCOMS in India is quite fragile.

Performance of DISCOMs

Financial Turnaround- After years of financial distress, these 72 DISCOMs have recorded a positive turnaround, with an aggregate Profit After Tax (PAT) of **Rs. 2,701 crore** in **FY 2024-25**, compared to a loss of **Rs. 67,962** crore in **2013-14**.

Improved Efficiency- Average Technical and Commercial (AT&C) losses have **declined** from **22.62% (FY14)** to **15.04% (FY25)**.

Outstanding Debt- Despite improvements, accumulated losses remained high, rising to **Rs. 6.47 lakh crore by 2024-25**, with outstanding debt at **Rs. 7.26 lakh crore**.

Challenges being faced by the DISCOMs in India

1 **High AT&C losses**

- The precarious financial position of DISCOMs is due to the **high level of aggregate technical and commercial (AT&C) losses**, the levy of **inadequate or lesser tariffs** when compared to the cost of power supply, and **insufficient subsidy support from state governments.**
- Their annual losses are estimated to be around 45,000 to 50,000 crore and the overall debt is around 6 lakh crore.
- The figure for AT&C loss in India, as per Ministry of Power, was **18-19%** in 2019. In countries such as the UK and US, it is about **6-7%**. However, in recent times, there has been an improvement as AT&C losses have been reduced to 15%.

Note: *AT&C loss reflects the loss due to energy loss during transmission and distribution (technical reasons), theft, and inefficiency in billing and commercial loss such as inefficiency in collection, and default in payment.*

2. **Determination of tariffs**- There are frequent delays in the determination of tariffs, which impacts the financial health of DISCOMs.
3. **Poor financial health (High ACS-ACR Gap)**
   - Consumer payment delays create a cash shortfall for Discoms, triggering a cycle of operational debt and unpaid dues to power generators.
   - This financial gap is filled by costly borrowings and subsidies, leading to higher interest expenses that are ultimately passed back to the consumer.

   Average Cost of Supply (ACS)- The expense incurred by a DISCOM to deliver a unit of electricity to consumers.

   Annual Revenue Requirement (ARR)- The revenue collected by the DISCOM as consumer tariffs and subsidy payments from the government.
4. **Lack of metering-** Even after 10 years of the launch of UDAY scheme, various levels in the distribution chain (the feeder, the distribution transformer (DT) and the consumer) have not been fully metered. This makes it difficult to isolate and identify loss-making areas and take corrective action.
5. **Emergence of alternative sources of energy and resultant decline in cross-subsidy tariff-** DISCOMs relied on **higher tariffs from commercial** and **industrial users** to subsidize agricultural and low-income households. However, these **high-paying consumers are shifting to alternatives like solar** due to **increased funding for clean energy** and **falling costs**. This migration is reducing DISCOM revenues and worsening their already weak financial position.
6. **Absence of political consensus at the state level to raise tariffs-** Many states report losses as they have not been able to eliminate the gap between power costs and revenue.

Implications of a weak DISCOM sector in India

A fragile state of DISCOMs in India will lead to various other problems which are listed below:

1. **Bailout Cycles-** Historically, states have had to take over **DISCOM debt** (like under the **UDAY scheme**). As of late 2025, outstanding DISCOM debt was approximately **₹7.26 lakh crore**, forcing states to divert funds from **healthcare**, **education**, and **infrastructure**.
2. **PPA Reluctance-** Many DISCOMs are locked into expensive, long-term **Power Purchase Agreements (PPAs)** with **old coal plants** and **are hesitant to sign new agreements for greener power**, even if it is cheaper.
3. **Payment Delays-** Financially stressed DISCOMs often **delay payments to renewable energy (RE) developers**. This **increases the risk profile of RE projects**, making it harder for developers to secure low-cost financing.
4. **Crowding Out Investment-** High borrowing costs for states (due to DISCOM liabilities) can **exceed the 3% GSDP fiscal deficit limit**, reducing the overall capital available for socio-economic development.
5. **Banking Sector Exposure-** Persistent DISCOM losses **create a systemic risk for public sector banks**, where power sector defaults contribute significantly to Non-Performing Assets (NPAs).
6. **Poor Service Quality & Vicious Cycle-** Financial weakness leads directly to **operational failure** as DISCOMs often resort to **load-shedding (power cuts)** to **avoid buying expensive power during peak hours**.
7. **Decarbonisation Bottleneck-** Weak DISCOMs struggle to meet India's target of **500 GW non-fossil fuel capacity by 2030**, as they are often **hesitant to sign new renewable energy agreements** while locked into **expensive**, **long-term coal-based contracts**.
8. **Delayed Project Execution-** Liquidity constraints **prevent DISCOMs from providing their share of funding** for central government modernization schemes, stalling national infrastructure goals.

Key initiatives taken by the Govt to address the DISCOM problem

Various steps have been taken by the govt to resolve the problems being faced by the DISCOMs:

UDAY Scheme	• Launched in November 2015, the Ujjwal DISCOM Assurance Yojana (UDAY) was designed to **turn around the financial position** of **state distribution companies** (DISCOMs). • The state governments took over **75% of the debt of their DISCOMs**, issuing lower-interest bonds to service the rest of the debt. • In return, **DISCOMs were given target dates (2017-19) to meet efficiency parameters** like reduction in power lost through transmission, theft and faulty metering.
Reforms-Linked, Result-Based Scheme for Distribution (RLRBSD)	• In budget 2021-22, the Union government had announced the launch of a "reforms-based and results-linked" scheme for **improving the financial health** and operational efficiency of discoms. • Under the scheme, AT&C losses will be brought down to **12-15%** by **2025-26**, from **21-22%**. • Operational efficiencies of discoms will be improved through **smart metering** and **upgradation of the distribution infrastructure**, including the segregation of agriculture feeders and strengthening the system.
Smart Metering Revolution	By late 2025, over **5.28 crore smart meters** have been installed across India. These help in **real-time billing**, **reducing theft**, and **enabling "prepaid" models** that ensure DISCOMs get paid upfront.

Electricity (Amendment) Bill	The Electricity (Amendment) Bill, 2025 is currently a major legislative focus aimed at modernising India's power sector- • **Cost-Reflective Tariffs-** It mandates that **tariffs must cover the actual cost of supplying power**, reducing the "ACS-ARR gap". • **Promoting Competition-** The bill proposes allowing multiple DISCOMs to **operate in the same area** using a **shared network**, effectively giving consumers **the choice to switch providers**- much like choosing a mobile network. • **Reducing Cross-Subsidies-** A plan to **phase out the practice of charging industries** more to **subsidize domestic users over a five-year period**.

Way forward

1. **Privatisation of DISCOMs-** Privatisation of DISCOMs has yielded positive results in many cities, including **Delhi**, **Mumbai**, **Kolkata** and **Ahmedabad**.

   Before it was privatised in 2002, **AT&C losses in the national capital were at a high 53%** and the **government was subsidising discoms** to the extent of **Rs 12,000 crore every year**. After privatisation losses came down, and today **Delhi has one of the lowest AT&C losses among DISCOMs** in the country at just **8 per cent**.
2. **DISCOM Restructuring-** While only **~10% of India's population is currently served by private distribution licensees**, **increased private participation** and **enhanced corporate governance** are **central** to **solving the DISCOM's chronic inefficiencies**.
3. **Regulatory Reforms-** The state governments should promote **autonomy, competence and transparency** of the **State Electricity Regulatory Commission (SERC)**. Regulatory reforms must aim at "**depoliticisation of DISCOMs"**.
4. **Operational Reforms-** Improve the billing efficiency of DISCOMs through **better** and **smart metering**.
5. **Increase share of solar power in power mix- A low-cost robust solar panel manufacturing industry** in India will lead to lower cost of power for DISCOMs. As the share of solar power increases, the cost for DISCOMs will come down.

Conclusion

Strengthening the **financial** and **operational health** of Power distribution companies in India (DISCOMs) is crucial for ensuring reliable and sustainable electricity supply in India. Effective reforms, improved governance, and initiatives such as the **Revamped Distribution Sector Scheme** can enhance efficiency, reduce losses, and support the country's long-term energy transition.

■■■

MGP/BP/GS3/011

EFFECTS OF LIBERALIZATION ON INDIAN ECONOMY

1. INDIA'S PRE-1991 ECONOMIC MODEL

From Independence (1947) to 1991, **India followed a mixed economy** model heavily influenced by **Fabian Socialism and Soviet-style** central planning. The dominant features were:

- **License Raj-** Industrial activity required government licenses at every stage — production, expansion, location, technology imports. Approximately 80 industries were reserved exclusively for the public sector.
- **Import Substitution Industrialisation (ISI):** High tariff walls and import restrictions to protect domestic industry, resulting in inefficiency and lack of global competitiveness.
- **MRTP Act, 1969:** Firms with assets above ₹100 crore classified as "MRTP firms" — unable to diversify without prior government approval, stunting private sector growth.
- **Financial Repression:** Interest rates set by RBI, not markets. Capital allocation through plan priorities, not returns.
- **Foreign Exchange Regulation Act (FERA), 1973:** Draconian restrictions on foreign exchange; treated violations as criminal offences.

Incremental Liberalisation (1980s)

Limited reforms began under Indira Gandhi (post-1980) and Rajiv Gandhi (1985–89): automobile sector opened to private players **(Maruti Suzuki, 1983);** import of capital goods partly eased; **Industrial Policy of 1985** tried to de-license some sectors. However, these were supply-side stimuli without correcting underlying structural distortions, resulting in growth financed by rising debt.

2. THE 1991 CRISIS- TRIGGERS & CAUSATION

Immediate Triggers

Immediate Trigger	Details
Gulf War (1990–91)	Oil prices spiked; remittances from Gulf fell sharply — India's key forex source dried up
Forex Reserves	Fell to **$1.2 billion** by June 1991 (barely 2–3 weeks of imports)
Fiscal Deficit	Reached **8.5% of GDP in 1990–91**; internal debt surged from **35% to 53% of GDP** between 1985–91

External Debt	Stood at **USD 70 billion**; debt service ratio critical
Inflation	Peaked at **17% in 1991**
Gold Pledge	India pledged **67 tonnes of gold with Bank of England and Union Bank of Switzerland** to raise $600 million as emergency collateral
Political Instability	Three governments in three years (1989–91) delayed decisive policy response

IMF CONDITIONALITY

India secured USD **2 billion from the IMF** under a Structural Adjustment Programme (SAP). The IMF conditioned the loan on sweeping economic reforms — devaluation of rupee, removal of import controls, reduction in fiscal deficit, and liberalisation of trade & investment.

"Let the whole world hear it loud and clear. India is now wide awake." — Dr. Manmohan Singh, Budget Speech, July 24, 1991

3. STRUCTURE OF REFORMS: THE LPG FRAMEWORK

The New Economic Policy (NEP) 1991 **rested on three pillars — collectively called LPG reforms:**

L — Liberalisation

- **Industrial De-Licensing:** Only 6 industries retained licensing requirements (strategic/hazardous: defence, atomic energy, etc.) which went down from 80+ industries.
- **Abolition of MRTP Threshold:** Pre-entry scrutiny for large companies abolished; companies free to expand
- **Deregulation of Interest Rates:** Commercial banks given freedom to set deposit and lending rates
- **Import Liberalisation:** Positive list replaced by negative list from 1992; intermediate and capital goods freely importable
- **Rupee Convertibility:** Made partially convertible on current account (1993); full current account convertibility by 1994 (Article VIII, IMF)
- **FERA → FEMA (1999):** FERA replaced by FEMA — foreign exchange violations treated as civil, not criminal offences

P — Privatisation

- **Disinvestment Policy:** Government began selling equity stakes in PSUs; National Investment Fund (NIF) created to channel disinvestment proceeds
- **Sick PSUs:** BIFR (Board for Industrial and Financial Reconstruction) established; unviable PSUs identified for closure or restructuring
- **Entry of Private Players:** Sectors like telecom, civil aviation, insurance, banking opened to private sector
- **Strategic Disinvestment:** Later govts pursued strategic sales — VSNL to Tata (2002), Balco to Sterlite.

G — Globalisation

- **Trade Policy Reform:** Import tariffs reduced progressively; average tariff fell from ~87% to under 15%
- **FDI Liberalisation:** Equity limit for foreign investment raised from 40% to 51%, then 74%, then 100% in many sectors

- **Export Promotion:** Export Processing Zones (EPZs) upgraded to Special Economic Zones (SEZs) under SEZ Act 2005
- **WTO Membership (1995):** India became founding member; committed to multilateral trade rules
- **Exchange Rate:** Moved from fixed to market-determined (Liberalised Exchange Rate Management System — LERMS, 1992)

4. EFFECTS ON GDP & MACROECONOMIC INDICATORS

GDP Growth Trajectory

Period	Average GDP Growth	Remarks
1950s–1970s	~3.5%	"Hindu Rate of Growth" — protectionist era
1980–1991	5.6%	Debt-driven growth; pre-reform partial liberalisation
1991–2003	6.2%	Post-reform transition; reforms starting to show effect
2003–2008	8.4%	Peak growth phase; "India Shining"
1991–2020 (avg)	6.7%	Nearly doubled compared to preceding three decades (3.5%)
2022–23	7.2%	Post-COVID recovery; world's fastest major economy

Other Key Macro Indicators Post-1991

Indicator	Pre-1991	Post-Reform
FDI Inflows	$97 million (1990–91)	$81.72 billion (2020–21); cumulative ~$1.12 trillion (2000–2025)
Forex Reserves	<$1.2 billion (June 1991)	$687 billion (2025) — world's 4th largest
Inflation	17% (1991)	Averaged 5.5% (2013–2023) under inflation targeting
GDP (Nominal)	~$270 billion (1991)	₹357.14 lakh crore (approximately $4.51 trillion) in FY 25-26 becoming 4th largest economy
Per Capita Income	~$300	~$2,600 (2024) — ~8× increase

Industrial & Manufacturing Sector

Positive Effects

- **End of License Raj:** Industrial production no longer constrained by government-set capacity limits; firms free to respond to market signals
- **Competition and Efficiency:** Domestic firms modernised technology to survive import competition; productivity in manufacturing rose by 6.8% annually between 1991–2006 (RBI study)
- **Automobile Industry Boom:** Entry of Hyundai, Honda, Ford, BMW etc. transformed India into a global auto-manufacturing hub; car prices fell dramatically for consumers
- **Capital Goods Import Freedom:** Industries freely import machinery and raw materials, reducing production costs
- **MSMEs:** Investment ceiling for SSI raised to ₹1 crore; later expanded further — enabled MSME ecosystem growth

- **FDI in Manufacturing:** Make in India, PLI schemes attract global supply chain relocation

Limitations & Concerns

- **Manufacturing's Share Stagnant:** Manufacturing as % of GDP remained around 12–14%; India skipped the manufacturing-led development path followed by East Asian economies
- **Premature Deindustrialisation:** Services grew faster than manufacturing — unusual for a developing economy at India's stage
- **Labour Law Rigidity:** Industrial Disputes Act, 1947 — requirement of government permission for laying off workers in firms with 100+ employees discouraged large-scale formal manufacturing
- **Land Acquisition Bottleneck:** Land Acquisition Act of 1894 (amended 2013) created protracted disputes, delaying industrial projects
- **Infrastructure Deficit:** Power, ports, roads lagged behind manufacturing needs

Critical Observation:

Unlike China and East Asian tigers, India **leapfrogged manufacturing to services**. While services productivity is high, manufacturing's inability to absorb low-skilled labour remains a structural deficit. Jobless growth in manufacturing persists.

Services Sector & IT Revolution

The IT/ITES Success Story

Liberalisation's most spectacular outcome was the rise of India's IT and software services sector. Key enablers:

- **Software Technology Parks (STPs) established** — tax benefits + fast data communication infrastructure
- **NASSCOM (est. 1988)** standardised practices and promoted Indian IT globally
- **Telecom deregulation** enabled internet backbone; private telecom operators entered post-1994 NTP
- **Rupee depreciation** made Indian IT exports price-competitive globally
- **Y2K crisis (1999)** gave Indian IT firms massive global exposure

Telecom Revolution

- **Mobile subscribers:** negligible pre-1991 → 1.1 billion+ today
- **Spectrum auctions, private operators** (Airtel, Vodafone, Jio etc.) created intense competition; call rates among world's lowest
- **Jio's 2016 entry democratised data access** — India became world's largest mobile data consumer

Financial Services

- **Private and foreign banks permitted to operate** — HDFC, ICICI, Axis Bank born in reform era
- **SEBI strengthened (1992)** — capital markets professionalised; BSE SENSEX from ~1,000 in 1991 to ~75,000+ in 2024
- **Insurance sector opened** to private players via IRDA (2000)

Agriculture

Negative Impacts on Agriculture

- **Import Competition:** Removal of agricultural import restrictions caused price crashes for domestic produce; farmers faced subsidised foreign imports
- **Subsidy Cuts:** Neoliberal conditionalities pressured subsidy reduction — fertilisers, water, electricity subsidies targeted; hit small farmers hardest
- **Farmer Distress & Suicides:** Debt trap, volatile commodity prices, low MSP coverage; NCRB data showed over 3 lakh farmer suicides between 1995–2015

- **Declining Agriculture's GDP Share:** Agriculture fell from ~35% of GDP (1990–91) to ~16% (2022–23), while still employing 40–45% of workforce — structural imbalance
- **Growth Rate Stagnation:** Agricultural sector grew at sluggish 2.5% annually post-1991 — far below industry (6%) and services (7%)
- **Withdrawal of Institutional Credit:** As banks went commercial, flow of cheap institutional credit to rural areas declined; farmers driven to informal money-lenders

Limited Positives

- India became world's largest milk producer, 2nd in fruits and vegetables — Green Revolution legacy + private dairy chains (Amul model)
- Agricultural exports rose — rice, spices, marine products — boosting some farmers
- Contract farming and agri-processing investments attracted private capital

Amartya Sen and Jean Drèze argue that liberalisation created a "two-speed India" — booming urban enclaves vs stagnant rural hinterland. The agricultural-industrial terms of trade worsened for farmers in the post-reform decades.

Foreign Trade & External Sector

Trade Expansion

- India's share of world merchandise exports rose from 0.5% (1991) to ~1.8% (2023)
- Total exports: $437.6 billion (2024); imports: $720.2 billion
- Trade-to-GDP ratio increased from ~15% to over 40% — deeper global integration
- Export diversification: services exports (especially IT) emerged as key earner

FDI & Capital Flows

- FDI: From $97 mn (1990–91) → $83 bn (2021–22); cumulative $1.12 trillion (2000–2025)
- Major FDI sources: Singapore, Mauritius, USA, Netherlands, Japan
- Key FDI-receiving sectors: Services, IT, telecom, pharmaceuticals, automobile
- FPI (Foreign Portfolio Investment): Capital markets opened; but volatile "hot money" creates exchange rate risks

Social Sector

Dimension	Positives (Post-Reform Achievements)	Negatives / Challenges
Poverty & Income	Poverty ratio fell from ~45% (1993–94) to ~21% (2011–12), lifting ~270 million people out of poverty	Rising inequality: Gini coefficient increased from 0.32 (1991) to 0.38 (2018); rural-urban income gap widened
Human Development	Improvement in literacy rates and life expectancy; HDI steadily rose	Public healthcare spending remains low at 1.5% of GDP (global average ~6%), limiting equitable access
Middle Class & Consumption	Expansion of middle class (~350–400 million in 2024); wide consumer choice; quality-of-life goods more affordable	Consumption-driven growth often concentrated in urban areas; rural and informal sectors lag behind

Education	Higher education enrollment rose from 6 million (1991) to 40 million (2023), boosting skilled workforce	Skills mismatch persists; access and quality still uneven across regions
Employment & Workforce	Economic growth created diverse opportunities, particularly in services and tech sectors	~90% of workforce remains informal; high GDP growth not translating into proportionate formal job creation ("jobless growth")
Lifestyle & Standard of Living	Significant rise in living standards, better access to goods, services, and technology	Benefits unevenly distributed; large segments of population still face deprivation

Employment

- Initially: Liberalisation accelerated employment generation in IT, telecom, retail, finance
- Over time: Employment growth slowed; manufacturing failed to absorb the demographic dividend
- Informalisation: Globalisation pressured firms to reduce permanent workers; contract/casual workers increased — "informalisation of the formal sector"
- Unemployment Rate: CMIE data shows youth unemployment (~23%) and graduate unemployment as persistent concerns

Regional Disparities

- **FDI concentrated** in Maharashtra, Karnataka, Delhi NCR, Tamil Nadu, Gujarat
- Eastern and central states (Bihar, Jharkhand, UP, Odisha) received minimal FDI
- IT hubs (Bengaluru, Hyderabad, Pune) flourished while BIMARU states stagnated

5. SECOND & THIRD GENERATION REFORMS:

Second Generation Reforms (2000s–2010s)

- Fiscal Responsibility & Budget Management Act (FRBM), 2003: Institutionalised fiscal discipline; targets for deficit reduction
- VAT Reform: Replaced sales tax with state-level VAT (2005) — precursor to GST
- Competition Act, 2002: Replaced MRTP Act; established CCI to promote market competition
- SEZ Act, 2005: Export-oriented manufacturing zones with tax incentives
- NREGA, 2005: Largest employment guarantee scheme — attempt at inclusive redistribution
- RTI Act, 2005: Transparency reform to reduce rent-seeking in liberalised economy

Third Generation Reforms (2014–present)

- **GST (2017):** Unified indirect tax; simplified compliance, widened tax base, boosted formalisation
- **IBC (2016):** Insolvency & Bankruptcy Code — time-bound resolution of NPAs; major financial reform
- **Corporate Tax Cut (2019):** Base rate cut from 30% → 22%; new manufacturing firms: 15% — to attract investment
- **PLI Schemes (2020–):** Production Linked Incentives in 14 sectors to boost manufacturing — ₹1.97 lakh crore outlay
- **Labour Codes (2019–20):** 44 labour laws consolidated into 4 codes — intended to ease hiring
- **India Stack & UPI:** Digital public infrastructure democratising financial access

- **PM GatiShakti:** National Master Plan integrating multi-modal infrastructure
- **FDI Further Liberalised:** FDI allowed in defence (100%), insurance (74%), retail (single-brand 100%)

Critical Analysis

What Worked:

- BoP crisis resolved; never repeated since 1991 — key stabilisation success
- IT/services sector became global leader — $254 bn industry
- Middle class creation and consumption-driven growth
- Foreign exchange reserves from $1.2 bn → $687 bn
- GDP growth nearly doubled in post-reform era
- Financial markets deepened; capital allocation improved

What Didn't Work / Incomplete Reforms

- **Labour Market Reform:** Rigid laws prevented manufacturing scale-up; informalization increased
- **Land Reforms:** Land acquisition remained contentious; industrial land availability constrained
- **Agricultural Liberalisation:** APMC Acts, MSP regime, farm subsidies unreformed — structural distortions persist
- **Public Health & Education:** Liberalisation did not improve state capacity in social services; public spending remained low
- **Environmental Cost:** Industrialisation without adequate regulation; Delhi's air quality, Yamuna pollution, deforestation
- **Ease of Doing Business:** Despite improvements, India ranked 63rd in World Bank Doing Business Index (2020)
- **Crony Capitalism:** Privatisation without strong regulatory institutions enabled rent-seeking and capture

Way Forward:

- **Inclusive Liberalisation:** Reforms must extend to agriculture, education, health — not just industry and finance
- **State Capacity:** Markets work best with strong regulatory institutions — SEBI, CCI, RBI need strengthening, not weakening
- **Employment-First Manufacturing:** Labour-intensive sectors (textiles, leather, food processing) need PLI-scale support
- **Regional Equity:** Fiscal transfers, targeted investment in lagging states to address spatial inequality
- **Social Protection Floor:** Universal health coverage, pension for informal workers — market failures in welfare need state intervention
- **Sustainable Liberalisation:** Environmental regulations must accompany economic openness

Conclusion

Liberalisation **was necessary but not sufficient.** It created the conditions for growth, but growth without equity, without manufacturing employment, and without agricultural reform remains structurally fragile. India's **challenge is to move from liberalisation 1.0 (remove controls) to liberalisation 2.0** (build institutions, invest in human capital, and make growth inclusive)

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MGP/BP/GS3/012

ENVIRONMENT AND ECOLOGY

1. CORAL BLEACHING

The Coral Triangle, also known as the "Amazon of the seas," faces mounting extinction dangers due to growing carbon emissions, destructive fishing, air, water, and soil pollution, and the accelerating effects of climate change, which are all driving coral bleaching. In one form of resistance against these threats, the Philippines is preparing to host Southeast Asia's first coral larvae cryobank to help restore and protect these coral reefs.

Coral and Coral Reefs

Corals- Corals are small (0.25-12 inches), soft-bodied marine invertebrates (not possessing a spine) which **belong to the group cnidaria**. These are **sessile animals** which permanently attach themselves to the ocean floor.

Each individual coral animal is known as a **polyp** and it lives in groups of hundreds to thousands of genetically identical polyps that form a **'colony'**. They **use their tiny tentacle-like hands to catch food** from the water and sweep into their mouth.

Mutualistic relation between Corals and Algae

These coral polyps have **microscopic algae called zooxanthellae** living within their tissues. The corals and algae have a mutualistic relationship. The coral provides the zooxanthellae **with the compounds necessary for photosynthesis.** In return, the **zooxanthellae supply the coral with organic products** of photosynthesis, like **carbohydrates,** which are utilized by the coral polyps for **synthesis of their calcium carbonate** skeletons. In addition to providing corals with essential nutrients, **zooxanthellae** are responsible for the **unique and beautiful colours of corals.**

Types of Corals

There are two types of corals- Hard coral and soft coral.

a. **Hard corals-** These corals also called **hermatypic** or '**reef building'** corals which extract **calcium carbonate from the seawater** to build **hard, white coral exoskeletons**.
b. **Soft Corals-** Soft corals borrow their appearance from plants, and attach themselves to such skeletons and older skeletons built by their ancestors. Soft corals add their own skeletons to the hard structure over the years and these growing multiplying structures gradually form coral reefs. They are the **largest living structures on the planet.**

Coral Reefs in India

Coral reefs in India are located in seven regions namely **Goa coast, Kerala coast, Palk Bay, Gulf of Kutch, Gulf of Mannar, Lakshadweep islands** and **Andaman and Nicobar islands.**

Significance of Corals

- **Rainforests of the sea-** Coral reefs **support over 25% of marine biodiversity,** including fish, turtles and lobsters, while taking up 1% of the seafloor.

  For ex- The Great Barrier Reef contains over **400 coral species, 1,500 fish species, 4,000 mollusc species** and **six of the world's seven sea turtle species.**
- **Economic value-** Coral structures provide **economic goods and services worth about $375 billion each year**. More than **500 million people** across the world depend on coral reefs for food and income.
- **Coastal protection-** Coral reefs absorb up to **97% of the energy from waves, storms, and floods,** which prevents loss of life, property damage, and soil erosion. The absence of coral reefs results in severe ramifications for marine life and humans.

Coral Bleaching and Mass Coral Bleachings

Coral Bleaching- When **corals are stressed** by **changes in conditions** such as **temperature, light,** or **nutrients**, they expel the symbiotic algae **(zooxanthellae)** living in their tissues, causing them to turn completely white. This is called **coral bleaching**.

Coral Death- Bleached corals can survive depending on the levels of bleaching and the recovery of sea temperatures to normal levels. If heat-pollution subsides in time, over a few weeks, the zooxanthellae can come back to the corals and restart the partnership. But in cases of **severe bleaching and prolonged stress in the external environment** can lead to **coral death.**

Mass Coral Bleaching Events

First Mass Bleaching	It occurred in **1998** when the El Niño weather pattern caused sea surfaces in the **Pacific Ocean to heat up.** This event led to the **death of 8% of the world's coral.**
Second Mass Bleaching	This event took place in **2010.** It led to increased **destruction of 35% coral reefs.**

Third Mass Bleaching	The event took place between **2014-17** and affected reefs in Guam in the **Western Pacific region, the North, South-Pacific,** and **the Indian Ocean.** It affected 56% of coral reefs around the world.

Fourth global bleaching event is currently underway. Nearly 54 countries, territories and local economies- from Florida, the US, Saudi Arabia to Fiji- have confirmed bleaching. Bleaching has been confirmed in the **Western Indian Ocean,** including **Tanzania, Kenya, Mauritius, Seychelles**, and **off the western coast of Indonesia.**

Causes Responsible for Coral Bleaching

- **Rise in Sea Temperature-** The rise in global warming has led to increase in the sea temperatures which has resulted in the destruction of corals.
- **Ocean Acidification-** Due to rise in carbon dioxide levels, oceans absorb more carbon dioxide. This **increases the acidity of ocean water** and **inhibits the coral's ability to create calcareous skeletons,** which is essential for their survival.
- **Solar radiation and ultraviolet radiation- Changes in tropical weather patterns** result in less cloud cover and more radiations which induce coral bleaching.
- **Infectious Diseases-** Penetration of bacterium like **vibrio shiloi** inhibits **photosynthesis of zooxanthellae.** These bacteria become more potent with elevated sea temperatures.
- **Chemical Pollution- Increased nutrient concentrations affect corals by promoting phytoplankton growth,** which in turn supports increased numbers of organisms that compete with coral for space.
- **Human Induced Threats- Over-fishing, pollution from agricultural and industrial runoff, coral mining, development of industrial areas** near coral ecosystems also adversely impact corals.

Consequences of coral bleaching

- **Affecting the Food Chain-** Changes in coral communities can **affect the species that depend on them, such as the fish and invertebrates** that rely on live coral for food, shelter. Loss of such marine animals can disturb the entire food chain.
- **Loss of Biodiversity-** Declines in genetic and species diversity occurs when corals die as a result of bleaching.
- **Economic Decline-** Bleached and degraded reefs can discourage tourism, which can affect the local economy.
- **Affects Food Availability-** Coral bleaching can **cause large shifts in fish communities.** This can translate into reduced catches for fishers, which in turn impacts food supply and associated economic activities.
- **Impact on Coastal Protection-** Coral reefs protect coastlines by absorbing constant wave energy from the ocean, thereby protecting people living near the coast from **increased storm damage, erosion** and **flooding.**

Initiatives to Protect Corals

A number of global initiatives are being taken to address the issues-

Institution of International organisations	The International Coral Reef Initiative, Global Coral Reef Monitoring Network (GCRMN), Global Coral Reef Alliance (GCRA) and The Global Coral Reef R&D Accelerator Platform have been established to deal with the issue.

Biological restoration	It uses methods for coral growth and transplanting to assist the restoration of a coral reef. It includes **Asexual propagation** and **Coral Gardening**.
Structural restoration	It involves the **construction of artificial reefs**, or **relocation of rocks/ dead coral heads.** The goal is to **increase the amount of reef structure** and **habitat available for the corals and other reef organisms to grow on. For ex- Bio rock technology**
Physical restoration	It addresses the conditions in which the corals are growing to improve their health, growth rates, or reproductive ability.

Way forward

- **Halting unplanned development-** Halting unplanned coastal development would play a significant role in reversing the decline of reefs in some locations.
- **Promotion of sustainable fishing-** Promoting sustainable fishing and **providing opportunities for ecotourism** can help conserve corals.
- **Minimal use of chemical fertilisers-** There is a need to **minimise the use of chemically enhanced fertilizers, insecticides, pesticides, and herbicides which are non degradable** and harm corals.
- **Measures for prevention of global warming-** Taking all possible measures to **prevent actions that worsen global warming** since Climate change is the greatest global threat to coral reef ecosystems.

Conclusion

Coral bleaching is a stark indicator of the growing stress on marine ecosystems caused by climate change, ocean warming, and human activity. If left unchecked, it threatens not only marine biodiversity but also coastal livelihoods and global ecological balance.

2. MANGROVE ECOSYSTEM IN INDIA

Recently, the **IUCN Red List of Mangroves Ecosystems** has indicated that about **50% of the ecosystems examined** were classified as vulnerable, endangered, or critically endangered.

Mangroves forests and their key features

Mangrove forests are coastal ecosystems composed of salt-tolerant trees and shrubs that thrive in intertidal zones of tropical and subtropical regions.

Key features of Mangrove forests

Habitat & Growth Conditions: Mangroves thrive in **tidal flats**, **estuaries**, and **deltas** with high silt deposition, experiencing bi-daily tidal flooding.

- They adapt to high solar radiation, anaerobic mud, and can extract freshwater from saline water.

Adaptation to Saline Water: Mangrove trees have developed special adaptations, such as **salt-filtering roots** and **salt-excreting leaves**, to survive in saltwater environments.

Pneumatophores: Aerial roots that grow upwards from the soil, allowing the roots to breathe (e.g., Avicennia species).

Prop Roots/Stilt Roots: Roots that grow from the trunk and branches, providing stability in soft, muddy substrates (e.g., Rhizophora species).

Vivipary: Seeds germinate while still attached to the parent tree, increasing their chances of survival in the harsh environment.

Global Mangrove Distribution

Mangrove thrive only in **tropical and subtropical latitudes** near the equator, as they cannot withstand freezing temperatures. As per **FAO** (2023), the global mangrove extent in 2020 was 14.8 million hectares, covering less than 1% of all tropical forests globally.

- Largest mangrove areas are in South and Southeast Asia, followed by South America, Africa, North and Central America, and Oceania.
- Indonesia, Brazil, Nigeria, Mexico, and Australia hold 47% of the global mangrove cover.

Mangrove Forest in India

As per the **Indian State of Forest Report (ISFR) 2023**, India's mangrove cover is around **4,992 sq. km**, constituting **0.15% of the country's total geographical area**.

- Major mangrove ecosystems are found in **Odisha** (Bhitarkanika), **Andhra Pradesh** (Godavari-Krishna delta), **Gujarat**, **Kerala**, and the **Andaman Islands**.
- The Sundarbans is the largest contiguous mangrove forest in the world, while Bhitarkanika is the second largest in India.

Importance of Mangroves

- **Biodiversity Hotspots-** They are rich in diverse flora and fauna, supporting unique species adapted to the intertidal zone.
  - Mangroves are home to **5,700+ species** across **21 phyla** in India, including Bengal tigers, estuarine crocodiles, Indian pythons, and 260+ bird species.
- **Coastal Protection-** Mangroves act as natural barriers, protecting coastlines from erosion, storm surges, tsunamis, and high winds, thereby reducing the impact of natural disasters.
  - Mangroves lower flood depths by **15-20%** and up to **70% in certain areas**, playing a crucial role in disaster risk reduction.
- **Carbon Sinks-** These ecosystems are highly effective at sequestering large amounts of **"blue carbon"** (carbon stored in coastal and marine ecosystems) in their biomass and sediments.
  - Mangrove ecosystems sequester approximately **11 billion tonnes of carbon**, which is almost **three times** the amount stored by **tropical forests** of the same size.
- **Nurseries for Marine Life-** Their intricate root systems provide sheltered breeding and nursery grounds for a wide array of **fish**, **crabs**, **shrimp**, and **other invertebrates**, which in turn supports coastal fisheries.
- **Livelihood Support-** Mangroves directly support the livelihoods of millions of coastal communities through activities like **fisheries**, **honey collection**, and **sustainable forestry products**.

Threats to the Mangrove ecosystem

Anthropogenic Pressures

- **Encroachment and Conversion:** According to the "**State of the World's Mangroves 2024**" report, **aquaculture** (26%), along with **oil palm plantations** and **rice cultivation**, has been a major driver of mangrove loss.
- **Pollution:** Discharge of **industrial effluents**, **domestic sewage**, and **agricultural runoff**. **Oil spills**, particularly in areas like the **Niger Delta**, threaten mangrove regeneration and ecosystem health.
- **Over-exploitation of natural resources:** Unsustainable harvesting of timber, fuelwood, and non-timber forest products, threaten the mangrove ecology.

Climate Change

- **Sea-Level Rise:** Mangroves are highly sensitive to sea-level rise, which can inundate them beyond their tolerance levels.
- **Increased Frequency of Extreme Weather Events:** Cyclones and storms can severely damage mangrove forests.
- **Reduced Freshwater Flow:** Diversion of river water for irrigation and other uses reduces freshwater flow to deltas, increasing salinity and stressing mangroves.
- **Invasive Species:** Non-native species may outcompete or disrupt native mangrove species, reducing biodiversity and changing ecosystem dynamics.
- The spread of Prosopis juliflora, an aggressive invasive species found in the mangroves of Tamil Nadu and Sri Lanka, has destroyed the mangrove ecosystem.

Initiatives that have been undertaken for mangrove ecosystem restoration

MISHTI Scheme (Mangrove Initiative for Shoreline Habitats & Tangible Incomes)	The program aims to restore and reforest approximately **540 km² of mangroves** across **9 states** and **3 Union Territories** over five years. It promotes **best management practices** and seeks to **make local communities stakeholders** in conservation and income generation.

National Coastal Mission Programme	Through its '**Conservation and Management of Mangroves and Coral Reefs**' initiative, the government provides guidance and financial assistance to states for preparing and implementing **Management Action Plans** (MAPs) focusing on survey, demarcation, afforestation, and awareness.
Coastal Regulation Zone (CRZ) Notification (2019)	Issued under the **Environment (Protection) Act, 1986**, this regulation strictly regulates development activities in mangrove-adjacent coastal zones.
Co-Management Committees	In states like Maharashtra, co-management approaches involving **local communities**, **gram panchayats**, and **women's self-help groups** have successfully restored mangroves while improving climate resilience and livelihoods.
NGO and Public Engagement	Projects led by organizations like **WWF India** and the **Aga Khan Rural Support Programme** actively involve locals in restoration and stewardship, promoting education and sustainable aquaculture within mangrove ecosystems.
Mangrove Alliance for Climate (MAC)	India became a member of the **Mangrove Alliance for Climate (MAC)** in **2022**, enhancing international knowledge-sharing and scaling up best conservation practices.

Way forward

1. **Enhance and enforce national and state-level policies:** Update and rigorously enforce laws (like **India's CRZ Notification**, **Forest Acts**, and **EIA norms**) to prohibit destructive activities, enable restoration, and define clear frameworks for ownership and access rights.
2. **Integrate mangroves into climate and disaster adaptation plans:** Recognize mangroves as "**natural coastal infrastructure**" and prioritize their protection in all national and local planning.
3. **Empower local communities, women's groups, and indigenous peoples:** Capacity-building and community-led restoration have repeatedly proven effective in sustainable mangrove management. **For ex-** Engagement platforms like '**Mangrove Mitras**'.
4. **Emphasize natural regeneration: Move beyond monoculture mass-planting** towards methods that facilitate the **natural recovery of mangrove ecosystems**, with targeted planting only where necessary.
5. **Innovative restoration techniques:** Employ methods like the "**trenched hydrology**" approach for better survival and ecosystem function of restored mangroves.
6. **Integrated ownership models-** Integrated ownership models like **Eco-development Communities** & **Joint Forest Management Committees** must be mobilized for conservation of urban mangroves.

Conclusion

Policymakers & businesses must recognise mangroves as vital climate & economic infrastructure rather than mere biodiversity zones. Protecting & restoring these ecosystems demands a coalition where science, business & community voices work in harmony & tandem – each bringing their unique strengths to secure their future.

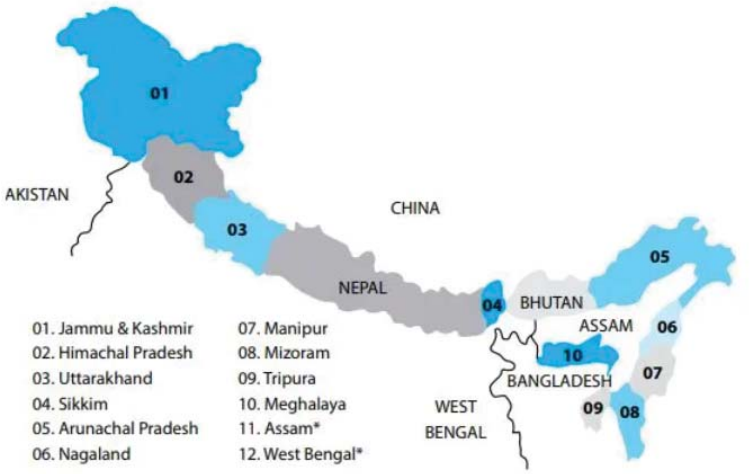
3. FRAGILE HIMALAYAN ECOSYSTEM

Since the onset of the monsoon season in June 2025, the Himalayas have been faced with relentless rains, with rainfall in Uttarakhand reported at 421 per cent above average. At least 280 deaths have been reported

due to flash floods in India and Pakistan. The deluge has also buried homes,hotels, shops, and roads under tons of mud and debris.

Reasons behind increased vulnerability of Himalayan Ecosystem

The Indian Himalayan Region (Source: National Mission on Himalayan Studies)



- **Climate Change induced amplification of monsoon intensity-** The warming of the Indian Ocean and Arabian Sea has increased atmospheric moisture as the warmer air holds more moisture (**up to 7 per cent more per 1 degree celcius rise**). This has led to amplification of monsoon intensity.
- **Glacial retreat and permafrost thawing-** Glacier retreat and permafrost thawing have destabilised mountain slopes, exacerbating landslides and flash floods.
- **Lopsided Unplanned Development- Unregulated Urbanisation** with rampant building constructions on vulnerable slopes without geological consultation has resulted in choking of drainages leading to worsening of the region's ecological vulnerability.
- **Infrastructure projects-** Heavy infrastructure projects like, **hydroelectric dams** and **four lane highways** have led to improper slope cutting, exacerbating the Himalayan Vulnerability.
- **Floating Tourist Population-** The floating population in the Hilly Tourist states of **Uttarakhand** and **Himachal Pradesh** (**5 times of the original population**), has stretched the resources of these hilly himalayan states.

Significance of Himalayan Ecosystem

Ecological Significance	**Forest Wealth-** The Himalayan forests provide fuel wood and a large variety of raw materials for forest based industries. **For Ex- Medicinal plant based pharmaceutical industry.** **Biodiversity Hotspot-** The Himalayan region is recognized as one of the world's biodiversity hotspots and contributes to global ecological balance. **For Ex- Eastern Himalayan Biodiversity hotspot.**
Socio-Economic Significance	**Energy Security-** Almost **33%** of the country's thermal electricity and **52%** of its hydro power is dependent on river waters originating in the Himalayan Regions. **Cultural and Spiritual Significance-** The Hindu Kush Himalayas are regarded as a sacred and spiritual centre by numerous cultures and religions, including **Hinduism**, **Buddhism** and **Jainism**. **Livelihoods-** Millions in India and neighboring countries depend on Himalayan agriculture, forests, and tourism.

Geopolitical and Strategic Significance	**Border security-** The Himalayas form a natural boundary with China, Nepal, Bhutan, and Pakistan; ecological degradation could affect border stability. **Transboundary water security-** Preservation ensures continued water sharing and cooperation with downstream countries.

Initiatives Taken for the restoration of the Himalayan Ecosystem

National Mission on Sustaining Himalayan Ecosystem	Focuses on conservation, sustainable development, and climate-resilient livelihoods in the Himalayan region. It is part of India's National Action Plan on Climate Change (NAPCC).
SECURE Himalaya Project	It is a part of "Global Partnership on Wildlife Conservation and Crime Prevention for Sustainable Development" (Global Wildlife Program) funded by the Global Environment Facility (GEF). It promotes sustainable management of alpine pastures and forests in the high range Himalayan ecosystems.
Mishra Committee Report 1976	The committee recommended placing restrictions on heavy construction work, blasting and felling of trees in the Himalayan region.
Regional Cooperation Framework	The International Centre for Integrated Mountain Development (ICIMOD) through its regional cooperation framework aims to **enhance biodiversity and ecosystem services**, and **promote transboundary cooperation** for sustainable development.

Further measures that need to be taken to protect the Himalayan ecosystem's sanctity

- **Real Time weather Monitoring-** Deploy **Doppler radars**, **micro-weather stations**, and **real-time satellite monitoring** to improve weather forecasting during monsoon months.
- **Ecology and Geology based zonal development plans-** Prepare ecology and geology based zonal development plans with the help of experts.
  - Undertake measures to shift people from vulnerable slopes.
- **Maintenance of natural flow-** Efforts must be undertaken to ensure that **no natural flow of water/ tributaries/sub-surface water channels** is obstructed.
  - The drainage must be maintained throughout the year in almost all areas.
- **Finetuned forest Management-** Forest management and afforestation programmes must be **finetuned** to ensure **slope stabilisation** through planting of shrubs before tree planting, recharge of aquifers and reduction in runoff.
- **Community resilience-** Build community resilience through establishment of **block level disaster rescue centres** and frequent and regular drills by locals.
  - Nominate a nodal officer for each block to oversee the weather pattern and prepare for climate induced disaster.

Conclusion

In Indian culture, the Himalayas are revered as the abode of gods, and rivers like the Ganges are divine. The reckless exploitation of this fragile ecosystem betrays the principle of non-violence and respect for nature embedded in Indian traditions.

The Union Environment Ministry must take action and fix responsibility and accountability for this repeated assault on the Environment by a combined force of nature's retribution through climate change and man's greed.

4. WETLAND CONSERVATION IN INDIA

Globally, **wetlands cover 6% of the Earth's land surface** and **support 40% of the world's species**, as per the Ramsar Convention. However, since 1970, 35% of global wetland areas have been lost due to human activities like urbanization, agriculture, and pollution.

In India, wetlands are equally vital but face similar threats. The National Wetland Decadal Change Atlas (2017-18) reveals that **India has 27,403 wetlands, covering 18.4% of the country's area**. Despite their ecological and economic significance, wetlands are disappearing at an **alarming rate of 2-3% annually**, with **50% of India's wetlands already lost**.

Present status of wetlands

Global Scenario	Wetlands are declining at a rate of **(-)0.78% annually**, with **81% of inland wetland species** and **36% of coastal species** experiencing **population declines since 1970**, as per the Wetland Extent Trends (WET) index. Since 1900, **50% of global wetland area has been lost**, with a decline of **35% between 1970 and 2015**.
Indian Context	• **India has 98 Ramsar sites** (wetlands of international importance), covering 1.36 million hectares. However, this represents only a fraction of the total wetland area. • Wetlands **cover 18.4% of India's land area** (excluding rivers); 70% of these are used for paddy cultivation. • The coastal wetlands span **6,750 sq. km**, with **"mangrove ecosystems"** concentrated in Sundarbans, Andaman & Nicobar Islands, Odisha, and Gujarat. • Southern peninsular wetlands are largely man-made reservoirs, locally known as **Yeris (tanks)**, essential for water storage and biodiversity conservation. • **50% of India's wetlands** have been **lost due to encroachment**, **land-use change**, and **pollution**. E.g. **Pallikaranai Marsh** and **East Kolkata Wetlands** have shrunk drastically due to urbanization. • The Wildlife Institute of India reports that **70-80% of freshwater marshes in the Gangetic floodplains** have been **lost** in the **last five decades**. • Urban areas like Mumbai, Chennai, and Kolkata have lost **71%**, **85%**, and **36% of their wetlands**, respectively, due to urbanization and pollution. • Indian mangrove areas have shrunk from **700,000 hectares in 1987** to **453,000 hectares in 1995**, with only **4,000 sq. km remaining today**, as per remote sensing data.

Significance of Wetlands

- **Ecological Importance**- Wetlands are biological supermarkets, supporting a wide range of species, nutrient cycling, hydrological regulation and habitat heterogeneity. They act as carbon sinks, mitigating climate change by sequestering carbon.

  They also support biodiversity, **hosting 40% of the world's species**, as per the Ramsar Convention. **For ex-** The Sundarbans, a Ramsar site, is home to the endangered Bengal tiger and acts as a natural barrier against cyclones.

- **Economic Value**- Wetlands contribute significantly to livelihoods. The East Kolkata Wetlands treat wastewater and support fisheries, generating an **estimated $100 million annually.**
- **Climate Resilience**- Wetlands act as carbon sinks, sequestering carbon and mitigating climate change. They also regulate floods and droughts, providing **"nature-based solutions"** for climate adaptation.
- **Cultural & Recreational Value-** Wetlands like **Chilika Lake**, **Loktak Lake**, and **Wular Lake** have spiritual, tourism, and aesthetic value.
- **Wetlands as Natural Disaster Buffers**- Economic Survey (2020-21) recognizes wetland conservation as a **Nature-Based Solution (NbS)**. Mangrove forests reduce storm surge heights by up to **50% (World Bank, 2021)**.

Progress has been made so far in wetland conservation

Policy Framework	India is a **signatory to the Ramsar Convention (1971)** and has **designated 85 Ramsar sites**. The **Wetlands (Conservation and Management) Rules, 2017**, provide a **legal framework for wetland conservation**. The **National Committee on Wetlands, Mangroves, and Coral Reefs** has identified **93 wetlands** for priority conservation.
Government Initiative	National Plan for Conservation of Aquatic Ecosystems (NPCA), ₹900 crore allocated. **AMRUT 2.0 (2021)**, focuses on "urban blue-green infrastructure" for wetlands. **Namami Gange Programme**, wetland rejuvenation along the Ganga River Basin.
Judicial Interventions	The **National Green Tribunal (NGT)** and **the Meghalaya High Court** have emphasized the need for wetland conservation through suo motu PILs and directives.
Global Initiatives	**Ramsar COP14 (2022) in Wuhan & Geneva** highlighted the need to link **wetland conservation with global sustainability goals**, including the **SDGs**, **IPCC targets**, and **UN Decade on Ecosystem Restoration, Global Biodiversity Framework**.

Challenges in Wetland Conservation

Wetlands are among the most threatened ecosystems because they sit at the intersection of land, water, & development.

- **Urbanization and Land Use Change**: Urban expansion has led to the loss of wetlands in cities like Mumbai and Chennai. The **National Wetland Decadal Change Atlas** shows a **decline in natural wetlands** due to **infrastructure development**.
- **Pollution**: Industrial effluents and untreated sewage have **degraded wetlands like the Bellandur Lake in Bengaluru**, which famously caught fire due to pollution.
  - Growing pollution results in **eutrophication of water bodies when untreated sewage**, **industrial effluents**, **agricultural runoff**, and **solid waste** are pushed into **wetlands**. When a **wetland becomes the dumping ground**, its **biodiversity collapses**, along with its ability to buffer floods & purify water.
- **Climate Change**- Rising temperatures and erratic rainfall patterns threaten wetland ecosystems. **For ex-** The **Sundarbans face salinity intrusion** due to **sea-level rise**.
- **Lack of Awareness and Governance**- **Weak enforcement of conservation laws** and **limited public awareness** hinder **effective wetland management**. State wetland authorities are often underfunded & understaffed.

- **Biodiversity Decline**- **81% of inland wetland species populations have declined** since 1970 (WWF, 2022).

Way Forward

- **Integrated Management**- Adopt an **ecosystem-based approach to wetland management**, as advocated during the Ramsar COP14. Integrate wetland conservation into urban planning and development policies.
- **Strengthening Legal Frameworks**- **Amend the Wetlands (Conservation and Management) Rules, 2017**, to include **stricter penalties for violations**. Ratify and implement global agreements like the **ILO's Recommendation 204** on **transitioning to sustainable ecosystems**.
- **Community Participation**- Engage local communities in wetland conservation through programs like **Community-Based Resource Management (CBRM).** Promote eco-tourism to generate livelihoods while conserving wetlands.
- **Technological Interventions**- Use **Geographic Information Systems (GIS)** and **remote sensing** for **real-time monitoring of wetlands**. Develop blue-green infrastructure in urban areas to restore wetland ecosystems.
- **International Best Practices**- Learn from the **Netherlands' Room for the River Program,** which **integrates wetlands** into **flood management**. Adopt **China's Sponge Cities Initiative,** which uses wetlands for urban water management.
- **Financial Mechanisms**- Allocate funds for wetland conservation in the Union Budget and State Budgets. Leverage international funding mechanisms like the **Green Climate Fund (GCF).** Wetland Conservation Bonds to raise funds for restoration.
- **Wastewater Treatment-** Treat the wastewater before it meets the wetlands. For urban & peri-urban wetlands, a key action is to ensure treated inflows. Wetlands cannot be substituted for sewage treatment plants.
- **Coastal & Riparian Wetlands-** Make **coastal & riparian wetlands** central to disaster risk reduction. Planning authorities should treat them as risk buffers - worthy of investment as compared to '**grey infrastructure**'.
- **Build Skills & Institutions-** A **national capacity mission for wetland managers** with **accredited training in hydrology**, **restoration ecology**, **GIS/remote sensing**, environmental law & community-led governance.

Conclusion

India must adopt a holistic approach, combining legal, technological, and community-driven strategies, to conserve its wetlands. As **Mahatma Gandhi** said, "**The earth provides enough to satisfy every man's needs, but not every man's greed**." Let us **strive to protect our wetlands**, ensuring a sustainable future for generations to come.

5. NATIONAL GREEN TRIBUNAL

The **National Green Tribunal (NGT)** is a specialized judicial body in India established to handle cases related to environmental protection and the conservation of forests and other natural resources.

It was established in 2010, under the **National Green Tribunal Act**, India became only the third country in the world (after Australia and New Zealand) to have a dedicated environmental tribunal.

It is guided by the principles of **sustainable development**, the **precautionary principle**, and the **polluter pays principle.**

The NGT has a **Principal Bench** in New Delhi and four zonal benches to ensure accessibility across India:

- **Central Zonal Bench**: Bhopal
- **Eastern Zonal Bench**: Kolkata
- **Southern Zonal Bench**: Chennai
- **Western Zonal Bench**: Pune

Composition of NGT

	Qualification	Appointment
Chairperson	Must be a retired Judge of the Supreme Court or a retired Chief Justice of a High Court.	Appointed by the Central Government in consultation with the Chief Justice of India.
Judicial Members	Must be a Judge of a High Court (or have been one), or qualified to be a Judge of a High Court.	Judicial and Expert members are appointed by a committee formed by the Central Government.
Expert Members	Must have professional qualifications and experience (minimum 15 years) in environmental science, forestry, biodiversity, climate change, or related fields, or administrative experience in environmental governance.	

- **Strength of the Tribunal:**
  - Minimum 10 and maximum 20 Judicial Members
  - Minimum 10 and maximum 20 Expert Members
- **Term of Office:** Members are appointed for a term of **5 years**
- **Non-Reappointment:** To ensure unbiased decision-making during their tenure, members are **not eligible for re-appointment** once their term ends.

Objectives & functions of NGT

- **Expeditious Justice-** To ensure environmental cases are settled within **6 months** of filing, preventing long-drawn-out legal battles that often allow environmental damage to continue.
- **Specialized Adjudication-** To provide a forum where technical environmental issues are handled by people who actually understand the science (Expert Members) alongside legal professionals.
- **Enforcement of Rights-** To protect the citizen's right to a healthy environment, which the Supreme Court of India considers a part of the "Right to Life" under **Article 21** of the Constitution.
- **Reducing Judicial Burden-** To take the weight of thousands of complex environmental cases off the shoulders of the High Courts and the Supreme Court.

Jurisdiction of the National Green Tribunal (NGT)

- **Original Jurisdiction:** The NGT only has the power to hear civil cases that involve a "substantial question relating to the environment." These questions must arise from the implementation of the following **seven laws** listed in Schedule I of the NGT Act:
  - The Water (Prevention and Control of Pollution) Act, 1974
  - The Water (Prevention and Control of Pollution) Cess Act, 1977
  - The Forest (Conservation) Act, 1980

- The Air (Prevention and Control of Pollution) Act, 1981
- The Environment (Protection) Act, 1986
- The Public Liability Insurance Act, 1991
- The Biological Diversity Act, 2002

- **Appellate Jurisdiction**: NGT also acts as an appellate body. It can hear appeals against specific orders or decisions made by various authorities (like CPCB, SPCBs, State governments) under the laws mentioned above.

  For example, if a government authority grants or refuses an environmental clearance, or if a State Pollution Control Board issues certain directions, an aggrieved party can appeal that decision to the NGT.

Powers of NGT

- **Power of a Civil Court:** It has the same powers as a civil court under the Code of Civil Procedure (CPC) for summoning witnesses, requiring discovery of documents, and receiving evidence on oath.
- **Power to Grant Relief & Compensation:** It can award **monetary damages** to victims of pollution and order the **restitution** of damaged property or **restoration** of the environment.
- **Regulatory Independence:** It is not strictly bound by the "Indian Evidence Act" or "CPC." Instead, it is guided by the **Principles of Natural Justice**, allowing it to be more flexible and faster than regular courts.
- **Penalties for Non-Compliance**: If a person or organization fails to comply with an NGT order, the Tribunal can impose severe punishments like Imprisonment up to 3y, fine (Individual) up to **₹10 crore,** fine (Company) up to **₹10 crore**.
- **Suo Motu Powers:** The NGT has the unique authority to take up cases on its own based on media reports or public information, even if no formal complaint has been filed by an affected party.

Significance of NGT

- **Faster Justice Delivery:** Before the NGT, environmental cases would languish in civil courts for decades. The NGT is mandated to deliver a final decision within **6 months.**
- **Scientific Decision Making**: Environmental law is complex and involves data on carbon emissions, effluent levels, and biodiversity. Since the NGT bench includes scientists and environmental experts alongside judges, the rulings are based on **technical accuracy** rather than just legal technicalities. This reduces the reliance on external committees, which often delayed court proceedings in the past.
- **Upholding Global Environmental Principles**: The NGT has been a pioneer in enforcing three major international legal doctrines:
  - **Polluter Pays:** It has slapped massive fines on major corporations and state governments for dumping waste or polluting rivers.
  - **Precautionary Principle:** It can stop a project *before* damage occurs if there is a high risk to the ecosystem.
  - **Intergenerational Equity:** Ensuring that today's development doesn't rob future generations of clean air and water.
- **Self-Initiative:** It can take up cases on its own (**Suo Motu**) based on a news report or even a letter from a concerned citizen, ensuring that even marginalized communities have a voice against powerful industrial lobbies.
- **Enhancing Access to Justice:** Relaxed procedures, suo motu powers, and broad "aggrieved person" definition make it easier for citizens and groups to seek environmental justice.
- **Institutional Primacy & Trust:** NGT has been recognized by the Supreme Court as the primary body for environmental oversight, entrusted with monitoring complex national issues like river pollution.

- **Expanding the Right to Life**: The NGT has played a crucial role in expanding the interpretation of the Indian Constitution. Its proactive interventions have effectively extended the scope of the **right to a clean environment** as an integral part of the fundamental **Right to Life (Article 21).**

Landmark judgments of NGT

Shivalik Hills "Vanishing Hills" Case (Punjab)	• Landmark intervention to protect ecologically fragile Shivalik range (groundwater recharge zone). Case taken up suo motu based on media reports. • Closure of 13 stone crushers in Hoshiarpur and Rupnagar; recovery of over **₹180 crore** in environmental compensation.
Morbi Ceramic Units Case (Gujarat)	• Upheld a **₹122 crore** fine against 337 ceramic units for using prohibited coal gasifiers.
Bhopal Adampur Waste Plant Case (Madhya Pradesh)	• Imposed **₹1.80 crore** penalty on Bhopal Municipal Corporation for flouting solid waste management rules.
Ban on Old Vehicles in Delhi (2015)	• The NGT banned all **diesel vehicles older than 10 years** and **petrol vehicles older than 15 years** from plying on Delhi-NCR roads.
Yamuna Floodplain Case (2016)	• The NGT fined the **Art of Living Foundation ₹5 crore** as "environmental compensation" for damage caused to the Yamuna floodplains during the World Culture Festival.

Limitations/criticisms of NGT

- **Persistent Vacancies:** By law, the NGT should have a minimum of 10 Judicial and 10 Expert members. However, it often operates with far fewer, leading to a massive backlog of cases.
  - Lack of appointments has occasionally forced regional benches (like Chennai or Kolkata) to shut down temporarily, forcing litigants to travel to the Principal Bench in New Delhi or attend hearings via video conferencing.
- **Limited Jurisdiction:** Despite being the primary environmental court, the NGT cannot hear cases under two very important laws: Wildlife **(Protection) Act, 1972** & **Indian Forest Act, 1927**.
  - This creates a fragmented system where a person might have to go to the NGT for a pollution issue but to a regular civil court for a wildlife-related grievance in the same area.
- **Appeals to Higher Courts:** Almost every major NGT order is immediately challenged in the Supreme Court. This often leads to "**litigation fatigue**," where the environmental damage continues while the case is stuck in the superior court's appeals process.
- **Lack of "Teeth" for Execution:** The NGT relies on state pollution control boards and local authorities to enforce its orders. If these bodies are corrupt or inefficient, the NGT's ruling remains only on paper.
- **Arbitrary Fines:** Often, fines ranging from ₹5 crore to ₹100 crore are imposed based on a company's **revenue** rather than a scientific assessment of the actual **environmental damage** caused. This has led to many NGT orders being stayed or overturned by the Supreme Court on the grounds that the penalties were "vague" or "unscientific."
- **Conflict of Interest in "Expertise":** Out of 13 expert members appointed since 2016, 9 were part of decision making process for granting various green clearances at the highest level of the govt.

- **Dismissal on "Technical Grounds":** Public grievances are increasingly being **silenced by procedural technicalities** rather than being heard on merit. The NGT follows a strict **90 days appeal filing deadline**.
  - A significant number of citizen appeals are being dismissed on technical grounds, labelled as **'time barred'** for being filed after more than 90 days.

Way forward

- **Mandatory Full Strength:** The government should be legally mandated to maintain the minimum strength of **10 Judicial and 10 Expert members**. Delays in appointments should be treated as a violation of the NGT Act.
- **Academic & Independent Experts:** Instead of almost exclusively appointing retired Indian Forest Service (IFS) or government officers, the NGT should include **active environmental scientists, ecologists, and independent researchers** from top universities and NGOs.
- **Transparency in Selection:** The selection process for members should be made public, with clear criteria to prevent the appointment of "pro-project" individuals who might have a conflict of interest.
- **Monitoring Wing:** Currently, the NGT depends on State Pollution Control Boards (SPCBs) to enforce its orders. A dedicated monitoring wing within the NGT could conduct independent "surprise audits" to ensure compliance.
- **Uniform Compensation Framework:** As suggested by the CPCB in late 2025, there should be a standardized formula for calculating environmental fines. This would move away from "arbitrary" fines and make the orders harder to challenge in the Supreme Court.
- **Include Wildlife and Indian Forest Acts:** Bringing the **Wildlife (Protection) Act, 1972** and the **Indian Forest Act, 1927** under the NGT's umbrella would create a "one-stop shop" for environmental justice, preventing litigants from having to run between different courts for the same project.
- **Relaxing the "90-Day" Rule:** The strict timeline for filing appeals often penalizes rural communities who may not find out about a project's "Environmental Clearance" until it's too late. The Tribunal should have more flexibility to condone delays in public interest cases.
- **Multilingual Proceedings:** Making judgments and notices available in regional languages (not just English) would empower local communities to participate in the judicial process more effectively.

Conclusion

The National Green Tribunal has played a significant role in strengthening environmental governance in India by ensuring speedy and effective disposal of environmental cases. Strengthening its institutional capacity, improving enforcement of its orders, and enhancing coordination with regulatory authorities will be crucial for promoting sustainable development and environmental justice.

6. HUMAN WILDLIFE CONFLICT

Human-Wildlife Conflict (HWC) refers to the direct, adverse interactions between people and wild animals, increasingly driven by habitat fragmentation, agricultural expansion, and human encroachment into natural corridors. In a highly biodiverse and densely populated country like India, this friction manifests as severe crop raiding, livestock depredation, and tragic losses of both human and animal lives.

Major Human-Wildlife Conflict Zones in India

- **Western Ghats Landscape (Karnataka, Kerala, Tamil Nadu):** Severe crop raiding and human casualties occur in districts like Wayanad, Hassan, and Kodagu due to the fragmentation of contiguous elephant corridors.

- **Central Indian & Eastern Ghats Landscape (Vidarbha, Odisha, Chhattisgarh):** The Tadoba-Andhari and Chandrapur belts experience heavy conflict as dispersing young tigers stray into human-dominated mining and agricultural zones.
- **Terai & Shivalik Arc Landscape (Uttarakhand, Uttar Pradesh):** Agricultural expansion right up to the borders of Corbett and Dudhwa tiger reserves forces big cats to utilize standing sugarcane fields as substitute nurseries.
- **Brahmaputra Valley & Northeast Hills (Assam):** Linear infrastructure projects across the Kaziranga-Karbi Anglong landscape trap elephant herds, forcing them into intense seasonal clashes over ripening paddy crops.
- **Sundarbans Mangrove Delta (West Bengal):** Traditional honey-collectors (Mouli) and fishermen frequently face fatal encounters with Royal Bengal Tigers adept at swimming across tidal channels.
- **Arid and Semi-Arid Landscape (Gujarat, Rajasthan):** As lions spill over from Gir National Park into the revenue areas of Amreli and Bhavnagar, conflicts over livestock depredation are rising.
- **Blackbuck & Nilgai Crop Damage:** Large populations of blue bulls (Nilgai) and blackbucks in western Rajasthan trigger extensive crop losses, testing the traditional tolerance of local communities.

Causes of Human-Wildlife Conflict in India

- **Habitat Fragmentation & Encroachment**
  - **Linear Infrastructure Intrusion:** Construction of highways, railways, and canals cuts directly through historic wildlife corridors, trapping animals in small pockets.

    As per wildlife trust of India, **two-thirds** (**~66%**) of validated elephant corridors intersect with National or State Highways, structurally disrupting wildlife migration pathways.
  - **Agricultural Expansion:** Conversion of natural buffer forests into standing crop fields and commercial plantations brings livestock and crops directly to the forest edge.

    As per MoEFCC, Over **58.4%** of crucial animal movement passages fall under active settled cultivation, eliminating historical forest edge buffers.
  - **Encroachment of Elephant Corridors:** Over **70% of India's identified elephant corridors** feature active human settlements and cultivation, forcing herds onto public roads.
  - **Mining and Industrialization:** Large-scale open-cast mining in resource-rich states like Odisha and Jharkhand destroys and fragments contiguous forest tracts.
- **Ecological & Environmental Drivers**
  - **Degradation of Forest Quality:** Proliferation of aggressive invasive weeds (like *Lantana camara*) smothers native grasses, wiping out food for wild herbivores.

    Invasive weed Lantana camara has heavily choked the understory of over 44% (300,000 sq. km) of total Indian forest areas
  - **Drying of Waterholes:** Climate-induced prolonged droughts and dipping water tables force wild herds to leave protected areas in search of water.
  - **Straying of Dispersing Carnivores:** Successful conservation drives have increased tiger and leopard populations, forcing young adults out of saturated reserves into human zones.
  - **Behavioral Adaptations:** Wild animals like leopards and wild boars are adapting to human landscapes, using sugarcane fields and urban scrub as substitute habitats.
- **Institutional & Anthropogenic Drivers**
  - **Unprotected Forest Boundaries:** Less than **5% of India's geographical area** lies within strictly Protected Areas, meaning most wildlife routinely roams through shared, multi-use landscapes.

- **Changes in Cropping Patterns:** Cultivating highly attractive cash crops like sugarcane, banana, and paddy right along forest fringes lures wild herbivores out of the woods.

  High-calorie crop cultivating shifts (paddy, sugarcane) within a 5 km periphery of forest fringes induce regular crop-raiding by mega-herbivores

- **Unsafe Waste Disposal:** Open garbage dumps on the edges of hill stations and sanctuaries attract leopards, sloth bears, and wild boars, habituating them to human food.
- **Inadequate Early Warning Systems:** A lack of real-time tracking and localized communication networks prevents forest departments from alerting villages before an animal enters.

  As per Project elephant reports, less than **15%** of vulnerable forest-fringe villages have **automated real-time SMS** or **SMS-integrated hooter warning systems**.

Impacts of Human-Wildlife Conflict in India

- **Ecological & Conservation Impacts**
  - **Retaliatory Killings of Wildlife:** Stressed communities resort to illegal poisoning, electrocution, or lynching of endangered species like tigers, leopards, and elephants.

    According to the **MoEFCC**, 1,160 elephants were killed due to non-natural causes (like poisoning and poaching) over a 10-year period.

  - **Erosion of Community Support:** Chronic conflict turns local populations against forest departments, severely undermining grass-roots conservation efforts.
  - **Disruption of Wildlife Behavior:** Constant human presence and artificial barriers alter natural migration patterns, gene flow, and feeding habits of wild animals.
  - **Accidental Infrastructure Fatalities:** Fragmented paths lead to tragic wildlife deaths, such as elephants being killed by speeding trains or hit by high-voltage power lines.

    According to **MoEFCC data**, electrocution claimed 741 elephant lives, while train hits caused 186 elephant deaths over a decade.

- **Economic & Livelihood Impacts**
  - **Devastating Crop Depredation:** Resource-poor marginal farmers face sudden, catastrophic financial ruin when herds of wild elephants or boars wipe out entire seasonal harvests.

    According to **Proceedings of the National Academy of Sciences (PNAS)**, human-elephant conflict in India **destroys an estimated 1 million hectares of crops annually**.

  - **Loss of Productive Livestock:** Big cat attacks on cattle deplete the primary source of milk income and draft power for vulnerable forest-fringe households.

    According to the **Centre for Wildlife Studies (CWS)**, livestock predation accounts for **approximately 20% of all reported human-wildlife conflict incidents** nationwide.

  - **Diversion of Household Labor:** Farmers are forced to spend sleepless nights guarding fields in hazardous conditions, leading to physical exhaustion and lost productivity.
  - **High Fiscal Cost of Compensation:** State exchequers face a mounting financial burden due to escalating yearly payouts for crop damage, injuries, and human deaths.
- **Socio-Psychological & Governance Impacts**
  - **Loss of Human Lives and Injuries:** Fatal encounters tear families apart, taking away primary breadwinners and pushing rural households into deep poverty.

    According to **Lok Sabha data (2026)** provided by the MoEFCC, elephant attacks alone caused over **600 human deaths** in a single calendar year.

- **Severe Post-Traumatic Stress:** Families living along volatile forest borders suffer from constant fear, anxiety, and a reduced quality of life.

  According to the **National Institute of Mental Health and Neurosciences (NIMHANS)**, over **60% of forest-fringe families** show chronic anxiety and fear symptoms due to wildlife threats.
- **Public Outrage and Law & Order Extremes:** Tragic wildlife attacks frequently spark violent local protests, localized lockouts, and targeted attacks on forest department field offices.
- **Deepening Trust Deficits:** Delayed compensation deliveries and perceived institutional apathy alienate rural citizens from regional administrative bodies.

Measures Taken for Human-Wildlife Conflict in India

Statutory, Policy & Financial Interventions	• **National Wildlife Action Plan (2017-2031):** Prioritizes the integration of human-wildlife conflict mitigation directly into national development planning and forest management. • **MoEFCC Advisory on HWC Management (2021):** Empowers Gram Panchayats under the Wildlife (Protection) Act to swiftly manage problematic wild animals and streamline local ex-gratia relief. • **Enhanced Ex-Gratia Compensation:** Substantially increased and standardized financial payouts across states for crop loss, livestock depredation, and human casualties. • **Agri-Insurance Interventions:** Includes crop damage by wild animals under the **Pradhan Mantri Fasal Bima Yojana (PMFBY)** on a pilot basis in select high-conflict states.
Technological & Infrastructure Measures	• **AI-Powered Early Warning Systems:** Deploys **seismic sensors**, **thermal cameras**, and **automated SMS alerts** to track elephant movements in volatile belts like West Bengal and Assam. • **Radio-Collar Tracking:** Fits satellite radio-collars on dispersing **alpha carnivores** and **matriarch elephants** to monitor real-time proximity to human settlements. • **Eco-Fencing & Barriers:** Installs **low-cost solar-powered fences**, **hanging power fences**, and **bio-fencing (like chili-smoke and beehives)** along forest fringes. • **Rail Barrier Projects:** Constructs **physical barricades out of used train tracks in states like Karnataka** to block elephants from crossing onto dangerous railway lines.
Community-Led & Field Operations	• **Rapid Response Teams (RRTs):** Establishes **highly trained, well-equipped forest department task forces** to swiftly tranquilize, capture, or safely guide straying wildlife back to safety. • **Primary Response Teams (PRTs):** Mobilizes **village-level youth volunteers** trained to handle crowd control and act as immediate first responders during encounters. • **Voluntary Village Relocation:** Coordinates financial packages to peacefully relocate core-area forest villages, effectively restoring historic wildlife corridors. • **Project RE-HAB (Khadi and Village Industries Commission):** Deploys honeybee boxes along forest pathways to naturally deter elephants from invading agricultural fields.

Way Forward for Human-Wildlife Conflict in India

- **Landscape-Scale Corridor Planning**
  - **Secure and Restore Elephant Corridors:** Fast-track the legal notification and structural restoration of critical migratory paths by removing illegal resorts and infrastructure.
  - **Design Smart Eco-Infrastructures:** Mandate the construction of dedicated wildlife underpasses and overpasses across all national highways and rail lines traversing forests.
  - **Implement Strict Eco-Sensitive Zone (ESZ) Norms:** Enforce regulated land-use zoning around protected areas to prevent hazardous industrial or mining expansion.
- **Technological Expansion & Real-Time Warning**
  - **Deploy Geo-Fenced SMS Alerts:** Scale up integrated networks of thermal camera traps and AI drones to broadcast automated, localized safety alerts to fringe villages.
  - **Expand Satellite Telemetry:** Increase the radio-collaring of dispersing carnivores and matriarch elephants to closely track and manage herd movements near borders.
  - **Create Integrated Emergency Dashboards:** Set up unified district-level command centers to optimize the dispatch and routing of forest Rapid Response Teams (RRTs).
- **Community Empowerment & Socio-Economic Co-Management**
  - **Institutionalize Village Primary Response Teams:** Formally train, equip, and insure local youth volunteer groups (like Gaja Bandhus) to lead crowd management.
  - **Fast-Track Instant Digital Compensation:** Streamline insurance and ex-gratia payouts through a dedicated, paperless online portal to eliminate administrative delays.
  - **Promote Conflict-Resilient Agriculture:** Incentivize farmers along forest fringes to switch from appetizing crops like sugarcane to unpalatable alternatives like lemongrass, ginger, or citrus fruits.
- **Institutional, Behavioral & Policy Reforms**
  - **Recognize HWC as a Disaster:** Integrate severe human-wildlife conflict into **State Disaster Response Funds (SDRF)** to mobilize rapid financial and relief resources.
  - **Enhance Inside-Forest Habitat Quality:** Enrich protected reserves by aggressively removing invasive weeds and restoring natural watering holes to minimize animal straying.
  - **Foster Coexistence Education:** Launch targeted awareness drives in forest-fringe schools and gram sabhas to teach safe behavioral protocols during animal encounters.

Best Practices for Human-Wildlife Conflict (HWC) Mitigation

Best Practices from India	• **Sonitpur Model (Assam):** Utilizes **community-driven anti-depredation squads** paired with low-cost, single-strand solar fencing to secure agricultural lands from elephant herds. • **Gajraj System (Indian Railways):** Deploys AI-enabled, fiber-optic acoustic sensors along critical railway tracks to detect elephant footsteps and alert loco-pilots in real time. • **WildSeve Digital Platform (WCS India):** Uses an **automated mobile-based logging system** to help forest-fringe farmers instantly report crop losses and receive transparent, paperless compensation. • **Malyagiri Elephant Hanging Fences (Odisha):** Replaces **conventional ground fences with overhead wire fences** that allow small wildlife to pass through while keeping elephants out.

Best Practices from Abroad	• **Community-Based Natural Resource Management (Namibia/Botswana):** Shares a direct percentage of ecotourism and wildlife revenues with local pastoral communities to turn wildlife into a financial asset. • **Integrated Ecological Spatial Planning (Costa Rica):** Interweaves contiguous wildlife migratory corridors directly into national infrastructure and urban land-use development maps. • **Real-Time Biosurveillance Networks (Finland):** Uses predictive tracking models and rapid-payout insurance systems to lower both public risk and herder resentment toward wild carnivores. • **Predator-Proof Community Enclosures (Bhutan & Nepal):** Builds galvanized wire mesh pens backed by stable local micro-finance funding to completely prevent nocturnal livestock depredation by snow leopards. • **Kwando Carnivore Project Lion Cages (Namibia):** Installs low-cost, mobile lion-proof bomas (enclosures) coupled with a community-monitored livestock insurance program. • **Certified Wildlife-Friendly Enterprises (Global):** Markets premium, certified commodities like **Elephant-Friendly Tea** from India or **Jaguar-Friendly Coffe**e from Colombia to directly fund grass-roots coexistence initiatives.

Conclusion

Managing human-wildlife conflict requires a decisive paradigm shift from reactionary crisis management toward an institutionalized model of peaceful coexistence. By seamlessly integrating smart landscape-scale infrastructure with community-led economic incentives, India can protect both vulnerable rural livelihoods and endangered biodiversity.

7. ENVIRONMENTAL IMPACT ASSESSMENT

As per United Nations Environment Programme (UNEP), **Environmental Impact Assessment (EIA) is a tool used to identify the environmental**, **social** and **economic impacts of a project prior to decision-making**. It aims to predict environmental impacts at an early stage in project planning and design, find ways and means to reduce adverse impacts, shape projects to suit the local environment and present the predictions and options to decision-makers.

Food and Agricultural Organization (FAO) defines it as, a formal process to predict the environmental consequences of human development activities and to plan appropriate measures to eliminate or reduce adverse effects and to augment positive effects.

Evolution of Environmental Impact Assessment (EIA) in India

Administrative Phase (Pre-Statutory)	• **1976–77 (Initial Step):** Planning Commission directs the Department of Science and Technology to review river valley projects from an environmental perspective. • **1980 (Institutionalization):** Establishment of the dedicated Department of Environment, which later expanded into the Ministry of Environment and Forests (MoEF) in 1985. • **1986 (Legal Bedrock):** Enactment of the Environment (Protection) Act (EPA), 1986, providing the overarching statutory authority to regulate and restrict industrial activities.

Codified & Statutory Phase	• **1994 (First Formal Notification):** MoEF promulgates the first statutory EIA Notification under the EPA 1986, making environmental clearance mandatory for 30 selected categories of developmental projects. • **1997 (Public Participation):** Amendments introduce mandatory public hearings into the EIA process, empowering local communities to voice environmental concerns. • **2006 (Decentralization & Re-engineering):** Replaced the 1994 regime to decentralize clearances into Category A (National level via EAC) and Category B (State level via SEIAA) based on spatial scale and potential impacts.
Modern Rationalization & Digital Governance Phase	• **2020 (Draft Controversy):** Introduction of the Draft EIA Notification 2020, sparking widespread debate over provisions like post-facto clearances and reduced public consultation windows. • **2022–2024 (Strategic Exemptions):** Amendments exempt strategic defense infrastructure within 100 km of international borders and biomass-based thermal plants from prior environmental clearance. • **2025–2026 (Digital Integration & Parivesh 2.0):** Complete transition to the single-window PARIVESH 2.0 web portal, embedding automated geographic information system (GIS) mapping to track compliance and accelerate green clearances.

EIA notifications

Central govt has the power to issue EIA notifications under Section 3 of the Environment Protection Act, 1986, wherein it can impose restrictions on setting up new projects or expansion or modernisation of existing projects. The section stipulates that such measures must benefit the environment.

Under the Environment (Protection) Act of 1986, the first EIA notification was issued in 1994. Later, it was replaced by a modified draft in 2006.

Objectives of Environmental Impact Assessment (EIA)

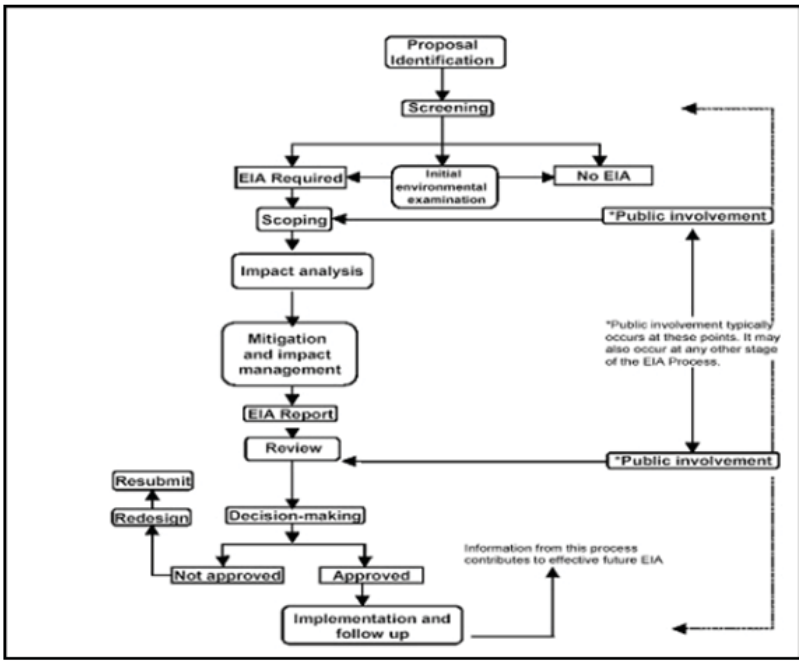
- **Sustainable Development & Ecological Balance**
  - **Harmonize Growth and Ecology:** Ensures that long-term macroeconomic development goals do not cause irreversible damage to the natural environment.
  - **Uphold the Precautionary Principle:** Identifies, predicts, and prevents severe ecological risks right at the project conceptualization stage before investments are made.
  - **Integrate the Carrying Capacity Concept:** Assesses whether regional ecosystems can physically handle the pollution and resource extraction load of a proposed project.
- **Analytical & Technical Decision-Making**
  - **Formulate Mitigation Strategies:** Mandates a structured Environmental Management Plan (EMP) to avoid, minimize, or offset expected environmental damage.
  - **Evaluate Feasible Alternatives:** Compares alternative project sites, layouts, and manufacturing technologies to choose the least environmentally damaging option.
  - **Quantify Cumulative Impacts:** Analyzes the combined, long-term environmental footprint of multiple industrial projects operating in the same geographic region.
- **Democratic Governance & Stakeholder Inclusion**
  - **Institutionalize Public Participation:** Empowers local community members, indigenous groups,

and civic bodies to voice concerns during mandatory public hearings.

- **Ensure Administrative Transparency:** Provides a formal, public portal (PARIVESH) to scrutinize technical data and minimize arbitrary bureaucratic clearances.
- **Secure Environmental Justice:** Protects the livelihood and health rights of vulnerable, frontline populations living adjacent to heavy industrial or mining sites.

Need for Environmental Impact Assessment (EIA)

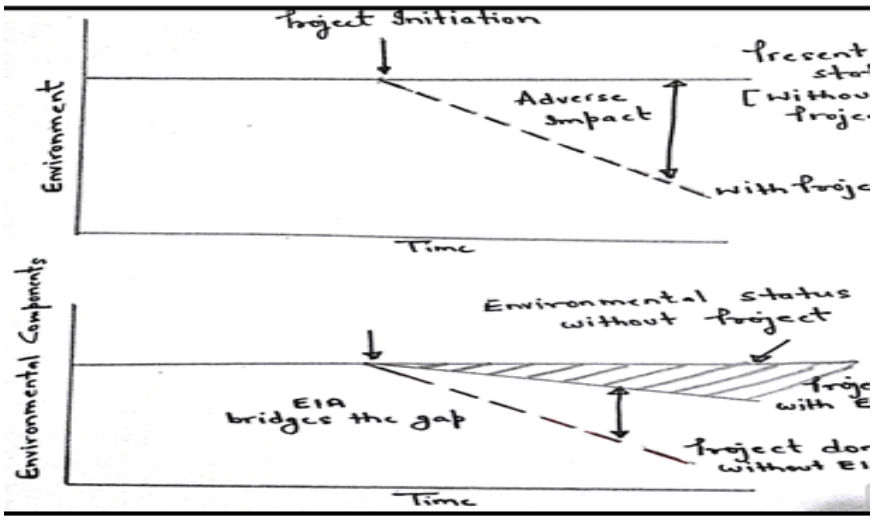
- **Mitigating Ecological Degradation & Climate Risks**
  - **Arrest Rapid Biodiversity Loss:** Prevents developmental projects from fragmenting critical wildlife corridors and destroying ecologically sensitive zones (ESZs).
  - **Climate Change Adaptation:** Ensures massive infrastructure works are structurally designed to withstand extreme, climate-induced weather anomalies.
  - **Prevent Natural Disasters:** Checks unscientific construction in fragile terrains like the Western Ghats and Himalayas to avert landslides and flash floods.
- **Safeguarding Public Health & Common Resources**
  - **Control Cumulative Pollution:** Evaluates regional carrying capacity to prevent heavy industries from pushing air and water quality into toxic zones.
  - **Protect Finite Water Ecosystems:** Prevents industrial effluents from permanently contaminating groundwater resources and critical river basins.
  - **Minimize Public Health Costs:** Curbs the spread of respiratory and waterborne illnesses by restricting toxic setups near dense human settlements.
- **Socio-Economic Security & Governance Stability**
  - **Tribal Displacement Conflicts:** Protects indigenous livelihoods by evaluating the social impact of diverting forest lands for mining or industrial use.
  - **Enforce Inter-Generational Equity:** Secures essential natural resources for future generations by checking reckless, short-term extraction practices.
  - **De-Risk Corporate Investments:** Identifies environmental and legal liabilities early, preventing costly mid-project halts and resource litigation.



Steps involved in Environmental Impact Assessment (EIA)

Core Process of EIA in India

Pre-Clearance Stage: Project Definition & Assessment	• **Screening:** Determines if the project requires a full EIA based on its scale and sector, classifying it into Category A (Central clearance) or Category B (State clearance). • **Scoping:** Defines the key environmental issues and geographical boundaries of the study to draft the sector-specific Terms of Reference. • **Baseline Data Collection:** Gathers primary and secondary data on regional air, water, soil, noise, and biodiversity parameters across a specific season. • **Impact Prediction:** Uses scientific models to forecast the expected biological, social, and economic changes caused by the project's construction and operation.
Consultation & Management Stage	• **Environmental Management Plan (EMP):** Outlines specific mitigation steps, clean technology upgrades, and financial allocations to handle anticipated ecological risks. • **EIA Report Preparation:** Compiles the scientific findings and the EMP into a comprehensive draft environmental impact assessment report. • **Public Hearing:** Conducts a mandatory open forum at the project site to record concerns, objections, and feedback from local residents and stakeholders.
Decision-Making & Post-Clearance Stage	• **Appraisal:** A multidisciplinary Expert Appraisal Committee (EAC or SEAC) reviews the final EIA report and public hearing transcripts to make a decision. • **Decision-Making:** Regulatory authority (MoEFCC or SEIAA) grants or rejects the formal Environmental Clearance (EC) based on the appraisal advice. • **Post-Project Monitoring:** Mandates regular, six-monthly compliance reports through the PARIVESH portal to check that the project stays within its legal environmental limits.



Issues with Environmental Impact Assessment (EIA) in India

- **Structural & Integrity Flaws**
  - **Inherent Conflict of Interest:** Project proponents directly hire and pay the EIA consultants, which often leads to biased, overly favorable assessment reports.

    As per Centre for Science and Environment evaluation, nearly **80%** of sampled EIA reports exhibit pro-industry bias.
  - **Poor Data Credibility:** Widespread issues with plagiarized, copy-pasted data and a reliance on single-season baseline metrics rather than thorough, multi-season studies.
  - **Exclusion of Cumulative Impacts:** Assessments focus almost entirely on individual projects, completely ignoring the combined, long-term ecological load on a single region.

    As per vidhi centre for legal policy, Over **95%** of clearances are granted based on isolated project assessments
- **Dilution of Democratic Checkpoints**
  - **Compromised Public Hearings:** Consultations are often reduced to mere administrative formalities, frequently marred by local intimidation, lack of safety, and severe language barriers.
  - **Shrinking Notice Timelines:** EIA Notification amendments structurally reduced the minimum public notice window from **30 days to 20 days**, crippling the community's time to review complex tech.
  - **Post-Facto Clearance Loophole:** Granting retrospective environmental clearances to already-operational, violating industries directly violates the globally accepted Precautionary Principle.
- **Monitoring, Enforcement & Bureaucratic Gaps**
  - **Weak Post-Clearance Tracking:** Regulatory bodies rely heavily on self-certified, six-monthly compliance reports submitted by the companies, with almost zero independent on-ground field checks.
  - **Chronic Institutional Capacity Deficits:** State-level authorities suffer from severe budget constraints, major staff vacancies, and a lack of specialized environmental scientists.
  - **Expanding Blanket Exemptions:** Broad definitions of **strategic** and linear projects have removed critical highway, defense, and border infrastructure works from open public scrutiny.

Provisions of the Draft EIA Notification 2020

- **Introduction of Post-Facto Clearances**
  - **Retrospective Approvals:** Allows projects that started operating without mandatory environmental clearance to secure legal regularisation after paying late fees and remediation fines.
  - **Cognizance of Violations:** Restricts the reporting of environmental violations exclusively to government authorities and the project proponent themselves, completely excluding the public.
- **Dilution of Public Consultation Mechanisms**
  - **Reduced Notice Period:** Drops the mandatory time window for public response during localized public hearings from 30 days down to 20 days.
  - **Exemption of Modernization Projects:** Removes the public consultation requirement for projects expanding or modernizing their production capacity up to 50%.
- **Expansion of Clearance Exemptions**
  - **Strategic Project Classification:** Grants the Central Government absolute authority to tag any project as "strategic," completely removing its data from the public domain.
  - **Bypassing B2 Category Projects:** Places a long list of chemical, construction, and linear projects into the "B2 Category," exempting them from both a full EIA report and public scrutiny.

- **Inland Waterways and National Highways:** Completely waives the requirement for prior environmental clearances for major inland waterway and wide national highway expansions.

- **Relaxed Compliance Tracking and Validity Extensions**
  - **Lengthened Compliance Reporting:** Relaxes the requirement for submitting mandatory environmental compliance reports from once every six months to just once a year.
  - **Extended Clearance Validity:** Increases the lifespan of primary environmental clearances for core infrastructure sectors, stretching mining project validity up to 50 years.

Way Forward for EIA

- **Structural Reforms & Ensuring Data Integrity**
  - **Establish an Independent Regulatory Body:** Set up a centralized National Environmental Protection Authority (NEPA) to decouple environmental monitoring from bureaucratic ministries.
  - **Create an Independent Consultant Pool:** Mandate that consultants be assigned randomly to projects through an automated portal, completely breaking the corrupt corporate-consultant funding link.
  - **Mandate Multi-Season Baseline Studies:** Transition from inadequate single-season data collection to mandatory multi-season ecological assessments to map real year-round environmental impacts.
  - **Institutionalize Strategic Environmental Assessment (SEA):** Scale up from narrow, project-specific appraisals to regional-level SEA to account for the combined, cumulative load on ecosystems.
- **Revitalizing Public Participation & Environmental Democracy**
  - **Ensure Inclusive Public Hearings:** Conduct consultations in local vernacular languages, provide easily digestible non-technical summaries, and video-record all site proceedings for absolute transparency.
  - **Restore Adequate Notice Timelines:** Maintain a strict minimum 30-day notice window for public review to give vulnerable, remote communities ample time to evaluate technical impacts.
  - **Abolish Post-Facto Clearances:** Eliminate retrospective approvals entirely to uphold the core Precautionary Principle and prevent industries from presenting ecological destruction as a done deal.
- **Advancing Technology & Enforcement Capabilities**
  - **Deploy AI and Continuous Satellite Monitoring:** Leverage remote sensing data and real-time drone imagery to automatically cross-verify baseline facts and catch illegal construction instantly.
  - **Strengthen Third-Party Compliance Auditing:** Replace corporate self-certification with random, mandatory environmental audits conducted by certified third-party academic and research bodies.
  - **Build State-Level Institutional Capacity:** Fill critical scientific vacancies within SEIAAs and State Pollution Control Boards (SPCBs) while upgrading state lab infrastructures.
  - **Enforce Strict Penalty Regimes:** Implement the Polluter Pays Principle by imposing heavy financial penalties, blacklisting fraudulent consultants and canceling clearances for chronic compliance failures.

Conclusion

Environmental Impact Assessment framework must transition from a mere regulatory hurdle into a proactive statutory tool that upholds the precautionary principle and environmental democracy. By balancing industrial growth with regional carrying capacity and transparent public participation, it ensures that developmental needs do not compromise ecological preservation.

8. INVASIVE SPECIES IN INDIA

Invasive Alien Species (IAS) are non-native plants, animals, or microorganisms introduced deliberately or accidentally into an ecosystem, where they rapidly outcompete native species and disrupt local ecological balances. As per the **Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES)** Global Assessment, invasive species stand as one of the top five direct drivers of global biodiversity loss.

Major Invasive Alien Species (IAS) in India

Common Name	Scientific Name & Origin	Ecological & Economic Impact
Lantana / Wild Sage	Lantana camara (Tropical America)	• Smothers native undergrowth; releases allelopathic chemicals that inhibit native flora. • Reduces palatable forage for herbivores, driving human-wildlife conflict.
Water Hyacinth	Eichhornia crassipes (Amazon Basin)	• known as the "Terror of Bengal"; forms dense floating mats that block sunlight. • Triggers eutrophication, depletes dissolved oxygen, and chokes aquatic biodiversity.
Vilayati Kikar	Prosopis juliflora (South/Central America)	• Aggressive, deep-rooted xerophyte that severely depletes local groundwater tables. • Destroys indigenous pastoral grasslands and native Acacia ecosystems.
Congress Grass / Carrot Weed	Parthenium hysterophorus (North America)	• Infests standing crops, causing severe agricultural yield drops. • Produces toxic parthenin, triggering dermatitis and respiratory allergies in humans and livestock.
Siam Weed	Chromolaena odorata (South/Central America)	• Highly flammable weed that intensifies forest fires during dry seasons. • Rapidly colonizes cleared forest areas, entirely blocking natural tree regeneration.
African Apple Snail	Achatina fulica (East Africa)	• Voracious agricultural pest that destroys food crops, vegetables, and plantation sprouts. • Acts as a vector for the rat lungworm parasite, posing public health risks.
African Catfish	Clarias gariepinus (Africa & Middle East)	• Carnivorous, air-breathing predator that directly preys on indigenous small fish. • Introduced illegally; poses a severe threat to native riverine fish populations.

Causes of Invasive Species Spread in India

- **Intentional Anthropogenic Introductions**
  - **Colonial Forestry & Greening:** Historic introduction of fast-growing exotic species (like *Prosopis juliflora*) by the British for rapid timber and fuelwood generation.

- **Ornamental & Horticultural Escapes:** Decorative plants (like *Lantana camara* and *Water Hyacinth*) escaping botanical gardens and private estates into wild ecosystems.

  According to research in the **Journal of Environmental Management**, Lantana camara alone has infested more than 154,000 square kilometers of Indian forests.

- **Commercial Aquaculture:** Illegal or poorly regulated introduction of high-yield, aggressive exotic species (like the *African Catfish*) for rapid commercial farming.

- **Unintentional & Trade-Driven Spread**
  - **Contaminated Food Imports:** Accidental entry of exotic weed seeds mixed with imported agricultural shipments.

    As per **Indian Council of Agricultural Research (ICAR)**, Parthenium hysterophorus (Congress grass) accidentally entered India during the 1950s within millions of tons of wheat imported from the US under the PL-480 **Food for Peace** program.

  - **Maritime Ballast Water:** Discharge of untreated ballast water from international cargo ships, releasing marine invasive microorganisms and alien aquatic species into Indian coastal waters.
  - **Expanding Transport Corridors:** Modern highway and railway construction deep into pristine forests acting as physical vectors for rapid seed dispersal.
- **Ecological Degradation & Climate Change**
  - **Habitat Fragmentation:** Deforestation and land-use changes create stressed, open-canopy ecosystems that are easily colonized by highly adaptable aggressive weeds.
  - **Climate-Induced Range Expansion:** Rising global temperatures and erratic rainfall patterns are expanding the habitable geographic ranges for tropical invasive species.
  - **Extreme Weather Dispersion:** Increased frequency of floods and cyclones physically sweeping and accelerating the spread of invasive seeds across interconnected river basins.
- **Institutional & Regulatory Gaps**
  - **Fragmented Legal Framework:** Absence of a dedicated, unified national policy to preemptively identify and eradicate invasive species before they establish.

    According to an economic assessment of **Mongabay-India**, lack of a unified national biosecurity policy has contributed to India losing $127.3 billion (₹8.3 trillion) over 60 years.

  - **Weak Border Bio-Security:** Inadequate phytosanitary and quarantine infrastructure at seaports and international borders failing to intercept smuggled alien flora and fauna.

Impacts of Invasive Species Spread in India

- **Ecological & Biodiversity Impacts**
  - **Loss of Indigenous Flora:** Aggressive weeds outcompete native species, completely altering the structure of critical forest understories.

    According to a study by the **Wildlife Institute of India (WII)**, fast-spreading invasive plants are expanding across 15,500 square kilometers of India's natural areas every year.

  - **Destruction of Wildlife Habitats:** Species like *Lantana* smother palatable grass, leaving herbivores without forage and destabilizing the food web.

    Lantana camara *alone has severely invaded more than 40% of India's total tiger range, choking out primary forage for herbivores.*

  - **Disruption of Aquatic Ecosystems:** Water Hyacinth mats deplete dissolved oxygen, triggering mass fish kills and rapid wetland eutrophication.

According to the **National Environmental Engineering Research Institute (NEERI)**, dense Water Hyacinth mats can block up to 95% of sunlight penetration.

- **Allelopathic Suppression:** Invasive plants secrete toxic biochemicals into the soil, permanently halting the germination of native seeds.

- **Economic & Agricultural Impacts**
  - **Severe Crop Yield Reductions:** Weeds like *Parthenium* aggressively choke agricultural fields, driving up farmers' weeding and management costs.
  - **Depletion of Groundwater Reserves:** Deep-rooted exotics like *Prosopis juliflora* aggressively drain water tables in already water-scarce, arid zones.
  - **Loss of Livestock Pastureland:** Infestations transform highly productive grasslands (like the Banni grasslands) into unpalatable, barren scrublands.

    According to **Space Applications Centre (ISRO)**, invasive scrub has caused the loss of roughly 25% of native grassland areas in key pastoral biomes like the Banni grasslands.
  - **Collapse of Inland Fisheries:** Choked waterways block navigation and destroy native fish breeding grounds, hitting the livelihoods of traditional fishing communities.

    According to the **National Fisheries Development Board (NFDB)**, excessive weed infestation in public wetlands correlates with up to a 40% drop in seasonal fish catch yields.
- **Socio-Cultural & Public Health Impacts**
  - **Escalation of Human-Wildlife Conflict:** Forage degradation forces wild herbivores and carnivores out of protected reserve areas and into human settlements.
  - **Triggering of Chronic Allergies:** Airborne pollen from Congress Grass causes widespread contact dermatitis and severe respiratory issues in rural populations.
  - **Spread of Vector-Borne Diseases:** Stagnant water pockets trapped under dense aquatic weed mats create ideal breeding grounds for malaria and dengue vectors.
  - **Threat to Tribal Livelihoods:** loss of indigenous non-timber forest produce directly undermines the economic survival of forest-dwelling tribal groups.

Measures Taken Against Invasive Species in India

Statutory & Regulatory Frameworks	• **Plant Quarantine (Regulation of Import into India) Order, 2003:** Mandates strict phytosanitary checks at entry ports to intercept accidental alien flora introductions. • **Biological Diversity Act, 2002:** Empowers the National Biodiversity Authority (NBA) to regulate, access, and protect indigenous biological resources from exotic bio-piracy. • **Wildlife (Protection) Amendment Act, 2022:** Introduces provisions enabling the Central Government to strictly regulate or ban the import, possession, and proliferation of invasive alien species.
Institutional & Field-Level Interventions	• **Project Elephant & Project Tiger Guidelines:** Mandates targeted funds within protected areas for the systematic physical uprooting and manual clearing of *Lantana camara*. • **NITI Aayog National Action Plans:** Formulates multi-sectoral macro roadmaps focusing on the ecological restoration of the degraded Banni grasslands by replacing *Prosopis juliflora* with native *Cenchrus ciliaris* grasses.

	• **Indian Council of Agricultural Research (ICAR) Bio-Control Initiatives:** Deploys targeted host-specific biological agents, such as using the Mexican beetle (*Zygogramma bicolorata*) to suppress *Parthenium* weeds.
Innovation, Commercialization & Global Alignments	• **Weed-to-Wealth Community Programs:** Promotes rural livelihoods by training local artisans and self-help groups (SHGs) to harvest *Lantana* wood for making eco-friendly furniture. • **The 'Lantana Elephants' Initiative:** Commercially utilizes uprooted *Lantana* biomass to craft life-sized elephant sculptures, raising global conservation funds and driving local tribal employment. • **National Green Tribunal (NGT) Directives:** Issues binding statutory orders to state municipal corporations for the mandatory, time-bound clearance of toxic African Apple Snails and Water Hyacinth from urban water bodies.

International Measures Taken Against Invasive Species

- **Convention on Biological Diversity (CBD - Article 8h):** Obligates contracting parties to prevent the introduction of, control, or eradicate alien species that threaten indigenous ecosystems and habitats.
- **Kunming-Montreal Global Biodiversity Framework (KMGBF - Target 6):** Commits nations to reduce the rate of introduction and establishment of priority invasive alien species by **at least 50% by 2030**.
- **International Plant Protection Convention (IPPC, 1951):** Sets global phytosanitary standards and quarantine certification protocols to prevent international trade and shipping from spreading plant pests and weeds.
- **CITES (1975):** Monitors and regulates international trade pathways to ensure exotic wildlife transport does not inadvertently spark biosecurity risks or wild population collapses.
- **IMO Ballast Water Management Convention (2017):** Mandates international commercial vessels to mechanically treat, filter, or exchange ballast water mid-sea to neutralize transboundary marine bio-invasions.
- **ICAO & IATA Guidelines:** Enforces global aviation bio-security screening for commercial aircraft hulls, passenger baggage, and cargo holds to block invasive hitchhikers.
- **FAO Code of Conduct for Exotic Biocontrol Agents:** Establishes strict international safety checklists for the transboundary movement and release of non-native organisms used to combat agricultural weeds.
- **IPBES Thematic Assessment Report (2023):** Delivers a comprehensive global scientific baseline on biological invasions, projecting that biological invasions impose a global economic toll exceeding **$423 billion annually**.
- **IUCN Invasive Species Specialist Group (ISSG):** Anchors global field research, maintaining the **Global Invasive Species Database (GISD)** to issue early alerts and trace propagation pathways.
- **Global Register of Introduced and Invasive Species (GRIIS):** Coordinates an open-access, verified country-wise taxonomic inventory to assist nations in developing localized border-control strategies.

Difference between Alien Species and Invasive Species

Parameter	Alien Species	Invasive Species
Definition	Any species introduced outside its natural historic geographic range by human activity, whether deliberately or accidentally.	An alien species that establishes itself, multiplies rapidly, and actively displaces native biodiversity.

Ecological Impact	Most are benign or integrated; they do not automatically disrupt local food webs or choke native habitats.	Highly destructive; they cause severe ecological imbalance, habitat fragmentation, and native species extinction.
Economic Implication	Often highly beneficial to the host economy (e.g., boosting agricultural yields or aesthetic landscaping).	Imposes massive economic losses by destroying crops, depleting groundwater, and clogging vital waterways.
Characteristic	Defined solely by its geographic origin (being non-native to the ecosystem).	Defined by its aggressive behavior and capacity to inflict multi-dimensional harm on the host ecosystem.
Examples in India	Potato, Tomato, Maize, and Eucalyptus (vital for food security and commercial plantation forestry).	Lantana camara, Water Hyacinth, and African Catfish (ecologically toxic and economically damaging bio-pollutants).

Way Forward for Managing Invasive Species in India

- **Policy & Regulatory Strengthening**
  - **Enact a Comprehensive IAS Act:** Create a single, unified national law dedicated exclusively to biosecurity, replacing fragmented sectoral rules with a clear mandate.

    Australia's Biosecurity Act 2015 provides a single, unified legal framework managing all national biosecurity risks and invasive threats.
  - **Establish a National Invasive Species Authority:** Set up a specialized nodal agency under MoEFCC to coordinate early warning, quarantine, and rapid eradication protocols.

    New Zealand Environmental Protection Authority (EPA) acts as the specialized nodal body regulating and approving all alien species entries.
  - **Update the Harmonized Master List:** Maintain a dynamic, science-backed black list of high-risk alien species to block their import at maritime and aviation borders.

    Directorate General of Foreign Trade (DGFT) maintains an import restriction list, which researchers urge updating to include highly invasive exotic aquarium fish.
- **Early Detection & Data Systems**
  - **Deploy AI-Driven Geospatial Mapping:** Use satellite imagery and drone tracking to map the real-time spread of aggressive canopy weeds like *Lantana* across tiger reserves.
  - **Establish National Bio-Surveillance Networks:** Set up digital dashboards at major ports to dynamically screen and quarantine international shipping ballast water.

    National Institute of Oceanography (NIO) runs genetic bio-surveillance at Mumbai and Visakhapatnam ports to detect exotic bio-fouling organisms.
  - **Launch Citizen Science Portals:** Develop mobile apps allowing forest guards, farmers, and trekkers to instantly geo-tag and report fresh bio-invasion outbreaks.

    'iNaturalist' platform and India Biodiversity Portal are used by Indian researchers to crowdsource community observations of the destructive Fall Armyworm.
- **Scientific & Field Interventions**
  - **Adopt Integrated Weed Management (IWM):** Combine physical uprooting with host-specific bio-control agents (like *Zygogramma bicolorata*) for sustainable containment.
  - **Mandate Ecological Restoration:** Ensure that the removal of invasive species is immediately followed by the dense replanting of indigenous grasses and trees.

The Periyar Tiger Reserve model follows mechanical Lantana clearance with the immediate intensive replanting of native grasses and bamboo.

- **Enforce Biosafety Protocols in Aquaculture:** Implement strict zoning, geo-fencing, and heavy penalties to stop the illegal farming of carnivorous exotics like African Catfish.

- **Socio-Economic Models & Community Action**
  - **Scale Up Weed-to-Wealth Initiatives:** Incentivize the commercial conversion of *Lantana* and *Water Hyacinth* into eco-friendly furniture, handicrafts, and bio-gas.

Theosophical Society and **tribal cooperatives in BR Hills** actively harvest invasive Lantana stalks to manufacture premium eco-friendly handicraft furniture.

  - **Converge with MGNREGA:** Mobilize rural workforce by linking local invasive clearance and eco-restoration drives with assured wage employment schemes.
  - **Empower Gram Sabhas & Joint Forest Management Committees:** Devolve financial and technical resources to local communities to spearhead native habitat restoration.

Conclusion

Effectively managing invasive alien species is vital to safeguard India's rich ecological heritage, ensure rural food security and protect agricultural livelihoods from biological pollution. By integrating robust early-warning technologies with community-led **weed-to-wealth** initiatives, India can successfully restore its degraded native ecosystems.

9. WILDLIFE CONSERVATION IN INDIA

Wildlife conservation in India is a critical ecological imperative rooted in Constitutional mandates like **Article 48A** and **Article 51A(g)**, which direct the state and citizens to protect the natural environment. India is home to nearly **8% of global biodiversity** packed into just 2.4% of the world's landmass, India faces the immense challenge of balancing rapid infrastructure growth with habitat preservation.

Regulatory Framework for Wildlife Conservation in India

- **Constitutional Provisions**
  - **Article 48A (Directive Principles):** Directs the State to protect, improve, and safeguard the country's forests and wildlife resources.
  - **Article 51A(g) (Fundamental Duties):** Mandates every Indian citizen to show compassion for living creatures and protect natural environments.
  - **Concurrent List (7th Schedule):** Places "Protection of Wild Animals and Birds" under entry 17B, enabling both Union and State governments to legislate.
- **Statutory Acts & National Policies**
  - **Wildlife (Protection) Act (WPA), 1972:** Establishes six distinct legal schedules of protection and prohibits the hunting of endangered species.
  - **WPA Amendment Act, 2022:** Streamlines conservation schedules into four clear lists and integrates provisions for CITES compliance directly into domestic law.
  - **Forest (Conservation) Amendment Act, 2023:** Regulates the diversion of forest land for non-forest strategic or security purposes while promoting eco-tourism.
  - **Biological Diversity Act, 2002:** Enforces decentralized decentralization of access and benefit-sharing (ABS) mechanisms for genetic resources through local committees.
  - **Environment (Protection) Act, 1986:** Authorizes the Central Government to notify Eco-Sensitive Zones (ESZs) as shock absorbers around wildlife sanctuaries.

- **Institutional Authorities**
  - **National Board for Wildlife (NBWL):** Apex statutory body chaired by the Prime Minister that reviews and approves all development projects inside protected zones.
  - **National Tiger Conservation Authority (NTCA):** Statutory body under MoEFCC tasked with strengthening tiger reserves via anti-poaching forces and scientific tracking.
  - **Wildlife Crime Control Bureau (WCCB):** Statutory intelligence body mandated to gather intelligence on transboundary illegal wildlife trade and coordinate enforcement actions.
  - **Central Zoo Authority (CZA):** Regulates and oversees national zoo management standards, animal registration, and ex-situ endangered species breeding programs.
- **International Commitments**
  - **CITES (Washington Convention):** Regulates international cross-border trade in specimens of wild fauna and flora to prevent species endangerment.
  - **Convention on Migratory Species (CMS):** Guides international cooperative action to protect transboundary migratory species along global flyways.

Major Wildlife Conservation Projects in India

Project Name	Launch Year	Key Objective
Project Tiger	1973	Statutory protection via NTCA utilizing M-STrIPES digital patrol mapping to secure core habitats.
Project Crocodile	1975	Captive breeding and rear-and-release protocols to restore populations, primarily launched in Odisha.
Project Rhino	2005	Expand distribution across 7 protected areas in Assam under Indian Rhino Vision to prevent concentration risks.
Project Vulture	2006	Action plan to ban veterinary Diclofenac and establish Conservation Breeding Centres (VCBCs) to counter toxicity.
Project Snow Leopard	2009	High-altitude landscape conservation across the Himalayas utilizing community-based conservation models.
Project Elephant	1992	Secure designated Elephant Corridors and deploy AI seismic alerts (Gajraj system) to minimize rail-kill conflict.
Project Dolphin	2021	Conserve riverine biodiversity indicators using modern sonar tracking and stopping direct wastewater discharge.
Project Lion	2020	Disease surveillance, habitat restoration in Gir, and developing relocation sanctuaries like Barda in Gujarat.
Project Cheetah	2022	World's first intercontinental wild carnivore translocation to re-establish functional grassland biomes in Kuno.
Project Hangul	1970	Habitat stabilization inside Dachigam National Park to reverse severe genetic isolation and inbreeding.

Issues with Wildlife Conservation in India

- **Habitat Fragmentation & Connectivity Loss**
  - **Linear Infrastructure Encroachment:** Rapid construction of national highways, railway networks, and power transmission lines cuts directly through critical wildlife landscapes.

According to **Land Use Policy**, the rapid development inside India's dense forests has caused a 6% increase in patchiness, resulting in a severe 71.5% reduction in large forest core zone.

- **Corridor Vulnerability:** Traditional elephant and tiger migratory corridors lack official legal protection, leaving them open to agricultural encroachment and commercial resorts.

According to **Wildlife Trust of India (WTI)**, over **35%** of traditional, functional elephant and tiger migratory corridors completely lack designated legal protection.

- **Monoculture Plantation Focus:** Expanding state forest covers often prioritize commercial monoculture trees instead of complex and biodiverse natural forests.

- **Anthropogenic & Biotic Pressures**
  - **Escalating Human-Wildlife Conflict (HWC):** Shrinking buffer zones bring megafauna into frequent, fatal direct contact with fringe villages, leading to high annual human casualties.

According to **MoEFCC**, human-wildlife encounters result in an annual average of over 500 human fatalities.

  - **Unregulated Livestock Grazing:** Excessive cattle grazing inside Protected Areas (PAs) depletes grass reserves for wild herbivores and risks cross-species zoonotic disease (Canine Distemper Virus).
  - **Invasive Alien Species (IAS):** Proliferation of aggressive non-native weeds like *Lantana camara* and *Senna spectabilis* chokes native flora and starves local wildlife populations.

According to **Wildlife Institute of India (WII)**, the aggressive Central American weed Lantana camara has heavily invaded over 44% of India's total forest area.

- **Regulatory, Enforcement & Administrative Gaps**
  - **Dismal Conviction Rates:** Despite stringent laws like the amended Wildlife (Protection) Act, convictions for complex wildlife crimes hover at a low **2–3%** due to weak forensic collection.
  - **Fragmented Institutional Authority:** Inefficient inter-departmental coordination between the forest, revenue, agriculture and infrastructure ministries hampers cohesive, long-term spatial layout planning.
  - **Porous Borders & Organized Poaching:** High transnational black-market demand for pangolin scales, tiger parts, and star tortoises exploits vulnerable border checks.
  - **Community Alienation:** Protracted delays in recognizing Community Forest Resource (CFR) rights under the Forest Rights Act (FRA), 2006, alienate indigenous groups and stifle local conservation support.

According to the **Ministry of Tribal Affairs**, long delays in processing and approving Community Forest Resource titles leave over 60% of potential tribal claims uncertified.

Measures Taken for Wildlife Conservation in India

- **Recovery and Translocation Projects**
  - **Project Tiger & Project Elephant Consolidation:** Integrated administrative operations to manage contiguous landscapes, pushing the wild tiger population up to over 3,600 individuals.
  - **Project Cheetah (Kuno National Park):** Initiated the world's first intercontinental wild carnivore translocation to restore functional grassland ecosystems in central India.
  - **Project Lion (Asiatic Lion Landscape):** Expanding intensive genetic management and creating satellite habitats in Gujarat to secure the population against localized epidemic outbreaks.
  - **Integrated Development of Wildlife Habitats (IDWH):** Provides direct central funding for 22 identified critically endangered species, including the Great Indian Bustard and the Hangul.

- **Technological Interventions & Digital Surveillance**
  - **M-STrIPES Digital Patrol Monitoring:** Uses an Android-based GIS application across all national tiger reserves to log patrol routes, map ecological data, and pinpoint poaching vulnerabilities.
  - **Thermal Drone and AI-Based Tracking:** Deploys UAV night-vision surveillance and automated camera traps along forest perimeters to stop unauthorized human entry and predict herd movements.
  - **e-Eye Electronic Surveillance System:** Uses intelligent long-range, high-definition thermal cameras mounted on towers within sensitive zones (like Corbett) to catch poachers in total darkness.
- **Structural Infrastructure & Landscape Mitigation**
  - **Eco-Friendly Linear Infrastructure Guidelines:** Mandates structural animal crossings, such as dedicated underpasses and overpasses, on all new national highways crossing forest zones.
  - **Acoustic Rail Alerts & Underpasses:** Installs optical-fiber seismic detection sensors (*Gajraj system*) along high-risk rail tracks to automatically alert train pilots of nearby megafauna.
- **Legal, Community, and Financial Initiatives**
  - **WPA Sanctions Escalation (2022):** Significantly increased the statutory financial penalties and prison sentences for hunting Schedule-I protected wildlife.
  - **Community Reserve Expansion:** Notified over 200 highly flexible **Community and Conservation Reserves** to legally protect private and community-owned lands acting as migratory wildlife corridors.
  - **Voluntary Village Relocation Funding:** Elevated compensation packages to 15 lakh rupees per family to support completely voluntary community relocations outside core tiger reserves, creating undisturbed wildlife spaces.

Way Forward

- **Shift toward Landscape-Level & Ecosystem-Based Management**
  - **Transition Away from Isolated Fortress Conservation:** Shift focus from isolated protected areas to a continuous "Landscape-Level Approach" to safely secure transboundary wildlife habitats.
  - **Legalize and Restore Migratory Corridors:** Grant formal, statutory protection to threatened ecological corridors to ensure genetic exchange among fragmented megafauna populations.
  - **Prioritize Grassland and Wetland Rejuvenation:** Shift compensatory afforestation targets from commercial tree monocultures toward restoring native grasslands and fragile riverine basins.
- **Deepen Community Co-Management & Economic Incentives**
  - **Mainstream Community-Led Conservation Models:** Accelerate the formal recognition of Community Forest Rights (CFR) under the Forest Rights Act (FRA) to empower indigenous populations as frontline protectors.
  - **Institutionalize Micro-Ecotourism and Green Livelihoods:** Create community-owned homestay networks and eco-tourism co-operatives to link village financial growth directly with local biodiversity health.
  - **Scale Up Smart Conflict-Resistant Agriculture:** Incentivize perimeter fringe farmers to cultivate unpalatable buffer crops like lemongrass, chilly, and honeybee hives (*Project RE-HAB*) to deter wildlife.
- **Advance Technological & Intelligence Infrastructures**
  - **Establish a National Wildlife Forensic Database:** Set up advanced DNA sequencing facilities to standardise wildlife forensic evidence collection and improve low statutory conviction rates.
  - **Deploy AI-Driven Early Warning Systems:** Expand the integration of fiber-optic seismic sensor networks (*Gajraj system*) and thermal drones to intercept animal movements near rail lines.

- **Optimize Inter-Departmental Spatial Planning:** Mandate automated, digital environmental impact clearings using GIS mapping to ensure infrastructure projects bypass core breeding grounds.

Best Practices for Wildlife Conservation

- **Indian Best Practices & Grassroots Models**
  - **International Big Cat Alliance (IBCA):** Led a treaty-based global alliance in 2025 across 27 countries to institutionalize multi-national funding, capacity building, and range-country intelligence sharing.
  - **Indore and Udaipur Wetland City Accreditations:** Pioneered Asia's first Ramsar-accredited Wetland Cities in January 2025, blending municipal zoning with the ecological restoration of Urban Wildlife Habitats.
  - **Community-Led Hornbill Nest Adoption:** Program in Arunachal Pradesh pays local Nyishi tribesmen structural financial stipends to actively guard and monitor nesting trees of vulnerable hornbills.
  - **Gajraj AI-Driven Elephant Interception:** Deployed fiber-optic acoustic and seismic sensor arrays along high-risk railway lines to automatically transmit real-time early warning telemetry to train pilots.
  - **Sacred Groves (Orans & Devrais) Mapping:** Leveraged spatial AI under MoEFCC guidelines to map indigenous community-managed forests, preventing corporate land-use conversions.
- **Global Best Practices & Interventions**
  - **US Endangered Species Act (ESA) Success Model:** Saved apex avian predators like the Bald Eagle from localized extinction through strict statutory pesticide bans paired with state-led captive breeding programs.
  - **Namibia's Communal Conservancies Framework:** Devoted over 20% of national land to local community trusts, granting indigenous populations complete legal ownership and revenues from wildlife tourism.
  - **Costa Rica's Payments for Environmental Services (PSA):** Reversing historic deforestation by using state fuel-tax allocations to directly pay landowners for preserving wildlife habitats and natural corridors.
  - **Europe's Transboundary Emerald Network:** Standardized contiguous, cross-border biological corridors across dozens of sovereign European states to protect wide-ranging, migratory carnivore populations.
  - **Norway's High-Tech Smart Salmon Fences**: Integrates underwater computer-vision scanners and automated mechanical gates within rivers to automatically isolate and block invasive fish species from native breeding grounds.

Committee recommendation for wildlife conservation

- **T.S.R. Subramanian Committee (2014):** Recommended creating a National Environment Management Authority (NEMA) and State Environment Management Authorities (SEMA) for faster, tech-driven ecological clearances.
- **Madhav Gadgil Committee / WGEEP (2011):** Recommended classifying the entire Western Ghats landscape as an Ecologically Sensitive Area (ESA), divided into three graded zones (ESZ-1, ESZ-2, ESZ-3).
- **K. Kasturirangan Committee (2013):** Diluted the Gadgil report by recommending that only 37% (the biologically richest patches) of the Western Ghats be legally demarcated as an Ecologically Sensitive Area.
- **Sunita Narain Committee / Tiger Task Force (2005):** Recommended the immediate statutory establishment of the National Tiger Conservation Authority (NTCA) to overhaul tiger reserve administration and funding.

- **M.K. Ranjitsinh Committee:** Recommended the mandatory undergrounding of low-voltage power transmission lines in the priority habitats of Rajasthan and Gujarat to halt Great Indian Bustard (GIB) electrocutions.

Conclusion

Success of wildlife conservation in India lies in moving beyond administrative boundaries to treat entire landscapes as single, connected living systems. By integrating advanced digital tools like AI-driven tracking with the traditional wisdom of indigenous communities, India can effectively minimize human-animal friction.

10. ONE HEALTH APPROACH

One Health approach is an integrated framework that recognizes the inseparable link between the health of humans, animals, plants, and their shared environment. By breaking down institutional silos, it drives cross-sectoral collaboration to tackle complex global threats like antimicrobial resistance (AMR) and emerging zoonotic pandemics at their ecological source.

Pillars of the One Health Approach

Human-Animal-Environment Pillars	• **Human Health & Epidemiology:** Managing public health vectors, food safety systems, and clinical diagnostics to protect communities from emerging pathological threats. • **Animal Health & Veterinary Care:** Enhancing livestock healthcare, monitoring domestic poultry biosecurity, and managing wild fauna populations to intercept viral mutations early. • **Environmental & Ecosystem Health:** Conserving natural habitats, managing water basins, and protecting forest ecosystems to maintain the biological balances that suppress disease vectors.
Operational & Collaborative Pillars (Four Cs)	• **Communication:** Building open data pipelines to ensure real-time medical, veterinary, and environmental information flows instantly across different scientific sectors. • **Collaboration:** Aligning multi-agency forces—such as the **Quadripartite alliance (WHO, WOAH, FAO, UNEP)**—to tackle complex health issues that no single group can solve alone. • **Coordination:** Organizing joint emergency response drills and shared field logistics to quickly contain local disease outbreaks before they escalate. • **Capacity Building:** Training a new generation of professionals in cross-disciplinary skills, while upgrading diagnostic labs across the medical, animal, and environmental sectors.
Structural & Strategic Frameworks	• **Antimicrobial Resistance (AMR) Mitigation:** Enforcing coordinated restrictions on antibiotic overuse in livestock farming and human medicine to preserve global drug efficacy. • **Zoonotic Disease Surveillance:** Tracking pathogens at the critical human-animal-wildlife boundaries to catch potential spillover events before they trigger pandemics. • **Food Safety and Security System:** Monitoring the entire food supply chain—from agricultural fields and livestock farms to the consumer's plate—to eliminate foodborne toxins and illnesses.

Significance of the One Health Approach

- **Pandemic Preparedness & Biosecurity**
  - **Intercepts Zoonotic Spillover:** Catches over 60% of known human infectious diseases and 75% of emerging pathogens directly at their animal source.

    According to the **World Health Organization (WHO)**, roughly 60% of known human infectious diseases and 75% of emerging infectious pathogens are zoonotic.
  - **Prevents Mass Economic Losses:** Saves trillions of dollars in global GDP by intercepting catastrophic pandemics before they trigger nationwide lockdowns.

    According to the **World Bank**, investing in an operational One Health framework globally yields an estimated annual return of up to $37 billion by preventing catastrophic pandemic lockouts.
  - **Curtails Antimicrobial Resistance (AMR):** Preserves life-saving antibiotic efficacy by systematically stopping drug overuse in commercial livestock and fisheries.
- **Ecological & Environmental Protection**
  - **Arrests Habitat Fragmentation:** Mitigates disease risks by identifying how deforestation and wildlife displacement drive animals closer to human settlements.
  - **Preserves Vital Ecosystem Services:** Protects clean water basins, native vegetation, and soil biomes that naturally filter out toxic environmental pathogens.
  - **Tackles Climate-Driven Health Risks:** Monitors how rising global temperatures cause deadly disease vectors like mosquitoes and ticks to migrate into new regions.
- **Food Security & Governance Efficiency**
  - **Secures Global Food Supply Chains:** Eradicates highly contagious livestock diseases like Foot-and-Mouth or Avian Flu to protect low-income farmers' livelihoods.
  - **Breaks Down Institutional Silos:** Unifies isolated medical, veterinary, and environmental departments under a single, cost-effective governance framework.
  - **Accelerates SDG Achievement:** Directly drives multiple United Nations Sustainable Development Goals, specifically Zero Hunger (SDG 2), Good Health (SDG 3), and Life on Land (SDG 15).

Issues of the One Health Approach

- **Institutional & Bureaucratic Roadblocks**
  - **Departmental Silos:** Strict administrative separation between medical, agricultural, and environmental ministries prevents integrated policy action.
  - **Asymmetrical Funding:** Human clinical sector receives the vast majority of financial resources, leaving veterinary and environmental wings severely underfunded.
  - **Statutory Authority:** Lack of a single, centralized legislative body with the legal teeth to command cross-departmental accountability.
- **Operational & Data Deficits**
  - **Fragmented Surveillance Infrastructure:** Absence of a unified, cross-sectoral digital platform prevents real-time sharing of animal-to-human outbreak data.
  - **Neglect of the Environmental Pillar:** Ecological component is routinely ignored, with policies treating the environment as a passive background rather than an active driver.

    According to the **United Nations Environment Programme (UNEP)**, the environmental component receives less than **5%** of total global One Health project allocations.
  - **Acute Ground-Level Capacity Deficits:** Severe shortages of specialized wildlife veterinarians, field biometric labs, and local environmental trackers in remote areas.
- **Socio-Economic & Global Hurdles**

- **Livelihood Conflicts with Local Communities:** Enforcing strict livestock biosecurity measures causes heavy financial distress for smallholder, low-income farmers.

  According to the **Food and Agriculture Organization (FAO)**, enforcing strict livestock quarantine triggers up to a **40% seasonal income collapse** for low-income smallholder farmers.
- **Geopolitical and Data-Sharing Skepticism:** Nations often delay reporting local zoonotic flare-ups due to fears of immediate international trade bans and tourism losses.
- **Inadequate Academic Integration:** Higher education programs remain heavily isolated, producing medical doctors and veterinarians who rarely train together.

Way Forward

- **Institutional Alignment & Policy Integration**
  - **Establish a Unified Statutory Framework:** Create a powerful, overarching National One Health Directorate to directly bind the operations of human health, animal husbandry, and environmental ministries.
  - **Operationalize State-Level Governance:** Rapidly roll out the centralized Model Governance Framework to decentralize multi-sectoral health responses across all States and Union Territories.
  - **Secure Institutional Budget Pools:** Move away from isolated, sector-specific funding loops by writing explicit, combined One Health line items directly into national and state annual budgets.
- **Technological Innovation & Early Warning**
  - **Deploy AI-Driven Metagenomic Surveillance:** Leverage artificial intelligence and Next-Generation Sequencing (NGS) to automatically map and spot novel pathogens at critical ecological boundaries.
  - **Build a Unified Data Exchange Portal:** Create an open, real-time digital pipeline connecting clinical, veterinary, and wildlife field databases to instantly flags zoonotic spillover risks.
  - **Strengthen a Joint BSL Laboratory Network:** Connect existing Biosafety Level 3 and 4 (BSL-3/4) testing facilities across diverse scientific departments into a single, synchronized outbreak task force.
- **Community Engagement & Grassroots Capacity**
  - **Integrate Community and Indigenous Knowledge:** Use the hands-on ecological insights of tribal populations and forest dwellers to monitor wildlife health anomalies and ecosystem disruptions.
  - **Enforce Strict Antimicrobial Stewardship:** Strictly penalize the unregulated, non-therapeutic use of critical antibiotics in commercial aquaculture and industrial poultry farming.
  - **Uphold Joint Outbreak Mock Drills:** Conduct regular, mandatory inter-agency emergency simulation exercises to perfect ground-level logistics before a pathogen spreads into the public domain

Global Models for the One Health Approach

Global Intergovernmental & Treaty Models	• **Quadripartite Alliance:** A unified institutional model between WHO, WOAH, FAO, and UNEP that co-authored the *One Health Joint Plan of Action (2022–2026)* to harmonize global health governance. • **WHO Pandemic Agreement (2025):** The first international treaty legally binding signatory nations to embed One Health surveillance directly into their national pandemic prevention and preparedness laws. • **G20 Deccan High-Level Principles:** Formulated under India's G20 Presidency to create a global blueprint for institutionalizing pandemic funds and cross-border digital health data tracking.

	• **G7 Shared Understanding Action Tracks:** A financial and policy framework focused on six collaborative vectors, primarily prioritizing the reduction of antibiotic misuse in global agrifood chains.
Regional & Country-Level Implementation Models	• **USA (CDC's One Health Office):** A multi-agency national model featuring real-time data sharing across the CDC, USDA, and EPA to catch cross-species outbreaks (e.g., *Salmonella* transfers between wild birds and humans). • **Africa CDC Framework:** A continental model utilizing a dedicated *One Health Coordinating Group on Zoonotic Diseases* to streamline data pipelines across asymmetric national public health institutes. • **PREZODE Initiative (France/Global):** An international framework focusing on upstream preventative field research to map and mitigate zoonotic spillover risks before pathogens adapt to human hosts. • **Kenya's Zoonotic Disease Unit (ZDU):** A highly successful, country-level structural model that physically co-locates medical epidemiologists and veterinarians under one roof for joint outbreak tracing. • **Zambia's Interdisciplinary Pandemic Plan:** A localized governance blueprint using integrated veterinary surveillance to rapidly crush transboundary avian influenza vectors before they hit human poultry handlers.

Committee recommendation for One health approach

High-Level Expert Panel for One Health (OHHLEP)	• **Definition of One Health:** Formulated the definitive global definition of One Health, integrating humans, animals, and ecosystems into a unified, balanced framework. • **One Health Joint Plan of Action:** Collaborated with the Quadripartite (FAO, UNEP, WHO, WOAH) to create a 5-year global operational roadmap targeting zoonotic spillovers and AMR. • **Early Warning Integration:** Recommended syncing environmental degradation indicators (like deforestation rates) with veterinary and public health surveillance systems to predict spillovers.
K. VijayRaghavan Committee	• **Frontier Bio-Surveillance:** Recommended strengthening genetic bio-surveillance networks at international airports and maritime ports to catch exotic pathogens in ballast water and cargo. • **Wildlife Health Integration:** Advocated for integrating the Ministry of Environment, Forest and Climate Change (MoEFCC) directly into pandemic prevention budgeting, elevating the environmental pillar.

	• **Community-Level Capacity Building:** Proposed training frontline workers (like ASHA workers and paravets) to collaboratively identify and report early signs of livestock-to-human clusters.

Conclusion

Operationalizing the One Health approach, as emphasized in NITI Aayog's strategic roadmaps, is imperative to transition India from a reactive outbreak management model to a proactive, robust biosecurity regime. By permanently dismantling institutional silos and integrating human, veterinary, and environmental surveillance pipelines, this framework reinforces public health equity at the grassroots level.

■■■

MGP/BP/GS3/013

POLLUTION

1. AIR POLLUTION

Air pollution in India has mutated from a seasonal environmental hazard into a perennial public health emergency and a major macroeconomic drag. According to the **2025 World Air Quality Report by IQAir**, India ranks as the **6th most polluted country globally**, with a population-weighted PM 2.5 concentration of **48.9 ug/m3**, nearly ten times higher than the World Health Organization's safe guideline.

Current Status of Air Pollution in India

- **Global Standings & Particle Concentrations**
  - **Sixth Most Polluted Nation:** As per IQAir World Air Quality Report, India ranks 6th globally out of 143 countries in terms of atmospheric toxicity.
  - **Widespread Non-Compliance:** As per Centre for Research on Energy and Clean Air, Nearly half of monitored Indian cities (123 out of 248) routinely log monthly average concentrations higher than India's own National Ambient Air Quality Standards.
- **Urban Pollution Hotspots**
  - **Dominance in Global Rankings:** India accounts for 66 out of the world's 100 most polluted cities, with the worst concentrations within the landlocked Indo-Gangetic Plain.
  - **World's Most Toxic Capital:** New Delhi holds the title of the world's most polluted capital city, logging an annual average concentration.
  - **New Global Peak:** Loni in Ghaziabad, Uttar Pradesh, is recorded as the most polluted city globally, with an average annual PM2.5 level of 112.5 ug/m^3—exceeding WHO guidelines by over 22 times.

Major Air Pollutants under CPCB Guidelines

S.No.	Pollutant Name	Core Anthropogenic Sources
1	**PM2.5**(Fine Particulate)	Stubble burning, vehicular exhaust, coal plants
2	**PM10** (Coarse Particulate)	Construction activities, road dust, fly ash
3	**Sulphur Dioxide (SO2)**	Coal thermal plants, oil refineries, metal smelting
4	**Nitrogen Dioxide (NO2)**	Automobile exhaust, heavy diesel fleets, boilers
5	**Carbon Monoxide (CO)**	Incomplete fuel combustion, old vehicular fleets
6	**Ozone (O3)** (Ground-Level)	Photochemical reaction of NOx + VOCs in sunlight
7	**Ammonia (NH3)**	Nitrogenous fertilizers, livestock waste, sewage
8	**Lead (Pb)**	Battery recycling, industrial smelting, aviation fuel
9	**Benzene (C6H6)**	Crude oil processing, petrol evaporative emissions
10	**Benzo(a)pyrene (BaP)**	Biomass combustion, incomplete burning of waste

Major Causes of Air Pollution in India

- **Primary Anthropogenic & Industrial Sources**
  - **Coal-Fired Thermal Power:** Heavily relies on sub-critical coal plants lacking Flue Gas Desulfurization units, emitting massive volumes of SO2 and NOx.

    As per **Centre for Science and Environment,** Over **90%** of coal plants missed the initial Flue Gas Desulfurization (FGD) installation deadlines.
  - **Industrial Emissions:** Unregulated combustion in MSMEs, brick kilns, and chemical units using high-sulfur petcoke and furnace oil.

    Acc. to TERI study, industrial sector contributes **25–30%** of total ambient PM2.5 emissions across major Indian cities
  - **Vehicular Exhaust:** High density of older, poorly maintained commercial fleets emitting toxic black carbon, hydrocarbons, and nitrogen oxides.

    As per SAFAR, the industrial sector contributes **25–30%** of total ambient PM2.5 emissions across major Indian cities.
  - **Rapid Construction Dust:** Unmitigated infrastructure expansion generating vast amounts of coarse fugitive dust (PM10) due to lack of wind barriers.
- **Agricultural & Rural Activities**
  - **Stubble Burning:** Massive seasonal open-field burning of paddy straw residue in Punjab, Haryana, and UP during the narrow October–November transition window.

    As per CPCB, Farm fires contribute up to **42%** of Delhi's daily PM2.5 concentrations during peak winter days in November
  - **Household Biomass Use:** Widespread combustion of wood, crop residue, and dung cakes for cooking in rural kitchens lacking clean LPG access.

    According to Lancet Planetary Health, Solid biomass cooking fuel causes roughly **25–30%** of India's outdoor ambient PM 2.5 concentrations
  - **Ammonia from Agriculture:** Heavy application of nitrogenous chemical fertilizers releases ammonia (NH3), a key precursor to secondary inorganic aerosols.

    As per **Indian Agricultural Research Institute,** the agricultural sector (fertilizers and livestock) is responsible for over **80%** of total ammonia emissions in India.
- **Geographical & Meteorological Amplifiers**
  - **Indo-Gangetic Topography:** Landlocked, low-lying northern plain terrain acts as a massive geographic trap, collecting and concentrating pollutants against the Himalayas.

    As per **World Bank Report on South Asian Airsheds**, Over **60%** of the Indo-Gangetic Plain population lives in areas exceeding India's own national safe air limits.
  - **Winter Thermal Inversion:** Cool winter air traps heavy pollutants close to the ground beneath a low layer of warm air, completely blocking vertical dispersion.
  - **Dust Storm Migrations:** Seasonal transport of coarse desert dust particles carried by strong summer winds from western Rajasthan and the Arabian Peninsula.

Major Impacts of Air Pollution in India

- **Public Health & Epidemiological Toll**
  - **Drastic Life Expectancy Drops:** Cuts the average Indian's life expectancy by **5.3 years**, with Delhi residents losing up to **11.9 years** of life.
  - **Severe Chronic Diseases:** Triggers high rates of ischemic heart disease, stroke, COPD, lung cancer, and acute lower respiratory infections across demographics.

- **Cognitive and Neonatal Damage:** Linked directly to low birth weight, premature births, infant mortality, and impaired cognitive development in children.

  According to Lancet Regional Health, Exposure to toxic particulate matter is responsible for **113 deaths per 100,000 people** from household and neonatal air hazards.

- **Macroeconomic & Labor Liabilities**
  - **Staggering Fiscal Drain:** As per Lancet study, Inflicts direct public health costs and labor welfare losses equivalent to **9.5% of India's GDP**.
  - **Severe Labor Productivity Loss:** Causes massive numbers of lost workdays due to employee illness, physical fatigue, and frequent sick leaves.

    According to Global Climate Risk Index, Ambient air pollution and extreme heat exposure resulted in a record high loss of **247 billion potential labor hours** per year

  - **Suppressed Foreign Investment:** Discourages international corporations from setting up regional headquarters in heavily polluted urban centers like New Delhi.
- **Agricultural & Environmental Degradation**
  - **Reduced Crop Yields:** Surface-level ozone (O3) and heavy smog blankets damage plant tissues, cutting rice and wheat yields by up to 20-30%.

    As per IARI, Rising surface-level ozone concentrations have caused relative wheat yield losses to escalate between **25.2% and 35.3%.**

  - **Acid Rain and Soil Toxicity:** High atmospheric SO2 and NOx loads trigger acid deposition, accelerating the acidification of vital soil and water bodies.

    As per CPCB, Chronic nitrogen oxide and sulfur dioxide emissions have driven localized soil pH drops, decreasing vital soil microbial biomass by **20–30%** in heavily industrialized belts.

  - **Accelerated Glacier Melting:** Black carbon deposits darken Himalayan glaciers, drastically increasing solar heat absorption and speeding up glacial retreat.

    As per the **Intergovernmental Panel on Climate Change**, Black carbon deposition causes localized radiative forcing that accelerates glacial retreat by up to **0.5–1.0 meter per year**.

Measures taken for Air Pollution in India

Central Legal & Institutional Frameworks	• **National Clean Air Programme (NCAP):** Sets a targeted, outcome-based goal of a **40% reduction** in particulate matter (PM10 and PM2.5) concentrations by **2026** across 131 non-attainment cities. • **CAQM Act, 2021:** Established the Commission for Air Quality Management as a supreme statutory body for coordinated, airshed-level regulation across Delhi-NCR and adjoining states. • **Graded Response Action Plan (GRAP):** Enforces emergency institutional curbs (e.g., bans on diesel generators, industrial shutdowns, construction halts) based on fluctuating AQI severity stages. • **National Ambient Air Quality Standards (NAAQS):** Sets legal, ambient concentration ceilings for 12 critical air pollutants under the monitoring mandate of the Central Pollution Control Board.
Transport & Clean Energy Interventions	• **Leapfrogging to Bharat Stage VI (BS-VI):** Shifted the entire automobile manufacturing ecosystem directly from BS-IV to BS-VI emission standards nationwide to slash vehicular sulfur outputs.

	• **FAME-India Scheme:** Accelerates rapid electric mobility transition by funding the manufacturing and public adoption of commercial EV fleets and fast-charging networks. • **PM Ujjwala Yojana (PMUY):** Targets household indoor air pollution by delivering over 10 crore free clean LPG cooking connections to rural, below-poverty-line families. • **SATAT Initiative:** Promotes the production and vehicular adoption of Compressed Bio-Gas by setting up commercial waste-to-energy processing plants.
Industrial & Agricultural Interventions	• **Mandatory FGD Installation:** Legally forces thermal power plants to deploy Flue Gas Desulfurization units to chemically strip sulfur dioxide from their stack gaseous emissions. • **Online Continuous Emission Monitoring (OCEMS):** Mandates polluting industrial clusters to install automated, real-time telemetry systems linked directly to CPCB servers. • **Subsidized Agro-Machinery:** Provisions massive fiscal subsidies for in-situ crop crop residue management equipment like Happy Seeders, Smart Seeders and PUSA Decomposers to curb open field stubble burning. • **Corporate Green Crackers Mandate:** Enforces the exclusive domestic manufacture and sale of CSIR-certified **Green Crackers** with low barium formulations and reduced particulate emission loads.
Technological & Municipal Actions	• **Expanded CAAQMS Infrastructure:** Scaled up the installation of fully automated Continuous Ambient Air Quality Monitoring Stations across urban industrial centers to eliminate manual data tampering. • **Mechanized Urban Dust Management:** Deploys automated high-efficiency mechanical road sweepers, localized anti-smog water cannons, and mandatory high-rise green-net sheeting at construction sites. • **Real-Time Geo-Tagged Portals:** Implements AI-driven, geo-tagged mobile monitoring portals to track and suppress illegal localized municipal solid waste and garbage burning events.

Major Indices and Platforms Tracking Air Pollution in India

- **National Air Quality Index (AQI)**
  - **Core Slogan and Launch:** Launched in September 2014 under the Swachh Bharat Abhiyan with the guiding principle of *"One Number, One Color, One Description"*.
  - **Administrative Monitor:** Formulated and tracked hourly by the Central Pollution Control Board in coordination with State Pollution Control Boards.
  - **Six-Tier Severity Scale:** Converts complex concentration data into six color-coded public health categories: *Good (0-50), Satisfactory (51-100), Moderate (101-200), Poor (201-300), Very Poor (301-400), and Severe (401-500)*.
- **National Ambient Air Quality Standards (NAAQS)**
  - **Statutory Legal Mandate:** Institutionalized by the CPCB under the explicit legislative powers of the *Air (Prevention and Control of Pollution) Act, 1981*.
  - **Twelve-Pollutant Coverage:** Enforces regulatory concentration limits across 12 specific pollutants (the 8 core AQI indicators plus *Benzene, Benzo(a)pyrene, Arsenic, and Nickel*).

- **Long-Term Exposure Focus:** Mandates stricter, localized air quality criteria measured over annual and 24-hour schedules rather than real-time hourly spikes.

- **Advanced Forecasting Systems**
- **SAFAR System Framework:** Developed by the Ministry of Earth Sciences to offer high-resolution, location-specific 1-to-3 day advanced air quality forecasting.
- **SAMEER Mobile Application:** Official, public-facing digital platform designed by the CPCB to stream real-time hourly AQI updates and facilitate crowdsourced, geo-tagged pollution grievance mapping.
- **Continuous Ambient Air Quality Monitoring (CAAQMS):** Fully automated field station infrastructure that automatically telemeters raw, real-time particulate data directly to central government servers to eliminate manual intervention.

Judicial Interventions for Air Pollution in India

Constitutional Jurisprudence & Fundamental Rights	• **Expansion of Article 21:** Elevated the **Right to Clean Air** into an absolute fundamental right under the umbrella of the Right to Life (Subhash Kumar v. State of Bihar, 1991). • **Public Trust Doctrine:** Ruled that the State acts as a legal trustee of vital natural resources (like air and water) and cannot delegate them to private exploitation (M.C. Mehta v. Kamal Nath, 1997). • **Precautionary and Polluter Pays Principles:** Legally mandated that environmental protection must anticipate damage and polluters must bear the complete cost of reversal (Vellore Citizens' Welfare Forum, 1996).
Concrete Structural & Regulatory Directives	• **Taj Trapezium Case:** Enforced a complete ban on coal combustion and mandated a rapid switch to Natural Gas (CNG) across local industries to protect heritage architecture (M.C. Mehta v. Union of India, 1996). • **Delhi Public Transport Transition:** Directed the comprehensive, historic conversion of Delhi's entire public commercial fleet (buses and auto-rickshaws) from diesel to CNG fuel (Vehicular Pollution Case). • **Institutionalization of GRAP and CAQM:** Utilized the tool of Continuing Mandamus to order the operationalization of emergency intervention measures like the Graded Response Action Plan. • **Ban on Heavily Polluting Commercial Fuels:** Prohibited the open domestic storage, distribution, and industrial use of highly toxic unrefined petcoke and furnace oil.

Way Forward for Mitigating Air Pollution

- **Governance, Regulatory, and Structural Reforms**
  - **Transition to NCAP 2.0:** Shift performance metrics and funding priorities from coarse dust (PM10) directly to finer, high-toxicity combustion particles (PM2.5).
  - **Enforce a Trans-Boundary Airshed Approach:** Shift planning away from isolated municipal limits to establish synchronized regional airshed management committees across the Indo-Gangetic Plain.
  - **Legally Bind NCAP:** Grant the National Clean Air Programme formal legislative backing to ensure legal enforceability of its regional compliance targets.
  - **Adopt Health-Based Success Metrics:** Gauge policy success by tracking declines in pollution-triggered clinical admissions and lung-function drop rates, rather than just atmospheric counts.

- **Multi-Sectoral & Technical Interventions**
  - **Implement Source-Apportionment Funding:** Allocate central municipal clean-air grants strictly based on real-time, local source-apportionment scientific findings.
  - **De-Risk Industrial Fuel Transitions:** Provide fiscal subsidies and tax breaks to accelerate the adoption of clean natural gas across all small-to-medium manufacturing clusters.
  - **Scale Up Ex-Situ Biomass Commercialization:** Build massive, institutionalized public-private supply chains to process farm stubble into commercial bio-pellets for thermal power co-firing.
  - **Deploy Zero-Emission Public Transit:** Completely electrify public transit fleets across all 131 non-attainment cities to sharply reduce urban vehicular NOx loads.
- **Urban Planning & Citizen Defense**
  - **Incorporate Wind Ventilation Corridors:** Design green urban pathways and configure high-rise geometry to prevent the formation of static, landlocked urban heat islands and stagnant micro-smog traps.
  - **Deploy Targeted Micro-Zone Air Filtration:** Install high-efficiency localized particulate filters inside highly vulnerable public spots like state hospitals, government schools, and busy transport hubs.
  - **Subsidize Citizen Protective Gear:** Formulate public welfare procurement channels to distribute certified N95 respirator gear to exposed outdoor workers during heavy seasonal smog spikes.
- **Success Models for Air Pollution Mitigation**

Global Proven Models & Best Practices	• **London ULEZ Model (UK):** Ultra Low Emission Zones utilizing dynamic automatic number-plate recognition cameras to charge highly polluting vehicles, slashing toxic NO2 by nearly 50%. • **Beijing Command & Control Model (China):** Replaced coal-fired boilers with natural gas, aggressively restricted new vehicle registrations via lottery, and systematically co-located polluting heavy industries outside city lines. • **LRTAP Treaty Model (Europe):** Convention on Long-Range Transboundary Air Pollution created a legal framework for cross-border airshed management, cutting regional sulfur emissions by over 30-80%. • **Bogotá Active Mobility Model (Colombia):** Scaled up the **Ciclovía** infrastructure network to support over 1 million daily bicycle trips while simultaneously electrifying its massive public Bus Rapid Transit fleet. • **Seoul Eco-Friendly Boiler Mandate (South Korea):** Enforced a blanket legal ban on diesel cars within public sector fleets, while deploying over 61,000 certified low-NOx domestic heating boilers. • **California Cap-and-Trade Model (USA):** Combines strict market-based industrial carbon pricing with the Good Neighbor Plan to legally limit interstate, transboundary industrial smog migrations.
Emerging Indian Replication Models	• **Surat Emissions Trading Scheme (Gujarat):** India's first market-based cap-and-trade system for industrial particulate matter, successfully cutting total emissions by 24% while sustaining economic growth. • **Delhi-NCR Airshed Governance Model:** Institutionalized through the statutory Commission for Air Quality Management to legally bind unified regulatory actions across four distinct state jurisdictions.

	• **Mumbai Urban Greening Model (Maharashtra):** Implemented the **Miyawaki** dense urban forestry method along public spaces to act as organic particulate sinks and mitigate micro-level heat islands. • **Punjab Bio-Pellet Co-Firing Initiative:** Creates a circular market by forcing thermal power plants to blend and co-fire 5-10% agricultural crop stubble pellets alongside conventional coal stock. • **PUSA Decomposer Microbial Model (ICAR):** Uses a specialized low-cost liquid fungal cocktail to rapidly bioconvert paddy straw into rich in-situ soil manure, eliminating open-field stubble burning. • **Bengaluru Public Transit Electrification Model:** Scales up clean urban mass transit infrastructure by operating over 1,000 commercial electric buses alongside continuous metro rail expansion tracks.

Committee recommendations for air pollution

- **Bhure Lal Committee:** Recommended the complete conversion of all public transport buses, auto-rickshaws, and taxis in Delhi to Compressed Natural Gas to curb vehicular smoke.
  - **Phasing Out Commercial Diesels:** Mandated a strict ban on commercial diesel vehicles older than 15 years from entering the National Capital Region.
  - **Introduction of Euro Norms:** Accelerated the time-bound adoption of Euro-I and Euro-II equivalent emission standards for private passenger cars ahead of the national timeline.
- **Madhav Chitale Committee (2016):** Advised systematic river de-silting and reservoir maintenance to prevent heavy silt drying, which contributes significantly to airborne particulate matter (PM10).
- **Justice Lokur Committee:** Recommended 100% subsidization and deployment of crop-residue machinery like Happy Seeders to eliminate open paddy stubble burning.

Conclusion

Combating air pollution requires transitioning from fragmented, administrative responses to a unified **Airshed Approach** that aligns with the core vision of **Sustainable Development Goal 3 (Good Health and Well-being)**. By strengthening the statutory powers of the **Commission for Air Quality Management (CAQM)** and scaling up market-driven models like the **Surat Emissions Trading Scheme**, India can decouple its macroeconomic growth from environmental degradation.

2. AIR POLLUTION IN DELHI

The choking air pollution has led to the invocation of Graded Response Action Plan (GRAP) in Delhi-NCR region. The Delhi government, in partnership with **IIT Kanpur**, has recently implemented **artificial rain experiments (cloud seeding)** as an emergency measure to combat its chronic air pollution crisis.

Air pollution and its measurement in India

Air Pollution: Air pollution is the introduction of chemicals, particulates or biological materials into the atmosphere that cause discomfort, disease, or death to humans.

Measurement of Air Pollution In India:

- In India, air pollution is measured according to the National Air Quality Index developed by Central Pollution Control Board (CPCB) in 2014.

- The measurement of air quality in the NAQI framework is based on eight pollutants, namely- **Particulate Matter (PM10), Particulate Matter (PM2.5), Nitrogen Dioxide (NO2), Sulphur Dioxide (SO2), Carbon Monoxide (CO), Ozone (O3), Ammonia (NH3) and Lead (Pb).**

Categorization of Air Quality under AQI:

AQI	Remark	Colour Code	Possible Health Impacts
0-50	Good		Minimal Impact
51-100	Satisfactory		Minor breathing discomfort in sensitive people
101-200	Moderate		Breathing discomfort to people with asthma and heart disease.
201-300	Poor		Breathing discomfort to most people on prolonged exposure.
301-400	Very Poor		Respiratory illness on prolonged exposure
401-500	Severe		Affects healthy people and seriously impacts those with existing diseases.

Source- CPCB

Reasons behind the rise in air pollution in Delhi

- **Stubble Burning:** Stubble burning in Punjab, Rajasthan, and Haryana emits large amounts of toxic pollutants in the atmosphere. These pollutants contain **harmful gases** like methane (CH4), carbon monoxide (CO), volatile organic compounds (VOC) and carcinogenic polycyclic aromatic hydrocarbons.

  - The **IIT consortium report** have estimated that stubble burning contributed up to **35% of Delhi's PM 2.5 levels** during the peak October-November season.
- **Reduced Wind Speed:** Low-speed winds in winters are unable to disperse these pollutants effectively. Further, Delhi lies in a **landlocked region** and does not have the geographical advantage of sea breeze to disperse the suspended pollutants.
- **Effect of Northwesterly winds:** After the withdrawal of monsoons, the predominant direction of winds in northern India is northwesterly. These northwesterly winds bring the **dust from the gulf region**, **northern Pakistan** and **Afghanistan**.
- **Dip in Temperatures lowers the inversion height:** With the decrease in temperature, **inversion height lowers down, leading to concentration of pollutants in the lower atmosphere**. (Inversion height is the layer beyond which pollutants cannot disperse into the upper layer of the atmosphere).
- **Vehicular Pollution:** Delhi has one of the **highest number of registered private vehicles** in India. Official emissions inventories of 2018 show that vehicles emit about 40 per cent of the particulate load in the city of Delhi.
- **Construction Activities and Open Waste Burning:** Pollution due to **landfill burning and construction debris** enhances the pollution levels in the Delhi NCR region.
- **Firecrackers:** Firecrackers burning during Diwali further adds to the increase in air pollution levels.
- **Urban development strategy in Indi**a: The current strategy focuses on real estate development, a widening of roads and allowing large fuel guzzling vehicles which are major reasons for increased pollution.
- **Expansion of 'Grey' infrastructure:** Water bodies, urban forests, green cover, and urban agriculture have all reported shrinkage, and "grey" infrastructure has seen rapid expansion.
- **Land use change:** The handing over of open spaces to real estate developers and lack of any meaningful afforestation affects the city's ecology.
- **Promotion of car sales:** Widening roads induces people to buy more cars, thus leading to more pollution levels.
- **Construction activities:** It contributes roughly 10% of air pollution in Delhi-NCR. There are hardly any steps being taken to monitor construction activities.

Harmful effects of air pollution

The harmful effects of air pollution have been tabulated below:

Economic effects	• Leads to **loss of labour productivity, GDP and per capita income levels.** (The Confederation of Indian Industry estimates that air pollution costs Indian businesses $95 billion, or 3 per cent of India's GDP every year). (Poor air amounts to about Rs 7 lakh crore of annual economic loss, which is more than a third of our annual GST collection) • Air pollution **reduces agricultural crop yields** and commercial forest yields.
Human Health Effects	• Air pollution leads to multiple health conditions including **respiratory infections**, **heart disease** and **lung cancer**. • As per the Global Burden of Disease comparative risk assessment for 2015, air pollution exposure contributes to approximately **1.8 million premature deaths** and loss of **49 million disability adjusted life-years (DALYs)** in India.
Environment	• **Acid Rain:** Damages crops, natural vegetation, soil chemistry and leads to damage to ancient monuments (Taj Mahal Yellowing). • **Eutrophication of water bodies:** Increases nitrogen intake of freshwater bodies leading to Eutrophication.

Govt initiatives that have been taken to reduce air pollution in Delhi

- **Cloud Seeding (Artificial Rain):** Piloted with IIT Kanpur, three rounds were conducted in 2025 to temporarily reduce airborne particulate matter, although results were mixed due to weather constraints. The key reason for limited results was **insufficient cloud moisture (only 15–20%)**, while successful cloud seeding typically requires 50% or more. High cloud bases (near 10,000 feet) also reduced effectiveness.
- **Crop Residue Management Scheme:** Crop Residue Management (CRM) scheme which provides subsidy to farmers for buying **'Turbo Happy Seeder', 'Super SMS attachment', 'rotavators'** and **'superseeder'.**
- **Commission for Air Quality Management (CAQM):** CAQM is a statutory body formed under the Commission for Air Quality Management in National Capital Region and Adjoining Areas, Act 2021. CAQM has provided a framework to tackle the problem of air pollution due to stubble burning.
- **Initiatives to reduce Vehicular Pollution:** The **shift from BS-IV to BS-VI**, push for **Electric Vehicles (EVs), Odd-even Policy** have all been implemented to reduce Vehicular pollution.
- **Graded Response Action Plan (GRAP):** GRAP measures like shutting down thermal power plants and a ban on construction activities are implemented to curb air pollution.
- **Dust and construction controls:** Mandatory deployment of anti-smog guns and water sprinklers at large buildings, construction sites, and hotspots; mist sprayers installed on electric poles at 13 major air pollution hotspots.

Way ahead

A start has been made to recognise the severity of air pollution on the health of the population and the economy.

- **Increase the AQI monitoring stations:** The adequate numbers of AQI monitoring stations as mandated by the National Clean Air Programme (NCAP) must be installed. Also, **sensor based AQI monitoring units** that give hyperlocal data must be set up.
- **Enhanced powers to the authorities to take action according to the hyperlocal data:** It will help the authorities take **pre-emptive actions** and allow **denser monitoring**. For example, if it is found that construction is taking place in a certain pocket without following norms, the authorities can identify the location and immediately penalise the violators.
- **National Nodal Authority for Air pollution:** India needs a **nodal authority with constitutional powers** to ensure collaborative pre-emptive action on air pollution with timelines for all stakeholders.
- **Setting up Independent commissions for management of AQI:** Independent commissions like the **Commission for Air Quality Management (CAQM) for NCR and adjoining regions,** should be set up in other major cities like Mumbai and Chennai.
- **Airshed Approach- A regional or airshed approach** must be used to counter the geographical challenge to air pollution management as done in Los Angeles, Mexico City and many mega-urban regions in China.
- **Stricter guidelines for industrial emissions: SEBI's Business Responsibility and Sustainability Report (BRSR) framework** can lay down tighter guidelines to ensure uniformity in the unit of reporting pollutant emissions, declaration of air pollution mitigation targets (like companies do for carbon emissions), and reporting of disaggregated emissions data.
- **Making 'Clean air' an investment sector:** The push for substitution of fossil fuels will increase investment opportunities in the clean energy transition sector like **green mobility, clean cooking.** This will open up a new sector for investment and will help in reducing the air pollution simultaneously.
- **Funds and Manpower Training to combat pollution:** The Sixteenth Finance commission should provide finance to urban local bodies for climate change and air pollution reduction interventions. The urban local bodies manpower must be properly trained and the dysfunctional **State pollution Control Boards** must be empowered to take punitive actions.
- **Increased awareness and incentive for civil society:** Different stakeholders need to know why cleaning the air will benefit their livelihoods and businesses.

  - For instance, **farmers will not curb stubble burning until a viable circular economy** for alternative uses of biomass emerges. We must adopt sustainable lifestyles which are in line with the government's LiFE initiative.
- **Increased use of Public transport and reduced dependence on private vehicles: The Delhi Master Plan target of 80 per cent of motorised trips** by public transport by 2020 has not been met yet. The deadline has been shifted to 2041. This deadline needs to be met by augmenting the public transport system. The private vehicles use must be disincentivized by introducing **ward-wise parking management area plans** and **parking tax.**
- **Limit crop residue burning:** We must implement known solutions like shifting to less water-intensive crops, **altering irrigation arrangements**, timing, harvesting, baling practices and building a wider year-round market for straw.
- **End-to-end construction and waste management:** It is pivotal to reducing tonnes of dust and waste released in the air and water bodies.
- **Learnings from London, China, Singapore, Hong Kong must be incorporated:** London does not allow private vehicles in areas well-connected by public transport. China uses '**fixed number of car sales per year**' in Beijing. We must improve public transport like London, China, Singapore and Hong-Kong.

Conclusion

Air pollution in Delhi remains a severe environmental and public health crisis, demanding urgent, sustained, and collective action to ensure breathable air and a livable city.

3. WATER POLLUTION

Water pollution in India has escalated from an environmental concern into a severe socio-economic crisis, with NITI Aayog's *Composite Water Management Index* revealing that nearly **70% of the country's freshwater resources are contaminated**. This widespread degradation places India at a dismal **120th out of 122 nations** in global water quality rankings.

India's Water Pollution Crisis

- **According to the Central Pollution Control Board (CPCB) Periodic Review:** A total of **296 polluted river stretches** have been identified across India following a comprehensive water quality evaluation of 2,116 distinct monitoring locations.
- **As per NITI Aayog's Composite Water Management Index:** India is critically ranked at **120th out of 122 countries** on the global Water Quality Index.
- **According to National Mission for Clean Ganga (NMCG) Infrastructure Tracking:** 105 Ganga riverfront towns collectively generate **3,558.5 MLD** (Million Liters per Day) of urban sewage; however, only 72% (2,561.7 MLD) of this volume is actively treated, allowing **~258 MLD of raw sewage** to enter the river stem daily.
- **As per the National Green Tribunal (NGT) Water Quality Assessment:** At Triveni Sangam (Prayagraj), the Biochemical Oxygen Demand (BOD) regularly breached the safe threshold of 3 mg/L on multiple days, accompanied by an alarming **Faecal Coliform count of 49,000 MPN/100 ml** against the standard safe bathing limit of 2,500 MPN/100 ml.
- **According to the Ministry of Jal Shakti (Parliamentary Submission):** A massive **402.67 MLD of industrial effluent** is systematically discharged directly into the Ganga and Yamuna river networks by 3,186 designated Grossly Polluting Industries.
- **As per the Delhi Pollution Control Committee (DPCC) Water Quality Report:** The Yamuna encounters acute biological stress at its Delhi exit point (Asgarpur), registering a **Faecal Coliform peak of 79 lakh MPN/100 ml** due to an ongoing deficit that allows **641 MLD of completely untreated urban sewage** to infiltrate the river daily.

- **According to the Central Ground Water Board (CGWB) Global Inventory:** India stands as the **world's largest consumer of groundwater**, extracting more underground water than any other nation, with **87%** of this highly depleted volume diverted exclusively for agricultural irrigation and **11%** dedicated to domestic consumption.

Classification of Water Pollution Types

- **Based on Emission Source**
  - **Point Source Pollution:** Contamination discharged from a single, identifiable origin, such as an industrial effluent pipe or a municipal sewage outfall.
  - **Non-Point Source Pollution:** Diffuse, unidentifiable contamination originating from broad areas, such as agricultural fertilizer runoff or urban stormwater drainage.
- **Based on Pollutant Characteristics**
  - **Biological Pollution:** Infiltration of pathogenic micro-organisms (bacteria, viruses, protozoa) primarily through untreated domestic faecal sludge.
  - **Chemical Pollution:** Introduction of synthetic organic/inorganic compounds, including industrial heavy metals, agricultural pesticides, and pharmaceutical residues.
  - **Thermal Pollution:** Discharge of excessively heated coolant water from power plants, which abruptly raises ambient temperature and severely depletes Dissolved Oxygen (DO).
  - **Radioactive Pollution:** Leaching of radioactive isotopes from uranium mining, nuclear power plant waste, and discarded medical diagnostics.
  - **Physical (Suspended) Pollution:** Accumulation of non-biodegradable plastics, micro-plastics, and heavy soil sediments that block sunlight and disrupt photosynthesis.
- **Based on Affected Ecosystem**
  - **Surface Water Pollution:** Contamination of rivers, lakes, and wetlands, primarily leading to nutrient-induced eutrophication and algal blooms.
  - **Groundwater (Aquifer) Pollution:** Deep soil percolation of agricultural nitrates, industrial leachates, and toxic geogenic elements like Arsenic and Fluoride.
  - **Marine and Coastal Pollution:** Oceanic contamination driven by localized crude oil spills, ship ballast water discharge, and coastal industrial runoff.

Water Quality Parameters: DO, BOD, and COD

- **Dissolved Oxygen (DO):** The measure of free, non-compound oxygen gas dissolved in water, essential for the respiration of all aerobic aquatic organisms.
- **Biochemical Oxygen Demand (BOD):** The amount of dissolved oxygen required by aerobic microorganisms to break down the biodegradable organic matter present in a water sample at a specific temperature over a 5-day period.
  - **Relationship with DO:** Displays an inverse relationship with DO; a high organic waste load triggers explosive microbial growth, rapidly consuming and depleting available dissolved oxygen.
- **Chemical Oxygen Demand (COD):** The total amount of oxygen required to chemically oxidize all organic matter in water (both biodegradable and non-biodegradable/toxic fractions) using a strong chemical oxidant like potassium dichromate.

Major Causes of Water Pollution in India

- **Domestic and Municipal Drivers**
  - **Massive Sewage Treatment Deficit**: Urban centers generate over 72,000 million liters of sewage daily, discharging nearly 60% of it entirely untreated into natural water bodies.
  - **Unregulated Open Solid Waste Dumping:** Mismanaged municipal plastic wastes and landfill garbage accumulate along riverbanks, leaching toxic chemical micro-pollutants during monsoonal rains.

- **Unsewered Urbanization & Slum Runoff:** Rapid growth of peripheral colonies lacking central drainage pipelines channels raw greywater directly into nearby open streams.

  According to **NITI Aayog**, less than 35% of peripheral urban expansions are connected to centralized sewage grid networks

- **Mass Cultural and Religious Activities:** Large-scale seasonal immersions of non-biodegradable clay idols containing heavy lead-based paints severely damage local lake beds.

  According to **State Pollution Control Boards (SPCBs)**, heavy metal spikes—particularly lead concentrations—increase by up to 20 to 50 times baseline levels in urban water bodies

- **Industrial and Chemical Drivers**
  - **Untreated Effluent Discharge:** Highly toxic industrial units (tanneries, distilleries, paper mills) routinely bypass regulatory treatment plants to dump chemical waste.

    According to **CPCB**, nearly 30% of highly polluting "red category" industrial units fail to consistently run their primary treatment facilities.

  - **Heavy Metal Surface Leaching:** Industrial clusters release high loads of carcinogenic chromium, cadmium, and lead directly into urban drainage channels.
  - **Inadequate Common Effluent Infrastructure:** Small-scale industrial areas frequently face absolute structural failure or severe operational undercapacity of shared treatment units.
  - **Thermal Pollution from Power Units:** Coal-fired thermal power installations discharge highly heated cooling water into river streams, severely depleting dissolved oxygen.

    According to **MoEFCC**, water discharged from coal-fired cooling units can raise ambient river temperatures by 5-8 degree celsius.

- **Agricultural and Agro-Chemical Drivers**
  - **Excessive Chemical Fertilizer Runoff:** Heavy application of synthetic nitrogenous and phosphatic fertilizers washes into rural water bodies, causing severe ecological eutrophication.
  - **Persistent Pesticide Accumulation:** Unregulated use of toxic agrochemicals infiltrates local agricultural soil, leading to the bioaccumulation of organochlorines in aquatic life.

    According to **Journal of Environmental Management**, over 65% of tested freshwater fish samples in heavy farming belts show persistent bioaccumulated organochlorine pesticide residues.

  - **Intensive Aquaculture and Livestock Waste:** Unmanaged organic animal wastes and concentrated chemical feed from commercial fish farms induce heavy nutrient loading in rivers.
- **Ground-Level Contaminants**
  - **Widespread Arsenic Seepage:** Geologic formations in the landlocked Ganga-Brahmaputra basin leach high levels of toxic arsenic into the shallow rural aquifer system.
  - **Widespread Fluoride Infiltration:** Over-exploitation of deep groundwater in arid regions alters underground rock chemistry, triggering widespread geogenic fluoride contamination.
  - **Salinity Ingress in Coastal Zones:** Excessive over-pumping of coastal groundwater tables induces destructive landward seawater intrusion, ruining local drinking wells.

Major Impacts of Water Pollution in India

- **Public Health**
  - **High Infant Mortality Rate:** Remains a leading cause of childhood diarrheal deaths, claiming over 2 lakh lives annually due to waterborne pathogens.
  - **Widespread Waterborne Epidemics:** Triggers seasonal outbreaks of acute cholera, typhoid, amoebiasis, and hepatitis A/E infections across low-income urban settlements.

According to the **National Center for Disease Control (NCDC)**, seasonal outbreaks of acute cholera, typhoid, and hepatitis infections account for over 10-12 million cases.

- **Crippling Fluorosis Defect:** Chronic geogenic fluoride ingestion inflicts permanent dental discoloration and crippling skeletal deformities across rural communities.

According to official groundwater quality logs by the **Ministry of Jal Shakti**, chronic geogenic fluoride ingestion beyond the safe 1.5mg/L limit inflicts permanent dental problems.

- **Carcinogenic Arsenicosis Burden:** Prolonged exposure to arsenic-laced groundwater causes severe hyperkeratosis, painful skin lesions, and multi-organ systemic cancers.

- **Ecological & Biodiversity Destruction**
  - **Severe Freshwater Eutrophication:** Nutrient-rich runoff sparks toxic algal blooms, depleting dissolved oxygen (DO) and creating **aquatic dead zones**.

According to the **Central Pollution Control Board (CPCB)**, intensive nutrient runoff pushes more than 50% of nearby rural lakes into advanced ecological eutrophication.

  - **Mass Riverine Bioaccumulation:** Persistent heavy metals and microplastics bioaccumulate across aquatic food webs, threatening apex species like the Ganges River Dolphin.

According to **Journal of Environmental Management**, over 65% of tested freshwater organisms show bioaccumulated heavy metal and organochlorine residues.

  - **Disruption of Marine Nurseries:** Heavy industrial and plastic effluents choke estuarine mangrove systems, destroying critical spawning grounds for marine fish.
- **Economic & Food Security Vulnerabilities**
  - **Severe Macroeconomic Drag:** Public health expenditures and reduced labor productivity from water contamination drain nearly 1.5-2% of India's annual GDP.
  - **Crashing Inland Fishery Yields:** Toxic chemical spikes induce sudden, large-scale fish kills, decimating the primary livelihoods of traditional riverine fishing communities.
  - **Agricultural Soil Toxicity:** Irrigating farmlands with polluted, saline, or industrial wastewater alters soil pH, reducing long-term crop productivity.
  - **Escalating Urban Water Treatment Costs:** Rapidly deteriorating raw river quality forces municipal utilities to invest heavily in expensive, multi-stage filtration technologies.

Measures taken for Water Pollution in India

Central Legal & Statutory Frameworks	• **Water Act, 1974:** Provides the primary statutory framework for preventing and controlling water pollution, establishing the CPCB and SPCBs. • **Environmental Protection Act, 1986:** Empowers the central government to set strict nationwide industrial effluent discharge standards and regulate ecologically sensitive basins. • **National Water Policy, 2012:** Mandates the overarching management of water resources with a core focus on ecological flows, recycling, and pricing untreated waste.
Flagship Missions & Basin Interventions	• **Namami Gange Programme:** Employs an integrated river basin approach to clean the Ganga via sewage infrastructure creation, riverfront development, and biodiversity conservation. • **National River Conservation Plan (NRCP):** Extends central financial and technical assistance to states to intercept, divert, and treat raw municipal sewage across major non-Ganga river basins. • **Jal Jeevan Mission (JJM):** Supplies safe, piped drinking water to rural households while explicitly prioritizing the installation of community water purification plants in contaminated zones.

	• **Atal Bhujal Yojana (ABY):** Drives community-led sustainable groundwater management to prevent the over-extraction that triggers toxic geogenic arsenic and fluoride spikes.
Industrial Regulation & Infrastructure Upgrades	• **Zero Liquid Discharge (ZLD):** Mandates highly polluting industrial sectors (tanneries, pulp & paper) to completely treat, recycle, and reuse 100% of their operational wastewater. • **Common Effluent Treatment Plants (CETPs):** Establishes shared, state-subsidized effluent processing infrastructure to treat chemical waste generated by clusters of small-scale industrial units (MSMEs). • **Online Continuous Effluent Monitoring (OCEMS):** Enforces the installation of real-time telemetry systems at large industrial discharge points to transmit water quality data directly to CPCB servers.
Technological & Governance Platforms	

Way Forward for Mitigating Water Pollution

- **Governance, Regulatory, and Legislative Reforms**
  - **Transition to an Integrated River Basin Approach:** Shift planning from fragmented administrative boundaries to unified river basin authorities managing entire aquatic ecosystems.
  - **Enforce Strict Polluter Pays Penalties:** Upgrade the statutory fine structures under the Water Act to make industrial illegal dumping financially prohibitive.
  - **Mandate Third-Party Environmental Auditing:** Require independent, accredited third-party labs to verify industrial effluent data before submission to State Pollution Control Boards.
  - **Establish Statutory Minimum Ecological Flows:** Legally mandate a minimum volume of continuous river flow to maintain natural self-purification and dilution capacities.
- **Infrastructure, Technology, and Circular Economy**
  - **Scale Up Decentralized Sewage Treatment:** Deploy localized, low-cost decentralized treatment units rather than relying solely on capital-intensive centralized grids.
  - **Mandate Wastewater Recycling for Heavy Industry:** Force non-potable commercial sectors to meet 100% of their operational water needs using recycled municipal sewage.
  - **Adopt Specialized Bioremediation Techniques:** Utilize engineered wetlands, phytoremediation, and microbial strains to treat low-intensity urban drain runoff naturally.
  - **Expand Community-Scale Purification in Rural Areas:** Install targeted, low-cost community water purification systems (e.g., Ion-Exchange filters) in arsenic- and fluoride-hit aquifers.
- **Agricultural Shift and Community Action**
  - **Promote Precision Agriculture and Direct Seeded Rice:** Incentivize micro-irrigation and alternative farming methods to drastically reduce chemical and nutrient runoff.
  - **Establish Dynamic Participatory Groundwater Management:** Empower village-level Pani Samitis to map, monitor, and regulate local extraction to prevent geogenic toxin spikes.
  - **Incentivize Buffer Strip Afforestation:** Plant dense, deep-rooted native vegetation belts along riverbanks to naturally filter agricultural and urban surface runoff.

Success Models for Water Pollution Mitigation

- **Global Proven Models & Best Practices**
  - **Rhine Action Programme Model (Europe):** Transformed Europe's biggest sewer via the International Commission for the Protection of the Rhine (ICPR), binding 6 nations to a cross-border river basin framework that cut heavy chemical loads by 50-70%.

- **Singapore NEWater Circular Model:** Uses a multi-stage closed-loop recycling system (microfiltration, reverse osmosis, and UV disinfection) to turn industrial and municipal wastewater into high-purity, potable grade water.
- **Chesapeake Bay Total Maximum Daily Load (USA):** Implemented a strict "pollution diet" capping the exact tonnage of nitrogen, phosphorus, and sediment runoff allowed into the watershed across six distinct US states.
- **Cheonggyecheon Stream Restoration (South Korea):** Demolished an elevated urban highway to uncover and ecologically naturalize a buried, highly polluted stream, drastically improving local biodiversity and cutting urban heat effects.
- **Tokyo Sewerage Master Plan (Japan):** Deployed massive subterranean stormwater storage tunnels alongside automated split-drain networks to handle flash monsoonal flows and prevent raw urban sewage overflows.
- **European Union Water Framework Directive:** Enforces a mandatory statutory framework that shifts water quality tracking away from localized political borders to natural, cross-boundary river basin districts.

- **Emerging Indian Replication Models**
  - **Surat Emissions Trading Scheme (Gujarat):** Pioneered a highly successful, market-based cap-and-trade market mechanism to regulate and financially penalize industrial particulate and chemical effluent discharges.
  - **Mithi River Circular Economy Initiative (Mumbai):** Employs large-scale daily boom barriers to trap floating river plastics, utilizing localized pyrolysis technology to convert low-value waste directly into industrial fuel oil.
  - **Puttenahalli Lake Community-Led Model (Bengaluru):** Driven entirely by neighborhood citizen trusts (*Pani Samitis*) to divert urban greywater through localized, artificial wetlands and biological root-zone filtration systems.
  - **Gautam Buddha Nagar Rural Pond Revival (Uttar Pradesh):** Rejuvenated 15 degraded rural waterbodies by utilizing natural de-siltation, stabilizing native embankments, and generating livelihoods through sustainable lotus and chestnut farming.
  - **Sabarmati Riverfront Reimagination (Ahmedabad):** Re-engineered an ephemeral, highly polluted urban riverbed into a perennial channel by diverting city sewage lines to central treatment plants and building public green embankments.
  - **Namami Gange Hybrid Annuity Model (HAM):** Restructured public infrastructure funding by tying 60% of private developer payouts directly to the long-term, verifiable performance of local Sewage Treatment Plants (STPs).

Committee Recommendations for Water Pollution in India

- **Mihir Shah Committee (2016):** Recommended replacing CPCB and CGWB with a single statutory body, the **National Water Commission (NWC)**, to unify surface and groundwater pollution management.
- **Kasturirangan Committee on Western Ghats (2013):** Urged a complete prohibition on highly polluting "Red Category" industries within Ecologically Sensitive Areas to safeguard pristine headwaters.
- **Parliamentary Standing Committee on Water Resources (2020):** Directed mandatory geo-tagging of all urban water bodies and called for a legally enforceable national policy on wastewater recycling.
- **Justice Pritam Pal Committee (NGT, 2019):** Advised levying heavy, daily environmental compensation penalties on municipal bodies failing to meet decentralized sewage treatment deadlines.
- **Madhav Chitale Task Force on Ganga Siltation:** Recommended comprehensive river desiltation and natural flood-plain zoning to restore a river's self-purification velocity and dilution capacity.

Conclusion

Rejuvenating India's aquatic ecosystems requires transitioning from fragmented, administrative responses to a unified **Integrated River Basin Management (IRBM)** approach that secures crucial **Ecological Flows (e-flows)**. By implementing market-driven regulatory mechanisms like the **Hybrid Annuity Model (HAM)** for sewage infrastructure and enforcing **Zero Liquid Discharge (ZLD)** mandates, India can effectively decouple macroeconomic growth from severe ecological degradation.

4. PLASTIC POLLUTION

Plastic pollution has accelerated into a major ecological crisis, driven by the generation of over **400 million tonnes of plastic waste globally each year**, of which less than 10% is successfully recycled. In India, the situation is increasingly acute, with the country generating roughly **3.4 million tonnes of plastic waste annually**, causing immense stress on municipal dump sites and marine networks.

Classification of Plastic Pollution Types

Classification Based on Particle Size

Category	Size Dimensions	Environmental Impact
Macroplastics	Greater than 20 mm	Entangles marine megafauna; physically chokes urban drainage networks.
Mesoplastics	5 mm to 20 mm	Readily ingested by coastal birds; accumulates fast on beach surfaces.
Microplastics	1 micrometer to 5 mm	Bioaccumulates in aquatic chains; bypasses standard drinking water filters.
Nanoplastics	Less than 1 micrometer	Penetrates cellular membranes; translocates directly into vital human organs.

Classification Based on Origin & Ecosystem Target

Category	Definition & Scope	Key Core Examples
Primary Plastics	Manufactured directly at micro-size for specific industrial uses.	Cosmetic microbeads, industrial virgin plastic pellets (nurdles).
Secondary Plastics	Formed via environmental weathering and fragmentation of large items.	Broken down water bottles, degraded synthetic clothing fibers.
Terrestrial Pollution	Accumulation of synthetic polymers across land-based networks.	Landfill leachates, plastic-choked agricultural soils.
Marine Pollution	Accumulation of synthetic debris across oceanic systems.	Great Pacific Garbage Patch, abandoned nylon ghost nets.

Causes of Plastic Pollution in India

- **Production and Structural Market Drivers**
  - **Low Cost of Virgin Polymers:** Fossil-fuel subsidies make producing fresh virgin plastic cheaper than collecting, sorting, and processing recycled polymers.
  - **FMCG Reliance on Single-Use Packaging:** Fast-moving consumer goods brands favor cheap, single-use multi-layered plastics (MLPs) for sachet marketing to maximize retail reach.

According to the Council on Energy, Environment and Water, packaging is the primary driver of plastic waste in India, accounting for nearly 59% of total plastic consumption.

- **E-Commerce Packaging Surges:** Exponential growth in digital logistics has generated massive amounts of single-use protective bubble wraps and plastic films.

According to CPCB, exponential growth in digital logistics has led to India generating roughly 26,000 tonnes of plastic waste daily.

- **Widespread Industrial Microbead Integration:** Commercial cosmetics, facial scrubs, and industrial abrasives explicitly incorporate primary micro-polymers for texture.

India emits an estimated 9.3 million tonnes of plastic into the environment annually, heavily accelerated by primary micro-polymers.

- **Waste Management and Structural Deficits**
  - **Low Rates of Municipal Waste Segregation:** Household failure to source-separate wet food waste from dry recyclables contaminates plastics, making recycling economically unviable.
  - **Severe Under-Capacity of Recycling Units:** Municipalities lack the specialized infrastructure required to process complex polymers, leading to heavy landfill dumping.

According to CEEW research, high transportation costs and mixed waste streams leave India's existing recycling units at only 50–60% capacity.

  - **High Informal Sector Inefficiencies:** Plastic collection relies heavily on informal ragpickers, who ignore low-value thin films and sachets due to poor market returns.

According to Mongabay India, informal ragpickers largely ignore multi-layered plastics (MLPs) because the market return rate is extremely low at just ₹1 per kilo.

  - **Uncontrolled Open-Landfill Dumping:** Massive urban dumpsites (**Ghazipur, Deonar**) allow accumulated macro-plastics to break down continuously under solar radiation.

According to CPCB landfill data, nearly 94% of the waste accumulated in major urban dumpsites is thermoplastic, which breaks down continuously under environmental exposure.

- **Consumption and Regulatory Weaknesses**
  - **High Consumer Demand for Single-Use Products:** Modern dependency on takeaway food, disposable cutlery, and bottled water drives continuous, high-volume plastic generation.
  - **Weak Enforcement of SPCB Regulations:** State Pollution Control Boards face chronic staffing and resource shortages, limiting their ability to penalize banned plastic vendors.
  - **Lagging Extended Producer Responsibility (EPR):** Implementation loopholes allow many manufacturers to delay setting up effective, nationwide take-back channels for their plastic waste.
  - **Inadequate Port and Maritime Oversight:** Ports rarely enforce strict regulations against international cargo ships discharging broken synthetic fishing gear or ballast plastics.

Impacts of Plastic Pollution

- **Human Health & Physiological Threats**
  - **Cellular and Organ Translocation:** Ingested nanoplastics cross the human blood-brain and placental barriers, triggering chronic cellular inflammation and tissue damage.
  - **Endocrine System Disruption:** Chemical plasticizers like **Bisphenol A** and **phthalates** leach into food networks, mimicking hormones and causing reproductive disorders.
  - **Pathogen Carrier Vectoring:** Environmental microplastics act as **floating microscopic hubs**, binding toxic heavy metals and deadly waterborne pathogens to their surfaces.
- **Ecological & Marine Ecosystem impacts**
  - **Severe Marine Megafauna Mortality:** Endangered whales, sea turtles, and pelagic birds die from intestinal blockages after mistaking macro-plastics for prey.

According to the **United Nations Environment Programme**, marine plastic debris causes the death of more than 1 million seabirds and 100,000 marine mammals globally each year.

- **Ghost Fishing Ecosystem Toll:** Abandoned nylon fishing nets drift through ocean currents for decades, continuously entangling and killing coral reefs and marine life.

According to global marine litter estimates by the **Food and Agriculture Organization (FAO)**, abandoned, lost, or discarded fishing gear makes up roughly 10% of all marine waste.

- **Aquatic Dead Zone Acceleration:** Concentrated layers of microplastics block essential sunlight from marine phytoplankton, crippling global carbon sequestration capacities.

According to **Nature Communications**, ingestion of microplastics by marine zooplankton and phytoplankton can reduce their photosynthesis efficiency by up to 15%.

- **Economic, Agricultural & Urban Vulnerabilities**
  - **Aggravated Urban Flash Flooding:** Discarded single-use plastics choke city drainage networks, turning regular monsoon rains into destructive urban floods.

According to **Ministry of Housing and Urban Affairs (MoHUA)**, single-use plastic waste chokes up to 80% of urban storm-water drainage capacities

  - **Agricultural Soil Degradation:** Plastic accumulations in farmland disrupt soil porosity and water-holding capacity, stunting crop root development and yields.
  - **Crashing Coastal Tourism Economies:** Heavily littered beaches and marine plastic debris cause steep financial losses for coastal tourism and local fishing industries.

Plastic Waste Management (Amendment) Rules, 2026

- **Mandatory Recycled Content Targets (FY 2026–27)**
  - **Rigid Plastics (Category I):** Mandates a minimum of **40% recycled content**, escalating to 60% by FY 2028–29.
  - **Flexible Packaging (Category II):** Sets a **10% recycled content target**, rising to 20% by FY 2027–28.
  - **Multi-Layered Plastics (Category III):** Requires **5% recycled content**, climbing to 10% by FY 2027–28.
  - **Statutory Exemptions:** Allowed only where secondary plastic use is restricted by other regulators (e.g., FSSAI for food-contact items or CDSCO for medical packaging).
- **Market-Driven EPR and Flexibility Provisions**
  - **Tradable EPR Certificates:** Introduces a formalized commercial trading platform managed via MSTC Ltd to exchange surplus compliance credits.
  - **Strict Category Ring-Fencing:** Bans inter-category certificate trading; surplus credits in rigid plastics cannot offset deficits in flexible or multi-layered plastics.
  - **Carry-Forward Window:** Permits PIBOs to defer unfulfilled FY 2025–26 targets for up to 3 years, provided at least one-third (1/3rd) of the backlog is cleared annually.

Measures Taken Against Plastic Pollution in India

Central Legislative and Regulatory Frameworks	• **Plastic Waste Management Rules, 2016:** Formulates the primary national framework for plastic classification, handling, and municipal enforcement. • **Nationwide Ban on Selected Single-Use Plastics (SUP):** Prohibits the manufacture, import, stocking, distribution, sale, and use of 19 high-littering, low-utility single-use items.

	• **Mandatory Extended Producer Responsibility (EPR):** Legally obligates producers, importers, and brand owners to fulfill binding, target-based collection and recycling quotas via a centralized CPCB portal. • **Minimum Plastic Thickness Regulations:** Mandates the gradual increase of plastic carry-bag thickness to 120 microns to enhance structural reusability and recycling economics.
National Missions and Interventions	• **Swachh Bharat Mission 2.0 (Urban & Rural):** Prioritizes achieving complete "Garbage Free Cities" through mandatory source segregation and setting up Material Recovery Facilities. • **National Dashboard on Elimination of Single Use Plastic (SUP):** Integrates central ministries and state authorities to monitor the real-time enforcement of plastic bans and commercial compliance. • **Prakriti Mascot and Green Ribbons Campaign:** Launched a national mass-awareness drive by the MoEFCC to promote sustainable behavioral modifications and eco-friendly alternatives.
Technological, Industrial, and Infrastructure Upgrades	• **IRC Guidelines for Plastic Roads:** Mandates the compulsory blending of shredded waste plastic into bituminous hot mixes for all national highway construction within 50 km of urban hubs. • **BIS Standards for Compostable Plastics:** Establishes rigid certification protocols (IS/ISO 17088) to verify and label genuinely biodegradable and compostable alternative polymers. • **Unnat Bharat Abhiyan Inventions:** Funds premier academic institutions to scale up decentralized plastic pyrolysis plants that convert low-value waste into industrial-grade fuel oil.
Global Commitments and Treaties	• **UNEA Resolution 5/14 (Global Plastic Treaty):** Active Indian participation in negotiating a legally binding international treaty to end plastic pollution across its entire lifecycle by late 2024–2026. • **GloLitter Partnerships Project:** Implemented by FAO and IMO with Indian cooperation to prevent, reduce, and control marine plastic litter from the maritime transport and fishing sectors.

Way Forward for Mitigating Plastic Pollution

- **Regulatory Frameworks and Enforcement Overhauls**
  - **Implement Strict Upstream Lifecycle Caps:** Shift regulatory focus from downstream waste management to placing mandatory limits on raw virgin polymer production.
  - **Close EPR Loopholes with Blockchain Tracking:** Integrate end-to-end blockchain tracing on the CPCB portal to prevent fraudulent recycling credits by producers.
  - **Impose Progressive Environmental Tax on Virgins:** Levy an environmental cess on raw plastic production to make recycled polymers economically competitive.
  - **Standardize Global Cross-Border Ship Inspection:** Enforce strict port-state controls to penalize international vessels discarding synthetic ghost nets into ocean basins.
- **Infrastructure, Technological Innovation, and Circularity**
  - **Scale Up Centralized Material Recovery Facilities (MRFs):** Mandate automated, optical-sorting sorting hubs at municipal dumpsites to process complex multi-layered plastics.

- **Subsidize Advanced Chemical Recycling and Pyrolysis:** Provide fiscal incentives to convert low-value, unrecyclable plastic waste directly into industrial-grade fuel oils.
- **Accelerate R&D in Marine-Degradable Biopolymers:** Fund public-private research to scale up cost-effective, non-toxic alternatives derived from seaweed, starch, and agricultural waste.
- **Enforce Decentralized Twin-Pit Segregation Models:** Mandate absolute 100% household-level sorting of wet and dry waste through strict municipal penalty bylaws.

- **Market formalization, Economic Shifts, and Community Action**
  - **Formalize Informal Ragpicker Network:** Integrate informal waste pickers into the formal green economy with fixed wages, health safety gear, and digital bank IDs.
  - **Launch Nationwide Institutional Buy-Back Hubs:** Incentivize citizens by installing reverse-vending machines providing monetary rewards for deposited plastic bottles at transit hubs.
  - **Establish Transboundary Marine Boom Barriers:** Deploy macro-debris river traps across high-yield estuarine channels to intercept urban plastics before they breach coastal waters.

Success Models for Plastic Pollution Mitigation

- **Global Proven Models & Best Practices**
  - **Rwanda Absolute Ban Success:** Implemented a strict nationwide prohibition on plastic bags, utilizing rigorous border checks, commercial fines, and mandatory community cleanup (*Umuganda*).
  - **German Pfand Deposit-Return System:** Charges a temporary financial deposit on plastic bottles, achieving a **98% recycling rate** by returning cash to consumers at automated reverse-vending hubs.
  - **Swedish Waste-to-Energy (WTE) Matrix:** Employs advanced, low-emission incineration systems that convert unrecyclable residual plastic waste into electricity and regional district heating grids.
  - **Ellen MacArthur New Plastics Economy:** A collaborative global framework that pushes multinational brands to redesign packaging for 100% circular reuse, elimination, or composting.
  - **Norwegian EPR Clean Ocean Model:** Levies a variable environmental tax on producers that drops to zero when the collective national bottle collection rate breaches 95%.
  - **Interceptor River Cleaning Model:** Deploys automated, solar-powered catamaran barriers in high-emission rivers to extract floating macro-plastics before they reach ocean basins.
- **Emerging Indian Replication Models**
  - **Alappuzha Clean Kerala Model:** Eliminated centralized landfills by deploying decentralized household composting units and empowering women-led *Haritha Karma Sena* networks to collect dry plastics.
  - **Indore Source Segregation Paradigm:** Achieved status as India's cleanest city by enforcing a strict, multi-bin waste segregation system at the doorstep, backed by automated sorting facilities.
  - **Himachal Pradesh "Plastic-Waste Buy-Back" Scheme:** Offers a fixed minimum support price to rural citizens and ragpickers for collecting single-use plastic films, which are then diverted to road construction.
  - **Tamil Nadu Meendum Manjapai Movement:** A state-led public campaign reviving the use of traditional, eco-friendly yellow cloth bags to replace single-use plastic carry bags at commercial retail hubs.
  - **Ambikapur Garbage Cafe (Chhattisgarh):** Provides free hot meals to marginalized citizens and waste collectors in direct exchange for delivering specified weights of collected plastic waste.
  - **Plastic-to-Roads Industrial Blueprint:** Spearheaded by the Indian Roads Congress, this model mixes shredded plastic waste into bituminous road overlays, enhancing highway durability.

Conclusion

Combating plastic pollution requires transitioning from a linear "take-make-waste" approach to a robust **circular economy** driven by strict **Extended Producer Responsibility (EPR)** and innovations in eco-friendly alternatives. This technical shift must be supported by grassroots behavioral change, turning plastic waste management into a true **Jan Andolan (people's movement)** under **Swachh Bharat Mission 2.0**.

5. LAND DEGRADATION AND DESERTIFICATION

Land degradation and desertification present a **severe ecological crisis**, driven by unsustainable agricultural practices, deforestation, and accelerating climate change. Reversing this trend through **Sustainable Land Management (SLM)** is imperative to achieve land degradation neutrality, preserve vital ecosystems, and fulfill **Sustainable Development Goal 15 (Life on Land)**.

Major Causes of Land Degradation and Desertification

- **Anthropogenic and Agricultural Drivers**
  - **Unsustainable Irrigation Practices:** Excessive groundwater use in low-rainfall areas leads to widespread soil salinization and waterlogging.

    According to FAO, India is the largest user of ground water for irrigation purposes.
  - **Overcultivation and Monoculture:** Intensive cropping without adequate fallow periods depletes essential soil micronutrients and organic matter.
  - **Indiscriminate Chemical Usage:** Overuse of chemical fertilizers and pesticides disrupts the soil microbiome, altering its physical structure.
  - **Accelerated Deforestation:** Clearing forests for agriculture and urbanization exposes vulnerable topsoil to severe wind and water erosion.
  - **Uncontrolled Overgrazing:** Livestock overpopulation removes protective vegetative cover, compounding soil compaction and erosion.

    According to the **Indian Grassland and Fodder Research Institute (IGFRI)**, India faces a green fodder deficit of nearly 30%, forcing livestock to overgraze fragile lands.
- **Climate and Hydrological Shifts**
  - **Changing Rainfall Regimes:** Frequent, high-intensity downpours wash away loose topsoil, causing deep gully erosion.
  - **Prolonged Severe Droughts:** Extended dry spells deplete soil moisture reserves, accelerating the aridification of vulnerable drylands.
  - **Rising Global Temperatures:** Elevated ambient temperatures increase soil evapotranspiration rates, destroying organic carbon stores.

    According to the **Intergovernmental Panel on Climate Change (IPCC)**, **every 1 degree Celsius rise in temperature** accelerates soil organic matter decomposition.
- **Industrial and Structural Factors**
  - **Open-Cast Mining Operations:** Large-scale mineral extraction leaves behind toxic tailing dumps and completely destabilizes local topography.
  - **Unplanned Urban Sprawl:** Expanding concrete infrastructure permanently seals fertile agricultural land under layers of bitumen and cement.
  - **Industrial Effluent Discharge:** Untreated heavy metals and toxic chemicals from factories poison and sterilize adjacent soils.

Impacts of Land Degradation and Desertification

- **Ecological and Climatic Consequences**

- **Depletion of Soil Organic Carbon (SOC):** Strips the soil of its organic matter, turning fertile lands into absolute net sources of carbon emissions.

  According to the **ICAR**, degraded soils across India have lost over 50% of their original soil organic carbon pool, turning vast agricultural tracts into net sources of carbon emissions.

- **Accelerated Biodiversity Loss:** Destroys natural habitats, triggering a rapid decline in subterranean microbes and native plant species.

  According to studies by the **National Biodiversity Authority (NBA)**, intensive land degradation causes up to a 60% decline in essential subterranean microbial biomass.

- **Altered Hydrological Cycles:** Reduces the soil's water-holding capacity, which lowers groundwater recharge rates and worsens flash floods.

  According to the **Central Ground Water Board (CGWB)**, degraded soils experience a massive reduction in localized groundwater recharge by up to 30%.

- **Expansion of Dust and Sandstorms:** Lack of vegetative cover allows heavy winds to lift loose topsoil, intensifying regional dust storms.

- **Socio-Economic and Food Security Crises**
  - **Crashing Agricultural Productivity:** Severe loss of topsoil fertility leads to falling crop yields, directly threatening national food security.

    According to the **National Academy of Agricultural Sciences (NAAS)**, progressive land degradation is causing average yield losses of up to 20-30% for major staple crops.

  - **Massive Rural Distress and Out-Migration:** Shrinking arable land forces dependent farmers and pastoralists into distress migration toward cities.
  - **Escalating Resource Conflicts:** Dwindling fertile lands and water scarcity trigger localized conflicts over shrinking grazing grounds.
  - **Compounding Rural Poverty:** Declining farm incomes trap vulnerable smallholders and marginal farmers in deep, structural poverty cycles.

    According to the **National Bank for Agriculture and Rural Development (NABARD)**, declining soil fertility cuts smallholder farm incomes by over 25% on average.

- **Public Health and Macroeconomic Costs**
  - **Widespread Malnutrition:** Falling crop yields and micro-nutrient deficiencies in degraded soils directly cause widespread hidden hunger.
  - **Rise in Airborne Respiratory Illnesses:** Increased dust and particulate matter (PM10) in desertified zones spark chronic respiratory diseases.
  - **Heavy Economic Output Losses:** Causes significant annual GDP contractions due to lost agricultural value and high land reclamation costs.

    According to **The Energy and Resources Institute (TERI)**, land degradation and desertification cost India roughly 2.5% of its annual Gross Domestic Product (GDP).

Measures taken Against Land Degradation and Desertification

Global Commitments and Frameworks	• **UNCCD (1994):** Operates as the sole legally binding international agreement linking environment and development to sustainable land management. • **Land Degradation Neutrality (LDN) Targets:** Commits global signatories to achieve a state where the quantity and quality of land resources remain stable.

	• **The Bonn Challenge:** Aims to restore 350 million hectares of the world's deforested and degraded lands by 2030. • **G20 Global Land Initiative:** Seeks to achieve a 50% reduction in degraded land globally by the year 2040.
India's Domestic Legislative and Policy Measures	• **National Afforestation Programme (NAP):** Focuses on the ecological restoration of degraded forests and adjoining lands through community participation. • **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY-WDC):** Implements watershed development projects to improve soil moisture retention and reduce water run-off. • **Soil Health Card Scheme:** Provides customized nutrient status reports to farmers to prevent the irrational overuse of chemical fertilizers. • **National Mission for Green India (GIM):** Aims to protect, restore, and enhance India's diminishing forest cover under the climate change action plan. • **Desertification and Land Degradation Atlas of India:** Published by ISRO to geo-spatially track, map, and prioritize degraded zones for targeted intervention.
Technological and Grassroots Approaches	• **Subsidized Micro-Irrigation Adoption:** Promotes drip and sprinkler systems via central schemes to arrest rising soil salinization and waterlogging. • **Paramparagat Krishi Vikas Yojana (PKVY):** Promotes organic and bio-fertilizer-driven farming to naturally restore soil microbial health. • **Shelterbelt Plantation Drives:** Plants dense tree rows across arid regions (e.g., Thar desert margins) to check sand dune movement.

Way Forward

- **Agro-Ecological and Technical Interventions**
  - **Scale Up Regenerative Agriculture:** Mandate crop rotation, cover cropping, and agroforestry to naturally restore Soil Organic Carbon.

    Andhra Pradesh Community-Managed Natural Farming (APCNF) utilizes multi-cropping and bio-inputs to restore soil organic carbon for over 800,000 farmers.
  - **Promote Precision Irrigation Technologies:** Expand subsidized drip and sprinkler networks to halt soil salinization and waterlogging in drylands.

    Pradhan Mantri Krishi Sinchayee Yojana (PMKSY - "Per Drop More Crop") provides up to an 80% subsidy for micro-irrigation systems to check waterlogging in drylands.
  - **Enforce Watershed-Based Micro-Planning:** Implement ridge-to-valley treatments and check-dams to maximize localized groundwater recharge and arrest runoff.

    China's Loess Plateau Watershed Project deployed massive terracing and sediment check-dams, successfully reversing land degradation across 3.5 million hectares.
  - **Deploy Geo-Spatial Monitoring Systems:** Utilize real-time satellite data to track topsoil erosion and prioritize hyper-degraded zones for restoration.

FAO's Open Foris Collect Earth tool leverages high-resolution satellite arrays to monitor real-time global dryland degradation and vegetation loss.

- **Policy and Regulatory Reforms**
  - **Align Policies Toward Land Degradation Neutrality (LDN):** Institutionalize the Bonn Challenge targets into regional and state-level agricultural master plans.
  - **Rationalize Input Subsidies:** Restructure power and fertilizer subsidies to curb the over-extraction of groundwater and indiscriminate chemical use.
  - **Create Integrated Land-Use Frameworks:** Strictly regulate the conversion of prime, fertile agricultural land for industrial and urban sprawl.
  - **De-Risk Restorative Agriculture Financially:** Provide direct fiscal incentives and carbon credits to smallholders practicing Sustainable Land Management.
- **Community and Institutional Enablers**
  - **Empower Gram Sabhas in Joint Forest Management:** Decentralize execution powers to local bodies for community-led pasture and wasteland restoration.
  - **Establish Social Forestry and Shelterbelt Corridors:** Plant multi-layered green walls along desert margins to check wind erosion and sand dune migration.
  - **Bridge the Lab-to-Land Knowledge Divide:** Leverage rural extension centers to train farmers in soil-health management and organic composting techniques.

Vietnam's Field Schools Network utilizes peer-to-peer localized training modules to quickly scale down eco-friendly soil health management protocols to rice farmers.

Successful Models for Land Restoration

Global Proven Models & Best Practices	• **African Great Green Wall (GGW):** Plants an 8,000 km mosaic of trees and vegetation across the Sahel to halt Sahara desert expansion. • **Loess Plateau Model (China):** Transformed a highly eroded ecosystem into a productive agricultural zone through massive terracing and sediment dams. • **Regreening of the Sahel (Niger):** Leverages Farmer-Managed Natural Regeneration (FMNR) to protect and manage sprouting native trees on farmlands. • **Al Baydha Project (Saudi Arabia):** Uses ancient water harvesting, swales, and check dams to restore hyper-arid desert landscapes into permaculture. • **Zero Budget Natural Farming (ZBNF) Global Model:** Replaces chemical fertilizers with indigenous microbial inoculants to restore degraded soil microbiomes.
Successful Indian Replication Models	• **Hiware Bazar Model (Maharashtra):** Banned tube-wells and grazing, shifting to watershed management to revive groundwater and reverse land aridity. • **Ralegan Siddhi Model (Maharashtra):** Built contour trenches and check dams under Anna Hazare to restore biomass in a highly drought-prone village. • **Aravali Green Wall Project:** Establishes a 1,400 km green ecological corridor across four states to block Thar desertification toward Delhi-NCR. • **Jhabua Watershed Model (Madhya Pradesh):** Empowered Bhil tribal communities to restore highly degraded hilly terrains through social fencing and afforestation. • **Kadwanchi Watershed Model (Maharashtra):** Deployed plastic-lined farm ponds and drip irrigation to check soil salinity and drought vulnerability.

Committee Recommendations on Land Degradation and Desertification

- **Ashok Dalwai Committee (Doubling Farmers' Income):** Recommends massive scaling of watershed management and agroforestry to restore organic carbon in degraded soils.
- **Swaminathan Commission (NCF):** Advocates for the mandatory setting up of soil-testing labs in every block to check indiscriminate chemical fertilizer overuse.
- **Mihir Shah Committee:** Proposes restructuring central water bodies into a unified National Water Commission to manage waterlogging and land salinity.
- **Kasturirangan Committee:** Emphasizes strict protection of ecologically sensitive areas in the Western Ghats to prevent slope destabilization and landslide-led degradation.
- **Planning Commission Working Group (12th Plan):** Urges the convergence of MGNREGS funds directly with wasteland development and social forestry projects.

Conclusion

Reversing land degradation requires a structural transition toward regenerative agro-ecology, robust watershed management, and community-driven land restoration models. Restoring soil health is a developmental necessity to ensure long-term food security, alleviate rural distress, and build climate resilience across vulnerable drylands.

6. MARINE POLLUTION

Marine pollution is an escalating global crisis caused by the entry of land-based agricultural runoff, industrial chemicals, and over **12 million tonnes of plastic waste into our oceans annually**. This widespread contamination destabilizes critical saltwater ecosystems, accelerates coral reef degradation, and chokes marine megafauna from coastal estuaries to the deepest oceanic trenches.

Marine Pollution vs. Water Pollution

- **Water Pollution:** A broad, umbrella term encompassing the contamination of all hydrological bodies, including inland freshwater (rivers, lakes, wetlands) and groundwater aquifers.
- **Marine Pollution:** A specific subset of water pollution confined strictly to tidal, saltwater oceanic environments, including seas, estuaries, coastal zones, and deep-ocean basins.

Major Causes of Marine Pollution

- **Land-Based and Terrestrial Sources**
  - **Agricultural Runoff:** Excess nitrogen and phosphorus fertilizers wash into rivers, causing coastal algal blooms and marine dead zones (*hypoxia*).

    According to the **United Nations Environment Programme (UNEP)**, excess nutrient runoff from global agriculture has contributed to 500 marine "dead zones" globally.
  - **Untreated Municipal Sewage:** Billions of liters of domestic wastewater are dumped daily into coastal waters, depleting dissolved oxygen levels.

    According to **CPCB**, Indian urban centers generate over 72,000 million liters of sewage per day, with more than 70% discharged entirely untreated into water bodies
  - **Industrial Heavy Metals:** Coastal manufacturing units discharge untreated mercury, cadmium, and lead, which bioaccumulate in marine food webs.

    The Ministry of Environment has identified **320 high-probability contaminated sites across India** heavily polluted with industrial toxic metals like lead, chromium, and mercury.
  - **Macro and Micro-Plastics:** Urban plastic waste enters rivers and estuaries, generating micro-polymers that gather in oceanic gyres.

According to data compiled by the **CSIRO and CPCB**, India generates approximately 26,000 metric tonnes of plastic waste every single day

- **Maritime, Shipping, and Offshore Activities**
  - **Operational Oil Discharges:** Commercial ships illegally flush bilge water and fuel oil residues into open sea lanes during transit.
  - **Catastrophic Tanker Spills:** High-impact accidents of crude oil tankers (e.g., historical *Exxon Valdez* or regional spills) smother marine life.
  - **Ballast Water Discharges:** Large cargo vessels release ballast water at destination ports, introducing destructive invasive alien species.
  - **Ghost Fishing Equipment:** Abandoned, lost, or discarded nylon fishing gear drifts endlessly, entangling marine megafauna.
- **Atmospheric and Deep-Sea Extraction**
  - **Atmospheric Deposition:** Burning fossil fuels releases carbon dioxide, which dissolves into seawater to drive oceanic acidification.
  - **Offshore Drilling Blowouts:** Deep-sea oil and gas extraction failures cause uncontrollable underwater hydrocarbon leaks.
  - **Deep-Sea Mining Disturbances:** Emerging polymetallic nodule extraction destroys benthic habitats and creates toxic underwater sediment plumes.

Impacts of Marine Pollution

- **Ecological and Biological Disruption**
  - **Marine Megafauna Mortality:** Endangered whales, sea turtles, and seabirds die from intestinal blocks after swallowing macro-plastic debris.

    According to the **United Nations Environment Programme (UNEP)**, marine plastic debris causes the death of more than 1 million seabirds and 100,000 marine mammals globally each year.
  - **Ghost Fishing Ecosystem Toll:** Abandoned nylon nets drift through ocean currents, continuously killing corals and marine species.

    According to the **Food and Agriculture Organization (FAO)**, abandoned, lost, or discarded fishing gear makes up roughly 10% of all marine waste.
  - **Oceanic Dead Zone Expansion:** Nutrient runoffs trigger massive algal blooms, creating hypoxic (oxygen-depleted) zones where marine life suffocates.
  - **Coral Reef Degradation:** Heavy sedimentation, chemical toxins, and rising ocean acidity stress and bleach sensitive coral ecosystems.

    According to the **IPCC**, rising ocean acidity and chemical pollution threaten to degrade or kill up to 70-90% of the world's remaining coral reefs.
- **Trophic and Physiological Threats**
  - **Microplastic Bioaccumulation:** Micro-polymers move up marine food chains, transferring toxins from tiny plankton to apex marine predators.
  - **Endocrine Disruption in Marine Life:** Industrial heavy metals and chemical leachates mimic hormones, causing reproductive failures in fish.
  - **Disruption of the Biological Carbon Pump:** Toxic microplastics lower the photosynthesis rates of marine phytoplankton, harming global carbon absorption.

Research published in **Nature Communications**, pointed out that ingestion of microplastics by marine zooplankton and phytoplankton can reduce their photosynthesis and carbon absorption efficiency by up to 15%.

- **Socio-Economic and Public Health Impacts**
  - **Threats to Human Food Security:** Toxic heavy metals like methylmercury bioaccumulate in commercial fish, directly threatening human consumers.
  - **Crashing Coastal Tourism Economies:** Heavily polluted coastal zones and plastic-littered beaches trigger sharp financial losses for global tourism.

    According to the **World Bank**, severe plastic littering and coastal water pollution cause global tourism economies to suffer estimated losses of over $13 billion annually.
  - **Livelihood Losses for Artisanal Fisheries:** Deteriorating marine health leads to crashing wild fish stocks, threatening the livelihoods of coastal fishing communities.

Measures against Marine Pollution

Global Conventions and Legal Frameworks	• **UNCLOS (1982):** Establishes the primary global legal architecture mandating all signatory states to protect and preserve the marine environment. • **MARPOL Convention:** Restricts and regulates international shipboard pollution from oil, noxious liquids, sewage, garbage, and air emissions. • **London Convention (1972):** Prohibits the deliberate, high-volume dumping of hazardous industrial and radioactive wastes into deep-ocean basins. • **Global Plastic Treaty:** Establishes a legally binding UN framework addressing the full lifecycle of plastic pollution, including marine inflows.
India's Domestic Legislative and Policy Measures	• **Merchant Shipping Rules, 2023:** Aligns Indian maritime commercial laws with MARPOL to control single-use plastics and waste discharge from commercial vessels. • **Coastal Regulation Zone (CRZ) Notifications:** Restricts polluting industrial setups and regulates infrastructure development near sensitive coastal and estuarine zones. • **National Marine Litter Management Strategy:** Formulates a systematic roadmap for tracking and trapping land-based plastic inflows before they reach open waters. • **Swachh Sagar, Surakshit Sagar Campaign:** Spearheads massive, multi-agency public beach cleanups along India's 7,500+ km coastline to remove macro-plastics.
Technological and Regional Collaborations	• **Project GloLitter Partnerships:** Collaborates with FAO and IMO to reduce marine plastic litter arising from the international fishing and shipping sectors. • **South Asia Co-operative Environment Programme (SACEP):** Unites regional nations to implement joint oil spill contingency plans across the Indian Ocean. • **Deployment of Estuarine Trash Booms:** Places automated river-mouth barriers across high-yield urban channels to intercept floating debris before oceanic entry.

Way forward

- **Source Control and Upstream Regulatory Reforms**
  - **Enforce Strict Source-to-Sea Land Interceptions:** Trap industrial effluent and urban solid waste at river mouths before they breach coastal waters.
  - **Impose Nutrient Credit Systems in Agriculture:** Penalize excessive chemical fertilizer use to curb agricultural runoff and coastal eutrophication.
  - **Establish Zero-Discharge Shipboard Port Policies:** Mandate 100% digital tracking and mandatory offloading of ship bilge, ballast, and plastic waste at ports.
  - **Ratify and Implement the Global Plastic Treaty:** Enforce legally binding, cross-border polymer lifecycle caps to reduce oceanic plastic inflows.
- **Technological and Ecological Restorations**
  - **Deploy Automated Riverine Trash Booms:** Scale up floating barrier technology across major estuarine channels to intercept macro-plastics.
  - **Expand Marine Protected Areas (MPAs):** Designate strict ecological zones to allow degraded coral reefs and depleted marine food webs to recover naturally.
  - **Subsidize Marine-Degradable Fishing Gear:** Provide fiscal incentives to transition commercial fisheries toward bio-based, non-ghost-fishing equipment.
  - **Utilize Bioremediation for Offshore Oil Spills:** Scale up the application of oil-degrading bacterial formulations (*Oilzapper*) for rapid spill containment.
- **Institutional Coordination and Community Models**
  - **Establish an Ocean Governance Authority:** Create an apex national agency to bridge the jurisdictional gaps between coastal states, ports, and environmental ministries.
  - **Institutionalize Coastal Community Stewardship:** Integrate traditional fishing communities into the formal blue economy to lead local beach and mangrove restoration drives.
  - **Mandate Transboundary Oil Spill Contingency Exercises:** Conduct frequent joint-naval operations across the Indian Ocean Region (IOR) to optimize disaster response.

Success Models for Marine Pollution Control

Global Proven Models & Best Practices	• **Ocean Cleanup Interceptor Model:** deploys solar-powered, automated catamaran barriers in high-emission rivers to trap floating plastic before it enters the sea. • **MARPOL Special Areas Matrix:** Designates highly sensitive oceanic zones (e.g., Mediterranean and Baltic Seas) where any shipboard waste discharge is strictly illegal. • **North Sea Nutrient Reduction Model:** Unites European nations to enforce strict binding limits on agricultural runoff, successfully shrinking coastal dead zones. • **Global Ghost Gear Initiative (GGGI):** Partners with international commercial fisheries to track, report, and recover lost or abandoned nylon fishing nets. • **Australian Great Barrier Reef Marine Park Model:** Uses zoning maps and satellite tracking to strictly isolate vulnerable coral ecosystems from commercial shipping lanes.
Emerging Indian Replication Models	• **Swachh Sagar, Surakshit Sagar Model:** Spearheads coordinated, multi-agency beach cleanups across India's 7,500 km coastline, removing thousands of tonnes of marine debris.

	• **Suchitwa Sagaram Project (Kerala):** Trains artisanal fishing communities to collect and bring back sea-tossed plastic waste, which is then recycled for road construction. • **Mumbai Mithi River Trash Boom Model:** Deploys floating mesh barriers across heavy urban river mouths to block city plastic waste from reaching the Arabian Sea. • **Net-Fish MPEDA Incentive Model:** Provides direct financial rewards and infrastructure support to coastal fishers who hand over torn, end-of-life commercial fishing nets. • **Gulf of Mannar Biosphere Restoration Model:** Combines artificial reef deployment with community-led mangrove plantation to naturally filter out coastal land sediment.

Conclusion

Mitigating marine pollution requires an integrated "source-to-sea" framework that strictly intercepts terrestrial plastic, chemical, and sewage runoffs before they enter fragile coastal waters. This structural transition must combine global maritime agreements with grassroots, community-led coastal conservation models to foster a truly sustainable and resilient Blue Economy.

7. MICRO PLASTICS

Microplastics are tiny plastic particles measuring less than 5 millimeters in size, arising from both the direct manufacturing of commercial products and the fragmentation of larger plastic waste. In the context of India's rapid urbanization, these persistent pollutants have deeply infiltrated marine, freshwater, and terrestrial ecosystems, posing severe threats to biodiversity and entering the human food chain.

Major Causes of Microplastics

Primary Microplastics

- **Microbeads in Personal Care Products:** Industrial addition of polyethylene spheres to facial scrubs, toothpastes, and cosmetics for exfoliating properties.

  According to the **United Nations Environment Programme (UNEP)**, a single bottle of exfoliating shower gel can release roughly 300,000 plastic microbeads into wastewater systems with every use.

- **Industrial Plastic Pellets (Nurdles):** Accidental spillage and loss of raw pre-production plastic resins during cross-border maritime and overland transport.

  According to global plastic pollution data from the **IUCN**, raw pre-production pellets (nurdles) are the second-largest primary source of microplastics in the ocean

- **Synthetic Textiles and Laundering:** Sloughing off of thousands of polyester, nylon, and acrylic microfibers from clothing during domestic washing cycles.

  According to a study by the **University of Plymouth**, a single 6 kg domestic wash cycle of synthetic clothing can release over 700,000 individual polyester and acrylic microfibers into sewage networks.

- **Blasting Media and Industrial Abrasion:** Use of plastic grit in specialized sandblasting operations to strip paint and rust from commercial ship hulls.

Secondary Microplastics

- **Photodegradation of Mismanaged Waste:** Solar ultraviolet radiation breaking down abandoned plastic bottles, bags, and packaging materials on land and beaches.

- **Mechanical Marine Weathering:** Continuous wave action and physical abrasion against rocks that fragment floating macro-plastic debris inside oceanic gyres.
- **Vehicular Tire Wear and Tear:** Physical friction between moving vehicle tires and road surfaces that releases tons of rubber-plastic dust into the atmosphere.

> According to the **International Union for Conservation of Nature (IUCN)**, tire wear particles generated by road friction account for roughly 28% of the total microplastics entering the world's oceans.

- **Weathering of Agricultural Mulch Films:** Fragmentation of thin polyethylene sheets used for soil moisture retention under intense field heat and agricultural tilling.
- **Degradation of Commercial Fishing Gear:** Gradual shedding of micro-fragments from synthetic nylon fishing nets, longlines, and buoys abandoned at sea.

> According to the **Food and Agriculture Organization (FAO)**, the degradation of abandoned, lost, or discarded fishing contributes directly to the estimated 10% share of global marine plastic waste.

Impacts of Micro plastic

- **Ecological & Environmental Impacts**
  - **Bioaccumulation in Marine Food Webs:** Ingestion by low-trophic organisms like zooplankton and small fish leads to the biomagnification of toxins up to apex predators.
  - **Adsorption of Persistent Organic Pollutants (POPs):** Microplastics act as chemical sponges, attracting and concentrating toxic waterborne pollutants like DDT and PCBs.

> According to the **Marine Pollution Bulletin**, microplastics act as chemical sponges, concentrating persistent waterborne toxins like DDT and PCBs up to 1 million times higher than the surrounding seawater.

  - **Disruption of Soil Ecosystems:** Accumulation in agricultural soils alters soil porosity, restricts root growth, and severely impacts earthworm mortality.

> According to an environmental soil study in **Global Change Biology**, high microplastic contamination in agricultural soils decreases earthworm survival rates by up to 15%

  - **Altered Marine Carbon Cycling:** Ingested microplastics accelerate the fragmentation of marine snow, disrupting the oceans' natural biological carbon pump.
- **Human Health & Toxicological Impacts**
  - **Infiltration of the Human Food Chain:** Microplastics have been detected in commercial seafood, tap water, table salt, and even human placentas and breast milk.
  - **Endocrine Disruption:** Chemical additives like Phthalates and Bisphenol A (BPA) leach into tissues, potentially disrupting hormonal regulation and fertility.
  - **Cellular Stress and Inflammation:** Inhaled or ingested microparticles can trigger localized tissue inflammation, cellular toxicity, and oxidative stress.
  - **Vector for Pathogenic Microbes:** The surface of floating microplastics serves as a structural breeding ground for hazardous bacterial colonies and pathogens.
- **Economic & Developmental Impacts**
  - **Threat to Coastal Fisheries:** Depleted fish stocks and toxic contamination risks damage local coastal economies and threaten traditional seafood export markets.
  - **Escalating Water Purification Costs:** Municipal water treatment plants face steep financial burdens to upgrade filtration infrastructure to trap sub-micron particles.
  - **Impact on Coastal Ecotourism:** Heavy deposition of plastic fragments on pristine beaches undermines the aesthetic value and tourism revenues of coastal states.

Measures Taken against Micro plastics

- **Domestic Measures**
  - **Formal Legal Definition:** Formally recognized microplastics for the first time under the **Plastic Waste Management (Amendment) Rules, 2024**, defining them as solid plastic particles between 1 and 1,000 microns.
  - **Extended Producer Responsibility (EPR 2.0):** Mandates minimum recycled content targets for plastic packaging (up to **40% rigid plastic recycling for fiscal year 2026–27**) backed by environmental compensation fees.
  - **Strict Biodegradability Certification:** Requires manufacturers of compostable and biodegradable plastics to pass Central Pollution Control Board (CPCB) testing to prove zero microplastic residue.
  - **Digital Supply-Chain Tracking:** Introduces mandatory QR codes and barcodes under the 2025 guidelines to track plastic waste life cycles via a centralized online portal.
  - **National Plastic Pollution Reduction Campaign (NPPRC):** Targets rural gram panchayats to halt the degradation of agricultural mulch films into terrestrial soil systems.
  - **Ban on Single-Use Plastics:** Prohibits the manufacture, import, stocking, and sale of identified low-utility, high-littering single-use plastics to cut off secondary fragmentation sources.
- **International & Global Measures**
  - **Global Plastics Treaty Negotiations:** Advancing a legally binding international framework through **UN Intergovernmental Negotiating Committee (INC)** sessions to govern the entire life cycle of polymers.
  - **Kunming-Montreal Global Biodiversity Framework (Target 6):** Sets a global target to significantly minimize plastic waste leakage and pollution risks by 2030.
  - **Operation Clean Sweep (OCS):** A global voluntary industry initiative adopted by manufacturers to enforce **Zero Pellet Loss** protocols, preventing raw resin nurdles from entering waterways.
  - **Un-Plastic Collective (UPC):** A voluntary collaborative initiative launched by UNEP-India, WWF-India, and the CII to guide corporate supply chains toward plastic neutrality.
  - **G20 Osaka Blue Ocean Vision:** Commits member nations to voluntarily reduce additional marine plastic litter to zero by the year 2050 through optimized waste management systems.

Way Forward for Mitigating Micro plastics

- **Regulatory Precision & At-Source Accountability**
  - **Enforce Tightened Recycled Content Mandates:** Implement the strict targets of the **Plastic Waste Management (Amendment) Rules, 2026**, forcing Category-I rigid plastics to achieve up to **60% recycled content by 2028–29**.
  - **Plug EPR Compliance Loopholes:** Eliminate the misuse of plastic credit trading and restrict the **carry-forward** of unfulfilled recycling deficits to ensure immediate operational accountability for brand owners.
  - **Standardize Global Zero Pellet Loss Protocols:** Mandate international industrial safety frameworks like Operation Clean Sweep to legally prevent raw pre-production resin nurdles from escaping factories into public drains.
  - **Implement Consumer Tech Manufacturing Mandates:** Enforce advanced design criteria (similar to Euro-7 standards) for tire manufacturers to structurally minimize synthetic rubber road-friction wear and dust emission.
- **Infrastructure Retrofitting & Technological Interventions**
  - **Upgrade Municipal Wastewater Facilities:** Mandate tertiary treatment technologies, such as **Membrane Bioreactors (MBRs)** and deep sand filtration, to trap up to **99% of microplastics** before effluent discharge.
  - **Mandate Microfiber Filters in Washing Machines:** Legislate factory-installed, built-in structural microfilters for all new domestic and commercial washing machines to intercept synthetic clothing shards.

- **Deploy AI-Driven Supply-Chain Traceability:** Leverage the centralized CPCB digital portal to enforce tracing via mandatory QR codes, ensuring counterfeit-free auditing of plastic components across vendors.
- **Incentivize Biomimetic Green Chemistry:** Provide strategic state grants to clean-tech startups developing marine-compostable alternatives derived from starch, seaweed, and agricultural waste.

- **Institutionalization & Global Governance**
  - **Update National Air and Water Quality Standards:** Formally integrate microplastic concentration thresholds into the **National Ambient Air Quality Standards (NAAQS)** and the **Bureau of Indian Standards (BIS)** for safe drinking water.
  - **Finalize a Binding Global Plastics Treaty:** Actively participate in the **UN Intergovernmental Negotiating Committee (INC)** rounds to seal a legally binding global pact regulating the absolute transboundary lifecycle of plastic polymers.
  - **Mainstream Weed-to-Wealth Circular Models:** Foster localized, decentralized recycling infrastructure in rural gram panchayats to collect macro-plastics before solar UV radiation fragments them into soil ecosystems.

Conclusion

Combating the invisible crisis of micro plastics demands a systemic transition from a linear **take-make-waste** approach to a robust, technology-driven circular economy. By stringently enforcing the latest plastic waste management rules and scaling up indigenous filtration innovations like the IISc hydrogels, India can safeguard its fragile ecosystems and public food security

8. NOISE POLLUTION

Noise pollution is an escalating environmental and public health hazard driven by unregulated urbanization, industrial operations, and high-density transportation networks. Far from a mere transient public nuisance, chronic exposure to high decibel levels acts as a pervasive physiological stressor that alters wildlife behavior and masks ecological communication.

Major Causes of Noise Pollution

- **Urban Infrastructure and Transportation Drivers**
  - **Expanding Road Traffic:** Unregulated honking, lack of engine maintenance, and high vehicle density generate high ambient decibels.

    According to the **Central Pollution Control Board (CPCB)**, average traffic noise levels in major Indian metros consistently exceed 75 to 80 decibels (dB).
  - **Railway Operations:** High-speed locomotive engines, metal-on-metal track friction, and marshaling yards disturb adjoining urban pockets.

    According to the **National Physical Laboratory (NPL)**, residential areas situated within 100 meters of major railway tracks experience **85 to 90 dB during locomotive transits**.
  - **Aviation Noise:** Low-altitude flights, jet engine takeoffs, and supersonic sonic booms severely affect residential zones near airports.

    Aircraft takeoffs and low-altitude flights generate localized peak noise levels between 100 and 110 dB, severely affecting surrounding residential zones.
  - **Unplanned Construction:** Continuous operation of heavy earthmovers, pneumatic drills, jackhammers, and concrete mixers produces intense ambient noise.
- **Industrial and Commercial Factors**
  - **Heavy Machinery Operations:** Mechanical equipment like stamping presses, grinding mills, and cooling towers run without acoustic insulation.

- **Commercial Diesel Generators:** Widespread use of un-attenuated backup generators in markets and residential areas spikes local noise levels.

  Un-attenuated diesel generator sets running as power backups emit direct noise levels of up to 85 to 90 dB.

- **Commercial Hub Activities:** High-density open-air markets, outdoor public address systems, and localized exhaust ventilation create urban noise clusters.

- **Societal and Socio-Cultural Drivers**
  - **Festivals and Public Gatherings:** High-decibel loudspeakers, public amplifiers, and uncontrolled firecracker use during celebrations exceed permissible limits.

    Deployment of high-decibel loudspeakers and illegal firecrackers pushes localized noise levels past 110 to 120 dB

  - **Domestic and Household Appliances:** Daily use of high-power mixers, vacuum cleaners, old air conditioners, and loud entertainment systems increases indoor noise.
  - **Unregulated Nightlife:** Late-night operations of open-air restaurants, discotheques, and community halls near residential zones disrupt quiet hours.

Impacts of Noise Pollution

- **Physiological and Physiological Health Hazards**
  - **Noise-Induced Hearing Loss:** Chronic exposure to over 85 decibels causes irreversible damage to inner ear cochlear hair cells.
  - **Cardiovascular Complications:** Elevated ambient noise triggers continuous adrenaline release, raising blood pressure and risk of ischemic heart disease.

    According to a study in **The Lancet**, chronic exposure to elevated road and aircraft noise correlates with an estimated 8-10% increase in the risk of arterial hypertension.

  - **Severe Sleep Fragmentation:** Disrupts deep sleep cycles, causing chronic fatigue, cognitive decline, and metabolic disorders.
  - **Elevated Psychological Stress:** High ambient decibels induce constant cortisol spikes, accelerating anxiety, panic disorders, and aggressive behavior.
- **Ecological and Wildlife Disruptions**
  - **Acoustic Communication Masking:** Drowns out mating calls and territorial warnings of birds and amphibians, crippling their reproductive success.
  - **Cetacean Disorientation:** Marine sonar and shipping noise disrupt echolocation in whales and dolphins, causing mass beaching events.
  - **Altered Predator-Prey Dynamics:** Prevents nocturnal predators from hearing prey movements, destabilizing local food webs.
  - **Avian Habitat Abandonment:** Forces sensitive bird species to permanently migrate out of urban forests, disrupting seed dispersal.

    According to a study by the **Wildlife Institute of India (WII)**, continuous infrastructure noise causes a 15-20% decline in avian species richness in urban forests.

- **Socio-Economic and Productivity Losses**
  - **Reduced Cognitive Performance:** High noise levels in schools and offices impair concentration, short-term memory, and learning capacity.

    According to the **WHO's Environmental Noise Guidelines**, a 10 dB increase in continuous aircraft noise exposure is directly linked to a 1-to-2 month delay in memory development.

  - **Depreciation of Real Estate Values:** Residential properties situated near major highways, airports, and industrial tracks face steep market value drops.

- **Escalating Healthcare Expenditures:** Increases the public economic burden due to rising treatments for chronic hypertension, deafness, and insomnia.

Measures taken for Noise Pollution

- **Legislative and Policy Frameworks**
  - **Air (Prevention and Control of Pollution) Act, 1981:** Formally includes "noise" within the legal definition of an air pollutant.
  - **Noise Pollution (Regulation and Control) Rules, 2000:** Prescribes distinct day and night ambient noise standards for Industrial, Commercial, Residential, and Silence zones.
  - **The Environment (Protection) Act, 1986:** Empowers authorities to mandate strict manufacture-level noise ceilings for diesel generators, firecrackers, and motor vehicles.
  - **Central Pollution Control Board (CPCB) Guidelines:** Lays down clear standard operating procedures for the confiscation of illegal public address systems and high-decibel amplifiers.
- **Monitoring and Enforcement Systems**
  - **National Ambient Noise Monitoring Network (NANMN):** Operates real-time automated noise monitoring stations across major Indian metro cities to track compliance.
  - **Silence Zone Enforcement:** Mandates a strict 100-meter acoustic buffer zone around all hospitals, educational institutions, and courts where honking is illegal.
  - **Judicial Directives (Supreme Court):** Imposes a nationwide ban on the use of public loudspeakers and high-decibel amplifiers between 10:00 PM and 06:00 AM.
- **Technological and Urban Interventions**
  - **Mandatory Acoustic Enclosures:** Requires all commercial diesel generators to be fitted with soundproof acoustic canopies before deployment.
  - **Eco-Innovative Urban Buffers:** Deploys multi-layered green belts (trees like Neem and Ashoka) along highways to act as natural acoustic barriers.
  - **Acoustic Camera Trials:** Emerging deployment of automated sound-detecting cameras in select metros to automatically fine vehicles with modified, loud exhaust pipes.

Way Forward for Mitigating Noise Pollution

- **Urban Planning and Engineering Interventions**
  - **Construct Structural Noise Barriers:** Install high-efficiency acoustic panels and transparent sound walls along elevated expressways, flyovers, and metro corridors.

    Germany's Autobahn Network installs miles of high-efficiency wood-concrete and transparent acrylic acoustic panels directly alongside urban expressway stretches.
  - **Design Multi-Layered Urban Green Belts:** Plant dense, sound-absorbing tree species (e.g., Neem, Tamarind, Ashoka) along highway margins to act as natural acoustic buffers.
  -
  - **Mandate Acoustic Building By-Laws:** Enforce the use of double-glazed soundproof windows, insulation pads, and acoustic drywalls in high-density residential zones.

    UK Building Regulations enforces strict acoustic laboratory testing and sound-insulation benchmarks for all newly built multi-family apartments.
  - **Optimize Heavy Traffic Routing:** Redesign urban freight corridors to divert commercial heavy vehicles completely away from core residential and silent zones.

    Eastern and Western Peripheral Expressways successfully divert over 50,000 commercial transit trucks daily completely outside Delhi's core urban center.
- **Technological and Enforcement Reforms**

- **Deploy Automated Acoustic Cameras:** Install sound-activated cameras at major intersections to automatically detect vehicles with modified exhausts or illegal horns.

  City of Paris, France deploys automated "Medusa" noise radar cameras that scan, trace, and instantly fine vehicles exceeding decibel cutoffs.

- **Transition to Electric Mobility:** Accelerate public and private adoption of Electric Vehicles to drastically lower engine noise in congested cities.

  FAME-II Scheme and Delhi EV Policy aggressively subsidize electric bus fleets and three-wheelers to lower high-frequency transit friction.

- **Upgrade Low-Noise Pavement Materials:** Use porous asphalt and rubberized bitumen on highways to significantly reduce tire-to-surface rolling friction noise.
- **Enforce Real-Time Digital Mapping:** Integrate automated noise-monitoring sensors with city dashboards to enable immediate police dispatches to high-decibel violations.

  European Environment Agency (EEA) maintains automated Noisewatch dashboards that map live environmental decibel spikes across major member capitals.

- **Regulatory and Socio-Cultural Adjustments**
  - **Adopt Smart No-Honking Zones:** Transition silent zones into automated fine corridors using intelligent transport systems and license plate recognition.

    Mumbai Traffic Police "Punishing Signal" Campaign linked automated decibel meters to traffic lights, resetting red timers if drivers honked over 85 db.

  - **Strictly Enforce Time and Decibel Caps:** Upgrade local police mandates to immediately seize illegal public amplifiers used past the 10:00 PM curfew.
  - **Promote Community-Led Quiet Zones:** Incentivize Resident Welfare Associations to establish "Silent Hours" and monitor internal commercial activities.
  - **Launch Nationwide Civic Awareness Campaigns:** Conduct target-driven public sensitizations like No-Honk Days to anchor civic discipline against recreational noise.

Global Proven Models for Noise Mitigation

- **Barcelona's Superblocks Model (Spain):** Restricts motorized traffic to peripheral roads, reclaiming interior grid streets for pedestrians to reduce decibel levels.
- **HOSANNA Toolbox Model (European Union):** Deploys optimal combinations of green roofs, vegetated facades, and recycled acoustic barriers to reduce transport noise by up to 10 dB.
- **Low-Noise Pavement Engineering (Germany & Japan):** Mandates widespread use of porous asphalt and rubberized bitumen on highways to significantly absorb tire-surface friction.
- **Array of Things Sensor Network (Chicago, USA):** Deploys interactive IoT sound sensors to dynamically track noise spikes, successfully reducing localized noise complaints by 30%.
- **Medusa Acoustic Camera Model (France):** Uses multi-microphone automated radar systems to instantly identify, track, and fine high-decibel vehicles in real time.
- **London Smart Acoustic Mapping Initiative (UK)**: Integrates 1,000 active IoT noise sensors into city dashboards to crowdsource spatiotemporal soundscapes for urban planning.

Conclusion

Mitigating noise pollution requires a paradigm shift that integrates acoustic management into core urban planning, recognizing a quiet environment as an essential component of public health and human dignity. Achieving this transition demands a strict technological and regulatory convergence, leveraging real-time IoT monitoring, automated enforcement cameras, and community-led civic discipline.

9. WASTE MANAGEMENT IN INDIA

India's rapid urbanization and demographic expansion have escalated its municipal solid waste generation to an estimated 62 million tonnes annually. This exponential surge presents an acute ecological and public health challenge, severely straining the country's municipal infrastructure and landfill capacities.

Classification of waste types in India

Waste Type	Major Source
Municipal Solid Waste (MSW)	Everyday household and commercial trash, comprising organic matter, recyclables, and inert debris.
E-Waste	Discarded electrical or electronic devices like phones, computers, and household appliances.
Plastic Waste	Non-biodegradable polymer packaging and single-use products generating massive long-term pollution.
Biomedical Waste	Infectious and hazardous materials generated from hospitals, clinics, and diagnostic labs.
Hazardous Waste	Industrial byproducts possessing toxic, chemical, flammable, corrosive, or explosive characteristics.
Construction & Demolition (C&D)	Debris from infrastructure building, remodeling, and demolition, predominantly concrete, soil, and bricks.
Liquid Waste / Sewage	Domestic wastewater and industrial effluents requiring effluent treatment plants before discharge.

Constitutional provisions relevant to waste management in India

- **State List (List II), Entry 6:** Mandates Public Health and Sanitation as a primary state responsibility (the core legal basis for local waste management).
- **Article 48A:** Directs the State to protect and improve the environment, safeguarding the nation's forests and wildlife.
- **Article 51A(g):** Imposes a duty on every citizen to protect and improve the natural environment, including forests, lakes, rivers, and wildlife.
- **Eleventh Schedule (Article 243G):** Empowers Panchayats to manage rural sanitation and environmental protection.
- **Twelfth Schedule (Article 243W):** Explicitly tasks Urban Local Bodies (Municipalities) with public health, sanitation, conservancy, and solid waste management.
- **Article 21 (Right to Life):** Judicially expanded by the Supreme Court to include the Right to a clean, pollution-free environment and proper waste disposal.

Regulatory framework for waste management in India

- **Environment (Protection) Act, 1986:** The overarching framework enabling the central government to notify all specific waste management rules.
- **Waste Management Rules (MoEFCC)**
  - **Solid Waste Management (SWM) Rules:** Mandates source segregation into distinct streams (wet, dry, sanitary, and special-care) and prioritizes decentralized processing over landfills.
  - **Plastic Waste Management Rules:** Enforces a ban on identified single-use plastics and introduces mandatory recycling targets through digital Extended Producer Responsibility (EPR) portals.
  - **E-Waste (Management) Rules:** Implements a formalized trading system for EPR certificates and mandates hazardous substance reductions (lead, mercury) in electronics.

- **Biomedical Waste Management Rules:** Categorizes bio-waste into four color-coded bins (Yellow, Red, White, Blue) to strictly segregate infectious waste from municipal streams.
- **Hazardous & Other Wastes Rules:** Regulates the safe generation, storage, and transboundary movement of toxic industrial byproducts via strict manifest tracking.
- **Construction & Demolition (C&D) Waste Rules:** Mandates bulk generators to submit waste utilization plans, emphasizing debris recycling into secondary materials like pavement blocks.
- **Battery Waste Management Rules:** Enforces a mandatory collection-and-recycle mechanism for all types of discarded batteries under an updated EPR ecosystem.

- **Regulatory & Enforcement Bodies**
  - **CPCB (Central Pollution Control Board):** Coordinates national policies, sets technical disposal standards, and manages centralized EPR portal registries.
  - **SPCBs (State Pollution Control Boards):** Issues operational authorizations, monitors industrial emission/effluent compliance, and enforces penalties.
  - **Urban Local Bodies (ULBs):** Functions as the frontline implementation authority responsible for direct door-to-door collection, transport, and setup of processing facilities.

Waste management methods

Method	Process	Ideal Application
Composting	Aerobic biodegradation of organic waste into nutrient-rich manure.	Biodegradable (wet) waste.
Vermicomposting	Using earthworms to accelerate organic waste decomposition.	Household/small-scale organic.
Incineration	High-temperature combustion converting waste into ash and gas.	Biomedical/Hazardous waste.
Pyrolysis	Thermochemical decomposition of organics in the absence of oxygen.	Plastics and rubber waste.
Recycling	Material recovery and reprocessing into new secondary products.	Glass, paper, metals, plastics.
Sanitary Landfills	Engineered disposal in lined pits to prevent groundwater seepage.	Inert and non-recyclable residue.
Waste-to-Energy	Energy recovery via thermal combustion or anaerobic digestion.	High-caloric non-recyclable waste.

Current status of waste management in India

- **Current Landscape**
  - **Daily Generation:** India generates approximately **160,000 metric tonnes (MT)** of municipal solid waste per day. (Ministry of Housing and Urban Affairs - MoHUA)
  - **Processing Leap:** The national waste processing rate has surged to nearly **80%**, a major improvement from a mere 18% in 2014 under the Swachh Bharat Mission legacy. (SBM-Urban)
  - **Dumpsite Remediation:** Under SBM 2.0's "Garbage-Free Cities" mandate, roughly **19% of the nation's 2,424 identified legacy dumpsites** have been completely reclaimed. (Swachh Bharat Mission 2.0 Progress Data)
  - **Disposal Gap:** Landfills remain the dominant end-point, absorbing **44%** of the country's collective untreated or mixed waste streams. (IMARC Waste Management Market Report)
  - **E-Waste Horizon:** India ranks **3rd globally**, generating **2.2 million metric tonnes** of electronic waste annually, with formal processing mechanisms capturing less than half. (Central Pollution Control Board)

Significance of waste management

- **Environmental & Ecological Significance**
  - **Combating Climate Change:** Reduces potent **methane emissions** from decomposing landfill waste, directly supporting India's net-zero targets and COP commitments.
  - **Preserving Ecosystems:** Prevents toxic **leachate** from contaminating groundwater aquifers and blocks plastic debris from degrading marine ecosystems.
- **Economic & Resource Significance**
  - **Driving a Circular Economy:** Transitions the economic model from **take-make-dispose** to **resource recovery**, extracting valuable secondary materials from urban waste streams.
  - **Energy Security:** Boosts domestic energy alternatives through **Waste-to-Energy (WtE)** tech, converting organic municipal waste into Compressed Biogas (CBG).
- **Health & Social Significance**
  - **Public Health Security:** Eliminates breeding grounds for vectors, drastically reducing the transmission of waterborne and airborne diseases like dengue and cholera.
  - **Inclusive Development:** Formally integrates and safeguards the livelihoods of India's **informal waste pickers**, providing social security and dignified labor conditions.
- **Governance & Urban Excellence**
  - **Achieving SDGs:** Serves as the primary operational vehicle to achieve **SDG 11** (Sustainable Cities and Communities) and **SDG 12** (Responsible Consumption and Production).
  - **Urban Livability:** Promotes the **Garbage-Free Cities** mandate under Swachh Bharat Mission 2.0, directly enhancing the ease of living in India's expanding urban centers.

Issues with India's waste management sector

- **Upstream & Social Bottlenecks**
  - **Source Segregation Deficit:** Chronic failure in primary segregation at the household level leads to contaminated mixed waste and advanced downstream recycling issue.

    As per MoHUA, national urban door-to-door collection stands high at **97%**, the actual processing rate gets limited to **80.31%** due to massive gaps in daily home-level wet vs. dry segregation.
  - **Social Stigma & Marginalization:** As per **NITI Aayog**, India's informal waste sector comprises over 1.5 million waste pickers who handle structural recycling volumes without institutional recognition.
  - **Culture of Linear Consumption:** A rising urban consumerist lifestyle increasingly prioritizes quick single-use convenience over circular Reduce and Reuse practices.

    Urban India generates 1,62,000 tonnes of municipal solid waste every single day
- **Midstream & Infrastructure Gaps**
  - **Inefficient Budgetary Allocation:** Capital-starved Urban Local Bodies (ULBs) drain 60–70% of their sanitation budgets strictly on secondary transport and collection logistics.
  - **Underdeveloped Recycling Markets:** Suboptimal domestic market linkages keep the commercial demand for recycled secondary products severely low.

    As per NITI Aayog, current recycling infrastructure captures less than **20% of the massive ₹51,000 crore circular economy potential** embedded in annual domestic waste.
  - **Wastewater Deficit:** Glaring infrastructure backlogs leave nearly 60% of municipal wastewater completely untreated, choking urban river systems.

    As per CPCB, India generates **72,368 Million Liters per Day of urban wastewater**, but a glaring infrastructure gap leaves **72% flowing entirely untreated** into rivers .
- **Regulatory & Governance Flaws**

- **Technocratic Centralization over Subsidiarity: Solid Waste Management Rules, 2026** face strong criticism for imposing identical, compliance-heavy digital dashboards on structurally underfunded rural Gram Panchayats and metropolises alike.
- **Fragmented Institutional Accountability:** Severe coordination gaps persist among the Central Pollution Control Board (CPCB), State Boards, and municipal line-departments.
- **Lax EPR Enforcement:** Partial implementation of Extended Producer Responsibility (EPR) mandates allows manufacturers to bypass end-of-life lifecycle liabilities.

> As per CPCB, Extended Producer Responsibility net has expanded, yet dynamic baseline auditing shows a massive **35% to 40% lag between digital registrations and ground-level physical plastic waste recycling collection.**

- **Downstream & Ecological Fallout**
  - **Legacy Dumpsite Crisis:** Thousands of hectares of high-value urban land remain locked under expanding mountain landfills, continuously emitting volatile **methane** gas.
  - **Severe Air & Water Pollution:** Widespread open burning of unsegregated refuse drives hazardous seasonal smog, while toxic runoff aggressively degrades local water bodies.
  - **Monsoon Vulnerability:** Dense plastic litter continuously clogs city drains, instantly transforming mild seasonal rainfall into disruptive urban flash floods.

> As per NDMA, Plastic debris blocks urban waterways, lowering the discharge capacity of stormwater drainage networks **by up to 50% to 60%.**

Impact of poor waste management in India

- **Environmental & Climate Impact**
  - **Methane Footprint:** Decomposing landfill mass accelerates toxic greenhouse gas emissions, directly derailing India's **2070 Net-Zero targets**.
  - **Leachate Contamination:** Untreated liquid runoff infiltrates ground aquifers, causing long-term chemical and heavy-metal water poisoning.
  - **Microplastic Crisis:** Open dumping fuels macro-plastic breakdown, choking aquatic food chains and causing irreversible soil toxification.
- **Economic & Financial Drain**
  - **Urban Land Lock:** Expanding legacy dumpsites permanently freeze thousands of hectares of high-value municipal real estate.
  - **Tourism Deterioration:** Visual pollution, persistent smog, and landscape degradation aggressively damage international and domestic tourism revenues.
  - **Logistical Friction:** Unsegregated plastic refuse systematically blocks urban drainage networks, triggering destructive **urban flash floods** (e.g., Chennai, Mumbai).
- **Public Health & Social Impact**
  - **Vector-Borne Epidemics:** Unmanaged organic waste creates massive breeding zones for vectors, driving endemic spikes in dengue, malaria, and cholera.
  - **Respiratory Illnesses:** Routine, unauthorized open burning of urban waste releases highly carcinogenic dioxins and furans, worsening respiratory crises.
  - **Generational Marginalization:** Entrenches extreme socio-economic vulnerability and health hazards for India's 1.5 million un-formalized waste pickers.

Measures taken for waste management in India

- **Overarching Flagship Schemes**
  - **Swachh Bharat Mission (SBM-Urban 2.0):** Drives the national mandate to transition all Indian cities into **Garbage-Free Cities** via intensive source segregation.

- **Dumpsite Remediation Accelerator Programme (DRAP):** Targets the core structural goal of **Lakshya: Zero Dumpsites**, prioritizing the accelerated bio-mining and reclamation of legacy urban wasteland.
- **AMRUT 2.0:** Scales up extensive municipal infrastructure funding to achieve **100% wastewater treatment** and safe septage recycling across statutory towns.

- **Technology & Bio-Energy Incentives**
  - **SATAT & GOBARdhan Scheme: I**ncentivizes commercial infrastructure to convert domestic agricultural, cattle, and organic urban waste into **Compressed Biogas** .
  - **National Critical Mineral Mission (NCMM):** Deploys a financial incentive scheme explicitly targeted at boosting secondary recycling capacities for **e-waste and spent lithium-ion batteries**.
  - **CITIIS 2.0:** Provides dedicated multi-sectoral funding to municipal bodies to design, integrate, and pilot highly localized **circular economy urban innovations**.
- **Structural & Market-Linked Reforms**
  - **Extended Producer Responsibility Portals:** Establishes a transparent, centralized online credit trading registry across the plastic, e-waste, battery, and tyre sectors.
  - **Industrial RDF Substitution Mandate:** Legally compels high-combustion industrial units (like cement plants) to steadily substitute a portion of their conventional fossil fuel intake with **Refuse-Derived Fuel**.
  - **National Clean Air Programme Synergy:** Mandates the setting up of specialized **Construction & Demolition (C&D) waste recycling facilities** in non-attainment cities to suppress ambient urban dust pollution.

Way forward

- **Upstream**
  - **Behavioral Nudge:** Scale **Mission LiFE** to institutionalize mandatory, household-level triple-stream source segregation.
  - **Circular Design:** Enforce strict “Design for Recyclability” to eliminate multi-layered, non-recyclable plastics at the blueprint stage.
- **Midstream**
  - **Smart Logistics:** Deploy IoT-enabled bins, GPS tracking, and blockchain-verified EPR portals to eliminate data leakages.
  - **Decentralized Processing:** Establish localized **Material Recovery Facilities (MRFs)** to drastically slash municipal transport costs.
- **Downstream**
  - **Bio-Methanation:** Expand anaerobic digestion under the **GOBARdhan** scheme to convert wet waste into Compressed Biogas (CBG).
  - **Bio-Mining Acceleration:** Scale mechanized bio-remediation across legacy dumpsites to systematically achieve **“Lakshya: Zero Dumpsites”**.
- **Socio-Economic & Governance**
  - **Formalization:** Integrate informal waste pickers into municipal frameworks as recognized, salaried **Environmental Service Providers**.
  - **Fiscal Autonomy:** Empower Urban Local Bodies (ULBs) by levying graded user fees and issuing market-linked **Green Bonds**.
  - **Paradigm Shift:** Transition the policy outlook from a disposal crisis to a **USD 45 billion circular economy opportunity** by 2030.

Global and Indian waste management models

- **Global Models**
  - **Sweden (Waste-to-Energy Mastery):** Achieved a near **Zero Landfill status (<1%)** by pioneering high-efficiency incineration that converts 50% of domestic waste into district heating and electricity.
  - **Japan (Kamikatsu Zero-Waste Model):** Implements meticulous, community-driven **45-category source segregation**, avoiding municipal incineration via absolute local upcycling.
  - **South Korea (Pay-As-You-Throw - PAYT):** Deploys volume-based food waste pricing using **RFID smart bins**, charging citizens by weight to behaviorally force waste minimization.
  - **Germany (The Green Dot / Duales System):** A pioneer in structural **Extended Producer Responsibility (EPR)**, forcing manufacturers to financially fund the recycling ecosystem of their own packaging.
- **Successful Indian Models**
  - **Indore Model (Biomethane & 100% Segregation):** India's cleanest city utilizes a strict **six-stream door-to-door source segregation** loop, feeding wet waste into Asia's largest Bio-CNG plant (Gobardhan).
  - **Ambikapur Model (Chhattisgarh's Decentralized Hub):** Uses a self-sustaining, women-led cooperative model that treats waste as a resource via **Material Recovery Facilities (MRFs)**, eliminating landfills entirely.
  - **Pune Model (SWaCH Cooperative Integration):** Formally integrates over 3,500 **informal waste pickers into a pro-poor municipal pro-rata collection system**, institutionalizing dignity of labor.
  - **Alappuzha Model (Kerala's Decentralized Composting):** Driven by the slogan *"Clean Home, Clean City"*, it completely bypassed centralized transport logistics via **mandatory household composting** and aerobic pipe units.
  - **Vellore Model (Tamil Nadu's Sol:** Utilizes decentralized Wealth Centers powered by self-help groups to turn organic waste into organic fertilizer via eco-friendly cattle-rearing integration.

Recommendations from committees on waste management

- **Asim Burman Committee (1999)**
  - **Door-to-Door Collection:** Recommended mandatory daily, door-to-door collection of municipal solid waste across all urban local bodies.
  - **Prohibition of Open Dumping:** Advocated for an absolute ban on open dumping, emphasizing the transition to scientifically engineered sanitary landfills.
- **Kasturirangan Committee**
  - **PPP in Waste-to-Energy:** Mandated Public-Private Partnerships to mobilize capital for high-efficiency Waste-to-Energy installations.
  - **Viability Gap Funding (VGF):** Recommended providing targeted VGF to make municipal waste-processing projects commercially attractive for private players.
  - **Refuse-Derived Fuel (RDF) Mandate:** Advised making it legally obligatory for heavy industries (like cement kilns) to substitute coal with high-calorific RDF.

Conclusion

Effective waste management is no longer just a sanitation challenge, but a vital cornerstone of India's sustainable urban future and climate resilience. Transitioning from passive dumping to a true circular economy requires strict source segregation, formalizing waste pickers, and empowering local bodies with green financing.

10. SOLID WASTE MANAGEMENT IN INDIA

The **Solid Waste Management in India** remains an Achilles heel for India. The Supreme Court of India **recently criticized the solid waste management in New Delhi.** There are more than **3,800 tones of untreated solid waste** in Delhi alone. This waste reaches landfills and threatens public health and the environment. Proper steps need to be undertaken for safe disposal and treatment of solid waste in India.

Solid Waste

- **Solid Waste:** Solid waste refers to any **unwanted or discarded material** that is not in a liquid or gaseous state. The solid waste includes a **wide range of materials** generated from various sources such as **households, industries, commercial establishments, construction sites**, and **institutions**.
- **Types of Solid Waste:**
  - **Municipal Solid Waste (MSW)** – Household, commercial, market waste.
  - **Biomedical Waste** – Hospitals, clinics (requires special handling).
  - **Electronic Waste (E-waste)** – Phones, laptops, appliances.
  - **Construction & Demolition (C&D) Waste** – Debris, bricks, tiles.
  - **Industrial Waste** – By-products from factories, often hazardous.
  - **Plastic Waste** – Single-use plastics, packaging material.

Status of Solid Waste generation in India

Status of Solid Waste Generation in India:

- According to a study published in 'Nature' - India is the biggest plastic polluter in the world - releasing 9.3mT of plastic waste annually - which is equivalent to around 20% of global plastic emission.
- According to CPCB report, only ~50% of total solid waste generated in the country is treated. The processing of solid waste in India has improved significantly, from **19% in 2015-16** to **~50% in 2020-21**. In the corresponding period, the proportion of solid waste landfilled has **fallen from 54% to 18.4%**.

The total quantity of Solid waste generated in India per Day	~**1,60,000** Metric Tonnes Per Day (TPD)
Waste Collection per day	~**1,53,000** Metric Tonnes Per Day (TPD) Waste Collection efficiency is ~**96%**.
Waste treatment per day	~**80,000** Metric Tonnes Per Day (TPD) Only **50%** of the total waste is treated.
Waste Landfilled per day	~**30,000** Metric Tonnes Per Day (TPD) **18.4%** of the total waste generated ends in landfill.
Unaccounted Waste Generation	~**50,000** Metric Tonnes Per Day (TPD) **31.2%** of the total waste generated remains unaccounted.

About 50-55% of the waste generated in Indian cities is biodegradable wet waste, about 35% is non-biodegradable wet waste and 10% is an inert component.

Challenges with Solid Waste Management in India

- **Rising Waste Generation:** Rapid economic growth **has raised the consumption levels in the economy**, which has in turn increased the waste generation. Further, **the expansion of digital**

economy is leading to a **multifold increase in e-waste generation**. Rising plastic waste generation in eco-sensitive regions like Himalayas are choking the fragile ecosystems present there. For ex- A Planning Commission Report had estimated that India will generate 165 million tonnes by 2030.

- **Lack of proper Waste Management:** India lacks proper waste management and disposal techniques.
  - **Inadequate Infrastructure:** Many urban and rural areas lack proper infrastructure for waste collection, segregation, transportation, processing, and disposal.
  - **Collection & Treatment:** While collection rates are improving, a significant portion of the generated waste remains uncollected.
  - **Poor Processing:** Only 50% of the waste produced is actually processed in India. **~30% of waste is not accounted** and **~20% ends up in landfills**, reflecting poor waste disposal method.
  - **Incorrect and Inadequate Segregation Techniques:** There is poor segregation of waste at source. Hazardous waste and e-waste is not sealed and labelled leading to improper disposal. For ex- Valuable materials like aluminum and plastics end up in landfills instead of being recycled.
  - **Reuse/recycling of waste:** Reuse and recycling of waste is predominantly an informal economy, lacking access to advanced technology.
  - **Financial Constraints:** Local municipal bodies often face budget limitations hindering investments in modern waste management systems.
- **Littering and Illegal Dumping:** Due to poor disposal methods, **almost half of waste is placed in uncontrolled dumps and landfills**. A substantial amount of untreated waste, approximately 24%, ends up in landfills, many of which are unscientific and overflowing. These landfills are the **source of generation of methane gases, leachates**, and **landfill fires**, adversely affecting the surrounding environment
- **Lack of land resources:** The urban areas in India lack adequate land resources to set up waste processing plants. For ex- Waste processing plants in Delhi need large land parcels, of about 30-40 acres each for treatment.
- **Lack of public awareness:** Lack of **public awareness regarding proper waste management practices**, contributes to littering and improper disposal habits.
- **Lack of regular waste collection services:** The **lack of regular waste collection services** adds up to the **building up of waste as well as littering**. Illegal dumping in open areas and waterbodies increases the pressure on the municipal body, warranting more resources for clean-up.
- **Lack of proper data:** Lack of data regarding the quantity & quality of waste generated & processed in India is a major roadblock in its management. The data regarding the rate of waste generation in India is under-estimated & of waste collection is over-estimated.
  - **For e.g.** according to the official estimates, the plastic waste generation rate in India is 0.12 kg/capita/day, while according to the study published in 'Nature' , it is as high as 0.54 kg/capita/day. The agencies in India claim to collect 95% of the waste generated, however, these official statistics do not include rural areas, open burning of uncollected waste or the waste recycled by the informal sector.
- **Informal Sector:** The informal sector, consisting of ragpickers and recyclers, plays a crucial role in managing and extracting value from waste, though often under hazardous conditions.
- **Waste Composition:** A large percentage of Indian waste is organic, offering potential for composting and bio-methanation. However, the increasing proportion of non-biodegradable waste like plastics and e-waste presents management challenges.

Harmful impacts of poor Waste Management

- **Health Issues:** The improper waste management leads to several health issues such as:
  - Open burning of waste leads to formation of harmful particles which can cause **lung diseases**.
  - Poor collection of solid waste leads to garbage dumps which act as **breeding ground for rats and mosquitoes** etc. Mosquitoes act as carriers of diseases like malaria and dengue.

- **Environmental Issues:** Improper waste management techniques lead to various environmental problems such as:
  - **Unscientific dumping in landfill** leads to **formation of harmful chemicals** which permeate into soil and groundwater. This renders groundwater unfit for drinking and cause multiple diseases
  - Waste in landfills leads to formation of harmful gases leading to air pollution. For ex- Around 90-98% of landfill gases are made up of methane and carbon dioxide, remaining 2-10% includes nitrogen, oxygen, ammonia, sulphides, hydrogen and various other gases.
  - A lot of land-based waste eventually **ends up in sea leading to marine pollution.**
- **Economic Impacts:** Improper waste management usually has grave economic impacts such as:
  - **Expansion of landfills occupy useful land**, leading to wasteful utilization of an economic resource.
  - Poor waste collection leads to clogging of drains, which has become a factor in urban flooding, leading to economic losses.
  - **Poor waste management leads to general filth in cities**, which impacts tourism potential.

Government interventions for Solid Waste Management

Policy and Legal Framework for Waste Management in India	The Government of India (GoI) has formulated various Rules and Regulations. These rules are updated periodically and have been formulated under the **Environment Protection Act, 1986**. These include: a. Solid Waste Management Rules b. e-Waste Management Rules c. Plastic Waste Management Rules
Extended Producer Responsibility (EPR) Mechanism	EPR is a policy approach in waste management that **makes producers responsible for the entire lifecycle of their products**, including their collection, recycling, and disposal. In 2022, EPR initiatives utilizing market mechanisms were implemented for plastic packaging, **E-waste, battery waste**, and **used oil**.
Swachh Bharat Mission for Solid Waste Management	Central assistance is provided under Swachh Bharat Mission for solid waste management, including plastic waste management in urban and rural areas.
Compost Banao, Compost Apnao Campaign	It is a multi-media campaign launched by MoHUA on waste-to-compost under SBM-(U). The aim is to **encourage people to convert their kitchen waste into compost to be used as fertilizer** and to reduce the amount of waste getting to landfill sites.
Promotion of Waste to Energy	The **Ministry of New and Renewable Energy (MNRE)** launched Program on Energy from Urban, Industrial, Agricultural waste/residues and Municipal Solid Waste to promote setting up of **Waste-to-Energy projects and to provide central financial assistance**.
GOBAR-Dhan Scheme	This scheme promotes the conversion of cattle dung and organic farm waste into biogas and organic compost in rural areas.
National Action Plan for Municipal Solid Waste Management	This plan by the Central Pollution Control Board (CPCB) outlines strategies for waste minimization, utilization, recycling, processing, and environmentally sound disposal.

Mission LiFE	Mission LiFE (Lifestyle for Environment) is an India-led global mass movement to encourage individuals and communities to adopt sustainable, climate-friendly lifestyles and reduce mindless consumption. Mission LiFE actions are organised around themes such as energy conservation, water saving, waste reduction, sustainable food systems, reduced single-use plastics, healthy lifestyles, and e-waste management.

Way Forward

- **Scientific Waste Management:** The waste management planning should be based on **sound scientific and engineering studies**. They should consider **waste composition**, **capital and long-term operating costs, transport distances**, and the **geographical location of waste processing and disposal facilities.**
- **Smart Waste Management System:** In the long term, technology like (Internet of Things) can be integrated into waste management. For ex- RFID-enabled door-to-door waste collection monitoring can enhance collection efficiency and GPS based vehicle tracking can help in real time monitoring.
- **Emphasis on Recycling, Resource recovery & Processing:** Policies supporting recycling and recovery of resources from waste must be implemented stringently. Waste processing methods like **composting, vermicomposting and bio-methanation** should be adopted for treating organic waste. Establish efficient material recovery facilities (MRFs) and support the formalization of the recycling sector.
- **Scaling up Waste-to-Energy:** Bio-methanation (anaerobic digestion) which uses microorganisms to convert the organic waste into methane, can be used as fuel. **Bio-methanation plants** should be scaled up. Also, **Refuse-Derived Fuel (RDF) which consists of plastics, paper, and textile waste**, having good calorific value, can be used to generate power in waste-to-energy projects.
- **Polluter Pays Principle:**
  - Waste Management Rules which have incorporated '**Polluter Pays Principle**', need to be stringently implemented to penalize non-compliance.
  - Polluter pays principle casts absolute liability on the polluter for the harm caused to the environment & extends not only to compensate the victims of pollution but also the cost of restoring environmental degradation.
  - While the liability is clear under this principle, but the process of determining an equitable compensation is difficult as it must account for both tangible & intangible damages inflicted on environment & the affected communities. To overcome this, the Courts have modified the principle into **'Government Pays Principle'** under which it is the government which has to pay the compensation to the affected individuals & recover the same from the polluters, until the damage caused to the ecology is fully reversed.
- **Increasing Public Awareness: Self- help groups, residents' welfare associations, and community-based organizations** should be encouraged to educate and acquaint people with beneficial waste management strategies, including separation, recycling modes, and drop off centers for recyclables, as well as composting.
- **Data collection:** There is an urgent need to collect & provide reliable data about waste generation & its composition in the country for its effective management. We need to know how much of the waste is being generated, where & how it is being managed for finding an effective solution. We also need to have data regarding the infrastructure that has been built over the years for waste management & such infrastructure needs to be geotagged to help in proper disposal of waste.
- **Extended Producer Responsibility (EPR):** To effectively operationalize the EPR, the producers, importers & and brand owners that have a legal obligation to collect the waste, can collectively form kiosks across the country to collect the waste from local bodies - so that all the waste that is covered under EPR can be deposited & effectively managed.
- **Circular Economy Model:** The circular economy model underlines waste as a resource. India needs

to move away from a linear to circular mode of waste management - with the twin objectives of minimizing waste & recovering energy & other resources.

Conclusion

According to the SC of India, environmental protection is not only a regulatory obligation but also a constitutional imperative which aims to safeguard the fundamental rights of the individuals & preserve the ecological balance. Thus, it is the right time to hold the waste management system in the country accountable to the people whose health is impacted by the land, water & air pollution caused by unmanaged & mismanaged waste all across the country.

11. E- WASTE MANAGEMENT IN INDIA

As one of the world's fastest-growing digital economies, India faces a massive challenge with rapidly accumulating electronic waste, or e-waste. This crisis is fueled by short device lifespans and low public awareness, with over 90% of discarded electronics ending up in the hazardous, informal recycling sector.

E-Waste Volumes and Processing in India

- **Annual Waste Quantum:** According to the **Central Pollution Control Board (CPCB)**, India's baseline generation reached **13.9 lakh tonnes** annually, making it one of the largest global accumulators of electronic scrap.
- **Structural Horizon Tipping Point:** As per **NITI Aayog**, India's electronic waste is projected to more than double, escalating from a baseline of 6.19 million metric tonnes (MMT) to over **14 MMT by 2030.**
- **Embedded Resource Valuation:** According to **NITI Aayog's 2026 Circular Economy reports**, the total economic value embedded inside India's annual e-waste is estimated at **₹51,000 crore**, with nearly 33% composed of critical precious metals and minerals.
- **Component-Wise Generation Weight:** As per **CPCB analysis**, obsolete computer equipment accounts for the primary structural chunk at **68% of the total volume**, followed by telecommunication devices (12%) and traditional electrical appliances (8%).
- **EPR Target Progression:** Under the **E-Waste (Management) Rules**, the Extended Producer Responsibility (EPR) mandatory collection target has been aggressively scaled up to **70% of historical sales volume**, moving toward an 80% cap.
- **Registered Recycler Footprint:** According to official **CPCB EPR Portal statistics**, there are **386 registered e-waste recyclers** operational across the country, boasting a total institutional processing capacity of 34.65 lakh metric tonnes per annum.
- **Emerging Critical Waste Streams:** As per the **Ministry of Environment, Forest and Climate Change (MoEFCC)**, the inclusion of solar PV panels and Lithium-ion batteries has expanded the compliance net, as EV battery demand is projected to surge from 29 GWh to **248 GWh**.

Primary drivers of e-waste generation in India

- **Technological & Market Drivers**
  - **Planned Obsolescence:** Manufacturers intentionally design hardware with limited lifespans and restrict software updates, forcing premature upgrades.
  - **Rapid Technological Innovations:** Fast-paced rollouts of 5G infrastructure, AI-driven processors, and smarter consumer tech render older devices obsolete overnight.

    As per the **Ministry of Electronics and Information Technology (MeitY)**, exponential rollout of 5G infrastructure has accelerated premature retirement of nearly 110 million 4G handsets.
  - **Falling Electronics Costs:** Aggressive market competition and mass production have made buying new electronic gadgets cheaper than repairing old ones.

- **Socio-Economic & Consumer Drivers**
  - **Explosive Digital Adoption:** India's massive digital economy push has drastically increased the household ownership of smartphones, laptops, and smart appliances.

    According to the **Telecom Regulatory Authority of India (TRAI)**, India's active mobile telephone subscription base has breached **1.19 billion.**
  - **Rising Disposable Income:** A growing middle class fueled by corporate expansion drives a high-consumption lifestyle, accelerating device replacement cycles.

    As per the **Ministry of Statistics and Programme Implementation (MoSPI)**, a rise in urban per capita disposable income has led the average smartphone replacement cycle down to just **18 to 24 months**.
  - **Right to Repair Deficit:** Monopolistic repair ecosystems and non-availability of authentic spare parts compel consumers to discard fixable items.
- **Institutional & Regulatory Gaps**
  - **Weak Take-Back Networks:** Formal collection centers are deeply inadequate, leaving consumers with no accessible options to return end-of-life electronics.

    According to a national **CPCB Audit**, less than 15% of designated producers have operational, consumer-accessible physical collection centers
  - **Flawed Lifecycle Tracking:** Poor baseline data collection and a lack of real-time inventory tracking allow massive volumes of industrial e-waste to slip under the radar.

Impact of E- Waste in India

- **Environmental & Ecological Degradation**
  - **Heavy Metal Poisoning:** Toxic leakages of lead, mercury, cadmium, and arsenic permanently contaminate groundwater aquifers and local food chains.
  - **Persistent Air Pollution:** Informal open-air burning of circuit boards and plastic casings releases highly carcinogenic dioxins, furans, and fine particulate matter PM2.5.
  - **Soil Toxification:** Unscientific dumping destroys soil microbiomes, rendering land surrounding informal processing clusters completely infertile.
- **Public & Occupational Health Hazards**
  - **Severe Organ Damage:** Chronic exposure to e-waste toxins causes neurological disorders, kidney failures, and respiratory damage among informal handlers.

    According to the **World Health Organization (WHO)**, chronic occupational exposure to electronic waste recyclers releases up to 1,000 harmful chemical substances.
  - **Maternal & Child Vulnerability:** High concentration of lead in recycling hubs causes congenital birth defects, premature births, and impaired neurodevelopment in children.
  - **Lack of Safety Gear:** Over 90% of workers dismantle hazardous electronics with bare hands, leading to direct chemical burns and chronic occupational illness.

    According to **NITI Aayog**, over 90% of electronic waste dismantling is managed by the unregulated informal sector.
- **Economic & Resource Deprivation**
  - **Critical Resource Loss:** Dumping or crude burning destroys rare precious metals like gold, silver, and palladium instead of capturing them through urban mining.

    According to **NITI Aayog's Circular Economy Reports**, India loses roughly **₹21,250 crore** worth of valuable secondary metals annually due to primitive sorting.

Escalating E-Waste Import Burden: Low domestic recovery forces India to rely heavily on volatile, carbon-intensive imports for primary electronic raw materials.

- **Municipal Fiscal Strain:** Inefficient e-waste disposal clogs municipal landfill sites, driving up local administrative and environmental cleanup costs.

Constituents of e-waste

Source	E-Waste Component	Envi Hazard	Effects on Human
Desktops / CRT Monitors	Lead (solder, glass) Sulphuric Acid (lead-acid batteries)	Soil contamination Groundwater pollution	Nervous system damage Developmental issues in children
Laptops / LCD Backlights	Mercury (switches, backlights)	Air pollution (when incinerated) Accumulates in water bodies	Chronic kidney damage Neurological disorders
Mobile Phones / Rechargeable Devices	Cadmium (contacts, chip resistors, batteries)	Bioaccumulation in wildlife and crops Soil toxicity	Severe respiratory problems Bone fragility and kidney disease
Circuit Boards (PCBs)	Beryllium (motherboards, connectors)	Plant damage and growth inhibition Persistent soil toxicity	Skin rashes and warts Berylliosis (chronic lung disease)
Televisions / Plastic Castings	Brominated Flame Retardants (BFRs in plastics/cases)	Leaches into air and dust Highly persistent environmental pollutant	Potential carcinogen Endocrine and hormone disruption

Regulatory framework governing e-waste in India

- **Legal framework**
  - **Environment (Protection) Act, 1986:** Serves as the primary parent legislation empowering the central government to enact specific electronic waste mandates.
- **Institutional Enforcement Matrix**
  - **Central Pollution Control Board (CPCB):** Functions as the national regulatory anchor, auditing material flows and overseeing the EPR certificate portal.
  - **State Pollution Control Boards (SPCBs):** Conducts mandatory physical inspections and issues the necessary **Consent to Operate (CTO)** licenses based on verified factory machinery capacity.
  - **Bulk Consumer Obligations:** Legally binds large entities (corporates running over 250 devices) to channel end-of-life tech exclusively to authorized entities and file mandatory annual logs by **June 30th**.
  - **Environmental Compensation Regime:** Establishes a strict Polluter Pays penalty system, levying heavy statutory fines for non-fulfillment of annual recycling targets.

E-Waste (Management) Rules, 2016

- **Extended Scope:** Brought manufacturer, dealer, refurbisher, and Producer Responsibility Organisation (PRO) under the ambit of the rules to eliminate regulatory loopholes.
- **Extended Producer Responsibility (EPR):** Mandated a single EPR authorization for producers nationwide, setting targets to collect and channelize a specific percentage of their end-of-life products.
- **Collection Mechanism:** Shifted the focus from establishing collection centers to a more flexible choice of collection mechanisms, including take-back systems and deposit-refund schemes.
- **Simplification of Authorization:** Centralized and streamlined the process by empowering the Central Pollution Control Board (CPCB) to grant pan-India EPR authorizations instead of individual State PCBs.
- **Restriction of Hazardous Substances (RoH&S):** Mandated that all new electrical and electronic equipment must comply with the permissible threshold limits for hazardous constituents like lead, mercury, and cadmium.
- **Integration of Informal Sector:** Provided clear provisions for the state governments to register, recognize, and skill informal waste pickers to formally integrate them into the recycling chain.
- **Liability & Penalty Clause:** Introduced financial penalties and liability clauses for environmental damage caused due to improper handling, transport, or disposal of e-waste.

Major initiatives for e-waste management in India

- **Core Digital & Regulatory Initiatives**
  - **Centralized E-Waste EPR Portal:** Developed a single-window digital platform by the CPCB for mandatory registration of producers, manufacturers, refurbishers, and recyclers.
  - **Tradeable EPR Certificate Market:** Launched a formalized online credit trading exchange where registered recyclers generate and sell EPR certificates to producers to fulfill compliance quotas.
  - **Common EPR Single Sign-On (SSO):** Recent system upgrades integrated multi-stream waste platforms into a unified tracking stack to prevent data manipulation and double-counting of credits.
- **Informal Sector Integration & Infrastructure**
  - **MeitY Capacity Building Scheme:** An initiative by the Ministry of Electronics and IT to upgrade informal waste pickers into the formal economy by forming dedicated recycling clusters under MSME frameworks.
  - **National Critical Mineral Mission:** Deploys targeted fiscal incentives to scale up domestic recycling plants for extracting valuable critical minerals (like cobalt, lithium, and rare earths) from e-waste.
  - **Authorized Collection & Take-Back Channels:** Mandated electronics brands to establish transparent toll-free helplines, buy-back deposit schemes, and designated drop-off points nationwide.
- **Enforcement & Monitoring Frameworks**
  - **Strict RoHS Testing:** Implemented rigorous laboratory audits to certify that new electrical and electronic equipment complies with reduction limits for hazardous heavy metals.
  - **Mandatory SPCBs Joint Action Plan:** Compels State Pollution Control Boards to conduct periodic enforcement drives to track material flows and shut down toxic, unauthorized open-air breaking yards.
  - **Environmental Compensation Matrix:** Codified a strict Polluter Pays financial penalty mechanism, imposing direct fines on bulk consumers and brands that fail to file their annual electronic disposal logs.

Way forward

- **Upstream & Product Design**
  - **Mandating Green Electronics:** Enforce eco-design and Design for Disassembly parameters to phase out hazardous heavy metals and non-recyclable polymers early.
  - **Right to Repair Legislation:** Codify a legally binding Right to Repair framework to expand component accessibility and combat planned obsolescence.
- **Regulatory Compliance**
  - **Cross-Portal API Integration:** Seamlessly link the CPCB e-waste portal with GST networks and Customs (ICEGATE) to curb fraudulent data trails and fake EPR certificate trading.
  - **Granular EPR Pricing Tariffs:** Introduce tier-based EPR credit pricing to incentivize the recycling of low-value, toxic chemistries like lithium-iron-phosphate (LFP).
- **Technology & Infrastructure Upgradation**
  - **Transitioning to Advanced Metallurgy:** Shift infrastructure funding toward automated material recovery units to capture high-value elements like Palladium, rather than just copper.
  - **Setting up Strategic E-Waste Parks:** Establish certified, specialized recycling clusters equipped with Zero-Liquid Discharge tech across major IT corridors.
- **Socio-Economic Integration & Market Reforms**
  - **Formalizing 'E-Waste Heroes':** Train, certify, and absorb informal waste pickers into urban collection channels under organized MSME or Self-Help Group frameworks.
  - **Deposit Refund Schemes:** Introduce clear commercial buy-back models and financial incentives to encourage consumers to safely surrender end-of-life devices.
  - **Promoting Urban Mining:** Shift the national policy outlook to treat electronic scrap as a **strategic domestic raw material resource**, reducing import reliance for critical energy-transition minerals.

E-waste management models

- **Switzerland (SWICO Model):** Operates an industry-driven, voluntary model funded via an **Advance Recycling Fee** charged on new electronics, ensuring completely free and sustainable end-of-life disposal for consumers.
- **Japan (Home Appliance Recycling Law):** Mandates a strict split-cost structure where consumers pay visible recycling/transport fees at the point of disposal, while retailers and manufacturers manage physical take-back logistics.
- **Germany (Clearing House Model):** Uses a centralized registry (*Stiftung EAR*) to coordinate between local municipal collection centers and manufacturers, allocating recycling duties based on a company's real-time market share.
- **South Korea (Eco-Assurance System):** Integrates product design with recycling, legally penalizing manufacturers if their electronics fail to meet strict recyclability and hazardous substance reduction targets.
- **Karo Sambhav Model:** An authorized Producer Responsibility Organisation (PRO) that utilizes an integrated app platform to provide transparent tracking, formalize small aggregators, and manage nationwide compliance for tech conglomerates.
- **E-Parisaraa Model:** India's pioneer formal recycling plant (Bengaluru) that demonstrates eco-friendly, zero-liquid-discharge manual dismantling paired with chemical recovery of precious metals.

Conclusion

Effective e-waste management is a strategic necessity for India to secure its technological future and achieve a sustainable circular economy. Moving from hazardous informal processing to organize **urban mining** is vital to recover rare critical minerals and eliminate toxic heavy metal pollution.

12. BIOMEDICAL WASTE MANAGEMENT

Biomedical waste includes all potentially infectious and hazardous materials generated during healthcare, research, and immunization activities. Biomedical waste creates severe risks of cross-contamination, environmental pollution, and accidental needle-stick injuries for frontline workers.

Causes of biomedical waste generation

- **Institutional & Policy Drivers**
  - **Infrastructure Expansion:** Rapid scaling of public healthcare access under schemes like Ayushman Bharat boosting hospital admissions.

    According to the **CPCB**, the exponential scaling of healthcare access has driven a generation of waste from 619 Tonnes/day to over 743 TPD within a five-year window
  - **Medical Tourism Influx:** Growth of corporate healthcare hubs attracting international patients for high-end complex surgeries.

    As per the **Ministry of Tourism**, India's corporate medical hubs handle more than 2 million international patient visits annually
- **Clinical & Technological Shifts**
  - **Single-Use Monopolization:** Mass adoption of disposable plastic equipment over reusables to eliminate hospital-acquired cross-infections.

    According to the **WHO**, non-hazardous general waste and single-use disposables accounting for up to 85% of a hospital's total daily waste footprint.
  - **Diagnostic Proliferation:** Massive expansion of standalone commercial pathology chains generating high volumes of liquid chemical waste.
  - **Home Healthcare Surge:** Proliferation of self-administered therapeutics (like insulin) injecting infectious syringes directly into municipal streams.

    According to **CPCB Guidelines**, the rise of home isolation and self-administered therapeutics injects approximately 100 Tonnes/day of infectious domestic bio-waste
- **Operational & Behavioral Gaps**
  - **Segregation Deficit:** Routine mixing of non-hazardous general ward waste with infectious bio-waste, multiplying the total toxic volume.

    According to a national **CPCB Waste Audit**, a systematic gap of 6% to 14% exists between the quantity of biomedical waste generated and what is safely treated.
  - **Pre-Treatment Shortage:** Near-total absence of localized autoclaves in tier-3 clinics, allowing active pathogens to leave medical facilities.

Regulatory framework governing biomedical waste in India

Primary Statutory & Core Legal Pillars	• **Environment Act, 1986:** Serves as the foundational umbrella legislation empowering the center to enforce medical waste rules. • **BMW Management Rules, 2016:** Replaced older 1998 norms to widen statutory coverage, simplify waste classes, and phase out chlorinated plastic bags.
Institutional Accountability & Compliance Mechanisms	• **Barcoded Tracking Mandate:** Enforces real-time GPS and barcode tracking logs on all waste bags to eliminate illegal dumping in municipal areas. • **State Pollution Control Boards:** Functions as the key monitoring authority responsible for granting **Consent to Operate** licenses.

	• **Common Bio-Medical Waste Treatment Facilities:** Shifts the treatment burden from individual clinics to centralized, audited regional processing hubs. • **Environmental Compensation Matrix:** Imposes a strict **Polluter Pays** regime with financial penalties and closure notices for non-compliant hospitals.

Coding Standards of biomedical waste

- **Yellow Category:** Mandates strict incineration or deep burial for highly infectious human tissues, anatomical waste, and soiled dressings.
- **Red Category:** Restricts to recyclable plastic items (tubing, bottles, syringes) destined for mandatory autoclaving and shredding.
- **White Category:** Dedicates exclusively to puncture-proof containers for metal sharps, needles, and blades to prevent stick injuries.
- **Blue Category**: Reserves for glass vials, ampoules, and broken metallic body implants requiring specialized sodium hypochlorite treatment.

Significance of biomedical waste management

- **Public Health & Epidemiological Security**
  - **Interrupting Disease Transmission:** Prevents the spread of highly infectious blood-borne pathogens like HIV, Hepatitis B, and Hepatitis C.
  - **Containing Hospital Acquired Infections:** Lowers cross-contamination rates within clinical wards, safeguarding vulnerable, immunocompromised patients.
  - **Curbing Anti-Microbial Resistance:** Prevents active antibiotic and pharma residues from entering local water tables and fueling superbug mutation.
- **Environmental & Ecological Protection**
  - **Preventing Aquifer Toxification:** Stops highly toxic heavy metals, chemical reagents, and cytotoxic drugs from leaching into groundwater.
  - **Reducing Carcinogenic Air Emissions:** Ensures controlled, high-temperature incineration to prevent the release of toxic dioxins and furans into urban airsheds.
  - **Ecosystem Preservation:** Protects local urban soil biomes and downstream aquatic life from enduring pharmaceutical pollution.
- **Occupational Safety & Socio-Economic Value**
  - **Protecting Frontline Sanitation Workers:** Minimizes accidental needle-stick injuries and hazardous chemical exposure for ragpickers and waste handlers.
  - **Eliminating Black-Market Rackets:** Strict barcoded tracking prevents the illegal washing, repackaging, and resale of used syringes and medical plastics.
  - **Unlocking High-Value Circularity:** Promotes the safe autoclaving and shredding of non-infectious medical plastics into high-grade recyclable raw materials.

Bio-Medical Waste Management Rules, 2016

- **Expanded Scope:** Includes temporary health camps, vaccination drives, blood donation drives, and surgical camps under its compliance umbrella.
- **Simplified Categorization:** Streamlines waste segregation into four distinct, color-coded bags: Yellow (infectious/anatomical), Red (recyclable contaminated plastics), White (sharps/needles), and Blue (glassware/implants).

- **Mandatory Pre-treatment:** Requires on-site sterilization or disinfection of microbiological waste, laboratory waste, and blood bags before final transport.
- **Chlorinated Plastic Phase-out:** Bans chlorinated plastic bags, gloves, and blood bags to stop toxic dioxin emissions during incineration.
- **Barcoding and GPS Tracking:** Mandates barcode systems and GPS tracking for waste bags to monitor transport and prevent illegal dumping.
- **Stricter Emission Standards:** Sets higher incinerator operational standards by lowering allowable particulate matter and raising secondary combustion chamber time.
- **Occupational Safety Focus:** Obligates healthcare operators to systematically provide safety training and mandatory immunizations (Hepatitis B and Tetanus) for waste handlers.
- **Enhanced Operator Accountability:** Increases the legal responsibility of Common Bio-medical Waste Treatment Facility (CBMWTF) operators to timely collect waste from small, remote clinics.

Measures taken for biomedical waste management in India

Technological & Digital Interventions	• **National Barcode Tracking System:** Mandated real-time tracking of bio-waste bags using barcodes and GPS to eliminate illegal municipal dumping. • **Online Compliance Monitoring:** Launched centralized SPCB portals for online submission of annual biomedical waste logs by healthcare facilities. • **Non-Burn Technology Deployment:** Promoted advanced eco-friendly alternatives like autoclaving, microwaving, and hydroclaving over traditional incineration.
Infrastructure & Structural Scaling	• **CBWTF Network Expansion:** Established hundreds of Common Bio-Medical Waste Treatment Facilities to offer centralized regional processing hubs. • **Phasing Out Chlorinated Plastics:** Enforced a strict nationwide ban on chlorinated plastic bags and gloves to prevent toxic dioxin emissions during burning. • **Universal Color-Coded Segregation:** Streamlined hospital waste management by implementing a standardized 4-color bin framework across all states.
Institutional Capacity & Enforcement	• **Mandatory Staff Training:** Integrated bio-waste segregation protocols into regular continuing medical education for nurses and sanitation staff. • **Strict Licensing Links:** Connected the issuance of hospital **Consent to Operate** certificates directly to verified bio-waste compliance. • **Joint Enforcement Drives:** Deployed joint task forces of Central and State Pollution Control Boards to conduct surprise audits and fine violators under the **Polluter Pays** principle.

Way Forward

- **Regulatory Policy & Governance Reforms**
  - **Decentralized Hub-and-Spoke Infrastructure:** Construct mini-CBWTFs (Centralized Treatment Facilities) in tier-3 cities and aspirational districts to bridge deep geographical processing gaps.

  - **Legalizing Home Healthcare Handlers:** Expand the statutory definition of the 2016 Rules to legally bind urban home-care services, pathology chains, and self-treatment waste into municipal streams.
  - **Strict Cross-Portal Data Integration:** Interlink the CPCB tracking application seamlessly with hospital diagnostic registration portals to arrest illegal black-market plastic leakage.
- **Upstream Innovations & Technology Upgradation**
  - **Transitioning to Green Alternative Logistics:** Shift processing priorities toward non-burn combination technologies (like hydroclaving and plasma pyrolysis) to completely avoid carcinogenic incinerator smoke.
  - **Mandating Bio-Degradable Medical Polymers:** Incentivize the indigenous manufacturing of non-toxic, starch-based compostable gloves and aprons to cut back on virgin medical plastics.
  - **AI-Powered Automated Segregation Systems:** Deploy machine-learning sorting systems at the hospital yard level to automatically filter hazardous mixed clinical waste into correct color bins.
- **Socio-Economic Integration & Capacity Building**
  - **Formalizing 'Green Sanitation' Cadres:** Design specialized certificate programs to upskill informal collectors into specialized, fully insured hazardous material handlers.
  - **Fiscal Incentives for Eco-Startups:** Provide tax credit links for domestic small enterprises that convert safely autoclaved medical plastic scrap into high-grade engineering materials.
  - **Mandatory Biosecurity Risk Auditing:** Incorporate rigorous bio-waste protocol training into the standard professional curriculum of all medical, nursing, and dental students nationwide.

Global Models

French Decentralized On-Site Model: Mandates large hospitals to install compact, automated on-site shredder-sterilizers to render infectious waste neutral before leaving premises.

German Dual-System (DSD) Model: Employs a strict commercial "Polluter Pays" model where healthcare facilities pay dynamic tariffs based on the precise volume and weight of hazardous waste generated.

American Cradle-to-Grave Track Model: Utilizes a highly regulated manifest system tracking medical waste from generation, through transport, to ultimate destruction to ensure absolute legal liability.

Japanese Manifest System for Infectious Waste: Relies on an electronic manifest framework where waste producers, transporters, and disposal firms must digitally log and verify processing status in real-time.

CBWTF Hub-and-Spoke Model: Centers regional compliance around a public-private partnership (PPP) where a single centralized facility processes hazardous waste collected from multiple small clinics within a **75-km** radius.

BIMAA' Digital Tracking Model: Implements a localized, app-based barcode tracking ecosystem that monitors real-time waste bag movement from clinical wards to regional treatment sites.

Kerala IMAGE IMA Model: A highly successful state-wide model fully owned and operated by the Indian Medical Association (IMA), ensuring self-sustaining, cooperative institutional bio-waste disposal.

Conclusion

Transitioning India's expanding healthcare landscape from a compliance-heavy reporting model to an efficient **Zero-Leakage Bio-Circular Economy** is critical to safeguard national epidemiology, mitigate AMR (Anti-Microbial Resistance) vectors.

13. HAZARDOUS WASTE MANAGEMENT

Hazardous waste comprises industrial, chemical, and manufacturing byproducts that exhibit toxic, ignitible, corrosive, or reactive characteristics that pose immediate environmental threats. Hazardous

substances leads to irreversible aquifer contamination, severe ecological degradation, and long-term public health crises through bioaccumulation.

Classification of Hazardous Waste

Sub-Type	Major Source	Impact and Management Protocol
Ignitable Waste	Flashpoint less than 60°C; easily catches fire.	High thermal destruction or energy recovery.
Corrosive Waste	Highly acidic or basic	Mandatory chemical neutralization before land disposal.
Reactive Waste	Unstable; reacts violently or explodes with water/heat.	Pre-treatment to chemically stabilize compounds.
Toxic Waste	Contains heavy metals (Lead, Mercury) or pesticides.	Secured Landfills (TSDF) with leachate collection.
Listed Waste	Explicitly cataloged industrial processes (e.g., oil sludge).	Strict statutory "cradle-to-grave" manifest tracking.
Universal Waste	Common household hazards (Batteries, e-waste, bulbs).	Streamlined collection channels for recycling.

Hazardous Waste Treatment & Disposal Methods

Functional Sub-Type	Operational Method	Primary Target Waste
Physical Treatment	Air Stripping & Carbon Adsorption	Extracts volatile organic compounds (VOCs) from industrial wastewater.
	Sedimentation & Flocculation	Aggregates and separates toxic heavy metal sludges from liquids.
Chemical Treatment	Acid-Base Neutralization	Adjusts extreme pH levels of corrosive chemicals to neutral status.
	Chemical Oxidation/Reduction	Converts toxic compounds (e.g., cyanide) into harmless, stable gases.
Thermal Treatment	High-Temp Incineration	Destroys complex organic hazards and medical toxins via combustion.
	Plasma Gasification	Uses high-energy plasma arcs to convert waste into clean syngas.
Biological Treatment	Bioremediation	Deploys microbes to break down surface oil spills and hydrocarbons.
	Phytoremediation	Uses engineered plants to absorb heavy metals from contaminated soil.
Ultimate Disposal	Secured Landfills (TSDFs)	Encapsulates non-biodegradable toxic ash using high-density liners.
	Deep Well Injection	Pumps liquid hazardous waste into deep, isolated geologic formations.

Significance of hazardous waste management

- **Ecological & Environmental Preservation**
  - **Preventing Aquifer Contamination:** Stops carcinogenic heavy metals and chemical leachates from infiltrating precious groundwater reserves.
  - **Preserving Soil Health:** Eliminates long-term chemical degradation of topsoil, protecting agricultural productivity and soil biomes.
  - **Combating Biomagnification:** Prevents persistent organic pollutants (POPs) from entering food chains and accumulating in apex predators.
- **Public Health & Epidemiological Safety**
  - **Reducing Carcinogenic Risks:** Minimizes community exposure to airborne toxins (like dioxins) that trigger respiratory and oncological illnesses.
  - **Preventing Neurological Damage:** Restricts the release of neurotoxic heavy metals (such as Lead and Mercury) near urban population centers.
  - **Ensuring Clean Water Supply:** Reduces the toxic chemical load on public water filtration infrastructure, preventing large-scale waterborne toxicity.
- **Economic & Industrial Sustainability**
  - **Promoting Urban Mining:** Enables the recycling and recovery of high-value rare earths and precious metals from industrial byproducts.
  - **Averting Environmental Litigation:** Protects industrial enterprises from facing heavy financial penalties under the judiciary's Polluter Pays doctrine.
  - **Boosting Green Export Potential:** Aligns Indian manufacturing with global ESG (Environmental, Social, and Governance) norms, easing access to Western markets.

Issues in hazardous waste management in India

- **Regulatory & Enforcement Gaps**
  - **Ineffective Manifest Tracking:** Widespread data gaps and falsification in tracking waste movement from factories to disposal sites.
  - **Porosity of SPCBs:** Severe manpower shortages and technical deficits within State Pollution Control Boards crippling routine physical audits.

    As per **Parliamentary Standing Committee Reports**, State Pollution Control Boards face structural staff shortages of up to 40% to 50% in technical cadres
  - **Lax Polluter Pays Implementation:** Low penalty rates failing to act as a financial deterrent against illegal, night-time dumping.
- **Infrastructure & Technical Deficits**
  - **Acute TSDF Shortage:** Highly uneven geographical distribution of Treatment, Storage, and Disposal Facilities, causing high interstate transport risks.

    According to the **CPCB**, the distribution of common Treatment, Storage, and Disposal Facilities (TSDFs) is highly uneven, with over 80% of hazardous waste generated in just 7 industrialized states.

    Legacy Waste Accumulation: Massive volumes of historic, unscientific toxic dumps lying untended near older industrial estates.
  - **Substandard Incineration Tech:** Continued reliance on low-temperature, inefficient burning that releases highly toxic dioxins and furans.

    As per CPCB, Nearly **40%** of generated toxic elements are left unmonitored or handled with insufficient thermal destruction metrics.
- **Socio-Economic & Supply-Chain Challenges**

- **Informal Sector Dominance:** Unregulated, illegal backyard processing of hazardous e-waste and lead-acid batteries by untrained workers.

  > According to **NITI Aayog Material Circularity Papers**, more than 90% of spent lead-acid batteries slip into unorganized backyard processing hubs.

- **NIMBY (Not In My Backyard) Syndrome:** Intense local community resistance against setting up new hazardous landfills or incinerators near residential zones.
- **Illegal Transboundary Smuggling:** Continued illicit entry of global toxic scrap masked as cheap secondary raw material imports.

Measures taken for hazardous waste management in India

- **Statutory & Policy Frameworks**
  - **HOWM Rules, 2016:** Standardized the core regulatory architecture for Hazardous and Other Wastes by linking management directly to the waste hierarchy.
  - **Non-Ferrous Metals EPR Amendment (2026):** Enforced target-based **Extended Producer Responsibility (EPR)** for recyclers and bulk consumers handling aluminum, copper, and zinc alloys.
  - **National Green Tribunal (NGT) Directives:** Strengthened active oversight by ordering immediate clean-ups of toxic industrial clusters and imposing heavy environmental compensation.
- **Infrastructure & Spatial Interventions**
  - **Centralized TSDF Network:** Established a nationwide grid of audited Treatment, Storage, and Disposal Facilities to securely consolidate heavy industrial residues.
  - **Co-Processing in Cement Kilns:** Promoted the utilization of high-calorific hazardous waste as alternative fuel in cement industries to bypass traditional landfilling.
  - **Dedicated Inter-State Transit Corridors:** Regulated bulk transit operations to manage the logistical flow of hazardous materials between processing-deficit states safely.
- **Digital & Enforcement Mechanisms**
  - **CPCB National EPR Portal:** Rolled out a centralized dashboard for the mandatory registration of producers and transparent trading of recycling certificates.
  - **Manifest-Based GPS Tracking:** Mandated real-time vehicle tracking logs to prevent late-night, illicit dumping of liquid chemical effluents in open fields.
  - **OCMMS Integration:** Unified State Pollution Control Boards under the **Online Consent Management & Monitoring System** to tie industrial license renewals strictly to verified waste disposal logs.

Way Forward

- **Regulatory & Governance Reforms**
  - **Transition to Blockchain Manifests:** Replace vulnerable paper manifests with tamper-proof blockchain ledgers to secure **cradle-to-grave** industrial waste logs.
  - **Third-Party Biosecurity Auditing:** Institutionalize mandatory, independent environmental audits for SPCBs to eliminate deep enforcement gaps and corruption.
  - **Empowering District Task Forces:** Decentralize punitive powers to district magistrates to execute immediate seizures of illegal night-dumping units.
- **Technical & Upstream Innovations**
  - **Mandating Green Chemistry Protocols:** Incentivize domestic chemical industries to adopt zero-waste atom-economy designs that eliminate toxic byproduct creation.
  - **Scaling Up Plasma Pyrolysis:** Provide capital subsidies to transition away from traditional incinerators to ultra-high-temperature, zero-emission plasma gasification.
  - **AI-Driven Leachate Monitoring:** Deploy smart internet-of-things (IoT) sensors around all operational TSDFs to detect groundwater toxic plumes in real-time.

- **Socio-Economic Integration & Financing**
  - **Formalizing Informal E-Waste Guilds:** Integrate backyard recyclers into authorized public-private partnership (PPP) collection hubs through targeted financial credits.
  - **Floating Sovereign Green Waste Bonds:** Raise dedicated long-term infrastructure capital specifically to finance the biological remediation of legacy toxic dumpsites.
  - **Implementing Dynamic Pigouvian Taxes:** Levying steep, volume-based hazard manufacturing fees to automatically cross-subsidize eco-friendly structural alternatives.

Models for hazardous waste management

- **US 'Superfund' (CERCLA) Model:** Employs a strict retroactive liability framework that forces polluting industries to finance the multi-million dollar clean-up of legacy toxic sites.
- **EU Circular Economy Package:** Enforces high land-disposal taxes coupled with strict statutory bans on sending organic hazardous waste to conventional landfills.
- **Swiss Closed-Loop Chemical Model:** Mandates chemical manufacturers to design processes where byproducts are instantly recycled as raw inputs within the same industrial cluster.
- **German Duale System Extended EPR:** Compiles a comprehensive liability matrix forcing producers to finance the absolute collection and eco-safe destruction of all industrial chemicals.
- **Industrial Symbiosis Model (Kalundborg):** Interlinks diverse factories so that the hazardous toxic byproduct of one industry becomes the cheap operational fuel or raw material for another.
- **Cement Kiln Co-Processing Model**: Utilizes the ultra-high temperatures of operational cement kilns to cleanly incinerate high-calorific toxic waste with zero residual ash.

Conclusion

Securing a robust hazardous waste management framework is vital to prevent **groundwater contamination** and achieve **sustainable development**. Transitioning from simple landfills toward **industrial symbiosis** and eco-friendly **co-processing** will eliminate toxic **legacy waste** accumulations. Ultimately, success demands strict enforcement of **Extended Producer Responsibility (EPR)** and real-time digital tracking to build a safe, **circular economy**.

MGP/BP/GS3/014

CLIMATE CHANGE

1. CLIMATE CHANGE AND GLOBAL WARMING

Climate change and global warming, driven primarily by anthropogenic greenhouse gas emissions, represent the defining ecological and existential crises of the 21st century. This rapid warming accelerates extreme weather events, shifts global climate zones, and threatens global food and water security across vulnerable populations.

About global warming

Global warming refers to the long-term, human-induced rise in Earth's average surface temperature, driven primarily by the excessive accumulation of atmospheric greenhouse gases. This thermal imbalance acts as the core driver behind broader global climate change, accelerating rapid glacial melt and triggering unprecedented extreme weather events.

Theories of climate change

Theory	Core Driving Mechanism
Astronomical / Milankovitch Cycles (M. Milanković)	Alterations in Earth's orbital shape (eccentricity), axial tilt (obliquity), and axial wobble (precession).
Atmospheric Composition (T.C. Chamberlin)	Fluctuations in gaseous thermal trapping capacities (CO_2, CH_4, water vapor) regulating the planetary greenhouse blanket.
Solar Output / Sunspot Theory	Periodic 11-year cycles of intense magnetic solar storms changing the total volume of electromagnetic energy emitted.
Volcanic Dust Veil Theory	High-intensity explosive eruptions injecting heavy SO_2 gas and fine ash aerosols straight into the stratosphere.
Continental Drift & Tectonics (A. Wegener)	Long-term physical displacement of massive landmasses modifying ancient global oceanic and atmospheric current patterns.
Oceanic Circulation / Conveyor Belt	Disruption or weakening of the deep global thermohaline conveyor belt driven by massive polar freshwater melt.

Causes of global warming

- **Anthropogenic Emissions**
  - **Fossil Fuel Combustion:** Burning coal, oil, and natural gas for electricity and transport accounts for the largest share of global CO_2 emissions.
  - **Deforestation and Land Clearing:** Cutting down tropical carbon sinks releases stored CO_2 and reduces the planet's capacity to absorb future emissions.

As per the **IPCC Working Group III**, net Land Use, Land-Use Change, and Forestry (LULUCF) generating roughly **5.9 GtCO2/year** due to tropical deforesting.

- **Methane-Intensive Agriculture:** Intensive paddy cultivation and enteric fermentation from livestock release vast volumes of potent short-lived CH4 gas.

  According to the **Global Methane Budget**, agriculture is contributing **145 million tonnes/year**, primarily driven by enteric fermentation 32%.

- **Industrial Chemical Releases:** Manufacturing processes emit high-warming synthetic fluorinated gases (HFCs, PFCs, SF6) into the atmosphere.

  As per **IPCC emissions data**, synthetic fluorinated gases have shown the highest relative growth among all greenhouse gases, surging by **254% since 1990**.

- **Terrestrial and Land-Use Feedback Loops**
  - **Rapid Urbanization (Heat Islands):** Replacing natural vegetation with concrete and asphalt traps thermal radiation, raising local and regional ambient temperatures.

    According to the **IPCC Synthesis Report**, share of global emissions directly attributable to urban areas has escalated to between **67%and 72%.**

  - **Permafrost Thawing:** Melting Arctic tundra creates a dangerous feedback loop by releasing gigatons of trapped methane and carbon dioxide back into the atmosphere.

    As per NOAA Climate Assessments, Arctic permafrost containing an estimated 1,400 to 1,600 gigatons of trapped carbon.

  - **Albedo Loss from Glacial Melt:** Shrinking ice sheets expose dark ocean waters and land masses, which absorb more solar heat instead of reflecting it.

    According to the **National Snow and Ice Data Center (NSIDC)**, the Arctic Ocean has lost nearly **40% of its summer sea ice extent** over the last four decade.

- **Natural Forcing Mechanisms**
  - **Milankovitch Cycles:** Long-term, cyclical shifts in Earth's orbital path and axial tilt naturally alter the distribution of solar energy reaching the planet.
  - **Volcanic Outgassing:** Major volcanic eruptions release episodic plumes of CO2, though their long-term warming effect is far outweighed by human activity.
  - **Solar Irradiance Variations:** Periodic fluctuations in solar energy output and sunspot activity slightly influence baseline planetary temperatures over multi-decade cycles.

Global Warming Potential

Greenhouse Gas	Atmospheric Lifetime	GWP (100-Year Horizon)	Primary Anthropogenic Source
Carbon Dioxide (CO2)	100–300 years	**1**	Fossil fuel combustion & deforestation.
Methane (CH4)	~12 years	**28–36**	Paddy farming, livestock, & landfills.
Nitrous Oxide (N2O)	~114 years	**265–298**	Chemical fertilizers & industrial processes.
Hydrofluorocarbons (HFCs)	1–270 years	**138–12,400**	Refrigeration, ACs, & foam blowing.

Perfluorocarbons (PFCs)	2,600–50,000 years	**6,500–9,200**	Aluminum smelting & semiconductor manufacturing.
Sulfur Hexafluoride (SF6)	3,200 years	**23,500**	Electrical transmission grid insulators.

Effects of global warming

- **Climatological & Meteorological Shifts**
  - **Frequency of Extreme Weather:** Triggers more intense heatwaves, unpredictable cloudbursts, and severe, frequent cyclonic storms.

    According to the **IPCC Sixth Assessment Report (AR6)**, human-induced warming has made hot extremes surging significantly over **80% of all land regions** since 1950.
  - **Altered Monsoon Patterns:** Induces spatial and temporal variations in Indian monsoons, causing a simultaneous increase in both floods and droughts.

    As per the **Ministry of Earth Sciences (MoES)**, India's surface temperature rise of **0.7^C (1901–2018)** has directly caused a concurrent increase in localized heavy rain-cloudburst.
  - **Jet Stream Disturbances:** Disrupts high-altitude wind patterns, leading to prolonged spells of freezing weather or intense heat blocks.

    According to **WMO Climate Assessments**, a disproportionate **3 times faster warming rate in the Arctic** relative to the global average has weakened the polar jet stream
- **Cryosphere & Marine Ecosystem Impacts**
  - **Accelerated Glacial Retreat:** Melts Himalayan glaciers rapidly, threatening the perennial water supply of major North Indian river basins.

    According to the ICIMOD Cryosphere Reports, glaciers across the Hindu Kush Himalaya (HKH) lost 12% of their total area and 9 % of ice reserves from 1990 to 2020.
  - **Thermal Sea Level Rise:** Expands ocean volumes and melts polar ice sheets, risking the inundation of low-lying coastal cities like Mumbai and Chennai.

    As per the India Meteorological Department (IMD), mean sea levels along the Indian coast are rising at an accelerated average rate of 1.3 mm/year.
  - **Marine Heatwaves & Bleaching:** Raises ocean temperatures, causing widespread coral bleaching in sensitive ecosystems like the Lakshadweep reefs.
- **Terrestrial Ecology & Biodiversity**
  - **Habitat Fragmentation:** Shifts ecological niches upward, forcing alpine and montane species toward higher altitudes where space is limited.
  - **Wildfire Escalation:** Creates prolonged dry, high-temperature windows that intensify forest fires across the Western Ghats and Himalayas.

    As per **Forest Survey of India (FSI) data**, high-temperature anomalies have placed more than **36% of India's total forest cover** under frequent forest fire vulnerabilities.
  - **Phenological Mismatch:** Disrupts synchronized natural cycles, such as the timing of plant flowering and the arrival of migratory birds.

    According to **WWF**, rising ambient spring temperatures have triggered a mismatch of **10 to 14 days** in the synchronized arrival dates of long-distance migratory birds

- **Socio-Economic & Geopolitical Vulnerabilities**
  - **Agricultural Yield Drops:** Lowers the output of climate-sensitive crops like wheat and rice due to heat stress and unpredictable rainfall.

    According to the **IARI**, an unmitigated temperature increase of 1^C is projected to cause an immediate structural reduction of **4% to 10% in national wheat yields**.
  - **Rising Climate Refugees:** Triggers large-scale internal and transboundary human migration from vulnerable coastal zones and delta regions.
  - **Public Health Emergencies:** Expands the geographical reach and breeding seasons of vector-borne vectors carrying diseases like Dengue and Malaria.

Polar Stratospheric Clouds (PSCs)

Polar Stratospheric Clouds (PSCs), also known as **nacreous** or **mother-of-pearl** clouds, occur high up in the stratosphere (about 15 to 25 km high) during the extreme cold of polar winters.

- **Stratospheric Cooling Paradox**
  - **Greenhouse trapping:** Greenhouse gases (like CO2 and methane) trap heat close to the Earth's surface in the lower atmosphere (troposphere).
  - **Stratospheric cooling:** Because this heat is trapped below, less of it escapes upward, which causes the upper layer—the stratosphere—to cool down significantly.
  - **More clouds form:** PSCs can only form when stratospheric temperatures drop below critical thresholds. Therefore, **global warming at the surface paradoxically increases the formation of PSCs in the upper atmosphere.**
- **Destruction of the Ozone Layer**
  - **Chemical breeding grounds:** PSCs provide ice crystal surfaces that convert harmless chlorine reservoir compounds (like hydrogen chloride and chlorine nitrate) into highly reactive, unstable chlorine gas.
  - **Springtime destruction:** When sunlight returns to the poles in spring, UV radiation splits this Cl2 into free chlorine radicals, which rapidly destroy ozone molecules.
  - **The impact:** A depleted ozone layer allows more ultraviolet radiation to reach the Earth's surface.
- **Alteration of Global Wind and Climate Patterns**
- **Southern Annular Mode (SAM):** The ozone hole created by PSCs cools the polar stratosphere even further. This temperature drop strengthens the polar vortex (the band of powerful winds circling Antarctica).
- **Jet stream shifting:** A stronger polar vortex pulls the Southern Hemisphere jet stream further south toward Antarctica.
- **Regional weather impacts:** This shift alters rainfall patterns and temperature distributions across Australia, New Zealand, Southern Africa, and South America—directly changing regional climates.

Measures taken to combat global warming

- **Global Legal Treaties & Multilateral Frameworks**
  - **Paris Agreement (2015):** Commits nations to holding global temperature rise well below 2 degree Celsius and pursuing efforts to limit it to 1.5 degree Celsius.
  - **Kigali Amendment:** Mandates a global phase-down of high-GWP Hydrofluorocarbons (HFCs) to avoid up to 0.5 degree Celsius of warming by 2100.
  - **UNFCCC COP Presidencies:** Operationalized the **Loss and Damage Fund** to financially assist vulnerable developing nations facing climate disasters.

- **India's Domestic Policy & Frameworks**
  - **Panchamrit Targets:** Pledged a non-fossil energy capacity of 500 GW and achieving structural **Net-Zero emissions by 2070**.
  - **NAPCC Architecture:** Guides domestic climate resilience through 8 core national missions, including the National Solar and Water Missions.
  - **Energy Conservation (Amendment) Act:** Institutionalized the framework for a domestic **Indian Carbon Market (ICM)** to trade carbon credits.
- **Sectoral & Operational Interventions**
  - **National Green Hydrogen Mission:** Aims to produce 5 MMT of green hydrogen per annum to decarbonize hard-to-abate heavy industries.
  - **PM-KUSUM Scheme:** Deploys solar-powered agricultural water pumps to phase out diesel-based irrigation on farms.
  - **FAME India Scheme:** Subsidizes the manufacturing and rapid adoption of electric vehicles (EVs) to decarbonize public and private transit.
- **Global Alliances Led by India**
  - **International Solar Alliance (ISA):** Interlinks solar-rich nations between the Tropics to mobilize over $1 trillion in solar energy investments.
  - **Coalition for Disaster Resilient Infrastructure (CDRI):** Promotes the upgrading of global infrastructure to withstand climate-induced extreme weather.
  - **Mission LiFE (Lifestyle for Environment):** Introduces a global mass movement shifting consumer behavior from mindless consumption to mindful utilization.

Way Forward

- **Energy Transition & Grid Decarbonization**
  - **Scaling Up Battery Energy Storage Systems (BESS):** Invest heavily in grid-scale battery storage to manage the structural intermittency of solar and wind power.
  - **Aggressive Phase-Down of Coal Dependency:** Establish a time-bound roadmap to retire older, sub-critical thermal power plants while halting new coal approvals.
  - **Accelerating Deep Decarbonization via Green Hydrogen:** Subsidize green hydrogen production to completely substitute fossil fuels in hard-to-abate sectors like steel, cement, and fertilizers.
- **Climate-Resilient Infrastructure & Adaptation**
  - **Mandating Climate-Proof Infrastructure Design:** Integrate multi-hazard climate risk assessments into the construction of all new highways, railways, and urban transit systems.
  - **Expanding Urban Sponge-City Infrastructure:** Develop nature-based urban wetlands and permeable pavements across cities to counter rapid urban heat island effects and flash floods.
  - **Upgrading to Hyper-Local Early Warning Systems:** Deploy AI-driven meteorological systems to deliver block-level, real-time alerts for cloudbursts, heatwaves, and cyclones.
- **Sustainable Land-Use & Carbon Sequestration**
  - **Transitioning to Climate-Smart Agriculture:** Incentivize direct-seeded rice (DSR) and precision farming to drastically drop agricultural methane and nitrous oxide emissions.
  - **Building Native Bio-Shields and Micro-Forests:** Scale up localized afforestation models to maximize natural carbon sinks and prevent coastal erosion.
  - **Enforcing Global Circular Economy Mandates:** Establish a robust statutory framework for complete material recycling to lower upstream industrial carbon footprints.
- **Climate Finance & Global Governance**

- **Operationalizing Substantive Sovereign Green Bonds:** Broaden domestic green bond markets to unlock low-cost, long-term international capital for clean-tech startups.
- **Enforcing Strict Global Carbon Pricing:** Transition from voluntary credits to a mandatory, high-tariff domestic carbon market integrated with global trading architectures.
- **Demanding Equitable Technology Transfer:** Use multilateral forums like COP to hold developed nations accountable for delivering predictable, grant-based climate finance to the Global South.

Conclusion

Effectively mitigating global warming is paramount to preventing irreversible **ecological tipping points** and securing **climate justice** for vulnerable populations. Transitioning from fossil fuel dependence toward **deep decarbonization** and scaling up nature-based **carbon sinks** will anchor India's target of achieving **Net-Zero by 2070**.

2. NET ZERO EMISSIONS

At the 2021 UN Climate Change Conference in Glasgow (COP26), 120 world leaders gathered to turn political will into real action based on climate science. The final agreement, known as the Glasgow Climate Pact, was the result of intense negotiations among nearly 200 countries working together for almost two years. At this historic summit, Prime Minister Narendra Modi announced India's ambitious five-point plan, the Panchamrit, to help limit global warming to 1.5 degree Celsius.

Net Zero Target

Net zero means cutting carbon emissions to a small number of residual emissions that can be absorbed and durably stored by nature and other carbon dioxide removal measures, leaving zero in the atmosphere.

Significance of Net-Zero for India

- **Geopolitical Leadership & Global Standing**
  - **Championing Climate Justice:** Solidifies India's role as the voice of the Global South by balancing aggressive local decarbonization with calls for equitable climate finance.
  - **Strategic Panchamrit Alignment:** Translates India's high-level Glasgow COP26 commitments into concrete domestic industrial and economic policies.
  - **Bypassing Unilateral Trade Barriers:** Safeguards export competitiveness against protectionist Western policies like the EU's Carbon Border Adjustment Mechanism (CBAM).
- **Macro-Economic & Energy Security**
  - **Slashing Fossil Fuel Import Bills:** Reduces fiscal deficits by curbing India's massive 85% dependency on foreign crude oil and coking coal imports.
  - **Spurring Green Capital Inflows:** Acts as a magnet for trillions in global green bonds, foreign direct investment (FDI), and private transition finance.
  - **Boosting Green Employment:** Creates millions of job opportunities in emerging frontier sectors like Green Hydrogen, solar manufacturing, and EV infrastructure.
- **Technological Transformation & Growth**
  - **Decarbonizing Hard-to-Abate Sectors:** Forces structural technology overhauls and efficiency gains in carbon-heavy industries like steel, cement, and chemicals.
  - **Modernizing the National Power Grid:** Drives rapid upgrades in grid resilience, AI-based energy load forecasting, and grid-scale battery storage infrastructure.
  - **Fostering Domestic R&D Ecosystems:** Incentivizes local innovation in critical technologies like Carbon Capture, Utilization, and Storage (CCUS).

- **Ecological & Socio-Economic Resilience**
  - **Protecting Vulnerable Communities:** Shields climate-exposed regions from catastrophic losses driven by flash floods, erratic monsoons, and extreme heatwaves.
  - **Enhancing Natural Carbon Sinks:** Accelerates large-scale afforestation and eco-restoration, which simultaneously restores degraded lands and protects biodiversity.
  - **Mitigating Severe Air Pollution:** Lowers public health expenditures by systematically cutting toxic industrial emissions and heavy particulate air pollution.

India's Climate Commitments (Panchamrit)

500 GW of non-fossil fuel electricity capacity by 2030.

- 50% of energy requirements from renewable sources by 2030.
- Reduction of 1 billion tonnes of projected carbon emissions by 2030.
- Reduction in emissions intensity of GDP **by 45% by 2030 compared to 2005 levels.**
- Achieve net-zero emissions by 2070.

Issues associated with India's Net-Zero transition

- **Financial & Investment Impediments**
  - **Astronomical Funding Bottlenecks: According to the Council on Energy, Environment and Water** India requires an estimated **$10.1 trillion** in total investments for 2070 Net-Zero target
  - **Prohibitive Green Capital Costs:** High interest rates and perceived investment risks inflate borrowing costs for domestic renewable energy developers.

    As per the RBI, High interest rates and perceived investment risks inflate borrowing costs, requiring an additional annual green investment equivalent to **2.5% of GDP** until 2030.
  - **Acute Climate Finance Deficits:** Developed nations continue to delay delivering the promised concessional grants and low-interest transition finance.

    According to the UNFCCC COP29 Baku Declaration, Developed nations have delayed delivering equitable grants, committing to a new target of mobilizing just **$300 billion** annually by 2035.
- **Energy Security & Infrastructure Vulnerabilities**
  - **High Baseline Coal Dependency:** Coal still generates over 70% of India's electricity, making a rapid phase-down highly risky for baseline grid stability.
  - **Severe Grid Integration Strains:** Upscaling intermittent solar and wind power creates immense voltage instability without expensive, grid-scale battery storage.

    As per the IEA, Upscaling solar and wind power creates immense voltage instability, requiring India to deploy **140 gigawatts (GW)** of grid-scale battery storage by 2040.
  - **Critical Mineral Supply Vulnerabilities:** India relies heavily on imports from monopolized supply chains for lithium, cobalt, and rare earth elements needed for EVs.

    According to the Ministry of Mines, India relies heavily on imports for **100%** of its lithium, cobalt, and nickel via monopolized Chinese supply chains.
- **Industrial & Macro-Economic Trade-Offs**
  - **Hard-to-Abate Sector Bottlenecks:** Heavy industries like steel, cement, and chemicals lack commercially viable, scaled alternatives to fossil fuels.

    According to NITI Aayog's Steel and Cement Decarbonization Blueprints, Heavy industries account for nearly **30%** of India's total industrial emissions.

- **Regressive Impact on MSMEs:** Small and medium enterprises lack the upfront capital and technical skills required to comply with green regulations.

  As per SIDBI, Indian MSMEs account for over **40%** of manufacturing emissions but require an unbacked **$50 billion** in green tech investments to comply with emerging carbon markets.

- **Severe Fiscal Revenue Losses:** State and central governments risk losing massive tax revenues currently generated from fossil fuel royalties and petroleum taxes.

- **Socio-Economic & Regional Disparities**
  - **Just Transition Livelihood Crisis:** Phasing out coal threatens to displace over 10 to 15 million workers in India's central and eastern coal-belt economies (e.g., Jharkhand, Odisha).
  - **Stranded Industrial Assets:** Indian banks face rising Non-Performing Assets (NPAs) as older coal power plants and carbon-heavy factories are prematurely shut down.

    As per the Reserve Bank of India (RBI), Indian banks face rising systemic risks and Non-Performing Assets (NPAs) as over **$25 billion** in older coal-fired power assets

  - **Land-Use and Agricultural Conflicts:** Setting up giant utility-scale solar parks requires vast tracts of land, creating direct conflicts with local farming communities.

Measures taken by India to achieve its 2070 Net-Zero emissions target

Renewable Energy & Grid Decarbonization	• **Scaling Up Non-Fossil Generation Capacity:** Reached over 260 GW of non-fossil power capacity (accounting for 52.5% of total capacity), meeting early NDC milestones years ahead of time. • **PM Surya Ghar:** Launched an expansive ₹75,021 crore solar rooftop subsidization drive targeting 1 crore low-and-middle-income households. • **Green Energy Corridor (GEC) Project:** Deploying dedicated synchronized transmission networks across 8 states to seamlessly integrate large-scale intermittent clean energy into the national grid. • **PM-KUSUM Scheme Expansion:** Solarizing millions of agricultural diesel water pumps to simultaneously de-carbonize farming and secure rural grid-independence.
Clean Mobility & Infrastructure Transitions	• **FAME & Electric Vehicle Promotion Policies:** Implementing multi-phase Faster Adoption and Manufacturing of Electric Vehicles (FAME) subsidies paired with targeted Production Linked Incentive (PLI) schemes for advanced chemistry cell (ACC) batteries. • **Indian Railways Mission 100% Electrification:** Rapidly transitioning the national rail backbone to hit net-zero operational emissions by 2030 through regenerative breaking tech and rooftop solar. • **National Biofuels Policy Targets:** Mandated aggressive blending thresholds, achieving a 20% ethanol-blended petrol milestone across major fuel outlets to curb reliance on fossil imports.
Industrial Transformation & Efficiency Market Mechanisms	• **National Green Hydrogen Mission:** Committed an initial ₹19,744 crore outlay to scale domestic green hydrogen production to 5 MMT annually, targeting heavy, hard-to-abate sectors.

	• **Carbon Credit Trading Scheme (CCTS):** Codified the regulatory structural framework under the Bureau of Energy Efficiency (BEE) to transition heavy industries from energy saving credits to absolute carbon value trading. • **Perform, Achieve, and Trade (PAT) Compliance:** Imposed legally binding energy intensity consumption reduction targets across 13 core carbon-intensive sectors.
Carbon Sinks & Agro-Ecological Conservation	• **Updated Nationally Determined Contributions (NDC):** Codified a binding intermediate target to create an additional carbon sink of 3.5 to 4.0 billion tonnes of CO2 equivalent via forest restoration. • **MISHTI Mangrove Initiative:** Committed institutional and corporate CSR funding to systematically restore vulnerable coastal mangrove ecosystems along India's coastline. • **LiFE (Lifestyle for Environment) Movement:** Mainstreamed a global public advocacy framework to nudge mass public consumer habits toward sustainable, circular resource utilization

Role of Carbon Sinks in achieving carbon net zero

- **Bridging the Residual Deficit:** Sinks neutralize the remaining **10% to 20% of stubborn baseline emissions** that cannot be technically eliminated by renewable energy or electrification.
- **Time-Buying Mechanism:** Scaling up Nature-based Solutions (NbS) offers an immediate, scalable mitigation potential of roughly **10 gigatonnes of CO2 per year** while long-term industrial green tech matures.
- **Reversing Historical Loads:** Beyond achieving net-zero, ultra-efficient sinks are the only mechanism available to eventually achieve *negative emissions*, drawing down legacy carbon accumulated since the Industrial Revolution.

Way Forward

- **Grid De-risking & Clean Baseload Management**
  - **Ramping Up Battery Energy Storage Systems (BESS):** Scale up pumped hydro storage and grid-scale lithium-ion battery banks to absorb intermittent solar and wind generation surges.
  - **Expanding Clean Firm Power via SMRs:** Accelerate the deployment of Small Modular Nuclear Reactors (SMRs) to create low-carbon, continuous baseload capacity as coal reliance scales down.
  - **Enhancing Thermal Grid Flexibility:** Incentivize the modernization of the existing coal fleet to undergo rapid cyclic loading and deep flexing to stabilize renewable input variations.
- **Hard-to-Abate Sector Interventions**
  - **Mandating Green Hydrogen Procurement:** Enforce minimum Green Hydrogen purchase obligations for the domestic fertilizer, oil refining, and petrochemical industrial sectors.
  - **Transitioning to Commercial Green Steel:** Incentivize integrated steel plants to shift away from blast furnaces toward scrap-based electric arc furnaces and hydrogen direct reduction (H-DRI) pathways.
  - **Codifying Lifecycle Assessment (LCA) Norms:** Introduce strict cradle-to-grave carbon-accounting standards across building materials to push construction toward low-clinker cement.
- **Transition Finance & MSME Support**
  - **De-risking MSME Capital Inflows:** Establish specialized credit-guarantee funds through SIDBI to lower the borrowing costs for micro-enterprises adopting energy-efficient technologies.

- **Broadening the Sovereign Green Bond Basket:** Expand green bond eligibility criteria to cover early-stage transition projects like Advanced Chemistry Cell (ACC) value chains.
- **Creating a Blended Finance Ecosystem:** Utilize public capital or multilateral grants as first-loss default guarantees to attract massive pools of commercial global private equity.

- **Supply Chain & Mineral Security**
  - **Securing Global Critical Mineral Corridors:** Deepen strategic overseas joint ventures through KABIL to secure sovereign long-term access to essential cobalt, lithium, and rare earth deposits.
  - **Institutionalizing Battery Recycling Rules:** Strictly enforce Extended Producer Responsibility (EPR) targets to build a local, circular supply chain for recovered battery materials.
  - **Developing Domestic Processing Clusters:** Setup local processing and refining hubs for strategic minerals to break deep import reliance on single-country monopolies.
- **Policy Governance & Forest Sinks**
  - **Integrating Absolute Caps into the CCTS:** Transition the newly launched compliance carbon market from simple energy-intensity targets to binding, absolute emission-cap reduction limits.
  - **Enforcing Blockchain-Based MRV Registries:** Deploy localized digital tracing systems to systematically monitor, report, and verify national carbon sink metrics without double counting.
  - **Updating the Sub-National Climate Action Plans:** Synchronize the State Action Plans on Climate Change (SAPCC) directly with the national 2035 target of reducing emission intensity by 47%.

Macro-Economic & Integrated Energy Assessment Models

- **NITI Aayog's IESS 2047 Model:** The India Energy Security Scenarios simulation tool that evaluates demand-supply pathways across electricity, transport, and industry sectors.
- **CSTEP's SAFARI Model:** The Sustainable Alternative Futures for India macroeconomic engine that matches long-term GDP growth, urbanization, and material footprints with net-zero bounds.
- **CEEW's GCAM-India Model:** A highly localized Global Change Analysis Model that tracks structural inter-linkages between water scarcity, agricultural land-use, energy tech, and economic policies.
- **NITI Aayog's Low-Carbon Development Model:** A centralized coordination cell setup under the PMCCC to align central five-year sectoral budgets with state-level execution frameworks.
- **Deep Electrification Demand Model:** A demand-side model built to triple electricity's share in Total Final Energy Consumption (TFEC) to 60% by 2070 via EVs and induction cooking.
- **Sovereign Blended Finance Model:** Strategic utilization of public and concessional multilateral capital as first-loss guarantees to de-risk and crowd-in commercial private equity.
- **Unified Climate Finance Taxonomy Model:** A statutory reference standard designed by the RBI to prevent greenwashing and systematically direct domestic banking capital into green assets.
- **Interoperable Digital Infrastructure Models:** The deployment of the Unified Energy Interface (UEI) layer to standardize transaction logging across decentralized EV charging and smart power grids.

Recommendations from key high-level panels and expert groups

- **Tarun Kapoor Panel**
  - **Ban on Urban Diesel 4-Wheelers:** Phasing out all diesel-powered four-wheelers by 2027 in cities with a population of over 1 million, replacing them with electric and gas vehicles.
  - **100% Clean Urban Public Transport:** Eliminating new diesel city bus procurement within urban transit networks to achieve fully electrified or CNG/LNG fleets by 2030.
  - **Banning ICE 2/3 Wheelers by 2035:** Imposing a definitive compliance ban on Internal Combustion Engine (ICE) two and three-wheelers by 2035.

- **Shyamala Gopinath Panel**
  - **De-risking Private Capital Inflows:** Establishing centralized sovereign de-risking funds to provide first-loss default guarantees to clean-tech startups.
  - **Expanding the Corporate Bond Market:** Deepening the domestic corporate green bond market from 16% of GDP to 30% by 2070 to unlock long-term retail capital.
  - **Preventing Systemic Stranded Asset Risks:** Mandating banks to run rigorous, periodic climate stress tests on loan portfolios exposed to coal and fossil infrastructure.

Conclusion

Achieving Net-Zero by 2070 is not just an environmental obligation for India, but a structural opportunity to ensure long-term energy security and macroeconomic resilience. By successfully balancing green technology deployment with a "Just Transition" for fossil-fuel dependent regions, India can pioneer a unique, equitable development model for the Global South.

3. NET ZERO EMISSIONS

A carbon market is a policy tool that allows countries and industries to buy and sell carbon credits, effectively putting a market price on carbon emissions to incentivize reduction. By treating greenhouse gas reductions as a tradeable commodity, it channels private investments directly into clean technology and renewable energy projects.

Climate Finance Needs of India

- **Short-Term NDC Targets (By 2030):** India requires an estimated **₹162.5 trillion ($2.5 trillion)** by 2030 to successfully meet its Nationally Determined Contributions (NDC) targets.
- **Long-Term Net-Zero Targets (By 2070):** The cumulative financial requirement scales up to **$10.1 trillion** by 2070 to achieve India's ultimate target of net-zero greenhouse gas emissions.
- **Annual Macroeconomic Capital Requirement:** To bridge the green investment gap, India needs an additional annual green investment equivalent to **2.5% of GDP** until 2030, **according to Reserve Bank of India (RBI) estimates**.

Benefits of climate finance and carbon markets

- **Macro-Economic & Investment Dynamics**
  - **Mobilizing Private Capital:** Channels global private equity and corporate social responsibility (CSR) funds directly into high-risk, clean-technology startups.
  - **Cost-Effective Mitigation:** Allocates financial resources to industries where greenhouse gas emission reductions are the cheapest to achieve.
  - **New Revenue Streams:** Generates vital sovereign revenue for developing nations through the export of international carbon credits.
  - **De-risking Green Tech:** Lowers the cost of capital for capital-intensive projects like Green Hydrogen and offshore wind via viability gap funding.
- **Technological Innovation & Industrial Transition**
  - **Accelerating Technology Transfer:** Facilities the flow of advanced proprietary clean technologies (like CCUS) from the Global North to the Global South.
  - **Incentivizing Low-Carbon R&D:** Places a clear market price on carbon emissions, forcing heavy industries to invest in energy-efficient machinery.
  - **Phasing Out Fossil Subsidies:** Creates an economic disincentive for coal and oil usage by making carbon-intensive operations increasingly expensive.

- **Environmental & Social Co-Benefits**
  - **Securing Climate Justice:** Delivers predictable funding via mechanisms like the **Loss and Damage Fund** to help highly vulnerable communities adapt.
  - **Promoting Nature-Based Solutions:** Funds large-scale afforestation and mangrove restoration projects by converting conserved ecosystems into tradeable carbon offsets.
  - **Improving Public Health:** Drives down localized particulate matter and industrial air pollution as factories clean up their energy mixes.

Issues associated with climate finance and carbon markets

- **Structural & Integrity Flaws**
  - **Widespread Greenwashing & Junk Credits:** Over-allocation of voluntary carbon credits to standard, "business-as-usual" projects compromises the core environmental integrity of global offsets.
  - **Flawed Baseline & Additionality Metrics:** Discrepancies in establishing project baselines reward projects that would have naturally occurred even without carbon finance.

    As per the Berkeley Carbon Trading Project, Up to **70%** of projects in the global voluntary carbon market fail strict additionality metrics.
  - **Glacial Project Registration Timelines:** Onerous compliance pathways mean registering a domestic voluntary carbon project in India takes upwards of 1,600 days.
- **Global Climate Finance Disparities**
  - **Pronounced Mitigation-Adaptation Imbalance:** Over 90% of global climate finance targets mitigation, leaving critical adaptation projects severely underfunded.

    As per the Climate Policy Initiative, Over **95%** ($1.9 trillion) of global tracked climate finance targets mitigation, while critical adaptation investment plateaus at just **$64 billion**.
  - **Debt-Trap Loan Proclivity:** Developed nations fulfill their funding pledges primarily via high-interest commercial loans rather than equitable, grant-based resource pools.
  - **MDB Risk Aversion Overhauls:** Multilateral Development Banks (MDBs) continuously refuse to absorb early-stage transition risks, blocking capital flows to the Global South.

    As per the G20 Independent Expert Group,Multilateral Development Banks mobilize a highly risk-averse **$0.26** of private capital for every dollar of their own operational funding
  - **Staggering Real-World Funding Gaps: According to the CPI Global Landscape of Climate Finance 2026 Report,** Tracked global funding runs at **$2 trillion** annually, whereas the actual requirement scales over **$6.2 trillion** annually by 2035.
- **Geopolitical & Unilateral Trade Friction**
  - **Protectionist Carbon Border Taxes:** Unilateral systems like the EU's Carbon Border Adjustment Mechanism (CBAM) penalize emerging economies under the guise of halting carbon leakage.
  - **Sovereignty Trust Deficit:** failure of developed nations to meet their $300 billion New Collective Quantified Goal (NCQG) commitments strains international negotiations.
  - **Sanction Alignment Vulnerabilities:** Recent controversies, like issuing UN carbon credits for projects linked to sanctioned global regimes, create volatile investment climates.
- **Domestic Compliance & Market Bottlenecks**
  - **MSME Capital & Technology Starvation:** Small and medium enterprises lack the upfront financial capital to transition and benefit from India's Carbon Credit Trading Scheme.

As per SIDBI, Indian MSMEs account for over **40%** of manufacturing emissions but require **$50 billion** in upfront tech investments to successfully adapt to India's CCTS.

- **Absence of a Settled Green Taxonomy:** lack of a legally codified green taxonomy limits the local banking sector's capacity to systematically scale up Priority Sector Lending.
- **Pronounced Price Volatility Risks:** A lack of robust price adjustment or supply-stabilization mechanisms exposes new compliance markets to severe price crashes.

UNFCCC Mandated Financial Mechanisms

- **Green Climate Fund (GCF):** Formally designated as the largest dedicated operating entity under the UNFCCC financial mechanism, it aims for a 50:50 allocation balance between mitigation and adaptation interventions in developing countries.
- **Global Environment Facility (GEF):** Operating as an independent financial organization since the 1992 Rio Earth Summit, it acts as a joint institutional focal mechanism for multiple conventions including the UNFCCC, CBD, and UNCCD.
- **Adaptation Fund (AF):** Established originally under the Kyoto Protocol to address climate-resilience infrastructure, it is partially self-sustained through a statutory 2% levy on Certified Emission Reduction (CER) credits

Measures taken to advance climate finance and carbon markets

- **Global Market & Regulatory Architecture**
  - **Paris Agreement Article 6 Rules:** Provisional issuance of the first international carbon credits under the Paris Agreement Crediting Mechanism.
  - **Expansion of Emissions Trading Systems (ETS):** Widened global compliance pricing platforms to cover 29% of all greenhouse gas emissions.
  - **New Collective Quantified Goal (NCQG):** Initiated international frameworks to scale public-private climate funding for developing countries to $1.3 trillion annually by 2035.
  - **Global Green Taxonomy Harmonization:** Doubled the deployment of formal green investment taxonomies across emerging markets to prevent capital misalignment.
- **India's Domestic Carbon Market Framework**
  - **Carbon Credit Trading Scheme (CCTS):** Operationalized India's formal domestic compliance carbon market across 7 heavy industrial sectors and 490 specific industries.
  - **BEE Dual-Market Mechanism:** Finalized the structural procedures for both the mandatory compliance mechanism and the voluntary offset mechanism under the Bureau of Energy Efficiency (BEE).
  - **Transitioning the PAT Infrastructure:** Phased the existing Perform, Achieve, and Trade (PAT) energy savings scheme systematically into a liquid carbon credit trading framework.
- **Sovereign Financial & Banking Instruments**
  - **Sovereign Green Bond Issuances:** Scaled up domestic funding channels via fresh issuances of Sovereign Green Bonds worth ₹15,000 crore in FY26, bringing cumulative totals past ₹72,000 crore.
  - **SEBI BRSR Core Disclosures:** Mandated the Business Responsibility and Sustainability Reporting (BRSR) framework to ensure high data integrity for green corporate bonds.
  - **Specialized Institutional Lending:** Leveraged specialized state financial entities like IREDA, SIDBI, and NABARD to clear funding backlogs for clean-tech startups and MSMEs.

Way Forward

- **Market Standardization & Data Integrity**

- **Codifying a Formal National Green Taxonomy:** Finalize and legally enforce India's draft climate finance taxonomy to prevent greenwashing and systematically guide bank lending.
- **Tightening Baseline & Additionality Benchmarks:** Transition India's Carbon Credit Trading Scheme (CCTS) from simple energy intensity baselines to absolute, time-bound emission reduction caps.
- **Enforcing Blockchain-Backed MRV Architectures:** Deploy decentralized ledger technologies to build a foolproof Monitoring, Reporting, and Verification (MRV) registry.

- **Supply and Price Stabilization Mechanisms**
  - **Integrating Financial Intermediaries for Liquidity:** Open up secondary market trading within the domestic CCTS to banks and institutional funds to deepen capital availability and smooth out price discovery.
  - **Formulating Dynamic Supply Adjustment Pools:** Establish a centralized market-stability reserve to regulate carbon credit supply and buffer the market against sudden price crashes.
  - **Providing Long-Term Forward Guidance on Benchmarks:** Announce clear, multi-year tightening trajectories for carbon benchmarks to give heavy industries the investment confidence needed for long-term R&D.
- **Alternative Instruments & Frontier Technology Scaling**
  - **Designing a National Carbon Removal Strategy (NCRS):** Formulate a dedicated regulatory pathway to integrate durable Carbon Dioxide Removal (CDR) tech like biochar and enhanced rock weathering into voluntary offsets.
  - **Expanding Corporate Social Responsibility (CSR) Pipelines:** De-risk early-stage green tech projects by allowing corporate entities to safely channel mandatory CSR obligations into bankable carbon-removal pilots.
  - **Mainstreaming Carbon Capture (CCUS) Incentives:** Broaden the domestic carbon trading framework to directly incentivize industrial Carbon Capture, Utilization, and Storage (CCUS) operations in steel and cement.
- **Global Equity & Adaptation Refocusing**
  - **Balancing the Mitigation-Adaptation Funding Split:** Mandate that a fixed percentage of sovereign green bond revenues and domestic market fees be locked explicitly for local climate adaptation projects.
  - **Demanding Grants Over Debt-Trap Instruments:** Use multilateral forums like COP to pressure developed nations into providing grant-based and concessional funding.
  - **Building Regional Alliances Against Border Taxes:** Lead coalitions of emerging economies to challenge unilateral trade barriers like the EU's Carbon Border Adjustment Mechanism (CBAM) under the principle of CBDR-RC.

Several models for carbon market

- **Blended Finance Model:** Strategic public or multilateral funds are deployed to de-risk green investments, effectively crowding-in large pools of private commercial capital.
- **Grant-Based Fiscal Model:** Non-repayable financial resource transfers dedicated strictly to non-revenue generating adaptation projects in vulnerable Least Developed Countries (LDCs).
- **Perform, Achieve, and Trade (PAT) Model:** India's baseline model assigning Specific Energy Consumption targets to heavy sectors, issuing tradeable Energy Saving Certificates (ESCerts).
- **Carbon Credit Trading Scheme (CCTS) Model:** India's evolving compliance marketplace aimed at directly pricing and trading absolute carbon emission metrics across core industries.
- **Sovereign Green Bond Model:** State debt instruments structured specifically to raise low-cost market capital earmarked for public eco-friendly infrastructure.

Conclusion

Developing an integrated **Indian Carbon Market (ICM)** and securing predictable **climate finance** are vital to achieve **deep decarbonization** across hard-to-abate industries. By pricing carbon emissions and de-risking clean tech investments, these economic tools bridge the funding gap needed to build a sustainable **circular economy**.

4. LOSS AND DAMAGE

Loss and Damage Fund is a global financial mechanism designed specifically to provide financial rescue and assistance to vulnerable developing nations suffering from the destructive impacts of climate change.

Evolution of the Concept of Loss and Damage

- **Origin:** The concept was heavily pushed by Small Island Developing States (SIDS). The decision to establish the fund was made at COP27 (2022) in Egypt, and it was formally operationalized at COP28 (2023) in Dubai.
- **Host and Management:** The fund is hosted by the World Bank as an interim trustee for a four-year period. Its independent Secretariat is based in the Philippines.
- **Board Structure:** It is managed by a 26-member governing board with a majority from developing countries (14 developing, 12 developed) to ensure fair representation.

Features of the Loss and Damage Fund (LDF)

- **Dual Classification:** It covers both economic losses (destruction of infrastructure, property, crops) and non-economic losses (loss of life, cultural heritage, biodiversity, and community displacement).
- **Interim Trustee:** The World Bank hosts and administers the fund as an interim trustee for an initial four-year period.
- **Contribution Nature:** Financial contributions from developed countries are completely voluntary and do not imply legal liability or compensation obligations.
- **Eligibility Criteria:** The primary beneficiaries are defined broadly as "particularly vulnerable" developing countries, with a heavy emphasis on Small Island Developing States (SIDS) and Least Developed Countries (LDCs).
- **Legal Status:** It operates as an independent legal entity under the financial mechanism of the UNFCCC and the Paris Agreement.
- **Funding Diversity:** While relying mainly on public grants from developed countries, its charter allows for funding from philanthropic organizations and innovative sources (like carbon pricing).

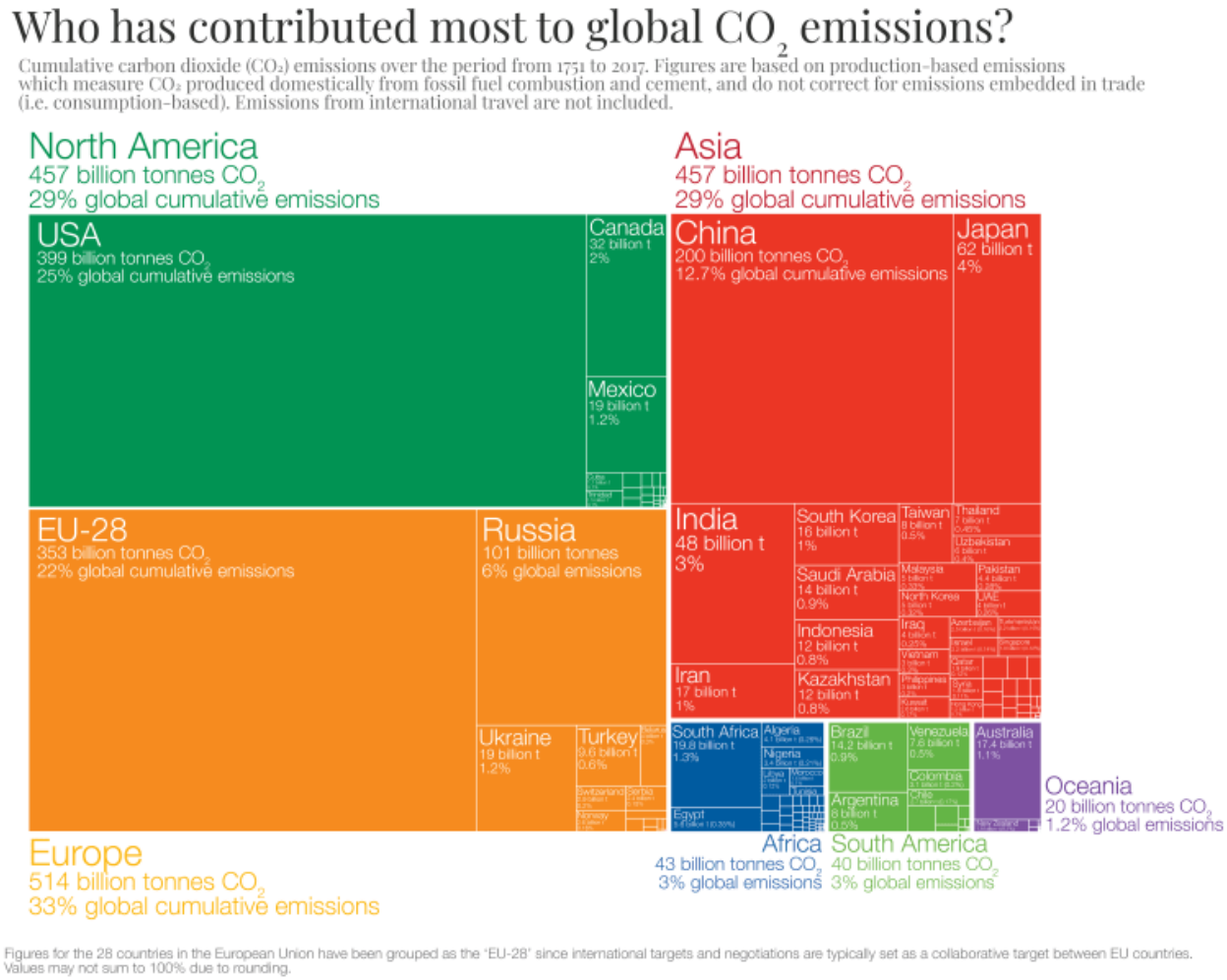
Types of Loss and Damage

Broad Type	Dimension	Major Impacts & Specific Examples
Economic Losses	**Business & Income**	Loss of agricultural revenue from ruined crops. Disrupted industrial business supply chains and operational income.
	Physical Property	Destruction of residential houses and commercial properties.
	Public Infrastructure	Heavy damage to transport links (roads, bridges, ports). Destabilization of power grids and energy distribution networks.

Non-Economic Losses (NELs)	**Individual / Human**	Direct loss of human lives. Chronic physical morbidity and direct post-disaster psychological trauma.
	Social / Cultural	Loss of ancestral territories and geographic habitability. Destruction of sacred architectural marvels, heritage sites, and traditional knowledge. Breakdown of long-standing community cohesion due to forced displacement.
	Environmental	Irreversible mass extinction of terrestrial and marine species. Degradation of vital ecosystem services (e.g., natural flood-mitigation barriers).

Significance of the Fund for responding to Loss and Damage

- **Global Equity & Climate Justice**
  - **Operationalizing Polluter-Pays Principle:** Formally forces affluent, historically emitting nations to acknowledge financial accountability for global environmental degradation.
  - **Institutionalizing Climate Reparations:** Moves the global climate discourse beyond mere mitigation and adaptation into the realm of compensating for irreversible damages.
  - **Empowering the Global South**: Solidifies the collective bargaining power of Least Developed Countries (LDCs) and Small Island Developing States (SIDS).
- **Socio-Economic Rehabilitation & Safety Nets**
  - **Rebuilding Critical Infrastructure:** Provides targeted liquidity to reconstruct public transport, hospitals, and homes destroyed by catastrophic extreme weather events.
  - **Compensating Non-Economic Losses:** Recognizes and financially accounts for intangible impacts like psychological trauma, displacement, and loss of cultural heritage.
  - **Preventing Regressive Debt Traps:** Delivers grant-based post-disaster relief, preventing vulnerable economies from relying on high-interest commercial emergency loans.
- **Comprehensive Risk & Slow-Onset Management**
  - **Mitigating Slow-Onset Environmental Hazards:** Earmarks dedicated resources to tackle long-term climate transformations like rising sea levels, desertification, and ocean acidification.
  - **Complementing Humanitarian Aid Pipelines:** Fills a critical post-disaster timeline gap by funding intermediate and long-term economic rehabilitation after immediate emergency aid recedes.
  - **Preserving Global Biodiversity Sinks:** Finances localized eco-restoration and conservation efforts for endangered ecosystems pushing past their natural adaptive thresholds.
- **Macro-Economic & Geopolitical Security**
  - **Averting Climate-Induced Migration Crises:** Stabilizes border security by funding the rehabilitation of populations displaced by chronic agricultural failures or land loss.
  - **De-risking Sub-National Economies:** Protects highly exposed domestic regional belts (e.g., India's cyclone-prone coastlines or fragile Himalayan landslide zones like Wayanad) from fiscal shocks.
  - **Strengthening UNFCCC Multilateral Credibility:** Restores trust in global climate negotiations by showing concrete progress on hard-fought, consensus-driven structural finance pledges.



Figures for the 28 countries in the European Union have been grouped as the 'EU-28' since international targets and negotiations are typically set as a collaborative target between EU countries. Values may not sum to 100% due to rounding.

Difference between mitigation, adaptation and addressing Loss and Damage

- **Mitigation:** Focuses on the root cause of climate change by reducing greenhouse gas (GHG) emissions or enhancing carbon sinks to limit global temperature rise.
- **Adaptation:** Focuses on the consequences of climate change by adjusting systems, infrastructure, and behaviors to coexist with actual or expected climate impacts.
- **Addressing Loss and Damage:** Focuses on the residual impacts after climate change has already occurred, handling the destruction that happens when adaptation limits are breached or bypassed.

Issues with Loss and Damage

- **Financial Capital & Scale Deficits**
  - **Asymmetrical Funding Disconnect:** Accumulated global pledges reach under $1 billion, covering less than 0.1% of the estimated $400 billion annual loss suffered by the Global South.
  - **Absence of Mandatory Obligations:** Contributions from historic emitters remain strictly voluntary, completely diluting legal liability for historical carbon pollution.
  - **Risk of Rapid Fund Depletion:** Current capital pools face complete exhaustion by 2027 without a structured, predictable replenishment mechanism built into global climate pacts.
- **Institutional Governance & Geopolitical Friction**
  - **World Bank Interim Hosting Bias:** Developing nations strongly oppose World Bank management due to high administrative fees, strict bureaucratic conditionalities, and heavy Western influence.

- **Severe Eligibility Exclusion Risks:** Mounting diplomatic pressure to restrict payouts exclusively to SIDS and LDCs threatens to leave out highly vulnerable, massive developing economies like India.
- **Erosion of the CBDR Principle:** Advanced economies consistently attempt to decouple fund operations from the foundational principle of Common But Differentiated Responsibilities.

- **Operational & Methodological Bottlenecks**
  - **Deficit in Quantification Metrics:** Absolute lack of a universally accepted definition for "Loss and Damage" creates severe accounting friction between sudden disasters and slow-onset events.
  - **Marginalization of Non-Economic Losses:** No clear legal or economic framework exists to compensate for intangible losses like psychological trauma, cultural heritage and biodiversity decay.
  - **Sluggish Fund Disbursement Windows:** Complicated procedural pipelines delay the rapid release of liquidity, failing to bridge the immediate post-disaster gap before standard aid arrives.
- **Sub-National & Local Structural Gaps**
  - **Centralized Top-Down Access:** The current funding architecture lacks a direct sub-national or territorial access window, sidelining frontline municipal bodies.
  - **Double-Counting and MRV Weaknesses:** Inadequate Monitoring, Reporting, and Verification (MRV) systems create transparency risks, allowing existing developmental aid to be rebranded as loss and damage finance.
  - **Neglect of Climate-Induced Displacement:** The operational framework fails to provide dedicated legal or financial safety nets for millions of internal climate refugees.

Santiago Network on Loss and Damage (SNLD)

The Santiago Network is a network to enable Loss and Damage fund flow from developed to developing countries. It will work towards the implementation of relevant approaches at the local, national and regional level, in developing countries that are particularly vulnerable to the adverse effects of climate change.

Way Forward

- **Financial Scalability & Replenishment Predictability**
  - **Transitioning to Mandatory Resource Targets:** Shift from volatile, voluntary donor pledges to a binding, rule-based mandatory replenishment system aimed at reaching $400 billion annually by 2030.
  - **Capitalizing via Innovative Global Taxation:** Explore alternative international funding streams such as global maritime fuel taxes, aviation levies, and windfall carbon taxes on fossil fuel companies.
  - **Linking to the New Collective Quantified Goal (NCQG):** Formally institutionalize loss and damage allocations as a core, distinct pillar within the post-2035 global climate finance frameworks.
- **Operational Autonomy & Direct Governance Reforms**
  - **Decoupling from the World Bank Interim Framework:** Construct a fully independent, sovereign UNFCCC secretariat after the initial four-year interim hosting cycle to eliminate heavy Western institutional bias.
  - **Simplifying the Disbursement Architecture:** Create fast-tracked, direct-access budgetary windows to radically compress the time between disaster impact and liquid capital deployment.
  - **Expanding Direct Sub-National Access Windows:** Allocate dedicated small-grant pipelines directly to municipal local bodies and frontline communities, bypassing sluggish central bureaucracy.
- **Methodological Upgrades & Accounting Frameworks**
  - **Quantifying Non-Economic Losses (NELs):** Codify globally standardized, non-monetary valuation metrics to financially account for psychological trauma, biodiversity decay, and cultural heritage loss.

- **Standardizing Slow-Onset Evaluation Systems:** Design rigorous, AI-driven predictive frameworks to systematically track and fund gradual ecological degradation like sea-level rise and desertification.
- **Institutionalizing the Global Loss and Damage Gap Report:** Mandate the regular publication of the newly proposed scientific tracking framework under the Warsaw International Mechanism (WIM) to maintain global visibility on finance deficits.

- **Legal Protections & Inclusive Climate Justice**
  - **Enforcing Human Rights Preamble Anchors:** Align all fund disbursement criteria directly with the Paris Agreement's human rights obligations to protect indigenous and displaced communities.
  - **Resolving the Strategic Legal Liability Impasse:** Re-negotiate text limitations to ensure that receiving vital emergency aid does not require vulnerable nations to waive their sovereign legal rights to claim historical climate reparations.
  - **Providing Comprehensive Protection for Climate Refugees:** Create dedicated cross-border legal and economic security safety nets for millions of internally and externally displaced climate migrants.

Conclusion

Operationalization of the Loss and Damage Fund marks a watershed moment for climate justice, turning historical accountability into concrete financial support for vulnerable nations. However, its ultimate success depends on replacing voluntary pledges with mandatory, multi-billion-dollar funding streams that can be disbursed rapidly during climate crises.

5. CLIMATE RESILIENT DEVELOPMENT

Climate resilient development is a comprehensive strategy that integrates adaptation measures and emission cuts directly into sustainable economic planning. Rather than treating climate change as a standalone environmental issue, it treats climate safety as a baseline requirement for all future public infrastructure and policy decisions.

Significance of climate-resilient development

- **Infrastructure & Urban Development**
  - **Preventing Critical Infrastructure Failures:** Safeguards vital communication networks, transport corridors, and power grids from catastrophic extreme weather breakdowns.
  - **Optimizing Multi-Decadal Public Investments:** Future-proofs capital-intensive national mega-projects (e.g., high-speed rail, deep-water ports) against rising sea levels and intense thermal stress.
  - **Mitigating Urban Heat Islands & Flooding:** Reduces localized urban temperatures and flash-flood damage through smart features like permeable pavements and green roofs.
- **Agriculture & Food Security**
  - **Insulating Crop Yields from Monsoon Shocks:** Stabilizes aggregate national food grain production against erratic rainfall distribution using drought-tolerant seeds.
  - **Securing Smallholder Farm Livelihoods:** Minimizes rural household income shocks and debt traps by preventing total crop failures.
  - **Optimizing Dwindling Water Resources:** Enhances water-use efficiency through climate-adaptive micro-irrigation networks and groundwater recharge structures.
- **Macroeconomic Stability & Fiscal Resilience**
  - **Slashing Post-Disaster Emergency Expenditures:** Lowers fiscal deficits by minimizing the capital required for emergency rehabilitation and reconstruction.

- **Protecting Sovereign Credit Ratings:** Lowers borrowing costs by demonstrating structural systemic resilience to international credit rating agencies.
- **Unlocking Specialized Global Capital Pipelines:** Attracts dedicated institutional green bonds and private transition finance seeking low-risk, resilient infrastructure assets.

- **Socio-Ecological & Public Health Security**
  - **Radically Reducing Human Mortality:** Lowers casualties from climate hazards like heatwaves and flash floods through climate-resilient housing and early warning shelters.
  - **Averting Mass Climate-Induced Displacement:** Prevents internal migration crises by securing the climate habitability of vulnerable coastal and hill regions.
  - **Preserving Critical Ecosystem Services:** Safeguards native biodiversity and natural carbon sinks from irreversible degradation during severe droughts and wildfires.

Steps taken by India toward Climate-Resilient Development

Sector	Measure Taken
Agriculture & Food Security	**NICRA & NMSA:** Rolled out over 2,900 climate-resilient, stress-tolerant crop varieties alongside 448 model Climate-Resilient Villages.
	PM-KUSUM & Micro-Irrigation: Solarized over 9.7 lakh farm pumps and brought 109 lakh hectares under "Per Drop More Crop" drip systems.
Urban & Rural Infrastructure	**CDRI Initiative:** Led the global Coalition for Disaster Resilient Infrastructure and launched telecom resilience frameworks across vulnerable states.
	CSCAF (AMRUT & Smart Cities): Institutionalized the Climate Smart Cities Assessment Framework to embed flood and heat metrics into city layouts.
Water Resource Management	**Jal Jeevan Mission**: Provided climate-safe tap water connections to over 11.9 crore households to decouple rural life from groundwater shocks.
	Catch the Rain Campaign: Mandated decentralized, structural rainwater harvesting and artificial aquifer recharge across all districts.
Institutional & Policy Frameworks	**NAPCC & SAPCC Overhauls:** Operates 9 core national climate missions while legally binding state climate budgets directly to regional vulnerability maps.
	Sovereign Green Bonds: Issued over ₹72,600 crore in green capital bonds since FY23 to exclusively fund public adaptation and resilient infrastructure.
Ecosystem & Coastal Protection	**MISHTI & Forest Sinks:** Deployed corporate CSR and CAMPA funds to systematically restore coastal mangrove belts and hit NDC carbon sink targets.

Issues hindering Climate-Resilient Development in India

- **Climate Finance & Adaptation Funding Gaps**
  - **Severe Mitigation Bias:** Climate finance is heavily skewed toward mitigation, leaving adaptation projects underfunded by an estimated 5.6% of GDP annually.

    According to Economic Survey 2025-26, India's internal adaptation-related domestic spending has actually scaled up to **5.6% of GDP**.
  - **Fiscal Starvation of Local Bodies:** Urban Local Bodies (ULBs) lack the credit ratings and financial autonomy to float local municipal or green bonds independently.

    As per the Reserve Bank of India, Less than **1% of GDP** accounts for the total expenditure of Urban Local Bodies (ULBs) in India.

  - **Absence of Blended Capital Models:** Persistent lack of sovereign risk-guarantee mechanisms discourages private institutional equity from entering long-term, low-return adaptation projects.
- **Infrastructure Vulnerability & Urban Planning Flaws**
  - **Outdated Engineering Benchmarks:** Critical public structures (dams, bridges, power stations) are built on historical weather averages that fail to withstand modern extreme anomalies.
  - **Systemic Destruction of Blue-Green Buffers:** Rapid real estate speculation and illegal construction routinely choke natural urban wetlands and floodplains (e.g., Chennai, Bengaluru).
  - **High Post-Disaster Maintenance Friction:** Sluggish budgetary release windows delay the structural strengthening of public assets after extreme weather strikes.
- **Agrarian Vulnerability & Marginalization**
  - **Low Grassroots Agri-Tech Diffusion:** Small and marginal farmers (comprising 86% of total holdings) lack the liquidity to purchase high-cost, climate-resilient seed varieties.
  - **Chronic Over-Reliance on Rainfed Farming:** More than 50% of India's net sown area is entirely rainfed, keeping national food security hypersensitive to monsoon delays.
  - **Distorted Agricultural Subsidies:** Power and fertilizer subsidies actively incentivize water-guzzling crops (paddy) in critically water-stressed, drought-prone aquifers.
- **Governance Deficits & Institutional Silos**
  - **Bureaucratic Overlap and Fragmentation:** Disconnected operational lines between central ministries, state pollution boards, and municipalities block coordinated disaster management.
  - **Superficial State Climate Action Plans:** State Action Plans on Climate Change (SAPCC) function largely as static policy papers rather than legally binding, funded development masterplans.
  - **Deficit in Hyper-Local Vulnerability Data:** Severe shortage of high-resolution, block-level meteorological forecasting models makes local climate-risk mapping highly imprecise.

Way Forward

- **Urban Infrastructure & Housing Climate-Proofing**
  - **Mandating Disaster-Resistant Building Codes:** Enforce strict, localized building norms that dynamically account for rising thermal stress, seismic zones, and coastal storm surges.
  - **Expanding Green-Blue Infrastructure:** Legally protect and aggressively restore urban wetlands, lakes, and mangrove belts to act as natural buffers against flash floods.
  - **Promoting Compact City Designs:** Integrate Transit-Oriented Development (TOD) and mixed land-use into municipal master plans to reduce spatial footprint and energy loads.
- **Agrarian & Food System Shock-Proofing**
  - **Scaling Up "Mission Mausam" Advisories:** Leverage AI-driven, hyper-local weather forecasting under the recently expanded national mission to provide real-time last-mile sowing alerts.
  - **Codifying a National CRA Roadmap:** Align biotechnology research under the BioE3 framework to accelerate the field deployment of climate-adapted, salinity-tolerant crop varieties.
  - **Institutionalizing District Climate Cells:** Set up dedicated Climate Resilience Cells across all agricultural departments to systematically audit cropping patterns through a climate lens.
- **Institutional Governance & Legislative Anchors**
  - **Enacting an Omnibus Climate Enabling Law:** Move beyond fragmented regulations to create a comprehensive framework legislation that links all economic choices to resilience metrics.
  - **Operationalizing Sub-National Autonomy:** Empower Panchayati Raj Institutions (PRIs) and Urban Local Bodies (ULBs) with scientific data and dedicated fiscal resources as first responders.
  - **Mainstreaming Judicial "Climate Rights":** Integrate the Supreme Court's landmark *Ranjitsinh* ruling into environmental impact assessments, establishing the right against climate change under Article 21.

- **Transition Finance & Budgetary Innovation**
  - **Enforcing Climate-Tagging in Budgeting:** Mandate all central ministries and state departments to climate-tag public expenditures to ensure capital explicitly supports adaptive infrastructure.
  - **Deepening the Municipal Green Bond Market:** Upgrade the credit-rating frameworks of tier-2 and tier-3 city corporations to let them independently pool long-term global adaptation equity.
  - **De-risking Private Adaptation Capital:** Create blended finance mechanisms where state capital acts as a first-loss guarantee, crowding private players into low-yield resilience projects.

Best practices for Climate-Resilient Development in India

- **Odisha Millets Mission (OMM) Model:** A globally acclaimed structural shift to climate-hardy, drought-tolerant indigenous millets, securing tribal nutrition and rainfed farm income.
- **Zero-Budget Natural Farming (ZBNF) Model:** Andhra Pradesh's landscape-level shift to bio-inputs, drastically slashing water requirements and chemical input costs for marginal smallholders.
- **Marathwada's "One-Acre" Farming Model:** A Swayam Shikshan Prayog (SSP) initiative empowering women farmers with land titles to practice multi-cropping of drought-resilient food crops over cash crops.
- **Meenangadi Carbon-Neutral Panchayat Model:** Kerala's local-body landmark initiative utilizing a tree-banking system to incentivize agroforestry and achieve complete local net-zero status.
- **Tamil Nadu's Climate Resilient Villages Programme:** A place-based, holistic village-unit model combining localized rainwater harvesting, flood buffers, and solar infrastructure highlighted by the Economic Survey.
- **East Kolkata Wetlands (EKW) Natural Model:** A classic Nature-based Solution (NbS) utilizing peri-urban wetland ecosystems for low-cost, organic municipal wastewater treatment and natural flood absorption.
- **Surat Heat Action Plan (HAP) Framework:** A pioneering civic-health model utilizing cool-roof painting and real-time thermal early warnings to protect vulnerable informal settlements.
- **Ahmedabad Sponge-City Master Plan:** Integrating localized permeable pavements and urban green corridors to transform city catchments into active groundwater recharge blocks.
- **KABIL Strategic Mineral Consortium:** A central joint-venture framework to secure long-term overseas acquisition of transition minerals (lithium/cobalt) for green tech supply-chain safety.

Conclusion

Mainstreaming climate-resilient development is indispensable for India to insulate its macroeconomic trajectory from escalating climate vulnerabilities. By harmonizing nature-based solutions with robust sub-national governance, India can successfully bridge the gap between developmental imperatives and environmental sustainability.

6. UNFCCC AND COP 30

United Nations Framework Convention on Climate Change (UNFCCC) is the foundational 1992 global treaty established to stabilize atmospheric greenhouse gas concentrations and prevent dangerous human interference with the Earth's climate system. Its supreme decision-making body, the Conference of the Parties (COP), serves as the premier annual multilateral summit where 198 member nations gather to negotiate, monitor, and enforce global climate pacts like the Paris Agreement.

About UNFCC (United Nations Framework Convention on Climate Change)

- **Background**
  - Adopted in 1992 at the Rio Earth Summit (UN Conference on Environment and Development).
  - Officially entered into force on March 21, 1994.
  - **Headquarters:** Secretariat is located in Bonn, Germany (initially Geneva).
  - **Membership:** Near-universal participation with 198 Parties (197 States + European Union).
- **Primary Objective & Principles**
  - **Ultimate Goal:** Stabilize greenhouse gas (GHG) concentrations to prevent **dangerous anthropogenic (human-induced) interference with the climate system.**
  - **Ecosystem Protection:** Achieve stabilization within a timeframe that allows ecosystems to adapt naturally, preserves food security, and enables sustainable economic growth.
  - **CBDR-RC Principle:** Grounded in Common But Differentiated Responsibilities and Respective Capabilities, requiring developed nations to bear the primary historical and financial burden.
  - **Annex Classification:** Divides countries into Annex I (industrialized/developed nations and economies in transition) and Non-Annex I (predominantly developing countries).
- **Governance & Mechanisms**
  - **Supreme Body:** Conference of the Parties (COP) is the highest decision-making authority and meets annually.
  - **Subsidiary Bodies:** Supported by SBSTA (Scientific and Technological Advice) and SBI (Implementation) which meet semi-annually.
  - **Reporting Tool:** Mandates National Communications and GHG inventories to monitor transparency and compliance.
  - **Financial Flow:** Channels climate finance from developed to developing countries, managed primarily via the Global Environment Facility (GEF) and Green Climate Fund (GCF).
- **Major Milestone Achieved**
  - **Kyoto Protocol (1997):** First major offshoot; introduced legally binding emission reduction targets strictly for developed (Annex I) countries.
  - **Paris Agreement (2015):** Landmark legally binding pact shifting all nations toward Nationally Determined Contributions (NDCs) to cap warming well below 2°C (ideally 1.5°C).
  - **Global Stocktake (GST):** A mandatory five-year review mechanism under Paris to assess collective progress; the first GST concluded in 2023 (COP28) yielding the UAE Consensus to transition away from fossil fuels.

Outcomes of UNFCCC COP 30 (Belém, Brazil)

Outcome	Primary Purpose	Launched By / Under
Global Implementation Accelerator (GIA)	Bridge the gap between national climate commitments and the 1.5°C pathway over a two-year timeline.	COP 30 Presidency
Global Mutirão Decision	Establish a high-level political consensus text focused heavily on balanced climate adaptation and finance.	UNFCCC Parties
Belém Mission to 1.5	Foster international cooperation and accountability across mitigation, adaptation, and green investment.	COP29–COP31 Troika

NCQG Trillion Target	Scale up global climate finance for developing nations to at least $1.3 trillion annually by 2035.	New Collective Quantified Goal (NCQG)
Expanded Adaptation Finance Goal	Increase specific adaptation funding to $120 billion per year as part of the broader $300 billion fast-start framework.	Baku-to-Belém Finance Roadmap
GGA Assessment Matrix	Operationalize a package of 60 concrete indicators to measure and track global adaptation progress.	Global Goal on Adaptation (GGA)
Just Transition Mechanism (JTM)	Boost technical cooperation, capacity-building, and equitable funding to ensure inclusive energy shifts.	COP 30 Action Agenda
The Forest and Climate Roadmap	Chart clear global milestones to completely halt and reverse tropical deforestation by 2030.	COP 30 Action Agenda
Fossil Fuels Transition Roadmap	Provide voluntary national pathways for countries to steadily transition away from fossil fuel dependencies.	COP 30 Action Agenda
NAPs Implementation Alliance	Support national adaptation plans and accelerate public-private investment mobilization pipelines.	Plan to Accelerate Solutions (PAS)
Tropical Forest Investment Fund (TFIF)	Manage a $125 billion mechanism to provide results-based payments to nations conserving standing tropical forests.	Tropical Forests Forever Facility (TFFF)
IP&LC Direct Finance Mandate	Legally protect forest communities by ensuring at least 20% of all TFIF payments go directly to local populations.	Under the TFIF / TFFF framework
JREDD+ Coalition	Mobilize $3 billion to $6 billion annually by 2030 through specialized jurisdictional forest preservation frameworks.	Govts, Indigenous Groups & Investors
FINI Initiative	Unlock a USD 1 trillion pipeline of investible adaptation projects within the next three years.	Atlantic Council & NRDC
Belém Health Action Plan	Target and mitigate climate-driven global public health threats through synchronized cross-border relief.	COP 30 Action Agenda
Enhanced Gender Action Plan	Promote gender-responsive budgeting and bolster leadership roles for rural and Indigenous women.	COP 30 Action Agenda
Power Grids Expansion Plan	Coordinate global stakeholders to upgrade and expand transmission grids for renewable energy integration.	Green Grids Initiative (GGI)
Plan to Accelerate Coal Transitions (PACT)	Fast-track structured phase-outs from coal power generation while securing regional energy stability.	Powering Past Coal Alliance (PPCA)
Belém 4x Pledge	Quadruple the global production and deployment of sustainable fuels (hydrogen, biofuels, e-fuels) by 2035.	Co-led by India, Brazil, Italy & Japan

Information Integrity Commitments	Counter science-related climate disinformation and promote accurate, evidence-based communication frameworks.	COP 30 Action Agenda
CREWS 2030 Strategy	Build advanced multi-hazard early warning systems for highly climate-vulnerable and island nations.	Climate Risk and Early Warning Systems
Heavy Industries Decarbonization	Accelerate deep emission cuts and pioneer green industrialization pathways within heavy manufacturing.	35 countries alongside UNDP & UNIDO
UN Trade Forum Dialogue	Ensure unilateral green trade regulations (like carbon tariffs) do not act as hidden barriers to developing markets.	COP 30 in coordination with UNCTAD

Significance of UNFCCC COP 30

- **Paradigm Shift in Climate Governance**
  - **Transition to Implementation:** Officially designated as the **COP of Action** or **COP of Truth**, shifting the global focus from making new pledges to tracking existing commitments.
  - **Belém Package:** Adopted a comprehensive suite of 29 negotiated decisions to legally operationalize the final phase of the Paris Agreement.
  - **Global Mutirão Framework:** Established a community-driven, collective effort approach to narrow the 1.5°C implementation gap without imposing top-down mandatory cuts.
- **Climate Finance Architecture**
  - **Baku-to-Belém Roadmap:** Streamlined the operational pathways to scale global climate finance to at least $1.3 trillion annually by 2035.
  - **Adaptation Funding Mandate:** Formally committed to a political agreement to triple adaptation finance from public sources by 2035.
  - **Unlocking Private Capital:** Launched the FINI (Fostering Investible National Implementation) initiative to mobilize a $1 trillion pipeline for adaptation projects.
- **Adaptation & Vulnerability Metrics**
  - **Quantifying the Global Goal on Adaptation (GGA):** Finalized a breakthrough consensus on 59 voluntary metrics to scientifically evaluate water, food, and infrastructural resilience.
  - **Belém Health Action Plan:** Unveiled the world's first international climate adaptation plan specifically targeting climate-induced public health vulnerabilities.
  - **Operationalizing Loss & Damage:** Advanced structural rules and fiduciary standards for the Loss & Damage Fund, raising total pledges close to $1.2 billion.
- **Forest, Nature, and Energy Sub-Systems**
  - **Tropical Forest Forever Facility (TFFF):** Launched a flagship $125 billion performance-based endowment fund that directly incentivizes nations for preserving standing rainforests.
  - **Direct Equity for Indigenous Communities:** Mandated that a minimum of 20% of all TFFF conservation payouts flow directly to local and indigenous populations.
  - **The Belém 4x Pledge:** Secured a political commitment from over 20 nations to quadruple the domestic deployment of sustainable liquid and gaseous alternative fuels by 2035.
- **Geopolitical Dimensions & India's Strategic Alignment**
  - **Institutionalizing Equity:** Mainstreamed the "Global Ethical Stocktake" to integrate intergenerational equity and Common but Differentiated Responsibilities (CBDR) into compliance audits.

- **Climate-Trade Nexus:** Initiated formal multilateral dialogues to counter unilateral green trade barriers like the EU's Carbon Border Adjustment Mechanism (CBAM).
- **India's Leadership Vector:** Enabled India to consolidate its position as the voice of the Global South, emphasizing climate justice and demanding predictable, concessional finance over rhetoric.

How is India Positioning Itself at COP30?

- **Grounding in Equity & Core Principles**
  - **CBDR-RC Anchor:** Strongly anchored negotiations around Common But Differentiated Responsibilities (CBDR-RC), demanding historical emitters lead emission cuts.
  - **Protecting Carbon Space:** Exercised pushback against shifting the mitigation burden to developing states, safeguarding the economic growth space of the Global South.
  - **Preserving Sovereignty:** Strongly opposed top-down mandatory emission caps, insisting on a rules-based global order that respects national sovereignty.
- **Demanding True Financial Architecture**
  - **Article 9.1 Mandate:** Asserted that providing predictable, concessional, and grant-based climate finance is a binding legal obligation of developed countries, not a voluntary act.
  - **Push for the $1.3 Trillion Target:** Demanded accountability from the Global North to fulfill the post-Baku annual $1.3 trillion climate finance goal by 2035.
  - **Calling out Financial Backsliding:** Formally flagged data showing developed nations cut their active financial commitments by 51% to 100% in recent reporting cycles.
  - **Standardizing Definitions:** Called for an official, universally agreed UN definition of "climate finance" to eliminate corporate greenwashing and high-interest private loans.
- **Prioritizing Adaptation & Just Transition**
  - **COP of Adaptation Pivot:** Urged the global community to treat climate adaptation as an absolute operational priority over pure mitigation tracking.
  - **Flexible GGA Indicators:** Supported the 59 voluntary metrics under the Global Goal on Adaptation (GGA) while rejecting any extra reporting burdens on developing states.
  - **Action-Oriented Just Transition:** Spearheaded the operationalization of the *Belém Action Mechanism (BAM)* to safeguard carbon-dependent workforces.
- **Technology, Trade & Industry Alliances**
  - **Technology as a Sovereign Right:** Stated that clean technology transfer is an absolute right, demanding the removal of restrictive Intellectual Property (IP) and market barriers.
  - **Rejecting Protectionist Green Tariffs:** Co-led coalitions (via BASIC and LMDC) to formally oppose unilateral, trade-restrictive measures like the EU's Carbon Border Adjustment Mechanism (CBAM).
  - **Driving Sectoral Partnerships:** Co-launched the *Belém 4x Pledge* alongside Brazil, Italy, and Japan to quadruple the deployment of sustainable alternative fuels by 2035.

Issues and Challenges with UNFCCC COP 30 (Belém)

- **Ambition & Mitigation Gaps**
  - **Fossil Fuel Language Omission:** final text completely omitted a legally binding roadmap for a phased-out transition from fossil fuels, caving to petro-state lobbying.
  - **1.5°C Overshoot Acknowledgment:** Officially acknowledged the high probability of overshooting the 1.5^C Paris target, reflecting a deep failure to bridge the global emissions gap.
  - **Inadequate NDC 3.0 Alignment:** Voluntary actions and updated 2035 targets submitted deliver less than 15% of the total emission cuts required by science.

  - **Lack of Long-Term Strategies:** The summit over-indexed on near-term targets while completely ignoring the updates needed for long-term decarbonization pathways.
- **Finance Bottlenecks**
  - **Quality vs. Quantum Conflict:** Left open the critical dispute of counting high-interest private loans and market leverages instead of predictable, public grants.
  - **BAM Institutional Dilution:** Belém Action Mechanism (BAM) for a Just Transition was structurally diluted into a non-binding framework postponed for COP 31 review.
  - **Adaptation Financing Shortfalls:** Despite a political signal to triple funding, the text failed to secure legally binding resource commitments from developed nations.
- **Nature & Land-Use Loopholes**
  - **Deforestation Backsliding:** Failed to establish a binding global compliance matrix to enforce the pledge to halt and reverse tropical deforestation by 2030.
  - **TFFF Voluntary Structure:** $125 billion Tropical Forest Forever Facility relies heavily on voluntary endowments, risking chronic underfunding.
  - **Friction Over Direct IP&LC Payouts:** Bureaucratic friction between state frameworks and indigenous councils delayed the operationalization of the 20% direct funding rule.
- **Geopolitical & Trade Rifts**
  - **Emergence of Parallel Blocs:** Deepening fractures emerged as an alliance of major powers directly blocked standard consensus texts on mitigation binding metrics.
  - **Unilateral Trade Friction:** Creation of a 3-year trade dialogue highlights growing retaliatory friction against green tariffs like the EU's Carbon Border Adjustment Mechanism.

Way Forward for UNFCCC COP 30

- **Mitigation & Energy**
  - **Enforce NDC Absolute Caps:** Mandate economy-wide, absolute emission cuts in NDC 3.0 updates rather than intensity-based pledges.
  - **Codify Fossil Fuel Phase-out:** Formally bind the "transitioning away from fossil fuels" text into the core UNFCCC legal agreements.
  - **Grid Infrastructure Scaling:** Synchronize renewable generation capacity additions with immediate grid upgrading targets.
- **Finance Architecture**
  - **Standardize Green Accounting:** Restrict the definition of climate finance strictly to public grants and highly concessional loans.
  - **Fast-track Adaptation Funds:** Cut bureaucratic delays under the Baku-to-Belém roadmap to accelerate localized adaptation capital.
  - **Blended Finance De-risking:** Deploy sovereign guarantees to de-risk and unlock private investment through the FINI platform.
- **Forests & Equity**
  - **Bind Deforestation Targets:** Convert voluntary forest and climate roadmap targets into legally binding non-deforestation laws by 2030.
  - **Mandatary TFFF Funding:** Transition the $125 billion Tropical Forest Facility from a voluntary endowment to a mandatory public fund.
  - **Direct Tribal Payouts:** Bypass state bureaucracies to route the mandated 20% IP&LC funds via direct digital transfers to forest dwellers.
- **Trade & Just Transition**

- **Multilateralize Carbon Tariffs:** Use the UN Trade Forum Dialogue to regulate protectionist trade barriers like the EU's CBAM.
- **Institutionalize BAM:** Upgrade the Belém Action Mechanism into a well-funded, binding framework to protect transition workforces.
- **Circular Economy Integration:** Embed resource circularity goals into global manufacturing and supply chains to tackle hard-to-abate sectors.

Conclusion

UNFCCC COP 30 in Belém concluded as a critical **Implementation COP** that operationalized vital mechanisms like the $125 Billion Tropical Forest Forever Facility (TFFF) and structured indicators for the Global Goal on Adaptation (GGA). However, the Belém Package highlighted deep North-South fractures, failing to secure legally binding resource commitments for the $1.3 Trillion annual climate finance goal or a definitive global roadmap to phase out fossil fuels.

7. CONVENTION ON BIOLOGICAL DIVERSITY (CBD)

Convention on Biological Diversity (CBD) is a legally binding multilateral treaty adopted at the 1992 Rio Earth Summit to protect global ecosystems, species, and genetic resources. It operates under three main goals: conserving biological diversity, ensuring the sustainable use of its components, and fairly sharing the benefits arising from genetic resources.

About Convention on Biological Diversity (CBD)

- **Origin**
  - Adopted at the 1992 Rio Earth Summit; entered into force on December 29, 1993.
  - **Legal Nature:** It is a legally binding multilateral international treaty.
  - **Secretariat:** Headquartered in Montreal, Canada.
  - **Membership Exceptions:** Near-universal participation (196 parties); notably, the United States has signed but never ratified it.
- **Three Core Pillars**
  - **Conservation:** Preserving global biological diversity across ecosystems and species.
  - **Sustainable Use:** Ensuring the components of biological diversity are used sustainably.
  - **Fair & Equitable Benefit Sharing:** Ensuring the equitable distribution of benefits arising from genetic resources.
- **Supplementary Protocols**
  - **Cartagena Protocol (2000):** Governs Biosafety, specifically the transboundary movement, handling, and use of Living Modified Organisms (LMOs).
  - **Nagoya Protocol (2010):** Focuses on Access and Benefit Sharing (ABS) for genetic resources and associated traditional knowledge.
  - **Nagoya-Kuala Lumpur Supplementary Protocol (2010):** Focuses on international liability and redress for damages caused by LMOs.
- **Recent Targets & Mechanisms (COP 15 & COP 16)**
  - **Successor Framework:** Kunming-Montreal Global Biodiversity Framework (KMGBF) replaced the failed Aichi Targets, mapping 4 goals (for 2050) and 23 targets (for 2030).
  - **30x30 Target:** Explicitly mandates conserving at least 30% of the world's terrestrial, inland water, coastal, and marine ecosystems by 2030.
  - **Global Environment Facility (GEF):** Acts as the official designated financial mechanism channeling funds to developing states.

Recent Developments at the Convention on Biological Diversity (CBD)

Outcome	Primary Purpose	Launched By / Under
Operationalization of the Cali Fund	Ensure industries using digital genetic data (Digital Sequence Information - DSI) pay into a global conservation fund.	Finalized at the Resumed Session of COP 16 (Rome, 2025).
Establishment of Permanent Financial Mechanism	Create a long-term, democratic UN-governed structure to replace the interim Global Environment Facility (GEF) by 2030.	Formally adopted under Article 21 of the Convention.
Subsidiary Body on Article 8(j)	Grant Indigenous Peoples and Local Communities (IP&LCs) a permanent, institutional role in all future UN biodiversity decisions.	Adopted by the General Plenary of COP 16.
KMGBF Monitoring Framework	Finalize a core package of indicators to quantitatively track national progress against the 23 global targets for 2030.	Decision 16/31 under the Kunming-Montreal Framework.
Headline Indicator 22.1 Integration	Track the safety and legal security of land tenure directly held by indigenous populations worldwide.	Formally integrated under Target 22 of the KMGBF.
EBSA Identification Breakthrough	Streamline the scientific process to identify Ecologically or Biologically Significant Marine Areas across international waters.	Finalized under the CBD Marine Conservation track.
7th and 8th National Reporting Template	Mandate strict, standardized templates for country compliance reports due ahead of the upcoming mid-term review.	Decision 16/32 to guide the upcoming COP 17 (2026).
Kunming Biodiversity Fund (KBF)	Distribute a specialized $200 million grant to fast-track ecosystem restoration projects across developing countries.	Formally launched with capital support from the Government of China.
Synthetic Biology Thematic Action Plan	Provide technical guidelines to help developing nations safely evaluate, regulate, and use gene editing technologies.	Structured under the CBD Synthetic Biology Open-Ended Forum (2026).
Invasive Alien Species Cross-Border Guidelines	Harmonize international trade and e-commerce rules to halt the transboundary spread of ecologically destructive species.	Adopted as global trade guidelines aligned with Target 6 of KMGBF.
Renewed ITTO Partnership to 2029	Coordinate global technical projects to enhance landscape restoration and expand protected tropical forest reserves.	Joint Memorandum of Understanding between CBD Secretariat and ITTO (2026).

MDB Biodiversity Disclosures	Press Multilateral Development Banks and large companies to fully disclose financial dependencies and risks related to nature loss.	Adopted under the Revised Strategy for Resource Mobilization.

Significance of CBD

- **Global Environmental Governance & Sovereignty**
  - **Umbrella Architecture:** Serves as the premier, legally binding framework anchoring global ecosystem conservation, sustainable usage, and bio-resource equity.
  - **Affirming Resource Sovereignty:** Displaced the **common heritage of mankind** doctrine, establishing nations' absolute sovereign rights over domestic biological resources.
  - **Anti-Biopiracy Safeguards:** Created the legal mechanism (via the Nagoya Protocol) requiring Prior Informed Consent before extracting domestic genetic resources.
- **Socio-Economic and Indigenous Rights**
  - **Rights-Based Custodianship:** Explicitly mandates the preservation of traditional environmental knowledge, positioning indigenous communities as central conservation stakeholders.
  - **Direct Equity Devolution:** Codified through recent mechanisms (like the Cali Fund) that corporate commercial profits from digital genetic data must be financially shared with local dwellers.
  - **Gender Mainstreaming:** Formally integrates and safeguards women's leadership roles in rural landscape management and local resource governance.
- **Financial and Scientific Integration**
  - **Quantified Conservation Targets:** Shifted biodiversity management from generic aims into highly scientific, time-bound objectives like the *30x30 Kunming-Montreal Target*.
  - **Bridging the Finance Gap:** Establishes multilateral funding tracks (such as the Global Biodiversity Framework Fund) to mobilize $200 Billion annually by 2030 for the Global South.
  - **One Health vector:** Legally links ecosystem vitality directly to global public health to prevent the emergence of zoonotic diseases.
- **Institutional Compliance & Value for India**
  - **Enabling Act Enactment:** Provided the foundational global obligation that led to India enacting its statutory *Biological Diversity Act, 2002*.
  - **Decentralized Local Policing:** Enabled India to build its unique three-tier governance system down to local *Biodiversity Management Committees*.
  - **Global South Consolidation:** Empowers India to lead mega-biodiverse coalitions against protectionist green trade barriers while protecting traditional medicine patents.

Issues and Challenges with the Convention on Biological Diversity (CBD)

- **Structural and Enforcement Gaps**
  - **Non-Binding Nature:** Lacks top-down penalizing mechanisms or a compliance court, making national target implementation purely voluntary.
  - **Aichi credibility:** Suffered severe institutional loss of credibility after member states failed to fully meet a single one of the 20 Aichi Targets by 2020.
  - **NBSAP Deficit:** Over 70% of participating parties failed to submit their finalized National Biodiversity Strategies and Action Plans on time.
  - **Geopolitical Incompleteness:** The continuous refusal of the United States to ratify the treaty limits global legal and financial leverage.

- **Finance Architecture Bottlenecks**
  - **Massive Funding Chasm:** Faces an estimated $700 Billion annual conservation funding gap, with no clear global strategy to bridge it.
  - **Global North Foot-Dragging:** Developed nations are lagging behind the target to mobilize $20 Billion annually for developing states.
  - **Voluntary Capitalization Flaws:** New frameworks (like the Cali Fund and Kunming Fund) rely on voluntary or non-binding corporate donations.
  - **Interim Mechanism Friction:** Developing countries oppose the dominance of the Global Environment Facility (GEF), demanding a decentralized fund.
- **Target Implementation & Metric Friction**
  - **30x30 Exclusion Risk:** Rush to protect 30% of land and sea by 2030 risks triggering the forced displacement of forest-dwelling communities.
  - **Species Recovery Bias:** National reporting disproportionately focuses on high-profile, charismatic flagship species while neglecting micro-flora and insects.
  - **Incomplete Monitoring Matrices:** Contentious debates over indicators for genetic resources have left global tracking frameworks functionally incomplete.
  - **Invasive and Chemical Neglect:** Lacks rigorous, quantitative tracking to measure pesticide reductions or control the spread of invasive alien species.
- **Socio-Economic & Technical Divides**
  - **Digital Sequence Information (DSI) Loopholes:** Open-access digital gene mapping allows corporations to bypass physical benefit-sharing laws.
  - **Bureaucratic IP&LC Barriers:** Strict state reporting standards often exclude or ignore independent conservation data gathered by local tribes.
  - **Synthetic Biology Rift:** Deep North-South divides persist over regulating advanced gene-editing technologies and synthetic biology applications.
  - **Land Degradation Outpacing:** Rapid expansion of commercial monoculture agriculture is driving global land degradation faster than restoration projects can repair it.

Way Forward for the Convention on Biological Diversity (CBD)

- **Tightening Enforcement & Target Compliance**
  - **Codifying Penal Enforcement:** Transition from purely voluntary targets to legally binding domestic compliance matrices with penalties for environmental degradation.
  - **Streamlining NBSAP Submissions:** Mandate strict, time-bound deadlines for updating National Biodiversity Strategies and Action Plans with independent review tracking.
  - **Institutionalizing Universal Audits:** Establish a mandatory peer-review compliance dashboard similar to the UNFCCC's transparency frameworks.
- **Restructuring the Financial Architecture**
  - **Closing the $700 Billion Gap:** Phase out global harmful subsidies (like chemical fertilizer and fossil fuel incentives) by at least $500 billion annually to redirect funds.
  - **Mandating the Cali Fund Payouts:** Transition corporate contributions for using digital genetic data (DSI) from voluntary percentages to mandatory statutory levies.
  - **Decentralizing Fund Access:** Establish an autonomous, democratic UN Biodiversity Fund to bypass bureaucratic roadblocks within the Global Environment Facility (GEF).
- **Protecting Human Rights & Local Equity**
  - **Securing Rights-Based Conservation:** Implement a strict, non-negotiable legal framework ensuring "Free, Prior, and Informed Consent" to prevent land grabs under the 30x30 target.

- **Direct Financial Devolution:** Create frictionless digital banking channels to route at least 20% of global conservation funds directly to indigenous and tribal councils.
- **Mainstreaming Gender Parity:** Legally mandate equal representation and leadership roles for rural women within community-led conservation committees.

- **Advancing Science & Technical Integration**
  - **Regulating Synthetic Biology:** Create an international, legally binding bio-safety protocol specifically designed for tracking gene drives and synthetic biology.
  - **Curbing Invasive Species:** Harmonize international e-commerce and maritime trade regulations to implement real-time biosecurity screening at national borders.
  - **Scaling One Health Frameworks:** Embed ecosystem monitoring directly into national epidemiological dashboards to identify and prevent zoonotic disease spillovers.

Conclusion

Convention on Biological Diversity (CBD) remains the premier global legal framework for reversing ecosystem degradation and establishing bio-resource equity across 196 nations. While structural challenges like the $700 Billion annual funding gap and the digital exploitation of genetic data persist, frameworks like the Kunming-Montreal Global Biodiversity Framework (KMGBF) offer a highly quantifiable roadmap through targets like “30x30.”

MGP/BP/GS3/015

WATER AND RIVER CONSERVATION

1. WATER SECURITY

Water security is the reliable availability of an adequate quantity and acceptable quality of water to sustain livelihoods, human well-being, and socio-economic development. It demands safeguarding ecosystems against water-borne pollution while ensuring absolute protection of lives and property from water-related hazards like floods and droughts.

Current status of water security in India

- **Global Mismatch:** India supports nearly **18% of the global population** but possesses only **4% of the world's freshwater resources** (NITI Aayog).
- **Plummeting Per-Capita Availability:** Annual per-capita water availability has dropped from 5,000 m^3 in 1947 to approximately **1,400 m^3**, pushing India dangerously close to the official water-scarcity threshold of 1,000 m^3 (Ministry of Jal Shakti).
- **Stress Footprint:** Around **60 crore people** across India currently experience high-to-extreme water stress in their daily lives (NITI Aayog).
- **Supply-Demand Gap:** India's aggregate water demand is projected to be **twice the available supply by 2030**, requiring a massive structural infrastructure overhaul (NITI Aayog).
- **Overall Extraction Intensity:** Stage of Ground Water Extraction (SoE) stands at **60.63%** for the country as a whole (Central Ground Water Board - CGWB Report).
- **Total Volumetric Footprint:** The total annual groundwater recharge is **448.52 BCM** (Billion Cubic Meters), while annual extraction sits at **247.22 BCM** (CGWB).
- **Critical Blocks Breakdown:** Out of 6,762 assessed hydrological units in India, **10.8% are classified as "Over-Exploited"** (extraction exceeds 100%, 2.97% are "Critical", and 11.21% are "Semi-Critical" (CGWB).
- **Agrarian Consumption Load:** Agriculture remains the largest consumer, monopolizing **85 - 90% of India's total freshwater resources**, primarily due to flood-irrigated paddy and sugarcane (Ministry of Agriculture).

Significance of water security

- **Agrarian Stability & Macro-Food Security**
  - **De-risking Rainfed Farm Systems:** Insulates over 50% of India's net sown area from monsoon failure, preventing localized farm-income collapses.

    As per ministry of agriculture, Rainfed agriculture constitutes **51%** of India's net sown area and accounts for nearly 40% of national food production
  - **Securing Aggregate Grain Production:** Stabilizes national food output to sustainably feed 1.4 billion people without relying on volatile global import markets.

As per economic survey 2024-25, India's total foodgrain production hit a record **332.2 million tonnes** in FY 2023-24, anchoring macro-food security for **1.4 billion** citizens.

- **Protecting Rural Allied Livelihoods:** Preserves livestock viability, inland fisheries, and dairy operations that serve as primary safety nets for marginal farmers.

As per ministry of fisheries, Fisheries and aquaculture support over **30 million** livelihoods, contributing **7.43%** to agricultural GVA, with total fish output reaching **197.75 lakh tonnes** in FY 2024-25

- **Public Health, Nutrition & Demographic Dividends**
  - **Eliminating Water-Borne Disease Burden:** Drastically reduces public healthcare spending by preventing outbreaks of cholera, typhoid, and diarrheal vectors.

As per WHO, Diarrheal diseases cause over **9%** of all under-5 deaths in India; safe water access is estimated to prevent up to **400,000** of these deaths annually.

  - **Curbing Inter-generational Malnutrition:** Ensures clean water access to break the direct cyclical link between poor sanitation, gut infections, and child stunting.

According to NFHS-5, **35.5%** of children under 5 are stunted; access to clean sanitation and safe water reduces environmental enteropathy (gut infections) by **25-30%**

  - **Unlocking Female Economic Agency:** Saves millions of daily multi-hour labor journeys for rural women, freeing time for education and micro-enterprises.

Jal Jeevan Mission has achieved **82.07%** rural household tap connection coverage as of 2026, saving women an average of **1.5 to 2.5 hours daily**

- **Industrial Growth & Urban Sustainability**
  - **Powering Manufacturing Infrastructure:** Secures critical cooling and processing lines for heavy industries like steel, chemicals, and data centers.

As per CWC, Industrial sector water demand is projected to jump to **92 billion cubic meters (BCM)** by 2050, accounting for nearly **9%** of India's total water usage

  - **Sustaining Thermal & Green Energy:** Prevents costly operational shutdowns at thermal power stations and green hydrogen production plants due to dry spells.
  - **Preventing Urban Metros from Collapsing:** Future-proofs massive urban economic engines (e.g., Bengaluru, Chennai) from catastrophic groundwater depletion crises.

As per CGWB, Over **60%** of urban India's water requirements are met by groundwater, yet **25%** of assessed groundwater blocks in major urban metros are categorized as over-exploited.

- **Ecological Integrity & Geopolitical Peace**
  - **Preserving Riverine & Wetland Ecosystems:** Maintains minimum ecological river flows necessary to save native biodiversity, aquatic biomes, and natural carbon sinks.
  - **Averting Sub-National Inter-State Conflicts:** Minimizes political friction and water-sharing disputes across critical river basins like the Cauvery, Krishna, and Mahanadi.
  - **Strengthening Transboundary Diplomacy:** Anchors regional stability and reinforces India's upper-riparian credibility under agreements like the Indus Waters Treaty.

Issues undermining water security in India

- **Resource Depletion & Hydro-Geological Stress**
  - **Catastrophic Groundwater Depletion:** Over-exploitation has placed nearly 30% of India's assessment blocks in semi-critical, critical, or over-exploited categories.
  - **Rapid Encroachment of Catchments:** Real estate speculation and unplanned urban growth routinely choke natural floodplains, lakes, and traditional Stepwells.

As per **First Census on Water Bodies**, Across India, 1.6% of total enumerated water bodies (38,496 structures) are officially reported as encroached.

- **Intensifying Climate Change Anomalies:** Erratic monsoon behavior accelerates spatial-temporal water imbalances, triggering simultaneous floods and droughts across regions.

As per **NITI Aayog ,**Average annual per capita water availability has dropped below 1,486 cubic meters, pushing India into a structurally water-stressed state.

- **Agrarian Distortion & Efficiency Deficits**
  - **Highly Inefficient Irrigation Footprint:** Flood irrigation remains the dominant practice, trapping agricultural water-use efficiency at a low 30-38%.

As per **Department of Agriculture & Farmers Welfare**, Traditional surface/flood irrigation retains field-level water-use efficiency at a low 30–38%.

  - **Subsidized Cropping Disconnection:** Power and fertilizer subsidies incentivize water-guzzling crops like paddy and sugarcane in critically water-scarce aquifers.

As per NITI Aayog, Agriculture consumes over 80% of India's total extracted freshwater, heavily skewed toward water-intensive paddy and sugarcane crops.

  - **Low Micro-Irrigation Adoption Penetration:** High initial capital costs discourage small and marginal farmers from transitioning to drip and sprinkler systems.

As per Ministry of Agriculture, 2025, Total micro-irrigation coverage under the Per Drop More Crop *scheme spans 11.09 million hectares, accounting for just 7.98% of India's net sown area.*

- **Pollution, Quality & Public Health Hazards**
  - **Widespread Geogenic Contamination:** Toxic levels of arsenic, fluoride, and salinity plague deep groundwater aquifers across the Indo-Gangetic plains and arid belts.

As per CGWB, Groundwater in 440 districts is contaminated with excessive nitrates, alongside severe arsenic and fluoride concentrations across the Indo-Gangetic plains.

  - **Untreated Domestic Sewage Dumping:** Over 60% of urban municipal wastewater is discharged raw into rivers, turning major water bodies into toxic ecological dead zones.

As per CPCB, Out of 72,368 million liters per day (MLD) of urban sewage generated, only 28% (20,236 MLD) gets actually treated, leaving 72% to flow untreated into water bodies.

  - **Heavy Industrial Effluent Discharges:** Ineffective enforcement of zero-liquid discharge norms allows untreated chemical runoff to infiltrate shallow community aquifers.

Non-compliance with Zero Liquid Discharge (ZLD) norms causes heavy metal contamination, leaving nearly 300 river stretches severely polluted

- **Policy, Governance & Transboundary Friction**
  - **Fragmented Administrative Jurisdictions:** Water management is crippled by a lack of coordination between the Ministries of Jal Shakti, Agriculture, and Urban Development.
  - **Escalating Inter-State Water Conflicts:** Constitutional delays in resolving river-basin disputes (e.g., Cauvery, Krishna, Mahadayi) stall cohesive watershed management.
  - **Geopolitical Vulnerabilities:** Upper-riparian hydro-hegemony and unilateral damming on transboundary rivers (e.g., China on the Brahmaputra) threaten downstream water flows.

Water stress

Water stress occurs when the demand for water exceeds the available amount during a certain period or when poor quality restricts its use. It is a measure of the "tightness" of the water situation in a specific region, calculated as the ratio of total water withdrawals to the available renewable supply.

Steps taken by the government to ensure water security

- **Jal Jeevan Mission (JJM):** Provided functional tap connections to over 75% of rural households, ensuring decentralized, safe drinking water security.
- **Mission Amrit Sarovar:** Developed and rejuvenated over 75,000 localized water bodies across districts to boost surface water storage capacity.
- **Jal Shakti Abhiyan (Catch the Rain):** Driven nationwide annual campaigns to mandate decentralized rainwater harvesting and artificial aquifer recharge.
- **Interlinking of Rivers (ILR):** Operationalized the landmark Ken-Betwa Link Project to mitigate chronic water scarcity in the drought-prone Bundelkhand region.
- **Atal Bhujal Yojana (ABY):** Implemented a ₹6,000 crore community-led scheme to scientifically manage groundwater across water-stressed blocks.
- **National Aquifer Mapping Program (NAQUIM):** Completed comprehensive 3D mapping of major pan-India aquifers to regulate sustainable groundwater extraction.
- **Crop Diversification Programmes:** Incentivized low-water alternative crops (like pulses and millets) to transition away from water-guzzling paddy and sugarcane.
- **AMRUT 2.0 Framework:** Mainstreamed the Sponge Cities concept and mandated circular water economy principles across all urban local bodies.
- **National Mission for Clean Ganga (NMCG):** Created a massive network of Sewage Treatment Plants (STPs) under Namami Gange to curb riverine pollution.
- **Zero Liquid Discharge (ZLD) Norms:** Enforced strict statutory recycling mandates for highly polluting, water-intensive heavy industries.
- **Unified Jal Shakti Ministry:** Merged fragmented water departments into a single ministry to ensure integrated, holistic water governance.
- **National Water Awards Institutionalization:** Created competitive federal incentives to encourage states, panchayats, and industries to adopt best water practices.
- **National Water Quality Sub-Mission:** Monitored and successfully eliminated geogenic contamination (arsenic/fluoride) across targeted rural habitations.

Water budgeting

Water budgeting is just like tracking your bank account, but with water instead of money. It counts how much water comes into a specific area through rain or pipes, and tracks how much goes out through usage, evaporation, or leaks. Knowing this balance helps communities ensure they never use more water than they actually have available.

Way Forward

- **Decentralized Planning & Digital Governance**
  - **Scaling Up Block-Level Water Budgeting:** Roll out NITI Aayog's *Varuni* application nationwide to calculate local demand-supply gaps across human, livestock, and industrial sectors.
  - **Expanding High-Resolution Aquifer Mapping:** Deploy the NAQUIM 2.0 framework down to the Gram Panchayat level to precisely track and regulate localized groundwater tables.
  - **Institutionalizing the Sujalam Bharat Platform:** Use unique *Sujal Gaon IDs* to track the real-time functionality and asset health of village water supply schemes in a central digital network.
- **Transitioning to a Circular Water Economy**
  - **Mandating Urban Wastewater Recycling:** Enforce 100% treatment of municipal sewage under AMRUT 2.0 to convert urban wastewater into recovered income for industrial use.

- **Enforcing Strict Zero Liquid Discharge (ZLD):** Tighten statutory compliance for heavy, water-intensive industries to eliminate the chemical contamination of freshwater bodies.
- **Mainstreaming Sponge Cities Infrastructure:** Integrate permeable pavements, urban wetlands, and green corridors into municipal layouts to maximize local stormwater absorption.

- **Demand-Side Management & Agricultural Reforms**
  - **Decoupling Agriculture from Crop Subsidies:** Restructure Minimum Support Prices (MSP) and power incentives to actively pivot farmers away from water-guzzling paddy and sugarcane in dry tracts.
  - **Harnessing Green Water as a Strategic Asset:** Launch a focused national mission for regenerative farming (mulching, zero-tillage) to maximize the moisture retention capacity of agricultural soils.
  - **Accelerating Bureau of Water Use Efficiency (BWUE) Mandates:** Enforce strict field targets to improve national irrigation water-use efficiency by 20% through micro-drip networks.
- **Policy Integration & Legislative Reforms**
  - **Transitioning to Integrated Water Resource Management (IWRM):** Shift water administrative boundaries away from rigid state borders to follow natural hydrological and river-basin boundaries.
  - **Codifying Common-Pool Resource Laws:** Overhaul legacy property laws by adopting the Model Groundwater Bill to treat aquifers as public trusts rather than private landlord assets.
  - **Institutionalizing Jal Arpan Governance:** Formally hand over completed rural water assets to Gram Panchayats to ensure lasting community-led ownership, maintenance, and democratic equity.

Best practices for water security

- **Hiware Bazar Model (Maharashtra):** A legendary village-scale water-budgeting framework that banned water-guzzling crops to achieve complete drought resilience.
- **Tarun Bharat Sangh Johad Model:** Rejuvenated the Arvari river basin in Rajasthan by constructing thousands of low-cost earthen check dams (*johads*).
- **Paani Foundation "Water Cup" Model:** A highly successful gamified, community-driven decentralized watershed management movement across rural Maharashtra.
- **Lapodia Chauka System (Rajasthan):** A traditional pasture-land management model utilizing interconnected square dykes to optimize rainwater retention and grass growth.
- **Telangana's Mission Kakatiya Model:** A massive structural program focused on de-silting and restoring traditional minor irrigation tanks to boost groundwater tables.
- **Maharashtra's Jalyukt Shivar Abhiyan:** Eradicated water scarcity across thousands of villages by deepening streams, building farm ponds, and constructing concrete bunds.
- **Gujarat's Sujalam Sufalam Abhiyan:** A public-private partnership model for large-scale de-silting of rivers, canals, and reservoirs ahead of monsoons.
- **Rajasthan's MJSY Framework:** *Mukhyamantri Jal Swavlamban Abhiyan* integrated four-waters concept (rain, runoff, ground, soil moisture) for watershed saturation.
- **Puri Sujal Mission (Odisha):** India's first urban center to achieve a 24x7 "Drink from Tap" reality, drastically cutting plastic waste and water losses.
- **Andhra Pradesh's Natural Farming Model (APCNF):** Uses organic bio-inputs and multi-cropping to enrich soil organic carbon, maximizing "green water" (soil moisture) storage.
- **Varuni Digital Model (NITI Aayog):** A real-time data application piloting block-level demand-supply balancing across select water-stressed districts.

Conclusion

Water security is a critical pillar of Viksit Bharat. India's experience shows that drinking water supply, sanitation, groundwater recharge, river conservation, wastewater management, and climate resilience

cannot be addressed separately. Sustained investment, community participation, and integrated water management are essential for securing water resources, improving quality of life, and building long-term national resilience.

2. WATER CONSERVATION IN INDIA

Water conservation in India is the systematic practice of capturing, storing, and efficiently managing hydrological resources to prevent depletion and ecological degradation. Faced with highly erratic monsoon cycles and rapidly falling groundwater tables, conserving water is no longer a choice but a survival strategy.

Significance of Water conservation

- **Macroeconomic & Energy Security**
  - **Sustaining Heavy Industrial Value Chains:** Prevents multi-billion dollar operational bottlenecks in water-intensive sectors like steel, semiconductor fabrication, and data centers.
  - **Insulating Power Grids from Shutdowns:** Guarantees uninterrupted cooling water supplies for thermal reactors and high-volume green hydrogen production plants.
  - **Mitigating Climate-Induced GDP Losses:** Counteracts NITI Aayog's projections of a potential 6% contraction in national GDP driven by acute systemic water scarcity.
- **Agrarian Stability & Food Sovereignty**
  - **Immunizing the Rainfed Farming Belt**: Protects over 50% of India's net sown area from immediate structural crop failures during prolonged monsoon dry spells.
  - **Buffers Against Global Food Shocks:** Ensures self-sufficiency in core cereal production, preventing reliance on volatile and expensive international import markets.
  - **Preserving Rural Allied Farm Incomes:** Stabilizes the livestock, dairy, and aquaculture ecosystems that serve as primary financial shock-absorbers for marginal smallholders.
- **Public Health & Demographic Dividends**
  - **Slashing the Out-of-Pocket Medical Burden:** Curtails public and private healthcare spending by preventing outbreaks of toxic water-borne pathogens.
  - **Breaking the Cycle of Child Stunting:** Eliminates the intestinal infections that directly drive chronic undernutrition and stunting across vulnerable rural districts.
  - **Unlocking Female Economic and Educational Capital:** Ends multi-hour water-fetching journeys, allowing rural women and girls to pursue formal education and micro-enterprises.
- **Hydrological Integrity & Strategic Diplomacy**
  - **Maintaining Essential Ecological Flows:** Protects critical riverine biomes, inland wetlands, and natural carbon sinks from collapsing due to dry-season desiccation.
  - **Preventing Sub-National Civil Friction:** Mitigates escalating political polarization and inter-state water disputes over shared river basins like the Cauvery and Krishna.
  - **Anchoring Transboundary Upper-Riparian Credibility:** Strengthens India's strategic leverage and geopolitical position under existing international treaties like the Indus Waters Agreement.

Issues plaguing water conservation in India

- **Traditional & Nature-Based Infrastructure Neglect**
  - **Decay of Historic Catchments:** Legacy structures like *Ahar-Pynes*, *Stepwells*, and *Johads* are structurally neglected, filled with garbage, or built over.

    As per 1st Water Bodies Census, Out of 24.24 lakh water bodies enumerated in India, 16.3% (approx. 3.94 lakh) are currently non-functional due to structural decay, pollution, or drying up.

- **Siltation of Local Reservoirs:** Unchecked topsoil erosion and lack of periodic de-silting reduce the actual holding capacity of village ponds and tanks.

  As per Ministry of jal Shakti, A substantial 45.2% of surveyed local water bodies across India have never undergone any structural repair or de-silting maintenance.

- **Destruction of Natural Floodplains:** Urban expansion concrete-jackets natural recharge areas, blocking rainwater from seeping back into deep aquifers.

- **Technological & Field-Level Implementation Deficits**
  - **High Initial Micro-Irrigation Capital Costs:** Small and marginal farmers struggle to afford the upfront costs of drip and sprinkler system installations.

    As per ICAR, Only 15.85% of India's ultimate micro-irrigation potential of 105.6 million hectares has been realized due to high initial capital barriers.

  - **Poor Maintenance of Public Rainwater Systems:** Rooftop rainwater harvesting units in urban buildings routinely fail due to choked filters and lack of maintenance.
  - **Low Focus on Evaporation Control:** Shallow open storage structures suffer massive water losses from evaporation during extreme summer heatwaves.

    A massive 72.4% of all surface water resources in India have a spread area of less than 0.5 hectares, making them shallow and extremely vulnerable to high evaporation losses.

- **Socio-Economic & Behavioral Bottlenecks**
  - **Absence of Community Ownership:** A top-down, subsidy-driven administrative approach has weakened traditional community-managed water systems (*Jal Sanskruti*).

    As per water census, Over 44.8% of water bodies are public owned (with 62.4% managed top-down by Panchayats), indicating a drop in community-driven traditional management.

  - **Perception of Water as a Limitless Free Commodity:** Highly subsidized rural power and free urban water supplies discourage conservation behavior.
  - **Lack of Localized Hydro-Geological Awareness:** Panchayats often build check-dams without analyzing local rock strata, causing water to evaporate rather than recharge aquifers.

- **Policy, Regulatory & Institutional Gaps**
  - **Fragmented Administrative Frameworks:** Conservation funds are split across separate ministries (Jal Shakti, Rural Development, Agriculture), preventing integrated watershed development.
  - **Weak Legal Teeth for Water Harvesting Laws:** Municipal laws mandate rainwater harvesting on paper, but enforcement is weak and penalties for violations are rarely applied.

    As per CPCB, 72% of urban sewage (approx. 52,132 million liters per day) flows untreated into water systems due to weak urban structural enforcement.

  - **Absence of Dedicated Maintenance Budgets:** State schemes fund the creation of new water conservation assets but ignore long-term funding for repair and maintenance.

Major interventions implemented by the government for water conservation

- **Mission Amrit Sarovar:** Successfully constructed and restored over 75,000 localized water bodies across districts to expand surface retention.
- **Jal Shakti Abhiyan (Catch the Rain):** Launched annual pre-monsoon campaigns to scale up check-dam repairs and rooftop rainwater structures.
- **Namami Gange & Clean River Missions:** Combined biological conservation with effluent monitoring to restore floodplain wetlands along the Ganga basin.

- **PMKSY - Watershed Development Component:** Financed structural contour trenching, gully plugs, and check-dams across rainfed farm blocks.
- **National Interlinking of Rivers (ILR):** Advanced execution on the Ken-Betwa link to move surplus monsoon run-off to drought-prone reservoirs.
- **Amrit Generation Campaign:** Engaged youth and local communities to map, clean, and protect micro-catchment zones around village schools.
- **Atal Bhujal Yojana (ABY):** Deployed community-led water budgeting across seven water-stressed states to restrict extraction to recharge levels.
- **PMKSY - Per Drop More Crop:** Financed heavily subsidized micro-irrigation systems to reduce agricultural water consumption.
- **Crop Diversification Interventions:** Provided financial incentives for transitioning away from water-guzzling paddy and sugarcane to millets and oilseeds.
- **AMRUT 2.0 (Water-Secure Cities):** Mandated circular economy practices including dual-piping systems and stormwater recycling across urban local bodies.
- **Model Groundwater Bill Enactment:** Pushed states to legislate mandatory rainwater harvesting systems as a prerequisite for building plan approvals.
- **National Bureau of Water Use Efficiency (BWUE):** Set up a dedicated agency to formulate industrial and domestic water-efficiency labeling protocols.

Framework for managing wastewater

- **Decentralized & Nature-Based Treatment Systems (NbS)**
  - **Constructed Wetlands (Root Zone Technology):** Deploy reed-bed ecosystems to filter municipal greywater naturally using marsh plants and microbes.
  - **Phytorid Technology Implementation:** Scale up CSIR-NEERI's patented plant-based system to treat sewage in open drains before it reaches major rivers.
  - **Decentralized Wastewater Treatment Systems (DEWATS):** Install localized, gravity-driven modular units in rural clusters to avoid expensive pumping infrastructure.
- **Urban Circular Economy & Reuse Models**
  - **Mandatory Dual-Piping Infrastructure:** Legislate separate plumbing lines in new urban layouts to reuse treated greywater exclusively for flushing and gardening.
  - **Industrial Wastewater Co-Location Clusters:** Route treated municipal sewage directly to thermal power and manufacturing plants to preserve primary freshwater.
  - **3R Urban Water Mandate:** Embed "Reduce, Recycle, Reuse" into municipal bylaws under AMRUT 2.0 to turn wastewater into a sellable utility asset.
- **Industrial Efficiency & Technology Interventions**
  - **Strict Zero Liquid Discharge (ZLD) Enforcement:** Force heavy, water-intensive factories to completely purify and reuse their entire effluent volume on-site.
  - **Upgrading to Membrane Bioreactors (MBR):** Transition municipal treatment plants from basic filtration to high-tech MBR systems for ultra-clean water recovery.
  - **Common Effluent Treatment Plants (CETPs):** Establish shared, state-of-the-art purification hubs to handle chemical runoffs for micro, small, and medium enterprises (MSMEs).
- **Policy, Regulatory & Economic Reforms**
  - **Setting National Water Quality Standards:** Empower the CPCB to enforce strict, uniform purity benchmarks for discharging treated water into aquifers.
  - **Introducing Polluter Pays Fiscal Penalties:** Levy heavy statutory fines on municipal bodies and industries that fail to meet minimum effluent treatment quotas.
  - **Financing via Green Municipal Bonds:** Issue specialized public bonds to raise low-cost capital for building advanced city-wide sewage treatment networks.

Way Forward

- **Technological Interventions & Digital Mapping**
  - **Expanding Smart Hydro-Informatics:** Deploy IoT-based meters and AI sensors across canal networks to rapidly detect and plug distribution leakages.

    Karnataka's Hubbali-Dharwad 24x7 water supply project uses automated metering to cut non-revenue water.
  - **Scaling Gram Panchayat Water Budgeting:** Roll out digital estimation tools nationwide to balance crop-water demand with local aquifer capacity.

    Australia's Murray-Darling Basin uses digitized local water accounting tools for community allocation.
  - **Upgrading High-Tech Aquifer Models:** Leverage high-resolution NAQUIM 2.0 mapping to design precision artificial recharge structures based on specific sub-surface rock strata.

    CGWB's heliborne geophysical surveys in Rajasthan's arid zones map precise paleochannels for targeted recharge
- **Nature-Based Solutions & Structural Infrastructure**
  - **Mainstreaming Sponge Cities Mandates:** Integrate permeable concrete, urban wetlands, and bioswales into municipal master plans to capture flash stormwater.

    China's Wuhan Sponge City program utilizes urban wetlands and green roofs to absorb 70% of rainwater.
  - **Reclaiming Riverine Floodplains:** Enforce strict eco-sensitive "No-Construction Zones" along river basins to safeguard natural monsoon riparian recharge.

    Netherlands' Room for the River program deliberately lowers floodplains to safely accommodate high water levels.
    - **Restoring Traditional Hydrological Sinks:** Allocate dedicated institutional funding to systematically de-silt and interconnect legacy structures like *Ahar-Pynes* and *Johads*.

    Bihar's revival of Ahar-Pynes and Rajasthan's rejuvenation of Johads via the Tarun Bharat Sangh movement.
- **Agri-Economic Policy & Regulatory Reforms**
  - **Restructuring Irrigation Energy Subsidies:** Transition from flat power subsidies to Direct Benefit Transfers (DBT) directly tied to verifiable groundwater savings.

    Mexico's Procampo restructured agricultural subsidies to decouple production volume from resource extraction.
  - **Aggressively Pushing "Green Water" Farming:** Incentivize zero-tillage, mulching, and agroforestry to maximize the organic moisture-retention capacity of farm soils.
  - **Enforcing Water-Efficiency Star Ratings:** Task the Bureau of Water Use Efficiency (BWUE) with making consumption labels mandatory for appliances and industrial pumps.

    Australia's mandatory Water Efficiency Labelling and Standards (WELS) scheme on consumer appliances.
- **Decentralized Governance & Behavior Change**
  - **Transforming Conservation into a *Jan Andolan*:** Expand participatory grassroot movements to revive traditional, community-led water conservation ethos (*Jal Sanskruti*).
  - **Empowering Women-Led Water Committees:** Mandate 50% female representation in *Pani Samitis* to leverage local knowledge for managing neighborhood water security.

- **Codifying Aquifers as Common Pool Resources:** Implement the Model Groundwater Bill across states to legally treat sub-surface water as a public trust rather than private land property.

Committee recommendations for water conservation in India

Committee	Core Recommendation
Mihir Shah Committee	Merge CWC & CGWB into a unified National Water Commission for integrated basin-wide water conservation.
NITI Aayog (CWMI)	Establish competitive federal ranking via the Composite Water Management Index to push states toward conservation.
Model Groundwater Bill	Legally separate groundwater rights from land ownership to treat aquifers as a Common Pool Resource.
Ashok Dalwai Committee	Replace flat agricultural power subsidies with Direct Benefit Transfers (DBT) to curb groundwater over-pumping.
Swaminathan Commission	Mandate localized, crop-specific water budgeting via community-led Aquifer Water User Associations.
Kasturirangan Committee	Enforce statutory bans on water-intensive commercial projects within Ecologically Sensitive Areas (ESAs).

Conclusion

Transforming water conservation from a bureaucratic mandate into a technology-driven *Jan Andolan* (people's movement) is the ultimate key to unlocking India's hydrological resilience. By integrating nature-based solutions like "Sponge Cities" with cutting-edge digital aquifer mapping, the nation can effectively decouple economic growth from groundwater depletion.

3. RIVER CONSERVATION IN INDIA

River conservation in India requires a structural shift from treating rivers as passive economic dumping grounds to respecting them as living, dynamic ecosystems. By blending traditional community stewardship with cutting-edge, nature-based engineering like riverfront wetlands, we can restore essential ecological flows and self-purifying capacities.

Current status of river health and water quality in India

- **Identified Polluted Stretches:** Central Pollution Control Board has mapped **296 polluted river stretches across 271 distinct rivers** nationwide ***(Source: CPCB Report)*.**
- **Critical Biological Collapse:** Number of "Priority I" critically polluted river stretches (where Biochemical Oxygen Demand (BOD) exceeds an alarming 30 mg/Lhas **dropped from 45 to 37** ***(Source: CPCB)*.**
- **Regional Risk Concentration: Maharashtra leads the country with 54 polluted river stretches**, heavily driven by dense industrial corridors and urban discharge networks ***(Source: CPCB)*.**
- **Domestic Sewage Gap:** India generates over 72,000 million liters of sewage per day (MLD), but municipal systems leave nearly **70% of urban sewage completely untreated**, directly polluting river streams ***(Source: Central Pollution Control Board)*.**
- **Metal Contamination:** Long-term hydrological monitoring reveals a significant presence of heavy metals like **iron, lead, and nickel exceeding permissible BIS limits** across multiple river basins ***(Source: Central Water Commission - CWC Trace & Toxic Metals Report)*.**

Strategic significance of river conservation

- **Ecological Integrity & Climate Resilience**
  - **Preserving Riverine Biomes:** Safeguards unique, threatened aquatic wildlife like the Ganges River Dolphin and Gharials from local extinction.
  - **Maintaining Critical Eco-Flows:** ensures Rivers retain their natural self-purifying capacity to dilute and process localized pollutants.

    As per CPCB Guidelines, Maintaining a baseline minimum environmental flow (E-flow) of **15% to 20%** ensures rivers dilution capacity to naturally self-purify localized biological waste.
  - **Sustaining Blue Carbon Sinks:** Protects vital deltaic and estuarine ecosystems like the Sundarbans that act as massive coastal carbon shields.

    Protecting riverine flows sustains vital blue carbon sinks like the Sundarbans, which sequester **4.71 to 6.54 Mg C per hectare annually** and act as natural coastal barriers.
- **Macroeconomic & Food Security**
  - **Anchoring Alluvial Agrarian Belts:** Secures permanent surface-water irrigation networks for the Indo-Gangetic and major peninsular fertile plains.
  - **Driving Inland Water Transport:** Stabilizes dry-season water depth channels to support low-cost, fuel-efficient inland cargo navigation.

    As per the Inland Waterways Authority of India, Reliable dry-season river depths support fuel-efficient inland cargo transport, which moved a record **145.84 MMT** of cargo via National Waterways in FY25.
  - **Powering Sustainable Grid Energy:** Guarantees predictable, silt-free run-of-the-river flow patterns required for reliable hydroelectric energy generation.

    Total operational large-scale hydroelectric capacity stands at **51,664.66 MW** (approx. 51.6 GW). Of this, the state-run NHPC holds a **16.98%** share (**8,771.20 MW** across 24 stations).
- **Civilizational, Social & Cultural Capital**
  - **Shielding Public Health Metrics:** Lowers the incidence of lethal water-borne epidemics by eliminating toxic pathogens from primary water channels.
  - **Sustaining Traditional Blue Economies:** Protects the livelihoods of millions of inland artisanal fishing communities and river-dependent populations.
  - **Preserving Sacred Heritage Spaces:** Safeguards the deep cultural identity, ritual landscapes, and historical roots of India's civilizational river banks.
- **Hydrological Balance & Geopolitics**
  - **Recharging Adjoining Rural Aquifers:** Promotes continuous lateral sub-surface water seepage to naturally replenish collapsing regional groundwater tables.
  - **Curbing Downstream Marine Degradation:** Restores healthy freshwater-sediment discharges to prevent sea-water intrusion and coastal delta erosion.
  - **Anchoring Riparian Treaty Leverage:** Strengthens transboundary hydro-diplomatic status and compliance metrics with lower-riparian neighboring nations.

Issues undermining river conservation in India

- **Hydrological & Flow Deficits**
  - **Severe Lean-Season Flow Depletion:** Massive upstream dam diversions for agriculture Leave Rivers without the basic ecological flows (*E-flows*) needed to self-purify.

According to NGT, 15% to 20% baseline minimum environmental flow (E-flow) is legally mandated during lean seasons but routinely breached.

- **Aggressive Siltation of Riverbeds:** Deforestation in upper catchments triggers heavy soil erosion, raising riverbeds and magnifying monsoon flood risks.

As per ISRO, Catchment deforestation has accelerated riverbed siltation, slashing the water-carrying capacity of primary channels by up to **30%**.

- **Unregulated Riverbed Sand Mining:** Illegal mechanised mining destroys natural river geometry, lowers groundwater tables, and erodes stable banks.

- **Anthropogenic & Industrial Pollution**
  - **Massive Municipal Sewage Deficit:** Urban centers release nearly 70% of their municipal wastewater raw, turning rivers into open sewer channels.

According to CPCB, India generates 72,368 MLD of urban sewage, but treatment capacity stands at just **44%**, leaving over **55%** to enter rivers raw.

  - **Lax Industrial Effluent Enforcement:** Flagrant violations of Zero Liquid Discharge (ZLD) by factories dump toxic heavy metals into river catchments.
  - **Non-Point Agricultural Runoff:** Heavy chemical pesticide and fertilizer residues wash from farm fields into streams, causing widespread eutrophication (oxygen choking).
- **Structural & Encroachment Challenges**
  - **Concrete-Jacketing of Floodplains:** Unplanned urban real estate encroaches on natural floodplains, blocking lateral aquifer recharge and choking river lifelines.

According to NITI Aayog, Unplanned real estate expansion concrete-jackets critical floodplains, blocking up to **60%** of lateral sub-surface aquifer recharge.

  - **Destructive Micro-Dam Fragmentation:** Proliferation of run-of-the-river hydro projects fragments river paths, blocking the migration of aquatic species.
  - **Disappearance of Feeder Wetlands:** Encroachments systematically eliminate the localized lakes and marshes that naturally feed rivers during dry seasons.

According to the Ministry of Jal Shakti, Over **38,400 water bodies (1.6%)** nationwide are explicitly recorded as encroached

- **Policy, Legal & Institutional Gaps**
  - **Fragmented Administrative Jurisdictions:** Split control between the center, states, and ministries prevents cohesive, single-entity river basin management.
  - **Paralysis via Inter-State Disputes:** Political polarization over water sharing blocks neighboring states from executing collaborative river clean-up plans.

As per ISRWD Tribunal Logs, More than **8 major trans-boundary river systems** remain locked in protracted legal stalemates, paralyzing inter-state cleanup plans.

  - **Weak Punitive Enforcement Powers:** Environment protection bodies lack swift, criminal enforcement mechanisms to permanently shut down major polluters.

Measures taken river conservation in India

- **Macro-Level Infrastructure & Pollution Abatement**
  - **Namami Gange Programme (Phase II):** Scaled up sewage infrastructure to create over 6,600 MLD of operational treatment capacity across the Ganga basin.
  - **National River Conservation Plan (NRCP):** Extended central-sector funding to treat municipal sewage across major non-Ganga rivers like the Musi, Mahanadi, and Godavari.
  - **National Framework for Safe Reuse of Treated Water:** Mandated states to recycle municipal sewage for industrial and thermal cooling to protect primary freshwater flows.

- **AMRUT 2.0 and SBM-Urban 2.0 Integration:** Combined urban renewal funds to block city drains from discharging raw waste directly into local river basins.

- **Basin-Level Governance & Hydrological Restoration**
  - **River Basin Management (RBM) Scheme:** Extended the operational scheme with an enhanced multi-crore outlay to manage river basins as integrated hydrological units.
  - **National Interlinking of Rivers (ILR) Project:** Advanced field execution on the Ken-Betwa link while finalizing Detailed Project Reports for other inter-basin transfers.
  - **Mandatory Ecological Flow (E-Flows) Notification:** Institutionalized statutory minimum flow requirements for the Ganga and major tributaries to maintain lean-season survival.
  - **National Clean Air & River Inter-Sectoral Mission:** Linked riverfront development with green urban belts to capture atmospheric and surface runoff simultaneously.
- **Aquatic Biodiversity & Eco-Restoration**
  - **Project Dolphin Initiatives:** Launched high-tech, range-wide population mapping for the endangered Gangetic and Indus river dolphins.
  - **Large-Scale River Ranching Programs:** Released millions of native fish fingerlings into major river systems to biologically restore food webs and ecology.
  - **Catchment Area Forestry Interventions:** Funded extensive, targeted tree-plantation drives along riverbanks to control severe topsoil erosion and siltation.
  - **Aquatic Ecotoxicology Monitoring:** Established dedicated research centers under the Wildlife Institute of India to track microplastics and chemical toxicity in river life.
- **Digital Technology & Grassroots Engagement**
  - **PRAYAG Real-Time Monitoring Dashboard:** Operationalized an online IoT platform to continuously track water quality and sewage treatment plant efficiencies.
  - **Nature-Based In-Situ Filtering:** Scaled up eco-friendly Constructed Wetlands and Phytorid systems to treat toxic drain water naturally before river confluence.
  - **District Ganga/River Committees:** Activated localized district-level administrative panels to execute monthly, monitored cleanup activities.
  - **Ganga Praharis & Community Stewardship:** Trained thousands of village volunteers to serve as ground-level guardians for reporting illegal dumping and poaching.

Way Forward

- **Institutional Framework & Basin Re-Engineering**
  - **Enacting the River Basin Management Bill:** Transition from rigid, fragmented state administrative boundaries to a single, unified Integrated River Basin Management (IRBM) model.
  - **Codifying Dynamic E-Flow Thresholds:** Legally enforce seasonal Ecological Flow (*E-flows*) baselines that fluctuate based on real-time monsoon variations to maintain self-purification.
  - **Establishing a Federal River Bureaucracy:** Merge independent oversight groups into a powerful, statutory National River Authority with direct environmental policing powers.
- **Nature-Based Solutions & Catchment Defense**
  - **Mandating Green River Corridors:** Implement wide, continuous afforestation buffer strips along riverbanks to naturally filter agricultural pesticide run-off and trap silt.
  - **Restoring Urban Riparian Floodplains:** Declare strict No-Development Eco-Zones on river floodplains to sustain natural absorption during heavy seasonal floods.
  - **Reviving Tributary Springsheds:** Fund targeted ridge-to-valley watershed treatments in upper catchments to keep micro-tributaries and feeding streams flowing year-round.

- **Industrial Enforcement & Circular Economy**
  - **Pushing Strict Zero Liquid Discharge (ZLD):** Mandate closed-loop recycling systems across all heavily polluting sectors like leather, textiles, and chemical manufacturing.
  - **Scaling Decentralized Nature-Based STPs:** Deploy cost-effective Constructed Wetlands and Phytorid plants to naturally treat city sewers before they hit major water channels.
  - **Creating Municipal Treated Reuse Markets:** Monetize treated urban wastewater by feeding it directly to nearby industries and thermal plants to spare raw river water.
- **Legal Protections & Community Stewardship**
  - **Granting Statutory Rights to Living Rivers:** Explore robust legal frameworks that grant specific, endangered river systems legal personhood to fiercely penalize toxic dumping.
  - **Expanding the Arth Ganga Revenue Model:** Link river conservation with local livelihoods by actively promoting sustainable river tourism, organic farming, and medicinal plantations.
  - **Institutionalizing the 4M District Matrix:** Empower local citizens by standardizing **Monthly, Mandated, Minuted, and Monitored** field cleanup panels across every river-facing district.

Committee recommendations for river conservation in India

Committee	Core Recommendation
Mihir Shah Committee	Replace CWC and CGWB with a single *National Water Commission* to manage river basins as unified hydrological units.
Madhav Chitale Committee	Formulate a National Sediment Management Policy to systematically de-silt riverbeds and prevent severe monsoon flooding.
Kasturirangan Committee	Impose absolute bans on hydro-projects and polluting industries within the *Ecologically Sensitive Areas (ESAs)* of Western Ghats rivers.
National Green Tribunal (NGT)	Mandate a strict 100-meter "No-Development Zone" from the high-flood line of major rivers to preserve natural floodplains.
Central Water Commission (CWC)	Enforce legal, basin-specific minimum *Ecological Flows (E-flows)* during lean summer months to prevent river drying.
Parliamentary Stand. Committee	Link all river clean-up grants directly to the installation of real-time, online effluent monitoring sensors at industrial discharge points.

Best practices for river conservation

Case Study	Best Practice
Whanganui River Model (New Zealand)	Granted full legal personhood to the river, allowing indigenous guardians to sue polluters directly for environmental damage.
River Rhine Action Plan (Europe)	Established transboundary data-sharing and uniform penalty structures across 9 riparian nations to successfully eliminate industrial toxins.
Room for the River (Netherlands)	Shifted from building rigid concrete dykes to deliberately lowering floodplains, giving rivers space to safely expand during monsoons.
Thames Re-oxygenation (UK)	Deployed specialized "Oxygen Bubbler" boats to inject compressed oxygen directly into dead urban river zones during hot summer spells.

See in a New Light (Seoul, South Korea)	Completely demolished a multi-lane highway to un-earth and restore the buried, natural Cheonggyecheon river ecosystem.
Yalong River Cascade Model (China)	Integrated pumped-storage hydro with solar-wind grids to keep river flow dynamics stable without massive dam blockages.
Tarun Bharat Sangh (Alwar, India)	Mobilized village communities to build thousands of traditional mud check-dams (Johads), successfully reviving 5 dead seasonal rivers.
Kalyani River Phytorid Project (India)	Deployed CSIR-NEERI's plant-based wetlands inside open city drains to naturally strip out heavy metals before they hit major rivers.

Conclusion

Transitioning from an anthropocentric paradigm of linear exploitation to an ecocentric "living river" framework is paramount to securing India's hydrological and civilizational future. By intertwining the socio-economic incentives of the *Arth Ganga* model with cutting-edge nature-based solutions, the nation can effectively restore the natural self-purifying pulse of its fragmented basins.

4. GROUND WATER CONSERVATION

Groundwater is India's invisible hydrological lifeline, acting as a massive subterranean shield that sustains over 60% of irrigated agriculture and 85% of rural drinking supplies. However, our pursuit of agrarian growth has triggered an "aquifer bankruptcy," transforming this strategic common-pool resource into a vanishing safety net under the weight of free power and indiscriminate pumping.

Current status of groundwater conservation in India

- **Total Hydrological Inflow:** The nationwide annual groundwater recharge stands at **448.52 BCM (Billion Cubic Meters)**, showing a marginal baseline increase driven by aggressive artificial recharging (Central Ground Water Board - CGWB)**.**
- **Net Extractable Volume:** After accounting for statutory allocations toward natural eco-flows and downstream safety buffers, India's total annual extractable resource is pegged at **407.75 BCM** (Ministry of Jal Shakti)**.**
- **Annual Consumption Quantum:** India's combined annual groundwater extraction for all industrial, domestic, and agrarian sectors has reached **247.22 BCM** (Dynamic GW Resource Assessment Report).
- **National Stage of Extraction (SoE):** The overarching baseline metric for utilization vs. natural replenishment stands at **60.63% at a national scale** (CGWB)**.**
- **Granular Unit Mapping:** Out of the 6,762 localized assessment blocks analyzed across the country, **73.14% (4,946 blocks) are officially classified under the 'Safe' category** (Ministry of Jal Shakti)**.**
- **Critically Over-Exploited Zones:** A total of **730 blocks (10.80%) are designated as 'Over-exploited'**, meaning their annual abstraction rate dangerously exceeds their natural monsoon recharge rate (CGWB)**.**
- **Sectoral Demand Skew:** Agriculture continues to act as the primary structural driver of aquifer depletion, consuming **87% of all extracted groundwater** due to crop choices like paddy and sugarcane (CGWB Annual Quality Report)**.**

Significance of groundwater conservation

- **Macroeconomic Stability & Food Security**

  - **Sustaining Agrarian Food Bowls:** Anchors the critical irrigation network that feeds over 60% of India's multi-crop agricultural production.
  - **Insulating Against Monsoon Shocks:** Acts as a strategic subterranean safety buffer to shield farm outputs from climate-induced rainfall variability.
  - **Securing Rural Domestic Supply:** Guarantees decentralized, low-cost drinking water infrastructure for over 85% of the rural population.
  - **Reducing National Subsidy Burdens:** Lowering groundwater tables requires deeper, high-power pumping; conserving water helps curb state power subsidy drains.
- **Ecological Balance & Hydrological Health**
  - **Maintaining Lean-Season River Flows:** Feeds essential base-flows to peninsular and alluvial rivers during dry summer months via lateral subsurface discharge.
  - **Arresting Geogenic Toxin Spikes:** Halts the natural chemical concentration of lethal Arsenic, Fluoride, and Uranium triggered by falling water tables.
  - **Preventing Coastal Saltwater Intrusion:** Maintains strong freshwater hydrostatic pressure to stop marine saltwater from invading and ruining fragile coastal aquifers.
  - **Halting Urban Land Subsidence:** Preserves internal underground pressure to prevent structural soil compaction and the sinking of heavy urban areas.
- **Policy Commitments & Rights-Based Governance**
  - **Anchoring Global SDG Metrics:** Serves as the primary operational indicator to fulfill global commitments under SDG 6 (Clean Water and Sanitation).
  - **Upholding Constitutional Article 21:** Fulfills the state's legal obligation to safeguard the fundamental right to clean, toxin-free life and water.
  - **Driving Decentralized Water Budgeting:** Forms the foundational bedrock for village-level block water mapping under the *Viksit Bharat @ 2047* vision.

Issues undermining groundwater conservation in India

- **Structural, Legal & Regulatory Failures**
  - **Anachronistic Legal Framework:** Colonial-era *Easement Act of 1882* gives absolute extraction rights to landowners, blocking state regulation of private wells.
  - **Perverse Power Subsidy Regime:** Free or heavily subsidized electricity leads to competitive, round-the-clock pumping and massive aquifer depletion.
  - **Fragmented Institutional Control:** Split regulatory oversight between central, state, and local water bodies delays a unified response to the crisis.

    As per CGWA 2024, Splitting responsibilities hinders critical management of India's **7,213 assessment units**, where **11.13% (736 units)** remain chronically overexploited.
- **Anthropogenic & Agrarian Pressures**
  - **Highly Skewed Cropping Patterns:** Minimum Support Price (MSP) incentives favor water-guzzling crops like paddy and sugarcane in water-stressed regions.

    As per NITI Aayog, Agriculture consumes **89%** of India's total extracted groundwater, largely to sustain water-guzzling crops like paddy and sugarcane in water-scarce regions.
  - **Rapid, Unplanned Urbanization:** Pervasive concrete waterproofing across cities blocks natural stormwater infiltration and chokes recharge zones.

    As per NITI Aayog, Concrete waterproofing reduces natural urban recharge zones, aggravating water stress for the **40% of India's urban population.**
  - **Indiscriminate Borewell Proliferation:** Absence of strict licensing has led to millions of deep tube-wells drilling directly into confined, non-renewable aquifers.

As per Minor irrigation census, over **20 million private tube wells/borewells** operating unregulated across India.

- **Environmental & Quality Degradation**
  - **Severe Geogenic Contamination:** Dropping water tables alter underground geochemistry, triggering sharp spikes in toxic Fluoride, Arsenic, and Uranium.

    Acc to CGWB, deep drilling alters geochemistry, causing **132 districts** to exceed the safe limit (30 ppb) for Uranium and **263 districts** to exceed safe limits (1.5 mg/L) for Fluoride
  - **Unregulated Industrial Effluent Injection:** Illegal deep-well injection of untreated factory waste causes irreversible chemical damage to local aquifers.
  - **Unchecked Non-Point Farm Pollution:** Heavy chemical pesticide and fertilizer runoff leaches into shallow water tables, driving widespread nitrate contamination.

    Heavy fertilizer runoff leaches into shallow tables, resulting in **19.8% of all tested groundwater samples** across **443 districts** exceeding the permissible limit (45 mg/L) for Nitrates
- **Technical & Implementation Barriers**
  - **Absence of Local Water Budgeting:** Most village panchayats lack the micro-level hydrological data needed to practice real-time community water budgeting.
  - **Poor Maintenance of Recharge Infrastructure:** Millions of rainwater harvesting systems lie dysfunctional due to siltation and a lack of community ownership.
  - **Subterranean Data Blindspots:** India lacks a widespread, high-density piezometric sensor network to monitor deep aquifer fluctuations in real-time.

Major initiatives for groundwater conservation

- **Jal Sanchay Jan Bhagidari (JSJB) Initiative:** Scaled up community-led construction to complete over 39.6 lakh artificial groundwater recharge and storage structures nationwide.
- **Master Plan for Artificial Recharge to Groundwater:** Formulated a massive layout targeting the setup of 1.42 crore custom structures to capture and channel 185 BCM of monsoon runoff underground
- **Mission Amrit Sarovar**: Executed a countrywide network model to construct or deeply rejuvenate at least 75 high-capacity water ponds per district to boost localized aquifer percolation
- **Jal Shakti Abhiyan (Catch the Rain Campaign):** Deployed multi-phase central funds across all districts to enforce mandatory urban-rural rooftop rainwater harvesting and rapid watershed check-dam building
- **Atal Bhujal Yojana (Atal Jal):** Allocated ₹6,000 crore across 7 highly water-stressed states to implement bottom-up, community-led water budgeting and cropping modifications
- **Model Groundwater Management Bill Adoption:** Drafted and circulated a comprehensive structural framework, successfully convincing 21 states and UTs to legally regulate indiscriminate private borewell drilling
- **Central Ground Water Authority (CGWA) Mandate:** Enforced strict, compulsory "No-Objection Certificates" (NOCs) and digital water-audits for all groundwater-abstracting industrial units
- **District Jal Shakti Kendras (JSKs):** Set up over 712 permanent district-level technical guidance hubs to advise local farming and housing communities on scientific aquifer recharge methods

Improvements (2017–2024)

- Groundwater recharge increased by 15 BCM.
- Extraction decreased by 3 BCM.
- Safe assessment units increased from 62.6% (2017) to 73.4% (2024).
- Over-exploited units declined from 17.24% to 11.13%.
- **(Source: National Compilation on Dynamic Ground Water Resources of India, 2024)**

Way Forward

- **Legal & Regulatory Re-alignment**
  - **Decoupling Land and Water Rights:** Reform the colonial *Easement Act of 1882* to establish groundwater as a national public trust rather than an unrestricted private resource.
  - **National Universalization of the Model GW Bill:** Legally bind all remaining states to create independent state-level water regulators with strong penal enforcement powers.

    Maharashtra's Groundwater Development and Management Act establishes a dedicated authority to strictly regulate deep wells and protect public drinking sources.
  - **Dynamic Ground-Water Extraction Fees:** Scale up the Central Ground Water Authority's (CGWA) consumption fees for heavy commercial and industrial borewells.

    Denmark's green water taxes charge heavy volumetric abstraction fees to commercial and industrial users, drastically reducing industrial groundwater reliance.
- **Fiscal Incentives & Agrarian Demands**
  - **Scaling Up "Paani Bachao, Paisa Kamao" Schemes:** Replicate Punjab's model nationwide by offering direct cash transfers to farmers who consume less electricity.

    Mexico's PEUA program decoupled agricultural electricity subsidies from flat rates, giving cash incentives to farmers who reduced pumping energy.
  - **MSP Realignment with Agro-Climatic Zones:** Shift price-support incentives away from water-guzzling crops like paddy and sugarcane in historically dry, over-exploited blocks.

    Haryana's "Mera Pani-Meri Virasat" provides direct financial assistance to farmers who switch from water-guzzling paddy to maize or pulses.
  - **Mandating Smart Sensor-Based Irrigation:** Expand PMKSY to incentivize IoT-driven soil moisture sensors and subsurface drip networks.

    Israel's Netafim systems integrate automated subsurface drip networks with real-time soil moisture AI to optimize localized root-zone watering.
- **Supply-Side Re-engineering & Urban Defense**
  - **Scaling Village Participatory Water Budgeting:** Integrate Atal Bhujal Yojana's community-led *Water Security Plans* into the mandatory annual budget layout of every Gram Panchayat.

    Brazil's Paraíba do Sul River Basin Committees empower local municipalities to collectively vote on seasonal groundwater and surface water allocations.
  - **Enforcing Urban Sponge City Urbanization:** Mandate permeable concrete pavement, localized bioswales, and urban wetland restoration to capture rapid stormwater runoff underground.
  - **Constructing Deep Subterranean Aquifer Shafts:** Transition from simple surface ponds to building deep recharge shafts that can directly bypass impermeable clay layers to replenish deep confined aquifers.
- **Technological Surveillance & Quality Shields**
  - **Deploying High-Density Telemetric Networks:** Install automated digital water level recorders across all *NAQUIM* mapped zones to transmit real-time groundwater updates to a public portal.

    US High Plains (Ogallala) Aquifer Monitoring Network uses a dense grid of automated telemetry stations to upload real-time water-level data publicly.
  - **Subterranean Geogenic Dilution Injections:** Use intensive localized monsoon-water injection wells to dilute hazardous geogenic spikes of toxic Arsenic, Fluoride, and Uranium.
  - **Mandating Industrial Greywater Cycling:** Enforce strict zero-liquid-discharge norms to ensure treated urban-industrial greywater is injected for groundwater recharge.

Tootukudi (Tamil Nadu) industrial cluster utilizes treated municipal greywater from local sewage treatment plants instead of extracting fresh groundwater.

Community-Led & Governance Models for ground water conservation

- **APFAMGS Model (Andhra Pradesh):** Pioneered participatory groundwater management where thousands of farmers use simple tools to measure their own aquifers and self-regulate crop choices.
- **Tarun Bharat Sangh (Rajendra Singh/Waterman of India):** Rejuvenated dead rivers in arid Rajasthan by building over 11,000 decentralized, community-managed *Johads* (earthen check dams).
- **Managed Aquifer Recharge (MAR) - Israel:** Uses large-scale infiltration basins to systematically inject treated urban wastewater back into underground aquifers for long-term storage and reuse.
- **AEM Mapping Model (Nebraska, USA):** Employs Airborne Electromagnetic imaging to create real-time 3D subsurface profiles to regulate irrigation quotas dynamically.
- **Bhungroo Technology (Gujarat):** An innovative social enterprise model that injects excess monsoon floodwater underground into saline aquifers, storing it for winter irrigation while reducing land salinity.
- **WELS Regulation Model (Australia):** The mandatory Water Efficiency Labelling and Standards scheme that uses consumer-facing star ratings to drive the adoption of water-saving appliances and industrial pumps.
- **Subhash Palekar Natural Farming / SPNF (Andhra Pradesh):** Promotes *Acchadan* (mulching) and bio-inputs to organically retain soil moisture (green water), heavily reducing groundwater extraction.

Committee recommendations for groundwater conservation in India

Committee	Core Recommendation
Mihir Shah Committee	Restructure CWC and CGWB into a single National Water Commission to unifiedly manage surface and groundwater.
Kirit Parikh Committee	Rationalize farm electricity pricing and replace free power with direct cash transfers to end competitive pumping.
NITI Aayog (Composite Water Index)	Institutionalize mandatory annual Water Budgeting across all Gram Panchayats to match crop selection with aquifer limits.
Central Ground Water Board (CGWB)	Transition the National Aquifer Mapping Programme (NAQUIM) from macro-mapping to village-level micro-aquifer management.
Planning Commission (12th Plan)	Amend the Indian Easements Act, 1882 to strip land ownership of absolute subterranean water extraction rights.
Swaminathan Commission (NCRF)	Promote low-water millets and incentivize subsurface drip irrigation to dramatically lower agriculture's groundwater footprint.
G.S. Jha Committee	Create a dedicated "National Aquifer Rejuvenation Fund" to finance deep subsurface recharge shafts in dry alluvial zones.
Madhav Chitale Task Force	Implement a strict regulatory ban on drilling deep borewells that penetrate deep, non-renewable confined aquifers.

Conclusion

To secure our water future, India must transition from competitive groundwater extraction to cooperative aquifer management, treating water as a shared, life-sustaining asset. By blending traditional rainwater harvesting structures with cutting-edge satellite monitoring and smart-metering, we can build community-led water resilience.

MGP/BP/GS3/016

FOREST AND NATURAL RESOURCES MANAGEMENT

1. FOREST CONSERVATION IN INDIA

From the ancient ethos of *Aranyakas* (forest texts) to the modern mandate of Article 51A(g), safeguarding green cover has always been foundational to India's civilization. Today, forest conservation is no longer just a choice for ecological balance, but a critical imperative to ensure the resource security of 1.4 billion people.

Forest conservation in India

- **Headline Statistics**
  - **Total Green Cover:** 8,27,357 sq km, which constitutes **25.17%** of India's total geographical area ***(Source: India State of Forest Report - ISFR, MoEFCC)*.**
  - **Core Forest Cover:** Spans 7,15,343 sq km, representing **21.76%** of the country's land area ***(Source: ISFR, MoEFCC)*.**
  - **Standalone Tree Cover:** Accounts for 1,12,014 sq km, contributing **3.41%** of the national total ***(Source: ISFR, MoEFCC)*.**
  - **Mangrove Footprint:** Extends across 4,991.68 sq km, making up **0.15%** of India's geography ***(Source: ISFR, MoEFCC)*.**
- **State-Wise Leaders & Trends**
  - **Largest Area:** Madhya Pradesh holds the largest forest area (77,073 sq km), followed by Arunachal Pradesh and Chhattisgarh ***(Source: ISFR, MoEFCC)*.**
  - **Highest Percentage:** Lakshadweep leads at 91.33% cover, followed by Mizoram at 85.34% ***(Source: ISFR, MoEFCC)*.**
  - **Top Net Green Gainers:** Chhattisgarh recorded the highest increase (+684 sq km), closely followed by Uttar Pradesh (+559 sq km**)** ***(Source: ISFR, MoEFCC)*.**
  - **Regional Decline:** Ecologically sensitive North-Eastern region faced a net loss of 327 sq km in total forest cover over the decade ***(Source: ISFR, MoEFCC)*.**
- **Global Rankings & Carbon Target Assets**
  - **Global Forest Area Standing:** India ranks **9th globally** in total forest area, accounting for roughly 2% of the world's forest cover ***(Source: FAO Global Forest Resources Assessment - GFRA)*.**
  - **Net Annual Gain Rank:** India holds the **3rd position worldwide** for net annual forest area gains driven by strategic afforestation ***(Source: FAO GFRA)*.**
  - **Global Carbon Sink Standing:** India ranks **5th globally** among top carbon sinks, actively removing 150 million tonnes of CO_2 annually ***(Source: FAO GFRA Brief)*.**
  - **National Carbon Stock:** Total forest carbon stock has touched 7,285.5 million tonnes (30.43 billion tonnes of CO_2 equivalent) ***(Source: ISFR, MoEFCC)*.**

Need for Forest Conservation in India

- **Climate & Ecological Resilience**

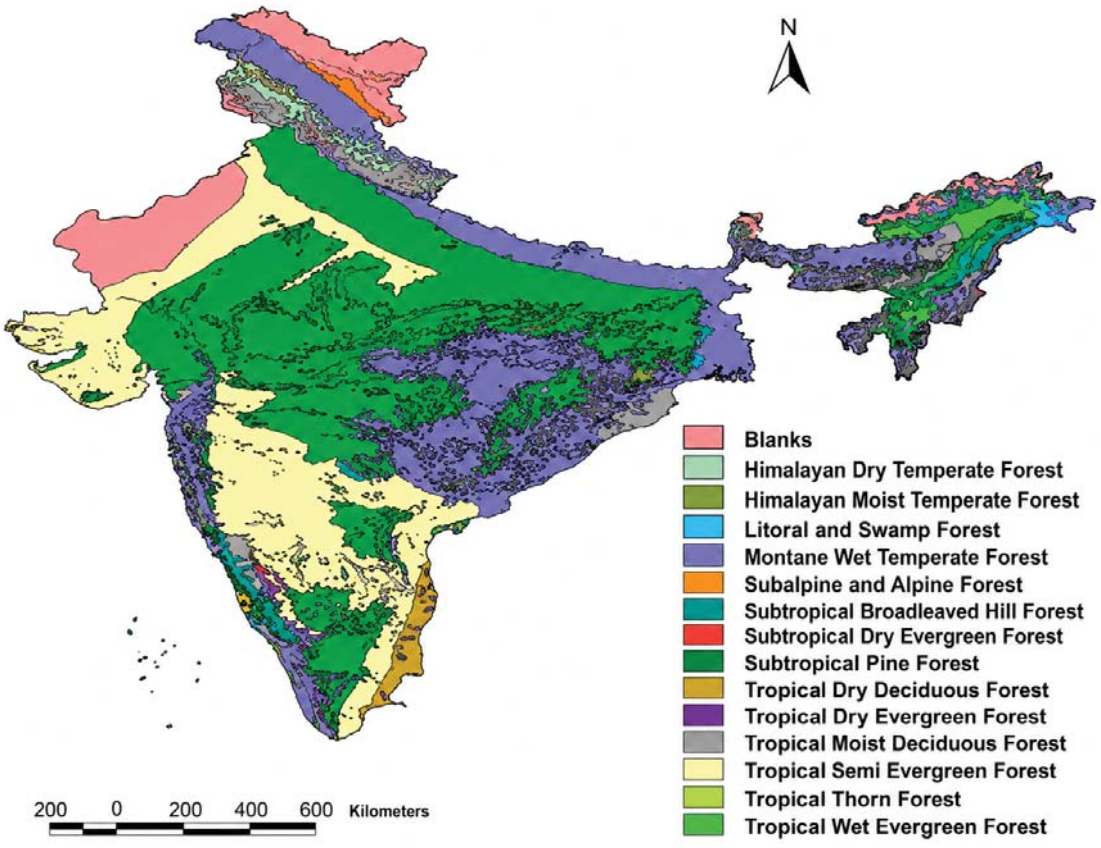
- **Carbon Sink:** Crucial for achieving India's Net-Zero target by 2070 via carbon sequestration (absorbing atmospheric carbon).
- **Monsoon Anchor:** Regulates local precipitation cycles and maintains the stability of the vital Indian Monsoon.
- **Disaster Mitigation:** Acts as a natural shield against extreme weather events like landslides in the Himalayas and cyclones along coasts.
- **Soil & Water Security:** Prevents massive topsoil erosion and recharges critical underground aquifers and river basins.

- **Economic & Resource Security**
  - **Livelihood Support:** Sustains over 275 million forest-dependent rural and tribal populations across India.
  - **NTFP Economy:** Supplies essential Non-Timber Forest Products (NTFPs) like tendu leaves, honey, and medicinal herbs.
  - **Timber & Fuel:** Meets industrial timber demands and fulfills basic fuelwood needs for domestic rural energy.
  - **Ecotourism Hubs:** Drives regional green economies through wildlife sanctuaries and national park tourism.
- **Biodiversity & Genetic Capital**
  - **Megadiverse Safeguard:** Protects India's unique wildlife across four global biodiversity hotspots.
  - **Human-Wildlife Peace:** Minimizes rising Human-Wildlife Conflicts (HWC) by preventing habitat fragmentation.
  - **Genetic Reservoir:** Preserves wild crop relatives essential for climate-resilient future agriculture.
- **Constitutional & Ethical Mandates**
  - **State Duty:** Fulfills Article 48A of the DVP (Directive Principles of State Policy) to protect and improve the environment.
  - **Citizen Duty:** Matches Article 51A(g), which mandates every citizen to protect forests, lakes, and wildlife.
  - **Intergenerational Equity:** Secures a healthy, resource-rich environment for future generations.

Classification of forests in India

Ecological Classification	• **Tropical Wet Evergreen:** Heavy rainfall (greater than 250 cm); multi-layered canopy; found in Western Ghats and NE India (e.g., Rosewood, Mahogany). • **Tropical Semi-Evergreen:** Transitional zones (200–250 cm rainfall); mix of evergreen and deciduous species (e.g., White Cedar, Kail). • **Tropical Moist Deciduous:** Moderate rainfall (100–200 cm); sheds leaves in dry spring; dominates Central India and foothills (e.g., Teak, Sal). • **Tropical Dry Deciduous:** Lower rainfall (70–100 cm); extensive open tracts across plains (e.g., Tendu, Palas, Bel). • **Tropical Thorn:** Arid regions (less than 50 cm rainfall); xerophytic adaptations with deep roots and spines (e.g., Babool, Kair, Khair). • **Subtropical Broadleaved Hill:** Sub-montane zones (1000–2000m altitude); evergreen character (e.g., Oak, Chestnut in Nilgiris/Himalayas). • **Montane Wet Temperate:** High altitude (1500–3000m) with cold winters; dense evergreen stunted trees (e.g., Deodar, Chinar). • **Alpine & Sub-Alpine:** High-altitude meadows near snowlines (above 3000m); stunted, small leaf shrub vegetation (e.g., Juniper, Birch, Rhododendron). • **Littoral & Swamp (Mangroves):** Intertidal deltaic mudflats; salt-tolerant halophytes with pneumatophores (e.g., Sundari, Rhizophora).

Legal Status Classification (Indian Forest Act, 1927)	• **Reserved Forests:** Highest protection level; no public entry or resource exploitation allowed unless explicitly permitted by the government. • **Protected Forests:** Gradated protection; all activities are allowed unless explicitly prohibited by the state forest department. • **Village Forests:** Forest lands assigned to local communities or village Panchayats for regulated collective resource extraction. • **Unclassed Forests:** Unrecorded or unnotified lands with minimal statutory legal restrictions on public access.
Canopy Density Classification	• **Very Dense Forest (VDF):** Lands with a tree canopy density of **70% and above**. • **Moderately Dense Forest (MDF):** Lands with a tree canopy density ranging between **40% and 70%**. • **Open Forest (OF):** Lands featuring a sparse tree canopy density between **10% and 40%**. • **Scrub:** Degraded forest land pockets showing a canopy density of less than 10%.



Issues with Forest Conservation in India

- **Legal & Regulatory Loopholes**
  - **Narrowed Definition:** Forest (Conservation) Amendment Act (FCAA) 2023 excludes unnotified/ deemed forests from legal safety, diluting the landmark 1996 *Godavarman* verdict.
  - **Blanket Exemptions:** Removing clearance mandates for strategic projects within 100 km of international borders risks ecologically fragile zones like the Northeast.

Reports highlight that the 100-km border exemption directly covers **nearly 100% of ecologically hyper-fragile North-Eastern states**,

- **Commercial Overuse:** Broad classifications allowing zoos, safaris, and eco-tourism structures inside forests weaken anti-diversion checks.
- **Diluted Local Consent:** The 2023 amendment rules drastically bypass the mandatory prior consent of tribal *Gram Sabhas* for land diversion.

- **Implementation & Policy Bottlenecks**
  - **CAMPA Mismanagement:** Compensatory Afforestation Fund Management (CAMPA) funds are underutilized or spent on monoculture plantations rather than biodiverse restoration.

    As per CAG, States average a direct underutilization and misallocation rate of **30% to 35%** out of the multi-crore accumulated CAMPA corpus.
  - **Paper Protection:** Large swathes of **Recorded Forest Areas** exist only on official maps without having actual dense canopy cover.

    As per FSI, Over **30% of India's ~77 million hectares** of Recorded Forest Area (RFA) features degraded scrub or lacks actual tree canopy cover.
  - **Land Records Friction:** Conflicting revenue and forest department land data create widespread litigation and boundary disputes.
  - **Greying Greenery:** Focus heavily tilts toward counting tree density (plantations) instead of protecting genuine, complex forest ecosystems.

    As per ISFR, Over **1,500 sq km of natural, moderate-to-dense interior forests** were lost in recent cycles while plantation "tree cover" artificially expanded.
- **Anthropogenic & Socio-Economic Pressures**
  - **Developmental Diversion:** Massive forest land diversion for linear infrastructure like highways, mining, and hydroelectric dams.

    As per MoEFCC Parivesh, An average of **15,000 to 20,000 hectares** of pristine forest land are legally diverted annually for mining, dams, and highways.
  - **Encroachment & Grazing:** Sustained pressure from illegal agricultural expansion and livestock overgrazing beyond carrying capacity.

    As per MoEFCC, Over **1.3 million hectares** are under active illegal encroachment, with **100+ million livestock** grazing far beyond local carrying capacity.
  - **Forest Fires:** Highly inadequate modern firefighting infrastructure to tackle seasonal, anthropogenic forest fires.

    As per FSI Van Agni, Over **3.4 lakh thermal fire points are logged annually**, with **35.46%** of India's total forest cover highly prone to seasonal, man-made fires.
  - **FRA Friction:** Ongoing implementation tussles between state forest acts and individual/community forest rights under the Forest Rights Act (FRA), 2006.
- **Ecological & Climate Vulnerabilities**
  - **Habitat Fragmentation:** Breaking large forests into small pockets isolates wildlife populations, causing severe inbreeding depression.

    As per wildlife institute of India, Over **70% of India's protected areas** are isolated ecological islands, breaking crucial wildlife gene flows.
  - **Human-Wildlife Conflict:** Dwindling buffer zones push wildlife into human settlements, damaging crops and lives.

    Shrinking buffer zones lead to an average of **500+ human fatalities** and **100+ wildlife deaths** annually.

- **Invasive Species:** Aggressive spread of exotic weeds like *Lantana camara* suppresses natural native forest regeneration.
- **Climate Distress:** Rapid shifting of forest vegetation types due to rising temperatures and localized drought stresses.

Measures taken for forest conservation in India

- **Constitutional & Legislative Frameworks**
  - **Statutory Shield:** Forest (Conservation) Act, 1980 (amended in 2023) regulates non-forest use of forest lands via strict central approvals.
  - **Wildlife Safeguard:** Wildlife (Protection) Act, 1972 establishes protected areas like National Parks and Wildlife Sanctuaries.
  - **Decentralized Rights:** Forest Rights Act (FRA), 2006 empowers local tribal communities and *Gram Sabhas* with community forest management.
  - **Judicial Overwatch:** The Supreme Court's *Godavarman Case* (1996) institutionalized dictionary-meaning protection and continuous judicial monitoring.
- **Core Government Schemes & Missions**
  - **Green India Mission (GIM):** A core NAPCC mission aimed at increasing forest cover by 5 million hectares and improving quality on another 5 million.
  - **Nagar Van Yojana:** Focuses on urban forestry by developing 200-plus urban forests to enhance micro-climates in cities.
  - **MISHTI Scheme:** Specifically targets the intensive restoration and conservation of ecologically sensitive coastal mangrove habitats.
  - **Sub-Mission on Agroforestry:** Promotes "Trees Outside Forests" (TOF) to reduce industrial extraction pressures on natural forests.
- **Financial & Market-Based Incentives**
  - **CAMPA Act, 2016:** Channels funds paid by industries for forest diversion into state-level compensatory afforestation and restoration.
  - **Green Credit Programme (GCP):** Incentivizes individuals, communities, and private sectors to undertake afforestation on degraded lands for tradable credits.
  - **Devolution Incentives:** 15th Finance Commission linked 10% of tax pool allocation to states based on their dense forest cover.
  - **Van Dhan Yojana:** Promotes value addition of Non-Timber Forest Products (NTFPs) to make sustainable forest utilization economically viable for tribals.
- **Technological & Monitoring Interventions**
  - **PARIVESH 2.0:** A single-window, integrated portal ensuring transparent, geo-tagged processing of environmental and forest clearances.
  - **FSI Real-Time Alerts:** Forest Survey of India (FSI) utilizes satellite-based remote sensing data (MODIS/SNPP-VIIRS) for near real-time forest fire alerts.
  - **e-Green Watch:** An online tracking system ensures transparency by monitoring the physical and financial progress of CAMPA plantations.
  - **Biomass Mapping:** Uses advanced Synthetic Aperture Radar (SAR) data to estimate above-ground forest biomass and report precise carbon stocks.

Way Forward

- **Legal & Policy Refinements**
  - **Clarify Definitions:** Define "Deemed Forests" explicitly to plug protection gaps left by the 2023 Amendment Act.
  - **Gram Sabha Empowerment:** Mandate prior, informed consent from local bodies for any linear infrastructure project diversion.

  - **Landscape-Level Planning:** Shift from rigid administrative boundaries to eco-sensitive landscape management for wildlife corridors.
  - **Strict Eco-Audits:** Standardize mandatory, independent Environmental Impact Assessments (EIAs) for all border-area strategic projects.
- **Institutional & Financial Reforms**
  - **Revamp CAMPA:** Transition funds away from artificial monocultures toward restoring native, climate-resilient mixed species.
  - **Harness Green Credits:** Scale up the Green Credit Programme via strict verification to attract genuine private investment.
  - **Frictionless FRA:** Expedite the pending recognition of Community Forest Rights (CFR) to turn dwellers into active guardians.
  - **Reward Performance:** Increase the Finance Commission's tax devolution weightage for states maintaining high dense forest canopy.
- **Technological & Ecological Upgrades**
  - **AI-Driven Monitoring:** Deploy AI, drones, and real-time satellite data to pre-empt illegal logging and land encroachments.
  - **Early Fire Warning:** Upgrade to predictive, satellite-linked early warning grids for community-led seasonal firefighting.
  - **Eradicate Invasives:** Launch localized, mission-mode programs to clear aggressive exotic weeds like *Lantana* and *Senna*.
  - **Smart Linear Infrastructure:** Build mandatory eco-ducts, underpasses, and canopy bridges to completely eliminate habitat fragmentation.
- **Socio-Economic & Community Integration**
  - **Scale Van Dhan:** Deepen the value-addition chain for tribal-sourced Non-Timber Forest Products (NTFPs) to boost green livelihoods.
  - **Promote Agroforestry:** Incentivize farmers to grow Trees Outside Forests (TOF) to ease industrial timber pressure on natural wood.
  - **Co-Management Models:** Expand Joint Forest Management (JFM) into true partnerships, sharing tourism revenue directly with locals.
  - **Urban Miyawaki:** Mandate Miyawaki-style micro-forests in urban planning to combat the rising heat island effect.

Best practices and models for forest conservation in India

- **Community-Led & Indigenous Models**
  - **Malkangiri Model (Odisha):** Women-led Self-Help Groups (SHGs) spearhead forest protection patrols, directly curbing illegal logging and timber smuggling.
  - **Naga Customary Governance (Nagaland):** Indigenous Village Councils utilize customary laws to strictly regulate clan-owned forests and rotational *Jhum* (shifting cultivation) fallows.
  - **Sacred Groves (Devrai/Orans):** Traditional, community-protected forest pockets preserved through socio-religious beliefs, acting as critical gene banks for endemic species.
  - **Chipko & Appiko Legacy:** Grassroots, non-violent community movements that institutionalized tree-hugging as a powerful defense against commercial felling.
- **Institutional & Institutionalized Co-Management Models**
  - **Joint Forest Management (JFM):** A co-management model sharing minor forest produce and timber revenue between village Forest Protection Committees and State Forest Departments.
  - **Arabari Model (West Bengal):** The historic precursor to JFM, proving that provisioning fuel and employment to locals completely reverses rapid forest degradation.

- **Van Panchayat Model (Uttarakhand):** A unique, century-old statutory framework mapping local forest governance directly to democratically elected village councils.
- **Eco-Development Committees (EDCs):** Promotes community-led ecotourism and alternative livelihoods in Periyar Tiger Reserve to eliminate local pouching and illegal grazing.

- **Agroforestry & Eco-Restoration Models**
  - **Miyawaki Method:** Deploying ultra-dense, multi-layered native plantations to create self-sustaining urban forests within 2 to 3 years.
  - **Assisted Natural Regeneration (ANR):** Focusing on fencing and protecting natural rootstocks to regenerate native canopies instead of planting artificial commercial monocultures.
  - **Wadi Model (NABARD):** Converting small, degraded tribal landholdings into highly productive tree-based orchard farming systems to reduce natural forest extraction.
  - **Araku Valley Model (Andhra Pradesh):** Massive tribal-led afforestation combining ecological restoration with profitable, shade-grown organic coffee plantations.
- **Technological & Innovative Administrative Models**
  - **Eco-Task Forces (ETF):** Deploying dedicated Territorial Army units of ex-servicemen for high-altitude afforestation and rugged terrain ecosystem restoration.
  - **Green Credit Market Model:** A market-driven mechanism incentivizing private entities to fund open forest restoration to earn tradable green credits.
  - **Finance Commission Performance Model:** Linking 10% of the national tax devolution pool directly to a state's success in maintaining dense forest canopy cover.
  - **AI & Geospatial Fire Alerting:** FSI's use of real-time SNPP-VIIRS satellite data to instantly map and broadcast forest fire alerts down to local beat guards.

Conclusion

Reclaiming the Vedic philosophy of *Vasudhaiva Kutumbakam* and the core ethos of "enoughness," India's path to forest conservation must rely on treating forests as living natural assets rather than mere economic resources. Achieving our ambitious Net-Zero target by 2070 demands a structural overhaul that accelerates native biodiversity restoration, rewards states for dense tree covers, and halts habitat fragmentation.

2. AFFORESTATION AND REFORESTATION

Afforestation and reforestation are essential nature-based strategies used to combat climate change, restore ecosystems, and halt deforestation. Afforestation involves planting trees on land that has not been forested for a long period, while reforestation focuses on replanting trees in recently deforested or degraded areas.

Afforestation

- **Definition:** The human-induced creation of forests on lands that have historically not been forested or have remained treeless for a prolonged period (often at least 50 years).
- **Primary Use:** Transforming abandoned agricultural land, degraded grasslands, or barren areas into thriving, wooded ecosystems.
- **Benefits:** Prevents soil erosion, stops desertification, and establishes new carbon sinks to absorb CO_2 from the atmosphere.
- **Considerations:** It is critical to carefully select native plant species to avoid altering or disrupting the original grassland ecosystems.

Reforestation

- **Definition:** The re-establishment of forest cover in areas where forests existed in the recent past but were cleared or depleted.
- **Primary Use:** Restoring woodlands lost to commercial logging, agricultural expansion, wildfires, or disease.
- **Benefits:** Rapidly restores local biodiversity, maintains historical ecosystems, and secures timber and wood resources sustainably

Significance of afforestation and reforestation

- **Climate Change Mitigation & Carbon Dynamics**
  - **Carbon Sinks:** Directly assists India in meeting its Bonn Challenge commitment to restore 26 million hectares of degraded land by 2030.

    According to the MoEFCC, India has already brought over **14 million hectares** under restoration against its global Bonn Challenge commitment of 26 million hectares by 2030.
  - **Net-Zero Pathway:** Accelerates the creation of an additional 2.5 to 3 billion tonnes of CO2 equivalent carbon sink required by 2030.
  - **Microclimate Regulation:** Lowers extreme local ambient temperatures and curbs the intense "Urban Heat Island" effect in expanding cities.

    As per MoEFCC, the **Nagar Van Yojana** aims to develop 400 Urban Forests and 200 Nagar Vatikas nationwide to lower city ambient temperatures by **2°C to 5°C**
  - **Albedo & Energy Balance:** Modifies surface reflectivity and regional moisture levels, helping to stabilize erratic localized rainfall patches.
- **Ecological Restoration & Disaster Shields**
  - **Bio-Corridor Revival:** Reforestation reconnects fragmented, isolated forest patches, ensuring safe migratory corridors for endangered wildlife.
  - **Hydrological Rejuvenation:** Enhances root-induced water infiltration, effectively recharging drying aquifers and perennial river basins.

    According to the **MoEFCC,** rejuvenation of 13 major river basins (e.g., Ganga, Yamuna) are projected to increase groundwater recharge by up to **20%**.
  - **Soil Salvage:** Binds loose topsoil to halt rapid desertification and prevents massive siltation in downstream river reservoirs.

    According to the ISRO Desertification and Land Degradation Atlas of India, land degradation currently impacts 29.7% of India's Total Geographical Area (TGA).
  - **Disaster Buffering:** Acts as a living geo-engineering shield against landslides in the Himalayas and cyclonic storm surges along coasts.
- **Socio-Economic & Livelihood Security**
  - **Green Employment:** Generates millions of informal rural man-days of employment through nursery management, planting, and upkeep.

    According to MGNREGA data, forestry, nursery development and plantation works under **MGNREGA** generate over **10% to 12% of total rural man-days** of employment annually.
  - **NTFP Safety Net:** Boosts the sustainable harvesting of tribal-dependent Non-Timber Forest Products like honey, resin, and medicinal herbs.

    According to **TRIFED**, afforestation directly provides a livelihood safety net to over **5 crore (50 million) tribal forest-dwellers**.
  - **Agroforestry Buffers:** Integrating trees with agricultural fields reduces farmer vulnerability to crop failure during severe drought years.

  - **Import Substitution:** Eases the extraction pressure on natural forests while slashing India's massive timber and pulp wood import bills.
- **Policy, Constitutional & Global Commitments**
  - **SDG Target Anchor:** Directly aligns with and helps fulfill UN Sustainable Development Goals like **SDG 13** (Climate Action) and **SDG 15** (Life on Land).
  - **Constitutional Compliance:** Manifests the state's duty under Article 48A and the citizens' core environmental mandate under Article 51A(g).
  - **Global Standing:** Projects India as a proactive, climate-responsible leader at global multilateral forums like the UNFCCC and CBD.

Issues with afforestation and reforestation programs in India

- **Ecological & Biodiversity Flaws**
  - **Monoculture:** Heavy institutional reliance on fast-growing exotic commercial species (e.g., *Eucalyptus*, *Acacia*) creates green deserts lacking complex biodiversity.
  - **Grassland Destruction:** Mass afforestation drives often blindly target natural grasslands and semi-arid scrubs, destroying unique open-ecosystem wildlife habitats.

    MoEFCC data indicates that blind afforestation targets have led to the loss of **over 31% of India's natural semi-arid grasslands** and open scrub ecosystems
  - **Low Survival Rates:** Poor post-planting maintenance, zero tracking, and harsh climatic stresses cause mass mortality of planted saplings.

    CAG Performance Audit revealed that the long-term survival rate of planted saplings under massive afforestation drives stands at an alarming **30% to 40%** due to poor post-care.
  - **Faunal Displacement:** Replanted single-tier tree blocks fail to mirror the multi-layered canopy of natural forests, worsening human-wildlife conflicts.

    WII (Wildlife Institute of India) studies link single-tier monocultures to a **40-50% reduction in local herbivore carrying capacity.**
- **Institutional & Financial Inefficiencies**
  - **CAMPA Misdirection:** Compensatory Afforestation Fund resources are frequently underutilized, diverted for non-forestry use, or spent on clearing native brush for plantation blocks.

    The Central Empowered Committee (CEC) revealed that while 85% of compensatory targets are physically logged, **only 67.5% of total accumulated CAMPA funds** are officially utilized.
  - **Quantity vs. Quality:** Bureaucratic metrics reward the *number of saplings planted* rather than monitoring long-term canopy density or ecological functionality.
  - **Top-Down Administration:** Program execution heavily favors rigid forest department mandates, sidelining Joint Forest Management Committees (JFMCs).
  - **Fund Release Bottlenecks:** Chronic bureaucratic delays in transferring funds to grassroots implementation agencies cripple seasonal planting cycles.
- **Socio-Economic & Land Conflicts**
  - **Competing Land Pressures:** Severe domestic scarcity of non-forest land leads to massive legal litigation and boundary friction with agriculture and urbanization.
  - **Tribal Displacement:** Afforestation drives on community lands frequently violate individual and group land rights guaranteed under the Forest Rights Act (FRA), 2006.

    Ministry of Tribal Affairs (MoTA, 2026) progress report shows that over **47,901 Community Forest Resource (CFR) claims** have been rejected nationwide.
  - **Fodder & Fuel Deprivation:** Fencing off newly afforested village commons cuts off poor livestock-dependent rural communities from essential grazing and fuel resources.

- **Private Sector Friction:** Sharp regulatory ambiguity surrounding the Green Credit Programme limits large-scale corporate investments in land restoration.

- **Technical & Climatic Challenges**
  - **Exotic Invasive Spread:** Newly planted zones are highly prone to aggressive overtaking by stubborn invasive weeds like *Lantana camara* or *Senna spectabilis*.

    FSI inventory assessments estimate that more than **36% of India's forest landscapes** are severely infested with stubborn, aggressive invasive species.
  - **Poor Site Mapping:** Inadequate pre-planting soil-moisture profiling leads to matching the wrong tree species with unsuitable soil terrains.

    National Bureau of Soil Survey and Land Use Planning data indicates that **over 50% of public block plantations** suffer from ecological mismatching.
  - **Climate & Fire Vulnerability:** Uniform monoculture blocks act as highly combustible fuel tubes, surging the risk and intensity of seasonal forest fires.

    FSI Forest Fire Alerts System (Van Agni) records over **3.4 lakh fire points annually**, confirming that **more than 36% of India's overall forest cover** is highly fire-prone.
  - **Biomass Overestimation:** Lack of high-resolution Synthetic Aperture Radar (SAR) data causes inaccurate reporting of actual carbon stocks sequestered.

    ISRO research reports show that relying strictly on optical remote sensing overestimates actual carbon stocks by up to **20% to 25%** compared to Synthetic Aperture Radar data.

Measures taken for afforestation and reforestation in India

Sub-Type Category	Government Measure	Major points
1. Policy & Institutional Drivers	National Forest Policy, 1988	Establishes the national target of achieving 33% geographical area under forest and tree cover.
	Green India Mission (GIM)	A core NAPCC initiative targeting the restoration and enhancement of 10 million hectares of forest/non-forest land.
	Compensatory Afforestation Fund Act (CAMPA), 2016	Unlocked over ₹40,000 crore to ensure strictly audited, mandatory afforestation whenever industrial diversion occurs.
2. Target-Specific Schemes	Nagar Van Yojana (Urban Forest Scheme)	Aims to develop 200 urban forests across cities to enhance urban microclimates and create green lungs.
	MISHTI Scheme (Budget 2023)	Focuses on intensive, community-led mangrove afforestation along India's coastline and salt-pan lands.
	Sub-Mission on Agroforestry (SMAF)	Promotes "Trees Outside Forests" (TOF) to help farmers plant timber/ fruit trees alongside traditional crops.

3. Community & Economic Incentives	Green Credit Programme (GCP)	Market-based mechanism allowing private/public entities to invest in degraded land afforestation to earn tradable credits.
	National Afforestation Programme (NAP)	A decentralized scheme utilizing village-level Joint Forest Management Committees (JFMCs) for eco-restoration.
	15th Finance Commission Mandate	Assigns 10% weightage in tax devolution formula to states based on their success in expanding dense forest cover.
4. Technology & Infrastructure	PARIVESH 2.0 Portal	Single-window tech portal ensuring geo-tagged, transparent mapping and tracking of afforestation plots.
	e-Green Watch & FSI Satellites	Uses remote sensing data to run real-time audits on survival rates and biomass levels of CAMPA plantations.
	Eco-Task Forces (ETF)	Deploys specialized Territorial Army battalions of ex-servicemen for mass afforestation in hyper-rugged terrains.

Way forward

- **Ecological & Biodiversity Reforms**
  - **Phase Out Monocultures:** Mandate a multi-layered planting mix of 100% native, site-specific species to completely replace exotics like Eucalyptus.

    Miyawaki Urban Forests in Kerala and Mumbai using 100% native multi-layered species (e.g., Neem, Peepal) to phase out Eucalyptus blocks.
  - **Protect Non-Forest Ecosystems:** Exclude natural open grasslands, semi-arid scrubs, and cold deserts from blind, target-driven tree plantation drives.

    Great Indian Bustard (GIB) Habitat Protection in Rajasthan, legally halting massive tree-planting drives in natural, semi-arid grasslands.
  - **Prioritize Natural Healing:** Pivot budgets toward Assisted Natural Regeneration (ANR) by fencing surviving native rootstocks instead of forcing artificial plantations.

    Brazil's Atlantic Forest Restoration Pact, using natural regeneration over artificial planting to restore degraded cattle pastures cost-effectively.
  - **Focus on Bio-Corridors:** Prioritize the reforestation of fragmented forest patches to reconstruct safe migratory corridors for endangered wildlife.

    Kaziranga-Karbi Anglong Elephant Corridors (Assam), where focused canopy rebuilding reconnected critical fragmented tracks.

Institutional & Governance Overhauls

- **Decentralize CAMPA Funds:** Devote at least 50% of CAMPA allocations directly to Gram Sabhas and local Forest Rights Committees for ground execution.

Odisha's Ama Jungle Yojana, which directly channels CAMPA financial allocations to village-level Vana Surakshya Samitis (VSS) for execution.

- **Strict FRA Compliance:** Legally link all afforestation site selections to the prior, written, and informed consent of tribal bodies to eliminate land conflicts.

Canada's First Nations Land Management Framework, legally requiring prior, informed consent from indigenous bodies for all regional ecological projects.

- **Independent Green Audits:** Replace self-reporting with mandatory, third-party ecological audits evaluating canopy density and biodiversity index.
- **Cadre Specialization:** Implement the Subramanian Committee recommendation to create an Indian Environment Service (IES) for professional institutional governance.

US Federal Senior Executive Service (SES) Environment Track, which recruits specialized environmental scientists to lead conservation bodies instead of generalists.

- **Socio-Economic & Market Models**
  - **Tighten Green Credit Metrics:** Refine the Green Credit Programme to ensure corporate credits are only awarded based on verifiable 3-year survival rates.
  - **Aggressive Agroforestry:** Simplify tedious inter-state timber transit rules to incentivize smallholder farmers to seamlessly adopt Trees Outside Forests.
  - **Value-Chain Integration:** Link reforested zones to tribal cluster micro-enterprises under Van Dhan Vikas Kendras to make standing trees economically valuable.
  - **Reward Performance:** Increase the Finance Commission's tax devolution weightage for states that show measurable improvements in their dense forest canopy.
- **Technical & Smart Interventions**
  - **AI-Driven Site Matching:** Deploy AI algorithms to analyze high-resolution ISRO satellite data, automatically matching soil-moisture profiles with correct native species.
  - **Drone-Assisted Seeding:** Use specialized seed-bombing drones to rapidly reforest hyper-rugged, inaccessible terrains across the Western Ghats and Himalayas.
  - **Geo-Fenced Survival Grids:** Enforce mandatory e-Green Watch geo-tagging for every single plantation block to run strict time-series monitoring of canopy growth.

e-Green Watch Portal (MoEFCC), which enforces mandatory geo-fencing and time-series satellite tracking for all compensatory afforestation plots.

- **Predictive Fire Mapping:** Establish satellite-linked, predictive early warning grids to prevent human-induced seasonal forest fires from destroying young plantations.

Major committee and judicial recommendations

- **T.S.R. Subramanian Committee (2014):** Recommended increasing the mandatory compensatory afforestation ratio to **2:1 on revenue land** and **3:1 on degraded forest lands** to factor in sapling mortality.
- **Madhav Gadgil Committee (WGEEP - 2011):** Strongly advocated for a complete ban on planting exotic, thirsty monocultures (like *Eucalyptus*) in eco-sensitive zones, recommending **100% native species** instead.
- **National Forest Commission (2006):** Urged that any afforestation target must be legally coupled with an independent **Forest Impact Assessment** to prevent the destruction of natural grasslands.
- **National Forest Commission (2006):** Recommended establishing a statutory **National Board for Forestry**, chaired by the Prime Minister, to centrally unify and monitor state-level reforestation targets.
- **Kasturirangan Committee (HLWG - 2013):** Recommended financial incentives and infrastructural support for **"Trees Outside Forests" (TOF)** to promote agroforestry in buffer areas surrounding natural landscapes.

Best practices for afforestation and reforestation

- **Bonn Challenge Model:** A global framework targeting the ecological restoration of **26 million hectares** of degraded and deforested Indian lands by 2030.
- **Green Credit Programme (GCP) Model:** A market-driven model allowing public/private entities to fund degraded land restoration to earn tradable green credits.
- **Arabari Co-Management Model:** The historic West Bengal model proving that providing fuel and employment to locals completely reverses rapid forest degradation.
- **Van Panchayat Model (Uttarakhand):** A unique, century-old statutory framework mapping local forest governance directly to democratically elected village councils.
- **Sacred Groves (Orans/Devrai):** A traditional, ethno-forestry model protecting patches of native forest through deep socio-religious and cultural taboos.
- **Poplar-Eucalyptus Agroforestry (Punjab/Haryana):** A successful commercial farm model integrating fast-growing timber trees with seasonal crops to maximize farmer income.
- **Eco-Task Force (ETF) Model:** Deploys dedicated Territorial Army units of ex-servicemen for mass afforestation in hyper-rugged, hostile terrains like the Himalayas.
- **Drone Seed-Bombing Model:** An automation model utilizing specialized aerial drones to drop pelletized native seeds across inaccessible slot canyons and degraded hillsides.
- **Mizoram Jhum Fallow Management:** A landscape model planting nitrogen-fixing trees (like *Alnus nepalensis*) on abandoned slash-and-burn patches to fast-track natural soil recovery.

Conclusion

Healing the fragmented lungs of our sub-continent requires a profound shift toward the restorative philosophy of *Prakriti Rakshati Rakshita*—protecting the nature that protects us. Modern afforestation and reforestation serve as our most critical multi-layered shields against intensifying hydro-meteorological shocks while fulfilling our core constitutional mandates for environmental care.

3. COMMUNITY FOREST MANAGEMENT

Community forest management (CFM) empowers local and Indigenous communities to govern, protect, and sustainably utilize nearby forest resources. This decentralized approach combines traditional knowledge with formal decision-making to generate local income, regenerate biodiversity, and bolster climate resilience while meeting the community's daily livelihood needs.

Principles and Mechanisms of Community forest management

- **Devolution of Rights:** CFM shifts control from centralized state forestry departments to local governance structures—such as local village assemblies or community forest associations. This grants communities the autonomy to set harvesting rules and regulate access
- **Livelihood Generation:** Rather than focusing solely on large-scale timber extraction, CFM emphasizes sustainable practices that support local economies, such as harvesting non-timber forest products (medicinal plants, resins, fibers, and fruits).
- **Ecological Stewardship:** Communities often enforce strict conservation and management plans to protect ecologically sensitive zones, regenerate depleted forests, and preserve sacred sites.

Benefit of Community forest management

- **Ecological & Biodiversity Benefits**
  - **Halts Degradation:** Local vigilance effectively eliminates illegal commercial logging, timber smuggling, and encroachment.

- **Fire Mitigation:** Traditional knowledge ensures early detection, clearance of dry biomass, and creation of seasonal firelines.
- **Native Recovery:** Favors the natural regeneration of native, multi-layered flora over artificial, thirsty exotic monocultures.
- **Wildlife Safeguard:** Reduces human-wildlife conflict by restoring local bio-corridors and controlling unchecked forest fragmentation.

- **Socio-Economic & Livelihood Security**
  - **NTFP Safety Net:** Secures regular income via sustainable collection of minor forest produce (e.g., honey, tendu leaves, bamboo).
  - **Direct Wealth Sharing:** Grants village committees direct ownership over timber and ecotourism revenues, bypassing middle-men bottlenecks.
  - **Agroforestry Synergy:** Interlocks forest protection with community grazing lands and minor orchard cultivation (*Wadi* systems).
  - **Poverty Alleviation:** Lowers distress seasonal migration by generating local green jobs in nursery management and eco-restoration.
- **Institutional & Governance Strengths**
  - **Democratic Decentralization:** Empowers local Gram Sabhas, actively fulfilling the true intent of the **73rd Amendment** and **PESA Act, 1996**.
  - **Gender Inclusion:** Prioritizes women-led forest protection sub-committees, bringing frontline gender equity to climate action.
  - **Cost-Effective Auditing:** Drastically slashes state expenditure on deployment of forest guards through voluntary community patrolling.
  - **Customary Law Integration:** Utilizes time-tested indigenous taboos and community-enforced penalties to maintain resource limits.
- **Climate Change & Policy Alignment**
  - **FRA Compliance:** Provides a functional mechanism to implement Community Forest Resource (CFR) rights under the **Forest Rights Act, 2006**.
  - **Resilient Carbon Sinks:** Sustains higher canopy survival rates than top-down drives, accelerating India's **Net-Zero 2070** target.
  - **SDG Localization:** Translates global mandates like **SDG 13** (Climate Action) and **SDG 15** (Life on Land) into actionable grassroots targets.

Issues with Community forest management

- **Bureaucratic & Institutional Hurdles**
  - **Departmental Hegemony:** Forest departments remain reluctant to devolve true administrative and silvicultural powers to local Gram Sabhas.
  - **Institutional Friction:** Overlapping jurisdictions and conflicts exist between state-controlled Joint Forest Management (JFM) committees and autonomous FRA bodies.

    As per FSI, **1.12 lakh** JFM committees directly clash and overlap spatially with autonomous FRA committees over 22 million hectares of land.
  - **Tokenistic Inclusion:** Women and marginalized factions are often given nominal positions on paper without real executive decision-making power.

    As MoTA, more than **10%** of women members actively participate in executive decisions despite a **33%** mandatory reservation on paper.
  - **Working Plan Control:** State bureaucracies retain veto power over Forest Working Plans, effectively paralyzing autonomous community management.

As per CAG, 100% of commercial timber harvests remain locked under state-designed 10-year Working Plans, allowing Gram Sabhas **0%** independent flexibility.

- **Legal & Implementation Bottlenecks**
  - **CFR Recognition Delays:** Bureaucratic lethargy leads to high rejection rates and massive backlogs in verifying Community Forest Resource (CFR) rights under FRA, 2006.

    As per MoTA, **42.4%** backlog/rejection trajectory recorded out of 2,11,609 community forest resource claims filed nationwide.
  - **Weak PESA Implementation:** Poor operationalization of the PESA Act, 1996, leaves village councils without the statutory teeth to halt major mining or industrial land diversions.
  - **Boundary Contentions:** Lack of high-resolution digital mapping and updated land records triggers severe inter-village and state-village land disputes.
  - **Regulatory Roadblocks:** Tedious, state-enforced transit permit rules discourage communities from freely trading sustainably harvested timber.
- **Socio-Economic & Financial Weaknesses**
  - **Elite Capture:** Dominant local factions and wealthy village elites frequently monopolize forest revenues and resource distribution.

    Top **15-20%** of wealthy, dominant-caste households monopolize **80%** of financial revenues from tendu leaf and bamboo sales.
  - **Financial Starvation:** Village forest committees lack independent budgetary allocations and suffer from chronic delays in state revenue-sharing payouts.
  - **Exploitative Value Chains:** Lack of local processing infrastructure forces communities to sell minor forest produce (NTFPs) to middlemen at distressed rates.

    An **85% deficit** in local primary processing infrastructure forces gatherers to capture only **15% to 30%** of final NTFP consumer market value.
  - **Internal Resource Friction:** Rising population pressures lead to internal community conflicts over equitable grazing and fuel-wood extraction quotas.

    As per NSSO, Average household fuelwood availability in fringe villages dropped by **~40%** over two decades, escalating internal quota disputes.
- **Technical & Eco-Management Gaps**
  - **Capacity Deficits:** Local communities generally lack advanced training in data-driven silviculture, pest management, and inventory tracking.
  - **Invasive Weed Crisis:** Gram Sabhas lack the massive funding and technical tools needed to eradicate aggressive invasive species like *Lantana camara*.

    Invasive Lantana camara aggressively chokes out **>40%** of core forest understories, far outstripping local manual clearance budgets
  - **Over-Extraction Pressures:** Acute rural poverty can inadvertently drive unsustainable, short-term over-harvesting of fragile non-timber resources.
  - **Climate Vulnerability:** Communities often struggle to combat massive, climate-induced seasonal forest fires using only rudimentary traditional tools.

    As per FSI, Community forests experience a **70% higher vulnerability** to extended seasonal fire damage.

Measures taken by the government for Community Forest Management (CFM) in India

Sub-Type Category	Major Measure	Pointers
1. Statutory & Legal Safeguards	Forest Rights Act (FRA), 2006	Legally recognizes Community Forest Resource (CFR) rights, vesting management powers directly in Gram Sabhas.
	PESA Act, 1996	Empowers Scheduled Area Gram Sabhas with mandatory ownership and minor forest produce regulation rights.
	MoTA CFR Guidelines	Standardizes the formation of District Level Committees to map boundaries and fast-track unresolved community claims.
2. Decentralized & Joint Frameworks	Joint Forest Management (JFM)	Pairs state forest departments with village committees to co-manage, protect, and share timber revenues.
	CFR Management Committees	Mandates Gram Sabhas under FRA Rule 4(1)(e) to draft statutory localized sustainable forest use plans.
	Van Panchayats (Uttarakhand)	Keeps alive a century-old democratic village council framework dedicated to handling local community forest lands.
3. Financial & Livelihood Backing	Van Dhan Vikas Kendras	Aggregates tribal forest-gatherers into value-addition clusters for better minor forest produce (NTFP) monetization.
	MSP for MFP Scheme	Insulates local gatherers against exploitative market middlemen by providing a Minimum Support Price for minor forest items.
	CAMPA Devolution Rules	Channels compensatory fund portions toward village-level forest regeneration and employment programs.
4. Technology & Policy Upgrades	Amended Forest Guidelines	Encourages state-supervised partnerships with non-government entities to help local bodies restore degraded scrub lands.
	Geo-Spatial Claims Audit	Deploys satellite-based geographic tracking to cross-verify, correct, and clear historically rejected community titles.
	Meri LiFE Portal Integration	Digitally monitors large-scale public forestry drives to maintain transparency in community plantation survivals.

Way Forward

- **Legal & Structural Harmonization**
  - **Enforce FRA Primacy:** Formally establish the legal supremacy of the Forest Rights Act (FRA), 2006, over conflicting colonial-era clauses of the Indian Forest Act, 1927.
  - **Launch a National FRA Mission:** Implement a time-bound mission to eliminate massive backlogs in verifying Community Forest Resource (CFR) titles.

    Odisha's 'Mo Jungle Jami Yojana' (2023-24), which deployed dedicated district-level task forces to systematically clear massive backlogs of Community Forest Resource (CFR) claims.
  - **Standardize Dispute Resolution:** Set up fast-track, inter-departmental tribunals at the district level to quickly resolve overlapping boundary disputes between villages.
  - **Secure Transit Freedom:** Deregulate and simplify state-enforced transit permit rules, allowing Gram Sabhas to freely transport and trade sustainably harvested timber.
- **Institutional & Executive Empowerment**
  - **Dismantle Departmental Veto:** Amend forest guidelines to strip the state forest bureaucracy of its unilateral veto power over village-drafted CFR Working Plans.
  - **Institutional Synergy:** Integrate state Joint Forest Management Committees (JFMCs) into statutory Gram Sabha sub-committees to eliminate administrative friction.
  - **Mandatory Gender Quotas:** Statutorily mandate a minimum of 50% representation for women in all local community forest resource management executive councils.
  - **Informed Industrial Consent:** Legally enforce that no corporate or mining land diversions can occur without the prior consent of the affected Gram Sabha.

    The landmark **2013 Niyamgiri Gram Sabha referendums**, where 12 tribal village councils completely rejected Vedanta's bauxite mining project proposal.
- **Financial Capacity & Value-Chain Backing**
  - **Direct Fiscal Devolution:** Devote a dedicated, non-divertible share of the national CAMPA and state forestry budgets directly to Gram Sabha banking accounts.

    Scheduled Tribes Component (STC) funds under the Tribal Sub-Plan being routed directly to village accounts in select blocks of Chhattisgarh.
  - **Broaden Price Safety Nets:** Expand the Minimum Support Price (MSP) framework to cover all regional non-timber forest products (NTFPs) to insulate tribals from middle-men shocks.
  - **Value-Addition Clusters:** Scale up Van Dhan Vikas Kendras within community forests to build local solar-drying, oil-extraction, and packaging micro-industries.
  - **Direct Green Credit Payouts:** Ensure that financial returns generated through the Green Credit Programme for ecological restoration flow directly to village committees.

    Global voluntary carbon market pilots in Central India, where international carbon credits directly finance village-level water harvesting structures.
- **Technical & Climate Upgrades**
  - **Provide Digital Mapping Kits:** Equipped Gram Sabhas with handheld GPS tools and open-source GIS software to precisely map their traditional boundary claims.

    CFR Mapping Project by civil society in Jharkhand, using handheld GPS and open-source tools like QGIS to map traditional tribal boundaries.
  - **Scientific Silviculture Training:** Provide institutional training to forest-dwelling youths in carbon stock calculation, data-driven soil health monitoring, and scientific nursery management.

    IIFM Bhopal / KVK extension programmes, which train local tribal youths as "Van Mitras" to measure carbon sequestration and monitor local soil quality.
  - **Modern Firefighting Grids:** Equip local protection committees with satellite-linked early warning receivers and modern, lightweight firefighting gear to combat seasonal forest fires.

- **Invasive Eradication Funds:** Allocate specific technical funds to community councils to execute systematic, bio-secure removals of aggressive weeds like *Lantana camara*.

Bandipur Tiger Reserve peripheral eco-development projects, which fund local groups to manually clear Lantana camara and restore indigenous grasslands.

Global and Indian Success Models

- **Nepal's Community Forest User Groups (CFUGs):** A pioneering model where 30% of national forests are legally handed to local user groups, successfully reversing mountain deforestation while funding local public infrastructure.
- **Guatemala's Maya Biosphere Concessions:** A highly successful model granting 25-year community forest concessions that permit sustainable timber/non-timber extraction and ecotourism, maintaining lower deforestation rates than adjacent protected national parks.
- **Mexico's Ejidos Forestales:** A unique, pioneer model of collective land management where communally farmed lands (*ejidos*) operate community-run forest enterprises, directly reinvesting profits into local health and education.
- **Namibia's Communal Conservancies:** An integrated ecosystem framework that grants rural residents statutory powers to co-manage, protect, and financially profit from forest wildlife and local natural assets.
- **Nigeria's Ekuri Community Model:** A rare, completely indigenous-led initiative where a minority tribe independently drafted a formal land-use plan to protect 33,600 hectares of primary rainforest from state logging concessions.
- **Nagaland's Community Conservation Areas (CCAs):** An autonomous model rooted in uncodified customary tribal laws, where village councils voluntarily convert shifting cultivation (*jhum*) fallows into strictly guarded biodiversity reserves.
- **Odisha's Thenkapallu Patrolling Model:** A gender-inclusive grassroots initiative where village women rotate daily patrolling shifts using wooden sticks (*thenkapallu*) to protect local forest patches from illegal timber mafias.
- **Maharashtra's Gadchiroli CFR Model (Mendha Lekha):** frontline face of FRA execution, where the Gram Sabha became the first in India to win statutory rights to sustainably harvest, chart, and commercially auction local bamboo blocks.

Conclusion

Community Forest Management serves as India's premier institutional mechanism for fostering localized climate resilience and reversing landscape degradation. By legalizing community tenure, it incentivizes frontline populations to aggressively defend native biodiversity while sustainably driving the rural **Non-Timber Forest Produce (NTFP)** value chain.

4. AGROFORESTRY

Agroforestry is a land management system that deliberately integrates trees or shrubs with agricultural crops and/or livestock on the same parcel of land. It functions as an ecologically friendly polyculture that boosts soil fertility, prevents erosion, and provides farmers with diverse, resilient income streams.

Practices and methods of agroforestry

Sub-Type Classification	Specific Method	major feature
1. Agrosilvicultural Systems *(Crops + Trees)*	Alley Cropping (Hedgerow)	Growing food crops in alleys between wide rows of nitrogen-fixing tree hedges to block wind and enrich soil.

	Multi-tier Tree Cropping	Layering different-canopy crops simultaneously (e.g., coconut overhead, banana mid-tier, black pepper ground).
	Taungya System	Allowing landless laborers to grow seasonal food crops between young rows of newly established timber plantations.
	Windbreaks / Shelterbelts	Planting dense, multi-row protective tree lines along farm boundaries to shield delicate crops from dry, high-speed winds.
2. Silvopastoral Systems *(Trees + Animals/Pasture)*	Protein Bank (Fodder Trees)	Cultivating high-protein leguminous trees (*Subabul*) on pasture margins to harvest nutrient-rich fodder during lean seasons.
	Living Fences	Planting closely spaced, thorny leguminous trees to create living, self-maintaining barriers against stray cattle.
	Horti-Pastoral System	Integrating commercial fruit tree orchards with a continuous ground cover of nutritious perennial forage grasses.
3. Agrosilvopastoral Systems *(Crops + Trees + Animals)*	Home Gardens (Multipurpose)	Highly complex, multi-tiered urban-rural backyard systems fusing fruit trees, vegetables, fuel wood, and small livestock.
	Woody Hedgerows for Browse	Utilizing fast-regrowing woody hedge boundaries that simultaneously provide green mulch for crops and animal fodder.
	Aqua-forestry (Silvo-fishery)	Planting specialized water-tolerant trees (mangroves/willows) along the bunds of aquaculture ponds to clean fish wastewater.
4. Specialized & Traditional	Shift-Fallow Integration	Interventions introducing rapid nitrogen-fixing trees into shifting cultivation (*Jhum*) fallow cycles to speed up soil recovery.
	Apiculture-Silviculture	Strategic placement of honeybee hives inside commercial flowering tree plantations to boost pollination and timber yields.
	Riparian Forest Buffers	Establishing permanent strips of native trees along agricultural river banks to intercept chemical fertilizer runoff.

Advantages of Agroforestry

- **Ecological & Soil Health Benefits**
  - **Enhances Carbon Sequestration:** Acts as a vital decentralized carbon sink, directly assisting India in achieving its Net-Zero 2070 targets.

    NITI Aayog G.R.O.W. Report states that agroforestry holds a carbon mitigation potential of **25.7 tonnes of CO2 equivalent per hectare per year** across India.
  - **Controls Soil Erosion:** Deep tree roots lock soil matrices firmly, reducing topsoil erosion caused by wind and heavy rainfall.

    ICAR-CAFRI field trials show that tree-based contour barriers reduce topsoil erosion by **45% to 63%** on sloping agricultural land.
  - **Boosts Nutrient Cycling:** Deep-root tree systems pull nutrients from deep subterranean layers and deposit them on topsoil via leaf litter.

    Indian Institute of Soil Science (IISS) show that leaf-litter decomposition increases topsoil organic carbon by **20% to 38%** over a 5-year cycle.
  - **Maximizes Microclimate Control:** Reduces ambient land temperatures and extreme evaporation losses, preserving soil moisture for adjacent field crops.
  - **Filters Chemical Runoff:** Acts as an active biological riparian filter, trapping hazardous chemical fertilizers before they reach freshwater streams.

    Environmental monitoring confirms that riparian tree buffers trap up to **70% of nitrogen** and **65% of phosphorus** from agricultural runoff before it reaches water bodies.
- **Economic & Farm Income Diversification**
  - **Insulates Income Shocks:** Provides a reliable secondary income source via timber, fruits, and fuelwood during unexpected crop failures.

    NABARD economic surveys indicate that agroforestry models increase net farm income by **40% to 60%.**
  - **Supplies Off-Season Fodder:** Yields continuous high-protein green fodder (*Subabul*) for farm livestock during harsh lean summer months.

    National Livestock Mission reports that leguminous fodder trees (e.g., Subabul) bridge the **20% green fodder deficit** during lean summer months.
  - **Reduces Input Expenditures:** Decreases expensive commercial chemical fertilizer costs through biological nitrogen fixation by leguminous trees.

    ICAR data reveals that integrating nitrogen-fixing trees (e.g., Khejri) reduces commercial urea/ fertilizer input costs by **₹4,000 to ₹6,500 per hectare**.
  - **Generates High-Value Timber:** Creates long-term capital assets for smallholder farmers through the systematic cultivation of high-value timber woods.
  - **Eases Rural Energy Pressures:** Provides local, sustainable farm-sourced fuel wood, eliminating long hours spent foraging in natural forests.

    NITI Aayog energy briefs show that farm-sourced agroforestry fulfills up to **40% of local rural fuelwood needs.**
- **Climate Resilience & Biodiversity**
  - **Alleviates Natural Forest Stress:** Shifting timber and fuelwood extraction to private farmlands drastically reduces structural degradation pressures on wild forests.

Forest Survey of India (FSI) confirms that Trees Outside Forests (TOF) now fulfill over 80% of India's industrial timber demands.

- **Establishes Eco-Corridors:** Creates vital stepping-stone bio-corridors across fragmented agricultural zones for migratory birds and beneficial insects.
- **Pests Natural Regulation:** Attracts natural insect predators like birds and spiders to farm canopies, reducing chemical pesticide dependencies.
- **Mitigates Extreme Weather:** Works as functional physical shelterbelts that protect tender crops from devastating hailstorms and heatwaves.
- **Localizes Hydrological Cycles:** Aids rapid groundwater recharge by improving the water infiltration capacity of compacted agricultural soils.

Disadvantages of Agroforestry

- **Agronomic & Crop Production Trade-offs**
  - **Sunlight Competition:** Broad tree canopies intercept solar radiation, casting deep shade that significantly lowers the yields of understory photophilic (sun-loving) crops.

    Broad tree canopies cut incident light by 40-50%, triggering a 15-25% drop in understory crop yields
  - **Moisture Depletion:** Shallow-rooted tree species compete aggressively with adjacent crops for limited soil moisture, causing severe moisture stress during dry spells.

    Soil moisture profiling data shows that fast-growing species can exhaust up to **35% of available topsoil moisture** within a 5-meter radius.
  - **Nutrient Robbing:** Non-leguminous, fast-growing trees can deplete essential topsoil macronutrients, outcompeting seasonal food crops.
  - **Allelopathic Inhabitation:** Certain tree species (like *Eucalyptus*) exude toxic allelopathic chemicals from roots and leaf litter that suppress nearby seed germination.
  - **Harvesting Disruption:** Tall tree trunks and expanding root networks hinder the smooth maneuvering of modern agricultural machinery and tractors.
- **Economic & Financial Bottlenecks**
  - **Delayed Financial Returns:** Trees require long gestation periods (5 to 15 years) before yielding timber revenue, creating severe cash-flow crises for smallholder farmers.

    NABARD sector analyses highlight that popular agroforestry timber species have a long gestation period of **5 to 8 years (Poplar) and 10 to 15 years (Teak).**
  - **High Initial Capital:** Setting up complex systems requires high upfront expenditures on quality saplings, fencing, and specialized drip irrigation setups.
  - **Price Volatility Risk:** Long-term timber investments leave farmers highly vulnerable to market crashes and cheap wood import shocks at the time of harvest.
  - **Lack of Insurability:** Standard crop insurance schemes (like PMFBY) do not cover multi-year agroforestry systems, leaving farmers exposed to climate disasters.

    Over 90% of agroforestry timber remains completely excluded from PMFBY multi-year climate risk insurance.
- **Institutional & Policy Barriers**
  - **Stringent Transit Rules:** Tedious state-level felling and inter-state timber transit permit regulations discourage farmers from freely harvesting their own trees.
  - **Unclear Land Tenure:** Sharecroppers and tenant farmers are legally barred or disincentivized from planting long-gestation trees on rented farmlands.

- **Credit Access Deficits:** Commercial banks remain hesitant to provide low-interest agricultural loans for multi-year timber-based farming models.

  As per RBI, Multi-year agroforestry secures less than 1.5% of total bank credit due to rigid annual recovery cycles.

- **Extension Support Gaps**: Krishi Vigyan Kendras (KVKs) often lack specialized agroforestry extension workers to train farmers in advanced canopy management.

  As per ICAR, Fewer than 15% of operational Krishi Vigyan Kendras feature a dedicated agroforestry specialist.

- **Ecological & Biological Risks**
  - **Pest and Vector Harbors:** Dense tree canopies can inadvertently host crop-damaging birds, rodents, and fungal pathogens close to seasonal fields.
  - **Water Table Depletion:** Blindly planting thirsty exotic tree species as monocultures can over-extract deep aquifers and accelerate local groundwater depletion.

    As per CGWB, Exotic block plantations cause localized water tables to drop by 0.5 to 1 meter annually in semi-arid zones.

  - **Invasive Species Threat:** Selecting aggressive non-native species can lead to their escape into nearby natural ecosystems, choking out native biodiversity.

Measures taken for Agroforestry

Policy & Institutional Frameworks	• **National Agroforestry Policy (2014):** Made India the first nation in the world to adopt a comprehensive policy framework to remove bottlenecks and expand tree-based farming. • **National Agroforestry Mission (Sub-Mission):** Launched to operationally fast-track the policy by providing financial assistance, quality planting material, and institutional capacity building. • **National Forest Policy Target Alignment:** Structured agroforestry as the primary driver to expand "Trees Outside Forests" (TOF) to help achieve the mandate of 33% national green cover. • **Subramanian Committee Integration:** Adopted key governance insights to streamline environmental clearances, easing constraints for farmers growing commercial timber.
Regulatory & Legal Simplification	• **Felling and Transit Liberalization:** Promoted the deregulation of tedious state-level transit permits, exempting popular agroforestry species like Poplar, Eucalyptus, and Bamboo. • **Indian Forest Act Amendment (2017):** Omitted Bamboo grown on non-forest lands from the legal definition of a "tree," completely lifting strict cutting and transit restrictions for farmers. • **Model Agricultural Land Leasing Act:** Drafted to encourage long-term land security, enabling tenant farmers to invest confidently in long-gestation agroforestry trees.
Financial & Market-Driven Incentives	• **Green Credit Programme (GCP):** Operationalized market-linked incentives allowing private and public entities to trade credits generated through verifiable farm-forestry and eco-restoration. • **Restructured National Bamboo Mission:** Upgraded to develop robust, localized value chains, setting up bamboo processing clusters and linking farmers directly to artisan markets.

	• **NABARD's Wadi Model Funding:** Scaled up institutional financial backing to establish small-orchard agroforestry systems on degraded tribal lands. • **MSP for MFP Extension:** Expanded the price safety net to cover tree-borne oilseeds and minor timber elements, insulating agro-foresters from middlemen exploitation.
Technical, Digital & Extension Support	• **Bhuvan Agroforestry Portal:** Deployed ISRO's high-resolution satellite mapping to precisely monitor tree canopy growth and assess potential lands for agroforestry expansion. • **Har Medh Par Ped Campaign:** Integrated into central schemes to provide direct subsidies to smallholders for planting high-yield woodlots on agricultural field bunds. • **KVK Capacity Upgrades:** Equipped Krishi Vigyan Kendras with specialized agroforestry extension tools to train farmers in advanced canopy pruning and multi-tier cropping. • **Clonal Nursery Networks:** Established state-certified nurseries to supply high-yielding, pest-resistant, and climate-resilient clonal saplings to farmers.

Way forward

- **Regulatory & Legal Reforms**
  - **Universal Transit Deregulation:** Enact a uniform, pan-India green channel policy to completely eliminate inter-state transit permits for all major agroforestry species.
  - **Legalize Tree Tenancy:** Amend state land leasing laws to guarantee long-term tenancy rights, securing tree-ownership incentives for sharecroppers and tenant farmers.
  - **Delineate Forest vs. Farm Wood:** Create a distinct digital classification for Trees Outside Forests (TOF) to insulate farmers from rigid forest conservation laws.
- **Financial & Risk-Mitigation Innovations**
  - **Launch Composite Insurance:** Expand the PM Fasal Bima Yojana (PMFBY) into a long-term, multi-year insurance matrix covering timber gestation periods against weather shocks.
  - **Direct GCP Monetization:** Streamline the Green Credit Programme (GCP) registry to let smallholders directly trade carbon credits via simplified, local mobile apps.
  - **Provide Gestation Credit:** Design specialized, low-interest bank loans with 5-to-7-year moratoriums tailored to the growth cycle of commercial timber trees.
- **Supply Chain & Market Integration**
  - **Establish Digital Timber Mandis:** Launch online trading platforms to bypass middlemen, directly connecting agro-foresters with paper, plywood, and furniture industries.
  - **Scale Value-Addition Hubs:** Set up decentralized wood-processing micro-enterprises at the block level under the One District One Product (ODOP) scheme.
  - **Promote Certified Clonal Seeds:** Scale up public-private partnerships to establish accredited local nurseries supplying low-cost, pest-resistant clonal saplings.
- **Agro-Technological & Extension Upgrades**
  - **Deploy AI Canopy Management:** Use mobile-based AI tools to guide farmers on precise tree pruning schedules, minimizing shade-induced field crop losses.
  - **Mainstream Allelopathic Mapping:** Standardize Krishi Vigyan Kendra (KVK) soil testing to recommend native, non-allelopathic tree species that match specific field crops.

- **Expand Har Medh Par Ped:** Intensify boundary-plantation campaigns by leveraging MGNREGA funds for digging trenches and providing initial sapling care.

Best models for agro forestry

- **Malabar Neem (Melia dubia) Model (Western Ghats/Deccan):** A rapid-yielding commercial model pairing fast-growing plywood timber with understory cash crops like turmeric, ginger, and papaya.
- **Khasis Alder-based Agroforestry (Nagaland/Meghalaya):** An indigenous model integrating nitrogen-fixing *Alnus nepalensis* trees into shifting cultivation (*Jhum*) fields to accelerate soil fertility recovery.
- **Dehesa Silvopastoral Model (Spain/Portugal):** A historic Mediterranean model combining widely spaced oak trees with native pastures for high-value livestock grazing and sustainable cork harvesting.
- **Taungya Structural Model (Myanmar Origin):** An administrative system allowing landless forest-dwellers to raise short-term food crops between rows of establishing commercial timber plantations.
- **Quesungual Slashing Model (Honduras):** A climate-smart alternative to slash-and-burn farming where smallholder farmers prune native trees selectively rather than clearing them completely.
- **Acacia-Millet Alley Model (Sahel/Africa):** An arid-zone adaptation model planting rows of drought-resilient *Acacia* trees to fix nitrogen and shade intercropped pearl millet fields from scorching sun.
- **Three-Tier Multi-Crested Model (Kerala/Karnataka):** A high-density tropical canopy system layering tall coconut/arecanut trees, mid-tier cocoa or bananas, and ground-level black pepper vines.
- **Araku Shade-Grown Coffee Model (Andhra Pradesh):** A massive tribal-led cooperative model using silver oak and native fruit trees to shade organic coffee plants while generating structural carbon sinks.
- **Silvo-Fishery (Aqua-Forestry) Model (Sundarbans):** An estuarine restoration model planting protective mangrove barriers along brackish aquaculture ponds to filter organic effluent and shield farm bunds.
- **Homegarden (Kaavu/Bari) Model (Pan-India):** A highly resilient, traditional multi-tier backyard system fusing fuel-wood trees, fruit orchards, vegetables, and small livestock for total family self-sufficiency.
- **ITC Bhadrachalam Clonal Model:** A pioneering industry-linked model distributing high-yielding clonal eucalyptus saplings to buy back mature pulpwood directly from farmers.

Conclusion

The true success of India's agroforestry transition hinges on combining grassroots regulatory freedom—such as streamlined felling and transit green channels—with new-age market frameworks like the Green Credit Programme. Fusing traditional intercropping methods with precision technology and certified high-yielding clonal varieties can turn vulnerable smallholdings into highly productive, self-sustaining carbon sinks.

5. BLUE ECONOMY

Blue Economy represents the sustainable use of ocean and coastal resources for economic growth, improved livelihoods, and job creation, while preserving the health of marine ecosystems. It encompasses vital sectors such as fisheries, deep-sea mining, offshore renewable energy, and maritime trade.

Core objectives of the Blue Economy

- **Sustainable Growth:** Promoting economic development via marine activities like aquaculture, tourism, and biotechnology while preserving ocean health.
- **Resource Optimisation:** Harnessing living and non-living marine resources, including deep-sea minerals and hydrocarbons, within ecological limits.

- **Renewable Energy Generation:** Scaling up offshore wind, tidal energy, and wave power to reduce dependence on fossil fuels.
- **Maritime Security & Connectivity:** Enhancing coastal security, expanding modern port infrastructure, and improving shipping corridors (e.g., SAGAR initiative).
- **Livelihood & Food Security:** Creating green-blue jobs and enhancing fish production to support the nutritional needs of coastal communities.
- **Coastal Resilience & Climate Mitigation:** Protecting marine biodiversity, restoring mangroves, and using blue carbon sinks to combat rising sea levels.
- **Marine Spatial Planning (MSP):** Implementing data-driven zoning to balance industrial development, conservation efforts, and traditional fishing rights.

Need for the Blue Economy

- **Land Resource Exhaustion:** Relieves critical pressure on saturated terrestrial land resources by pivoting toward vast oceanic spaces for food and energy.
- **Geopolitical and Strategic Leverage:** Strengthens maritime diplomacy, counters regional dominance, and secures lines of communication under India's **SAGAR** vision.
- **Energy Security Transition:** Provides massive untapped capacity for offshore wind, tidal, and wave energy to accelerate India's green energy shift.
- **Economic Diversification:** Unlocks multi-billion dollar growth engines through deep-sea mining (e.g., polymetallic nodules), marine biotech, and coastal tourism.
- **Livelihood Upgradation:** Supports food security and uplifts over 4 million traditional fishers and coastal communities facing economic distress.
- **Climate Adaptation and Mitigation:** Leverages massive "blue carbon" sinks like mangroves and seagrasses to absorb emissions and defend coasts from extreme weather.
- **Trade and Logistics Efficiency:** Modernizes maritime infrastructure to cut high logistics costs and handle over 95% of India's international trade volume smoothly.

Components of Blue Economy

Renewable Energy	Renewable and Sustainable marine energy resources like offshore wind and tidal energy constitutes an important component of Blue Economy.
Fisheries	Sustainable management of fisheries ensures a continuous supply of fish for greater revenue while simultaneously contributing to the restoration of fish stocks. This supports both economic and environmental goals.
Maritime Transport	Sustainable, safe and secured maritime transport is an important component of the global economy, as it connects nations and facilitates trade.
Tourism	Ocean and coastal tourism offer recreational opportunities, contribute to job creation and economic growth, making it a key component of the blue economy.
Climate Change	Oceans act as crucial carbon sinks for the absorption and storage of carbon dioxide.

Issues and challenges associated with the Blue Economy

- **Ecological & Environmental Issues**

- **Marine Pollution:** Severe degradation of ocean ecosystems due to plastic waste dumping, agricultural run-off, and industrial chemical effluents.

  As per UNEP, India dumps over **0.6 million tonnes** of plastic waste into the ocean annually.

- **Unregulated Fishing:** Overexploitation of marine stocks by commercial trawlers, leading to rapid biodiversity depletion and habitat destruction.
- **Climate Vulnerability:** Acidification of oceans, warming sea temperatures, and rising sea levels that damage critical coral reefs and breeding grounds.

  As per IPCC, ocean acidification has increased by **26%** since the pre-industrial era, triggering bleaching in over **60%** of Gulf of Mannar coral reefs

- **Geopolitical & Jurisdictional Issues**
  - **Maritime Border Disputes:** Friction over Exclusive Economic Zones (EEZs) and overlapping claims that hinder collaborative deep-sea exploration.

    As per MEA, India's **2.02 million sq km** Exclusive Economic Zone (EEZ) faces active maritime boundary friction and fishing rights standoffs with Pakistan and Sri lanka.

  - **Piracy and Non-State Threats:** Security vulnerabilities along key trade routes from maritime piracy, smuggling, and illicit trafficking.
- **Economic & Technological Issues**
  - **High Capital Requirements:** Deep-sea mining and offshore renewable energy projects demand massive, high-risk capital with long gestation periods.

    Commissioning a deep-sea mining block or offshore wind farm requires an upfront capital investment **2.5 to 3 times higher** than onshore equivalents.

  - **Technological Deficit:** Inadequate domestic capability in manufacturing advanced deep-submergence vehicles and automated marine recovery tools.
  - **Poor Infrastructure:** Fragmented cold-chain facilities and outdated processing centers at domestic fishing harbors, causing high post-harvest losses.

    As per the Ministry of Fisheries, Lack of integrated automated cold-chains at domestic fishing harbors causes a staggering **20-25% post-harvest value loss** in marine catches annually.

- **Socio-Economic Issues**
  - **Marginalization of Local Fishers:** Large-scale commercial corporate fishing encroaching upon the traditional livelihoods of small-scale coastal communities.

    As per CMFRI Census, Large commercial mechanized trawlers grab over **75%** of near-shore economic value, directly sidelining **4 million** traditional, small-scale coastal fishers.

  - **Displacement Risks:** Coastal development projects and port expansions frequently cause the displacement and economic alienation of native populations.
  - **Occupational Hazards:** Lack of adequate social security, weather insurance, and safety gear for fishers facing increasingly volatile severe weather.

    As per NDMA, Over **90%** of traditional marine fishers lack comprehensive multi-year weather insurance.

Measures taken to develop the Blue Economy

- **Legislative & Policy Frameworks**

- **Draft National Blue Economy Policy:** Formulated by the Ministry of Earth Sciences to optimize coastal assets and integrate maritime spatial planning.
- **Coastal Regulation Zone (CRZ) Notification 2019:** Streamlined coastal development rules to balance economic growth with active marine ecosystem conservation.
- **Deep Ocean Mission (DOM):** Launched a central sector scheme to explore deep-sea resources like polymetallic nodules and develop underwater mining technologies.

- **Infrastructure & Trade Connectivity**
  - **Sagarmala Programme:** Upgrading domestic ports, improving port connectivity, and developing coastal economic zones (CEZs) to slash logistic costs.
  - **National Maritime Domain Awareness (NMDA) Project:** Integrated real-time tracking systems to enhance coastal safety, commercial navigation, and security.
  - **Inland Waterways Development:** Transforming key national rivers into active commercial cargo corridors to reduce heavy cargo load on roads and rails.
- **Fisheries & Marine Resource Management**
  - **Pradhan Mantri Matsya Sampada Yojana (PMMSY):** Investing ₹20,000+ crore to modernize the fisheries supply chain, boost exports, and insure fishers.
  - **Blue Revolution Scheme:** Focused on the comprehensive development and multi-layered management of inland and marine aquaculture assets.
  - **Fisheries and Aquaculture Infrastructure Development Fund (FIDF):** Provides concessional credit to state entities and private entrepreneurs for modern fishing harbor creation.
- **Environmental & Climate Actions**
  - **National Coastal Mission (NCM):** Dedicated to mapping marine vulnerabilities, restoring damaged coral reefs, and expanding dense mangrove cover.
  - **Marine Spatial Planning (MSP) Framework:** Partnered with Norway to execute pilot coastal zoning projects in Lakshadweep and Puducherry.
  - **India-Norway Task Force on Blue Economy:** Joint bilateral initiative established to advance sustainable ocean management and combat marine plastic litter.

- **Potential for the World Economy:** Oceans contribute to around 3-5% of Global GDP and over 80% of international goods transportation. The UN has declared the period 2021-2030 as the 'UN Decade of Ocean Science for Sustainable Development'.
- **Extent of Blue Economy in India:** India has a coastline of more than ~11,000 km spread across 9 states. India also has an Exclusive Economic Zone (EEZ) of more than 2.2 million sq km. India's blue economy accounts for ~4% of the GDP and is estimated to increase over the period of time.

Way forward

- **Policy & Governance Interventions**
  - **Finalize Blue Economy Policy:** Expedite the formal notification of the Draft National Blue Economy Policy to establish a unified maritime regulatory authority.
  - **Scale Up Marine Spatial Planning (MSP):** Implement data-driven ocean zoning nationwide to eliminate usage conflicts between ports, conservationists, and fishers.
  - **Strengthen Global Treaties:** Actively ratify and implement international frameworks like the UN High Seas Treaty (BBNJ) to safeguard ocean governance.
- **Technological & Infrastructural Upgrades**
  - **Indigenize Deep-Sea Tech:** Accelerate the development of indigenous human-submersible vehicles and remote mining tools under the Deep Ocean Mission.

- **Modernize Fishing Infrastructure:** Invest heavily in solar-powered cold chains, specialized processing parks, and deep-sea tuna long-liners at landing centers.
- **Deploy AI and IoT Systems:** Install smart marine sensor networks for real-time tracking of illegal fishing, oil spills, and coastal erosion dynamics.

- **Financial & Market Reforms**
  - **Introduce Blue Bonds:** Launch dedicated sovereign and corporate "Blue Bonds" to channel long-term institutional investment exclusively into marine projects.
  - **Expand Priority Sector Lending:** Direct commercial banks to offer concessional credit for sustainable aquaculture, seaweed farming, and marine startups.
  - **De-risk Offshore Energy:** Provide viability gap funding (VGF) to accelerate high-capEx offshore wind, tidal energy, and wave power installations.
- **Social & Ecological Safeguards**
  - **Promote Community-Led Conservation:** Involve traditional coastal communities in mass mangrove restoration (e.g., MISHTI scheme) and ecotourism management.
  - **Scale Sustainable Aquaculture:** Heavily incentivize climate-resilient seaweed cultivation and open-sea cage farming to diversify fisher livelihoods safely.
  - **Enforce Strict Waste Interception:** Build river-mouth plastic traps and upgrade municipal effluent treatment to completely halt land-based ocean pollution.

Committee recommendations for India's Blue Economy

- **Ramesh Chand Report (NITI Aayog, 2025/2026):** Recommends complete fleet modernization and cluster-based cooperative ownership to safely transition traditional fishers into high-value, deep-sea tuna long-lining.
- **Shailesh Nayak Committee (MoEFCC):** Recommended a comprehensive overhaul of the Coastal Regulation Zone (CRZ) rules to permit ecologically sustainable tourism and strategic infrastructure development within vulnerable coastal lines.

Conclusion

The blue economy is India's ultimate frontier to transform its vast coastline into an engine of sustainable wealth, turning oceans into corridors of green energy and food security. By seamlessly anchoring advanced deep-sea technology with traditional coastal wisdom, India can balance high-value economic growth with active marine conservation.

6. CIRCULAR ECONOMY

A **circular economy** is an economic model focused on minimizing waste by sharing, leasing, reusing, repairing, refurbishing, and recycling existing materials and products for as long as possible. Unlike the traditional **take-make-waste** linear model, it aims to create a closed-loop system where products and resources are maintained at their highest value.

Seven Pillars of the Circular Economy

- **Materials Design:** Engineering products using non-toxic, bio-based, or highly recyclable materials right from the start.
- **Product Journey:** Tracking and managing products through their entire lifecycle to optimize reuse and prevent them from reaching landfills.
- **Business Models:** Shifting from selling ownership to selling services, such as leasing, sharing, or pay-per-use platforms.

- **Ecosystem Thinking:** Designing local, interconnected systems where the waste or byproduct of one industry becomes the raw material for another.
- **Energy and Water:** Transitioning entirely to renewable energy sources and establishing closed-loop water purification and reuse systems.
- **Social & Cultural Capital:** Engaging communities and creating fair, inclusive jobs that support sustainable consumption and regional development.
- **Policy and Metrics:** Enacting supportive government regulations and using data-driven indicators to measure true resource efficiency.

Significance of the circular economy

- **Economic Significance**
  - **Import Substitution:** Cuts reliance on expensive foreign imports of critical minerals and crude oil via aggressive recycling.
  - **Cost Efficiency:** Maximizes manufacturing margins by using cheaper, recovered secondary raw materials instead of virgin resources.
  - **New Market Creation:** Fuels the growth of high-potential sectors like e-waste processing, vehicle scrapping, and refurbishing tech.
- **Environmental Significance**
  - **Decarbonization:** Directly supports India's Net-Zero 2070 target by lowering industrial manufacturing emissions.
  - **Landfill Mitigation:** Relieves severe urban land pressure by diverting municipal, plastic, and electronic waste from saturation points.
  - **Biodiversity Protection:** Minimizes habitat destruction caused by open-pit mining, logging, and extensive resource extraction.
- **Socio-Strategic Significance**
  - **Formalization of Labor:** Integrates millions of informal waste-pickers into the formal economy with safer, dignified green jobs.
  - **Global Leadership:** Drives India's international **LiFE (Lifestyle for Environment)** mission, showcasing sustainable development models to the Global South.
  - **Supply Chain Resilience:** insulates domestic manufacturing from volatile global commodity prices and geopolitical disruptions.

Challenges and issues in adopting a circular economy

- **Technological & Infrastructural Issues**
  - **Rudimentary Recycling Tech:** Domestic recycling is largely confined to low-value downcycling due to a lack of advanced sorting and recovery technologies.

    As per Ministry of Housing and Urban Affairs, Out of the municipal solid waste collected, only about **20-25%** is recycled or processed.
  - **Infrastructure Deficit:** India lacks integrated waste collection hubs, common effluent treatment plants, and specialized e-waste or vehicle scrapping yards.

    As per NITI Aayog, India generates over **3.4 million tonnes (MT)** of plastic waste and **1.6 MT** of e-waste annually, but leave over **80%** of cities without specialized processing hubs.
  - **Design Limitations:** Most consumer products are inherently designed for obsolescence, making disassembly, component recovery, and repair difficult.

- **Financial & Economic Issues**
  - **High Initial CapEx:** Upgrading factories to zero-waste or closed-loop systems requires massive upfront capital investments with long payback periods.
  - **Absence of Green Finance:** Credit mechanisms, venture capital, and low-interest bank loans tailored specifically for circular startups remain highly limited.

    As per RBI, Less than **3-5%** of the total priority sector lending and domestic venture capital investments are allocated toward specialized circular economy.
  - **Virgin vs. Recycled Cost Disparity:** Raw virgin materials are often cheaper than high-quality recycled alternatives, disincentivizing market adoption.

    As per Ministry of Petroleum and natural gas, High-quality food-grade recycled plastics cost up to **15-30% more** to produce cleanly in India than creating virgin raw alloys.
- **Regulatory & Institutional Issues**
  - **Informal Sector Dominance:** Over 80% of waste management is run by an unorganized, unregulated informal sector, making standard enforcement difficult.

    As per CPCB, Over **85-90%** of electronic and plastic waste management is driven by the unorganized, unregulated sector.
  - **Weak EPR Enforcement:** Extended Producer Responsibility (EPR) regulations suffer from fragmented monitoring, data gaps, and low compliance by manufacturers.
  - **Sub-national Policy Lacunae:** While central guidelines exist, state pollution control boards and urban local bodies often lack the administrative capacity to execute them.

    More than **60%** of State Pollution Control Boards (SPCBs) and Urban Local Bodies (ULBs) operate with severe staff shortages (vacancies over **40-50%**)
- **Socio-Cultural Issues**
  - **Use-and-Throw Mindset:** Rising urbanization and disposable incomes have rapidly accelerated consumerism and a preference for single-use conveniences.

    As per MoEFCC, Urban India's per capita waste generation is growing at a compound annual rate (CAGR) of **5%**.
  - **Stigma of Refurbished Goods:** A strong cultural preference for brand-new products limits the mainstream market demand for second-hand or repaired items.

Measures and initiatives taken to transition toward a circular economy

- **Legislative & Regulatory Frameworks**
  - **Comprehensive EPR Rules:** Notified mandatory Extended Producer Responsibility (EPR) targets across distinct categories including Plastics, E-Waste, Battery Waste, Waste Tyres and Used Oil.
  - **National Critical Mineral Mission:** Launched with a dedicated framework to heavily incentivize the extraction and recycling of lithium, cobalt, and nickel from spent electronic scrap.
  - **Vehicle Scrappage Policy:** Drives the Voluntary Vehicle-Fleet Modernization Program, offering registration concessions and scrap incentives to phase out unfit vehicles systematically.
- **Sectoral & Industrial Initiatives**
  - **Steel Scrap Recycling Policy:** Promotes organized scientific processing hubs to utilize domestic scrap steel, lowering the carbon intensity of primary steel manufacturing.
  - **Business Responsibility and Sustainability Reporting (BRSR):** Mandated by SEBI for top corporations to transparently disclose material consumption, product life cycles, and recycling metrics.

- **Urban & Waste-to-Wealth Interventions**
  - **GOBARdhan Scheme:** Successfully operationalized over 1,000 biogas plants nationwide to convert cattle residue and organic agricultural waste into commercial bio-CNG.
  - **Swachh Bharat Mission (U) 2.0:** Prioritizes 100% source segregation of urban waste, aiming for garbage-free cities by completely bioremediating legacy dumpsites and saturated landfills.
  - **Smart Cities Clean Initiatives:** Financed mechanical material recovery facilities (MRFs) and automated sorting systems across selected municipal corporations to streamline waste processing.
- **Behavioral & Financial Subventions**
  - **Mission LiFE (Lifestyle for Environment):** A global, citizen-led mass mobilization campaign to shift individual consumption patterns away from throw-away culture toward mindful utilization.
  - **SASCI Capital Assistance:** Central government allocates targeted infrastructure grants to state governments to expedite the setup of Registered Vehicle Scrapping Facilities (RVSFs).
  - **Green Credit Programme (GCP):** Establishes market-based tradable certificates to reward voluntary environmental actions, such as wasteland afforestation and industrial water recycling.

Way forward for a circular economy

- **Technological & Design Interventions**
  - **Mandate 'Right to Repair':** Institutionalize strict eco-design standards forcing manufacturers to make consumer electronics modular, easily repairable, and simple to disassemble.
  - **Invest in R&D:** Provide central funding and tax breaks for advanced sorting, AI-driven waste segregation, and high-value chemical recycling technologies.
  - **Develop Digital Product Passports:** Deploy blockchain-based tracking tags on industrial goods to monitor material composition and facilitate lifecycle tracing.
- **Economic & Financial Reforms**
  - **Rationalize GST Structure:** Lower the tax rates on recycled materials and refurbished goods to make them price-competitive with virgin raw materials.
  - **Scale Up Green Financing:** Integrate circular economy metrics into the RBI's priority sector lending and issue dedicated Sovereign Green Bonds for waste infrastructure.
  - **Incentivize Secondary Markets:** Create public procurement mandates that require government departments to buy products with a minimum percentage of recycled content.
- **Institutional & Structural Reforms**
  - **Formalize the Waste-Picking Sector:** Integrate informal waste collectors into municipal networks via micro-entrepreneurship models, offering them social security and proper safety gear.
  - **Strengthen EPR Enforcement:** Create a unified digital monitoring portal to eliminate data gaps, verify green credits, and penalize non-compliant producers swiftly.
  - **Empower Urban Local Bodies (ULBs):** Build municipal administrative capacity and fund decentralized material recovery facilities (MRFs) through local public-private partnerships.
- **Socio-Cultural Alignment**
  - **Institutionalize Mission LiFE:** Embed circular principles into school curriculums and launch mass media campaigns to eliminate the social stigma around using refurbished goods.
  - **Promote Product-as-a-Service (PaaS):** Foster shared-economy platforms for tools, automobiles, and machinery to transition society from an ownership mindset to a service-use model.

Global models and best practices for the circular economy

- **Legislative & National Framework Models**
  - **Japan's Sound Material-Cycle Society:** A strict top-down legislative framework prioritizing a comprehensive **3R (Reduce, Reuse, Recycle)** system with high urban compliance.
  - **EU's Circular Economy Action Plan (CEAP):** A pioneer model introducing **Digital Product Passports** and a legally binding **Right to Repair** to reduce electronic waste.
  - **China's Circular Economy Promotion Law:** The world's first national legislation integrating circular targets directly into macro-level multi-year industrial economic planning.
- **Industrial Symbiosis & Cluster Models**
  - **Denmark's Kalundborg Symbiosis:** A world-renowned private-sector model where the waste, steam, and byproducts of one factory serve as the direct raw input for an adjacent industry.
  - **South Korea's Eco-Industrial Parks (EIPs):** Government-backed regional industrial clusters designed to completely eliminate toxic wastewater and thermal emissions via inter-factory energy sharing.
  - **South Korea's Volume-Based Waste Fee (VBWF):** A municipal financial model charging citizens per unit of trash generated while leaving separated recyclables completely free of cost.
  - **Danish Deposit-Return System (Dansk Retursystem):** A high-efficiency public-private partnership achieving over 93% recycling rates for aluminum and plastic beverage containers via automated cash-back kiosks.

Conclusion

Transitioning to a circular economy is the definitive pivot needed to decouple India's economic growth from resource depletion, transforming waste from an environmental liability into a strategic asset. By blending the traditional ethos of *Mission LiFE* with advanced technology, India can pioneer a unique, community-led model of sustainable development for the entire Global South.

■■■

MGP/BP/GS3/017

DISASTER MANAGEMENT

1. DISASTER MANAGEMENT

"By failing to prepare, you are preparing to fail." — Benjamin Franklin

It is often said that while **"Disasters are natural; disasters turning into catastrophes are often man-made."** The objective of disaster management is not merely to respond to disasters, but to build a society that anticipates, withstands and recovers stronger from them.

Disaster management refers to the organized effort to reduce risk, prepare for, respond to, and recover from disasters — both natural (floods, cyclones, earthquakes) and man-made (industrial accidents, fires). It is not just about relief after a calamity strikes, but about **prevention, mitigation and building resilience** so that a hazard does not turn into a full-blown human tragedy. Given India's geography, climate vulnerability, and dense population, disaster management has become an urgent governance priority rather than an optional add-on.

Need for Disaster Management Strategy in India

1. **Rising Frequency and Intensity of Disasters**: Climate change is making hazards more frequent and severe, straining traditional response systems.

   Between 2008 and 2024, floods and storms displaced over 45 million and 15 million people respectively in India **(IDMC data).**
   Events like the **Wayanad landslides (2024) and Sikkim glacial lake outburst flood (2023)** show how erratic rainfall and glacial melt are creating new categories of risk.

2. **Massive Economic Losses**: Disasters wipe out productive assets and divert scarce public resources from development to relief.

   India lost about $79.5 billion economically to natural disasters between 1990 and 2023 **(UN Global Assessment Report, 2023)**, while disasters cause an estimated $31.59 billion in average annual losses to infrastructure assets alone **(CDRI Report**).

3. **Low Financial Protection**: Without adequate risk transfer mechanisms, the financial burden of disasters falls almost entirely on the state.

   Only about 5% of disaster-related damage in India is covered by insurance, compared to nearly 40% in the USA, leaving households and governments to absorb almost the entire shock.

4. **High Physical Vulnerability of the Country**: India's geography makes it naturally exposed to multiple hazards across regions.

   Nearly two-thirds of India's states and Union Territories are disaster-prone — the Himalayan belt to earthquakes and landslides, the coastline to cyclones and tsunamis, and the Gangetic plains to recurring floods **(NDMA, Annual Report 2021-22)**.

5. **Disproportionate Impact on the Poor**: Disasters push vulnerable and low-income households deeper into poverty, undoing years of development gains. With India's per capita GDP still below USD 3,000, a single flood or drought can devastate small farmers and daily-wage workers who have little capacity to recover on their own.
6. **Protecting Human Life and Preventing Displacement**: The core objective of disaster management is saving lives and minimising forced migration.

> As per the **IDMC's 2024 Global Report,** about 5,28,000 people were displaced in India due to natural disasters in a single year, underlining the human scale of the challenge.

7. **Safeguarding Critical Infrastructure**: Roads, power grids, and railways represent huge public investment that disasters can destroy overnight. A CDRI study (in partnership with the Department of Economic Affairs) found that the Roads, Power, and Rail sectors are among the most climate-exposed, threatening India's broader infrastructure pipeline.
8. **Rising Urban Risk**: Rapid, unplanned urbanisation is creating new disaster hotspots that older, district-centric systems were not designed to handle

> Recurring urban flooding events — Chennai (2015), Bengaluru (2022), Mumbai waterlogging — prompted the Disaster Management (Amendment) Act, 2025 to create dedicated **Urban Disaster Management Authorities (UDMAs)** for state capitals and large municipal cities.

9. **Constitutional and Welfare Obligation**: The State's duty to protect life under **Article 21** extends to protecting citizens from foreseeable hazards, not just responding after the fact. This is why the legal framework has evolved from a relief-based approach to a rights-based, **prevention-first approach under the Disaster Management Act, 2005** and its 2025 amendment.
10. **International Commitments and Global Leadership**: As a signatory to global frameworks, India has obligations to reduce disaster risk and share expertise with vulnerable nations.

> India is committed to the **Sendai Framework for Disaster Risk Reduction (2015–2030)** and launched the **Coalition for Disaster Resilient Infrastructure (CDRI)** in 2019, which now has over 50 member countries.

Conclusion

Disasters are no longer rare, isolated events — climate change and unplanned development are making them a recurring feature of India's growth story. Effective disaster management is therefore essential not just to save lives, but to protect the economy, infrastructure, and hard-won development gains, ensuring India's growth is resilient and inclusive rather than repeatedly set back by disasters.

> **"Prevention is better than cure."** *This timeless proverb should guide India's philosophy of disaster management .*

2. DISASTERS AND DISASTER RISK IN INDIA

The **Disaster Management Act, 2005** defines a disaster as a catastrophe, mishap or grave occurrence — natural or man-made — that causes substantial loss of life, property or environmental damage, beyond the coping capacity of the affected community.

> There is an Old Chinese proverb often cited -: "Dig the well before you are thirsty"

Key Disasters Faced by India & The Extent of Their Risk

1. **High Seismic Exposure**: A large part of the country lies in active earthquake zones due to the Himalayan tectonic collision and intra-plate faults.

As per the Bureau of Indian Standards (BIS) seismic zoning, nearly **58-59% of India's landmass** is susceptible to moderate-to-very-high seismic activity (Zones III, IV and V), with the Himalayan belt and the North-East being the most vulnerable.

2. **Recurring Flood Risk**: Erratic monsoon patterns and river systems originating in the Himalayas make floods India's most frequent disaster.

Over **40 million hectares (about 12% of India's land)** are flood-prone, with floods affecting an average of **7.5 million hectares every year**, concentrated in the Gangetic and Brahmaputra plains.

3. **Cyclone and Coastal Vulnerability**: A long coastline exposes eastern and western states to frequent tropical cyclones and storm surges.

About **5,700 km of India's coastline** is vulnerable to cyclones, and recent events like **Cyclone Montha (2025)** showed how shifting cyclone tracks and intensities are creating fresh preparedness challenges for coastal states.

4. **Drought and Agrarian Distress**: Uneven rainfall distribution and dependence on the monsoon leave large parts of the country chronically water-stressed.

Nearly **68% of India's cultivable land is drought-prone**, especially across western and central India, deepening agrarian distress in states like Maharashtra and Rajasthan.

5. **Himalayan Fragility — Landslides and GLOFs**: Young, unstable mountain geology combined with unregulated construction increases landslide and glacial risk.

The **Sikkim Glacial Lake Outburst Flood (2023)** and the **Wayanad landslides (2024)** highlighted how glacial melt and deforestation are turning the Himalayas and Western Ghats into high-risk zones.

6. **Emerging Heatwave Risk**: Rising temperatures linked to climate change are turning heatwaves into a recognised disaster category rather than a seasonal inconvenience.

The **2024 heatwave** saw record-high temperatures across northern India, leading to thousands of heat-related hospitalisations, prompting States to notify **Heat Action Plans** under the Disaster Management Act.

7. **Man-Made and Industrial Disasters**: Rapid industrialisation without matching safety regulation adds a second, human-driven layer of disaster risk.

The **Vizag gas leak (2020)** exposed gaps in India's chemical disaster preparedness, prompting renewed focus on NDMA's industrial and chemical (CBRN) safety guidelines.

8. **Urban Disaster Risk**: Unplanned urban growth, poor drainage, and encroachment on wetlands are turning ordinary rainfall into urban flooding disasters.

Cities like **Chennai (2015 floods)**, **Bengaluru (2022 floods)** and **Mumbai (annual waterlogging)** illustrate why the government created dedicated **Urban Disaster Management Authorities (UDMAs)** under the 2025 amendment.

9. **Socio-Economic Vulnerability**: Poverty, informal housing, and weak insurance penetration convert even moderate hazards into severe disasters for the poor.

With **only about 5% of disaster losses insured** in India (compared to 40% in the USA), low-income households and daily-wage workers bear a disproportionate share of every disaster's cost.

10. **India's Standing in Global Risk Assessments**: International indices confirm that India's disaster risk is among the highest globally, driven by both exposure and vulnerability.

The **World Risk Report 2025** places India among the countries facing the **highest overall disaster risk** globally (alongside the Philippines and Indonesia), while the **Global Climate Risk Index** ranks **India 9th most climate-vulnerable nation**.

Conclusion

India's disaster risk stems from a rare combination of almost every natural hazard . Understanding disaster risk as an interplay of hazard, exposure and capacity, rather than as an act of nature alone, is what has pushed India's policy shift from post-disaster relief towards pre-disaster risk reduction and resilience-building.

3. DISASTER MANAGEMENT IN INDIA

Transition from relief centric to preventive approach

There is an Old proverb that : "A stitch in time saves nine"

Before 2005, India's approach to disasters was **relief-centric** — the government stepped in only after a calamity struck, and disaster management sat under the Ministry of Agriculture, a colonial-era hangover from droughts and famines.

The **2004 Indian Ocean Tsunami**, which killed over 10,000 people in India, exposed this gap badly — there was no unified law, no dedicated force, and chaotic centre-state coordination.

This led to the **Disaster Management Act, 2005**, which built a permanent, multi-tier institutional structure. It was substantially updated recently through the **Disaster Management (Amendment) Act, 2025** (assented on 29 March 2025, in force from 9 April 2025) to fix gaps exposed over two decades, including during COVID-19.

Existing Institutional Framework (Including latest amendments to Disaster Management Act)

1. **National Disaster Management Authority (NDMA)**: The apex body, chaired by the **Prime Minister**, responsible for laying down policies, plans, and guidelines for disaster management at the national level.

   NDMA's push to shift from relief to **prevention and mitigation** is credited with India's near "zero-casualty" approach in recent cyclones — Cyclone Fani (2019) caused only 64 deaths compared to the 1999 Odisha super-cyclone's 10,000+ deaths.

2. **State and District Disaster Management Authorities (SDMA/DDMA)**: A **three-tier federal structure** (national–state–district) under Sections 14 and 25 of the Act, ensuring national guidelines are adapted to local hazards.

   An SDMA in a seismic state like Uttarakhand focuses on earthquake and landslide preparedness, while a coastal DDMA in Tamil Nadu focuses on cyclone evacuation — same law, locally tailored plans.

3. **National Disaster Response Force (NDRF)**: A specialised, uniformed disaster response force under the Act, currently structured to be expanded — the 2025 Amendment enables scaling **NDRF up to 16 battalions**.

   NDRF teams led rescue operations during the **Wayanad landslides (2024)** and the **Sikkim GLOF (2023)**, becoming India's go-to responders for large-scale disasters

4. **National Institute of Disaster Management (NIDM)**: The country's premier institute for training, research, and capacity-building in disaster management, working closely with NDMA.

NIDM trains state and district officials on the **Sendai Framework for Disaster Risk Reduction (2015-2030)**, keeping India's local disaster planning aligned with global best practice.

5. **National Crisis Management Committee (NCMC) and High-Level Committee (HLC)**: Previously informal administrative arrangements, these have now been given **statutory status** by the 2025 Amendment for faster, legally-backed decision-making during emergencies.

The HLC is the body that approves financial assistance to states from the **National Disaster Response Fund (NDRF-fund)** and the newly created **National Disaster Mitigation Fund (NDMF)**

While NCMC is chaired by the cabinet secretary , the HLC is chaired by the Union Home Minister.

6. **Urban Disaster Management Authorities (UDMAs)**- **2025 Amendments allowed** the creation of a dedicated city-specific authority, headed by the Municipal Commissioner, created to tackle urban-specific hazards like flooding and heatwaves that district-centric planning missed.
7. **State Disaster Response Force (SDRF)**: The 2025 Amendment formally empowers state governments to raise their own **SDRFs**, giving states first-responder capacity instead of relying solely on the central NDRF.

This addresses a long-standing complaint that 16 NDRF battalions are too thin for a country of 1.4 billion people spread across flood, cyclone, and earthquake zones simultaneously

8. **National Disaster Database**: A new, mandatory database under the 2025 Amendment consolidating **risk assessments, mitigation plans, and real-time disaster data** for evidence-based planning.

This is meant to work alongside tech upgrades like **Mission Mausam (launched 2025)**, which uses AI/ML models for 7-day advance flood and cyclone forecasting.

9. **Financial Architecture — NDRF, SDRF, and Mitigation Funds**: A structured funding mechanism, guided by **Finance Commission recommendations**, ensures disaster relief and mitigation money reaches states.

The government has stated in the parliament that funds have been released to states **as per or above** the 15th Finance Commission's prescribed disaster relief formula — though **unspent SDRF balances** in several states remain a flagged concern.

10. **Sendai Framework Alignment**: India's National Disaster Management Plan (first issued 2016) was the **first national plan in the world** explicitly aligned with the Sendai Framework's priorities for global disaster risk reduction.

4. KEY ISSUES WITH EXISTING FRAMEWORK OF DISASTER MANAGEMENT

Though , India built an elaborate, three-tier disaster management architecture through the **Disaster Management Act, 2005**, and strengthened it further via the **2025 Amendment**. Yet, as multiple **CAG performance audits** have repeatedly flagged since 2013, having institutions on paper is not the same as having them work effectively on the ground. Nearly two decades in, the framework still struggles with implementation gaps, financing disputes, and outdated planning tools — issues that surface painfully every time a major disaster strikes.

Key Issues

- **NDMA's Weak Supervisory Role**: The apex body was designed as the country's planning and

supervising authority, but audits have found it lacks real control over state-level implementation.

> The **CAG's 2013 Performance Audit of Civil Disaster Preparedness** found NDMA had **neither information nor control** over the progress of work at the state level, and was unsuccessful in implementing many of its own projects.

- **Non-Functional District Disaster Management Authorities (DDMAs)**: DDMAs are legally constituted almost everywhere but frequently exist only on paper, with plans never actually formulated or reviewed.

> The CAG's **2016 audit of Jammu & Kashmir** found DDMAs, though constituted, were **non-functional**, and the District Disaster Management Plan of Leh — approved back in May 2011 — had **never been implemented or reviewed** even years later.

- **Weak Enforcement of Building Codes**: Disaster risk reduction depends heavily on enforcing safe construction standards, but this vetting is routinely skipped at the local level.

> A **2025 CAG audit of Mizoram** found the State Executive Committee had **not vetted construction projects** for compliance with the **National Building Code, 2005** and relevant BIS codes — despite Mizoram's high landslide and earthquake risk.

- **Advisory Bodies Falling Dormant**: Statutory consultative bodies meant to guide disaster policy often stop meeting altogether, weakening the entire planning process upstream.
- **Slow, Politically Contested Fund Disbursal**: The classification of a disaster as "severe" (which unlocks bigger central funding) is discretionary, leading to repeated Centre-State disputes over the amount and timing of relief.

> During the **Wayanad landslides (2024)**, Kerala sought **₹2,219 crore** in relief, while Centre-State exchanges over classification and memoranda dragged on for months — Home Minister Amit Shah citing delayed submission of damage memoranda, and Kerala citing inadequate and delayed central assistance.

- **Unspent and Poorly Tracked Disaster Funds**: Ironically, even when money is released, poor financial tracking at the state level means it often isn't spent on the ground when it's needed most.

> In the Wayanad case, the **Kerala High Court** itself reprimanded the state government in December 2024 for failing to clarify how much of its ₹677-782 crore SDRF balance was actually available for immediate relief — and separately, ₹529.50 crore released by the Centre for Wayanad rebuilding remained **completely unspent as of August 2025**, per a Lok Sabha reply.

- **Risk-Blind Fund Allocation Formula**: The Finance Commission's method for distributing disaster funds relies on **population and geographical area**, not actual, scientific hazard exposure.
- **NDRF Personnel and Capacity Shortage**: India's specialist disaster response force remains stretched thin relative to the scale of the country and the frequency of disasters.

> Analysts have long pointed out that NDRF personnel often **lack sufficient training, equipment, and residential accommodation**, and even after the 2025 Amendment's push to scale up to 16 battalions, that is a thin force for 1.4 billion people.

- **Man-Made Contributions to "Natural" Disasters**: Poor urban planning and encroachment on natural drainage often turn ordinary rainfall into major disasters, which audits classify as governance failures, not just weather events.

> The **CAG's report on the 2015 Chennai floods** explicitly termed the disaster **"man-made,"** holding the Tamil Nadu government responsible for poor drainage management and encroachment on the Adyar and Cooum river floodplains.

- **Slow Implementation of the 2025 Reforms Themselves**: Even the newest fixes — like Urban Disaster

Management Authorities — are struggling to move from law to practice.

More than a year after the 2025 Amendment mandated UDMAs for all state capitals and municipal corporation cities, **only Karnataka** has actually constituted one (for Bengaluru), showing how slowly reforms translate into ground-level institutions.

Key Recommendations : Way Forward

Committee	Key Recommendations
High Powered Committee (HPC) (1999–2001)	• Draft a **National Calamity Management Bill** and a **Model State Disaster Management Bill**. • Establish dedicated institutional mechanisms for managing both **natural and man-made disasters**. • Shift from an ad hoc relief approach to an institutional disaster management framework. • *(On the basis of its recommendations, the Disaster Management Act 2005 was passed)*
National Commission to Review the Working of the Constitution (NCRCW)	• Include **"Management of Disasters and Emergencies"** in the **Concurrent List** of the Constitution. • Strengthen Centre–State coordination in disaster management through a constitutional framework.
2nd Administrative Reforms Commission (2006) – Crisis Management	• Shift from **relief-centric** to **prevention, mitigation and preparedness**. • Integrate **Disaster Risk Reduction (DRR)** into development planning. • Invest more in mitigation, recognising that prevention is more cost-effective than relief.
2nd ARC : On Institutional Reforms	• Clearly delineate the roles of the **Centre, States and Districts**. • Decentralise disaster management by empowering **States and District Collectors**. • Strengthen institutional coordination and crisis response mechanisms.
2nd ARC : On building Community Resilience	• Integrate **traditional and indigenous knowledge** with modern disaster science. • Promote **Community-Based Disaster Management (CBDM)** • Build awareness, training and local capacity for disaster preparedness.
15th Finance Commission (2020–25)	• Create **National Disaster Mitigation Fund (NDMF)** and **State Disaster Mitigation Fund (SDMF)**. • Increase financial support for **pre-disaster mitigation and preparedness**. • Reduce excessive dependence on post-disaster relief funding.

15th Finance Commission – Disaster Financing	• Adopt **risk-based allocation** of disaster funds instead of relying mainly on population and geographical area. • Incorporate hazard vulnerability into fund distribution criteria.
Sendai Framework for Disaster Risk Reduction (2015–2030) (Global Framework)	• Prioritise **Disaster Risk Reduction (DRR)** over disaster response. • Adopt the principle of **"Build Back Better"** during recovery and reconstruction. • Strengthen resilience through risk-informed development planning.

Conclusion

Disasters are becoming **less "natural" and more a test of governance** — how well a nation plans, warns, builds, and responds decides whether a hazard becomes a headline or a catastrophe. India has moved from a **relief-driven** to a **prevention and mitigation-driven** approach, but as Wayanad, Mizoram, and repeated CAG audits show, laws on paper still outrun action on the ground. The real shift needed now is from **managing disasters to managing risk** — investing before the crisis, not just responding after it.

As the saying goes: "The best time to fix the roof is before it rains." India's disaster management journey will be judged not by how fast it recovers, but by how much it prevents in the first place.

5. DISASTER VULNERABILITY AND RISK ASSESSMENT

Vulnerability refers to the physical, social, economic, and environmental factors that make a community susceptible to the damaging effects of a hazard, while **risk assessment** is the scientific process of measuring how likely a hazard is to cause loss, given that vulnerability.

The assessment of disaster vulnerability is important because "You cannot manage what you do not measure."

Current Disaster Vulnerability Assessment mechanisms in India

Mechanism	Key Features
Vulnerability Atlas of India	• Maps **district-wise vulnerability** to earthquakes, floods, cyclones/wind and landslides. • Assesses the vulnerability of **housing stock and built infrastructure** • Serves as a planning tool for **hazard-resistant construction, land-use planning and retrofitting**.
National Disaster Risk Index (MHA, 2019)	• Ranks **640 districts** based on disaster risk • Uses **14 hazard indicators, 14 vulnerability indicators and 2 exposure indicators**.
Disaster Resilience Index	• Assesses the **coping capacity and resilience** of States and UTs • Evaluates **institutional preparedness, early warning systems and mitigation capacity** • Highlights gaps between **hazard exposure and disaster preparedness**.

Multi-Hazard District Risk Mapping	• Prepares **hazard-specific risk maps** for heat waves, cold waves, floods, forest fires, landslides and industrial hazards. • Uses scientific datasets from **IMD, IITM, CWC and NDMA**. • Supports preparation of **Heat Action Plans** and other hazard-specific mitigation strategies.

An Assessment Of India's Multidimensional Vulnerability To Natural And Manmade Disasters

Vulnerability is the set of physical, social, economic, and environmental conditions that increase a community's susceptibility to a hazard's impact. India's disaster vulnerability cannot be assessed through **geography alone** — being in a flood plain or seismic zone is only half the story. The other half is **who lives there**: their income, housing, gender, caste, and access to warning systems. A full assessment therefore needs to look at both physical exposure and the socio-economic layers that decide who actually suffers when a hazard strikes.

Dimensions of India's Vulnerability

1. **Physical/Geographic Vulnerability**: India's geography exposes it to nearly every major hazard type simultaneously — seismic, hydro-meteorological, and climatic.

   As per BIS's revised seismic code (2025), **61% of India's landmass** falls in moderate-to-high earthquake risk zones, while roughly **40 million hectares** remain flood-prone — a rare double exposure most countries don't face together.

2. **Social Vulnerability- Declining but Uneven**: A national-scale academic study using two decades of data found India's overall **Social Vulnerability (SoV) has declined**, largely due to rising female literacy and more people moving from marginal to main occupations. However, this decline is not even across the country.

3. **Gendered Vulnerability**: Women face disproportionately higher risk due to occupational exposure, mobility restrictions, and unequal access to relief and resources.

   During the **2004 Indian Ocean Tsunami**, women made up **70% of the fatalities in coastal Tamil Nadu** — a figure widely cited in disaster gender studies, reflecting lower swimming ability, caregiving duties that delayed evacuation, and restrictive social norms.

4. **Vulnerability of Marginalised Communities (SC/ST and Urban Poor)**: Disaster recovery capacity is not uniform — Scheduled Castes, Scheduled Tribes, and the urban poor consistently take longer to recover from the same disaster event.

   Studies flag that these groups, often living in the **most hazard-exposed and lowest-quality housing**, face compounding barriers — limited savings, weak insurance access, and social exclusion from relief networks — meaning disaster policy needs group-specific measures, not a one-size-fits-all approach.

5. **Occupational & Livelihood Vulnerability**: A large share of India's workforce depends directly on climate-sensitive sectors, transmitting hazard impact straight into household income loss.

   Roughly **65% of Indian women workers are employed in agriculture** (2011 Census), often as farm labourers with no formal contracts — meaning a single drought or flood can wipe out their livelihood overnight with no institutional safety net.

6. **Housing and Infrastructure Vulnerability**: A large share of India's housing stock is structurally unfit to withstand disaster-scale hazards, multiplying casualties from moderate-intensity events.

7. **Economic Vulnerability and Weak Risk Transfer**: Low insurance penetration means disaster losses are absorbed directly by already-poor households rather than spread through financial systems.

> India's disaster insurance penetration remains around **5%**, compared to roughly 40% in the USA, meaning most families and small farmers rebuild from personal savings or debt after every disaster — deepening long-term economic vulnerability.

8. **Climate Change as a Vulnerability Multiplier**: Rising temperatures and erratic rainfall are intensifying hazards faster than India's adaptive capacity is improving, widening the vulnerability gap over time.

> A 2026 policy analysis by **NIDM** flagged that states like **Odisha, Andhra Pradesh, Kerala, and Assam** face rising future risk from intensifying cyclones and extreme rainfall, yet remain **underserved by current disaster resilience formulas** that don't account for this changing risk profile.

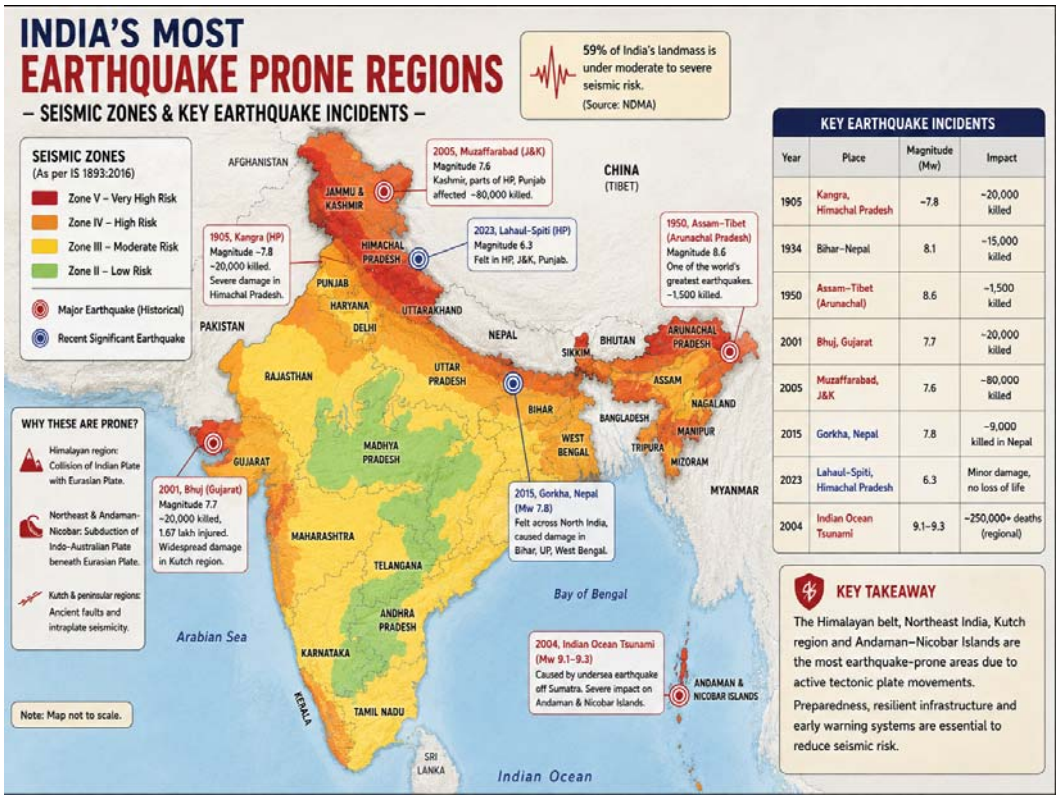
9. **Institutional/Governance Vulnerability**: Weak last-mile implementation , non-functional DDMAs, unenforced building codes, dormant advisory committees — adds a governance layer of vulnerability on top of physical and social factors.

> CAG audits in **Jammu & Kashmir and Mizoram** have shown that even well-designed institutions fail to translate into ground-level resilience if District Disaster Management Plans are never implemented or reviewed for years at a stretch.

6. EARTHQUAKES IN INDIA

India sits partly on the boundary of the **Indian and Eurasian tectonic plates**, which continue to collide at roughly **5 cm per year** — one of the most active convergent zones on Earth. This constant tectonic stress makes large parts of the country genuinely earthquake-prone, not just in the Himalayas but across the northeast and parts of the peninsula too.

As per **NDMA**, India has witnessed **10 major earthquakes in the last 15 years, causing over 20,000 casualties** — making earthquake risk one of the most life-threatening hazards the country faces.



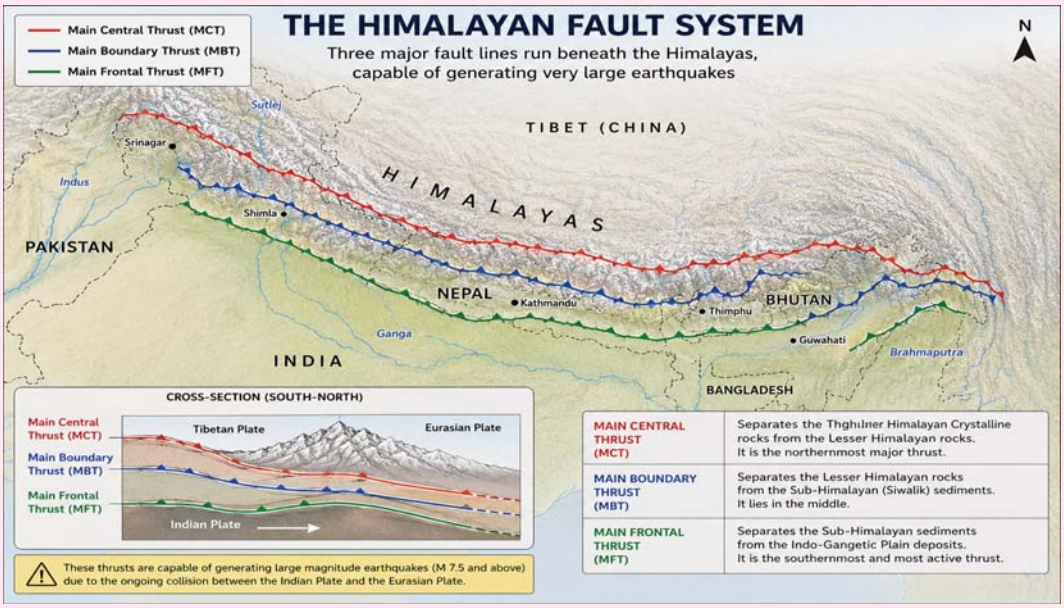
Key Dimensions of India's Earthquake Vulnerability

1. **Currently Applicable Seismic Zonation** : India is officially divided into **four seismic zones (II, III, IV, V)**, with Zone V being the most active and Zone II the least, based on this **currently applicable** BIS code.

> As per this 2016 code, roughly **59% of India's landmass** falls into moderate-to-high seismic hazard zones (Zones III–V) — this remains the legally binding classification for construction today.

2. **The Himalayan Fault System**: Three major fault lines — the **Main Central Thrust (MCT)**, **Main Boundary Thrust (MBT)**, and **Main Frontal Thrust (MFT)** — run beneath the Himalayas, capable of generating very large earthquakes.

> Scientists warn some of these fault segments have **not ruptured in 500-700 years**, meaning stored tectonic energy could eventually trigger a "great earthquake" (magnitude 8+), similar to the pattern behind the **2015 Nepal earthquake (M7.8)**, which also caused damage in Bihar and North India.



3. **High Frequency of Tremors**: The Himalayan and northeastern regions experience frequent low-to-moderate tremors, a sign of ongoing tectonic stress build-up. India recorded **159 earthquakes in just four months** between November 2024 and February 2025 — most were too small to cause damage, but they were clear indicators of continuous seismic activity in these zones.
4. **Unregulated Hill-Town Construction**: Rapid, poorly planned urbanisation in Himalayan towns often ignores slope stability and building codes, converting a natural hazard into a much bigger human disaster.

> The **Joshimath land subsidence crisis (2023)**, while triggered by multiple factors, exposed how unregulated construction and drainage failure in a seismically sensitive Himalayan town can cause large-scale structural damage even without a major earthquake.

5. **Northeast India's Risk**: The northeastern states sit at the intersection of the Himalayan and Indo-Burman tectonic systems, making them nearly as vulnerable as the Himalayan belt itself, though it receives comparatively less national attention.

> The region has already experienced two of the most powerful earthquakes recorded in India — **Assam, 1897 (M8.1)** and **Assam-Tibet, 1950 (M8.6)** — both were among the largest ever recorded on land globally.

6. **Human and Economic Costs** : Past major earthquakes illustrate the scale of damage possible when hazard meets high population density and weak construction.

> The **2001 Bhuj (Gujarat) earthquake (M7.7)** killed close to **20,000 people** and caused massive economic losses, remaining one of the deadliest earthquakes in independent India's history and a key trigger for India's post-2001 disaster management reforms

7. **Growing Urban Exposure**: As India urbanises, more people and more expensive infrastructure are being concentrated in seismically active cities, raising the potential scale of future losses

> A World Bank research projects that by **2050, around 200 million urban Indians** will be vulnerable to earthquakes and storms combined — a scale that makes retrofitting and code enforcement an urgent, not optional, priority.

Key Measures Taken Up for Earthquake Mitigation and Management in India

Measure	Key Features
National Centre for Seismology (NCS)	• Nodal agency for **24×7 earthquake monitoring**. • Operates the **National Seismological Network** across India. • Provides real-time earthquake detection and scientific data.
National Building Code (NBC) & BIS Seismic Codes	• Prescribes **earthquake-resistant construction standards** • Promotes safe construction and retrofitting of vulnerable buildings.
National Retrofitting Programme	• Strengthens **lifeline buildings** such as hospitals, schools and government offices. • Encourages retrofitting of old structures in high seismic zones. • Promotes compliance through financial and regulatory measures.
National Earthquake Risk Mitigation Project (NERMP)	• Builds institutional capacity in **earthquake-prone cities** • Trains **engineers, architects, masons and urban planners**.Promotes earthquake-resistant construction practices.
Earthquake Disaster Risk Index (EDRI)	• Assesses **hazard, vulnerability and exposure** of cities. • Helps prioritise mitigation investments based on scientific risk assessment • .Supports evidence-based urban planning.
Real-Time Public Alert Systems	• Disseminates **verified earthquake information** to the public. • Uses digital platforms like the **BhooKamp App**. • Reduces panic and misinformation during disasters.
Institutional Framework (NDMA–SDMA–NDRF)	• NDMA formulates national policies and guidelines. • SDMAs coordinate state-level preparedness. • NDRF undertakes specialised **search, rescue and relief operations**.
Aapda Mitra Scheme	• Trains **community volunteers** as first responders • Conducts mock drills and public awareness programmes. • Strengthens community-based disaster preparedness.

NDMA Earthquake Management Guidelines (2007)	• Promote **earthquake-resistant construction** and retrofitting. • Strengthen regulation, enforcement and capacity building. • Enhance emergency preparedness and public awareness.
International Cooperation & Regional Exercises	• Conducts joint disaster response exercises with partner countries. • Shares seismic knowledge and best practices. • Strengthens regional preparedness through initiatives like **SCOJtEx** and **BIMSTEC Disaster Management Exercises**.

National/ International Best Practices

Case Study	Key Learnings
Bhuj Earthquake Reconstruction, Gujarat (2001)	• Established **Gujarat State Disaster Management Authority (GSDMA)** within two weeks. • Adopted **Owner-Driven Reconstruction (ODR)** with financial and technical support to homeowners. • Achieved **rapid reconstruction** of damaged houses. • Enacted the **Gujarat Disaster Management Act, 2003**, which inspired the **DM Act, 2005**. • Demonstrated **"Build Back Better"**, leading to strong economic recovery in Kutch.
Japan's Earthquake Early Warning System (J-Alert / EEW)	• Uses **4,200+ seismometers** for real-time earthquake detection. • Issues **seconds-early warnings** before destructive S-waves arrive. • Automatically halts **bullet trains, elevators, factories and gas supply**. • Strengthens **public preparedness** through technology and awareness. • Serves as the **global benchmark** for earthquake early warning systems.

Way Forward

India already has an earthquake management system in place — the NDMA, seismic monitoring, building codes, and retrofitting programmes. But experts and government reviews themselves point out that the real gaps are still there in **enforcement, technology, and financing**,and policy design.

Key Measures Needed

1. **Dedicated Earthquake Early Warning System (EEWS)**: India currently has strong seismic monitoring but no functioning public early warning system like Japan's, which gives citizens precious seconds before shaking begins.

   Though the government has confirmed that an **EEWS for the Himalayan region is still in the research phase**, experts have repeatedly flagged this as the single biggest technological gap compared to countries like Japan and Mexico.

2. **Dedicated Annual Budget for Retrofitting of Critical infrastructure** : Policy analysts recommend earmarking **around ₹50,000 crore per year** specifically for retrofitting, seismic audits, and risk-reduction projects — turning retrofitting from a one-off scheme into a sustained national programme.
3. **Incentive-Based Retrofitting Compliance**: Rather than relying only on legal enforcement, experts

suggest making retrofitting financially attractive so private owners actually comply.

> A proposed model includes **tax breaks or development-rights incentives** for owners who retrofit their buildings to current seismic codes

4. **Stronger Urban Planning Integration**: Experts point out that even flagship programmes like the **Smart Cities Mission** have not built earthquake preparedness into urban planning, treating seismic safety as a separate, legal-enforcement issue rather than a core design principle.
5. **Learning from Japan's Building Technology**: Experts recommend India study and adapt specific Japanese engineering practices — such as their **Shin-Taishin seismic regulations and base-isolation techniques** — to strengthen India's own BIS codes
6. **Multi-Hazard Planning, Not Just Earthquake-Specific Plans**: Since a major earthquake often triggers secondary disasters like landslides, fires, or tsunamis, experts argue India's plans need to account for these cascading effects together.

> The **2011 Great East Japan Earthquake**, which triggered both a tsunami and a nuclear crisis, is often cited as the reason India's **National Disaster Management Plan (2016)** needs to more explicitly integrate multi-hazard scenarios, especially for the Himalayan and coastal regions.

7. **Expanded Public Awareness and School Curricula**: Experts consistently flag that earthquake preparedness is not deeply embedded in Indian public consciousness, with drills and awareness campaigns remaining sporadic rather than routine.

> The suggested measures include embedding **regular, mandatory school earthquake drills** (similar to Japan's culture of preparedness) and making programmes like Doordarshan's **"Aapda ka Samna"** a consistent, scaled-up feature rather than an occasional broadcast.

8. **Faster and Graded Retrofitting of Critical Lifeline Buildings**: Rather than trying to retrofit all buildings at once, experts suggest prioritising hospitals, schools, and emergency-response buildings first, since their failure has the highest human cost.

> Following the Bhuj earthquake, **AIIMS New Delhi** and hospitals in Bhuj were among the first structures retrofitted — experts want this **"critical infrastructure first" model** formally extended nationwide, rather than left to individual institutions to initiate.

9. **Simplified, More Enforceable Building Codes**: Overly technical codes are harder for smaller towns and local municipal bodies to actually implement and check, so experts recommend simplifying compliance requirements without diluting safety standards.
10. **Resolving the Regulatory Uncertainty from the 2025 Code Rollback**: Given that BIS's revised 2025 seismic code (with the new Zone VI) was withdrawn in March 2026 over cost concerns, experts and geologists are pushing for a **phased, cost-sensitive rollout** rather than an outright reversal, so India doesn't lose the benefit of improved seismic science.

Conclusion

With India still lacking a functioning early warning system and facing a genuine **20-crore-building retrofitting backlog**, the biggest suggested shift is treating earthquake resilience as a **sustained, budgeted national mission**, not a one-time scheme revived only after the next big quake.

7. TSUNAMI HAZARD

A **tsunami** is a series of large ocean waves triggered by the sudden, large-scale displacement of seawater — most commonly caused by an undersea earthquake, though volcanic eruptions, landslides, and (rarely) meteorite impacts can also generate one.

Causes of Tsunami in the Indian Region

Unlike the Pacific Ocean, which sees around **75% of the world's tsunamigenic activity**, the Indian

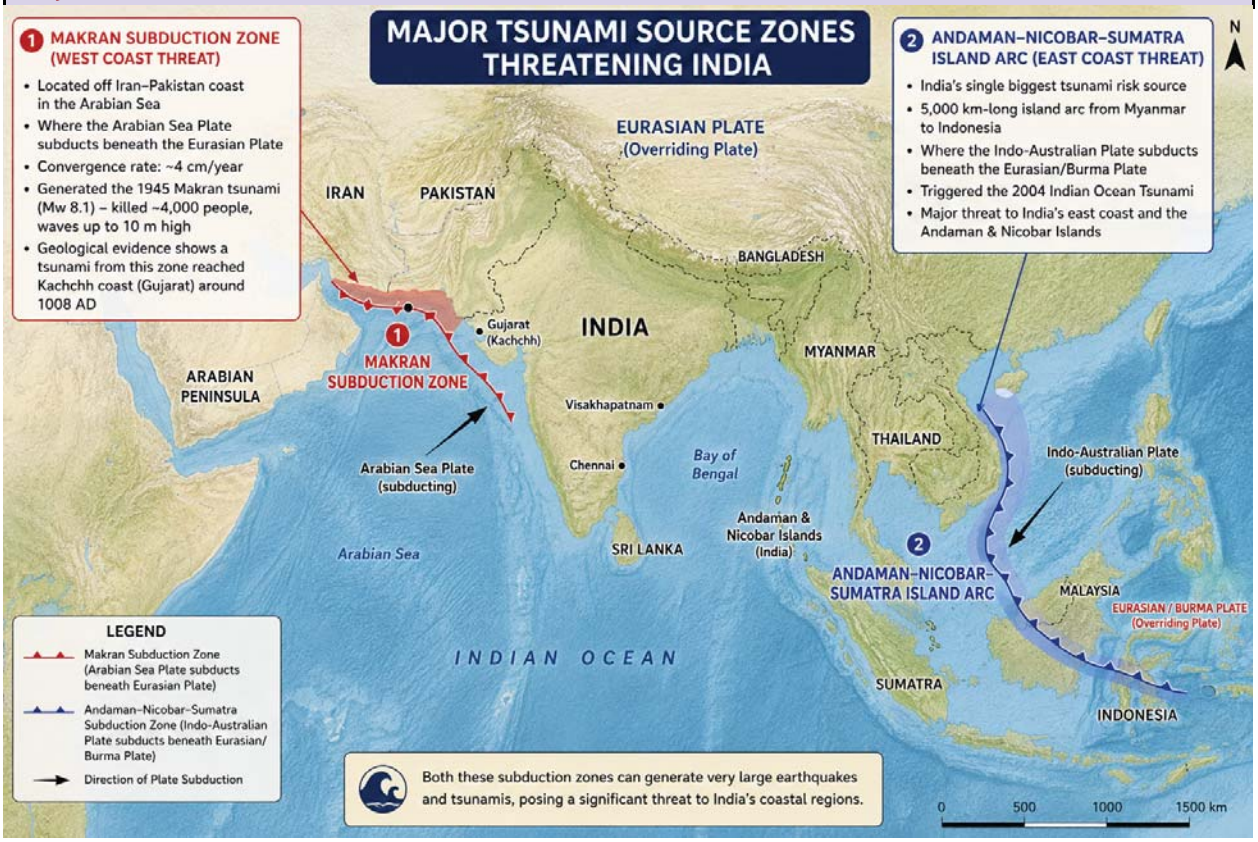
Ocean accounts for a much smaller share — but as the devastating **2004 Indian Ocean Tsunami** proved, even a rare event here can be catastrophic.

As per **INCOIS** (Indian National Centre for Ocean Information Services), all of India's coastal states and Union Territories are tsunami-prone.

1. **Undersea Earthquakes at Subduction Zones**: The single most common cause — when one tectonic plate is forced beneath another, the sudden vertical jolt of the seafloor displaces a massive column of water above it.

> On **26 December 2004**, when a subduction earthquake off the west coast of Sumatra caused the seafloor to rise several metres, displacing an estimated **30 cubic km of water** and triggering waves that killed over **2,50,000 people** across the Indian Ocean rim, including India.

Key Subduction Zones in India



Tsunami Source	Key Features
Andaman–Nicobar–Sumatra Island Arc	• **Primary tsunami source** for India. • Formed by **subduction of the Indo-Australian Plate beneath the Eurasian/Burma Plate** • Extends over **~5,000 km** from Myanmar to Indonesia. • Triggered the **2004 Indian Ocean Tsunami**. • Poses a major threat to **India's east coast and Andaman & Nicobar Islands**.

Makran Subduction Zone	• Major tsunami source for **India's west coast** • Located off the **Iran–Pakistan coast** in the Arabian Sea. • Formed by **subduction of the Arabian Plate beneath the Eurasian Plate**. • Generated the **1945 Makran Tsunami (Mw 8.1)** • Historical evidence indicates a tsunami reached **Kachchh (Gujarat) around 1008 AD**.

2. Volcanic Eruptions: Underwater or coastal volcanic explosions can displace huge volumes of water instantly, generating tsunamis even without a major earthquake.

For example, The **1883 Krakatoa eruption** in the Sunda Strait, Indonesia, generated a **30-metre tsunami** that killed around 36,000 people — a stark historical reminder that India's eastern neighbourhood carries volcanic tsunami risk too, given the active volcanic arc along the Sunda region.

3. **Submarine Landslides**: Underwater landslides, sometimes triggered by an earthquake itself, can displace water just as violently as the initial seismic event, occasionally amplifying a tsunami's local impact.

These are considered a secondary risk multiplier in India's tsunami hazard assessments, particularly along steep undersea slopes near the Andaman-Nicobar arc.

4. **India's Long, Low-Lying Coastline:** While not a "cause" of tsunami generation itself, India's exposure comes from having a **7,500 km coastline**, much of it low-lying and densely populated, which converts a distant seismic event into a major domestic disaster.

This is why the 2004 tsunami caused significant damage in Tamil Nadu, Andhra Pradesh, Kerala, and the Andaman & Nicobar Islands, even though the earthquake itself occurred near Indonesia, thousands of kilometres away.

Existing Tsunami Warning and Mitigation Framework in India

Measure	Key Features
Indian Tsunami Early Warning Centre (ITEWC)	• Operates under **INCOIS** (Hyderabad). • Issues tsunami warnings **within 10 minutes** of a major undersea earthquake. • Provides **24×7 monitoring** for the Indian Ocean.
Multi-Layered Detection Network	• Integrates **seismic stations, tsunami buoys and tide gauges**. • Enables cross-verification before issuing alerts. • Minimises false alarms.
DART Buoys & Bottom Pressure Recorders (BPRs)	• Detect **pressure changes** on the ocean floor. • Estimate **tsunami height, speed and arrival time**. • Enable early offshore detection.
Regional Tsunami Service Provider	• India is a **UNESCO-IOC accredited** Tsunami Service Provider. • Issues warnings to **28 Indian Ocean Rim countries**. • Strengthens regional disaster cooperation.
UNESCO 'Tsunami Ready' Programme	• Promotes **community preparedness** and evacuation planning. • Conducts awareness campaigns and mock drills. • Builds coastal community resilience.

IOWave Mock Exercises	• Conducted **biennially** across the Indian Ocean. • Tests the **end-to-end warning and evacuation system**. • Identifies operational gaps.
Multi-Channel Alert Dissemination	• Alerts issued through **SMS, email, GTS, websites and NAVIC**. • Ensures **rapid and wide communication**. • Reduces dependence on a single platform.
Satellite & Numerical Modelling	• Uses **satellite data and numerical models** for forecasting. • Improves reliability during adverse weather. • Enhances real-time monitoring.
Next-Generation Sub-Sea Cable Network	• Deploys **sub-sea acoustic sensor cables** along subduction zones. • Provides **faster and more reliable detection** than surface buoys. • Enhances India's tsunami early warning capability.

Best Case Studies For Tsunami Warning & Response

Case Study	Key Learning
Indian Tsunami Early Warning Centre (ITEWC), India	• Following the **2004 Indian Ocean Tsunami**, India established the **ITEWC** under INCOIS in 2007. • Today, it issues tsunami warnings within **10 minutes** and serves as a **UNESCO-accredited Tsunami Service Provider** for **28 Indian Ocean Rim countries**, making India a regional leader in tsunami early warning.
Simeulue Island, Indonesia – "Smong" Tradition	• During the **2004 tsunami**, the people of Simeulue immediately fled to higher ground after feeling the earthquake and seeing the sea recede, following the traditional **"Smong"** folklore passed down since 1907. • As a result, only a handful of people died despite the island being close to the epicentre.
"Miracle of Kamaishi", Japan (2011)	• During the **2011 Tōhoku Tsunami**, nearly **3,000 school children** survived because years of disaster education had taught them to evacuate immediately without waiting for instructions. • Their prompt action became a global model for **school-based disaster preparedness**.

Way Forward: Strengthening India's Tsunami Response and Preparedness

India's tsunami warning system has come a long way since 2004 — but recent events, like the **magnitude-8.8 Kamchatka earthquake (29 July 2025)**, which triggered precautionary alerts across the Pacific and reminded India's coastal states of their own vulnerability, show that preparedness must keep evolving rather than resting on the 2007 system's success.

1. **Complete the Next-Generation Seabed Sensor Cable Network**: India's current DART buoys can be damaged or lost, and satellite links introduce delays — a direct seabed cable network would fix this at the source.

DART Buoys or Deep-ocean Assessment and Reporting of Tsunamis buoys are critical ocean-based monitoring systems that detect and measure tsunamis in the deep ocean.

Developed and operated by NOAA, they use seafloor pressure sensors to record passing tsunami waves, relaying critical real-time data via satellite to warning centers to protect coastal populations

2. **Mandatory Tsunami-Resistant Construction Standards**: NDMA's own Tsunami Risk Management Guidelines flagged this gap years ago, but the actual construction code has still not been finalised.

NDMA has repeatedly urged the **Bureau of Indian Standards (BIS) to roll out its pending "Criteria for Tsunami-Resistant Design of Structures"** — until this is notified, new coastal buildings, including hospitals and schools, have no binding seismic-tsunami design standard to follow.

3. **Scale Up Mangrove and Bio-Shield Restoration**: Nature-based coastal defences are proven to physically weaken a tsunami's force before it reaches populated areas, making them a cost-effective complement to technology.

IIT-Bombay's 2025 hydrodynamic simulations found that a **100-metre mangrove belt can cut peak tsunami pressure by up to 90%**, particularly along the Sundarbans — experts now recommend expanding India's mangrove stewardship programme to **2,000 sq km by 2030**, offering blue-carbon credits to coastal Gram Sabhas as an incentive.

4. **Universal Mobile Alert Systems**: **mandate bilingual Common Alerting Protocol (CAP) push alerts on all mobile networks**, with text-to-speech conversion for basic feature phones and dedicated VHF marine radio channels for fishermen at sea — closing a real gap in who actually receives a warning in time.
5. **Vertical Evacuation Infrastructure**: Flat coastal terrain often leaves communities with nowhere nearby to run to when a tsunami warning gives only minutes of lead time, unlike hilly regions where "running to higher ground" is straightforward.

Experts recommend **retrofitting at least one public building per coastal panchayat as a vertical-evacuation refuge**, adapting FEMA's (US Federal Emergency Management Agency) vertical-evacuation design codes to Indian conditions and budgets.

6. **Dedicated Disaster Risk Financing for Coastal Resilience**: Tsunami mitigation infrastructure competes for funding with many other priorities, so a dedicated, ring-fenced revenue stream has been proposed.

One specific suggestion is a **"Coastal Resilience Cess" on port cargo and beachfront real estate**, with proceeds earmarked specifically for multi-hazard shelters and the upkeep of coastal warning sirens — since sirens that aren't maintained are as good as not having them at all.

7. **Institutionalise Regular, Transparent Mock Drills**: India already conducts the biennial IOWave exercises, but experts want home-grown accountability built in as well, not just participation in the international drill calendar.

The recommendation is to **institutionalise annual mock drills on National Tsunami Awareness Day (5 November)**, with public dashboards tracking participation metrics — turning preparedness drills from a one-off compliance exercise into a measurable, improving national habit.

8. **Community-Level Education, Kamaishi and Simeulue-Style:** Technology gives India minutes of warning at best; genuine preparedness depends on what people actually do in that window, which requires sustained, repeated education rather than one-time awareness campaigns

India's own **Tsunami Ready programme** (piloted with UNESCO-IOC) is a step in this direction, but experts argue it needs to be scaled from a handful of pilot communities to **all vulnerable coastal panchayats**, drawing lessons from Japan's Kamaishi schools and Indonesia's Simeulue oral traditions.

9. **Cross-Agency and Cross-Border Coordination**: Since tsunamis generated near Indonesia or even as far as Russia's Kamchatka Peninsula can send waves across the entire Indian Ocean, India's preparedness cannot be a purely domestic exercise.

India's role as a **UNESCO-accredited Tsunami Service Provider for 28 Indian Ocean Rim countries** should be leveraged further — experts suggest deeper coordination with the Pacific Tsunami Warning Center and regional partners, especially as events like the 2025 Kamchatka quake show threats can originate far outside the Indian Ocean basin.

Conclusion

"The best time to plant a tree was 20 years ago; the second-best time is now"

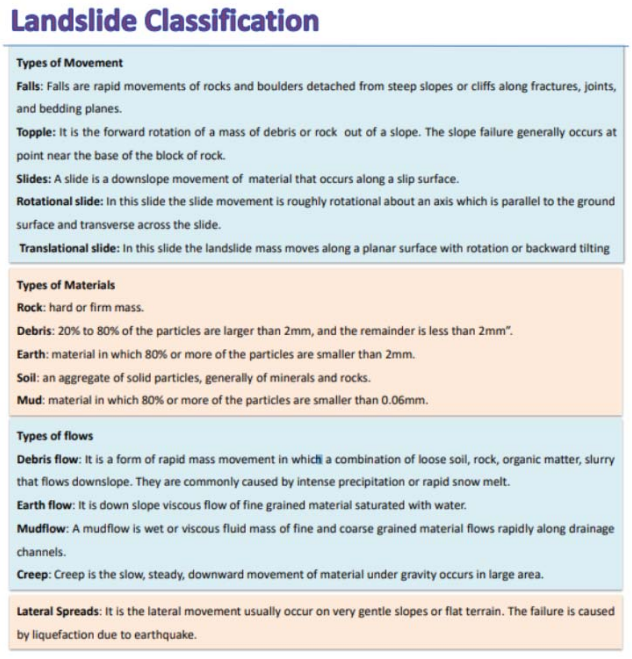
India's tsunami framework doesn't need to be rebuilt — it needs to be **completed and hardened**. The pending BIS construction code, the still-developing seabed cable network, and the gap between pilot-scale community programmes and nationwide coverage are the clearest unfinished pieces.

As the 2025 Kamchatka earthquake reminded coastal nations worldwide, the next tsunami-triggering event could come from anywhere in the Pacific Rim — meaning India's way forward must combine faster detection, stronger nature-based defences, and genuine last-mile community preparedness, not just wait for the next disaster to expose the next gap.

8. LANDSLIDES IN INDIA

Landslide: It is a natural phenomenon which is **usually observed in steep mountainous regions** where large quantities of material, composed of rock, boulders, loose mud, soil and debris, are driven down the slope with force.

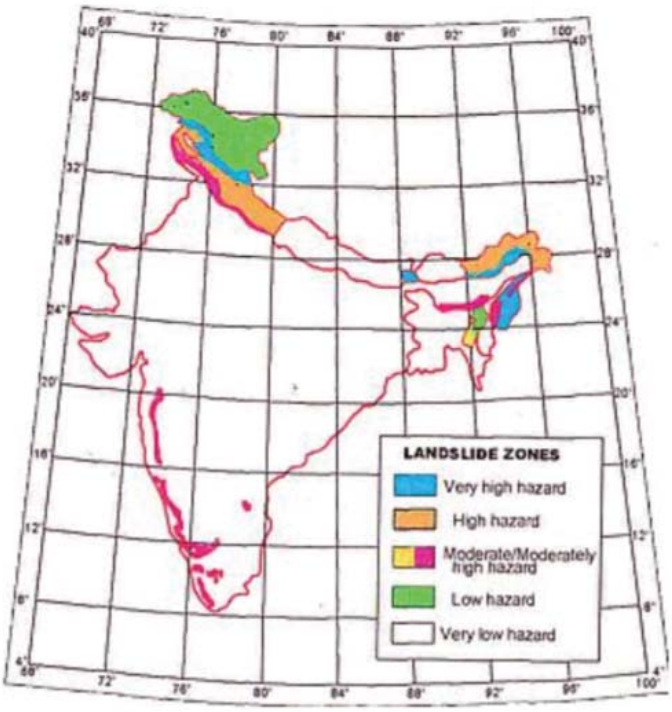
Types of Landslides:



Source- ISRO Landslide Atlas of India

Vulnerability of India to Landslides:

ISRO's Landslide Atlas of India	According to ISRO's Landslide Atlas of India, **India is among the top four countries with highest landslide risk.** It has been estimated that **loss of life per 100 km2** is **greater than one.**
India Landslide Susceptibility Map (ILSM) by IIT Madras	India accounts for about **8% of global fatalities** due to landslides. **13.17% of the country** is susceptible to landslides, and **4.75%** is 'very highly susceptible'. Kerala is the most vulnerable state with over **14%** of its land mass in the 'very high susceptibility' category.



Source- NDMA

Causes of landslides in India

Natural Causes	• **Heavy Rainfall:** The occurrence of heavy or continuous rainfall may lead to heavy landslides in the areas of steep slopes. **Water works as a lubricating agent and loosens up the material. For ex: The 2013 Uttarakhand Cloudburst brought immense landslides in the region.** • **Earthquakes:** The sudden shaking of earth's crust creates stress on materials resulting in their flow downwards. **For ex: Tectonically active Himalayas** enhance the probability of earthquakes, resulting in an increased vulnerability. • **Nature of Rivers:** The youthful and mighty rivers increase the downcutting of earth on the mountain slopes. **For ex:** The **increased flow in the Chaliyar river** resulted in heavy sediment erosion, resulting in the Wayanad landslide catastrophe.

Anthropogenic Causes	• **Heavy Infrastructure Development:** The creation of roads, railways, dams, linear infrastructure development in the mountainous region, has resulted in increased landslides. **For ex: The development of Konkan Railways in western ghats i**s considered a significant reason behind the frequent occurrences of the disaster in the region. • **Mining activities:** It involves removal of the surface of the earth for mineral exploration, which loosens the grip of material. **For ex: Iron ore, bauxite, manganese, chromite, and limestone mining** in the western ghats has increased the risks of this disaster. • **Increased Deforestation:** The cutting down of trees for agricultural and industrial activities, loosens the soil grip and makes the region more vulnerable to landslides. **For ex: Plantation agriculture in Western ghats** has resulted in loss of native trees and increased the disaster risk. • **Unsustainable Tourism:** The rising tourist influx in the Himalayan region and Western Ghats puts greater pressure on land, as more infrastructure and amenities are needed to accommodate them. This leads to more construction and greater landslide risks. • **Climate change:** Anthropogenic activities has accelerated the climate change events, leading to increased glacial melts and rainfall. **For ex: Warming of Arabian sea** has resulted in **increased rainfall in the Western Ghats region**, leading to enhanced landslides.

Impacts of Landslides in India

- **Loss of Lives:** The most traumatizing impact of a landslide is the loss of precious human and animal lives. For e.g. In the **Wayanad landslide, 298 people died.**
- **Economic Losses:** Landslide leads to enhanced economic losses as, **significant amount** is **spent** on **reconstructing the lost infrastructure, rehabilitating the masses**, and **providing relief support** to the impacted people.
- **Infrastructural Damage and restricted Movement:** The mud, rocks, and debris moving down the slope result in infrastructural damage to houses, buildings, roads and create a **barrier on critical transportation routes.** This restricts the movement of goods and people.
- **Jeopardized Water Availability:** The accumulated mass of debris results in the blockage of a river by **creating a Landslide dam.** It impairs the availability of water for nearby people.
- **Displacement and Migration:** Landslides force communities to evacuate and migrate. This leads to **social disruption** and **psychological stress** among affected populations

Initiatives have been taken by the Government to mitigate the risks due to landslides

National Landslide Susceptibility Mapping (NLSM)	Geological Survey of India initiated the National Landslide Susceptibility Mapping (NLSM) program in 2014 to **map the landslide areas on a Macro Scale** (1:50,000). The goal is to **map the 0.42 million sq. km landslide-prone areas** across the country.
National Landslide Risk Management Strategy (2019)	It **addresses all the components of landslide disaster risk reduction** and management, which includes: a. Hazard mapping, monitoring, and early warning system, awareness programs. b. Capacity building, training, regulations, and policies, stabilization and mitigation.

National Disaster Management Authority (NDMA) Guidelines on Landslide Hazard Management (2009)	It proposes measures that should be taken to avoid or mitigate the risk posed by landslide hazards. Some of the guidelines are as follows: a. **Delineating areas** susceptible to landslide hazards. b. Encouraging **implementation of successful remediation** and mitigation technologies. c. **Developing institutional capacity** and **training** for geoscientists, engineers, and planners is necessary for effective management.
National Institute of Disaster Management (NIDM)	It is a premier institute that **provides capacity building support** to various National and State level agencies in the field of Disaster Management & Disaster Risk Reduction.

Challenges in implementation of measures proposed for landslide Mitigation

- **Non implementation of Gadgil and Kasturirangan committee reports:** The Gadgil committee report proposed to **regulate developmental activities** in the **entire 1,60,000 sq km stretch of the Western Ghats.** K Kasturirangan report recommended **demarcation of 37% of the Western Ghats as an ecologically sensitive area** (ESA) and complete ban on mining and quarrying in these protected zones. These reports have not been implemented with resistance by all states and parties across the political spectrum in the biodiverse region.
- **Lack of landslide mitigation plan:** Many **States vulnerable to landslides including Kerala** lack landslide mitigation plan. These maps and plans help to explain the route a landslide can take once it has been triggered.
- **Insufficient Monitoring Systems:** Many regions lack effective monitoring and early warning systems, which are crucial for timely evacuation and disaster preparedness. The **absence of real-time data** led to increased casualties and damage during landslide events.
- **Policy and Regulatory Gaps: Weak enforcement of land-use regulations** and **building codes** often exacerbates vulnerability to landslides.
- **Low Community Awareness and Participation:** Low levels of public awareness regarding landslide risks and preparedness measures impede community involvement in disaster management.

Way Forward

- **Drawing up of Route Maps:** Route maps help in **marking out areas which should be out of bounds for settlements or any activities**. These maps also map out the routes that landslides are likely to take after they have been triggered. The route maps need to be drawn at the earliest to complement the susceptibility maps.
- **Proper Environment Impact Assessment (EIA):** Proper implementation of EIA norms must be carried out **before initiating mining or dam construction**, in the ecologically fragile mountainous regions.
- **Landslide Micro-zoning procedure:** There must be **adoption of landslide micro zoning procedures** for mountainous and other highly vulnerable regions. This will help the states in restoring denuded flora and soil cover.
- **Involvement of locally trained manpower:** There should be involvement of locally available trained manpower to intensify the hazard reduction and public awareness programs.

Differing Reasons Behind The Landslides In Himalayas And Peninsula

Basis	Himalayas	Peninsular India
Geological Setting	Young **fold mountains** that are still undergoing uplift and deformation.	Ancient **stable shield** with comparatively mature landforms.

Rock Characteristics	Composed of **highly fractured and weathered sedimentary & metamorphic rocks**, making slopes inherently unstable.	Dominated by **hard crystalline rocks**; failures mainly occur due to weathered surface layers.
Tectonic Activity	**Active plate collision** continuously creates faults, fractures and slope instability.	**Minimal tectonic activity**; landslides are rarely earthquake-induced.
Nature of Slope Failure	Primarily caused by **structural instability** of young rocks.	Mainly caused by **weathering and saturation** of lateritic or residual soils.
Slope Gradient	Characterised by **very steep slopes and deep valleys**, increasing gravitational instability.	Features **moderate slopes and escarpments**, making failures more localised.
Climatic Trigger	Triggered by **cloudbursts, intense rainfall, snowmelt and glacial processes**.	Triggered mainly by **prolonged monsoon rainfall** and occasional cyclonic precipitation.
Hydrological Factors	**Glacial melt and Glacial Lake Outburst Floods (GLOFs)** further destabilise slopes.	Absence of glacial influence; failures occur mainly due to **waterlogging of weathered soils**.
Human Activities	**Road widening, tunnelling, hydropower projects and hill urbanisation** aggravate instability.	**Quarrying, plantations, slope cutting and unplanned tourism infrastructure** weaken slopes.
Frequency & Magnitude	**Frequent and large-scale** landslides due to combined geological and climatic factors.	Generally **less frequent and localised**, though extreme rainfall can trigger major events.

9. FLOODS IN INDIA

A **flood** occurs when water overflows onto normally dry land, most often when a river's carrying capacity is exceeded by heavy rainfall, poor drainage, or upstream releases.

Floods are India's **most frequent and deadliest natural disaster** — as per **NDMA**, out of India's total geographical area of 329 million hectares, **more than 40 million hectares are flood-prone**.

Major Causes of Floods in India

Natural Causes

1. **Monsoon Concentration**: India receives the overwhelming bulk of its annual rainfall in a very short window, leaving little time for the land to absorb it gradually.

   India gets **75-90% of its annual rainfall in just four months (June-September)**, and the **Brahmaputra basin alone receives over 2,500mm annually** — when this volume hits degraded catchments, flooding becomes almost inevitable regardless of any human factor.

2. **Excess River Silt and Reduced Carrying Capacity**: Natural erosion upstream deposits silt in riverbeds downstream, gradually shrinking how much water a river channel can safely hold.
   - This silting effect is a key reason why some rivers that were not traditionally flood-prone now experience severe inundation even from relatively normal rainfall years.

3. **Cloudbursts and Cyclonic Rainfall**: Extremely intense, short-duration rainfall events over a small area can overwhelm any river or drainage system, no matter how well-maintained.
   - **Cyclones, cyclonic circulations, and cloudbursts** are specifically flagged by NDMA as causing sudden **flash floods** distinct from the slower, more predictable river floods of the regular monsoon season.
4. **Glacial Melt and Snowmelt**: In the Himalayan river systems, melting snow and glaciers add an extra source of water on top of monsoon rainfall, a factor absent in peninsular rivers.
   - This is why Himalayan-origin rivers like the **Ganga and Brahmaputra** face a genuinely different, compounded flood risk compared to purely rain-fed peninsular rivers.
5. **Cross-Border River Origins**: A portion of India's flood water doesn't even originate within its borders, adding a natural geographic complication no domestic policy alone can fully solve.

> NDMA notes that **some rivers causing major damage in India originate in neighbouring countries** — meaning India's flood risk is partly shaped by rainfall and river behaviour outside its own territory.

Man-Made Causes

1. **Deforestation in Catchment Areas**: Forests act like a natural sponge, absorbing rainfall and slowing runoff — removing them sends far more water rushing into rivers, far faster.

> UN Office of Disaster Risk Reduction Research shows that **every 10% increase in deforestation raises flood risk by up to 28%**, and the **Himalayas have lost 25% of forest cover in vulnerable zones since 1980** — directly worsening the region's already-high natural flood risk.

2. **Encroachment on Floodplains and Wetlands**: Rivers and wetlands need space to naturally absorb excess water — building over that space removes the very buffer that used to prevent flooding.

> **Chennai lost 85% of its wetlands between 1980 and 2015**, with the Pallikaranai marshland shrinking from 50 sq km to just 5 sq km — a key reason the **2015 Chennai floods** killed 269 people and caused **₹20,000 crore** in damage despite "only" three days of intense rain.

3. **Dam and Reservoir Mismanagement**: Dams are meant to control water flow, but poorly timed or poorly managed water releases during heavy rain can turn a controlled reservoir into a flood amplifier.

> The alleged mismanagement of the **Mullaperiyar dam** on the Tamil Nadu-Kerala border was widely blamed for worsening the **2018 Kerala floods**, while sudden reservoir releases have similarly worsened flooding along the **Godavari and Krishna rivers**.

4. **Unplanned Urbanisation and Poor Drainage**: Converting natural, water-absorbing land into concrete and asphalt removes the ground's ability to soak up rainfall, forcing all of it into already-inadequate drainage systems.

> Recurring flooding in cities like **Bengaluru** is now widely attributed less to unusually heavy rainfall and more to **poor drainage maintenance and unchecked urban sprawl** over natural water channels.

5. **Unregulated Mining and Construction in Fragile Hills**: Blasting hillsides for roads, tourism infrastructure, and mining destabilises slopes, sending loosened debris straight into rivers during heavy rain and worsening downstream flooding.

> A Supreme Court bench remarked in **July 2025** that unchecked illegal mining and deforestation in Himachal Pradesh had so destabilised the terrain that continued unchecked activity could see the state "**vanish in thin air from the map**" — directly linking human activity in the hills to the flash floods that devastated downstream Punjab in both 2023 and 2025.

India's Current Flood Mitigation Framework

Institution	Key Features
NDMA & Disaster Management Act, 2005	• Provides the **overall policy framework** for flood disaster management. • Coordinates preparedness, mitigation and response. • Flood control works remain primarily a **State Government responsibility**.
Central Water Commission (CWC)	• Nodal agency for **flood forecasting and hydrological monitoring**. • Operates a nationwide **flood forecasting network**. • Issues **real-time and advisory flood forecasts**.
Flood Management & Border Areas Programme (FMBAP)	• Provides **Central financial assistance** for flood protection works. • Supports **embankments, drainage improvement and anti-erosion projects**. • Covers flood-prone and border States.
SACHET Portal	• Disseminates **real-time flood alerts** using the Common Alert Protocol (CAP). • Integrates warnings from multiple agencies for faster communication.
GLOF Monitoring System	• Monitors **glacial lakes** using satellite-based remote sensing. • Supports early warning against **Glacial Lake Outburst Floods (GLOFs)**. • Operates under NDMA-CWC-MoES coordination.
Flood Risk Mitigation Scheme (FRMS)	• Supports **community flood shelters** and local early warning systems. • Promotes **basin-specific flood mitigation** and resilience.

Urban Flooding Causes

Urban flooding is the inundation of built-up city land by water faster than the drainage system can clear it. Unlike riverine flooding, which unfolds over days, urban flooding is characterised by **short-duration, high-intensity rainfall** causing rapid runoff over impervious surfaces — turning even a couple of hours of intense rain into a waterlogged city.

As per the **Ministry of Home Affairs**, over **5,161 Urban Local Bodies** in India are now considered prone to this hazard, and cities from Mumbai to Bengaluru to Delhi have all seen this shift from a "monsoon inconvenience" into a genuine structural disaster.

Key Causes

1. **Outdated Drainage Infrastructure**: Most Indian cities' storm-water drains were designed decades ago for much lower rainfall intensities than they now actually face.

   Mumbai's storm drains, built in the 1860s, were designed for 25mm/hour of rain — but the city now regularly sees over **100mm/hour**, while **Delhi's drainage plan (from 1976) was designed for 50mm/day**, yet the city recorded **185.9mm in a single day in 2023**.

2. **Loss of Lakes and Wetlands**: Urban water bodies once acted as natural sponges, absorbing excess rainwater before it could flood streets — their disappearance removes this buffer entirely.

   Bengaluru has lost around 65-80% of its historic lakes to encroachment, while **Chennai lost over 62% of its wetlands between 1893 and 2017**, including its once-vast **Pallikaranai marsh**, directly removing the city's natural flood-absorption capacity.

3. **Unregulated Urbanisation and Concretisation**: Rapid construction converts naturally permeable land into concrete and asphalt, which cannot absorb rainwater the way soil and vegetation once did.
4. **Intensifying, Climate-Linked Rainfall**: Climate change is making short bursts of extreme rain more frequent, overwhelming infrastructure that was never designed for today's rainfall patterns.

> **Delhi recorded over 100mm of rain in under an hour in 2023**, while Bengaluru's July rainfall has increasingly exceeded **100mm in a single day** — both far beyond what 20th-century drainage systems were engineered to handle.

5. **Encroachment on Natural Drainage Channels**: Storm-water drains and natural stream channels are routinely narrowed, built over, or blocked, cutting off the city's own water-escape routes.

> Bengaluru's traditional **"rajakaluves" (storm-water drains)** are widely **choked, buried, or encroached upon**, drastically reducing their water-carrying capacity exactly when the city needs it most during heavy rain.

6. **Solid Waste Mismanagement**: Unregulated dumping of garbage physically blocks drains and natural waterways, turning what should be a clear water channel into a clogged, overflowing one.

> During the **2023 Himachal Pradesh floods**, plastic waste clogging both natural and artificial waterways significantly worsened the flooding, while the **2015 Chennai floods** were similarly aggravated by silt and waste blocking estuary canals.

7. **Low-Lying and Undulating City Terrain**: Some cities are simply built on naturally depressed land, which makes them structurally predisposed to waterlogging regardless of how well the drainage is managed.

> **Bengaluru sits on undulating terrain with natural low-lying basins** like the Hebbal and Koramangala-Challaghatta valleys, meaning water naturally collects there even before accounting for any human-made drainage failures.

8. **Uncoordinated Urban Governance**: Multiple civic agencies — water boards, municipal corporations, urban development authorities — often work in silos without a shared master plan, delaying both prevention and emergency response.
   - This lack of coordination is repeatedly flagged as a root cause behind why cities keep experiencing the same flooding pattern year after year, despite each agency technically having some role in flood management.
9. **Destruction of Natural Buffers Beyond Water Bodies**: It isn't just lakes and wetlands — surrounding natural landforms that once slowed or absorbed runoff are also being built over.

> In Delhi, the **encroachment and filling of the Aravalli hills, Ghata Lake, and Najafgarh Jheel** have removed natural buffers that once slowed water flow into the city, compounding the drainage system's existing inadequacy.

10. **Vulnerable Settlements in Flood-Prone Zones**: The human cost of urban flooding is magnified because the most exposed communities are often those with the least capacity to relocate or rebuild.
    - **Slum settlements concentrated in low-lying zones** near encroached water bodies face disproportionately higher risk, since they typically lack the elevated construction or resources available to relocate away from the most flood-prone parts of a city.

Global Case Studies on Flood Management

Case Study	Key Learning For India
"Room for the River" Programme, Netherlands	After the devastating floods of **1993 and 1995**, the Netherlands shifted from merely building higher embankments to **giving rivers more space** by widening floodplains, relocating dykes and restoring wetlands. It is now a global model for **nature-based flood management**.

"Sponge Cities" Initiative, Wuhan (China)	To tackle recurrent urban flooding, China transformed cities into **"sponges"** using permeable pavements, rain gardens, wetlands and green roofs. The initiative demonstrates how **nature-based urban planning** can effectively reduce flood risk.
G-Cans Underground Flood Diversion System, Tokyo (Japan)	Facing severe urban floods, Japan built the **world's largest underground flood diversion tunnel**, which channels excess floodwater into the Edo River. The project showcases how **advanced engineering infrastructure** can protect densely populated cities where land is scarce.

Empowering India's Flood Management Architecture : Way Forward

1. **Implement the Rashtriya Barh Ayog's Recommendations**: India's first dedicated flood commission (1980) recommended a shift toward **floodplain zoning** to regulate man-made activity in vulnerable areas — but this recommendation has sat largely unimplemented for over four decades.

   Enforcing strict floodplain zoning would directly address encroachment problems like **Delhi's Yamuna floodplain**, which has been built over systematically for decades, and would finally act on advice the government itself accepted back in 1980.

2. **Act on the R. Rangachari Committee's Priority List (2003)**: Set up specifically to review why RBA's 1980 recommendations weren't working, this committee narrowed **207 recommendations down to 25 priority actions** — majority of which are still not paid attention to.
3. **Introduce Staggered Dam Release Protocols Nationwide**: The 2018 Kerala floods showed how simultaneous, poorly-timed dam releases can turn a manageable rainfall event into a catastrophe.
4. **Shift Priority from Structural to Non-Structural Measures**: Given that embankments have visibly failed to shrink India's flood-prone area over 50 years, the evidence strongly favours prioritising forecasting, zoning, and insurance over more concrete.

   In Assam, embankments along the Brahmaputra **regularly breach and then trap water inside** when they do — worsening rather than preventing damage —

   This is why the Assam State Disaster Management Authority has now pivoted toward **flood-resilient housing and community-based early warning** instead of relying solely on more embankments.

5. **Complete and Enforce Floodplain Zoning Regulations**: NDMA has already issued guidelines classifying flood-prone land into prohibited, restricted, regulated, and free zones, but very few states have actually turned these guidelines into enforceable state-level laws.

   Making floodplain zoning **legally binding and locally enforced** — rather than an advisory guideline municipal bodies can ignore is the most-repeated recommendation across every flood commission from 1980 to today.

6. **Scale Up Nature-Based, "Sponge City"-Style Solutions**: Since India's own urban flooding crisis mirrors what China faced before its Sponge City programme, adapting similar nature-based principles domestically could reduce dependence on expensive hard infrastructure.

   This means restoring lost urban wetlands (like **Chennai's Pallikaranai marsh** or **Bengaluru's lost lakes**) and increasing permeable surfaces in cities, rather than only building bigger stormwater drains that will eventually be overwhelmed again by more intense rainfall.

7. **Modernise Flood Forecasting with AI and Satellite Data**: While CWC's forecasting network is genuinely strong technically, converting raw data into faster, hyper-local warnings remains a gap, especially for flash floods and cloudbursts.

Building on CWC's existing 350-station network and its new **seven-day advisory forecast portal**, the next step is integrating **AI-based prediction models**, similar to what **Tokyo's post-2022 Resilience Project** has adopted, to give more precise, location-specific advance warning rather than broad regional alerts.

8. **Strengthen Institutional Coordination Across States and Agencies**: A recurring criticism across committee reports is fragmented, siloed decision-making between state water departments, NDMA, CWC, and municipal bodies during an actual flood emergency.
9. **Increase Flood Insurance Penetration**: With flood insurance coverage still minimal across India, households and farmers bear almost the entire financial burden of flood losses themselves, delaying recovery and deepening rural distress.
    - Expanding accessible, subsidised flood insurance products — beyond just crop insurance schemes like PMFBY — would reduce the pressure on state disaster relief funds and speed up household-level recovery after each flood event.

Conclusion

As the Dutch would say: you don't beat the water, you make peace with it. The day India starts planning with its rivers with the "Basin Based Approach" instead of just negating the danger posed by their mismanagement , floods will stop being an annual headline and start becoming a managed risk.

10. DROUGHTS IN INDIA

Drought is a prolonged period of below-normal rainfall leading to water shortage, crop failure, and stress on livelihoods. Unlike sudden disasters like earthquakes or floods, drought develops **slowly and silently**.

Types of Droughts

Drought Type	Key Features
Meteorological Drought	• Occurs due to a **significant deficiency in rainfall** compared to the long-term average. • Represents the **earliest stage** of drought. • Assessed using **IMD rainfall criteria**. As per **IMD**, a drought exists when seasonal rainfall falls below **75% of the long-term average**, with a **25% deficit triggering "moderate" drought** classification and a **50% deficit triggering "severe" drought** — this is the earliest warning signal, often visible weeks before any crop or water-supply impact appears.
Agricultural Drought	• Develops when **soil moisture becomes insufficient** for normal crop growth. • Caused by deficient or poorly distributed rainfall. • Leads to **crop stress and reduced agricultural productivity**.
Hydrological Drought	• Occurs when **rivers, reservoirs, lakes and groundwater** decline below normal levels. • It usually follows prolonged meteorological drought. • Affects irrigation, drinking water and hydropower generation.

Socio-Economic Drought	• Arises when **water scarcity disrupts livelihoods and economic activities**. • Results in shortages of **food, fodder, drinking water and employment**. • Represents the socio-economic impact of prolonged drought.
Ecological Drought	• Occurs when **ecosystems fail to sustain ecological functions** due to prolonged moisture stress. • Causes **forest degradation, biodiversity loss and increased forest fires**.
Flash Drought	• Develops **rapidly within weeks** due to high temperatures, low humidity and sudden rainfall deficits. • Intensified by **climate change and rising evapotranspiration**. • Leaves limited time for preparedness and response.

Reasons Behind India's Vulnerability to Droughts

Natural and Geographic Reasons

1. **Overwhelming Dependence on a Single Monsoon System**: Unlike countries with rainfall spread evenly across the year, India gets the overwhelming bulk of its water supply from one seasonal weather system, making any disruption to it catastrophic nationwide.
   - Any delay, weakening, or early withdrawal of the southwest monsoon directly threatens **sowing, crop growth, groundwater recharge, and drinking water supply simultaneously** — a single bad monsoon year can trigger drought stress across multiple sectors at once, not just agriculture.
2. **The Rain Shadow Effect**: Mountain ranges like the Western Ghats and Himalayas block moisture-laden monsoon winds, leaving the land on the other side chronically starved of rain even in otherwise normal monsoon years.

   This effect is why regions like **Rajasthan, parts of Gujarat, and the Deccan Plateau** experience consistently low rainfall — and interior Tamil Nadu districts like **Karur, Erode, and Dharmapuri** show unfavourable long-term drought index trends purely due to this rain-shadow positioning.
3. **Uneven Spatial and Temporal Rainfall Distribution**: Even in years with "normal" total rainfall, the way that rain is distributed across regions and across the season matters just as much as the total amount.

   India averages **110cm of annual rainfall** — a healthy figure on paper — but its erratic and concentrated nature means some regions flood while others simultaneously face long dry spells, effectively creating drought conditions despite adequate national totals.
4. **Topography and Poor Natural Water Retention**: Undulating terrain in many parts of the country causes rainwater to run off quickly rather than soaking into the ground, limiting the natural groundwater recharge that would otherwise buffer against dry spells.
   - This rapid runoff effect is a key reason why regions without dedicated water-harvesting infrastructure lose most of their rainfall to rivers and drainage almost immediately, leaving little in reserve for the dry months ahead.

Human and Policy-Driven Reasons

1. **Groundwater Over-Extraction**: Groundwater is meant to act as a buffer during dry years, but decades of unchecked pumping have drained this buffer faster than nature can refill it.

A **2024 peer-reviewed study (Earth's Future, Wiley)** found India lost a net **450 cubic km of groundwater between 2002-2021**, depleting at roughly **1.5cm per year** — meaning the very resource meant to protect India during droughts is itself running out.

2. **High Dependence on Rain-Fed Agriculture**: A huge share of India's farmland has no irrigation backup at all, meaning a weak monsoon translates almost directly into crop failure with no buffer in between.

Rain-fed agriculture accounts for approximately 56% of India's total cropped area, and these regions can see agricultural productivity fall by **up to 40% during severe drought years** — a direct line from monsoon failure to farmer distress.

3. **Cultivation of Water-Intensive Crops in Water-Scarce Regions**: Farmers continue growing thirsty crops like paddy and sugarcane in areas that simply don't have the water to sustain them, often because of assured government procurement and subsidies rather than actual water availability.
   - This mismatch between **crop choice and regional water availability** is a key reason drought-prone states keep depleting their groundwater even in years of relatively normal rainfall — the crop pattern itself is manufacturing scarcity.
4. **Inefficient Irrigation and Water Management**: Large amounts of water meant for agriculture and drinking supply are lost before they even reach their intended use, due to outdated infrastructure and poor maintenance.
   - **Leaky canals, unregulated borewells, and inefficient flood irrigation** all drain water reserves faster than natural recharge can replace them — meaning India loses water to its own infrastructure gaps even before drought conditions set in.
5. **Deforestation and Loss of Natural Water-Retention Capacity**: Forests act as natural sponges that hold moisture in the soil and support groundwater recharge — removing them directly weakens a region's natural resilience to dry spells.
   - Loss of forest cover **reduces groundwater recharge and increases soil erosion**, compounding drought conditions especially in already-vulnerable regions like Bundelkhand and parts of central India.
6. **Climate Change Compounding Every Existing Vulnerability**: Rising temperatures aren't just making droughts more frequent — they're actively worsening every other factor on this list simultaneously.

The same 2024 Wiley study found that **summer monsoon rainfall in north India declined by about 8.5% between 1951-2021**, while winter warming is simultaneously **increasing irrigation water demand** — meaning India needs more water at exactly the moment climate change is making less of it available.

Global Case Studies on Drought Management

Case Study	Key Learning
Israel – Technology-led Water Security	Following the **1998–2002 drought**, Israel transformed its water sector by investing in **large-scale desalination, wastewater recycling and drip irrigation**. Today, most of its drinking water comes from desalination, making it a global model for **technology-driven drought resilience**.
Australia – Murray–Darling Basin Reform	The **Millennium Drought (1997–2009)** prompted Australia to establish the **Murray–Darling Basin Authority** and adopt basin-wide water governance. The reforms balanced **agricultural needs with environmental sustainability**, demonstrating the importance of **integrated river basin management**.

Cape Town, South Africa – Demand-side Water Management	During the **2015–2018 "Day Zero" crisis**, Cape Town drastically reduced water consumption through **real-time public information, water-use restrictions, tiered pricing and citizen participation**. The city became a global example of **behavioural change and demand management** in drought mitigation.

Way Forward

Drought in India is not just a rainfall problem anymore. It is a **governance and planning problem** — of how we store water, share it, and use it.

With nearly **68% of India's land area drought-prone** (India Meteorological Department) and groundwater tables falling across large parts of the north-west and peninsular India, the way forward requires moving away from short-term relief measures towards a system that is resilient by design.

1. **Institutional Reform - Merging the different bodies:** India currently manages surface water (Central Water Commission) and groundwater (Central Ground Water Board) as separate silos, even though the same water cycle links them.

   The **Mihir Shah Committee (2016)**, later built upon in the **Draft National Water Policy (2021)**, recommended merging both bodies into a single **National Water Commission (NWC)**, with dedicated divisions for water security, groundwater mapping, and drought-flood management.

   This "**One Water Approach**" would end the current fragmentation where the same aquifer is drained for irrigation and drinking water without joint accounting.

2. **Groundwater Management Through Community Participation:** Nearly 62% of India's irrigation and 85% of rural drinking water depends on groundwater, yet its use remains largely unregulated at the local level.

   The **Atal Bhujal Yojana** (completed October 2025) shifted from a **pure supply-side approach** to **community-led water budgeting** across 8,203 gram panchayats in 7 water-stressed states — constructing over 83,800 recharge structures and bringing nearly 6.7 lakh hectares under efficient irrigation, recharging an estimated 642 million cubic metres of groundwater (Ministry of Jal Shakli, 2025 data).

   The model shows that **behavioural change at the village level**, backed by real-time data disclosure, works better than top-down control.

3. **Watershed Development and Springshed Rejuvenation:** In hilly and rain-fed regions, treating the entire watershed - and not just individual farms - helps hold rainwater where it falls.

   Under **PMKSY-Watershed Development Component 2.0**, NITI Aayog's recommendation to add **springshed rejuvenation** as a core activity has been incorporated for mountain states, alongside mandatory GIS/remote-sensing-based project planning.

   This builds on the older success of the **Hiware Bazar model in Maharashtra**, where watershed treatment and groundwater regulation turned a drought-hit village into a water-surplus one within a decade.

4. **Climate-Resilient and Diversified Agriculture:** Water-guzzling crops like paddy and sugarcane continue to be grown in naturally water-scarce regions such as Marathwada and parts of Punjab, deepening the drought stress.

5. **Real-Time Drought Monitoring and Early Warning:** Delayed declaration of drought often delays relief, worsening distress migration and agrarian stress.

   The **NDMA's Drought Management Manual** mandates the use of remote sensing, soil moisture indices, and rainfall deviation data for faster, more objective drought declaration by states, replacing the earlier, more subjective "crop-cutting experiment" method.

   This needs to be scaled up using the **India Meteorological Department's** improving climate and rainfall forecasting capacity.

6. **Convergence of Rural Schemes for Employment and Water Security:** Drought hits daily-wage and small farm households hardest, making guaranteed rural employment a critical buffer.

> **MGNREGA (VB-G-RAM-G)** works — such as farm ponds, check dams and afforestation — are being converged with Atal Bhujal Yojana and PMKSY at the gram panchayat level, ensuring that the same worksite builds both **water security assets** and **local livelihoods**, rather than running as disconnected schemes.

7. **Long-Term Supply Augmentation via River Linking:** For chronically water-deficit basins, moving surplus water from water-rich basins remains part of the long-term strategy, though it needs careful ecological scrutiny.

> The **Ken-Betwa Link Project**, India's first river-linking project under implementation, aims to transfer surplus water from the Ken basin to the drought-prone Bundelkhand region, though experts caution that such projects must be paired with local water conservation rather than treated as a standalone fix.

Conclusion

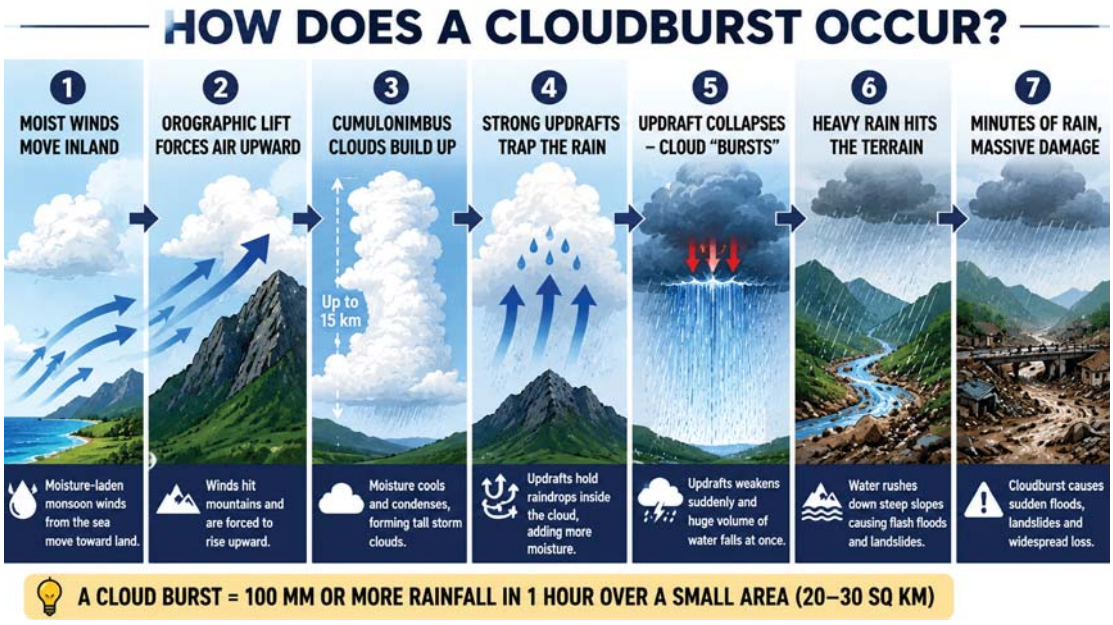
As the proverb says, "Dig the well before you are thirsty." India's long-term drought resilience lies not in emergency relief but in conserving every drop through sustainable water management, efficient agriculture and proactive planning.

11. CLOUDBURSTS IN INDIA

The **India Meteorological Department (IMD)** defines a cloudburst as rainfall of **100 mm or more in one hour over a small area of 20-30 sq km**. It is essentially nature's version of a burst pipe - a sudden, violent release of accumulated moisture from a cloud, usually a cumulonimbus formation, triggered by strong updrafts that suddenly collapse.

Cloudbursts are most common in the **Himalayan belt and the Western Ghats**, where steep slopes, narrow valleys and rising terrain (orography) force moist air upward rapidly, but their destructive potential — flash floods, landslides and debris flows - has made them one of the deadliest weather events in India's hill states, especially as climate change and unplanned construction combine to worsen the damage.

How Does A Cloudburst Occur



1. **Moisture-Laden Winds Move Inland:** It all begins with monsoon winds picking up huge amounts of water vapour from the sea and carrying it toward land.
   - During the **southwest monsoon**, moist winds travel from the Arabian Sea and Bay of Bengal into the Indian subcontinent, and when they reach hilly terrain like the **Himalayas or the Western Ghats**, they have nowhere to go but up.
2. **Orographic Lift Forces the Air Upward:** As these moisture-heavy winds hit a mountain slope, they are forced to rise sharply along the terrain — a process called **orographic lift**.
   - This rising air cools rapidly as it climbs, and the cooling causes water vapour to condense into cloud droplets much faster than over flat land — which is why cloudbursts are concentrated between **1,000-2,500 metres altitude**, the typical valley-fold zone of the Himalayan southern rim.
3. **Cumulonimbus Clouds Build Up:** The condensing moisture keeps stacking vertically, forming towering **cumulonimbus clouds** that can grow as high as 15 km into the troposphere.
   - Scientists note these events are often linked to **westward-moving cyclonic circulations** over the Tibet-Ladakh region during active monsoon spells, which intensify the convection driving these clouds even higher.
4. **Strong Updrafts Trap the Rain Inside the Cloud:** Instead of falling immediately, the raindrops are held up by powerful upward air currents (updrafts) inside the storm.
   - This is the crucial "loading" phase — new droplets keep forming and merging with existing ones, so the cloud keeps accumulating an enormous volume of water instead of releasing it gradually, almost like water building up behind a dam.
5. **The Updraft Suddenly Collapses — and the Cloud "Bursts":** Eventually the upward current weakens or breaks down, and it can no longer support the accumulated weight of water.
   - All that trapped moisture then falls at once as an intense, short burst rather than a normal spread-out shower — for a 20 sq km area, a 100mm/hour cloudburst means roughly **2 billion litres of water dumped in just one hour** (IITM Climate Research Lab estimate).

Reasons Behind Frequent Cloudburst Occurrences In India

1. **A Warmer Atmosphere Holds More Moisture:** Rising global temperatures allow air to carry significantly more water vapour before it condenses, setting up the fuel for more intense downpours.

   Every **1°C rise in temperature lets the air hold about 7% more moisture**, and the warming of the **Indian Ocean and Arabian Sea** is pushing more moisture inland during the monsoon, directly feeding the kind of extreme, short-duration rainfall spikes that trigger cloudbursts.

2. **Glacier and Snow Melt Altering Local Weather Patterns:** As Himalayan glaciers retreat due to warming, they add extra moisture and destabilise the region's usual weather behaviour.

   This melt-driven moisture, combined with local convection, makes rainfall events more **erratic and extreme** rather than spread out — a pattern experts link to the growing overlap between cloudburst events and other hazards like Glacial Lake Outburst Floods (GLOFs), such as the **South Lhonak GLOF in Sikkim (2023)**.

3. **Deforestation and Loss of Green Cover:** Trees and vegetation normally intercept rainfall and slow down runoff; removing them leaves hillsides directly exposed to sudden downpours.

   Widespread **deforestation, road-cutting and slope blasting** across the Himalayan belt has reduced natural water absorption, so even a moderate cloudburst now triggers faster and more destructive runoff, worsening landslide and flash-flood risk .

4. **Unplanned Construction in Fragile Hill Areas:** Hotels, roads, hydropower projects and homes are increasingly built on unstable slopes and riverbeds, right in the path of cloudburst-triggered flash floods.

> The **Dehradun cloudburst (September 2025)** flooded the Tapkeshwar Mahadev Temple after the Tamsa river overflowed, while the **Kishtwar (J&K) cloudburst (August 2025)**, which killed dozens, struck an area where encroachment on natural drainage channels had already narrowed the stream's capacity to carry excess water.

5. **Rising Tourist and Population Pressure in Hill Towns:** Growing footfall in Himalayan tourist and pilgrimage towns has pushed unplanned urban expansion faster than drainage and infrastructure can keep up.
6. **Orographic and Convective Atmospheric Conditions Specific to the Himalayas:** Beyond human factors, the region's natural geography itself concentrates the conditions needed for a cloudburst.

> Moisture-laden monsoon winds are forced upward rapidly along steep slopes (**orographic lift**), and this, combined with **western disturbances interacting with local topography**, creates strong convective instability that IMD scientists say is especially difficult to forecast, since it plays out over a very small area and time window.

7. **Blocked Natural Drainage and Poor Urban Planning:** Even where rainfall itself isn't unusual, blocked drains, inadequate culverts, and construction over natural streams turn heavy rain into destructive flash floods.

Way Forward

1. **Strengthen Forecasting Through Mission Mausam:** India's weather models currently work at a coarse 12 km resolution, too broad to catch a cloudburst confined to a few square kilometres.

> The **Ministry of Earth Sciences' Mission Mausam** aims to sharpen resolution from 12 km to 6 km and cut nowcast lead time from three hours to one hour, directly targeting hard-to-predict events like cloudbursts and thunderstorms.

2. **Expand Doppler Radar and Observation Networks in Gaps:** Remote Himalayan valleys, where most cloudbursts originate, remain the most poorly monitored.

> IMD's Doppler Weather Radar network now covers close to 92% of India's area (Rajya Sabha reply, February 2026), but coverage of the country is not the same as covering every valley — closing this last gap, especially in the Eastern Himalayas and North-East, remains an ongoing priority under Mission Mausam.

3. **Follow NDMA's Hazard Zonation and Land-Use Guidelines:** NDMA's guidelines on floods call for identifying flood/cloudburst-prone zones and regulating habitation in low-lying areas along rivers and drains through State and District Disaster Management Authorities.
4. **Use the National Disaster Management Plan's Multi-Hazard Framework:** The **NDMP (2016, revised 2019)** brings cloudbursts under a broader flood-and-landslide response framework, aligning India's approach with the Sendai Framework for Disaster Risk Reduction and the Paris Agreement.

> IMD's Doppler Weather Radar network now covers close to 92% of India's area, but coverage of the country is not the same as covering every valley — closing this last gap, especially in the Eastern Himalayas and North-East, remains an ongoing priority under Mission Mausam.

5. **Build NDMA's Digital Decision-Support Tools for Local Use:** NDMA has developed a web-based **Dynamic Composite Risk Atlas (Web-DCRA)** and **Decision Support System**, currently used for cyclones, that can be extended to cloudburst-prone river basins for real-time risk visualisation at the

district level.

6. **Strengthen Community-Level Preparedness:** Since forecasting will never fully close the warning gap, **NDMA's disaster management exercises (DMEx)** — like the flood mock drill conducted across 118 tehsils of Uttar Pradesh in June 2025 — show the value of regularly testing local evacuation plans before the monsoon arrives, rather than only after a disaster strikes.

Conclusion

Cloudbursts cannot be prevented, but their impacts can be significantly reduced through better forecasting, hazard mapping, resilient infrastructure and strict regulation of vulnerable mountain areas. The focus must shift from disaster response to disaster risk reduction.

12. HEAT WAVES IN INDIA

Once considered a seasonal discomfort, heat waves in India have now become a **recurring public health and economic emergency** — 2024, 2025 and 2026 have each broken previous heat records in succession, turning what used to be a few bad weeks into a two-month ordeal for large parts of the country.

IMD Criteria for Declaring a Heat Wave

- **Temperature Threshold**
  - **≥ 40°C** in **plains**
  - **≥ 37°C** in **coastal areas**
  - **≥ 30°C** in **hilly regions**
- **Departure from Normal**
  - Maximum temperature is **4.5°C or more above the normal** for that location.

OR

- **Absolute Temperature Criterion**
  - Maximum temperature **exceeds 45°C**, **irrespective of the normal temperature**.

India's Vulnerability Assessment For Heat Waves

1. **Geographic Spread of High-Risk Districts:** Heat risk in India is no longer confined to a few desert states — it now covers most of the country.

   A **2025 study by the Council on Energy, Environment and Water (CEEW)** found that **57% of Indian districts, home to over three-quarters of the population, are now classified as high to very high risk** from extreme heat, with risk defined as a combination of hazard, exposure and vulnerability together.

2. **Heavy Dependence on Outdoor, Informal Work:** A large share of India's workforce simply cannot avoid the sun, because their livelihood depends on being outdoors.

   Agriculture, construction, transport, sanitation and street vending employ a huge share of India's labour force, and the **2025 Lancet Countdown on Health and Climate Change** found that heat exposure caused India to lose **247 billion potential labour hours in 2024** — a record high and a 124% increase compared to 1990s levels.

3. **Poor Housing and Urban "Heat Traps":** The materials people live in often decide whether they can recover from the day's heat at night.

Overcrowded housing, tin-roofed or poorly ventilated informal settlements, and a shortage of parks, tree cover and water bodies (**blue-green infrastructure**) mean cities trap heat rather than release it — for instance, tree-cutting for redevelopment projects like **T T Nagar in Bhopal** is expected to intensify local heat further (IDR/urban heat research).

4. **Homeless and Slum Populations Facing Round-the-Clock Exposure:** For those without secure shelter, heat stress doesn't end when the sun goes down.
5. **Weak and Fragmented Death Reporting:** India's inability to consistently count heat deaths makes it harder to direct resources to the most vulnerable.

The **National Centre for Disease Control (NCDC)** reported 7,192 suspected heatstroke cases between 1 March and 24 June 2025 but confirmed only 14 deaths

6. **Gendered Vulnerability:** Women, especially in agriculture and informal work, face heat risks that are often invisible in standard heat data..
7. **Gaps Between Policy and Implementation:** Even where Heat Action Plans (HAPs) exist, they haven't always been designed around the people most at risk.

Experts note that earlier HAPs identified vulnerable groups but lacked **tailored measures for street vendors, construction workers and waste pickers**. In response, the **National Disaster Management Authority (NDMA)** issued an advisory in 2025 urging better recognition of informal workers, and **Delhi's 2025 HAP** now requires uninterrupted water supply and shaded cooling shelters specifically in informal settlements .

Reasons Behind Growing Heatwaves In India

1. **Human-Induced Global Warming:** Rising worldwide temperatures are the biggest single driver, making extreme heat events more frequent and intense everywhere, including India.

The **IPCC's Sixth Assessment Report** confirms that heatwaves have become more frequent and intense globally due to anthropogenic warming, and identifies India as one of the global heatwave risk regions where both apparent temperature and wet-bulb temperature are projected to exceed safe human tolerance limits.

2. **Loss of Natural Buffers — Deforestation and Green Cover Decline:** Forests and wetlands help regulate local temperatures; losing them removes a natural shield against extreme heat.

Forest Survey of India reports show that urban green cover in several metropolitan regions has declined due to infrastructure expansion, weakening the land's natural ability to absorb heat rather than reflect it back.

3. **Unplanned Urbanisation and the Urban Heat Island Effect:** Concrete, glass and dense construction absorb heat during the day and release it slowly at night, keeping cities hotter than surrounding rural areas.

Rapid, unplanned urban expansion combined with high population density is creating **"urban heat islands"** in Indian metros, meaning cities like Delhi, Kolkata and Ahmedabad now experience noticeably higher night temperatures than their outskirts, reducing the body's chance to recover overnight.

4. **Specific Atmospheric Conditions During Peak Summer:** Certain weather patterns directly set up the conditions for extreme heat to build and persist.

Factors like **clear skies allowing direct solar radiation, reduced snow cover across the Himalayas and Eurasia (which normally reflects heat back), warming of the Pacific Ocean, neutral ENSO conditions, and dry northwesterly winds preventing cloud formation** all combine to trap heat over the Indian subcontinent (India Water Portal, citing IMD data, 2026).

5. **Early and Unstable Seasonal Onset:** Heat is no longer confined to the traditional summer months — it is arriving earlier and lingering longer than historical norms.

   For example , In **February 2025**, Goa and Maharashtra recorded India's first-ever winter heatwave — an event the IMD itself described as unprecedented — while 2026 also saw an unusually early onset of extreme heat, weeks ahead of historical patterns.

6. **Groundwater Depletion and Himalayan Glacier Melt:** Falling water tables and melting glaciers are linked to rising heat in a two-way relationship — heat accelerates melt and depletion, which in turn worsens local warming.

   Rising regional temperatures are accelerating **Himalayan glacier melt**, altering flow patterns in rivers like the Ganga, Yamuna and Brahmaputra, while falling groundwater levels reduce the land's natural cooling capacity, compounding the heat problem in both mountain and plains regions.

7. **India's Rising Baseline Vulnerability:** Even at the same temperature, India is becoming less able to cope with heat than it was a few decades ago.

   According to the **Lancet Countdown on Health and Climate Change**, India has become **15% more vulnerable to extreme heat compared to 1990**, reflecting how population growth, urban crowding and changing work patterns have outpaced the country's adaptive capacity even as the climate itself keeps warming.

Issues Caused by Heat Waves in India

Issue	Key Features
Health Impacts	• Causes **heat exhaustion, heatstroke and deaths**. • Worsens **cardiovascular and kidney diseases**. • Most vulnerable: **elderly, children and outdoor workers**.
Loss of Labour Productivity	• Reduces **working hours** in heat-exposed sectors. • Lowers productivity in **agriculture, construction and informal sectors**. • Results in significant **economic losses**.
Agriculture & Food Security	• Causes **crop failure and livestock stress**. • Reduces farm incomes and raises **food prices**. • Intensifies **drought and water scarcity**.
Pressure on Power Infrastructure	• Sharp rise in **electricity demand** for cooling. • Increases risk of **grid failures and power shortages**. • Raises energy costs.
Water Stress	• Increases **evaporation** from rivers and reservoirs. • Raises demand for **drinking water and irrigation**. • Worsens groundwater depletion.
Impact on Renewable Energy	• Reduces **solar power efficiency** and grid reliability. • Increases renewable energy curtailment during peak demand.
Social & Economic Inequality	• Disproportionately affects **daily-wage earners and informal workers**. • Deepens existing **income and health inequalities**. • Increases vulnerability of poor urban and rural households.

Way Forward

1. **Notify Heat Waves as an Official Disaster:** Since heat waves are not formally notified as a disaster nationally, many heat deaths remain outside disaster relief funding.

   States like **Rajasthan, Tamil Nadu and Kerala** have already declared heat waves a **state-specific disaster**, unlocking State Disaster Response Fund (SDRF) money for relief and preparedness — a model experts recommend extending nationwide.

2. **Strengthen and Scale Up Heat Action Plans (HAPs):** HAPs remain India's core policy tool, but many still lack dedicated funding and tailored measures for the most exposed groups.

   Delhi's 2025 HAP now mandates **installing around 3,000 water coolers, shaded footpaths, cool roofs, and dedicated heat-wave hospital wards**, while central hospitals have been directed to set up **"heat-stroke rooms"** for rapid cooling treatment — but a review of 37 HAPs found only two had identified clear funding sources, showing implementation gaps remain the biggest challenge .

3. **Expand Cool Roof and Urban Greening Programmes:** Since roughly 35% of India's urban population lives in low-rise, tin-roofed housing that traps heat, low-cost cooling retrofits can directly cut indoor temperatures.

   Ahmedabad's Cool Roofs Programme reduces indoor temperatures by up to 5°C using reflective paint,

   while **Tamil Nadu's Urban Greening Policy** mandates at least 15% green cover in cities, and its "Green Schools" initiative has cut classroom temperatures from 31°C to 27°C.

4. **Build Financial Safety Nets for Outdoor and Informal Workers:** Since a large share of India's workforce cannot avoid working outdoors, income protection during extreme heat is as important as physical cooling.

   The **Self-Employed Women's Association (SEWA)**, in partnership with Arsht-Rock and Blue Marble, launched an **Extreme Heat Income Micro-Insurance** scheme — in May 2024 alone, it delivered around USD 340,000 to 46,000 women whose income was hit by extreme heat, letting them skip dangerous outdoor work without losing pay.

5. **Improve Localised Early Warning and Vulnerability Mapping:** National heatwave thresholds often miss local variations caused by humidity, urban heat islands, or housing type.

   NDMA, working with IMD, has developed **city-specific threshold assessments using 30 years of temperature and mortality data**, allowing area-specific advisories rather than one-size-fits-all warnings — an approach now being extended to more of the 23 heat-prone states.

6. **Introduce Worker Protection Measures During Peak Heat:** Simple administrative steps — rescheduling work hours or mandating breaks — can directly cut heat exposure for outdoor labourers.

   Delhi declared paid breaks for outdoor labourers during the peak afternoon heat, while **Rajasthan's Heat Action Plans** now mandate shaded worksites, drinking water points, and rescheduled outdoor work hours in high-risk districts.

7. **Strengthen Institutional Coordination and Long-Term Planning:** NDMA has begun pushing states beyond reactive measures toward integrated, sector-wide heat mitigation frameworks.

   At its 2024 national workshop on **"Developing Framework for Heat Wave Mitigation,"** NDMA officials called for integrating land-cover and infrastructure adaptation into HAPs, alongside sector-specific plans covering health, energy, and urban infrastructure — recognising that cooling shelters alone cannot substitute for heat-resilient city design.

13. MARINE HEATWAVES

A **marine heatwave (MHW)** is a prolonged period of abnormally high sea surface temperatures, typically defined as temperatures exceeding the **90th percentile of the historical average for at least five consecutive days**

Unlike land heatwaves, marine heatwaves can last from days to even months or years, and India's own coastal waters — the Arabian Sea and Bay of Bengal — are now among the regions experiencing this warming most intensely.

Causes of Increasing Marine Heatwaves

1. **Oceans Absorbing Excess Greenhouse Heat:** The single biggest driver of marine heatwaves is that oceans act as the planet's main heat sink, soaking up most of the extra warming caused by greenhouse gases.

   > Oceans absorb close to **90% of the excess heat trapped by greenhouse gases** , and in the Indian Ocean specifically, research shows about **90% of the rising sea surface temperature trend is linked to human-caused emissions**.

2. **The Indian Ocean Warming Faster Than the Global Average:** Some ocean basins are heating up quicker than others, and the Indian Ocean happens to be one of the fastest-warming in the world.

   > Studies note the **Indian Ocean has warmed by about 1°C**, compared to a global average rise of 0.7°C, making it one of the most rapidly warming ocean basins on the planet.

3. **Anticyclonic Eddies and Ocean Circulation Patterns:** Beyond long-term warming, short-term swirling ocean current patterns can trap warm water in one place, intensifying a heatwave locally.

   > Research on the Bay of Bengal found that **anticyclonic eddies (large rotating water masses linked to elevated sea surface height)** were a key driver behind its most intense recorded marine heatwave — a 69-day event in 2016 that reached a peak intensity of 5.29°C above the historical average.

4. **Natural Climate Variability — La Niña and El Niño Cycles:** Alongside long-term warming, natural ocean-atmosphere cycles can push sea temperatures even higher in certain years.

   > The **India Meteorological Department** projected the Indian Ocean would stay warmer than normal from March to May 2026, even during a **La Niña phase** — and warned that if a warming **El Niño** phase develops later in 2026, temperatures could climb even further.

5. **Weak Net Heat Flux Regulation:** Reduced heat exchange between the ocean surface and atmosphere allows warmth to build up in surface waters instead of dispersing.

6. **Wider Ocean-Atmosphere Feedback with the Monsoon System:** Marine heatwaves don't just sit in the ocean — they interact with and are reinforced by broader regional climate systems like the monsoon.

   > A 2026 study in Frontiers in Climate examined marine heatwave drivers in the North Indian Ocean and their links to South Asian monsoon rainfall variability, showing that ocean heating and monsoon behaviour are increasingly interconnected rather than separate phenomena.

Issues Caused by Marine Heatwaves

Issue	Key Features

Coral Bleaching	• Causes **corals to expel symbiotic algae**, leading to bleaching. • Weakens reef ecosystems and increases coral mortality. • Threatens marine biodiversity.
Loss of Fish Nursery Habitats	• Damages **coral reefs and seagrass beds**, the breeding grounds for fish. • Reduces fish recruitment and long-term fish stocks.
Fish Migration	• Forces fish to migrate towards **deeper and cooler waters**. • Reduces fish catch and affects traditional fishing grounds.
Threat to Fisheries & Livelihoods	• Reduces incomes of **coastal fishing communities**. • Threatens food security and employment in the fisheries sector. • Increases fish prices.
Cyclone Intensification	• Warmer seas provide **more energy for tropical cyclones**. • Increases the likelihood of **rapid cyclone intensification**.
Disruption of Monsoon	• Alters **ocean-atmosphere interactions**. • Leads to erratic rainfall, floods and droughts across India.
Ocean Deoxygenation & Acidification	• Aggravates **oxygen depletion and ocean acidification**. • Increases stress on marine ecosystems and biodiversity.
Decline in Ocean Productivity	• Reduces **plankton productivity**, disrupting the marine food chain. • Lowers overall fish production.
Increasing Frequency & Duration	• Marine heatwaves are becoming **more frequent, intense and prolonged**. • Increases long-term climate and ecological risks.
Governance & Preparedness Gaps	• Limited integration of **INCOIS warnings** into fisheries management. • Weak local preparedness and adaptive planning increase vulnerability.

Way Forward

- **Integrate INCOIS Alerts Directly into Fishing Advisories**: INCOIS's warnings currently exist somewhat separately from the day-to-day advisories fishers actually rely on, creating a gap between good science and actual on-the-water decisions.

  Experts recommend fully **integrating INCOIS's marine heatwave alerts into Potential Fishing Zone (PFZ) advisories**, building on lessons from the **2023 Bay of Bengal extreme warming event**, so fishers get combined, actionable guidance rather than two separate data streams to interpret themselves.

- **Expand Coral Reef and Mangrove Restoration as Natural Infrastructure**: Healthy reefs and mangroves act as natural buffers that both protect coastlines and sustain fish nurseries — rebuilding them is a direct, nature-based response to heat-driven habitat loss.

  With the **Gulf of Mannar having lost 85% of its coral cover** in past heat stress events, restoring these reefs alongside mangrove belts is being pushed as essential blue-economy infrastructure, not just an environmental afterthought.

- **Promote Deep-Sea Fishing and Climate-Resilient Aquaculture**: Since traditional coastal stocks like sardine and mackerel are moving to deeper, cooler waters, fishing livelihoods need to follow that shift rather than keep targeting increasingly scarce shallow-water catch.
  - This means investing in **deep-sea fishing capacity and climate-resilient aquaculture** so fishers aren't left chasing a resource that has physically relocated — directly relevant given Kerala's sardine catch has already fallen **57% over the past decade**.
- **Diversify Livelihoods for Coastal Fishing Communities**: With over **70% of fishers dependent on capture fisheries alone**, this concentration leaves almost no financial cushion when a heatwave disrupts catch for an entire season.
  - Building alternate income streams — from aquaculture to coastal eco-tourism — is recommended specifically to reduce this dangerous over-reliance on a single, increasingly climate-vulnerable livelihood source.
- **Strengthen Local Governance and Climate Awareness**: Even world-class alerts are useless if the panchayat or fishing cooperative receiving them doesn't know how to act on the information in time. Experts specifically flag **low local governance awareness and poor uptake of early warning systems** as a key gap — meaning training coastal administrators and fishing cooperatives to actually use INCOIS data is as important as generating the data itself.
- **Expand Satellite-Based Real-Time Monitoring**: More granular, frequent ocean-temperature data allows both scientists and fishers to react faster and more precisely as heat stress develops.

  To **expand satellite-based monitoring and real-time INCOIS alerts** further, building on the six-basin coverage already active as of 2026, so smaller or more localised heat pockets don't go undetected between broader regional assessments.
- **Factor Marine Heatwaves into Monsoon and Cyclone Forecasting**: Since MHWs are now scientifically linked to weakened monsoon winds and rapid cyclone intensification, treating ocean heat data as a core input for weather forecasting — not a separate marine-only concern — would improve prediction accuracy for both.
- **Invest in Longer-Range Predictive Climate Models**: Since marine heatwaves are being driven by long-term ocean warming trends, not just short-term weather variability, better long-range models would give fisheries and coastal planners more lead time to adapt.

  Improving **advanced climate models and statistical forecasting approaches** for ocean heat extremes would allow policymakers to plan fishing seasons and coastal infrastructure investment further in advance, rather than reacting only once a heatwave has already begun.
- **Pursue International Cooperation on Indian Ocean Monitoring**: Since Indian Ocean warming affects multiple coastal nations simultaneously, and India already plays a regional leadership role through its tsunami warning system, extending similar cooperation to marine heatwave data would strengthen the whole region's response.

  For example, sharing monitoring data and best practices with **Indian Ocean Rim neighbours now** — rather than waiting for that projection to fully materialise — would build the kind of regional resilience India has already demonstrated through its tsunami-warning leadership.

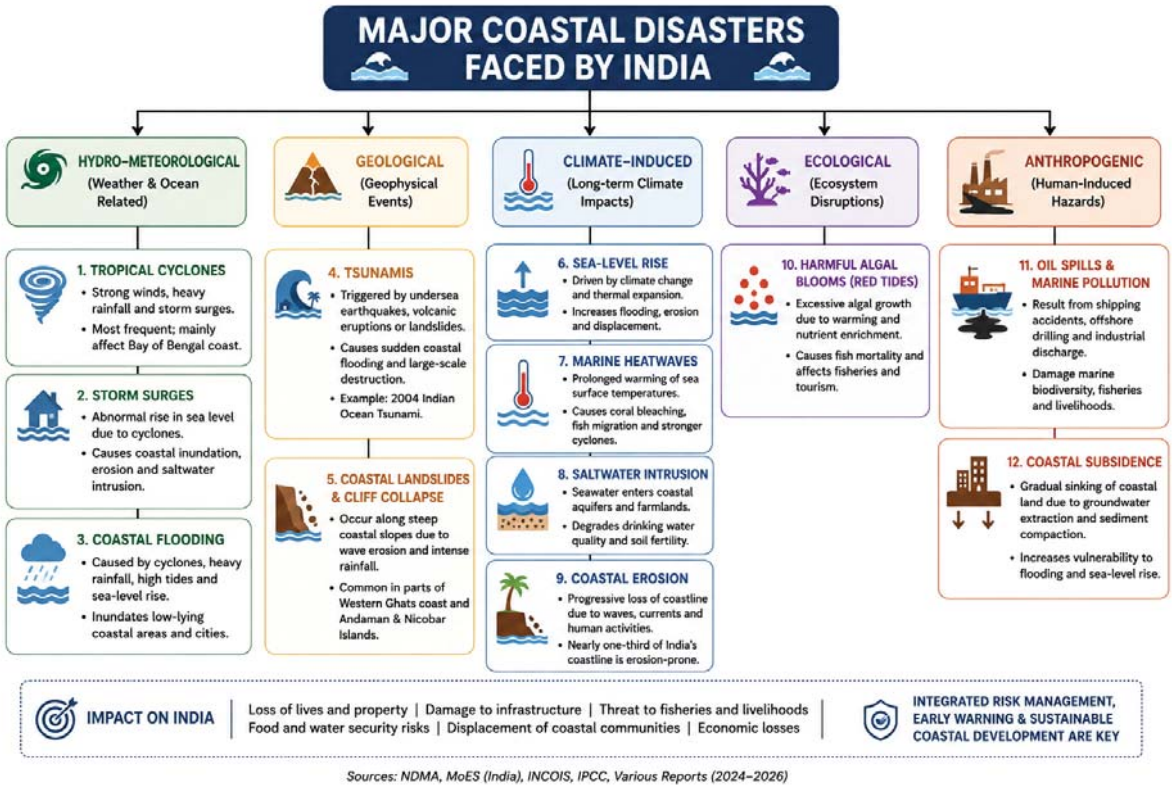
Conclusion

As India pursues a sustainable Blue Economy, addressing marine heatwaves through scientific monitoring, adaptive fisheries management and conservation of coral reefs and mangroves must become a national priority.

14. COASTAL DISASTERS IN INDIA

India has a **7,500 km coastline** spanning 13 coastal states and Union Territories, home to over **4 million marine fishers and fish farmers** — but this same coastline is now facing a **compounding set of hazards**: stronger cyclones, rising seas, and eroding shorelines, all intensifying together rather than as separate, isolated problems.

What makes coastal disasters distinct from other hazards is this **layering effect** — a single cyclone today doesn't just bring wind and rain. It damages infrastructure , hurts the economies - which are increasingly turning "blue" and also creates a food security risk - along with many other consequences .



Sources: NDMA, MoES (India), INCOIS, IPCC, Various Reports (2024–2026)

Key Coastal Disasters and Their Dimensions

- **Tropical Cyclones**: An intense circular storm originating over warm tropical oceans, needing sea surface temperatures above **27°C to a depth of 60-70m** to form and intensify.

> The **eastern coast bears the brunt** — Odisha, Andhra Pradesh, and West Bengal saw 98, 79, and 69 cyclones respectively between 1981-2002 — and **Cyclone Montha (2025)** most recently exposed continuing vulnerability, causing widespread damage across Andhra Pradesh and Odisha despite decades of improved forecasting.

- **Rapid Intensification of Cyclones**: Warmer sea surfaces are now allowing cyclones to strengthen unusually fast, sometimes within hours, making them far harder for evacuation planners to keep pace with.

> **Cyclones Tauktae (2021) and Yaas (2021)** both saw this "rapid intensification" phenomenon, driven by higher sea surface temperatures providing more latent heat — Tauktae alone killed **104 people, the highest cyclone death toll in a decade**.

- **Sea Level Rise**: Global sea levels are climbing at roughly **3.3mm per year**, gradually submerging low-lying coastal land and amplifying every other coastal hazard on top of it.

  - This directly threatens the **Sundarbans, Mumbai, and Kolkata** through permanent inundation and land loss — and studies confirm these same three cities face the **highest future flood risk** in India due to their low elevation and dense populations.
- **Coastal Erosion**: The gradual wearing away of shoreline by wave action, tides, and storm surges, worsened when sea levels rise and natural buffers like mangroves are removed.

  This disaster has seen largely policy level ignorance in India.For example, The government's disaster corpus (2021-2026) allocates just **₹1,500 crore specifically for erosion mitigation** — a strikingly small figure given that just three megacities (Mumbai, Kolkata, Chennai) face a combined **$6 billion in annual disaster-related losses**.
- **Storm Surge**: A sudden rise in sea level caused by a cyclone's strong winds pushing water toward the shore, often the single deadliest element of a cyclone since it can inundate low-lying coastal areas within minutes.

  Rising baseline sea levels mean storm surges now reach further inland than they would have decades ago for the same cyclone strength — compounding flood risk in port cities like **Visakhapatnam**, where insurers have reported **25% rate increases** tied specifically to this cyclone-sea-level-rise combination.
- **Salinity Intrusion**: Rising seas push saline water into freshwater aquifers and agricultural land, quietly destroying farmland productivity long after any single storm has passed.

  This is a growing concern particularly in the **Sundarbans, Gujarat, and the Krishna-Godavari delta**, where farmers are seeing freshwater sources turn increasingly saline, threatening both drinking water and crop yields.
- **Loss of Fishing Livelihoods Beyond Direct Disaster Damage**: Even when a cyclone doesn't make landfall, the disruption to fishing communities is real and repeated, since fishers cannot safely go to sea once any warning is issued.
- **Industrial and Oil Spills**: Coastal industrial hubs concentrate refineries and ports right next to fragile marine ecosystems, meaning any accident or negligence causes damage that spreads fast through tides and currents.

  The **2017 Ennore oil spill**, caused by a collision between two tankers near Chennai, spilled nearly **9.9 million gallons of oil**, affecting **34,000 square metres** of coastline
- **Coral Reef and Mangrove Degradation**: The natural buffers that would otherwise absorb storm energy and support fisheries are themselves being destroyed — by warming seas, pollution, and coastal construction — reducing India's built-in resilience to every other disaster on this list.

  Studies have found that Mangroves can **reduce cyclone wave energy by up to 70-90%**, yet India's total mangrove cover remains limited to roughly **4,992 sq km** nationally — meaning much of the coastline still lacks this natural first line of defence.

Coastal Erosion

Coastal erosion is the progressive and continuous loss of land along a shoreline due to the removal of sediment by waves, currents, tides, or human interference.

As per the **National Centre for Coastal Research (NCCR)**, under the Ministry of Earth Sciences, a comprehensive satellite-based study of India's **6,632 km mainland coastline (1990-2018)** found that 33.6% of India's coastline is eroding, while 26.9% is actually growing (accretion) and only 39.6% remains stable — meaning erosion is a serious but geographically uneven problem, not a uniform national crisis.

Key Causes

- **Wave Action and Wave Energy**: The most fundamental, ever-present driver of coastal erosion — the continuous mechanical force of waves striking the shoreline naturally wears away sediment over time.

- **Sea Level Rise**: As global temperatures rise, thermal expansion of seawater and melting glaciers are gradually pushing sea levels higher, allowing waves to reach further inland than before.

  Compounding effect : This climate-driven factor compounds every other cause on this list, since even a stable shoreline today becomes progressively more vulnerable as the baseline sea level itself keeps climbing year after year.

- **Cyclones and Storm Surges**: Intense tropical storms deliver a concentrated, high-energy erosion event within just a few days — often causing more shoreline loss than months of ordinary wave action combined.

  For example, The **Paradip coast in Odisha** has seen significant erosion linked to the increased frequency of cyclones in the region, compounding the effect of nearby port construction on the same stretch of coastline.

- **Interception of Littoral Drift**: Sand naturally moves along the coastline through wave-driven currents called littoral drift — when structures like ports and harbours block this movement, sand piles up on one side while the other side starves and erodes.

  This is a key reason the **Paradip Port in Odisha** disrupted natural sediment flow along the coast, leading to displacement of local communities and loss of agricultural land in areas downstream of the port.

- **Sand and Coral Mining**: Removing sand and coral directly from beaches and nearshore waters strips away the very material that would otherwise naturally replenish an eroding shoreline.

  NCCR studies specifically flag **dredging, sand mining, and coral mining** as causing sediment deficit and altered wave refraction.

- **Dam Construction on Rivers**: Rivers naturally carry sediment down to the coast, replenishing beaches — but dams trap this sediment upstream, starving the coastline of the material it needs to stay stable.

  NCCR's own classification of anthropogenic causes explicitly includes **dam building on rivers** alongside harbour construction as a major human contributor, since blocked river sediment never reaches the coast to compensate for natural erosion.

- **Changes at River Mouths**: Alterations to how rivers meet the sea — whether from engineering or natural shifts — can disrupt the delicate sediment balance that keeps a nearby coastline stable..

- **State-Wise Variation Confirms Human Activity's Role**: Regional differences in erosion severity across similar coastal geography suggest local human activity, not just uniform natural forces, is driving the disparity.

  West Bengal (60.5%), Puducherry (57%), Kerala (46.4%), and Tamil Nadu (42.7%) show the highest erosion rates, while **Odisha uniquely shows over 51% accretion (coastline growth)** — a stark contrast that points to local factors like port placement and sediment management, not rainfall or wave energy alone, driving the difference.

- **Unplanned Coastal Construction and Development**: Building resorts, roads, and settlements too close to the shoreline removes natural vegetation and dunes that would otherwise buffer wave impact, leaving the coast directly exposed.

  This is why the Ministry of Environment, Forest and Climate Change has had to delineate a "Hazard Line" **for India's entire coastline** — a boundary based on shoreline change and sea-level rise data, specifically meant to guide where construction should and shouldn't happen.

Impacts of Coastal Erosion

- **Loss of Land and Property**: The most direct and visible impact — erosion permanently removes usable land, destroying homes, public spaces, and property value along the coast.

  The loss of land along **Chennai's Marina Beach** has severely affected nearby property and public spaces, showing that even a well-known, urban stretch of coastline isn't immune to this slow but permanent land loss.

- **Displacement of Coastal Communities**: As land physically disappears, families and entire villages are forced to relocate, often losing generational roots to a place they've lived for decades.
- **Destruction of Fishing Livelihoods**: Erosion doesn't just take homes — it takes away the physical working space fishing communities need for boats, nets, and daily operations, directly hitting their income.

  A study of the **Alappad fishing community in Kerala** found coastal erosion — worsened by sand mining and the 2004 tsunami — degraded groundwater quality and caused **decreased fishing productivity**, with the once-vibrant beach area significantly shrinking.

- **Damage to Critical Infrastructure**: Roads, bridges, and coastal highways are constantly undermined by wave action, since much of India's coastal transport network was built without anticipating today's erosion rates. Coastal highways and transport embankments are regularly compromised by wave action, and it's now common to hear of **roads collapsing after heavy rains** once their foundations have been weakened by years of erosion underneath.
- **Agricultural Land Loss Through Salinity**: As the sea encroaches, saltwater increasingly contaminates nearby farmland, turning once-productive fields unusable for regular crops.
  - This saltwater encroachment is specifically flagged as ruining agricultural fields along eroding stretches of coast, adding a food-security dimension to what might otherwise look like a purely geographical problem.
- **Destruction of Coastal Ecosystems**: Mangroves, salt marshes, coral reefs, and sand dunes — all of which act as natural buffers — are themselves destroyed by the same erosion they would otherwise help prevent, creating a worsening feedback loop.

  In the **Sundarbans region of West Bengal**, erosion has led to significant loss of mangrove forests — a serious concern given the Sundarbans **support the livelihoods of over 1.3 million people** and also naturally moderate the physical impact of cyclones like Amphan on the surrounding region.

  Increased Flooding Risk: Once natural barriers like dunes and mangroves are eroded away, coastal areas lose their built-in protection against storm surge and heavy rain, making flooding both more frequent and more severe.

  In coastal **Kerala**, erosion has measurably increased flooding risk, worsening the impact of heavy rains and storms on already low-lying areas that previously had more natural buffering.

  Decline in Tourism and Fisheries Revenue: Beyond subsistence fishing, erosion also damages the broader coastal economy — beaches that draw tourists and fishing grounds that support commercial fleets both shrink as the shoreline retreats.
  - This decline directly increases the financial burden on state governments, which then have to spend more on coastal protection measures just to preserve what economic activity remains along an already-shrinking coastline.
- **Social Unrest and Resource Conflicts**: As land and fishing grounds shrink, competition over the remaining usable coastal space can create friction between communities, and between communities and developers or the state.

Techniques to prevent Coastal Erosion

Technology	How it works ?
Seawalls (Hard Engineering)	• Concrete/stone walls built **parallel to the coastline** to absorb and reflect wave energy. • Protects coastal settlements and infrastructure. • **Case Study:** Marina Beach, Chennai constructed these seawalls and the experiment was highly effective. • **Limitation:** Expensive and may increase erosion on adjacent unprotected coasts due to wave reflection.
Groynes	• Structures built **perpendicular to the shoreline**. • They Trap sand transported by **littoral drift**, creating wider beaches on the updrift side. Widely used on India's east and west coasts. • **Limitation:** Causes **downdrift erosion** by interrupting sediment flow.
Geotextile Tubes (Geotubes)	• Large sand-filled synthetic fabric tubes placed along the shoreline. • Absorb wave energy instead of reflecting it (soft engineering). • Cost-effective alternative where natural rock is scarce. • **India has over 1,200 km of eroding coastline.** • **Case Studies:**– Hamla & Dahanu (Maharashtra): Achieved stable **equilibrium beach profile**. – *Pentha (Odisha) & Uppada (Andhra Pradesh):* Combined with **gabions and rock armour** for improved wave dissipation and structural stability.
Beach Nourishment (Soft Engineering)	• Artificial addition of sand to replenish eroding beaches. • Restores natural coastal buffer and supports tourism/ecosystems. • Enhances storm-energy absorption without rigid structures. • Often combined with geotubes to reduce long-term maintenance costs.
Dune Restoration & Coastal Vegetation	• Restoration of sand dunes and plantation of deep-rooted vegetation. • Stabilises sand and reduces erosion naturally. • **Seagrasses and coastal plants** anchor sediments and dissipate wave energy. • Low-cost and self-sustaining solution.
Mangrove Restoration – MISHTI Scheme (Nature-based Solution)	• **MISHTI (Mangrove Initiative for Shoreline Habitats & Tangible Incomes)** launched on **5 June 2023** under **MoEFCC** after India's participation in the **Mangrove Alliance at COP27**. • Target: **540 sq km** mangrove restoration across **11 coastal States/UTs by 2028**. • Funding: **CAMPA (70%) + MGNREGS (30%)**. • **Case Study:** Gujarat planted **19,020 ha** of mangroves within two years, surpassing its proportional target.

Integrated Coastal Zone Management Project (ICZMP)	• World Bank-assisted integrated coastal management programme. • Combines **structural, regulatory and ecosystem-based approaches**. • Implemented in **Gujarat, Odisha and West Bengal**.Integrates mangrove conservation, monitoring and regulated coastal development.
CRZ Notification, 2019 (Regulatory Measure)	• Provides legal backing for sustainable coastal management. • Conserves ecologically sensitive coastal stretches. • Permits scientifically designed erosion-control measures. • Protects livelihoods of fishing communities while regulating coastal construction.

Harmful Algal Blooms

An **algal bloom** is a natural phenomenon where phytoplankton (microscopic algae like diatoms and cyanobacteria) suddenly multiply in unusually large numbers, often visible as a discolouration of seawater.

While some blooms are harmless or even beneficial to fisheries, a **Harmful Algal Bloom (HAB)** produces toxins or depletes oxygen, turning a natural event into a genuine coastal hazard.

Causes Behind The Formation Of HABs

- **Eutrophication — Excess Nutrient Enrichment**: The single biggest driver of HABs is an oversupply of nitrogen and phosphorus in the water, which acts like fertiliser for algae, letting them multiply far beyond normal levels.

  Agricultural runoff carrying fertiliser and animal manure is a primary contributor globally, and along India's own coastline, research has flagged a clear pattern of increasing eutrophication linked to **population growth, urbanisation, and industrial waste**.

- **Sewage and Wastewater Discharge**: Untreated or poorly treated urban sewage dumps large amounts of nutrient-rich waste directly into rivers, wetlands, and coastal waters, feeding algae growth right where human settlements are densest.

  This was the major driver behind the recent **Puducherry red-tide event**, where reports attributed the algal bloom directly to **unregulated wastewater flow** combined with pollution and climate change.

  Industrial Pollution and Land-Use Change: Beyond sewage, industrial discharge and changing land-use patterns near water bodies add further nutrient and chemical loading, compounding the eutrophication problem from multiple directions at once

  In Assam, the **Deepor Beel wetland in Guwahati** — the city's major stormwater storage basin — has come under mounting stress from unregulated urban runoff, land-use change, and solid waste dumping, all combining to trigger harmful algal blooms that now threaten the wetland's biodiversity.

- **Coastal Upwelling**: A natural oceanographic process where wind pushes surface water offshore, allowing deep, nutrient-rich water to rise and replace it — while natural, this process delivers a concentrated nutrient surge that can directly trigger bloom formation.

  Research on India's own coastline confirms that **seasonal upwelling, monsoonal forcing, and nutrient-rich riverine discharge** are all key factors driving the spread and intensification of algal blooms in both the Arabian Sea and Bay of Bengal.

- **Rising Sea Surface Temperatures**: Algae generally grow faster and more aggressively in warmer water, meaning ocean warming directly creates more favourable conditions for blooms to form and persist longer than they historically would have.

  - Warmer waters and enclosed or semi-enclosed seas are particularly prone to HABs, which is part of why marine heatwave conditions — already increasingly common in the Arabian Sea and Bay of Bengal — are considered a compounding risk factor alongside nutrient pollution.
- **Climate Change–Driven Rainfall Extremes**: Both unusually heavy rainfall and prolonged drought can each independently favour bloom formation, just through different mechanisms — heavy rain flushes in more nutrients, while drought lets those nutrients accumulate for longer.

  Increased rainfall intensity flushes higher nutrient loads from land into waterways, while if this is followed by extended dry periods, water bodies **retain those nutrients longer**, giving algae extra time to exploit the nutrient-rich conditions.
- **Monsoon-Linked River Discharge**: India's seasonal monsoon dramatically increases river flow, which in turn carries large pulses of land-based nutrients into coastal waters during a fairly narrow window each year.

  Studies on the Indian coast note that blooms are most frequently reported during the **Southwest Monsoon (SWM) season**, directly linking the timing of India's rainiest months to a corresponding spike in coastal algal bloom activity.
- **Aquaculture and Ballast Water Introduction**: Fish and shrimp farming can add excess nutrients to nearby waters through feed and waste, while ships discharging ballast water can accidenta\lly introduce non-native algal species into new coastal areas entirely.

  This is precisely why India's **Ballast Water Management Programme (BAMPI)**, which monitors ports like Mumbai, exists specifically to reduce this risk — since even a single ship's ballast water can transplant a harmful algal species from one part of the world to a previously unaffected Indian port.

Case Studies on Algal Bloom Management

Case Study	Key Learning
Qingdao Green Tide Cleanup, China (2008)	Before the **Beijing Olympics**, Qingdao faced a massive algal bloom threatening the sailing events. China deployed thousands of personnel, boats and satellite monitoring to clear the bloom in time, demonstrating the importance of **rapid emergency response and real-time monitoring**.
Chesapeake Bay, USA	Recurrent algal blooms were tackled through **long-term nutrient reduction** by controlling nitrogen and phosphorus from agriculture and urban runoff. The programme highlights the importance of **watershed-based pollution management** over reactive clean-ups.
INCOIS Algal Bloom Information Service (ABIS), India	India developed **ABIS**, a satellite-based early warning system that detects algal blooms and issues advisories to fishermen in major coastal hotspots. It demonstrates the value of **early detection and preventive management** using space technology.

Way Forward to Deal with Algal Blooms

Measure	Key Features
Reduce Nutrient Pollution	• Reduce **nitrogen and phosphorus** from sewage and agricultural runoff. • Addresses the root cause of algal blooms.

Strengthen Sewage Treatment	• Improve **wastewater treatment** in coastal cities. • Prevents untreated sewage from entering rivers and coastal waters.
Promote Sustainable Agriculture	• Encourage **judicious fertiliser use** and buffer strips. • Minimises nutrient runoff into water bodies.
Restore Wetlands & Riparian Buffers	• Conserve **wetlands, mangroves and riverbank vegetation**. • Naturally filter nutrients before they reach the sea.
Expand INCOIS-ABIS Coverage	• Extend **Algal Bloom Information Service (ABIS)** to all identified bloom-prone coastal hotspots. • Strengthens early warning systems.
Integrate Alerts with SAMUDRA App	• Provide **real-time HAB alerts** through the existing **SAMUDRA App**. • Improves last-mile communication with fishers.
Real-Time Water Quality Monitoring	• Deploy sensors to monitor **temperature, chlorophyll and nutrient levels**. • Enables early detection of bloom formation.
Emergency Bloom Removal	• Use **mechanical removal and floating booms** during severe bloom events. • Limits damage to critical coastal areas.
Adopt Bioremediation Technologies	• Pilot **phosphorus-locking compounds and modified clay technologies**. • Helps control blooms in enclosed bays and estuaries.
Community Awareness & Monitoring	• Promote **citizen reporting and fisher participation** in bloom monitoring. • Complements satellite-based early warning systems.

Saltwater Intrusion

Saltwater (or seawater) intrusion is the movement of saline water into freshwater aquifers, typically occurring when groundwater is over-extracted or when rising seas increase pressure along the coast.

As per the **Ghyben-Herzberg principle**, freshwater and saltwater in a coastal aquifer naturally maintain a stable balance based on their density difference — but once that balance is disturbed, saline water pushes inland, permanently contaminating what was once usable freshwater.

With **nearly 7,500 km of coastline**, India faces this as a genuinely serious and growing concern.

Causes of Saltwater Intrusion

- **Excessive Groundwater Extraction**: Over-pumping freshwater faster than natural recharge can replace it lowers the water table, allowing denser seawater to move in and fill the resulting gap.

  In **Chennai and Nagapattinam**, over-pumping for agriculture and urban needs has already led to noticeable salinity in local borewells, while **Saurashtra and Kutch (Gujarat)** experience saline ingress specifically due to excessive groundwater withdrawal for irrigation.

- **Sea Level Rise Linked to Climate Change**: Rising seas increase the hydrostatic pressure pushing against the coastline, forcing saline water further into freshwater aquifers than it would naturally reach.

  As per **Various studies** , the **Sundarbans and Odisha coasts** have reported increasing salinity directly linked to sea-level rise

- **Tidal Fluctuations and Storm Surges**: High tides and cyclone-driven storm surges can force seawater to flow inland and infiltrate shallow aquifers directly, sometimes causing sudden, one-time-but-lasting salinity spikes..

  The **2004 Indian Ocean tsunami** caused severe seawater ingress in parts of **Tamil Nadu**, showing how even a single extreme event can leave a long-term mark on coastal groundwater quality

- **Geological and Hydrogeological Conditions**: Sandy soils, porous rock formations, and low-lying coastal depressions allow seawater to percolate into aquifers far more easily than they would through denser, less permeable ground.

  The **Mangrol-Chorwad coast in Saurashtra, Gujarat** faces salinity largely due to this natural hydrogeology combined with regular tidal inundation, as documented by the **Central Ground Water Board (CGWB)**.

- **Destruction of Natural Barriers - Mangrove Loss**: Mangroves and coastal wetlands act as a natural buffer that slows seawater's inland movement and supports freshwater recharge — clearing them removes this protection entirely.

  The **Sundarbans has lost around 40% of its mangrove cover**, directly worsening the region's vulnerability to saltwater intrusion — a clear example of how ecological degradation and water security are directly linked.

- **Coastal Development and Infrastructure**: Ports, roads, and urban construction near the coast can disrupt natural groundwater recharge patterns, indirectly making aquifers more vulnerable to saline contamination.
- Unregulated coastal construction reduces the natural land area available for rainwater to seep into and replenish freshwater aquifers, compounding the pressure already caused by over-extraction elsewhere.

Key Problems Caused by Saltwater Intrusion

Issues	Key Features
Contamination of Drinking Water	• Saline groundwater becomes **unfit for drinking and domestic use**. • For example, In **Munroe Island (Kerala)**, wells have shown changes in taste and colour, forcing residents to depend on alternative water sources.
Agricultural Land Degradation	• Salt accumulation reduces **soil fertility and agricultural productivity**. • For example, In **Saurashtra (Gujarat)**, salinity has affected **0.712 million hectares**, impacting **534 villages** and over **1.88 million people**.
Decline in Crop Productivity	• High salinity damages **crop roots and nutrient absorption**, lowering yields. • **Pulses, oilseeds and paddy** are particularly vulnerable, as observed in the **Sundarbans**.

Mobilisation of Secondary Pollutants	• Salinity can release **nitrogen and other pollutants** previously trapped in soils. • This compounds groundwater contamination beyond salt alone, making treatment more difficult.
Public Health Risks	• Long-term consumption of saline water increases the risk of **hypertension and cardiovascular diseases**. • Coastal communities relying on groundwater are particularly vulnerable.
Livelihood & Economic Losses	• Reduced farm productivity lowers incomes and increases **farmer indebtedness**. • In **Saurashtra**, deteriorating groundwater quality has significantly affected the regional economy.
Damage to Coastal Ecosystems	• Disturbs the **freshwater–saline water balance** in wetlands and estuaries. • Leads to habitat degradation, fish mortality and declining fisheries.
Decline in Land Value & Habitability	• Persistent salinity makes land **unsuitable for farming and human settlements**. • In **Munroe Island**, prolonged salinity and waterlogging have made habitation increasingly difficult.
Deterioration of Soil Structure	• Irrigation with saline water degrades **soil structure and water-holding capacity**. • Even after freshwater irrigation resumes, recovery of soil productivity is slow.
Compounding Environmental Risks	• Saltwater intrusion often occurs alongside **arsenic contamination and groundwater depletion**. • For example, The **Sundarban Delta** faces both salinity and arsenic pollution, creating a complex water-quality challenge.

Way Forward

1. **Strengthen Aquifer Mapping for Targeted Action:** We cannot fix what we haven't properly mapped — many intrusion-hit stretches still lack detailed local groundwater data.

   The **Central Ground Water Board (CGWB)**'s **National Aquifer Mapping and Management Programme (NAQUIM 2.0)** is now generating high-resolution, panchayat-level groundwater data, allowing site-specific recharge and management plans instead of generic state-wide rules.

2. **Regulate and Control Groundwater Extraction:** Since over-pumping is the single biggest trigger for intrusion, controlling how much water is drawn out — and from where — is needed.

   CGWB's own reviews of **Saurashtra and Kutch in Gujarat** show that excessive groundwater withdrawal for agriculture and industry has driven acute seawater ingress in the region, which is why state groundwater regulation acts now increasingly require permits for new borewells in identified saline-risk zones.

3. **Build Artificial Recharge Structures to Create a Freshwater Barrier:** Adding more fresh water back into the aquifer helps push back against advancing seawater rather than just reducing extraction.

Percolation ponds, check dams, recharge shafts and injection wells are being used along the Gujarat and Tamil Nadu coasts to build up a freshwater hydraulic barrier, physically resisting the inland movement of saline water.

4. **Deploy Subsurface Barriers in Critical Stretches:** In severely affected pockets, physical underground structures can directly block the path of intruding seawater.

Underground cut-off walls and subsurface dykes, used experimentally along parts of India's western coast, work like an invisible underground wall — stopping saline water from moving further inland even where recharge alone isn't enough.

5. **Promote Community-Led Groundwater Management:** Since coastal groundwater is used daily by thousands of small farmers and households, sustainable management works best when the community itself takes ownership.

Atal Bhujal Yojana empowers gram panchayats to prepare Water Security Plans and carry out participatory water budgeting, extending the same demand-side model that worked for drought management to coastal groundwater protection.

6. **Push for Institutional Reform in Water Governance:** India's split management of surface water and groundwater under two different bodies weakens coordinated action against a problem that involves both.

The **Mihir Shah Committee (2016)** recommended merging the **Central Water Commission (CWC)** and **CGWB** into a single **National Water Commission**, which would allow integrated, basin-level decisions — though this reform, as of January 2026, is still pending implementation.

7. **Protect and Restore Natural Coastal Buffers:** Mangroves, wetlands and coastal vegetation naturally slow down both storm surges and salinity ingress, but their steady loss has removed this buffer in many stretches.
   - Coastal afforestation and wetland restoration programmes are being promoted alongside engineering solutions, since natural buffers reduce the pressure on artificial recharge and barrier structures to do all the work alone.
8. **Maintain Adequate River Freshwater Flow:** Global research shows that reduced river discharge into estuaries — from upstream dams and diversions — allows salinity fronts to shift further inland, independent of groundwater pumping.

INDIA'S OVERALL COASTAL DISASTER MANAGEMENT FRAMEWORK

1. **Institutional and Legal Backbone:** The **Disaster Management Act, 2005** created a clear chain of responsibility — from the **National Disaster Management Authority (NDMA)** at the top, down to **State Disaster Management Authorities (SDMAs)** and **District Disaster Management Authorities (DDMAs)** — giving coastal disaster response a legal structure rather than ad-hoc coordination.
   - The **National Disaster Management Plan (NDMP)** builds on this by assigning IMD and the Central Water Commission joint responsibility for flood and cyclone forecasting, ensuring different central agencies work under one coordinated plan.
2. **Structural Mitigation Through NCRMP:** Since cyclones remain India's most frequent coastal hazard, the **National Cyclone Risk Mitigation Project (NCRMP)** — a World Bank-assisted, NDMA-implemented scheme — has built physical protection infrastructure across the most vulnerable coastal states.

3. **Early Warning Systems:** Getting warnings to people fast enough is treated as being just as important as predicting the hazard itself.
   - NCRMP's **Component A — Early Warning Dissemination System (EWDS)** — was specifically designed to strengthen last-mile warning delivery, while NDMA's separate **Guidelines on Management of Tsunamis**, developed after extensive expert consultations following the 2004 tsunami, laid down a dedicated national framework for tsunami risk that didn't exist before.
4. **Regulation of Coastal Land Use:** Uncontrolled construction along the coast increases exposure to every hazard from cyclones to erosion, which is why land-use regulation is treated as a core disaster-reduction tool, not just an environmental one.
   - The **Coastal Regulation Zone (CRZ)** notification governs what can and cannot be built close to the shoreline, and is periodically updated by the **Ministry of Environment, Forest and Climate Change** to balance coastal development with disaster risk reduction.
5. **Integrated Coastal Zone Management (ICZM):** Beyond just regulating construction, India has pushed for a broader, ecosystem-based approach to managing the entire coastal zone as one system.
   - Under the **ICZM project**, coastal states prepare integrated management plans covering mangrove and wetland protection, fisheries, and hazard mitigation together — recognising that healthy natural ecosystems like mangroves reduce storm-surge impact more effectively than concrete barriers alone.
6. **Community-Based Preparedness and Shelters:** Since even the best early warning is only useful if people know where to go, physical safety infrastructure at the community level remains central to the framework.
   - NCRMP's **Component B (Cyclone Risk Mitigation Infrastructure)** funded the construction of **multi-purpose cyclone shelters and cattle mounds** across vulnerable coastal districts, giving communities a designated safe location during a cyclone rather than relying on last-minute evacuation alone.
7. **Capacity Building and Institutional Training:** Effective disaster response also depends on training the people and institutions who implement it on the ground.
   - NCRMP's **Component C** provides technical assistance for capacity building in disaster risk management, working alongside the **National Institute of Disaster Management (NIDM)**, which trains state officials and local functionaries in coastal hazard response.

15. FOREST FIRES IN INDIA

A **forest fire** is an uncontrolled fire that spreads through forest vegetation, destroying tree cover, wildlife habitats and, increasingly, nearby human settlements.

In India, forest fires are tracked by the **Forest Survey of India (FSI)** using MODIS and SNPP-VIIRS satellite sensors, and reported biennially through the **India State of Forest Report (ISFR)**.

According to the **Forest Survey of India (FSI)**, close to **36% of India's forests are prone to frequent fires**, and the country lost about **1.65 million hectares of forest to fires between 2001 and 2023** (Global Forest Watch data)

Causes Of Forest Fires In India

1. **Overwhelming Human Origin of Fires:** Unlike the popular image of forest fires being mostly natural, in India the vast majority are actually started by people, whether deliberately or by accident.

As per the **Forest Fires in India** report, **approximately 95% of forest fires in the country are attributed to human activities**, with only about 5% occurring due to natural causes like lightning strikes

2. **Traditional Agricultural Practices — Shifting Cultivation:** In parts of Northeast India, the traditional practice of clearing forest patches by burning for farming remains a significant fire trigger.

Jhum (shifting) cultivation, along with deforestation and firewood-related burning, is repeatedly cited as a major human-driven cause of forest fires, particularly in the hill states of the Northeast.

3. **Deliberate Burning for Forest Produce Collection:** Local communities sometimes set small fires intentionally, believing it helps regeneration or eases the collection of forest products, but these often spread out of control.

Villagers frequently use fire to clear undergrowth for easier collection of **mahua flowers, tendu leaves and other minor forest produce**, or to encourage fresh grass growth for grazing — a practice that, without proper control, is a recurring accidental-fire trigger across central Indian forests.

4. **Unregulated Grazing on Forest and Common Land:** Large-scale, unmanaged grazing dries out and thins forest floor vegetation, making it easier for any spark to catch and spread rapidly.
   - With a large share of India's livestock relying on common and forest land for grazing without proper regulation, the resulting dry undergrowth adds significantly to fuel load, intensifying fire spread once ignited.
5. **Rising Temperatures and Erratic Rainfall (Climate Change):** Hotter, drier weather conditions directly increase how easily forests catch fire and how intensely they burn once alight.

A systematic review published in **Environmental Sciences Europe (2025)** found that India is witnessing a rise in forest fire frequency and intensity due to climate factors such as **increasing temperatures, erratic rainfall, and prolonged droughts**, which are altering forest fuel dryness and fire-prone seasons.

6. **Sharp Rise in Fire Detections Despite Forest Cover Growth:** Even as India's total forest area has grown slightly, the number of fires detected has increased far more sharply — pointing to worsening fire-conducive conditions rather than more forest simply existing.

As per the **FSI State of Forest 2021 Report**, while total forest cover increased by 0.48% between 2013 and 2021, the number of forest fires detected in the same period rose by a striking **186%**.

7. **Accumulation of Dry Litter and Natural Fuel Load:** Certain forest types naturally accumulate highly flammable ground litter, making them especially prone to fast-spreading fires once triggered by any of the above human causes.

Pine forests in particular, (common across parts of the Himalayan foothills) - shed dry needles that form a thick, easily combustible layer on the forest floor — meaning even a small human-lit spark in these forests can escalate into a large, fast-moving fire.

Case Studies on Forest Fire Prevention

Case Study	Key Learning
Van Panchayats, Uttarakhand (India)	Uttarakhand's **Van Panchayats** empower local communities to manage forests. Villagers regularly remove dry biomass, maintain water sources and monitor forests before summer, demonstrating that **community ownership is one of the most effective tools for preventing forest fires**.

Tech–Community Hybrid Model, Uttarakhand (India)	Uttarakhand integrated the **"Forest Fire Uttarakhand" app**, WhatsApp groups and Gram Panchayat Fire Safety Committees with community vigilance. The model shows how **technology combined with local participation enables faster detection and early response**.
Arnhem Land Fire Abatement Project, Australia	Indigenous communities carry out **controlled early dry-season burns** to reduce fuel loads before peak summer. The project has significantly reduced large wildfires while generating **carbon credits**, making fire prevention both ecologically and economically sustainable.
Kimberley Region, Australia	Traditional Indigenous fire management creates a **mosaic of burnt and unburnt patches**, preventing large-scale wildfires. Scientific studies have validated this approach as an effective method of **fuel management and biodiversity conservation**.
Xingu Indigenous Territory, Brazil	Indigenous fire brigades combine **traditional fire knowledge with satellite monitoring** to detect and control fires before they spread. The model highlights the importance of **integrating indigenous practices with modern technology** for forest fire prevention.

Way Forward

India's forest fire management already rests on a national framework — the **National Action Plan on Forest Fires (NAPFF, 2018)** and the **Forest Fire Prevention and Management (FPM) Scheme** — but with fire incidents spiking sharply in recent years, the way forward lies in strengthening implementation, community trust, and technology together, rather than introducing entirely new policies.

1. **Strengthen the Satellite-Based Alert System Further:** Early detection remains the single most effective way to stop a small fire from becoming a large one.

   The **Forest Survey of India (FSI)**, under the Ministry of Environment, Forest and Climate Change, already runs a **satellite-based Forest Fire Monitoring and Alert System**, sending near-real-time alerts via SMS and email to registered field staff, and the Ministry maintains a **24x7 Disaster Management Control Room** to track fire incidents nationally .

2. **Deepen Community Participation Through Existing Institutions:** Since a large share of India's fires are human-caused, the same communities can be turned into the first line of defence rather than being seen as a source of risk.

   Joint Forest Management Committees (JFMCs), Eco-Development Committees (EDCs), and Van Panchayats already form the backbone of ground-level participation (MoEFCC), but the **National Action Plan on Forest Fires** itself is designed to strengthen this by informing, enabling and incentivising forest fringe communities to work directly with State Forest Departments.

3. **Complete Pre-Fire Season Preparatory Work Consistently:** Much of fire risk can be reduced simply by preparing the forest before the fire season even begins, rather than reacting once a blaze has started.
   - Standard measures already in use — such as **creating and maintaining fire lines, removing excess dry grass around sensitive zones, and pre-season maintenance of firefighting equipment** — need to be carried out uniformly across all fire-prone states, not just in the better-resourced ones .
4. **Deploy Fire Watchers and Mobile Patrol Units:** Human vigilance on the ground remains necessary even with satellite technology, since satellites detect fires only after they've started.

States like Maharashtra already engage **dedicated fire watchers and mobile squad units** for regular patrolling during the fire season — a model that needs consistent central funding support .

5. **Expand the Use of Drones and Modern Firefighting Technology:** Technology can reach terrain and detect fires in places human patrols cannot easily access.
6. **Address the Funding Gap in the National Action Plan:** A good plan needs dedicated money behind it, not just the promise of coordination.

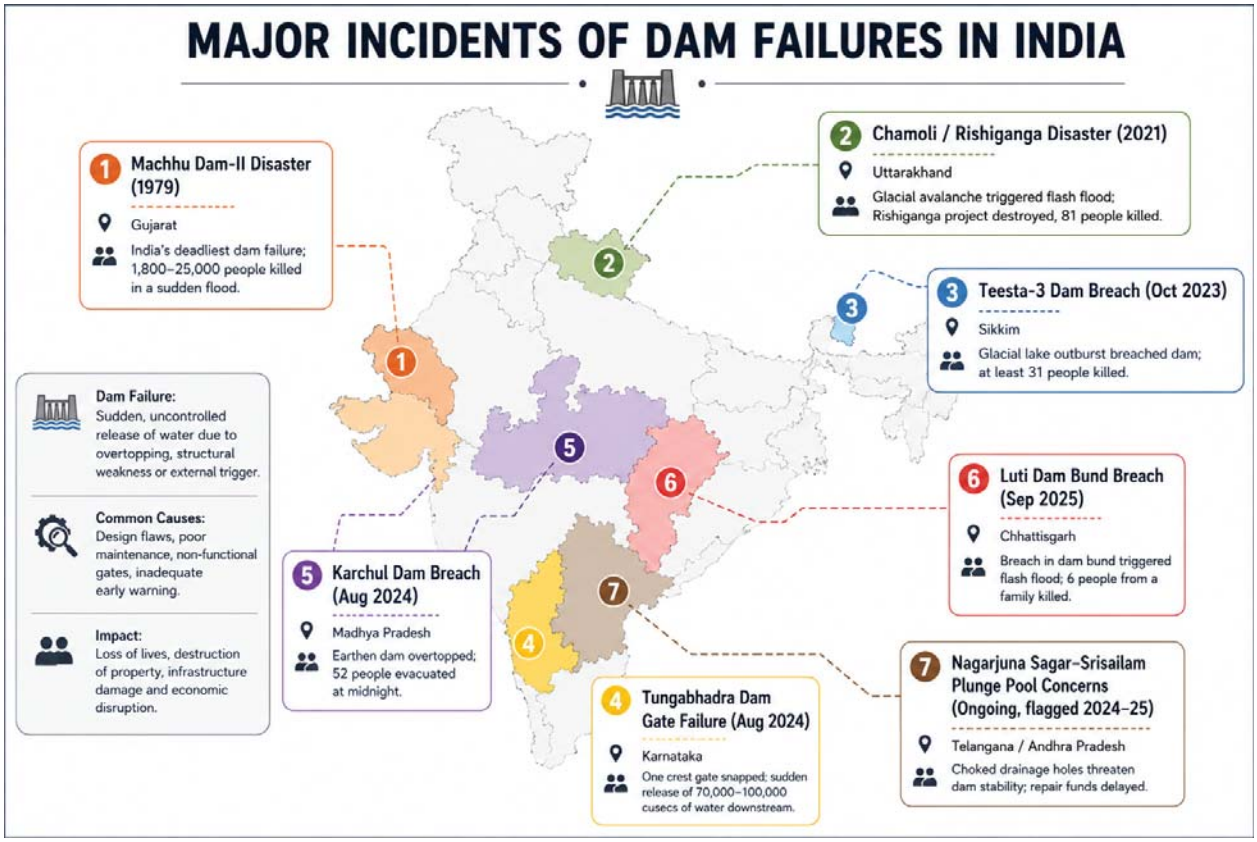
Researches point out that the NAPFF still awaits **dedicated funding of its own**, relying instead on allocations from the **Compensatory Afforestation Fund Management and Planning Authority (CAMPA)** — securing an independent budget line for the plan would give states more predictable, long-term support .

7. **Encourage Sustainable Biomass Management in Fire-Prone Forests:** Certain forest types accumulate highly flammable litter every year, and turning that litter into a resource rather than just fuel can reduce risk while creating local income.
   - Encouraging the **collection of dry biomass like pine needles for productive use** reduces the forest floor's "tinderbox effect," directly cutting fuel load in high-risk zones like the Himalayan foothills.

16. DAM SAFETY AND DAM FAILURES IN INDIA

Dam safety refers to the systems, laws and institutions that ensure a dam is properly designed, monitored, maintained, and does not fail under stress from floods, earthquakes, or structural ageing.

India, with over 6,600 large dams (National Register of Specified Dams, 2025), ranks **third in the world after China and the USA** in the number of large dams — and with a large share of them now decades old, dam safety has become a pressing governance issue rather than a purely technical one.



Factors Behind India's Vulnerability to Dam Failures

1. **A Rapidly Ageing Dam Stock:** Many of India's dams were built decades ago under engineering standards suited to a different climate, and they are now reaching or crossing their intended structural lifespan.

   A **2025 peer-reviewed study** found that **over half of India's major dams have now crossed 50 years of age**, escalating the risk of structural deterioration and reduced functionality — since many of these dams were never designed to handle today's more extreme and erratic rainfall patterns.

2. **Reservoir Sedimentation Reducing Structural Margins:** Over time, silt accumulation inside a reservoir doesn't just reduce storage capacity — it also increases pressure on the dam structure itself..

3. **Rapidly Growing and Unstable Glacial Lakes:** In the Himalayan belt, dams face a unique climate-driven risk — sudden flood surges from glacial lakes that are expanding faster than they used to.

   A **2024 Central Water Commission report found a 33.7% increase in the surface area of glacial lakes in India between 2011 and 2024**, with **67 glacial lakes showing more than a 40% increase**, placing them in the high-risk category for a Glacial Lake Outburst Flood (GLOF).

4. **High Seismic Vulnerability in Dam-Building Regions:** Many of India's dams, especially in the Himalayas, sit in tectonically active zones where earthquakes can destabilise both the dam structure and nearby glacial moraine dams.

   States like **Uttarakhand & Himachal Pradesh fall in Seismic Zone IV and V**, meaning even a moderate earthquake can weaken loose moraine or dam structures, a risk compounded further when combined with unregulated hydropower construction nearby

5. **Institutional Delay and Weak Enforcement:** Even where problems are identified, India's record shows that reports and repair recommendations often sit unimplemented for years due to bureaucratic and inter-agency delays.

6. **Climate Change Intensifying Extreme Rainfall Events:** Dams designed decades ago for a certain range of flood flows are now facing rainfall intensities well beyond those original assumptions.

   More frequent and severe weather events are stressing dams "not originally designed for such variability" .This is A pattern already visible in incidents like the 2024 Tungabhadra gate failure and the 2025 Punjab floods, where dam operations struggled to keep pace with sudden, intense inflows.

7. **Emerging Cyber-Physical Vulnerabilities:** As dam operations become more automated and digitally connected, they are also opening up to a newer category of risk that Indian dam safety frameworks are yet to fully address.

   The **World Economic Forum's Global Risks Report 2026** flags that ageing critical infrastructure is increasingly vulnerable to cyberattacks, citing an April 2025 cyber-physical attack on the Bremanger dam in Norway that caused an unplanned water release — a warning sign for India as its own dam monitoring systems become more digitised .

Global Case Studies on Dam Safety & Failure Management

Case Study	Key Learning
National Dam Safety Program (USA)	Following the **1976 Teton Dam failure**, the USA launched a nationwide dam inspection programme, which later evolved into the **National Dam Safety Program (NDSP)** under FEMA. It demonstrates the importance of **regular safety inspections, emergency action plans and dedicated institutional oversight**.

Post-Banqiao Dam Reforms (China)	After the catastrophic **Banqiao Dam failure (1975)**, China undertook a nationwide programme to **rehabilitate ageing dams, upgrade design standards and strengthen dam safety regulations**. The reforms significantly improved the resilience of dams against extreme floods.
Post-Brumadinho Dam Reform (Brazil)	The **2019 Brumadinho tailings dam disaster** led Brazil to launch a time-bound programme to **eliminate unsafe upstream tailings dams** and strengthen regulatory oversight. The case highlights the importance of **phasing out high-risk infrastructure and ensuring accountability through rehabilitation and compensation**.

INDIA'S CURRENT FRAMEWORK FOR DAM SAFETY AND REHABILITATION

1. **The Dam Safety Act, 2021:** Before this law, dam safety in India rested on non-binding guidelines, meaning there was no real legal consequence for ignoring safety recommendations.
   - The Act, effective from **30 December 2021**, applies to all **"specified dams"** — those above 15 metres in height, or 10-15 metres with specific technical hazards — and makes dam owners **legally responsible** for safe construction, operation and maintenance, with penalties including **imprisonment of up to two years** in case of loss of life due to negligence.
2. **A Four-Tier Institutional Structure:** The Act ptovides for an entire chain of accountability from the Centre down to individual dam sites.
   - At the national level, the **National Committee on Dam Safety (NCDS)** sets policy, while the **National Dam Safety Authority (NDSA)** implements it and resolves inter-state disputes; at the state level, the **State Committee on Dam Safety** and **State Dam Safety Organisation (SDSO)** carry out actual surveillance, inspection and monitoring of each dam.
3. **Regular Risk Categorisation of Dams:** Rather than treating all dams the same, the framework actively grades dams by the severity of deficiencies found during inspection, allowing resources to be targeted where the risk is highest.
   - Post-monsoon inspections in 2025 found **3 dams categorised under "Category I" (critical deficiencies needing immediate action)** and **188 dams under "Category II" (major deficiencies needing prompt remedial action)**, giving authorities a clear, prioritised action list rather than a vague sense of overall risk.
4. **Dam Rehabilitation and Improvement Project (DRIP) — Physically Fixing Ageing Dams:** Since a legal framework alone can't repair a crumbling structure, India runs a dedicated, externally-funded rehabilitation programme alongside the Act.
   - **DRIP Phase-I (2012-2021)** rehabilitated 223 dams across 7 states at a cost of ₹2,567 crore.
   - **DRIP Phase II & III**, jointly funded by the **World Bank and Asian Infrastructure Investment Bank**, cover **736 dams across 19 states and 3 central agencies** (Central Water Commission, Bhakra Beas Management Board, Damodar Valley Corporation) — of which **43 dams have already completed rehabilitation**.
5. **Technology-Driven Monitoring:** Beyond physical repairs, India is increasingly using digital tools to track dam health in real time rather than relying only on periodic manual inspections.
   - The **Dam Health and Rehabilitation Monitoring Application (DHARMA)** is being used to digitise dam inspection data and track rehabilitation progress across states, allowing central authorities to monitor thousands of dams without waiting for state-level paper reports.
6. **Academic and Research Support:** Recognising that good dam safety needs skilled engineers as much as good laws, the government has built dedicated research capacity into the system.

- Centres of Excellence have been established at **IIT Roorkee** (focusing on seismic hazard mapping and reservoir sedimentation) and **IISc Bangalore** (focusing on advanced construction materials and risk assessment), alongside dedicated M.Tech programmes to build India's specialised hydraulic engineering workforce.

7. **Persistent Implementation Gaps Despite the Framework:** Even with strong laws on paper, the framework's effectiveness is undercut when known problems aren't fixed fast enough.
   - Independent reviews note that major safety concerns at dams like **Kaleshwaram, Polavaram and Hirakud** have remained unresolved for years despite being flagged, showing that the framework's biggest current weakness is not a lack of rules, but **slow inter-agency follow-through** on the rules that already exist.

Way Forward

Dam-related disasters include both structural failures and disasters caused by poor dam operation (sudden water release triggering downstream floods, as seen in Punjab in 2025). Since India's dam safety framework already exists on paper through the **Dam Safety Act, 2021**, the way forward now lies in strengthening implementation, forecasting, and downstream coordination together.

1. **Strengthen Mandatory Emergency Action Plans (EAPs):** A dam can be structurally sound and still cause a disaster if there's no clear plan for what to do when something goes wrong.

   Under the **Dam Safety Act, 2021**, dam owners are legally required to prepare **Emergency Action Plans, risk assessments, and expert safety evaluations**, but enforcement needs to be made uniform — many disaster reviews show dams that lacked functioning EAPs at the time of failure, showing the gap between the legal requirement and ground reality .

2. **Expand and Modernise Flood Forecasting Coverage:** Since dam-triggered floods move fast, giving downstream communities more advance notice is one of the most direct ways to save lives.

   The **Central Water Commission (CWC)**, India's nodal flood forecasting agency, now operates **350 forecasting stations nationwide** providing **7-day advance advisories** alongside short-range 24-hour alerts.

3. **Deploy Next-Generation, High-Resolution Forecasting Tools:** Traditional forecasting models often can't capture fast-changing, hyper-local flood dynamics around a dam breach or sudden release.

   The **C-FLOOD system**, jointly developed by CWC, C-DAC and NRSC and launched in July 2025, uses high-performance computing to achieve **15-20 cm accuracy in water depth estimation**, offering a **48-hour flood prediction window with just a 2-hour simulation turnaround** — a major leap from older, slower forecasting models.

4. **Address Glacial Lake Risk to Himalayan Dams Directly:** Since several recent dam disasters (like Teesta-3 in 2023) were triggered by glacial lake outburst floods, dam safety and glacier monitoring need to be planned together, not separately.

   CWC has completed **flow-path mapping of 100 glacial lakes** in India, finished risk assessment of these 100 lakes (with 581 more in progress), and published a dedicated **Guideline for Structural Measures to Mitigate GLOF Impact on Dams/Barrages in August 2025**

5. **Make Dam Operations More Transparent and Accountable:** Several recent floods, including the 2025 Punjab disaster, were worsened not by structural failure but by poorly coordinated water release from dams during heavy rainfall.

6. **Adoption of Simplified Risk-Screening and Dam-Break Assessment Tools:** With thousands of dams to monitor, India needs practical tools to quickly identify which dams need urgent attention, rather than treating every dam with equal priority.

- The **2026 International Conference on Dam Safety** recommended simplified risk screening and dam-break assessment tools to help authorities prioritise safety measures across India's large dam portfolio, rather than spreading limited resources thinly across all dams equally.

7. **Strengthen Basin-Level, Forecast-Informed Reservoir Operations:** Since a single river basin often has multiple dams operated by different agencies, coordinated operation across the whole basin — not just individual dams — is key to preventing cascading floods.
   - Experts emphasise on **forecast-informed and basin-level coordinated reservoir operations**, using dynamic reservoir rules that adjust water release based on incoming rainfall forecasts, rather than each dam operator acting independently based only on their own reservoir levels.

17. AC FIRE HAZARDS

An **AC fire hazard** refers to the risk of fire or explosion caused by air conditioners, usually triggered by electrical overload, poor installation, or lack of maintenance rather than the appliance's core cooling technology itself.

With India's air conditioner usage rising sharply and summers growing longer and hotter, this has emerged as a serious **urban electrical safety concern**, not just a technical appliance issue.

Reasons Behind India's Vulnerability to AC Fire Hazards

1. **Legacy Electrical Infrastructure Was Not Designed for Today's Loads:** Much of India's residential wiring was installed decades ago for basic lighting and fans, not for today's high-power appliances.

   Buildings wired in the **1980s and earlier** are now expected to run **1.5-tonne inverter ACs, geysers and induction cooktops** on the same circuits, and this fundamental mismatch between old wiring capacity and new electrical demand is a root structural vulnerability that simply adding more ACs makes worse each year.

2. **Fire Services as a State Subject, Leading to Uneven Enforcement:** Because fire safety isn't centrally enforced, the same electrical fault can be treated very differently depending on which city or state it occurs in.

   Under India's constitutional scheme, **"Fire Services" fall under the State List (Seventh Schedule)**, meaning the Centre can only issue advisory guidelines and Model Bills, while actual enforcement depends entirely on individual state governments and municipal bodies — creating wide inconsistency in how strictly electrical safety norms are applied across the country.

3. **Building Codes Aren't Uniformly Binding:** India already has a detailed national safety standard, but it doesn't automatically apply everywhere.

   The **National Building Code (NBC) 2016**, published by the **Bureau of Indian Standards (BIS)**, lays out detailed requirements for electrical installations, including flame-retardant wiring, but it only becomes legally enforceable once a **state government formally adopts it into local law** — and many states are still working with outdated versions of their own local building bye-laws.

4. **Severe Shortage of Fire Infrastructure and Trained Personnel:** Even where hazards are identified, India's capacity to inspect, prevent and respond to fires remains critically thin.

   Data from the **Directorate General of Fire Services, Civil Defence and Home Guards** shows India faces a **97.6% shortage of fire stations, a 96.3% shortage of firemen, and an 80% shortage of fire tenders and rescue vehicles** nationally — meaning the country is under-resourced not just for large fires, but for the far more common, smaller electrical fires that ACs increasingly trigger.

5. **Rapid AC Growth Outpacing Safety Regulation:** India's cooling appliance market is expanding at a pace that regulatory and inspection systems haven't caught up with.

The **International Energy Agency** projects India's installed AC base will grow from **93 million units in 2024 to around 240 million by 2030**, but this growth is happening without any matching mandatory requirement for periodic electrical safety inspection of homes adding these high-load appliances.

6. **Absence of Mandatory Periodic Electrical Inspection:** Unlike some countries, India doesn't require homes to be electrically re-certified as appliance loads grow over time.
   - India currently has **no mandatory periodic inspection regime for domestic electrical installations**, unlike models used in some East Asian countries, meaning ageing wiring behind walls can silently degrade for years, undetected, until an AC's high start-up current finally exposes the weak point.
7. **Weak Data and Under-Reporting Masking the True Scale:** Without accurate classification of what actually causes a fire, it's difficult for policymakers to target the right fix.

The **National Crime Records Bureau (NCRB)**'s Accidental Deaths and Suicides in India (ADSI) report recorded **7,435 fire accident deaths in 2022** out of over 7,500 total fire accidents, but electrical fires — including AC-linked incidents — are often recorded under broad, generic categories rather than being separately tracked, understating how large the AC-driven share of this problem actually is.

Way Forward

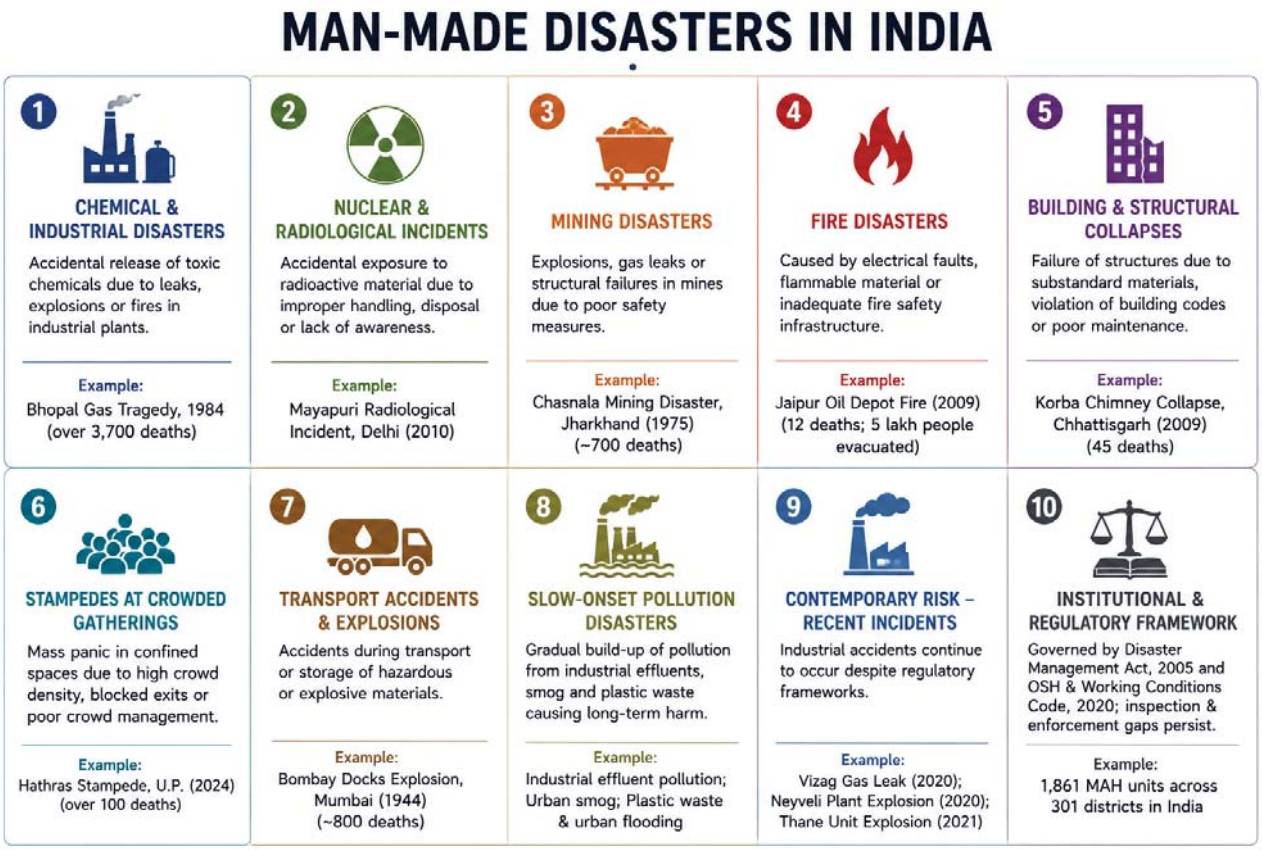
Measure	Key Features
Mandate RCCBs for AC Circuits	• Make **Residual Current Circuit Breakers (RCCBs)** mandatory for AC installations. • RCCBs detect **leakage currents** and disconnect power before a fire can occur, unlike MCBs which only protect against overloads and short circuits.
Dedicated Electrical Circuits	• Ensure every AC unit has a **separate, correctly rated electrical circuit**. • Prevents circuit overloading and overheating caused by sharing high-load appliances on the same wiring.
Strengthen BIS Enforcement	• Enforce strict use of **ISI-certified wires, switches and circuit breakers**. • Crack down on counterfeit electrical components that significantly increase fire risk.
Periodic Electrical Safety Audits	• Mandate **regular inspection of household electrical systems**, especially after installing ACs or other high-load appliances. • Helps identify ageing wiring, loose connections and overloaded circuits before accidents occur.
Uniform Adoption of National Building Code (NBC)	• Ensure all States implement the latest **National Building Code** provisions on electrical and fire safety. • Promotes uniform safety standards in residential and commercial buildings.
Smart Fire Detection Systems	• Promote **IoT-based energy monitors, thermal sensors and smart alarms**. • Enables early detection of overheating, electrical faults and potential fire hazards.

Public Awareness & Preventive Maintenance	• Encourage **annual AC servicing**, proper installation and use of certified electrical components. • Awareness campaigns through RWAs and local bodies can significantly reduce preventable fire incidents.
Strengthen Fire Services	• Improve the availability of **fire stations, trained personnel and modern firefighting equipment**. • Faster emergency response reduces casualties and property damage during electrical fires.

18. MAN MADE DISASTERS

A **man-made disaster** is any catastrophic event triggered directly or indirectly by human activity rather than natural forces.

As per **NDMA**, these are classified under **technological and human-induced hazards**, covering industrial accidents, fires, building collapses, stampedes, oil spills, and nuclear or radiological incidents.



Chemical & Industrial Disasters In India

A **chemical or industrial disaster** is a sudden, unplanned event — a gas leak, explosion, or fire — arising from the manufacture, storage, handling or transport of hazardous chemicals, causing serious harm to human life, property or the environment.

These disasters often emerge from mishandling of Industrial processes .Bhopal Gas tragedy (1984) was one of the most significant and criticised Industrial disasters in India's History.

India's Vulnerability To Chemical & Industrial Disasters

1. **Residential Areas Growing Around Old Industrial Clusters:** Many hazardous plants were originally built on the outskirts of cities decades ago, but unplanned urban growth has since brought dense residential neighbourhoods right up to their boundaries.

   The **National Disaster Management Authority (NDMA)** itself notes that enforcement remains especially uneven in **older industrial clusters established before modern zoning norms**, where dense residential areas have since grown up around the plants — the exact pattern that turned the Bhopal leak into a mass-casualty event rather than a contained industrial accident.

2. **Sheer Scale of Hazardous Units Versus Regulatory Capacity:** With thousands of hazardous units to monitor, India's inspection machinery is stretched thin relative to the scale of what it needs to oversee.

   NDMA data shows India has **1,861 Major Accident Hazard (MAH) units spread across 301 districts** in 25 states and 3 Union Territories, alongside thousands of additional registered factories and a large unorganised sector — a scale that makes uniform, regular safety audits genuinely difficult to sustain nationwide.

   Major Accident Hazard (MAH) units are industrial facilities or isolated storage sites that handle, manufacture, or store hazardous chemicals equal to or in excess of specified threshold quantities

3. **Weaker Safety Compliance in Small and Medium Enterprises:** Large multinational plants tend to have dedicated safety systems, but a big share of India's chemical-handling capacity sits in smaller units with far less rigorous oversight.

   The Micro, Small, and Medium Enterprises (MSME / SME) sector contributes **31.1% to India's total Gross Domestic Product (GDP)** - making a large part of the Industrial Production chain susceptible to these hazards.

4. **Ageing Equipment and Deferred Maintenance:** Plants and pipelines built decades ago are still operating well past the point where routine wear and tear becomes a genuine safety risk.
   - Corrosion of **decades-old chemical conduits and pressure vessels** remains a recurring cause behind pipe bursts and reactor failures at older industrial estates, showing that infrastructure age, not just regulation, is a direct driver of risk.

5. **Fragmented Regulatory Architecture Across Multiple Ministries:** Chemical safety in India is currently governed by several different laws and ministries working somewhat independently, rather than one unified regulator.

   Current oversight is split across the **Ministry of Environment, Forest and Climate Change, Ministry of Labour and Employment, Ministry of Petroleum and Natural Gas**, and others, each administering separate rules (EPA 1986, MSIHC 1989, Factories Act 1948) — a structure that experts argue would benefit from a single, independent regulator, similar to the **Atomic Energy Regulatory Board (AERB)** model used for nuclear safety.

6. **Limited Real-Time Monitoring and Digital Oversight:** Much of India's chemical safety compliance still relies on periodic manual inspections rather than continuous, technology-based monitoring.

   Unlike more automated international models, most Indian hazardous units still lack **IoT-based sensors or digital monitoring systems** that could flag pressure build-up, leaks or temperature anomalies in real time — meaning many failures are detected only after they've already begun, rather than through early warning.

India' Current Legal & Regulatory Framework To Deal With Chemical & Industrial Disasters

Legal / Administrative Provision	Key Features
Environment (Protection) Act, 1986	• Umbrella law enacted after the **Bhopal Gas Tragedy (1984)**. • Empowers the Central Government to regulate hazardous substances. • Prescribes environmental standards, inspections and preventive measures. • Parent legislation for chemical safety rules such as **MSIHC Rules, 1989**.
MSIHC Rules, 1989 (Manufacture, Storage & Import of Hazardous Chemicals Rules)	• Core regulation for industries handling hazardous chemicals. • Requires **Safety Reports** and **Safety Data Sheets (SDS)**. • Mandates **On-site Emergency Plan** (to be prepared by the industry) and **Off-site Emergency Plan** (by District Authority). • Major chemical accidents must be reported to the Government within the prescribed timeframe.
Chemical Accidents (EPPR) Rules, 1996	• Establishes the emergency response mechanism. • Creates **Central, State and District Crisis Groups**. • Ensures coordinated response among Environment, Home, Health, Labour departments and local administration during chemical accidents.
Public Liability Insurance Act, 1991 (PLIA)	• Provides **immediate no-fault compensation** to victims of hazardous accidents. • Makes insurance compulsory for industries handling hazardous chemicals. • Ensures quick relief without waiting for lengthy court proceedings.
Disaster Management Act, 2005 & NDMA	• Brings chemical disasters under the national disaster management framework. • **NDMA** formulates policies and guidelines. • Issued **Guidelines on Chemical Disaster Management (2007)** and the **National Action Plan on Chemical Industrial Disaster Management (NAP-CIDM)**.
Occupational Safety, Health & Working Conditions (OSH) Code, 2020 (Implemented in Nov. 2025)	• Consolidates **13 labour laws**, including the **Factories Act, 1948**. • Introduces competency-based certification for hazardous workers. • Mandates creation of Safety Committees. • Promotes digital and self-reporting based compliance for industries.
National Green Tribunal (NGT) Act, 2010	• Provides speedy environmental justice. • Can award compensation and order environmental restoration. • Holds polluting industries accountable through specialised adjudication.

Best Practices

Case Study	Explanation
Seveso Directive, European Union (1976–Present)	• Following the **1976 Seveso (Italy) chemical accident**, the EU introduced the **Seveso Directive** (now **Seveso III**). It covers **11,000+ hazardous industries**, mandating a **Major Accident Prevention Policy (MAPP)**, **Safety Management System (SMS)**, periodic risk assessments, emergency planning and public information. • The framework has been continuously strengthened by incorporating lessons from disasters such as **Bhopal** and later global industrial accidents.
GIS-Based Off-Site Emergency Planning, India	• Developed by the **Ministry of Environment & Forests** with **NIC**, this GIS-based system maps **Major Accident Hazard (MAH) industries** along with nearby **fire stations, hospitals, police stations, schools and other vulnerable locations**. • Implemented across **14 states**, it enables faster emergency response and realistic district-level off-site emergency planning.
APELL Project, India *(Awareness and Preparedness for Emergencies at Local Level)*	• India adopted the **UNEP's APELL framework** to improve preparedness around hazardous industries. • It promotes **community awareness, stakeholder coordination, emergency drills and local participation**, ensuring that residents living near industrial units know how to respond during a chemical emergency rather than relying solely on official responders.
Gujarat Chemical Industrial Disaster Risk Management (CIDRM) Programme	• Implemented by the **Gujarat Institute of Disaster Management (GIDM)**, the programme strengthens preparedness in India's largest chemical manufacturing state through **capacity building, simulation exercises, mock drills and specialised training** for officials and industries. • It addresses Gujarat's combined risks of **chemical accidents, earthquakes and cyclones**, making it a leading state-level institutional model.

These case studies can be used in various other man made disasters as well.

Way Forward

Measure Needed	Explanation
Comprehensive Risk Mapping and Zoning	• Geo-tag all **Major Accident Hazard (MAH)** units and define a scientifically determined **Zone of Impact (ZOI)**. • Prohibit residential settlements and public infrastructure within mandatory **buffer zones**. • **Case Study:** The **2020 Vizag (LG Polymers) styrene gas leak** occurred because a residential colony had expanded up to the plant boundary with **no buffer zone**, resulting in **12 deaths** and **585 hospitalisations**.

Modernising a Fragmented Legal Framework	• Improve convergence among the **Factories Act, 1948**, **Environment (Protection) Act, 1986**, and **MSIHC Rules, 1989**. • Eliminate overlapping jurisdictions and strengthen enforcement. • **Case Study:** The **High-Power Committee on the LG Polymers accident** identified regulatory overlap between the **Factories Department** and the **Pollution Control Board** as a major reason safety violations remained unchecked.
Strengthening Inspection and Enforcement Capacity	• Increase staffing and technical capacity of **DISH (Directorate of Industrial Safety and Health)**. • Ensure regular inspections of hazardous industries. • **Case Study:** The **High-Power Committee** found that the **Andhra Pradesh Pollution Control Board** permitted LG Polymers to expand without valid environmental clearance, leading to suspension of **two APPCB officials** and the **Deputy Chief Inspector of Factories**.
Mandatory On-Site and Off-Site Emergency Plans	• Every MAH unit should maintain an **On-Site Emergency Plan**, while district authorities prepare an **Off-Site Emergency Plan**. • Conduct joint mock drills at least **twice annually**. • Mandated under **NDMA Guidelines on Chemical (Industrial) Disasters** and the **Chemical Accidents (Emergency Planning, Preparedness and Response) Rules, 1996**, which also establish **Crisis Groups** at the Central, State and District levels.
Real-Time Hazard Monitoring and Early Warning	• Install gas leak sensors, automated shutdown systems, and functional alarm/siren networks. • Ensure continuous monitoring and rapid public warning mechanisms.
Fixing Ageing Industrial Infrastructure	• Periodically audit and replace ageing reactors, boilers and pipelines. • Introduce mandatory integrity assessments before restarting idle plants. • **Case Study:** The **2024 Dombivli (Thane) blast** at **Amudan Chemicals** involved an ageing boiler that exploded after the unit resumed operations, killing **10 people**.
Strict Financial Liability and Insurance Coverage	• Strengthen implementation of the **Public Liability Insurance Act, 1991**. • Increase insurance coverage and compensation limits for victims and environmental restoration.
Institutional Convergence through the Disaster Management (Amendment) Act, 2025	• Provide statutory backing to the **National Crisis Management Committee (NCMC)** and promote **Urban Disaster Management Authorities (UDMAs)**. • Establish a **National Disaster Database** for hazard mapping and accident records. • Addresses recommendations made by the **LG Polymers Committee** regarding the absence of a central inventory of hazardous industries.
Right-to-Know and Community Participation	• Adopt the **HAZCHEM** approach by informing nearby communities about hazardous chemicals and emergency procedures. • Promote public awareness and preparedness. • **Case Study:** Residents around **LG Polymers** were reportedly unaware that **styrene** was stored at the plant, delaying appropriate emergency response.

Worker Safety and Training	• Provide mandatory safety training, particularly for contractual workers handling hazardous operations. • Conduct regular emergency drills and certification programmes.
Accountability and Fast-Tracked Prosecution	• Ensure prompt investigation and prosecution under the **Polluter Pays Principle**. • Fast-track industrial disaster cases to strengthen deterrence.

19. NUCLEAR AND RADIOLOGICAL DISASTERS

In India, the term **"nuclear disaster"** is technically reserved for large-scale events, such as a nuclear weapon attack or a massive release of radioactivity (similar to Chernobyl), that result in mass casualties and exceed local coping capacities.

To date, India has not experienced a large-scale nuclear disaster at its power plants, maintaining what the National Disaster Management Authority (NDMA) describes as an **"enviable and impeccable record of safety"**.

However, there have been recorded radiological and industrial incidents related to nuclear materials:

Historical Radiological and Nuclear Incidents

- **Mayapuri Radiological Accident (2010):** This is the most significant radiological incident mentioned in the sources. A university irradiator containing **cobalt-60** was sold as scrap and dismantled by dealers in a scrap yard who were unaware of the hazardous material. The incident resulted in **one fatality** and several injuries.
- **Hyderabad Fuel Complex Explosion (2002):** A chemical explosion occurred at the Natural Uranium Oxide Fuel Plant in Hyderabad. While it occurred at a nuclear fuel facility, it was classified as **"out of scale"** on the International Nuclear and Radiological Event Scale (INES) because it posed no radiological hazard.
- **Nanda Devi (1965):** During a CIA-led expedition, a **nuclear-powered telemetry listening device** was abandoned on the mountain.

India's Nuclear Disaster Response Framework

Component	Key Provisions
Nodal Agencies	• **Department of Atomic Energy (DAE):** Technical nodal agency for nuclear safety and radiation management. • **Ministry of Home Affairs (MHA):** Nodal ministry for managing nuclear/ radiological emergencies affecting the public.
Regulatory Oversight	• **Atomic Energy Regulatory Board (AERB):** Regulates nuclear and radiation safety.Frames and enforces safety standards. • Empowered to inspect facilities and order shutdowns for non-compliance.
Response Forces	• **National Disaster Response Force (NDRF)** maintains specialized **CBRN (Chemical, Biological, Radiological and Nuclear)** battalions. • Trained for detection, decontamination, rescue, evacuation, and first response during radiological or nuclear emergencies.

Emergency Response Centres (ERCs)	• **18 Emergency Response Centres (ERCs)** established by **BARC** across India. • Provide round-the-clock radiation monitoring, technical guidance, and emergency support during nuclear/radiological incidents.
Recent Legislative Reform – SHANTI Act, 2025	• Introduced a unified legal framework for nuclear energy governance. • Replaced older fragmented legislations. • Granted **statutory status** to the **AERB**. • Recalibrated the **exclusive operator liability**, capping it at **₹3,000 crore** for large nuclear power plants.

Key Issues In India's Nuclear Disaster Response Framework

Challenge	• Explanation
Weak Institutional Coordination and Supervisory Control	• Although the **National Disaster Management Authority (NDMA)** is the apex body for disaster planning, it has limited authority to enforce implementation at the state level. • Several recommendations and projects remain inadequately executed due to weak Centre–State coordination and monitoring mechanisms. • This creates a gap between policy formulation and actual disaster preparedness.
Non-Functional Local Disaster Management Authorities	• Many **District Disaster Management Authorities (DDMAs)** exist only formally, with limited operational capacity. • In several states, approved **District Disaster Management Plans (DDMPs)** have remained unimplemented, outdated, or unreviewed for years. • Weak district-level preparedness significantly delays response during nuclear or radiological emergencies.
Slow Implementation of Institutional Reforms	• Recent reforms, including the establishment of **Urban Disaster Management Authorities (UDMAs)** in state capitals, have witnessed slow adoption. • Urban areas housing research facilities, hospitals, and radioactive material therefore continue to face institutional preparedness gaps.
Limited Response Capacity	• The **National Disaster Response Force (NDRF)** has specialised **CBRN (Chemical, Biological, Radiological and Nuclear)** units, but their numbers remain insufficient for India's large population and geographical spread. • A major nuclear emergency could quickly overwhelm existing response capabilities.
Inadequate Emergency Monitoring Infrastructure	• India currently operates only **18 Emergency Response Centres (ERCs)** established by **BARC** for radiation monitoring and technical assistance. • Experts consider this network inadequate for a country of India's size, limiting rapid detection and technical support during widespread emergencies.

Shortage of Radiation Detection Equipment	• Adequate **radiation detection instruments**, contamination monitors, dosimeters, and **Personal Protective Gear (PPG)** are largely confined to **Department of Atomic Energy (DAE)** establishments. • State agencies, hospitals, police, and district administrations often lack essential equipment for effective first response.
Weak Evacuation and Logistics Planning	• Many off-site emergency plans suffer from inadequate transportation facilities, insufficient evacuation vehicles, and poor road connectivity. • These logistical shortcomings could delay mass evacuation and increase radiation exposure during emergencies.
Insufficient Medical Preparedness	• India faces a severe shortage of doctors trained in radiation medicine and nuclear trauma management. • The country also lacks a dedicated network of hospitals equipped to treat large-scale radiation injuries and contamination cases.
Low Public Awareness and Risk Communication	• Limited public awareness regarding radiation risks has created a widespread **"fear of the unknown."** • Even minor radiological incidents are often perceived as disasters comparable to **Chernobyl disaster** or **Atomic bombings of Hiroshima and Nagasaki**, leading to panic and misinformation.
Absence of Disaster Education	• Nuclear and radiological disaster management is not adequately integrated into mainstream education or community awareness programmes. • Consequently, India has not yet developed a strong **culture of preparedness**, reducing community resilience during emergencies.
Concerns over Liability Framework	• Under the **SHANTI Act, 2025**, the operator liability for large nuclear power plants is capped at **₹3,000 crore**. • Several experts argue that this amount may be insufficient to cover the enormous economic, environmental, and rehabilitation costs associated with a catastrophic nuclear accident such as **Fukushima Daiichi nuclear disaster**.
Physical Security of Nuclear Facilities	• Growing geopolitical tensions have raised concerns regarding the physical protection of nuclear installations, particularly proposed **Small Modular Reactors (SMRs)** located near populated areas. • Recent conflicts demonstrate that international norms protecting civilian nuclear infrastructure may not always be respected during warfare, increasing strategic vulnerability.

Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025

The **Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025**, represents a significant shift in India's nuclear energy governance by replacing the Atomic Energy Act, 1962, and the Civil Liability for Nuclear Damage Act, 2010.

The following are the key provisions of the Act:

1. **Private Sector Participation**

   The Act significantly broadens the category of entities eligible to apply for a nuclear license, ending the total central government monopoly established by the 1962 Act.

   - **Eligible Entities:** It allows **"any other company"** (incorporated in India) or any other **"person"** (including individuals and joint ventures) to apply for a license to set up nuclear facilities.
   - **Permitted Activities:** Licensed private entities can **build, own, operate, and decommission** nuclear power plants or reactors. They can also engage in limited nuclear fuel fabrication, transportation, and storage of fuel.
   - **State Control:** The government retains strict control over strategically sensitive domains like spent fuel reprocessing, enrichment, and heavy water production.

2. **Overhaul of Liability Regime**

   The SHANTI Act aligns India's liability framework with the international **Convention on Supplementary Compensation for Nuclear Damage (CSC)**.

   - **Exclusive Operator Liability:** Liability is strictly and exclusively channeled to the nuclear operator.
   - **Graded Liability Caps:** It introduces graded operator liability limits based on thermal power capacity, ranging from **₹100 crore to ₹3,000 crore** (approximately 300 million SDRs).
   - **Removal of Supplier Liability:** The Act effectively rules out automatic supplier liability for "latent defects," which was a major point of departure in the 2010 Act.
   - **Territorial Extension:** Coverage for compensation is extended to include nuclear damage in the territory of a **foreign state** resulting from an incident in India.

3. **Statutory Status for the AERB**

   The Act grants statutory status to the **Atomic Energy Regulatory Board (AERB)**, which was previously an executive body.

   - **Institutional Independence:** This shift enhances its functional independence from the Department of Atomic Energy (DAE), clearly defining its powers and organizational structure.
   - **Enforcement Powers:** The AERB is empowered to issue binding directions, suspend or cancel licenses, and has authority for **inspection, search, and seizure**.

4. **Dispute Redressal and Claims**

   - **Layered Mechanism:** It creates the **Atomic Energy Redressal Advisory Council (AERAC)** to review AERB decisions.
   - **Appellate Forum:** Appeals against AERAC decisions lie with the **Appellate Tribunal for Electricity**, with final jurisdiction resting with the **Supreme Court of India**.
   - **Claims Commission:** The government must appoint a **Claims Commissioner** within 30 days of a notified incident. For large incidents, a **Nuclear Damage Claims Commission** may be constituted.

5. **Promotion of Innovation and Research**

   - **Licensing Exemptions:** Clause 9 exempts research, development, and innovation activities from licensing requirements, provided they are not of a "sensitive nature" reserved for the government.
   - **Patent Advancement:** It delineates procedures for granting patents for inventions in the peaceful nuclear energy sector, aiming to foster technological advancement.

Key Issues In The Provisions Of The Act

While the **Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025**, modernizes India's nuclear governance, the sources identify several critical concerns regarding its liability regime, physical safety, and financial implications.

1. **Inadequacy of Liability Caps**

   A primary concern is the **adequacy of contingency funds** allocated under the Act.

   - **Comparison to Global Disasters:** The operator's liability is capped at **₹3,000 crore** for large plants, which is viewed as insufficient when compared to major incidents like the Fukushima disaster, where total liability costs exceeded **₹18,000 crore**.
   - **Taxpayer Burden:** Because the government acts as the **ultimate insurer**, any claims exceeding the operator's cap, the Nuclear Liability Fund (NLF), or international funds must be covered by the state, placing a significant **financial burden on public funds and taxpayers**.

2. **Flaws in the Graded Liability Architecture**

   The Act introduces graded liability limits based on the thermal power capacity of the reactor.

   - **Lack of Correlation:** Critics argue this architecture requires review because there is **no direct correlation between reactor size and the extent of damage** an accident might cause.
   - **Oversight of Other Factors:** The actual impact of a disaster is often more influenced by **site location, population density, and the nature of the incident** rather than the reactor's megawatt capacity.

3. **Physical Safety and Security during Conflict**

   The Act's push for a proliferation of reactors, including **Small Modular Reactors (SMRs)**, raises security concerns.

   - **Proximity to Civilians:** SMRs are likely to be built near **densely populated civilian areas**, raising questions about their ability to withstand conventional military attacks.
   - **Disregard for International Norms:** Although international protocols (like the Geneva Conventions) prohibit attacking nuclear plants, these norms are **frequently ignored during wars**, as seen in attacks on the Zaporizhzhia plant in Ukraine and other facilities historically.
   - **Feasibility of Protections:** While measures like underground construction could improve resilience, their **practical feasibility remains uncertain** due to high costs and land-use constraints.

4. **Environmental and Waste Management**

   While the Act vests the Central Government with the power to frame policies for the **safe disposal of radioactive waste**, managing the by-products of nuclear energy remains a persistent environmental challenge. The sources note that while India's record is robust, the long-term management of spent fuel is a **universal concern** that the Act must continuously address as the sector expands.

5. **Shift in Public Accountability**

   The Act aligns India with the international **Convention on Supplementary Compensation (CSC)**, which effectively **rules out automatic supplier liability** for latent defects. While this facilitates the entry of foreign technology and private players, it represents a shift away from the **enhanced public accountability** provisions found in the earlier 2010 Act, which were originally influenced by lessons from major industrial disasters like the Bhopal Gas Tragedy.

Way Forward

Classification of measures	Provisions Needed
Strengthen Regulatory Independence	• Fully operationalise the **SHANTI Act, 2025**, which grants **statutory status** to the **Atomic Energy Regulatory Board (AERB)**. • Ensuring that the AERB functions independently of the **Department of Atomic Energy (DAE)** with adequate legal, financial, and administrative autonomy

Empowerment of District Disaster Management Authorities (DDMAs)	• Transform **DDMAs** from largely paper-based institutions into fully functional operational agencies. • Designate the **District Collector** as the non-delegable Incident Commander for off-site nuclear and radiological emergencies
Expand the Emergency Response Centre (ERC) Network	• Increase the present network of **18 Emergency Response Centres (ERCs)** established by **BARC**. • Establishing more ERCs in major metropolitan cities, industrial hubs, and vulnerable regions. • Co-locating ERCs with **District Police Control Rooms** can significantly improve emergency response time and inter-agency coordination.
Develop a Dedicated National Disaster Communication Network	• Establish a secure **National Disaster Communication Network (NDCN)** with **EMP-hardened** communication systems and reliable last-mile connectivity. • Dedicated communication infrastructure will ensure uninterrupted coordination between district, state, and national authorities even during large-scale emergencies when conventional networks fail.
Operationalise GIS-Based Decision Support Systems	• Replace conventional paper-based planning with **GIS-enabled databases**, digitised hazard maps, and real-time plume modelling systems. • Such systems enable rapid radiation dispersion modelling, impact assessment, evacuation planning, and resource deployment during emergencies.
Expand Real-Time Radiation Monitoring (IERMON)	• Strengthen and expand the **Indian Environmental Radiation Monitoring Network (IERMON)** to cover all major cities, nuclear installations, ports, and vulnerable locations. • Continuous monitoring will provide early warning of abnormal radiation levels and support timely public safety measures.
Establish Radiation Injury Treatment Centres (RITCs)	• Designate specialised hospitals with **Radiation Injury Treatment Centres (RITCs)** equipped with burns units, blood banks, intensive care facilities, and **Bone Marrow Transplant (BMT)** capabilities. • These centres would significantly improve survival during large-scale radiation exposure incidents.
Create a National Medical Response Network	• Develop an integrated network of primary, secondary, and tertiary healthcare institutions for radiological emergencies. • Maintain a national database of trained doctors, radiation specialists, toxicologists, and **Radiological Safety Officers (RSOs)** to enable rapid deployment during crises.
Stockpiling the Radiation Countermeasures	• Maintain strategic state-level reserves of essential **de-corporation agents** such as **DTPA** and **Prussian Blue**, along with radio-protective drugs like **Amifostine**. • Domestic production and stockpiling would reduce dependence on imports and improve emergency preparedness.
Build a Culture of Preparedness through Education	• Integrate nuclear and radiological disaster management into school, college, and professional education. • Public education can reduce misinformation, eliminate the "fear of the unknown," and build a culture of preparedness instead of panic during emergencies.

Strengthen CBRN (Chemical, Biological, Radiological, and Nuclear defense) Training for First Responders	• Expand specialised **CBRN** training for the **NDRF**, police, fire services, civil defence personnel, healthcare workers, and local administrators. • Training should include personal protection, radiation detection, decontamination, evacuation, triage, and incident command procedures.
Develop Comprehensive Technical Standards	• The **DAE** and **AERB** should establish scientifically validated **Intervention Levels** (radiation dose thresholds for evacuation and sheltering) and **Action Levels** (acceptable contamination limits for food and water). • Standard operating procedures should also be developed for scenarios involving **dirty bombs**, nuclear terrorism, and wartime radiological emergencies.
Strengthen Border and Transit Security	• Install advanced **radiation portal monitors** at airports, seaports, land borders, cargo terminals, and major metal scrap facilities. • Enhanced surveillance will help detect **orphan radioactive sources**, prevent illicit trafficking of nuclear materials, and strengthen national nuclear security.
Mainstream Disaster Risk Financing	• Integrate nuclear and radiological disaster preparedness into national and state development planning with dedicated budgetary allocations. • Stable financial support is essential for infrastructure creation, capacity building, equipment procurement, and regular preparedness exercises.
Promote Public–Private Partnerships (PPP) and CSR Participation	• Encourage **Public–Private Partnerships (PPP)** and utilise **Corporate Social Responsibility (CSR)** funds to strengthen disaster preparedness infrastructure. • Private sector participation can support the establishment of mobile hospitals, specialised treatment facilities, emergency shelters, training centres, and advanced monitoring infrastructure, thereby enhancing overall national resilience.

20. STAMPEDE DISASTER IN INDIA

Stampedes in India are primarily classified as **man-made disasters** that frequently occur at venues of mass gatherings, such as religious shrines, railway stations, and cultural festivals. These incidents are of significant national concern due to India's high population density and a cultural landscape characterized by numerous large-scale events.

In 2025, over 11 people died during RCB parade in Bengaluru - drawing nationwide criticism against crowd mismanagement.

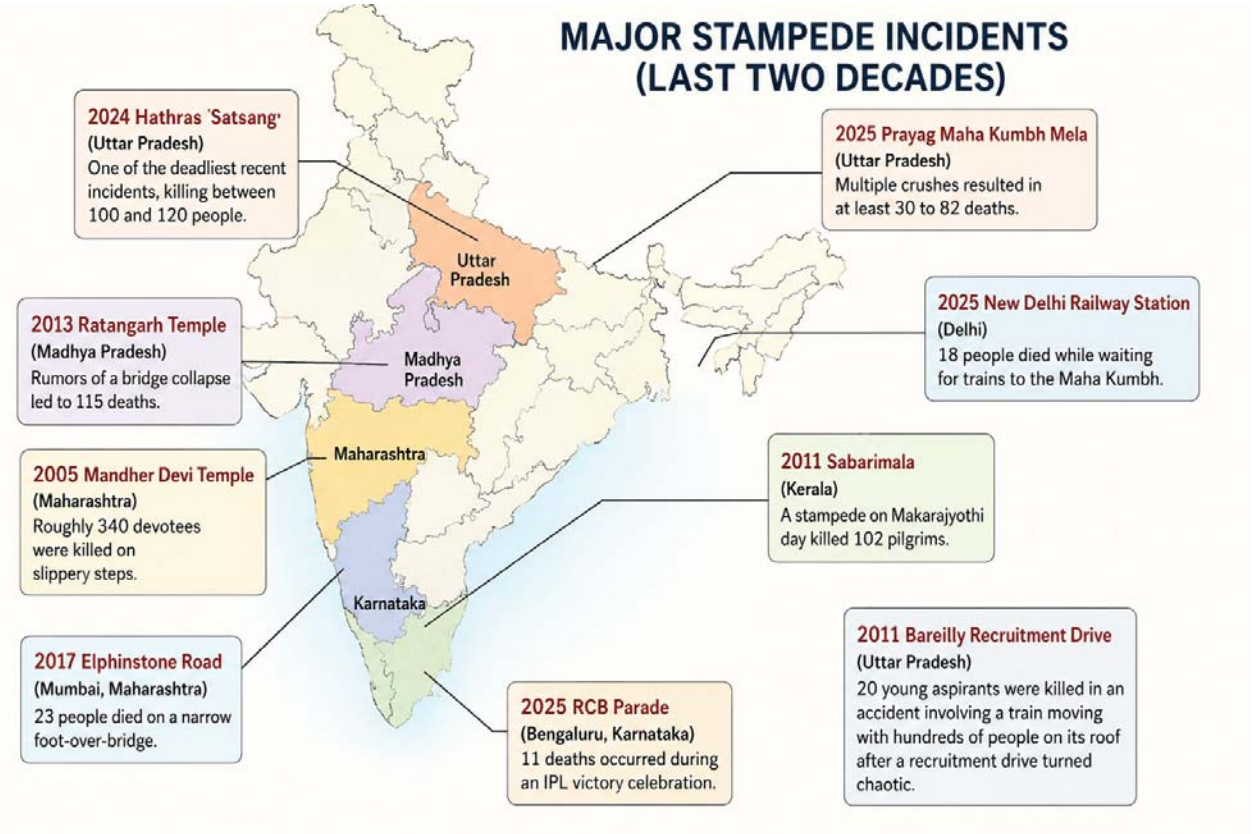
Major Causes

The causes of stampedes in India are classified by the National Disaster Management Authority (NDMA) into three main categories: **structural failures, human errors, and natural causes**, often occurring at religious shrines, railway stations, and venues of mass gatherings.

1. **Structural Failures**

   Inadequate infrastructure or the failure of makeshift structures often provides the physical catalyst for a crush.

   - **Bridge and Slab Collapses:** Narrow bridges or temporary structures that cannot handle the weight of a surge are frequent causes.



In **Indore (2023)**, 36 people died when a slab constructed over an ancient well collapsed during a temple program. Similarly, a narrow bridge in **Mumbai (2017)** connecting railway stations led to 23 deaths.

- **Infrastructure Infringements:** Fixed structures with insufficient clearances can lead to chaos.

In **Shahjahanpur (2011)**, a railway overbridge with only three feet of vertical clearance from a train roof crushed 14 young men who had climbed atop the coaches, leading to a subsequent chaotic situation at the site.

2. Human Errors and Management Deficiencies

Human behavior and organizational failures are identified as the most common factors in crowd disasters.

- **Rumors and Panic:** Sudden, unfounded rumors can trigger a deadly rush.

In **Ratangarh (2013)**, rumors of a bridge collapse led to 115 deaths. Similar tragedies occurred at **Naina Devi (2008)** due to rumors of rockslides and at **Jodhpur (2008)** due to rumors of a bomb.

- **Overcrowding and Unauthorized Entry:** When crowds far exceed a site's carrying capacity, management becomes impossible..

During a **Bareilly recruitment drive (2011)**, more than two lakh aspirants arrived for 416 posts, causing officers to suspend the drive and resulting in mass chaos as crowds rushed back to rail terminals

- **VIP Culture and Mismanagement:** Prioritizing the movement of high-profile guests can obstruct the flow of regular attendees.

> In the **2025 Prayag Maha Kumbh Mela**, mismanagement and a focus on VIP movement were cited as factors that contributed to a crush killing at least 30 people.

- **Ritual Rigidity:** The need to perform specific rituals at precise, auspicious times can lead to unmanageable surges.

> Millions of pilgrims jostling for space during **'Amrit Snan'** at the **2025 Maha Kumbh** is a primary example of this trigger.

3. **Natural and Environmental Causes**

The physical landscape and sudden environmental changes create inherent risks at many mass gathering sites.

- **Difficult and Steep Terrain:** Many prominent shrines are located on hilltops with narrow, winding pathways that hinder crowd regulation.

> Examples include the **Mansa Devi temple** in Haridwar, the **Sabarimala shrine** in Kerala, and the **Mata Vaishno Devi shrine**.

- **Slippery Surfaces:** Rituals can inadvertently make pathways dangerous.

> In **2005**, roughly 340 devotees were killed at the **Mandhardevi temple** when some fell on steps made slippery by devotees breaking coconuts.

Way Forward

To prevent stampedes in India, authorities focus on an integrated approach involving proactive planning, infrastructure upgrades, and the application of modern technology. The National Disaster Management Authority (NDMA) notes that these disasters can be completely prevented through **proactive and holistic planning** and flawless execution.

1. **Planning and Coordination**
   - **Vulnerability Assessments:** Authorities must undertake hazard, risk, and vulnerability assessments of the site before an event. This includes considering potential natural disasters and the inherent risks of the terrain.
   - **Standard Operating Procedures (SOPs):** States are advised to prepare their own context-specific SOPs for crowd management using templates provided by the NDMA and the Bureau of Police Research & Development (BPR&D).
   - **Collaborative Planning:** Plans should be developed collaboratively, involving event organizers, local and district administration, surrounding communities, and disaster management authorities.
   - **Capacity Management:** A critical step is understanding and regulating the **carrying capacity** of a venue to manage inflow and outflow accordingly.

> For example, after a 2011 tragedy, the ITBP limited recruitment cycle candidates to 600 per day.

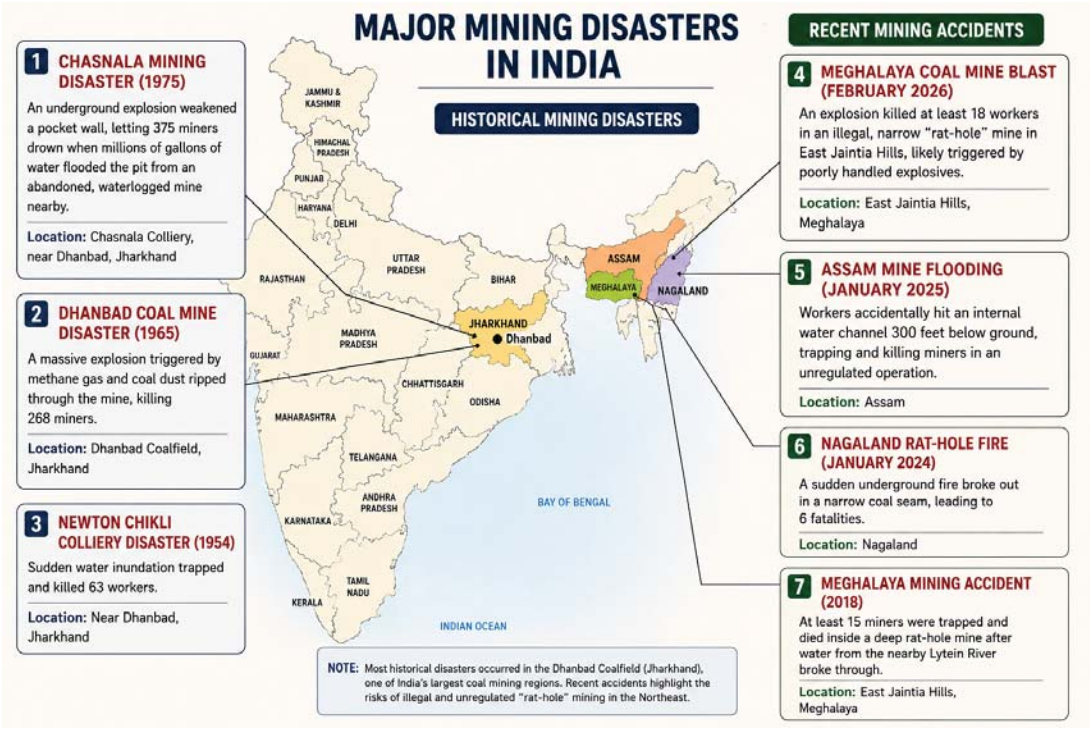
2. **Infrastructure Improvements**
   - **Removing Infringements:** Following a Supreme Court judgment, Indian Railways was directed to identify and remove structure infringements, such as low-hanging foot-over-bridges (FOBs), that could cause accidents during surges.
   - **Expansion of Facilities:** For example, To manage massive crowds during the 2025 Maha Kumbh, Railways built seven additional platforms and 17 new permanent **Yatri Ashryas** (shelters), increasing holding capacity from 21,000 to over 110,000.
   - **Unidirectional Movement:** Implementing single entry and exit points and maintaining unidirectional movement on platforms, ramps, and FOBs helps prevent opposing crowd flows.

3. **Use of Technology**
   - **Real-time Monitoring:** Extensive use of **CCTV, drones, and Face Recognition Systems (FRS)** allows for constant surveillance of tracks and approach roads.
   - **Smart Analytics:** Modern approaches involve integrating multiple data sources into a single operational picture using video analytics to improve public safety and mitigate risks.
   - **Virtual Queues:** Online portals for devotees to book darshan slots, such as the virtual queue system used at **Sabarimala,** help predict and regulate pilgrim flow.
   - **Communication Equipment:** Equipping staff with high-tech walkie-talkies and digital announcement systems ensures clear information dissemination during crowd situations.
4. **Tactical Management Measures**
   - **VIP Movement Regulation:** Reducing focus on "VIP culture" during peak crowd times is essential, as prioritizing high-profile guests can obstruct regular attendees and cause mismanagement.
   - **Ritual Flexibility:** Managing the rigidity of religious rituals—where millions attempt to reach a specific spot at a precise time—is a major challenge that **requires constant dialogue with religious authorities**.
   - **Security Deployment:** Large-scale events like the Maha Kumbh require massive personnel deployment; For example, in 2025, over 15,000 RPF and GRP personnel were deployed specifically for security and crowd management.
   - **Medical Preparedness:** Establishing medical outposts and ensuring free flow for ambulances 3–4 months in advance of an event is crucial for rapid response if an incident occurs.

21. MINING RELATED DISASTERS IN INDIA

Mining-related disasters in India have historically resulted in significant loss of life and continue to be a major concern, particularly in the coal sector.

While fatality rates have declined over the long term, accidents remain frequent due to factors like geological disturbances, neglect of safety principles, and unscientific mining practices.



Causes behind Mining Disasters in India

Mining-related disasters in India are driven by a variety of technical, geological, and human factors.:

- **Gas and Dust Explosions**: These disasters are often caused by the ignition of explosive natural gases, particularly firedamp (methane), or coal dust in underground workings.

  A major historical example is the **1965 Dhanbad Coal Mine Disaster** at the Ghori Dhori colliery, where an explosion followed by a fire killed 375 miners. Another catastrophic event was the **1975 Chasnala mining disaster**, which began with a massive explosion in a coal mine near Dhanbad.

- **Inundation (Flooding)**: This occurs when mining operations cause a breach that allows water from surface bodies or adjoining abandoned workings to rapidly flood the mine.

  In the **2001 Bagdigi colliery disaster**, 38 miners were trapped and killed when routine explosives caused a breach that led to water from an adjoining pond flooding the mine. The **Chasnala disaster of 1975** also involved massive flooding that followed the initial explosion.

- **Mine Fires**: These typically result from the spontaneous heating of coal through oxidative and thermal processes.

  Approximately **10% of India's total national coal resources** are located in fire-affected areas, causing steady destruction of energy resources and land damage through subsidence. Long-standing mine fires in the **Jharia coalfields** continue to endanger lives and the environment.

- **Neglect of Basic Safety Principles**: Reports highlight that many tragedies are avoidable and point to a "long neglect of workers and workplaces" by the industry.

  For example, During the **1996 Kusunda Colliery accident**, workers had warned the Mining Sirdar that the roof condition was precarious, but their concerns were dismissed, and they were ordered to continue working, leading to the fatal collapse.

- **Unscientific and Unsystematic Mining**: This involves mining without regard for environmental norms or approved mining plans, often seen in illegal or indiscriminate operations.

  Fr example, In **Sahibganj**, such practices resulted in "flattened hills" and improper dump management, where huge, steep piles of crushed stone chips created a risk of slope failure for men and machinery.

- **Machinery and Transportation Accidents**: Heavy machinery used in mines is a major source of fatal accidents.
  - Data shows that **Dumpers and Trucks** are the biggest contributors to fatal accidents in both coal and non-coal mines. These disasters can also stem from general mechanical errors or the use of malfunctioning equipment.
- **Land Subsidence**: This phenomenon is a major hazard in coal mining regions like **Jharia and Raniganj**, where underground extraction causes the surface to sink.
- Subsidence leads to significant damage to surface infrastructure, buildings, and the environment.
- **Mining-Induced Seismicity**: Deep mining operations can trigger seismic hazards, including rockbursts and anomalous stress zones.
- **Improper Handling and Use of Explosives**: Accidents can occur due to faulty blasting techniques or the use of improper explosives underground.

  For example, In the **Bagdigi disaster (2001)**, it was specifically the use of "routine explosives" that triggered the catastrophic breach and subsequent inundation of the mine.

India's Current Mining Safety & Management Framework

Component	Explanation
Constitutional & Legislative Framework	• **Mining safety** is a **Union subject** under **Entry 55 of the Union List**, empowering the Central Government to legislate on the regulation of labour and safety in mines. • The legal framework aims to ensure worker safety, accident prevention, environmental protection, and scientific mining practices across the country.
The Mines Act, 1952	• Serves as the principal legislation governing the **health, safety, and welfare of mine workers**. • Prescribes standards for working conditions, ventilation, inspections, mine management, emergency preparedness, accident reporting, and occupational health.
Sector-Specific Safety Regulations	• Different categories of mines are governed through specialised regulations tailored to their operational risks. • These include the **Coal Mines Regulations (CMR), 2017**, **Metalliferous Mines Regulations (MMR), 1961**, **Oil Mines Regulations (OMR), 2017**, and the **Mines Rescue Rules, 1985** • .These regulations prescribe detailed standards relating to mine design, ventilation, blasting, roof support, rescue operations, fire protection, and emergency response.
Recent Legislative Reforms	• The **Mines and Minerals (Development and Regulation) (MMDR) Amendment Act, 2025** seeks to accelerate operationalisation of auctioned mineral blocks. • Introduces provisions for inclusion of newly identified minerals, facilitates faster mineral development, and permits one-time lease extensions to improve continuity in mining operations while maintaining regulatory oversight.
Directorate General of Mines Safety (DGMS)	• The **DGMS**, functioning under the **Ministry of Labour & Employment**, is India's principal mine safety regulator. • Conducts inspections of mines, investigates fatal and serious accidents, grants statutory approvals, certifies mining operations, conducts competency examinations for mine managers, and enforces compliance with safety regulations. • Acts as the primary authority for preventing mining accidents through regulatory enforcement.
Indian Bureau of Mines (IBM)	• The **Indian Bureau of Mines (IBM)**, under the **Ministry of Mines**, regulates the scientific development and conservation of mineral resources (excluding coal and petroleum). • Enforces the **Mineral Conservation and Development Rules (MCDR)**, monitors sustainable mining practices, approves mining plans, oversees mine closure plans, and promotes environmental protection during mining operations.
Geological Survey of India (GSI)	• The **Geological Survey of India (GSI)** provides scientific and geotechnical support for mining safety. • Conducts geological investigations, slope stability studies, landslide hazard assessments, and geotechnical mapping to minimise geological risks associated with mining activities.

Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)	• Implemented through **District Mineral Foundations (DMFs)** using contributions collected from mining lease holders. • Focuses on improving healthcare, education, drinking water, livelihood opportunities, environmental restoration, and infrastructure in mining-affected areas to reduce the social impacts of mining activities.
Mining Surveillance System (MSS)	• A satellite-based monitoring platform that continuously monitors a **500-metre buffer** around mining lease boundaries. • Detects unauthorised excavation, encroachment, and illegal mining activities using remote sensing technology, thereby improving regulatory enforcement.
Star Rating System for Sustainable Mining	• Introduced under **Rule 35 of the Mineral Conservation and Development Rules (MCDR), 2017**. • Evaluates mines based on compliance with the **Sustainable Development Framework (SDF)**, covering environmental management, scientific mining, worker safety, community development, and mine closure practices. • Provides ratings of up to **7 stars**, encouraging responsible and sustainable mining.
Safety Management Plans (SMPs)	• Every mine is required to prepare a **Safety Management Plan (SMP)** based on detailed risk assessment. • SMPs identify operational hazards, establish preventive measures, define emergency procedures, and prescribe Standard Operating Procedures (SOPs) for routine and emergency situations.
Mines Rescue Services	• Public Sector Undertakings such as **Coal India Limited (CIL)** and **Singareni Collieries Company Limited (SCCL)** maintain dedicated **Mine Rescue Stations** and **Rescue Rooms**.
Accidental Insurance Coverage	• Central Public Sector Enterprises (CPSEs) under the **Ministry of Mines** provide financial protection to mine workers. • Insurance coverage extends up to **₹1 crore** for regular employees and up to **₹20 lakh** for contractual workers, providing financial support to families in case of fatal accidents.
Drone and Satellite-Based Digital Monitoring	• Mining lease holders are required to submit **annual drone surveys and satellite imagery** to the **IBM** and State Governments • .Remote sensing enables continuous monitoring of mining activities, lease compliance, environmental restoration, and mine closure implementation without relying solely on physical inspections.
Real-Time Geotechnical Monitoring	• Deep underground mines increasingly deploy **microseismic monitoring systems** to detect ground movement, stress accumulation, and rockburst risks. • Real-time monitoring helps operators refine **Trigger Action Response Plans (TARPs)**, enabling early evacuation and preventive action before catastrophic failures occur.

Way Forward

I. Technological and Digital Transformation

A primary strategy is the adoption of advanced technology to move from reactive accident investigation to proactive hazard prevention.

- **Real-time Monitoring and Data Analysis:** The **National Coal Mines Safety Report Portal**, developed under the guidance of the **High-Level Expert Committee on Safety in Coal Mines**, is a central pillar.

  It uses an "Accident Module" for near-time reporting and a "Safety Audit Module" to ensure coal companies act on inquiry recommendations.

- **Deep Mining Safety:** For deep mines like Rampura Agucha, the we need to extend **real-time microseismic monitoring** to evaluate seismic hazards like rockbursts and refine Trigger Action Response Plans (TARPs).
- **Remote Surveillance:** The **Mining Surveillance System (MSS)** and the mandatory submission of **drone and satellite imagery** (under the 2021 MCDR amendments) provide "eyes from the sky" to curb unauthorized mining and ensure scientific mine closure.
- **Digital Connectivity:** Pilot projects, such as establishing **5G private networks** and video calls at 400-meter depths, are critical for improving emergency communication and operational transparency.

II. Committee Recommendations and Strategic Frameworks

- **Paul Committee Recommendations:** This committee emphasized the necessity of **roof bolting** to support Freshly Exposed Roof (FER) in underground mines, which is now being augmented by mechanized drilling and resin capsules.
- **NGT Joint Committee (Sahibganj Matter):** Following an in-depth study of unscientific mining in Jharkhand, this committee recommended a **cluster approach** for environmental clearances, immediate closure of units without valid CTOs, and the development of **dedicated roads** for heavy machinery to reduce community exposure to dust and accidents.

  Safety Management Plans (SMPs): Current strategy mandates site-specific, risk-assessment-based SMPs and **Principal Hazards Management Plans (PHMPs)** to systematically address issues like inundation and explosions.

III. Legislative and Regulatory Reforms

Modernizing the legal framework is essential for aligning Indian mining with global standards:

- **Consolidation of Laws:** The **Occupational Safety, Health, and Working Conditions Code, 2020** subsumes the Mines Act, 1952, aiming to provide a more cohesive safety regime across all sectors.
- **Vision Zero:** The Directorate General of Mines Safety (DGMS) has set a goal to achieve **Zero Harm Potential (ZHP)** in Indian mines by 2047.
- **Incentivizing Excellence:** The **Star Rating System** serves as a governance tool, where mines must achieve at least a three-star rating to avoid suspension, thereby compelling companies to adopt sustainable and safe practices.

IV. Environmental and Social Sustainability

The long-term safety of the mining sector is inextricably linked to the welfare of affected communities and the management of waste.

- **Circular Economy:** Under the **National Critical Mineral Mission (NCMM)**, a major focus is on recovering critical minerals (like Nickel and Cobalt) from **legacy dumps and mine tailings**, reducing environmental waste while securing resource supply.
- **Occupational Health:** Strategies include mandatory **Personalized Safety Counseling**, annual medical examinations for early detection of dust-related diseases (like silicosis), and the use of specialized equipment like **LED vests** for underground workers.

Conclusion

India's mining sector requires a shift from **compliance-driven regulation to a zero-harm safety culture**, where scientific risk assessment, real-time monitoring, strict enforcement, and worker welfare become integral to mining operations. Sustainable mineral development can be achieved only when **economic growth is balanced with the protection of human life and environmental safety.**

■■■

MGP/BP/GS3/018

SPACE TECHNOLOGY

1. SPACE SECTOR IN INDIA

India's space sector has achieved remarkable progress, exemplified by the success of Chandrayaan-3, which highlights its advancing technological capabilities. Over the next two decades, the Government of India has set ambitious goals, including the development of powerful, reusable rockets like ISRO's upcoming Next Generation Launch Vehicle (NGLV). These advancements are critical for achieving self-reliance and global competitiveness in space exploration.

Recent Developments in the Space Sector in India post the successful Chandrayaan-3 mission

New Space Launches

Aditya-L1 Mission	The Aditya-L1 spacecraft has been **launched to study solar radiation** from the Earth-Sun Lagrange point (L1), aboard the **Polar Satellite Launch Vehicle (PSLV)**. It reached its orbit around L1 by January 6, 2024, and completed its first orbit on July 2, 2024. In May 2024, it tracked a **solar storm in collaboration** with ground observatories and lunar spacecraft.
Gaganyaan TV-D1 Test Flight	The test successfully **separated the crew module from the Test Vehicle** (TV), ensuring it safely descended and was recovered by the **Indian Navy vessel INS Shakti**. This test is a key part of **ISRO's human spaceflight program**.
XPoSat Launch	It studies the polarization of radiation from celestial objects and **follows NASA's IPEX mission.**
RLV-TD Tests	ISRO tested its Reusable Launch Vehicle (RLV), Pushpak, with **landing experiments that mimicked conditions from space**. These successful tests provided key data and set the stage for the upcoming Orbital Return Flight Experiment.
SSLV Development	ISRO successfully completed the final test flight of the Small Satellite Launch Vehicle (SSLV). This milestone **confirms the SSLV's readiness for commercial use**. The payloads included earth observation tools and an ultraviolet dosimeter for the Gaganyaan mission.

Regulatory and Institutional Developments

NewSpace India Ltd. (NSIL)	NSIL now **manages commercial activities** like Indian Remote Sensing satellite data. On May 1, 2024, NSIL signed a **launch deal with SpaceX** for the GSAT-20/GSAT-N2 satellite. It has also sought qualifications for LVM-3 production and signed a launch agreement with an Australian company for the SSLV.

Private Sector Contributions	Private space companies are progressing with their missions- **Agnikul Cosmos** launched its SoRTeD-01 vehicle, **Skyroot Aerospace** is developing the Vikram 1 rocket, and **Dhruva Space and Bellatrix Aerospace** contributed to the PSLV-C58 mission.
Regulatory Developments	India's space regulator, **IN-SPACe**, has **updated its policies and issued new licences,** including the first satellite broadband license to Eutelsat OneWeb and the first ground station license to Dhruva Space. The government has amended its foreign direct investment (FDI) policy to allow 100% FDI in most space sectors. However, there are some limits on **satellite manufacturing** (74%) and **launch infrastructure** (49%).

Future Roadmap and Initiatives

Gaganyaan Program	**a.** ISRO is advancing its Gaganyaan program, which will send an Indian crew to space for the first time, showcasing India's human spaceflight capabilities. **b.** By 2035, ISRO aims to establish India's own space station, the **'Bharatiya Antariksh Station' (BAS).** The revised Gaganyaan Programme includes the development of the first module of BAS and four missions to demonstrate and validate various technologies for BAS by December 2028.
Next-Generation Launch Vehicle	**a.** ISRO is working on a **Next Generation Launch Vehicle** (NGLV). It will be a **three-stage rocket** using **semi-cryogenic**, **liquid**, and **cryogenic engines**. **b.** The NGLV will replace some existing launch vehicles, such as the GSLV. The NGLV will **triple the payload capacity of the LVM3** (Geosynchronous Satellite Launch Vehicle Mk III), India's most powerful rocket. Additionally, ISRO is upgrading the LVM3 with a new semi-cryogenic engine. **c.** In contrast to India's existing rockets, which are expendable and designed for single use, a **significant portion of the NGLV will be reusable.** Reusability requires that the rocket keep some of its fuel for controlled descent back to the earth's surface.

Significance of Space Sector in India

1. **Boost to 'Space Industrialisation'-** Currently, India manages to occupy only 2%, or US$ 7 billion, of the global space economy. The development of the space sector will **promote space Industrialisation in India**, **provide a boost to the space-tech ecosystem** and **help India's space economy** to **reach $100 billion** by **2040**.
2. **Low Cost Missions-** The Indian space sector has the **potential of launching space vehicles at a much lower cost.** This will help in getting many foreign contracts. **For ex- Mars Orbiter Mission was 10 times cheaper than western missions**.
3. **Presence of Budding Entrepreneurs-** According to a global report published in June 2021, **India has 368 private space firms**, **placing it 5th in the world** in size after the US, the United Kingdom, Canada, and Germany. With these many firms, India is leading China (288), France (269), and Spain (206) in the private space industry.
4. **Enhancement of India's role and position in the Artemis accord-** India is now a member of the Artemis Accords. With the further growth and development of the space sector, India has an opportunity to lead the other Artemis countries alongside the U.S.

Artemis Accord- It is the U.S.-led **multilateral effort to place humans on the moon by 2025** and thereafter to expand human space exploration to the earth's wider neighbourhood in the solar system.

5. **Expansion of cooperation in outer space-** While geo-political rivalry is a reality, India's space sector provides India an **opportunity to limit competition** and expand cooperation in outer space. However, **it also allows India to gain military advantage in space** over their geopolitical rivals on Earth.

Challenges in further development of Space Infrastructure in India

1. **Budgetary Challenges-** Space sector in India faces budget constraints despite its successes in launching missions. There has been an **8% decline in budget allocations to ISRO in 2023-2024** with respect to **2022-2023.** The funds allocated to the space sector are very less in comparison to other countries. The US spent 10 times and China 6 times more than India in the space sector in 2019-20.
2. **Manpower Challenges-** ISRO, the foundational pillar of the Indian space sector, faces a manpower challenge due to the **problem of Brain Drain** and **fewer students pursuing advanced spaced space studies**.
3. **Absence of a Clear Legislative Framework-** The draft Space Activities bill, which was **introduced way back in 2017 but has not yet been passed**. This has hindered the further growth and development of the space sector in India.
4. **Lack of robust Dispute Settlement Mechanism-** This discourages private investment in the space sector in India. The void was seen in **Antrix-Devas cancelled satellite deal**. The Government of India owes nearly **$1.2 billion to Devas Multimedia** as per an order of a tribunal of the International Chamber of Commerce.
5. **Technological Challenges-** ISRO faces **challenges in upgrading its technology**, such as **developing more powerful launch vehicles** with higher payload capacities. **For example**, Chandrayaan-3 took about six weeks to reach the Moon, while the failed Russian mission Luna-25 took only one week. Currently, India uses two LVM3 rockets for uncrewed lunar missions and **relies on foreign rockets like SpaceX's Falcon 9 for heavier payloads.**
6. **Government funding driven sector-** Some critics have argued that such **massive spending by the government alone in the space sector** curtails the **Indian government's spending capability in social sectors** like poverty alleviation, education, and health which must be the priority for developing countries like India.

Steps have been taken by the Government to address the challenges of the Space Sector in India

Indian Space Policy, 2023	Indian Space Policy 2023 is a **landmark policy for the development of India's space sector**. It offers an opportunity to build upon past achievements and harness the potential of the evolving space sector.
IN-SPACe	It aims to provide a **level-playing field for private companies to use the Indian space architecture**. IN-SPACe will act as a channel between ISRO and any private players that want to participate in space activity, thereby culling lengthy bureaucratic procedures.
FDI Policy	The government has amended its **foreign direct investment** (FDI) policy to allow 100% FDI in most space sectors. However, there are some limits on **satellite manufacturing** (74%) and **launch infrastructure** (49%).

New Space India Limited (NSIL)	It is a **Central Public Sector Enterprise** under the Department of Space that was established in 2019. It has been mandated to **transfer the technologies emanating out of the Indian space programme** and enable Indian industry to scale up high-technology manufacturing base.

Way Forward

1. **Push for greater privatization-** India should create **space policies that boost private sector investment** and **focus on commercial growth**. The Department of Space can **award contracts to Indian companies to develop reusable, heavy-lift rockets**. A milestone-based funding mechanism, where payments are made to private players after meeting specific objectives at each stage, would ensure accountability and reduce cost overruns.
2. **Passage of Space Activities Bill-** The passage of the **Space Activities Bill** should also be done in order to give private players greater clarity and protection. This should involve proper consultation and discussions with the concerned stakeholders.
3. **Setting up Space Dispute Tribunal-** The plan to set up **an independent tribunal to adjudicate disputes among private space entities** should be implemented promptly.
4. **Enhanced International Collaboration-** India must do more collaboration and research with pioneer countries like the **US & Russia**, who have already ramped up their space infrastructure.

The Indian Space sector possesses huge untapped potential which can be realized with adequate policy measures by the government. This would boost the confidence of the private sector and deliver optimum results, thereby helping the country acquire the top spot in the global space industry.

2. INDIAN SPACE POLICY 2023

Indian Space Policy 2023 is a comprehensive and dynamic framework enacted by the Government of India to institutionalize and expand the role of the private sector across the space economy. It clearly delineates the operational mandates of key public entities—shifting ISRO's focus entirely to advanced research and development while delegating commercial activities to NSIL and regulatory approvals to IN-SPACe.

Objectives of the Indian Space Policy 2023

- **Private Sector & Market Expansion**
  - **Boost Global Market Share:** Transform India into a competitive global hub to capture a significantly higher share of the multi-billion-dollar global space economy.
  - **Promote Startups & MSMEs:** Create a thriving domestic commercial ecosystem by handholding tech startups and small-to-medium industries.
- **Institutional Re-engineering & Roles**
  - **Refocus ISRO's Mandate:** Shift ISRO away from routine satellite/rocket production to prioritize advanced R&D, deep-space exploration, and human spaceflight.
  - **Empower IN-SPACe:** Operationalize a single-window, autonomous regulator to authorize, monitor, and level the playing field for private operators.
  - **Commercialize via NSIL:** Mandate NewSpace India Limited to serve as the chief commercial arm for manufacturing, leasing, and transferring public space assets.
- **Infrastructure & Resource Maximization**
  - **Share Public Infrastructure:** Provide private space companies non-discriminatory, fee-based access to publicly funded ISRO launchpads and testing facilities.

- **Democratize Satellite Data:** Establish an open data policy making low-resolution archived satellite data freely accessible for civilian research and commercial use.
- **Encourage Space Mining:** Provide a clear framework for private players to engage in the commercial extraction, ownership, and sale of asteroid or space resources.

- **Socio-Economic & Strategic Development**
  - **Enhance Communication & Navigation:** Leverage space applications to improve rural connectivity, weather forecasting, disaster management, and governance.
  - **Strengthen National Security:** Expand advanced remote-sensing and defense-space capabilities to protect national borders and critical infrastructure.
  - **Secure Space Assets (SSA):** Develop robust domestic capabilities in Space Situational Awareness to track space objects, manage debris, and avoid collisions.

Salient features of the Indian Space Policy 2023

- **End-to-End Private Operations:** Permits Non-Governmental Entities (NGEs) to independently build, launch, and operate satellites, rockets, spaceports, and tracking stations.
- **Single-Window Clearance:** Establishes **IN-SPACe** as an autonomous, single-window regulatory body to authorize and promote all space activities by both public and private entities.
- **Research-Centric ISRO:** Restructures ISRO's role to focus primarily on advanced research and development, deep-space exploration, and human spaceflight programs.
- **Commercial Execution via NSIL:** Mandates **NewSpace India Limited (NSIL)** to handle commercial manufacturing, leasing, and procurement of space assets based on sound market principles.
- **Open Satellite Data Policy:** Makes low-resolution archived satellite data freely accessible for research, while high-resolution data (GSD under 30 cm) requires strict safety clearances.
- **Public Infrastructure Sharing:** Guarantees private space companies a level playing field to utilize publicly funded ISRO infrastructure and launchpads for a nominal fee.
- **Next-Gen Space Mining:** Explicitly allows private players to engage in the commercial recovery, ownership, and sale of asteroid or space resources.
- **Space Situational Awareness (SSA):** Encourages private entities to develop capabilities for tracking space objects, ensuring collision avoidance, and managing space debris.

Structural components of the Indian Space Policy, 2023

- **Institutional Components**
  - **IN-SPACe:** Serves as the single-window, autonomous regulatory and promotional agency for both government and private space activities.
  - **ISRO:** Transitions out of commercial manufacturing to focus strictly on advanced R&D, deep-space exploration, and technology proving.
  - **NewSpace India Limited (NSIL):** Functions as the commercial arm to manufacture, lease, and trade space assets developed via public funds.
  - **Department of Space (DoS):** Acts as the nodal agency overseeing overall policy guidance, international diplomacy, and dispute resolution.
- **Operational & Economic Components**
  - **Non-Governmental Entities (NGEs):** Empowers startups and private industries to conduct end-to-end space activities independently.
  - **Infrastructure Sharing:** Provides an institutional setup for private players to access public ISRO facilities and launchpads for a fee.

- **Open Data Access:** Establishes a framework for the seamless, free dissemination of low-resolution satellite data for civilian research.
- **New Frontiers Framework:** Sets guidelines for emerging domains like private space mining, resource ownership, and Space Situational Awareness (SSA).

Institutional roles defined by the Indian Space Policy 2023

- **IN-SPACe (Indian National Space Promotion and Authorisation Centre)**
  - **Single-Window Regulator:** Acts as the autonomous, single-window nodal agency to authorize and oversee all space activities for both government and private players.
  - **Private Promoter:** Mandated to handhold, promote, and guide Non-Governmental Entities (NGEs) while facilitating industry clusters and technology transfers.
  - **Infrastructure Bridge:** Functions as the clearinghouse ensuring private industries get equitable, fee-based access to publicly funded ISRO facilities.
- **ISRO (Indian Space Research Organisation)**
  - **R&D Powerhouse:** Transitions out of manufacturing operational systems to focus exclusively on cutting-edge research and development.
  - **Deep-Space Pioneer:** Spearheads high-risk, long-term national prerogatives like deep-space exploration, planetary science, and human spaceflight (e.g., Gaganyaan).
  - **Technology Transferor:** Mandated to share its mature tech, data, and processes with the domestic industry for commercial scale-up.
- **NSIL (NewSpace India Limited)**
  - **Commercial Executioner:** Functions as the primary Public Sector Undertaking (PSU) responsible for commercializing space technologies built via public funds.
  - **Market-Driven Procurement:** Handles the manufacturing, procurement, leasing, and trade of space assets based strictly on commercial principles.
- **Department of Space (DoS)**
  - **Policy Architect:** Serves as the apex oversight body responsible for high-level policy guidelines, resource distribution, and strategic planning.
  - **Global Diplomat:** Acts as the nodal department coordinating international cooperation and shaping global space governance with foreign ministries.
  - **Dispute Resolver:** Responsible for establishing appropriate administrative mechanisms to resolve conflicts among various space stakeholders.
- **NGEs (Non-Governmental Entities / Private Sector)**
  - **End-to-End Operators:** Legally empowered to independently build, own, and launch satellites, rockets, and ground tracking stations.
  - **Resource Owners:** Allowed to engage in commercial space mining and legally own, transport, or sell recovered asteroid resources.

Issues associated with the Indian Space Policy (ISP) 2023

- **Legal and Statutory Deficiencies**
  - **Lack of Legislative Backing:** ISP 2023 is merely an executive policy framework rather than a statutory law, reducing long-term regulatory certainty.
  - **Absence of a National Space Act:** Missing dedicated law leaves critical issues like international liability, insurance caps, and intellectual property rights uncodified.

- **Liability Allocation Ambiguity:** Under the international Outer Space Treaty (OST), the Indian state remains fully liable for damages caused by private space objects, with no clear domestic law to assign risk or indemnity limits to startups.

- **Regulatory & Institutional Overlap**
  - **Conflicts of Interest:** IN-SPACe serves as the regulator but functions directly under the Department of Space (DoS), casting doubts on its structural autonomy from ISRO.
  - **Agnikul/Skyroot Delay Risks:** Startups face single-window processing bottlenecks as state-sponsored national missions often take structural priority over commercial private launches.
  - **Vague Dispute Resolution:** Administrative mechanism for arbitrating commercial disputes between NGEs (Non-Governmental Entities) and state bodies like NSIL/ISRO remains poorly defined.
- **Financial & Market Bottlenecks**
  - **Capital Deficit vs. Global Giants:** Indian space startups lack the massive scale of venture capital funding compared to global behemoths like SpaceX or Blue Origin.
  - **Low Global Market Share:** Despite reforms, India controls less than 2-3% of the multi-billion-dollar global space market, leaving a steep mountain to climb to reach its 10% target.

    IN-SPACe confirms India controls roughly 2% ($8.4 billion) of the global space market, with a strict policy road map targeting an increase to 8% ($44 billion) by 2033
  - **High Insurance Premiums:** Global and commercial satellite launch insurance remains prohibitively expensive for domestic startups due to unmanaged liability frameworks.
- **Technical & Structural Limitations**
  - **Heavy Infrastructure Dependence:** Private entities remain completely reliant on ISRO's test facilities and launchpads, creating acute infrastructure queues.

    The Indian domestic supply chain remains dependent on foreign vendors for over 70% of highly specialized space-qualified materials.
  - **Import Dependency for Core Tech:** India continues to rely on foreign supply chains for over 70% of space-qualified components, such as high-end semiconductors and carbon fibers.
  - **Reusability Technology Gap:** India's Next-Generation Launch Vehicle (NGLV) with reusable tech is still in early R&D, leaving local launch costs economically less competitive internationally.
- **Security & Sovereignty Concerns**
  - **Weaponization & Geopolitical Competition:** Policy does not adequately address defensive space postures against rapid counter-space advancements by regional rivals like China.
  - **Cybersecurity Vulnerabilities:** Increased private commercial nodes in space data handling exponentially widen the attack surface for state-sponsored cyber strikes.
  - **Orbital Slot Scarcity:** Slow frequencies in domestic ITU (International Telecommunication Union) filings mean India risks losing out on prime low-Earth orbit (LEO) real estate to global constellation networks.

Way Forward for the Indian Space Policy 2023

- **Legislative and Statutory Strengthening**
  - **Enact a National Space Act:** Pass comprehensive legislation to convert executive guidelines into statutory law, providing permanent legal certainty to global investors.
  - **Codify Sovereign Liability Rules:** Establish clear domestic frameworks defining limits for financial risk-sharing and insurance caps between the government and private operators.
  - **Secure Intellectual Property Rights (IPR):** Formulate transparent patent and commercial licensing guidelines to protect private technology innovations during public-private transfers.

- **Institutional and Operational Streamlining**
  - **Ensure Structural Autonomy for IN-SPACe:** Gradually decouple the single-window regulator from the Department of Space (DoS) to prevent any perceived conflict of interest or bias toward ISRO.
  - **Accelerate Approval Processes:** Streamline defense and dual-use clearances to eliminate operational launch bottlenecks for space startups.
  - **Institutionalize Dispute Arbitration:** Create dedicated commercial tribunals under NSIL/IN-SPACe to guarantee fast-track resolution of industrial space disputes.
- **Financial and Market Enhancements**
  - **Leverage Liberalized FDI Norms:** Effectively channel the 100% Foreign Direct Investment automatic/government routes into long-term infrastructure and assembly pipelines.
  - **Expand Risk Capital & Anchor Orders:** Scale up state-backed venture funds (like the ₹1,000 crore SIDBI pool) and guarantee minimum government procurement contracts to steady startup revenue.
  - **Create Affordable Space Insurance Pools:** Partner with domestic financial syndicates to build dedicated consortia capable of underwriting high-risk space ventures at competitive rates.
- **Infrastructure and Technological Scaling**
  - **Build Dedicated Commercial Spaceports:** Construct private-access launchpads to decongest ISRO's test installations and support aggressive launch cadences.
  - **Fast-Track Reusability R&D:** Prioritize developmental milestones for the Next-Generation Launch Vehicle (NGLV) and the Pushpak Reusable Launch Vehicle (RLV) to slash per-kilogram orbit costs.
  - **Indigenize Critical Tech Supply Chains:** Implement targeted production-linked incentives (PLI) to eliminate deep reliance on foreign space-grade semiconductors and carbon-fiber composites.
- **Geopolitics, Security, and Sustainability**
  - **Operationalize Defense-Space Synergies:** Strengthen the integrated pipeline between private space tech providers and the Defence Space Agency (DSA) to augment borders and maritime surveillance.
  - **Pioneer Active Debris Mitigation:** Build competitive commercial capabilities in Space Situational Awareness (SSA) to clean orbital lanes and secure crucial slots in Low Earth Orbit (LEO).
  - **Lead Global Space Governance:** Leverage high-profile forums and strategic alliances (e.g., Artemis Accords, iCET) to place India at the forefront of crafting international space rules.

Conclusion

Indian Space Policy 2023 represents a paradigm shift that successfully transforms India's space sector from a state-monopolized domain into a collaborative, market-driven ecosystem. By establishing a clear separation of powers between regulation, commercialization, and core research, it effectively unlocks private-sector innovation while freeing ISRO to pioneer deep-space exploration.

3. HUMAN SPACE FLIGHT PROGRAM

India's human space flight program, primarily known as the Gaganyaan Mission, is led by the Indian Space Research Organisation (ISRO). Its goal is to indigenously launch three astronauts (Vyomanauts) to a Low Earth Orbit (~400 km) for up to three days, placing India as the fourth nation to achieve independent human spaceflight capability.

Objectives of India's Human Spaceflight Programme (Gaganyaan)

- **Core Technical Objectives**

- **Demonstrate Human Space Capability:** Launch a 3-member crew to a 400 km Low Earth Orbit (LEO) for a 1-to-3 day mission.
- **Master Safe Return Technologies:** Achieve perfect mastery over critical re-entry physics, heat shield performance, and safe ocean splashdown recovery.
- **Validate Crew Safety Infrastructure:** Test and deploy a reliable Crew Escape System (CES) to protect astronauts during any launch-pad or flight emergencies.
- **Indigenize Life Support Systems:** Build and operationalize native Environmental Control and Life Support Systems (ECLSS) to manage cabin pressure, oxygen, and waste.

- **Scientific & Research Objectives**
  - **Conduct Microgravity Experiments:** Perform unique scientific payloads in zero-gravity across medicine, materials science, and biotechnology.
  - **Study Human Physiology in Space:** Gather crucial biomedical data on how the human body reacts to long-term space environments and radiation.
  - **Pioneer Future Deep-Space Tech:** Lay down the fundamental technical building blocks for long-duration human missions to the Moon, Mars, and beyond.
- **Strategic & Geopolitical Objectives**
  - **Elite Space Club:** Position India as only the fourth nation globally to independently launch humans into space (after Russia, USA, and China).
  - **Establish the Bharatiya Antariksha Station:** Use the program as a direct stepping stone to launch India's own indigenous space station by 2035.
  - **Enhance Global Diplomatic Leverage:** Strengthen India's position in international space governance, opening doors for collaborative deep-space treaties.

Features of India's Human Spaceflight Programme (Gaganyaan)

- **Launch Vehicle and Orbital Parameters**
  - **Human-Rated Launcher:** Utilizes the specially modified LVM3 rocket, redesignated as **HLVM3** (Human Rated Launch Vehicle Mark 3).
  - **Target Orbit Profile:** Designed to inject the orbital crew module into a Low Earth Orbit (LEO) at an altitude of approximately **400 km**.
  - **Mission Duration:** Planned to sustain a 3-member crew in space for a duration ranging from **1 to 3 days**.
- **Spacecraft and Modular Architecture**
  - **Two-Module Spacecraft:** Consists of a **Crew Module (CM)** housing the astronauts in a habitable environment and a **Service Module (SM)** containing propulsion systems.
  - **Native ECLSS System:** Equipped with an indigenous Environmental Control and Life Support System to maintain pressure, remove carbon dioxide, and provide oxygen.
  - **Thermal Protection Shield:** Integrated with advanced carbon-phenolic heat shields to withstand extreme re-entry temperatures exceeding 1,500 degrees Celsius.
- **Crew Safety and Emergency Systems**
  - **Crew Escape System (CES):** Features a high-thrust solid motor mechanism to pull the crew module safely away from the launch vehicle in case of a pad or ascent abort.
  - **Multi-Stage Parachute System:** Utilizes a cluster of redundant parachutes to decelerate the crew module smoothly prior to touchdown.
  - **Splashdown Recovery Network:** Managed primarily in the Indian Ocean (Arabian Sea) via close coordination with the Indian Navy and Indian Coast Guard.

- **Training and Support Infrastructure**
  - **Astronaut Training Hub:** Headquartered at the **Astronaut Training Facility (ATF)** in Bengaluru for physical, simulator, and survival training.
  - **Space Medicine Support:** Backed by specialized medical wings, including IAF's Institute of Aerospace Medicine, to monitor real-time crew health metrics.
  - **Vyommitra Humanoid:** deploys a female half-humanoid robot in uncrewed precursor flights to validate microgravity and life-support efficiency.

Challenges associated with India's Human Spaceflight Programme

- **Technological and Engineering Challenges**
  - **Stringent Human-Rating Thresholds:** Upgrading the LVM3 rocket to a **human-rated** vehicle (**HLVM3**) requires an exceptionally high safety reliability margin of over 99%.
  - **Complex Environmental Control (ECLSS):** Mastering the indigenous tech to continuously recycle oxygen, scrub carbon dioxide, and manage temperature/humidity within a small capsule.
  - **Precision Atmospheric Re-entry:** Designing thermal protection shields to endure friction-induced temperatures soaring up to 1,500 degrees Celsius during re-entry.
  - **Flawless Crew Escape System (CES):** Perfecting quick-acting solid motors to instantaneously jettison the crew module away during catastrophic launch or ascent failures.
- **Physiological and Psychological Human Factors**
  - **Zero-Gravity Physiological Degradation:** Mitigating bone density loss, muscle atrophy, and cardiovascular deconditioning caused by prolonged microgravity exposure.
  - **Space Radiation Hazards:** Protecting astronauts from intense solar flares and cosmic radiation without adding excessive structural weight to the module.
  - **Isolation and Psychological Stress:** Managing cabin fever, sleep disorders, and cognitive fatigue due to confinement in an extremely restricted orbital space.
- **Logistical and Infrastructure Bottlenecks**
  - **Nascent Training Infrastructure:** Historically heavy reliance on foreign agencies (like Russia's Glavkosmos) due to a lack of complete domestic simulators and centrifuges.
  - **Complex Marine Recovery Networks:** Establishing rapid, real-time tracking and naval retrieval protocols across unpredictable open sea waters (Arabian Sea/Indian Ocean).
  - **Sustained Ground Tracking Architecture:** Maintaining uninterrupted deep-space communications requires a globally dispersed network of ground stations and tracking ships.
- **Financial and Operational Constraints**
  - **High Capital Requirements:** Managing a massive budget (~₹9,000–10,000 crores) where project delays systematically escalate overall economic costs.
  - **Opportunity Cost Dilemma:** Balancing monumental budgets for human exploration against pressing socioeconomic developmental and civilian space applications.

Way Forward for India's Human Spaceflight Programme

- **Phased Mission Execution and Technology Validation**
  - **Sustain Precursor Milestones:** Systematically complete the scheduled uncrewed flights (including the *Vyommitra* humanoid tests) to mathematically validate system redundancies.
  - **Achieve Flawless Crewed Baseline:** Execute the maiden human launch into Low Earth Orbit (LEO) with zero safety anomalies to establish baseline operational confidence.
  - **Master Orbital Rendezvous & Docking:** Prioritize the development of automated orbital docking mechanisms essential for module connection and supply runs.

- **Infrastructure and Industrial Scaling**
  - **Indigenize Life Support Systems (ECLSS):** Transition completely from foreign sub-components to 100% home-grown life-support and cabin-pressurization technology.
  - **Incentivize Private Aerospace Participation:** Handhold domestic private consortia to mass-manufacture spacecraft modules, structural assemblies, and crew-support equipment.
- **Institutional Training and Space Medicine**
  - **Expand Native Astronaut Facilities:** Fully operationalize Bengaluru's Astronaut Training Facility (ATF) to cut down long-term training dependence on international space agencies.
  - **Deepen Space Medicine Collaborations:** Institutionalize formal research frameworks (like the ISRO-AIIMS partnership) to discover medical countermeasures for long-term space radiation and muscle atrophy.
  - **Standardize Space Psychology Protocols:** Formulate rigorous domestic psychological training matrices to shield crews from deep isolation distress.
- **Next-Generation Strategic Horizons**
  - **Phased Deployment of BAS:** Execute the five-module layout for the **Bharatiya Antariksha Station (BAS)**, targeting the first unit by 2028 and completion by 2035.
  - **Develop the Next-Gen Launcher (NGLV):** Fast-track the R&D of heavy-lift, reusable cryogenic launch vehicles to drastically decrease per-mission operational budgets.
  - **Chart the 2040 Manned Lunar Roadmap:** Establish a rigorous cascading pipeline of robotic sample-returns and lunar flybys culminating in an Indian crewed landing on the Moon by 2040.

Conclusion

Human Spaceflight Programme (Gaganyaan) marks a historic transition for India from basic satellite application to advanced crewed deep-space capabilities. By mastering critical life-support, emergency safety, and re-entry technologies, it propels India into an elite group of spacefaring nations while driving massive domestic innovation across science and industry.

4. BHARATIYA ANTARIKSH STATION

Bharatiya Antariksha Station (BAS) is India's planned indigenous space station, spearheaded by ISRO to establish a permanent human presence in Low Earth Orbit by 2035. It will serve as a microgravity research laboratory to conduct advanced scientific experiments in biology, space medicine, and materials science.

Historical background of Bharatiya Antariksh Station

- **Early Announcements**
  - **Initial Conception (2019):** Officially conceptualized and announced by ISRO in 2019 as the next logical milestone following the standalone Gaganyaan mission.
  - **Amrit Kaal Vision 2047 Alignment:** Embedded formally into India's long-term space vision to drive self-reliance during the country's centenary era.
- **Policy & Statutory Approvals**
  - **Gaganyaan Expansion (2024):** Formally absorbed under an expanded, revised scope of the multi-crore Gaganyaan Programme by the Union Cabinet in September 2024.
  - **Cabinet Fiscal Clearance:** Granted independent regulatory and budgetary sanctions to support the launch of its earliest base unit.
  - **National Review Finalization (2025):** Overall five-module structural architecture successfully cleared formal evaluation by the National Level Review Committee by late 2025.

- **Technological Milestones**
  - **Space Recovery Mastery (2007):** Rooted in early tech breakthroughs, starting with the Space Capsule Recovery Experiment (SRE-1) validating re-entry physics.
  - **Orbital Platform Testing (POEM):** Leveraged the PSLV Orbital Experimental Module (POEM) series to test persistent power generation and solar arrays in low Earth orbit.
  - **Fuel Cell Breakthrough (2024):** Validated vital life-support systems through successful zero-gravity fuel cell power system (FCPS) tests onboard POEM-3.
  - **SpaDeX Autonomous Docking (2025):** Successfully executed the Space Docking Experiment (SpaDeX) mission to prove the precise alignment and autonomous locking of distinct orbital vehicles.

Features of the Bharatiya Antariksha Station (BAS)

- **Structural and Physical Architecture**
  - **Modular Layout Design:** Comprises a five-module configuration (Base, Core, Science, Lab, and Common Working Modules) launched sequentially in distinct phases.
  - **Total Mass and Footprint:** Engineered to weigh a combined mass of ~52 tonnes upon completion, with final dimensions spanning 27 meters by 20 meters.
  - **Volumetric Capacity:** Provides an estimated total pressurized volume of 265 m^3 with a dedicated internal habitable living space of around 105 m^3.
  - **Phased Launch Timeline:** Initial base module (BAS-01, weighing ~10 tonnes) is slated for launch by 2028, with full operational capability targeted for 2035.
- **Orbital and Trajectory Characteristics**
  - **Low Earth Orbit Placement:** Designed to stabilize in a circular Low Earth Orbit (LEO) at an altitude of 400 to 450 km above the Earth's surface.
  - **Global Interoperable Inclination:** Positioned at an orbital inclination of 51.5° to 51.6°, allowing seamless tracking and accessible launch windows from global spaceports.
  - **Persistent Velocity Orientation:** Features an optimized velocity vector orientation to reduce atmospheric drag and ensure highly efficient station-keeping operations.
- **Crew Capacity and Life Support Systems**
  - **Nominal Habitation Range:** Scaled to accommodate a nominal crew size of 3 to 4 astronauts for extended stays lasting three to six months.
  - **Short-Duration Surge Capacity:** Built to comfortably sustain a maximum surge capacity of up to 6 astronauts during brief crew-handover intervals.
  - **Indigenous Closed-Loop ECLSS:** Integrates home-grown Environmental Control and Life Support Systems to automatically manage cabin pressure, recycle water, and scrub carbon dioxide.
  - **Dedicated Escape Sanctuary:** Features a permanently docked safety crew-escape module to guarantee immediate astronaut extraction during extreme emergency scenarios.
- **Technical and Interoperable Subsystems**
  - **Universal Docking Framework:** Uses the native Bharat Docking System (BDS), which incorporates international standards to allow interoperability with global spacecraft.
  - **Common Berthing Mechanisms:** Equipped with multiple dedicated Common Berthing Mechanism (CBM) ports on Module 5 to dock cargo haulers and orbital crew crafts.
  - **Robotic Manipulation Assembly:** Integrates an external robotic manipulator arm (tested via precursor POEM missions) to assist in autonomous module assembly and spacewalks (EVA).
  - **Decentralized Solar Array Networks:** Outfitted with independent, adjustable solar panel wings across 4 out of the 5 modules to guarantee redundant power generation.

- **Research and Mission Functionality**
  - **Microgravity Scientific Lab Hub:** Designed with dedicated experimental bays to fast-track microgravity research in muscle atrophy, fluid dynamics, and advanced materials processing.
  - **Deep Space Stepping Stone:** Built explicitly as a technical testing playground to evaluate the precise deep-space technologies needed to land an Indian on the Moon by 2040.

Modules configuration of the Bharatiya Antariksha Station (BAS)

- **BAS-01: Base Module**
  - **Target Launch Timeline:** Scheduled to be the premier module launched into Low Earth Orbit (LEO) by December 2028.
  - **Structural Dimensions:** Measures approximately **3.8 meters in diameter and 8 meters in length**, weighing around 9 to 10 tonnes.
  - **Launch Vehicle:** Configured to be launched using India's standard human-rated LVM3 vehicle.
- **BAS-02: Core Module**
  - **Station Governance Function:** Acts as the primary architectural nucleus for overall station control, command, and structural telemetry integration.
  - **Mass Profiling:** Engineered with an envelope of 3.8 meters in diameter and 9.25 meters in length, scaling to a mass of **~10 tonnes**.
  - **Launch Configuration:** Optimized for orbital delivery via the heavy-lift LVM3 platform.
- **BAS-03 & BAS-04: Science and Laboratory Modules**
  - **BAS-03 Scientific Mandate:** Dedicated explicitly to microgravity experimental racks across biotechnology, human physiology, and materials science.
  - **BAS-04 Operational Mandate:** Configured for long-term observation equipment, advanced space-manufacturing technology demonstrations, and permanent storage.
  - **Mass and Launcher Scaling:** Each module expands up to ~10.6 to 10.9 tonnes, requiring the upscaled semi-cryogenic LVM3-SC rocket for deployment.
- **BAS-05: Common Berthing Mechanism (CBM) Module**
  - **Universal Interoperability Node:** Integrates the standardized Bharat Docking System (BDS) to enable cross-compatible locking with global spacecraft.
  - **Multi-Port Architecture:** Serves as a central junction point featuring multiple berthing slots to concurrently anchor cargo crafts and crew transit modules.
  - **Heaviest Modular Segment:** Represents the densest single structural unit of the station configuration, scaling to a total mass of ~11 tonnes.
  - **Robotic Manipulator (RM) Arm:** Functions as an external mechanical limb to assist in autonomous modular layout alignment and structural maintenance.

Significance of the Bharatiya Antariksha Station (BAS)

- **Scientific and Technological Advancements**
  - **Microgravity Research Frontier:** Establishes a permanent, home-grown platform to conduct advanced zero-gravity experiments in biology, metallurgy, and chemistry.
  - **Pharmaceutical Innovation Hub:** Accelerates cutting-edge crystallization research to formulate life-saving drugs and vaccines under microgravity conditions.
  - **Testing Ground for Deep-Space Subsystems:** Serves as a low-Earth orbit test bed to evaluate indigenous life-support loops (ECLSS), power grids, and radiation shields.
- **Strategic and Geopolitical Importance**

- **Sovereign Space Autonomy:** Eliminates strategic dependence on international platforms like the ISS, ensuring absolute data security and national mission privacy.
- **Global Space Seat:** Elevates India into an elite tier of spacefaring powers, providing heavy diplomatic leverage to co-author future international space laws.
- **Lunar Exploration Stepping Stone:** Acts as an essential orbital logistics checkpoint, providing the docking and refueling benchmarks needed for a 2040 crewed Moon landing.
- **Countering Space Monopolies:** Balances geopolitical dynamics in Asia by establishing a competitive counterweight to China's operational Tiangong space station.

- **Economic and Industrial Transformation**
  - **Expanding Global Market Share:** Serves as a vital driver to scale up India's contribution to the global space economy from 2% toward its targeted 10%.
  - **Catalyzing Private Space Consortia:** Generates high-volume, precision manufacturing orders for domestic MSMEs and private aerospace startups.
  - **High-Tech Employment Generation:** Creates thousands of direct and indirect niche jobs across structural engineering, data analytics, and space medicine.
- **Human Capital and Socio-Cultural Impact**
  - **Pioneering Space Medicine Knowledge:** Gathers vital biometric data on long-term physiological changes, bone density loss, and radiation impacts on Indian body types.
  - **National STEM Inspiration Catalyst:** Galvanizes millions of domestic students to pursue advanced research careers by providing accessible microgravity science slots.
  - **Fostering Academic Integration:** Promotes deep structural partnerships between premier public research bodies (like IISc and IITs) and global space science faculties.

Issues related to the Bharatiya Antariksha Station (BAS)

- **Technological and Engineering Complexities**
  - **Unproven In-Orbit Refueling:** Mastering automated fuel transfer in a zero-gravity vacuum remains a highly complex, unvalidated technology for ISRO.
  - **Robotic Manipulation Assembly:** Operating heavy external robotic arms for autonomous structural attachment demands flawless, high-precision engineering algorithms.
  - **Environmental Control Loops (ECLSS):** Transitioning from short-duration capsules to permanent, multi-year closed loops that recycle water and oxygen seamlessly is a massive leap.
  - **Long-Term Structural Integrity:** Mitigating the cumulative atmospheric drag and constant micro-vibrations over decades requires specialized modular stress management.
- **Financial and Budgetary Constraints**
  - **Low National R&D Spend:** India's stagnant gross expenditure on R&D (~0.7% of GDP) fundamentally restricts aggressive long-term space lab funding.

    > According to the Ministry of Science & Technology, India's Gross Expenditure on Research and Development (GERD) remains largely flat at 0.6% to 0.7% of national GDP.

  - **Operational Costs**: Unlike one-off satellite launches, a space station imposes relentless, multi-decade financial obligations via recurring supply logistics.
  - **Opportunity Cost Balance:** Convincing policymakers to allocate massive capital (~₹20,000+ crore overall program) over urgent terrestrial socio-economic priorities.
- **Astronaut Health and Biomedical Hazards**
  - **Severe Physiological Hazards:** Sustained microgravity exposure risks acute bone density degradation, muscular atrophy, and cerebral fluid displacement.

  - **Cosmic Radiation Exposure:** Shielding crew members from lethal solar particle events and deep-space cosmic rays without adding massive, prohibitive weight.
  - **Psychological Isolation Confinement:** Managing severe sensory deprivation, depression, and long-term interpersonal friction within highly restricted modular cabins.
- **Space Environment and Operational Risks**
  - **Hyper-Velocity Space Debris:** Navigating an increasingly cluttered Low Earth Orbit (LEO) where paint flecks traveling at 28,000 km/h can breach station hulls.
  - **Launch Vehicle Liftoff Bottlenecks:** Heavy module deployment completely depends on scaling up the Next Generation Launch Vehicle (NGLV) to handle massive modular components economically.
  - **Uninterrupted Tracking Logistics:** Ensuring seamless, real-time telemetry links requires building a vast, expensive constellation of data-relay satellites and global maritime tracking ships.
- **Geopolitical and Regulatory Friction**
  - **Navigating Strategic Alignments:** Balancing autonomous national control with the deep technological insights needed from established space agencies like NASA or Roscosmos.
  - **Regional Space Race Dynamics:** Intensifying counter-space capabilities and geopolitical tensions with neighbors like China, which already runs the operational Tiangong station.
  - **Outer Space Treaty Compliance:** Managing strict international governance compliance regarding orbital slot allocation and operational liability in LEO zones.

Way Forward for the Bharatiya Antariksha Station (BAS)

- **Technology Demonstration and Incremental Testing**
  - **Execute SPADEX Missions:** Prioritize the scheduled SPADEX-2 and SPADEX-3 missions to perfect high-precision orbital docking and pressurized module connectivity.
  - **Validate Uncrewed Tests:** Conduct the scheduled uncrewed modular test runs between 2026 and 2028 to gather critical baseline telemetry before launching the base module.
  - **Perfect In-Orbit Refueling:** Fast-track experimental development of automated, zero-gravity fuel-transfer protocols necessary for maintaining multi-decade station orbits.
- **Heavy-Lift Infrastructure and Logistics**
  - **Scale Up the Next Generation Launch Vehicle:** Accelerate the development of the high-capacity, reusable NGLV to economically transport heavy 10-to-15 tonne modules.
  - **Deploy a Heavy Data-Relay Constellation:** Launch a dedicated network of Indian Data Relay Satellites (IDRSS) to secure uninterrupted, 24/7 ground communication loops with the station.
- **Financial Sustainability and Private Integration**
  - **Utilize Expanded Gaganyaan Allocations:** Strategically deploy the revised ₹20,193 crore funding pool to achieve early construction targets for the BAS-01 base module by 2028.
  - **Incentivize Private Manufacturing Consortia:** Transition routine structural fabrication, solar array production, and launch hardware to private industries and MSMEs via IN-SPACe.
  - **Monetize Through Microgravity Services:** Formulate international commercial frameworks to lease out future lab space to foreign nations, pharmaceutical companies, and private researchers.
- **Human Capital and Space Medicine**
  - **Indigenize Life Support Loops (ECLSS):** Pivot entirely from imported parts to domestic manufacturing for 100% reliable, closed-loop oxygen and water recycling tech.
  - **Institutionalize Space Bio-Medicine**: Expand partnerships between ISRO, national premier research labs, and medical institutions to study long-term zero-gravity physiological changes.
  - **Democratize Research Payloads:** Open up dedicated payload slots on the station for domestic universities to build a robust pipeline of young space-tech scholars.

- **Geopolitics, Debris Management, and Alliances**
  - **Maximize Strategic Alliances:** Leverage frameworks like the Artemis Accords for astronaut cross-training while preserving India's absolute strategic autonomy over BAS operations.
  - **Reinforce Debris Defenses:** Strengthen Space Situational Awareness (SSA) networks (e.g., Project NETRA) to insulate the station from LEO hyper-velocity junk threats.

International Space Station (ISS)

- **Institutional Framework**
  - **Joint Collaborative Venture:** Co-managed by five premier space agencies: NASA (USA), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and CSA (Canada).
  - **Intergovernmental Agreement (1998):** Governed by the 1998 Space Station Intergovernmental Agreement establishing international jurisdictional laws in space.
  - **Human Presence:** Maintained a persistent, uninterrupted human presence in orbit for over 25 consecutive years since November 2000.
- **Orbital and Structural Specifications**
  - **Low Earth Orbit (LEO): Maintained** at a circular orbital altitude of approximately 400 to 420 km above the Earth.
  - **High-Velocity Trajectory:** Travels at a velocity of ~27,600 km/h, completing one full orbit around Earth roughly every 93 minutes.
  - **Modular Assembly Architecture:** Comprises interconnected pressurized modules, solar arrays, and a large central truss backbone.
- **Scientific and Research Mandate**
  - **Microgravity Research Frontier:** Serves as a unique zero-gravity laboratory to study fluid dynamics, biotechnology, and complex materials processing.
  - **Deep Space Testing Bed:** Evaluates critical life-support, communication, and radiation-shielding subsystems needed for long-duration Moon and Mars voyages.
  - **Global Environmental Monitoring:** Captures continuous, high-resolution global telemetry data to monitor climate change, ocean health, and natural disasters.

Conclusion

Bharatiya Antariksha Station (BAS) is a landmark project that will establish India's permanent, independent foothold in Low Earth Orbit by 2035. By driving breakthrough microgravity research and pioneering indigenous life-support technologies, it will transform domestic scientific industries and aerospace startups.

5. INDIA'S LUNAR EXPLORATION PROGRAM

India's Lunar Exploration Programme, known as the **Chandrayaan program**, is a series of ongoing outer-space missions spearheaded by ISRO to systematically map and study the Moon. The program achieved a historic global milestone with Chandrayaan-3, which made India the very first nation to successfully soft-land a spacecraft near the lunar south pole.

Significance of India's Lunar Exploration Programme

- **Scientific Discovery and Planetary Evolution**
  - **Pioneering Southern Polar Geography:** Yielded the world's premier in-situ data from the unexplored high-latitude lunar south pole via Chandrayaan-3.
  - **Definitive Lunar Water Confirmation:** Chandrayaan-1's Moon Impact Probe (MIP) definitively confirmed the existence of sub-surface hydroxyl and water ice molecules.

- **Sub-Surface Regolith Stratigraphy:** Upcoming sample-return tasks (Chandrayaan-4) aim to excavate ancient rock lines to decode 4.3-billion-year-old lunar magma ocean history.
- **Seismic and Thermal Profiling:** Payloads like ChaSTE and ILSA mapped unprecedented thermal conductivity and localized moonquakes to chart crustal behavior.

- **Technological Leap and Core Indigenization**
  - **Mastering Deep-Space Logistics:** Validates complex, multi-leg orbital operations, including high-precision cryogenic rocket steering and autonomous soft-landing metrics.
  - **Orbit-to-Orbit Robotics Prowess:** Transitions ISRO toward advanced mechanisms, including robotic sample scooping, autonomous container sealing, and moon-surface liftoffs.
  - **Validating Critical Docking Math:** Drives the operational deployment of the Space Docking Experiment (SPADEX) for dual-spacecraft locking maneuvers in transit orbits.
  - **High-Heat Atmospheric Re-entry:** Perfects the structural thermal engineering required to shield sample returns from friction during ballistic terrestrial re-entry.
- **Geopolitical Alignment and Strategic Leverage**
  - **Elite Global Space Positioning:** Established India as the 4th nation to achieve a soft landing and the absolute pioneer at the lunar south pole.
  - **Shaping Lunar Governance Treaties:** Enhances India's diplomatic weight within the Artemis Accords framework to co-author equitable rules for outer-space resource mining.
  - **Securing High-Value Space Contracts:** Boosts Antrix/NSIL's commercial standing to attract international satellite launching contracts onto Indian heavy lifters like LVM3.

List of India's Chandrayaan Mission

Mission Name	Launch Year	Launch Vehicle	Prime Configuration Components	Objective & Primary Result
Chandrayaan-1	2008	PSLV-C11	Lunar Orbiter & Moon Impact Probe (MIP)	Discovered Lunar Water: Definitively mapped and confirmed hydroxyl (OH) and water-ice presence.
Chandrayaan-2	2019	GSLV Mk III-M1	Orbiter, Vikram Lander, and Pragyan Rover	High-Res Mapping: Lander crash-landed, but the Orbiter remains active, delivering critical terrain and water data.
Chandrayaan-3	2023	LVM3-M4	Propulsion Module, Vikram Lander, Pragyan Rover	Historic South Pole Landing: Made India the 1st country to land at the lunar South Pole; discovered local surface sulphur.
Chandrayaan-4	~2028	Dual LVM3 Rockets	5 Modules (Propulsion, Descender, Ascender, Transfer, Re-entry)	Lunar Sample Return: Designed to drill, collect, and bring back up to 3 kg of lunar soil autonomously to Earth.

Chandrayaan-5 (LUPEX)	2028–2029 (Financial Sanctioned)	JAXA H3 Rocket	ISRO Lander & Heavy JAXA Rover (~350 kg)	Joint Polar Prospecting: Deep drilling (up to 1.5m) to evaluate the state, quality, and

Features of India's Lunar Exploration Programme

- **Mission Architecture and Configuration**
  - **Two-Module Basic Stack:** Standardized the use of an orbital Propulsion Module to ferry the heavy landing stack from Earth-injection lines into a 100 km circular lunar orbit.
  - **Five-Module Return Stack (Chandrayaan-4):** Integrates five distinct modules (Propulsion, Descender, Ascender, Transfer, and Re-entry) spread across two separate rocket launches.
  - **Dual Earth-Orbit Docking (SPADEX):** Relies on autonomous in-orbit rendezvous and modular locking math to assemble the complete sample-return craft directly in Earth orbit before heading to the Moon.
- **Targeting and Landing Framework**
  - **High-Latitude South Pole Focus:** Specifically optimized to target the Moon's southern polar crust (e.g., Mons Mouton for Chandrayaan-4).
  - **Autonomous Hazard Avoidance (LHDAC):** Outfitted with high-speed processing Hazard Detection and Avoidance Cameras to evaluate real-time terrain slopes during terminal descent.
  - **Throttleable Cryogenic Propulsion:** Utilizes clusters of high-precision, throttleable liquid engines (typically 800 Newtons each) to ensure smooth deceleration and soft-landing execution.
- **Surface Exploration and Subsurface Robotics**
  - **Multi-Modal Surface Sampling:** Employs a combination of mechanical surface scoops and deep rotary drills on the lander to harvest and store lunar soil intact.
  - **Deep Subsurface Core Augers (LUPEX):** Scales up drill depths significantly to excavate deep ice mixed with rock up to 1.5 meters down.
  - **Sealed Vacuum Containerization:** Features automated sealing mechanisms to isolate returned lunar samples in pristine vacuum conditions, preventing Earthly contamination.
- **International Interoperability and Alliances**
  - **Joint Hardware Integration (Chandrayaan-5):** Divides engineering roles seamlessly, where ISRO builds the high-capacity lander and Japan's JAXA supplies the heavy-duty 350 kg rover.
  - **Global Launch Port Delivery:** Transitions from indigenous heavy launchers (like LVM3) to international rockets (such as Japan's H3 rocket) for heavily loaded joint missions.

Challenges and issues facing India's Lunar Exploration Programme

- **Advanced Technical and Multi-Launch Complexities**
  - **High-Risk In-Orbit Docking:** Upcoming sample-return architecture (**Chandrayaan-4**) requires absolute precision to dock separate modules in Earth orbit via unproven SPADEX math.
  - **Insufficiency of Current Rocket Power:** Current heavy-lifter (LVM3) lacks the single-launch capability to lift a complete five-module return vehicle, forcing complex, dual-launch split missions.
  - **Untested Lunar Lift-Off Mechanics:** Launching an Ascender Module autonomously from the Moon's surface using a landed structure as an improvised launchpad presents severe stabilization risks.
  - **High-Heat Atmospheric Re-entry:** Designing sample capsules that can survive intense thermodynamic friction (~11 km/s) during ballistic re-entry into Earth's atmosphere without burning up.

- **Environmental and Topographical Hazards**
  - **Extreme Polar Temperatures:** Shadowed craters at the lunar South Pole plunge to lethal levels down to **230 ^C**, threatening to instantly crack structural metals and freeze electronics.
  - **Highly Rugged Terrain Navigation:** The southern polar landscape is densely packed with steep slopes, deep craters, and large boulder fields, making landing margins extremely narrow.
  - **Abrasive Lunar Dust (Regolith):** Jagged, electrostatically charged lunar dust easily sticks to moving parts, threatening to scratch camera lenses, degrade solar panels, and jam sample drills.
  - **Limited Solar Power Windows:** Operating in permanently shadowed regions (PSRs) cuts off traditional solar energy lines, requiring heavy battery payloads or risky nuclear power alternatives.
- **Financial and Institutional Constraints**
  - **Relatively Stagnant Budget Allocations:** Despite high-profile successes, ISRO's funding remains a small fraction of NASA's or China's space budgets.

    India's Department of Space was allocated ₹13,705 crore ($1.64 billion) for the 2026–27 fiscal year, a minor fraction of NASA's $25+ billion budget.
  - **Absence of a Comprehensive Space Law:** The long-pending National Space Activities Bill leaves regulatory holes regarding international liability, commercial mining rights, and asset insurance.
  - **High Academic and Industry Brain Drain:** Niche skill shortages persist because premier engineering talent frequently migrates abroad rather than entering domestic research lines.
- **Sample Integrity and Planetary Protection**
  - **Strict Vacuum Contamination Risk:** Sealing excavated soil and ice samples flawlessly in absolute vacuum conditions to prevent volatile gases from escaping or mixing with Earth's atmosphere.
  - **Microbial Bio-Safety Protocols:** Establishing specialized, high-security Lunar Sample Receiving Facilities (LSRFs) domestically to handle raw alien materials without risk of planetary contamination.

Way Forward for India's Lunar Exploration Programme

- **Near-Term Technology Demonstrations**
  - **Execute the SPADEX Benchmark:** Prioritize the upcoming Space Docking Experiment to perfect the precise orbital locking math required for multi-module assembly.

    ISRO successfully validated its autonomous docking algorithm in mid-January 2025 via the Space Docking Experiment (SpaDeX).
  - **Validate Sample-Return Systems:** Fast-track the development of autonomous mechanical scoops and vacuum-sealed container units for **Chandrayaan-4**.
  - **Develop Lunar Ascent Capabilities:** Design and test light-weight, highly stabilized rocketry algorithms to smoothly launch the Ascender Module off the Moon's surface.
- **Heavy-Lift Infrastructure Scaling**
  - **Operationalize the NGLV Rocket:** Accelerate development of the Next Generation Launch Vehicle to shift from complex dual-launch missions to single heavy-lift deployments.
  - **Construct Advanced Terrestrial Curation Facilities:** Establish state-of-the-art Lunar Sample Receiving Facilities (LSRFs) with high biosafety ratings to analyze returned regolith cleanly.
  - **Expand the Deep Space Communication Net:** Upgrade India's Deep Space Network (IDSN) by placing maritime tracking vessels to remove line-of-sight communication gaps.
- **Bilateral and Strategic Alliances**
  - **Deepen the ISRO-JAXA LUPEX Pact:** Solidify the structural interface and weight optimizations for **Chandrayaan-5 (LUPEX)** ahead of its planned 2028 launch window.

- **Maximize Artemis Accords Leverage:** Utilize international frameworks to secure advanced astronaut training slots with NASA while maintaining complete sovereign mission control.
- **Lead Multilateral Lunar Governance:** Actively spearhead global space forums to advocate for equitable resource-sharing and non-conflict zones at the overcrowded lunar south pole.

- **Financial and Industrial Reforms**
  - **Enact a Comprehensive Space Activities Bill:** Expeditiously pass a national space law to fix ambiguous rules regarding private liability, commercial space mining, and asset insurance.
  - **Deepen Private Vendor Integration:** Utilize IN-SPACe to hand over standard rocket stages and satellite buses to private aerospace consortia, allowing ISRO to focus solely on deep-space R&D.
  - **Incentivize Sovereign Venture Funds:** Launch dedicated public-private tech funds to prevent engineering brain drain by financing home-grown space-tech startups and MSMEs.

Conclusion

India's lunar program, highlighted by the upcoming dual-launch Chandrayaan-4 mission, represents a strategic leap that cements the nation's position within elite global space alliances. While deep technical gaps in re-entry and extreme polar environmental hazards present steep hurdles, the programmatic shift toward public-private partnerships offers a sustainable pathway forward.

6. SOLAR AND OBSERVATORY MISSIONS

India's Solar and Space Observatory Programme represents a specialized arm of ISRO dedicated to tracking stellar dynamics, space weather, and deep-space cosmic phenomena from beyond Earth's atmosphere. Platforms like AstroSat, XPoSat, and Aditya-L1 have successfully transitioned India from passive reliance on international telemetry to a primary provider of global space science data.

Solar Missions from India and around the World

Mission Name	Space Agency	Core Objective & Primary Result
Aditya-L1	ISRO (India)	**First Indian Solar Lab:** Tracks coronal heating, Coronal Mass Ejections (CMEs), and real-time space weather dynamics.
Parker Solar Probe	NASA (USA)	Flew directly through the Sun's upper corona to study solar wind acceleration up close.
Solar Orbiter	ESA & NASA (Europe/ USA)	**First Solar Polar Images:** Captures the first-ever close-up images of the Sun's unmapped polar regions.
SOHO	ESA & NASA (Europe/ USA)	**Legacy Observatory**: Longest-running solar tracker; revolutionized understanding of the solar interior and discovered thousands of comets.
STEREO (A & B)	NASA (USA)	**3D Solar Imaging:** Provided the first-ever stereoscopic, three-dimensional images of the entire Sun and CMEs.
Proba-3	ESA & ISRO (Europe/ India Partnership)	**Artificial Eclipse Formation:** Uses precise formation-flying satellites to block the solar disk and reveal the inner corona.
Advanced Space-based Solar Observatory (ASO-S / Kuafu-1)	CNSA (China)	**Solar Magnetic Network:** Simultaneously maps the solar magnetic field, solar flares, and CMEs to forecast space hazards.

Hinode (Solar-B)	JAXA (Japan) / NASA / ESA	**Magnetic Energy Transfer:** Investigates how the Sun's microscopic magnetic fields generate violent explosive flares.

- **Lagrange Point 1 (L1):** A gravitationally stable spot where the pull of the Earth and Sun balance out. Spacecraft like Aditya-L1 and SOHO use it to get a 100% uninterrupted view of the Sun without Earthly eclipses or blockages.
- **The Coronal Heating Paradox:** The surface of the Sun is only about 5,500^C, but its outer atmosphere (corona) burns at over 1,000,000^C. Solving this temperature paradox is the common baseline objective for nearly all global solar missions

Objectives of global and domestic solar missions

- **Solar Atmospheric Dynamics & The Heating Paradox**
  - **Deciphering Coronal Heating:** Investigating why the outermost solar atmosphere (corona) burns at over 1,000,000^C while its surface (photosphere) is just 5,500^C.
  - **Tracking Layered Coupling:** Observing the thermodynamic physics and plasma flows connecting the photosphere, chromosphere, and outer corona.
  - **Mapping Flare Expirations:** Detailing the chronological processes in sub-coronal layers that violently trigger solar flares and sudden UV radiation bursts.
- **Solar Wind and Particle Acceleration Mechanics**
  - **Tracing Solar Wind Origins:** Identifying the exact magnetic structures that accelerate solar wind plasma from subsonic to supersonic speeds.
  - **Analyzing Particle Composition:** Measuring the real-time density, directional energy, and ionic velocity of solar wind electrons, alphas, and protons.
  - **Investigating Transport Prowess:** Decoding how solar energetic particles (SEPs) are injected and transported across interplanetary magnetic highways.
- **Magnetic Field Topology & Eruptive Phenomena**
  - **Quantifying Coronal Magnetism:** Measuring the strength, layout, and rapid topology changes of magnetic fields directly in the outer solar atmosphere.
  - **Understanding Reconnection Events:** Studying how twisting solar magnetic lines snap and explosively realign, releasing immense kinetic energy.
  - **Tracking Coronal Mass Ejections (CMEs):** Examining the early-stage velocity, acceleration, and driving forces behind massive plasma clouds blasted into deep space.
- **Space Weather Forecasting & Planetary Protection**
  - **Real-Time Hazard Alerting:** Building predictive models for Earth-bound geomagnetic storms to shield communication satellites and regional power grids from blackouts.
  - **Evaluating Ionospheric Triggers:** Profiling solar near-UV radiation flux to study its direct impact on Earth's upper atmospheric chemistry and shortwave radio propagation.
  - **Establishing Baseline Exoplanet Signatures:** Using specialized spectro-polarimetry (like SHAPE on Chandrayaan-3) to view Earth as an exoplanet, helping identify habitable solar systems elsewhere.

Why is there a Need for Solar Mission?

- **Space Weather Mitigation and Planetary Protection**
  - **Predicting Geomagnetic Storms:** Provides critical early warnings (up to 1 hour via L1 stations) of incoming Coronal Mass Ejections (CMEs) before they strike Earth.
  - **Shielding Terrestrial Power Grids:** Allows power operators to balance voltage lines to prevent massive solar-induced transformer blowouts and grid failures.
  - **Securing Global Navigation (GPS/GNSS):** Mitigates sudden radio signal degradation in Earth's ionosphere caused by extreme X-ray flare radiation.
  - **Protecting Space Infrastructure Assets:** Gives mission control time to put multi-billion dollar communication satellites into safe mode to prevent electrical circuit shorting.
- **Resolving Fundamental Astrophysics Mysteries**
  - **Decoding the Coronal Heating Paradox:** Investigates why the Sun's outer atmosphere (corona) burns at over 1,000,000^C while its surface is just 5,500^C.
  - **Tracking Solar Wind Accel Math:** Directly observes how solar energetic particles accelerate from subsonic to supersonic speeds inside interplanetary space.
  - **Understanding Magnetic Reconnection:** Documents how twisting magnetic loops snap and explosively realign to generate violent solar eruptions.

Issues facing India's solar missions

- **Advanced Technical & Orbital Constraints**
  - **Unstable L1 Trajectory:** Maintaining a halo orbit around **Lagrange Point 1** requires continuous, high-precision thrust adjustments to prevent gravity drifts.
  - **Mass vs. Fuel Balancing:** Payload instruments are strictly capped due to severe weight and fuel constraints on long-distance cruise routes.

    Due to launch vehicle weight constraints, ISRO's Aditya-L1 was capped at a launch mass of 1,500 kg, carrying only seven lightweight.
  - **Thermal & Radiation Shields:** Sensors demand highly complex, specialized materials to survive continuous exposure to unshielded solar flare radiation.
  - **Single-Vantage Blind Spots:** Observing from one position (L1) provides static views, making it impossible to capture the true 3D shape of multi-directional solar explosions.
- **Environmental & Operational Hazards**
  - **Instrument Blinding Overload:** High-energy particle bursts during heavy solar storms risk permanently blinding or damaging sensitive optical cameras.
  - **Extreme Simulation Caps:** The hyper-intense magnetic fields of the Sun's corona are too volatile to replicate or test accurately in Earth laboratories.
  - **Data Telemetry Gaps:** Severe deep-space solar noise frequently corrupts or interrupts the real-time transmission of research data back to Earth stations.
- **Institutional & Policy Bottlenecks**
  - **Infrastructure Crowding:** Running solar missions alongside heavy lunar (Chandrayaan) and manned (Gaganyaan) timelines overburdens ISRO's active launch pads.
  - **Private Supply Chain Delays:** Domestic aerospace ecosystem struggles to produce space-grade, radiation-hardened components locally.
  - **Regulatory Voids:** A lingering lack of a finalized National Space Law leaves legal gray areas regarding space asset insurance and international liability.

Way Forward for India's solar missions

- **Architectural Scaling & Deep-Space Engineering**
  - **Reusable Next-Gen Rockets (NGLV):** Fast-track the 80% cost-reducing reusable launch vehicle to deploy future multi-spacecraft constellations affordably.
  - **Deploy Formation-Flying Observatories:** Scale up precise dual-satellite formation technology (validated via Proba-3) to block the bright solar disc and expose the faint inner corona continuously.
  - **Build Interplanetary Solar Links:** Equip upcoming deep-space platforms, like the **Venus Orbiter Mission (2028)**, with solar wind analyzers to track radiation trends from different vantage points in the solar system.
  - **Indigenize Radiation-Hardened Circuits:** Build localized semiconductor lines to manufacture space-grade, radiation-shielded electronics capable of surviving harsh solar flares.
- **Multi-Agency Alliances & Global Data Sovereignty**
  - **Expand Open-Source Data Pipelines:** Standardize and share the massive 27+ Terabytes of data harvested by Aditya-L1 globally to establish India as a primary hub for space-weather science.
  - **Standardize Interoperable Networks:** Align India's Deep Space Network (IDSN) tracking frequencies with ESA and NASA protocols to secure uninterrupted fallback telemetry tracking.
  - **Co-Author Lagrangian Defense Accords:** Lead multilateral diplomatic efforts to formulate clear zoning and anti-collision guidelines for the increasingly crowded **Sun-Earth L1** point.
- **Institutional Modernization & Commercial Reforms**
  - **Pass the National Space Activities Bill:** Expeditiously enact a comprehensive national space law to remove legal ambiguities surrounding commercial asset liability and space mining rights.
  - **Utilize the IN-SPACe Venture Capital Fund:** Leverage the newly approved ₹1,000-crore VC fund to nurture private aerospace startups capable of building commercial solar-radiation trackers.
  - **Delegate Routine Telemetry to NSIL:** Outsource standard orbital calculations and station-keeping commands to commercial arms, freeing up ISRO's core team to focus entirely on advanced solar R&D.

Conclusion

India's solar missions, led by platforms like Aditya-L1, have transformed the nation from a consumer of space telemetry into a global leader in space-weather forecasting. By mastering complex lagrangian orbits and providing open-source cosmic data, ISRO is playing a vital role in shielding global communication networks and critical infrastructure from solar hazards.

7. SPACE EXPLORATION MISSIONS

Global space exploration missions have evolved from highly competitive, nationalistic races into deeply collaborative efforts aimed at discovering resources, finding habitable worlds, and understanding our place in the universe. Global space exploration missions are led by pioneering space agencies like ISRO, NASA, and ESA, these missions are pushing beyond Earth's orbit into deep-space domains like the Moon, Mars, and the outer solar system.

Indian and Global Space Exploration Missions

Mission Name	Space Agency / Country	Objective
Chandrayaan-3	ISRO (India)	Historic first-ever soft landing on the Lunar South Pole, discovering surface sulphur.

Mangalyaan (MOM)	ISRO (India)	Made India the first nation to successfully enter Martian orbit on its maiden attempt.
Aditya-L1	ISRO (India)	India's first solar observatory stationed at Lagrange Point 1 to track coronal heating.
Gaganyaan	ISRO (India)	India's flagship upcoming crewed human spaceflight mission into a 400 km orbit.
NISAR	ISRO & NASA	Joint dual-frequency radar mission to monitor global crustal deformations and ecosystem changes.
Artemis Program	NASA (USA)	Multi-nation initiative aimed at returning humans (including the first woman) to the Moon.
James Webb (JWST)	NASA / ESA / CSA	Infrared space observatory capturing the earliest star formations and exoplanet atmospheres.
Perseverance Rover	NASA (USA)	Actively searching for ancient microbial life and collecting core samples in Jezero Crater.
JUICE	ESA (Europe)	Investigating the habitability of Jupiter's icy moons: Ganymede, Callisto, and Europa.
Europa Clipper	NASA (USA)	Conducting high-resolution flybys to confirm the presence of a sub-surface liquid ocean.
Tianwen-1	CNSA (China)	China's first successful all-in-one mission containing an Orbiter, Lander, and Zhurong Rover.
Psyche	NASA (USA)	Exploring a unique, metal-rich asteroid hypothesized to be an exposed iron-nickel protoplanet core.
BepiColombo	ESA / JAXA	Dual-orbiter mission tracking Mercury's magnetic field and core composition under extreme heat.

Significance of space exploration missions

- **Geopolitical & Strategic Dominance**
  - **Enhancing Global Soft Power:** Establishes technological leadership and elevates a nation's negotiating clout in international diplomatic and space forums.
  - **Securing Dual-Use Capabilities:** Drives breakthrough advancements in rocketry, missile guidance, and secure telemetry that naturally strengthen national defense.
  - **Fostering Strategic Alliances:** Cultivates deep transnational partnerships through major collaborative frameworks like the Artemis Accords and joint planet tracking.
- **Socio-Economic & Commercial Yields**
  - **Expanding the Global Space Economy:** Feeds a rapidly growing multi-billion dollar private industrial ecosystem spanning satellite broadband, imaging, and logistics.
  - **Generating High-Value Spin-off Tech:** Spawns critical consumer breakthroughs in everyday sectors like water purification, medical imaging, and solar energy storage.
  - **Securing Long-Term Resource Lines:** Lays the operational groundwork for extraterrestrial mining of rare materials like lunar Helium-3 and platinum-group metals on asteroids.
- **Scientific Innovation & Planetary Security**
  - **Decoding Universe Evolution:** Investigates extreme cosmic environments to answer fundamental questions regarding planetary origin, dark matter, and alien life.

- **Strengthening Planetary Defense:** Builds observation platforms to detect, catalog, and intercept hazardous Near-Earth Objects (NEOs) before potential collisions.
- **Refining Terrestrial Climate Models:** Leverages advanced deep-space sensors to better monitor Earth's changing atmosphere, ocean currents, and dynamic weather patterns.

Issues facing global and domestic space exploration missions

- **Environmental Hazards & Orbital Overcrowding**
  - **Kessler Syndrome Risks:** Explosive growth of mega-constellations drastically increases low-Earth orbit (LEO) crowding, threatening a runaway chain reaction of cascading debris collisions.

    As of mid-2026, the European Space Agency (ESA) tracks over 28,800 orbital objects larger than 10 cm in space.
  - **Ozone Footprint:** Heavy kerosene and solid-fuel rocket launches emit high levels of black carbon and soot directly into the upper stratosphere, depleting the ozone layer.

    According to NOAA, global rocket launches inject roughly 1,000 metric tons of black carbon (soot) into the stratosphere annually.
  - **Non-Cooperative Active Debris Removal:** Capturing and clearing dead satellites remains extremely difficult due to erratic, uncontrolled tumbling and a lack of standard docking handles on older spacecraft
- **Geopolitical Friction & Legal Lacunae**
  - **Emerging Polar Geopolitical Blocs:** Deep space is forcing nations to choose between the US-led **Artemis Accords** and the China-Russia **International Lunar Research Station (ILRS)**.

    Over 40 nations have signed the US-led Artemis Accords, while the China-Russia ILRS bloc has independently secured a dozen country partnerships.
  - **Outdated Outer Space Treaty:** Foundational 1967 international space treaty lacks teeth or clear rules regarding commercial resource ownership, asteroid mining, and property rights.
  - **Threat of Space Militarization:** Rapid testing of direct-ascent anti-satellite (ASAT) weapons, jammer platforms, and orbital cyber warfare compromises peaceful global payloads.

    Anti-satellite (ASAT) strikes, like Russia's 2021 test against the Cosmos 1408 satellite, instantly produced over 1,500 pieces of trackable, mission-threatening debris.
- **Financial, Supply Chain, & Technical Barriers**
  - **Prohibitive Deep-Space Cost Curves:** Constructing high-payload sample-return crafts (like Chandrayaan-4) requires massive public capital, competing with urgent domestic socioeconomic spending.
  - **Critical Vulnerabilities in Advanced Electronics:** Global space programs face persistent export bottlenecks and supply shocks for radiation-hardened semiconductors and specialized carbon fiber components.
  - **Launch Pad Infrastructure Constraints:** A high global reliance on a few single geographical spaceports creates operational traffic bottlenecks as launch frequencies spike.

    Over 65% of all successful orbital launches are bottlenecked through just five major spaceports worldwide (such as Kourou, Cape Canaveral, and Jiuquan.

Way forward for global and domestic space exploration missions

- **Regulatory Modernization & Structural Reforms**
  - **Enact the National Space Activities Bill:** Fast-track the pending comprehensive space law to clarify liability, spectrum allocation, and intellectual property rights for private firms.

US Commercial Space Launch Competitiveness Act (2015) provides clear private ownership rights for asteroid/space resources.

- **Establish Private Aerospace Corridors:** Set up a second dedicated national launch facility to ease scheduling blockages on ISRO's infrastructure and double the annual launch frequency

Kulasekarapattinam Spaceport (Tamil Nadu) is being developed exclusively for private and Small Satellite Launch Vehicles (SSLVs) to de-congest Sriharikota.

- **Deploy IN-SPACe Capital Pools:** Effectively utilize the ₹1,000-crore venture capital fund to incubate deep-tech startups and graduate them from prototypes to commercial scales.

IN-SPACe & SIDBI's ₹1,000-Crore Venture Capital Fund (2025) directly finances and scales local deep-tech space startups.

- **Space Sustainability & Environmental Safeguards**
  - **Codify Space Traffic Management (STM):** Lead international efforts through the UN Committee on the Peaceful Uses of Outer Space to standardise orbital lanes and tracking protocols.

ISRO's IS4OM (System for Safe and Sustainable Space Operations Management) monitors space assets and actively coordinates orbital safety with global tracking nodes.

  - **Mandate Active Debris Removal (ADR):** Incentivise the development of magnetic and robotic capture systems to clear dead satellites and reverse Kessler Syndrome hazards.

ESA's ClearSpace-1 mission is the first commercial multi-million dollar contract to capture and de-orbit a piece of space debris.

  - **Adopt Eco-Friendly Propellants:** Prioritise the transition to green, non-toxic liquid propellants (like methalox) to minimise chemical soot emissions in the stratosphere.

ISRO's green propulsion initiatives (e.g., test-firing ISROENE kerosene variants and eco-friendly monopropellants for Gaganyaan thrusters).

- **Diplomatic Interoperability & Tech Collaboration**
  - **Leverage Multiparty Alliances:** Use frameworks like the Artemis Accords and joint platforms (e.g., NISAR) to co-develop planetary protection defenses and sample-return tech.
  - **Formulate Lunar Heritage Governance:** Co-author multilateral treaties to prevent territorial friction and protect zones of special scientific interest on the Moon.
  - **Bridge the Global South Tech Gap:** Offer cost-effective launch facilities and share open-source climate and hydro-informatics data to champion space-for-development diplomacy.

Conclusion

Space exploration missions have advanced from geopolitical races into vital tools for global technological growth, economic development, and planetary defense. By providing critical data on climate change and space hazards, these missions directly protect modern communications and infrastructure on Earth.

8. INDIAN REGIONAL NAVIGATION SATELLITE SYSTEM (IRNSS)

The Indian Regional Navigation Satellite System (IRNSS), commercially known as **NavIC** (Navigation with Indian Constellation), is India's independent, homegrown regional satellite navigation system developed by ISRO. It is designed to provide highly accurate position and timing information over India and an extended region stretching up to **1,500 km** beyond its geographical borders.

Satellite-Based Navigation Systems

- **Concept & Global Architectures**
  - Satellite navigation relies on measuring the precise time delay of radio signals from at least four satellites to determine a user's exact latitude, longitude, altitude, and time.
  - **Global Systems (GNSS):** Only four systems offer full global coverage: GPS (USA), GLONASS (Russia), Galileo (European Union), and BeiDou (China).
  - **Regional Systems:** Two systems provide localized, high-precision coverage: India's NavIC (IRNSS) and Japan's QZSS (Michibiki).
- **Satellite-Based Augmentation Systems (SBAS)**
  - SBAS uses a network of ground stations to measure ionospheric signal delays and broadcasts correction messages via geostationary satellites to improve standard GPS accuracy.
  - **India's GAGAN System**: Developed jointly by ISRO and the Airports Authority of India (AAI) to provide safety-of-life navigation services over the Indian Flight Information Region (FIR).
  - GAGAN made India the fourth nation in the world to deploy an indigenous satellite-based landing assistance system, enabling high-precision Localizer Performance with Vertical Guidance (LPV) approaches at regional airports.
  - **Global SBAS Counterparts:** GAGAN operates alongside equivalent systems worldwide, including WAAS (USA), EGNOS (Europe), and MSAS (Japan).

Features of IRNSS (NavIC)

- **Constellation Architecture**
  - **Hybrid Orbit Design:** Comprises a structural baseline of 7 active satellites working together in space.
  - **Orbit Split:** Uses 3 Geostationary (GEO) satellites over the Indian Ocean and 4 inclined Geosynchronous (GSO) satellites to maintain constant line-of-sight over India.
  - **Sovereign Timing Elements:** Employs high-precision, space-grade Rubidium atomic clocks to calculate nanosecond signal delays.
- **Signal & Coverage Footprint**
  - **Geographical Range:** Covers the entire Indian mainland and an extended zone stretching up to 1,500 km beyond national borders.
  - **Dual-Frequency Advantage:** Broadcasts on both the L5 and S-bands, which helps automatically correct ionospheric delays for sharper accuracy.
  - **Consumer-Standard L1 Band:** Integrated into newer generations (NVS-series) to allow standard, commercial smartphones to use NavIC directly.
- **Service Profile & Interoperability**
  - **Standard Positioning Service (SPS):** An open, unencrypted signal provided for general civilian use, offering an accuracy of under 20 meters.
  - **Restricted Service (RS):** An encrypted, secure signal reserved exclusively for the Indian Armed Forces and strategic agencies, offering an accuracy of under 10 meters.
  - **Messaging Payload Feature:** Possesses a unique, independent capability to broadcast emergency text messages and disaster warnings to areas with zero cellular network.

List of all satellites launched under the IRNSS-NavIC program

First Generation Series (IRNSS-1 Baseline)

Satellite Name	Launch Year	Orbit Type	Current Operational Status
IRNSS-1A	2013	Geosynchronous (GSO)	**Partial Failure**; atomic clocks failed, used only for short broadcast text messaging.
IRNSS-1B	2014	Geosynchronous (GSO)	**Operational**; active and fully providing critical navigation tracking metrics.
IRNSS-1C	2014	Geostationary (GEO)	**Partial Failure**; clock failed, degraded to basic one-way public warning broadcasts.
IRNSS-1D	2015	Geosynchronous (GSO)	**Decommissioned**; reached the end of its design life and taken out of active network service.
IRNSS-1E	2016	Geosynchronous (GSO)	**Partial Failure**; onboard timing system failed, restricted to baseline broadcast messages.
IRNSS-1F	2016	Geostationary (GEO)	**Partial Failure**; successfully completed its 10-year design lifespan before its clock failed.
IRNSS-1G	2016	Geostationary (GEO)	**Partial Failure**; replaced by NVS-01 for main positioning data; handles secondary text alerts.
IRNSS-1H	2017	Geosynchronous (GSO)	**Launch Failed**; heat shield failed to separate, trapping the satellite inside the rocket fairing.
IRNSS-1I	2018	Geosynchronous (GSO)	**Operational**; successfully launched to replace the dead components of the initial 1A satellite.

Second Generation Series

Satellite Name	Launch Year	Orbit Type	Operational Status
NVS-01	2023	Geostationary (GEO)	**Operational**; first 2nd-Gen satellite featuring the civilian **L1 band** and a **homegrown atomic clock**.
NVS-02	2025	Geostationary (GEO)	**Partial Failure**; propulsion failure stranded the craft in orbit before it reached its target slot.
NVS-03	2026 (Planned)	Geosynchronous (GSO)	**Planned Replacement**; designated to restore full 3D positioning capabilities to the network.

Applications of IRNSS (NavIC)

- **Strategic & Military Applications**
  - **Sovereign Missile Guidance: Provides** jam-resistant, precise coordinate tracking for ballistic and cruise missiles without risk of foreign service blockades.
  - **Border Surveillance & Logistics:** Tracks military troop deployments, tactical drone maneuvers, and border patrols across rugged Himalayan and desert terrains.
  - **Encrypted Command Networks:** Utilizes the secure Restricted Service (RS) band to safeguard classified military communications from electronic jamming.
- **Civil, Commercial & Infrastructure Applications**
  - **Terrestrial Fleet Logistics:** Powers mandatory vehicle tracking systems (AIS-140 standard) on public buses, commercial trucks, and mining fleets for safety.
  - **Deep-Sea Maritime Safety**: Broadcasts real-time emergency text alerts, cyclone updates, and international boundary warnings directly to artisanal fishing boats beyond cellular range.

- **Critical Utility Synchronization**: Delivers nanosecond-accurate timing signals required to balance national power grid frequencies and secure banking transactions.
- **Disaster Management Logistics**: Guides search-and-rescue teams directly into active flood, landslide, or earthquake zones using independent satellite messaging.
- **Precision Farming**: Coordinates automated drone paths for targeted pesticide spraying and fuels land-mapping under developmental schemes like SVAMITVA.

Comparison between NavIC (IRNSS) and GPS

Feature Dimension	NavIC (IRNSS)	Global Positioning System (GPS)
Origin & Agency	India (Developed and managed by ISRO)	United States (Developed and managed by US Air Force)
Operational Scale	Regional system (India + 1,500 km beyond its borders)	Global system (Provides uninterrupted full-planet coverage)
Satellite Count	7 active satellites in its baseline network	Minimum 24 active satellites (typically 31 to 32 operational)
Orbital Layout	High altitude: 3 GEO (Geostationary) + 4 GSO (Geosynchronous)	Medium altitude: All in MEO (Medium Earth Orbit)
Signal Constancy	Satellites are fixed relative to India, maintaining constant local visibility	Satellites move across the sky, orbiting Earth twice daily
Frequency Bands	Operates on L5, S, and L1 bands (Dual-frequency balances local errors)	Operates primarily on L-bands (L1, L2, L5)
Position Accuracy	Highly accurate regionally (<10–20 meters); superior in local valleys	Standard global accuracy (<5–10 meters globally)
Messaging Feature	Includes a built-in warning feature to blast text alerts to users	No direct text broadcast feature to user receivers

Advantages of Having an Indigenous Regional Navigation System

- **Strategic & Sovereign Advantages**
  - **Guarantees Strategic Autonomy:** Prevents critical navigation blackouts during wartime conflicts, eliminating reliance on foreign networks that can be restricted at will.
  - **Secures Sensitive Telemetry:** Protects military troop deployments, missile trajectory data, and defense movements from foreign surveillance or data logging.
  - **Anchors Critical Infrastructure:** Provides an un-spoofable, sovereign timing frequency to safely synchronize national banking networks, telecom routing, and power grids.
- **Technical & Localized Advantages**
  - **Optimized Regional Accuracy:** Utilizes specific dual-frequency bands (like L and S bands) that penetrate local tropical canopies and deep mountain valleys far better than global systems.
  - **Direct Emergency Broadcasting:** Features built-in text-messaging capabilities to send instant, localized disaster alerts directly to users without relying on cellular networks.
- **Economic & Geopolitical Advantages**
  - **Conserves Foreign Exchange:** Ends heavy recurring licensing costs and technology royalty payments made to international navigation providers.
  - **Drives the Domestic Tech Ecosystem:** Stimulates local industrial growth, commercial chip manufacturing, and private deep-tech startups.

- **Projects Regional Soft Power:** Establishes the host nation as a primary "net security provider" by offering free navigation and tracking support to friendly neighboring states.

Global Navigation Satellite Systems (GNSS)

- **GPS (Global Positioning System):** United States
- **GLONASS (Global Navigation Satellite System):** Russian Federation
- **Galileo:** European Union
- **BeiDou (BDS):** China

NavIC vs. GAGAN

NavIC is an *independent regional navigation constellation* (it generates its own positioning data). GAGAN is an *augmentation system* (it corrects and improves existing GPS signals for aviation safety).

Conclusion

NavIC provides India with absolute strategic autonomy by eliminating dangerous dependencies on foreign satellite networks during critical security operations. By delivering highly accurate, localized data, it serves as an indispensable tool for national disaster mitigation, modern commercial transport infrastructure, and military guidance systems.

9. SPACE DEBRIS AND SPACE SUSTAINABILITY

Space debris refers to the millions of pieces of defunct human-made objects, such as dead satellites and rocket fragments, cluttering Earth's orbits at hazardous speeds. Space sustainability is the practice of managing these orbital paths safely to prevent catastrophic collisions that could permanently destroy our global communication networks.

International Regulation of Space Debris

- **International Frameworks**
  - **Outer Space Treaty (1967):** Establishes the foundational rule that space is the province of all mankind, barring weaponization but lacking explicit penalties for creating debris.
  - **Liability Convention (1972):** Holds the launching nation completely liable to pay financial compensation if its space objects cause physical damage on Earth or to aircraft in flight.
  - **UNCOPUOS Guidelines (2007/2025):** Provides a 7-point voluntary roadmap targeting the restriction of normal mission trash and the prevention of on-orbit breakups.
  - **IADC Coordination Forum (1993):** An inter-governmental body that sets technical standards for post-mission disposal, such as safely venting unused propellant.
  - **ESA's Zero Debris Charter (2023):** A multi-partner initiative driving space actors to achieve zero new orbital debris creation by the year 2030.
  - **FCC 5-Year Deorbit Mandate (2024):** A major domestic regulation forcing operators seeking US market access to safely deorbit Low Earth Orbit (LEO) satellites within 5 years of mission completion.

Issues with Space Debris

- **Operational & Safety Hazards**
  - **Kessler Syndrome Risk:** Triggers a cascading chain reaction of orbital collisions that could render critical low-Earth orbits completely unusable.

As of mid-2026, the European Space Agency (ESA) tracks over 28,800 objects larger than 10 cm.

- **Hypervelocity Collision Energy:** Debris traveling at 28,000 km/h turns millimeter-sized fragments into lethal objects capable of obliterating active satellites.
- **Threat to Human Spaceflight:** Poses a continuous risk of lethal hull decompression for astronauts on the International Space Station (ISS) and crewed missions.

- **Economic & Infrastructure Vulnerabilities**
  - **Critical Service Disruptions:** Risks destroying satellites responsible for global banking synchronization, weather forecasting, and defense communication networks.

    Over 90% of global telecommunications, weather tracking, and GPS synchronized services rely on vulnerable low-Earth orbit satellites.
  - **Wasted Satellite Lifespan:** Forces active satellites to repeatedly expend expensive onboard propellant on emergency collision avoidance maneuvers, shortening their mission life.

    SpaceX reported that its Starlink constellation had to perform 355,848 emergency collision avoidance maneuvers in a single year.
  - **Mega-Constellation Congestion:** Exponential launch of commercial internet satellite fleets heavily crowds the safest orbital paths and exponentially raises daily collision risks.

    As per NASA, Commercial fleets have more than tripled active orbiters since 2019, with over 14,500 active satellites severely crowding space lanes.
- **Legal & Geopolitical Challenges**
  - **Tragedy of the Commons:** Current UN space guidelines are voluntary, meaning states face no binding international legal penalties for leaving junk in space.
  - **Dual-Use Technology Friction:** Developing robotic arms or nets for active debris removal raises geopolitical suspicion, as these technologies can also be weaponized to disable enemy satellites.
  - **Toxic Atmospheric Re-entry:** Large debris pieces surviving atmospheric re-entry risk raining toxic chemicals or structural fragments over civilian areas on Earth.

Measures Taken for Space Debris Mitigation

- **International Regulations & Policy Frameworks**
  - **UNCOPUOS Guidelines:** Provides global, voluntary frameworks focusing on restricting on-orbit debris creation and end-of-life passivation.
  - **FCC 5-Year Deorbit Rule:** Mandates commercial satellite operators to safely deorbit defunct Low Earth Orbit (LEO) crafts within five years of completing missions.
  - **ESA's Zero Debris Charter:** Establishes a multinational collaborative target to entirely eliminate new orbital debris creation by the year 2030.
  - **Space Debris Conference (2026):** Solidifies inter-governmental alliances to transition voluntary tracking guidelines into binding space sustainability frameworks.
- **Technological & Active Removal Measures**
  - **Active Deorbit Passivation:** Mandates venting remaining propellant and draining onboard batteries post-mission to stop accidental pressure explosions.
  - **Active Debris Removal (ADR):** Deploys robotic arms, magnetic harpoons, and nets to capture and drag down heavy dead upper rocket stages.
  - **Collision Avoidance Manoeuvres (CAM):** Uses real-time radar networks to automatically adjust working satellite paths away from closing fragments.

- **India's Indigenous Initiatives**
  - **Debris Free Space Mission (DFSM 2030):** Imposes a strict domestic goal for all public and private Indian space missions to hit zero debris creation by 2030.
  - **IS4OM Operational Command Centre:** Functions as India's central hub to handle global Space Situational Awareness (SSA) and actively protect domestic space assets.
  - **Project NETRA Network:** Installs dedicated indigenous early-warning tracking radars and optical telescopes to trace incoming space hazards.
  - **SpaDeX Mission Milestone:** Demonstrated successful autonomous space docking and proximity maneuvers, a critical step toward active debris removal technology.

Way Forward for Space Debris and Space Sustainability

- **Hardening Global Governance & Legal Frameworks**
  - **Transition to Binding Regulations:** Convert voluntary UNCOPUOS guidelines into mandatory international space laws equipped with financial penalties for non-compliance.

    EU Space Law (EUSL) proposed a framework to establish legally binding safety, security, and sustainability thresholds for all satellites operating over Europe.
  - **Implement Global Space Traffic Management (STM):** Establish a unified, real-time international traffic control body to manage satellite paths and prevent close-approach crises.

    US Department of Commerce's TraCSS (Traffic Coordination System for Space) acts as a modern civil real-time database to manage commercial space situational safety.
  - **Formulate "Deorbit-by-Design" Mandates:** Make stringently short end-of-life deorbit plans (like the 5-year LEO rule) a compulsory requirement to secure commercial launch licenses.

    US Federal Communications Commission (FCC) 5-Year Rule mandates that all low-Earth orbit satellites must deorbit within 5 years of mission completion.
- **Technological Innovations & Best Practices**
  - **Scale Active Debris Removal (ADR):** Incentivize public-private development of orbital clean-up technologies like robotic capture arms, magnetic harpoons, and deorbiting nets.

    JAXA & Astroscale's ADRAS-J mission successfully completed safe, non-cooperative flybys to inspect and assess a massive discarded rocket upper stage.
  - **Adopt Reusable & Green Systems:** Standardize multi-restart upper stages and reusable launch vehicles to minimize operational hardware dropped into permanent orbits.

    SpaceX's Falcon 9 & Starship utilize multi-restart engines and vertical landing technology to eliminate discarded booster junk in oceans and low-Earth orbit.
  - **Expand In-Orbit Servicing (IOS):** Standardize refueling and repair missions (using experiments like ISRO's SpaDeX) to prolong satellite lifespans and avoid active hardware abandonment.
- **Strategic & Cooperative Blueprints**
  - **Enhance Space Situational Awareness:** Bridge geopolitical divides to openly share cross-border radar data and maximize global collision avoidance accuracy.

    India-US SSA Data Sharing Agreement signed under the iCET framework enables real-time sharing of defensive space radar data to protect national assets.
  - **Enact Domestic Space Legislation:** Pass dedicated national laws (like India's pending Space Activities Act) to enforce domestic accountability over private mega-constellation operators.
  - **Create a Global Space Sustainability Fund:** Pool financial contributions from major spacefaring nations to fund international cleaning missions targeting non-attributable legacy junk.

India advocates for a **G20 Space Economy Leaders Meeting (SELM) framework** to pool multilateral resources for clearing non-attributable heritage junk.

Conclusion

Managing space debris is no longer a choice but a strategic necessity to prevent orbital collisions from crippling our global communication and defense infrastructure. Achieving true space sustainability requires a swift transition from voluntary guidelines to binding international laws alongside collaborative investments in active cleanup technologies.

10. OUTER SPACE GOVERNANCE

Outer space governance refers to the collective framework of international treaties, national laws, and voluntary guidelines designed to regulate human activities beyond Earth's atmosphere. It aims to prevent the weaponization of space, manage traffic, and resolve disputes among nations to ensure that exploration remains peaceful and cooperative.

Regulatory Framework for Outer Space Governance

Instrument	Core Objective
Outer Space Treaty (1967)	Declares space the **province of all mankind**; bans WMDs in orbit and prevents claims of national sovereignty.
Liability Convention (1972)	Establishes absolute state liability for damage caused by space objects to ground targets or aircraft.
Registration Convention (1975)	Mandates that states must register all space objects launched into orbit with the UN for transparency.
UNCOPUOS	The UN Committee on the Peaceful Uses of Outer Space; provides the forum for developing global guidelines.
Space Debris Mitigation Guidelines	Voluntary norms from the IADC and UN to minimize orbital debris and prevent on-orbit breakups.
Domestic Space Acts	National laws (e.g., India's Space Policy 2023) regulating private participation, licensing, and compliance.
Moon Agreement 1979	Agreement Governing the Activities of States on the Moon and Other Celestial Bodies.
Artemis Accords	Led by the US to build common, civilian governance principles for lunar exploration and resource extraction, expanding its reach to 68 signatory nations

Need for Outer Space Governance

- **Security & Geopolitical Drivers**
  - **Preventing Outer Space Weaponization:** Halts the dangerous deployment of anti-satellite (ASAT) weapons and prevents space from turning into an active theater for great-power conflict.
  - **Managing Dual-Use Technologies:** Controls high-risk technologies (like robotic capture arms) that can serve peacefully as satellite repair tools or aggressively as weapons to disable enemy satellites.
  - **Mitigating Strategic Trust Deficits:** Reduces the risks of sudden geopolitical miscalculations caused by unannounced military satellite launches or secretive orbital maneuvers.

- **Environmental & Operational Drivers**
  - **Averting Kessler Syndrome:** Prevents runaway orbital collisions by enforcing strict, standardized clean-up rules for dead hardware and rocket fragments.
  - **Resolving Orbital Slotted Crowding:** Regulates the distribution of scarce, prime orbital slots (especially in Low Earth Orbit) to prevent rich nations from hoarding them.
  - **Protecting Global Ground Infrastructure:** Safeguards critical global systems like GPS navigation, weather tracking, and banking networks from sudden orbital destruction.
- **Commercial & Legal Drivers**
  - **Regulating Commercial Mega-Constellations:** Enforces operational traffic rules on private internet fleets (like Starlink) that launch thousands of satellites into orbit annually.
  - **Codifying Space Mineral Rights:** Resolves deep legal gaps regarding ownership, mining rights, and revenue sharing for celestial resources extracted from the Moon or asteroids.
  - **Updating State Liability Laws:** Adapts older, state-centric treaties to clearly hold nations financially responsible for damages caused by their private commercial space companies.

UNOOSA (United Nations Office for Outer Space Affairs)

- **Profile & Institutional Setup**
  - **Origin & Headquarter:** Established in 1958 by the UN General Assembly; currently located at the **United Nations Office at Vienna, Austria**.
  - **Secretariat Role:** Functions as the official secretariat for **COPUOS** (UN Committee on the Peaceful Uses of Outer Space).
  - **Primary Objective:** Tasked with promoting international cooperation for the **peaceful uses and exploration of outer space**.
- **Major Mandates & Statutory Duties**
  - **Space Object Registry:** Manages the United Nations **Register of Objects Launched into Outer Space** on behalf of the UN Secretary-General to ensure transparency.
  - **Treaty Enforcement:** Assists member states in implementing and developing national policies aligned with the core **5 UN Space Treaties**.
  - **Bridging the Space Divide:** Runs the *"Access to Space for All"* initiative to help developing nations build satellite engineering capabilities and technical know-how.
- **Dedicated Interventions & Programs**
  - **UN-SPIDER Programme:** Acts as a specialized platform that provides direct, free satellite data and images to nations struggling with **disaster management and climate emergency responses**.
  - **UN-Space Forum:** Coordinates space-related workflows across various UN branches to eliminate resource duplication.
  - **Space Sustainability Focus:** Formulates standard frameworks to address contemporary risks like **rapid space debris multiplication** and Near-Earth Object impacts.

Issues with Outer Space Governance

- **Legal Loopholes & Outdated Regimes**
  - **Anachronistic Treaty Backbone:** Outer Space Treaty (1967) was designed for slow-moving, state-centric Cold War programs and completely fails to address modern private sector operations
  - **Absence of Binding Enforcement:** Core international guidelines on traffic and debris remain entirely voluntary, leaving the UN with no legal mechanisms to penalize non-compliant nations.

According to the **UNOOSA**, the Inter-Agency Space Debris Coordination Committee (IADC) mitigation guidelines remain **100% voluntary** with zero enforcement teeth.

- **Regulatory Arbitrage Risks:** Companies can exploit the lack of uniform international laws by registering their launches in nations with weak safety and liability standards.
- **The Resource Exploitation Blindspot:** Existing treaties ban countries from claiming sovereignty over celestial bodies but leave a massive legal gray area regarding commercial mining of lunar or asteroid minerals.

- **Geopolitical Blocs & Fragmentation**
  - **Bifurcation of Space Blocs:** Rise of competitive, exclusive coalitions—specifically the US-led **Artemis Accords** vs. the China-Russia **International Lunar Research Station (ILRS)**—is fracturing universal UN-based diplomacy.
  - **De Facto Sovereignty Concerns:** The concept **of safety zones** or **operational areas** on the Moon risks allowing early-moving nations to establish monopolistic control over vital lunar resources.
  - **Militarization & Verification Deficit:** Proliferation of dual-use orbital technologies (like satellite-servicing robotic arms) combined with restricted national security data creates deep international mistrust.

    SIPRI highlights that over **60% of global satellites** launched are "dual-use," masking anti-satellite (ASAT) capabilities as civilian servicing tools.
- **Commercial & Traffic Congestion**
  - **Mega-Constellation Crowding:** Rapid, uncoordinated deployment of commercial internet fleets creates an orbital "Tragedy of the Commons" that exponentially multiplies collision risks.

    SpaceX's **Starlink** alone operates over **10,700 active low-Earth orbit satellites**, accounting for roughly **60% to 65% of all active satellites** currently tracking in space.
  - **Tracking & Information Gap:** Geopolitical rivalries and commercial secrecy prevent nations from openly sharing critical tracking data, leading to severe information asymmetry in space safety.

    According to **ESA's Space Debris Office**, out of an estimated **130 million debris objects** larger than 1 mm, global tracking networks can actively monitor **fewer than 35,000 objects**.
  - **Widening Global Space Divide:** Developing and Global South nations lack the capital to secure premium orbital slots or participate in rule-making, leading to intense technological inequality in space access.

    Union of Concerned Scientists (UCS) Satellite Database reveals that over 80% of all operating satellites belong to just the Global North.

Way Forward for Outer Space Governance

- **Hardening Legal Frameworks & Verification**
  - **Codify the UN Pact for the Future Actions:** Transition the newly adopted multilateral space commitments into binding international legislation to legally bar an orbital arms race.

    India actively champions the **UN PAROS (Prevention of an Arms Race in Outer Space)** initiative to transition the Pact's pledges into legally binding restrictions.
  - **Draft a Unified Outer Space Treaty Amendment:** Update the aging 1967 convention to explicitly regulate modern commercial mega-constellations and private deep-space mining.

Artemis Accords (Section 10) establishes a modern minilateral legal precedent for commercial space resource extraction outside the rigid 1967 treaty.

- **Establish a Global Space Monitoring System (GSMS):** Deploy an independent, AI-driven digital diplomacy platform to track orbital compliance and guarantee state accountability.

ESA's Space Surveillance and Tracking (SST) segment operates a digital network to autonomously track debris and verify international orbital compliance.

- **Operational & Tactical Coordination**
  - **Standardize Space Traffic Management (STM):** Institutionalize a centralized, real-time global traffic control body under UNOOSA to systematically prevent satellite close-approaches.

US Bureau of Space Commerce (TraCSS) serves as a real-time, public civil space traffic management body to coordinate commercial satellite paths.

  - **Enforce Uniform National Space Legislation:** Mandate all spacefaring states to pass domestic laws that legally bind private actors to safe end-of-life deorbit rules.
  - **Implement Deorbit-by-Design Licensing:** Make the integration of passive safety venting and active debris-removal hooks a mandatory prerequisite for granting commercial launch permits.

ESA's Zero Debris Charter (2024/2025) forces commercial entities to build satellites with automated passivation valves and passive deorbit sails to clear licensing.

- **Geopolitical Bridging & Resource Equity**
  - **Bridge Polarized Interplanetary Blocs:** Build direct diplomatic hotlines between competing coalitions—specifically the US-led Artemis Accords and the China-Russia ILRS.

Inter-Agency Space Debris Coordination Committee (IADC) acts as a rare technical bridge between NASA, ESA, Roscosmos, and CNSA to prevent orbital overlap.

  - **Pool an International Space Sustainability Fund:** Collect mandatory financial levies from commercial operators to actively fund multilateral cleaning missions for legacy orbital junk.
  - **Institutionalize Open-Source Radar Sharing:** Eradicate military data hoarding by building open, global Space Situational Awareness (SSA) networks to optimize early warning systems.

Which role India can play in outer space governance?

- **Diplomatic Bridge between Polarized Blocs**
  - **Neutral Interlocutor:** As an **Artemis Accords** signatory with deep historical space ties to Russia, India can prevent a Western vs. China-Russia (ILRS) structural split.
  - **Codifying UN-led Consensus:** India can champion the transition of voluntary UNCOPUOS guidelines into mandatory rules for space mining and private mega-constellations.
  - **Preventing Weaponization:** Leveraging its **Mission Shakti ASAT** capability, India can authoritatively push for binding **PAROS** (Prevention of an Arms Race in Outer Space) treaties.
- **Championing Global South Equity**
  - **Democratizing Launch Access:** Using low-cost systems like the **SSLV** (Small Satellite Launch Vehicle), India can help developing nations secure space access before orbits get crowded out.
  - **Space-for-Development Data:** Expanding on initiatives like the **SAARC Satellite**, India can institutionalize open-source climate and hydro-informatics sharing for vulnerable nations.
- **Operationalizing Space Sustainability**
  - **Setting Debris-Free Norms:** By targeting its own **Debris Free Space Missions (DFSM) by 2030**, India can set a regulatory benchmark for global **Deorbit-by-Design** licensing.

- **Expanding Global Traffic Monitoring:** Scaling **ISRO's IS4OM** and the India-US SSA pact creates a decentralized, unclassified grid for international collision avoidance.
- **Standardizing In-Orbit Servicing:** Leveraging **SPADEX** (Space Docking Experiment), India can help draft international technical safety standards for automated docking, refueling, and robotic debris removal.

Conclusion

Outer space governance must rapidly evolve from fragmented, voluntary norms into a binding, internationally verifiable legal framework to prevent militarization and orbital chaos. Bridging the geopolitical divide between competing space blocs is essential to ensure equitable access to celestial resources and protect global digital infrastructure.

11. SATELLITE LAUNCH VEHICLES AND REUSABLE LAUNCH VEHICLES

Satellite launch vehicles are specialized rocket systems designed to transport payloads, such as satellites and scientific instruments, from Earth into their designated orbital trajectories. Reusable launch vehicles represent a paradigm shift in space exploration by allowing key components like first-stage boosters to return safely to Earth for recovery, refurbishment, and subsequent relaunch.

Major Satellite Launch Vehicles Portfolio

Launch Vehicle	Operational Status	Stage Configuration & Propulsion	Mission Capability	Significance
SLV-3	Historical (Retired)	4-Stage All-Solid Propulsion	40 kg to Low Earth Orbit (LEO)	India's first experimental launcher; successfully placed the Rohini satellite in orbit (1980).
ASLV	Historical (Retired)	5-Stage Solid Fuel with Strap-ons	150 kg to Low Earth Orbit (LEO)	Developed to evaluate critical technologies like closed-loop guidance systems and strap-on boosters.
PSLV	Active (Operational)	4-Stage Alternating Solid & Liquid	1,750 kg to Sun-Synchronous Polar Orbit.	Dubbed the **"Workhorse of ISRO"**; launched landmark deep-space missions like Chandrayaan-1 and Mangalyaan (MOM).
GSLV (Mk II)	Active (Operational)	3-Stage Solid / Liquid / Cryogenic	2,250 kg to Geosynchronous Transfer Orbit (GTO)	Integrated with India's fully indigenous **Cryogenic Upper Stage (CE-7.5)** engine to achieve self-reliance.

LVM3 (GSLV Mk III)	Active (Operational)	3-Stage Heavy Solid / Liquid / Cryogenic	4,000 kg to GTO 10,000 kg to LEO	ISRO's heaviest launcher; chosen for the crewed **Gaganyaan** human spaceflight program.
SSLV	Active (Operational)	3-Stage Solid + Velocity Trimming Stage	500 kg mini/micro satellites to LEO	Engineered for a rapid 72-hour turnaround time to tap into the booming commercial small-satellite launch market.
RLV-TD	Technology Demonstrator	Two-stage-to-orbit (TSTO) spaceplane	Experimental Scaled Prototype Testing	Flying testbed validating autonomous landing, thermal protection tiles, and cruise flight mechanics for future reuse.
NGLV (Soorya)	Under Development	3-Stage Liquid (LOX-Methane) & Cryo	30,000 kg to LEO with partial reusability.	Next-Gen heavy-lifter intended to establish the **Bhartiya Antriksh Station (BAS)** and power human lunar landings.
Falcon 9 (SpaceX)	Active (Global Benchmark)	2-Stage Liquid Oxygen (LOX) & Kerosene	22,800 kg to LEO (Fully Expendable mode)	Set the global benchmark for orbital reusability by routinely landing first-stage boosters upright on land and sea barges.

Significance of Satellite Launch Vehicles (SLVs) and Reusable Launch Vehicles (RLVs)

- **Strategic & Geopolitical Significance**
  - **Ensures Sovereignty and Access:** SLVs guarantee uninterrupted, independent national access to outer space, removing reliance on foreign powers during strategic crises.
  - **Bolsters Defense & Dual-Use Tech:** Rocket navigation and thermal shielding insights directly advance country-specific hypersonic cruise missiles and military defense frameworks.
  - **Enables Critical Deep-Space Science:** Heavy expendable lifters serve as the literal bedrock for flagship interplanetary voyages, lunar landings, and national security surveillance constellations.
- **Economic & Commercial Significance**
  - **Slashes Launch Costs:** RLVs aim to decrease the steep "cost-to-orbit" metric by up to 80% by salvaging the most expensive engine and avionics hardware.
  - **Captures Global Launch Market Share:** Low-cost, agile systems (like India's SSLV and upcoming reusable NGLV Soorya) maximize multi-billion dollar commercial launch revenues via NSIL.
  - **Powers Megaconstellation Turnarounds:** Rapid reuse minimizes turnaround manufacturing timelines from months to days, permitting the rapid deployment of massive internet broadband networks.

- **Space Sustainability & Future Sovereignty**
  - **Mitigates Kessler Syndrome Risks:** Reusing core rocket stages curbed the physical abandonment of spent boosters in permanent orbit, actively shrinking high-altitude space junk accumulation.
  - **Acts as Logistics Core for Space Stations:** Serves as the primary cost-effective freight vehicle for cargo delivery and human rotation schedules on upcoming orbital stations.
  - **Enables In-Orbit Satellite Servicing:** Advanced RLV flight configurations pave the path toward autonomous space mechanics capable of refueling, upgrading, or capturing dying satellites.

Did you know?

- India's primary spaceport is the Satish Dhawan Space Centre in Sriharikota, Andhra Pradesh, which houses two active operational launch pads that handle all major LVM3 and PSLV missions.
- To maximize payload efficiency for smaller rockets, ISRO is constructing a second major spaceport at Kulasekarapattinam, Tamil Nadu, which offers a direct southward launch trajectory over the Indian Ocean.
- India also utilizes a dedicated Sounding Rocket Launching Station at Thumba, Kerala, which sits almost perfectly on Earth's magnetic equator and has been used for upper-atmospheric research since 1963.

Issues with reusable vehicles

- **Technical & Engineering Barriers**
  - **Extreme Re-entry Stress:** Vehicles must survive friction-induced temperatures exceeding 1500°C, requiring heavy and costly Thermal Protection Systems (TPS).
  - **Severe Structural Fatigue:** Repeated high-velocity launches and landings cause material micro-fractures, strictly limiting the safe operational lifespan of reusable components.
  - **Diminished Initial Payload Capacity:** Reusable systems must carry massive additional deadweight—including landing gear, grid fins, and retro-propulsion fuel—reducing net payload capacity.
  - **Complex Guidance and Navigation:** Demands autonomous, sub-meter precise landing capabilities that can adjust in real-time to erratic atmospheric winds and turbulence.
- **Economic & Operational Challenges**
  - **Diminishing Returns on Refurbishment:** High post-flight inspection, cleaning, and recertification costs can completely negate the economic benefits of not building a new rocket.
  - **Exorbitant Capital Expenditure:** Developing cutting-edge engine technologies (like multi-restart throttlable engines) requires massive, high-risk upfront financial investments.
  - **Hardware Trust Deficit:** Recertifying previously flown engines and components for high-stakes payloads like crewed human spaceflight presents severe insurance and regulatory safety hurdles.
  - **Infrastructural Deficits:** Many nations lack dedicated downrange ocean recovery barges, automated drone ships, and specialized high-speed tracking networks.
- **Safety & Geopolitical Risks**
  - **Dual-Use Technology Suspicions:** The precise guidance, proximity maneuvering, and robotics required for RLV recovery can be easily militarized into offensive anti-satellite (ASAT) weapons.
  - **Risk of Catastrophic Ascent Failures:** Traditional expendable launch vehicles carry inherent failure risks, but any RLV failure double-exposes high-value assets during both launch and recovery phases.
  - **Widening Technology Divide:** Monopolization of reusable tech by advanced nations risks priced-out exclusion of Global South countries from accessing low-cost commercial orbits.

Why We Launch Rockets Near the Equator?

- **Maximum Surface Spin:** The equator is the widest part of the planet and spins eastward at roughly 1,670 km/h.
- **The Slingshot Effect:** Rockets launching eastward from the equator inherit this natural speed, getting a free kinetic boost right off the pad.
- **Massive Fuel Savings:** Utilizing Earth's spin lowers the amount of propellant the rocket has to burn to achieve orbit.
- **Higher Payload Capacity:** Burning less fuel means the exact same rocket can carry much heavier satellites, rovers, or cargo.
- **Direct Access to Orbit:** Most communications satellites live in geostationary orbits directly above the equator; launching from there eliminates the need for expensive, fuel-wasting steering maneuvers mid-flight.
- **Ocean Downrange Safety:** Equatorial spaceports are strategically chosen with vast oceans to their east, ensuring spent rocket stages drop safely into the water instead of over populated cities.

Way Forward for Satellite Launch Vehicles (SLVs)

- **Technology Advancements & Indigenization**
  - **Fast-Track Heavy-Lift Reusable Rockets:** Accelerate development of ISRO's **NGLV (Project Soorya)** with a reusable first stage to scale payload capacity to 30 tonnes.
  - **Master Methane-Based Propulsion:** Perfect cleaner-burning Liquid Methane (LOX-Methane) engines to eliminate carbon buildup and minimize post-flight engine refurbishment times.
  - **Transition to Next-Phase Aerothermal Testing:** Graduate from horizontal landing experiments (LEX) to real-world **Orbital Return Flight Experiments (OREX)** to master 1500°C re-entry heat shielding.
- **Commercial, Structural & Policy Reforms**
  - **Offload Mature Fleet to Public-Private Partnerships (PPPs):** Shift production of established, expendable workhorses like the PSLV to industry consortiums (e.g., HAL-L&T), freeing ISRO to focus strictly on R&D.
  - **Enact a Statutory National Space Act:** Pass comprehensive space legislation to clearly define insurance, third-party liability, and licensing rules for private rocket startups
  - **Deploy High-Risk Venture Capital Funds:** Leverage IN-SPACe's ₹1,000-crore VC fund to actively supply capital to domestic deep-tech launch startups (like Agnikul and Skyroot)
- **Geopolitical, Logistical & Sustainability Alignments**
  - **Build Dedicated Downrange Infrastructure:** Invest in automated ocean drone ships and marine recovery barges to establish reliable vertical take-off, vertical landing (VTVL) landing zones.
  - **Integrate Launch Upgrades with Orbit Logistics:** Sync RLV capabilities with future mega-projects, utilizing them as the low-cost cargo and crew backbone for the upcoming **Bhartiya Antriksh Station (BAS)**.
  - **Mandate Debris-Free Staging Standards:** Align future launch system milestones with the **Debris-Free Space Missions (DFSM) 2030** target to guarantee that discarded rocket components do not turn into permanent orbital junk.

Conclusion

Transition from traditional expendable launchers to reusable vehicles is essential to slash the high cost of space access and minimize orbital debris. For India, developing indigenous reusable systems will secure commercial competitiveness in the global market while cementing strategic self-reliance.

MGP/BP/GS3/019

BIOTECHNOLOGY

1. BIOTECHNOLOGY

Biotechnology is the branch of science that uses living organisms, cells, or their components (like enzymes and genes) to develop products and technologies that improve human life. It combines biology with technology, using tools like **genetic engineering**, **recombinant DNA technology**, **cell culture**, and **fermentation** to create solutions in health, agriculture, industry, and environment.

In India, biotechnology is treated as a **sunrise sector** with the potential to transform the economy through the concept of a **Bioeconomy** — the use of renewable biological resources to produce food, energy, and industrial goods sustainably.

India's Bioeconomy - An Overview

- India's **Bioeconomy** reached **US$ 195.3 billion in 2025**, growing at **18%**, the highest growth rate recorded so far (India Bioeconomy Report - IBER 2026, BIRAC/ABLE).
- The sector has grown almost **20-fold in a decade** — from just **US$ 10 billion in 2014**.
- Biotechnology now contributes close to **5% of India's GDP**.
- Government target: a **US$ 300 billion Bioeconomy by 2030**, moving towards a longer-term **US$ 1 trillion** vision.
- Over **11,800 biotech startups** are currently operational, up from just about 50 in 2014.
- Segment-wise (2025): **BioIndustrial** is the largest at **US$ 90.2 billion**, followed by **BioPharma (US$ 64.5 billion)**, **BioServices (US$ 26 billion)**, and **BioAgri (US$ 14.6 billion)**.

Applications of Biotechnology in India

(a) Healthcare (Medical Biotechnology)

- **Vaccines and Biologics:** India is often called the **'vaccine hub of the world'**, producing over **60% of global vaccine supply** through firms like Serum Institute of India and Bharat Biotech.
- **Biosimilars and Generics:** India accounts for **44% of global Abbreviated New Drug Applications (ANDAs)** and hosts over **1,400 WHO-compliant manufacturing plants**, cutting costs of life-saving drugs.
- **Cell and Gene Therapy:** For example, DBT and BIRAC are supporting indigenous **CAR-T cell therapy** and gene-editing research to treat cancers and genetic disorders.
- **Diagnostics:** Rapid, low-cost diagnostic kits (e.g., for COVID-19, TB) developed using biotechnology have improved disease surveillance.

(b) Agricultural Biotechnology

- **Genetically Modified (GM) Crops: Bt Cotton** remains India's only commercially cultivated GM crop, credited with reducing pest damage and boosting yields since its introduction.

- **Climate-Resilient Crops:** For example, Development of drought-tolerant chickpea variety **SAATVIK (NC-9)** and genome-edited rice lines (e.g., **DEP1-edited MTU-1010**) for higher yield under climate stress.
- **Genomic Tools:** India's first **90K SNP arrays** — **IndRA** (rice) and **IndCA** (chickpea) — enable precise variety identification and faster breeding.
- **Biofertilizers and Biopesticides:** Reduce dependence on chemical inputs; e.g., nano-formulations from Myrothecium verrucaria for eco-friendly pest control.

(c) Industrial and Environmental Biotechnology

- **Bio-manufacturing:** Production of **bio-based chemicals, biopolymers, and enzymes** to replace petroleum-based products, supporting India's **net-zero** commitments.
- **Biofuels:** Second-generation ethanol and biodiesel produced from agricultural waste support the **Ethanol Blending Programme**.
- **Bioremediation:** Use of microbes to clean polluted soil and water, and to treat industrial effluents.
- **Carbon Capture:** Emerging biotech-based carbon capture and marine/space research form part of the newly identified strategic thematic areas.

3. GOVERNMENT INITIATIVES AND INSTITUTIONAL FRAMEWORK

BioE3 Policy (2024)	'Biotechnology for Economy, Environment and Employment' — India's first comprehensive national biotech policy, approved by the Union Cabinet in August 2024. It identifies six thematic areas: bio-based chemicals, smart proteins, precision biotherapeutics, climate-resilient agriculture, carbon capture, and marine/space research.
Biopharma SHAKTI Scheme (Budget 2026-27)	A five-year mission with an outlay of Rs. 10,000 crore (US$ 1.20 billion) to build India as a global hub for biologics and biosimilars.
Bio-RIDE Scheme	Merged and relaunched in 2024, with an allocation of about Rs. 9,200 crore (US$ 1.09 billion) for Research, Innovation and Entrepreneurship in biotechnology.
Research, Development and Innovation (RDI) Fund	A new Rs. 1 lakh crore fund to support long-term, deep-tech biotech ventures.
BIRAC (Biotechnology Industry Research Assistance Council)	A not-for-profit PSU under DBT that acts as an interface between academia and industry; supports 73 bio-incubators and a growing network of biofoundries and Bio-AI hubs across the country.
National Biopharma Mission	Industry-academia collaboration to accelerate development of vaccines, biosimilars, and medical devices.
BioE3 Challenge for Youth (2025)	Programme for school students (Classes VI–XII) to encourage young minds to design biotech solutions.
Regulatory bodies	Institutional Biosafety Committee (IBSC), Review Committee on Genetic Manipulation (RCGM), and the Genetic Engineering Appraisal Committee (GEAC) under the Ministry of Environment, Forest and Climate Change regulate GM research and release.

Significance of Biotechnology for India

Dimension	How Biotechnology Helps
Health Security	Affordable vaccines, biosimilars and diagnostics reduce import dependence and improve public health outcomes.
Food Security	Climate-resilient, high-yield crop varieties help sustain food production amid erratic monsoons and shrinking farmland.

Dimension	How Biotechnology Helps
Employment	Once fully operational, BioE3 Centres are projected to support **over 250 startups/MSMEs** and create nearly **10,000 skilled jobs**.
Environment	Bio-manufacturing and carbon capture support India's **net-zero by 2070** target and reduce dependence on petrochemicals.
Strategic Autonomy	Domestic biomanufacturing of APIs and biologics reduces reliance on imports, especially from China.

Challenges Facing Biotechnology in India

- **Regulatory Logjam over GM Crops:** Despite **GEAC** clearance in 2022, commercial release of **GM Mustard (DMH-11)** remains stayed by the Supreme Court amid an ongoing PIL. **Bt Brinjal** has remained under a moratorium since 2010.
- **Absence of a Comprehensive Biosafety Law:** The **Biotechnology Regulatory Authority of India (BRAI) Bill** has not been enacted, leaving the regulatory landscape fragmented and less transparent.
- **Illegal Cultivation:** Unapproved **HT-Bt Cotton** is estimated to cover **15-25% of cotton acreage** in states like Gujarat and Maharashtra, exposing gaps in enforcement.
- **Ethical and Social Concerns:** Fears over impact of GM crops on biodiversity, pollinators (bees in mustard fields), and farmer dependence on seed companies.
- **Capital-Intensive Infrastructure:** Startups and academic institutions struggle to afford pilot-scale manufacturing facilities, though shared biofoundries are now easing this.
- **Skill and Talent Gaps:** Shortage of trained manpower in emerging areas like **synthetic biology**, **bioinformatics**, and **precision fermentation**.
- **IP and Data Protection:** Weak enforcement of intellectual property rights discourages private R&D investment in the sector.
- **Trade Pressure:** India faces pressure in trade talks (e.g., with the US) to open its market to GM crop imports, which it has resisted, calling agriculture and dairy **'sacrosanct red lines'**.

Way Forward

- **National GM Crop Policy:** Acting on the Supreme Court's **March 2025** directive, the government should frame a clear, science-based national policy on regulation and adoption of GM crops.
- **Enact a Biosafety Law:** Revive and pass a version of the **BRAI Bill** to create a single-window, transparent regulator for genetic engineering research and products.
- **Scale up Shared Infrastructure:** Expand the network of **biofoundries** and **Bio-AI hubs** (like the DBT-ICGEB Biofoundry, New Delhi) to lower entry barriers for startups.
- **Strengthen Skilling:** Align biotech education and vocational training with emerging areas through platforms like **iGOT-Karmayogi**-style modules for scientists.
- **Public Awareness and Consultation:** Build trust through transparent, science-based public communication on the safety of GM and gene-edited products.
- **Deepen Industry-Academia Linkages:** Strengthen BIRAC-style interfaces to convert lab research into market-ready, export-worthy products faster.

Conclusion

Biotechnology has emerged as a genuine **sunrise sector** for India, straddling health security, food security, environmental sustainability, and economic growth. With the **BioE3 Policy**, **Biopharma SHAKTI**, and a fast-expanding startup ecosystem, India is well placed to move towards its **US$ 300 billion Bioeconomy target by 2030**.

However, the sector's true potential can only be realised if governance keeps pace with science — through a **clear regulatory framework for GM crops**, a **dedicated biosafety law**, and sustained investment in **skills and infrastructure**. A well-regulated, innovation-friendly biotechnology ecosystem will be central to India's aspirations of becoming a **Viksit Bharat by 2047**.

2. GENETIC ENGINEERING

Genetic engineering, also referred to as **genetic modification** or **genetic manipulation**, is a process that utilizes laboratory-based technologies to alter the DNA makeup of an organism. It involves the direct "act of modifying the genetic makeup" of a living being to achieve specific results, such as producing a desired trait or correcting a genetic defect.

Genetic engineering typically involves one or more of the following actions:

- **Changing a single base pair:** Modifying individual A-T or C-G pairings within the DNA sequence.
- **Deleting DNA:** Removing a specific region of an organism's DNA.
- **Adding DNA:** Inserting a new segment of DNA, which may include taking a gene from one species and adding it to another.

Key Techniques

Genetic engineering is distinguished from traditional methods like selective breeding, which is not considered a form of genetic engineering. Instead, it relies on advanced molecular techniques, including:

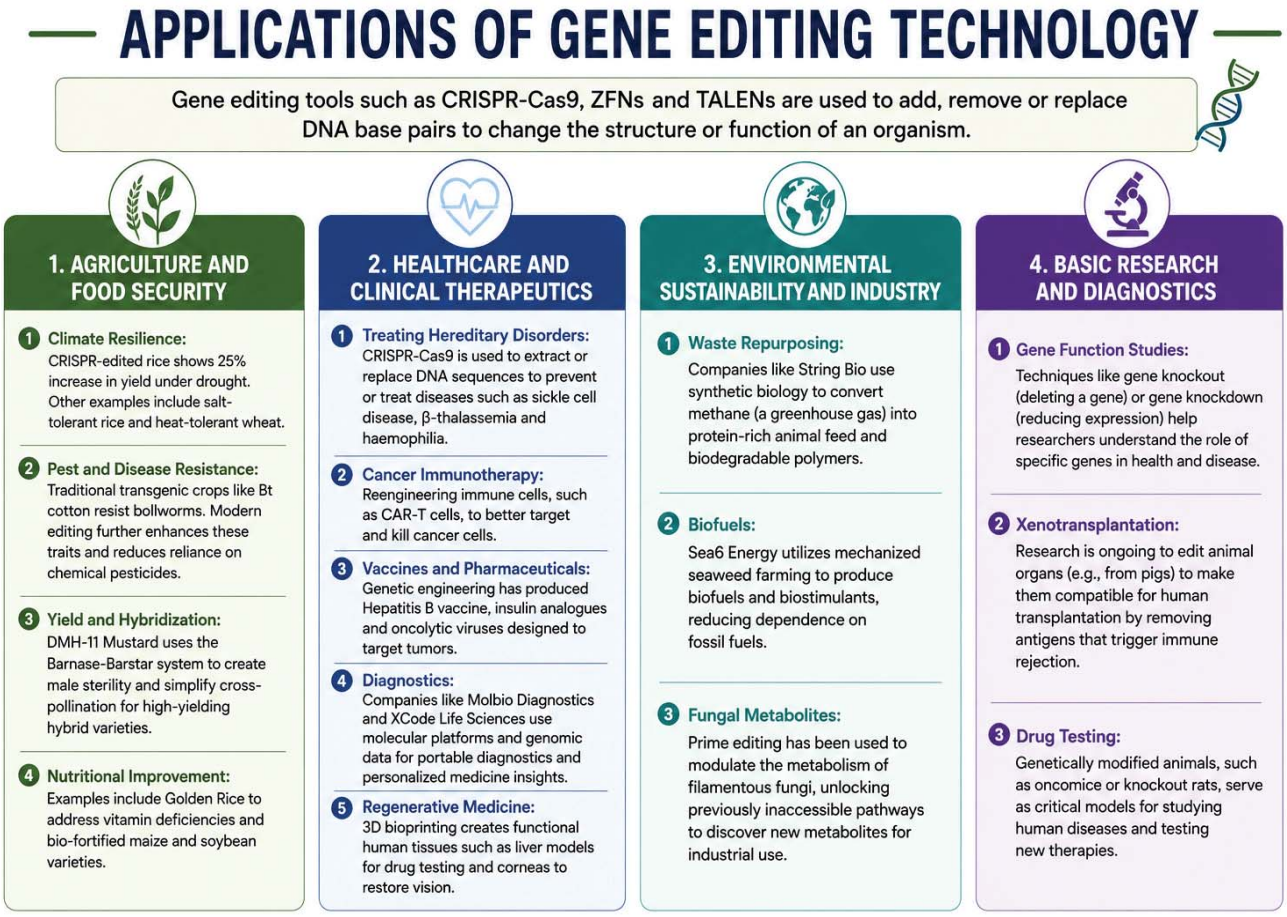
- **Recombinant DNA Technology:** This foundational method involves transferring specific genes from one organism to another.
- **Genome Editing:** A more recent and precise form of genetic engineering that uses tools like **CRISPR-Cas9**, **ZFNs**, and **TALENs** to target and modify specific sequences within a genome. CRISPR-Cas9, in particular, is noted for being faster, cheaper, and more accurate than earlier techniques.
- **Gene Therapy:** A medical application of genetic engineering that introduces, removes, or changes an individual's genetic material to treat diseases or potentially achieve long-term cures.
- **Other Methods:** These include gene targeting, nuclear transplantation, and the use of viral vectors to insert genetic material into cells.

Gene Editing vs Genetic Modification

The terms **genetic modification** (GM) and **gene editing** are often used interchangeably to describe the process of altering an organism's DNA, but they represent different technical approaches with distinct regulatory implications.

Aspect	Genetic Modification (GM) / Genetic Engineering	Gene Editing (Genome Editing)
Meaning	Broad umbrella term for laboratory techniques that alter an organism's genetic makeup.	A **precise subset** of genetic engineering that modifies specific DNA sequences.
Precision	Comparatively less precise; often requires screening of many samples to identify successful modifications.	Highly **precise and targeted**; enables addition, deletion, or replacement of specific DNA bases.

Foreign DNA	Often involves **transgenesis**—insertion of genes from another species.	Can modify genes **without introducing foreign (exogenous) DNA**, especially using SDN-1 and SDN-2 approaches.
Technology Used	Primarily based on **recombinant DNA technology** and transgenic methods.	Uses advanced genome-editing tools such as **CRISPR-Cas9, ZFNs (Zinc Finger Nucleases), and TALENs (Transcription Activator-Like Effector Nucleases)**.
Nature of Genetic Change	May introduce entirely new genes or traits from unrelated organisms.	Alters existing genes by making **targeted edits** within the organism's own genome.
Regulatory Framework (India)	Governed by the **Rules, 1989** under the **Environment (Protection) Act, 1986**, with stringent biosafety and regulatory oversight.	Certain genome-edited plants (**SDN-1 & SDN-2**) are exempt from some provisions of the 1989 Rules if they **do not contain foreign DNA**.
Regulatory Rationale	Treated as **Genetically Modified Organisms (GMOs)** due to introduction of foreign genetic material.	Recognized as distinct from transgenic GMOs because the edits **could occur naturally or through conventional breeding**.



Key Issues With Gene Editing In India

1. **Regulatory & Legal Challenges**
   - **Fragmented regulation:** Multiple agencies make approvals slow and complex.
   - **Need for a single regulator:** The proposed **Biotechnology Regulatory Authority of India (BRAI) Bill** seeks to streamline approvals.
   - **Limited transparency:** In the **DMH-11 mustard** case, concerns were raised over non-disclosure of complete biosafety data.
   - **Lengthy approvals:** Commercial approval of a GM crop may take **8–10 years**, increasing costs and discouraging innovation.
   - **Patent uncertainty:** Ongoing global disputes over **CRISPR-Cas9** patents create legal and commercial uncertainty.
2. **Environmental & Ecological Challenges**
   - **Genetic pollution:** Cross-pollination may spread edited genes to wild relatives, especially **Brassica (mustard)** species.
   - **Pest resistance:** for example, the **pink bollworm** evolved resistance to **Bt cotton**, reducing its long-term effectiveness.
   - **Secondary pest outbreaks:** Bt cotton cultivation has been associated with higher infestations of **jassids and thrips**, requiring pesticide use.

     Thrips and jassids (leafhoppers) are highly damaging, sap-sucking agricultural pests
   - **Super weeds:** Herbicide-tolerant crops such as **DMH-11** may encourage excessive herbicide use, leading to resistant weeds.
   - **Impact on biodiversity:** for example Possible adverse effects of increased use of pesticides on **honeybees**, beneficial insects, and soil microbial diversity.
3. **Socio-Economic Challenges**
   - **High seed prices:** Gene-edited seeds may remain unaffordable for many **small and marginal farmers**.
   - **Corporate monopoly:** Patent-protected technologies may increase dependence on multinational seed companies.
   - **Loss of rural employment:** Herbicide-resistant crops reduce manual weeding, affecting employment of **rural women labourers**.
   - **Unequal access:** Benefits may be concentrated among resource-rich farmers with better access to technology and markets.
4. **Ethical & Medical Challenges**
   - **Human germline editing:** India prohibits heritable genome editing due to ethical concerns.

Somatic Cell Gene Editing	Germline Gene Editing
Edits genes in **body (somatic) cells**; changes affect only the treated individual and **are not inherited**.	Edits genes in **embryos, sperm, or eggs**; changes are **heritable** and passed to future generations.

   - **Designer babies:** Fear that technology could be misused for **human enhancement** rather than disease treatment.
   - **Biopiracy:** Genetic information of **indigenous communities** may be exploited without informed consent.

- **Medical risks:** Human gene therapy may cause **off-target mutations** and severe immune reactions such as **Cytokine Release Syndrome (CRS)**.
- **Laboratory biosafety:** e.g Incidents such as **vaccinia virus exposure in a Pune laboratory (2020–25)** highlight the need for stronger biosafety standards.

Way Forward

Pillar	Policy suggestions
1. Building a Simple and Strong Regulatory System	• **Create a single regulator (BRAI):** A unified biotechnology regulator can reduce delays, improve coordination, and ensure faster approvals. • **Adopt risk-based regulation:** Continue separate rules for low-risk gene-edited crops (like SDN-1 & SDN-2) while maintaining strict oversight for transgenic GM crops. • **Strengthen expert review:** Bodies like **GTAEC** should ensure scientific, clinical, and ethical evaluation of gene therapies.
2. Ensuring Safe and High-Quality Research	• **Thorough testing before approval:** Conduct proper animal studies and phased clinical trials before human use. • **Long-term safety monitoring:** Track patients for several years to identify delayed side effects and ensure lasting benefits. • **Upgrade research infrastructure:** Expand certified **BSL laboratories** and **GMP-compliant facilities** for safe research and production. **GMP (Good Manufacturing Practices):** A set of quality standards that ensure medicines, vaccines, and biotechnology products are **consistently manufactured, tested, and controlled** to meet safety, quality, and efficacy requirements.
3. Increasing Transparency and Public Trust	• **Share scientific evidence:** Make biosafety and trial data publicly available to improve transparency and build confidence. • **Create public awareness:** Educate farmers, patients, and consumers about the benefits, risks, and limitations of gene editing to reduce misinformation.
4. Promotion of Inclusive Innovation	• **Support Indian startups and research:** Expand schemes such as **BIG, BioNEST, Bio-RIDE, and BioE3** to encourage indigenous innovation. • **Make technology affordable:** Provide seed subsidies, fair pricing, and extension support so that small and marginal farmers also benefit from gene-editing technologies.
5. Strengthening the Ethical and Legal Safeguards	• **Continue ban on germline editing:** Allow gene editing only for therapeutic purposes and prohibit heritable genetic modification or “designer babies.” • **Improve patent and data protection:** Reform patent laws to encourage innovation while preventing monopolies, and protect the genetic resources of indigenous communities from biopiracy.

The challenge is not whether India should adopt gene editing, but **how to harness it responsibly**. A science-based, transparent, and people-centric approach can unlock its full potential for sustainable development.

3. CRISPR CAS-9 TECHNOLOGY

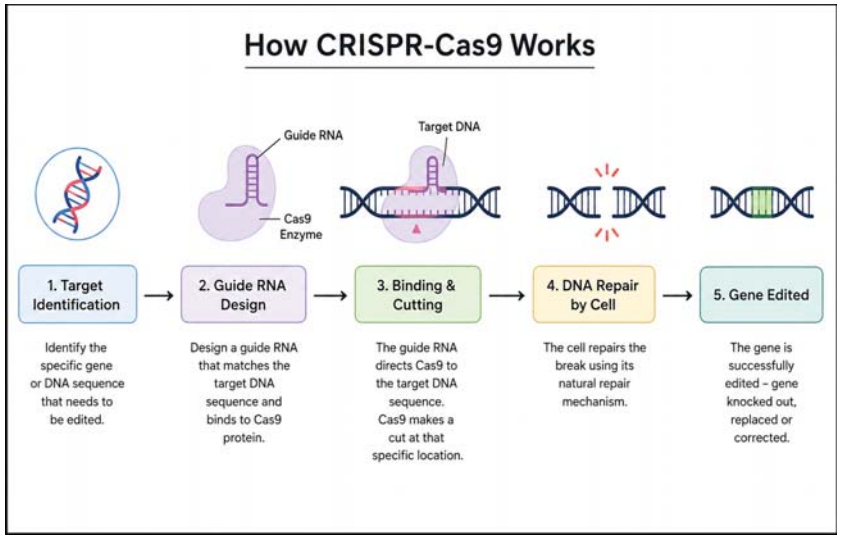
CRISPR-Cas9 stands for **'Clustered Regularly Interspaced Short Palindromic Repeats'** associated with the **Cas9** protein. It is a precise **genome-editing** tool that allows scientists to cut DNA at an exact target location and then delete, correct, or insert a gene — much like using guided 'molecular scissors'.

The system was adapted from a natural **bacterial immune mechanism**, where bacteria store snippets of viral DNA to recognise and destroy the same virus if it attacks again.

Jennifer Doudna and **Emmanuelle Charpentier** won the **Nobel Prize in Chemistry, 2020** for developing this into a programmable gene-editing tool.

How does the CRISPR CAS-9 Technology Works

- A **guide RNA (gRNA)** is designed to match a specific DNA sequence in the target gene.
- The **Cas9 enzyme**, guided by this RNA, locates and precisely cuts the DNA at that exact point.
- The cell's natural repair machinery then fixes the cut — either **disabling the gene**, **correcting a mutation**, or **inserting a new sequence**, depending on what is required.
- Unlike older recombinant DNA technology, CRISPR usually edits an organism's **own genes** rather than inserting foreign DNA, which is why it is called **genome editing (GE)** rather than genetic modification (GM).



Advantages Of CRISPR Technology Over Other Techniques:

- **Precision:** Targets a specific gene with far greater accuracy than older tools like **ZFNs** and **TALENs**.
- **Low Cost and Speed:** Much cheaper and faster to design than earlier gene-editing systems, democratising research.
- **No Foreign DNA (in SDN-1/SDN-2 edits):** The final edited organism can be free of any foreign gene, making it closer to a naturally occurring mutation.

CRISPR-Cas9 technology in the Indian Context

(a) Agriculture

- **Genome-Edited Rice:** Using CRISPR-Cas9 and Cas12a, ICAR scientists edited native genes in **Samba Mahsuri** and **MTU-1010** rice varieties; the resulting lines showed about **19% higher yield** and better tolerance to saline and alkaline soils in 2025 multi-location trials.

- **Water-Saving Rice: DRR Dhan 100 (Kamala)** and **Pusa DST Rice 1** are India's first genome-edited rice varieties, engineered to use less water while maintaining or increasing yield — a potential 'game changer' for water-stressed regions.
- **Edited Mustard:** A CRISPR-edited mustard line with **low pungency** and resistance to major fungal pathogens and pests is under second-year trials, with possible release around **2026**.
- **Indigenous Gene-Editing Platform: ICAR** scientists have unveiled a **native, indigenous genome-editing tool** to reduce dependence on foreign, patent-heavy platforms like CRISPR-Cas9.

(b) Healthcare

- **BIRSA 101 - India's First Indigenous CRISPR Gene Therapy:** Launched in **November 2025** by **CSIR-Institute of Genomics and Integrative Biology (IGIB)**, in partnership with the **Serum Institute of India**, for **Sickle Cell Disease**. Built on an indigenous CRISPR platform called **enFnCas9**, developed under the **Atmanirbhar Bharat** push for medical self-reliance.
- **Named after Birsa Munda:** The therapy is dedicated to the tribal freedom fighter, reflecting its focus on India's **tribal communities** (Gond, Munda, Bhil, Santal, and others), where Sickle Cell Disease is most prevalent.
- **Affordability Goal:** Global gene therapies for the same disease can cost **Rs 20-25 crore**; the indigenous platform aims to cut this cost sharply, supporting the **National Sickle Cell Anaemia Elimination Mission (2023-2047)** and the goal of a **'Sickle-Cell-Free India' by 2047**.
- **Diagnostics:** CRISPR-based diagnostic kits (using Cas12a/Cas13 detection) enable rapid, low-cost, point-of-care testing for infectious diseases.

New Regulatory Framework for CRISPR in India

CRISPR-based research and products are regulated under the **Environment (Protection) Act, 1986** and the **1989 Rules** on genetically engineered organisms, through the same multi-tier system used for GM crops — **IBSC**, **RCGM**, and the apex **Genetic Engineering Appraisal Committee (GEAC)**.

- **2022 GEAC Guidelines (a turning point):** Genome-edited plants falling under **SDN-1** and **SDN-2** categories — where the final product carries **no foreign DNA** — were exempted from the stringent GMO regulatory pathway applicable to transgenic crops.
- **Effect of the Exemption:** This has significantly **accelerated research and field trials** of CRISPR-edited crops in India compared to conventional GM crops, which continue to face a long regulatory and legal logjam (e.g., **GM Mustard DMH-11**, stayed by the Supreme Court).
- **Human Applications:** CRISPR-based therapies for humans additionally require clearance from the **Central Drugs Standard Control Organisation (CDSCO)** and must follow **ICMR's National Ethical Guidelines for Biomedical Research**.

SDN Type	Key technique used
SDN-1	Creates **small mutations (insertions/deletions)** without using any DNA template or foreign DNA.
SDN-2	Makes **precise DNA changes** using a short DNA template, **without inserting a foreign gene**.
SDN-3	**Inserts an entire gene or large DNA sequence** (often foreign DNA), producing a **transgenic GMO**.

Significance of CRISPR-Cas9 for India

Dimension	Role of CRISPR-Cas9
Food Security	Faster development of climate-resilient, water-saving, high-yield crop varieties without the long GM regulatory delay.
Health Security	Affordable indigenous therapies like BIRSA 101 reduce dependence on gene therapies costing crores abroad.
Strategic/Tech Autonomy	Indigenous platforms (**enFnCas9**, ICAR's native gene-editing tool) cut reliance on foreign, patent-protected CRISPR systems.
Inclusive Development	Targeted treatment of Sickle Cell Disease directly benefits **tribal populations** in central and eastern India.
Ease of Research	The **2022 SDN-1/SDN-2 exemption** has made India a comparatively faster and cheaper place to research genome-edited crops.

Challenges and Concerns

- **Regulatory Ambiguity:** While SDN-1/SDN-2 edits are exempted, classification disputes can arise over whether a specific edit truly has 'no foreign DNA', creating uncertainty for developers.
- **Absence of a Comprehensive Biosafety Law:** The **Biotechnology Regulatory Authority of India (BRAI) Bill** remains unenacted, so CRISPR governance still relies on decades-old 1989 Rules.
- **Off-Target Effects:** CRISPR edits can sometimes unintentionally affect unrelated parts of the genome, raising safety concerns that need rigorous long-term study.
- **Germline Editing Concerns:** Editing human embryos would create heritable changes, raising the spectre of unregulated **'designer babies'** — a line India's ethical guidelines strictly prohibit for reproductive purposes.
- **High Initial Cost of Human Therapies:** Even indigenous platforms need scale-up before treatment costs fall meaningfully for patients.
- **IP Dependence:** Foundational CRISPR patents are largely held abroad, raising licensing costs for widespread commercial use unless indigenous alternatives (like enFnCas9) scale up.
- **Public Trust:** Past controversies around GM crops (Bt Brinjal moratorium) mean CRISPR products may also face public scepticism without transparent risk communication.

Way Forward

1. Codification of Clear Classification Norms	Issue clear guidelines for classifying **SDN-1 and SDN-2** products to reduce regulatory uncertainty and enable faster approvals.
2. Enactment of a Dedicated Biosafety Law	Revive the **BRAI Bill** to establish a single, science-based regulator for GM and genome-edited products.
3. Scaling Up Indigenous Gene-Editing Platforms	Expand funding for Indian technologies such as **enFnCas9** and **ICAR's native gene-editing tools** to reduce costs and dependence on foreign patents.
4. Strengthening Bioethics Oversight	Strictly enforce **ICMR guidelines** on human germline editing and periodically update ethical standards.
5. Investment in Skills and Research Infrastructure	Expand **biofoundries, CRISPR research centres, and skilled manpower** beyond premier institutes like **CSIR-IGIB**.

6. Enhancing Public Awareness and Transparency	Improve public communication and disclose safety data on genome-edited crops and therapies to build public trust.

Conclusion

The true potential of CRISPR-Cas9 relies on establishing clear, science-based national biosafety policies and addressing ethical concerns. Safely harnessed, it remains an indispensable tool for achieving global health security, agricultural sustainability, and driving the transition toward a trillion-dollar bioeconomy.

4. GENOME SEQUENCING TECHNOLOGY

Genome sequencing is the process of determining the complete, precise order of nucleotide bases — **Adenine (A), Thymine (T), Cytosine (C), and Guanine (G)** — that make up an organism's **DNA**.

The human genome contains about **3.2 billion base pairs** and roughly **23,500 genes**, and reading this sequence is like decoding the complete biological 'instruction manual' of a person.

Whole Genome Sequencing (WGS) reads the entire genome in one process, unlike older methods that sequenced only specific genes. It relies on high-throughput **DNA sequencers**, which read tagged DNA fragments, followed by computational reassembly of the full sequence.

From Decades to Days

- The first human genome, sequenced under the international **Human Genome Project**, took **13 years** and cost about **US$ 3 billion**, and was completed in **2003**.
- India announced its **first complete human genome** in 2009.
- Today, thanks to advances in sequencing technology, an entire human genome can be sequenced and quality-checked in just about **5 days**, at a fraction of the earlier cost.

Genome India Project (GIP) - India's Flagship Initiative

The **Genome India Project**, launched by the **Department of Biotechnology (DBT)** on **3rd January 2020**, is India's most ambitious genomics initiative — often described as inspired by the global Human Genome Project.

Key Data and Milestones

- **Completion:** Announced by PM Modi at the **Genomics Data Conclave in January 2025**, marking completion of Phase 1.
- **Scale: Whole Genome Sequencing (WGS)** completed for **10,074 individuals** from **83-99 diverse Indian population groups** — about **2% of India's roughly 4,600 anthropologically defined population groups**.
- **Collaboration:** A nationwide effort involving **over 20 institutions**, led by the **Centre for Brain Research, Indian Institute of Science (IISc), Bengaluru**, along with CSIR-CCMB Hyderabad, CSIR-IGIB New Delhi, and NIBMG Kolkata, among others.
- **Genetic Variations Identified:** Over **135 million genetic variations** were identified, including rare variants unique to Indian sub-groups.
- **Biobank:** A biobank of **20,000 blood samples** has been established at the Centre for Brain Research, IISc, Bengaluru, to support future research.
- **Data Size:** The project generated about **8 petabytes** of data, with roughly **80 GB per individual genome sequence**.

- **Data Access:** All data is archived at the **Indian Biological Data Centre (IBDC), Faridabad** — India's first national life-science data repository — and is accessible via the **Framework for Exchange of Data (FeED)** protocol under the **BIOTECH-PRIDE Guidelines**.
- **Future Target:** The government has announced an ambitious future goal of sequencing **10 million genomes** to accelerate personalized medicine in India.

Applications of Genome Sequencing

(a) Healthcare and Medicine

- **Precision/Personalised Medicine:** A reference Indian genome allows treatments to be tailored to an individual's genetic makeup, especially for **rare and hereditary diseases**.
- **Disease Diagnostics:** Helps identify disease-causing mutations linked to conditions like **cancer, diabetes, and neurological disorders**, enabling earlier and more accurate diagnosis.
- **Pharmacogenomics:** Predicts how an individual will respond to specific drugs based on their genetic profile, helping optimise drug choice and dosage.
- **Prenatal and Newborn Screening:** Enables early detection of genetic disorders before or shortly after birth.

(b) Public Health

- **Epidemiology and Outbreak Tracking:** Genome sequencing of pathogens (e.g., during COVID-19) allows real-time tracking of variants and better public health responses.
- **Vaccine Development:** Pathogen genome data supports the design of targeted and updated vaccines.

(c) Agriculture and Beyond

- **Crop Improvement:** Identifies genes responsible for desirable traits like **disease resistance, drought tolerance, and higher yield**, speeding up breeding programmes
- **Livestock Genomics:** Genetic insights help improve livestock breeding and disease management.
- **Forensics:** DNA sequencing aids criminal investigations and identification in mass disasters.

Significance of Genome Sequencing for India

Dimension	How Genome Sequencing Helps
Genetic Diversity	India, with over **4,600 population groups**, is often called the **'largest genetic lab in the world'** — a reference genome captures this unique diversity, which global databases have historically under-represented.
Health Security	A reference Indian genome enables **India-specific diagnostic tools** instead of relying on genetic markers developed for other populations.
Digital Public Good	Data hosted at the **IBDC** functions as an open, national resource, fostering research collaboration and innovation across institutions.
Bioeconomy Growth	Genomic data underpins new diagnostics, therapeutics, and precision-medicine start-ups, contributing to India's fast-growing **Bioeconomy**.
Strategic Self-Reliance	Reduces dependence on sending Indian biological samples abroad for sequencing, strengthening **data sovereignty**.

Challenges Facing Genome Sequencing in India

- **Data Privacy and Regulation Gaps:** Existing regulations still allow commercial export of biological samples abroad for sequencing, raising **data security and privacy** concerns; a dedicated genomic data protection law is still awaited.
- **Ethical Concerns:** Risks of **genetic discrimination** (e.g., by insurers or employers), questions around **informed consent**, and long-standing fears around **eugenics**.

- **Cost and Accessibility:** Even though costs have fallen sharply, large-scale sequencing projects remain expensive, limiting access in low-resource public health settings.
- **Limited Coverage So Far:** The completed 10,000-genome sample covers only about **2% of India's population groups**, so much of India's genetic diversity is still unmapped.
- **Data Accuracy:** Sequencing technologies, especially **long-read sequencing**, still face error-correction challenges that need continuous refinement.
- **Fragmentation of Data:** Multiple private genetic-testing companies hold data in separate silos, limiting the value of a unified national resource.
- **Translating Data into Treatment:** Only a small fraction of diseases are **monogenic** (caused by a single gene); most conditions are complex, so sequencing data alone does not guarantee new cures.
- **Massive Data Management:** The **8-petabyte** dataset from Phase 1 alone poses major challenges of storage, computing capacity, and long-term security.

Way Forward

1. Enactment of a Genomic Data Protection Law	Establish strong legal safeguards against misuse of genetic data and regulate the export of biological samples.
2. Expansion of Phase II of the Genome India Project (GIP)	Scale up sequencing towards the target of **10 million genomes** to ensure wider population coverage.
3. Strengthening Domestic Sequencing Capacity	Invest in indigenous sequencing facilities and high-performance computing to reduce dependence on foreign laboratories.
4. Strengthening the Informed Consent Framework	Ensure participants clearly understand how their genomic data will be collected, stored, shared, and used.
5. Enhancing Data Interoperability	Align public and private genomic databases with national standards such as **IBDC** and **FeED** to reduce data fragmentation.
6. Translating Genomic Research into Diagnostics	Promote applied research to develop affordable, India-specific diagnostic kits, precision medicines, and therapies.

Conclusion

With initiatives such as the **Genome India Project**, India is laying the foundation for a **genomics-driven bioeconomy**. The next step is to translate genomic data into affordable diagnostics, targeted therapies, and better public health outcomes.

5. GM CROPS

Genetically Modified (GM) crops are plants whose **DNA has been altered using genetic engineering**, typically by inserting a gene from an unrelated organism — a plant, animal, or soil bacterium — to introduce a desirable trait such as **pest resistance, herbicide tolerance, or higher yield**.

As per the **World Health Organization (WHO)**, this counts as genetic material altered in a way that does not occur naturally through mating or natural recombination.

In India, GM crops are regulated as **'Genetically Engineered Organisms'** under the **Environment (Protection) Act, 1986**, with the **Genetic Engineering Appraisal Committee (GEAC)** under the **Ministry of Environment, Forest and Climate Change** as the apex approving authority for their environmental release

Status of GM Crops in India

Crop	Status	Key Facts
Bt Cotton	**Only commercially approved GM crop** (since 2002)	Uses a gene from soil bacterium *Bacillus thuringiensis* for resistance to the bollworm pest; adoption is now near-universal among Indian cotton farmers.
Bt Brinjal	Approved by GEAC in 2009, but under an **indefinite moratorium since 2010**	Faced massive public backlash over biosafety and health concerns before commercial release.
GM Mustard (DMH-11)	GEAC-cleared (Oct 2022) but **release stayed**; Supreme Court gave a **split verdict in July 2024**	• India's first indigenously developed transgenic **food crop**, using 'barnase-barstar' genes from Bacillus amyloliquefaciens; • claims **~28% higher yield** than the national check.
GM Maize, Chickpea, Tomato	Various stages of **research and trials**	None released commercially so far.

The GM Mustard (DMH-11) Case

- **Background: DMH-11 (Dhara Mustard Hybrid-11)**, developed at Delhi University, is a cross between the Indian variety **'Varuna'** and East European **'Early Heera-2'**, engineered for **Herbicide Tolerance (HT)**.
- **GEAC Approval:** Environmental release was approved by GEAC in **October 2022**, but the Supreme Court asked the Centre to **maintain status quo** in November 2022 following petitions in *Gene Campaign & Anr. v. Union of India*.

  3. Split Verdict (July 2024):
  - A two-judge bench delivered differing opinions — **Justice B.V. Nagarathna** invalidated the approval, citing violations of **Article 21**, the **precautionary principle**, and the **doctrine of public trust**; **Justice Sanjay Karol** upheld GEAC's process, finding no illegality.
- **Common Directive:** Despite disagreeing on the outcome, **both judges directed the Centre to frame a comprehensive national policy on GM crops** — covering research, cultivation, trade and commerce — after consulting all stakeholders.
- **Current Status:** The matter now awaits the constitution of a **larger bench** by the Chief Justice of India, and the government's promised **national GM crop policy** is still awaited.

Arguments in Favour of GM Crops	Arguments Against GM Crops
Higher agricultural productivity: DMH-11 mustard recorded **~28% higher yield** than the national check and **~37% higher** than zonal checks, with potential yields of **3–3.5 t/ha**.	**Biosafety concerns:** Long-term human health and ecological impacts remain uncertain, with criticism over limited indigenous safety studies.
Reduced import dependence: GM oilseed crops can increase domestic edible oil production and reduce India's dependence on costly imports.	**Transparency gaps:** Concerns over non-disclosure of biosafety data and conflict of interest in the approval process (highlighted in the **GM Mustard case**).

Lower cultivation costs: Pest-resistant and herbicide-tolerant traits reduce pesticide use, manual weeding, and overall input costs.	**Threat to biodiversity:** Risk of gene flow to wild relatives and possible adverse effects on pollinators such as **honeybees**.
Food & nutritional security: Higher productivity and biofortified crops can strengthen **food security** and contribute to **SDG-2 (Zero Hunger)**.	**Herbicide resistance:** Excessive herbicide use may lead to the emergence of **herbicide-resistant "super weeds."**
Proven success of Bt Cotton: Helped increase cotton production, making India one of the world's largest cotton producers.	**Farmer dependence:** High seed costs and patent protection may increase dependence on private seed companies and farmer indebtedness.
	Illegal cultivation: Unapproved **HT-Bt cotton** is estimated to cover **15–25%** of cotton acreage in states like Gujarat and Maharashtra, indicating weak enforcement.
	Public trust deficit: Controversies surrounding **Bt Brinjal** and **GM Mustard** have reduced public confidence in GM technology.

Regulatory Framework for GM Crops in India

- **Environment (Protection) Act, 1986** and the **1989 Rules** on hazardous microorganisms/genetically engineered organisms form the legal basis for regulation.
- **Institutional Biosafety Committee (IBSC):** Institution-level body for initial biosafety oversight.
- **Review Committee on Genetic Manipulation (RCGM):** Reviews research-stage and field-trial-level proposals under the Department of Biotechnology.
- **Genetic Engineering Appraisal Committee (GEAC):** The apex regulator that approves large-scale use and **environmental release** of GM crops; approvals are typically valid for **4 years**, renewable for **2 years** at a time.
- **Absence of a Standalone Law:** The proposed **Biotechnology Regulatory Authority of India (BRAI) Bill**, meant to create a single-window regulator, has still not been enacted, leaving GM crop governance fragmented across multiple committees.

GM Crops and India's Trade Policy

GM crops are also a live issue in India's **trade negotiations**, especially with the United States, which has pushed for greater market access for GM soybean, corn, and dairy products.

India has consistently resisted opening its **agriculture and dairy sectors**, calling them **'sacrosanct red lines'** in trade talks, citing concerns for the livelihoods of small and marginal farmers and India's strict food-safety regime under the **Food Safety and Standards Authority of India (FSSAI)**.

Way Forward

- **Frame the Directed National GM Crop Policy:** Fulfilling the Supreme Court's **2024 directive** by adopting a transparent, stakeholder-consulted national policy on GM crop research, cultivation, and trade.
- **Enact a Comprehensive Biosafety Law:** Revive a version of the **BRAI Bill** to create a single, accountable regulator, replacing today's fragmented committee structure.

- **Mandatory Public Disclosure of Safety Data:** Make all risk-assessment and biosafety studies publicly available, as directed by courts, to build trust and scientific accountability.
- **Independent, Indigenous Research:** Invest in India-specific, peer-reviewed biosafety and environmental-impact studies rather than relying primarily on foreign data.
- **Crack Down on Illegal Cultivation:** Strengthen enforcement against unapproved HT-Bt cotton to preserve the integrity of the regulatory system.
- **Differentiate GM from Genome Editing:** Use the lighter-touch regulatory pathway already available for **genome-edited (SDN-1/SDN-2) crops** to fast-track safer innovations, while keeping GM crops under rigorous scrutiny.
- **Sustained Public Engagement:** Build long-term, transparent communication with farmers and citizens on the science, benefits, and risks of GM crops.

Conclusion

As climate change and food demand intensify, the debate should shift from **"whether to adopt GM crops"** to **"how to deploy them safely, transparently, and responsibly."**

A transparent, science-based, and legally sound regulatory framework will be essential to resolving this long-running debate as India works towards **Viksit Bharat by 2047**.

6. STEM CELL TECHNOLOGY

Stem cells are unique, unspecialised cells with two defining properties: **self-renewal** (the ability to divide and produce more stem cells) and **potency** (the ability to differentiate into various specialised cell types such as blood, muscle, nerve, or bone cells).

Because of this dual ability, stem cells are seen as the foundation of **regenerative medicine** — repairing, replacing, or regenerating damaged tissues and organs.

Types of Stem Cells

Type of Stem Cell	Explanation
Embryonic Stem Cells (ESCs)	• Derived from **early-stage embryos**. • **Pluripotent**— meaning,can develop into almost any cell type. • Their use raises **ethical concerns** due to destruction of embryos.
Adult (Somatic) Stem Cells	• Found in tissues such as **bone marrow and adipose (fat) tissue**. • **Multipotent**—can produce only related cell types (e.g., blood cells). • Widely used in established therapies such as **bone marrow transplantation**.
Induced Pluripotent Stem Cells (iPSCs)	• Created by **reprogramming adult cells** in the laboratory. • **Pluripotent**, similar to embryonic stem cells. • They are made to Avoid ethical issues associated with embryos. • Discovered by **Shinya Yamanaka** (Nobel Prize, 2012).
Cord Blood (Perinatal) Stem Cells	• Collected from **umbilical cord blood** at birth. • These are Stored in **cord blood banks** for future therapeutic use. • Used mainly in the treatment of blood-related disorders.

Mesenchymal Stem Cells (MSCs)	• Found in **bone marrow, fat, and other tissues**. • **Multipotent** cells - used in regenerative medicine research • used in Many commercial regenerative therapies

Applications of Stem Cell Technology

- **Haematopoietic Stem Cell Transplantation (HSCT):** The only category of stem cell therapy currently recognised as **'standard care'** in India — used to treat blood cancers like **leukaemia**, and blood disorders like **thalassemia** and **sickle cell disease**.
- **Regenerative Medicine:** Long-term research goal of growing replacement tissues and even whole **organs** to reduce dependence on organ transplants.
- **Drug Testing and Disease Modelling:** Stem-cell-derived tissue models allow new medicines to be tested for safety and effectiveness before human trials, and help researchers study disease mechanisms in the lab.
- **Burns and Corneal Repair:** Skin and corneal stem cells are used clinically to treat severe burns and certain eye injuries.
- **Research Frontier:** Experimental research continues into stem-cell-based treatments for conditions like **Parkinson's disease, spinal cord injury, and diabetes**, though most remain at the trial stage.

Regulatory Framework in India

Stem cell research and therapy in India have evolved through a series of guidelines and are now shaped by a landmark **2026 Supreme Court judgment**.

- **Early Guidelines (2001-2007):** The **Department of Biotechnology (DBT)** issued the first guidelines in 2001 in response to misuse of stem cell science; **ICMR** followed with a supportive policy in 2002. DBT and ICMR jointly issued the first **national-level guidelines in 2007**.
- **National Guidelines for Stem Cell Research, 2017 (NGSCR):** Classified stem cell interventions into permissible clinical practice (**HSCT only**) versus research requiring **clinical trial approval**

  It created the **National Apex Committee for Stem Cell Research and Therapy (NAC-SCRT)** for oversight.
- **Legal Classification as 'Drugs':** Stem cells fall under the **Drugs and Cosmetics Act, 1940** and the **New Drugs and Clinical Trial Rules, 2019**, meaning unapproved therapeutic claims can attract legal liability.
- **Yash Charitable Trust v. Union of India (30 January 2026):** A landmark Supreme Court ruling holding that stem cell therapy for **Autism Spectrum Disorder (ASD)** cannot be offered as a **routine or commercial clinical service**, since it lacks proven scientific evidence; it may only be administered within **approved, monitored clinical trials**. The Court also classified stem cells as **'drugs'** under the 1940 Act and held that **consent cannot substitute for proven safety and efficacy**.

Significance of Stem Cell Technology for India

Dimension	How Stem Cell Technology Helps
Health Security	**Haematopoietic stem cell transplants** offer a proven cure pathway for blood cancers and inherited blood disorders like thalassemia and sickle cell disease.
Future of Medicine	Long-term potential for **regenerative medicine** — repairing damaged tissue and, eventually, growing replacement organs.

Drug Development	Enables safer, faster, and more ethical **drug testing** using lab-grown tissue models instead of only animal or human trials.
Patient Protection	A clearer regulatory framework, reinforced by the **2026 Supreme Court judgment**, protects vulnerable patients from **unproven, costly 'stem cell tourism'**.
Research Ecosystem	A growing base of **cord blood banks**, biobanks and research institutions positions India as an emerging hub for legitimate stem cell science.

Challenges Facing Stem Cell Technology in India

- **Unproven, Commercialised Therapies:** Private clinics have long marketed unapproved **mesenchymal stem cell** treatments for conditions like autism and cerebral palsy as 'cures', exploiting the hopes of vulnerable families — the core issue flagged in the **Yash Charitable Trust** case.
- **Regulatory Fragmentation:** Oversight has historically been split across DBT, ICMR, CDSCO, and NMC(National Medical Commission).
- **Absence of Dedicated Legislation:** Stem cell therapy is currently governed only through **guidelines** and the general **Drugs and Cosmetics Act, 1940**. There is an absence of a specific law — a gap the Supreme Court has now asked Parliament to fill.
- **Ethical Concerns over Embryonic Stem Cells:** Using human embryos raises questions about the moral status of the embryo, requiring strict ethical review.
- **Informed Consent Challenges:** As the Supreme Court noted, **consent cannot be 'informed'** when the underlying treatment itself lacks established scientific evidence — making many past consent forms legally and ethically questionable.
- **High Costs and Financial Exploitation:** Families of patients with conditions like autism have incurred **huge, often futile, financial costs** pursuing unproven interventions.
- **Balancing Hope and Evidence:** Genuine tension exists between patients'/parents' desire to try any available option and the medical-legal requirement that treatments be backed by **robust clinical evidence**.

Way Forward

Way Forward	Broad steps needed
1. Enactment of Dedicated Stem Cell Legislation	Enact a comprehensive law to regulate stem cell research, clinical trials, and therapies, with clear penalties for violations.
2. Strengthening the Regulatory Framework	Reconstitute and empower **NAC-SCRT** as a single apex body for uniform and continuous oversight.
3. Expansion of Regulated Clinical Trials	Increase access to ethically approved clinical trials so patients can receive experimental therapies under proper supervision.
4. Strict Action Against Unproven Therapies	Crack down on clinics promoting unapproved stem cell treatments and misleading therapeutic claims, in line with **NMC guidelines**.
5. Public Awareness and Patient Education	Educate the public about the difference between **approved therapies (e.g., HSCT)** and unproven stem cell interventions.
6. Promotion of Indigenous Stem Cell Research	Support ethically approved Indian research and clinical trials to develop safe, effective, and evidence-based stem cell therapies.

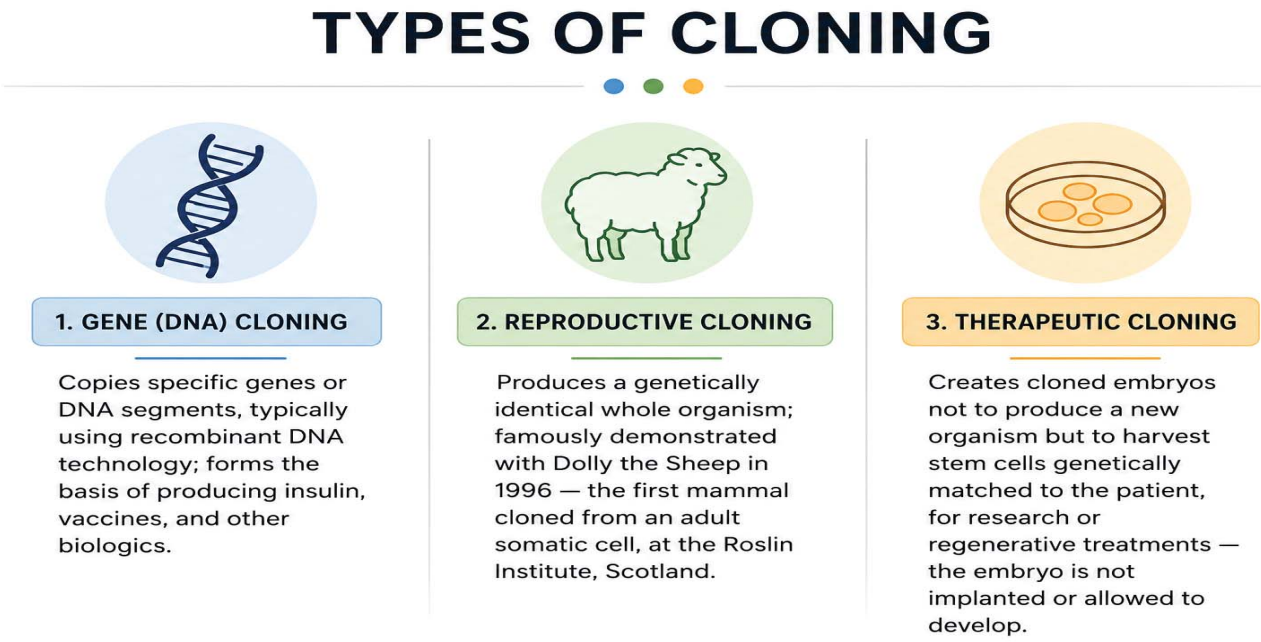
Conclusion

With strong regulatory oversight and sustained investment in indigenous research, stem cell technology can become a cornerstone of **precision medicine and affordable healthcare** in India.

7. CLONING

Cloning is the process of creating a genetically **identical copy** of a biological entity — a gene, cell, tissue, or entire organism — using biotechnology. The clone shares the exact same **DNA/genetic makeup** as the original, unlike sexual reproduction, which combines genetic material from two parents to produce a genetically unique offspring.

The most common technique for cloning whole animals is **Somatic Cell Nuclear Transfer (SCNT)** . In this technology, the nucleus of a somatic (body) cell from the animal to be cloned is inserted into an egg cell whose own nucleus has been removed. The egg is then stimulated to develop into an embryo genetically identical to the donor.



Applications of Cloning

(a) Agriculture and Animal Husbandry

- **Multiplying Elite Livestock:** Cloning allows rapid multiplication of high-yielding cattle and buffaloes with superior genetic traits (e.g., higher milk yield), instead of waiting generations through natural breeding.
- **India's Cloning Milestones: ICAR-National Dairy Research Institute (NDRI), Karnal** produced **'Ganga'**, India's first cloned **Gir cow calf**, and has cloned over **30 buffaloes** using its indigenous **'handmade cloning'** technique — a simpler, more economical variant of SCNT.
- **Conservation of Indigenous Breeds:** For example, NDRI (National Dairy research Institute) has extended cloning to indigenous breeds like **Sahiwal** and **Red Sindhi** to conserve valuable genetic resources.
- **Saving Endangered Species: For example,** In a pioneering case, NDRI scientists cloned **'Deepasha'**, a female wild buffalo, from the sole surviving female wild buffalo ('Asha') in Chhattisgarh's **Udanti**

Wildlife Sanctuary — the state animal of Chhattisgarh. It was listed under **Schedule I of the Wildlife Protection Act, 1972**.

(b) Medicine and Research

- **Therapeutic Cloning for Regenerative Medicine:** Could, in principle, generate patient-matched **stem cells** for tissue repair without risk of immune rejection, though this remains largely experimental.
- **Disease Modelling and Drug Testing:** Cloned cell lines allow researchers to study diseases and test drug responses on genetically defined material.
- **Gene Cloning in Biopharma:** The foundation of producing **recombinant insulin, growth hormone, and vaccines** at industrial scale.
- **Xenotransplantation Research:** Cloning combined with gene editing is being explored globally to develop genetically modified animal organs suitable for human transplantation.

(c) Conservation Biology

- **De-extinction Research:** Globally, cloning technology is being explored (controversially) to potentially revive extinct species using preserved genetic material.

  - For example, Scientists at Colossal Biosciences achieved a milestone in de-extinction by producing three engineered dire wolf-like pups.
  - Rather than a true clone, these are gray wolves with 20 modified genes designed to replicate the size, fur, and skull shape of the extinct Aenocyon dirus

- **Genetic Rescue:** For species reduced to critically low numbers, cloning can help preserve genetic diversity that would otherwise be lost.

Regulatory and Ethical Position in India and Globally

Aspect	Explanation
Global Position on Human Cloning	**No country permits human reproductive cloning**, reflecting a global scientific and ethical consensus.
UNESCO Declaration (1997)	Declares **human reproductive cloning** contrary to **human dignity** under the Universal Declaration on the Human Genome and Human Rights.
India's Legal Framework	India has **no separate cloning law**, but the **ART Act, 2021**, **Surrogacy Act, 2021**, and **ICMR bioethics guidelines** effectively prohibit it.
ICMR National Guidelines	Prohibit **human reproductive cloning** and allow embryo research only after **ethical approval**.
Animal Cloning	Permitted for **agriculture and conservation** under institutional **Animal Ethics Committee** approvals (e.g., **ICAR-NDRI**), despite the absence of a dedicated cloning law.

Significance of Cloning for India

Dimension	How Cloning Helps
Livestock Productivity	Rapid multiplication of high-yielding cattle/buffaloes supports India's position as the **world's largest milk producer**.
Conservation	Demonstrated ability to clone endangered/rare animals (e.g., **Deepasha**, the wild buffalo) offers a new tool for wildlife conservation.

Indigenous Breed Protection	Cloning of native breeds like **Gir and Sahiwal** helps preserve indigenous genetic resources threatened by cross-breeding.
Biopharma	**Gene cloning** underpins India's large-scale, affordable production of insulin, vaccines, and biosimilars.
Research Capacity	Institutions like **NDRI** have developed indigenous, cost-effective cloning techniques (**'handmade cloning'**), reducing dependence on costlier foreign methods.

Challenges and Ethical Concerns

- **Low Success Rates:** Cloning remains technically difficult, with **low birth rates**, higher rates of miscarriage, and health complications in cloned offspring compared to natural breeding.
- **Health and Longevity Concerns:** Cloned animals have historically shown a higher incidence of health problems and, in some studied cases, **reduced lifespan** — Dolly the sheep herself died relatively young.
- **Reduced Genetic Diversity:** Widespread cloning of 'elite' animals could **narrow the gene pool**, making livestock populations more vulnerable to disease outbreaks.
- **Human Cloning - Deep Ethical Concerns:** Reproductive cloning of humans raises fundamental questions about **identity, individuality, and human dignity**, and risks of exploitation, which is why it remains banned worldwide.
- **Therapeutic Cloning Debate:** Even non-reproductive, therapeutic cloning is ethically contentious because it involves creating and then destroying human embryos for research.
- **High Cost:** Cloning technology remains expensive and technically demanding, limiting its use to select research institutions rather than widespread farm-level adoption.
- **Absence of a Dedicated Law:** India regulates cloning-adjacent activities through **institutional ethics committees** and general bioethics guidelines, rather than a specific, comprehensive cloning law.

Way Forward

Way Forward	Explanation
1. Scaling Up Indigenous Cloning Technologies	Expand **NDRI's handmade cloning technique** to conserve indigenous cattle and buffalo breeds at lower cost.
2. Maintaining Genetic Diversity	Combine cloning with breeding programmes to prevent excessive narrowing of the genetic pool.
3. Strengthening Ethical Oversight	Ensure all cloning research undergoes rigorous review by **Institutional Ethics Committees** and **ICMR**.
4. Promoting Global Cooperation	Continue supporting international efforts, including the **UNESCO Declaration**, to prohibit human reproductive cloning.
5. Expanding Conservation Cloning	• Use cloning to conserve endangered wildlife • fOR example, building on the **Deepasha** cloning initiative in collaboration with wildlife authorities.
6. Enhancing Public Awareness	Promote public understanding of the science and ethics of cloning for informed policy and societal acceptance.

Conclusion

Cloning technology holds significant promise for **medicine, agriculture, and biodiversity conservation**, but its application must remain guided by **ethical principles, scientific evidence, and robust regulation**.

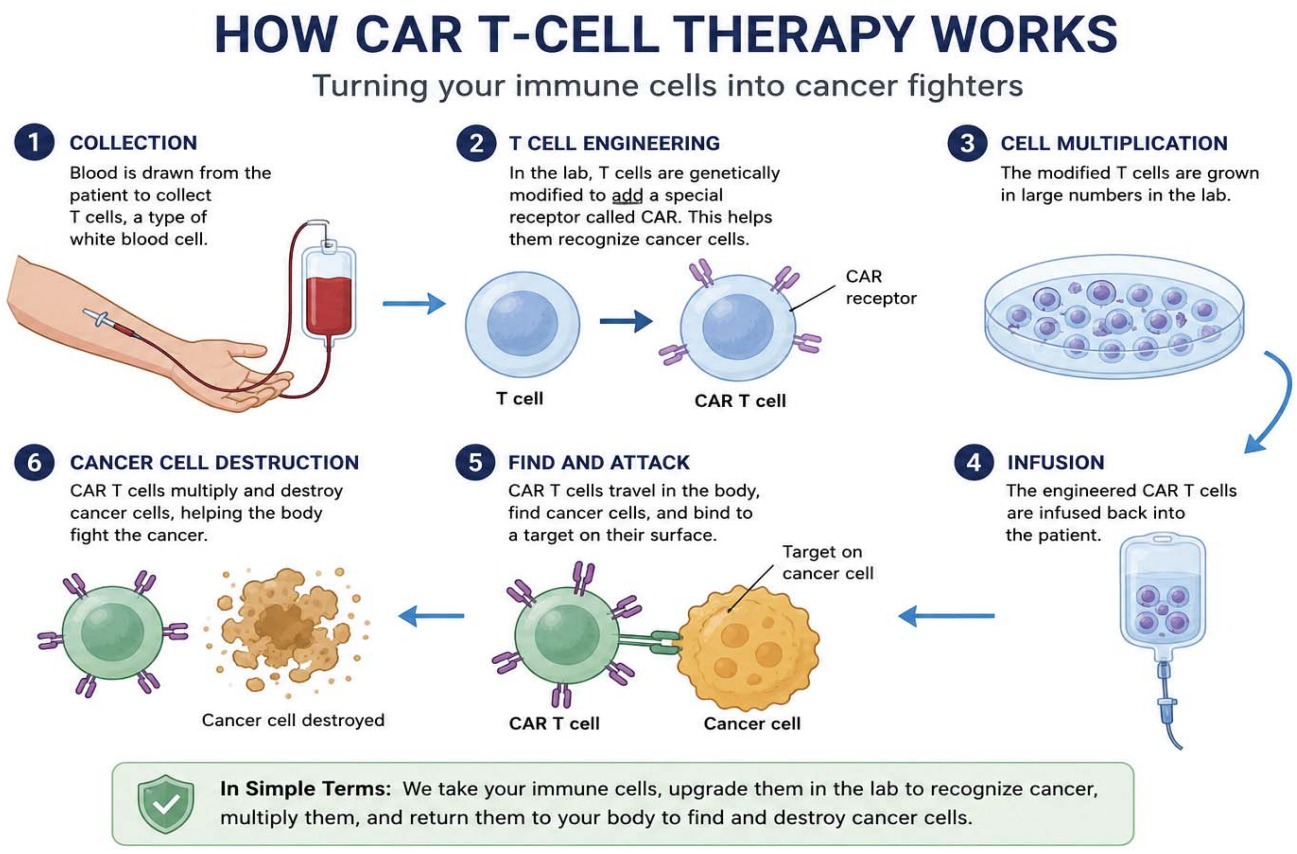
8. CLONING

CAR-T (Chimeric Antigen Receptor T-cell) therapy is a form of **personalised immunotherapy** that genetically re-engineers a patient's own immune cells to recognise and destroy cancer cells.

It is often called a **'living drug'**, since the modified cells continue to multiply and act inside the patient's body.

How it Works

- **Leukapheresis:** T-cells (a type of white blood cell central to immunity) are extracted from the patient's blood using a cell-separator machine, typically over about **4 hours**.
- **Genetic Engineering:** In the laboratory, these T-cells are modified using a **viral vector** to express a **Chimeric Antigen Receptor (CAR)** on their surface — a receptor designed to specifically recognise a cancer marker (most commonly **CD19 receptor**).
- **Expansion:** The engineered CAR-T cells are multiplied in the lab to reach the numbers needed for an effective dose.
- **Re-infusion:** The modified cells are infused back into the patient, where they seek out and destroy cancer cells bearing the target marker.



NexCAR19 – India's Indigenous CAR-T Cell Therapy

- **Developed by: ImmunoACT** (IIT Bombay spin-off) in collaboration with **Tata Memorial Centre**; first patient infusion in **June 2021**.
- **Regulatory approval:** Approved by **CDSCO (October 2023)** as **India's first indigenously developed CAR-T cell therapy**.
- **Clinical outcomes: 67–70% overall response rate** with ~50% complete remission; **lower Cytokine Release Syndrome (CRS)** and **no neurotoxicity** reported; results published in **The Lancet Haematology (2025)**.
- **Cost advantage:** Priced at **₹30–40 lakh**, approximately **one-tenth** the cost of comparable Western CAR-T therapies.
- **National recognition:** Showcased among **India's Top Three Innovations** at **ESTIC 2025**.

Government Support Behind the Breakthrough

- **BIRAC's BioNest Initiative:** Provided early-stage funding, mentorship, and incubation support to ImmunoACT through IIT Bombay's technology incubator, **SINE**.
- **DBT and BIRAC Funding:** The **Department of Biotechnology** and **Biotechnology Industry Research Assistance Council** backed the project through its translational research to commercial-scale stages.
- **BioE3 Policy Support:** Under the **Biomanufacturing** component of the **BioE3 Policy**, DBT funded a **200-litre GMP-grade lentiviral vector and plasmid production platform**, enabling manufacture of gene-delivery vectors for up to **1,000 patients annually**.
- **Regulatory Precedent:** Since India's drug regulatory framework was historically built around evaluating **generics**, NexCAR19's approval required regulators and industry to jointly build a **new evaluation pathway for cell and gene therapies (CGTs)** — a precedent that will ease approval of future indigenous therapies.

Applications and Emerging Uses of CAR-T Therapy

- **Blood Cancers:** Currently the primary approved use — relapsed/refractory **leukaemia, lymphoma, and (globally) multiple myeloma**, especially where CD19 or similar markers are expressed on 85-90% of cancer cells.
- **Autoimmune Diseases:** Emerging global and Indian research is exploring CAR-T cell therapy for severe **autoimmune conditions**, extending its use beyond cancer.
- **Future Targets:** DBT-supported research is working towards indigenous CAR-T solutions for **multiple myeloma, glioblastoma (brain cancer)**, and reducing therapy-related toxicities.
- **Solid Tumours (Research Stage):** Globally, extending CAR-T therapy to solid tumours remains a major research frontier, given the added complexity compared to blood cancers.

Significance of CAR-T Therapy for India

Dimension	How CAR-T Therapy Helps
Health Security	Offers a life-saving option for patients with relapsed/refractory blood cancers who have exhausted conventional treatments.
Affordability	At **1/10th to 1/13th** the global price, NexCAR19 makes advanced cell therapy accessible to a far wider set of Indian patients.
Strategic/Tech Autonomy	Demonstrates India's capacity to develop **complex, novel biologics indigenously**, not just generics, reducing dependence on costly imported therapies.

Global Impact	India's low-cost model is being seen as a template for making CAR-T therapy viable for other **low- and middle-income countries**.
Regulatory Capacity Building	NexCAR19's approval helped CDSCO build fresh regulatory experience in evaluating **cell and gene therapies**, easing the path for future innovations.

Challenges Facing CAR-T Therapy in India

- **Still Expensive for Most Patients:** Despite being far cheaper than global prices, **Rs 30-40 lakh** remains unaffordable for the majority of Indian households without insurance support.
- **Limited Insurance Coverage:** Most private insurance policies in India **do not fully cover** CAR-T therapy yet, leaving patients dependent on out-of-pocket payment or limited government/research assistance.
- **Concentration in Major Cities:** Specialised infrastructure and trained personnel remain largely limited to **tertiary cancer centres** in metros like Mumbai, Delhi, Bengaluru, and Chennai, limiting access for patients elsewhere.
- **Skilled Manpower Needs:** Administering CAR-T therapy requires specialised haemato-oncology teams trained to manage complications like **Cytokine Release Syndrome (CRS)**.
- **Manufacturing Complexity:** Each dose is **personalised** to the patient, making scale-up and quality-control far more complex than for conventional drugs.
- **Regulatory Learning Curve:** As India's first approved CGT, NexCAR19's journey highlighted gaps in a regulatory system historically built for generics rather than novel, gene-modified cell therapies.
- **Competition and Sustainability:** Emerging players (e.g., Immuneel Therapeutics) will need sustained government and private investment to keep expanding indigenous capacity.

Way Forward

Steps needed	Explanation
1. Expanding Insurance Coverage	Integrate CAR-T therapy into **Ayushman Bharat** and insurance schemes to improve affordability of high-cost curative treatments.
2. Decentralising Access	Expand the **centralised manufacturing–decentralised treatment** model to improve access beyond major metropolitan cities.
3. Strengthening Skilled Manpower	Increase specialised training for **haemato-oncology** teams to safely deliver CAR-T therapy across more hospitals.
4. Sustaining Indigenous R&D Funding	Continue government support through **DBT, BIRAC, and BioE3** to strengthen domestic cell and gene therapy manufacturing.
5. Streamlining the Regulatory Framework	Build on the **NexCAR19** approval to create a faster and more predictable regulatory pathway for future therapies.
6. Expanding Therapeutic Applications	Support research to extend indigenous CAR-T therapy to **multiple myeloma, glioblastoma, and autoimmune diseases**.

Conclusion

CAR-T cell therapy marks a paradigm shift in **personalised cancer treatment**, offering hope for patients with otherwise untreatable blood cancers. Its full potential in India will depend on **affordability, accessibility, and robust regulatory oversight**.

9. MRNA TECHNOLOGY

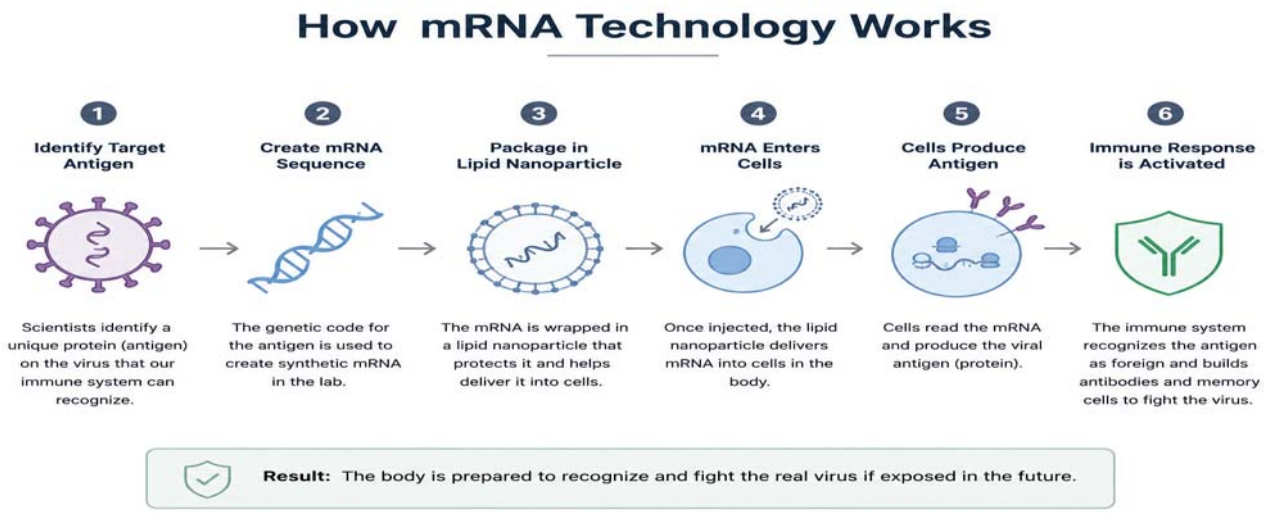
Messenger RNA (mRNA) is a molecule that carries genetic instructions from **DNA** to a cell's protein-making machinery (ribosomes).

mRNA technology in medicine works by delivering a synthetic mRNA strand — wrapped in a protective **lipid nanoparticle (LNP)** — into human cells. The cell's own machinery then reads these instructions to produce a specific protein (e.g., a viral spike protein), training the immune system to recognise and fight the real pathogen, without ever introducing the live virus itself.

Unlike traditional vaccines that use weakened or inactivated pathogens, mRNA vaccines only supply a **genetic 'recipe'**, making them faster to design and adapt — a strength dramatically demonstrated during the **COVID-19 pandemic** with vaccines like Pfizer-BioNTech and Moderna.

Two Main Types of mRNA Platforms

- **Conventional (non-replicating) mRNA:** Delivers a single, non-multiplying strand of mRNA; used in most first-generation global vaccines (e.g., Pfizer, Moderna).
- **Self-Amplifying RNA (saRNA):** Carries additional genetic machinery that lets the mRNA **replicate itself** inside the cell, so a much smaller dose can produce a strong, durable immune response — a technology India has notably pioneered.



Applications of mRNA Technology

- **Infectious Disease Vaccines:** Beyond COVID-19, mRNA platforms are being explored for **influenza, Nipah, Zika, rabies**, and other emerging infectious diseases, given the speed with which a new vaccine 'recipe' can be designed once a pathogen's genetic sequence is known.
- **Cancer Vaccines (Personalised Immunotherapy):** mRNA can be designed to instruct cells to produce **tumour-specific antigens**, training the immune system to attack cancer cells — an active global and Indian research frontier.
- **Protein Replacement Therapy:** For certain genetic disorders where the body cannot produce a specific protein, mRNA could direct cells to make the missing proteins (similar to gene therapy but without altering the DNA itself)
- **Rapid Outbreak Response ('Disease X' Preparedness):** Because mRNA vaccines can be redesigned quickly once a new pathogen's sequence is known, the technology is central to global efforts (including CEPI's **100 Days Mission**) to develop vaccines against unknown future pandemic threats.

Disease X is a placeholder name adopted by the **World Health Organization (WHO)** to represent a hypothetical, yet-to-be-identified pathogen with the potential to cause a severe global pandemic. It is a deliberate planning concept rather than an actual existing illness

- **Biosimilars and Therapeutics:** Gennova itself began its mRNA journey through work on biosimilars like **erythropoietin**, showing how mRNA/recombinant platforms can also support broader biopharmaceutical manufacturing.

Significance of mRNA Technology for India

Dimension	How mRNA Technology Helps
Pandemic Preparedness	An indigenous, adaptable mRNA/saRNA platform lets India respond faster to future outbreaks, including unknown 'Disease X' threats.
Health Security	**Thermostable** vaccines suited to India's climate and cold-chain constraints improve last-mile vaccine delivery, especially in rural areas.
Strategic Autonomy	Reduces dependence on imported mRNA vaccines and technology, aligning with **Atmanirbhar Bharat**.
Global Leadership	India's saRNA and thermostable innovations position it as a potential supplier of **affordable mRNA vaccines** for other developing countries.
Broader Bioeconomy	Success with Gennova's platform strengthens India's case for further investment in advanced **biopharma manufacturing** under the BioE3 Policy.

Challenges Facing mRNA Technology in India

- **High Development Costs:** Establishing mRNA/saRNA manufacturing capability requires significant investment in specialised infrastructure, limiting the number of Indian companies currently capable of it.
- **Cold-Chain and Distribution (for Non-Thermostable Variants):** While Gennova solved this for its own product, mRNA technology in general still often demands more sophisticated storage than conventional vaccines.
- **Public Perception and Hesitancy:** As a relatively new vaccine technology for the Indian public, mRNA vaccines faced some hesitancy and misinformation during the COVID-19 rollout.
- **Limited Domestic Players:** Unlike the broader vaccine manufacturing base built around conventional platforms, indigenous mRNA capability remains concentrated in a **small number of companies** like Gennova.
- **IP and Global Competition:** Foundational mRNA delivery technologies (like certain lipid nanoparticle formulations) are patent-protected globally, which can raise costs or limit flexibility for developers.
- **Regulatory Pathway for a New Modality:** As with CAR-T therapy, mRNA vaccines represented a **novel category** for Indian regulators (CDSCO), requiring new evaluation frameworks distinct from those for conventional generics.
- **Scaling Beyond COVID-19:** Translating early success in COVID-19/Nipah vaccine platforms into a broader, sustainable pipeline of mRNA-based products for other diseases remains a work in progress.

Way Forward

Way Forward	Broad Measures
1. Expanding Indigenous Manufacturing Capacity	Scale up investment in **mRNA/saRNA manufacturing infrastructure** across multiple Indian biotech companies.

2. Diversifying the Disease Pipeline	Develop indigenous **mRNA vaccines** for diseases such as **influenza** and other endemic or emerging infections.
3. Strengthening the Regulatory Framework	Enhance **CDSCO's** capacity to evaluate and approve novel RNA-based therapeutics more efficiently.
4. Enhancing Public Communication	Promote science-based awareness to improve public understanding and trust in **mRNA technology**.
5. Deepening International Collaboration	Strengthen partnerships with organisations such as **CEPI** for advanced technologies and pandemic preparedness.
6. Leveraging the BioE3 Policy	Use the **BioE3 Policy** to expand India's **mRNA/saRNA ecosystem** through biomanufacturing and precision biotherapeutics.

Conclusion

India's progress in indigenous **mRNA and saRNA platforms** reflects its growing biotechnology capabilities. Continued investment can position the country as a global hub for **next-generation vaccines and precision therapeutics**.

■■■

MGP/BP/GS3/020

DIGITAL TECHNOLOGIES

1. FOURTH INDUSTRIAL REVOLUTION (INDUSTRY 4.0)

The Fourth Industrial Revolution, or Industry 4.0, is about bringing digital technologies into physical production. This helps industries become smarter, more independent, and better connected. Technologies such as Artificial Intelligence (AI), Internet of Things (IoT), Big Data Analytics, Robotics, Cloud Computing, Cyber-Physical Systems (CPS), Blockchain, and Digital Twins work together to build smart factories that can self-monitor and make real-time decisions.

Industry 4.0 shifts manufacturing away from mass production toward mass customisation, enabling companies to make personalised products more efficiently.

Its use is also growing in sectors such as agriculture, healthcare, education, government, logistics, defence, and financial services, making it an important part of the digital economy.

The German Government first introduced the term "Industry 4.0" at the Hannover Fair in 2011. Klaus Schwab, the Founder of the **World Economic Forum (WEF),** later described it as a revolution that is *"blurring the boundaries between the physical, digital, and biological worlds."*

Evolution of Industrial Revolutions

Industrial Revolution	Time Period	Key Driver	Major Contribution
Industry 1.0	Late 18th Century	Steam Engine	Mechanisation of production
Industry 2.0	Late 19th Century	Electricity	Mass production and assembly lines
Industry 3.0	1970s onwards	Electronics & Computers	Automation using PLCs and Information Technology
Industry 4.0	21st Century	AI, IoT, Big Data, Cloud Computing	Smart factories, intelligent automation and connected production systems
Industry 5.0 (Emerging)	Future	Human-AI Collaboration	Sustainable, resilient and human-centric manufacturing

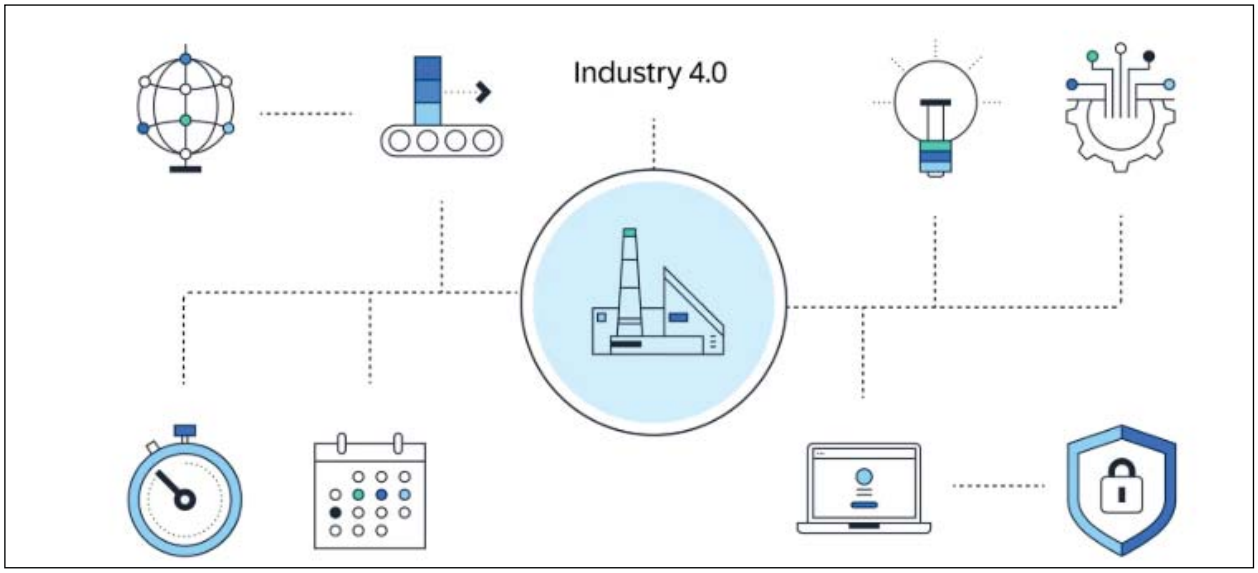
Why is Industry 4.0 Different?

Previous industrial revolutions aimed to boost production by adding machines, electricity, or automation. Industry 4.0 takes things further by enabling machines to communicate with one another, analyse data, learn from experience, and make their own decisions. As a result, factories become smart systems that can quickly adapt to market changes.

Features of Industry 4.0

- **Interconnectivity:** Machines, sensors, and production systems connect through the Internet of Things (IoT). This allows them to share information in real time and work together more efficiently across the whole production process.

- **Cyber-Physical Systems (CPS):** CPS brings together physical machines and digital technology. This makes it possible to monitor equipment in real time, control it automatically, and work more efficiently.
- **Artificial Intelligence and Machine Learning:** AI and ML look at large amounts of data to make production better, predict when machines need maintenance, improve quality checks, and help with smart decision-making.
- **Intelligent Automation:** Unlike older automation, smart machines can adjust to new production needs and improve how things run, all with little help from people.
- **Real-Time Data Analytics:** By constantly collecting and analysing data, companies can watch production closely, spot problems, improve quality, and reduce machine downtime by 30–50%usingh predictive maintenance.
- **Decentralised Decision-Making:** Smart machines and Cyber-Physical Systems can make their own decisions during operations. This helps reduce delays and makes production more flexible.
- **Mass Customisation:** Industry 4.0 lets manufacturers make custom products as quickly and cheaply as mass-produced ones, which makes customers happier.
- **Cloud and Edge Computing:** Cloud computing offers flexible storage and processing power. Edge computing processes data near its source, enabling faster, real-time decision-making.
- **Advanced Robotics:** Collaborative robots, or cobots, handle complex and repetitive tasks with great accuracy. They work safely alongside people, boosting productivity and enhancing workplace safety.
- **High System Integration:** Industry 4.0 connects product design, manufacturing, logistics, supply chains, and customer service through digital tools. This creates a smooth and efficient process from start to finish.



Impact of Industry 4.0

Industry 4.0 is changing how things are made by adding smart technology to manufacturing. It helps companies work more efficiently, be more creative, use resources better, and find new ways to grow sustainably.

- **Increases Productivity and Economic Growth:** Intelligent automation, AI, and real-time data analytics improve production efficiency, optimise resource utilisation, and reduce operational costs. According to the WEF, AI adoption could increase global productivity by 0.8–1.4% annually.
- **Promotes Smart Manufacturing:** Smart factories use AI, IoT, robotics, and Cyber-Physical Systems to automate production, monitor operations in real time, and make manufacturing more efficient.
- **Improves Product Quality:** AI, Machine Vision, and Digital Twins enable real-time quality inspection, reducing manufacturing defects by up to 50% and improving product reliability.
- **Reduces Manufacturing Costs:** Predictive maintenance and smart resource management help cut downtime, reduce waste, and use energy more efficiently. Predictive maintenance alone can lower equipment downtime by **30–50%.**

- **Enables Mass Customisation:** New manufacturing technologies let companies make custom products as quickly and cheaply as mass-produced ones, which makes customers happier.
- **Strengthens Supply Chains:** Tools like **IoT**, **RFID**, **Blockchain**, and **AI** help manage inventory, improve logistics, and make supply chains stronger by monitoring everything in real time.
- **Creates High-Skilled Employment:** Automation might replace some routine jobs, but it also opens up new roles in AI, robotics, data analytics, cybersecurity, cloud computing, and semiconductor design. The **WEF Future of Jobs Report 2025** says these changes could create **170 million new jobs** worldwide by **2030**.
- **Promotes Sustainable Manufacturing:** Smart manufacturing helps save energy, cut down on waste, lower carbon emissions, and support recycling and reuse.
- **Accelerates Innovation**: Tools like digital twins, AI, and simulation software help companies develop products faster, lower research costs, and speed up innovation.
- **Enhanced Consumer Experience:** Data analytics and AI make it possible to offer personalized products, predict after-sales needs, and give better customer support, which leads to happier customers.

Industry 4.0 and India

India has a strong foundation for Industry 4.0, thanks to its growing digital infrastructure, large manufacturing sector, active startup scene, and skilled workers. Using advanced technologies can help India become a global manufacturing leader and move closer to the Viksit Bharat 2047 vision.

- **Supports Make in India:** Industry 4.0 encourages smart manufacturing, boosts industrial competitiveness, and attracts valuable investments. This helps achieve the goals of the Make in India initiative.
- **Advances in Atmanirbhar Bharat:** Building local skills in AI, semiconductors, robotics, and electronics helps India rely less on imports and become more self-sufficient in technology. The India Semiconductor Mission is an important move toward this goal.
- **Boosts Manufacturing Growth:** Manufacturing makes up about **17% of India's GDP**, and the National Manufacturing Policy wants to **increase this to 25%**. Industry 4.0 can help by improving productivity, exports, and growth in manufacturing.
- **Strengthens Global Value Chains (GVCs):** Smart manufacturing helps Indian companies meet global quality standards, make supply chains more efficient, and join international value chains.
- **Leverages Digital Public Infrastructure (DPI):** India's digital systems like Aadhaar, UPI, DigiLocker, ONDC, and Account Aggregator give a solid base for using Industry 4.0 technologies.
- **Promotes Innovation and Startups:** As the world's third-largest startup ecosystem, India supports new ideas in AI, robotics, Industrial IoT, drones, and deep-tech. This accelerates the development of local Industry 4.0 solutions.

Government Initiatives Supporting Industry 4.0 in India

The Indian government has started many programs to support advanced manufacturing, new technologies, and digital infrastructure. These efforts aim to make Indian industry more competitive and ready for Industry 4.0.

- **IndiaAI Mission (2024)**: With a budget of **₹10,300 crore**, this mission supports local AI skills through high-performance computing, AI startups, research, AI datasets, and the creation of Indian **Large Language Models (LLMs)**.
- **India Semiconductor Mission (ISM): W**ith an incentive package of **₹76,000 crore**, this mission aims to build a local semiconductor industry and cut down on imports. Major approved projects include **Micron, Tata Electronics, CG Power, and Keynes Semicon**.
- **Production-Linked Incentive (PLI) Scheme: This scheme giv**es financial rewards for manufacturing in areas like electronics, semiconductors, cars, telecom, drones, and pharmaceuticals. It aims to grow local manufacturing and make India more competitive globally.
- **Digital India Programme: This program improve**s digital infrastructure, broadband access, digital governance, and digital skills. It lays the groundwork for AI, IoT, cloud computing, and smart manufacturing.
- **PMKVY 4.0:** Promotes industry-oriented skill development in AI, Robotics, IoT, Data Analytics, Coding, Drone Technology, and Mechatronics to build a future-ready workforce.

- **National Quantum Mission: With a budget** of **₹6,003 crore**, this mission funds research in quantum computing, communication, sensing, and related fields to boost India's next-generation innovation.
- **SAMARTH Udyog Bharat 4.0 promotes** awareness, training, pilot projects, and smart manufacturing adoption, particularly among MSMEs, through industry–academia collaboration.
- **Make in India 2.0: This program encourages high-tech** manufacturing in areas like electronics, semiconductors, electric vehicles, renewable energy, aerospace, and defence to strengthen India's manufacturing sector.
- **Startup India Initiative: This initiative help**s deep-tech startups with funding, incubation, and regulatory support. It encourages innovation in AI, robotics, IoT, cybersecurity, and other Industry 4.0 technologies.
- **Open Network for Digital Commerce (ONDC):** This network builds an open digital commerce system, helping MSMEs reach more markets and making digital value chains stronger.
- **National Logistics Policy (2022):** Promotes AI, IoT, GIS, and digital platforms to improve logistics efficiency, multimodal connectivity, and smart supply chain management.
- **National Manufacturing Mission (Budget 2025–26):** This mission focuses on advanced manufacturing, adopting new technologies, making business easier, developing the workforce, and joining global value chains. It also helps MSMEs use Industry 4.0 technologies.

Industry 4.0 and MSMEs

Micro, Small and Medium Enterprises (MSMEs) play a key role in India's industrial growth and in the adoption of Industry 4.0. They make up **31.1% of GDP**, **35.4% of manufacturing output**, and **48.6% of exports. There are over 7.47 crore MSMEs, providing jobs to about 32.8 crore people. Still, many MSMEs have not adopted** advanced digital technologies due to financial, technological, and skill-related challenges.

Significance of Industry 4.0 for MSMEs

- **Improves Productivity and Efficiency:** AI, IoT, and automation help MSMEs make production more efficient, reduce downtime, improve product quality, and lower costs.
- **Enhances Global Competitiveness:** Smart manufacturing helps MSMEs meet international quality standards, join Global Value Chains (GVCs), and become more competitive in exports.
- **Expands Digital Market Access:** Digital platforms like **ONDC, e-commerce, cloud services, and digital payments** help MSMEs reach more customers in India and abroad.
- **Promotes Innovation and Customisation:** Industry 4.0 supports flexible manufacturing, so MSMEs can make customised products to meet customer needs.
- **Improves Access to Finance:** Digital business records and transaction data make it easier for MSMEs to get credit and access formal finance.

Challenges in Adopting Industry 4.0

- **High Cost of Adoption: Many MSMEs find it hard to adopt new technology because it requires large** investments in AI, robotics, Industrial IoT, cloud infrastructure, and automation.
- **Skill Gap: There is a shortage** of skilled professionals in AI, robotics, data analytics, and digital technologies, which is a major barrier to digital transformation.
- **Inadequate Digital Infrastructure: Limited broadband, unreliable power, and not enough cloud infrastructure make it hard to adopt technology, especial**ly in rural areas.
- **Cybersecurity Risks: As MSMEs use more digital tools, they face more risks from cyberattacks, ransomware, and data breaches. Many are not fully prepared for these threats**.
- **Limited Awareness:** Many MSMEs think Industry 4.0 technologies are too expensive and only for big companies, which slows down adoption.

Challenges of Industry 4.0

Industry 4.0 brings many opportunities, but implementing it also poses significant economic, technological, social, and regulatory challenges.

- **Skill Gap and Workforce Readiness-** Rapid technological advancements have created a mismatch between workforce skills and industry requirements, particularly in AI, Robotics, Data Analytics, and Cybersecurity. According to the **WEF Future of Jobs Report 2025**, **39% of workers' existing skills** are expected to become outdated by 2030.
- **Job Displacement:** Automation could replace repetitive and low-skilled jobs, especially in manufacturing. Ongoing reskilling and a fair transition are needed to reduce job losses.
- **High Cost of Adoption-** Industry 4.0 technologies require significant investments in robotics, AI, IoT, cloud infrastructure, and cybersecurity, making adoption difficult for MSMEs.
- **Digital Divide: Not all regions and businesses have the same access to digital infrastructure and advanced technologies, which can increase** economic and social gaps.
- **Cybersecurity Risks: More use of** IoT and cloud platforms means industries face higher risks from cyberattacks, ransomware, industrial spying, and data breaches.
- **Data Privacy and AI Ethics: Collecting large amounts of data brings up concerns about** privacy, ownership, bias in algorithms, transparency, and the ethical use of AI.
- **Technological Dependence: Relying heavily** on imported semiconductors, industrial robots, and advanced software can create strategic weaknesses and supply chain risks.
- **Regulatory Challenges: Technology is changing faster than regulations for AI**, labour laws, cybersecurity, and industrial standards can keep up.
- **Environmental Concerns-** Growing deployment of digital infrastructure increases concerns over **e-waste, energy-intensive data centres, and demand for critical minerals**, necessitating sustainable manufacturing practices.

Global Best Practices

Several countries have used Industry 4.0 to boost their manufacturing and become technology leaders. India can learn important lessons from their experiences.

- **Germany (Industry 4.0):** Germany started the **Industry 4.0** strategy in 2011, encouraging smart factories, Cyber-Physical Systems, and industrial automation by working closely with government, industry, and universities.
- **Japan (Society 5.0):** Japan's **Society 5.0** brings together AI, IoT, and robotics to build a digital society focused on people. This transformation goes beyond manufacturing to include healthcare, government, and transport.
- **South Korea:** South Korea has encouraged smart manufacturing by offering financial incentives, building industrial clusters, and using robotics widely. This has made it one of the most automated manufacturing economies in the world.
- **China (Made in China 2025):** China has invested a lot in AI, semiconductors, robotics, electric vehicles, and advanced manufacturing as part of its **Made in China 2025** plan to become more self-reliant in technology.
- **United States-** The United States leads in industrial AI, cloud computing, semiconductor innovation, and advanced R&D through strong collaboration between universities, startups, and technology companies.

Way Forward

- **Build a Future-Ready Workforce: Increase training and retraining in AI, robotics, data analytics, cybersecurity, and semiconductor design by strengthening collaboration between industry and academia.**
- **Enable MSME Digital Transformation: Offer** financial incentives, technology vouchers, subsidised cloud services, and easier access to digital credit to speed up Industry 4.0 adoption.
- **Boost Indigenous R&D and Innovation: Invest more** in AI, semiconductors, quantum technologies, robotics, and deep-tech startups to reduce dependence on foreign technology.
- **Expand Digital Infrastructure-** Accelerate the rollout of BharatNet, 5G, cloud infrastructure, edge computing, and data centres to support smart manufacturing.

- **Strengthen Industrial Cybersecurity-** Promote AI-enabled threat detection, regular cyber audits, sector-specific security standards, and public-private collaboration to protect critical infrastructure.
- **Develop a Responsible AI Ecosystem- establish** adaptive regulatory frameworks to ensure AI ethics, transparency, accountability, and data privacy while fostering innovation.
- **Promote Sustainable Smart Manufacturing by using** renewable energy, circular economy practices, and energy-efficient technologies as part of Industry 4.0.
- **Deepen Global Value Chain Integration: Us**e the PLI Scheme, National Manufacturing Mission, FTAs, and logistics reforms to attract investment and make India's manufacturing more competitive.
- **Promote Industry–Academia Partnerships: Set up** Centres of Excellence in AI, robotics, IoT, and smart manufacturing to support research, innovation, and technology sharing.

Conclusion

The Fourth Industrial Revolution is transforming industries through smart automation, digital connectivity, and advanced technologies. For India, this is a chance to strengthen manufacturing, boost global competitiveness, and speed up economic growth. By supporting local innovation, helping MSMEs, investing in digital infrastructure, and building a skilled workforce, India can use Industry 4.0 to achieve the goals of **Atmanirbhar Bharat** and **Viksit Bharat 2047**.

2. ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is the ability of machines or computer systems to learn, reason, solve problems, make decisions, understand language, and recognise things.

AI systems rely on technologies such as Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision to process large amounts of data, find patterns, and improve over time.

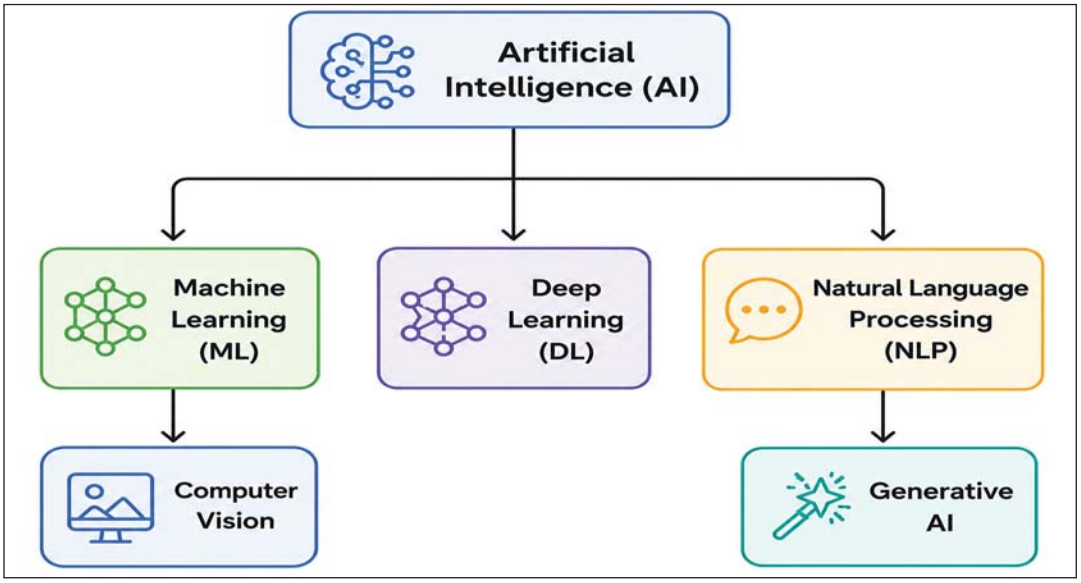
Today, AI is a general-purpose technology that is changing many fields, including healthcare, farming, education, manufacturing, finance, government, defence, and transportation.

New developments in **Generative AI**, especially **Large Language Models like** ChatGPT, Gemini, and Claude, have accelerated the pace at which people use AI by enabling machines to create text, images, audio, and code that appear human-made.

The phrase **"Artificial Intelligence"** was first used by **John McCarthy** at the **Dartmouth Conference in 1956, which many see** as the start of AI as a science.

AI Ecosystem

Artificial Intelligence comprises multiple technologies that enable machines to mimic human intelligence.



Difference between AI, ML and DL

Artificial Intelligence	Machine Learning	Deep Learning
Broad field of simulating human intelligence	Subset of AI that learns from data	Subset of ML using artificial neural networks
Performs reasoning, planning and decision-making	Identifies patterns and makes predictions	Learns complex patterns from massive datasets
Example: Virtual assistants	Spam detection	Image recognition, ChatGPT

Characteristics of Artificial Intelligence

- **Learning Ability:** AI systems use past data to learn and get better over time, even without being directly programmed for each improvement.
- **Reasoning and Decision-Making:** AI reviews information, considers different options, and suggests or takes the best actions based on set goals.
- **Pattern Recognition:** AI finds patterns and connections in all kinds of data, which helps make better predictions and gain new insights.
- **Automation:** AI takes over repetitive and rule-based tasks, making work more efficient and requiring less human effort.
- **Adaptability:** Modern AI systems keep updating as they get new data, so they can respond better to changes around them.
- **Natural Language Understanding:** AI can understand, interpret, and create human language using Natural Language Processing (NLP).
- **Computer Vision:** AI helps machines recognise objects, faces, medical images, and other visual patterns very accurately.

Types of Artificial Intelligence

Based on Capability

- **Narrow AI (Weak AI) :** Narrow AI is built to perform a single task well, but it lacks general intelligence. **Examples include** ChatGPT, Siri, Alexa, Google Maps, and recommendation systems.
- **General AI (Strong AI):** General AI refers to machines that can perform any intellectual task a human can. These systems are still mostly theoretical and do not yet exist.
- **Super AI (Hypothetical):** Super AI would be more intelligent than humans in every area, including creativity, reasoning, and decision-making. For now, this idea is only speculation.

AI Intensity Across Economic Sectors

Indicators of AI Intensity

- **AI Talent:** This is measured by the number of job openings that need AI skills, showing how much demand there is for AI professionals.
- **AI Innovation:** This is judged by how many AI-related patents there are, which shows how much a sector is helping to create new AI technologies.
- **AI Exposure:** This looks at how much AI could change the tasks in a sector, taking into account technology and rules that might limit it.
- **AI Adoption:** This shows what share of companies are already using AI in their work.

Sector-wise AI Intensity

High AI Intensity Sectors

- **IT Services, Media & Telecommunications, Scientific R&D**
- These sectors are ahead in **AI talent, innovation, adoption, and patents. They play an active role in** shaping the future of AI.

Medium AI Intensity Sectors

- **Pharmaceuticals, Electronics Manufacturing, Automobile Manufacturing, Machinery Manufacturing**
- These sectors have strong potential for AI, but they face challenges such as outdated systems, a shortage of skilled workers, and high costs of implementing new technology.

Low AI Intensity Sectors

- **Construction, Textiles, Food Processing, Wood & Paper Industries**
- AI adoption is still limited in these sectors because the work is physical and labour-intensive, not because they lack technology.

Applications of Artificial Intelligence in Different Sectors

Artificial Intelligence is changing many sectors of the economy. It can analyze large amounts of data, automate complex tasks, and help make better decisions. These abilities are making work more productive, efficient, and improving services.

1. Healthcare

AI helps with disease diagnosis, medical imaging, drug discovery, personalized treatment, and hospital management. Predictive analytics can find high-risk patients, and AI diagnostic tools make disease detection more accurate.

- **For example,** ICMR's *Ethical Guidelines for Application of AI in Biomedical Research and Healthcare* encourage responsible use of AI in health care.

2. Agriculture

AI makes precision farming possible by using satellite images, drones, IoT sensors, and predictive analytics. It helps farmers monitor crops, detect pests, manage irrigation, and predict yields, which improves productivity.

- **For example,** AI-powered advisory services in the **Digital Agriculture Mission** help farmers make decisions based on data.

3. Education

AI supports personalised learning, smart tutoring systems, automated assessments, multilingual education, and adaptive learning platforms. These tools help make education more inclusive and effective.

- **For example, the National AI Resource Platform (NAIRP), creat**ed by IIT Kharagpur and AWS, aims to improve learning with AI.

4. Manufacturing

AI helps smart manufacturing by enabling predictive maintenance, quality checks, industrial robots, digital twins, and better supply chain management. These uses reduce downtime and boost productivity.

- **For example, AI is a key technology behind Industry 4.0**.

5. Governance and Public Service Delivery

AI helps improve governance by using predictive analytics, handling complaints, detecting fraud, informing policy planning, and making public services more efficient.

- **For example, government departments are increasingly using AI chatbots to improve services for citizens.**

6. Financial Services

Banks and financial institutions use AI for fraud detection, credit scoring, algorithmic trading, customer support, and risk assessment. These uses make operations more efficient and help more people access financial services.

7. Judiciary

AI helps courts with legal research, document translation, case management, and court administration. This makes the justice system work more efficiently.

- **For Example-**
  - **SUPACE (Supreme Court Portal for Assistance in Court Efficiency)**
  - **SUVAS (Supreme Court Vidhik Anuvaad Software)**

8. Cybersecurity

AI enables real-time threat and anomaly detection, malware analysis, fraud prevention, and automated incident responses. These features make cyber systems stronger and more secure.

9. Defence and National Security

AI is used for autonomous drones, smart surveillance, border management, cyber defence, predictive maintenance of military equipment, and decision support systems.

10. Transportation and Logistics

AI helps with traffic management, self-driving vehicles, route planning, warehouse automation, and supply chain management. These uses lower logistics costs and make operations more efficient.

11. Commerce and Business

AI enables personalised marketing, recommendation systems, inventory management, customer relationship management, and demand forecasting. These tools help businesses stay competitive.

12. Scientific Research and Space

AI accelerates scientific discovery by analysing complex data, aiding climate modelling, genomic research, and space exploration.

- **For example, ISRO is increasingly using AI to analyse satellite images and plan missions**.

Government Initiatives for Promoting AI in India

The Government of India sees AI as a key technology and has launched several programs to support AI research, innovation, digital infrastructure, and responsible use across sectors.

IndiaAI Mission (2024)	Approved with an outlay of **₹10,300 crore**, the mission aims to develop indigenous AI capabilities through AI computing infrastructure, datasets, startups, skilled workforce, and indigenous Large Language Models (LLMs).
National Strategy for AI (AI for All)	Released by **NITI Aayog (2018)**, it identifies healthcare, agriculture, education, smart cities, and mobility as priority sectors while promoting inclusive and responsible AI.
INDIAai Portal	A national AI portal jointly developed by MeitY, NASSCOM, and NIC to promote AI research, innovation, startups, learning resources, and collaboration.
Digital India Programme	It strengthens digital infrastructure, broadband connectivity, Digital Public Infrastructure (DPI), and e-governance, creating an enabling ecosystem for AI adoption.
India Semiconductor Mission (ISM)	It promotes domestic semiconductor manufacturing, which forms the hardware backbone for AI systems and high-performance computing.
National Quantum Mission	It supports research in quantum computing, communication, and sensing, enabling next-generation AI applications.

FutureSkills PRIME	A joint initiative of **MeitY** and **NASSCOM** that provides skilling and reskilling in AI, Machine Learning, Cybersecurity, Cloud Computing, and other emerging technologies.
Responsible AI for All	NITI Aayog's framework promotes ethical, transparent, and inclusive AI while addressing concerns related to bias, privacy, accountability, and fairness.

Challenges of Artificial Intelligence

Even though AI has great potential, it also brings important challenges in technology, the economy, ethics, and governance.

1. **Ethical and Algorithmic Bias:** AI systems trained on biased or incomplete data can produce unfair results. This can affect hiring, lending, healthcare, and law enforcement.
2. **Data Privacy Concerns: AI uses a lot** of personal and sensitive data, which raises worries about privacy, surveillance, and misuse of information.
3. **Job Displacement: Automating** routine and repetitive tasks can lower the need for low-skilled workers. This means people will need to keep learning new skills and change jobs.
4. **Lack of Transparency:** Many AI models work like "black boxes," so it is hard to understand or explain how they make decisions.
5. **Cybersecurity Risks:** AI can be misused for deepfakes, phishing, creating malware, and cyber warfare. These risks are serious challenges for national security.
6. **High Development Cost:** Building AI systems needs a lot of investment in computing power, skilled people, good data, and semiconductors.
7. **Regulatory and Legal Challenges: AI is advancing quickly, often faster than current laws about** accountability, intellectual property, liability, and governance can keep up.
8. **Dependence on Foreign Technologies: India still relies a lot** on imported AI chips, cloud infrastructure, and advanced AI models. This creates technology risks.

Global Best Practices

Many countries have created special AI strategies to boost innovation, improve governance, and make their industries more competitive.

1. **The United States leads** in AI research, foundation models, semiconductor innovation, and private-sector investment through companies such as OpenAI, Google, Microsoft, and NVIDIA.
2. **China-** Through the **Next Generation AI Development Plan (2017)**, China has invested heavily in AI research, smart cities, autonomous vehicles, and facial recognition technologies.
3. **European Union-** The **EU AI Act** establishes the world's first comprehensive legal framework for AI, adopting a **risk-based approach** to ensure safe and trustworthy AI.
4. **United Kingdom:** Promotes a **sector-specific, innovation-friendly** approach to AI regulation while emphasising safety, transparency, and accountability.
5. **Singapore-** Through the **National AI Strategy 2.0**, Singapore focuses on responsible AI adoption, public-sector innovation, and AI talent development.

Way Forward

1. **Develop Responsible AI- promote** ethical, transparent, explainable, and accountable AI systems through robust regulatory frameworks.
2. **Strengthen Indigenous AI Ecosystem-** Invest in AI research, indigenous foundation models, semiconductor manufacturing, and high-performance computing infrastructure.

3. **Build AI-Ready Workforce-** Expand skilling and reskilling in AI, Machine Learning, Data Science, and Cybersecurity through stronger industry–academia collaboration.
4. **Improve Data Governance-** Establish secure, high-quality, and interoperable data-sharing frameworks while protecting privacy and individual rights.
5. **Promote AI Adoption among MSMEs-** Provide financial incentives, cloud infrastructure, digital public goods, and technical support to accelerate AI adoption.
6. **Strengthen AI Governance-** Develop comprehensive legal frameworks covering AI ethics, liability, intellectual property, transparency, and algorithmic accountability.
7. **Enhance International Cooperation-** Collaborate with global institutions on AI standards, safety, governance, and cross-border research to ensure responsible AI development.
8. **Promote Human-Centric AI- encourage** AI systems that augment human capabilities rather than replace them, ensuring inclusive and sustainable technological development.

Conclusion

Artificial Intelligence is becoming a general-purpose technology that can change every part of the economy by making work more productive, innovative, and smarter. For India, AI is a significant opportunity to accelerate digital transformation, improve public services, boost industry, and advance the goal of Viksit Bharat 2047. To get the most from AI, India needs to balance innovation with ethical rules, data privacy, inclusion, and a focus on people.

3. GENERATIVE AI AND LARGE LANGUAGE MODELS (LLMS)#

Recent progress in Artificial Intelligence (AI) has brought about Generative AI and Large Language Models (LLMs). These tools are changing how people create content, work with machines, and tackle difficult problems. They have sped up innovation in fields like healthcare, education, government, finance, cybersecurity, research, and creative work by allowing machines to produce text, images, audio, videos, and code that seem human-made.

Generative Artificial Intelligence (Generative AI)

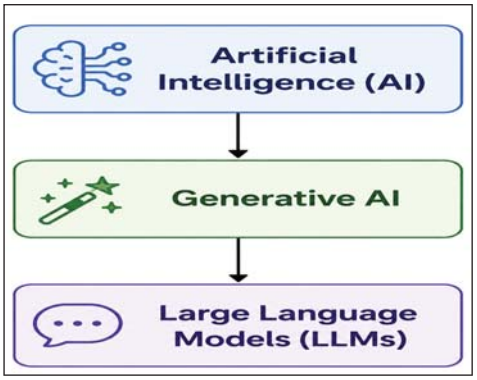
- Generative AI is a type of Artificial Intelligence that makes new content by learning from large amounts of data. While traditional AI usually analyses or sorts existing information, Generative AI creates new things like text, images, videos, audio, code, and 3D models when users ask for them.
- Generative AI uses advanced deep learning methods like Transformers, Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion Models.
- Examples: ChatGPT, Gemini, Claude, DALL·E, Midjourney, Stable Diffusion.

Large Language Models (LLMs)

- Large Language Models (LLMs) are deep learning systems trained on huge amounts of text. They can understand, create, summarize, translate, and reason with human language. Most LLMs today use the Transformer method, which helps them handle context well and produce clear, human-like answers.
- LLMs are the main technology behind conversational AI. They are used in chatbots, virtual assistants, search engines, coding tools, document summaries, and content creation.
- Examples: GPT-4.1/GPT-5 family, Gemini, Claude, Llama, DeepSeek.

Key Point: All LLMs are Generative AI, but not all Generative AI are LLMs. LLMs primarily generate text, whereas Generative AI can generate text, images, audio, video, code, and other digital content.

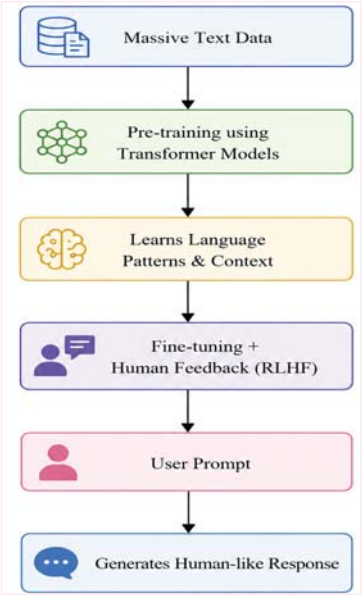
Relationship between AI, Generative AI and LLMs



Difference between AI, Generative AI and LLMs

Artificial Intelligence	Generative AI	Large Language Models
Simulates human intelligence	Generates new content	Generates and understands human language
Focuses on prediction, classification and decision-making	Creates text, images, audio, videos and code	Specialised in text generation and Natural Language Processing
Includes ML, Computer Vision, Robotics	Uses GANs, Diffusion Models, Transformers	Built mainly on Transformer architecture
Example: Face Recognition	DALL·E, Midjourney	GPT, Gemini, Claude, Llama

How do Large Language Models Work?



Features of Generative AI and LLMs

1. **Human-like Content Generation:** creates natural text, images, audio, videos, and computer code from simple prompts.
2. **Natural Language Understanding:** understands context, intent, and conversational language, which allows for more human-like interactions.
3. **Multi-modal Capabilities:** Modern AI models can process and generate different types of data, such as text, images, speech, and video.
4. **Context-Aware Responses:** LLMs analyse the context of a conversation to give coherent and relevant answers.
5. **Knowledge Extraction and Summarisation:** quickly summarises long documents, finds key insights, and answers questions from large datasets.
6. **Code Generation**: helps developers by writing, debugging, explaining, and improving software code.
7. **Continuous Learning through Fine-Tuning:** Performance gets better with domain-specific fine-tuning and learning from human feedback (RLHF).

Applications of Generative AI and LLMs

1. **Education**: supports personalised learning, content creation, automated assessment, language translation, and AI tutors.
2. **Healthcare:** helps with clinical documentation, medical report summaries, drug discovery, diagnosis support, and patient interaction.
3. **Governance:** improves citizen services with AI chatbots, multilingual translation, document summaries, and policy analysis.
4. **Software** Development: generates code, finds bugs, automates documentation, and boosts developer productivity.
5. **Finance:** enables fraud detection, financial report creation, customer support, risk assessment, and investment analysis.
6. **Media and Content Creation:** produces articles, ads, images, videos, music, and digital artwork, helping boost creativity and productivity.
7. **Legal Services:** summarises legal documents, helps with legal research, drafts contracts, and improves case management.
8. **Scientific Research:** speeds up literature reviews, hypothesis generation, data analysis, and research documentation.
9. **Cybersecurity:** supports threat intelligence, malware analysis, incident reporting, and automated security documentation.
10. **Business and Customer Service:** powers virtual assistants, smart chatbots, recommendation systems, and automated customer support.

Government Initiatives

1. **IndiaAI Mission**: supports local AI models, computing infrastructure, datasets, startups, and skilled workers with a budget of ₹10,300 crore.
2. **IndiaAI Compute Portal** gives researchers, startups, and academic institutions access to high-performance GPU computing.
3. **National Strategy for AI (AI for All):** NITI Aayog's plan aims for inclusive AI use in healthcare, agriculture, education, smart cities, and mobility.
4. **FutureSkills PRIME** develops AI and technology skills through industry-focused training.
5. **Bhashini** uses AI and LLMs to support multilingual communication and digital inclusion in Indian languages.

Challenges

1. **Hallucinations:** LLMs can sometimes give factually incorrect or misleading answers with high confidence.
2. **Data Privacy:** Training and use of these models need large datasets, which can raise concerns about privacy and sensitive information.
3. **Algorithmic Bias:** Biased training data can lead to unfair or discriminatory results.
4. **Copyright and Intellectual Property:** AI-generated content can create questions about ownership, plagiarism, and copyright infringement.
5. **Misinformation and Deepfakes:** Generative AI can make realistic fake content, which increases the risk of misinformation and cyber fraud.
6. **High Computing Cost:** Building and using foundation models require a lot of computing power and advanced semiconductors.
7. **Explainability:** It is often hard to understand how LLMs make decisions, which can reduce transparency and trust.

Way Forward

1. **Develop Responsible AI:** Set up ethical AI frameworks to ensure transparency, accountability, fairness, and human oversight.
2. **Strengthen Indigenous AI Ecosystem:** Invest in local foundation models, semiconductor manufacturing, and AI research infrastructure.
3. **Improve Data Governance:** Develop secure, high-quality datasets while protecting privacy and ensuring responsible data sharing.
4. **Promote AI Literacy:** Expand AI education, digital skills, and workforce training to prepare people for an AI-driven economy.
5. **Encourage Innovation with Regulation:** Use flexible regulations that support innovation while managing risks like misinformation, deepfakes, and copyright violations.

Conclusion

Generative AI and Large Language Models are changing the digital economy by transforming how we create knowledge, communicate, and make decisions. These technologies bring many opportunities for innovation, productivity, and better public services. However, using them responsibly means having strong governance, ethical rules, and local technology skills. With programs like the IndiaAI Mission and a growing AI ecosystem, India is ready to use Generative AI to help achieve Digital India and Viksit Bharat 2047.

4. ROBOTICS

- **Robotics** is a field that brings together science and engineering to design, build, and use robots. It combines areas such as mechanical engineering, electronics, computer science, artificial intelligence, machine learning, and control systems to create machines that can perform tasks on their own or with minimal human assistance.
- Today's robots often use **AI, sensors, computer vision, and smart control systems. These features help robots understand their environment, make choices, and work with people. Robots are now common in areas like manufacturing, healthcare, farming, logistics, defence, space missions**, and disaster response.
- Isaac Asimov first used the word "Robotics" in the 1940s. He also came up with the well-known **Three Laws of Robotics** to help guide the safe and ethical use of smart machines.

Isaac Asimov's Three Laws of Robotics

A robot shall not harm a human being or allow a human to come to harm through inaction.	**A robot must obey human instructions**, except where such instructions conflict with the First Law.	**A robot must protect its own existence**, provided such protection does not conflict with the First or Second Law.

Even though these laws are fictional, they started important conversations about AI ethics and how to use robots responsibly.

Components of a Robot

A robot is made up of different parts that help it sense things, process information, and carry out tasks.

1. **Controller (Brain) -** The controller is like the robot's brain. It processes sensor data and sends commands to other parts, managing decisions, movement, and the robot's overall operation.
2. **Sensors - Sensors gather details about the** environment, like light, temperature, pressure, distance, and obstacles. This helps robots interact smartly with what's around them.
3. **Actuators -** Actuators turn electrical, hydraulic, or air power into movement. Examples include motors, hydraulic cylinders, and pneumatic systems.
4. **End Effectors -** These are tools attached to the robot's arm, like grippers, welding guns, spray nozzles, surgical tools, or cameras. They let the robot do specific jobs.
5. **Power Supply - Robots run on** batteries, electricity, hydraulic systems, or air power, depending on what they need to do.
6. **Programming and AI -** Software, algorithms, and artificial intelligence help robots analyse information, make decisions, and do their tasks on their own.

Characteristics of Robots

1. **Automation -** Robots can do repetitive tasks with little or no help from people.
2. **Precision and Accuracy -** Robots can handle complex jobs with great consistency and very few mistakes.
3. **Programmability - You can change what robots do by updating their software, without needing to change their hardware.**
4. **Sensory Perception -** Sensors let robots notice changes around them and react as needed.
5. **Intelligence - Robots with AI** can learn from data, adjust to new situations, and make their own decisions.
6. **Continuous Operation -** Robots can work for long periods without getting tired, which helps boost productivity and efficiency.

Types of Robots

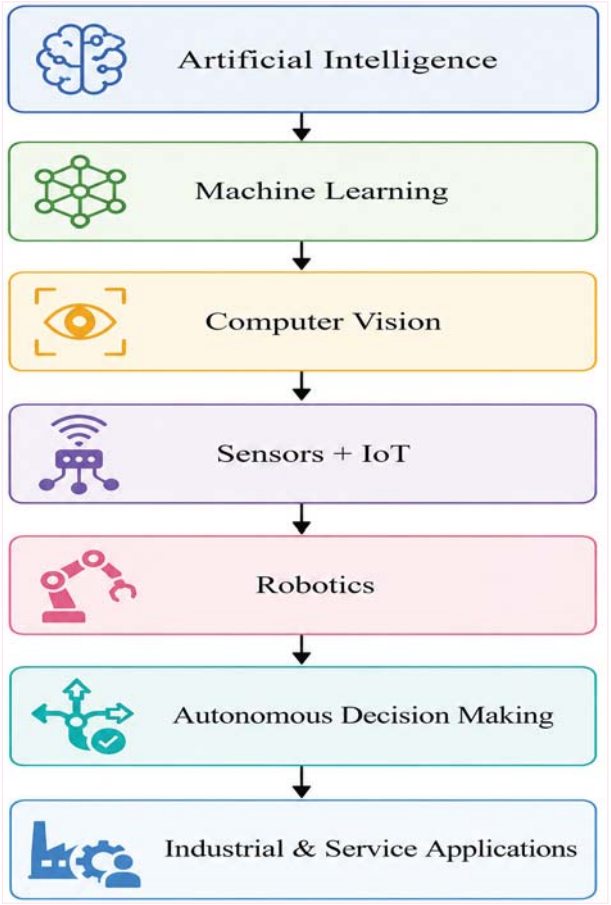
Robots are built based on what they need to do and where they will work.

1. **Industrial Robots - These are robotic arms used in factories for jobs like** welding, painting, assembling, packaging, and moving materials.
2. **Collaborative Robots (Cobots) - These robots are made to work safely with people, helping with repetitive, dangerous, or precise** tasks.
3. **Humanoid Robots - These robots look or act like humans and are main**ly used in research, healthcare, education, and customer service.

   Sophia, **MANAV** (India's first 3D-printed humanoid robot).
4. **Autonomous Mobile Robots (AMRs) - These robots move around on their own** using AI, sensors, GPS, LiDAR, and computer vision. They are often found in warehouses and logistics.
5. **Medical Robots - These robots help with** surgery, rehabilitation, patient care, and telemedicine.

6. **Military Robots - These are u**sed for surveillance, bomb disposal, scouting, and dangerous tasks.
7. **Agricultural Robots-** Perform precision farming activities such as planting, spraying, harvesting, and crop monitoring.
8. **Space Robots - These robots are used for exploring planets, servicing satellites, and carrying out** space missions. ISRO's **Vyommitra** for the **Gaganyaan Mission**.

Robotics Ecosystem



Applications of Robotics

Artificial Intelligence, Machine Learning, Computer Vision, and IoT have greatly expanded the use of robotics across different fields. Today, robots handle tasks in areas like manufacturing, healthcare, agriculture, defence, and space exploration.

1. Manufacturing and Industrial Automation

Robots handle repetitive, dangerous, and precise jobs like welding, painting, assembly, packaging, and quality checks. They help boost productivity, lower defects, and support **smart manufacturing (Industry 4.0)**.

2. Healthcare

Robots help with minimally invasive surgeries, rehabilitation, patient care, hospital logistics, and telemedicine. Surgical robots with AI make procedures more precise and help patients recover faster. The **Da Vinci Surgical System** is a common example of robotic-assisted surgery.

3. Agriculture

Agricultural robots help with precision farming by automating tasks like sowing, harvesting, spraying, weed control, crop monitoring, and soil analysis. This boosts productivity and reduces the need for manual labour.

4. Defence and National Security

Robots are used for bomb disposal, surveillance, reconnaissance, border monitoring, mine detection, and unmanned combat. These uses help keep defence personnel safer. DAKSHA, made by DRDO, is used to handle Improvised Explosive Devices (IEDs).

5. Space Exploration

Robots take on dangerous space missions, service satellites, explore planets, and carry out scientific experiments where it is hard for humans to go. ISRO is developing the Vyommitra humanoid robot for the **Gaganyaan Mission**.

6. Logistics and Warehousing

Autonomous Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs) help manage inventory, run warehouses, handle packaging, and move materials. This makes supply chains work more efficiently.

7. Disaster Management

Robots help with search and rescue, firefighting, earthquake response, and working in dangerous places like nuclear accidents and chemical spills.

8. Mining and Hazardous Industries

Robots do drilling, inspections, and move materials in mines and other dangerous workplaces where people could be at risk.

9. Household and Service Sector

Domestic robots perform cleaning, security surveillance, elderly care, and customer service applications in homes, hotels, airports, and shopping centres.

10. Education and Research

Educational robots support STEM learning, robotics training, lab automation, and advanced research.

5. ROBOTICS IN INDIA

India's robotics sector is growing quickly because of more automation, government support, new startups, and progress in AI. The World Robotics Report 2022 says India was 10th in the world for new industrial robot installations, and the number of robots in use doubled from **2016 to 2021**. Still, India has fewer robots per worker compared to top manufacturing countries.

Significance for India

1. **Boosts Manufacturing Competitiveness:** Robotics helps the **Make in India** initiative by raising productivity, improving quality, and making Indian manufacturing more competitive worldwide.
2. **Supports Industry 4.0: Combining** robotics with AI, IoT, and Big Data allows for smart automation and advanced factories.
3. **Addresses Labour Shortages-** Robots undertake repetitive and hazardous tasks, allowing human workers to focus on high-value activities.
4. **Promotes High-Tech Manufacturing: Robotics makes industries like** electronics, semiconductors, aerospace, automobiles, and defence stronger.
5. **Encourages Startup Innovation: More investment** in AI, robotics, drones, and automation is helping India's deep-tech startups grow.

Government Initiatives

The Indian government has launched several programs to support robotics research, innovation, skill-building, and industrial applications.

1. **National Strategy for Robotics (Draft)-** Aims to position India as a global leader in robotics by **2030** through research, innovation, manufacturing, and skilled workforce development.
2. **ARTPARK (AI & Robotics Technology Park)-** Established under the **National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)** to promote AI and robotics innovation.
3. **CAMRAS-** The **Centre for Advanced Manufacturing for Robotics and Autonomous Systems** supports indigenous development of robotics and automation technologies.

4. **I-HUB Foundation for Cobotics (IHFC), IIT Delhi, promotes** collaborative robotics research in healthcare, medical simulators, rehabilitation, and drone technologies.
5. **FutureSkills PRIME- a MeitY–NASSCOM** initiative providing training in AI, Robotics, Cybersecurity, Blockchain, and emerging technologies.
6. **e-YANTRA-** A Ministry of Education initiative encouraging robotics education and innovation among engineering students.
7. **Atal Innovation Mission (AIM)-** Atal Tinkering Labs provide students with robotics, AI, and IoT laboratories to promote innovation at the school level.
8. **ISRO's Vyommitra- an indigenous** humanoid robot being developed for the **Gaganyaan Human Spaceflight Mission**.

Challenges of Robotics

Even with rapid technological progress, using robots still presents technical, economic, and ethical challenges.

1. **High Initial Cost:** Industrial robots need a lot of money for hardware, software, upkeep, and skilled workers, which makes it hard for small and medium businesses to use them.
2. **Employment Concerns-** Automation of repetitive jobs may displace low-skilled workers, making reskilling and workforce transition essential.
3. **Cybersecurity Risks:** Connected robots can be hacked, attacked by malware, or controlled without permission, especially in important infrastructure.
4. **Ethical Issues:** AI-powered robots bring up questions about responsibility, privacy, autonomous weapons, and how decisions are made.
5. **Skill Gap:** India does not have enough professionals trained in robotics, AI, embedded systems, automation, and control engineering.
6. **Limited Indigenous Manufacturing: Relying on imported robot parts, sensors, actuators, and semiconductors makes it harder for India to be self-reliant in technology**.
7. **Regulatory Challenges: Without clear rules for safety, responsibility, and use, it is harder to adopt robotics on a large scale**.

Global Best Practices

Many countries have used robotics to boost industrial productivity and become technology leaders.

1. **Japan- a** global leader in industrial and humanoid robotics, with extensive use of robots in manufacturing, healthcare, and elderly care.
2. **South Korea has** one of the world's highest robot densities, supported by strong government incentives and smart manufacturing policies.
3. **Germany: Germany combines robotics with Industry 4.0 and encourages smart factories by working closely with universities and industry**.
4. **The United States leads** in AI-powered robotics, autonomous systems, and defence technologies through strong innovation ecosystems.
5. **China-** Through the **Made in China 2025** strategy, China has rapidly expanded domestic robot manufacturing and industrial automation.

Way Forward

1. **Promote Indigenous Robotics Manufacturing- strengthen** domestic production of robots, sensors, actuators, semiconductors, and automation systems under **Atmanirbhar Bharat**.
2. **Expand Robotics R&D-** Increase investment in AI, robotics, autonomous systems, and human–robot collaboration through industry–academia partnerships.
3. **Build a Skilled Workforce- strengthen** robotics education and reskilling through **FutureSkills PRIME, e-YANTRA, and Skill India**.
4. **Support MSME Automation-** Provide financial incentives, shared robotics facilities, and technology support to encourage adoption among MSMEs.

5. **Develop Robust Regulatory Frameworks- formulate** standards for robot safety, cybersecurity, ethical AI, liability, and human oversight.
6. **Strengthen Startup Ecosystem-** Promote deep-tech startups working in robotics, drones, medical robotics, and autonomous mobility through funding and incubation.
7. **Encourage Responsible Robotics- ensure** that robotics complements human capabilities while protecting employment, privacy, and safety.

Conclusion

Robotics is becoming a major part of the **Fourth Industrial Revolution**, transforming industries through smart automation, greater accuracy, and improved productivity. In India, robotics can help make manufacturing stronger, improve healthcare, modernize farming, and boost national security. By encouraging local innovation, building skills, and using robotics responsibly, India can reach the goals of **Atmanirbhar Bharat**, **Digital India**, and **Viksit Bharat 2047**.

6. INTERNET OF THINGS (IOT)

The **Internet of Things (IoT)** is a network of **physical objects, or "Things," that have sensors, software, processors, and communication tools. These let** them collect, share, and analyse data online with little help from people. IoT connects not just computers and phones, but also everyday items, machines, vehicles, appliances, and industrial systems. This enables real-time monitoring, automation, and smarter decision-making**.**

IoT is a key technology in the Fourth Industrial Revolution, also known as Industry 4.0. It is important in areas like smart manufacturing, healthcare, farming, transportation, energy management, and smart cities.

Components of an IoT System

An IoT system is made up of connected parts that collectively collect, send, process, and use data.

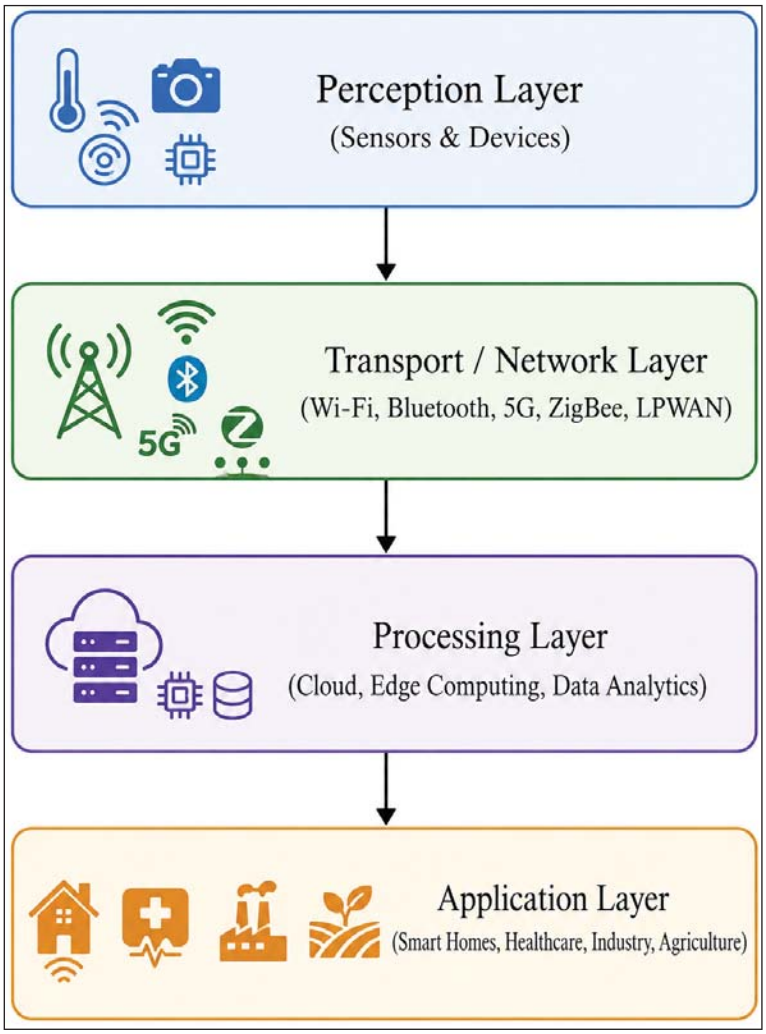
1. **IoT Devices (Things): These are physical items like** smart appliances, vehicles, wearables, industrial machines, and sensors that collect or create data.
2. **Sensors and Actuators:** Sensors gather real-world data such as temperature, humidity, motion, or pressure. Actuators turn digital commands into physical actions.
3. **Embedded Processors:** Microcontrollers and microprocessors handle sensor data, run control tasks, and manage how devices work.
4. **Connectivity-** Devices communicate using technologies such as **Wi-Fi, Bluetooth, ZigBee, Cellular Networks, LPWAN, and RFID**.
5. **IoT Gateway: This acts as a middle point, gather**ing data from many devices, changing data formats, and safely sending information to cloud or edge platforms.
6. **Cloud and Edge Computing:** Cloud platforms store and analyse lots of data. Edge computing processes data near the device, making responses faster and reducing delays.
7. **User Interface: Mobile app**s, dashboards, and web portals let users check devices, get alerts, and control IoT systems from anywhere.

Characteristics of IoT

1. **Connectivity**: IoT lets devices communicate easily using both wired and wireless networks.
2. **Real-Time Monitoring: Sensors constantly gather and send live data, helping people make quick decisions**.
3. **Automation:** IoT systems can do tasks on their own when certain conditions are met, needing little help from people.
4. **Scalability:** IoT networks can handle billions of connected devices in different fields.
5. **Interoperability:** Standard communication rules let devices from different makers work together smoothly.

6. **Remote Accessibility:** Users can check and control IoT devices from anywhere with an internet connection.
7. **Intelligence:** When IoT uses AI and data analytics, it can predict outcomes and make decisions on its own.

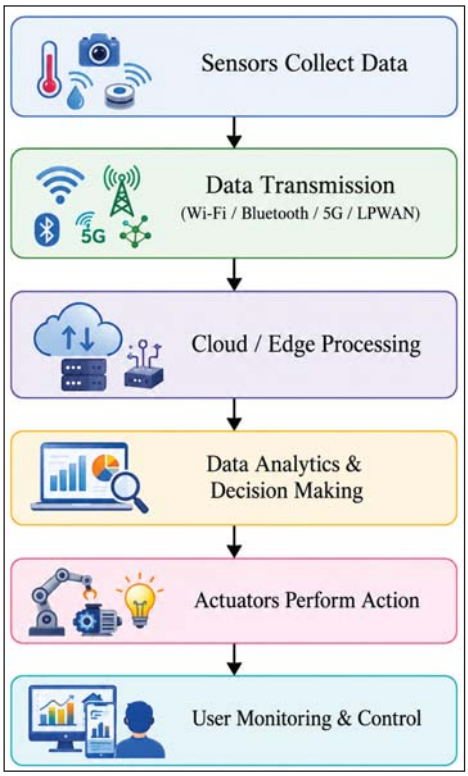
IoT Architecture



Enabling Technologies of IoT

1. Sensors and embedded systems collect data from the environment and help control how devices work.
2. RFID and NFC make it possible to identify, track, and communicate with devices over short distances automatically.
3. Wireless sensor networks allow many sensors placed in different locations to share information with each other.
4. Cloud computing and big data offer flexible storage and tools to analyse the huge amounts of data created by IoT devices.
5. Edge and Fog Computing process data close to where it is collected. This helps reduce delays and lowers the amount of data sent over the network.
6. Artificial intelligence and machine learning study IoT data to help predict problems, automate tasks, and make quick decisions.
7. 5G networks offer very fast connections and low delays, which are important for connecting many IoT devices at once.

Working of IoT



Applications of IoT

IoT has transformed many industries by enabling system monitoring, task automation, and data-driven decision-making.

1. Smart Manufacturing (Industrial IoT)

With IoT, factories can predict when machines need maintenance, monitor equipment, automate processes, manage inventory, and run more efficiently. This helps boost productivity and cut down on downtime. For example, Industrial IoT (IIoT) plays a key role in Industry 4.0.

2. Healthcare (Internet of Medical Things - IoMT)

IoT devices enable remote patient monitoring, wearable health devices, smart hospitals, medical equipment tracking, and telemedicine services.

- **For example, smartwatches can track heart rate, oxygen levels, and physical activity continuously**.

3. Smart Agriculture

IoT helps farmers by using soil moisture sensors, tracking the weather, automating irrigation, monitoring livestock, and checking crop health.

4. Smart Cities

IoT helps cities manage traffic, control street lighting, handle waste, monitor the environment, and improve public safety.

5. Transportation and Logistics

IoT facilitates fleet management, vehicle tracking, smart traffic systems, route optimisation, and real-time cargo monitoring.

6. Energy Management

Smart grids and smart meters optimise electricity generation, distribution, and consumption, improving energy efficiency.

7. Environmental Monitoring

IoT sensors monitor air and water quality, weather, pollution, and potential disaster risks.

8. Defence and Security

IoT helps with surveillance, border monitoring, connecting military systems, and managing battlefields using the **Internet of Military Things (IoMT)**.

9. Smart Homes

With IoT, people can use their smartphones to control lights, air conditioning, security systems, appliances, and energy use at home.

10. Retail and Supply Chains

IoT makes it easier to manage inventory, automate warehouses, track assets, and improve customer experience using RFID and smart sensors.

IoT and India

India is quickly using IoT in manufacturing, farming, healthcare, city management, and public infrastructure. Programs like **Digital India, 5G, BharatNet, Smart Cities Mission, and Industry 4.0** are speeding up IoT growth across the country.

1. **Supports Smart Manufacturing-** Enables automation, predictive maintenance, and intelligent production under **Make in India**.
2. **Promotes Precision Agriculture: IoT sensors help with better irrigation, fertilizer use, and crop yields**.
3. **Strengthens Smart Cities: IoT helps manage traffic, waste, water supply, and energy use in cities**.
4. **Improves Public Service Delivery-** Supports digital governance through real-time monitoring of utilities, healthcare, and transportation.
5. **Drives Digital Economy: Growth in** 5G, cloud computing, and Digital Public Infrastructure (DPI) is building a strong base for IoT in India.

Government Initiatives

1. **Draft National IoT Policy (2015)-** Promotes India as a global hub for IoT design, manufacturing, and innovation.
2. **IoT Centre of Excellence (CoE)-** Established by **MeitY, NASSCOM, and ERNET** to support IoT startups and innovation.
3. **Digital India Programme-** Expands digital infrastructure and connectivity to enable IoT adoption.
4. **National Digital Communications Policy (2018)-** Promotes broadband connectivity, 5G, Machine-to-Machine (M2M) communication, and IoT ecosystems.
5. **Smart Cities Mission uses** IoT for intelligent transportation, surveillance, smart utilities, and urban governance.
6. **FutureSkills PRIME builds** workforce capabilities in IoT, AI, Robotics, Cybersecurity, and Cloud Computing.
7. **Jal Jeevan Mission deploys IoT-based sensors** for real-time monitoring of drinking water quality and supply.

Challenges of IoT

1. **Cybersecurity Risks:** Connected devices are vulnerable to hacking, malware, ransomware, and Distributed Denial of Service (DDoS) attacks.
2. **Data Privacy Concerns-** Continuous collection of personal and operational data raises concerns regarding surveillance and misuse.
3. **Interoperability Issues-** Lack of common standards makes integration between devices from different manufacturers difficult.
4. **High Implementation Cost-** Large-scale deployment requires significant investment in devices, connectivity, cloud infrastructure, and maintenance.
5. **Connectivity Constraints-** Reliable broadband and 5G connectivity remain limited in many rural and remote regions.

6. **Energy Consumption:** Billions of connected devices increase power consumption and require energy-efficient embedded systems.
7. **Regulatory Challenges-** Evolving issues related to cybersecurity, data protection, spectrum management, and device standards require comprehensive policy frameworks.

Global Best Practices

1. **The United States leads** in Industrial IoT, cloud platforms, and smart manufacturing.
2. **Germany integrates** IoT with **Industry 4.0** to enable intelligent manufacturing.
3. **China- rapidly** expanding IoT through smart cities, 5G, and industrial automation.
4. **Singapore uses** IoT extensively under the **Smart Nation Initiative** for urban governance and public services.
5. **Japan integrates** IoT with **Society 5.0** to create a human-centric digital society.

Way Forward

1. **Expand Digital Infrastructure-** Accelerate **5G, BharatNet, cloud computing, and edge computing** to support large-scale IoT deployment.
2. **Strengthen IoT Security- develop** robust cybersecurity standards, encryption mechanisms, and secure-by-design IoT devices.
3. **Promote Indigenous IoT Manufacturing- encourage** domestic production of sensors, chipsets, embedded systems, and communication modules under **Atmanirbhar Bharat**.
4. **Develop Interoperability Standards-** Establish common protocols to ensure seamless communication across IoT platforms and devices.
5. **Enhance AI–IoT Integration-** Leverage Artificial Intelligence for predictive analytics, intelligent automation, and real-time decision-making.
6. **Build Skilled Workforce-** Expand skilling programmes in embedded systems, IoT, cloud computing, cybersecurity, and AI.
7. **Encourage Responsible IoT Adoption- formulate** comprehensive regulations for data privacy, device security, ethical use, and governance.

Conclusion

The **Internet of Things** is transforming the physical world into a smart, connected system by enabling real-time data collection, automation, and improved decision-making. As a major part of the **Fourth Industrial Revolution**, IoT could change manufacturing, healthcare, farming, government, and cities. By building strong digital infrastructure, encouraging local innovation, and ensuring IoT is used safely and responsibly, India can leverage IoT to accelerate **Digital India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**.

7. BLOCKCHAIN TECHNOLOGY

Blockchain is a **digital ledger** that records transactions across many computers in a network. Information is stored in blocks, and each block is securely linked to the one before it, creating a chain that cannot be easily altered. This setup makes blockchain transparent, secure, and hard to tamper with.

Blockchain was first created in 2008 by Satoshi Nakamoto as the technology behind Bitcoin. Since then, it has grown and is now used in areas like finance, supply chains, healthcare, government, digital identity, and Web 3.0.

Evolution of Blockchain

- **1982:** David Chaum proposed cryptographic protocols for secure digital transactions.
- **1991:** Stuart Haber and W. Scott Stornetta developed cryptographically secured timestamped records.
- **2008:** Satoshi Nakamoto introduced **Bitcoin** and the first decentralised blockchain.
- **2015:** Ethereum introduced **Smart Contracts**, expanding blockchain beyond cryptocurrencies.
- **Today, blockchain is used in many fields, including** finance, supply chain management, healthcare, government, NFTs, and Decentralised Finance (DeFi).

Components of Blockchain

1. **Blocks: These hold** transaction details, timestamps, and a secure code that links to the previous block.
2. **Distributed Ledger: Every participant, or node, keeps the same copy of the ledger. This removes** the need for a central authority.
3. **Nodes: These are computers in** the blockchain network that check, store, and share transaction records.
4. **Cryptography: This uses special codes** and digital signatures to keep transactions safe and make sure data stays accurate.
5. **Consensus Mechanism: Methods like Proof of Work (PoW)** and **Proof of Stake (PoS)** are used to check transactions before new blocks are added.
6. **Smart Contracts: These are digital agreements that carry out actions automatically when certain** conditions are met.

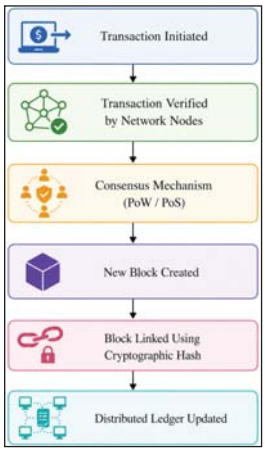
Features of Blockchain

1. **Decentralisation: No one person or group controls the network. Instead, data is shared across many** nodes.
2. **Immutability: Once data is added, it cannot be changed unless the whole network agree**s.
3. **Transparency: All authorised users can check transactions, which helps build** trust and accountability.
4. **Security: Special codes** and digital signatures protect data from being changed or faked.
5. **Consensus-Based Validation:** Transactions are only added after the network's nodes check and approve them.
6. **Traceability:** Every transaction can be followed from start to finish, which helps keep everyone accountable.
7. **Smart Automation:** Smart contracts let transactions happen automatically, without needing a middleman.

Types of Blockchain

1. **Public Blockchain: These are open** networks where anyone can join and take part.
   - **Examples:** Bitcoin, Ethereum.
2. **Private Blockchain-** Restricted networks controlled by a single organisation for enterprise applications.
   - **Examples:** Hyperledger Fabric, Corda.
3. **Consortium Blockchain:** This is managed by several organisations together, offering both shared control and limited access.
4. **Hybrid Blockchain:** These mix public and private blockchain features, so some data is open and some is restricted.

Working of Blockchain



Applications of Blockchain

Blockchain is changing many industries by making transactions more secure, transparent, and decentralized.

1. Banking and Finance

Blockchain makes digital payments, cross-border transactions, digital currencies, and trade finance more secure. It also helps reduce fraud by removing the need for intermediaries.**For example, blockchain is the technology behind cryptocurrencies like Bitcoin** and **Ethereum**.

2. Supply Chain Management

Blockchain helps track products in the supply chain, making it easier to verify their authenticity and reduce fraud or counterfeiting. **For example, Walmart** uses blockchain to follow food products from farms all the way to customers.

3. Healthcare

Blockchain can safely store electronic health records, manage medicine supply chains, and allow secure sharing of medical information.

4. Governance and Public Administration

Blockchain helps create secure digital identities, manage land records, deliver public services, verify documents, and make governance more transparent.

5. Smart Contracts

Smart contracts are self-executing agreements that automatically carry out contract terms, so there is no need for middlemen. This helps save time and money.

6. Digital Identity

Blockchain enables people to manage their digital identities securely and gain greater control over their personal data.

7. Voting Systems

Blockchain-based voting systems are more transparent, help prevent tampering, and make elections more trustworthy.

8. Insurance

In insurance, blockchain can automate claim settlements, detect fraud, verify policies, and help onboard new customers using smart contracts.

9. Internet of Things (IoT)

Blockchain helps IoT devices communicate securely, making data exchanges more trustworthy and lowering cybersecurity risks.

10. Intellectual Property and Media

Blockchain protects copyrights, digital assets, and royalty payments by creating records of ownership that cannot be changed.

Blockchain and India

India is increasingly using blockchain in areas such as governance, financial services, healthcare, logistics, and digital public infrastructure.

1. **Strengthens Digital Governance:** Blockchain makes public services more transparent, secure, and efficient.
2. **Secures Digital Records: Blockchain helps keep** land records, certificates, and government databases safe from tampering.
3. **Improves Supply Chains: Blockchain makes it easier to track products** in agriculture, pharmaceuticals, and logistics.
4. **Promotes Financial Innovation: Blockchain allows for** faster digital payments, smart contracts, and helps more people access financial services.
5. **Supports Digital Economy:** Blockchain works well with new technologies like AI, IoT, Web 3.0, and digital public infrastructure.

Government Initiatives

1. **National Strategy on Blockchain: This plan encourages the use of homegrown blockchain platforms** in governance, healthcare, agriculture, and logistics.
2. **Centre of Excellence (CoE) for Blockchain- established** by the **National Informatics Centre (NIC)** to develop blockchain-based e-governance solutions.
3. **RBI Regulatory Sandbox: This program let**s fintech startups test new blockchain financial ideas while being supervised by regulators.
4. **Digital India Programme: This initiative se**es blockchain as an important technology for secure digital governance.
5. **State Initiatives: States like Telangana, Tamil Nadu, Maharashtra, and Kerala** have started blockchain policies, innovation hubs, and pilot projects.

Challenges of Blockchain

1. **Scalability Issues:** Most blockchain networks process transactions slower than conventional payment systems.
2. **High Energy Consumption: Blockchains that use Proof of Work (PoW) need a lot of** computing power and electricity.
3. **Regulatory Uncertainty: Without clear legal rules, there is confusion about** digital assets and how blockchain can be used.
4. **Interoperability Challenges: Many** blockchain platforms do not share common standards, making it hard for them to work together.
5. **High Implementation Cost: Setting up blockchain for businesses can be expensive because it needs special** infrastructure and skilled workers.
6. **Cybersecurity Risks: There are still security concerns like smart contract bugs, stolen private keys, and 51% attack**s.
7. **Limited Awareness and Skilled Workforce-** Adoption is constrained by shortage of blockchain expertise and limited awareness among businesses.

Global Best Practices

1. **Estonia uses** blockchain extensively for digital identity, healthcare, and e-governance.
2. **Dubai (UAE)-** Aims to become one of the world's first blockchain-powered governments through the **Dubai Blockchain Strategy**.
3. **Singapore: Singapore encourag**es blockchain innovation in digital finance, trade, and cross-border payments.
4. **European Union: The EU s**upports blockchain with its **European Blockchain Services Infrastructure (EBSI)**.
5. **China: China is encouraging businesses to use blockchain and is build**ing the **Blockchain-based Service Network (BSN)**.

Way Forward

1. **Develop a Comprehensive Regulatory Framework: Create clear rules for blockchain**, digital assets, smart contracts, and data management.
2. **Promote Indigenous Blockchain Platforms: Support local** research, innovation, and business blockchain solutions as part of **Digital India** and **Atmanirbhar Bharat**.
3. **Enhance Blockchain Infrastructure-** Expand secure cloud infrastructure, digital identity systems, and high-performance computing to support blockchain adoption.
4. **Build Skilled Human Resources: Encourag**e education, research, and special training in blockchain, cryptography, and distributed systems.
5. **Encourage Public–Private Partnerships-** Foster collaboration among government, academia, startups, and industry to accelerate blockchain innovation.
6. **Improve Interoperability Standards: Create shared rules so different blockchain networks can work together easily**.

7. **Integrate Blockchain with Emerging Technologies: Use blockchain together with AI, IoT, Big Data, and Web 3.0** to create secure and smart digital systems.

Conclusion

Blockchain is changing the digital economy by making transactions **secure, transparent, decentralised, and tamper-proof** in many areas. It is now a key technology for governance, finance, healthcare, supply chains, and Web 3.0, not just for cryptocurrencies. By encouraging local innovation, strengthening regulations, and training skilled professionals, India can use blockchain to support **Digital India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**.

8. QUANTUM TECHNOLOGY

Quantum technology uses the principles of quantum mechanics, the branch of physics that explains how matter and energy behave at the atomic and subatomic levels. Unlike traditional technologies, quantum technologies leverage unique quantum effects such as superposition, entanglement, interference, and quantum tunnelling to enable capabilities that conventional systems cannot.

Quantum technology is seen as one of the most important advances of the 21st century. It has uses in **quantum computing, communication, sensing, materials, healthcare, defence, finance, cybersecurity, and scientific research**.

Fundamental Principles of Quantum Technology

Quantum technology is based on several special properties of quantum mechanics.

1. Superposition

Superposition means a quantum particle can exist in several states at once until it is measured. Unlike a regular bit, which is either 0 or 1, a **qubit** can be both 0 and 1 simultaneously. This allows for much faster computing.

2. Quantum Entanglement

Entanglement happens when two or more quantum particles become linked, so that the state of one instantly affects the other, no matter how far apart they are.

This property is the foundation for quantum c**ommunication** and **Quantum Key Distribution (QKD).**

3. Quantum Interference

Quantum interference occurs when quantum waves mix, either strengthening or cancelling each other out. Quantum algorithms use this effect to increase the number of correct answers and reduce the number of incorrect ones.

4. Quantum Measurement (Observation)

When you measure a quantum system, its state changes to a definite value. This makes quantum communication very secure because any attempt to intercept the information will disturb the quantum state.

Pillars of Quantum Technology

Quantum technology can be grouped into four main areas.

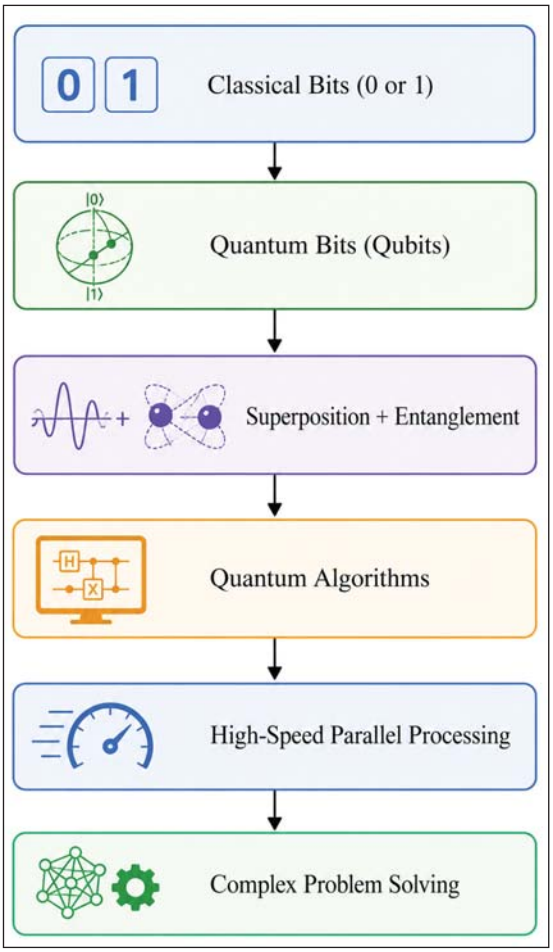
1. Quantum Computing	2. Quantum Communication	3. Quantum Sensing	4. Quantum Materials
Quantum computers use **qubits** instead of regular bits to solve complex problems much faster than traditional computers.	Quantum communication uses quantum effects to create extremely **secure communication** with **Quantum Key Distribution (QKD)**.	Quantum sensors can detect very small changes in magnetic fields, gravity, time, and movement with great accuracy.	Quantum materials have special electronic and magnetic properties because of quantum effects.

Applications • Drug discovery • Material science • Weather forecasting • Financial modelling • Artificial Intelligence • Cryptography	If someone tries to intercept the qubits being sent, their state changes right away, so eavesdropping can be detected.	**Applications** • Medical imaging • Navigation • Earth observation • Mineral exploration • Defence • Environmental monitoring	**Applications** • Superconductors • Quantum chips • Solar cells • Advanced batteries • Medical imaging devices

Difference between Classical and Quantum Computing

Classical Computing	Quantum Computing
Uses Bits (0 or 1)	Uses Qubits (0 & 1 simultaneously)
Sequential processing	Parallel computation through superposition
Limited by transistor architecture	Uses quantum mechanical principles
Suitable for general computing	Suitable for highly complex optimisation and simulation problems

Working of Quantum Computing



Features of Quantum Technology

1. **Massive Computational Power:** Quantum computers can solve complex optimisation and simulation problems much faster than traditional computers.
2. **Ultra-Secure Communication:** Quantum Key Distribution (QKD) makes communication almost tamper-proof by detecting any attempts to intercept messages.
3. **High Precision Measurement:** Quantum sensors offer very accurate measurements of time, gravity, magnetic fields, and movement.
4. **Parallel Processing:** Superposition lets quantum systems handle several possibilities at the same time.
5. **Advanced Simulation:** Quantum computers can model molecular interactions and quantum systems that are hard for traditional computers to handle.
6. **Next-Generation Materials:** Quantum materials help create advanced semiconductors, superconductors, and energy-saving devices.

Applications of Quantum Technology

Quantum technology could change many industries by solving problems that traditional technologies cannot handle.

1. Healthcare

Quantum computing speeds up drug discovery, protein folding analysis, disease modeling, and personalized medicine. Quantum sensors also improve medical imaging and diagnostics.

2. Cybersecurity

Quantum communication uses Quantum Key Distribution (QKD) to make communication very secure and improve cyber resilience.

3. Defence and National Security

Quantum technologies help with secure military communication, quantum radar, GPS-free navigation, intelligence gathering, and advanced cryptography.

4. Financial Services

Quantum computing improves portfolio optimisation, fraud detection, risk analysis, and financial modelling.

5. Climate Science

Quantum simulations improve climate modelling, weather prediction, disaster forecasting, and carbon capture research.

6. Artificial Intelligence

Quantum computing speeds up machine learning, optimization, and the processing of large datasets.

7. Space Research

Quantum sensors make satellite navigation, deep-space communication, and precise timing for space missions better.

8. Material Science

Quantum simulations speed up the development of superconductors, advanced batteries, semiconductors, and clean-energy technologies.

9. Telecommunications

Quantum communication enables next-generation secure internet and long-distance encrypted communication.

10. Energy Sector

Quantum simulations help improve battery design, renewable energy technology, smart grids, and energy storage systems.

Quantum Technology and India

India sees Quantum Technology as a key area for technological independence, economic growth, and national security.

1. **Strengthens National Security- enables** secure communication, advanced cryptography, and defence technologies.
2. **Promotes Technological Self-Reliance: Encourages local** development of quantum computers, sensors, communication systems, and semiconductor technologies.
3. **Accelerates Scientific Research: Boost**s research in healthcare, materials science, pharmaceuticals, and climate modelling.
4. **Supports Digital Economy- complements** AI, Cybersecurity, Semiconductors, and Digital Public Infrastructure.
5. **Builds Global Competitiveness: Helps India join the** leading nations in next-generation computing technologies.

Government Initiatives

1. **National Quantum Mission (2023)-** Approved with an outlay of **₹6,003 crore**, the mission aims to develop indigenous capabilities in **Quantum Computing, Quantum Communication, Quantum Sensing, and Quantum Materials** over eight years.
2. **Thematic Hubs (T-Hubs)-** Centres established in leading academic institutions to promote research in Quantum Computing, Communication, Sensing, and Materials.
3. **QuEST Programme-** The **Quantum-Enabled Science and Technology (QuEST)** initiative of the Department of Science & Technology supports research and innovation in quantum technologies.
4. **Quantum Lab (Indian Army)-** Established at the **Military College of Telecommunication Engineering (MCTE), Mhow**, to strengthen defence applications of quantum technologies.
5. **QSim Toolkit- an** indigenous quantum simulation platform that enables students and researchers to learn quantum programming and hardware design.
6. **C-DOT Initiatives:** The **Centre for Development of Telematics (C-DOT)** is developing indigenous **Quantum Key Distribution (QKD)** solutions for secure communication.
7. **Startup Ecosystem-** Indian startups such as **QpiAI, BosonQ Psi, and QRDLab** are developing quantum computing and quantum software solutions.

Challenges of Quantum Technology

1. **Decoherence:** Quantum states are very sensitive to outside disturbances, which can cause information loss during computation.
2. **Error Correction:** Quantum computations are prone to errors and need advanced error-correction methods.
3. **High Development Cost-** Building quantum computers requires expensive infrastructure, cryogenic systems, and specialised hardware.
4. **Limited Skilled Workforce: There are not enough** quantum physicists, engineers, and researchers, which slows down technology development.
5. **Cybersecurity Risks:** Future quantum computers might break current encryption systems, so **Post-Quantum Cryptography (PQC) is needed.**
6. **Technological Complexity: Creat**ing stable qubits and scalable quantum systems is still a big scientific challenge.
7. **Infrastructure Constraints:** Quantum systems need special labs and extremely low-temperature environments.

Global Best Practices

1. **The United States leads** in quantum computing research through **IBM, Google, Microsoft, and Quantinuum**, supported by the **National Quantum Initiative**.

2. **China invests** heavily in quantum communication, satellite-based QKD, and quantum computing for strategic applications.
3. **European Union- supports** research through the **Quantum Flagship Programme**, focusing on computing, sensing, and communication.
4. **Canada- home** to leading quantum companies such as **D-Wave** and **Xanadu**, with strong academic research ecosystems.
5. **Japan promotes** quantum technologies for industrial innovation, healthcare, and next-generation communication systems.

Way Forward

1. **Strengthen Indigenous R&D-** Increase investment in quantum computing, communication, sensing, and advanced quantum materials.
2. **Develop Quantum Infrastructure-** Establish national quantum laboratories, testing facilities, and high-performance research centres.
3. **Build Skilled Human Capital-** Expand education and specialised training in quantum science, engineering, and quantum programming.
4. **Promote Industry–Academia Collaboration-** Encourage collaboration among universities, startups, industry, and government laboratories.
5. **Develop Post-Quantum Cybersecurity-** Accelerate the adoption of **Post-Quantum Cryptography (PQC)** to secure future digital infrastructure.
6. **Encourage Global Collaboration-** Strengthen international partnerships in quantum research, standards, and innovation.
7. **Support Quantum Startups-** Provide funding, incubation, and venture support to accelerate indigenous quantum innovation.

Conclusion

Quantum technology is leading the way in a new era by changing computing, communication, sensing, and materials science. Because it can solve problems that traditional technologies cannot, it is set to drive future economic growth, scientific discovery, and national security. By investing in local research, skilled people, and strong quantum infrastructure, India can become a global leader in quantum technology and help achieve the goals of **Atmanirbhar Bharat** and **Viksit Bharat 2047**.

9. CLOUD COMPUTING

Cloud computing is a way to deliver services like servers, storage, databases, networking, software, analytics, and artificial intelligence over the Internet, often called "the Cloud." Instead of buying and maintaining their own hardware, users can get these resources when they need them and only pay for what they use. This approach makes it easier to scale, adapt, and save money.

Cloud computing is a key technology behind the **Fourth Industrial Revolution, also known as Industry 4.0.** It supports new technologies like artificial intelligence, big data analytics, the Internet of Things, blockchain, and digital public infrastructure**.**

Evolution of Cloud Computing

- **1960s:** John McCarthy proposed computing as a public utility.
- **1999:** Salesforce introduced one of the first **Software as a Service (SaaS)** platforms.
- **2006:** Amazon Web Services (AWS) launched cloud infrastructure services.
- **2010 onwards:** Rapid adoption of cloud computing by businesses, governments, and startups.
- **Present:** Cloud computing forms the backbone of AI, IoT, Digital Governance, Big Data, and Industry 4.0.

Components of Cloud Computing

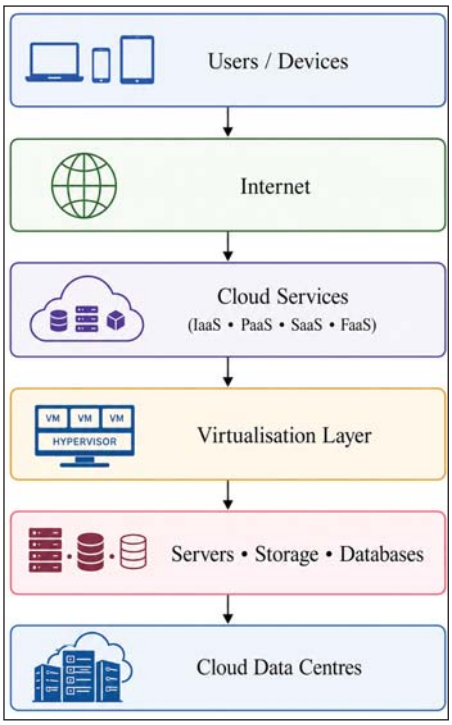
A cloud computing system is made up of different parts that work together to provide services efficiently.

1. **Front End (Client Interface): This includes** devices like computers, smartphones, tablets, web browsers, and cloud apps that people use to access cloud services.
2. **Back End (Cloud Infrastructure): This part includes servers, storage**, databases, virtual machines, and data centers that handle and manage cloud resources.
3. **Network: This is the** Internet or private networks that link users to the cloud and allow data to move between them.
4. **Cloud Management Software: This software handles things like assigning resources**, monitoring, security, billing, and delivering services.
5. **Virtualisation: This technology lets one physical server run several virtual machines, which helps use resources better and makes it easier to scale up or down**.

Characteristics of Cloud Computing

1. **On-Demand Self-Service:** Users can get computing resources whenever they need them, without having to contact the provider directly.
2. **Broad Network Access: People can use cloud services from anywhere as long as they have an internet-connected device**.
3. **Resource Pooling: Cloud providers share computing resources among many** users by using a system that lets multiple customers use the same hardware.
4. **Rapid Elasticity: Cloud resources can be increased or decreased quickly based on what users nee**d.
5. **Measured Service: Users only pa**y for the computing resources they actually use.
6. **High Availability:** Cloud providers keep services running all the time by using data centers in different locations.

Cloud Computing Architecture



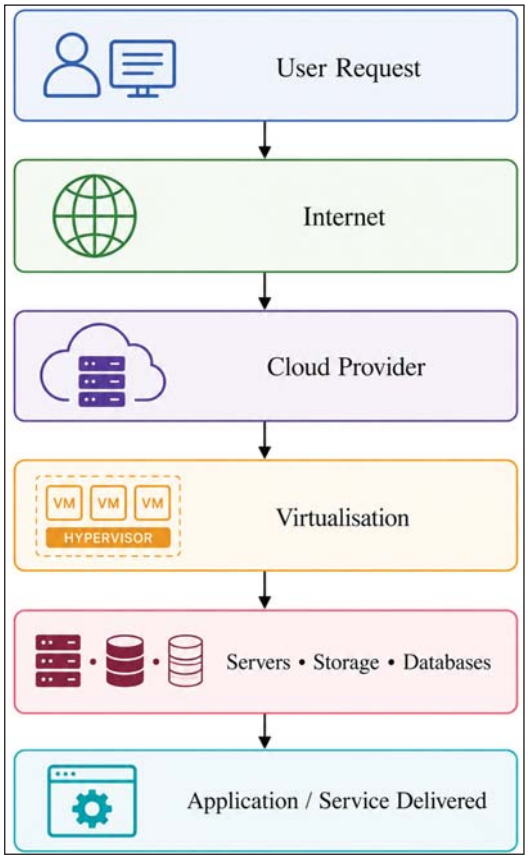
Types of Cloud Deployment

1. Public Cloud	2. Private Cloud	3. Hybrid Cloud	4. Multi-Cloud
Cloud infrastructure owned and managed by third-party providers and made available to the general public. **Examples:** AWS, Microsoft Azure, Google Cloud Platform.	Cloud infrastructure dedicated to a single organisation, offering greater control, privacy, and security.	Combines public and private cloud environments, allowing organisations to balance security with scalability.	Uses cloud services from multiple providers to improve flexibility and avoid vendor lock-in.

Cloud Service Models

1. Infrastructure as a Service (IaaS)	2. Platform as a Service (PaaS)	3. Software as a Service (SaaS)	4. Function as a Service (FaaS)
Provides virtual servers, storage, networking, and computing resources over the internet. **Example:** Amazon EC2.	Provides a platform for developing, testing, and deploying applications without managing infrastructure. **Example:** AWS Elastic Beanstalk, Microsoft Azure App Service.	Delivers software applications over the internet through web browsers. **Examples:** Google Workspace, Microsoft 365, Salesforce.	Allows developers to execute code without managing servers (Serverless Computing). **Example:** AWS Lambda.

Working of Cloud Computing



Features of Cloud Computing

1. **Scalability:** Resources can be increased or decreased instantly based on demand.
2. **Cost Efficiency: With a pay-as-you-go model, there is no need for large investments in physical infrastructure**.
3. **High Availability:** Distributed infrastructure helps keep services reliable and reduces downtime.
4. **Resource Sharing: Cloud resources can be shared efficiently among many user**s.
5. **Flexibility: Users can access** applications and data from any location.
6. **Automatic Updates:** Service providers take care of software upgrades, maintenance, and security patches.
7. **Disaster Recovery:** Cloud backup and redundancy help businesses continue operating after disruptions.

Applications of Cloud Computing

Cloud computing has changed many industries by making digital services more scalable, secure, and affordable.

1. Artificial Intelligence and Machine Learning

Cloud platforms provide high-performance computing infrastructure for AI model training, deployment, and data analytics.

2. Big Data Analytics

Cloud computing allows businesses to store and process large amounts of data to improve decision-making and predictive analytics.

3. Healthcare

Cloud computing supports telemedicine, electronic health records, medical imaging, and AI-powered diagnostics.

4. Education

Cloud-based platforms facilitate online learning, virtual classrooms, digital libraries, and collaboration.

5. E-Commerce

Cloud infrastructure helps online shopping platforms grow, supports digital payments, and simplifies inventory management.

6. Banking and Financial Services

Cloud computing supports digital banking, helps detect fraud, assists with risk analysis, and secures financial transactions.

7. Government Services

Cloud computing enables digital governance, citizen services, e-office functions, and secure public data management.

8. Internet of Things (IoT)

Cloud platforms collect, store, and analyse data generated by billions of connected IoT devices.

9. Media and Entertainment

Cloud computing supports video streaming, cloud gaming, content delivery networks (CDNs), and digital media storage.

10. Scientific Research

Cloud computing provides High-Performance Computing (HPC) for simulations, climate modelling, genomics, and space research.

Cloud Computing and India

India is among the fastest-growing cloud markets, thanks to **Digital India, AI adoption, startup growth, 5G, and digital governance**.

1. **Strengthens Digital India:** Cloud computing supports e-governance, Digital Public Infrastructure (DPI), and services focused on citizens.
2. **Supports Startup Ecosystem-** Cloud platforms reduce infrastructure costs and enable rapid innovation.
3. **Enables AI and Industry 4.0: Cloud computing offers scalable** infrastructure for AI, IoT, robotics, and data analytics.
4. **Promotes Digital Economy: Cloud computing speeds up** digital payments, e-commerce, fintech, and online services.
5. **Improves Public Service Delivery: Cloud-based platforms support** healthcare, education, agriculture, and smart city projects.

Government Initiatives

1. **MeghRaj (GI Cloud)-** National cloud initiative to accelerate e-governance and optimise ICT infrastructure across government departments.
2. **Digital India Programme: This program encourag**es cloud adoption for digital governance and better public services.
3. **DigiLocker: A cloud-based platform for securely storing and verifying** digital documents.
4. **Government e-Marketplace (GeM):** Uses cloud infrastructure to offer scalable digital procurement services.
5. **National Informatics Centre (NIC) Cloud:** Provides cloud infrastructure for government ministries and departments.
6. **RailCloud:** Indian Railways' cloud platform that supports digital services and improves operational efficiency.
7. **National Data Centres: These centres strengthen** secure cloud infrastructure for digital governance and citizen services.

Challenges of Cloud Computing

1. **Data Security and Privacy-** Storing sensitive information on third-party infrastructure raises concerns regarding data breaches and unauthorised access.
2. **Cybersecurity Threats-** Cloud environments remain vulnerable to cyberattacks, ransomware, malware, and API exploitation.
3. **Vendor Lock-In: Mov**ing applications and data between cloud providers can be hard and costly.
4. **Compliance and Data Sovereignty: Organizations must follow rules about where data is stored and how it is transferred across bord**ers.
5. **Downtime Risks:** Cloud service outages can interrupt important business operations.
6. **Skill Gap: There is a s**hortage of professionals with skills in cloud architecture, cybersecurity, and cloud-native development.
7. **Cost Management: If resources are not managed well,** unexpected operational costs can occur.

Global Best Practices

1. **United States: The US is home to top** cloud providers like **AWS, Microsoft Azure, and Google Cloud**, which drive innovation in AI and cloud services.
2. **European Union: The EU p**romotes secure cloud infrastructure with projects like **GAIA-X**, focusing on data sovereignty and interoperability.
3. **Singapore: Singapore u**ses cloud computing widely for digital governance as part of the **Smart Nation Initiative**.

4. **Japan: Japan combin**es cloud computing with AI, IoT, and Society 5.0 to modernise industries and public services.
5. **China: China is quick**ly expanding its cloud infrastructure with providers like **Alibaba Cloud** and **Tencent Cloud**.

Way Forward

1. **Strengthen Cloud Infrastructure:** Expand hyperscale data centres, cloud regions, and high-speed broadband connections.
2. **Enhance Cloud Security: Use** strong encryption, Zero Trust Architecture, Multi-Factor Authentication (MFA), and regular security audits.
3. **Promote Indigenous Cloud Ecosystem- encourage** domestic cloud service providers and reduce dependence on foreign infrastructure under **Atmanirbhar Bharat**.
4. **Develop Skilled Workforce: Increase** training in cloud computing, DevOps, cybersecurity, and cloud-native technologies.
5. **Encourage Hybrid and Multi-Cloud Adoption: Using multiple cloud environments can improve resilience, flexibility, and business continuity**.
6. **Strengthen Regulatory Frameworks: Make sure there are strong standards for** data protection, privacy, interoperability, and compliance.
7. **Integrate Cloud with Emerging Technologies-** Combine cloud computing with **AI, IoT, Big Data, Blockchain, and Quantum Computing** to accelerate digital transformation.

Conclusion

Cloud Computing has become the backbone of the modern digital economy by providing scalable, secure, and cost-efficient computing infrastructure. It enables governments, businesses, and individuals to leverage emerging technologies without investing in expensive physical infrastructure. With initiatives such as **MeghRaj**, **Digital India**, and expanding cloud infrastructure, India is well-positioned to harness cloud computing as a key driver of **Digital India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**.

10. CYBERSECURITY TECHNOLOGIES

Cybersecurity encompasses the tools, methods, and policies used to protect computer systems, networks, devices, apps, and data from attacks, unauthorized access, damage, theft, or disruption. Its main goal is to protect the confidentiality, integrity, and availability (CIA) of digital information and important infrastructure.

As Digital India, AI, cloud computing, IoT, 5G, and digital public infrastructure grow quickly, cybersecurity has become essential for national security, economic stability, and digital governance.

Objectives of Cyber Security

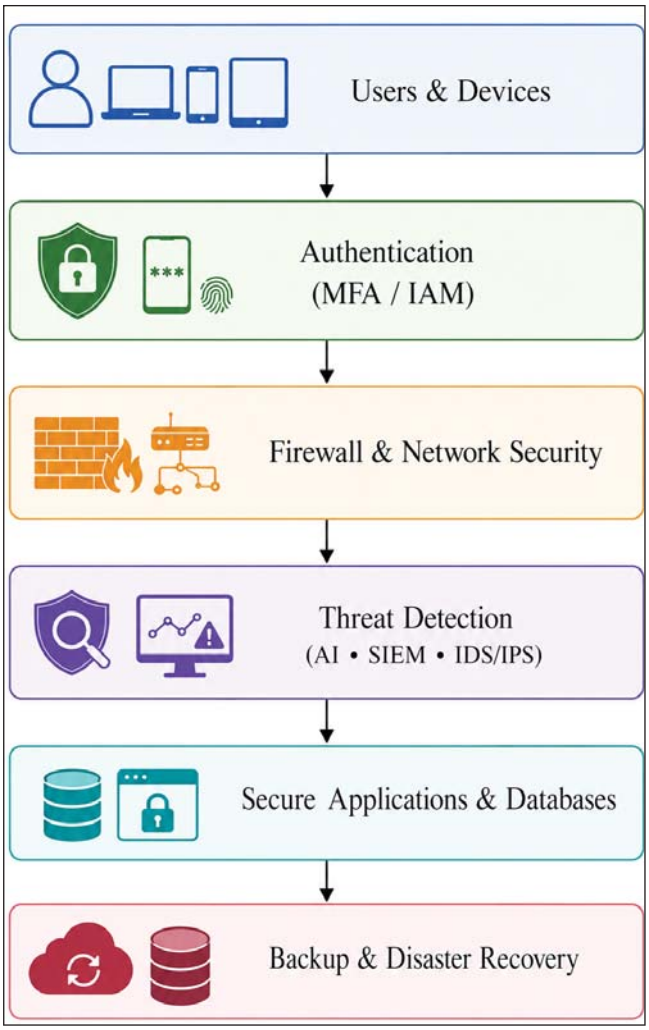
1. **Confidentiality prevents** unauthorized access to sensitive information.
2. **Integrity ensures** that data stays accurate, complete, and unchanged.
3. **Availability: makes sure** that systems, networks, and data are always accessible to authorised users when needed.
4. **Authentication:** Verifies the identity of users and devices before granting access.
5. **Non-Repudiation: stops** users from denying their actions by using digital signatures and audit trails.

Types of Cyber Security

1. **Network Security: Protects** computer networks from unauthorised access, malware, and cyberattacks.
2. **Application Security keep**s software applications secure during their development and use.
3. **Cloud Security: Protects** cloud infrastructure, applications, and data by using encryption, Identity and Access Management (IAM), and security monitoring.

4. **Information Security: Protects both** digital and physical information from unauthorised access, disclosure, or changes.
5. **Endpoint Security keeps** devices like laptops, desktops, servers, and smartphones safe when they are connected to an organisation's network.
6. **Critical Infrastructure Security: Protects important** sectors like power, banking, telecommunications, healthcare, transportation, and defence from cyber threats.
7. **Identity and Access Management (IAM): Makes sure** that only authorised users can access digital resources by using authentication and access controls.

Cyber Security Architecture



Types of Cyber Threats

Cyber threats can come from individuals, criminal organizations, terrorist groups, or even countries.

1. **Cybercrime: These are illegal activities done** for financial gain, fraud, identity theft, or to disrupt digital services.
2. **Cyber Espionage: This involves gaining unauthorize**d access to confidential government, military, or corporate information for a strategic advantage.
3. **Cyber Terrorism: These are cyberattacks meant to create fear, disrupt important** infrastructure, or achieve political goals.
4. **Cyber Warfare: This is cyberattacks sponsored by a country,** targeting another nation's critical infrastructure, defence systems, or important assets.

Common Types of Cyber Attacks

1. **Malware: This is harmful** software created to damage systems or steal information.
   - **Virus**
   - **Trojan Horse**
   - **Spyware**
   - **Adware**
   - **Botnets**
2. **Ransomware: This type of attack lock**s a victim's data and asks for payment to unlock it.
   - **Examples:** WannaCry, Petya
3. **Phishing: These are fake** emails, messages, or websites that try to steal passwords, bank details, or personal information.
4. **Distributed Denial of Service (DDoS): This attack flood**s servers with too much traffic, causing online services to stop working.
5. **SQL Injection: This attack takes advantage of weakness**es in databases to steal or change information.
6. **Man-in-the-Middle (MitM) Attack:** Intercepts communication between two parties to steal or modify information.
7. **AI-Powered Cyber Attacks: These use artificial intelligence to create phishing scams, deepfakes, malware, and trick people through** social engineering.

Need for Cyber Security

As more of India goes digital, the country faces a wider range of cyber threats.

1. **Protects Critical Infrastructure: Cybersecurity helps keep important sectors like** banking, power, defence, healthcare, transportation, and telecommunications safe.
2. **Secures Digital Economy: It b**uilds trust in digital payments, online shopping, fintech, and public services.
3. **Protects National Security- prevents** cyber espionage, cyber warfare, and attacks on strategic assets.
4. **Ensures Data Privacy-** Safeguards citizens' personal and financial information from unauthorised access.
5. **Supports Digital Governance: Cyber security allows government services to be delivered safely through digital platforms**.
6. **Prevents Financial Losses: It lowers the risk of losing money to** cyber fraud, ransomware, identity theft, and data breaches.

Applications of Cyber Security

1. Banking and Financial Services

Cybersecurity protects digital payments, ATMs, online banking, UPI transactions, and financial institutions from cyber fraud.

2. Healthcare

It keeps electronic health records, telemedicine platforms, and hospital information systems secure.

3. Defence and National Security

Cybersecurity protects military communication networks, intelligence systems, satellites, and important defence infrastructure.

4. Government Services

It keeps e-governance platforms, digital identities, tax systems, and public databases safe.

5. Cloud Computing

Cybersecurity protects cloud infrastructure, virtual machines, APIs, and stored data from cyber threats.

6. Internet of Things (IoT)

It keeps connected devices, industrial IoT systems, smart homes, and smart city infrastructure secure.

7. E-Commerce

Protects online transactions, customer information, and digital marketplaces.

8. Industrial Control Systems

It keeps manufacturing plants, power grids, oil refineries, and transportation systems safe from cyberattacks.

Cyber Security and India

India is one of the fastest-growing digital economies, so cybersecurity has become a top national priority.

1. **Supports Digital India:** Cyber security helps make digital governance and citizen services safe and reliable.
2. **Protects Digital Public Infrastructure (DPI: It keep**s platforms likes **Aadhaar, UPI, DigiLocker, and OND secure.C**.
3. **Strengthens National Security: Cyber security protects important sectors from cyber spying** and cyber warfare.
4. **Builds Digital Trust: It helps people feel confident using** digital banking, fintech, e-commerce, and cloud services.
5. **Promotes Economic Growth:** A secure cyber environment attracts investment and makes India's digital economy stronger.

Government Initiatives

1. **Information Technology (IT) Act, 2000-** India's primary legislation dealing with cybercrimes, electronic records, and cybersecurity.
2. **Digital Personal Data Protection (DPDP) Act, 2023-** Protects digital personal data while balancing innovation and privacy.
3. **Indian Computer Emergency Response Team (CERT-In)-** National nodal agency for responding to cybersecurity incidents and issuing security advisories.
4. **National Critical Information Infrastructure Protection Centre (NCIIPC)-** Protects Critical Information Infrastructure under the IT Act.
5. **Indian Cyber Crime Coordination Centre (I4C)-** Coordinates investigation, prevention, and capacity building related to cybercrime.
6. **National Cyber Coordination Centre (NCCC)-** Generates situational awareness and facilitates real-time threat monitoring.
7. **Cyber Swachhta Kendra** detects botnets and provides free malware removal tools.
8. **Defence Cyber Agency (DCA)-** Coordinates cyber defence operations for the Armed Forces.
9. **National Cyber Security Policy, 2013-** Provides the overall framework for strengthening India's cybersecurity ecosystem.
10. **National Cyber Crime Reporting Portal-** An online platform for reporting cybercrime complaints, including financial fraud.

Challenges of Cyber Security

1. **Rapidly Evolving Threats-** Cybercriminals continuously develop sophisticated attack techniques using AI and automation.
2. **Skill Shortage-** India faces a shortage of trained cybersecurity professionals.
3. **Increasing Ransomware Attacks-** Ransomware keeps targeting healthcare, government, banking, and important infrastructure.

4. **Cloud and IoT Vulnerabilities-** As cloud services and connected devices grow, there are more opportunities for cyber attacks.
5. **Cross-Border Cyber Attacks-** It is hard to identify and prosecute attackers because cyberspace is global.
6. **Weak Cyber Awareness-** Human error, phishing, and poor cyber hygiene remain major causes of cyber incidents.
7. **Regulatory Challenges-** Rapid technological advancements often outpace existing cybersecurity laws and standards.

Global Best Practices

1. **The United States strengthens** cyber resilience through the **Cybersecurity and Infrastructure Security Agency (CISA)** and Zero Trust Architecture.
2. **The European Union implements** the **NIS2 Directive** and the **Cyber Resilience Act** to improve cybersecurity across member states.
3. **Israel is recognised** as a global leader in cybersecurity innovation through strong government–industry collaboration.
4. **Singapore-** The **Cyber Security Agency (CSA)** promotes national cyber resilience through robust governance and public awareness.
5. **Estonia-** Developed one of the world's most secure digital governance ecosystems after the 2007 cyberattacks.

Way Forward

1. **Develop a National Cyber Security Strategy-** Update India's cybersecurity framework to address emerging threats such as AI, Quantum Computing, and IoT.
2. **Strengthen Critical Infrastructure Protection-** Make banking, energy, healthcare, transportation, and defence sectors more resilient to cyber threats.
3. **Build a Skilled Workforce-** Increase cybersecurity education, certifications, and specialised training programs.
4. **Promote Cyber Hygiene-** Teach people more about phishing, digital fraud, password safety, and staying safe online.
5. **Adopt Zero Trust Security-** Implement continuous authentication, least-privilege access, and multi-factor authentication across organisations.
6. **Enhance International Cooperation-** Work more closely with other countries on cybercrime investigations, sharing information, and building skills.
7. **Invest in Indigenous Cyber Technologies-** Support the development of homegrown cybersecurity software, encryption tools, AI-based threat detection, and secure hardware.

Conclusion

Cybersecurity is now a key part of the digital economy, national security, and governance. As India moves forward with Digital India, AI, cloud computing, IoT, and digital public infrastructure, keeping cyberspace safe and strong is crucial. By improving laws, building local skills, raising cyber awareness, and working with other countries, India can create a trusted digital world that supports **Atmanirbhar Bharat** and **Viksit Bharat 2047**.

11. DEEPFAKES

Deepfakes are synthetic media made by AI, such as images, videos, or audio, that make it look like people are saying or doing things they never did. These are created with **Artificial Intelligence** and **Deep Learning** to produce very realistic digital content.

The word **Deepfake** comes from combining **"Deep"** (from Deep Learning) and **"Fake" because it uses** advanced neural networks to make realistic but fake media. While deepfakes started in entertainment and creative fields, they are now often linked to misinformation, cybercrime, identity theft, election interference, and online abuse.

Evolution of Deepfakes

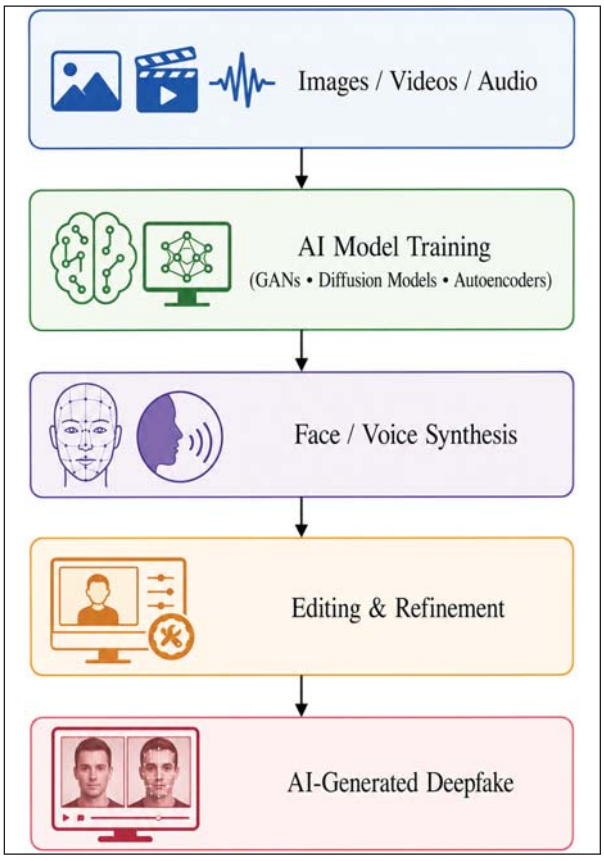
- **2014:** Generative Adversarial Networks (GANs) introduced by Ian Goodfellow, laying the foundation for realistic AI-generated content.
- **2017:** The term **"Deepfake"** emerged from a Reddit user who created AI-generated face-swapped videos.
- **From 2020 onwards, advance**s in Generative AI, diffusion models, and Large Language Models (LLMs) have made it much easier and more accessible to create deepfakes.
- Today deepfakes are used in areas like entertainment, education, healthcare, marketing, and cybersecurity. They are also used for harmful purposes, including fraud and spreading misinformation.

How Deepfakes Work

To create deepfakes, AI models are trained on large collections of images, videos, or voice recordings.

1. **Data Collection-** Gather many images, videos, and audio recordings of the person being targeted.
2. **AI Model Training-** Deep Learning models like GANs, Autoencoders, or Diffusion Models learn the person's facial expressions, voice, and movements.
3. **Content Generation-** The trained model creates realistic fake images, videos, or voices.
4. **Refinement-** AI improves realism by adjusting lighting, facial expressions, lip movements, and voice synchronisation.

Deepfake Generation Process



Features of Deepfakes

1. **Hyper-Realistic Content-** Deepfakes can create **images, videos, and voice recordings** that look and sound very real.
2. **AI-Driven-** Deepfakes use Deep Learning and Neural Networks to generate media.
3. **Face Swapping-** Replaces one person's face with another while preserving expressions.
4. **Voice Cloning-** Deepfakes can copy a person's speech, tone, and accent very accurately.
5. **Rapid Content Generation-** Today's AI tools can make deepfakes in just a few minutes.
6. **Difficult to Detect-** Advanced deepfakes are getting harder to spot without special detection tools.

Types of Deepfakes

1. **Image Deepfakes-** AI-generated or manipulated photographs.
2. **Video Deepfakes-** Videos showing people performing actions or making statements they never made.
3. **Audio Deepfakes-** AI-generated cloned voices used to imitate real individuals.
4. **Text Deepfakes-** AI-generated messages or conversations falsely attributed to individuals.

Positive Applications of Deepfakes

When used ethically, deepfake technology has several beneficial applications.

1. **Entertainment and Film-** It is used for visual effects, dubbing, making actors look younger, and creating digital characters.
2. **Education-** Deepfakes can bring historical figures to life and help create more engaging educational experiences.
3. **Healthcare-** Supports medical training, patient simulation, and AI-assisted communication tools.
4. **Accessibility-** Voice cloning assists individuals with speech disabilities and helps preserve voices affected by illness.
5. **Digital Preservation-** Deepfake technology can restore damaged historical photos, videos, and cultural heritage materials.
6. **Marketing and Media-** It is used to make personalised ads, multilingual campaigns, and virtual brand ambassadors.

Threats and Challenges of Deepfakes

Even though deepfakes have benefits, they also bring serious risks to society, democracy, and cybersecurity.

1. **Misinformation and Disinformation-** Deepfakes can spread false news and influence public opinion, especially during elections or crises.
2. **Financial Fraud-** AI-generated voice cloning is now often used for banking fraud, impersonation, and financial scams.
3. **Identity Theft-** Cybercriminals can pretend to be someone else by using fake voices and videos.
4. **Deepfake Pornography-** Most deepfake videos online are non-consensual and explicit, often targeting women and violating their privacy and dignity.
5. **Cybersecurity Threats-** Deepfakes make it easier for criminals to carry out phishing, social engineering, stealing credentials, and business email scams.
6. **National Security Risks-** Deepfakes can be used to spread propaganda, cause diplomatic problems, or interfere with military communications.
7. **Erosion of Public Trust-** Because deepfakes are so common, it is getting harder to tell real media from fake, which makes people trust digital information less.

Deepfakes and India

As India's digital world grows quickly, the country is especially at risk from the misuse of deepfakes.

1. **Threat to Electoral Integrity-** Deepfakes can affect elections by spreading false information and fake political content.

2. **Rise in Digital Fraud-** AI-generated voice cloning and impersonation are leading to more online financial scams.
3. **Risks to Women's Safety-** Deepfake content made without consent is adding to online harassment and gender-based violence.
4. **Challenges for Law Enforcement-** Detecting and investigating AI-generated media requires advanced forensic skills.
5. **Need for Digital Literacy-** People need to be aware so they can help stop the spread of fake or manipulated content.

Government Initiatives

1. **Information Technology (IT) Act 2000-** This law includes rules against impersonation, identity theft, and cyber fraud.
2. **Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021-** These rules require social media companies to remove illegal or impersonating content within set time limits.
3. **Digital Personal Data Protection (DPDP) Act, 2023-** This Act gives stronger protection for personal digital data and privacy.
4. **CERT-In-** This agency gives warnings about AI-related cyber threats and digital fraud.
5. **MeitY Advisories on Deepfakes-** These advisories tell social media platforms to quickly find, label, and remove harmful AI-generated content.

Global Best Practices

1. **European Union-** The **AI Act** states that AI-generated content, including deepfakes, must be clearly identified.
2. **China-** The law requires clear labels on AI-generated media and asks users to verify their identity.
3. **United States-** Some states have passed laws to address harmful deepfakes related to elections and those made without consent.
4. **UNESCO-** UNESCO supports ethical AI with its guidelines, focusing on transparency, accountability, and human rights.
5. **International Telecommunication Union (ITU)-** The ITU creates worldwide standards for watermarking, content authentication, and tracking digital origins.

Challenges in Regulating Deepfakes

1. **Rapid Technological Evolution-** AI models are advancing faster than the rules meant to control them.
2. **Detection Difficulties-** New deepfakes are getting better at avoiding current detection tools.
3. **Cross-Border Jurisdiction-** Deepfakes made in one country can spread around the world very quickly.
4. **Balancing Innovation and Regulation-** Too many rules could slow down useful AI developments.
5. **Platform Accountability-** Social media companies find it hard to manage large amounts of AI-generated content.

Way Forward

1. **Strengthen Legal Frameworks-** Create strong laws that directly address AI-generated media and harmful deepfakes.
2. **Promote AI Watermarking-** Require digital watermarks and tracking for AI-generated media.
3. **Develop Deepfake Detection Tools- S**upport the creation of AI tools that can spot fake or changed content.
4. **Enhance Platform Accountability-** Make social media companies quickly find, label, and remove harmful deepfake content.
5. **Improve Digital Literacy-** Teach people to check information before sharing and to spot AI-generated misinformation.

6. **Encourage Responsible AI Development-** Use safety-first ideas, transparency, and ethical standards at every stage of AI development.
7. **Strengthen International Cooperation-** Support worldwide teamwork on AI rules, standards, and stopping deepfake abuse across borders.

Conclusion

Deepfakes are a powerful and disruptive use of Generative Artificial Intelligence. They bring significant opportunities in areas such as entertainment, education, healthcare, and communication, but also pose serious risks to privacy, cybersecurity, democracy, and public trust. To get the benefits of deepfake technology and reduce its harms, we need a balanced approach that includes new technologies, strong regulations, ethical AI, digital education, and international collaboration.

12. EXTENDED REALITY (AR/VR/MR)

Extended Reality (XR) is an umbrella term that encompasses **Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)**. It combines the physical and digital worlds to create immersive, interactive, and intelligent experiences using devices such as **VR headsets, AR glasses, smartphones, and smart wearables**.

XR is considered a key enabling technology of the **Fourth Industrial Revolution (Industry 4.0)**, with applications across **gaming, healthcare, education, manufacturing, defence, retail, tourism, and smart cities**.

Evolution of Extended Reality

- **1968:** Ivan Sutherland developed the first head-mounted display (HMD).
- **1990:** The term **Virtual Reality (VR)** gained popularity.
- **1992:** The term **Augmented Reality (AR)** was coined by Tom Caudell.
- **2016:** Pokémon GO popularised AR for mass consumers.
- **2020 onwards:** Growth of Metaverse, AI, 5G, Digital Twins, and Spatial Computing accelerated XR adoption.
- **Present:** XR is transforming industries through immersive learning, simulation, remote collaboration, and digital interaction.

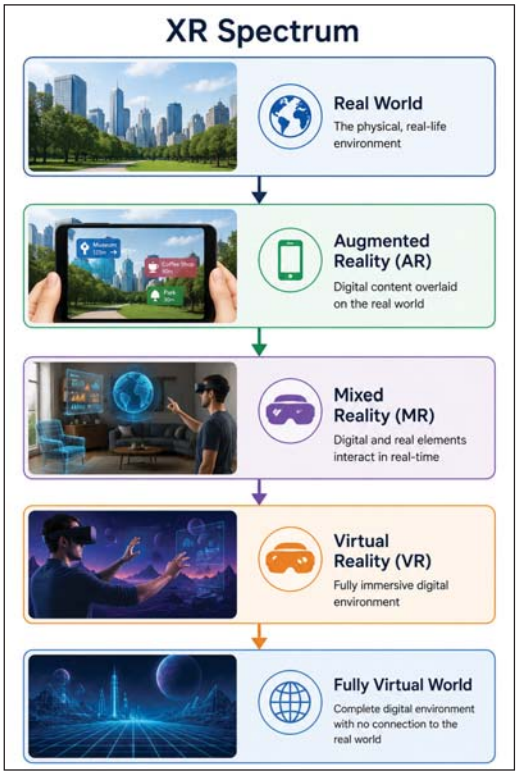
Components of Extended Reality

Extended Reality consists of three core technologies.

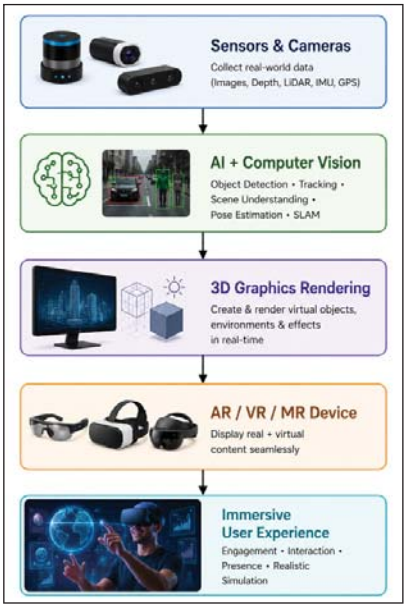
1. Virtual Reality (VR)	2. Augmented Reality (AR)	3. Mixed Reality (MR)
Virtual Reality creates a **fully computer-generated environment** that completely immerses users, replacing the real world. Users interact through **VR headsets**, controllers, motion sensors, and haptic devices.	Augmented Reality overlays **digital information** such as images, text, videos, or 3D models onto the real world without replacing it. Users remain aware of their physical surroundings while interacting with digital content.	Mixed Reality combines the real and virtual worlds, allowing digital objects to interact with the physical environment in real time. Unlike AR, virtual objects remain anchored to the real environment and respond to user interactions. **Examples** • Microsoft HoloLens • Magic Leap

Examples	Examples	
• Meta Quest • PlayStation VR • Flight simulators • Military training	• Pokémon GO • Snapchat filters • Google Maps Live View • IKEA Place	

XR Spectrum



Working of Extended Reality



Features of Extended Reality

1. **Immersive Experience:** Creates realistic digital environments that help users feel more engaged.
2. **Real-Time Interaction:** Lets users interact with digital objects right away.
3. **Spatial Computing:** Maps and understands the real world to place digital objects accurately.
4. **3D Visualisation:** Shows realistic 3D objects and environments.
5. **Multi-Sensory Experience:** Combines visuals, sound, movement, and touch feedback.
6. **AI Integration:** Uses Artificial Intelligence to recognise objects, track gestures, and make interactions feel natural.
7. **Remote Collaboration:** Makes it possible to hold virtual meetings, work together on designs, and team up in immersive ways.

Applications of Extended Reality

Extended Reality is transforming many industries by making it easier to visualise, simulate, and collaborate in new ways.

1. **Gaming and Entertainment:** XR provides immersive gaming, virtual concerts, interactive storytelling, and cinematic experiences.
2. **Healthcare:** Supports surgical simulation, medical training, rehabilitation, mental health therapy, and patient education.
3. **Education and Training:** Creates virtual laboratories, interactive classrooms, and skill-based simulations for immersive learning.
4. **Manufacturing:** Helps with product design, digital prototypes, predicting maintenance needs, training workers, and checking quality.
5. **Defence and Military:** Used for combat simulation, battlefield visualisation, mission planning, and pilot training.
6. **Retail and e-commerce:** Enables virtual shopping, product visualisation, virtual fitting rooms, and AR-based product demonstrations.
7. **Real Estate:** Allows virtual property tours, interior design visualisation, and remote property inspection.
8. **Tourism and Heritage:** Provides virtual museum tours, digital heritage preservation, and immersive travel experiences.
9. **Sports:** Supports athlete training, performance analysis, fan engagement, and virtual stadium experiences.
10. **Remote Work:** Facilitates virtual offices, immersive meetings, collaborative design, and remote technical assistance.

Advantages of Extended Reality

1. **Improves Learning:** Makes education and job training more interactive and effective.
2. **Enhances Productivity:** Helps people work together better, design more efficiently, and improve how things get done in industry.
3. **Reduces Costs:** Cuts down on travel, the need for physical prototypes, and training costs.
4. **Improves Safety:** Lets people practice dangerous tasks in a safe, simulated environment.
5. **Better Customer Experience:** Offers more personalised and immersive ways to shop and enjoy entertainment.
6. **Accelerates Innovation:** Makes it easier to quickly test new ideas and try out digital experiments.

Challenges of Extended Reality

1. **High Cost:** XR devices and the equipment needed to run them are still expensive.
2. **Hardware Limitations:** Headsets that are both light and powerful are still being developed.
3. **Privacy Concerns:** XR devices collect sensitive data like your biometrics, location, and behaviour.

4. **Cybersecurity Risks:** Immersive environments can make it easier for hackers to steal information or identities.
5. **Motion Sickness:** Some people feel uncomfortable or dizzy if they use XR for a long time.
6. **Digital Divide:** High costs and poor internet access make it hard for people in some areas to use XR.
7. **Ethical Concerns:** Problems like digital identity, addiction, misinformation, and online harassment need rules and oversight.

Extended Reality and India

India is becoming an important market for XR because of its growing digital economy, active startup scene, and expanding 5G networks.

1. **Supports Digital India-** Enhances digital education, governance, and citizen services.
2. **Boosts Skill Development-** Enables immersive vocational and industrial training.
3. **Strengthens Healthcare-** Supports telemedicine, medical simulation, and remote consultation.
4. **Accelerates Industry 4.0-** Improves manufacturing through Digital Twins, remote maintenance, and smart factories.
5. **Promotes Startup Innovation-** Encourages development of XR applications in education, gaming, healthcare, and retail.

Government Initiatives

1. **Digital India Mission-** Expands digital infrastructure that supports XR adoption.
2. **IndiaAI Mission-** Promotes AI technologies that enhance XR through computer vision and intelligent interaction.
3. **National Education Policy (NEP) 2020-** Encourages immersive digital learning using emerging technologies.
4. **National Digital Health Mission (ABDM)-** Supports digital healthcare applications that can integrate XR technologies.
5. **5G Rollout-** High-speed, low-latency networks enable real-time XR applications.
6. **Startup India-** Supports startups working in AR, VR, MR, gaming, and immersive technologies.

Global Best Practices

1. **United States-** Leads XR innovation through companies such as Meta, Apple, Microsoft, and Google.
2. **South Korea-** Integrates XR into smart manufacturing, education, and the Metaverse ecosystem.
3. **Japan-** Uses XR extensively in industrial automation, robotics, healthcare, and entertainment.
4. **China-** Invests heavily in XR for manufacturing, education, smart cities, and e-commerce.
5. **European Union-** Promotes XR research through Horizon Europe and Digital Europe programmes.

Way Forward

1. **Develop Affordable XR Devices-** Reduce hardware costs to improve accessibility.
2. **Expand Digital Infrastructure-** Strengthen 5G, cloud computing, and edge computing for seamless XR experiences.
3. **Promote Indigenous Innovation-** Support Indian startups and research institutions developing XR technologies.
4. **Build Skilled Workforce-** Introduce XR-focused education and professional training programmes.
5. **Strengthen Privacy and Cybersecurity-** Develop robust standards for biometric data protection and secure immersive environments.
6. **Encourage Industry Adoption-** Promote XR integration across manufacturing, healthcare, education, defence, and governance.
7. **Develop Ethical Frameworks-** Establish guidelines for responsible use, digital identity protection, and prevention of misuse.

Conclusion

Extended Reality (XR) is changing how people interact by blending the physical and digital worlds using **Augmented Reality, Virtual Reality, and Mixed Reality**. XR can help transform sectors such as education, healthcare, manufacturing, government, and entertainment. If India builds strong digital infrastructure, supports innovation, keeps data secure, and encourages responsible use, XR could become a key part of **Digital India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**.

13. FACIAL RECOGNITION TECHNOLOGY

Facial Recognition Technology (FRT) is an **Artificial Intelligence (AI)-based biometric technology** that identifies or verifies an individual's identity by analysing unique facial features from images, videos, or live camera feeds. It combines **Computer Vision, Machine Learning, and Deep Learning** to extract facial patterns and compare them with stored biometric data for authentication.

Unlike traditional authentication methods such as passwords or PINs, facial recognition uses a person's **facial biometrics**, making identity verification faster, contactless, and more secure. It is widely used in **smartphones, banking, airports, law enforcement, healthcare, surveillance, and digital governance**.

Evolution of Facial Recognition

- **1960s:** Early facial measurement techniques were developed manually.
- **1990s:** Eigenfaces and machine learning improved automated face recognition.
- **2010 onwards:** Deep Learning and Convolutional Neural Networks (CNNs) significantly increased recognition accuracy.
- **Present:** AI-powered facial recognition enables real-time authentication using **FaceNet, ArcFace, DeepFace, and Transformer-based models**.

Components of a Facial Recognition System

A facial recognition system consists of several interconnected stages.

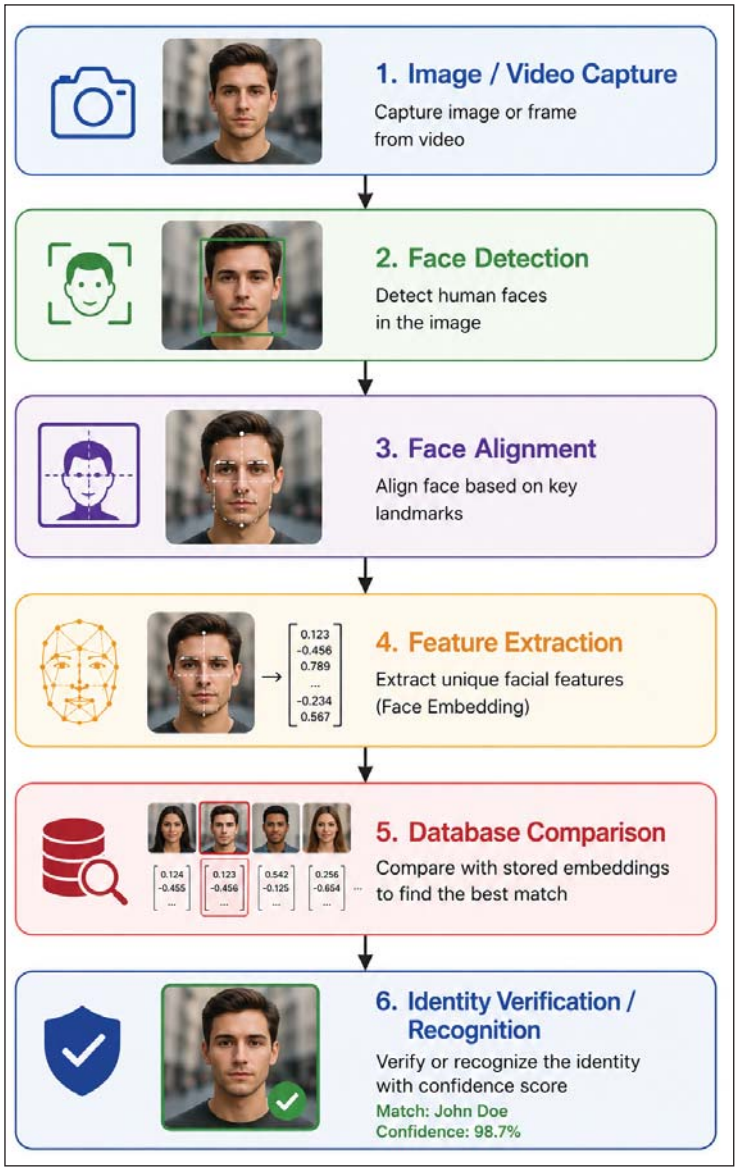
1. **Image Acquisition-** Captures facial images through cameras, smartphones, CCTV, or video streams.
2. **Face Detection-** AI algorithms identify and locate human faces within an image or video.
3. **Face Alignment-** Facial landmarks (eyes, nose, mouth) are aligned to compensate for pose, lighting, and facial expressions.
4. **Feature Extraction-** Deep learning models extract unique facial characteristics and convert them into numerical representations called face embeddings.
5. **Database Storage-** Face embeddings are securely stored in a biometric database.
6. **Face Matching-** The captured embedding is compared with stored embeddings using similarity algorithms.
7. **Decision Making-** The system either verifies (1:1) or identifies (1:N) the individual's identity.

Working of Facial Recognition

Features of Facial Recognition

1. **Contactless Authentication enables** identity verification without physical contact.
2. **Biometric Identification uses** unique facial characteristics that are difficult to replicate.
3. **AI-Based Recognition-** Deep learning continuously improves recognition accuracy.
4. **Real-Time Processing-** Identifies individuals from live video feeds within seconds.
5. **High Scalability-** Can compare faces against millions of stored identities.
6. **Multi-Factor Authentication-** Often used alongside passwords, OTPs, or fingerprints for enhanced security.

7. **Adaptive Learning-** Modern AI models recognise faces despite changes in lighting, pose, ageing, or facial expressions.



Face Detection vs Face Recognition

Face Detection	Face Recognition
Detects the presence of a face	Identifies or verifies whose face it is
Finds face location	Matches facial identity
No identity database required	Requires stored facial database
Used in camera autofocus and filters	Used for authentication and surveillance

Applications of Facial Recognition

Facial recognition technology is now widely used to securely verify identities in many different fields.

1. **Smartphone Authentication:** Facial recognition lets people unlock smartphones, digital wallets, and apps securely.

2. **Banking and Financial Services:** It helps with e-KYC, onboarding customers, preventing fraud, making digital payments safer, and verifying accounts.
3. **Law Enforcement:** It helps identify criminals, find missing people, support forensic investigations, and monitor CCTV footage.
4. **Airports and Border Security:** Facial recognition speeds up immigration, lets passengers use e-gates, verifies travelers, and makes airport check-in faster.
5. **Healthcare:** It helps identify patients, control access to hospitals, and keep medical records secure.
6. **Attendance Management:** Facial recognition automates attendance in schools, universities, offices, and factories.
7. **Smart Cities:** It improves public safety by helping with surveillance, managing crowds, and monitoring traffic.
8. **Retail:** Facial recognition helps stores offer personalised experiences, run loyalty programs, and make payments more secure.
9. **Defence and National Security:** It helps secure military sites, supports intelligence work, and improves border surveillance.
10. **Digital Governance:** Facial recognition makes it safer to deliver government services using biometric authentication.

Advantages of Facial Recognition

1. **Enhanced Security:** It uses strong biometric checks to help prevent identity fraud.
2. **Contactless Verification:** People can verify their identity in a clean and easy way, without touching anything.
3. **Faster Authentication:** It saves time compared to using passwords or checking IDs by hand.
4. **Improved User Experience:** People can access devices and digital services more smoothly.
5. **Operational Efficiency:** It automates identity checks in banks, airports, and public services.
6. **Supports Digital Governance:** It helps make access to government services more secure.

Challenges of Facial Recognition

1. **Privacy Concerns:** Collecting and storing facial data can lead to worries about surveillance and misuse.
2. **Algorithmic Bias:** If AI is trained on biased data, it may not work as well for people of different ages, genders, or backgrounds.
3. **Cybersecurity Risks:** Hackers may target facial databases, which could lead to identity theft.
4. **Deepfake Threats:** Fake faces and videos made by AI can trick facial recognition systems if there are no strong protections.
5. **Accuracy Issues:** Facial recognition may not work as well if someone is wearing a mask, the lighting is bad, or their face changes with age or expression.
6. **High Implementation Cost:** Setting up facial recognition on a large scale needs advanced cameras, strong AI systems, and secure databases.
7. **Ethical and Legal Concerns:** Too much surveillance can threaten people's rights and privacy.

Facial Recognition and India

India is increasingly using facial recognition technology in government, security, and digital services.

1. **Strengthens Law Enforcement:** It helps police investigate crimes and find missing people.
2. **Improves Digital Governance:** It helps make citizen services more secure through better authentication.
3. **Enhances Border Security:** It allows for biometric checks at airports and immigration points.
4. **Promotes Digital Payments:** It makes financial services safer by using biometric authentication.
5. **Supports Smart Cities-** Improves surveillance, public safety, and urban security management.

Government Initiatives

1. **Digi Yatra uses facial recognition technology** to enable paperless and contactless passenger processing at Indian airports.
2. **Crime and Criminal Tracking Network & Systems (CCTNS)-** Integrates digital databases to support criminal identification and investigation.
3. **National Automated Facial Recognition System (NAFRS):** This proposed project aims to help law enforcement find criminals and missing people.
4. **Digital Personal Data Protection (DPDP) Act, 2023-** Regulates the processing of personal digital data, including biometric information.
5. **Information Technology (IT) Act, 2000-** Provides the legal framework for electronic records, cyber offences, and digital security.

Global Best Practices

1. **United States-** Uses facial recognition for airport security, law enforcement, and smartphone authentication while several states regulate its public use.
2. **European Union:** The AI Act treats facial recognition as high-risk and sets strict rules to protect people.
3. **China:** Facial recognition is widely used in smart cities, transportation, and public safety.
4. **Singapore:** Facial recognition is part of Singpass and Smart Nation programs to make digital services more secure.
5. **Japan:** Facial recognition is used in transport, shops, healthcare, and big sports events.

Way Forward

1. **Strengthen Data Privacy-** Ensure facial biometric data is collected, stored, and processed with informed consent and strong safeguards.
2. **Reduce Algorithmic Bias:** Train AI with a wide range of data to make it fairer and more accurate.
3. **Develop Strong Anti-Spoofing: Improve systems to spot fake face**s, masks, deepfakes, and synthetic identities.
4. **Enhance Regulatory Frameworks-** Establish clear standards for ethical deployment, accountability, and oversight.
5. **Improve Cybersecurity: Protect** biometric databases with encryption, access controls, and regular security checks.
6. **Promote Responsible AI-** Ensure transparency, explainability, and human oversight in facial recognition systems.
7. **Encourage Indigenous Innovation-** Support Indian research and startups developing secure AI-based biometric technologies.

Conclusion

Facial recognition technology is a big step forward in AI, computer vision, and biometrics. It helps securely and easily verify identities across many areas. While it brings significant benefits to security, government, healthcare, and digital services, it is essential to balance its use with robust protections for privacy, fairness, and transparency. By improving laws, making AI more accurate, and using the technology responsibly, India can use facial recognition to support Digital India, Atmanirbhar Bharat, and Viksit Bharat 2047.

14. SATELLITE INTERNET

- **Satellite Internet** is a **wireless broadband technology** that provides internet connectivity via communication satellites orbiting Earth, rather than relying on terrestrial infrastructure such as fibre-optic cables, mobile towers, or DSL networks. It enables high-speed internet access in remote, rural, mountainous, island, maritime, and disaster-affected regions where conventional networks are unavailable or uneconomical.

- Modern satellite internet primarily uses **Low Earth Orbit (LEO)** constellations, which provide significantly lower latency and higher speeds than traditional Geostationary Earth Orbit (GEO) satellites. Companies such as **Starlink (SpaceX), OneWeb (Eutelsat), Project Kuiper (Amazon)** and Indian telecom operators like **Jio Satellite Communications** and **Bharti-backed OneWeb** are driving the next generation of satellite broadband.

Evolution of Satellite Internet

- **1965:** First commercial communication satellite (Intelsat I) launched.
- **1990s:** GEO satellites began offering commercial satellite internet.
- **2019:** SpaceX launched Starlink's first LEO satellites.
- **2020 onwards:** Rapid deployment of LEO mega-constellations enabled low-latency broadband.
- **Present:** India is transitioning towards LEO- and MEO-based satellite internet through regulatory reforms and private sector participation.

Components of Satellite Internet

A satellite internet network consists of three major segments.

1. Space Segment: Communication satellites in GEO, MEO, or LEO orbits receive and transmit internet signals.	**2. Ground Segment-** Gateway stations connect satellite networks with terrestrial fibre-optic internet infrastructure.	**3. User Segment-** User terminals (dish antennas, satellite modems, or future direct-to-device smartphones) receive internet signals from satellites.

Working of Satellite Internet

Features of Satellite Internet

1. **Global Coverage-** Provides internet connectivity almost anywhere on Earth.
2. **Wireless Broadband-** Eliminates dependence on underground fibre or mobile towers.
3. **Low Latency (LEO)-** Modern LEO satellites offer latency of around **20–50 milliseconds**, suitable for video calls and gaming.
4. **Rapid Deployment-** Internet access can be established quickly without extensive ground infrastructure.
5. **Disaster Resilience-** Continues functioning even when terrestrial communication networks fail.
6. **Mobility- supports** internet access on ships, aircraft, defence vehicles, and remote expeditions.

Types of Satellite Orbits

Orbit	Altitude	Advantages	Limitations	Applications
Geostationary Earth Orbit (GEO)	35,786 km	Extensive coverage, fewer satellites	High latency	TV broadcasting, weather, disaster alerts
Medium Earth Orbit (MEO)	2,000–35,786 km	Moderate latency and wider coverage	Requires multiple satellites	Navigation, aviation, maritime internet
Low Earth Orbit (LEO)	Below 2,000 km	Low latency, high speed	Requires thousands of satellites	Broadband internet, IoT, defence, telemedicine

Applications of Satellite Internet

Satellite internet is transforming connectivity across multiple sectors.

1. **Rural Connectivity:** Provides broadband access to villages, tribal areas, border regions, deserts, and islands.
2. **Disaster Management:** Maintains communication during floods, earthquakes, cyclones, and other disasters when terrestrial infrastructure is damaged.
3. **Defence and National Security:** Supports secure military communication, surveillance, border monitoring, and operations in remote locations.
4. **Healthcare:** Enables telemedicine, remote diagnostics, and emergency healthcare services in underserved regions.
5. **Education:** Supports online education and digital classrooms in remote and rural schools.
6. **Agriculture:** Facilitates precision farming, IoT-based irrigation, weather monitoring, and market access.
7. **Maritime and Aviation:** Provides uninterrupted internet connectivity on ships, aircraft, and offshore platforms.
8. **Internet of Things (IoT):** Connects remote sensors used in mining, oil exploration, logistics, and smart agriculture.
9. **Smart Cities:** Support disaster monitoring, emergency communication, and intelligent infrastructure.
10. **Digital Economy:** Strengthens e-commerce, fintech, digital governance, and remote working.

Advantages of Satellite Internet

1. **Bridges the Digital Divide-** Connects remote and underserved communities.
2. **Universal Coverage-** Delivers the internet irrespective of terrain or geography.
3. **Disaster Resilience-** Provides reliable communication during emergencies.
4. **Rapid Expansion-** Scales connectivity without laying fibre networks.
5. **Strategic Independence-** Enhances national communication resilience.

6. **Supports Emerging Technologies-** Enables IoT, autonomous systems, AI, and connected devices.

Challenges of Satellite Internet

1. **High Initial Cost-** User terminals and subscription costs remain expensive.
2. **Weather Interference-** Rain, snow, and storms may reduce signal quality.
3. **Limited Bandwidth-** Heavy network congestion may reduce speeds.
4. **Space Debris-** Large satellite constellations increase collision risks.
5. **Regulatory Challenges-** Spectrum allocation and licensing require coordinated policy frameworks.
6. **Security Concerns-** Satellite communications may be misused for unlawful activities if not properly monitored.
7. **Environmental Concerns-** Satellite launches and atmospheric burn-up contribute to space debris and environmental impacts.

Satellite Internet and India

India is rapidly expanding satellite communication to strengthen digital inclusion and strategic connectivity.

1. **Bridges Rural Connectivity- provides** broadband access to remote villages, islands, Himalayan regions, and border areas.
2. **Supports Digital India- complements** BharatNet and terrestrial broadband infrastructure.
3. **Enhances Disaster Preparedness- ensures** uninterrupted communication during natural disasters.
4. **Strengthens National Security- supports** defence communication in strategically important regions.
5. **Boosts Space Economy-** Encourages private participation and innovation in satellite communications.

Government Initiatives

1. **Indian Space Policy, 2023-** Enables greater private sector participation across the space value chain.
2. **IN-SPACe-** Promotes and authorises private participation in satellite communication and space activities.
3. **New Space India Limited (NSIL)-** Operates communication satellites and commercial satellite services.
4. **Telecommunications Act, 2023-** Provides the legal framework for assigning spectrum and regulating satellite communication.
5. **TRAI Recommendations (2025)-** Recommended a five-year satellite spectrum assignment with extension based on market conditions.
6. **National Broadband Mission 2.0-** Expands broadband connectivity, particularly in rural and underserved regions.
7. **BharatNet Project:** Integrates satellite connectivity with fibre infrastructure to improve rural broadband.
8. **Digital Bharat Nidhi (DBN)** Funds broadband expansion in remote and difficult areas.

Major Global Satellite Internet Projects

Project	Organisation	Key Feature
Starlink	SpaceX	World's largest LEO satellite broadband network
OneWeb	Eutelsat Group	Enterprise and government broadband services
Project Kuiper	Amazon	Affordable global LEO broadband constellation
Qianfan Constellation	China	National LEO satellite internet network

Way Forward

1. **Reduce User Costs-** Provide subsidies and public-private partnerships to make user terminals affordable.
2. **Strengthen Indigenous Capacity-** Expand India's communication satellite manufacturing and launch capabilities.
3. **Improve Regulatory Framework- balance** innovation with national security, spectrum management, and data governance.
4. **Promote Hybrid Networks- integrate** satellite internet with fibre, 5G, and mobile networks for seamless connectivity.
5. **Strengthen Space Sustainability- develop** effective debris mitigation, satellite tracking, and responsible space traffic management.
6. **Expand Rural Connectivity-** Prioritise satellite broadband for remote villages, islands, border regions, and disaster-prone areas.
7. **Encourage Private Investment-** Foster competition, innovation, and indigenous satellite communication startups.

Conclusion

Satellite Internet is emerging as a transformative technology capable of delivering **universal, resilient, and high-speed broadband connectivity** beyond the limitations of terrestrial infrastructure. For India, it offers a unique opportunity to bridge the digital divide, strengthen disaster resilience, improve national security, and accelerate the goals of **Digital India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**. With supportive regulations, indigenous innovation, and affordable access, satellite internet can become a cornerstone of India's future digital infrastructure.

15. BHARAT 6G

- The Bharat 6G Vision is India's national roadmap to become a global leader in the design, development, standardisation, and deployment of Sixth Generation (6G) wireless communication technology by 2030. Launched by the Government of India in March 2023, the vision is built on the principles of Affordability, Sustainability, and Ubiquity, aiming to develop secure, intelligent, and energy-efficient communication networks that support India's digital transformation.
- The vision seeks to position India not merely as a consumer of 6G technology but as a major innovator, patent holder, standards contributor, and exporter of telecom technologies, contributing significantly to Digital India, Atmanirbhar Bharat, and Viksit Bharat 2047.

6G Technology

- 6G is the sixth generation of wireless communication technology, succeeding 5G. It is expected to deliver ultra-high data speeds (up to 1 Tbps), latency below 100 microseconds, AI-native intelligent networks, integrated sensing, and seamless connectivity between humans, machines, and the physical world.
- The International Telecommunication Union (ITU) has designated 6G as IMT-2030, with commercial deployment expected around 2030.

Objectives of Bharat 6G Vision

1. **Global Leadership-** Position India among the world's leading developers of 6G technology.
2. **Indigenous Innovation**- Promote domestic R&D, intellectual property (IP), patents, and telecom manufacturing.
3. **Digital Inclusion-** Provide affordable and ubiquitous high-speed connectivity across urban and rural India.

4. **Sustainable Networks-** Develop energy-efficient and environmentally sustainable communication systems.
5. **International Standardisation- I**ncrease India's contribution to global telecom standards through ITU and other standards bodies.

Guiding Principles

1. **Affordability-** Ensure cost-effective communication technologies accessible to all citizens.
2. **Sustainability**- Develop green, energy-efficient, and environmentally responsible telecom infrastructure.
3. **Ubiquity-** deliver seamless connectivity across villages, cities, mountains, islands, oceans, and remote regions.

Key Features of Bharat 6G Vision

1. **Nine-Year Mission (2022–2031)-** A phased roadmap covering research, prototypes, field trials, and commercial deployment.
2. **Bharat 6G Alliance** brings together industry, academia, startups, research institutions, and government agencies to develop indigenous 6G technologies.
3. **Advanced Research Infrastructure-** Establishment of 6G Terahertz (THz) Testbed and Advanced Optical Communication Testbed for domestic innovation.
4. **Academic Capacity Building**- Establishment of 100 5G laboratories across educational institutions to create a 6G-ready research ecosystem.
5. **Global Standards Participation**- Active contribution to ITU's IMT-2030 framework and international telecom standardisation.

Six Pillars of Bharat 6G Vision

Emerging Technologies in 6G

The Bharat 6G Vision focuses on several advanced technologies.

1. **Artificial Intelligence (AI) & Machine Learning (ML)**- AI-native networks for intelligent resource allocation and self-optimising communication.

2. **Terahertz (THz) Communicatio**n- ultra-high-frequency spectrum enabling extremely high data rates.
3. **Intelligent Reflecting Surfaces (IRS**)- Smart surfaces that improve wireless signal propagation.
4. **Space-Terrestrial Integration:** Integration of satellites, High-Altitude Platform Systems (HAPS), and terrestrial networks.
5. **Tactile Internet enables** ultra-low latency for remote surgery, industrial automation, and robotics.
6. **Edge Computing** processes data closer to users for faster response times.
7. **Optical Networks**- High-capacity fibre-based backbone supporting future communication demands.

Applications of 6G

1. **Smart Cities:** Support intelligent traffic management, digital governance, and connected infrastructure.
2. **Healthcare:** Enables remote robotic surgery, AI-assisted diagnosis, and immersive telemedicine.
3. **Industry 5.0:** Supports intelligent manufacturing, Digital Twins, robotics, and autonomous factories.
4. **Agriculture:** Facilitates precision farming, AI-enabled crop monitoring, and smart irrigation.
5. **Education:** Enables immersive learning using Virtual Reality (VR), Augmented Reality (AR), and holographic classrooms.
6. **Transportation:** Supports autonomous vehicles, intelligent transport systems, and drone communication.
7. **Defence:** Strengthens secure military communication, surveillance, and AI-enabled battlefield management.
8. **Metaverse and Extended Reality:** Enables immersive virtual environments with ultra-low latency communication.

Significance of Bharat 6G Vision

1. **Strengthens Digital India-** Accelerates India's digital transformation through next-generation communication.
2. **Promotes Atmanirbhar Bharat-** Encourages indigenous telecom equipment manufacturing and innovation.
3. **Enhances Global Competitiveness-** Positions India as a global telecom technology exporter.
4. **Generates Employment-** Creates opportunities in research, semiconductor manufacturing, AI, electronics, and telecom industries.
5. **Reduces Digital Divide-** Provides high-speed connectivity to remote and underserved regions.
6. **Supports Sustainable Development Goals (SDGs)-** Enables inclusive digital access, smart infrastructure, and sustainable economic growth.

Government Initiatives

1. **Bharat 6G Vision (2023)-** National roadmap for 6G leadership by 2030.
2. **Bharat 6G Alliance-** National collaboration platform for 6G research and deployment.
3. **100 5G Labs-** Established across academic institutions to develop future telecom talent.
4. **IndiaAI Mission-** Supports AI technologies that will power intelligent 6G networks.
5. **Semiconductor Mission-** Promotes indigenous semiconductor manufacturing essential for 6G devices.
6. **Telecom Technology Development Fund (TTDF)**- Supports research and innovation in indigenous telecom technologies.
7. **Production Linked Incentive (PLI) Scheme-** Encourages domestic manufacturing of telecom equipment.

Challenges

1. **High R&D Investment-** Developing indigenous 6G technologies requires substantial long-term investment.

2. **Spectrum Availability-** Efficient allocation of new spectrum bands, including Terahertz frequencies, remains challenging.
3. **Semiconductor Dependence-** India still relies heavily on imported semiconductor chips.
4. **Skilled Workforce** requires large-scale capacity building in AI, telecom engineering, semiconductor design, and quantum technologies.
5. **Global Competition-** Countries such as the US, China, Japan, South Korea, and the EU are investing aggressively in 6G research.
6. **Standardisation Challenges-** Active participation in international standard-setting bodies is essential.

Way Forward

1. **Strengthen Indigenous Research-** Increase investment in telecom R&D, AI, semiconductors, and advanced communication technologies.
2. **Build Global Intellectual Property-** Encourage Indian patents and contributions to international telecom standards.
3. **Expand Industry**–Academia Collaboration- Strengthen partnerships among universities, startups, research institutions, and industry.
4. **Develop Skilled Human Capital-** Promote specialised education in wireless communication, AI, electronics, and semiconductor engineering.
5. **Strengthen Manufacturing Ecosystem-** Expand domestic production of telecom equipment, chips, and communication devices.
6. **Enhance International Cooperation-** Collaborate with global partners while protecting India's strategic interests.
7. **Promote Green 6G Networks-** Develop energy-efficient communication infrastructure with low carbon emissions.

Conclusion

The Bharat 6G Vision represents India's ambition to move from being a telecom technology adopter to becoming a global innovation leader in next-generation wireless communications. By integrating Artificial Intelligence, Terahertz communication, satellite networks, edge computing, and intelligent automation, 6G has the potential to revolutionise healthcare, education, manufacturing, governance, and digital services. Through sustained investment in research, indigenous innovation, global standardisation, and skilled human capital, Bharat 6G Vision can become a key pillar of Digital India, Atmanirbhar Bharat, and Viksit Bharat 2047.

16. 3D PRINTING AND 4D PRINTING

- **3D Printing (Three-Dimensional Printing)**, also known as **Additive Manufacturing**, is a manufacturing process that creates three-dimensional objects by depositing material layer by layer based on a digital design (CAD model). It enables the production of complex, customised, and lightweight components with minimal material waste.
- **4D Printing** is an advanced form of 3D printing that uses **smart materials** capable of changing their **shape, properties, or functionality over time** in response to external stimuli such as **heat, light, moisture, electricity, or magnetic fields**. The fourth dimension represents **time**, as the printed object transforms after fabrication.
- Both technologies are considered important pillars of **Industry 4.0**, supporting advanced manufacturing, healthcare, aerospace, defence, and sustainable production.

Evolution of 3D and 4D Printing

- **1980s:** 3D printing (Additive Manufacturing) developed using stereolithography.
- **2000s:** Rapid adoption in aerospace, automotive, and healthcare.
- **2013:** MIT researcher **Skylar Tibbits** introduced the concept of **4D Printing**.
- **Present:** 3D printing is commercially mature, while 4D printing remains in the research and early commercialisation stage.

Working of 3D Printing

The manufacturing process begins with a digital design.

1. **CAD Design-** A three-dimensional model is created using Computer-Aided Design (CAD) software.
2. **Slicing-** Software converts the model into hundreds or thousands of printable layers.
3. **Layer-by-Layer Printing-** The printer deposits material sequentially until the object is completed.
4. **Post-Processing-** Cleaning, polishing, curing, or finishing improves the final product.

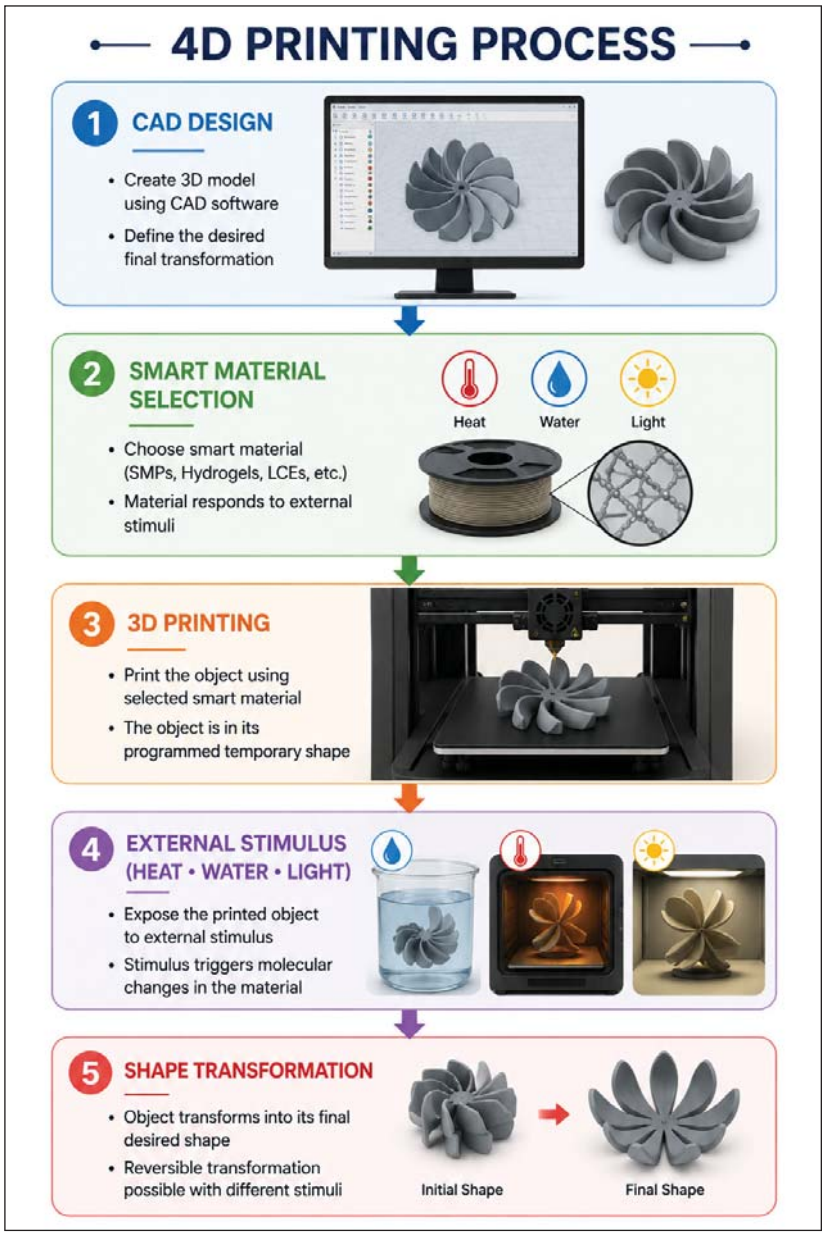
Working of 4D Printing

The manufacturing process is similar to 3D printing but uses smart materials.

1. **Digital Design-** A CAD model is created with programmed transformation behaviour.
2. **Smart Material Selection-** Shape-memory polymers, hydrogels, or smart alloys are selected.
3. **Layer-by-Layer Printing:** The object is printed using additive manufacturing techniques.
4. **External Stimulus:** Heat, moisture, light, electricity, or magnetic fields trigger the object to transform automatically.

3D Printing Process

4D Printing Process



Technologies Used

3D Printing Technologies

1. **Fused Deposition Modelling (FDM)-** Builds objects by melting and depositing thermoplastic filaments.
2. **Stereolithography (SLA)-** Uses ultraviolet lasers to cure liquid resin layer by layer.
3. **Selective Laser Sintering (SLS)-** Uses lasers to fuse powdered materials into solid objects.
4. **Direct Metal Laser Sintering (DMLS)-** Produces high-strength metal components for aerospace and healthcare.

4D Printing Materials

1. **Shape Memory Polymers (SMPs)-** Return to their original shape after external stimulation.
2. **Hydrogels-** Expand or contract when exposed to moisture.
3. **Shape Memory Alloys (SMAs)-** Metallic materials that recover their original form upon heating.
4. **Smart Composites-** Multi-material structures capable of adaptive behaviour.

Features

3D Printing

1. **Additive Manufacturing builds** objects layer by layer.
2. **High Customisation-** Easily produces personalised products.
3. **Complex Geometry-** Manufactures intricate designs impossible with conventional methods.
4. **Reduced Material Waste- uses** only the required material.
5. **Rapid Prototyping-** Accelerates product development.

4D Printing

1. **Smart Materials- Use** materials that respond to environmental stimuli.
2. **Self-Transformation-** Objects change shape without external mechanical intervention.
3. **Adaptive Structures:** Products automatically adjust to changing conditions.
4. **Self-Assembly:** Components can assemble themselves after printing.
5. **Self-Repair Potential-** Future materials may repair minor damage autonomously.

Comparison between 3D and 4D Printing

Feature	3D Printing	4D Printing
Principle	Layer-by-layer additive manufacturing	Additive manufacturing using smart materials
Materials	Plastics, metals, ceramics, composites	Shape-memory polymers, hydrogels, smart alloys
Behaviour	Static after printing	Changes shape over time
Fourth Dimension	Not applicable	Time
External Stimulus	Not required	Heat, water, light, electricity, and a magnetic field
Technology Status	Commercially mature	Emerging technology
Cost	Lower	Higher
Applications	Manufacturing, healthcare, aerospace	Smart implants, soft robotics, adaptive structures

Applications of 3D Printing

1. **Healthcare:** Custom prosthetics, dental implants, surgical models, and bioprinting research.
2. **Aerospace:** Lightweight aircraft components and complex engine parts.
3. **Automotive:** Rapid prototyping, customised spare parts, and lightweight vehicle components.
4. **Construction:** 3D-printed houses, bridges, and modular buildings.
5. **Defence:** Manufacture of customised equipment and on-demand spare parts.
6. **Education:** Engineering prototypes and STEM learning.
7. **Consumer Products:** Jewellery, footwear, toys, furniture, and customised products.

Applications of 4D Printing

1. **Smart Medical Implants:** Stents, tissue scaffolds, and implants that adapt inside the human body.

2. **Soft Robotics:** Flexible robots capable of adapting to changing environments.
3. **Aerospace:** Aircraft components that automatically adjust during flight for better efficiency.
4. **Construction:** Self-assembling structures and self-healing building materials.
5. **Defence:** Adaptive camouflage, intelligent protective equipment, and deployable structures.
6. **Fashion:** Clothing that adjusts according to body temperature or environmental conditions.
7. **Agriculture:** Smart irrigation systems and moisture-responsive materials.

Advantages

3D Printing

1. **Rapid Prototyping-** Shortens product development cycles.
2. **Design Flexibility enables** highly complex geometries.
3. **Reduced Waste- uses** material efficiently.
4. **Cost-effective for customisation- ideal** for low-volume production.
5. **Local Manufacturing- reduces** transportation and inventory costs.

4D Printing

1. **Self-Adaptive Products:** Automatically respond to environmental changes.
2. **Greater Customisation-** Tailored behaviour for specialised applications.
3. **Reduced Maintenance-** Self-adjusting structures require fewer interventions.
4. **Improved Sustainability- potentially** reduces material consumption through self-repair.
5. **Intelligent Manufacturing supports** next-generation smart products.

Challenges

3D Printing

1. **Slow Mass Production- less** suitable for high-volume manufacturing.
2. **Material Limitations-** Limited range compared to conventional manufacturing.
3. **High Equipment Cost-** Industrial-grade printers remain expensive.
4. **Quality Consistency-** Mechanical properties may vary.

4D Printing

1. **Limited Smart Materials-** Few commercially available responsive materials.
2. **High Cost-** Advanced materials significantly increase manufacturing costs.
3. **Complex Design requires** expertise in materials science and engineering.
4. **Early Development Stage:** Most applications remain experimental.
5. **Scalability Issues-** Large-scale commercial production is still limited.

3D & 4D Printing and India

India is increasingly adopting additive manufacturing to strengthen innovation and indigenous manufacturing.

1. **Supports Make in India-** Promotes advanced domestic manufacturing.
2. **Strengthens Healthcare-** Enables affordable prosthetics, implants, and customised medical devices.
3. **Boosts Defence Manufacturing-** Supports indigenous production of critical components.
4. **Encourages Startup Innovation-** Expands opportunities in medical devices, aerospace, and industrial design.
5. **Accelerates Industry 4.0-** Integrates additive manufacturing with AI, IoT, Robotics, and Digital Twins.

Government Initiatives

1. **National Strategy for Additive Manufacturing (NSAM), 2022-** Aims to position India as a global hub for additive manufacturing by promoting R&D, innovation, and industrial adoption.

2. **Make in India-** Encourages indigenous manufacturing using advanced technologies.
3. **Startup India-** Supports startups working in additive manufacturing and advanced materials.
4. **IndiaAI Mission-** Promotes AI integration with intelligent manufacturing technologies.
5. **Production-Linked Incentive (PLI) Scheme strengthens** advanced manufacturing across strategic sectors.

Global Best Practices

1. **The United States leads** in aerospace, defence, healthcare, and industrial additive manufacturing.
2. **Germany integrates** additive manufacturing into Industry 4.0 and advanced manufacturing.
3. **Japan uses** additive manufacturing for robotics, healthcare, and automotive industries.
4. **China:** Rapidly expanding large-scale industrial 3D printing capabilities.
5. **MIT (USA)-** Pioneer of 4D printing through research on programmable materials and self-assembling structures.

Way Forward

1. **Invest in Smart Materials- accelerate** research on shape-memory polymers, hydrogels, and intelligent composites.
2. **Strengthen Industry–Academia Collaboration-** Promote joint research in additive manufacturing and advanced materials.
3. **Expand Indigenous Manufacturing-** Encourage domestic production of 3D printers, smart materials, and specialised components.
4. **Develop Skilled Workforce-** Train professionals in additive manufacturing, CAD design, and material science.
5. **Promote Commercialisation-** Support startups in translating laboratory innovations into industrial products.
6. **Integrate with Industry 4.0- combine** additive manufacturing with AI, IoT, Robotics, Digital Twins, and Cloud Computing.
7. **Develop Standards and Regulations-** Establish quality, safety, and certification standards for additive manufacturing products.

Conclusion

3D Printing has revolutionised manufacturing by enabling rapid prototyping, customised production, and efficient use of materials, while **4D Printing** represents the next frontier by creating intelligent objects capable of **self-transformation, adaptation, and self-assembly**. Together, these technologies are reshaping industries such as healthcare, aerospace, defence, construction, and manufacturing. By investing in research, smart materials, indigenous innovation, and skilled human capital, India can leverage additive manufacturing to strengthen **Make in India**, **Atmanirbhar Bharat**, and **Viksit Bharat 2047**.

MGP/BP/GS3/021

INTELLECTUAL PROPERTY RIGHTS

INTELLECTUAL PROPERTY RIGHTS (IPR)

Intellectual Property Rights (IPR) are legal rights granted to individuals or creators to protect their unique inventions, artistic works, designs, and brand identities from unauthorized commercial use. In India, a robust IPR regime balances private innovation incentives with public welfare, driving economic growth while safeguarding access to essential medicines and technology.

Status of IPR in India

- **Global Rankings & Indices**
  - **Global Innovation Index (GII):** India ranked **38th** out of 139 economies in 2025 *(World Intellectual Property Organization - WIPO)*.
  - **International IP Index:** India ranked **42nd** out of 55 major global economies with an overall score of 38.64% *(U.S. Chamber of Commerce)*.
  - **Regional Leader:** Ranked **1st** among lower-middle-income economies and **1st** in the Central and Southern Asia region in innovation performance.
- **Patent & Trademark Filings**
  - **Annual Patent Filings:** Crosses over **82,000+** patent applications annually *(DPIIT Annual Report)*.
  - **Domestic Filing Dominance:** Resident Indian filings account for over **52%** of total patent applications, surpassing foreign non-resident filings *(DPIIT)*.
  - **Trademark Registrations:** Exceeds **4,60,000+** annual applications filed across regional registries *(Trade Marks Registry, DPIIT)*.
- **Geographical Indications & Revenue**
  - **GI Tag Registrations:** Over **600+** total products granted Geographical Indication status till date *(GI Registry, Chennai / DPIIT)*.
  - **IP Revenue Generation:** IP Office generated **₹1,185+ Crore** in revenue annually through application fees and maintenance *(Office of CGPDTM)*.

Objectives of Intellectual Property Rights

- **Encourage Innovation:** Protects original ideas and technical inventions, giving creators the incentive to invest time and money into research and development.
- **Economic Growth:** Facilitates commercialization, attracts Foreign Direct Investment (FDI), and drives industrial development through tech transfers.
- **Prevent Unfair Exploitation:** Provides exclusive legal rights to prevent unauthorized copying, piracy, counterfeiting, and commercial theft of intellectual work.
- **Promote Knowledge Sharing:** Grants temporary monopolies to inventors in exchange for public disclosure of technical details, enriching the public domain.

- **Ensure Public Welfare:** Balances private rights with societal needs by including safety nets like compulsory licensing for affordable medicines and agricultural access.
- **Global Trade Compliance:** Aligns national intellectual property laws with international frameworks like the WTO's TRIPS Agreement for smoother cross-border trade.

Nodal & Administrative Bodies for Intellectual Property Rights

- **DPIIT (Nodal Agency):** Department for Promotion of Industry and Internal Trade (Ministry of Commerce & Industry) acts as the central agency for all IPR administration, policies, and laws.
- **CGPDTM Office:** Controller General of Patents, Designs and Trade Marks functions under DPIIT to execute statutory operations across regional offices.
- **CIPAM:** Cell for IPR Promotion and Management functions under DPIIT to implement and monitor the National IPR Policy.

International conventions for IPR

- **Major multilateral treaties**
  - **TRIPS Agreement (1994):** Enforced by the WTO, it sets compulsory minimum standards across all IP types and links global trade rules with IP enforcement.
  - **Paris Convention (1883):** Covers industrial property—including patents, industrial designs, and trademarks—establishing the principle of **National Treatment**.
  - **Berne Convention (1886):** Protects literary and artistic copyrights automatically without requiring explicit registration in member nations.
- **Patent-Specific Treaties**
  - **Patent Cooperation Treaty (1970):** Streamlines international patenting by allowing inventors to file a single unified application for multiple countries.
  - **Budapest Treaty (1977):** Mandates international recognition of depositing micro-organisms at specialized authorities for biotechnology patent procedures.
- **Trademark & Design Treaties**
  - **Madrid Protocol (1989):** Simplifies global trademark registration through a centralized single-window application process.
  - **Hague Agreement (1925):** Provides a single international registration system to secure industrial design protection across multiple regions.
- **Special Purpose & Public Welfare Treaties**
  - **Marrakesh Treaty (2013):** Waives strict copyright rules to enable cross-border sharing of adapted books for blind and print-disabled individuals.
  - **WIPO Copyright Treaty (1996):** Extends traditional Berne Convention copyright protections to cover digital works and internet distribution.

Major Governing Acts & Statutes

Governing Act & Rules	Nodal Ministry	Key Provisions & Features	Duration of Protection
Patents Act, 1970 (Amended in 2005)	DPIIT, Ministry of Commerce & Industry	• Covers product and process patents. • Includes Section 3(d) to prevent evergreening. • Provides Compulsory Licensing (Section 84) for public health.	20 Years

Copyright Act, 1957 (Amended in 2012)	DPIIT, Ministry of Commerce & Industry	• Protects literary, musical, dramatic, artistic works, films, and computer code. • Aligned with WIPO Internet Treaties.	Lifetime of Author + 60 Years
Trade Marks Act, 1999 (Amended in 2010)	DPIIT, Ministry of Commerce & Industry	• Protects brand names, logos, slogans, and symbols. • Aligned with the Madrid Protocol for international registration.	10 Years (Renewable indefinitely)
Geographical Indications of Goods Act, 1999	DPIIT, Ministry of Commerce & Industry	• Protects agricultural, natural, or manufactured goods linked to a specific region (e.g., Darjeeling Tea). • Prevents misuse by unauthorized producers.	10 Years (Renewable every 10 years)
Designs Act, 2000	DPIIT, Ministry of Commerce & Industry	• Protects the aesthetic or visual appearance of manufactured products. • Does not cover functional/technical features.	10 Years (Extendable by 5 years)
Semiconductor Integrated Circuits Layout-Design Act, 2000	MeitY (Ministry of Electronics & IT)	• Protects 3D layout topographies of integrated circuits (microchips). • Essential for boosting domestic electronics manufacturing.	10 Years
Plant Varieties & Farmers' Rights Act, 2001	Ministry of Agriculture & Farmers Welfare	• Balances private plant breeder rights with traditional farmers' rights to save, use, and exchange seeds.	15 Years (Crops) 18 Years (Trees & Vines)

Advantages of IPR

- **For Innovators & Creators**
  - **Financial Returns:** Grants exclusive commercial rights, enabling inventors to monetize their research and recover costly R&D investments.
  - **Legal Protection:** Provides strong judicial remedies against unauthorized copying, counterfeiting, piracy, and market exploitation.
  - **Market Advantage:** Helps startups and businesses establish unique brand identities, leading to a competitive edge in global markets.
- **For Economic Growth & Trade**
  - **Attracts Investment:** High standards of IPR enforcement boost investor confidence, driving Foreign Direct Investment (FDI) into tech sectors.
  - **Employment Generation:** Encourages the growth of knowledge-based industries, creating high-skill jobs in technology, pharmaceuticals, and design.
  - **Global Trade Integration:** Harmonizes national standards with international trade bodies like the WTO, expanding export opportunities.
- **For Society & Public Welfare**
  - **Drives Technological Progress:** Encourages continuous innovation in critical sectors like healthcare, renewable energy, and agriculture.
  - **Protects Traditional Knowledge:** Safeguards indigenous practices and regional heritage using Geographical Indications (GIs) and databases like TKDL.
  - **Spurs Knowledge Sharing:** Requires inventors to publish technical details in exchange for patents,

expanding the public database for future researchers.

Disadvantages of IPR

- **Economic & Market Drawbacks**
  - **Monopoly Pricing:** Grants exclusive market control, allowing patent holders to charge exorbitant prices and exploit consumers.

    According to the **World Health Organization (WHO)**, patent monopolies allow originators to charge premium prices, whereas competitive generic entry reduces prices by up to **80–90%**.
  - **Higher Costs for Small Businesses:** Acquiring, maintaining, and defending patents involves heavy legal costs that burden small enterprises (MSMEs).

    As per **DPIIT data**, total official fees, attorney charges, and maintenance costs over a 20-year term can exceed **₹1–3 Lakhs per patent in India.**
  - **Impairs Fair Competition:** Big firms often build **patent thickets**—buying up numerous overlapping patents to block new competitors from entering the market.

    According to a WIPO Smartphone Patent Study, modern tech devices are choked by over 250,000+ active overlapping patents.
- **Health & Social Accessibility**
  - **Restricts Essential Medicines:** Long-term patent protection on life-saving drugs reduces access to affordable generic alternatives in developing nations.
  - **Aggravates Tech Inequality:** Wealthy countries hold the most advanced patents, deepening the digital and technological divide with the Global South.

    According to **WIPO's World Intellectual Property Indicators**, over **70%** of total global active patents are held by five high-income countries (USA, EPO, Japan, South Korea, and China).
  - **Reduces Public Domain Access:** Strict copyrights and digital access locks hinder open public access to educational materials and scientific literature.
- **Technological & Ethical Concerns**
  - **Patent Evergreening:** Companies make minor, trivial tweaks to existing inventions just to artificially extend their patent protection.

    According to **DPIIT**, nearly **70–80%** of pharmaceutical patent applications filed globally are for minor tweaks (secondary patents) rather than new chemical entities.
  - **Stifles Follow-Up Research:** Fear of patent infringement lawsuits deters independent scientists from experimenting on patented technologies.
  - **Biopiracy & Traditional Exploitation:** Foreign entities occasionally attempt to patent indigenous knowledge and natural resources without fair benefits to local communities.

    According to the CSIR (India), the Traditional Knowledge Digital Library (TKDL) has successfully defended and revoked over 2,500+ faulty foreign patent applications.

Way forward

- **Policy & Legislative Reforms**
  - **Enact Trade Secrets Law:** Pass dedicated legislation to protect confidential business processes, trade secrets, and test data to boost foreign R&D investments.
  - **Review National IPR Policy:** Update the 2016 policy framework to address emerging domain issues like AI-generated inventions, digital piracy, and green technology patents.
  - **Streamline Statutory Timelines:** Set strict timeframes for patent examination and trademark approvals to clear administrative backlogs.

- **Research, Development & Commercialization**
  - **Boost R&D Expenditure:** Raise national research spending from less than 0.7% of GDP toward 2% using private sector tax breaks and innovation bonds.
  - **Bridge Lab-to-Market Gap:** Establish dedicated IP commercialization cells in universities to help researchers monetize patented technology.
  - **Encourage Academic-Industry Links:** Build joint R&D hubs between research institutes and MSMEs to turn indigenous patents into marketable products.
- **Enforcement & Capacity Building**
  - **Dedicated IP Benches:** Strengthen specialized IP divisions within High Courts to handle complex corporate disputes faster.
  - **Digital Piracy Safeguards:** Improve coordination between law enforcement and customs officials using AI tools to fight counterfeit goods and cyber piracy.
  - **Strengthen Patent Offices:** Hire and train more technical examiners at the Controller General of Patents, Designs and Trade Marks (CGPDTM) to ensure faster processing times.
- **Balancing Public Interest & Global Compliance**
  - **Protect Public Health Safeguards:** Maintain flexibilities under TRIPS, using Section 3(d) against evergreening and Compulsory Licensing (Section 84) during health emergencies.
  - **Digitize & Expand Traditional Knowledge:** Expand the Traditional Knowledge Digital Library (TKDL) and integrate local communities to protect indigenous practices from global biopiracy.
  - **Democratize Grassroots IP Literacy:** Expand schemes like KAPILA to build IP awareness across rural schools, startups, and agricultural sectors.

National IPR Policy 2016

- **Vision**: "Creative India; Innovative India" — aims to foster creativity and integrate IPR into national economic development.
- **7 Major Objectives**
  - **IPR Awareness:** Create widespread awareness about the economic, social, and cultural benefits of IPR across all sections of society.
  - **Generation of IPRs:** Stimulate the creation and registration of patents, trademarks, and copyrights, especially among MSMEs and startups.
  - **Legal & Legislative Framework:** Update and strengthen IP laws to balance inventor rights with public interest (e.g., healthcare access).
  - **Administration & Management:** Modernize and digitize patent/trademark offices to drastically reduce administrative pendency.
  - **Commercialization of IPR:** Help creators monetize their IP through financial valuation, technology transfer, and market linkages.
  - **Enforcement & Adjudication:** Strengthen judicial and police mechanisms to combat piracy, counterfeiting, and IP theft effectively.
  - **Human Capital Development:** Build a pool of IP professionals by introducing IPR studies in schools, colleges, and training institutes.

Conclusion

By balancing strong patent protection with robust safeguards for public health, traditional knowledge, and affordable technology, India can foster a vibrant knowledge-driven economy. Ultimately, a streamlined and transparent IPR ecosystem is essential to realize the vision of an *Atmanirbhar Bharat* while fulfilling global trade commitments under the WTO.

2. TRADITIONAL KNOWLEDGE DIGITAL LIBRARY

Traditional Knowledge Digital Library (TKDL) is a pioneer digital database established in 2001 by India to protect its rich heritage of traditional medicine and prevent biopiracy. **TKDL** Jointly created by the **Council of Scientific and Industrial Research (CSIR)** and the **Ministry of AYUSH**, it translates ancient texts on Ayurveda, Unani, Siddha, Sowa Rigpa, and Yoga into five international languages (English, French, German, Japanese, and Spanish).

Features of Traditional Knowledge Digital Library

- **Joint Initiative:** Formed in 2001 as a collaboration between CSIR (Council of Scientific and Industrial Research) and the Ministry of AYUSH.
- **Prior Art Database:** Converts undocumented or traditional Indian medical knowledge into a searchable digital format to prove "prior art" to global patent offices.
- **Multi-Language Access:** Translates ancient Indian texts into 5 international languages—English, French, German, Japanese, and Spanish.
- **Massive Medical Repository:** Documents over 4.6+ lakh traditional formulations across Ayurveda, Unani, Siddha, Sowa Rigpa, and Yoga.
- **Non-Disclosure Access Agreements:** Shares access strictly with international patent examiners (like EPO, USPTO, JPO) for search and examination, preventing public domain commercial leaks.
- **Patent Cancellation Tool:** Enables India to legally challenge and cancel faulty foreign patent applications before they are granted.
- **Digitized Standards:** Converts complex Sanskrit, Arabic, and Persian terminology into standardized Patent Classification Codes (IPC) recognized worldwide.

Significance of Traditional Knowledge Digital Library

- **Defense Against Biopiracy**
  - Blocks Wrongful Patents: Prevents foreign commercial entities from claiming exclusive rights over age-old Indian remedies like neem, turmeric, and ashwagandha.
  - **Saves Litigation Costs:** Obviates expensive, multi-year court battles in foreign jurisdictions by stopping bad patents at the application stage.
  - **Preserves Cultural Heritage:** Ensures centuries-old traditional knowledge remains in the public domain rather than turning into private foreign monopolies.
- **Global IPR Leadership**
  - **Pioneers IP Defense Model:** Acts as a world-first benchmark for developing nations looking to protect indigenous systems under international law.
  - **Streamlines Global Patent Searches:** Equips top international patent offices (EPO, USPTO, JPO) with standardized data to identify prior art instantly.
  - **Reinforces Benefit-Sharing:** Protects local communities from commercial exploitation by upholding CBD (Convention on Biological Diversity) principles.
- **Boost to Domestic Healthcare & Science**
  - **Promotes AYUSH Systems:** Validates traditional systems like Ayurveda, Unani, Siddha, Sowa Rigpa, and Yoga on modern scientific parameters.
  - **Accelerates Drug Discovery:** Provides a structured, digitized scientific base to help Indian researchers convert traditional recipes into modern medicine.
  - **Drives Economic Value:** Safeguards India's wellness and herbal export market from unfair foreign non-tariff barriers and patent blocks.

Challenges with Traditional Knowledge Digital Library

- **Administrative & Accessibility Challenges**
  - **Restricted Public Access:** Database is closed to general researchers and the public due to secrecy agreements, limiting domestic innovation.

    According to the **CSIR**, the database is kept locked under strict Non-Disclosure Agreements (NDAs) signed with only **18 international patent offices.**
  - **Complex Inter-Ministry Coordination:** Operations are split between CSIR and the Ministry of AYUSH, often leading to bureaucratic delays.
  - **Resource Constraints:** Inadequate funding and technical staff slow down the digitization of thousands of remaining manuscripts.
- **Legal & IP Limitations**
  - **Lacks Defensive Legal Power:** TKDL only acts as a search database; it does not automatically grant legal ownership or patent protection to Indian communities.
  - **Excludes Non-Codified Knowledge:** Focuses mostly on written ancient texts, leaving out oral traditions and tribal remedies passed down verbally.
  - **Limited Scope of Coverage:** Does not cover non-medicinal traditional knowledge, such as indigenous agricultural practices, handicrafts, or local arts.
- **Global & Implementation Hurdles**
  - **Voluntary International Adoption:** Foreign patent offices are not legally bound to check TKDL, leading to occasional oversight of Indian prior art.
  - **Language & Translation Gaps:** Translating archaic Sanskrit, Persian, or Tamil terms into modern patent classification codes can result in loss of context.
  - **No Fair Benefit-Sharing Mechanism:** Fails to provide direct monetary compensation or royalties to the local communities who originated the knowledge.

Way forward

- **Policy & Legal Reforms**
  - **Enact Protection Legislation:** Pass dedicated national laws to legally protect non-codified traditional knowledge and grant community ownership rights.
  - **Mandatory International Checks:** Push WIPO to make searching TKDL legally mandatory for global patent examiners during application reviews.
  - **Establish Benefit-Sharing Models:** Create clear mechanisms to return financial benefits and royalties to indigenous communities when traditional knowledge is commercialized.
- **Database & Technical Expansion**
  - **Digitize Oral Traditions:** Expand the database beyond written ancient texts to include unwritten tribal remedies and folk practices.
  - **Integrate Emerging Technologies:** Use AI and machine learning to speed up natural language processing, translation, and ancient manuscript digitization.
  - **Broaden Beyond Medicine:** Include non-medicinal traditional knowledge like indigenous farming techniques, handicrafts, and local water management systems.
- **Access & Commercialization Balance**
  - **Tiered Public Access:** Grant controlled, tiered access to Indian universities and startups to encourage domestic R&D while keeping sensitive data secure.
  - **Boost AYUSH Commercialization:** Link TKDL with Indian MSMEs to convert validated traditional recipes into patented, global-grade wellness products.
  - **Capacity & Awareness Building:** Train local tribal communities and grassroots innovators on how to register and safeguard their traditional practices.

Similar Global Initiatives & Databases

- **WIPO WPKD (World Intellectual Property Organization Portal):** A global gateway integrating national traditional knowledge databases to help worldwide patent offices verify prior art.
- **South Africa (NKSAL / IKS Database):** Digital repository under the National Knowledge Management System that records indigenous knowledge to prevent biopiracy and safeguard community rights.
- **China (Traditional Chinese Medicine Patent Database):** A massive digital archive cataloging over 40,000+ ancient Chinese medical prescriptions to prevent foreign patent exploitation.
- **South Korea (Korean Traditional Knowledge Portal - KTKP):** A structured database documenting traditional Korean medicine, herbs, and patents to establish defensive prior art globally.
- **Peru (Registries of Traditional Knowledge):** National register protecting the collective bio-cultural knowledge and indigenous plant varieties of native Andean communities.

AI and tradition knowledge digital library

Integration of Artificial Intelligence (AI) with the Traditional Knowledge Digital Library (TKDL) marks a shift from a static digital archive to an active, intelligence-driven research and protection hub. Recognized by the World Health Organization (WHO), this convergence uses modern computational tools—such as machine learning, natural language processing (NLP), and optical character recognition (OCR)—to safeguard, translate and scientifically validate India's ancient heritage.

Core Applications of AI in TKDL

- **AI-Powered OCR & Manuscript Digitization:** Uses deep-learning OCR algorithms to scan, clean, and digitize ancient, decaying palm-leaf scripts, rare books and handwritten manuscripts.
- **Automated Translation & NLP:** Leverages Natural Language Processing to convert complex, archaic texts in Sanskrit, Persian, Arabic, and Tamil into structured patent classification codes across multiple international languages.
- **Automated Prior-Art Search & Biopiracy Detection:** AI search algorithms continuously scan global patent databases (USPTO, EPO, JPO) in real time to instantly detect and flag unauthorized biopiracy filings.
- **Cross-Disciplinary Ayurgenomics:** Merges traditional *Prakriti* (body constitution) classification with modern genomic datasets using machine learning to enable personalized medicine research.
- **Drug Discovery & Pattern Recognition:** Analyzes thousands of multi-herb classical formulations to identify active bio-molecules and predict novel therapeutic targets for modern diseases.

Conclusion

Traditional Knowledge Digital Library stands as India's pioneering shield, successfully safeguarding centuries of indigenous health wisdom from global biopiracy and unfair foreign patents. By bridging ancient Indian texts with modern digital patent databases, it has set a global benchmark for protecting cultural heritage under international IPR frameworks.

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ENERGY TECHNOLOGIES

1. NUCLEAR ENERGY IN INDIA

Nuclear energy has emerged as a cornerstone of India's quest for energy security, sustainable development, and technological leadership. With a growing population and rapid industrialization, India's energy demand is projected to double by 2040. Nuclear energy, with its low carbon footprint and high energy density, offers a viable solution to meet this demand while aligning with global climate goals.

Present status of Nuclear Energy in Indias

Civil Nuclear Power	**Number of Operational Nuclear Power Plants:** As of early 2026, India has **24 operational nuclear reactors** spread across seven major power plants, with a combined installed capacity of **8,780 MW**. These facilities are managed by the Nuclear Power Corporation of India Limited (NPCIL) and BHAVINI. **New Nuclear Projects:** Several new reactors are under construction, including **Kudankulam** (Units 3-6), **Gorakhpur Haryana Anu Vidyut Pariyojana**, and **Kaiga** Units 5 & 6, expected to add nearly 7,000 MW in the next decade which will add 8,000 MWe by 2031. **Indigenous Reactor Technology:** India has developed **Pressurized Heavy Water Reactors** (PHWRs), **Prototype Fast Breeder Reactor (PFBR)**, and is progressing towards thorium-based **Advanced Heavy Water Reactor (AHWR)**.
Nuclear in Defense	**Nuclear-Powered Submarines:** India has developed **INS Arihant,** the country's first indigenous nuclear-powered ballistic missile submarine (SSBN), and is working on INS Arighat and additional units. **Nuclear-Capable Missiles:** India has a range of nuclear-capable missiles, including **Agni (I-V)**, **Prithvi**, and **K-15** (Sagarika). **Nuclear Research and Collaborations:** India collaborates with the **IAEA**, **Russia** (Rosatom), and **the US (123 Agreement)** for nuclear technology exchange, while advancing indigenous research at BARC, IGCAR, and NPCIL.
Technologies in Use and Development	**PHWRs:** The backbone of India's nuclear program, using natural uranium as fuel. **Fast Breeder Reactors (FBRs):** The 500 MWe Prototype Fast Breeder Reactor (PFBR) is under development, aiming to use plutonium and thorium for fuel. **Molten Salt Reactors (MSRs):** It is still in the research phase, focusing on thorium utilization. **Small Modular Reactors (SMRs):** These reactors are being explored for deployment at retiring coal plant sites.

Significance of Nuclear Energy

1. **Energy Security:** The three-stage nuclear program aims to utilize thorium, ensuring long-term energy independence. Nuclear power accounts for about **3% of India's electricity generation** but is targeted to reach **22% by 2050**. Nuclear energy delivers base load power, which is essential to building modern economies.
2. **Clean Energy:** Nuclear power is a low-carbon energy source, contributing to **India's Net Zero by 2070 goal**. It complements renewable energy by providing base-load power, ensuring grid stability.
3. **Space Exploration:** Nuclear energy powers **radioisotope thermoelectric generators (RTGs)** for deep-space missions, such as Chandrayaan and Mangalyaan.
4. **Defence:** Nuclear-powered submarines and missiles enhance India's strategic deterrence and second-strike capability. Indigenous development of nuclear reactors and weapons reduces dependency on foreign technology.
5. **Medical Advancements:** BARC-developed radiopharmaceuticals aid in cancer treatment, with India producing over **50 radioisotopes for medical use**.
6. **Food Preservation:** Nuclear technology enhances food irradiation, increasing shelf-life and preventing spoilage.

Government initiatives for the promotion of Nuclear Energy

Early start	India was the 1st country in Asia to set up a nuclear research reactor, **APSARA,** in 1956. India was also the 1st in Asia to work on the nuclear power reactor at Tarapore in 1963.
Three-Stage Nuclear Program	**Stage 1:** PHWRs using natural uranium. **Stage 2:** Fast Breeder Reactors using plutonium and thorium. **Stage 3:** Thorium-based reactors for sustainable energy.
Bharat Small Reactors (BSR)	It aims to deploy small modular reactors in collaboration with the private sector.
Nuclear Legislation	**Atomic Energy Act:** Authorizes the establishment of nuclear power plants and R&D institutions. **Civil Liability for Nuclear Damage Act, 2010 (CLNDA):** Enacted to define **liability and compensation** in case of nuclear accidents. Aligns India with the **Convention on Supplementary Compensation (CSC), 1997.** **SHANTI Act:** The Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025, overhauls India's nuclear sector by allowing private companies to build and operate nuclear plants. It **repeals the Atomic Energy Act, 1962**, and the **Civil Liability for Nuclear Damage Act, 2010**, while **granting statutory status to the Atomic Energy Regulatory Board (AERB)**.
Bodies	**Nuclear Power Corporation of India Limited (NPCIL):** NPCIL is a govt-owned company that builds, owns & operates the PHWRs. **Atomic Energy Regulatory Board (AERB):** Certification & safety oversight is the responsibility of AERB.

International Collaborations	**Indo-US Nuclear Deal:** Facilitated access to global uranium markets and advanced technologies. **Civil Nuclear Cooperation Agreements:** Signed with Russia, France, USA, Japan, and others for fuel supply and technology transfer. **Nuclear Insurance Pool (2016):** Mitigates risks associated with nuclear energy projects.
Union Budget 2025-26	• Announced an ambitious target of 100GW nuclear power generation capacity by 2047, up from 8.18GW at present. • **Nuclear Energy Mission:** Aims to develop at least 5 indigenously designed & operational Small Modular Reactors (SMRs) by 2033.

Challenges associated with Nuclear Energy

1. **Technological and Infrastructure Challenges:**
   - **High Upfront Capital Costs:** Nuclear plants require significant capital investment compared to coal or renewables.
   - **Limited Indigenous Uranium Supply:** Dependence on imports hinders energy security.
   - **Delayed Project Completions:** Kudankulam and PFBR projects have faced repeated delays.
2. **Safety and Environmental Concerns:**
   - **Radiation Risks:** Safety concerns post-Fukushima impact public perception.
   - **Nuclear Waste Management:** India lacks robust long-term nuclear waste disposal mechanisms.
3. **Regulatory and Political Challenges:**
   - **Public Opposition:** Protests in Tamil Nadu and Maharashtra have stalled nuclear projects.
   - **Policy Uncertainty:** Frequent changes in regulations delay private sector participation.
4. **Geopolitical Constraints:**
   - **NSG Membership Roadblock:** India's non-signatory status to the NPT limits access to advanced nuclear technologies.
   - **Global Uranium Market Volatility:** Prices and supply chain disruptions affect India's nuclear fuel procurement.

Way Forward

1. **Accelerate Thorium-Based Reactors:** Given India's vast thorium reserves, fast-tracking **AHWR deployment** is crucial for the growth of nuclear energy.
2. **Enhancing Private Sector Involvement:** Allowing greater participation of companies like **L&T, Tata, and Reliance** in nuclear infrastructure can boost innovation.
3. **Expanding Public Awareness and Transparency:** Addressing misconceptions through public engagement can improve acceptance.
4. **Strengthening Global Collaborations:** Expanding partnerships with **France, Russia, and the US** for technology exchange and uranium supply.
5. **Developing Small Modular Reactors (SMRs):** Investing in **indigenous SMR designs** can complement large nuclear plants.
6. **Focus on Nuclear Waste Management:** Setting up **deep geological repositories** for spent nuclear fuel.

7. **Leveraging AI and Digital Tools:** Integrating **AI-driven safety systems** and predictive maintenance can enhance reactor efficiency.
8. **Nuclear energy as 'Renewable Power':** While nuclear energy is a low-carbon energy source, it is not classified as 'renewable', like solar & wind. Revising this classification would make nuclear power projects eligible for tax incentives & specially designed 'green financing' instruments.

Conclusion

With bold initiatives, technological innovation, and strategic collaborations, India is poised to harness the full potential of nuclear energy. As the world moves toward a low-carbon future, India's nuclear energy program stands as a beacon of sustainable development and technological leadership. **The dream of "Viksit Bharat" is not just a vision; it is a mission powered by the atom.**

2. NUCLEAR FUSION & ITER

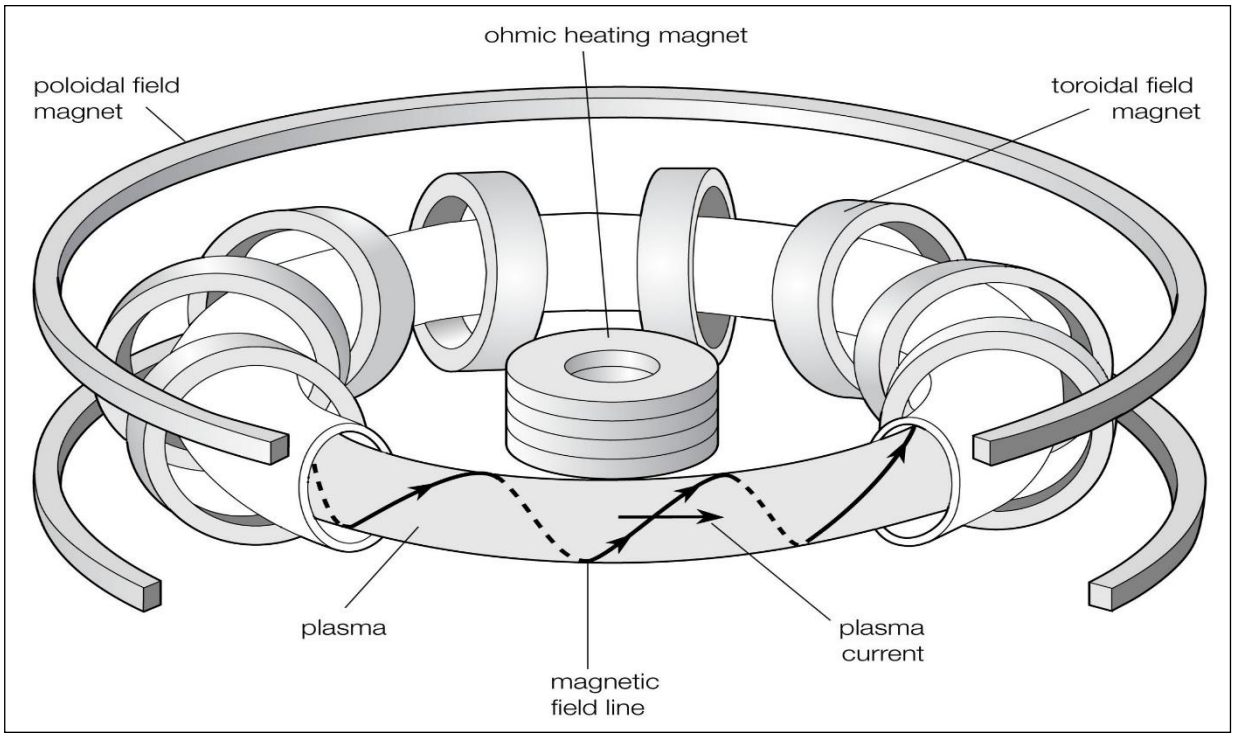
Nuclear fusion is the process that powers the sun, where light atomic nuclei combine to release massive, clean, and virtually limitless energy. The International Thermonuclear Experimental Reactor (ITER) is a megaproject in France—jointly funded by seven member entities including India—building the world's largest magnetic confinement tokamak.

Core Process of Nuclear Fusion

- **High Temperature & Pressure:** Light hydrogen isotopes (Deuterium and Tritium) are heated to extreme temperatures (over 150 million °C), causing atoms to strip their electrons and form a high-energy plasma.
- **Overcoming Repulsion:** At these super-hot, high-pressure states, positive nuclei overcome electrostatic repulsion (Coulomb force) and collide forcefully.
- **Nuclear Fusing:** The Deuterium and Tritium nuclei smash together to fuse into a heavier Helium nucleus and a free neutron.
- **Mass Defect & Energy Release:** A tiny fraction of mass is lost during the reaction; this lost mass is converted into massive kinetic energy via Einstein's equation (E = mc^2).
- **Neutron Energy Capture:** High-energy neutrons escape the plasma, strike the reactor's surrounding walls (blanket), and generate thermal heat to drive conventional steam turbines for power generation.

Objectives of ITER

- **Demonstrate Net Energy Gain:** Produce **500 MW** of fusion power from just **50 MW** of thermal input power.
- **Achieve Steady-State Burning Plasma:** Maintain a self-sustaining, controlled fusion reaction for extended operating periods (over 400 seconds).
- **Test Tritium Breeding Technology:** Validate the in-vessel breeding of Tritium fuel inside lithium blankets to ensure self-sufficient fusion fuel cycles.
- **Demonstrate Commercial Safety & Zero Carbon:** Prove that magnetic fusion produces no long-lived radioactive waste, carries zero risk of nuclear meltdown, and emits no greenhouse gases.
- **Integrate Core Engineering Systems:** Test and seamlessly run complex large-scale technologies together—including superconducting magnets, heating systems, and remote handling robotics.



Working Principle of International Thermonuclear Experimental Reactor (ITER)

- **Magnetic Confinement Fusion (MCF)**
  - **Tokamak Design:** Utilizes a doughnut-shaped vacuum chamber (**tokamak**) to house superheated fuel gases.
  - **Superconducting Magnets:** Powerful **toroidal and poloidal magnetic coils** generate massive magnetic fields to trap, shape, and suspend the superhot charged plasma away from the reactor walls.
- **Extreme Heating & Ignition**
  - **Auxiliary Heating Systems:** Combines neutral beam injection and high-frequency radio waves to heat the plasma fuel to **150 million °C** (10 times hotter than the sun's core).
  - **Controlled Fusion Reaction:** At extreme kinetic energies, Deuterium and Tritium nuclei overcome electrostatic repulsion and fuse together to release high-energy helium and neutrons.
- **Thermal Energy Capture**
  - **Blanket & Divertor Systems:** High-energy neutrons escape the magnetic cage, striking the lithium-lined **shielding blanket** to absorb heat and breed fresh Tritium fuel.
  - **Heat Transfer to Power:** Absorbed heat energy is transferred via water-coolants to drive steam turbines for clean electricity generation.

India's Contribution to the ITER

- **Organizational Framework**
  - **Nodal Agency:** Managed by **ITER-India**, an empowered domestic project under the **Institute for Plasma Research (IPR)**, Gandhinagar (Department of Atomic Energy).
  - **Share of Contribution:** India holds a **9.1% share** in ITER construction, primarily delivered through **in-kind** procurement of component packages.

- **Core Technical & Engineering Packages**
  - **World's Largest Cryostat:** Manufactured and supplied the 3,850-tonne stainless steel vacuum vessel to thermally insulate the superheated tokamak.
  - **In-Wall Shielding (IWS):** Delivered specialized neutron-shielding blocks installed inside the vacuum vessel walls to absorb high-energy radiation.
  - **Cryogenic & Distribution Lines:** Designed and supplied **over 11 km** of complex vacuum-jacketed cryolines and warm lines for liquid helium distribution.
  - **Cooling Water System:** Developed the massive Cooling Water and Heat Rejection System (CHWS) to dissipate thermal energy generated during fusion.
  - **RF & Plasma Heating Systems:** Contributed high-power **Ion Cyclotron (ICRF)** and **Electron Cyclotron (ECRF)** radio-frequency heating sources to bring plasma to 150 million °C.
  - **Diagnostic Systems:** Engineered the Diagnostic Neutral Beam (DNB) system and diagnostic tools for real-time monitoring of plasma behavior.

Significance of International Thermonuclear Experimental Reactor (ITER)

- **Geopolitical & Strategic Energy Independence**
  - **Global Energy Security:** Provides a path to end geopolitical dependence on volatile fossil fuels and finite mineral resources.
  - **Geopolitical Technology Cooperation:** Fosters unprecedented multilateral collaboration among seven global partners (covering over half the world's population).
  - **High-Tech Technology Transfer:** Enhances domestic industrial capacities of member nations like India through participation in megaproject manufacturing.
- **Climate & Environmental Sustainability**
  - **Zero Greenhouse Gas Emissions:** Emits zero carbon or greenhouse gases during operation, aiding global net-zero targets.
  - **Abundant & Accessible Fuel:** Uses Deuterium (extracted from seawater) and Tritium (bred from lithium), providing a virtually unlimited energy supply.
  - **No Long-Lived Nuclear Waste:** Produces no high-level, long-lived radioactive waste, avoiding the spent-fuel disposal challenges of fission reactors.
- **Operational & Economic Advantages**
  - **Inherently Safe (Zero Meltdown Risk):** Cannot suffer Fukushima-style meltdowns, as any plasma instability causes the reaction to stop within seconds.
  - **High Energy Density:** Fusing just 1 gram of DT fuel releases the energy equivalent of burning 8 tonnes of oil.
  - **Base-Load Power Capacity:** Operates continuously regardless of weather, unlike intermittent solar and wind energy sources.
- **Technological Spin-Offs & Scientific Breakthroughs**
  - **Advanced Superconductivity:** Accelerates innovations in large-scale superconducting magnets, cryogenics, and plasma physics.
  - **Robotics & Remote Handling:** Drives breakthroughs in radiation-hardened robotics and automated industrial maintenance systems.
  - **Materials Science Innovations:** Spurs development of advanced heat-resistant and low-activation structural alloys for future energy systems.

Issues with International Thermonuclear Experimental Reactor (ITER)

- **Timeline Delays & Massive Cost Overruns**
  - **Repeated Schedule Postponements:** Construction target for full operational **First Plasma** has slipped repeatedly, pushing full-power operations to nearly **2039**.
  - **Budget Tripling:** Overall project estimates have ballooned from an initial **€5 billion to over €22+ billion**, putting financial pressure on member states.
  - **Private Competition:** Rapid advances by agile, venture-backed private fusion startups threaten to make ITER's tech obsolete before it goes live.

    Over **$9.7 billion** in private capital has surged into **53+ private fusion startups** targeting grid power as early as 2030–2035
- **Technical & Engineering Bottlenecks**
  - **Material Degradation:** Reactor vessel walls suffer severe structural degradation and embrittlement caused by constant high-energy neutron bombardment.

    As per IAEA, Reactor walls face **14 MeV high-energy neutron bombardment**, causing material displacement damages up to **20 dpa (displacements per atom)** over operational lifespan.

    Component Defects & Repairs: Manufacturing defects—like cracks in thermal shield cooling pipes and misalignment in vacuum vessel sectors—require complex, time-consuming repairs.
  - **Tritium Fuel Shortage:** Global commercial supplies of **Tritium** are extremely scarce, making in-vessel self-breeding technology an unproven necessity.
  - **Plasma Instabilities:** Controlling magnetic disruption and preventing superhot plasma from touching and melting the reactor lining remains a major physics challenge.
- **Geopolitical & Governance Complexities**
  - **Multilateral Coordination Delays:** Decisions require consensus among 7 major international partners, creating bureaucratic bottlenecks and execution delays.
  - **Geopolitical Tensions:** Sanctions and international conflicts involving member nations complicate smoothly sharing technology and physical components.
  - **Strict Domestic Regulations:** Stricter nuclear safety guidelines and licensing delays mandated by the French Nuclear Safety Authority (ASN) hold up local construction progress.

Way forward for ITER

- **Technical & Structural Realignment**
  - **Implement Revised Baseline (2024 Baseline):** Re-sequence assembly to start scientific research operations around 2034, moving directly to full magnetic energy tests and Deuterium-Tritium operations by 2039.
  - **Switch First-Wall Materials:** Replace beryllium with **tungsten** armor tiles on the inner wall to better withstand heat fluxes and prevent hydrogen retention.
  - **Enhance Auxiliary Plasma Heating:** Boost initial heating capacity by increasing radio-frequency (ECRH/ICRH) power to ensure smooth plasma ignition under the updated operational setup.
- **Management & Governance Streamlining**
  - **Strengthen Quality Control:** Enforce rigid pre-assembly inspections to catch manufacturing defects (like cooling pipe cracks) before components reach the main assembly pit.
  - **Re-organize Leadership Structure:** Empower the ITER Director-General with unified oversight over domestic agencies to accelerate decision-making and clear project backlogs.
  - **Improve Regulatory Compliance:** Maintain proactive, continuous engagement with the French Nuclear Safety Authority (ASN) to prevent licensing and safety delays.

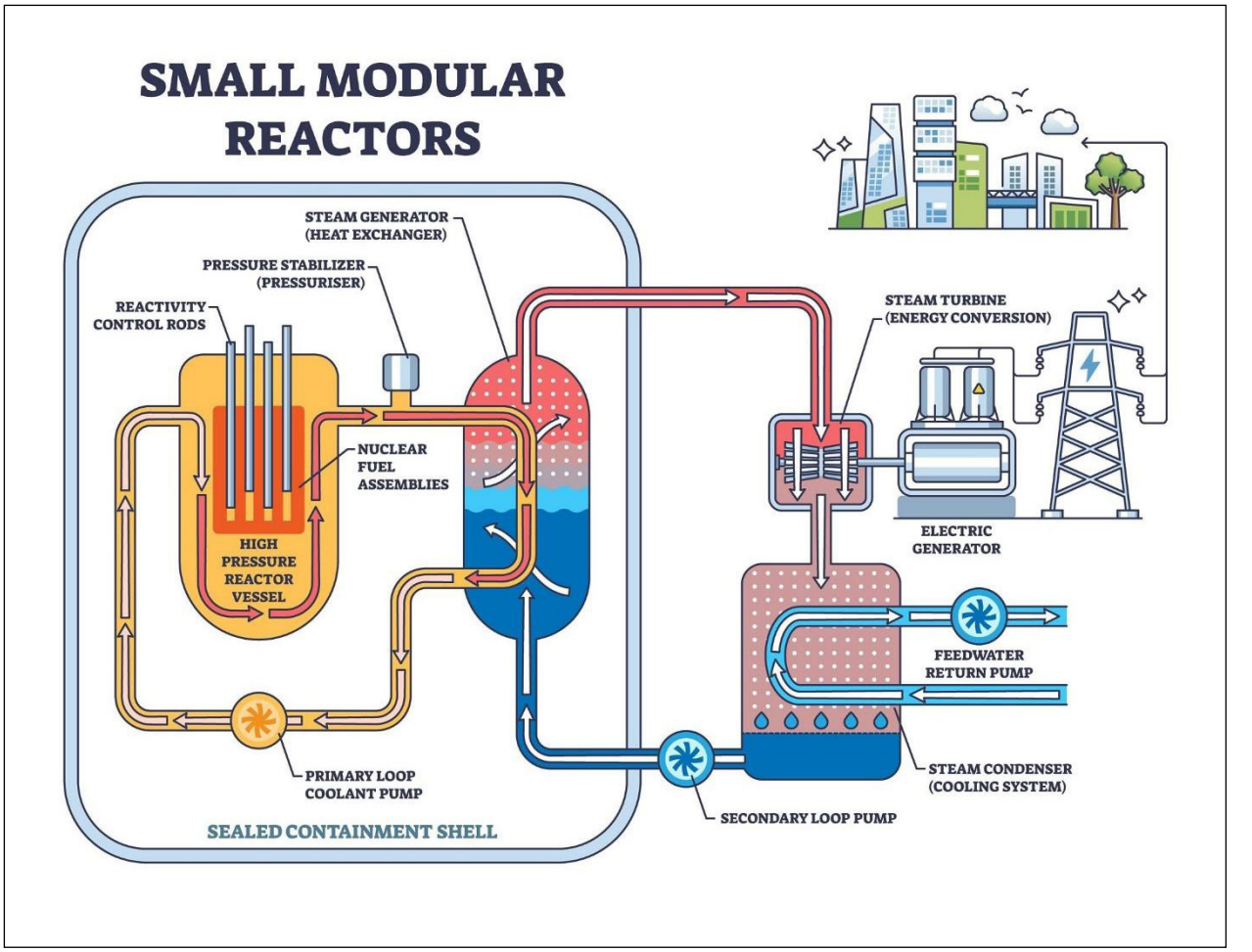
- **Strategic & Technological Synergies**
  - **Leverage National & Private Fusion Hubs:** Utilize insights from smaller national magnetic devices (like Japan's JT-60SA and Europe's ASDEX Upgrade) to de-risk plasma controls before ITER starts operation.
  - **Foster Public-Private Partnerships:** Collaborate with venture-backed private fusion startups to share breakthroughs in high-temperature superconductors and AI-driven plasma stabilization.
  - **Accelerate Tritium Self-Sufficiency R&D:** Prioritize testing and scaling lithium-breeding blanket modules to resolve global commercial shortages of tritium fuel.

Conclusion

To secure its long-term future, ITER must strictly execute its updated operational baseline while resolving component defect backlogs and switching to robust first-wall materials like tungsten. The management needs to streamline decision-making across all seven international partners to prevent further budget overruns and timeline delays.

3. SMALL MODULAR REACTORS (SMRS)

Small Modular Reactors (SMRs) are advanced nuclear power units with an electricity generation capacity of up to 300 MW per reactor, roughly one-third the size of traditional nuclear plants. **SMR are** designed with factory-assembled modular components, they can be easily transported and installed on-site, offering a flexible and lower-cost nuclear alternative.



Working of small modular rector

- **Core Fission Reaction**
  - **Nuclear Fission in Compact Core:** Low-enriched Uranium (or HALEU fuel) undergoes controlled fission in a small, compact reactor core to release immense heat energy.
  - **Integrated Primary Circuit:** Unlike traditional plants, key components like the steam generator and pressurizer are housed inside a single, factory-built **reactor pressure vessel**.
- **Heat Transfer & Power Generation**
  - **Coolant Circulation:** A primary coolant (water, liquid metal, or molten salt) absorbs heat from the reactor core and transfers it to a secondary loop via heat exchangers.
  - **Turbine Drive:** Secondary water boils into high-pressure steam, which expands to drive a steam turbine connected to an electric generator for grid power.
- **Advanced Safety Mechanisms**
  - **Passive Safety Systems:** Relies on natural physics—such as **gravity, natural circulation, and convection**—to cool the reactor automatically during power failures without human intervention.
  - **Underground Placement & Containment:** Frequently installed underground inside sealed containment vessels, significantly lowering the risk of external radiation leaks or physical attacks.

Types of small modular rector

SMR Type	Coolant Used	Primary Fuel Type	Key Feature
Land-based Water-Cooled SMRs	Light Water (pressurized or boiling)	Low-Enriched Uranium	Most mature tech; adapts proven large-scale reactor designs into compact modules.
High-Temperature Gas-Cooled Reactors (HTGR)	Helium Gas	TRISO fuel particles	Produces ultra-high heat (>700^C); ideal for industrial hydrogen production.
Liquid Metal-Cooled Fast Reactors (LMFR)	Liquid Sodium or Lead	Enriched Uranium / Plutonium	Operates at low pressure with fast neutrons; recycles spent fuel to reduce nuclear waste.
Molten Salt Reactors (MSR)	Liquid Fluoride / Chloride Salts	Liquid-dissolved Uranium / Thorium	Operates at near-atmospheric pressure; cannot melt down because the fuel is already liquid.
Microreactors	Helium / Heat pipes	HALEU (High-Assay Low-Enriched Uranium)	Ultra-compact (<20 MW); easily transportable by truck for remote off-grid locations.

Significance of small modular rector

- **Climate & Clean Energy Transition**
  - **24/7 Zero-Carbon Power:** Provides continuous, weather-independent clean electricity to complement intermittent solar and wind grids.
  - **Accelerates Decarbonization:** Replaces retiring coal-fired power plants using existing grid infrastructure and land sites.
  - **Industrial Decarbonization:** Delivers high-temperature process heat for hard-to-abate sectors like steel production, chemical refining, and hydrogen generation.
- **Financial & Economic Advantages**

- **Lower Initial Capital Cost:** Requires significantly less upfront capital investment than massive traditional nuclear plants, lowering financial risk.
- **Faster Construction Timelines:** Factory fabrication and standardized modular assembly reduce build times from decades to just 3–5 years.
- **Flexible Revenue Models:** Incremental capacity addition allows utilities to add modules one by one as power demand grows over time.

- **Grid & Geographical Flexibility**
  - **Small Physical Footprint:** Requires less land area and reduced emergency planning zones compared to conventional reactors.
  - **Grid Stability for Remote Areas:** Ideal for small or isolated grids, remote mining sites, island nations, and off-grid military bases.
  - **Water-Independent Cooling Options:** Advanced gas or molten salt cooling models can operate in arid regions without access to large water bodies.
- **Enhanced Safety & Operational Reliability**
  - **Built-in Passive Safety:** Operates on natural physical principles (convection, gravity) to shut down and cool safely without human intervention or electricity.
  - **Lower Radioactive Fuel Inventory:** Smaller reactor cores hold far less nuclear material, dramatically reducing the potential impact of an emergency leak.

Issues with Small Modular Reactor

- **High Electricity Costs & Economic Viability**
  - **Higher Cost Per Megawatt:** Loss of "economies of scale" means SMRs can be more expensive per unit of electricity generated compared to massive traditional reactors.
  - **Unproven Commercial Scale:** Factory production cost savings only kick in after hundreds of identical units are ordered and built, which hasn't happened yet.
- **Fuel Supply & Proliferation Risks**
  - **HALEU Supply Monopoly:** Many advanced SMRs require High-Assay Low-Enriched Uranium (HALEU), whose global supply chain is heavily controlled by Russia.

    As per IAEA, Over 80% of the global commercial supply capacity for High-Assay Low-Enriched Uranium is currently controlled by Russia's state firm **TENEX (Rosatom)**
  - **Nuclear Proliferation Concerns:** Distributing many small reactors across remote locations increases the risk of nuclear material theft or diversion.
- **Waste Management & Environmental Issues**
  - **More Waste Per Unit Energy:** Studies suggest some SMR designs may produce more radioactive waste per unit of energy generated than large nuclear plants.

    Stanford University study revealed that SMR designs will increase the volume of nuclear waste requiring management by factors of **2 times to 30 times** per unit energy generated

    Lack of Permanent Repositories: Like traditional nuclear energy, spent fuel management remains unsolved globally due to a lack of permanent deep geological repositories.

    Globally, only **1 country** (Finland - Onkalo) has completed an operational underground deep geological repository for spent nuclear fuel.
- **Regulatory & Licensing Hurdles**
  - **Outdated Regulatory Frameworks:** Nuclear safety laws worldwide are designed for large plants, making the licensing of new, multi-module designs slow and complex.

- **Public Perception & NIMBYism:** Local opposition ("Not In My Back Yard") persists regarding placing reactors closer to urban or industrial zones.

  As per Pew research centre, **over 55% of local populations** resist living within a 10-mile Emergency Planning Zone (EPZ) of micro-reactors.

Way forward

- **Policy & Regulatory Reforms**
  - **Streamline Licensing Procedures:** Regulatory bodies (like India's AERB) must create dedicated, agile safety frameworks for standardized factory-built modules.

    The US Nuclear Regulatory Commission (NRC) established a dedicated 10 CFR Part 53 regulatory framework to fast-track approval processes for factory-standardized SMR modules.
  - **Establish International Harmonization:** Work with the IAEA to align safety and licensing rules across countries to enable seamless global export of SMR designs.

    AERB actively collaborates through the IAEA's Nuclear Safety Standards Committees to align national nuclear safety codes with global baseline rules.
- **Financial & Business Model Support**
  - **Incentivize Private Sector Participation:** Amend domestic atomic energy acts (e.g., India's Atomic Energy Act, 1962) to allow private investment and PPP models in SMR deployment.

    SHANTI Act repeals the Atomic Energy Act (1962) to explicitly allow private companies and public-private joint ventures (PPPs) to build, own, and operate civil nuclear plants.
  - **Provide Financial Risk-Mitigation:** Offer state-backed green bonds, production-linked incentives (PLIs), and long-term Power Purchase Agreements (PPAs) to offset early high costs.

    The Union Budget allocated a ₹20,000 crore Nuclear Energy Mission fund to absorb early R&D costs and de-risk commercial private entry.
- **Fuel Supply Security & R&D Focus**
  - **Diversify HALEU Supply Chains:** Build domestic fuel enrichment capabilities and form international alliances to reduce reliance on foreign fuel suppliers.

    US DOE committed $2.7 billion under the HALEU Availability Program to build domestic commercial enrichment cascades (e.g., Centrus Energy in Ohio).
  - **Scale Factory Manufacturing:** Invest in dedicated, high-tech manufacturing hubs to achieve true **economies of scale** and drive down unit production costs.
- **Infrastructure & Public Acceptance**
  - **Repurpose Retiring Thermal Plants:** Install SMRs at sites of retiring coal-fired plants to reuse existing land, cooling infrastructure, and grid connections.
  - **Proactive Community Engagement:** Build public trust through transparent safety demonstrations and local economic benefit programs to overcome **NIMBY** concerns.

Conclusion

Small Modular Reactors (SMRs) represent a crucial bridge between traditional nuclear energy and a flexible, low-carbon power grid. While high initial costs, fuel supply bottlenecks, and regulatory hurdles remain, streamlined policies and private sector participation can unlock their full potential.

4. BIOFUELS

Biofuels are liquid or gaseous fuels produced from organic materials such as agricultural crops, plant waste, and microalgae. They serve as a clean, renewable alternative to fossil fuels, helping to reduce greenhouse gas emissions in transportation and industrial sectors.

Types of Biofuels

Biofuel Generation	Primary Source	Key Examples
First Generation (1G)	Food crops (sugarcane, corn, edible oilseeds)	Bioethanol, Biodiesel
Second Generation (2G)	Non-food biomass (paddy straw, bagasse, Jatropha)	Cellulosic Ethanol, Bio-CNG
Third Generation (3G)	Microalgae cultivated in wastewater or saline water	Algal Biodiesel, Bio-jet fuel
Fourth Generation (4G)	Genetically modified organisms (GMOs) + engineered crops	Synthetic Biofuels, Solar Fuels

Significance of biofuels

- **Energy Security & Foreign Exchange**
  - **Reduces Oil Import Dependence:** Cuts heavy crude oil import bills, preserving foreign exchange reserves for national development.

    As per Ministry of Petroleum & Natural Gas, biofuels substituted over **310 lakh metric tonnes** of crude oil imports via the Ethanol Blending Programme.
  - **Buffers Fuel Price Shocks:** Protects the domestic economy from volatile global oil price swings and geopolitical supply disruptions.

    Saved **₹1.90+ lakh crore** in foreign exchange while insulating domestic retail fuel prices from global Brent crude price spikes
- **Environmental & Climate Benefits**
  - **Lowers Carbon Emissions:** Burns significantly cleaner than fossil fuels, cutting tailpipe greenhouse gas emissions and urban air pollution.
  - **Mitigates Stubble Burning:** Utilizes crop residue (2G biofuels) for fuel, helping eliminate seasonal agricultural smog in northern India.

    As per NITI Aayog, Commercial 2G bio-ethanol refineries (e.g., IOCL Panipat unit) process **200,000+ tonnes** of agricultural crop residue annually to prevent seasonal air pollution.
- **Agriculture & Rural Economy**
  - **Boosts Farmers' Income:** Provides an extra revenue stream by converting agricultural waste and surplus food grains into cash crops.
  - **Generates Rural Employment:** Creates local job opportunities in biomass collection, processing plants, and supply chain logistics.
  - **Strategic & Industrial Growth**
  - **Promotes Circular Economy:** Transforms urban organic waste and municipal solid waste into compressed bio-gas (CBG).

Enforced a mandatory Compressed Bio-Gas (CBG) blending obligation starting at **1% in FY26, scaling to 5% by FY29** under the SATAT initiative.

- **Decarbonizes Aviation & Heavy Transport:** Serves as a key drop-in option for Sustainable Aviation Fuel (SAF) where electrification is difficult.

National Policy on Biofuels 2018

- **Primary Objectives & Target Timelines**
  - **Ambitious Blending Targets:** Envisaged 20% ethanol blending in petrol (E20) and 5% biodiesel blending in diesel by 2030 (later advanced to 2025–26 via the 2022 amendment).
  - **Import Reduction & Decarbonization:** Aimed to boost domestic clean fuel production to lower crude oil import bills and reduce greenhouse gas emissions.
- **Categorization & Expanded Feedstocks**
  - **Categorized Biofuels:** Classified biofuels into Basic (1G Bioethanol & Biodiesel), Advanced (2G Cellulosic Ethanol & Bio-CNG), and Next-Gen (3G/4G Algal & Synthetic fuels).
  - **Expanded Raw Materials for Ethanol:** Allowed sugarcane juice, sugar beet, sweet sorghum, and starch-containing materials like corn, cassava, and damaged food grains (rotten potatoes, broken rice).
  - **Surplus Grain Utilization:** Permitted using surplus food grains for ethanol production during bumper harvest years, subject to National Biofuel Coordination Committee (NBCC) approval.
- **Financial Incentives & Viability Gap Funding**
  - **Support for 2G Biorefineries:** Announced a Viability Gap Funding (VGF) scheme offering ₹6,000 crore over 6 years for 2G ethanol projects (later implemented via PM JI-VAN Yojana).
  - **Differential Pricing:** Introduced higher off-take purchase prices for non-food/2G ethanol to incentivize private investment in advanced processing plants.

Application of biofuels

- **Transportation Sector**
  - **Road Vehicles:** Blended directly with petrol (Bioethanol up to E20) and diesel (Biodiesel up to B20) to power cars, buses, and commercial trucks.
  - **Aviation Industry:** Processed into Sustainable Aviation Fuel (SAF) to cut carbon footprints in long-haul commercial flights.
  - **Maritime Transport:** Used as low-sulfur marine biofuels to power cargo ships and ocean liners, meeting international maritime emission standards.
- **Industrial & Thermal Energy**
  - **Industrial Boilers:** Burned as solid biomass pellets or compressed bio-gas (CBG) to supply high-temperature heat for factories, textiles, and brick kilns.
  - **Chemical Feedstock:** Replaces petroleum-based inputs to manufacture green chemicals, bioplastics, and bio-lubricants.
- **Rural & Decentralized Power**
  - **Clean Cooking Gas:** Compressed Bio-Gas (CBG/Bio-CNG) produced from agricultural and animal waste powers rural kitchens, replacing traditional firewood and LPG.
  - **Off-Grid Electricity Generation:** Used in localized diesel generators (biomass gasifiers) to supply steady electricity to remote villages and micro-grids.
- **Urban Waste Management**

- **Municipal Solid Waste (MSW) Processing:** Urban organic waste is converted into biomethane, cleaning up landfill sites while producing urban transit fuel.

Issues of biofuels

- **Food Security & Resource Strain**
  - **Food vs. Fuel Dilemma:** Diverting food crops (sugarcane, corn, food grains) to 1G biofuel production can inflate food prices and trigger food inflation.

    Diverted over **25–30 lakh metric tonnes** of sugar annually for ethanol production during peak surplus years.
  - **Intensive Water Depletion:** Cultivating water-heavy feedstocks like sugarcane drastically strains groundwater resources in water-stressed regions.

    Producing 1 liter of sugarcane-based 1G ethanol requires **2,500 to 3,000 liters of water**, severely straining groundwater in water-stressed states like Maharashtra and UP
- **Technological & Supply Chain Bottlenecks**
  - **High Production Cost of 2G/3G Biofuels:** Advanced biofuel technologies (cellulosic and algal) are capital-intensive and struggle with commercial viability.
  - **Supply Chain Disruption:** Collecting, aggregating, and transporting distributed crop residue (paddy straw) to biorefineries remains logistically challenging.
- **Environmental & Ecosystem Concerns**
  - **Indirect Land Use Change (ILUC):** Clearing forests and natural ecosystems to cultivate energy crops releases stored carbon, negating net climate benefits.

    As per US EPA studies indicate ILUC emissions can add **13-104g CO2e/MJ** to corn and sugarcane ethanol.
  - **Soil Health Deterioration:** Excessive removal of crop residue for 2G ethanol depletes vital organic carbon and nutrients from agricultural soil.
- **Vehicle & Infrastructure Compatibility**
  - **Engine Modification Requirements:** Higher ethanol blends (beyond E10/E20) require engine modifications and corrosion-resistant vehicle materials.
  - **Infrastructure Retrofitting Costs:** Transporting, storing, and dispensing higher biofuel blends requires costly updates to pipelines and fuel station pumps.

Measures taken for Biofuels

- **Core Policy Frameworks**
  - **National Policy on Biofuels (Amended 2022):** Advanced the target for 20% ethanol blending in petrol (E20) to 2025–26 and expanded allowable feedstocks.
  - **Ethanol Blended Petrol (EBP) Programme:** Mandates Oil Marketing Companies to sell ethanol-blended petrol to curb import dependency.
- **Financial & Infrastructure Support**
  - **Pradhan Mantri JI-VAN Yojana:** Provides Viability Gap Funding (VGF) to set up commercial and demonstration 2G ethanol projects using agricultural waste.
  - **SATAT Initiative:** Promotes Compressed Bio-Gas (CBG) production from organic waste and guarantees off-take by state oil marketing companies.
  - **Mandatory CBG Blending Obligations:** Phase-wise compulsory blending of Compressed Bio-Gas into domestic CNG and PNG grids.

- **Circular Economy & Waste-to-Wealth Schemes**
  - **GOBARdhan Scheme:** Supports converting cattle dung, crop residue, and farm waste into biogas and organic bio-slurry.
  - **RUCO (Repurpose Used Cooking Oil):** Enables the collection and conversion of used cooking oil into commercial biodiesel.
- **Global & Technological Alliances**
  - **Global Biofuel Alliance (GBA):** Co-founded at the G20 Summit to harmonize international biofuel standards, facilitate trade, and share technology.
  - **Priority Sector Lending:** RBI included Compressed Bio-Gas plants under priority sector lending to ease private bank credit access.

Way forward

- **Feedstock Diversification & Advanced Tech**
  - **Shift to 2G & 3G Feedstocks:** Transition from food-grain-based 1G biofuels to non-food biomass (crop residue, bagasse) and microalgae (3G) to eliminate "Food vs. Fuel" risks.
  - **Commercialize Advanced Biorefineries:** Accelerate funding for 2G ethanol plants (e.g., via *PM JI-VAN Yojana*) to lower processing costs through technology transfer.
- **Infrastructure & Supply Chain Efficiency**
  - **Establish Agrarian Supply Chains:** Set up decentralized biomass aggregation hubs and farm-machinery cooperatives to secure a reliable supply of crop residue.
  - **Mandate Engine & Pump Upgrades:** Require automakers to roll out Flex-Fuel Vehicles (FFVs) and update retail fuel station infrastructure to handle E20 and higher blends safely.
- **Policy, Governance & Global Synergy**
  - **Leverage the Global Biofuel Alliance (GBA):** Work with international partners via GBA to harmonize fuel quality standards and facilitate smooth global biofuel trade.
  - **Ensure Carbon-Neutral Farming:** Implement strict lifecycle assessments (LCA) to ensure biofuel production avoids deforestation, groundwater depletion, or soil degradation.

Conclusion

Biofuels serve as a vital cornerstone for India's clean energy transition by linking energy security, rural development, and environmental sustainability. Overcoming technology costs, supply chain bottlenecks, and resource constraints will be essential to ensure their long-term viability.

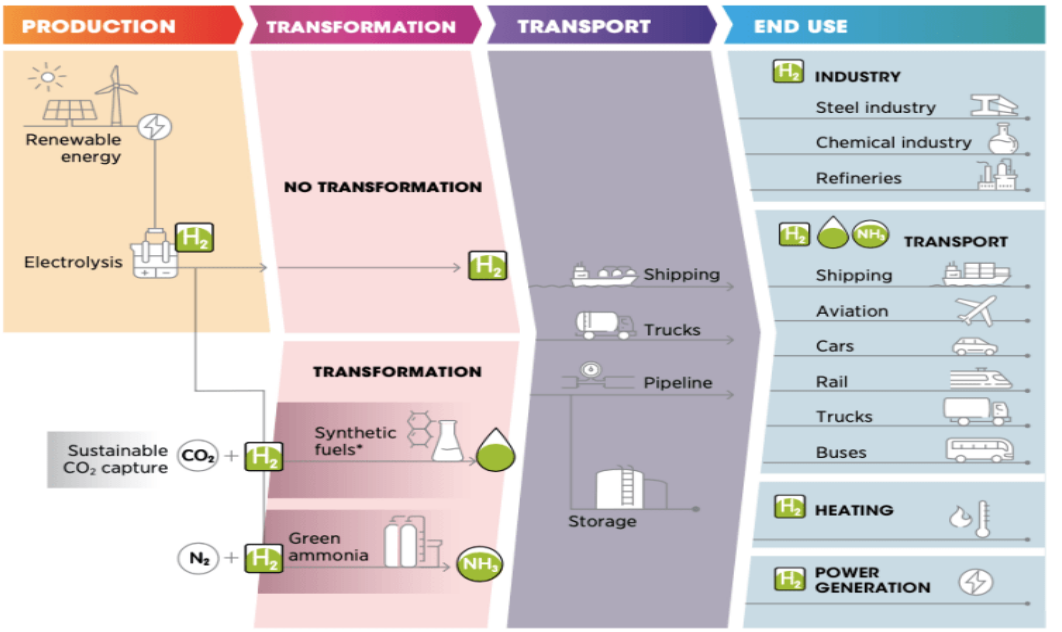
5. GREEN HYDROGEN

Green hydrogen is clean hydrogen gas produced through the electrolysis of water using renewable energy sources like solar and wind. Unlike traditional hydrogen production, this process releases zero greenhouse gas emissions, making it an eco-friendly energy carrier.

Green Hydrogen Production Methods

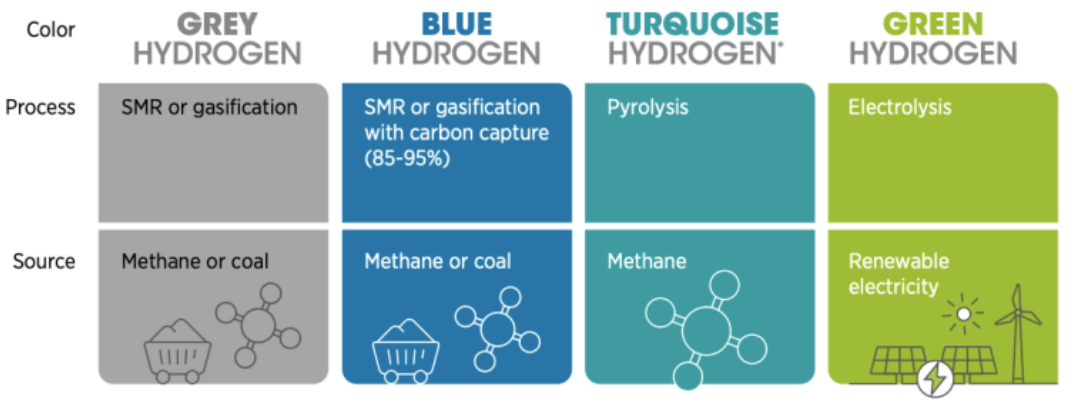
- **Water Electrolysis Technologies (Primary Route)**
  - **PEM (Proton Exchange Membrane) Electrolysis:** Uses a solid polymer electrolyte and pure water; offers fast response times ideal for intermittent renewable power.
  - **Alkaline Water Electrolysis (AWE):** Uses a liquid alkaline solution (KOH/NaOH); the most mature, low-cost commercial method, though less dynamic than PEM.

FIGURE 1.1 Green hydrogen production, conversion and end uses across the energy system



- **Solid Oxide Electrolysis Cells (SOEC):** Operates at high temperatures (700- 800^ C) using steam; highly efficient when integrated with industrial waste heat.
- **AEM (Anion Exchange Membrane) Electrolysis:** Combines the low cost of alkaline systems with the efficiency of PEM; currently emerging in commercial scale.

- **Thermochemical & Solar Thermal Methods**
  - **Solar Thermochemical Water Splitting:** Uses concentrated solar thermal heat (above 1000^C) with metal oxide catalysts to split water molecules directly.
  - **Nuclear-Assisted High-Temperature Electrolysis:** Uses high-temperature heat and power from nuclear reactors to drive steam electrolysis efficiently.
- **Biological & Biomass-Based Routes**
  - **Biomass Gasification with Carbon Capture:** Converts agricultural waste and biomass into syngas using steam at high heat, coupled with CCS to maintain a net-zero footprint.
  - **Photo-Biological Splitting (Dark Fermentation / Algae):** Employs specific green algae or photosynthetic bacteria to produce hydrogen gas directly from water or organic waste under sunlight.



Note: SMR = steam methane reforming.
** Turquoise hydrogen is an emerging decarbonisation option.*

Significance of green hydrogen

- **Decarbonizing Hard-to-Abate Sectors**
  - **Replaces Heavy Fossil Fuel Use:** Serves as a direct zero-emission substitute for coking coal and natural gas in steel, fertilizer, and chemical industries.
  - **Achieves Net-Zero Goals:** Crucial for helping nations meet global climate targets by eliminating emissions from industrial manufacturing and heavy transport.
- **Energy Security & Foreign Exchange**
  - **Reduces Fossil Fuel Import Dependence:** Decreases reliance on imported natural gas, coking coal, and crude oil, conserving valuable foreign exchange.
  - **Insulates Against Geopolitical Volatility:** Shields domestic industries and fertilizer production from global oil price fluctuations and international supply chain shocks.
- **Integration with Renewable Energy**
  - **Solves Renewable Intermittency:** Acts as a long-term energy storage medium by absorbing excess solar and wind electricity produced during peak hours.
  - **Enhances Grid Stability:** Balances national power grid loads by converting excess renewable power into storable hydrogen fuel.
- **Economic Opportunities & Clean Technology Leadership**
  - **Unlocks High-Value Clean Exports:** Positions nations with abundant solar/wind potential to export green hydrogen and green ammonia to energy-importing regions.
  - **Drives Industrial Manufacturing:** Boosts domestic manufacturing of electrolyzers, fuel cells, and high-pressure storage tanks, generating green jobs.

Applications of Green Hydrogen

- **Hard-to-Abate Heavy Industry**
  - **Green Steel Manufacturing:** Replaces coking coal with green hydrogen in Direct Reduced Iron (DRI) processing to produce zero-emission steel.
  - **Chemicals & Fertilizers:** Serves as a clean feedstock to produce green ammonia for agriculture and green methanol for chemical synthesis.
  - **Petroleum Refining:** Replaces fossil-fuel grey hydrogen in hydrocracking and desulfurization during oil refining processes.
- **Transportation & Mobility**
  - **Heavy-Duty Commercial Transport:** Powers long-haul trucks, buses, and heavy mining machinery using hydrogen fuel cells for fast refueling and high range.
  - **Sustainable Aviation Fuel (SAF):** Combined with captured carbon dioxide (CO_2) to synthesize zero-emission synthetic kerosene (e-kerosene) for aviation.
  - **Maritime Shipping:** Converts to green ammonia or liquid hydrogen to fuel zero-emission transoceanic cargo ships.
- **Power Sector & Energy Storage**
  - **Seasonal Grid Storage:** Stores excess seasonal renewable energy (solar and wind) as hydrogen gas for power generation during low-generation periods.
  - **Decentralized & Off-Grid Power:** Supplies clean electricity to micro-grids, remote data centers, and backup power units via stationary fuel cells.
- **Domestic & Thermal Energy**
  - **Gas Network Blending:** Blends directly into natural gas pipelines (up to 10-15%) for domestic cooking and commercial heating to lower carbon intensity.
  - **High-Temperature Industrial Heating:** Serves as a high-heat fuel for cement, glass, and ceramic manufacturing plants where electric heating is unviable.

Challenges with green hydrogen

- **High Production Costs & Economic Parity**
  - **Cost Gap with Grey Hydrogen:** Green hydrogen costs around $ 3.5--5.0 /kg, which is more than double the cost of fossil-fuel-based grey hydrogen ($ 1.5-2.5/kg).
  - **High Electrolyser CAPEX:** Electrolysers (alkaline, PEM, SOEC) remain capital-intensive to manufacture and deploy at industrial scales.
  - **Expensive Input Electricity:** Renewable power accounts for 50--70% of green hydrogen's overall levelized cost.
- **Infrastructure & Transportation Bottlenecks**
  - **Storage Challenges:** Requires high-pressure compression (350- 700 bar) or cryogenic liquefaction (-253^C), demanding energy-intensive infrastructure.
  - **Pipeline Infrastructure Deficit:** Specialized hydrogen-compatible or lined pipelines are lacking, and blending into existing natural gas grids poses embrittlement risks.
  - **Logistical Safety Risks:** Low volumetric density makes long-distance transport complex, expensive, and prone to leakage hazards.
- **Resource & Supply Chain Constraints**
  - **High Water Consumption:** Requires roughly 9-11 liters of demineralized, ultra-pure water per kilogram of hydrogen produced, straining water-scarce regions.
  - **Supply Chain Vulnerability:** Heavy reliance on imported critical minerals (like platinum, iridium, and nickel) for PEM electrolyser catalysts creates import bottlenecks.
- **Market & Demand Uncertainties**
  - **Lack of Mandatory Offtake:** Absence of strict, legally binding blending mandates across steel, refinery, and fertilizer sectors slows private investment.
  - **Incomplete Regulatory Standards:** Global harmonization of green hydrogen certification, safety norms, and carbon-accounting rules is still evolving.

Measures taken in India for green hydrogen

- **National Green Hydrogen Mission (NGHM):** Launched with an initial financial outlay of ₹19,744 crore to produce 5 MMT of green hydrogen annually by 2030.
- **Green Hydrogen Standard Defined:** Officially notified that emissions must not exceed 2 kg CO2 equivalent per kg of hydrogen produced over a 12-month average.
- **SIGHT Programme:** Allocated ₹17,490 crore for PLI-style financial incentives for domestic electrolyser manufacturing and green hydrogen production.
- **ISTS Transmission Charge Waiver:** Granted a 25-year waiver of Inter-State Transmission System (ISTS) charges for renewable energy used in green hydrogen production.
- **Targeted Industrial Pilots:** Budgeted ₹1,466 crore for demonstration pilot projects in hard-to-abate sectors like steel (₹455 Cr), transport (₹496 Cr), and shipping (₹115 Cr).
- **Port-Based Green Hydrogen Hubs:** Developing major ports (e.g., Kandla, Tuticorin, Paradip) as dedicated production, storage, and export hubs.
- **Natural Gas Grid Blending:** Permitted up to 10% green hydrogen blending into existing City Gas Distribution (CGD) networks.
- **SHIP Framework:** Facilitates public-private partnerships for targeted, goal-oriented R&D under the Strategic Hydrogen Innovation Partnership.
- **Skilling Framework:** Introduced dedicated guidelines for upskilling and reskilling workforce capabilities across the green hydrogen value chain.

Global measures for green hydrogen

- **EU REPowerEU Plan:** Targeted producing 10 MMT and importing 10 MMT of renewable hydrogen annually by 2030 to cut reliance on imported fossil fuels.
- **EU Hydrogen Bank:** Acts as an auction mechanism to provide fixed-premium subsidies per kg of green hydrogen produced, bridging the gap between supply and market demand.
- **Breakthrough Agenda (COP26/COP28):** Coordinated international initiative to make affordable, renewable, and low-carbon hydrogen globally available by 2030.
- **IPHE (International Partnership for Hydrogen and Fuel Cells in the Economy):** Facilitates multilateral coordination on safety, infrastructure, and technical fuel cell standards.
- **Clean Hydrogen Mission (Mission Innovation):** Aims to accelerate technological competitiveness and deliver 100 integrated "Hydrogen Valleys" worldwide by 2030.
- **IRENA & ISO Certification Frameworks:** Developing unified global standards for measuring life-cycle carbon intensity to validate "green" hydrogen imports and exports.

Way forward

- **Cost Reduction & Financial Incentives**
  - **Scale Up Domestic Electrolyser Manufacturing:** Fully implement SIGHT scheme incentives to build local manufacturing capacity and reduce electrolyser import costs.
  - **Reduce Input Power Costs:** Extend Inter-State Transmission System (ISTS) charge waivers and optimize round-the-clock (RTC) renewable energy tariffs.

    The Ministry of Power granted a **25-year waiver of Inter-State Transmission System (ISTS) charges** for green hydrogen/ammonia projects commissioned before Dec 2030.
- **Demand Creation & Mandatory Targets**
  - **Enforce Sectoral Offtake Mandates:** Introduce phased green hydrogen purchase obligations (GHPOs) for refineries, fertilizer plants, and steel mills.

    The National Green Hydrogen Mission sets a baseline target of producing at least **5 MMT (Million Metric Tonnes) per annum** by 2030.
  - **Establish Green Public Procurement:** Mandate the use of green steel and green chemicals in government-funded infrastructure projects.

    The Ministry of Steel established 13 task forces to integrate green steel standards into public infrastructure tenders, supporting India's overall **50 MMT/year GHG reduction target** by 2030.
- **Infrastructure Development & Storage**
  - **Develop Integrated Green Hydrogen Hubs:** Establish port-based clusters (e.g., Kandla, Tuticorin, Paradip) for efficient storage, processing, and export.
  - **Build Hydrogen-Ready Pipelines:** Retrofit existing natural gas networks for blending and construct dedicated hydrogen supply pipelines to cut transport costs.

    GAIL initiated pilot projects blending **up to 5% green hydrogen** into natural gas City Gas Distribution (CGD) networks in Indore
- **Innovation & Resource Management**
  - **Target High-Efficiency R&D:** Invest in next-generation electrolyser technologies (like Solid Oxide and AEM) through the SHIP framework to minimize energy loss.

    Allocated **₹400 crore** for R&D under the Strategic Hydrogen Innovation Partnership (SHIP) framework to advance SOEC/AEM electrolyser technologies.

- **Promote Desalination & Recycled Water:** Pair green hydrogen production plants with seawater desalination to avoid over-extracting local freshwater.

- **Standards & Global Synergies**
  - **Harmonize Global Certification:** Align the Green Hydrogen Certification Scheme of India (GHCI) with global standards to enable seamless international trade.
  - **Form Strategic Bilateral Alliances:** Leverage international partnerships (such as the EU-India Trade & Technology Council) for joint R&D and export contracts.

Conclusion

Shift to Green Hydrogen based economy requires a mission mode approach to ensure quick transition without any disruption. The Green Hydrogen Mission is a foresighted initiative in this regard. The focus should be on effective implementation with the target oriented approach. The Green Hydrogen Mission may be the single biggest initiative in transition of the economy to Net Zero.

6. HYDROGEN FUEL TECHNOLOGY

Hydrogen fuel technology utilizes clean-burning hydrogen gas to produce electricity or thermal power through chemical processes, releasing only water vapor as a byproduct. It serves as a zero-emission energy solution that can decarbonize heavy transportation, steel production, and power generation where battery electrification falls short.

Hydrogen fuel technologies can be broadly categorized into two main streams: **Energy Generation Technologies** (how we extract power from hydrogen) and **Production Technologies** (how we generate clean hydrogen fuel).

Major Hydrogen Fuel Technologies

- **Energy Generation & Application Technologies**
  - **Hydrogen Fuel Cells (Electrochemical Conversion):**
  - **Proton Exchange Membrane (PEM) Fuel Cells:** Uses a solid polymer membrane to combine hydrogen and atmospheric oxygen electrochemically, generating electricity and pure water. Primarily used in heavy vehicles (buses, trucks) and electric cars due to fast start-up times.
  - **Solid Oxide Fuel Cells (SOFC):** Operates at very high temperatures using ceramic materials. Highly efficient for stationary grid power, industrial microgrids, and backup power generation.
  - **Direct Methanol & Phosphoric Acid Fuel Cells:** Specialized fuel cell variants used for low-power mobile applications and medium-scale stationary power plants.
  - **Hydrogen Internal Combustion Engines (H2 ICE):** Burns pure hydrogen directly in a modified conventional internal combustion engine, emitting water vapor and trace nitrogen oxides (NOx). Offers an lower-cost immediate alternative for heavy transport and agriculture machinery.
  - **Direct Industrial Hydrogen Reduction:** Replaces coking coal with green hydrogen as a reducing agent in Direct Reduced Iron (DRI) furnaces, enabling direct zero-carbon production of **"Green Steel"**.
- **Clean Production Technologies (Electrolysis)**
  - **PEM Electrolysers:** Uses pure water and a solid membrane to split water molecules efficiently under variable renewable energy supply (ideal for solar and wind power coupling).
  - **Alkaline Electrolysers:** Uses a liquid alkaline solution (KOH) to split water; mature, highly durable, and cost-effective, though physically larger.
  - **Solid Oxide Electrolyser Cells (SOEC):** Uses high-temperature steam to split water, significantly reducing electricity consumption by leveraging industrial waste heat.

Significance of Hydrogen fuel technology

- **Decarbonizing Hard-to-Abate Heavy Sectors**
  - **Replaces Heavy Fossil Fuels:** Acts as a direct, clean substitute for coking coal and natural gas in heavy industries like steelmaking, refineries, and fertilizer manufacturing.
  - **Powers Long-Haul Transport:** Solves the weight and long range limits of lithium batteries by powering heavy-duty trucks, buses, commercial ships, and trains.
- **Boosting Energy Security & Economic Resilience**
  - **Reduces Oil Import Bills:** Reduces India's heavy financial dependency on foreign crude oil, natural gas, and LNG imports.
  - **Maximizes Domestic Energy Self-Reliance:** Enables localized energy generation by leveraging abundant domestic solar and wind energy to produce green hydrogen.
- **Supporting Climate Mandates & Grid Stability**
  - **Drives Net-Zero Targets:** Key technological driver for achieving India's goal of reaching net-zero carbon emissions by 2070.
  - **Enables Large-Scale Renewable Storage:** Functions as a long-duration chemical energy storage medium to store surplus seasonal solar and wind power.
- **Unlocking Export Growth & Economic Value**
  - **Builds Export Competitiveness:** Positions India as a global manufacturing and export hub for green hydrogen and derivative green fuels (like green ammonia).
  - **Creates High-Skilled Jobs:** Generates substantial direct and indirect employment opportunities across manufacturing, infrastructure, and R&D under initiatives like the National Green Hydrogen Mission.

Challenges with hydrogen fuel technology

- **High Capital Costs & Unfavorable Economics**
  - **Expensive Production:** Green hydrogen costs around **3.5 – 5 /kg**, making it vastly uncompetitive against fossil-fuel-based grey hydrogen (**$1.5 – 2 /kg**) without heavy subsidies.
  - **Capital Heavy Equipment:** Advanced electrolysers (PEM/SOEC) and specialized fuel cells rely on expensive, rare precious metals like **platinum and iridium**, driving up upfront equipment costs.
- **Severe Infrastructure & Logistics Constraints**
  - **Storage Hurdles:** Storing hydrogen requires either ultra-high compression (**350 -700 bar**) or cryogenic cooling to liquid form (**-253^C**), consuming significant extra energy.
  - **Lack of Transport Networks:** India lacks dedicated, non-corrosive pipeline grids, forcing reliance on expensive specialized tube trailers or conversion into derivatives like green ammonia for transport.
- **Technical & Environmental Challenges**
  - **High Leakage & Embrittlement:** As the smallest known molecule, hydrogen easily escapes through seals and causes **metal embrittlement** in standard steel pipelines and storage tanks.
  - **Massive Water Footprint:** Producing 1 kg of green hydrogen via water electrolysis requires **9-10 liters of highly purified demineralized water**, posing a major challenge in water-stressed regions.
  - **Energy Conversion Losses:** Conversion steps (electricity to hydrogen, and back to electricity via fuel cells) suffer round-trip efficiency losses of **40%**.
- **Safety & Regulatory Gaps**
  - **High Flammability:** Hydrogen has an extremely wide explosive limit in air and burns with an invisible flame, creating severe operational handling risks.

- **Incomplete Standards:** Lack of global and national harmonized standards for safety certification, pipeline blending thresholds, and carbon credit tracking.

Major Initiatives for Hydrogen Technology in India

- **Mission & Policy Framework**
  - **National Green Hydrogen Mission (NGHM):** Launched in 2023 with a budget outlay of ₹19,744 crore to produce 5 MMT of green hydrogen per year by 2030.
  - **Grid Policy & ISTS Waivers:** Provides a 25-year waiver on Inter-State Transmission System (ISTS) charges and open-access rights for renewable energy used in green hydrogen production.
- **Financial Incentives & Schemes**
  - **SIGHT Programme:** Strategic Interventions for Green Hydrogen Transition, allocating ₹17,490 crore for electrolyser manufacturing and green hydrogen production incentives.
  - **Fertilizer Offtake Procurement:** Long-term tenders managed by SECI to procure green ammonia for domestic fertilizer manufacturing units.
- **Infrastructure & Port-Led Hubs**
  - **Port Green Hydrogen Hubs:** Notified major ports at Kandla (Deendayal), Tuticorin (V.O.C.), and Paradip for hydrogen bunkering, storage, and export.
  - **Coastal Green Shipping Corridor:** Launched between Kandla and Tuticorin supported by green methanol/ammonia refueling facilities.
  - **City Gas Blending Pilots:** GAIL's pilot project in Indore blending green hydrogen into local City Gas Distribution (CGD) networks.
- **Sector-Specific Pilot Projects**
  - **Hydrogen Rail & Transport Mobility:** Pilot projects running hydrogen fuel-cell buses, heavy trucks, and retrofitted hydrogen trains (e.g., Northern Railway's Jind–Sonepat route).
  - **High-Altitude Mobility (Leh):** World's highest-altitude green hydrogen mobility project and fueling station established by NTPC in Leh.
  - **Green Steel Pilots:** Ministry of Steel funding pilot projects to evaluate direct green hydrogen injection for iron reduction in DRI furnaces.
- **R&D & Safety Regulations**
  - **SHIP Programme:** Strategic Hydrogen Innovation Partnership fostering public-private R&D for next-gen electrolysers and novel catalyst materials.
  - **Safety & Regulatory Oversight:** Safety protocols developed under the Petroleum and Explosives Safety Organisation (PESO) and Bureau of Indian Standards (BIS).

Way Forward for Hydrogen Fuel Technology

- **Policy & Financial Interventions**
  - **Scale Subsidies:** Expand SIGHT incentives to bridge the price gap between green and fossil-based hydrogen.
  - **Offer Grid Waivers:** Waive inter-state transmission charges (ISTS) and cross-subsidy surcharges to lower renewable power costs.

    EU Renewable Energy Directive (RED II/III) allowing grid-connected electrolysers to claim zero-emission electricity via temporal and geographic additionality rules.
- **Infrastructure & Logistics Development**
  - **Develop Coastal Hubs:** Build integrated Green Hydrogen Hubs at ports like Kandla, Tuticorin, and Paradip for storage and export.

Port of Rotterdam (Netherlands) developing a dedicated import/export hydrogen hub with a centralized distribution backbone across Europe.

- **Repurpose Gas Networks:** Blend green hydrogen into existing city gas networks to lower initial transport costs.

HyBlend Initiative (US DOE) and **HyDeploy Project (UK)**, testing safe injection thresholds into existing natural gas pipelines.

- **Indigenous Technology & R&D**
  - **Fund Low-Cost R&D:** Drive innovation via SHIP to produce noble-metal-free and high-efficiency electrolysers.

US Department of Energy's Hydrogen Shot Initiative, targeting a reduction in green hydrogen costs to $1 per 1 kg in 1 decade ("111 Goal").

  - **Develop Alternative Feedstocks:** Promote direct seawater and wastewater electrolysis to save precious freshwater resources.
- **Market Demand & Offtake Creation**
  - **Enforce Usage Mandates:** Introduce mandatory consumption obligations in refining, steel, and fertilizer manufacturing.
  - **Deploy Sector Pilots:** Expand hydrogen mobility and steel production pilot projects to test commercial feasibility.

HYBRIT Project (Sweden), operating the world's first fossil-free steel plant using green hydrogen, and **Coradia iLint (Germany)** hydrogen-powered passenger trains.

- **Safety, Regulations & Global Standards**
  - **Enforce Safety Protocols:** Frame comprehensive national safety and leak-detection standards for handling high-pressure gas.

US EPA & ISO/TC 197 Standards, governing safety systems, leak detection, and material compatibility for high-pressure gas handling.

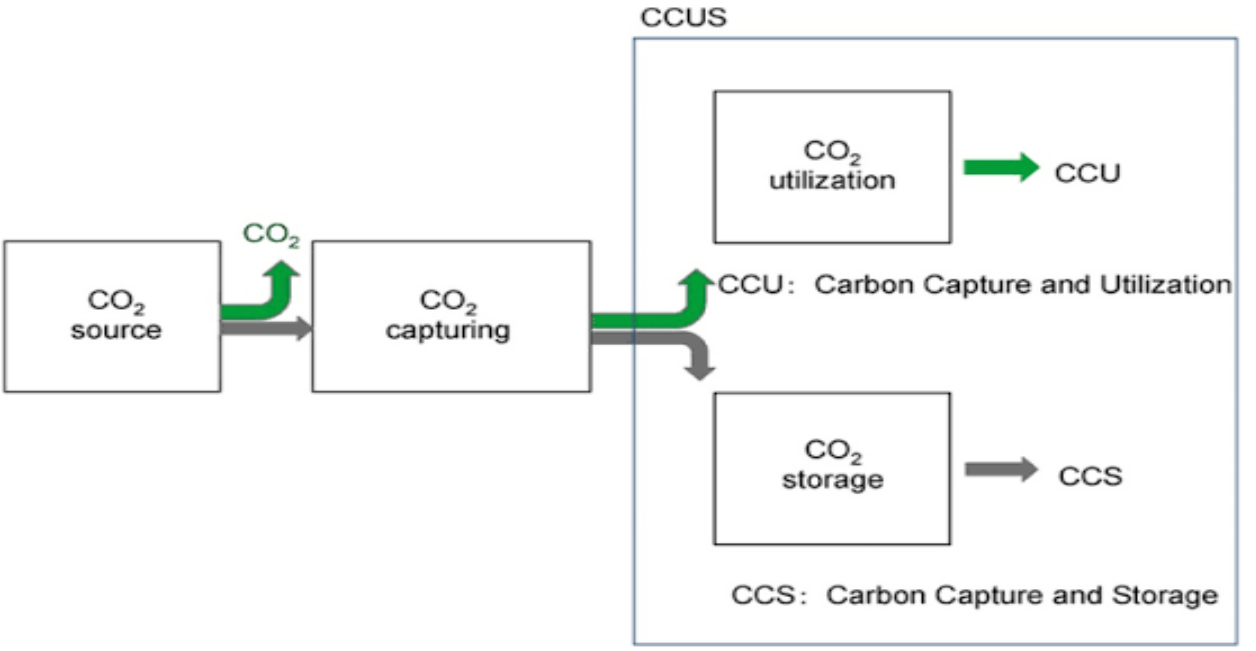
  - **Align Global Certification:** Establish international trade partnerships and harmonized carbon certification to boost exports.

Conclusion

Hydrogen fuel technology serves as a critical game-changer for decarbonizing hard-to-abate industrial sectors, strengthening India's national energy security, and achieving its net-zero target by 2070. By leveraging domestic renewable resources under the National Green Hydrogen Mission, India can reduce its heavy reliance on crude oil imports while transforming into a global manufacturing and export hub.

7. CARBON CAPTURE UTILIZATION AND STORAGE (CCUS)

Carbon Capture, Utilisation, and Storage (CCUS) is a strategic suite of technologies designed to capture carbon dioxide emissions directly from industrial smokestacks or the atmosphere. The captured emissions are subsequently transported and either converted into commercial products like fuels and concrete or permanently sequestered deep underground.



About Carbon capture utilization and storage

- **Capture:** Separates CO2 directly from industrial smokestacks using chemical solvents (post-combustion/pre-combustion) or directly from ambient air (Direct Air Capture).
- **Utilisation:** Converts captured carbon into commercial products like urea, methanol, synthetic fuels, building aggregates, and polymers.
- **Storage:** Injects remaining CO2 deep underground into depleted oil/gas fields or deep saline aquifers for permanent isolation.
- **India's Target:** 750 mtpa (Million Tonne Per Annum) of CO2 capture, utilization, and storage in NITI Aayog's report titled ‹Carbon Capture, Utilisation, and Storage Policy Framework.

Significance of CCUS for India

- **Decarbonizing Heavy Industries**
  - **Solves Process Emissions:** Captures irreducible, inherent chemical reaction emissions in hard-to-abate sectors like cement, steel, and refineries where fuel-switching is impossible.
  - **Avoids Stranded Assets:** Prevents the early retirement of relatively young, capital-intensive thermal power plants and industrial plants across India.
- **Supporting Climate & Net-Zero Mandates**
  - **Achieves Net-Zero Targets:** Serves as a vital technological pillar to bridge the gap toward India's target of achieving net-zero emissions by 2070.
  - **Meets NDC Commitments:** Direct lever to fulfill India's updated Nationally Determined Contribution (NDC) of reducing GDP emission intensity by 45% from 2005 levels by 2030.
- **Unlocking the Clean Energy**
  - **Enables Low-Cost Blue Hydrogen:** Allows India to produce low-carbon hydrogen from abundant domestic coal reserves via gasification before green hydrogen scales up.
  - **Transforms Carbon into Economic Assets:** Converts captured CO2 into high-value commercial chemicals (methanol, urea, soda ash) and synthetic fuels, reducing import bills.
- **Geological Potential & Employment Generation**

- **Utilizes Vast Domestic Storage Capacity:** Capitalizes on India's estimated **350+ Gigatonnes** of underground geological storage potential in saline aquifers and basalt formations.
- **Spurs High-Skilled Job Growth:** NITI Aayog projects that establishing a national CCUS industrial ecosystem could create **8-10 million** direct and indirect jobs by 2050.

Carbon Utilization Pathways for India's Self-Sufficiency

Areas	Use of Captured CO2
Green Urea	**Replaces traditional LNG/NG-based** production of ammonia and urea
Food & Beverage (F&B) Applications	Used in **carbonated drinks, dry ice**, and modified atmosphere packaging
Building Materials	Industrial waste (like steelmaking slag) is directly **injected into aggregates and concrete**
Chemicals	Manufacturing **methanol and ethanol** (vital for producing value-added chemicals like acetic acid)
Polymers	Manufacturing **bio-plastics** used in laptop packaging, cell phone casings, etc.
Enhanced Oil Recovery (EOR)	Producing **low-carbon oil** from aging and mature oil fields

Issues with CCUS

- **High Economic Costs**
  - **Capital Heavy:** As per *NITI Aayog CCUS Roadmap*, Carbon capture equipment alone accounts for **70-80%** of total project expenditure, requiring an estimated **$100 -150 billion** to scale up to 750 MTPA capacity by 2050.
  - **Raises Output Prices:** Implementing CCUS can increase the cost of electricity by **60-80%** and inflate green steel and cement prices, threatening export competitiveness.
- **Energy Penalty & Operational Efficiency Loss**
  - **Consumes Heavy Parasitic Power:** Capturing and compressing CO2 creates an **energy penalty**, consuming **15-25%** of a plant's total power generation and reducing overall efficiency.
  - **Retrofitting Complexity:** Retrofitting old, space-constrained thermal power and steel units is technically difficult and cost-prohibitive.
- **Geological & Environmental Risks**
  - **Complex Basaltic Formations:** A large portion of India's storage potential lies in basalt (Deccan Traps), where long-term CO2 mineralization is technically challenging and unproven at commercial scale.
  - **Seismic & Leakage Threats:** Injecting high-pressure CO2 risks triggering micro-seismicity or accidental leaks that can pollute local groundwater aquifers.
- **Infrastructure, Market & Regulatory Gaps**
  - **Missing Supply Network:** India lacks dedicated, high-pressure CO2 cross-country pipeline networks to link industrial source clusters with storage sinks.
  - **Limited Utilization Demand:** Industrial markets for CO2 utilization (urea, soda ash, fire extinguishers) absorb **less than 5%** of overall emissions.
  - **Legal Liability Vacuum:** Lack of clear long-term regulations defining legal responsibility, safety oversight, and carbon credit accounting in the event of subterranean leaks.

Way forward

- **Policy & Financial Support**
  - **De-Risk Upfront Capital:** Deploy Viability Gap Funding (VGF) and specialized Production Linked Incentives (PLI) using the Union Budget's ₹20,000 crore CCUS allocation to bridge high capital costs.

    US Inflation Reduction Act (IRA) expanding tax credits to lower private investment risks for carbon management.
  - **Implement Production Tax Credits:** Adopt a domestic carbon tax credit framework (similar to the US Section 45Q model) offering direct tax deductions per tonne of CO2 captured, stored, or utilized.
  - **Monetize via Carbon Markets:** Integrate CCUS offsets directly into the Indian Carbon Credit Trading Scheme (CCTS) to create tradeable financial incentives for early-adopting industries.
- **Infrastructure & Operational Efficiency**
  - **Develop Regional Industrial Clusters:** Establish shared **Hub & Cluster** models in high-emission industrial zones (e.g., Odisha, Gujarat) to pool transport pipelines and storage infrastructure, drastically lowering per-tonne costs.

    Planned CCUS Industrial Hubs & Clusters in high-emission zones like **Gujarat** (Hazira/Dahej) and **Odisha** (Kalinganagar).
  - **Prioritize Utilization Over Storage (CCU):** Focus on converting captured CO2 into commercial commodities like green urea, methanol, and carbonated concrete to drive immediate economic returns in price-sensitive markets.
  - **Build Shared CO2 Pipelines:** Construct dedicated, open-access trunk pipelines connecting inland industrial clusters to coastal ports or offshore geological sinks.

    Alberta Carbon Trunk Line (ACTL) in Canada, a 240 km multi-user open-access pipeline capable of transporting up to 14.6 million tonnes of CO2 annually.
- **Research, Technology & Environmental Safety**
  - **Scale Up Indigenous R&D:** Fund second-generation capture methods (e.g., advanced solid sorvents, membrane technologies, and Solid Oxide Electrolyser Cells) through National Centres of Excellence to reduce the operational energy penalty.
  - **Advance Basalt Mineralization:** Accelerate geological pilot research for safe, permanent CO2 mineralization within the Deccan Traps basalt formations.

    Carbfix project in Iceland, which dissolves CO2 in water and injects it into basaltic rock to turn it into stone within two years.
  - **Map Geological Storage Sinks:** Conduct high-resolution seismic mapping to accurately quantify and certify India's deep saline aquifers and depleted hydrocarbon fields for permanent storage.
- **Governance, Regulation & Global Collaboration**
  - **Enforce Regulatory Frameworks:** Establish clear Bureau of Indian Standards (BIS) and AERB/ASNR-style guidelines governing subterranean CO2 injection, leak monitoring, and long-term legal liability.
  - **Mandate Green Procurement:** Introduce mandatory "Green Steel" and "Green Cement" procurement quotas for public infrastructure projects to build sustained market demand for low-carbon products.
  - **Leverage Global Partnerships:** Engage in multilateral initiatives like *Mission Innovation* to pool technology transfer, lower learning curves, and attract global climate finance.

    Active participation in **Mission Innovation (Innovation Challenge 3)** and bilateral MoUs with countries like Norway and the US for CCUS technology transfer.

Conclusion

India is the world's third-largest carbon dioxide emitter after China and the United States, making large-scale decarbonization an urgent national priority. Adopting CCUS is essential for India to fulfill its Nationally Determined Contribution (NDC) commitments and progress toward long-term climate targets. However, the technology faces major hurdles, including steep initial setup costs, significant operational energy penalties, and a lack of dedicated regulatory frameworks.

8. FLEX FUEL VEHICLES

Flex fuel vehicles (FFVs) are modified automobiles capable of running on petrol, pure ethanol, or any blended combination of the two in a single fuel tank. Equipped with specialized engine sensors and corrosion-resistant components, they automatically adjust engine parameters based on the fuel mix. They play a critical role in reducing carbon emissions and lowering national crude oil import bills.

How Do Flexible Fuel Cars Work Using Ethanol?

- **Fueling & Storage**
  - **Single Fuel Tank:** Accepts petrol, pure ethanol, or any random mixture of both directly into a single, unified fuel tank.
  - **Corrosion-Resistant Components:** Features specialized rubber, plastic, and stainless steel fuel lines to prevent high-ethanol fuel from degrading the system.
- **Detection & Sensing**
  - **Fuel Composition Sensor:** Analyzes the exact alcohol-to-petrol ratio in real time as the fuel flows toward the engine.
  - **Instant Signal Transmission:** Sends continuous data regarding ethanol percentage to the vehicle's central computer.
- **Electronic Control Unit (ECU) Adjustment**
  - **Smart Parameter Tuning:** The engine control unit instantly recalculates and adjusts the air-fuel mixture, ignition timing, and fuel injection volume.
  - **Optimized Combustion:** Compresses and ignites the fuel mixture at the precise millisecond needed, compensating for ethanol's lower energy density and higher octane rating.
- **Engine Power & Exhaust**
  - **Specialized Fuel Injectors:** Spray larger volumes of fuel into the engine cylinders when running on high-ethanol blends to maintain performance.
  - **Tailpipe Emission Control:** Burns cleaner during combustion, releasing lower levels of carbon monoxide and hydrocarbons through the catalytic converter.

Significance of Flex fuel vehicles

- **Energy Security & Foreign Exchange Savings**
  - **Reduces Oil Import Dependence:** Cuts down expensive crude oil import bills by substituting petrol with domestically produced bioethanol.

    Full-scale FFV adoption running on high-blend ethanol (e.g., E85/E100) can replace over **302 lakh metric tonnes** of crude oil imports over time.
  - **Saves Foreign Exchange:** Conserves valuable foreign currency reserves, strengthening national balance-of-payments resilience.

As per ministry of petroleum, India's Ethanol Blending Programme (EBP) has already saved over **₹1.90+ lakh crore** in foreign exchange reserves since 2014.

- **Agricultural & Rural Economy Boost**
  - **Increases Farmers' Income:** Creates an assured market for cash crops like sugarcane, maize, and damaged food grains used in ethanol production.

    Ethanol procurement has routed **₹1.60+ lakh crore** directly to sugarcane, maize, and grain farmers.
  - **Promotes Rural Industrialization:** Drives investments in local bio-refineries, creating non-farm jobs in agrarian belts.
- **Environmental & Climate Benefits**
  - **Lowers Tailpipe Carbon Emissions:** Reduces greenhouse gas emissions (CO2) and toxic pollutants like carbon monoxide (CO) and hydrocarbons (HC).
  - **Stubble Burning Mitigation:** Utilizes agricultural crop residue (2G ethanol) as feedstock, helping curb seasonal stubble burning and air pollution.

    Advanced 2G ethanol commercial plants (e.g., IOCL Panipat unit) utilize **200,000 tonnes of paddy straw** per plant annually, directly preventing open-field crop residue burning.
- **Consumer Flexibility & Technology Transition**
  - **Fuel Choice Flexibility:** Allows consumers to seamlessly switch between petrol, high-blend ethanol (like E85), or pure ethanol based on fuel prices and availability.
  - **Bridge Technology:** Acts as an immediate, scalable step toward clean mobility without requiring massive changes to existing combustion engine manufacturing.

    Rolling out FFV infrastructure is estimated to be **10-15 times** cheaper in capital expenditure compared to building out a nationwide fast-charging EV network.

Ethanol blending program

- **Key Concept & Objective**
  - **Definition:** A government initiative to blend plant-derived ethanol into petrol to reduce fossil fuel dependency, cut emissions, and boost rural economies.
  - **Core Target:** Achieved the target of 20% ethanol blending in petrol (E20) well ahead of the original 2030 timeline.
- **Key Significance & Benefits**
  - **Reduces Oil Import Bills:** Saves foreign exchange reserves by replacing imported crude oil with domestically produced biofuel.
  - **Boosts Agricultural Economy:** Provides an assured income stream for sugarcane, maize, and grain farmers by monetizing excess crop yields.
  - **Lowers Carbon Emissions:** Reduces tailpipe greenhouse gas emissions (CO2) and toxic pollutants like carbon monoxide (CO).
  - **Mitigates Stubble Burning:** Utilizes agricultural waste and crop residue for 2G ethanol production, helping curb seasonal air pollution.

Challenges with flex- fuel vehicles

- **Technical & Vehicle Maintenance Issues**
  - **Corrosion & Component Wear:** High ethanol concentrations degrade rubber, plastic, and metal

parts in non-adapted fuel systems.
    - **Lower Fuel Efficiency:** Ethanol has a lower energy density than petrol (30% less energy per liter), resulting in reduced fuel mileage.
    - **Cold-Start Difficulties:** High-ethanol blends (like E85) struggle to vaporize at low temperatures, causing starting issues in cold climates.
- **Environmental & Resource Trade-offs**
    - **Food vs. Fuel Dilemma:** Diverting edible crops like sugarcane and maize for ethanol risks increasing food prices and threatening food security.
    - **High Water Footprint:** Cultivating water-intensive feedstock crops like sugarcane places severe stress on local groundwater reserves.
    - **Land-Use Change Emissions:** Clearing land for intensive biofuel crop farming can offset the overall net carbon savings of ethanol.
- **Economic & Infrastructure Bottlenecks**
- **Higher Upfront Vehicle Cost:** Integrating anti-corrosive materials and specialized engine sensors raises vehicle manufacturing costs for consumers.
- **Lack of Dispensing Infrastructure:** Fuel stations require specialized, corrosion-resistant underground tanks and pumps to handle high-ethanol blends.
- **Price Parity Challenge:** Lower fuel mileage makes FFVs economically unattractive unless ethanol retail prices are kept significantly cheaper than petrol.

Ethanol

Ethanol is a clear, colorless liquid alcohol produced naturally by fermenting sugars from organic matter like sugarcane, corn, and crop waste. It serves as a clean-burning biofuel that can be blended directly into petrol to reduce vehicle emissions and fossil fuel consumption. Because it is derived from renewable plant sources, ethanol plays a key role in reducing carbon footprints and promoting energy self-reliance.

Way forward

- **Policy Incentives & Tax Rationalization**
    - **Reduce GST on Flex Fuel Vehicles:** Lower the GST rate on FFVs (currently 28%) to narrow the price gap with conventional vehicles and boost buyer adoption.
    - **Maintain Differential Fuel Pricing:** Ensure retail prices for high-blend ethanol (like E85) are kept lower than petrol to compensate for reduced fuel mileage.
- **Feedstock Diversification & Sustainable Farming**
    - **Shift to 2G/3G Biofuels:** Accelerate transition toward non-food feedstocks like agricultural residue, bamboo, and municipal waste to resolve the food-vs-fuel trade-off.
    - **Promote Water-Efficient Crops:** Transition feedstock sourcing toward less water-intensive crops like sweet sorghum and maize to protect groundwater reserves.
- **Infrastructure Development & Retailing**
    - **Expand Dedicated E85 Dispensing:** Mandate Oil Marketing Companies (OMCs) to install ethanol-compliant pumps and underground tanks at fuel stations nationwide.
    - **Ensure Supply Chain Security:** Develop dedicated regional storage facilities and long-term procurement contracts to prevent seasonal ethanol supply shortages.
- **R&D & Technological Upgradation**
    - **Advance Cold-Start Solutions:** Invest in advanced engine technologies (e.g., dual-injection systems and heated fuel injectors) to solve cold-weather starting issues.

- **Indigenous Sensor Manufacturing:** Localize manufacturing of anti-corrosive engine materials and fuel-composition sensors to lower production costs for automakers.

Measures Taken in India

- **E85 Fuel Debut & Infrastructure Expansion:** Launched the country's first public E85 dispensing stations in major metropolitan corridors (Delhi-NCR, Mumbai-Pune-Nagpur), with plans targeting 5,000 high-blend pumps nationwide.
- **Prototypes & Product Rollouts:** Automakers (e.g., Maruti Suzuki, Hero MotoCorp, TVS) unveiled initial flex-fuel two-wheelers and passenger vehicles capable of handling fuel blends up to E85/E100.
- **Regulatory Fuel Standards Framework:** Notified official technical fuel and safety specifications for higher ethanol blends, including E22, E25, E27, E30, E85, and E100.
- **Tax Rationalization on Biofuel:** Reduced Goods and Services Tax (GST) on ethanol supplied for blending from 18% to 5% to lower raw material and production costs.
- **Pricing & Discount Strategies:** Introduced E85 fuel at a regulated discount of approximately ₹20/liter relative to petrol to compensate for ethanol's lower mileage.

Global Measures

- **Brazil (Pro-Álcool & Mass Market Adoption):** Pioneered mandatory ethanol blending since 1975, expanding E100 hydrous ethanol retail coverage to over 90% of filling stations and making FFVs 85% of new car sales.
- **Brazil (Fiscal & Regulatory Incentives):** Offered reduced industrial tax rates (IPI) for flex-fuel cars, mandated state oil company procurement, and adjusted fuel taxation to maintain ethanol price competitiveness.
- **United States (Inflation Reduction Act & Renewable Fuel Standard):** Leverages the RFS program alongside federal tax subsidies to fund E85 fueling infrastructure across agricultural belts.
- **US & European Conversion & Blend Regulations:** France and Sweden introduced streamlined legal certification and tax exemptions for E85 conversion kits on existing gasoline engines.
- **Global OEM Technology Investments:** Global automakers (Volkswagen, GM, Fiat, Ford) engineered advanced, low-cost ECU sensor maps and higher compression engines specifically for high-ethanol markets.

Conclusion

Flex Fuel Vehicles offer a viable bridge technology to accelerate India's transition toward low-carbon mobility and energy self-reliance. By aligning ethanol production with sustainable farming and robust retail infrastructure, FFVs can effectively balance environmental goals with economic growth.

MGP/BP/GS3/023

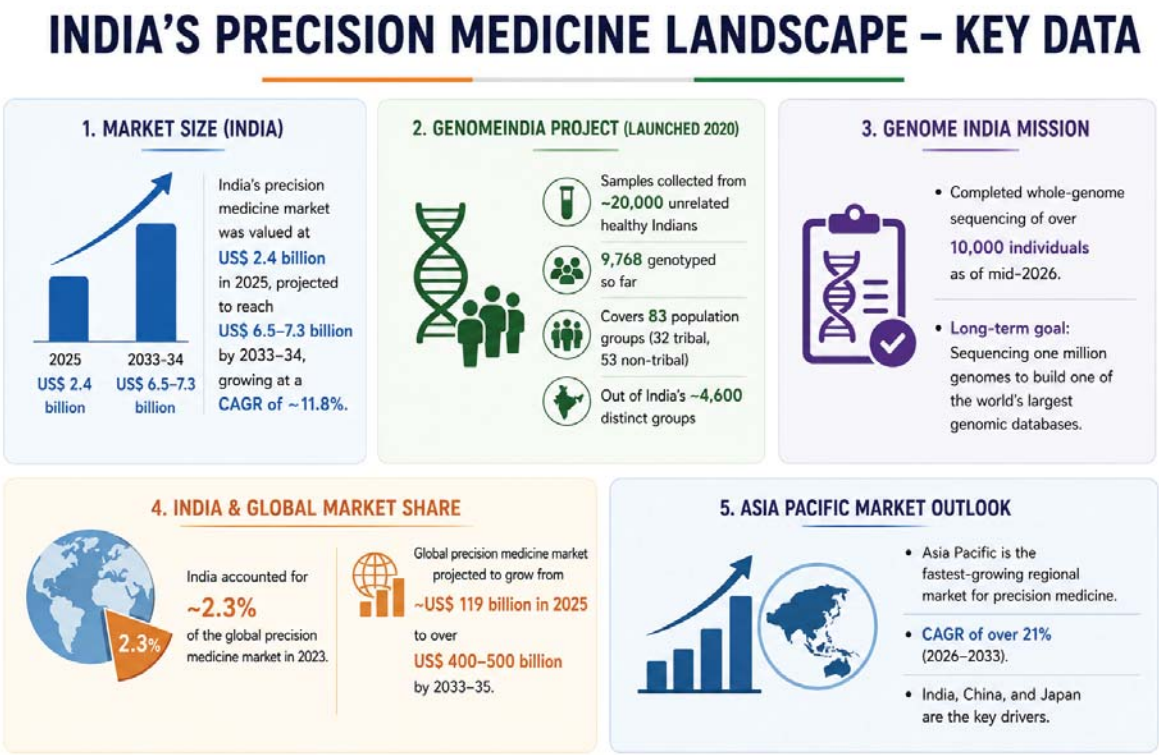
HEALTH SCIENCES

1. PRECISION MEDICINE

Precision medicine (also called **personalized medicine**) is a healthcare approach that tailors **prevention, diagnosis, and treatment** to an individual's **genetic makeup, lifestyle, and environmental factors**, instead of following a 'one-size-fits-all' model.

It relies on **genomics, molecular diagnostics, and data analytics** to identify the exact biological basis of a disease in a given patient.

India, with its enormous **genetic diversity** — over **4,000-5,000 distinct communities** and around **4,600 population groups** — is seen as uniquely positioned to both benefit from and contribute to global precision medicine research, especially for diseases with strong genetic and population-specific patterns.



Government Policy and Institutional Framework

- **BioE3 Policy (Biotechnology for Economy, Environment and Employment):** The Department of Biotechnology (DBT) has identified **Precision Biotherapeutics** as a priority thematic area, supporting research, development, and manufacturing of indigenous **mRNA therapeutics, monoclonal antibodies, and cell and gene therapies**.

- **GenomeIndia Project:** A pan-India genomic diversity initiative led by DBT, involving a consortium of **20 national institutes** including **CSIR-IGIB (Delhi)**, **CSIR-CCMB (Hyderabad)**, and **BRIC-NIBMG (Kalyani)**; data is shared through the **Indian Biological Data Centre (IBDC) portal**, governed by the **Biotech-PRIDE** guidelines.
- **UMMID Initiative (Unique Methods of Management and treatment of Inherited Disorders):** Integrates genetic diagnostics, prenatal and newborn screening, and genetic counselling into a unified public health model. It has benefited nearly **three lakh individuals** and set up nearly **30 NIDAN Kendras** for advanced diagnostics beyond metro cities.
- **National Biobank:** Aims to build a nationwide cohort of **10,000 individuals**, capturing genomic, lifestyle, and clinical data to support long-term research into disease patterns and personalized interventions.
- **ICMR's Role:** The **Indian Council of Medical Research** has issued **national ethical guidelines** on biomedical research, biobanking, genomic data use, and gene therapy product development, alongside facilities like **mPRAGATI at IIT Delhi** for precision diagnostic design and validation.
- **Ayushman Bharat Digital Mission (ABDM):** By creating **longitudinal digital health records**, ABDM lays the data infrastructure that can eventually support AI-driven, personalised treatment pathways at scale.

Applications of Precision Medicine

(a) Oncology (Cancer Care)

- **Precision Oncology:** Biomarker-based testing enables **targeted therapies** for specific cancer subtypes, improving outcomes and reducing side-effects;

> This is especially critical given India recorded over **1.4 million new cancer cases** in 2023 (ICMR estimate).

(b) Rare and Genetic Disorders

- **Gene Therapy for Haemophilia:** Indian researchers have successfully demonstrated an **indigenous gene therapy for haemophilia**, a landmark achievement showcased by the government in 2026 as proof of India's growing gene-therapy capability.
- **Newborn and Prenatal Screening:** Programmes like **UMMID** use genetic diagnostics to detect inherited disorders early, enabling timely intervention and counselling for families.

(c) Infectious Disease and Public Health

- **Pathogen Genomics:** Genomic surveillance tools, refined during the **COVID-19 pandemic**, are now being used to track viral evolution (e.g., **Zika virus** cases) and guide targeted public health responses.

(d) AI-Enabled Diagnostics and Telemedicine

- **Multi-omics and AI Integration:** Combining **genomics, transcriptomics, and proteomics** with AI and big-data analytics allows clinicians to design **personalized treatment plans** based on a patient's genetic makeup, lifestyle, and environment.
- **Hybrid Healthcare Models:** AI-enabled **telemedicine** is connecting patients in remote and underserved regions with specialists, extending precision diagnostics beyond metropolitan hospitals.

Significance of Precision Medicine for India

Dimension	Role of Precision Medicine
Disease Burden Management	Targeted therapies improve outcomes for **cancer, cardiovascular disease, and diabetes**, which together drive a large share of India's **non-communicable disease (NCD)** burden.
Rare Disease Care	**UMMID** and genetic screening enable early diagnosis of inherited disorders, reducing suffering through timely intervention and counselling.
Scientific Leadership	India's unmatched **genetic diversity** offers a scientific edge — findings from Indian genomic research can benefit global healthcare, not just domestic patients.
Atmanirbhar Bharat in Biotech	The **BioE3 Policy** promotes indigenous manufacturing of mRNA therapeutics, monoclonal antibodies, and gene therapies, reducing import dependence on costly foreign biologics.
Access and Equity	**NIDAN Kendras** and telemedicine extend precision diagnostics beyond metros to **Aspirational Districts** and underserved regions.

Challenges Facing Precision Medicine in India

- **Cost Barriers:** Genomic tests and targeted therapies remain **expensive**, limiting access for lower-income and rural populations; insurance reimbursement for genetic testing is still limited.
- **Data Privacy Concerns:** Handling sensitive **genetic and health data** at scale raises risks of misuse, discrimination, or breaches, requiring robust data-protection safeguards.
- **Skill and Infrastructure Gaps:** Shortage of trained genetic counsellors, bioinformaticians, and molecular diagnostic labs, especially outside major cities.
- **Limited Genomic Representation:** Despite progress, **only a small fraction of India's population diversity (83 of about 4,600 groups) has been sampled so far under GenomeIndia**, limiting the generalisability of findings.
- **Regulatory and Ethical Complexity:** Gene therapy, biobanking, and genomic data-sharing require clear, harmonised ethical and regulatory frameworks that are still evolving.
- **Rural-Urban Divide:** Advanced diagnostics and precision therapeutics remain concentrated in metropolitan hospitals, despite initiatives like NIDAN Kendras.

Way Forward	Measures Needed
1. Expanding the GenomeIndia Project	**Increase genome sequencing across diverse population groups** to build a comprehensive national genomic database.
2. Strengthening Genomic Data Protection	Develop robust privacy and cybersecurity safeguards for genetic data, building on the **Digital Personal Data Protection (DPDP) Act**.
3. Improving Affordability	Expand insurance coverage under **Ayushman Bharat (PM-JAY)** for genomic testing and targeted therapies.
4. Building Skilled Manpower	Train more **genetic counsellors, bioinformaticians, and precision diagnostics professionals** through dedicated programmes.
5. Promoting Indigenous Manufacturing	Strengthen domestic production of **mRNA therapeutics, monoclonal antibodies, and gene/cell therapies** under the **BioE3 Policy**.

6. Expanding Rural Access	Scale up **NIDAN Kendras** and **AI-enabled telemedicine** to deliver precision medicine in rural and underserved areas.

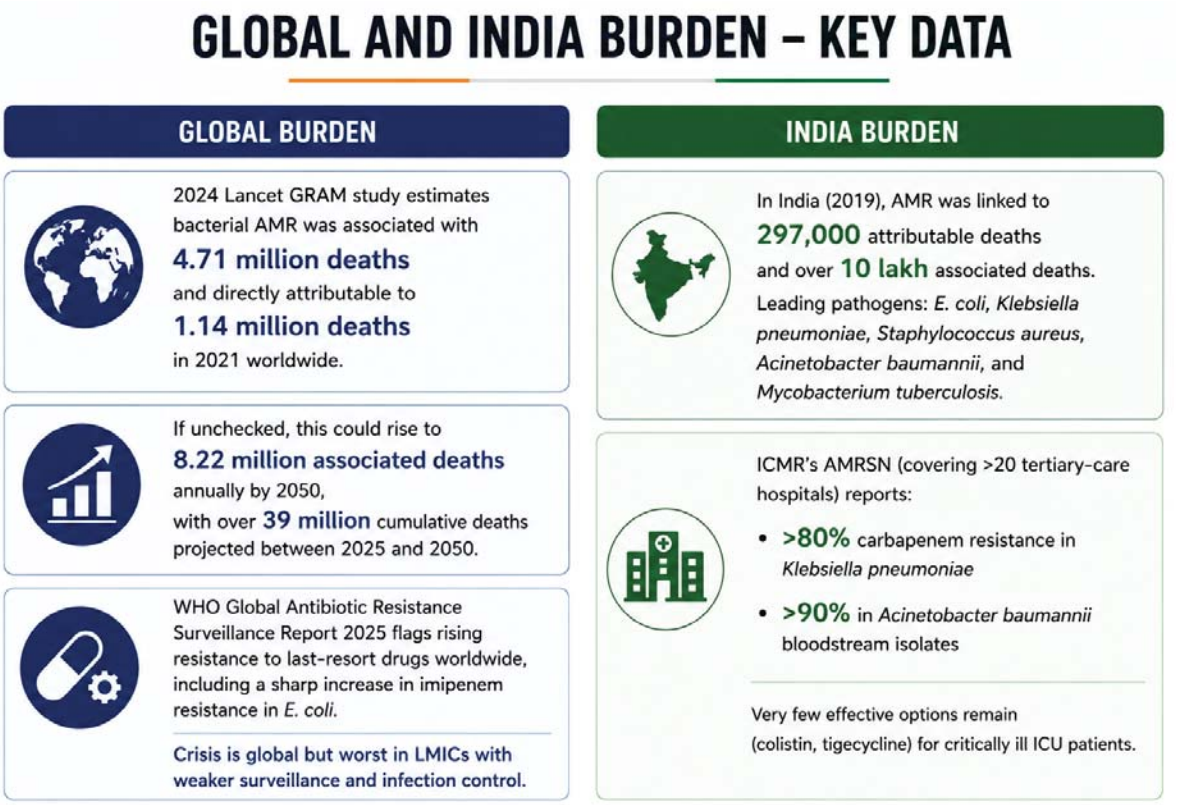
Conclusion

A strong ecosystem of **genomic databases, indigenous innovation, skilled manpower, and ethical regulation** will enable India to make precision medicine **accessible, affordable, and inclusive**.

2. ANTI-MICROBIAL RESISTANCE (AMR)

Antimicrobial Resistance (AMR) is the ability of **bacteria, viruses, fungi, and parasites** to resist antimicrobial medicines, making infections difficult to treat. It is a **"silent pandemic"** that threatens surgeries, cancer treatment, organ transplants, and neonatal care.

India has one of the world's highest AMR burdens due to **high antibiotic misuse, infectious disease load, and antibiotic use in livestock**. Addressing AMR requires a **One Health approach**, integrating **human, animal, agricultural, and environmental health**.



Government Policy and Regulatory Framework

- **National Action Plan on AMR (NAP-AMR), 2017:** India's first action plan, released in April 2017 in line with the WHO's 2015 Global Action Plan on AMR. It was built around a **One Health approach**. But it suffered from weak inter-ministerial coordination and limited state-level uptake — only a handful of states (Kerala, Madhya Pradesh, Delhi, Gujarat, and a few others) developed their own state action plans.
- **NAP-AMR 2.0 (2025–2029):** Launched on 18 November 2025 by the Union Health Minister during WHO's World AMR Awareness Week, this updated five-year plan involves over 20 ministries and departments and rests on six strategic objectives — raising awareness, strengthening surveillance

and laboratory capacity, reducing infections, optimising antimicrobial use, promoting research, and reinforcing governance.

  - Unlike the 2017 plan, each ministry now has its own budgeted, time-bound implementation roadmap for stronger accountability.

- **Red Line Campaign:** A public-awareness initiative requiring prescription-only antibiotics to carry a vertical red line on packaging, signalling that they must not be sold or consumed without a doctor's prescription.
- **Schedule H1 (Drugs and Cosmetics Rules):** Mandates that **pharmacies maintain a register of sales for specified antibiotics and habit-forming drugs**, restricting their sale without a valid prescription, though enforcement at the retail level remains weak.
- **Delhi Declaration (2017) and Chennai Declaration(2012):** Inter-ministerial and medical-society consensus statements committing India to a coordinated, One Health approach against AMR.
- **ICMR Antimicrobial Resistance Surveillance Network (AMRSN):** Generates annual national data on resistance patterns from a growing network of state medical college laboratories; India has also been part of the WHO's Global AMR Surveillance System (GLASS) since 2017.
- **India AMR Innovation Hub:** A collaborative platform supporting the development of new diagnostics and antimicrobial technologies, involving national and international partners.
- **State-Level Leadership:** Kerala's **PROUD/AMRITH** initiatives and Kerala and Gujarat's ban on over-the-counter antibiotic sales are often cited as pioneering state-level models for the rest of the country to emulate.

Key Drivers and Dimensions of AMR

(a) Human Health Sector

- **Irrational Antibiotic Use:** Widespread self-medication, over-the-counter sale of antibiotics without prescription, and incomplete dosage courses accelerate the development of resistant strains.
- **Hospital-Acquired Infections:** Weak infection prevention and control (IPC) practices in ICUs and tertiary hospitals allow resistant 'superbugs' such as carbapenem-resistant Klebsiella and Acinetobacter to spread among critically ill patients.
- **Diagnostic Gaps:** Limited access to rapid diagnostic testing forces doctors to prescribe broad-spectrum antibiotics empirically rather than pathogen-specific drugs.

(b) Agriculture, Livestock, and Environment

- **Antibiotic Use in Livestock and Aquaculture:** Antibiotics are widely used as growth promoters and for disease prevention in poultry, dairy, and fisheries, creating resistance that can transfer to humans through the food chain.
- **Pharmaceutical and Hospital Effluent:** Untreated waste from drug-manufacturing units and hospitals discharged into rivers and soil (a concern flagged around pharma hubs such as Hyderabad) creates 'resistance hotspots' in the environment.
- **Poor Sanitation and Waste Management:** Inadequate sewage treatment allows resistant genes to circulate widely in water bodies used for drinking, irrigation, and bathing.

(c) Economic and Innovation Gaps

- **'Discovery Void' in New Antibiotics:** Very few new classes of antibiotics have been developed in recent decades, as low commercial returns discourage pharmaceutical investment in antimicrobial R&D compared to chronic-disease drugs.
- **Weak Private-Sector Engagement:** NAP-AMR 2.0 specifically seeks deeper private-sector involvement, since a large share of antibiotic consumption in India occurs through private clinics and pharmacies outside direct government oversight.

Significance of Tackling AMR for India

Dimension	Role of AMR Measures
Public Health	Preserving **antibiotic effectiveness** is essential to **keep routine surgeries, cancer treatment**, organ transplants, and neonatal care safe from untreatable infections.
Economic Cost	**Rising AMR increases hospital stays, treatment costs, and productivity losses**; the World Bank has warned AMR could push millions into poverty and shave off significant global GDP by 2050 if unaddressed.
Food Security	Controlling antibiotic misuse in livestock and aquaculture protects both animal health and the safety of India's food exports.
Global Health Leadership	As a pioneer among developing countries in AMR containment **(Red Line Campaign, FDC ban, NAP-AMR 2.0)**, **India's actions influence the wider One Health agenda under WHO.**
Sustainable Development Goals	Reducing bloodstream infections from resistant organisms is a specific SDG indicator (3.d.2), linking AMR control directly to India's health and development commitments.

Challenges in Countering AMR in India

- **Weak Enforcement: Schedule H1 and the Red Line Campaign exist on paper**, but over-the-counter sale of antibiotics without prescription remains common across much of the country.
- **Fragmented Federal Implementation:** Health, veterinary care, pharmacy regulation, agriculture, and waste management are **largely state subjects**, so national guidelines alone cannot ensure uniform action; only a few states have operational AMR action plans.
- **Limited Surveillance Coverage:** ICMR's AMRSN network, though expanding, still draws mainly from tertiary hospitals rather than the wider network of district hospitals and private facilities, giving an incomplete national picture.
- **Low Public Awareness:** Studies show many pharmacists, school students, and even some healthcare workers remain unaware of the risks of antibiotic misuse.
- **Antibiotic Pipeline Crisis:** Very few new antimicrobial molecules are in the global development pipeline, meaning resistant infections may soon outpace available treatments.
- **Diagnostic and Infrastructure Deficit:** High cost and limited availability of rapid diagnostics in smaller towns forces continued reliance on empirical, broad-spectrum prescribing.

Way Forward

Way Forward	Explanation
1. Strengthening Enforcement	Strictly implement **Schedule H1** and the **Red Line Campaign**, with real-time monitoring of antibiotic sales.
2. Expanding AMR Surveillance	Extend **ICMR-AMRSN** and **NARS-Net** to district hospitals, private healthcare, and environmental surveillance.
3. Strengthening State Action Plans	Encourage all states to implement **NAP-AMR 2.0**, building on successful models like **Kerala** and **Gujarat**.
4. Promoting Diagnostics and Stewardship	Expand rapid diagnostic facilities and strengthen **hospital antimicrobial stewardship programmes** to reduce irrational antibiotic use.

5. Promoting Antibiotic Innovation	Support R&D through the **India AMR Innovation Hub**, public funding, and incentives for new antibiotics and diagnostics.
6. Regulating Agricultural Antibiotic Use	Enforce restrictions on antibiotics in **livestock and aquaculture** and strengthen pharmaceutical waste management.
7. Enhancing Public Awareness	Promote AMR awareness through nationwide campaigns and integrate it into **medical, veterinary, and school education**.

Conclusion

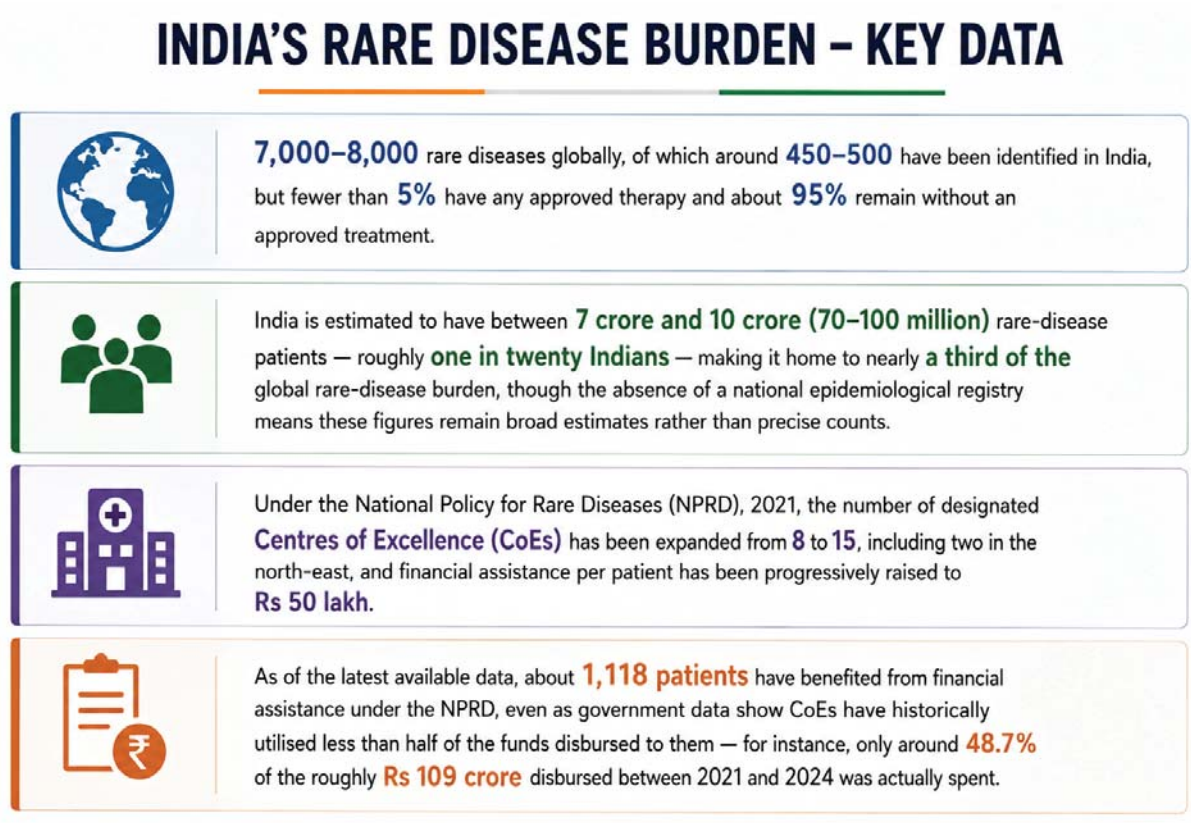
AMR is a **silent pandemic** that threatens decades of medical progress. Combating it requires a **One Health approach**, rational antibiotic use, and strong surveillance.

A robust ecosystem of **surveillance, antimicrobial stewardship, innovation, regulation, and public awareness** will enable India to effectively contain AMR and safeguard future generations.

3. RARE DISEASES

Rare diseases (orphan diseases) are **chronic, life-threatening, or debilitating disorders** affecting a **small proportion of the population**. Around **70–80% are genetic**, with **over 75% affecting children**.

Due to the small patient population, **effective treatments (orphan drugs)** are often unavailable or unaffordable. Examples include **Spinal Muscular Atrophy (SMA), Gaucher disease, Pompe disease, Haemophilia, Thalassaemia, and Duchenne Muscular Dystrophy**.



Government Policy and Regulatory Framework

1. National Policy for Rare Diseases (NPRD), 2021

- Replaced the **2017 policy** and serves as India's primary policy on rare diseases.
- Focuses on **prevention, early diagnosis, treatment, and financial assistance**.
- Funding is provided through the **Umbrella Scheme of Rashtriya Arogya Nidhi (RAN)**.

2. Disease Classification under NPRD

- **Group 1:** Diseases amenable to **one-time curative treatment** (financial assistance up to **₹50 lakh**).
- **Group 2:** Diseases requiring **long-term, relatively low-cost treatment** (financial assistance up to **₹20 lakh**).
- **Group 3:** Diseases requiring **lifelong, high-cost treatment** (e.g., Lysosomal Storage Disorders); managed through Centres of Excellence and other support mechanisms.

3. Institutional Mechanism

- **Centres of Excellence (CoEs):** 15 designated tertiary hospitals (e.g., **AIIMS Delhi, KEM Hospital Mumbai**) providing diagnosis, treatment, counselling, and patient support.
  - Each CoE receives a **one-time grant of up to ₹5 crore** for strengthening diagnostic facilities.
- **National Rare Diseases Committee (NRDC):**
  - Constituted in **2023** and chaired by the **Director General, ICMR**.
  - Oversees implementation of the NPRD and advises the Government on policy and programme design.

4. Patient Support & Financial Assistance

- **National Fund for Rare Diseases (NFRD):**
  - Established following **Delhi High Court** directions (e.g., Master Arnesh Shaw v. Union of India**).**
  - Courts have directed that **allocated funds should not lapse** due to non-utilisation.
- **Voluntary Crowdfunding Platform:**
  - Enables **individuals and CSR donors** to contribute towards treatment costs.

5. Diagnosis & Digital Infrastructure

- **National Rare Disease Information Portal:**
  - Developing a **national patient registry** and information repository on rare diseases and available therapies.
- **Nidan Kendras (DBT):**
  - Support **screening, diagnosis, and genetic testing** of rare diseases.

6. Promoting Affordable Treatment & Research

- **Customs Duty Exemptions:**
  - Exemption on **life-saving orphan drugs** and **Food for Special Medical Purposes (FSMP)** to reduce treatment costs.
- **Indigenous Research:**
  - **DBT** and **CSIR** are promoting **gene therapy, cell therapy (including CAR-T),** and **drug repurposing** through Centres of Excellence to improve affordability.

Key Challenges Underlying the Rare Disease Burden

(a) Diagnosis and Awareness

- **Diagnostic Delay:** Patients often wait several years and see multiple specialists before receiving an accurate diagnosis, by which time irreversible disease progression may have already occurred.

- **Low Clinical Awareness:** Many primary-care doctors are not trained to recognise the early signs of rare genetic disorders, leading to repeated misdiagnosis.

(b) Cost and Access to Treatment

- **Extremely High Treatment Costs:** Annual treatment for a 10 kg child with a rare disease can range from Rs 10 lakh to over Rs 1 crore, and costs rise as the child grows, making even the enhanced Rs 50 lakh support ceiling inadequate for many lifelong therapies.
- **'Orphan Drug' Market Failure:** Because patient numbers for any single rare disease are very small, pharmaceutical companies have little commercial incentive to invest in research or manufacturing, and India has almost no domestic manufacturers of specialised rare-disease drugs.

(c) Governance and Data Gaps

- **Fund Underutilisation:** Parliamentary and RTI data repeatedly show that a large share of the funds released to CoEs remains unspent even as patients are denied treatment, pointing to administrative bottlenecks rather than a genuine lack of resources.
- **Absence of a National Registry:** India still lacks systematic, population-based data on the incidence, prevalence, and regional spread of rare diseases, making policy planning largely reactive rather than evidence-based.
- **Treatment Discontinuity:** Because financial assistance is capped per patient rather than structured as continuing support, many patients on lifelong therapies lose access once their approved ceiling is exhausted.

Significance of Addressing Rare Diseases for India

Dimension	Why It Matters
Right to Health	Courts have held that denying treatment to rare-disease patients, especially children, purely on cost grounds violates the **Right to Life under Article 21.**
Equity in Healthcare	A dedicated policy ensures tha**t patients with low-prevalence conditions are not left behind by a health system** designed mainly around high-burden diseases.
Biotechnology and Genomics Leadership	I**ndia's large, genetically diverse population offers a strong base for indigenous gene-therapy and genomic research**, positioning the country as a potential hub for affordable orphan-drug innovation.
Reducing Import Dependence	Promoting domestic manufacturing of orphan drugs and diagnostics can substantially **lower treatment costs that currently rely on expensive imported therapies**.
Universal Health Coverage (UHC)	Strengthening rare-disease care is integral to India's broader commitment to UHC and the **Sustainable Development Goal** of ensuring healthy lives for all.

Way Forward

1. Establishing a Permanent Rare Disease Fund	Fully operationalise the **National Fund for Rare Diseases (NFRD)** to ensure uninterrupted, patient-centric financial support.
2. Ensuring Continuity of Care	Shift from a one-time funding cap to **lifelong financial support** for patients requiring continuous treatment.

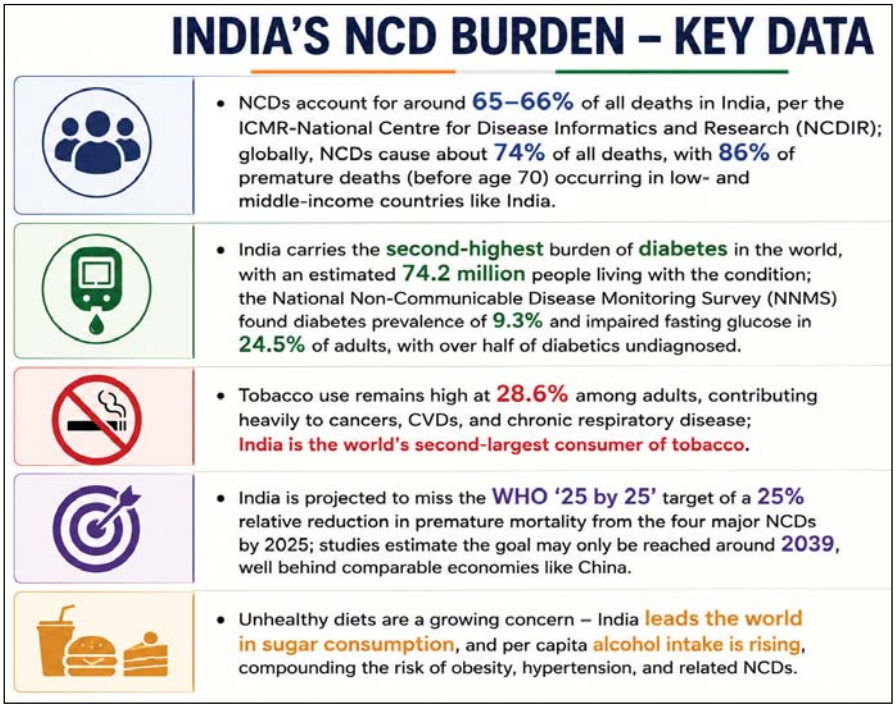
3. Building a National Rare Disease Registry	Operationalise the **National Rare Disease Information Portal** to generate reliable data for evidence-based policymaking.
4. Strengthening Centres of Excellence (CoEs)	Improve infrastructure, streamline approvals, and ensure transparent utilisation of funds at **CoEs**.
5. Promoting Indigenous Research and Manufacturing	Support domestic development of **gene therapies, biosimilars, and repurposed drugs** through **DBT, CSIR, and public-private partnerships**.
6. Encouraging State-Level Initiatives	Motivate states to establish **rare disease support programmes**, following the **Kerala and Karnataka** models.
7. Expanding Early Screening	Scale up **newborn and carrier screening** for early diagnosis and timely treatment of rare diseases.

Conclusion

Rare diseases may affect **few individually but millions collectively**. Ensuring **early diagnosis, affordable treatment, and sustained policy support** is essential for an inclusive healthcare system.

4. NON-COMMUNICABLE DISEASES

Non-Communicable Diseases (NCDs) - also called **chronic diseases**, are conditions that are not transmitted from person to person and tend to be of long duration, progressing slowly. They result from a combination of **genetic, physiological, behavioural, and environmental factors**, especially modifiable risk factors like **tobacco use, unhealthy diet, physical inactivity, and harmful alcohol use**.



The four major NCDs are **cardiovascular diseases (CVDs), cancers, chronic respiratory diseases, and diabetes**; mental health conditions are increasingly recognised as a fifth major component of the NCD burden.

India is undergoing a rapid **epidemiological transition**, with NCDs overtaking communicable diseases as the leading cause of death and disability, even as infectious diseases remain a challenge in several regions.

Government Policy & Regulatory Framework

1. National Programme for Prevention and Control of NCDs (NP-NCD)

- India's flagship NCD programme under **NHM**.
- Screens people **30+ years** for **hypertension, diabetes, and oral, breast & cervical cancers**.
- Target: **50 crore people**.

2. National Health Policy, 2017

- Targets **25% reduction** in premature deaths from major NCDs by **2025**, in line with WHO goals.

3. National Multisectoral Action Plan (NMAP)

- Coordinates action across ministries to reduce **tobacco use, unhealthy diet, alcohol consumption, and physical inactivity**.

4. '75 by 25' Initiative (2023)

- Aims to bring **7.5 crore** people with **hypertension and diabetes** under standard treatment through **Ayushman Arogya Mandirs (AAMs)**.

5. Tax & Fiscal Measures

- **GST reduced to 5%** on diagnostic kits and selected medical devices.
- **Higher taxes on cigarettes** and **National Security Cess on pan masala**.
- Proposal for a **Health Cess** on tobacco and sugary drinks.

6. Union Budget 2026–27

- **Biopharma SHAKTI** (₹10,000 crore) for biologics and biosimilars.
- **50% increase** in trauma care capacity.
- **NIMHANS-2** in North India.
- **Customs duty exemption** on **17 cancer medicines**.

7. National Tobacco Control Programme (NTCP)

- Promotes tobacco awareness, enforces tobacco-control laws, and provides **cessation support**.

8. Ayushman Bharat

- **Ayushman Arogya Mandirs (AAMs):** NCD screening and primary healthcare.
- **PM-JAY:** Financial protection for costly NCD treatments.

Challenges in Tackling NCDs

- **Missed Global Targets:** India is on track to miss the **WHO '25 by 25' target** and the **SDG target** of a one-third reduction in premature NCD mortality by 2030, with projections suggesting the 25% goal may only be met by **2039**.
- **Low Screening Coverage:** Despite scale-up, **NFHS-5 (2019-21)** found only **1.9% of women** had ever been screened for cervical cancer and **0.9%** for breast cancer, showing a large gap between programme rollout and actual population coverage.
- **Weak Fiscal Deterrents:** Ad valorem (according to value) GST on tobacco and sugar-sweetened beverages, and low taxation on **bidis (just 22%)**, blunt the impact of **'sin taxes'** compared to globally recommended specific, per-unit taxes.

- **Human Resource and Infrastructure Gaps:** Studies show only **1.1% of rural public facilities** are adequately equipped to manage major NCDs, compared to **9% of urban private facilities**, reflecting deep rural-urban disparities.
- **Rising Risk Factors:** India **leads the world in sugar consumption**, alcohol intake is rising, and **obesity** is increasing across socioeconomic groups, adding to the future NCD burden even as some risk factors like tobacco use decline.
- **Underspending and Implementation Gaps:** Programmes like Jan Aushadhi and NP-NCD have seen a persistent gap between **budgeted allocations and actual expenditure**, weakening on-ground delivery.
- **Mental Health Neglect:** Mental health conditions remain **under-treated and under-resourced** within the broader NCD framework, despite growing recognition of their scale.

Way Forward

1. Shifting to Preventive Healthcare	Strengthen **early screening** for all adults (30+) and increase public health investment to reduce the NCD burden.
2. Rationalising Health Taxes	Introduce a **Health Cess** on tobacco, pan masala, and sugary drinks to discourage unhealthy consumption and fund NCD programmes.
3. Strengthening Primary Healthcare	Build a dedicated public health workforce and empower **ASHAs, ANMs, and MPWs** for NCD screening and follow-up.
4. Expanding Digital Health Integration	Integrate the **National NCD Portal** with the **Ayushman Bharat Digital Mission (ABDM)** for seamless patient tracking and continuity of care.
5. Addressing Lifestyle & Environmental Risks	Reduce risk factors such as **air pollution, unhealthy diets, physical inactivity, and unsafe urban environments** through policy interventions.
6. Strengthening Mental Healthcare	Expand the **District Mental Health Programme**, tele-psychiatry, and crisis helplines as part of comprehensive NCD care.

Conclusion

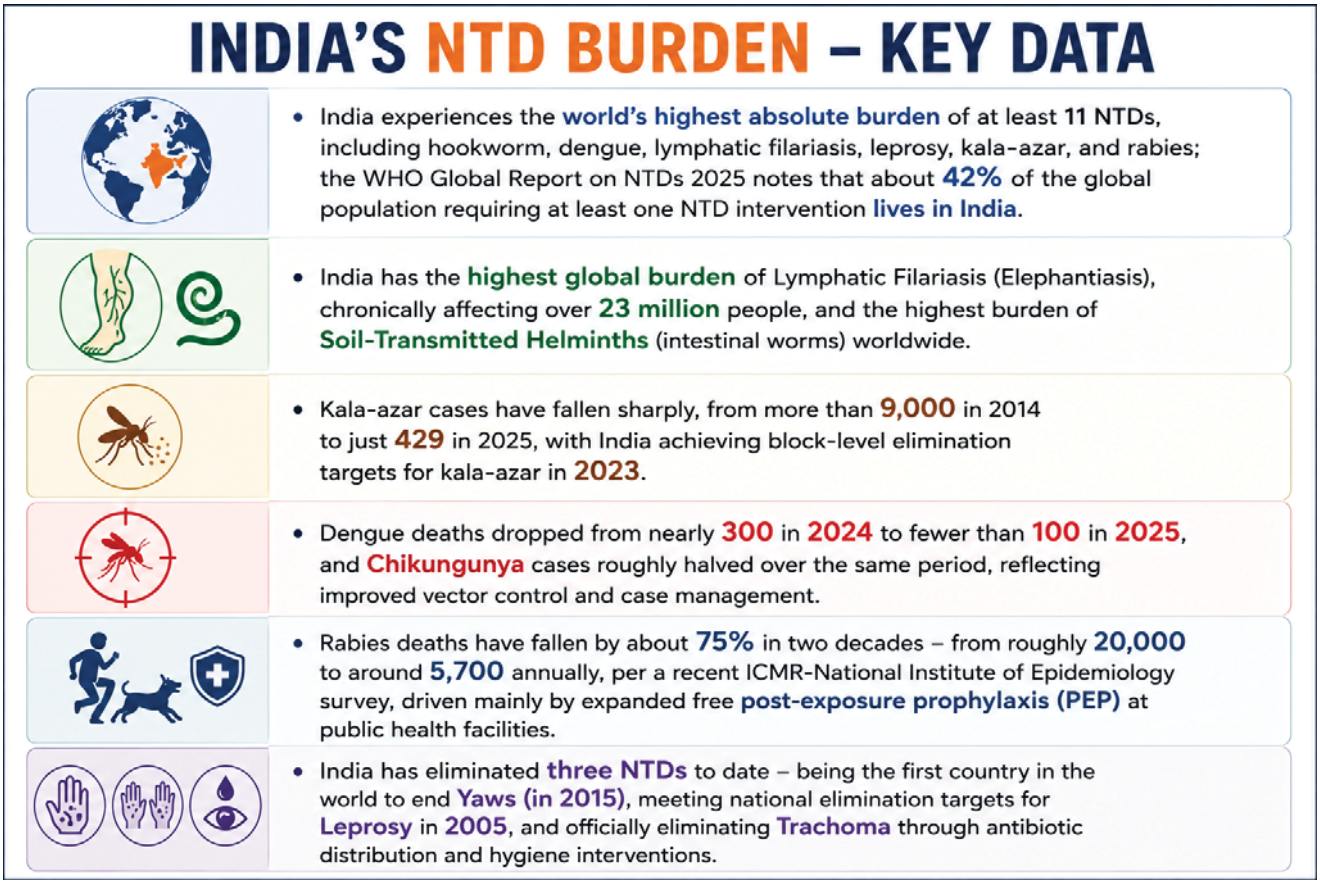
Non-Communicable Diseases are India's **largest public health challenge**, requiring a shift from **curative to preventive healthcare** through early screening, healthy lifestyles, and strong primary care.

A robust ecosystem of **preventive healthcare, digital health, fiscal measures, and multi-sectoral action** will enable India to effectively tackle the rising burden of non-communicable diseases.

5. NEGLECTED TROPICAL DISEASES

Neglected Tropical Diseases (NTDs) are a group of **21 infectious diseases** that mainly affect **poor populations in tropical and subtropical regions**. They spread through **vectors, animals, or contaminated water and soil** and are closely linked to **poor sanitation and hygiene (WASH)**.

Despite affecting **over one billion people**, they receive limited attention and funding. Examples include **Kala-azar, Lymphatic Filariasis, Leprosy, Dengue, Rabies, Trachoma, and Snakebite envenoming**.



Government Policy & Regulatory Framework

1. National Centre for Vector Borne Diseases Control (NCVBDC)

- Nodal programme for control of **Malaria, Dengue, Chikungunya, Kala-azar, Lymphatic Filariasis, and Japanese Encephalitis**.

2. Kala-azar Elimination Programme

- Provides **free diagnosis and treatment**.
- **99% of endemic blocks** have achieved the elimination target.
- Includes **wage compensation** for Kala-azar and **PKDL** patients.

3. Accelerated Plan for Elimination of Lymphatic Filariasis (APELF)

- Targets **elimination by 2027**.
- Implements **Mass Drug Administration (MDA)** and care for **lymphoedema** and **hydrocele**.

4. National Leprosy Eradication Programme (NLEP)

- Provides **free Multi-Drug Therapy (MDT)**.
- Focuses on **early detection** and **disability prevention**.
- Leprosy was eliminated as a public health problem in **82% of districts**.

5. National Rabies Control Programme

- Provides free **Anti-Rabies Vaccine (ARV)** and **Anti-Rabies Immunoglobulin (ARIg)**.
- Promotes **animal-bite management** and **dog vaccination** under the **One Health** approach.

6. Regional & Global Collaboration

- Collaborates with **Bangladesh and Nepal** for **filariasis elimination**.

- Aligned with the **WHO NTD Roadmap 2021–2030**.

7. World NTD Day

- Observed every year on **30 January** to promote awareness and accelerate elimination of **Neglected Tropical Diseases (NTDs)**.

Key Initiatives

(a) Mass Drug Administration (MDA) and Screening

- **Nationwide MDA rounds** for Lymphatic Filariasis are conducted annually across endemic districts, reaching hundreds of millions of people with free anti-filarial medicines, alongside **night blood surveys** to track microfilaria prevalence.
- **Active and passive case-finding camps** for kala-azar and leprosy help identify cases early, particularly in the endemic states of **Bihar, Jharkhand, Uttar Pradesh, and West Bengal** for kala-azar.

(b) Vector Control and 'One Health' Approach

- **Integrated Vector Management (IVM):** Combines indoor residual spraying, insecticide-treated bed nets (for kala-azar-endemic sandfly control), source reduction, and community engagement to curb dengue, chikungunya, and kala-azar transmission.
- **Animal Reservoir Management:** Since diseases like rabies and kala-azar involve animal or environmental reservoirs, coordinated dog-vaccination drives and sandfly-control measures are integral to sustained elimination.

(c) Water, Sanitation and Hygiene (WASH) Integration

- Programs like **Swachh Bharat Mission** indirectly support NTD control, since diseases such as **trachoma, soil-transmitted helminths, and lymphatic filariasis** are closely tied to poor sanitation and hygiene access.

(d) Treatment Access and Compensation

- **Wage compensation schemes** cushion the economic impact on daily-wage patients undergoing prolonged kala-azar or PKDL treatment, addressing a key barrier to treatment completion.
- **Free multi-drug therapy** under NLEP and free anti-rabies biologicals under the rabies programme ensure that treatment cost does not deter care-seeking among poor and rural populations.

Challenges Facing NTD Elimination in India

- **Weak Surveillance and Underreporting:** Cases are often missed in exactly the places where disease risk is highest - remote, poorly-mapped, and high-mobility informal settlements - clouding the true national burden.
- **Risk of Resurgence:** Elimination is not a permanent endpoint. for example, **kala-azar resurged in the 1970s** when vector control and surveillance weakened, and similar lapses in **WASH infrastructure** have caused localised trachoma resurgence elsewhere.
- **Post-Kala-Azar Dermal Leishmaniasis (PKDL):** Patients often feel well but continue to harbour parasites capable of sustaining transmission; the absence of a **safe, effective, and shorter PKDL treatment** keeps the risk of resurgence real.
- **Community Trust and MDA Compliance:** Mass Drug Administration can only interrupt transmission with **high coverage**, but participation depends heavily on community trust; even small coverage gaps can significantly weaken programme impact.
- **Urban Informal Settlements:** High population density, administrative fragmentation, and frequent population movement make both **treatment delivery and follow-up** difficult in expanding cities.

- **Climate and Systemic Shocks:** Vector control interruptions, extreme weather, and health-system shocks such as the **COVID-19 pandemic** can rapidly undo hard-won elimination gains.

Way Forward

1. Strengthening Disease Surveillance	Develop robust **real-time surveillance systems** for early detection and continuous monitoring of NTDs.
2. Integrating NTD Care into Primary Healthcare	Deliver **screening, treatment, and follow-up** through **Ayushman Arogya Mandirs (AAMs)**.
3. Enhancing Community Participation	Strengthen public awareness and community engagement to improve **Mass Drug Administration (MDA)** and treatment compliance.
4. Promoting Research and Innovation	Develop better treatments for **PKDL** and improve access to **anti-snake venom** and rural treatment facilities.
5. Sustaining Post-Elimination Surveillance	Continue **vector control and disease surveillance** even after elimination targets are achieved to prevent resurgence.
6. Strengthening the One Health Approach	Improve coordination among **human, animal, and environmental sectors** and expand **WASH** infrastructure to tackle NTDs at their source.

Conclusion

Neglected Tropical Diseases are diseases of **poverty and inequality**. Their elimination requires **strong primary healthcare, improved sanitation, vector control, and sustained community participation**.

6. TRADITIONAL MEDICINE

The **WHO** defines **Traditional Medicine** as the sum of knowledge, skills, and practices based on theories, beliefs, and experiences indigenous to different cultures, used to maintain health and to prevent, diagnose, or treat illness.

In India, this vast body of knowledge is institutionally represented by **AYUSH** - an acronym for **Ayurveda, Yoga & Naturopathy, Unani, Siddha, Homoeopathy, and Sowa-Rigpa** - each a distinct medical system with its own philosophy, pharmacopoeia, and diagnostic approach, but united by a common emphasis on **holistic, preventive, and personalised care**.

Unlike allopathic medicine, which largely targets pathogens or specific disease pathways, most traditional systems in India centre on restoring **balance within the body** (e.g., the tridosha framework of vata, pitta, and kapha in Ayurveda) and emphasise **lifestyle, diet, and mind-body practices** alongside herbal and mineral formulations.

Government Policy & Regulatory Framework

1. National AYUSH Mission (NAM)

- Launched in **2014** to strengthen **AYUSH infrastructure, education, and drug quality**.

2. National AYUSH Regulators

- **NCISM** and **NCH** regulate **education and professional standards** for Indian Systems of Medicine and Homoeopathy.

3. Drug Quality Standards

- **Pharmacopoeia Commission for Indian Medicine & Homoeopathy (PCIM&H)** sets quality standards for AYUSH medicines.
- **AYUSH Mark (2022)** certifies **authentic and quality-tested AYUSH products**.

4. WHO Global Centre for Traditional Medicine

- Located at **Jamnagar, Gujarat**.
- World's **first WHO centre** dedicated to traditional medicine research, training, and global collaboration.

5. AYUSH in Mainstream Healthcare Insurance

- **IRDAI (2024)** mandated insurance coverage for **170 standardised AYUSH therapies**, improving affordability and access.

6. Traditional Knowledge Digital Library (TKDL)

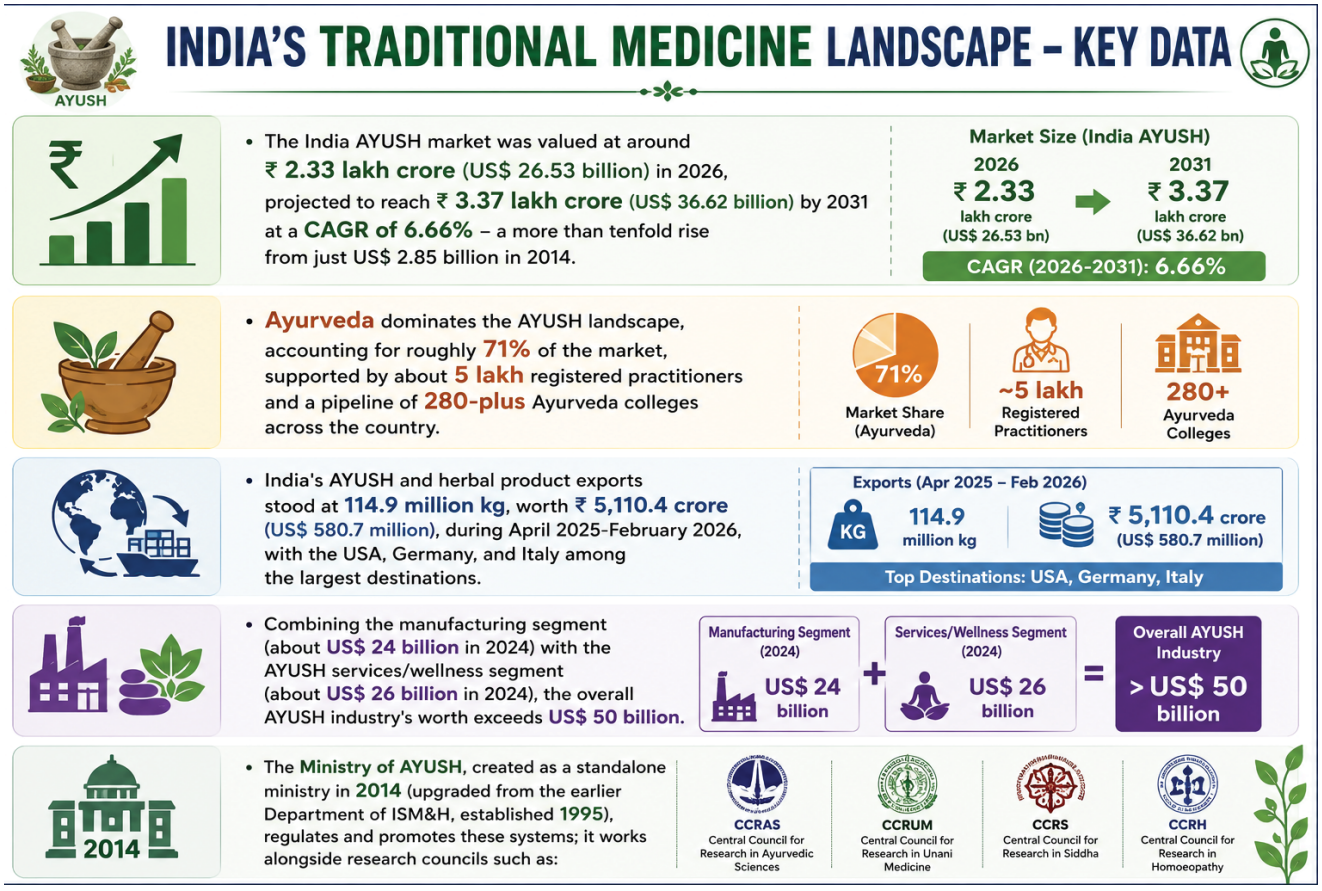
- Digital database protecting India's traditional knowledge from **biopiracy** (e.g., **turmeric** and **neem** patents).

7. Union Budget 2026–27.

- Announced **three new All India Institutes of Ayurveda (AIIAs)**, upgraded drug-testing laboratories, and promotion of **AYUSH medical tourism**.

8. India–EU Free Trade Agreement

- Facilitates **AYUSH services and wellness centres** in the EU.
- Recognises **TKDL**, strengthening protection against **biopiracy**.



Applications and Key Initiatives

(a) Mainstreaming into Public Healthcare

- **Ayushman Arogya Mandirs** and district hospitals increasingly co-locate AYUSH counters, integrating traditional care into **Comprehensive Primary Health Care** alongside allopathic services.
- Studies cited by the Ministry show **22% lower average episode costs** for chronic diabetes management under AYUSH-integrated pilots in **Karnataka and Kerala** compared with purely allopathic care, strengthening the case for integrative models.

(b) Global Outreach and Soft Power

- **International Day of Yoga (21st June):** Adopted by the **UN General Assembly in 2014** following India's initiative, it has become one of India's most successful instruments of **health diplomacy and soft power**, observed in nearly every country.
- India has committed long-term support to the **WHO GCTM** for global research and promotion of traditional medicine. In December 2025 INDIA announced plans to unveil the **world's first Digital Traditional Medicine Library** at a WHO Global Summit, aimed at consolidating traditional knowledge for global researchers.

(c) Research, Standardisation, and Industry Support

- A **Production-Linked Incentive (PLI)-style scheme** (around **US$ 288 million**) supports the medicinal-herb supply chain, helping secure raw material availability and stabilise prices for AYUSH manufacturers.
- **Good Manufacturing Practice (GMP)** compliance is being progressively enforced for AYUSH manufacturing units, improving quality control though many micro and small units still struggle to meet **WHO-GMP** thresholds.

(d) Wellness Tourism and Economic Value

- India's **wellness tourism** sector, valued at about **Rs 1,64,164 crore (US$ 19.43 billion) in 2024**, is closely tied to AYUSH, with Ayurveda retreats, Panchakarma centres, and yoga tourism drawing both domestic and international visitors.
- The new **AYUSH visa** and **Regional Medical Value Tourism Hubs** aim to formally combine traditional therapies with modern diagnostics and post-operative care, positioning India as an integrative-health destination.

Significance of Traditional Medicine for India

Dimension	How Traditional Medicine Helps
Affordable & Preventive Care	AYUSH offers a **low-cost, accessible** complement to allopathy, particularly suited to managing **chronic, lifestyle-linked NCDs** like diabetes and hypertension through diet, yoga, and herbal regimens.
Rural & Last-Mile Access	With a wide network of AYUSH dispensaries and practitioners, traditional medicine extends **basic healthcare access** to underserved rural and remote areas where allopathic infrastructure is thin.
Economic Growth & Exports	A **US$ 50-billion-plus industry** generating employment in cultivation, manufacturing, and services, while AYUSH **exports to over 200 countries** strengthen India's trade balance and rural livelihoods.
Soft Power & Diplomacy	**International Yoga Day** and the **WHO GCTM in Jamnagar** have made India the global reference point for traditional medicine, enhancing its cultural diplomacy and international standing.

Protecting Traditional Knowledge	The **TKDL** safeguards India's centuries-old formulations from **biopiracy**, ensuring traditional knowledge benefits Indian communities and industry rather than being unfairly patented abroad.

Challenges Facing Traditional Medicine Development in India

- **Weak Evidence Base:** Medical bodies such as the **Indian Medical Association (IMA)** argue many AYUSH therapies lack rigorous validation through **randomised controlled trials (RCTs)**, limiting their acceptance within evidence-based medicine and international regulatory systems.
- **Quality Control and Adulteration:** Despite AYUSH pharmacopoeial standards, enforcement remains weak; species-level adulteration (e.g.**, substitution of authentic herbs with cheaper look-alikes**) and inconsistent phytochemical quantification undermine product reliability.
- **Safety Concerns:** Some Ayurvedic formulations, particularly certain herbomineral (bhasma) products, have faced **international health advisories over heavy-metal contamination (lead, mercury)**, damaging global consumer confidence.
- **GMP Compliance Gaps:** Roughly **4,200 micro and small AYUSH manufacturing units** still fall short of **WHO-GMP** thresholds for air-handling, in-process validation, and residual solvent testing, straining smaller players who lack capital for costly upgrades.
- **'Mixopathy' Disputes:** Policy decisions permitting Ayurveda practitioners to perform certain surgical procedures or prescribe allopathic drugs have triggered **legal disputes and professional opposition**, exposing unclear regulatory boundaries between AYUSH and modern medicine.
- **Informal Practice and Quackery:** A large informal sector of unregistered or under-qualified practitioners persists alongside the formal, regulated system, posing risks to patient safety and complicating public trust.
- **Uneven Research Translation:** While research councils and new institutes are expanding, translating traditional formulations into internationally accepted, patent-protected, evidence-backed products remains slow relative to the scale of the industry.

Way Forward

1. Strengthening Scientific Evidence	Expand **clinical trials and outcome-based research** to build global credibility for AYUSH therapies.
2. Ensuring Uniform Quality Standards	Promote **Good Manufacturing Practices (GMP)** and strengthen quality control across AYUSH manufacturing units.
3. Clarifying Regulatory Framework	Clearly define the roles of **AYUSH and allopathy** to promote safe and evidence-based integrative healthcare.
4. Expanding Healthcare Integration	Increase insurance coverage and integrate **evidence-based AYUSH services** into **Ayushman Bharat** and primary healthcare.
5. Protecting Traditional Knowledge	Strengthen **Traditional Knowledge Digital Library (TKDL)** and intellectual property protection to prevent biopiracy and promote innovation.
6. Expanding Global Outreach	Leverage trade agreements and international partnerships to increase global recognition, exports, and acceptance of **AYUSH**.

Conclusion

India's traditional medicine systems are not merely a part of its cultural heritage but a **strategic healthcare and economic asset**. Their global success will depend on **scientific evidence, quality assurance, and responsible integration with modern medicine**.

■■■

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ACHIEVEMENTS OF INDIANS IN S&T

1. SIR MOKSHAGUNDAM VISVESVARAYA

Sir Mokshagundam Visvesvaraya was a monumental Indian engineer, statesman, and the 19th Diwan of Mysore whose pioneering work in water management revolutionized irrigation systems across British India. Widely revered as the father of modern Indian planning, he designed the iconic Krishna Raja Sagara dam and patented the automatic floodgate system to secure critical water resources.

Early life and academic background

- **Birth:** Born on September 15, 1860, in Muddenahalli, Kingdom of Mysore (present-day Karnataka), into a Telugu Brahmin family.
- **Early Hardship:** Faced severe financial constraints early in life following the untimely death of his father when Visvesvaraya was just 15 years old.
- **Primary Schooling:** Completed his early education in Chikballapur and later walked to Bangalore to attend Central College for high school.
- **Graduation:** Earned his Bachelor of Arts (BA) degree from Central College, Bangalore, in 1881, graduating with top honors.
- **Engineering Education:** Shifted to the College of Science, Pune (now COEP Technological University), with the help of a scholarship from the Mysore Government.
- **Academic Distinction:** Completed his License in Engineering (L.C.E.) in 1883, securing the absolute first rank across the entire University of Bombay.

Major contribution of Sir Mokshagundam Visvesvaraya

- **Water Resource Engineering**
  - **Automatic Floodgates (1903):** Patented automatic sluice gates at Khadakwasla Dam (Pune), which raised storage capacity without compromising structural safety.
  - **Krishna Raja Sagara Dam:** Designed and constructed Asia's largest reservoir of its time in Mysore, transforming arid lands into a fertile agricultural belt.
  - **Flood Protection System:** Designed a comprehensive embankment and drainage network for Hyderabad following the catastrophic Musi River floods of 1908.
  - **Block System of Irrigation:** Invented an efficient water-distribution model that prevented water wastage and ensured equitable distribution among farmers.
- **Industrialization & Economics**
  - **Mysore Iron and Steel Works:** Established the facility at Bhadravati (now VISL), pioneering independent heavy metallurgy in southern India.
  - **State Bank of Mysore (1913):** Founded the banking institution to mobilize local capital and extend low-interest credit to indigenous businesses.
  - **Economic Planning Blueprint:** Authored the seminal book "*Planned Economy for India*" (1934), laying the foundational blueprint for national planning models.

- **Institutional & Educational**
  - **University of Mysore (1916):** Co-founded the state's first university to democratize high-quality, professional higher education.
  - **Sri Jayachamarajendra Polytechnic:** Established this technical institute in Bangalore to train a skilled workforce for rapid industrialization.
  - **Century Club Bangalore:** Founded the elite socioeconomic hub to bring together administrators, visionaries, and technocrats.

Awards honored to Sir M Visvesvaraya

Honor / Award / Recognition	Year	Significance
Companion of the Order of the Indian Empire (CIE)	1911	Conferred by King Edward VII for early engineering excellence as Chief Engineer of Mysore.
Knight Commander of the Order of the Indian Empire (KCIE)	1915	Knighted (Sir) by King George V for his monumental public works as Dewan of Mysore.
President of the Indian Science Congress	1923	Honored with the leadership role to preside over the national scientific congregation.
Bharat Ratna	1955	Awarded India's highest civilian honor for his unparalleled contributions to industrialization and nation-building.
Honorary Membership, Institution of Civil Engineers	Lifetime	International professional recognition conferred by the premier London-based civil engineering body.
National Engineer's Day	Posthumous	His birth anniversary (September 15) was declared as national celebration day to honor engineering excellence.

Conclusion

Sir M. Visvesvaraya stands as the quintessential architect of modern Indian engineering and state-led economic planning, translating technical precision into impactful nation-building. His dual legacy as a visionary technocrat and clean administrator proved that indigenous industrialization could successfully thrive decades before independence.

2. JAGDISH CHANDRA BOSE

Sir Jagdish Chandra Bose was a pioneering Indian polymath whose groundbreaking research in the late 19th and early 20th centuries laid the foundations for modern radio communication and plant physiology. He famously invented the crescograph to prove that plants have life and react to stimuli, while also pioneering millimeter-wave optics well ahead of his time.

Early life and academic background

- **Birth:** Born on November 30, 1858, in Munshiganj, Bengal Presidency (now Bangladesh), to a family steeped in nationalist values.
- **Early Schooling:** Educated in a traditional vernacular *pathshala* (village school) in Faridpur to foster a deep connection with native language and culture.
- **Higher Secondary:** Moved to Kolkata (then Calcutta) and attended St. Xavier's School and College, where Jesuit priest Father Eugene Lafont ignited his passion for natural sciences.

- **Medical Studies:** Traveled to England in 1880 to study medicine at the University of London, but left due to health issues caused by the dissection room odors.
- **Cambridge Graduation:** Shifted to Christ's College, Cambridge, graduating with a Natural Sciences Tripos in 1884 alongside a Bachelor of Science from the University of London.
- **Influential Mentors:** Studied under legendary British scientists like Lord Rayleigh, Michael Foster, and Francis Darwin, who deeply shaped his empirical research methodology.

Contribution of Jagdish Chandra bose

- **Physics & Electronics**
  - **Microwave Communication:** Demonstrated millimeter-wave transmission in 1895, serving as the technological precursor to modern radio and Wi-Fi.
  - **Solid-State Innovation:** Invented the galena crystal detector, laying the early groundwork for modern semiconductor technology.
- **Plant Physiology**
  - **Crescograph:** Developed this high-precision instrument to quantify plant growth and their response to environmental stimuli.
  - **Biological Proof:** Demonstrated through empirical data that plants possess a nervous system and react to stimuli similarly to animals and metals.
- **Institutional & Ethical**
  - **Bose Institute (1917):** Established Asia's first multidisciplinary research center, institutionalizing indigenous scientific self-reliance.
  - **Open Science Ethos:** Refused to patent his breakthroughs, setting a precedent that scientific knowledge should remain free for humanity.

Awards honored to JC bose

Honor / Award / Recognition	Year	Significance
Companion of the Order of the Indian Empire (CIE)	1903	Imperial recognition for his early academic and scientific excellence.
First Asian US Patent (No. 755840)	1904	Granted for his galena crystal solid-state diode detector.
Companion of the Order of the Star of India (CSI)	1911	Awarded by the British Raj for his landmark contributions to experimental science.
Knight Bachelor (Sir)	1917	Conferred by the British Crown, institutionalizing his status as a pioneer of global science.
Fellow of the Royal Society (FRS), London	1920	First Indian fellow in the field of science, breaking absolute racial barriers in western academia.
IEEE Milestone Recognition	Posthumous	Recognized as a father of radio communication; his work is Asia's oldest IEEE milestone.
Lunar Crater 'Bose'	Posthumous	A crater on the far side of the Moon named in his honor, celebrating his global impact on physics.

Conclusion

Sir Jagdish Chandra Bose remains an enduring icon of Indian science, proving that groundbreaking global research could thrive despite colonial constraints. His legacy as a pioneer of wireless technology and plant physiology paved the way for self-reliant indigenous research. By choosing open science over personal profit, his ethical foundation continues to inspire generations of researchers worldwide.

3. C V RAMAN

Sir Chandrasekhara Venkata Raman was a towering Indian physicist whose groundbreaking discovery of light scattering in 1928 revolutionized modern spectroscopy and molecular physics. By proving that light changes its wavelength when traversing a transparent medium, he became the first Asian to win the Nobel Prize in Physics in 1930.

Early life and academic background

- **Birth:** Born on November 7, 1888, in Tiruchirappalli, Tamil Nadu, into a family of physics and mathematics academics.
- **Pre-cocious Schooling:** Completed his matriculation at age 11 and intermediate education at 13 from St. Aloysius High School, Visakhapatnam.
- **Graduation:** Earned his Bachelor of Arts from Presidency College, Chennai, in 1904, topping the university with a gold medal in Physics.
- **Post-Graduation:** Completed his Master of Arts in Physics from Presidency College in 1907, securing the highest marks on record.
- **Early Research:** Published his first independent scientific paper on the diffraction of light in London's *Philosophical Magazine* (1906) while still a student.
- **Civil Service Tenure:** Topped the Financial Civil Service (FCS) examination in 1907, joining the Indian Finance Department in Calcutta due to limited scientific careers.

Contribution of CV Raman

- **Optics & Quantum Physics**
  - **Raman Effect (1928):** Discovered inelastic light scattering, providing definitive empirical proof of the quantum nature of light.
  - **Color of the Sea:** Proved the Mediterranean's blue color is caused by molecular light scattering, rewriting Rayleigh's sky-reflection theory.
- **Acoustics & Wave Mechanics**
  - **Indian Percussion:** Conducted the first scientific analysis of the *tabla* and *mridangam*, explaining their unique harmonic overtones.
  - **String Instruments:** Developed mathematical models for stringed instruments, standardizing the mechanics of acoustic resonance.
- **Institutional & Mentorship**
  - **IASc Bangalore (1934):** Founded the Indian Academy of Sciences to build a self-reliant publishing platform for native scientists.
  - **Raman Research Institute (1948):** Established RRI to sustain pure, independent research in post-independence India.
  - **Mentoring Leaders:** Trained the next generation of top Indian scientists, including space pioneer Vikram Sarabhai.

Raman Effect

When a beam of monochromatic (single-color) light passes through a transparent chemical compound, a small fraction of the light shifts in color and emerges in directions other than the incoming beam.

- **Phenomenon:** Most light photons collide with molecules and bounce off with the exact same energy and color (**Rayleigh Scattering**). However, about one in ten million photons collides inelastically, exchanging energy with the molecules. This causes the scattered light to change its wavelength, frequency, and color (**Raman Scattering**).
- **Major Components: Stokes vs. Anti-Stokes**
  - **Stokes Lines:** Occur when the incoming photon gives up some of its energy to the molecule. The scattered photon emerges with **lower energy, lower frequency, and a longer wavelength** (e.g., green light shifting to red).
  - **Anti-Stokes Lines:** Occur when the photon absorbs energy from an already excited molecule. The scattered photon emerges with **higher energy, higher frequency, and a shorter wavelength** (e.g., green light shifting to blue).

Applications of Raman Effect

- **Medicine & Healthcare**
  - **Cancer Diagnostics:** Enables real-time, non-invasive identification of malignant tumors and cancerous cells during surgeries.
  - **Pharmaceutical Analysis:** Used to verify drug purity, monitor real-time chemical reactions, and detect counterfeit medicines.
- **Defense & Homeland Security**
  - **Explosives Detection:** Scans and identifies hidden military explosives, improvised explosive devices (IEDs), and gunpowder from a safe distance.
  - **Narcotics Interdiction:** Identifies illicit drugs, contraband, and hazardous chemical agents instantly through transparent plastic or glass packaging.
- **Industry & Material Science**
  - **Semiconductor Quality Control:** Analyzes stress, defects, and crystalline purity in silicon wafers used for manufacturing microchips.
  - **Carbon Nanomaterials:** Used to classify and evaluate the structural integrity of graphene and carbon nanotubes.
- **Science & Exploration**
  - **Space Missions:** Deployed on planetary rovers (like Mars rovers) to analyze extraterrestrial mineralogy and search for organic signatures of life.
  - **Environmental Monitoring:** Detects microplastics, chemical pollutants, and oil spills in aquatic ecosystems and water bodies.
- **Heritage & Forensics**
  - **Art Conservation:** Analyzes the chemical composition of pigments in ancient paintings and artifacts without causing any physical damage.
  - **Forensic Investigation:** Identifies trace evidence at crime scenes, such as unknown powders, fibers, and bodily fluids, without altering the sample.

Do you know?

The Government of India designated February 28 as National Science Day in 1986. The first National Science Day was celebrated on February 28, 1987. This specific date was chosen to commemorate Sir C.V. Raman's discovery of the **Raman Effect** on February 28, 1928, which led to his historic Nobel Prize in 1930.

Awards honored to CV Raman

Honor / Award / Recognition	Year	Significance
Fellow of the Royal Society (FRS)	1924	Early global recognition by the London scientific body for his contributions to optics.
Matteucci Medal	1928	Awarded by the Italian Society of Sciences for outstanding contributions to physics.
Knight Bachelor (Sir)	1929	Conferred by the British Crown before he received the Nobel Prize, acknowledging his status in physics.
Nobel Prize in Physics	1930	First Asian and non-white person to win a science Nobel Prize, awarded for his discovery of the Raman Effect.
Hughes Medal	1930	Awarded by the Royal Society of London for his landmark discovery of inelastic light scattering.
Franklin Medal	1941	International recognition presented by the Franklin Institute (USA) for his contributions to physics.
Bharat Ratna	1954	Recipient of India's highest civilian honor in its inaugural year for elevating national scientific prestige.
Lenin Peace Prize	1957	International recognition from the Soviet Union highlighting his global scientific standing.
Lunar Crater 'Raman'	Posthumous	A crater on the Moon named in his honor to memorialize his foundational impact on spectroscopy.

Conclusion

Sir C.V. Raman remains a foundational pillar of Indian science, demonstrating that world-class, Nobel Prize-winning research could be achieved using basic, indigenously designed equipment. His legacy extends far beyond optics, as he spent his life establishing robust institutional frameworks and mentoring the first generation of independent India's scientific leaders.

4. HOMI JEHANGIR BHABHA

Dr. Homi Jehangir Bhabha, universally revered as the **Father of the Indian Nuclear Programme**, was a brilliant theoretical physicist and visionary administrator who laid the baseline blueprint for India's atomic energy security. He founded premier institutions like TIFR and BARC, introducing a unique three-stage nuclear power plan designed to exploit India's vast domestic thorium reserves.

Early life of Homi Jehangir Bhabha

- **Parsi Background:** Born on October 30, 1909, into an affluent, highly educated Parsi family in Bombay with close familial ties to the industrialist Tata Group.
- **Cultured Upbringing:** Raised in an environment rich in classical music, fine arts, and literature, which deeply cultivated his lifelong passion for painting and opera.
- **Schooling Foundations:** Completed his primary and secondary schooling at the prestigious **Cathedral and John Connon School** in Bombay, demonstrating an early aptitude for advanced mathematics.

Education background

- **Higher studies:** Traveled to England in 1927 to join **Gonville and Caius College, Cambridge**, initially to pursue a degree in mechanical engineering to satisfy his father's wish to see him join the Tata steel mills.

- **Theoretical Physics:** Shifted his core focus to theoretical physics after discovering a deep fascination for the burgeoning field of quantum mechanics, earning his Mathematical Tripos.
- **Isaac Newton Studentship:** Awarded the highly competitive Isaac Newton Studentship in 1934, which allowed him to extend his high-energy cosmic ray research across Europe.
- **Doctoral Thesis (1935):** Earned his PhD from Cambridge with a groundbreaking doctoral thesis titled "*The Absorption of Cosmic Radiation*," which laid the groundwork for his future discoveries in particle physics.

Contributions of Homi Jehangir bhabha

- **Scientific Discoveries & Quantum Physics**
  - **Bhabha Scattering:** Discovered electron-positron scattering, a fundamental particle physics phenomenon that remains a cornerstone of quantum electrodynamics today.
  - **Cascade Theory of Cosmic Rays:** Co-developed the cascade theory with Walter Heitler, explaining how high-energy cosmic rays interact with the atmosphere to produce showers of subatomic particles.
  - **Predicting the Muon Lifecycle:** Conducted early theoretical work that helped identify and explain the behavior of the muon, a heavy subatomic particle found in cosmic radiation.
- **Strategic Policy & Nuclear Planning**
  - **Three-Stage Nuclear Power Programme:** Architected India's unique, long-term energy blueprint designed to use limited uranium reserves first to eventually unlock the energy of India's vast domestic thorium deposits.
  - **Establishing the Atomic Energy Commission (AEC):** Founded the AEC in 1948 alongside Prime Minister Jawaharlal Nehru, acting as its first chairperson to spearhead independent nuclear research.
  - **Advocating for Strategic Self-Reliance:** Championed the dual-use potential of nuclear technology, ensuring India retained the sovereign capability to develop both civilian energy and a strategic defense deterrent.
- **Iconic Institution Building & Diplomacy**
  - **Founding TIFR (1945):** Established the Tata Institute of Fundamental Research in Mumbai, creating the ultimate nursery for India's modern mathematical and physical sciences.
  - **Establishing BARC (1954):** Set up the Atomic Energy Establishment at Trombay (later renamed the Bhabha Atomic Research Centre) to serve as the core hub for nuclear technology and reactor design.
  - **Presiding over the Geneva Conference (1955):** Chaired the historic first United Nations Conference on the Peaceful Uses of Atomic Energy, cementing India's reputation as a responsible global scientific power.

Conclusion

Dr. Homi Jehangir Bhabha's enduring legacy lies in his uncanny ability to foresee that true national sovereignty in the modern era requires absolute energy independence and technological self-reliance. By architecting the three-stage nuclear plan and building resilient institutions like TIFR and BARC, he single-handedly insulated India from restrictive global technology regimes.

5. VIKRAM SARABHAI

Dr. Vikram Sarabhai, universally revered as the **Father of the Indian Space Programme**, was a visionary physicist and masterful institution-builder who laid the foundational bedrocks of ISRO, PRL, and IIM Ahmedabad. He pioneered the transformative philosophy that a developing nation must leverage advanced space technology not for aggressive geopolitical rivalry, but to solve grassroots societal challenges in agriculture, education, and communication.

Early Life and Education of Vikram Sarabhai

- **Birth**
  - **Industrial and Philanthropic Pedigree:** Born on August 12, 1919, in Ahmedabad into the ultra-wealthy, socially progressive Sarabhai family, a premier industrialist lineage deeply intertwined with India's freedom struggle.
  - **Mahatma Gandhi's Influence:** Raised in an elite home environment regularly visited by national stalwarts like Mahatma Gandhi and Rabindranath Tagore, instilling early values of national self-reliance and societal service.
  - **Alternative Experimental Education:** Completed his early schooling at the family-run **Retreat School**, an avant-garde institution that prioritized hands-on scientific experimentation and creative learning over rote curricula.
- **Academic Background & Cambridge Years**
  - **Undergraduate Education:** Matriculated at Gujarat College in Ahmedabad before moving to England to join **St. John's College, Cambridge**, where he earned his Natural Sciences Tripos in 1940.
  - **World War II impacts:** Forced to temporarily return to India during World War II, where he conducted foundational cosmic ray research at the Indian Institute of Science (IISc), Bangalore, under the direct guidance of Nobel Laureate **Sir C.V. Raman**.
  - **Doctoral Thesis and Degree:** Returned to Cambridge post-war to finish his PhD in 1947, presenting a highly acclaimed doctoral thesis on "*Cosmic Ray investigations in Tropical Latitudes*" that shaped his future space exploration trajectory.

Contribution of Vikram Sarabhai

- **Architectural Bedrocks of India's Space Program**
  - **Founding of INCOSPAR:** Formed the Indian National Committee for Space Research in 1962, which laid the direct operational and policy bedrock for the birth of **ISRO** in 1969.
  - **Establishing Thumba Launch Station:** Set up the Thumba Equatorial Rocket Launching Station (TERLS) near Thiruvananthapuram, strategically utilizing its alignment with Earth's magnetic equator.
  - **Initiating Indigenous Satellite Projects:** Conceived the blueprint for India's historic first domestic satellite, **Aryabhata**, establishing deep technical self-reliance before his sudden demise.
- **Grassroots Societal Interventions**
  - **Pioneering the SITE Project:** Formulated the landmark Satellite Instructional Television Experiment (1975-76), using American satellites to beam critical agricultural and health education to over 2,400 rural villages.
  - **Conceiving the STEP Project:** Designed the Satellite Telecommunications Experiments Project to test domestic television and mass communication delivery via geo-synchronous orbits.
  - **Philosophical Focus on Development:** Rejected the Cold War geopolitical arms race, asserting that space assets must be used primarily as socio-economic catalysts to uplift the common citizen.
- **Iconic Institution-Building Across Sectors**
  - **Physical Research Laboratory:** Established PRL in Ahmedabad in 1947 at his own residence, creating India's premier cradle for cosmic ray, space science, and atmospheric research.
  - **Management and Industry Leadership:** Co-founded the **Indian Institute of Management, Ahmedabad (IIMA)** in 1961 and the Ahmedabad Textile Industry's Research Association (ATIRA) to professionalize modern Indian business.
  - **Strategic Atomic Energy Command:** Succeeded Dr. Homi Bhabha as the Chairman of the **Atomic Energy Commission (AEC)** in 1966, aggressively advancing India's indigenous nuclear power plants and radar installations.

Institutes Established by Vikram Sarabhai

- **Space and Strategic Science Cradles**
  - **Physical Research Laboratory (PRL), Ahmedabad (1947):** Founded as the absolute "cradle of space sciences" in India to pioneer cosmic ray and atmospheric research.
  - **Indian Space Research Organisation (ISRO), Bengaluru (1969):** Evolved from his initial 1962 INCOSPAR framework into a unified national agency for autonomous space exploration.
  - **Space Applications Centre (SAC), Ahmedabad:** Formed by merging his early experimental research cells to design satellite payloads and telecommunication hardware.
  - **Thumba Equatorial Rocket Launching Station (TERLS), Kerala (1963):** Established as India's first operational civilian sounding rocket facility near Earth's magnetic equator.
- **Management, Industrial, and Public Sector Institutions**
  - **Indian Institute of Management (IIM), Ahmedabad (1961):** Co-founded alongside premier business leaders to institutionalize professional management education in modern India.
  - **Ahmedabad Textile Industry's Research Association (ATIRA) (1947):** Created to modernize and scale production technologies across India's traditional textile sector.
  - **Electronics Corporation of India Limited (ECIL), Hyderabad:** Setup to secure complete hardware and electronics self-reliance for the nation's nuclear and space installations.
  - **Uranium Corporation of India Limited (UCIL), Jharkhand:** Established during his tenure to streamline and secure domestic atomic fuel mining loops for nuclear power plants.
- **Public Science and Socio-Cultural Spaces**
  - **Vikram A. Sarabhai Community Science Centre (VASCSC), Ahmedabad (1966):** Created to popularize mathematics and experimental physics directly among grassroots students.
  - **Darpana Academy of Performing Arts, Ahmedabad (1949):** Co-founded alongside his wife Mrinalini Sarabhai to preserve and promote classical Indian dance and theater.

Indian Space Research Organisation (ISRO)

- **Introduction & Institutional composition**
  - **Genesis and Evolution:** Evolved from the Indian National Committee for Space Research (INCOSPAR, 1962) into the Indian Space Research Organisation (ISRO) on August 15, 1969, under the vision of Dr. Vikram Sarabhai.
  - **Administrative Governance:** Functions as the primary research and development arm under the **Department of Space (DOS)**, which reports directly to the Prime Minister's Office (PMO). The Chairman of ISRO concurrently holds the post of Secretary, DOS.
  - **Core Philosophy:** Uniquely distinct from Cold War space programs; prioritized **"societal applications"** (tele-education, remote sensing for agriculture, and disaster warning) over military or geopolitical dominance.
- **Core Operational Portfolios & Technologies**
  - **Launch Vehicle:** Transitioned from basic experimental rockets to a robust multi-tier fleet:
    - *PSLV (Polar Satellite Launch Vehicle):* Four-stage "Workhorse of ISRO" known for its highly reliable Sun-Synchronous Polar Orbit deliveries.
    - *LVM3 (Launch Vehicle Mark-3):* India's heaviest operational three-stage rocket utilizing the indigenous high-thrust Cryogenic upper stage, central to human spaceflight.
    - *SSLV (Small Satellite Launch Vehicle):* Designed for rapid, low-cost commercial launch turnaround times for small satellites.
  - **Satellite Constellations:** Manages one of the world's largest civilian fleets of Earth Observation Satellites (EOS, formerly IRS series) and communication networks (GSAT/INSAT series).

  - **NavIC (IRNSS):** An independent regional satellite navigation system providing highly accurate position positioning across India and up to 1,500 km beyond its land borders.
- **Landmark Achievements**
  - **Aryabhata (1975) & Rohini (1980):** First domestic satellite (launched via a Soviet rocket) followed by India's first successful indigenous orbital launch using the SLV-3 rocket.
  - **Chandrayaan Series:** *Chandrayaan-1* (2008) confirmed the presence of water molecules on the lunar surface. *Chandrayaan-3* (2023) achieved a historic landing near the South Pole of the Moon, making India the fourth nation to soft-land on the Moon.
  - **Mars Orbiter Mission / Mangalyaan (2013):** Made India the first nation globally to reach Martian orbit on its maiden attempt, executed at a fraction of Western mission budgets.
  - **Aditya-L1 (2023)**: India's first dedicated space-based solar observatory, successfully placed around the Lagrange Point 1 (L1) to study the dynamics of the solar corona.

Conclusion

Dr. Vikram Sarabhai's enduring legacy lies in his unique ability to marry elite scientific inquiry with real-world grassroots development, transforming space technology into a powerful engine for social equity. By building self-sustaining institutions like ISRO and IIM Ahmedabad, he created the permanent human and technological infrastructure that continues to drive India's economic growth.

6. MS SWAMINATHAN

Dr. M.S. Swaminathan, universally revered as the **Father of the Green Revolution in India**, was a legendary geneticist and administrator whose pioneering work rescued the nation from the brink of mass starvation and chronic food insecurity in the 1960s. By introducing high-yielding varieties of wheat and rice alongside innovative agronomic practices, he transformed India from a desperate, **ship-to-mouth** food importer into a self-reliant agricultural powerhouse.

Early life of M.S. Swaminathan

- **Birth and Heritage:** Born on August 7, 1925, in Kumbakonam, Tamil Nadu, into a cultured and socially conscious family led by a physician father dedicated to public welfare.
- **Impact of the Freedom Struggle:** Raised in an atmosphere deeply influenced by Mahatma Gandhi's Swadeshi movement, which instilled an early commitment to national self-reliance and social equity.
- **1943 Bengal Famine:** Deeply traumatized by the devastating human toll of the Bengal Famine during his student years, he consciously chose to abandon a promising medical path to dedicate his life to feeding the nation.

Educational background of M.S. Swaminathan

- **Undergraduate Foundations:** Completed his Bachelor of Science in Zoology from Maharajas College, Ernakulam, before pursuing his passion with a Bachelor of Science in Agricultural Sciences from the Agricultural College, Coimbatore.
- **Post-Graduate Research:** Joined the Indian Agricultural Research Institute (IARI), New Delhi, in 1947, where he specialized in cytogenetics (cell genetics) and plant breeding.
- **Civil Services Choice:** Successfully cleared the prestigious Indian Police Service (IPS) exam in 1949 but chose to reject the appointment to remain fully dedicated to agricultural research.

Doctoral Milestones of M.S. Swaminathan

- **UNESCO Fellowship:** Traveled to Europe on a UNESCO fellowship to conduct critical research on potato genetics at the Institute of Agriculture in Wageningen, Netherlands.
- **Cambridge PhD Achievement:** Earned his Doctor of Philosophy (PhD) from the School of Agriculture

at the University of Cambridge in 1952, presenting a thesis on species differentiation within the genus *Solanum* (potato).

- **Post-Doctoral Tenure:** Conducted high-yield research on radiation genetics at the University of Wisconsin, Madison, before choosing to return home to serve India's struggling agricultural sector.

Contribution of MS swaminathan

- **Architect of the Green Revolution**
  - **Semi-Dwarf Wheat:** Collaborated closely with Nobel Laureate Dr. Norman Borlaug to introduce Mexican semi-dwarf wheat varieties (like *Sonora 64*) to prevent lodging.
  - **Massive Crop Yield Turnaround:** Engineered the hybridization of dwarf genes into domestic wheat and rice, transforming India from a "ship-to-mouth" importer to a food-surplus nation
  - **Demonstration Camps:** Spearheaded thousands of live demonstration plots across Punjab and Haryana to physically convince skeptical traditional farmers to adopt high-yielding variety (HYV) seeds and chemical fertilizers.
- **Institution Building & Agricultural Administration**
  - **Directorship of IARI:** Served as the Director-General of the Indian Agricultural Research Institute (IARI), transforming it into a cutting-edge hub for genetic and agronomic research.
  - **Leading ICAR and IRRI:** Guided the Indian Council of Agricultural Research (ICAR) as its Director-General, and later became the first Asian Director-General of the International Rice Research Institute (IRRI) in the Philippines.
  - **National Commission on Farmers:** Chaired the historic National Commission on Farmers (NCF, 2004–2006) to comprehensively address the structural roots of rural agrarian distress.
- **Policy Innovations & Swaminathan Formula**
  - **C2 + 50 % MSP Formula:** Championed the landmark recommendation that the Minimum Support Price (MSP) must be fixed at at least 50% more than the comprehensive cost of production, a key demand of farmer unions.
  - **Transitioning to the "Evergreen Revolution":** Coined and popularized the philosophy of the "Evergreen Revolution," which demands scaling up productivity without causing ecological degradation, groundwater depletion, or soil erosion.
  - **MSSRF Foundation:** Established the M.S. Swaminathan Research Foundation in Chennai to bridge science and society by promoting sustainable, eco-friendly livelihood options for rural and tribal communities.

Major Awards and Recognitions of Dr. M.S. Swaminathan

Year	Award / Honour	Significance
2024	**Bharat Ratna**	India's highest civilian award, conferred posthumously for transforming national food security.
1987	**Inaugural World Food Prize**	Widely regarded as the "Nobel Prize for Agriculture"; recognized his leadership in the Green Revolution.
1989	**Padma Vibhushan**	India's second-highest civilian award for exceptional and distinguished public service.
1972	**Padma Bhushan**	India's third-highest civilian award for distinguished service of high order.
1967	**Padma Shri**	Awarded early in his career for outstanding contributions to genetic and agricultural sciences.
1971	**Ramon Magsaysay Award**	Conferred for Community Leadership, emphasizing his trust-building work among grassroots farmers.

1961	**Shanti Swarup Bhatnagar Prize**	Awarded for his core research in biological sciences and plant cytogenetics.
1986	**Albert Einstein World Award of Science**	International recognition for his pioneering global contributions to scientific development.
1994	**UNEP Sasakawa Environment Prize**	Acknowledged his landmark global work in sustainable development and economic ecology.
2000	**Indira Gandhi Prize**	Conferred for Peace, Disarmament, and Development, noting his work towards a hunger-free world.

Conclusion

Dr. M.S. Swaminathan's ultimate legacy lies in turning a famine-threatened India into a self-reliant, food-surplus nation, permanently securing the fundamental right to food for millions. Beyond the immediate success of the Green Revolution, his vision evolved into the "Evergreen Revolution," which championed structural, farmer-centric policies like fair minimum support prices to safeguard rural livelihoods.

7. SATYENDRA NATH BOSE

Satyendra Nath Bose was a legendary Indian physicist and mathematician whose ground-breaking research in the 1920s permanently revolutionized the world of quantum mechanics. By deriving a new statistical law for light particles without relying on classical physics, he provided the foundational framework for an entire class of subatomic particles, which were named "bosons" in his honor by Paul Dirac.

Early life of Satyendra Nath Bose

- **Birth and Heritage:** Born on January 1, 1894, in Calcutta (now Kolkata), West Bengal, as the eldest and only son among seven children in a middle-class Bengali family.
- **Father's Influence:** Raised by a father who was an accountant in the East Indian Railway Company and a passion-driven mathematician, who fostered Satyendra's early love for numbers.
- **Schooling Brilliance:** Completed his schooling at the historic **Hindu School** in Calcutta, where his mathematics teacher famously awarded him 110 out of 100 marks because he solved problems using multiple original methods.

Early Career

- **Self-Taught Quantum Scientist:** Entered academia when modern physics textbooks (on Einstein's relativity and Planck's quantum theory) were unavailable in English, forcing him to learn German and French to read original research papers.
- **Calcutta University Faculty:** Appointed as a research scholar and lecturer at the newly founded University College of Science by Sir Asutosh Mookerjee in 1916, where he began co-translating Einstein's papers into English.
- **Dhaka University Foundation:** Moved to the newly established **University of Dhaka** in 1921 as a Reader in Physics, setting up the advanced laboratories where he would eventually make his historic quantum discoveries.

Contribution of Satyendra nath bose

- **Quantum Statistics & Born of the Boson**
  - **Bose Statistics (1924):** Derived Max Planck's quantum radiation law entirely without relying on classical physics, introducing a radical new way to count and track identical subatomic particles.
  - **Einstein Collaboration:** Mailed his short research paper directly to Albert Einstein after British journals rejected it; Einstein recognized its genius, translated it into German, and got it published in the prestigious *Zeitschrift für Physik*.

- **Naming of the "Boson":** His foundational work led legendary physicist Paul Dirac to coin the term **"Boson"** to permanently classify all fundamental force-carrying particles (like photons and gluons) that follow Bose's statistical rules.

- **Deep Matter Research & Relativity**
  - **Bose-Einstein Condensate (BEC):** Collaborated theoretically with Einstein to predict a unique fifth state of matter, where atoms cooled to near absolute zero collapse into a single quantum "super-atom."
  - **Unified Field Theory:** Spent years working on complex mathematical equations to support Einstein's ambitious, unfinished Unified Field Theory, which aimed to merge gravity with electromagnetism.
  - **Co-translating Einstein:** Partnered with his colleague Meghnad Saha to publish the world's very first English translation of Albert Einstein's groundbreaking papers on Special and General Relativity.
- **Institutional Leadership & Swadeshi Science**
  - **Dhaka and Calcutta Leadership:** Served as the Head of the Physics Department at Dhaka University and later as a Professor at Calcutta University, building premier indigenous research laboratories from scratch.
  - **Promoting Science in Mother Tongue:** Strongly advocated for teaching higher sciences and cutting-edge physics in regional Indian languages, founding the *Bangiya Bijnan Parishad* (Bengali Science Society) to popularize scientific temper.
  - **National Professor Appointment:** Appointed as a National Professor by the Government of India in 1958, the highest academic honor in the country, recognizing his lifetime of service to independent Indian science.

Major Awards and Recognitions of Satyendra Nath Bose

Year	Awards	Significance
1954	Padma Vibhushan	India's second-highest civilian award; conferred in the inaugural year of the awards for outstanding contribution to physical sciences.
1958	National Professor of India	Appointed to the nation's highest academic chair, allowing him to pursue independent research for life.
1958	Fellow of the Royal Society (FRS)	Elected as a Fellow for his pioneering work in quantum statistics, cementing his place in global elite science.
1944	General President of ISC	Elected to preside over the 31st annual session, a position reserved for India's foremost scientific leaders.
1948	Founder President of BBP	Founded and led this body to popularize and teach deep science using regional vernacular languages.
1952–58	Nominated Member of Parliament	Nominated to the Rajya Sabha (Upper House) in recognition of his immense intellectual stature and service to society.
1959	Vice-Chancellor of Visva-Bharati	Appointed to head the unique university founded by Rabindranath Tagore, bridging scientific logic with cultural philosophy.
Continuous	The Physics Nobel Snub	Nominated multiple times for the Nobel Prize; though never awarded, the Committee explicitly acknowledged that the 2001 and 2013 Physics Nobels were built directly upon Bose's foundational discoveries.

Conclusion

Satyendra Nath Bose's profound legacy lies in redefining the foundational laws of quantum mechanics, ensuring that his name remains permanently etched in the annals of global physics through the naming of the **boson**.

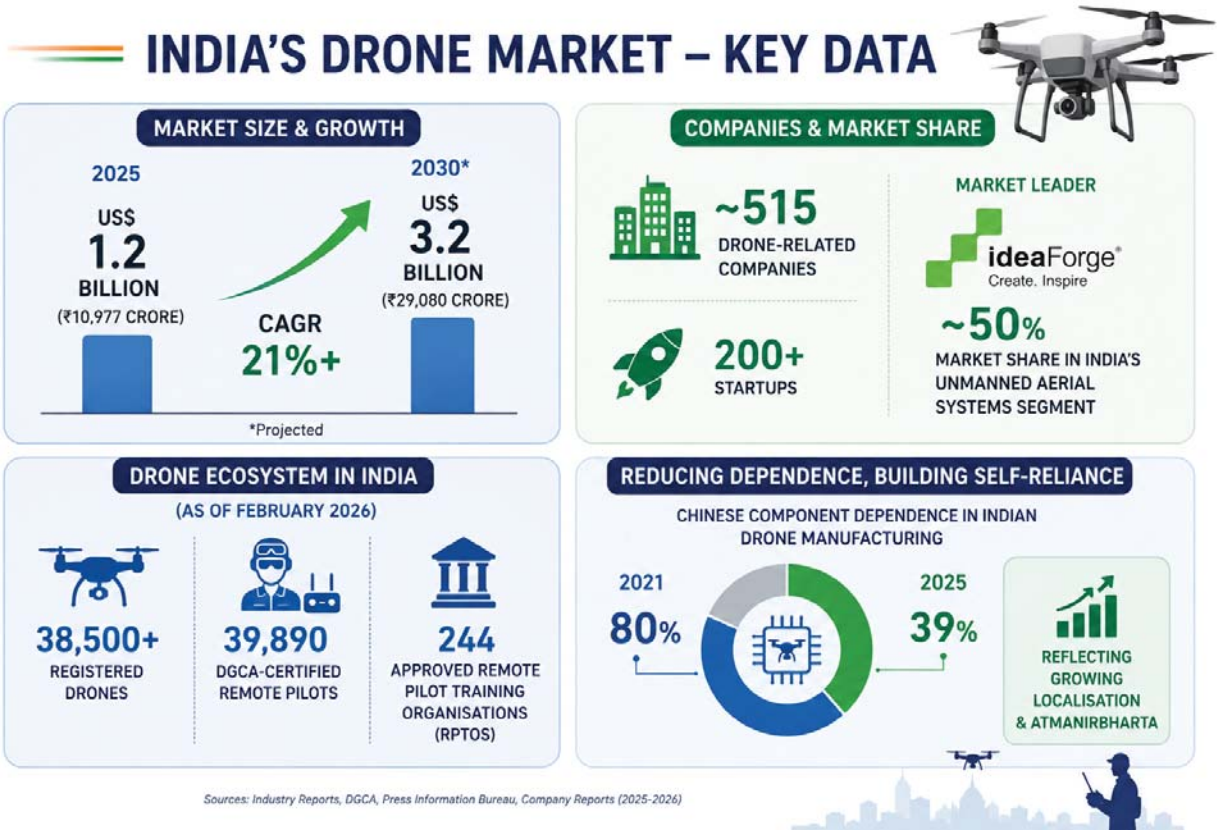
MGP/BP/GS3/025

EMERGING TECHNOLOGIES

1. DRONES

A **drone**, formally called an **Unmanned Aerial Vehicle/System (UAV/UAS)**, is an aircraft that operates **without a human pilot on board**. It is controlled either remotely by a ground-based operator or autonomously through onboard computers, GPS, and sensors.

India has emerged as one of the fastest-growing drone markets globally, with the government aiming to make India a **global drone hub by 2030**, driven by a mix of **regulatory reform, financial incentives, and rising strategic demand**.



Government Policy and Regulatory Framework

- **Drone Rules, 2021:** A liberalised, trust-based regulatory framework that replaced the earlier, stricter 2018 rules — reduced the number of approvals needed, expanded the 'green zone' for unrestricted flying, and removed the requirement for **security clearance** before registration.

- **Import Ban (February 2022):** Government banned import of foreign fully-built drones (except for defence, R&D, and security purposes) to boost domestic manufacturing under **Make in India** and **Atmanirbhar Bharat**.

  Component Freedom: While fully built drones face strict prohibitions, the import of individual drone parts (e.g., motors, sensors, flight controllers, batteries) is classified as 'Free,' requiring no prior DGFT approvals

- **Production Linked Incentive (PLI) Scheme:** Launched in September 2021 with an outlay of **Rs 120 crore** across 23 beneficiary firms. It is aimed at boosting drone and drone-component manufacturing; credited with triggering a **seven-fold rise** in domestic drone turnover.
- **Digital Sky / eGCA Platform:** An online platform for drone registration, remote pilot certification, and flight permissions; regulatory services have now migrated to the **eGCA (electronic Government Civil Aviation)** portal, while operational services like flight plans remain integrated with Digital Sky.
- **GST Rationalisation (September 2025):** GST on drones cut from the earlier **18%/28%** to a uniform **5%**, also extended to flight and motion simulators used for pilot training, lowering costs across the value chain.
- **Draft Drone Bill, 2025:** A new draft bill has been proposed, though several industry stakeholders view some of its provisions as **more restrictive** than the liberalised 2021 Rules, raising concerns about new entry barriers.
- **Capacity-Building Platforms:** Flagship events like **Bharat Drone Shakti**, the **Bharat Drone Mahotsav**, and the **Drone International Expo** promote Drone-as-a-Service (DaaS) startups and showcase indigenous technology; **SwaYaan** supports human resource development in unmanned aircraft systems.

Applications of Drone Technology

(a) Agriculture and Rural Development

- **Namo Drone Didi Scheme:** Launched in **November 2023** by the Ministry of Agriculture, it aims to provide drones to **15,000 Women Self-Help Groups (SHGs)** for precision spraying and rental drone services, boosting rural incomes.
- **Precision Agriculture:** Drones support **crop health monitoring, nano-fertiliser and pesticide spraying, and soil mapping**, cutting input costs and improving yields (studies show input cost reduction of about **20%** and yield gains of about **10%**).

(b) Governance and Land Administration

- **Reduced Survey Cost :** According to estimates , Drones reduce land survey costs by nearly **40%** compared to traditional satellite mapping.

  - **SVAMITVA SCHEME** was Launched in April 2020 by the **Ministry of Panchayati Raj**.
  - It uses drone-based mapping of rural inhabited (abadi) areas to issue legal **property cards**, helping resolve land disputes and improve access to bank credit.
  - By **December 2025**, drone surveys covered **3.28 lakh villages** (about 95% of the target), producing **2.76 crore property cards** for **1.82 lakh villages** across 31 states/UTs.

(c) Infrastructure and Disaster Management

- **Highway Monitoring:** The **National Highways Authority of India (NHAI)** mandates monthly drone-video recording of all highway projects for progress tracking and dispute resolution.
- **Disaster Response:** Drones assist in damage assessment, search-and-rescue, and relief coordination during floods, earthquakes, and other emergencies.

- **Smart Cities and Logistics:** Used for GIS-based urban surveillance, last-mile delivery pilots, and cold-chain logistics in remote and hard-to-reach areas.

(d) Defence and National Security

- **Operation Sindoor:** India's response to the Pahalgam terror attack showcased large-scale, tri-service deployment of **indigenous drones and loitering munitions** (e.g., **Nagastra-1**) for precision strikes, marking the first major India-Pakistan 'non-contact war'.
- **Integrated Air Defence:** The indigenous **Akashteer** command-and-control system, networked with **Akash missiles** and the **Integrated Air Command and Control System (IACCS)**, reportedly neutralised hostile drones and missiles with a near **100% kill rate** during the operation.
- **Counter-Drone Technology:** DRDO's **Anti-Drone D-4 System** and private-sector players (Zen Technologies, Big Bang Boom Solutions) are strengthening India's layered counter-UAS grid.
- **Post-Sindoor Push:** Emergency procurement of nearly **Rs 9,000 crore** was directed towards UAVs, loitering munitions, and counter-drone systems; the Army's **'Technology Roadmap for UAS and Loitering Munitions'** has identified **30 types across 80 variants** of drones needed across surveillance, strike, air defence, and logistics roles.

Significance of Drone Technology for India

Dimension	How Drone Technology Helps
Rural Livelihoods	**Namo Drone Didi** creates new income opportunities for women SHGs while modernising farming practices.
Land Governance	**SVAMITVA** brings legal clarity to rural land ownership, reducing disputes and improving credit access for villagers.
National Security	**Operation Sindoor** demonstrated that indigenous drones and counter-drone systems can deliver a decisive edge in modern, 'non-contact' warfare.
Ease of Doing Business	Liberalised **Drone Rules 2021**, GST cuts, and Digital Sky/eGCA integration have sharply reduced entry barriers for drone startups and MSMEs.
Atmanirbhar Bharat	The **2022 import ban** and PLI scheme have driven a steep fall in import dependence for drone components, from 80% to 39% in four years.

Challenges Facing Drone Technology in India

- **Fragile Industry Economics:** Despite rapid growth, the sector has seen heavy **consolidation**, reflecting thin margins and business-model uncertainty for many smaller players.
- **Import Dependence for Components:** Even with progress, a meaningful share of critical components (batteries, sensors, chips) remains imported, exposing supply chains to geopolitical risk.
- **Regulatory Uncertainty:** Industry concerns that the draft Drone Bill, 2025 could roll back some of the liberalisation achieved under the 2021 Rules, potentially slowing innovation.
- **BVLOS Limitations:** Beyond Visual Line of Sight (BVLOS) operations — essential for large-scale delivery and long-range surveillance — remain constrained pending fuller regulatory maturity.
- **Skill Shortage:** Rapid sectoral growth has outpaced the availability of trained remote pilots, technicians, and data analysts, despite expanding RPTO networks.
- **Counter-Drone Threats:** The rise of hostile drone swarms and loitering munitions (as seen in Operation Sindoor and global conflicts like Ukraine) necessitates continuous upgrades to counter-UAS technology.
- **Privacy and Airspace Management Concerns:** Wider civilian drone use raises questions of privacy, airspace congestion, and safety, especially in urban areas.

Way Forward

Measures needed	Broad Explanations
1. Calibrating the Drone Bill	Ensure the new **Drone Bill** preserves the liberalisation introduced by the **Drone Rules, 2021** while strengthening regulation through stakeholder consultation.
2. Deepening Component Localisation	Expand **PLI-style incentives** for critical components such as batteries, sensors, and flight controllers to reduce import dependence.
3. Enabling BVLOS Operations	Fast-track regulations for **Beyond Visual Line of Sight (BVLOS)** operations to unlock applications in logistics, agriculture, and surveillance.
4. Scaling Up Skilling and Training	Expand **RPTOs** and strengthen programmes like **SwaYaan** to develop expertise in AI-enabled drones, data analytics, and counter-drone technologies.
5. Strengthening Counter-Drone Capability	Invest in indigenous **Counter-UAS systems** by building on operational lessons from **Operation Sindoor**.
6. Balancing Innovation with Safety	Establish robust standards for **privacy, airspace management, and operational safety** to promote responsible drone adoption.

Conclusion

As drones evolve from niche tools to critical infrastructure, India must balance **technological innovation with privacy, security, and airspace safety** to enable responsible growth of the sector.

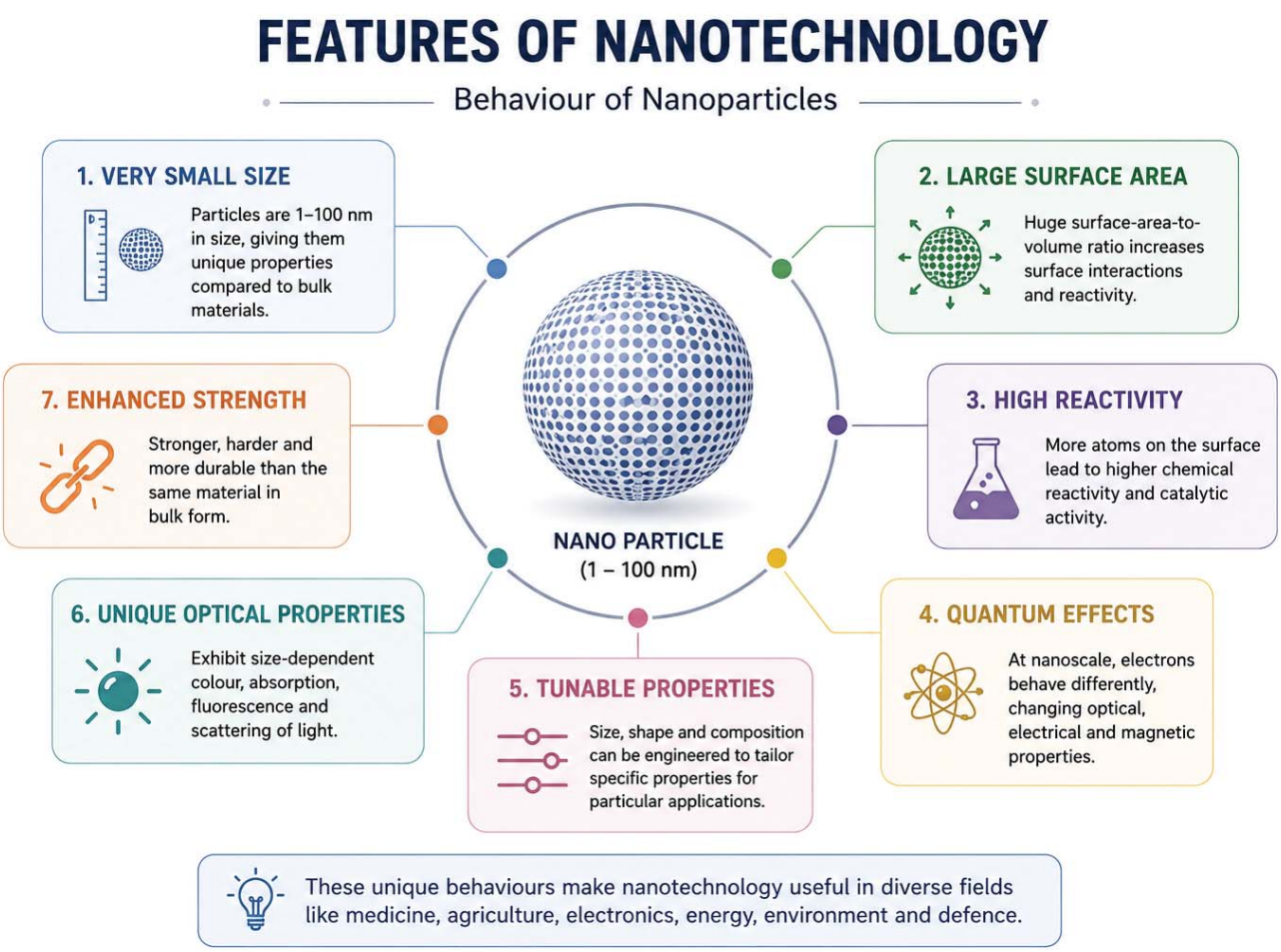
2. NANO-TECHNOLOGY AND ITS USE IN DIFFERENT SECTORS

Nanotechnology is the manipulation of matter at **1–100 nanometres**. In Nano state , materials exhibit unique properties such as **higher strength, reactivity, and conductivity**- enabling its applications across **healthcare, agriculture, energy, electronics, defence, and water treatment**.

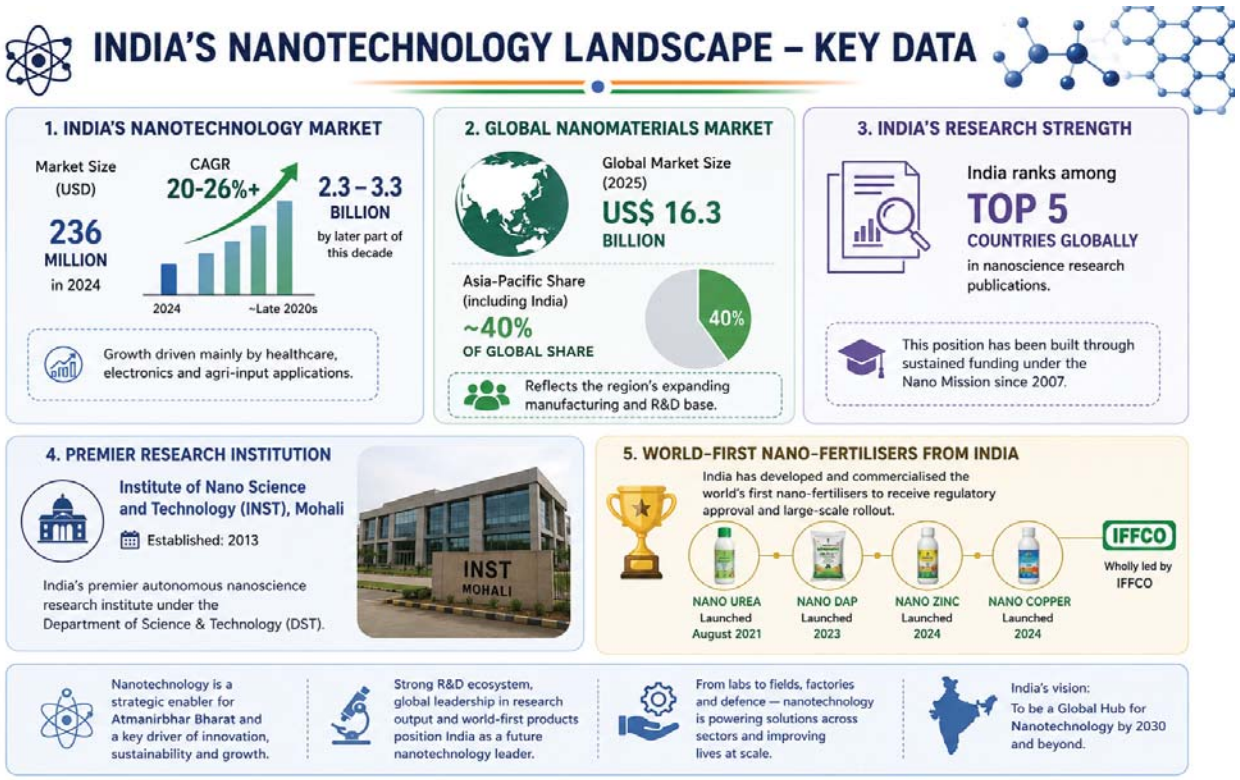
India has been building capacity in this field for close to two decades through the **Nano Mission**, and is now translating lab research into real products — from **Nano Urea** in farmers' fields to nano-coatings in defence gear — as it aims to become a serious global player in nanoscience and its applications.

Government Policy and Institutional Framework

- **Nano Mission (2007):** Launched in May 2007 by the **Department of Science and Technology (DST)** as an umbrella capacity-building programme. It is aimed at basic research, infrastructure and human resource development in nanoscience.
- **Nano Mission Phase-II (2014-2017):** Continued under the 12th Plan with a further allocation of about **Rs 650 crore**; the Mission formally stood completed on **31 March 2017**.
- **National Programme on Nano Science and Technology: After 2017, the Nano Mission was converted into this ongoing programme** — **no new projects** are sanctioned, but DST continues to monitor and support previously approved research.



- **Institute of Nano Science and Technology (INST), Mohali:** Set up in 2013 as a dedicated autonomous institute for advanced nanoscience research, training and international collaboration.
- **ICONSAT (International Conference on NanoScience and NanoTechnology):** A biennial international conference held under the Nano Mission to bring together researchers across energy, health, agriculture and materials science.
- **National Regulatory Framework Road-map for Nanotechnology (NRFR-Nanotech):** A DST-led dialogue process working towards common standards and a regulatory approach for nano-based products and materials.
- **Fertiliser Control Order (FCO), 1985 - Schedule VII:** Nano Urea, Nano DAP, Nano Zinc and Nano Copper have been notified under this order, giving them formal regulatory status as approved agricultural inputs.
- **PLI Scheme for Semiconductors:** While not nano-specific, the scheme (with over **US$ 1 billion** invested by early 2025) is driving adoption of nanomaterials such as carbon nanotubes and graphene in Indian chip-making and electronics.
- **BIRAC (Biotechnology Industry Research Assistance Council):** Supports nano-biotechnology start-ups and companies working on nanomedicine, diagnostics and drug-delivery innovation.



Applications of Nanotechnology

(a) Agriculture and Rural Livelihoods

- **Nano Urea and Nano DAP:** Developed and patented by **IFFCO**, these liquid nano-fertilisers deliver nitrogen and phosphorus in particles under 100 nanometres, giving far higher nutrient-use efficiency (over 80% for nano urea, against 30-50% for conventional urea) and cutting the quantity of fertiliser needed by roughly half.

> *A single 500 ml bottle of Nano Urea is designed to substitute one 45 kg bag of conventional urea, directly easing the government's fertilizer subsidy bill.*

- **Nano Zinc and Nano Copper (Liquid):** Government-approved micronutrient formulations that help correct zinc and copper deficiencies in soil, improving crop productivity.
- **Precision Agriculture: Nano-sensors** and **nano-encapsulated pesticides** enable slow, targeted release of nutrients and crop-protection chemicals, reducing runoff, leaching and environmental pollution.

(b) Healthcare and Nanomedicine

- **Targeted Drug Delivery:** Nanoparticles and nanocarriers are used to deliver drugs precisely to diseased cells (especially in cancer and diabetes treatment), improving effectiveness and reducing side effects.

> Use of Carbon Nano tubes for drug delivery

- **Diagnostics:** Nano-enabled diagnostic tools, including **quantum dots** and **nano-biosensors**, allow faster and more accurate early detection of disease.
- **Personalised Medicine:** Nanotechnology supports tailoring of drug formulations to individual patients, an area expected to grow strongly over the coming years with **rising chronic-disease burden in India**.

(c) Electronics, Semiconductors and Energy

- **Chips and Devices:** Nanomaterials such as **graphene, carbon nanotubes and nanowires** are improving transistor scalability, chip density and thermal management, supporting India's semiconductor manufacturing push.
- **Batteries and Solar Cells:** Nanotechnology is enabling next-generation **sodium-ion batteries** as a **cheaper, more sustainable alternative to lithium-ion**, along with more efficient nanomaterial-based solar cells.
- **Domestic Innovation:** Indian start-ups such as those producing single-walled carbon nanotubes are developing materials many times more conductive than copper for use in solar panels and thermal management.

(d) Water, Environment and Textiles

- **Water Purification:** Research institutes such as **IIT Madras** have used nanotechnology for **low-cost arsenic decontamination of drinking water**, a major public health concern in parts of India.
- **Smart Textiles: IIT Delhi** has developed nanotechnology-based, water-efficient self-cleaning fabric treatments; nano-coatings more broadly give textiles **antimicrobial, UV-resistant and flame-retardant properties**.
- **Pollution Sensing:** Nano-sensors are being explored for real-time detection of air and water pollutants at very low concentrations.

(e) Defence and Strategic Applications

- **Protective Gear:** DST-supported projects have developed nano-coatings and nano-based materials to strengthen **Personal Protective Equipment (PPE)** for soldiers and frontline personnel.
- **Advanced Materials:** Nanomaterials are being explored fo**r lightweight armour, stealth coatings and more durable components for defence platforms,** in line with the **Atmanirbhar Bharat** push in defence manufacturing.

Significance of Nanotechnology for India

Dimension	How Nanotechnology Helps
Agriculture & Food Security	**Nano Urea and Nano DAP** cut fertiliser over-use, reduce the subsidy burden, and raise nutrient-use efficiency, supporting the **Atmanirbhar Krishi** goal.
Healthcare	Nano-enabled **targeted drug delivery, diagnostics, and nanomedicine** improve treatment of cancer, diabetes and infectious disease while lowering side effects.
Electronics & Semiconductors	Nanomaterials like **graphene and carbon nanotubes** improve chip density and thermal management, complementing India's semiconductor and PLI push.
Energy	Nano-enabled **batteries, solar cells and fuel cells** support cheaper storage and cleaner power, aiding India's renewable energy and net-zero targets.
Water & Environment	Nano-filtration membranes provide low-cost removal of **arsenic, fluoride and heavy metals** from drinking water in rural and contaminated areas.
Defence & Strategic Security	Nano-coatings, lightweight armour and smart materials strengthen protective gear and platforms, supporting **Atmanirbhar Bharat** in defence manufacturing.

Global S&T Standing	India ranks among the **top five countries** worldwide in nanoscience publications, boosting its profile as an emerging knowledge economy.

Challenges Facing Nanotechnology Development in India

- **Weak Industry-Academia Translation:** A large volume of research and publications has not yet translated into a proportionate number of patents or commercialised products; private-sector contribution to nanotech R&D remains limited.
- **Regulatory Gaps:** India still lacks a comprehensive, dedicated regulatory framework for nano-based products;

  > Most nano-fertilisers have been approved case-by-case under the existing Fertiliser Control Order rather than a nanotechnology-specific law.

- **High Cost of Infrastructure:** Nanoscale characterisation and fabrication require expensive, specialised equipment (like electron microscopes and clean rooms), which are concentrated in a few premier institutes.
- **Skill and Manpower Shortage:** There remains a shortage of trained nano-scientists, technicians and interdisciplinary researchers able to work across physics, chemistry, biology and engineering.
- **Health and Environmental Safety Concerns:** The **long-term toxicit**y and **environmental impact of many nanomaterials** are still not fully understood, raising safety and disposal concerns as products scale up.
- **Import Dependence:** Critical nanomaterials, precursor chemicals and fabrication equipment are still substantially imported, exposing Indian manufacturers to global supply-chain risk.
- **Public Awareness and Trust Deficit:** Farmers, consumers and even some regulators have limited awareness of nano-based products (like Nano Urea), which can slow adoption despite proven benefits.

Way Forward

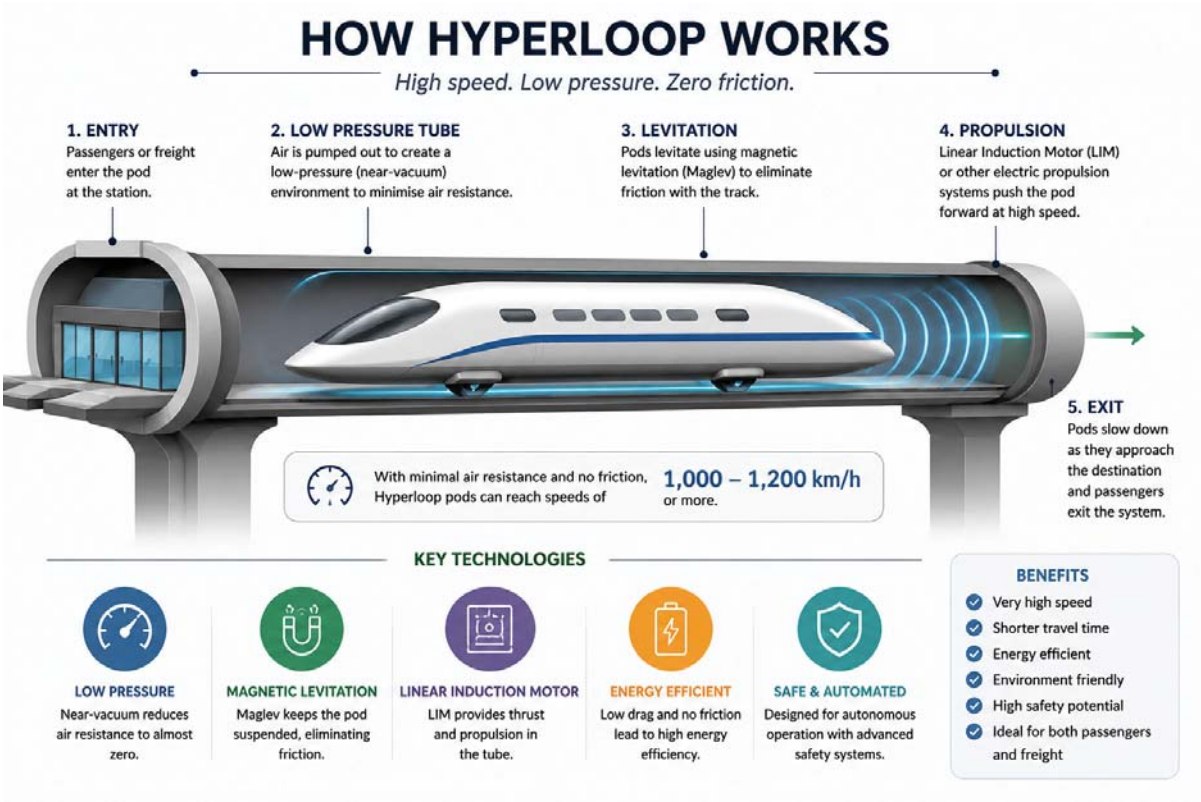
- **Build a Dedicated Nanotechnology Regulatory Framework:** Fast-track the proposed **National Regulatory Framework Road-map for Nanotechnology** so that approvals for new nano-products are faster and more predictable, without compromising safety.
- **Strengthen Industry-Academia Linkages: Create stronger incentives (tax breaks, co-funded labs)** for private companies to commercialise nanotech research coming out of institutions like INST Mohali and the IITs.
- **Scale Up Nano-Agri Inputs:** Expand production and last-mile distribution of Nano Urea, Nano DAP and micronutrient formulations, especially through cooperatives, to further cut the fertilizer subsidy burden.
- **Invest in Skilling:** Expand **dedicated nanoscience and nanotechnology courses and fellowships**, building on the Nano Mission's human-resource-development mandate, to widen the talent pipeline.
- **Deepen Component Localisation:** Encourage domestic production of nanomaterials, precursor chemicals and characterisation equipment to cut import dependence, in line with Atmanirbhar Bharat.
- **Mainstream Safety Research: Fund dedicated toxicology and environmental-impact studies** on nanomaterials so that safety standards keep pace with rapid commercialisation.
- **Raise Public Awareness:** Run structured outreach for farmers and consumers on nano-products (as IFFCO has done with Nano Urea) to build trust and accelerate adoption.

Conclusion

As the world moves towards the next industrial revolution, nanotechnology will be a key driver of **precision manufacturing, sustainable development, and the bioeconomy**, making it central to **Viksit Bharat@2047**.

3. HYPERLOOP

Hyperloop is a proposed **fifth mode of transport**. It uses **pods** that travel through **low-pressure (near-vacuum) tubes**. Pods are **levitated and propelled** using **Maglev** or a **Linear Induction Motor (LIM)**.



The technology can achieve speeds of **1,000–1,200 km/h**. It was popularised by **Elon Musk** through the **Hyperloop Alpha (2013)** paper. No commercial Hyperloop is operational yet. In India, **IIT Madras** and **TuTr Hyperloop**, supported by the **Ministry of Railways**, are leading its development.

Government Policy and Institutional Framework

- **NITI Aayog Expert Committee on Hyperloop (2020):** A high-level panel set up by the government think-tank to study the **technological, commercial and financial viability** of hyperloop technology (initially examined in the context of Virgin Hyperloop) and to assess the **safety standards and regulatory needs** for its introduction in India.
- **Ministry of Railways support to IIT Madras:** Railways has provided repeated **financial and technical backing** to the IIT Madras hyperloop programme and its Avishkar student team, treating it as the anchor institution for indigenous hyperloop R&D.
- **Integral Coach Factory (ICF), Chennai:** Designated to develop the **electronics and control systems** for the hyperloop project, leveraging ICF's existing expertise from building **Vande Bharat** trainsets.
- **State-level partnership (Maharashtra):** The JNPT-Vadhavan hyperloop cargo corridor MoU with TuTr Hyperloop, backed by technical inputs from **IIT Bombay and IIT Madras**, marks the first instance of a state government actively co-developing a hyperloop corridor in India.
- **Absence of a dedicated regulator:** Unlike aviation (DGCA) or railways (CRS), India currently has **no dedicated regulatory body or statute** for hyperloop systems; safety, certification and right-of-way questions are still being examined on a case-by-case, project-specific basis.

Applications of Hyperloop in the Indian Context

(a) Freight and Port Logistics

The **JNPT-Vadhavan** corridor is being designed primarily as a **cargo hyperloop**, since JNPT alone handles over half of India's containerised cargo and the upcoming Vadhavan Port is expected to handle roughly **250 million tonnes annually**.

(b) Passenger Mobility on High-Density Corridors

Over the long run, hyperloop is being explored as a possible complement to **high-speed rail (the Mumbai-Ahmedabad bullet train corridor)** for very high-density intercity routes

(c) Indigenous Deep-Tech and Start-up Ecosystem

TuTr Hyperloop, incubated at IIT Madras, exemplifies how a student research project (the Avishkar Hyperloop team) has grown into a **deep-tech start-up** attracting state government partnership.

(d) Industrial and Manufacturing Linkages

Large industrial partners such as **Larsen & Toubro** and steelmakers like **ArcelorMittal** have been associated with hyperloop tube and infrastructure development, which could give a boost to domestic **heavy engineering, tube-fabrication and precision manufacturing** capacity.

(e) Sustainable, Low-Emission Transport

Because hyperloop pods are designed to run on **electric propulsion** rather than fossil fuels, the technology is being positioned as consistent with India's **net-zero-by-2070** commitment

Challenges Facing Hyperloop Development in India

- **No Dedicated Regulatory Framework:** India lacks defined standards for hyperloop safety, tube-pressure management, electromagnetic systems, certification and emergency evacuation — a **gap flagged by both the NITI Aayog committee** and independent feasibility studies.
- **Technology Still Experimental Globally:** No country has yet operated a commercial hyperloop line.

Even pioneering foreign firms such as Virgin Hyperloop have shifted focus away from passenger service toward freight, underlining the technology's unresolved engineering and safety challenges.

- **High Capital Cost:** Building sealed, depressurised tubes over long distances, along with the propulsion and control infrastructure, **involves very high upfront capital cost** compared with conventional rail, raising questions about financial viability.
- **Land Acquisition:** Like other large linear infrastructure projects in India (highways, bullet trains), a hyperloop corridor would face the familiar challenge of **land acquisition and right-of-way** disputes along its alignment.
- **Limited Public Awareness and Trust:** As a wholly new mode of transport, hyperloop faces the task of building public confidence in its safety, especially for any future passenger application.
- **Competing Priorities:** With India already committing large resources to the Mumbai-Ahmedabad bullet train and to expanding Vande Bharat services, hyperloop must compete for policy attention and capital against more proven high-speed rail technology.

Way Forward

1. Establishing a Dedicated Regulatory Framework	Notify clear standards for **safety, certification, and right-of-way** before commercial deployment.
2. Prioritising Freight-First Deployment	Focus initial hyperloop projects on **cargo transport and port logistics**, where commercial viability is higher.
3. Strengthening Industry–Academia Collaboration	Expand collaboration among **IIT Madras, TuTr Hyperloop, ICF Chennai, and Indian Railways** to develop a full-scale demonstration corridor.
4. Expanding State Partnerships	Encourage more states to adopt the **Maharashtra model** through land support and co-funding for hyperloop projects.
5. Deepening Component Localisation	Promote indigenous manufacturing of **tubes, LIM propulsion systems, and vacuum technologies** to reduce import dependence.
6. Investing in Safety Research and Skill Development	Strengthen R&D on **tube safety, emergency evacuation, and electromagnetic compatibility**, while developing a skilled hyperloop workforce.

Conclusion

Hyperloop has the potential to redefine **high-speed, sustainable transportation** by significantly reducing travel time and logistics costs. Its success, however, depends on **technological maturity, robust regulation, and commercial viability**.

4. AUTONOMOUS VEHICLES

Autonomous Vehicles (AVs) are vehicles that can sense their environment and drive with little or no human input. They do so by using sensors like **LiDAR, radar and cameras** combined with AI.

Automation is graded on the **SAE 0-5 scale**: Level 0 is no automation and Level 5 is full self-driving in all conditions.

Advanced Driver Assistance Systems (ADAS) are the stepping-stone technologies (mostly **Level 1-2**) that assist rather than replace the driver. Features like **Automatic Emergency Braking (AEB), lane-keep**

assist, adaptive cruise control and blind-spot warning. India is currently at **Level 2** at best, with ADAS being rolled out commercially while full autonomy remains a distant goal.

Government Policy and Institutional Framework

- **MoRTH ADAS Mandate:** GSR 184(E) makes a five-part ADAS suite compulsory for heavy vehicles in a phased manner through 2026, marking India's first binding ADAS regulation.

  GSR 184(E) is a major road safety notification issued by the Ministry of Road Transport and Highways (MoRTH) in India, mandating Advanced Driver Assistance Systems (ADAS) and stability controls for medium and heavy commercial vehicles

- **Bharat NCAP (BNCAP):** India's own car-crash-rating programme, increasingly factoring in ADAS and active-safety features alongside crash-test performance.
- **Motor Vehicles Act, 1988:** Still framed around the assumption of a **human driver**; it does not clearly address liability, insurance or certification for autonomous systems.
- **Government stance on full autonomy:** Road Transport Minister has repeatedly said India **will not permit fully driverless vehicles**, citing the risk to an estimated 70-80 lakh driving-related jobs and India's chaotic, mixed-traffic road conditions.
- **IndiaAI Mission and NITI Aayog's National AI Strategy** provide the broader policy umbrella under which autonomous-mobility R&D is being funded and coordinated.

Applications in the Indian Context

(a) Passenger Vehicle Safety (ADAS)

Mainstream carmakers like **Maruti Suzuki, Hyundai, Kia, Tata, Mahindra** — are fitting Level 1-2 ADAS such as **lane-keep assist and adaptive cruise control** into mid-range and premium cars, driven by BNCAP ratings and consumer demand.

(b) Freight and Logistics Corridors

Start-ups like **Flux Auto** are piloting autonomous trucking for **mining, ports and controlled logistics corridors**. It is seen as India's most realistic near-term use case since these are geo-fenced, low-complexity environments.

(c) Campus and Last-Mile Shuttles

Autonomous shuttle pilots have been tested on **university and tech-park campuses** in cities like **Bengaluru and Hyderabad**, offering a controlled setting to trial Level 3-4 technology.

(d) Heavy Vehicle Safety Mandate

The MoRTH ADAS mandate for **buses and trucks** targets India's high commercial-vehicle fatality rate by making collision-avoidance and driver-alertness systems compulsory.

Even though commercial trucks comprise only about 4.4% of the country's total registered vehicle fleet, they are the perpetrating vehicle in roughly 20% of all national road accident fatalities.

(e) Software and Engineering Services

Firms like **Tata Elxsi and KPIT** have built a large **global AV software outsourcing business**, giving India a strong role in the AV value chain even without large-scale domestic deployment.

Challenges Facing AVs/ADAS Ecosystem in India

- **Chaotic, Mixed Traffic:** Indian roads mix cars, two-wheelers, pedestrians, animals and carts without strict lane discipline — **conditions AV algorithms trained abroad struggle with**.
- **Outdated Legal Framework:** The Motor Vehicles Act, 1988 does not clarify **liability, insurance or certification** for autonomous systems.

- **Job-Loss Concerns:** Government reluctance to permit full autonomy stems directly from the risk to **millions of driving-sector jobs**.
- **High Cost and Infrastructure Gaps:** LiDAR and sensor costs, though falling, remain high; India also lacks dedicated **test tracks and simulation infrastructure**.
- **Cybersecurity and Data Privacy:** Connected, always-tracked vehicles raise risks under the **DPDP Act, 2023**, requiring robust safeguards against hacking and misuse of location data.
- **Compressed Compliance Timelines:** OEMs face tight deadlines to integrate the new ADAS mandate (AIS-162 to 188) into vehicles already deep into their design cycles.

Way Forward

Way Forward	Explanation
1. Adopting an ADAS-First Approach	Prioritise **ADAS deployment** and phased automation before moving towards fully autonomous vehicles.
2. Strengthening the Legal Framework	Amend the **Motor Vehicles Act** to define liability, insurance, and certification norms for autonomous vehicles.
3. Expanding Testing Infrastructure	Develop dedicated **AV test tracks, simulation facilities, and BIS certification** systems to support safe deployment.
4. Promoting Controlled-Environment Deployment	Deploy autonomous vehicles first in **ports, mines, campuses, and industrial corridors** before public roads.
5. Strengthening Cybersecurity Standards	Build on **AIS-189** and align connected vehicle security with the **Digital Personal Data Protection Act (DPDPA) 2023** to protect vehicle data.
6. Reskilling the Workforce	Expand **NSDC-led training** for roles in fleet monitoring, cybersecurity, and autonomous vehicle maintenance.

Conclusion

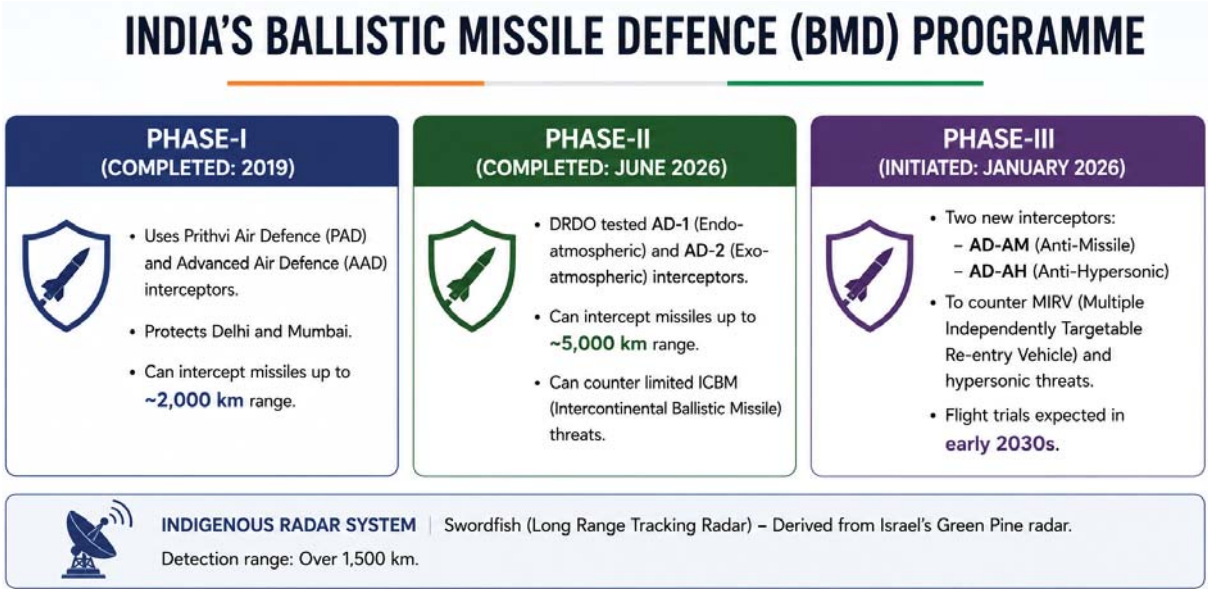
Though autonomous mobility has the potential to transform **road safety, logistics, and transport efficiency**.However, its adoption in India must be **gradual, safety-driven, and supported by robust regulation and digital infrastructure**.

5. BALLISTIC MISSILE DEFENSE

Ballistic Missile Defence (BMD) refers to a system of interceptor missiles, radars, and command centres designed to **detect, track, and destroy incoming ballistic missiles** before they hit their target.

Unlike cruise missiles, ballistic missiles follow a parabolic trajectory — launched by a rocket, they coast through space, and re-enter the atmosphere at very high speed, making interception technically difficult.

India's BMD programme was launched in **1999**, in the aftermath of **Pakistan's 1998 nuclear tests** and China's growing missile capability. It aims to give India a credible **multi-layered shield** against nuclear and conventional missile threats from its neighbourhood.



Government Policy and Institutional Framework

- **DRDO-led Indigenous Development:** The BMD programme is developed entirely in-house by DRDO. This reflects India's push for **strategic autonomy** in a domain where technology transfer is rarely available from other nations.
- **Mission Sudarshan Chakra:** An overarching plan to build a **comprehensive, multi-layered air and missile defence shield** for India by combining BMD interceptors, imported systems like the **S-400 ('Sudarshan Chakra')**, and indigenous air-defence systems.
- **Project Kusha (Extended Range Air Defence System):** A complementary indigenous **long-range air defence system**, often called India's 'S-400 killer', with three interceptor variants — **M1 (150 km), M2 (250 km), and M3 (350–400 km)**. This was approved by the Cabinet Committee on Security for five squadrons,in 2018.

- **Integrated Air Command and Control System (IACCS):** Acts as the digital backbone linking BMD radars, interceptors, the Akashteer battlefield air-defence network, and the IAF's command centres for real-time, network-centric engagement.
- **Indigenous Industry Role: Bharat Dynamics Limited (BDL)** manufactures the interceptor missiles while **Bharat Electronics Limited (BEL)** builds the radars and control systems. BEL is now committing over **Rs 562 crore** to a new production facility in **Chitrakoot, Uttar Pradesh**, under the UP Defence Industrial Corridor.
- **Import Route (Interim Cover):** Pending full indigenous rollout, India has procured the Russian **S-400 Triumf** system to provide near-term protection against aircraft, drones, and cruise missiles, alongside the indigenous BMD shield.

Key Systems and Recent Developments

(a) Ballistic Missile Defence Interceptors

- **AD-1 and AD-2:**
  - Core interceptors of **Phase-II BMD**
  - validated together in **June 2026** flight tests
  - capable of engaging missiles across endo- and exo-atmospheric altitudes
- **Phase-III Interceptors (AD-AM, AD-AH):**
  - Designed for **hypersonic glide vehicles** and **MIRV(Multiple Independently Targetable Re-entry Vehicle)-equipped missiles**
  - expected to use kinetic warheads and speeds exceeding Mach 7

(b) Complementary Air Defence Systems

- **Project Kusha (ERADS):** Under DRDO development to counter stealth aircraft, cruise missiles, drones, and some ballistic threats. It targets a single-shot kill probability of about **85%**, rising to **98.5%** with a two-missile salvo. Its induction is expected in phases between **2028 and 2030**.
- **Akash-NG and Akashteer:** The upgraded **Akash-NG** missile and the **Akashteer** command-and-control network provide shorter-range, mobile air-defence cover and reportedly achieved a near **100% kill rate** against hostile drones and missiles during **Operation Sindoor**.
- **S-400 'Sudarshan Chakra':** Russian long-range air defence system; its fourth squadron arrived in India in **June 2026**, with a **second deal** for additional units reportedly expected by late 2026.

(c) Operational Validation

- **Operation Sindoor:** India's response to the Pahalgam terror attack demonstrated the value of layered air and missile defence, with the S-400 and Akashteer-linked systems credited with neutralising Pakistani drones, missiles, and an AWACS-related asset during the conflict.

Significance of BMD for India

Dimension	Role of BMD
Strategic Deterrence	A credible BMD shield strengthens India's **no first use** nuclear doctrine by ensuring a second-strike capability even if adversary missiles are launched first.
National Security	Protects high-value targets like **Delhi-NCR** and other metros from missile attacks by hostile neighbours possessing nuclear-capable missiles.
Strategic Autonomy	Indigenous development (AD-1, AD-2, **Project Kusha**) reduces reliance on imported systems like the S-400, avoiding CAATSA-related sanctions risk.

Defence Manufacturing	Anchors production for **BDL and BEL**, creating a domestic ecosystem for missiles, radars, and command systems, with potential export value.
Regional Signalling	Demonstrated capability during **Operation Sindoor** signals resolve and technological maturity to both **Pakistan and China**.

Challenges Faced by India's BMD Programme

- **Saturation Attacks:** BMD systems can be overwhelmed by **missile swarms** or simultaneous multi-axis attacks, a tactic increasingly used in modern conflicts (e.g., Ukraine).
- **Hypersonic and MIRV Threats:** China's **DF-41 ICBM** and hypersonic glide vehicles, along with Pakistan's potential Multiple Independently Targetable Reentry Vehicle (MIRV) capability, are evolving faster than India's Phase-III interceptors, which are still years from deployment.
- **Import Dependence:** Despite indigenisation, India continues to rely on Russian **S-400** systems and some imported radar/sensor technology as an interim layer.
- **Cost and Timelines:** Big-ticket programmes like **Project Kusha (Rs 21,700 crore)** face long development-to-induction timelines (2028–2030), during which threat profiles may change further.
- **Limited Geographic Coverage:** Current deployment prioritises **Delhi and Mumbai**; extending a credible shield across all strategic and border cities remains a challenge.
- **Testing and Reliability Gaps:** Repeated but incremental testing (AD-1, AD-2) still needs to translate into full **operational deployment and production at scale**.

Way Forward	Broad Measures Needed
1. Accelerating Phase-III Development	Fast-track **AD-AM** and **AD-AH** interceptor trials to counter **hypersonic missiles** and **MIRV** threats.
2. Expanding BMD Coverage	Extend **Ballistic Missile Defence (BMD)** and **Project Kusha** beyond **Delhi and Mumbai** to other strategic cities.
3. Deepening Indigenisation	Increase production of **Project Kusha** and **Akash-NG** to reduce dependence on imported air defence systems.
4. Strengthening Integrated Air Defence	Integrate **BMD, Project Kusha, Akashteer, and IACCS** for a seamless, multi-layered missile defence network.
5. Enhancing Counter-Saturation Capability	Develop **cost-effective interceptors** and **directed-energy weapons** to counter missile and drone swarms.
6. Promoting Defence Exports	Leverage indigenous systems such as **Project Kusha** to strengthen India's position as a global defence exporter.

Conclusion

With the successful development of **BMD Phase-II** and **Project Kusha**, India is steadily emerging as a **self-reliant missile defence power** under **Atmanirbhar Bharat**.

■■■

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GOVERNMENT BUDGETING

1. UNION BUDGET OF INDIA - AN OVERVIEW

The Constitution of India has the following provisions when it comes to the Union Budget:

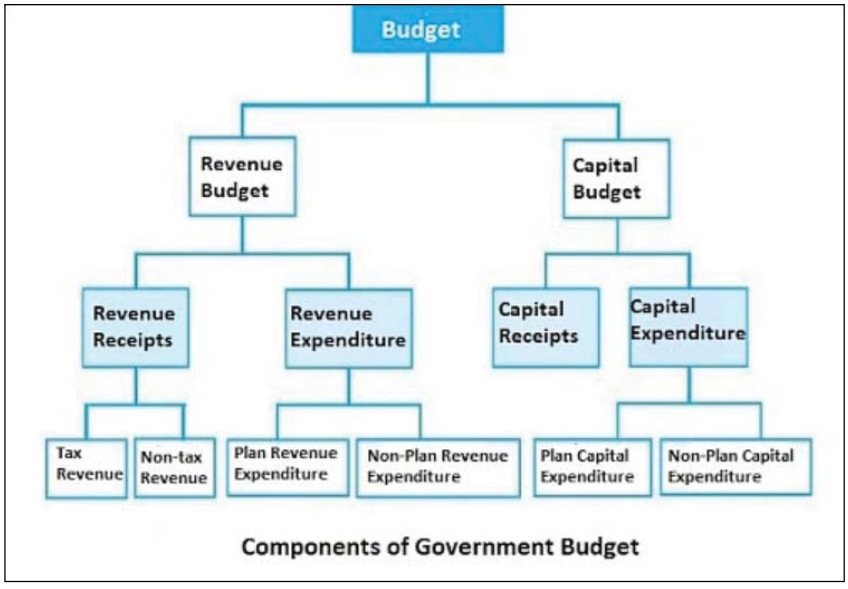
According to **Article 112** of the Indian Constitution, the Union Budget refers to the **Annual Financial Statement** of the Central Government. This statement provides details of estimated receipts and expenditures for the financial year.

The Union Budget is prepared by the **Department of Economic Affairs (DEA**) under the Ministry of Finance. It covers all aspects of government finances, including revenue generation, expenditure allocation, and policy announcements.

The Union Budget can be **sub divided into three categories**:

- **Budget Estimates (BE):** Projections for the upcoming fiscal year.
- **Revised Estimates (RE):** Updated estimates for the current fiscal year.
- **Provisional Actuals (PA):** Actual receipts and expenditures of the previous fiscal year.

Government Budget Components



Components of Government Budget

The Government Budget is **sub-divided** into two main components- **Revenue Budget and Capital Budget**

1. Revenue Budget

- The Revenue Budget details the government's **revenue receipts and revenue expenditure.**

- **Revenue Receipts**
  - Income received by the government that is not repayable.
  - Tax Revenue includes:
    - **Direct Taxes:** e.g., Income Tax, Corporation Tax
    - **Indirect Taxes:** e.g., Customs Duties, Excise Duties, Service Tax, GST
    - **Other Direct Taxes:** Wealth Tax, Gift Tax, etc.
  - **Non-Tax Revenue**
    - Interest receipts on loans
    - Dividends & profits from government investments
    - Fees & service charges
    - Spectrum revenue
    - Grants from foreign countries/institutions
- **Revenue Expenditure**
  - Revenue Expenditure neither creates assets nor generates future returns.
    - **Examples:** Salaries, pensions, subsidies, interest payments, grants to states/UTs, and daily functioning of government departments.

2. Capital Budget

- The Capital Budget reflects the assets and liabilities of the government and funds required for long-term development.

(a) Capital Receipts

- Funds that **either create liabilities or reduce assets**.
- **Debt-Creating:** Fresh loans and borrowings
- **Non-Debt Creating:** Recovery of loans, disinvestment proceeds

(b) Capital Expenditure

- Capital Expenditure is the spending that leads to creation of assets or investments.
  - **Examples:** Infrastructure (roads, schools, hospitals), equity in PSUs, loans to states/UTs, and repayment of loan principal.

Key Differences Between Revenue Budget and Capital Budget

Aspects	Revenue Budget	Capital Budget
Definition	Deals with the revenue receipts (tax & non-tax) and revenue expenditure of the government.	Deals with capital receipts (borrowings, recovery of loans, disinvestment, etc.) and capital expenditure (creation of assets, loan repayments).
Nature	Relates to the day-to-day functioning and maintenance of government services.	Relates to long-term investments and development expenditures.
Examples of Receipts	Tax revenues (Income Tax, GST, Excise), Non-tax revenues (Dividends, Fees, Interest).	Market borrowings, External loans, Recovery of loans, Disinvestment proceeds.
Examples of Expenditure	Salaries, pensions, subsidies, grants, interest payments, defence maintenance.	Building roads, railways, bridges, hospitals, buying machinery, equity infusion in PSUs.

Impact on Assets/ Liabilities	Does not create assets or reduce liabilities.	Leads to creation of assets or reduction of liabilities.
Fiscal Importance	High revenue deficit indicates the government is unable to meet its routine expenditure from its revenue.	High capital expenditure indicates focus on infrastructure and long-term growth.
Deficits Related	Revenue Deficit = Revenue Expenditure – Revenue Receipts.	Fiscal Deficit includes capital receipts & expenditures. Primary Deficit also related here.

Revenue Budget: Critical for immediate social needs (e.g., 80 crore beneficiaries under free grains scheme) but high revenue deficit (1.8% GDP, 2025-26) challenges FRBM targets (3% fiscal deficit by 2026).

- **Capital Budget:** Drives long-term growth (capex multiplier 1.5-2x, per Economic Survey 2024-25); Rs 11.21 lakh crore in 2025-26 aligns with Viksit Bharat, but debt-funded capex (Rs 14.82 lakh crore borrowings) risks sustainability.

Types of Budget Deficits

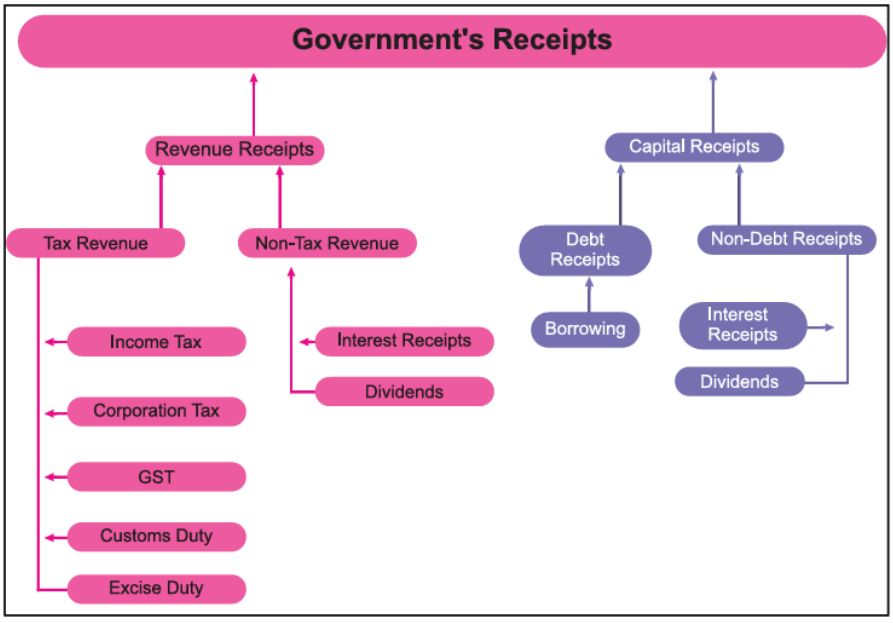
The Deficits in Budget is of the following types:

- **Budget Deficit =** Total Expenditure – Total Receipts (rarely used now).
- **Revenue Deficit =** Revenue Expenditure – Revenue Receipts (indicates non-asset spending).
- **Effective Revenue Deficit (ERD) =** Revenue Deficit – Grants for Capital Assets (introduced in 2012–13).
- **Fiscal Deficit** = Total Expenditure – (Revenue Receipts + Non-Debt Capital Receipts).
- **Primary Deficit** = Fiscal Deficit – Interest Payments.
- **Monetized Deficit =** Borrowings from RBI + Cash drawdown (leads to increase in money supply)

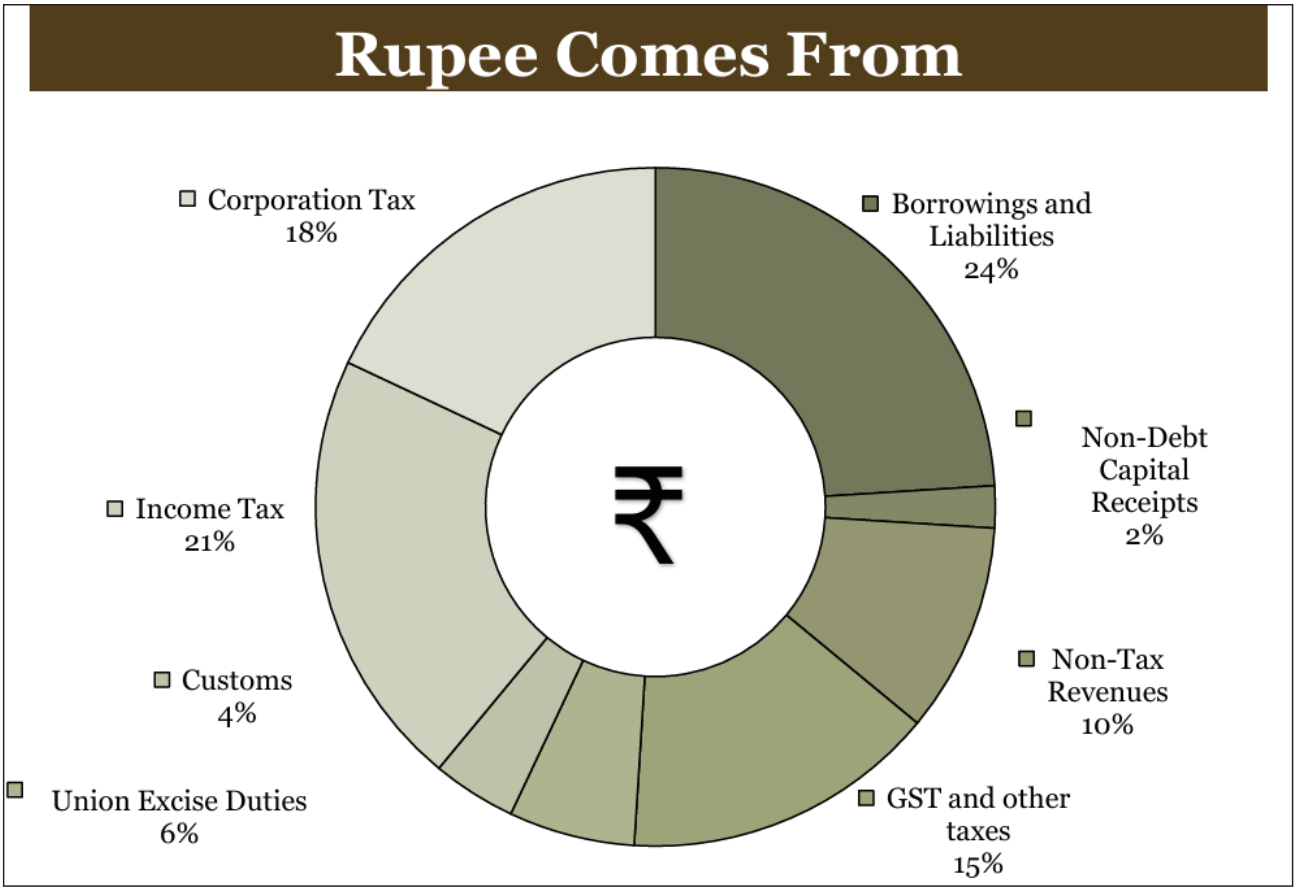
2. UNION BUDGET 2026-27- HIGHLIGHTS

Budget- Budget is essentially a statement of anticipated receipts and expenditure. The annual budget covers all aspects of government finances, including revenue generation, expenditure allocations, and policy announcements.

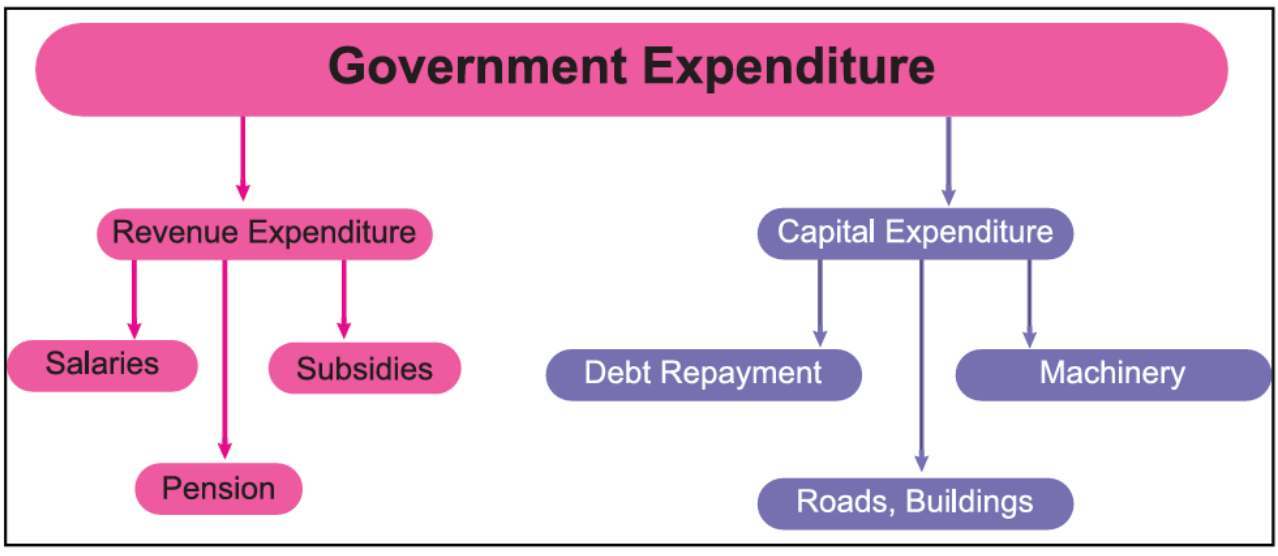
Government Receipts- The Government receipts are categorised into revenue and capital receipts.



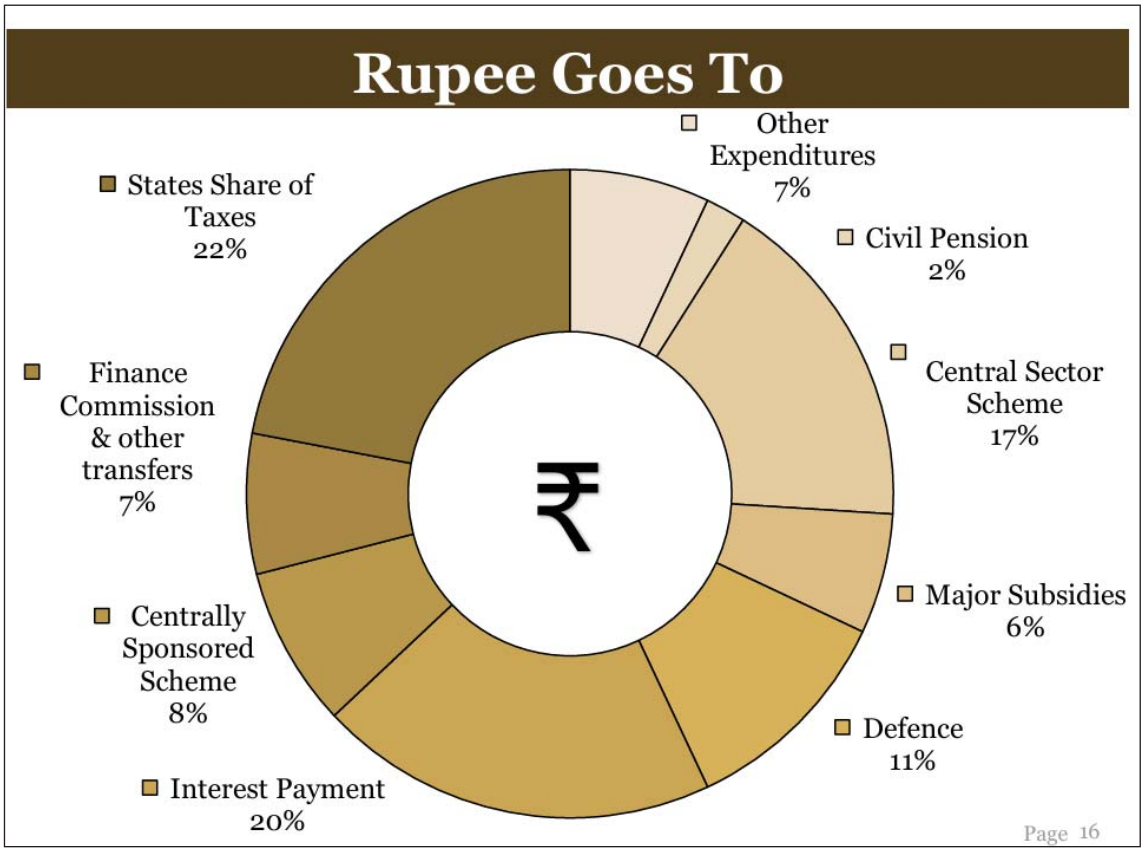
Government Receipts for 2026-27



Government Expenditure- The expenditure of the Government is also categorized into revenue and capital expenditure. The components of the expenditure are also shown below

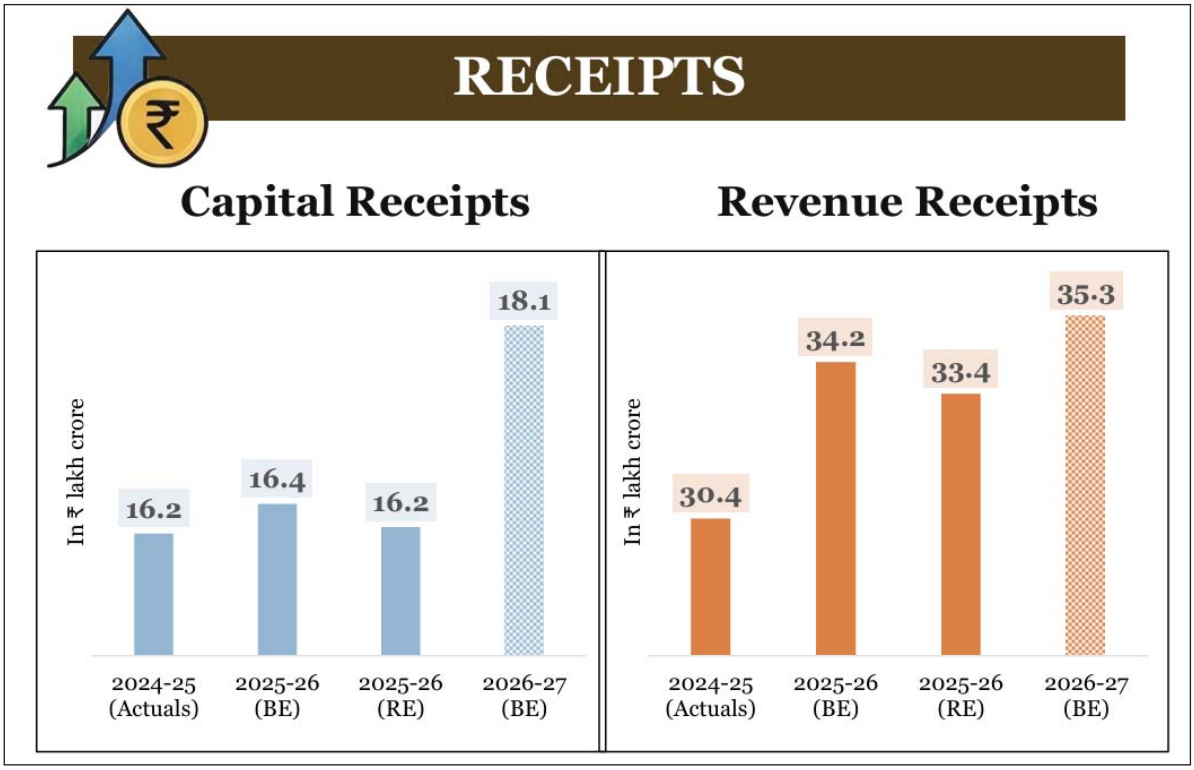


Government expenditure for 2026-27

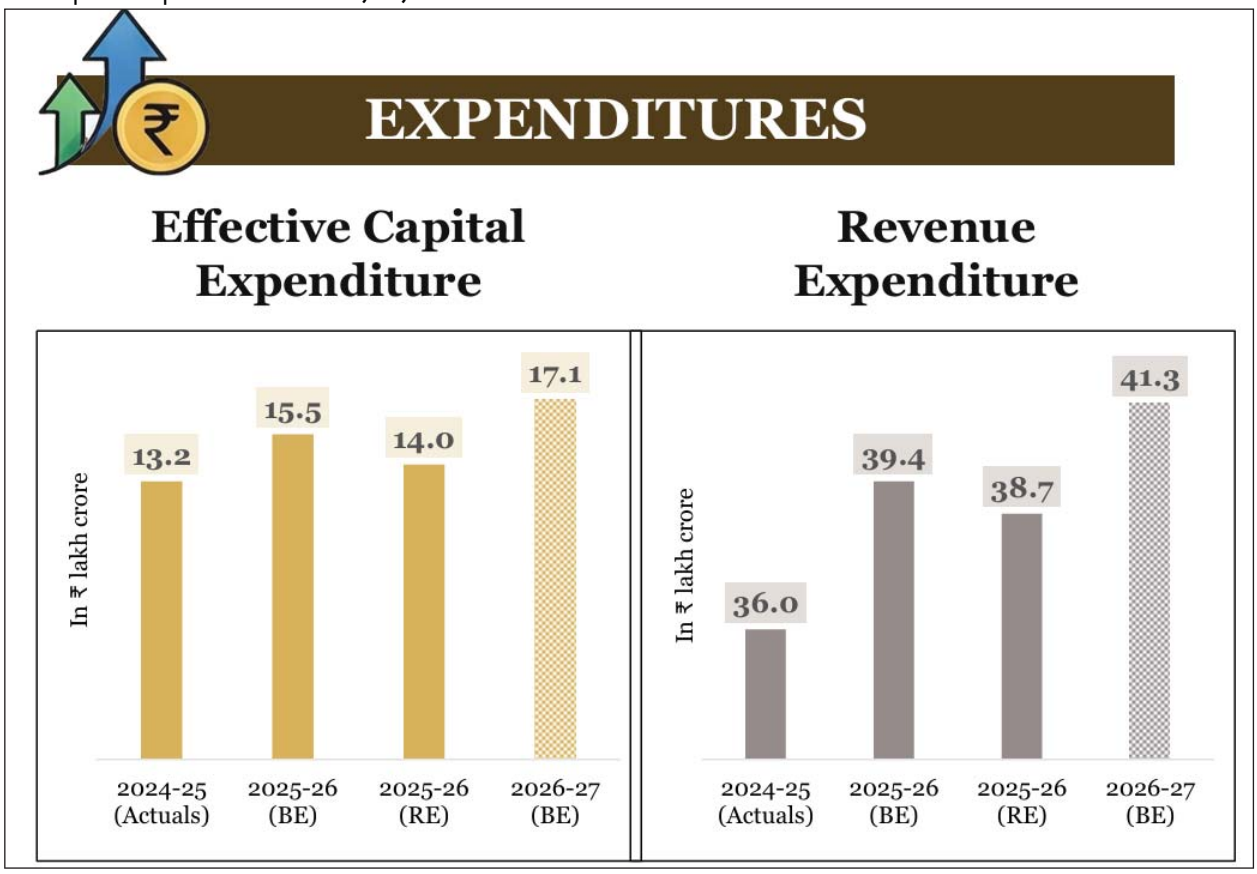


MACRO-ECONOMIC HIGHLIGHTS OF THE BUDGET 2026-27

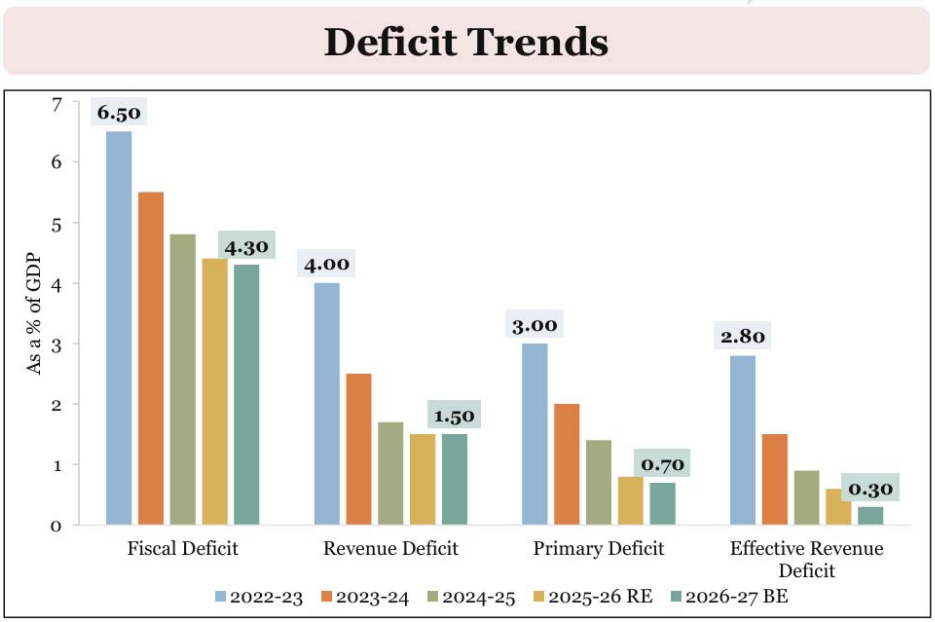
1. **Total Receipts for FY 2026-27-** The total receipts for FY 2026-27 is estimated at Rs.53.4 lakh crore.



2. **Total Expenditure for FY 2026-27-** The total expenditure in Budget Estimates (BE) 2026-27 is estimated at **₹53,47,315 crore** of which total capital expenditure is **₹12,21,821 crore** and effective capital expenditure is **₹17,14,523 crore**.



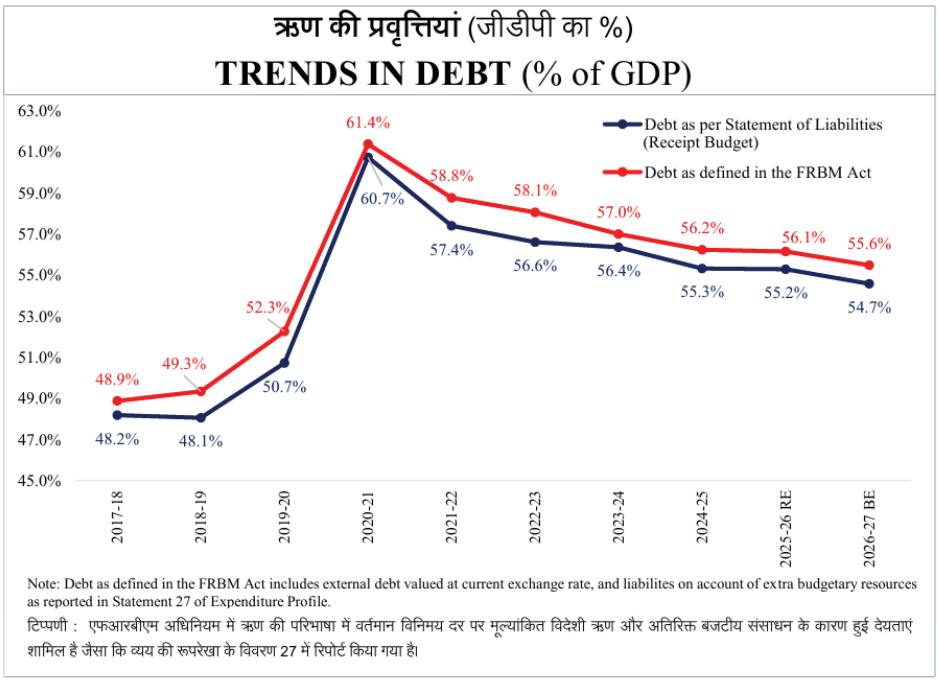
3. **Fiscal Deficit-** The fiscal deficit is estimated at 4.3 per cent of GDP.



4. The Government has accepted the recommendation of the 16th Finance Commission to retain the vertical share of devolution at **41%**. **₹1.4 lakh crore** has been provided to the States for the FY 2026-27 as Finance Commission Grants, which include Rural and Urban Local Body and

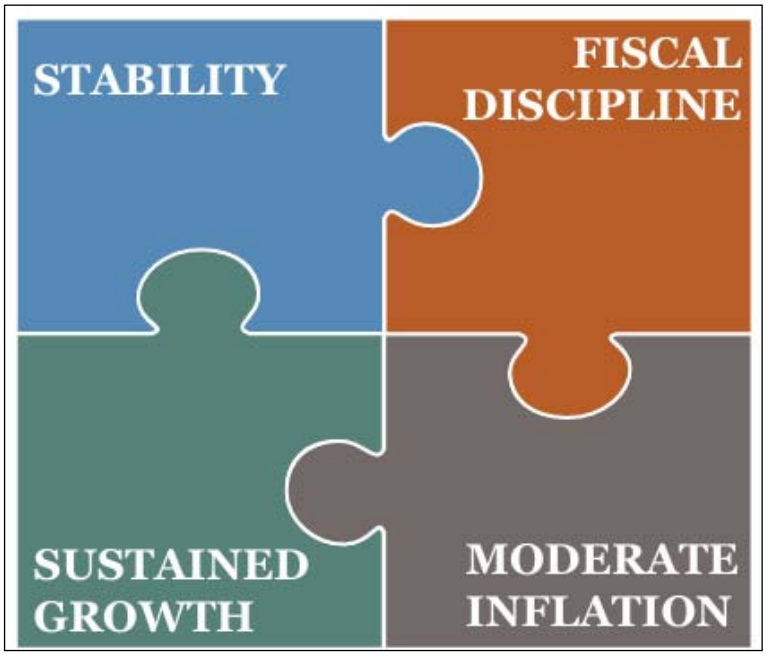
Disaster Management Grants.

5. The debt-to-GDP ratio is estimated to be **55.6 percent of GDP in BE 2026-27**, compared to 56.1 percent of GDP in RE 2025-26.



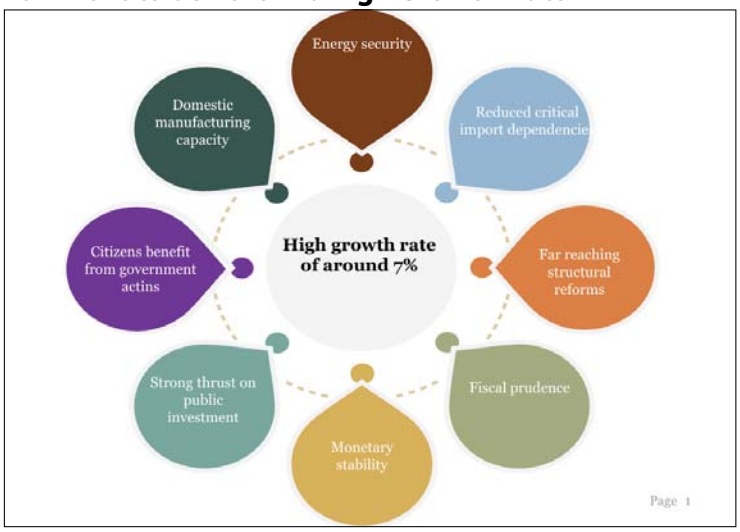
Budget Highlights 2026-27-Part A

India's economic trajectory has been marked by stability, fiscal discipline, sustained growth and moderate inflation.



India will continue to take confident steps towards Viksit Bharat, balancing ambition with inclusion. India will prefer action over ambivalence, reform over rhetoric and people over populism.

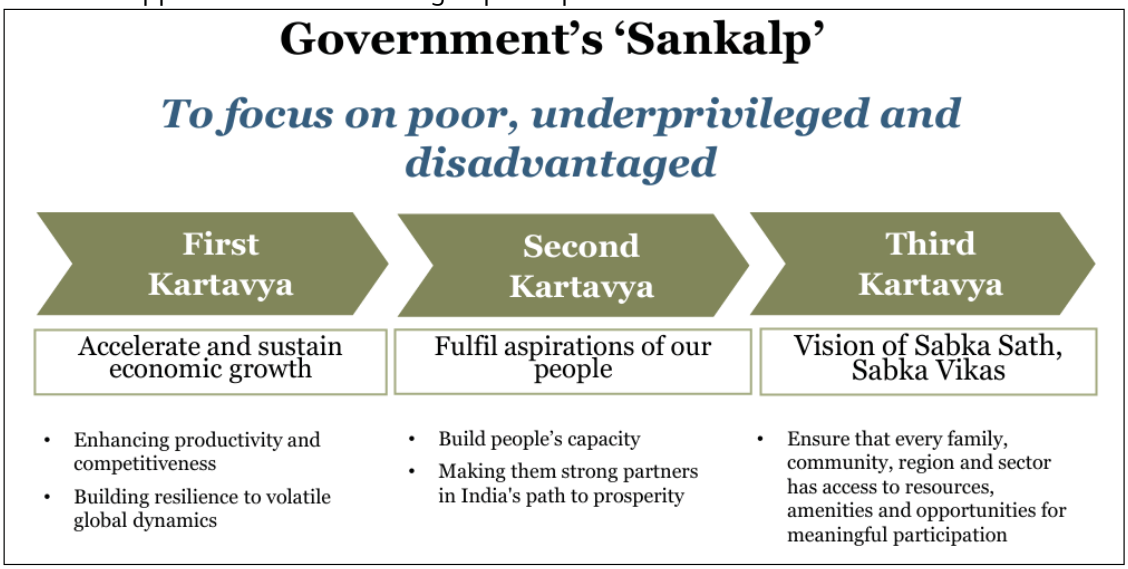
Steps taken by the Government to achieve 7% High Growth Rate



Government's Sankalp

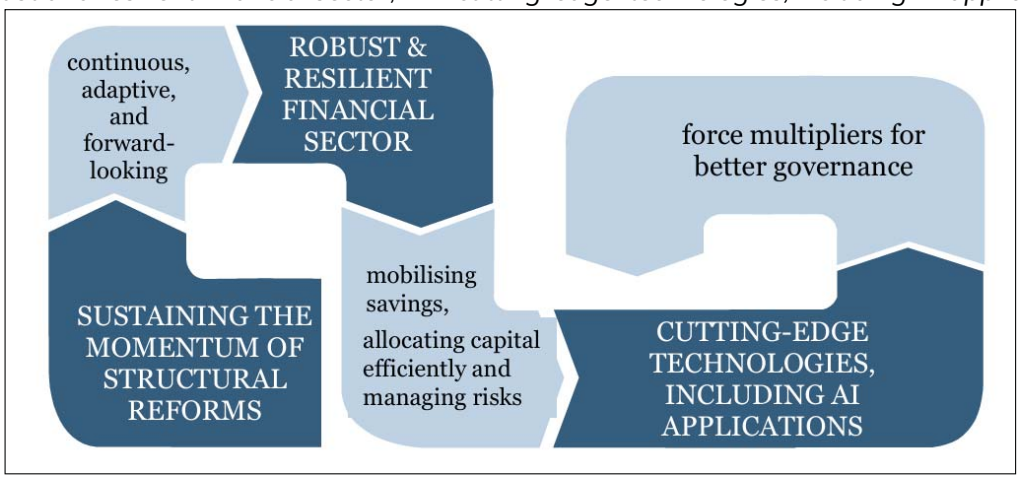
The budget reiterates Government's 'Sankalp' of its focus on poor, underprivileged and the disadvantaged. The Budget 2026-27 is inspired by ***3 kartavya***.

- ***First Kartavya-*** Accelerate and sustain economic growth, by enhancing productivity and competitiveness, and building resilience to volatile global dynamics.
- ***Second kartavya-*** To fulfil aspirations of our people and build their capacity, making them strong partners in India's path to prosperity.
- **Third kartavya-** To ensure that every family, community, region and sector has access to resources, amenities and opportunities for meaningful participation.



Ecosystem required for the achievement of the Sankalp

This threefold approach requires a **supportive ecosystem**- *Sustaining the momentum of structural reforms*, *robust and resilient financial sector*, and *cutting-edge technologies, including AI applications*.



India's Reforms Express

The Government has undertaken comprehensive economic reforms towards creating employment, boosting productivity and accelerating growth. Over 350 reforms have been rolled out. These include GST simplification, notification of Labour Codes, and rationalisation of mandatory Quality Control Orders.

First kartavya- Accelerate and sustain economic growth

Two achieve the *first kartavya, 7 key areas* have been identified:

Kartavya-1- Area-1 Scaling up manufacturing in 7 strategic and frontier sectors

1. **Biopharma SHAKTI** has been proposed with an outlay of ₹ 10,000 crores over the next 5 years, to develop India as a global Biopharma manufacturing hub. This will build the ecosystem for domestic production of biologics and biosimilars, for non-communicable diseases, like diabetes, cancer and autoimmune disorders.
   - Setting up **3 new National Institutes of Pharmaceutical Education and Research (NIPER)** and upgrading 7 existing ones.
   - Creation of a network of over **1000 accredited India Clinical Trials** sites.
2. **India Semiconductor Mission (ISM) 2.0** to be launched to produce equipment and materials, design fullstack Indian IP, and fortify supply chains, for expansion of India's semiconductor sector capabilities.
3. **The Electronics Components Manufacturing Scheme-** Outlay on the scheme to be increased to ₹40,000 crore.
4. **Establishment of dedicated Rare Earth Corridors** in the mineral-rich States of Odisha, Kerala, Andhra Pradesh and Tamil Nadu to promote mining, processing, research and Manufacturing.
5. **Scheme to support States in establishing 3 dedicated Chemical Parks** to enhance domestic chemical production and reduce import dependency.
6. **Building strong capital goods capability**:
   - **Hi-Tech Tool Rooms to be established by CPSEs at 2 locations** as digitally enabled automated service bureaus that locally design, test, and manufacture high-precision components at scale and at lower cost.
   - **Scheme for Enhancement of Construction and Infrastructure Equipment (CIE)** to be introduced to strengthen domestic manufacturing of high-value and technologically-advanced CIE.
   - **Scheme for Container Manufacturing** to create a globally competitive container manufacturing ecosystem, with a budgetary allocation of ₹10,000 crore over a 5 year period.

7. Labour- Intensive Textile Sector

- The **National Fibre Scheme** for self-reliance in natural fibres such as silk, wool and jute, man-made fibres, and new-age fibres.
- **Textile Expansion and Employment Scheme** to modernise traditional clusters .
- A **National Handloom and Handicraft programme** to integrate and strengthen existing schemes and ensure targeted support for weavers and artisans.
- **Tex-Eco Initiative** to promote globally competitive and sustainable textiles and apparels.
- **Samarth 2.0** to modernize and upgrade the textile skilling ecosystem.

Launch of **Mahatma Gandhi Gram Swaraj initiative** to strengthen khadi, handloom and handicrafts. This will help in global market linkage and branding.

Dedicated initiative for sports goods that will promote manufacturing, research and innovation in equipment design as well as material sciences.

Kartavya-1- Area-2 Rejuvenation of Legacy Industrial Clusters

Scheme to revive 200 legacy industrial clusters to improve their cost competitiveness and efficiency through infrastructure and technology upgradation.

Kartavya-1- Area-3 Creating "Champion SMEs" and supporting micro enterprises

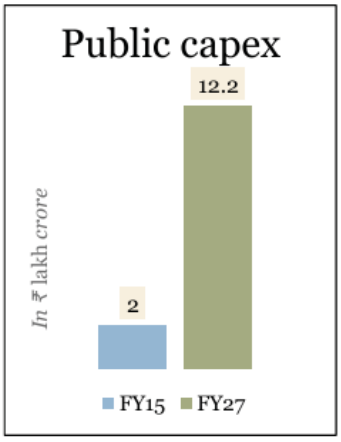
Three-pronged approach to help the MSME's grow as "champions".

Equity Support	Liquidity Support	Professional Support
Dedicated **₹10,000 crore SME Growth Fund**, to create future Champions. **Self-Reliant India Fund** to be topped up with **₹2,000 crore** to continue support to micro enterprises.	Mandate **TReDS as the transaction settlement platform** for all purchases from MSMEs by CPSEs. Introduce a **credit guarantee support mechanism** through CGTMSE. **Link GeM with TReDS** for sharing information with financiers. **Introduce TReDS receivables as asset-backed securities**, helping develop a secondary market.	Facilitate Professional Institutions such as ICAI, ICSI, ICMAI to design short-term, modular courses and practical tools to develop a cadre of '**Corporate Mitras**'. These accredited para-professionals will help MSMEs meet compliance requirements at affordable costs.

Kartavya-1- Area-4 Delivering a powerful push to Infrastructure

Continue to **focus on developing infrastructure in cities with over 5 lakh population** (Tier II and Tier III), which have expanded to become growth centres.

Public capex to be increased to ₹12.2 lakh crore, to continue the infrastructure momentum.



Setting up **Infrastructure Risk Guarantee Fund** to provide prudently calibrated partial credit guarantees to lenders.

Recycling of **real estate assets of CPSEs through the setting up of dedicated REITs**.

Promotion of environmentally sustainable movement of cargo:

- Establishment of **new Dedicated Freight Corridors** connecting Dankuni in the East, to Surat in the West.
- Operationalising **20 new National Waterways** connecting mineral rich areas, industrial centres and ports.
- Setting up a **ship repair ecosystem** catering to inland waterways at Varanasi and Patna.
- Launch a **Coastal Cargo Promotion Scheme** to increase the share of inland waterways and coastal shipping from 6% to 12% by 2047.

Launching a **Seaplane VGF Scheme** to indigenise manufacturing.

Kartavya-1- Area-5 Ensuring long-term energy security and stability

An outlay of **₹20,000 crore over the next 5 years**, is proposed for **CCUS technologies at scale** to achieve higher readiness levels in end-use applications across five industrial sectors, including, power, steel, cement, refineries and chemicals

Kartavya-1- Area-6 Developing City Economic Regions

Amplify the potential of cities to deliver the economic power of agglomerations by mapping **city economic regions (CER)**, based on their specific growth drivers. An allocation of **₹ 5000 crore per CER over 5 years** is proposed for implementing their plans.

Develop **seven High-Speed Rail corridors** between cities as 'growth connectors', namely i) Mumbai-Pune, ii) Pune-Hyderabad, iii) Hyderabad-Bengaluru, iv) Hyderabad-Chennai, v) Chennai-Bengaluru, vi) Delhi-Varanasi, vii) Varanasi-Siliguri.

Kartavya-1- Other areas:

Financial Sector	• "**High Level Committee on Banking for Viksit Bharat**", to comprehensively **review the banking sector** and align it with India's next phase of growth. • **Restructuring** the **Power Finance Corporation** and **Rural Electrification Corporation**. • Comprehensive review of the **Foreign Exchange Management (Non-debt Instruments) Rules** for user-friendly framework for foreign investments.
Corporate Bond Market	• **Market making framework** with **suitable access to funds** and derivatives on corporate bond indices. • Introduction of total return swaps on corporate bonds.
Municipal Bonds	• Incentive of **₹100 crore** for a single bond issuance of more than **₹1000 crore to encourage the issuance of municipal bonds** of higher value by large cities. • AMRUT which **incentivises issuances up to ₹200 crore**, to continue to support smaller and medium towns.
Ease of Doing Business	• **Individual Persons Resident Outside India (PROI)** to be permitted to **invest in equity instruments of listed Indian companies** through the Portfolio Investment Scheme. • **Increase in the investment limit for an individual PROI** under this scheme from **5% to 10%**, with an overall investment limit for all individual PROIs to 24%, from the current 10%.

Emerging technologies, including AI	• **Support to new technologies** through **AI Mission**, **National Quantum Mission**, **Anusandhan National Research Fund**, and **Research, Development and Innovation Fund**.

Second kartavya- To fulfil aspirations and build capacity

Second Kartavya- Area- Renewed emphasis on the Services Sector

To provide a pathway to fulfilling aspirations of a youthful India. The following measures are envisaged under the following pillars to achieve the second Kartavya.

Services Sector	**High-Powered 'Education to Employment and Enterprise'** **Standing Committee** to focus on the Services Sector as a core driver of Viksit Bharat. Committee to prioritise areas to optimise the potential for growth, employment and exports.
Health Sector	Upgradation of existing institutions for **Allied Health Professionals (AHPs)** and establishment of new AHP Institutions in private and Government sectors, to cover 10 selected disciplines.
Care Ecosystem	**Building a strong Care Ecosystem**, covering geriatric and allied care services. Training of **1.5 lakh caregivers**, covering core care and allied skills, such as, wellness, yoga and operation of medical and assistive devices.
Medical Value Tourism	Schemes to support States in establishing **Five Hubs for Medical Value Tourism** in partnership with the private sector.
AYUSH	**3 new All India Institutes of Ayurveda**, upgrading AYUSH pharmacies and Drug Testing Labs for higher standards of certification ecosystem, & upgrading the WHO Global Traditional Medicine Centre.
Animal Husbandry	Scale up availability of **veterinary professionals by more than 20,000**. **Loan-linked capital subsidy support** scheme for establishment of veterinary and paravet colleges, veterinary hospitals, diagnostic laboratories and breeding facilities in the private sector.
Orange Economy	Support to the **Indian Institute of Creative Technologies, Mumbai** in setting up **AVGC Content Creator Labs** in 15,000 secondary schools and 500 colleges.
Design	Setting up of a **new National Institute of Design** through the **challenge route** in the **eastern region of India**.
Education	• **5 University Townships** in the vicinity of major industrial and logistic corridors. • **A girls' hostel in Higher Education STEM institutions** in every district. • Setting up or upgrading of **four Telescope Infrastructure facilities**.
Tourism	• Setting up a **National Institute of Hospitality** as a bridge between academia, industry and the Government. • Pilot scheme for upskilling **10,000 guides in 20 iconic tourist sites**. • **National Destination Digital Knowledge Grid** to digitally document all places of significance.

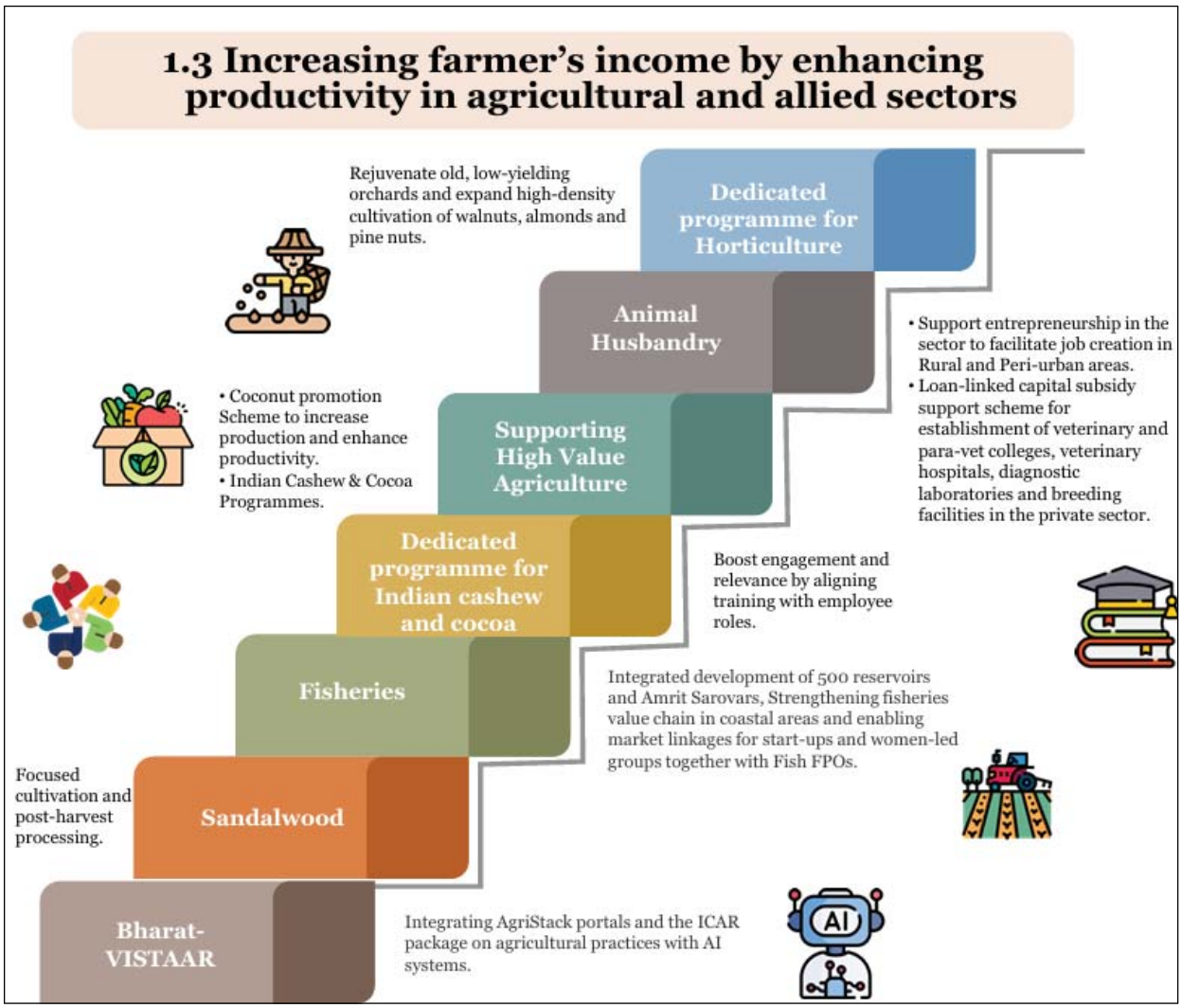
	• To develop ecologically sustainable Mountain trails, Turtle Trails and Bird watching trails- (a) **Mountain trails** in Himachal Pradesh, Uttarakhand and Jammu and Kashmir; Araku Valley in the Eastern Ghats and Podhigai Malai in the Western Ghats. (b) **Turtle Trails** along key nesting sites in the coastal areas of Odisha, Karnataka and Kerala. (c) **Bird watching trails** along the Pulikat lake in Andhra Pradesh and Tamil Nadu. • India to host the **first ever Global Big Cat Summit**. • Development of **15 archeological sites** (Lothal, Dholavira, Rakhigarhi, Adichanallur, Sarnath, Hastinapur, and Leh Palace) into vibrant, experiential cultural destinations. • Development of **Buddhist Circuits** in the North East Region.
Sports	Launch of **Khelo India Mission** for integrated talent development pathway, systematic coaching development, integration of science & technology and development of sports infrastructure.

Third kartavya- Sabka Sath, Sabka Vikas towards a Viksit Bharat

Third Kartavya- Area 1- Increasing farmer incomes

Fisheries	• Undertake initiatives for **integrated development of 500 reservoirs** and Amrit Sarovars. • Strengthen the **fisheries value chain in coastal areas** and enable market linkages.
Animal Husbandry	Entrepreneurship development through: • Credit-Linked Subsidy Programme • Scaling-up and modernisation of livestock enterprises • Enhanced creation of livestock, dairy and poultry-focused integrated-value chains • Creation of Livestock Farmer Producers Organisations.
High Value Agriculture	Support high value crops such as coconut, sandalwood, cocoa and cashew in our coastal areas. Support Agar trees in North East and nuts such as, almonds, walnuts and pine nuts in hilly regions. • **Coconut Promotion Scheme** to increase production and enhance productivity of coconuts. • **Enhance export competitiveness** and **transform Indian Cashew** and **Indian Cocoa** into premium global brands by 2030. • Partner with State Governments to promote focused cultivation and post-harvest processing to restore the glory of the **Indian Sandalwood ecosystem**. • Rejuvenate old, low-yielding orchards and expand high-density cultivation of walnuts, almonds and pine nuts, through a **dedicated programme to enhance farmer incomes**.

Bharat-VISTAAR	A multilingual AI tool that shall integrate the **AgriStack portals** and the **ICAR package** on agricultural practices with AI systems.
SHE-Marts for Rural Women-led Enterprises	**Community-owned retail outlets** within the cluster level federations through enhanced and innovative financing instruments.



Third Kartavya- Area 2-Empowering Divyangjan

Divyangjan Kaushal Yojana	Ensure dignified livelihood opportunities for divyangjans in IT, AVGC sectors, Hospitality and Food and Beverages sectors.
Divyang Sahara Yojana	• Support to **Artificial Limbs Manufacturing Corporation of India (ALIMCO)** to scale up production of assistive devices, invest in R&D and AI integration • Strengthening of **PM Divyasha Kendras** and support setting up of **Assistive Technology Marts** as modern retail-style centres where Divyangjans and senior citizens can see, try and purchase assistive products.

Third Kartavya- Area 3-Empowering the vulnerable to access mental health and trauma care

- Setting up of **NIMHANS-2 in North India** and **upgradation** of **National Mental Health Institutes** in **Ranchi** and **Tezpur** as Regional Apex Institutions.

- Strengthen and increase the **mental capacities by 50%** in **District Hospitals** by establishing Emergency and Trauma Care Centres.

Third Kartavya- Area 4- Focus on the Purvodaya States and the North-East Region

- Development of an integrated East Coast Industrial Corridor with a well-connected node at Durgapur, creation of 5 tourism destinations in the 5 Purvodaya States, and the provision of 4,000 e-buses.
- Scheme for Development of Buddhist Circuits in Arunachal Pradesh, Sikkim, Assam, Manipur, Mizoram and Tripura.

Budget Tax Proposals- Part B

Tax Reforms to Boost the Manufacturing Sector

- Exemption from income tax for five years to non residents providing capital goods, equipment or tooling, to any toll manufacturer in a bonded zone.
- Provision of safe harbour to non-residents for component warehousing in a bonded warehouse.
- Deferred duty payment window to trusted manufacturers.
- Increase the limit for duty-free imports of specified inputs used for processing seafood products for export, from the current 1 per cent to 3 per cent of the FOB value of the previous year's export turnover.
- Duty-free imports of specified inputs extended to export of shoe uppers in addition to leather or synthetic footwear.
- Extension of time for the export of final product from the existing 6 months to 1 year, for exporters of leather or textile garments, leather and synthetic footwear.
- Exemption from basic customs duty on specified parts used in the manufacture of microwave ovens.
- Exemption from basic customs duty on components and parts used in aircraft manufacturing.
- Exemption from basic customs duty on raw materials imported for manufacture of aircraft parts used in maintenance, repair, or overhaul requirements defence units.
- Regular importers with trusted longstanding supply chains to be recognized in the risk system.
- Export cargo using electronic sealing to be provided through clearance from the factory premises to the ship.
- A special one-time measure to facilitate sale in domestic tariff area at concessional rate of duty by eligible manufacturing units of SEZs.

Tax Proposal for MSMEs- Removal of the current value cap of ₹10 lakh per consignment on courier exports.

Tax Reforms to boost Services Sector

- Clubbing of services under a single category of information technology services with a common safe harbour margin of 15.5%.
- Safe harbour threshold for IT services increasing from ₹ 300 crore to ₹2,000 crore.
- Approval of safe harbour for IT services by an automated rule-driven process.
- Continuation of safe harbour for a period of five years at the company's choice.
- Fastracking unilateral APA process for IT services with an aim to conclude it within a period of two years. Can be extended by a further period of six months on the taxpayer's request.
- Extension of facility of modified returns for APA-availing entities to its associated entities.
- Provision of tax holidays until 2047 to foreign companies providing cloud services to global customers through India-based data centre services. Related Entities providing data center services from India to get a safe-harbour of 15% on cost.
- Exemption to global income of non-resident experts for a stay period of 5 years under notified schemes.

Tax Proposals for Financial Sector

- Raising the STT on Futures from 0.02% to 0.05%.
- STT on options premium and exercise of options to be raised to 0.15% from rate of 0.1% and 0.125%, respectively.

Tax Proposals for Agriculture Sector

- Fish catch by an Indian fishing vessel in Exclusive Economic Zone (EEZ) or on the High Seas to be made free of duty. Treating the landing of such fish on foreign port as export of goods.
- Deduction allowed to primary cooperative society engaged to include supply of cattle feed and cotton seed produced by members.
- Allowing inter-cooperative society dividend income as deduction under the new tax regime to the extent distributed to members.
- Exemption from tax dividend income received by a notified national co-operative federation from investments made in companies up to 31.1.2026 from tax for a period of three years. Exemption to be allowed only for dividends distributed to its member co-operatives.

Tax Proposals for Energy Sector

- Exemption of BCD on import of sodium antimonate for use in manufacture of solar glass.
- Exemption of BCD on import of capital goods required for the processing of critical minerals in India.
- Extending exemption of BCD to capital goods used for the manufacture of Lithium-Ion Cells for batteries to be used in battery energy storage systems.
- Extension of the existing basic customs duty exemption on imports of goods required for Nuclear Power Projects till the year 2035 and expand it for all nuclear plants irrespective of their capacity.
- Exclusion of entire value of biogas in Central Excise duty payable on biogas blended CNG.

Proposals for Trust Based Governance

- Enhancement of duty-deferral period for Tier 2 and Tier 3 Authorised Economic Operators (AEO) from 15 to 30 days. Eligible manufacturer importers to get the same duty deferral facility. Government agencies will be encouraged to leverage AEO accreditation.
- Extension of validity period of advance ruling, binding on Customs, from 3 years to 5 years.
- Transformation of Customs warehousing framework into a warehouse operator centric system with self-declarations etc.
- The filing of Bill of entry by a trusted importer, and arrival of goods to automatically notify Customs.
- Regular importers with trusted longstanding supply chains will be recognized in the risk system, so that the need for verification of their cargo every time can be minimized.

Proposals for Ease of Doing Business and Ease of Living

- Individual Persons Resident Outside India (PROIs) will be permitted to invest in equity instruments of listed Indian companies through the Portfolio Investment Scheme (PIS).
- Interest awarded by the motor accident claim tribunal to a natural person will be exempt from Income Tax, and any TDS on this account will be done away with.
- Reduce TCS rate on sale of overseas tour program package from 5% and 20% to 2% without any stipulation of amount.
- Reduce TCS for pursuing education and for medical purposes under the Liberalized Remittance Scheme (LRS) from 5% to 2%.
- TDS on Supply of manpower services to be at the rate of either 1% or 2%.
- Obtaining a lower or nil deduction certificate through a rule-based automated process for small taxpayers.
- Enable depositories to accept Form 15G or Form 15H from taxpayers holding securities in multiple companies.
- Time available for revising returns extended from 31st December to up to 31st March with the payment of a nominal fee.
- Individuals with ITR 1 and ITR 2 returns will continue to file till 31st July and non-audit business cases or trusts are proposed to be allowed time till 31st August.

- TDS on the sale of immovable property by a non-resident to be deducted and deposited through resident buyer's PAN instead of TAN.
- Introducing a one-time 6-month foreign asset disclosure scheme below a certain size for small taxpayers.
- Allow taxpayers to update their returns even after reassessment proceedings have been initiated at an additional 10 percent tax rate over and above the rate applicable for the relevant year.
- Framework for immunity from penalty and prosecution in the cases of underreporting extended to misreporting.
- Non-production of books of account and documents and requirement of TDS payment is decriminalised.
- Immunity from prosecution with retrospective effect from 1.10.2024 for non-disclosure of non immovable foreign assets with aggregate value less than ₹ 20 lakh.
- Exemption from Minimum Alternate Tax (MAT) to all non-residents who pay tax on presumptive basis.
- Constitute a Joint Committee of the Ministry of Corporate Affairs and Central Board of Direct Taxes for incorporating the requirements of Income Computation and Disclosure Standards (ICDS) in the Indian Accounting Standards (IndAS).
- Tax buyback for all types of shareholders as Capital Gains. However, promoters will pay an additional buyback tax.
- Set-off using available MAT credit to be allowed to an extent of 1/4th of the tax liability in the new regime.
- MAT is proposed to be made final tax.
- Exempt BCD on 17 drugs or medicines for cancer patients.
- Single and interconnected digital window for cargo clearance approvals.
- Customs Integrated System (CIS) to be rolled out in 2 years.
- Honest taxpayers willing to settle disputes will now be able close cases by paying an additional amount in lieu of penalty

3. FRBM ACT

- The **Fiscal Responsibility and Budget Management Act** (FRBM Act) was enacted in 2003. It aims to promote fiscal discipline, transparency, and accountability in the management of India's finances. The Fiscal Responsibility and Budget Management Act, 2003 is regulated by the Department of Economic Affairs, Ministry of Finance.
- Fiscal Responsibility and Budget Management Act, 2003 ensure inter-generational equity in fiscal management and long-term macro-economic stability by reducing fiscal deficit. It was enacted for effective conduct of monetary policy and prudential debt management consistent with fiscal sustainability.
- FRBM Act mandates the following statements to be laid before the Parliament along with the Budget:
  - Macro-economic Framework Statement.
  - Medium Term Fiscal Policy Statement.
  - Fiscal Policy Strategy Statement.
  - The Fiscal Responsibility and Budget Management Committee was constituted under the chairmanship of **N.K.Singh** to review the government's fiscal performance and make recommendations to FRBM Act.

FRBM Act: Provisions

The Central Government shall:

- Take appropriate measures to limit the fiscal deficit up to 3% of gross domestic product by the 31st March, 2021. However, following the pandemic-era disruption, the government transitioned to a revised target path of below 4.5% of GDP by FY 2025-26.
- **Endeavor** to ensure that-
  - The General Government (Centre + State) debt does not exceed 60%.
  - The Central Government debt does not exceed 40% of gross domestic product by the end of financial

year 2024-2025.

- **Annual Guarantees Ceiling-** The Central government cannot extend new loan guarantees backed by the Consolidated Fund of India exceeding 0.5% of GDP in any given fiscal year.
- **Deviation Allowance:** The government can breach the fiscal deficit target by a maximum of 0.5% of GDP in a single year.
- **Permitted Triggers:** This clause can only be invoked for specific emergency reasons, including **national security**, **acts of war**, **national calamities**, s**evere collapse of agriculture**, or **far-reaching structural economic reforms** with major fiscal costs.

Significance of FRBM Act

- **Fiscal discipline**: The FRBM Act promotes fiscal discipline by setting up the targets for reducing the fiscal deficit and by mandating the constitution of a Fiscal Responsibility and Budget Management Committee (FRBMC) to review the government's fiscal performance.
- **Fiscal transparency**: The FRBM Act promotes transparency and accountability of public finances through publication of various reports, such as the annual budget, the medium-term fiscal policy statement, and the Fiscal Policy Strategy Statement.
- **Macroeconomic stability**: The FRBM Act aims to maintain macroeconomic stability by ensuring that the government's fiscal policies are sustainable and do not lead to high levels of public debt.
- **Investor confidence**: The FRBM Act helps to boost investor confidence in the Indian economy by transparency and compliance and demonstrating the government's commitment to fiscal discipline and macroeconomic stability.
- **Long-term planning**: The FRBM Act requires the government to present a medium-term fiscal policy statement and an FPSS, which helps in long-term planning and ensures that the government's fiscal policies are aligned with its future economic goals.
- **Inter-Generational Equity:** FRBM Act emphasizes the responsibility of the government to maintain fiscal prudence so that future generations are not burdened by excessive debt.

FRBM Act: Amendments

Since the enactment of the FRBM act in 2003, it has been amended 4 times – 2004, 2012, 2015 & 2018. The FRBM Act and the Rules underwent more significant changes during the 4th Amendment, in 2018.

Table 1.1: Details of FRBM amendments made
(As percentage of GDP)

Fiscal Indicators		Target detail	Principal Act/ Rules	1st Amendment (in 2004)	2nd Amendment (in 2012)	3rd Amendment (in 2015)	4th Amendment (in 2018)
1.	Revenue Deficit	Target	Zero	Zero	2	2	Target for RD has been removed.
		Annual reduction	0.5	0.5	0.6	0.4	
		Beginning with FY	2004-05	2004-05	2013-14	2015-16	
		Sunset Target date	31.03.08	31.03.09	31.03.15	31.03.18	
2.	Fiscal Deficit	Target	3	3	3	3	**3**
		Annual reduction	0.3	0.3	0.5	0.4	**0.1**
		Beginning with FY	2004-05	2004-05	2013-14	2015-16	**2018-19**
		Sunset Target date	31.03.08	31.03.09	31.03.17	31.03.18	**31.03.21**
3.	Effective Revenue Deficit	Target	Introduced in 2012		Zero	Zero	Target for ERD has been removed.
		Annual reduction			0.8	0.5	
		Beginning with FY			2013-14	2015-16	
		Sunset Target date			31.03.15	31.03.18	
4.	Guarantee	Aggregate guarantees in any FY not to exceed 0.5 *per cent* of GDP, beginning with FY 2004-05					No additional guarantee for any loan on security of CFI, in excess of 0.5 *per cent* of GDP, in any financial year
5.	Liability/ Debt	Not to assume additional liabilities (including external debt at current exchange rate) in excess of 9 *per cent* of GDP for the financial year 2004-05 and progressively reduce the limit of 9 *per cent* of GDP by at least one *percentage* point of GDP in each subsequent financial year.					General Govt debt and Central Govt debt not to exceed 60 and 40 *per cent* of GDP respectively by the end of the FY 2024-25.

Source - Report of the CAG on Compliance of the FRBM Act, for year 2018-19

N.K.Singh Committee Recommendations

- **Debt to GDP ratio:** The Committee suggested using debt as the primary target for fiscal policy. A debt to GDP ratio of 60% should be targeted with a 40% limit for the centre and 20% limit for the states.
- To achieve the targeted debt to GDP ratio, it proposed yearly targets to progressively reduce the fiscal and revenue deficits till 2023.
- The Committee proposed to create an autonomous **Fiscal Council** with a Chairperson and two members appointed by the centre.
- The Committee noted that under the FRBM Act, the government can deviate from the targets in case of a national calamity, national security or other exceptional circumstances notified by it.
- **Escape Clause:** The government may be allowed to deviate from the specified targets upon the advice of the Fiscal Council in the following circumstances:
  - Considerations of national security, war, national calamities and collapse of agriculture affecting output and incomes.
  - Structural reforms in the economy resulting in fiscal implications.
  - Decline in real output growth of at least 3% below the average of the previous four quarters. These deviations cannot be more than 0.5% of GDP in a year.
- The Committee recommended that the 15th Finance Commission (which was also led by N.K.Singh) should be asked to recommend the debt trajectory for individual states.

Issues with FRBM Act

- **Frequent Deviations and Use of Escape Clause:**
  - The government often fails to meet deficit and debt targets, invoking broad escape clauses in times of economic stress, natural disasters, or revenue shortfalls, undermining the Act's credibility.
  - During major events (e.g., pandemic, GST introduction), targets have been postponed and reset several times.
- **Rigid Fiscal Targets v/s Growth Needs:** Fixed deficit targets (like 3% of GDP) can be too inflexible during downturns, limiting the government's ability to stimulate the economy with increased spending when needed.
- **Weak Enforcement & Accountability:**
  - No strong penalties or incentives for non-compliance; governments face limited consequences for missing targets.
  - Parliamentary and independent oversight are not always rigorous, allowing deviations to go unchecked.
- **Transparency & Classification Issues:**
  - Governments sometimes manipulate fiscal data by shifting expenditures off-budget, creative accounting, or reducing spending on critical social services to achieve targets.
  - Lack of clarity in what qualifies as "revenue expenditure" or "capital expenditure" can obscure true fiscal positions.
- **Impact on States and Fiscal Federalism:**
  - Centralized fiscal authority has led to tension with states, who argue that FRBM impinges on their fiscal autonomy and lacks sensitivity to their varying needs.
  - States are pressured to adhere to union-dictated deficit targets, even when their economic situations differ.

Way Forward

- **Introduce a Credible Debt Anchor:** Shift focus from rigid fiscal deficit targets to a debt-to-GDP ratio anchor (such as the 60% general government debt recommended by the N.K. Singh Committee), aligning with best international practices and ensuring inter-generational equity.
- **Strengthen Escape Clauses & Trigger Mechanisms:** Clearly define and limit the use of escape clauses only for major shocks (pandemics, disasters, financial crises), with transparent triggers and conditions to prevent frequent fiscal slippage or misuse.

- **Revise Fiscal Deficit Target Periodically:**
  - Make fiscal targets more flexible, revising them in response to macroeconomic changes, business cycles, and structural reforms while protecting spending on key priorities.
  - Allow phased fiscal consolidation to minimize adverse effects on the social sector and capital investment.
- **Improve Centre-State Coordination:** Evolve joint targets and monitoring frameworks for deficit and debt, tailored to state-specific needs, reducing centre-state tensions and fostering cooperative federalism.
- **Enhance Transparency & Accountability:** Ensure full disclosure of off-budget borrowings, guarantees, and contingent liabilities to avoid creative accounting and hidden fiscal risks.
- **Establish Fiscal Council:** Set up an independent Fiscal Council to assess compliance, review projections, and advise on fiscal policy, increasing credibility and transparency.
- **Prioritise the Quality of Expenditure:** The FRBM Act has traditionally focused on controlling the quantity of government spending, but the quality of that spending is just as important. The Act should distinguish between **revenue expenditure** (like salaries and subsidies) and **capital expenditure** (like building roads, ports, and power plants).

Conclusion

FRBM Act is central to India's fiscal discipline and macroeconomic stability. However, it should shift from a rigid rule-based system into a dynamic and credible tool that ensures both fiscal discipline and sustainable economic growth.

4. FISCAL CONSOLIDATION IN INDIA

Fiscal consolidation refers to the set of government policies aimed at reducing fiscal deficit and the accumulation of public debt while maintaining macroeconomic stability and creating space for growth. It involves a mix of revenue enhancement (tax reforms), expenditure rationalization (subsidy targeting), and structural reforms (disinvestment) to achieve intergenerational equity and macroeconomic stability.

Indicators: Key metrics include the **fiscal deficit-to-GDP ratio, revenue deficit, and debt-to-GDP ratio.**

Benefits of Fiscal Consolidation

- **Macro-Economic Stability:** Controls inflation, stabilizes currency, and supports sustainable long-term growth.
- **Reduced Debt Burden:** Contains unsustainable borrowing, lowering future interest obligations and freeing public resources for development.
- **Investor Confidence:** Sound fiscal management attracts foreign and domestic investment by signaling economic discipline.
- **Crowding-in Private Investment:** Reduced government borrowing creates space for private sector funding and entrepreneurship.
- **Resource Efficiency:** Ensures public resources are spent on growth-oriented and productive sectors, cutting wasteful expenditure.
- **Improved Credit Rating:** Lower deficits and debt ratios boost international credit rating, reducing borrowing costs.

Challenges in India's Fiscal Consolidation

- **Narrow Direct Tax Base:** Only about **7% to 9% of India's population files income tax returns**, forcing a heavy reliance on indirect taxes like GST.

- **Interest Burden:** Interest payments **consume roughly 40% of the Central Government's net revenue receipts**.
- **High subsidy burden-** The high subsidy burden of food, fertilizer, fuel increases the fiscal pressure. Further populist spending pressures such as loan waivers, social schemes complicate fiscal consolidation.
- **Low tax base**: India's combined tax-to-GDP ratio (center and states) is approximately 19.6%, whereas the average across OECD member countries reaches a record high of 34.1%.
- **Power Sector Bailouts:** State-owned electricity distribution companies (DISCOMs) are perennially loss-making due to power theft and politically motivated free electricity schemes.
- **Fiscal Stress-** State-level fiscal stress (**several states have fiscal deficit >3.5%**) and Interest payments burden is around ~40% of revenue receipts.

Key Measures undertaken for Fiscal Consolidation in India

Legislative/Institutional

- **FRBM Act (2003):** Sets targets for fiscal deficit (3% of GDP) and debt; revised by NK Singh Committee (2016) for flexibility and focus on debt.
- **NK Singh Committee (2017):** Recommended flexible glide paths post-pandemic and focus on debt sustainability.
- **Escape Clause:** Allows deviation from deficit targets for calamities, structural reforms, or security reasons.
- **Outcome-Based Budgeting:** Ensuring public money is tied to measurable results.

Revenue-Side Measures

- **Tax Reforms:** Implementation of GST (2017), corporate tax rationalization, digital tax administration, expanding direct tax base.
- **Disinvestment:** Monetizing public assets via sales, strategic partnerships, and listing of PSUs.
- **Digital Compliance:** E-filing, pre-filled forms, e-TDS, and use of Aadhaar for tax and subsidy disbursement.

Expenditure-Side Measures

- **Subsidy Rationalization:** Targeted subsidies using Direct Benefit Transfer (DBT) to cut leakage. For example- DBT for LPG, fertilizer, and scholarship payments**.**
- **Reduction in Revenue Expenditure:** Cutting non-productive spending and prioritizing essential services like health, education, infrastructure.
- **Prioritizing Capital Spending:** Shift from revenue to capital expenditure. The Union Budget 2026-27 raised public capital expenditure (capex) to a record ₹12.2 lakh crore (approximately ₹12.22 lakh crore).

Way Forward

Revenue Augmentation Strategies

- **Tax Digitization & Compliance-** Expanding formal tracking mechanisms to minimize tax evasion. For ex- India's implementation of automated e-way bills and the Goods and Services Tax (GST) network raised tax receipts from 10% of GDP in 2014-15 to 11.8% in recent fiscal cycles.
- **Asset Monetization & Divestment-** Selling non-core or loss-making public sector undertakings (PSUs) and monetizing brownfield infrastructure assets to generate non-debt capital receipts.

Expenditure Rationalization

- **Targeted Subsidies-** Replacing blanket price controls with direct cash transfers via biometric identification. **For Ex- Direct Benefit Transfer (DBT) implementation for cooking gas (LPG)** and **fertilizer subsidies** reduced leakages and ghost beneficiaries.
- **Protecting Capital Expenditure-** Prioritizing long-term infrastructure asset creation over recurring revenue expenses like **un-targeted consumption subsidies** or **unchecked administrative overheads**.

Institutional & Medium-Term Frameworks

- **Debt-to-GDP Anchors:** Moving away from rigid annual deficit numbers alone toward a medium-term debt glide path that offers economic stabilization flexibility. **For Ex-** The Union Budget projects a **reduction in the fiscal deficit down to 4.3% of GDP**, driving down the primary deficit while keeping a medium-term objective of lowering the central debt-to-GDP ratio toward 50%.

Conclusion

Fiscal consolidation in India is a multi-dimensional process, targeting reduced deficits, prudent public debt, and efficient resource use. Through legislative reforms, revenue mobilisation, subsidy rationalisation, and capex prioritisation, India seeks sustainable growth and investor confidence. Despite challenges from welfare imperatives and global uncertainties, India's gradual glide path and digital-oriented measures position it well for resilient fiscal management.

5. TAX TO GDP RATIO IN INDIA

Tax-to-GDP ratio- It is the ratio of tax revenue collected by all levels of government to the gross domestic product. It is a key indicator of how much of the economy is "captured" by the state via taxes.

Tax-to-GDP ratio = (Total annual tax revenue of a country/Gross Domestic Product) x 100

Importance of 'Tax-to-GDP Ratio'

- **Measurement and Indicator of Fiscal Health-** The tax to GDP ratio serves as a crucial indicator of a nation's fiscal health. A **higher Tax to GDP ratio** suggests a **more robust fiscal position**.
- **Higher Government Revenue Generation-** A higher Tax to GDP ratio underscores the government's proficiency in mobilizing revenue from various sources, including income tax, corporate tax, sales tax, and other levies.
- **Financial Stability Marker-** Tax to GDP Ratio offers valuable insights into the government's financial stability and its ability to meet expenditure obligations without resorting to excessive borrowing or deficit financing.
- **Economic Development Indicator-** The ratio reflects the level of economic development and income distribution within a country. Higher ratios are often associated with more developed economies with robust tax systems.
- **Public Service Provision-** A well-maintained Tax to GDP ratio enables governments to adequately fund essential public services, such as healthcare, education, infrastructure development, and social welfare programs.

Present Tax to GDP Ratio in India

The tax-to-GDP ratio in India was around **11.7%** in the fiscal year **2024-25**, marking a steady increase from **11.6%** in **2023-24** and **11.2%** in **2022-23**.

World Bank (WB) endorses Tax revenues exceeding 15% of GDP as pivotal for fostering economic growth and alleviating poverty. Some OECD countries like Sweden, Finland and Norway have a figure of over 40 percent in recent years.

Factors Driving Tax to GDP Growth In India

Dominance of Direct Taxes- The upsurge in the tax ratio is primarily attributed to the growth of direct taxes, renowned for their equitable distribution. The direct tax to GDP ratio for India stands at around **6.6%** (up from ~6.1% in 2022-23).

Reduction in Corporate and Personal Income Taxes- Recent reductions in corporate and personal income taxes have incentivized taxpayers to opt for the new tax regime, boosting compliance and widening the tax base.

Record GST Collection- The stabilisation and streamlining of the GST regime has resulted in record GST collections. GST collection reached a record high of **₹22.08 lakh crore** in the financial year 2024–25, **a 9.4% increase from the previous year**, with the average monthly collection standing at **₹1.84 lakh crore**.

Improved Tax Compliance- The **Wider use of technology** (**e-filing**, **pre-filled returns**, **faceless assessment**), **PAN–Aadhaar linking to** reduced duplicate/fake taxpayers, **tighter reporting norms for TDS/TCS** (tax deducted/collected at source), **amnesty schemes** like **Vivad se Vishwas Scheme** have improved the tax compliance.

Reasons for Low Tax to GDP Ratio in India

- **Presence of Informal Sector-** The substantial size of India's informal/unorganized sector poses challenges for tax collection. This leads to **greater tax evasion** and **reduced revenue inflows**.
- **Dominance of Agriculture Sector-** A significant portion of the population **engaged in agriculture**, which is often exempt from taxation. Hence a large segment of the economy remains outside the tax net.
- **Tax Disputes and Low Rate of Recovery-** High numbers of tax disputes coupled with low rates of tax arrears recovery have undermined the effectiveness of tax collection efforts. This has contributed to a lower tax to GDP ratio.
- **Generous Government Tax exemption Policies-** The presence of numerous tax exemptions and incentives, particularly favoring the richer private sector, diminishes tax revenues and erodes the tax base.
- **Low Per Capita Income and High Poverty-** India's low per capita income and widespread poverty limit the tax base, as a significant portion of the population falls below taxable income thresholds.

Implications of Low Tax to GDP Ratio

- **Limited Government Spending:** Decreased tax revenues **constrain** the **government's capacity to invest in crucial areas** such as **national security**, **welfare programs**, and **infrastructure development**.
- **Persistent Fiscal Deficit:** The reliance on borrowing to compensate for low tax revenues results in persistent fiscal deficits. This perpetuates a **deficit bias** in fiscal policy.
- **Political Incentives for Borrowing:** Political pressures to appease voters through spending rather than implementing effective tax reforms, exacerbate fiscal imbalances and hinder economic growth.
- **Unchecked Tax Evasion:** Widespread tax evasion undermines economic growth and places a disproportionate burden on high-productivity sectors, stifling overall development.
- **Impact on Welfare Schemes:** Reduced tax collections limit the government's ability to fund welfare schemes and social programs, exacerbating socio-economic disparities.
- **Dependence on Regressive Taxes:** Increased reliance on indirect taxes, which tend to be regressive, exacerbates income inequality and places a heavier burden on lower-income segments of society.
- **Social Inequality and Economic Disparities:** The asymmetric distribution of economic resources perpetuates social inequality, contributing to disparities in access to opportunities and services.

Various Measures to Increase the Tax to GDP Ratio

Widening Individual Taxpayer Base- Implement policies to expand the individual taxpayer base by targeting tax evasion.

- Bring more taxpayers into the formal tax net through improved compliance measures and outreach programs.

Organizational Restructuring- Consider the **merger** of the **Central Board of Direct Taxes (CBDT)** and the **Central Board of Indirect Taxes and Customs (CBIC)** as recommended by the **Tax Administration Reform Commission (TARC)** to streamline tax administration and enhance efficiency.

Reassessment of Exemptions- Reassess and rationalize tax exemptions provided under various provisions, such as **transfer pricing** and **base erosion and profit shifting (BEPS)**, to prevent misuse and ensure fair taxation while minimizing revenue leakages.

Effective Dispute Settlement Mechanisms- Strengthen dispute resolution mechanisms to expedite the resolution of tax disputes. This would **reduce tax litigation** and **enhance taxpayer confidence** in the tax system.

Cultural Change and Awareness- Foster a culture of tax compliance and civic responsibility by educating citizens about the importance of paying taxes for nation-building and development.

- Instill a sense of national pride and duty towards contributing to the country's progress through tax compliance.

Tax Buoyancy

Tax buoyancy stands as a pivotal measure, reflecting the responsiveness of tax revenue to fluctuations in the Gross Domestic Product (GDP) or national income.

It unveils how tax revenue adjusts to changes in the overall economic activity.

Indicator of Economic Health- Tax buoyancy not only reflects the health of the economy but also the efficacy of tax policies in generating revenue.

It acts as a vital tool for policymakers to evaluate the impact of economic measures on revenue generation.

Challenges Impacting Tax Buoyancy

Low Base Effect: The dip in tax buoyancy in 2022-23 can be attributed to a diminishing low base effect, which led to a slower growth rate of taxes relative to the national income.

Decline in Tax Collections Growth Rate: Despite a 17.79% growth rate in tax collections, it outpaced the nominal GDP growth rate of 15.11%, contributing to decreased tax buoyancy.

Increase in Tax Base: While the number of income tax return filers increased, a significant portion of taxpayers remains unaccounted for, possibly affecting tax buoyancy despite a broader tax base.

Conclusion

India's tax-to-GDP ratio is poised to reach a historic high, reflecting the government's efforts to bolster revenue generation through direct taxes and streamline tax policies. This surge underscores the nation's improving fiscal health and resilience amidst economic fluctuations. However, challenges such as the dominance of the informal sector, low tax base, and disparities in tax collections growth rate persist, impacting tax buoyancy.

6. DIRECT TAX IN INDIA

Direct Taxes- Direct tax is a tax which an individual or organisation pays direct taxes directly to the imposing body.

- The **incidence and impact of the tax** falls on the same entity. Its burden cannot be shifted to a different individual or entity.
- It is **progressive in nature** which means the tax obligation increases in proportion to an individual's or entity's income.

Historical background of Direct tax

- **Ancient & Medieval Foundations**

- **Kautilya's Arthashastra (3rd Century BCE):** Established early principles of taxation based on paying capacity, levying direct taxes on agricultural yield, land, and merchant profits.
- **Mughal Era System:** Standardized direct land revenue models, notably Raja Todar Mal's *Zabt* system under Akbar, which directly assessed crop yields and land quality.

- **Colonial Era & Introduction of Modern Income Tax**
  - **Income Tax Act of 1860:** Introduced modern income tax in India by Sir James Wilson (India's first Finance Member) to recover financial losses from the Revolt of 1857.
  - **Income Tax Act of 1886:** Replaced initial temporary measures with a permanent, structured direct tax framework across British India.
  - **Income Tax Act of 1922:** Created a dedicated tax administration framework, separating legislative tax policy from administrative tax assessment.
- **Post-Independence Evolution**
  - **Income Tax Act of 1961:** Replaced colonial-era laws with a comprehensive, modern direct tax statute that remains the primary governing legislation today.
  - **Direct Taxes Enquiry Committee / Wanchoo Committee (1970):** Recommended rationalizing high marginal tax rates, plugging tax evasion loopholes, and demonetization to curb black money.
  - **Chelliah Committee Recommendations (1991):** Initiated post-economic reform tax restructuring by lowering peak tax rates, simplifying rules, and broadening the tax base.
- **Modern Reforms & Digital Era**
  - **Central Board of Direct Taxes (CBDT) Restructuring:** Strengthened administrative mechanisms under the Revenue Department to enforce direct tax laws efficiently.
  - **Direct Taxes Code (DTC) Initiatives (2009–2019):** Multiple expert panels proposed replacing the 1961 Act with a simplified, modern Direct Taxes Code to reduce litigation.
  - **Faceless Assessment & Digital Integration (2020s):** Introduced faceless assessments, pre-filled returns, and e-filing portals to eliminate human bias and boost tax compliance.

Types of direct tax

Income Tax	It is imposed on income that is progressive in nature, with different slabs for various taxpayer categories.
Personal Income Tax	Personal Income Tax is the income tax levied on the incomes of Individuals, Households, Hindu Undivided Families (HUF), Partnership Firms, Sole Proprietorships. It is governed by the Income Tax Act, 1961 (IT Act, 1961).
Corporate Tax	Corporate Income Tax is the income tax levied on profits earned by the incorporated firms or companies. Corporate Tax is also a type of Income Tax. Since the contribution of Corporate Tax to the overall tax revenue is significantly high, it has been made a separate category of income tax. It is also governed by the Income Tax Act, 1961 (IT Act, 1961).
Capital Gains Tax	Capital Gains Tax (CGT) is a tax levied on the profit one makes from selling a **Capital Asset**. (Examples of Capital Assets include Land, Buildings, House Property, Vehicles, Patents, Trademarks, Leasehold Rights, Machinery, and Jewellery.)
Securities Transaction Tax	Securities Transaction Tax (STT) is a tax levied on every purchase or sale of securities listed on recognized stock exchanges such as **shares**, **bonds**, **mutual funds**.

Minimum Alternative Tax (MAT)	Minimum Alternate Tax (MAT) refers to the tax imposed on "**Zero Tax Companies**", which escape the Corporate Tax net or pay very low taxes by using the provisions of exemptions, deductions, incentives.
Dividend Distribution Tax (DDT)	Dividend Distribution Tax (DDT) is a tax imposed on dividends paid by a corporation to its shareholders. It is a **withholding tax**, meaning the company deducted the tax before paying out the dividends to shareholders. • In India, only domestic companies as defined under the **IT Act** had to pay **Dividend Distribution Tax (DDT)**. However, **DDT has been abolished** in India w.e.f. **1st April, 2020**.

Merits and Demerits of Direct tax in India

Merits and Demerits of Direct Tax

Merits of Direct Tax	Demerits of Direct Tax
Social and Economic Equity- Direct Tax are **progressive in nature**, as they are levied on the basis of ability to pay. Thus, they help in containing income inequalities and ensure social justice.	**Tax evasion-** Direct tax faces the issue of tax evasion as Individuals and companies often **conceal income, under-report profits**, or **exploit loopholes**.
Relatively Elastic- They are relatively elastic as an **increase in the income of individuals and companies** leads to an **increase in the revenue** from these taxes.	**Impacts Capital Formation-** Direct taxes **directly affect savings and investments** in the economy, thereby affecting capital formation.
Revenue Certainty- The Direct tax creates certainty for the taxpayers (how much tax to be paid) and the government (revenue from the taxes).	**Administrative complexity-** It requires extensive documentation, audits, and assessments. **Compliance costs** (time + money) are often high for both taxpayers and tax authorities.
Low Cost of Collection- Collection of these taxes is generally economical as compared to other forms of taxes. **For ex- Personal Income Tax** can be **deducted at source (TDS)** from the income or salaries of the individuals.	**Imbalance in Sectoral Taxation-** Certain sectors like the corporate sector are heavily taxed, whereas the agriculture sector is 100% tax-free.
Checks Inflation- They can help control inflation. When inflation is on the rise, the government may increase the direct tax rate. This reduces the money supply, and hence demand and inflation.	**Arbitrariness in Rates and Exemptions-** There are no objective criteria for determining the tax rates of direct tax, with the exemptions often being determined in arbitrary manner.

Major highlights of Direct Taxation in India

- **Direct Tax Growth:** Net direct tax collections stood at **₹23.40 lakh crore in FY 2025–26** (growing 5.12% YoY), up from ₹19.58 lakh crore in FY 2023–24.
- **Shift in Revenue Composition:** Personal Income Tax (PIT) at **₹10.44 lakh crore** surpassed Corporate Tax at **₹9.11 lakh crore** in FY 2023–24, shifting the bulk direct-tax contribution to individuals.
- **Direct-to-Indirect Tax Ratio:** Direct taxes contribute **over 54% of total central tax revenue**, reversing historical reliance on indirect taxes to build a progressive revenue profile.

- **Corporate Tax Rate Cut (2019):** Corporate tax rates for existing domestic companies were reduced from 30% to **22% (effective ~25.17% with surcharge/cess)** under Section 115BAA.
- **New Manufacturing Incentives:** Concessional corporate tax rate of **15%** (Section 115BAB) established for new domestic manufacturing setups.
- **New Tax Regime (Section 115BAC):** Simplified individual slab structure offering lower tax rates without 80C/80D deductions, making income effectively tax-free up to **₹12 lakh** via Section 87A rebate.
- **Abolition of Distortive Direct Taxes:** Wealth Tax abolished in 2015–16 and Dividend Distribution Tax (DDT) abolished in 2020 (shifting taxation to shareholders at slab rates).

Issues with Direct tax

- **Narrow Tax Base & Low Compliance**
  - **Extremely Small Taxpayer Base:** Less than 5% of India's total population actually pays income tax, placing a heavy tax burden on a small middle class.

    As per Ministry of finance, Only **6.68% of India's population** (8.09 crore taxpayers) filed Income Tax Returns (ITRs) in FY 2023–24. Out of these, **4.90 crore individuals reported zero tax liability.**
  - **Exemption of Agricultural Income:** High-income agricultural earners remain entirely tax-exempt under Section 10(1), leading to widespread misuse for tax evasion.
  - **Large Shadow Economy:** A significant portion of the cash-driven informal sector stays outside the formal tax net.

    India's informal/shadow economy is estimated at **15% to 20% of GDP**, leaving millions of cash-based commercial transactions outside the direct tax framework
- **High Tax Evasion & Black Money Generation**
  - **Under-Reporting of Personal Income:** Widespread under-reporting of income among self-employed professionals, traders, and businesses.
  - **Offshore Evasion & Tax Havens:** Complex corporate structures, transfer pricing manipulation, and offshore accounts are used to siphon money out of the country.

    As per Ministry of finance, Undisclosed foreign assets and income exceeding **₹20,000+ crore** were detected and taxed under the Black Money (Undisclosed Foreign Income and Assets) Act.
- **Structural & Complexity Bottlenecks**
  - **Outdated Legislative Framework:** The Income Tax Act of 1961 is cluttered with over 100 exceptions and deductions, creating loopholes and fueling litigation.
  - **Excessive Reliance on Corporate Tax:** Corporate tax and personal income tax contribute disproportionately, leaving non-corporate business entities under-taxed.

    Personal Income Tax (PIT) collections (**₹10.44 lakh crore**) surpassed Corporate Tax collections (**₹9.11 lakh crore**) in FY 2023–24
  - **Frequent Policy Volatility:** Annual budget amendments create regulatory uncertainty, dampening long-term business confidence.
- **Administrative & Dispute Resolution Challenges**
  - **Massive Tax Litigation Backlog:** Billions of rupees remain trapped in prolonged legal disputes across Income Tax Appellate Tribunals, High Courts, and the Supreme Court.

    As per Ministry of finance, Over **₹14 lakh crore** remains locked in direct tax litigation across CIT(A), ITATs, High Courts, and the Supreme Court.
  - **Harassment & Tax Terrorism:** Arbitrary tax demands and aggressive compliance checks by revenue authorities historically created fear among taxpayers.
  - **High Cost of Collection:** Administrative costs for assessing and collecting taxes from small taxpayers remain disproportionately high.

Central Board of Direct Taxes (CBDT)

Central Board of Direct Taxes (CBDT) is a statutory authority functioning under the Central Board of Revenue Act, 1963, and operates as an apex body under the Department of Revenue within the Ministry of Finance. It serves as the primary administrative authority responsible for formulating direct tax policies, overseeing the enforcement of direct tax laws—most notably the Income-tax Act, 1961—and supervising the Income Tax Department across India.

Reforms taken for Direct taxes in India

Recent Major Direct tax reforms Undertaken in India

Reduction in tax rates and Simplification of Direct Tax Laws	**Corporate Tax-** Through the Taxation Laws (Amendment) Act, 2019, an option has been provided to the corporates to pay tax at concessional rate of 22% (plus applicable surcharge and cess) if they do not avail certain exemptions or incentives. • This is in line with the stated policy of the government by removing exemptions and incentives while reducing the rates of taxes. **Personal Income Tax relief-** Those with total income up to **₹ 12,00,000 (plus standard deduction of Rs. 75,000)**, are exempted from paying Income tax. **Abolition of Dividend Distribution Tax(DDT)-** The Finance Act, 2020, removed the Dividend Distribution Tax.
Ease of Compliance for Taxpayers	**Faceless Assessment Scheme (2019)-** It provides for a new mechanism for making assessment by eliminating the interface between the assessing officer and assessee. **Faceless Appeals (2020)-** It eliminates the human interface in the appellate function of the department between the appellant and the Commissioner of Income-tax (appeals). **Pre-filling of Income Tax-** In order to make tax compliance more convenient, pre-filled Income Tax Returns (ITR) have been provided to the individual taxpayers.
Reduction in Litigation	**Vivad se Vishwas-** Direct Tax Vivad se Vishwas Act, 2020 was passed to provide for resolution of pending tax disputes. **Raising of Monetary Limit for filing of appeal-** The monetary threshold for filing of departmental appeals have been raised from **Rs. 20 lakh to Rs. 50 lakh** for appeal before **ITAT**, from **Rs. 50 lakh to Rs. 1 crore** for appeal before the **High Court** and from **Rs. 1 crore to Rs. 2 crore** for appeal before the **Supreme Court**.
Promotion of Digital Transaction	The existing rate of deemed profit under presumptive schemes for small businesses has been reduced from 8% to 6% in respect of digital turnover.
Curbing of Tax Evasion and widening of Tax Base	**Black Money Act-** In order to curb the flow of Black Money stashed abroad, the Black Money (Undisclosed Foreign Income and Assets) and Imposition of Tax Act, 2015 (the Black Money Act), has been enacted. **Benami Law-** The Benami Transactions (Prohibition) Act,1988, was comprehensively amended in 2016, to enable confiscation of Benami Property and prosecution of benamidar and the beneficial power. **Abolition of Equalisation Levy-** The Finance Act, 2025 has abolished the equalization levy introduced in 2016. It was a direct tax imposed on foreign digital service providers to tax income generated from digital transactions in India.

Way forward for Direct Tax Reforms in India

- **Broadening the Tax Base & Improving Equity**
  - **Taxation of Large Agricultural Incomes:** Tax high-income corporate farmers while protecting small and marginal farmers to plug evasion loopholes.

    USA & France tax commercial agribusinesses under standard corporate/business income tax structures while providing targeted exemptions only for small family farms.
  - **Rationalization of Capital Gains Framework:** Simplify holding periods and tax rates across equity, real estate, and debt to ensure equal tax treatment across assets.

    United Kingdom maintains a unified Capital Gains Tax (CGT) framework with standardized holding periods and tax rates across equities, real estate, and fixed-income assets.
  - **Effective Enforcement on High-Net-Worth Individuals (HNWIs):** Utilize global financial data sharing to curb offshore tax evasion and aggressive tax planning.

    USA (FATCA) and **OECD Common Reporting Standard (CRS)** enable automatic tax information exchange across 120+ jurisdictions to track offshore HNWI assets.
- **Administrative & Technological Modernization**
  - **Deeper AI & Data Analytics Integration:** Leverage big data to cross-check GST returns, property registrations, and bank transactions for automated discrepancy detection.
  - **Enhancing the Faceless Assessment Ecosystem:** Refine faceless grievance redressal mechanisms to solve technical glitches and ensure fair hearings.
  - **Continuous Simplification of Forms:** Streamline e-filing systems and pre-filled returns to make tax compliance effortless for first-time filers.
- **Policy & Legislative Stability**
  - **Smooth Transition to the New Direct Tax Code:** Ensure clear, stable rules under the updated tax framework to minimize litigation and build business trust.

    New Zealand operates a streamlined direct tax framework based on broad tax bases, minimal rate brackets, and zero complex exemptions to minimize litigation.
  - **Phasing Out Distortive Deductions:** Transition completely toward simple, low-rate tax structures that eliminate complex exceptions.
  - **Predictable International Tax Rules:** Align domestic digital tax rules with global OECD Inclusive Framework standards (Pillar 1 and Pillar 2).

    European Union & Japan have legislated the **OECD Pillar 2** rules, establishing a uniform 15% global minimum corporate tax rate across jurisdictions.
- **Dispute Resolution & Taxpayer Trust**
  - **Strengthening Alternative Dispute Mechanisms:** Fast-track administrative settlements through expanded Dispute Resolution Committees (DRCs) to reduce court backlogs.
  - **Promoting Voluntary Compliance:** Foster a culture of trust by honoring honest taxpayers and upholding the statutory promises of the Taxpayers' Charter.

    Australia (ATO Taxpayers' Charter) and **USA (Taxpayer Bill of Rights)** grant legally binding rights to taxpayers and offer expedited services to compliant filers.

Conclusion

A robust direct tax framework is the backbone of India's fiscal autonomy, enabling equitable wealth redistribution and sustainable funding for national development. By accelerating digital transformation, broadening the taxpayer base, and simplifying compliance, India can build a modern, transparent tax ecosystem.

7. DIRECT TAX CODE

Various committees have been constituted by the GOI, to consolidate, rationalise and simplify the direct tax regime, like **Raja Chelliah Committee** (early 1990s), **Vijay Kelkar Committee** (2002), **Arbind Modi Committee** (2014), and the **Easwar Panel** (2016).

Recently, with the constitution of the **Akhilesh Ranjan Panel on formulating a new Direct Tax Code (DTC) in 2017**, the Government seems to be moving firmly in the direction of Direct Tax reform.

Direct Taxes Code (DTC) aims to **revise**, **consolidate** and **simplify the structure of direct tax laws** (like **Incometax Act, 1961**; **Wealth Tax Act, 1957**) in India into a single legislation.

Need for Direct Tax Code

Simplification of Tax Laws- The current Income-tax Act, 1961 has become overly complex due to thousands of amendments, provisions, and case laws.

- A new code would consolidate, simplify, and use plain language thereby reducing ambiguity for taxpayers and tax officers.

Broader Tax Base & Improved Compliance- A simpler code with fewer exemptions can bring more individuals and businesses into the tax net by reducing evasion and making compliance easier.

Equity and Fairness- Direct Tax Code aims to rationalise exemptions and incentives, which currently benefit higher-income groups disproportionately.

Reduced Litigation & Disputes- A modern tax code with clearer definitions (like of "residence," "capital gains," "income source") reduces disputes.

Certainty & Investor Confidence- Clearer rules on **cross-border taxation**, **transfer pricing,** and **GAAR (General Anti-Avoidance Rules)** reduce uncertainty for businesses and foreign investors.

Challenges in Implementation of Direct Tax Code

Transition Costs & Uncertainty- Shifting from the Income-tax Act, 1961 to a new code may cause **financial uncertainty** for **households** (savings-related deductions) and **businesses** (investment-linked incentives).

Resistance from Political Economy- Multiple lobbies and interest groups that benefit from exemptions like real estate, SEZs, exporters, insurance, etc. may resist, dilute or delay reform.

Administrative Preparedness- Tax authorities will need extensive retraining, upgraded IT systems, and dispute resolution mechanisms. Without capacity building, the DTC could result in more confusion rather than clarity.

Increased Litigation during Transition- Change in definitions (e.g., **residence**, **capital gains**, **income source**) can lead to rise in fresh tax disputes. This will add to the already high backlog of tax cases.

Technological & Compliance Challenges- Effective rollout requires seamless integration with **PAN**, **Aadhaar**, **GST data**, and **digital payment ecosystems**. Data privacy, cyber-security, and taxpayer adaptability are key hurdles in the implementation of Direct Tax Code.

Major Recommendations of the Akhilesh Ranjan Panel for Direct Tax Code

Personal Income Tax (PIT)	**Rejigging of tax slabs:** to make it more favourable for middle-income and upper middle class, thereby reducing tax burden.
Corporate Tax	A roadmap to reduce corporate tax rate to **25%** for all companies, including those above **₹400 crore turnover**. **Equal treatment of domestic and foreign companies** by reducing the disparity in tax rates.

Compliance, Law Simplification & Certainty	**Simplify the Income-tax law:** reduce provisions, explanations, convoluted clauses, & simplify language. **Improve tax predictability & certainty** to encourage voluntary compliance. Use of technology, faceless assessments and reduction of compliance burden.
Litigation Management & Dispute Resolution	**Strengthen mechanisms to reduce litigation** (e.g. mediation, collegiums of officers for dispute settlement) rather than adversarial proceedings. **Clarify rules in contentious areas** (capital gains, dividend distribution tax, MAT etc.) to reduce ambiguity.
Tax Relief & Incentives	**Special guidelines/incentives for startups** to encourage entrepreneurship. **Relief for middle-class taxpayers** via lower effective tax rates or fewer surcharges.

8. NEW INCOME TAX ACT 2025

Income Tax Bill, 2025, has been passed by the parliament which seeks to simplify, rationalise, and shorten the existing Income Tax 1961 Act.

Need for a New Income Tax Law

Outdated Income Tax, 1961- The Income Tax Act, 1961 has become outdated after decades of multiple amendments. The legislation has become complex, lengthy, and difficult for citizens to understand.

Reduction in the number of Chapters and Sections- The new Income Tax Bill,2025 reduces the number of **chapters from 47 to 23** and **sections from 819 to 536**, cutting down on redundancy and confusion.

- The Bill expands the use of tables (57 vs. 18 earlier) and formulae (46 vs. six earlier).

Simplification of language- The new Income Tax Bill, 2025, simplifies the language of the Income Tax Act 1961. It **removes legal jargon** and **includes practical examples**, making the **law more accessible** to ordinary citizens.

Comparison between Old Income Tax Act 1961 and New Income Tax Act 2025

Parameters	Old Income Tax 1961	New Income Tax Act 2025
Tax Year	Uses terms "**Previous Year**" (when income is earned) and "**Assessment Year**" (when tax is assessed on that income).	Removes "**previous year**" and "**assessment year**". It introduces "**tax year**" which corresponds to the financial year starting 1 April, and simplifies time reference.
Virtual Assets	The old law had some provisions for capital gains. **Virtual digital assets (crypto/NFTs)** were less explicitly addressed.	The new bill makes explicit mention of **virtual digital assets being treated as "assets"**. It introduces clarified rules for treatment, and holding periods for Virtual assets.

Powers of CBDT	Under the old law, **many procedural changes required legislative amendments**. There was less delegation of rule-making powers to CBDT.	The new bill gives **more powers to CBDT to frame rules**, **introduce compliance measures** and **digital tax administration schemes** without the need of Parliamentary amendment every time.
Faceless / Digital Processes	Many processes **involved human interface**, leading to potential delays, and complexities.	The new law emphasizes **faceless assessment**, **digital-first compliance**, and reduction of **human interface**.
Deductions / Exemptions	Many deductions and exemptions across multiple sections like **Section 10**, **Section 80C-80U**.	Simplified and consolidated deductions. There were fewer redundant/explanatory provisions.
Tax Filing Process	Complex filing process, multiple forms required	Simplified filing with fewer forms and a more streamlined process
Corporate Tax Rates	Higher corporate tax rates	Reduced corporate tax rates to make India more competitive globally.

The New Income Tax Bill 2025 showcases a reform in India's tax system and replaces the Income Tax Act 1961. The new changes create a simpler more efficient taxation regime benefitting all individuals, businesses and the economy eventually driving economic growth.

9. GST AND GST 2.0

Goods and Services Tax (GST)

- Goods and Services Tax (GST) is an indirect, destination-based tax levied on the supply of goods and services in India.
- It was implemented on July 1, 2017, and replaced a complex web of multiple central and state taxes, aiming to create a single, unified national market.
- GST works on the principle of "value addition". It's a multi-stage tax collected at every step of the supply chain, from manufacturing to the final consumer. The key mechanism is the Input Tax Credit (ITC).
- GST had five main rate slabs: 0%, 5%, 12%, 18%, and 28%.
  - India follows a dual GST model:
    - CGST (Central GST) – for intra-state transactions
    - SGST (State GST) – for intra-state transactions
    - IGST (Integrated GST) – for inter-state or inter-country transactions

Challenges with the GST Regime

- **Complex Rate Structure-** Multiple GST slabs (0%, 5%, 12%, 18%, 28% + cess and their frequent changes created confusion for businesses.
  - Classification of items often led to confusion and disputes. **For ex-** Food item's treatment as "luxury" or "essential"
- **Inverted Duty Structure-** In several sectors like textiles, footwear, renewable energy), inputs were taxed higher than final products. These caused blocking of credits and cash-flow issues for MSMEs.
- **High Compliance Burden-** Small and medium enterprises (SMEs/MSMEs) faced difficulties like filing of multiple returns, reconciling invoices, and complying with e-invoicing.

- Small firms had to invest in technology and tax consultants, which raised their costs.

- **Impact on Federalism / State Revenues-** States autonomy to set indirect taxes were subsumed by the GST. The expiry of compensation cess (guaranteed for 5 years), created tensions upon its expiry.
  - States have complained of under-compensation or faced delayed payments from the Centre.
- **Burden on Consumers-** Multiple slabs created inequities as taxes on some essential goods like food items attracted higher taxes. Certain luxury goods benefitted from classification loopholes.
  - Services largely taxed at 18%, raised costs for common consumers.
- **Frequent Changes & Uncertainty-** Regular notifications, circulars, and amendments confused taxpayers. Businesses struggled to plan due to the unpredictability of tax treatment.
- **Anti-Profiteering Issues-** National Anti-Profiteering Authority (NAA's) intervention and enforcement was inconsistent. Full tax cuts benefits rarely reached the consumers.

Key reforms proposed under new GST Structure (GST 2.0)

Two Main Slabs	**5% GST:** Applies to essential goods and services like packaged food, footwear, medicines, daily-use items, agriculture inputs, and small household goods. Also includes items like hair oil, toiletries, and stationery. **18% GST:** Standard rate applied to most goods and services such as restaurant and telecom services, financial services, electronics (TVs, washing machines, laptops), motorcycles below 350cc, small cars below 1200cc, and industrial goods.
New 40% De-merit Rate	**Introduced for luxury and sin goods** such as **high-end cars**, **premium bikes**, **aerated beverages**, **tobacco products**, and **gambling services**. This replaces the previous highest slab of 28% plus cess.
Exemptions from GST	Some items remain exempt from GST (0% GST) such as **fresh fruits**, **vegetables**, **milk**, **bread**, along with **new exemptions** on **personal health and life insurance**, **exercise books,** and **notebooks**.
Simplified Compliance / Process Reforms	Faster refunds or provisional release for certain cases (exporters, inverted duty etc.) with only high-risk ones subject to full scrutiny. Removing or easing contentious rules like requirements for pre-sale agreements for post-sale discounts via credit notes.
Classification Clarity	More precise definitions / rules so that goods and services are more clearly fitted into slabs, reducing litigation and uncertainty.
Focus on Ease for MSMEs	Simplified compliance, support for technology, reducing compliance costs, working capital relief (through better refunds, fewer blocked ITCs)

Proposed benefits from the rationalized GST tax slabs

- **Simplification & Ease of Compliance:** Fewer slabs help minimize ambiguity in product/service classification, lowering risks of misinterpretation, non-compliance, and litigation.
  - **For ex-** Earlier a food item was taxed at 5% if it was "unbranded" but at 12% or 18% if it was "branded" or processed.
- **Boost to consumption & economic growth:** By reducing taxes on common goods, especially those currently in the 12% and 28% slabs, the government can make these items more affordable for the common person.
  - This would increase disposable income and potentially stimulate consumption and demand, leading to higher economic activity.

- **Improved Ease of Doing Business**: A simpler structure makes filing, invoicing, and input tax credit reconciliation easier for taxpayers, especially MSMEs.
  - Fewer slabs improve system efficiency and reduce technical glitches.
- **Reduction in Tax Burden:** The proposed reform to shift 99% of the items in the 12% slab to 5% tax rate, and 90% of the items in the 28% slab to 18%, will substantially reduce the tax burden on most consumers.
- **Prevent Tax Evasion:** With a large number of items set to be taxed at just 5%, the incentives for input tax credit scams & tax evasion will also be substantially removed.
- **In accordance with the International Practice:** Most successful GST systems worldwide have 1–3 rates, helping tax administrators track and audit transactions more efficiently.
- **MSME empowerment:** With simpler compliance, better handling of refunds / blocked credits, MSMEs may face lower costs and fewer bottlenecks.
- **Reduced litigation & certainty:** Clearer classification, fewer slabs, better rules reduce disputes and legal uncertainty.

Challenges to the rationalisation of tax slabs

- **Significant Revenue Losses-** The government anticipates a drop in GST revenues of around **₹48,000 crore**- which poses serious fiscal pressure risks for the government.
- **Revenue Uncertainty for States-** A reduction in GST slabs may decrease overall tax revenue in the short term, particularly for state governments that have already lost GST compensation after the initial five-year transition.
  - States whose citizens consume more high-rate goods might be adversely affected.
- **Input Tax Credit (ITC) Complications-** Businesses (automobiles and FMCG) are concerned about lapsing of their accumulated compensation cess credits. This may strain their working capital.
- **Technological and Compliance Burdens-** All GST Network (GSTN) systems, business ERP, and accounting software nationwide must be updated. This can cause significant short-term disruption and add compliance costs for MSMEs and small traders.
- **Exclusion of Key Sectors-** Major sectors like petroleum, real estate, and alcohol remain outside the GST framework.
- **Classification Disputes-** Disagreements over product categorization persist, potentially leading to litigation and uncertainty.

Way forward with regards to rationalisation of tax slabs

- **Managing Revenue Loss:** To address states' concerns about a potential revenue shortfall, the government must ensure that any reduction in rates is offset by a **rise in consumption** and **improved tax compliance**.
- **Correcting Inverted Duty Structure:** Rationalization offers a crucial opportunity to fix the inverted duty structure in many sectors where the tax on inputs is higher than on the final product.
  - Aligning these rates will prevent the accumulation of **Input Tax Credit (ITC)** and improve the cash flow for businesses.
- **Use of Technology:** Use technological improvements and compliance simplification to offset potential short-term revenue loss.
- **Ensure Transparency & Clarity:** Provide long-term clarity on rates and policy direction to confidently support business planning and investment. Communicate changes transparently, including timelines and potential impacts on businesses and consumers.
- **Monitor impact & Make data-driven adjustments:** Track demand, prices, compliance levels, and revenue after rationalisation. Make incremental or corrective changes if needed for growth and equity.

Conclusion

Reforms to simplify the current GST regime must not only be limited to reducing the multiplicity of rates but should also be about making it easier & less-time consuming for tax payers as well as addressing the issue of compliance & classification.

10. FISCAL HEALTH INDEX

About Fiscal Health Index (FHI):

- **Annual Publication**: The Fiscal Health Index report will be released every year.
- **Focus on Fiscal Health**: It will assess the fiscal health of Indian states.
- **Data-Driven Insights**: The report will provide insights based on data, guiding state-level policy decisions.
- The index ranks States on the basis of a composite fiscal index, which is based on five major sub-indices and nine minor sub-indices.
- States have been classified on the basis of the FHI score:
  - **Achiever:** Greater than 50
  - **Front Runner:** Greater than 40 & less than equal to 50.
  - **Performer:** Greater than 25 & less than equal to 40
  - **Aspirational:** Less than equal to 25
- It covers **18 major states** (excluding "special category / hilly / North-Eastern" states for better comparability) over the Financial Year **2022-23**.
- Data source is primarily the **Comptroller and Auditor General of India (CAG)**.

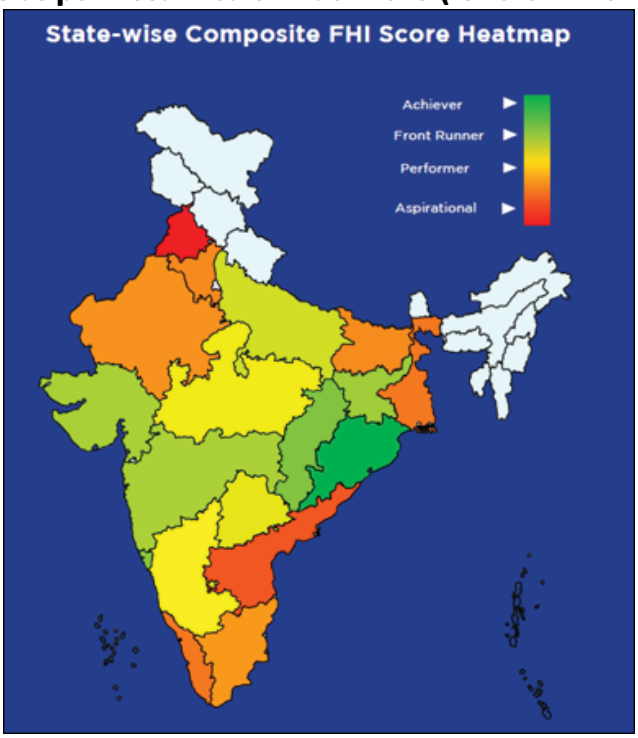
Criteria for ranking: The Fiscal Health Index 2025 is based on a **comprehensive set of indicators** that are grouped into five major sub-indices:

- **Quality of Expenditure:**
  - Measures the share and effectiveness of productive developmental spending.
  - Indicators include ratio of developmental expenditure to total expenditure, and capital outlay to GSDP.
- **Revenue Mobilization:**
  - Assesses states' ability to generate their own revenue independently.
  - Indicators include state own revenue relative to GSDP and to total expenditure.
- **Fiscal Prudence:**
  - Evaluates fiscal discipline through deficits.
  - Indicators include gross fiscal deficit to GSDP and revenue deficit to GSDP.
- **Debt Index:**
  - Captures debt burden and servicing capacity.
  - Indicators include interest payments as a proportion of revenue receipts and total outstanding liabilities to GSDP.
- **Debt Sustainability:**
  - Measures the difference between economic growth and debt servicing growth.
  - Indicator: Growth rate of GSDP minus growth rate of interest payments (a positive value indicates sustainable debt).

MAJOR SUB-INDICES	MINOR SUB-INDICES
1. Quality of Expenditure	1.1 Total Developmental Expenditure/Total Expenditure
	1.2 Total Capital Outlay/ GSDP*
2. Revenue Mobilization	2.1 State Own Revenue/ GSDP*
	2.2 State Own Revenue/ Total Expenditure
3. Fiscal Prudence	3.1 Gross Fiscal Deficit/ GSDP*
	3.2 Revenue Deficit/ GSDP*
4. Debt Index	4.1 Interest Payments/Revenue Receipts
	4.2 Outstanding Liabilities/ GSDP*
5. Debt Sustainability	5.1 Growth Rate of GSDP* – Growth Rate of Interest Payments

GSDP at current prices for the year 2022-23

Performance of the States as per Fiscal Health Index 2025 (for the FY 2023):



Achiever	Front Runner	Performer	Aspirational
Odisha (1)	Maharashtra (6)	Tamil Nadu (11)	Kerala (15)
Chhattisgarh (2)	Uttar Pradesh (7)	Rajasthan (12)	West Bengal (16)
Goa (3)	Telangana (8)	Bihar (13)	Andhra Pradesh (17)
Jharkhand (4)	Madhya Pradesh (9)	Haryana (14)	Punjab (18)
Gujarat (5)	Karnataka (10)		

Objectives of Fiscal Health Index:

- To provide a **comparative analysis of fiscal health** across Indian states through standardized metrics.
- To **identify areas of strength and concern** in states' fiscal management practices.
- To **promote transparency, accountability, and prudent fiscal management** through empirical assessment.

- To **assist policymakers in making informed decisions** aimed at enhancing fiscal sustainability and resilience.

Need/Importance of Fiscal Health Index:

- **Strengthen National Macroeconomic Stability:** Since states account for about two-thirds of public spending and a significant part of revenue, their fiscal health matters for national stability.
- **Competitive federalism:** FHI can promote healthy competition among states to align their fiscal strategies with national objectives, ensuring their contributions to the goal of a fiscally stable and prosperous India.
- **Transparency and accountability:** Index can promote transparency in fiscal practices by providing a public measure of how well states manage their finances.
- **Informed policy making:** By evaluating fiscal health through quantifiable metrics, the FHI can help policymakers identify where states need to focus reforms or allocate resources for better fiscal outcomes.
- **Diagnostic Tool:** The FHI acts as a diagnostic tool, highlighting structural fiscal risks and strengths and assisting in targeted interventions for improvement.

Conclusion:

The Fiscal Health Index 2025 offers a valuable tool for assessing the fiscal performance of Indian states. It highlights the need for continuous monitoring, prudent fiscal management, and proactive measures to enhance states' financial health. The Index underscores the importance of revenue generation, efficient expenditure management, debt control, and adherence to fiscal deficit targets for overall fiscal sustainability.

11. STATE FINANCES

Status of State Finances

Reserve Bank of India's (RBI) report, **State Finances- A Study of Budgets of 2024-25**, has highlighted the progress made by state governments in fiscal consolidation.

States Performance Post-Pandemic

Improved Tax Revenue	The **average tax buoyancy** (responsiveness of tax revenue to changes in the economic growth rate) increased to **1.4** (during 2021-25) from **0.86** (during 2013 to 2020). This reflects improved efficiency in tax collection. Higher tax revenues have enabled states to allocate greater funds for asset creation, including highways and bridges.
Increase in Capital Expenditure	States have consistently improved the quality of expenditure, with capital expenditure rising from **2.4% of Gross Domestic Product** (GDP) in 2021-22 to **2.8% in 2023-24**, and it is budgeted at **3.1% of GDP** in 2024-25. This indicates a sustained focus on improving the quality of expenditure with growth-enhancing investments.
Fiscal Discipline	States' gross fiscal deficit is budgeted at **3.2% of GDP** for 2024-25. This is a marginal increase from **2.9% of GDP** in 2023-24 (2.9%).
Dependence on Borrowing	States' dependence on market borrowings **has surged**, accounting for 79% of the gross fiscal deficit (GFD) in FY25. Gross market borrowings of states and Union Territories increased by 32.8% to Rs 10.07 trillion in FY 2023-24.

Elevated Debt Levels	States' debt-to-GDP ratio (relative measure of debt compared to economic output) decreased from **31.0% of GDP in March 2021** to **28.5% in March 2024**. However, it remains higher than the pre-pandemic level of **25.3% in March 2019**. State debt levels **exceed the Fiscal Responsibility and Budget Management committee's recommended debt-to-GDP ratio** of 60% by 2023 (with 40% for the Central Government and 20% for State Governments).
Interstate Variations	States like Arunachal Pradesh, Himachal Pradesh, Sikkim, and Tripura have forecast high fiscal deficits, while larger economies like Gujarat and Maharashtra have lower deficits as a share of GDP.
Distressed DISCOMS	Electricity distribution companies (DISCOMs) continue to strain State finances. The accumulated losses have reached **Rs 6.5 lakh crore** by **2022-23** (**2.4% of India's GDP**).

Concerns regarding State Budgets

- **High Debt Levels and Debt-to-GSDP Ratios-** Several states have very high debt relative to their economic output (Punjab has one of the highest debt-to-GSDP ratios at ~ 46.6%).
  - High debt means high interest payments, which reduce fiscal space for welfare, infrastructure, capital investment.
- **Weak Revenue Mobilization-** Several states struggle to raise their own tax and non-tax revenue. There is high dependence on the devolution of transfers from the centre.
  - GST rate changes (rate cuts) affect state revenues. For ex- West Bengal expects around ₹5,000 crore loss in 2025-26 due to GST reductions.
- **Rising Subsidy Burden:** Several states like Maharashtra, Madhya Pradesh, Karnataka have announced high subsidies such as "farm loan waivers, free/subsidised services (like electricity to agriculture and households, transport, gas cylinder and cash transfers to farmers, youth and women)".
  - These measures risk crowding out funds meant for critical social and economic infrastructure.
  - Subsidy schemes like income transfers for women (around Rs 2 lakh crore, ~0.6% of GDP) strain state finances.
- **Rise in off-budget Liabilities and Guarantees-** State's off-budget liabilities (loans or obligations of PSUs, corporations, etc.) have increased. Inadequate reporting of off-budget liabilities obscures the true fiscal position.
- **Regulatory & Institutional Constraints-** Borrowing limits from Centre (Ways and Means Advances), fiscal rules (FRBM limits) constrain states borrowing.

Recommendations by RBI on State Finances

- **Debt Consolidation:** Establish clear, transparent, and time-bound glide paths for debt reduction. Ensure uniform reporting of liabilities to improve fiscal accountability.
- **Expenditure Efficiency:** Focus on outcome-based and climate-sensitive budgeting.
  - Rationalise centrally-sponsored schemes (CSS) effectively to free up resources for state-specific needs and reduce fiscal stress.
- **Subsidy Rationalization:** Contain and optimize subsidies to prevent them from overshadowing more productive expenditures.
- **Efficient Market Borrowings:** Reduce over-reliance on market borrowings to manage fiscal deficits and minimize financial vulnerabilities.
- **Revenue Generation:** Strengthen collection mechanisms for SGST, stamp duty, and other key revenue sources. Increase non-tax revenues and grants to reduce dependency on market borrowings.

- **Enhanced devolution by the 16th Finance Commission-** The 16th Finance Commission (FC) must look to enhance the state's share in the net taxes from 41% (currently awarded by 15th FC). Further, the principles of Vertical and Horizontal devolution, must be relooked to ensure equitable distribution of taxes among the states.
- **Rationalisation of Public Expenditure by Central Govt-** A mechanism must be instituted for thorough financial rationalisation of the Central Sector and Centrally Sponsored schemes, in collaboration with state governments.
- **Addressing the GST related Concerns-** The anomalies in GST like the Integrated GST which favours the consuming states like UP and Bihar, rather than the producing states of TN, Gujarat must be corrected. Also, efforts must be undertaken to open more avenues for revenue generation by broadening the scope of GST to include petrol, diesel etc.
- **Reduction of Borrowing Constraints on States-** The Union government should revisit the borrowing constraints placed on state investment funds, as suggested by Kerala.
- **Reduction in Cesses and Surcharges-** The Union government should reduce the degree to which it uses cesses and duties to expand its share of tax collections.

Committee recommendations for State Finances

- **FRBM Review Committee (NK Singh Panel - 2017)**
  - **Golden Rule Implementation:** Recommended amending state-level FRBM Acts to ensure borrowing is strictly utilized for capital asset formation, completely eliminating revenue deficits.
  - **Escape Clause Framework:** Proposed strict, narrow operational windows (such as national security, war, or systemic agricultural collapses) under which a state may breach its 3% fiscal deficit cap by up to 0.5%.
- **NITI Aayog & RBI Working Group on State Finances (2023–2024)**
  - **Universalization of Fiscal Councils:** Recommended setting up independent, non-partisan State Fiscal Councils to provide unbiased macroeconomic forecasts and audit state budgetary figures.
  - **Strict Off-Budget Tracking:** Urged the immediate elimination of off-budget liabilities by factoring state-government-backed public corporate loans directly into the formal legislative debt limits.
- **2nd Administrative Reforms Commission (ARC)**
  - **Strengthening State Finance Commissions (SFCs):** Recommended mandatory, timely constitution of SFCs and automatic passing of their fiscal award models to prevent cash idling at Urban Local Bodies and Panchayats.
  - **Outcome-Driven Budgetary Frameworks:** Advised a total shift from an input-based budgeting structure to strict, time-bound, and measurable public outcome budgeting audits.

Conclusion

Balancing subsidies with fiscal discipline is crucial for sustaining India's growth trajectory. Efficient allocation of resources, along with strategic reforms in welfare spending and revenue generation, can pave the way for a stable and prosperous economy.

12. FREEBIES CULTURE IN INDIA

In Indian politics, giving "freebies" or following a "revadi culture" has become a common way for parties to win votes. During elections, parties promise free goods, services, or subsidies to attract voters. While these help the poor in the short term, they raise concerns about long-term effects on the economy, governance, and society.

Definition of Freebies: While there is no universally accepted definition of freebies, the Reserve Bank of India describes them as "a public welfare measure provided free of charge." In practice, freebies often take the form of promises made by political parties during elections and have become a significant feature of Indian politics.

Welfarism vs Freebies in India

Aspect	Welfarism	Freebies
Definition	Programs rooted in constitutional duty, designed for the long-term development of human potential	Short-term handouts like free electricity, water, debt waivers, or cash transfers
Objective	Focused on sustainable development and empowering citizens to improve their lives	Provide immediate benefits, often with a political angle
Impact	Strengthens skills, health, and education, creating lasting value	Can upset market balance, weaken credit culture, and discourage work (as noted by the Supreme Court)
Examples	Public Distribution System (PDS), MGNREGA, education and health schemes	Laptops, TVs, gold coins, cash handouts

Constitutional and Legal Framework

Directive Principles of State Policy (DPSP): The Indian Constitution guides the state through Articles 38, 39, and 41 to:

- Maintain a social order that ensures the welfare of all citizens
- Provide men and women with adequate means of livelihood
- Prevent excessive concentration of wealth
- Guarantee the right to work, education, and public assistance when needed

Supreme Court Judgements:

- **Subramaniyam Balaji v. Tamil Nadu (2013):** The Court observed that distributing items like TVs and laptops falls within the ambit of DPSP, and the judiciary should not interfere.
- **Ashwini Kumar Upadhyay v. Union of India (pending):** The Court is currently examining whether offering freebies during election campaigns aligns with the Constitution.

Election Commission Guidelines:

- To ensure accountability, the ECI has asked political parties to clearly disclose how they plan to fund electoral promises, especially freebies, promoting transparency in the democratic process.

Arguments in Favour of Freebies

- **Constitutional and Governance Backing-** The Constitution, through **DPSPs** (Articles 38, 39, 41), envisions India as a welfare state responsible for social justice, equitable wealth distribution, and public assistance in cases like unemployment, sickness, and old age.
- **Socio-Economic Upliftment-** Freebies like **Pradhan Mantri Garib Kalyan Anna Yojana** or **Subhadra Yojana (Odisha, 2024)** provide food grains or financial aid, reducing poverty and supporting marginalized communities.
- **Enhanced Access to Education and Digital Resources-** Schemes such as the **Free e-Scooty Scheme (Madhya Pradesh)** and **Gruha Jyoti (Karnataka)** promote education, reduce dropout rates, and support digital literacy.

- **Financial Inclusion and Transparency- Direct Benefit Transfer (DBT) schemes**, e.g., Haryana's ₹1.06 lakh crore transfers (2025), eliminate fake beneficiaries and reduce leakages.
- **Empowerment of Women and Marginalized Groups-** Freebies like female pensions, sewing machines, or bicycles (Bihar) and the **Yuva Nidhi Scheme (Karnataka)** aid women and youth, promoting social and economic empowerment.
- **Productivity and Disposable Income-** Meeting basic needs through healthcare, education, and cash transfers increases productivity and disposable income, potentially creating a multiplier effect in the economy.

Arguments Against Freebies

- **Dependency Syndrome-** Freebies may reduce self-reliance and discourage productive work, fostering long-term dependency on the state.
- **Environmental Unsustainability-** Free electricity for farmers in Punjab has led to groundwater depletion and stressed power utilities (DISCOMs).
- **Skewed Government Expenditure-** Large budget shares often favor a small section of society. **Example**: Rajasthan's old pension scheme consumed 56% of state revenues while benefiting only 6% of the population.
- **Low Productivity Impact-** Schemes like MNREGA or cash handouts provide short-term relief but often do not enhance long-term productivity or employment.
- **Fiscal Strain and Debt Burden-** States distributing significant freebies face rising fiscal deficits and reduced investment capacity. NITI Aayog's Fiscal Health Index indicates weaker fiscal parameters in such states.
- **Reduced Infrastructure Investment-** Resources diverted to freebies limit long-term infrastructure and asset creation, slowing economic growth.

Way Forward to address Freebie Culture in India

- **Regulation-** The government should introduce clear guidelines to regulate the distribution of freebies, ensuring they are **targeted, need-based, and aligned with long-term welfare goals** rather than mere electoral promises.
- **Electoral Reform-** The Election Commission must enforce stricter rules on the announcement and distribution of freebies during elections, **limiting excessive promises** and safeguarding the fairness of the electoral process.
- **Fiscal Responsibility-** Both the Centre and state governments should adopt **fiscally responsible policies**, ensuring welfare schemes are sustainable and do not lead to excessive debt or fiscal strain.
- **Empowerment over Handouts-** Promote skill development, employment, and self-reliance while expanding insurance for vulnerable groups to reduce dependency on freebies.
- **Voter Awareness and Political Consensus**– Raise voter awareness about the long-term economic and social impacts of freebies. Promote Centre-State coordination to prevent welfare schemes from being exploited for short-term electoral gains.
- **Judicial Oversight and Expert Evaluation-** Establish **expert committees** comprising members from NITI Aayog, RBI, and the Finance Commission to assess the **economic, social, and fiscal impacts** of freebies.
- **Learning from Global Experiences-** Avoid mistakes made by other countries, such as **Sri Lanka (2019 tax cuts)** and **Venezuela (loan waivers and populist handouts)**, which resulted in fiscal crises and prolonged economic instability.

13. 16TH FINANCE COMMISSION RECOMMENDATIONS

The Finance Commission (FC) is a Constitutional body constituted by the President every five years to make recommendations on centre-state fiscal relations. The Report of the **16th Finance Commission (Chairman: Dr. Arvind Panagariya)** was tabled in Parliament on February 1, 2026 for the five-year period between 2026-27 and 2030-31.

Key Recommendations of 16th Finance Commission

<table>
<tr><td>Vertical Devolution</td><td>The share of states in the divisible pool of central taxes has been recommended at 41%. This is the same as the share recommended by the 15th Finance Commission.</td></tr>
<tr><td>Horizontal Devolution</td><td>To provide for the distribution of central taxes among states, Finance Commissions define a formula with weightage for certain parameters - which include:

• Income distance
• Population (2011)
• Demographic performance
• Area
• Forest
• Contribution to GDP = The 16th FC has introduced this parameter to account for the contribution to national GDP. This replaces the Tax and Fiscal Efforts parameter used by the 15th FC which rewarded states with a higher tax collection efficiency.

<table>
<tr><th>Criteria</th><th>15th FC (2021-26)</th><th>16th FC (2026-31)</th></tr>
<tr><td>Income Distance</td><td>45%</td><td>42.5%</td></tr>
<tr><td>Population (2011)</td><td>15%</td><td>17.5%</td></tr>
<tr><td>Demographic Performance</td><td>12.5%</td><td>10%</td></tr>
<tr><td>Area</td><td>15%</td><td>10%</td></tr>
<tr><td>Forest</td><td>10%</td><td>10%</td></tr>
<tr><td>Tax and Fiscal Efforts</td><td>2.5%</td><td>-</td></tr>
<tr><td>Contribution to GDP</td><td>-</td><td>10%</td></tr>
<tr><td>Total</td><td>100%</td><td>100%</td></tr>
</table>
Sources: Reports of the 15th and 16th Finance Commissions; PRS.</td></tr>
<tr><td>Grants-in-Aid</td><td>• The 16th FC has recommended grants worth Rs 9.47 lakh crore over the five-year period. These comprise grants for:
  • Urban & Rural Local Bodies
  • Disaster Management
• The 16th FC has discontinued the following grants recommended by the 15th FC:
  • Revenue Deficit Grants
  • Sector-specific Grants
  • State-specific Grants</td></tr>
</table>

Grants for Local Bodies

- The 16th FC has recommended grants worth Rs 4.4 lakh crore and Rs 3.6 lakh crore for rural and urban local bodies, respectively.
- These grants are divided into **basic (80%)** and **performance-based (20%)** components:
  - **Basic grants:** 50% of the basic grant will be untied and the rest 50% will be tied to: (i) sanitation and solid waste management, and/or (ii) water management.
  - **Performance grants:** These grants for local bodies are further divided into state performance grants and local body performance grants. **State performance grants** will be made available upon meeting a minimum benchmark for transfers to local bodies from their own resources. **Local body performance grants** are linked to achievement of minimum targets specified by the Commission for own source revenue growth.
- Special Infrastructure Grants and Urbanisation Premium Grants have also been recommended for urban local bodies.
- All local body grants will be made available upon fulfilment of three entry-level criteria:
  - Constitution of the local bodies as per the Constitution.
  - Publication of provisional and audited accounts of the local bodies in the public domain.
  - Timely constitution of the State Finance Commission.

Grants	Amount
Local governments	**7,91,493**
Rural local bodies	***4,35,236***
Basic Grant	*3,48,188*
Performance Grant	*87,048*
Urban local bodies	***3,56,257***
Basic Grant	*2,32,125*
Performance Grant	*58,032*
Special Infrastructure Component	*56,100*
Urbanisation Premium	*10,000*
Disaster management	**1,55,916**
Total	**9,47,409**

Source: Report of the 16th Finance Commission; PRS.

Disaster Management Grants

- The Commission has recommended disaster management corpus of **Rs 2,04,401 crore for State Disaster Relief and Management Funds (SDRF and SDMF)**.
- The cost-sharing pattern between the centre and states is recommended to be: (i) **90:10** for north-eastern and Himalayan states, and (ii) **75:25** for all other states.

Fiscal roadmap	• The Commission has recommended that the Centre should bring down the fiscal deficit to **3.5% of GDP** by 2030-31. It recommended the annual fiscal deficit limit for states to be 3% of GSDP. • It also recommended strictly discontinuing the practice of **off-budget borrowings** for states and bringing all such borrowings onto their budgets. The definition of fiscal deficit and debt should be expanded to uniformly include all off-budget borrowings.
Power-sector Reforms	• The Commission recommended that states should **actively pursue privatisation of electricity distribution companies (DISCOMs)**. To shield the private investor from debt burden after discom takeover, a special purpose vehicle may be created to warehouse the debt. • Pre-payment or eventual repayment of this debt may be allowed using the **funds from the Special Assistance Scheme for Capital Investment**. It also recommended that states should be allowed to utilise this assistance only after the privatisation process is complete.
Subsidy Expenditure	• The Commission recommended states to review and **rationalise their subsidy expenditure**. It noted that schemes providing unconditional cash transfers tend to have large and untargeted beneficiaries. • It recommended **setting clear exclusion criteria** and **a rigorous review process to ensure effective targeting**. In addition, it recommended discontinuing financing of subsidies through off budget borrowings.
Public Sector Enterprise (PSE) Reforms	• The Commission recommended a review and closure of **308 inactive State Public Sector Enterprises (SPSEs)**. It recommended formulation of a state-level PSEs disinvestment policy to target inactive and underperforming SPSEs. • State or union PSEs, which **incur losses for three out of four consecutive years**, should be placed for the respective Cabinet's consideration. The Cabinet may decide closure, privatisation, or continuation depending on the strategic importance of the enterprise.

Positive aspects of the 16th Finance Commission's recommendations

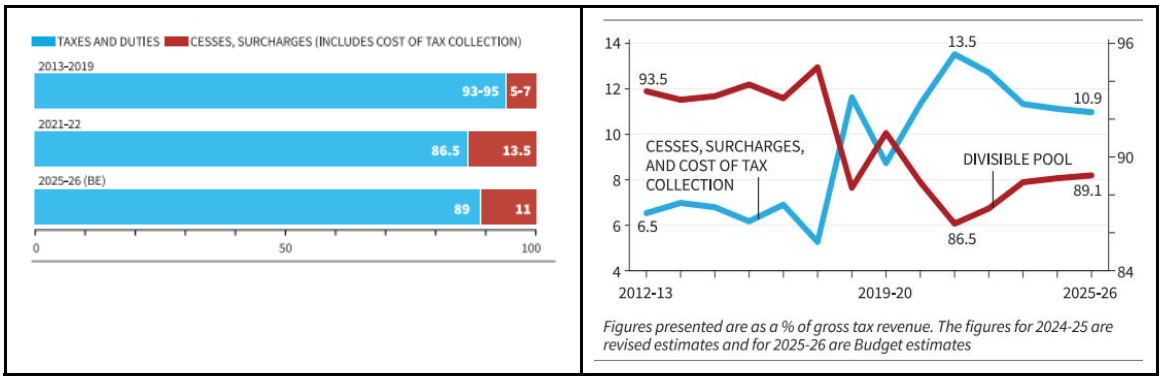
1. **Rewarding Economic Performance (The GDP Parameter)-** By **replacing the narrow *"Tax and Fiscal Effort" metric*** with ***"Contribution to GDP"***, the 16th FC rewards states that drive national economic growth. This **addresses long-standing grievances from industrialized states** (like Karnataka, Tamil Nadu, and Maharashtra) that felt penalized for their economic success under previous formulas.
2. **Strategic Urbanisation Focus-** The 16th FC acknowledged that India's future growth lies in its cities, introducing two innovative grants:
   - **Urbanization Premium-** A one-time incentive for states to **merge "peri-urban" (semi-urban) villages** into **larger Urban Local Bodies (ULBs)**. This encourages **planned urban expansion** rather than **chaotic sprawl**.
   - **Special Infrastructure Component-** Dedicated funding for **wastewater management in mid-sized cities (population 10–40 lakh)** which often lack the financial muscle of mega-metros.
3. **Enforcing Fiscal Discipline**
   - **End of Off-Budget Borrowings-** 16th FC called for a **strict discontinuation of off-budget borrowings**, requiring all such liabilities to be disclosed in the main budget. This provides a truer picture of public debt.

- **Rationalizing Subsidies-** It advised states to **move away from "unconditional cash transfers"** and **implement clear exclusion criteria**, ensuring welfare spending reaches the most vulnerable rather than becoming a general populist tool.

4. **Empowerment of Local Bodies-** While the 15th FC started the trend, the **16th FC has solidified the push for grassroots accountability** by recommending **Performance-linked grants**.
   - To even access basic grants, **states must ensure timely constitution of State Finance Commissions** and the **public disclosure of audited accounts** for all panchayats and municipalities.
5. **Transparency in "Net Proceeds"-** The commission recommended that the **Centre annually disclose tax data as certified by the CAG** to ensure states have full visibility into the "divisible pool."

Criticisms against the recommendations of the 16th finance commission

1. **Vertical Devolution Stagnation-** A major point of contention is the **decision to retain the states' share at 41%**. **18 out of 28 states** had formally requested an increase to **50%**.
   - States contend that **41% is insufficient** given that **states now bear the majority of "ground-level" expenditure responsibilities** in **health**, **education**, and **social welfare**, while their power to raise independent revenue has been curtailed post-GST.
2. **The "Cess & Surcharge" Loophole-** The commission has been criticized for not addressing the growing "leakage" of the divisible pool through cesses and surcharges. Cess & surcharge are collected by the Centre but are not shared with states.
   - Between **2013-2019**, for every **Rs. 100 collected by the Centre**, about **Rs. 93-95 was collected as taxes & duties** that form the divisible pool & the remaining **Rs. 5-7 was collected as cess & surcharge**. For 2025-26, the Centre is expected to collect **Rs. 89 as taxes & duties** & **Rs.11 as cesses & surcharges**.



3. **Impact on Poorer States-** States like **Bihar and Uttar Pradesh** have seen a marginal decline in their share because to accommodate the GDP parameter, the **weight of "Income Distance" was reduced** from **45%** to **42.5%**.
   - Critics argue that this parameter **benefits already industrialized states (the "richer" states)**, **potentially widening the developmental gap** and violating the principle of "**Economic Convergence**."
4. **Discontinuation of Revenue Deficit Grants (RDGs)- Complete scrapping of RDGs will impact the Hill & Special Category States like NE States, Himachal Pradesh, Uttarakhand.**
   - Revenue deficit of these states is due to **lack of favourable geography** & **industrialization**. Removing these grants without a transition plan could **cripple their public services**.
5. **"Freebie" Criticism-** The commission's warning against **"unconditional cash transfers"** (like *Majhi Ladki Bahin* or *Gruha Lakshmi*) is viewed as an **ideological intervention into state welfare politics**.

6. **Power Sector Privatization-** Linking capital assistance to the privatization of DISCOMs is **criticized as a "one-size-fits-all" approach** that ignores the specific political and social realities of different states.

Conclusion

The 16th Finance Commission plays a crucial role in strengthening fiscal federalism by ensuring fair distribution of financial resources between the Centre and States. Its recommendations aim to promote balanced economic growth, transparency, and cooperative governance across India.

14. PUBLIC DEBT IN INDIA

Public debt in India, comprising the consolidated liabilities of the Centre and States, serves as a vital macroeconomic tool for financing large-scale infrastructure and socio-economic welfare programs. As per the **Reserve Bank of India (RBI)**, the country's general government debt remains elevated at over **80% of GDP**, creating structural pressures through high interest-servicing obligations.

Status of Central Government Liabilities and Debt Management

- **Macro Status of Central Liabilities**
  - **Debt-to-GDP Trajectory:** According to the Union Budget 2026-27, the Central Government's outstanding liabilities are estimated at 55.6% of GDP for FY27, down from 56.1% in the Revised Estimates (RE) of FY26.
  - **Consolidation Target:** The government has set a structural medium-term anchor to reduce the central debt-to-GDP ratio to 50±1% by 2030-31.
  - **Fiscal Deficit Anchoring:** The fiscal deficit—the primary operational driver of fresh debt—has been reduced to 4.4% of GDP in RE FY26 and is budgeted at 4.3% of GDP for FY27.
- **Structural Composition of Central Debt**
  - **Internal Debt Dominance:** As per Union Budget Data (2026), India's public debt is overwhelmingly domestic; internal debt and other liabilities are estimated at ₹207.71 lakh crore for FY27, accounting for over 96% of total central obligations.
  - **Insulated External Risk:** According to the Ministry of Finance's Receipt Budget, external debt is strictly contained at a book value of ₹7.11 lakh crore for FY27, shielding India's public debt from severe global currency volatility shocks.
- **Debt Management and Interest Burden Risks**
  - **Revenue-Servicing Strain:** According to PRS India's Union Budget 2026-27 Analysis, interest payments have risen to consume 26% of total government expenditure and 40% of all revenue receipts.
  - **Committed Expenditure Lock-In:** As per PRS India (2026), approximately 65.3% of the Centre's revenue receipts are now locked under committed expenditure (salaries, pensions, and interest payments).

Significance of Public Debt

- **Financing Strategic Infrastructure:** Acts as a vital non-inflationary resource mobilizer to fund capital-intensive projects (e.g., National Infrastructure Pipeline) that possess a high long-term economic multiplier effect.
- **Counter-Cyclical Fiscal Stabilizer:** Enables the government to deploy stimulus packages and support vulnerable sections during economic contractions or black swan shocks without depleting active reserves.
- **Deepening Domestic Financial Markets:** Continuous issuance of government dated securities (G-Secs) creates a risk-free benchmark yield curve, helping price corporate bonds and commercial papers.

- **Bridging Intergenerational Infrastructure Gaps:** Distributes the financial burden of building permanent national assets (like high-speed rail and dams) across current and future generations who will benefit from them.
- **Preserving External Sovereignty:** Overwhelming reliance on domestic currency-denominated public debt (~95%) shields the national economy from global exchange rate volatility and external balance of payments shocks.
- **Inter-Temporal Consumption Smoothing:** Allows the state to maintain a steady flow of essential social sector spending (health, education, food security) even during periods of cyclical tax revenue shortfalls.

Public Debt Management and Associated Agencies in India

- **Core Regulatory Framework & Institutional Roles**
  - **The Reserve Bank of India (RBI):** Acts as the statutory investment planner and merchant banker to both the Central and State Governments under the RBI Act, 1934, regulating and executing all market-borrowing operations.
  - **Public Debt Office (PDO):** A specialized wing within the RBI that manages the actual issuance, registration, interest servicing, and systematic redemption of government dated securities and Treasury Bills.
  - **Internal Debt Management Department (IDMD):** The strategic cell of the RBI tasked with assessing market liquidity, setting the calendar for G-Sec auctions, and pricing domestic debt.
  - **Ministry of Finance (MoF):** Retains the ultimate sovereign constitutional authority to determine the annual net borrowing ceiling for the Centre and individual states under Articles 292 and 293.
  - **Public Debt Management Cell (PDMC):** An interim institutional setup housed within the Ministry of Finance, functioning as the statutory **Middle Office** to formulate long-term debt management strategies.
  - **External Debt Management Unit (EDMU):** A specialized unit within the Department of Economic Affairs (DEA) that tracks, audits, and maintains the statistical database of India's external commercial and sovereign liabilities.
  - **Proposed Public Debt Management Agency (PDMA):** A planned independent statutory body aimed at separating debt management from monetary policy, transferring the debt-issuance role from the RBI to an autonomous treasury under the MoF.
  - **Clearing Corporation of India Limited (CCIL):** The central clearing house providing guaranteed settlement and risk-mitigation infrastructure for the secondary trade of government debt.
  - **Negotiated Dealing System-Order Matching (NDS-OM):** The advanced electronic screen-based trading platform managed by the RBI for anonymous, electronic secondary market trading of G-Secs.

Classification of Government Debt and Liabilities in India

- **Public Debt**
  - **Internal Debt**
    - **Marketable Dated Securities:** Fixed-coupon long-term bonds issued by the government to institutional investors to finance the primary fiscal deficit.
    - **Treasury Bills (T-Bills):** Short-term debt instruments issued in three standard tenors (91, 182, and 364 days) to manage temporary cash flow mismatches.
    - **14-day Intermediate Treasury Bills:** Non-marketable short-term instruments issued exclusively to enable state governments to park their surplus cash balances.
    - **Special Securities Issued to International Financial Institutions:** Non-negotiable, non-interest-bearing rupee securities issued for India's quota contributions to the IMF, World Bank, and ADB.

- **Securities Issued Against Small Savings:** Government bonds issued directly to the National Small Savings Fund (NSSF) to utilize public small savings deposits.
- **Market Stabilization Scheme (MSS) Bonds:** Specialized securities issued by the government to assist the RBI in sterilizing excess liquidity from the financial system.

- **External Debt**
  - **Multilateral Agency Borrowings:** Sovereign loans contracted from international development institutions like the International Bank for Reconstruction and Development (IBRD), IDA, and Asian Development Bank (ADB).
  - **Bilateral Sovereign Loans:** Long-term development assistance contracted directly from individual foreign partner nations (such as Japan's JICA).
- **Other Liabilities**
  - **National Small Savings Fund (NSSF):** Public deposits pooled via post office savings schemes, recurring deposits, and Public Provident Fund (PPF) accounts.
  - **State Provident Funds:** Savings contributions held in trust by the government on behalf of public sector employees.
  - **Special Deposits and Reserve Funds:** Isolated reserve pools, including security deposits, departmental funds, and specific balancing reserves kept in the Public Account.
- **Contingent & Invisible Liabilities**
  - **Sovereign Guarantees:** Financial backing extended by the state to shield lenders against payment defaults by public sector undertakings (PSUs) or infrastructure SPVs.
  - **Off-Budget Liabilities:** Debts explicitly raised by state-owned enterprises (like FCI or NHAI) to fund public programs but serviced ultimately through the budgetary allocations of the government.

Factors Contributing to Increased Public Debt in India

- **Structural Revenue Mismatches**
  - **Persistent Fiscal Deficits:** A chronic macroeconomic imbalance where total state and central non-debt expenditures consistently outpace total revenue receipts, requiring continuous market borrowings.

    According to the **Union Budget 2026-27,** the Centre's fiscal deficit is budgeted at **4.3% of GDP** for FY27 (down from **4.4%** in RE FY26),
  - **Narrow Direct Tax Base:** Low tax-to-GDP elasticity and sub-optimal personal income tax compliance, forcing heavy reliance on debt to finance public expenditures.

    As per **PRS India (2026)**, India's gross tax-to-GDP ratio structurally hovers around a rigid **11.5% to 12%**, meaning a relatively small share of the economy is formally taxed
  - **Subnational Revenue Stagnation:** Rigid local taxation streams at the state level, creating a structural fiscal gap that must be bridged through market borrowings.

    According to the **Reserve Bank of India (RBI) State Finances Report**, States' Own Tax Revenue (SOTR) is stagnant at around **6.5% to 7.5% of GSDP**.
- **Expenditure Pressures & Welfare Overheads**
  - **Capital Outlay Drive:** A multi-year structural shift toward public-led growth, requiring massive capital investments in national corridors, railways, and public digital infrastructure.

    As per the **Union Budget 2026-27**, the central government has scaled up its public capital expenditure to a historic high of **₹12.2 lakh crore** (an 11.5% year-on-year increase)

- **Compounding Interest Servicing Trap:** A fiscal legacy where previous debts command massive interest obligations, consuming nearly 40% of the Centre's total revenue receipts.

  According to **PRS India's Budget Analysis (2026-27)**, interest payments have risen to **₹14.04 lakh crore**, consuming **26% of total government expenditure** and **40% of all revenue receipts**

- **Escalating Committed Expenditures:** Growing, inflexible budgetary outflows toward government salaries and public sector pensions that systematically crowd out developmental revenue.
- **Untargeted and Non-Merit Subsidies:** Heavy financial bleeding caused by open-ended agricultural, fertilizer, and power subsidies that carry very low structural fiscal multipliers (~0.9).

  According to **Union Budget Data (2026-27)**, major central subsidies (food, fertilizer, fuel) demand **₹4.55 lakh crore**.

- **Institutional & Systemic Disruption Factors**
  - **Pandemic Debt Legacy:** A sharp, permanent structural baseline shift in the public debt-to-GDP ratio due to emergency health and social safety-net spending executed alongside severe revenue contractions during the COVID-19 shock.
  - **Off-Budget Cash Shortfalls:** The forced regular onboarding of hidden liabilities—previously raised covertly by state power distribution companies (DISCOMs) or national bodies—into the formal legislative debt ledger.
  - **Susceptibility to Macroeconomic Shocks:** Sudden, geopolitical or climate-induced price shocks that trigger emergency fiscal spending on resource stabilization funds and infrastructure rebuilding.

Public Debt Management Cell (PDMC)

- **Core Mandate and Institutional Setup**
  - **Interim Statutory Nature:** Established in 2016 as an interim cell housed within the Department of Economic Affairs (DEA), Ministry of Finance, to transition India toward an independent debt management structure.
  - **Separation of Policy Powers:** Created to systematically decouple the government's debt management functions from the RBI's monetary policy operations, resolving inherent conflicts of interest in inflation handling.
  - **Path to PDMA:** Functions as the operational precursor to the proposed statutory Public Debt Management Agency (PDMA), which will eventually centralize all sovereign debt operations under one autonomous roof.
- **Strategic Operational Functions**
  - **Formulating the MTDS:** Mandated with drafting and executing the Medium-Term Debt Management Strategy (MTDS) for the central government, focusing on low-cost and low-risk borrowing profiles.
  - **Issuance Calendar Coordination:** Collaborates directly with the RBI to prepare the annual and half-yearly issuance calendars for Government Securities (G-Secs) and Treasury Bills.
  - **Cash Management Advisory:** Assists the government in cash forecasting and optimal timing of market borrowings to avoid structural cash idling or severe liquidity crunches.

Way forward

- **Consolidation & Growth Anchors**
  - **Target the 50% Debt Ceiling:** Enforce structural glide paths to lower the central debt-to-GDP ratio to 50±1% by 2030-31.
  - **Tighten State Borrowing Links:** Leverage 16th Finance Commission targets to enforce strict subnational debt stabilization caps (~28% of GSDP).

- **Revenue Mobilization & Asset Recycling**
  - **Broaden the Tax Net:** Deploy AI data analytics to expand tax buoyancy, reducing structural dependence on fresh market borrowings.
  - **Monetize Idle Infrastructure:** Scale up asset recycling and land-bank monetization to generate non-debt capital pools to retire legacy debts.
  - **Rationalize Utility User Charges:** Phase out un-targeted subsidies and update pricing for public utilities to boost non-tax revenues.
- **Institutional & Quality Reform**
  - **Enforce the Golden Rule:** Mandate that all fresh market borrowings are deployed strictly for capital creation, never for revenue expenditure.
  - **Fast-track the Independent PDMA:** Set up the Public Debt Management Agency to completely decouple sovereign debt operations from RBI's monetary policy.
  - **Cap Contingent Liabilities:** Impose statutory limits on state-issued guarantees and off-budget liabilities to prevent hidden subnational fiscal stress.

Conclusion

A prudent public debt strategy must balance growth-inducing capital creation with strict adherence to the 51% debt-to-GDP medium-term anchor by 2030-31. By sustaining a positive growth-interest differential, operationalizing an independent Public Debt Management Agency (PDMA), and checking subnational off-budget liabilities, India can safeguard its macroeconomic stability.

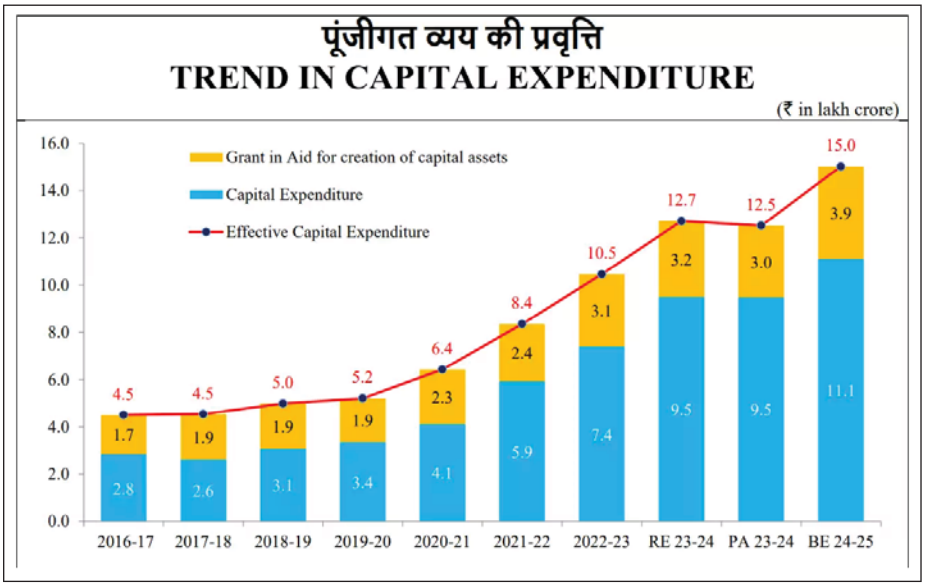
15. CAPITAL EXPENDITURE

What is Capital Expenditure?

- The **Union Budget** defines **capital expenditure** (CapEx) as the funds allocated and utilised by the government to **create assets or reduce liabilities** that contribute to a country's economic growth.
- This **includes long-term investments** in infrastructure, machinery, healthcare, education, and other essential sectors.
- It **covers expenses** to acquire fixed assets, upgrade or repair existing assets, repaying loans, and other government investments that yield future profits or dividends.
- **Repayment of loans** is also treated as **capital payment** since it reduces financial liabilities.
- Capital Expenditure v/s Revenue Expenditure:

Budget (Government) Expenditure

Revenue Expenditure	Capital Expenditure
(Components)	*(Components)*
(Expenditure which neither creates assets nor reduces liabilities)	*(Expenditure which creates assets or reduces liabilities)*
(i) Payment of interest	(i) Construction of roads, bridges, buildings
(ii) Payment of salaries and pensions	(ii) Purchase of land and machinery
(iii) Grants and subsidies	(iii) Investment in shares
(iv) Education and health services	(iv) Loans to states and foreign government
(v) Defence services	(iv) Repayment of loans



Importance of Capital Expenditure

- **Drives Economic Growth:** Investment in infrastructure (roads, power, transport) enhances productive capacity and efficiency, resulting in higher GDP growth rates. It generates a high multiplier effect, often with each ₹1 spent generating ₹2 or more in economic output.
- **Employment Generation:** Large capital projects (in construction, energy, logistics) create direct and indirect jobs, thus reducing unemployment and boosting incomes.
- **Crowds-in Private Investment:** Government capital spending signals confidence and can attract private investment by improving ease of doing business and lowering logistics costs. Better public infrastructure enables industries to flourish and deepen regional integration.
- **Enhances Productivity:** Investments in new technology and machinery increase the efficiency and productivity of both the public and private sectors.
- **Fiscal Health:** A government's ability to increase its capital expenditure while keeping its fiscal deficit in check is a strong sign of fiscal discipline and good financial management. It shows that the government is investing in future growth rather than just covering current expenses.
- **Creates Long-Term Assets:** Capital expenditure leads to the creation of assets that can generate returns and benefits for decades. These assets can also be monetized in the future to generate additional revenue.
- **Social & Human Capital Development:** Capital outlays on health, education, water, and sanitation create lasting societal benefits by improving human capital and living standards. Such investments are crucial for inclusive and sustainable development.

Various Government Initiatives to Boost the Capital Expenditure

- **Large Capital Outlays in Union Budgets:** The Union Budget 2025-26 proposes ₹11.2 lakh crore in capital expenditure (approx. 3.4% of GDP), with a continued focus on transport, railways, rural development, and urban infrastructure.
- **Scheme for Special Assistance to States for Capital Investment (SASCI):**
  - Launched in 2020-21, SASCI provides 50-year, interest-free loans to states exclusively for capital projects.
  - The FY26 allocation: ₹1.5 lakh crore, with cumulative support since FY21 crossing ₹5 lakh crore.
  - Loans are linked to reforms in municipal and financial management, urban/rural infrastructure, and digital public infrastructure for agriculture.

- **National Infrastructure Pipeline (NIP):** NIP envisages investments of ₹111 lakh crore during 2020-2025, with a coordinated pipeline of energy, transport, water, and digital projects, involving both government and private sector.
- **PM Gati Shakti National Master Plan:** Launched in 2021, it integrates multi-modal infrastructure planning and execution across ministries (rail, road, ports, airports), aiming for seamless logistics and higher capital efficiency.
- Urban Challenge Fund and Smart Cities Mission:
  - In FY26, a ₹1 lakh crore Urban Challenge Fund will finance urban redevelopment, creative city renewal, and water/sanitation projects, leveraging additional funds through bonds, bank loans, and PPPs.
  - The Smart Cities Mission continues to upgrade city infrastructure, digital services, and connectivity.

Challenges related to Capital Expenditure

- **Project Delays and Institutional Inefficiency:** Poor project preparation, delayed land acquisition, cumbersome approvals, and inefficient procurement lead to large time and cost overruns.
- **Inadequate Coordination:** Weak coordination among central, state, and local agencies hinders seamless implementation, especially in large infrastructure and social sector projects.
- **Limited Capacity and Skill Gaps:** Shortage of skilled workers and technical expertise, especially with complex, high-tech projects in new sectors.
- **Data Gaps and Misclassification:** Capex reporting often suffers from inconsistent definitions, inclusion of debt repayments and advances (not true asset creation), and lack of detail on project outcomes.
- **Underutilization of Funds:** Bureaucratic delays and late project clearances often mean allocated funds remain unspent by financial year-end.
- **Weak Private Investment Response:** Despite government push, the expected "crowding in" of private investment is often weak, especially outside a few large corporate groups.
- **Regulatory and Market Uncertainty:** Policy instability, global supply chain disruptions, and risk aversion due to past insolvency/bankruptcy cases deter greenfield investments.

Way Forward

- **Improve Project Planning and Execution:**
  - Strengthen pre-project feasibility studies, land acquisition processes, and timely environmental clearances to reduce delays and cost overruns.
  - Promote use of digital project management tools for real-time monitoring and accountability across ministries and states.
- **Enhance Institutional Capacity and Skills:**
  - Invest in training and capacity-building of officials handling infrastructure and social sector projects.
  - Foster public–private partnerships with best practices transfer, especially in high-tech and complex project areas.
- **Boost Transparency and Data Quality:**
  - Standardize capex reporting across government agencies and improve public disclosure on project status, expenditure, and outcomes.
  - Strengthen independent monitoring and promote citizen oversight for large infrastructure projects.
- **Ensure Timely Fund Utilization and Release:**
  - Rationalize approval and fund release systems to avoid year-end rush and underutilization.
  - Track fund disbursement and project completion via digital dashboards.
- **Focus on High-Impact Sectors and Regional Balance:**

- Prioritize capex in health, education, green energy, and logistics where returns are highest for growth and equity.
- Address regional disparities by ensuring more equitable capex deployment across states and rural-urban areas.

- **Finance Commission's Recommendations:**
  - The 15th Finance Commission strongly advocated for increased capital outlay in sectors like health, education, and defence.
  - Recommended that states increase health spending to more than 8% of their budgets, and established a dedicated **"Modernisation Fund for Defence and Internal Security"** for capital investments in these areas.
  - Extra annual borrowing up to 0.5% of GSDP permitted for states undertaking power sector reforms, incentivizing capital investment.

Conclusion

Capital expenditure is vital for sustainable growth, productivity, and employment, and is a central pillar of India's developmental fiscal strategy.

16. GENDER BUDGETING

Gender budgeting in India serves as a powerful fiscal strategy that translates gender commitments into budgetary allocations, ensuring that public resources are channeled equitably to bridge socio-economic gender gaps. As a critical component of inclusive development, it goes beyond creating a separate budget for women to systematically assess and restructure the differential impact of all government spending and revenue collection.

Highlights of the Gender Budget Statement (GBS) 2025-26

- **Macro Allocation & Fiscal Share**
  - **Record Quantum Expansion:** GBS 2025-26 allocated a historic ₹4.49 lakh crore for the welfare of women and girls, registering a massive 37.25% increase over the previous fiscal's allocation (₹3.27 lakh crore).
  - **Enhanced Budgetary Share:** As per the Ministry of Women and Child Development (MWCD), the share of the Gender Budget within the total Union Budget scaled up significantly to 8.86% in FY 2025-26 from 6.8% in FY 2024-25.
- **Structural Three-Part Classification**
  - **Part A (100% Women-Specific Schemes):** According to Union Budget Documents, Part A accounts for ₹1,05,535.40 crore (23.50% of the total GBS), covering exclusive women-centric schemes like Mission Shakti.
  - **Part B (30% to 99% Earmarked for Women):** Part B commands the share at ₹3,26,672.00 crore (72.75% of total allocation), driving large-scale developmental interventions with major female sub-components.
  - **Part C (Below 30% Indirect Women Allocations):** According to MWCD Guidelines, Part C accounts for ₹16,821.28 crore (3.75% of the total GBS), capturing institutional spending with a smaller, yet critical, indirect impact on women.

Significance of Gender Budgeting in India

- **Macroeconomic & Human Capital Growth**

- **Unlocking Demographic Dividend:** Acts as a fiscal engine to boost India's low Female Labor Force Participation Rate (FLFPR), which can add up to 1.4% to annual GDP growth.
- **Enhancing Human Capital Elasticity:** Channels targeted resource allocations into women's health and technical education, creating long-term, cross-generational multi-sectoral improvements.
- **Formalizing the Care Economy:** Directs public infrastructure spending toward daycares and safe public transit, reducing the unpaid domestic care burden that limits women's economic freedom.

- **Socio-Economic Equity & Empowerment**
  - **Correcting Historical Asset Imbalances:** Drives economic empowerment by prioritizing female ownership in welfare schemes, such as mandating registration under the **PM Awas Yojana**.
  - **Fostering Financial Inclusion:** Accelerates economic self-reliance by funding financial pipelines, notably the **Lakhpati Didi** initiative and interest subventions for women-led Self-Help Groups (SHGs).
  - **Plugging Grassroots Development Gaps:** Helps resource-poor rural women access essential sanitation and clean cooking energy through target programs like **Swachh Bharat** and **PM Ujjwala Yojana**.
- **Institutional & Policy Engineering**
  - **Translating Rhetoric into Fiscal Action:** Converts abstract, high-level constitutional and political gender-equality goals into concrete, legally binding budgetary outlays.
  - **Enforcing Inter-Ministerial Accountability:** Compels non-traditional departments (e.g., Electronics, Civil Aviation) to view policy through a gender lens to qualify for public funds.
  - **Fulfilling Global Commitments:** Serves as a vital domestic fiscal mechanism to help India meet its international obligations under **UN Sustainable Development Goal 5 (Gender Equality)**.

Challenges Faced by Gender Budgeting in India

- **Methodological & Reporting Anomalies**
  - **Input-Centric Bias over Outcomes:** Focuses overwhelmingly on financial tracking and output compliance rather than evaluating measurable on-ground qualitative changes in women's socio-economic indicators.
  - **Subjective Over-Reporting Inconsistencies:** Prominent schemes regularly list flat 40% allocations for women without providing precise statistical empirical justifications.
  - **Under-Reporting of Core Components:** Major programs like MGNREGS record high female labor workdays (~59%) but reflect low, diluted outlays in Part B of the Gender Budget Statement (GBS).
  - **Part C Tracking Inefficiencies:** Recent classification system introduces complex sub-30% indirect spending without clarifying a standardized, automated methodology for reporting ministries.

    According to **Ministry of Women and Child Development (MWCD) Specifications**, the newly integrated Part C (capturing sub-30% indirect outlays) commands **5.95%** of the GBS in FY27

- **Structural & Institutional Bottlenecks**
  - **Skewed Inter-Ministerial Concentration:** Nearly 90% of the aggregate gender budget is rigidly concentrated across just five core welfare ministries.
  - **Under-Equipped Gender Budgeting Cells (GBCs):** Internal ministerial GBCs routinely act as mere administrative checklist desks due to a severe deficit in technical gender-responsive analysis skills.
  - **Absence of Institutionalized Gender Audits:** Lacks a mandatory annual statutory auditing framework overseen by the Comptroller and Auditor General (CAG) to track actual fiscal leaks and deviations.
  - **Weak Decentralization at Grassroots:** Local panchayats and urban municipal bodies—who are closest to real operational needs—lack the data-handling capacity and financial autonomy to deploy localized gender metrics.

- **Data & Socio-Cultural Deficits**
  - **Severe Sex-Disaggregated Data Gaps:** Real-time, localized gender-segregated data collection systems remain absent across large multi-sectoral programs like Ayushman Bharat, blurring beneficiary visibility.
  - **Socio-Cultural Implementation Drag:** Deep-rooted patriarchy, safety issues, and household care burdens frequently block target beneficiaries from physically accessing or utilizing well-funded allocations.

Way forward

- **Methodological & Institutional Overhauls**
  - **Transition to Outcome Budgeting:** Shift the fiscal focus from mere financial allocations (inputs) to quantifiable on-ground progress via mandatory **Gender Outcome Indicators** for all participating ministries.
  - **Institutionalize Standalone Gender Audits:** Empower the **Comptroller and Auditor General (CAG)** to conduct regular, independent statutory gender audits to plug implementation leaks and ensure fiscal accountability.
  - **Strengthen Gender Budgeting Cells (GBCs):** Revitalize internal ministerial GBCs by providing continuous, expert-led capacity-building training in gender-responsive planning and economic analysis.
  - **Standardize Part C Accounting:** Develop an automated, algorithmic methodology for the newly added Part C to ensure uniform and realistic reporting of indirect, low-intensity gender spending.
- **Sectoral & Structural Expansion**
  - **Diversify Sectoral Outlays:** Democratize the Gender Budget by expanding resource allocations to non-traditional ministries like **Civil Aviation, Renewable Energy, and Electronics & IT** to foster non-farm female employment.
  - **Embed Grassroots Gender Budgeting:** Decentralize the framework by training and financially empowering Panchayati Raj Institutions (PRIs) and Urban Local Bodies (ULBs) to deploy localized, need-based gender plans.
  - **Mandate Sex-Disaggregated Data (SDD):** Build a centralized, digital real-time data repository requiring all welfare schemes (e.g., *Ayushman Bharat*) to log beneficiaries by gender.
- **Strategic Policy Alignments**
  - **Address the Care Economy Balance:** Use the Gender Budget to heavily finance child-care centers, smart public toilets, and safe public transit infrastructure to directly lower the domestic burden on women.
  - **Harmonize via the Knowledge Hub:** Leverage the newly launched **Gender Budgeting Knowledge Hub** to continuously cross-reference national fiscal outlays with global best practices and academic research.

Global Best Practices for Gender Budgeting

- **Statutory and Constitutional Backing (Austria):** Embedded gender equality directly into its Federal Constitution, making Gender-Responsive Budgeting (GRB) a non-negotiable statutory mandate across all federal organs.
- **Insulation from Political Cycles (Spain):** Enacted dedicated regional and national legislative frameworks to ensure gender budgeting parameters remain structurally insulated from shifting political transitions.
- **Centralized Treasury Leadership (Iceland):** Integrated the GRB framework entirely within the overarching Ministry of Finance operations, preventing it from becoming a marginalized **welfare-only** exercise.

- **Rigorous Ex-Ante and Ex-Post Impact Assessments (Sweden):** Mandates that all fresh budgetary bills pass through a rigorous Ex-Ante (upfront) and Ex-Post (retrospective) gender impact test before legislative clearance.
- **Pioneer Women's Budget Statement (Australia):** Established the world's first formal Women's Budget Statement, creating a globally replicated structural model to cross-examine general fiscal policy impacts on women.
- **Independent External Quality Assurances (Ireland):** Deployed a robust system of continuous **Equality Budgeting** monitored via external civil society peer reviews and parliamentary oversight committees.

Conclusion

India's gender budgeting framework must transition from an outlay-driven compliance exercise to an outcome-oriented macroeconomic strategy institutionalized via mandatory CAG gender audits and intersectional data tracking. By systematically embedding a gender-responsive lens into capital-intensive, non-traditional sectors, fiscal policy can effectively dismantle deep-rooted systemic barriers and accelerate female labor force participation.

17. GREEN BUDGETING

Green budgeting is an innovative public finance tool that integrates environmental and climate goals into a government's fiscal framework and national budget process. It involves systematically identifying, measuring, and evaluating standard budget allocations to ensure they align with sustainable development targets and climate commitments.

As per OECD, Green budgeting refers to the use of budgetary policy-making tools to better understand environmental impact of spending choices, and ensure public budgets are aligned with climate and environmental objectives.

Significance of Green budgeting

- **Achieves Panchamrit & Net-Zero Targets:** Directly aligns government spending with India's goal to reach net-zero carbon emissions by 2070.
- **Mobilizes Climate Finance:** Attracts international green funds and climate investments by bringing transparency to public environmental spending.
- **Reduces Environmental Risks:** Mitigates economic losses from climate hazards like floods, droughts, and heatwaves through proactive funding.
- **Accelerates Clean Energy Transition:** Prioritizes budget allocations for renewable energy, electric mobility, and green hydrogen projects.
- **Protects Vulnerable Communities:** Ensures funding reaches rural and marginalized populations who suffer the most from climate impact.
- **Prevents Environmentally Harmful Subsidies:** Helps identify and gradually phase out fiscal support for polluting industries and fossil fuels.
- **Promotes Sustainable Agriculture:** Drives public investments toward natural farming, soil health conservation, and efficient water management.
- **Strengthens Policy Accountability:** Tracks whether government ministries are actually meeting their green spending promises each year.

Tools for green budgeting

Tool	Mechanism
Green Budget Tagging (GBT)	Classifies and codes budget line items as green (favorable), neutral, or brown (harmful) before allocation.
Environmental Impact Assessment (EIA)	Evaluates potential ecological costs and benefits of new major public infrastructure projects.
Climate Cost-Benefit Analysis (CBA)	Integrates the shadow price of carbon and ecosystem valuation into public investment planning.
Environmental Fiscal Reform (EFR)	Imposes carbon/pollution taxes on brown activities while offering tax incentives for clean technology.
Subsidy Rationalization	Identifies, reduces, and redirects fossil fuel and chemical subsidies toward green alternatives.
Sovereign Green Bonds (SGrBs)	Issues dedicated government debt instruments where capital is strictly earmarked for green projects.
Green Outcome Budgeting	Links financial spending directly to measurable environmental Key Performance Indicators (KPIs).
Green Expenditure Reviews (GER)	Conducts periodic baseline audits of existing sectoral spending to eliminate eco-harmful outlays.
CAG Environmental Audits	Mandates independent post-implementation audits to verify ground-level ecological outcomes.
Natural Capital Accounting (SEEA)	Measures national natural resource stocks and ecological depletion alongside standard GDP.

Issues with green budgeting for India

- **Institutional & Structural Framework Issues**
  - **Absence of Central Union-Level Statement:** India lacks a consolidated, mandatory national Green Budget Statement (similar to Statement 13 for Gender Budgeting).
  - **Institutional Silos:** Poor inter-ministerial coordination leads non-environmental line departments (like Transport, Steel, Agriculture) to treat climate goals as secondary priorities.

    Over **88% of total climate mitigation allocations** remain concentrated within just two ministries—Ministry of New and Renewable Energy (MNRE) and Ministry of Power.
  - **Lack of Standardized Methodology:** Absence of uniform, objective metrics across states and union departments to classify, tag, and measure green vs. non-green spending.

    Uttar Pradesh tags **26% of scheme outlays** as green expenditure while Bihar tags **~4.5% to 7.5%**, reflecting mismatched classification metrics rather than actual spending parity.
- **Operational & Execution Bottlenecks**
  - **Risk of Greenwashing:** Standard tagging exercises often become superficial administrative paperwork without forcing structural spending shifts.
  - **Capacity Constraints:** Severe lack of technical expertise, data systems, and trained personnel within line departments to conduct environmental audits.
  - **Underutilization of Allocated Funds:** Earmarked green funds (e.g., Clean Energy Fund, CAMPA, Green India Mission) often suffer from chronic under-spending or administrative diversion.

CAMPA: Out of **₹51,300 crore** transferred to States between 2019–20 and 2021–22, States utilized only **₹13,690 crore (27%)**, leaving over **₹37,000 crore unspent.**

- **Policy & Fiscal Contradictions**
  - **Contradictory Fiscal Signals:** Government simultaneously funds clean energy expansion while continuing heavy subsidies for fossil fuels, coal, and chemical fertilizers.
  - **Mitigation Bias over Adaptation:** Budgetary allocations heavily favor large-scale renewable energy (mitigation) while neglected sectors like climate-resilient agriculture and disaster preparedness (adaptation) face funding gaps.

    More than **85% of tracked climate public finance** in India flows into climate mitigation (renewable energy, EVs).
  - **Fiscal Stress at the State Level:** Most states lack independent financial bandwidth and remain heavily reliant on central grants to implement state climate action plans.
- **Data & Accountability Gaps**
  - **Absence of Outcome-Based Auditing:** Tracking focuses mostly on initial financial outlays rather than verifying actual post-implementation environmental impact.
  - **Missing Indicators in Performance Frameworks:** Core ecological metrics—such as biodiversity loss, soil carbon, and coastal resilience—are frequently absent from standard outcome-budgeting frameworks.

Way forward

- **Institutional & Structural Reforms**
  - **Mandate a Union Green Budget Statement:** Introduce a dedicated Green Budget Statement (similar to Statement 13 for Gender Budgeting) in the annual Union Budget.
  - **Formulate Standardized Tagging Guidelines:** Develop clear, uniform criteria to categorize expenditures as green, climate-neutral, or brown across all government departments.
  - **Establish Dedicated Green Budget Cells (GBCs):** Set up GBCs in every ministry to review spending through an environmental lens and ensure cross-sectoral alignment.
- **Operational & Capacity Enhancement**
  - **Build Technical Expertise:** Train administrative officers, finance teams, and line department staff in climate risk assessment and environmental auditing.
  - **Shift to Outcome-Based Auditing:** Evaluate programs based on measurable post-spending environmental outcomes (like carbon reduced or soil health restored), not just money spent.
  - **Involve CAG in Environmental Auditing:** Empower the Comptroller and Auditor General to regularly conduct independent green audits of public programs.
- **Fiscal & Policy Realignment**
  - **Rationalize Harmful Subsidies:** Gradually phase out subsidies for polluting fuels and chemical fertilizers, redirecting funds toward renewables and natural farming.
  - **Prioritize Climate Adaptation:** Rebalance budgetary focus to increase allocations for climate-resilient agriculture, coastal protection, and flood defense mechanisms.
  - **Strengthen State-Level Financial Capacity:** Increase central grants and incentive-based funding for states that actively publish and execute green budgets.
- **Financial Mobilization & Data Integration**
  - **Leverage Sovereign Green Bonds:** Expand the pool of green spending by blending budget outlays with proceeds from Sovereign Green Bonds and private ESG investments.

- **Integrate Real-Time Digital Tracking:** Utilize data tools to monitor green spending line items in real time, preventing superficial greenwashing and project delays.

Conclusion

Green budgeting is a vital public finance tool that transforms climate commitments into tangible fiscal action. By aligning government revenues and expenditures with ecological goals, it ensures every rupee spent drives sustainable, low-carbon growth.

18. OUTCOME BUDGETING

Outcome budgeting in India acts as a transformative fiscal management tool that shifts the focus of public financial administration from traditional financial inputs and outlays to tangible, measurable physical outcomes. Outcome budgeting was first introduced by the Union Government in 2005, it serves as a performance-linked contract between the state and implementing agencies, establishing strict accountability for every rupee of public money spent.

Framework of Outcome Budgeting in India

- **Institutional Architecture**
  - **DMEO Engine:** Anchored by the Development Monitoring and Evaluation Office (DMEO) under NITI Aayog, which drives independent, data-backed performance tracking across ministries.
  - **OOMF Blueprint:** Operates via the standardized Output-Outcome Monitoring Framework (OOMF), mapping every rupee of financial outlay directly to quantifiable progress metrics.
  - **Finance Ministry role:** Jointly steered by the Department of Expenditure (Ministry of Finance) to match subsequent quarterly cash rollouts with proven physical performance.
- **Major Conceptual Elements**
  - **Output vs. Outcome Bifurcation:** Differentiates clearly between Outputs (immediate, physical tangibles; e.g., kilometers of highway laid) and Outcomes (long-term, qualitative impacts; e.g., travel time reduction).
  - **Key Performance Indicators (KPIs):** Employs explicit, scheme-specific baseline and target KPIs, leaving zero room for subjective administrative assessments.
- **Operational Guidelines & Scale**
  - **Universal CS and CSS Coverage**: Enforces mandatory OOMF tracking for all Central Sector (CS) and Centrally Sponsored Schemes (CSS), ensuring uniform accountability.
  - **₹500 Crore Parliamentary Mandate:** Mandates that a unified Outcome Budget statement for all major schemes with outlays ₹500 crore be tabled directly in Parliament alongside the Union Budget.
  - **Decentralized Ministry Ownership:** Requires line ministries to track smaller programs (outlays less than ₹500 crore) via their internal Detailed Demands for Grants under the same OOMF parameters.

Significance of Outcome Budgeting in India

- **Enhancing Fiscal Discipline & Governance**
  - **Shifting Administrative Culture:** Transition public spending behavior from a rigid **outlay-spending chase** to a progressive, result-oriented **value-for-money** delivery model.
  - **Combating the March Rush:** Eliminates the unscientific, hurried year-end utilization of public funds by tracking physical milestone completions on a quarterly basis.
  - **Plugging Fiscal Leaks: I**ncreases grassroot transparency by reducing systemic room for rent-seeking, target-shifting, and administrative execution delays.

  - **Evidence-Based Policy Feedback**: Feeds real-time performance indicators directly back into subsequent Union Budget cycles to systematically prune out redundant or underperforming public schemes.
- **Improving Developmental & Social Outcomes**
  - **Securing Qualitative Impact:** Forces a clear focus on the ultimate societal impact (e.g., lower infant mortality) rather than just immediate, physical milestones (e.g., hospital buildings constructed).
  - **Optimizing Flagship Infrastructure:** Maximizes execution efficiency across capital-heavy national programs like PM Gati Shakti and Jal Jeevan Mission through rigorous, data-backed timeline checks.
  - **Empowering Parliamentary Oversight:** Equips the Parliament and standing committees with objective, quantifiable data to scrutinize the Executive's performance during Demands for Grants.
  - **Dismantling Ministerial Silos:** Compels multi-sectoral ministries to align and cooperate on overarching national outcomes instead of working within isolated vertical operational channels.

Issues with Outcome Budgeting in India

- **Methodological & Measurement Flaws**
  - **Output-Centric Data Inertia:** Focuses overwhelmingly on easily measurable outputs (e.g., number of skill-training centers opened) rather than tracking true outcomes (e.g., long-term sustainable formal employment rates).
  - **Attribution Challenge Paradox:** Suffers from a causality dilemma, where it is structurally difficult to isolate whether a socio-economic improvement is a direct result of a specific central scheme or external macroeconomic factors.
  - **Subjective KPI Selection:** Suffers from line ministries deliberately designing soft, easily achievable Key Performance Indicators (KPIs) to camouflage systemic implementation deficiencies.
  - **Absence of Uniform Definitions:** Lacks a strict, uniform conceptual taxonomy across different ministries, leading to overlapping, inconsistent definitions of what constitutes a **final outcome.**
- **Structural & Institutional Bottlenecks**
  - **Capacity Deficit Dilemma:** Internal ministerial units and department desks lack specialized data-analytics training and econometric expertise.

    Human resource audits reveal that **less than 15% of administrative staff** positioned inside department-level internal statistical cells possess formal, specialized certifications or training.
  - **Asymmetric Subnational Devolution:** Centrally Sponsored Schemes (CSS) face execution bottlenecks because final-mile state departments and Panchayati Raj Institutions (PRIs) lack the financial autonomy and technical infrastructure to log local outcome indicators.

    According to **NITI Aayog's Voluntary National Review (VNR)** indices, over **60% of Gram Panchayats** and Urban Local Bodies lacking the necessary ICT infrastructure or technical manpower
  - **Weak Post-Spending Audit Linkages:** framework remains disconnected from a mandatory, retrospective statutory evaluation by the Comptroller and Auditor General (CAG), reducing its legal enforcement weight.
  - **Fragmented Real-Time Tracking:** Lacks a unified, inter-ministerial digital dashboard, causing critical data lags that prevent immediate course correction during active fiscal quarters.

    Development Monitoring and Evaluation Office (DMEO) highlight Centrally Sponsored Schemes frequently suffer from lags, creating a data-reporting delay of **2 to 3 financial quarters** that prevents real-time course correction.
- **Operational & Behavioral Distortions**

- **Spend-Speed vs. Result Friction:** Administrative survival is still heavily evaluated on year-end financial spending speed rather than physical outcome qualities.

  20-30% of total annual allocations are rushed out the door within the final fiscal quarter, prioritizing immediate output velocity over outcome quality.

- **Neglect of Long-Gestation Capital:** Disincentivizes ministries from proposing vital, complex infrastructure projects with long-term gestation horizons.

Way Forward for Outcome Budgeting in India

- **Methodological Refinements & Data Systems**
  - **Shift toward High-Quality Outcome Indicators:** Shifting Swachh Bharat Mission (Grameen) metrics away from just tallying Individual Household Latrines (IHHLs) constructed to verifying long-term reduction in infant diarrhea and childhood stunting cases
  - **Integrate Real-Time Digital Dashboards:** Build a centralized, public platform that syncs line-ministry accounts with the **Public Financial Management System (PFMS)** to allow real-time tracking of physical milestones.

    PM-KISAN portal with the **Public Financial Management System (PFMS)** to directly link transaction-level DBT outlays with real-time land record verifications

  - **Overcome the Attribution Challenge:** Deploy robust econometric modeling and independent **Control Group Evaluations** to accurately isolate the direct impact of individual flagship schemes from general macro market changes.
  - **Standardize KPI Taxonomies:** Develop an objective, uniform set of performance indicator definitions across all ministries to eliminate self-serving or easily achievable target-setting by individual departments.
- **Institutional & Structural Strengthening**
  - **Mandate Retroactive CAG Outcome Audits:** Formally empower the **Comptroller and Auditor General (CAG)** to conduct annual, statutory post-spending outcome audits.
  - **Establish Dedicated Data-Analytics Units:** Build technical capacity across line ministries by embedding specialized economists and data scientists into internal budgeting desks.
  - **Onboard Long-Gestation Frameworks:** Shift the OOMF layout from a rigid annual timeline to rolling multi-year tracking structures, protecting long-term capital investments from being downsized.
- **Subnational Devolution & Global Alignments**
  - **Role of Local Bodies:** Incentivize and financially support State Governments, Panchayati Raj Institutions (PRIs) and Urban Local Bodies (ULBs) to adopt localized outcome budgeting practices.

    Kerala's People's Plan Campaign to financially reward Gram Panchayats that successfully present a localized, village-level Maternal and Child Nutrition Outcome Budget.

  - **Mapping with SDGs:** Explicitly link individual scheme-level OOMF target indicators with the **UN Sustainable Development Goals (SDG 2030)** to systematically monitor and evaluate India's progress.

Key Committee Recommendations on Outcome Budgeting

- **First Administrative Reforms Commission (1st ARC, 1966):** Recommended the initial introduction of a Performance Budget, shifting focus from line-item inputs to functional programs and physical achievements.
- **Second Administrative Reforms Commission (2nd ARC, 2009):** Advised a transition from **Performance Budgeting** to a strict **Outcome Budgeting** framework, demanding clear vertical linkages between financial inputs, physical outputs, and ultimate socio-economic outcomes.

- **B.K. Chaturvedi Committee (2011):** Recommended restructuring Centrally Sponsored Schemes (CSS) into fewer, umbrella categories to ease the monitoring of multi-state outcome indicators.
- **NITI Aayog Outcome Budgeting Architecture (2017–Present):** Established the modern framework requiring ministries to map progress on a standardized online dashboard, linking annual outlays directly to specific Sustainable Development Goals (SDGs).

Conclusion

Outcome budgeting is essential to shift India's fiscal architecture from mere input-centric utilization to high-multiplier asset creation and concrete socio-economic delivery. By embedding robust monitoring frameworks like the NITI Aayog dashboard, it enforces maximum accountability, minimizes administrative leakages, and ensures optimum value for public money.

MGP/BP/GS3/027

LAND REFORMS IN INDIA

1. LAND REFORMS IN INDIA

Land reforms refer to various reforms in regulations regarding land ownership and property rights for the benefit of the community as a whole. The main objective of the land reforms is to do away with the existing inequalities in the system of landholding and to increase agricultural productivity.

Evolution of land reforms in india

Pre-Independence Period (Before 1947): Colonial Land Systems	• **Permanent Settlement (1793)**: Introduced by Lord Cornwallis in Bengal, Bihar, and Odisha, it created Zamindars (landlords) who collected revenue from peasants but led to tenant exploitation. • **Ryotwari System (Madras & Bombay)**: Direct revenue settlement between British and cultivators (Ryots), but with high taxation. • **Mahalwari System (North India)**: Village-based revenue collection, yet exploitation remained.
First Phase: Abolition of Intermediaries (1947–1950s)	The government prioritized land reforms to address historical injustices. **Key Measures**: • **Abolition of Zamindari System**: States enacted laws to abolish Zamindari and other intermediary systems. **E.g**. Uttar Pradesh Zamindari Abolition and Land Reforms Act, 1950. • **First Five-Year Plan (1951–1956):** Emphasized the need for land reforms to achieve social justice and equitable distribution
Second Phase: Tenancy Reforms (1950s–1960s)	Aimed to protect tenants and provide them with ownership rights. **Key Measures**: • **Regulation of Rent**: Laws were enacted to regulate rent and prevent exploitation. • **Security of Tenure**: Tenants were given permanent rights over the land they cultivated. **E.g.** Operation Barga in West Bengal (1978) registered sharecroppers and gave them legal rights. • **Second Five-Year Plan (1956–1961):** Focused on tenancy reforms and redistribution of surplus land.
Third Phase: Land Ceiling Laws (1960s–1970s)	Introduced to limit land ownership and redistribute surplus land to the landless. **Key Measures**: • **Land Ceiling Acts**: States imposed limits on land ownership. **E.g.** Kerala Land Reforms Act, 1963 successfully redistributed land to tenants.

	• **Redistribution of Surplus Land**: Surplus land was taken from landlords and distributed to landless farmers. **E.g.** Over 2 million acres were redistributed nationwide by the 1970s. • **Third Five-Year Plan (1961–1966):** Highlighted the need for effective implementation of land ceiling laws.
Fourth Phase: Green Revolution and Shift in Focus (1960s–1980s)	The Green Revolution shifted focus from land redistribution to increasing agricultural productivity. **Key Measures:** • **Bhoodan and Gramdan Movements**: Led by Vinoba Bhave, these movements encouraged voluntary land donations. **E.g.** Over 4 million acres were donated, but the impact was limited. • **Consolidation of Land Holdings**: Aimed to consolidate fragmented land holdings for better productivity. **E.g.** Successful in states like Punjab and Haryana. • **Report of the Task Force on Agrarian Relations (1973):** Highlighted the need for effective implementation of land reforms.
Fifth Phase: Stagnation and Challenges (1980s–1990s)	Land reforms lost momentum due to political resistance and bureaucratic inefficiency. **Key Issues**: • Ineffective implementation of land ceiling laws. • Growing commercialization of agriculture marginalized small farmers. • **Fourth and Fifth Five-Year Plans**: Emphasized the need for land reforms but faced implementation challenges
Sixth Phase: Contemporary Developments (2000s–Present)	Recent land reforms focus on addressing new challenges like urbanization, industrialization, and land acquisition. **Key Measures**: • **Forest Rights Act, 2006**: Recognized the rights of tribal communities and forest dwellers over forest land. **E.g.** Implementation in states like Odisha and Chhattisgarh. • **Land Acquisition and Rehabilitation Acts**: Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013: Ensured fair compensation for displaced farmers. **E.g.**protests against land acquisition in Singur (West Bengal) and Nandigram. • **Digital Land Records**: Initiatives like the Digital India Land Records Modernization Programme (DILRMP) aimed to digitize land records. **E.g**. Bhoomi Project in Karnataka. • **National Commission on Farmers (2004–2006)**: Chaired by Dr. M.S. Swaminathan recommended land reforms to protect small and marginal farmers. • **Land Acquisition Act (2013)**: Replaced 1894 Act, ensuring fair compensation & rehabilitation. • **NITI Aayog's Land Leasing Model (2016)**: Encouraged lease markets for small farmers.

Significance of Land Reforms for India

- **Ensuring Food Security & Agricultural Productivity:** With India's population projected to reach ~1.6 billion by 2047, increasing food production requires land consolidation, digital land records, and sustainable farming. Land leasing reforms can encourage corporate farming, contract farming, and agri-tech investments, while precision farming and agroforestry can improve yields.
  - **For Ex- Punjab and Haryana's Green Revolution (1960s)** demonstrated how land reform-linked mechanization transformed productivity**.**
- **Reducing Rural Poverty & Strengthening Livelihoods:** Around 58% of rural households remain landless (SECC 2011), and land distribution can uplift millions from poverty. Strengthening tenancy rights, tribal land security, and women's land ownership ensures equitable growth. Secure land tenure improves access to credit, increasing investment in agriculture and MSMEs.
  - **For ex- West Bengal's Operation Barga (1978)** successfully enhanced sharecropper rights, boosting productivity**.**
- **Facilitating Industrialization & Infrastructure Growth:** With India aiming for a $30 trillion economy by 2047, streamlined land acquisition and digitization can unlock land for industrial corridors, logistics hubs, and SEZs. Transparent land policies will reduce disputes, attract FDI, and ease business expansion.
  - **For Ex- Gujarat's Industrial Development Corporation (GIDC)** efficiently allocated land to industries, driving manufacturing growth.
- **Accelerating Urbanization & Smart Cities Development:** By 2047, India's urban population will surpass 50%, necessitating planned urbanization, land pooling, and affordable housing projects. Mixed-use development, transit-oriented planning, and slum rehabilitation can ensure inclusive cities.
  - **For Ex-** The **Navi Mumbai Airport land pooling model** enabled equitable compensation and stakeholder participation, offering a replicable model for urban expansion.
- **Reducing Land Conflicts & Improving Legal Frameworks:** As per World Bank land disputes cost India $10 billion annually, delaying infrastructure projects and investment. Digitizing land records, streamlining land titling, and legal reforms can eliminate disputes, ensuring faster economic growth and efficient governance.
  - **For ex-** The **PM-SVAMITVA scheme (2020) is successfully mapping rural land digitally**, enhancing property rights and dispute resolution.
- **Ensuring Environmental Sustainability & Climate Resilience:** According to State of India's Environment 2023 with 30% of India's land degraded, reforms should focus on sustainable land use, afforestation, and climate-smart urban planning. Integrating reforestation, soil conservation, and carbon farming can restore degraded land.
  - **For Ex- Rajasthan's agroforestry projects** reclaimed over 10 lakh hectares, boosting soil health and livelihoods.

Challenges in realizing the full potential of Land Reforms

- **Land Fragmentation & Smallholding Crisis:** According to Agricultural Census 2015-16 with 86% of Indian farmers owning less than 2 hectares, excessive land fragmentation reduces productivity, increases operational costs, and limits mechanization. The average farm size has declined from 2.3 hectares in 1970 to 1.08 hectares in 2015, affecting economies of scale.
- **Outdated & Complex Land Laws:** Land governance is fragmented across over 300+ state and central laws, leading to conflicting regulations, legal disputes, and delays in land transactions. Inconsistent land acquisition policies create investor uncertainty, hampering infrastructure projects.
  - **For Ex-** The **failure of the 2015 Land Acquisition Bill** due to compensation and consent issues exemplifies legislative roadblocks.

- **Land Record Digitization & Titling Issues:** According to NCAER, 2023 report, despite efforts like DILRMP only 89% of rural land records are digitized, with over 60% of land holdings lacking clear titles. Poor land record maintenance fuels litigation, land grabbing, and corruption.
  - **For Ex-** The PM-SVAMITVA scheme (2020) has made progress, but full implementation remains slow.
- **Land Acquisition & Rehabilitation Bottlenecks:** India faces frequent protests over land acquisition, delaying critical projects like bullet trains, industrial corridors, and highways. Compensation disputes, inadequate resettlement, and opposition from farmers lead to conflicts.
  - **For Ex-** POSCO's Odisha project stalled due to resistance over displacement and rehabilitation concerns.
- **Urban Land Shortage & Unaffordable Housing:** Rapid urbanization is increasing slum proliferation, land encroachment, and housing unaffordability. India needs ~25 million affordable homes by 2047, but land hoarding, speculation, and unclear land-use policies restrict urban land availability.
  - **For Ex-** Mumbai's Dharavi Redevelopment Plan has struggled due to unclear property rights and relocation challenges.
- **Weak Women's Land Rights & Marginalized Communities' Exclusion:** According to Agricultural Census 2015-16 despite policy efforts, women hold only 14% of land in rural India, and tribal communities face displacement due to mining and industrial projects.
- **For Ex- Weak implementation of the Forest Rights Act (2006)** has led to repeated evictions of Adivasi populations. Jharkhand's Pathalgadi movement highlighted local resistance to land dispossession.

Way forward for Land Reforms in India

- **Digital Land Records & Transparent Governance:** Digitizing land records ensures transparency, reduces disputes, and minimizes corruption in land transactions.
  - **For Ex- Estonia's blockchain-based land registry** provides a global benchmark for secure and tamper-proof records.
- **Land Consolidation for Productivity:** Pooling fragmented agricultural lands enhances mechanization, irrigation efficiency, and overall farm output.
  - **For Ex- China's Collective Farming Model**, which transformed small landholdings into high-yield farms.
- **Inclusive Land Rights & Gender Equity:** Ensuring land ownership for marginalized groups, especially women, strengthens economic independence and social equity.
  - **For Ex- Odisha's 'Vasundhara Scheme'** has provided land rights to over 200,000 landless families, while **Rwanda's Land Tenure Regularization Program (LTRP)** increased women's land ownership to 86%.
- **Fair Land Acquisition & Rehabilitation:** A just and participatory land acquisition process ensures fair compensation and safeguards livelihoods while enabling infrastructure expansion.
  - **For Ex-** Amaravati Land Pooling Scheme (Andhra Pradesh) attempted a balanced approach, inspired by Germany's Land Pooling Model, which ensures equitable returns for landowners.
- **Community-Based Forest Rights & Tribal Empowerment:** Recognizing indigenous communities' land rights improves conservation and socio-economic development.
  - **For Ex-** Maharashtra's Gadchiroli tribal communities have successfully managed forest land under the Forest Rights Act (2006), similar to Mexico's Ejidos System, where collective land ownership ensures sustainable forest management.
- **Promoting Sustainable Land Use:** Integrating environmental considerations into land management policies and practices. Protecting natural resources, mitigates climate change, and ensures long-term agricultural productivity.

- **For Ex- Costa Rica's Payment for Ecosystem Services (PES) program**, incentivizing landowners to conserve forests, has contributed to reforestation and biodiversity protection.

2. LAND TITLING SYSTEM IN INDIA

The **land titling system** in India primarily operates on a presumptive model governed by the Registration Act of 1908, where the state registers only the transaction of a property sale deed rather than verifying absolute ownership, leaving titles perpetually prone to legal disputes. To fix this structural vulnerability, the government is driving a paradigm shift toward a **conclusive land titling system**, anchored by the Digital India Land Records Modernization Programme (DILRMP).

Current Status of Land Titling in India

- **Record of Rights (RoR) Saturation:** Reached near-total computerization with over 95.5% of village RoRs digitized and dynamically accessible online across the country.
- **Cadastral Map Modernization:** Successfully digitized more than 60% of all rural cadastral maps, linking spatial plot shapes directly to text-based ownership data (Record of Rights).
- **Massive Bhu-Aadhaar (ULPIN) Rollout:** Assigned a 14-digit geo-referenced Unique Land Parcel Identification Number to over 36 crore land parcels across 29 States and UTs.
- **Registration Office Integration:** Achieved over 95.7% computerization of Sub-Registrar Offices (SROs), with 88.5% of them completely integrated with central revenue land databases.
- **Urban Expansion (NAKSHA Pilot):** Extended the DILRMP canvas into complex urban habitations through the newly launched NAKSHA program, mapping 150 cities via high-resolution drone imagery.
- **Linguistic Barrier Elimination:** Implemented automated C-DAC transliteration systems to convert local language land records into any of the 22 Scheduled VIII languages for universal accessibility.
- **Residual Legislative Lag:** Despite technical milestones, the core legal system remains presumptive as most states have yet to pass statutory bills enforcing absolute state-guaranteed conclusive titles.

Presumptive vs. Conclusive Titling

Evaluation Criteria	Presumptive Titling System	Conclusive Titling System
Definition	state registers only the transaction of property (sale deeds) rather than verifying the actual ownership of the title itself.	state officially certifies and guarantees the absolute validity of the land title ownership.
Legal Status of Title	Regarded as prima facie evidence only; ownership remains presumptive and can be challenged at any time in a civil court.	Title is legally unassailable and conclusive; cannot be challenged in court except under strictly defined fraud exceptions.
Onus of Proof & Verification	**On the Buyer:** The buyer must independently verify past transactions (typically up to 30 years) to check for encumbrances.	**On the State:** The government takes responsibility for verifying past records before entering the title into the official register.
State Guarantee	**No State Guarantee:** The government provides zero financial protection or indemnity if the registered title is later found to be defective.	**State Guaranteed:** The state provides a statutory guarantee and compensates the title-holder from an indemnity fund if a title defect emerges.

Underlying Legislation	Governed by legacy laws like the Registration Act (1908), Transfer of Property Act (1882), and state revenue codes.	Enabled via state-specific adoptions of the NITI Aayog Model Conclusive Land Titling Bill.
Impact on Judiciary	Breeds massive civil litigation; land disputes account for 60–70% of all pending cases in Indian civil courts.	Drastically slashes judicial backlog by legally blocking vexatious, retrospective ownership claims.
Socio-Economic Leverage	Restricts formal agricultural credit access as banks are hesitant to accept opaque, presumptive titles as collateral.	Unlocks fast, institutional credit flow, accelerates land pooling, and slashes transactional friction for infrastructure projects.

Issues with the Current Presumptive Land Titling System in India

- **Legal & Regulatory Vulnerabilities**
  - **Opaque Transaction Registration:** The **Registration Act of 1908** mandates registering only the *execution of deeds*, not verifying the legal validity of the title being transferred.
  - **Absence of State Guarantee:** Revenue department accepts no statutory liability for title correctness, leaving buyers completely vulnerable to hidden encumbrances or past fraud.
  - **Presumptive Onus on Buyers:** Forces buyers to independently conduct costly, retrospective title searches across multiple decades to ensure a clean chain of ownership.
  - **Fragmented Legal Architecture:** Governed by an overlapping web of legacy colonial laws (e.g., Transfer of Property Act, 1882) and disjointed, non-uniform state revenue codes.
- **Administrative & Technological Gaps**
  - **Deep Institutional Silos:** Land records are critically fractured across three unlinked departments: the **Revenue Department** (textual RoRs), **Survey Department** (cadastral maps), and **Registration Department** (SRO transaction deeds).
  - **Delayed Real-Time Mutations:** Updating of land records (*mutation*) after a property sale or inheritance takes months, creating a massive discrepancy between ground reality and official registries.
  - **Outdated and Inaccurate Cadastral Surveys:** Many rural boundaries rely on spatial maps drawn during colonial-era *survey and settlement* exercises, leading to persistent physical encroachment disputes.
  - **Weak Urban Record Systems:** While rural digitization (DILRMP) has advanced, urban and semi-urban municipal property records remain highly unorganized, fragmented, and prone to duplication.
- **Socio-Economic & Judicial Impacts**
  - **Massive Judicial Bottlenecks:** Land and property title ownership disputes clog the Indian judiciary, accounting for roughly **60% to 70% of all pending civil litigations**.
  - **Impediment to Institutional Credit:** Farmers struggle to secure low-interest bank loans as financial institutions reject opaque, dispute-prone presumptive titles as valid collateral.
  - **Infrastructure Acquisition Standoffs:** Linear projects (highways, railways) face severe delays, project cost overruns, and litigation due to untraceable or highly contested title claims.
  - **Informalization and Land Vulnerability:** Opaque titles allow local elites to easily manipulate records, leading to the illegal dispossession of lands belonging to poor, illiterate SC, ST, and women farmers.

Government Initiatives for Land Titling and Records

- **Core Umbrella & Structural Programs**

- **Digital India Land Records Modernization Programme (DILRMP):** A 100% centrally-funded central sector scheme (extended up to 2026) aimed at moving land administration entirely online to achieve state-guaranteed conclusive titling.
- **NITI Aayog's Model Conclusive Land Titling Bill (2020):** Provides a statutory blueprint for state governments to introduce localized legislations that substitute the legacy presumptive deed system with absolute title verification.
- **National Generic Document Registration System (NGDRS):** Implements a unified "One Nation, One Registration" pipeline across states to standardize e-registration processes, reducing citizen-bureaucrat interface and transaction friction.

- **Geospatial Identity & Technology Interventions**
  - **ULPIN / Bhu-Aadhaar Framework:** Assigns a 14-digit alphanumeric, geo-coordinated unique identity to individual land parcels across India to eliminate boundary duplicity and permanently block *benami* transactions.
  - **SVAMITVA Yojana:** Leverages high-resolution drone technology to map inhabited rural residential lands (*Abadi* areas), distributing undisputed "Property Cards" to grant collateral value to villagers.
  - **NAKSHA Pilot Programme:** Extends geospatial mapping into urban areas across more than 150 cities, deploying high-resolution aerial photography to clear up complex urban real estate title gridlocks.
  - **Unified Land Stack Initiative:** Deploys an interoperable, GIS-based platform that dynamically integrates land data, building structures, municipal registries, and agricultural parameters into a single source of truth.
- **Judicial & Linguistic Harmonization Measures**
  - **e-Courts Integration Pipeline:** Links localized revenue and registry databases directly with the judicial ecosystem to feed real-time title data to judges, expediting the resolution of civil land disputes.
  - **Automated Transliteration Modules:** Employs advanced machine tools to automatically translate land records across the **22 Scheduled VIII languages**, breaking linguistic barriers to inter-state land investments.
  - **Glossary of Revenue Terms (GoRT) Rollout:** Standardizes highly localized, legacy jargon (e.g., *Khasra, Dag, Pula*) into multi-script vernacular and English entries to ensure absolute data interoperability across state borders.

Way forward

- **Legislative & Structural Reforms**
  - **Enact State-Level Conclusive Land Titling Bills:** Accelerate the adoption of state-specific legislations modeled on the **NITI Aayog Conclusive Land Titling draft framework** to systematically transition away from presumptive deeds.
  - **Amend Legacy Colonial Registries:** Substantively amend the **Registration Act of 1908** to mandate that sub-registrars formally verify the legal ownership chain of a title rather than merely recording transactions.
  - **Institute State Title Guarantee & Indemnity Funds:** Create a statutory state-backed financial pool to provide absolute title insurance and compensate any genuine third-party claimants without disturbing the actual landholder's ownership.
  - **Set Up Specialized Land Titling Tribunals:** Establish dedicated **Land Dispute Resolution Officers (LDRO)** and quasi-judicial fast-track appellate tribunals to rapidly adjudicate property disputes outside overloaded civil courts.

- **Technological Integration & Administration**
  - **Unify Institutional Silos into a Single Registry:** Deeply integrate the separate databases of the **Revenue Department** (textual RoRs), **Survey Department** (spatial maps), and **Registration Department** (SRO transaction deeds) into a seamless, unified window.
  - **Establish Real-Time Automated Mutations:** Link property registration databases directly with revenue records to trigger instant, biometrically verified mutations upon the completion of a transaction.
  - **Scale Up Drone-Based Spatial Mapping:** Fully leverage high-resolution drone photography and 3D GIS tools under the **SVAMITVA Scheme** and **DILRMP 2.0** to eliminate inaccuracies in legacy cadastral maps.
  - **Deploy Immutable Blockchain Ledgering:** Build a decentralized, cryptographic blockchain ledger infrastructure for land registries to systematically block backend data tampering and unauthorized manual record updates.
- **Grassroots Inclusivity & Formal Markets**
  - **Decentralize Title Verification via Gram Sabhas:** Empower local panchayats to hold community-led peer reviews of draft title registries, ensuring oral family partitions and historical inheritance claims are captured accurately without bureaucratic exclusion.
  - **Legalize Agricultural Leases:** Adopt the **NITI Aayog Model Land Leasing Act** to provide full legal protection to tenant cultivators, decoupling actual operational land use from underlying title ownership disputes.

Conclusion

Transitioning to a state-guaranteed conclusive land titling system is not merely an administrative upgrade, but a fundamental economic imperative for building a modern, friction-free Indian economy. By leveraging robust digital infrastructures like ULPIN (Bhu-Aadhaar) and blockchain ledgers alongside the NITI Aayog Model Bill, the state can decisively unlock trillions in dead capital, eliminate debilitating judicial backlogs, and safeguard marginalized communities from asset insecurity.

3. LAND LEASING

Land leasing is a legal agreement where a landowner allows a tenant or farmer to use their land for cultivation or commercial purposes for a specified period in exchange for a fixed rental payment or a share of the crop harvest. Unlike a sale, the ownership of the land remains securely with the landlord, while the tenant gains temporary operational and usage rights.

Significance of Land Leasing

- **Agricultural Productivity & Economic Significance**
  - **Prevents Land Fallow Frictional Waste:** Allows absentee landowners to securely lease out fields, drastically reducing the volume of uncultivated, non-fallow arable land.
  - **Achieves Economies of Scale:** Enables small and marginal farmers to lease adjacent plots, consolidating fragmented holdings into operationally viable sizes.
  - **Incentivizes Capital Infusion:** Provides secure, long-term tenure that encourages tenants to invest in high-yield inputs like micro-irrigation, tractors, and fertilizers.
  - **Unlocks Formal Agricultural Credit:** Allows legal tenants to access institutional bank loans, crop insurance, and KCC (Kisan Credit Card) benefits instead of high-interest informal loans.
- **Socio-Economic Equity & Livelihood Security**

- **Empowers Landless and Marginal Farmers:** Opens secure livelihood avenues for Scheduled Castes (SCs), Scheduled Tribes (STs), and agricultural laborers who lack formal land titles.
- **Facilitates Rural Structural Transformation:** Secures land ownership rights, allowing landlords to safely migrate to urban areas for non-farm jobs without the fear of losing their land to adverse possession.
- **Accelerates Rural Poverty Alleviation:** Boosts the incomes of sharecroppers by formally eliminating arbitrary, exploitative crop-sharing arrangements.
- **Promotes Occupational Mobility:** Creates a dynamic rural land market where inefficient cultivators can smoothly exit agriculture without losing their primary asset.

- **Policy Alignment & Sustainable Governance**
  - **De-clogs Civil Judicial Pipelines:** Minimizes widespread land disputes and ownership litigation by replacing oral, informal contracts with legally binding, written leases.
  - **Ensures Targeted Welfare Delivery:** Helps the state directly route agricultural subsidies, PM-KISAN benefits, and disaster compensations straight to the actual tiller.

Framework governing land leasing in India

Regulatory Pillar	Core Operational Framework
Constitutional Status	State Subject (List II, Entry 18): State governments retain complete, exclusive power to enact and modify land leasing laws.
NITI Aayog Model Act	Model Agricultural Land Leasing Act (2016): Offers a template to legalize and standardize written leasing contracts nationwide.
Legal Backbone	Transfer of Property Act (1882): Defines standard rights, leases, and statutory liabilities for immovable assets.
Contract Registration	Registration Act (1908): Mandates that any land lease exceeding eleven months must be officially registered.
Welfare Linkage	PM-KISAN & KCC Integration: Moves toward linking central farming benefits directly to the actual cultivator.
Digital Alignment	DILRMP Grid: Connects lease registries directly to the Digital India Land Records Modernization Programme.

Issues with Land Leasing in India

- **Legal & Regulatory Fragmentation**
  - **Highly Restrictive State Laws:** Archaic tenancy laws in states like Bihar, Kerala, and UP either completely ban or place rigid caps on agricultural land leasing.
  - **Slow Adoption of NITI Aayog's Model Act:** Most states have refused or delayed adopting the Model Agricultural Land Leasing Act, 2016 due to local political resistance.

    Less than 25% of Indian states (such as MP, Maharashtra, and Uttarakhand) have modified local land codes to align with the Model Agricultural Land Leasing Act, 2016.
  - **Adverse Possession Legal Fear:** Landlords routinely refuse to sign written lease agreements out of fear that long-term tenants will legally claim permanent ownership of the property.
  - **Informal Oral Tenancy Dominance:** Over 90% of land tenancy in India remains entirely informal and unrecorded (NITI Aayog), leaving agreements out of the ambit of law.

    According to the National Sample Survey Office (NSSO) and World Bank briefs, **only 12.8% of the total leased-in agricultural area** in India is formally registered

- **Socio-Economic & Credit Bottlenecks**
  - **Systemic Exclusion from Institutional Credit: Lack** of a formal, written lease prevents tenant farmers from obtaining standard bank loans and Kisan Credit Cards (KCC).
  - **Welfare Benefit Diversion:** State financial incentives like PM-KISAN subsidies and disaster relief payouts go to the absentee landlord instead of the actual cultivator.
  - **Pre-emptive Eviction and Short Leases:** Landlords deliberately rotate tenants every 11 months to prevent them from building legal tenancy claims, destroying operational continuity.
  - **Incentive Deficit for Land Capitalization:** Because tenants have zero long-term security of tenure, they avoid investing in sustainable practices like micro-irrigation or soil conservation.
- **Operational & Data Deficits**
  - **Complete Absence of Tenancy Records:** The Digital India Land Records Modernization Programme (DILRMP) tracks land ownership details but completely omits recording active lease/tenancy data.

    DILRMP portal records **greater than 95% of rural land ownership (Record of Rights)** across major states, it captures **0% of operational land-use rights** (active leases) on its main dashboards (DoLR).
  - **Exploitative Crop-Sharing Frameworks:** Weak legal protection forces marginal sharecroppers to surrender up to 50-70% of their total harvest to landlords as rent.
  - **Severe Vulnerability to Climate Shocks:** Unregistered tenant tillers bear 100% of the crop loss risk from extreme weather events but receive 0% of the government insurance compensation.

Way Forward for Land Leasing in India

- **Legislative & Structural Reforms**
  - **Expedite NITI Aayog Model Act Adoption:** Incentivize states to harmonize local laws with the Model Agricultural Land Leasing Act, 2016 to completely legalize tenancy.
  - **Remove the Adverse Possession Clause:** Explicitly remove clauses that allow tenants to claim land ownership over time, completely eliminating the landlord's fear of losing property titles.
  - **Institutionalize Standardized Written Leases:** Replace informal oral agreements with simple, legally recognized written lease contracts to create transparent tenant-landlord dynamics.
  - **Derick Land Titles via Conclusive Titling:** Shift from presumptive to conclusive land titling frameworks to minimize title-related litigation and secure private lease investments.
- **Financial & Welfare Inclusivity**
  - **Decouple Credit from Land Ownership:** Amend RBI and NABARD guidelines to allow banks to issue Kisan Credit Cards (KCC) and crop loans solely based on registered lease agreements.
  - **Digital Tenant Verification Portals:** Integrate active tenancy registries with the DILRMP database to safely track land cultivation details without affecting ownership rights.
  - **Ensure Direct Benefit Routing:** Use Aadhaar-linked bank accounts to route PM-KISAN subsidies and disaster compensations directly to the actual cultivator instead of the absentee landlord.
  - **Expand Tenant-Centric Crop Insurance:** Modify Pradhan Mantri Fasal Bima Yojana (PMFBY) to enable tenant tillers to independently insure crops using their lease agreements.
- **Operational & Market Enhancements**
  - **Encourage Land Pooling and Consolidation:** Promote community-based land leasing models through Farmer Producer Organizations (FPOs) to build operational economies of scale for smallholders.
  - **Scale Up Special Land Courts:** Set up fast-track revenue tribunals and dispute resolution cells at the tehsil level to resolve landlord-tenant friction within a fixed timeline.
  - **Formalize Land-Leasing Market Registries:** Create localized, transparent digital marketplaces where landowners and potential tenants can easily match demand and lease rates.

Committee Recommendations on Land Leasing

Committee / Body	Recommendation Pillar	Strategic Action / Operational Mechanism
NITI Aayog Expert Committee (T. Haque, 2016)	**Legalization of Tenancy**	Harmonize state laws to completely legalize land leasing across all states without any ceiling caps.
	Title De-risking	Explicitly secure land ownership titles of landlords to remove the fear of losing land through adverse possession.
	Credit & Subsidy De-coupling	Enable tenant farmers to independently access institutional credit, KCC, and insurance using written lease agreements.
Arvind Subramanian (Chief Economic Advisor, 2016)	**Land Market Dynamism**	Implement transparent, digital land-leasing frameworks to facilitate operational farm consolidation and increase productivity.
Dalwai Committee (Doubling Farmers' Income, 2018)	**FPO-Led Leasing Integration**	Promote land leasing through Farmer Producer Organizations (FPOs) to pool fragmented smallholdings into viable economies of scale.
National Commission on Farmers (M.S. Swaminathan, 2006)	**Tenure Security for Smallholders**	Ensure minimum long-term lease tenures to incentivize tenants to invest in land development and sustainable micro-irrigation.
	Asset Equity for Women	Issue joint land lease registries to formally empower and recognize women sharecroppers and agricultural laborers.

Global case studies on land leasing reforms

- **China: The Household Responsibility System (HRS) & Three-Rights Separation**
  - Separated land ownership (retained by village collectives) from contract rights (given to households) and management/usage rights (open for leasing).
- **Vietnam: The 1993 Land Law (Doi Moi Reforms)**
  - Granted households 20-to-50-year Land Use Rights Certificates (LURCs) that explicitly allowed five core freedoms: the right to exchange, transfer, lease, inherit, and mortgage.
- **Ethiopia: Second-Level Land Certification (SLLC)**
  - Launched a massive, localized digital mapping program in 2013 to register state-owned lands and issue geo-referenced, digital land certificates to farmers.
- **Kenya: Mobile-Enabled Digital Lease Registries (ArdhiSasa)**
  - Launched a fully decentralized, mobile-accessible online land management platform to track ownership, register leases, and process payments instantly.

Conclusion

Legalizing and formalizing land leasing is an indispensable structural land reform needed to secure the property rights of landlords while extending formal credit and crop insurance to tenant-cultivators. Swiftly adopting the **NITI Aayog Model Act** across states will eliminate informal tenancy traps and drive operational farm consolidation to achieve economies of scale.

4. LAND CEILING POLICY

The **land ceiling policy** in India refers to a legal instrument of institutional land reform that establishes an upper statutory limit on agricultural holding size an individual or family can legally possess. **Land ceiling policy was** initiated primarily during the Second and Third Five-Year Plans to dismantle residual colonial feudalism, the policy mandates the state to seize **surplus land** above the threshold for redistribution to landless laborers, Scheduled Castes (SCs), and Scheduled Tribes (STs).

Evolution of Land Ceiling Policy in India

- **Phase I: Post-Independence & Initial Legislations (1950s–1960s)**
  - **Kumarappa Committee Recommendation (1949):** Formulated the blueprint for agrarian restructuring by recommending a statutory maximum limit on land holdings to eliminate absentee landlordism.
  - **First and Second Five-Year Plans:** Recommended that state governments enact land ceiling laws, but left the definition of a **family unit** and ceiling limits loosely defined.
  - **State-Level Legislative Failures:** Suffered widespread evasion as weak laws allowed landlords to exploit loopholes by transferring land titles through fictitious, fraudulent sales (benami transactions).
  - **Exemptions for Commercial Exploitation:** Excluded cash-crop plantations (tea, coffee, rubber), orchards, and specialized mechanized farms from the ceiling, which landlords used to shield vast holdings from state seizure.
- **Phase II: National Standardization & Revision (1970s)**
  - **Chief Ministers' Conference (1972):** Formulated the National Guidelines on Land Ceilings under the Indira Gandhi administration to enforce nationwide uniformity and lower the maximum permissible limits.
  - **Lowering the Upper Thresholds:** Standardized the legal limits to 10–18 acres for high-yielding perennial irrigated land, and up to a maximum of 54 acres for un-irrigated drylands.
  - **Redefining the Basic Holding Unit:** Shifted the legal benchmark of ownership from an **individual** to a **family unit** (husband, wife, and minor children) to systematically curb intra-family land division evasions.
  - **Ninth Schedule Constitutional Shield:** Placed major state land ceiling acts under the Ninth Schedule via constitutional amendments to immunize them from judicial review on grounds of fundamental rights violations.
- **Phase III: Consolidation, Post-Liberalization & Digital Shifts (1980s–Present)**
  - **Shifting Focus to Physical Redistribution:** Prioritized the formal physical handover of previously seized surplus land parcels specifically to landless Scheduled Caste (SC) and Scheduled Tribe (ST) households.
  - **Economic Liberalization Pressure (1990s onward):** Witnessed several state governments (e.g., Karnataka, Maharashtra) relaxing ceiling limits to allow large agribusiness corporate entities to acquire land for modern contract farming.
  - **Digital Modernization Wave (2000s–Present):** Shifted operational focus to the Digital India Land Records Modernization Programme (DILRMP) to digitize land maps and transparently expose hidden or unaccounted corporate and individual land concentrations.

Features of the Land Ceiling Policy in India

- **Legal & Conceptual Framework**

- **Statutory Limit Fixation:** Establishes a maximum legal limit on agricultural land ownership, varying by state based on soil fertility, water availability, and crop types.
- **Family Operational Unit:** Evaluates ownership thresholds based on a defined **Family Unit** (husband, wife, and minor children) rather than an individual to prevent legal evasion.
- **Variable Ceiling Thresholds:** Standardizes lower limits for perennially irrigated double-cropped land (**10–18 acres**) and higher limits for non-irrigated drylands (up to **54 acres**).

- **Acquisition & Redistributive Mechanisms**
  - **Surplus Land Vesting:** Dictates that any agricultural land held in excess of the statutory ceiling automatically lapses and vests with the State Government.
  - **Targeted Asset Redistribution:** Mandates the priority allocation of seized surplus land to landless agricultural laborers, Scheduled Castes (SCs), and Scheduled Tribes (STs).
  - **Regulated State Compensation:** Provides for the payment of a regulated, non-market compensation (*solatium*) to original landlords for the acquired surplus land.
- **Constitutional Shields & Exemptions**
  - **Ninth Schedule Immunization:** Enjoys protection from judicial review by placing state-level ceiling legislations under the **Ninth Schedule** of the Constitution.
  - **Specific Sectoral Exemptions:** Excludes plantations (tea, coffee, rubber), sugarcane research farms, religious/charitable trusts, and orchards to prevent industrial disruption.
  - **Restrictions on Land Alienation:** Imposes strict legal bans on the resale or transfer of redistributed surplus land by beneficiaries to protect them from land sharks.

Issues with the Land Ceiling Policy in India

- **Legislative & Administrative Loopholes**
  - **Proliferation of Benami Transactions:** Suffered severe evasion as landlords transferred surplus land titles to fictitious entities or relatives (benami) to bypass the family unit ceiling.
  - **Exploitation of Blanket Exemptions:** Landowners intentionally reclassified standard agricultural fields as exempted categories like orchards, plantations, or religious trusts to shield vast holdings from state acquisition.
  - **Individual vs. Family Delay:** The delay by states in adopting the 1972 guideline shifting the unit from **individual** to "family" gave landlords years to legally partition and dilute their holdings.
  - **Corrupt & Inefficient Revenue Machinery:** low-level revenue administration (Patwaris/Tehsildars) colluded with powerful local elites to manipulate village land records and conceal surplus holdings.

    As per Appu committee, Field evaluations indicate that nearly **one-third (33%) of all rural tenant households** were summarily evicted by landlords to erase records before official verification
- **Operational & Redistributive Failures**
  - **Poor Quality of Surplus Land:** Vast majority of land surrendered by landlords was uncultivable waste, hilly terrain, or barren dryland, yielding little to no agricultural value for beneficiaries.
  - **Dismal Actual Redistribution Rates:** Out of the total land declared surplus across India, a significant portion remained locked in bureaucratic delays or was never physically handed over to the poor.

    As per MoRD, Out of **7.34 million acres** of land officially declared surplus under revised ceiling laws, only **5.39 million acres** were physically distributed to around 5.64 million households
  - **Chronic Judicial Litigation:** Wealthy landowners clogged the judiciary with endless civil disputes challenging acquisition notices, effectively stalling the physical possession of land for decades.

By the mid-1980s, over **1.6 million acres** of declared surplus land was locked up in active litigation, completely stalling physical possession.

- **Paradox of Land Fragmentation:** Forcing large, integrated estates into tiny, sub-economic plots reduced the scale of mechanization, contributing to India's structural challenge of unviable smallholder farming.

As **Agricultural Censuses**, small and marginal holdings (less than 2 hectares) constitute **over 86% of all operational holdings** in India, operating on less than 47% of the total area.

- **Socio-Economic & Post-Reform Structural Bottlenecks**
  - **Social Eviction and Retaliation:** Landless Scheduled Caste (SC) and Scheduled Tribe (ST) beneficiaries who received surplus land often faced violent social boycotts or physical eviction by dominant-caste former landlords.

According to MoRD data, **Scheduled Castes (SCs) comprised 36%** and **Scheduled Tribes (STs) comprised 15%** of the total land ceiling beneficiaries.

  - **Lack of Forward Credit Linkages:** State distributed land but failed to provide matching access to institutional credit, seeds, and irrigation, forcing poor beneficiaries to lease the land back to elites or leave it fallow.
  - **Incompatibility with Modern Agribusiness:** Rigid ceiling limits discourage corporate investments and large-scale contract farming, which are vital for modern agro-processing.

Way forward

- **Digitization & Fraud Elimination**
  - **Universal Bhu-Aadhar Rollout:** Accelerate the Unique Land Parcel Identification Number (ULPIN) under the DILRMP to map every land boundary geophysically, instantly exposing hidden benami (proxy) structural holdings.
  - **Aadhaar-Linked Land Mutations:** Enforce mandatory, biometrically verified land registry processes to shut down fraudulent, informal family land partitions designed to bypass family ceiling unit laws.
  - **Integrating Revenue Records with e-Courts:** Wire local land registration servers directly to the e-Courts pipeline to expedite the clearance of pending land-ceiling litigations currently clogged in civil tribunals.
- **Modern Agribusiness & Statutory Flexibilities**
  - **Adopt the NITI Aayog Model Land Leasing Act:** Legalize and formalize tenant farming markets to allow smallholders to lease out land securely without the fear of permanent asset loss or eviction.
  - **Encourage Land Pooling Models:** Replicate successful corporate-cooperative aggregation models (e.g., Amul, industrial corridors) to scale up farm mechanization without physically altering statutory individual ceiling limits.
  - **Calibrated Ceiling Relaxations for High-Tech Agribusiness:** Permit highly monitored, state-approved threshold exemptions specifically for capital-intensive global investments in seed-R&D, food processing, and protected tissue culture.
- **Socio-Economic Support & Grassroots Delivery**
  - **Mandated Post-Redistribution Credit Linkages:** Hardwire immediate institutional credit, micro-irrigation subsidies, and PM-KISAN cash flows to marginalized SC/ST beneficiaries of surplus land to prevent plots from falling fallow.
  - **Empowering Panchayati Raj Institutions (PRIs):** Decentralize local land registry verification powers directly to **Gram Sabhas** to leverage community-backed peer transparency against high-level bureaucratic corruption.

- **Prioritize Women's Asset Joint-Ownership:** Enforce mandatory joint-titling or sole female-ownership over newly redistributed state surplus land parcels to bypass local patriarchal title-inheritance bottlenecks.

Best Practices on Land Ceiling and Reforms from Indian States

- **Historical State-Led Success Models**
  - **West Bengal's Operation Barga Model:** Mobilized Gram Sabhas and peasant unions to dynamically record 1.5 million sharecroppers, successfully securing tenure rights and accounting for nearly 25% of India's total surplus land pool.
  - **Kerala's Radical Land to the Tiller Act (1963):** Enforced a zero-allowance family ceiling unit that completely dismantled absentee landlordism, directly transitioning hutment dwellers into absolute property owners.
  - **Jammu & Kashmir's Big Landed Estates Abolition Act (1950): Executed** a highly restrictive statutory ceiling limit (**22.75 acres**) without paying a single rupee of fiscal compensation to landlords, pioneering decisive, asset-equalizing redistributive justice.
- **Technology & Modern Administrative Practices**
  - **Karnataka's Bhoomi & Geo-Spatial Mapping Integration:** Digitized millions of rural land records to structurally match land titles with remote sensing data, freezing illegal, fragmented corporate configurations.
  - **Telangana's Dharani Portal Framework:** Developed a unified, cryptographic single-source-of-truth registry that links automated mutation with sub-registrar databases, preventing hidden proxy (*benami*) land accumulation.

Conclusion

India's land ceiling policy must transition from a legacy framework of rigid land fragmentation to a dynamic model of technology-driven asset consolidation and tenure security. By seamlessly integrating digital tracking systems like ULPIN with progressive statutory reforms like the NITI Aayog Model Land Leasing Act, the state can effectively bridge the gap between smallholder equity and large-scale agribusiness efficiency.

5. LAND ACQUISITION

Land acquisition is the statutory process by which the government takes over privately owned land for public purposes, such as infrastructure development, industrialization, or urbanization. It is Governed by the **RCFTLARR Act of 2013** in India, legally mandates the state to provide fair financial compensation, rehabilitation, and resettlement to affected landowners and livelihood dependents.

Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement (RFCTLARR) Act, 2013

- **Core Institutional & Procedural Features**
  - **Replaced Colonial Legislation:** Repealed the archaic, state-centric *Land Acquisition Act of 1894* to move toward a rights-based framework for affected communities.
  - **Mandatory Social Impact Assessment (SIA):** Compulsory independent evaluation for all acquisitions to determine public purpose, assess social impact, and minimize food security risks.
  - **Public Purpose Definition:** Strictly limits acquisition to defined public uses like national security, infrastructure, rural utilities, and government-administered PPP projects.
  - **Gram Sabha Consultation:** Mandates statutory consultation with local Gram Sabhas in Scheduled Areas (Panchayat/Tribal belts) before initiating any land acquisition process.

- **Consent & Food Security Safeguards**
  - **Rigid Consent Thresholds:** Mandates explicit, prior consent from **70%** of affected families for Public-Private Partnership (PPP) projects and **80%** for purely private projects.
  - **Exempted Public Sector Projects:** Requires zero prior citizen consent if the acquisition is entirely executed, owned, and controlled by the government.
  - **Multi-Crop Land Protections:** Restricts the acquisition of irrigated multi-crop agricultural land to a strict state-defined ceiling cap to protect national food security.
  - **Lapse of Acquisition Clause (Section 24):** Stipulates that legacy acquisition proceedings under the 1894 Act lapse if physical possession is not taken or compensation is not paid within 5 years.
- **Compensation, Welfare & R&R Features**
  - **Enhanced Multiplied Compensation:** Mandates market-value payouts multiplied by **2x in urban areas** and up to **4x in rural regions**.
  - **Statutory Solatium Award:** Enforces a mandatory **100% solatium** (additional payment on the total market value) for the distress caused by compulsory acquisition.
  - **Inseparable R&R Package:** Legally links land acquisition with mandatory **Rehabilitation and Resettlement (R&R)** benefits, treating them as a single statutory right.
  - **Livelihood Protection for Non-Owners:** Explicitly includes landless agricultural laborers, sharecroppers, and traditional artisans as beneficiaries entitled to R&R packages.
  - **Land Acquisition Monitoring Authorities:** Establishes dedicated, state-level fast-track *LARR Authorities* to resolve disputes regarding valuation and resettlement within a fixed timeline.

Significance of Land Acquisition

- **Infrastructure & Macroeconomic Growth**
  - **Catalyzes Public Infrastructure Multiplier:** Enables the timely execution of mega-projects like bullet trains, industrial corridors, and the National Infrastructure Pipeline (NIP).

    *The Ministry of Statistics (*MoSPI - PAIMANA Portal*) tracks **1,980+ mega central infrastructure projects worth over ₹42.50 lakh crore**, requiring prompt land assembly to prevent capital lock-up.*
  - **Unlocks Strategic Industrialization:** Facilitates the aggregation of large contiguity land parcels for Special Economic Zones (SEZs) and manufacturing clusters under Make in India.

    Department for Promotion of Industry and Internal Trade (DPIIT) reports over **300+ operational SEZs** and massive national industrial corridors
  - **Accelerates Planned Urbanization:** Provides the physical space required for smart cities, high-density mass transit corridors (Metros), and affordable public housing schemes.
  - **Secures Critical Connectivity Networks:** Streamlines the construction of national highways, dedicated freight corridors, and multi-modal logistics parks to lower high logistics costs.

    National Highways Authority of India (NHAI) *capital expenditure touched **₹2.44 lakh crore**, sustaining a rapid network expansion pace of nearly **34 km per day** to actively slash logistics overheads*
- **Social Welfare & Public Utility Creation**
  - **Enables Public Welfare Asset Creation:** Allows the state to secure land for essential non-commercial public utilities like AIIMS-level hospitals, IITs, and rural schools.

    Ministry of Health and Ministry of Education to secure large land packages for **22 functional AIIMS** and **23 IITs** to expand national tertiary care and higher education grids.
  - **Drives Energy Security and Transition:** Secures extensive land areas required for massive solar parks, wind farms, and strategic green hydrogen manufacturing hubs.

Essential to meet India's COP26 *commitment to scale up non-fossil fuel capacity to* ***500 GW by 2030***

- **Strengthens Defense and Border Security:** Empowers the state to rapidly acquire land for strategic border roads, forward airbases, and critical defense infrastructure.

- **Institutional & Economic Equity**
  - **Institutionalizes Fair Value Monetization:** Replaces arbitrary land-grabbing with standard, legally mandated market-linked compensations (up to 4x in rural areas).
  - **Guarantees Statutory Livelihood Security:** Mandates formal Rehabilitation & Resettlement (R&R), legally converting displaced agricultural laborers into skilled, non-farm workers.

Issues with Land Acquisition in India

- **Financial & Compensatory Bottlenecks**
  - **Exorbitant Acquisition Cost:** Rural payout multiples of 4x under the *RFCTLARR Act 2013* drastically inflate overall project capital expenditures, making infrastructure projects unviable.

    As per NHAI, Land acquisition alone now accounts for a massive **25-30% of total project costs** for major highway networks
  - **Flawed Market-Value Calculations:** Outdated, deflated circle rates on official stamp registries fail to reflect actual market valuations, sparking intense legal friction.

    As per Dept of land resources, official stamp registries routinely capture **less than 50% of the actual market transaction value** in rapidly commercializing rural-urban fringes
  - **Compensation Delivery Delays:** Arbitrary bureaucratic loops in revenue departments lead to prolonged multi-year holdups in the direct disbursement of compensation awards.
- **Administrative & Procedural Delays**
  - **Stalled Social Impact Assessments (SIA):** Complicated, multi-tiered independent SIA processes and mandatory public hearings routinely delay initial project approvals by 12 to 18 months.
  - **Rigid Democratic Consent Thresholds:** Securing the mandatory, explicit consent of 70-80% of affected families remains near-impossible in highly fragmented, politicized rural ecosystems.
  - **Severe Judicial Chokepoints:** An overwhelming backlog of land valuation disputes clogs special *LARR Authorities* and high courts, completely paralyzing project execution.

    Land-related litigations account for **nearly 25% of all cases pending before the Supreme Court** and over 60% of civil suits in lower courts.
- **Socio-Economic & Mapping Discrepancies**
  - **Incomplete Rehabilitation and Resettlement (R&R):** State agencies routinely default on the statutory mandate to provide vocational training and sustainable non-farm livelihoods to displaced communities.
  - **Neglect of Livelihood Dependents:** Landless agricultural laborers, sharecroppers, and artisans are systematically bypassed in compensation loops due to a lack of formal land titles.

    Over **55% of the rural agricultural workforce** is comprised of landless laborers and sharecroppers (Census of India)
- **Legacy Mapping Discrepancies:** Outdated, siloed paper records trigger immediate ownership disputes and spatial mismatches when high-resolution drone mapping begins.

Way Forward for Land Acquisition in India

- **Legislative & Procedural Streamlining**

- **Optimize Social Impact Assessments (SIA):** Standardize and digitize the SIA timeline to complete evaluation procedures within a strict, fast-tracked 6-month statutory window.
- **Introduce Flexible State Exemptions:** Strategically utilize Section 10A to exempt critical public infrastructure, linear projects, and defense corridors from tedious consent clauses.
- **Adopt the Land Pooling Model:** Replicate the successful *Amaravati/Gujarat* models where landowners voluntarily partner with the state, receiving developed commercial plots instead of outright displacement.
- **Transition to Conclusive Land Titling:** Shift from presumptive to conclusive titles under the *DILRMP* framework to eliminate title-related litigation before the acquisition phase begins.

- **Financial & Valuation Reforms**
  - **Rationalize Circle Rates Real-Time:** Update registry circle rates dynamically using GIS and market data to reflect actual property values and prevent compensation disputes.
  - **Establish Land Value Capture Finance (VCF):** Implement VCF mechanisms to fund high compensation costs by monetizing the steep increase in land value surrounding new infrastructure.
  - **Automate Compensation via Escrow:** Mandate the use of digital Escrow accounts linked with Aadhaar for the direct, zero-delay transfer of compensation and *solatium* awards.
- **Inclusive Rehabilitation & Digital Governance**
  - **Institutionalize Skill-Based Resettlement:** Replace one-time cash payouts with mandatory, structured vocational training through the *National Skill Development Corporation (NSDC)* for livelihood dependents.
  - **Include Non-Titleholder Dependents:** Create localized, verified digital registries of sharecroppers and landless laborers during the SIA to guarantee statutory R&R benefits.
  - **Deploy Single-Window LARR Portals:** Integrate text, spatial maps, and judicial dispute data onto a unified digital dashboard to track acquisition, compensation, and R&R status in real-time.

Best practices and models for land assembly in India

- **Gujarat Town Planning Scheme (TPS) Model:** Landowners voluntarily contribute a fixed portion (typically 35–40%) of their land to local urban bodies for public utilities and roads.
- **Amaravati Land Pooling Scheme (LPS) Model:** Assembled over 33,000 acres of contiguous greenfield land through an integrated voluntary contract framework.
- **Magarpatta City Model (Pune):** A unique corporatized model of "Farmers' Direct Investment" (FDI) where a massive 430-acre integrated township was built by pooling adjacent ancestral farms.
- **Greater Noida / Naya Raipur Negotiated Acquisition Model:** Direct, bilateral price negotiations between public development authorities and local landowners instead of rigid statutory notifications.
- **Mumbai Transferable Development Rights (TDR) Model:** The state acquires congested private land for public amenities by issuing a certificate giving the owner additional building rights (Floor Space Index) elsewhere.

Conclusion

Balancing the urgent demands of industrialization with the socio-economic rights of displaced communities remains the core challenge of land acquisition in India. Overcoming procedural delays under the **RFCTLARR Act 2013** requires shifting toward progressive alternatives like voluntary land pooling and direct, market-negotiated price discovery.

6. LAND RECORDS IN INDIA: DIGITIZATION AND MODERNIZATION

The digitisation and modernisation of land records in India represents a structural paradigm shift from an archaic, presumptive framework toward a transparent, data-driven ecosystem. Land records are anchored by the Digital India Land Records Modernization Programme (DILRMP) and powered by foundational pillars like **Bhu-Aadhaar (ULPIN)** and the **SVAMITVA** drone mapping scheme, this transformation targets the core of agrarian and urban property security.

Significance of Digitized and Modernized Land Records in India

- **Economic & Financial Significance**
  - **Unlocks Dead Capital:** Converts opaque, undocumented property into a liquid financial asset, enabling citizens to formalize land equity for wealth generation.
  - **Accelerates Institutional Credit:** Provides banks with verified, dispute-free digital collateral, freeing poor farmers from the trap of predatory local moneylenders.
  - **Reduces Infrastructure Transaction Friction:** Speeds up public land acquisition for highways, railways, and industrial corridors, dramatically reducing project gestation periods and cost overruns.
  - **Boosts Municipal & State Revenue:** Enhances local government fiscal health by broadening property tax bases and eliminating leakage in stamp duty collection.
- **Judicial & Administrative Significance**
  - **De-clogs the Judiciary:** Decongests overloaded civil courts by systematically targeting the source of 60–70% of all pending litigations in India.
  - **Eliminates Corruption:** Cuts out the discretionary, exploitative power of low-level revenue officials (Patwaris/Tehsildars) through automated, transparent e-mutation pipelines.
  - **Data-Driven Policy Implementation:** Enables targeted, leakage-free delivery of direct benefit transfers like PM-KISAN, crop insurance (PMFBY), and fertilizer subsidies based on precise operational holding size.
- **Socio-Ecological Significance**
  - **Secures Marginalized Asset Rights:** Protects vulnerable Scheduled Castes (SCs), Scheduled Tribes (STs), and women from illegal land dispossession by dominant-caste elites.
  - **Fosters Agriculture Scale Economies:** Facilitates secure, formal land leasing markets (NITI Aayog Model Act) without the fear of title loss, promoting corporate investment and contract farming.
  - **Climate Change & Disaster Resilience:** Integrates spatial cadastral maps with GIS data to model flood-drought risk zones, enabling precise, plot-level crop damage compensation and hazard planning.

Issues with the Digitization and Modernization of Land Records

- **Structural Data & Legacy Glitches**
  - **Garbage In, Garbage Out Risk:** Digitizing existing inaccurate, outdated, or manipulated paper records simply formalizes and hardcodes legacy errors onto digital platforms.
  - **Ground-to-Record Area Mismatch:** High-resolution GPS and drone surveys frequently reveal steep area discrepancies compared to text records, triggering fresh boundary disputes.
  - **Unrecorded Spatial Mutations:** Informal family partitions, oral land successions, and internal boundary adjustments are missing from localized revenue updates.
- **Technological & Operational Bottlenecks**
  - **Interoperability and Silo Cracks:** The text registry (Revenue), spatial maps (Survey), and sale deeds (Registration) remain digitally disconnected in several states.

- **Federal Divergence & Legislative Lags:** Since land is a **State Subject (List II)**, non-standardized formats, distinct local land laws, and variable technical speeds break national uniformity.
- **Language and Revenue Jargon Barriers:** Automated transliteration tools struggle with highly localized colonial-era vocabulary (*Khasra, Khatauni, Dag*), causing recording errors.

- **Socio-Legal & Cybersecurity Risks**
  - **Exclusion of Vulnerable Smallholders:** Low digital literacy and lack of formal papers among marginalized groups (SCs, STs, and women) lead to administrative exclusion during fast-tracked digitizations.
  - **Cyber Vulnerabilities & Data Tampering:** Centralized databases lack robust, immutable cryptographic protections, rendering vital land records vulnerable to unauthorized backend manipulation or cyberattacks.
  - **Continuous Presumptive Paradox:** Even after full digitization, the legal nature of titles remains *presumptive* under the **Registration Act of 1908**, failing to legally shield buyers from structural court challenges.

Measures Taken for the Digitization and Modernization of Land Records in India

- **Core Umbrella Frameworks & Portals**
  - **DILRMP Central Shift:** Revamped the *Digital India Land Records Modernization Programme* into a 100% centrally funded Central Sector Scheme to build an Integrated Land Information Management System.
  - **NGDRS Implementation:** Deployed the **National Generic Document Registration System** to establish a uniform "One Nation, One Registration" interface for e-registration of deeds.
  - **Integrated Land Stack Pilot:** Rolled out a GIS-based, interoperable **"Land Stack" platform** (piloted in Chandigarh/Tamil Nadu) to seamlessly bridge data among the Revenue, Survey, and Registration departments.
- **Tech-Driven Mapping & Unique Identification**
  - **ULPIN (Bhu-Aadhaar) Rollout:** Assigned a 14-digit alphanumeric, geo-coordinated **Unique Land Parcel Identification Number** to over 36 crore land parcels to eliminate boundary duplicity.
  - **SVAMITVA Drone Surveys:** Leveraged high-resolution drone mapping across rural inhabited (*Abadi*) zones to issue legal property cards, unlocking rural property collateral value.
  - **NAKSHA Urban Expansion:** Launched the *National Geospatial Knowledge-based Land Survey of Urban Habitations* to map over 150 cities using aerial photography for complex urban real estate records.
- **Institutional & Structural Integrations**
  - **e-Courts Database Pipeline:** Linked localized property registries directly with the e-Courts ecosystem to provide judicial benches with authentic title histories for swift case disposal.
  - **Revenue Court Computerization:** Integrated rural revenue courts directly into the primary land records database to trigger automatic updates immediately following dynamic judicial orders.
  - **Linguistic Harmonization:** Deployed C-DAC automated transliteration systems across 22 Schedule VIII languages alongside a centralized **Glossary of Revenue Terms (GoRT)** to resolve regional vocabulary barriers (*Khasra, Dag, Pula*).

Way forward

- **Legislative & Statutory Interventions**

- **Enact Pre-Registration Verifications:** Amend the *Draft Registration Bill / Legacy Acts* to legally mandate sub-registrars to verify underlying title ownership instead of passively recording deeds.
- **Statutory Protected / Negative Lists:** Create institutional digital firewalls to automatically flag and block illegal registrations involving ceiling-surplus, communal, forest, and ecologically sensitive zones.
- **Transition to Conclusive Statutory Frameworks:** Urgently pass state-specific legislations modeled on the **NITI Aayog Model Conclusive Land Titling Bill** to give state-backed guarantees to digitized records.

- **Technological & Data Quality Enhancements**
  - **Implement Strict Data Hygiene (Clean In, Clean Out):** Enforce rigorous physical field verification prior to database entry to prevent the digitization and hardcoding of legacy manual errors and fraud.
  - **Deep Architectural Integration:** Fully unify the database silos of the **Revenue** (textual RoRs), **Survey** (spatial maps), and **Registration** departments into a singular, dynamically updating mechanism.
  - **Scale Up Cryptographic Security:** Deploy decentralized **Blockchain Ledgers** across state land registries to permanently block unauthorized backend data tampering and unauthorized manual overwrites.
  - **Expand High-Resolution Drone Auditing:** Continuous, institutionalized ground-truthing via **SVAMITVA** and **NAKSHA** drone frameworks to bridge physical-to-digital area discrepancies.
- **Socio-Administrative & Federal Adaptations**
  - **Customized Communal Frameworks:** Develop flexible, non-individualistic digital templates for the Northeast and Fifth/Sixth Schedule areas to accommodate historical community-held land patterns.
  - **Institutionalize Gender Equity:** Ensure full field implementation of the newly mandated **Gender Column in Record of Rights (RoRs)** to protect women's co-ownership rights and direct targeted agri-benefits.
  - **Decentralized Disputes Resolution:** Set up dedicated online grievance portals linked with ***Revenue Court Case Management Systems (RCCMS)*** to resolve spatial data conflicts transparently at the grassroots.

Conclusion

Comprehensive digitization and modernization of land records is not merely a technological upgrade, but the foundational pillar for a friction-free, inclusive, and formal Indian economy. Seamlessly transitioning to a state-guaranteed conclusive titling model via **Bhu-Aadhaar (ULPIN)**, blockchain ledgers, and structural legal reforms will decisively unlock trillions in dead capital while clearing a clogged judicial backlog.

7. MODERN LAND GOVERNANCE INITIATIVES

Modern land governance marks a transformative shift from extractive, colonial-era presumptive tracking toward an integrated, tech-driven, and rights-based ecosystem. **Modern land governance** anchored by the concept of a unified **Land Stack** and driven by frameworks like NITI Aayog's Model Conclusive Land Titling Bill, it transitions land from a friction-heavy administrative bottleneck into a fluid socio-economic asset.

Significance of Modern Land Governance in India

- **Economic Growth & Financial Inclusion**
  - **Mobilizes Dead Capital:** Converts opaque, undocumented real estate into a liquid financial asset, unlocking trillions in trapped economic equity.
  - **Secures Institutional Credit:** Replaces predatory informal moneylenders by providing banks with dispute-free, digital collateral for seamless agri-loans.

- **Eases Infrastructure Bottlenecks:** Slashes project gestation periods and cost overruns by accelerating transparent, litigation-free public land acquisition.
- **Optimizes Fiscal Revenues:** Plugs revenue leakages and broadens municipal tax bases through automated GIS-based property tax mapping.

- **Judicial De-clogging & Administrative Efficiency**
  - **Decongests Civil Courts:** Systematically resolves the root cause of land disputes, which currently account for **60–70% of all pending civil litigations**.
  - **Eliminates Rent-Seeking:** Cuts out the discretionary, exploitative power of low-level revenue officials through automated, transparent e-mutation pipelines.
  - **Targeted Welfare Delivery:** Enables accurate, leakage-free delivery of direct benefit transfers like **PM-KISAN** based on precise, verified land holding sizes.
  - **One Nation, One Registration:** Establishes cross-border data interoperability, allowing seamless inter-state land transactions and investments.
- **Social Equity & Climate Resilience**
  - **Empowers Vulnerable Groups:** Protects marginalized smallholders, Scheduled Castes (SCs), Scheduled Tribes (STs), and women from illegal land dispossession.
  - **Formalizes Land Leasing:** Drives corporate investment and contract farming by enabling secure leasing markets without the fear of title loss.
  - **Enhances Disaster Resilience:** Integrates spatial cadastral maps with climate data to model flood-drought risk zones for plot-level crop insurance payouts.

Issues with Modern Land Governance in India

- **Structural, Legal & Regulatory Friction**
  - **Presumptive Legal Trap:** Despite modern digitization, the **Registration Act of 1908** still registers only the *deed*, leaving the legal nature of titles fundamentally presumptive and open to court challenges.

    As per NITI Aayog, Land disputes account for **66% of all civil litigation** in India, with the average case taking 20 years to resolve.
  - **Federal Divergence and Lags:** Because land is a **State Subject (List II)**, varying state laws, local land revenue codes, and uneven technical speeds break national standardization.

    NCAER Land Records Services Index (N-LRSI) reveals top states scoring **greater than 85%** in digitization efficiency while several northeastern states score **less than 20%**.
  - **Inflexible Customary Clashes:** Standardized digital templates often clash with the complex, non-individualistic, and orally held community land tenure systems of the Northeast and Fifth/Sixth Schedule regions.

    According to the Ministry of Tribal Affairs, **over 50% of community forest resource rights claims** under the FRA remain unmapped or pending due to incompatible digital templates.
  - **Gender-Disparity in Registries:** Deeply entrenched socio-cultural biases persist in blocking the structural record-updating of women's co-ownership and inheritance rights

    As per Agriculture Census, over 70% of agricultural labor, women own only **13.9% of operational landholdings** in India.
- **Technological & Operational Bottlenecks**
  - **Hardcoding Legacy Errors:** Digitizing unverified, inaccurate, or historically manipulated paper records creates a dangerous **Garbage In, Garbage Out** cycle on digital portals.

CAG Report highlighted that up to **30-40% of legacy paper entries** digitized in initial phases contained unverified mutation records.

- **Ground-to-Record Area Mismatches:** High-resolution drone and GPS mapping frequently reveal steep spatial discrepancies compared to old text records, triggering fresh boundary conflicts.
- **Persistent Institutional Silos:** Textual records (Revenue), spatial maps (Survey), and transaction deeds (Registration) remain digitally isolated and unlinked in several states.
- **Linguistic and Jargon Roadblocks:** Automated systems struggle to accurately translate highly localized, medieval-era revenue vocabulary (*Khasra, Khatauni, Dag*) across the 22 Scheduled languages.

- **Socio-Economic & Cybersecurity Vulnerabilities**
  - **Digital Exclusion of Smallholders:** Low digital literacy, poor rural connectivity, and lack of formal papers systematically exclude marginalized SC/ST farmers.

    As per Agriculture Census, Small and marginal farmers constitute **86.2% of all Indian cultivators**, yet less than 15% have independent digital access to download certified land records
  - **Data Privacy and Cyber Threats:** Centralized land databases lack robust, immutable cryptographic firewalls, rendering vital property records vulnerable to data breaches and backend manipulation.
  - **Displaced Agrarian Tenants:** Modernizing title clarity without formalizing tenant protection often drives landlords to evict informal sharecroppers to prevent them from claiming occupancy rights.

    As per NITI Aayog, 90% of land tenancy in India is completely informal and unrecorded.

Measures Taken for Modern Land Governance in India

Initiative Name	Mechanism
DILRMP Saturation Phase	Transitioned into a 100% centrally funded scheme, achieving near-saturation in Records of Rights (RoRs) digitization.
DILRMP 3.0 Real-Time Updates	Initiated the next-generation phase focusing on dynamic, automated real-time record updates across state systems.
NITI Aayog Model Bill (2020)	Provided a statutory blueprint for states to introduce localized laws that substitute presumptive titles with absolute state guarantees.
NGDRS Rollout	Deployed a uniform **One Nation, One Registration** pipeline for online, standardized deed registrations.
Bhu-Aadhaar (ULPIN)	Assigned a 14-digit alphanumeric, geo-coordinated identity to over 36 crore land parcels to eliminate boundary duplicity.
SVAMITVA Yojana	Leveraged high-resolution drone mapping across rural inhabited (Abadi) zones to distribute legal **Property Cards.**
Digital Public Land Stack	Built an interoperable layer integrating land registers, municipal mutations, land-use logs, and agri-credit delivery.
NAKSHA Urban Mapping	Extended 3D geospatial mapping to major cities, replacing legacy physical municipal maps with aerial registries.
e-Courts Interoperability	Tied property registries directly into the e-Courts judicial pipeline to give benches instant access to title histories.
Revenue Court Computerization	Linked rural revenue courts directly to the main database to trigger automated record updates post-judicial decrees.

Linguistic Harmonization	Deployed AI-driven C-DAC tools to automatically transliterate records across 22 Scheduled VIII languages.
Bhoomi Samman Awards	Institutionalized a national reward framework for districts achieving over 99% digitization completion to foster competitive federalism.

Way forward

- **Legislative & Statutory Reforms**
  - **Adopt Conclusive Titling Legislations:** Enact state-level laws modeled on the **NITI Aayog Model Bill** to replace the legacy presumptive deed system with absolute, state-guaranteed property titles.
  - **Amend the Registration Act of 1908:** Mandate sub-registrars to legally verify underlying land ownership and historical title chains rather than merely recording transaction deeds.
  - **Legal Recognition for Land Leasing:** Implement formal, state-backed land leasing frameworks to protect tenant cultivators' rights while assuring landowners against illegal occupation.
- **Technical Enhancements & Data Integration**
  - **Enforce a Clean In, Clean Out Protocol:** Mandate rigorous physical ground-truthing before uploading data to ensure legacy errors and historic frauds are not permanently hardcoded onto digital portals.
  - **Bridge Departmental Silos:** Fully integrate the disconnected databases of **Revenue** (textual RoRs), **Survey** (spatial maps), and **Registration** into a singular, dynamically updating open-API pipeline.
  - **Deploy Blockchain Ledgers:** Scale up decentralized cryptographic ledgers across all state land registries to permanently block unauthorized backend data tampering and manual overrides.
  - **Synchronize Ground-to-Record Areas:** Institutionalize continuous drone-based audits through **SVAMITVA** and **NAKSHA** frameworks to settle spatial area discrepancies dynamically.
- **Socio-Administrative & Judicial Adaptations**
  - **Respect Customary Tribal Tenures:** Develop flexible, non-individualistic digital registry templates for **Northeast and Fifth/Sixth Schedule regions** to safely accommodate community-held land patterns.
  - **Institutionalize Gender Budgeting in RoRs:** Enforce a mandatory gender-disaggregated column in land registries to systematically secure and track women's co-ownership and inheritance rights.
  - **Establish Fast-Track Land Tribunals:** Set up dedicated online grievance portals integrated with *Revenue Court Case Management Systems (RCCMS)* for the time-bound, localized resolution of boundary conflicts.
  - **Link Land Stack with Financial Pipelines:** Interconnect the national Land Stack with the **Account Aggregator framework** to enable instant, algorithmic agri-credit flow to smallholders based on verified digital assets.

Conclusion

The modernization of land records marks a structural paradigm shift from an archaic, presumptive framework toward a transparent, data-driven Digital Public Infrastructure ecosystem. By integrating departmental silos through an interoperable **Land Stack** and scaling up immutable blockchain security, India can decisively dismantle legacy title disputes and administrative corruption.

■■■

MGP/BP/GS3/028

INFRASTRUCTURE

1. ROAD INFRASTRUCTURE IN INDIA

Road infrastructure serves as an important means of communication that provides last mile connectivity. It is the basic infrastructure and an important driver of rapid economic growth.

Introduction:

- India's road infrastructure in 2025 is one of the largest and fastest-growing in the world, playing a crucial role in economic development, national connectivity, and logistics efficiency.
- As of March 2025, India has over 63 lakh km of road network, making it the second-largest globally. This includes 1,46,204 km of National Highways, 1,79,535 km of State Highways, and a vast rural road network.
- The National Highway length increased by 60% in the past decade, from 91,287 km in 2014 to 1,46,195 km in 2024.

Development of road infrastructure in India:

- **Nagpur Road Plan:** In 1943, the Indian Roads Congress proposed a 20-year plan to increase the road density in the country, with a target of 16 km per 100 km2 of land. This massive construction of roads was to be enhanced through the duties imposed on petrol. This plan is popularly known as the Nagpur Plan. Its targets were achieved in the 1950s.
- **Bombay Road Plan:** To further expand the road network in the country, another 20-year plan was proposed for the period 1961-81. It laid special emphasis on the construction of National and State Highways. It had an ambitious target of increasing the road density in the country to 32 km per 100 km2.
- In 1988, the **National Highways Authority of India (NHAI)** was established as an autonomous body by an Act of the Parliament. NHAI was entrusted with the development, maintenance, and management of India's National Highways network.
- In 1998, the **National Highways Development Project (NHDP)** was started. It aimed at developing the Golden Quadrilateral that links four major cities of the nation (Mumbai, New Delhi, Kolkata, and Chennai) in its first phase and in subsequent phases, improving and developing national highways. In 2018, NHDP was subsumed in Bharatmala Pariyojana.
- After independence the First Five Year Plan gave a four-fold classification of roads in the country:
  - **National Highways (NH):**
    - Roads built and maintained by the Central Government (NHAI) that link major cities and states.
    - They are essential for interstate travel, logistics, and military mobilisation.
    - They account for just 2% of the total road network in the country but carry around 40% of the total traffic.

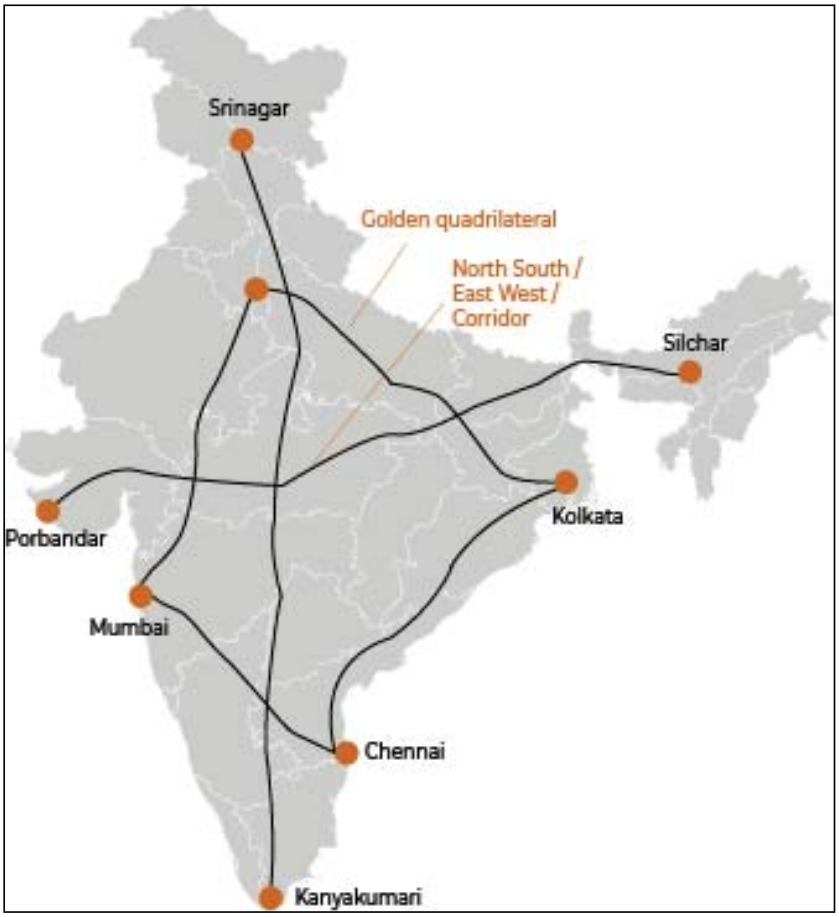
- **State Highways (SH):**
  - These are roads that are built and maintained by the state government.
  - They link state capitals, major cities, and district headquarters.
  - They often eventually connect to the NHs.
  - SHs constitute around 4% of the total road network.
- **Major District Roads and Other District Roads:**
  - These roads are an important link between the district and other important nodes of the district.
  - They account for around 14% of the total road network in the country.
- **Village Roads:**
  - They provide linkages between various villages and district headquarters.
  - More than 70% of the roads are classified as rural roads.

Steps that led to rapid road development:

- **Technological changes:** In the initial stages of development, employment generation was linked to road development. This meant that labour-intensive road construction techniques were deployed that were cost inefficient and of low quality. This linkage was abandoned, and newer technological changes were introduced. High-tech road construction equipment was deployed, which led to faster, better, and more cost efficient road construction.
- **Leveraging PPP:** The government has a limited capacity in terms of financial resources. This was a major roadblock to any infrastructure development. To overcome this, Public Private Partnership as a possible solution was proposed. Private players had the technical know-how and the financial resources to execute the project in cooperation with the government.
- **Organisational changes:** Over the years, various organisational changes were made to bring in efficiency in the system and to follow a more focused approach. For example, NHAI was tasked with highway development and maintenance, NHIDCL was tasked with road development in border areas, etc.
- **Viability Gap Funding:** Leveraging PPP in road development was seen as an important strategy. But the private player shied away from taking part in large road construction projects due to the risk involved. To mitigate this risk and make the projects more attractive to private players, the government offered viability gap funding (VGF). VGF has attracted interest from the private players and expedited infrastructure development.
- **Focus on Rural Roads:** PMGSY has been the most successful road construction project in the country. Many rural roads were constructed based on objective criteria. The project was placed under the Ministry of Rural Development rather than the Ministry of Road Transport and Highways to focus on the demand side rather than the supply. The project was overseen by various independent agencies for better realisation of goals.
- **New contracting models:**
  - Infrastructure development has been done through the Engineering, Procurement, and Construction (EPC) model or the PPP through the Build, Operate, and Transfer (BOT) model.
  - The Hybrid Annuity Model (HAM) and the Toll, Operate, Transfer (TOT) have emerged as preferred options among private players. The HAM offers better risk allocation, as 40% of the capital cost is borne by the authority and 60% is paid to the private player in annuities over the years. In TOT, private players maintain a built road and collect tolls.
- **Electronic Toll Collection (ETC):** ETC penetration was around 96% by March 2022. ETC has ensured that there is better revenue and fewer leakages. In addition to this, now the vehicles need not stop for a long time at the toll plaza, which leads to energy savings and cuts carbon footprints. Recently the government has announced that toll will now be calculated only for the distance travelled on the road through a satellite tracking and computing mechanism to eliminate toll booth choke points.

Government schemes/projects for development of road infrastructure:

- **Pradhan Mantri Gram Sadak Yojana (PMGSY):**
  - It was started in the year 2000. It aims to provide all-weather road connectivity to unconnected rural habitations, significantly improving rural access to markets and services.
  - It is implemented under the overall authority of the Ministry of Rural Development.
  - It is a fully funded centrally sponsored scheme i.e. 60% Centre and 40% states except for Northeastern states and Himalayan states (90:10 share).
  - 75 paise/litre of the cess levied on high speed diesel is earmarked for this scheme.
- **Bharatmala Pariyojana:**
  - A flagship umbrella program focusing on the construction of economic corridors, border and coastal roads, national corridors, inter-corridor and feeder routes, and improvement of rural roads.
  - Targets building ~83,677 km of new highways and improving existing ones with an investment of over ₹5 lakh crore.
- **National Highways Development Project (NHDP):**
  - Initiated in **1998**, implemented by **NHAI**.
  - Key components:
    - **Golden Quadrilateral (GQ)**: 5,846 km connecting Delhi–Mumbai–Chennai–Kolkata.
    - **North–South & East–West Corridors**.
    - Upgradation of high-density corridors.
  - It has now been subsumed under Bharatmala Pariyojana.



- **National Infrastructure Pipeline (NIP):** Prioritizes and allocates funds for upgrading and expanding the road network including national highways and state highways.
- **Gati Shakti National Master Plan:** An integrated planning initiative for infrastructure projects to ensure convergence and reduce delays across different modes of transport like roads, rail, ports, and airports.

Significance of Road infrastructure:

- **Driver of Economic Growth:** Roads are vital for the smooth movement of goods, agricultural produce, and raw materials, lowering transportation costs and boosting productivity.
- **Employment Generation:** Construction, maintenance, and expansion of roads directly generate large-scale employment, including semi-skilled and unskilled jobs, fostering economic empowerment at the grassroots.
- **Balanced Regional Development:** Roads improve connectivity for remote and rural areas, enabling access to markets, healthcare, education, and government services. Enhanced road links reduce regional inequalities, promote inclusive growth, and help alleviate poverty.
- **National Security & Strategic Mobility:** Strategic road corridors along borders and coasts enable swift movement of defense personnel, supplies, and disaster relief, reinforcing India's national security. Border road development is critical for sovereignty, military readiness, and regional stability.
- **Social Mobility & Inclusion:** Roads serve as lifelines for millions, connecting people to economic opportunities, schools, and health centers, particularly in underserved areas. Improved road access enhances social mobility and boosts living standards for rural populations.
- **Urbanisation:** Urban road development supports decongestion, planned expansion, public transport integration, and the creation of smart cities.
- Sustainable Development: Modern road projects incorporate green technologies and promote sustainable growth.

Challenges related to road infrastructure:

- **Land Acquisition & Clearances:**
  - Delays in land acquisition due to high land prices, legal disputes, and rehabilitation issues slow project timelines and inflate costs.
  - Environmental, forest, and regulatory clearances are often time-consuming and complex.
- **Funding & Cost Overruns:** Escalating costs of construction materials, slow fund disbursal, and dependence on public finances or limited PPP investments can cause project delays and cancellations.
- **Project Execution & Quality Control:**
  - Weak contractor management, delays by executing agencies, and poor oversight lead to construction delays and substandard quality.
  - Insufficient adoption of advanced technologies and global best practices in design, monitoring, and maintenance reduce road longevity.
- Focus on new construction often comes at the cost of neglecting maintenance and repair of existing roads, causing premature deterioration and safety hazards.
- **Urban Congestion:** Rapid urbanization strains existing city road networks, leading to chronic congestion, pollution, and unsafe travel conditions.
- **Road Safety Issues:** India's roads report some of the world's highest traffic fatalities, with poor signage, inadequate lighting, and unsafe crossings contributing to risks.
- **Environmental Concerns:** Construction activities and expanding road networks sometimes negatively impact the environment, including deforestation and pollution.

- **Regional connectivity:** Many rural and hilly regions still lack reliable all-weather roads due to topographical challenges and limited local funding.

Way forward:

- **Streamline Land Acquisition & Regulatory Clearances:**
  - Simplify and expedite land acquisition processes with transparent compensation and dispute resolution mechanisms.
  - Fast-track environmental, forest, and statutory clearances using digital platforms and single-window clearances.
- **Financial Models:** Increase use of public-private partnerships (PPP), infrastructure investment trusts (InvITs), and asset monetization to attract private capital.
- **Quality Enhancement:** Adopt advanced construction technologies such as mechanization, drones for monitoring, and smart pavement materials to improve quality and durability.
- **Urban Mobility:**
  - Enhance urban road design with smart traffic management, non-motorized transport infrastructure, and pollution control measures.
  - Develop multimodal transport hubs integrating road, rail, metro, and last-mile connectivity.
- **Road Safety:** Implement stringent road safety standards, install proper signage, lighting, and pedestrian crossings.
- **Regional Development & Inclusion:** Prioritize rural and difficult terrain connectivity via targeted schemes such as Pradhan Mantri Gram Sadak Yojana (PMGSY).
- **Sustainable Practices:** Promote green construction practices, use recycled materials, and encourage adoption of electric and hydrogen fuel vehicles.

Conclusion:

Road infrastructure plays a significant role in the growth of any economy. Road development brings prosperity, generates employment, and helps sustain growth in the long term. However, the challenges in the sector need to be addressed with high priority. To achieve SDG 9, the pace of development in the sector needs to be sustained so that functioning and resilient infrastructure are made available to all.

2. RAILWAY SECTOR IN INDIA

The Indian Railways (IR), often called the lifeline of India, is central to the country's socio-economic development. Since its establishment in 1853, it has grown into one of the world's largest and busiest railway networks. **Covering over 68,000 km of track**, it serves more than 23 million passengers daily and ranks as the **fourth-largest railway system** globally, after the US, China, and Russia.

Significance of Indian Railways

- **Economic Significance**
  - **Freight Transport:** Railways are indispensable for moving bulk commodities like coal, cement, iron ore, petroleum products, fertilizers, and agricultural goods. In **FY25**, Indian Railways loaded **1,617 million tonnes (MT)** of freight, generating revenue of **₹1.75 lakh crore (US$ 20.5 billion).**
  - **Revenue Contribution:** Overall revenue in FY25 reached **₹2.70 lakh crore (US$ 31.57 billion)**, with **freight accounting for approximately 65%** and passenger services making up the rest.
  - **Industrial Development:** Railways ensure smooth industrial and port connectivity, enabling **efficient supply chains** for FMCG, automobiles, containerized cargo, and consumer durables thus enhancing India's economic competitiveness.

- **Social Significance**
  - **Mass Connectivity:** With over **13,000 passenger trains**, including Mail, Express, suburban, and local services, Indian Railways links urban centers and remote villages alike, bringing people together across regions.
  - **Affordability:** For millions of Indians, rail transport remains the **most cost-effective mode** of long-distance travel, ensuring accessibility and social inclusion.
  - **Employment Generation:** As one of the **largest employers** in the country, Indian Railways provides livelihoods to millions directly and indirectly, supporting national socio-economic stability.
- **Environmental Significance**
  - **Energy Efficiency:** Rail transport consumes significantly less energy per **tonne-km for freight** and per **passenger-km** compared to road or air transport, making it an environmentally friendly option.
  - **Electrification:** Nearly **97% of broad-gauge lines are electrified**, aligning with India's sustainability goals under the Paris Agreement and reducing dependency on fossil fuels.

Market Size and Economic Impact

Passenger Traffic	Reached 715 crore passengers in FY25, a growth of 67 crore from the previous year.
Revenue from Passenger Traffic	₹94,927 crore (US$ 11.10 billion) in FY25, up from ₹70,693 crore in FY24.
Revenue from Freight	₹1.75 lakh crore (US$ 20.50 billion) in FY25, up from ₹1.68 lakh crore in FY24.
FDI in Railway Sector	₹9,155 crore (US$ 1.42 billion) between April 2000 and December 2024.
Capital Outlay	Budget 2024-25 allocated ₹2.52 lakh crore (US$ 30.3 billion) for infrastructure and modernization.

Structure and Governance of Indian Railways

The **Ministry of Railways** oversees policy-making, budget allocation, project approvals, and strategic guidance, while the **Railway Board** manages daily operations and long-term planning.

As of September 2025, Indian Railways comprises 18 operational zones, with the South Coast Railway (SCoR) officially becoming the 18th zone. Each zone is further divided into divisions, headed by **Divisional Railway Managers**, who supervise local operations, track maintenance, safety, and revenue collection.

Key Responsibilities:

- **Ministry of Railways:** Policy formulation, budget allocation, project approvals, facilitation of FDI.
- **Railway Board:** Operational oversight, modernization initiatives, safety regulation, and coordination among zones.
- **Zonal Railways:** Maintenance of tracks, rolling stock management, and daily passenger and freight operations.

Modernization and Technological Advancements

Indian Railways has embraced **cutting-edge technology** to enhance operational efficiency, passenger experience, and safety.

- **Semi-High-Speed Trains**
  - **Vande Bharat Express:** Operating at speeds of up to **160 kmph**, these trains are equipped with onboard infotainment, GPS-based passenger information, automatic doors, retractable footsteps, vacuum bio-toilets, and CCTV cameras. By **December 2024**, **136 Vande Bharat trains** were operational.
  - **Global Export Plans:** India aims to export these semi-high-speed trains to **Europe, South America, and East Asia** by 2025-26.
- **Safety Technologies-**
  - **KAVACH:** An indigenous **Train Collision Avoidance System** being rolled out over **37,000 km of track**, reducing the likelihood of collisions.
  - **Automatic Signalling:** Approximately **15,000 km of track** have been upgraded to automatic signalling, minimizing human error.
  - **Safety Improvements:** Consequential rail accidents declined **by 80% between 2016-21**, though **95% of accidents** are still linked to driver errors, highlighting the importance of further technological interventions and comprehensive training programs.
- **Digitisation and Connectivity**
  - **Wi-Fi:** Available at **6,045 stations**, connecting millions of passengers to the digital world.
  - **PM Gati Shakti Initiative:** Integrates **multi-modal logistics** to streamline freight and passenger movement.
  - **Predictive Analytics & Digital Twins:** AI-driven simulations help in **proactive maintenance, capacity optimization, and real-time decision-making**.

Challenges and Issues in India's Railway Sector

- **Infrastructure Overload** – Many routes operate at over **150% of their designed capacity**, causing congestion, delayed maintenance, and higher accident risks.
- **Safety Concerns** – Outdated technology, human error, unmanned level crossings, and inadequate driver training contribute to accidents. Committees such as **Kakodkar (2012), Bibek Debroy (2014), Vinod Rai (2015), and Rakesh Mohan (2010)** have recommended comprehensive reforms in safety, regulation, and human resources.
- **Financial Constraints** – Populist fare policies and limited private investment strain finances, affecting infrastructure development, modernization, and technological upgrades.
- **Organizational Bottlenecks** – Excessive centralization and political interference slow down decision-making and project implementation. Experts emphasize the need for **corporatization, decentralization, and autonomous zonal operations**.
- **Human Resource and Service Quality Issues** – Staff shortages, inadequate training, long working hours, and gaps in service quality—including cleanliness, food, and punctuality—affect operational efficiency and passenger satisfaction.

Government Initiatives and Reforms in Indian Railways

- **Financial and Administrative Reforms**
  - **Dynamic Fare Pricing:** Helps optimize revenue from passenger services while balancing affordability.
  - **Merger of Rail and Finance Budgets:** Ensures a unified approach to funding major infrastructure projects.
  - **Rail Infrastructure Development Fund (RIDF):** Provides long-term financing for rail network development.
  - **Rail Drishti Portal:** Promotes transparency and accountability by offering real-time operational insights.

- **Organizational Restructuring**
  - **Indian Railway Management Service (IRMS):** A unified management cadre to improve operational efficiency.
  - **Digital HR Systems:** Streamlined employee monitoring and management for better administrative control.
- **Infrastructure Development**
  - **National Rail Plan (NRP) 2030:** Lays out a roadmap to meet traffic demands projected for 2050.
  - **Vande Bharat Trains:** India's semi-high-speed trains, designed to meet global standards.
  - **Dedicated Freight Corridors (WDFC & EDFC):** Supported by Japan and the World Bank to decongest passenger lines and enhance freight efficiency.
- **Safety and Security**
  - **Railway Safety Fund (RSF):** Targets elimination of unmanned level crossings to reduce accidents.
  - **Zero Accident Mission & KAVACH:** Advanced systems aimed at preventing collisions.
  - **MeriSaheli Initiative:** Focused on ensuring women's safety on trains and stations.
  - **CORAS Railway Commando Battalion:** Special units for counter-terrorism and anti-Naxal operations.
- **Freight and Logistics**
  - **Freight on Priority Policy:** Ensures timely and customer-focused freight services.
  - **Kisan Rail:** Dedicated trains for transporting agricultural produce efficiently.
  - **Gati Shakti Cargo Terminal Policy:** Encourages development of cargo terminals to support multi-modal logistics.
- **Tourism Promotion**
  - **Bharat Gaurav Scheme:** Theme-based trains designed to promote tourism and enrich passenger experience.
- **Railways (Amendment) Bill, 2024**
  - Provides statutory backing to the Railway Board.
  - Modernizes outdated provisions from the Indian Railway Board Act of 1905.
  - Introduces independent regulators for tariffs, safety, and private sector participation.
  - Grants greater autonomy to individual railway zones for efficient functioning.

Recommendations from Expert Committees

- **Kakodkar Committee (2012):** Statutory Railway Safety Authority, non-lapsable safety fund, advanced technology adoption.
- **Bibek Debroy Committee (2014):** Outsourcing non-core activities, separating budget, Railway Infrastructure Authority.
- **Vinod Rai Committee (2015):** Independent Railway Safety Authority, Railway Accident Investigation Board, performance-linked incentives.
- **Rakesh Mohan Committee (2010):** Accounting reforms, expansion into FMCG, containerized cargo, and inter-city high-speed corridors.

Measures to Improve the Railways Sector

- **Modernisation of Infrastructure** – Tracks, stations, locomotives, and signalling systems need significant upgrades. Emphasis should be placed on **high-speed trains and LHB coaches**, while **ICT integration** can enable real-time monitoring, mobile ticketing, and enhanced passenger services.
- **Safety Enhancement** – Establishing an **independent Railway Safety Regulatory Authority** is crucial to enforce strict safety standards. Efforts should also focus on eliminating manned and unmanned level

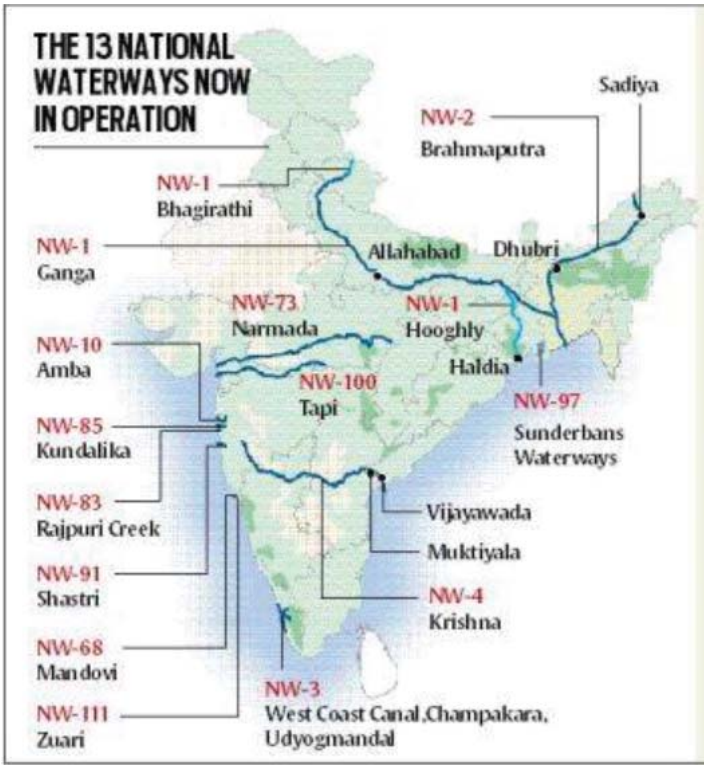
crossings, upgrading signalling systems, and implementing advanced technologies like KAVACH to prevent collisions.

- **Integrated Multimodal Logistics** – Developing logistics parks, intermodal hubs, and door-to-door freight solutions will enhance supply chain efficiency and reduce congestion on passenger routes.
- **Public-Private Partnerships (PPP) and FDI** – Encouraging private operators and foreign investment in both freight and passenger services can bring in global technology, expertise, and improved service standards, boosting overall efficiency.
- **Human Resource Reforms** – Addressing **staff shortages**, improving **training programs**, professionalizing recruitment, and focusing on employee welfare and productivity will help reduce errors and operational delays.
- **Commercial Utilisation of Railway Land** – Optimizing railway-owned land and airspace for retail, real estate, and industrial projects can generate non-fare revenue, which can be reinvested into infrastructure and modernization.
- **Customer Service Improvements** – Enhancing punctuality, cleanliness, and food quality, along with digital ticketing and real-time train tracking, will significantly improve the overall passenger experience and satisfaction.

3. INLAND WATERWAYS OF INDIA

Inland Water Transport refers to the transportation of people, goods, and materials via waterways such as **rivers, canals, lakes, and other navigable bodies** of water that are located within a country's boundaries.

Inland Waterways in India: Under the **National Waterways Act, 2016, 111 waterways** (including **5 existing and 106 new**) spread over **24 States** have been declared **National Waterways (NWs)** for development of **Inland Water Transport (IWT)** in the country. On the basis of the outcome of techno-economic feasibility study and Detailed Project Report (DPR), **26 NWs** which were found viable for cargo and passenger movement.



THE LINKS AND THE LENGTHS

NW-1	Ganga–Bhagirathi–Hooghly (Haldia-Allahabad)	1,620 km
NW-2	Brahmaputra river	891 km
NW-3	West Coast Canal–Champakara Canal–Udyogmandal Canal	205 km
NW-4	Krishna (Muktiyala–Vijayawada)	82 km
NW-10	Amba river	45 km
NW-83	Rajpuri Creek	31 km
NW-85	Revadanda Creek–Kundalika river	31 km
NW-91	Shastri river-Jaigad Creek System	52 km
NW-68	Mandovi river (Usgaon Bridge–Arabian Sea)	41 km
NW-111	Zuari river (Sanvordem Bridge–Marmugao Port)	50 km
NW-73	Narmada river	226 km
NW-100	Tapi river	436 km
NW-97	Sunderbans Waterways	172 km

Some recent data related to Inland Waterways in India

- India achieved a record **145.5 million tonnes** cargo movement on inland waterways in **FY 2024–25**, up from 18.1 MMT in FY 2013–14, registering a CAGR of **20.86%.**
- The **number of National Waterways increased** from **5 to 111,** with the operational **length** growing from **2,716 km (2014–15) to 4,894 km (2023–24).**
- **Massive infrastructure development** including Multi-Modal Terminals (**MMTs**), Inter-Modal Terminals (**IMTs**), community jetties, floating terminals, and green tech like Hybrid Electric and Hydrogen Vessels.
- Launch of **Jalvahak Scheme** with **₹95.42 crore** budget offering 35% operating cost incentive for cargo owners and scheduled services on key routes (NW-1, NW-2, NW-16).
- **Future aims:** India **aims** to **increase IWT modal share from 2% to 5%,** and **raise traffic to 200+ MMT by 2030 and 500+ MMT by 2047** under **Maritime Amrit Kaal Vision.**

Advantages of Inland Waterways in India

- **Lower Transportation Costs:** Inland water transport (IWT) is considerably cheaper than road and rail transport. It is reported to be 60% cheaper than road transport and 20-30% cheaper than rail.
- **Fuel and energy efficiency:** It is a fuel and energy efficient medium of transportation as compared to the other modes of transport like rail and road. A/c to the Integrated National Waterways Transportation Grid Study states that one litre of fuel will move 24 tons through 1km on road, 85km on rail and 105 km on inland water transport.
- **Environment Friendly:** IWT emits 10 times less carbon dioxide per ton-kilometer compared to road transport, contributing to lower greenhouse gas emissions.
- **Less Land Requirement:** Utilization of waterways minimizes the need for extensive land acquisition. Land acquisition is often a contentious and time-consuming process associated with road and rail projects.
- **Employment Generation:** The development of inland waterways is labor-intensive, creating jobs in various sectors such as river conservancy, vessel operation, terminal management, and tourism.
- **Better connectivity:** IWT helps in creating seamless interconnectivity connecting hinterlands along navigable river coasts and coastal routes. Further, riverine routes are likely to play a crucial role in connecting the north-eastern states to the mainland.
- **Tourism Development:** Inland waterways can boost tourism through initiatives like river cruises. This helps to enhance passenger movement and also promote cultural and heritage tourism along riverbanks.

Steps taken by the Government for the development of Inland Waterways

Measures taken	Explanations
Maritime India Vision 2030	Inland Water Waterways Authority of India (IWAI) aims to increase the modal share of freight movement through IWT from 2% to 5% and traffic volume to more than 200 MMT in line with the Maritime India Vision 2030.
Jal Marg Vikas Project	Jal Marg Vikas Project (JMVP) aims at capacity augmentation of navigation on National Waterway-1 (NW-1).The project is being implemented by GOI with technical assistance and investment support from the World Bank.
Sagarmala Programme	It aims at enhancing multi-modal connectivity including rail, inland water, coastal and road services.

Interlinking of Rivers Programme	The project is expected to offer potential benefits to the inland waterways transport sector through navigation.
Jalvahak – Cargo Promotion Scheme	Launched in December, 2024; the scheme covers cargo movement on NW-1, NW-2, and NW-16, benefiting surrounding regions and building trust in waterway transport.
Extension of Tonnage Tax to Inland Vessels	The tonnage tax regime has been extended to inland vessels registered under the Indian Vessels Act, 2021. **Benefit:** Provides a stable and predictable tax regime based on vessel tonnage rather than profits, thereby lowering the tax burden and encouraging broader adoption of inland shipping.
Regulatory Framework for Private Investment	The National Waterways (Construction of Jetties/Terminals) Regulations, 2025 have been notified, enabling private investment in inland waterways infrastructure by establishing a clear legal and operational framework for the construction and management of jetties and terminals.

Challenges in Inland Waterways in India

IWT is the most economical mode of transportation, especially for bulk cargo like coal, iron ore, cement, food grains and fertilizer. However presently the IWT sector remains underutilized at a share of 2% in India's modal mix due to the following challenges-

- **Infrastructural/Technical challenges:** There is a challenge of Inadequate depths in Indian rivers, especially northern rivers which suffer from high siltation. This necessitates extensive dredging of Indian rivers**.**
- **Lack of adequate Terminals:** There is a shortage of modern terminals and maintenance, repair, and overhaul (MRO) facilities for vessels, impacting their operational efficiency.
- **Environmental Concerns:** Dredging activities necessary for maintaining navigability can lead to environmental degradation (like riverbed damage, aquatic species damage) and community resistance due to fears of displacement or ecological harm.
- **Technical Limitations:** Low vertical clearances from bridges obstruct the passage of bigger vessels and the lack of Night Navigation Facilities such as Differential Global Positioning Systems (DGPS) hampers safe navigation at night.
- **Institutional and Regulatory challenges:** The IWT sector is often burdened by regulatory complexities and overlapping jurisdictions among various government agencies. For ex- Multiple authorities including the Central Inland Water Corporation Limited (CIWTC Ltd), port authorities and state governments.
- **Lack of funds:** Dredging as well as infrastructure for IWT requires huge investments. However, both public and private funding in the sector is low.

Way Forward

NITI Aayog has recommended the following measures for the development of Inland Waterways in India.

- **Streamline the governance of inland waterways:** NITI Aayog recommends streamlining the regulatory structure and bringing an overarching body to oversee Inland Water Transport.
- **Develop measures for year-round navigation:** Efforts should be made to develop deeper stretches of the river, i.e., at least 2.5 m to 3 m to achieve year-around navigation. Further the adequate maintenance of rivers, including continuous dredging to maintain adequate water depth for servicing shipping lines should be ensured.

- **Ease restrictions on river-sea movement:** Utilizing a single vessel for both inland and coastal waters, lowers transport costs and minimizes handling. The state authorities should draw up coordinates for inland vessel limits under the Inland Vessel Act for their coastal waters
- **Creation of Inland Waterways transport hub in the North East:** State governments should commence work on dredging and channel stabilization to create about 20 new ports in the Brahmaputra and Barak rivers. The protocol for Inland Waterways between Bangladesh and India should be extended for at least 10 years to reduce uncertainty.
- **Strengthen public-private partnership:** Private players can undertake terminal development, cargo and passenger handling, and building low-draft vessels and related repair facilities.
- **Increasing economic activities along the Inland waterways:** PM Mitra parks, Mega food parks, should be located along the inland waterways route to encourage use of National Waterways (NWs).

Conclusion

India's concerted efforts in developing its inland waterways have yielded significant results, with record cargo movements and expanded infrastructure. The combination of strategic investments, policy initiatives, and digital innovations positions the country to further enhance its IWT sector, contributing to sustainable transportation and economic development. Continued focus on these areas will be crucial in achieving the ambitious targets set for the coming decades.

4. AVIATION SECTOR IN INDIA

The Aviation Sector in India has experienced significant growth in recent years. India has become the third-largest domestic aviation market in the world. According to the International Air Transport Association (IATA), by the year 2030, India is expected to overtake China and the United States as the world's largest air passenger market. However, the aviation sector in India also faces numerous challenges, as highlighted by the recent insolvency of Go First Airlines and grounding of Spice jet aircrafts on account of safety concerns.

Status of Aviation Sector in India

- India's aviation industry has experienced significant growth in the past 9 years.
- India is the third-largest domestic aviation market globally, with domestic air passenger traffic reaching 131.7 lakh in August 2025.
- Indian aviation also contributed 5% of the GDP, creating a total of 4 million jobs. In addition to it, there is a 72 billion dollars gross value-added contribution to GDP by this industry.
- The Indian aviation fleet consists of over 860 aircraft, accounting for around 2.4% of the global fleet, and total operational airports increased from 74 in 2014 to 162 in 2025.
- Passenger traffic from January to April 2025 showed significant growth (nearly 10% year-on-year), led by low-cost carriers (64.3% market share).

Potential of the Aviation sector in India

- **Increased Market Size of Indian Aviation Sector:** According to IATA, India is expected to surpass the aviation sector of the United States and China by 2030. This will make India a lucrative market for airlines and related businesses.
- **Promotes Balanced Economic Growth:** Passenger airlines and air cargo overcome geographic barriers by connecting remote areas which are alienated from the mainstream. For e.g. Development of North-East due to enhanced airline connectivity.
- **Growth of Tourism sector-** Aviation industry functions as a growth pole by promoting spill-over & trickling-down of economic growth for e.g. Aviation sector gives a boost to tourism sector which in

turn drives the supporting infrastructure in a region, like roads, railways, hotels, markets. This helps in providing employment opportunities to the locals.

- **Boost to manufacturing sector:** India's expanding aviation sector offers potential for the growth of maintenance, repair, and overhaul (MRO) facilities, as well as the development of a domestic aerospace manufacturing industry. For e.g. Enhanced employment opportunities in aerospace engine maintenance.
- **FDI in the Expansion of Infrastructure:** Booming aviation sector has attracted significant FDI to the tune of around ~$3 bn in the development of aerospace infrastructure like airports, arrow bridges, airstrips. For e.g. Greenfield airport development like Navi Mumbai, Noida (Jewar) airport and expansion of Bengaluru airport.
- **Increased Employment opportunities:** The growth of the aviation sector in India has created a need for skilled professionals, including pilots, cabin crew, and maintenance staff. For e.g. Indian scheduled operators are likely to require 10,900 additional pilots by FY30 (IATA projection).

Challenges With India's Aviation Sector

- **Increase in number of Grounded Unsafe Aircrafts:** Airlines like Air India, Spice Jet, Go Air, and IndiGo face issues of poor financial performance due to grounded unsafe aircrafts. For e.g. Over 160 aircraft are currently grounded which represents about a quarter of the total fleet size of Indian carriers.
- **Supply Chain Disruptions:** Delays in aircraft deliveries and supply chain issues with original equipment manufacturers (OEMs) has hindered the industry's capacity to meet growing demand.
- **Duopoly in Indian Aviation Market:** India's domestic aviation market is heading in the direction of a duopoly of market leader IndiGo (60% market size) and the Tata group airlines (20% market size).
- **Heavy Financial Losses:** Indian airlines are projected to record a consolidated loss of $1.6 to 1.8 billion in FY24, due to heavy financial bleeding of Go first, Spice Jet and Jet Airways.
- **Operational Disruptions due to crew shortage:** Lack of skilled pilots, maintenance engineers, and cabin crew members have led to operational disruptions like increase in turn-around time of airlines.
- **Low per-capita penetration of domestic air travel:** India's per capita penetration of domestic air travel (0.13 seats deployed per capita) remains significantly lower than countries like China (0.49) and Brazil (0.57). This indicates the failure of the aviation industry in India to tap the maximum potential of the domestic air market.
- **Inflated Projections:** Airlines in India often announce ambitious growth plans without adequately analysing their financial security, infrastructural and personnel requirements. For e.g. Failure of Kingfisher, Jet Airways and Go First on account of inflated projections.
- **Regulations acting as barriers:** Tough entry barriers for new entrants, high fuel prices on account of high taxes on ATF (Air Turbine Fuel) and monopoly of inefficient public sector airports have all acted as barriers in the rapid growth of the airlines sector.
- **Policy Lacunae:** The Aircraft Act, 1934 and Aircraft Rules, 1937 have not kept pace with modern technology in aerospace. This has led to increased costs of the industry's operation and ultimately affected passenger growth.
- **Poor rural connectivity:** With mega airports controlling air and ground space, there has been a challenge of enhancing the rural air connectivity. For e.g. Less number of flights to tier 2 and tier 3 towns despite the UDAN scheme.
- **Environmental Concerns:** The Indian aviation industry faces increasing pressure to reduce its carbon footprint and adopt sustainable practices (The Carbon Offsetting and Reduction Scheme for International Aviation or CORSIA). This has also posed a challenge for growth and expansion of the airline sector.
- **Aviation Safety:** The recent crash of Air India Airline in Ahmedabad into a residential area that led to the killing of more than 240 people & is considered as one of the worst airline disasters.

Government Initiatives For the Aviation Sector In India

National Civil Aviation Policy, 2016	This aims to improve the international footprint of India-based airline services. Airlines can commence international operations, provided they deploy 20 aircrafts or 20% of their total capacity (whichever is higher) for domestic operations.
UDAN Scheme	This aims to expand access to air travel for Tier 2 and Tier 3 cities and shift the traffic pattern away from Metro routes.
DigiYatra	Paperless, biometric-enabled air travel to enhance passenger convenience.
Open Sky Policy	Aims to liberalise the aviation sector in India by opening the airport sector to private participation. Currently, 6 PPP airports are being developed and 60% of airport traffic is handled under PPP.
Open Sky Air Service Agreement	Open Sky Air Service Agreement allows for airlines from the two countries to have an unlimited number of flights as well as seats to each other's jurisdictions. India has signed these agreements with multiple nations like the US, Greece, Jamaica, Japan, Finland, Sri Lanka.
FDI Policies, Tax and Duty cuts	100% FDI is being allowed under the automatic route for greenfield projects, whereas 74% FDI is allowed under automatic route for brownfield projects. 100% tax exemption has been provided for airport projects for a period of 10 years. Indian aircraft Manufacture, Repair and Overhaul (MRO) service providers have been completely exempted from customs and countervailing duties.
Bhartiya Vayuyan Adhiniyam, 2024	This is a landmark piece of legislation that came into effect on January 1, 2025, replacing the nearly century-old Aircraft Act of 1934. It modernizes India's aviation system, aligning it with contemporary needs and global standards (like the Chicago Convention and ICAO).
GAGAN (GPS-Aided Geo-Augmented Navigation)	Developed jointly by AAI and ISRO, GAGAN is India's own Satellite-Based Augmentation System (SBAS). Operational since 2015, it enhances the accuracy and integrity of GPS signals, improving navigation, especially for approach and landing, and enabling precision approaches at non-instrumented airports, thereby significantly enhancing safety, particularly in challenging terrains.

Way Forward

- **Capacity Expansion:**
  - Continue development of new airports, modernization of existing ones, and upgrading air traffic management systems.
  - Strengthen regional connectivity under the UDAN scheme by incentivizing flights to underserved and remote areas.
- **Boost Cargo & Logistics:** Develop dedicated air cargo hubs and multimodal logistics parks to support India's export-import trade.
- **Initiate reforms in the Directorate General of Civil Aviation (DGCA):** DGCA should be modernized, well-staffed and incentivised. DGCA should be headed by aviation professionals rather than bureaucrats.
- **Promote 'Start-up India' initiative in the aviation sector:** Entrepreneurship must be promoted in the maintenance, repair, and overhaul (MRO) facilities of the aviation industry.
- **Rationalisation of taxes:** Tax rationalisation must be initiated in aviation fuel taxes (State and Central, which in India are among the highest in the world), air cargo and airport operations.

- **Safety Management Systems (SMS) Effectiveness:** Ensure that all aviation service providers (airlines, airports, MROs, ATC) have fully functional and effective SMS that are integrated into their daily operations and decision-making. Regulators should audit the effectiveness of these SMS, not just their presence.

Conclusion

By addressing these challenges and implementing the suggested reforms, India can pave the way for a thriving aircraft leasing industry, making the country a global leasing hub and bolstering the aviation sector.

5. MARITIME SHIPPING AND PORTS (MAJOR & MINOR) OF INDIA

India has **12 major and 200+ notified minor and intermediate ports**. Under the **National Perspective Plan for Sagarmala**, **six new mega ports** will be developed in the country. India is the **sixteenth-largest maritime country** in the world with a coastline of **7,516.6 kms.** According to the Ministry of Shipping, around **95% of India's trading by volume and 70% by value** is done through maritime transport.

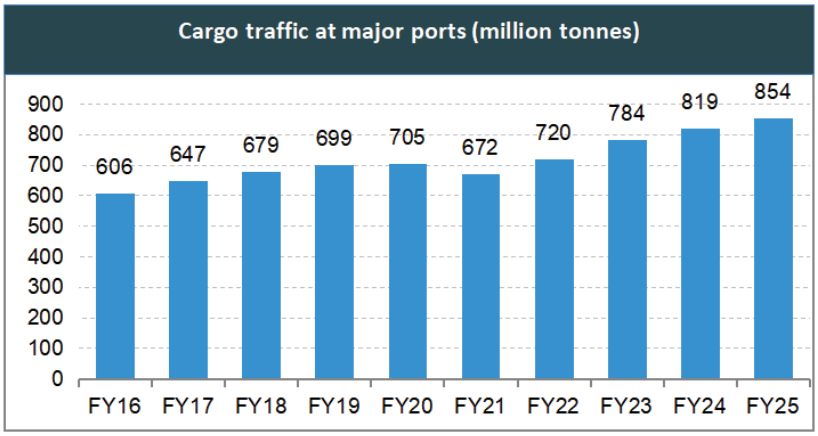
Major Ports of India

There are **12 major ports** in the country; **6 on the Eastern coast and 6 on the Western** coast. These are governed by the provisions of the **Major Port Authorities Act, 2021.**

Western Coast	Eastern Coast
1. Kandla	2. Kolkata (Riverine) Known for twin dock systems viz., Kolkata Dock System (KDS) on the eastern bank and Haldia Dock Complex (HDC) on the western bank of river Hooghly.
3. Mumbai (Busiest and Biggest in India)	4. Paradip
5. Jawaharlal Nehru/ Nhava Seva	6. Vishakhapatnam (oldest shipyard and natural harbour)
7. Marmugao (Natural Harbour)	8. Chennai (oldest and artificial harbour)
9. Mangalore	10. Ennore
11. Cochin	12. Tuticorin (artificial deep sea harbour)

Cargo Major Ports recorded a **4.3% growth in cargo handling**, reaching approximately **855 million tonnes, up from 819 million tonnes in FY24**.

For the first time, **Paradip Port and Deendayal Port** surpassed the **150 mt** cargo handling mark.



Source: Ministry of Shipping

Non-Major Ports of India

Non-major ports come under the jurisdiction of the **respective state Governments' Maritime Boards (GMB).**

Functional Major and Non-Major Ports in India

Sr. No.	State / UT	Non-Major Ports	Major ports
1	Andhra Pradesh	15	1
2	Goa	5	1
3	Gujarat	48	1
4	Karnataka	13	1
5	Kerala	17	1
6	Maharashtra	48	2
7	Odisha	14	1
8	Tamil Nadu	17	3
9	West Bengal	1	1
10	Andaman and Nicobar Islands	24	-
11	Daman & Diu	2	-
12	Puducherry	3	-
13	Lakshadweep	10	-
(As of July 26, 2024)		Total = 217	12

Current Status of India's Shipping Industry

Trade handled by Maritime Sector	95% by volume and 70% by value of India's trade.
Market share of exported ships	India is a leading exporter of shipping vessels with a 33% market share (Economic Survey 2024).
Share in Global shipbuilding and Ship ownership	India has a tiny presence in the global shipbuilding market, with just 0.07% share, and owns only about 1.2% of the world's ships.
Ship Recycling	3rd largest ship recycling by tonnage with 30% global market share in shipbreaking with the world's largest ship-breaking facility located in Alang
Port Infrastructure	12 major ports and 200+ notified minor & intermediate ports.

Recent Developments and Initiatives

- **Sagarmala Programme:** The Sagarmala Programme is a flagship initiative by the Ministry of Ports, Shipping, and Waterways **aimed** at driving port-led development across India.

- **Maritime India Vision (MIV) 2030:** This comprehensive blueprint **aims** to accelerate and harmonize the growth of India's maritime sector over the next decade. MIV 2030 encompasses **more than 150 initiatives** across ten key themes, addressing all areas of the maritime sector, from ports and shipyards to inland waterways and trade bodies.
- **Green Tug Transition Program (GTTP):** It **aims** to **phase out conventional, fuel-based harbour tugs** at Indian major ports **by 2040.**
- **Multi Modal Logistics Parks (MMLP):** Ministry of Road Transport and Highways, the Ministry of Ports, Shipping & Waterways, and the Ministry of State for Road Transport & Highways signed a **tripartite agreement** for the swift development of modern **Multi Modal Logistics Parks (MMLP) under Bharatmala Pariyojna** across the country.
- **EoDB:** The **Indian Ports Act, 2025** is enacted with a forward-looking, integrated framework tailored to the demands of a modern economy.
- **"Jalvahak" Cargo Promotion Scheme:** To encourage a modal shift to **Inland Water Transport (IWT),** the Government introduced the "Jalvahak" Cargo Promotion Scheme on December 15, 2024, **offering 35% reimbursement** on operational costs and launching scheduled cargo services on key NW routes.
- In 2023-24, **major** Indian ports **reduced container turnaround time to 22.57 hours,** exceeding global benchmarks.
- India plans to establish a **new shipping company** to **expand its fleet by at least 1,000 ships within a decade**. This initiative **aims** to **cut foreign freight costs by one-third by 2047** and **increase trade revenue**, with joint ownership involving state-run corporations and foreign firms.
- **Paradip Port** became **India's largest major port** by cargo **volume in FY24**, handling 145.38 million tonnes. It **surpassed Deendayal Port** Authority due to enhanced operational efficiency, record coastal shipping traffic, and a surge in thermal coal shipments.
- India has outlined **investments of US$ 82 billion in port infrastructure** projects by **2035** to bolster the maritime sector.
- In **June 2024**, the Government **approved** the establishment of a **major port** at **Vadhavan, Maharashtra**, with an estimated cost of ₹76,220 crore (US$ 9.14 billion). This project **aims** to expand **EXIM trade capacity** and attract public-private partnerships for infrastructure development.
- In July 2024, the Government updated the **Shipbuilding Financial Assistance Policy (SBFAP),** providing ₹337 crore (US$ 40.40 million) in **financial aid** to boost India's shipbuilding competitiveness.
- The **'Panch Karma Sankalp,'** announced in May 2024, includes **five major announcements** focusing on **green shipping and digitization**.
- **DFCs:** Introduction of **dedicated freight corridors** to reduce logistics cost and time.

Challenges in India's Maritime and Shipbuilding Sector

- **Financing & Infrastructure Constraints:** Ships are not classified as infrastructure, limiting financing options, despite shipyards having infrastructure status since 2016. Also, the SARFAESI Act 2002 prevents banks from offering long-term loans, as ships cannot be mortgaged.
- **Lack of Port Infrastructure & Lower Operational Efficiency:** Indian ports handled 1.4 billion tonnes of cargo (2022-23) but **lag in efficiency**.
  - **Depth limitations** at major ports restrict ultra-large container vessels, increasing dependency on transshipment hubs in other countries.
- **Skilled Workforce Deficit & Technology Gaps:** India supplies 10-12% of global seafarers but faces a shortage of specialized shipbuilding skills.
- **Regulatory & Policy Outrage:** Port expansion and maritime infrastructure projects face 2–3-year approval delays due to land acquisition and Coastal Regulation Zone compliance.

- **Lower Performance in Global Competition & Market Positioning:** China dominates 46.6% of global shipbuilding, creating high entry barriers for India. Indian shipyards operate at 60-70% capacity, limiting economies of scale and global competitiveness.
- **Slow Coastal Shipping Development:** Despite 7,500 km of coastline, coastal shipping accounts for only 6% of domestic freight movement. Lack of dedicated freight corridors raises logistics costs by 15-20%.
- **Discriminatory provisions for Indian vessels:**
  - Foreign vessels are exempt from duty on bunker fuel while Indian vessels have to pay this duty
  - Seafarers aboard Indian flag vessels are subject to Indian income tax while those working aboard foreign vessels are not subject to this tax.

Way Forward

- **Strengthening Domestic Shipbuilding**:
  - Create a National Shipbuilding Mission with direct financial support.
  - Introduce a Production-Linked Incentive (PLI) scheme for shipbuilding.
  - Improve access to long-term financing at concessional rates.
    - **For example,** South Korea's Korea Ocean Business Corporation (KOBC) provides low-cost financing to domestic shipbuilders, helping them dominate global markets.
- **Enhancing Indian Shipping Competitiveness**:
  - Provide tax incentives to Indian shipowners to increase fleet size.
  - Reintroduce the Cargo Reservation Policy to increase the share of Indian-flagged vessels in EXIM trade.
  - Reduce taxes on ship leasing and ship financing.
- **Boosting Coastal Shipping & Inland Waterways**:
  - Improve last-mile connectivity between ports and industrial hubs.
  - Expand multimodal logistics hubs under PM Gati Shakti.
  - Incentivize ferry and roll-on/roll-off (RoRo) services.
- **Enhancing Port Infrastructure & Operational Efficiency**:
  - Develop ultra-large container vessel (ULCV)-friendly ports to reduce transshipment dependency on hubs like Singapore and Colombo.
  - Invest in automation, blockchain, IoT, and AI for faster port operations.
    - **For example,** Jebel Ali Port (UAE) operates with AI-driven efficiency, reducing ship turnaround times to just 4-6 hours.
  - **Green Shipping & Sustainability Initiatives**: Invest in hydrogen-powered ships, LNG bunkering facilities, and solar-powered port operations.
    - **For example,** Norway's electric ferry fleet has drastically reduced maritime emissions.
  - **Public-Private Partnerships (PPP)**: Encourage **foreign investment** in shipbuilding and port development.
    - **For example,** Singapore's PSA Port Expansion through PPPs has made it a global transshipment hub.

6. LOGISTICS SECTOR OF INDIA

"Strengthening of the logistics sector will not only make the life of a common man easier but will also help in increasing the respect of labor and workers". **- Prime Minister Narendra Modi**

India is the world's fourth-largest economy, driven by strong growth in services, manufacturing, and agriculture. After the COVID-19 pandemic, the economy grew 15.3% in 2021–22, and it remains the fastest-growing major economy, with a projected GDP growth of 6.5% in 2024–25.

A key factor behind this growth is India's logistics sector, which efficiently connects producers with consumers, supports trade, and strengthens manufacturing and e-commerce. With government focus on infrastructure and digitalization, India is aiming to become a global logistics hub, boosting its economic resilience and competitiveness in Asia.

Overview of the Logistics Landscape in India

The logistics sector encompasses transportation, warehousing, storage, inventory management, and distribution. It plays a critical role in:

- **Manufacturing**: Ensuring timely delivery of raw materials and finished goods.
- **Retail and E-Commerce**: Enabling efficient last-mile delivery and reverse logistics.
- **International Trade**: Supporting seamless export-import (EXIM) supply chains.

In July 2017, a **dedicated logistics unit** was established under the Department of Commerce to coordinate the integrated development of the logistics sector.

Market Size and Growth

- **Valuation**: The sector was valued at USD 215 billion in 2021 and is projected to reach **USD 357.3 billion by 2030**, growing at a compound annual growth rate (CAGR) of 7.7% from 2025 to 2030.
- **Infrastructure Status**: Granting infrastructure status enables logistics companies to access cheaper, long-term funding, similar to the treatment of roads and railways.
- **Contribution to GDP**: Approximately 13–14% (National Council of Applied Economic Research (NCAER), 2021–22).

Strategic Significance of the Logistics Sector in India

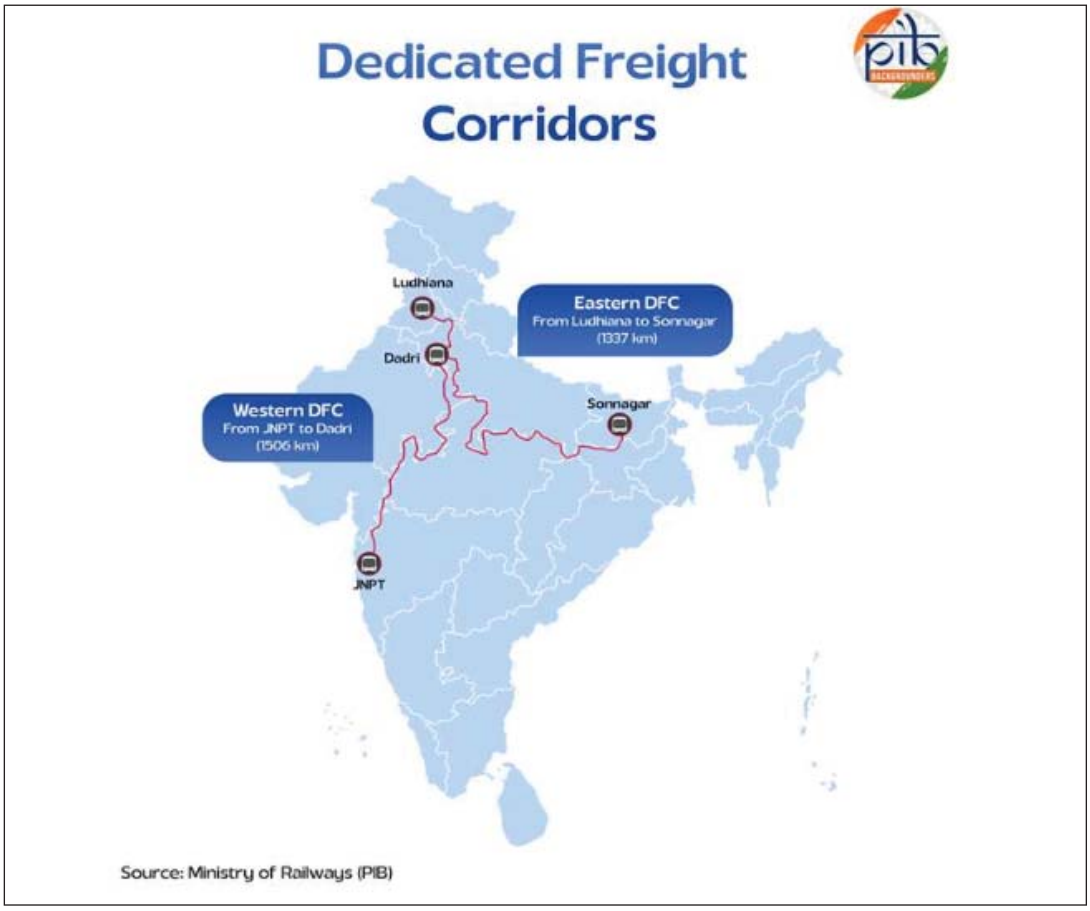
Source: IBEF and PIB

- **Economic Growth and Competitiveness**
  - **Logistics Cost Reduction**: Reducing logistics costs by 1% of GDP could save approximately $15 billion annually, enhancing India's economic competitiveness.
  - **Government Initiatives**: The logistics sector supports key government initiatives such as Make in India, Ease of Doing Business, and Export Competitiveness, contributing to overall economic development.
- **Infrastructure and Urbanization**
  - **Major Hubs**: Key logistics hubs include Mumbai, Chennai, the National Capital Region (NCR), and Pune.
  - **Emerging Centers**: Tier-2 and Tier-3 cities like Patna, Lucknow, and Coimbatore are emerging as significant warehousing and cold chain centers, reflecting the sector's expansion beyond traditional hubs.
- **Employment Generation**
  - **Current Employment**: Over 22 million people are employed in the logistics sector.
  - **Skill Development**: In 2025, the government announced the establishment of five National Centres of Excellence for skill development in logistics and warehousing, aiming to enhance workforce capabilities.
- **Environmental Sustainability**
  - **Decarbonization Goals**: Decarbonization is critical for achieving Net Zero emissions by 2070.
  - **Mode Shift**: Shifting freight transport from road to rail and inland waterways can significantly reduce greenhouse gas emissions, contributing to environmental sustainability.
- **Sectoral Backbone-** The logistics sector supports various industries, including e-commerce, agriculture, pharmaceuticals, and retail, ensuring efficient supply chains and timely delivery of goods.

- **Inclusive Development-** Expanded logistics networks enhance rural access, helping Micro, Small, and Medium Enterprises (MSMEs) reach new markets and integrating remote regions into the national economy.
- **Geopolitical Significance-** Infrastructure projects like the Delhi–Mumbai Industrial Corridor (DMIC) and the Sagarmala initiative enhance India's geo-economic leverage and national security by improving connectivity and promoting port-led development.

Key Government Initiatives in Logistics

- **National Logistics Policy (NLP)** – Launched in September 2022, NLP aims to lower logistics costs and create a seamless supply chain. Digital platforms like Unified Logistics Interface Platform (ULIP) and Logistics Data Bank (LDB) help track EXIM cargo, improve transparency, and boost efficiency.
- **PM GatiShakti Master Plan** – Introduced in October 2021, this plan connects road, rail, air, and waterways for coordinated infrastructure planning. It involves 57 central ministries, all states and UTs, and over 1,700 data layers to enable faster, smarter multimodal connectivity.
- **Maritime Amrit Kaal Vision 2047** – A long-term plan to modernize ports, enhance efficiency through digitization, promote green shipping, develop maritime skills, and make India a global maritime and shipbuilding hub, supported by over ₹10 lakh crore in investment commitments.
- **Dedicated Freight Corridors (DFCs)** – The Eastern and Western DFCs, covering 2,843 km, are dedicated rail lines for heavy freight. They reduce congestion, cut transport costs, improve energy efficiency, and support industrial growth. As of March 2025, 96.4% of the corridors are operational.



- **Multi-Modal Logistics Parks (MMLPs)** – These hubs combine warehousing, storage, and transport facilities. Thirty-five locations, including Chennai, Bengaluru, and Nagpur, are approved, with five expected to be operational by 2027, aiming to reduce logistics costs and improve supply chain efficiency.

- **Logistics Data Bank (LDB)** – LDB tracks EXIM cargo in real-time, with over 45 lakh container searches per month, increasing transparency, efficiency, and trust in India's logistics network.
- **Unified Logistics Interface Platform (ULIP)** – ULIP brings together data from multiple ministries for real-time shipment tracking and inventory management. It has recorded over 100 crore API transactions, showing India's push for a smart, globally competitive logistics ecosystem.
- **Goods and Services Tax (GST)** – Implemented in 2017, GST simplified interstate goods movement by removing checkpoints and streamlining taxation, reducing transport time by over 33% and improving logistics productivity.
- **E-Way Bill** – A mandatory digital document for goods over ₹50,000 transported interstate, it reduces paperwork, ensures compliance, and speeds up interstate transport.
- **Gati Shakti Vishwavidyalaya (GSV)** – India's first university for transport and logistics, GSV trains skilled professionals to support a faster, more efficient supply chain, with MoUs signed with 40 industrial and academic partners.
- **Logistics Ease Across Different States (LEADS)** – An annual report by the Ministry of Commerce & Industry, LEADS evaluates states and UTs on infrastructure, services, regulation, and sustainability, guiding policy and investment decisions.

Milestones and Achievements

- **World Bank Logistics Performance Index (2023)**: India **ranked 38th** out of 139 countries, up 6 places from 2018.
- **Inland Waterway**s: Cargo movement reached 145.5 million tonnes (2024–25), with 29 operational national waterways.

Challenges in India's Logistics Sector

- **High Logistics Costs-** India's logistics costs are around 7.97% of GDP, higher than the global average of 8%. Inefficiencies in transport, warehousing, and supply chains drive these costs.
- **Fragmented Supply Chains-** The sector has a mix of large companies and numerous small informal players, making standardization and technology adoption difficult, which leads to inefficiencies.
- **Inadequate Last-Mile Connectivity-** Last-mile delivery makes up nearly 40% of logistics costs. Poor rural roads and limited warehouse availability delay deliveries and raise expenses.
- **Reliance on Traditional Technologies-** Many SMEs still use manual systems for inventory and transport management, limiting visibility, speed, and responsiveness in operations.
- **Regulatory Compliance Variations-** Even with GST, state-level differences in rules and taxes cause delays and increase costs across the logistics network.
- **Workforce Skill Deficit-** There is a shortage of trained manpower in logistics, including supply chain management, cold chains, and technology-driven operations.

Way Forward

- **Shift Freight to Rail and Waterways-** To ease road congestion and cut transport costs, India should move more goods via rail and waterways. The Dedicated Freight Corridors (DFCs) and Ganga waterways provide efficient routes for bulk cargo, reduce highway wear, and improve delivery speed.
- **Digital Integration-** Wider use of real-time tracking, single-window systems, and ULIP-based coordination will make logistics more transparent and efficient. Digital platforms allow businesses to monitor shipments, optimize routes, and respond quickly to delays.
- **Skill Development at Scale-** A skilled workforce is essential for a modern logistics system. Training youth in logistics technology, cold chains, and electric vehicle (EV) delivery, especially in smaller city hubs, will strengthen supply chain efficiency.

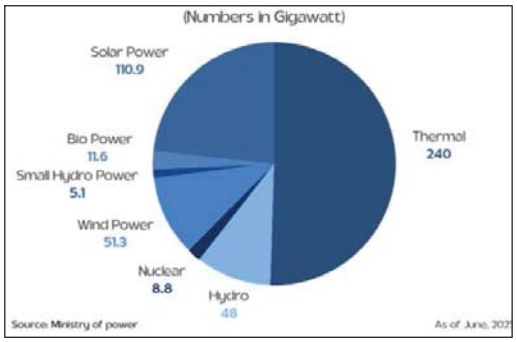
- **Promote Green Logistics-** Sustainability is critical. Supporting electric trucks, solar-powered warehouses, and low-emission zones will reduce the sector's carbon footprint and help India achieve its net-zero targets.
- **Urban Freight Planning-** Cities like Pune and Bengaluru can benefit from mini freight hubs to ensure faster, more reliable last-mile delivery. These hubs help decongest urban areas and support growing e-commerce and retail demand.
- **Encourage Private Sector Innovation-** Startups and private players should leverage government data and platforms to create innovative logistics solutions, improving efficiency, competitiveness, and technology adoption.
- **Policy and Regulatory Reforms-** A Unified National Logistics Regulatory Framework will improve coordination across ministries. Using the Logistics Performance Index (LPI) for benchmarking can identify gaps and guide targeted improvements.

7. ENERGY SECURITY

- Energy security refers to the uninterrupted availability of energy sources at an affordable price. It involves ensuring a reliable supply of energy to meet the growing demands of a nation, while managing risks such as geopolitical conflicts, supply chain disruptions, and environmental challenges.
- For India, energy security is crucial for sustaining economic growth, reducing dependency on imports, and achieving long-term sustainability.
- India currently relies heavily on fossil fuels, primarily coal, for its energy needs, which poses environmental challenges and increases dependence on imports. Renewable energy (RE), including solar, wind, and hydropower, accounts for 44% of the installed capacity but contributes only 23% to energy generation. Despite progress, the need for a diversified energy mix is critical to address rising demand and reduce carbon emissions.

Current status of India's Energy Sector:

- Total installed power capacity reached 476 GW by June 2025.
- Thermal power accounts for 50.52% of total installed capacity.
- Per capita electricity consumption increased to 1,395 kWh in 2023–24 from 957 kWh in 2013–14.
- India achieved 100% village electrification by April 2018.
- Non-fossil fuel sources contribute 49% of total capacity by June 2025.
- India ranks 4th globally in Renewable Energy Installed Capacity, 4th in Wind Power, and 3rd in Solar Power capacity.
- Solar capacity increased to 110.9 GW.
- Installed wind increased to 51.3 GW.
- Installed nuclear capacity grew to 8,780 MW, across 25 reactors.
- Hydro capacity increased to 48 GW.



Significance of energy security

- **Economic Stability & Growth:**
  - Reliable and affordable access to energy fuels industrial production, transportation, agriculture, and services, supporting GDP growth and job creation.

  - Energy security shields the economy from global supply shocks, volatile prices, and supply disruptions, ensuring sustained development.
- **National Security & Strategic Autonomy:**
  - Reduces vulnerability to geopolitical risks, embargoes, and external supply disruptions by diversifying energy sources and suppliers.
  - Maintaining strategic reserves (petroleum, gas, critical minerals) enhances resilience during international crises or conflicts.
- **Technological Advancement & Competitiveness:**
  - Stable energy supply encourages investment in advanced manufacturing, digital infrastructure, and innovation in sectors like AI, data centers, and green tech.
  - Promotes Make in India initiatives and supports export competitiveness.
- **Sustainable & Inclusive Development:**
  - Ensures universal access to clean and affordable energy, vital for alleviating poverty, improving healthcare, and providing education in rural and remote areas.
  - Drives energy transition towards renewables and cleaner fuels, supporting environmental sustainability and meeting climate commitments.
- **Social Security & Quality of Life:** Universal, reliable access to electricity and clean cooking fuels enhances health, reduces indoor pollution, and improves standards of living, particularly for vulnerable populations.

Challenges to India's energy security

- **Import Dependency:** India imports over 85% of its crude oil and more than 50% of its natural gas, over 25% of the total import bill in FY24, puts pressure on rupee, inflates the trade deficit, and compromises macroeconomic stability.
- **Strategic Imbalance:** India's reliance on Russia for 35%-40% of its crude imports, up from 2% before the Ukraine war, highlights the vulnerability of single-sourced reliance. Diversification, not substitution, defines true sovereignty.
- **Global Market Volatility:** Geopolitical conflicts, such as the Russia-Ukraine war and the Gaza crisis, and international sanctions cause sharp fluctuations in global oil prices, impacting India's oil import bill, fiscal balance, and foreign exchange reserves.
- **Financing Needs for Energy Transition:** According to the International Energy Agency (IEA), India requires $160 billion per year to meet its energy transition goals by 2070.
- **Rising Energy Demand:** Driven by economic growth, population increase, urbanization, and industrialization, India's energy demand is projected to double by 2040; increase imports if domestic production cannot keep pace.
- **Renewable Energy Intermittency:** Renewable energy sources, like solar and wind, pose challenges due to their inherent variability, uncertainty, and concentration, require continuous balance to maintain grid stability.
- **Infrastructure Gaps:** Storage options for excess renewable capacity have not developed as quickly as renewable generation, leading to market stabilization issues.
- **Inadequate Strategic Petroleum Reserves:** Limited buffer against supply disruptions, and their coastal concentration makes them susceptible to attacks and natural disasters.
- **Access to Critical Inputs:** Securing critical minerals (e.g., lithium) is challenging due to their geographic concentration and long mining gestation periods.

India's Plan for Energy Security

- **Diversification of Energy Sources:** India aims to diversify its energy portfolio by increasing the share

of renewables like solar, wind, and hydropower, along with exploring other alternatives such as nuclear energy, biomass, and waste-to-energy. The goal is to achieve 50% of installed energy capacity from renewables before 2030.

- **Expansion of Nuclear Energy:** India is expanding its nuclear energy capacity as a clean and efficient alternative. With 25% of the world's thorium deposits, India is exploring thorium-based nuclear reactors and small modular reactor technology to enhance operational flexibility and safety.
- **Boosting Hydropower and Cross-Border Energy Trade:** Hydropower remains a key part of India's energy strategy, with plans to expand cross-border energy cooperation with neighboring countries like Nepal and Bhutan. By importing hydropower, India seeks to enhance its energy mix and ensure stable supply, particularly during peak demand periods.
- **Strengthening Transmission Networks:** India is focused on enhancing transmission networks to efficiently absorb and distribute increased renewable capacity. This includes expanding inter-state transmission lines and developing energy corridors to connect renewable-rich states with energy-deficient regions.
- **Promoting Distributed Energy Generation:** India is investing in distributed energy solutions such as rooftop solar, biogas, and small-scale wind projects. These decentralized energy systems can reduce transmission losses, support rural electrification, and increase energy resilience.

Various Government Initiatives for ensuring Energy Security

- **Target of 500 GW non-fossil fuel capacity by 2030:** Includes solar, wind, hydro, and nuclear sources. As of 2025, installed non-fossil fuel capacity crossed 225 GW (solar 97.9 GW, wind 48.2 GW, hydro 46.9 GW, nuclear 8.2 GW). India ranks 4th globally in renewables.
- **Solar Parks Scheme & PM-KUSUM:** Boosts grid-connected and decentralized solar power, with 50 solar parks sanctioned and solar pumps for farmers.
- **National Green Hydrogen Mission:** Aims for 5 million tonnes annual production of green hydrogen by 2030, with incentives for electrolyser manufacturing and use in refineries, transport, and fertiliser.
- **National Bioenergy Mission & SATAT Scheme:** Promotion of biogas, CBG, waste-to-energy plants to diversify energy sources and utilize agricultural waste.
- **Strategic Petroleum Reserves (SPR):** India maintains reserves at Mangalore, Padur, and Vizag (5.33 million tonnes), with expansion plans at Bikaner, Mangalore (additional), Bina, Padur, and Chandikhol. These will buffer supply disruptions and build resilience. Private participation in SPR policy is encouraged.
- **Nuclear Power Expansion:** Ambitious plan to increase nuclear capacity to 100 GW by 2047, operationalizing Bharat small modular reactors, and strengthening long-term baseload supply.

Way forward

- **Further Diversify the Energy Sources & Mix:**
  - India should reduce dependence on any single source or supplier by promoting alternative fuels and increasing sourcing from various global regions (e.g., Russia, Africa, U.S., Latin America).
  - India should continue to explore underutilized energy sources such as tidal, geothermal, and hydrogen fuel, which can provide sustainable alternatives. Expanding research and development in these areas can unlock new opportunities for energy security.
  - Secure the supply of critical minerals through international collaboration, long-term stockpiles, and financial strengthening of entities like KABIL.
- **Invest in Energy Storage Solutions:** Improving energy storage technologies, like advanced batteries and pumped hydro storage, is crucial for balancing supply and demand, especially with the increasing share of renewables. Enhanced storage capacity can stabilize the grid and ensure a reliable power supply during fluctuations.

- **Focus on Energy Efficiency and Conservation:** Adopting energy-efficient technologies and practices across industries, buildings, and transportation can significantly reduce energy demand. Policies promoting energy conservation, retrofitting, and smart grids can optimize energy use and lower dependency on imports.
- **Strengthen International Energy Cooperation:** India should strengthen its partnerships with global energy leaders to secure access to advanced technologies, invest in joint ventures, and develop cross-border energy projects. Collaborating on research, sharing best practices, and participating in global energy markets can enhance India's energy resilience.
- **Expand Infrastructure for Alternative Fuels:** To reduce reliance on conventional fuels, India should invest in infrastructure for alternative fuels like compressed natural gas (CNG), liquefied natural gas (LNG), and biofuels. This includes setting up fueling stations, pipelines, and processing facilities to support the adoption of cleaner alternatives.
- **Expand Biofuels Potentials:** Ethanol blending program reduces crude imports and transfers over ₹92,000 crore to farmers, foreign exchange savings.
- **Strengthen Nuclear energy Roadmap:** Revive thorium roadmap, secure uranium partnerships, and localize Small Modular Reactor (SMR) technologies. Nuclear power offers zero-carbon energy.
- **Advance Utility Reforms:** Strengthen governance in distribution companies by empowering boards, enhancing financial independence, and listing state-owned utilities on stock exchanges to attract private investment.
- **Strengthen Ecosystem:** Adopt a "whole-of-government" approach with enhanced inter-ministerial coordination and a high-level committee on resilient supply chains for energy transition.

Conclusion

Energy security is vital for India's economic growth and environmental sustainability. By diversifying its energy mix, enhancing infrastructure, and exploring new technologies, India can build a resilient energy system that meets growing demand while reducing carbon emissions. Continued investment in renewables, nuclear energy, and cross-border cooperation will be key to securing a sustainable energy future.

8. WIND ENERGY SECTOR IN INDIA

Status of Wind Energy in India

Fourth globally in installed wind capacity	India ranks fourth globally for installed wind energy capacity. According to the **National Institute of Wind Energy (NIWE)**, India has a wind potential of **1,163.86 GW at 150 meters above ground level.**
Wind Energy as percentage of India's installed renewable energy capacity	As of September 2024, India's total installed renewable energy capacity is **approximately 200 GW**, including large hydro. Wind Power contributes to **nearly 47 GW** of installed renewable energy capacity in India.
States with wind capacity generation potential	Key states for wind capacity include **Gujarat, Tamil Nadu, Karnataka, Maharashtra, Rajasthan, and Andhra Pradesh**. These states collectively account for over **93% of India's wind power capacity**. With 10,603.5 MW of installed wind capacity, Tamil Nadu holds the second-highest capacity in the country.

	Category		Installed Generation Capacity (MW)	% Share in Total
Fossil Fuel	Coal		2,10,970	46.8%
	Lignite		6,620	1.5%
	Gas		24,818	5.5%
	Diesel		589	0.1%
	Total Fossil Fuel :		2,42,997	53.9%
Non-Fossil Fuel	RES (Incl. Hydro)		1,99,583	44.3%
	Hydro	46,928		10.4%
	Wind, Solar & Other RE	1,52,654		33.9%
	Wind	*47,192*		*10.5%*
	Solar	*89,432*		*19.8%*
	BM Power/Cogen.	*10,355*		*2.3%*
	Waste to Energy	*604*		*0.1%*
	Small Hydro Power	*5,071*		*1.1%*
	Nuclear		8,180	1.8%
	Total Non-Fossil Fuel :		2,07,763	46.1%
	Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)		4,50,760	100%

Benefits of Wind Energy

- **Renewable and Sustainable Energy Source:** Wind energy is inexhaustible and it replenishes naturally. This makes it a sustainable power source compared to fossil fuels. **For ex- Denmark** generates nearly half of its electricity from wind energy, thereby securing a steady, renewable energy supply.
- **Reduced Greenhouse Gas Emissions:** Wind turbines generate electricity without producing CO_2 emissions. This helps to reduce the carbon footprint and mitigate climate change. **For ex-** In 2021, wind energy in the **United States helped avoid approximately 189 million metric tons of CO_2 emissions**, equivalent to taking 41 million cars off the road.
- **Energy Independence and Local Economic Growth:** Wind power reduces dependence on imported fuels, bolstering energy independence and enhancing energy resilience.
- **Job Creation:** Wind energy projects create jobs in manufacturing, installation, maintenance, and operations. **For ex-** India's wind energy sector has created thousands of jobs, with **Tamil Nadu alone providing numerous opportunities** for local technicians and engineers.

Challenges with the Wind Energy Sector in India

- **Land Acquisition Issues-** Acquiring large tracts of land for wind farms is a complex and time-consuming process in India. This often leads to delays in project implementation and increased costs due to bureaucratic hurdles and land disputes.

- **Intermittency and Unpredictability-** Wind energy generation depends on weather conditions. This makes it unpredictable and variable during rough weather conditions such as monsoons.
- **Inadequate Transmission and Grid Infrastructure-** India's grid infrastructure is not equipped to handle large-scale integration of wind energy. **For ex- Curtailment issues in Gujarat and Tamil Nadu** due to limited transmission lines connecting wind farms to the main grid.
- **Lack of Updated Policies and Incentives-** Older policies do not always accommodate modern technological advancements or the current needs of the sector. **For ex- Opposition to TN's wind energy policy from wind energy** investors for not accounting for older turbines in the wind energy policy.
- **High Initial Costs and Financial Viability-** Tamil Nadu turbines installed post-2018 **lack a banking facility**. This means that energy produced by repowered turbines cannot be banked, affecting the financial feasibility of these projects.
- **Inconsistent Wind Mapping and Resource Assessment-** India lacks consistent, high-quality wind resources leading to suboptimal site selection and underutilized resources. **For ex-** In **parts of Andhra Pradesh**, **inadequate wind mapping** has resulted in underperforming wind farms.
- **Increased Urbanisation:** Habitation growth near wind farms has also impacted the financial viability of the wind energy projects in India.
- **Local Ecological Concerns:** The construction of wind farms can disrupt local ecosystems and wildlife habitats. **For ex-** Rajasthan has faced resistance to wind energy installations due to concerns over the impact on bird migratory routes and local wildlife habitats.
- **Cybersecurity threat-** The power converters used in wind turbines & solar inverters in solar projects are software-defined assets & they depend upon **SCADA (Supervisory Control & Data Acquisition) system** – which could be compromised subtly & invisibly. A compromised SCADA system or remote-access breach could disable a wind farm, or multiple sites, in mere seconds.

Government's policies for Wind Energy Sector in India

National Wind-Solar Hybrid Policy (2018)	The policy aims to encourage hybrid projects that integrate wind and solar power, improving grid stability and reducing dependency on one energy source.
Repowering Policy for Wind Power Projects	This policy focuses on repowering older wind turbines (under 2 MW) with more efficient, higher-capacity turbines to enhance output from existing sites.
Guidelines for Tariff-Based Competitive Bidding (2017)	This policy promotes competitive bidding to drive down wind energy tariffs, encouraging transparency and cost-efficiency in power procurement from wind projects.
National Offshore Wind Energy Policy (2015)	This policy provides a framework to develop offshore wind energy in India's Exclusive Economic Zone. It grants the National Institute of Wind Energy (NIWE) the authority to identify potential offshore sites and support project implementation.
Green Energy Corridor Project	The Green Energy Corridor Project focuses on creating transmission infrastructure to facilitate renewable energy integration into the national grid, including wind energy.

Localisation of manufacturing	According to a proposed amendment, Original Equipment Manufacturers (OEMs) are required to establish R&D facilities in India – this change signals a decisive shift from passive assembly to active domestic innovation in India. The amendment is proposed to eliminate vulnerabilities to external cyberthreats. Moreover, mandating local R&D isn't just about job creation or investment but about creating wind turbine technology designed for India's grid, climate & challenges.

Way Forward

- **Comprehensive and beneficial policy:** Commercially beneficial, field-responsive policy should be implemented for long-term investment in wind energy.
- **Promoting Land Use Efficiency:** Wind farms can coexist with agricultural or grazing land, allowing dual use and enabling farmers to earn additional income through land leases for wind turbines. **For ex-** In **Iowa, USA**, many farmers lease portions of their land for wind turbines, supplementing their agricultural income without significantly disrupting farming activities.
- **Rapid Deployment and Scalability:** Wind farms should be built quickly and scaled to meet local electricity demand. **For ex- UK's recent rapid expansion of offshore wind energy plants** in recent years.
- **Promotion of Hybrid Renewable Projects**: Wind-solar hybrid systems can help overcome intermittency issues by ensuring energy availability even during low-wind periods. Such hybrid projects will also help in the maximization of land use and improvement of grid reliability.
- **Cybersecurity:**
  - Mandatory audit of software stacks & hardware backdoors, especially from OEMs in adversarial jurisdiction (such as China).
  - To operate in India, vendors/OEMs must obtain certification & clearance from India authorities & adhere to Govt of India regulations. These regulations should include provisions for force majeure situations, where vendor/OEM-supplied equipment may fail to receive support & patches, potentially leading to significant security breaches & national security risks. This ensures that even in times of geopolitical uncertainty, India's energy infra remains operable, supported & secure.

Conclusion

India's wind energy sector is poised for robust growth, backed by strong policy support, competitive tariffs, and ambitious targets. Addressing infrastructure, grid, and cybersecurity challenges will be crucial for sustainable expansion and for meeting the country's clean energy goals.

9. SOLAR ENERGY IN INDIA

"Green Future, Net Zero" aren't just fancy words but reflect India's need and commitment, making it the best destination for investment and innovation in renewable energy**"** **- Prime Minister Narendra Modi**

Just a decade ago, India's solar energy landscape was in its infancy, with solar panels limited to a few rooftops and isolated desert installations. Today, India has emerged as a global solar power leader, surpassing Japan to become the world's **third-largest solar energy producer**. According to the **International Renewable Energy Agency (IRENA)**, India generated **1,08,494 GWh**, outpacing Japan's 96,459 GWh.

About Solar Energy

Solar energy harnesses sunlight and converts it into thermal or electrical energy through two primary methods:

- **Photovoltaic (PV) Cells** – Semiconductor devices in solar panels that directly convert sunlight into electricity.
- **Concentrating Solar Power (CSP) Systems** – Use mirrors to concentrate sunlight onto receivers, producing heat that drives turbines to generate electricity.

India's solar progress is propelled by strategic policies, long-term planning, and international climate commitments. Under its **COP26 pledge**, India aims to achieve **500 GW of non-fossil fuel electricity capacity by 2030**, marking a critical step toward clean energy transition.

Current State of Solar Energy in India

As of **July 2025**, India's cumulative solar power capacity stood at **119.02 GW**, distributed as follows:

- **Ground-mounted solar plants:** 90.99 GW
- **Grid-connected rooftop systems:** 19.88 GW
- **Hybrid projects:** 3.06 GW
- **Off-grid solar installations:** 5.09 GW

India's geographical advantage—lying along the **Tropic of Cancer**—ensures abundant solar radiation. The **total solar sector potential** is estimated at **748 GW**.

Growth of India's Solar Sector

Decadal Expansion (2014–2025)	India's solar power capacity witnessed an extraordinary surge of 3,450%, rising from 2.82 GW in 2014 to 100 GW by 2025.
Leadership in Renewable Energy	Solar energy **constitutes 47% of the country's total renewable energy capacity**, establishing it as the dominant renewable sector in India
Solar Installations in 2024	**Utility-scale projects**: 18.5 GW, representing nearly 2.8 times growth over 2023 **Rooftop installations**: 4.59 GW, reflecting a 53% increase from the previous year
Top Performing States (Utility-Scale Solar)	Rajasthan, Gujarat, Tamil Nadu, Maharashtra, Madhya Pradesh
Trajectory of Solar Manufacturing	India's solar module production capacity expanded from 2 GW in 2014 to 60 GW in 2024, with a target of achieving 100 GW by 2030.

Notable Achievements:

- India's solar capacity increased **4,000% in a decade**.
- **Palli village, Jammu & Kashmir** became India's **first carbon-neutral panchayat**, running entirely on solar power.

India's Renewable Sector

India's **energy mix** (2025):

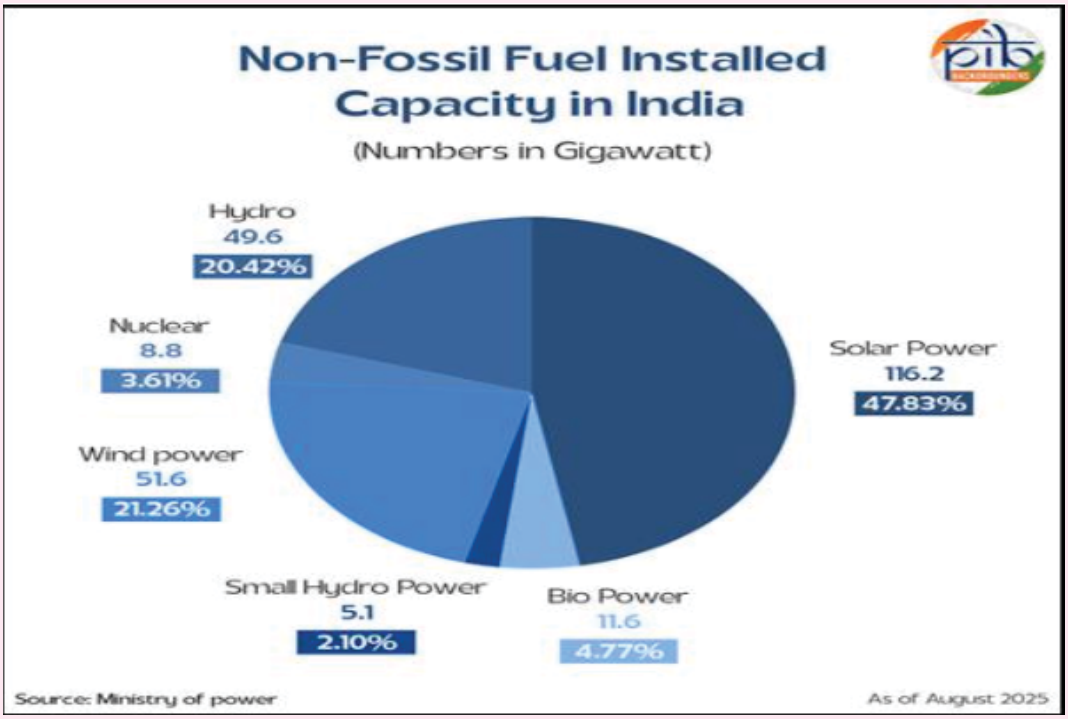
- Total power capacity: **~485 GW**

- Thermal: 242 GW
- Solar: 116 GW
- Wind: 51.6 GW

Non-fossil fuel capacity has grown from 76.37 GW (2014) → 233.99 GW (2025), accounting for **50.07%** of total installed power. India added **29.52 GW renewable energy in FY 2024–25**, moving closer to the **500 GW target by 2030**.

Global rankings (2025):

- 4th in **Renewable Energy Installed Capacity**
- 4th in **Wind Power Capacity**
- 3rd in **Solar Power Capacity**



Key Flagship Initiatives in Solar Energy

- **PM Surya Ghar: Muft Bijli Yojana-** Provides free electricity via rooftop solar to 1 crore households (₹75,021 crore). Offers up to 300 units/month and subsidies of ₹30,000–78,000 depending on system size.
- **PM-KUSUM-** Helps farmers switch from diesel to solar with 30–50% subsidies on pumps. Farmers can set up solar plants up to 2 MW and sell electricity, supporting energy security and India's 2030 goal of 50% non-fossil fuel power.
- **Solar Parks Scheme-** Develops large grid-connected solar plants, targeting 40 GW by 2026. 53 parks (39,323 MW) approved; 18 fully developed (10,856 MW) and 13,896 MW operational in 26 parks.
- **PM JANMAN: Solar Electrification for PVTG Communities-** Under the Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan (PM JANMAN), the New Solar Power Scheme aims to electrify 1 lakh un-electrified households in tribal areas across 18 states.
- **Increase in Solar PV Manufacturing Capacity-** Since 2014, India's solar PV cell production capacity has grown 21 times, from 1.2 GW to 25 GW by March 2025. PV module production has increased over 34 times, from 2.3 GW to 78 GW.

Other Important Initiatives

- **Floating Solar Projects:** Projects like the **Omkareshwar Floating Solar Park** in Madhya Pradesh (600 MW) use water surfaces for solar generation, saving land and improving efficiency.
- **Agrivoltaics:** Solar panels are installed above crops, allowing farming underneath and boosting farmer income. **Examples** include the Sun Master Plant in Delhi and a 105 KW system by ICAR in Jodhpur.

India's Global Solar Leadership

- **International Solar Alliance (ISA):** Launched by India and France in 2015, ISA unites 100+ countries to promote solar energy, mobilize $1 trillion by 2030, reduce costs, and support Least Developed Countries (LDCs) and Small Island Developing States (SIDS).
- **One Sun, One World, One Grid (OSOWOG):** Initiated in 2018, this global solar grid concept connects regions from South Asia to Africa and Europe, ensuring continuous solar energy availability worldwide.

Significance of Solar Energy

- **Climate Change Mitigation-** Solar energy helps India reduce emissions and increase clean electricity. By 2030, India aims to cut emissions intensity by 45% and generate half of its electricity from non-fossil fuel sources.
- **Enhancing Energy Security-** With energy demand growing rapidly, solar power reduces dependence on imported fuels and supports a stable power supply.
- **Empowering Rural Communities-** Solar energy brings electricity to remote areas. Programs like PM-KUSUM install solar water pumps, improving farming and water access in villages.
- **Driving Economic Growth-** The solar sector creates jobs in manufacturing, installation, and maintenance. It also lowers electricity costs and encourages domestic production through schemes like PLI.
- **Environmental Benefits-** Solar power cuts greenhouse gas emissions and air pollution. Large projects like the 2,245 MW Bhadla Solar Park in Rajasthan show its positive environmental impact.
- **Cost-Effectiveness-** Falling installation costs and government subsidies make solar energy affordable and accessible, especially for farmers.
- **Versatile Applications-** Solar energy can be used for homes, businesses, rural electrification, and industries, making it highly adaptable.

Factors Accelerating Solar Energy Growth

- **Abundant Solar Potential-** India enjoys approximately 300 sunny days per year, with solar insolation ranging from 4 to 7 kWh/m²/day, making it ideal for solar energy generation.
- **Supportive Government Policies-** Initiatives like the National Solar Mission and the Solar Park Scheme have streamlined the development of solar infrastructure. The government has also waived interstate transmission charges for solar projects to encourage investment.
- **Financial Incentives-** The allowance of 100% Foreign Direct Investment (FDI) in solar projects and the implementation of the PLI scheme have bolstered domestic manufacturing and attracted international investments.
- **Infrastructure Development-** The Green Energy Corridor Scheme aims to enhance transmission capacity, facilitating the integration of renewable energy into the national grid.
- **International Collaboration-** Partnerships such as the International Solar Alliance (ISA) and the Indo-German Solar Energy Partnership (IGSP) promote knowledge exchange and joint ventures in solar technology.

Challenges Facing the Expansion of Solar Energy in India

- **Land Acquisition-** Large-scale solar farms need significant land, which can compete with agriculture or

other uses. Securing land can be difficult due to regulatory hurdles.

- **Grid Integration-** Solar power is intermittent, depending on sunlight. Integrating it into existing grids requires technological upgrades and infrastructure improvements to maintain stability.
- **Environmental Concerns-** Mining materials like lithium, cobalt, and nickel for solar panels can harm the environment. Manufacturing and disposal of panels also involve energy use and potentially hazardous materials.
- **Domestic Manufacturing Limitations-** India's solar manufacturing sector faces challenges in research and development and relies heavily on imports for critical components.
- **Policy and Regulatory Hurdles-** Complex, inconsistent, or changing regulations can delay project approvals and affect long-term investment decisions.
- **High Initial Costs and Efficiency Issue-** Installing solar panels requires substantial upfront investment, though costs are declining. Panels convert only 15–20% of sunlight into energy, limiting efficiency.
- **Energy Storage and Intermittency-** Current battery technologies are expensive and have limited capacity, making solar less reliable during cloudy days or at night.
- **Dependence on Weather and Location-** Solar energy generation is less effective in regions with low sunlight or frequent overcast conditions.

Way Forward

- **Upgrading the Power Grid-** Implementing smart grids and microgrids can improve the efficiency and reliability of solar energy distribution, particularly in rural and remote areas.
- **Efficient Land Utilization-** Techniques like agrivoltaics and floating solar installations on reservoirs and water bodies can optimize land use while increasing energy output.
- **Streamlining Policies-** Aligning central and state regulations and simplifying approval procedures will help accelerate solar project implementation.
- **Advancing Technology-** Enhancing solar panel efficiency, improving energy storage solutions, and adopting hybrid systems combining solar and wind energy will boost overall performance and reliability.
- **Promoting a Circular Economy-** Setting up effective recycling programs for solar panels and batteries will minimize waste and encourage sustainable practices.
- **Boosting Domestic Manufacturing-** Supporting "Make in India" initiatives for solar components, cells, and storage solutions will reduce import dependence and drive economic growth.

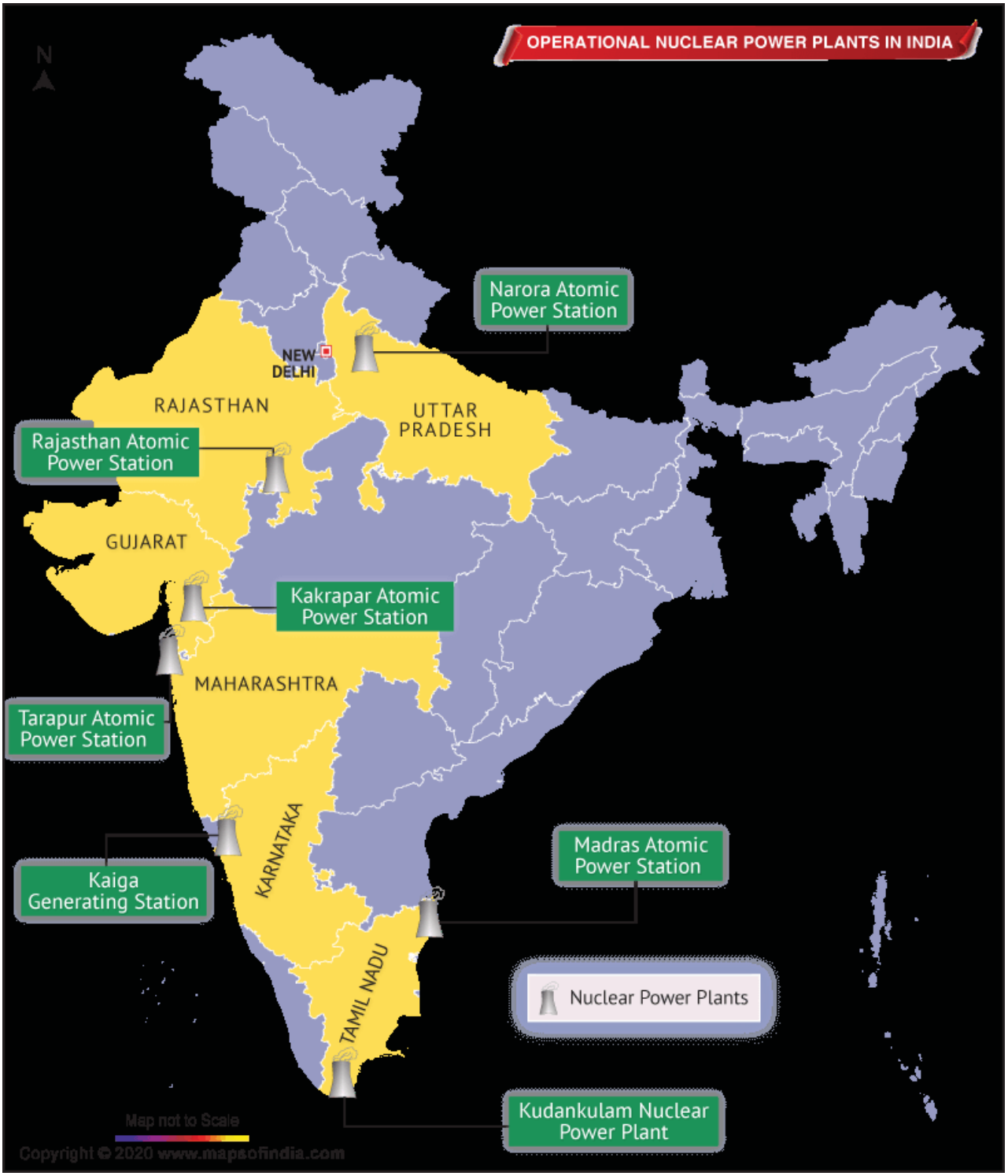
10. NUCLEAR ENERGY

Nuclear energy has emerged as a cornerstone of India's quest for energy security, sustainable development, and technological leadership. With a growing population and rapid industrialization, India's energy demand is projected to double by 2040. Nuclear energy, with its low carbon footprint and high energy density, offers a viable solution to meet this demand while aligning with global climate goals.

Introduction

- India's three-stage nuclear program, centered on uranium, plutonium, and thorium, is a testament to its long-term vision for energy independence. From powering homes to propelling submarines and advancing space exploration, nuclear energy is integral to India's strategic, economic, and scientific aspirations.
- India's nuclear energy program is not just about powering homes, it is about powering the nation's aspirations. From ensuring energy security to advancing space exploration and strengthening national defence, nuclear energy is a strategic asset for India. The journey from uranium to thorium symbolizes India's transition from energy dependence to energy independence.

Operational Nuclear Power Plants in India



Present status of Nuclear Energy in India

Civil Nuclear Power	• **Number of Operational Nuclear Power Plants:** India has 22 operational nuclear reactors with a total installed capacity of 7,480 MW (as of 2024).

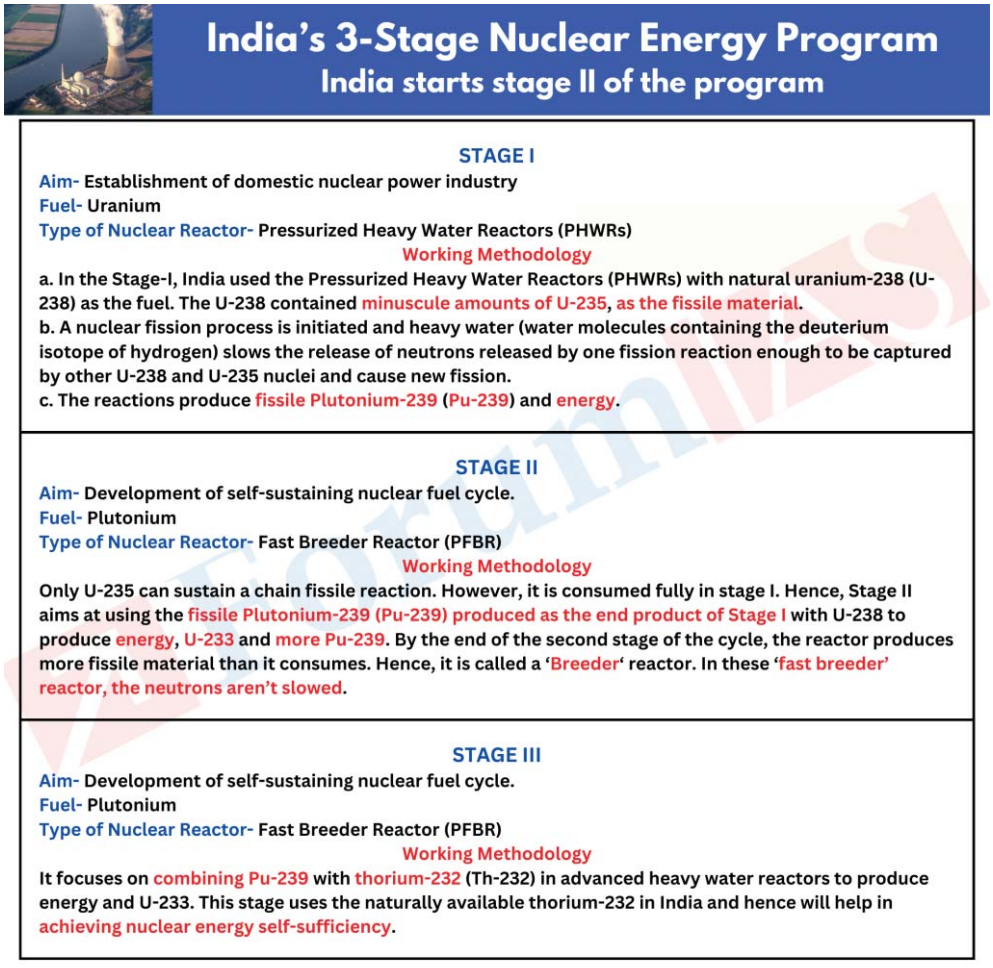
	• **New Nuclear Projects:** Several new reactors are under construction, including **Kudankulam** (Units 3-6), **Gorakhpur Haryana Anu Vidyut Pariyojana**, and **Kaiga** Units 5 & 6, expected to add nearly 7,000 MW in the next decade which will add 8,000 MWe by 2031. • **Indigenous Reactor Technology:** India has developed Pressurized Heavy Water Reactors (PHWRs), Prototype Fast Breeder Reactor (PFBR), and is progressing towards thorium-based Advanced Heavy Water Reactor (AHWR).
Nuclear in Defense	• **Nuclear-Powered Submarines:** India has developed INS Arihant, the country's first indigenous nuclear-powered ballistic missile submarine (SSBN), and is working on INS Arighat and additional units. • **Nuclear-Capable Missiles:** India has a range of nuclear-capable missiles, including Agni (I-V), Prithvi, and K-15 (Sagarika). • **Nuclear Research and Collaborations:** India collaborates with the IAEA, Russia (Rosatom), and the US (123 Agreement) for nuclear technology exchange, while advancing indigenous research at BARC, IGCAR, and NPCIL.
Technologies in Use and Development	• **PHWRs:** The backbone of India's nuclear program, using natural uranium as fuel. • **Fast Breeder Reactors (FBRs):** The 500 MWe Prototype Fast Breeder Reactor (PFBR) is under development, aiming to use plutonium and thorium for fuel. • **Molten Salt Reactors (MSRs):** It is still in the research phase, focusing on thorium utilization. • **Small Modular Reactors (SMRs):** These reactors are being explored for deployment at retiring coal plant sites.

Significance of Nuclear Energy

- **Energy Security:** The three-stage nuclear program aims to utilize thorium, ensuring long-term energy independence. Nuclear power accounts for about 3% of India's electricity generation but is targeted to reach 22% by 2050. Nuclear energy delivers base load power, which is essential to building modern economies.
- **Clean Energy (Decarbonisation of Power Sector):**
  - Nuclear power is a low-carbon energy source, contributing to India's Net Zero by 2070 goal. It complements renewable energy by providing base-load power, ensuring grid stability.
  - India's electricity generation capacity at present is 480GW. It will have to grow 5-fold for India to achieve developed economy status by 2047. On the other hand, India also has the target to become net zero by 2070, thus India will not be able to rely on fossil fuels for its growth.
  - Renewable energy (including solar, hydro, wind & biomass) is estimated to provide 20% of the demand & up to 25% with investments in battery & pumped storage. Therefore, the obvious candidate to fuel India's energy growth is nuclear power.
- **Limitations attached with other renewable energy sources:** Solar energy is land intensive, wind energy requires energy storage systems. Also, they require imported technologies and materials such as photovoltaic cells, batteries, and storage equipment. On the other hand, indigenous nuclear reactors have reduced dependency on critical imports.
- **Cheaper to Operate:** Nuclear power plants are cheaper to operate than coal or gas plants, despite the cost of managing radioactive fuel and disposal. According to estimates, nuclear plants cost only 33-50% of a coal plant and 20-25% of a gas combined-cycle plant.
- **Space Exploration:** Nuclear energy powers radioisotope thermoelectric generators (RTGs) for deep-space missions, such as Chandrayaan and Mangalyaan.

- **Defence:** Nuclear-powered submarines and missiles enhance India's strategic deterrence and second-strike capability. Indigenous development of nuclear reactors and weapons reduces dependency on foreign technology.
- **Medical Advancements:** BARC-developed radiopharmaceuticals aid in cancer treatment, with India producing over 50 radioisotopes for medical use.
- **Food Preservation:** Nuclear technology enhances food irradiation, increasing shelf-life and preventing spoilage.

Government initiatives for the promotion of Nuclear Energy

<table>
<tr><td>Early start</td><td>• India was the 1st country in Asia to set up a nuclear research reactor, APSARA, in 1956.
• India was also the 1st in Asia to work on the nuclear power reactor at Tarapore in 1963.</td></tr>
<tr><td>Three-Stage Nuclear Program</td><td>• Stage 1: PHWRs using natural uranium.
• Stage 2: Fast Breeder Reactors using plutonium and thorium.
• Stage 3: Thorium-based reactors for sustainable energy.
 
</td></tr>
<tr><td>Bharat Small Reactors (BSR)</td><td>• It aims to deploy small modular reactors in collaboration with the private sector.</td></tr>
</table>

Nuclear Legislation	• **Atomic Energy Act:** Authorizes the establishment of nuclear power plants and R&D institutions. • **Civil Liability for Nuclear Damage Act, 2010 (CLNDA):** Enacted to define **liability and compensation** in case of nuclear accidents. Aligns India with the **Convention on Supplementary Compensation (CSC), 1997.**
Bodies	• **Nuclear Power Corporation of India Limited (NPCIL):** NPCIL is a govt-owned company that builds, owns & operates the PHWRs. • **Atomic Energy Regulatory Board (AERB):** Certification & safety oversight is the responsibility of AERB.
International Collaborations	• **Indo-US Nuclear Deal:** Facilitated access to global uranium markets and advanced technologies. • **Civil Nuclear Cooperation Agreements:** Signed with Russia, France, USA, Japan, and others for fuel supply and technology transfer. • **Nuclear Insurance Pool (2016):** Mitigates risks associated with nuclear energy projects.
Union Budget 2025-26	• Announced an ambitious target of 100GW nuclear power generation capacity by 2047, up from 8.18GW at present. • **Nuclear Energy Mission:** Aims to develop at least 5 indigenously designed & operational Small Modular Reactors (SMRs) by 2033.

Challenges associated with Nuclear Energy

- **Technological and Infrastructure Challenges:**
  - **High Upfront Capital Costs:** Nuclear plants require significant capital investment compared to coal or renewables.
  - **Limited Indigenous Uranium Supply:** Dependence on imports hinders energy security.
  - **Delayed Project Completions:** Kudankulam and PFBR projects have faced repeated delays.
  - **Use of outdated Technology:** Currently operational Indian nuclear reactors have become outdated and suffer from multiple operational problems. **For e.g.** 6 VVER (water-water energy reactor) design reactors encountering operational problems at Kudankulam.
- **Safety and Environmental Concerns:**
  - **Radiation Risks:** Local communities in India have been resisting nuclear reactors due to fears of nuclear disasters like Chernobyl, 1986 or Fukushima, 2011. **For e.g.** Locals protesting against the Mithi virdi nuclear project in Gujarat.
  - **Nuclear Waste Management:** India lacks robust long-term nuclear waste disposal mechanisms.
- **Regulatory and Political Challenges:**
  - **Public Opposition:** Protests in Tamil Nadu and Maharashtra have stalled nuclear projects.
  - **Policy Uncertainty:** Frequent changes in regulations delay private sector participation.
  - **Nuclear Liability:** India's Civil Liability for Nuclear Damage Act 2010, has been a contentious issue for foreign suppliers. Foreign suppliers have been reluctant to invest in India's Nuclear Energy Programs due to fears of being held accountable for accidents beyond their control.
- **Geopolitical Constraints:**
  - **NSG Membership Roadblock:** India's non-signatory status to the NPT limits access to advanced nuclear technologies.
  - **Global Uranium Market Volatility:** Prices and supply chain disruptions affect India's nuclear fuel procurement.

Way Forward

- **Accelerate Thorium-Based Reactors:** Given India's vast thorium reserves, fast-tracking **AHWR deployment** is crucial for the growth of nuclear energy.
- **Enhancing Private Sector Involvement:** Allowing greater participation of companies like **L&T, Tata, and Reliance** in nuclear infrastructure can boost innovation.
- **Expansion of indigenous PHWR reactors:** The Indigenous 700 MWe PHWR, must be expanded in fleet mode to add to the installed nuclear power capacity in India.
- **Push to the Stage-3 of Nuclear Power Program:** The second and third stages of the nuclear-power programme must be propelled to utilise the existing thorium energy potential in the country.
- **Development of Nuclear Fusion technology:** The development of nuclear fusion technology must be explored, which is safer than nuclear fission. The vast reserves, in the form of ocean water, will be an added advantage for India.
- **Augmentation of safety of nuclear facilities:** There must be constant updation of safety skills of nuclear operators. Further, masses must be comprehensively sensitised about the functioning of nuclear power plants using highly intellectual individuals having mass appeal. **For e.g.** Dr. APJ Abdul Kalam sensitizing the masses before the establishment of the Kudankulam nuclear power plant.
- **Ensuring Regulatory Autonomy:** The AERB, India's nuclear regulatory body, must be provided functional autonomy by removing its reporting from the Department of Atomic Energy (DAE).
- **Strengthening Global Collaborations:** Expanding partnerships with **France, Russia, and the US** for technology exchange and uranium supply.
- **Developing Small Modular Reactors (SMRs):** Indigenous Small Modular Reactors (SMRs) must be built at coal plant sites which would be retiring in the coming decades. SMRs offer the advantages of being safe, economical, compact and adaptable. Partnerships with NTPC and other thermal plant owners must be explored.
- **Focus on Nuclear Waste Management:** Setting up **deep geological repositories** for spent nuclear fuel.
- **Leveraging AI and Digital Tools:** Integrating **AI-driven safety systems** and predictive maintenance can enhance reactor efficiency.
- **Nuclear energy as 'Renewable Power':** While nuclear energy is a low-carbon energy source, it is not classified as 'renewable', like solar & wind. Revising this classification would make nuclear power projects eligible for tax incentives & specially designed 'green financing' instruments.

Conclusion

With bold initiatives, technological innovation, and strategic collaborations, India is poised to harness the full potential of nuclear energy. As the world moves toward a low-carbon future, India's nuclear energy program stands as a beacon of sustainable development and technological leadership. **The dream of "Viksit Bharat" is not just a vision; it is a mission powered by the atom.**

■■■

MGP/BP/GS3/029

INVESTMENT MODELS

1. PUBLIC PRIVATE PARTNERSHIP (PPP)- NEED AND CHALLENGES

Public-Private Partnership (PPP) is a **collaboration** between a **government or government-owned entity** & a **private sector entity** for the provision of public assets and services.

The **Private Investment Unit** in the **Department of Economic Affairs** is responsible for policy-level matters concerning PPPs.

Various Types of PPP Models

Build Operate Transfer (BOT)	The private entity **finances**, **builds**, and **operates** the facility for a **specified concession period,** then transfers it back to the government. Revenue is earned from **user charges** (for ex- tolls on highways). **Variants of BOT:** **BOT (Toll)-** Private partner collects tolls directly from users. **BOT (Annuity)-** Government pays fixed annual annuities to the private partner.
Build-Own-Operate (BOO)	The private sector **builds**, **owns**, and **operates** the facility indefinitely. There is no transfer back to the government. It is more common in power projects.
Build–Own–Operate–Transfer (BOOT)	The Private partner **owns** and **operates** the project during the **concession period**, then transfers it back at the end. It is similar to BOT, but the private player holds ownership rights during the concession period.
Engineering Procurement and Construction (EPC)	Under EPC, the government fully funds the project (through budgetary resources, borrowings, or grants). The **private contractor's role** is limited to: • **Designing** the project (engineering). • **Procuring** raw materials, equipment, and services. • **Constructing** the facility as per agreed specifications and timelines. Once construction is complete, the contractor hands it over to the government. Operation and maintenance (O&M) usually remain with the government.
Hybrid Annuity Model (HAM)	• It has been applied specifically in the highways construction in India by the **National Highways Authority of India** (**NHAI**). • It is a combination of the **Build-Operate-Transfer** (**BOT**) and **Engineering Procurement and Construction** (**EPC**).

	• The government contributes **40% of project cost** during **construction**, while the **private player** invests the remaining **60%**. The private player recovers its investment through annuities from the government (not tolls). • It balances the risks between public and private sectors.

Need for PPP Models

- **Public budget constraints-** Infrastructure projects require substantial capital investment, which governments often struggle to provide. PPPs enable private sector participation, complementing public resources and ensuring project feasibility.
  - A 2023 RBI report highlights that many states face fiscal deficits of over 3%, limiting their infrastructure budgets.
- **Plugging the Infrastructure Gap-** India's infrastructure needs are massive in roads, ports, power, water, urban transport. Joint partnership between the government and private players, help in plugging the huge infrastructure gap.
  - According to the PPP in India portal, there are over **9,242 PPP projects** across states with a total project cost of approximately **₹6,813,008 crore**.
- **Operational Efficiency**- The private sector's expertise, innovative technology, and management practices lead to better execution, improved quality, & higher operational efficiency in infrastructure projects.
  - **For ex**- **PPP modelled developed airports** have **high operational efficiency** like the IGI Delhi.
- **Risk Mitigation-** PPPs distribute risks between public and private entities, thereby improving project sustainability.
- **Faster Execution-** Private sector involvement helps overcome bureaucratic delays, ensuring timely completion of projects and reducing overall costs.
- **Economic Multiplier Effect-** PPP-driven infrastructure projects stimulate economic activity across sectors such as manufacturing and agriculture. This leads to creation of jobs and fostering long-term growth.

Challenges of PPP in India

- **Regulatory Issues in granting clearances-** According to the **Ministry of Statistics and Programme Implementation (MoSPI)**, **449 infrastructure projects** faced cost overruns totalling over **Rs 5.01 lakh crore** due to regulatory delays like land acquisition issues, environmental clearance delays.
- **High Gestation Period & Cost Overruns-** Long construction timelines and market volatility result in significant cost and time overruns, affecting project efficiency.
  - A **CAG (Comptroller and Auditor General)** report on **PPPs in India** found that several highway projects faced cost escalations of **25–30%** due to delays in land acquisition and utility shifting.
- **Financing Constraints-** High capital needs and perceived risks make securing financing difficult.
  - **For ex- Mundra Ultra Mega Power Project (Tata Power)** was hit hard by rising imported coal prices. It made the project financially unviable under the original tariff structure.
- **Long-Term Contract Issues-** Public Private Partnership (PPP) Projects lasting for **20-30 years** create "**obsolescing bargains**," where the private sector loses negotiating power due to economic or policy changes.
- **Lack of Effective Dispute Resolution Mechanisms-** Lack of effective frameworks **hampers project integrity** and **stakeholder commitment** leading to cost overrun and delays.
- **Involvement of State-Owned Entities-** State-owned enterprises are often seen as **government entities**, discouraging their involvement in PPPs.
- **Inclusivity & Affordability Issues-** Ensuring affordability and inclusivity in sectors like railways, water, education, and healthcare remains a major challenge.

- The **Delhi Airport Metro Express Line** (**PPP with Reliance Infra**) became unaffordable for the public due to incurrence of high costs.

Government Steps to Improve PPP in India

Viability Gap Funding (VGF)	Provides up to **40% capital grant to make projects financially viable**, supporting infrastructure development.
India Infrastructure Project Development Fund (IIPDF)	Offers financial support for project development activities like **feasibility reports** & **project structuring for PPPs** at central, state, and local levels.
PPP Appraisal Committee	Recommended 77 Central sector projects worth **Rs. 2.4 lakh crore** (FY15-FY24)
National Monetisation Pipeline (NMP)	Targets **Rs.6.0 lakh crore** by monetising government assets (FY22-FY25)
IIFCL Support	Provides **long-term debt financing** for infrastructure projects with long gestation periods, addressing the capital needs of such initiatives.
Foreign Direct Investment (FDI)	Allows up to **100% FDI** in **Special Purpose Vehicles (SPVs)** under the automatic route for most sectors, encouraging private investment in PPPs.

Way Forward to Improve PPP in India (According to Vijay Kelkar Committee)

- **Legal Revisions-** Amend the **Prevention of Corruption Act, 1988** to distinguish between **genuine errors** in decision-making and acts of corruption.
  - Enact a National PPP Policy, establish a **PPP Institute** and **implement a National Facilitation Committee** for timely project approvals.
- **Expert Mechanism for Dispute Resolution-** Establish a **PPP Project Review Committee** and **PPP Adjudication Tribunal** to resolve complex issues and streamline projects awaiting clearances.
- **Focus on Service Delivery-** Contracts should prioritize service delivery rather than just fiscal benefits.
- **Independent Regulation-** Establish independent regulators for various sectors, and enhance project development in roads and ports.
- **Risk Management-** Assess risk management efficiency and cost-effectiveness with advanced modeling.

2. INFRASTRUCTURE FINANCING IN INDIA

Infrastructure financing in India refers to the mobilization of long-term capital required to build and modernize core sectors like transport, energy, and urban utilities. Historically dependent on public exchequer and commercial banks, India is rapidly pivoting toward innovative market instruments like InvITs and asset monetization to sustainably fund its expansion.

Major Infrastructure Financing Institutions in India

Institution	Core Objective
NaBFID (National Bank for Financing Infrastructure and Development)	Acts as the principal statutory DFI (established in 2021) to coordinate, anchor, and extend long-term debt to the National Infrastructure Pipeline (NIP).

IIFCL (India Infrastructure Finance Company Limited)	A wholly owned Government of India enterprise (est. 2006) providing long-term financial assistance via direct lending, refinancing, and take-out financing to viable greenfield and brownfield projects.
NIIF (National Investment and Infrastructure Fund)	India's collaborative sovereign wealth fund (est. 2015) that channels global and domestic institutional "patient capital" into commercial equity across core energy, transport, and social infrastructure projects.
PFC (Power Finance Corporation Ltd.)	A specialized Maharatna Central Public Sector Enterprise (CPSE) focused exclusively on providing comprehensive financial backing and liquidity to India's power sector value chain.
REC (Rural Electrification Corporation Ltd.)	Serves as the key financial nodal agency dedicated to funding and developing power infrastructure, rural electrification, and clean energy distribution grids across states.
IREDA (Indian Renewable Energy Development Agency)	Operates as a specialized non-banking financial institution under the Ministry of New and Renewable Energy to promote, develop, and extend financial assistance specifically for green energy and energy-efficiency projects.
NABARD (National Bank for Agriculture and Rural Development)	Spearheads the Rural Infrastructure Development Fund (RIDF) to channel commercial banking shortfalls into critical state-level rural connectivity, irrigation, and agricultural assets.

Methods of Infrastructure Financing in India

- **Public-Private Partnerships (PPP):** Collaborative contractual frameworks (like BOT, HAM, and EPC) that leverage private sector capital, efficiency, and execution expertise to build public assets.
- **Infrastructure Investment Trusts (InvITs):** Mutual-fund-like instruments that pool institutional and retail capital to invest in income-generating operational infrastructure, freeing up developer equity.
- **Asset Monetization (NMP):** Shifting operational public assets (like roads or transmission lines) to private operators via concession periods to generate upfront revenues for building new greenfield projects.
- **Development Financial Institutions (DFIs):** Specialized statutory agencies (such as NaBFID) that provide long-term, low-cost debt tailored to navigate the high gestation periods of mega-projects.
- **Sovereign Wealth & Pension Funds:** Mobilizing global, long-term "patient capital" via entities like India's National Investment and Infrastructure Fund (NIIF) to back equity requirements.
- **External Commercial Borrowings (ECBs):** Enabling domestic infrastructure firms to raise low-cost foreign currency loans or bonds directly from international capital markets.
- **Infrastructure Debt Funds (IDFs):** Investment vehicles (structured as trusts or non-banking finance companies) that refinance existing post-COD (Commercial Operation Date) bank debt to lower project borrowing costs.
- **Masala Bonds:** Rupee-denominated bonds listed on international exchanges (like the London Stock Exchange) that allow Indian companies to raise global funds without bearing exchange-rate risk.
- **Green Bonds:** Specialized debt securities dedicated exclusively to raising eco-friendly capital for renewable energy, clean transport, and climate-resilient infrastructure.

Significance of Infrastructure Financing in India

- **Macroeconomic Drivers**
  - **Accelerates Capital Formation:** Boosts Gross Fixed Capital Formation (GFCF), creating the foundation for a $5 trillion economic path.

According to the Ministry of Statistics and Programme Implementation (MoSPI), focused infrastructure push pushed India's Gross Fixed Capital Formation (GFCF) around 34-35% of GDP in recent fiscal cycles.

- **Triggers Multiplier Employment:** Ignites a cross-sectoral supply chain that generates millions of direct construction and indirect service jobs.

  According to National Skill Development Corporation (NSDC), every ₹1 Crore invested in infrastructure creates an employment multiplier effect of 2.5x to 3.0x in allied sectors.

- **Fiscal & Banking Stability**
  - **Attracts Global Equity:** Channels deep international **patient capital** (pension and sovereign funds) to relieve pressure on the public exchequer.

    According to DPIIT data, the infrastructure development sector ranks among the top 5 sectors for Foreign Direct Investment, pulling in over $33 Billion in cumulative FDI.

  - **Prevents Banking Stress:** Uses dedicated DFIs and InvITs to prevent Asset-Liability Mismatches (ALM) and Non-Performing Assets (NPAs) in commercial banks.
- **Structural Competitiveness**
  - **Lowers Logistics Costs:** Modernizes transport and port networks, cutting domestic transaction overheads to make exports globally competitive.

    According to the Ministry of Commerce and Industry, multi-modal financing under the LEADS framework aims to compress India's overall logistics cost from 13-14% of GDP down to a competitive 8-9%.

  - **Enhances Resource Efficiency:** Finances smart grids, automated logistics, and green energy arrays to maximize resource life cycles and asset longevity.
- **Inclusive Development**
  - **Bridges Regional Divides:** Funds large-scale spatial connectivity projects (highways, broadband corridors) to link rural hinterlands with urban markets.

    BharatNet project laid optical fiber cables to connect over 2,50,000 Gram Panchayats, linking rural crop markets (e-NAM) with urban commodity hubs.

  - **Supports Social Infrastructure:** Enables sustainable financing for clean water, urban waste systems, and healthcare facilities to elevate human development indices.

Issues in Infrastructure Financing in India

- **Structural & Banking Constraints**
  - **Asset-Liability Mismatch (ALM):** Commercial banks rely on short-term deposits but fund long-gestation (20–30 years) infra projects, triggering structural liquidity stress.
  - **Legacy NPA Risk Aversion:** Past non-performing asset (NPA) crises in corporate lending have made scheduled commercial banks heavily cautious about backing greenfield projects.

    RBI show that the legacy infrastructure credit boom of the previous decade caused Gross Non-Performing Assets (GNPAs) to peak heavily at 11.46% in 2018

  - **Shallow Corporate Bond Market:** India's corporate bond market stays small (around 16–17% of GDP), leaving developers without deep alternatives for long-term domestic debt.

    According to the IMC Chamber of Commerce Industry Report (2026), India's corporate bond market stands at just 18% of the country's GDP.

- **Regulatory & Execution Risks**

- **Land & Clearance Bottlenecks:** Delays in land acquisition and environmental clearances stall execution, causing upfront project costs to spiral.

  According to MoSPI, out of 1,987 monitored ongoing central infrastructure projects, regulatory delays have accumulated a staggering cumulative cost overrun of about ₹5.4 Lakh Crore.
- **Weak Contract Enforcement:** Prolonged legal resolution timelines and overlapping regulatory bodies lock up private investor capital in endless litigation.
- **Policy & Tariff Inconsistency:** Sudden regulatory flip-flops or retrospective revisions in user fees and power tariffs damage long-term investor confidence.

- **Market & Investment Barriers**
  - **High Cost of Hedging:** Heavy currency fluctuation risk and expensive hedging costs discourage foreign pension and sovereign wealth funds from committing long-term debt.
  - **Flawed Risk Allocation:** Traditional Public-Private Partnership (PPP) contracts push uneven traffic and commercial risks onto private entities instead of balancing them.

    Economic Survey reveals that under Build-Operate-Transfer frameworks, 100% of the commercial traffic and revenue risk was pushed onto private players.
  - **Choked Exit Options:** A lack of smooth, institutional secondary market routes makes it difficult for initial developers to exit mature assets and recycle capital into new setups.

Measures Taken for Infrastructure Financing in India

- **Institutional Anchoring & Capital Infusion**
  - **Setting up NaBFID:** Established the National Bank for Financing Infrastructure and Development (NaBFID) as a dedicated statutory DFI to provide long-term, low-cost debt.
  - **Expanding NIIF Platforms:** Capitalized the National Investment and Infrastructure Fund (NIIF) across specialized wings like the Master Fund and India-Japan Fund to anchor global equity.
  - **Record Public Capex Allocation:** Scaled the Union Budget capital expenditure target to a record ₹12.22 Lakh Crore to crowd-in secondary private investments.
  - **Interest-Free Loans to States:** Continued the 50-year interest-free loan scheme for capital expenditure to support decentralized, state-level infrastructure creation.
- **Regulatory & Risk Mitigation Re-engineering**
  - **Launching the Risk Guarantee Fund:** Introduced the Infrastructure Risk Guarantee Fund to guarantee private lenders against non-commercial risks and policy shifts.
  - **Unified Project Finance Directions:** Enforced RBI's unified project finance regulations to regularize stage-based asset disbursements and check banking stress.
  - **Partial Credit Enhancement (PCE):** Deployed institutional PCE mechanisms via NaBFID to elevate infrastructure bond ratings and unlock bond-market access.
  - **Standardizing PPP Contracts:** Rolled out outcome-oriented model concession agreements (like the shift to HAM and TOT) to ensure balanced risk distribution.
- **Capital Market & Asset Monetization Frameworks**
  - **National Monetisation Pipeline (NMP):** Executed a systemic brownfield asset recycling drive to free public capital and rotate it directly into greenfield projects.
  - **Deepening InvITs & REITs:** Liberalized SEBI norms for Infrastructure Investment Trusts ($InvITs$) to transition mature post-construction assets off bank balance sheets.
  - **Small & Medium REITs (SM REITs):** Relaxed the minimum asset threshold limit down to ₹50 Crore to expand the universe of monetizable, mid-tier localized properties.
  - **Sovereign Green Bonds:** Issued specialized, rupee-denominated sovereign green debt instruments to exclusively capture global capital for clean energy transitions.
- **Technological & Spatial Synchronization**
  - **PM GatiShakti Master Plan:** Institutionalized a GIS-enabled multi-modal planning platform to de-risk projects upfront and prevent cost overruns from delayed coordination.

- **GatiShakti Public Data Access:** Opened up regulated geospatial data to private developers to optimize feasibility assessments and streamline bidding models.
- **City Economic Regions (CERs):** Allocated targeted financial outlays under a reform-linked model to strengthen municipal bond structures in Tier-II and Tier-III cities.

Way Forward for Infrastructure Financing in India

- **Market Diversification & Structural Shifting**
  - **Deepen the Corporate Bond Market:** Offer Partial Credit Enhancement (PCE) facilities which are recently introduced by specialized institutions to uplift infrastructure bond ratings and bypass bank credit limits.
  - **Scale InvITs & REITs:** Embed Infrastructure Investment Trusts (InvITs) deeper into public asset management to move post-construction assets off bank balance sheets and eliminate Asset-Liability Mismatches (ALM).
  - **Catalyze Municipal Bond Markets:** Drive structural reforms in municipal governance to allow Tier-II and Tier-III cities to independently issue bonds for localized urban development.
- **Innovative Fiscal & Risk Management**
  - **Deploy Risk Mitigation Funds:** Capitalize the **Infrastructure Risk Guarantee Fund** to offer prudently calibrated partial credit guarantees to institutional lenders.
  - **Accelerate Asset Monetization:** Optimize the updated National Monetization Pipeline (NMP) targets to systematically recycle capital from brownfield public utilities directly into greenfield expansions.
  - **Adopt Flexible Fiscal Frameworks:** Align national accounting targets with big investment pushes by adopting flexible budgetary deficit targets that favor high-quality, high-multiplier capital expenditure.
- **Resource & Operational Optimization**
  - **Mobilize Long-Term Patient Capital:** Ease regulatory investment norms for sovereign wealth funds and pension networks via anchor platforms like the National Investment and Infrastructure Fund (NIIF).
  - **Enforce Upfront Project De-Risking:** Fast-track statutory, land, and environmental clearances prior to final bidding to prevent execution delays from trapping private finance.
  - **Integrate Planning Frameworks:** Leverage the **PM Gati-Shakti National Master Plan** to ensure multi-modal synchronization across ministries, avoiding disconnected projects that hurt revenue generation.

Global Models for Infrastructure Financing

- **Asset Recycling Model (Australia):** Monetizes mature, operational public assets (like ports or roads) to lease them to private players, directly reinvesting the unlocked proceeds into new greenfield infrastructure.
- **Availability Payment PPP Model (Europe):** The government pays the private developer fixed, regular fees based on asset upkeep and performance metrics, fully absorbing the underlying consumer commercial and traffic risks.
- **Regulated Asset Base (RAB) Model (UK):** Gives private monopolies a predictable, legally guaranteed rate of return on their capital investments, adjusted routinely by independent regulators to shield investors from market shifts.
- **Blended Finance Model (World Bank / MDB Model):** Combines concessional capital or grants from multilateral banks with private commercial debt, acting as a strategic de-risking cushion to pull capital into low-margin social projects.
- **Land Value Capture Model (Hong Kong / East Asian Model):** Finances public transit networks by developing, leasing, or selling the high-value commercial real estate situated directly around transit nodes and stations.
- **Sovereign Wealth Co-Investment Model (Singapore / Temasek Model):** Uses state-backed equity platforms to match domestic funds directly with global pension networks, co-investing as commercial equity partners in critical national utilities.

Committee Reports on Infrastructure Financing in India

- **Rakesh Mohan Committee (1996):** Authored the seminal India Infrastructure Report, pioneering the commercialization of infrastructure and recommending the entry of private players into core public utilities.
- **Deepak Parekh Committee (2012–14):** Recommended the rationalization of user tariffs, strict restructuring of state power distribution companies (DISCOMs), and the creation of dedicated Infrastructure Debt Funds (IDFs) to bridge equity shortages.
- **Vijay Kelkar Committee (2015):** Focused on Revisiting & Revitalizing the PPP Model, advising the elimination of regular BOT contracts in favor of the Hybrid Annuity Model (HAM), banning retrospective policy changes, and setting up independent sectoral regulators.
- **H.R. Khan Committee (2016):** Formulated deep structural roadmaps to develop India's corporate bond market, easing investment limits for Foreign Portfolio Investors (FPIs) and introducing electronic book building platforms.
- **K.V. Kamath Committee (2020–21):** Outlined the operational blueprint and financial architecture required to set up a new-age development financial institution, directly culminating in the establishment of **NaBFID**.
- **Injeti Srinivas Committee (IFSCA, 2021–22):** Designed the global regulatory framework to turn GIFT City into an international hub for sustainable infrastructure financing, promoting offshore green bonds and alternative investment funds (AIFs).

Conclusion

Transforming India's infrastructure financing landscape requires a decisive shift away from bank-led credit toward a mature corporate bond market and scalable asset monetization models. By leveraging modern frameworks like NaBFID, InvITs, and the PM GatiShakti platform, the country can de-risk projects and smoothly channel deep global patient capital.

3. NATIONAL MONETISATION PIPELINE

National Monetisation Pipeline (NMP)

- **National Monetisation Pipeline (NMP)** is a Government of India initiative launched in **August 2021** by the **NITI Aayog**.
- It is a **4-year plan (2021–2025)** to **monetise core public sector brownfield assets**.
- The aim is to **unlock value from underutilised government assets** by leasing them to private players, while the **ownership remains with the government**.
- The fundamental idea behind the NMP is not to sell off government assets but to transfer their management and operation to private entities for a specified period, typically between 20 to 30 years. The ownership of the assets remains with the government.

Objectives of NMP

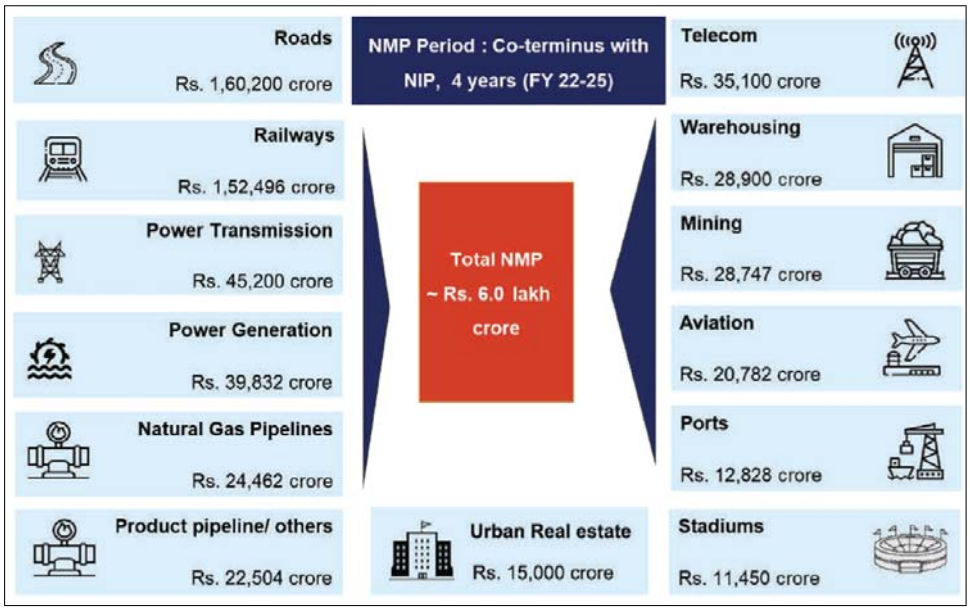
- **Capital Generation:** The primary purpose is to raise capital without increasing the public debt. The funds generated are intended to be reinvested into building new, modern infrastructure for the country.
- **Enhanced Efficiency:** The private sector is expected to bring greater efficiency and operational expertise to these assets, leading to better service delivery and maintenance.
- **Value Unlocking:** Many government assets are either underutilized or have not been maintained properly. Monetizing them helps unlock their hidden value and turn them into productive sources of income.

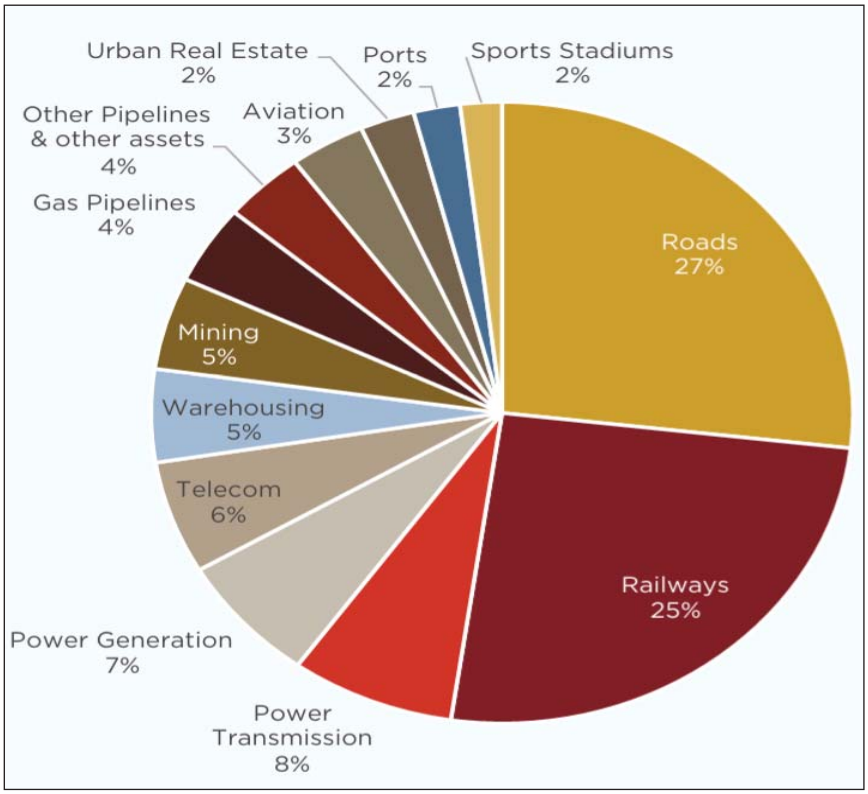
Features of NMP

- **Assets Leased, Not Sold**: Monetisation under NMP focuses on 'rights' rather than permanent transfer, with assets returned to government control after the contracted period.
- **Private Sector Participation:** The pipeline encourages public-private partnerships (PPPs) using models like Operate-Maintain-Transfer (OMT), Infrastructure Investment Trusts (InvITs), and Real Estate Investment Trusts (REITs).
- **Monetisation Target:** The first phase (FY 2022-2025) seeks to monetize core assets worth ₹6 lakh crore across sectors such as roads, railways, power, oil and gas pipelines, telecom, ports, airports, and warehousing. These are brownfield, revenue-generating assets, not non-core assets like land or real estate.
- **Second Phase (2025-2030):** Building on initial success, a higher target of ₹10 lakh crore for monetisation has been announced for highways, mining, power, and metro rail networks from FY26 to FY30.
- **Revenue for Infrastructure:** Monetised funds are earmarked for new and upgradation of existing infrastructure under the National Infrastructure Pipeline (NIP), worth ₹111 lakh crore.
- **Top Sectors:** Roads (27%), Railways (25%), Power (15%), Oil & Gas Pipelines (8%), Telecom (6%) are key focus areas, constituting over 80% of total potential.
- **Progress:** By early 2025, 90% of the monetisation target for the first phase had reportedly been achieved, with Rs 3.85 lakh crore raised by leasing assets such as highways, coal mines, and airports.

Key Sectors Covered

- **Roads:** A significant portion of national highways and expressways are identified for monetization.
- **Railways:** This includes railway stations, passenger trains, freight corridors, and railway colonies.
- **Power and Pipelines:** Power generation, transmission assets, and gas and petroleum product pipelines are on the list.
- **Telecommunications:** The government aims to monetize assets like optical fiber networks and telecom towers.
- **Ports and Aviation:** Select ports, berths, and airports are identified for monetization to improve their operational efficiency.
- **Stadiums and Urban Real Estate:** Other assets like sports stadiums, hotels, and urban development properties are also part of the plan.





Benefits of NMP

- **Mobilization of Capital for Infrastructure Development:** Generates substantial revenue (~₹6 lakh crore in phase 1) which the government reinvests into new and upgraded infrastructure projects, accelerating development under the National Infrastructure Pipeline (NIP).
- **Efficient Asset Utilization and Management:** Private sector participation promotes better operation and maintenance of brownfield assets, ensuring their optimum utilization and reducing wastage.
- **Reduced Fiscal Burden:** By monetizing existing assets instead of selling them outright, NMP reduces pressure on government finances and debt, allowing more budgetary space for social and economic priorities.
- **Boost to Economic Growth and Job Creation:** Monetization encourages investment, stimulates industrial activity and infrastructure-related job creation, supporting broad-based economic recovery and growth post-pandemic.
- **Promotion of Public-Private Partnerships (PPP):** Encourages structured collaborations between the public and private sectors, combining government oversight with private sector efficiency and innovation.
- **Improved Ease of Doing Business:** Standardized contracts and increased transparency reduce delays, litigation, and risks for private investors, making India more attractive for infrastructure investment.
- **Enhanced Quality of Public Services and Infrastructure:** Private sector management often leads to improved services, better technology adoption, and faster completion and upgradation of infrastructure assets.

Challenges/limitations faced by NMP

- **Valuation and Realising Adequate Value:**
  - Ensuring that assets are monetized at fair market value is difficult due to the complexity of valuation and potential lack of competitive bidding.

  - Low investor interest in assets with limited revenue potential or poor condition hampers value realization.
- **Investor Participation and Competition:**
  - Limited participants due to high capital requirements and risk perceptions could lead to cronyism where a few business houses corner assets.
  - Attracting diverse, long-term investors remains a challenge, especially for sectors with regulated tariffs or operational constraints.
- **Regulatory Issues:**
  - Fear of price hikes for end-users by private operators fuels opposition and social resistance.
  - Suboptimal judicial and contractual enforcement frameworks can result in implementation delays or disputes.
- **Political & State-level issues:**
  - Some state governments exhibit reluctance toward participation due to political considerations or concerns over loss of control.
  - Coordination between central and state governments is essential but challenging.
- **Monopoly Risks:** Monetisation of essential infrastructure assets like highways or railways can lead to monopolies, giving excessive market power to private players with potential negative effects on pricing and service.
- **Execution Risk:** Large-scale monetisation programs face risks in execution timelines, project cost overruns, and maintaining operational standards.
- **Social Concern:** Questioning the fairness of monetizing public assets that taxpayers have already paid for, and concerns over handing over essential services to private entities.

Way forward

- **Expansion & Diversification:**
  - **Including State-Level Assets:** Extending the pipeline to incorporate assets owned by state governments to significantly broaden the monetisation base.
  - **New Sectors:** Adding urban infrastructure, digital assets, sports infrastructure, and other public utilities to diversify asset classes and investor appeal.
- **Strengthening Investor Confidence:**
  - Creating transparent, standardized bidding processes and contracts with well-defined Key Performance Indicators (KPIs) and service-level agreements (SLAs).
  - Enhancing legal and regulatory frameworks to ensure dispute resolution, contract enforcement, and predictability for investors.
- **Enhancing Private Sector Participation:**
  - Developing co-investment models involving sovereign wealth funds, pension funds, and multilateral agencies to attract long-term, stable funding.
  - Addressing concerns about monopoly risks and ensuring competitive market environments.
- **Leveraging Asset Recycling:** Promoting "asset recycling" where leased assets generate funds that are reinvested back into new infrastructure projects, creating a sustainable monetisation cycle aligned with the National Infrastructure Pipeline.
- **Capacity Building & Technical Assistance:** Assisting Public Sector Undertakings (PSUs) and government bodies with technical expertise, capacity building, and digital tools to effectively manage the monetisation process and optimize asset value.

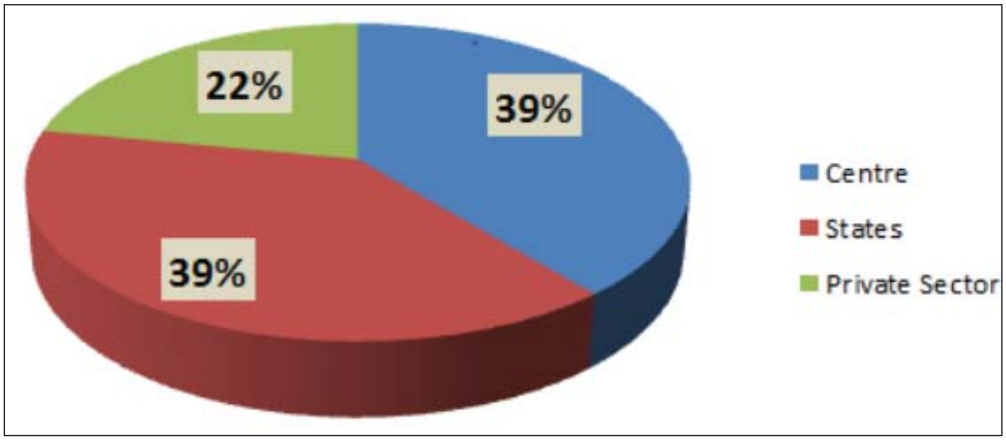
4. NATIONAL INFRASTRUCTURE PIPELINE

National Infrastructure Pipeline

- The National Infrastructure Pipeline (NIP) is a flagship initiative by the Government of India aimed at providing a comprehensive roadmap for infrastructure development across the country over a period of five years (2020-2025).

Key Features of NIP:

- **Massive Investment Outlay:** The NIP initially projected a total investment of **₹111 lakh crore** for infrastructure projects over a five-year period (2020-2025).
- **Comprehensive Scope:** The NIP covers a wide range of social and economic infrastructure projects. These include both **greenfield** (new projects) and **brownfield** (existing projects) and are spread across various sectors.
- **Multi-Stakeholder Funding:** The NIP is designed with a joint funding model to ensure financial viability. The capital expenditure is planned to be shared among the **Centre, States, and the private sector**. The original target for this split was roughly 39%, 39%, and 22% respectively, with a strong emphasis on attracting private investment.

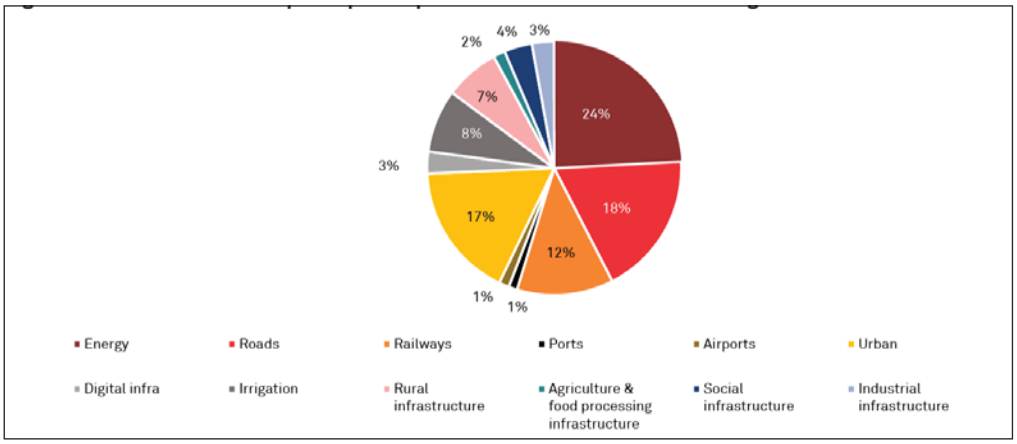


- **Project Monitoring:** To ensure timely completion and to avoid cost overruns, a task force was established to monitor the progress of the NIP. The India Investment Grid (IIG) was also launched to provide real-time updates on the implementation of these projects, enhancing transparency and attracting domestic and foreign investors.
- **Complementary Initiatives:** The NIP is closely linked with other government initiatives like the **National Monetization Pipeline (NMP)**, which aims to generate funds by monetizing existing public infrastructure assets, and the **PM Gati Shakti** program, which provides a digital platform for integrated and seamless planning and execution of projects.

Sectors Covered

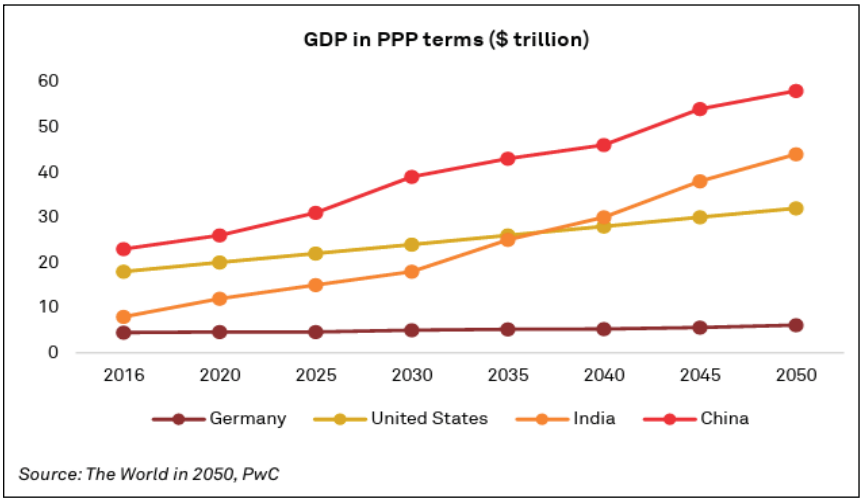
- Energy- 24%
- Roads- 19%
- Railways- 13%
- Ports- 1%
- Airports-1%
- Urban-16%

- Digital Communication- 3%
- Irrigation-8%
- Rural Infrastructure-8%
- Agriculture and Food Processing Infrastructure-1%
- Social Infrastructure-3%
- Industrial Infrastructure-3%

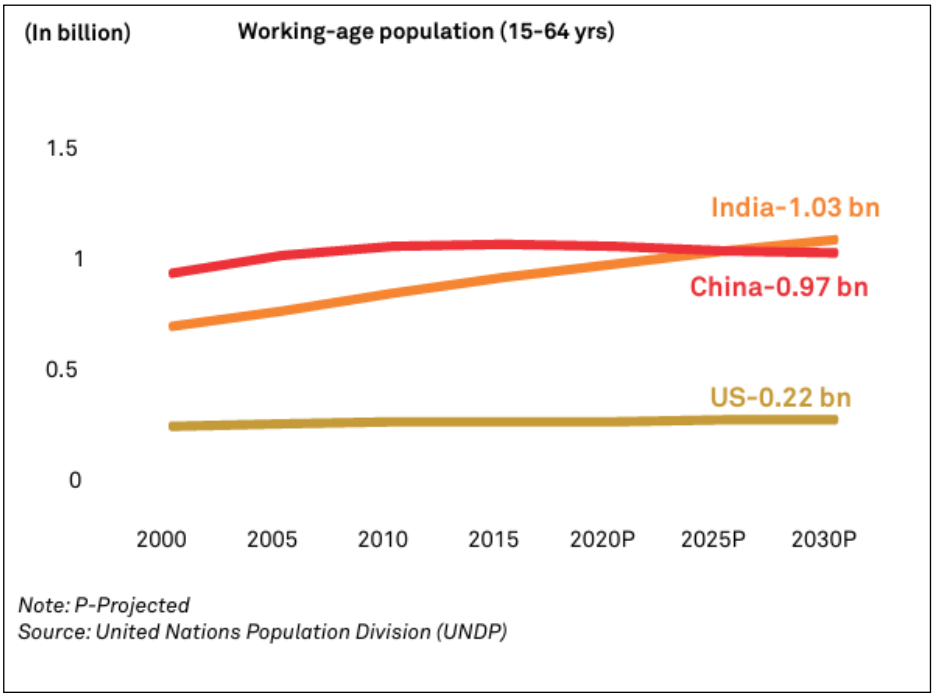


Benefits of NIP

- **Boost to Economic Growth:**
  - Infrastructure development under NIP is critical to achieving India's goal of becoming a $5 trillion economy by 2025 & 2nd largest economy in the world in PPP terms.



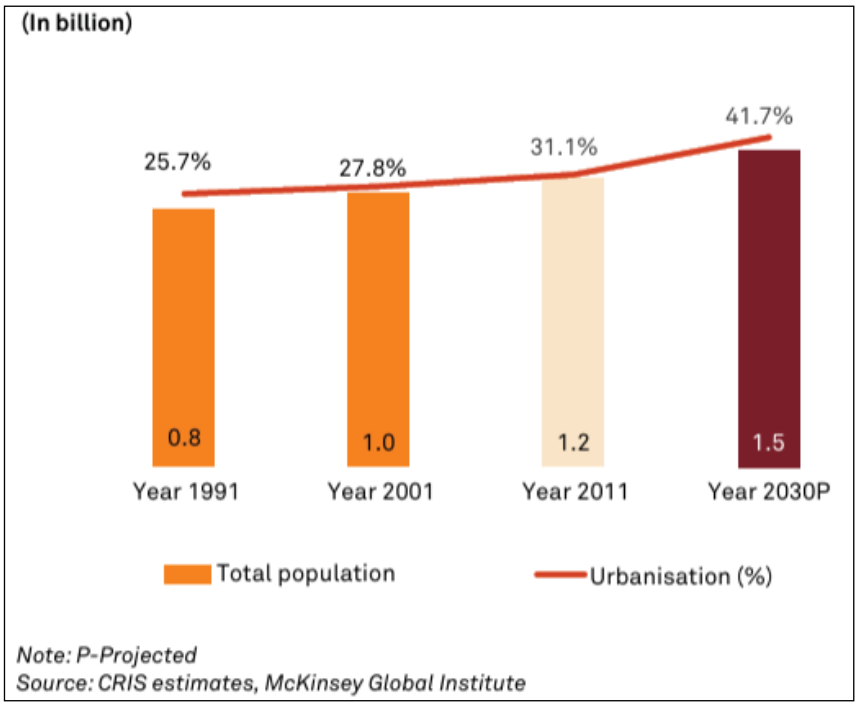
  - Improves competitiveness by enhancing transportation, energy, and urban infrastructure, promoting industrial and commercial activities.
  - Expected to increase GDP growth through better connectivity and efficient resource utilization.
- **Job Creation & Employment:**
  - The large-scale infrastructure projects generate millions of direct and indirect jobs across sectors such as construction, manufacturing, and services.



Note: P-Projected
Source: United Nations Population Division (UNDP)

- Employment growth supports increased income and domestic demand, further boosting economic expansion.

- **Improved Quality of Life & Inclusive Growth:**
  - Enhances access to essential services including roads, energy, healthcare, education, and urban amenities.



Note: P-Projected
Source: CRIS estimates, McKinsey Global Institute

- Facilitates rural connectivity and infrastructure, promoting inclusive development across regions.

- **Investor confidence & Project Efficiency:**
  - NIP's well-defined pipeline improves project preparation, timely execution, and monitoring, reducing risks and failures.

  - Attracts private sector and foreign investment by providing a transparent, accessible project database via platforms like the India Investment Grid (IIG).
- **Fiscal Space & Revenue Enhancement:**
  - Better infrastructure improves economic activity, expanding the government's revenue base and enabling productive expenditure.
  - Asset monetization of completed projects under NIP creates additional fiscal resources.
- **Agriculture & Rural Development:** Significant allocation to irrigation and rural infrastructure strengthens the agricultural sector and boosts rural livelihoods. It also enhances supply chains and market access for farmers.

Challenges/limitations of NIP

- **Funding & Financial Constraints:** The biggest challenge is securing the massive investment of **₹111 lakh crore**. While the plan relies on a multi-stakeholder funding model (Centre, states, and the private sector), there are significant hurdles:
  - **State Finances:** Many states have limited fiscal space and are already burdened with high debts, making it difficult for them to contribute their share of the funding.
  - **Attracting Private Investment:** While the NIP aims to attract private capital, a number of factors could deter investors, including regulatory uncertainties, land acquisition issues, and long gestation periods for projects. The global economic slowdown could also make investors more cautious.
  - **Inflation and Cost Overruns:** Large-scale infrastructure projects are highly vulnerable to inflation and cost overruns, which could push the final bill far beyond the initial estimates.
- **Implementation & Bureaucratic Hurdles:**
  - **Land Acquisition:** This is a persistent bottleneck for large-scale projects in India. Delays in acquiring land can stall projects, leading to significant cost overruns and public protests.
  - **Lack of Timely Clearances:** Obtaining various clearances—environmental, forest, and local—is a time-consuming and complex process. Delays in these approvals can severely impact the project timeline.
  - **Inadequate Project Management:** A lack of skilled project managers and poor coordination between different government agencies can lead to inefficiency and delays. While initiatives like **PM Gati Shakti** aim to address this, on-ground implementation remains a challenge.
- **Political & Governance Issues:**
  - **Centre-State Coordination:** The NIP's multi-stakeholder model requires seamless cooperation between the central government and state governments, regardless of political affiliation. Any political friction can lead to delays in approvals and funding.
  - **Uneven Distribution:** There is a risk that the allocation of funds may be politically motivated, leading to a skewed distribution of projects across states, with some regions receiving a disproportionately high share of investment while others are neglected.
  - **Regulatory Challenges:** A lack of consistent and transparent regulatory frameworks can make it difficult for investors to operate. Changes in policy, particularly regarding public-private partnerships, can create uncertainty and deter investment.

Way forward

- **Enhance Financial Innovation:** India needs to move beyond traditional bank-based financing for infrastructure projects. The government should encourage the development of the corporate bond market and promote innovative financing instruments, such as infrastructure investment trusts (InvITs), which allow a wide range of investors to participate in infrastructure projects and provide stable long-term funding.

- **Asset Monetization:** The **National Monetization Pipeline (NMP)** is a crucial part of the strategy. By monetizing existing public assets like roads, railways, and power transmission lines, the government can generate new funds that can be reinvested into greenfield and brownfield projects under the NIP.
- **Use of PM Gati Shakti:** The **PM Gati Shakti** platform is the central tool for integrated infrastructure planning and execution. PM Gati Shakti should fully utilize its potential to ensure seamless coordination between different ministries and state governments. The platform provides a digital, data-driven approach to planning, which can help in reducing delays, preventing cost overruns, and ensuring that projects are executed on time.
- **Dispute Resolution:** A fast and effective dispute resolution mechanism is vital for investor confidence. Establishing a dedicated **GST Appellate Tribunal** and other such specialized bodies can help in resolving contractual disputes efficiently, reducing the burden on courts.
- **Focus on States:** Since states are expected to contribute a significant portion of the funding, the central government must work with them to improve their fiscal health and help them in the effective planning and execution of projects.
- **Inclusive & Social Infrastructure:** Greater focus should be put on social infrastructure (health, education, water), rural connectivity, and climate-resilient projects to ensure equitable growth.

■■■

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AGRICULTURAL SYSTEMS

1. INDIAN AGRICULTURE: STATUS & CHALLENGES

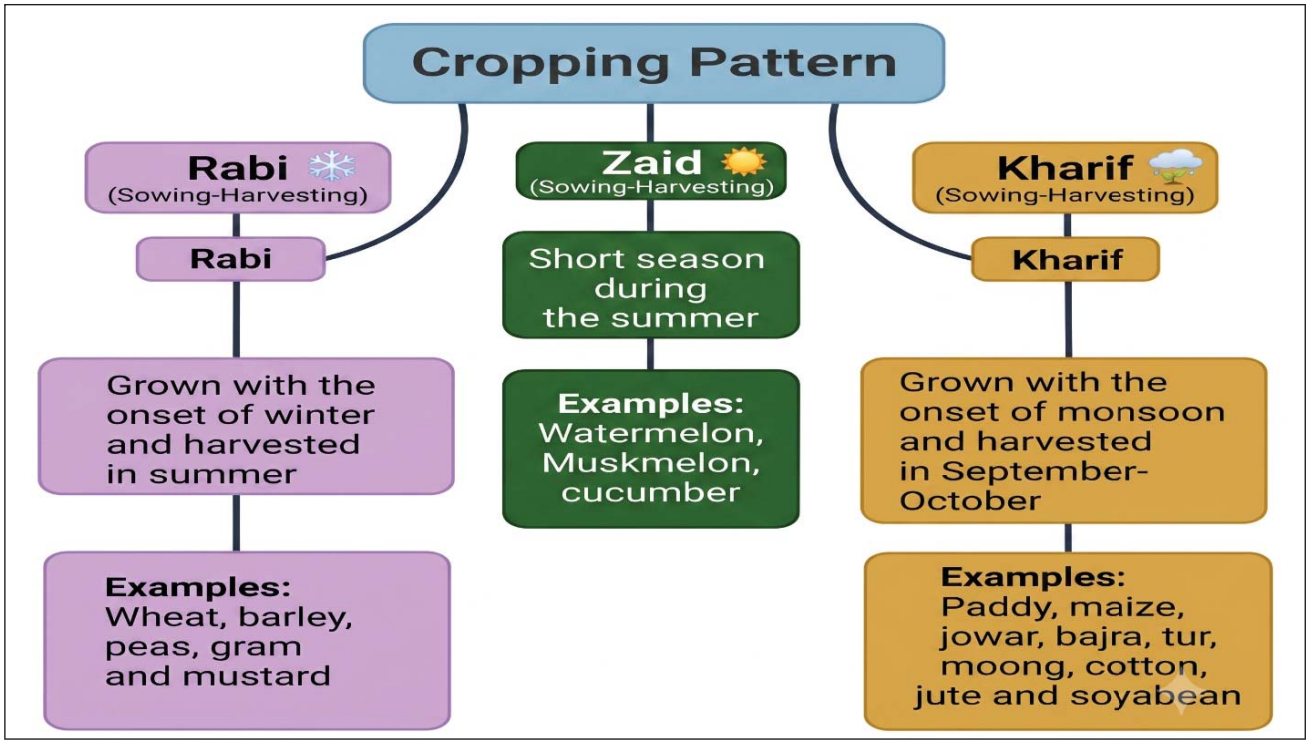
Introduction:

Agriculture has always been central to India's growth. It feeds the nation, supports the livelihoods of millions of people, and remains the main source of income in rural areas. Although the economy has expanded into industry and services, farming continues to be crucial for rural development and environmental sustainability.

Evolution of Indian Agriculture

- **Post-Independence Phase (1947–1965)**
  - **Pre-conditions:** During Independence, India faced frequent food shortages, low agricultural productivity, and depended heavily on food grain imports from the United States under the PL-480 programme, leading to a **ship-to-mouth** situation.
  - **Institutional reforms:** The government introduced land reforms to abolish the Zamindari system, protect tenant farmers, and impose land ceiling laws to make land ownership more equitable.
  - **Infrastructure development:** Large irrigation and multipurpose river valley projects such as **Bhakra Nangal** and **Hirakud** were built to expand irrigation and reduce dependence on uncertain monsoon rains.
- **Green Revolution Phase (1965–1980s)**
  - **Technological breakthrough:** India introduced **High-Yielding Variety (HYV)** seeds for wheat and rice, with major contributions from **M. S. Swaminathan** and **Norman Borlaug**.
  - **Modern farming practices:** Farmers began using a package of HYV seeds, chemical fertilizers, pesticides and tube-well irrigation, which significantly increased production per hectare.
  - **Government support:** To encourage farmers, the government introduced the **Minimum Support Price (MSP)** system and set up institutions to ensure fair prices, procure food grains and maintain buffer stocks.
  - **Impact:** Green Revolution turned India from a food-deficit country into a food-surplus nation. However, its benefits were largely concentrated in states like Punjab, Haryana, and western Uttar Pradesh.
- **Sectoral Diversification Phase (1980s–2000s)**
  - **White Revolution:** India promoted dairy cooperatives through the **Amul model**, helping millions of farmers increase their incomes and making the country the world's largest milk producer.
  - **Yellow Revolution:** Special efforts were made to increase the production of oilseeds so that India could reduce its dependence on imported edible oils.
  - **Golden Revolution:** The focus expanded beyond food grains to horticulture, including fruits, vegetables, flowers and beekeeping, giving farmers new and more profitable sources of income.

- **Commercialization Phase (2000s–2015)**
  - **Biotech Adoption:** Approved Bt Cotton for commercial cultivation, turning India into a major global cotton exporter.
  - **Policy Push:** Launched national programs like the National Food Security Mission (NFSM) and Rashtriya Krishi Vikas Yojana (RKVY) for decentralized farm development.
  - **Early Market Reforms:** Introduced model APMC Acts to encourage private investment, contract farming, and direct marketing channels.
- **Sustainable Phase (2015–Present)**
  - **Shift to Income Security:** Reoriented policy focus from mere crop production to boosting overall net farm income.
  - **Direct Income:** Launch of PM-KISAN for direct income transfers and PM Fasal Bima Yojana (PMFBY) for crop insurance coverage.
  - **Resource Conservation:** Promoted sustainable practices through Soil Health Cards, micro-irrigation under PMKSY, Paramparagat Krishi Vikas Yojana (PKVY) for organic farming and Integrated Farming Systems (IFS).
  - **Digital Support:** Integrated farm markets via e-NAM, supported Farmer Producer Organizations (FPOs), and deployed agritech solutions (drones, AI, and precision farming).



Major Types of Agricultural Systems in India

Agricultural Type	Characteristics	Primary Crops & Regions
Intensive Subsistence Farming	High labor input and heavy fertilizer/ irrigation use on small, fragmented plots; aim is maximum yield for family consumption and local markets.	**Crops:** Rice, Wheat, Pulses, Vegetables. **Regions:** Densely populated plains of UP, Bihar, West Bengal, and Punjab.

Primitive Subsistence (Shifting / Jhuming)	Slash-and-burn method on forest patches using basic tools (hoes, digging sticks)	**Crops:** Coarse Rice, Maize, Millets, Yams. **Regions:** Tribal belts of North-East India (Assam, Meghalaya, Nagaland).
Plantation Agriculture	Capital-intensive, single-crop (monoculture) commercial farming introduced during the colonial era	**Crops:** Tea, Coffee, Rubber, Sugarcane, Coconut. **Regions:** Assam, West Bengal, Nilgiri hills (Karnataka, Kerala, Tamil Nadu).
Mixed Farming	Simultaneous cultivation of crops alongside livestock rearing, poultry, or dairy on the same farm footprint.	**Crops:** Food grains, fodder crops, milk, eggs, meat. **Regions:** Indo-Gangetic plains, Western India, parts of AP and Karnataka.
Dryland / Rainfed Farming	Practiced in semi-arid regions receiving less than 75 cm annual rainfall	**Crops:** Millets (Jowar, Bajra, Ragi), Pulses, Oilseeds, Cotton. **Regions:** Rain-shadow areas of Deccan Plateau, Rajasthan, Gujarat, Marathwada.
Wetland Farming	Practiced in high-rainfall, flood-prone, or well-irrigated coastal/river delta regions.	**Crops:** Paddy (Rice), Jute, Sugarcane. **Regions:** Coastal Odisha, Andhra Pradesh, Kerala, West Bengal, Brahmaputra valley.
Natural & Organic Farming	Zero-chemical farming relying on local natural inputs (cow dung/urine formulations, vermicomposting, cover crops) to restore soil microbes.	**Crops:** Organic Grains, Pulses, Fruits, Vegetables. **Regions:** Andhra Pradesh (ZNF), Sikkim (100% Organic state), Himachal Pradesh.

Status of Indian Agriculture

- **Contribution & Growth**
  - **Contribution to Gross Value Added (GVA):** Agriculture and allied sectors contribute approximately **16.2%–17.5%** to India's total GVA.
  - **Sectoral Growth Rate:** Registered **3.1% (FY26 projection)** and **3.6% growth (H1 FY26)**. The 5-year average annual growth rate stands at around **4.4%** at constant prices. **(**Economic Survey 2025–26)
- **Production Performance**
  - **Total Foodgrain Production:** Reached an all-time record high of 357.73 Million Metric Tonnes (MMT).
    - **Rice Output:** Record **150.18 MMT** (1,501.84 LMT).
    - **Wheat Output:** Record **117.94 MMT** (1,179.45 LMT).
    - **Pulses: 25.68 MMT**.
    - **Oilseeds: 42.9 MMT** (~43 MMT).
  - **Horticulture Output:** Reached **362.08 Million Tonnes (MT)**, continuing to surpass traditional foodgrain production.

Significance of Indian Agriculture

- **Source of Livelihood and Employment**
  - **Main source of livelihoods:** Agriculture is the largest source of employment in India, directly or indirectly supporting the livelihoods of nearly **45–48%** of the population.
  - **Acts as a safety net:** During economic crises, natural disasters, or slowdowns in urban areas, many workers return to their villages and depend on agriculture for work and income.
- **Food & Nutritional Security**
  - **Ensuring food security:** Agriculture produces over **330 million tonnes of food grains** every year, providing enough food to meet the needs of India's **1.4 billion** people and strengthening the country's food security.
  - **Supporting welfare programmes:** Food grains produced by farmers are used in major government schemes such as the **National Food Security Act (NFSA)** and the **PM Garib Kalyan Anna Yojana**.
- **Contribution to Economy & National Income**
  - **Contributor to the economy:** Agriculture contributes to India's **Gross Value Added (GVA)** and plays a key role in supporting economic growth.

    As per the National Accounts Statistics released by MoSPI (23-24), the Agriculture and Allied Sector contributes approximately 18% to the total national Gross Value Added (GVA)
  - **Boosts rural demand:** Higher farm incomes increase spending in rural areas, leading to greater demand for everyday consumer goods, tractors, two-wheelers, automobiles and construction materials.
- **Linkage with the Industrial Sector**
  - **Provides raw materials:** Agriculture supplies essential raw materials to many industries, including cotton and jute textiles, sugar mills, edible oil processing and food processing industries.
- **Foreign Exchange & Global Trade**
  - **Source of export earnings:** Agricultural products such as **basmati and non-basmati rice, spices, tea, sugar, coffee and marine products** make a significant contribution to India's exports and earn valuable foreign exchange.

    As per APEDA (2024 -25), India's total agricultural and allied commodity exports stood at USD 51.91 billion during the 2024–25 fiscal year, accounting for about 11.86% of the country's total merchandise exports
  - **Presence in global markets:** India is one of the world's leading exporters of several agricultural products, helping strengthen trade ties with countries in **Africa, the Middle East, and South Asia**.

    India ranks among the top 10 agricultural exporters globally (holding a 2.5% share in global agricultural trade).

Conservation Agriculture

Conservation Agriculture is a farming approach that helps protect fertile farmland from degradation while restoring damaged soil. It focuses on conserving soil, water, biodiversity, and farm resources. The system is based on practices such as **zero or minimum tillage**, **crop rotation**, and **intercropping**, which improve soil health, reduce cultivation costs, and make farming more sustainable over the long term.

Factors Influencing Indian Agriculture

- **Natural & Environmental Factors**
  - **Monsoon rainfall:** Timely and well-distributed rainfall helps farmers achieve good harvests, while delayed rainfall can lead to droughts, floods and lower crop production.

- **Agro-climatic diversity:** India's diverse climate, with different temperatures, rainfall patterns, and growing conditions, allows farmers to cultivate a wide variety of tropical, subtropical, and temperate crops across different regions.
- **Soil type and fertility:** The type and quality of soil play an important role in deciding which crops can be grown.

  Black soil is well suited for cotton because it retains moisture, while alluvial soil is highly fertile and ideal for crops such as wheat, rice, and sugarcane.

- **Physical Factors**
  - **Irrigation Facilities:** Access to canal networks, tubewells, or drip irrigation protects crops against draught and enables multi-cropping on the same plot.
  - **Rural Power & Storage:** Dependable electricity for running pumpsets and local cold storage facilities helps store perishable produce like fruits, milk, and vegetables.
  - **Road & Transport Connectivity:** All-weather village roads ensure quick transport of harvested produce to nearby markets.
- **Economic & Market Factors**
  - **Government price support (MSP): Minimum Support Price (MSP)** plays a major role in farmers' crop choices. Since crops like wheat and paddy receive assured procurement, many farmers prefer growing them instead of crops that have little or no price support.
  - **Credit Access:** Availability of low-interest institutional loans through Kisan Credit Cards (KCC) prevents farmers from falling into informal debt traps.

    As per NABARD, Over 7.3+ crore active Kisan Credit Cards are currently in operation across the country, covering crop cultivation as well as allied activities

- **Technical Factors**
  - **Hybrid Seeds:** Access to pest-resistant and climate-resilient seed varieties directly increases crop productivity per hectare.

    As per ICAR, High-Yielding Varieties (HYV) and climate-resilient hybrid seeds contribute 20%–25% to total crop yield enhancement per hectare.

  - **Farm Mechanization:** Use of tractors, harvesters, and seed drills reduces labor shortages during peak seasons and improves overall efficiency.

    India's overall farm mechanization level stands at 40%–45% (compared to over 80% in developed nations like the USA and 70% in China).

Major Challenges Facing Indian Agriculture

- **Land-Related Issues**
  - **Small and fragmented landholdings:** Most farmers in India own very small pieces of land, making it difficult to use large machines and reduce production costs.

    As per Agriculture Census 2015-16, small and marginal farmers account for 86.08% of total operational holdings, with overall average farm size declining to 1.08 hectares.

  - **Insecure land leasing:** In many villages, land is leased through informal oral agreements instead of written contracts. As a result, it becomes difficult for them to access bank loans, crop insurance, and government compensation or subsidy schemes.

    As per NITI Aayog, Tenant farmers account for ~10%–12% of operational holdings nationally, with over 80% operating on informal oral leases without written contracts.

- **Input & Technology Deficits**
  - **Excessive Fertilizer Use:** Unrestricted subsidies on urea have disbalanced the ideal N-P-K ratio, leading to severe soil degradation, micro-nutrient deficiencies and falling crop response to inputs.
  - **Low Seed Replacement Rate:** High costs and limited availability of certified High-Yielding Variety (HYV) and climate-resilient seeds force small farmers to reuse low-quality farm-saved seeds.

    As per ICAR, Seed Replacement Rate (SRR) for crucial pulse and oilseed crops remains low at 20%–30%, whereas cereals like hybrid maize and paddy range from 35%–45%.
- **Water Management issue**
  - **Heavy dependence on the monsoon:** Nearly **half of India's net sown area** depends on rainfall. As a result, delayed monsoon rains can seriously affect crop production and farmers' incomes.
  - **Declining groundwater levels:** The excessive use of groundwater to grow water-intensive crops like **paddy** and **sugarcane**, has led to a sharp fall in groundwater levels.

    As per CGWB, Agriculture accounts for 89%–90% of total groundwater extraction in India. Over 30% of assessed assessment units are categorized as 'Over-exploited', 'Critical', or 'Semi-Critical'.
- **Financial Bottlenecks**
  - **Limited access to affordable credit:** Many small farmers still cannot access formal bank loans and depend on local moneylenders for credit. These loans carry very high interest rate than formal bank loans.
  - **Limited crop diversification:** Since crops like **rice and wheat** receive MSP support, many farmers continue to grow them. In contrast, pulses, oilseeds, fruits, and vegetables often face uncertain market prices, discouraging farmers from shifting to these crops.

    As per FCI, Paddy and wheat alone account for over 70%–75% of total government procurement expenses under MSP.
- **Climate Change Vulnerability**
  - **Frequent Extreme Weather Events:** Rising global temperatures, unpredictable dry spells, flash floods, and unseasonal rains during harvest seasons cause massive yield losses each year.

    As per Centre for Science and Environment, Climate variability causes an estimated average loss of 1.5% to 2% of agricultural GDP annually.
  - **Limited crop insurance coverage:** Although schemes like Pradhan Mantri Fasal Bima Yojana (PMFBY) provide crop insurance, many farmers still do not benefit fully due to delays in claim settlement and disputes over crop loss assessment.

Major Policy Measures & Government Initiatives in Indian Agriculture

- **Financial Support**
  - **PM-KISAN (Pradhan Mantri Kisan Samman Nidhi):** Provides direct income support of **₹6,000 per year** in three equal installments to landholding farmer families.
  - **PM-KMY (Pradhan Mantri Kisan Maandhan Yojana):** A voluntary pension scheme assuring a monthly payout of **₹3,000** to small and marginal farmers upon attaining 60 years of age.
- **Risk Management**
  - **PMFBY (Pradhan Mantri Fasal Bima Yojana):** Provides comprehensive crop insurance against non-preventable natural risks (droughts, floods, localized storms) at low, standardized premium rates (1.5% for Rabi, 2% for Kharif, 5% for commercial crops).
  - **PM-AASHA (Annadata Aay Sanrakshan Abhiyan):** Protects farmers from sudden market price crashes through the Price Support Scheme (PSS), Price Deficiency Payment Scheme (PDPS) and Market Intervention Scheme for horticultural produce.

- **Resource Conservation**
  - **Soil Health Card Scheme:** Provides customized nutrient status reports to farmers every 2–3 years, encouraging balanced fertilizer application and cutting unnecessary urea usage.
  - **PMKSY – "Per Drop More Crop":** Promotes micro-irrigation technologies (drip and sprinkler systems) to maximize water-use efficiency and reduce groundwater overdraft.
  - **PKVY & Natural Farming Push:** Paramparagat Krishi Vikas Yojana promotes cluster-based organic farming and natural, chemical-free input management.
- **Post-Harvest Infrastructure**
  - **Agriculture Infrastructure Fund (AIF):** A **₹1 lakh crore** financing facility offering 3% interest subvention for setting up cold storages, primary processing units, and farm-gate supply chains.
  - **e-NAM (National Agriculture Market):** A pan-India electronic trading portal that connects existing APMC mandis to build a unified national market for transparent price discovery.
  - **Formation of 10,000 FPOs:** Encourages smallholders to aggregate through Farmer Producer Organizations to improve their collective bargaining power for inputs and market access.
- **Technology, Digitalization & Modern Initiatives**
  - **AgriStack & Digital Public Infrastructure:** Builds a unified digital ecosystem featuring digital crop surveys, electronic land record integration and targeted advisory services.
  - **Sub-Mission on Agricultural Mechanization (SMAM) & Namo Drone Didi:** Subsidizes farm machinery, custom hiring centers and agricultural drones operated by rural women self-help groups for precise spraying.

Way forward

- **Promoting Sustainable & Climate-Resilient Agriculture**
  - **Adopt integrated farming:** Encourage farmers to follow **Integrated Farming Systems (IFS)** by combining crops with dairy, poultry, fisheries, and agroforestry.

    Rice-Duck farming (Japan/Vietnam) where ducks control weeds and pests naturally while oxygenating the soil.
  - **Promote natural and organic farming:** Expand **Natural Farming** and the **Paramparagat Krishi Vikas Yojana (PKVY)** to reduce dependence on chemical fertilizers and pesticides.
  - **Improve water management:** Expand micro-irrigation systems such as **drip** and **sprinkler irrigation** under **PMKSY's "Per Drop More Crop"** programme. Community-based water budgeting can also help conserve water in drought-prone regions .

    Israel's micro-irrigation and wastewater grid, recycling over 85% of municipal wastewater for agriculture.
  - **Encourage climate-resilient crops:** Promote the cultivation of **pulses, oilseeds, and millets (Shree Anna)** in water-scarce areas instead of water-intensive crops like paddy and sugarcane.

    Odisha Millet Mission (OMM) promoting millets (Shree Anna) in dryland areas through direct procurement and PDS supply.
- **Marketing Reforms**
  - **Strengthen agricultural markets:** Improve the functioning of **e-NAM** by linking it more effectively with local **APMC mandis** and providing facilities such as sorting, grading, quality testing and quick dispute resolution.
  - **Strengthen Farmer Producer Organizations:** Expand and support **Farmer Producer Organizations** so that small farmers can work together. This will help them sell directly to large buyers and food processing companies.

Sahyadri Farmers Producer Company (Maharashtra), aggregating smallholders to directly export grapes to international markets.

- **Improve storage and logistics:** Use the **Agriculture Infrastructure Fund (AIF)** to build cold storages, warehouses, primary processing units, and drying yards closer to villages.

- **Institutional & Policy reforms**
  - **Improve coordination:** Bring the **Agriculture, Animal Husbandry, Fisheries, and Forest** departments together at the district level through a single coordination mechanism.
  - **Legalise land leasing:** Encourage states to adopt the **Model Agricultural Land Leasing Act** so that tenant farmers receive legal recognition.
- **Technological reforms**
  - **Use digital technology:** Strengthen **AgriStack** by linking farmer databases, land records, and satellite-based crop monitoring. This will help deliver credit more efficiently, speed up crop insurance settlements under **PMFBY**.

    Karnataka's FRUITS portal linking farmer IDs, land records, and satellite surveys for instant loans and insurance payouts.
  - **Promote modern farming:** Encourage the use of agricultural drones, **Custom Hiring Centres**, and technologies such as IoT-based sensors for precision farming.
  - **Strengthen agricultural services:** Make **Krishi Vigyan Kendras (KVKs)** more practical by focusing on field demonstrations instead of only classroom training.

    Krishi Vigyan Kendras (KVKs) organizing practical "Field Days" for on-ground crop residue management in Punjab.

Major Agriculture Committees & Recommendations

Committee & Year	Key Recommendations
National Commission on Farmers (M.S. Swaminathan Committee, 2004–2006)	• Set MSP at cost of production + 50% (popularly known as the C2 + 50% formula). • Move Agriculture to the Concurrent List of the Constitution. • Issue Soil Health Cards to all farmers to check over-fertilization. • Distribute surplus ceiling land to landless agricultural labor.
Committee on Doubling Farmers' Income (Ashok Dalwai Committee, 2016–2018)	• Treat farming as a business enterprise rather than a mere subsistence activity. • Promote Integrated Farming Systems (IFS)—combining crops, dairy, poultry, and trees to ensure regular daily cash flows. • Reform APMC mandis, encourage private contract farming, and expand e-NAM. • Focus on non-farm income generation through rural agro-processing hubs.
High-Level Committee on FCI Restructuring (Shanta Kumar Committee, 2014–2015)	• Unbundle FCI operations into three distinct arms: Procurement, Storage, and Distribution. • Hand over procurement of wheat and paddy to states with sufficient infrastructure and shift FCI focus to Eastern states. • Introduce direct cash transfers in place of physical grain distribution under PDS where markets are well-connected. • Rationalize coverage under NFSA from 67% down to 40%.

Indian Agricultural Case Studies & Success Models

Model & Region	Working Mechanism
Andhra Pradesh Community-Managed Natural Farming (APCNF) (Andhra Pradesh)	• Eliminates synthetic fertilizers/pesticides using natural bio-stimulants (Jeevamrutha, Beejamrutha) and continuous soil mulching (Achaadana). • Promotes year-round 365-day green cover and multi-cropping to revive soil microbes and conserve moisture. • Driven entirely by community cadres and women's Self-Help Groups (SHGs) for peer training and village-level extension.
Amul Cooperative Model	• Collects milk twice daily with automated fat-testing machines, ensuring transparent, immediate payouts to farmers. • Operates a unified cold-chain processing network, returning the vast majority of the consumer rupee directly back to dairy farmers.
Solar Pump Water Leasing (IWMI-TAAG Model) (Dhundi, Gujarat)	• Farmers form Solar Pump Irrigators' Cooperative Societies (SPICE) to share solar-powered irrigation infrastructure. • Solar panels power water pumps during daylight hours to irrigate crops without diesel fuel. • Excess solar power generated is sold directly back to the local electricity grid (DISCOM) at a fixed feed-in tariff.
Hiware Bazar Watershed & Water Budgeting (Ahmednagar, Maharashtra)	• Enforces a strict community ban on water-intensive crops (like sugarcane or banana) in favor of pulses and millets. • Builds continuous contour trenches, check dams, and percolation tanks to capture surface runoff. • Mandates a village **Water Audit** every year to calculate available groundwater before deciding winter crop choices.

Conclusion:

India needs to promote **Integrated Farming Systems (IFS)**, strengthen **Farmer Producer Organizations (FPOs)**, and develop better storage, processing, and cold-chain facilities close to farms. At the same time, better use of digital technologies like **AgriStack**, along with climate-smart farming practices, can help farmers cope with unpredictable weather and fluctuating market prices. Improving the lives of farmers is essential for building a self-reliant agricultural sector (**Atmanirbhar Krishi**), ensuring inclusive rural development and achieving the vision of **Viksit Bharat @2047**.

2. CROPPING PATTERN IN INDIA

Introduction:

Cropping pattern refers to the **proportion of area under different crops at a given point of time**, along with the yearly sequence and spatial arrangement in which those crops are grown on a piece of land. India's post-Green Revolution cropping pattern successfully delivered foodgrain self-sufficiency, but it did so by locking large tracts into a **water-intensive, cereal-dominated rice–wheat cycle** that is now colliding with groundwater depletion, soil fatigue, nutritional imbalance and climate stress. Reshaping this pattern without sacrificing food security or farm incomes is among the central policy challenges of Indian agriculture today.

Basic Concepts:

- **Cropping Pattern:** The **proportion of area under various crops** in a unit area at a point of time, and the yearly sequence and spatial arrangement of crops or crops-and-fallow on a given area.

  - **Temporal arrangement (time):** The yearly sequence of crops on the same land — e.g. growing only rice year after year is **mono-cropping**.
  - **Spatial arrangement (space):** How crops are arranged on the land — e.g. two crops in alternate rows is **row inter-cropping**.
- **Cropping System:** A **broader concept** — the sum total of all crops plus the practices, resources and technology used to grow them, along with their interaction with soil, water, livestock and the surrounding environment.
  - **Simple cropping system:** One crop variety per year with fertiliser replenishment.
  - **Complex cropping system:** Grains, fodder, vegetables, tree crops and livestock integrated on one farm with internal nutrient recycling.
- **Cropping Intensity (CI):** The number of crops raised from the same field in one agricultural year.
  - **CI = (Gross Cropped Area ÷ Net Sown Area) × 100**
  - Reflects **land-use efficiency**; India shows wide regional variation, from under 125% in rainfed tracts to over 200% in irrigated deltas.
- **Crop Diversification:** Shifting area from a dominant crop or crop group towards **a wider basket** — horizontally (more crops in the same season) or vertically (value addition and processing).

Types of Cropping Systems

- **Mono-cropping:** One crop on the same land year after year (e.g. paddy in floodplains).
- **Crop rotation:** Crops changed in a planned sequence from year to year — the most effective known method of weed suppression and nutrient balance.
- **Inter-cropping:** Two or more crops grown simultaneously in a definite row arrangement.
- **Mixed cropping:** Two or more crops grown together without a definite row pattern — a traditional risk-hedging device.
- **Relay cropping:** The succeeding crop is planted before the preceding one is harvested.
- **Sequence (multiple) cropping:** One crop grown after another on the same field within the year.
- **Ratoon cropping:** Regrowth from the stubble of a harvested crop (common in sugarcane).
- **Alley cropping / agroforestry systems:** Crops grown between rows of trees or shrubs.

Cropping Pattern vs Cropping System:

Basis	Cropping Pattern	Cropping System
Meaning	Proportion of area under various crops in a locality at a given point of time.	Cropping pattern **plus its management** to derive benefits from a given resource base.
Focus	Type and arrangement of crops in **time and space**.	Interaction of crops with **farm resources, enterprises and technology**.
Nature	**Descriptive** — what is grown, where and when.	**Systemic** — how it is grown, with what tillage, manuring, irrigation, etc.
Example	Rice–wheat dominance in the Indo-Gangetic Plain.	Rice–wheat system managed through irrigation, fertilizers, mechanisation, crop rotation, residue management, etc.

Factors Affecting Cropping Pattern:

- **Physical and Agro-Climatic Factors**
  - **Climate and rainfall:** Bajra dominates arid Rajasthan; rice dominates the high-rainfall Brahmaputra valley and eastern deltas.

  - **Soil type:** Cotton on the black regur soils of Maharashtra and Gujarat; wheat, rice and sugarcane on the alluvial loams of Punjab, Haryana and western UP.
  - **Topography and drainage:** Terraced and shifting cultivation in the hilly North-East; deep-water rice in flood-prone lowlands.
- **Technological and Input Factors**
  - **Irrigation:** The single most powerful transformer — assured water permits water-guzzling crops and double or triple cropping.
  - **Seed technology:** HYVs, hybrids and now genome-edited varieties reshape what is viable in a given zone.
  - **Mechanisation and labour:** Crops needing intensive manual operations lose ground as rural wages rise.
- **Economic Factors**
  - **Relative profitability and price signals:** Farmers respond to expected net returns, not just yield.
  - **Market access and post-harvest infrastructure:** Perishables need cold chains; without them farmers default to storable cereals.
  - **Access to credit and risk-bearing capacity:** Cash crops such as cotton require high capital and are avoided by the most marginal farmers.
- **Policy Factors**
  - **MSP and assured procurement:** Effective price support concentrated in paddy, wheat and sugarcane skews acreage decisively.
  - **Input subsidies:** Free or near-free power for tubewells, canal water pricing and urea subsidy all tilt choices toward water- and nitrogen-intensive crops.
  - **Trade policy:** Import duties on edible oils and export restrictions on rice and onion alter relative returns.
- **Farmer-Level and Institutional Factors**
  - **Size of holding:** With small and marginal farmers dominating, shrinking holdings limit diversification and entrench subsistence mono-cropping.
  - **Household food and fodder self-sufficiency:** Subsistence needs anchor a minimum area under staple cereals.
  - **Literacy and extension access:** Awareness of intercropping, rotation and new varieties determines adoption.
  - **Land tenure and tenancy:** Insecure tenure discourages investment in long-gestation crops such as orchards and plantations.

Do You Know?

- **Land use base:** India's **gross cropped area has risen to about 219.16 million hectares**, with **cropping intensity at 155.85% in 2022-23** — up from roughly **111% in 1950-51**.
- **Irrigation: Gross irrigated area stood at 122.29 million hectares** and net irrigated area at **79.31 million hectares (2022-23)**; the **Economic Survey 2025-26** notes gross irrigated area has reached **55.8% of gross cropped area**, up from 41.7% in 2001-02.
- **The rice-wheat footprint:** In 2024-25, **rice occupied 51.3 million hectares and wheat 32.8 million hectares** — together about **61% of the total foodgrain area** of 137.6 million hectares.
- **The horticulture crossover:** For 2025-26, horticulture output is estimated at a record **377.78 million tonnes from just 301.51 lakh hectares**, marginally **overtaking foodgrain production of 376.56 million tonnes** grown on more than four times the area.

- **Millets in retreat:** Millets accounted for around **40% of India's cultivated grains before the Green Revolution and about 20% today**; per capita consumption collapsed from **32.9 kg (1962) to 3.87 kg (2022)**. India nonetheless remains the world's largest producer, with a **38.4% share of global output**.
- **The water bill: Punjab's stage of groundwater extraction reached 156.36% in 2025 — the highest in India** — with **111 of 153 assessment blocks over-exploite**d, against annual extraction of 26.27 bcm versus an extractable resource of 16.80 bcm.

Changes in India's Cropping Pattern Since Independence

- **Rising Cropping Intensity**
  - Cropping intensity has climbed from about **111% (1950-51) to 155.85% (2022-23)**, driven by population pressure on a nearly static net sown area.
  - **Enablers:** expansion of irrigation, short-duration and photo-insensitive varieties, laser levelling, the furrow-irrigated raised bed (FIRB) system, and mechanised sowing and harvesting.

    Systematic intercropping with compatible crop combinations has the potential to raise cropping intensity to 400–500% even in rainfed systems, which still account for a very large share of India's net cropped area.
- **Gradual Decline in the Dominance of Foodgrains**
  - At Independence, **over 80% of gross cropped area** was under food crops; the share of foodgrains has since declined to roughly **two-thirds of gross cropped area**.
  - **Reason:** Commercialisation of agriculture and changing consumer demand have pushed farmers toward non-food crops with better price realisation. Yet within the food group, **rice and wheat have consolidated their dominance** because of better prices, lower production risk and superior seed availability.
- **Consolidation of Rice and Wheat**
  - As per the Economic Survey 2025-26, gross area under rice expanded from 43.5 million ha (2015-16) to 51.3 million ha (2024-25) and wheat from 30.4 to 32.8 million ha over the same period, while jowar contracted from 6.1 to 4.0 million ha
  - The concentration is not just in area but in procurement: **92% of wheat procurement comes from three states**(Punjab, Haryana, Madhya Pradesh) and **65% of paddy from five states**.
- **Decline of Coarse Cereals:**
  - Area under jowar, bajra, ragi and small millets has shrunk drastically since 1970-71 as irrigation spread and dietary preferences shifted to fine cereals.
  - **Maize is the exception within the coarse-cereal group** — driven by poultry feed, starch and ethanol demand.

    Maize area rose from 8.8 million ha (2015-16) to 12.1 million ha (2024-25), and 2025-26 production touched an all-time high of 55.09 million tonnes, an increase of 11.68 MT in a single year. Total nutri-cereal area, in contrast, has been broadly stagnant at around 24–26 million hectares.
  - **Renewed millet push:** The International Year of Millets (2023), the rebranding as **Shree Anna**, the Nutri-Cereals Sub-Mission and MSP hikes (ragi received the second-highest absolute increase in 2025-26) have arrested but not reversed the decline.
- **Persistent Stagnation in Pulses**
  - Pulse area has hovered between **25 and 31 million hectares** for a decade, peaking at 30.7 million ha in 2021-22 and slipping to **27.7 million ha in 2024-25**.
  - India remains simultaneously the **largest producer, largest consumer and a major importer** of pulses

Around 84% of pulse area is rainfed and largely on marginal, low-fertility land, since the Green Revolution pushed pulses out of irrigated tracts. Effective MSP procurement has historically bypassed pulse growers, who often realise prices below MSP.

- **Expansion of Oilseeds — but Import Dependence Persists**
  - Oilseed area rose from **26.1 million ha (2015-16) to 30.4 million ha (2024-25)**; 2025-26 output is estimated at **43.06 million tonnes**, with record rapeseed-mustard (13.77 MT) and groundnut (13.07 MT).
- **Shift Toward Horticulture and High-Value Crops**
  - Horticulture has been the fastest-growing segment, with production overtaking foodgrains **on roughly one-seventh of the area**.

    Horticulture production is estimated at 377.78 million tonnes in 2025-26 from 30.15 lakh hectares, against foodgrains of 376.56 MT from over 137 million hectares — a productivity of well over 12 tonnes/ha versus about 2.3 tonnes/ha for foodgrains. India is the world's second largest producer of fruits and vegetables.
  - **Drivers:** rising incomes and health awareness, demand from food processing, export potential, and better returns per hectare.
- **Growth of Water-Intensive Commercial Crops**
  - **Sugarcane area** stood at about **5.5 million hectares in 2024-25**, with 2025-26 production at a record **500.06 million tonnes**, further boosted by the **ethanol blending programme** (about 20% blending achieved).
  - Assured FRP, the mill-catchment obligation to purchase cane, and now ethanol demand have made sugarcane structurally attractive even in water-stressed Maharashtra and Karnataka.
  - **Cotton area has contracted** from 12.3 million ha (2015-16) to **11.5 million ha (2024-25)**, reflecting pest pressure (pink bollworm), stagnant Bt technology and price volatility.
- **Changes Driven by Climate Change**
  - Climate stress is already forcing adaptive shifts in cropping pattern — from kharif to rabi crops in flood-hit Assam, apple cultivation moving to higher altitudes in Himachal Pradesh, a shift from grapes to pomegranate in drought-prone Maharashtra, and from agriculture to aquaculture in salinity-affected coastal Gujarat.

    An ICAR climate change impact assessment projects that, without adaptation, rainfed rice yields could fall by 20% by 2050 and 47% by 2080; irrigated rice by 3.5% and 5%; wheat by 19.3% by 2050 and up to 40% by 2080; and kharif maize by 18% and 23% respectively. Crops such as soybean, groundnut and chickpea may benefit, implying a long-run reallocation of area.
- **Technology-Driven Change**
  - **Transgenics: Bt cotton remains the only commercially cultivated GM crop** in India. GM mustard **DMH-11** received GEAC clearance in 2022, but the Supreme Court delivered a **split verdict in** Gene Campaign v. Union of India**.**
  - **Genome editing:** In **May 2025 India became the first country to release genome-edited rice varieties**, **DRR Dhan 100 (Kamala)** and **Pusa DST Rice 1**, developed by ICAR using CRISPR-Cas.

Regional Cropping Patterns in India

Region	Dominant System	Character
North-Western Plains (Punjab, Haryana, W-UP)	Rice–wheat	Highest cropping intensity and irrigation; Green Revolution core; acute groundwater stress.

Middle & Lower Gangetic Plains(E-UP, Bihar, West Bengal)	Rice–wheat, rice–rice, rice–pulses	High population density, small holdings, flood-prone, low mechanisation; large untapped yield gap.
Deccan Plateau (Maharashtra, Karnataka, Telangana)	Jowar/bajra-based, cotton–wheat, sugarcane	Semi-arid, rainfed dominance, with irrigated sugarcane islands creating severe water conflict.
Western Dry Region (Rajasthan, Kutch)	Bajra, mustard, gram, guar	Arid, low and erratic rainfall, single cropping, high dependence on groundwater.
Central Highlands (Madhya Pradesh, Chhattisgarh)	Soybean–wheat, rice-based, gram	Soybean belt of Malwa; MP is now a leading wheat and gram procurer.
Coastal & Deltaic Regions (Andhra Pradesh, Tamil Nadu, Odisha)	Rice–rice, rice–pulses, plantation	High cropping intensity, cyclone and salinity exposure, aquaculture competition.
North-Eastern Hills	Shifting cultivation (jhum), rice, horticulture	Ecologically fragile, subsistence-oriented, high organic and horticultural potential.
Himalayan Region	Maize, rice, apple, temperate horticulture	Altitude-determined; visible upward migration of orchards under warming.

Issues Arising from India's Current Cropping Pattern

- **Ecological and Resource Issues**
  - **Groundwater exhaustion in the Green Revolution belt:** Assured paddy procurement in a semi-arid state has produced the country's sharpest aquifer decline.

    In its latest assessment the Centre identified 111 of Punjab's 153 groundwater assessment blocks as over-exploited, with a stage of extraction of 156.36% in 2025 — the highest of any state. Ludhiana alone has 14 over-exploited blocks, Jalandhar 12 and Amritsar 10. Paddy occupies roughly 87% of Punjab's kharif area.
- **Soil degradation and skewed nutrient use:**

  The Economic Survey 2025-26 notes that fertiliser use efficiency has not improved and the NPK ratio has deteriorated because subsidy-driven price distortions favour nitrogenous fertilisers, contributing to declining soil health alongside salinity and micronutrient depletion.
  - **Stubble burning and air pollution:** The narrow window between paddy harvest and wheat sowing in Punjab and Haryana — itself a consequence of the rice-wheat cycle and delayed transplanting under groundwater-preservation laws — drives seasonal residue burning across the Indo-Gangetic Plain.
  - **Biodiversity loss:** Displacement of landraces and traditional millet and pulse varieties by a narrow set of HYVs erodes the genetic base needed for future climate adaptation.
- **Economic and Policy Issues**
  - **Policy-induced acreage distortion:** Although MSP is announced for 23 crops, effective procurement is concentrated in paddy, wheat and sugarcane, so relative price signals remain skewed.
  - MSP payments to paddy farmers alone totalled ₹16.08 lakh crore between 2014-15 and 2025-26, against ₹18.99 lakh crore for all 14 kharif crops combined — a measure of how heavily the price-support system leans on a single crop.
  - **The diversification paradox:** Farmers will shift away from paddy only against **assured returns**, but assured returns exist only for paddy — a circularity that has defeated diversification programmes since the **Johl Committee (1986)** first recommended shifting area out of paddy in Punjab.

- **Fragmented holdings limit diversification:** With small and marginal farmers dominating and average holding size shrinking, farmers gravitate to low-risk, storable, subsistence-compatible cereals rather than capital-intensive commercial crops.
- **Yield gaps below global benchmarks:** The Economic Survey 2025-26 records that although India's agricultural growth (a decadal average of 4.5%, and 4.6% in 2024-25) exceeds the global average of 2.9%, yields of cereals, maize, soybean and pulses remain below global averages.

Nutritional Issues

- **Cereal-centric diets:** The cropping pattern feeds directly into the nutrition problem — **ICMR-NIN Dietary Guidelines (2024)** recommend capping cereals at **45% of daily calorie intake**, while the actual share is about **48% in rural India**.
- **Protein deficit:** Stagnant pulse output keeps pulse-derived protein at around 9% of rural protein intake, even as cereal-derived protein has fallen.
- **Nutrient-poor per drop of water:** Among Indian cereals, **rice consumes the most water per tonne of output while delivering the least iron, zinc and protein** — replacing part of the rice crop with nutri-cereals would cut irrigation demand substantially while improving dietary quality.

Structural and Infrastructure Issues

- **Rainfed neglect:** A very large share of net cropped area remains rainfed, with irrigation coverage **below 15% for millets against nearly 67% for paddy** — the very crops best suited to dryland farming get the least support.
- **Missing post-harvest chain:** Diversification to fruits, vegetables and pulses requires cold storage, grading, processing and market linkage that remain thin outside a few clusters, so perishable crops carry disproportionate risk.
- **Weak seed system for non-cereals:** Low seed replacement rates and poor availability of quality millet and pulse seed keep productivity — and hence profitability — low.

Technology and Regulatory Issues

- **Regulatory paralysis in biotechnology:** Two decades after Bt cotton, no transgenic food crop has been commercialised; the **GM mustard case remains before a larger bench** of the Supreme Court, and a promised national GM policy is still awaited.
- **Illegal cultivation:** Weak enforcement has allowed unapproved HT cotton and other illegal GM varieties to spread, defeating both biosafety oversight and farmer protection.
- **Government Initiatives:**
  - **Crop-Specific Missions**
    - **Mission for Aatmanirbharta in Pulses (2025):** Area and productivity expansion in tur, urad and masoor, backed by certified seed, processing units and a commitment to **procure registered farmers' produce at MSP**.
    - **NMEO – Oilseeds (2024) and NMEO – Oil Palm (2021):** Cluster-based oilseed development and oil palm expansion to cut edible oil import dependence; introduced a **Viability Price** shielding oil palm growers from global price swings.
    - **Sub-Mission on Nutri-Cereals / Shree Anna:** Millet promotion through cluster demonstrations and seed hubs, with inclusion in **PDS, PM POSHAN and ICDS** following the International Year of Millets (2023).
  - **Area and District-Based Programmes**
    - **PM Dhan-Dhaanya Krishi Yojana (2025):** Converged development of **100 low-productivity districts**, with crop diversification and sustainable practices as an explicit objective alongside irrigation, credit and block-level storage.

  - **Bringing Green Revolution to Eastern India (BGREI):** Raises cropping intensity and rice-fallow utilisation in seven eastern states where water is abundant but yields lag.
- **Price and Market Support**
  - **MSP for 23 crops** at **A2+FL plus 50%**, with **PM-AASHA** enabling procurement of pulses and oilseeds through NAFED and NCCF.
  - **Operation Greens, MIDH and the Agriculture Infrastructure Fund:** Price stabilisation for TOP crops, horticulture development, and cold chain, grading and processing infrastructure.
  - **eNAM and FPO promotion:** Aggregation and market reach, giving diversifying farmers scale and bargaining power.
- **Water and Soil Sustainability**
  - **PMKSY – Per Drop More Crop**, **direct-seeded rice incentives** and the **Punjab Preservation of Subsoil Water Act, 2009** to cut agricultural water use through micro-irrigation and regulated transplanting dates.
  - **National Mission on Natural Farming, PKVY, Soil Health Cards and PM-PRANAM** to restore soil health and reduce chemical fertiliser dependence.
- **Research and Technology**
  - **NICRA and the ICAR varietal programme** for climate-resilient technologies and new varieties, including India's first **genome-edited rice** (2025).
  - **Krishonnati Yojana (umbrella scheme)** consolidating NFSM, NMSA, MIDH, extension, seeds, mechanisation, marketing and agri-statistics under one head.

Way Forward:

- **Realign the Incentive Structure**
  - **Make procurement crop-neutral:** Extend effective, time-bound MSP procurement to pulses, oilseeds and millets in over-exploited blocks, so switching away from paddy carries no income penalty.
  - **Reform input subsidies gradually:** Move from free farm power toward **direct benefit transfer for electricity**, so farmers who save water retain the savings.

    Punjab's Paani Bachao Paisa Kamao pilot paid farmers directly for electricity units saved on tubewells, showing that compensating conservation works better than penalising consumption.
- **Strengthen the Agronomic Foundation**
  - **Close the seed gap for non-cereals** through seed hubs and seed villages, raising seed replacement rates for pulses, millets and oilseeds.
  - **Promote scientifically designed cropping systems** — inter-cropping, relay cropping and legume-inclusive rotations that raise intensity, fix nitrogen and diversify income within the same holding.
  - **Adopt water-saving agronomy at scale:** Direct-seeded rice, alternate wetting and drying, laser levelling, micro-irrigation and short-duration varieties that free the field earlier for a rabi crop.
- **Build the Market for Diversified Crops**
  - **Guarantee demand through institutional channels** — absorb millets and pulses into PDS, PM POSHAN, ICDS and defence procurement; assured offtake is the cheapest diversification subsidy

    Odisha's Millet Mission pairs MSP-backed ragi procurement with inclusion in the state PDS and school meals — the clearest demonstration that a demand-side guarantee can revive a crop markets had abandoned.
  - **Invest in the post-harvest chain:** Cold storage, grading, primary processing and millet milling near production clusters, since diversification without processing simply transfers risk to the farmer.

- **Develop value-added products** to address the demand-side constraint of long cooking times and lost culinary knowledge.

- **Use Technology and Data**
  - **Settle the biotechnology question:** Frame the **national GM policy** the Supreme Court called for, with transparent biosafety protocols, while keeping genome editing on its lighter, non-transgenic track.
  - **Deploy geospatial planning:** Use **FASAL, NADAMS and CHAMAN** for agro-ecologically matched crop planning rather than uniform state-wide advisories.
  - **Deliver location-specific advisories** through KVKs, Kisan Call Centres, Agristack and the Digital Crop Survey.
- **Adapt to Climate**
  - **Redraw agro-climatic zoning** using updated climate data rather than 1980s baselines, given projected losses in rice, wheat and maize and possible gains in soybean, groundnut and chickpea.
  - **Scale conservation and regenerative agriculture:** Zero tillage with residue retention and rotation reduces the stubble-burning window, sequesters carbon and lowers cost.
  - Adoption of the Happy Seeder and Super Seeder in Punjab and Haryana allows wheat to be drilled directly through paddy residue — an agronomic fix that also reduces a public-health externality.
  - **Strengthen risk transfer:** Faster PMFBY settlement and weather-index insurance for pulses, millets and horticulture, since the absence of insurance is itself a barrier to diversification.

Address the Structural Base

- **Aggregate small holdings** through FPOs, cooperatives and contract farming to give small farmers scale, machinery and market access.
- **Realise eastern India's potential:** Bihar, eastern UP, Odisha, Chhattisgarh and West Bengal have water and land but the largest yield gaps — raising intensity here is the least ecologically costly way to add output while relieving Punjab and Haryana.

Conclusion:

India's cropping pattern is the accumulated residue of policy choices made for a food-deficit country that no longer exists. The rice–wheat system solved the crisis of the 1960s; it now generates the crises of the 2020s. The task is to **shift the guarantee, not merely the advice** — pairing procurement, water pricing reform, seed systems and post-harvest infrastructure so that the ecologically rational choice becomes the economically rational one.

3. CROP DIVERSIFICATION:

Introduction

Crop diversification means **growing different types of crops instead of depending on just one or two crops**. It helps farmers **reduce risks, earn better income and use land and water more efficiently**. It helps farmers **reduce income and climate risks, improve soil health and use water more wisely**. For India, crop diversification is therefore an important step towards **sustainable, profitable and climate-resilient agriculture**.

Methods of Crop Diversification

Method	About	Examples
Crop Rotation	• Growing different crops one after another on the same land. Helps maintain soil fertility and reduces pests.	Rice → Wheat → Pulses

Intercropping	• Growing two or more crops together in the same field. Makes better use of land, water and nutrients.	Maize + Pigeon pea
Mixed Cropping	• Growing different crops together to reduce the risk of total crop failure. Useful when rainfall is uncertain.	Wheat + Gram
Shifting from Water-intensive Crops	• Replacing crops that need a lot of water with less water-intensive crops. Helps save groundwater.	Paddy → Millets
Horticulture Diversification	• Moving partly towards fruits, vegetables, flowers and spices. Can provide higher returns per hectare.	Wheat → Vegetables
Diversification towards Pulses & Oilseeds	• Increasing the cultivation of pulses and oilseeds instead of depending mainly on cereals. Improves nutrition and reduces import dependence.	Rice → Pulses / Oilseeds
Millets-based Diversification	• Bringing millets into areas dominated by rice or wheat. Millets generally need less water and are climate-resilient.	Paddy → Bajra / Jowar
Integrated Farming	• Combining crops with activities like dairy, poultry, fisheries or livestock. Gives farmers more than one source of income.	Crop + Dairy + Fishery

Advantages of Crop Diversification

- **Economic Benefits**
  - **Higher income:** High-value crops such as fruits, vegetables, spices and flowers can give better returns than traditional cereals.

    According to the Economic Survey 2025-26, horticulture accounts for around 33% of India's agricultural Gross Value Added (GVA).
  - **More market opportunities:** Farmers can sell different products to different markets instead of depending on a single crop.
- **Environmental Benefits**
  - **Saves water:** Replacing water-intensive crops with millets, pulses and oilseeds can reduce pressure on groundwater.
  - **Reduces chemical use:** A wider mix of crops can break pest cycles and reduce excessive dependence on pesticides and fertilisers.
- **Climate Resilience**
  - **Reduces crop failure:** Failure of one crop does not necessarily mean complete loss for the farmer.
  - **Supports climate-smart farming:** Crops such as millets and pulses can perform relatively well under water and heat stress.

    India's pulse production increased from about 19.25 million tonnes in 2013-14 to 25.68 million tonnes in 2024-25.

- **Food and Nutrition Security**
  - **More diverse food basket:** It increases the production of pulses, millets, fruits and vegetables.
  - **Reduces cereal dependence:** It helps move the food system beyond excessive dependence on rice and wheat.
- **Employment Generation**
  - **More farm work:** Horticulture and other diversified activities often require more labour for planting, harvesting and processing.
  - **Reduces seasonal migration:** More year-round activities can provide regular rural livelihoods.

Issues with Crop Diversification

- **Market-Related Problems**
  - **Uncertain prices:** Prices of fruits, vegetables, pulses and other crops can change sharply, making farm income uncertain.
  - **Limited markets:** Farmers may not have easy access to nearby markets, buyers or organised value chains.
- **Infrastructure Gaps**
  - **Poor storage:** Lack of cold storage and warehouses leads to post-harvest losses, especially in horticulture.

    According to the NABCONS 2022 study commissioned by MoFPI, post-harvest losses were estimated at 6.02–15.05% for fruits and 4.87–11.61% for vegetables.
  - **Weak processing facilities:** Limited local processing means farmers often sell raw produce at lower prices.
  - **Poor transport:** Weak rural roads and transport facilities make it difficult to take perishable produce to distant markets.
- **Policy and MSP Issues**
  - **Rice–wheat bias:** MSP procurement and government support are much stronger for rice and wheat than for many alternative crops.

    As per FCI(2024 - 25), paddy procurement reached 813.88 LMT, worth about ₹1.9 lakh crore, benefiting 1.15 crore farmers.
  - **Policy uncertainty:** Frequent changes in trade, export and import policies can affect farmers' willingness to diversify.
- **Financial Constraints**
  - **High initial investment:** New crops may require better seeds, irrigation, machinery, storage or specialised equipment.
  - **Limited access to credit:** Small and marginal farmers may find it difficult to arrange additional working capital.

    NABARD's report found that small and marginal farmers accounted for about 75.5% of commercial-bank loan accounts but only 47.3% of loan disbursement in 2019-20.
- **Knowledge and Technology Gaps**
  - **Lack of awareness:** Farmers may not have enough information about suitable alternative crops.
  - **Limited extension services:** Scientific advice on seeds, pests, irrigation and cultivation practices may not reach every farmer.
  - **Technology gap:** Modern practices such as protected cultivation, precision farming and micro-irrigation may be costly or difficult to adopt.

From 2015-16 onwards, the Per Drop More Crop scheme had covered around 96.97 lakh hectares under micro-irrigation by March 2025.

Way Forward for Crop Diversification

- **Make Diversification Profitable**
  - **Assured prices:** Strengthen price support and procurement for pulses, oilseeds, millets and other alternative crops.

    Haryana's Bhavantar Bharpayee Yojana, which pays farmers the price difference when market rates for vegetables, millets, and pulses drop below fixed floor prices.
  - **Better market access:** Connect farmers with organised retailers, processors, exporters and institutional buyers.

    PepsiCo's contract farming model, where farmers in Punjab and Gujarat grow special potato varieties (such as FC-5) for making potato chips.
  - **Reduce price risk:** Expand crop insurance and develop better price-risk management mechanisms.

    Farmers can use put options on commodity exchanges like NCDEX to protect themselves if the market price of chana, mustard or soybean falls before harvest.
- **Market-Centric Support**
  - **Beyond rice and wheat:** Gradually expand procurement and market support for all crops suited to local conditions.
  - **Demand-based farming:** Encourage farmers to grow crops according to **market demand, agro-climatic suitability and water availability**.
- **Build Strong Value Chains**
  - **Storage facilities:** Expand warehouses, cold chains and pack houses, especially in horticulture-producing regions.

    AIF (Agriculture Infrastructure Fund): Financing post-harvest infrastructure such as pack-houses, cold stores, and primary processing centers in Andhra Pradesh and Maharashtra.
  - **Food processing:** Promote local processing of fruits, vegetables, pulses, oilseeds and millets.
  - **Better transport:** Improve farm-to-market roads and efficient logistics to reduce wastage and costs.

    Kisan Rail: Temperature-controlled train routes (e.g., transporting pomegranates and bananas from Maharashtra directly to North Indian mandis).
- **Promote Farmer Collectives**
  - **Strengthen FPOs:** Help small farmers come together for bulk purchase, production and marketing.

    MahaFPC, Maharashtra: Farmer Producer Companies work together to market pulses collectively, improving their bargaining power and reducing dependence on middlemen.
- **Promote Region-Specific Diversification**
  - **One-size-fits-all approach:** Diversification should be based on the **soil, rainfall, irrigation and climate** of each region.

    Promoting high-value saffron, walnuts, and organic temperate fruits across Jammu & Kashmir and Himachal Pradesh.
  - **District-level planning:** Prepare crop plans based on local agro-climatic conditions and market opportunities.

- **Improve Water Management**
  - **Expand micro-irrigation:** Promote drip and sprinkler irrigation for efficient water use.
  - **Link water with crop choice:** Incentives should encourage farmers to grow crops according to **water availability** rather than only assured procurement.

    Punjab's Mera Pani Meri Virasat scheme, providing ₹7,000 per acre to farmers who switch from water-intensive paddy to pulses, maize, or cotton.

Successful Case Studies of Crop Diversification

Model	Primary Mechanism	Impact
Israel	Closed-loop drip irrigation, fertigation, brackish water farming, and growing date palms, jojoba, & cherry tomatoes.	**Water Productivity:** World leader in crop yield per drop of water; turned arid desert into an agricultural export hub.
Brazil	**Integrated Crop-Livestock-Forestry (ILPF):** Rotating cash crops with cattle pasture and planted eucalyptus trees.	**Soil Carbon & Deforestation Control:** Restored 15+ million hectares of degraded lands without cutting Amazon rainforest.
European Union (e.g., France / Netherlands)	EU Common Agricultural Policy (CAP): Subsidies tied to crop rotation, legume inclusion	**Reduced Fertilizer Dependence:** Legumes fix natural nitrogen, reducing reliance on synthetic chemical fertilizers.
Vietnam (Mekong Delta)	Adapting to climate change and coastal saline intrusion by alternating rice in wet season with brackish aquaculture in dry season.	**Climate Adaptation & High Returns:** Reduced vulnerability to sea-level rise while doubling household farm profits

Conclusion

Crop diversification is not just about growing more crops; it is about **making farming safer, more profitable and sustainable**. By giving farmers better markets, technology, irrigation and price support, diversification can reduce risks while protecting **soil, water and livelihoods**. Ultimately, India needs a **right crop, right region, right market** approach for a resilient agricultural future.

4. HIGH VALUE AGRICULTURE

Introduction

High-value agriculture (HVA) refers to non-traditional agricultural sectors such as fruits, vegetables, livestock, poultry and aquaculture that yield higher returns per unit of land than traditional food grains. As land fragmentation limits physical expansion, HVA serves as a crucial pathway to double farmers' income, boost rural employment and meet India's evolving nutritional needs.

Significance of High-Value Agriculture

- **Economic & Income Benefits**
  - **Higher Returns:** Crops like fruits, vegetables, flowers and spices can generate more revenue per hectare compared to traditional cereal crops like paddy or wheat.
  - **Higher Export:** HVA commodities such as fruits, vegetables, spices, marine products and processed foods have high demand in global markets, helping India earn valuable foreign exchange.

According to APEDA, India's agricultural exports rose from USD 34.5 billion (FY20) to USD 51.1 billion (FY25).

- **Agrarian & Smallholder Viability**
  - **Livelihood Security:** Over 85% of Indian farmers are small and marginal with shrinking land sizes. HVA allows them to maximize profitability on small land parcels.
  - **Employment Generation:** High-value agriculture, especially horticulture, dairy and food processing, creates more year-round jobs in rural areas.
- **Environmental & Resource Efficiency**
  - **Water-Use Efficiency:** Drip-irrigated horticulture and micro-irrigated cash crops produce more economic output per drop of water compared to flood-irrigated cereals.

    According to ICAR, micro irrigation achieves 80% to 95% water-use efficiency (compared to only 30%–50% in traditional flood irrigation).
  - **Soil Health and Diversification:** Rotating traditional staples with leguminous vegetables or agroforestry breaks pest cycles, restores soil nutrients, and reduces chemical fertilizer overuse.

    According to ICAR (Indian Council of Agricultural Research), incorporating leguminous crops into cereal rotations biologically fixes 30 to 50 kg of atmospheric nitrogen per hectare.
- **Nutrition & Consumer relation**
  - **Shifting Dietary Patterns:** As incomes rise, Indian diets are shifting from cereal-heavy meals toward protein, vitamin, and micronutrient-rich diets (milk, eggs, fruits, vegetables). HVA aligns supply with this structural demand shift.

    According to the NSSO Household Consumption Expenditure Survey 2022–23, the share of food expenditure spent on cereals dropped significantly to under 3.6% in urban India and 4.9% in rural India.
  - **Addressing Malnutrition & Stunting:** Local production of nutrient-dense foods improves dietary diversity in rural communities, aiding national nutritional security.

    According to the National Family Health Survey, 35.5% of children under 5 years in India are stunted (height-for-age), while 32.1% are underweight.
- **Supply Chain Integration**
  - **Backward & Forward Linkages:** High-value produce feeds directly into the food processing industry, driving investments in cold storage, pack-houses and refrigerated transport.
  - **Direct Market Access:** HVA encourages models like contract farming and Farmer Producer Organizations (FPOs), bridging the gap between field and supermarket shelf by cutting out intermediaries.

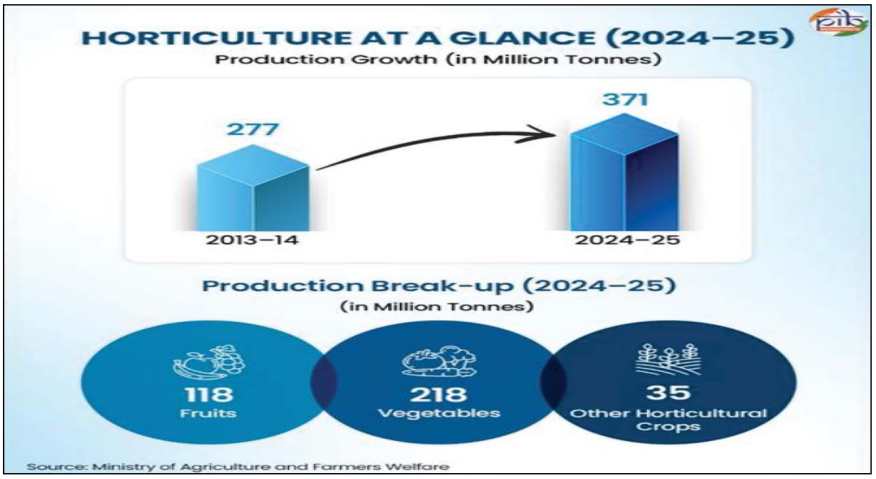
India's Global Leadership in High-Value Agriculture (HVA)

Sector / Commodity	Global Rank	Production Volume & Year	Major Producing States
Sugarcane	**2nd**	454.61 Million Tonnes (2024–25)	Uttar Pradesh, Maharashtra, Karnataka
Cotton	**2nd**	~5.05 Million Tonnes (2024–25)	Gujarat, Maharashtra, Telangana, Karnataka
Tea	**2nd**	1.203 Million Tonnes (Apr–Dec 2024–25)	Assam, West Bengal, Tamil Nadu, Kerala, Karnataka

Spices	**1st**	12.0 Million Metric Tonnes (2023–24)	Madhya Pradesh, Gujarat, Andhra Pradesh, Rajasthan
Coconut	**1st**	~21.3 Billion Nuts (Annual Output)	Kerala, Tamil Nadu, Karnataka, Andhra Pradesh
Coffee	**Top 8**	~0.36 Million Tonnes (Annual Output)	Karnataka, Kerala, Tamil Nadu
Milk (Dairy)	**1st**	~230+ Million Tonnes (Contribution: ~24% to global milk output)	Uttar Pradesh, Rajasthan, Madhya Pradesh, Gujarat, Punjab
Aquaculture & Fisheries	**2nd** (Inland) **3rd** (Overall)	~17.5+ Million Metric Tonnes	Andhra Pradesh, West Bengal, Gujarat, Tamil Nadu
Poultry & Eggs	**2nd** (Eggs) **5th** (Broiler Meat)	~138+ Billion Eggs; ~4.8 MMT Meat	Andhra Pradesh, Tamil Nadu, Telangana, Maharashtra
Fruits	**2nd** (Overall) **1st** (Mango, Banana, Papaya)	~110+ Million Metric Tonnes	Andhra Pradesh, Maharashtra, Uttar Pradesh, Gujarat
Vegetables	**2nd** (Overall) **1st** (Okra, Onion, Potato)	~212+ Million Metric Tonnes	West Bengal, Uttar Pradesh, Bihar, Madhya Pradesh

Examples of High Value Crops

Fruits	Pomegranate, dragon fruit, black rice, baby corn, mango, grapes, guava, oranges.
Vegetables	Tomatoes, onions, okra, capsicum, baby corn.
Flowers & Ornamentals	Marigold, rose, orchids
Spice & Condiments	Saffron, cardamom, turmeric, ginger, black pepper, chili.
Medicinal & Aromatic Plants	Ashwagandha, shatavari, mushrooms (esp. Ganoderma), ginseng
Oilseeds	Sunflower, groundnut (peanut), almond
Specialty or Export Crops	Cotton, organic herbs, exotic vegetables



Challenges Facing High-Value Agriculture in India

- **High Perishability & Cold Chain Deficits**
  - **Post-Harvest Losses:** India loses its horticulture produce annually due to fragmented cold storage networks and lack of refrigerated transport.

    According to NABARD, annual post-harvest losses stand at 6.0% to 15.1% for fruits and 4.9% to 11.6% for vegetables.
  - **Lack of Processing Capabilities:** Only a small fraction of India's high-value produce undergoes processing, limiting value addition and shortening shelf life.

    According to the Ministry of Food Processing Industries, India processes less than 10% of its total agricultural produce, compared to over 60–70% in developed economies.
- **Market Inefficiencies & Price Risk**
  - **Exploitative Value Chains:** Despite APMC reforms, long chains of intermediaries continue to siphon off profits, leaving the actual grower with a tiny share of the consumer rupee.

    According to Reserve Bank of India Study, farmers receive only 31% to 37% of the consumer rupee for key fruits and vegetables (TOP crops), with intermediaries capturing nearly two-thirds.
  - **Thin Local Markets:** Many high-value crops lacks access to urban or export markets. Rural farmers without direct logistics links struggle to sell their produce at fair prices.

    According to NITI Aayog, nearly 60% of rural farm produce is sold in unregulated local markets or at the farm gate due to a lack of cold-chain connectivity to urban hubs.
- **Capital & Credit Constraints**
  - **High Capital Requirements:** Setting up green houses, drip irrigation, polyhouses, or buying high-yield livestock demands major upfront investment.

    According to Mission for Integrated Development of Horticulture Guidelines, establishing protected cultivation (like polyhouses) requires a heavy upfront capital cost of ₹840 to ₹1,000 per square meter.
  - **Institutional Credit Gap:** Small and marginal farmers often lack collateral to secure bank loans for HVA setup, pushing them toward informal moneylenders with higher interest rates.
- **Climate Vulnerabilities**
  - **Weather Hazards:** High-value crops and livestock are far more sensitive to unseasonal rains, hail, heatwaves, and pest outbreaks than traditional grains.

    According to the Ministry of Agriculture, extreme weather events damage over 5 to 7 million hectares of standing crops in India annually.
  - **Inadequate Crop Insurance:** Existing schemes are primarily tailored for standard field crops, with weak coverage and slow claim settlements for specialized high-value or perishable crops.

    Pradhan Mantri Fasal Bima Yojana (PMFBY) insured 4.19 crore farmers during 2024-25, covering 6.2 crore hectares
- **Fragmented Landholdings & Quality Standards**
  - **Lack of Scale:** Individual smallholders struggle to produce uniform, high-volume yields required by retail chains or international buyers.

    According to the Agriculture Census (2015-16), 86.1% of operational holdings in India belong to small and marginal farmers, with an average farm size of just 1.08 hectares.
  - **Strict Export Barriers:** International markets mandate strict Sanitary and Phytosanitary (SPS) standards regarding pesticide residues and hygiene which discourage Indian growers.

High-Value Agriculture

- **Production:** Horticulture production (~367+ Million Tonnes) consistently exceeds total foodgrain production (~357 MT) in India.
- **GVA Share:** Horticulture alone accounts for nearly **33% of Agri-GVA**, despite utilizing less than 20% of the total cropped area.
- **Decadal Growth Engine:** According to the Economic Survey, while overall crop sector GVA grew at 3.5%, allied high-value sectors grew much faster (**Livestock at ~7.1%** and **Fisheries at ~8.8%**).

Reforms taken for High-Value Agriculture in India

- **Dedicated Missions**
  - **Mission for Integrated Development of Horticulture (MIDH):** Serves as the primary umbrella initiative driving production, post-harvest management, and technology adoption in fruits, vegetables, spices, and flowers.
  - **National Livestock Mission (NLM):** Focuses on breed improvement, fodder availability, and expanding formal milk-procurement networks through dairy cooperatives.
  - **Pradhan Mantri Matsya Sampada Yojana (PMMSY):** Aims to boost the fisheries sector by investing heavily in aquaculture, modernizing harbor infrastructure, and expanding inland fisheries.
- **Infrastructure & Post-Harvest Strengthening**
  - **Agriculture Infrastructure Fund (AIF):** A financing facility providing long-term debt with interest subvention to build cold stores, primary processing units and pack-houses.
  - **PM Kisan SAMPADA Yojana:** Focuses on creating modern supply chains by connecting farm gates directly to retail outlets, funding mega food parks, integrated cold chains and processing clusters.
  - **Kisan Rail & Kisan Udaan:** Special trains and air cargo services to transport fruits, vegetables, and perishables rapidly to distant urban and export markets, with up to 50% transportation subsidy.
- **Market Reforms**
  - **Farmer Producer Organizations (FPOs):** Groups small and marginal farmers together so they can aggregate high-value produce, buy inputs at bulk rates, and negotiate directly with retail buyers and exporters.
  - **National Agriculture Market (e-NAM):** A digital trading platform linking mandis nationwide to enable better price discovery and reduce dependency on local intermediaries.
  - **Operation Greens:** Extended from initial crops (Tomatoes, Onions, Potatoes - TOP) to all perishable fruits and vegetables (TOTAL) to stabilize prices and prevent distress sales during peak harvest seasons.
- **Input Efficiency & Technology Adoption**
  - **PM Krishi Sinchayee Yojana:** Promotes drip and sprinkler systems crucial for high-value horticulture, boosting yield while saving water and fertilizer.
  - **Subsidies for Protected Cultivation:** Government offers capital support for setting up polyhouses, shade nets, and drip-fertigation systems to protect high-value crops from climate shocks.
  - **Agri-Tech & Digital Agriculture:** Initiatives supporting drone usage (for precision spraying), soil mapping, and startup funding under **RKVY-RAFTAAR** to modernize farm management.
- **Risk Mitigation**
  - **Weather-Based Crop Insurance:** A specialized crop insurance component launched to cover climatic risks for delicate horticultural crops.

Way forward

- **Building Last-Mile Cold Chains**

- **Solar-Powered Farm-Gate Storage:** Expand decentralized, micro-cold storages and pack-houses right at the village level to give smallholders holding capacity during seasonal price crashes.

  Ecozen's "Ecofrost", Solar-powered micro cold rooms installed right at farm gates in Maharashtra and Karnataka to prevent perishable spoilage.

- **Private Sector Investment in Logistics:** Utilize the Agriculture Infrastructure Fund (AIF) with viability gap funding to attract private logistics providers for temperature-controlled transport (reefer trucks).

  WayCool Foods & Ninjacart – Agri-logistics startups utilizing tech-enabled refrigerated supply chains to transport fresh produce directly to urban hubs.

- **Market Reforms**
  - **Strengthening FPOs:** Enable Farmer Producer Organizations (FPOs) to transition from simple input suppliers to enterprise managers that aggregate produce, grade it, and directly supply to retail chains.

    Rabobank FPO Model (Netherlands) – Farmer-owned agricultural cooperatives running large-scale processing, marketing, and global retail distribution.

  - **Expanding Digital Platforms:** Modernize e-NAM to handle quality-grading standards online and introduce a transparent, standardized contract farming framework to ensure fair, pre-agreed pricing for growers.
- **Improve Credit and Insurance**
  - **Perishable-Centric Insurance:** Redesign crop insurance scheme to include price-trigger clauses and satellite-based loss assessment for high-value horticultural products.
  - **Flexible Capital Credit:** Increase Kisan Credit Card (KCC) limits and streamline credit flow through Agri-Fintechs for specialized HVA equipment like polyhouses, drip-fertigation systems, and shade nets.

    Jai Kisan & Samunnati (Agri-Fintechs) – Tech-driven credit platforms providing fast working capital loans to farmers specifically for micro-irrigation and polyhouses.
- **Promote Climate-Smart Farming**
  - **Precision Farming:** Fast-track micro-irrigation and fertigation coverage to optimize water usage, along with protected cultivation to protect high-value crops against heatwaves and heavy unseasonal rains.

    Netafim Drip Fertigation (Israel) – Advanced micro-irrigation systems combining water delivery with precise nutrient dosing to maximize crop yields in arid zones.

  - **Agronomic Advisory:** Leverage digital extension platforms to train farmers in pest management, soil health, and optimum chemical application to prevent crop damage.
- **Export Diversification**
  - **District Export Clusters:** Establish specialized, cluster-based export hubs equipped with testing laboratories to check pesticide residues and meet international Sanitary and Phytosanitary (SPS) standards.
  - **Brand Building and Certification:** Accelerate Geographical Indication tagging, organic certifications, and global branding for premium Indian produce to capture high-margin export markets.

    Alphonso Mango (Ratnagiri GI Tag) & Sikkim 100% Organic State – Leveraged GI tags and state organic certification to command high-value premium export prices.
- **Committees Recommendations for High-Value Agriculture**
  - **Ashok Dalwai Committee:** Advocated a structural shift from basic cereal cultivation to high-value horticulture, livestock, and fisheries to increase per-hectare farm profitability.

- **Sanjay Kaul Committee, 2018:** Recommended establishing crop-specific export clusters for high-value perishables (fruits, spices, vegetables) with dedicated cold-chain infrastructure to meet global quality standards.
- **M.S. Swaminathan Commission:** Highlighted secondary agriculture (apiculture, mushroom farming, agro-processing) as essential for supplementing smallholder income and creating non-farm rural jobs.

Case Studies

- **MahaFPC Onion & Vegetable Aggregation Model (Maharashtra):** A federation of 300+ Farmer Producer Companies aggregated smallholder vegetable and onion growers to bypass local mandis and sell directly to processors, boosting farmer income by 15–20%.
- **Exotic Crop Diversification in Mizoram (Dragon Fruit & Kiwi):** Mizoram shifted tribal farmers away from traditional Jhum (slash-and-burn) shifting cultivation to high-value dragon fruit and kiwi farming.
- **Sahyadri Farmers Producer Company (Nashik, Maharashtra):** Built a integrated, cold-chain-enabled grape and pomegranate export corridor, transforming small grape growers into India's largest fresh-fruit exporter hub.
- **Apple Cluster Development in Himachal Pradesh:** Shifted traditional apple growers toward high-density plantations and anti-hail net technology, doubling yields per acre and protecting crops against extreme weather events.
- **Kerala's Subicsha Coconut Producer Company:** Moved smallholders away from selling raw coconuts toward manufacturing high-value products like virgin coconut oil, neera, and packaged chips.

Conclusion:

High-value agriculture holds the key to transforming Indian farming from a subsistence activity into a profitable, commercial enterprise. However, its long-term success hinges on moving **from production-centric policies to market-linked infrastructure**. By combining FPO-led scale, climate-resilient technology, and robust processing networks, India can stabilize rural incomes while positioning itself as a major supplier in the global agri-food chain.

5. INTEGRATED FARMING SYSTEM

Introduction

An Integrated Farming System (IFS) is an agricultural approach where different farm components like crops, livestock, poultry, fish and trees are managed together on the same piece of land. Integrated farming emphasizes the use of natural processes and ecological principles to enhance productivity and sustainability.

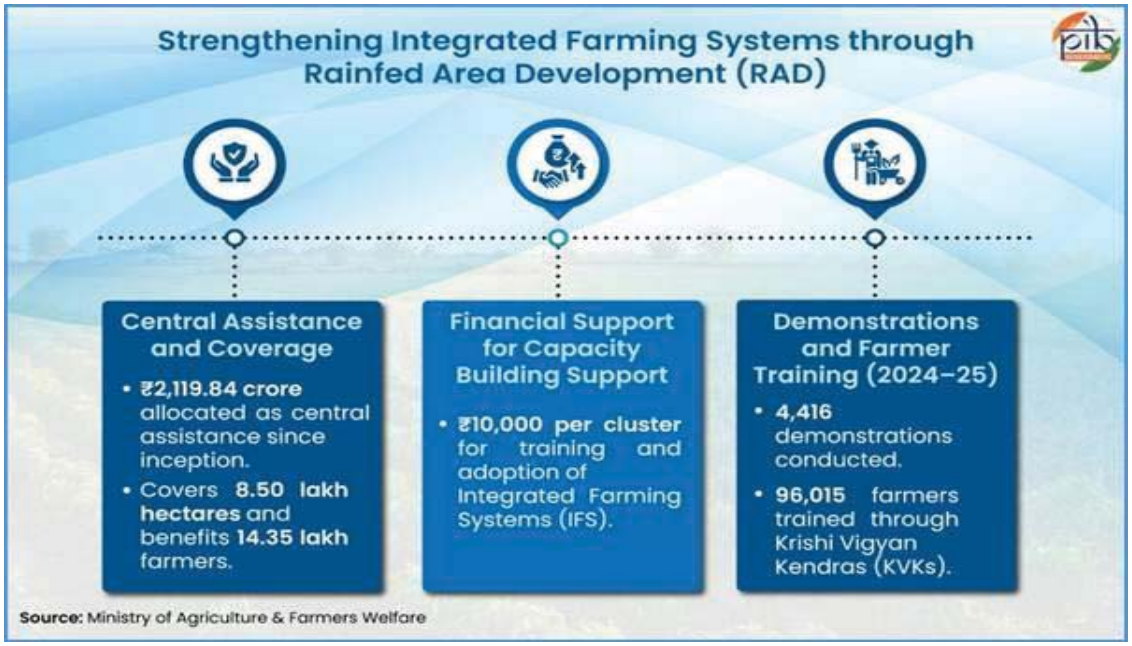
Significance of Integrated farming system

- **Crop–Livestock System**
  - **Daily Cash Flow:** Milk sales provide daily household income, removing total dependence on seasonal crop harvests.
  - **Cost Reduction:** Cattle manure replaces expensive shop-bought chemical fertilizers, while crop leftover serves as free cattle feed.
- **Crop–Livestock–Fishery System**
  - **Waste Recycling:** Animal manure fuels plankton growth in fish ponds, turning animal waste into high-value protein.
  - **Maximum Land-Water Utility:** Generates grains milk and fish from the exact same farm footprint without extra land requirements.

- **Agri-Horticulture System**
  - **Dual Income source:** Crops generate fast cash flow in the short run, while the fruit trees act as a long-term investment.
  - **Better Space Utilization:** Intercropping makes full use of empty space between tree rows and maximizes sunlight absorption.
- **Agri-Silviculture System**
  - **Asset Creation:** Trees act as a growing bank account that can be harvested for timber or fuel during financial emergencies.
  - **Soil Protection:** Deep tree roots stabilize topsoil, control erosion and act as windbreakers during heavy storms.
- **Rice–Fish Culture System**
  - **Natural Pest Control:** Fish feed on harmful insects, weeds and larvae in paddy fields, reducing the need for chemical pesticides.
  - **Soil Enrichment:** Fish movement aerates the water, while their droppings naturally add nitrogen to the paddy roots.
- **Poultry–Fishery System**
  - **Zero Feed Expense:** Poultry sheds built over ponds drop manure directly into the water, eliminating external fish feed costs.
  - **High-Protein Yield:** Delivers simultaneous harvests of meat, eggs and fish from a small physical footprint.

Major Components of Integrated farming system

- **Field Crops (Agriculture):** Farmers grow food grains, pulses and cash crops for food and income. The leftover crop residue is used as fodder for animals or as organic manure.
- **Livestock (Dairy & Cattle):** Cows and buffaloes provide milk, giving farmers a regular source of income. Their dung and urine are used to make organic manure and biogas.
- **Poultry Farming:** Poultry farming provides extra income through the sale of eggs and meat. Poultry waste is also used as manure for crops or fish ponds.
- **Aquaculture (Fish Farming):** Fish are reared in farm ponds, where animal waste can be used as natural fish feed. The pond also stores water that can be used to irrigate crops when needed.
- **Horticulture (Fruits & Vegetables):** Farmers grow fruits and vegetables to earn higher income, make better use of their land and improve the family's nutrition.
- **Agroforestry & Silviculture:** Trees are planted on farm boundaries to provide timber, fuelwood, and fodder. They also protect the soil from erosion and become a valuable asset over time.
- **Mushroom Cultivation:** Mushrooms are grown using crop waste like wheat or paddy straw. This helps farmers earn extra income with very little land.
- **Apiculture (Beekeeping):** Bees help pollinate crops, which increases crop production. Farmers also earn additional income by selling honey and beeswax.
- **Vermicomposting & Biogas:** Animal waste and crop residues are turned into organic manure and biogas, which improve soil fertility and provide clean fuel for cooking.
- **Sericulture (Silk Farming):** Farmers grow mulberry plants to rear silkworms and produce silk. The leftover plant material is recycled into compost, reducing waste.



Major Bottlenecks faced by Integrated farming system

- **Financial & Economic Challenges**
  - **High Initial Investment:** Setting up fish ponds, cattle sheds, or poultry units requires a lot of money, which many small and poor farmers cannot easily afford.

    As per NABARD, Setting up an integrated farm unit (crop plot, cattle shed, farm pond and vermicompost pit) requires an upfront investment of ₹1.5 lakh to ₹3.5 lakh per hectare.
  - **Slow Income from Trees:** Trees grown under agroforestry and fruit orchards take 4–7 years to start giving good returns, so farmers have to wait a long time to earn income.
  - **Limited Access to Credit:** Banks usually provide separate loans for crops or livestock, but there are very few loan schemes that support the entire integrated farming system under one plan.

    According to the Reserve Bank of India, over 75% of institutional farm loans are disbursed strictly as short-term crop loans (KCC).
- **Operational & Technical Constraints**
  - **Needs More Skills:** Farmers need proper knowledge and training for managing crops, animals, fish, and other farm activities
  - **Risk of Disease Spread:** When animals, poultry, and fish are kept close together, diseases can spread quickly from one activity to another if proper care is not taken.

    African Swine Fever (ASF) outbreaks in states like Assam, Arunachal Pradesh and Mizoram caused heavy losses to pig farmers, especially where pigs were kept in close contact.
- **Supply Chain & Market Hurdles**
  - **Perishability & Storage Issues:** Products like milk, eggs, fish, and vegetables rot quickly without cold storage facilities, forcing farmers into distress sales at low market prices.

    As per the Ministry of Food Processing Industries (MoFPI), over 75% of India's cold storage capacity is concentrated solely on potatoes.
  - **Lack of Integrated Supply Chains:** Local mandis and markets are often organized around specific grains rather than handling diverse daily outputs like fish, eggs and fruit together.

As per NITI Aayog, over 90% of regulated APMC mandis in India are structurally designed and equipped purely for bulk cereal and grain trading.

- **Ground-Level Policy Gaps**
  - **Lack of Coordination:** Different government departments like Agriculture, Fisheries and Animal Husbandry usually work separately. As a result, farmers often do not get all the support they need from one place.
  - **Small Landholdings:** Most farmers in India own less than 1 hectare of land, making it difficult to keep crops, fish ponds, livestock and trees together on the same farm.

According to Agriculture Census 2015-16, small and marginal farmers constitute 86.2% of total operational landholdings in India, with an average landholding size of just 0.6 to 1.08 hectares

Do you know?

Kuttanad Farming System (Kerala) is recognized by the Food and Agriculture Organization (**FAO)** as a **Globally Important Agricultural Heritage System (GIAHS)**. The Kuttanad farming system combines paddy cultivation with fish farming, duck rearing and coconut plantations on bunds, making efficient use of the wetland ecosystem.

Measures Taken by the Government

- **Dedicated Missions**
  - **Rainfed Area Development (NMSA-RAD):** Acts as the main driver for IFS by providing subsidies to small farmers in rainfed regions to set up integrated crop-livestock-tree clusters.
  - **Rashtriya Krishi Vikas Yojana (RKVY-RAFTAAR):** Offers flexible grants to states to build common IFS infrastructure like cattle sheds, farm ponds and fodder units based on local needs.
- **Allied Sector Integration**
  - **Pradhan Mantri Matsya Sampada Yojana (PMMSY):** Subsidizes farm pond construction and fish seed to promote fish farming alongside paddy and livestock rearing.
  - **National Livestock Mission (NLM):** Provides financial assistance for backyard poultry, goatry, and dairy units to help crop farmers add steady daily income.
  - **Sub-Mission on Agroforestry (Har Med Par Ped):** Incentivizes planting timber and fruit trees along field boundaries to protect topsoil and build long-term assets.
- **Organic Farming**
  - **Paramparagat Krishi Vikas Yojana (PKVY):** Supports organic waste recycling, vermicompost production, and natural manure usage to reduce reliance on chemical fertilizers.
  - **GOBAR-Dhan Scheme:** Financial assistance to convert cattle dung and organic waste into biogas and bio-slurry for cleaner cooking fuel and natural farm manure.
- **Research & Field Demonstrations**
  - **ICAR Model Development:** Indian Council of Agricultural Research (ICAR) developed over 45 climate-resilient, location-specific IFS models tailored for different agro-climatic zones.
  - **KVK Live Model Hubs:** Krishi Vigyan Kendras across districts maintain live, working IFS demonstration farms to give hands-on training to local farmers.

Way forward

- **Financial Support**
  - **Single Loan for All Farm Activities:** Banks should provide one loan that covers crops, livestock, fish farming and other integrated farming activities. So, farmers do not have to apply for separate loans.

NABARD's Composite Kisan Credit Card (KCC) allows farmers to get one loan for crops, livestock, and fish farming instead of applying for separate loans for each activity.

- **More Support for Initial Investment:** Government should provide financial assistance to help small farmers build fish ponds, cattle sheds, biogas plants, and other basic facilities needed to start integrated farming.

  Under Rainfed Area Development, the Ministry of Agriculture provides up to 50% direct subsidy for setting up integrated farm units.

- **Institutional Reform & Policy Alignment**
  - **Coordination Between Departments:** Government should create a **District-Level Integrated Farming Cell** so that the Agriculture, Animal Husbandry Fisheries, and Forest Departments can work together
  - **Promote Land Pooling & FPOs:** Encourage Farmer Producer Organizations (FPOs) to pool small, fragmented land parcels into larger integrated farming blocks.

    Dharmapuri Integrated Farmer Producer Company (Tamil Nadu): Small farmers joined together through an FPO and combined their small landholdings to run shared dairy units, a biogas plant and a fish pond.

- **Capacity Building**
  - **Training via KVKs:** Shift extension work from classroom lectures to hands-on training at live Krishi Vigyan Kendra demonstration units.

    FAO's Farmer Field Schools: FAO trains farmers to learn practical methods such as growing rice and fish together and controlling pests naturally.
  - **Train Local Farm Volunteers:** Train village youth as **IFS Pashu-Krishi Sakhis** to guide farmers on basic animal care, fish pond management, and organic compost preparation.
- **Post-Harvest and Market Support**
  - **Better Storage and Processing:** Set up affordable cold storage and small processing units at the Gram Panchayat level so that farmers can safely store milk, fish, eggs, and other perishable products instead of selling them at low prices.

    Ecozen's "Ecofrost", Solar-powered micro cold rooms installed right at farm gates in Maharashtra and Karnataka to prevent perishable spoilage.
  - **Direct Access to Markets:** Help farmers sell their produce directly to nearby towns, weekly markets (haats), schools, hospitals, and other buyers.
- **Technology and Environmental Support**
  - **Region-Specific Farming Models:** Promote integrated farming models that suit the local climate and farming conditions.
  - **Digital Support for Farmers:** Provide simple mobile apps that help farmers manage animal feed, use farm waste efficiently, keep farm records and track their income and expenses easily.

Best Practices & Real-World Field Success Stories

Field	Ground-Level Case Study
Integrated Rice–Fish Farming	**Ziro Valley (Arunachal Pradesh):** Apatani tribe follows an integrated farming system where rice, fish, and poultry are raised together. This method increases paddy and fish production while reducing the need for chemical fertilizers and pesticides.

Piggery–Crop–Aquaculture Integration	**Longleng District (Nagaland):** An ICAR-KVK developed a **6-hectare integrated farming model** managed by women, combining crops, piggery, and fish farming. This model generated an annual income of about **₹12.9 lakh**, mainly from livestock and allied activities.
Hilly & Island Ecosystem Models	**Car Nicobar (Nicobar Islands):** An ICAR-KVK developed a **2,000 m² integrated farming model** combining vegetables, fruits, piggery, poultry and coconut (copra). The model achieved a **Benefit-Cost (B:C) ratio of 2.55** and generated a **net annual income of about ₹2.43 lakh**.
Rainfed & Dryland IFS Models	**Ranchi (Jharkhand):** KVK-supported dryland transformation where small farmers converted single-crop upland rice land into multi-crop systems integrated with piggery, poultry and dairy farming.
Bio-Waste Recycling & Circular Economy	**ICAR-CSSRI Model (Karnal, Haryana):** Developed a **1-hectare integrated farming model** combining dairy, crops, horticulture and vermicomposting. The model recycled more than **80% of farm waste** into useful inputs, reducing the need for chemical fertilizers and cutting fertilizer costs by nearly **50%**.

Conclusion:

An Integrated Farming System (IFS) is not merely a combination of agricultural practices. It is a strategic paradigm shift necessary to ensure climate resilience, ecological sustainability and long-term income security for small and marginal farmers. In alignment with **SDG 2**, **SDG 6**, and **SDG 13**, Integrated Farming System continues to promote **sustainable agriculture** while ensuring **long-term food and livelihood security** in India.

6. PRECISION FARMING

Introduction

Precision Farming is a modern farming approach that uses technology to provide the right input, at the right place, at the right time, and in the right quantity. It helps farmers use water, fertilizers, pesticides, and seeds more efficiently based on the specific needs of each part of the field. This improves crop yield, reduces costs, and promotes sustainable agriculture.

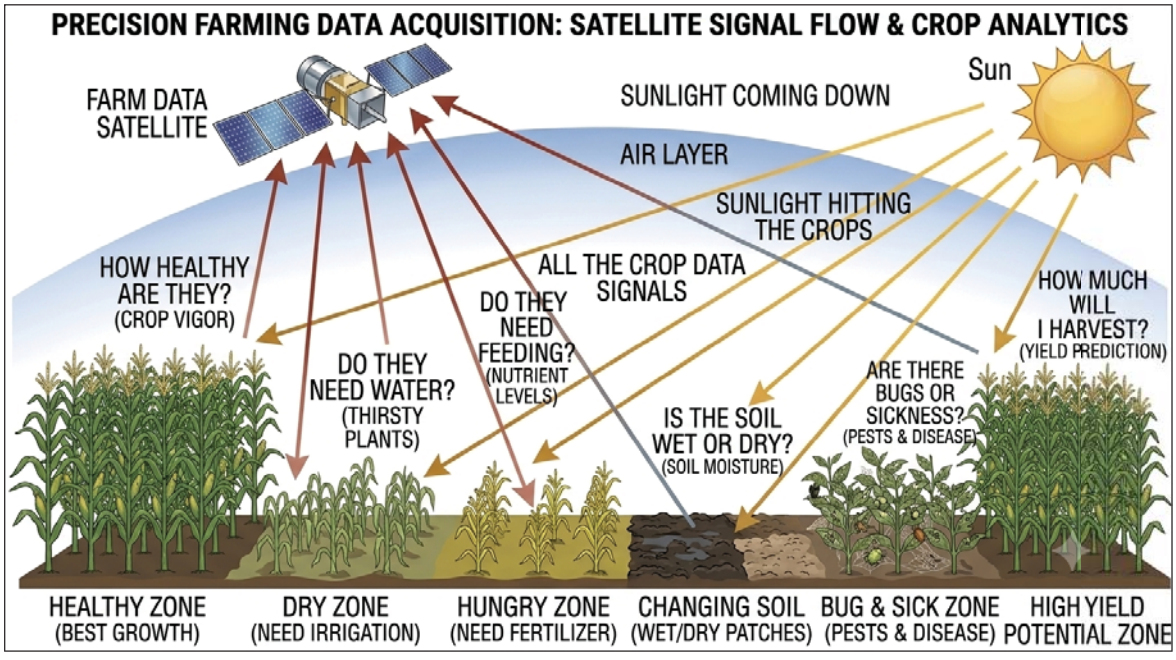
Tools Used in Precision Farming

- **GPS (Global Positioning System):** Helps farmers accurately map fields and carry out precise farming operations.
- **GIS (Geographic Information System):** Collects and analyses field data to support better farm planning and decision-making.
- **Remote sensing:** Uses satellites and drones to monitor crop health, soil conditions, and water stress.
- **Drones (UAVs):** Capture high-resolution images, monitor crop growth, and assist in spraying fertilizers and pesticides.
- **Soil sensors:** Measure soil moisture, nutrients, temperature, and pH to guide efficient input use.
- **Weather stations:** Provide local weather information to help farmers plan irrigation, sowing, and crop protection.
- **Variable Rate Technology (VRT):** Applies seeds, fertilizers, and pesticides only where and in the quantity they are needed.

- **Internet of Things (IoT) devices:** Connect sensors and farm equipment to provide real-time data and automated farm management.
- **Artificial Intelligence (AI) and Data Analytics:** Analyse farm data to predict crop health, pest attacks, and input requirements.
- **Mobile apps and farm management software:** Help farmers monitor field conditions, maintain records, and make timely farming decisions.

Benefit of Precision Farming

- **Economic Efficiency**
  - **Reduces farming costs:** Inputs like fertilizers, pesticides, seeds, and water are used only where needed, reducing waste and lowering cultivation costs.
  - **Increases crop yield and income:** Proper use of nutrients and timely farm operations improve crop productivity, helping farmers earn higher incomes.
- **Environmental Sustainability**
  - **Reduces pollution:** Uses fertilizers and pesticides only where needed, reducing soil and water pollution caused by excessive chemical use.
  - **Saves water:** Smart irrigation systems and soil moisture sensors provide water only when crops need it, reducing wastage and conserving water resources.
- **Climate Resilience and Risk Mitigation**
  - **Early Warning Systems:** Real-time weather data and analytics help farmers anticipate pest attacks, heatwaves, or unseasonal rain, allowing them to take preventive action before damage occurs.
  - **Better soil health management:** Regular monitoring of soil nutrients and organic matter helps maintain soil fertility and supports sustainable farming in the long run.



- **Better Use of Resources and Labour**
  - **Reduces labour requirement:** Drones and automated machines help with tasks like crop monitoring and spraying, reducing manual labour and saving time.
  - **Saves time and improves efficiency:** Farmers can monitor fields and manage irrigation using mobile phones and digital tools, making farm management easier and more efficient.

- **Better Decision-Making**
  - **Improves market access:** Digital records of farming practices help farmers meet quality standards, obtain certifications, and sell their produce in premium domestic and export markets.
  - **Better access to credit and insurance:** Reliable farm data helps banks and insurance companies assess crops more accurately, making it easier for farmers to get loans and faster insurance claim settlements.

Challenges with Precision Farming

- **Small and Fragmented Landholdings**
  - **Small farm size:** Most Indian farmers have small and scattered landholdings, which makes advanced technologies difficult to use on small size of lands.

    According to the 10th Agricultural Census, small and marginal farmers account for 86.2% of all operational landholdings, with an average farm size of just 1.08 hectares.
  - **Scattered and irregular fields:** Uneven field shapes and fragmented plots make it difficult to operate drones, GPS-guided machines, and other precision farming technologies efficiently.
- **High Cost of Technology**
  - **High initial investment:** Precision farming tools such as drones, sensors and smart irrigation systems are expensive, making them difficult for small farmers to afford.

    ICAR studies shows that baseline precision tools like agricultural drones cost ₹1.5 lakh to ₹2 lakh+, which is unaffordable for average smallholder farming households.
  - **Financial risk for farmers:** Many farmers are hesitant to invest in costly technologies because the returns are uncertain and the risk of financial loss is high.

    NABARD reports highlight that over 80% of marginal farmers lack formal institutional credit access for high-tech capital expenditure
- **Digital and Technical Challenges**
  - **Limited digital skills:** Many farmers find it difficult to use and understand digital tools, mobile apps, sensors and data used in precision farming.

    NSSO (2025) findings reveal that only 8% of rural households have broadband connections, leaving most farmers without the digital literacy
  - **Lack of technical support:** Rural areas often do not have enough trained technicians or repair facilities to maintain advanced farming equipment, leading to delays and higher costs.
- **Poor Rural Infrastructure**
  - **Weak internet and power supply:** Frequent power cuts and poor internet connectivity make it difficult to use GPS, IoT devices, and other digital tools effectively.
  - **Lack of local farm data:** Precision farming requires accurate field-level weather and soil data, but such detailed information is often not available to farmers.
- **Farming Challenges**
  - **Diverse cropping patterns:** Indian farms often grow multiple crops together, making it difficult for many precision farming technologies that are designed mainly for single-crop farming.
  - **Variation in soil and climate:** Soil and weather conditions can vary even within the same field, making it difficult to collect accurate data and apply technology uniformly.

Government Reforms to Promote Precision Farming

- **Digital Agriculture Mission (2021–2025):** Promotes the use of AI, drones, GIS, IoT, and digital databases to modernise agriculture and improve farm-level decision-making.

- **Sub-Mission on Agricultural Mechanization (SMAM):** Provides financial assistance for modern farm machinery and supports **Custom Hiring Centres (CHCs)**, enabling small farmers to access advanced equipment at affordable rates.
- **Drone Promotion in Agriculture:** Government provides financial assistance for purchasing agricultural drones and encourages their use for crop monitoring and spraying fertilizers.
- **Soil Health Card Scheme:** Provides farmers with information on soil nutrient status, helping them apply the right type and quantity of fertilizers.
- **PM Krishi Sinchayee Yojana (PMKSY):** Promotes micro-irrigation systems such as drip and sprinkler irrigation to improve water-use efficiency.
- **National Mission on Sustainable Agriculture (NMSA):** Encourages climate-smart and resource-efficient farming practices, including precision farming technologies.
- **Agristack and Digital Public Infrastructure (DPI) for :** Builds digital databases of farmers, land records and crop information to support personalised services and better policy implementation.
- **Kisan Drones Initiative:** Encourages the use of drones for crop surveys, pesticide spraying, nutrient application, and monitoring crop health, especially for small and marginal farmers.
- **Support for Agri-Tech Start-ups:** Through initiatives like the **Agriculture Accelerator Fund** and **Innovation and Agri-Entrepreneurship Development Programme (RKVY-RAFTAAR)**, the government promotes affordable digital solutions for farmers.
- **Digital Advisory Services:** Platforms such as **Kisan Suvidha**, **mKisan**, and AI-based advisory systems provide weather forecasts, crop advisories, market prices, and pest alerts directly to farmers.
- **National e-Governance Plan in Agriculture (NeGPA):** Strengthens digital agricultural services by improving farmers' access to information, advisory services, and government schemes through digital platforms.

Way forward

- **Make Precision Farming Affordable**
  - **Promote shared use of technology:** Strengthen **Farmer Producer Organisations (FPOs)** and **Custom Hiring Centres (CHCs)** so, farmers can rent drones, sensors, and other modern equipment at affordable rates.

    Kisan Drone Scheme and CHC-Farm Machinery mobile app allow FPOs and Custom Hiring Centres to rent out high-cost agricultural drones and laser land levelers.
  - **Encourage collective farming:** Support FPOs in bringing together nearby small farms for activities like drone spraying and digital mapping, making precision farming more practical and cost-effective.

    Maharashtra's MAHAVAP (FPO Network) pools contiguous smallholder plots to conduct joint drone spraying for sugarcane and cotton.
- **Develop Affordable Technologies**
  - **Promote low-cost technologies:** Encourage the development and use of affordable, locally made precision farming tools that are suitable for small and marginal farmers.

    Startups like Fasal and Agri-G8 develop low-cost, solar-powered IoT soil sensors and micro-weather stations tailored specifically for budget-conscious Indian farmers.
  - **Design solutions for Indian farming:** Develop technologies that support mixed cropping, intercropping, and horticulture, keeping in mind the diverse farming practices followed in India.
- **Strengthen Digital Infrastructure**
  - **Improve digital connectivity:** Expand high-speed internet, reliable electricity and digital infrastructure in rural areas so farmers can easily use precision farming technologies.

BharatNet initiative, which is actively linking Gram Panchayats with high-speed fiber-optic internet to support real-time IoT sensors and cloud-based farm advisories.

  - **Build digital agriculture system:** Connect land records, soil health data and weather information on a common digital platform to provide farmers with advisories and services.
- **Strengthen Training and Farmer Support**
  - **Build local technical capacity:** Train **Krishi Vigyan Kendras (KVKs)**, **Drone Didis**, and local agri-entrepreneurs to help farmers understand precision farming technologies.

    Israel's MASHAV extension program, which deploys field technicians into rural communities to train local farmers directly on calibrating smart drip-irrigation sensors.
  - **Promote demonstration farms:** Establish model precision farms in different agro-climatic regions so farmers can see the benefits of modern technologies before adopting them.
- **Ensure Data Security**
  - **Protect farmers' data:** Establish clear rules to ensure that farmers' data is collected, stored, and used securely, with their consent and protection from misuse.

Case studies for precision farming

Region	Impact
Anantapur (Andhra Pradesh, India)	30% reduction in water use, 25% drop in pesticide spending, and 20% higher crop yields.
Nanded & Jalna (Maharashtra, India)	90% reduction in water used for spraying, 40% lower chemical costs, and completed spraying in minutes instead of hours.
Punjab & Haryana (India)	20-25% water savings due to uniform land leveling, improved soil organic carbon, and reduced air pollution.
Mekong Delta (Vietnam)	Enabled over 30,000 smallholders to adjust planting schedules and avoid crop loss from saltwater intrusion.
Almería (Spain)	Transformed an arid region into Europe's largest vegetable exporter while using half the water of traditional farming.
Embu County (Kenya)	Reduced soil preparation costs by 50% and allowed smallholders to plant on time during short rainy seasons.

Conclusion:

Precision farming is helping Indian agriculture become **more productive, resource-efficient and climate-resilient** by enabling farmers to use water, fertilizers and pesticides more wisely. As digital technologies become affordable and accessible, they can improve both farm incomes and environmental sustainability.

7. ORGANIC FARMING

Introduction

Organic farming is a natural way of farming that reduces the use of chemical fertilisers and pesticides and relies more on compost, manure, bio-fertilisers and natural pest control. It helps keep the soil healthy, protects the environment and supports safer food production, making it an important part of sustainable agriculture.

Principles of Organic Farming

- **Healthy soil:** Build soil fertility naturally through **compost, manure, crop residues and bio-fertilisers**.

- **Natural pest control:** Manage pests mainly through **biological and natural methods** instead of chemical pesticides.
- **Crop diversity:** Grow **different crops and use crop rotation** to maintain soil health and reduce pest attacks.
- **Efficient use of resources:** Make better use of **local resources, farm waste and natural inputs**, reducing dependence on external chemicals.
- **Protect biodiversity:** Encourage **beneficial insects, soil organisms and diverse plant life** to maintain a healthy farm ecosystem.
- **Safe and sustainable farming:** Produce food while **protecting soil, water, biodiversity and human health** for the long term.

Tools Used in Organic Farming

- **Compost:** Uses farm and kitchen waste to add **organic matter and nutrients** back to the soil.
- **Farmyard manure (FYM):** Animal dung and crop residues are used to **improve soil fertility and structure**.
- **Bio-fertilisers:** Helpful microorganisms such as **Rhizobium and Azotobacter** naturally increase nutrient availability to plants.
- **Green manure:** Crops like **sunhemp and dhaincha** are grown and mixed into the soil to improve fertility.
- **Bio-pesticides:** Natural products such as **neem-based pesticides and microbial agents** help control pests without heavy chemical use.
- **Crop rotation:** Different crops are grown one after another to **maintain soil health and reduce pests and diseases**.
- **Mulching:** Crop residues, straw or leaves are spread over the soil to **conserve moisture and control weeds**.
- **Biological pest control: Ladybirds, parasitoids and other beneficial organisms** are used to naturally control harmful pests.
- **Traps: Light traps, pheromone traps and sticky traps** help monitor and control insect pests.
- **Vermicomposting:** Earthworms convert organic waste into **nutrient-rich compost** that improves soil fertility.

Do you know?

- As on 30.01.2025, **₹2,265.86 crore** released under **Paramparagat Krishi Vikas Yojana** (2015–25).
- **₹205.46 crore** released for **Paramparagat Krishi Vikas Yojana** under Rashtriya Krishi Vikas Yojana in FY 2024–25.
- Around **15 lakh hectares** under organic farming; **52,289 clusters** formed; **25.30 lakh** farmers benefitted (as of Feb 2025).
- By Dec 2024, **6.23 lakh farmers**, **19,016 local groups**, **89 input suppliers**, and **8,676 buyers** registered on Jaivik Kheti portal**.**

Significance of Organic Farming

- **Soil & Environmental Benefits**
  - **Improves soil health:** Organic manure and compost add nutrients and organic matter, keeping the soil fertile for longer.
  - **Supports biodiversity:** Organic farms provide a better environment for **earthworms, bees, birds and other beneficial organisms**.
- **Economic Benefits for Farmers**

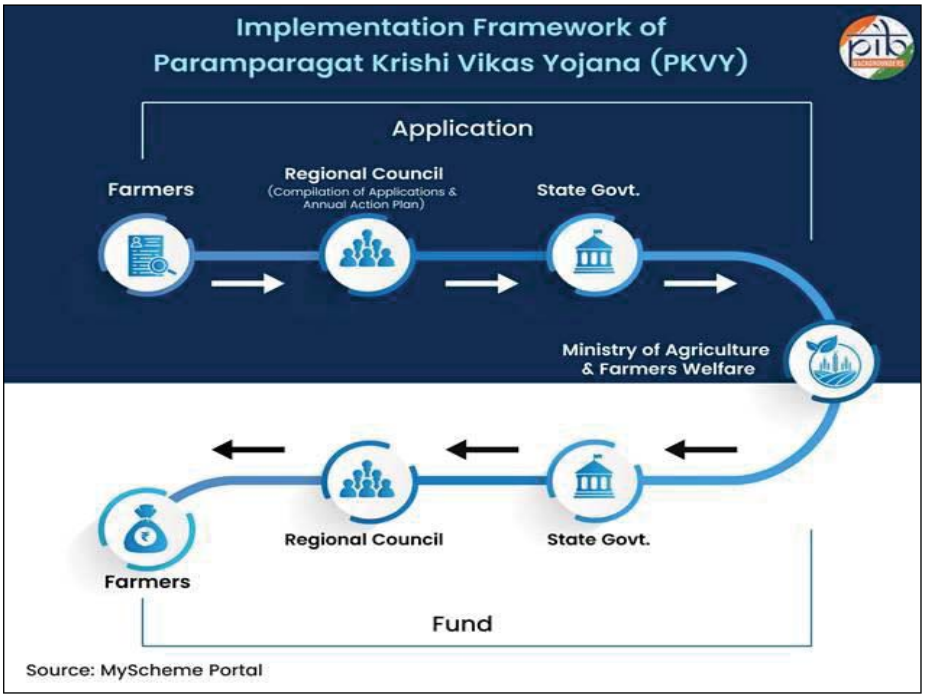
- **Reduces input costs:** Farmers can use **farm waste, manure and locally available natural inputs**, reducing dependence on costly chemicals.
- **Creates rural opportunities:** Organic farming supports activities such as **composting, processing, packaging and direct marketing**, creating additional rural jobs.

- **Health & Nutritional Benefits**
  - **Safer food:** Lower dependence on synthetic pesticides and fertilisers can reduce **chemical residues in food**.
  - **Better food quality:** Organic farming promotes the production of **fresh and naturally grown food** with greater attention to soil and crop health.
- **Climate & Resource Benefits**
  - **Reduces carbon footprint:** Organic farming relies more on **local and natural inputs**, reducing dependence on energy-intensive chemical fertilisers.
  - **Uses resources efficiently:** Recycling crop residues and animal waste helps **reduce farm waste and improve resource use**.
- **Social & Rural Benefits**
  - **Promotes traditional knowledge:** It brings back practices such as **mixed cropping, composting, seed conservation and natural pest control**.
  - **Supports sustainable villages:** Local production of manure, bio-inputs and farm-based resources can make rural farming systems **more self-reliant**.

Organic Farming vs Conventional Farming vs ZBNF

Basis	Organic Farming	Conventional Farming	Zero Budget Natural Farming (ZBNF)
Basic idea	Uses **natural inputs** and avoids synthetic chemicals as far as possible.	Relies on **modern inputs** such as chemical fertilisers, pesticides and improved seeds.	Tries to farm using **local, natural resources with very little external input**.
Fertilisers	Uses **compost, FYM, vermicompost and bio-fertilisers**.	Mainly uses **chemical fertilisers** such as urea, DAP and MOP.	Uses **farm-made preparations** such as Jeevamrut and Ghanajeevamrut.
Pest control	Uses **neem, bio-pesticides, traps and biological control**.	Uses **chemical pesticides** for quick pest control.	Focuses on **natural pest management** and farm-made preparations.
Seeds	Prefers **organic or untreated seeds** where available.	Commonly uses **HYV, hybrid and other improved seeds**.	Uses **locally suitable seeds**, with emphasis on traditional varieties.
Soil health	Builds soil health through **organic matter, compost and crop rotation**.	Can maintain productivity with fertilisers, but **excessive chemical use may degrade soil**.	Focuses strongly on **soil cover, natural microbes and minimum disturbance**.
Water use	Promotes **efficient and careful use of water**.	Can involve high irrigation use, especially for water-intensive crops.	Uses techniques like **mulching and soil moisture conservation** to reduce irrigation needs.

Cost of farming	Can involve **higher labour and certification costs**, especially initially.	Often has higher **external input costs** for fertilisers, pesticides and other inputs.	Aims for **very low external input costs** by using locally available farm resources.
Yield	May be **lower during the transition period**, but can improve with good management.	Often gives **higher and quicker yields** with adequate inputs and irrigation.	Yield can vary widely depending on **crop, soil, climate and farmer practices**.
Certification	Usually requires **organic certification** for selling produce as certified organic.	No organic certification is required.	ZBNF is a farming approach, not the same as certified organic farming.



Issues of Organic Farming

- **Production Challenges**
  - **Lower yields initially:** Organic farms may produce **less in the first few years**, especially when the soil is shifting away from chemical inputs.

    ICAR-Network Project on Organic Farming (NPOF) long-term studies show an initial yield drop of 15% to 30% during the 2-to-3-year conversion period.
  - **More pest and disease risk:** Without quick chemical control, farmers may find it **harder to manage sudden pest attacks**.
  - **Labour-intensive farming:** Practices like **weeding, composting and manual pest control** often require more labour and time.

    CACP (Commission for Agricultural Costs and Prices) cost studies reveal organic cultivation requires 25% to 40% more manual labor-days per hectare.
- **Economic Challenges**

- **Uncertain premium prices:** Farmers do not always get a **higher price for organic produce**, especially in local markets.
- **Small and scattered farms:** Fragmented landholdings make **large-scale organic production and marketing** difficult.

  As per the 10th Agriculture Census (MoA&FW), small and marginal farmers operate 86% of total landholdings (average size <1.08 hectares).

- **Certification & Market Challenges**
  - **Costly certification:** Small farmers may find **organic certification expensive and complicated**.

    APEDA reports show third-party international-grade certification costs between ₹10,000 to ₹50,000 per year

  - **Weak market linkages:** Many farmers struggle to connect directly with **reliable buyers and organised organic markets**.
  - **Fake organic products:** Poor monitoring and misleading labels can **reduce consumer trust** in genuine organic products.
- **Input & Infrastructure Challenges**
  - **Limited availability of organic inputs:** Quality **bio-fertilisers, bio-pesticides and organic seeds** are not easily available everywhere.

    NITI Aayog Study on Bio-inputs estimates an annual supply-demand deficit exceeding 40% for quality-certified bio-fertilizers.

  - **Limited local facilities:** Many villages lack proper **composting, testing and certification support**.
- **Knowledge & Technology Gaps**
  - **Lack of technical knowledge:** Farmers may not know the right methods for **organic pest control, nutrient management and crop planning**.
  - **Limited research:** More research is needed to develop **high-yielding, locally suitable varieties and affordable organic farming technologies**.
  - **Consumer & Awareness Issues**
  - **Higher retail prices:** Organic products can be **more expensive**, making them difficult for many ordinary households to buy regularly.

    FSSAI Surveys show certified organic food products carry a price markup 30% to 100% higher in urban supermarkets compared to conventional produce.

  - **Low consumer awareness:** Many consumers still **cannot clearly distinguish certified organic products from conventionally grown products**.

Measures taken to promote organic farming in India

- **Schemes & Financial Push**
  - **Paramparagat Krishi Vikas Yojana:** Promotes cluster-based organic farming. Farmers receive **₹31,500/hectare over 3 years** (with ₹15,000 sent via Direct Benefit Transfer for organic inputs).
  - **Mission Organic Value Chain Development for NE Region:** Launched for North-Eastern states to build end-to-end organic value chains. Offers **₹46,500/hectare over 3 years** for FPOs and input support.
  - **PM-PRANAM:** Incentivizes states to cut down on chemical fertilizers. States receive **50% of the saved fertilizer subsidy** as a grant to reinvest in organic alternatives.
  - **National Mission on Natural Farming (NMNF):** Setting up **10,000 Bio-Input Resource Centres** for easy access to bio-fertilizers.

- **Certification & Quality Standards**
  - **Participatory Guarantee System:** A peer-review certification run by farmers, for farmers. It keeps certification cheap, local, and accessible for smallholders selling domestically.
  - **National Programme for Organic Production:** Operated under APEDA for strict, third-party international-grade certification required for high-value exports.
- **Market Access & Marketing Support**
  - **Jaivik-Kheti Portal:** A dedicated digital marketplace connecting millions of organic farmers directly with buyers, cutting out middlemen and ensuring better prices.
  - **Market Development Assistance (MDA):** Provides **₹1,500 per Metric Ton** subsidy on Fermented Organic Manure (FOM) under the GOBARdhan initiative to make organic fertilizers affordable.
- **Regional & Ecosystem Models**
  - **State Success Models: Sikkim** established itself as India's first 100% organic state, followed by **Lakshadweep** achieving 100% organic certification.
  - **Ganga Belt Green Corridor:** Special organic farming clusters developed along a 5-km corridor on both sides of River Ganga to curb chemical runoff into the river.

Way Forward for Organic Farming

- **Make Organic Farming More Productive**
  - **Improve organic yields:** Invest in research for **better organic seeds, bio-fertilisers and natural pest-control methods** suited to local conditions.
  - **Strengthen farmer training:** Provide simple, practical training on **soil health, composting, pest control and organic nutrient management**.

    Himachal Pradesh's Subhash Palekar Natural Farming (SPNF) initiative trains lakhs of state farmers directly through practical, field-level workshops on local bio-inputs.
- **Make Organic Farming More Affordable**
  - **Reduce input costs:** Promote **local production of compost, bio-fertilisers and bio-pesticides** so farmers do not have to buy costly inputs from outside.

    Andhra Pradesh Community Managed Natural Farming (APCNMF) empowers smallholders to prepare on-farm botanical concoctions (Jeevamrutha, Neemastra).
  - **Support small farmers:** Use **FPOs and cooperatives** to help farmers buy inputs, process produce and market it together.
- **Improve Certification & Trust**
  - **Simplify certification:** Make organic certification **cheaper, faster and easier**, especially for small and marginal farmers.

    PGS-India (Participatory Guarantee System) offers a peer-review, community-based certification tailored for local smallholder clusters, bypassing high third-party fees.
  - **Stop false labelling:** Improve monitoring to prevent **fake organic claims** and protect both farmers and consumers.

    FSSAI's Jaivik Bharat logo and unified QR-code tracking system allow consumers to scan packaged goods to verify NPOP or PGS certification validity.
- **Build Better Markets**
  - **Create assured markets:** Connect organic farmers with **government institutions, supermarkets, processors and exporters** to provide stable demand.

- **Build an Indian organic brand:** Promote Indian organic products through **quality standards, branding and export support** to capture premium global markets.

  APEDA's "India Organic" master brand, actively showcased at global trade expos to position Indian organic tea, basmati, and spices in premium international markets.

- **Strengthen Infrastructure**
  - **Improve storage and processing:** Set up more **warehouses, cold storages, grading, packaging and processing units** near organic farming clusters.
  - **Develop local input centres:** Establish village-level facilities for producing **compost, bio-fertilisers and bio-pesticides**.
- **Make Organic Farming Climate-Friendly**
  - **Promote mixed farming:** Combine **crops, livestock, trees and allied activities** to recycle farm resources and reduce risks.

    Deccan Development Society (Telangana) promotes traditional Akkadi Saalu (polycropping millets, pulses, and oilseeds paired with livestock) for closed-loop farm recycling.
  - **Improve soil health:** Encourage **crop rotation, green manure, mulching and composting** to build healthy and water-retaining soils.

    Bio-village projects in Tripura, where whole villages transition to green manuring, mulching, and Azolla cultivation to restore degraded soils naturally.

Case Studies of Natural Farming in India

Case study	Explanation
Himachal Pradesh – Prakritik Kheti Khushhal Kisan Yojana	Small farmers adopted **natural farming methods using locally available inputs** such as cow dung, urine and biomass.
Karnataka – Natural Farming Programme	Farmers were trained in **Jeevamrut, mulching, mixed cropping and natural pest management**.
Sikkim – Organic Farming Model	The state gradually moved away from chemical fertilisers and pesticides and promoted **organic and natural farming practices**.
Maharashtra – Subhash Palekar Natural Farming	Farmers adopted the **four basic principles of ZBNF**, including Jeevamrut, Bijamrut, mulching and moisture conservation.
Kerala – Natural Farming & Mixed Farming	Farmers combined **crops, livestock, mulching and biological pest control**, making better use of farm waste and resources.
Himachal Pradesh – Prakritik Kheti Khushhal Kisan Yojana	Small farmers adopted **natural farming methods using locally available inputs** such as cow dung, urine and biomass.
Karnataka – Natural Farming Programme	Farmers were trained in **Jeevamrut, mulching, mixed cropping and natural pest management**.

Conclusion

Organic farming is not just about reducing chemicals; it is about building **healthy soils, resilient farms and safer food systems**. With better **technology, markets, certification and farmer support**, it can

become a practical and profitable choice for Indian farmers. The real goal should be **sustainable agriculture with higher farmer incomes, environmental protection and long-term food security**.

8. HORTICULTURE IN INDIA

Introduction

Horticulture is a branch of agriculture that focuses on the cultivation of fruits, vegetables, flowers, spices, plantation crops, and medicinal and aromatic plants. In India, horticulture plays a vital role in improving nutritional security, increasing agricultural exports, diversifying farm income, and making agriculture more sustainable and profitable.

Major Segments of Horticulture in India

Segment	Covers	Status
Fruits	Mango, Banana, Papaya, Citrus, Guava, Apple, Grape, Pineapple	India is the 2nd largest producer of fruits globally (after China) and ranks 1st in Banana, Mango, and Papaya. Production stands at over 117 Million Metric Tonnes (MMT).
Vegetables	Potato, Onion, Tomato (TOP crops), Brinjal, Okra, Cabbage, Cauliflower	Vegetables form the largest chunk of horticulture, crossing 218 MMT. India is the 2nd largest producer worldwide and leads in Okra and Pea production.
Plantation Crops	Coconut, Arecanut, Cashewnut, Cocoa, Oil Palm	India is a global powerhouse in Coconut (ranking among the top 3) and Arecanut. Production hovers around 15 MMT. Government is giving a massive push to Oil Palm via the NMEO-OP scheme to cut import bills.
Spices	Black Pepper, Cardamom, Turmeric, Ginger, Garlic, Cumin, Coriander	Known as the "Spice Bowl of the World", India is the largest producer, consumer, and exporter of spices. Annual production exceeds 11-12 MMT, bringing in significant foreign exchange.
Aromatic & Medicinal Plants	Mentha, Ashwagandha, Senna, Aloe Vera, Lemongrass, Isabgol	A rapidly growing sector driven by global demand for Ayurveda and wellness. India produces over 0.8 MMT and is a global leader in Mentha mint oil extraction.
Flowers (Floriculture)	Loose flowers (Marigold, Jasmine) and Cut flowers (Rose, Orchid, Gladiolus)	Production crosses 3 MMT (loose) and billions of cut flower stems. Driven by urban celebrations, religious demand, and high-value exports to Europe and the Middle East.
Honey & Sericulture (Allied)	Natural Honey (Apiculture), Silk production (Muga, Eri, Mulberry, Tasar)	Under the National Beekeeping & Honey Mission (NBHM), honey production has reached ~140,000 Tonnes, making India a major exporter to the US and Europe.
Honey & Sericulture (Allied)	Natural Honey (Apiculture), Silk production (Muga, Eri, Mulberry, Tasar)	Under the National Beekeeping & Honey Mission (NBHM), honey production has reached ~140,000 Tonnes, making India a major exporter to the US and Europe.

Second Advance Estimates of 2025-26 of Area and Production of various Horticultural crops:

Total Horticulture	2024-25 (Final)	2025-26 (First Advance Estimates)	2025-26 (Second Advance Estimates)
Area (in Lakh Ha)	301.36	301.31	301.51
Production (in Lakh Tonne)	3707.38	3708.46	3777.76

Significance of Horticulture

- **Economic Boost**
  - **Higher income from less land:** Horticulture gives farmers much higher returns than crops like wheat and paddy. It contributes about one-third of India's agricultural GVA.
  - **Boosts export earnings:** High-value horticultural products such as mangoes, grapes, spices, cashew nuts, and flowers earn valuable foreign exchange.
  - **Supports the food processing industry**: Fruits, vegetables, and spices provide raw materials for industries producing juices, jams, pickles, essential oils, spices, and dried foods..
- **Farmer Well-Being & Jobs**
  - **Regular income for farmers:** Many fruits, vegetables, and other horticultural crops are harvested more frequently, helping farmers earn a **steady income throughout the year**.
  - **Creates year-round employment:** Horticulture needs regular activities such as **planting, pruning, irrigation, harvesting, grading, and packaging**, generating continuous employment.
  - **More opportunities for women:** Activities like **nursery management, harvesting, sorting, grading, and food processing** provide significant employment opportunities for women.
- **Health & Environment**
  - **Improves nutrition:** Fruits and vegetables are rich in **vitamins, minerals, and antioxidants**, helping reduce malnutrition and promote healthier diets.
  - **Makes better use of difficult land:** Many horticultural crops such as **spices, cashew, and berries** grow well in dry, hilly, or less fertile areas, making productive use of land.
  - **Promotes efficient water use:** Horticulture is increasingly supported by **drip and micro-irrigation**, which helps save water while improving crop productivity.
- **Climate Resilience**
  - **Reduces climate-related risks:** Growing a mix of fruits, spices, medicinal plants, and other horticultural crops helps farmers reduce losses when weather conditions affect traditional crops.

Major takeaways from horticulture in India

- **Horticulture is growing fast:** Production rose from **280.70 MT in 2013–14 to 367.72 MT in 2024–25**.
- **More specialised centres:** By July 2025, **58 Centres of Excellence** had been approved across States, along with **55 horticulture clusters**.
- **Better planting material:** The government is setting up **9 Clean Plant Programme centres and 4 Post-Entry Quarantine facilities** to provide healthier planting material.
- **Post-harvest support is expanding:** Over the last five years, **55,748 post-harvest facilities and 11,140 marketing infrastructures** have been created under MIDH.
- **Productivity is improving:** Horticulture productivity increased from **12.10 tonnes/ha in 2019–20 to 12.56 tonnes/ha in 2024–25**.

Issues with horticulture sector in India

- **Infrastructure & Post-Harvest Gaps**
  - **High post-harvest losses:** Poor storage, improper handling, and lack of temperature control lead to **10–25% losses** in many fruits and vegetables before they reach consumers.

    NABARD Studies report that over 85% of smallholder horticulture growers rely on local intermediaries due to shortage of farm-gate primary processing and grading hubs.
  - **Weak value-addition facilities:** Insufficient food processing, grading and packaging infrastructure reduces farmers' earnings and limits the export potential of horticultural produce.
- **Market & Price Volatility**
  - **Frequent price fluctuations:** Prices of perishable crops such as **tomato, onion, and potato (TOP)** often crash during bumper harvests because they cannot be stored for long durations.

    The production of Tomato is expected to be 214.61 Lakh Tonne, which is higher by 8.62 Lakh Tonne (4.19%) as compared to 205.99 Lakh Tonne in 2024-25.
  - **Limited market access:** Inadequate storage, grading, and direct marketing facilities force many farmers to sell their produce at unfavourable prices.
- **Production & Technical Barriers**
  - **Limited quality planting material:** Many farmers do not get **certified seeds, disease-free saplings, and quality planting material** on time, leading to lower productivity.
  - **Low level of mechanisation:** Most horticultural activities such as **planting, harvesting, pruning, and grading** are still done manually because affordable machines for small farms are limited.
  - **Knowledge and technology gaps:** Limited access to modern cultivation practices, extension services, and technical training affects crop quality and overall productivity.

    NITI Aayog (Evaluation of Extension Services) reports that fewer than 12% of horticulture farmers receive regular, crop-specific extension advisories from Krishi Vigyan Kendras (KVKs)
- **Climate & Financial Risks**
  - **Vulnerable to extreme weather:** Horticultural crops are easily damaged by **heatwaves, unseasonal rainfall, hailstorms, frost, and cyclones**, leading to heavy yield losses.

    India Meteorological Department indicates that extreme weather events cause annual localized yield damages of 20% to 50% in perishable fruit.
  - **High financial risk:** Horticulture requires significant spending on **quality planting material, irrigation, fertilizers, and protected cultivation**, making farmers more vulnerable to financial losses if crops fail.

Government Measures to Promote Horticulture in India

- **Mission for Integrated Development of Horticulture (MIDH):** Supports farmers in **planting material, new orchards, protected cultivation, post-harvest management, and marketing**, covering fruits, vegetables, spices, flowers and other horticultural crops.
- **Horticulture Cluster Development Programme (HCDP):** Develops major horticulture clusters with better **production, storage, processing, branding and market linkages**.
- **PMKSY – Per Drop More Crop:** Promotes **drip and sprinkler irrigation**, helping horticulture farmers use water more efficiently and improve productivity.
- **National Horticulture Board (NHB):** Provides support for **commercial horticulture, cold storage, post-harvest infrastructure and quality planting material**, helping farmers reduce losses and improve returns.

- **Agriculture Infrastructure Fund (AIF):** Provides financing support for **cold chains, warehouses, pack houses, sorting and grading units**, helping address major post-harvest gaps in horticulture.
- **Operation Greens:** Supports the **TOP crops (Tomato, Onion and Potato**), to reduce price fluctuations and post-harvest losses
- **PM Formalisation of Micro Food Processing Enterprises (PMFME):** Helps small food-processing units with **credit, technology, branding and training**, allowing farmers and local entrepreneurs to add value to fruits, vegetables and spices.
- **Quality planting material:** Government programmes promote **nurseries, planting material development and improved varieties**, helping farmers get healthier and more productive plants.
- **Export promotion:** Through **APEDA and other export-support measures**, the government promotes horticultural exports by improving **quality standards, packaging and certification.**
- **Crop insurance under PMFBY:** Notified horticultural crops can be covered under **Pradhan Mantri Fasal Bima Yojana**, providing financial support against specified crop losses caused by natural risks.
- **Farmer Producer Organisations (FPOs):** Government support for FPOs helps small farmers **pool their produce, buy inputs together, access technology and negotiate better with buyers**,
- **Cold-chain development:** Schemes such as **PM Kisan SAMPADA Yojana** support integrated cold-chain and food-processing infrastructure, helping fruits and vegetables move from **farm to market with less wastage**.
- **Clean Plant Programme:** A central sector scheme that focuses on providing high-quality, disease-free planting material to strengthen India's position in global horticulture trade.

Way forward

- **Build Cold Chains**
  - **Cold storage at the farm level:** Set up **small, solar-powered cold rooms near villages** so farmers can safely store freshly harvested fruits and vegetables.

    Startups like Ecofroz and CoolCrop install decentralized, solar-powered cold rooms directly at farm gates in states like Bihar and Odisha.
  - **Better cold transport:** Connect farms with markets through **refrigerated vans and trucks**, keeping perishable produce fresh during transportation.

    Kisan Rail initiative, operating temperature-controlled trains to ship perishables like bananas from Andhra Pradesh and grapes from Maharashtra to North Indian metros.
  - **More village-level processing:** Expand **PM Kisan SAMPADA Yojana** to support small processing units near farms, where surplus fruits and vegetables can be turned into **juices, jams, pickles and dried products**.
- **Boost High-Tech & Protected Farming**
  - **Expand polyhouses and greenhouses:** Use support under **MIDH** to help farmers set up **polyhouses, greenhouses and shade-net houses**, allowing them to grow high-value crops.

    Himachal Pradesh and Uttarakhand's wide adoption of polyhouses under MIDH, enabling farmers to grow off-season capsicum, exotic flowers, and tomatoes.
  - **Promote smart irrigation:** Combine **drip irrigation with sensors and fertigation** to give plants the right amount of water and nutrients directly to their roots.
  - **Encourage high-density planting:** Promote **high-density orcharding**, where more plants are grown in a smaller area, helping farmers get **higher yields from limited land**.

Jammu & Kashmir's High-Density Apple Orcharding Scheme, which increased tree density from 250 trees/hectare to over 2,000 trees/hectare.

- **Fix the Supply Chain**
  - **Strengthen FPOs:** Bring small horticulture farmers together through **Farmer Producer Organisations (FPOs)** so they can buy inputs in bulk, reduce costs, and negotiate better prices.

    Sahyadri Farmers Producer Company, India's largest horticulture FPO, which aggregates over 18,000 grape and tomato growers to export directly to European markets.
  - **Develop horticulture clusters:** Expand the **Horticulture Cluster Development Programme (CDP)** to create specialised clusters for crops such as **Nagpur oranges and Nashik grapes**.
  - **Connect farmers directly to markets:** Strengthen platforms such as **e-NAM** and other digital marketplaces so farmers can reach more buyers and reduce dependence on middlemen.
- **Better Quality Planting Material**
  - **Provide clean planting material:** Strengthen Clean Plant Centres to supply farmers with disease-free, healthy and high-quality saplings for fruits and other high-value crops.
  - **Promote organic and natural farming:** Encourage sustainable cultivation of spices, fruits and medicinal plants to meet growing demand for chemical-free products in domestic and export markets.

    Sikkim, designated as India's first 100% Organic State, which built global export demand for organic large cardamom, ginger, and turmeric.

Conclusion

Horticulture plays a vital role in strengthening India's agricultural growth and ensuring nutritional security. The wide variety of crops, ranging from fruits and vegetables to spices, flowers and plantation crops, reflects the rich diversity of the sector. With sustained government support and adoption of modern practices, horticulture has the potential to further boost rural livelihoods, enhance exports and contribute significantly to the nation's economy.

9. MILLETS IN INDIA

Introduction

Millets are **traditional, nutritious and climate-resilient crops** that have been grown in India for centuries. They need **less water, can grow in poor soils, and provide good nutrition**, making them especially useful for small farmers. Millets were rebranded as "Shree Anna" in the Union Budget 2023-24 to promote production, consumption, and nutritional awareness.

Types of Millets in India

- **Major Millets**

Millet Type	Local Name	Soil & Climate Conditions	Leading Producing States
Pearl Millet	Bajra	Warm/dry climate; <50 cm rainfall; sandy, well-drained, or black soils.	Rajasthan, Uttar Pradesh, Gujarat, Haryana.
Sorghum	Jowar	Semi-arid zones; 45–100 cm rainfall; deep black soils or loams.	Maharashtra, Karnataka, Rajasthan, Madhya Pradesh.

Finger Millet	Ragi	Warm, dry climate; 50–100 cm rainfall; red loamy, sandy, or light black soils.	Karnataka, Tamil Nadu, Uttarakhand, Andhra Pradesh.

- **Minor (Small) Millets**

Millet Type	Local Name	Soil & Climate Conditions	Leading Producing States
Foxtail Millet	Kangni / Kakun	Tropical to temperate regions; short duration crop (80–100 days); minimal rain.	Andhra Pradesh, Karnataka, Tamil Nadu, Telangana.
Barnyard Millet	Sanwa / Jhangora	Harsh, high-altitude or flood-prone fields; withstands waterlogging & drought.	Uttarakhand, Madhya Pradesh, Uttar Pradesh, Tamil Nadu.
Kodo Millet	Kodon	Shallow, stony, marginal dryland soils; Deccan plateau to Himalayan foothills.	Madhya Pradesh, Chhattisgarh, Tamil Nadu, Karnataka.
Little Millet	Kutki	Tribal belts, high altitudes (up to 2,100m); survives extreme rain deficits.	Madhya Pradesh, Chhattisgarh, Odisha, Tamil Nadu.
Proso Millet	Chena	Warm/dry conditions; very short crop cycle (60–75 days); low water footprint.	Bihar, Uttar Pradesh, Madhya Pradesh, Andhra Pradesh.
Browntop Millet	Hari Kangni	Shade-tolerant; thrives on poor, dry, degraded soils.	Karnataka, Andhra Pradesh, Tamil Nadu.

Significance of Millets in India

- **Climate Resilience & Environmental Security**

- **Saves water:** Millets need less water than paddy, roughly **200–300 litres of water per kg**. **(Paddy need 3,000–5,000 litres of water per kg)**.
- **Handles heat better:** Millets are **C4(four carbon) crops**, so they can tolerate **high temperatures, dry conditions and poor soils** better than many conventional cereals.
- **Lower environmental impact:** Millets generally need **less irrigation and fewer chemical inputs**. Unlike flooded paddy fields, they do not create the same conditions for **methane emissions**.

- **Nutritional Security**
  - **Rich in essential nutrients:** Millets provide good amounts of iron, calcium, zinc, fibre and B-vitamins, helping address micronutrient deficiencies.

    National Institute of Nutrition (NIN-ICMR) show Finger Millet (Ragi) contains 344 mg of calcium per 100g.
  - **Helps manage lifestyle diseases:** Being rich in dietary fibre and generally having a lower glycaemic response, millets can support better blood-sugar control.
  - **Naturally gluten-free:** Millets are gluten-free, making them a useful cereal alternative for people who need to avoid gluten.

    FSSAI certifies millets as 100% naturally gluten-free, serving as an ideal cereal alternative for populations affected by celiac disease.
- **Economic & Farmer Security**
  - **Lower farming costs:** Millets generally need less water, fertiliser and pesticides, so farmers can grow them at a lower cost than water-intensive crops like paddy.

    Commission for Agricultural Costs and Prices (CACP) Reports highlight that cultivation costs for millets are 30%–40% lower per hectare than irrigated paddy.
  - **Safer for small farmers:** Most millet-growing areas are rain-fed and drought-prone. Since millets can survive with limited water and poorer soils, they offer small farmers a more reliable crop when rainfall is uncertain.
  - **Supports livestock too:** Crops like jowar (sorghum) provide both food grain and nutritious fodder, helping farmers earn from crops as well as support their dairy animals.
- **Policy & Global Leadership**
  - **India's global leadership:** India is the world's largest producer of millets. By taking the lead in the UN International Year of Millets 2023, India highlighted millets as a solution for food security, nutrition and climate resilience.

    In 2024–25, India exported a total of 89,164.96 tonnes of millets. The total value of these exports stood at 37 million USD.

Issues:

- **Consumer Challenges**
  - **Rice-Wheat Habit:** Decades of cheap rice and wheat under the PDS made our palates used to refined grains.

    According to Department of Food & Public Distribution) Allocations, over 98% of total PDS foodgrains distributed annually consist of rice and wheat.
  - **Difficult to cook:** Millets are gluten-free, so making soft rotis can be a little difficult. Some millet grains also need more preparation or cooking time.
- **Processing & Supply Chain Bottlenecks**

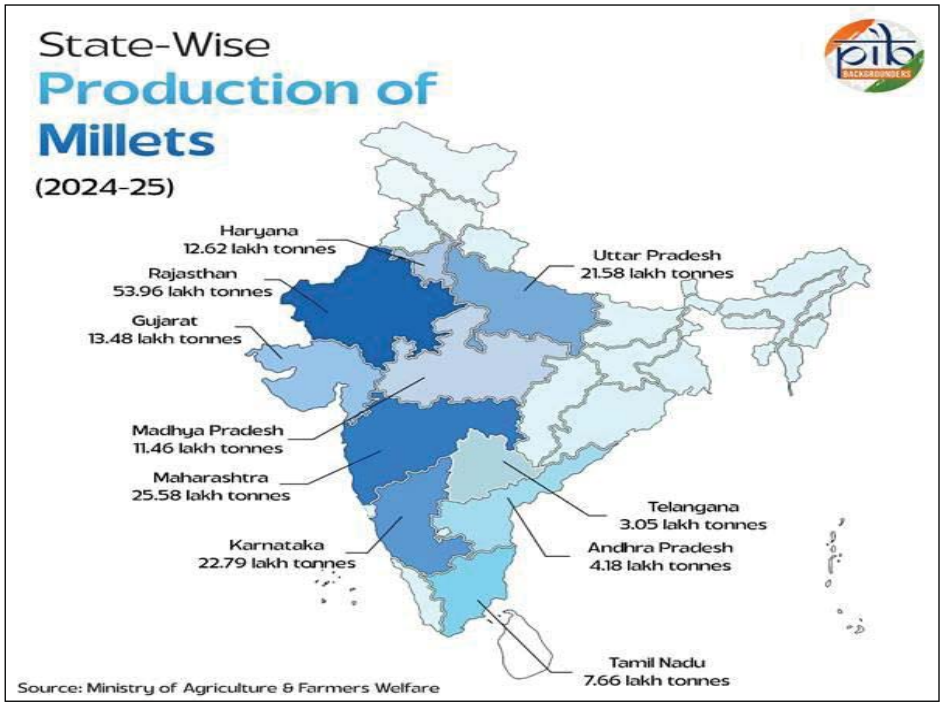
- **Lack of processing facilities:** Many villages do not have **small, affordable machines for cleaning, de-husking and processing millets**, making processing difficult for farmers.
- **Scattered production:** Millet farming is often spread across **small and remote farms**, making it difficult to collect produce in bulk and connect farmers with larger markets.
- **Weak market linkages:** Poor transport, storage and processing facilities often mean farmers have to **sell locally at lower prices**.

NITI Aayog Evaluation Studies note that over 70% of small millet farmers are forced to sell their produce to local village traders (Arhtiyas) at 20–30% below MSP.

- **Farm & Economic challenges**
  - **Lower yield:** Some millets give **lower grain yields per hectare**, which can make farmers hesitant to shift from more established crops.

Directorate of Economics and Statistics (DES) crop data reveals an average national yield gap: paddy yields ~2,800 kg/ha and wheat ~3,500 kg/ha, whereas small millets yield only ~1,000–1,500 kg/ha.

  - **Labour-intensive work:** Activities such as **weeding, harvesting, threshing and protecting crops from birds** often require more manual labour.
  - **Weak market returns:** Where processing and organised markets are limited, farmers may struggle to get **stable and attractive prices**, further reducing the appeal of millet cultivation.



Government Measures to Promote Millets:

- **National Millets Mission:** The government has promoted millets through the **National Food Security Mission (NFSM)** by supporting farmers with better seeds, demonstrations, training and
- **International Year of Millets 2023:** India took the lead in promoting **2023 as the International Year of Millets**, taking Indian millets to the global stage and highlighting their role in **nutrition, food security and climate resilience**.
- **MSP support:** Millets such as **jowar, bajra and ragi** are covered under **Minimum Support Price (MSP)**, giving farmers some protection against very low market prices.

As of May 2025, the Cabinet approved MSP for Ragi for the 2025–26 marketing season, with the second-highest absolute increase of ₹596/quintal.

- **Promotion of value addition:** Schemes such as **PMFME** support small businesses in making **millet flour, snacks, ready-to-eat foods and other processed products**, helping farmers earn more from their produce.
- **Support for FPOs:** Farmer Producer Organisations are being encouraged to help millet farmers **buy inputs together, process their produce, access better markets and bargain for better prices**.
- **Promotion through branding and awareness:** Campaigns such as **"Shree Anna"** have helped create greater awareness about millets as **healthy, nutritious and climate-friendly foods**.
- **Research and innovation:** Institutions such as **ICAR and IIMR** are working on improved varieties, processing technologies and new millet-based products to make cultivation more attractive.

Do you know?

- India produced **180.15 lakh tonnes of millets in 2024–25**, an increase of **4.43 lakh tonnes** over the previous year. **Rajasthan** was the largest producer, followed by **Maharashtra and Karnataka**.
- Among all millet varieties, **Bajra accounted for the largest share of production at 60.3%**.
- ICAR–Indian Institute of Millets Research (IIMR), Hyderabad, is the nodal agency for millet research in India under the Indian Council of Agricultural Research (ICAR). Recognised as the Global Centre of Excellence on Millets during the International Year of Millets 2023.
- India has spearheaded a global millet movement through various platforms such as the G20 presidency, Millet Culinary carnival, and International Trade Events.
- The Global Millets (Shree Anna) Conference held in March 2023 at IARI, Pusa Campus, New Delhi.
- According to the **UN Food and Agriculture Organization (FAO)**, India produces over **80% of Asia's and 20% of the world's millets**

Way forward

- **Boost Production & Processing**
  - **Set up village-level mills:** Create **small and affordable de-husking and processing units** near millet-growing areas so farmers can process their produce locally.

    Odisha Millets Mission (OMM) provides decentralized, solar-powered de-husking and primary processing units directly to rural Self-Help Groups (SHGs) and tribal clusters.
  - **Develop better seeds:** Strengthen **ICAR research** to develop millet varieties that give **higher yields and can withstand drought, heat, pests and diseases**.
  - **Improve shelf life:** Support research into **better milling, packaging and storage methods** so millet flour stays fresh for longer and can reach distant markets.

    ICAR have engineered cold-grinding techniques and specialized modified atmosphere packaging (MAP) to extend raw millet flour shelf-life from days to several months.
  - **Promote local value addition:** Encourage FPOs and rural enterprises to make **millet flour, snacks, ready-to-cook mixes and other products**, creating more income and jobs locally.

    Millet Shakti Cafés in Odisha, operated by local women's SHGs, convert raw millets into ready-to-eat snacks, bakery items, and local dishes.
- **Fix Markets & Strengthen Policy Support**
  - **Use millets in public programmes:** Increase the regular use of millets in **PDS, PM POSHAN and Anganwadis**, creating a steady market while improving nutrition.

Karnataka and Odisha introduced Ragi (Finger Millet) balls and Ragi laddoos into the Public Distribution System (PDS) and Integrated Child Development Services (ICDS / Anganwadi).

- **Make millet foods affordable:** Encourage **local processing, better supply chains and affordable packaging** so processed millet products become affordable for ordinary households.
- **Connect farmers with buyers:** Strengthen **FPOs, e-NAM and direct marketing channels** so farmers can reach buyers more easily and receive a better share of the final price.

- **Empower Farmers & Strengthen Aggregation**
  - **Strengthen millet FPOs:** Bring small and rain-fed farmers together through **FPOs** so they can buy inputs at lower prices, process produce together and sell in bulk.

    Nari Pragati Farmer Producer Company (Odisha), a 100% women-led FPO that aggregates smallholder millet harvests to sell in bulk to government and private buyers.
  - **Support women-led SHGs:** Train **Self-Help Groups (SHGs)** to take up local millet processing, packaging and marketing, creating **income and jobs for rural women**.
  - **Promote intercropping:** Encourage farmers to grow **millets with pulses and oilseeds**, which can improve soil health and spread crop risks.

    Deccan Development Society (Telangana) promotes traditional Akkadi Saalu (intercropping millets with pulses and oilseeds), improving soil nitrogen and reducing crop risk.
- **Change Consumer Mindset & Expand Global Reach**
  - **Make millets easier to use:** Promote **ready-to-cook mixes, millet-based flour and simple recipes** so busy households can include millets in their daily meals.

    Japan and South Korea, where pre-blended multi-grain rice packs (containing foxtail and proso millet) are sold in convenient, easy-to-cook pouches.
  - **Change the perception:** Use **food festivals, school programmes and awareness campaigns** to present millets as a tasty, nutritious and modern food rather than a "poor man's food".
  - **Build a strong global brand:** Use **APEDA and the "Shree Anna" identity** to improve branding, quality and export for global markets.

Conclusion

Millets or Shree Anna, hold the promise of a healthier, more resilient and sustainable future. With their rich nutritional profile and climate-smart qualities, they offer a practical solution to many challenges faced by Indian agriculture today. The collective efforts of central ministries, state governments, research bodies, farmer organisations, and private players are helping restore millets to their rightful place in farms, markets, and diets.

10. CROP INSURANCE SCHEMES IN INDIA

Crop insurance schemes in India are designed to protect farmers against financial losses due to crop failure caused by natural calamities, pests, and diseases.

Aim/Objective of the Crop Insurance Schemes in India

- To provide financial support to the farmers in the event of natural calamities, pests and diseases.
- To restore the credit worthiness of farmers
- To encourage the farmers to adopt progressive farming practices, high value in-puts and higher technology in Agriculture.
- Crop insurance could mitigate the risk faced by farmers.

Types of Crop Insurance Schemes in India

National Agricultural Insurance Scheme (NAIS) 1999 and Modified National Agricultural Insurance Scheme (MNAIS) 2010 **(These have now mostly been replaced by PMFBY).**

Comparison Between RWBCIS and PMFBY

Aspect	RWBCIS	PMFBY
Basis of Claims	Weather parameters as proxies for crop yields.	Yield-based, relying on Crop Cutting Experiments (CCEs).
Claim Settlement Speed	Faster claim settlements due to real-time weather data.	Delays in payouts due to dependence on CCEs and verification processes.
Coverage	Weather-specific risks like rainfall, temperature, and humidity.	Comprehensive coverage, including pests, diseases, and post-harvest losses.
Risk Focus	Focuses on weather variability and its impact on crops.	Covers broader agricultural risks, including yield and external damages.
Targeted Needs	Suitable for farmers facing frequent weather fluctuations.	Suitable for farmers needing broader risk protection across crop cycles.

Challenges with Crop Insurance Schemes in India

- **Low Farmer Awareness & Participation-** Small and marginal farmers **lack awareness regarding enrollment** and **claim of benefits**. Further, literacy gaps and poor outreach reduce their participation.
- **Delayed Claim Settlements-** Settlement of claims often takes months due to **slow crop-cutting experiments (CCEs)**, **bureaucratic delays**, and **disputes** between insurers and states.
- **Dependence on Government Subsidy-** The insurance claims depend heavily on state subsidy, which on being delayed, leads to delays in the insurance settlement claims.
- **Technology & Data Issues**- Crop-cutting experiments (for yield estimation) remain **manual**, **outdated**, and prone to manipulation.
- **Exclusion of Tenant & Sharecroppers-** Tenant farmers, sharecroppers, and landless cultivators (who are most vulnerable) remain outside the coverage net.
- **Moral Hazard and Adverse Selection-** Farmers often **under-invest in crop care** on the assumption of insurance covering their losses. Insurers often **avoid high-risk areas**, leading to partial coverage.

Way Forward:

- **Improve Awareness & Inclusivity**
  - Launch intensive awareness campaigns (local languages, panchayat-level outreach).
  - Ensure coverage for tenant farmers, sharecroppers, and landless laborers, not just landowners.
- **Timely Settlement of Claims**
  - Use **technology (drones, satellite imagery, remote sensing, AI)** for faster and more transparent yield estimation.
  - Introduce **direct benefit transfer (DBT)** to ensure farmers get money directly in their bank accounts.
  - Fix **strict timelines** for insurers and states to settle claims.
- **Financial Sustainability**
  - Create a **national risk pool** so that high losses in one region don't bankrupt the scheme.
  - Encourage **public–private partnerships (PPPs)** with better accountability of insurers.

- Explore **index-based insurance (weather/area yield index)** to reduce disputes.

- **State & Centre Coordination-** Allow states to design **crop-specific localized insurance products** under the PMFBY framework.
- **Leverage Digital Infrastructure-** Integrate crop insurance with **Digital India initiatives** like **land records digitization**, **PM-KISAN database**, and **Aadhaar**

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MGP/BP/GS3/031

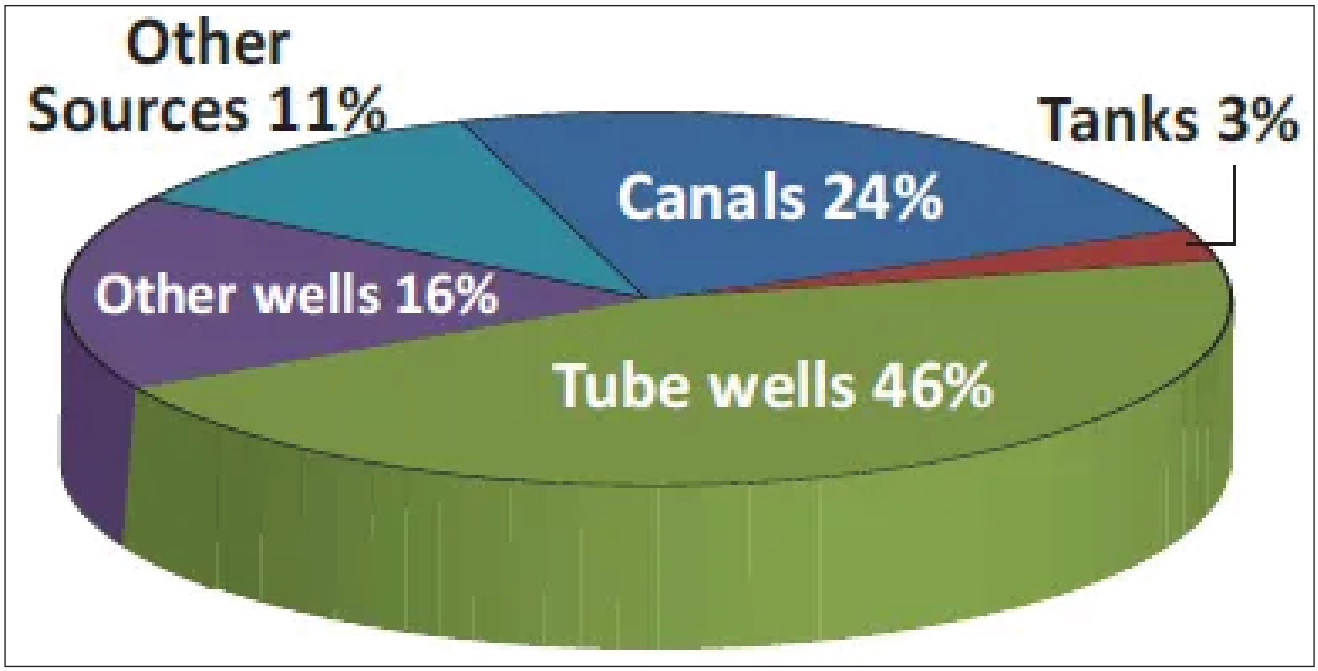
IRRIGATION & WATER MANAGEMENT

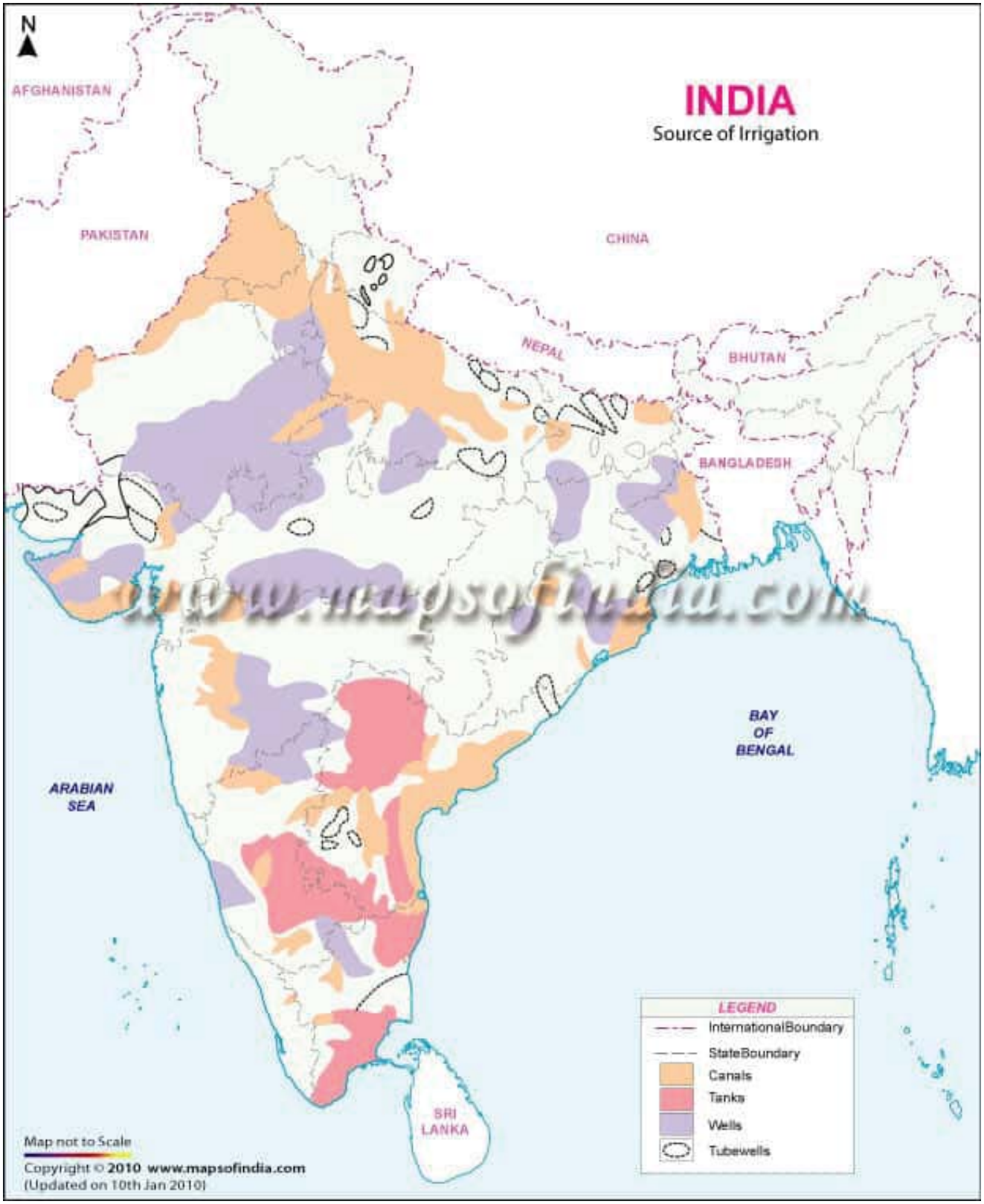
1. IRRIGATION: OVERVIEW

Irrigation is a key agricultural practice that entails the regulated and intentional application of water to plants or crops. It involves adding water to the soil in order to promote plant development and raise agricultural yield. Since ancient times, the method has been utilized to boost crop production and lessen the effects of dry spells or droughts.

Current status of Irrigation in India:

- India is one of the world's largest irrigated areas, with roughly 68 million hectares (MHa) of gross irrigated area as of 2025.
- About 48% of India's net sown area is under irrigation, with the remainder depending on rainfall.
- Major crops like wheat and sugarcane have over 90% irrigation coverage, while pulses and oilseeds rely more on rainfed farming.
- **Sources of Irrigation:**
  - Groundwater accounts for nearly 63% of irrigated area, making India the largest user of groundwater for irrigation globally.
  - Canals contribute about 24%, while tanks, wells, and other sources provide the rest.





Types of Irrigation Systems:

- **Surface Irrigation**: This irrigation technique, which involves flooding the entire field or a section of it with water, is the most popular. It is possible for the water to permeate the soil and get to the roots of the

crops. Surface irrigation is simple to set up and maintain, but it is inefficient since a lot of water is wasted to evaporation and runoff.

- **Sprinkler Irrigation**: In this technique, water is applied to the crops using sprinkler heads. Droplets of water are sprayed and land on the soil's surface and leaves. Sprinkler irrigation is effective because it minimises water loss from runoff and evaporation. Although it may be used for a variety of crops, it uses a lot of energy to operate.
- **Drip Irrigation**: This is a low-pressure irrigation technique that involves the application of water directly to the roots of the crops through a network of tubes and emitters. Drip irrigation is highly efficient as it delivers water directly to the plant roots, minimizing water loss due to evaporation and runoff. It is particularly useful for crops with shallow roots and in areas with limited water availability.
- **Sub-surface Irrigation**: Irrigation below the soil's surface using buried pipes or porous tubes is known as sub-surface irrigation. Sub-surface irrigation is effective because it minimises water loss from evaporation and runoff by delivering water to the plant roots directly. It helps to lessen soil salinity and waterlogging, making it especially beneficial for places with high water tables or saline soils.
- **Center Pivot Irrigation**: This method involves the use of a large sprinkler system mounted on a pivot that rotates around a central point to distribute water to crops in a circular pattern. The water is supplied through a pipeline running along the pivot, and the sprinklers distribute water evenly over a large area of land. Center pivot irrigation is efficient as it reduces water loss due to evaporation and runoff and is suitable for crops such as corn, soybeans, and wheat that are planted in rows. However, it requires a significant investment in infrastructure and energy to operate.

Need for Irrigation in India:

- **Uneven Distribution of Rainfall:** The arrival of the monsoon is uncertain and variable, especially in areas of low rainfall, e.g., Punjab, Haryana, and Western Uttar Pradesh (High coefficient of variability). There is also variation in the spatial distribution of rainfall e.g. Meghalaya has much more rainfall than the Thar Desert.
- **Inadequate Coverage:** Only 30% of cultivated land receives sufficient rainfall of more than 100cm. In areas of high rainfall, irrigation is necessary to increase farm productivity further. Monsoon gaps (no rain for two or more weeks during the sunny season) may damage crops without irrigation facilities.
- **Seasonal Nature:** The monsoon is "seasonal." 75% of rainfall happens in 3-4 months of the year, and the remaining 8-9 months are marked by a dry season when irrigation is badly needed for growing crops (5 months dry in Kerala, nine months dry in North Western India).
- **Type of Rainfall:** Rainfall in most parts of India is torrential and therefore there is less opportunity for soil to absorb water and surface water goes waste. Also, rainwater flows down very quickly along the hill slopes. Certain crops, such as rice, sugarcane, jute, and cotton, require more water and need irrigation even in areas of heavy rainfall.
- **Problem with Monoculture:** Irrigation is necessary to end monoculture cropping practices in Indian agriculture. It is also necessary for the socio-economic transformation of rural India by making agriculture variable.
- **Green Revolution**: Since the Green Revolution, the introduction of HYV seeds and heavy doses of chemical fertilisers have made irrigation necessary. Sandy and loamy soil can't retain water like Alluvial and black soil.

Significance of Irrigation:

- **Enhanced agricultural productivity:** Regular and reliable water supply promotes optimal plant growth, increases land utilization, and raises per acre yields for food and cash crops. It supports cultivation of water-intensive and high-value crops like wheat, sugarcane, and vegetables, improving farm incomes.
- **Ensure Stable Crop Production:** Irrigation shields crops from erratic monsoons and rainfall variability, reducing dependence on rainfed agriculture and stabilizing yields.

- **Improved food security:** Irrigation mitigates the risk of crop failure caused by drought or insufficient rainfall, thereby ensuring a stable food supply for local populations.
- **Stimulated economic growth:** Through increased agricultural output and productivity, irrigation can create job opportunities and stimulate economic growth in rural areas.
- **Water storage:** Building dams and reservoirs that may be used to store water for irrigation and other uses is a common part of irrigation projects. This will boost infrastructural advancements.
- **Climate resilience:** By providing a source of water when rainfall patterns become unpredictable, irrigation can aid farmers in adjusting to climate change. Also prevents deforestation and famine.

Challenges/Limitations related to Irrigation:

- **Over-dependence on Groundwater:** Excessive extraction of groundwater—now around 63% of total irrigation use—has led to rapidly falling water tables in several states (Punjab, Haryana, Rajasthan, UP). Unsustainable groundwater use threatens long-term availability and increases costs for small farmers.
- **Inefficient water usage & Water loss:** Traditional canal and surface irrigation systems suffer high water loss due to seepage, evaporation, and poor maintenance. Low efficiency means less water reaches crops, especially at tail ends of canal systems.
- **Soil degradation:** Overuse of irrigation can cause soil salinization and waterlogging, which can lead to a decline in soil quality and ultimately reduce crop yields over time.
- **Water pollution:** Irrigation can lead to contamination of nearby waterbodies due to the runoff of fertilizers and pesticides, causing ecological degradation and water pollution.
- **Impact on biodiversity:** Irrigation can modify natural water flows, which can disrupt wetlands and other natural habitats, leading to a reduction in biodiversity.
- **Regional Disparities:** Western and southern regions enjoy better irrigation coverage, while eastern, central, and northeastern states remain largely rainfed.
- **Socio-economic inequality:** While irrigation can benefit some farmers, it can also create disparities between those who have access to irrigation and those who do not, leading to social and economic inequality within the region.

Various government initiatives related to irrigation:

- **Pradhan Mantri Krishi Sinchai Yojana (PMKSY):**
  - Focuses on “Har Khet Ko Pani” (water to every field) to enhance irrigation coverage.
  - Promotes efficient water use with the slogan “Per Drop More Crop” by encouraging micro-irrigation (drip and sprinkler) and on-farm water management.
  - Integrates various schemes like Accelerated Irrigation Benefits Programme (AIBP), Command Area Development (CAD), and other water conservation measures.
- **Accelerated Irrigation Benefits Programme (AIBP):** Provides central assistance for accelerating ongoing major, medium, and minor irrigation projects to improve water storage and distribution capacity.
- **Micro Irrigation Fund (MIF):**
  - Set up under NABARD with a corpus of ₹5,000 crore to promote adoption of drip and sprinkler irrigation nationwide.
  - Encourages private investment and subsidizes equipment to improve water use efficiency.
- **Atal Bhujal Yojana (ABY):** A recent scheme focusing specifically on sustainable groundwater management in water-stressed areas through community participation and demand-side management.
- **Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyan (PM-KUSUM):** This initiative promotes the installation of **solar pumps** for irrigation. It has three components:
  - Installing standalone solar pumps.
  - Solarizing existing grid-connected pumps.
  - Setting up small solar power plants on agricultural land for farmers to sell excess power to the grid.

- **Irrigation Projects:**
  - **The Indira Gandhi Canal**: The Indira Gandhi Canal, which is located in Rajasthan, is one of India's biggest irrigation projects. The Sutlej River's water is diverted through the 600+ km long canal and into the Thar desert. Increased agricultural production is now possible in the desert region thanks to the project's creation of irrigated land covering more than 1.8 million hectares.(CCA of 5,28,000 ha.)
  - **Narmada Valley Project**: In Gujarat, Madhya Pradesh, and Maharashtra, the Narmada Valley Project uses water diverted from the Narmada River to irrigate more than 1.7 million hectares of land. Millions of people receive irrigation and drinking water from the project's numerous dams, canals, and reservoirs. (CCA is 1,46,800 ha.)
  - **Project Krishna Godavari**: The Krishna Godavari Basin Project, which is situated in Andhra Pradesh, is one of the country's greatest river basin projects. In addition to irrigating more than 5 million hectares of land and producing hydroelectric power, the project includes a number of dams, canals, and reservoirs. As part of the project, various cities in the area receive drinking water.(CCA is 3,45,000-5,04,000 ha.)
  - **Sardar Sarovar Project**: Located on Gujarat's Narmada River, the Sardar Sarovar Project serves a variety of purposes. Over 1.8 million hectares of land are irrigated by the project's several dams, canals, and reservoirs, which also provide hydroelectric power. A number of the region's cities also receive drinking water from the project.(CCA of 18,45,000 ha).

Way forward:

- **Promote Water-Use Efficiency and Micro-Irrigation:**
  - Scale up adoption of drip and sprinkler irrigation systems to conserve water and improve crop water productivity.
  - Provide subsidies, credit support, and technical training to farmers for efficient irrigation technologies.
- **Sustainable Groundwater Management:**
  - Implement strict monitoring and regulation of groundwater extraction, especially in overexploited regions.
  - Encourage adoption of water-saving cropping patterns and recharge techniques like rainwater harvesting.
- **Modernise Canal System:**
  - Upgrade and maintain canal infrastructure to reduce seepage and improve equitable water delivery.
  - Implement automation and remote sensing technologies for better water control and scheduling.
- **Expand Irrigation Coverage & Equity:**
  - Focus on neglected and rainfed regions, especially in Eastern and Northeastern India, to reduce regional disparities.
  - Support small and marginal farmers through community-managed irrigation and participatory water governance.
- **Integrate Climate Resilient Practices:**
  - Promote drought-tolerant and less water-intensive crops along with climate-smart water management.
  - Use climate forecasting and advisories to optimize irrigation scheduling.
- **Promote Renewable Energy for Irrigation:** Expand use of solar-powered pumps to reduce dependence on grid electricity and diesel, ensuring sustainable, cost-effective irrigation.

Conclusion:

The development and management of irrigation systems in India are pivotal for the sustained growth of the agricultural sector and the livelihoods of millions of farmers. By prioritising sustainable irrigation practices, India can enhance agricultural productivity, secure rural livelihoods, and contribute to the nation's economic growth.

2. MICRO-IRRIGATION

Introduction

Micro-irrigation means **delivering water directly to the root zone of crops through systems like drip and sprinkler irrigation**. It helps farmers **"grow more with less water"** by reducing water wastage, improving crop productivity and enabling better use of fertilizers. In a water-stressed country like India, micro-irrigation is therefore an important tool for **water security, sustainable agriculture and climate-resilient farming**.

Types of micro- irrigation

Type	Process	features / Uses
Drip Irrigation	• Water is delivered drop-by-drop directly to the root zone through emitters.	• Very high water efficiency • Ideal for fruits, vegetables, sugarcane, cotton etc. • Reduces evaporation and runoff • Allows fertigation (fertiliser through irrigation water)
Sprinkler Irrigation	• Water is sprayed over the field through nozzles under pressure, much like artificial rainfall.	• Suitable for wheat, pulses, oilseeds, vegetables etc. • Useful in uneven or sandy land • Reduces water loss compared with flood irrigation
Micro-Sprinkler Irrigation	• Uses small sprinklers that distribute water over a limited area around the plant.	• Provides gentle and uniform watering • Useful for orchards, nurseries and plantations • Covers a larger area than drip but uses less water than conventional sprinklers
Bubbler Irrigation	• Water is released through low-pressure bubblers near the base of plants.	• Suitable for fruit trees and widely spaced crops • Delivers water directly to the root zone • Simple and relatively easy to maintain
Sub-surface Drip Irrigation (SDI)	• Drip pipes are placed below the soil surface, allowing water to reach the roots directly.	• Very low evaporation loss • Keeps the soil surface relatively dry • Useful for water-scarce regions and high-value crops

Current Status of Micro-Irrigation in India

- **Growing Coverage**: As per the **Ministry of Agriculture & Farmers Welfare**, **96.97 lakh hectares** have been covered under **PDMC since 2015–16**, including **46.37 lakh ha drip** and **50.60 lakh ha sprinkler**.
- **Drip vs sprinkler:** Sprinkler accounts for around **52%** and drip around **48%** of PDMC coverage, showing the importance of sprinkler systems for **field crops and dryland areas**.
- **Untapped Potential:** The **2004 Task Force on Micro-Irrigation** estimated India's potential at **69.5 million ha**, including **27 million ha for drip** and **42.5 million ha for sprinkler**.
- **Potential vs coverage:** The latest PDMC coverage of around **9.7 million ha** remains much below the **69.5 million ha potential estimated by the 2004 Task Force**.

Benefits of Micro-Irrigation

- **Saves Water**
  - **Direct watering:** Water goes straight to the root of the plant and very less water is lost through evaporation and run-off.
  - **Better efficiency:** Farmers can grow the same or more crop with less water, which is crucial in water-stressed regions.
- **Improves Crop Yield**
  - **Right water, right time:** Crops get a steady and controlled supply of water according to their needs.
  - **Better growth:** Uniform moisture improves crop productivity and quality of crop.
- **Reduces the Cost of Farming**
  - **Less labour:** Automated systems of sprinkler and drip reduce the need for manual irrigation.
  - **Less fertiliser wastage:** Fertigation supplies nutrients closer to the roots, improving their use.
- **Raises Farmers' Income**
  - **Higher productivity:** Better water management can translate into higher yields and better-quality produce.
- **Encourages Crop Diversification**
  - **Supports high-value crops:** Fruits, vegetables, spices and flowers can be cultivated more efficiently.
  - **Reduces water-intensive farming:** It can make a shift towards less water-intensive crops easier in suitable regions.
- **Protects Soil and Land**
  - **Less soil erosion:** Slow and controlled watering stops water from **washing away fertile soil**.
  - **Prevents waterlogging:** Only the required water is given, so **extra water does not collect in the field**.

Issues with Micro-Irrigation

- **High Initial Cost**
  - **Expensive setup:** Drip pipes, sprinklers, pumps and filters require a large initial investment, which can be difficult for small farmers.

    According to the NABARD Research Study on Micro-Irrigation, the average capital setup cost for a drip system ranges between ₹40,000 to ₹1,00,000 per hectare
- **Maintenance Problems**
  - **Clogging:** Small holes in drip systems can get blocked by mud, salts and algae.

    According to study by the NITI Aayog (Evaluation of PMKSY-PDMC), over 22% of installed micro-irrigation units in hard-water or groundwater-depleted regions face operational downtime.
  - **Regular care:** Filters and pipes need cleaning and timely repair.
- **Subsidy Delays**
  - **Delayed payments:** Farmers may face delays in receiving subsidies, making the system costlier to install.

    From FY16 to FY25 (up to December 2024), the government released ₹21,968.75 crore to States under PDMC, covering 95.58 lakh hectares.

    Lack of Awareness
  - **Limited knowledge:** Many farmers may not know how to install, operate and maintain the system properly.

According to the NSSO 77th Round Survey, only about 10.2% of agricultural households accessed technical advice through official extension services like Krishi Vigyan Kendra.

- **Small and Fragmented Landholdings**
  - **Installation difficulty:** Scattered and very small plots can make the system costly and difficult to manage.
- **Poor Water and Power Quality**
  - **Water quality:** Muddy or salty water can block pipes and damage equipment.

As per the Central Ground Water Board (CGWB) Report, over 250 districts across India report high salinity/hardness or heavy siltation in groundwater.

  Power supply: Irregular electricity can affect pumping and timely irrigation.
- **Unequal Adoption**
  - **Regional gap:** Adoption remains higher in some states, while rainfed and poorer regions often lag behind.

Karnataka accounted for about 21.59% of total PDMC coverage, while Bihar accounted for only 0.34% and Punjab 0.18%.

- **Not Suitable for Every Situation**
  - **Crop limitation:** Every crop does not need the same type of irrigation system.
  - **Local conditions:** Soil type, land shape and water quality can affect its effectiveness.

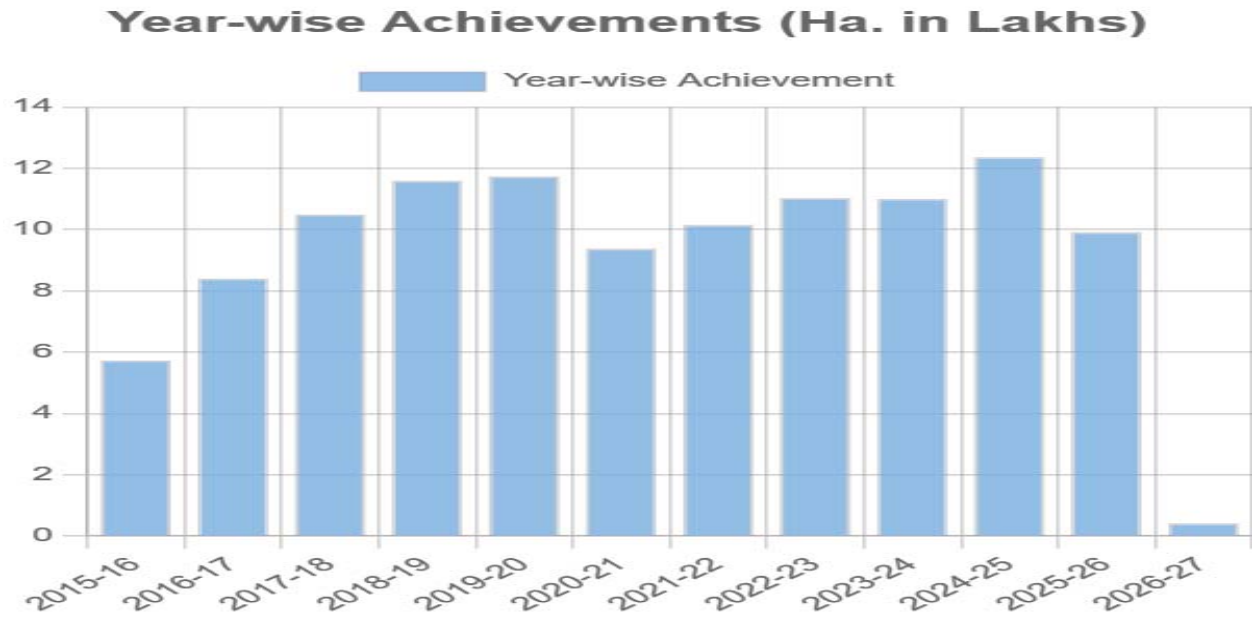
Key Government Initiatives & Policy Measures

- **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)**
  - **Per Drop More Crop (PDMC):** The core pillar focused specifically on drip and sprinkler irrigation. It offers financial assistance and subsidies to farmers to make hardware affordable.
  - **Micro Irrigation Fund (MIF):** A dedicated corpus created with NABARD (initial corpus of ₹5,000 Crore, later expanded) to help states access concessional loans to expand coverage.
- **Financial Assistance & Subsidies**
  - **Targeted Subsidies:** Direct financial support ranging from 45% to 55% of the installation cost, prioritizing small and marginal farmers who struggle with upfront costs.
  - **Direct Benefit Transfer (DBT):** Subsidies are routed directly to farmers' bank accounts via DBT portals to cut out middlemen and reduce leakages.
  - **Agricultural Infrastructure Fund (AIF):** AIF was launched with a **₹1 lakh crore corpus** to improve agricultural infrastructure. It provides **3% interest subsidy** on eligible loans for micro-irrigation projects taken up by **FPOs, cooperatives and agri-entrepreneurs**.
- **Institutional & Technological Infrastructure**
  - **Bhuvan PDMC Portal:** Uses GIS technology and satellite imagery for geotagging micro-irrigation assets, ensuring accountability and preventing phantom installations.
  - **Standardization by BIS:** Quality guidelines set by the Bureau of Indian Standards (BIS) to ensure companies supply durable pipes, pumps, and emitters.
- **Water-Energy Nexus & Integration**
  - **PM-KUSUM Scheme:** Solarization of agricultural pumps allows off-grid farmers to power drip/sprinkler systems smoothly without relying on erratic electricity.
  - **Command Area Development (CAD):** Linking canal networks directly to micro-irrigation systems (**pipe supply** instead of open flood channels).
- **Structural & Field-Level Actions**
  - **Integration with Fertigation:** Encouraging fertigation (supplying liquid fertilizers directly through drip systems), which cuts down fertilizer waste by up to 30% while saving labor.

- **Focus on Water-Stressed Areas:** Special push in rain-fed regions and dark blocks (severely depleted groundwater zones) across states like Maharashtra, Rajasthan, Gujarat, and Tamil Nadu.
- **Capacity Building & Awareness:** Krishi Vigyan Kendras (KVKs) run field demonstrations and training camps to teach farmers how to clean filters, prevent clogging, and maintain lines.
- **Har Medh Par Ped (HMPP):** The initiative encourages farmers to **plant trees along the edges of their fields**. Trees provide shade, **reduce water loss through evaporation and help retain soil moisture**.

State-Level Initiatives for Micro-Irrigation in India

- **Gujarat – Gujarat Green Revolution Company (GGRC):** Created a dedicated single-window agency to manage subsidies, vendor registration, and quality checks directly.
- **Andhra Pradesh – AP Micro Irrigation Project (APMIP):** A pioneer state-level dedicated mission exclusively managing drip and sprinkler expansion across all districts.
- **Tamil Nadu & Andhra Pradesh** offer up to 100% subsidy for Small and Marginal Farmers and around 75% for Other Farmers to ensure zero upfront cost burden.
- **Karnataka & Maharashtra:** Provide additional state top-up subsidies specifically for water-guzzling crops like sugarcane to accelerate drip adoption.
- **Maharashtra – Sugarcane Mandate:** Made drip irrigation **mandatory** for all sugarcane cultivation under canal command areas to curb groundwater depletion.
- **Karnataka – Ramthal Community Drip Irrigation Project:** Asia's largest automated community drip irrigation project in Hungund (Bagalkot district). It uses a single centralized system to deliver drip water to thousands of individual smallholdings without individual pump setups.



Way forward

- **Policy & Financial Reforms**
  - **Single-Window Portal:** Streamline state and central fund releases onto a single digital platform to avoid payment delays that scare off small farmers.

    Gujarat Green Revolution Company (GGRC) Portal: *Acts as a single-window digital hub for Gujarat farmers, unifying subsidy approvals and equipment vendor selection.*

- **Flexible Financing:** Offer low-interest, collateral-free credit through NABARD and Regional Rural Banks (RRBs) to cover the farmer's out-of-pocket share (40–50%).

  Kisan Credit Card (KCC) Collateral-Free Limit: *KCC credit limit up to ₹1.6 lakh (and up to ₹3 lakh through tie-ups with FPOs/RRBs) allows smallholders to cover upfront equipment payments without pledging land.*

- **Rationalizing Power & Water Tariffs**
  - **De-link Free Power:** Unrestricted or free electricity leads to over-pumping, defeating the purpose of drip systems. Shift toward metered power or solar-linked incentives.

    Paani Bachao, Paisa Kamao (Punjab): *A direct benefit transfer (DBT) scheme where farmers who use less electricity for pumping groundwater receive cash credits.*

  - **Incentivize Water Saving:** Reward farmers who save water under PMKSY with direct cash transfers or extra crop insurance coverage.
- **Customized & Low-Cost Solutions**
  - **Affordable Tech:** Promote low-pressure drip systems and DIY gravity-fed drip kits designed specifically for smallholders (0.5 to 1 hectare).

    IIT Madras Low-Pressure Drip Innovations: *Indigenous, affordable drip emitters engineered to work under low water pressure in off-grid rural farms.*

  - **Automation & IoT:** Encourage smart fertigation and sensor-based drip systems that deliver water based on real-time soil moisture.
- **Quality Control & Maintenance Support**
  - **After-Sales Services:** Mandate equipment vendors to provide 3–5 years of free maintenance and anti-clogging support.
  - **Water Quality Solutions:** Install basic filtration and water-softening units in hard-water belts to prevent line blockages.
- **Promoting Water User Associations (WUAs)**
  - **Community Micro-Irrigation:** Group small farms under FPOs (Farmer Producer Organizations) or WUAs to share a single high-capacity pump and filter unit, bringing down per-acre costs.
- **Cropping Pattern Realignment**
  - **Crop Diversification:** Mandate or heavily subsidize Micro Irrigation for water-guzzling crops like sugarcane and paddy, shifting away from flood irrigation in water-stressed states.

    Direct Seeded Rice (DSR) + Sprinkler Combo (Punjab): *Promoting micro-sprinklers alongside DSR to eliminate traditional field flooding in water-stressed paddy belts.*

- **Capacity Building & Extension Services**
  - **Demonstration Hubs:** Set up operational **Drip Model Farms** at the Panchayat level through Krishi Vigyan Kendras (KVKs) so farmers can see the yield gains firsthand.

    Precision Farming Development Centres (PFDCs): *22 state-level research hubs established across India to train local farmers on crop-specific micro-irrigation techniques.*

Best Micro-Irrigation Case Studies

Project	Mechanism
Ramthal Model (Karnataka)	**Asia's Largest Automated Community Drip Project:** Piped supply from canal direct to small farm gates, managed by a single automated system.
GGRC Model (Gujarat)	**Special Purpose Vehicle (SPV):** Created Gujarat Green Revolution Company as a single-window hub for direct subsidies and vendor quality control.

APMIP Model (Andhra Pradesh)	**Targeted Equity Subsidies:** Offered 90–100% subsidies exclusively for Small & Marginal Farmers (SMFs) and SC/ST farmers.
Sugarcane Mandate (Maharashtra)	**Regulatory Enforcement:** Made drip irrigation mandatory for sugarcane in canal command areas; linked crushing licenses to compliance.
Israel (Global Benchmark)	**National Closed-Loop Ecosystem:** 100% drip adoption paired with treated wastewater (shafdan) and precision fertigation.

Major Committee Recommendations on Micro-Irrigation (MI) in India

Committee / Body	Key Recommendations
Task Force on Micro-Irrigation (2004)	Recommended **3 million ha during the 10th Plan** and another **14 million ha during the 11th Plan** under micro-irrigation. Recommended stronger **subsidy, credit and institutional support** to encourage adoption, particularly among farmers facing high initial costs.
Swaminathan Commission (NCF, 2006)	Emphasised improving productivity and resilience in **rainfed and dryland farming**. Called for better management of water resources and greater emphasis on **efficient irrigation and water conservation**.
Mihir Shah Committee (2016)	Recommended restructuring the **Central Water Commission (CWC) and Central Ground Water Board (CGWB)** into a unified **National Water Commission (NWC)**. Emphasised integrated management of **surface water and groundwater** rather than treating them separately.
Ashok Dalwai Committee (DFI, 2018)	Create a Micro Irrigation Fund (MIF) under NABARD with corpus support. Emphasised greater use of **drip and sprinkler irrigation** to increase water productivity Mandate drip for water-guzzlers (paddy/sugarcane).

Conclusion

India needs to move from simply **installing more systems to ensuring that farmers can afford, use and maintain them properly**. Greater focus should be given to **small farmers, local repair services, farmer training and timely subsidies**. Micro-irrigation should also be linked with **crop diversification, rainwater harvesting and groundwater management** for true **"More Crop per Drop"**.

3. WATERSHED DEVELOPMENT

Introduction:

Watershed development means **managing rainwater, soil and land together within a natural drainage area** so that water is saved and used wisely. It helps **reduce soil erosion, recharge groundwater and improve farm productivity**, especially in rainfed areas.

Watershed Development Fund (WDF)

- **About WDF and its composition**
  - **Establishment:** Set up in **1999–2000** by the Government of India based on the recommendations of the **EAS Sarma Committee**.
  - **Nodal Managing Body:** Maintained and administered by the **National Bank for Agriculture and Rural Development (NABARD)**.
  - **Initial Corpus:** Created with an initial corpus of **₹200 crore**, contributed equally (50:50) by the **Ministry of Agriculture** and **NABARD**.
  - **Core Objective**: To promote participatory watershed development in rainfed regions by providing financial support, capacity building, and low-cost credit to rural communities.

Principles of Watershed development

- **Ridge-to-Valley Approach**
  - **Top-down treatment:** Interventions start at the highest point (ridge/hilltop) and move down to the lowest point (valley/stream).
  - **Prevents destruction:** Treating slopes first slows down run-off velocity, preventing downstream check dams in the valley from silting up or bursting.
- **In-Situ Moisture Conservation**
  - **Catch water where it falls:** Maximizing the infiltration of rainwater into the ground right where it drops using continuous contour trenches, bunding, and mulching.
  - **Reduces runoff:** Minimizes surface run-off velocity, giving soil time to absorb moisture for local crops.
- **Safe Run-Off Management**
  - **Controlled discharge:** Drainage paths are designed so excess rainwater flows out safely without stripping topsoil or carving deep gullies.
  - **Silt traps:** Structures like loose boulder check dams trap heavy sediment while letting clear water pass through.
- **Bio-Engineering**
  - **Integration of Structures:** Physical engineering structures (check dams, gabions) are always combined with biological measures (planting deep-rooted grasses, shrubs, and native trees).
  - **Root binding:** Vegetation binds the soil permanently, ensuring long-term stability after concrete structures age.
- **Participatory Approach (People's Involvement)**
  - **Bottom-up planning:** Projects succeed only when local villagers, Gram Sabhas, and Watershed Committees design and maintain them—not bureaucratic top-down orders.
  - **Local knowledge:** Blends modern hydrological science with traditional indigenous knowledge of local terrain and rainfall.
- **Equity and Inclusivity**
  - **Focus on the landless & women:** Ensures benefits aren't limited only by rich landowners along the riverbed. Landless households get access to common lands, pasture grounds, and SHG-led micro-enterprises.

Integrated Watershed Management Programme (IWMP)

- **About IWMP**
  - **Launch & Nodal Agency:** Launched in **2009–10** as a Centrally Sponsored Scheme by the **Department of Land Resources (DoLR)**, Ministry of Rural Development.
  - **Merger of Earlier Schemes:** Created by integrating three legacy area-development programs:
    - Drought Prone Areas Programme (DPAP)
    - Desert Development Programme (DDP)
    - Integrated Wastelands Development Programme (IWDP)
  - **Subsumed under PMKSY (2015):** In 2015–16, IWMP was restructured and renamed as the **Watershed Development Component of Pradhan Mantri Krishi Sinchayee Yojana (WDC-PMKSY)**. It is currently being implemented under **WDC-PMKSY 2.0 (2021–2026)**.
- **Primary Objectives**
  - **Restoration of Degraded Natural Capital:** Prevent soil erosion, arrest surface water runoff, and harvest rainwater to recharge dry aquifers.
  - **Enhancing Agricultural Productivity:** Increase double-cropping (Rabi crop coverage) and crop yields across rainfed/dryland belts.
  - **Livelihood Security:** Create sustained off-farm and non-farm employment for landless rural households, women's Self-Help Groups (SHGs), and marginal farmers.
  - **Climate Resilience:** Build village-level eco-buffers against frequent droughts, erratic rainfall, and land degradation.
- **Core Principles & Key Features**
  - **Ridge-to-Valley Approach:** Treatments strictly begin at the top of the hill slopes (ridge) to check water velocity before building water storage structures downstream in the valley beds.
  - **Institutional Architecture (4-Tier Structure):**
    - **State Level:** State Level Nodal Agency (SLNA).
    - **District Level:** Watershed Cell-cum-Data Centre (WCDC).
    - **Project Level:** Project Implementing Agency (PIA) and Watershed Development Team (WDT).
    - **Village Level:** Watershed Committee (WC) elected by the Gram Sabha.
  - **Funding & Cost Sharing:** Cost shared **60:40** between Centre and States (90:10 for North-East & Himalayan states; 100% for UTs).
  - **Participatory Planning:** Heavy emphasis on bottom-up micro-watershed plans created with local villager involvement.

Objectives of Watershed Development

- **Soil & Land Management**
  - **Prevent Soil Erosion:** Controlling topsoil loss caused by runoff through contour bunding, check dams, and terracing.
  - **Reclaim Degraded Lands:** Restoring gullied, ravinous, and degraded wastelands to make them productive again for agriculture or forestry.
- **Water Harvesting & Recharge**
  - **In-situ Moisture Conservation:** Retaining rainwater right where it falls to increase soil moisture for rainfed crops.
  - **Groundwater Recharging:** Capturing surface runoff to replenish dry wells and underground aquifers, raising local water tables.

- **Agricultural Productivity & Risk Reduction**
  - **Droughtproofing Drylands:** Reducing crop vulnerability in rainfed areas (which make up ~50% of India's net sown area) during dry spells.
  - **Cropping Intensity:** Enabling farmers to grow a second crop (Rabi) due to improved winter moisture and water availability.
- **Livelihood Diversification & Poverty Alleviation**
  - **Promoting Allied Activities:** Creating sustainable water sources for dairy, animal husbandry, agro-forestry, and inland fisheries.
  - **Generating Local Employment:** Providing direct employment through watershed civil works (check dams, trench digging) under PMKSY-Watershed Development Component and MGNREGA.
- **Community Ownership & Empowerment**
  - **Participatory Governance:** Involving local communities through Watershed Committees, Self-Help Groups (SHGs), and User Groups (UGs).
  - **Social Equity:** Ensuring marginalized groups, landless laborers, and women get access to common property resources (like village ponds and pasture lands).
- **Ecological Balance & Climate Resilience**
  - **Afforestation & Biomass Building:** Planting native trees and grasses along slopes to build natural carbon sinks and improve eco-restoration.
  - **Climate Adaptation:** Building natural village-level buffers against climate shocks like erratic monsoons and localized flash droughts.

Status of watershed in India

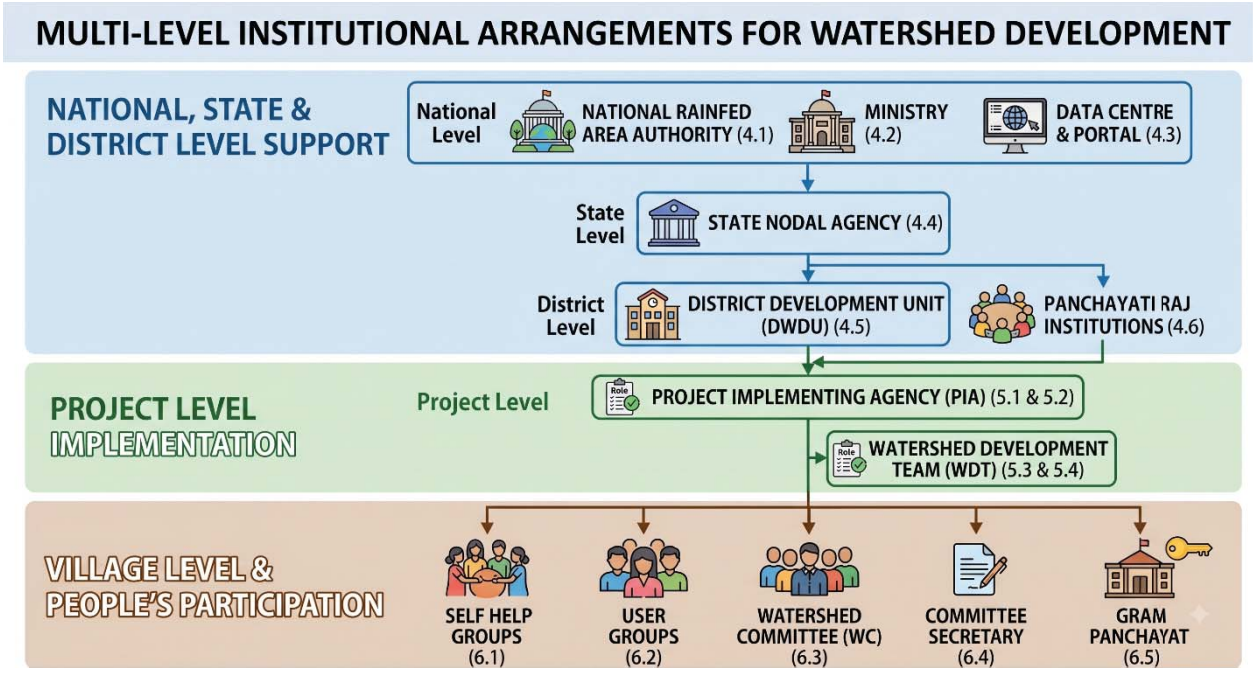
- **Macro Status & Governance Framework**
  - **WDC-PMKSY 2.0 (2021–2026):** Watershed development is primarily executed through the **Watershed Development Component of Pradhan Mantri Krishi Sinchayee Yojana (WDC-PMKSY 2.0)** managed by the Department of Land Resources (DoLR).
  - **Target & Outlay:** Targets covering **49.5 lakh hectares** of degraded rainfed land with a financial outlay of **₹8,134 Crore**.
  - **Upward Revision of Unit Costs**: Per-hectare cost allocations were increased from ₹12,000/ha to **₹22,000/ha for plain areas** and up to **₹28,000/ha for hilly, difficult, and LWE regions** to meet rising material costs.
  - **Shorter Project Cycles**: Reduced implementation timelines from the earlier 4–7 years down to **3–5 years** to speed up completion and avoid cost overruns.
  - **Classification of area into smaller units based on area:**
    - Sub-watershed: 10,000 to 50,000 hectares
    - Milli-watershed: 1,000 to 10,000 hectares
    - Micro-watershed: 100 to 1,000 hectares
    - Mini-watershed: 1 to 100 hectares
- **Major Quantitative Achievements**
  - **Physical Coverage & Land Restoration**: Tens of thousands of projects have restored over **5+ lakh hectares of degraded land** and rainfed areas through active soil-moisture conservation work.
  - **Water Structure Creation**: Over **3 lakh water harvesting structures** (check dams, farm ponds, percolation tanks) have been newly created or rejuvenated, expanding protective irrigation to over 3+ lakh hectares.
  - **Socio-Economic Outcomes**: Generated over **2.8 crore local employment**. Plan directly benefited over **11+ lakh farmers**, increasing cropping intensity and dryland incomes.

Issues with Watershed Development

- **Poor Community Participation**
  - **Limited involvement:** Projects often focus more on construction than on **involving local people in planning and decision-making**.

    *Government launched the **Watershed Yatra in 26 States and 2 UTs** across **2,045 locations** in 2025 to increase **Jan Bhagidari** in watershed projects.*
  - **Weak ownership:** When people do not feel ownership, maintaining the structures becomes difficult.
- **Water Conflicts**
  - **Upstream–downstream conflict:** Water conservation in one area can sometimes affect **water availability in another area**.
- **Poor Maintenance**
  - **Neglected structures:** Check dams, farm ponds and trenches may become **damaged or filled with silt**.
  - **No regular upkeep:** Lack of funds and local responsibility reduces their long-term usefulness.
- **Groundwater Paradox**
  - **More water, more extraction:** Improved groundwater recharge can encourage farmers to **pump more water**.
  - **Water-intensive crops:** Farmers may shift towards crops like **paddy and sugarcane**, increasing water demand again.
- **Small and Fragmented Landholdings**
  - **Difficult implementation:** Small and scattered farms make it harder to plan and implement **large-scale soil and water conservation measures**.

    According to the Ministry of Agriculture's 10th Agriculture Census, 86.2% of all operational holdings in India are small and marginal (below 2 hectares).
  - **Unequal capacity:** Small farmers may not have enough resources to invest in complementary activities.
- **Weak Local Institutions**



- **Inactive committees:** Watershed committees or user groups may exist on paper but remain **weak in practice**.
- **Limited capacity:** Local institutions often lack the **skills, funds and technical support** needed for management.

*According to the **NSSO 77th Round Survey**, less than **10% of rural local bodies/committees** received formal technical training for hydrological data monitoring*

- **Dependence on Rainfall**
  - **Limited impact in poor rainfall years:** Watershed structures depend on receiving enough rainfall to **capture and store water**.

According to the National Rainfed Area Authority (NRAA), nearly 51% of India's net cultivated area is rainfed.

  - **Climate uncertainty:** Erratic rainfall and extreme rainfall events can reduce the effectiveness of traditional designs.

Measures taken by the Indian government for watershed development

- **Core Flagship Programs**
  - **WDC-PMKSY (Watershed Development Component):** It focuses on restoring degraded land, building check dams, and capturing every drop of rainwater where it falls.
  - **Mission Amrit Sarovar:** A major nationwide drive to construct or rejuvenate at least 75 local water bodies (ponds) in every district across India to boost groundwater recharge.
  - **MGNREGA Integration:** A huge chunk of rural employment under MGNREGA is dedicated specifically to water conservation, such as digging trenches and clearing traditional silted tanks.
- **Springshed Development:** In hilly regions like the Himalayas and the North-East, special programs focus on restoring natural mountain springs (*dharas*) which serve as the primary drinking water source for local villages.
- **Handing Control to Local Communities**
  - **Watershed Committees:** local Gram Panchayats and village Watershed Committees are given the power and funds to plan, construct, and maintain their own water structures.
  - **Atal Bhujal Yojana (Atal Jal):** A unique scheme where villagers are taught how to measure their own groundwater levels and create a **Village Water Budge**t.
- **Using Modern Technology**
  - **Bhuvan-Srishti Portal (ISRO):** The government uses satellite imagery and space technology through ISRO to monitor watershed projects in real-time.
  - **GIS Mapping:** Maps created with Geographic Information Systems (GIS) help planners identify the exact spots where building a check dam or farm pond will harvest the most rainwater.
- **Blending Forestry with Farming**
  - **National Agroforestry Policy:** Encourages farmers to plant fruit and timber trees along field borders alongside their usual crops.

Way Forward for Watershed Development

- **Put People at the Centre**
  - **Community-led planning:** Involve local people in **planning, implementation and monitoring**.

***Hiware Bazar Model (Maharashtra)**, where the Gram Sabha collectively plans, executes, and audits all watershed structures without relying on outside contractors.*

- **Local ownership:** Give communities responsibility for **maintaining ponds, check dams and other structures**.

- **Ridge to Valley Approach**
  - **Move towards valleys:** Treat upper slopes first to **slow down runoff and soil erosion**. Then develop **farm ponds, check dams and recharge structures** downstream.

    DHARA Vikas Program (Sikkim) uses ridge-to-valley mapping to recharge dried-up mountain springs (dharas) from hilltops down to valley villages.

- **Focus on "Catch the Rain, Where It Falls"**
  - **Groundwater recharge:** Build small structures to **capture rainwater near its source**, which increase the amount of water that **seep into the ground**.

    Catch the Rain Campaign, which focuses on building low-cost, decentralized structures like check dams, rooftop rainwater harvesting units

- **Link Water Conservation with Crop Planning**
  - **Avoid water-intensive crops:** Saving water should not lead to increased cultivation of **paddy and sugarcane** in water-stressed areas. Promote crops according to **local water availability**.

- **Strengthen Local Institutions**
  - **Ensure transparency:** Use social audits and community monitoring to ensure **proper use of funds and assets**.

    MGNREGA Social Audits, where local village assemblies publicly review all expenditure, labor registers, and physical assets built under watershed works.

- **Ensure Fair Sharing of Benefits**
  - **Include vulnerable groups:** Ensure benefits reach **small farmers, women, landless households and tribal communities**.

    Pani Panchayat Model (Maharashtra), which decouples water rights from land ownership, granting landless households and women equal water shares to lease land for farming.

  - **Fair water sharing:** Develop clear rules for using **common water resources**.

- **Use Science and Technology**
  - **GIS and satellite mapping:** Identify suitable locations for **water harvesting and soil conservation**.
  - **Data-based planning:** Use rainfall, groundwater and soil data to design **location-specific solutions**.

- **Build Climate Resilience**
  - **Prepare for extremes:** Design structures to handle **both heavy rainfall and long dry periods**.

    Jalyukt Shibar Abhiyan 2.0 (Maharashtra), which builds deepened, flood-resilient stream channels and desilted percolation tanks to handle unseasonal cloudbursts and droughts.

  - **Increase green cover:** Promote **afforestation, agroforestry and pasture development** to improve soil and water retention.

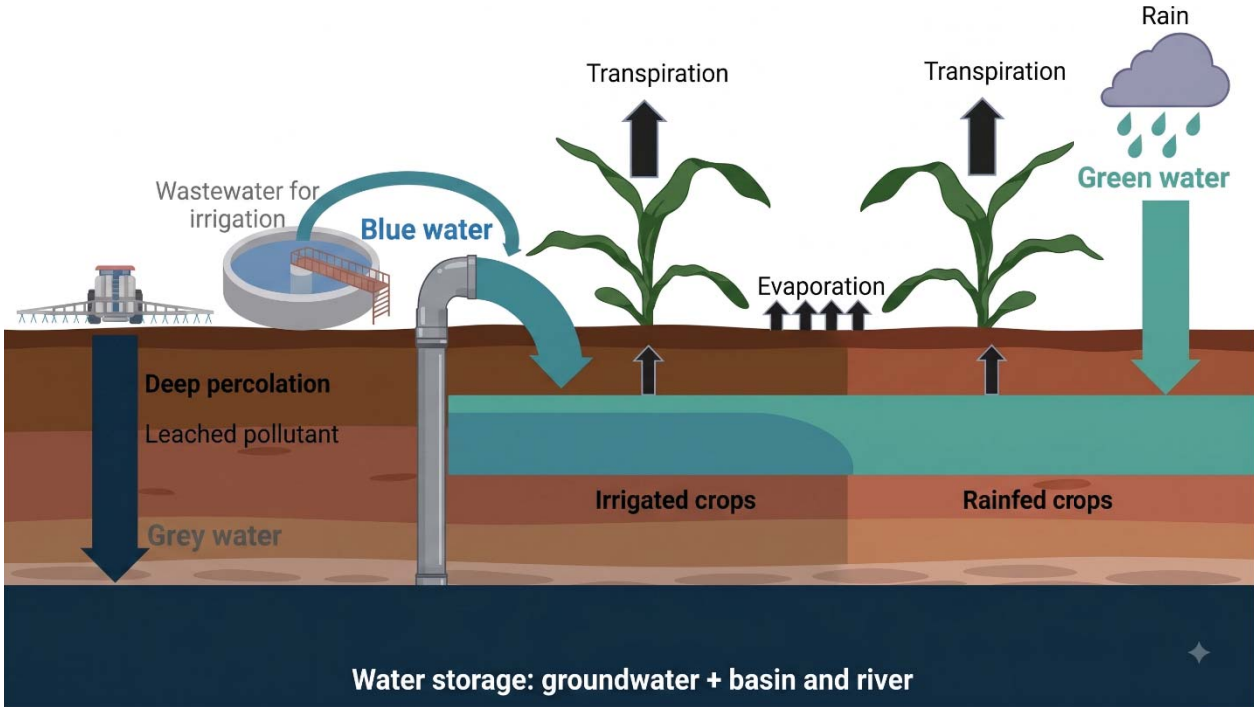
Conclusion

Watershed development is not just about building ponds and check dams; it is about **using water, soil and land wisely**. With **community participation, better planning and proper maintenance**, it can improve farming and rural livelihoods. Ultimately, it can help India move towards **water security, climate-resilient agriculture and sustainable rural development**.

4. WATER USE EFFICIENCY IN AGRICULTURE

Introduction

Water Use Efficiency means **getting more crop output from every drop of water used in farming**. It is important for India because agriculture uses a large share of available water, while **groundwater is under growing pressure**. Improving water efficiency through **micro-irrigation, rainwater harvesting and better crop choices** can make Indian agriculture more productive and sustainable.



Significance of Water Use Efficiency in Agriculture

- **Saves Water**
  - **Less wastage:** Helps reduce water lost through evaporation, runoff and over-irrigation.
  - **More crop per drop:** The same amount of water can produce more agricultural output.
- **Protects Groundwater**
  - **Less extraction:** Efficient irrigation reduces the need to pump large amounts of groundwater.
  - **Long-term security:** Helps slow down groundwater depletion in water-stressed regions.
- **Improves Farm Productivity**
  - **Right water, right time:** Crops receive water according to their actual needs.
  - **Better yield:** Proper moisture management can improve crop growth, productivity and quality.
- **Reduces Farming Costs**
  - **Saves energy:** Less water pumping means lower electricity and diesel costs.
  - **Saves inputs:** Better irrigation can also reduce fertiliser and labour costs.
- **Supports Farmers' Income**
  - **Higher returns:** Better productivity with lower input costs can improve farm profitability.
  - **More cropping opportunities:** Water saved can support additional crops, wherever conditions permit.
- **Promotes Crop Diversification**
  - **Right crop, right place:** Encourages farmers to choose crops according to local water availability.

  - **Reduces water pressure**: Can reduce excessive dependence on water-intensive crops in stressed regions.
- **Protects Soil**
  - **Less waterlogging:** Avoids unnecessary irrigation and keeps soil conditions healthier.
  - **Less erosion and nutrient loss:** Controlled watering reduces soil erosion and leaching.
- **Strengthens Food Security**
  - **More output from limited water:** Helps increase agricultural production without continuously increasing water use.
  - **Future-ready agriculture:** Important for meeting rising food demand under water and climate stress.

Bureau of Water Use Efficiency (BWUE)

- **About and composition**
  - **Nodal Wing & Ministry:** Functions as a specialized arm under the **National Water Mission (NWM)**, Department of Water Resources, River Development & Ganga Rejuvenation, **Ministry of Jal Shakti**.
  - **Setup Purpose:** Established to address structural water stress by improving water productivity across agriculture, industry, and domestic sectors.
  - **Core Mandate:** Serves as an overarching facilitator, policy planner, and regulator to achieve NWM's target of **increasing Water Use Efficiency (WUE) by 20%** across sectors.
- **Key Functions & Scope of Operations**
  - **Sectoral Coverage:**
    - **Agriculture:** Drives micro-irrigation expansion, pressurized pipe schemes (e.g., Modernized Command Area Development / MCAD), and precision irrigation.
    - **Industry:** Promotes water audits, Zero Liquid Discharge (ZLD) technologies, circular water economies, and wastewater recycling.
    - **Domestic:** Partners with bodies like the Indian Plumbing Association (IPA) to mandate low-flow fixtures, rainwater harvesting, and greywater reuse.
  - **Standardization & Water Footprinting:** Works toward marking water footprints on industrial products in coordination with the Bureau of Indian Standards (BIS).
  - **Data & Digital Monitoring:** Encourages real-time water usage data collection, digital SCADA monitoring, and automated irrigation management systems.
  - **Inter-State Knowledge Sharing:** Organizes regional conferences and platforms to replicate successful state models (e.g., SCADA in Karnataka, *Sahi Fasal* & *Mera Pani Meri Virasat* in Haryana).

Measures Taken for Water Use Efficiency (WUE)

- **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)**
  - **Per Drop More Crop (PDMC):** Promotes micro-irrigation systems (drip and sprinkler) to ensure water reaches crop roots directly without loss through evaporation or seepage.
  - **Command Area Development & Water Management (CADWM):** Focuses on completing "last-mile connectivity" by building field channels to carry canal water straight to farm gates.
- **National Water Mission (NWM):** Sets a specific target to increase water use efficiency by 20% across sectors, with a major focus on agriculture (which consumes over 80% of India's fresh water).
- **Sahi Fasal Campaign:** Launched by the National Water Mission to nudge farmers in water-stressed areas away from water-guzzling crops (like paddy and sugarcane) toward crops like millets, pulses, and oilseeds.
- **Micro-Irrigation & Fertigation**
  - **Drip & Sprinkler Systems:** Delivers water directly to the root zone, boosting water efficiency up to 80–90% compared to traditional flood irrigation (which has ~30–40% efficiency).

  - **Fertigation:** Mixing liquid fertilizers into drip lines so plants absorb nutrients directly with water, reducing both fertilizer and water waste.
- **Resource-Conserving Farming Techniques**
  - **System of Rice Intensification (SRI):** Reduces water requirement in paddy cultivation by 30–50% by keeping fields moist rather than continuously flooded.
  - **Direct Seeded Rice (DSR):** Bypasses nursery transplanting and continuous field flooding, saving significant irrigation water and labor.
  - **Alternate Wetting and Drying (AWD):** Periodic drying of paddy fields instead of keeping them submerged, lowering water footprint without hitting crop yields.
- **Precision Farming & Digital Tech**
  - **Laser Land Levelling:** Evens out field surfaces using laser-guided equipment, ensuring uniform water spread and preventing waterlogging or wasteful deep percolation.
  - **Soil Health Cards** & **Soil Moisture Sensors:** Helps farmers time their irrigation based on real-time soil moisture levels rather than routine flooding schedules.
- **Community-Led Water Management**
  - **Participatory Irrigation Management (PIM):** Handing over operational maintenance of local canal distributaries to Water User Associations (WUAs) to curb transit losses and leakage.
  - **Atal Bhujal Yojana (Atal Jal):** Community-led groundwater management scheme active across water-stressed districts, training villagers to create crop-water budgets.
- **Power & Electricity Rationalization**
  - **Direct Benefit Transfer for Electricity (DBTE):** Schemes like Punjab's *Paani Bachao, Paisa Kamao* credit cash directly into farmers' bank accounts for every unit of power/water they save.
  - **PM-KUSUM Scheme**: Solarization of agricultural pumps paired with micro-irrigation so farmers feed surplus solar power back into the grid for income.

Issues in Water Use Efficiency in Agriculture

- **Water-Wasting Irrigation Methods**
  - **Flood irrigation:** Farmers often flood the entire field, causing large losses through **evaporation, runoff and seepage**.

    *According to the **Ministry of Jal Shakti**, the overall irrigation efficiency of major and medium irrigation projects is estimated at only **around 38%**.*
  - **Low efficiency:** A lot of water is used on a single crop without necessarily increasing crop output.
- **Overdependence on Groundwater**
  - **Easy access:** Cheap or subsidised electricity encourages farmers to **pump more groundwater**.

    *According to the **CGWB National Compilation on Dynamic Ground Water Resources, 2024**, groundwater contributes nearly **62% of India's irrigation needs**.*
  - **Falling water tables:** Excessive pumping is causing serious **groundwater depletion** in many regions.

    *As per CGWB, Out of 7,089 administrative assessment units, over **14% are classified as 'Over-Exploited'**, heavily concentrated in states with high power subsidies like Punjab, Haryana, and Rajasthan.*
- **Water-Intensive Crops**
  - **Wrong crop choice:** Crops such as **paddy and sugarcane** are often grown in areas where water is already scarce.

    Research by ICRIER highlights that Paddy and Sugarcane together consume more than 60% of total irrigation water in India.

  - **Policy influence:** MSP, procurement and electricity subsidies can encourage farmers to continue such cultivation.
- **Low Adoption of Micro-Irrigation**
  - **High initial cost:** Drip and sprinkler systems can be expensive for **small and marginal farmers**.

    *A **NABARD Research Study** indicates that the average capital setup cost for drip systems ranges from **₹40,000 to ₹1,00,000 per hectare.***
  - **Maintenance issues:** Clogging, repairs and lack of technical support can discourage adoption.
- **Poor Water Management**
  - **Uneven distribution:** Some farms receive excess water while others face **water shortages** throughout the year.
  - **Weak planning:** Irrigation is often planned around **water supply rather than actual crop needs**.
- **Lack of Awareness**
  - **Limited knowledge:** Farmers may not know about **efficient irrigation methods, soil moisture management or suitable crop choices**.
- **Fragmented Landholdings**
  - **Small farms:** Small and scattered plots make it difficult to install and manage **modern irrigation systems**.

    *According to the **Agriculture Census 2015–16**, small and marginal operational holders constituted about **86% of total operational holders** in India.*

Way Forward for Water Use Efficiency in Agriculture

- **Adopt "More Crop per Drop"**
  - **Efficient irrigation:** Promote drip and sprinkler systems so that water reaches the crop **with minimum wastage**.
  - **Better practices:** Encourage mulching, timely irrigation and soil-moisture management.
- **Right Crop in the Right Place**
  - **Crop planning:** Choose crops according to the **local soil, climate and water availability**.

    ***Sahi Fasal Campaign** by the National Water Mission nudges farmers away from water-guzzling crops toward climate-resilient options based on local agro-climatic zones.*
  - **Diversification:** Encourage millets, pulses, oilseeds and suitable horticultural crops in **water-stressed regions**.
- **Make Micro-Irrigation Affordable**
  - **Timely subsidy:** Ensure that subsidies for micro-irrigation project should reach farmers **quickly and easily**.

    ***Gujarat Green Revolution Company (GGRC)** acts as a single-window digital clearinghouse for micro-irrigation subsidies, preventing payment delays to farmers.*
  - **Collective adoption:** FPOs and cooperatives can help small farmers **share the cost and infrastructure**.
- **Save Water in Canals**
  - **Repair infrastructure:** Repair of old infrastructure reduces water losses caused by **leakage and seepage** in old canals.

    ***Command Area Development & Water Management (CADWM)** program focuses on lining canals and constructing last-mile field channels to check seepage losses.*
  - **Better distribution:** Ensure that water reaches **tail-end farmers** fairly and efficiently.
- **Harvest Every Drop of Rain**

- **Local storage:** Promote **farm ponds, check dams and recharge structures**.

  Jalyukt Shibar Abhiyan (Maharashtra) and Mission Amrit Sarovar focus on constructing farm ponds, check dams, and desilting water bodies to trap surface runoff.

- **Recharge groundwater:** Allow rainwater to **enter the ground** instead of simply flowing away.

- **Use Technology Wisely**
  - **Smart irrigation:** Soil-moisture sensors and weather advisories can tell farmers **when and how much to irrigate**.

    *AgTech platforms like **Fasal** and **Cropin** deploy IoT soil moisture sensors and hyper-local weather advisory apps to tell farmers precisely when and how much to irrigate.*

- **Reduce Unnecessary Groundwater Use**
  - **Better electricity policy:** Move towards incentives that reward **saving electricity and water**, rather than excessive pumping.
  - **Community management:** Local communities can monitor and manage **groundwater extraction**.

    ***Andhra Pradesh Farmer Managed Groundwater Systems (APFAMGS)** trains local village committees to perform crop-water budgeting before every sowing season.*

- **Build Farmer Awareness**
  - **Demonstration farms:** Seeing the benefits in nearby fields can encourage farmers to **adopt new practices**.

    ***Krishi Vigyan Kendras (KVKs)** set up local Drip Model Farms and conduct field days (Krishi Mela) at the Panchayat level to demonstrate actual yield gains to farmers.*

Best International Case Studies on WUE

Region / Project	Strategy / Mechanism
Israel (Global Gold Standard)	**National Closed-Loop Water Framework:** 100% drip adoption paired with treated municipal wastewater (Shafdan) and precision fertigation.
Murray-Darling Basin (Australia)	**Piped Distribution & Water Markets:** Replaced open leaky canals with pressurized pipes + established **tradable water entitlement rights**.
Xinjiang Drip-Film Model (China)	**Plastic Film Mulching + Drip:** Combines subsurface drip lines beneath transparent plastic film mulching in dryland cotton fields.
California Central Valley (USA)	**Automated Canal SCADA Systems:** Automated canal gates and sensor networks adjust surface water distribution based on real-time soil demand.
Highland Water Policy (Jordan)	**Groundwater Caps & Crop Realignment:** Strict volumetric metering on tubewells + heavy penalties for water-guzzlers in desert aquifers.

Conclusion

Sustainable irrigation is important for **food security, healthy ecosystems and better rural livelihoods**. As PM Modi says, **"Per drop, more crop"** should not remain just a slogan but become a commitment for future generations. With growing water scarcity and changing climate, India needs to use every drop wisely and move towards **inclusive and sustainable agricultural growth**.

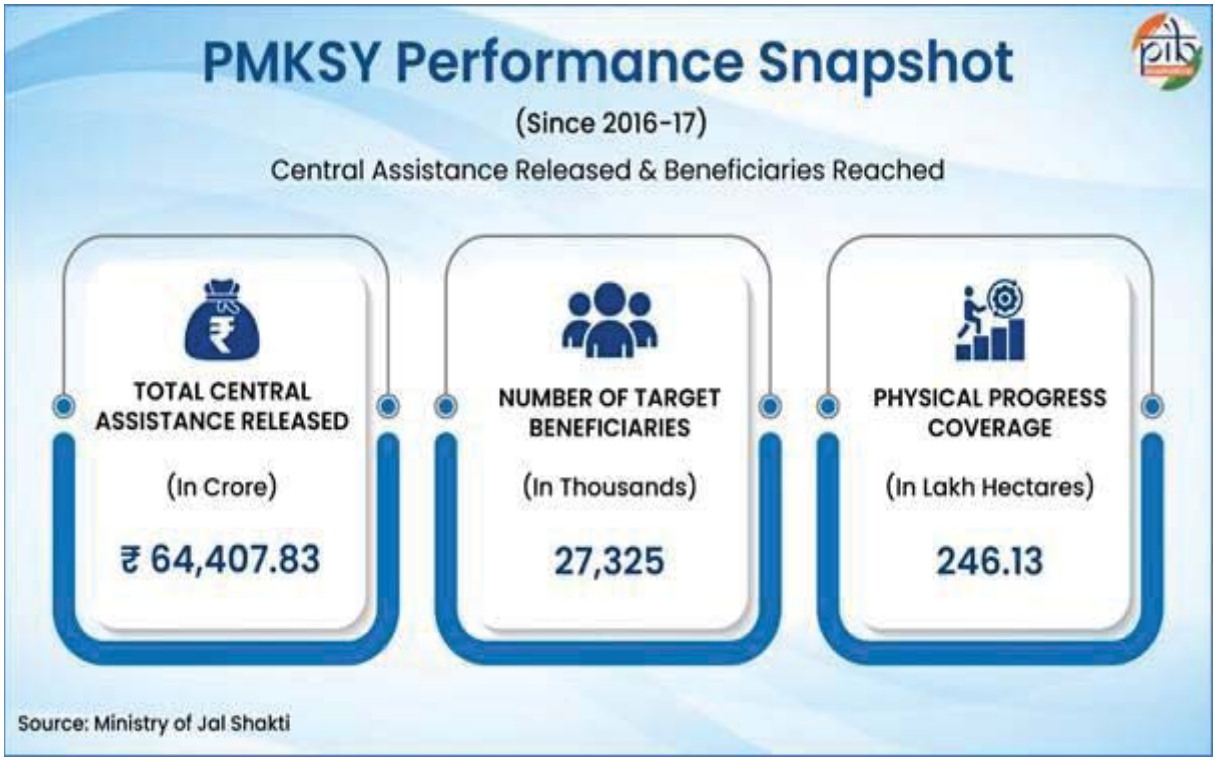
5. PRADHAN MANTRI KRISHI SINCHAI YOJANA

Introduction

Pradhan Mantri Krishi Sinchai Yojana (PMKSY) is a flagship government scheme launched in 2015 to improve irrigation and ensure that every farm gets access to water. It promotes the efficient use of water through methods like drip and sprinkler irrigation while encouraging better water conservation. The scheme aims to increase farm productivity, reduce dependence on rainfall, and improve farmers' incomes.

Objectives of PMKSY

- **Water for every field (*Har Khet Ko Pani*):** Expand irrigation facilities so that farms across the country, especially in dry areas, have a reliable supply of water.
- **More crop per drop:** Promote smart watering techniques like drip and sprinkler systems, helping farmers save water, reduce costs and boost crop yields.
- **Smart water management:** Focus on harvesting rainwater, recharging groundwater and preventing soil erosion to keep farm ecosystems healthy for the long run.
- **Stop water waste:** Upgrade delivery channels and canals so water doesn't leak away before reaching the fields.
- **Attract investment:** Encourage both central and state governments, along with private partners, to invest heavily in modernizing farming water systems.



Components of PMKSY

- **Accelerated Irrigation Benefit Programme (AIBP):** Speeds up the completion of long-pending major and medium irrigation projects so that canal water reaches farmers' fields without further delay.

  Since 2016–17, it has received ₹21,023 crore in central assistance, benefiting 17.30 million farmers. It makes it the largest PMKSY component by beneficiary coverage.

- **Har Khet Ko Pani (HKKP):** Works to ensure that every farm gets irrigation by creating small irrigation systems, reviving village ponds, and making better use of available groundwater.

  Since inception, 3,462 Surface Minor Irrigation and Repair, Renovation & Restoration schemes have been completed. It has also created 5.93 lakh hectares of irrigation potential.
- **Per Drop More Crop (PDMC):** Promotes the efficient use of water by encouraging drip and sprinkler irrigation and training farmers to use water wisely.
- **Watershed Development:** Improves soil and water conservation by harvesting rainwater, building check dams, and reducing soil erosion, especially in rain-fed areas.

  Since 2016–17, it has received ₹12,432.09 crore in central assistance, benefiting 1.34 million farmers and supporting sustainable natural resource management.

Micro Irrigation Fund

Micro Irrigation Fund (MIF) was launched in **2018–19** to help States expand the use of **drip and sprinkler irrigation** across the country. Managed by the **National Bank for Agriculture and Rural Development (NABARD)** with an initial corpus of **₹5,000 crore**, later increased to **₹10,000 crore**, the fund provides concessional loans to State Governments for micro-irrigation projects. It supports the **'Per Drop More Crop'** component of PMKSY by promoting efficient use of water, reducing irrigation costs, improving crop productivity, and encouraging sustainable farming.

Challenges with PMKSY

- **Implementation and Bureaucratic Bottlenecks**
  - **Delays in irrigation projects:** Many large canal and dam projects under AIBP are delayed because of land acquisition problems, environmental approvals and rising construction costs.

    According to the CAG performance audit report on the AIBP, out of 118 sampled major projects surveyed, 105 projects suffered time overruns ranging up to 18 years.
  - **Poor coordination between departments:** PMKSY brings together several ministries but weak coordination and slow decision-making often delay work at the ground level.
- **Financial challenges**
  - **High initial cost of micro-irrigation:** Installing drip and sprinkler systems requires upfront investment, which many small and marginal farmers find difficult to afford.

    According to NABARD & ICAR studies, installing drip irrigation systems costs between ₹45,000 to ₹1,000,000 per hectare.
  - **Delay in subsidy payments:** Farmers often have to wait months to receive subsidy amounts, reducing their interest in adopting modern irrigation systems.
- **Regional challenges**
  - **Uneven implementation across states:** States with better infrastructure have benefited more from the scheme, while many less-developed regions continue to lag behind.
  - **Unreliable power supply:** Drip and sprinkler systems depend on pumps, but frequent power cuts in rural areas often disrupt irrigation.

    As per Central Electricity Authority (CEA), rural agricultural feeders average only 6 to 8 hours of daily power supply.
- **Environmental Challenges**
  - **Maintenance problems:** Drip and sprinkler systems often get clogged due to silt or hard water, and many farmers do not have easy access to repair and technical support.
  - **Groundwater overuse:** In some areas, efficient irrigation has encouraged the cultivation of water-intensive crops such as sugarcane, increasing pressure on depleting groundwater resources.

According to the World Bank, India extracts over 250 billion cubic meters (bcm) of groundwater annually—accounting for more than 25% of the total global groundwater extraction.

How Sprinkler Irrigation Transformed a Farming Enterprise?

Brothers **Uma Shankar Verma** and **Subhash Verma** from **Barabanki, Uttar Pradesh**, had been farming since the early 1990s. They grew crops such as rice, wheat, mustard, banana, and mentha using traditional flood irrigation, which consumed large amounts of water and labour and often led to uneven crop growth.

In **2016**, they adopted **sprinkler irrigation** under the **Pradhan Mantri Krishi Sinchai Yojana (PMKSY)**. The new system used water more efficiently and improved crop growth, increasing yields by **25–30%**. It also reduced the use of **water, electricity, labour, and fertilizers by nearly 50%**, bringing down cultivation costs significantly.

As a result, the farmers earned profits of around **₹60,000–70,000 per hectare**. Seeing these benefits, many neighbouring farmers also started adopting sprinkler irrigation. Their experience clearly shows how PMKSY's **Per Drop More Crop** approach helps save water, improve farm productivity, and increase farmers' incomes.

Way forward

- **Infrastructure and Financial Support**
  - **Speed up project approvals:** Introduce a single-window system for land acquisition and environmental clearances to complete irrigation projects on time.
  - **Ensure timely subsidy payments:** Transfer micro-irrigation subsidies directly to farmers' bank accounts without delays, making modern irrigation systems more affordable.

    Direct Benefit Transfer (DBT) under PMKSY (Per Drop More Crop) – Sends micro-irrigation subsidies directly to farmers' bank accounts to prevent payment delays.
- **Strengthening Regional and Ground-Level Support**
  - **Focus on lagging regions:** Provide dedicated funds and technical support to eastern and rain-fed states so they can improve irrigation facilities and catch up with better-performing states.
  - **Expand village-level technical support:** Train local youth as irrigation technicians to help farmers maintain drip and sprinkler systems and provide quick repair services when needed.

    Jal Doot / Krishi Sakhi Cadre – Trains local rural youth and women as technicians to install, repair, and maintain micro-irrigation hardware in villages.
- **Promoting Smart Water and Energy Solutions**
  - **Expand the use of solar-powered pumps:** Combine micro-irrigation with solar-powered pumps to reduce dependence on unreliable electricity and lower irrigation costs for farmers.

    Off-Grid Solar Irrigation (Kenya & Bangladesh) – Pairs pay-as-you-go solar pumps with drip lines to reduce diesel reliance on small farms.
  - **Promote crop diversification:** Encourage farmers to grow less water-intensive crops, such as pulses and millets, along with drip irrigation to conserve groundwater.

    Mera Pani Meri Virasat (Haryana) – Gives financial cash incentives to farmers who switch from water-heavy paddy to pulses, maize, or millets using drip systems.
- **Community Participation & Governance**
  - **Empower Water User Associations:** Involve local farmer groups in the operation and maintenance of canals and ponds to improve upkeep and ensure better use of irrigation assets.
  - **Use digital monitoring:** Adopt satellite imagery and mobile applications to track project progress in real time, improve transparency and ensure irrigation projects are completed effectively.

Bhuvan-PMKSY Satellite Tracking – Uses ISRO satellite imagery and mobile geo-tagging apps to track canal construction in real time.

Do you know?

- As of August 2025, PMKSY benefited over 27 million farmers through irrigation, micro-irrigation, watershed, and groundwater interventions, supported by ₹64,407 crore. Collectively, these initiatives have covered or developed more than 24.61 million hectares, strengthening water security and promoting sustainable agricultural growth.
- Coverage of irrigation area has increased between FY16 and FY21 from 49.3 per cent to 55 per cent of the gross cropped area (GCA), while irrigation intensity has risen from 144.2 per cent to 154.5 per cent.

Conclusion

Pradhan Mantri Krishi Sinchai Yojana (PMKSY) has played an important role in expanding irrigation and promoting efficient water use in agriculture. With sustainable water management and modern irrigation practices, PMKSY can strengthen India's agricultural productivity and improve farmers' livelihoods.

6. JAL SHAKTI ABHIYAN

Introduction

Jal Shakti Abhiyan is a nationwide water conservation campaign launched by the Government of India to tackle water scarcity through community participation and better water management. It focuses on rainwater harvesting, groundwater recharge, and restoration of traditional water bodies.

The Jal Shakti Abhiyan was initiated in 2019 to address the acute water stress in various parts of the country. The campaign was divided into two phases: the first phase was during the monsoon season from **July 1 to September 15, 2019**, and the second phase from **October 1 to November 30, 2019**, for states receiving the Northeast retreating monsoons.

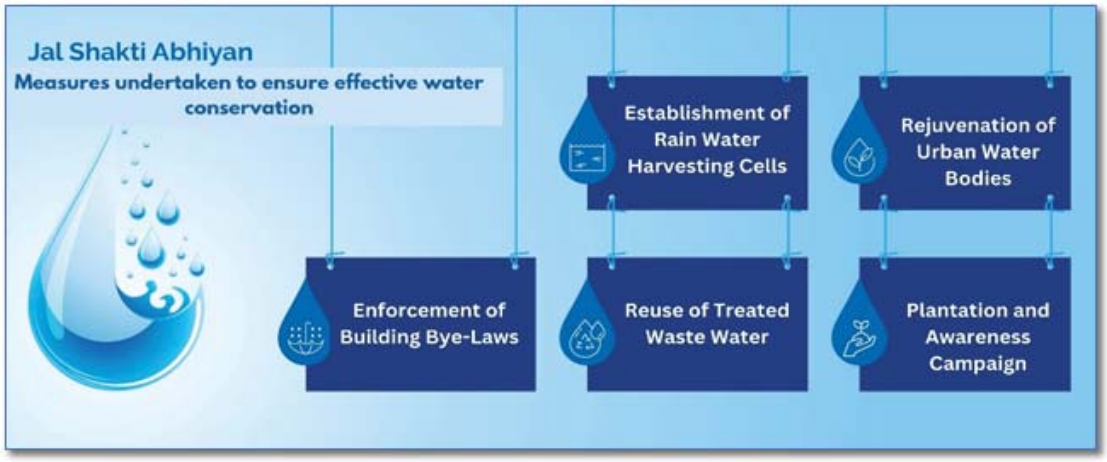
Objectives of Jal Shakti Abhiyan

- **Conserve water:** Encouraging the collection and storage of rainwater to increase water availability.
- **Recharge groundwater:** Increase groundwater levels through rainwater harvesting and recharge structures.
- **Revive water bodies:** Restore ponds, lakes, tanks, and other traditional water sources.
- **Increase people's participation:** Encourage communities to actively take part in water conservation efforts.
- **Ensure water security:** Improve the availability of water for drinking, farming, and daily needs.
- **Improve watershed management:** Treat catchment areas to reduce runoff and enhance water retention.
- **Promote sustainable water use:** Encourage long-term and climate-resilient management of water resources.
- **Converge government schemes:** Bring together different ministries and programmes for better water conservation outcomes.
- **Create awareness:** Build public awareness about the importance of saving every drop of water.

Features of Jal Shakti Abhiyan

- **Campaign-based approach:** A nationwide mission that encourages mass participation in water conservation.
- **Focus on water-stressed areas:** Gives priority to districts facing severe water scarcity.

- **Rainwater harvesting:** Promotes the construction of structures to capture and store rainwater.
- **Groundwater recharge:** Creates recharge pits, check dams, and other structures to replenish groundwater.
- **Revival of water bodies:** Restores ponds, lakes, tanks, wetlands, and traditional water sources.
- **Watershed development:** Treats catchment areas to improve soil moisture and reduce surface runoff.
- **Afforestation:** Encourages tree plantation to improve groundwater recharge and protect watersheds.
- **Convergence of schemes:** Brings together programmes of different ministries, states, and local bodies for better outcomes.
- **Community participation:** Involves Panchayats, local communities, NGOs, and volunteers, making water conservation a **Jan Andolan**.
- **Use of technology:** Uses GIS mapping, remote sensing, and online monitoring to track progress and improve planning.



Significance of Jal Shakti abhiyan

- **Community-Led Water Management**
  - **People's Movement (Jan Andolan):** campaign encourages local communities, Panchayats and youth to take responsibility for conserving and managing their own water resources.
  - **Focus on creating water assets:** Gives priority to on-ground work such as building check dams, farm ponds, and rainwater harvesting structures.
- **Targeting Water-Stressed Districts**
  - **Priority to water-stressed areas:** The campaign gives special attention to districts and blocks facing severe water scarcity and falling groundwater levels.
  - **Catch the Rain Campaign:** Promotes the idea of **Catch the Rain, Where it Falls, When it Falls** by encouraging rainwater harvesting and water conservation before the monsoon, so that every drop of rain is used effectively.
- **Convergence of Government Programmes**
  - **Convergence of government schemes:** Brings together programmes like **MGNREGA, PMKSY, and the Smart Cities Mission** so that different departments work together for better water conservation.
  - **Cost-effective implementation:** Uses existing government staff, funds, and infrastructure, making the campaign economical while ensuring faster execution.
- **Climate Resilience and Ecological Balance**
  - **Groundwater recharge:** Encourages rainwater to seep into the ground, helping refill underground water sources and reducing dependence on deep borewells.
  - **Tree plantation:** Promotes planting trees in catchment areas to reduce soil erosion, improve water retention, and support long-term environmental sustainability.
- **Use of Technology**

- **GIS mapping and Jal Shakti Kendras:** Uses GIS technology to map water bodies and provides technical guidance through Jal Shakti Kendras for rainwater harvesting.
- **Geo-tagging of assets:** Newly created and restored water conservation structures are geo-tagged to ensure transparency, proper monitoring and timely maintenance.

Achievements of Jal Shakti Abhiyan

- **Expanded nationwide monitoring:** Under **Jal Shakti Abhiyan (2019)**, Central teams visited **1,592 water-stressed blocks** through **706 field visits**, supported by **256 Central Nodal Officers, 446 Block Nodal Officers, and 446 Technical Officers**.
- **Regular review of water-stressed districts: Catch the Rain (CTR) 2022**, Central and Technical Officers reviewed implementation in **527 districts**, strengthening on-ground monitoring.
- **Focused support to vulnerable areas:** Under **JSA: CTR 2023**, **121 of the 150 identified water-stressed districts** were visited to improve implementation and water conservation efforts.
- **Renovation of Traditional Water Bodies**: **25,397 ponds, tanks and other traditional water bodies** have been restored.
- **Water Reuse and Groundwater Recharge**: **71,096 recharge and water reuse structures** have been completed.
- **Intensive Afforestation**: **89,408 afforestation projects** have been completed to improve groundwater recharge and ecological balance.
- **Strengthening Rural Drinking Water Sources**: **6.31 lakh piped water supply sources** have been identified, of which **4.27 lakh sources** have been geotagged.

Challenges with jal Shakti abhiyan

- **One-Size-Fits-All Approach**
  - **One-size-fits-all approach:** The same water conservation strategy is often applied across different regions, even though each area has unique geographical needs.
  - **Limited use of traditional knowledge:** Local water harvesting practices and indigenous knowledge are sometimes overlooked, reducing the effectiveness of water conservation efforts.
- **Quantity Over Quality**
  - **Poor maintenance of water structures:** Main focus is often on building new water conservation structures, while their regular maintenance receive less attention.

    CAG Audit Report on Water Resources highlighted that up to 30–40% of newly built rainwater harvesting and recharge structures become silted or non-functional within two years.
- **Limited Community Participation**
  - **Seasonal approach:** Most activities are carried out before the monsoon, whereas effective water conservation requires continuous efforts throughout the year.
  - **Declining public participation:** Community involvement often reduces after awareness campaigns end, making it difficult to sustain water conservation as a long-term Jan Andolan.
- **Weak Coordination**
  - **Poor coordination among departments:** Different government departments often face difficulties in working together and coordinating funds, slowing the effective implementation of water conservation projects.
  - **Lack of technical expertise:** Many Panchayats and local bodies do not have enough engineers or water experts, leading to poorly planned water conservation structures.
- **Limited Focus on Efficient Water Use**
  - **Focus on water conservation than water use:** Campaign mainly emphasizes storing and harvesting rainwater, while giving less attention to reducing water demand.

- **Water-intensive cropping patterns:** Continued cultivation of crops like paddy and sugarcane in water-scarce areas puts pressure on water resources and reduces the long-term benefits of water conservation efforts.

- **Challenges in Urban Areas**
  - **Rapid urbanisation:** Expansion of concrete roads and buildings reduces natural recharge areas and water bodies, limiting groundwater recharge despite conservation efforts.

Way forward

- **Focus on Maintaining Water Assets**
  - **Year-round implementation:** Make Jal Shakti Abhiyan a continuous programme managed by Panchayats throughout the year instead of limiting activities to the pre-monsoon period.

    Hiware Bazar (Maharashtra), where the Gram Panchayat actively maintains watershed structures, enforces bans on water-intensive crops, and manages water budgets.
  - **Measure long-term impact:** Assess success based on improvements in groundwater levels and water availability over time, rather than only on the number of structures created.

    Groundwater Impact Assessment framework under the Atal Bhujal Yojana (Atal Jal), which uses piezometers to reward Panchayats based on actual rise in groundwater levels
- **Promote Efficient Water Use**
  - **Encourage crop diversification:** Motivate farmers in water-scarce areas to shift from water-intensive crops like sugarcane to millets, pulses, and oilseeds, which require less water.

    Haryana's Mera Pani Meri Virasat scheme, which offers financial incentives of ₹7,000 per acre to farmers who shift from paddy to millets, pulses, or maize.
  - **Promote micro-irrigation:** Expand the use of drip and sprinkler irrigation under PMKSY so that every drop of stored water is used efficiently.
- **Strengthen Community Participation (Jan Andolan)**
  - **Empower local communities:** Equip Gram Sabhas with simple tools to monitor rainfall and groundwater levels, enabling villagers to actively participate in water management.
  - **Revive traditional water systems**: Promote traditional water harvesting methods along with modern technology to ensure sustainable and locally suitable water conservation.

    Mission Kakatiya in Telangana (restoring traditional Cheruvu tanks) and Rajasthan's restoration of traditional Johads led by Tarun Bharat Sangh.
- **Strengthen Technical Capacity**
  - **Use scientific planning:** Plan water conservation structures using GIS mapping, satellite imagery, and groundwater surveys to ensure effective groundwater recharge.

    ISRO's Bhuvan-Bhujal portal, which uses GIS mapping and satellite imagery to identify optimal locations for check dams and artificial recharge structures under MGNREGA.
  - **Provide technical support:** Deploy trained engineers and water experts at the block level to help Panchayats design and implement durable and effective water conservation projects.
- **Improve Water Management in Urban Areas**
  - **Ensure compliance rules:** Conduct regular inspections to make sure buildings implement and maintain rainwater harvesting systems as required.
  - **Protect urban water bodies:** Restore and safeguard lakes, wetlands, and their catchment areas from encroachment, pollution, and sewage discharge.
- **Strengthen Local Planning**
  - **Promote local water planning:** Encourage Panchayats to prepare Village Water Security Plans based on local water needs and priorities, ensuring better planning and effective use of funds.

Water Security Plans (WSPs) developed under Atal Bhujal Yojana, where Panchayats map water availability versus demand and allocate MGNREGA and Finance Commission funds accordingly.

Conclusion

The Jal Shakti Abhiyan represents a significant step toward sustainable water management in India. By engaging citizens, leveraging technology, and implementing targeted interventions, the campaign aims to address water scarcity and promote the conservation, restoration, recharge, and reuse of water resources. As the campaign continues to evolve, it remains focused on building a water-secure future for all, with a special emphasis on the pivotal role of women in driving these efforts forward.

MGP/BP/GS3/032

STORAGE, TRANSPORT & AGRICULTURAL MARKETING

1. STORAGE AND BUFFER STOCK MANAGEMENT

Introduction:

Storage and buffer stock management means **storing enough food grains safely so that they can be supplied when needed**. It helps India deal with **food shortages, price rise, natural disasters and poor harvests**, while also supporting the PDS.

Objectives of Storage and Buffer Stock Management

- **Ensure Food Security**
  - **Food availability:** Keep enough food grains in stock so that people have access to food even during shortages.
  - **Support PDS:** Ensure regular supply of grains through the Public Distribution System (PDS).
- **Handle Emergencies**
  - **Crisis support**: Maintain food stocks to deal with droughts, floods, pandemics and other emergencies.
  - **Quick supply:** Stored grains can be released quickly when normal supply is disturbed.
- **Control Price Rise**
  - **Market intervention:** Release stored grains when prices rise sharply. It helps protect consumers from sudden increases in food prices.
- **Support Farmers**
  - **MSP procurement:** Government procurement at MSP gives farmers a price guarantee for important crops.
  - **Income protection:** It reduces the risk of farmers having to sell produce at very low prices after harvest.
- **Maintain Buffer Stocks**
  - **Minimum stock:** Keep adequate food grains available throughout the year. It also ensure supply during periods when fresh procurement is low.
- **Reduce Food Wastage**
  - **Scientific storage:** Proper warehouses, silos and handling help protect grains from pests, moisture and spoilage.
- **Ensure Smooth Food Distribution**
  - **Regular movement**: Move food grains from surplus areas to deficit areas.
  - **Last-mile supply:** Ensure grains reach consumers through the PDS and welfare programmes.
- **Strengthen National Food Security**

- **Stable supply:** Helps India maintain a reliable food supply even during poor harvests or global disruptions.

Food Grain Storage Systems in India

India uses different storage systems to keep food grains safe, reduce losses and ensure their timely availability:

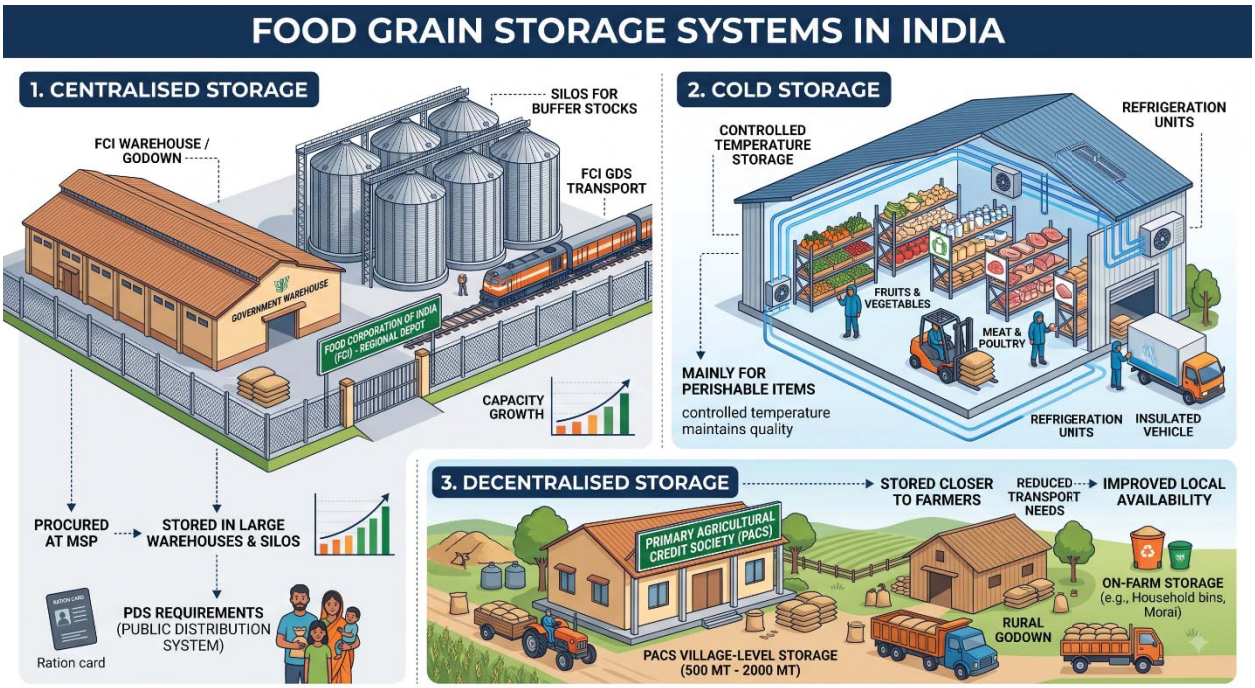
- **Centralised Storage:** Food grains are procured and stored mainly through agencies such as FCI in large warehouses and silos for buffer stocks and PDS requirements.
- **Cold Storage:** Used mainly for perishable food items such as fruits, vegetables, dairy products and meat, where controlled temperature helps maintain quality.
- **Decentralised Storage:** Food grains are stored closer to farmers through rural godowns, PACS and on-farm storage, reducing transport needs and improving local availability.

Measures Taken for Storage and Buffer Stock Management

- **World's Largest Grain Storage Plan in the Cooperative Sector**
  - **Cooperative Warehouses:** Interlinking Primary Agricultural Credit Societies (PACS) to build localized godowns, custom hiring centers, and processing units at the village level.
- **Steel Silos & Hub-and-Spoke Model**: Shifting from traditional, leaky covered godowns to modern, temperature-controlled **Steel Silos** through Public-Private Partnerships (PPP).
- **Private Entrepreneur Guarantee (PEG) Scheme:** FCI leases modern storage facilities built by private investors with a long-term storage guarantee.
- **Digitization & Transparency Measures**
  - **Anna Darpan Portal (FCI 2.0 Digitization)**: An end-to-end supply chain management micro-system replacing legacy systems like Depots Online System (DOS). It tracks foodgrains in real time from farm procurement, depot entry, transit via railways, to final PDS delivery.
  - **Depot Online System (DOS) & Geotagging**: Digitized operations across thousands of FCI and CWC (Central Warehousing Corporation) godowns to monitor exact stock availability, stack quality, and storage duration.
  - **Negotiable Warehouse Receipt (NWR) System**: Warehousing Development and Regulatory Authority (**WDRA**) registers scientific warehouses and issues **electronic NWRs (e-NWRs)**.
- **Operational & Policy Reforms (Buffer Stock Management)**
  - **Revision of Foodgrain Stocking Norms**: The Cabinet Committee on Economic Affairs (CCEA) fixes quarterly **Foodgrain Stocking Norms** (Operational Stock + Strategic Reserve) for wheat and rice (as of 1st Jan, 1st April, 1st July, 1st Oct).
  - **Open Market Sale Scheme (OMSS) Interventions**: Offloading surplus central buffer stock in the open market via bulk e-auctions during lean seasons.
  - **Diversification into Pulses & Oilseeds Buffer**: Created a dedicated **Buffer Stock of Pulses** (managed by NAFED, SFAC, and FCI) to prevent severe price volatility in pulses.
- **Quality Control & Wastage Reduction**
  - **Phase-Out of Cover and Plinth (CAP) Storage**: Systematic phase-out of open, tarpaulin-covered outdoor storage (CAP) which leaves foodgrains vulnerable to rain, rot, and pest damage.
  - **Food Safety Standards (FSSAI Integration)**: Mandatory periodic chemical treatment, aeration, and scientific pest management across state and central godowns to prevent degradation.
  - **Irradiation & Cold Chain Technologies**: Scaling up gamma irradiation technology for perishable buffer items (like onions and potatoes) to inhibit sprouting and extend shelf-life.

Major Takeaways

- India achieved record foodgrain production of 353.96 million tonnes in 2024-25 (As per third advance estimates).
- Food Corporation of India (FCI) and State agencies hold 917.83 LMT Covered and CAP Storage Capacity for central pool grains.
- 8,815 cold storages with 40.21 million MT capacity preserve perishables nationwide.
- Primary Agricultural Credit Societies (PACS) decentralised storage is expanding, with 5,937 new PACS registered and 73,492 computerized by June 2025.

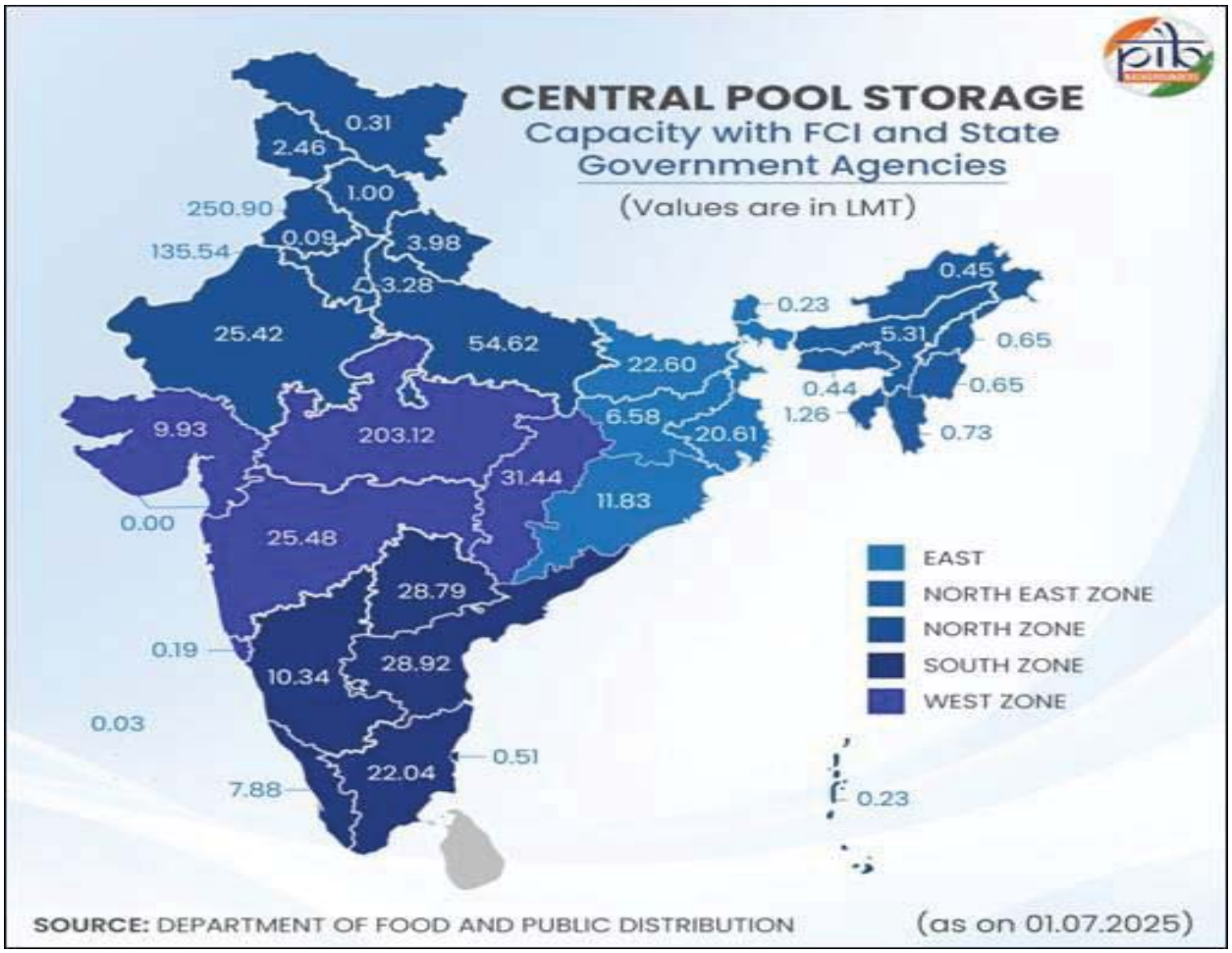


Challenges of Storage and Buffer Stock Management

- **High Storage Costs**
  - **Expensive maintenance:** Building and maintaining warehouses, silos and cold storage needs large investment.
  - **Rising costs:** Handling, transportation and storage add to the overall food subsidy burden.
- **Lack of Modern Storage**
  - **Poor facilities:** In some places, lack of proper ventilation, temperature control and scientific storage leads to grain damage.
- **Foodgrain Wastage**
  - **Spoilage:** Grains can be damaged by moisture, insects, rats and fungi.
  - **Poor handling:** Losses can occur during loading, unloading, transportation and storage.
- **Regional Imbalance**
  - **Surplus–deficit gap:** Moving grains from surplus states to deficit areas increases transport costs and delays.
- **Excessive Stock Holding**
  - **Stocks above requirement:** Government may sometimes hold more grain than the required buffer level and holding large stocks increase storage, handling and interest costs.

- **Quality Deterioration**
  - **Long storage**: Grains kept for a long time can lose quality and nutritional value.
  - **Poor rotation:** Delays in releasing older stocks can increase the risk of damage and wastage.
- **Overdependence on Rice and Wheat**
  - **Narrow procurement basket:** Buffer stocks mainly focus on rice and wheat. This can encourage farmers to continue growing water-intensive crops instead of pulses, oilseeds and millets.

    *According to the **FCI Annual Report 2024–25**, FCI purchased **222.33 LMT of wheat and 339.05 LMT of rice**, compared with only **0.40 LMT of coarse grains** and **zero pulses**.*
- **Climate and Weather Risks**
  - **Extreme weather:** Heatwaves, floods and heavy rainfall can damage stored and transported grains.
  - **Changing storage needs:** Climate change makes scientific and climate-resilient storage systems increasingly important.



Way forward

- **Infrastructure & Physical Modernization**
  - **Replace CAP Storage:** Phasing out open-air Cover and Plinth (CAP) storage by replacing it with modern, temperature-controlled **Steel Silos** built under Public-Private Partnerships (PPP).
  - **Bulk Handling Rail Sidings:** Integrating silos directly with railway sidings to enable automated, bulk-grain loading, eliminating gunny bag requirements and transit spillage.
- **Decentralized Storage at Village Level**

  - **Empowering PACS:** Scaling up the *World's Largest Grain Storage Plan in the Cooperative Sector* by building 500–2,000 MT capacity godowns at the Primary Agricultural Credit Society (PACS) level.
  - **Reducing Freight Costs:** Storing grains near production centers cuts unnecessary two-way transport between mandis and distant central depots.
- **Financial & Operational Restructuring**
  - **Restructure FCI Roles:** Outsourcing stocking, movement, and PDS grain handling to state governments or private logistics agencies in states with strong administrative capacity (e.g., Andhra Pradesh, Tamil Nadu, Haryana).
  - **Rationalize Economic Cost:** Streamlining state levies and mandi fees (capping them at a uniform 3–4%) to curb the ballooning economic cost borne by FCI.
  - **Clear Food Subsidy Dues Timely:** Ensuring predictable, scheduled releases of food subsidy funds to FCI to reduce its dependence on high-interest short-term borrowings.
- **Policy & Market Reforms**
  - **Capping Procurement Volumes:** Capping central procurement of wheat and rice at mandatory buffer requirements plus PDS needs, leaving surplus produce for open market trade.
  - **Dynamic Buffer Norms:** Using algorithmic, real-time stock-tracking to trigger automatic Open Market Sale Scheme (OMSS) releases when market retail prices cross specified thresholds.
- **Diversifying Buffer Stock Composition**
  - **Pulses, Oilseeds & Millets:** Expanding price-stabilization buffer stocks (managed by NAFED and SFAC) for pulses, oilseeds, and coarse grains (Shree Anna).
  - **Encouraging Crop Diversification:** Incentivizing farmers in water-stressed belts (Punjab/Haryana) to shift away from water-intensive paddy toward crops backed by guaranteed buffer procurement.
- **Digital Governance & Farmer Integration**
  - **Credit via e-NWRs:** Linking WDRA-accredited warehouses with the e-NAM portal so small farmers can pledge stored grain for low-interest bank loans.
  - **Standardizing Local Warehouses:** Simplifying registration norms for private and gram-panchayat level godowns to meet WDRA accreditation standards.

Conclusion:

Efficient storage and buffer stocks are essential for **food security, price stability and farmer welfare**. Modern silos, better stock management and decentralised storage can reduce wastage while keeping food **available, affordable and accessible**. A strong food-storage system will not only support **Atmanirbhar Bharat**, but also help build a **Samriddh Bharat** with greater resilience and sustainable growth

2. FOOD CORPORATION OF INDIA

Introduction

The **Food Corporation of India** is the government's main agency for **procuring, storing and distributing food grains**. It buys food grains mainly at **MSP**, maintains buffer stocks and supplies them through the **Public Distribution System (PDS)**. FCI acts as a **bridge between farmers, food stocks and consumers**, helping India ensure food security and price stability.

Food Corporation of India (FCI)

- **Basic Mandate**
  - **Establishment:** Set up on **14th January 1965** under the **Food Corporations Act, 1964**.
  - **Nodal Ministry:** Functions as a statutory public sector undertaking under the **Ministry of Consumer Affairs, Food and Public Distribution**.
  - **Headquarters:** Located in **New Delhi**, operating through 5 Zonal Offices and 26 Regional Offices across India.
  - **Core Purpose:** Acts as the main operational arm of the Government of India for executing the **National Food Policy**.
- **Core Objectives & Functions**
  - **Effective Price Support (MSP Procurement):** Procures wheat, paddy/rice, and coarse grains directly from farmers at the government-announced **Minimum Support Price (MSP)** to prevent distress sales.
  - **Public Distribution System (PDS Supply):** Transports and supplies foodgrains to state governments for subsidized distribution to beneficiaries under the **National Food Security Act (NFSA)** and schemes like **PM-GKAY**.
  - **Maintaining Buffer Stocks:** Stores operational and strategic reserves of foodgrains to cushion against crop failures, droughts, flash floods, or price shocks.
  - **Price Stabilization (OMSS Interventions):** Offloads surplus grains in the open market via the **Open Market Sale Scheme (OMSS)** during lean seasons to control retail food inflation.

Significance of Food Corporation of India:

- **Ensures Food Security**
  - **Food availability:** FCI maintains adequate food stocks so that people have access to food even during shortages or emergencies.
  - **Supports PDS:** It supplies food grains for programmes such as the Public Distribution System (PDS).
- **Supports Farmers**
  - **MSP procurement:** FCI buys mainly wheat and rice at MSP, giving farmers a degree of price security.
  - **Reduces distress sales:** Farmers are less likely to sell immediately at very low market prices after harvest.
- **Maintains Buffer Stocks**
  - **Emergency reserve:** FCI maintains stocks that can be used during droughts, floods, poor harvests or other crises.
  - **Regular supply:** It helps ensure that food grains remain available throughout the year.
- **Controls Food Prices**
  - **Market intervention:** FCI can release food grains into the market when prices rise sharply. This helps protect consumers from sudden food-price increases.
- **Strengthens the PDS**
  - **Movement of grains:** FCI transports food grains from surplus-producing areas to deficit regions. This enables states to receive food grains for distribution to eligible households.
- **Reduces Regional Food Imbalance**
  - **Surplus to deficit:** Food grains can be moved from states with high production to areas facing shortages.
- **Provides Food During Emergencies**

  - **Crisis response:** FCI stocks can be used during natural disasters, pandemics and other emergencies.
- **Reduces Dependence on Imports**
  - **Domestic food reserve:** Adequate stocks strengthen India's ability to manage global supply disruptions.
  - **Food self-reliance**: It contributes to India's broader goal of food security and Atmanirbhar Bharat.

Commission for Agricultural Costs and Prices (CACP)

- **About CACP:** CACP is an attached office of the Ministry of Agriculture & Farmers Welfare that recommends Minimum Support Prices (MSPs) for major crops.
- **Established:** It was set up in 1965 as the Agricultural Prices Commission (APC) and was renamed CACP in 1985.
- **Main Role**
  - **MSP recommendation:** It recommends MSPs for 22 mandated crops, along with a fair and remunerative price (FRP) for sugarcane being determined separately by the government.
  - **Price advice:** It advises the government on pricing policy for agricultural commodities.

National Food Security Act (NFSA), 2013

- **About NFSA**
  - **Enactment Date:** Passed in July 2013 to mark a paradigm shift in food security—from a *welfare-based approach* to a rights-based approach.
  - **Nodal Ministry:** Administered by the Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution.
  - **Core Objective**: To provide for food and nutritional security in human life cycle approach, by ensuring access to adequate quantity of quality food at affordable prices to people to live a life with dignity.

Challenges of Food Corporation of India:

- **High Food Subsidy Burden**
  - **Rising costs:** FCI spends heavily on procurement, storage, transport and distribution of food grains.

According to the **Union Budget 2025–26**, food subsidy was **₹2,11,814 crore in 2023–24** and is budgeted at **₹2,03,420 crore for 2025–26**.

- **Financial pressure:** High costs increase the government's food subsidy burden.

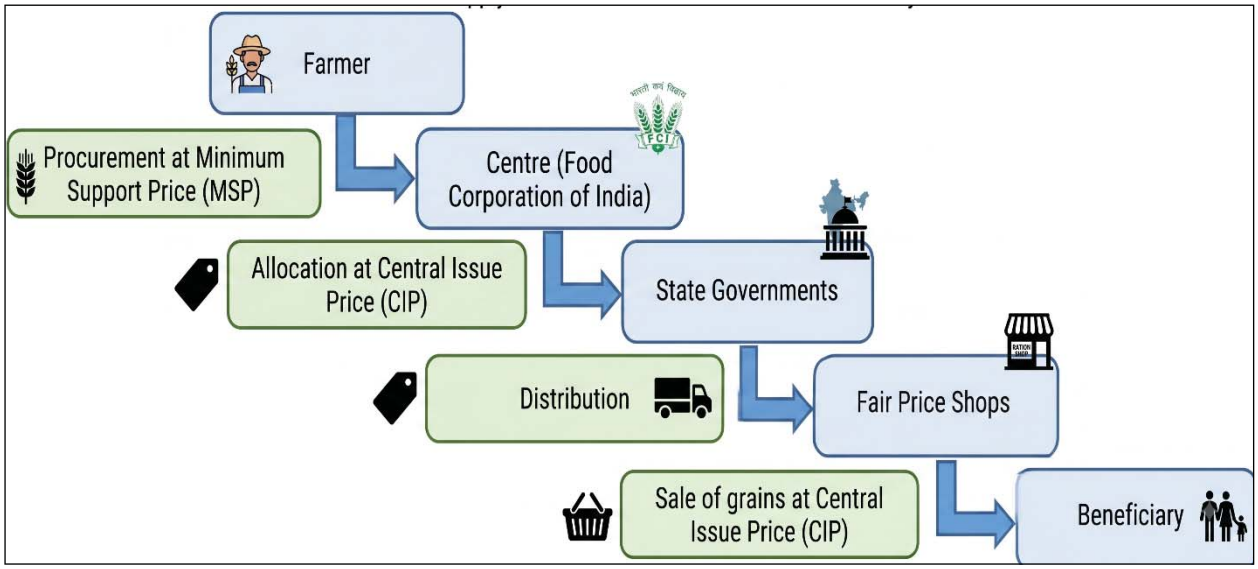
- **Excessive Dependence on Wheat and Rice**
  - **Narrow procurement:** FCI mainly procures wheat and rice, while pulses and oilseeds receive less attention.

    As per **FCI Annual Report 2024–25**, FCI purchased **222.33 LMT of wheat and 339.05 LMT of rice**, compared with only **0.40 LMT of coarse grains** and **zero pulses**.

  - **Cropping imbalance:** This can encourage farmers to continue growing water-intensive crops, especially paddy.
- **High Buffer Stocks**
  - **Stocks above requirement:** Procurement can sometimes remain higher than the buffer stock and Keeping excess grain for long periods increases storage and handling costs.

    As per FCI reports, Central-pool rice stocks were **over 490 LMT**, while annual rice requirement was around **400 LMT** and the prescribed July buffer norm was only **135 LMT**.
- **Storage Problems**
  - **Limited modern facilities:** Dependence on traditional godowns can lead to moisture, pest and quality problems.
  - **Need for silos:** Modern bulk storage and scientific warehouses are still required at a larger scale.

    As of **1 April 206**, FCI and State agencies together had 952 LMT **of covered storage capacity**, against **609.02 LMT of stored central-pool food grains**.
- **Regional Imbalance**
  - **Concentration of procurement:** Procurement is heavily concentrated in a few states such as Punjab, Haryana and Madhya Pradesh.
  - **Uneven benefits:** Farmers in many eastern and northeastern states have less access to effective procurement.
- **Quality and Wastage Issues**
  - **Storage losses:** Poor storage and handling can cause damage due to insects, rodents and moisture.
- **Financial Management Issues**
  - **Delayed subsidy payments:** Delays in receiving government food subsidy payments can increase FCI's working-capital needs.
  - **Borrowing burden:** FCI may have to borrow to meet its day-to-day expenses, increasing financial costs.
- **Procurement and Distribution Inefficiencies**
  - **High operational costs:** Multiple stages of procurement, storage and movement make the system costly and complex.



Government Reforms in FCI

- **Financial Strengthening**
  - **₹10,700 crore equity infusion:** Government approved this for FCI's working capital in **2024–25**, helping reduce its dependence on short-term borrowing and **interest burden**.
  - **Lower interest cost:** FCI reduced interest expenses from **₹3,679 crore in 2022–23 to ₹3,268 crore in 2023–24**, saving **₹411 crore**.
  - **Asset monetisation:** FCI monetised assets worth **₹6,088 crore by FY 2023–24**, helping improve financial efficiency.
- **Modernisation of Storage**
  - **From traditional to modern storage:** Traditional **Cover and Plinth (CAP)** storage, which stood at **30.25 LMT in 2014**, is being replaced with scientific storage facilities.
  - **Steel silos: 22.75 LMT** of silo capacity is operational, while another **41 LMT** is under development.
  - **Depot upgradation:** Of **489 FCI-owned depots** assessed by a third party, most have been upgraded to meet quality standards.
- **Digital Procurement**
  - **Computerised procurement:** Foodgrain procurement operations have been comprehensively digitised to make the process **faster and more transparent**.
  - **Central Food Procurement Portal (CFPP):** MSP procurement data across India is integrated into a **single digital platform**, improving transparency and data availability.
- **End-to-End Supply Chain Digitisation**
  - **ANNA DARPAN:** Integrates FCI operations from **depots to headquarters**, creating a more seamless supply-chain system.
  - **WINGS:** Helps manage **mill–depot linkages and warehouse space**, improving inventory management.
  - **Vehicle Location Tracking System (VLTS):** GPS-based tracking helps monitor **long-distance movement of food grains**.
  - **Depot Online System:** Implemented across around **1,900 warehouses**, improving digital monitoring of storage operations.

- **Faster Payments**
  - **Bill Tracking System:** Digitisation reduced average payment-processing time from **23 days to just 8 days**.
  - **Direct online payment:** The government has promoted direct digital payment mechanisms to improve **transparency and timely payments to farmers**.
- **Better Quality Control**
  - **AI-based grain testing: Automatic Grain Analysers (AGA)** reduce human intervention and make grain procurement more **objective and transparent**.
  - **Rice-age testing:** A scientific method for determining the age of custom-milled rice has been mandatory since **1 October 2021**.
  - **Digital laboratories:** Quality-control labs are connected through a **Quality Management System (QMS)**, enabling real-time monitoring of testing and rejection rates.
- **Reducing Storage and Transit Losses**
  - **Storage efficiency:** Combined wheat and rice storage losses improved from a **0.17% loss in 2013–14 to a 0.22% net gain in 2023–24**.
  - **Lower transit losses:** Transit losses declined from **0.46% to 0.16%** during the same period.
  - **High-security seals:** Tamper-proof seals introduced from **January 2023** helped reduce rakes with transit losses above 0.5% by **96%**.
- **Expansion of Decentralised Procurement**
  - **More states involved:** DCP participation increased from **10 to 17 states for rice** and **5 to 9 states for wheat** between 2014 and 2023–24.

Way Forward:

- **Modernise Storage**
  - **More silos:** Shift from old godowns to modern steel silos and scientific warehouses to reduce storage losses.

    World's Largest Grain Storage Plan in Cooperative Sector: Building 500–2,000 MT steel/concrete godowns at the Primary Agricultural Credit Society (PACS) level.
  - **Better technology:** Use sensors and digital monitoring to track grain quality and stock levels.
- **Diversify Procurement**
  - **Beyond rice and wheat:** Increase procurement of pulses, millets and oilseeds where suitable. This can encourage crop diversification and reduce pressure on groundwater.

    PM-AASHA (Pradhan Mantri Annadata Aay Sanraksan Abhiyan): *Price Support Scheme (PSS) and Price Deficiency Payment Scheme (PDPS) executed by* ***NAFED & SFAC*** *to procure pulses (chana, tur, urad) and oilseeds (mustard, groundnut).*
- **Reduce Excess Stocks**
  - **Better forecasting:** Use production and demand data to procure only what is actually needed.
  - **Quick stock rotation:** Older stocks should be released first to reduce spoilage and carrying costs.

    Bharat Atta & Bharat Rice Interventions: *Offloading surplus Central Pool stocks via the* ***Open Market Sale Scheme (OMSS)*** *directly to retail consumers through NAFED, NCCF, and Kendriya Bhandar.*
- **Decentralise Procurement**
  - **Closer to farmers:** Strengthen procurement centres in eastern, northeastern and other under-served regions.

  - **Lower transport cost:** Local procurement can reduce the need to move grain over long distances.
- **Improve Financial Management**
  - **Timely subsidy:** Ensure regular release of food subsidy to reduce FCI's dependence on borrowing.
  - **Lower carrying cost:** Better stock planning can reduce interest, storage and handling expenses.
- **Strengthen Logistics**
  - **Efficient movement:** Use railways, dedicated freight corridors and modern handling systems for faster grain movement.

    PM Gati Shakti & Dedicated Freight Corridors (DFC): *Mapping FCI depots along the Eastern and Western DFCs with direct railway sidings for rapid bulk-grain movement.*
  - **Better last-mile supply:** Improve coordination with states so that food grains reach beneficiaries on time.

    One Nation One Ration Card (ONORC): *Automated integration of FCI regional supply nodes with state* ***e-POS (Electronic Point of Sale)*** *devices at Fair Price Shops.*
- **Use Technology**
  - **Data-based decisions:** Use technology to predict production, demand and storage requirements more accurately.
- **Promote Climate-Resilient Storage**
  - **Weather-safe facilities:** Warehouses should be designed to handle heat, floods and heavy rainfall.
  - **Reduce climate losses:** Better infrastructure can protect food stocks from extreme weather events.

    Gamma Irradiation Units: *Scaled up in pilot potato/onion storage centers to inhibit sprouting during unseasonal heatwaves and high humidity.*
- **Support Small Farmers**
  - **Wider procurement network:** Make procurement centres more accessible to small and marginal farmers.
  - **FPO participation:** Farmer Producer Organisations can help small farmers aggregate produce and access procurement markets.

Shanta Kumar Committee Recommendations

The Shanta Kumar Committee (2015) was set up to suggest reforms in the functioning of FCI, food procurement, storage and the Public Distribution System (PDS). Recommendations are given below:

- **Reform the Role of FCI:** FCI should focus mainly on procurement, buffer stock and movement of food grains.
- **Reduce Excessive Procurement:** FCI should avoid procuring more grain than required for buffer stocks and welfare schemes.
- **Change the MSP–Procurement System:** MSP should not automatically lead to unlimited government procurement.
- **Expand procurement:** Procurement facilities should reach more states and more farmers, rather than remaining concentrated in a few regions.

Conclusion

FCI needs to move from a **"procure, store and distribute" model to a more efficient, technology-driven and diversified food-management system**. This can reduce costs, support farmers, protect consumers and contribute to **Atmanirbhar Bharat and a Samriddh Bharat**.

3. NATIONAL AGRICULTURE MARKET – E-NAM

"I firmly believe that now my farmers would decide where, when and at what price their produce will be sold. And it is my belief that there would be no burden on the consumer."

- Prime Minister Narendra Modi

The National Agriculture Market (e-NAM) is a flagship initiative of the Government of India aimed at **creating a unified national market for agricultural commodities**. Launched on April 14, 2016, by Prime Minister Narendra Modi, e-NAM seeks to eliminate barriers to inter-state trade and promote a transparent, efficient, and competitive marketplace for farmers and traders.

e-NAM is implemented by the **Small Farmers' Agribusiness Consortium (SFAC)** under the Ministry of Agriculture and Farmers' Welfare. The platform **integrates existing Agricultural Produce Market Committees (APMCs)** into a digital trading system, facilitating online trading of agricultural produce.

Objectives of e-NAM

- **Market Integration**: To integrate markets at the state level and eventually across the country through a common online market platform, facilitating pan-India trade in agricultural commodities.
- **Streamlining Procedures**: To streamline marketing and transaction procedures, making them uniform across all markets to promote efficient functioning.
- **Better Marketing Opportunities**: To promote better marketing opportunities for farmers and sellers through online access to more buyers and markets, removal of information asymmetry, better price discovery, and online payment facilities.
- **Quality Assurance**: To establish quality assaying systems for quality assurance to promote informed bidding by buyers.
- **Consumer Benefits**: To promote stable prices and availability of quality produce to consumers.

Key Features of e-NAM

- **Single Window Services**: e-NAM provides a single window service for all APMC-related information and services, including commodity arrivals, quality and prices, buy and sell offers, and e-payment settlement directly into farmers' accounts.
- **Transparent Price Discovery**: The platform facilitates online competitive and transparent price discovery systems, ensuring fair prices for farmers.
- **Online Payments**: Farmers receive payments directly into their bank accounts, reducing delays and ensuring transparency.
- **Quality Assaying**: e-NAM integrates quality assaying systems to promote informed bidding by buyers.
- **Mobile Application**: The e-NAM mobile app provides real-time information on commodity prices, arrivals, and allows traders to participate in online bidding.

Scheme Components

- **Selection of APMCs**: States and Union Territories select APMCs for integration into e-NAM after mandatory reforms in their APMC Acts and approval of their Detailed Project Reports (DPRs).
- **Financial Assistance**: The central government provides financial assistance up to Rs. 30 lakh per market for the purchase of hardware, internet connections, assaying equipment, and related infrastructure.
- **Technical Assistance**: Small Farmers' Agribusiness Consortium (SFAC), through strategic partners, deploys Mandi Analysts at each market for day-to-day support.
- **Capacity Building**: Orientation programs for APMC staff and awareness campaigns for farmers are conducted to facilitate smooth implementation.

Pre-requisites for e-NAM

To propose mandis for integration with e-NAM, a state must fulfill **three basic criteria**:

- The **state APMC Act** should include a provision for **e-auction or electronic trading** as a mode of price discovery.
- There should be a **single trading license** that is valid across the entire state or Union Territory.
- A **single-point levy of market fe**e must be applied uniformly across the state or UT.

New Features and Innovations

- **Platform of Platforms (POP)**: Launched in July 2022, POP facilitates inter-state trade by connecting multiple online platforms, increasing farmers' digital access to various markets and buyers.
- **Warehouse-Based Trading**: This feature allows farmers to trade their produce from warehouses based on electronic Negotiable Warehouse Receipts (e-NWRs), providing flexibility and reducing the need for immediate sale.
- **FPO Trading Module**: Introduced during the COVID-19 pandemic, this module enables FPOs to trade their produce from collection centers without bringing it to APMCs, ensuring continuity of business during restrictions.
- **AI-Based Assaying Equipment**: Integration of Artificial Intelligence in quality testing equipment enhances the accuracy and efficiency of produce quality assessments.

Stakeholders of the Scheme

Achievements (as of June 30, 2025)

- **Farmers Registered:** Over **1.79 crore** farmers have registered on the e-NAM platform, expanding their access to digital agricultural markets.
- **Traders Registered:** Approximately **2.68 lakh** traders are active on the platform, facilitating a broader trading network.
- **Farmer Producer Organizations (FPOs): 4,518 FPOs** have been onboarded, enhancing collective bargaining power and market access for small farmers.
- **Mandis Integrated:** A total of **1,522 mandis** across **27 states and UTs** have been integrated into the e-NAM platform, promoting a unified national market

- **Trade Value:** The cumulative value of agricultural produce traded on the e-NAM portal stands at approximately **₹4.39 lakh crore**.
- **Commodity Coverage:** The platform now includes **231 commodities**, with the addition of 10 new items, providing a broader market for various agricultural products.

Benefits of Trading on e-NAM:

- Transparent online trading with enhanced accessibility to the market.
- Real-time price discovery for better and stable price realization for producers.
- Reduced transaction cost for buyers.
- Availability of information on the e-NAM mobile app about commodity prices. The details of the price of the commodity sold along with quantity are received through SMS.
- Quality certification.
- More efficient supply chain and warehouse-based sales.
- Online payment directly to the bank accounts of the farmers.

Challenges faced by e-NAM

- **Infrastructure Limitations:** Scarcity of electricity, internet facilities, and quality testing infrastructure in some regions hampers the effective implementation of e-NAM.
- **Resistance to Change:** Resistance from traders and stakeholders accustomed to traditional systems poses challenges in transitioning to the digital platform.
- **Awareness and Training:** Lack of awareness and digital literacy among farmers limits their participation and benefits from the platform.
- **Quality Assurance:** Ensuring consistent and accurate quality assessment across diverse commodities remains a challenge.
- **Inter-State Logistics and Transportation:** Limited cold storage, transport facilities, and logistical support for moving produce across states can restrict pan-India trade.

Way Forward

- **Infrastructure Development:** Investing in electricity, internet connectivity, cold storage, transport, and quality testing facilities is crucial for the success of e-NAM.
- **Capacity Building:** Conducting training programs for farmers and traders to enhance digital literacy and platform usage will increase participation.
- **Policy Reforms:** Implementing uniform policies across states regarding market fees, licensing, and quality standards will streamline operations.
- **Public-Private Partnerships:** Encouraging collaboration between government agencies and private players can enhance service delivery and innovation.
- **Strengthening Logistics:** Developing robust inter-state logistics, including warehousing and cold chain networks, will enable seamless movement of produce and support pan-India trade.

4. FARMER PRODUCER ORGANIZATIONS

Introduction

Farmer Producer Organisations (FPOs) bring small and marginal farmers together so they can **buy inputs, sell produce and access markets collectively**. FPOs are regulated under the **Companies Act,**

2013 (as Producer Companies) or the **Cooperative Societies Act**. FPOs operate as business entities owned, managed, and driven by the farmers themselves.

Significance:

- **Better Bargaining Power:** By selling collectively, small farmers can negotiate **better prices** with traders, processors and retailers.
- **Lower Input Costs:** Bulk purchase of **seeds, fertilisers, pesticides and machinery** can reduce the cost of cultivation.
- **Better Market Access:** FPOs help farmers reach **larger markets, organised retailers and exporters** instead of depending only on local traders.
- **Value Addition:** FPOs can undertake **grading, sorting, packaging, processing and branding**, helping farmers earn more from the same produce.
- **Access to Credit:** Collective organisations can improve farmers' access to **institutional credit and financial services**.
- **Economies of Scale:** Pooling land-related resources, machinery and produce allows small farmers to enjoy some benefits of **large-scale farming**.
- **Reduced Post-Harvest Losses:** Shared **warehouses, cold storage and transport** can reduce wastage and improve the shelf life of farm produce.
- **Technology & Knowledge:** FPOs can provide members with access to **better technology, training, extension services and market information**.
- **Export Opportunities:** By supplying larger and more uniform quantities, FPOs can help small farmers meet the **quality and quantity requirements of export markets**.

Status of FPO in India

- **Large-scale expansion:** Under the **Central Sector Scheme for Formation and Promotion of 10,000 FPOs**, **10,000 FPOs have been formed** across the country.
- **Strong women participation:**
  - **1,175 FPOs** have been formed with **100% women members**.
  - Women account for around **40% of total FPO shareholders**.
  - **Gender representation:** Every FPO is required to have **at least one woman member on its Board of Directors/Governing Body**.
- **Financial & institutional support:**
  - **9,865 FPOs** have received **management-cost support**.
  - **8,357 FPOs** have received **equity grants**.
  - **3,140 FPOs** have accessed **credit guarantee support**.
- **Growing economic footprint:** The **cumulative turnover of FPOs formed under the Central Sector Scheme reached ₹20,358 crore as of 31 July 2026**.

Issues with Farmer Producer Organisations:

- **Financial & Business Challenges**
  - **Limited working capital:** Many FPOs struggle to arrange funds for procurement, storage, processing and marketing.

*As per the Ministry of Agriculture, the FPO scheme specifically provides **₹18 lakh per FPO for management costs, equity support up to ₹15 lakh.***

  - **Dependence on grants:** Some FPOs remain dependent on government support and find it difficult to become financially self-sustaining.
- **Management & Governance Issues**
  - **Lack of professional management:** Shortage of skilled people affects accounting, business planning, branding and market management.
  - **Weak governance:** Poor participation of members and limited transparency can weaken decision-making.
- **Market Linkage Problems**
  - **Limited buyers:** Many FPOs still depend on local traders and mandis instead of directly accessing processors, retailers and exporters.
  - **Weak bargaining:** Small volumes and limited market knowledge can reduce their ability to negotiate better prices.
- **Technology & Knowledge Gaps**
  - **Limited digital capacity:** Many FPOs lack access to digital marketing, price information, traceability and modern business tools.
  - **Low technical knowledge:** Farmers may need more support in quality standards, packaging, processing and export requirements.
- **Trust & Participation Issues**
  - **Low member participation:** If farmers do not actively use the FPO for buying and selling, its collective strength remains limited.
  - **Lack of awareness:** Some farmers are still not fully aware of the long-term benefits of collective marketing.

Way Forward for Farmer Producer Organisations (FPOs)

- **Strengthen Financial Capacity**
  - **Easy access to credit:** Improve access to working capital so FPOs can procure, store and market farmers' produce at the right time.
  - **Credit guarantees:** Expand credit-guarantee support to help FPOs borrow without depending heavily on collateral.
- **Build Professional Management**
  - **Skilled professionals:** Provide FPOs with trained managers in finance, marketing, accounting, logistics and business planning.
  - **Better governance:** Promote transparent accounts, regular audits and greater participation of farmer-members.
- **Improve Market Linkages**
  - **Direct buyers:** Connect FPOs with processors, retailers, exporters, organised buyers and e-commerce platforms.

    Almería Producer Organizations (Spain): *Fruit and vegetable producer organizations (APOs) directly negotiate long-term supply contracts with European supermarket chains (like Aldi and Carrefour)*

- **Digital markets:** Encourage FPO participation in platforms such as e-NAM to widen their buyer base and improve price discovery.
- **Promote Aggregation & Economies of Scale**
  - **Cluster approach:** Bring FPOs working in the same crop or geographical area together for bulk procurement and collective marketing.
  - **Assured supply:** Develop systems for planned collection so FPOs can provide buyers with regular quantity and consistent quality.

DKV & Danish Crown (Denmark): *Pig and livestock producers operate integrated cluster-based collection routes, using specialized logistics systems to guarantee a continuous, standardized daily supply*

- **Improve Technology & Market Intelligence**
  - **Digital tools:** Equip FPOs with digital systems for inventory, payments, traceability and accounting.
  - **Market information:** Give real-time information on prices, demand, quality standards and export opportunities.
- **Increase Farmer Participation**
  - **Build trust:** Ensure transparent pricing and timely payments so farmers see the FPO as their own business organisation, not merely another government scheme.

*As of February 2025, around **30 lakh farmers** were connected with FPOs, with women accounting for about **40**% of participating farmers.*

Best Case Studies of FPOs in India

FPO / Location	features
Sahyadri Farmers Producer Company – Maharashtra	• Brings together grape and other horticulture farmers. • Supports grading, packaging, processing and marketing. • Connects farmers with domestic as well as export markets.
Vrutti FPOs – Karnataka	• Organise small farmers and improve access to inputs, technology and markets. • Help farmers collectively market their produce.
Pragati FPO – Odisha	• Farmers collectively procure inputs and market agricultural produce. • Focus on improving market access for smallholders.
Mulkanoor Cooperative Rural Bank & Marketing Society – Telangana	• Provides farmers with credit, inputs and marketing support. • Collective procurement and marketing improve farmers' bargaining position.
Dharmapuri Integrated Farmer Producer Company – Tamil Nadu	• Farmers collectively undertake activities around integrated farming and marketing. • Pooling resources helps small farmers access better markets.

Committee Recommendations for Farmer Producer Organisations (FPOs)

Committee / Report	Recommendations for FPOs
Y.K. Alagh Committee (2000)	• Promote producer companies so farmers can combine the benefits of cooperatives with professional business management. • Give farmer organisations greater autonomy and commercial orientation.

National Commission on Farmers (Swaminathan Commission)	• Promote group-based approaches to overcome the disadvantages of small and fragmented holdings. • Strengthen farmers' access to markets, credit, technology and value addition through collective institutions.

"Unity is strength. And with this collective power, the farmers will also progress towards prosperity. The collective strength of the farmers will now be used to get fair prices to the farmers. Today the same idea runs behind the new FPOs i.e. Farmer Producer Organizations started in Chitrakoot."

-**Prime Minister Narendra Modi**

Conclusion

FPOs can help small farmers **pool their strength, improve incomes and participate more effectively in agricultural value chains**. The focus should now be on **collective action, professional management, market integration and value addition**. Strong FPOs can become a foundation for **farmer-led growth, rural transformation and inclusive prosperity**.

5. AGRICULTURAL MARKETING IN INDIA

Context:

India has solved production but not price realisation — e-NAM now covers 1,656 mandis and ₹4.84 lakh crore of trade, yet a farmer still loses 20–25% of output post-harvest and pays over 8% in mandi levies in some states. The unresolved question is whether the fix is a GST-Council-style national market, as the **2024 draft policy proposes**, or state-specific strengthening of APMCs.

Introduction

Agricultural marketing is the **bridge between the farmer and the consumer**, helping farm produce reach the right market at the right time and price. A strong marketing system can **reduce middlemen, cut post-harvest losses and improve farmers' income**.

Components of Agricultural Marketing:

- **Marketable Surplus:** The quantity of farm produce left after meeting the farmer's own household and farm needs and available for sale in the market.
- **Collection & Aggregation:** Produce from small farmers is collected at **village-level centres, FPOs, cooperatives and mandis** to create larger marketable lots.
- **Grading & Standardisation:** Produce is **cleaned, sorted, graded and classified** according to quality, size and other standards to ensure fair pricing.
- **Storage & Warehousing:** Warehouses, silos and cold storages allow farmers to **store produce safely and avoid distress sales** immediately after harvest.
- **Transportation:** Roads, railways and other transport systems help move agricultural produce **from farms and collection centres to markets and consumers**.
- **Market Information:** Information on **prices, demand, quality and market conditions** helps farmers decide where and when to sell.
- **Price Discovery:** Platforms such as **APMC markets, e-NAM, cooperatives and direct marketing** help determine the price of agricultural produce.

- **Financing & Credit:** Timely credit helps farmers **meet immediate needs and avoid selling produce at low prices due to urgent cash requirements**.
- **Processing & Value Addition:** Food processing, packaging and branding increase the **value and shelf life** of agricultural products.
- **Marketing Channels:** Produce can move through **APMC mandis, FPOs, cooperatives, processors, retailers, exporters or directly to consumers**.

Significance of Agricultural Marketing:

- **Better Price for Farmers**
  - **Fair returns:** Efficient markets help farmers get a better and more transparent price for their produce.
  - **Less distress sale:** Farmers can avoid selling immediately after harvest when prices are usually low.
- **Wider Market Access**
  - **More buyers:** Platforms like **e-NAM, FPOs and direct marketing** connect farmers with buyers beyond the local market.
  - **Better opportunities:** Farmers can reach larger markets and sell according to demand.
- **Reduces Middlemen Dependence**
  - **Fewer layers:** Direct links between farmers, processors and consumers can reduce unnecessary intermediaries.
  - **Higher farmer share:** A more efficient chain can help farmers receive a larger share of the final price.
- **Encourages Crop Diversification**
  - **New crops:** Better markets encourage farmers to grow **fruits, vegetables, pulses, oilseeds and other high-value crops**.
  - **Less cereal dependence:** Assured markets can help reduce excessive dependence on rice and wheat.
- **Promotes Value Addition**
  - **Processing:** Agricultural produce can be processed, packaged and branded before reaching consumers.
  - **Higher income:** Value addition increases the economic value of farm produce and creates new income opportunities.
- **Strengthens Food Processing**
  - **Regular supply:** Efficient marketing provides industries with a steady supply of quality raw materials.
  - **Rural industries:** It supports the growth of food-processing units closer to production areas.
- **Creates Rural Employment**
  - **New jobs:** Activities such as **grading, packaging, storage, transportation and processing** generate rural employment.

Primary bodies governing and facilitating agricultural marketing in India

- **Policy, Regulatory & Administrative Bodies**
  - **National Institute of Agricultural Marketing:** An autonomous organization under the Ministry of Agriculture based in Jaipur. It handles policy research, capacity building, advisory services, and specialized training for agricultural marketing personnel and farmers.
- **State-Level Regulatory Bodies**
  - **State Agricultural Marketing Boards (SAMBs):** Apex state-level bodies (e.g., Punjab Mandi Board, MSAMB in Maharashtra) that oversee, build and maintain physical marketing infrastructure across district and regional levels.

  - **Agricultural Produce Market Committees (APMCs):** Statutory market committees established by state governments under their respective APMC Acts. They administer and regulate physical wholesale markets (*mandis*) to ensure fair auctions, transparent weighing, and timely payments.
- **Price Support & Market Intervention Bodies**
  - **Food Corporation of India (FCI):** A statutory corporation under the Ministry of Consumer Affairs, Food and Public Distribution. It conducts bulk open-market operations, procures food grains (wheat and paddy) at **Minimum Support Prices (MSP)**.
  - **Commission for Agricultural Costs and Prices (CACP):** An advisory body that evaluates production costs, market demand, and supply dynamics to recommend MSPs for 22 mandated agricultural crops.
  - **Cotton Corporation of India (CCI) & Jute Corporation of India (JCI):** Central public sector undertakings responsible for conducting price support operations and procurement for cotton and raw jute, respectively, when market prices fall below MSP.
- **Cooperative & Trade Aggregation Bodies**
  - **National Agricultural Cooperative Marketing Federation of India (NAFED):** The apex organization of marketing cooperatives in India. It serves as a nodal procurement agency under the **PM-AASHA** scheme for pulses, oilseeds, and copra.
  - **National Cooperative Development Corporation (NCDC):** A statutory corporation that provides financial and technical assistance to cooperative societies for setting up marketing, processing, storage, and cold chain infrastructure.
  - **Small Farmers' Agribusiness Consortium (SFAC):** A specialized society under the Ministry of Agriculture that promotes agribusiness development, acts as the nodal agency for implementing the **10,000 FPO Scheme**.
- **Export Promotion & Specialized Commodity Boards**
  - **APEDA (Agricultural and Processed Food Products Export Development Authority):** A statutory body under the Ministry of Commerce and Industry responsible for promoting the export of fresh fruits, vegetables, processed foods, and livestock products.
  - **MPEDA (Marine Products Export Development Authority):** Oversees market promotion, cold chain logistics, and quality standardization for seafood and marine exports.
  - **Statutory Commodity Boards:** Specialized boards functioning primarily under the Ministry of Commerce and Industry that promote the production, domestic marketing, and export of specific plantation and cash crops:
    - *Tea Board of India*
    - *Coffee Board of India*
    - *Spices Board of India*
    - *Rubber Board*
    - *Tobacco Board*
    - *National Turmeric Board & National Makhana Board*

Issues:

- **Poor Market Infrastructure**
  - **Inadequate mandis:** Many markets lack proper storage, grading, weighing and basic facilities.
  - **Limited facilities:** Farmers may have to travel long distances to access a well-functioning market.

*According to the **National Centre for Cold-Chain Development (NCCD)**, India faces a **75% deficit in reefer trucks** (fewer than 15,000 available against a national requirement of 62,000+).*

- **Too Many Intermediaries**
  - **Long supply chain:** Produce often passes through several traders before reaching consumers.
  - **Lower farmer share:** Each layer can add costs, while the farmer may receive a smaller share of the final price.
- **Distress Sale**
  - **Weak bargaining power:** Lack of storage and limited buyers can force farmers to accept low prices.
- **Weak Transportation & Cold Chain**
  - **High transport cost:** Poor roads and expensive logistics increase the cost of taking produce to markets.

    *As per **NITI Aayog**, agricultural logistics costs in India account for **13–14% of the total product value** (compared to 8–9% in developed economies).*

  - **Perishable produce:** Fruits, vegetables, dairy and fish can lose quality quickly without a proper cold chain.

    *According to **Reserve Bank of India (RBI) studies**, farmers receive only **28% to 35%** of the consumer rupee for high-value perishable crops.*

- **Lack of Market Information**
  - **Price gap:** Farmers may not know the prices available in different markets.
  - **Information delay:** Limited access to timely information on demand and quality reduces their bargaining power.
- **Unequal Access to APMC Markets**
  - **Regional imbalance:** Well-developed markets are concentrated in certain regions, while many areas have weaker market infrastructure.
  - **Small farmers:** Marginal farmers may find it difficult to transport small quantities to distant mandis.
- **Quality & Grading Problems**
  - **Poor standardisation:** Lack of proper grading and quality testing can lead to different prices for similar produce.

Major Initiatives for Agricultural Marketing:

- **e-NAM (National Agriculture Market):** Connects APMC mandis through an online platform and enables more transparent price discovery. It helps farmers access buyers beyond their local mandi.
- **Agricultural Produce Market Committee (APMC) Reforms:** States are encouraged to improve mandi infrastructure and make agricultural trade more competitive.
- **Agriculture Infrastructure Fund (AIF):** Provides financing support for warehouses, cold stores, grading units and other agricultural infrastructure.
- **PM Kisan SAMPADA Yojana:** Supports cold chains, processing and preservation infrastructure. It helps agricultural produce move from the farm to consumers with less wastage.
- **Operation Greens:** Supports better marketing and supply-chain infrastructure for key horticultural crops.
- **AGMARK:** Provides standards for grading and quality certification of agricultural products.
- **Agricultural Marketing Infrastructure (AMI):** Provides assistance for developing and upgrading agricultural marketing infrastructure.
- **Gramin Agricultural Markets (GrAMs):** Aims to improve rural market infrastructure and connect farmers with larger markets.
- **National Cooperative Development Corporation Support :** Supports cooperatives involved in agricultural marketing, storage and processing.

Way Forward:

- **Strengthen Market Infrastructure**
  - **Better mandis:** Improve facilities for weighing, grading, storage, packaging and transparent trading.
  - **More markets:** Develop more regulated and well-connected markets closer to farming areas.
- **Promote Farmer Producer Organisations (FPOs)**
  - **Collective selling:** Help small farmers pool their produce and sell in larger quantities.
  - **Better bargaining:** Strong FPOs can negotiate better prices and reduce dependence on middlemen.
- **Expand Digital Marketing**
  - **Wider markets:** Strengthen platforms like **e-NAM** so farmers can reach buyers beyond their local mandi.

    *Karnataka – **Unified Market Platform (UMP),** pioneering digital platform that connected APMCs and enabled **online bidding and transparent price discovery**.*
  - **Price information:** Provide timely information on **prices, demand and market conditions** through mobile and digital platforms.
- **Build Strong Cold Chains**
  - **Farm-to-market link:** Connect farms with **collection centres, cold storage, refrigerated transport and markets**.

    ***Nashik grape cluster, Maharashtra:** Farmers and exporters use **pack-houses, pre-cooling, cold storage and refrigerated logistic**s to maintain grape quality from farm to international markets.*
- **Improve Rural Transport**
  - **Lower logistics cost:** Promote shared transport through FPOs and cooperatives to make movement cheaper.

    ***Kisan Rail**, provides faster movement of agricultural and perishable produce from production regions to distant markets.*
- **Promote Grading and Standardisation**
  - **Quality-based pricing:** Strengthen scientific grading and testing so farmers are rewarded for better-quality produce.
  - **Consumer trust:** Standardisation can improve transparency and make Indian produce more competitive.

Best Practices in Agricultural Marketing (India & Global)

Best Practice Model	Strategy
Rythu Bazars (Andhra Pradesh/Telangana) & Uzhavar Sandhai (Tamil Nadu)	Bypasses commission agents and middlemen by establishing direct farmer-to-consumer markets managed by state authorities.
Sahyadri Farms Model (Nashik, Maharashtra)	A farmer-owned company uniting 18,000+ smallholders to build a shared, world-class cold chain, grading hub, and export supply network.
Bhavantar Bhugtan Yojana (Madhya Pradesh)	A Price Deficiency Payment Scheme (PDPS) where the state pays farmers the direct cash difference between MSP and the actual lower market selling price.

e-Choupal Network (ITC)	Places internet-enabled kiosks in villages to provide real-time local mandi prices, weather forecasts, and direct procurement services at farm gates.
Almería Greenhouse Cooperative Model (Spain)	Thousands of smallholders aggregate output through producer organizations (APOs) to own shared logistics, auction hubs, and cold chains.
Amul's aggregator model of cooperative farming	The Amul model of cooperative farming was based on establishing a number of collection centers which collected milk from the dairy farmers, aggregated them and sold it to the consumers.

Conclusion

A strong agricultural marketing system can help farmers **get better prices, reduce losses and reach wider markets**. The focus should be on **FPOs, digital markets, better storage, transport and value addition**. This will make Indian agriculture more **competitive, farmer-friendly and supportive of a prosperous Bharat**.

6. AGRICULTURAL PRODUCE MARKET COMMITTEES

Introduction:

APMCs are **regulated agricultural markets** where farmers can sell their produce through an organised system. They aim to provide farmers with **fair price discovery, transparent trading and protection from unfair practices**. APMCs act as an important **link between farmers, traders and consumers** in India's agricultural marketing system.

Objectives of Agricultural Produce Market Committees:

- **Ensure Fair Prices:** APMCs aim to help farmers get a **fair and competitive price** for their produce through regulated market transactions.
- **Protect Farmers from Exploitation:** They seek to protect farmers from **unfair traders, under-weighing, delayed payments and other malpractices**.
- **Promote Transparent Trading:** APMCs provide a regulated platform where **auctioning, weighing and pricing** can happen in a more transparent manner.
- **Provide Better Market Facilities:** They provide basic facilities such as **auction yards, weighing centres, storage, loading and unloading areas** to make trading easier.
- **Improve Price Discovery:** By bringing buyers and sellers together, APMCs help in **discovering a more competitive market price** for agricultural produce.
- **Ensure Timely Payments:** A key aim is to ensure that farmers **receive payment properly and on time** after selling their produce.
- **Regulate Market Participants:** APMCs regulate **traders, commission agents and other market intermediaries** to maintain orderly agricultural trade.
- **Provide Market Information:** They can help farmers understand **prevailing prices, demand and market conditions**, enabling better selling decisions.
- **Build Trust in Agricultural Trade:** Overall, APMCs aim to create a **more organised, transparent and farmer-friendly agricultural marketing system**.

Issues with APMCs:

- **Limited Competition**
  - **Few buyers:** In some mandis, a small number of traders dominate buying, leaving farmers with less bargaining power.
- **Too Many Intermediaries**
  - **Long chain:** Produce may pass through several traders and commission agents before reaching consumers.
  - **Lower farmer share:** Each layer can add costs and reduce the farmer's share of the final price.
- **High Marketing Costs**
  - **Fees and commissions:** Market fees, commissions and other charges can increase the cost of selling produce.
- **Limited Market Access**
  - **Distance problem:** Small farmers may have to travel long distances to reach major APMC markets.
  - **High transport cost:** Transporting small quantities can make selling in distant mandis less profitable.
- **Regional Imbalance**
  - **Uneven development:** Some states have well-developed APMC networks, while others have weaker market infrastructure.
  - **Farmer disadvantage:** Farmers in poorly connected regions may have fewer selling options.
- **Technology Gap**
  - **Limited digitalisation:** Some mandis still have limited use of **digital auctions, online payments and real-time price information**.
  - **Information gap:** Farmers may not have access to the real time information of commodities in other market price before selling.

Measures taken for APMCs

- **Legal & Legislative Reforms**
  - **Model APMC Act (2003):** Introduced by the Centre as a template for states to allow private markets and permit direct sales from farmers to bulk buyers.
  - **Model APLM Act (2017):** Upgraded the 2003 model by declaring the entire state as a "single unified market," capping mandi fees, and separating the regulatory role of the board from its commercial operations.
- **Digital & Technological Integration**
  - **National Agriculture Market (e-NAM):** Launched in 2016, this pan-India electronic portal connects physical APMC mandis online, allowing buyers anywhere in the country to bid for a farmer's produce based on quality.
  - **Digital Assaying & Automated Payments:** Equipping mandis with third-party quality testing labs and integrating electronic payment gateways.
- **Post-Harvest & Storage Reforms**
  - **Warehouses as Virtual Mandis:** Amending state laws to declare WDRA-accredited warehouses, silos, and cold storages as **Market Sub-Yards.**
  - **Agriculture Infrastructure Fund (AIF):** Providing ₹1 Lakh Crore in concessional credit to build sorting, grading, assaying, and cold chain infrastructure directly inside or near APMC yards.
  - **e-NWR Pledging:** Enabling farmers to store crops in registered godowns, receive Electronic Negotiable Warehouse Receipts

Way Forward:

- **Increase Competition**
  - **More buyers:** Allow more traders, processors, retailers and private buyers to participate in APMC markets.
  - **Better price discovery:** Greater competition can help farmers get a more competitive price.
- **Modernise Mandi Infrastructure**
  - **Cold-chain support:** Add cold storage and refrigerated facilities for fruits, vegetables and other perishables.

    Vashi APMC, Maharashtra: *One of India's major agricultural markets, with organised facilities for* ***trading, storage and distribution***
- **Promote Digital Markets**
  - **e-NAM integration:** Connect more APMCs with **e-NAM** to expand the number of buyers and markets available to farmers.

    Guntur APMC (Andhra Pradesh) e-NAM Model: *The Guntur Chili APMC—one of Asia's largest dry-chili trading hubs—successfully digitized operations through* ***e-NAM integration****.*
  - **Digital payments:** Encourage transparent online bidding and quick digital payments.
- **Give Farmers More Selling Choices**
  - **Direct marketing:** Allow farmers to sell directly to **processors, retailers, cooperatives and consumers** where suitable.
  - **Multiple channels:** APMC should become **one option among many**, rather than the only route to market.
- **Strengthen FPOs**
  - **Collective selling:** Help FPOs aggregate produce from small farmers and sell larger quantities through organised markets.
  - **Better bargaining:** Collective marketing can improve farmers' negotiating power with traders and buyers.
- **Improve Market Information**
  - **Real-time prices:** Provide farmers with simple and timely information on prices across nearby markets.
  - **Demand information:** Information on demand and quality requirements can help farmers decide **when and where to sell**.

    AGMARKNET: *Provides market-related information on* ***commodity prices and arrivals****, helping farmers compare prices across different markets.*

Best Practices in Agricultural Produce Market Committees (APMCs)

Best Practice Model & Location	Mechanism
Sirsa APMC (Haryana) & Nizamabad APMC (Telangana)	Mandates rapid AI-based and chemical assaying laboratories at mandi entry gates to test moisture, oil content, and purity.
FPO Trading Module on e-NAM (Pan-India)	Allows Farmer Producer Organizations (FPOs) to upload digitized crop images and quality assay parameters directly from rural aggregation centers.

Vashi & Private Horticultural Mandis (Maharashtra)	Permits private entities to set up alternative modern wholesale market yards alongside traditional state APMCs.
Uzhavar Sandhai (Tamil Nadu) & Rythu Bazars (Telangana/AP)	Dedicated government-managed urban yards where smallholders sell directly to consumers, bypassing traders and commission fees.
Azadpur Mandi Cold Chain Hub (Delhi)	Integrated controlled-atmosphere (CA) cold stores, temperature-controlled ripening chambers, and electronic weighbridges at the mandi core.

Conclusion:

APMCs can help farmers get **fair prices, transparent trade and better market access**. The way forward is to **modernise mandis, increase competition and give farmers more choices to sell**. A strong and farmer-friendly APMC system can make agricultural marketing more **efficient, inclusive and supportive of Samriddh Bharat**.

7. TRANSPORTATION AND LOGISTICS OF AGRICULTURAL PRODUCE

Introduction:

Efficient transportation and logistics help move agricultural produce **from farms to markets quickly, safely and at low cost**. A strong logistics system reduces **post-harvest losses, transport delays and unnecessary costs**, while helping farmers get better prices. For India, better farm-to-market connectivity is key to **higher farmer incomes, food security and a stronger agricultural value chain**.

Major Components of Agri-Logistics

- **First-Mile Collection & Aggregation:**
  - **Farm-Gate Handling:** Initial harvesting, cleaning, basic sorting, and grading right at the farm.
  - **Pack-Houses & Aggregation Hubs:** Local collection centers where smallholders bring their produce to consolidate small batches into larger bulk shipments.
- **Storage & Warehousing:**
  - **Dry Storage / Godowns:** Ventilated structures and bulk steel silos for non-perishables like grains, pulses, and oilseeds.
  - **Cold Storages:** Temperature-controlled chambers and Controlled Atmosphere (CA) stores to extend the shelf-life of fruits, vegetables, and flowers.
- **Cold Chain & Temperature Management:**
  - **Pre-Cooling Facilities:** Rapidly removing field heat right after harvest to slow down decay.
  - **Reefer Transport:** Refrigerated trucks, container vans, and specialized rail cars (*e.g., Kisan Rail*) to keep temperature-sensitive goods cool during transit.
- **Multi-Modal Transportation Networks:**
  - **Road Freight:** Small commercial vehicles for local village-to-mandi links; heavy trucks for long-haul inter-state movement.
  - **Rail & Waterways:** Used for low-cost, bulk movement of grains over long distances.

  - **Air Freight**: Reserved for high-value, highly perishable exports like fresh flowers, berries, and exotic vegetables.
- **Packaging & Cargo Handling:**
  - **Protective Packaging:** Use of ventilated plastic crates, corrugated boxes, jute bags, and modified atmosphere packaging to prevent physical bruising and moisture loss.
  - **Material Handling Equipment:** Conveyor belts, forklifts, and grain elevators to reduce manual labor damage during loading and unloading.
- **Market & Distribution Infrastructure:**
  - **Primary Mandis & Wholesale Markets:** Physical distribution centers where price discovery occurs.
  - **Processing & Retail Hubs:** Final routing to food processing plants, urban hypermarkets, or export seaports/airports.

Significance of Logistics of Agricultural Produce

- **Economic & Farmer-Centric Significance**
  - **Drives Farmer Income:** Enables farmers to access distant, higher-paying markets instead of relying on localized distress sales at low farm-gate prices.
  - **Reduces Post-Harvest Losses:** Prevents heavy financial leakage by moving delicate, perishable produce (fruits, vegetables, dairy) rapidly through cold corridors.
  - **Lowers Transaction Costs:** Streamlined logistics reduce fuel consumption, transit delays, and multi-handling charges, cutting the overall cost of doing business for smallholders.
  - **Eliminates Middlemen Dependence:** Gives farmers direct physical and digital access to bulk buyers, food processors, and urban wholesale mandis.
- **Food Security & Consumer Well-Being**
  - **Price Stabilization:** Smooth supply movement absorbs seasonal production gluts and prevents sharp food inflation spikes in consumer markets.
  - **National Food Distribution:** Essential for moving bulk food grains (wheat, rice) from surplus regions (e.g., Punjab, Haryana) to deficit states via networks like Indian Railways.
  - **Preserves Nutritional Quality:** Fast, temperature-controlled logistics ensure fresh, nutrient-rich food reaches urban plates without losing quality.
- **Macro-Economic & Industry Linkages**
  - **Feeds the Food Processing Sector:** Supplies a steady, reliable flow of raw agricultural materials to food processing industries, agro-exporters, and retail chains.
  - **Encourages Crop Diversification:** Reliable cold transport encourages farmers to shift away from low-value water-guzzling cereals toward high-value cash crops (horticulture, floriculture, dairy).
  - **Boosts Agricultural Exports:** Meets strict international delivery timelines and phytosanitary temperature requirements, driving high-value agri-exports.

Issues with Transportation and Logistics of Agricultural Produce

- **Poor Rural Connectivity**
  - **Bad roads:** Poor-quality rural roads make transportation slow, costly and sometimes damaging for farm produce.

    *According to **Ministry of Rural Development**, over **1.5 lakh rural habitations** still lack all-weather road connectivity.*
  - **Weak last-mile connectivity:** Many farmers find it difficult to move produce from the farm to the nearest market, warehouse or collection centre.
- **High Transportation Costs**

- **Rising expenses:** Fuel, vehicle maintenance and freight costs reduce the farmer's final income.

  *As per **NITI Aayog**, agricultural logistics costs in India account for **13–14% of the total product value** (compared to 8–9% in developed economies).*

- **Small quantities:** Small farmers often have limited produce, making individual transportation expensive.

- **Weak Cold-Chain Facilities**
  - **Perishable produce:** Fruits, vegetables, milk, fish and meat can spoil quickly if they are not transported under proper temperature conditions.
  - **Limited refrigerated transport:** Lack of enough refrigerated vehicles and storage facilities increases wastage.

  *According to the **National Centre for Cold-Chain Development (NCCD)**, India faces a **75% deficit in reefer trucks** (fewer than 15,000 available against a national requirement of 62,000+).*

- **Post-Harvest Losses**
  - **Transport delays:** Long waiting and transit times can reduce the quality and freshness of agricultural produce.
  - **Poor handling:** Improper loading, unloading, packaging and storage can cause physical damage and spoilage.

  As per MoFPI-NABCONS Study, transit delays cause annual post-harvest quantitative losses of 6.02%–15.05% in fruits, 4.87%–11.61% in vegetables, and 3.89%–6.74% in cereals.

- **Fragmented Supply Chain**
  - **Poor coordination:** Farmers, traders, warehouses, transporters and processors often work separately, creating delays.
  - **Multiple intermediaries:** Too many stages between the farmer and consumer can increase costs and reduce the farmer's share.

  According to the RBI Report on Agricultural Produce Marketing, the presence of 4 to 6 intermediaries in traditional APMC supply chains eats up 30% to 50% of the consumer rupee.

- **Seasonal Transport Pressure**
  - **Harvest-time rush:** Large quantities of produce reach markets during the same period, creating pressure on transport facilities.
  - **Higher freight rates:** Shortage of vehicles during peak harvest seasons can increase transportation costs

  As per the Ministry of Agriculture & Farmers Welfare, peak harvest seasons witness an estimated 30% to 50% spike in local freight rates.

Measures taken:

- **Rail & Multi-Modal Freight Initiatives**
  - **Kisan Rail Services:** Special multi-commodity, temperature-controlled trains operated by Indian Railways to transport perishables like fruits, vegetables, milk, and fish from farms to markets.
  - **Dedicated Freight Corridors (DFCs):** The construction of high-speed Eastern and Western Dedicated Freight Corridors significantly reduces transit time and congestion for long-distance agricultural cargo.
  - **Freight Subsidies under Operation Greens:** A 50% freight transport subsidy provided by the government to prevent distress sales and reduce transit costs for notified fruits and vegetables.
- **Air & Cargo Expansion Measures**
  - **Krishi UDAAN Scheme:** A targeted initiative to provide time-bound, cost-effective air freight and cargo handling facilities for agricultural, horticultural, and organic produce.

  - **Focus on Remote & Hilly Belts:** Special priority given under Krishi UDAAN 2.0 to connect farmers in the North-Eastern Region, tribal belts, and Himalayan states to national and international airport cargo terminals.
  - **Air Cargo Hubs with Cold Storage:** Setting up dedicated perishable cargo centers and cold storage infrastructure at major airports to support high-value horticultural exports.
- **First-Mile & Rural Infrastructure Interventions**
  - **PM Gati Shakti National Master Plan:** A multi-modal digital platform integrating roads, railways, ports, and aviation to map and eliminate physical connectivity gaps.
  - **Agriculture Infrastructure Fund (AIF):** A ₹1 Lakh Crore financing facility providing interest subvention and credit guarantees to build farm-gate pack-houses and primary collection hubs.
  - **Gramin Agricultural Markets (GrAMs):** Physical and digital upgrading of rural periodic markets to serve as modern aggregation centers
- **Digital & Tech-Enabled Logistics**
  - **Kisan Rath Mobile Application:** A digital platform connecting farmers and Farmer Producer Organizations (FPOs) directly with network providers of trucks, tractors, and refrigerated transport vehicles.
  - **Unified Logistics Interface Platform (ULIP):** Integrates digital data from railways, highways, ports, and aviation onto a single platform to enable real-time tracking and faster freight movement.
  - **e-NAM Logistics Integration:** Linking the electronic National Agriculture Market (e-NAM) with private logistics service providers so farmers can arrange transport directly upon selling produce online.
- **Policy & Institutional Frameworks**
  - **National Logistics Policy (NLP):** Aimed at reducing overall logistics costs, eliminating regulatory bottlenecks, and shifting freight from road-dominated transport toward efficient rail and multi-modal networks.
  - **Multi-Modal Logistics Parks (MMLPs):** Setting up integrated logistics hubs across the country that combine warehousing, automated cargo handling and direct highway-rail access.

Way Forward:

- **Improve Rural Connectivity**
  - **Better roads:** Strengthen rural roads so farmers can move produce quickly and safely to markets and storage centres.

    China's "Taobao Villages" & Rural Highway Expansion Project *constructed paved all-weather roads directly linking remote agricultural villages to logistics hubs.*
  - **Last-mile connectivity:** Connect farms with collection centres, warehouses, cold stores and mandis.
- **Reduce Transport Costs**
  - **Shared transport:** Encourage FPOs and cooperatives to use common vehicles and logistics services to reduce costs.
- **Strengthen Cold-Chain Infrastructure**
  - **Farm-to-market chain:** Connect farm, collection centre, cold storage, refrigerated transport and market in one smooth system.
- **Promote Digital Logistics**
  - **Real-time information:** Use digital platforms to provide farmers with information on prices, demand and transport availability.

    USA's FreightWaves & Uber Freight Agri-Logistic*s platforms match farm shipments with empty returning trucks in real-time, providing dynamic pricing and GPS temperature tracking via mobile apps.*
- **Support Farmer Producer Organisations**
  - **Collective marketing:** FPOs can aggregate produce from many small farmers and arrange transport at lower cost.

  - **Better bargaining:** Collective selling can improve farmers' bargaining power and market access.
- **Improve Handling and Packaging**
  - **Scientific packaging:** Promote proper grading, sorting and packaging to reduce damage during transportation.
  - **Modern loading:** Use better loading and unloading systems to minimise physical losses.
- **Promote Multimodal Transport**
  - **Better connectivity:** Link road, rail, waterways and ports wherever suitable to reduce logistics costs.
  - **Faster movement:** Use the most suitable transport mode according to the type and distance of produce.
- **Build Local Logistics Hubs**
  - **Near production areas:** Develop collection and aggregation centres close to farms.
  - **Integrated facilities:** Combine storage, grading, packaging, processing and transport at these hubs.

Recommendations from various committee

- **Ashok Dalwai Committee (Doubling Farmers' Income, 2017)**
  - **Focus on Agri-Logistics Corridors:** Recommends developing dedicated **Agri-Logistics Corridors** and **Perishable Express Freights** (which paved the way for *Kisan Rail*) to connect high-density production clusters directly to major urban consuming hubs.
  - **Essential Infrastructure Status:** Recommended declaring farm-gate pack-houses, reefer fleets, and cold hubs as **Essential Infrastructure**, granting them priority power tariffs and long-term concessional credit.
- **Shanta Kumar Committee (Restructuring of FCI, 2015)**
  - **Containerization & Bulk Grain Handling:** Recommended moving away from traditional, labor-intensive bag-storage transport to **bulk handling via automated steel silos and containerized rail wagons**.
  - **Outsourcing to Private Players:** Advised outsourcing long-haul grain transport and multi-modal handling to private logistics operators under Public-Private Partnership (**PPP**) models.

Conclusion:

Efficient transportation and logistics can ensure that **farm produce reaches the consumer fresh, quickly and at a fair cost**. Better roads, cold chains, digital logistics and FPO-led transport can reduce **wastage and improve farmers' income**. A strong **farm-to-market network** can make Indian agriculture more competitive, resilient and farmer-friendly.

8. AGRICULTURAL EXPORTS

Context:

India's agricultural exports touched a **record USD 52.55 billion in FY 2025-26**, up from USD 51.12 billion in FY 2024-25 and USD 32.08 billion in FY 2014-15. Growth was, however, only -2.8%, **signalling a plateau** despite the record.

Introduction

Agricultural exports connect **Indian farmers to global markets**, creating opportunities to earn more from farm produce. They help increase **farmer income, foreign exchange earnings and value addition** in agriculture. With better quality, processing and global market access, India can move from **farm to market** towards **farm to world.**

Current status of Agricultural Exports

- **Share in Total Merchandise Exports:** Agri-exports contribute roughly **11% to 12%** of India's total merchandise export basket.
- **Share in Global Trade:** India accounts for ~2.2% to 2.5% of global agricultural trade, placing it among the Top 10 global agri-exporters (alongside the US, EU, Brazil, and China).
- **Top Destinations:** United States, United Arab Emirates (UAE), Saudi Arabia, China, Bangladesh, Vietnam, Iraq, Malaysia, and Indonesia.

Major Agricultural Export Commodities

Commodity/Category	India's Global Export Rank	Major Markets & Destinations
Rice *(Basmati & Non-Basmati)*	#1 Globally *(>40% Global Market Share)*	Saudi Arabia, Iran, Iraq, UAE, Benin, Bangladesh, West Africa.
Marine Products	Top 5 Globally	USA, China, European Union, Japan, Vietnam.
Spices	#1 Globally	USA, China, UAE, Bangladesh, UK, Malaysia.
Buffalo Meat *(Carabeef)*	Top 2 Globally	Vietnam, Malaysia, Egypt, Indonesia, Saudi Arabia.
Sugar	#2 Globally	Indonesia, Bangladesh, Sudan, UAE, Malaysia.
Processed Foods & Preparations	Fastest Growing	USA, Middle East, UK, Nepal, Australia.
Raw Cotton	Top 3 Globally	Bangladesh, Vietnam, China, Indonesia.
Fresh Fruits & Vegetables	Top 10 Globally	UAE, Bangladesh, Saudi Arabia, UK, Netherlands.
Coffee	Top 7 Globally	Italy, Germany, Russia, Belgium, UAE.

Significance of Agricultural Exports

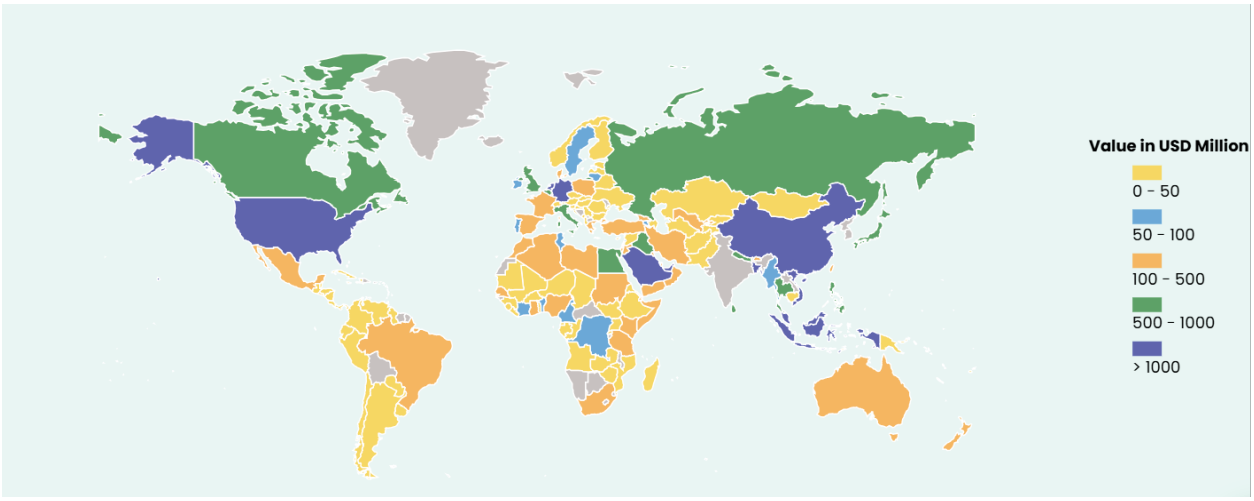
- **Higher Farmer Income**
  - **More buyers:** Exports open up larger markets and create more opportunities for farmers to earn better returns.
  - **Better demand:** Strong global demand can improve the value of Indian agricultural produce.
- **Foreign Exchange Earnings**
  - **Export income:** Agricultural exports bring foreign exchange into the country.
  - **Economic strength:** They contribute to India's trade earnings and overall economic growth.
- **Wider Market Access**
  - **Global reach:** Farmers can benefit from demand beyond local and domestic markets.
  - **Market diversification:** Selling to different countries reduces dependence on a single market.
- **Promotes Value Addition**
  - **Processing:** Export demand encourages grading, processing, packaging and branding.
  - **Higher returns:** Value-added products can generate more income than selling raw produce.
- **Encourages Crop Diversification**
  - **High-value crops:** Export opportunities encourage production of spices, fruits, vegetables, tea, coffee and other high-value products.

- **Better choices:** Farmers get incentives to move beyond traditional cereal-based farming.

- **Creates Rural Employment**
  - **Farm jobs:** Higher production for exports can increase employment in agriculture. Processing, packaging, storage, transport and logistics create employment.

Major Institutional Drivers of Agricultural Exports

- **APEDA:** Promotes exports of agricultural and processed food products and helps Indian exporters access global markets.
- **Ministry of Commerce & Industry:** Frames export policies and works to improve India's overall agricultural export ecosystem.
- **Ministry of Agriculture & Farmers Welfare:** Supports better production, productivity and quality of agricultural commodities needed for export.
- **MPEDA:** Promotes exports of **marine products** and supports quality, processing and international market access.
- **FIEO:** Helps exporters, including agri-exporters, connect with international buyers and understand global markets.
- **Export Inspection Council (EIC):** Supports **quality control, inspection and certification**, which are important for meeting international standards.



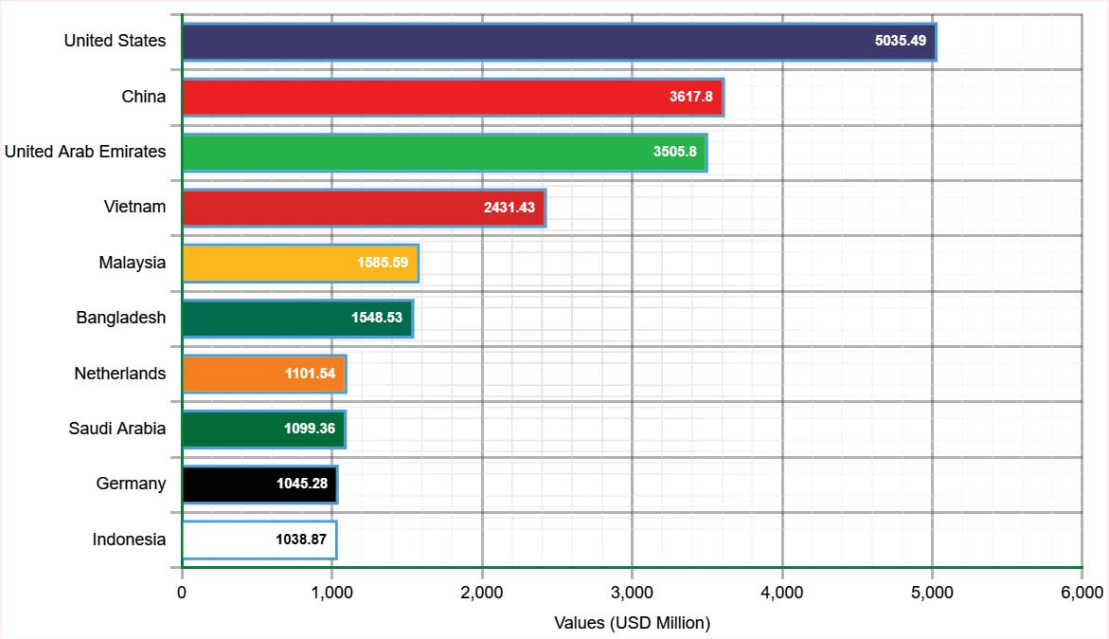
Issues with Agricultural Exports

- **Unstable Global Prices**
  - **Price fluctuations:** Changes in global prices can quickly affect the income of Indian farmers and exporters.
  - **Market uncertainty:** Farmers may hesitate to produce export-oriented crops when prices are unpredictable.
- **Export Restrictions**
  - **Policy changes:** Sudden bans, export duties or minimum export prices can create uncertainty for farmers and traders.
  - **Loss of markets:** Frequent changes can make foreign buyers look for more reliable suppliers.
- **Poor Quality Standards**
  - **Quality gap:** Some Indian produce may not consistently meet international standards for quality and food safety.

  - **Rejection risk:** Export consignments may face rejection due to excessive pesticide residues or contaminants, such as **tricyclazole in Basmati rice, ethylene oxide in spices, and aflatoxins in groundnuts**.
- **High Logistics Costs**
  - **Expensive transport:** Freight, handling and port-related costs can make Indian agricultural products less competitive.

    *As per **NITI Aayog & Economic Survey Data**, logistics costs in Indian agriculture account for **13% to 14% of the total product value** (compared to 8% to 9% in developed nations)*
  - **Last-mile problems:** Poor connectivity from farms to collection centres increases the overall cost.
- **Small and Fragmented Farms**
  - **Low exportable surplus:** Small farmers often produce limited quantities, making direct export difficult.

    *According to the **10th Agriculture Census**, **86.1% of operational holdings in India** are small and marginal (average size 1.08 ha)*
  - **Aggregation problem:** Weak farmer aggregation makes it harder to ensure uniform quality and regular supply.
- **Limited Value Addition**
  - **Raw produce focus:** A large share of agricultural exports remains in relatively low-value or less-processed forms.
  - **Missed opportunity:** Limited processing, branding and packaging reduces the potential value earned from exports.



Agriculture Export Policy (AEP), 2018

The **Agriculture Export Policy (AEP), 2018** was launched by the Government of India to **increase India's farm exports and help Indian agricultural products reach global markets**.

- **Major Objectives**
  - **Double Agri-Exports:** Aimed to boost agricultural exports from ~$30 Billion to $60 Billion (and eventually target $100 Billion in the following years).

- **Double Farmers' Income:** Connect Indian farmers directly to high-paying international markets so they get better price realization.
- **Diversify the Basket & Destinations:** Move beyond just rice and marine products to export value-added items, organic foods, and indigenous crops to new countries.
- **Stable Policy Regime:** Eliminate sudden, unpredictable export bans so foreign buyers can trust Indian suppliers over the long run.

Way Forward for Agricultural Exports

- **Ensure Stable Export Policy**
  - **Policy certainty:** Avoid sudden export bans, duties and restrictions so farmers and exporters can plan production with confidence.
- **Improve Quality Standards**
  - **Better quality:** Strengthen testing, grading and certification to meet international food-safety standards.

    *APEDA's Traceability Systems, such as **HortiNet**, help monitor the production and export of horticultural products*
  - **Traceability:** Build a clear **farm-to-export traceability system** to improve trust among global buyers.

    ***Basmati rice traceability under APEDA** helps track produce from farms through processing to export.*
- **Promote Value Addition**
  - **Process more:** Encourage food processing, packaging, branding and certification instead of exporting only raw produce.
- **Strengthen FPOs**
  - **Aggregate produce:** FPOs can bring together small farmers and create larger, uniform lots for export.

    ***Sahyadri Farmers Producer Company, Maharashtra**, has successfully aggregated grapes from farmers and connected them with domestic and international markets.*
  - **Better bargaining:** Collective marketing can improve farmers' bargaining power with exporters and processors.
- **Promote Export-Oriented Clusters**
  - **Crop-specific areas:** Develop clusters around products such as **mango, grapes, spices, tea and marine products**.

    ***Ratnagiri–Sindhudurg** for Alphonso mango, **Nashik** for grapes, and **Karnataka/Kerala** for coffee and spices.*
  - **Common facilities:** Provide shared facilities for **grading, testing, packaging, storage and processing**.

    ***Agricultural Export Clusters under the Agriculture Export Policy** can provide common infrastructure and connect farmers with exporters.*

Conclusion:

Agricultural exports can help Indian farmers **reach bigger markets, earn better and create more value from their produce**. The focus should be on **stable policies, better quality, strong logistics, value addition and wider global markets**. With the right support, India can move **from being a major producer to a trusted global supplier**, strengthening **food security.**

MGP/BP/GS3/033

E-TECH IN AID OF FARMERS

1. E-TECHNOLOGY TO THE AID OF FARMERS: OVERVIEW

Introduction

E-technology in agriculture refers to the use of digital tools such as mobile phones, the internet, satellites, drones, sensors, and Artificial Intelligence (AI) to support farming. It helps farmers make better decisions, improve productivity, reduce costs, and connect directly with markets, making agriculture more efficient and sustainable.

Significance of E-Technology in Agriculture

- **Improves Farm Productivity:** Farmers receive timely information on weather, soil health, irrigation, and crop management, helping them achieve better yields.
- **Enables Precision Farming:** Technologies like drones, GPS, sensors, and AI ensure the right amount of water, fertilizers, and pesticides are used, reducing waste and improving efficiency.
- **Reduces Cost of Cultivation:** Better planning and efficient use of inputs lower production costs while increasing farm profitability.
- **Strengthens Risk Management:** Early warnings about weather changes, pest attacks, and crop diseases help farmers take preventive measures and reduce losses.
- **Improves Market Access:** Digital platforms connect farmers directly with buyers, provide real-time price information, and reduce the role of middlemen.
- **Expands Access to Financial Services:** Online platforms make it easier for farmers to access crop insurance, institutional credit, subsidies, and Direct Benefit Transfers (DBT).
- **Promotes Sustainable Agriculture:** Efficient use of water, fertilizers, and pesticides conserves natural resources and reduces environmental damage.
- **Supports Better Governance:** Digital land records, satellite monitoring, and online beneficiary databases improve transparency and ensure schemes reach the intended farmers.
- **Increases Farmers' Income:** Higher productivity, lower input costs, better market access, and informed decision-making together improve farmers' earnings.

Digital Agriculture Mission (DAM)

- **Basics & Core Objective**
  - **Launch & Nodal Ministry:** Approved in September 2024 as an overarching umbrella scheme under the Ministry of Agriculture and Farmers Welfare (with an outlay of over ₹2,800 Crore).
  - **Core Vision:** To build a modern, technology-driven agricultural ecosystem in India by creating Digital Public Infrastructure (DPI) for agriculture—similar to what UPI did for digital payments.

- **Primary Objective:** Deliver seamless, real-time services to farmers (advisories, credit, insurance, and market access) using AI, remote sensing, GIS mapping, and big data.

- **Three Core Pillars (Digital Public Infrastructure for AgTech)**
  - **AgriStack (Farmer-Centric Registry)**
    - **Farmer ID:** Creates a unique digital identity (like Aadhaar) for every farmer, linked to their land records, crop details, and demographic data.
    - **Digital Crop Survey:** Replaces manual state-level crop reporting with real-time, geotagged crop survey data collected via mobile apps and satellite imagery.
  - **Krishi Decision Support System (Krishi-DSS)**
    - Integrates spatial data on weather, soil health cards, drought conditions, groundwater levels, and reservoir capacities.
    - Acts as a central intelligence hub for governments and research bodies to issue early warnings (droughts, pest attacks, heatwaves).
  - **Soil Profile Mapping**
    - Generates high-resolution digital soil maps across agricultural land to provide targeted fertilizer and micro-nutrient recommendations.

Issues

- **High Upfront Costs**
  - **Expensive Technology:** Devices like drones, GPS-enabled tractors, IoT sensors, and smart farming equipment are costly, making them difficult for many farmers to purchase.
- **Poor Rural Connectivity**
  - **Weak Internet Connectivity:** Many villages still have poor or unreliable internet, making it difficult for farmers to use digital farming tools and online services.

    *As per the **Telecom Regulatory Authority of India**, rural tele-density stands at approximately **59%** compared to over **140% in urban areas.***
  - **Frequent Network Failures:** Smart farming technologies depend on stable internet and mobile networks. Poor connectivity can interrupt their functioning and reduce their effectiveness.
- **Steep Learning Curve**
  - **Difficult to Use:** Many digital farming apps and devices are complicated, making them hard to use for farmers who are not familiar with technology.

    *According to **NSSO,** Only about 24% to 25% of rural households or individuals demonstrate functional computer ownership or advanced digital proficiency.*
  - **Lack of Training:** Many farmers do not receive proper training, making it difficult for them to use new technologies effectively and benefit from them.
- **Data Privacy Concerns**
  - **Uncertainty Over Data Ownership:** Farmers are often unsure who owns the data collected from their farms and how it will be used.
  - **Fear of Misuse of Data:** Many farmers worry that companies may use or share their farm data for commercial purposes without their knowledge or consent.

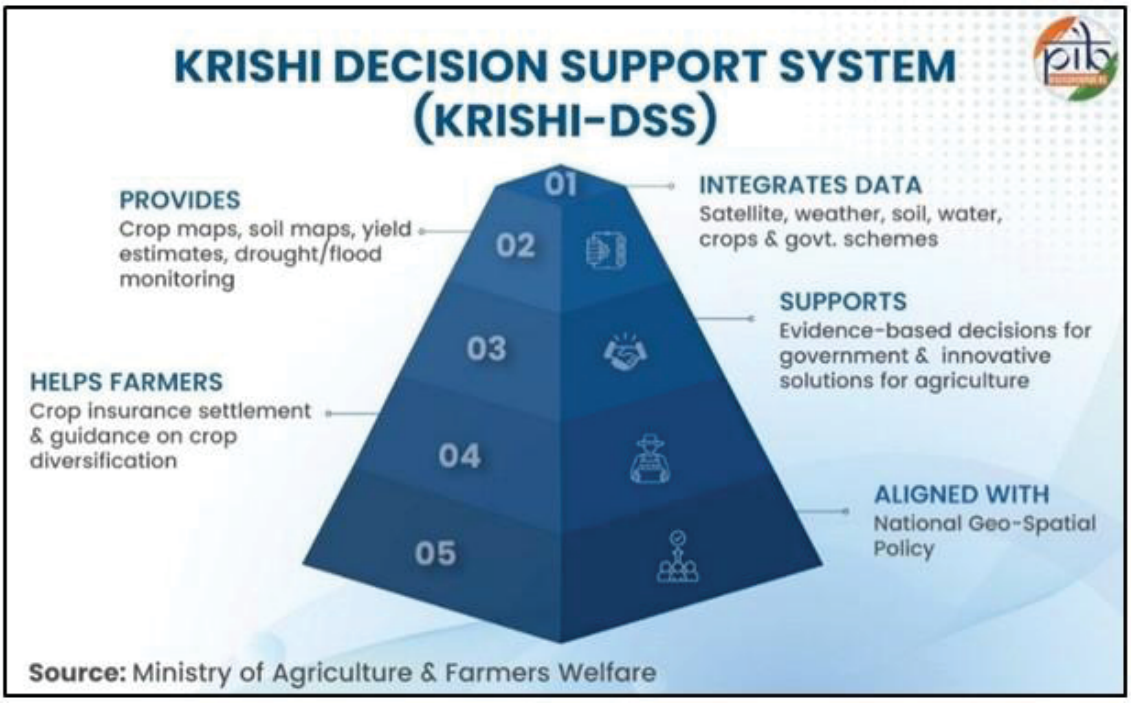
    ***John Deere's repair and software policies** have raised concerns globally over farmers' control and use of data generated by smart farm machines.*
- **Limited Right to Repair**

- **Restricted Repairs:** Many modern farm machines use locked software and proprietary parts, preventing farmers from repairing their own equipment.
- **Higher Costs and Delays:** Farmers often have to wait for authorized service centres, increasing repair costs and causing delays during crucial sowing or harvesting seasons.

- **Government Measures to Promote E-Technology in Agriculture**
  - **Digital Agriculture Mission (2021–2025):** Promotes AI, drones, GIS, IoT and digital databases in agriculture.

    The Digital Agriculture Mission, launched in 2024 with a total outlay of ₹2,817 crore and an allocation of ₹54.972 crore for FY 2025–26
  - **AgriStack:** Builds a digital ecosystem with unique digital identities for farmers and seamless service delivery.

    Over 7.63 crore IDs have been generated as of 27 November 2025, including 1.93 crore IDs for women farmers.
  - **National e-Governance Plan in Agriculture (NeGPA):** Provides digital advisory, weather updates, market information, and online services.
  - **Digital Public Infrastructure (DPI) for Agriculture:** Integrates farmer, land, and crop data to improve policy implementation and service delivery.
  - **e-NAM (National Agriculture Market):** Connects mandis through an online trading platform for better price discovery.
  - **Kisan Suvidha & Meghdoot Apps:** Offer weather forecasts, market prices, crop advisories, and pest management information.
  - **Drone Promotion Scheme:** Financial assistance for the use of drones in crop monitoring, spraying, and precision farming.
  - **Soil Health Card Scheme:** Uses digital soil testing to provide crop-specific nutrient recommendations.
  - **PM-Kisan & DBT:** Digital transfer of financial assistance directly into farmers' bank accounts.
  - **Common Service Centres (CSCs):** Help farmers access digital agricultural services in rural areas.



Way forward

- **Bridging the Digital Divide**
  - **Improve Rural Internet Connectivity:** Expand broadband and mobile network coverage in villages so farmers can easily use digital farming tools and online services.

    BharatNet Project *– Connecting over 200,000 Gram Panchayats with optical fiber to bring high-speed broadband to rural farm belts.*
  - **Develop Offline-Friendly Apps:** Promote applications that can work even without an internet connection and automatically update when the network becomes available.
- **Lowering Financial Barriers**
  - **Promote Rental Services:** Expand custom hiring centres and digital rental platforms so small farmers can rent drones, tractors, and other modern equipments.

    FARMS App (Custom Hiring Centres) – A government-backed platform enabling smallholders to hire tractors, harvesters, and drones on a pay-per-use basis.
  - **Provide Better Financial Support:** Offer targeted subsidies and easy credit for affordable technologies such as soil sensors, drip irrigation systems, and other smart farming tools.
- **Building Local Capacity and Digital Skills**
  - **Strengthen Local Training:** Train village youth and agricultural extension workers to help farmers use digital technologies and solve basic technical problems.

    Digital Farmer Ambassadors (Vietnam/FAO) – Rural youth are trained as digital champions to educate local rice farmers on using smart Agri-apps.
  - **Promote Simple, Local-Language Tools:** Develop voice-based digital services and AI assistants in regional languages so farmers can easily access information without needing advanced digital skills.
- **Ensuring Data Protection**
  - **Protect Farmers' Data:** Create clear laws to ensure that farmers have ownership and control over their farm data and that it is not misused without their consent.

    EU Code of Conduct on Agricultural Data Sharing – A non-binding agreement guaranteeing that farmers retain rights over the data generated on their farms.
  - **Strengthen the Right to Repair:** Ensure that farmers and local mechanics can repair modern farm machinery easily, reducing repair costs and delays during farming seasons.
- **Designing Farmer-Friendly Technology**
  - **Develop Technology with Farmers:** Involve farmers in designing and testing digital tools so that they are simple, practical and meet real farming needs.

    Living Labs Initiative (Canada/EU) – Collaborative innovation hubs where farmers, scientists, and software developers co-design precision agriculture tools in actual field conditions.
  - **Focus on Everyday Problems:** Give priority to useful features such as weather alerts, pest and disease warnings, and real-time market prices.

Case studies for E-Technology in Agriculture

Case Study	Result
Namo Drone Didi Scheme	Trained rural women SHG members to fly drones. Reduced pesticide usage by up to 20–30% and water usage by 80–90%, while creating new livelihoods for women.

Plantix App (Telangana / Andhra Pradesh)	Farmers take a photo of a sick plant on their phone; AI diagnoses the disease in seconds and prescribes exact, cheap treatments in local languages.
Digital Green Video Models (Bihar / Odisha)	Local community members film short videos of successful local farmers sharing best practices, then show them at village meetings via pico-projectors.
MahaAgriTech Project (Maharashtra)	Uses ISRO satellites and drones to track crop health, estimate yield, and assess disaster damage accurately in days instead of months.
Kisan e-Mitra Chatbot (Central Govt)	A multi-lingual AI voice bot that lets farmers ask questions about scheme payouts (like PM-KISAN) in their spoken local dialects via phone calls or WhatsApp.
CropIn SmartFarm Platform (Madhya Pradesh)	Geotags farms and uses weather sensors to send customized alerts on when to sow, irrigate, or harvest.

Conclusion

E-technology has the potential to transform Indian agriculture by making farming more productive, profitable, and sustainable. However, its success depends on affordable access, strong digital infrastructure, and farmer-friendly solutions. **Tools like Bharat-VISTAAR and Kisan e-Mitra provide multilingual, real-time advisory services, making expert knowledge accessible even in remote areas and further enhancing decision-making.** .

2. BIOTECHNOLOGY IN AGRICULTURE

Introduction:

Biotechnology in agriculture uses **modern biological techniques** to make farming more productive, resilient and sustainable. It helps develop **better seeds, improve crop nutrition, control pests and reduce dependence on chemicals**, especially under changing climatic conditions.

Applications of Biotechnology in Agriculture

- **Development of Improved Crop Varieties:** Biotechnology helps develop varieties with better yield, quality and nutritional value.
- **Disease resistance:** Biotechnology can help develop varieties resistant to **viral, bacterial and fungal diseases**, reducing crop losses.
- **Biofortification:** Crops can be enriched with essential **vitamins, minerals and proteins**.
- **Tissue Culture & Rapid Multiplication**: Tissue culture allows large numbers of **disease-free and genetically uniform plants** to be produced quickly.
- **Production of Biofertilisers**: Microorganisms such as **Rhizobium, Azotobacter and phosphate-solubilising bacteria** improve nutrient availability to plants.
- **Biopesticides**: Microbial agents such as **Trichoderma, Bacillus thuringiensis (Bt) and NPV** can control pests and diseases.
- **Marker-Assisted Breeding**: DNA markers help scientists identify desirable traits at an early stage.
- **Genome Editing**: Technologies such as **CRISPR-Cas** can make targeted changes in plant genes. It can help develop crops with **higher productivity, better nutrition, stress tolerance and improved shelf life**.

- **Agricultural Waste Management**: Biotechnology can convert crop residues and other agricultural waste into **biogas, biofuels, compost and other useful products**.

Significance of Biotechnology in Agriculture

- **Higher Farm Productivity**
  - **Better crop varieties:** Biotechnology helps develop crops with higher yield potential, better quality and improved nutritional value.

    HD-3226 (Pusa Yashasvi) *wheat variety yielding up to 7 tonnes/ha with multi-disease resistance.*
  - **Faster crop improvement:** Techniques like marker-assisted selection and genetic engineering can speed up the development of improved varieties.

    MAS-derived Basmati varieties (Pusa Basmati 1885/1886) *developed via Marker-Assisted Selection for rapid gene pyramiding against bacterial leaf blight and blast.*
- **Climate-Resilient Farming**
  - **Stress tolerance:** Crops can be developed to withstand drought, salinity, heat and floods, which are becoming more frequent due to climate change.

    Swarna Sub1 *rice (survives 14 days of total submergence/floods) and* ***CR Dhan 108 / CSR 101*** *(drought and coastal salinity tolerant).*
  - **Stable production:** Such crops can reduce yield losses during weather shocks and strengthen climate-resilient agriculture.

    Short-duration climate-resilient pulses like ***DAPM-1*** *mungbean maturing in under 60 days to evade terminal heat stress*
- **Reduces Pest and Disease Losses**
  - **Natural protection:** Technologies such as Bt crops provide resistance against specific insect pests.

    Bt Cotton (Bollgard II) *expressing Cry1Ac and Cry2Ab genes providing native resistance against American Pink Bollworm.*
  - **Lower chemical use:** Better pest resistance can reduce dependence on chemical pesticides, lowering cultivation costs and environmental damage.

    Pusa Basmati 1718*, engineered for resistance against Bacterial Leaf Blight, reducing farmer expenditure on chemical bactericides by over 30%.*
- **Better Nutritional Security**
  - **Biofortification:** Biotechnology can improve the nutritional content of crops by increasing vitamins, minerals and proteins.
  - **Hidden hunger:** It can help tackle micronutrient deficiencies, especially among vulnerable rural populations.

    Pusa Biofortified Maize Hybrids 1, 2 & 4 *containing high Lysine and Tryptophan amino acids to address rural protein-energy malnutrition.*
- **Efficient Use of Resources**
  - **Less water and chemicals:** Stress-tolerant and input-efficient varieties can help farmers use water, fertilisers and pesticides more efficiently.
  - **Sustainable agriculture:** This supports the shift from input-intensive farming towards resource-efficient agriculture.

Do you know?

- **Bt Cotton:** Remains the **only commercially approved GM crop** in India (approved in 2002).
- **Acreage & Market Penetration:** According to **ICAR & Ministry of Agriculture Data**, Bt Cotton covers over **93% of India's total cotton cultivation area**.
- **GM Mustard (DMH-11):** Recommended for environmental release for seed production by the GEAC in October 2022; commercial rollout remains pending under judicial review in the Supreme Court.
- **Nano-Fertilizers:** Developed via public-private biotech initiatives, **Nano Urea** and **Nano DAP** (liquid formulations by IFFCO) have received official approval under the **Fertilizer Control Order (FCO**) to reduce bulk synthetic chemical usage.

Achievements of biotechnology in India

- **Transgenic Crops**
  - **Bt Cotton Revolution:** Introduced in 2002 using Cry1Ac and Cry2Ab genes, it cut pesticide use by ~40% and turned India into one of the world's largest cotton exporters.
- **Genome Editing**
  - **First Commercial Genome-Edited Crops:** India released its first non-transgenic genome-edited rice varieties, DRR Dhan 100 Kamala and Pusa DST Rice 1.
- **Marker-Assisted Selection (MAS) & Biofortification**
  - **Combatting Hidden Hunger:** ICAR has developed over 100+ biofortified crop varieties using non-transgenic molecular markers:
    - **High-Zinc & Iron Rice:** DRR Dhan 45 and Pusa 45.
    - **High-Protein Maize:** HQPM-1 and HQPM-5.
    - **Biofortified Pearl Millet:** Pusa 1201 (rich in iron and zinc).
- **Climate-Resilient & Abiotic Stress Adaptations**
  - **Climate smart crops:** Developed SAATVIK chickpea and Arun rice, tailored specifically to maintain yields under heat and moisture stress.
  - **Submergence Tolerance:** Introduced the Sub1A gene into popular rice varieties (e.g., Swarna-Sub1, ADT 39-Sub1), allowing plants to survive 14+ days completely underwater during severe flooding.
- **Indigenous Bio-Inputs & Diagnostics**
  - **Biopesticides & Bio-fertilizers:** Mass-produced native strains like Trichoderma (fungal control) and Rhizobium/Azotobacter cultures to reduce chemical dependence.
  - **Diagnostic Kits:** Developed low-cost DNA strip assays and ELISA kits for rapid, field-level identification of plant viruses and bacterial blights.
  - **Kisan-Kavach:** Kisan Kavach is an anti-pesticide protective suit designed to address the widespread issue of pesticide-induced toxicity in agriculture.
- **Issues with Biotechnology in Agriculture**
  - **Safety & Environmental Concerns**
    - **Ecological impact:** Genetically modified crops may have unintended effects on biodiversity, beneficial insects and soil ecosystems.

      *Post-Bt Cotton adoption, non-target pests like whiteflies and mealybugs caused severe infestations affecting up to **20–30% of cotton acreage** in Northern India (Punjab and Haryana).*

  - **Gene flow:** Modified genes may spread to wild relatives or nearby crops, creating concerns about genetic contamination.

- **Health Concerns**
  - **Food safety:** Some people remain concerned about the possible allergenicity and long-term health effects of genetically modified foods.
  - **Need for evidence-based regulation:** Safety assessment must be based on scientific testing, transparency and continuous monitoring rather than either blind acceptance or outright rejection.
- **Dependence on Private Companies**
  - **Seed concentration:** Patented biotechnology can increase farmers' dependence on a limited number of seed companies.
  - **Intellectual Property Rights:** Strong IPR protection may raise the cost of seeds and create concerns about farmer autonomy and seed sovereignty.
- **Weak Regulatory Capacity**
  - **Complex regulation:** Biotechnology involves agriculture, health, environment and trade, making regulation difficult and fragmented.
  - **Slow approvals:** Lengthy approval processes can delay the adoption of potentially useful technologies.

    *India has approved **only 1 GM crop for commercial cultivation** (Bt Cotton in 2002) in over 20 years.*
  - **Need for stronger institutions:** India needs science-based, transparent and predictable biosafety regulation.

    *As per economic survey, State-level enforcement bodies like State Biotechnology Coordination Committees (SBCCs) and District Level Committees (DLCs) remain non-functional in **over 70% of districts.***
- **Biodiversity Concerns**
  - **Monoculture risk:** Large-scale adoption of a few high-performing varieties can reduce agrobiodiversity.

    *Cultivation of a few commercial Bt cotton hybrids expanded to cover **>93% of India's total cotton area***
  - **Loss of traditional varieties:** Indigenous seeds and traditional crop varieties may gradually disappear, weakening India's genetic resource base.
- **Unequal Access to Technology**
  - **Regional divide:** Advanced biotechnology is more likely to reach farmers with better infrastructure, irrigation, extension services and market access.

    As per NITI Aayog Agri-Index,** Adoption of advanced agricultural biotechnology is concentrated in **5–6 well-irrigated states
  - **Digital and knowledge gap:** Farmers may lack the technical knowledge needed to understand and properly use new technologies.

    ***As per NSSO 77th Round, Over 88% of farming households** do not receive technical guidance on agricultural technology management*

Genetic Engineering Appraisal Committee

Genetic Engineering Appraisal Committee (GEAC) is the statutory apex body in India responsible for the appraisal and approval of activities involving large-scale use of hazardous microorganisms and recombinants in research, including the commercial release of **Genetically Modified (GM) organisms and crops**. It operates under the **Ministry of Environment, Forest and Climate Change (MoEFCC)**.

- **Statutory & Governance Framework**
  - **Statutory Basis:** Constituted under the **Rules for the Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells, 1989** (popularly called Rules, 1989), framed under the **Environment (Protection) Act, 1986**.

- **Nomenclature Shift:** Originally named the Genetic Engineering Approval Committee, it was renamed the Genetic Engineering Appraisal Committee in 2010.
- **Composition:** Chaired by a Special Secretary / Additional Secretary of MoEFCC, co-chaired by a representative from the Department of Biotechnology (DBT), and comprises experts from ICAR, ICMR, CSIR, CPCB, and various ministries.

- **Major Functions & Mandate**
  - **Commercial Release Approval:** Evaluates proposals for environmental release and commercial clearance of GM crops (e.g., Bt Cotton, GM Mustard, Bt Brinjal).
  - **Industrial & Large-Scale Approvals:** Authorizes large-scale use, import, export, and production of genetically engineered organisms and products in research, healthcare, and industrial sectors.
  - **Risk & Biosafety Assessment:** Reviews bio-safety data, field trial results, environmental impact assessments, and food/feed safety evaluations submitted by developers.
  - **Power to Direct & Stop Activities:** Holds statutory authority to inspect, investigate, and issue directives or prohibition orders if GM activities violate biosafety norms under the EPA, 1986.
- **Multi-Tiered Regulatory Architecture in India**
  - **Institutional Biosafety Committee (IBSC):** Operates at the institute/research lab level.
  - **Review Committee on Genetic Manipulation (RCGM):** Under DBT; monitors small-scale research and contained field trials.
  - **Genetic Engineering Appraisal Committee (GEAC):** Under MoEFCC; apex body for environmental release and large-scale applications.
  - **State Biotechnology Coordination Committee (SBCC):** State-level monitoring and enforcement body.
  - **District Level Committee (DLC):** District-level monitoring and compliance verification.
  - **Recombinant DNA Advisory Committee (RDAC):** Advisory body on global biotech developments.

Measures taken by government for biotechnology in agriculture

- **ioE3 Policy (2024):** Short for Biotechnology for Economy, Environment, and Employment, this policy sets up **Biofoundries** and **Bio-AI Hubs** to scale biomanufacturing. It prioritizes climate-resilient farming, bio-fertilizers, and smart proteins.
- **Bio-RIDE Scheme:** Consolidates DBT initiatives under a unified ₹1,500 crore budget to bridge the gap between academic lab research and commercial farming tools.
- **National Biotechnology Development Strategy:** Sets clear targets to expand India's bioeconomy toward $300 billion, focusing on modern crop genetics and bio-based farm
- **Eased Rules for Gene Editing (2022 Guidelines):** Exempted non-transgenic crops (SDN-1 and SDN-2 categories—which make precise tweaks without adding foreign DNA) from multi-year GEAC trials.
- **Stacked Events Guidelines:** Streamlined safety assessments when multiple useful traits (like drought resistance and high yield) are combined into a single seed line.
- **BRIC-NABI Institute:** Created the National Agri-Food Bio-Manufacturing Institute to convert bench research into scalable agricultural products faster.
- **Biotech-KISAN Scheme:** Connects agricultural scientists directly with small and marginal farmers, especially in drought-prone and aspirational districts.
- **BIRAC Seed Grants:** The Biotech Industry Research Assistance Council funds agri-biotech startups developing low-cost biopesticides, bio-fertilizers, and field-level diagnostic kits.
- **ICAR National Missions:** Funding major projects like the ICAR Genome Editing Project, involving 24 ICAR institutes across India working on 24 distinct food crops.

Way Forward:

- **Strengthen Science-Based Regulation**
  - **Clear approval system:** Make biotechnology approvals transparent, time-bound and science-based without compromising safety.
  - **Strong biosafety checks:** Conduct rigorous assessment of human health, soil, biodiversity and ecological impacts before large-scale adoption.
- **Invest in Indian Research**
  - **Public R&D:** Strengthen institutions such as ICAR, agricultural universities and public research laboratories to develop technologies suited to Indian conditions.
  - **Focus on local problems:** Research should prioritise drought, salinity, heat stress, pests, diseases and nutritional deficiencies affecting Indian farmers.
- **Keep Farmers at the Centre**
  - **Affordable access:** Ensure that improved seeds and biotechnology-based inputs remain affordable for small and marginal farmers.
  - **Farmer participation:** Involve farmers in field trials and technology development so that innovations reflect ground-level needs.
- **Protect Biodiversity**
  - **Gene banks & native varieties:** Conserve traditional seeds and indigenous crop varieties through community seed banks and national gene banks.
  - **Prevent gene flow:** Maintain appropriate buffer zones, monitoring systems and refuge strategies wherever required.
- **Improve Extension & Awareness**
  - **Simple communication:** Farmers should receive clear information about benefits, limitations, usage and possible risks of new technologies.
  - **Demonstration farms:** Krishi Vigyan Kendras and extension agencies can use field demonstrations and farmer training to build confidence.

Conclusion

Biotechnology can make Indian agriculture **more productive, climate-resilient and nutrition-secure**. The focus should be on using science not just to increase yields, but also to **reduce farmers' costs and protect natural resources**. With **strong safety checks, affordable access and farmers at the centre**, biotechnology can become an important pillar of **sustainable agriculture**.

3. ARTIFICIAL INTELLIGENCE IN AGRICULTURE

Why in the news?

The **Union Budget 2026-27** proposed Bharat-VISTAAR, a multilingual AI tool to integrate the AgriStack portals and the ICAR package with AI system.

Introduction:

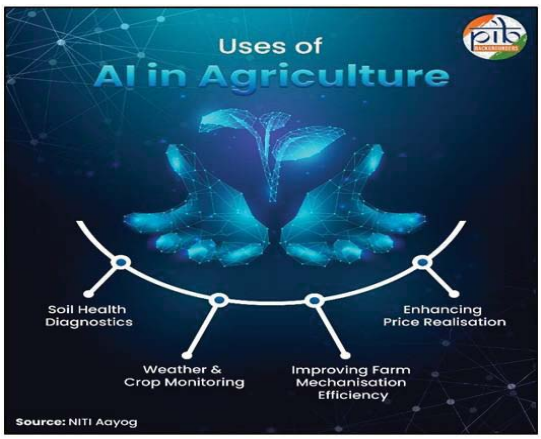
Artificial Intelligence (AI) in agriculture means using smart technologies to make farming decisions **faster, more accurately and with less waste**. It can help farmers with **weather forecasting, pest detection, crop monitoring, irrigation and market-price information**. AI can make Indian agriculture **more productive, efficient and climate-resilient**, while supporting better farm incomes.

What is Artificial Intelligence?

Artificial Intelligence (AI) is the ability of machines to perform tasks that normally require human intelligence. It enables systems to learn from experience, adapt to new situations, and solve complex problems independently. AI uses datasets, algorithms, and large language models to analyse information, recognise patterns, and generate responses. Over time, these systems improve their performance, allowing them to reason, make decisions, and communicate in ways similar to humans.

Components of Artificial Intelligence in Agriculture

- **Smart Crop Monitoring**
  - **Satellite & drone images:** AI analyses images to monitor crop health, growth and stress.
  - **Early detection:** It can identify diseases, pests and nutrient deficiencies at an early stage.
- **Weather & Climate Prediction**
  - **Weather forecasting:** AI uses weather and historical data to provide more accurate rainfall, temperature and drought predictions.
- **Precision Farming**
  - **Right input, right amount:** AI helps decide the precise amount of water, fertiliser and pesticides needed.
  - **Less wastage:** This improves input-use efficiency and can reduce cultivation costs.
- **Smart Irrigation**
  - **Water management:** AI-based systems use soil moisture, weather and crop data to decide when irrigation is needed.
  - **Water saving:** This helps avoid both over-irrigation and water stress.
- **Pest & Disease Management**
  - **Image-based diagnosis**: AI can identify crop diseases and pest attacks using mobile or camera images.
  - **Targeted action:** Farmers can take early and more focused measures instead of spraying chemicals everywhere.
- **AI-Based Farm Machinery**
  - **Automation:** AI can support autonomous tractors, robotic harvesters, drones and smart sprayers.
  - **Labour support:** It can reduce dependence on manual labour, especially during sowing and harvesting peaks.
- **Market & Price Intelligence**
  - **Price prediction**: AI can analyse market trends and help farmers understand prices, demand and supply.
  - **Better selling decisions:** Farmers can make better choices about when, where and to whom to sell.



Issues with Artificial Intelligence in Agriculture

- **High Cost of Technology**
  - **Expensive tools:** Drones, sensors, smart machines and AI-based systems can be costly for small and marginal farmers.

    *A DGCA-type certified commercial spraying drone costs between **₹4.5 Lakh and ₹10 Lakh** (e.g., Garuda's Kisan drone starting around ₹4.5 Lakh; Marut AG 365 listing up to ₹9.89 Lakh)*
  - **Maintenance cost:** Regular repair, software updates and technical support can further increase the cost.
- **Digital Divide**
  - **Poor connectivity:** Weak internet and mobile connectivity in rural areas can limit the use of AI-based services.

    *According to **TRAI (Q1 2026 data)**, internet subscriber density in rural areas sits at **48.31 per 100 people** (440.87 million subscribers) compared to **126.80 per 100 people** in urban areas.*
  - **Limited digital access:** Many farmers may not have smartphones or other devices needed to use advanced technologies.
- **Lack of Skills & Awareness**
  - **Technology gap:** Farmers may not have the technical knowledge needed to understand and use AI tools.

    According to the NSSO 77th Round (Land & Livestock Survey), over 88% of farming households receive no formal technical extension or training.
  - **Weak extension support:** Lack of trained local personnel makes it difficult to provide **farm-level guidance**.
- **Poor Quality of Data**
  - **Limited farm data:** AI needs reliable data on **soil, weather, crops and markets**, which is often incomplete or scattered.
  - **Local accuracy:** A model trained using data from one region may not work equally well in another region.
- **Small & Fragmented Landholdings**
  - **Low affordability:** Individual investment in AI tools may not make economic sense on very small farms.

    According to the 10th Agriculture Census, 86.1% of operational landholdings are small and marginal (averaging 1.08 hectares).
  - **Need for collective use:** Technologies such as drones and sensors may need to be shared through **FPOs or Custom Hiring Centres**.
- **Data Privacy & Ownership**
  - **Ownership concerns:** Farmers may have limited control over how their data is stored, shared or commercially used.
- **Employment Concerns**
  - **Labour displacement:** Greater automation may reduce demand for some **routine agricultural jobs**.

Major Takeaways

- India has created a large-scale digital foundation for agriculture with over 7.63 crore Farmer IDs and 23.5 crore crop plots surveyed under the Digital Agriculture Mission.
- National Pest Surveillance System supports 66 crops and over 432 pest types, providing real-time advisories to more than 10,000 extension workers for early pest detection.

- As of December 2025, the Kisan e-Mitra chatbot has answered more than 93 lakh queries, handling over 8,000 farmer queries daily in 11 regional languages.
- An AI-based pilot for local monsoon onset forecasting for Kharif 2025 reached 3.88 crore farmers across 13 states via SMS, with 31–52% of surveyed farmers adjusting sowing and land preparation decisions based on the forecasts.

Reforms taken for Artificial Intelligence (AI) into Indian agriculture.

- **Building Digital Public Infrastructure (DPI)**
  - **Digital Agriculture Mission (DAM):** Approved with an outlay of **₹2,817 Crore** as an umbrella framework to deploy big data, remote sensing, and AI/ML across state farm networks.
  - **AgriStack ("Kisan ki Pehchaan"):** A national digital registry that creates unique **11-digit Farmer IDs** linked with geo-referenced village land maps and crop records.
  - **BharatVistaar:** A multilingual, AI-powered Digital Public Infrastructure platform that offers real-time location-specific crop advisories in over 10 regional languages via web, mobile, and chatbots.
- **AI-Driven Smart Farmers' Advisories**
  - **National Pest Surveillance System (NPSS):** An AI- and machine-learning-driven platform that allows field workers and farmers to upload smartphone images of crops.
  - **Kisan e-Mitra Chatbot:** An AI-powered, voice-enabled multi-lingual assistant developed to answer farmer queries instantly regarding scheme benefit transfers like PM-KISAN and Kisan Credit Cards (KCC).
  - **AI-Based Monsoon & Weather Advisories:** Integration of Indian Meteorological Department (IMD) historical data with AI algorithms to send hyper-local rainfall and sowing timeline alerts.
- **Smart Crop Insurance & Yield Estimation**
  - **YES-TECH & CROPIC (Under PMFBY):** Utilization of AI algorithms, satellite imagery, and drone observations to estimate crop damage and yield cuts rapidly.
  - **Digital Crop Survey (DCS):** Deployment of AI image-recognition technology to verify crop sowing types across millions of farm.
- **Precision Farming & Agritech Ecosystem**
  - **Subsidies for AI-Powered Drones:** Grants under the **Kisan Drone Scheme** and **Namo Drone Didi** program to encourage drone-based ultra-low volume precision spraying of nano-fertilizers and pesticides.
  - **MeiTY & Academic Agritech Hubs:** Setting up specialized Centers of Excellence (CoEs)—such as the **AgriHub at IIT Indore**—to mentor agri-tech startups developing AI computer-vision systems, smart irrigation sensors, and automated weed-removal robotics.
  - **State-Level AI Policies:** Initiatives like Maharashtra's **MahaAgri-AI Policy** that deploy localized voice assistants and AI crop-disease diagnostic tools.

AI-Enabled Precision Farming

The experience of Rajaratnam Kanakarajan illustrates the practical and scalable application of artificial intelligence in Indian agriculture. By adopting an AI-enabled precision farming system developed by Farm Again, a Tamil Nadu-based startup, he leveraged solar-powered sensors to monitor soil moisture, irrigation, and fertilizer use in real time through a mobile platform. The system automated farm operations, reduced over-irrigation and input use, and optimised crop conditions, resulting in a doubling of coconut yields.

Way Forward for Artificial Intelligence in Agriculture

- **Make AI Affordable**

- **Lower the cost:** Provide targeted support and easy credit for **AI tools, sensors, drones and smart machinery**.
- **Shared access:** Promote **FPOs and Custom Hiring Centres (CHCs)** so small farmers can use expensive technologies without buying them individually.

- **Improve Rural Digital Infrastructure**
  - **Better connectivity:** Strengthen **internet, mobile networks and electricity** in rural areas.

    *Using support from the **Universal Service Obligation Fund (USOF)** to extend 4G mobile connectivity to **uncovered rural areas**.*
  - **Last-mile access:** Take AI-based agricultural services to villages through **mobile and voice-based platforms**.

    ***Hello Tractor (Kenya/Nigeria)** uses AI algorithms to aggregate smallholder demand and rent out tractor-sharing services via a mobile pay-per-use platform.*
- **Build Farmer Skills**
  - **Digital literacy:** Train farmers to understand and use AI-based advisories and farm technologies.

    Training women Self-Help Groups (SHGs) as drone pilots and agritech operators under the Namo Drone Didi Scheme.
  - **Local support:** Strengthen **KVKs, extension workers and FPOs** to provide hands-on guidance.
- **Protect Farmer Data**
  - **Data ownership:** Clearly define who owns and controls farmers' **land, crop and production data**.
  - **Privacy by design:** Ensure farmers know **what data is collected, why it is collected and how it will be used**.
- **Promote Farmer-Friendly AI**
  - **Simple technology:** Develop AI tools that are **easy to use, affordable and available in local languages**.

    Kisan e-Mitra Chatbot, a voice-enabled multi-lingual AI assistant, answers farmers' scheme and crop queries in over 10 regional languages.
  - **Human + AI approach:** AI should **support farmers' decisions**, rather than completely replacing their knowledge and experience.
- **Strengthen AI-Based Advisory Services**
  - **Timely advice:** Provide farmers with personalised information on **weather, pests, irrigation, fertilisers and crop management**.
  - **Early warning:** Use AI with satellite and weather data to provide **early alerts for droughts, floods, pests and diseases**.

Conclusion:

AI can help Indian agriculture move towards **precision farming, higher productivity, efficient resource use and climate resilience**. With **digital inclusion, quality data, farmer-centric innovation and responsible AI**, technology can reach even small and marginal farmers. The ultimate goal should be **"AI-enabled, farmer-led agriculture"** that strengthens **farm incomes, food security and sustainable development**.

4. DRONES IN AGRICULTURE

Introduction:

Agricultural drones are unmanned aerial vehicles used to monitor crops and carry out farm activities with greater speed and precision. They can help farmers in **crop monitoring, spraying, disease detection, irrigation planning and assessing crop health**.

Benefits of Drones in Agriculture:

- **Better Crop Monitoring:** Drones can quickly cover large fields and identify crop stress, nutrient deficiency, pests and diseases.
- **Precise Spraying:** Drones can apply fertilisers, pesticides and other inputs precisely, reducing unnecessary use and wastage.
- **Saves Time & Labour:** Large areas can be surveyed or sprayed in a short time, reducing dependence on manual labour.
- **Saves Water & Inputs:** Drone-based monitoring helps farmers identify where crops actually need water, nutrients or plant protection, improving resource-use efficiency.
- **Early Detection of Problems:** Regular aerial images can help detect pest attacks, diseases, water stress and crop damage at an early stage.
- **Higher Farm Productivity:** Timely monitoring and targeted application of inputs can improve crop health, yield and farm efficiency.
- **Useful for Difficult Areas:** Drones can monitor waterlogged, hilly or otherwise difficult-to-reach fields without requiring farmers to physically enter them.
- **Supports Precision Farming:** Combining drones with GPS, satellite data, sensors and AI enables more accurate, data-driven farm management.
- **Reduces Farmer Exposure:** Drone-based spraying can reduce farmers' direct exposure to harmful pesticides and chemicals.

Table 1: Classification of Drones

Sr. No	Types	Weight(kg)	Wingspan(m)
1	Nano	> 0.25	0.25
2	Micro	0.25 to 2	0.50
3	Small	2 to < 25	1
4	Medium	25 to 150	5.10
5	Large	>150	30

Issues with Drones in Agriculture

- **High Cost**
  - **Expensive technology:** Buying drones, cameras, sensors and related equipment can be costly for small and marginal farmers.
  - **Maintenance cost:** Batteries, repairs, software and regular servicing add to the overall expense.
- **Skill & Training Gap**
  - **Technical knowledge:** Operating drones safely requires training in **flight control, mapping and spraying**.
  - **Limited local support:** Rural areas may lack trained operators and technicians for quick assistance.
- **Infrastructure Constraints**
  - **Poor connectivity:** Weak internet and digital infrastructure can limit advanced drone-based applications.
  - **Battery & charging issues:** Lack of reliable electricity and charging facilities can affect regular use.
- **Regulatory & Safety Concerns**
  - **Airspace restrictions:** Drone operations have to follow **DGCA rules and permitted airspace requirements**.

- **Safety risks:** Poorly operated drones can cause accidents, damage property or interfere with other aircraft.

- **Weather Dependence**
  - **Limited usability:** Strong winds, heavy rain and poor visibility can make drone operations difficult.
  - **Seasonal challenge:** Farmers may not be able to use drones during certain extreme weather conditions when monitoring is most needed.

Reforms taken for Drones in Agriculture

- **Kisan Drone Scheme (SMAM Guidelines):** Under the Sub-Mission on Agricultural Mechanization, the government provides generous grants to buy drones.
- **Namo Drone Didi Scheme:** Launched to train and equip 15,000 women Self-Help Groups (SHGs) under the Day-NRLM scheme with agricultural drones, turning rural women into certified drone pilots and commercial service providers in their villages.
- **Agriculture Infrastructure Fund (AIF):** Provides low-interest institutional loans to FPOs, cooperatives, and rural youth to set up drone-based Custom Hiring Centres (CHCs).
- **Simplified Drone Rules:** The Ministry of Civil Aviation notified liberalized **Drone Rules**, removing cumbersome approvals, reducing pilot license fees, and creating digital airspace maps.
- **Integration with PMFBY:** Drones are used alongside satellite imagery under the Pradhan Mantri Fasal Bima Yojana to conduct fast and accurate crop-damage assessments.
- **BharatVistaar & Krishi-DSS Integration:** Drone-captured imagery feeds into national digital platforms like the Krishi Decision Support System to generate precise maps of soil health, crop stress, and pest infestations.
- **PLI Scheme for Drones:** A **₹120 Crore Production-Linked Incentive (PLI)** scheme was introduced to boost local manufacturing of agricultural drones and components under the Make in India initiative.

Way Forward:

- **Make Drones Affordable**
  - **Lower the cost:** Provide easier access to subsidies, credit and rental services for agricultural drones.
  - **Shared use:** Expand **Custom Hiring Centres (CHCs), FPOs and cooperatives** so small farmers can use drones without purchasing them.
- **Strengthen Drone-as-a-Service**
  - **Pay-per-use model:** Farmers should be able to hire drones for **spraying, crop monitoring and field mapping** instead of owning them.

  - **Local service providers:** Encourage rural youth and agri-startups to run **village-level drone services**.
- **Build Skills & Training**
  - **Train rural youth:** Develop more trained **drone pilots, technicians and service providers** in rural areas.

    *Central **Drone Shakti** scheme integrates drone operation and repair courses into Industrial Training Institutes (ITIs) across rural districts to upskill rural youth.*
  - **Farmer awareness:** Demonstrate the actual benefits of drones through **Krishi Vigyan Kendras (KVKs), FPOs and field demonstrations**.
- **Improve Rural Infrastructure**
  - **Charging facilities:** Develop reliable **electricity and battery-charging facilities** in rural areas.

    *Agri-tech startups like **Marut Drones** and **IoTechWorld** deploy mobile charging van stations and establish rural "Drone Hubs" to support field operations.*
  - **Repair network:** Create local centres for **maintenance, spare parts and technical support**.
- **Promote Precision Agriculture**
  - **Better integration:** Combine drones with **AI, GPS, satellite imagery and soil sensors** for precise farm decisions.

    Fertilizer companies like IFFCO deploy Kisan Drones for targeted variable-rate spraying of Liquid Nano Urea
  - **Targeted application:** Use drones for **spot spraying and precise application of fertilisers and pesticides** to reduce wastage.
- **Strengthen Regulations & Safety**
  - **Simple compliance:** Make regulatory procedures easier for legitimate agricultural use while maintaining safety standards.
  - **Safe operations:** Ensure proper training, equipment standards and responsible flying to prevent accidents and misuse.
- **Encourage Innovation & Local Manufacturing**
  - **Indian solutions:** Support startups and research institutions to develop **low-cost, lightweight and crop-specific drones**.
  - **Make in India:** Promote domestic manufacturing of **drones, sensors, batteries and related components**.

Conclusion:

Drones can make Indian farming **more precise, efficient and resource-efficient** by improving crop monitoring and targeted input use. With **affordable access, trained operators, strong regulations and Drone-as-a-Service**, their benefits can reach small and marginal farmers. Thus, drones can become an important tool for **precision farming, higher farm incomes and climate-smart agriculture**.

5. CLIMATE SMART AGRICULTURE

Introduction

Climate-Smart Agriculture (CSA) means farming in a way that can **cope with climate change while maintaining or increasing farm productivity**. It focuses on **using water, soil and other resources wisely**, while reducing agriculture's environmental impact. CSA aims to make Indian agriculture **more productive, climate-resilient and sustainable**.

Three Pillars of CSA

- **Productivity:** Boost farm incomes and food security safely without overexploiting resources.
- **Adaptation:** Build resilience against climate shocks like droughts, erratic monsoons, and floods.
- **Mitigation:** Lower carbon footprints and enhance carbon sequestration in soils and crops.

Significance of Climate-Smart Agriculture

- **Climate Resilience:** Helps farmers cope with droughts, floods, heatwaves and changing rainfall patterns.
- **Higher Productivity:** Promotes better farming practices and technology to maintain or increase crop yields even under changing climatic conditions.
- **Water Conservation:** Encourages drip irrigation, sprinkler irrigation, rainwater harvesting and efficient water management, reducing pressure on groundwater.
- **Soil Health:** Practices such as crop rotation, mulching, cover crops and balanced fertiliser use help maintain soil fertility and reduce degradation.
- **Lower Emissions:** Better management of fertilisers, livestock, crop residues and water can reduce greenhouse-gas emissions from agriculture.
- **Higher Farmer Income:** Reduces input losses and crop damage while creating opportunities for diversification, value addition and climate-resilient crops.
- **Food Security:** Climate-resilient farming helps maintain a stable food supply despite extreme weather and rising climate risks.
- **Resource Efficiency:** Promotes the principle of "more crop per drop" and efficient use of land, water, energy and other farm inputs.
- **Biodiversity Conservation:** Encourages crop diversification, agroforestry and integrated farming, reducing dependence on a single crop.

Components of Climate-Smart Agriculture

- **Climate-Resilient Crops**: Promote **drought-tolerant, flood-tolerant, heat-resistant and short-duration crop varieties**.
- **Efficient Water Management**: Promote **drip irrigation, sprinkler irrigation, rainwater harvesting and watershed development**.
- **Climate-Smart Soil Management** : Use **mulching, conservation tillage, crop rotation and organic matter** to improve soil health.
- **Crop Diversification**: Reduce excessive dependence on a single crop and promote **millets, pulses, oilseeds, horticulture and agroforestry**.
- **Agroforestry & Integrated Farming**: Combine **trees with crops and livestock** to improve farm resilience.
- **Precision & Digital Agriculture**: Use **drones, satellite data, sensors and AI-based advisories** for better decisions on irrigation, fertilisers and pest management.
- **Low-Carbon Farming**: Promote efficient use of **fertilisers, energy and water** to reduce agricultural emissions.
- **Climate-Smart Livestock & Fisheries**: Promote **heat-tolerant livestock breeds, better fodder management and improved animal health**.

Issues with Climate-Smart Agriculture

- **Financial & Economic Barriers**
  - **High initial cost:** Drip irrigation, precision equipment, solar pumps and climate-resilient technologies can be expensive for small farmers.

As per NABARD, High-capital interventions like micro-irrigation systems cost around ₹50,000--80,000 per hectare.

- **Limited credit:** Small and marginal farmers may find it difficult to access affordable finance for these investments.

*As per RBI, Small and marginal farmers receive only **~30% of total institutional credit.***

- **Technology & Infrastructure Gaps**
  - **Digital divide:** Poor internet connectivity, limited access to smartphones and lack of digital skills can restrict the use of smart farming technologies.

*As per TRAI, Rural internet penetration stands at **~37–40%**, with female rural smartphone ownership below **30%.***

  - **Weak infrastructure:** Inadequate **storage, cold chains, irrigation and rural connectivity** can reduce the benefits of climate-smart practices.
- **Knowledge & Skill Gaps**
  - **Limited awareness:** Farmers may not know which climate-smart practice is suitable for their **crop, soil and local climate**.

*As per NSSO, Only **~10–12% of agricultural households** access formal technical advisory services from government extension officers or Krishi Vigyan Kendras (KVKs).*

  - **Weak extension services:** Lack of trained field-level workers makes it difficult to provide regular technical guidance.

*India's extension worker-to-farmer ratio is approximately **1:1,200** (against the FAO recommended norm of **1:500**), leaving field guidance severely understaffed.*

- **Small & Fragmented Landholdings**
  - **Limited economies of scale:** Small and scattered farms make the adoption of **drones, machinery, precision irrigation and other technologies** less economical.

As per 10th Agriculture Census, Small and marginal operational holdings make up 86.1% of total farms (averaging 1.08 hectares).

  - **Collective action needed:** FPOs and cooperatives are often required to make shared technologies affordable.
- **Water & Resource Constraints**
  - **Groundwater dependence:** Excessive groundwater extraction can undermine climate-resilient water management.

As per Central Ground Water Board, Agriculture accounts for ~89% of total groundwater extraction in India, with over 30% of assessment units classified as over-exploited.

- **Risk & Uncertainty**
  - **Changing climate:** Increasingly unpredictable **heatwaves, floods, droughts and rainfall** can reduce the effectiveness of traditional farming practices.

As per IPCC AR6, Climate change and extreme weather events (heatwaves, droughts, unseasonal rain) cause an estimated 1.5% to 2% annual GDP loss.

  - **Investment risk:** Farmers may hesitate to invest in new technologies when the returns are uncertain or take time to materialise.

Measures Taken for Climate-Smart Agriculture in India

- **PMKSY – Per Drop More Crop:** Promotes **drip and sprinkler irrigation** to improve water-use efficiency.
- **Watershed development:** Supports **rainwater harvesting, groundwater recharge and soil-moisture conservation** in rainfed areas.
- **National Mission on Sustainable Agriculture (NMSA):** Promotes farming practices that improve **resilience to climate risks**.
- **National Mission on Natural Farming:** Encourages practices that reduce dependence on external inputs and improve **soil health and farm resilience**.
- **National Innovations in Climate Resilient Agriculture (NICRA):** Led by **ICAR**, it develops and demonstrates technologies and crop varieties suited to **drought, floods, heat and other climate stresses**.
- **Soil Health Card Scheme:** Provides farmers with information on **soil nutrients and fertiliser requirements**, encouraging balanced fertiliser use.
- **National Food Security Mission (NFSM):** Supports production of **pulses, millets and other crops**, helping reduce excessive dependence on a few crops.
- **Millet promotion:** The government has promoted **Shree Anna (millets)** as climate-resilient and nutrition-rich crops requiring comparatively fewer resources.
- **Efficient fertiliser use:** Better nutrient management helps reduce unnecessary fertiliser application and associated emissions.
- **Pradhan Mantri Fasal Bima Yojana (PMFBY):** Provides crop insurance against losses caused by **droughts, floods, cyclones and other climatic risks**.
- **PM-KUSUM:** Promotes **solar-powered agricultural pumps**, helping reduce dependence on diesel and supporting cleaner irrigation.
- **ICAR/KVKs:** Krishi Vigyan Kendras conduct **demonstrations, training and awareness programmes** on climate-resilient farming practices.

World Bank's Role in Promoting Climate-Smart Agriculture

- **Country Climate and Development Reports (CCDRs)**
  - **Climate diagnosis:** CCDRs assess how climate change can affect agriculture through **lower crop yields, water stress and rising food prices**.
  - **Country-specific solutions:** They identify suitable **technologies, investments and policy reforms** for climate adaptation and emission reduction.
  - **Development focus:** The aim is to balance **climate resilience, productivity and broader development goals**.
- **Climate-Smart Agriculture (CSA) Country Profiles**
  - **Country-specific assessment:** These profiles identify the major **agricultural and climate challenges** faced by individual countries.
  - **Practical guidance:** They explain how CSA practices can help farmers **adapt to climate change and reduce emissions**.
  - **Knowledge support:** They provide a common understanding of **CSA concepts, practices and their local relevance**.
- **Climate-Smart Agriculture Investment Plans (CSAIPs)**
  - **Mainstreaming CSA:** CSAIPs help countries integrate **climate-smart practices into national agricultural policies and plans**.

- **Identify investments:** They identify priority areas such as **climate-resilient infrastructure, technology, capacity building and knowledge sharing**.
- **Financial & technical support:** The World Bank provides **technical assistance and financial support** to help countries prepare and implement these plans.
- **Country examples:** CSAIPs have been developed or are under preparation in countries such as **Bangladesh, Ethiopia, Ghana, Kenya, Nepal and Zimbabwe**.

Way Forward for Climate-Smart Agriculture

- **Make Climate-Smart Farming Affordable**
  - **Lower the cost:** Provide targeted subsidies and easy credit for **drip irrigation, solar pumps, precision farming and climate-resilient seeds**.
  - **Shared facilities:** Promote **FPOs and Custom Hiring Centres (CHCs)** so small farmers can use costly machinery and technology collectively.
- **Strengthen Water Management**
  - **Save water:** Expand **micro-irrigation, rainwater harvesting and watershed development**.
  - **Right crop, right place:** Encourage farmers to shift from water-intensive crops to **millets, pulses and oilseeds** in water-stressed areas.
- **Promote Climate-Resilient Crops**
  - **Better seeds:** Develop and distribute varieties that can withstand **drought, floods, heat and salinity**.
- **Improve Soil Health**
  - **Conservation practices:** Encourage **mulching, crop rotation, conservation tillage and balanced fertiliser use**.
  - **Build soil carbon:** Promote **agroforestry and organic matter management** to improve soil moisture retention and fertility.
- **Strengthen Climate Information & Technology**
  - **Timely advisories:** Provide farmers with **local weather forecasts, pest alerts and sowing advisories** through mobile and digital platforms.
  - **Smart farming:** Promote **satellite data, drones, sensors and AI-based tools** where they are affordable and useful.
- **Reform Incentives & Markets**
  - **Better policy signals:** Align **MSP, procurement, electricity and irrigation policies** with water conservation and climate resilience.
  - **Assured markets:** Create stronger markets and procurement systems for **millets, pulses, oilseeds and other climate-resilient crops**.

Case Studies of Climate-Smart Agriculture

Case Study	Features
1. Farmer-Managed Natural Regeneration (FMNR) – Niger	• Farmers allowed indigenous trees to naturally regenerate on their farms. • Trees were managed along with crops rather than being removed.
2. Sustainable Rice Intensification – Vietnam	• Farmers adopted less seed, less nitrogen fertiliser and less water in rice cultivation. • Farmer Field Schools and farmer-to-farmer learning supported adoption.

3. Kenya Agricultural Carbon Project – Kenya	• Small farmers adopted reduced tillage, cover crops, mulching, green manure, targeted fertiliser use and agroforestry. • Farmers were involved in monitoring carbon benefits.
4. Green Growth Agreement – Denmark	• Improved manure management, reduced fertiliser use and promoted biogas, organic farming and wetlands. • Focused on making agriculture both competitive and environmentally sustainable.
5. Carbon Farming Initiative – Australia	• Farmers earn carbon credits by adopting practices that reduce emissions or increase carbon storage. • Activities include better livestock and fertiliser management, soil-carbon improvement and reforestation.

Conclusion

Climate-smart agriculture can make Indian farming **more productive, resilient and sustainable** in the face of climate change. The focus should be on **efficient use of water and soil, climate-resilient crops, better technology and stronger farmer support**. Ultimately, the aim is to build **climate-resilient farms and secure farmer incomes while ensuring long-term food security**.

■■■

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AGRICULTURAL SUPPORT POLICIES & INPUTS

1. FARM SUBSIDIES

Introduction:

Farm subsidies are financial transfers by the government that reduce the cost of agricultural inputs or assure a remunerative output price, with the objective of making farming viable, keeping food affordable and ensuring national food security. In India they operate mainly through underpriced inputs such as fertiliser, power, irrigation and credit, and through price and income support such as MSP procurement and PM-KISAN.

Two-fold classification for answer writing:

- **Input Subsidies:** Reduce the cost of cultivation by supplying fertiliser, electricity, irrigation water, seeds and credit below their economic cost.
- **Output and Income Subsidies:** Support the price received or the income earned, through MSP-backed procurement, crop insurance premium support and direct income transfers such as PM-KISAN.

Scale in the Union Budget 2026-27:

- Fertiliser subsidy of **Rs 1,70,944 crore**, comprising **Rs 1,16,805 crore for urea** and **Rs 54,000 crore for nutrient-based subsidy** on phosphatic and potassic fertilisers.
- Combined food, fertiliser and fuel subsidy budgeted at about **Rs 4.54 lakh crore**, against a revised estimate of Rs 4.69 lakh crore for 2025-26.
- Agriculture and Farmers' Welfare allocation of **Rs 1,32,561 crore**, about **2.48 per cent of the total Budget** of Rs 53.50 lakh crore.

Significance of Farm Subsidies

- **Viability of Farming and Farmer Welfare**
- **Reduces Cost of Cultivation:** Subsidised urea at Rs 242 per 45 kg bag against an economic cost of roughly Rs 2,200 keeps input costs within reach of small farmers who cannot absorb global price swings.
- **Protects Against Global Price Volatility:** With fertiliser imports of nearly 27.2 billion dollars in 2025-26, subsidy acts as a buffer that insulates Indian farmers from geopolitical disruptions in phosphate, potash and gas markets.
- **Sustains Small and Marginal Holdings:** With over 86 per cent of holdings below two hectares and minimal marketable surplus, subsidies are often the difference between cultivation remaining viable and land being abandoned.

Food Security and Macroeconomic Stability

- **Underpinned the Green Revolution:** Subsidised fertiliser, irrigation, power and assured prices were the package that made high-yielding varieties attractive, taking foodgrain output from about 50 million tonnes at Independence to over 330 million tonnes.
- **Enables Buffer Stocking and the PDS:** Food subsidy finances procurement and free distribution under the National Food Security Act and PMGKAY, with rice and wheat stocks of 58.40 million tonnes on 1 January 2026 against a buffer norm of 21.41 million tonnes.
- **Moderates Food Inflation:** Assured domestic supply and calibrated release of buffer stocks reduce reliance on imports during international price spikes, protecting urban consumers as much as farmers.

Equity and Strategic Autonomy

- **Instrument of Distributive Justice:** In a sector employing nearly half the workforce but contributing a far smaller share of GDP, subsidies partly correct the terms of trade that run against agriculture.
- **Reduces Import Dependence:** Targeted support for pulses and oilseeds under missions and higher MSP is aimed at cutting India's large edible oil and pulses import bill.
- **Counterweight to Developed-Country Support:** India provides roughly 260 dollars per farmer per year, a fraction of what OECD countries provide, making subsidy withdrawal without reciprocal global reform inequitable.

Institutional Architecture

Fertiliser Subsidy: The Two-Track System

- **Urea Under Statutory Price Control:** Urea is kept outside the nutrient-based framework, with its maximum retail price frozen at Rs 242 per 45 kg bag since March 2018 and the government absorbing roughly 85 to 90 per cent of its cost.
- **Nutrient Based Subsidy (NBS) Scheme, 2010:** A fixed per-kilogram subsidy is announced for phosphatic and potassic nutrients each season, leaving the retail price to be set by companies, which is why P and K prices move with global markets while urea does not.
- **Direct Benefit Transfer in Fertilisers:** Subsidy is released to companies only after actual retail sale, verified through Point of Sale devices with Aadhaar or Kisan Credit Card authentication, making it a payment-on-sale system rather than a cash transfer to farmers.
- **National Investment Policy for Urea (NIPU) 2026:** Offers a return on equity band of 12 to 16 per cent and foreign exchange risk mitigation to attract fresh investment in gas-based urea plants and cut import dependence.

Food Subsidy

- **Route of Delivery:** The Food Corporation of India procures at MSP, bears carrying and distribution costs, and issues grain at heavily subsidised central issue prices, the gap being met as food subsidy.
- **Scale:** Food subsidy for 2025-26 was revised upward by about 12 per cent to roughly Rs 2.28 lakh crore, driven by higher procurement volumes and continued free distribution under PMGKAY.

Other Major Subsidies

- **Power and Irrigation Subsidy:** Provided by state governments through free or heavily subsidised electricity for agricultural pumpsets, it is largely off the Union Budget and shows up instead as discom losses and state fiscal stress.
- **Credit Subsidy:** The Modified Interest Subvention Scheme, allocated Rs 22,600 crore in 2026-27, lowers the effective interest rate on short-term crop loans through Kisan Credit Cards.
- **Insurance Subsidy:** Pradhan Mantri Fasal Bima Yojana, allocated Rs 12,200 crore in 2026-27, subsidises the bulk of the actuarial premium so farmers pay only a nominal share.
- **Income Supp**ort: PM-KISAN, allocated Rs 63,500 crore in 2026-27, transfers Rs 6,000 per year directly, and is the largest crop-neutral component of the subsidy architecture.

Issues with Farm Subsidies:

- **Fiscal Sustainability**
  - **Crowding Out Investment:** Subsidies consume a large share of the agriculture budget, squeezing capital expenditure on irrigation, research, extension and post-harvest infrastructure, which studies consistently show yield higher returns per rupee spent.
  - **Rigidity of the Bill:** Because urea's retail price has not been revised since 2018, any rise in global gas or phosphate prices is absorbed entirely by the exchequer, as seen when the fertiliser subsidy overshot to Rs 1.86 lakh crore in 2025-26 against a budget estimate of Rs 1.67 lakh crore.
  - **Arrears and Working Capital Stress:** Delayed release of subsidy dues to fertiliser companies strains their finances and can disrupt timely supply during the sowing window.
- **Distorted Price Signals and Nutrient Imbalance**
  - **Skewed NPK Ratio:** Because urea is far cheaper relative to its nutrient value, the all-India NPK consumption ratio has worsened from 4.3:2:1 in 2009-10 to about **10.9:4.4:1**, against the agronomic benchmark of 4:2:1.
  - **Low Nutrient Use Efficiency:** Only about 35 to 40 per cent of applied nitrogen is taken up by crops, the rest being lost through volatilisation, leaching and denitrification.
  - **Output and Input Growth Mismatch:** Since Independence foodgrain production has risen about 6.6 times while fertiliser consumption has risen roughly 464 times, indicating sharply diminishing returns to chemical inputs.
- **Environmental Damage**
  - **Soil Degradation:** Continuous urea overuse causes acidification, depletion of soil organic carbon and spreading deficiencies of zinc and sulphur, undermining the very productivity the subsidy is meant to raise.
  - **Groundwater Depletion:** Free or near-free farm power removes any cost to pumping, driving over-extraction in Punjab, Haryana, western Uttar Pradesh and parts of the Deccan where a majority of assessment blocks are over-exploited.
  - **Water and Air Pollution:** Nitrate leaching contaminates drinking water sources, while nitrous oxide released from nitrogenous fertiliser has a global warming potential many times that of carbon dioxide.
- **Inequity in Distribution**
  - **Capture by Larger Farmers:** Input subsidies are proportional to the quantity of input used and therefore to landholding size, so the largest transfers accrue to the largest farmers.
  - **Regional Skew:** Benefits concentrate in irrigated, procurement-rich states such as Punjab, Haryana and western Uttar Pradesh, while rainfed eastern and central India receives far less despite greater need.
  - **Exclusion of Tenants and the Landless:** Because both power connections and land records are held by owners, tenant farmers and sharecroppers who bear the actual production risk are largely excluded from input and income subsidies alike.
- **Leakage and Targeting Failure**
  - **Diversion to Industry:** Subsidised urea has historically been diverted to industrial users and across borders, a problem neem coating has reduced but not eliminated.
  - **Subsidy Paid to Companies, Not Farmers:** Under the present fertiliser DBT, money flows to manufacturers on proof of sale, so the farmer never sees the transfer and has no incentive to economise on quantity.
  - **Absence of Soil Linkage:** Subsidy entitlement is unconnected to soil test results, so a farmer with nitrogen-saturated soil receives the same support as one whose land genuinely needs nitrogen.

- **External and Trade Constraints**
  - **Amber Box Classification:** MSP-linked procurement is trade-distorting support under the WTO Agreement on Agriculture, capped at a de minimis of 10 per cent of the value of production for developing countries.
  - **Recurring Challenge at the WTO:** India's notification of about 48 billion dollars in farm input subsidies for 2022-23 drew questions from the United States, the United Kingdom, Canada, Brazil, Australia, Japan and the European Union at the Committee on Agriculture.
  - **Asymmetry in the Rules:** Developed countries hold over 90 per cent of global AMS entitlements worth nearly 160 billion dollars, while India and most developing countries have access only to the de minimis, a structural inequity India raises at every ministerial including MC14 at Yaoundé in March 2026.
  - **Unresolved Public Stockholding Issue:** India continues to rely on the interim Bali Peace Clause of 2013 and presses for a permanent solution that updates the 1986-88 external reference price for inflation.
- **Government Measures**
  - **Nutrient Based Subsidy Scheme (2010):** Fixes a per-nutrient subsidy on phosphatic and potassic fertilisers to encourage balanced application, though urea's continued exclusion has blunted its effect.
  - **Neem-Coated Urea (2015):** Mandatory neem coating slows nitrogen release, improves uptake and sharply reduces diversion of subsidised urea to industrial use.
  - **Direct Benefit Transfer in Fertilisers:** Subsidy is released to companies only after Aadhaar-authenticated sale through PoS machines, curbing bogus claims and enabling real-time tracking of movement.
  - **Soil Health Card Scheme:** Provides farmers with crop-specific nutrient recommendations based on soil testing, creating the informational basis for any future soil-linked subsidy.
  - **Nano Urea and Nano DAP:** Developed by IFFCO to raise nutrient use efficiency and cut transport and storage costs, though field efficacy remains contested and adoption is uneven.
  - **National Investment Policy for Urea (NIPU) 2026:** Offers an assured return on equity of 12 to 16 per cent and forex risk cover to attract investment in gas-based urea plants and reduce import dependence.
  - **PM-KISAN:** A crop-neutral income transfer of Rs 6,000 per year, allocated Rs 63,500 crore in 2026-27, representing the shift from product subsidy towards direct income support.
  - **PM-KUSUM:** Subsidises solar pumps and grid-connected solar capacity for farmers, addressing the power subsidy burden and groundwater over-extraction together.
  - **PM-AASHA and PMFBY:** Allocated Rs 7,200 crore and Rs 12,200 crore respectively in 2026-27, they deliver price assurance for pulses and oilseeds and subsidised crop insurance premiums.

Case Studies:

Case	What was done	Result and Lesson
Neem-Coated Urea (2015)	Mandatory neem coating of all subsidised urea.	Slowed nitrogen release and sharply reduced industrial diversion, showing that a low-cost technical fix can plug leakage without touching price. It did not, however, correct the underlying nutrient imbalance.

Punjab Free Farm Power	Free electricity for agricultural pumpsets since the late 1990s.	Sustained paddy cultivation in a semi-arid state but produced severe groundwater depletion and heavy discom losses. The standard illustration of a subsidy whose ecological cost exceeds its income benefit.
Paani Bachao Paise Kamao (Punjab)	Pilot paying farmers directly for electricity units saved against a monthly entitlement.	Demonstrated that farmers respond to positive incentives for conservation, but scaling stalled on metering costs and political resistance to touching free power.
Rythu Bandhu (Telangana)	Flat per-acre investment support decoupled from crop and input use.	Delivered timely liquidity without distorting crop choice, but excluded tenant farmers entirely since payment followed land title. The model and its flaw are equally instructive.
DBT in Fertilisers	Subsidy released to companies only after Aadhaar-authenticated PoS sale.	Improved traceability and curbed bogus claims, but because the money still reaches the company and not the farmer, it has not reduced the quantity of urea applied.
Sri Lanka's Fertiliser Ban (2021)	Abrupt nationwide prohibition on chemical fertiliser imports.	Rice output collapsed and the country faced food shortages within a year, a cautionary case for why subsidy reform must be gradual, sequenced and paired with alternatives.

Way Forward:

- **Correct the Price Signal**
  - **Bring Urea Under NBS:** The CACP and the Fertiliser Association of India have both recommended including urea in the nutrient-based framework so that all three nutrients face comparable subsidy treatment.
  - **Rationalise Rather Than Cut:** ICRIER suggests lowering support for nitrogen while raising it for phosphorus and potassium, restoring correct relative prices without increasing the overall subsidy bill.
  - **Adopt the Economic Survey Formula:** The Economic Survey 2025-26 recommends raising the urea price while transferring an equivalent amount directly to farmers on a per-acre basis, preserving purchasing power while making nitrogen reflect its true cost.
- **Move from Product Subsidy to Direct Transfer**
  - **Soil-Linked DBT:** Route fertiliser subsidy to the farmer as a per-hectare entitlement calibrated to Soil Health Card data, so that efficient users gain and over-appliers face a real cost.
  - **Consolidate Fragmented Transfers:** Merge overlapping input subsidies into an augmented PM-KISAN framework, which is crop-neutral, harder to divert, and classifiable as non-trade-distorting at the WTO.
  - **Include Tenant Cultivators:** Link entitlement to cultivator status rather than land title, on the model of Andhra Pradesh's Crop Cultivator Rights Cards, so the person bearing production risk receives the support.
- **Rebalance Towards Investment**
  - **Shift the Subsidy-Investment Ratio:** Redirect savings from rationalisation towards irrigation, agricultural research, extension, cold chains and market infrastructure, where the return per rupee is demonstrably higher.
  - **Reform Power Pricing:** Move from free power to metered supply with direct benefit transfer, as attempted through Punjab's Paani Bachao Paise Kamao scheme, paying farmers for electricity and water saved.

- **Scale Solar Substitution:** Expand PM-KUSUM so that solar pumps replace subsidised grid power, with buy-back of surplus solar energy creating an incentive against over-pumping.
- **Manage the External Front**
  - **Press for a Permanent Solution:** Continue negotiating at the WTO for an updated external reference price that accounts for inflation, and for post-2013 programmes to be covered by the Peace Clause.
  - **Shift Support to the Green Box:** Reclassify support towards research, extension, infrastructure, insurance and decoupled income transfers, which are permitted without limit and are also better economics.

Conclusion:

India's farm subsidy regime succeeded at the task it was designed for, which was to buy food security in an era of scarcity by making inputs cheap and prices assured. That success has now produced its own pathologies. The reform question is not whether to withdraw support but how to change its form, moving from **underpriced products to direct, soil-linked and crop-neutral transfers** that preserve the farmer's purchasing power while restoring honest price signals for nitrogen, water and power.

2. MINIMUM SUPPORT PRICE

Context:

Recently, the Government has approved an increase in MSP for 14 Kharif Crops for Marketing Season 2026-27.

Introduction:

Minimum Support Price is a price-floor mechanism through which the Government of India **commits to purchase specified crops** from farmers at a pre-announced rate, insulating them from **distress sales** when market prices crash. It is **recommended by the Commission for Agricultural Costs and Prices (CACP)** — an attached office of the Ministry of Agriculture, not a statutory body and approved by the Cabinet Committee on Economic Affairs (CCEA).

MSP Calculation:

MSP has **no statutory backing.** It is an administrative price policy, not a legally enforceable right. It is announced for 22 mandated crops (14 Kharif + 6 Rabi + 2 commercial — jute and copra). The Commission for Agricultural Costs & Prices (CACP) details three major formula to arrive at MSP.

- A2: Costs incurred by the farmer in production of a particular crop. It includes several inputs such as expenditure on seeds, fertilisers, pesticides, leased-in land, hired labour, machinery and fuel.
- A2+FL: Costs incurred by the farmer + the value of family labour.
- C2: A comprehensive cost, which is A2+FL cost + imputed rental value of owned land + interest on fixed capital, rent paid for leased-in land.

At present, the CACP adds both A2 and FL to determine the MSP. The government adds 50 percent of the value obtained by adding A2 and FL only, i.e. 1.5 times A2 + FL cost to fix the MSP.

Significance of MSP:

- **Income and Price Security for the Farmer**
  - **Floor Against Distress Sale:** MSP protects farmers from price crashes during glut years and post-harvest arrival pressure, when small farmers without storage are forced to sell immediately at whatever the market offers.

  - **Benchmark for Private Trade:** Even where the government does not physically procure, the announced MSP anchors bargaining and serves as a reference rate in mandi negotiations with private traders.
  - **Credit and Investment Confidence:** An assured price improves creditworthiness with formal lenders and encourages farmers to invest in quality seed, fertiliser and irrigation at the time of sowing.
- **Food Security and Macro-Stability**
  - **Foundation of the Public Distribution System:** MSP-based procurement of rice and wheat is the physical basis of the National Food Security Act, 2013 entitlements covering about 81 crore people, and of the central buffer stocking norms.
  - **Price Stabilisation for Consumers:** Buffer stocks built through MSP procurement are released via the Open Market Sale Scheme and subsidised retail channels such as Bharat Atta, Bharat Dal and Bharat Rice to moderate food inflation.
  - **Insulation from Global Volatility:** Domestic procurement reduces dependence on imports during international price spikes, as seen when India restricted rice exports while maintaining domestic supply.
- **Policy Signalling and Crop Steering**
  - **Steering Cropping Patterns:** Differential MSP hikes nudge farmers towards deficit crops, as in Kharif 2026-27 when sunflower seed received the highest increase at Rs 622 per quintal against just Rs 72 for paddy.
  - **Import Substitution:** Higher MSP for pulses and oilseeds, backed by assured procurement, supports the goal of reducing India's large edible oil and pulses import bill.
  - **Instrument of Regional Equity:** Extending procurement to states outside Punjab and Haryana through decentralised procurement allows MSP to support farm incomes in eastern and central India.

PM-AASHA:

Launched September 2018 and continued through the 15th Finance Commission cycle with an outlay of Rs 35,000 crore, with the procurement guarantee raised from Rs 16,000 crore to Rs 45,000 crore. *(Status beyond 2025-26 should be verified against the latest Budget documents.)*

- **Price Support Scheme (PSS):** NAFED and NCCF physically procure notified pulses, oilseeds and copra at MSP with state agencies, limited to Fair Average Quality produce and capped at 25 per cent of national production.
- **Price Deficiency Payment Scheme (PDPS):** Pre-registered farmers are paid the gap between MSP and the realised auction price directly into their accounts, with coverage raised from 25 to 40 per cent of a state's oilseed production.
- **Market Intervention Scheme (MIS):** Extends support to perishables such as tomato, onion and potato, triggered when prices fall at least 10 per cent below the previous normal season.
- **Price Stabilisation Fund (PSF):** Administered by the Department of Consumer Affairs, not Agriculture, it maintains buffer stocks of pulses and onion for calibrated release to check volatility and hoarding.
- **Private Procurement and Stockist Scheme (PPSS):** A pilot allowing private stockists to buy notified oilseeds from registered farmers at MSP in selected APMCs, without expanding government storage.

Implementing Agencies

- **Food Corporation of India (FCI):** Nodal agency for procurement, storage and distribution of rice and wheat for the central pool.
- **NAFED and NCCF:** Central Nodal Agencies for procurement of pulses, oilseeds and copra under the Price Support Scheme.
- **State Agencies and Cooperative**s: Carry out procurement on the ground, including under the Decentralised Procurement Scheme.

Issues

- **Narrow reach — MSP benefits a small minority**
  - **Shanta Kumar Committee (2015): only about 6% of farmers actually sell to procurement agencies at MSP.**The remaining ~94% transact in open markets, often below MSP.
  - NSO Situation Assessment Survey data shows very low awareness of MSP among cultivators, and lower still awareness of the procurement agency for their crop.
- **Skewed by crop and geography**
  - Procurement is overwhelmingly **rice and wheat**, concentrated in **Punjab, Haryana, Madhya Pradesh, Chhattisgarh, Telangana and Andhra Pradesh**.
  - Farmers of horticulture, coarse cereals and non-notified crops — the bulk of India's cultivators — are effectively outside the system despite MSP being announced for 22 crops.
- **Ecological cost of the paddy–wheat lock-in**
  - Assured procurement of water-intensive paddy in semi-arid Punjab and Haryana has driven **groundwater over-exploitation** (a majority of Punjab's assessment blocks are classified over-exploited by the Central Ground Water Board), soil salinity, monoculture, and **paddy stubble burning** contributing to North India's winter air pollution.
  - MSP thus operates at cross-purposes with the water and climate goals in the same syllabus.
- **Fiscal burden and stock overhang**
  - Rising MSP plus open-ended procurement produces buffer stocks far above norms, high FCI carrying and storage costs, and mounting food subsidy — crowding out capital investment in irrigation, research and extension, which have higher returns per rupee.
- **The formula dispute**
  - The government uses **A2+FL × 1.5**; unions demand **C2 + 50%**. Because C2 imputes land rent and capital interest, C2-based MSP would be materially higher — and the government has held the line on the formula while using headline absolute increases to signal commitment.
- **WTO constraints**
  - Under the **Agreement on Agriculture**, MSP-linked procurement counts as **Amber Box** (trade-distorting) support, capped at a **de minimis 10% of the value of production** for developing countries.
  - India has repeatedly invoked the **Bali Peace Clause (2013)** to shield its public stockholding for food security from challenge, and continues to press for a **permanent solution** on public stockholding at the WTO. The external-account exposure of a *legally guaranteed* MSP is a serious argument against it.
- **Exclusion of tenants and the landless**
  - Procurement is routed through land records, mandi registration and state portals. **Tenant farmers, oral lessees, sharecroppers and women cultivating without title are structurally excluded**, because the person who bears the production risk is often not the person on the record of rights.
  - This is the same design flaw that limits AgriStack's Farmer Registry — worth linking across topics.
  - **MSP functions as a ceiling, not a floor**
  - In practice, private traders in many mandis treat MSP as the reference *maximum* rather than minimum, especially where procurement infrastructure is thin and the government is not a credible competing buyer. Announcing a price without the capacity to buy at it can *depress* the market rather than support it.
- **No legal enforceability**
  - Since MSP is not statutory, a farmer sold below MSP has no remedy. This is the crux of the ongoing **legal-guarantee demand**, pressed through the 2020-21 and 2024-25 farmer agitations, and now before the **Supreme Court-appointed High Powered Committee headed by Justice Nawab Singh**.

Government Measures:

- **MSP at 1.5 Times A2+FL:** Since the Union Budget 2018-19, MSP for all 22 mandated crops is fixed at a minimum of 1.5 times the A2+FL cost of production, assuring farmers a margin of at least 50 per cent over paid-out costs and imputed family labour.
- **PM-AASHA (2018):** An umbrella price assurance scheme with four components, PSS, PDPS, MIS and PSF, carrying an outlay of Rs 35,000 crore and extending price support beyond rice and wheat to pulses, oilseeds and copra.
- **Full Procurement of Key Pulses:** The 25 per cent national production ceiling under PSS was waived for tur, urad and masur, permitting procurement of 100 per cent of state production to achieve self-sufficiency and reduce import dependence.
- **Decentralised Procurement Scheme (DCP):** States procure, store and distribute foodgrains for the central pool on the Centre's behalf, cutting FCI's transport and handling costs and widening MSP's geographic reach beyond the north-western states.
- **e-NAM:** Transparent electronic price discovery across 1,656 mandis in 23 states and 4 union territories reduces the information asymmetry that allows traders to bid below the announced MSP.
- **eSamridhi and eSamyukti Portals:** Operated by NAFED and NCCF respectively, these platforms allow farmers to pre-register for pulses procurement, creating a verifiable beneficiary list for direct benefit transfer of MSP payments.
- **National Mission on Edible Oils and Mission on Pulses:** Supply-side missions provide seed, extension and productivity support, ensuring the MSP price signal for import-substituting crops is matched by the capacity to respond to it.
- **PM-KISAN:** A direct income transfer of Rs 6,000 per year in three instalments, representing the parallel policy shift from price support towards income support decoupled from crop choice.

Case Studies:

Case	What was done	Result / Lesson
Bhavantar Bhugtan Yojana (Madhya Pradesh, 2017)	India's first large-scale price deficiency payment pilot — farmers sold in the open market and were paid the MSP–modal price gap.	**Cautionary.** Traders reportedly depressed mandi prices knowing the state would cover the gap, transferring the subsidy to buyers; fiscal outgo overshot and the scheme was scaled back. Shows PDPS needs a credible reserve buyer to work.
Rythu Bandhu (Telangana, 2018)	Flat per-acre investment support paid to landowners ahead of each season, decoupled from crop and price.	Delivered timely liquidity at sowing and avoided cropping distortion — **but excluded tenant farmers entirely**, since payment followed land title. The model and its flaw both instructive.
Punjab paddy procurement	Open-ended assured procurement of paddy since the Green Revolution.	Guaranteed farmer incomes but produced **severe groundwater depletion, monoculture and stubble burning** — the standard illustration of MSP's ecological externality.

Decentralised Procurement (Chhattisgarh, Odisha)	States procure, store and distribute directly for the central pool.	Widened MSP's geographic reach beyond the north-west, cut FCI transport costs, and improved payment timeliness — the most replicable fix for skewed procurement.
Kerala vegetable base price (2020)	First state to fix a floor price for 16 vegetables at ~20% above production cost.	Demonstrated that MSP-type support can be extended to **horticulture**, the fastest-growing but least price-protected segment. Sustainability depends on state fiscal capacity.
100% tur/urad/ masur procurement under PSS	Ceiling of 25% of national production waived for three pulses.	Assured-buyer signal raised pulses sowing confidence — evidence that **procurement guarantee, not MSP level, drives cropping response**.

Way Forward

- **Reform price discovery**
  - Give **CACP statutory status and greater autonomy**, with transparent publication of cost data and state-wise cost variation, so MSP is insulated from electoral timing.
  - Move towards **C2-based reference costs at least for disclosure**, even if the 1.5× multiplier continues to be applied to A2+FL, so that the gap is visible and debated on evidence.
- **Shift from price policy to income policy**
  - Scale **direct income support decoupled from crop choice** (PM-KISAN model, Telangana's Rythu Bandhu, Odisha's KALIA) — this supports incomes without distorting cropping patterns or attracting WTO Amber Box classification.
  - Expand **PDPS** as an alternative to physical procurement where storage and logistics are weak — but only with safeguards against trader collusion (see the Bhavantar case study below).
- **Rebalance towards diversification**
  - Use **assured procurement, not just higher announced MSP**, for pulses, oilseeds and millets as an MSP without a buyer changes nothing.
  - Pair with **crop-neutral incentives**: Punjab-style *Paani Bachao Paise Kamao* (direct benefit transfer for electricity saved), and crop diversification payments for shifting out of paddy.
- **Fix the procurement architecture**
  - **Widen the buyer base:** empower FPOs and PACS as procurement points, aggregate small holdings, and integrate them with e-NAM so price discovery and procurement are the same transaction.
  - **Include tenants:** create a mechanism for registration of tenancy or cultivator status (as Andhra Pradesh attempted through Crop Cultivator Rights Cards) so procurement follows the cultivator, not the title-holder.
  - **Extend warehousing and negotiable warehouse receipts** so farmers can hold produce past the arrival glut rather than sell into a depressed market.
- **Manage the external front**
  - Press for a **permanent solution on public stockholding** at the WTO, and design any legal MSP guarantee with Amber Box exposure explicitly modelled, not assumed away.

Conclusion:

MSP was designed for a scarcity economy, to induce farmers to adopt an unfamiliar technology under a guaranteed floor. India now **faces the opposite problem** — surpluses in cereals, deficits in pulses and oilseeds, depleting groundwater, and a fiscal envelope that cannot indefinitely absorb open-ended procurement. The reform question is therefore not whether MSP should exist, but whether it should remain a *price* instrument at all, or evolve into an assured-income and assured-market architecture that reaches the 94% of farmers it currently misses.

3. PESTICIDES

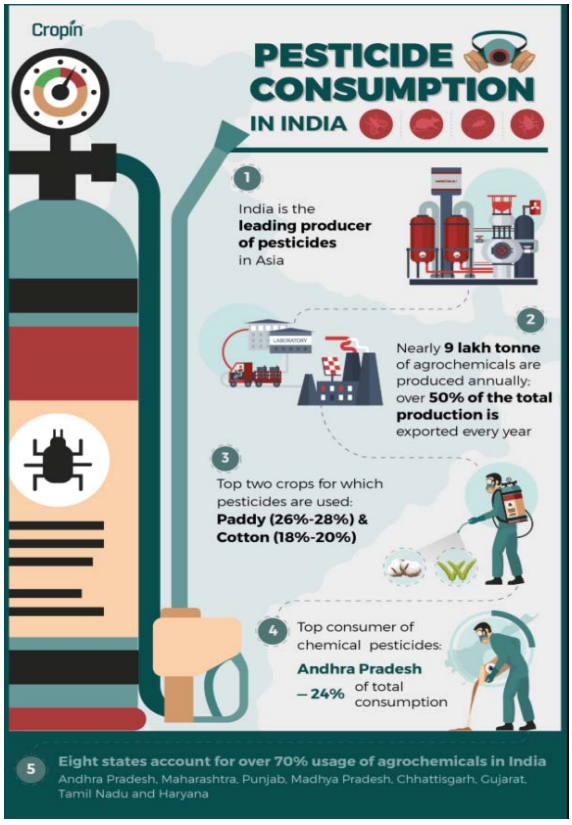
Introduction:

Pesticides represent a broad category of chemicals formulated to control, repel, or eliminate pests that can inflict harm on crops, transmit diseases, or impact livestock. Pests in this context encompass a range of entities, including insects, fungi, rodents, and weeds. The multifaceted nature of pesticides plays a crucial role in modern agriculture, safeguarding crops and ensuring global food security.

Types of pesticides:

- **Insecticide** = Engineered to target and eliminate insects.
- **Fungicide** = Developed to combat fungal infections in plants.
- **Herbicide** = Designed to manage and eliminate unwanted vegetation.
- **Rodenticide** = Formulated to control rodent populations.

India's domestic crop protection chemicals market is valued at Rs 24,000cr. Insecticide is the largest segment, followed by herbicide & fungicide.



Source: Cropin

Significance/Need of Pesticides:

- **Protecting Crops from Damage:** Pests, including insects, weeds, and pathogens, can cause extensive damage to crops, leading to reduced yields and lower quality produce. Weeds compete with crops for essential resources like water, nutrients, and sunlight. Insects can consume crops, and fungi and other pathogens can cause diseases that destroy plants. Without pesticides, significant losses of fruit (up to 78%), vegetables (54%), and cereals (32%) would occur due to pest damage.
- **Ensuring Food Security:** Pesticides play a vital role in increasing crop yields and preventing post-harvest losses. By protecting crops from a wide range of pests, they help ensure a stable and affordable food supply for the growing population. Without pesticides, crop yields could drop significantly, leading to food shortages and higher prices.
- **Controlling Disease Vectors:** Many times weeds harbor pests & pathogens that can cause damage to the standing crops. Herbicides are essential tools for controlling these weeds and preventing them from causing further harm.
- **Labour Shortage:** Traditionally, agricultural operations like weed control have been performed manually by the agricultural labourers - either by hand or using simple tools like *Khurpi.* But manual weeding is time consuming, costly & most importantly, labours are not available when the farmers need it. On the other hand, using modern tools like power weeders significantly reduce the time to remove the weeds but are not very effective against weeds with deep roots or growing within densely planted crops. Hence, a more effective solution to the problem is the use of herbicides.
- **Enhancing Economic Stability:** The use of pesticides supports agricultural economies by protecting the investments of farmers and ensuring predictable harvests. This stability helps to maintain food prices and provides a reliable source of income for agricultural communities. Furthermore, by protecting the crops from pests, pesticides help in reducing the cost of production & increasing the farmer's income.

Challenges of the Use of Pesticides & to the Pesticide Sector:

- **Health Risks and Poisoning:** Farm workers and communities face acute and chronic health issues due to exposure, including respiratory problems, neurological disorders, developmental delays in children, cancers, and—tragically—high rates of both accidental and intentional (suicide) poisonings, especially with highly hazardous pesticides.
- **Environmental Pollution:** Pesticide residues are commonly found in soil, water, and air, contaminating drinking water, food products, and harming biodiversity such as beneficial insects, fish, and wildlife. Certain regions, like Punjab, Maharashtra, and Uttar Pradesh, report pesticide levels in water and crops that exceed permissible limits, impacting ecosystems and public health.
- **Regulatory and Enforcement Gaps:** Although the Insecticides Act governs pesticide use, enforcement is weak. Sales of banned or restricted pesticides are still common due to inadequate oversight, corruption, and regulatory loopholes. Monitoring infrastructure and regular residue testing are limited, especially for small farmers.
- **Farmer Knowledge and Practice:**
  - Many small-scale farmers lack training in safe and appropriate pesticide use. There is widespread off-label and excessive application, often guided more by vendor recommendations than scientific advice. Proper use of protective gear and safe storage are frequently absent - which exposes the farmers to the harmful effects of pesticides poisoning.
  - Moreover, farmers in India generally use pesticides when they physically observe & assess the pest population or disease incidence to be significant enough to impact crop yield & quality (i.e. post-emergence) - which is not an effective way to use the pesticides.
- **Increase in Cost of Crop Production:** The financial burden of repeated pesticide applications, coupled with the loss of crop yields due to pest resistance and climate change, increases indebtedness and

distress among farming communities. Overdependence on chemical solutions is compounded by lack of access to affordable, effective alternatives.

- **Monopoly of MNCs:** Unlike seeds & fertilisers, where there are enough Indian public & private sector players, the crop protection chemicals industry in India is practically a multinational monopoly. A significant part of India's herbicide market is controlled by MNCs. For e.g. German Bayer AG has a market share of 15%, China's state-owned Syngenta has 12% & ADAMA has 10% market share.

Various government initiatives related to pesticides:

- **Integrated Pest Management (IPM):**
  - The most significant initiative has been the widespread promotion of IPM. This is a comprehensive, eco-friendly approach that aims to keep pest populations below a level where they cause economic damage, without a heavy reliance on chemical pesticides.
  - Under IPM, the government has established a network of Central Integrated Pest Management Centres (CIPMCs) across the country. These centers are tasked with pest and disease monitoring, producing and releasing **bio-control agents**, and providing training to farmers and agricultural extension officers.
- **The Insecticides Act, 1968:**
  - This is the primary legislation that governs the import, manufacture, sale, and use of pesticides in India.
  - It created the **Central Insecticides Board & Registration Committee (CIB&RC)**, which is responsible for registering pesticides after a thorough evaluation of their efficacy and safety.
  - The CIB&RC also has the power to ban or restrict the use of certain pesticides if they are found to be harmful to human health or the environment.
- **Pesticide Management Bill, 2020:**
  - This proposed legislation is intended to replace the outdated 1968 Act.
  - The Bill aims to modernize regulations, promote the use of **bio-pesticides**, and hold companies accountable for the quality and safety of their products.
  - It also includes provisions for compensating farmers who suffer losses due to spurious or substandard pesticides.
- **Increasing footprint of Indian pesticide manufacturers:** In recent years, some of the Indian companies have acquired the rights to active ingredients & brands from big global majors or are introducing innovative formulations. For e.g. Indian pesticide manufacturer, CCPL (Crystal Crop Protection Ltd.) has collaborated with USA & Japan based companies to develop a new paddy herbicide called **SIKOSA.**
- **Biopesticides and Botanical Alternatives:** There is a strong push to promote **neem-based pesticides** and other botanical alternatives. Research projects are focused on developing and promoting these non-chemical options as effective tools for pest management.
- **"Grow Safe Food" Campaign:** This initiative was started to create awareness among farmers and other stakeholders about the importance of judicious pesticide use and the proper methods of application.

Way forward:

- **Expand Integrated Pest Management (IPM):** Government and industry should mainstream IPM through extension, training, and demonstration programs, aiming to drastically reduce chemical dependence.
- **Accelerate Adoption of Biopesticides:** Simplify the registration process for eco-friendly **biopesticides** and provide financial incentives to manufacturers and farmers for their production and use.
- **Implement the Pesticide Management Bill, 2020:** The swift implementation of this Bill is crucial to replace the outdated Insecticides Act, 1968. This new framework will bring more stringent rules for registration, sales, and quality control.
- **Curb Spurious Pesticides:** Enforce strict penalties and strengthen state-level monitoring to combat the sale of counterfeit and substandard products, which pose significant risks to farmers and the environment.

- **Boost investment in R&D:** Prioritize innovation in climate-resilient, locally formulated crop protection solutions. Encourage patents and public-private partnerships to reduce dependence on imports and multinationals.
- **Empower farmers through extension:** Scale up training on safe pesticide handling, label comprehension, use of protective gear, and responsible disposal.
- **Precision Agriculture:** Encourage the use of technology like drones, sensors, and satellite imagery to enable targeted and efficient pesticide application, minimizing waste and environmental contamination.
- **Improve Extension Services:** Strengthen agricultural extension services and Krishi Vigyan Kendras (KVKs) to educate farmers on the correct and judicious use of pesticides, dosage, and safety precautions.
- **Move from 'post-emergence' to 'pre-emergence' pesticide use:** Farmers should be encouraged to shift from post-emergence pesticide application to prophylactic application of pre-emergent herbicides around or just after crop sowing. This will stop the weeds from coming out, helping keep the field clean from the very start - thus, reducing the losses as well as cost of application of pesticides.

Conclusion:

Overall, India should aim to develop its own Sinochem - which owns MNCs like Syngenta & ADAMA - to reduce the monopoly of MNCs & import dependence of the pesticide sector. However, in the long term, the agriculture sector should be encouraged to move towards adopting more sustainable practices that rely less on chemical pesticides.

4. SEEDS

In agriculture, the fundamental input is the seed, which is more than just a mere grain. Unlike grains, seeds encompass living embryos nestled within supporting or food storage tissues. Seeds prioritize the biological essence, emphasizing the living organism within, whereas grains emphasize the economic yield of the supporting tissue.

Introduction:

- Seed plays an important role in farming due to its characteristics, Seeds are not just inert grains; they contain living organisms crucial for plant growth.
- As per **Seed Act 1966**, Seed means any of the following classes of seeds:
  - Used for sowing or planting, seed of food crops including edible oil seeds and seeds of fruits and vegetables,
  - Cotton seeds;
  - Seeds of cattle fodder;
  - Jute seeds include seedlings, tubers, bulbs, rhizomes, roots, cuttings, all types of grafts and other vegetatively propagated material, of food crops or cattle fodder.
- **Seed Replacement Rate (SRR):**
  - Measures how much of the total cropped area was sown with certified seeds in comparison to farm saved seeds.
  - Higher the Seed Replacement Ratio, higher is production as well as productivity.
- Regulation ensures the quality, safety, and integrity of seeds, safeguarding farmers' investments and ensuring consistent yields.

Types of Seeds:

Nuclear Seeds	• This is the hundred percent genetically pure seed with physical purity and produced by the original breeder/Institute /State Agriculture University (SAU) from basic nucleus seed stock. • A pedigree certificate is issued by the producing breeder.
Breeder Seeds	• Progeny of nucleus seed of a variety and is produced by the originating breeder or by a sponsored breeder. • Breeder seed production is the mandate of the Indian Council of Agricultural Research (ICAR).
Foundation Seeds	• Progeny of breeder seed is required to be produced from breeder seed or from foundation seed which can be clearly traced to breeder seed. • The responsibility for agricultural productivity of foundation seed has been entrusted to the NSC, SFCI, State Seeds Corporation, State Departments of Agriculture and private seed producers. • Foundation seed is required to meet the standards of seed certification prescribed in the Indian Minimum Seed Certification Standards, both at the field and laboratory testing.
Certified Seeds	• Progeny of foundation seed and must meet the standards of seed certification prescribed in the Indian Minimum Seeds Certification Standards, 1988. • Its production shall be so handled as to maintain specific genetic identity and purity. • Certified seed may be the progeny of certified seed provided this reproduction does not exceed three generations beyond foundation seed stage-I.
Labelled Seeds	• The seed notified under Section 5 of the Seeds Act, 1966, such seed sold in the market has to be labelled as prescribed under Section 6(a) and (b) of the Seeds Act Such seed is called Labelled Seed.
Hybrid Seeds	• Hybrid(certified) seed is the first generation resulting from the cross of two approved inbred lines, one of which is male sterile.

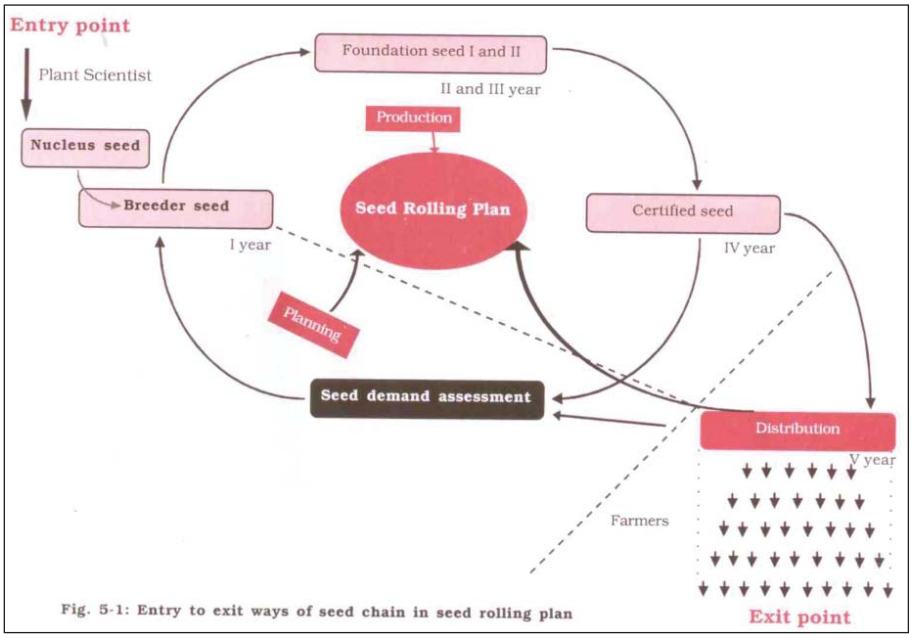
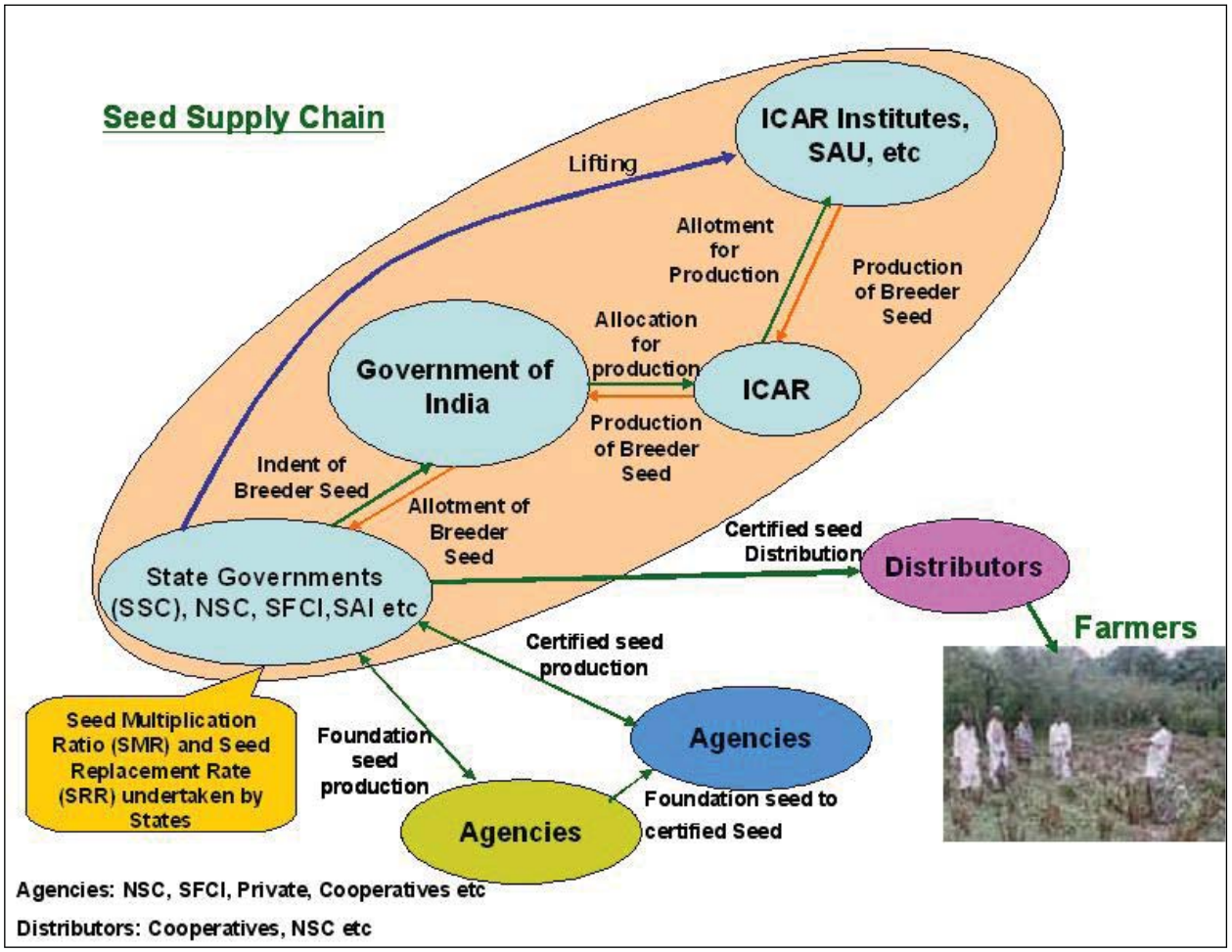


Fig. 5-1: Entry to exit ways of seed chain in seed rolling plan

Seed Supply Chain:



Significance of Seeds for Agriculture:

- **Determinant of Crop Yield & Quality:**
  - High-quality seeds contribute 20–25% directly to crop yields and, when combined with proper management, can boost total productivity up to 45%.
  - Improved and certified seeds lead to better plant vigor, uniformity, and enhanced resistance against pests and diseases, ultimately resulting in higher output and quality produce.
- **Driver of Agricultural Innovation:**
  - Use of hybrid, genetically improved, and biofortified seeds enables adoption of high-yielding and stress-tolerant varieties.
  - Seeds are central to the dissemination of new technologies such as drought-tolerant, pest-resistant, and climate-resilient crop varieties, supporting sustainable agriculture.
- **Food Security:** Timely and affordable access to quality seeds is critical in ensuring food security for India's large population.
- **Risk Management:** Quality seeds help farmers manage risks from biotic and abiotic stresses—like droughts, floods, and pest outbreaks—by providing greater yield stability.
- **Input Cost:** Better seeds mean less need for chemical inputs (fertilizers, pesticides), lowering cultivation costs and minimizing environmental impact.
- **Environmental Benefits:** Improved seed systems support diversified cropping, value-added crops, and market-oriented production, boosting farm income and export potential.

- **Foundation of Rural Prosperity:** The availability of quality seeds is the bedrock of farm success, enabling small and marginal farmers to participate in productivity gains, poverty reduction, and rural development.

Challenges related to seeds:

- **Limited Access to Quality Seeds:** Many farmers, especially smallholders, rely on farm-saved or uncertified seeds, leading to poor yields, low resistance to pests/diseases, and inconsistent quality. The availability of certified, hybrid, or improved seeds is uneven across regions and crops.
- **High Cost (Affordability Issue):** Hybrid and biofortified seeds often cost significantly more than traditional varieties, creating affordability barriers for resource-poor farmers. Lack of credit and input subsidies further restricts access for marginal farmers.
- **Counterfeit & Substandard Seeds:** The market is sometimes flooded with counterfeit, spurious, or low-quality seeds, particularly in high-demand seasons. Poor enforcement of seed quality standards and lack of awareness among farmers increases risk of crop failure.
- **Storage Challenges:** Poor seed storage infrastructure leads to loss of viability, higher rates of decay, and wastage—especially among small producers.
- **Distribution Bottlenecks:** Inadequate seed distribution networks (especially in remote/rainfed areas) cause shortages and seasonal mismatch.
- **Regulatory Issues:** Delays in variety approval, seed certification, and complex regulatory procedures impede market entry for new, improved seeds.
- **Intellectual Property & Legal Hurdles:**
  - Limited access to information, extension services, and demonstration plots hampers adoption of stress-resilient, hybrid, and climate-smart seed varieties.
  - Research-extension gaps mean many advances in seed technology do not reach farmers in time.

Government initiatives for seeds:

- **Seed Legislation:** The seed industry in India has been governed by several legislative & policy frameworks such as Seed Act (1966), Seed Rules (1968), Seed (Control) Order (1983), New Policy on Seed Development (1988), Plants, Fruits & Seeds (Regulation of Import into India) Order (1989), Protection of Plant Varieties and Farmers' Right Act (2001), and the Essential Commodities Act, 1955 including Seeds (1955), National Seed Policy (2002), and Seed Bill (2004).
- **National Seed Policy 2002:**
  - Sets the framework for quality assurance, production, certification, and distribution of seeds across varieties and crops.
  - Focuses on protecting farmers' rights, encouraging private innovation, and improving availability of high-yielding and resilient seeds.
- **National Mission on Seeds & Planting Material:**
  - Aims to increase production and supply of certified/hybrid seed, promote breeder and foundation seed multiplication, and strengthen public sector seed farms.
  - Supports seed hubs and demonstration plots to fast-track adoption of improved varieties.
- **PM Fasal Bima Yojana Synergy:** While primarily a crop insurance scheme, it complements quality seed initiatives by reducing farmers' risk in trying new or improved seed varieties.
- **Seed Mini-Kit Programme:** Distributes free/highly subsidized minikits of improved seed varieties (especially in pulses, oilseeds, and coarse cereals) to showcase performance and encourage adoption, especially among smallholders and in rainfed/difficult areas.
- **Seed Certification & Distribution:**

- The government, through State Seed Certification Agencies and the National Seed Corporation, strictly enforces certified seed production, testing, labeling, and market surveillance to curb substandard and spurious seeds.
- Digital tools and the National Seed Traceability System (NSTS) improve transparency and track the seed supply chain.

- **Protection of Plant Varieties and Farmers' Rights (PPV&FR) Act, 2001:** Initiatives for faster variety registration, IPR protection under Protection of Plant Varieties and Farmers' Rights (PPV&FR) Act, and simplified field trial procedures.
- **Indian Seed Vault:**
  - It is the second such seed bank in the world after the one at Norwegian Arctic island of Svalbard.
  - The gene bank was created at Chang-La, 75 km from Leh in Ladakh
  - Another Seed Bank in India for long term storage of seeds is maintained in New Delhi by the Indian Council of Agricultural Research (ICAR) but that runs a huge electricity bill for refrigeration.
  - Currently holds 5,000 seeds of vegetables and crops developed and grown by the defence scientists.

Way Forward:

- **Enhance Quality Assurance and Enforcement:**
  - Strengthen certification systems and market surveillance to eliminate substandard and counterfeit seeds.
  - Expand digital platforms like National Seed Traceability System for transparent tracking from production to sale.
- **Promote Innovation and Research:**
  - Invest in development of climate-resilient, biofortified, and hybrid seeds suited for diverse agro-ecological zones.
  - Facilitate collaboration between public research institutions and the private sector for rapid variety development and adoption.
- **Increase Seed Replacement Rate and Adoption:**
  - Set ambitious seed replacement targets and incentivize adoption through minikits, demonstrations, and targeted subsidies, especially in rainfed/remote regions.
  - Encourage adoption of new varieties via extension services and capacity-building initiatives.
- **Strengthen Distribution and Accessibility:**
  - Improve last-mile distribution channels and infrastructure to ensure timely, affordable access of quality seeds to all regions, especially for small and marginal farmers.
  - Leverage digital ordering and agri-tech platforms for seed availability updates and farmer support.
- **Support Farmer Awareness and Training:**
  - Scale up farmer training, field demonstrations, and digital extension to increase knowledge about benefits of improved and certified seeds.
  - Promote awareness of seed quality, proper storage, and the risks of using uncertified seeds.
- **Policy and Regulatory Reform:**
  - Streamline variety registration, approval procedures, and IPR protection to encourage private innovation and rapid market entry.
  - Enforce effective grievance redressal mechanisms and simplify availing government schemes.

Conclusion:

Recognizing the significance of seeds as living entities and implementing effective regulations is essential for sustainable agriculture and food security.

5. FERTILISER SECTOR

Fertilizers are essential for India's agricultural productivity, helping to sustain high crop yields for a large and growing population. India is one of the world's largest producer, consumer & importer of fertilisers in the world.

Significance of fertiliser sector in India:

- **Ensure the availability of essential soil nutrients:** The fertiliser sector has played a crucial role in sustaining India's agricultural output by ensuring the timely availability of essential nutrients such as nitrogen (N), phosphate (P), and potash (K) through both domestic production and imports.
- According to recent reports Indian soils suffer from acute deficiencies in essential nutrients which are listed below. These nutrient deficiencies severely affect soil fertility and crop yields:

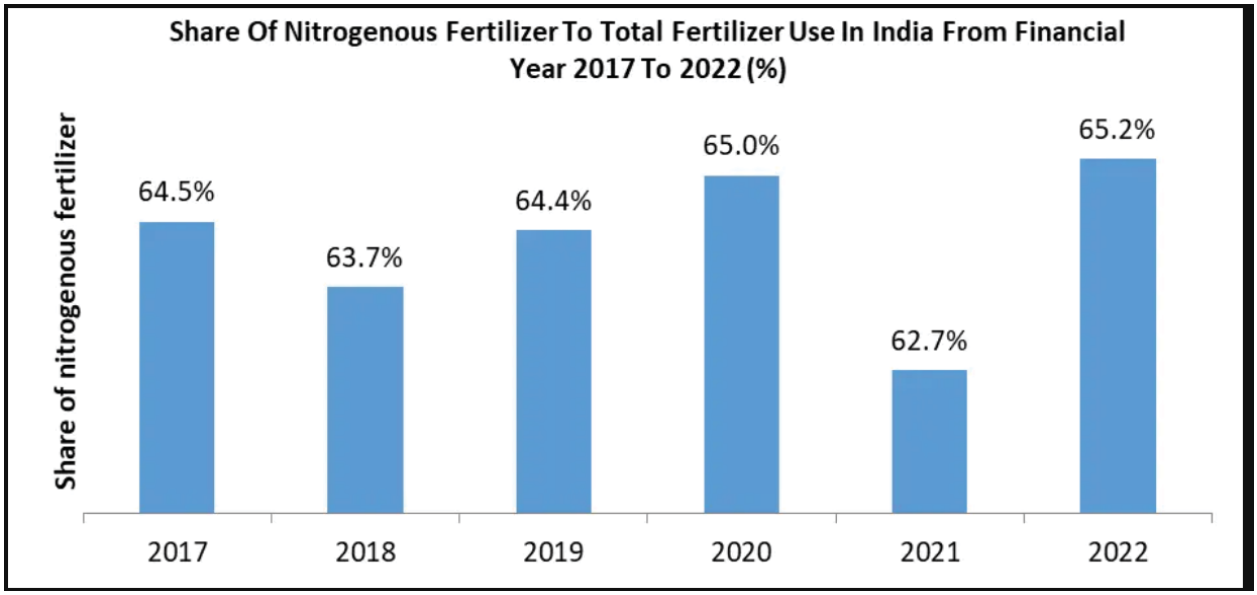
Nitrogen Deficiency	Less than 5% of Indian soils have adequate nitrogen levels, essential for healthy plant growth.
Phosphate Deficiency	Only 40% of soils are sufficient in phosphate, an important nutrient for root development and flowering.
Potash Deficiency	32% of soils are deficient in potash, which is crucial for plant disease resistance and overall growth.
Organic Carbon Deficiency	Just 20% of soils have adequate organic carbon, which is necessary for improving soil structure and water retention.
Micronutrient Deficiencies	Soils in India also suffer from deficiencies of micronutrients such as Zinc, Iron, Sulphur, and Boron, which are critical for optimal plant growth.

- **Driving Agriculture Productivity:**
  - The fertilizer sector has been a key driver in India's Green Revolution and sustained agricultural growth, enabling higher cereal production to feed the growing population.
  - Between 2020-21 and 2022-23, India exported 85 million tonnes of cereals and ensured food security for hundreds of millions during crises.
- **Economic Growth & Employment Generation:**
  - India is the second-largest consumer and third-largest producer of fertilizers globally, with domestic production steadily rising.
  - The sector supports significant rural employment directly and indirectly in production, distribution, and retail networks.
- **Driving India's Agricultural Dominance:** The efforts of the fertiliser industry in supplying critical nutrients have helped India maintain its status as a global agricultural powerhouse. Between 2020-21 and 2022-23, the country exported 85 million tonnes of cereals while providing near-free grain to over 813 million people during the pandemic.
- **Supporting Sustainable Farming:** The fertiliser sector encourages the use of advanced products like slow-release fertilisers and those fortified with micronutrients to enhance efficiency and reduce environmental impact.
- **Education and Awareness:** The sector works closely with farmers to educate them about proper fertiliser application techniques, soil testing (Soil Health Card), and nutrient management for maintaining soil vitality.

Challenges related to fertiliser sector in india:

- **Imbalanced Fertiliser Use:** A major issue in Indian agriculture is the overuse of nitrogen (N) fertilisers, especially urea, while other essential nutrients like phosphorus (P) and potassium (K) are underused. For e.g. in Punjab, nitrogen use is 61% higher than recommended, but under uses potash by 89% and

phosphate by 8%. This imbalance results in green fields but lower crop yields, as plants need all three nutrients for optimal growth.



- **Low Nutrient Use Efficiency (NUE):** The efficiency of fertiliser use in India is very low, with only 35-40% of fertilisers being absorbed by crops. The rest is wasted or lost to the environment, such as nitrogen escaping as nitrous oxide, a harmful greenhouse gas. This inefficiency leads to higher costs for farmers and environmental harm.
- **Issues related to fertiliser subsidy:**
  - **Nutrient Imbalance:** By focusing heavily on urea, the subsidy system has encouraged an imbalanced use of fertilisers, with nitrogen being overused and phosphate and potash underused. This imbalance harms soil health and reduces agricultural productivity in the long run.
  - **Financial Strain on the Government:** Fertiliser subsidies are a significant financial burden on the government, amounting to ₹1.88 lakh crore, or nearly 4% of the Union budget. This high expenditure diverts resources away from other critical sectors, such as health and education.
- **Environmental Harm:** Excessive and imbalanced fertilizer use has severe environmental consequences:
  - **Groundwater Contamination:** Leaching of excess nitrates into groundwater, posing health risks (e.g. “blue baby syndrome”).
  - **Water Body Eutrophication:** Runoff of nitrogen and phosphorus into surface water bodies (lakes, rivers) leads to algal blooms, depleting oxygen and harming aquatic life.
  - **Greenhouse Gas Emissions:** Nitrous oxide (N2O), a potent greenhouse gas, is emitted from agricultural soils due to nitrogenous fertilizer use, contributing to climate change. Ammonia volatilization also contributes to air pollution and acid rain.
  - **Soil Degradation:** Long-term overuse of chemical fertilizers can alter soil pH, increase salinity in some areas, and negatively impact soil microbiology and structure.
- **Fertiliser Diversion and Smuggling:** A significant portion (20-25%) of subsidised urea is diverted for non-agricultural uses or smuggled out of the country. This deprives farmers of necessary fertilisers and strains government finances.
- **Neglect of Micronutrients:** Micronutrients like zinc, boron, and iron are often overlooked, despite their crucial role in plant growth. Their deficiency is widespread and contributes to the decline in soil health and agricultural productivity.
- **High import dependency & Global volatility:** India does not produce enough fertilizers or their raw materials to meet its domestic demand:

- **DAP:** DAP is India's 2nd most consumed fertiliser (Annual avg sale = 103.4 lt). A significant portion of the consumption comes from imports (57 lt) (because India has limited rock phosphate deposits). China has been one of the top suppliers of phosphate fertiliser to India. However, China has put curbs on its exports - to ensure that its farmers have access to the product first & also to meet the growing demand for phosphate in production of EV batteries. It resulted in shortage of key fertiliser DAP in India.
- **MOP (Muriate of Potash):** India is 100% dependent on imports for MOP, as there are no domestic potash reserves. Similarly, India imports about 90% of its phosphatic fertilizers.
- **Vulnerability to Global Price Fluctuations:** This heavy import dependence makes India highly vulnerable to geopolitical events (like the Ukraine-Russia conflict, Gaza crisis), supply chain disruptions, and fluctuations in international prices of fertilizers and their raw materials (e.g., natural gas, phosphoric acid, rock phosphate). This directly impacts the cost of fertilizers for farmers and the government's subsidy bill.

Government initiatives related to fertilisers:

- **Fertiliser Subsidy:** The government provides a subsidy to fertiliser producers so farmers can buy fertilisers at lower prices. The subsidy covers the difference between the cost of making or importing the fertiliser and the price farmers pay. Subsidy on 3 basic fertilisers in India- Urea, DAP and Muriate of Potash (MOP) is discussed below:
  - **Subsidy on Urea:** Urea is the most widely produced and used fertiliser in India. It is subsidised only for agricultural use. The government pays a subsidy based on the cost of production at each plant, and urea is sold at a fixed Maximum Retail Price (MRP).The subsidized MRP of 45 kg bag of urea is Rs.242 per bag (exclusive of charges towards neem coating and taxes applicable).
  - **Subsidy on Non-Urea Fertilisers:** Non-urea fertilisers, such as DAP and MOP, are generally sold at market prices, but the government has recently regulated them due to global price increases (especially after the Russia-Ukraine war). These fertilisers are covered under the Nutrient-Based Subsidy (NBS) Scheme. The MoP is being sold at Rs 1,500-1,600 per bag, while the price of DAP is Rs 1,350 per bag.
- **Fertiliser subsidy schemes:**

Urea Subsidy Scheme	• Under the Urea Subsidy Scheme, urea is sold at a statutorily notified uniform MRP (Maximum Retail Price). Farmers pay a subsidised price of ₹242 per 45 kg bag of urea, significantly lower than the market price. • The difference between the cost of production/ importation and the retail price is paid to the urea manufacturer/importer by the government as a subsidy.
Nutrient-Based Subsidy (NBS) Scheme	• The NBS scheme was introduced in 2010 to address the nutrient imbalance in Indian agriculture. • Under this scheme, fertilisers are provided at subsidised rates based on the nutrients they contain, namely nitrogen, phosphate, potash, and Sulphur. • Fertilisers fortified with secondary and micronutrients are also given additional subsidies. However, urea is excluded from the NBS scheme.

  - **Direct Benefit Transfer (DBT) of Fertilizer Subsidy:** 100% subsidy on various fertilizer grades is released to fertilizer companies after actual sales are made to farmers by retailers through Point of Sale (PoS) machines. Farmers are identified through their Aadhaar card at the retail outlet. The aim is to improve transparency, reduce diversion, ensure the subsidy reaches the intended beneficiaries, and track real-time sales data.
- **Boosting Domestic Production and Self-Sufficiency:**
  - **Revival of Closed Fertilizer Units:** A major focus has been on reviving defunct or sick urea fertilizer plants to increase domestic production capacity and reduce import reliance for e.g. Hindustan Urvarak

& Rasayan Limited (HURL), Ramagundam Fertilizers and Chemicals Ltd (RFCL), Talcher Fertilizers Limited (TFL).

- **New Urea Policy (NUP) – 2015:** For existing gas-based urea units, this policy aimed at maximizing indigenous urea production, promoting energy efficiency, and rationalizing the subsidy burden. It has led to a significant increase in indigenous urea production.
- **Potash Derived from Molasses (PDM):** To reduce 100% import dependence for Potash, indigenous production of Potash Derived from Molasses has been encouraged and included under the NBS scheme.

- **Promoting Balanced Fertilizer Use and Soil Health:**
  - **Neem-Coated Urea (NCU):** The government mandated 100% neem-coating of all indigenously produced and imported urea.
  - **Soil Health Card (SHC) Scheme:** Promotes soil test-based and balanced use of fertilizers, reduces overuse of chemical fertilizers, saves costs for farmers, and improves soil health and productivity.
  - **Promotion of Bio-fertilizers and Organic Manures:** Schemes like Paramparagat Krishi Vikas Yojana (PKVY), National Project on Organic Farming (NPOF), Mission Organic Value Chain Development for North-Eastern Region (MOVCDNER), and Bhartiya Prakritik Krishi Padhati (BPKP) provide assistance and promote the use of organic fertilizers and bio-fertilizers.
- **PM Programme for Restoration, Awareness Generation, Nourishment, and Amelioration of Mother-Earth (PM-PRANAM):** States/UTs receive a grant equivalent to 50% of the fertilizer subsidy saved by reducing chemical fertilizer consumption (Urea, DAP, NPK, MOP) compared to the average of the previous three years. This grant is to be used by states to promote alternative fertilizers and related assets.
- **One Nation, One Fertilizer (ONOF) / Pradhan Mantri Bhartiya Jan Urvarak Pariyojana (PMBJP):** Standardize fertilizer brands, reduce farmer confusion, reduce criss-cross movement of fertilizers (thus saving freight subsidies), and ensure a uniform appearance and quality perception for subsidized fertilizers.
- **Nano Fertilizers:** The government is actively promoting the adoption of nano-fertilizers, particularly Nano Urea Liquid e.g. IFFCO Nano Urea Liquid.

Way Forward:

- **Rationalizing the Subsidies:** The government should bring urea under the Nutrient-Based Subsidy (NBS) scheme to balance the prices of nitrogen, phosphate, and potash. This will encourage balanced fertiliser use and reduce reliance on urea.
- **Deregulating Fertiliser Prices:** Allowing market forces to set fertiliser prices can remove distortions caused by price controls. Farmers should receive direct income support, such as digital coupons or cash transfers, to purchase fertilisers based on their needs.
- **Promoting Micronutrient Use:** To address micronutrient deficiencies, the government should promote fertilisers enriched with micronutrients. Encouraging balanced fertiliser use, including micronutrients, is crucial for better soil health and crop yields.
- **Improving Nutrient Use Efficiency (NUE):** Improving NUE is essential, which can be achieved through precision farming, better soil testing , and using technology to apply fertilisers more efficiently.
- **Direct Income Support to Farmers:** Explore replacing product-specific subsidies with a system of direct income support to farmers (like PM-KISAN, but specifically tied to fertilizer use or nutrient management). This would empower farmers to choose the right fertilizers based on their soil needs, reduce distortions, and prevent diversion.
- **Diversify Import Sources:** Reduce over-reliance on a few countries for raw materials (rock phosphate, phosphoric acid) and finished fertilizers (MOP, DAP). Forge long-term agreements and joint ventures with resource-rich countries beyond traditional partners, including those in West Asia, Africa, Latin America, and other regions.

- **Integrated Nutrient Management (INM):** Vigorously promote INM, which combines chemical fertilizers with organic manures, bio-fertilizers, and green manures, to improve soil health and reduce reliance on synthetic inputs.

Conclusion:

India needs to build a fertilizer sector that is not only robust and self-sufficient but also environmentally responsible, fiscally sustainable, and ultimately contributes to long-term agricultural prosperity and food security.

6. MECHANISATION IN AGRICULTURE

Introduction

Agricultural mechanisation means using **machines, tools and modern equipment** for activities such as ploughing, sowing, irrigation, harvesting and processing. It helps farmers **save time and labour, reduce production costs and improve productivity**, especially when labour is costly or scarce. Mechanisation can make Indian farming **faster, more efficient and less dependent on manual labour**.

Components:

- **Land Preparation:** Use of tractors, power tillers, ploughs and cultivators for ploughing, levelling and preparing the soil.
- **Sowing & Planting:** Seed drills, planters and transplanters help farmers sow seeds at the right depth and spacing.
- **Irrigation:** Use of electric/solar pumps, drip and sprinkler systems to make irrigation faster and more efficient.
- **Crop Management:** Power weeders, sprayers and drones help in weeding, pest control and applying farm inputs.
- **Harvesting:** Combine harvesters, reapers and threshers reduce the time and labour needed for harvesting.
- **Post-Harvest Management:** Machines for cleaning, grading, sorting, drying, milling and packaging reduce post-harvest losses and improve quality.
- **Storage & Processing:** Mechanised cold storage, grain handling and food-processing equipment help farmers preserve and add value to their produce.
- **Digital & Smart Mechanisation:** GPS, sensors, drones, satellite technology and AI-based machinery help farmers make more precise decisions and use inputs efficiently.
- **Custom Hiring Services:** Custom Hiring Centres (CHCs) allow small farmers to rent expensive machinery instead of purchasing it, making mechanisation more accessible.

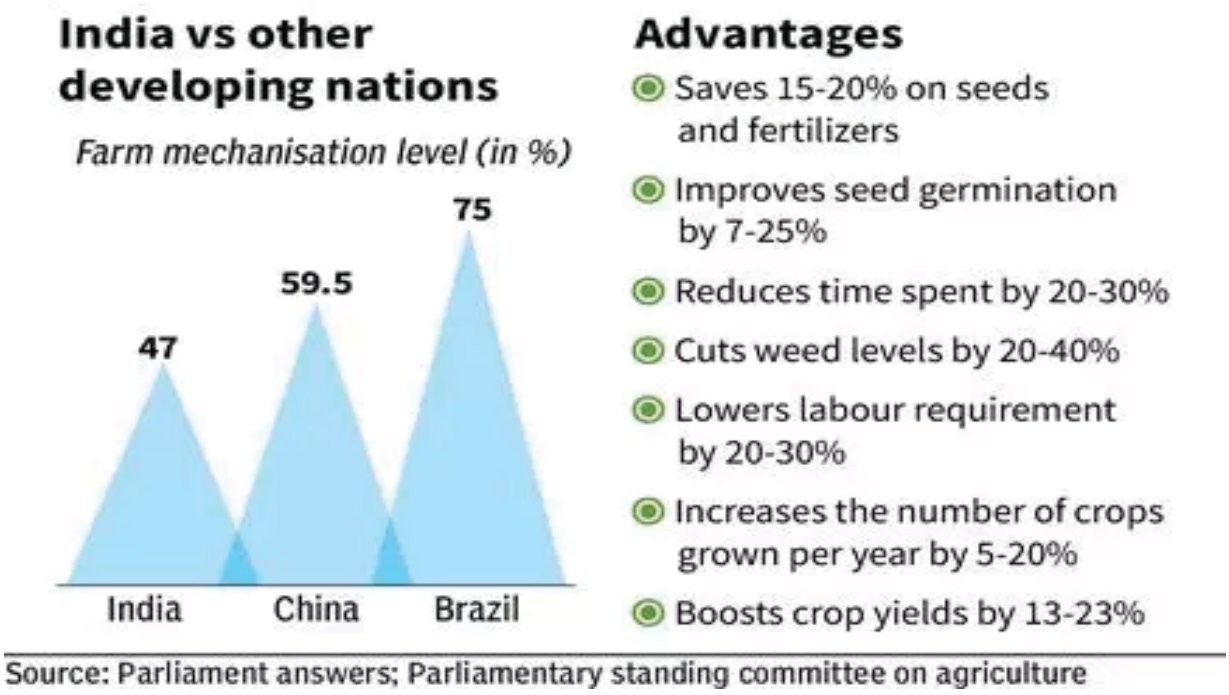
Significance:

- **Economic Benefits**
  - **Reduces costs:** Machines can reduce dependence on costly manual labour and improve the efficiency of farm operations.

    According to an ICAR report, farm mechanization reduces input costs by saving 15–20% on seeds and fertilizers while increasing cropping intensity by 5–20%.
  - **Higher productivity:** Timely sowing, harvesting and better input use can **increase output per unit of land**.
- **Labour & Time Benefits**

- **Saves time:** Machines can complete operations such as **ploughing, sowing and harvesting** much faster than manual labour.
- **Addresses labour shortage:** Mechanisation helps farmers cope with **seasonal labour shortages**, especially during peak agricultural seasons.

- **Productivity & Resource Efficiency**
  - **Reduces wastage:** Mechanised harvesting and post-harvest operations can reduce **crop losses and damage**.
  - **Multiple cropping:** Faster farm operations can help farmers use the same land for **more than one crop cycle** where conditions permit.
- **Social & Structural Benefits**
  - **Improves working conditions:** Mechanisation reduces dependence on physically demanding manual farm work.
  - **Supports small farmers: Custom Hiring Centres (CHCs)** allow small and marginal farmers to rent machinery instead of buying expensive equipment.
  - **Modernises agriculture:** Mechanisation helps shift Indian agriculture towards **efficient, technology-driven and climate-resilient farming**.



Challenges of Mechanisation in Agriculture

- **High Cost of Machinery**
  - **Expensive equipment:** Tractors, harvesters, drones and precision machines can be costly, especially for small and marginal farmers.
  - **High maintenance cost:** Repairs, fuel, spare parts and servicing add to the overall cost of farming.

    *As per CACP, Fuel costs (diesel) and operational maintenance account for nearly **25% to 30% of total variable cultivation costs** during peak sowing and harvesting seasons.*
- **Small & Fragmented Landholdings**
  - **Limited suitability:** Large machines are difficult to use on small and scattered farms.

*As per 10th Agriculture census, **86.1% of operational holdings** classified as small and marginal and an average holding size of just **1.08 hectares.***

- **Low economies of scale:** Farmers may not find it profitable to purchase costly machinery for limited land.

- **Labour & Employment Concerns**
  - **Displacement risk:** Excessive mechanisation can reduce demand for manual agricultural labour, particularly for routine operations.
  - **Skill gap:** Operating modern machinery requires technical skills and training that many rural workers may lack.

*As per skill India, Over **85% of rural agricultural workers** lack formal technical training in operating, calibrating, or repairing modern precision tools*

- **Poor Rural Infrastructure**
  - **Weak connectivity:** Poor roads and transport facilities make it difficult to move large machines between farms.

*According to **Ministry of Rural Development**, over **1.5 lakh rural habitations** still lack all-weather road connectivity.*

  - **Limited service centres:** Lack of nearby repair facilities and spare parts can cause delays during critical farming periods.

*SMAM reveal that **over 60% of rural blocks** suffer from an acute deficit of certified Farm Machinery Repair Workshops and spare-part centers.*

- **Reforms taken:**
  - **Sub-Mission on Agricultural Mechanization (SMAM):** The flagship scheme that provides **40% to 50% financial subsidies** to farmers for buying tractors, power tillers, harvesters, and specialized implements.

SMAM has supported the distribution of 21.61 lakh agricultural machines to individual farmers across India.

  - **Special Subsidies for Vulnerable Groups:** Provides higher subsidies for small/marginal farmers, women farmers, and farmers in SC/ST categories or the North-Eastern states to ensure inclusive technology adoption.
  - **Custom Hiring Centres (CHCs):** Government subsidies help FPOs, cooperatives, and rural entrepreneurs establish village-level CHCs to rent out equipment at nominal hourly rates.
  - **"FARMS" Mobile App:** A digital platform launched by the Ministry of Agriculture that connects farmers directly with nearby CHC owners, allowing them to book farm machinery on-demand from their smartphones.
  - **Advanced Machinery Promoted:** Happy Seeders, Super Seeders, Smart Seeders, and Paddy Straw Choppers allow farmers to sow wheat directly into standing paddy stubble without burning it.
  - **Kisan Drone Scheme:** Introduced dedicated guidelines and financial support to promote agricultural drones for precision spraying of liquid fertilizers (like Nano Urea) and pesticides, as well as for crop health monitoring.
  - **Namo Drone Didi Scheme:** Empowers women's Self-Help Groups (SHGs) under the Day-NRLM scheme by providing them with drones and training to become commercial drone service providers in their villages.

Over 40,928 drone demonstrations covering 40,918 hectares were conducted with Rs.52.5 crore financial support, promoting precision agriculture adoption nationwide.

- **Testing & Quality Assurance:** Mechanization policies mandate that machinery undergo testing at designated Farm Machinery Training and Testing Institutes (FMTTIs) to ensure safety, durability, and fuel efficiency.
- **Make in India in Agri-Equipment:** Incentivizing local manufacturing of specialized implements (like zero-till drills, rotavators, and micro-irrigation systems) to reduce dependence on expensive imports.

Rashtriya Krishi Vikas Yojana (RKVY)

The Rashtriya Krishi Vikas Yojana (RKVY), launched in 2007, is a flagship scheme that incentivizes states to increase public investment in agriculture and allied sectors. It provides states with flexibility and autonomy to design and implement projects tailored to local needs. The schemes encompass a broad range of interventions, including crop production, infrastructure development, mechanization, and value addition. In 2017-18, the Scheme was restructured as the RKVY-RAFTAR (Remunerative Approaches for Agriculture and Allied Sector Rejuvenation), with an increased focus on strengthening both pre-and post-harvest infrastructure.

Way Forward:

- **Make Machinery Affordable**
  - **Easy finance:** Provide easier credit and targeted subsidies for tractors, harvesters, drones and other farm equipment.
  - **Shared ownership:** Promote Custom Hiring Centres (CHCs) so small farmers can rent machines instead of buying them.
- **Promote Small & Suitable Machines**
  - **Farm-size friendly:** Develop lightweight and affordable machines suitable for small and fragmented landholdings.

    *Development of mini-tractors (15 –25 HP), power tillers, and walk-behind weeders designed by institutes like **ICAR-CIAE (Central Institute of Agricultural Engineering, Bhopal)** specifically for small, fragmented plots (< 1 hectare).*
  - **Crop-specific tools:** Promote machinery designed for crops such as paddy, pulses, cotton, sugarcane and horticulture.
- **Use FPOs & Cooperatives**
  - **Collective access:** Encourage FPOs and cooperatives to purchase and share expensive machinery.
  - **Lower costs:** Bulk purchase, shared use and collective maintenance can make mechanisation more economical.
- **Promote Smart & Precision Mechanisation**
  - **Better technology:** Encourage drones, GPS, sensors, AI and precision equipment for efficient use of seeds, fertilisers, pesticides and water.
  - **Right machine, right job:** Focus on technologies that solve specific farm problems rather than promoting mechanisation for its own sake.
- **Build Rural Infrastructure**
  - **Better connectivity:** Improve farm roads, electricity and storage facilities to support mechanised agriculture.

    *Expansion of all-weather rural roads under **Pradhan Mantri Gram Sadak Yojana (PMGSY-III)** to connect remote agricultural fields directly to main roads.*

- **Repair network:** Develop village-level service centres, spare-parts facilities and trained mechanics to reduce machine downtime.

- **Encourage Sustainable Mechanisation**
  - **Clean energy:** Promote electric and solar-powered farm machinery where technically suitable.
  - **Protect soil**: Encourage appropriate machinery and practices that minimise soil compaction, erosion and fuel use.

Promotion of Laser Land Levelers and Zero-Till Seed Drills under the National Innovations in Climate Resilient Agriculture (NICRA) program to ensure uniform water coverage and prevent soil erosion.

Global Trends & Models in Agricultural Mechanization

Model	Countrie	Core Mechanism
Machinery-as-a-Service (MaaS) / "Uber for Tractors"	Kenya & Nigeria (*Hello Tractor*) China (*YTO Group*)	Digital booking platforms connect tractor owners with nearby smallholders for pay-per-use tilling, planting, and harvesting services.
Equipment Co-ownership / Machine Rings	Germany (*Maschinenringe*) Japan (*JA Cooperatives*)	Local farmer associations voluntarily pool, cross-rent, and share heavy equipment to optimize machine usage and cut idle capital.
Autonomous Swarm Robotics & Automation	UK (*Hands Free Hectare*) Japan & Australia	Uses small driverless tractors, AI-driven laser weeding robots, and autonomous rice transplanters to handle field operations without human drivers.
Precision Drone Operations	China (*DJI Agriculture*) South Korea	Widespread adoption of agricultural drones for ultra-low volume pesticide spraying, nano-fertilizer application, and multispectral crop mapping.
Controlled Environment & Vertical Automation	Netherlands & Israel	Automated glasshouses with robotic fruit pickers, climate sensors, and automated drip fertigation delivering high yield per square meter.

Conclusion

Agricultural mechanisation can make farming **faster, more productive and less labour-intensive**, while reducing costs and post-harvest losses. The focus should be on **affordable, appropriate and sustainable mechanisation**, especially for small and marginal farmers. With **Custom Hiring Centres, FPOs and modern technology**, mechanisation can help build a **more efficient and resilient Indian agriculture**.

MGP/BP/GS3/035

FOOD SECURITY & PUBLIC DISTRIBUTION

1. FOOD SECURITY IN INDIA: OVERVIEW

Introduction:

Food security means that all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs for an active and healthy life. India has moved from the ship-to-mouth dependence of the 1960s to a position of record foodgrain surplus, yet the challenge has changed rather than disappeared — the country now stores more grain than it needs while a large share of its population still cannot afford a nutritious diet.

Dimensions of Food Security

- **Availability:** Sufficient quantity of food through **domestic production, imports and government stocks** (buffer stocks with FCI).
- **Accessibility:** Food must be **physically and economically within reach** of every person without discrimination — depends on PDS network, roads, income and employment.
- **Affordability:** Households must have **enough purchasing power** to buy safe and nutritious food, not merely cereals.
- **Utilisation (Absorption):** Food must be **biologically absorbed** — requires clean water, sanitation, healthcare and dietary knowledge.
- **Stability:** Access and availability must be **stable over time**, resilient to droughts, floods, price shocks, pandemics and conflicts.

Note: The first three are the classic Indian framing (used in our NCERTs); FAO's four-pillar framework uses **Availability – Access – Utilisation – Stability**. Recent FAO work adds **Agency** and **Sustainability** as emerging pillars.

Framework of Food Security in India

- **Constitutional and Legal Basis**
  - **Article 21:** No explicit right to food, but the **right to life with dignity** has been judicially read to include the right to food — *PUCL v. Union of India (2001)*, the "Right to Food" case, which converted mid-day meals and ICDS into entitlements.
  - **Article 47 (DPSP):** Duty of the State to raise the **level of nutrition and standard of living** and improve public health.
  - **National Food Security Act, 2013:** Shifted food security from a **welfare approach to a rights-based approach**.
  - **Eleventh Schedule (Article 243G):** Panchayats entrusted with poverty alleviation and PDS-related functions.

- **National Food Security Act, 2013: Core Provisions**
  - **Coverage:** Up to **75% of rural** and **50% of urban** population — about **81.35 crore persons** as per Census 2011.
  - **Antyodaya Anna Yojana (AAY):** "Poorest of the poor" households — **35 kg of foodgrains per household per month**.
  - **Priority Households (PHH): 5 kg of foodgrains per person per month**.
  - **Women-centric provision:** The **eldest woman aged 18 or above** is deemed head of household for issuing ration cards.
  - **Nutritional support:** Free meals for **children aged 6 months–14 years** through Anganwadi Centres and schools; **maternity benefit of ₹6,000** for pregnant and lactating women.
  - **Grievance redressal: District Grievance Redressal Officers** and **State Food Commissions**; food security allowance if entitlements are not supplied.

Do You Know?

- **Record output:** Foodgrain production estimated at an all-time high of **376.56 million tonnes in 2025-26** (Third Advance Estimates)
- **Stocks far above norms:** As on **1 April 2026**, central pool stocks stood at **604.02 lakh tonnes** against a buffer norm of **210.40 lakh tonnes**.
- **Budget scale: ₹2,35,047 crore** allocated to the Department of Food and Public Distribution in **2026-27** — about **4.4% of total central expenditure**;
- **Cost of free grain:** The five-year free foodgrain commitment is estimated to cost around **₹12 lakh crore**.
- **Digital PDS: 99% of ration cards Aadhaar-seeded** (Dec 2025). ONORC live in **all 36 States/UTs**; **6.77 crore ration cards deleted** cumulatively.
- **The nutrition gap:** India ranks **102 out of 123** in the **Global Hunger Index 2025** with a score of **25.8 ("serious")**

Significance of Food Security:

- **Economic Significance**
  - **Stabilises the macroeconomy:** Food constitutes a large weight in the CPI basket; assured supply through buffer stocks **contains food inflation** and reduces the risk of monetary tightening.
  - **Protects farm incomes:** MSP-backed procurement provides an **assured price floor**, insulating farmers from distress sales and market crashes.
  - **Raises productivity of labour:** A well-nourished workforce translates into **higher human capital returns**— malnutrition is estimated to cost developing economies significant GDP each year through lost productivity.
- **Social and Human Development Significance**
  - **Reduces poverty burden:** Free foodgrains act as an **implicit income transfer**, freeing household budgets for health, education and housing.
  - **Nutritional floor for vulnerable groups:** Children, pregnant women and lactating mothers are covered through **PM POSHAN, Saksham Anganwadi and POSHAN 2.0**.
  - **Empowers women:** Making the eldest woman the head of household for ration purposes strengthens **intra-household bargaining power**.
- **Political and Strategic Significance**
  - **Sovereignty in a volatile world:** The Russia–Ukraine conflict and pandemic-era export bans demonstrated that **import dependence in staples is a strategic vulnerability**.

- **Crisis-response capacity:** PMGKAY during COVID-19 fed over 80 crore people and is regarded as among the **world's largest social protection responses**.
- **Bargaining position at the WTO:** Public stockholding underpins India's demand for a **permanent solution on** food security subsidies.

- **Developmental and SDG Significance**
  - Directly serves **SDG-2 (Zero Hunger)** and indirectly SDG-1 (No Poverty), SDG-3 (Good Health) and SDG-10 (Reduced Inequalities).
  - Anchors the **demographic dividend** — stunting in early childhood permanently reduces cognitive and earning potential.

In-Kind PDS vs Cash Transfer vs Food Coupons — The Reform Debate

Basis	In-Kind Transfer (Current PDS)	Direct Cash Transfer (DBT)	Food Coupons / Vouchers
Basic idea	Physical foodgrains supplied free through Fair Price Shops.	Cash equivalent of food subsidy credited to the beneficiary's bank account.	Restricted-use vouchers redeemable only against notified food items.
Nutritional effect	Guarantees calories, but locks consumption into rice and wheat.	Allows dietary diversification, but cash may be diverted to non-food use.	Preserves choice within food, while preventing diversion to non-food items.
Administrative cost	**Very high** — procurement, transport, storage and handling.	**Low** — leverages the JAM Trinity (Jan Dhan, Aadhaar, Mobile).	**Moderate** — requires a network of accredited retailers.
Leakage risk	Diversion, ghost cards and transit losses are possible.	Lower diversion, but risks exclusion due to banking-access gaps.	Lower than in-kind transfers; however, a grey market in coupons may emerge.
Effect on farmers	Sustains open-ended MSP procurement and assured demand.	Weakens the procurement link, which may erode price support.	Neutral to procurement, depending on scheme design.
Inflation exposure	Beneficiary is insulated as the quantity is fixed.	Real value erodes unless indexed to food inflation.	Value erodes unless periodically revised.
Indian experience	Nationwide; default model under **NFSA**.	Pilots in **Chandigarh, Puducherry and Dadra & Nagar Haveli** since 2015; **Nagri (Ranchi) pilot, 2017** was withdrawn within 10 months.	Limited trials in **Andhra Pradesh, Bihar and Jammu & Kashmir**.
Committee / Expert view	Retained as the legal entitlement under NFSA.	**Shanta Kumar Committee (2015)** favoured it for farmer-beneficiaries and estimated savings of **₹30,000–35,000 crore**.	**World Bank (2017)** noted that vouchers can address the concern of **cash being diverted to non-food uses**.

Issues and Challenges in India's Food Security System:

- **Targeting and Coverage Challenges**
  - **Frozen coverage based on outdated Census:** NFSA coverage is still calculated on **Census 2011**, though the Act requires the "latest Census". India's population is estimated to have grown by about **18% between 2011 and 2026**, leaving genuinely eligible households outside the net.

    In March 2026, the Government informed the Lok Sabha that there is scope to add about 1.95 crore beneficiaries under existing NFSA coverage norms; as of August 2025, states had identified 80.6 croreof the permissible 81.35 crore. The Supreme Court (2021) had directed the Centre to re-determine coverage.
  - **Inclusion and exclusion errors coexist:** Non-poor households retain cards while migrants and the urban homeless are excluded.

    ˇInclusion error under PDS is estimated to have risen from 29% (2004-05) to 37% (2011-12). Meanwhile, poverty at the $2.15 (2017 PPP) line fell from 16.2% in 2011-12 to 2.4% in 2022-23, yet coverage remains at 67% — the Shanta Kumar Committee (2015) had argued that 40% coverage would comfortably cover BPL households.
  - **Reduction in per-capita entitlement:** The move to 5 kg per person under NFSA left BPL households worse off than the earlier TPDS norm of **7 kg per person**. The **draft National Food Security (Amendment) Bill, 2026**proposes shifting AAY from a flat 35 kg per household to **7 kg per person, capped at 35 kg per household**

Fiscal and Subsidy Challenges

- **Ballooning and rigid subsidy bill:** The Central Issue Price has been **zero since January 2023**, while procurement costs keep rising with MSP, so the entire gap is absorbed as subsidy.

  ˇFCI's economic cost in 2024-25 stood at ₹28.5 per kg for wheat and ₹40.4 per kg for rice, growing at about 3% annually over the previous decade. Roughly 80% of this cost is the pooled cost of grain, i.e. MSP paid to farmers.
- **Crowding out of capital investment:** A large share of the agriculture-related budget flows into subsidies and DBT rather than **irrigation, mechanisation, storage and R&D**, which have far higher long-run multipliers.

Procurement and Production Distortions

- **Geographic concentration of procurement:** Despite NFSA's mandate for geographical diversification, procurement remains confined to a handful of states, lengthening transport chains and inflating costs.

  ˇIn 2024-25, 92% of wheat procurement came from just three states — Punjab, Haryana and Madhya Pradesh — while 65% of paddy came from five states (Punjab, Chhattisgarh, Telangana, Odisha, Uttar Pradesh). Punjab and Haryana alone accounted for 64% of wheat and 27% of paddy.
- **Open-ended procurement and ecological cost:** Assured MSP procurement of paddy and wheat has entrenched a **water-intensive rice–wheat monoculture** in the north-west, driving groundwater depletion, soil salinity, declining NPK balance and stubble burning.
- **Slow adoption of decentralised procurement:** DCP covered only about **65% of wheat and rice procurement in 2024-25**.
- **High statutory charges:** Mandi fees, market fees and *arhatia* commission are levied as a **percentage of MSP**, so high-procurement states such as Punjab and Haryana levy charges well above the Centre's suggested cap of **2% of MSP**.

Storage, Movement and Stock Management

- **Chronic over-stocking:** Stocks routinely run at multiples of the norm, raising carrying costs without any prescribed **maximum** stock ceiling.

  ˅The CAG (2023) found that average stocks between 2017-18 and 2022-23 were nearly twice the required level, and that liquidation through OMSS and exports was inadequate. As on 1 January 2026, wheat stocks were about two times and rice about four times the norms.

- **Heavy dependence on hired storage:** FCI owned only **154 LMT** of capacity as of December 2025 against **327 LMT hired** — hired capacity is **68% of FCI's total**. The Standing Committee (2025) noted there has been **no systematic cost-benefit assessment** of renting versus owning.
- **Transit losses and logistics inefficiency:** *CAG (2023) valued **transit losses at ₹1,652 crore** over 2017-18 to 2021-22 and noted that FCI forwent a rail freight rebate of about **₹1,736 crore** by not availing the Railways' Long-Term Tariff Contract policy of 2017.*
- **Diversion from the pipeline:** Leakage occurs through exclusion, diversion to the open market, and pilferage during handling.

  ˅ICRIER estimated leakage at 47% of offtake in 2011-12, improving to about 28% in 2022-23 with digitisation. The Department of Food and Public Distribution has contested this methodology, arguing that offtake also includes stocks in transit, buffer allocations and grain for other welfare schemes.

Nutritional Challenges:

- **Cereal-centric basket:** PDS has not diversified beyond rice and wheat despite NFSA's mandate.

  ˅In 2025-26, rice (65%) and wheat (33%) accounted for almost the entire PDS allocation, with coarse grains at just 1.6%. Pulses were supplied for only about eight months during COVID-19 and were not continued.

- **Double burden of malnutrition:** Undernutrition now coexists with rising obesity — overweight children rose from **2.7 million (2012) to 4.2 million (2024)**, and adult obesity from 33.6 million to 71.4 million.
- **Fortification gaps:** Fortified rice distribution is approved up to **December 2028** but **FSSAI has notified only 56 testing labs across 15 states**; **13 states and 8 UTs have none**

  ˅A NITI Aayog–ICMR-NIN white paper (2023) concluded fortification would produce only a modest reduction in anaemia.

Climate and Sustainability Challenges:

- **Yield vulnerability:** Erratic monsoons, heat stress during grain-filling, unseasonal rain and hailstorms increasingly threaten output stability.
- **Irrigation and input inefficiency:** *As per the **Economic Survey 2025-26**, gross irrigated area rose to **55.8% of gross cropped area (2022-23)** from 41.7% in 2001-02, but coverage is skewed — under **15% for millets** against nearly **67% for paddy**. The **NPK ratio has deteriorated** due to price distortions favouring nitrogenous fertilisers, and yields of cereals, maize, soybean and pulses remain **below global averages**.*
- **Land and soil constraints:** Fragmented landholdings, declining soil organic carbon, salinisation and land-use conversion shrink the productive base.
- **Pest and biosecurity shocks:** The **Fall Armyworm** outbreak (2018) and **desert locust** invasions in Rajasthan and Gujarat (2020) demonstrate how quickly a surplus position can be strained.

External and Trade-Related Challenges

- **Unresolved WTO stalemate:** Under the Agreement on Agriculture, product-specific support for developing countries is capped at **10% of the value of production**.

- **Import dependence in pulses and edible oils:** Despite being the largest producer and consumer of pulses, India continues to import; edible oil import dependence remains a structural vulnerability.
- **Global supply-chain fragility:** The COVID-19 pandemic and the Russia–Ukraine war showed how quickly export restrictions abroad transmit into domestic price shocks, and how India's own export bans (broken rice, wheat, non-basmati rice, onion) affect food-deficit partners.

Measures Taken to Strengthen Food Security in India

- **Legal and Entitlement Framework**
  - **National Food Security Act, 2013:** Converted food access into a **justiciable entitlement** for about two-thirds of the population.
  - **Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY):** Launched in **April 2020**; merged with NFSA from January 2023 to make the entire entitlement free; extended for five years to **December 2028** covering **81.35 crore persons**.
  - **Draft NFSA (Amendment) Bill, 2026:** Proposes **per-person entitlement of 7 kg for AAY members**, capped at 35 kg per household, to remove intra-AAY inequity; public consultation closed on **13 July 2026**.
- **Production and Productivity Push**
  - **PM Dhan-Dhaanya Krishi Yojana (PMDDKY):** Launched **11 October 2025** with an outlay of **₹24,000 crore**over six years (to 2030-31); converges existing schemes across **100 low-productivity districts**, targeting **1.7 crore farmers**; monitored through a **NITI Aayog dashboard** on the Aspirational Districts model.
  - **Mission for Aatmanirbharta in Pulses:** Outlay **₹11,440 crore**; aims to raise pulse area from **27.5 to 31 million hectares** and production from **24.2 to 35 million tonnes by 2030-31**; provides **1.26 crore quintals of certified seed**, **88 lakh free mini-kits**, and **100% MSP procurement of registered farmers' produce**.
  - **National Mission on Natural Farming and PKVY:** Reduce input cost and restore soil health for long-term production sustainability.
  - **Seed and research backbone:** ICAR released **339 crop varieties in 2025-26** suited to different agro-climatic zones; the **Sub-Mission on Seeds and Planting Material** has created **6.85 lakh seed villages** benefiting over 2.5 crore farmers.
- **Procurement, Stocking and Price Stabilisation**
  - **MSP for 23 crops** with a formula of at least **1.5 times the A2+FL cost** since 2018-19.
  - **Decentralised Procurement Scheme** to localise procurement and diversify the grain basket.
  - **Price Stabilisation Fund** and buffer stocks of pulses and onion; retail interventions through **Bharat Atta, Bharat Rice and Bharat Dal**.
  - **Open Market Sale Scheme (OMSS)** to offload surplus stocks and cool open-market prices.
- **PDS Technology and Governance Reforms**
  - **End-to-end computerisation and ePoS automation: 5.41 lakh of 5.43 lakh** Fair Price Shops are ePoS-enabled; **97.9% of transactions** used Aadhaar authentication in December 2025.
  - **One Nation One Ration Card:** Operational across **all 36 States/UTs**, enabling portability for migrant workers.
  - **Smart-PDS:** An integrated national system for standardised, interoperable PDS operations.
  - **Aadhaar seeding and eKYC: 99% seeding** achieved; **6.77 crore** duplicate, ghost or ineligible cards deleted cumulatively.
  - **Annavitran and Mera Ration app** for transparency in allocation and offtake.

- **Nutrition-Sensitive Interventions**
  - **Mission Saksham Anganwadi and POSHAN 2.0:** Merges ICDS, POSHAN Abhiyaan, Scheme for Adolescent Girls and the National Creche Scheme; over **1.29 lakh Anganwadi Centres upgraded**; supplementary nutrition norms revised in January 2023 for diet diversity.
  - **Poshan Tracker:** Real-time ICT platform covering roughly **9 crore beneficiaries**, enabling dynamic identification of stunting, wasting and underweight prevalence.
  - **PM POSHAN (Mid-Day Meal):** Hot cooked meals for schoolchildren, with fortified ingredients and **Poshan Vatikas** (nutrition gardens).
  - **Rice fortification:** Universalised across PDS and welfare schemes, extended to **December 2028** with an outlay of **₹17,082 crore**.
  - **Pradhan Mantri Matru Vandana Yojana** for maternal nutrition; **Anaemia Mukt Bharat** for micronutrient deficiency.
- **Risk Management and Infrastructure**
  - **Pradhan Mantri Fasal Bima Yojana:** Over **4.19 crore farmers insured in 2024-25** against natural calamities, pests and adverse weather.
  - **Agriculture Infrastructure Fund** and **Pradhan Mantri Kisan Sampada Yojana** for storage, cold chains and processing.
  - **World's largest grain storage plan in the cooperative sector**, creating godowns at the PACS level to decentralise storage.
  - **eNAM** for market integration; **PM-KISAN** and **MGNREGA** to strengthen purchasing power, which underpins economic access.
- **International Positioning**
  - India champions a **permanent solution on public stockholding** at the WTO, seeking amendment of the outdated **1986-88 external reference price** and inclusion of programmes introduced after 2013 under the Peace Clause. India also leads on **millets diplomacy** following the International Year of Millets (2023).

Way Forward:

- **Targeting and Entitlements**
  - **Update the Denominator:** Recalibrate NFSA coverage as soon as Census 2027 results are available, so that eligible households are not excluded by an outdated population base.

    Coverage still rests on Census 2011 estimates, which is the single largest source of exclusion error in the system.
  - **Dynamic Registries:** Replace static beneficiary lists with periodic re-verification and self-declaration backed by random audits and linked to socio-economic databases.
  - **Tiered Entitlements:** Differentiate the basket rather than de-listing the beneficiary, with free grain for AAY and the poorest and partial subsidy for the near-poor.
- **Nutrition Security**
  - **Diversify the Basket:** Progressively include millets, pulses and edible oil in the PDS, using local procurement so that the grain matches regional dietary preference.

    Odisha's Millet Mission has brought ragi into the state PDS and the ICDS and PM POSHAN supply chain, backed by procurement at MSP, the clearest working proof that diversification is administratively feasible.
  - **First 1,000 Days:** Concentrate resources on maternal nutrition, exclusive breastfeeding and complementary feeding, since stunting is largely irreversible after the age of two.

  - **Fortification Quality:** Expand FSSAI-notified testing laboratories to states that currently have none, and run public awareness campaigns to counter misperceptions.
- **Subsidy and Procurement**
  - **Revisit Central Issue Price:** Raise the issue price for better-off beneficiaries while fully protecting AAY and priority households.

    Recommended by both the Fifteenth Finance Commission and the Economic Survey 2020-21.
  - **Shift MSP Geographically:** Extend assured procurement for pulses, oilseeds and millets in rainfed regions, easing the north-western groundwater crisis and import dependence at the same time.

    The Mission for Aatmanirbharta in Pulses commits to procuring 100 per cent of registered farmers' produce at MSP, and its record in Madhya Pradesh, Maharashtra and Rajasthan will decide whether diversification becomes real.
  - **Universalise Decentralised Procurement:** Extend DCP to all states with central assistance for funds, storage and manpower, shortening transport chains and matching grain to local taste.
- **Storage and Movement**
  - **Cap Stocks, Not Only Floor Them:** Prescribe a maximum stock ceiling alongside the buffer norm, with an automatic liquidation trigger through OMSS or exports once it is breached.
  - **Cut Logistics Cost:** Use the Railways' Long Term Tariff Contract, formalise direct issue of grain from railhead to state godowns, and cap state statutory charges at the suggested 2 per cent of MSP.
- **Technology and Delivery**
  - **Complete eKYC and ONORC:** Run focused drives in lagging states and awareness campaigns for migrants, who are the intended beneficiaries of portability.

    Inter-state portability still accounts for under 1 per cent of transactions, with Manipur, Meghalaya and Uttarakhand lagging furthest on eKYC completion.
  - **Geospatial Planning:** Deploy FASAL for crop forecasting, NADAMS for drought monitoring, CHAMAN for horticulture assessment and NDVI-based indices for early warning and targeted allocation.
  - **Pilot Choice-Based Delivery:** Test food vouchers or conditional cash transfer in urban areas with dense retail and high banking penetration, retaining in-kind PDS in remote and tribal blocks where markets are thin.
- **Climate Resilience**
  - **Stress-Tolerant Varieties:** Mainstream ICAR's climate-resilient releases and expand micro-irrigation, direct-seeded rice and natural farming in over-exploited groundwater blocks.
  - **Strengthen Risk Transfer:** Improve PMFBY claim settlement timelines and expand weather-index insurance for rainfed crops, where yield-based assessment works poorly.
  - **Geographic Diversification:** Develop the eastern India rice belt and the rainfed Deccan as secondary production and procurement hubs to reduce dependence on a single region.
- **External Environment**
  - **Permanent WTO Solution:** Press for amendment of the 1986-88 external reference price, coverage of post-2013 programmes under the Peace Clause, and easier notification requirements on public stockholding.
  - **Predictable Trade Policy:** Replace abrupt export bans with calibrated, rules-based instruments such as export duties and quotas with notice periods.

    Sudden bans erode India's standing as a reliable supplier, which matters when the country accounts for a large share of the global rice trade.

Conclusion:

India has solved the problem of growing enough food and not the problem of feeding people well. Record output of 376.56 million tonnes and stocks near three times the buffer norm sit alongside a "serious" hunger ranking and the world's highest burden of child wasting. The binding constraint has moved **from production to distribution, nutrition and ecological sustainability.** Food security must therefore be measured not by the height of the grain stack, but by the health of the child at the end of the ration queue.

2. PUBLIC DISTRIBUTION SYSTEM

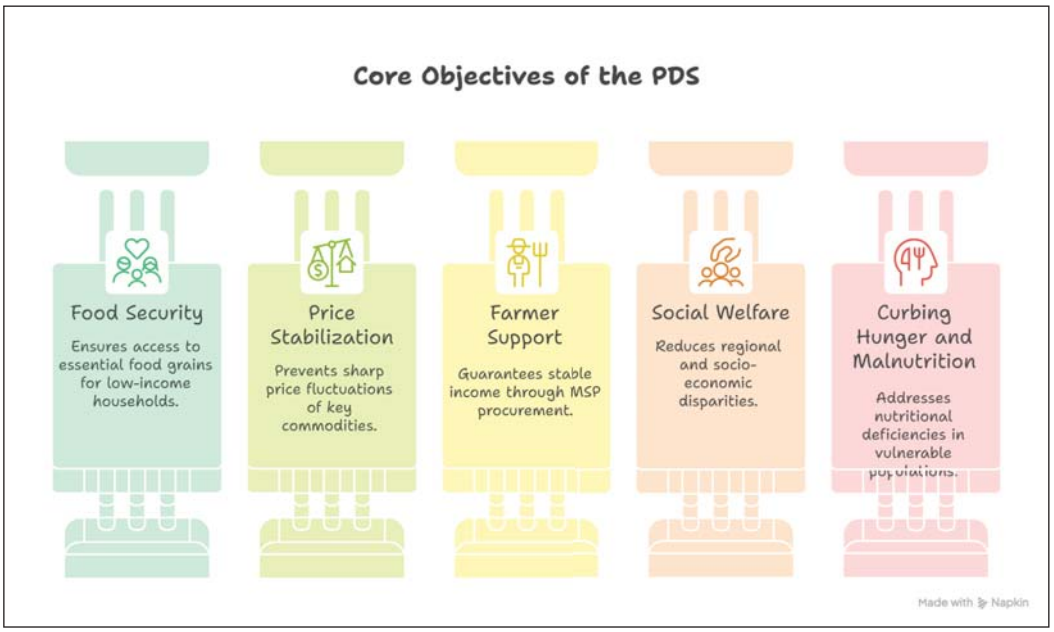
Introduction:

India's Public Distribution System (PDS) is a cornerstone of the nation's food security framework, striving to ensure that essential food grains reach economically vulnerable populations. Managed by the Ministry of Consumer Affairs, Food, and Public Distribution, the PDS aims to provide subsidized food through a vast network of Fair Price Shops (FPSs), playing a pivotal role in stabilizing market prices, supporting farmers, and promoting social welfare.

Structure and Functioning of the PDS

The PDS operates through a decentralized framework involving both **central and state governments**, ensuring that national objectives meet local needs:

- **Central Government**: Through the Food Corporation of India (FCI), the center is responsible for **procurement, storage, transportation, and bulk allocation** of food grains. Procurement at **Minimum Support Prices (MSP)** protects farmers' income, particularly for wheat and rice.
- **State Governments**: States take charge of **allocating grains locally, identifying eligible households, issuing ration cards, and supervising FPS operations**.



Evolution of the PDS

- **1940s**: Initiated during World War II to counter food shortages.
- **1960s**: Expanded with the creation of the Agricultural Prices Commission and the Food Corporation of India (FCI) to manage procurement and distribution.

- **1992**: Revamped Public Distribution System (RPDS) introduced to cover remote areas.
- **1997**: Targeted Public Distribution System (TPDS) introduced, focusing on Below Poverty Line (BPL) and Above Poverty Line (APL) families.
- **2000**: Antyodaya Anna Yojana (AAY) implemented, providing highly subsidized food grains to the poorest households.
- **2013**: National Food Security Act (NFSA) passed, legally guaranteeing food entitlements for **over 82 crore people**.

Significance of the PDS:

- **Poverty Alleviation-** By providing subsidized grains, the PDS ensures that low-income families get basic nutrition, reducing hunger and improving living standards.
- **Food Security and Price Stabilization-** The PDS protects poor households from sudden price spikes by supplying essential grains at subsidized rates, especially during inflation.
- **Farmer Support and Stable Income-** Procurement at Minimum Support Prices (MSP) guarantees farmers a steady market and income, encouraging production in states like Punjab and Haryana.
- **Crisis Mitigation-** During emergencies, such as the COVID-19 pandemic, the PDS acts as a safety net. The PM Garib Kalyan Anna Yojana (PMGKAY) provided free food grains to around 800 million people.
- **Nutritional Support-** Some states, like Tamil Nadu, distribute pulses, fortified rice, and edible oils through the PDS to improve diets and combat malnutrition among children and women.
- **Regional Equity and Social Inclusion-** Grains are sent from surplus areas to deficit regions, ensuring food equity across the country. The One Nation One Ration Card (ONORC) scheme allows migrant workers to access their entitlements anywhere.

Challenges and Issues in India's Public Distribution System:

- **Quality and Nutritional Gaps**: The system primarily distributes wheat and rice, with limited availability of pulses, oils, or fortified foods. This restricts dietary diversity and leaves nutritional needs, particularly for children and pregnant or lactating women, largely unmet.
- **Leakages and Diversions**: A significant portion of allocated grains—around 28%—does not reach the intended beneficiaries, resulting in annual losses exceeding ₹69,000 crore. Illegal diversion to the open market continues to undermine the program's purpose.
- **Targeting Errors**: Some genuinely needy households are excluded, while certain non-poor households access subsidized food. **For example**, only about 57% of the population benefits under Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY), even though 12.9% of Indians live in extreme poverty (World Bank, 2022).
- **Supply Chain Inefficiencies**: Weak storage, transport, and distribution networks cause delays and shortages, especially in remote areas. Poor storage also contributes to quality deterioration and food wastage.
- **Monocropping and Limited Agricultural Sustainability**: Heavy reliance on wheat and rice restricts crop diversity, reduces resilience in farming systems, and does not fully address nutritional security.
- **Rising Subsidy Burden**: Government spending on food subsidies has surged from ₹21,200 crore in 2002–03 to ₹2 lakh crore in 2024–25, highlighting the need for improved efficiency and cost management.

Recent PDS Reforms and Technological Innovations

- **Digitization of Ration Cards-** All States and Union Territories have digitized ration cards and beneficiary data under the NFSA. About **20.5 crore ration cards**, covering **80.5 crore beneficiaries**, are now available online for easy verification and real-time monitoring.

- **Aadhaar Seeding and Biometric Verification-** Over **99.8% of ration cards** are linked to Aadhaar, and more than **97% of food grain transactions** are verified biometrically. This helps prevent fraud, improve targeting, and reduce leakages.
- **Automation of Fair Price Shops (FPSs)-** Almost **99.6% of FPSs** (5.41 lakh out of 5.43 lakh) use **electronic Point of Sale (ePoS) devices**, ensuring transparent and accountable distribution of food grains.
- **Computerization and GPS Tracking-** Online stock management, web-based truck challans, and GPS tracking in states like **Chhattisgarh and Tamil Nadu** help monitor food grain movement from depots to FPSs, preventing diversion and ensuring timely delivery.
- **One Nation One Ration Card (ONORC)-** The **ONORC scheme** allows beneficiaries to access PDS entitlements from **any FPS across India**, helping migrant workers and mobile populations. By mid-2023, the scheme was implemented nationwide.
- **Promotion of Nutri-Cereals-** States are distributing **millets and other nutritious grains** through the PDS to improve dietary diversity and address nutritional gaps.
- **Social Audits and Community Participation-** Local communities participate in **social audits**, which reduces corruption, improves accountability, and builds public trust in the system.

Way Forward for Strengthening India's PDS

- **Universal Coverage-** The PDS should aim to reach every citizen, ensuring that no one is left without access to essential food supplies.
- **Aadhaar Integration & Direct Benefit Transfer (DBT)-** Linking ration cards with Aadhaar can help eliminate ghost beneficiaries. Where feasible, cash transfers can be used to reduce leakages and make the system more efficient.
- **Digitization & Smart Fair Price Shops-** FPS operations should be fully computerized with GPS tracking, digital stock updates, and SMS alerts. Fair Price Shops can be transformed into "smart shops" featuring biometric authentication, electronic weighing, QR-coded grain certificates, and digital payments.
- **Portable Benefits & Migrant Support-** Strengthening the **One Nation One Ration Card (ONORC)** scheme will allow migrants to access their PDS entitlements anywhere in the country. Temporary ration cards for seasonal workers and automatic benefit transfers can further support mobile populations.
- **Modern Storage Infrastructure-** Warehouses should be upgraded to modern silos with climate control and IoT-enabled monitoring. Hub-and-spoke storage models and public-private partnerships can improve storage efficiency and reduce losses.
- **Nutritional Security-** PDS should go beyond cereals by including pulses, oils, fortified foods, and nutrient-rich millets.
- **Public Participation & Accountability-** Social audits, community monitoring, and involvement of NGOs can strengthen transparency and build trust in the system.
- **Shanta Kumar Committee Recommendations (2015)**
  - **Direct Procurement by States:** States like Andhra Pradesh and Chhattisgarh procure grains directly from farmers for better pricing and timely procurement.
  - **Private Sector Involvement:** Engage private players in storage, transport, and distribution to improve efficiency.
  - **Diversified Procurement:** Include nutritious crops like millets to reduce dependence on rice and wheat and promote dietary diversity.
  - **End-to-End Computerization:** Implement ePOS at Fair Price Shops (FPSs) and GPS tracking of PDS trucks to minimize leakages and ensure accountability

3. NATIONAL FOOD SECURITY ACT

Introduction:

The National Food Security Act of 2013 was enacted with the objective of providing food and nutritional security by ensuring access to adequate quantity of quality food at affordable prices for people to live a life with dignity. The Act **covers up to 75% of the rural population and up to 50% of the urban population** for receiving subsidized good grains under the Targeted Public Distribution System (TPDS), thus covering about two-thirds of the population.

Provisions:

- Under NFSA, the government is obliged to give subsidized foodgrains every month to the beneficiaries identified by each state government based on the economic status of households.
- Identification of Beneficiaries: The identification of eligible households is done by the State Governments, adhering to inclusion/exclusion criteria set by the central government.
- Two categories of beneficiary households:
  - **Antyodaya Anna Yojana (AAY):** AAY households receive 35 kg of foodgrains monthly.
  - **Priority Households (PHH):** Each PHH member gets 5 kg of foodgrains monthly, including rice at Rs 3/kg, wheat at Rs 2/kg, and coarse grain at Rs 1/kg.
- **Nutritional Support for Children:** Children between 6 months and 14 years receive nutritious, free meals through the Integrated Child Development Services (ICDS) and Mid-Day Meal (MDM) schemes.
- **Maternity Entitlement:** Pregnant women and lactating mothers are entitled to a maternity benefit of at least ₹6,000 for six months to partially compensate for wage loss and ensure nutritional support.
- **Institutional Framework:**
  - Establishes State Food Commissions for monitoring and implementation.
  - Creates a grievance redressal mechanism at district level.
  - Mandates transparency and accountability through social audits.

Significance:

- **Philosophical Shift: From Welfare to Rights:**
  - The most significant contribution of the NFSA is the paradigm shift it mandated in India's approach to public assistance.
  - By explicitly stating that food access is a legal right for two-thirds of the population, the NFSA ensures that the government is legally obligated to provide subsidized food grains. This allows citizens to seek judicial recourse if entitlements are denied, fundamentally changing the relationship between the state and the beneficiary.
- **Food & Nutritional Security:**
  - Through the Public Distribution System (PDS), the Act ensures monthly entitlements of rice, wheat, and coarse grains at highly subsidized rates to over 810 million beneficiaries.
  - By guaranteeing affordable access to staples, the NFSA combats undernutrition and food insecurity among the poor.
  - It reduces vulnerability to food price shocks, safeguarding low-income households against hunger.
- **Enhanced Social Inclusion:**
  - Prioritizes marginalized sections, including SC/STs, women, and children, helping to address inequality.
  - Ensures women's empowerment with provisions for female household heads as ration card holders.

- **Poverty Reduction:** The assured supply of foodgrains acts as a form of income transfer. By reducing household expenditure on essential food items, it frees up a significant portion of the income of poor families, allowing them to spend on education, health, and other needs. This indirectly aids in poverty reduction.
- **Crisis Management:** During national crises, such as the COVID-19 pandemic, the NFSA mechanism allowed the government to immediately scale up distribution (e.g. through schemes like PMGKAY), ensuring no person went hungry when livelihoods were lost. The existing buffer stock and distribution network were instrumental in this massive relief effort.
- **Driver of PDS Reforms & Accountability:** The NFSA necessitated and accelerated long-pending reforms in the Public Distribution System (PDS), drastically improving its efficiency and reducing leakage such as:
  - **Technology Integration:** The mandate for effective delivery led to the widespread adoption of e-PDS (electronic PDS), including Aadhaar linking and the use of Point of Sale (PoS) devices at Fair Price Shops. This has significantly eliminated "ghost" beneficiaries and reduced the diversion of grains.
  - **Grievance Redressal:** The requirement for state-level and district-level Grievance Redressal Officers and mandatory Vigilance Committees has built a layer of institutional accountability into the food distribution system, enhancing transparency.
  - **Portability (ONORC):** The success of the NFSA is closely tied to the One Nation, One Ration Card (ONORC) scheme, which ensures that NFSA entitlements are portable. This means a beneficiary can collect their subsidized food grains from any Fair Price Shop (FPS) across the country, hugely benefiting migrant workers.

Challenges/Limitations:

- **Leakages and Inefficiency in PDS:**
  - Persistent leakages, diversion of foodgrains, and corruption in the Public Distribution System (PDS) undermine efficient delivery of benefits.
  - Errors in beneficiary identification and exclusion/inclusion issues result in deserving households missing out or ineligible ones receiving benefits.
- **Quality of Foodgrains:**
  - Beneficiaries often receive poor-quality or spoiled food grains, undermining the nutritional objectives of the Act.
  - There is little provision within the Act for pulses, oils, or more nutritious foods.
- **Fiscal Burden and Sustainability:**
  - **Food Subsidy:** The difference between the Minimum Support Price (MSP) paid to farmers and the Central Issue Price (CIP) charged to beneficiaries (₹1, ₹2, ₹3) is borne by the government as the food subsidy. This subsidy is one of the largest items of non-developmental expenditure, putting pressure on the national fiscal deficit.
  - **Buffer Stocks:** Maintaining adequate buffer stocks for distribution requires significant investment in storage and incurs heavy costs related to interest, storage, and wastage.
- **Supply Chain and Infrastructure Gaps:**
  - Shortcomings in logistics, transportation, and storage infrastructure lead to wastage and inefficiency.
  - Old and insufficient warehouses, especially in remote regions, make timely and quality delivery challenging
- **Exclusion and Inclusion Errors:**
  - **Exclusion Errors:** Genuinely poor and eligible households are left out of the list (often due to outdated data or complex documentation requirements), meaning those who need the food the most don't receive it.

  - **Inclusion Errors:** Non-poor or ineligible households manage to acquire subsidized ration cards, leading to wastage of public resources.
  - Migration, urbanization, and inter-state movement are not addressed adequately.
- **Limited Nutrition Focus:**
  - The Act focuses mainly on calorie security through grains, neglecting dietary diversity and micronutrient requirements.
  - Provisions for pulses, oils, and fresh foods are minimal or absent.
- **Functioning of Fair Price Shops (FPS):** While the implementation of PoS devices has improved transparency, challenges remain. These include internet connectivity issues in remote areas, resulting in transaction failures, and residual corruption by FPS dealers regarding timing, weighing, and behavior.
- **Weak Enforcement:** State-level Grievance Redressal Officers (GROs) often lack the authority, resources, or political will to effectively penalize violators or ensure timely resolution of complaints. The mandated Vigilance Committees are often defunct or non-functional.

Way Forward:

- **Updating Population Coverage:** The Act must be amended to use current population estimates instead of the rigid 2011 Census figures. This change is crucial to include families who have moved into poverty or who were born after 2011, thereby reducing exclusion errors.
- **Dynamic Exclusion Criteria:** Instead of relying solely on old BPL (Below Poverty Line) lists, states should implement dynamic exclusion criteria using technology and data points (e.g., vehicle ownership, government employment status, electricity consumption) to regularly update and clean beneficiary lists.
- **Universalizing ONORC:** While the One Nation, One Ration Card (ONORC) scheme has been implemented, continuous investment in seamless biometric and internet infrastructure is needed. The goal should be to make ration portability a frictionless experience, especially for internal migrant workers.
- **Mandating Fortification:** Central and state governments should mandate the fortification of all rice and wheat supplied through the PDS with essential micronutrients like Iron, Folic Acid, and Vitamin B12. This is a highly cost-effective public health measure.
- **Integrating with Local Produce:** Pilot programs should be launched to link Fair Price Shops (FPS) with local self-help groups (SHGs) or farmer producer organizations (FPOs) to provide seasonal, locally sourced, and subsidized fresh fruits and vegetables to beneficiaries, particularly through ICDS and Mid-Day Meal schemes.
- **Smart Procurement and Storage:** Modernize the food supply chain by shifting from traditional bag storage to steel silos and implementing End-to-End Computerization of the PDS. This drastically reduces storage costs and physical wastage (spoilage and theft).

Conclusion:

The NFSA can evolve from a basic food distribution program into a dynamic, transparent, and nutrition-sensitive social security pillar for the nation if its challenges are addressed & next-gen reforms are implemented properly.

MGP/BP/GS3/036

LIVESTOCK & ALLIED SECTORS

1. ECONOMICS OF ANIMAL REARING

Introduction:

Animal rearing covers **dairying, poultry, meat and wool production, piggery, apiculture and allied activities**, and functions in India less as a standalone industry than as the risk-diversifying arm of the farm household. It converts crop residue and common grazing land into daily cash income, which is why it matters **most precisely to those with the least land**.

Position of the sector:

- Livestock contributes about **5.5 per cent of total GVA in 2023-24**, up from 4.4 per cent in 2014-15, and roughly **31 per cent of agriculture and allied GVA**, having grown at a CAGR of about **12.77 per cent** at current prices since 2014-15.
- India ranks **first in milk production with about 25 per cent of global output, second in eggs and fourth in meat**, and accounts for roughly 43 per cent of world buffalo meat production.
- **Milk output reached 247.87 million tonnes in 2024-25**, up 3.58 per cent from 239.30 million tonnes, with egg production of 149.11 billion, meat of 10.50 million tonnes and wool of 34.57 million kg.
- **Nodal ministry:** Department of Animal Husbandry and Dairying under the Ministry of Fisheries, Animal Husbandry and Dairying, with data from the quinquennial **Livestock Census** and the annual **Integrated Sample Survey** published as Basic Animal Husbandry Statistics.

Significance of Animal Rearing:

- **Income and Livelihood**
  - **Daily Cash Flow:** Unlike crops, which pay once or twice a year, livestock yields income every day, which smooths household consumption and reduces dependence on moneylenders between harvests.
  - Dairying alone directly supports more than eight crore farmers, making it the single largest rural livelihood after crop cultivation.
  - **Pro-Poor Asset:** Ownership is inversely related to landholding, so the sector transfers income to precisely the households that crop policy reaches least.

    ˇAs per DAHD, more than 80 per cent of low-producing indigenous animals are held by small and marginal farmers and landless labourers.
  - **Landless Participation:** Livestock is the only agricultural asset that can be owned and operated without owning land, using common grazing and crop residue.
- **Risk Diversification**
  - **Buffer Against Crop Failure:** Livestock earnings are far less sensitive to monsoon variability than crop income, so the herd functions as insurance during drought and flood years.

- **Liquid Asset:** Animals can be sold at short notice to meet medical, marriage or input expenses, serving as a savings instrument where formal banking penetration is thin.
- **Countercyclical Growth:** The sector has grown steadily even in years when crop growth stalled, which is why it now accounts for nearly a third of agricultural GVA.

- **Nutrition and Food Security**
  - **Protein Access:** Milk, eggs and meat are the cheapest sources of animal protein and micronutrients for households facing widespread anaemia and stunting.

    Per capita milk availability has risen from 319 grams per day in 2014-15 to about 485 grams per day in 2024-25, well above the ICMR recommendation.
  - **Nutritional Reach of the Poor:** Backyard poultry and small ruminants deliver protein directly to the household without requiring a market transaction.
- **Women's Empowerment**
  - **Female-Dominated Work:** Animal rearing is carried out largely by women, so productivity gains in this sector translate more directly into women's earnings than in crop agriculture.
  - **Institutional Entry Point:** Women's dairy cooperatives have proved to be one of the few large-scale channels for putting cash directly into women's hands.

    Amul's women-run village societies and Kerala's Kudumbashree dairy units demonstrate the model at scale.

Exports and Circular Economy

- **Export Earnings:** Livestock products form a significant and growing part of the agri-export basket.

  Livestock product exports reached Rs 66,249 crore in 2024-25, driven largely by bovine meat and edible offal, with UAE, Vietnam, Egypt, Malaysia and Saudi Arabia the leading destinations.
- **Integration with Cropping:** Manure returns organic carbon to soil, draught animals supply farm power, and crop residue becomes feed, closing the nutrient loop.

Recent Budgetary Support

- **Kisan Credit Card Expansion:** The loan limit for animal husbandry farmers has been raised from Rs 3 lakh to Rs 5 lakh, recognising that working capital rather than land is the binding constraint in this sector.
- **Entrepreneurship Support:** Rs 500 crore allocated for an Integrated Scheme for Entrepreneurship Development to support livestock FPOs and startups.
- **Veterinary Infrastructur**e: A credit-linked capital subsidy scheme for private establishment of veterinary and para-veterinary colleges, hospitals, diagnostic laboratories and mobile veterinary units.

Issues:

- **Low Productivity**
  - **Yield Gap:** Average milk yield per animal remains far below global levels, because the bulk of the herd consists of low-producing indigenous and non-descript animals.

    India holds one of the world's largest cattle populations yet produces less milk per animal than countries with a fraction of the herd size, which is why output growth has come from herd expansion rather than productivity.
  - **Slow Genetic Improvement:** Artificial insemination coverage and conception rates remain limited, and elite germplasm reaches only a small share of the herd.

The very design of the Nationwide Artificial Insemination Programme, targeting 605 districts where coverage was below 50 per cent, is an admission of how uneven penetration has been.

- **Feed and Fodder Deficit**
  - **Persistent Shortage:** India faces a structural deficit in green fodder, dry fodder and concentrates, which caps productivity regardless of breed quality.
  - **No Land for Fodder:** Fodder competes with food and cash crops for scarce land, and receives neither MSP nor procurement support, so farmers have little incentive to grow it.
  - **Shrinking Commons:** Encroachment and diversion of village grazing land has removed the free input on which landless livestock rearing depended.
  - **Disease Burden**
  - **Recurring Epidemics:** Foot and Mouth Disease, Brucellosis and Peste des Petits Ruminants cause chronic productivity loss, while sudden outbreaks can devastate herds.

    The Lumpy Skin Disease outbreak of 2022 killed lakhs of cattle across Rajasthan, Gujarat and Punjab, exposing how thin surveillance and vaccine readiness were.
  - **Weak Veterinary Reach:** Shortage of veterinarians, para-veterinary staff and diagnostic laboratories means most rural animals never receive professional care.
  - **Antimicrobial Resistance:** Unregulated use of antibiotics in dairy and poultry, often as growth promoters, poses a public health risk and threatens export access.
- **Marketing and Value Chain Gaps**
  - **Dominance of the Unorganised Segment:** The greater share of milk and almost all meat is handled outside the organised sector, so producers receive neither assured prices nor quality-linked premiums.
  - **Cold Chain Dependence:** Milk works only because a chilling network exists, and meat and fish do not because it does not.
  - **Regional Concentration:** Production and processing cluster in a few states, leaving large parts of the country outside the organised value chain.

    Uttar Pradesh, Rajasthan, Madhya Pradesh, Gujarat and Maharashtra together account for about 54 per cent of India's milk output.
- **Credit, Insurance and Ownership**
  - **Small Share of Agricultural Credit:** Despite contributing nearly a third of agricultural GVA, livestock receives a disproportionately small share of institutional agricultural credit.

    The recent doubling of the Kisan Credit Card limit for animal husbandry farmers from Rs 3 lakh to Rs 5 lakh is a response to precisely this mismatch.
  - **Negligible Insurance Cover:** Most animals are uninsured, so the death of a single milch animal can wipe out a landless household's entire productive asset base.
  - **Gendered Ownership Gap:** Women perform the labour but ownership, bank accounts and cooperative membership typically rest with men, so the income does not follow the work.
  - **Environmental and Social Costs**
  - **Methane Emissions:** Enteric fermentation in ruminants is among India's largest sources of agricultural greenhouse gas emissions, creating tension between herd expansion and climate commitments.
  - **Heat Stress:** Rising temperatures reduce feed intake, conception rates and milk yield, particularly in crossbred cattle bred for yield rather than heat tolerance.
  - **Stray Cattle Problem:** Restrictions on the sale of unproductive animals have raised the cost of maintaining them, producing large stray populations that damage standing crops.
- **Government Measures**
- **Rashtriya Gokul Mission:** Genetic upgradation and conservation of indigenous breeds with a total outlay of Rs 3,400 crore for the Fifteenth Finance Commission cycle, including 22 IVF laboratories and sex-sorted semen production.

- **Heifer Rearing Centres:** One-time assistance of 35 per cent of capital cost for 30 facilities housing 15,000 heifers, with 3 per cent interest subvention for farmers purchasing high genetic merit IVF heifers.
- **Nationwide Artificial Insemination Programme:** Free doorstep artificial insemination in 605 low-coverage districts, having reached over 8.39 crore animals.
- **National Livestock Mission:** Entrepreneurship support in poultry, small ruminants and piggery, feed and fodder development and livestock FPO formation.
- **National Animal Disease Control Programme:** Mass free vaccination targeting eradication of Foot and Mouth Disease and Brucellosis.
- **Animal Husbandry Infrastructure Development Fund:** Rs 15,000 crore with interest subvention for private and cooperative investment in processing and feed infrastructure.
- **Bharat Pashudhan and National Digital Livestock Mission:** Unique animal identification and a national database linking breeding, health and productivity records.
- **Kisan Credit Card for Animal Husbandry:** Working capital access extended to dairy, poultry and fisheries farmers, with the limit raised to Rs 5 lakh.
- **Dairy Processing and Infrastructure Development Fund:** Concessional finance for modernising cooperative milk processing and chilling capacity.
- **GOBARdhan and Market Development Assistance:** Support of Rs 1,500 per tonne for organic manure from biogas plants, monetising dung as a third income stream.

Case Studies:

Case	What was done	Result and Lesson
Operation Flood and Amul	Three-tier cooperative structure with village societies, district unions and a state federation owning the entire chain from chilling to retail.	Made India the world's largest milk producer with per capita availability rising from 319 grams a day in 2014-15 to about 485 grams in 2024-25, and milk losses under 1 per cent. Producer ownership is the durable solution to fragmentation.
Nationwide Artificial Insemination Programme	Free doorstep artificial insemination in 605 districts with baseline coverage under 50 per cent.	Over 8.39 crore animals covered and 5.21 crore farmers benefited, showing that delivery to the doorstep, not technology availability, was the binding constraint.
Lumpy Skin Disease Outbreak, 2022	Rapid spread across Rajasthan, Gujarat and Punjab, followed by indigenous development of a vaccine.	Lakhs of cattle died before containment, exposing thin surveillance and reactive rather than pre-positioned vaccine capacity. Disease preparedness is an economic question, not only a veterinary one.
Contract Broiler Farming, Andhra Pradesh and Tamil Nadu	Integrators supply chicks, feed and veterinary care, and buy back birds at agreed rates.	Converted poultry into an organised, bankable activity where about a quarter of producers supply over 70 per cent of broiler meat. Shows integration can formalise a sector faster than regulation.

Brazil's Gir Herd	Systematic selection and record-keeping applied to cattle of Indian Gir origin.	Produced some of the world's highest-yielding tropical dairy animals from Indian genetics, demonstrating that the constraint is breeding systems rather than the breed itself.
GOBARdhan	Conversion of cattle dung into compressed biogas and organic manure with market development assistance of Rs 1,500 per tonne.	Turns a disposal cost into a revenue stream and links animal rearing to clean energy and soil health, though scale remains limited relative to dung availability.

Way Forward:

- **Raising Productivity**
  - **Scale Genetic Improvement:** Extend sex-sorted semen, IVF and embryo transfer beyond pilot districts so that herd quality rather than herd size drives output growth.

    Rashtriya Gokul Mission's 22 IVF laboratories provide the technical base, but the delivery gap lies in last-mile artificial insemination coverage.
  - **Breed for Heat Tolerance:** Combine yield traits with the climate resilience of indigenous breeds such as Gir, Sahiwal and Tharparkar rather than importing yield alone.

    Brazil built a world-leading dairy herd from Indian Gir genetics, showing the potential of indigenous stock when systematically selected.
  - **Record and Reward Performance:** Universal animal identification linked to milk recording allows farmers to be paid for quality and to select breeding stock on evidence.
- **Closing the Fodder Gap**
  - **Treat Fodder as a Crop:** Extend seed support, extension and area coverage to fodder crops, which currently receive neither MSP nor procurement backing.
  - **Promote Silage and Feed Blocks:** Conserving surplus green fodder as silage and densified blocks addresses the seasonal deficit that limits yield in lean months.
  - **Restore and Manage Commons:** Regenerate village grazing land through community management, which restores the free input base for landless rearers.

    Watershed and pasture regeneration under MGNREGA offers a ready funding route for common land restoration.
- **Strengthening Animal Health**
  - **Universal Vaccination and Surveillance:** Complete Foot and Mouth Disease and Brucellosis coverage and build early-warning surveillance for emerging diseases.
  - **Expand Veterinary Reach:** Scale mobile veterinary units and para-veterinary cadres so that care reaches the animal rather than requiring the animal to reach a clinic.
  - **Adopt a One Health Approach:** Regulate antibiotic use in dairy and poultry, since antimicrobial resistance is simultaneously a public health risk and an export barrier.
- **Building the Value Chain**
  - **Replicate the Cooperative Model:** The three-tier Anand pattern that transformed milk should be extended to meat, poultry, small ruminants and wool.

Amul's structure keeps milk losses under 1 per cent and pays producers a share of the consumer rupee no private intermediary matches.

- **Formalise Meat and Poultry:** Modern abattoirs, cold chain and hygiene certification would raise both domestic safety and export realisation from a largely informal trade.
- **Deepen Credit and Insurance:** Expand livestock insurance coverage and collateral-free working capital so that the death of one animal does not destroy a household's asset base.

- **Sustainability and Equity**
  - **Cut Methane Intensity:** Improved feed, balanced rations and feed supplements reduce emissions per litre of milk without reducing output.
  - **Monetise Waste:** Scale biogas and organic manure production so that dung becomes revenue rather than a disposal problem.
  - **Put Assets in Women's Names:** Link scheme benefits, cooperative membership and payments to the woman who does the work, rather than to the male head of household.

2. POULTRY SECTOR

Introduction:

Poultry farming is the practice of **domesticating and raising birds, primarily chickens, ducks, turkeys**, and geese, for the purpose of producing meat and eggs. It is a significant component of the agricultural sector worldwide.

Key Segments:

- **Layer Farming (Eggs):** Layer birds are raised specifically for egg production. This segment is characterized by high levels of organization, mechanization, and high input costs (feed). Automation is common in large commercial layer farms.
- **Broiler Farming (Meat):** Broilers are chickens bred for fast growth to be harvested for meat.

Current Status of Poultry Sector of India:

- **Global Ranking and Production:** India is the 3rd largest producer of eggs and 4th largest producer of broiler meat globally, producing nearly 5.6 million tonnes of poultry meat and 153.5 billion eggs annually as of 2025.
- Broiler meat production in the country is estimated at around 5 million tonnes (MT) annually.
- Industry growth is robust, with market value projected to reach INR 8,430 billion by 2033 at a CAGR of 12.6%.
- **Top Egg-Producing States:** Andhra Pradesh (20.13%), Tamil Nadu (15.58%), Telangana (12.77%), West Bengal (9.93%), and Karnataka (6.51%).

Key Factors Responsible for Rapid Growth of Poultry Industry in India:

- **Changing Dietary Habits:** Increased awareness of protein intake and the shift away from purely vegetarian diets, particularly among the youth and urban populations.
- **Urbanization and Rising Income:** As household incomes rise and more people move to cities, demand for convenient, easily prepared, and affordable protein sources like eggs and chicken increases.
- **Affordability:** Poultry meat and eggs remain significantly cheaper than other sources of animal protein, such as mutton or fish, making them the protein of choice for the masses.
- **Minimal Taboos:** Compared to beef and pork, poultry is accepted across most regions, communities, and religious groups in India, giving it a much wider market.

- **Technological Advancements:** The introduction of modern housing (climate control) and advanced genetics has drastically reduced the production cycle time and improved feed conversion ratio (FCR). The use of **Environmentally Controlled (EC) sheds with automated feeding**, and climate control systems has improved growth efficiency and reduced mortality rates.
- **Government Support and Policies:** Government initiatives, subsidies, and improved infrastructure for **transportation and cold storage have bolstered investment** and growth in the poultry sector, enhancing supply chain management.

Significance of Poultry Industry:

- **Export Potential:** India has the potential to become a major poultry product exporter, especially in Middle East & Southeast Asia.
- **Protein and Nutrition Security:** Poultry products—eggs and chicken meat—are among the most affordable and accessible sources of high-quality animal protein for the Indian population, addressing widespread nutritional deficiencies. Protein from poultry supports child development, maternal health, and overall population wellness.
- **Employment & Rural Livelihood:** The poultry sector directly and indirectly employs millions, including farmers, laborers, feed manufacturers, transporters, and processors. Rural and backyard poultry farming especially empower women and smallholders, promoting inclusive rural growth.
- **Economic Growth & Value Addition:** India is one of the world's largest producers of eggs and broiler meat, contributing significantly to agricultural GDP and rural prosperity. The sector encourages invested value chains (feed, hatcheries, farming, processing, logistics), supports exports, and drives innovation in cold chains and food safety.
- **Diversification & Risk Reduction:** Poultry provides a reliable income source during crop failures and droughts, buffering farm households against agricultural risks. Integration with livestock, fishery, and horticulture businesses improves resilience of rural economies.
- **Women Empowerment & Social Welfare:** Backyard poultry schemes are a crucial livelihood and nutritional support for women, landless, and marginal farmers, improving social equity.

Challenges/Limitations of Poultry Sector:

- **High Feed Costs:** Feed, primarily composed of Maize and Soybean meal, accounts for 65-75% of the total cost of production. Price volatility of these commodities directly impacts farm profitability.
- **Disease Management (Biosecurity):** The risk of highly pathogenic diseases, particularly **Avian Influenza (Bird Flu)**, remains a constant threat. Outbreaks lead to mass culling, trade restrictions, and significant economic losses.
- **Contaminated Environments:** High-density confinement of chickens in battery cages results in poor air quality, waste management problems, and greenhouse gas emissions.
- **Cruel Treatment of Animals:** Industrial poultry operations often involve inhumane practices such as mutilation, starvation, and overcrowding, violating the Prevention of Cruelty to Animals (PCA) Act, 1960.
- **Financial and Operational Challenges:** The industry faces **issues such as large debts, reliance on informal facilities**, and complex contract farming arrangements. Farmers often incur **significant losses due to market volatility** and the pressures from industry giants.
- **Supply Chain Inefficiencies:** Inadequacies in the supply chain, including transportation, **cold storage, and distribution networks, can lead to wastage** and affect the quality of poultry products, hindering market growth.
- **Waste Management Issues:** The poultry industry produces significant waste, including methane, **CO2, water effluents, and solid waste, causing soil and water pollution.**

Various Government Initiatives for Poultry Sector:

- **Animal Husbandry Infrastructure Development Fund (AHIDF):**
  - ₹15,000 crore fund to support infrastructure for modern poultry units, feed mills, cold chains, slaughter, and processing facilities.
  - Provides interest subsidies and credit guarantees to encourage private and cooperative investment.
- **National Livestock Mission (NLM):**
  - Focuses on entrepreneurship development, capacity building, and formation of Farmer Producer Organizations (FPOs).
  - Supports rural/backyard poultry through financial assistance, training, and input supply (chicks, feed, vaccination).
- **Poultry Farming Subsidy Schemes:** States offer direct capital and operational subsidies for both commercial and backyard poultry units, especially for marginalized groups and women.
- **Poultry Development Schemes:** Region- and state-specific schemes for expanding infrastructure, improving genetics, supplying improved chicks, and supporting feed quality and veterinary access.
- **Poultry Venture Capital Fund (PVCF):** The Department of Animal Husbandry and Dairying is implementing it under "Entrepreneurship Development and Employment Generation" (EDEG) of the National Livestock Mission.

Way Forward:

- **Promote Technology and Innovation:**
  - Accelerate adoption of climate-controlled sheds, automation, artificial intelligence, and biotech for improved productivity and disease monitoring.
  - Encourage research on feed optimization, breeder genetics, and value-added products to boost yields and profitability.
- **Strengthen Biosecurity and Disease Surveillance:**
  - Establish robust disease detection, control, and vaccine infrastructure to minimize losses from outbreaks like avian influenza and coccidiosis.
  - Enhance training programs for farmers and veterinarians in scientific health management and hygiene standards.
- **Develop Infrastructure and Cold Chains:**
  - Invest in cold storage, logistics, processing units, and integrated supply chains for better product quality and reduced post-harvest loss.
  - Leverage government schemes (e.g. AHIDF) to enable modernization and smallholder participation.
- **Boost Market Access and Value Addition:**
  - Diversify into processed, organic, and antibiotic-free products to meet rising health-conscious consumer demand.
  - Expand direct farmer linkages with urban markets, retail chains, and export promotion programs for global competitiveness.
- **Ensure Feed Efficiency and Price Stability:**
  - Foster domestic production of feed components like maize and soybean through policy support and innovation.
  - Support alternative feed research to manage cost pressures and sustainability.
- **Empower Small and Marginal Farmers:**
  - Expand financial inclusion, skill development, and collective farmer organizations (FPOs) for backyard and small-scale poultry.

- Strengthen rural poultry schemes for women, SC/ST, and landless households to improve livelihoods and nutrition.

- **Enforce Regulatory and Sustainability Standards:**
  - Update food safety, environmental, and welfare regulations for improved consumer confidence and export potential.
  - Promote environmentally sustainable practices, waste management, and circular economy models.

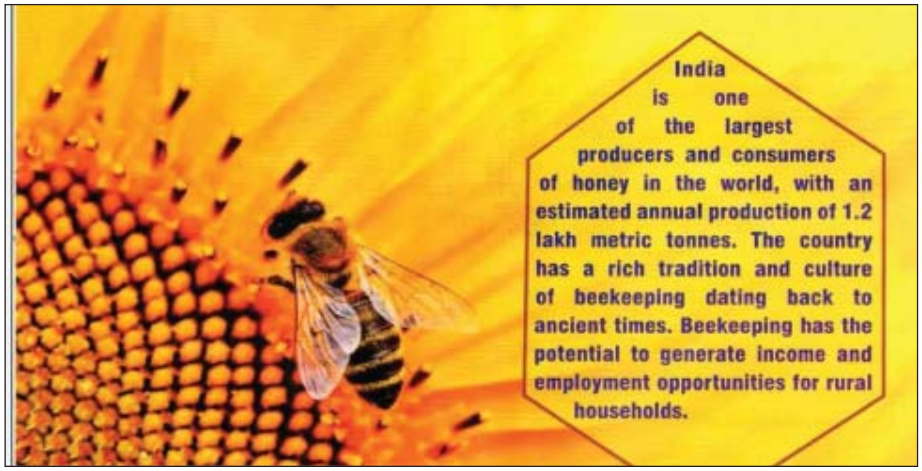
Conclusion:

Thus, India's poultry sector needs a multi-pronged approach focusing on innovation, sustainability, quality enhancement, and expanded market access to ensure continued growth and equitable benefits.

3. APICULTURE

Introduction:

Apiculture is the s**cience and culture of honeybees and their management**. Beekeeping is the practice of intentional maintenance of honey bee colonies, commonly in hives, by humans. Globally, there are more than 20,000 species of wild bees, many of which are solitary or which rear their young in burrows and small colonies, like mason bees.



Different types of honey bee species in India:

Sl. No.	Common name	Scientific name	Honey yield per year per hive (kg)
1	Indian bee	*Apis cerena*	8-10
2	European bee	*Apis mellifera*	25-30
3	Rock bee	*Apis dorsata*	30-35
4	Little bee	*Apis florae*	<1
5	Himalayan bee	*Apis laboriosa*	40-45

Source: AICRP on Honey Bee, 2018

Advantages of Apiculture:

- **Source of Important forest based Outputs:** Apiculture is important because it directly contributes to the outputs produced, such as honey, beeswax, queen, and bee colonies, and other products such as pollen, royal jelly, bee venom, and propolis in cosmetics and medicine.
- **Maintenance of ecological sanctity:** Apiculture plays a role in providing nutrition, economic, and ecological security, as bees are valuable pollinators of both agricultural crops and natural ecosystems.
- **Less capital requirement:** Apiculture business is less capital intensive as it requires less land and less initial capital.
- **Source of employment opportunities:** Apiculture is a major source of employment opportunities for people of all age groups.

Challenges related to Apiculture:

- **Lack of Awareness and Training:** Many potential beekeepers lack adequate knowledge about beekeeping practices, including hive management, disease prevention, and honey extraction techniques.
- **Limited Access to Apiculture Resources:** Access to beekeeping equipment, such as hives, protective gear, and bee colonies, is limited in certain regions of India. This lack of resources hampers the expansion of beekeeping activities, particularly in rural areas.
- **Climate Variability:** Erratic weather patterns, such as unseasonal rains or prolonged droughts, have been affecting the honeybees foraging behavior, disrupting flowering patterns, and ultimately impacting honey production.
- **Pests and Diseases:** Bee colonies are susceptible to various pests and diseases, such as Varroa mites, nosema, and foulbrood.
- **Chemical Use in Agriculture:** Pesticides and other agricultural chemicals pose a threat to bee populations by contaminating their food sources and weakening their immune systems.
- **Market Challenges:** Price fluctuations, competition from imported honey, and lack of market infrastructure have been hindering the profitability of beekeeping ventures.

Various government initiatives for Beekeeping Development in India:

- **National Beekeeping and Honey Mission (NBHM):** Launched in 2017 under the National Horticulture Mission, NBHM aims to promote scientific beekeeping practices, enhance honey production, and improve the livelihoods of beekeepers.
- **'Sweet Revolution' as part of Aatmanirbhar Bharat Abhiyaan:** NBHM aims for the overall promotion and development of scientific beekeeping in the country to achieve the goal of 'Sweet Revolution', which is being implemented through the National Bee Board (NBB).
- **National Bee Board (NBB):** The National Bee Board, established in 2000 under the Ministry of Agriculture & Farmers Welfare, serves as the apex body for the promotion and development of beekeeping in India.
- **Rashtriya Krishi Vikas Yojana (RKVY):** RKVY supports various agricultural and allied sector development activities, including beekeeping.

Way Forward:

- **Proper Training:** Proper training and awareness programs are essential to equip beekeepers with the necessary skills.
- **Adequate Pest Management:** Proper pest management and disease management must be readily available to all beekeepers.
- **Sustainable farming practices:** Encouraging sustainable farming practices that minimize chemical use is essential for the health of bee colonies.
- **Preservation of natural habitats:** Preserving natural habitats and promoting agroforestry practices can help mitigate the effects of land use change and provide additional forage for bee colonies.

Conclusion:

Apiculture in India is vital for agricultural growth, rural prosperity, and environmental sustainability. Thus, the sector must focus on achieving higher quality and efficiency through science, infrastructure, and targeted support to fully realize the potential of the **Sweet Revolution**.

4. AQUACULTURE

Introduction:

Aquaculture refers to the farming of aquatic organisms such as fish, crustaceans, molluscs, and aquatic plants under controlled conditions. Unlike capture fisheries, aquaculture is resource-efficient and can be undertaken in freshwater, brackish, or marine environments.

Types of Aquaculture in India

- **Mariculture:** Conducted in seawater or oceanic environments; involves cultivation of prawns, molluscs, shellfish, and seaweed.
- **Fish Farming:** The most common form, focusing on species such as carp, catfish, tilapia, and trout for human consumption.
- **Algaculture:** Cultivation of micro and macro algae for industrial, pharmaceutical, and biofuel applications.
- **Integrated Multi-Trophic Aquaculture (IMTA):** A modern system where waste from one species serves as feed for another, ensuring nutrient recycling and near-zero waste.
- **Recirculating Aquaculture Systems (RAS):** Closed-loop systems with water filtration, oxygenation, and temperature control—eco-friendly and water-efficient.
- Flow-through or Raceway Systems: Commonly used for trout culture; water flows through channels with proper waste disposal mechanisms.
- Cage and Pen Culture: Mesh enclosures placed in open water bodies; enable high production but may cause ecological risks if poorly managed.

India's Progress and Global Standing

- **India is the 3rd largest aquaculture produce**r globally and 2nd in prawn production, after China.
- Leading Aquaculture States: Andhra Pradesh, West Bengal, Tamil Nadu, Odisha, and Gujarat.
- Export Strength: Frozen shrimp is India's top seafood export, primarily to the U.S., EU, Japan, and China.
- Notable Species: Penaeus monodon (Black Tiger Shrimp) and Litopenaeus vannamei (Whiteleg Shrimp) dominate export-oriented brackish water aquaculture.
- The sector contributes **around 1.1% to India's GDP and 7% to agricultural GDP,** reflecting its crucial economic role.

Significance of Aquaculture in India

- **Food & Nutrition Security**
  - Provides high-quality protein, omega-3 fatty acids, and essential micronutrients.
  - Ensures affordable and accessible animal protein for rural and coastal populations.
  - Enables year-round food availability through continuous production cycles.
- **Livelihoods & Employment**
  - Supports over 30 million livelihoods directly and indirectly in farming, processing, transportation, and trade.
  - Strengthens rural and coastal economies and provides supplementary income to farmers.

  - Empowers women through their participation in post-harvest, processing, and marketing activities.
- **Economic Growth & Exports**
  - Contributes about 1% to GDP and 5–7% to agricultural GVA.
  - Major foreign exchange earner — seafood exports exceed ₹60,000 crore (~US$7 billion) annually (2023-24).
  - Fish production rose from ~5.6 million tonnes in 2000–01 to ~17.5 million tonnes in 2022–23, and further to ~18.40 million tonnes in 2023–24.
- **Resource Utilisation & Potential**
  - Vast inland water resources (ponds, tanks, reservoirs, wetlands) remain underutilized.
  - Huge potential for coastal and brackish-water aquaculture and mariculture of high-value species such as shrimp.
- **Poverty Alleviation & Equity**
  - Provides supplementary income to small and marginal farmers.
  - Offers livelihood options for landless and vulnerable communities through community-based or cage farming.
- **Environmental & Ecosystem Benefits**
  - Supports aquatic biodiversity when managed sustainably.
  - Promotes carbon sequestration and coastal resilience through mangrove-linked systems.
  - Reduces pressure on wild fish stocks, aiding marine conservation.

Major Challenges in Aquaculture

- **Trade Policy Risk:** U.S. tariffs on shrimp exports have impacted competitiveness. Overdependence on limited markets increases vulnerability.
- **Infrastructure & Cold Chain Gaps:** Despite progress, several producing states still face deficits in cold storage, processing, and logistics.
- **Quality & Residue Compliance:** Export rejections due to stricter residue standards underscore the need for traceability and biosecure practices.
- **Access to Credit & Inputs:** Small farmers struggle with high upfront costs and limited access to quality seed and feed despite subsidies.
- **Environmental Externalities:** Mangrove destruction, effluent discharge, and soil/water salinisation remain significant concerns.
- **Data Gaps & Monitoring:** Lack of disaggregated data on employment, antibiotic use, and environmental impact hinders effective policymaking.

Government Initiatives & Institutional Support:

Pradhan Mantri Matsya Sampada Yojana (PMMSY)	Launched in 2020, PMMSY aims to double seafood exports to ₹1,00,000 crore by FY 2024–25, create around 55 lakh jobs, and develop modern aquaculture infrastructure like hatcheries, aqua parks, disease labs, and brood banks.
Integrated Aquaparks	11 aquaparks have been approved with ₹459.18 crore funding to serve as hubs for seed production, feed, processing, logistics, and marketing
Blue Revolution Scheme	Supports sustainable growth by integrating the development and management of marine and inland fisheries.
Fisheries and Aquaculture Infrastructure Development Fund (FIDF)	Provides funds to build infrastructure such as fishing harbours, cold storage, and processing units for both marine and inland fisheries

Regional Projects	Aquaculture is being expanded in Northeast India with new aqua parks, hatcheries, and feed plants. Traditional systems like pokkali and kayal farming in Kerala's Kuttanad region are being revived to promote sustainable and climate-resilient practices

Way Forward:

- **Diversify Exports and Species**
  - Reduce over-reliance on shrimp and a few export markets (United States, European Union, and China).
  - Promote diversification into other seafood such as shellfish, molluscs, and seaweed.
  - Explore regional trade agreements and new market opportunities to broaden export access.
- **Strengthen Regulation and Monitoring**
  - Ensure all brackish water farms are licensed and regularly inspected by the Coastal Aquaculture Authority (CAA).
  - Enforce residue standards more strictly; expand testing laboratories and strengthen the National Residue Control Programme (NRCP).
  - Monitor environmental compliance with Coastal Regulation Zone (CRZ) norms, including mangrove and wetland protection at the local level.
- **Adopt Sustainable and Climate-Resilient Technologies**
  - Promote the adoption of Biofloc Technology (BFT), Recirculating Aquaculture Systems (RAS), and Integrated Multi-Trophic Aquaculture (IMTA) along with zero-discharge or low-water systems.
  - Invest in Research and Development (R&D) for breeding disease-resistant and climate-tolerant species.
- **Upgrade Infrastructure and Value Chains**
  - Expand cold chain, processing, logistics, and packaging infrastructure to minimize post-harvest losses.
  - Strengthen fish farmer organisations (FFOs) and producer companies to enhance scale, reduce costs, and improve bargaining power.
- **Enhance Financial and Input Access**
  - Simplify access to institutional credit and streamline financing under existing schemes.
  - Ensure availability and quality regulation of critical inputs such as seed, feed, and probiotics.
  - Provide targeted subsidies or grants to encourage adoption of modern, sustainable aquaculture systems.
- **Empower Communities and Women**
  - Enhance women's participation in aquaculture through ownership, training, and leadership roles.
  - Integrate traditional and local knowledge with scientific practices in small-scale aquaculture systems for better sustainability and inclusiveness.
- **Invest in Research, Data, and Innovation**
  - Fill data gaps on employment, AMR, and environmental impacts.
  - Encourage digital innovations—IoT, AI, blockchain—for real-time monitoring and traceability.

5. DAIRY SECTOR IN INDIA

Status of Dairy Sector in India

Milk Production Data	India is the **world's leading milk producer**. The milk production reached **230.58 million tonnes** in **2022-23**, up from **17 million tonnes** in **1951-52**. The per capita availability of milk in India is **459 grams per day**, surpassing the global average of **323 grams per day**.

Milk Sector contribution to Agri GDP	The milk sector (milk consumed or sold in liquid form, ghee, butter, and lassi produced by producer households) contributed almost **40%** (Rs 11.16 lakh crore) to agriculture and related sectors.
Top Five Milk producing states in India	The top five milk-producing states- **Uttar Pradesh, Rajasthan, Madhya Pradesh, Gujarat**, and **Andhra Pradesh**– contribute over 53% of the country's total milk production.
National Average Yield and production data	The national average yield is **8.55 kg per animal per day** for crossbred animals and **3.44 kg for indigenous ones**. Indigenous buffaloes account for **31.94% of production,** followed by **crossbred cattle at 29.81%.**
Handling of milk	The organized sector, led by cooperatives, manages about one-third of the marketable milk. The unorganized sector handles the remaining two-thirds.

Significance of Dairy Sector

- **Tackling agricultural uncertainties-** Milk animals provide income and livelihood support for farmers, especially during drought and flood. Dairy is a perennial occupation, unlike agriculture which is seasonal in nature.
- **Employment Generation-** Significant contributor to farmers' income as approximately **70 million farmers** are directly involved in dairying.
- **Reduced Import Bill-** Operation Flood (also called as White Revolution) converted India from a milk importer to the world's largest producer.
- **Women Empowerment-** Female population comprises around 69% of the sector's workforce, which are dependent on the sector for their livelihood. Therefore, the dairy sector's development automatically augments women's empowerment.
- **Nutritional Support-** The milk and associated products have immensely helped India in reducing the malnutrition and undernourishment levels in the country.
- **Boost to other sectors-** The dairy sector provides cow dung which is used as an organic manure for the agricultural sector. Further, the sector provides raw materials to manufacture processed foods, like cheese and protein powder.

Challenges with Dairy Sector

- **Fragmented Supply Chain-** Due to its perishable nature, dairy requires more **complex supply chain operations** and **logistics** to ensure freshness and safety.
- **Price Sensitivity-** Milk producers are highly susceptible to even minor shocks. Small changes in the employment and income of consumers often leaves a significant impact on milk demand.
- **Unorganised Industry-** The majority of cattle raisers are unorganised unlike sugarcane, wheat, and rice-producing farmers. This nature further inhibits the creation of political clout to advocate for their rights.
- **Poor returns-** There is **no MSP (Minimum Support Price)** for milk unlike 24 major agricultural commodities in the country including **wheat** and **rice**.

  ˇThe cooperatives adopt a fat-based pricing policy which is 20 to 30 % less than the price in the open market. Further, dairy cooperatives buy more than 75% of milk at its lower price band.
- **Competition from alternatives-** The traditional cow and buffalo milk is being shunned by some consumers for more eco-friendly alternatives like '**Soy Milk**' or '**Almond milk**'.
- **Data Deficiency-** There is a **lack of official and periodical estimate** of the cost of milk production, even though the value of milk produced outweighs the combined value of the output of wheat and rice in India.

Government schemes for the development of dairy sector

Rashtriya Gokul Mission (RGM)	It is being implemented for development and conservation of indigenous bovine breeds since December 2014. Its aim is to enhance milk production and to make it more remunerative to the farmers.
National Livestock Mission (NLM)	National Livestock Mission (NLM) scheme has been restructured for 2021-22 to 2025-26. The scheme focuses on entrepreneurship development and breeds improvement in poultry, sheep, goat and piggery, including feed and fodder development.
National Artificial Insemination Programme	To suggest novel methods of bringing about impregnation in female breeds and prevent the spread of certain diseases which are genital in nature.
National Cattle and Buffalo Breeding Project	To genetically upgrade important indigenous breeds on a priority basis with a focus on development and conservation.
National Animal Disease Control Programme	Implemented to control FMD and Brucellosis by completely vaccinating cattle, buffalo, sheep, goat and pig populations against Foot & Mouth Disease (FMD) and bovine female calves of 4-8 months of age against Brucellosis.
Animal Husbandry Startup Grand Challenge	To appreciate innovations coming from the villages to expand the dairy sector in India.

Way Forward

- **Support to Start-ups-** The government should support start-ups that come up with a solution-oriented approach.

  For ex- Country Delight is a Haryana-based dairy-tech company that is providing quality milk at consumers' doorstep. The milk undergoes 26 quality tests and farmers deliver it directly. This ensures good quality and optimum pricing.

- **Value-added product segment-** The producers should be given the requisite support to enter into value-added segments such as ice cream, yogurt, cheese, and whey. These segments show profit margins of 20%, which is much higher than the 3-5% margin in the case of simple milk produce.
- **Remunerative pricing-** The dairy farmers must be given a stable market and remunerative price for the milk. For the price, a greater weightage should be accorded to the quantity of milk than its fat content.\
- **Focus on Hub and spoke Model-** The government should focus on a hub and spoke model. Under this, the main farm (hub) should have all integrated facilities for milking, feed production, and milk processing.

6. WHITE REVOLUTION 2.0

Context:

Recently, Union Home and Cooperation Minister Amit Shah announced plans for **'White Revolution 2.0'**, which aims to further revolutionize the dairy industry. The Ministry of Cooperation unveiled standard operating procedure (SOPs) for White Revolution 2.0. It will be on the line of **Operation Flood**, which was launched in **1970** catalysed the White Revolution, and transformed India's dairy sector.

White Revolution 2.0 focuses on expanding the cooperative model, which was the backbone of Operation Flood. The government plans to increase milk procurement by cooperatives from **660 lakh kg per day** in **2023-24** to **1,007 lakh kg per day** by **2028-29.**

Government's roadmap for achieving White Revolution 2.0:

Expansion of Cooperative reach	• Presently, there are **1.7 lakh dairy cooperative societies (DCSs)** covering **2 lakh villages** (30% of the country's villages, and 70% of India's districts). • However, only **10-20% of villages are covered in states** like Uttar Pradesh, Uttarakhand, and Madhya Pradesh, while states like West Bengal, Assam, and Odisha have less than 10% coverage. • The **National Dairy Development Board (NDDB)** plans to establish **56,000 new multipurpose DCSs** and strengthen **46,000 existing ones over the next five** years, particularly in Uttar Pradesh, Odisha, Rajasthan, and Andhra Pradesh.
Planned Expansion	• A pilot project was launched in February 2023, with a budget of Rs 3.8 crore, to set up dairy cooperatives in **uncovered gram panchayats** across Haryana, Madhya Pradesh, and Karnataka.
Funding	• **National Programme for Dairy Development (NPDD) 2.0** will provide primary funding for White revolution 2.0. This scheme will provide financial assistance for village-level milk procurement systems, chilling facilities, and capacity-building, targeting 1,000 **Multipurpose Primary Agricultural Credit Cooperative Societies (MPACSs)** with Rs 40,000 each.

Significance of White Revolution 2.0

- **Increase in the share of agri and allied sector GDP-** White revolution 2.0 will lead to enhanced milk production, which will add to the growth of the agri and allied sector in India. **For ex-** Currently, the milk sector (milk and its derivative products) contribute around **40% (Rs 11.16 lakh crore)** to agriculture and related sectors.
- **Generation of rural employment and women empowerment-** White revolution 2.0 will generate employment and contribute to the empowerment of women in the process. **For instance-** The Indian dairy industry today, **directly or indirectly**, supports over **8.5 crore people, predominantly women.**
- **Improving Market Access and Income for Farmers-** White Revolution 2.0 will enable more farmers to participate in the organized dairy market by **providing market access to farmers in areas previously uncovered by cooperatives**. This will ensure better prices for their produce and more stable incomes.
- **Reducing Regional Disparities-** White Revolution 2.0 aims to address regional disparities in cooperative coverage. The initiative will target the underserved regions of **Uttar Pradesh, Odisha,** and **West Bengal,** and hence will ensure more equitable development across the country.
- **Food and nutritional security-** The initiative aims to increase daily milk procurement by dairy cooperatives from **660 lakh kg** to **1,007 lakh kg by 2028-29.** This expansion will **enhance the availability of milk,** contributing to both food security and nutritional needs in the country.

Challenges in achieving White Revolution 2.0:

- **Uneven Presence of Cooperative-** While states like Gujarat and Kerala have well-established cooperative networks, others like **West Bengal, Assam**, and **Jharkhand** have **less than 10% coverage**. This imbalance limits the ability to scale up production uniformly across the country.
- **Declining annual growth rate of milk production-** The annual rate of milk production has declined from **6.47% in 2018-19** to **3.83% in 2022-23.**

- **Variability in yield and per capita availability across states-** States like West Bengal and Odisha have much lower milk yields compared to states like Punjab and Gujarat. **For ex-** While Punjab boasts a high yield of **13.49 kg** per animal per day**, West Bengal's yield** is only **6.30 kg.**
- **Low cattle Productivity-** As per Integrated Sample Survey, average annual productivity of cattle in India during 2019-20 is **1777 kg per animal per year**, as against the world average of **2699 kg per animal per year during 2019.**
- **Prevalence of Diseases-** There has been a surge in the prevalence of contagious diseases among cattles. For ex- Outbreak of **Lumpy Skin Disease (LSD), Foot and Mouth Disease (FMD), Black Quarter infection.**
- **Feed and fodder scarcity-** The Jhansi-based Indian Grassland and Fodder Research Institute has estimated that there is a **deficit of 12% green fodder, 23% dry fodder,** and **30% grains-based concentrated animal feeds.**
- **Inadequate policy support-** The bulk of the agricultural subsidies by the central and state governments goes to the crops sector.
  - **For ex-** In the **2023-24 Union Budget**, the allocation for the Department of Animal Husbandry and Dairying is merely **Rs 4,328 crore** while well over **Rs 4 trillion** has been **allocated to food**, **fertiliser**, and **other agricultural subsidies**.
- **Lack of marketing support-** Milk and milk products lack price and marketing support that are enjoyed by the crop-based commodities in the form of MSP and official state procurement by FCI.
- **Lack of Institutional finance-** The share of livestock in the total agricultural credit (short-term, medium-term and long-term) is only around 4%.
- **Greenhouse gas contribution-** The dairy sector is a major contributor to the greenhouse gases like Methane, which have a very high global warming potential (GWP).
- **Lack of advancements in technology-** Fewer artificial inseminations, lower-quality germplasm and insufficient technical staff have inhibited the development of high quality cattle breed.

Way Forward:

- **Expansion of Cooperative led procurement in the organised sector-** Presently, about two-thirds of the marketable milk is in the unorganised sector, where informal middlemen dominate the supply chain. The potential for expanding organized sector's share in marketable milk (which is led by the cooperative sector) must be explored.
- **Providing sufficient feed and fodder resources-** Providing sufficient feed & fodder and drinking water is the need of the hour to increase productivity of cattle stock in India, particularly in the rain-shadow region.
- **Value addition and Marketing-** The milk producers should be given the requisite support to enter into value-added segments such as ice cream, yogurt, cheese, and whey. These segments have profit margins of 20%, which is much higher than the 3-5% margin in the case of simple milk produce.
- **Promoting indigenous breeds-** Our indigenous breed of cattle must be promoted, like the genetic enhancement of Badri Cow.
- **Research and development-** Government should focus on Research & Development in the livestock sector to increase per livestock productivity to provide more benefits to small & marginal farmers.

8. AGRICULTURE REVOLUTIONS

Introduction:

India has undergone several distinct Agricultural Revolutions since independence, each aimed at solving a specific challenge, from food scarcity to nutritional deficiency. These revolutions have been crucial in transforming India from a country dependent on food aid to a major global agricultural exporter.

Various types of revolutions in Agriculture after Independence in India:

Green Revolution	• This revolution led to a tremendous rise in production of food grains, especially wheat, by the use of high-yielding varieties of seeds, fertilisers and pesticides. • In the agriculture sector in the decade of 1960 India started the green revolution, and high yielding variety seeds were used in wheat, rice, and maize. This experiment was successful and extended to most parts of the country.
White Revolution	• Operation Flood (1970), an initiative of the National Dairy Development Board has led to revolution in milk production in India. • The world's largest dairy development programme transformed India from a milk deficient nation to world's largest milk producer. • It introduced new breeding technologies, high quality fodder, vaccination, and improved veterinary facilities.
Blue Revolution	• Blue revolution focussed on the management of the fisheries sector. • It led to a phenomenal increase in both fish production and productivity from aquaculture and fisheries resources of the inland and marine fisheries.
Other revolutions	• **Yellow revolution:** Introduced to achieve self-reliance in the production of edible oils. • **Golden fibre revolution:** For increasing the production of jute. • **Golden revolution**→ Horticulture (Fruits, Honey, Flowers) • **Silver fibre revolution**→ Cotton production • **Silver revolution**→ Egg production • **Red revolution**→ Meat production

Significance of agriculture revolutions:

- **Poverty:** These innovations in agriculture have lifted millions of people out of poverty by generating rural income opportunities for farmers, farm labourers, and also reduced prices for consumers.
- **Food security:** India has become self-sufficient in food grain production with the help of the green revolution. Green revolution increased the amount of food grains enormously, with this India came out of PL-480 agreements, through which India imports food grains from USA.
- **Nutritional security:** The exponential rise in milk production has led to nutritional security among the masses. Milk production in India has increased from 22 million tonnes in 1970 to 156 million tonnes in 2015-16. As a result, the per capita availability of milk in India is 337 gram/day as compared to average world per capita availability of 229 gram/day. The Blue revolution increased sea food production. With this fishermen community get benefited and also provide food and nutritional security.
- **Women empowerment:** These revolutions also provided income and employment to women leading to women empowerment helping millions to come out of poverty.

Green Revolution:

- The Green Revolution was a period of rapid agricultural transformation in the mid-20th century that significantly increased food grain production (especially wheat and rice) in countries like India.
- It involved the widespread adoption and transfer of new agricultural technologies and practices to developing countries, primarily aimed at combating food shortages and famine.
- Green Revolution was driven by a combination of scientific advancements and policy changes such as:
  - High-Yielding Varieties (HYVs) of Seeds

- Increased use of chemical fertilizers
- Extensive irrigation facilities
- Application of pesticides and herbicides
- Mechanization of agriculture
- Improved farming practices and management such as multiple cropping, better crop management techniques etc.

- Green Revolution in India was seeded by CIMMYT & IRRI (International Rice Research Institute).

Positive impacts of Green Revolution:

Social impact	• **Attainment of Food Self-Sufficiency and Food Security:** Before the Green Revolution, India was heavily dependent on food imports, particularly from the US (under PL-480). This left the nation vulnerable to external pressures and food shortages. The Green Revolution transformed India from a food-deficient nation to one of the world's leading agricultural producers. • **Socio-Economic Development:** Improved agricultural incomes stimulated local economies, leading to better standards of living and development of rural infrastructure such as roads and storage facilities.
Economic impact	• **Massive Increase in Agriculture Production & Productivity:** The introduction of High-Yielding Varieties (HYVs) of wheat (like Sonara 64 and Lerma Rojo) and rice (like IR8), coupled with increased use of fertilizers, irrigation, and pesticides, led to unprecedented increases in crop yields per hectare. Also, the shorter maturity period of HYVs allowed farmers to grow two or even three crops in a year on the same land, significantly increasing overall output. • **Increased Farmer Incomes:** Farmers, particularly those in the well-irrigated regions of Punjab, Haryana, and Western Uttar Pradesh who could afford the new inputs, experienced a substantial rise in their incomes. This allowed them to invest more in farming and improve their living standards. • **Shift to Commercial Farming:** The increased profitability encouraged a shift from subsistence farming to commercial farming, with farmers producing for the market rather than just for their own consumption. • **Stimulus to Allied Industries:** The demand for modern farm machinery (tractors, threshers, pump sets), chemical fertilizers, and pesticides boosted the growth of associated manufacturing industries. • **Strengthening of Agro-Based Industries:** The increased availability of raw materials from agriculture also spurred the growth of agro-processing and food-processing industries.

Negative Impacts of Green Revolution:

Environmental impacts	• **Environmental Degradation:** • **Loss of Fertility:** The intensive monoculture (growing the same crop year after year) and heavy reliance on chemical fertilizers (like urea) have depleted essential micronutrients from the soil, leading to a decline in soil fertility. Farmers are forced to use even more fertilizers to maintain yields.

	• **Soil Salinization and Alkalinization:** Over-irrigation, especially with groundwater containing dissolved salts, has led to waterlogging and the accumulation of salts on the soil surface in many Green Revolution areas (e.g., Punjab, Haryana). This renders the land unproductive (known as "reh" or "kallar" problem). • **Soil Erosion:** Intensive cultivation, removal of crop residues, and heavy machinery can increase soil erosion. • **Water Depletion and Pollution:** • **Groundwater Depletion:** HYVs are water-intensive crops, leading to excessive pumping of groundwater for irrigation, especially through tube wells. This has caused a drastic decline in groundwater levels in states like Punjab and Haryana. • **Water Pollution:** Runoff from fields containing chemical fertilizers and pesticides contaminates surface water bodies (rivers, lakes, ponds) and percolates into groundwater. This leads to eutrophication (excessive nutrient enrichment causing algal blooms and oxygen depletion, harming aquatic life) and makes water unsafe for drinking and other uses. • **Loss of Biodiversity:** The focus on a few high-yielding varieties of wheat and rice led to the widespread replacement and abandonment of thousands of diverse traditional, local crop varieties (landraces). This significantly reduced agricultural genetic diversity, making crops more vulnerable to new pests and diseases.
Social Impact	• **Increased Inequality:** The benefits of the Green Revolution largely accrued to larger farmers who had the capital to invest in HYV seeds, fertilizers, pesticides, irrigation facilities, and machinery. Small and marginal farmers often struggled to afford these inputs, leading to: • **Indebtedness:** Many small farmers took loans for inputs, and if crops failed or market prices were low, they fell into a debt trap, sometimes leading to suicides. • **Landlessness:** Some small farmers, unable to compete or manage debt, were forced to sell their land to larger, more prosperous farmers, increasing landlessness and the number of agricultural laborers. • **Regional Disparities:** The Green Revolution was most successful in areas with assured irrigation and existing infrastructure (e.g., Punjab, Haryana, Western Uttar Pradesh). Other regions, particularly rain-fed or drier areas, did not benefit significantly, leading to increased inter-state and intra-state regional disparities in agricultural development and prosperity. • **Rural-Urban Migration:** The decline in rural employment opportunities, coupled with the allure of urban jobs, led to increased migration from rural areas to cities. • **Hidden Hunger (Nutritional deficiency):** The focus on calorie-rich wheat and rice led to a decline in the cultivation and consumption of other nutritious crops like millets, pulses, and traditional vegetables, potentially leading to micronutrient deficiencies in the diet.

Way Forward:

- **Promoting Sustainable Agricultural Practices:**
  - **Reduced Chemical Dependence:** Shifting away from excessive reliance on synthetic fertilizers and pesticides. This involves promoting the use of organic manures (farmyard manure, compost, vermicompost), bio-fertilizers, and bio-pesticides.
  - **Integrated Nutrient Management (INM):** A balanced approach that combines organic and inorganic fertilizers based on soil testing and crop needs to maintain soil health and reduce chemical overuse.

  - **Integrated Pest Management (IPM):** Utilizing a combination of biological, cultural, physical, and chemical methods (as a last resort) to control pests, minimizing the use of harmful pesticides. This includes encouraging natural predators, crop rotation, and resistant varieties.
- **Precision Agriculture:**
  - **Precision Irrigation:** Widespread adoption of water-saving irrigation technologies like drip irrigation and sprinkler systems, especially for water-intensive crops.
  - **Soil Health Cards:** Scaling up the Soil Health Card scheme to provide every farmer with regular, precise information on their soil's nutrient status, enabling them to apply fertilizers judiciously and replenish deficient nutrients.
- **Diversification of Cropping Systems:**
  - **Shift from Monoculture:** Encouraging farmers to move away from the dominant rice-wheat cropping system in certain regions towards a more diverse range of crops.
  - **Promotion of Pulses, Oilseeds, and Millets:** These crops are often less water-intensive, improve soil fertility (pulses fix nitrogen), and are nutritionally rich. Boosting their production can address nutritional security and reduce import dependence.
  - **Horticulture and High-Value Crops:** Promoting the cultivation of fruits, vegetables, flowers, and medicinal plants, which often fetch better market prices and can diversify farmer incomes.
- **Strengthening Farmer Support Systems:**
  - **Farmer Producer Organizations (FPOs):** Promoting and strengthening FPOs to enable small and marginal farmers to collectively access inputs, technology, credit, and markets, reducing their reliance on middlemen.
  - **Post-Harvest Management:** Investing in robust post-harvest infrastructure, including cold storage, warehousing, and efficient transportation, to reduce food loss and enable farmers to get better prices.
  - **Credit and Financial Inclusion:** Ensuring timely and affordable credit for farmers, especially for adopting new sustainable technologies and diversifying crops.
- **Research and Development:** Continued investment in agricultural research by institutions like ICAR and State Agricultural Universities (SAUs) to develop sustainable solutions, new crop varieties, and climate-smart technologies relevant to diverse agro-climatic zones. Recently, India has got the opportunity to increase its funding in international agriculture research institutions like CIMMYT - which will be significant - because India has a vital stake in the institution & having a greater say in its running will help India in developing new varieties which are tolerant to the effects of climate change.

Conclusion:

To further carry on the momentum of these revolutions and assure food security in the long run in the face of ever increasing population, there is an urgent need for an 'evergreen revolution' that should focus on making agriculture sector more sustainable, resilient, inclusive & profitable for all farmers, while safeguarding the environment for future generations. This holistic approach is essential for long-term food and nutritional security.

9. AGROFORESTRY

Introduction

Agroforestry means growing **trees along with crops and/or livestock** on the same piece of land. It helps farmers **diversify their income** while improving soil health, conserving water and reducing climate risks. Thus, agroforestry connects **farm productivity with environmental sustainability**, making agriculture more resilient in the long run.

Types of Agroforestry

Type	Explanation
1. Agrisilviculture	• Trees + crops are grown together on the same land. • Trees provide timber, fuelwood, fruits or other benefits while crops provide seasonal income.
2. Silvopastoral System	• Trees + pasture/livestock are managed together. • Trees provide shade and fodder, while animals graze on the pasture.
3. Agrosilvopastoral System	• Combines trees + crops + livestock on the same farm. • It creates multiple sources of income and makes better use of farm resources.
4. Agrisilviculture with Horticulture	• Trees + fruit crops are grown together. • Farmers get income from both perennial trees and horticultural crops.
5. Silvofishery	• Trees + fish farming are integrated around ponds or wetlands. • Trees can provide additional products while the pond supports fish production.
6. Agri-horticulture	• Agricultural crops + fruit trees are grown together, especially during the early years of orchard development. • It provides farmers with income before fruit trees reach full production.
7. Multipurpose Tree System	• Farmers grow trees that provide fodder, fuel, fruits, timber and other products along with agricultural activities. • Particularly useful for small farmers and dryland areas.

Significance of Agroforestry

- **Economic Benefits**
  - **Multiple incomes:** Farmers can earn from **crops, fruits, timber, fodder and other tree products**, reducing dependence on a single crop.
  - **Income stability:** Trees can provide income during difficult crop years, acting as a **financial cushion** for farmers.
- **Soil & Land Management**
  - **Improves soil health:** Tree leaves add organic matter and help maintain **soil fertility**.
  - **Controls erosion:** Tree roots hold the soil together and reduce **wind and water erosion**.
- **Water Conservation**
  - **Groundwater recharge:** Trees and improved soil structure can support **better infiltration of rainwater**.
- **Climate Change Benefits**
  - **Carbon sink:** Trees absorb and store **carbon dioxide**, helping reduce atmospheric greenhouse gases.
  - **Climate resilience:** Tree cover can protect crops from **heat, strong winds and changing rainfall**, making farms more resilient.
- **Resource Efficiency**
  - **Better use of land:** Agroforestry allows farmers to produce **food, fodder, fuel and timber** from the same farm.
  - **Livestock support:** Trees can provide **fodder and shade** for animals, strengthening integrated farming systems.

- **Social & Rural Benefits**
  - **Fuelwood and fodder:** Trees can meet part of rural households' needs for **fuel, fodder and small timber**.
  - **Employment:** Nursery raising, tree management, harvesting and processing can create **additional rural employment**.

National Agroforestry Policy, 2014

- **Purpose:** Introduced to promote trees on farms along with crops and livestock, while improving farmers' income and environmental sustainability.
- **Farmer-friendly approach:** Aims to make tree cultivation easier by addressing issues related to credit, insurance, markets and regulations.
- **Climate focus:** Encourages agroforestry for soil conservation, water management, carbon sequestration and climate resilience.
- **Better coordination:** Seeks coordination among agriculture, forestry, environment and rural development departments.
- **Overall aim:** Make agroforestry a viable farming option that supports both farmer livelihoods and ecological security.

Issues with Agroforestry

- **Economic & Financial Challenges**
  - **High initial investment:** Planting trees requires money for saplings, fencing, irrigation and maintenance, while returns from timber may come only after several years.

*As per NABARD, Establishing tree plantations requires initial upfront capital of **₹50,000 to ₹1,500,000 per hectare** for quality saplings, fencing, and drip irrigation.*

- **Delayed returns:** Farmers often prefer seasonal crops because trees involve a long gestation period.

*As per ICAR, Commercial agroforestry species involve a long gestation lag of **5 to 15 years** before timber yields lump-sum cash flow.*

- **Land & Farm-Level Challenges**
  - **Competition for resources:** Trees may compete with crops for water, nutrients, sunlight and space if species are not selected properly.
  - **Small landholdings:** On very small farms, farmers may hesitate to allocate productive land to trees because it can reduce the area available for seasonal crops.

*As per 10th agriculture census, Small and marginal farmers make up **86.2% of operational landholdings** in India, averaging just **1.08 hectares**, making them reluctant to sacrifice land area for trees*

- **Market-Related Challenges**
  - **Uncertain prices: Prices** of timber, fruits, bamboo and other tree products can fluctuate.
  - **Weak market linkages:** Farmers may lack easy access to organised markets, processing facilities and reliable buyers.
- **Policy & Regulatory Challenges**
  - **Felling restrictions:** State-level rules on tree cutting and transportation can discourage farmers from planting trees on agricultural land.

*Farmers face bureaucratic hurdles because over **25 major tree species** still require complex felling and transit permits across various Indian states*

  - **Limited policy clarity:** Different rules across states can create uncertainty about harvesting and selling farm-grown trees.
- **Knowledge & Technical Gaps**
  - **Lack of awareness:** Farmers may not know which tree–crop combinations are suitable for their soil and climate.

*As per NITI Aayog, Over **70% of smallholder farmers** report having no formal technical training on optimal species selection for their local agro-climatic zones.*

  - **Limited extension support:** There is often inadequate guidance on spacing, pruning, pest management and tree harvesting.
- **Infrastructure Challenges**
  - **Poor planting material:** Availability of quality and locally suitable saplings can be limited in some areas.
  - **Weak processing facilities:** Lack of nearby sawmills, storage, processing and transport facilities can reduce the value farmers receive.

Government Measures for Promoting Agroforestry

- **Sub-Mission on Agroforestry (SMAF):** Provides support for **quality planting material, tree plantation and agroforestry models**.
- **National Mission for a Green India:** Encourages **tree cover, ecosystem restoration and climate-resilient landscapes**, including agroforestry.
- **PM-KISAN / Agricultural Credit:** Helps farmers access financial support that can also support **long-term farm investments** such as agroforestry.

- **National Mission on Sustainable Agriculture (NMSA):** Promotes **resource-efficient and climate-resilient farming practices**, including agroforestry.
- **Har Medh Par Ped:** Encourages farmers to plant trees on **farm boundaries and bunds**, generating additional income without significantly reducing cropped area.
- **Promotion of Bamboo:** Government support under the **National Bamboo Mission** promotes bamboo cultivation, processing and marketing as an additional farm-based livelihood.
- **Research & Extension: ICAR and State Agricultural Universities** develop suitable **tree–crop combinations** and provide technical guidance through extension networks.
- **Van dhan yojna:** Launched in 2018 by the Ministry of Tribal Affairs and TRIFED, the Pradhan Mantri Van Dhan Yojana (PMVDY) aims to improve the livelihood of tribal communities by enhancing the value of forest produce.

Way Forward:

- **Make Agroforestry Economically Attractive**
  - **Easy finance:** Provide affordable credit and financial support for saplings, irrigation, fencing and other initial costs.

    NABARD Agroforestry Sub-Mission offers direct credit line linkages and capital subsidies (covering up to 40–50% of costs) for farm pond irrigation, fencing, and purchasing certified saplings.
  - **Early income:** Promote fruit trees, fodder trees and short-duration intercrops so farmers earn while trees are still growing.
- **Reform Policies & Regulations**
  - **Simplify rules:** Make rules for tree planting, harvesting and transportation simple and farmer-friendly.

    National Timber Management System (NTMS) operates a digital portal for issuing online transit permits
  - **Stable policy:** Provide clear state-level guidelines so farmers have confidence that they can legally harvest and sell farm-grown trees.

Promote Suitable Tree–Crop Combinations

  - **Local solutions:** Select tree species according to soil, rainfall, water availability and local farming practices.

    Traditional Khejri-Bajra system in Rajasthan pairs Prosopis cineraria (deep-rooted, nitrogen-fixing) with pearl millet, surviving harsh desert conditions without depleting topsoil moisture.
  - **Avoid competition:** Promote proper spacing and pruning so trees do not unnecessarily compete with crops for water, nutrients and sunlight.

    Alley Cropping with Subabul/Gliricidia in semi-arid regions requires side-branch pruning twice a year to prevent light competition with pulses and supply green leaf manure to the soil.
- **Improve Quality Planting Material**
  - **Better saplings:** Strengthen nurseries to provide high-quality, locally suitable and climate-resilient planting material.

    Central Agroforestry Research Institute (CAFRI, Jhansi) certifies high-yielding, disease-resistant clones of Eucalyptus and Poplar that mature faster with higher timber volume.
  - **Research support:** Encourage ICAR, agricultural universities and state departments to develop suitable tree–crop combinations.

ICAR-CAFRI & State Agricultural Universities (SAUs) developed region-specific models, such as the Alder-Large Cardamom system in Sikkim, which fixes atmospheric nitrogen while supporting shade-loving spice crops.

- **Strengthen Farmer Institutions**
  - **FPO-led agroforestry:** Use FPOs and cooperatives for bulk purchase of saplings, technical support, processing and collective marketing.
  - **Cluster approach:** Develop agroforestry clusters around products such as bamboo, timber, fruits and medicinal plants.

Best case studies for agro forestry

Region & Model	Major Tree & Crop Pairings	Impact & Results
Thar Desert (Rajasthan)	• Tree: Khejri (Prosopis cineraria) • Crops: Pearl Millet (Bajra), Pulses, Cluster Beans	• **Soil Fertility:** Boosted soil Nitrogen by 90% and Organic Carbon dramatically. • **Drought Buffer:** Leaves provide critical green fodder for livestock during harsh dry summers when regular grass dies.
Indo-Gangetic Plains (Punjab, Haryana, UP)	• Tree: Poplar (Populus deltoides) & Eucalyptus • Crops: Wheat, Sugarcane, Turmeric, Potato	• **Income Boost:** Provides high seasonal revenue from wheat plus a major cash lump sum every 5–7 years from timber. • **Industry Supply:** Powers local plywood, pulp, and paper industries while saving natural forests.
Eastern Himalayas (Sikkim & Northeast)	• Tree: Alder (Alnus nepalensis) • Crops: Large Cardamom, Ginger	• **Eco-Service:** Alder trees fix atmospheric nitrogen into the soil without competing with shade-loving cardamom. • **Erosion Control:** Deep root networks hold fragile mountain slopes together during heavy monsoon rains.
Western Ghats (Karnataka, Kerala, TN)	• Trees: Silver Oak, Rosewood, Mango • Crops: Coffee, Black Pepper vines, Cardamom	• **Climate Shield:** Shade trees lower canopy temperatures, protecting coffee plants from heatwaves and erratic rain. • **Diversified Income:** Coffee provides nnual cash, pepper vines yield spice income, and timber acts as long-term savings.
Eastern Ghats (Koraput, Odisha)	• Trees: Teak, Gmelina, Eucalyptus • Crops: Turmeric, Pineapple, Bananas, Lemongrass	• **Livelihood Transformation:** Lifted smallholder tribal farmers out of poverty by ending reliance on single rain-fed crops. • **Land Restoration:** Restored degraded hilly terrain while generating year-round cash flow from fruits and timber.

Conclusion

India's commitment to forest conservation and sustainable development is evident through various initiatives like the National Agroforestry Policy, Green India Mission, Forest Fire Prevention & Management Scheme, and Van Dhan Yojana. On International Day of Forests 2025, it is crucial to reaffirm our dedication to preserving forests as vital resources for future generations. By integrating conservation efforts with community participation and sustainable policies, India continues to pave the way for a greener, healthier, and more prosperous future.

■■■

MGP/BP/GS3/037

FOOD PROCESSING AND RELATED INDUSTRIES

1. FOOD PROCESSING SECTOR: OVERVIEW

Introduction:

Food processing covers the set of operations that transform raw agricultural produce into consumable or storable products, ranging from primary activities such as cleaning, grading, milling and packaging to secondary and tertiary value addition such as ready-to-eat meals, frozen products and beverages. Administered by the **Ministry of Food Processing Industries (MoFPI)**, it is the bridge between agriculture and industry, and the principal instrument for raising farm incomes without raising farm output.

Position of the sector:

- India possesses one of the world's strongest raw material bases, being the **largest producer of milk, spices and tea, the second largest producer of fruits and vegetables, and the third largest producer of fish**, with foodgrain output of about 358 million tonnes and horticulture output of about 370.7 million tonnes in 2024-25.
- The sector's **Gross Value Added rose from Rs 1.34 lakh crore in 2014-15 to Rs 2.24 lakh crore in 2023-24**, and it contributes roughly **8.80 per cent of manufacturing GVA and 8.39 per cent of agriculture GVA**.
- **MoFPI has been allocated Rs 4,064 crore in the Union Budget 2026-27**, with Rs 1,700 crore for PMFME, Rs 1,200 crore for the PLI scheme and Rs 915 crore for PMKSY

Significance of Food Processing Sector:

Farmer Income and Demand Creation

- **Raises Realisation Without Raising Output:** Value addition allows the farmer to capture a larger share of the consumer rupee, which is the only route to higher income for a cultivator whose land cannot expand.

  Sahyadri Farmers Producer Company (Maharashtra) aggregates smallholder grape growers and exports directly, bypassing three layers of intermediaries.

- **Creates an Assured Alternative Market:** A processor is a buyer who does not disappear during a glut, which reduces distress sales in perishables where mandi prices collapse within days of peak arrival.

  Operation Greens was designed precisely for this, stabilising prices for tomato, onion and potato before being extended to a wider basket of perishables.

- **Enables Diversification into High-Value Crops:** Assured processing demand justifies the shift from cereals to horticulture, dairy, poultry and marine products, which carry far higher margins per hectare.
  - Contract-grown processing potato in Gujarat and Punjab has created an entire frozen French fry industry where none existed two decades ago.

Employment and Rural Industrialisation

- **Labour-Intensive and Rurally Located:** Processing units are sited near the raw material base, so the jobs created are non-farm jobs in rural and semi-urban areas, precisely where underemployment is concentrated.

  As per MoFPI, the PLI Scheme for Food Processing Industries had generated about 3.39 lakh direct and indirect jobs by February 2026, against a target of 2.5 lakh by 2026-27.

  Absorbs Surplus Agricultural Labour: It offers a transition path out of agriculture that does not require migration to metropolitan centres.
  - Operational PMKSY projects have generated about 7.22 lakh direct and indirect jobs across cold chains, agro-processing clusters and food parks.
- **Supports Women and Micro-Entrepreneurs:** Micro food enterprises and self-help groups form a large share of the sector's base, making it an instrument of women's economic participation.

  Kudumbashree (Kerala) and Lijjat Papad demonstrate how women-led collectives can scale food processing into national brands; under PMFME, seed capital of Rs 240.92 crore has reached 63,108 SHG members.

Reduction of Post-Harvest Loss and Food Security

- **Converts Waste into Supply:** Processing and cold chain extend shelf life, so produce that would have rotted becomes available food, easing the paradox of high wastage alongside high undernutrition.

  As per the NABCONS study commissioned by MoFPI for the reference period 2020-22, India loses about Rs 1.53 lakh crore worth of agricultural produce annually to post-harvest losses.
- **Stabilises Seasonal Availability:** Freezing, canning, dehydration and irradiation smooth the sharp seasonality of Indian horticulture, moderating both farmgate crashes and consumer price spikes.
  - The frozen pea industry around Jalandhar and the mango pulp cluster at Chittoor allow a six-week harvest to be sold across twelve months.
- **Improves Nutritional Outcomes:** Fortification, standardised packaging and safe processing raise the nutritional quality and safety of what reaches the consumer.

  Universal fortification of rice distributed through the PDS, ICDS and PM POSHAN is a processing intervention delivering an anaemia outcome.

Exports and Foreign Investment

- **Shifts the Export Basket Up the Value Chain:** Exporting processed products rather than raw commodities earns more per tonne and insulates the country from raw commodity price cycles.
  - As per MoFPI, the share of processed foods in India's agri-food exports has risen from **13.7 per cent to 20.4 per cent over the last decade**.
- **Attracts Foreign Capital and Technology:** Liberal FDI norms have brought in capital along with cold chain, packaging and food safety technology.

  World Food India 2025 drew investment interest of over Rs 1 lakh crore, with Reliance Retail, Coca-Cola, Amul and Olam among those signing agreements; cumulative FDI equity inflow into the sector stands at about 7.33 billion dollars.
- **Builds Global Brands:** Scheme-supported branding has expanded the international presence of Indian food products.

  Cumulative export sales of PLI beneficiaries reached Rs 89,053.44 crore between April 2021 and September 2025, growing at a CAGR of 13.23 per cent over 2019-20.

Institutional Architecture: Major Schemes

Pradhan Mantri Kisan SAMPADA Yojana (PMKSY)

- **Umbrella Infrastructure Scheme:** Covers integrated cold chain and value addition infrastructure, creation and expansion of processing and preservation capacities, agro-processing clusters, backward and forward linkages, food safety infrastructure and Mega Food Parks.
- **Enhanced Outlay:** In July 2025 the Cabinet approved an additional Rs 1,920 crore, taking the total to **Rs 6,520 crore** for the Fifteenth Finance Commission cycle up to 31 March 2026, including **Rs 1,000 crore for 50 multi-product food irradiation units**.

Production Linked Incentive Scheme for Food Processing Industries (PLISFPI)

- **Outlay and Period:** Rs 10,900 crore, implemented from 2021-22 to 2026-27, targeting processed food output of Rs 33,494 crore.
- **Three Components:** Incentives for manufacturing in four segments, namely ready-to-cook and ready-to-eat foods including millet-based products, processed fruits and vegetables, marine products and mozzarella cheese; a separate window for innovative and organic products of SMEs; and support for branding and marketing abroad.
- **Performance:** About 169 applications approved as of January 2026, cumulative investment of Rs 9,207 crore against a committed Rs 7,000 crore, and roughly 35 lakh MT per annum of capacity added.

Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME)

- **Design:** Launched in June 2020 with an outlay of Rs 10,000 crore on a One District One Product approach, offering credit-linked capital subsidy, seed capital to SHG members and support for common infrastructure and branding.
- **Purpose:** To formalise the vast unorganised micro-enterprise segment, giving it access to institutional credit, FSSAI registration and organised market linkages.

Supporting Framework

- **Operation Greens:** Price stabilisation and value chain development for tomato, onion and potato, later extended to a wider basket of fruits and vegetables.
- **Agriculture Infrastructure Fund:** Long-term concessional debt for post-harvest management infrastructure at the community level.
- **FSSAI and Eat Right India:** Standard setting, licensing, fortification and food safety compliance, which underpin both domestic trust and export acceptance.
- **Mega Food Parks and Agro-Processing Clusters:** Cluster model providing common facilities, primary processing centres and collection centres linked to a central processing unit.

Issues with Food Processing in India:

Low Level of Processing, Especially in Perishables

- **Value Addition Remains Shallow:** Most Indian processing is primary, meaning milling, grading and packing, rather than the secondary and tertiary conversion that generates real margin.

  The Netherlands, with a fraction of India's cultivated area, ranks among the world's largest agri-food exporters purely on the strength of processing, logistics and branding.

- **Perishables Bear the Worst Losses:** The commodities with the highest value per tonne are precisely those the supply chain handles worst.

  As per the NABCONS study (2022), losses range from 6.02 to 15.05 per cent in fruits and 4.87 to 11.61 per cent in vegetables, against 3.89 to 5.92 per cent in cereals.

- **Recurring Glut and Crash Cycle:** Without processing capacity near production clusters, a good harvest becomes a price collapse.

Tomato and onion prices in Nashik and Kolar routinely swing from farmgate crashes to consumer spikes within a single season, the classic symptom of missing processing and storage.

Cold Chain and Infrastructure Gaps

- **Infrastructure Divide Between Commodities:** Loss rates map almost exactly onto the presence or absence of temperature-controlled infrastructure, showing the problem is addressable rather than inevitable.

As per the NABCONS study, guava records the highest loss at about 15.05 per cent while milk records under 1 per cent, the latter the result of decades of cooperative chilling investment.

- **Broken Rather Than Absent Chains:** Cold storage capacity exists but pack houses, pre-cooling units and reefer transport are missing, so the chain breaks at the farm gate and again in transit.
  - India's cold storage capacity is heavily skewed towards potato and concentrated in a handful of states, leaving most horticultural belts effectively unserved.
- **Wide Interstate Variation:** Identical crops record very different losses across states, indicating a gap in logistics and handling rather than agronomy.

As per the NABCONS study, paddy losses stand at 6.07 per cent in Assam against 2.87 per cent in Uttarakhand, and sugarcane at 8.99 per cent in Bihar against 1.23 per cent in Punjab.

Fragmented and Unreliable Raw Material Supply

- **Aggregation Problem:** Processors need large volumes of uniform quality, while supply comes from millions of tiny holdings with no grading discipline.

As per the Agriculture Census 2015-16, small and marginal farmers account for 86.08 per cent of operational holdings, with average farm size down to 1.08 hectares.

- **Varietal Mismatch:** Indian cultivars are bred for table consumption and yield rather than processing traits such as dry matter content and low reducing sugars.

PepsiCo had to develop and register its own FC5 potato variety for chips, and HyFun Foods had to supply tissue-culture Santana and Frysona seed, because no suitable public variety was available at scale.

- **Seasonality and Capacity Underutilisation:** Plants run far below capacity outside the harvest window, raising unit costs and deterring investment.

Mango pulp units in Chittoor and Krishnagiri operate intensively for only a few weeks a year, which is why single-commodity plants struggle to be viable.

Food Safety and Export Compliance

- **Consignment Rejections Abroad:** Failure to meet phytosanitary and residue standards costs export value and reputational damage.

India recorded 527 rejection incidents at European Union ports between 2020 and 2024, and the EU's near-zero residue limit for tricyclazole in basmati rice has repeatedly disrupted shipments.

- **Testing Infrastructure Deficit:** Accredited laboratory capacity is insufficient for the volume and range of exports.

Structural and Regulatory Constraints

- **Dominance of the Unorganised Segment:** A very large share of units are micro-enterprises operating informally, without FSSAI registration, institutional credit or modern packaging, which caps quality and scale.

The entire premise of PMFME, with its Rs 10,000 crore outlay, is that these units cannot grow until they are formalised.

- **Multiplicity of Compliance:** Units face overlapping requirements from FSSAI, legal metrology, state APMC regulation, pollution control and labour law, which weighs disproportionately on small processors.
- **Credit and Skill Deficits:** Micro units lack collateral for term loans, and the sector faces a shortage of trained personnel in food technology, quality assurance and cold chain operations.
- **Scheme Execution Gap:** A significant share of sanctioned projects never becomes operational, indicating problems of land acquisition, promoter capacity and anchor investor commitment.

Only 1,185 of 1,618 approved PMKSY projects are operational, and several Mega Food Parks were cancelled or delayed over land acquisition and weak anchor investors

Government Initiatives:

- **Pradhan Mantri Kisan SAMPADA Yojana:** Umbrella scheme for cold chain, processing capacity, agro-processing clusters, backward and forward linkages and food safety infrastructure, with an enhanced outlay of Rs 6,520 crore for the Fifteenth Finance Commission cycle.
- **Production Linked Incentive Scheme (PLISFPI):** Rs 10,900 crore outlay from 2021-22 to 2026-27, incentivising ready-to-eat and millet-based products, processed fruits and vegetables, marine products and mozzarella cheese, along with branding abroad.
- **PM Formalisation of Micro Food Processing Enterprises:** Rs 10,000 crore scheme on a One District One Product basis, providing credit-linked subsidy, SHG seed capital and common infrastructure to formalise micro units.
- **Multi-Product Food Irradiation Units:** Rs 1,000 crore earmarked for 50 units to extend shelf life and address the phytosanitary rejections that limit export access.
- **Operation Greens:** Price stabilisation and value chain development, extended from tomato, onion and potato to a wider basket of perishables.
- **Agriculture Infrastructure Fund:** Long-term concessional debt for farm-gate and community-level post-harvest infrastructure.
- **FSSAI, Eat Right India and Fortification:** Standard setting, licensing and fortification of staples to raise safety and nutritional quality across the chain.
- **Liberal FDI Regime:** Automatic route permission in food processing and permission for trading of food products manufactured in India, which has drawn about 7.33 billion dollars of equity inflow.
- **World Food India:** Flagship investment platform, with the 2025 edition drawing investment interest of over Rs 1 lakh crore.
- **SPICED Scheme and National Turmeric Board:** Quality certification, post-harvest upgrading and market linkage support to move spice exports up the value chain.

Case Studies:

Case	What was done	Result and Lesson
Amul and the Dairy Cooperative Chain	Producer-owned three-tier structure with village-level chilling, assured daily procurement and integrated processing.	Milk records post-harvest losses of under 1 per cent, the lowest of any commodity in the NABCONS study, against about 15 per cent for guava. Proves that producer ownership of the processor is the most durable form of backward linkage.

Sahyadri Farms, Nashik	Farmer producer company aggregating smallholders with its own pack house, cold chain and export desk.	Smallholder grape growers reached EU retail shelves directly, showing that the aggregation gap can be closed by producer institutions rather than by corporate intermediaries.
Lasalgaon Irradiation Facility	BARC-designed irradiation unit for mango and onion.	Enabled Indian mango exports to the United States, which had been barred on phytosanitary grounds. Demonstrates that compliance infrastructure is market-access infrastructure.
HyFun Foods, Gujarat	Contract farming with tissue-culture processing potato seed, linked to a frozen French fry plant.	Procured about 3 lakh tonnes from 6,000 farmers in 2023-24, underpinning frozen French fry exports of 1,35,877 tonnes worth Rs 1,478.73 crore. Shows varietal supply is the binding constraint, not plant capacity.
PLI Scheme, Millet-Based Products	Targeted incentive for millet-based ready-to-eat products under Category I.	Sales rose from about Rs 345 crore in 2022-23 to about Rs 1,845 crore in 2024-25, showing a well-defined incentive can create a value-added category from a standing start.
Mega Food Park Scheme	Cluster model with a central processing unit fed by primary processing and collection centres.	Several parks were delayed or cancelled over land acquisition and weak anchor investors, and only 1,185 of 1,618 PMKSY projects are operational. Approval is not the same as execution.

Way Forward

Fix the Backward Linkage

- **Make FPOs the Aggregation Unit:** Producer organisations solve the processor's volume problem and the farmer's bargaining problem simultaneously, and should be the default counterparty in procurement contracts.
- **Develop Processing-Grade Varieties:** ICAR and state agricultural universities should prioritise breeding for processing traits, since without the right cultivar no amount of plant capacity delivers quality output.
- **Formalise Land Leasing:** Secure written tenancy would let tenant cultivators enter supply contracts and access institutional credit.

States should adopt the NITI Aayog Model Agricultural Land Leasing Act, 2016, since over 80 per cent of tenants currently operate on informal oral leases.

Build the Chain, Not the Node

- **Fund End-to-End Cold Chains:** Support should be conditional on an integrated package of pack house, pre-cooling, reefer transport and retail-end storage, since a standalone cold store cannot recover produce already spoiled at the farm gate.

The Amul model is the domestic proof: village-level chilling plus assured daily collection keeps milk losses under 1 per cent while guava, with no such chain, loses about 15 per cent.

- **Prioritise the Highest-Loss Commodities and States:** Target investment where the loss differential is largest rather than spreading it uniformly.

- **Leverage Rail and Multimodal Logistics:** Expand dedicated perishable cargo movement to cut transit time and handling damage.

  Kisan Rail demonstrated that time-tabled multi-commodity perishable trains can move produce from surplus to deficit regions at scale.

Strengthen Quality and Export Compliance

- **Scale Irradiation and Accredited Testing:** Complete the 50 irradiation units and 100 NABL-accredited laboratories so that compliance stops being the constraint on market access.

  The BARC-designed irradiation facility at Lasalgaon is what made Indian mango exports to the United States possible, a direct illustration of compliance infrastructure creating a market.

- **Build Digital Traceability from Plot to Pack:** Importing countries increasingly require batch-level provenance, which India has already shown it can deliver.

  APEDA's GrapeNet traceability system for grape exports to the EU is the working template, and can be extended to other commodities using AgriStack plot-level data.

Support the Small Processor

- **Ease the Compliance Burden:** A single-window clearance and a simplified FSSAI regime for micro units would formalise more enterprises than any subsidy.
- **Deepen Credit Access:** Extend credit-linked subsidy and collateral-free lending to micro food enterprises, which constitute the bulk of the sector by number but the smallest share by value.

  Under PMFME, 56,543 loans have been sanctioned since January 2025, indicating both latent demand and considerable headroom.

- **Skill the Workforce:** Expand food technology, quality assurance and cold chain training rather than leaving skilling to on-the-job improvisation.

Deepen Value Addition

- **Move from Primary to Tertiary Processing:** Incentives should be weighted towards ready-to-eat, frozen and functional foods rather than milling and packing, which capture the least margin.

  Sales of millet-based products under the PLI scheme rose from about Rs 345 crore in 2022-23 to about Rs 1,845 crore in 2024-25, showing that a targeted incentive can create a value-added category almost from scratch.

- **Learn from Comparable Exporters:** Thailand converts a modest agricultural base into a major processed food export industry through canned tuna, pineapple and ready meals, while Brazil dominates orange juice concentrate.
- **Fix Execution, Not Just Allocation:** With over a quarter of approved PMKSY projects still not operational, project appraisal, promoter due diligence and time-bound monitoring matter more than fresh outlays

Conclusion:

The food processing sector is pivotal for India's economic transformation, farmer income enhancement, reduction of food waste, and rural employment generation. It has the most profound impact on the rural economy. With **sustained focus on infrastructure, technology, formalization, and market linkages**, India is well-positioned to unlock the full potential of this "sunrise industry" and emerge as a global food processing hub.

2. SUPPLY CHAIN MANAGEMENT IN THE FOOD PROCESSING INDUSTRY

Introduction:

Supply chain management in food processing is the **coordinated movement of agricultural produce** from the farm gate to the final consumer, covering aggregation, primary processing, storage, transport, secondary processing, distribution and retail, along with the flow of information and payment that runs alongside. Because agricultural produce is perishable, seasonal, bulky and variable in quality, the food supply chain is harder to manage than almost any manufacturing chain.

Two directions of linkage, both examinable:

- **Upstream or Backward Linkage:** Everything that secures raw material of the right variety, volume and quality, including contract farming, FPO aggregation, collection centres, pre-cooling and pack houses.
- **Downstream or Forward Linkage:** Everything that moves the processed product to the consumer, including cold storage, reefer transport, distribution networks, retail cold display, branding and export compliance.

Significance of Supply Chain Management

Reducing Post-Harvest Loss

- **Logistics-Driven Loss:** Wastage occurs at handling, grading, transport and storage rather than in the field, which makes it fixable by logistics rather than by breeding or irrigation.

  As per the NABCONS study commissioned by MoFPI for 2020-22, India loses about Rs 1.53 lakh crore of agricultural produce annually to post-harvest losses.

- **Commodity-Wise Divide:** Losses track the presence or absence of a temperature-controlled chain almost exactly.
- **Interstate Variation:** The same crop records very different losses across states, proving the problem is infrastructural rather than inherent to the crop.

  Sugarcane losses stand at 8.99 per cent in Bihar against 1.23 per cent in Punjab, and paddy at 6.07 per cent in Assam against 2.87 per cent in Uttarakhand.

Higher Farmer Realisation

- **Fewer Intermediaries:** Every handling layer removed transfers margin back to the producer, which matters more than a higher headline price.
- **Freedom from Distress Sale:** Farm-gate storage lets the farmer decide when to sell rather than being forced to sell into the arrival glut.
  - Negotiable warehouse receipts financed under the Agriculture Infrastructure Fund allow a farmer to borrow against stored produce instead of dumping it.
- **Standing Demand Source:** A processor with committed capacity keeps buying through a glut, unlike a trader who withdraws when prices fall.

Price Stability

- **Seasonal Smoothing:** Storage and processing convert a six-week harvest into twelve months of supply, moderating both farmgate crashes and consumer spikes.
- **Safety and Traceability:** A managed chain allows batch-level tracking, which is now a condition of market access rather than a value addition.

  APEDA's GrapeNet system provides end-to-end traceability for grape exports to the European Union.

Export Competitiveness

- **Chain-Wide Compliance:** Residue limits, phytosanitary standards and cold chain integrity are verified across the whole journey, so a single weak link disqualifies the consignment.

India recorded 527 rejection incidents at European Union ports between 2020 and 2024.

- **Landed Cost Advantage:** In commodity exports, margins are thin enough that transport and handling costs decide competitiveness.

As per the first official NCAER and DPIIT assessment, India's logistics cost stands at about 7.97 per cent of GDP, or roughly Rs 24 lakh crore.

Components of the Agri-Food Supply Chain

The Cold Chain Continuum

- **Pre-Cooling and Pack House:** Removes field heat immediately after harvest and performs sorting, washing, grading and packaging, and is the most critical link because heat not removed at the farm gate cannot be removed later.
- **Cold Storage, Bulk and Hub:** Bulk storage holds seasonal crops such as potato for months, while hub storage holds mixed perishables for short periods near consumption centres.
- **Reefer Transport:** Refrigerated trucks, vans and rail wagons that maintain the chain in transit, without which storage at either end is wasted.
- **Ripening Chambers:** Controlled temperature and humidity units that ripen fruit such as banana and mango in a staged manner near the market.
- **Retail Cold Display:** The final link, absent in most Indian wet markets and small retail, where the chain most commonly breaks at the last mile.

Aggregation and Primary Processing

- **Collection Centres:** Village-level nodes where produce from scattered small holdings is pooled, cleaned and given basic treatment before onward movement.
- **Farmer Producer Organisations:** The institutional answer to fragmentation, allowing a processor to contract once for a volume that would otherwise need hundreds of separate transactions.
- **Agro-Processing Clusters:** Common infrastructure shared by several units, reducing the fixed cost that no single small processor could bear alone.

Information and Finance Flows

- **Market Intelligence:** Price and arrival data that allow production and processing decisions to be planned rather than reactive.
- **Traceability Systems:** Digital records linking a consignment back to the plot, increasingly mandatory in export markets.
- **Payment and Credit:** Warehouse receipt financing, direct transfer of payments and working capital for aggregators.

Issues with Supply Chain Management

Broken Cold Chain

- **Investment in One Link:** Cold storage attracted the bulk of investment while the links on either side were neglected, so produce is spoiled before arrival and after despatch.

As per the NCCD and NABCONS All India Cold-chain Infrastructure Capacity assessment, the gap stood at about 99 per cent in pack houses, 91 per cent in ripening chambers and 85 per cent in reefer vehicles, against under 10 per cent in cold storage.

- **Reefer Shortage:** Without refrigerated movement, cold storage merely delays spoilage rather than preventing it.

NCCD estimates about 10,000 reefer vehicles in operation against a requirement of at least 62,000, with the fleet projected to reach only around 33,000 by 2031.

- **Skewed Capacity:** Existing storage serves potato in Uttar Pradesh, West Bengal, Gujarat and Punjab, leaving most horticultural belts effectively unserved.

Supply-Side Fragmentation

- **Aggregation Constraint:** A processor needs uniform volume, while supply comes from millions of tiny holdings with no grading discipline or common standard.

As per the Agriculture Census 2015-16, small and marginal farmers account for 86.08 per cent of operational holdings, with average farm size down to 1.08 hectares.

- **Excess Intermediation:** Multiple handling points between farm and processor add cost, time and physical damage without adding value.
  - Every additional transfer of a crate of tomatoes is a bruising event, which is why loss rises with the number of handling stages rather than with distance alone.
- **Varietal Mismatch:** Cultivars bred for table sale rather than processing force firms to run their own seed programmes.

Logistics Inefficiency

- **Road Dependence:** Road is the costliest mode per tonne-kilometre yet carries the bulk of freight, inflating the landed cost of low-margin perishables.

Road carries about 60 to 65 per cent of freight against 27 to 28 per cent by rail and under 2 per cent by waterways, while costing roughly Rs 2.5 to 3.0 per tonne-kilometre against Rs 1.5 to 1.8 by rail.

- **Poor Fleet Utilisation:** Long transit times and empty return runs raise cost and reduce the shelf life left at the point of sale.

Indian trucks cover about 300 kilometres a day against a global average of 400 to 500, with empty running estimated at around 40 per cent.

- **Weak First and Last Mile:** Investment has gone into trunk corridors while village-to-collection-centre and warehouse-to-retail links remain unimproved.

Institutional Frictions

- **Fragmented Oversight:** Cold chain development is supported by multiple ministries and agencies with no single accountable authority, producing overlapping schemes and unmatched links.
- **Compliance Burden:** FSSAI licensing, legal metrology, APMC levies, pollution control and transport permits fall disproportionately on micro processors and small aggregators.
- **Energy Cost and Reliability:** Cold chain is energy-intensive and rural power supply is erratic, so operating cost and equipment risk both rise, discouraging private investment.

Execution and Financing Gaps

- **Approval Without Operation:** A significant share of sanctioned infrastructure never becomes functional owing to land, promoter capacity and anchor investor problems.

Only 1,185 of 1,618 approved PMKSY projects are operational, and several Mega Food Parks were delayed or cancelled over land acquisition.

- **Working Capital Shortage:** FPOs and small processors lack collateral for the credit needed to buy at harvest and sell across the year, which is precisely the function that stabilises prices.

Government Initiatives for Supply Chain Management:

- **Pradhan Mantri Kisan SAMPADA Yojana:** Umbrella scheme whose Integrated Cold Chain and Value Addition Infrastructure component funds pre-cooling, pack houses, reefer vehicles and ripening chambers as a package rather than in isolation.

- **Multi-Product Food Irradiation Units:** Rs 1,000 crore earmarked for 50 units to extend shelf life and meet phytosanitary requirements in export markets.
- **Agriculture Infrastructure Fund:** Long-term concessional debt for farm-gate and community post-harvest infrastructure including warehouses, cold stores, grading units and primary processing centres.
- **Operation Greens:** Support for transport and storage of perishables alongside price stabilisation, initially for tomato, onion and potato and later extended to a wider basket.
- **Kisan Rail:** Time-tabled multi-commodity perishable trains connecting surplus production belts to deficit consumption centres, with subsidised freight for notified produce.
- **PM Gati Shakti and the National Logistics Policy, 2022:** Integrated infrastructure planning and the Unified Logistics Interface Platform to reduce transit time and improve multimodal coordination.
- **Gati Shakti Cargo Terminals:** Indian Railways has approved 306 terminals with a combined capacity of 192 million tonnes per annum to improve first and last mile rail connectivity.
- **e-NAM:** Electronic price discovery across 1,656 mandis, with assaying and grading facilities that standardise quality at the point of first sale.
- **APEDA Traceability Platforms:** GrapeNet, HortiNet and related systems providing plot-to-consignment traceability for export commodities.
- **PMFME and One District One Product:** Formalisation of micro processors with credit-linked subsidy and common infrastructure, creating aggregation nodes close to production.

Way Forward

Funding the Whole Chain

- **Integrated Packages Only:** Subsidy should be released only for an end-to-end package of pre-cooling, pack house, reefer transport and hub storage, since a standalone cold store cannot recover produce already spoiled at the farm gate.
- **Targeted Investment:** Concentrate spending where the loss differential is largest rather than distributing it uniformly across states.
- **Last Mile Focus:** Extend the chain into retail through cold display in mandis, municipal markets and small shops, where the chain most often breaks after all upstream investment.

Solving Aggregation

- **FPOs as Counterparty:** Producer organisations cut the processor's transaction cost and raise the farmer's bargaining power in the same transaction.

  Amul remains the strongest domestic proof, since producer ownership of the chilling and processing infrastructure is what keeps milk losses under 1 per cent.

- **Village-Level Nodes:** Basic cleaning, grading and pre-cooling near the farm removes the largest share of loss at the lowest cost per tonne saved.
- **Processing-Grade Varieties:** ICAR and state agricultural universities should breed for dry matter content, uniformity and shelf life so firms need not run private seed programmes.

Logistics and Digital Reforms

- **Modal Shift:** Moving perishables from road to rail and water releases large savings because rail costs roughly half of road per tonne-kilometre.

  Kisan Rail showed that time-tabled perishable services are operationally feasible, and Gati Shakti Cargo Terminals supply the first and last mile that made them usable.

- **Real-Time Visibility:** Temperature and location sensors on reefer movement allow breaches to be corrected in transit rather than discovered at delivery.

The Unified Logistics Interface Platform provides the data backbone on which such visibility tools can be built.

- **Wider Traceability:** GrapeNet has already proved the model for grapes and can be replicated using AgriStack plot-level records and Digital Crop Survey data.

Institutional and Financial Reforms

- **Single-Point Accountability:** Consolidate the schemes spread across ministries under one authority responsible for the chain as a whole rather than for individual components.
- **Working Capital Access:** Expand warehouse receipt financing and collateral-free credit so FPOs and small processors can buy at harvest and release across the year.
- **Cheaper, Cleaner Cooling:** Promote solar-powered cold rooms and energy-efficient refrigeration to cut the operating cost that deters private investment in rural locations.

Learning from Models

- **Cooperative Logic Beyond Dairy:** The three-tier structure that transformed milk can be applied to fruit, vegetables and fisheries, where producer numbers are similarly large and individual volumes similarly small.
- **Comparable Exporters:** The Netherlands converts a fraction of India's cultivated area into among the world's largest agri-food exports on the strength of logistics, cold chain and auction infrastructure rather than land.

3. CONTRACT FARMING

Introduction

Contract farming is an arrangement in which a farmer agrees in advance to produce and deliver a specified agricultural commodity to a sponsor, usually a processor, exporter or organised retailer, at a pre-agreed price, quantity and quality standard, often with the sponsor supplying inputs, technology and extension support.

Constitutional and legal position:

- **Agriculture is a State subject** under Entry 14 of the State List, but **contracts fall under Entry 7 of the Concurrent List**, which is why the Centre can circulate a model law while only states can enact it.
- Contract farming agreements have always been enforceable under the **Indian Contract Act, 1872**, so the model laws are about registration, dispute resolution and farmer safeguards rather than about legality as such.

Evolution in India:

- **Origins:** Modern contract farming in India dates to the tomato processing plant set up by Pepsi Foods at Zahura in Hoshiarpur district, Punjab, in 1989, following the **Johl Committee (1986)** recommendation that at least 20 per cent of the area under wheat and paddy be diversified.
- **Model APMC Act, 2003:** Introduced the first statutory provisions, including compulsory registration of sponsors, dispute settlement machinery, market fee exemption and protection of farmers' land ownership.
- **Model Contract Farming Act, 2018:** A dedicated, standalone model law circulated to states.
- **Farm Laws, 2020:** The Farmers (Empowerment and Protection) Agreement on Price Assurance and Farm Services Act, 2020 created a national framework, and was **repealed in 2021** after sustained protests.

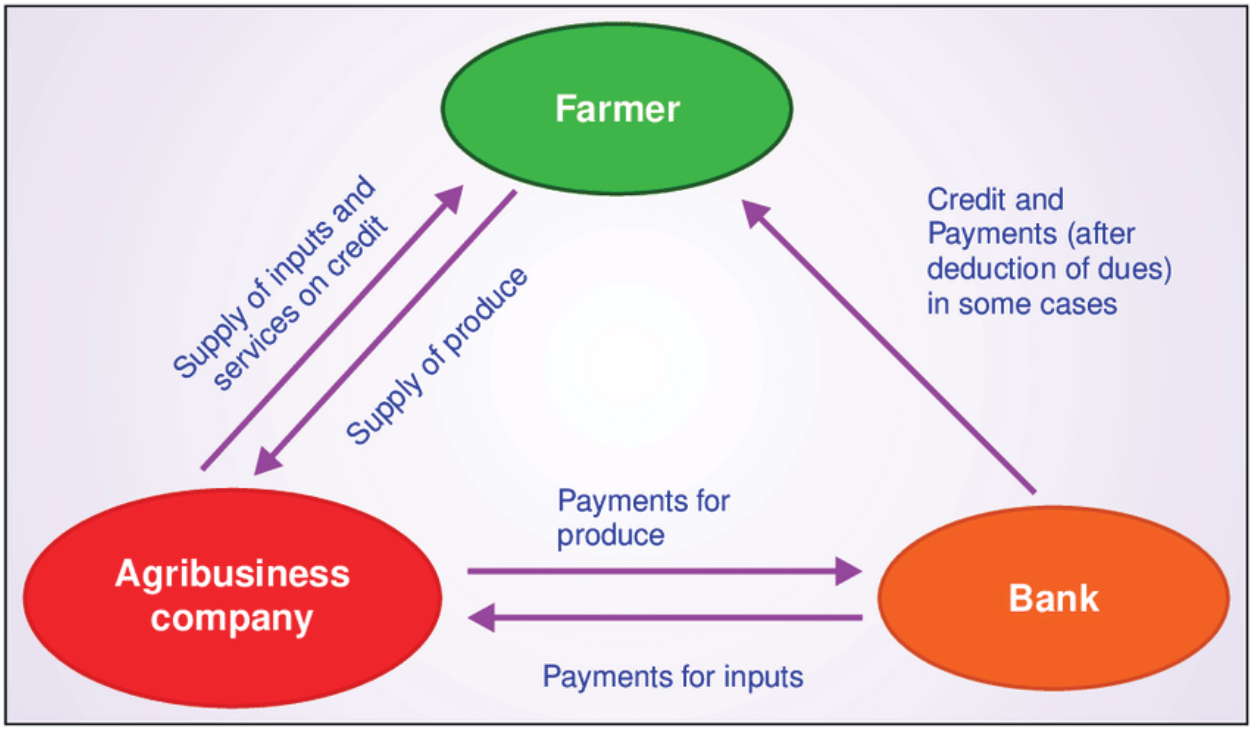
Salient features of the Model Contract Farming Act, 2018:

- The Act lays special emphasis on protecting the interests of the farmers, considering them as the weaker of the two parties entering into a contract.

- In addition to contract farming, services contract all along the value chain including pre-production, production and post-production have been included.
- "Registering and Agreement Recording Committee" or an "Officer" for the purpose at district/block/taluka level for online registration of sponsor and recording of agreement is provided.
- Contracted produce is to be covered under crop/livestock insurance.
- Contract framing is to be outside the ambit of the APMC Act.
- No permanent structure can be developed on farmers' land/premises.
- Promotion of Farmer Producer Organizations (FPOs)/Farmer Producer Companies (FPCs) to mobilise small and marginal farmers has been provided.
- It ensures buying of the entire pre-agreed quantity of one or more of agricultural produce, livestock or its product of the contract farming producer as per the contract.
- Contract Farming Facilitation Group (CFFG) is being created for promoting contract farming and services at village/panchayat the level provided.
- An accessible and simple dispute settlement mechanism at the lowest level possible is provided for the quick disposal of disputes.

How does Contract Farming work?

- Under contract farming, farmers can be given seeds, credit, fertilizers, machinery and technical advice so that their product is tailor-made for the requirements of the companies.
- There would be **no middlemen** involved and farmers would get a predetermined sale price from the companies.
- The farmer does not have to make trips to the mandis nor worry about getting seeds and credit for farming operations.
- By entering into a contract, the farmer reduces the risk of fluctuating market demand and prices for his produce and the companies reduce the risk of non-availability of raw materials.



Significance of Contract Farming

Income Security and Risk Transfer

- **Assured Market and Price:** A pre-agreed buyback price shields the farmer from post-harvest price crashes and removes the uncertainty of finding a buyer at the time of arrival glut.
- **Transfers Market Risk to the Sponsor:** The company absorbs price volatility and demand risk, which a small farmer with no storage and no working capital is least equipped to bear.
- **Reduces Dependence on Intermediaries:** Direct farm-to-firm linkage cuts out layers of commission agents, raising the share of the consumer rupee that reaches the producer.

Access to Technology, Inputs and Credit

- **Embedded Extension Services:** Sponsors typically supply improved seed, agronomic advice and pest management support, effectively substituting for a weak public extension system.
- **Input Supply on Credit:** Inputs advanced against future delivery give farmers working capital without collateral, which is decisive for those outside the formal credit net.
- **Quality and Traceability Upgrade:** Contract discipline enforces grading, maximum residue limit compliance and traceability, which are prerequisites for export markets that reject Indian consignments on residue grounds.

Diversification and Value Chain Development

- **Shift to High-Value Crops:** Contracts have driven cultivation of processing-grade potato, tomato, gherkin, green chilli, basmati rice and seed crops, moving farmers out of the low-margin cereal cycle.
- **Enables Food Processing Investment:** Assured raw material of consistent variety and quality is the precondition for processing capacity, which India needs to raise its low level of agricultural value addition.
- **Builds Export Capability:** Contract-grown processing potato allowed India to emerge as a frozen French fry exporter, shipping about **1,35,877 tonnes worth Rs 1,478.73 crore in 2023-24** to Southeast Asia, West Asia, Japan and Taiwan.

Issues

Asymmetry of Power and Information

- **Unequal Bargaining Position:** A single small farmer negotiating with a corporate buyer has neither legal literacy nor the ability to walk away, so terms on price, grading and delivery are effectively dictated.
- **Prevalence of Oral Contracts:** Field studies show most arrangements remain informal, with a study of small farmers in Baghpat district of Uttar Pradesh finding only about **20 per cent of contracts were written**, leaving farmers with no enforceable claim.
- **Cost of Dispute Resolution:** Even where machinery exists, pursuing a claim costs the farmer time and money disproportionate to the amount at stake, while the sponsor treats litigation as a routine business expense.

Contract Breach on Both Sides

- **Rejection on Quality Grounds:** When market prices fall below the contract price, sponsors have an incentive to reject deliveries on grading or quality pretexts, leaving farmers with an unsold crop grown to a specification nobody else wants.
- **Side-Selling by Farmers:** Conversely, when open market prices rise above the contract price, farmers divert produce to the market, which is the sponsor's principal complaint and a documented reason for early contract farming ventures failing in Punjab.
- **Delayed Payment:** Late payment is among the most frequently reported grievances, and it defeats the very liquidity advantage the contract was meant to provide.

Exclusion of the Most Vulnerable

- **Bias Towards Larger Farmers:** Sponsors minimise transaction costs by contracting with fewer, larger cultivators, so the smallest farmers who most need assured markets are the least likely to be offered a contract.
- **Exclusion of Tenants and Sharecroppers:** Since contracts are anchored to land records, oral lessees and sharecroppers who actually cultivate the land cannot enter agreements in their own name.
- **Regional Concentration:** Contract farming clusters around processing plants and good road connectivity, reinforcing rather than reducing regional disparity.

Agronomic and Food Security Concerns

- **Monocropping and Soil Depletion:** Contracts specify a single variety over large contiguous areas, encouraging monoculture, intensive chemical use and loss of on-farm biodiversity.
- **Shift Away from Food Crops:** Diversion of area to export-oriented and processing crops can reduce local foodgrain availability, a concern in districts with weak public distribution reach.
- **Technology Lock-In:** Dependence on sponsor-supplied proprietary seed and inputs reduces the farmer's autonomy and can raise costs once the initial support package is withdrawn.

Intellectual Property and Seed Rights

- **Conflict with Farmers' Rights:** The **PepsiCo case of 2019**, in which the company sued four Gujarat farmers for cultivating its registered FC5 potato variety before withdrawing amid public outcry, exposed the tension between plant variety protection and **Section 39 of the Protection of Plant Varieties and Farmers' Rights Act, 2001**, which preserves the farmer's right to save, use and exchange seed.

Political and Trust Deficit

- **Legacy of the Repealed Farm Laws:** Because the 2020 Act on price assurance and farm services was one of the three repealed laws, contract farming carries strong political charge, and any fresh central push is read by farmer unions as reintroduction by another route.
- **Weak State Uptake:** The near-absence of state adoption of the 2018 Model Act reflects this political caution more than any technical objection to its provisions.

Government Measures

- **Model APMC Act, 2003:** Introduced compulsory registration of contract farming sponsors, dispute settlement provisions and explicit protection of farmers' land ownership.
- **Model Contract Farming Act, 2018:** A dedicated model law taking contract farming outside APMC control, mandating online registration of agreements and barring any transfer of land rights to the sponsor.
- **Promotion of FPOs:** The 10,000 FPO scheme with an outlay of Rs 6,865 crore enables collective contracting, giving small farmers the scale and legal capacity to negotiate on equal terms.
- **Agriculture Infrastructure Fund:** Financing for post-harvest infrastructure, warehousing and primary processing creates the physical base without which contract farming cannot function beyond the farm gate.
- **PM Formalisation of Micro Food Processing Enterprises (PMFME):** Supports small processors who can act as local sponsors, reducing dependence on large corporate buyers alone.
- **PM Dhan-Dhaanya Krishi Yojana:** Targets agriculturally lagging districts and explicitly promotes contract farming as one instrument for raising productivity and market access.
- **Operation Greens and Mega Food Parks:** Build the processing demand for perishables that makes contract arrangements commercially viable for horticulture.
- **e-NAM and AgriStack:** Provide transparent reference prices and verified cultivator identity, both of which are preconditions for fair and enforceable contracts.

Case Studies:

Case	What was done	Result and Lesson
Pepsi Foods, Punjab (1989 onwards)	Tomato and chilli contracts around a processing plant at Zahura, later extended to processing-grade potato.	Pioneered the model and demonstrated real yield and quality gains, but early ventures were undermined by side-selling when open market prices rose, showing that a flat fixed price is fragile.
PepsiCo v. Gujarat Farmers (2019)	The company sued four farmers for growing its registered FC5 potato variety, then withdrew after public outcry.	Exposed the collision between plant variety protection and Section 39 of the PPVFR Act, and showed how a single legal action can destroy the trust an entire contracting model depends on.
HyFun Foods, Gujarat (2023-24)	Supplied disease-free tissue-culture seed of processing potato varieties and contracted for buyback.	Procured about 3,00,000 tonnes from 6,000 farmers with plans to reach 20,000 farmers across 80,000 acres by 2027-28, underpinning India's emergence as a frozen French fry exporter. The clearest success case for export-oriented contracting.
Amul and Dairy Cooperatives	Assured procurement of milk at announced rates through a three-tier cooperative structure.	Achieved what corporate contract farming has not, namely assured offtake for millions of very small producers, because the producers collectively own the processor. The strongest argument for the FPO route.
Punjab Contract Farming, 2003 to 2011	State-backed diversification contracts to move area out of the paddy and wheat cycle.	Widely adopted then abandoned, owing to weak financial backing, poor coordination between state and Centre and absent dispute machinery. Shows that a law alone cannot sustain contracting.
Baghpat District Study, Uttar Pradesh (2025)	Survey of 200 small farmers in potato and sugarcane contracts.	Incomes and living standards rose, but only about 20 per cent of contracts were written, and farmers reported delayed payment, produce rejection and limited legal awareness. Benefits are real but unevenly captured.

Way Forward

Rebalance Bargaining Power

- **Make FPOs the Default Contracting Party:** Contracts negotiated by a producer organisation on behalf of hundreds of members equalise bargaining power and reduce the sponsor's transaction cost at the same time.
- **Mandatory Written and Registered Agreements:** Require online registration of every contract with a simple standard-form template in the local language, so that oral and one-sided arrangements lose legal standing.
- **Farmer Legal Literacy:** Use extension machinery and Krishi Vigyan Kendras to explain contract terms, grading standards and grievance routes before signing rather than after a dispute.

Design Contracts That Survive Price Swings

- **Reference-Price-Linked Pricing:** Replace a flat fixed price with a floor price plus a share of any upside above a reference market price, which removes the farmer's incentive to side-sell and the sponsor's incentive to reject.
- **Independent Quality Arbitration:** Grading disputes should be settled by an accredited third-party assayer rather than by the sponsor's own inspector, closing the most common route to arbitrary rejection.
- **Time-Bound Payment Obligations:** Statutory payment deadlines with automatic interest for delay, on the model applied to sugarcane arrears.

Strengthen Dispute Resolution

- **Local, Low-Cost and Time-Bound:** Disputes should be resolved by the registering authority at block or district level within a fixed period, without pushing the farmer into civil courts.
- **Sponsor Registration and Security Deposit:** Registration should carry a modest security deposit or bank guarantee that can be drawn upon to settle proven farmer claims.

Widen Inclusion

- **Recognise Cultivator Status:** Permit tenants and sharecroppers to contract on the basis of a cultivator registration rather than land title, as attempted through Andhra Pradesh's Crop Cultivator Rights Cards.
- **Incentivise Small-Farmer Coverage:** Link any state support or infrastructure concession for sponsors to the proportion of small and marginal farmers in their contract base.

Safeguard Agronomy and Seed Rights

- **Crop Rotation Clauses:** Build rotation and soil health obligations into registered contracts so that monocropping is not the price of assured income.
- **Clarify the IPR Boundary:** Issue guidance reconciling plant variety protection with Section 39 of the PPVFR Act, so that the PepsiCo episode is not repeated.

Conclusion:

Contract farming is therefore seen as a key strategy to modernize India's supply chains and reduce agricultural risk, provided the regulatory framework remains strong enough to protect the interests of the small farmer.

MGP/BP/GS3/038

LINKAGES BW DEVELOPMENT

1. INTERNAL SECURITY AND ITS CHALLENGES IN INDIA

Internal security forms the backbone of national stability, ensuring peace, public order, and sovereignty within a country's borders. In India, it remains a multidimensional challenge shaped by historic, socio-political, and geopolitical factors.

Internal Security and External Security

- **Internal security** refers to the protection and maintenance of law, order, sovereignty, and integrity of the state from threats arising within the national boundaries: civil unrest, insurgency (Khalistan movement in Punjab), terrorism, organized crime, communal violence, etc.
- **External security**, in contrast, deals with threats emanating from outside the country- foreign aggression, cross-border incursions, wars, external espionage, cross-border terrorism, and broader geo-strategic threats.

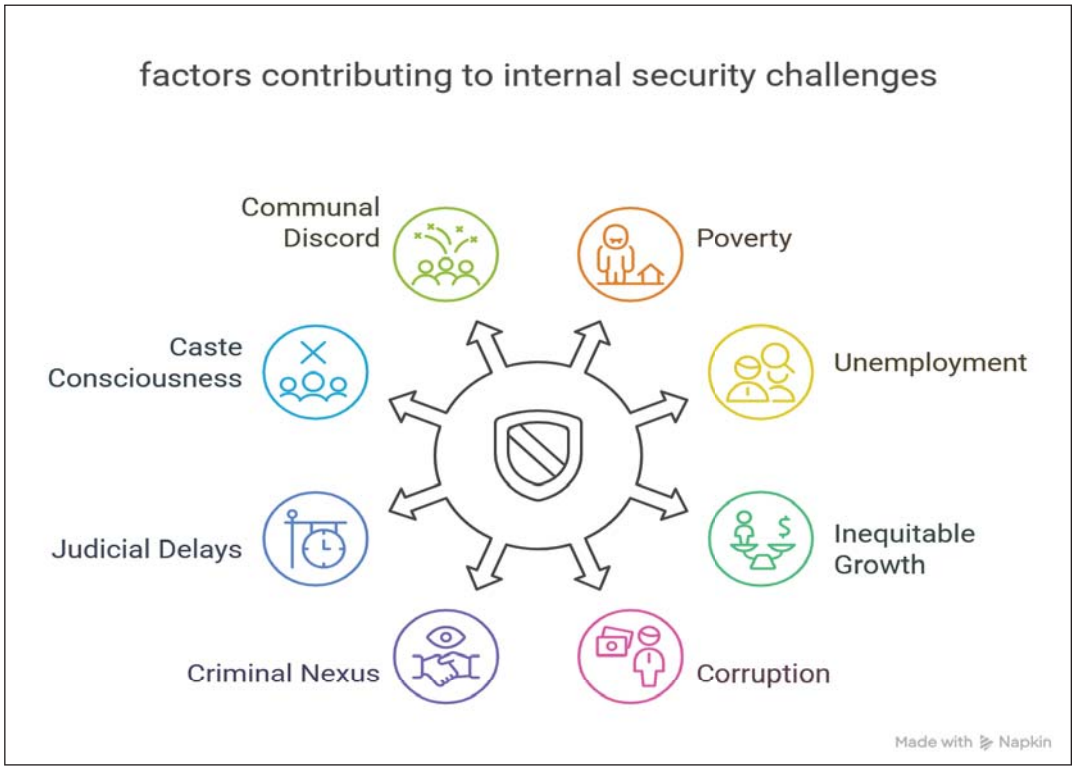
Difference between Internal Security and External Security

Aspect	Internal Security	External Security
Typical Threats	Insurgency, terrorism, communal riots, organized crime, drug trafficking, cyber-crime, internal unrest, border infiltration	War, foreign aggression, cross-border terrorism, border disputes, external espionage, maritime threats
Key Agencies	Police, Central Armed Police Forces (CAPFs), domestic intelligence agencies, investigation agencies	Armed forces (Army, Navy, Air Force), external intelligence agencies, border security forces
Admin Responsibility	Ministry of Home Affairs (MHA)	Ministry of Defence (MoD), external intelligence, foreign policy apparatus

Internal and external security are not entirely independent — external threats (e.g. cross-border terrorism, infiltration, smuggling) often impact internal security. The two overlap and can exacerbate each other.

Key Challenges to Internal Security in India

- **Terrorism and Radicalisation**- India faces domestic and cross-border terrorism, seen in the **26/11 Mumbai attacks** and repeated militant strikes in Jammu & Kashmir. Newer issues like **ISIS-inspired online radicalisation**, encrypted apps, and Hawala-based digital financing further complicate security.
- **Left-Wing Extremism (LWE) / Naxalism-** LWE persists in the **Red Corridor** (Chhattisgarh, Jharkhand, Odisha), where Maoist groups conduct IED ambushes like the 2021 Sukma-Bijapur attack, extort contractors, and sabotage roads and telecom towers, disrupting governance and affecting tribal communities.



- **Insurgency and Separatism in the North-East**- Groups such as NSCN-IM and ULFA factions draw strength from historical grievances and ethnic identities. The problem deepens due to sanctuaries in Myanmar's Sagaing region, porous borders with Myanmar and Bangladesh, and ethnic clashes like Manipur (2023).
- **Communal, Ethnic, and Sectarian Violence**- Events like the **Delhi riots (2020), caste clashes in Haryana**, and tribal conflicts in Tripura and Manipur disrupt social harmony. Such flashpoints are often exploited or amplified through misinformation, increasing polarisation.
- **Organized Crime and Illegal Trafficking**- Criminal networks engage in drug trafficking via Punjab and the '**Golden Crescent**', fake currency smuggling across the Bengal border, and human trafficking involving Nepal and Bangladesh. Many maintain links with terrorists or insurgents, especially in border states or coastal regions like Gujarat and Kerala.
- **Cybersecurity Threats and Digital Disinformation**- Cyberattacks are rising, including ransomware on hospitals and banks, hacking of power grids, and defence-related espionage. AI-driven misinformation — such as election deepfakes or communal rumours — can quickly trigger unrest.
- **Border Infiltration, Illegal Migration & Maritime Vulnerabilities**- Militant infiltration across the LoC and IB in J&K, **illegal migration from Bangladesh** into Assam and West Bengal, and narcotics smuggling through the Arabian Sea (e.g., heroin off Gujarat) pose major risks. The **7,500 km coastline**, exposed during 26/11, remains vulnerable.
- **External Influence Over Internal Conflicts**- External actors offer training, weapons, and funding to extremist groups, including those in Kashmir and NE groups using Myanmar safe havens. Instability in Afghanistan and tensions with Pakistan often spill over, linking internal security with regional geopolitics.

Key Measures Taken to Strengthen Internal Security

- **Legislative Strengthening-** India has strengthened its security framework through laws like **UAPA**, **NSA**, **PMLA**, **IT Act**, **NIA Act**, and **NDPS Act** to counter terrorism, organised crime, and money laundering.
- **Police and CAPF Reforms-** Prakash Singh reforms, **Assistance to States for Modernisation of Police**

(ASMP), improved training, ICT tools, welfare steps, and modern equipment strengthen policing and CAPFs for hybrid and internal threats.

- **Counter-Terrorism Strengthening-** The NIA now has wider jurisdiction, including cyber-terrorism and trafficking. **Multi-Agency Centre (MAC)** enables 24×7 intelligence sharing, while **NATGRID** links key databases. India also targets terror financing via the **Terror Funding and Fake Currency (TFFC) Cell** and cooperation with the Financial Action Task Force (FATF).
- **Border & Coastal Security-** Smart fencing, digital sensors, and optical fibre systems enhance border control. Coastal security is boosted through the Indian Coast Guard (ICG), coastal police stations, fishermen support schemes, and Border Security Force (BSF) modernisation to counter drones and narco-terrorism.
- **Tackling Insurgency-** The **National Policy and Action Plan on Left Wing Extremism (LWE), 2015** integrates security, rights, and development. Programmes like **Aspirational Districts Programme (ADP)** and **PM Gati Shakti** improve infrastructure, while Northeast peace accords have enabled mass insurgent surrenders.
- **Social & Economic Measures-** Laws like **Bharatiya Nyaya Sanhita (BNS) Section 192** act against hate speech. Targeted welfare schemes reduce poverty and alienation. Development efforts in tribal regions and **Civic Action Programmes** of CAPFs build trust.
- **Combating New-Age Crime-** State cyber units, National Crime Records Bureau (NCRB) cyber monitoring, and Crime and Criminal Tracking Network System (CCTNS) strengthen digital policing. Crackdowns on extremist groups like Popular Front of India (PFI) disrupt radical networks.
- **Technology-Driven Modernisation-** Use of drones, AI surveillance, Geographic Information System (GIS) mapping, predictive policing, and Integrated Command and Control Centres (ICCCs) ensures real-time monitoring and quicker responses across cities and borders.

Conclusion

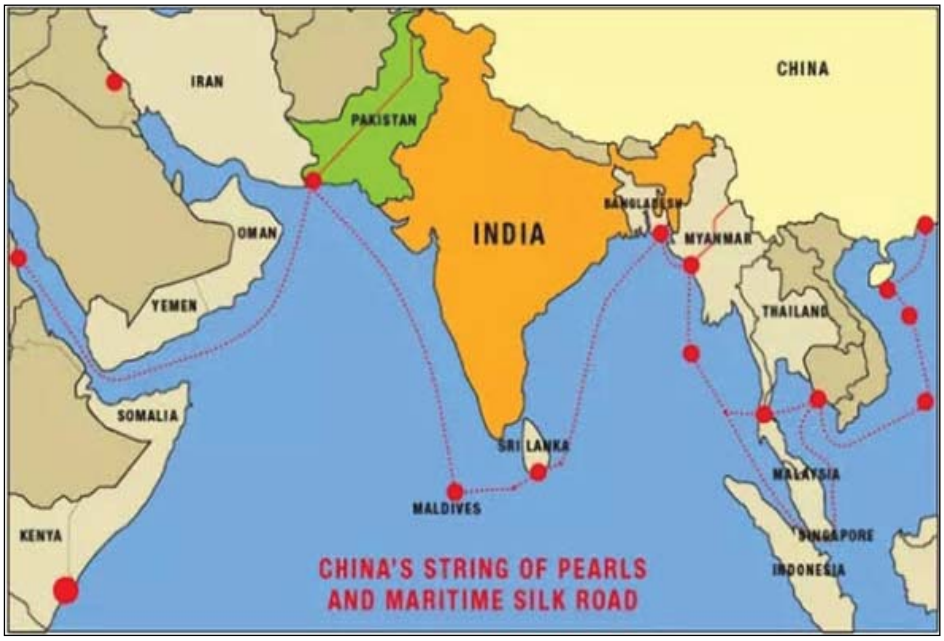
Internal security remains central to India's broader national security framework. Despite diverse and evolving threats, India's strong institutions, legal framework, and strategic focus ensure that challenges are managed effectively, reinforcing its stability and democratic resilience.

2. ROLE OF EXTERNAL STATE IN CREATING INTERNAL SECURITY CHALLENGES IN INDIA

The internal security of India is increasingly being shaped by external state actors whose strategic ambitions, covert operations and regional manoeuvres pose direct and indirect threats. From proxy warfare to demographic pressures, China, Pakistan, Bangladesh, Myanmar — and increasingly Nepal — all play varying roles in influencing instability within India.

Threats from China

- **Military and Border Pressure:** China's 3,488 km disputed border with India remains a key flashpoint. Guided by **Mao's Five Finger Policy**—seeking influence over Ladakh, Nepal, Sikkim, Bhutan and Arunachal Pradesh. Beijing uses "**grey-zone" tactics** like incremental incursions, salami-slicing and rapid infrastructure buildup along the LAC. The **Doklam (2017)** and **Eastern Ladakh (2020) crises** showed how quickly these moves can escalate, underscoring China's long-term strategic pressure on India.
- **Water as a Strategic Lever:** Chinese dams on the **Yarlung Zangbo (Brahmaputra)** raise fears that it could manipulate water flow, affecting ecology and livelihoods in Northeast India.
- **Strategic Encirclement & Maritime Expansion:** Through the **Belt and Road Initiative (BRI),** China has deepened its influence across South Asia. The "**String of Pearls**" — port access in India's neighbourhood — challenges India's maritime security and its traditional influence.



- **Nexus with Pakistan**: China and Pakistan describe their ties as an "all-weather" partnership — strengthened further by the **China–Pakistan Economic Corridor (CPEC).** Despite ups and downs, India remains the central factor shaping this relationship. As **Andrew Small** notes in The China-Pakistan Axis, India is the "strategic glue" binding the China–Pakistan partnership.
- **Support to Insurgency in the Northeast:** Indian analysts have long raised concerns that China covertly supports insurgent groups in the Northeast with arms, funding, ideological backing and sanctuary through third-country channels.
- **Overall:** China's challenge operates across border, water, maritime and hybrid-warfare domains, heightening India's internal vulnerabilities.

Threats from Pakistan

- **Proxy Terrorism & Insurgency:** ISI-backed groups like Lashkar-e-Taiba (LeT) and Jaish-e-Mohammed (JeM) have carried out major attacks — including the 2001 Parliament attack, **2008 Mumbai attacks**, **2019 Pulwama bombing**, and the 2025 Pahalgam attack.

- **LoC Tensions & Two-Front Risk:** Cease-fire violations, infiltration attempts and cross-border terrorism continue along the LoC. The **2025 Pahalgam attack** underscores the persistent threat.
- **Economic and Criminal Warfare:** Pakistan also destabilizes India through narcotics trafficking (via the "**Golden Crescent**"), fake Indian currency (FICN), and crime-terror networks that fund extremist groups.
- **Deterioration of Trust:** Reliance on proxies gives Pakistan plausible deniability, complicating diplomatic responses and fueling communal tensions and social instability within India.
- **Overall:** Terrorism, infiltration and crime-terror nexuses make Pakistan a key internal security threat.

Threats from Bangladesh

- **Illegal Migration & Demographic Changes:** The porous border has allowed undocumented migration for decades, especially affecting Assam, West Bengal, Tripura and other Northeastern states.
- **Ethnic and Social Tensions:** Demographic change has intensified concerns over land, identity, language and local resources, often causing local-migrant friction.
- **Crime, Smuggling & Illegal Trade:** The border is prone to human trafficking, drug and arms smuggling, cattle smuggling and fake currency circulation — all of which can be exploited by extremist or criminal groups.
- **Past Insurgent Sanctuary:** Earlier, several Northeast insurgent groups took shelter in Bangladesh. While crackdowns have reduced this threat, **Political instability** in Bangladesh could revive such risks.
- **Overall:** Bangladesh's challenges stem from migration pressures and border vulnerabilities rather than intentional hostile actions.

Threats from Myanmar

- **Insurgent Safe Havens:** Groups like **ULFA-I, NSCN-K** and other tribal/Meitei outfits have long used Myanmar's border regions (Sagaing, Kachin, Chin states) for shelter, recruitment and logistics.
- **Arms & Narcotics Trafficking:** Myanmar is part of the drug-producing "Golden Triangle." Weapons and narcotics flow into India, strengthening insurgency and organized crime.
- **Refugee Influx:** Since the 2021 military coup, refugees — especially into Mizoram, Manipur and Nagaland — have increased, creating humanitarian and security challenges.
- **Impact on Connectivity Projects:** Instability has delayed key India–Myanmar projects such as the **Kaladan Multi-Modal Transit Transport Project** and the India–Myanmar–Thailand Trilateral Highway, affecting India's Act East policy.
- **Overall:** Myanmar's turmoil fuels insurgency, trafficking, refugee pressures and strategic setbacks for India.

Threats from Nepal

- **Open Border & Smuggling:** The 1,850 km open border allows free movement but also enables trafficking of narcotics, gold, fake currency, small arms and contraband. Nepal is often used as a transit point for illicit trade.
- **Terrorist and Extremist Misuse:** Some terrorist and insurgent groups have attempted to use the border to move cadres, weapons, narcotics and fake currency.
- **Cross-Border Crime & Human Trafficking:** Weak enforcement enables human trafficking, drug smuggling and illegal trade.
- **Political Instability in Nepal:** Governance instability creates opportunities for criminal and extremist networks to exploit the border.
- **Strategic Concerns:** Instability also affects cooperation in security, energy, hydropower and trade — while allowing external powers more space to influence Nepal, indirectly affecting India.

- **Overall:** Nepal's threat is subtle, rooted in open borders, smuggling, cross-border crime and political instability.

Why External Influence Matters for India's Internal Security

- **Proxy warfare:** External powers can destabilize India through terrorism, hybrid tactics and demographic manipulation without direct conflict.
- **Demographic stress:** Illegal migration and refugee flows intensify communal and ethnic tensions.
- **Resource diversion:** Internal security operations divert resources from development, especially in border regions.
- **Strategic encirclement:** External powers gaining influence in India's neighbourhood weaken India's strategic position.
- **Porous borders:** Unregulated or open borders with Nepal, Bangladesh and Myanmar create entry points for terrorists, trafficking networks and contraband.

Way Forward

- **Strengthen Border Management:** Improve fencing, surveillance, technology-enabled monitoring and patrolling along Bangladesh, Nepal and Myanmar borders while ensuring humane treatment of refugees.
- **Boost Intelligence & Cooperation:** Enhance intelligence-sharing and joint border-management mechanisms with Nepal, Bangladesh and Myanmar.
- **Develop Border Regions:** Improve infrastructure, education and livelihoods in border and Northeastern areas to reduce alienation and insurgency.
- **Strategic & Diplomatic Engagement:** Strengthen ties through platforms like **QUAD and BIMSTEC** to counter strategic encirclement.
- **Clear Migration Policies:** Create humane but secure migration and refugee frameworks, ensuring registration, identity verification and prevention of illegal infiltration.
- **Community Awareness:** Conduct campaigns in border regions to counter radicalization and demographic vulnerabilities.
- **Strengthen Bilateral Mechanisms:** Improve border-security pacts, joint patrols, customs controls, extradition treaties and data sharing — especially with Nepal — to curb smuggling, trafficking, fake currency and extremist movement.

Conclusion

India's internal security is shaped by external pressures—proxy terrorism, China's strategic moves, migration-led demographic shifts, and porous borders. A comprehensive mix of stronger border management, regional cooperation, diplomacy, and border-region development is essential to safeguard national security and social stability.

3. ROLE OF NON-STATE ACTORS IN CREATING INTERNAL SECURITY CHALLENGES

Non-state actors (NSAs) refer to **individuals, groups, or organizations** operating **independently** of or **against state control**, including terrorists, insurgents, separatists, organized crime syndicates, cybercriminals, and radicalized NGOs.

Unlike state-sponsored threats, NSAs exploit **India's socio-economic fault lines**, **porous borders, and digital vulnerabilities** to undermine internal security.

The **Ministry of Home Affairs' 2024 Annual Report** notes a **15% rise** in NSA-linked incidents (1,200+), with Left-Wing Extremism (LWE), terrorism, and cybercrimes accounting for 70% of casualties.

Reasons for Rise of Non-State Actors

- **Globalisation:** Globalisation has allowed NSAs to **develop alliances and generate support** outside the immediate area of their operations.
- **Increased funding:** Rise of the **illicit global economy** due to globalisation, has provided increased funding opportunities for NSAs.
- **Increased use of social media:** The social media knows no international boundaries. Thus, it has become a **preferred media of communication** for NSA's making it difficult for security agencies to keep a tap on them.
- **Urbanisation:** The sheer size of such cities will generate immense law and order and security problems, especially in poorer areas, and impose additional burdens on urban infrastructures that are already under stress.

Threats/Challenges Posed by Non-State Actors

- **Left-Wing Extremism (LWE/Naxalism):** Naxalism is a form of left-wing extremism that seeks to **overthrow the government through armed struggle.**
  - **Example -** 2021 Sukma Ambush: 23 security personnel killed by CPI(Maoist).
- **Terrorism and Extremism:** Terrorist organizations like Lashkar-e-Taiba (LeT), Jaish-e-Mohammed (JeM), and Indian Mujahideen (IM) have carried out numerous attacks across India, targeting civilians, security forces, and critical infrastructure.
- **Drug Trafficking:** Proximity to the largest producers of heroin and hashish-the **Golden Triangle and Golden Crescent (Afghanistan-Pakistan-Iran)** - has made India's border vulnerable to drug trafficking.
- **Human Trafficking:** Children and women trafficking takes place via Bangladesh and Nepal.
- **Insurgency and Separatist Movements:** Insurgent groups, such as those in the North-Eastern states (like the United Liberation Front of Assam (ULFA) and National Socialist Council of Nagaland (NSCN)), demand autonomy or independence, challenging the territorial integrity of India.
- **Communalism/Ethnic violence:** Propaganda are run and funded by enemy countries and other non-state actors (NGOs and CSOs) to destabilize India by damaging the socio-religious fabric and ensuring riots.
  - **Example -** Manipur conflict
- **Cyberattacks:** Non-state actors increasingly exploit cyberspace for **illegal activities, including hacking, data theft, cyber espionage,** and online radicalization.
  - **Example -** in 2020, Hacking of pharmaceutical companies working on COVID-19 vaccines in India.
- **Cross-Border Infiltration and Proxy Wars:** Non-state actors, such as **militant groups** supported by neighboring countries, engage in cross-border infiltration to carry out attacks, sabotage infrastructure, and disrupt peace.

Steps Taken by India

- **Legislative:** UAPA (amended 2019) – terrorist tag to individuals; NIA powers enhanced.
- **Institutional:** Multi-Agency Centre (MAC) strengthened; NATGRID operationalised 2024 for real-time data sharing.
- **Border management:** 100% drone fencing pilot on Punjab border (2025); Comprehensive Integrated Border Management System (CIBMS).
- **Financial tracking:** ED/FIU-IND seized ₹2,500 crore terror assets (2020-25).
- **De-radicalisation: "Operation Samaritan"** in J&K surrendered 800+ youth (2020-25).
- **Cyber:** Indian Cyber Crime Coordination Centre (**I4C**) 2024 blocked 3.5 lakh SIMs used for terror funding.

Way Forward

- **Integrated Internal Security Doctrine 2030:** Single policy document merging counter-terror, counter-LWE, cyber and organised crime strategies.
- **National Counter-Radicalisation Grid:** AI-powered early-warning system using social media analytics (similar to Israel's model).
- **Public-Private Partnership in Cyber Defence:** Mandatory industry CERTs and real-time threat sharing.
- **Crypto Regulation Bill 2026:** Ban anonymity, mandate KYC on all wallets to choke terror finance.
- **Community Policing 2.0:** Expand "Surrender-cum-Rehabilitation" policy to urban Naxals and cyber radicals.
- **Regional NSA Disruption:** Deepen BIMSTEC and Act East Policy to deny sanctuaries in Myanmar/ Bangladesh.
- **Judicial Fast-track Courts:** Special benches for NSA cases with 6-month disposal mandate.
- **Human Intelligence Revival:** Recruit from surrendered cadres (like Ikhwan in 1990s Kashmir).

Conclusion

NSAs remain India's gravest internal security peril, morphing from ideological insurgents to hybrid threats amid digital globalization. While kinetic successes reduced violence by 70% (2014-24), holistic strategies blending hard power, development, and diplomacy are imperative. Strengthening multi-agency synergy and societal resilience will fortify India's internal security architecture, ensuring Viksit Bharat by 2047.

4. LEFT WING EXTREMISM IN INDIA

The Naxal footprint has considerably decreased in India, in line with the government's vision of **eliminating naxalism by March 31, 2026.** In 2025 alone, 270 Naxalites were killed and 680 arrested, with another 1,225 having surrendered to authorities.

Current status of Maoist Insurgency

- **Leadership weakened:** Maoist leadership has been decimated, and the People's Liberation Guerrilla Army is severely weakened.
- **Cadre losses and surrenders:** This year till October 29, the South Asia Terrorism Portal (SATP) has recorded- 333 Maoists kills, 398 arrests, and 1,787 surrenders.
- **Territorial contraction:** The number of LWE-affected districts has shrunk to 11 (MHA), marking a sharp contraction of operational space.
- **Decreasing Violence trend:** Incidents fell from 1,936 (2010) to 374 (2024) (decrease of 81%). Deaths fell from 1,005 (2010) to 150 (2024) (decrease of 85%).

Maoist insurgency- Maoist insurgency, which is also known as Left-wing Extremism (LWE)/Naxalism, is an armed movement rooted in political ideologies that seek to bring about socio-political and economic changes through violent means.

The Maoist Insurgency/Left Wing Extremism has been a long standing security threat which started in 1967 and gradually spread to different parts of the country. The Maoist insurgency peaked in the early mid-2000s.

Reasons that led to the growth of Left-Wing Extremism in India

- The **2006 D Bandopadhyay Committee** identified **governance gaps** and **extensive discrimination against tribals** in economic, socio-political, and cultural spheres as the primary causes of Naxalism's proliferation. Some of the reasons are enumerated below-

- **Inequitable Socio-economic development-** Increasing Socio-economic inequities and unemployment in the hinterland region nudge the youth to take up arms. There has been **disruption of traditional occupations** and lack of alternative work opportunities.
- **Governance failure- Governance deficit** in the remote parts of Red Corridor regions led to the growth of Left Wing Extremism. There was development apathy and huge corruption in the red corridor area. **For ex- Lack of food security due to corruption in the PDS scheme.**
- **Political Marginalization-** The tribal groups were politically marginalized by the self-serving dominant groups. This led to **political deprivation among the marginalized groups** leading to hopelessness and a sense of powerlessness.
- **Land Alienation and Displacement-** There was **failure of land reforms** especially in **land redistribution** after independence. The **development projects and acquisition of land for industrial purposes** have sometimes led to the **displacement of local communities without adequate compensation or rehabilitation**. This has been a focal point of Naxalite agitation.
- **Discrimination against tribals-** There has been poor implementation of laws prohibiting transfer of tribal land to non-tribals in the Fifth Schedule areas. **Non-regularisation** and **hasty rejections of land grants of traditional land rights under FRA, 2006** have also led to the growth of left-wing extremism.

Challenges posed by Left-wing extremism in India

- **Threat to Internal Security-** Left-wing extremism poses significant internal security threats to India, **leading to violent clashes** and **loss of lives** among security forces and civilians. **For ex- Maoist rebels engaging in deadly attacks** on security forces **in remote areas like Dantewada.**
- **Social and Economic Disruption-** LWE in India disrupts social and economic development in affected regions, **hindering infrastructure and welfare projects. For ex- Maoists targeting their roads, schools, and other vital infrastructure** to maintain control over their territories.
- **Forced Recruitment and Child Soldiers-** Maoist groups have been known to force vulnerable individuals, including children, to join their ranks, exploiting them for their cause. This practice **violates human rights** and **disrupts the lives of countless innocent people.**
- **Extortion and Illegal Activities-** LWE groups often engage in extortion and other illegal activities to fund their operations. **For ex-** Demands of '**protection money**' from local businesses, leading to economic stagnation in affected areas.
- **Political Instability-** Maoist insurgency contributes to political instability in India, as the affected regions struggle with governance issues and **weakened trust in democratic institutions**. This can undermine the overall stability of the country.
- **Human Rights Violations-** LWE groups have been known to commit human rights violations, including extrajudicial killings, abductions, and torture. These actions not only **harm innocent civilians** but **also tarnish India's international reputation.**
- **Alienation of Tribal Populations:** Maoist insurgents often exploit and **alienate tribal population** in their stronghold, creating a cycle of violence and mistrust between the communities and the government. This **makes it difficult to address the root causes of extremism and to foster long-term peace and stability.**

Government policies to control Maoist insurgency in India

The Indian government has employed various approaches to control Left-wing extremism (LWE), **focusing on a combination of security, development,** and **political initiatives.** Some of these approaches include:

Security Schemes for LWE

National Policy and Action Plan to address Left Wing Extremism 2015	This **comprehensive action plan** aims to **enhance the capabilities of security forces** to combat LWE. This action plan **ensures that the rights and entitlements of local communities are safeguarded** and focuses on socio-economic development in affected regions.
SAMADHAN Doctrine	SAMADHAN Doctrine encompasses the **entire strategy of government from short-term policy to long-term policy** formulated at different levels. **SAMADHAN** stands for **S–** Smart Leadership, **A–** Aggressive Strategy, **M–** Motivation and Training, **A–** Actionable Intelligence, **D–** Dashboard Based KPIs (Key Performance Indicators) and KRAs (Key Result Areas), **H**– Harnessing Technology, **A–** Action plan for each Theatre, **N–** No access to Financing.
Security Related Expenditure (SRE) Scheme	Under the Security Related Expenditure (SRE) Scheme, the **Central Government reimburses the Security Related Expenditure of 10 LWE affected States.** These security related **expenditures are related to training and operational needs of security forces, ex-gratia payment** to the family of civilians/security forces killed/injured in LWE violence, **rehabilitation of surrendered** LWE cadres, **community policing, village defence committees** and **publicity materials**.
Special Infrastructure Scheme (SIS)	Under this scheme, funds are being provided to States for strengthening the infrastructure related to Security. Under this scheme 250 Fortified Police Stations are being constructed.

Development Schemes for LWE

Road Connectivity Project for LWE affected areas (RCPLWE)	The Government introduced this scheme in 2016 **for improving road connectivity in LWE affected States.** The roads included under the scheme have been identified by the Ministry of Home Affairs in consultation with the State Governments and the security agencies. **8124 km roads and 306 bridge works have been completed under this scheme.**
LWE Mobile Tower Project	This project has been **launched to improve mobile connectivity in the LWE affected areas.** The aim is to **improve the communication network** for better governance delivery.
Aspirational District Programme	The Ministry of Home Affairs has been **tasked with the monitoring of Aspirational districts programme in 35 LWE affected districts.** The government has implemented various welfare and development schemes to address the root causes of LWE, such as poverty, unemployment, and social inequality in these aspirational districts.

Socio-Political Schemes for LWE

Panchayat Extension to Scheduled Areas (PESA) Act, 1996	Through this act, **the government aims to strengthen local governance in LWE-affected areas** by promoting democratic participation and empowering local institutions.
Civic Action Programme (CAP)	This Scheme aims to **bridge the gaps between Security Forces and local people** through personal interaction and bring the human face of Security Forces before the local population. The **CAPFs deployed in LWE affected areas are provided funds for conducting civic activities** for the welfare of the local people.

Media Plan	The government is implementing this Scheme in LWE affected areas to counter the false propaganda of Maoists. Under this scheme activities like **Tribal Youth Exchange programmes** are organised by NYKS along with radio jingles, documentaries and pamphlet distribution.
Inter-State Coordination	The government is promoting coordination among states affected by LWE to ensure a unified response. **For Ex- Regular meeting of the Union Home minister with the CMs and DyCMs of all LWE states.**
Surrender and Rehabilitation Policies	The government has introduced surrender and rehabilitation policies for LWE cadres willing to **renounce violence and join mainstream society.** These policies offer financial incentives, vocational training, and support for reintegration into society.

Issues associated with government measures to control LWE

- **Human Rights Violations-** Security operations against Maoist insurgency have sometimes led to **human rights violations,** including **extrajudicial killings, arbitrary arrests,** and **harassment of civilians.** These actions can alienate local communities and create resentment against the government.
- **Inadequate Implementation of Development Schemes-** Despite numerous welfare and development programs, their implementation in LWE-affected areas is often hampered by **corruption, bureaucratic inefficiencies**, and **a lack of coordination between various agencies.** As a result, the intended benefits may not reach the affected communities.
- **Insufficient Capacity Building of Security Forces-** The security forces sometimes lack specialized training, equipment, and intelligence support to effectively tackle Maoist insurgency. This sometimes leads to **operational failures** and **increased casualties** among security personnel.
- **Reliance on Force-** Excessive focus on security operations overshadows the need for political engagement and dialogue with LWE groups. A **purely militaristic approach could prolong the conflict** and hinder efforts to find a peaceful resolution.
- **Incomplete Rehabilitation Programs-** While surrender and rehabilitation policies exist, they are not effectively implemented. There are certain cases of **Inadequate support for reintegration** such as insufficient vocational training or financial assistance.
- **Lack of Trust and Cooperation-** Mistrust between the government and affected communities **hinders the implementation of development initiatives** and security operations. Building trust is crucial for fostering cooperation and addressing the root causes of LWE.
- **Political Instability and Opportunism:** Political instability and opportunism undermines efforts to address LWE. Politicians often **exploit the issue for electoral gains**. This further complicates the government's response to the problem.

Way Forward

- **Focus on Development-** The **successful implementation of development projects** in **Andhra Pradesh** and **Telangana** reduced Maoist insurgency. Thus we must focus more on the effective implementation of development projects.
- **Strengthen Local Governance-** Empowering local governments to address grievances and deliver public services effectively, as seen in the **Panchayati Raj system's success in reducing Naxalism in West Bengal**, can help build trust between communities and authorities.
- **Enhance Security Forces' Capabilities:** Provide specialized training and better equipment to security personnel, similar to the **Greyhounds force in Andhra Pradesh,** which proved effective in countering Maoist insurgency due to their specialized training and local knowledge.

- **Encourage Dialogue and Peaceful Resolution:** Engage in meaningful dialogue with Maoist groups to address their grievances, similar to the **peace talks between the Indian government and Naga insurgents**, which led to a ceasefire and reduced violence in the region.
- **Foster Community Engagement:** Involve local communities in decision-making processes and development projects, as seen in the success of the **Janmabhoomi program in Andhra Pradesh**, where community participation helped build trust and reduce Maoist influence.
- **Ensure Accountability and Transparency-** Strengthen measures to prevent corruption and human rights abuses, as demonstrated by the **National Human Rights Commission's interventions** in cases of alleged human rights violations by security forces, which can help restore public trust in government institutions.

5. JAMMU AND KASHMIR MILITANCY

The **November 2025 Delhi Red Fort car bomb blast** has been linked by Indian authorities to militancy networks operating out of Jammu & Kashmir (J&K). The bomber, Umar Un Nabi, was a Pulwama-based doctor with alleged ties to militant groups Jaish-e-Mohammed (JeM) and Ansar Ghazwat-ul-Hind, highlighting active terror module linkages between J&K and urban centers like Delhi.

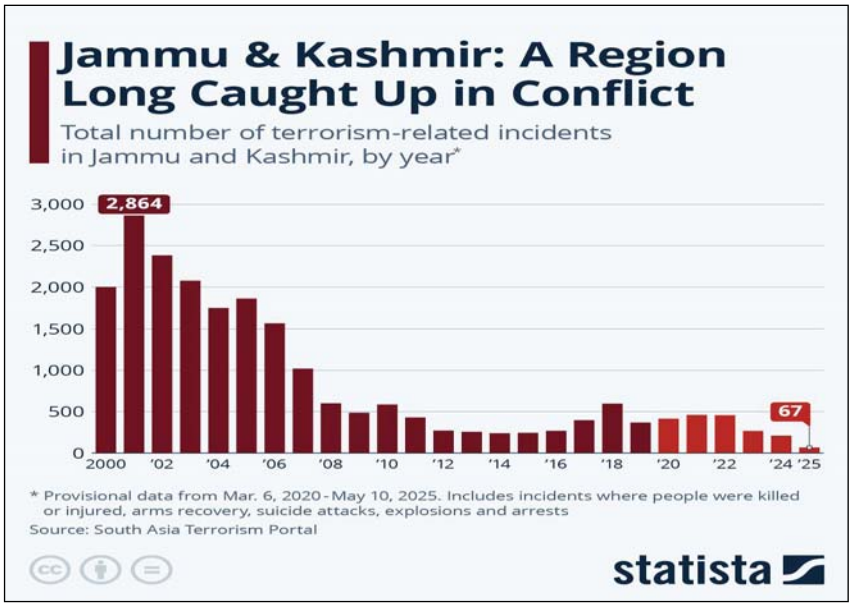
Introduction

Jammu & Kashmir militancy refers to the **ongoing armed insurgency and terrorist violence** in Jammu & Kashmir aimed at challenging Indian sovereignty over the region, either by **seeking secession/independence or accession to Pakistan**. It combines local separatist militancy with Pakistan-supported cross-border jihadist groups and has both political and religious dimensions.

Key phases of J&K Militancy

Phase 1: The Rise (Late 1980s – Mid-1990s)	• **Indigenous Start:** Initially, the movement was dominated by groups advocating for independence (Azaadi) or merger with Pakistan. The **Jammu and Kashmir Liberation Front (JKLF)**, which fought for independence, was prominent. • **Escalation:** The period saw massive infiltration, targeted killings of government officials and security personnel, and the migration of the Kashmiri Pandit minority community from the Valley. • **Shift in Ideology:** Over time, the more nationalist JKLF was overshadowed by groups that were religiously conservative and heavily supported by Pakistan, such as the **Hizbul Mujahideen (HM)**.
Phase 2: The Foreign Jihadi Wave (Late 1990s – Early 2000s)	• **De-Indigenization:** As the native Kashmiri militant leadership was decimated by counter-insurgency operations, foreign fighters (mainly from Pakistan, Afghanistan, and other countries) began dominating the ranks. Groups like **Lashkar-e-Taiba (LeT)** and **Jaish-e-Mohammed (JeM)** became the primary actors. • **Major Incidents:** This phase was marked by high-profile attacks, including the Kargil War (1999) and the attack on the Indian Parliament in 2001, which brought India and Pakistan to the brink of war. • **Decline in Local Support:** The increasing brutality and hardline religious ideology of foreign groups led to a dip in local support for the armed struggle.

Phase 3: Post-2010 Resurgence and the 'New Militancy'	• **Shift to Homegrown Militancy:** A new wave of militancy emerged, primarily fueled by young, highly educated, and radicalized local Kashmiris, often through social media and online propaganda. • **Mass Public Support:** The killing of popular militant commander Burhan Wani in 2016 triggered widespread public uprisings and a culture of youth rushing to encounter sites to disrupt operations by security forces. • **Hybrid Terrorism:** The focus shifted from conventional attacks to **"hybrid terrorism,"** where local, often non-listed militants carry out targeted, small-scale attacks before disappearing back into the civilian population.



Factors behind the militancy in J&K

- **Political and Constitutional Grievances (The Internal Catalyst):** The primary driver of alienation and subsequent militancy stems from a breakdown of trust between the people of the Kashmir Valley and the Central Government of India.
  - **The Disputed Accession and Autonomy:** While J&K acceded to India in 1947, the subsequent agreement (Article 370) promised a high degree of internal autonomy. Over the decades, this autonomy was progressively eroded by the central government, leading to a strong feeling that the constitutional relationship was being diluted unilaterally.
  - **Political Disenfranchisement and Rigged Elections:** The most crucial trigger was the widespread perception of political manipulation, particularly the widely believed rigging of the **1987 State Assembly Elections**. Many young political activists who had faith in the democratic process (like those in the Muslim United Front) saw this event as definitive proof that legitimate political avenues were closed to them, pushing them toward armed insurgency.
  - **Lack of Responsive Governance:** Poor governance, corruption, and a lack of accountability in the local political system compounded the sense of betrayal. The absence of credible local leadership exacerbated the disconnect with the mainstream Indian political system.
  - **Human Rights Concerns:** The deployment of large numbers of security forces and the implementation of laws like the Armed Forces (Special Powers) Act (AFSPA) led to allegations of human rights abuses, arbitrary detentions, and extrajudicial killings. These incidents became powerful rallying points for militant recruitment.

- **Socio-Economic Pressures:**
  - **High Unemployment:** Despite high literacy rates, the region consistently suffered from high rates of educated unemployment, particularly among the youth in the Valley. The lack of viable economic opportunities made extremist narratives appealing.
  - **Stagnant Tourism and Industry:** Political instability and recurring violence severely damaged the vital tourism industry and prevented large-scale industrial investment, thus limiting employment prospects.
  - **Development Deficit:** Many areas felt neglected in terms of infrastructural development, contrasting sharply with perceived central spending on security and administrative apparatus.
- **External and Geopolitical Intervention (The Primary Enabler):**
  - **Pakistan's Active Role:** Pakistan views J&K as "unfinished business" from the 1947 Partition. Its intelligence agency, the Inter-Services Intelligence (ISI), began actively supporting, arming, funding, and training militant groups in the late 1980s. This policy is often termed **"proxy warfare."**
  - **The Afghan Factor (1980s):** The Soviet-Afghan War created a pool of battle-hardened, religiously motivated foreign fighters and an abundance of weapons. Following the Soviet withdrawal, these resources and fighters were diverted toward the Kashmir conflict, escalating the violence dramatically.
  - **Ideological Shift (Global Jihad):** The initial, more nationalist and pro-Independence groups (like the JKLF) were systematically marginalized and replaced by Pakistan-backed, religiously focused organizations like Lashkar-e-Taiba (LeT) and Jaish-e-Mohammed (JeM). This shift globalized the ideology, moving the conflict beyond local political grievances to a broader jihadist narrative.
  - **Propaganda and Radicalization:** Pakistan uses vast propaganda networks (both media and online) to fuel anti-India sentiment and radicalize Kashmiri youth, often exploiting historical and religious narratives to encourage armed recruitment.

Challenges in tackling the J&K militancy:

- **Porous Borders and Terrain:** Difficult LoC/IB (740 km LoC, 192 km IB in Jammu) with forests/mountains enables infiltration and arms smuggling despite fencing.
- **Hybrid Militancy & Support Networks:** Educated "hybrid militants," Over Ground Workers (OGWs), and locals aiding logistics/funding via encrypted apps (Telegram) sustain operations without direct combat.
- **Intelligence & Coordination Gaps:** Fragmented intel-sharing between Army, CRPF, J&K Police; militants use locals' phones and online radicalisation, evading detection.
- **Geographical Shift:** Militancy moved to Jammu's Pir Panjal forests (Poonch, Rajouri, Kathua); less dense security grid and redeployment to LAC/China border weaken coverage.
- **Radicalisation & Local Sympathy:** Online propaganda, unemployment, grievances fuel youth vulnerability; community distrust hampers HUMINT despite de-radicalisation efforts.
- **Pakistan's Proxy Role:** Cross-border arms (drones), training, info warfare by LeT/JeM proxies; hard to deter non-state actors.
- **Manpower & Community Issues:** Extensive ops require heavy troops; reviving Village Defence Guards (VDGs) faces past abuse allegations and communal tensions.

Measures taken to tackle the issue of J&K militancy

Way forward to resolve the issue of J&K Militancy

- **Enhanced Security Measures with Human Rights Focus:**
  - Continue robust counter-terror operations and intelligence sharing to disrupt militant networks, while strictly adhering to human rights to rebuild trust with locals.
  - Reform laws and ensure accountability to prevent excesses that fuel alienation.
- **Political Dialogue and Reconciliation:**
  - Resume meaningful dialogue with all stakeholders, including separatists willing to renounce violence and mainstream political actors, with guarantees for autonomy and participation.
  - Consider calibrated political reforms to address local aspirations without compromising national integrity.

- **Community Engagement and Empowerment:**
  - Empower grassroots governance, develop local leadership, and implement de-radicalisation programs targeting vulnerable youth and communities.
  - Revive Village Defence Guards and community policing with training and safeguards.
- **Socio-Economic Development and Opportunities:**
  - Accelerate employment generation, skill development, education, infrastructure, and healthcare, especially for youth to address root causes like poverty and alienation.
  - Promote cross-community economic integration and cultural dialogue.
- **Diplomatic and Regional Stability Initiatives:**
  - India should utilize international platforms like UNSC (e.g., highlighting role of Pakistan in cross-border terrorism), FATF and take support of like-minded countries (e.g., Saudi Arabia and UAE) to put pressure on Pakistan to curb non-state actors operating from its soil.
  - Strengthen regional cooperation for security and development.

Conclusion

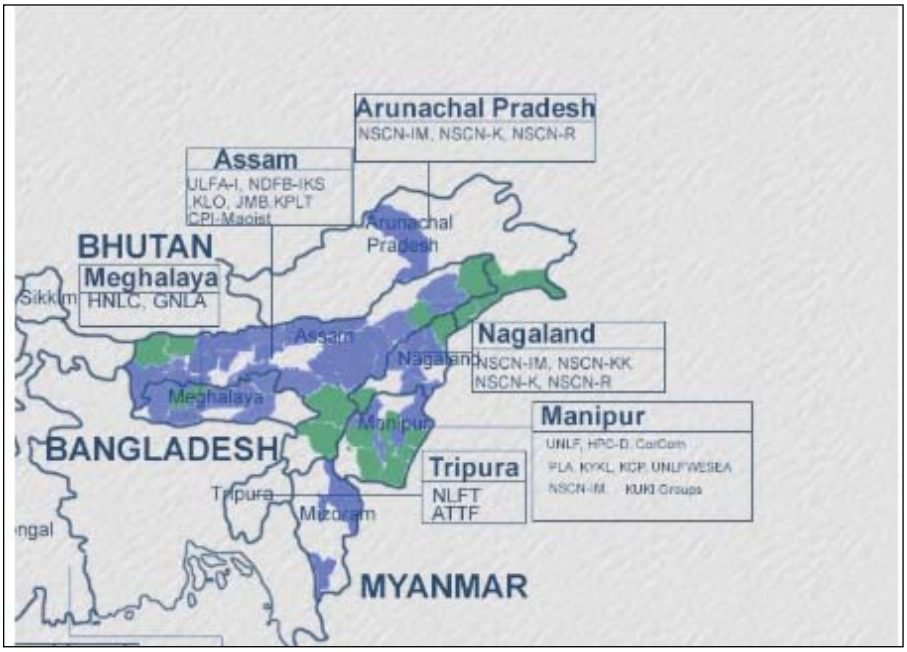
Achieving peace in J&K and tackling the militancy depends on synergizing the above mentioned dimensions through sustained political will, inclusive governance, and people-centered development.

6. NORTH EAST INSURGENCY

Manipur continues to remain one of the most volatile theatres of insurgency in India. In 2024 alone, the Northeast recorded **266 insurgency-related incidents**, of which **203 occurred in Manipur**, leading to **over 258 fatalities** and the **displacement of nearly 60,000 people**. This persistent cycle of violence underscores the deep-rooted ethnic tensions, governance challenges, and complex security dynamics that shape the region's instability.

North East Insurgency: Insurgency broadly means an internal armed rebellion by a section of citizens seeking secession, autonomy, or challenging state authority. The North East Insurgency refers to long-standing armed movements across India's northeastern states, driven by historical grievances, ethnic tensions, and socio-economic marginalisation.

Major Insurgent groups in North East



Causes of Insurgency in Northeast India

- **Colonial Causes**
  - **Colonial Administrative Policies which led to "Hill-valley identities"-** British measures of "restricted contact" such as the **Inner Line Regulation (1873) also** known as **Bengal Eastern Frontier Regulation (BEFR)** and the categorisation of **Excluded and Partially Excluded Areas,** resulted in psychological and administrative isolation of the North-East population.
    - This reinforced the "hill-valley" identities (hill communities were distinct and separate from the rest of India), giving rise to separatist movements in the hills.
  - **Indirect Rule and Limited Political Integration:** British rule largely followed indirect administration, giving tribal chiefs considerable autonomy. This preserved traditional systems but hindered political integration.
    - After Independence, the sudden shift to a centralised governance model created fears among hill communities of losing their autonomy and customary authority.
    - **Colonial ethnography hardened ethnic lines**: By fixing tribal identities into rigid categories, deepening divisions and later fuelling ethnic-based political mobilisation and insurgent movements.
- **Post-Independence Drivers of Insurgency**
  - **First Insurgency and Sovereignty Demand:** In Nagaland, the **Naga National Council (NNC**) rejected integration with India, demanding an independent homeland; the state's refusal triggered the earliest and longest-running insurgency, spawning splinter groups like **NSCN (IM)** and **NSCN (K)**, inspiring similar movements across the Northeast.
  - **Political and economic marginalisation:** It left tribal communities feeling neglected, with **poor infrastructure, weak representation, and limited development**, fostering **alienation** and creating fertile ground for **insurgent mobilisation**.
  - **Identity Insecurity and Ethnic Assertion:** Groups like **Bodos, Kukis, Meiteis, Karbis, and Dimasas** feared loss of land, culture, and authority, driving militant nationalism with demands for autonomy, new states, and constitutional safeguards.
  - **Territorial Overlaps and Ethnic Rivalries:** Overlapping ancestral claims from historical migrations fueled violent rivalries such as **Naga- Kuki, Meitei- Kuki, Bodo- Muslim, and Dimasa- Karbi**, with insurgents justifying violence on the basis of historical entitlement and ethnic survival.
  - **Land Alienation and Cultural Erosion:** Migration from plains and neighbouring regions raised fears of land loss and demographic imbalance, sparking movements like the **Assam Agitation** and strengthening insurgent calls to protect indigenous land and identity.
- **Inter-Ethnic Competition and Territorial Aspirations**
  - **Overlapping territorial claims** such as **Greater Nagalim, Kuki statehood, and Bodo statehood** create inter-community friction, while **ethnic clashes** like **Naga- Kuki, Meitei- Kuki, and Bodo-Muslim** deepen **mistrust** and strengthen **insurgent mobilisation**.
- **Political Marginalization and Governance Deficit**
  - **Weak state presence**, coupled with **distrust of central policies** and issues like **corruption, patronage, and poor service delivery**, creates a **governance vacuum** that insurgent groups exploit to run **parallel administrations** and assert de facto authority.
  - **Socio-Economic Underdevelopment and Infrastructure Deficit**
  - **Economic underdevelopment**, marked by **poor infrastructure, low industrialisation, limited jobs, and weak market integration**, drives youth into insurgency, with militancy becoming an **economic enterprise** through **extortion, smuggling, and illegal taxation**, reinforcing feelings of exclusion from India's development.

- **Illegal Migration and Demographic Changes**
  - **Challenging terrain** of mountains and forests, combined with **migration-driven demographic changes** from Bangladesh, fosters **identity-based tensions** and fears of marginalisation which insurgent groups exploit to strengthen narratives of **outsiders versus indigenous communities**.
- **External Linkages and Cross-Border Challenges**
  - **Porous international borders**, with 90% of the region adjoining **Myanmar, Bangladesh, and the Golden Triangle**, provide insurgents with **safe havens, training grounds, and escape routes**, facilitate **arms, drugs, and illegal trade**, and have at times been exploited by **foreign intelligence agencies** for strategic purposes.
- **Security-Related Causes:**
  - Many insurgent groups have shifted from ideological goals to **criminal activities** like **drug trafficking, arms smuggling, extortion, kidnapping, and illegal trade**, creating **self-sustaining finances** that make militancy difficult to resolve through negotiations alone.
- **Legal & Human Rights-Related Causes**
  - While **AFSPA** aids counterinsurgency, its **excessive powers and human rights concerns** deepen civilian alienation; despite partial withdrawals, **mistrust persists**, and insurgents exploit these grievances to **mobilise support against the state**.
- **Proliferation of Small Arms and Criminal Networks**
  - **Easy access to small arms** from **Myanmar and Southeast Asia**, combined with **smuggling networks** and links to **drug trafficking, extortion, and illegal taxation**, sustains insurgent operations and **prolongs violence** even after political grievances are addressed.
- **Slow Peace Processes and Fragmentation of Groups**
  - **Internal divisions and multiple splinter factions** within insurgent groups, like **NSCN and NDFB offshoots**, coupled with **overlapping demands and shifting political alignments**, make peace negotiations slow and allow **sporadic violence** to continue, hindering lasting conflict resolution.

Significance of the North East for India

- **Geo-strategic Gateway:** The Northeast linking India to ASEAN and BIMSTEC markets, enhances India's Act East Policy and bolsters regional connectivity.
  - **Example:** India-Myanmar-Thailand Trilateral Highway; Sittwe Port, Myanmar.
- **Renewable Energy Powerhouse:** With over **218 GW** of solar, hydro, wind, and biomass potential, the region can drive **India's green transition**.
- It holds nearly 40% of India's hydropower capacity, ideal for clean energy investments.
  - **Example:** Champhai Solar Park, Mizoram; 20 MW Solar Project, Mizoram.
- **Cultural Capital:** Home to 200+ ethnic groups, the Northeast fosters **cultural diplomacy** and strengthens India's soft power.
  - **Example: Hornbill Festival**, Nagaland; **Ziro Music Festival**, Arunachal Pradesh.
- **Biodiversity Hotspot:** The Northeast is a **biodiversity hotspot**, rich in **endemic flora and fauna**, and vital for **conservation and eco-tourism**.
  - **Example: Kaziranga National Park** (one-horned rhinos); Loktak Lake, Manipur.
- **Agricultural and Organic Hub:** Diverse agro-climatic zones make it ideal for organic farming, horticulture, and floriculture.
  - **Example: Sikkim's organic farming model**; Assam's tea plantations.
- **Human Capital Potential:** With high literacy, English fluency, and a young population, the region is ripe for skilling and innovation.

  - **Example:** IIT Guwahati graduates; 5G Telemedicine in Arunachal Pradesh.
- **Tourism and Wellness Destination:** Its scenic beauty, spiritual sites, and natural resources support eco-tourism and Ayurveda-based wellness.
  - **Example: Living Root Bridges**, Meghalaya; **Loktak Lake**, Manipur.
- **Industrial and Connectivity Boost:** Improved roads, digital networks, and manufacturing hubs are transforming the region's economy. New-age industries like semiconductors and bamboo processing are emerging.
  - **Example: First semiconductor plant in Assam**; Kaladan Multi-Modal Project.

Government Initiatives to Address Insurgency in Northeast India

- **Proportionate & Calibrated Use of Force**
  - **Coordinated counter-insurgency operations:** Regional security operations, including **Bhutan's Operation All Clear (2003)** against ULFA, NDFB, and KLO, **India–Myanmar's Operation Sunrise**, and post-2009 **cooperation with Bangladesh** to dismantle insurgent sanctuaries, have significantly targeted Northeast militant camps.
  - **Strengthening security architecture:** Increased deployment of **Army, Assam Rifles, and CAPFs**, strengthened border posts and intelligence networks, and the modernization of police and counter-insurgency infrastructure have bolstered the region's internal security.
  - **Humanitarian & rehabilitative approach: Surrender-cum-rehabilitation schemes** provide financial support, training, and counselling, emphasizing **reintegration of former militants** over punitive measures.
  - **Rationalization of AFSPA:** Progressive withdrawal from major districts of Assam, Manipur and Nagaland based on improved law and order.
- **Dialogue, Peace Accords & Political Negotiations**
  - **Major peace accords**
  - **Mizo Accord (1986):** Ended insurgency; creation of Mizoram; MNF into politics.
  - **Assam Accord (1985):** Addressed anti-immigration concerns; identity protections.
  - **Bodo Accord (2020):** Integration of NDFB factions; creation of BTR.
  - **On-going dialogues**
    - NSCN-IM Framework Agreement (2015) acknowledging unique Naga identity.
    - Ceasefires with NSCN factions and other groups.
  - **Recent agreements**
    - Karbi Anglong Peace Accord (2021) mainstreaming Karbi groups.
    - Bru Repatriation Agreement (2020) resolving two-decade displacement issue.
- **Constitutional, Administrative & Structural Reforms**
  - **State restructuring for ethnic aspirations:** Creation of Nagaland (1963), Meghalaya (1972), Mizoram (1987), Arunachal Pradesh (1987).
  - **Autonomy and tribal safeguards:** Autonomy safeguards include **Sixth Schedule District Councils** with legislative and financial powers, and **Articles 371(A) and 371(G)** protecting **customs, land, and resources** in Nagaland and Mizoram.
  - **Administrative empowerment: Increased fund devolution** to tribal districts and **strengthened grassroots institutions** empower local decision-making.
- **Development & Infrastructure Push (MDoNER-Led)**
  - **Institutional Mechanisms: Ministry of DoNER** coordinates centrally, **NEC** drives regional planning, and **NESIDS** addresses infrastructure gaps in **roads, water, energy, and health**.

- **Connectivity & Mega Projects:** Key projects include **Kaladan Multi-Modal Transit**, **Bharatmala**, **Trans-Arunachal Highway**, **Bogibeel** and **Dhola–Sadiya Bridges**, and **UDAN** schemes enhancing air links.
- **Economic & Industrial Initiatives: North East Industrial Corridor (Dawki–Moreh)**, major **Japanese investments** in highways, tunnels, and waterways boost industrial growth.
- **Livelihood Generation:** Programs like **NERLP** support **SHGs, rural value chains, and women-led enterprises**, while **TRIFED and** tribal craft initiatives promote income diversification.
- **Socio-Political & Community Integration Measures**
- **Youth-Centric Policies:** Development of **sports infrastructure, entrepreneurship schemes, and PMKVY** skill centres, along with **cultural exchange** and exposure programmes to promote integration.
- **Rehabilitation & Reintegration:** Provision of **education, training, counselling, and livelihood grants** for ex-militants, supported by community-led acceptance programmes to prevent re-radicalisation.
- **Democratic Participation:** Strengthening **village councils, municipal bodies, and tribal councils**, while encouraging **civil society engagement and community policing**.

- **Managing Migration, NRC & Citizenship-Related Issues**
  - **NRC as identity-security mechanism:** Curbs illegal migration, eases demographic anxieties, and strengthens border management and citizenship clarity.
  - **Border management reforms: Smart fencing, riverine patrols, and floodlights**, along with **border haats**, secure borders and regulate cross-border movement.
  - **Debates on CAA: Balancing humanitarian needs with ethnic and cultural protection** through ongoing stakeholder consultations to maintain harmony.

Challenges in Resolving Insurgency in the Northeast

- **Fragmentation of Insurgent Groups:** Major groups like **NSCN, NDFB, and ULFA** have split into multiple factions with differing demands, complicating comprehensive peace agreements.
- **Slow and Complex Peace Processes**
  - **Prolonged negotiations** (e.g., Naga peace talks ongoing since 1997)**, frequent leadership changes, and lack of a single interlocutor** create fatigue, mistrust, and fragile, contested agreements in insurgency talks.
- **Inter-Ethnic Rivalries & Competing Territorial Claims**
  - Many conflicts are **inter-ethnic**, such as **Naga–Kuki, Meitei–Kuki, Karbi–Dimasa, and Bodo–Muslim**, where **overlapping territorial claims** turn one group's settlement into another's grievance, triggering **fresh mobilization and insecurity**.
- **Drug Trafficking & Arms Smuggling Nexus:** Proximity to the **Golden Triangle** fuels drug trade, illegal arms from **Myanmar–Thailand** routes sustain militancy.

- **Governance Deficits in Remote & Tribal Areas:** Challenging terrain, poor infrastructure, and limited state presence allow insurgent groups to fill governance voids through **parallel administrations, taxation, and dispute resolution.**
- **Border Management Constraints: Porous borders** covering 90% of the region, with forests and cross-border ethnic populations like **Nagas, Kukis, and Mizos**, provide insurgents **sanctuaries in Myanmar and previously Bangladesh**, complicating enforcement.
- **Persistence of Alienation, Mistrust & Historical Grievances: Colonial legacies, AFSPA grievances, and socio-cultural marginalization**, combined with perceptions of **mainland apathy** and fear of **demographic dilution**, undermine trust and cause periodic relapses into violence.
- **Infrastructure Deficits:** Despite increased investments, the Northeast faces **inadequate roads, railways, healthcare, and education**, limiting economic growth
- **Limited Industrial Base and Unemployment:** Drives **high unemployment and youth migration**, undermining local economies.
- **Connectivity Gaps: Rugged terrain** and **complex geography** delay infrastructure projects, limiting intra-regional mobility and market access.
- **Environmental and Climate Vulnerabilities**: Frequent natural disasters like floods, earthquakes, and landslides threaten infrastructure and fragile ecosystems.
  - **Example: 2022 Assam floods** causing displacement and crop loss.
- **Resource Underutilization:** Vast renewable and mineral resources remain untapped due to policy gaps, poor access, and low private participation.
  - **Example:** Less than 7% of 129 GW hydro potential harnessed.
- **Financial and Skill Inclusion Gaps:** Limited credit access and skill mismatch restrict entrepreneurship and economic empowerment, especially in rural areas.

Way Forward

- **Comprehensive Border Management:** Smart fencing, drones, surveillance, integrated check-posts.
- **Accelerated Connectivity:** Highways, railways, telecom, waterways.
- **Inclusive Development:** Focus on education, health, livelihoods, youth skill development.
- **Cultural Sensitivity in Governance:** Local languages, institutions, customary laws.
- **Dialogue-Based Approach:** Bring all factions into peace talks.
- **Cooperative Federalism:** Stronger Centre-State coordination.
- **International Cooperation:** Myanmar and Bangladesh for dismantling terror sanctuaries.
- **Rehabilitation of Cadres:** Skill programmes, employment, reintegration.
- **Human Rights Balance:** Reforms in AFSPA implementation with accountability.
- **Integrated Infrastructure and Connectivity:** Fast-track multimodal connectivity via road, rail, air, and waterways to link the NER with national and ASEAN trade corridors.
  - **Example:** Completion of Trans-Arunachal Highway; operationalization of ICP Moreh on India-Myanmar border.
- **Industrial Promotion and Investment Facilitation:** Develop Special Economic Zones (SEZs), agro-processing clusters, and one-stop investor portals to boost employment and regional output.
  - **Example: First semiconductor plant in Assam,** Agri-processing SEZ in Assam; DoNER's Invest North East portal.
- **Peacebuilding and Border Management:** Sustain dialogue with insurgent groups, enhance border policing, and resolve disputes to foster a secure and cooperative environment.
  - **Example:** Ongoing Assam-Meghalaya border settlement talks; enhanced deployment near Indo-Myanmar border.
- **Environmental Sustainability and Climate Resilience:** Implement sustainable development policies, promote green energy, and build disaster-resilient infrastructure to safeguard fragile ecosystems.
  - **Example:** Community forest conservation in Arunachal Pradesh; World Bank-funded resilience infrastructure.

- **Skill Development and Human Capital Formation:** Align training with regional industries, expand vocational institutes, and bridge skilling gaps for youth employment.
  - **Example: NER Skill Plan 2022**; handloom training centers in Nagaland.
- **Tourism and Cultural Diplomacy:** Develop eco- and cultural tourism circuits and promote festivals to generate revenue and build soft power.
  - **Example:** Living root bridge circuit in Meghalaya; Hornbill Festival in Nagaland.
- **Digital and Financial Inclusion:** Expand broadband under BharatNet, promote e-governance, and improve credit access and digital banking in rural zones.
  - **Example:** Digital land records in Tripura; BHASHINI real-time translation tool.
- **Healthcare and Educational Transformation:** Upgrade medical facilities through telemedicine and enhance higher education with regional research institutes.
  - **Example: Telehealth services** in remote Arunachal; **North East Regional Institute of Education.**

Conclusion:

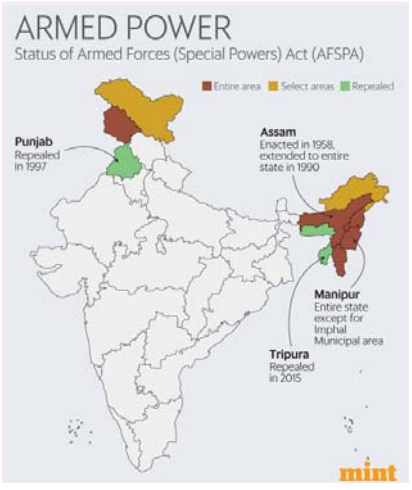
Insurgency in the Northeast stems from layered historical, socio- economic and governance challenges demanding inclusive dialogue, equitable development and effective border management. A holistic sensitive approach is essential to secure lasting peace. With its geostrategic location and rich resources, the NER can emerge as a dynamic hub of growth, connectivity, and cultural vitality strengthening India's overall development trajectory.

7. ARMED FORCES (SPECIAL POWERS) ACT (AFSPA)

Introduction

- AFSPA is a central law that gives special powers to the armed forces in areas officially declared as "disturbed" to help them tackle insurgency and serious internal security threats.
- The primary objective of AFSPA is to provide the armed forces (including the Army, Air Force, and paramilitary forces) with special legal immunity and powers necessary to control internal conflict, militancy, and insurgency in regions where the civil administration and police are deemed incapable of maintaining law and order.
- AFSPA was first promulgated in 1942, by Linlithgow, in response to the Quit India movement in 1942. Its aim was "to confer special powers upon certain officers of the armed forces.
- After Independence, the Act was retained by the ordinance enacted in 1958, to control increasing violence in the North-eastern States, which the State governments found difficult to control. In simple terms, AFSPA gives armed forces the power to maintain public order in "disturbed areas".

Status of AFSPA



Important Provisions of AFSPA

- **Declaration of "disturbed area" (Section 3):**
  - The Governor of a state, the Administrator of a Union Territory, or the Central Government can declare any area "disturbed" if the situation requires deployment of the armed forces to aid civil power.
  - Once so notified, the special powers under AFSPA apply in that area.
- **Special powers of the armed forces (Section 4):** In a notified disturbed area, any commissioned, warrant, non-commissioned officer or equivalent rank can:
  - Use force, even to the extent of causing death, after giving due warning, against persons:
    - acting in contravention of law prohibiting assembly of five or more persons, or
    - carrying weapons or things capable of being used as weapons or explosives.
  - Destroy arms dumps, fortified positions, shelters or training camps from which armed attacks are made or likely to be made.
  - Arrest without warrant anyone who has committed, is suspected of having committed, or is about to commit a cognisable offence, using necessary force.
  - Enter and search any premises without warrant to make such arrests or recover arms, ammunition, or explosives.
- **Duties after arrest (Section 5):** Any person arrested and any seized property must be handed over to the nearest police station "with the least possible delay," so that normal criminal procedure can then follow.
- **Legal protection / immunity (Section 6):**
  - No prosecution, suit or other legal proceeding can be instituted against any person acting under AFSPA, except with the prior sanction of the Central Government.
  - This clause is intended to protect personnel acting in good faith during operations, but is also the main focus of human-rights criticism.

Significance/Need for AFSPA

- **Enables deployment in high-risk "disturbed areas":** AFSPA allows the Centre or Governor to declare an area "disturbed," enabling the armed forces to aid civil authorities where police and local administration cannot control violence or insurgency.
- **Provides operational flexibility to forces:** Powers to use force after warning, arrest without warrant, and conduct searches and seizures allow the army/paramilitary to react quickly and pre-empt threats such as ambushes, armed mobs, or militant movements.
- **Strengthens state capacity in conflict zones:** In border and remote regions with difficult terrain and weak policing, AFSPA is seen as necessary to secure roads, infrastructure, elections, and civilian life against insurgent or terrorist groups.
- **Legal protection for personnel:** Requirement of prior central sanction for prosecution under AFSPA is argued to protect soldiers from vexatious litigation and allow them to act decisively in life-threatening situations.
- **National security and territorial integrity:** The Act is justified by governments as essential to preserve sovereignty and integrity in regions facing secessionist, insurgent or cross-border militant movements.

Criticisms of AFSPA

- **Human rights violations:**
  - Wide powers to shoot, arrest without warrant, and search are seen as enabling extrajudicial killings, torture, enforced disappearances, rape, and harassment of civilians in states like Manipur, Nagaland, J&K and others.
  - Human rights groups, UN experts and civil society describe AFSPA as "draconian" and incompatible with international human-rights norms.

- **Legal immunity and lack of accountability:**
  - Section 6 requires prior central government sanction before prosecuting armed forces personnel, which is rarely granted, leading to near-impunity even in serious abuse cases.
  - This undermines rule of law and victims' access to justice, and is seen as violating fundamental rights under Articles 14 and 21.
- **Violation of fundamental rights and militarisation:**
  - Critics say AFSPA effectively suspends normal constitutional protections in "disturbed areas," normalising prolonged militarisation and eroding civil liberties such as the right to life, liberty, movement and expression.
  - Long-term deployment of the army for internal policing is argued to blur civil–military boundaries and weaken civilian institutions.
- **Misuse, excess force and distrust:**
  - Supreme Court judgments have noted that any use of excessive or retaliatory force is illegal even under AFSPA, but multiple inquiries have documented fake encounters and misuse, deepening local resentment and distrust of the state.
  - Critics argue that such practices fuel alienation and may actually worsen insurgency instead of resolving it.
- **Committee recommendations for repeal/reform:**
  - The Justice B.P. The Jeevan Reddy Committee and the Second Administrative Reforms Commission both recommended repeal of AFSPA, with key provisions to be shifted into a more rights-compliant law like the Unlawful Activities (Prevention) Act.
  - Other bodies, like the Santosh Hegde Commission and Justice Verma Committee, have called for strong review, tighter oversight and removal of blanket immunity.

Various SC Judgements & Committees Recommendations on AFSPA

- **Supreme Court Judgements on AFSPA:**
- A five-judge bench upheld the constitutional validity of AFSPA but clarified that its powers are not absolute. It restricted Section 4 to "minimal force for effective action," referencing army guidelines, and emphasized that excessive force is illegal even under AFSPA.
- **Extra Judicial Execution Victim Families Association (EEVFAM) v. Union of India (2016):** The Supreme Court ruled that every encounter death must be investigated by the state police, regardless of AFSPA. It held that excessive or retaliatory force is unlawful, even in disturbed areas, and consolidated prior jurisprudence on AFSPA's limits.
- **Naga People's Movement case follow-up and others (various):** SC has repeatedly stressed that AFSPA does not grant a "licence to kill" and that prior central sanction under Section 6 is mandatory for prosecution but not a blanket immunity.

Key Committees and Recommendations

- **Justice B.P. Jeevan Reddy Committee (2004):**
  - Repeal AFSPA entirely; shift powers to Unlawful Activities (Prevention) Act (UAPA).
  - Set up district grievance cells; limit army deployment to 6 months with parliamentary review.
- **Second Administrative Reforms Commission (2nd ARC) (2007):** Repeal AFSPA after wide consultations; replace it with humane legislation balancing security and rights.
- **Santosh Hegde Commission (2013):** Investigate fake encounters in Manipur; regular review of AFSPA (not full repeal); called it a "symbol of oppression."

- **Justice J.S. Verma Committee (2013):**
  - Review AFSPA urgently, especially for sexual offences by armed forces.
  - Try such cases under ordinary criminal law, not military courts.

Way Forward

- **Phased Withdrawal and Review:**
  - Withdraw AFSPA from areas where normalcy has returned (e.g., parts of J&K, Northeast) after security assessments, as done in Tripura (2015) and Meghalaya (2018).
  - Mandate periodic 6-month reviews by Parliament or an independent body, with automatic sunset clauses for notifications.

Approach	Pros	Cons
Full repeal	Ends impunity, restores rights	May weaken counter-insurgency in active conflict zones.
Targeted Reform	Balances security & rights.	Requires political will, implementation capacity.
Phased Withdrawal	Builds trust, tests normalcy.	Risk of resurgence if premature.

- **Reforms to Existing Law:**
  - Amend Section 6 to make prior sanction faster and more transparent (e.g., time-bound decisions, judicial oversight), and allow prosecution for grave human rights violations without sanction.
  - Strengthen Section 5 to ensure arrested persons are handed to police within 24 hours, with mandatory video recording of operations.
  - Incorporate Supreme Court guidelines: mandatory FIRs and investigations for all deaths in encounters, independent magisterial inquiries.
- **Institutional Strengthening:**
  - Create a National Human Rights Commission (NHRC)-monitored oversight mechanism for AFSPA operations, with local grievance cells as recommended by the Jeevan Reddy Committee.
  - Enhance training for armed forces on human rights, minimal force, and distinction between combatants and civilians.
- **Long-term Alternatives:**
  - Shift non-combat powers (arrest, search) to police and paramilitary under UAPA or CrPC, reserving army for active combat only.
  - Invest in development, policing capacity, and peace-building in disturbed areas to reduce need for military deployment.

Conclusion

A pragmatic path involves judicial enforcement of existing safeguards, committee-inspired reforms, and gradual handover to civilian policing, aligning with Supreme Court directives and 2nd ARC recommendations.

8. TERRORISM IN INDIA

Recently, it was reported that **ISI-backed terror group Shahzad Bhatti Network** was destroyed ahead of Independence Day and thwarted its plans for subversive attacks by arresting more than 200 operatives

Introduction

Terrorism in India is a serious internal security challenge that threatens **life, national unity and social harmony**. It is driven by factors such as **extremism, cross-border support, radicalisation and socio-economic grievances**. Tackling terrorism therefore requires not only strong security measures but also **intelligence, development, community participation and effective governance**.

Section 113(1) defines a terrorist act as any action, involving dangerous weapons, substances, or materials, that threatens national security or public safety by causing death, destruction, or disruption to essential services.

Types of Terrorism in India

Type	Explanation
Cross-Border Terrorism	Terrorism where support, training, funding or direction comes from across India's borders.
Left-Wing Extremism (LWE)	Violence carried out by extremist groups claiming to fight for the rights of the poor and marginalised.
Separatist Terrorism	Violence aimed at creating a separate nation or breaking away from India.
Religious / Ideological Terrorism	Violence justified in the name of a particular religious or extreme ideology.
Ethno-Nationalist Terrorism	Violence based on ethnic, linguistic or cultural identity, often demanding greater autonomy or a separate homeland.
Narco-Terrorism	Use of illegal drug trade to finance terrorism or use of terror networks to facilitate drug trafficking.
Cyber Terrorism	Use of digital technologies to attack critical systems or create widespread fear and disruption.
State-Sponsored Terrorism	When a state provides funds, weapons, training or safe havens to terrorist organisations operating against another country.
Urban Terrorism	Terrorist activities carried out in cities to create maximum fear, casualties and disruption.
Lone-Wolf Terrorism	An individual carries out a terrorist attack without being directly controlled by a formal terrorist organisation.

Do you know?

In the **Global Terrorism Index (GTI) 2026**, released by the **Institute for Economics & Peace (IEP)**, India is ranked **13th** globally with an overall score of **6.428**.

Causes of Terrorism

- **Political Causes**
  - **Political exclusion:** When people feel that their concerns are ignored and they have no meaningful political voice.

    Widespread perception of rigged assembly elections in 1987 alienated local youth, directly fueling alienation and triggering armed militancy in the Kashmir Valley.
  - **Demand for autonomy:** Some groups use violence when they want a separate state, greater autonomy or political rights.

    Armed groups like the NSCN-IM (Nagaland) and ULFA (Assam) resorted to violence demanding sovereign homeland status (Nagalim) or complete secession from India.
  - **Weak governance:** Corruption, poor administration and lack of justice can create anger against the government.

Administrative vacuums and corrupt local bureaucracy in remote tribal belts of Chhattisgarh and Jharkhand allowed Maoists to establish parallel "Jan Adalats" (people's courts).

- **Economic Causes**
  - **Unemployment:** Lack of jobs, especially among young people, can increase frustration and make extremist recruitment easier.
  - **Poverty:** Extreme poverty can make vulnerable people more dependent on extremist groups for money or support.

    Severe poverty and lack of basic livelihood in the Dandakaranya forest region made local tribal populations dependent on Left-Wing Extremist outfits for basic survival.
  - **Economic injustice:** Perceived unfair distribution of land, resources and opportunities can create resentment.
- **Social Causes**
  - **Social exclusion:** People who feel isolated or discriminated against may become more vulnerable to extremist influence.
  - **Identity crisis:** Young people who feel disconnected from their community or society may search for a stronger identity.

    Hundreds of young Muslims from Western nations who felt disconnected from both their parents' traditional cultures and Western society joined ISIS to find a sense of belonging
  - **Community tensions:** Long-standing ethnic, communal or social tensions can be exploited by terrorist organisations.

    Decades of ethnic tension and violence gave rise to the Arakan Rohingya Salvation Army (ARSA).
- **Ideological Causes**
  - **Extremist ideology:** Radical ideas can convince people that violence is an acceptable way to achieve political goals.

    Outfits like Al-Qaeda and ISIS use radical interpretations to justify global terror acts and attack democratic values.
  - **Ethnic extremism:** Extreme ethnic or nationalist thinking can create hostility towards other communities.

    Groups like the NDFB (National Democratic Front of Bodoland) used violent ethnic nationalism to target non-Bodo communities in Assam.
- **Historical Causes**
  - **Historical grievances:** Memories of past discrimination, violence or displacement can be used to mobilise people.

    Operation Blue Star (1984): Exploited by Khalistani extremist networks operating in Canada, the UK, and Europe to keep anti-India sentiment and funding alive for decades.
  - **Unresolved injustice:** Long-pending grievances can be exploited by extremist groups to gain support.
- **External Causes**
  - **Cross-border infiltration:** Porous borders can make the movement of terrorists, weapons and illegal funds easier.

    Pakistan-based terror launchpads like Lashkar-e-Taiba and Jaish-e-Mohammed push armed militants into Jammu & Kashmir.
  - **Regional instability:** Wars and political instability in neighbouring countries can provide safe spaces for terrorist groups.

The Taliban's takeover of Afghanistan created a safe space for terror groups like the TTP (Tehrik-i-Taliban Pakistan) and ISKP (Islamic State – Khorasan Province).

- **International networks:** Terrorist organisations can use global networks for funding, recruitment and propaganda.

- **Technology and Online Radicalisation**
  - **Online recruitment:** Use of encrypted messaging platforms like Telegram and dark web forums to recruit urban Indian youth (e.g., the ISIS **Kalyan Module** or **Ranchi Module**).
  - **Rapid spread of propaganda:** Terror groups releasing AI-generated radical videos, digital magazines (Dabiq, Voice of Khorasan), and social media campaigns to incite lone-wolf attacks across the world.

Impacts of Terrorism

- **Human Impact**
  - **Loss of lives:** Terrorist attacks cause the death of innocent people, security personnel and public officials.

    As per MHA, Over 21,000 civilians and 6,000+ security personnel have lost their lives to terrorism and insurgency in India over the last three decades.
  - **Mental trauma:** Repeated violence creates fear, anxiety and a sense of insecurity among people.
- **Economic Impact**
  - **Damage to property:** Attacks can destroy roads, bridges, public buildings, businesses and other infrastructure.
  - **Loss of investment:** Continuous insecurity can discourage private investment and tourism.

    As per J&K Tourism dept, A drop in terror incidents post-2019 led to a record-breaking 20+ million tourist arrivals annually
  - **Higher security costs:** Governments have to spend more on policing, intelligence, border security and counter-terrorism.

    Ministry of Home Affairs allocates substantial budget shares of over ₹2 Lakh Crore toward central armed police forces, border fencing and high-tech intelligence machinery.
- **Social Impact**
  - **Communal tensions:** Attacks may create anger against entire communities, even when they are not responsible.
  - **Disruption of daily life:** Schools, markets, transport and other normal activities may be affected.

    Frequent hartals (strike calls) by separatist groups in the Kashmir Valley historically resulted in 100+ lost working/school days per year in last decade.
- **Political Impact**
  - **Weakens democracy:** Fear and violence can reduce people's confidence in democratic institutions.

    Militant groups in J&K and LWE belts historically issued death threats to voters and targeted local Panch/Sarpanch leaders.
  - **Pressure on civil liberties:** Governments may introduce stronger security measures, sometimes creating concerns about privacy and individual freedoms.
- **Impact on Development**
  - **Slows development:** Government resources may shift from development programmes towards security expenditure.

- **Regional backwardness:** Conflict-affected areas may struggle to attract businesses, infrastructure and skilled workers.

  Districts like Sukma, Dantewada, and Bijapur suffered decades of development lag due to contractors and civil workers refusing to operate in active Maoist red zones.

- **Migration:** People may move away from affected regions in search of safety and better opportunities.

  Mass violence and targeted threats in 1989–1990 led to the internal displacement of over 3,00000 Kashmiri Pandits from the Valley.

- **Impact on National Security**
  - **Border security challenges:** Cross-border terrorism can increase pressure on border forces and intelligence agencies.
  - **Growth of organised crime:** Terrorist groups may use illegal networks for weapons, drugs and money.

    D-Company and Pakistan-based handlers use drug smuggling networks across the Punjab border (heroin trafficking) to fund local terror cells (Golden Crescent route).

  - **Radicalisation:** Continued terrorist propaganda can create new recruits and prolong the cycle of violence.
- **International Impact**
  - **Damages India's image:** Frequent terrorist incidents can affect perceptions of India's safety and stability.
  - **Strains diplomatic relations:** Cross-border terrorism can create tensions between neighbouring countries.

    Major cross-border attacks (Uri 2016, Pulwama 2019) led to the suspension of bilateral talks, trade closure, and military standoffs between India and Pakistan.

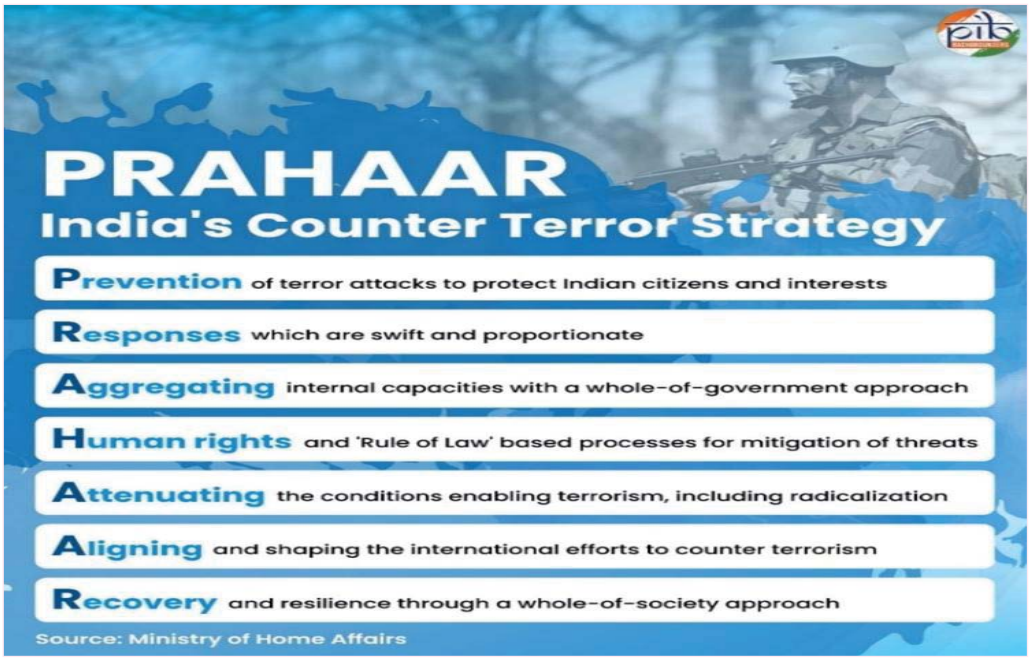
Major Counter-Terrorism & Security Agencies in India and Their Roles

Measures taken by India to tackle terrorism

- **Strong Legal Measures**
  - **UAPA Law:** The Unlawful Activities (Prevention) Act allows the government to ban terror groups, freeze their funds, and label individuals as terrorists.

- **New Criminal Law (BNS):** The Bharatiya Nyaya Sanhita clearly defines a "terrorist act" for the first time in ordinary criminal law, making prosecution easier.
- **Strict Anti-Money Laundering Laws:** The Prevention of Money Laundering Act allows authorities to seize properties and bank accounts linked to terror networks.

- **Measures to Stop Terror Funding**
  - **Tracking Financial Transactions:** The Financial Intelligence Unit (FIU-IND) monitors suspicious bank transfers to catch terror money flows early.
  - **Stopping Fake Currency:** Dedicated police units track and destroy counterfeit currency networks used to fund illegal activities.
- **Technology & Cyber Security**
  - **Cyber Crime Coordination Centre (I4C):** Monitors radical online content, blocks extremist channels, and coordinates cyber safety responses.
  - **Anti-Drone Technology:** Security forces deploy specialized radar and jammer systems along borders to stop drones from dropping weapons or money.
  - **High-Tech Surveillance:** Agencies like the National Technical Research Organisation (NTRO) use satellites and advanced electronics to track threat activities.
- **Community Engagement & Stopping Radicalization**
  - **Counter-Radicalization Division:** A specialized unit under the Home Ministry focuses specifically on blocking online propaganda and extremist recruitment.
  - **Regional Development:** Expanding roads, telecom networks, schools, and employment schemes in vulnerable areas to prevent local youth from joining extremist groups.
- **International Cooperation & Diplomacy**
  - **Pushing for a Global Definition:** India actively petitions the United Nations to pass the Comprehensive Convention on International Terrorism (CCIT) to define terrorism globally.
  - **Global Police Alliances:** Working closely with Interpol, G20 alliances, and partner nations to share intelligence and extradite hiding terrorists.
  - **Hosting Global Coalitions:** Hosting international platforms like the "No Money for Terror" conference to build worldwide consensus against state-sponsored terror financing.



Way Forward to Tackle Terrorism

- **Strengthen Intelligence and Security**
  - **Stronger policing:** State police should be given better training, modern equipment and adequate manpower.

    State-level specialized elite units like Maharashtra ATS, UP ATS, and Telangana's OCTOPUS trained specifically for urban counter-terror operations.
  - **Modern surveillance:** Drones, CCTV, data analysis and other technologies should be used for better monitoring of vulnerable areas.

    Using high-altitude long-endurance (HALE) UAVs and thermal sensors along sensitive forest corridors in LWE regions (Bastar) and border sectors.
  - **Better coordination:** Central and State agencies must work together through regular information sharing and joint operations.
- **Secure India's Borders**
  - **Stronger border management:** Vulnerable border areas should have better fencing, surveillance and patrolling.
  - **Stop illegal movement:** Effective checks are needed to prevent the movement of terrorists, weapons and illegal funds.
  - **Cooperation with neighbours:** India should improve intelligence sharing and security cooperation with neighbouring countries.

    Intelligence-sharing agreements and coordinated border patrols with friendly neighbours like Nepal (APF-SSB cooperation) and Bangladesh (RAB-BSF accords).
- **Stop Terror Financing**
  - **Break funding networks:** Hawala, fake businesses, drug trafficking and other sources of illegal funding must be targeted.

    Strict enforcement of Know-Your-Customer (KYC) norms and monitoring non-profit organizations (NPOs) to meet global Financial Action Task Force standards.
  - **Stronger financial intelligence:** Agencies should work together to identify suspicious transactions quickly.
- **Counter Radicalisation**
  - **Early intervention:** Young people showing signs of extremist influence should be provided counselling and positive alternatives.

    Maharashtra Police's Deradicalization Programme and Kerala's Operation Purity, which use family counseling and community elders to rehabilitate misguided youth.
  - **Counter online propaganda:** Extremist content and recruitment networks on social media should be identified and disrupted within the law.

    National Cybercrime Threat Analytics Unit (TAU) under MHA tracking and removing extremist recruitment channels on Telegram, Dark Web, and social media.
  - **Promote critical thinking:** Education should help young people identify fake news, hate propaganda and extremist narratives.
  - **Community involvement:** Families, teachers and local communities should play a role in preventing radicalisation.
- **Improve Governance and Development**

- **Reduce alienation:** People should feel that government institutions are accessible, fair and responsive.
- **Better public services:** Education, healthcare, roads, jobs and basic services should reach conflict-affected areas.
- **Inclusive development:** Development should reach marginalised regions and communities so that people do not feel left behind.

- **Strengthen Community Participation**
  - **Encourage community policing:** Local people can help identify suspicious activities when they trust the police.

    Initiatives like Friends of Police (Tamil Nadu), Janamaithri Suraksha (Kerala), and Gram Rakshek Dal (Gujarat) that build mutual trust between locals and law enforcement.
  - **Promote social harmony:** Dialogue and community programmes can reduce mistrust and communal tensions.

    Mohalla Committees: Community-level peace committees formed in urban areas to handle local disputes peacefully
- **Use Technology**
  - **Artificial Intelligence:** AI and data analytics can help identify unusual patterns and possible security threats.
  - **Cyber security:** India needs stronger systems to prevent terrorist groups from misusing digital platforms.
  - **Integrated databases:** Relevant agencies should be able to securely share information and identify links between cases.

Conclusion

Terrorism is not just a security problem. It also affects **people, development, social harmony and national unity**. India needs a balanced approach of **strong security, good governance, development and social inclusion** to tackle both the threat and its root causes. Ultimately, **a secure society is built not only by defeating terrorists, but also by creating trust, opportunity and hope for people**.

MGP/BP/GS3/039

BASICS OF CYBER SECURITY MONEY-LAUNDERING

1. CYBERSECURITY

Introduction

- Cybersecurity refers to the techniques of protecting computers, networks, programs and data from unauthorised access or attacks. It means securing cyberspace from attack, damage, misuse and economic espionage.
- The internet is a system of inter-connected devices using standardised communication protocols. Cyberspace is a connected internet ecosystem.
- Interconnectedness of the cyberspace, multiple entry points to the internet and dependency of critical infrastructure on the cyberspace makes it vulnerable to cyber threats.
- Cyber threats are anonymous and borderless. Thus, it makes the tracking of criminals difficult and cybersecurity a national security issue.

Cybersecurity Threats

As per the American political scientist Joseph Nye, there are four primary threats to cyberspace:

Cyber Warfare	• Cyber Warfare refers to a virtual conflict initiated as a politically motivated attack on the enemy's computer and information systems waged via the internet. • The aim of such an intrusion usually is to disrupt the systems, destroying them partially or entirely. • For instance, In 2010, Stuxnet, a malicious computer worm, was designed to attack industrial programmable logic controllers of the Iranian nuclear programme.
Cyber Espionage	• Use of computer networks to gain illicit access to confidential information, typically held by a government or other organizations. • For instance, in Operation Shady RAT, the hackers had access to seventy government and private agencies around the world.
Cybercrime	• **Hacking:** It is an illegal invasion into a computer system or network. • **Child Pornography:** It refers to the pornography showing children on the internet. • **Cyber Stalking**: The use of the internet, email or other electronic communications to stalk a person is called cyber stalking. • **Denial of Service**: A denial-of-service attack is a technology-driven attack that occurs when an attacker prevents legitimate users from accessing specific computer systems and networks.

	• **Malware**: It is a malicious software designed to perform an unwanted illegal act via the computer network. Some of the common malwares are - viruses, worms, trojans, hoax etc. • **Phishing**: It is a cybercrime in which targets are lured by mail to provide sensitive information(personal information, bank details) by someone posing as a legitimate website. • **Botnet**: Botnets are a number of internet-connected devices, with each one running its own bots. They are used to perform DoS attacks, steal data, send spam mails etc. Various other types of cybercrimes are Trojans, Ransomware, Wiper Attacks, Intellectual Property Theft, Data Manipulation, data destruction, spyware, Network traveling worms etc.
Cyber Terrorism	• Politically motivated use of computers and IT to cause severe disruption or widespread fear in society is called cyber terrorism. • It includes activities such as websites spreading extremist propaganda, recruiting terrorists, promoting the propaganda of terrorists etc.

Elements of cybersecurity

- **Application Security**: Use of software or hardware to protect applications from external threats.
- **Information Security:** Set of strategies for managing the processes, tools and policies for securing digital information.
- **Network Security**: Protecting networks against internal and external threats.
- **Disaster recovery plan:** Structured plan which guides the response to unplanned incidents; It enables an organisation to resume critical functions.
- **Operational Security**: Analytical processes that classify information assets and determine controls required to protect these assets.
- **End-user education:** Policies to guide users for secure use of an organisation's systems.

Indian scenario of cybersecurity

- **Security Aspect**: Cyberspace has emerged as the 5th arena of war(after land, air, water and space). Further, cyberspace has also allowed the terrorist groups to stay anonymous and propagate their nefarious activities.
- **Privacy of the users**: As per the Kantar report, India's internet base has breached the 600 million mark in 2023. It is likely to reach 900 million by the end of 2025. This will generate huge volumes of data.
- **Increasing cybercrimes:** As per a report by the internet solutions provider Symantec, India is the third most vulnerable country in the world in terms of cybersecurity threats.
- **Economy**: India aims to be a digital economy. India also has the third largest ecosystem for technology-driven startups in the world.
- Promotion of innovation and entrepreneurship through the startups and enabling services (e.g.- Aadhaar, MyGov, Government e-Market, Digi-Locker) through cyberspace requires a secure cyberspace.
- **E-commerce boom:** The Indian E-commerce market is expected to reach US$ 200 billion by 2026. This increases the vulnerability of the consumers to online frauds (identity thefts, credit card frauds).
- **Protecting the critical infrastructure:** Critical infrastructure systems such as finance, defence, transport, telecommunication etc requires security of networks to function efficiently.

- **Age of IoT:** As we move into the 4th Industrial Revolution, the rise of Internet of Things, artificial intelligence, machine learning etc would require the communication systems to be secure.

Cybersecurity: Architectural Issues

Dealing with data colonization	India is a net exporter of information, however data servers of majority of digital service providers are located outside India. India still lacks a comprehensive data protection law.
Policy Issues	The National Cyber Security Policy does not lay down a comprehensive broad framework for a robust cyber security ecosystem.
Legal Systems	A single IT Act is inadequate to deal with the different aspects of cybersecurity: Cyber Law, Cyber Security, Cyber Crimes etc.
No unified national security architecture	Each law enforcement agency/armed force has its own cyber security agencies which makes it difficult to act in a coordinated manner.
Lack of R&D in cyberspace	Basic research, technology demonstration, and skill enhancement of manpower has been lacking in the field of cyberspace.
Absence of a nodal body	India has a number of institutions to deal with cybersecurity, such as : MeiTY, NCIIPC, Ministry of Home Affairs. However, it has lacked an overarching nodal body to deal with cyber security issues in a coordinated manner.
Lack of awareness	Digital illiteracy in the Indian population has made them vulnerable to cyber-attacks.
Heavy import dependence	Lack of a indigenous electronic manufacturing base has led to heavy dependence on imports for electronics and IT hardware. Further, there is no coherent certification mechanism to check the standards of the imported products. This directly poses a threat to national security.

Cyber Security: Steps taken

Institutional Measures	• Indian Computer Emergency Response Team (CERT-in) functions under MEITY to secure India's Communications and Information Infrastructure. • National Intelligence Grid (NATGRID) is a central agency which aims to have an integrated intelligence grid that will source sensitive personal information on citizens from different government agencies for counter-terrorism investigations. • Cyber Swachhta Kendra, a Botnet Cleaning and Malware Analysis Centre. It helps in detecting malicious programmes and free tools to remove such programmes. • National Technical Research Organisation (NTRO) is a technical intelligence gathering agency. It undertakes hi-tech surveillance jobs such as: satellite monitoring, data gathering and processing, software development. • National Critical Information Infrastructure Protection Centre (NCIIPC) national nodal agency for the protection of Critical Information Infrastructure. It was created under the IT Act, 2000. • National Cyber Coordination Centre (NCCC) under the Ministry of Home Affairs, scans the internet traffic coming into the country and provides real time information regarding cyber threats to various security agencies.

Legislative Measures	• Information Technology Act, 2000 (amended in 2008) provides legal recognition to electronic documents. It also provides a legal framework for e-commerce transactions and to deal with cybercrimes. • National Cyber Security Policy 2013 aims to create a secure cyberspace; develop suitable indigenous security technologies and enhance global cooperation. • National Encryption Policy, 2015 aims to secure transactions in Cyber Space for individuals, businesses, Government including nationally critical information systems and networks.
Programmes and Initiatives	• Cyber Surakshit Bharat Initiative is the first public-private partnership where the various IT officers across the central and state government departments, banks, PSUs are trained by the IT industry. • Indian Cyber Crime Coordination Centre Scheme(IC4 Scheme)acts as a nodal point in the fight against cybercrime and misuse of cyberspace for furthering the cause of extremist and terrorist groups. • Cyber Warrior Police Force(CWPF) operates under the National Information Security Policy and Guidelines wing of MHA's CIS division. • Pradhan Mantri Gramin Digital Saksharta Abhiyan(PMGDISHA) is a component of the Digital India Programme and aims to make 6 crore rural households digitally literate.
International measures	• Cyber-diplomacy-Indian government has entered into cybersecurity collaborations with countries such as the USA, European Union and Malaysia. For example, U.S.-India Cyber Relationship Framework. • Global Centre for Cybersecurity launched by the World Economic Forum (WEF) to serve as laboratory and early-warning think tank for future cybersecurity scenarios and help build a secure global cyberspace.

Way forward for cybersecurity in India

- **Governance, Policy, and Regulatory Clarity:** A strong legal and institutional foundation is the essential first step for driving compliance and accountability:
  - **Data Protection and Sovereignty:** Implement and enforce the provisions of the **Digital Personal Data Protection (DPDP) Act, 2023** comprehensively. This standardizes data processing, localization, and breach reporting, fostering public trust.
  - **National Cybersecurity Strategy (NCS) Implementation:** The long-pending NCS must be fully operationalized with dedicated budget allocation, clear roles for central and state governments, and defined metrics for assessing cyber readiness.
  - **Critical Information Infrastructure (CII) Protection:** Strengthen the mandates and operational capabilities of the National Critical Information Infrastructure Protection Centre (NCIIPC). Mandate regular, independent security audits and simulations (red teaming) for entities in the finance, power, telecom, and health sectors.
  - **Standardized Compliance:** Develop a simplified, risk-based National Cyber Security Standard (NCSS) framework that all government and private entities must adhere to, moving away from disparate guidelines.
- **Technological Advancement and Cyber Defense:** Investment in indigenous capabilities and adopting advanced defense mechanisms are crucial to countering sophisticated persistent threats (APTs):

  - **Promoting Indigenous Technology:** Offer significant R&D grants, tax incentives, and preferential procurement policies (Make in India) for homegrown cybersecurity products, especially in niche areas like secure hardware, quantum-resistant cryptography, and deception technology.
  - **Adopting Zero Trust Architecture (ZTA):** Mandate phased adoption of Zero Trust principles across all government networks and CII, where no user or device is trusted by default, regardless of location.
  - **Enhanced Threat Intelligence and Sharing:** Bolster the technical and analytical capabilities of the Indian Computer Emergency Response Team (CERT-In). Establish industry-specific ISACs (Information Sharing and Analysis Centers) where private and public sectors can share real-time threat intelligence anonymously.
  - **Cyber Forensics and Attribution:** Upgrade cyber forensic labs and training for law enforcement agencies to quickly trace and attribute cyberattacks, enhancing the deterrence factor.
- **Skill Development and Capacity Building:** Cybersecurity is fundamentally a human-driven field. India must leverage its demographic dividend to close the skill gap:
  - **Integrating Cyber Education:** Introduce mandatory cybersecurity training modules from the school level (digital literacy) up to university curricula (specialized degrees).
  - **Certification and Skill Ecosystem:** Establish a government-backed national certification framework for cyber professionals to standardize competencies and recognize talent.
  - **Cyber Workforce Retention:** Develop incentives (competitive salaries, fast-track promotions, and research funding) to attract and retain top cybersecurity talent within government and defense institutions.
  - **Cyber Wargames and Exercises:** Conduct regular national and sector-specific cyber drill exercises to test the resilience and coordination capabilities of CII, CERT-In, and private companies under simulated attack scenarios.
- **International Collaboration and Diplomacy:** Cyber threats are borderless, necessitating strong global partnerships to combat state-sponsored and transnational criminal activity:
  - **Bilateral and Multilateral Partnerships:** Intensify cooperation with strategic partners (e.g., QUAD nations, EU) on threat intelligence sharing, capacity building, and developing norms of behavior in cyberspace.
  - **Cyber Diplomacy:** Actively participate in global forums (UN, ITU) to shape international laws and protocols governing cyberspace, focusing on preventing the weaponization of digital infrastructure.
  - **Mutual Legal Assistance Treaties (MLATs):** Accelerate and simplify the process for seeking and providing digital evidence from foreign jurisdictions to prosecute cyber criminals operating across borders.

Conclusion:

Thus, focus on moving from fragmented efforts to a unified, nationally coordinated approach that views cybersecurity not just as an IT issue, but as a core component of national security and economic growth.

2. DIGITAL PERSONAL DATA PROTECTION ACT, 2023

Introduction:

The **Digital Personal Data Protection Act, 2023 (DPDP Act)** creates a legal framework to protect people's personal data while allowing its lawful use for digital services and economic growth. It aims to create a balance between **"data protection and data-driven development."**

Objectives of the Digital Personal Data Protection Act, 2023

- **Protect personal data:** To protect individuals from misuse, unauthorised access and leakage of their personal data.

- **Protect privacy:** To give people greater control over how their personal data is collected, used and shared.
- **Give rights to individuals:** To allow people to access, correct, update and, where applicable, erase their personal data.
- **Ensure responsible data use:** To make organisations collect and use personal data only for lawful and specified purposes.
- **Improve data security:** To encourage organisations to take reasonable safeguards against data breaches and cyber threats.
- **Make organisations accountable:** To fix clear responsibilities on organisations that collect and process personal data.
- **Protect children:** To provide stronger safeguards for children's personal data and restrict harmful forms of processing.
- **Provide grievance redressal:** To give individuals a mechanism to raise complaints regarding misuse or mishandling of their personal data.
- **Create deterrence:** To impose financial penalties for serious violations and encourage organisations to follow data-protection rules.
- **Build trust in the digital economy:** To make citizens more confident in using digital services, e-commerce, fintech and other online platforms.
- **Balance privacy with innovation:** To protect personal data while allowing its legitimate use for **economic growth, governance and technological innovation**.

Provisions of the Digital Personal Data Protection Act, 2023

- **Consent-based data processing:** Personal data should generally be processed with the individual's **consent**, which should be informed, specific and clear.
- **Clear notice:** Before collecting data, organisations must tell individuals **what data is being collected and why it is being used**.
- **Data Fiduciary:** The organisation that decides **why and how personal data is processed** is called a Data Fiduciary and has responsibility for protecting the data.
- **Purpose-based processing:** Personal data should be processed for **lawful purposes** and in accordance with the Act.
- **Rights of individuals:** Individuals, called **Data Principals**, have rights to access information about their data, seek **correction and erasure** and obtain grievance redressal.
- **Right to nominate:** A person can nominate another individual to exercise their data rights in case of their **death or incapacity**.
- **Data security:** Data Fiduciaries must take **reasonable security safeguards** to prevent personal-data breaches.
- **Protection of children:** Processing children's personal data requires **verifiable parental consent**, with additional restrictions on harmful forms of processing.
- **Significant Data Fiduciaries:** The government can classify certain entities as **Significant Data Fiduciaries** based on factors such as the volume and sensitivity of data and their potential impact on India's sovereignty, security and public order.
- **Data Protection Board of India:** The Act provides for a **Data Protection Board of India** to deal with breaches and enforce the provisions of the law.
- **Financial penalties:** Organisations can face significant monetary penalties for non-compliance, creating a **deterrent against poor data protection**.
- **Exemptions:** Certain processing activities are exempted under specified conditions, including some situations related to **legal, regulatory, security and law-enforcement purposes**.
- **Duties of citizens:** Individuals also have certain duties, such as providing **authentic information** and not filing false or frivolous complaints

Issues with the Digital Personal Data Protection Act, 2023

- **Privacy Concerns**
  - **Wide exemptions:** Section 17 allows the central government to exempt any state agency from the Act's provisions in the name of national security, public order or sovereign interest.
  - **Risk of misuse:** Greater access to personal data by powerful institutions can create concerns about excessive surveillance or misuse.
- **Limited Individual Control**
  - **Limited awareness:** Many citizens may not fully understand what data they are sharing or how it is being used.
  - **Consent fatigue:** People often click "I agree" without reading long privacy notices, making consent less meaningful.
  - **Uncertain Implementation:** Essential definitions and operational boundaries remain unclear until the secondary rules are fully notified and tested.
- **Challenges for the Data Protection Board**
  - **Need for independence:** The effectiveness of the Data Protection Board will depend on its ability to take decisions fairly and independently.
  - **Capacity challenge:** The Board will need sufficient experts, staff and technical capacity to handle large numbers of complaints and data-breach cases.
- **Compliance Burden on Businesses**
  - **Technical requirements:** Organisations need better cybersecurity systems, trained staff and proper data-management practices.
  - **Disproportionate Compliance Costs:** While big tech firms can easily afford Data Protection Officers and periodic audits, smaller Indian startups and MSMEs struggle under heavy operational and legal overheads.
- **Lack of Compensation for Victims**
  - **Fines Go to the Government:** Fines (up to ₹250 crore) levied on defaulting companies are deposited directly into the Consolidated Fund of India.
  - **No Relief for Citizens:** the 2023 Act contains no provision to directly compensate individual victims whose data was leaked, misused, or stolen.
- **Uniform approach**
  - **One-Size-Fits-All Approach:** The Act does not separate personal data into categories like "Sensitive" (health, biometrics, financial info) or "Critical." All personal data is treated under a uniform regulatory standard.
  - **Blanket Age Threshold:** Treating everyone under 18 as a child requiring verifiable parental consent creates high compliance friction for EdTech platforms, gaming apps, and social media sites.

Way Forward:

- **Operational & Regulatory Measures**
  - **Independent DPBI:** Keep the Data Protection Board free from political interference by establishing fixed tenures and transparent appointment procedures.
  - **Draft Clear, Simple Rules:** Define ambiguous terms, such as **verifiable parental consent** and **data breach reporting timelines** through consultation with industry and civil society.
  - **Introduce Tiered Compliance:** Offer lower compliance burdens and longer transition periods for startups, MSMEs, and small businesses to avoid stifling innovation.
- **Balance Privacy & State Functioning**
  - **Incorporate Judicial Oversight:** Require independent judicial warrants for state exemptions to prevent arbitrary mass surveillance under **national security.**

  - **Harmonize with RTI Act:** Clarify that routine public data (like welfare beneficiary lists and official asset declarations) remains accessible to preserve transparency.
  - **Define Sensitive Data Categories:** Introduce specific, higher-level safeguards for sensitive information like health records, biometric data and financial transactions.
- **Citizen-Centric Protections**
  - **Provide Victim Compensation:** Amend rules to allow a portion of corporate penalties to directly compensate citizens whose data is leaked or stolen.
  - **Age-Appropriate Consent:** Shift from a rigid under-18 cutoff to a graduated age threshold (e.g., lower limits for teenagers) matching global standards like the EU GDPR.
  - **Launch Data Literacy Drives:** Educate citizens on digital privacy rights, reporting channels, and simple cyber safety practices.

Global Initiatives for Data Protection

- **EU – General Data Protection Regulation (GDPR):** The GDPR is one of the strongest data-protection frameworks in the world. It gives individuals greater control over their personal data and places strong responsibilities on organisations handling such data.
- **OECD Privacy Guidelines:** The OECD has developed internationally recognised principles such as **data minimisation, purpose limitation, security safeguards, transparency and individual participation**. These guidelines have influenced data-protection laws across the world.
- **Council of Europe – Convention 108:** Adopted in **1981**, it was the first legally binding international treaty specifically dealing with personal data protection. It promotes lawful and fair data processing and gives individuals rights over their data.
- **G20 Digital Economy Initiatives:** G20 discussions promote **trusted data flows**, digital cooperation and responsible data governance while trying to balance privacy, security and economic growth.
- **Data Free Flow with Trust (DFFT):** Promoted by Japan and supported in international forums, DFFT seeks to allow data to move across borders while maintaining **privacy, security and trust**. The OECD identifies it as an important approach to cross-border data flows
- **APEC Privacy Framework:** The Asia-Pacific Economic Cooperation (APEC) framework promotes responsible handling of personal information and helps facilitate **cross-border data flows** while protecting privacy.

Conclusion

The DPDP Act, 2023 is an important step towards **protecting citizens' personal data in India's growing digital economy**. Its success will depend on **strong implementation, better cybersecurity, citizen awareness and accountable data handling**. India needs a balance between **privacy, innovation and digital growth** to build a trusted digital society.

3. CYBER FRAUDS AND SAFEGUARDING INDIA'S DIGITAL ECONOMY

Context:

India's cyberspace has expanded rapidly with over 86% of households now connected to the internet, reflecting the transformative progress under the Digital India initiative. However, the rise in digital penetration has also led to an unprecedented increase in cyber frauds such as phishing, unauthorized access, data theft, and online scams targeting individuals and institutions.

The number of cybersecurity incidents has surged from **10.29 lakh in 2022** to **22.68 lakh in 2024**, highlighting both increased reporting and growing vulnerability in the digital ecosystem.

Emerging trends in cyber frauds:

- **AI-driven deepfakes-** The pattern of frauds has become more sophisticated, with the rise of AI-driven deepfakes and social engineering scams.
  - **Phishing**, **pig-butchering**, **phantom hacking**, and **instant loan app scams** are among the fastest-growing digital crimes.
- **Exploitation of UPI-** Fraudsters exploit Unified Payments Interface (UPI) by **cloning or misusing virtual numbers**, leading to unauthorized fund transfers.
- **Transnational cyber-frauds-** Organised crime networks, often based in Southeast Asian "fraud factories," are increasingly involved in transnational cyber-fraud operations.

Need for Safeguarding India's Digital Economy from Cyber Frauds:

Protecting Financial Inclusion & UPI Adoption: The digital economy's greatest strength is also its biggest vulnerability- its penetration into rural and lower-income segments:

- **Erosion of Trust:** Millions of citizens have entered the formal financial system solely through mobile apps and UPI. If these users—who may have lower digital literacy—become victims of phishing or impersonation scams, their trust in digital payments collapses. This setback could reverse years of hard-won financial inclusion progress.
- **Scale of Transactions:** India processes billions of UPI transactions monthly. Even a small fraction of successful frauds translates into astronomical financial losses, often impacting small merchants and everyday users most severely.
- **Vulnerability of Endpoints:** The weakest link is often the user's endpoint (mobile phone). Frauds rely heavily on social engineering, phishing links, and remote access apps rather than just technical network breaches, making user education and strong fraud-detection algorithms essential.

Mitigating Economic Losses & Investment Risks: Unchecked cyber fraud directly impacts capital, discourages investment, and strains enforcement agencies:

- **Direct Capital Loss:** Fraud results in the immediate loss of capital, both for individuals and the financial institutions that absorb or reimburse certain losses. This drains liquidity from the formal banking sector.
- **Discouraging FDI:** Foreign Direct Investment (FDI) depends on confidence in the security and regulatory environment. **High-profile cyber breaches** or **rampant fraud signal** a weak defense perimeter, making international investors hesitant to commit capital to digital enterprises.
- **Cost of Compliance and Security:** Businesses must divert increasing amounts of resources—time, money, and skilled personnel—from innovation into compliance and security patches, slowing down organic economic growth.
- **Securing Critical Infrastructure & Data Sovereignty:** As banking, energy, telecom, and transportation are now interconnected through digital systems, the scope of a cyber attack extends far beyond financial loss:
- **Attacks on Critical Infrastructure:** Malicious actors targeting the systems that underpin the digital economy (e.g., banking cores, power grids, or telecom networks) could lead to systemic paralysis, crippling national services and posing a direct threat to national security.
- **Data Exploitation:** Fraudulent activities often serve as precursors for large-scale data theft and identity compromise. Protecting user data—which underpins the success of the Digital India stack—is crucial for maintaining individual sovereignty and preventing misuse by foreign or adversarial entities.

Challenges in curbing cyber frauds:

- **Rapid Technological Evolution:** The fast pace of artificial intelligence, deepfakes, and quantum technologies outpaces the capability of enforcement and regulation.
- **Low Cyber Awareness:** A large segment of citizens lacks digital literacy and are unaware of basic cyber hygiene practices.

- **Jurisdictional and Cross-Border Barriers:** Many frauds originate overseas, complicating investigation and extradition.
- **Under-Reporting of Cases:** Victims often refrain from reporting due to fear of embarrassment or lack of faith in grievance systems.
- **Limited Skilled Workforce:** Shortage of trained digital forensic professionals in small towns and districts delays timely response to cyberfrauds.
- **Fragmented Institutional Coordination:** Overlaps between agencies like CERT-In, I4C, and NCIIPC can hinder real-time information exchange.
- **Implementation Gaps:** Enforcement of the Digital Personal Data Protection Act, 2023 and related frameworks remains nascent.
- **Financial Inclusion Risks:** First-time digital users and senior citizens remain the most vulnerable to UPI and SMS-based scams.

Legal and institutional safeguards to counter cyber frauds:

- The **Information Technology Act, 2000** forms the legal backbone of India's cyber law framework, covering offences like identity theft, impersonation, and cyber fraud.
- The **Intermediary Guidelines and Digital Media Ethics Code, 2021** ensures accountability of digital intermediaries and curbs the dissemination of harmful content.
- The **Digital Personal Data Protection Act, 2023** mandates lawful, consent-based processing of personal data, protecting citizens from misuse and unauthorized access.
- The **National Cyber Crime Reporting Portal (www.cybercrime.gov.in)** enables citizens to lodge online complaints and facilitates fund freezing in financial frauds.
- The **Indian Computer Emergency Response Team (CERT-In)** acts as the national nodal agency for cyber incident response. It has conducted 109 mock drills and engaged 1,438 organizations to assess cyber preparedness.
- The **National Critical Information Infrastructure Protection Centre (NCIIPC)** safeguards critical sectors such as banking, telecom, and power through sector-specific security audits.
- The **Indian Cybercrime Coordination Centre (I4C)** under the Ministry of Home Affairs coordinates law-enforcement efforts and has successfully blocked 3,962 Skype IDs and 83,668 WhatsApp accounts linked to cyber fraud.
- The **Citizen Financial Cyber Fraud Reporting and Management System (CFCFRMS)** has strengthened the grievance-redressal mechanism for online financial crimes.
- Through the **Cybercrime Prevention against Women and Children (CCPWC)** scheme with an outlay of ₹132.93 crore, cyber forensic labs have been set up across 33 States and Union Territories, and over 24,600 personnel have been trained in cyber investigation.

Way forward:

- **Strengthen Legal Architecture:** Amend and modernize the IT Act, 2000 to include AI-related, blockchain-based, and cross-border cyber offences.
- **Integrated Cyber Defence:** Establish a unified National Cyber Defence Architecture to synergize CERT-In, I4C, and NCIIPC operations.
- **Capacity Building:** Incorporate cyber law, digital forensics, and ethical hacking modules in police and judicial academies.
- **Public Awareness Campaigns:** Launch a nationwide Cyber Suraksha Abhiyan focusing on rural areas and vulnerable populations.
- **Technological Innovation:** Employ AI-driven fraud detection, blockchain for transaction traceability, and predictive analytics for threat forecasting.

- **International Collaboration:** Strengthen partnerships with INTERPOL, ASEAN, and the Budapest Convention on Cybercrime for intelligence sharing and extradition mechanisms.
- **Gender and Child-Sensitive Approach:** Expand the CCPWC programme and create safe online environments through parental control tools and awareness education.
- **Regular Cyber Audits:** Mandate annual cybersecurity audits for all government departments and financial institutions.

Conclusion

India's Prime Minister envisions **"a Digital India where cybersecurity becomes an integral part of national security."** With comprehensive frameworks like CERT-In, I4C, and the Data Protection Act, India is building robust digital defence mechanisms. The convergence of strong laws, institutional coordination, public awareness, and technological innovation will help India emerge as a digitally secure and resilient nation, ensuring that the benefits of digitalization are not undermined by the perils of cyber fraud.

4. MONEY LAUNDERING

Introduction:

Money laundering, as defined under **Section 3 of Prevention of Money Laundering Act (PMLA)**, is an act through which processes or activities connected to the proceeds of crimes are concealed, possessed, acquired, or used & projected as untainted property or claiming to be untainted property. Money laundering is the illegal process of concealing the origins of money obtained from illicit activities, making it appear to have come from legitimate sources.

Money laundering primarily involves 3 distinct stages:

PLACEMENT	This is the initial stage where "dirty" money (proceeds of crime, often in cash) is introduced into the legitimate financial system. The goal here is to get the money into a financial institution or a legitimate business to move it away from its direct criminal source. It is done by breaking up large amounts of cash into smaller sums (a process called **smurfing).**
LAYERING	Once the money is in the financial system, the layering stage involves creating multiple layers of complex financial transactions to obscure the audit trail and distance the funds from their illegal origin. The aim is to make it extremely difficult for authorities to trace the money back to its source.
INTEGRATION	This is the final stage where the "cleaned" money is reintroduced into the financial system through real estate, business or asset formation etc, in a way that makes it appear legitimate. At this point, the funds are virtually indistinguishable from legal funds, and the criminals can use them freely.

Impacts of Money Laundering:

- **Undermining Financial System Integrity and Stability:** Money laundering can compromise the integrity and stability of financial institutions. Illicit funds entering the financial system can distort market mechanisms, leading to economic imbalances and even crises.
- **Reduced Tax Revenues:** Money laundering allows individuals and businesses to conceal their true income, leading to a significant reduction in tax revenues for governments. This loss of public funds hinders the government's ability to finance essential public services, infrastructure projects, and social welfare programs.

- **Impact on Monetary Stability:** The sudden inflow and outflow of large, illicit funds can create unpredictable changes in money demand & supply, cause large fluctuations in international capital flows and exchange rates, and destabilize financial markets & monetary stability of the country.
- **Fueling Criminality and Organized Crime:** Money laundering is the lifeblood of criminal enterprises. By "cleaning" illicit profits, it enables drug trafficking, human trafficking, arms dealing, corruption, and other organized criminal activities to continue and expand. This directly leads to increased violence, drug abuse, and other social problems.
- **Corruption and Weakened Governance:** Money laundering often involves corruption at various levels of government and private institutions. This erodes public trust in the effectiveness and fairness of governance, contributing to a sense of injustice and disillusionment among citizens. It undermines the rule of law and democratic institutions.
- **Terror Financing:** Money laundering techniques are increasingly been used to finance terrorist organizations and activities. By obscuring the source and destination of funds, it allows terrorist groups to acquire resources, plan attacks, and sustain their operations, posing a direct threat to national and international security.

Initiatives taken to tackle the issue of Money Laundering:

- **Prevention of Money Laundering Act (PMLA), 2002:**
  - PMLA was enacted to prevent money laundering and provide for the confiscation of property derived from money laundering.
  - In line with the UN Political Declaration & Global Programme for Action (adopted by UNGA in 1990), PMLA aims to combat money laundering related to illegal activities such as drug trafficking, smuggling, and terrorism financing & to confiscate the property involved or obtained.
  - No FIR is required to initiate proceeds under the Act, ECIR is sufficient to initiate proceedings. The only requirement as per the SC was that a scheduled offence be essential for the offence of money laundering.
- **Establishment of Financial Intelligence Unit - India (FIU-IND):** Established in November 2004, FIU-IND is the central national agency responsible for receiving, processing, analyzing, and disseminating information related to suspect financial transactions. FIU-IND perform functions like:
  - **Collection of Information:** Acts as the central reception point for various reports (CTRs, STRs, Cross Border Wire Transfer Reports - CBWTRs, etc.) from reporting entities.
  - **Analysis of Information:** Analyzes collected information to uncover patterns suggesting money laundering and related crimes.
  - **Sharing of Information:** Disseminates valuable financial intelligence to national intelligence/law enforcement agencies (like ED), national regulatory authorities (RBI, SEBI, IRDAI), and foreign FIUs.
  - **Central Repository:** Maintains a national database of financial transactions.
- **Role of Enforcement Directorate (ED):** ED, under the Department of Revenue, Ministry of Finance, is the primary legal authority responsible for investigating and prosecuting money laundering offenses under the PMLA. Its mandate include:
  - Conducting investigations to trace assets derived from proceeds of crime.
  - Provisionally attaching properties involved in money laundering.
  - Ensuring prosecution of offenders and confiscation of property by Special Courts.
  - Enforcing the **Foreign Exchange Management Act (FEMA), 1999**, and the **Fugitive Economic Offenders Act (FEOA), 2018**, which often overlap with money laundering activities.
- **Regulatory Measures by Financial Regulators:** Regulatory bodies like the Reserve Bank of India (RBI) for banking, the Securities and Exchange Board of India (SEBI) for capital markets, and the Insurance Regulatory and Development Authority of India (IRDAI) for insurance, issue comprehensive Anti-Money Laundering (AML) and Combating the Financing of Terrorism (CFT) guidelines for entities under their purview.

- **KYC Norms:** These guidelines mandate strict Know Your Customer (KYC) norms, customer due diligence (CDD), enhanced due diligence (EDD) for high-risk clients, and ongoing transaction monitoring.
- **International Cooperation and FATF Compliance:**
  - **Implementation of FATF Recommendations:**
    - India became the 34th member of the Financial Action Task Force (FATF) in 2010, demonstrating its commitment to global AML/CFT standards.
    - India actively works to implement the 40 FATF Recommendations, which provide a comprehensive framework for combating money laundering, terrorist financing, and proliferation financing.
    - Regular mutual evaluations by FATF assess India's compliance and effectiveness.
  - **Mutual Legal Assistance:** India engages in international cooperation through Mutual Legal Assistance Treaties (MLATs) to facilitate the exchange of information and assistance in investigations and prosecutions of money laundering cases across borders.
  - **Double Taxation Avoidance Agreement (DTAA):** India has signed DTAA with 85 countries, which helps to check money laundering. These agreements promote exchange of financial & tax-related information between tax authorities of participating countries. This facilitates enforcement of tax regulations & helps prevent illegal activities like tax evasion & money laundering.

Concerns regarding the PMLA Act:

- **Inclusion of minor and less serious offences:** Inclusion of other less serious offences in the schedule dilutes the original intent of the law. The schedule of the Act has been expanded over the years, to include even minor and non-serious offences such as copyright and trademark infringements.
- **Equates punishment for ordinary crimes with serious crimes:** PMLA equates the punishment under an ordinary crime with a serious economic offence. **For e.g.** With the addition of The Prevention of Corruption Act, 1988 to the schedule of offences, PMLA applies with all its rigour to public servants. Thus, a public servant charged with corruption and a hard-core drug trafficker are treated alike.
- **Broad Definition of 'Proceeds of Crime':** The broad definition of 'proceeds of crime, provides considerable discretion to the authorities. Critics fear that this discretion, in determination of 'proceeds of crime', can be misused by the investigating authorities.
- **Stringent Bail Conditions:** An accused is denied bail by the entire hierarchy of courts because the bail provision under PMLA states that a judge can give bail only when he is satisfied that the accused is innocent. This is against the Anglo-Saxon jurisprudence, which presumes a person innocent until proven guilty.
- **Burden of Proof of innocence on the accused:** The burden of proof of innocence, on the accused, presents a prominent challenge in ensuring a free and fair trial.
- **Against the Federal principles and Basic structure Doctrine:** Under the PMLA, the ED, can carry out investigation without the prior consent of the concerned State. This is unlike other central police organisations, (like CBI), which are required to obtain the consent of the state before carrying out any policing/investigating activity. This impinges the principle of federalism, which is part of the Basic Structure of the Constitution.
- **Violation of the Fundamental Rights of Accused:**
  - **Violation of Article 21:** Under PMLA, the ED does not require disclosing the details of the Enforcement **Case Information Report (ECIR)**, which contains the allegations against the accused person. This is against the fundamental right of the accused to be informed of the charges and allegations, which is a universally recognised right, and is part of the right to life and liberty under Article 21 of the Constitution.
  - **Violation of Article 14:** Equating the punishment of accused of minor crimes and serious economic offence, is violation of the fundamental right to equality under Article 14 of the constitution.

- **Violation of Article 20(3):** The power of the authorities to issue summons to 'any person' (including the accused), to give evidence or produce records during the course of an investigation is in violation of the right against self-incrimination, which is a fundamental right under Article 20(3) of the Constitution.

- **Extensive powers to the authorities:** The Act grants the Enforcement Directorate (ED) extensive powers of summons, arrest, and raids. This excessive power to the authorities, can potentially lead to its misuse and overreach. SC held that to initiate prosecution under Section 3 of PMLA, registration as scheduled offence is a prerequisite, but for initiating attachment of property under Section 5, there need not be a pre-registered criminal case. This provision has often been misused by authorities with politically motivated intentions.
- **Low Conviction Rate:** According to the Report submitted by the FM in the Rajya Sabha, 5892 cases were taken up by the ED under PMLA since 2015. Of these cases, only 15 convictions have been ordered by the special courts.

Observations of the SC:

- **Strict bail conditions:**
  - **Nikesh Tarachand Shah vs Union of India (2018):** SC held that the bail provision of the PMLA Act was unconstitutional as it was violation of Article 14 and Article 21.
  - **Restoration of the Provision by the Parliament:** Parliament restored the strict bail provisions with certain amendments.
  - **Vijay Madanlal Choudhary vs Union of India (2022):** The SC upheld that the strict bail provision is reasonable and has direct nexus with the purposes and objects of the PMLA Act. SC upheld the constitutionality of the PMLA.
- **ED's Overreach:**
  - **Pankaj Bansal vs Union of India-** Supreme Court highlighted inconsistencies and lack of transparency in its operations. SC emphasized the need for the ED to act with fairness.
- **Procedural Violations:**
  - **Pavana Dibbur vs The Directorate of Enforcement (2023)-** SC observed procedural violations and misuse of the PMLA. It pointed out the need for strict adherence to legal standards by the ED and other authorities.

Way forward:

- **Amend PMLA:**
  - **Precise Definition of 'Proceeds of Crime':** A precise definition of 'Proceeds of Crime' under PMLA must be incorporated to mitigate the potential abuse of its definition by the authorities.
  - **Reassessment of Burden of Proof:** An amendment to PMLA which provides a more equitable distribution of the burden of proof between the prosecution and the accused, can be brought.
  - **Safeguards Against Overreach by Officers:** An independent oversight mechanism to review and monitor the actions of law enforcement officers must be established at the earliest.
  - **Review of the Stringent Bail Conditions:** The stringent bail conditions for minor and non-serious economic offence must be done away with.
- **Harmonization of Laws:** Ensure better harmonization and synergy between PMLA and other relevant laws (e.g., FEMA, Income Tax Act, Black Money Act, Fugitive Economic Offenders Act) to create a seamless legal framework for combating financial crime.
- **Capacity Building of Enforcement Agencies:** Significantly enhance the human and technical capacity of the Enforcement Directorate (ED), FIU-IND, CBI, and other investigating agencies. This includes:
  - **Specialized Training:** Provide advanced training to investigators, prosecutors, and forensic auditors in financial forensics, cyber forensics, blockchain analysis, and international cooperation.

  - **Adequate Staffing:** Ensure sufficient staffing levels to handle the increasing volume and complexity of money laundering cases.
  - **Technological Upgradation:** Equip agencies with cutting-edge tools for data analytics, artificial intelligence, and cyber intelligence to trace complex financial trails across jurisdictions.
- **Enhanced Independence and Transparency of ED:** Measures like regular reporting and disclosure of cases handled, convictions secured, and actions taken, must be incorporated to enhance the transparency in ED's functioning. These measures will ensure confidence among the public that ED is not a politicized institution.
- **Strengthening Regulatory Oversight:** Empower financial regulators (RBI, SEBI, IRDAI) to impose stricter penalties and conduct more frequent and thorough audits of reporting entities to ensure compliance with AML/CFT guidelines.
- **Leveraging Technology and Data Analytics:**
  - **Advanced Analytics and AI:** Encourage and mandate financial institutions to deploy advanced AI and machine learning tools for real time transaction monitoring, customer risk profiling, network analysis etc.
  - **Blockchain Forensics:** Develop expertise and tools for tracing illicit transactions on blockchain networks, given the increasing use of cryptocurrencies in money laundering.
  - **Data Standardization and Sharing:** Promote standardization of financial data across various entities to facilitate easier analysis and sharing of information while ensuring data privacy.

Conclusion:

It is important that the authorities follow the recommendations of the FATF & ensure that money laundering cases are handled with care & caution so that misuse could be checked & genuine cases reported & investigated properly to enhance the rate of conviction & prevent the misuse of PMLA for political motives.

5. PREVENTION OF MONEY LAUNDERING ACT

The Supreme Court has recently endorsed the **sweeping powers of the Enforcement Directorate (ED) under the Prevention of Money Laundering Act (PMLA)**. Recently, a SC bench headed by Justice Bela M. Trivedi, **castigated four Tamil Nadu District Collectors for failing to appear in person in response to a summons issued to them under PMLA,** by the anti-money laundering body. The SC bench reaffirmed ED's powers to call "anybody for any information" under the PMLA.

However, Prevention of Money Laundering Act 2002 (PMLA) has been under scrutiny for inclusion of certain offences which are not directly aimed at combating money laundering. Recently, the Delhi Court **slammed the Enforcement Directorate (ED) for a faulty Prevention of Money Laundering Act (PMLA) probe** and said the ED must introspect steps to ensure expeditious and fair investigations. Opposition parties in India have **alleged the misuse of PMLA provisions by the ED to stifle democracy** in India. It is in this context, a **detailed introspection of PMLA becomes necessary.**

What is the Prevention of Money Laundering Act (PMLA)? What is the Historical Background of its enactment?

PMLA- The Prevention of Money Laundering Act, 2002 (PMLA) is an Act of the Parliament of India enacted to **prevent money laundering** and **provide for the confiscation of property derived from money laundering**. It aims to combat money laundering related to illegal activities such as drug trafficking, smuggling, and terrorism financing.

Objectives of PMLA:

- **Prevention-** To prevent money laundering by implementing stringent measures and monitoring financial transactions.

Prevention of Money Laundering Act Salient Features

Section 3 (Defines money laundering)	Section 3 of the PMLA defines money laundering as any attempt, assistance, or involvement in processes connected to the proceeds of crime to project it as untainted property
Offences under PMLA	Offences mentioned under Part A, B and C of the Schedule of the Act. Part A- Includes money laundering offences under the following acts– Indian Penal Code, Prevention of Corruption Act, Narcotics Drugs and Psychotropic Substances Act, Antiquities and Art Treasures Act, Trademark Act, Wildlife Protection Act, Copyright Act and Information Technology Act Part B- Includes money laundering offences that are mentioned in Part A, but are of a value of Rs 1 crore or more. Part C- Includes money laundering offences under Trans-border crimes
Section 4 (Punishment for Money Laundering)	The offence of money-laundering shall be punishable with rigorous imprisonment for a term not be less than three years which can be extended upto seven years and shall also be liable to fine.
Agencies powers under the Act	1. Enforcement Directorate (ED)- ED is responsible for investigating the offences of money laundering and attachment of properties. 2. Financial Intelligence Unit-India (FIU-IND)- It is the central national agency responsible for receiving, processing, analyzing, and disseminating information relating to suspect financial transactions. 3. Other Agencies- The scheduled offences are separately investigated by the agencies mentioned under respective acts. For ex- The local police, CBI, customs departments, SEBI, or any other investigative agency, as the case may be.
Obligations under the Act	1. Maintenance of Records- Imposes obligations on banking companies, financial institutions, and intermediaries to verify and maintain client identity records. 2. Reporting of transactions- These financial firms are required to report their financial transactions to the Financial Intelligence Unit-India (FIU-IND).
Setting up of Authority	1. Adjudicating authority- PMLA envisages the setting up of an Adjudicating Authority to exercise jurisdiction, power and authority conferred by it. 2. Appellate Tribunal- It also envisages the setting up of an Appellate Tribunal to hear appeals against the order of the Adjudicating Authority and the authorities like Director FIU-IND.
Special Courts	One or more courts of sessions are designated as Special Courts to try offences under PMLA and other related offences.

- **Detection-** To detect and investigate instances of money laundering through proper enforcement and regulatory mechanisms.
- **Confiscation-** To confiscate properties derived from money laundering activities to deter offenders and disrupt illicit financial flows.
- **International Cooperation- T**o facilitate international cooperation in combating money laundering and terrorist financing activities.

Historical background of enacting PMLA

The enactment of PMLA finds its genesis in the various conventions to combat money laundering.

United Nations Convention against Illicit Traffic in Narcotic Drugs and Psychotropic Substances (1988)	This convention **urged all countries to take urgent steps to prevent the laundering of the proceeds of drug crimes** and other connected activities.
Establishment of FATF	FATF was established to **examine the problem of money laundering and make recommendations to combat it**. India adopted the FATF recommendations for enactment of PMLA.
UNGA's Political Declaration and Global Programme of Action	This resolution urged all **member-countries to enact suitable pieces of legislation** to effectively **prevent the laundering of drug money.**
UN special session 'Countering World Drug Problem Together'	A declaration was adopted on the urgent need to combat money laundering
UN Convention against Transnational Organized Crime of 2000 (Palermo Convention)	It advocated legislative and other measures to combat organized crime, and specifically called for '**criminalizing the laundering of proceeds of crime**'.

Salient Features of Prevention of Money Laundering Act

- **Section 3 (Defines money laundering)-** Section 3 of the PMLA defines **money laundering** as any **attempt, assistance, or involvement in processes connected to the proceeds of crime** to project it as untainted property.
- **Offences under PMLA-** Offences mentioned under Part A, B and C of the Schedule of the Act.

Part A	**Includes money laundering offences under the following acts**– Indian Penal Code, Prevention of Corruption Act, Narcotics Drugs and Psychotropic Substances Act, Antiquities and Art Treasures Act, Trademark Act, Wildlife Protection Act, Copyright Act and Information Technology Act
Part B	Includes money laundering offences that are mentioned in Part A, but are of a **value of Rs 1 crore or more**.
Part C	Includes money laundering offences under **Trans-border crimes**

- **Section 4 (Punishment for Money Laundering)-** The offence of money-laundering shall be punishable with **rigorous imprisonment for a term not be less than three years extending to seven years** and shall also be liable to fine.
- **Agencies powers under the Act**
  - The **Enforcement Directorate (ED)** in the Department of Revenue, Ministry of Finance, is responsible for investigating the offences of money laundering and attachment of properties.
  - The **Financial Intelligence Unit-India (FIU-IND)**, under the Department of Revenue, is the central national agency responsible for receiving, processing, analyzing, and disseminating information relating to suspect financial transactions.

  - The **scheduled offences are separately investigated by the agencies** mentioned under respective acts. For ex- The local police, CBI, customs departments, SEBI, or any other investigative agency, as the case may be.
- **Obligations under the Act**
  - Imposes obligations on banking companies, financial institutions, and intermediaries to verify and maintain client identity records.
  - These financial firms are required to **report their financial transactions to the Financial Intelligence Unit-India (FIU-IND).**
- **Setting up of Authority**
  - PMLA envisages the setting up of an Adjudicating Authority to **exercise jurisdiction, power and authority** conferred by it.
  - It also envisages the **setting up of an Appellate Tribunal** to hear appeals against the order of the Adjudicating Authority and the authorities like Director FIU-IND.
- **Special Courts under PMLA- One or more courts of sessions are designated as Special Courts** to try offences under PMLA and other related offences.

Concerns regarding the PMLA Act:

- **Inclusion of minor and less serious offences-** Inclusion of other less serious offences in the schedule dilutes the original intent of the law. The schedule of the Act has been expanded over the years, to include even **minor and non-serious offences** such as **copyright and trademark infringements.**
- **Equates punishment for ordinary crimes with serious crimes-** PMLA equates the punishment under an ordinary crime with a serious economic offence. For ex- With the **addition of The Prevention of Corruption Act, 1988** to the schedule of offences, **PMLA applies with all its rigour to public servants. Thus, a public servant charged with corruption and a hard-core drug trafficker are treated alike.**
- **Broad Definition of 'Proceeds of Crime'-** The broad definition of 'proceeds of crime, **provides considerable discretion to the authorities**. Critics fear that this discretion, in determination of 'proceeds of crime', can be misused by the investigating authorities.
- **Stringent Bail Conditions-** An accused is **denied bail by the entire hierarchy of courts** because the **bail provision under PMLA** state that a **judge can give bail only when he is satisfied that the accused is innocent**. This is **against the Anglo-Saxon jurisprudence**, which presumes a person innocent until proven guilty.
- **Burden of Proof of innocence on the accused-** The burden of proof of innocence, on the accused, presents a prominent **challenge in ensuring a free and fair trial.**
- **Against the Federal principles and Basic structure Doctrine-** Under the PMLA, **the ED, can carry out investigation without the prior consent of the concerned State.** This is unlike other central police organisations, (like CBI), which are required to obtain the consent of the state before carrying out any policing/investigating activity. This **impinges the principle of federalism**, which is part of the Basic Structure of the Constitution.
- **Violation of the Fundamental Rights of Accused**
  - **Violation of Article 21-** Under PMLA, the ED does not require disclosing the details of the **Enforcement Case Information Report (ECIR),** which contains the allegations against the accused person. This is against the fundamental **right of the accused to be informed of the charges and allegations,** which is a universally recognised right, and is **part of the right to life and liberty** under Article 21 of the Constitution.
  - **Violation of Article 14-** Equating the punishment of accused of minor crimes and serious economic offence, is **violation of the fundamental right to equality** under Article 14 of the constitution.
  - **Violation of Article 20(3)-** The power of the authorities to issue summons to 'any person' (including the accused), to **give evidence or produce records** during the course of an investigation is in **violation of the right against self-incrimination,** which is a fundamental right under Article 20(3) of the

Constitution.

- **Extensive powers to the authorities-** The Act grants the Enforcement Directorate (ED) extensive powers of summons, arrest, and raids. This excessive power to the authorities, can potentially lead to its misuse and overreach.

Observations of the SC:

Strict bail conditions	Nikesh Tarachand Shah vs Union of India (2018)- SC held that the bail provision of the PMLA Act was **unconstitutional** as it was in **violation of Article 14 and Article 21.** Restoration of the Provision by the Parliament: Parliament restored the strict bail provisions with certain amendments. Vijay Madanlal Choudhary vs Union of India (2022)- The SC upheld that the **strict bail provision is reasonable** and **has direct nexus with the purposes and objects of the PMLA Act.** SC upheld the constitutionality of the PMLA.
ED's Overreach	Pankaj Bansal vs Union of India- Supreme Court highlighted inconsistencies and lack of transparency in its operations. SC emphasized the **need for the ED to act with fairness.**
Procedural Violations	Pavana Dibbur vs The Directorate of Enforcement (2023)- SC **observed procedural violations** and misuse of the PMLA. It pointed out the need for strict adherence to legal standards by the ED and other authorities.

Way Forward:

- **Precise Definition of 'Proceeds of Crime'-** A precise definition of 'Proceeds of Crime' under PMLA must be incorporated to **mitigate the potential abuse of its definition** by the authorities.
- **Reassessment of Burden of Proof-** An amendment to PMLA which provides a **more equitable distribution of the burden of proof between the prosecution and the accused,** can be brought.
- **Safeguards Against Overreach by Officers-** An **independent oversight mechanism to review and monitor the actions of law enforcement officers** must be established at the earliest.
- **Review of the Stringent Bail Conditions-** The stringent bail conditions for **minor and non serious economic offenses must be done away with.**
- **Enhanced Independence and Transparency of ED-** Measures like **regular reporting and disclosure of cases handled, convictions secured, and actions taken**, must be incorporated to enhance the transparency in ED's functioning. These measures will ensure confidence among the public that ED is not a politicised institution.

6. FATF & OTHER GLOBAL EFFORTS AGAINST MONEY LAUNDERING

India has been elected to the vice presidency of the Financial Action Task Force (FATF), a key global institution combatting money laundering and terrorism financing.

Introduction:

Money laundering allows **illegal money from crime, corruption, drugs and terrorism to enter the legal financial system**, making it a serious threat to both security and the economy. The **Financial Action**

Task Force (FATF) leads global efforts by setting common standards to prevent money laundering and terrorist financing.

Causes of money laundering

- **Criminal Activities**
  - **Drug trafficking:** Large profits from illegal drugs create a strong incentive to hide and convert the money into apparently legal income.
  - **Corruption:** Bribes and illegal gains can be routed through businesses, properties or financial accounts to hide their real source.
  - **Smuggling:** Money earned through gold, weapons, wildlife and other illegal trade may be introduced into the formal economy.
  - **Fraud and cybercrime:** Online fraud and other financial crimes generate illegal funds that criminals try to disguise.
- **Weak Financial Systems**
  - **Poor KYC checks:** Weak verification of customers makes it easier to open and operate accounts using false or misleading identities.
  - **Poor transaction monitoring:** Suspicious transactions may go unnoticed when banks and financial institutions have weak monitoring systems.
  - **Cash-heavy economy:** Large cash transactions can make the movement and origin of money harder to trace.
  - **Weak regulation:** Poor implementation of anti-money laundering rules creates opportunities for criminals.
- **Use of Shell Companies**
  - **Fake transactions:** Shell or front companies can be used to move illegal money through apparently genuine business transactions.
  - **Multiple layers:** Money can be moved through several companies and countries, making its original source difficult to identify.
- **Cross-Border Factors**
  - **Tax havens and secrecy:** Jurisdictions with limited transparency can make it harder to identify the real owners of assets.
  - **Weak international cooperation:** Poor information sharing allows criminals to move money beyond the reach of one country's agencies.
- **Informal Financial Networks**
  - **Hawala-type systems:** Informal money-transfer networks can be misused to move funds outside normal banking channels.
  - **Unregulated intermediaries:** Weakly regulated money-transfer businesses can become channels for illegal funds.

Financial Action Task Force (FATF)

The **Financial Action Task Force (FATF)** is an independent inter-governmental global watchdog that sets international standards to prevent money laundering, terror financing, and weapons proliferation.

- **Basic Facts & Evolution**
  - **Origin:** Established in **1989** during the G7 Summit held in Paris.
  - **Headquarters:** Secretariat based at the OECD headquarters in **Paris, France**.
  - **Membership:** Consists of **40 members** (including India and regional bodies like the European Commission and GCC). Over 200 jurisdictions globally have committed to implementing FATF standards through regional bodies.
- **Evolution of Mandate:**
  - **1989:** Focus began strictly on **Anti-Money Laundering (AML)**.
  - **2001:** Mandate expanded to Countering Financing of Terrorism (CFT) post-9/11 attacks.
  - **2008 / 2012:** Expanded to cover **Countering Proliferation Financing (CPF)**—stopping funds meant for Weapons of Mass Destruction (WMDs).
  - **2019 Onward:** Added regulation for **Virtual Assets / Cryptocurrencies**.

FATF Monitoring Lists (Grey vs. Black List)

- **Grey List:** This list includes countries that are actively working with the FATF to address strategic deficiencies in their regimes to counter money laundering, terrorist financing, and proliferation financing. Being placed under increased monitoring means the country has committed to resolve the identified strategic deficiencies within agreed timeframes Jurisdictions under Increased Monitoring.
  - As on June 13, 2025 countries placed in this list are Algeria, Angola, Bolivia, Bulgaria, Burkina Faso, Cameroon, Côte d'Ivoire, Democratic Republic of Congo, Haiti, Kenya, Lao People's Democratic Republic, Lebanon, Monaco, Mozambique, Namibia, Nepal, Nigeria, South Africa, South Sudan, Syria, Venezuela, Vietnam, Virgin Islands (UK), Yemen.
- **Blacklist:** This identifies countries or jurisdictions with serious strategic deficiencies to counter money laundering, terrorist financing, and financing of proliferation and calls for application of enhanced due diligence and counter measures.
  - In the current list jurisdictions subject to a Call for Action include Democratic People's Republic of Korea, Iran & Myanmar, as on June 13, 2025.

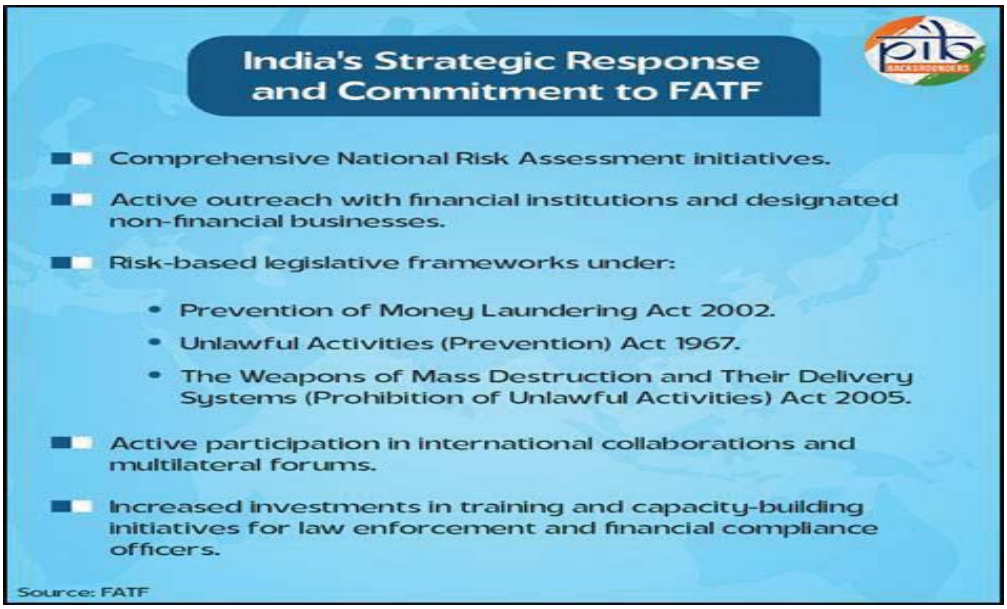
Mandates of FATF:

- **Set global standards:** FATF develops common international standards through its **40 Recommendations** to fight money laundering, terrorist financing and proliferation financing.
- **Identify financial threats:** It studies new methods, trends and risks through which criminals and terrorists misuse the financial system.
- **Assess countries:** FATF conducts **mutual evaluations and peer reviews** to check whether countries are effectively implementing its standards.
- **Track high-risk countries:** It identifies jurisdictions with serious weaknesses in their AML/CFT systems and encourages them to improve their safeguards.
- **Fight terrorist financing:** FATF helps countries **detect, disrupt and prosecute terrorist financing** and supports implementation of relevant UN Security Council measures.
- **Promote beneficial ownership transparency:** FATF works to prevent criminals from hiding illegal money behind **shell companies and complex ownership structures**.
- **Strengthen international cooperation:** It promotes sharing of financial intelligence and cooperation between countries because illegal money often moves across borders.

- **Address new financial risks:** FATF adapts its standards to emerging threats such as **virtual assets, cybercrime and new forms of terrorist financing**.
- **Work with regional bodies:** FATF works through **FATF-style Regional Bodies (FSRBs)** to promote implementation of its standards across the world.
- **Support capacity building:** It works with organisations such as the **IMF and World Bank** to provide technical assistance and strengthen countries' AML/CFT systems.

India at FATF – Timeline

- **2006:** India becomes an **Observer at FATF**.
- **November–December 2009:** FATF/APG team visits India for **on-site assessment of compliance with FATF's 40+9 Recommendations**.
- **23–25 June 2010:** FATF Plenary in Amsterdam discusses **India's Mutual Evaluation Report and membership**.
- **25 June 2010:** India is admitted as the **34th member of FATF**.
- **19 September 2024:** India is placed in the **"regular follow-up"** category, which is FATF's **highest rating category**.



Measures taken by FATF to tackle money laundering

- **Core Policy & Legal Measures**
  - **Risk-Based Approach:** Requires countries and banks to identify and assess their specific money laundering risks, allocating more resources to high-risk sectors.
  - **Criminalizing Money Laundering:** Mandates that member nations pass strict domestic laws defining money laundering as a serious crime.
  - **Confiscation & Asset Seizure:** Empowers law enforcement authorities to freeze, seize, and confiscate illegal proceeds and property tied to money laundering.
- **Preventive Measures**
  - **Customer Due Diligence KYC:** Mandates strict **Know Your Customer (KYC)** verification to confirm identity before opening bank accounts.
  - **Checks on Politically Exposed Persons:** Forces banks to conduct background checks and continuous monitoring on politicians, judges, and senior officials prone to corruption.

- **Suspicious Transaction Reporting:** Requires financial entities to immediately report suspicious or unusual financial movements to central authorities.
- **Travel Rule for Virtual Assets:** Mandates crypto exchanges and digital asset providers to gather and share sender and recipient details for virtual currency transfers.

- **Enforcement Mechanisms**
- **Country Mutual Evaluations:** Conducts thorough, peer-reviewed assessments to evaluate whether a country's anti-money laundering laws work effectively in practice.
- **Grey List:** Places countries with strategic deficiencies in their financial system on a public warning list.
- **Black List:** Applies severe economic and financial countermeasures against non-cooperative nations with major systemic failures (e.g., North Korea, Iran, Myanmar).
- **International Cooperation**
- **Mutual Legal Assistance:** Mandates cross-border cooperation between judicial authorities to freeze illegal assets, share evidence and extradite financial criminals.
- **Financial Intelligence Unit Networking:** Connects national Financial Intelligence Units (like India's FIU-IND) to exchange transaction data and financial intelligence seamlessly across borders.

Global efforts to combat global money laundering:

- **United Nations Conventions**
  - **Vienna Convention (1988):** The first major global treaty that mandated countries to make money laundering a criminal offense, initially focusing on illicit drug money.
  - **Palermo Convention (UNTOC, 2000):** Expanded money laundering offenses beyond drug crimes to cover all forms of organized crime, human trafficking, and extortion.
  - **UN Convention Against Corruption (UNCAC, 2003):** Focuses on tracing, freezing and recovering stolen state assets and bribe money hidden in offshore foreign banks.
  - **UN Office on Drugs and Crime (UNODC):** Operates the Global Programme against Money Laundering (GPML) to provide training, legal aid, and technical tools to developing nations.
- **Intelligence & Law Enforcement Networks**
  - **Egmont Group:** An international network connecting over 170 Financial Intelligence Units (FIUs) including India's FIU-IND, to securely share real-time financial intelligence across borders.
  - **INTERPOL's Financial Crime & Anti-Corruption Centre (IFCACC):** Coordinates global operations against transnational financial crimes, issues Red Notices against money launderers and helps track stolen assets.
  - **Europol & CARIN (Camden Asset Recovery Network):** Specialized regional networks that track illegal assets, seize properties and target dark-web money laundering schemes.
- **Banking & Financial Sector Initiatives**
  - **Basel Committee on Banking Supervision (BCBS):** Formulates global risk-management guidelines for central and commercial banks to prevent their systems from being exploited for money laundering.
  - **The Wolfsberg Group:** An association of major global commercial banks (e.g., HSBC, Citi, Barclays) that drafts voluntary, high-level Anti-Money Laundering industry standards for private banking.
- **Multilateral Financial Institutions**
  - **International Monetary Fund (IMF) & World Bank:** Provide technical assistance, run financial sector assessment programs and offer conditional loans to strengthen member states' domestic financial security systems.
  - **G20 High-Level Principles:** Promotes international tax transparency, digital finance regulation, and global implementation of Ultimate Beneficial Ownership (UBO) registers.

Way forward:

- **Regulating Digital Finance**
  - **Enforcement of the Travel Rule for Crypto:** Enforce full implementation of FATF Recommendation 15 across all Virtual Asset Service Providers (VASPs) to track senders and receivers of crypto transfers.

    UK FCA Travel Rule requires UK-registered crypto exchanges (e.g., Coinbase UK) to collect, verify and transmit sender/receiver identity data for all virtual asset transfers.

    o
  - **Regulate Unhosted Wallets:** Develop regulatory standards for Decentralized Finance platforms, stablecoins and unhosted crypto wallets that operate outside traditional banking systems.
  - **Combating Cyber Scams:** Develop targeted roadmaps to tackle transnational scams, deepfake financial fraud and AI-driven social engineering.

    Operation First Light (INTERPOL): A global crackdown across 61 countries targeting illicit call centers and deepfake voice-phishing leading to the freezing of $257 million in criminal assets.
- **Enhancing Ownership & Corporate Transparency**
  - **Mandatory Digital Ownership Registers:** Require all member countries to maintain verified, publicly accessible central registers for **Ultimate Beneficial Ownership**.

    Ministry of Corporate Affairs mandates the Indian companies to disclose individuals holding a "Significant Beneficial Ownership" (SBO) of 10% or more.
  - **Cracking Down on Offshore Havens:** Close legal loopholes that allow criminals to hide assets behind complex shell corporations and secrecy havens in the real estate sectors.
- **Policy Reforms in FATF**
  - **Depoliticizing FATF Listings:** Ensure Grey and Black List decisions remain purely evidence-based, avoiding geopolitical bias or selective targeting.
  - **Focus on real results:** Assess countries not just on laws, but on **actual convictions and seizure of illegal assets**.
- **Use of Technology**
  - **Deploying AI & Big Data Analytics:** Encourage national Financial Intelligence Units and banks to use AI and predictive algorithms to flag suspicious transactions in real time.

    FIU-IND's FINNET 2.0: India's Financial Intelligence Unit utilizing AI tools to process millions of Suspicious Transaction Reports and auto-flag high-risk money-laundering clusters.
  - **Cross-Border Data Sharing:** Create secure frameworks that allow financial institutions to share fraud data across borders without violating domestic data protection laws.
- **Stronger Multilateral Coordination**
  - **Tripartite Cooperation:** Build closer operational synergies between policy regulators (FATF), global police networks (INTERPOL) and legal assistance bodies (UNODC) to speed up asset recovery and extradition.
  - **Regulate Informal Networks:** Target cross-border Hawala systems, trade-based money laundering and underground banking channels through intelligence operations.

Conclusion:

Money laundering is a **global problem that needs a global response**. FATF and other international efforts can make financial systems **more transparent and harder for criminals to misuse**. The focus should be on **follow the money**, seize illegal assets and ensure that financial crimes have real consequences.

8. COUNTERFEIT CURRENCY NOTES AND BLACK MONEY AS A CHALLENGE TO INDIA'S INTERNAL SECURITY

Counterfeit currency and black money pose serious threats to India's **internal security** by undermining **economic stability,** fueling **organised crime** and **terror financing,** and **eroding public trust** in institutions. Their interplay strengthens **parallel economies, weakens governance,** and challenges **law-enforcement capabilities,** making them critical issues for **national security management.**

What is Counterfeit Currency?

Counterfeit currency refers to money **illegally produced to imitate genuine currency** with the intent to deceive individuals and institutions. The creation or circulation of **fake money** is a criminal offence globally, as it can destabilize **national economies** and finance criminal activities. Advances in **modern technology** have further enabled the production of **high-quality,** more sophisticated counterfeit notes.

What is Black Money?

In general, it denotes **income earned through illegal means** or **legitimate income that is deliberately concealed from tax authorities.** It represents **unaccounted wealth** on which mandatory **taxes have not been paid** to the government. Such money remains **outside official records** and is **absent from GDP calculations,** national income, and other key economic indicators.

Factors Responsible for Counterfeit Currency & Black Money

- **High cash dependency in informal sectors:** A predominantly **cash-based informal economy** increases reliance on high-value notes, making it easier for counterfeit currency to circulate undetected and providing fertile ground for **unaccounted transactions** that generate black money.
- **Technological advancements in printing & materials:** Access to **sophisticated printing technologies**, scanners, and specialized inks enhances the quality of fake notes, challenging **currency security systems** and enabling criminal networks to produce near-authentic counterfeit currency.
- **Weak financial compliance & benami practices:** Gaps in **financial regulation**, poor enforcement of tax norms, use of **benami accounts/assets**, and loopholes in disclosure systems help criminals hide illegal income, contributing to the growth of **black money ecosystems**.
- **Porous borders & smuggling routes:** Weak border surveillance and established **cross-border smuggling corridors** facilitate infiltration of counterfeit currency, often pushed by foreign networks aiming to destabilize India's **financial and internal security** environment.
- **Demand for anonymity & uneven digital penetration:** Preference for **anonymous transactions** in illegal trades, combined with limited access to digital payment infrastructure in rural and informal sectors, sustains the **cash-dominant parallel economy**, enabling both counterfeit circulation and black-money generation.

Counterfeit Currency Notes and Black Money as a Challenge to India's Internal Security:

- **Funding of Organised Crime & Insurgency**
  - **FICN** and **black money** fuel **criminal syndicates**, **drug trafficking**, and **hawala networks**, strengthening parallel power structures.
  - Layered and underground financial flows support **terror financing**, **arms procurement**, **logistics**, and **recruitment** for insurgent and extremist groups.
- **Economic Terrorism & Cross-Border Threats**
  - Circulation of FICN functions as **economic terrorism**, aimed at **destabilising markets**, **weakening the rupee**, and **hurting public confidence**.

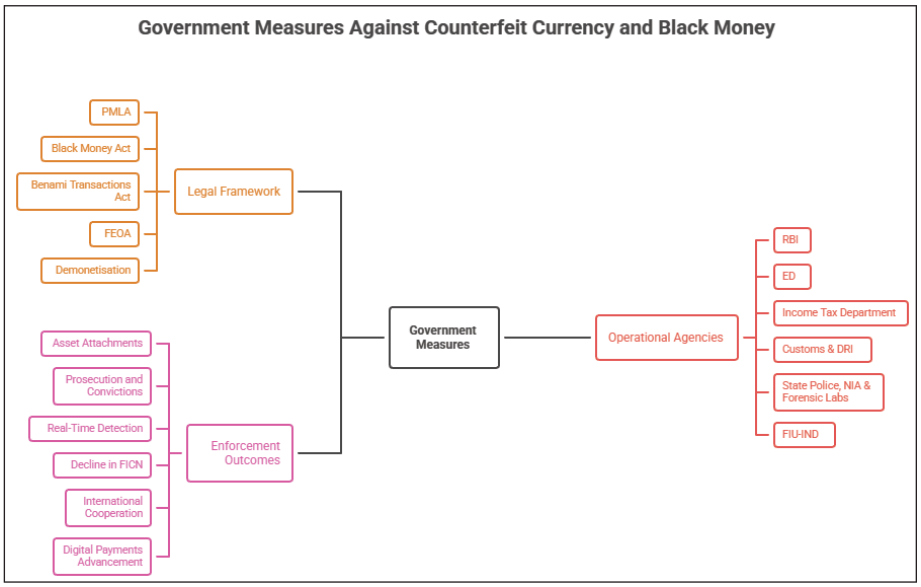
  - External state and non-state actors use counterfeit networks to fund covert operations against India.
- **Erosion of Governance & Institutional Integrity**
  - **Unaccounted cash** drives **corruption** in elections, procurement, policing, real estate, and licensing.
  - Weakens **rule of law**, reduces institutional credibility, and undermines democratic processes.
- **Economic Distortions & Parallel Economy**
  - Black money creates a large **informal, cash-dominant economy**, undermining fiscal policy and formal sector growth.
  - High-value transactions in **real estate, gold, benami assets**, and offshore accounts fuel speculative bubbles and reduce economic transparency.
- **Loss of Revenue & Fiscal Stress**
  - Massive tax evasion reduces **revenue mobilisation**, widening the **fiscal deficit**.
  - Forces the government to **increase taxes**, **borrow more**, or **cut developmental expenditure**, impacting welfare schemes.
- **Inflation, Inequality & Social Fragmentation**
  - Excess unaccounted cash increases demand for **luxury goods, property, gold**, causing **asset inflation**.
  - Hits poor households hardest by reducing purchasing power and widens **income and wealth inequality**, which extremist groups exploit for recruitment.
- **Operational Challenges for Law Enforcement**
  - **Money-laundering networks**, shell companies, **benami structures**, and **hawala** complicate tracing illicit flows.
  - High-quality counterfeits challenge detection systems; forensic gaps and coordination issues hinder prosecution.
- **Public Confidence & Market Trust Deficit**
  - Circulation of fake currency leads to **loss of trust** in cash transactions, increased black marketing, and illegal hoarding.
  - Non-reimbursement of FICN by banks causes recurring losses for businesses dependent on cash.

Government Measures against Counterfeit Currency and Black money

- **Legal Framework**
  - **Prevention of Money-Laundering Act (PMLA)**: Enables **attachment**, **freezing**, **prosecution**, and **confiscation** of proceeds of crime; strengthens the legal basis for targeting **money-laundering networks**.
  - **Black Money (Undisclosed Foreign Income and Assets) Act, 2015**: Addresses **illicit foreign holdings**, imposes stringent **penalties** and **prosecution**, and improves disclosure compliance.
  - **Benami Transactions (Prohibition) (Amendment) Act**: Expands powers to seize **benami properties**, strengthens adjudication, and deters concealment of black money through layered ownership.
  - **Strengthened penal provisions** under IPC, RBI Act, and Special Laws to curb **high-quality counterfeiting**, particularly involving cross-border networks.
  - **Fugitive Economic Offenders Act (FEOA)**: Enables confiscation of properties of offenders who flee the country to evade prosecution.
  - **Demonetisation (2016)**: Withdrawal of high-denomination notes aimed at disrupting **fake currency networks**, curbing **unaccounted cash**, and pushing the economy toward **digitisation and formalisation**.
- **Operational Agencies**
  - **Reserve Bank of India (RBI)**

  - Designs banknote **security features** (intaglio printing, security threads, watermark, latent images).
  - Issues **counterfeit detection protocols**, conducts surveillance on fake currency trends.
  - Mandates reporting of counterfeit detection by banks.
- **Enforcement Directorate (ED)**: Investigates **money-laundering**, attaches properties, and undertakes prosecution under PMLA and FEMA.
- **Income Tax Department**: Conducts searches, surveys, and detection of **undisclosed income**; analyses suspicious transactions via **data analytics**.
- **Customs & DRI**: Monitor **cross-border smuggling** of counterfeit currency, gold, narcotics, and hawala-related transactions.
- **State Police, NIA & Forensic Labs**: Detect counterfeit currency, dismantle local networks, enhance forensic capacity for note verification.
- **Financial Intelligence Unit (FIU-IND)**: Collects and analyses **Suspicious Transaction Reports (STRs)** and **Cash Transaction Reports (CTRs)** to identify laundering networks.
- **Terror Funding and Fake Currency (TFFC) Cell:** A Terror Funding and Fake Currency (TFFC) Cell has been constituted in **National Investigation Agency (NIA)** to conduct focused investigation of terror funding and fake currency cases.

- **Enforcement Outcomes**
  - **Massive asset attachments** under PMLA and Benami laws, reflecting enhanced enforcement capability.
  - Growth in **prosecution complaints**, **convictions**, and coordinated multi-agency raids on laundering networks.
  - Improved capability for **real-time detection** of counterfeit notes and penalties for banks failing to follow RBI norms.
  - Post-demonetisation, significant decline in **Fake Indian Currency Notes (FICN)** of old series; transition to redesigned notes with better security features.
  - Improved **international cooperation** through:
    - Mutual Legal Assistance Treaties **(MLATs)**
    - **FATF compliance**
    - **Information sharing** with foreign financial agencies
  - Advancement in **digital payments**, GST integration, and PAN-Aadhaar linkage helping reduce anonymity in financial transactions.



Way Forward:

- **Strengthen Currency Security & Detection:** Wider adoption of **digital authentication tools** like the **MANI app** to help citizens and merchants identify fake notes.
- **Improve Intelligence Integration:** Establish a **Counterfeit & Illicit-Cash Fusion Centre** bringing together **RBI, ED, Customs etc.,** enabling **real-time information sharing**.
- Target Hawala Networks & High-Risk Corridors
  - Strengthen **KYC/AML norms** across forex dealers, remittance agents, deploy **real-time monitoring systems** for cross-border remittances and high-value transactions.

    Intensify surveillance in border hotspots linked to FICN inflows (Indo-Nepal, Indo-Bangladesh sectors).
  - **Strengthen Prosecution & Enforcement:** Establish **fast-track PMLA and economic offence courts** for quicker trials.
  - **Promote Financial Inclusion & Formalisation:** Expand **digital payment infrastructure**, especially UPI, AEPS, and PoS machines in rural and unbanked regions.
  - **Public Awareness & Merchant Training:** Conduct sustained **awareness campaigns** on counterfeit detection and responsible cash handling.
  - **Legislative & Regulatory Refinements:** Periodic review of **AML/CFT frameworks** to plug emerging financial crime loopholes.

Conclusion:

Counterfeit currency and black money are not just economic concerns, they are critical **national security** threats that **weaken governance, empower criminal networks, and distort the economy**. A balanced strategy combining strong **legal frameworks, coordinated enforcement, technological upgrades**, and deeper **financial inclusion** is essential to safeguard India's internal security.

■■■

MGP/BP/GS3/040

COMMUNICATION NETWORKS/ROLE OF MEDIA

1. COMMUNICATION NETWORKS AND INTERNAL SECURITY

Communication Networks and Internal Security

Context:

India's first National Counter-Terrorism Policy and Strategy (PRAHAAR), released in February 2026, identifies the misuse of **drones, encryption and emerging technologies** by terrorist handlers, particularly in Punjab and Jammu and Kashmir, as a central operational challenge.

Introduction:

Communication networks are simultaneously India's most valuable infrastructure and its most exploited attack surface. With over **1.2 billion telecom subscribers**, the same networks that deliver welfare payments, banking and governance also carry the command, financing and propaganda traffic of terrorism, insurgency and organised crime.

What Counts as a Communication Network?

Hardware and software resources including mobile devices, laptops, sensors, servers, web applications, satellites, **SCADA industrial control systems**, LAN, WAN and optical fibre, providing the communication path between users, processes, applications, services and the internet.

Common Attack Types

Packet sniffing, **denial of service and distributed denial of service**, IP spoofing, man-in-the-middle and session hijacking, phishing including spear phishing and whale phishing, pharming, SQL injection, cross-site scripting, brute force and dictionary password attacks, eavesdropping, drive-by attacks, ransomware, and increasingly **AI-powered attacks**.

The statutory anchor: Section 70 of the IT Act, 2000 defines **Critical Information Infrastructure** as a computer resource "the incapacitation or destruction of which shall have debilitating impact on national security, economy, public health or safety." Communication networks are today treated not merely as carriers of CII but as CII themselves.

Why this has become the dominant vector:

- **Asymmetry of cost:** A hostile actor inflicts disproportionate damage at negligible expense, since a drone costs a fraction of the fence it flies over and a disinformation campaign costs less than a single armed operation.
- **Erasure of distance: Perpetrators operate from Cambodia or Pakistan** while the victim, the money and the harm are all inside India, which defeats a territorially organised police structure.
- **Deniability by design:** End-to-end encryption, cryptocurrency and layered proxies remove the attribution that conventional deterrence requires.

- **Speed of diffusion:** A rumour reaches a district faster than the district magistrate's clarification, so the response is structurally behind the threat.
- **Redistribution of power:** Because the technology is widely available and largely in private hands, small groups and individuals can now act against far larger and more complex adversaries.

Four Security Requirements: Any assessment of network security is organised around four properties, and each maps to a class of attack.

Requirement	Attack Class	Examples
Confidentiality	Unauthorised access to information	Eavesdropping, traffic analysis, packet sniffing, electromagnetic interception
Integrity	Unauthorised modification or theft	Man-in-the-middle, masquerade, replay, virus, worm, trojan horse, trapdoor
Availability	Denial of service	DDoS, resource exhaustion, equipment or software failure
Non-Repudiation	Denial or false claim of action	Service spoofing, after-the-fact denial

Classification of Threats:

- **Natural Threats:** Floods, earthquakes, tsunamis and volcanic activity can physically sever towers and cable, and **solar storms** can damage communication satellites, disabling weather forecasting, mobile services, DTH and telemedicine simultaneously.
  - In an ICT-driven smart city, a single earthquake severing local towers cascades into failure of power, water supply, ATMs and banking together.
- **Human-Induced Threats:** Insiders including disgruntled or socially engineered employees, criminal syndicates and terrorist outfits, and adversary nation states.

Threat Actor Taxonomy

- **Terrorists and Non-State Actors:** Seek psychological impact as much as physical damage, and now have access to educated recruits with working knowledge of networks and programming. Groups including ISIS and Lashkar-e-Taiba have developed their own secure communication applications.
- **Criminal Syndicates:** Motivated by monetary gain, and readily leveraged by any adversary state or terror group that wishes to rent their capability.
- **Nation States:** The most potent threat, given resources and persistence, and operating in the absence of any binding global norm restraining states from targeting each other's critical infrastructure.
- **Advanced Persistent Threats (APTs):** State-sponsored campaigns that are **sophisticated, targeted and prolonged**, using SQL injection, malware, spyware and phishing to infiltrate sensitive systems and remain undetected for years, leaving little forensic trace.
  - APT36 or Transparent Tribe, linked to Pakistan, is the most active APT against Indian defence and government targets.

Significance of Communication Networks:

- **Backbone of Critical Infrastructure**
  - **Interdependence Across Sectors:** Transport, government services and communications all depend on power, while the power sector itself depends on transport for fuel and on communications for grid data transmission

- A failure in one critical information infrastructure cascades rapidly across others, converting a technical fault into political, economic or national instability.
- **Financial Stability:** Net banking, ATM networks, payment systems and stock exchanges run entirely on communication infrastructure, so any breach becomes a question of financial stability rather than of IT.
- **Smart Systems:** Smart grids, smart cities and smart agriculture are by definition interconnected systems in which the network is the product, not an accessory.

- **Instruments of Security and Governance**
  - **Force Multiplier for Agencies:** Networked command, real-time intelligence sharing and digitised records let a smaller force cover more ground.

    NATGRID, MAC and the CCTNS-linked Inter-operable Criminal Justice System exist only because a communication backbone exists to carry them
  - **Chain of Command:** Security forces exchange intelligence horizontally and vertically over wireless systems, so an attack on that infrastructure can paralyse decision-making at the moment it is most needed.
  - **Border and Coastal Surveillance:** The Comprehensive Integrated Border Management System is fundamentally a communications architecture with sensors attached.
  - **Disaster Response:** Early warning, evacuation coordination and relief logistics all run on telecom, which is why network restoration is a first-order task after any disaster.
- **Development and Counter-Insurgency**
  - **Connectivity as Counter-Extremism:** Extending coverage into remote districts reduces the isolation that insurgent groups exploit and brings welfare delivery, banking and grievance redress within reach.
    - Dedicated mobile tower projects in Left Wing Extremism districts were conceived explicitly as a counter-insurgency measure rather than a telecom one.
  - **Digital Sovereignty and Public Trust:** Citizens will use DigiLocker, UPI and e-governance only if they believe the underlying networks are secure, so security is a precondition for digital adoption itself.

Challenges to Internal Security Through Communication Networks:

- **Terrorism and Insurgency**
  - **Encrypted Command and Control:** Handlers direct operations in real time over applications effectively invisible to lawful interception.

    A disguised application named “Calculator” recovered from the phones of terrorists in Jammu and Kashmir was being used to stay in contact with handlers in Pakistan-occupied Kashmir.
  - **Online Radicalisation and Recruitment:** Groups recruit remotely through curated propaganda and closed channels, producing self-radicalised operatives with no physical contact with any organisation.

    Encrypted Telegram channels were used by Islamic State sympathisers in South India to evade surveillance, and NIA cases increasingly involve cells assembled and financed entirely online.
  - **State-Linked Espionage Against Defence Personnel:** Malware delivered through socially engineered applications targets serving personnel directly.
    - APT36 deployed **CapraRAT** Android malware through fake dating applications to compromise military personnel and activists after the Pahalgam attack.
  - **Drone-Enabled Delivery:** Cheap, low-signature drones now deliver weapons, explosives and narcotics across the western border, and the same platforms serve for surveillance and attack.
  - **Terror Financing Through Digital Rails:** Cryptocurrency, hawala coordinated over messaging apps, crowdfunding fronts and mule accounts finance operations while defeating conventional financial surveillance.

- **Attacks on Critical Information Infrastructure:**
  - **Ransomware Against Public Institutions:** Attacks have shifted from small-scale extortion of individuals to targeted operations against entire organisations for larger payoffs.

    The AIIMS Delhi ransomware attack crippled hospital systems for days, and the Cosmos Bank cyber-attack demonstrated the exposure of the financial system.
  - **National-Scale Network Attacks:** A sufficiently large coordinated attack can bring a country's institutions to a standstill.

    Estonia, 2007, one of the world's most connected states with 98 per cent of banking transactions online, saw the networks of its banks, broadcasters, police, parliament and ministries brought down simultaneously.
  - **Industrial Control System Attacks:** SCADA systems governing power, water and nuclear facilities are reachable through networks and can be physically damaged remotely.

    Stuxnet and its role in disrupting Iran's nuclear programme remains the defining case of a cyber weapon producing physical destruction.
  - **Mass Data Exposure:** Health, identity and financial databases aggregate enormous quantities of sensitive information in single repositories.

    A German security firm reported medical records of over 120 million Indian patients exposed and freely accessible online.
- **Information Warfare and Social Media**
  - **Disinformation Triggering Violence:** Fabricated content spreads faster than verification and has repeatedly caused deaths and displacement.

    A fake video circulated on WhatsApp contributed to the 2013 Muzaffarnagar riots in western Uttar Pradesh, in which 62 people died and about 50,000 were displaced, and during the Manipur conflict of 2023-24 false content on WhatsApp and Instagram intensified ethnic violence.
  - **Mob Violence From Rumour:** Viral messages about child kidnappers on messaging platforms led to the lynching of around two dozen innocent people across several states.
  - **Bot Networks and Coordinated Inauthentic Behaviour:** Fake profiles and automated networks manipulate public opinion at scale, particularly around elections.

    Meta has reported networks of fake accounts originating in China spreading false content about India, and bot-driven hate campaigns targeting minorities were flagged in the pre-election period of 2024.
  - **Deepfakes and Synthetic Media:** Generative tools now permit convincing impersonation of officials and public figures, corroding the evidentiary value of audio and video itself.
  - **Scale of Exposure:** A Microsoft study found that **64 per cent of Indians had encountered fake news online**, among the highest rates recorded globally.
- **Hybrid and Fifth-Generation Warfare**
  - **The Doctrine:** Hybrid warfare fuses conventional and unconventional instruments of power and subversion, and was mapped as early as **1999 in *Unrestricted Warfare***, a publication of China's People's Liberation Army, which described a shift in the arena of violence from the military to the political, economic and technological.
  - **Five Domains:** Political warfare through interference in domestic politics, technological warfare, military warfare through irregular means, economic warfare through supply chain disruption and counterfeit currency, and social warfare exploiting existing cleavages through propaganda.

- **Data as a Weapon:** The collection of individually harmless granular information about persons, assembled into a relational database, enables targeted manipulation later.

  The Zhenhua Data case, in which a Chinese firm claiming to work with intelligence and military agencies was found profiling Indian individuals and institutions across politics, government, business, media and civil society, illustrates the model.

- **Espionage Against Strategic Sectors:** Chinese-linked group **Stone Panda** targeted Bharat Biotech and Serum Institute of India during COVID vaccine development, showing that commercial and scientific assets are now security targets.
- **Spyware Against Individuals: Pegasus**, developed by an Israeli cyber-arms firm, was reported to have compromised phones of activists, journalists and lawyers in India, extracting messages, calls and browser history including end-to-end encrypted data.

- **5G and Emerging Technology Risks**
  - **Expanded Attack Surface:** Security assessments indicate that 5G networks present roughly **200 times more attack vectors** than 4G.
  - **Software-Defined Vulnerability:** 5G moves from centralised hardware switching to distributed, software-defined digital routing, so an attacker who gains control of the managing software can control the entire network.
  - **Shared Infrastructure Risk:** Network slicing and shared infrastructure create the potential for cascading failure across multiple linked networks from a single successful intrusion.
  - **New Attack Types:** 5G is vulnerable to **mobile network mapping or MNmap attacks**, in which device information sent in plain text is harvested to build a map of everything connected to the network.
  - **Edge Computing Exposure:** Distributing computation to the network edge multiplies the number of points that must be defended.
- **Structural and Institutional Weaknesses**
  - **Cyber Security Is Not in the Seventh Schedule:** Cyberspace transcends geography and cyber security appears in none of the three Lists, so Union and state powers are undemarcated and central initiatives can attract state resistance.
    - NATGRID, conceived as an integrated intelligence database, was opposed by several state governments on precisely this federal ground.
  - **Supply Chain Dependence:** A large share of network hardware and software is imported and may carry backdoors, and the Indian Electrical and Electronics Manufacturers Association has flagged security risk in the power grid from foreign automation and communication systems.
  - **Private Ownership of Public Risk:** Telecom, banking, stock exchanges and energy utilities are largely privately operated, and firms treat auditing and compliance as an operating cost rather than a security obligation.
    - Losses from major breaches often amount to under one per cent of annual revenue, and less again after insurance, so the market provides weak incentive to invest in even low-cost security.
  - **Agencies Working in Silos:** Ministries, departments and private associations each maintain cyber units aligned to their own mandates, producing a fragmented approach with no single body assessing national threat.
  - **Inadequate Understanding of Interdependencies:** There is limited scientific analysis of inter-sector and intra-sector dependency, which is the primary reason preparedness lags complexity.
  - **Encryption Versus Lawful Access:** The first-originator traceability requirement under the IT Rules, 2021 is contested by platforms as incompatible with end-to-end encryption, and remains judicially unsettled.

- **Jurisdictional Fragmentation:** A single fraud can involve a call from one state, a victim in another, a mule account in a third and withdrawal in a fourth, requiring coordination among multiple police forces, banks and NPCI.
- **Underinvestment and Skill Deficit:** Cyber spending is scattered across MeitY, MHA and DoT with no trackable line, and state police lack cyber-forensic capacity, trained investigators and specialist prosecutors.
- **Outdated Doctrine:** The **National Cyber Security Policy dates from 2013**, and a National Cyber Security Strategy has been under preparation for years without release.

- **Rights and Federal Concerns**
  - **Internet Shutdowns:** India records among the highest numbers of shutdowns globally, imposing large economic and civil costs while evidence of security benefit remains contested.

    In Anuradha Bhasin v. Union of India (2020) the Supreme Court held that indefinite suspension is impermissible, that orders must be published and proportionate, and that they require periodic review
  - **Breadth of Executive Power:** The Telecommunications Act, 2023 carries an expansive definition of telecommunication and wide powers of interception and suspension exercised on executive authorisation without judicial warrant.
  - **Surveillance Creep:** The Telecom Cyber Security Rules introduced the **Telecommunication Identifier User Entity** concept permitting personal identification through phone numbers, and the late-2025 order mandating pre-installation of a government application on all handsets was withdrawn after public backlash.

Legal and Institutional Architecture:

Legislation

- **Telecommunications Act, 2023:** Replaces the 1885 and 1933 statutes, introduces an authorisation-based regime in place of licensing, and provides for **interception and temporary suspension** on grounds of public emergency or public safety in the interest of state security and public order.
- **Information Technology Act, 2000: Section 69** for interception, monitoring and decryption, **Section 69A** for blocking, **Section 70** defining critical information infrastructure and protected systems, and **Sections 70A and 70B**establishing NCIIPC and CERT-In. The **2008 amendment** introduced the cyber terrorism provision under Section 66F.
- **IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021:** Framed under Section 87(2), superseding the 2011 Rules, with due diligence obligations, grievance officers and the contested first-originator traceability requirement.
- **Digital Personal Data Protection Act, 2023 and Rules, 2025:** Governs personal data processing, with wide state exemptions on security grounds.
- **UAPA and Bharatiya Nyaya Sanhita:** Substantive offences for terrorism, terror financing and organised crime committed through digital means.

Rules and Directions

- **Telecommunications (Telecom Cyber Security) Rules, 2024:** The first rules under the 2023 Act, made under Sections 22(1) and 56(2)(v), superseding the 2017 IMEI tampering rules. Require a cyber security policy, a **Chief Telecommunication Security Officer**, incident reporting within prescribed timelines, and provision of traffic data to authorised agencies.
- **CERT-In Directions, 2022:** Mandatory reporting of specified incidents within six hours, 180-day log retention, and five-year retention of subscriber records by VPN, cloud and data centre providers.
- **National Security Directive on the Telecommunication Sector (2021):** Created the **trusted source and trusted product** regime through the Trusted Telecom Portal, with the National Cyber Security Coordinator as designated authority.

- **National Digital Communications Policy, 2018**: Seeks a data protection regime for digital communications, institutional accountability, security testing and certification, an encryption and data retention policy, and lawful interception capability.

Institutions:

Body	Parent	Function
Indian Cyber Crime Coordination Centre (I4C)	MHA	Nodal body for cybercrime response, National Cybercrime Reporting Portal, 1930 helpline, **Samanvay** platform for inter-state coordination and the Citizen Financial Cyber Fraud Reporting and Management System
CERT-In	MeitY	National incident response agency, threat advisories and cyber crisis management, with sectoral CSIRTs beneath it
NCIIPC	NTRO	Identification of all CII elements, strategic leadership across government, and monitoring, analysis and forecasting of national-level threats to CII
Cyber Swachhta Kendra	MeitY	Botnet Cleaning and Malware Analysis Centre, detecting botnet infections and providing free removal tools
National Cyber Security Coordinator	NSCS	Coordinates cyber security across ministries and administers the trusted telecom regime
Department of Telecommunications	Communications	Sanchar Saathi, Central Equipment Identity Registry, Digital Intelligence Platform, Financial Fraud Risk Indicator, spoofed international call detection
National Counter Ransomware Task Force	MHA	Dedicated response to the ransomware threat against Indian institutions

Government Measures:

- **Telecommunications Act, 2023 and Rules:** Modernised authorisation-based framework with explicit cyber security obligations, phased into force through 2024 to 2026.
- **Telecom Cyber Security Rules, 2024:** Mandatory security policy, designated security officer, timely incident reporting and data provision to authorised agencies.
- **Sanchar Saathi and CEIR:** Citizen platform for blocking lost or stolen handsets through the Central Equipment Identity Registry, reporting suspected fraud communication and identifying connections issued in one's name.
- **Digital Intelligence Platform and Financial Fraud Risk Indicator:** Secure coordination among telecom operators, banks and law enforcement, with mobile numbers classified by likelihood of association with fraud.
- **I4C Enforcement:** Blocking of around **seven lakh SIMs** used in fraud and over **1.11 lakh** items of suspicious online content during 2024-25.
- **Inter-Ministerial Committee on Southeast Asian Cybercrime:** Under the Special Secretary for Internal Security, addressing fraud originating from Cambodia, Myanmar and Laos.

- **Trusted Telecom Regime:** Trusted source and trusted product certification under the 2021 National Security Directive, alongside mandatory local certification and domestic equipment testing laboratories.
- **Cyber Swachhta Kendra:** Botnet detection and free malware removal tools for end users.
- **Blocking Powers:** Section 69A used to block applications, websites and accounts assessed as threats to sovereignty, security or public order.
- **Connectivity in Affected Areas:** Mobile tower programmes in Left Wing Extremism districts and BharatNet expansion.
- **PRAHAAR, 2026:** Whole-of-government framework covering technological capability, critical infrastructure protection, counter-radicalisation and financial disruption.

International Frameworks:

- **UNSC Counter-Terrorism Committee Delhi Declaration (2022):** On countering the use of new and emerging technologies for terrorist purposes, adopted at the special meeting held in India.
- **FATF Guidance on Virtual Assets (2018):** Extended anti-money laundering obligations to virtual asset service providers.
- **Global Counterterrorism Forum, Berlin Memorandum:** Good practices on countering terrorist use of unmanned aerial systems.
- **Christchurch Call to Action:** Commitment against terrorist and violent extremist content online, signed by India.
- **Budapest Convention:** The principal international instrument on cybercrime, which India has not acceded to, citing concerns over data sharing and sovereignty.

Way Forward:

- **Build Legal and Doctrinal Clarity**
  - **Issue a National Cyber Security Strategy:** Replace the 2013 policy with a current strategy on the people, processes and technology framework, covering critical infrastructure, offensive capability, incident response and institutional responsibility.
  - **Resolve the Encryption Question Openly:** Settle traceability through law and judicial scrutiny rather than through contested rules, since operational certainty benefits both agencies and platforms.
  - **Introduce Proportionality Safeguards:** Bring interception and suspension under independent authorisation with published reasons and periodic review, consistent with *Anuradha Bhasin* and *Puttaswamy*.
  - **Consider a Fake News Statute Carefully:** Comparative models such as Singapore's **Protection from Online Falsehoods and Manipulation Act, 2019** exist, but any Indian equivalent must be measured against Article 19(2) and the *Shreya Singhal* standard on vagueness.
- **Match Resources to the Threat**
  - **Create a Dedicated and Trackable Cyber Budget:** Consolidate allocations currently scattered across MeitY, MHA and DoT into an identifiable line so spending can be measured against loss.
  - **Scale Cyber Forensics to the District:** Establish district-level cyber-forensic units and expand National Forensic Sciences University capacity, since evidence collection is the binding constraint on conviction.
  - **Build Specialist Cadres:** Recruit cyber investigators, analysts and prosecutors as career streams rather than rotational police postings.
- **Strengthen Technical Capability**
  - **Adopt a Predict, Prevent, Detect and Respond Cycle:** Exposure analysis, defensive deployment to reduce attack surface, continuous monitoring for intrusion, and structured breach analysis, rather than reacting only after compromise.

  - **Counter-Drone Systems on Priority:** Institutionalise detection, jamming and interception along the western border where drone delivery is now routine.
  - **Deploy AI for Anomaly Detection:** Move from tracking known bad actors to flagging anomalous behaviour in traffic, transactions and content at scale.
  - **Indigenise Critical Equipment:** Extend Make in India to core network components, with 5G rollout built on domestic equipment and software wherever feasible, and servers for critical data located in India.
  - **Mandate CII Audits and Drills:** Sectoral red-teaming and incident exercises for power, banking, telecom, health, transport and defence networks.
- **Reform Public-Private Responsibility**
  - **Share the Defence Burden:** Private operators know their own architecture best while government agencies know more about state-sponsored malware, so responsibility for defending the most sensitive systems must be jointly held.
  - **Move Government From Regulator to Partner:** Build trust and confidence sufficient for genuine information sharing, since compliance-driven relationships produce minimum disclosure.
  - **Correct the Incentive Structure:** Introduce cyber insurance frameworks, disclosure obligations and liability rules so that under-investment carries a cost.
- **Address the Transnational Dimension**
  - **Treat Cybercrime as Foreign Policy:** Pursue extradition arrangements, mutual legal assistance reform and joint operations with states hosting scam compounds, since domestic enforcement cannot reach them.
  - **Deepen Platform Cooperation:** Establish binding, time-bound response obligations for intermediaries on lawful requests, with transparency reporting on both sides.
  - **Shape Global Norms:** Engage the Budapest Convention debate, the UN cybercrime process, FATF and Quad technology working groups to write rules rather than only respond to them.
- **Build Societal Resilience**
  - **Digital Literacy at Scale:** Awareness is the cheapest defence against social engineering, and the sharp fall in digital arrest cases followed public campaigns as much as enforcement.
  - **Counter-Narrative Rather Than Only Takedown:** Blocking removes content but does not address the grievance that made it persuasive.
  - **Institutionalise Rapid Rebuttal:** Fact-checking and official clarification must move at the speed of the rumour, particularly in communally sensitive districts.

Case Studies:

Case	What happened	Result and Lesson
Estonia, 2007	Coordinated attacks brought down the networks of banks, broadcasters, police, parliament and ministries in one of the world's most connected states	Practically brought a country to a standstill without a shot fired, and prompted NATO to establish its cyber defence centre in Tallinn. The founding case for treating networks as critical infrastructure
Muzaffarnagar Riots, 2013	A fake video circulated on WhatsApp inflamed communal tension in western Uttar Pradesh	62 people died and about 50,000 were displaced. Establishes that disinformation is not a nuisance but a proximate cause of mass violence

APT36 and CapraRAT (2025)	Pakistan-linked group ran cyber-espionage against Indian defence personnel after Pahalgam, using Android malware delivered through fake dating applications	Showed that state-linked actors now target individual soldiers directly through consumer platforms, bypassing every perimeter defence the institution maintains
Digital Arrest Scams (2022-2025)	Syndicates impersonating police and central agencies extorted victims over video calls, with roughly 2.41 lakh cases and Rs 3,012 crore lost	Nearly 46 per cent traced to Cambodia, Myanmar and Laos, and cases fell about 86 per cent in 2025 after enforcement and awareness. Awareness works, but the source lies outside Indian jurisdiction
Anuradha Bhasin v. Union of India (2020)	Challenge to the communications shutdown in Jammu and Kashmir	Indefinite suspension held impermissible, orders must be published, proportionate and periodically reviewed. Network access is a constitutional question, not purely an executive prerogative
AIIMS Ransomware Attack	Hospital information systems crippled, disrupting patient services for days	Demonstrated that public health infrastructure is critical information infrastructure, and that recovery capability matters as much as prevention
Zhenhua Data	A Chinese firm claiming links with intelligence and military agencies was found maintaining relational profiles of Indian individuals and institutions	Illustrates hybrid warfare through aggregation of individually harmless public data, and the near-impossibility of enforcing privacy law across jurisdictions
Sanchar Saathi Pre-Installation Order (2025)	Order requiring the government application to be pre-installed and non-removable on all handsets sold in India	Withdrawn after public backlash. Security measures without public consent and proportionality invite reversal, whatever their technical merit

Conclusion

Communication networks have collapsed the distance between an adversary's intent and India's vulnerability, so that a handler in one country, a scam compound in another and a victim in an Indian district now occupy the same operational space. India's response has been substantial in law, with a new telecom statute, cyber security rules, a trusted equipment regime and a published counter-terror doctrine, but it remains thin in capability, The problem is also **constitutional along with technical.** Every instrument that lets the state see hostile traffic also lets it see the citizen, and the legitimacy of the security architecture will **ultimately rest on whether interception, traceability and suspensio**n are exercised with the legality, necessity and proportionality the Constitution now requires.

2. ROLE OF SOCIAL MEDIA IN INTERNAL SECURITY

Introduction:

Social media refers to **internet-based platforms and applications** that allow people and organizations to create, share, and interact with content and with each other in virtual communities and networks. Social

media essentially empowers individuals to be both consumers and creators of content, leading to dynamic, interconnected online communities.

Examples include **social networking sites (Facebook, LinkedIn),** media-sharing platforms (Instagram, YouTube, TikTok), microblogging (X/Twitter), forums (Reddit), and review sites (TripAdvisor, Yelp).

Features of Social Media:

- **User-Generated Content**: Users produce and upload text, photos, videos, stories, live streams, and other media, forming the core of platforms.
- **Profiles and Networking:** Customizable personal/organizational profiles with unique web addresses; connect via friends, followers, groups, or lists to build social networks.
- **Interactivity and Engagement:** Two-way communication through likes, comments, shares, reactions, direct messaging, group chats, and forums for conversations and feedback.
- **Real-Time Sharing and Notifications:** Instant posting with timestamps; push notifications and activity feeds keep users updated on interactions and new content.
- **Virality and Sharing:** Content spreads rapidly via shares, reposts, and algorithms promoting popular material for exponential reach.
- **Privacy and Security Controls:** Customizable settings for visibility (public/private), secure login (e.g., MFA), and data protection options.
- **Analytics and Insights:** Built-in tools for tracking performance, audience demographics, engagement metrics, and trends.
- **Multimedia and Versatility:** Support for diverse formats (text, images, audio, video); responsive design across devices (mobile/desktop).

Importance of Social Media as a Tool for Internal Security Agencies:

- **Real-Time Intelligence Gathering and Surveillance:**
  - **Open Source Intelligence (OSINT):** Agencies constantly monitor public posts, trends, hashtags, and geographic check-ins to gain real-time insights into potential threats, public sentiment, and developing unrest.
  - **Predictive Policing and Threat Mapping:** Analyzing patterns of communication, association, and location data helps security forces map out criminal networks and predict potential hotspots for violence or protest.
  - **Digital Footprint Analysis:** Profiles provide crucial forensic evidence and intelligence on suspects, their location, contacts, movements, and future plans.
- **Crisis Management and Communication:**
  - **Rapid Dissemination of Facts:** During natural disasters, security breaches, or law-and-order situations, government and police departments use social media to swiftly broadcast verified information, counter rumors, and issue public safety warnings.
  - **De-escalation:** Security forces can use dedicated social media channels to interact with concerned citizens, address grievances publicly, and proactively work to de-escalate tensions before they turn violent.
  - **Emergency Response Coordination:** Platforms serve as vital communication links for coordinating disaster relief and emergency services when traditional networks might be overloaded or compromised.
- **Public Engagement and Image Building:**
  - **Citizen Feedback:** Agencies use social media for direct interaction with citizens, allowing for anonymous tips, faster complaint resolution, and gathering feedback on policing efforts.

  - **Transparency and Trust:** Regular updates on police action, successful arrests, and community outreach programs help build public trust and enhance the overall image of law enforcement agencies.
- **Counter-Narrative Strategy:**
  - **Fighting Extremism:** Security agencies and governments develop sophisticated counter-narrative campaigns designed to expose the hypocrisy of extremist groups, highlight the positive value of democratic institutions, and discourage youth radicalization.
  - **Truth Verification:** Establishing official handles for fact-checking and debunking misinformation immediately limits the viral spread of malicious content.

Risks & Threats to Internal Security:

- **Radicalization and Extremism:**
  - **Rapid Indoctrination:** Terrorist and extremist organizations (e.g., state-sponsored actors, insurgents) use platforms to rapidly expose susceptible individuals to extremist ideologies, bypassing traditional gatekeepers.
  - **Lone Wolf Recruitment:** Social media facilitates direct, private communication between recruiters and isolated individuals, leading to the self-radicalization and planning of "lone wolf" attacks, which are difficult to detect.
  - **Propaganda Dissemination:** High-quality, emotive propaganda (videos, manifestos) is shared instantly, generating sympathy, and glorifying violence among potential recruits.
- **Coordination of Unlawful Activities:**
  - **Logistical Planning:** Platforms are used for encrypted communication (Direct Messages, private groups) to coordinate the logistics of illegal activities, including protests, riots, arms/drug trafficking, and human smuggling.
  - **Mobilization and Velocity:** They enable the immediate mobilization of large crowds, giving little reaction time for law enforcement to deploy counter-measures during flash mobs, violent protests, or communal riots.
- **Misinformation, Disinformation, and Psychological Warfare (IW):**
  - **Fomenting Communal Tensions:** Malicious actors deliberately spread fabricated stories or manipulated videos (deepfakes) to incite hatred, distrust, and violence between different communities.
  - **Erosion of Public Trust:** Disinformation campaigns targeting government institutions (police, military, judiciary) are launched to undermine public faith, legitimacy, and stability.
  - **Hybrid Warfare:** State and non-state actors use coordinated campaigns to interfere in domestic political processes, influence elections, and create internal chaos.
- **Cyber and Infrastructure Vulnerabilities:**
  - **Phishing and Espionage:** Social engineering tactics are widely employed via social media to target critical government personnel, extracting credentials or sensitive information that compromises national networks.
  - **Data Leakage:** Inadvertent sharing of sensitive location data or routine work details by government employees can be exploited by foreign intelligence agencies for physical or cyber espionage.

Various government initiatives to regulate social media:

- **Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021:** The IT Rules, 2021 are the most comprehensive and controversial set of regulations targeting social media platforms (defined as "Intermediaries") and digital news publishers. They create a layered framework of obligations based on the size and nature of the platform:

- **Mandatory Due Diligence for all Intermediaries:** All social media companies must observe certain due diligence requirements, including:
  - **Clear User Agreements:** Informing users not to host, display, or share prohibited content (e.g., content that is obscene, defamatory, harmful to minors, or threatens national unity).
  - **Grievance Redressal:** Publishing the name and contact details of a Grievance Officer, who must acknowledge a complaint within 24 hours and resolve it within 15 days.
- **Enhanced Obligations for Significant Social Media Intermediaries (SSMIs):** SSMIs (platforms with over 50 lakh, or 5 million, registered users) face stricter compliance requirements:
  - **Appointment of Key Personnel:** SSMIs must appoint the following India-based officers:
    - **Chief Compliance Officer (CCO):** Responsible for ensuring compliance with the IT Act and Rules.
    - **Nodal Contact Person:** For 24x7 coordination with law enforcement agencies.
    - **Resident Grievance Officer (RGO):** For handling user complaints.
  - **Proactive Monitoring of Content:** Platforms must deploy technology-based tools to proactively identify and remove content related to child sexual abuse material and content depicting rape.
  - **Voluntary User Verification:** Platforms must offer users a mechanism to voluntarily verify their accounts.
  - **Tracing the Originator (First-Message Identification):** For messaging platforms (like WhatsApp or Telegram), the rules mandate tracing the originator of a message deemed unlawful by a court or competent government authority. This requirement is subject to ongoing legal challenges due to privacy concerns.
- **Regulation of Digital News and OTT Platforms (Digital Media Ethics Code):** This section extends regulation to over-the-top (OTT) streaming services (like Netflix and Amazon Prime) and digital news media, mandating a three-tier regulatory structure:
  - **Self-Regulation by Publishers:** Adherence to a specified "Code of Ethics."
  - **Self-Regulatory Bodies:** Formation of industry bodies to oversee the first tier.
  - **Oversight Mechanism:** The Ministry of Information and Broadcasting (MIB) acts as the final oversight body.
- **Digital Personal Data Protection Act (DPDP Act, 2023):** Although the DPDP Act applies broadly across all sectors, it fundamentally redefines how social media platforms (as "Data Fiduciaries") handle the data of Indian citizens ("Data Principals"):
  - **Consent Mandate:** Requires platforms to obtain clear, explicit, and informed consent from users before processing their personal data.
  - **Data Minimisation:** Platforms can only collect data that is necessary for a specific, lawful purpose.
  - **Right to Erasure:** Grants users the right to request the deletion or correction of their personal data.
  - **Data Breach Notification:** Mandates timely reporting of data breaches to the Data Protection Board of India and affected users.
  - **Global Transfer Restriction:** While allowing cross-border data transfer, it retains the power for the government to restrict transfers to specific countries deemed unsafe.
- **The Proposed Digital India Act (DIA):** The DIA is the proposed successor to the decades-old IT Act, 2000, and is intended to create a future-ready regulatory framework specifically tailored for the Web 3.0 era:
  - **New Regulatory Categories:** It proposes to classify online intermediaries into new categories (e.g., social media intermediaries, e-commerce, and search engines) with tailored obligations.
  - **Harmful Content Focus:** Expected to explicitly define and mandate quick action against deepfakes, algorithmic bias, and cyber-security threats.

  - **Open Internet Principles:** Aims to promote innovation while ensuring user rights and safety, potentially addressing issues like net neutrality and competition.
- **Blocking Orders:** Under Section 69A of the IT Act, the government retains the power to issue content-blocking orders to platforms for reasons of national security, sovereignty, or public order. This power has been frequently used to block specific accounts or content during periods of unrest or border tension.
- **Financial Scrutiny:** The Ministry of Finance often issues advisories to social media platforms to crack down on financial scams, crypto-related fraud, and illegal lending apps that proliferate on their sites.

Way forward:

- **Regulatory and Legal Enhancements:**
  - Institutionalize a National Social Media Policy with clear guidelines on content moderation, traceability, and accountability for platforms (e.g., expand IT Rules 2021 to mandate AI-driven proactive detection of extremist content).
  - Update cybersecurity policies (e.g., National Cyber Security Policy) for real-time monitoring of radicalization, fake news, and disinformation; make group admins liable for viral harmful content.
- **Technological and Intelligence Measures:**
  - Deploy advanced OSINT tools, AI analytics, and surveillance (e.g., CERT-In enhancements) for 24/7 monitoring of terror propaganda, honey-traps, and foreign influence operations.
  - Collaborate with platforms for algorithm transparency and swift takedowns under Section 69A IT Act.
- **Public Awareness and Capacity Building:**
  - Launch nationwide digital literacy campaigns in schools/colleges on fact-checking, cyber ethics, and recognizing misinformation/radicalization.
  - Train security agencies and build specialized cyber units; promote responsible user behavior via incentives.
- **International and Ecosystem Cooperation:**
  - Negotiate global standards for cross-border regulation; work with tech firms on “open and agile” anti-terror strategies.
  - Balance privacy with security through judicial oversight to avoid overreach.

Conclusion:

Thus, social media can strengthen internal security & turn into a security asset while mitigating risks like propaganda & terror recruitment when regulated effectively through surveillance, fact-checking, and collaboration with platforms.

3. ROLE OF MEDIA AND INTERNAL SECURITY

Context:

During Operation Sindoor (May 2025), The Ministry of Defence issued an advisory **prohibiting all media channels, digital platforms and individuals** from live or real-time coverage of defence operations and movement of security forces, warning that premature disclosure could assist hostile elements and endanger personnel. Coverage was restricted to periodic briefings by a designated officer until operations concluded.

Introduction:

The Constitution contains **no separate provision for press freedom**. It flows from **Article 19(1)(a)**, and is therefore subject to the same reasonable restrictions under **Article 19(2)**, which expressly include the

sovereignty and integrity of India, the security of the State, public order and incitement to an offence. Press freedom in India is thus constitutionally protected but never absolute, and internal security is one of the grounds on which it may be limited.

- **Positive Role of Media:**
  - **Watchdog Over the Security Establishment:** Investigative reporting has exposed procurement irregularities, intelligence lapses and operational failures that internal review would not have surfaced.

    Post-Kargil scrutiny of equipment shortages and intelligence gaps contributed to the political pressure that produced the Kargil Review Committee.
  - **Building Public Resolve:** Sustained, accurate coverage sustains national morale and public support for difficult security decisions.

    Kargil in 1999 was India's first televised conflict, and coverage of the campaign built broad public backing and recognition for the armed forces.
  - **Countering Adversary Narratives:** Credible domestic reporting and official briefings shape international perception, which matters when an adversary is running a parallel information campaign.
  - **Denying Space to Rumour:** Where verified information reaches the public quickly, unverified claims lose their audience, so a functioning press is a counter-misinformation instrument.
  - **Crisis and Disaster Communication:** Broadcast media remains the fastest channel for evacuation instructions, mock drill guidance and public safety advisories, particularly where internet access is limited.
  - **Voice for Affected Populations:** Reporting from conflict-affected regions brings grievance into public debate, which is a precondition for political rather than purely military resolution.
  - **Accountability for Excess:** Independent reporting on the conduct of security forces protects both civilians and the institutional legitimacy of the forces themselves.

Challenges:

- **Compromising Live Operations**
  - **Real-Time Intelligence to the Adversary:** Continuous telecast of an ongoing operation hands the opposing side a live feed of force positions and tactics.

    During 26/11, live coverage of the NSG operations was monitored by handlers in Pakistan who relayed information to the attackers inside the buildings.
  - **Endangering Personnel and Hostages:** Disclosure of assault positions, entry points and rescue attempts places both operators and hostages at direct risk.
  - **Persistence of the Problem:** The reissuing of the advisory in May 2025, seventeen years after 26/11, indicates that self-restraint has not become institutional habit.
- **Sensationalism and the Commercial Incentive**
  - **Ratings-Driven Coverage:** The competition for attention rewards speed, drama and volume over verification, which is structurally at odds with security reporting.

    During 26/11, live coverage of the NSG operations was monitored by handlers in Pakistan who relayed information to the attackers inside the buildings.
  - **Speculation Presented as Fact:** Source-based claims and unattributed speculation during an active operation can force the state into premature disclosure or public denial.
  - **The Oxygen of Publicity:** Terrorism is designed for an audience, so disproportionate coverage delivers precisely the psychological effect the perpetrator sought.

Social Cohesion and Communal Risk

- **Polarising Formats:** Prime-time debate formats structured around confrontation can deepen communal and regional cleavages, particularly during and after a security incident.
- **Identification of Communities:** Framing an incident around the identity of perpetrators rather than the act itself risks collective attribution and reprisal.
- **Amplification of Unverified Claims:** Mainstream repetition converts a fringe rumour into a national controversy, which is more damaging than the original claim.

Justice and Individual Rights

- **Trial by Media:** Parallel adjudication in the studio prejudices investigation and can compromise the accused's right to a fair trial under Article 21.
- **Exposure of Identities:** Publication of the identities of witnesses, informers, victims, undercover personnel or the families of security personnel creates direct physical risk.

Structural Weaknesses

- **Fragmented Regulation:** Print, broadcast and digital news are governed by three different regimes with different standards and different enforcement capacity.
- **A Toothless Print Regulator:** The Press Council of India has jurisdiction only over print and can do little beyond admonition, censure or warning.
- **Access and Dependence in Conflict Zones:** Journalists in insurgency-affected areas depend on either security forces or armed groups for access, which compromises independence in both directions.
- **Journalist Safety:** Reporters in conflict regions face intimidation and violence from multiple sides, which produces self-censorship as much as courage.
- **Paid News and Media Capture:** Ownership concentration, planted stories and commercial dependence weaken the independence on which the watchdog function rests.

Regulatory Framework

Constitutional and Statutory

- **Article 19(1)(a) and 19(2):** Press freedom is derived from free speech and restricted on grounds including sovereignty and integrity, security of the State, public order and incitement to an offence.
- **Cable Television Networks (Regulation) Act, 1995 and Rules:** The **Programme Code** governs broadcast content, and **clause 6(1)(p)**, introduced in response to 26/11, restricts live coverage of anti-terrorist operations to **periodic briefings by an officer designated by the appropriate government** until the operation concludes.
- **Press Council of India Act, 1978:** Establishes the PCI as a statutory self-regulatory body for print, with the Norms of Journalistic Conduct and powers limited to warning, admonishing or censuring.
- **IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021:** Part III applies a three-tier structure to digital news publishers, comprising self-regulation, a self-regulatory body and an oversight mechanism.
- **Official Secrets Act, 1923:** Penalises disclosure of prohibited information relating to defence and security.
- **Bharatiya Nyaya Sanhita:** Offences including acts endangering the sovereignty, unity and integrity of India, promoting enmity between groups, imputations prejudicial to national integration, and statements conducing to public mischief.
- **Prasar Bharati Act, 1990:** Governs the public broadcaster and its autonomy.

Self-Regulation

- **News Broadcasting and Digital Standards Authority:** Industry self-regulatory body for television news, issuing guidelines on coverage of terrorism, communal incidents and sting operations.
- **Government Advisori**es: Ministry of Information and Broadcasting and Ministry of Defence advisories, as issued during Operation Sindoor, operate alongside the statutory framework.

Judicial Position:

- **Romesh Thappar v. State of Madras (1950):** Established that press freedom is implicit in Article 19(1)(a) and that restrictions must fall within Article 19(2).
- **Bennett Coleman v. Union of India (1973):** Held that measures affecting circulation and volume can amount to restrictions on press freedom, even without direct censorship.
- **Sahara India Real Estate v. SEBI (2012):** Recognised a **doctrine of postponement** of publication, permitting courts to defer reporting temporarily where it would prejudice the administration of justice, as a proportionate alternative to prohibition.
- **Mohammed Ajmal Kasab v. State of Maharashtra (2012):** The Supreme Court held that the unrestricted live telecast of the 26/11 operations caused immeasurable harm and endangered lives, and observed that the coverage was of no public interest that could outweigh that risk.
- **Anuradha Bhasin v. Union of India (2020):** Affirmed that freedom of the press is protected and that restrictions on communication must be published, proportionate and subject to periodic review.
- **Kunal Kamra v. Union of India, Bombay High Court (2024):** Struck down the amendment to the IT Rules empowering a government Fact Check Unit to identify fake news about government business, holding it unconstitutional, which established that the state cannot be the sole arbiter of truth about itself.

Case Studies:

Case	What happened	Result and Lesson
Kargil, 1999	India's first televised conflict, with sustained coverage from the front	Built national resolve and public recognition for the forces, while raising concerns about disclosure of positions. Established that media is a strategic actor in conflict, not an observer
26/11 Mumbai, 2008	Continuous live telecast of NSG operations, monitored by handlers in Pakistan directing the attackers	The Supreme Court in Kasab held the coverage caused immeasurable harm, and clause 6(1)(p) restricting live coverage of anti-terror operations entered the Programme Code. The defining case in this topic
Operation Sindoor, May 2025	MoD advisory of 9 May prohibiting live coverage, alongside channels airing old Israeli footage as Indian strikes and misidentifying an Indian citizen as a terrorist	Showed both that the regulatory lesson of 26/11 has been codified and that it has not been internalised. Regulation without professional culture does not produce restraint

Sahara v. SEBI, 2012	Question of whether courts may restrain reporting that prejudices proceedings	Produced the doctrine of postponement, a proportionate middle path between free reporting and prohibition. The template for balancing security and speech
Kunal Kamra v. Union of India, 2024	Challenge to the government Fact Check Unit under the IT Rules	Struck down as unconstitutional, establishing that the state cannot be the sole arbiter of truth about its own functioning. A limit on how far counter-misinformation powers may extend

Way Forward:

- **Regulatory Reform**
  - **Create a Unified Media Council:** Replace the print-only Press Council with a single body covering print, broadcast and digital news, with genuine powers of enforcement and statutory independence from both government and industry.
  - **Codify an Operations Protocol:** Move from episodic advisories to a standing, published protocol on reporting anti-terror and military operations, specifying a designated briefing officer, pool arrangements, delayed telecast and an agreed embargo mechanism.
  - **Introduce Proportionate Consequence:** Provide for graded, appealable penalties for demonstrably false broadcast during a security operation, framed narrowly enough to survive Article 19(2) scrutiny.
- **Building Professional Capacity**
  - **Train for Conflict Reporting:** Institutionalise training on operational security, verification standards, trauma-sensitive reporting and identity protection, in partnership with the forces and journalism institutions.
  - **Adopt Structured Embedding:** Develop an Indian embedded-journalism framework with agreed ground rules, which gives access in exchange for operational discipline rather than leaving both to negotiation under pressure.
  - **Strengthen Self-Regulation:** Give the NBDSA and equivalent bodies real sanction capability, since credible self-regulation is the strongest argument against statutory control.
- **Getting Government Communication Right**
  - **Fill the Vacuum First:** The most effective control on speculation is timely, credible official information, and the structured daily briefings during Operation Sindoor showed this works better than restriction alone.
  - **Institutionalise a Crisis Communication Cell:** Provide a single, trained and empowered point of contact during any security incident, so that journalists are not driven to unverified sources.
  - **Protect Journalists:** Ensure the safety of reporters in conflict areas, since intimidation-driven self-censorship damages the state's own credibility.
- **Public Resilience**
  - **Media Literacy:** Build critical consumption into school and college curricula, so that audiences discount sensationalism rather than reward it.
  - **Transparency on Ownership and Funding:** Disclosure requirements reduce the risk of capture and let audiences weigh what they are reading.

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MGP/BP/GS3/041

ORGANISED CRIME & BORDER MANAGEMENT

1. KEY CHALLENGES IN INDIA'S BORDER MANAGEMENT

Introduction:

Border management refers to the **coordinated implementation of national policies by various government agencies** at and beyond a country's land, sea, and air frontiers. It is a critical component of national sovereignty and security. It is the s**et of policies, laws, institutions, and operational practices used by a country** to secure and regulate its borders while allowing legitimate movement of people and goods.

Core Functions of border management:

Modern border management involves balancing security needs with economic and humanitarian obligations, requiring integration across several key areas:

- **Security & Control:** This is the fundamental function, focused on preventing threats from entering or leaving the territory.
  - **Customs & Immigration:** Screening and processing travelers (visa checks, passport verification) and cargo to detect illegal entry, unauthorized goods, and persons of interest.
  - **Preventing Smuggling and Trafficking:** Interdicting illicit trade (drugs, weapons, counterfeits) and human trafficking operations.
  - **Terrorism Prevention:** Utilizing intelligence and technology (like facial recognition and behavioral analysis) to identify and stop potential security threats before they cross the border.
- **Trade and Travel Facilitation:** A country's economic vitality depends on the efficient flow of legitimate goods and people.
  - **Trade Logistics:** Implementing simplified customs procedures, electronic documentation, and risk-based inspections to speed up the passage of commercial cargo.
  - **Travel Efficiency:** Employing technologies (like automated gates and pre-clearance programs) to minimize wait times for low-risk travelers while maintaining security standards.
  - **Compliance:** Ensuring that all imports and exports adhere to international trade agreements, sanitation standards, and domestic tax laws.
- **Public Health and Safety:** Borders are the first line of defense against global threats like pandemics and agricultural diseases.
  - **Quarantine and Inspection:** Screening imports of food, plants, and animals to prevent the entry of invasive species or contagious diseases (e.g., foot-and-mouth disease, avian flu).
  - **Health Checks:** Implementing health screenings and documentation checks for travelers during global health crises.

- **Sovereignty and Territorial Integrity:** Upholding the physical boundaries and legal jurisdiction of the state.
  - **Border Demarcation:** Physically marking and maintaining the internationally recognized boundary line.
  - **Patrol and Surveillance:** Monitoring the border zone (especially porous land and maritime borders) to detect unauthorized crossings and illegal settlements.

India's international border:

- India shares borders with seven countries, totaling approximately 15,100 kilometers of land border and a coastline of ~11,000 kilometers.
- The complexity of these borders stems from diverse terrains (high mountains, deserts, plains, and marshlands), historical disputes, and geopolitical interests.

Neighboring Country	Length (km)	Key Indian States Bordered	Key Border Feature/Dispute
Bangladesh	4,096.7	West Bengal, Assam, Meghalaya, Tripura, Mizoram	Longest land border; highly porous; includes the *Teen Bigha Corridor*.

China	3,488	Ladakh (UT), Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh	Known as the **Line of Actual Control (LAC)**; major disputes over Aksai Chin and Arunachal Pradesh.
Pakistan	3,323	Gujarat, Rajasthan, Punjab, Jammu & Kashmir, Ladakh (UT)	Divided into the **International Border (IB)** and the **Line of Control (LoC)** in J&K/ Ladakh.
Nepal	1,751	Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Sikkim	Mostly open border facilitating free movement; some minor territorial disputes (e.g., Kalapani).
Myanmar	1,643	Arunachal Pradesh, Nagaland, Manipur, Mizoram	Hilly, rugged terrain; largely managed by the **Free Movement Regime (FMR)** (currently under review).
Bhutan	699	Sikkim, West Bengal, Assam, Arunachal Pradesh	Peaceful and friendly border; major security focus is on preventing insurgency and smuggling.
Afghanistan	106	Ladakh (UT) (through Pakistan-occupied Kashmir)	Shortest border; disputed as it lies entirely within the territory currently controlled by Pakistan.

Key challenges in India's border management:

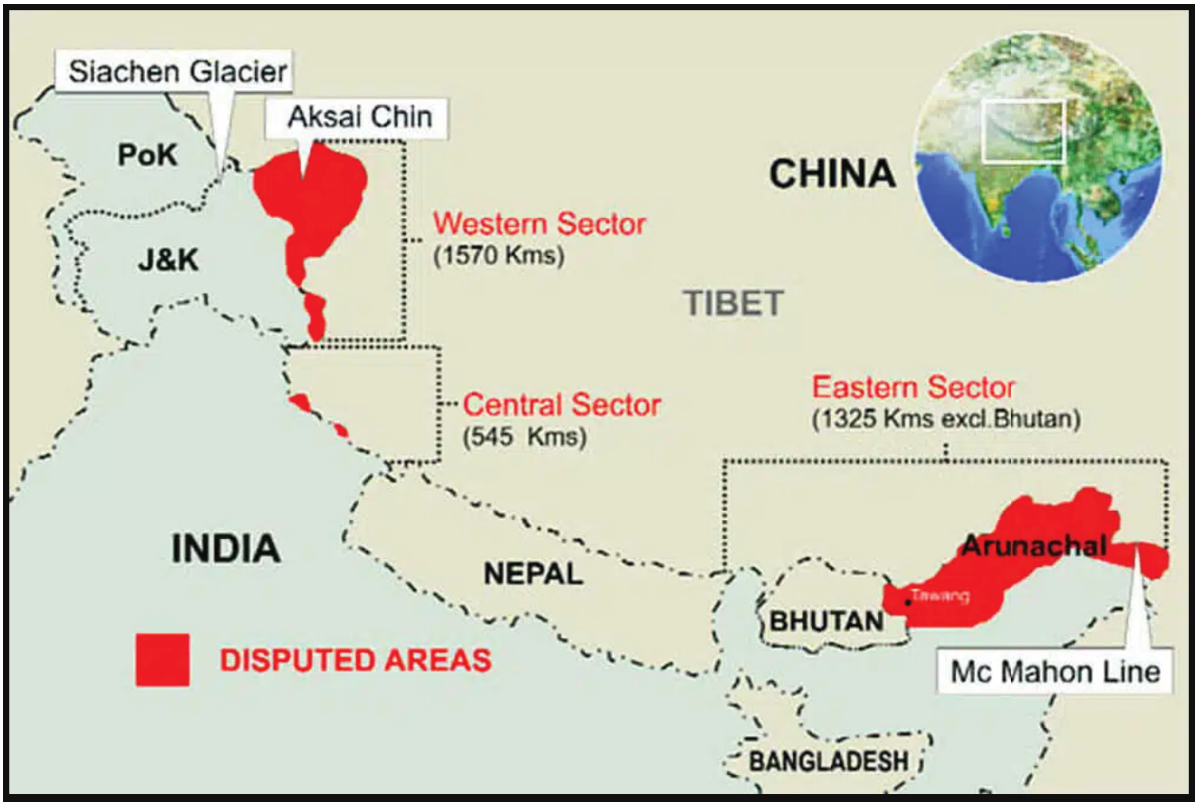
Security and Geopolitical Challenges	These challenges stem directly from historical conflicts, territorial disputes, and the constant threat of infiltration and external aggression.
	• **Active Border Disputes:**
	  • **Line of Actual Control (LAC) with China:** The LAC is largely undemarcated, leading to frequent standoffs and military build-ups, particularly in the Ladakh and Arunachal Pradesh sectors. This forces India to maintain a constant, high-alert military presence in inhospitable, high-altitude environments.
	  • **Line of Control (LoC) with Pakistan:** The LoC is highly volatile, characterized by cross-border terrorism, infiltration attempts, and cease-fire violations, requiring extensive use of counter-infiltration barriers and continuous surveillance.
	• **Cross-Border Terrorism and Insurgency:**
	  • **Infiltration:** Terrorist groups frequently attempt to cross the LoC and the International Border (IB) to destabilize internal security.
	  • **Insurgency:** Certain border regions, especially in the Northeast (along the Myanmar border) and Kashmir, struggle with local insurgency movements that receive support or safe haven from across the border.
	• **Narco-Terrorism and Smuggling:**
	• **Golden Crescent and Golden Triangle:** India is strategically located between the "Golden Crescent" (Afghanistan, Pakistan, Iran) and the "Golden Triangle" (Myanmar, Laos, Thailand), making it a significant transit point for narcotics trafficking, which often funds terrorist activities.

Geographical and Environmental Challenges	India's borders traverse some of the most difficult and diverse terrains on Earth, complicating effective policing and infrastructure development.
	• **Varied and Difficult Terrain:**
	  • **High Altitudes:** The Himalayas (bordering China, Nepal, Bhutan) are prone to extreme weather, avalanches, and landslides. Deployment and logistical support are exceedingly difficult and costly.
	  • **Riverine/Marshy Terrain:** The border with Bangladesh includes riverine areas and the Sundarbans Delta, which are often prone to flooding and are highly porous due to water bodies and dense vegetation.
	  • **Desert and Forests:** The Thar Desert (Pakistan border) and dense forests (Myanmar border) limit visibility and mobility, but also provide cover for infiltrators and smugglers.
	• **Porous and Unfenced Stretches:**
	  • **Open Borders:** The borders with Nepal and Bhutan are historically open due to deep cultural and economic ties. While mutually beneficial, this open nature is frequently exploited for smuggling, arms trafficking, and the movement of anti-national elements.
	  • **Unfenced Segments:** Due to topographical limitations (mountains, rivers, deep jungles), many critical stretches remain unfenced, requiring constant manual patrolling.

Administrative and Logistical Challenges	These issues relate to the implementation, coordination, and human resources required to maintain security.
	• **Multiplicity of Border Guarding Forces (BGFs):** India utilizes different BGFs for different borders, sometimes leading to command and coordination issues:
	  • **BSF:** Pakistan and Bangladesh borders.
	  • **ITBP:** China border.
	  • **SSB:** Nepal and Bhutan borders.
	  • **Assam Rifles:** Myanmar border.
	• While specialization is beneficial, a lack of unified command and coordination mechanisms can sometimes lead to gaps in intelligence sharing and resource allocation.
	• **Infrastructure and Technological Gaps:**
	  • **Fencing Maintenance:** Fences, especially along the Bangladesh border, suffer damage from weather, erosion, and vandalism, requiring continuous and expensive repair.
	  • **Integrated Surveillance:** Full operationalization of the **Comprehensive Integrated Border Management System (CIBMS)**—which uses smart fences, sensors, radars, and thermal imagers—is an ongoing process and faces challenges related to terrain complexity and system integration.
	• **Socio-Economic Issues in Border Areas:**
	  • **Smuggling as a Livelihood:** Poverty in remote border regions often drives locals into cross-border smuggling of goods (cattle, daily commodities) or participation in illegal activities, complicating BGFs' efforts.
	  • **Humanitarian Concerns:** Managing migration, especially on the western and eastern fronts, requires balancing security mandates with humanitarian responsibilities.

Specific border challenges & various initiatives for border management:

- **India-China Border:**



- **Challenges:**
  - **The Undemarcated Nature of the LAC:**
    - **Differing Perceptions:** Both sides have different perceptions of where the LAC runs, leading to frequent "transgressions" (military patrols entering the perceived territory of the other side).
    - **Conflict Flashpoints:** This ambiguity is the primary cause of conflict, particularly in key friction areas like Depsang Plains, Hot Springs, Pangong Tso, and Tawang.
  - **China's Superior Infrastructure:**
    - **Rapid Deployment:** Extensive road networks, high-speed rail links, and fully operational airfields near the LAC allow the Chinese military (PLA) to rapidly mobilize and deploy troops and heavy equipment to forward positions.
    - **All-Weather Connectivity:** China has built numerous all-weather tunnels and bridges, whereas Indian forces often rely on difficult mountain roads prone to seasonal closure.
  - **Geographical and Climatic Extremes:**
    - **Logistical Bottleneck:** Sustaining troops at altitudes above 15,000 feet requires complex and expensive logistical supply chains, often relying on airlifts.
    - **Harsh Conditions:** Extreme cold, low oxygen levels, and the constant threat of avalanches and blizzards lead to high costs in specialized equipment, habitat construction, and medical care for personnel.
  - **Grey Zone and Salami Slicing Tactics:**
    - **Incremental Encroachment:** This involves slowly pushing the limits of the LAC perception, often through short-duration patrols, construction of temporary structures, or the use of civilian herdsmen, forcing Indian forces to react defensively to minor changes.

- **Psychological Warfare:** The use of information warfare and sustained diplomatic pressure to influence international opinion also forms a tactical challenge.

- **Initiatives:**
  - **Front-Loaded Infrastructure Development:**
    - **Prioritization of Strategic Roads:** The **Border Roads Organisation (BRO)** has accelerated projects, including the construction and upgrade of key roads like the Darbuk-Shyok-Daulat Beg Oldie (DSDBO) road in Ladakh.
    - **Tunnel Construction:** Projects like the **Sela Tunnel** (Arunachal Pradesh) and **Zojila Tunnel** (Jammu & Kashmir/Ladakh) aim to provide all-weather, year-round connectivity to forward areas, bypassing treacherous, snowbound passes.
    - **Bridge Strengthening:** Rapid construction of double-lane, high-load bridges capable of supporting heavy artillery and tank movements.
  - **Technological and Surveillance Upgrades:**
    - **Integrated Surveillance:** Deployment of sophisticated surveillance equipment, including high-resolution cameras, radars, and thermal imagers, to cover difficult-to-patrol "gaps" and provide round-the-clock visibility.
    - **Drone Capabilities:** Massive procurement and indigenization of surveillance (ISR) and combat drones to monitor vast, uninhabited stretches and provide rapid reconnaissance.
    - **Fiber Optic Networks:** Laying robust, high-bandwidth fiber optic cables to forward posts to ensure secure and fast communication, replacing older satellite-based systems.
  - **Strategic Deployment and Force Modernization:**
    - **Mountain Strike Corps:** Strengthening and operationalizing the specialized **Mountain Strike Corps (MSC)**, designed for offensive operations in mountainous terrain.
    - **ITBP Modernization:** The **Indo-Tibetan Border Police (ITBP)**, the primary BGF for the LAC, is being re-equipped with specialized clothing, vehicles, and advanced weapons suitable for high-altitude deployment.
    - **Forward Post Upgrades:** Building permanent, insulated, and structurally sound habitats for troops in forward locations to improve living conditions and minimize non-battle casualties.
  - **Diplomatic and Institutional Mechanisms:**
    - **Working Mechanism for Consultation and Coordination (WMCC):** A diplomatic framework used to discuss and resolve immediate issues on the border and coordinate military disengagement.
    - **Strict Adherence to Protocols:** While military readiness is paramount, India emphasizes adherence to border protocols and Confidence-Building Measures (CBMs) to ensure that standoffs do not spiral into larger conflicts.
    - **International Outreach:** Leveraging diplomatic forums to highlight the importance of respecting international law and territorial integrity in the region.

- **India-Pakistan Border:**
  - **Challenges:**
    - **Cross-Border Terrorism and Infiltration (LoC and IB):**
      - **LoC Volatility:** The LoC is the primary route for the infiltration of foreign terrorists supported by state and non-state actors. Ceasefire violations, though reduced, are intended to provide cover fire for these infiltrations.
      - **Targeting Civilians:** Terrorist groups target security forces and civilian infrastructure in border states, necessitating counter-insurgency and counter-terror operations deep within Indian territory.
    - **The Threat of Cross-Border Tunnelling (IB):** In the Jammu, Punjab, and Rajasthan sectors of the IB, the flat, agricultural, or sandy terrain is exploited for digging sophisticated cross-border tunnels.

    - **Detection Difficulty:** These tunnels, often deep and wide enough for movement, bypass the physical fence and are extremely difficult to detect using conventional methods.
    - **Smuggling Route:** Tunnels serve as conduits for arms, ammunition, and narcotics, posing a direct threat to internal security.
  - **Geographical and Environmental Extremes:**
    - **High Altitude (LoC):** The LoC runs through high mountains (Kargil, Siachen) that require specialized winter warfare training, expensive logistical support, and pose high risks due to avalanches and extreme cold.
    - **Riverine and Marshy Gaps (IB):** Areas in Punjab and Gujarat (like the Rann of Kutch) have riverine crossings and marshlands that cannot be permanently fenced, creating porous entry points exploited by smugglers.
  - **Drone and UAV Incursions:**
    - **Payload Delivery:** Drones are used to drop arms, drugs, and cash over the fence into India, particularly in the Punjab and Jammu sectors, circumventing ground patrols.
    - **Counter-Technology Gap:** Defending against small, low-flying drones requires expensive, multi-layered anti-drone technology, which is still being fully deployed.
  - **Narco-Smuggling (IB):** Due to India's proximity to the 'Golden Crescent,' the Pakistan border is a critical entry point for narcotics, often linked to the funding of terrorist activities (Narco-Terrorism).

- **Initiatives:**
  - **Physical Obstacles and Fence Construction:** The physical security barrier is the most visible initiative along the International Border (IB).
    - **Barrier System:** The construction of a robust physical barrier, often reinforced with concertina wire, sensor equipment, and floodlights, has significantly reduced infiltration across the IB.
    - **Anti-Tunneling Technology:** Deployment of advanced ground-penetrating radar (GPR) and other subsurface surveillance systems to detect and neutralize tunnels before they become operational.
  - **Comprehensive Integrated Border Management System (CIBMS):** CIBMS is a flagship initiative to replace constant human patrolling with a technological grid in difficult stretches.
    - **"Smart Fencing":** This involves using a mix of sensors, including laser fences, unattended ground sensors, thermal imagers, and specialized battlefield surveillance radars.
    - **Real-time Response:** Data from these sensors is transmitted in real-time to a central command and control center, enabling rapid response to any breach or attempted infiltration.
  - **Dedicated Forces and Command Structure:**
    - **Indian Army (LoC):** The Army is responsible for security along the LoC, focusing on counter-infiltration and tactical operations in mountainous terrain.
    - **Border Security Force (BSF) (IB):** The BSF is the primary Border Guarding Force (BGF) along the IB, responsible for policing, manning border posts, and anti-smuggling/anti-tunnelling duties.
  - **Coastal and Riverine Security:**
    - **Marine Wings:** The BSF has established specialized marine wings and utilizes high-speed boats and patrol craft to secure riverine and marshy areas, particularly in Gujarat and West Bengal.
    - **Coastal Radar Chain:** Deployment of advanced radar chains along the coastline to prevent infiltration or movement by sea.
  - **Counter-Drone Measures:**
    - **Anti-Drone Systems:** Deployment of indigenous and imported anti-drone technology, including jammers, kinetic interception systems, and software to take over and land hostile UAVs.
    - **SOPs:** Establishing detailed Standard Operating Procedures (SOPs) for the BSF to react swiftly and effectively to drone sightings and incursions.
- **India-Bangladesh Border:**
  - **Challenges:**

- **Trans-Border Crimes and Smuggling:** The sheer length and population density make this frontier highly vulnerable to organized criminal activities.
  - **Cattle Smuggling:** This is one of the most prominent crimes, often involving violence and contributing to the illegal cross-border economy.
  - **Contraband and Counterfeiting:** Smuggling of narcotics (like Phensedyl), fake Indian currency notes (FICN), gold, and essential commodities is rampant.
  - **Human Trafficking:** The border is a major hub for trafficking women and children, often driven by economic disparity and exploitation.
- **Illegal Immigration and Demographic Pressure:** Although the flow has reduced over the decades, illegal migration remains a significant concern, particularly in the border states of West Bengal and Assam
  - **Porous Nature:** The lack of comprehensive fencing and the presence of riverine gaps facilitate unauthorized crossings.
  - **Socio-Political Impact:** High levels of undocumented migration place pressure on resources and lead to complex political and social tensions in the receiving states.
- **Fencing Gaps and Riverine Terrain:**
  - **Unfenced Segments:** Due to land acquisition issues, disputed pockets, and civilian resistance, several critical stretches remain unfenced, creating easy access points for infiltrators and smugglers.
  - **Riverine Gaps:** Approximately 250 kilometers of the border is riverine or marshy (especially in West Bengal and Tripura), making physical fencing impossible. These gaps require intensive water-based patrolling and technological solutions.
- **Border Haats and Enclaves (Pre-2015):** While the Enclaves issue (111 Indian enclaves in Bangladesh and 51 Bangladeshi enclaves in India) was resolved in 2015, challenges persist:
  - **Residual Issues:** While the Enclaves were exchanged, managing the newly demarcated boundary and integrating the populations into the respective systems requires continued administrative effort.
  - **Border Haats (Markets):** These bilateral markets, established for local trade, can sometimes be misused for facilitating minor illegal activities and require careful BSF monitoring.

- **Initiatives:**
  - **Comprehensive Border Fencing and Floodlighting:**
    - **Phased Fencing:** India is implementing a phased program to complete the border fencing, using double-row barbed wire fences and physical barriers, especially in highly populated areas.
    - **Floodlighting:** Continuous floodlighting along the completed stretches of the border allows the Border Security Force (BSF) to maintain visibility 24/7, deterring illegal activities at night.
  - **Upgrading Infrastructure and Checkpoints:**
    - **Integrated Check Posts (ICPs):** The development of modern, streamlined ICPs (like Petrapole) provides state-of-the-art facilities for customs, immigration, cargo handling, and trade, making legal crossings faster and more efficient.
  - **Technological Solutions (CIBMS and Riverine Patrol):**
    - **CIBMS (Comprehensive Integrated Border Management System):** Select vulnerable and unfenced stretches are being equipped with CIBMS technology, including thermal imagers, battlefield surveillance radars, and laser barriers to detect intrusion in real-time.
    - **Riverine Management:** Deployment of high-speed interception and patrol boats, combined with floating Border Out Posts (BOPs), to establish effective control over the riverine sections of the border.
  - **Bilateral Cooperation and Institutionalization:**
    - **Coordinated Border Management Plan (CBMP):** India and Bangladesh signed the CBMP to jointly tackle common challenges, including smuggling and infiltration, through simultaneous coordinated patrols by the BSF and the Border Guard Bangladesh (BGB).
    - **Confidence Building Measures (CBMs):** Regular flag meetings between the BSF and BGB at various levels (Sector Commander, Director General) ensure prompt communication and conflict resolution on the ground.

- **India-Nepal Border:**

- **Challenges:**
  - **Exploitation of the Open Border by Non-State Actors:** The primary concern is the use of this porous, unfenced border by criminal and hostile elements as a transit route.
    - **Third-Country Nationals:** The open border can be exploited by terrorists, extremists, or individuals from third countries to illegally enter Indian territory, bypassing regular immigration checks at airports and seaports.
    - **Insurgent Movement:** The route has historically been used by various militant/insurgent groups operating in India's Northeast and North to travel to foreign bases or seek safe haven.
  - **Smuggling and Economic Crime:** The absence of comprehensive border checks makes the border a major conduit for illicit trade and economic crimes.
    - **Contraband Smuggling:** Smuggling of high-value items, gold, narcotics, and forest products is rampant, particularly in remote areas.
    - **Counterfeit Currency:** The border is often used for the entry of Fake Indian Currency Notes (FICN) into the Indian financial system.
  - **Human Trafficking and Exploitation:** The open nature and the socio-economic vulnerabilities in both countries make this border a key hub for human trafficking.
    - **Vulnerable Populations:** Women and children are frequently trafficked across the border for exploitation, particularly from the more remote regions of Nepal.
  - Border Demarcation Disputes:
    - **Major Disputed Areas:** The most prominent disputes concern the **Kalapani-Limpiyadhura-Lipulekh** tri-junction area (involving Nepal, India, and China) and the **Susta** region.
    - **Local Resistance:** Disputes and political sensitivity often lead to resistance from local populations against any efforts to fence, patrol, or set up surveillance infrastructure in disputed zones.
  - **Infrastructure and Connectivity Gaps:** Poor infrastructure and communication networks in the remote border areas hinder effective deployment and rapid response capabilities for India's border forces.
- **Initiatives:**
  - **Specialized Border Guarding Force (SSB):**
    - **Sashastra Seema Bal (SSB):** The SSB is the primary Border Guarding Force (BGF). Their role is to patrol the border, monitor entry/exit points, and conduct civic action programs to build trust with local communities, relying heavily on intelligence gathering rather than physical barriers.
    - **Vulnerability Mapping:** The SSB regularly updates the vulnerability profile of border segments, focusing patrol intensity on known transit routes for criminals and smugglers.
  - **Integrated Check Posts (ICPs):**
    - **Modernizing Entry Points:** ICPs are established at major crossing points (like Raxaul, Jogbani, Sonauli) to integrate all necessary checks—customs, immigration, security, and quarantine—into a single, state-of-the-art facility.
    - **Facilitating Trade:** These posts streamline the legitimate flow of goods and people while allowing for the effective inspection of cargo and verification of identity documents, thereby curbing the entry of FICN and high-value smuggling.
  - **Surveillance and Technological Monitoring:**
    - **CCTV and Intelligence:** Deployment of CCTV cameras and advanced surveillance equipment at identified sensitive crossing points and known criminal transit areas.
    - **Data Sharing:** Enhanced intelligence sharing mechanisms between Indian security agencies (like R&AW and IB) and the SSB to track potential threats attempting to use the open border.
  - **Strengthening Bilateral Mechanisms:**
    - **Coordination Meetings:** Regular high-level meetings (Home Secretary-level talks, Director General-level talks between BSF/SSB and their Nepali counterparts) are held to discuss border issues, coordinated patrols, and anti-trafficking efforts.
    - **Joint Teams:** Establishment of Joint Technical Teams (JTTs) to physically survey and fix missing or damaged boundary pillars to prevent localized disputes from escalating.
- **India-Myanmar Border:**
  - **Challenges:**

- **The Free Movement Regime (FMR):** Until its recent decision to suspend the FMR, this regime was the biggest hurdle to effective border management.
  - **FMR History:** The FMR allowed tribes living along the border to travel up to 16 km into the other country without a visa, facilitating kinship ties and local trade.
  - **Exploitation:** This regime was heavily exploited by cross-border criminals, insurgents, and drug traffickers, allowing them to easily blend in with legitimate border residents and evade security checks.
- **Insurgency and Safe Havens:** The mountainous terrain and lack of effective border fencing have been crucial factors enabling India's internal security problems.
  - **Cross-Border Sanctuary:** Various Indian Insurgent Groups (IIGs), particularly those active in Manipur and Nagaland, operate camps and safe havens deep inside the territory of Myanmar, making counter-insurgency operations difficult.
  - **Infiltration and Exfiltration:** Insurgents frequently use the porous border for movement, procurement of arms, and communication.
- **Drug Trafficking (Proximity to the Golden Triangle):** The border serves as a major entry point for illicit drugs into India, posing severe societal and security threats.
  - **Source of Narcotics:** Due to its proximity to the **Golden Triangle** (the junction of Myanmar, Laos, and Thailand), the border is flooded with heroin, opium, and synthetic drugs (like methamphetamine tablets, or Yaba).
  - **Arms and Drug Nexus:** The drug trade provides significant funding for the Indian insurgent groups operating in the region, creating a dangerous nexus.
- **Refugee and Migration Crisis:** The political and civil unrest in Myanmar, especially after the 2021 military coup and subsequent civil war, has led to significant refugee influxes.
  - **Mass Movement:** India's border states, particularly Mizoram, have seen tens of thousands of refugees (mainly from the Chin community, who share ethnic ties with the Mizos) crossing over, straining local resources and administrative capacity.
- Rugged Terrain and Lack of Infrastructure:
  - **Dense Forest and Mountains:** The thick jungle and steep mountains make effective fencing, road building, and the establishment of permanent Border Out Posts (BOPs) extremely difficult and costly.
  - **Lack of Visibility:** The terrain hinders manual surveillance, necessitating reliance on expensive and complex technological solutions.

- **Initiatives:**
  - **Suspension of the Free Movement Regime (FMR):**
    - **Decision to Fence:** The Indian government has announced its decision to suspend the FMR and initiate a comprehensive plan to fence the entire 1,643 km border.
    - **Enhanced Security Checks:** This move will introduce compulsory visa and passport requirements and establish permanent check points at all legal entry locations, bringing the border management closer to the systems used on the western and eastern borders.
  - **Dedicated Border Guarding Force:**
    - **Assam Rifles:** The paramilitary force operates under the operational command of the Indian Army and is the primary force deployed along this frontier.
    - **Counter-Insurgency Role:** The Assam Rifles performs the dual task of counter-insurgency within the states and border guarding, often coordinating closely with the Army and local police.
  - **Infrastructure Development:**
    - **Border Roads:** India is investing heavily in building new roads and upgrading existing infrastructure (like the Kaladan Multi-Modal Transit Transport Project, though focused on connectivity, impacts security logistics) to ensure swift troop and equipment movement.
  - **Bilateral Security Cooperation:**
    - **Coordination Against IIGs:** India seeks the cooperation of the Myanmar military (Tatmadaw) to act against IIG camps operating in Myanmar's territory.
    - **Intelligence Sharing:** Maintaining intelligence pipelines to monitor the movement of arms, drugs, and insurgents across the border.

- **India-Bhutan Border:**
  - **Challenges:**
    - **Smuggling and Trans-Border Crime:**
      - **Wildlife and Forest Products:** The border runs through ecologically sensitive regions, leading to the smuggling of endangered wildlife, timber, and medicinal plants.
      - **Narcotics:** The route is sometimes used for the trafficking of low-intensity drugs and contraband, exploiting the lack of physical fencing in the difficult terrain.
    - **Strategic Vulnerability and Tri-Junction Point:**
      - **Doklam Plateau Proximity:** The western end of the Bhutan border is near the sensitive tri-junction area of Doklam, where India, Bhutan, and China's interests converge. India maintains a strong military presence to ensure the security of its strategic Siliguri Corridor (Chicken's Neck) which is relatively close to the border.
      - **Mountainous Access:** The high-altitude, rugged terrain of the Eastern Himalayas makes surveillance and infrastructure development extremely challenging.
    - **Historical Presence of Insurgent Groups:**
      - **Safe Havens:** Groups like the United Liberation Front of Asom (ULFA) and the National Democratic Front of Bodoland (NDFB) utilized the dense forests of Bhutan as safe havens and training grounds to launch attacks into Assam. While this challenge was addressed successfully, vigilance remains paramount.
    - **Open Border and Regulatory Gaps:**
      - **Unregulated Movement:** Similar to Nepal, the ease of movement, though primarily beneficial, requires careful management at formal crossing points to prevent the entry of undesirable elements or the circulation of counterfeit currency.
  - **Initiatives:**
    - **Dedicated Border Guarding Force:**
      - **SSB:** The SSB is deployed on the Bhutan border, operating under the operational principle of "Maitri" (Friendship).
      - **Community Engagement:** SSB personnel frequently engage in civic action programs and coordinate with Bhutan's Royal Bhutan Police (RBP) to monitor the border areas.
      - **Intelligence Focus:** The strategy here is primarily intelligence-based and relies on joint patrols and sharing information rather than creating physical barriers.
    - **High-Level and Field-Level Bilateral Security Coordination:**
      - **Security Meetings:** Regular bilateral security meetings at the Secretary and field commander levels ensure real-time sharing of threat assessments and coordination of anti-smuggling/anti-poaching efforts.
      - **Operation All Clear (2003):** This landmark joint operation, where the Royal Bhutan Army (RBA) successfully flushed out all Indian insurgent camps from Bhutanese soil with intelligence support from India, is the gold standard for border security cooperation in the region.
    - **Integrated Check Posts (ICPs) and Trade Facilitation:**
      - **Trade Infrastructure:** ICPs at major entry points (like Jaigaon/Phuentsholing) streamline customs, immigration, and cargo handling, making legal trade efficient and simultaneously discouraging illicit activities.
      - **Border Haats (Local Markets):** The establishment of regulated Border Haats aims to formalize and secure the local, small-scale movement of goods, providing a livelihood to border populations and reducing the incentive for smuggling.
    - **Infrastructure and Connectivity:**
      - **Road Network:** Indian funding and technical assistance for road development in Bhutan benefit both countries by improving accessibility for border forces during crises and facilitating commerce.
      - **First cross-border rail links:** Two new railway projects—Kokrajhar–Gelephu (Assam–Bhutan) and Banarhat–Samtse (West Bengal–Bhutan), about 89–90 km in total—will give Bhutan direct

access to India's 1.5 lakh km rail network, lowering logistics costs and boosting trade, tourism, and access to Indian ports.

- **Maritime Border:**

- **Limitations:**
  - **Vastness, Porosity, and Geographical Complexity:**
    - **Length and Diversity:** The sheer length of the coastline, featuring diverse terrain (mudflats, mangroves, sandy beaches, rocky cliffs), makes physical patrolling and fencing impossible.
    - **Island Gaps:** The thousands of inhabited and uninhabited islands in the Bay of Bengal and the Arabian Sea create numerous gaps that are difficult to monitor continuously, especially the critical sea lanes near the Malacca Strait.
    - **Continental Shelf and EEZ:** Monitoring the entire Exclusive Economic Zone (EEZ), which extends 200 nautical miles from the coast, is resource-intensive and requires high-end naval and aerial assets.
  - **Vulnerability to Non-State Actors:**
    - **Asymmetric Threats:** The primary threat comes from non-state actors, exemplified by the **26/11 Mumbai terror attacks** where terrorists used the sea route with relative ease.
    - **Smuggling Routes:** The coastline is a high-traffic corridor for the smuggling of arms, narcotics (especially from the West via the 'Golden Crescent'), gold, and counterfeit currency, often utilizing small, unregistered fishing vessels.
  - **Jurisdictional and Coordination Complexity:**
    - **Multi-Agency Framework:** Agencies involved include the Indian Navy (overall security), Indian Coast Guard (ICG—coastal surveillance and enforcement), State Marine Police (onshore and territorial waters up to 12 nautical miles), Customs, Directorate of Revenue Intelligence (DRI), and Port authorities.
    - **Lack of Interoperability:** Differences in standard operating procedures (SOPs), equipment, and training levels between the central (ICG, Navy) and state-level forces (Marine Police) create synchronization issues.
  - **Fisheries and Socio-Economic Challenges:**
    - **High Traffic Density:** Millions of traditional fishing vessels operate daily, making it extremely difficult to distinguish legitimate vessels from those used by criminals or terrorists.

- **Vulnerability to Exploitation:** The poverty and low wages of fishermen make them susceptible to recruitment by smugglers or anti-national elements for illegal activities.
- **Trans-Border Incidents:** Disputes over fishing grounds often lead to cross-border arrests of Indian fishermen by neighboring countries (e.g., Sri Lanka, Pakistan).

- **Initiatives:**
  - Following 26/11, India initiated a comprehensive overhaul of its coastal security architecture, focusing on a three-pronged approach: **Physical Patrols, Technological Surveillance, and Institutional Framework.**
    - **The Three-Tier Coastal Security System:**
      - **First Tier (Close to Shore): State Marine Police**—Patrolling territorial waters (0-12 nautical miles) and landward security.
      - **Second Tier (Mid-Range): Indian Coast Guard (ICG)**—Patrolling territorial waters and the Contiguous Zone (up to 24 nautical miles), with primary responsibility for search and rescue, anti-smuggling, and coastal surveillance.
      - **Third Tier (Deep Sea): Indian Navy**—Overall responsibility for maritime security and patrolling the Exclusive Economic Zone (EEZ) up to 200 nautical miles.
    - **Technological Surveillance and Monitoring Systems:**
      - **Coastal Surveillance Network (CSN) / Coastal Radar Chain:** This project involves installing a chain of modern radar stations and electro-optic sensors along the entire coastline and island territories. The feed from these radars is integrated into centralized control rooms.
      - **National Automatic Identification System (AIS) Network:** AIS transponders are mandatory for vessels above a certain tonnage. India is expanding the network to include all motorized fishing vessels, allowing them to be tracked through the control centers.
      - **Vessel Traffic Management Systems (VTMS):** These systems are installed in major ports and critical areas to monitor and manage maritime traffic flow.
      - **National Command Control Communication and Intelligence (NC3I) Network:** This is the nerve center of maritime security, linking all coastal stations, operational centers, and surveillance data into a single, comprehensive maritime domain awareness picture.
    - **Institutional and Legislative Frameworks:**
      - **Coastal Security Scheme (CSS):** Launched in phases, the CSS focuses on strengthening the State Marine Police. This includes providing modern boats, vehicles, arms, communication equipment, and constructing new coastal police stations and jetties.
      - **National Committee for Strengthening Maritime and Coastal Security (NCSMCS):** Chaired by the Cabinet Secretary, this committee ensures seamless coordination among all central and state agencies involved in maritime security.
      - **Identification Card Mandate:** Issuance of biometric identity cards to all legitimate fishermen and registration of all fishing boats to create a verifiable database and track movement.
    - **Community Engagement:**
      - **"Eyes and Ears" Scheme:** Fishermen are officially designated as the **"eyes and ears"** of the security apparatus, trained to report suspicious activities or movement of unknown vessels to the nearest ICG station or Marine Police post.
      - **Coastal Security Awareness Programs:** Regular programs are conducted in fishing villages to build trust and educate the community on security protocols and threat reporting.

2. LINKAGE OF ORGANISED CRIME WITH TERRORISM

Introduction:

India is confronting a deepening **security threat** as the boundaries between **organised crime and terrorism** increasingly blur. Most recently, the **National Investigation Agency (NIA)** launched sweeping raids across the country targeting the banned **Popular Front of India (PFI),** acting on allegations that the group's **criminal infrastructure** is being leveraged for **terror-related activities.**

Organized Crime:

According to the **UNODC,** an organised criminal group is defined as a **structured association** of at least **three individuals** that operates over a period of time and works together to carry out **serious offences** for **financial or material benefit.**

Similarities between Organized Crime and Terrorism:

- **Recruitment & Psychology:** Members often emerge from **marginalised groups,** attracted to **risk, thrill,** and defiance of social norms, creating similar recruitment pools.
- **Legality & Punishment:** Both **operate outside the law,** engage in serious criminal offences, and face strict national and international penalties.
- **Technology & Methods:** Each relies increasingly on **modern technology, digital tools,** high-tech communication, **financing,** and concealment systems.
- **Organisation & Discipline:** Both maintain **hierarchical structures,** enforce strict codes of conduct with internal punishment, and carry out **well-planned operations.**
- **Secrecy & Violence:** Activities are shrouded in **secrecy, and violence or intimidation** is used to achieve objectives or influence the surroundings.
- **Operational Nexus:** Terror groups often **finance operations** through organised crime, while criminal syndicates may pursue political goals using terror tactics, creating a blurred and **overlapping boundary.**

Differences between Organized Crime and Terrorism

- **Motives & Objectives:** Terrorism is driven by **ideological or political goals** aimed at transforming society, whereas **organized crime** seeks **financial gain and profit.**
- **Nature of Goals: Terrorist acts** pursue **specific political** or social change **using violence,** while **organized crime** focuses on **economic benefits** and legalizing illicit wealth, with minimal political intent.
- **Visibility & Messaging: Organized crime** maintains secrecy and **avoids publicity,** whereas terrorist groups openly declare their objectives to gain **support and demonstrate determination.**

- **Responsibility for Acts:** Terrorist organizations often **claim responsibility for attacks to maximize impact,** while organized crime groups **avoid any admission of involvement.**
- **Relationship with the State:** Terrorism confronts and **challenges the state to alter political order,** while organized crime i**nfiltrates state institutions** through **corruption** and selective cooperation.

Linkage Of Organized Crime With Terrorism

Understanding how **organized crime and terrorism** connect is crucial for **strong security policies.** Even though they often use **similar methods and structures,** they **pursue different goals.** Their relationship can involve **cooperation or convergence,** and most commonly, criminal activities are used to **fund terrorist operations.**

- **Financing Terrorism**
  - The **UNODC** has highlighted that **profits** from criminal activities increasingly **finance terrorism,** especially through **arms trafficking and money laundering.**
  - Examples include **Taliban** funding from **opium production in Afghanistan,** and FARC financing through cocaine trafficking, kidnapping, and extortion**.**
  - **Technology, communication, transport, and finance** networks now enable easier international cooperation between terrorists and criminal groups.
- **Narco Terrorism**
  - The **US Drug Enforcement Administration (DEA)** defines **narco-terrorism** as a form of **terrorism involving direct or indirect participation in the cultivation, production, transport, or distribution of drugs,** and using the profits to **support terrorism.**
  - The concept has a **dual nature,** depending on whether the focus is on **drug trafficking or terrorism.**
- **Financing Terrorism through Money Laundering**
  - **Money laundering** is used to **generate and move funds** for terrorism, by collecting money or property and using it fully or partially to support terrorist acts.
  - Terrorist financing involves **multiple channels,** including **banking systems, cross-border cash smuggling,** alternative transfer networks, and front organizations such as humanitarian or non-profit groups.
  - The major consequences include **destabilizing financial systems, economic instability,** declined investment, loss of state credibility, and threats to national security.
- **Human Trafficking and Terrorism**
  - Human trafficking is increasingly used by terrorist groups, not only for **raising funds** but also to **increase manpower, enable sexual exploitation,** and intimidate or harass populations.
  - It supports the formation of **military units,** with **child soldiers and girls recruited** and later exploited for sex, logistics, or intelligence, as highlighted in **UN reports.**
- **Intellectual Property Crime and Terrorism**
  - **Intellectual Property Crime (IPC)** involves the **production, transport, storage, and sale** of counterfeit or pirated goods without the **rights holder's consent,** often through trademark, patent, trade dress, or copyright infringement.
  - **In 2016,** terrorist organizations in **North Africa** were linked to a **$1 billion** contraband tobacco trade.

Linkages in the Indian Context:

Crime and terrorism are closely connected in many regions of India, even though the relationship may not be universal. **Case studies from India** show clear patterns where **terrorist activities depend on organized crime**.

- **North-East India**
  - **Extortion is the primary source of funding** for terrorism in the region.
  - **Kidnapping** is widely used to **spread fear and raise money**.
  - Other **criminal activities include Human trafficking, Drug trafficking, Gun running, etc.**
  - **Government officials in conflict areas** are often t**hreatened or bribed** to give contracts to groups supported by militants.
  - Essential commodities like **rice and fuel** are diverted to militant groups, who **resell them at higher prices**, demonstrating a **direct link between organized crime and terrorism**.
- **Jammu & Kashmir**
  - Unlike the Northeast, **extortion and kidnapping are less common**.
  - **External funding** plays a major role, especially from **Pakistan and Gulf countries.**
  - These funds are transferred through **Money Laundering and Hawala networks**, eventually reaching militant groups.
  - **Counterfeit currency** is a major source of financing- t**errorists act as couriers** of fake Indian currency in Kashmir. This currency later **spreads across India**.
- **Maoist-Affected Regions**
  - **Extortion is common** among Maoist groups.
  - They have also been involved in b**ank robberies, forcing cuts on drug-producing crops** to generate income.

Steps Taken by Government to Counter Organized Crime

- **Anti-Trafficking Nodal Cell (MHA)** acts as the **central coordination** point for decisions and follow-up with states.
- **MWCD** funds rehabilitation through **Ujjawala (sex-trafficking victims) and Swadhar (women in distress),** along with bilateral agreements with countries like Bangladesh, Nepal and Bahrain to curb trafficking.
- **NDPS Act, 1985 and Prevention of Illicit Trafficking Act, 1988** provide the legal backbone to combat drug production, transport and abuse.
- **Operation Dhvast (2023)** by NIA with state police targeted a terrorist–gangster–drug–arms nexus.
- **India** is a **signatory** to major **UN conventions** on narcotics, supporting global action through **UNODC.**
- **Trafficking of Persons (Prevention, Protection & Rehabilitation) Bill, 2021** proposes investigation units, rehabilitation committees, protection homes, and special courts to ensure faster trials.
- **Special courts** in each **district** aim to complete trafficking **trials within a year.**
- **MCOCA, 1999 (used in Maharashtra and Delhi)** has helped reduce gang and syndicate operations, despite the absence of a national organized crime law**.**

Global Initiatives to Tackle Organised Crime

- The **UNTOC,** adopted by UN General Assembly Resolution **55/25,** is the primary **international legal instrument** to combat **transnational organized crime.** It marks a major global commitment to address the growing threat of criminal networks and promotes strong international cooperation. By ratifying the Convention, States agree to:
  - Create **domestic criminal laws** targeting **organized criminal groups, money laundering, corruption, and obstruction of justice.**
  - Establish mechanisms for **extradition, mutual legal assistance, intelligence sharing, and joint investigations.**

  - **Strengthen law enforcement** capacities through training and technical support.
- Countries must become parties to the Convention before they can adopt its three supplementary Protocols, which focus on specific forms of organized crime:
  - **Trafficking** in Persons, Especially **Women and Children**
  - **Smuggling of Migrants by Land, Sea, and Air**
  - **Illicit Manufacturing and Trafficking** of Firearms, Parts, Components, and Ammunition
- These Protocols promote **victim protection, border controls,** arms tracing, and criminalization of trafficking and smuggling networks.
- The Convention plays a key role in **disrupting illicit financial flows,** asset laundering, and transnational criminal networks, and supports the seizure **and confiscation of criminal proceeds.** It also helps safeguard the global economy from criminal infiltration and corruption.

Challenges in Tackling Organized Crime in India

- **Weak Legal Framework:** Laws focus on **individual offenders**, not the **criminal networks**, making it hard to dismantle entire syndicates.
- **Evidence Barriers: Hierarchical and insulated structures** protect masterminds, making it difficult to gather **direct proof** against top leaders.
- **Slow Trials & Low Convictions: Delays in investigation, court procedures and poor conviction rates** reduce deterrence and discourage public cooperation.
- **Resource Constraints: Limited funds, manpower, training and technology** weaken policing capacity, especially in states with poor fiscal health.
- **Power Nexus: Collusion between criminals, politicians and officials** enables syndicates to **operate with impunity**, influencing decisions and evading action.

Way Forward

- **Institutional & Legal Reforms**
  - **Create specialised units at all levels** for **intelligence, documentation and investigation**, along with a **national body for data collection, analysis and information exchange** on domestic and international gangs.
  - **Strengthen legal mechanisms** through **speedy extradition, Mutual Legal Assistance (MLA) treaties, and strict implementation of laws such as MCOCA and the Gangster Act**, with **regular reviews and updates**.
- **Operational & Intelligence Strengthening**
  - Build **military, policing and intelligence capabilities** to disrupt criminal and terror networks.
  - **Share intelligence, exchange personnel, conduct joint operations** with other states and countries.
  - Enhance **Interpol cooperation** to break cross-border crime–terror links and track fugitives.
- **International Cooperation**
  - Promote **global coordination**, similar to **anti-human trafficking, anti-drug efforts** and the **Kimberley Process**.
  - Joint actions with **UN bodies, regional blocs** and neighbouring countries to counter **funding, trafficking and smuggling networks**.
- **Community & Media Engagement**
  - **Public awareness campaigns**, media outreach, workshops, and community initiatives to build **societal resistance**.
  - Encourage **citizen participation in reporting, prevention and social boycott** of crime syndicate figures.

- **Technology & Capacity Building**
  - Invest in **modern surveillance, analytics, cyber tools, forensics and training** of police forces.
  - Improve **documentation, databases and early-warning systems** on gang activities and terror financing.
- **Unified Global Strategy**
  - Adopt a **common framework**, with **data sharing and unified approach** to break the **crime–terror nexus**.
  - Follow **UN guidance**, ensuring countries **criminalize participation in organized crime** and enforce the convention effectively.

Conclusion:

Organized crime and terrorism are increasingly **interconnected, sharing resources, networks, and funding channels.** Criminal activities such as **drug trafficking, extortion, smuggling, and human trafficking** provide vital financial support to terror groups. **Weak law enforcement, porous borders, and corruption** allow these networks to thrive. Breaking this nexus requires **robust intelligence, strong legal frameworks, and global cooperation.** A coordinated and multi-dimensional approach is essential to safeguard national and international security.

3. DRONE THREATS AND COUNTER-DRONE TECHNOLOGIES

Introduction:

Unmanned Aerial Vehicles (UAVs), commonly called **drones**, have transformed the security environment by combining relatively low cost, low visibility, precision, persistence and reduced risk to the operator. Their applications now extend from surveillance and reconnaissance to intelligence gathering, logistics, weapon delivery, loitering munitions and precision strikes.

For India, the security challenge is particularly important along the **Pakistan border**, where drones have been increasingly associated with narcotics, weapons and ammunition smuggling. The challenge, however, now extends to terrorism, military operations, critical infrastructure and future autonomous warfare.

The **Chief of Defence Staff (CDS), in March 2025**, highlighted the disruptive potential of Unmanned Aerial Systems because of advances in robotics, Artificial Intelligence and the economics of low-cost, high-impact warfare.

The drone has democratised aerial power: a relatively inexpensive platform can impose disproportionate security costs.

India's Emerging Drone Ecosystem

India's security challenge must be viewed alongside its rapidly expanding legitimate drone ecosystem. According to the **Press Information Bureau (PIB), in February 2026**, India had **38,575 registered drones, 39,890 Remote Pilot Certificates and 244 approved Remote Pilot Training Organisations**. Drones had also been used to survey **3.28 lakh villages under the SVAMITVA Scheme**.

This creates a dual-use challenge.

India cannot secure its airspace simply by restricting drones; it must secure the legitimate drone economy against hostile exploitation.

The growing ecosystem also means that the drone sector can become a source of indigenous defence innovation, provided civilian and military requirements are appropriately integrated.

Why Drones Pose a Distinct Security Threat

Characteristic	Security implication
Low cost	Makes aerial capability accessible to non-state actors and criminal networks.
Small size and low altitude	Makes detection more difficult, particularly against background clutter.
Remote operation	Allows the operator to remain physically distant from the target.
Precision	Enables targeted delivery of weapons, explosives or contraband.
Persistence	Allows prolonged surveillance and target acquisition.
Modularity	Commercial platforms can potentially be modified for hostile purposes.
Swarm potential	Multiple platforms can create saturation challenges for conventional defences.
Dual-use nature	The legitimate civilian ecosystem can also create security vulnerabilities.
Rapid technological evolution	Countermeasures can become obsolete relatively quickly.

Major Drone Threats to India

- **Cross-Border Smuggling:** Drones can deliver narcotics, weapons, ammunition, explosives and other contraband across the border without requiring a human carrier to physically cross the border.

  In **May 2026**, the Union Home Minister directed the Border Security Force to intensify action against the smuggling of narcotics and weapons and emphasised deployment of anti-drone systems along the border.

- **Terrorism and Military Attacks:** Drones can be used for reconnaissance, target acquisition, explosive delivery, precision attack and loitering-munition operations.

  Operation SINDOOR in May 2025 *provided a major Indian example. During operation sindoor, Pakistan attempted to engage multiple Indian military targets* ***using drones and missiles****. These threats were neutralised through India's* ***Integrated Counter-Unmanned Aerial Systems Grid and Air Defence systems****.*

- **Surveillance and Intelligence Gathering:** Even an unarmed drone can collect photographs, video, terrain information, troop movements and details of military or critical infrastructure.
- **Swarm Attacks:** A drone swarm involves multiple drones operating in coordination. Its principal security advantage lies in creating saturation, thereby placing pressure on sensors, operators, communications and interceptor inventories.

  The Indian Air Force's **Mehar Baba Competition-3 in April 2026** has focused on collaborative drone-based surveillance radar capabilities.

- **Threats to Critical Infrastructure:** Potential targets include airports, ports, power plants, oil installations, military bases, nuclear facilities, railway infrastructure and large public gatherings. The threat is no longer confined to military establishments.

In **February 2026**, the **V.O. Chidambaranar Port Authorit**y became the first Indian port to initiate an integrated anti-drone system.

Drone Threat Matrix

Threat	Possible use	Principal vulnerability	Security response
Border smuggling	Drugs, weapons and ammunition can be delivered across the border.	Physical barriers can be bypassed.	Detection and interception need to be integrated with border surveillance.
Terror attack	Explosives or armed drones can be used against selected targets.	Low-cost precision delivery can create disproportionate damage.	Layered counter-drone and air-defence systems are required.
Surveillance	Military installations and critical infrastructure can be photographed or monitored.	Small, low-flying drones may be difficult to detect.	Multi-sensor surveillance and identification are required.
Swarm attack	Large numbers of drones can simultaneously approach a target.	Conventional interception capacity may be saturated.	Automated detection and scalable interception are required.
Critical infrastructure attack	Airports, ports and energy facilities can be targeted.	Numerous sensitive civilian installations create a wide security surface.	Site-specific counter-drone architecture is required.
Cyber-enabled drone threat	Communication and navigation systems may be disrupted or manipulated.	Drones depend on communication and navigation systems.	Cybersecurity and resilient communication architecture are necessary.
Psychological warfare	Repeated drone incursions can generate fear and uncertainty.	Public perception may amplify the security impact.	Rapid detection and credible official communication are important.
Dual-use misuse	Commercial drones can potentially be adapted for hostile purposes.	Large civilian ecosystems make complete physical control difficult.	Registration, identification and risk-based regulation are required.

- **Counter-Drone Technology**
  - **Counter-Unmanned Aircraft Systems (C-UAS)** are systems designed to detect, identify, track and neutralise unauthorised or hostile drones. An effective C-UAS system must first detect the drone, then identify and track it, assess the threat and finally select an appropriate method of neutralisation.

*The **DRDO D4 Counter-Drone System** for Indian VVIP security exemplifies the complete C-UAS chain. Its 3D radar **detects** micro-targets kilometers away, while integrated EO/IR cameras **identify** the drone model and **track** its flight path. Command-and-control software then **assesses** threat levels and **executes the optimal response**—using soft-kill radio jamming or hard-kill laser weapons.*

- **Detection and tracking technologies:** These include radar, Radio Frequency sensors, electro-optical and infrared cameras and other passive or active sensors.

  *To counter stealthy plastic quadcopters along the western international border, the **Border Security Force (BSF) deploys passive RF sensors with micro-Doppler radars**.*

- **Neutralisation technologies**
  - **Soft-kill systems** attempt to disrupt the drone's communication, navigation or control systems.
  - **Hard-kill systems** physically destroy or disable the drone through kinetic or directed-energy technologies.
  - The operational difference between neutralisation methods is starkly visible in modern conflict zones: Soft-kill systems, like the **handheld Smart Shooter SMASH Jammer,** project directional electronic microwave beams to flood a drone's receiver, breaking its GPS and radio links to force a harmless crash landing without firing a shot. Conversely, **hard-kill systems, such as the Raytheon Coyote Block 2 kinetic interceptor**, physically destroy the threat by launching a small explosive drone directly into an enemy swarm and detonating mid-air, using kinetic shrapnel to violently rip the targets apart.

    DRDO** had demonstrated an indigenous counter-drone system incorporating detection, soft-kill jamming and hard-kill laser capability by **July 2021**, with technology transferred to **Bharat Electronics Limited

Institutional Architecture

Domain	Principal institutions	Core role
Military defence	**Indian Army, Indian Air Force and Indian Navy**	Countering battlefield, aerial and maritime drone threats.
Border security	**Border Security Force, Indo-Tibetan Border Police, Sashastra Seema Bal and Assam Rifles**	Border surveillance, detection and interdiction of hostile drones.
Technology development	**Defence Research and Development Organisation, Bharat Electronics Limited, private defence companies and start-ups**	Development of sensors, C-UAS, electronic warfare and directed-energy technologies.
Critical infrastructure	**Central Industrial Security Force, airport authorities and port authorities**	Protection of sensitive civilian infrastructure.
Civil aviation regulation	**Directorate General of Civil Aviation**	Regulation of civilian drones and safe integration into Indian airspace.
Intelligence and investigation	**Intelligence Bureau, State Police and other intelligence agencies**	Threat intelligence, attribution and investigation.

Major Indian Counter-Drone Initiatives:

Initiative	Significance	Latest evidence
DRDO Counter-Drone System	Demonstrated indigenous detection, electronic disruption and laser-based hard-kill capability.	Technology demonstrated and transferred to BEL in **July 2021**.

Integrated Counter-UAS Grid	Provides networked protection against drone and other aerial threats.	Operationally demonstrated during **Operation SINDOOR in May 2025**.
Integrated Drone Detection and Interdiction System	Provides dedicated military counter-drone capability.	Included in emergency defence procurement during **June 2025**.
Integrated Drone Detection and Interdiction System Mk-II	Represents further development of Army counter-drone capability.	Acceptance of Necessity accorded in **December 2025**.
AKASH TARANG	Provides anti-UAV electronic-warfare protection to Army formations.	Defence Acquisition Council approved its procurement in **July 2026**.
Port anti-drone system	Extends C-UAS protection to critical maritime infrastructure.	V.O. Chidambaranar Port initiated the system in **February 2026**.
Advanced anti-drone training	Develops specialised human capacity for critical-infrastructure security.	Training initiatives involving security personnel were conducted in **March 2026**.
Mehar Baba Competition-3	Encourages indigenous development of advanced collaborative drone technologies.	IAF initiative launched in **April 2026**.
Private-sector C-UAS development	Expands India's counter-drone capabilities beyond traditional defence establishments.	Bhargavastra development by Solar Industries highlighted in **January 2026**.

Case Studies: How Drone Warfare Has Changed

Case study	What happened	Result / significance	Lesson for India
Nagorno-Karabakh War, Azerbaijan–Armenia, 2020	Azerbaijan extensively integrated Turkish Bayraktar TB2 drones with artillery, intelligence, surveillance and reconnaissance capabilities. Drones were used against tanks, artillery, air-defence systems and logistics.	The war proved drones enhance combined-arms operations and expose forces lacking camouflage, dispersion, and air defence.	**India needs integrated drone capabilities alongside strong camouflage, dispersion, passive defence and layered air defence.**
Russia–Ukraine War, 2022 onwards	Both sides massively expanded the use of reconnaissance drones, First-Person View drones, one-way attack drones and increasingly autonomous systems. By 2025, the battlefield had become characterised by very large-scale deployment of FPV and one-way attack drones.	Drone warfare moved from being a specialised capability to becoming a **routine battlefield system for reconnaissance, targeting, strike and attrition**. It also demonstrated that rapid battlefield innovation can be as important as possessing sophisticated platforms.	**India needs rapid procurement, decentralised innovation and continuous battlefield feedback rather than long technology-development cycles alone.**

Ukraine's deep-strike drone campaign against Russia, 2024–26	Ukraine increasingly used relatively inexpensive long-range drones against Russian energy and military infrastructure deep inside Russian territory. Recent analysis shows that these systems have been adapted for long-range operation, low observability and greater resistance to electronic warfare.	The case demonstrated that drones can impose **strategic effects beyond the tactical battlefield**, including pressure on critical infrastructure and the adversary's economic and military logistics.	**India's counter-drone doctrine should consider the protection of strategic infrastructure, not merely troops and border posts.**
Ukraine's Black Sea drone campaign, 2022 onwards	Ukraine used unmanned surface vessels against the Russian Black Sea Fleet and associated infrastructure. The campaign forced Russian naval assets to operate more cautiously and altered the operational environment in the Black Sea.	Demonstrated that relatively inexpensive unmanned systems can impose serious operational constraints on technologically superior naval forces.	**Maritime security must incorporate unmanned surface and aerial systems into both offensive and defensive planning.**
Red Sea and Bab-el-Mandeb, 2023–25	The Houthi movement used drones and missiles against commercial and military shipping. The threat later expanded to explosive unmanned surface vessels. IISS noted that the attacks created significant challenges for Western naval forces operating around the Red Sea, Bab-el-Mandeb and Gulf of Aden.	A non-state actor demonstrated the ability to impose substantial security and economic costs on global maritime trade using relatively inexpensive unmanned systems.	**Low-cost drone threats can generate high-cost defensive responses and disrupt global supply chains. India therefore needs an integrated maritime C-UAS and unmanned-systems doctrine.**
Iranian Shahed drone proliferation, Ukraine and Middle East	Iran-developed Shahed-type one-way attack drones have been employed by Russia in Ukraine and by Iran-aligned groups in the Middle East. Their relatively low cost, long range and ability to attack fixed targets have made them an important component of asymmetric warfare.	Demonstrated the **cost-asymmetry problem** in modern air defence, where the defender may have to expend considerably more resources than the attacker.	**India needs affordable countermeasures and layered defence so that low-cost drones do not impose unsustainable interception costs.**
Operation Spider's Web, Ukraine, June 2025	Ukraine used covertly positioned drones to attack Russian strategic aviation assets at multiple airbases deep inside Russia. The operation demonstrated the potential of small unmanned systems to reach highly protected targets through unconventional methods.	It reinforced the idea that **geographical distance and conventional air-defence barriers do not necessarily guarantee protection from drone attacks**.	

Operation SINDOOR as a Case Study

Operation SINDOOR in May 2025 is the strongest contemporary Indian case study for this topic. On **7 and 8 May 2025**, Pakistan attempted attacks against multiple Indian military locations using drones and missiles. India's Integrated Counter-UAS Grid and air-defence systems neutralised these threats. The Ministry of Defence subsequently highlighted the use of a combination of Counter-UAS, Electronic Warfare and Air Defence capabilities in multiple defensive layers.

The operation demonstrated four important lessons.

- First, **drone warfare has become part of conventional military operations**.
- Second, counter-drone systems cannot operate in isolation from wider air defence.
- Third, electronic warfare, sensors, command-and-control and physical interception must operate as an integrated system.
- Fourth, indigenous technology has strategic value because sustained conflict requires reliable access to critical defence capabilities.

In **May 2026**, the Raksha Mantri described Operation SINDOOR as an example of India's joint and technologically enabled military response, while emphasising the need to strengthen capabilities in **Artificial Intelligence, autonomous systems, data analytics and secure communication networks** for future warfare.

Major Challenges:

- **Detection of Small and Low-Flying Drones:** Small drones may have limited radar signatures and can exploit terrain and background clutter. The problem is therefore not merely interception; it begins with **reliable detection and identification**.

  The Threat Perception Report: *According to battlefield reports from the Eastern Ukraine Theater and the Israel-Lebanon border, standard military air defense radars fail to register First-Person View (FPV) drones due to their microscopic Radar Cross-Sections (RCS) of just 0.01 to 0.1 square meters—roughly the size of a large bird.*

- **Rapid Technology Arms Race:** Hostile actors can respond to counter-drone systems through autonomous navigation, frequency-hopping, encrypted communication,
- **Cost Asymmetry:** A relatively inexpensive drone may force the defender to employ a much more expensive interceptor. India therefore needs a balanced mix of low-cost electronic countermeasures, scalable interceptors and high-end hard-kill systems.

  A perfect recent example of cost asymmetry is the US Naval defense against Iranian-backed Houthi drone attacks in the Red Sea (2024–2026), where US destroyers routinely fired $2 million to $4 million SM-2 and SM-6 missiles to intercept cheap, ***Iranian-designed Shahed-136 drones costing just $20,000 to $50,000 each****.*

- **Swarm Saturation:** Large numbers of drones can overwhelm sensors, operators, communication networks and interceptor inventories. This increases the importance of automation and Artificial Intelligence-assisted threat classification.
- **Civilian Drone Ecosystem:** India's expanding civilian drone industry creates a regulatory dilemma. Excessive restrictions could discourage innovation, while weak regulation could create security vulnerabilities. The policy objective should therefore be: **Regulate the risk without suppressing the technology.**
- **Indigenous Technology Gaps:** The challenge extends beyond manufacturing the drone itself. India must develop domestic capabilities in sensors, processors, batteries, Radio Frequency components, navigation systems, software and electronic-warfare technologies.

- **Fragmented Protection of Critical Infrastructure:** Airports, ports, refineries, power plants and other sensitive facilities have different threat profiles. India therefore needs common standards for detection and response while allowing individual installations to adopt systems suited to their risk profile.

  *A prime example of India addressing the problem of Fragmented Protection of Critical Infrastructure is the **Ministry of Home Affairs' decision to designate the CISF as the central nodal agency for drone security***

Way Forward:

- **Build a National Layered C-UAS Architecture:** India should integrate radar, Radio Frequency sensors, electro-optical and infrared systems, electronic warfare, soft-kill systems and hard-kill systems within interoperable command-and-control networks.

  Example: The Indian Army's Saksham C-UAS Grid. It seamlessly links distinct radars, RF sensors, and EO/IR cameras with a unified Command-and-Control (C2) software layer to orchestrate both soft-kill (RF jamming) and hard-kill weapons at the tactical frontline

- **Reduce the Detection-to-Response Time:** Sensor fusion and Artificial Intelligence-assisted classification should allow security agencies to identify genuine threats rapidly and select the appropriate response.

  ***Example:** Grene Robotics' **Indrajaal autonomous defense dome**. Driven by AI modules like WeaponFusion and HyperSensing, it automatically correlates multi-sensor data to isolate true micro-drones from birds or clutter, instantly calculating a trajectory prediction to execute automated countermeasures*

- **Address Cost Asymmetry:** India should develop affordable countermeasures alongside sophisticated systems so that the defence response remains economically sustainable against mass deployment of inexpensive drones.

  ***DRDO's DURGA-2 (Directionally Unrestricted Ray-Gun Array) laser system.** Instead of expending a multi-lakh rupee missile to intercept a cheap cross-border smuggling drone, this Directed Energy Weapon achieves a hard-kill using electricity, dropping the cost per shot to just a few rupees*

- **Pursue Full-Spectrum Indigenisation:** Indigenisation should cover the complete technology chain, including sensors, electronics, communications, navigation, software, propulsion and countermeasure systems.

  Lessons from Operation Sindoor. This operation exposed supply-chain risks, pushing India to strictly mandate domestic components—ranging from indigenous 12 GHz RF front-ends to local navigation and tracking software—eliminating reliance on foreign tech chains.

- **Strengthen the Private Sector:** The Government should deepen links between the Armed Forces, DRDO, start-ups, MSMEs and established private defence companies through iDEX, testing facilities, rapid procurement and operational feedback.

  ***iDEX (Innovations for Defence Excellence) contracts:** Through iDEX, the Ministry of Defence fast-tracked procurement cycles with private firms like Astra Microwave and L&T, utilizing direct operational feedback from forward deployments to rapidly upgrade drone-hunting software*

- **Prepare for Autonomous and Swarm Warfare:** India needs greater investment in Artificial Intelligence, robotics, autonomous systems, swarm detection, secure communications and electronic warfare.

  The Defence Ministry in **May 2026** specifically emphasised AI, autonomous systems, data analytics and secure communication networks as critical capabilities for future warfare.

- **Integrate C-UAS with Border Security:** Since hostile drones can carry drugs, weapons and ammunition, counter-drone systems should be integrated with the Border Security Force, intelligence agencies, State Police and the Narcotics Control Bureau.

- **Extend Protection to Critical Infrastructure:** Ports, airports, energy installations and other sensitive facilities should have risk-based counter-drone assessments, trained personnel, standard response protocols and interoperability with local security agencies.

  *The **V.O. Chidambaranar Port anti-drone system introduced in February 2026** provides an important model for extending this capability to civilian strategic infrastructure.*

- **Maintain a Balance Between Regulation and Innovation:** India should strengthen drone identification, registration, certification and airspace management while continuing to encourage start-ups, indigenous manufacturing, research and exports.

Conclusion:

Drones have altered India's security calculus by making surveillance, precision attack, intelligence gathering and asymmetric warfare accessible at relatively low cost. Their use in cross-border smuggling, terrorism, military operations and potential attacks on critical infrastructure makes them simultaneously a **border-security, internal-security and defence challenge**.

India's response is consequently moving from isolated anti-drone weapons towards a **networked and layered C-UAS architecture** integrating sensors, electronic warfare, command-and-control, soft-kill, hard-kill and conventional air defence.

4. MARITIME AND COASTAL SECURITY IN INDIA

Introduction:

Maritime security covers the protection of India's **maritime interests** across the Indian Ocean Region, including **sea lines of communication, the Exclusive Economic Zone and offshore assets.** Coastal security is the narrower near-shore component, concerned with preventing infiltration, smuggling and attack from the sea across territorial waters and the coastline itself.

Scale of the challenge:

- India has a coastline of **7,516 km**, comprising 5,422 km on the mainland and 2,094 km along the islands, spanning **nine coastal states and four Union Territories**.
- The littoral includes about **1,382 islands, 3,331 coastal villages, 12 major and roughly 200 minor ports** and 95 landing centres, alongside an Exclusive Economic Zone of over 2 million square kilometres.
- Roughly **90 per cent of India's trade by volume and 70 per cent by value moves by sea**, along with the bulk of its energy imports, which makes maritime disruption an economic threat rather than only a military one.

Significance of Maritime and Coastal Security

- **Economic Security**
  - **Trade Lifeline:** Almost all of India's external trade moves through sea lanes, so a disruption at a chokepoint transmits directly into domestic prices and supply.

    The Red Sea crisis forced Indian shipping to reroute around the Cape of Good Hope, adding transit time and freight cost across the entire export basket.

  - **Energy Dependence:** A large share of India's crude imports transits the Strait of Hormuz, making the security of a single narrow waterway a national economic exposure.

    Nearly 62 per cent of India's oil imports originate in the Persian Gulf, which is why Operation Sankalp was launched in June 2019 to protect that route.

  - **Offshore and Digital Assets:** Oil rigs, platforms, pipelines and submarine fibre-optic cables lie beyond the coast but within the EEZ, and carry enormous economic value with minimal physical protection.

India's digital economy runs on submarine cables landing at a handful of coastal points, making them a strategic vulnerability disproportionate to their physical size.

- **National Security**
  - **Sea as an Infiltration Route:** The coast has repeatedly proved the softest entry point for weapons, explosives and armed attackers.

    The RDX used in the 1993 Mumbai blasts was landed on the Raigad coast by fishing boat, and the 26/11 attackers entered aboard a hijacked trawler in 2008.
  - **Hostile Proximity:** India's coasts lie close to politically volatile and unfriendly neighbours, which shortens the crossing available to hostile actors.

    The Sir Creek area of Kutch, Bangladesh's riverine boundary and the Palk Strait each present short, contested and difficult-to-police approaches.
  - **Convergence of Crime and Terror:** Narcotics, arms and human trafficking networks use the same routes and facilitators as terror logistics, so maritime crime is a terrorism indicator.

    Large narcotics seizures off Gujarat and Kerala have repeatedly involved vessels using fishing cover, the same modus operandi as 26/11.
  - **Strategic Competition:** The Indian Ocean is now contested space, with extra-regional naval deployments, survey vessels and port access agreements altering the security environment.
- **Regional Leadership**
  - **Net Security Provider:** India's capacity to respond first to incidents in the Indian Ocean underpins its claim to regional leadership and shapes its partnerships.

    The Navy responded to 18 incidents within the first 100 days of the renewed Operation Sankalp, establishing India as first responder and preferred security partner in the region.
  - **Doctrinal Framework:** The SAGAR vision of 2015, expanded into MAHASAGAR in 2025, positions maritime security as the basis of India's engagement with littoral neighbours.
  - **Diaspora and Seafarers:** A large share of the world's commercial seafarers are Indian nationals, so attacks on merchant shipping anywhere become a consular and protection responsibility for India.

Institutional Architecture:

The Three-Tier Structure

Instituted after 26/11, this is the single most examinable element of the topic.

- **Indian Navy:** Designated the authority responsible for **overall maritime security**, including coastal and offshore security, operating seaward and coordinating the whole structure.
- **Indian Coast Guard:** Responsible for **coastal security in territorial waters**, and additionally designated to coordinate between central and state agencies within those waters.
- **Marine Police:** State forces patrolling shallow waters close to shore and operating coastal police stations, forming the innermost and land-facing layer.

Doctrine and Strategy

- **Indian Maritime Security Strategy, 2015:** Titled *Ensuring Secure Seas*, it sets out the Navy's framework for securing sea lines of communication, coordinating maritime agencies and expanding multilateral engagement.
- **SAGAR and MAHASAGAR:** Security and Growth for All in the Region, articulated in 2015 and expanded in 2025, links India's own security to capacity building across the littoral.

Coordination Bodies

- **National Committee for Strengthening Maritime and Coastal Security:** Headed by the Cabinet Secretary, it coordinates all maritime and coastal security matters across ministries, though it functions without statutory backing.
- **National Maritime Security Coordinator:** Created in 2022 under the National Security Council Secretariat, with retired Vice Admiral G Ashok Kumar as first incumbent, to bridge more than fifteen agencies operating in this space.
- **Joint Operations Centres:** Command and control hubs at **Mumbai, Visakhapatnam, Kochi and Port Blair**, run by the Navy, functioning as the regional nodes of the coastal security network.
- **Coastal Security Operations Centres:** Coast Guard-run centres that link marine police, customs, fisheries and port authorities at the state level.

Surveillance and Domain Awareness

- **National Command Control Communication and Intelligence Network:** Links the operational centres and lower formations of the Navy and Coast Guard across the coastline and island territories into a single network.
- **Information Management and Analysis Centre, Gurugram:** The nodal hub of NC3I, fusing coastal radar, AIS and agency inputs into a common operating picture.
- **Information Fusion Centre for the Indian Ocean Region:** Established in 2018 at Gurugram, hosting international liaison officers and sharing maritime domain awareness with partner navies.
- **Coastal Surveillance Network:** A chain of static radar stations, electro-optical sensors and Automatic Identification System receivers along the coast and island territories.
- **National Maritime Domain Awareness Pro**ject: Seeks to integrate inputs from all agencies, satellites and sensors into one comprehensive maritime picture.

Schemes, Infrastructure and Exercises

- **Coastal Security Scheme:** Ministry of Home Affairs scheme, with Phase I from 2005 to 2011 and Phase II from 2011, under which **204 coastal police stations have been operationalised**, along with 204 boats, 37 jetties, 97 check-posts, 58 out-posts and 30 barracks.
- **Project Seabird:** Development of the Karwar naval base, where **INS Kadamba** was commissioned, giving the Navy a third major operating base free of the congestion of Mumbai and Visakhapatnam.
- **Regulation of Maritime Activity:** Multi-purpose biometric identity cards for fishermen, uniform licensing, colour-coding and GPS or transponder fitment on fishing craft.
- **Sagar Suraksha Dal:** Formalises the surveillance role of the fishing community in coastal states, converting fishermen into a reporting network.
- **Exercises and Drills: Sea Vigil**, the biennial national coastal defence exercise led by the Navy, **Sagar Kavach**, the biannual regional exercise led by the Coast Guard, and **Sajag**, the Coast Guard's routine drill on fishing craft compliance.

Maritime Anti-Piracy Act, 2022

- **Definition:** Covers any illegal act of violence, detention or depredation committed for private ends on the high seas, including inciting or facilitating such acts and voluntary participation in pirate operations.
- **Jurisdiction:** Applies to the high seas beyond India's territorial waters, including the EEZ and waters outside the jurisdiction of any state.
- **Enforcement Provisions:** Makes piracy an extraditable offence, establishes designated courts for speedy trial, and restricts arrest and seizure of pirate vessels to authorised personnel.

- **Criticisms**: The death penalty provision is argued to sit uneasily with the rarest of rare doctrine and with Articles 14 and 21, the scope of what constitutes aiding piracy remains ambiguous, and certain presumptions of guilt may complicate extradition.

Issues:

- **Fragmented Coordination**
  - **Too Many Agencies, No Single Authority:** Over fifteen central and state entities operate in the coastal space, none with authority over the others, so accountability dissolves at the seams.

    The Navy and Coast Guard have operational responsibility but no power of direction over customs, port trusts, fisheries departments or state police.
  - **Legislation Still Pending:** The draft Coastal Security Bill, prepared jointly by the Navy and Coast Guard to delineate stakeholder responsibilities and create a National Maritime Authority, has never been passed.

    The proposal for an apex coastal authority has been under consideration since the Kargil Review Committee of 1999, which is over two decades of institutional drift.
  - **Centre-State Friction:** Most coastal states treat maritime security as a central responsibility, which weakens their commitment to funding, staffing and running the marine police tier.
  - **Jurisdictional Confusion:** Inter-agency and intra-agency ambiguity over areas of operational jurisdiction produces overlapping claims in some waters and gaps in others.
- **Topography and Boundaries**
  - **Creeks and Rivulets:** The coastline is broken by creeks, mudflats and tidal channels that offer concealed entry points no radar chain covers well.

    Harami Nala in the Sir Creek area of Kutch, which originates in India and flows into Pakistan, has become a preferred route for infiltrators and smugglers.
  - **Unsettled Maritime Boundaries:** Disputed or undelimited maritime boundaries constrain patrolling and surveillance, since enforcement in contested waters carries diplomatic risk.

    The Sir Creek dispute with Pakistan remains unresolved, leaving the maritime boundary in the Kutch sector undefined.
  - **Proximity to Volatile Neighbours:** Short crossings from Pakistan, Bangladesh and Sri Lanka give hostile actors and trafficking networks limited exposure at sea.
- **Weak Marine Police Tier**
  - **Land Police at Sea:** Coastal police are drawn from ordinary state police cadres with no seafaring background, no dedicated cadre and frequent rotation out of maritime postings.
  - **Assets Without Operators:** Boats sanctioned under the Coastal Security Scheme are frequently underutilised owing to shortages of crew, fuel budgets, jetties and maintenance support.
  - **Infrastructure Still Incomplete:** Office buildings, weapons, jetties and vessels remain short in several states more than a decade and a half after 26/11.

    A CAG report tabled in Parliament found that the coastal security force raised after the 2008 Mumbai attacks still lacked full infrastructure, with Gujarat and Maharashtra among the states affected.
  - **Central Force Proposal Stalled:** The Ministry of Home Affairs proposal for a Central Marine Police Force remains pending with the Cabinet, leaving the weakest tier the least reformed.
- **Manpower and Capability Shortfalls**
  - **Coast Guard Understrength:** Recruitment difficulties, a demanding selection process and limited platform availability leave the Coast Guard stretched across an enormous area of responsibility.

  - **Training Deficit:** Marine police and customs personnel deployed on coastal patrolling frequently lack training in sea combat, boat handling and interdiction.
- **Surveillance and Detection Gaps**
  - **The Small Boat Problem:** With roughly three lakh registered fishing vessels operating in Indian waters, distinguishing a legitimate boat from a hostile one is the central technical challenge of coastal security.

    Crafts under 20 metres are not required to carry transponders, and these are precisely the vessels used in both the 1993 and 2008 attacks.
  - **Exercises Reveal the Gap:** Simulated attacks during coastal defence exercises have repeatedly succeeded in penetrating the multi-layered structure.

    Roughly a tenth of mock attacks launched during Exercise Sea Vigil breached coastal defences, an official audit finding rather than a critic's estimate.
  - **Emerging Technology Threats:** Armed drones, unmanned surface vessels and low-signature craft are cheap, hard to detect on conventional radar and already in use in the region.
- **Shifting External Threats**
  - **Threats Moving Closer to the EEZ:** Attacks once confined to distant waters now occur within striking distance of the Indian coast.

    The drone attack on MV Chem Pluto occurred about 220 nautical miles southwest of Porbandar, and the hijacking of MV Ruen about 700 nautical miles from the Indian coast.
  - **Piracy Resurgence:** Somali piracy revived as international naval attention shifted to the Red Sea and coalition resources were reallocated, placing the burden on regional navies.
  - **Extra-Regional Presence:** Sustained deployment of foreign warships, submarines and survey vessels, alongside port access arrangements across the littoral, has compressed India's strategic space.
- **Port and Infrastructure Vulnerability**
  - **Uneven Port Security:** Major ports have CISF cover and ISPS Code compliance, but roughly 200 minor ports and private terminals operate under widely varying standards.
  - **Undefended Critical Infrastructure:** Undersea cables, offshore platforms and single point moorings carry enormous economic value with minimal physical protection.
  - **Non-Traditional Threats Neglected:** Attention has concentrated on terrorism while illegal fishing, trafficking, marine pollution and disaster response receive less institutional focus despite occurring far more often.
- **Government Measures**
  - **Three-Tier Security Structure:** Navy for overall maritime security, Coast Guard for territorial waters, and state marine police for shallow waters, instituted after 26/11.
  - **National Maritime Security Coordinator:** Created in 2022 within the National Security Council Secretariat to unify the efforts of more than fifteen agencies.
  - **Coastal Security Scheme:** Funds coastal police stations, boats, jetties, check-posts and vehicles, with 204 coastal police stations operationalised so far.
  - **NC3I, IMAC and the National Maritime Domain Awareness Project:** Link radar stations, AIS feeds and agency inputs into a single national maritime picture.
  - **Information Fusion Centre for the Indian Ocean Region:** Established in 2018 with international liaison officers, extending awareness beyond national means.
  - **Joint Operations Centres:** Regional command hubs at Mumbai, Visakhapatnam, Kochi and Port Blair coordinating naval, Coast Guard and state assets.

- **Project Seabird and Platform Modernisation:** Commissioning of INS Kadamba at Karwar, alongside induction of ships, submarines, helicopters, radars and satellites across the Navy and Coast Guard.
- **Sagar Suraksha Dal and Fisherman Registration:** Biometric identity cards, uniform licensing, colour-coding and community watch groups converting fishermen into a surveillance network.
- **Exercises Sea Vigil, Sagar Kavach and Sajag:** Regular audits of preparedness and inter-agency coordination involving Navy, Coast Guard, marine police, customs, CISF, NSG, NCB and port authorities.
- **Operation Sankalp:** Launched June 2019 and re-initiated in December 2023, deploying destroyers, frigates, P-8I aircraft and Sea Guardian drones to escort merchant shipping.
- **Maritime Anti-Piracy Act, 2022:** Domestic legal framework for prosecuting piracy on the high seas through designated courts.
- **International Engagement:** SAGAR and MAHASAGAR, the Indian Ocean Naval Symposium, the Indian Ocean Rim Association, the Colombo Security Conclave, and exercises such as MILAN and coordinated patrols with neighbours.

Way Forward:

- **Fixing the Institutional Gap**
  - **Enact the Coastal Security Legislation:** Pass the long-pending Bill to define the statutory duties of every stakeholder from port trusts to state maritime boards, and create a National Maritime Authority with powers of direction.
  - **Empower the Coordinator:** Give the National Maritime Security Coordinator a dedicated secretariat, budget and authority to task agencies rather than only to convene them.
  - **Create a Central Marine Police Force:** Move the proposal pending with the Cabinet forward, with its own cadre, rules and training, and clarity on powers to register offences and investigate.
  - **Strengthening the Innermost Tier**
  - **Build a Dedicated Marine Cadre:** Recruit and retain coastal police as a specialised seafaring cadre rather than rotating general police into maritime postings.
  - **Fund Operations, Not Only Assets:** Provide for crew, fuel, jetties and maintenance alongside boats, since an unoperated patrol vessel adds nothing to security.

    The CAG's finding of incomplete infrastructure years after sanction shows that procurement without operational support does not produce capability.
  - **Address Coast Guard Shortfalls:** Expand recruitment, training capacity and platform availability to match the Coast Guard's area of responsibility.
- **Closing the Surveillance Gap**
  - **Universal Vessel Identification:** Complete registration, biometric cards, colour-coding and transponder fitment for all fishing craft including those under 20 metres.

    This is the exact gap the 1993 and 2008 attacks exploited, and it remains the most direct lesson still unimplemented.
  - **Layered Technology:** Combine the coastal radar chain with satellite surveillance, long-endurance drones and AI-based anomaly detection that flags unusual vessel behaviour rather than merely tracking vessels.
  - **Cover the Creeks:** Deploy shallow-draft craft, hovercraft and persistent aerial surveillance in creek areas such as Sir Creek, which conventional patrol vessels cannot access.
  - **Counter Unmanned Threats:** Invest in detection and interception of small aerial and surface unmanned systems, which conventional maritime radar handles poorly.
- **Enlisting the Coastal Community**

- **Fishermen as Eyes and Ears:** Strengthen Sagar Suraksha Dal with incentives, dedicated reporting channels and village-level watch groups, since fishermen are permanently at sea where no agency can be.
- **Community Policing on the Coast:** Extend intelligence gathering to the 3,331 coastal villages, where human intelligence is cheaper and faster than radar.

- **Securing Ports and Infrastructure**
  - **Standardise Minor Port Security:** Extend ISPS Code standards, vessel traffic management systems and minimum security requirements to non-major ports and private terminals.
  - **Protect Offshore and Digital Assets:** Develop a dedicated framework for platforms, pipelines and submarine fibre-optic cables, with ocean property mapped and periodically refreshed.
  - **Engage the Private Sector:** Bring shipping companies, port operators and blue economy enterprises into security planning, since they own much of the infrastructure being protected.
- **Regional and Doctrinal Steps**
  - **Implement the UNSC Five-Point Agenda:** India's framework presented at the United Nations Security Council covers barrier-free maritime trade, peaceful settlement of disputes under international law, responsible maritime connectivity, collective action against non-state and natural threats, and preservation of the marine environment.
  - **Deepen Regional Mechanisms:** Sustain leadership in the Indian Ocean Naval Symposium, MILAN, coordinated patrols with neighbours, the Colombo Security Conclave and Quad maritime domain awareness initiatives.
  - **Balance the Threat Picture:** Give institutional weight to illegal fishing, trafficking, pollution and disaster response, which occur constantly, alongside terrorism, which occurs rarely but catastrophically.
  - **Consider a Maritime Theatre Command:** Extend the Andaman and Nicobar Command model to integrate Navy, Coast Guard and supporting assets under unified operational command.

Conclusion:

India's coastal security architecture was built in the shadow of a single attack, and it has been shaped ever since by that attack's assumptions.. The threat has since moved outward and upward, into drone strikes within the Exclusive Economic Zone, hijackings hundreds of nautical miles offshore and contested presence across the Indian Ocean. The task now is less to add sensors than to **add authority**, by **giving the coordinating institutions statutory power,** the marine police a professional cadre, and every fishing boat an identity, so that the country stops relying on the Navy to compensate for gaps that were never naval to begin with.

5. NARCO-TERRORISM AND DRUG TRAFFICKING IN INDIA

Introduction:

Drug trafficking is no longer merely a **law-and-order or public-health problem**. It has evolved into a multidimensional internal-security challenge involving organised crime, border insecurity, money laundering, technology-enabled trafficking and, in certain cases, financing of terrorism.

The **United Nations Office on Drugs and Crime (UNODC)** describes the convergence between organised crime and terrorism as a growing security concern, with terrorist groups potentially benefiting from drug trafficking and other illicit activities. In India's context, the **Narcotics Control Bureau (NCB)** has also noted investigations indicating that proceeds of drug trafficking are being used to fund terrorism and armed groups.

Narco-terrorism

Narco-terrorism refers to the **nexus between narcotics trafficking and terrorism**, particularly where drug proceeds finance terrorist/insurgent activities or where terrorist groups facilitate and protect the drug trade.

Drug Trafficking and Narco-Terrorism:

Aspect	Drug trafficking	Narco-terrorism
Primary objective	Generation of illicit profit through production, transportation and distribution of narcotics.	Financing, facilitating or strengthening terrorist activity through narcotics-related networks.
Principal actors	Drug traffickers, organised criminal groups, couriers and financiers.	Terrorist organisations, terror-linked networks and their criminal partners.
Primary motivation	Financial gain.	Terrorist financing, logistical support and strategic objectives.
Security impact	Organised crime, corruption, addiction and violence.	Terror financing, arms procurement, infiltration, destabilisation and violence.
Nature of threat	Primarily criminal, although it can have wider security consequences.	Direct national-security and counter-terrorism concern.

Narco-Terrorism: a Serious Threat to India

- **Financing terrorism:** Drug profits provide terrorist organisations with an alternative source of financing that can be generated outside conventional banking channels.

  State Investigation Agency (SIA) Kashmir cracked a major transnational narco-terror network operating out of **Parimpora, Srinagar.** Large shipments of high-grade heroin were moved from across the international border into J&K to directly subsidise the operational costs of active terrorist modules.

- **Cross-border destabilisation:** Narcotics can be deliberately pushed across India's borders to finance terrorist networks while simultaneously creating social and economic vulnerabilities inside India.
- **Convergence of criminal and terrorist networks:** Terrorist organisations may exploit existing criminal networks for transportation, money laundering, safe houses, procurement and recruitment.
- **Arms and narcotics nexus:** The same smuggling channels can facilitate the movement of drugs, weapons, ammunition and other contraband.

  In August 2026, the **Delhi Police Special Cell and Federal Agencies** neutralised the **Shahzad Bhatti Network (SBN)**, a sprawling Pakistan-backed terror-crime syndicate, leading to over 200 arrests across 14 states

- **Youth and social vulnerability:** Drug abuse weakens human capital, increases crime and can create conditions for recruitment by criminal and extremist networks.
- **Corruption:** High-value narcotics profits create incentives for corruption among officials, intermediaries and logistics networks.
- **Money laundering:** Drug proceeds can be layered through legitimate businesses, informal financial channels, cryptocurrencies and other mechanisms to conceal their origin.

  The Narcotics Control Bureau (NCB) completely dismantled an elite, hidden pan-India drug network operating under the moniker **'Team Kalki'**.

- **Technology-enabled trafficking:** Traffickers increasingly leverage digital tools and advanced hardware—including dark web marketplaces, cryptocurrency transactions, encrypted messaging apps, and unmanned aerial vehicles (drones)—to conduct operations anonymously and bypass conventional border checkpoints.

The Government identified **dark web platforms, cryptocurrencies, online marketplaces and drones** as emerging challenges in India's fight against narcotics trafficking in **January 2025**.

India's Geographical Vulnerability:

Region / route	Significance for India	Major concern
Golden Crescent	Afghanistan, Pakistan and Iran lie to India's west and northwest.	Heroin and other opiate trafficking, with cross-border networks affecting Punjab and Jammu and Kashmir.
Golden Triangle	Myanmar, Laos and Thailand lie to India's east.	Opium and increasingly methamphetamine and other synthetic drugs entering through the Northeast.
Maritime routes	Drugs can move through the Arabian Sea, Indian Ocean, Bay of Bengal and commercial shipping networks.	Large consignments can exploit legitimate maritime trade and transnational logistics.
Land borders	India has extensive borders with major trafficking regions.	Difficult terrain and porous stretches facilitate clandestine movement.
Air and logistics networks	Airports, courier networks and commercial cargo provide alternative routes.	Small consignments and distributed trafficking networks are difficult to detect.
Digital routes	Dark-web marketplaces and encrypted communications facilitate transactions.	Physical interception becomes harder when transaction and coordination occur online.

Threat Matrix:

Threat	Mechanism	Security consequence	Appropriate response
Cross-border narcotics	Drugs are moved through land, maritime or air routes.	Financing of criminal and terrorist networks.	Integrated border surveillance and intelligence-led interdiction.
Narco-terrorism	Drug profits are channelled into terrorist activities.	Terror financing and internal security threats.	Joint counter-terrorism and narcotics investigation.
Synthetic drugs	Drugs are manufactured in clandestine laboratories using precursor chemicals.	Distributed production makes detection difficult.	Precursor monitoring, forensic capability and financial investigation.
Drone trafficking	Small consignments are delivered across borders without physical infiltration.	Difficult detection and direct linkage with cross-border networks.	Counter-drone systems, intelligence and border surveillance.
Dark-web trafficking	Buyers and sellers interact through concealed digital marketplaces.	Traditional physical investigation becomes less effective.	Cyber investigation and digital intelligence.
Cryptocurrency financing	Digital assets can be used to transfer or conceal illicit proceeds.	Difficult tracing and cross-border financial flows.	Financial intelligence and blockchain analytics.
Maritime trafficking	Large consignments exploit commercial shipping and maritime routes.	High-volume trafficking and international criminal networks.	Port intelligence, container profiling and maritime domain awareness.
Money laundering	Drug proceeds enter legitimate businesses and financial channels.	Criminal networks acquire economic influence.	Asset tracing, freezing and forfeiture.

Institutional Architecture: The institutional response has evolved from conventional drug enforcement towards a **multi-agency national-security architecture**.

Institution / agency	Principal role
Ministry of Home Affairs	Overall policy coordination, internal-security framework and inter-agency coordination.

Narcotics Control Bureau (NCB)	Apex coordinating and enforcement agency for drug-law enforcement, investigation of major cases and international coordination.
State Police and Anti-Narcotics Task Forces (ANTFs)	State-level investigation, intelligence, enforcement and local network disruption.
Central Bureau of Investigation (CBI)	Investigation of cases assigned under its jurisdiction and international cooperation through established mechanisms.
Intelligence Bureau (IB)	Intelligence inputs relating to national-security dimensions of narcotics and organised crime.
Directorate of Revenue Intelligence (DRI)	Detection and investigation of smuggling through customs and international trade channels.
Central Board of Indirect Taxes and Customs (CBIC)	Customs enforcement at ports, airports and land customs stations.
Enforcement Directorate (ED)	Investigation of money laundering and attachment of proceeds derived from crime under the Prevention of Money Laundering Act.
Financial Intelligence Unit–India (FIU-IND)	Collection and analysis of financial intelligence relating to suspicious transactions.
Border Security Force (BSF)	Border surveillance and action against narcotics trafficking along designated international borders.
Indian Coast Guard (ICG)	Maritime surveillance and interception of narcotics trafficking at sea.
Indian Navy	Maritime security and support to maritime interdiction operations.
Railway Protection Force (RPF)	Enforcement against narcotics trafficking through the railway network under applicable legal provisions.
Directorate General of Health Services / Central Drugs Standard Control Organisation (CDSCO)	Regulatory control relevant to narcotic and psychotropic substances, medicines and precursor-related issues.
Ministry of Social Justice and Empowerment	Demand reduction, prevention, treatment, rehabilitation and social responses to substance abuse.

Technology and Digital Enforcement driven government initiatives:

Initiative	Purpose	Significance
NCORD Portal	Provides a common information gateway for stakeholders involved in drug-law enforcement.	Improves inter-agency information sharing.
NIDAAN Portal	National Integrated Database on Arrested Narco-Offenders containing information on offenders involved in NDPS cases.	Enables offender profiling and network identification.

CCTNS	Links police stations through a common digital platform.	Supports investigation and data analytics.
MANAS	Citizen-centric platform for reporting drug-related activity and accessing counselling and rehabilitation support.	Converts citizens into participants in drug-control efforts.
Financial intelligence systems	Track suspicious financial transactions and illicit proceeds.	Attacks the economic backbone of trafficking networks.
Cyber investigation	Tracks dark-web marketplaces, digital communication and cryptocurrency-related activity.	Addresses the technology-enabled evolution of trafficking.

The **NIDAAN portal**, according to the Ministry of Home Affairs in **March 2025**, was developed to provide Drug Law Enforcement Agencies and other investigation agencies with information on narcotics offenders under the Narcotic Drugs and Psychotropic Substances Act, 1985.

The **MANAS helpline, 1933**, was launched in **July 2024** and expanded as a secure platform for reporting trafficking, illicit cultivation and related offences, while also providing counselling and rehabilitation support.

- **Legal and Policy Framework:**
  - • **Narcotic Drugs and Psychotropic Substances Act, 1985:** The **Narcotic Drugs and Psychotropic Substances Act, 1985** forms the principal legal framework for controlling narcotic drugs and psychotropic substances in India. It provides the legal basis for prohibition, regulation, investigation, seizure, prosecution and punishment relating to specified narcotic and psychotropic substances.
  - • **Prevention of Illicit Traffic in Narcotic Drugs and Psychotropic Substances Act, 1988:** The **PITNDPS Act, 1988** provides for preventive detention of persons involved in illicit trafficking of narcotic drugs and psychotropic substances. Its importance is increasing because repeat offenders and organised networks cannot always be addressed effectively through individual seizure cases alone.
  - **Prevention of Money Laundering Act, 2002:** Financial investigation is essential because the objective should be to **dismantle the financial architecture of the network**, rather than merely arresting couriers.

The Ministry of Home Affairs reported that financial investigations resulting in property seizure or freezing increased from **587 cases worth ₹276 crore in 2023 to 1,356 cases worth ₹836 crore in 2025**.

International Case Studies

Case study	What happened	Result	Lesson for India
Colombia	Drug trafficking became deeply intertwined with armed groups and insurgent financing.	Narcotics profits contributed to the persistence of armed violence and territorial criminal control. UNODC has identified Colombia as a major example of the drug-conflict nexus.	**Drug trafficking can evolve from organised crime into a broader challenge to state authority when armed groups acquire financial and territorial power.**

Afghanistan	Opium production historically provided a major illicit economic base, with links between narcotics and armed actors. UNODC continues to monitor the relationship between cultivation, trafficking and regional security.	Afghanistan became a major illustration of how illicit economies can interact with conflict and instability.	**Supply reduction must be accompanied by alternative livelihoods and regional cooperation; otherwise illicit economies can re-emerge through new routes.**
Myanmar and the Golden Triangle	Synthetic-drug production, particularly methamphetamine, has become a major regional concern.	The shift demonstrates how synthetic drugs can alter traditional trafficking geography because production can be geographically flexible.	**India's Northeast requires greater focus on synthetic drugs, precursor chemicals and clandestine laboratories, not only traditional opiates.**

Major Challenges:

- **Precursor chemicals:** India's legitimate chemical and pharmaceutical industries create a dual-use challenge because chemicals with legitimate applications can potentially be diverted towards illicit production.

  The Government specifically highlighted India's position as a major producer of precursor chemicals as a concern in January 2025.

- **Supply-side enforcement alone is insufficient:** Seizures remove particular consignments but do not necessarily dismantle the organisation behind them.

  Despite the Indian Narcotics Control Bureau making a record seizure of 3,300 kg of drugs from a dhow off the Gujarat coast, the international kingpins operating the cartel from overseas remained untouched.

- **Criminal-terrorist convergence:** The distinction between organised crime and terrorism becomes blurred when criminal networks provide logistics and finance to terrorist organisations.

  Shahzad Bhatti Network crackdowns exposed a deep nexus where local interstate gangsters and drug smugglers directly managed safe houses and weapons logistics for Pakistan-backed terror handlers

- **Synthetic drugs and clandestine laboratories:** Synthetic drugs reduce dependence on traditional cultivation geography and can move production closer to consumer markets.

- **Technology-enabled trafficking:** Dark-web marketplaces, cryptocurrencies, encrypted communications and drones make conventional policing less effective.

  The Narcotics Control Bureau's bust of the **'Team Kalki' syndicate** revealed an operation running entirely on darknet forums and the Session app, using cryptocurrency wallets to hide multi-crore payments

- **Maritime trafficking:** Large consignments can exploit India's extensive coastline, commercial shipping and transnational maritime routes.

  A joint Indian Navy and NCB operation intercepted a single **transnational cartel vessel carrying 700 kg of pure meth** via the Indian Ocean "Southern Route" bound for the Indian coastline.

- **Financial complexity:** Trafficking proceeds may pass through multiple accounts, shell entities, informal channels and digital assets before entering the legitimate economy.
- **International jurisdiction:** Drug networks frequently separate production, financing, transportation and distribution across different countries, creating investigation and extradition challenges.

  India faces complex diplomatic hurdles trying to **extradite high-profile narco-terror handlers hiding in Pakistan and the UAE**, who run Indian syndicates completely by remote control
- **Demand reduction:** A purely enforcement-oriented approach cannot address addiction and social vulnerability.

  The government expanded the **Nasha Mukt Bharat Abhiyaan** across hundreds of districts, recognizing that jail time alone cannot fix the deep-rooted addiction crisis in vulnerable border youth

Way Forward:

- **Shift from seizure-centric to network-centric policing:** The objective should be to identify the **financier, organiser, logistics provider and money-laundering architecture**, not merely the carrier.

  Operation WHITE STRIKE provides a useful example of this approach
- **Follow the money:** Financial investigation, asset freezing, forfeiture and money-laundering investigation should accompany every major trafficking case.
- **Strengthen border intelligence:** Border forces, NCB, State Police and intelligence agencies should combine narcotics intelligence with information on drones, suspicious logistics networks, arms trafficking and terror financing.

  Border Security Force (BSF) and Punjab Police integrated the **NIDAAN database** to automatically map cross-border drone crash sites alongside local phone logs of known weapons and narcotics receivers.
- **Strengthen precursor control:** India needs better monitoring of legitimate chemical supply chains, suspicious procurement patterns and clandestine manufacturing.

  Narcotics Control Bureau (NCB) mandated strict digital tracking on the sale of **acetic anhydride and ephedrine** to prevent industrial chemicals from being diverted to clandestine labs.
- **Build synthetic-drug forensic capacity:** Specialised laboratories and trained investigators are required to identify emerging synthetic drugs and New Psychoactive Substances quickly.

  Narcotics Control Bureau (NCB) mandated strict digital tracking on the sale of **acetic anhydride and ephedrine** to prevent industrial chemicals from being diverted to clandestine labs.
- **Deepen cyber-narcotics capabilities:** Law-enforcement agencies require specialised capabilities in dark-web investigation, cryptocurrency tracing, digital forensics and online undercover operations.

  The NCB's cyber wing deployed advanced blockchain analytics tools to crack the anonymous **'Team Kalki' syndicate**, which operated entirely via darknet marketplaces and the Session chat application
- **Strengthen maritime interdiction:** Indian Coast Guard, Indian Navy, Customs, Directorate of Revenue Intelligence and port authorities should strengthen intelligence-led profiling of suspicious maritime consignments..
- **Strengthen international cooperation:** Information exchange, extradition, mutual legal assistance, Red Notices, joint investigations and controlled deliveries are important because trafficking networks operate across jurisdictions.

  Indian enforcement agencies collaborated with Interpol to issue **Red Corner Notices** against Dubai- and Pakistan-based kingpins who orchestrate domestic drug distribution networks remotely
- **Make NCORD genuinely intelligence-driven:** NCORD should evolve beyond coordination meetings into a continuously updated national narcotics intelligence architecture that links Central agencies, State Police, border forces, financial intelligence and cyber-investigation systems.

- **Use citizens as an intelligence multiplier: MANAS** provides an opportunity to convert citizens into an additional source of intelligence against local trafficking and illicit cultivation while simultaneously providing rehabilitation support.
- **Implement the Vision Document 2026–2029:** The **Vision Document on Drug Control 2026–2029**, released in **June 2026**, provides a current policy framework based on a whole-of-government and whole-of-society approach. It should be used to align enforcement, demand reduction, rehabilitation, financial investigation and inter-agency intelligence.

Conclusion:

Narco-terrorism represents the **convergence of drugs, organised crime, terrorism, technology and illicit finance**. India's geographical location between major drug-producing regions, combined with its extensive borders, maritime connectivity and expanding digital economy, makes the threat multidimensional.

India has consequently moved towards an increasingly integrated response involving the **Narcotics Control Bureau, State Police, Anti-Narcotics Task Forces, Border Guarding Forces, Customs, financial-intelligence agencies, intelligence agencies and international partners**. The four-tier NCORD mechanism, NIDAAN, MANAS, financial investigations and the **Vision Document on Drug Control 2026–2029** indicate this shift.

The next stage must move from **"catching consignments" to "dismantling ecosystems"**, combining intelligence, border management, financial investigation, cyber capabilities, precursor control, international cooperation and demand reduction.

■■■

MGP/BP/GS3/042

VARIOUS SECURITY FORCES & AGENCIES AND THEIR MANDATE

1. VARIOUS SECURITY FORCES AND THEIR MANDATE

Context:

In a Rajya Sabha reply of July 2026, the MHA disclosed **61,738 vacancies across the CAPFs and Assam Rifles against a sanctioned strength of 10,73,728**, with actual strength at 10,04,482. CRPF leads with 18,134 vacancies, followed by CISF with 17,930.

Introduction:

India's security is delivered through a layered structure. The **Armed Forces under the Ministry of Defence** defend against external aggression, the **Central Armed Police Forces under the Ministry of Home Affairs** guard the borders and manage internal security, and the **state police** remain the first responder, since police and public order are State subjects under Entries 1 and 2 of the State List.

The scale of the task:

- India has a **land border of 15,106.7 km** with seven neighbours and a **coastline of 7,516.6 km**, covering terrain that ranges from glaciated passes above 18,000 feet to riverine char lands, desert, dense forest and marshland.
- Border-wise: **Bangladesh 4,096.7 km, China 3,488 km, Pakistan 3,323 km, Nepal 1,751 km, Myanmar 1,643 km, Bhutan 699 km and Afghanistan 106 km.**
- The CAPFs, at over ten lakh personnel, constitute one of the largest uniformed forces in the world outside the regular military.

A terminology correction that matters

The term **"paramilitary forces" is officially obsolete for the CAPFs.** In March 2011 the Ministry of Home Affairs directed that the expression not be used for these forces, which are correctly designated **Central Armed Police Forces**. Where "paramilitary" is still used with precision, it refers to forces commanded by Armed Forces officers, principally the **Assam Rifles, the Special Frontier Force and the Indian Coast Guar**d.

The seven CAPFs

The MHA recognises exactly seven: **Assam Rifles, BSF, CISF, CRPF, ITBP, NSG and SSB.** The NDRF is raised from CAPF battalions but functions separately under the NDMA, and the RPF is an Armed Force of the Union under the Ministry of Railways, not a CAPF.

Classification of India's Security Forces

Category	Ministry or Authority	Constituents
Armed Forces	Defence	Indian Army, Indian Navy, Indian Air Force
Central Armed Police Forces	Home Affairs	Assam Rifles, BSF, CISF, CRPF, ITBP, NSG, SSB
Other Armed Forces of the Union	Defence and Railways	Indian Coast Guard, Railway Protection Force
Specialised Central Forces	Cabinet Secretariat and MHA	Special Protection Group, Special Frontier Force, National Disaster Response Force
Army-Raised Internal Security Forces	Defence	Rashtriya Rifles for counter-insurgency in Jammu and Kashmir, Territorial Army as second line
State Forces	State Governments	State police, State Armed Police battalions, Home Guards, and specialist units such as Greyhounds, Special Operations Group and Thunderbolts

The Armed Forces

Service	Core Mandate
Indian Army	Land defence, mountain and high-altitude warfare, counter-insurgency, and aid to civil authority during internal disturbance and disaster
Indian Navy	Maritime security, protection of sea lines of communication, sea-based nuclear deterrence and blue-water operations across the Indian Ocean Region
Indian Air Force	Air dominance, strategic strike, aerial surveillance and reconnaissance, transport and rapid strategic lift
Indian Coast Guard	An armed force of the Union under the Ministry of Defence, established provisionally on 1 February 1977 and formally under the Coast Guard Act, 1978. Handles coastal surveillance, anti-smuggling, marine safety, fisheries protection and marine environment protection

The Central Armed Police Forces:

Force	Raised and Statute	Border or Domain	Distinguishing Feature
Assam Rifles	1835 as Cachar Levy; Assam Rifles Act, 2006	Indo-Myanmar, 1,643 km	Oldest central force; unique **dual control**, administrative under MHA and operational under the Army
Border Security Force	1 December 1965; BSF Act, 1968	Indo-Pakistan and Indo-Bangladesh	World's largest border guarding force, with its own air, marine and artillery wings
Central Industrial Security Force	CISF Act, 1968; raised 1969	Critical infrastructure	Only force with a dedicated **Fire Wing**; operates on a **compensatory cost basis**
Central Reserve Police Force	1939 as Crown Representative's Police; CRPF Act, 1949	Internal security nationwide	Largest CAPF, with CoBRA, Rapid Action Force and Mahila battalions

Indo-Tibetan Border Police	24 October 1962; ITBP Act, 1992	Indo-China, 3,488 km	High-altitude specialists operating up to 18,750 feet
National Security Guard	Approved 1984; NSG Act, 1986	Counter-terrorism	Federal contingency force with Army and CAPF components
Sashastra Seema Bal	1963 as Special Service Bureau; renamed 2003	Indo-Nepal 1,751 km and Indo-Bhutan 699 km	Community mobilisation mandate on open borders

Force Profiles:

- **Assam Rifles**
  - **Origin and Evolution:** Raised in 1835 as the **Cachar Levy** with 750 men to protect tea estates and settlements from tribal raids, renamed the Frontier Force, then Assam Military Police, and finally **Assam Rifles in 1917** in recognition of service in Europe and West Asia during the First World War.
  - **Mandate:** Counter-insurgency in the North East, guarding the Indo-Myanmar border under the One Border One Force policy since 2002, and rear area security during conflict.
  - **Dual Control:** Administrative control including salaries and infrastructure rests with the **MHA**, while operational control rests with the **Army**, and the Director General is a serving Lieutenant General.
  - **Distinction:** Known as the **"Friends of the North East"** for its heart-and-mind approach, and among the most decorated of India's central forces.
  - **Operations:** Sino-Indian War of 1962, **Operation Pawan** with the Indian Peace Keeping Force in Sri Lanka in 1987, and sustained counter-insurgency in Nagaland, Manipur and Assam.
  - **Live Issue:** Successive Parliamentary Standing Committee reports have urged that the dual control question be settled, since divided command affects postings, promotions, pensions and accountability.
- **Border Security Force**
  - **Origin:** Raised on **1 December 1965** after Pakistani attacks on **Sardar Post, Chhar Bet and Beria Bet** in the Rann of Kutch demonstrated that State Armed Police could not hold an international border. Created on the recommendation of the Committee of Secretaries, with **K. F. Rustamji** as first Director General.
  - **Motto and Mission:** *Jeevan Paryant Kartavya*, duty unto death, with the mission statement "Any Task, Anytime, Anywhere."
  - **Peacetime Tasks:** Building a sense of security among border populations, preventing infiltration, smuggling and trans-border crime, patrolling and protecting the border fence, and collecting border intelligence.
  - **Wartime Tasks:** Holding ground in less threatened sectors under Army control, protecting vital installations such as airfields, limited offensive action against enemy irregulars, guarding prisoners of war, maintaining order in occupied territory, and anti-infiltration and refugee control.
  - **Specialised Capability:** Operates its **own air wing, a marine wing including the largest riverine deployment in the Sundarbans, an artillery regiment**, and camel and dog breeding establishments.
  - **Operations:** Distinguished service in the 1971 war leading to the liberation of Bangladesh, in Kargil, and in anti-Naxal operations, elections and United Nations peacekeeping.
  - **Live Issue:** The **October 2021 extension of BSF jurisdiction from 15 km to 50 km** inside Punjab, West Bengal and Assam was contested by those states as an encroachment on policing powers, and Punjab moved the Supreme Court under Article 131.
- **Central Industrial Security Force**

- **Statutory Basis:** Established under the **CISF Act, 1968** and raised in 1969 with **3,129 personnel**, it is an Armed Force of the Union and has grown into a force of about two lakh sanctioned strength.
- **Deployment:** Airports, metro systems, seaports, atomic energy and space establishments, petroleum, coal, steel, power and mining installations, government buildings, heritage monuments and the Parliament House complex.
- **Compensatory Cost Model:** Client organisations reimburse the cost of security, which makes the force largely self-financing and distinguishes it from every other CAPF.
- **Fire Wing:** The only central force with a dedicated fire service, providing industrial fire protection and disaster response.
- **Private Sector and Consultancy:** Deployed at select private installations, and offers professional security consultancy covering **threat perception, risk analysis, access control, perimeter protection and security audits**.
- **VIP Protection:** Its Special Security Group provides Z Plus, Z, Y and X category protection to designated protectees.

- **Central Reserve Police Force**
  - **Origin:** Raised on **27 July 1939 at Neemuch** as the Crown Representative's Police, and reconstituted under the **CRPF Act, 1949** after Sardar Patel gave it its present name and character.
  - **Scale and Role:** The **largest CAPF**, with roughly 247 battalions, deployed across Jammu and Kashmir, the North East and the Left Wing Extremism belt, and used for riot control, election duty and VIP security.
- **Specialised Wings:**
  - **CoBRA (2008):** Commando Battalion for Resolute Action, trained in jungle warfare and guerrilla tactics for Left Wing Extremism operations.
  - **Rapid Action Force (1992):** Raised after the communal violence of that period, distinguished by blue camouflage, for riot control and communal situations.
  - **Mahila Battalions (1986):** The world's first all-women contingent in a force of this kind, now expanded to several battalions.
  - **Parliament Duty Group and VIP Security Wing** for high-value static and personal protection.
  - **Historic Significance:** The action at **Hot Spring in Ladakh on 21 October 1959**, where a CRPF patrol was ambushed by Chinese forces, is commemorated annually as **Police Commemoration Day**.
- **Indo-Tibetan Border Police**
- **Origin:** Raised on **24 October 1962** in the aftermath of the Sino-Indian conflict, with statutory backing through the **ITBP Act, 1992** and Rules of 1994. Motto *Shaurya, Dridhata, Karma Nishtha*.
- **Border Mandate:** Guards the entire **3,488 km India-China border** from the Karakoram Pass in Ladakh to Jachep La in Arunachal Pradesh, and replaced the Assam Rifles in Sikkim and Arunachal Pradesh in 2004 under One Border One Force.
- **Operating Conditions:** Border outposts sit between **9,000 and 18,750 feet**, with temperatures falling to minus 45 degrees Celsius, and the force maintains trained mountaineers, skiers and high-altitude medical capability.
- **Additional Duties:** Security to the **Kailash Mansarovar Yatra since 1981**, protection of the Indian embassy and consulates in Afghanistan, security of sensitive installations, anti-Naxal deployment in Chhattisgarh, and United Nations peacekeeping in Angola, Cambodia, Bosnia, Kosovo and elsewhere.
- **Disaster Response:** First responder in Himalayan disasters through Regional Response Centres, having rescued **over 33,000 pilgrims during the 2013 Uttarakhand flash floods**, and contributes battalions to the NDRF.
- **Civic Action:** Medical camps in remote border villages and large-scale afforestation and Himalayan ecology programmes.

- **National Security Guard**
  - **Origin:** Approved by the Union Cabinet in **1984** following the security crises of that year, and constituted under the **NSG Act, 1986**. Popularly the **Black Cats**.
  - **Character:** A **Federal Contingency Deployment Force** for counter-terrorism, hostage rescue, counter-hijacking and bomb disposal, employed only in exceptional situations rather than routine deployment.
- **Two Components:**
  - **Special Action Group (SAG):** Drawn from the Army, forming the strike element.
  - **Special Ranger Group (SRG):** Drawn from CAPFs and state police, providing support, cordon and VIP protection.
- **Doctrine:** Swift strike followed by immediate withdrawal once the objective is achieved, described institutionally as a zero-error force.
- **Major Operations:** Operation Black Thunder at the Golden Temple in 1988, the resolution of the 1993 Amritsar hijacking, the Akshardham temple operation of 2002, the Mumbai operations of November 2008, and Pathankot in 2016.
- **Post-26/11 Reform: Regional hubs** established at Mumbai, Chennai, Hyderabad, Kolkata and Gandhinagar to cut the response time that proved fatal in 2008.
- **Sashastra Seema Bal**
  - **Origin:** Raised in **March 1963 as the Special Service Bureau** after the 1962 war, with the mandate of building **Total Security Preparedness** in border areas by organising the local population, rather than by conventional guarding.
  - **Transformation:** Following the Kargil Review Committee and the Group of Ministers report, it was declared a **border guarding force under the MHA in January 2001** and renamed **Sashastra Seema Bal in December 2003**.
  - **Borders Guarded:** The **Indo-Nepal border of 1,751 km and the Indo-Bhutan border of 699 km**, both open borders with visa-free movement under long-standing treaty arrangements.
  - **Distinctive Mandate:** Beyond interdiction, it trains border residents in self-defence, vigilance and community response, converting the population into what the force describes as the eyes and ears of the border.
- **Core Challenge:** Open borders with shared kinship, cultural and economic ties are structurally porous, and cannot be sealed, which is why the community mobilisation model was designed for this frontier.

Other Important Forces:

- **Special Protection Group:** Raised in 1985 on the recommendation of the **Birbal Nath Committee** and governed by the **SPG Act, 1988**. Provides proximate security to the Prime Minister. The **2019 amendment** restricted cover to the Prime Minister and immediate family residing at the official residence, and to former Prime Ministers and their immediate family for **five years** after demitting office. The Director is an IPS officer designated as Secretary in the Cabinet Secretariat, with personnel drawn from CAPFs and the RPF.
- **Special Frontier Force:** Raised in **November 1962** as **Establishment 22**, operating under the Cabinet Secretariat with Armed Forces officers in command, and recruited historically from the Tibetan refugee community. A paramilitary force in the strict sense, it has seen action in 1971 and in the Ladakh operations of 2020.
- **National Disaster Response Force:** Constituted under the **Disaster Management Act, 2005**, with battalions drawn from BSF, CRPF, CISF, ITBP, SSB and Assam Rifles. The world's largest dedicated disaster response force, functioning under the NDMA.

- **Rashtriya Rifles:** Raised in 1990 as the Army's dedicated counter-insurgency force, deployed principally in Jammu and Kashmir through Counter Insurgency Force formations, and funded and controlled by the Ministry of Defence.
- **Territorial Army:** Raised under the **Territorial Army Act, 1948** as a citizen second line of defence, including Ecological Task Force battalions engaged in afforestation and environmental restoration.
- **Railway Protection Force:** An Armed Force of the Union under the Ministry of Railways, protecting railway property, passenger areas and passengers, running specialised operations against human trafficking, child rescue and passenger safety.
- **State Armed Police and Specialist Units:** State Armed Police battalions supply the states' own reserve, and specialist counter-insurgency units such as **Greyhounds** in Andhra Pradesh and Telangana, the **Special Operations Group** in Jammu and Kashmir and **Thunderbolts** in Kerala are frequently more effective locally than central deployment.
- **Home Guards**: A voluntary auxiliary force assisting the police in maintaining order, traffic regulation and disaster relief.

Contribution of the Central Forces:

- **Border Management**
  - **Vigil on Terrain the State Cannot Hold:** Central forces occupy ground where state police have neither the training nor the sustainment capacity to operate.

    ITBP holds border outposts up to 18,750 feet where temperatures fall to minus 45 degrees Celsius, and BSF maintains riverine deployment in the Sundarbans.
  - **Interdiction of Trans-Border Crime:** Patrols and surveillance intercept narcotics, fake currency, arms and trafficked persons before they reach the interior.

    Repeated seizures of heroin and weapons on the Punjab border, increasingly delivered by drone, are made almost entirely by BSF rather than by state police.
  - **Confidence Among Border Populations:** Civic action, medical camps and self-defence training convert border residents into a first line of information.

    SSB's community mobilisation model was designed for exactly this on the visa-free Nepal and Bhutan borders where physical sealing is impossible.
- **Internal Security**
  - **Counter-Insurgency:** CRPF and its CoBRA battalions conduct area domination and jungle operations in the Left Wing Extremism belt, while Assam Rifles carries the North East.
  - **Communal and Public Order Situations:** The Rapid Action Force provides a neutral, visibly non-local armed presence during riots and large agitations without displacing state police authority.
  - **Election Security:** Central deployment is what makes free polling possible in disturbed districts, since local police are perceived as partisan.
  - **Counter-Terror Response:** NSG regional hubs cut response time from hours to minutes, addressing the specific failure exposed in November 2008.
- **Protection of Critical Assets**
  - **Industrial and Aviation Security:** CISF secures airports, metros, ports, refineries, nuclear and space installations and the Parliament complex, and provides consultancy that raises security standards beyond its own deployments.
  - **VIP Protection:** SPG, CISF's Special Security Group and CRPF's VIP wing collectively cover protectees whom state police cannot secure to the required standard.

- **Disaster Response**
  - **First Responders in the Himalaya:** ITBP's high-altitude teams and regional response centres reach disaster sites ahead of any other agency.

    ITBP rescued over 33,000 pilgrims during the 2013 Uttarakhand flash floods, and central forces provided logistics and medical aid during the Kerala floods.
  - **National Reach:** NDRF battalions, drawn from CAPFs, deploy for floods, cyclones, earthquakes, industrial accidents and building collapses across the country and abroad.

Issues:

- **Manpower and Overstretch**
  - **Persistent Vacancies:** Chronic shortfalls force commanders to deny leave and extend duty hours in order to maintain operational strength.

    As of July 2026 there were 61,738 vacancies against a sanctioned strength of 10,73,728, with CRPF and CISF together accounting for over half.
  - **No Peace Postings:** Continuous rotation between counter-insurgency, election duty, VIP security and law and order leaves little recovery time between deployments.
  - **Mission Creep:** Forces raised for a defined mandate are routinely diverted, so CRPF battalions guard VIPs and border-guarding forces are drawn into anti-Naxal and election duty far from their assigned frontier.
- **Welfare and Morale**
  - **Attrition and Suicide:** The human cost of overstretch appears in exit data rather than in operational statistics.

    55,555 personnel took voluntary retirement or resigned between 2020 and 2024, and 730 died by suicide, with an MHA task force citing harassment, humiliation and leave denial among the triggers.
  - **No Tribunal or Grievance Mechanism:** CAPF personnel have **no access to the Armed Forces Tribunal**, and **Article 33** permits Parliament to restrict their fundamental rights, leaving very limited avenues for redress.
  - **Living and Working Conditions:** Housing shortfalls, sanitation, ration quality, potable water and family accommodation remain inadequate at many border outposts.
  - **Legacy Practices:** The sahayak or buddy system, in which jawans perform personal tasks for officers, persists in places and corrodes professional dignity.
  - **Family Separation:** Prolonged separation, compounded by insecurity at home in rural areas, is a documented contributor to stress and mental health vulnerability.
- **Cadre and Command**
  - **IPS Dominance in Senior Ranks:** Deputation of IPS officers to senior CAPF appointments limits promotion prospects for cadre officers and depresses morale.

    Litigation on Organised Group A Service status and Non-Functional Financial Upgradation for CAPF officers ran for years before being settled in the cadre's favour, which is itself evidence of how contested career progression has been.
  - **Assam Rifles Dual Control:** Divided administrative and operational control produces conflicting incentives on postings, promotions and pensions, and successive parliamentary committees have urged resolution.
  - **Parity Deficit with the Army:** Differences in Military Service Pay, hardship allowances, canteen benefits and martyr recognition for personnel serving in identical terrain generate persistent grievance.

  - **Opaque Promotion Pathways:** Absence of published, metric-based promotion criteria produces stagnation and a sense of arbitrariness.
- **Capability and Modernisation**
  - **Low Capital Allocation:** The overwhelming share of CAPF budgets goes to salaries and establishment, leaving limited room for weapons, protective equipment, surveillance and mobility.
  - **Ageing Equipment:** Modernisation plans have repeatedly slipped, so personnel operate with dated weapons and inadequate night-fighting, counter-IED and protective capability.
  - **Technology Asymmetry:** Drone-based weapons and narcotics delivery on the Punjab and Jammu borders has outpaced counter-drone capability, giving adversary networks a cheap technological edge.
  - **Intelligence Integration:** Border forces generate substantial ground intelligence that is not always integrated into the national picture through MAC and the state Subsidiary MACs.
- **Structural and Federal Issues**
  - **Jurisdiction Disputes:** Extension of BSF's operational jurisdiction to 50 km inside Punjab, West Bengal and Assam in 2021 was resisted by those states as an encroachment on policing, and became a constitutional dispute.
  - **Overlapping Mandates:** Duplication between central forces, and between central forces and state police, produces both redundancy in some sectors and gaps in others.
  - **Open and Riverine Borders:** Visa-free movement on the Nepal and Bhutan borders, and shifting char lands on the Bangladesh riverine stretch, limit what physical guarding alone can achieve.
  - **Civil-Military Understanding:** Limited familiarity with operational realities within the civilian decision chain contributes to delayed procurement and unrealistic deployment demands.

Committees and Judicial Directions

- **Kargil Review Committee (1999):** Identified fragmented border guarding as a systemic weakness and set in motion the reorganisation of border management.
- **Group of Ministers (2001), Task Force on Border Management under Madhav Godbole:** Produced the **One Border, One Force** principle and recommended a single ministry accountable for each border.
- **Padmanabhaiah Committee (2000) on Police Reforms:** Recommended restructuring of the police and clearer separation between law and order and investigation, which shapes what central forces are called upon to substitute for.
- **Second Administrative Reforms Commission, Fifth Report on Public Order:** Addressed the pattern of central forces being used to compensate for state police deficits.
- **Parliamentary Standing Committee on Home Affairs:** Has repeatedly reported on CAPF vacancies, housing satisfaction, medical facilities, attrition and the unresolved Assam Rifles command question.
- **Seventh Central Pay Commission:** Examined and partially addressed CAPF grievances on allowances and career progression, while parity on Military Service Pay remains unresolved.
- **Judicial Intervention on Cadre Status**: Courts upheld Organised Group A Service status and Non-Functional Financial Upgradation for CAPF officers, correcting a long-standing career disadvantage.

Government Measures

- **Reforms in the Armed Forces**
  - **Chief of Defence Staff and Department of Military Affairs:** Created in January 2020 for single point military advice and integration of tri-service planning, training and procurement.
  - **Theatreisation:** Integration of Army, Navy and Air Force assets under unified operational commands, with the final proposal submitted in May 2026 and awaiting Cabinet Committee on Security approval.

- **Agnipath Scheme:** Introduced in 2022 to lower the average age profile of the forces and create a leaner, technology-ready structure.
- **SRIJAN Portal and Indigenisation:** Enables Indian industry and MSMEs to substitute imported defence components, alongside positive indigenisation lists and the Uttar Pradesh and Tamil Nadu defence corridors.

- **Measures for the CAPFs**
  - **Recruitment Reform:** Over **71,000 new posts** created in recent years, recruitment for Constable General Duty routed through the Staff Selection Commission under a memorandum of understanding, a designated nodal force for General Duty recruitment, and **RFID technology** deployed in physical and medical testing to compress selection timelines.
  - **Women's Representation: 33 per cent reservation at constable rank in CRPF and CISF**, and **15 per cent in BSF, SSB and ITBP**, alongside Mahila battalions and all-women border outposts.
  - **Welfare Interventions:** Yoga and meditation programmes, professional counselling, regulated duty hours, transparent leave and transfer policy, choice posting after hard-area service, and regular Sainik Sammelans for grievance airing.
  - **Health and Housing: Ayushman CAPF** extending health cover to personnel and dependants, and an e-Awas housing allotment system aimed at raising the housing satisfaction ratio.
  - **Career Progression:** Modified Assured Career Progression at 10, 20 and 30 years, and pay upgradation for Group A officers following Pay Commission recommendations.
  - **Recognition:** Extension of gallantry awards and martyr status to central force personnel killed in action, and the Bharat Ke Veer platform for direct public contribution to families of the fallen.
- **Border Infrastructure and Technology**
  - **Comprehensive Integrated Border Management System (CIBMS):** Combines thermal imagers, radars, sensors, command and control and manpower for real-time situational awareness, piloted on riverine stretches in Assam and in Jammu.
  - **BOLD-QIT:** Border Electronically Dominated Quick Reaction Team Interception Technique, deployed on the riverine Brahmaputra stretch where physical fencing is not feasible.
  - **Vibrant Villages Programme:** Development of frontier settlements along the northern border to arrest out-migration and create a resident population that supports border security.
  - **Border Area Development Programme and Border Infrastructure Schemes:** Roads, fencing, floodlighting, border outposts and integrated check posts.

Way Forward:

- **Fix Manpower and Deployment**
  - **Fill Vacancies in a Time-Bound Manner:** Recruitment against the 61,738 vacant posts is the single measure that would simultaneously ease duty hours, leave denial and stress-related attrition.
  - **Raise Additional Battalions:** Sanctioned strength itself needs revision, since the mandate has expanded faster than the establishment.
  - **Guarantee Peace Postings:** Institutionalise a fixed rotation cycle so every unit receives a defined period at a peace station after hard-area deployment.
  - **End Mission Creep:** Restrict each force to its raised mandate and build dedicated capacity for VIP security and election duty rather than drawing down operational battalions.
- **Address Welfare and Justice**
  - **Create a Dedicated Tribunal:** Establish a statutory grievance redressal mechanism and a tribunal for CAPF personnel, since the absence of any equivalent to the Armed Forces Tribunal is the sharpest institutional inequity they face.

  - **Institutionalise Mental Health Support:** Make counselling, stress management and recreation a standard establishment at every unit rather than a periodic campaign.
  - **Adopt One Area, One Allowance:** Pay identical hardship and hard-area allowances to Army and CAPF personnel deployed in the same terrain, determined by conditions rather than by force.
  - **Build Composite Border Outposts:** Standardise border outposts with medical facilities, communication, potable water, sanitation, power and family accommodation.
  - **End Legacy Practices:** Formally abolish the sahayak system and enforce the prohibition through accountability at unit command level.
- **Reform Cadre and Command**
  - **Resolve Assam Rifles Dual Control:** Settle the administrative and operational command question, which parliamentary committees have repeatedly flagged as a source of institutional drift.
  - **Rationalise IPS Deputation:** Cap deputation at senior levels and ensure timely, criteria-based promotion for cadre officers.
  - **Extend Service Pay Equivalence:** Consider Military Service Pay equivalence for CAPF personnel performing comparable operational roles.
- **Modernise Capability**
  - **Protect Capital Allocation:** Ring-fence a fixed share of the CAPF budget for modernisation so funds are not diverted to revenue expenditure.
  - **Scale CIBMS and Counter-Drone Systems:** Extend smart border management beyond pilot stretches and institutionalise counter-drone capability on the western border.
  - **Deepen Private Participation:** Open equipment procurement to Indian industry through iDEX-type mechanisms to shorten timelines and reduce import dependence.
  - **Integrate Ground Intelligence:** Formalise the flow of border force intelligence into MAC and the state Subsidiary MACs so tactical inputs feed the national picture.
- **Work With the States**
  - **Resolve Jurisdiction Through Consultation:** Settle questions such as BSF's operational jurisdiction through a consultative framework rather than unilateral notification, since deployment without state cooperation is operationally weaker.
  - **Strengthen State Police Capacity:** Implement the *Prakash Singh* directions, since central forces are repeatedly used to compensate for state police deficits that only the states can fix.

Case Studies

Case	What happened	Result and Lesson
Sardar Post, Kutch (1965)	Pakistani attacks on border posts held by State Armed Police exposed the inadequacy of state forces on an international border	Led directly to the creation of the BSF on 1 December 1965 under a Committee of Secretaries recommendation. Border guarding requires a dedicated central force, not an extension of policing
Hot Spring, Ladakh (1959)	A CRPF patrol was ambushed by Chinese forces in eastern Ladakh	Commemorated annually as Police Commemoration Day, and a reminder that central police forces were holding the northern frontier before ITBP existed
26/11 Mumbai (2008)	NSG took hours to reach Mumbai from Manesar because it had no regional presence	Produced **regional NSG hubs** at Mumbai, Chennai, Hyderabad, Kolkata and Gandhinagar. Response time, not capability, was the binding constraint

Uttarakhand Flash Floods (2013)	ITBP high-altitude teams deployed ahead of every other agency across the affected Himalayan districts	Over 33,000 pilgrims rescued, establishing border forces as the default first responder in mountain disasters and shaping the Regional Response Centre model
BSF Jurisdiction Extension (2021)	Operational jurisdiction extended from 15 km to 50 km inside Punjab, West Bengal and Assam	Contested by those states as an encroachment on policing powers, with Punjab moving the Supreme Court under Article 131. Shows that central force deployment has a federal dimension that notification alone cannot settle
One Border, One Force (2001)	Each international border assigned to a single designated force on the Godbole Task Force recommendation	Ended overlapping jurisdiction and fixed accountability, and remains the organising principle of Indian border management. The clearest example of a committee recommendation being fully implemented

Conclusion:

India's security architecture rests on a **division of labour** that works well on paper. The Armed Forces face outward, the Central Armed Police Forces hold the borders and the interior, and the state police remain the first responder. The reform agenda is therefore not to raise new forces but to treat the **existing ones as institutions with careers, grievances and limits** rather than as an inexhaustible reserve of manpower, because a force that loses eleven thousand personnel a year to resignation is losing capability faster than any recruitment drive can replace it.

2. AGNIPATH SCHEME

Context:

The Agnipath scheme, launched on June 14, 2022, is currently operational. The initial batch of **Agniveers** has finished their training and joined different units of the armed forces. The scheme has faced **criticism**, especially from veterans, for various reasons.

It also became a major political issue in the recently concluded **Lok Sabha election**, especially in states like **Punjab, Haryana, Himachal Pradesh, Uttar Pradesh, Bihar, and Rajasthan,** where the Armed Forces are a popular choice for employment. The Principal opposition block in the elections campaigned against this recruitment scheme, which seemed to strike a chord with **voters** of the above-mentioned states.

Introduction:

- It is a **short-service manpower model** under which around 45,000 to 50,000 soldiers will be recruited **annually.** Of these, **75%** will leave the service in four years. 25% will be allowed to continue for another 15 years under **permanent commission.** At present, all sailors, airmen, and soldiers, except the technical cadre of the medical branch, are recruited to the services under the scheme, which has also opened doors for the recruitment of **women** to the IAF and the Navy.

Broad objectives of Agnipath scheme:

- To enhance the **youthful profile** of the Armed Forces so that they are at their best fighting abilities, and increased risk taking ability, at all times.
- To bring in young talent from society who can effectively use new technologies and have higher technical skills, using the country's technical institutions.

- To provide an opportunity to the youth who may be keen to **serve the Nation** in uniform albeit for a short period of time.
- To imbibe the Armed Forces **ethos, camaraderie, commitment and teamwork** in the youth.
- To provide abilities and qualities such as **discipline, dynamism, motivation and work-skills** so that the youth remains an asset.

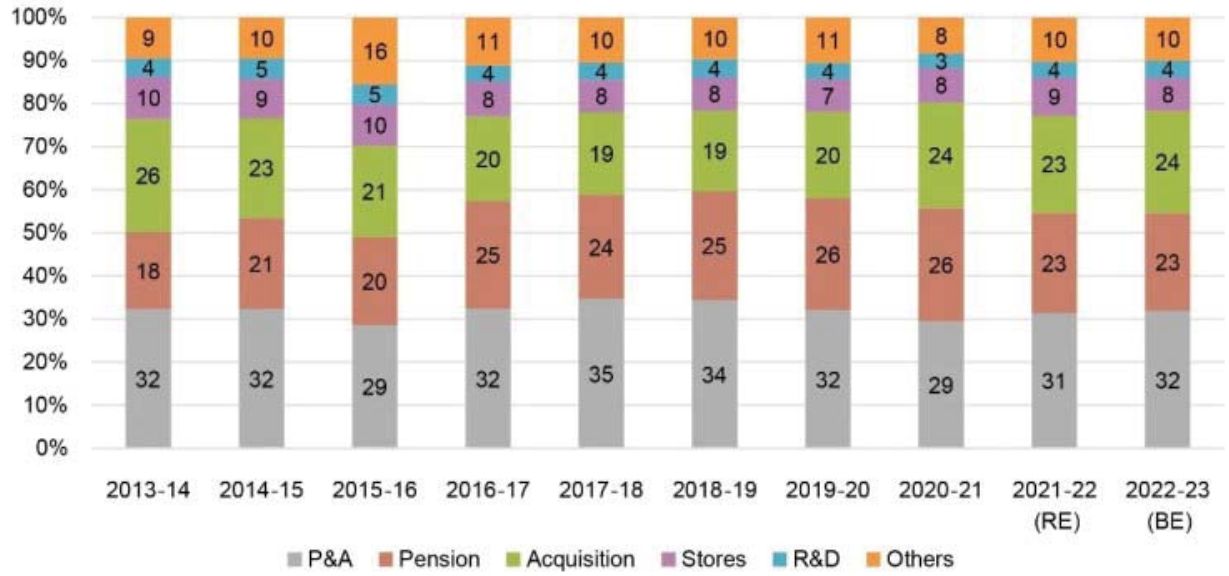
Post Demobilisation Employment Opportunities-

Central / State Government	Incentive
Ministry of Home Affairs	10% reservations in vacancies of the Central Armed Police Forces (CAPF) and Assam Rifles (AR) Relaxation of the upper age limit by three years (five years for the first batch of Agniveers) in recruitments to the CAPF and AR
Ministry of Defence	10% reservation in jobs in the Indian Coast Guard, defence civilian posts, and 16 Defence Public Sector Undertakings (DPSUs)
Ministry of Ports, Shipping and Waterways	Job opportunities for Agniveers in six sectors of the merchant navy
Ministry of Education	Special educational programme to enable Agniveers who finished 10th Class to obtain 12th Class pass certificate
Government of Assam	Assam Police to absorb Agniveers who are permanent residents of the State
Governments of Uttar Pradesh, Haryana and Madhya Pradesh	Priority in recruitment in state police forces

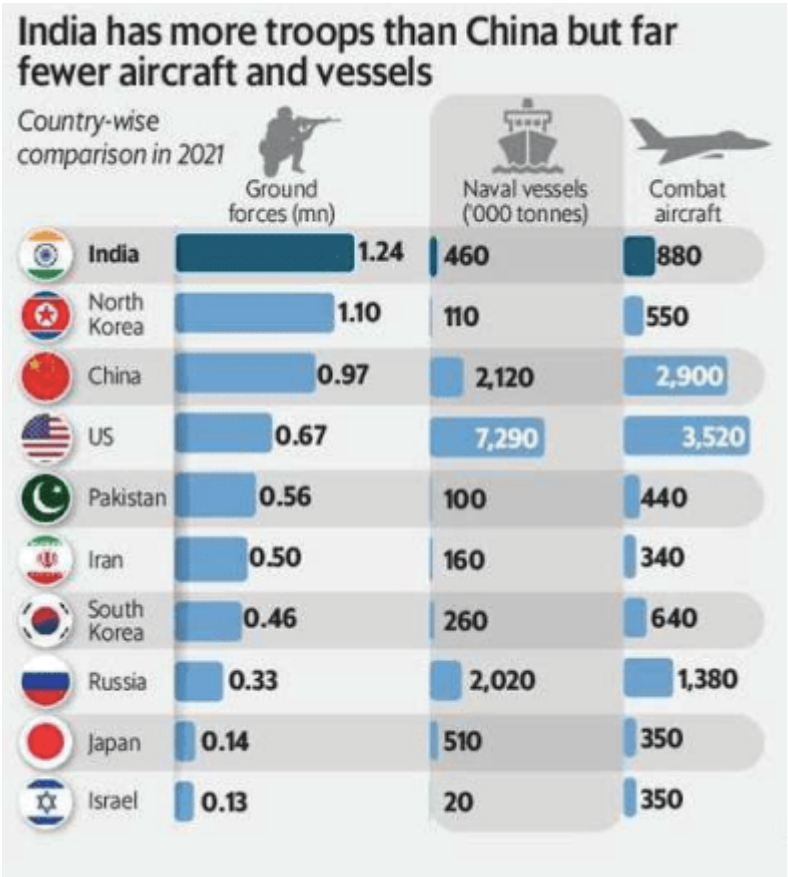
Significance of the Agnipath scheme:

- **Leaner and Younger Force-**It will reduce the **permanent force size** significantly for the country's armed forces, which currently consist of over 13 lakh personnel. Only 25% of recruits will be allowed to continue for another 15 years under permanent commission. This change is expected to lower the **average age in the forces from 32 to 26** within the next 6 to 7 years.
- **Reducing Defense Expenditure:** A leaner military and reduced benefits will significantly lower the **defense budget,** which has been a major concern for governments for many years. Over the past decade, Pay and Allowances (P&A) and Pension expenses have increased from 50% to 55%, while the portion allocated to acquisition and stores expenditure has decreased. **Research and Development**, which is critical for self-reliance and has been aggressively pushed by the government through its '**Make in India**' initiative and **Atmanirbhar Bharat Abhiyan**, hovers at a meagre **3-4 percent** . In 2022-23, P&A and Pension will make up **79% of the Ministry of Defence's total revenue expenditure.**

Broad Distribution of the MoD's Expenditure, 2013/14-2022/23



- **Investment in Modern Warfare-**The saved money can be utilized to buy **state-of-art technology** and **equipment** which are the backbone of modern warfare. With its implementation, the military will become a **future-ready fighting force,** capable of meeting multiple challenges across the full spectrum of conflict



- **All India, All Class Recruitment:** This will make recruitment into the armed forces more **inclusive** by allowing enrolment from all parts of the country and all sections of society. This is important for the Army, which currently follows a **regiment system** based on region and caste.

- **Ensuring rotation of troops-**The scheme aims to rotate soldiers more frequently by having shorter terms. This prevents **stagnation** and brings in **new talent and skills regularly**, ensuring a disciplined rotation of troops.
- **Adhoc Buffer Force:** The ex-agniveers could act as an **adhoc buffer force** who may be called to serve again for boosting national security in times of external/internal threats.
- **Providing skilled workforce:** After serving for 4 years, the 'Agniveers' (the temporary soldiers) would be discharged, and it is expected that they would contribute to the **skilled workforce** in various sectors, given their **disciplined training** and **exposure**
- **Following the Trend Towards Personnel Reduction and Increased Capital Expenditure-**There is a trend towards reduction in the number of personnel and emphasis on increasing capital expenditure on modern weapons and equipment in major countries. **For ex-** The **People's Liberation Army (PLA)** underwent a massive demobilization from the 1980s onwards, bringing down total numbers from 4.5 million to about 2 million, with the focus on.

Concerns associated with the Agnipath scheme:

- **Protests and opposition:** The Agnipath scheme has faced widespread protests and opposition from various sections, **including potential recruits, ex-servicemen, and political parties.** There are concerns about the j**ob security, pension benefits**, and future prospects of the "Agniveers" after their four-year tenure.
- **Increase in training capacities and infrastructure:-**With a constant inflow of new Agniveers every year, there will be a need to significantly enhance **training capacities and infrastructure** to accommodate their basic and advanced training requirements.
- **Augmentation of administrative setup-**The scheme will require an **augmentation** and **strengthening of the administrative machinery** to handle the increased recruitment, periodic release after 4 years, and retention of 25% of the Agniveers for longer service.
- **Impact on Armed Forces Structure and Soldier Motivation–** The change from **Class-Based Recruitment Replaced with All-India All-Class Recruitment** could harm the armed forces because it affects their managerial and operational efficiency. Indian Army soldiers, though trained well, also find motivation in their **social identity,** caring about their reputation among **peers in their caste, village,** or social group.
- **Weaken Unit Cohesion and Combat Effectiveness-**Technology and advanced weapon systems alone **cannot guarantee success** in combat. As seen in the ongoing Russia-Ukraine conflict and the Israel-Hamas conflict, technological superiority can be countered by the adversary's technological measures. Thus, **the human element, conventional combat tactics,** and **interpersonal bonding among soldiers** can not be overlooked to prepare the army for modern warfare.
- **Capability Drawbacks-**Agniveers undergo limited training of **six months.** Veterans and analysts doubt whether units would entrust **Agniveers** with tasks like **handling ammunition, ordnance, weapons,** and **classified stores,** or undertaking repairs on combat vehicles and artillery.
- **Unhealthy Competition Among Agniveers-Training, integrating,** and deploying soldiers with varying experience levels and motivations could pose significant challenges. Selecting the 25% of short-term contracted soldiers to be retained may lead to **unhealthy competition.**If not addressed, this could become a serious issue and harm the **unit's health** in the long run.
- **Recruitment by Inimical Elements-**There are concerns that after their discharge, Agniveers could be vulnerable to recruitment by **inimical elements like Naxalites, insurgents, militants,** and gangsters for nefarious purposes
- **Personnel Shortages and Resource Utilization-**The Army, along with the other two services, is short of about **1,55,000 personnel,** with the Army having the most vacancies at 1,36,000. The Agnipath scheme

aims to fill these positions, which are mostly **non-officer combat roles. Releasing 75% of trained recruits** after four years of service in this context would be a **waste** of resources.

Way forward:

- **Raising Age Limit and Permanent Retention Quota:-**The Indian Army proposed extending the service **period from 4 years to 7-8 years for Agniveers.** It suggested raising the entry age for Agniveers in technical roles to 23 years and increasing the percentage of Agniveers transitioning to regular service after 4 years from **25 percent to 60-70 percent**.
- **Nurturing the Human Element-**The facets of the human element in Agniveers need to be addressed by the leaders responsible for their training and assessment. **Value-based nurturing** based on **unit ethos n**eeds to commence from the outset, irrespective of technological advancements in warfare.
- **Incorporation of Psychological Testing-**There is a need to introduce **psychological test** as part of the recruitment process, similar to the methodology used for officer selection in the military. This will help unit commanders manage **human resources** better and facilitate the **grooming** and **assessment** of Agniveers.
- **Continuous Preparation for Combat and Integration of Agniveers-**In combat, a military unit must perform well. Preparation for facing the enemy happens continuously and doesn't wait for actual combat. Likewise, **unit commanders should** focus on integrating **Agniveers** into the unit and ensuring that they work well as part of the team.
- **Fostering Unit Cohesion through Character Development:-T**o enhance unit pride and cohesion, there should be focus on **individual character development**, as it forms the foundation of **discipline, motivation**, and teamwork, essential for battlefield strength and unit camaraderie.
- **Bridging the Military-Civilian Divide-**The military is highly respected for its role in national security. The beginning of Agniveer recruitment marks a move towards creating a **citizen's army**. Agnipath should help young people learn more about the military, bridging the gap between **military and civilian life.**
- **Enhanced Entitlements & Benefits**
  - They should be covered under **contributory pension and generous gratuity scheme**.Ex-gratia should be provided for disability during training. Moreover, if an Agniveer dies in war, their family should receive a **subsistence allowance.**
  - They should get opportunities in **other security forces,** veteran status, and preference for **government jobs.** Around 25-30% of soldiers should be retained through a transparent, merit-based system, with the option for voluntary extension.
- **Implement robust skilling and resettlement programs:-**Developing robust skilling and resettlement programs in **collaboration** with the **private sector** and other **government agencies** could help ensure a smooth transition for "Agniveers" into civilian life after their tenure.The government may bring a **legislation** that mandates compulsory absorption of agniveers by private employers and corporates.
- **Enhancing Educational Standards –**Educational requirements should be raised to 10+2, and a more rigorous national entrance exam should be implemented.

3. POLICE REFORMS IN INDIA

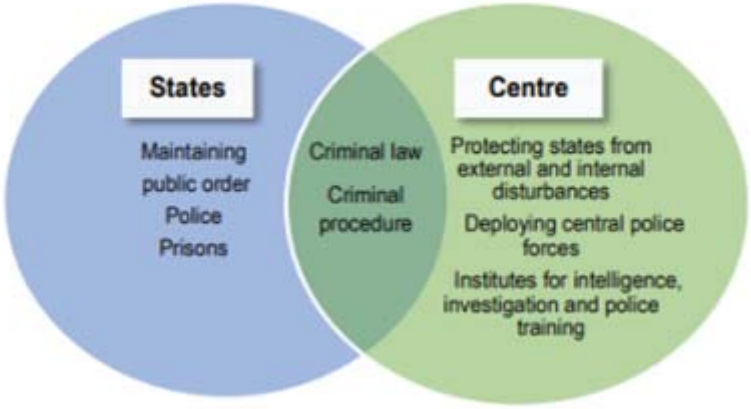
Introduction:

Police reforms refer to **systemic changes intended to professionalize the police force,** enhance its accountability, ensure political neutrality, and improve its overall efficiency in maintaining law and order while protecting human rights. They aim to transform the values, culture, policies and practices of police organizations. It envisages police to perform their duties with respect for democratic values, human rights and the rule of law.

The necessity for reform stems from the outdated framework of the Police Act of 1861 (a colonial-era law) and deeply entrenched structural weaknesses.

Constitutional & Legal Basis of Police System in India:

- Police is an exclusive subject under the State List (List II, Schedule 7 of the Constitution). However, the centre is also allowed to maintain its own police forces to assist the states with ensuring law and order. **For ex-** CRPF, CISF.
- The present Indian police system is largely based on the Police Act of 1861. State legislation on policing is based on this Act. **For Ex-** Bombay Police Act, 1951, Kerala police Act 1960, Delhi Police Act 1978.



Need for Police Reforms:

- **Overburdened Police Force:** Police-public ratio in India stands at 152.80 per lakh person, against the sanctioned public-police ratio of 196.23 per lakh person (as on 1st January 2022). This is way too low when compared with the UN's recommended standard of 222 police per lakh persons. This has led to a state of an overburdened police force and a huge backlog of cases.
- **Constabulary Issues:** The constabulary constitutes ~86% of the state police forces. However, the constabulary suffers from issues such as limited promotion opportunities (only once in his lifetime) and harsh working conditions (Lack of housing facilities). This has weakened their incentive to perform well.
- **Issues with IPS System:** The 'New Delhi conceived and managed' IPS has emerged as 'a permanent irritant' between the centre and states. States look upon the IPS as unreliable intruders and prefer to have their own recruits. **For ex-** Tussle over the promotions of Police officers at important positions.
- **Centre-State Tussle:** Even though police is a state subject, the public prefers CBI investigations over state police interventions as they lack faith in their own police forces, leading to centre-state tussles. **For ex-** Tussle between WB Police and CBI, Withdrawal of general consent to CBI by WB, TN.
- **Lack of Diversity in Police forces:** As per Status of Policing in India Report, 2022, the Indian police lacks diversity on the basis of caste, class, religion and gender. Representation of women in the police force is just 11.7% (Jan 2022).
- **Accountability to the Political Executives vs Operational Freedom:** The Second ARC (2007) has noted that political interference has interfered with professional decision-making by the police and resulted in a biased performance of duties. **For Ex-** Politicisation of DGP appointment.
- **Police Infrastructure Issues:** The Police suffers from issues like:
  - Weaponry shortage: Rajasthan-75% shortage and West Bengal- 71% shortage, according to CAG audit report 2015-16.
  - Surveillance vehicle shortage: 30.5% deficiency according to BPRD.

  - Underutilisation of Funds for modernisation of state police forces: Just 6 % of the Rs 620 cr fund for modernisation of state police forces was used in FY23.
  - Non-functioning of POLNET (Police Communication Network): A satellite-based network to facilitate faster communication between police and paramilitary forces (CAG report highlights the failure of states like Gujarat, Uttar Pradesh in setting up POLNET).
- **Colonial Legacy:** The Indian Police is still governed by the Police Act of 1861, which is a colonial law aimed at creating a 'suppressor police force' rather than an 'enabler police force'.
- **Police misconduct and Corruption:** In 2016, the vigilance department had conducted 55% more inquiries against its own men. India lacks institutional mechanism to complain against the erring police officials.
- **Deteriorating Police-Public Relations-** The Second ARC (2007) has noted that police-public relations is rapidly deteriorating as people view the police as corrupt, inefficient, politically partisan and unresponsive. According to a 2018 survey by CSDS, less than 25% of Indians expressed trust in the police.
- **Changing nature of Crimes-** The nature of crimes is becoming more complex like the emergence of various kinds of organized crimes, economic fraud, deepfakes. Police lack modern technological capabilities necessary to perform quality investigations.
- **Botched up Crime Investigation:** The 22nd Law Commission has pointed out that conviction rate for crimes recorded under the Indian Penal Code, 1860 is less than 50%. It has pointed out that the police lack training in essential skills like forensic laws, cybercrimes and legal knowledge of admissibility of evidence.
- **Custodial Deaths:** As per Asian Centre for Human Rights (ACHR) report, 144 custodial deaths took place from 2017 to 2018. The large number of custodial deaths point towards wide use of torture in police investigations.

Benefits of Police Reforms:

- **Shedding the Colonial Legacy:** The reform of Indian Police Act will help in shedding the colonial legacy like it was done in the case of IPC, CrPC and Evidence Act being replaced by Bhartiya Nyaya, Surkasha and sakshya Samhitas.
- **Help in tackling Modern day crimes:** The reform will help in tackling the modern day crimes like cyberespionage, deepfakes and online financial frauds.
- **Increase in social sensitivity:** Diversified workforce like greater representation of women in police force will increase gender sensitivity towards women in the society. **For ex-** Help in decreasing cases of crimes against women.
- **Reduction in custodial Deaths and Violence:** It will help in reducing inhumane tortures in police custody especially of the downtrodden and the disadvantaged. **For ex-** Reduction in cases like Thoothukudi district custodial death.
- **Bolster Internal security:** Modernised Police force will help in bolstering national security and containing crimes like naxalism and separatism.

Recommendations of Various Committees Regarding Police Reforms in India:

Committees constituted for the Police Reform:

Source: PRS.

Gore Committee 1971-1973	Enlarge the content of police training from law and order and crime prevention to a greater sensitivity and understanding of human behaviour.
National Police Commission 1977	Insulate the police from illegitimate political and bureaucratic interference.
Padmanabhaiah Committee 2000	Standardisation of recruitment procedures for the police force & training. Bring changes in police officers' behaviour, police investigations and prosecution. Change the criterion for constabulary recruitment and improve pay & perk.
Soli Sorabjee Committee 2005	A new model police bill to replace the colonial 1861 Police Act.
Second ARC 2007	Separation of crime investigation from other police functions like maintenance of law and order. Establishment of State police boards for welfare and grievances redressal mechanisms for police personnel.
Malimath Committee 2003	A complete revamp of the entire criminal procedure system. Creation of fund to compensate victims turning hostile under pressure of culprits. Separate national level authority to deal with crimes threatening the country's security.

Seven Directives of The Supreme Court in Prakash Singh vs Union of India:

In the Landmark Prakash Singh vs Union of India, for police reforms in India, SC has given the following observations:

- **State Security Commission (SSC)**: Constitute a State Security Commission (SSC) so that the state government does not exercise unwarranted influence or pressure on the police.
- **Merit based appointment of DGP**: Ensure that the DGP is appointed through the merit-based transparent process and secure a minimum tenure of two years of service.
- **Security of tenure:** Police officers on operational duties (including SP and SHO) are also provided a minimum tenure of two years.
- **Criminal Investigation and law and order separation**: Separate the investigation and law and order functions of the police.
- **Police Establishment Board (PEB)**: Set up a Police Establishment Board (PEB) to decide transfers, postings, promotions and other service related matters of police.
- **Police Complaints Authority (PCA)**: Set up a Police Complaints Authority (PCA) at state level to inquire into public complaints against police officers of above the rank of Deputy Superintendent of Police.
- **National Security Commission (NSC):** Set up a National Security Commission (NSC) at the union level to prepare a panel for selection and placement of Chiefs of the Central Police Organizations (CPO) with a minimum tenure of two years.

NITI Aayog Suggestions on Police Modernisation:

- **State level legislative reforms:** States should be encouraged, with fiscal incentives, to introduce 'The Model Police Act of 2015' as it modernises the mandate of the police.
- **Taskforce under MHA:** To identify non-core functions that can be outsourced to save on manpower and help in reducing the workload of the police. **For Ex-** Address verification for passport applications can be outsourced to private agents or government departments.
- **Greater representation of women:** The states should be encouraged to ensure that the representation of women in the police force is increased. For ex- Bihar's 33% women in police force rule.

- **Place police and public order in concurrent List-** Police as well as public order must be moved to the Concurrent List to tackle increasing inter-state crime and terrorism under a unified framework.

Way Forward:

- **Reform the 'Thana':** The thana system is the most basic unit of police functioning. Through the thana, the police force remains in contact with the common man. Thus, improving the working conditions of thana-level constables should be a priority.
- **Enhance Police Budget and upgrade Police Infrastructure-** More than 90% of the police budget in the states are used for salary and establishment costs. Allocation for training, procurement and technology deployment should increase. States must fully use the funds for police modernisation.
- **Reform the criminal justice system:** Focus must be on the effective implementation of the Malimath Committee recommendations to reform the criminal justice system.
- **Training and Capacity Building of Policemen:** Training the police forces should be a continuous process along with the evolving nature of the crime space. **For ex-** The cyber cell of the Delhi Police to deal with cyber criminals.
- **Upgrade technology and Research in Policing:** Reform the Bureau of Police Research and Development (BPRD) by making it a more inclusive institution. **For ex-** Expertise of social science experts for better research on criminology, NCRB's Automated Facial Recognition System(AFRS) to identify criminals.
- **Improving public perception through community policing:** Community policing initiatives like 'Janamaithri Suraksha Project-Kerala', 'Meira Paibi-Assam' must be undertaken to improve ground level policing.
- **Learning from international best practices:** Japan's separate police commission for recruitment, New York Police's specialisation of police model, must be used to reform our police and make it a global Police.
- **Increase women representation:** 33% women reservation in police should be implemented in all the states. States like Bihar have taken a step in this direction.

Conclusion:

Police reforms are not merely administrative changes; they are essential for strengthening the rule of law and safeguarding democracy. While the 2006 Supreme Court judgment provided a clear legal framework, true transformation requires sustained political commitment from state governments.

However, all reforms must be enunciated in the direction of development of **SMART Police**- Sensitive, Mobile, Alert, Reliable and Techno-savvy.

4. PERMANENT COMMISSION FOR WOMEN IN THE ARMED FORCES

There has been a long history of struggle for ensuring the Permanent Commission for Women in the Armed Forces. There has been much progress in recent times, with women progressively getting more responsibilities in the armed forces. Yet, women officers continue to face multiple challenges and equality in true sense has still not been achieved. Further reforms in the implementation are necessary for overcoming these challenges.

What is the meaning of Permanent Commission?

- A Permanent Commission (PC) means **continuing a career in the armed forces until retirement**.
- Personnel selected in the Armed Forces through Permanent Commission have the option of serving in the Force till the age of retirement.

- For a long time, the option of Permanent Commission was offered only to the male officers. The Permanent Commission for Women was not allowed. **Female Officers were offered only the Short Service Commission (SSC) having a tenure of 10 + 4 years**.
- Under the SSC system, the officers are enrolled for 10 years then they can continue their service for 4 years. After this completion of 10+4 years, the women officers were compulsorily made to opt out, while male officers could continue and take up permanent commission or they can also opt-out. Permanent Commission is granted to SSC officers subject to **service requirement** and **availability of vacancies**.
- This pattern has now changed and has become liberal towards female officers. Now, Women officers also have the option of permanent commission, on par with the male officers. The Supreme Court has played a major role in this colossal change that leads society one step closer to equality.

How has the Permanent Commission for Women in the Armed Forces evolved over time?

- The **Indian Military Nursing Services** originated in 1888 and women were given an opportunity to serve in the Armed Forces. The nurses of the Indian Army first served with distinction in World War I. Formation of **Women's Auxiliary Corps** allowed them to serve in primarily **non-combatant roles** like communications, accounting, administration etc. The opportunities for women officers were little.
- After Independence, the participation of women in the Armed Forces remained limited. The Acts governing the Armed Forces restricted the role of women e.g., **Section 12 of the Army Act, 1950**; **Section 12 of the Air Force Act, 1950**; and **Section 9(2) of the Navy Act, 1957** explicitly **make women ineligible to participate in the respective services** except in such bodies/departments/corps as notified by the Union Government.
- In 1991-92, the Government issued notifications regarding the role of women in the Armed Forces. The roles were limited e.g., women were allowed to serve only in **Logistics**, **Law** and **Education** roles within the **Indian Navy**. Similarly the roles notified in the Indian Army were support roles (rather than core) like the **Army Postal Service**, **Army Education Corps**, Army Ordinance Corps and **Army Service Corps** (Food Scientists and Catering Officers) etc.
- Even in these limited roles, Women were limited only to the Short Service Commission (initially for 5 years only, later extended to 10+4 system). After completion of 14 years, women were asked to leave while eligible male officers were granted Permanent Commission.
- The discriminatory provision led to numerous litigations.
- In 2008, while litigations were sub judice, the Union Government granted a chance of obtaining a PC in the Armed Forces on a restricted basis. There were two major restrictions:
  - The policy was prospective (not retrospective) i.e., women will be offered Permanent Commission only after September 2008.
  - Permanent Commission will be offered only to **specific cadres and branches** (and not universal PC).
- The Judiciary played a proactive role in ensuring permanent commission. The **Indian Air Force was first among the three services to grant PC to SSC women officers** post the Delhi High Court Judgment in ***Jasmine Kaur vs. Union of India WP (C) 8492/2009*** in favour of granting PC to women officers.
- Similarly in the ***Secretary, Ministry of Defence vs Babita Puniya (2020)*** case, the Supreme Court granted equal rights to women with respect to the Permanent Commission. The SC held that all the women officers presently on SSC service are eligible to PCs. All the **choices of specialization shall be available to the women officers** at the time of opting for the grant in PCs, on the same terms as their male counterparts. All the women officers who are eligible and granted PCs through SSC **should be entitled to all consequential perks** including pension, promotion, and financial incentives.
- The SC also noted that although **Article 33** of the Indian Constitution did **allow for restrictions on Fundamental Rights in the Armed Forces**, it is also clearly mentioned that it could be restricted only to

the extent that it was necessary to ensure the proper discharge of duty and maintenance of discipline. Hence, denial of PC to women is violative of their fundamental rights guaranteed under Article 14.

What are the benefits of the Permanent Commission for Women?

- The Permanent Commission is central to **removing structural gender discrimination** in the armed forces.
- It will provide **increased job security** and extended tenure for women officers. It will help **improve the economic opportunities** and improve social conditions and dignity of women. It will also ensure that women officers are not left unemployed after serving many years in the armed forces.
- The Supreme Court Judgment ensures that women officers also get **equal entitlements as men** and **boost their morale** to serve in the services. Traditionally, women officers received no pension as they were contractually mandated to retire within 14 years of service, while defence personnel are entitled to benefits after 20 years in service.
- Equality in opportunity and entitlements will ensure that **more women are attracted to serve** in the Armed Forces. It will **enlarge the talent pool** of officers available for the Armed Forces. As of 2020, women officers in the Indian army (excluding the medical corps) numbered just about 3% compared to 16% in the US, 15% in France and 10% in both Russia and the UK.
- In response to a question in Rajya Sabha in December 2021, the Minister of State for Defence said the Army has a shortage of 7,476 officers. Permanent Commission to women can help bridge the gap.

What are the challenges faced by Women in the Armed Forces?

- **Skewed Gender Ratio**: Low number of women in the Armed Forces (3%) acts as a deterrent for many aspiring women to join the forces.
- **Work-Life Balance**: A career in the armed forces is very challenging and demanding with odd work-locations and work hours. Women have dual responsibility of professional duties as well as **domestic care work**. The responsibility of childcare disproportionately falls on women. It is difficult to balance professional obligations with domestic responsibilities.
- **Limited Roles**: The roles offered to women officers are still limited. Women are still not allowed for combat roles in the Infantry/Mechanised Infantry/Armoured Corps/Artillery or in the Ships/Submarines. (The Indian Navy has, in-principle, accepted the proposal, subject to gender-specific facilities being created).
- **Gender Stereotypes**: The Supreme Court in its Judgement in the Babita Puniya case had criticized the observation made by the Government in the affidavit submitted before the Court. The SC order castigated the Government for portraying women as **physiologically unfit for answering the "call beyond duty" of the Army**. The note had observed that "isolation and hardships would eat into their resolve and that they would have to heed to the call of pregnancy, childbirth and family". The SC had criticized this patriarchal notion that domestic obligations rested only with women.
- **Discrimination**: After the 2020 SC Judgment, the Army established additional conditions for female officers to be eligible for permanent commission. Most of them related to physical fitness, which many officers contended that the new criteria would inevitably keep women from gaining permanent tenures.
- **Biological limitations**: According to studies carried out by the **Centre for Military Readiness in the United States**, female soldiers, on an average, are shorter and smaller than men, with **45-50% less upper body strength** and **25-30% less aerobic capacity**, which is **essential for endurance**. There is an argument that the **physical standards for women must be equivalent to the minimum/satisfactory standards for men** in supporting arms/services. For **fighting arms and Special Forces**, these should be at par with males to withstand the rigours of combat. Thus, on physical standards alone, the employment of women in armed forces gets restricted to specific roles.

Way forward:

- **Equality**: Equal rights and opportunities imply equal physical fitness standards, which must be based on the role and not gender. There are no separate battlefields for women.
- **Accept Gender-specific Rights**: The Forces will have to accept physiological and gender-specific rights like maternity and child care leave.
- **No Preference/Discrimination**: The armed forces are **merit-driven organisations** and the women should compete on equal grounds to rise in ranks in a pyramidical set-up. There is no scope for reservations for promotion.
- **Frame a Pragmatic Policy**: The forces need to **shed their patriarchal attitude** and frame a pragmatic policy for **gradual induction of women in all ranks and disciplines**. **Necessary infrastructure must be created for women soldiers**.
- **Codify Rules and Regulations**: The terms and conditions of service for women must be codified, keeping in view the organisational interests. Military law, rules and regulations to **address gender crimes** and related problems must be laid down. Only volunteer women officers/soldiers should be allowed to join fighting arms, subject to meeting the psychological and physical fitness standards. This is necessary as in the US military, **gender-related offences** are by far the **most common type of discipline problem**.

Conclusion:

Permanent Commission is not just a matter of welfare or right to equal entitlements like pension. It is the idea to "**give women an equal chance at nation-building".** The Supreme Court's Judgment in 2020 was only the first step in a long journey towards ensuring equal opportunity to women in the armed forces. Women aspirants and serving officers should also step up their resolve to meet the exacting physical, psychological and performance standards, and the conditions of service. The Armed Forces should also reform their policies to select the best talent for the role irrespective of gender.

5. INDIGENISATION AND MODERNISATION OF DEFENCE

Context:

India recorded its **highest-ever annual defence production** of ₹1.78 lakh crore in FY 2025-26, a 15.6% rise over the previous fiscal. Indigenous production has grown nearly fourfold from ₹43,746 crore in FY 2013-14. Defence exports also touched an all-time high of ₹38,424 crore in FY 2025-26. This has been backed by a **rise in the defence budget from ₹2.53 lakh crore in 2013-14 to ₹7.85 lakh crore in 2026-27.**

What is Indigenisation and Modernisation of Defence?

- **Indigenisation** focuses on building domestic **industrial and technological capabilities,** enabling India to manufacture critical weapons, platforms and systems, thereby **reducing dependence on foreign suppliers** and ensuring **self-sufficiency** in national security.
- **Modernisation,** on the other hand, involves **continuous upgradation** of military hardware, infrastructure, training and digital capabilities to meet evolving threats**.** Together, they strengthen **combat readiness,** foster a robust **defence manufacturing ecosystem,** and enhance India's strategic autonomy.

Background of Defence Indigenisation in India

India's path towards self-reliance in defence is marked by a considerable defence budget, a robust defence industrial sector, and a legacy of defence manufacturing.

- **Early Recognition of Need (1950s-1970s):** India acknowledged the necessity to lessen reliance on foreign sources and began efforts toward indigenisation, motivated by the **Wars of 1962, 1965, and 1971.**

- **Integrated Guided Missile Development Programme (IGMDP), 1983:** Its aim was to develop **strategic missile systems**. It was completed in **2008 under Abdul Kalam's** guidance, enhancing India's homegrown missile capabilities.
- **Self-Reliance Index and the initiative of Abdul Kalam (1992):** In 1992, Abdul Kalam established a committee to devise a decade-long strategy aimed at achieving an **SRI of 0.7 by 2005**, in light of the low SRI of 0.3 at that time (the ratio of indigenous content in defence procurements to the total expenditure on defence procurements within a financial year).
- **Transfer of Responsibility (2000s):** The responsibility for indigenisation moved from the **Department of Defence Production (DDP) and Directorate General of Quality Assurance (DGQA) to the Ordnance Factories Board (OFB) and Services**. The Army and Navy created 15-year Perspective Plans aimed at Indigenisation.
- **Defence Procurement Procedure (DPP):** DPP has seen ongoing developments and changes **since 2001**.
- **Project 75 (1997):** It concentrated on developing **indigenous submarines**, stressing the importance of **self-sufficiency** in submarine construction to lessen reliance on foreign sources.

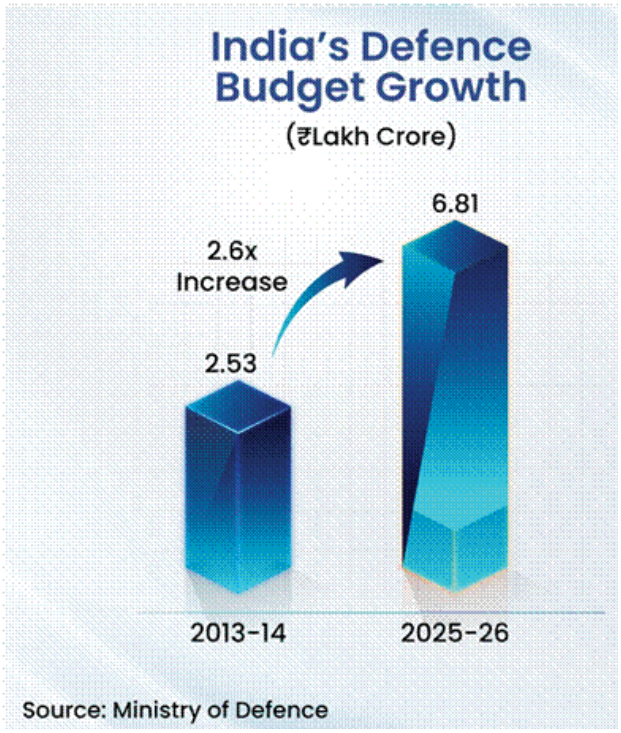
Need & Significance of Defence Indigenisation and Modernisation

- **Strategic Autonomy & National Security**
  - Reduces dependence on foreign suppliers and ensures uninterrupted access to critical platforms even during sanctions or crises.
  - Enhances operational readiness through **modern, upgraded systems** suited to emerging threats.
  - Strengthens India's ability to take **independent strategic decisions** without external pressure.
- **Economic Growth & Industrial Capability**
  - Drives industrial expansion through **local manufacturing, MSMEs and start-ups**.
  - Generates **employment**, boosts **skill development** and stimulates innovation in aerospace, electronics, metallurgy and AI.
  - **Defence Industrial Corridors** have attracted **over ₹8,600 crore** investment, promoting decentralised growth in regions like **UP and Tamil Nadu.**
- **Export Potential & Global Defence Diplomacy**
  - Defence exports have grown manifold, extending India's footprint to **100+ countries**.
  - Indigenous platforms such as **Tejas aircraft, Akash missile systems, artillery guns and interceptor boats** are gaining global interest.
- **Resilience Against Supply Chain Shocks**
  - Indigenisation **reduces vulnerability to global disruptions**, sanctions and geopolitical uncertainties.
  - Positive Indigenisation Lists now cover **over 5,500 items**, with **3,000+ already indigenised**, ensuring consistent availability of spares and components.
- **Technological Innovation & R&D Ecosystem**
  - Platforms like **iDEX** are driving breakthroughs in AI, unmanned systems, advanced missiles, quantum communication and electronic warfare.
  - **Start-ups and academia** are integrated into defence innovation, accelerating future-ready capability development.
- **Inclusive & Regional Industrial Development**
  - Defence corridors and clusters spur investment in **Tier-2 and Tier-3 cities**, enabling balanced development.
  - Generate employment and encourage local entrepreneurship in emerging hubs like **Lucknow, Coimbatore, Hosur and Cochin**.

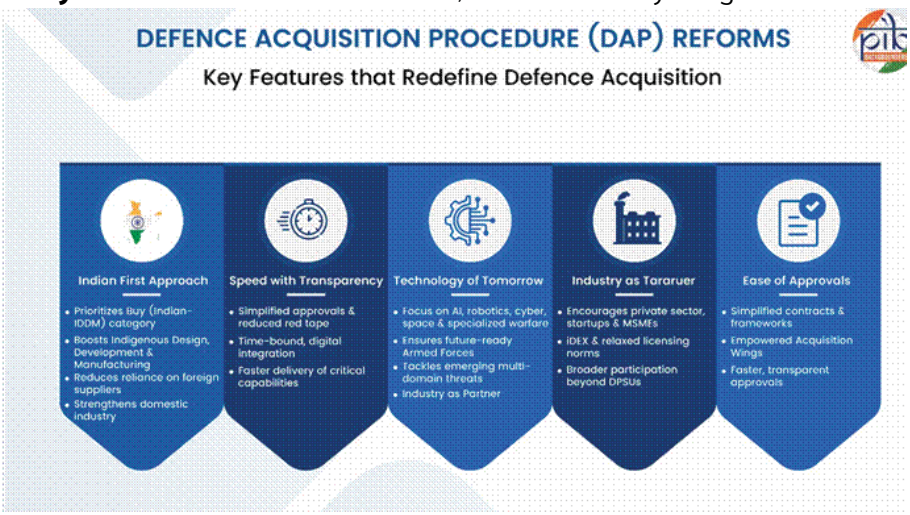
- Strengthen **supply chains** and create a **multiplier effect** across supporting industries.

Steps Taken by Government for Indigenisation & Modernisation of Defence

- **Policy Reforms & Procurement Framework**
  - The **rise in the defence budget, from ₹2.53 lakh crore in 2013-14 to ₹6.81 lakh crore in 2025-26,** underlines the government's determination to strengthen the country's military infrastructure.

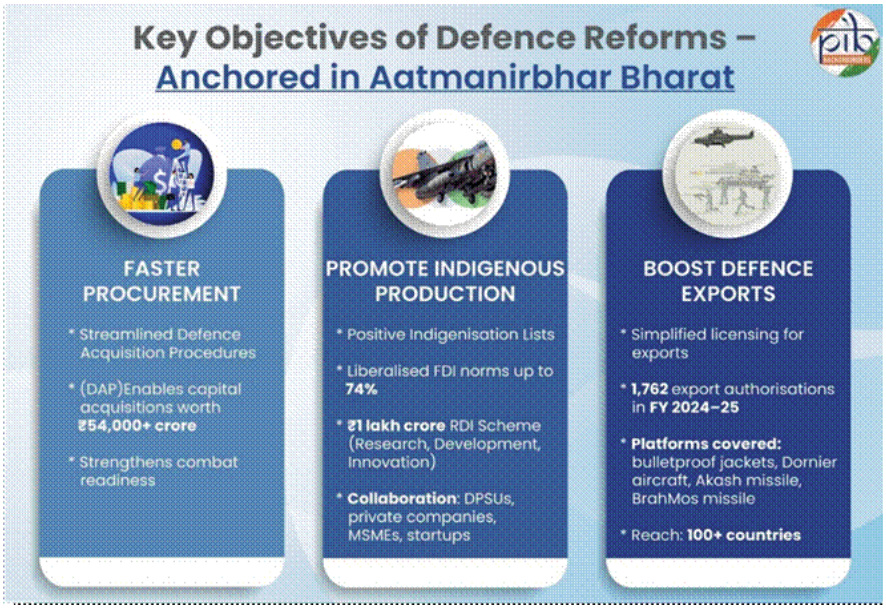


- Declared as the **Year of Reforms in 2025**, the initiatives aim to transform the Armed Forces into a **technologically advanced, combat-ready force** capable of **multi-domain integrated operations**, while expanding **defence production to ₹3 lakh crore** and achieving **export targets of ₹50,000 crore by 2029**.
- **Make in India (Defence)** promotes domestic manufacturing, reducing import dependence and pushing **self-reliance** in platforms, components and systems.
- **Defence Acquisition Procedure (DAP) 2020** prioritises domestic procurement through **Buy Indian (IDDM)** and **Buy Global–Manufacture in India**, with mandatory indigenous content.



- **Defence Procurement Policy 2016** introduced **Buy (Indian-IDDM)**, fast-track procurement via the **Defence Acquisition Council**, and preferential treatment for local industry.
- **Positive Indigenisation Lists (5 lists)** prohibit import of **over 5,500 items**, ensuring procurement only from Indian sources.

- **Industrial Ecosystem & Infrastructure Development**
  - **Defence Production & Export Promotion Policy (DPEPP) 2020** aims for **₹1.75 lakh crore turnover and ₹35,000 crore exports** by 2025, integrating production, technology, and markets.
  - **Defence Industrial Corridors (UP & TN)** create manufacturing hubs with upgraded logistics and incentives; **₹53,439 crore investment potential**, **₹8,600+ crore already realised**.
  - **Indigenous manufacturing reforms** include easier licensing, **74% automatic FDI**, uniform customs duty.
  - **E-Biz Portal** was for the digitization of the application process for Industrial License (IL) and Industrial Entrepreneur Memorandum (IEM) through the E-Biz Portal.
- **Innovation & Technology Development**
  - **iDEX and iDEX Prime** connect start-ups, MSMEs, academia and R&D bodies for defence solutions; **600+ startups engaged**, with grants, procurement support and challenges.
  - **Technology Development Fund (TDF)** supports MSMEs with enhanced funding- **₹50 crore per project** from FY25.
  - **SPRINT initiative (2022)** aims to integrate **75 indigenous technologies** into the Indian Navy through NIIO (Naval Innovation and Indigenisation Organisation,) and DIO (Defence Innovation Organisation).
- **Digital Tools & Indigenisation Platforms**
  - **SRIJAN portal** lists import-substitutable items for domestic industry; **14,000+ items indigenised** till 2025.
  - **Vendor Development Guidelines** empower **SMEs** through outsourcing and partnerships with DPSUs/ OFB.
  - **Acquisition of Advanced Weaponry**
  - Major inductions for combat capability include **Rafale jets**, **S-400 air defence systems**, **Arjun Mk-1A battle tanks**, coastal surveillance radars, and long-range artillery, strengthening air, land and maritime dominance.
- **Indigenous Defence Platforms**



  - **LCA Tejas** replaces **MiG-21 fleet** with multirole capability, mid-air refuelling and AESA radar integration.
  - **INS Arihant & SSBN programme** enhance **nuclear triad** and underwater deterrence.
  - Indigenous systems like **Akash, Pinaka, ATAGS, Prahar, and BrahMos** expand strike capability.
- **Network-Centric & Technological Modernisation**
  - **Network-centric warfare** enabled through **Project Sanjay, E-Sitrep**, joint situational awareness platforms and secure battlefield communication.
  - **AI, Robotics & Autonomous Systems** integrated for ISR and combat roles.
  - Cyber resilience strengthened via **National Cyber Security Policy, CERT-In**, and defence cyber cells.
- **Exports & Defence Diplomacy**
  - Record growth in exports under **Atmanirbhar Bharat** from **₹686 crore (2013-14) to ₹21,083 crore (2023-24)**, supplying **Tejas, BrahMos, naval craft, and radars** to **100+ nations**.
  - **Boosting defence exports** with simplified licensing, covering platforms such as bulletproof jackets, Dornier aircraft, Chetak Helicopters, and fast interceptor boats, and lightweight torpedoes.
- **Border Infrastructure & Rapid Mobilisation**
  - **Border Infrastructure & Management (BIM) Scheme** and **CIBMS(Comprehensive Integrated Border Management System)** improve border roads, bridges, tunnels and surveillance for faster troop mobilisation and effective border control.
  - Strategic connectivity expanded through **BRO projects, Himalayan tunnels (Atal, Sela) and forward airfields**.
- **Theaterisation of Commands**
  - Ongoing reforms seek **jointness and interoperability** among Army, Navy and Air Force through **theaterised commands** for integrated planning, logistics, communication and combat efficiency.
  - Joint theatre commands aim to reduce duplication, increase resource-sharing, and ensure faster response in multi-front scenarios.

Key Roadblocks in Defence Indigenisation & Modernisation

- **Technological Gaps:** Persistent shortfalls in critical technologies such as **engines, semiconductors, AESA radars and propulsion systems** keep India dependent on foreign suppliers, slowing project timelines and limiting full-spectrum self-reliance.
- **Low Private Sector Participation:** Private industry remains a marginal player due to **long gestation projects,** uncertain procurement pipelines and limited R&D support, resulting in DPSUs dominating production and restricting innovation and competition.
- **Bureaucratic & Procurement Delays: Multi-layered approvals, risk-averse decision-making and slow acquisition cycles** continue to delay modernisation, weakening operational readiness, as seen in the long gestation of the LCA programme.
- **Insufficient R&D Investment:** Low expenditure on defence R&D and **weak industry–lab integration** hinder rapid technology development, with limited funding only recently being expanded for programmes under TDF and iDEX.
- **Import Dependence for Strategic Systems:** Despite progress, India still relies on foreign platforms like **fighter jet engines, submarines and air defence systems,** exposing vulnerabilities during geopolitical shocks and curbing Atmanirbhar Bharat ambitions.
- **Testing & Certification Bottlenecks:** Inadequate test ranges, **slow validation cycles and limited certification** infrastructure delay deployment of indigenous systems, as DTIS facilities are yet to cater to the rising demand in UAVs, EW and electro-optics.

Way Forward

- **Strengthen R&D and Innovation Ecosystem:** India must scale up **defence R&D** funding with **collaborative research** among DRDO, private industry, academia and start-ups, supported by Defence Technology Incubation Hubs to drive innovation in AI, hypersonics, stealth, directed-energy weapons and space systems.
  - Ensure **faster commercialisation** of lab technologies through **PPP models** for quicker field deployment.
- **Deepen Private Sector & MSME Integration:**Predictable procurement frameworks with **assured orders, faster payments, single-window clearances,** and MoD support cells for certification will boost private sector and MSME participation.

  Decentralising manufacturing to Tier-2 and Tier-3 cities will widen the supply chain and unlock grassroots innovation.
- **Reform Defence Acquisition for Timely Delivery:** Shortened procurement cycles through **integrated project teams, digital monitoring,** and fixed DAP approval calendars, along with stronger lifecycle costing.
- **Enhance Testing, Trials & Certification Infrastructure:** Modern testing and certification infrastructure for **UAVs, EW, cyber, communications and autonomous systems,** created through **dual-use public–private facilities** in Defence Corridors and supported by fast-track, requirement-based certification, will accelerate validation and deployment cycles.
- **Forge Strategic Partnerships and Co-Development:** India should pursue **co-development and joint production,** focusing on **defence diplomacy** with Quad, I2U2.
- **Institutionalise Monitoring and Accountability: A real-time Defence Indigenisation Dashboard** should track localisation, progress and delays across contracts.
- **Streamline Procurement and Funding:** Simplified, AI-enabled procurement systems can shorten decision timelines and **increase transparency.**
- **Boost Indigenous Manufacturing & Exports: Convergence of Make in India with PLI schemes** will expand the manufacturing base and cut imports.
- **Focus on Emerging Domains:** Dedicated **cyber and space commands** are essential to counter new-age threats and secure strategic assets. For example, modernisation of the Navy.

Conclusion:

India's drive towards **defence indigenisation and modernisation** is central to securing **strategic autonomy, economic strength, and global defence competitiveness**. Sustaining progress requires **stronger R&D**, **faster and transparent acquisition**, and **integrated collaboration with industry and startups**. Building a **self-reliant defence ecosystem** will boost national security and elevate India as a **leading hub for innovation and defence exports**.

7. INTEGRATED THEATRE COMMAND

Army, Navy, and Air Force have divergent views over the structure of theatre commands, which was evident at the tri-service seminar "Ran Samwad 2025" at the Army War College, Mhow.

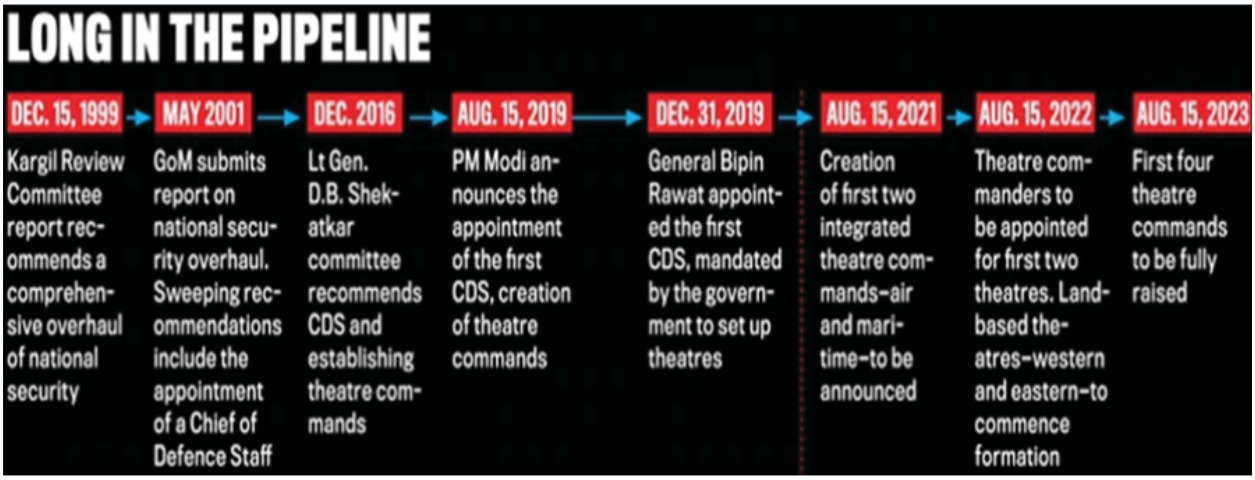
Integrated Theatre Command and its Structure

- **Integrated Theatre Command-** It is a **military structure** which seeks to integrate the capabilities of the three services– **army, air force** and **navy**– in order to optimally utilize their resources for wars and operations.

- **Organisation of Theatre Command-** These '**unified combat commands**' are organized either on a **geographical basis** (with a defined mission in a specific 'area of responsibility' somewhere on the globe) or on a **functional basis**.
- **International operation of Theatre command-** Almost all major countries like **China, Russia, the US, the UK** and **France** work on a theatre command concept. **For example-** China's Western Theatre Command looks after the entire borders with India.

Existing Command Structure in India	• India has **17 single service commands** and only two tri-service commands. • **Single Service Command-** The 17 single service commands are divided as- **Army – 7 commands; Airforce – 7 commands; Navy – 3 commands**. • **Tri-Service Command-**The two tri-service commands are- Andaman and Nicobar Command (ANC), and The Strategic Force Command (SFC). The Andaman and Nicobar command is a full-fledged command raised as a tri-service command in 2001. The Strategic Force Command (SFC) handles nuclear weapons.

Committee recommendations on Integrated Theatre commands in India



Committee Recommendations	**Kargil Review Committee**, **Naresh Chandra Committee** (2012), and **Shekatkar Committee** (2016), recommended creating a tri-service commander role, which will help in the creation of integrated theatre commands.
Appointment of CDS	The appointment of CDS in 2019 and the decision to create the **Department of Military Affairs (DMA)** within the Ministry of Defence have all been steps in the direction of achieving jointness between the three branches of the armed forces.

Recent steps taken by the Government on Integrated Theatre Command

Proposal to create a national defence committee	There is a proposal for theatre commanders under a national defence committee likely to be **headed** by the **Defence Minister**.

Inter-Services Organisation (Command, Control and Discipline) Act 2023	The act empowers the **commander-in-chief** and the **officer-in-command of inter-services organisations** with disciplinary and administrative powers over personnel from the other services serving in them.

Proposed Organizational structure for Integrated Theatre Command

Appointment of a Vice Chief of Defence Staff (CDS)	The Vice CDS is likely to be an officer of the rank of General or equivalent. He is expected to look after **strategic planning, capability development** and **procurement-related matters.**
Appointment of a Deputy CDS	The Deputy CDS is likely to be an **officer of the rank of Lieutenant General or equivalent**. He will be responsible for operations, intelligence and will coordinate the allocation of assets between theatres.
Proposed commands	The overall plan is to establish **three adversary-based theatre commands**– one facing Pakistan, another opposite China, and a maritime theatre command to tackle maritime threats outside the coastal borders of the country.
Role of three service chiefs	The three service chiefs will be responsible for raising, training and sustaining the individual services. No additional manpower is likely to be enrolled. Only readjustments within organisations will be carried out to meet the requirements of the theatre commands.

Benefits of Integrated Theatre Commands:

- **Increased efficiency-** It will lead to **rationalization of war fighting resources**, and **efficiency of executions** resulting in an integrated action to counter threats.
- **Rationalization of the command structure-** The present command structure in the Indian military is pretty uneven. The **Indian Army has seven commands**, while a much **smaller Indian Air Force** has a similar number of commands and the **Indian Navy** has three commands. This structure will be rationalized under the theatre command concept.
- **Effective neutralization of Chinese threat-** China's Western Theater Command looks after the entire borders with India, unlike India where it has multiple Commands and structures to respond with different officers at the top.
- **Cost-effective in the long run-** Theaterization will help in better planning and military response and also bring down costs. **While the cost may go up in the immediate future since all theaters** would have to be armed with sufficient systems, it will prove to be cost-effective in the long term as all acquisitions will be a unified one.
- **Operational Freedom-** The integrated theatre commander **will not be answerable to individual Services**, and will be free to train, equip and exercise his command to make it a cohesive fighting force capable of achieving designated goals.

Challenges in the establishment of Integrated Theatre Command:

- **Inter-Service Competition-** The **intense focus of each service on its own assets** and influence can hinder the creation of synergy among the services.

- **Massive funding-** Creation of infrastructure for the Theatre commands will require **huge amounts of funding.**
- **Different service cultures-** The service culture and way of functioning of the three services is very different. The Indian Army has regimental affiliations and is bound by its legacy. **Finding the right mix of service culture** remains a challenge.
- **Perception of Army Dominance-** There are concerns that the integration of the commands may perpetuate the perceived dominance of the army and provide it with greater operational control.
- **Infrastructure and Logistical coordination issues-** Coordinating and synchronizing the infrastructure and logistics requirements across different services could be complex and resource-intensive.
- **Lack of National Security Strategy (NSS)-** Many retired military professionals have criticised implementing theatre commands without having a coherent National Security Strategy (NSS).

Way Forward:

- **A supportive ecosystem-** In order for Theatre Commands to be effective, there needs to be a **supporting ecosystem**. India still remains the **2nd largest arms importer in the world**. Hence, indigenous technology and hardware by self-reliance should be a priority.
- **Managing internal & external security-** There needs to be an **assessment of how the internal and external security will be managed under the theatre commands.** It is because there are lakhs of paramilitary personnel and the Indian Coast Guard.
- **Parliament's role-** The legislature needs to play a far more pro-active role. A **Dedicated Standing Committee of Parliament should be created**. It needs to be staffed with military advisers and other professionals to independently monitor the transition very minutely.
- **Geography-based theatres-** Keeping in mind the changing nature of threats, India can also **aim for four geography-based theatres—Northern, Southern, Eastern, and Western**—each equipped to use land, sea, air, space, and cyber power to handle all threats in their areas.
- **Talent based appointments- No theatre should 'belong' to any service**, and **command appointments should be decided based on talent** and experience in joint operations.
- **Clear working out of Budgetary allocations- Budgetary allocations** and the **distribution of funds** need to be clearly worked out to enable the setting up of a seamless theatre command system.
- **Modular structure-** In the theaterization debate, **India must not forget to modularize its armed forces**. The present big hierarchical formations need to be divided into many small networked brigades. Multiple studies have shown that these are superior to the division-based structure in terms of deployability, employability, and sustainability.
- **Establishment of Joint Planning & Coordination Centre**- Establish Joint Planning & Coordination Centre in Delhi under the Chiefs of Staff Committee to enhance jointness.

Conclusion:

India operates the 4th largest military in the World, and with each service acting independent of each other, the formation of theatre commands is indeed a need of the hour. But, the successful launch of the Theatre Commands should not be rushed. Issues and concerns of all the stakeholders must be resolved first.

8. INTELLIGENCE AND INVESTIGATIVE AGENCIES

Introduction:

Intelligence agencies collect and assess information to warn of threats before they materialise, while investigative agencies gather evidence to secure convictions after an offence has occurred. The distinction

matters institutionally, because intelligence requires secrecy and speed while investigation requires disclosure and due process, and Indian bodies increasingly straddle both.

Evolution of India's Intelligence Architecture

- **Colonial Origins (1887):** The Central Special Branch was created after the 1857 revolt to monitor political activity, and became the Intelligence Bureau, making it India's oldest surviving intelligence institution.
- **Post-Independence Overload:** The IB handled both internal and external intelligence until the wars of 1962 and 1965 demonstrated that a single agency could not do both.
- **Creation of R&AW (1968):** External intelligence was hived off into a dedicated agency under the Cabinet Secretariat, with **R. N. Kao** as founding chief, following the recommendations that emerged from the 1962 and 1965 reverses.
- **Kargil Review Committee (1999):** Chaired by **K. Subrahmanyam**, it identified the absence of an integrated intelligence assessment mechanism as the central failure and recommended systemic restructuring.
- **Group of Ministers on National Security (2001):** Chaired by L. K. Advani with four task forces, on **intelligence apparatus, internal security, border management and defence management**, it produced the Multi-Agency Centre, the Defence Intelligence Agency and the recommendation that became NTRO.
- **Post-26/11 Expansion (2008-09):** Produced the **National Investigation Agency**, expanded MAC with Subsidiary MACs in state capitals, conceptualised **NATGRID**, and created regional NSG hubs.
- **Doctrinal Consolidation (2026): PRAHAAR** brings prevention, intelligence sharing, response, financial disruption and resilience into a single published framework for the first time.

Why the architecture matters:

- **Prevention over reaction:** Intelligence is the only instrument that can stop an attack rather than punish it, which is why the Kargil and 26/11 failures were framed as intelligence failures first.
- **Federal reach:** Police and public order are State subjects, so central agencies supply the capability that individual states cannot build, particularly in terror finance, cyber and transnational crime.
- **Evidentiary strength:** Investigative agencies convert intelligence into prosecutable cases, without which enforcement collapses at the trial stage.
- **Convergence of threats:** Terrorism, narcotics, counterfeit currency and organised crime now share the same routes and financiers, so no single agency can see the whole picture alone.

Intelligence Agencies

Agency	Year and Parent	Mandate
Intelligence Bureau (IB)	1887 lineage, Ministry of Home Affairs	India's oldest intelligence organisation, responsible for internal intelligence, counter-intelligence and counter-terrorism. Hosts MAC, SMAC and the Joint Task Force on Intelligence
Research and Analysis Wing (R&AW)	1968, Cabinet Secretariat	External intelligence, created after the intelligence failures of 1962 and 1965, with R. N. Kao as founding chief. Reports to the Prime Minister through the NSA
National Technical Research Organisation (NTRO)	2004, under the NSA	Technical intelligence covering satellite imagery, signals, cyber monitoring and critical infrastructure protection. A Kargil Review Committee outcome

Defence Intelligence Agency (DIA)	2002, Ministry of Defence	Coordinates the intelligence directorates of the three services and manages defence attachés and imagery processing. A Group of Ministers 2001 outcome
Service Directorates	Ministry of Defence	Directorate of Military Intelligence, Directorate of Naval Intelligence and Directorate of Air Intelligence, handling service-specific tactical intelligence

Investigative Agencies

Agency	Statute and Parent	Mandate
Central Bureau of Investigation (CBI)	DSPE Act, 1946, under DoPT	Anti-corruption, economic offences and special crimes. Requires state consent under Section 6, which is the source of its present difficulties
National Investigation Agency (NIA)	NIA Act, 2008, Ministry of Home Affairs	India's federal counter-terror investigation agency. The 2019 amendment allows suo motu takeover of scheduled offences and investigation of offences committed abroad affecting Indian interests
Enforcement Directorate (ED)	PMLA, FEMA and FEOA, Department of Revenue	Money laundering, foreign exchange violations and attachment of proceeds of crime
Narcotics Control Bureau (NCB)	NDPS Act, 1985, Ministry of Home Affairs	Nodal agency for drug law enforcement and international narcotics cooperation
Directorate of Revenue Intelligence (DRI)	Under CBIC	Smuggling of gold, narcotics, arms and wildlife, and commercial fraud at the border
Serious Fraud Investigation Office (SFIO)	Companies Act, 2013, Ministry of Corporate Affairs	Complex corporate fraud with multi-disciplinary investigation teams

Coordination and Technical Platforms

Platform	Parent	Function
Multi-Agency Centre (MAC)	Intelligence Bureau	Daily platform for sharing terrorism-related inputs across agencies. Created on Kargil Review Committee recommendation, expanded after 26/11 with **Subsidiary MACs** in state capitals
NATGRID	Ministry of Home Affairs	Secure query platform allowing authorised agencies to search multiple databases simultaneously across immigration, banking, telecom and tax records
Financial Intelligence Unit (FIU-IND)	Department of Revenue	Receives and analyses suspicious transaction reports under PMLA and shares them with enforcement agencies
Indian Cyber Crime Coordination Centre (I4C)	Ministry of Home Affairs	Coordinates cybercrime response, hosts the National Cybercrime Reporting Portal and the citizen financial fraud helpline

CCTNS and ICJS	NCRB, Ministry of Home Affairs	Digitised police records linked across police, courts, prisons, prosecution and forensics
NCIIPC and CERT-In	NTRO and MeitY respectively	Protection of critical information infrastructure, and national incident response for cyber security

Issues:

- **Absence of Legal Foundation**
  - **No Statutory Charter:** Neither the IB nor R&AW is created or governed by an Act of Parliament, so their powers, limits and accountability rest on executive orders alone.
  - **No Parliamentary Oversight:** India has no dedicated committee with security clearance to review intelligence performance or budgets, unlike most comparable democracies.
  - **Surveillance Without a Framework:** NATGRID aggregates highly sensitive personal data without dedicated legislation governing access, purpose limitation, retention or grievance redress.
- **Federalism and Autonomy**
  - **Consent Withdrawal Has Hollowed the CBI:** Ten states including West Bengal, Punjab, Kerala, Tamil Nadu, Telangana, Rajasthan, Jharkhand, Chhattisgarh, Meghalaya and Mizoram have withdrawn general consent under Section 6 of the DSPE Act.

    A Parliamentary Standing Committee has urged a new law granting the CBI wider powers, while the government has declined to amend Section 6.
  - **Perception of Political Use:** The Supreme Court's description of the CBI as a caged parrot in the coal blocks matter continues to shape public and judicial scepticism about its independence.
  - **The NCTC Precedent:** The National Counter Terrorism Centre proposed in 2011 and 2012 was shelved after states objected that powers of arrest and search in a body carved out of the IB infringed the constitutional distribution of policing.
- **Coordination Failures**
  - **Sharing Rather Than Fusion:** MAC circulates inputs but does not produce a single national assessment, so responsibility for connecting fragments remains diffuse across agencies.
  - **Turf and Trust Deficits:** Agencies reporting to different ministries have incentives to withhold rather than share, since credit for a lead accrues to whoever acts on it.
  - **Uneven State Participation:** Subsidiary MAC engagement, CCTNS completion and forensic capacity vary sharply across states, so the national picture is only as good as its weakest contributor.
- **Capacity Constraints**
  - **Human Intelligence Deficit:** Technical collection has expanded far faster than the cultivation of human sources, which remains the only reliable way to learn intent rather than activity.
  - **Deputation-Based Staffing:** Heavy reliance on officers on deputation from the police and other services limits the development of specialist career intelligence and forensic cadres.
  - **Prosecution Bottleneck:** Investigative agencies file cases faster than the courts and prosecution can process them, so deterrence weakens even where investigation is competent.
- **Reforms and Judicial Directions:**
  - **Kargil Review Committee, 1999 and Group of Ministers, 2001:** Produced MAC, the Subsidiary MACs, the Defence Intelligence Agency and the recommendation for a technical intelligence body that became NTRO.
  - **Vineet Narain v. Union of India (1997):** Directed statutory status for the Central Vigilance Commission and insulation of the CBI Director through a fixed tenure and a selection process.

- **Subramanian Swamy v. Director, CBI (2014):** Struck down Section 6A of the DSPE Act, which had required prior central approval to investigate senior officers, as violative of Article 14.
- **State of West Bengal v. Union of India (2024):** A Constitution Bench held the state's suit against the CBI maintainable, affirming that consent under the DSPE Act carries real constitutional weight.
- **NIA Amendment, 2019:** Extended jurisdiction to offences committed abroad against Indians or Indian interests, and to human trafficking, cyber terrorism and arms offences.
- **PRAHAAR, 2026:** Names MAC and the Joint Task Force on Intelligence as central intelligence platforms and creates the annual NIA-organised **Anti-Terror Conference** as a review mechanism.

Way Forward:

- **Legal and Accountability Reform**
  - **Enact a Statutory Charter:** Place the IB and R&AW on a legislative footing defining mandate, powers and limits, which protects the agencies as much as it constrains them.

    The Intelligence Services (Powers and Regulation) Bill introduced as a private member's bill in 2011 remains the template most often cited.
  - **Create Parliamentary Oversight:** Establish a standing committee with security clearance to review budgets and performance in camera, on the model followed in the United Kingdom and the United States.
  - **Legislate for NATGRID:** Provide a statutory basis for access, retention and redress, which is also the surest way to make the platform survive constitutional challenge after *Puttaswamy*.
- **Institutional Reform**
  - **Give the CBI Its Own Law:** Replace the 1946 DSPE Act with dedicated legislation clarifying jurisdiction, appointment, tenure and the consent requirement, as the Parliamentary Standing Committee has recommended.
  - **Move from Sharing to Fusion:** Empower MAC and the Joint Task Force on Intelligence to issue single national assessments rather than circulate agency inputs.
  - **Build Specialist Cadres:** Create direct-entry career streams in intelligence analysis, cyber forensics and financial investigation instead of relying wholly on deputation.
  - **Strengthen Prosecution:** Develop a dedicated cadre of federal prosecutors trained in anti-terror law and cyber forensics, which PRAHAAR itself identifies as a gap.
- **Working With Federalism**
  - **Build Consent, Not Bypass It:** Central capacity is more durable when routed through structures states have accepted, namely the NIA, NSG hubs, MAC and PRAHAAR's mechanisms, than through bodies states have resisted.
  - **Invest in State Capacity:** Tie central assistance to measurable improvement in SMAC participation, CCTNS completion and forensic laboratory capability, since the state police remain the first responder.

9. NATIONAL SECURITY ARCHITECTURE IN INDIA

Introduction:

National security architecture is the **ensemble of institutions, doctrines, laws and processes** through which a state identifies threats, takes decisions and delivers responses. India's version has been built almost entirely reactively, with each major layer added after a crisis exposed a gap rather than in anticipation of one.

The three crises that built the architecture:

- **Kargil, 1999:** The **Kargil Review Committee** under K. Subrahmanyam, followed by the **Group of Ministers on National Security (2001)** with its four task forces on intelligence, internal security, border management and defence management, produced most of the present structure.

- **Mumbai, 2008:** Produced the **National Investigation Agency**, the expansion of the **Multi-Agency Centre**, the conceptualisation of **NATGRID**, regional NSG hubs and the coastal security framework.
- **Recent conflict: Operation Sindoor** of May 2025 has given fresh urgency to theatreisation and multi-domain integration, with 2025 declared the **Year of Reforms** in the Ministry of Defence.

The persistent gap: India still has **no publicly articulated National Security Strategy**, so the architecture operates without a stated hierarchy of threats or objectives against which agencies can be held accountable.

Significance of a Coherent Architecture:

- **Unity of Command and Decision**
  - **Single Point of Advice:** A unified structure removes the confusion of multiple channels reaching the political leadership with competing assessments during a crisis.

    The Chief of Defence Staff, created in 2020, serves as single point military adviser to the Defence Minister and Prime Minister on tri-service matters.
  - **Speed of Response:** Modern conflicts compress decision timelines from weeks to hours, which siloed structures cannot absorb.

    Operation Sindoor was a four-day engagement, a timeframe in which inter-service coordination must already exist rather than be arranged.
- **Efficient Use of Resources**
  - **Ending Duplication:** Integrated planning prevents three services from procuring separately for the same operational requirement.

    India currently operates 17 single-service commands, namely seven each of the Army and Air Force and three of the Navy, each with its own logistics and support tail.
  - **Prioritised Capability Building:** A stated strategy allows scarce capital budget to be allocated against agreed threats rather than distributed by service seniority.
- **Whole-of-Government Response**
  - **Threats Cross Ministries:** Terrorism, cyber intrusion, narcotics and radicalisation cut across Home, Defence, External Affairs, Finance and IT, so no single ministry can respond alone.

    PRAHAAR, launched in February 2026, is explicitly framed as a whole-of-government and whole-of-society strategy for precisely this reason
  - **Federal Coordination:** Since police and public order are State subjects, the architecture must deliver central capability without displacing state authority.
- **Deterrence and Credibility**
  - **Signalling Capability:** A visibly integrated structure raises the cost of adventurism for an adversary, independent of whether force is actually used.
  - **Alliance Value:** Partner states engage more readily with a counterpart whose decision structures they can identify and work with.

Institutional Architecture:

- **Apex Political and Strategic Tier**
  - **Cabinet Committee on Security:** Chaired by the Prime Minister, it is the highest decision-making body on national security, defence acquisitions and senior appointments.
  - **National Security Council:** Established in 1998, it is the apex advisory body, serviced by the National Security Adviser and operating through three supporting tiers.
  - **Strategic Policy Group:** Chaired by the Cabinet Secretary, it is the principal mechanism for inter-ministerial coordination and undertakes the Strategic Defence Review.

- **National Security Advisory Board:** Non-official experts drawn from academia, media, the services and the civil services, who provide long-term assessments outside the bureaucratic line.
- **National Security Council Secretariat:** The permanent staff arm under the NSA, which prepares assessments and now also houses the National Maritime Security Coordinator.
- **Defence Planning Committee:** Constituted in 2018 under the NSA to prepare a national security strategy, capability development plans and defence manufacturing roadmaps.

- **Military Tier**
  - **Chief of Defence Staff:** Created in January 2020, functioning simultaneously as Permanent Chairman of the Chiefs of Staff Committee and Secretary of the Department of Military Affairs.
    - **General Bipin Rawat** was the first CDS, **General Anil Chauhan** the second, and **General N. S. Raja Subramani** took charge as the third CDS on **31 May 2026**.
  - **Department of Military Affairs:** Created as the fifth department of the Ministry of Defence, handling tri-service procurement, joint training, human resources and veterans' welfare, and placing serving officers inside the ministry for the first time.
  - **Headquarters Integrated Defence Staff:** Headed by the Chief of Integrated Staff to the Chairman COSC, it coordinates joint planning and will coordinate operational issues between theatres and service headquarters during and after transition.
  - **Existing Joint Structures:** The **Andaman and Nicobar Command**, India's only tri-service theatre command, and the **Strategic Forces Command**, which manages nuclear delivery assets.
  - **Tri-Service Agencies (2019):** The **Defence Cyber Agency**, **Defence Space Agency** and **Armed Forces Special Operations Division**, created as precursors to full commands in the emerging domains.
- **Intelligence Tier**
  - **Intelligence Bureau:** The internal intelligence agency under the Ministry of Home Affairs, which also hosts the principal coordination platforms.
  - **Research and Analysis Wing:** The external intelligence agency, functioning under the Cabinet Secretariat.
  - **Defence Intelligence Agency and NTRO:** Military intelligence coordination and technical or signals intelligence respectively.
  - **Multi-Agency Centre:** Created on Kargil Review Committee recommendation and expanded after 26/11, it is the daily platform for sharing terrorism-related intelligence, with **Subsidiary Multi-Agency Centres** in state capitals pushing inputs to district police.
  - **Joint Task Force on Intelligence:** Housed in the IB and, alongside MAC, designated under PRAHAAR as a central platform for real-time intelligence sharing.
  - **NATGRID:** Conceptualised after 26/11 and operational since the early 2020s, it links about **21 data providers**across immigration, banking, telecom and tax records, handles roughly **45,000 queries a month**, and is developing an **Organised Crime Network Database**.
- **Internal Security and Legal Tier**
  - **Ministry of Home Affairs:** Nodal ministry for internal security, controlling the Central Armed Police Forces and the counter-terror legal framework.
  - **National Investigation Agency:** Created by the NIA Act, 2008, empowered by the 2019 amendment to take over scheduled offences suo motu and to investigate offences committed abroad against Indians or Indian interests.
  - **National Security Guard:** The federal contingency force for counter-terrorism and hostage rescue, with regional hubs created after 26/11 to cut response time.
  - **CCTNS and ICJS:** Police record digitisation under the NCRB, linked into an inter-operable criminal justice system connecting police, courts, prisons, prosecution and forensics.

- **Legal Framework:** The Unlawful Activities (Prevention) Act as the principal counter-terror statute, now operating alongside the Bharatiya Nyaya Sanhita.

Theatreisation: The Reform in Progress

- **Concept:** Replace service-wise commands with unified operational entities in which Army, Navy and Air Force assets serve under a single commander for a geography or function.
- **Proposed Structure:** Three theatres are the emerging consensus, namely a **Northern Theatre** oriented to China and the LAC, a **Western Theatre** oriented to Pakistan, and a **Maritime Theatre** covering the Indian Ocean Region and coastal defence.
- **Current Status:** General Anil Chauhan submitted the final report before demitting office in May 2026, and the roadmap now moves from the Defence Minister to the Cabinet Committee on Security for approval.
- **Process**: A dedicated tri-service team under the Chief of Integrated Staff is studying comparable global structures, with outstanding issues resolved in the Chiefs of Staff Committee.

Doctrine: PRAHAAR, 2026

India's **first comprehensive National Counter-Terrorism Policy and Strategy**, unveiled by the Ministry of Home Affairs on **23 February 2026**. The acronym sets out its seven pillars.

- **P for Prevention** of terror attacks to protect Indian citizens and interests.
- **R for Response** that is swift and proportionate to the threat posed.
- **A for Aggregating internal capacities** to achieve synergy through a whole-of-government approach.
- **H for Human rights** and rule of law based processes in mitigating threats.
- **A for Attenuating** the conditions that enable terrorism, including radicalisation.
- **A for Aligning** and shaping international efforts to counter terrorism.
- **R for Recovery and resilience** through a whole-of-society approach.

Its oversight runs partly through the annual **Anti-Terror Conferenc**e organised by the NIA, and it names MAC and the Joint Task Force on Intelligence as the central intelligence-sharing platforms.

Issues:

- **Absence of a Stated Strategy**
  - **No Published Security Strategy:** India has never issued a National Security Strategy document, so agencies operate without an authoritative statement of threat priorities or capability objectives.

    The Defence Planning Committee was constituted in 2018 partly to draft one, and its output has not been placed in the public domain.
  - **Reactive Institution Building:** Nearly every major body was created after a failure rather than in anticipation, which produces architecture shaped by the last crisis rather than the next.
  - **Budgeting Without Priorities:** Without a stated strategy, capital allocation tends to follow service claims and legacy commitments rather than an agreed threat hierarchy.
- **Slow Military Integration**
  - **Theatreisation Still Unapproved:** More than six years after the CDS post was created, the theatre command structure remains at the proposal stage awaiting Cabinet Committee on Security clearance.
  - **Inter-Service Divergence:** The Air Force in particular has raised concerns about dividing scarce air assets across theatres, given that aircraft can be switched between fronts within hours.
  - **Risk of Cosmetic Reform:** Critics warn that renaming headquarters without genuinely transferring command authority and resources would exchange one bureaucracy for another.

  - **Civil-Military Gap:** Despite the Department of Military Affairs, the interface between the uniformed leadership and the civilian bureaucracy remains a recurring source of delay in procurement and planning.
- **Intelligence Deficits**
  - **No Statutory Charter:** Neither the Intelligence Bureau nor the Research and Analysis Wing is governed by a parliamentary statute defining its powers, limits or accountability.
  - **No Parliamentary Oversight:** Unlike most democracies, India has no dedicated legislative committee overseeing intelligence agencies, which weakens both accountability and institutional protection.
  - **Coordination Over Fusion:** MAC is a forum for sharing inputs rather than a body producing single fused assessments, so responsibility for connecting dots remains diffuse.
  - **Privacy Without a Framework:** NATGRID aggregates highly sensitive financial, travel and telecom data without a dedicated legislative framework governing access, retention and redress.
- **Federalism Frictions**
  - **The NCTC Failure:** The National Counter Terrorism Centre proposed in 2011 and 2012 was shelved after states objected that powers of arrest and search vested in a body carved out of the IB infringed the constitutional distribution of policing.
  - **Police Is a State Subject:** The first responder to almost every internal security incident is the state police, which the central architecture can support but not direct.
  - **Uneven State Capacity:** Subsidiary MAC participation, CCTNS completion and modernisation vary widely across states, so the weakest link determines national effectiveness.
- **Capacity and Modernisation Gaps**
  - **Police Reforms Unimplemented:** The directions in **Prakash Singh v. Union of India (2006)** on security of tenure, separation of investigation from law and order and police establishment boards remain largely unimplemented.
  - **CAPF Strain:** Central Armed Police Forces face vacancies, prolonged deployment away from home, and stress-related attrition, while being used for tasks ranging from counter-insurgency to election duty.
  - **Emerging Domains Under-Institutionalised:** Cyber, space, artificial intelligence and drones are handled by agencies rather than full commands, and the national cyber security policy dates from 2013.
  - **Import Dependence:** Despite indigenisation lists and defence corridors, dependence on foreign platforms in critical segments constrains strategic autonomy during a crisis.
- **Government Measures:**
  - **Chief of Defence Staff and Department of Military Affairs:** Created in January 2020 to provide single point military advice and place serving officers within the Ministry of Defence.
  - **Theatreisation Process:** Final report submitted in May 2026 and now moving through the Defence Minister to the Cabinet Committee on Security, with 2025 designated the Year of Reforms.
  - **Tri-Service Agencies:** Defence Cyber Agency, Defence Space Agency and Armed Forces Special Operations Division established in 2019 for the emerging and special operations domains.
  - **PRAHAAR, 2026:** India's first comprehensive National Counter-Terrorism Policy and Strategy, providing a seven-pillar whole-of-government and whole-of-society framework.
  - **National Investigation Agency:** Strengthened by the 2019 amendment granting suo motu takeover and extraterritorial jurisdiction, with a high conviction rate in completed cases.
  - **NATGRID:** Integrated query platform across about 21 data providers with roughly 45,000 queries a

month, now extending access to state police and building an Organised Crime Network Database.

- **Multi-Agency Centre and Subsidiary MACs:** Daily intelligence-sharing platform under the IB extending to state capitals and district police.
- **Defence Planning Committee:** Constituted in 2018 under the NSA for strategy, capability development and defence manufacturing planning.
- **Defence Indigenisation:** Positive indigenisation lists, the Defence Acquisition Procedure 2020, iDEX for startups, and defence industrial corridors in Uttar Pradesh and Tamil Nadu.
- **CCTNS and Inter-operable Criminal Justice System:** Digitisation and linking of police, court, prison, prosecution and forensic records.

Case Studies

Case	What happened	Result and Lesson
Kargil, 1999	Intelligence and coordination failures exposed by intrusion across the Line of Control	Produced the Kargil Review Committee and the Group of Ministers report of 2001 with four task forces, which built most of today's architecture including MAC and the Defence Intelligence Agency. Reform in India follows crisis rather than anticipation
26/11 Mumbai, 2008	Coordinated attack revealing gaps in coastal security, intelligence fusion and federal response capacity	Produced the NIA, expanded MAC with state-level Subsidiary MACs, conceptualised NATGRID and created regional NSG hubs. Shows that institutional creation is the easy part and inter-agency fusion the hard part
NCTC Proposal, 2011-12	Proposal for an operational counter-terror centre carved out of the IB with powers of arrest and search	Shelved after sustained objection from states that it infringed the constitutional distribution of policing powers. The definitive case study on the limits of centralisation in Indian internal security
Creation of the CDS, 2019-20	Post created in January 2020 alongside the Department of Military Affairs, with General Bipin Rawat as first incumbent	Ended two decades of resistance to a single point military adviser and put serving officers inside the ministry. Demonstrates that structural reform is possible when political will is present
Operation Sindoor, May 2025	A short, intense engagement with Pakistan conducted across domains within a four-day window	Gave decisive urgency to theatreisation and multi-domain integration, and preceded the designation of 2025 as the Year of Reforms. Compressed conflict timelines make pre-existing jointness a precondition rather than an aspiration

PRAHAAR, 2026	India's first comprehensive National Counter-Terrorism Policy and Strategy, released in February 2026	Provides a seven-pillar framework combining prevention, proportionate response, capacity aggregation, human rights compliance, de-radicalisation, international alignment and societal resilience. Shows that doctrine can be published, which strengthens the case for a National Security Strategy

Way Forward:

- **Publish a Strategy**
  - **Issue a National Security Strategy:** A public document stating threat priorities, capability objectives and the division of institutional responsibility would give the architecture a purpose against which performance can be judged.
  - **Periodic Strategic Defence Review:** Institutionalise the review already assigned to the Strategic Policy Group, with a fixed cycle and reporting to Parliament in summary form.
- **Complete Military Integration**
  - **Approve and Sequence Theatreisation:** Move the pending proposal through the Cabinet Committee on Security with a phased rollout rather than a single transition.
  - **Resolve Command Authority Clearly:** Specify who controls scarce air and naval assets across theatres, since ambiguity here is the substantive objection rather than institutional turf.
  - **Build Joint Culture First:** Expand joint training, cross-posting and a common staff college stream, since structures without a joint officer cadre revert to service loyalties in practice.
- **Reform Intelligence**
  - **Enact a Statutory Charter:** Place the intelligence agencies on a legislative footing defining mandate, powers and limits, which protects the agencies as much as it constrains them.
  - **Create Parliamentary Oversight:** Establish a standing committee with security clearance to review intelligence performance and budgets, as most comparable democracies do.
  - **Move from Sharing to Fusion:** Give MAC and the Joint Task Force on Intelligence the capacity to produce single national assessments rather than only circulate inputs.
  - **Legislate for NATGRID:** Provide a statutory basis governing access, purpose limitation, retention and grievance redress to make the platform sustainable against constitutional challenge.
- **Work With Federalism, Not Against It**
  - **Revive Counter-Terror Capacity Cooperatively:** Rather than reviving NCTC in its original form, build central capability through structures states have accepted, namely NIA, NSG hubs, MAC and PRAHAAR's coordination mechanisms.
  - **Implement Police Reforms:** Give effect to the Prakash Singh directions, since the quality of the state police determines the effectiveness of everything built above it.
  - **Fund State Capacity:** Tie central assistance to measurable improvements in SMAC participation, CCTNS completion and forensic capability.
- **Build for Emerging Threats**
  - **Upgrade Agencies to Commands:** Convert the Defence Cyber Agency and Defence Space Agency into full commands with dedicated cadres and budgets.
  - **Refresh Cyber Doctrine:** Replace the 2013 policy with a current national cyber security strategy covering critical infrastructure, offensive capability and incident response.

- **Counter-Drone and Counter-Technology Capacity:** Institutionalise counter-drone capability, given documented use of drones for weapons and narcotics delivery in Punjab and Jammu and Kashmir.
- **Deepen Indigenisation:** Sustain iDEX, indigenisation lists and defence corridors so that capability is not hostage to external supply during a crisis.

Conclusion:

India's national security architecture has been **assembled crisis by crisis.** The military tier is now closer to genuine integration than at any point since Independence, with the **theatre command proposal awaiting only political clearance,** while the intelligence tier still operates without a statutory charter and the internal security tier still rests on a state police force. The unfinished task is therefore not to add more bodies but to **give the existing ones a stated purpose, a legal basis and an accountable chain of command,** so that the next reform is driven by foresight rather than by the next inquiry commission.

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